

Original Mixed Cohomology Periods, Kernels and Arithmetic Control

Split-Zero research programme

15 September 2026

Full receiving calculation. This local successor preserves the complete accepted 811-page source and adds the full original tail through conductor transport, the proper-source value and the complete action. The new calculation is in Section 237, with the proper-source maps in Section 238 and the action in Section 239. Human-source citations remain attached to their uses. Earlier edition statements below retain their historical dates and scope. This local successor has no newly assigned deposited DOI.

Programme origin and human mathematical sources. The Split-Zero construction and programme were developed in KokunoYumeto's project with AI assistance, with the originating notes documented in [H19, H20]. This uses the expressly authorized public pseudonym and the documented project provenance; it makes no assertion of global priority. The historical deposited creator fields are preserved in the citations. The present continuation was derived and checked with AI assistance within that programme. Its classical polynomial-modification input is Szegő's Theorem 2.5, with his credit to Christoffel retained; its scalar orthogonal-polynomial formulas use the authored DLMF chapter and exact parameter dictionary; its Schur minima use the directly read Boyd–Vandenberghe derivations with the complex Hermitian specialization proved here. These human sources are cited at their actual uses. Project-provenance citations do not replace those mathematical source citations.

Local continuation and source coverage. This local edition adds the complete correlated-tail calculations to the accepted 782-page proof corpus through its supplied historical citation derivatives. Human-source citations and exact dictionaries are preserved in both the new calculations and their selected prerequisites. The broader historical provenance work remains open with its assigned owner; inclusion of the supplied citation layer makes no claim of complete retrospective coverage. The accepted 782-page source is locally sealed; the public predecessor remains the earlier 765-page edition. No new deposited DOI is assigned to this local edition.

Complete proof edition. The chapters below retain the original periods, amplitude units, primary multiplicities, cyclic maps, source masses and four endpoint metrics. Each chapter supplies its full calculation and proof. Earlier finite bounds remain recorded in their original form; subsequent chapters give their proved refinements. In particular KAF retains the factorial denominator of OKV and improves the absolute kernel bound, while AMT evaluates the source differences left in OMV.

Sources and scope. The complete preceding Gamma and observation proofs, including their historical analytic providers, are supplied in the source closure. The original all-multiplicity cyclic maps remain in OCS. The evaluated five-member quadric orbit, exact kernel dimension, and its individual volume conclusions state their simple-quartet period domain explicitly. The full all-multiplicity operator estimate is given in KAF. OCF, OKM and OKA calculate the actual kernel

frame, its original metrics and the boundary action. OMG proves the full all-multiplicity Segre–Veronese geometry and retains the actual cyclic stabilizer.

Current refinement. AKS sharpens the complete kernel bound to $32 \log(25 \log 3 / (4\pi))$ at the kq scale. AKJ evaluates the original residue inverse and conductor degree flag. AKP proves the precise theta representative-change and minimum-section maps. RUI now proves all-rank exterior control of the original finite conductor, and CEP carries it through the exact source ideals, four endpoint quotients and original compressed covariances. The resulting conductor error is $o(kq)$ for each fixed actual period. CTV now gives the exact outer-root quotient and its controlled norm tails. ROQ calculates the actual residual observation, all its positive increments and full graph metric. OSP proves a full-norm comparison with explicit moments of the original Gamma sources. OAR and PMR carry these finite errors into the arithmetic return. OFV now evaluates the outer contribution at the kq scale, using the full EIQ equilibrium and QLG finite partition estimate supplied below. FMC constructs fixed original coefficient frames with their exact Gamma and polynomial-degree maps. OVR propagates the evaluated outer term into the original kernel, residual observation, arithmetic boundary and signed mixed return. Its remaining analytic quantity is the original residual Gamma compression. OOO controls every direction of the complete original outer quotient. GIP gives the full residual source and relation graph, with the exact kernel–cokernel sequence of its outer-coordinate map. PCL evaluates the original conductor limit and proves its pivot restriction. ROG straightens the complete low-degree relation graph with a proved subleading exterior cost. RDS supplies the exact dual differential realization and original Gamma coefficient map of that relation quotient. FEX gives the complete fixed-period exterior transfer; FGC proves its exact comparison with the original graph, retaining both low-degree metric corrections and the actual residual observation. ECL evaluates the original equilibrium coefficient with its explicit certified interval. EQR carries that calculation into the original earlier centre estimate while preserving its full error radius. RWC proves strict original full-window residual growth and every exterior coefficient. OER propagates the complete fixed-column and window bounds into the earlier kernel, boundary, signed and same-class control equations. MCE computes the same arithmetic class’s action control, including every kernel, boundary and mixed exterior term. MCP gives the exact marked-pairing action and source-derivative identities. RFX proves the full arithmetic and metric reflection, evaluates the total phase at the original source, and retains the individual mixed pairings. OMR carries these calculations into the earlier receiving equations and the complete arithmetic control-radius budget. RCF calculates the complete common factor of the actual polynomial relations and its exact quotient sequence. RCM carries the resulting quotient through both original Gamma source orders and all four windows, retaining the actual residual subspace and its metric. DNE realizes that weighted relation metric by its minimum dual extension with the complete numerator, source coefficients and masses. PEC controls its positive exterior expansion and transports the exact negative spectral deficits into the original kernel, residual and arithmetic action bounds. Every remaining term is kept in its original space with the full coordinate maps. ACR gives finite formulas for the actual common-factor map on the original kernel, including every coefficient-support case. LBRE proves the complete boundary exclusions at the original period limit, and ASR gives an explicit support radius and finite exceptional-period set. Their zero common-factor conclusion retains the entire complementary residual space and its metric; all exceptional cases keep their exact maps. PBR composes these full calculations through the original kernel, residual, low-image and arithmetic maps, retaining both spectral deficits and the original centred finite-radius estimate. LJR proves the exact low-jet quotient Jacobian and the leading term of both low-window deficits, with the complete finite remainder. LRR installs that result into the same four-window signed return and arithmetic action calculation. HJR proves the compatible later-jet graph, both native conditioning denominators and their exact positive row increments. It transports the original finite total bound and cancels the common low-window term in the signed deficits. HCR substitutes the two complete row sums in the original mixed-return interval and action budget, retaining every finite flag, low-image and arithmetic error in that same

calculation. All preceding chapters remain in full, with their proved refinements.

Current primitive-source and residual-boundary calculation. The complete PDS and PDI constructions retain the original primitive quotient, its three missing low directions, and both native increments. CIT gives its full connecting-image target and its place in the original invariant boundary chain. PSG evaluates the full monic input band modulo all lower polynomials; SGT proves the exact Gamma translation and its uniform small-rank exterior estimate. PFG evaluates the four actual primitive-row Gram with a proved moment minor. PSR carries these through the full relation and dual source determinants, exact common Taylor coordinates, and both original connecting source-order pairs. RQB computes the surviving boundary quotient, its nonreduced branch constraint, complete arithmetic rows, and directed finite target budget. The defining low-cutoff source determinants are now evaluated with every frame, mass and factorial retained. The connecting comparison keeps both actual independent-gauge terms, and its common later return retains the full residual and missing target quotients.

Contents

1	The literal cyclic sectors and the original mixed kernel	24
1.1	Original data and the Fourier basis of the literal matrix	24
1.2	The ordered-to-symmetric map and the exact sector projector	25
1.3	Every primary period coefficient of the projected curve	25
1.4	The actual finite observation matrix and its complete kernel	26
1.5	Exact invariant dimensions and unconditional forced losses	27
1.6	The actual simple-quartet quadric and a stronger rank bound	29
1.7	A second exact finite matrix for the simple-quartet kernel	30
1.8	All primary action directions remain in the mixed calculation	31
2	The original quadric orbit and an explicit invariant lifting map	32
2.1	The actual coefficient ring and the original observation	32
2.2	The complete orbit decision from the actual coefficients	33
2.3	An explicit invariant lift for the five-member orbit	34
2.4	The induced original dual map and its complete action defect	35
3	Exact original rank from the complete quadric orbit	36
3.1	Every invariant vanishing polynomial contains the full orbit	36
3.2	The exact invariant quotient and the restriction map	36
3.3	All original degrees and the exact kernel dimension	37
4	The actual period quadric and its cyclic continuation	38
4.1	Original roots, entire unit and centred phase	38
4.2	The original periods as an absolutely convergent series	38
4.3	The actual transformed quadric and every deciding coefficient	40
4.4	The cyclic action is the actual original continuation in u	41
4.5	The large-parameter geometry and a positive-ray obstruction	41
5	The original period quadric as a residue commutator	43
5.1	A flat alternating pairing with all constants	43
5.2	The full arithmetic unit and the exact quadratic form	44
5.3	An explicit original-parameter radius forcing orbit five	45
6	A finite bound for the original kernel volume from its actual dimension	48

7	The actual simple-quartet kernel and boundary at the original volume scale	53
7.1	The original domain and the proved period range	53
7.2	A uniform finite bound on the actual kernel volume	53
7.3	The two individual leading coefficients	54
7.4	Exact propagation to the arithmetic identity and the finer scale	55
8	The original arithmetic kernel and boundary receive the full source comparison	56
8.1	The complete original source and the same observation	56
8.2	All constants of the actual full-polynomial inequality	57
8.3	Exact remainder, minimum-section, kernel and boundary maps	57
8.4	The exact nested deficits and their rank factors	58
8.5	All four original endpoints and the correlated mixed return	59
8.6	The full original multiplier and its transported observation	60
8.7	The evaluated original simple-quartet arithmetic kernel	62
8.8	The original boundary receives the full baseline coefficient	63
8.9	The same comparison evaluates the mixed Gamma return on the kq scale .	64
8.10	The refined penalty in the complete arithmetic action	66
8.11	The sharp canonical-band constants in the same original action	67
9	Factorial control of the original common kernel at the kq scale	70
9.1	Original coordinates, all primary multiplicities, and source norms	70
9.2	A degree-dependent constant in the unchanged remainder estimate	71
9.3	Retaining the factorial denominator	72
9.4	All four actual kernel and boundary Grams	72
9.5	The actual simple quartet and the arithmetic source	73
10	An exact conductor frame and a fixed invariant-generator recursion	75
10.1	The original Segre coefficient isomorphism	75
10.2	Every leading-coefficient case gives an exact strip frame	76
10.3	Twenty-five fixed invariant monomials generate all degrees	76
10.4	Exact multiplication on the strip and the invariant recursion	77
10.5	Exact ranks, a finite pivot atlas, and the original quotient map	78
10.6	The same recursion with fourteen proved generators	80
11	The exact common kernel frame and its original Gamma metrics	81
11.1	The exact dual and primal maps of the strip quotient	81
11.2	The complete original source and quotient metric	82
11.3	Christoffel–Darboux with the exact source mass	82
11.4	Every pivot change and the four original endpoints	83
12	The original kernel graph, boundary basis and complete action	85
12.1	The coefficient quotient gives a graph in the original primary basis	85
12.2	An explicit basis of the actual boundary	85
12.3	The whole original action and its retained defect	86
13	The complete homogeneous geometry of the original multiplicity curve	88
13.1	The actual primary, unit, period and Fourier coefficient map	88
13.2	The complete homogeneous ideal, with quadratic generators	89
13.3	The projective model and its complete chart inverses	90
13.4	The exact graded source-dual algebra and every factorial	91
13.5	The full semisimple and nilpotent action in this geometry	92
13.6	The exact original cyclic observation as invariant restriction	93

13.7	The actual orbit ideal, including every stabilizer	93
13.8	The complete component conductor and an explicit finite witness	94
13.9	The exact action defect and all subsequently observed directions	95
14	The absolute common kernel: the new finite result	97
15	An exact coefficient realization of the actual lost directions	98
15.1	The period matrix and invariant restriction, with their full coefficients . . .	98
15.2	A determinant formula requiring no assumed nonzero pivot	99
16	A bounded exact remainder operator on the original source	100
16.1	The complete source norms and coefficient map	100
16.2	Newton interpolation with all roots and their orders	101
16.3	Cauchy extraction in the actual Gamma generating function	101
17	The four original kernel Grams and the evaluated scale	102
18	The complete arithmetic and mixed receivers	104
18.1	The unchanged finite arithmetic allowances	104
18.2	The original Gamma multiplier, not a replacement higher source	105
18.3	Boundary, QGQ and arithmetic receiving identities	105
19	Actions, support faces, and the actual arithmetic class	107
20	Exact period domain and the interior alternative	108
21	Verification scope and the remaining absolute coefficient	108
22	The new ceiling in the original coefficient frame	110
23	Residue detection expressed in every original primary coefficient	111
24	The actual conductor confines the kernel's polynomial degrees	112
25	The exact conductor translations in the original source norm	114
26	The original theta class and the checked metric interface	116
26.1	The complete minimum-section calculation at each endpoint	117
27	The finite conductor inverse in the two original Gamma norms	119
27.1	A coefficient-controlled disk for the literal finite symbol	119
27.2	The exact coefficient map and its norm	120
27.3	The infinite symbol differs by an exact low-degree map	121
27.4	An explicit right inverse of the original conductor	121
27.5	The exact quotient and exterior transfer supplied by this bound	122
28	Subleading exterior control from the actual finite conductor	124
28.1	The forward polynomial bound and exact determinant	124
28.2	Complementary singular values retain the full exterior norm	125
28.3	The actual conductor quotient inherits the sharper volume control	125
29	The original conductor at every source endpoint	127
29.1	The full source, remainder and primary-coordinate square	127
29.2	The literal finite inverse at the larger cutoffs	128
29.3	The exact source operator matrix and its full determinant	129

29.4	The induced original quotient metric and its correction	130
29.5	An exact graph quotient with controlled exterior metric error	132
29.6	All ideal graph entries as original polynomial products	134
29.7	The compressed conductor correction is subleading at every endpoint	135
30	The total-source difference as an outer-strip quotient	137
30.1	The exact nested root polynomials in the original real coordinate	137
30.2	The full monic source and relation determinant identities	137
30.3	The exact quotient of the outer-strip polynomial	138
30.4	Finite bounds for all norm tails in both original sources	139
30.5	The proved first-variation reduction and its remaining matrix	140
31	The actual observation remaining inside the conductor kernel	142
31.1	An explicit residual map and its actual boundary basis	142
31.2	All four induced Gamma metrics and their fixed determinant factor	143
31.3	The exact constrained covariance in the original source	144
31.4	Exact increments and a positive residual four-endpoint return	145
31.5	Finite scalar enclosures and propagation into CEP49	146
31.6	The complete residual quotient in the conductor graph	146
32	A quantitative power comparison for the actual outer-strip quotient	148
32.1	The two original source masses and a proved interval lower bound	148
32.2	A complete coefficient-space estimate for the inner interval	149
32.3	The exact original-root map and a finite full-norm comparison	150
32.4	Uniform bounded logarithmic cost at every required dimension	151
32.5	Every relation determinant and the exact outer quotient	152
32.6	The resulting same-source moment target	153
33	The outer-root and actual residual bounds in the original arithmetic return	154
33.1	The two finite error endpoints retain their signs	154
33.2	Original positive increments give explicit kernel caps	154
33.3	The correlated arithmetic kernel and boundary	155
33.4	The signed mixed return with its complete finite error	155
34	The same-source Gamma moment comparison in the arithmetic return	156
35	Exact equilibrium for the square-root logarithmic potential	158
35.1	Existence and uniqueness of the two endpoints	158
35.2	A positive density and its exact Cauchy transform	159
35.3	The Euler equality, its strict exterior inequality, and uniqueness	160
35.4	An explicit endpoint integral for the logarithmic moment	161
35.5	Parameter continuity and the variational derivative	162
36	A uniform finite partition estimate from the original equilibrium	164
37	The leading first variation of the actual outer-strip quotient	169
37.1	Exact original-order recurrence and a small metric error	169
37.2	A uniform finite Gamma-to-exponential determinant estimate	170
37.3	Direct low-dimension estimates, outside the compact equilibrium range . . .	171
37.4	High-dimension partitions and their controlled first variation	171
37.5	All seven determinants and the resulting leading constant	173

38	Fixed coefficient control of the original residual module	174
38.1	A fixed free module containing the entire invariant image	174
38.2	Finite reduction data with only fixed actual denominators	175
38.3	A fixed weight bounds every homogeneous reduction path	176
38.4	The two exact maps back to the original conductor strip	177
38.5	An identity pivot block for the entire residual image	178
38.6	Exact relation to every earlier certificate frame	179
38.7	The unchanged Gamma source in the fixed degree model	180
38.8	The exact map to the original polynomial degree flag	182
39	The evaluated outer contribution in the original mixed return	184
39.1	The exact remainder and the sourcewise leading identity	184
39.2	Transfer between the two original Gamma sources	184
39.3	Arithmetic kernel and its correlated boundary	184
39.4	The signed return retains its original Gamma correction	185
39.5	Bounds on the remaining residual coefficient	185
40	All directions of the original outer quotient at the leading scale	187
40.1	Uniform equilibrium constants and a polynomial evaluation bound	187
40.2	A matching polynomial lift from actual equilibrium quantiles	188
40.3	The entire analytic remainder map	189
40.4	High source degrees: a full operator estimate	189
40.5	Low source degrees, including the extra ideal column	191
40.6	Transport back to both complete original Gamma sources	191
40.7	Relative operators and all fixed subquotients of this actual target	192
41	The complete residual map on the original conductor graph	194
41.1	The full graph denominator and its metric	194
41.2	The exact natural map to the CTV outer divisor	195
42	The original mixed return and its same-class action control	197
42.1	The original arithmetic class and all three source metrics	197
42.2	The exact rank-two control from the original recurrence	198
42.3	The same nonzero arithmetic exterior class	198
42.4	Complete kernel, boundary and mixed pieces of the control	199
42.5	Degree-step volumes determine the squared component sizes	200
42.6	The signed return gives an exact original control-radius budget	201
42.7	Uniform contraction control in the original minimum allowance	203
43	The next marked pairing as an original action and source derivative	205
44	The exact original reflection and the retained mixed phase	207
44.1	The entire arithmetic unit and the source involution	207
44.2	The quotient, primary jets and arithmetic class	207
44.3	The full metric and the total endpoint pairing	208
44.4	The original mixed blocks remain present	209
44.5	Propagation through the same arithmetic control budget	210
45	The outer operator, the full residual graph, and original action control	211
45.1	Original objects, domains and the same-target operator interval	211
45.2	The full graph denominator and its exact outer morphism	212
45.3	Backward propagation through the original kernel and signed return	213
45.4	The original action, its mixed components, and its positive class	214
45.5	The exact arithmetic radius budget and its backward bound	215

45.6	The exact marked derivative and original reflection specialization	217
45.7	Propagation through the original mixed action receiver	219
46	The literal original conductor limit and an invertible pivot restriction	220
46.1	An entire coefficient matrix in the inverse original period	220
46.2	All eighty-one actual conductor coefficients and their limit	221
46.3	An exact positivity slice for the literal positive period ray	222
46.4	A complete invertible restriction supplied by the original pivot	222
46.5	The exact inverse-period reflection of the central section	223
46.6	The evaluated zero-order domain in the original period	224
47	The original relation operator and its finite-rank graph transport	225
47.1	The complete positive-shift relation operator	225
47.2	An explicit finite inverse with coefficient control	225
47.3	Complete original-source bounds for the low graph map	227
47.4	The exact graph shear and its complementary singular pairs	228
47.5	The unchanged residual target and its four endpoint metrics	229
48	The exact differential realization of the original relation quotient	231
48.1	A finite-dimensional quotient and its analytic dual	231
48.2	Every possible singularity and its original multiplicity	232
48.3	An explicit original-data analytic disk and source coefficients	233
49	The fixed-period problem and the result proved here	234
50	A full-flag determinant bound	235
51	Polynomial forward control replaces the exponential exterior loss	235
52	The literal conductor at every source endpoint	237
53	The residual observation and the smaller coefficient space	238
54	A finite error for the actual compressed determinant	239
55	The original arithmetic and signed receiving identities	240
56	Actions, representatives, and the retained mixed-support maps	242
57	Scope of verification and the next exact quantity	243
58	The fixed-column conductor quotient and the full GIP graph	244
58.1	The precise projection and its complete kernel	244
58.2	The actual fixed columns and the Schur metric	245
58.3	Both original low metric terms and their signed identity	246
59	The exact closed equilibrium coefficient	247
59.1	Arcsine identities with their original constants	247
59.2	Differentiation with the moving endpoints retained	248
59.3	The moment and energy formulas	250
59.4	The original outer coefficient and the two scale conventions	250
59.5	An integer-only outward certificate	251
60	The closed coefficient in the original QGQ return centre	253

61	Strict full-window growth of the original residual covariance	255
61.1	Every consecutive full block is invertible	255
61.2	The simultaneous conditioned update	255
61.3	All exterior coefficients and finite bounds	256
61.4	Return to the original four endpoints	257
62	The fixed-period determinant and closed coefficients in the original receiver	258
62.1	The existing outer coefficient has an exact endpoint expression	258
62.2	The actual period coefficients and the pivot morphism	258
62.3	The full low graph has its own finite-rank transport	259
62.4	The fixed columns arise from an exact quotient of that graph	260
62.5	Both low terms and all one-sided errors enter the current interval	260
62.6	The relation quotient and the earlier finite Gamma centre	261
62.7	Full-window residual bounds in these same receivers	262
63	The complete common factor of the original polynomial relations	264
63.1	Every original finite degree window and its exact metric sequence	265
64	Original Gamma control on the actual common-factor quotient	267
64.1	The exact original divisor, windows, phases and source	267
64.2	Complete finite bounds at the low source windows	267
64.3	The high windows with the actual analytic remainder and lift	268
64.4	Both full Gamma sources and the four-return coefficient	269
64.5	The resulting original residual quotient and its retained complement	270
65	The original weighted relation metric as a minimum dual extension	272
65.1	A fixed original exponential quotient frame	272
65.2	The exact original source extension problem	272
65.3	Complex-linear duality and the actual fixed kernel columns	273
65.4	A finite source bound before the high-order derivative	274
66	Subleading positive exterior expansion of the original dual metric	276
66.1	The actual symbol-zero jet subspaces	276
66.2	A complete finite Gamma bound on each cancellation subspace	277
66.3	The whole positive logarithmic spectrum	278
66.4	Every fixed restriction and quotient keeps this bound	278
66.5	The exact negative spectral return on the actual kernel	279
66.6	The complementary original residual and its low subspace	280
66.7	Backward and forward return to the same arithmetic class	281
67	The actual common-factor map on the original kernel	282
67.1	Support and its precise extended coefficient rectangle	282
67.2	The residual map evaluated on every original primary coordinate	283
67.3	The exact retained residual quotient in biform coordinates	284
67.4	A smaller invariant recursion evaluating every exceptional rank	285
68	Exact exclusion of all four boundary rulings at the original period limit	287
68.1	The exact even and odd evaluation maps	287
68.2	The four complete quadratic restrictions	288
68.3	A nonzero exact eliminant for every original amplitude	288

69	Actual boundary multiplicities and a finite full-support radius	290
69.1	Every finite-period boundary multiplicity	290
69.2	The exact symplectic map underlying the coefficient tests	291
69.3	An explicit radius from the actual limiting corner calculation	292
69.4	Consequence for the actual common factor and residual	293
69.5	Only finitely many support exceptions on the original exterior domain . . .	293
69.6	The second radius and its exact native and polar consequences	294
70	The two exact dual subquotients in the original receiver	295
70.1	Finite positive-spectrum allowance and every small eigenvalue	295
70.2	The original metric on the low image	297
70.3	Both low terms and the full one-sided kernel errors	297
70.4	The complete current finite interval	298
70.5	The signed return and the complete same-class action budget	299
70.6	Actual support, the common-factor map, and the retained residual	300
71	Exact low-jet reconstruction and its original quotient spectrum	302
71.1	The exact finite reconstruction and its inverse	302
71.2	Both exact quotient kernels and their metrics	303
71.3	The complete full-space and quotient Jacobians	303
71.4	Finite reconstruction bounds on the actual grid	304
71.5	A direct estimate of the nested determinant difference	305
71.6	The complete remaining native Gram	306
71.7	The exact remainder and every paired exterior volume	307
72	First-endpoint reconstruction in the actual kernel and annihilator	309
73	Compatible high jets and the original positive return	311
73.1	The exact compatible high-jet graph	311
73.2	One lower-evaluation minimum over the complete window	312
73.3	The original kernel and annihilator updates	312
73.4	Exact one-row updates and their allocation	313
73.5	The four original signs and an exterior bound at scale kq	314
73.6	Exact cancellation of the low leading scale	315
74	The compatible high rows in the original receiving interval	318
74.1	Native multiplier return at the original compatible-row site	320
74.2	The original low metrics after native window and polar transport	320
75	Sharp bounds from the actual annihilator components	321
75.1	The exact native metric component	321
75.2	The weighted aggregate and its joint tail	322
75.3	Bounds from the actual native coefficient columns	322
76	A full-form comparison for the original allocation	323
76.1	A polynomial dilation bound with its exact coefficient map	324
76.2	Full original Gamma and root comparisons	325
76.3	The same actual relations in an explicit moment matrix	326
76.4	A subleading error for each original allocation	327

77	Exact mixed half-line volumes for the original allocation	327
77.1	Both half-lines in one explicit coefficient map	328
77.2	The original kernel and residual as consecutive volumes	329
77.3	Every mixed minor and a finite maximal-minor error	329
77.4	Exact return to the arithmetic and mixed quantities	330
78	A deterministic calculation of the original mixed volumes	331
79	Original threshold spectral flow and compatible-row returns	333
79.1	Original source forms and threshold-kernel maps	333
79.2	A pencil assembled from all primitive original columns	334
79.3	The exact original one-row spectral flow	335
79.4	Exact deficit, positive-part and full-row integrals	336
79.5	Explicit finite spectral cutoffs from original source bounds	336
79.6	Finite brackets for both actual row allocations	337
79.7	Return to the same finite receiving intervals	338
80	The actual component, mixed-volume and threshold controls together	339
80.1	Five simultaneous finite descriptions of the same two scalars	340
80.2	Simultaneous intervals with exact total coupling	341
80.3	The unchanged low metrics and all original finite errors	342
80.4	Arithmetic, signed return and the entire action	342
80.5	Deterministic full-row pivots in the same interval	343
80.6	The attained multiplier-image interval in the same allocation pool	344
80.7	The native boundary continuation in the simultaneous allocation	345
81	The actual branch correspondence between the two row allocations	345
81.1	The complete conductor dual and the low annihilator	345
81.2	Exact exchange on the four original intersection components	346
81.3	The induced original kernel–residual correspondence	348
81.4	Both induced Gamma metrics, without assigning an isometry	348
81.5	Exact allocation contrast and all unmatched directions	349
81.6	The full degree-four invariant exchange in the original coordinates	350
82	A bounded-degree invariant presentation of the original target	351
82.1	A finite family in the original coordinates	351
82.2	The exact original primary and symbol maps	352
82.3	Coupled and individual low-jet conditions	352
82.4	The actual branch map on these generators	353
82.5	The proved first-moment bound on the identical coupled domain	354
83	Original endpoint control for every finite-rank remainder	354
83.1	Two exact metric maps give one endpoint comparison	354
83.2	Every original paired restriction and quotient	355
83.3	The exact image and its native unmatched quotient	356
84	Native transport of the bounded original invariant multipliers	356
84.1	The literal finite translation sum and its two centres	356
84.2	The complete source-order change	357
84.3	Every original lower evaluation in the quotient map	358
84.4	The finite family reaches the native kernel image	358
84.5	The exact displaced source endpoints	359
84.6	The original displaced windows with their achieved finite bound	360

85	The bounded multiplier image and its exact original receiving maps	360
85.1	Both low-jet domains and the exact image quotient	361
85.2	The native direct sum and its full source kernel	361
85.3	Matched relative determinants and the finite unmatched return	363
85.4	The full original branch correspondence in this interval	364
85.5	Return through SAR, HCR and every original finite allowance	364
85.6	Backward propagation of the achieved windows and complete polar flag . .	366
86	Scope, source audit, and the original finite relation	367
87	Audit of the common factor and its full finite complement	368
88	Sharper full-metric control of this particular divisor	369
88.1	Every equilibrium and source constant	369
88.2	Low and high errors with the smaller disk	370
89	The common quotient visible to the actual residual rows	371
90	All-rank control and actual RWC exterior coefficients	372
91	The whole remaining complement, including its covariance update	374
92	Which dual directions the common factor can and cannot recover	374
93	Original receivers, action, and exact remaining quantity	376
94	Original domain and the precise new result	377
95	An explicit nonvanishing calculation at the actual period limit	378
96	A finite original-period radius and the exact visible rank	379
97	The full complement has a finite principal-part detector	380
98	The full Gamma metric and an evaluated action-leakage rank	382
99	Leading-scale receiving statement and remaining analytic expression	383
100	The original rings and every degree-four map	384
101	The exact branch identity and every elementary quotient kernel	386
102	Exact calculation on the full nonreduced intersections	386
103	Joint injectivity and the exact preceding descent defect	388
104	Finite matrices for the invariant and low-jet kernels	389
105	The complete homogeneous identity	391
106	Original exponential and coefficient formulas	392
107	The exact original lower-gauge kernel	393
108	Original conductor gauges and the full ambient target	394
109	A declared exact algebraic fixture	395

110	A finite native inverse and the exact compatible lower gauges	395
110.1	The original finite coefficients of a left inverse	396
110.2	Both native source norms and the exact finite matrices	396
110.3	The exact gauges before and after their compatibility condition	398
110.4	The two attained native minima and their exact correction	399
110.5	Restriction to the original conductor and arithmetic maps	400
111	The original compatibility operator and its native determinant	402
111.1	The complete finite compatibility homotopy	402
111.2	Exact native gauge and coefficient spaces	403
111.3	The actual source restriction and its complementary projection	404
111.4	The four original endpoints	405
112	The original grid, source norm and exact enlarged denominator	406
113	The exact conditional boundary Gram and every original numerator	407
114	The high-endpoint kernel is the full chart-boundary module	408
115	Explicit boundary determinants in both rank regimes	409
116	Original Gamma control at all displaced cutoffs	409
117	A scalar low-boundary estimate with a proved subleading error	411
118	The identical original conductor in the two continuations	412
119	Exact support exceptions and two quantitative radii	413
120	The high-window boundary kernel and its complete native metric	414
121	The exact first-moment refinement and its backward consequences	415
122	The exact map from independent gauges to the enlarged boundary	415
123	The full polar observation and its original low-jet flag	425
123.1	The pole map, without discarding the origin jets	425
123.2	The exact native-to-pole coefficient matrix	426
123.3	The full conditioned pole Gram and its low Schur factor	427
123.4	The native Gamma map and its explicit exterior error	428
123.5	The algebraic correspondences in these same coordinates	429
124	The unchanged source flag at its displaced cutoffs	429
125	Every finite native and outer endpoint error	430
126	The asymmetric correction and the original source-kernel return	432
127	The finite bounds returned through all earlier receiving maps	434
128	The proved subleading window cost and the remaining exact center	435
129	The original domains and the simultaneous boundary refinement	436
130	The degree-four tuple before and after the invariant quotient	437

131	The original enlarged boundary as a quantitative source loss	438
132	The complete polar metric and the low-volume return	439
133	Propagation through the identical allocation and entire action	440
134	Original objects and the finite conclusion	441
135	Potential bounds and the strict side separation	443
136	A uniform finite Riemann estimate at every original boundary node	444
137	Finite top sums, the full corner, and proof of the estimate	445
138	The actual finite ranking and its threshold	446
139	An explicit one-dimensional integral for the coefficient	448
140	Full original data and an explicit finite volume interval	448
141	The complete logarithmic leading expression	450
142	The complete original complex and its oriented matrices	451
143	Local sheaf exactness and the full Čech argument	451
144	Every rank and every coefficient kernel	453
145	Determinant-line torsion in the original coefficient frames	453
146	The original exponential realization and its finite jet kernels	456
147	The complete original-grid Jacobian in the graded compatibility complex	457
148	The five original native Gram volumes at the high endpoints	459
148.1	Every monic factor and source mass in the exact identity	459
148.2	The complete Gamma-to-moment comparison at these cutoffs	460
148.3	The exact parameter shifts and their moment cancellation	460
148.4	The complete finite Gamma recurrences	462
148.5	The exact native relation to all companion maps	462
149	The original proper source and the companion Jacobians	463
149.1	The exact proper-source determinant and its finite error	463
149.2	Every nested source and its actual Schur quotient	464
149.3	The four-endpoint transport with both low terms retained	465
149.4	The accepted shared-update bound at the actual difference rank	466
150	The complete native complex on the actual conductor	467
150.1	Both exact sequences and all low-jet connecting maps	467
150.2	The literal finite native square	468
150.3	Every rank lost at the original finite cutoffs	469
150.4	The exact determinant line and original kernel/residual forms	470
150.5	Quantitative forward control with every inverse term retained	472
151	The exact finite complexes and all term dimensions	473

152	Two successive connecting maps for the full germ quotient	474
153	The analytic image and its complete two-term quotient	475
154	The actual four-row differential has full rank	475
155	The entire long sequence, connecting ranks and zero cases	477
156	Literal connecting representatives and native matrix orientation	478
157	Three-component late-kernel control and the exact low return	480
158	The original low conductor cohomology and its two native metrics	481
158.1	The two exact ambient sequences	481
158.2	The actual conductor image and its last missing jet	483
158.3	All original low cohomology groups	484
158.4	The complete native cohomology Grams	485
158.5	The exact morphism to the original kernel and residual spaces	488
158.6	The exact next-jet inclusion and its bounded missing rank	489
158.7	The missing low directions have a proved subleading return	490
159	The original low mixed flags and their contraction returns	491
159.1	Original objects and the literal four flags	491
159.2	A full determinant identity for the two nested low flags	493
159.3	The entire difference of contraction returns is subleading	493
159.4	The actual adjacent common-space contraction is also controlled	494
160	Original spaces, maps and two positive forms	495
161	An explicit original coefficient determinant for the target loss	496
162	Return of both lower-gauge corrections, including the actual kernel	497
163	Exact cancellation of the common compatibility denominator	498
164	Branch defect and all four original endpoint signs	498
165	The identical native update sharpens the error to the image rank	499
166	The original finite domain and coefficient flags	501
167	The original arithmetic rows, without an unspecified evaluator	502
168	Both quotient operations as the same raw Gram cross ratio	503
169	The exact four-row update and its two positive terms	504
170	The actual long four-endpoint sum	505
171	Exact continuation without replacing the native metric	506
172	Original invariant-target loss: signed flag flow and finite image-rank updates	506
172.1	IF1. Original domain, coefficients, metrics, and fixed subspaces	506
172.2	IF2. Positive comparison path and its original endpoint forms	507
172.3	IF3. Intrinsic lift derivatives and the signed determinant differential	508
172.4	IF4. Exact principal angles of the two original quotient lifts	508

172.5	IF5. Full generalized-spectrum differential, including crossings	509
172.6	IF6. A full-interval angle comparison with an explicit endpoint condition ratio	510
172.7	IF7. Exact finite numerator and denominator updates of size r	511
172.8	IF8. Directed integral bounds and endpoint-only enclosures	512
172.9	IF9. The four original endpoints and exact scope	513
172.10	Source record and verification scope	513
173	The same quotient lifts and their individual paired planes	514
174	Simultaneous variation costs one ambient native volume	515
175	The trace budget is itself a scalar four-row arithmetic quantity	515
176	The same original sources at every receiving map	516
177	The evaluated boundary enters through the entire numerator map	517
178	All companion maps and the exact proper-source term	517
179	The complete low groups replace the earlier Euler-only calculation	519
180	The residual/kernel metric maps and both computed contractions	519
181	The native target loss in the earlier matched quotient	522
182	Every original row and both signs in the long return	522
183	The simultaneous native budget and the exact signed receiver	523
184	The original primitive source and its adjacent Gamma metric	526
184.1	Original objects and the polynomial quotient map	526
184.2	The full original Gamma dual metric	527
184.3	The exact codimension-three truncation	528
184.4	One rank-one update, with every frame retained	529
184.5	An explicit coefficient-product upper estimate	529
185	The original connecting image and its full native target determinant	532
185.1	The literal primitive domain and target coefficient matrix	532
185.2	Complete connecting kernel and attained target norm	533
185.3	The original source-to-image determinant at its low cutoff	534
185.4	The fixed connecting flag at every original target row	537
185.5	Its exact place in the common low invariant loss	537
186	Primitive-source transport inside the original connecting image	543
186.1	The exact source coordinate map and its target naturality	543
186.2	The two low metric increments in common coordinates	544
186.3	The two fixed connecting invariant losses	546
187	The original monic source band and its low-polynomial quotient	547
187.1	Original coordinates and finite parameters	547
187.2	The complete Gamma density	547
187.3	Finite determinant bounds in all original roots	548
187.4	The exact original low-polynomial quotient	549
187.5	The original factorial band	550

188	The original finite Gamma translation on every small-rank plane	552
188.1	Full Gamma measure, recurrence and coefficient isometry	552
188.2	The original quotient map and its exact Toeplitz matrix	553
188.3	A proved original-data zero-free disk	554
188.4	Every actual pole and its finite-rank Toeplitz contribution	554
188.5	Finite exterior bounds, including the expensive directions	556
188.6	Uniform small-rank volume distortion and the actual PDS frame	557
188.7	Receiving the conductor estimate on the complete native quotients	558
189	The surviving boundary quotient and its original arithmetic kernel	560
189.1	The exact coefficient kernel, rank and original kernel quotient	560
189.2	A branch constraint for arbitrary source numerators	561
189.3	Full native and raw Gamma realizations	562
189.4	The exact original source denominator and compatibility transport	563
189.5	Four original rows and the complete retained target budget	564
189.6	The directed original kernel cap and its source correction	566
190	The four original primitive rows and their full Gamma Gram	569
190.1	Full native frame and the exact PDS matrix	569
190.2	The four actual moment pivots	570
190.3	Full finite Gram bounds	571
190.4	The derived factorial scale and the primitive correction	572
191	The original relation determinant and its connecting-image return	574
191.1	Finite bounds on the actual translated relation Gram	574
191.2	Absolute expansion with the full factorial band retained	575
191.3	Dual source and both primitive coefficient maps	575
191.4	Return to the original connecting-image calculation	576
191.5	The four-row estimate in both original primitive sources	576
191.6	Exact common coordinates for the two original source orders	577
192	The original Gamma source order on the native conductor and connecting flags	579
192.1	The established whole-source estimate with its complete constants	579
192.2	The exact original functional and Taylor coordinate square	580
192.3	Complete attained fibres and the original conductor quotient	580
192.4	Determinants of the surviving native flags	582
192.5	The literal primitive source and connecting-image receiving	583
192.6	The finite comparison of the actual compatibility losses	584
193	Source-order intervals in the original connecting and invariant returns	586
193.1	The finite native intervals and their exact value spaces	586
193.2	The two low source and image determinants at their actual ranks	587
193.3	Exact connecting and full-cohomology contraction return	588
193.4	The complete boundary ranks, metrics and positive returns	590
193.5	The common contraction in CIT30 with all boundary factors	591
193.6	Both source-order intervals in the matched invariant return	592
193.7	The existing arithmetic allocation comparison and its exact map	593
194	The connecting image inside the full remaining target quotient	595
194.1	All coefficient maps and their kernels	595
194.2	The two original source and target minima	596
194.3	Positive returns and the formerly signed remainder	596

194.4	Actual row formulas for each new factor	597
194.5	Back substitution into both source orders and the allocation	598
194.6	The actual endpoint inclusion and the numerator return	600
195	A single spectral budget for the original common long loss	601
195.1	The original objects and the precise scope	601
195.2	The common product is one projection determinant	602
195.3	Variation of the actual common loss	602
195.4	The exact original four-row update	603
195.5	The common long interval and its evaluated budget	604
195.6	The complementary terms and the allocation-side constraint	605
195.7	What is new and what is still uncomputed	605
196	An explicit four-endpoint enclosure from the initial common angle	608
196.1	The original common-angle inequality	608
196.2	Exact extrema of the four-angle constraint	609
196.3	The complete allocation receiver	610
197	Native Gamma-order transport and its equilibrium-support refinement	611
197.1	Original data and the result's scope	611
197.2	A complete power-weight comparison	612
197.3	Finite constants in the original Gamma comparison	614
197.4	The two-half-line transport is literal	615
197.5	An evaluated allowance at scale q_k	615
197.6	The defining primitive source and connecting image	616
197.6.1	The exact scalar cancellation and its compatibility return	617
197.7	Source-order control for the original common long loss	618
197.8	Residual boundary and original arithmetic receivers	620
197.9	Exact constants, verification, and remaining calculation	620
197.10	Original quantities, inherited inputs and exact comparison domains	622
197.10.1	The complete inherited finite logarithmic constants	623
197.11	The original equilibrium and an explicit exterior gap	623
197.12	Uniform polynomial localization and a growing moment shift	624
197.13	Both square-map sheets and the finite improved native interval	625
197.14	Evaluated asymmetric constants at scale q_k	627
197.15	The original common long loss and its fixed-frame receiver	627
197.15.1	The actual rows, half-lines and all remaining terms	628
197.16	Certified constants and the precise remaining task	629
198	The complete original action relative to its arithmetic total	631
198.1	Domain, unchanged arithmetic objects and exact minimum metrics	631
198.2	The actual rank-two control, reflection and exterior benchmark	632
198.3	Exact cancellation before any estimates	633
198.4	All-degree multiplication bounds on the actual arithmetic blocks	634
198.5	Total-degree tensor compression, including the output face	635
198.6	Both original contraction penalties on their complete window	636
198.7	The four-norm factorial window and its finite remainder	636
198.8	Complete finite interval, action mean and exact receiving scope	637
199	A smaller threshold-profile error from the original nested flags	641
199.1	The evaluated first row and the full transported profile	642

200	The original connecting image as a polynomial translation relation space	644
200.1	A full polynomial model of the original target kernel	644
200.2	The exact polynomial image of the connecting transpose	645
200.3	Eliminating the conductor division and the primitive complement	646
200.4	The full low-cohomology excess and what remains in the metric	648
201	The complete original barycentric low-tail calculation	649
201.1	Original objects and the coefficient kernel	649
201.2	Exact barycentric tail coordinates	649
201.3	Irreducibility from maximal contact	650
201.4	Independence of the four unchanged fourth powers	651
201.5	Exact vanishing when each tail has dimension at most two	651
201.6	An exact lower-degree polynomial presentation for every tail	652
201.7	Successive original moments and the improved uniform bound	652
202	The complete original strip-to-pole inverse	654
202.1	Exact inverse between the original conductor strip and all pole observations	654
203	The original pole detector, connecting image, and sharper low-tail return	655
203.1	A full quotient-value isomorphism supplied by the poles	655
203.2	Transport of the actual connecting map, including its scalar	655
203.3	The original metrics carried through these same coordinates	657
203.4	The sharper tail bound in every affected receiving expression	657
204	Original objects and the finite comparison data	659
205	Exact intersection before any restriction or quotient	661
206	Four endpoints and equal-rank contractions	662
207	The five original CPQ quotients and their full frames	662
208	The common return and the complete allocation intervals	663
209	The degree-k source at the displaced NFR cutoffs	665
210	The defining primitive source: exact finite intersection	666
211	Consequences and exact scope	670
212	Uniform exponential suppression of the original adjacent ratios	671
212.1	Source and review	671
212.2	1. Actual source and original relation norm	671
212.3	2. A two-band lower bound for every monic direction	672
212.4	3. Restore the original roots, arithmetic density and mass	673
212.5	4. The penalties are exponentially small	674
212.6	5. This also controls the minimum, not only the action mean	674
212.7	6. Extension to any fixed proportional window	675
212.8	Verification and scope	675
213	Uniform canonical-band review and a concentrated penalty certificate	677
213.1	0. Provenance, scope, and result	677
213.2	1. Original quotient and adjacent factors	678
213.3	2. Small relation rank: the new patch is valid	678
213.4	3. Independent proof of the uniform bulk monic-norm estimate	679

213.4.1	Lower norm	679
213.4.2	Upper norm and quantile uniformity	679
213.4.3	The square substitution and a separate finite constant	680
213.5	4. The profile derivative, positivity, and the sharp constant	681
213.6	5. A sharper finite penalty bound	681
213.7	6. Penalty concentration near the upper endpoint	684
213.8	7. The next degree regime: a reference-profile fact, not an arithmetic extension	685
213.9	8. Execution and formal scope	689
213.10	References	689
213.11	An explicit constant for the complete compact-rectangle norm bound	690
214	Original shared-flag projector threshold and joint receiving cancellation	691
214.1	2026-09-16: incoming order transport, shared flags, and a leading-scale next calculation	691
214.1.1	Incoming results and revised assignment	691
214.1.2	Cancellation-first receiver check	691
214.1.3	Additional finite reduction for formalization/review	692
214.1.4	Proposed complementary work	693
215	The complete original paired-root comparison	694
215.1	Original packet and coordinate for the retained paired-root provider	694
215.2	Audit of the complete paired-root norm comparison	695
216	Complete small-rank mechanism in the original Gamma reference	697
216.1	2. A new monic-norm bound controls all small ranks uniformly	697
217	A root-sensitive full-form comparison at growing polynomial degree	700
217.1	Original arithmetic input and a finite comparison theorem	700
217.2	The complete inner-region bound	701
217.3	Outer region, finite proof and degree range	703
217.4	A non-equilibrium reference lower bound at every relation degree	703
217.5	Transport through the original arithmetic monic and quotient maps	705
217.6	A genuinely super-proportional range	706
217.7	Derive the changed four endpoints and complete action	706
217.8	Support, interpretation, and remaining work	707
218	The complete root-form comparison at growing degree ratios	708
218.1	Original objects and exact dependency statements	708
218.2	The complete root polynomial	709
218.3	The exact parity coordinate and moving evaluation bound	710
218.4	The entire inner region and the finite full-form theorem	712
218.5	Proof of the broad growing-ratio range	713
218.6	Source pins and exact audit boundary	715
219	The reference norm at its actual moving parameters	716
219.1	Exact Robin and logarithmic-moment identities	716
219.2	Original Gamma constants, mass, and monic factorials	717
219.3	A global density bound and a full tail integral	718
219.4	Arcsine lower bound and explicit quantile upper bound	719
219.5	Exact parity map, shifts, and factorial cancellation	720
219.6	Uniform size of the retained errors	721
220	Uniform elliptic endpoint estimates with the original quarter shift	723

221	Original arithmetic transfer and the complete shifted action	728
221.1	Objects, coefficient maps, and all-degree arithmetic constants	728
221.2	Full forms, fixed affine fibres, and arithmetic monic minima	729
221.3	Original quotient, full-jet support, and adjacent allowance	730
221.4	Growing ratios and uniformity on every shifted index	732
221.5	Exact shifted two-window identities and finite enclosures	733
221.6	Complete asymptotics and the unchanged mixed receiver	735
222	An evaluated original-period domain with zero conductor vanishing order	737
222.1	The exact limiting value, including both exceptional amplitude phases . . .	737
222.2	A finite original-period domain and the complete zero-order test	738
223	The two exceptional unit phases have exact second inverse-period order	739
223.1	Finite coefficients obtained from the complete original series	740
223.2	The phase $r = 1$	740
223.3	The phase $r = (2X - Y)/(2Y - X)$	741
223.4	One finite radius for every actual amplitude and the original receiving maps	742
224	The original three-equation transfer of every connecting return	745
224.1	The actual three rows in the original period coefficients	745
224.2	An exact sequence of these three directions	746
224.3	Evaluation of the difference of the actual CPQ returns	747
224.4	Directed finite bounds with the original masses	748
225	The first original row: exact lower cancellation and its finite bound	749
225.1	The actual annihilator and the full first-row leverage	749
225.2	The exact lower ideal norm	750
225.3	An explicit finite bound from the original root products	751
225.4	Evaluation of the first-row scale and the cancelled leading term	752
226	The first original mixed row in the full Gamma polynomial ideal	754
226.1	The exact original row and its two coupled leverages	754
226.2	A polynomial minimum for the complete lower denominator	755
226.3	All eight compressed matrices in exact polynomial coordinates	756
226.4	Finite evaluation from the unchanged Gamma moments	757
226.5	A conductor-moment witness inside the actual residual	757
226.6	Position in the full return	758
226.7	Receiving the proved finite endpoint estimate in every signed term	758
227	The original conductor on every native mixed subquotient	760
227.1	The exact dual conductor square	760
227.2	All numerator observations and all conductor columns	761
227.3	Rank-dependent exterior bounds on arbitrary subquotients	762
227.4	The uniform limit and the exact DNE scalar conventions	765
227.5	All original mixed flags and all four-endpoint signs	766
228	The original numerator kernel and the complete mixed spectral return	768
228.1	The actual finite numerator observations	768
228.2	Evaluation by two original recurrence polynomials	769
228.3	The original eight flags and their numerator row update	769
228.4	Finite and limiting transfer of the complete signed profile	770
228.5	All complementary returns and the original arithmetic action	771

229	Original-grid interpolation, relation norms, and an interior mixed row	773
229.1	The full original evaluation map and its attained cardinal norm	773
229.2	All conductor columns, vanishing moments, and quotient duals	774
229.3	The full relation Christoffel row and the original observation row	775
229.4	The retained low-moment correction and conductor correction	776
229.5	A finite calculation at the second original target row	777
229.6	The exact interface to the older arithmetic norm results	778
230	The original native interior rows and a growing initial segment	780
230.1	The complete original annihilator at every subsequent row	780
230.2	The exact corrected native row at every interior position	781
230.3	All finite constants for the original later row	782
230.4	Uniform initial interior segment and all complementary terms	783
230.5	Uniform bound on the full original row range and a superlinear prefix . . .	784
231	Human sources for the original Gamma polynomial metric	785
232	The controlled original prefix and its complete remaining return	787
232.1	Exact prefix and tail maps on the original evaluation sites	787
232.2	Both original allocations and all four complementary quotients	787
232.3	Exact transport of the two partial returns	789
232.4	Receiving the estimate in the existing source and action maps	789
233	The original native tail as an attained relation Gram matrix	793
233.1	Original data and the classical tools being used	793
233.2	Finite evaluation of every coefficient and every pairing	794
233.3	The actual relation graph and its full attained Gram	795
233.4	The literal analytic quotient metric and the native phase	796
233.5	Joint minimization of the complete later relations	797
233.6	Finite analytic improvement from the retained cross pairings	798
233.7	The surviving original tail and its precise receiving scope	799
233.8	Exact accepted source locators	799
234	Szegő's confluent modification formula on the full original lower grid	800
234.1	The primary source and its exact original-data dictionary	800
234.2	The full confluent determinant and its leading cofactor	800
234.3	The norm, the cofactor sign and every polynomial coefficient	802
234.4	Evaluation of every original relation coefficient	802
234.5	The full original NTG graph and analytic frame receive the formula	804
234.6	Primary-source read receipt and accepted inputs	804
235	The simultaneous original tail and its eight coefficient flags	806
235.1	Two matrices for the entire original tail	806
235.2	Each original quotient, with its cross terms	807
235.3	The eight flags and the exact allocation identity	808
235.4	Finite comparison with the original row-threshold profile	808
236	The original relation graph supplies the simultaneous mixed flags	810
237	The full original tail through conductor transport and all receiving values	812
237.1	The original data and the two exact tail matrices	812
237.2	Every original quotient and both allocations	813
237.3	Directed conductor transport at the actual evaluation sites	814
237.4	The complete transported threshold profile from the common matrices . . .	816

238	The original invariant value fibre and its full proper-source return	818
238.1	The unchanged domain and actual quotient values	818
238.2	All three full metrics and their attained lifts	819
238.3	Literal two-endpoint matrices on those same values	820
238.4	The full relative return, with the degree-k source intact	821
238.5	The connecting receiver and all compatibility terms	822
239	The analytic tail inside the complete original action	823
239.1	Both stages of the analytic conductor quotient	823
239.2	The exact subquotient and its two complementary factors	825
239.3	Whole-volume return with the native factor inserted	828
239.4	Exact complete action and finite receiving interval	830
239.5	Why the allocation enters twice and cancels once	831
239.6	The same finite joint control in the complete action	831
240	The established low-polynomial bound in the complete packet action	834
240.1	The original metric and the finite bound	834
240.2	The exact dual map and propagation through the receiving equations	836

1 The literal cyclic sectors and the original mixed kernel

This calculation evaluates the cyclic projector already present in the original period observation. Its coefficients retain the original period integrals, amplitude unit, and all primary nilpotents. The result includes an exact finite matrix for the actual rank, as well as unconditional rank bounds obtained from the original symmetric tensor geometry.

1.1 Original data and the Fourier basis of the literal matrix

Use the following data of OPG1–5, including the same contours and their orientations:

$$\begin{aligned} \alpha_{\epsilon,\eta} &= \frac{1}{2} + \epsilon\delta + i\eta\gamma, \quad \epsilon, \eta \in \{+1, -1\}, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \\ h(z_1) &= \prod_{\epsilon,\eta} (z_1 - \alpha_{\epsilon,\eta})^m, \quad n = 4m, \quad N = n + 1, \quad E_h = \mathbb{C}[z_1]/(h), \\ k &= 4l + 1, \quad l, m \geq 1, \quad c = k/2, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \\ \beta_{a,b} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \beta_{a,b})^e, \quad E = \mathbb{C}[S]/(\chi). \end{aligned} \tag{OCS1}$$

Here A_h and A are multiplication by z_1 and S on their respective algebras, and $Y = (A - cI)/i$. The original amplitude unit is $U = M_{j_h v_h}$, with $v_h = (2\xi)/h$ and every original $v_h(\alpha) \neq 0$. For the specified nonzero u and its branch, $\Phi'_h = h$, $\Phi_h(0) = 0$, and

$$\Gamma_j = -\ell_0 + \ell_j, \quad \arg \ell_j = \frac{\arg u + \pi + 2\pi j}{N}, \quad (\Pi[P])_j = \int_{\Gamma_j} e^{\Phi_h(z_1)/u} \text{rem}_h P(z_1) dz_1. \tag{OCS2}$$

The representative has degree less than n . The convergence and invertibility of this original period map are the complete period theorems retained in OPG4. No period value or unit derivative is chosen in the present calculation. The original injection and observation are

$$\begin{aligned} \iota[P] &= [P(z_1 + \cdots + z_k)], \quad \mathcal{V} = \Pi^{\otimes k} U^{\otimes k} \iota, \\ P_k &= \frac{1}{N} \sum_{a=0}^{N-1} (C^{-a})^{\otimes k}, \\ L &= P_k \mathcal{V} = j\Lambda, \quad B = \text{im } L, \quad K = \ker \Lambda. \end{aligned} \tag{OCS3}$$

The map $j : B \hookrightarrow (\mathbb{C}^n)^{\otimes k}$ is the original image inclusion. The injection ι has precisely the defining polynomial in OCS1: on each ordered primary tensor the sum of its k independent nilpotents has index $e = 1 + k(m - 1)$, its top coefficient being $(k(m - 1))!/((m - 1)!)^k \neq 0$, and its distinct eigenvalues are the displayed $\beta_{a,b}$. Thus $\mathcal{V}$ is injective.

On a period-coordinate column $x \in \mathbb{C}^{N-1}$ the literal matrix in OPG4–5 is

$$(Cx)_j = x_{j+1} - x_1 \quad (1 \leq j < N - 1), \quad (Cx)_{N-1} = -x_1. \tag{OCS4}$$

Set $\zeta = \exp(2\pi i/N)$ and, for $1 \leq t \leq N - 1$, let

$$(f_t)_j = \zeta^{jt} - 1, \quad F = [f_1 \ \cdots \ f_{N-1}], \quad (F^{-1}x)_t = \frac{1}{N} \sum_{j=1}^{N-1} \zeta^{-tj} x_j. \tag{OCS5}$$

The last expression is the exact inverse. Indeed $\sum_{j=1}^{N-1} \zeta^{-tj} (\zeta^{uj} - 1) = N \mathbf{1}_{t=u}$ for $t, u \neq 0$ modulo N , by the finite geometric sum. The first $N - 2$ rows of OCS4 give $\zeta^{(j+1)t} - \zeta^t = \zeta^t (\zeta^{jt} - 1)$, and the last row gives $1 - \zeta^t = \zeta^t (\zeta^{(N-1)t} - 1)$. Consequently $Cf_t = \zeta^t f_t$, proving the orientation of every inverse power in OCS3, as well as $C^N = I$. The original Euclidean Gram of this basis is

$$F^* F = N(I_{N-1} + \mathbf{1}\mathbf{1}^*). \tag{OCS6}$$

To verify it, expand $\sum_j (\zeta^{-jt} - 1)(\zeta^{ju} - 1)$ and use the same geometric sums; its value is $N(\mathbf{1}_{t=u} + 1)$. Thus the complete coefficient and metric transformations are both specified.

1.2 The ordered-to-symmetric map and the exact sector projector

Let

$$\mathcal{A}_k = \{\nu \in \mathbb{Z}_{\geq 0}^{N-1} : \sum_t \nu_t = k\}, \quad \text{wt}(\nu) = \sum_{t=1}^{N-1} t\nu_t \pmod{N}, \quad \mathcal{I}_{k,\tau} = \{\nu \in \mathcal{A}_k : \text{wt}(\nu) = \tau\}. \quad (\text{OCS7})$$

For each ν define the following unscaled symmetric tensor:

$$S_\nu = \sum_{\substack{(t_1, \dots, t_k) \in \{1, \dots, N-1\}^k \\ \#\{j: t_j = t\} = \nu_t \text{ for each } t}} f_{t_1} \otimes \dots \otimes f_{t_k}. \quad (\text{OCS8})$$

Every distinct ordered word occurs once. Since the ordered tensors in the f_t form a basis, their disjoint word supports show that the S_ν are linearly independent. A permutation-invariant tensor has equal coefficients on each word orbit, so they span exactly $\text{Sym}^k(\mathbb{C}^{N-1})$, realized as that invariant subspace of the full ordered tensor product. In particular, for $x = (x_1, \dots, x_{N-1})$,

$$(Fx)^{\otimes k} = \sum_{\nu \in \mathcal{A}_k} x^\nu S_\nu, \quad (C^{-a})^{\otimes k} S_\nu = \zeta^{-a \text{wt}(\nu)} S_\nu, \quad P_k(Fx)^{\otimes k} = \sum_{\nu \in \mathcal{I}_{k,0}} x^\nu S_\nu. \quad (\text{OCS9})$$

The finite average in OCS3 is exactly one on weight zero and zero on every other weight, by its geometric sum. No multinomial coefficient is omitted: the multiplicity $k! / \prod_t \nu_t!$ is already the number of summands in OCS8.

Write $\mathcal{F}_k : \mathbb{C}^{\mathcal{A}_k} \rightarrow \text{Sym}^k(\mathbb{C}^{N-1})$ for the basis isomorphism $e_\nu \mapsto S_\nu$. Let J_0 be the coordinate inclusion of $\mathbb{C}^{\mathcal{I}_{k,0}}$ into $\mathbb{C}^{\mathcal{A}_k}$, π_0 the coordinate restriction in the opposite direction, and let $\mathcal{F}_0 = \mathcal{F}_k J_0$, with its target the original ordered tensor space. The exact projector factorization is

$$P_k \mathcal{F}_k = \mathcal{F}_0 \pi_0. \quad (\text{OCS10})$$

All images of $\mathcal{V}$ are symmetric: polynomial evaluation at a sum is invariant under every factor permutation, and the same map ΠU is applied in all factors. Therefore OCS10 applies on the entire original image $\mathcal{V}(E)$.

For completeness, the full inherited metric is retained in these coordinates. Put $G = F^* F$ as in OCS6. The Gram of the basis OCS8 is

$$\mathcal{G}_{\nu, \mu} = \sum_{\substack{a_{tu} \geq 0 \\ \sum_u a_{tu} = \nu_t, \sum_t a_{tu} = \mu_u}} \frac{k!}{\prod_{t,u} a_{tu}!} \prod_{t,u} G_{tu}^{a_{tu}}. \quad (\text{OCS11})$$

Indeed the coefficient of $\bar{x}^\nu y^\mu$ in $(\bar{x}^T G y)^k$ is the right side by the multinomial theorem. Taking the inner product of the two pure tensors in OCS9 gives the same coefficient as $\langle S_\nu, S_\mu \rangle$. This proves OCS11, including its off-diagonal entries; it is positive definite because OCS8 is a basis. On the full symmetric coordinates the projector matrix is $J_0 \pi_0$, and its adjoint for this exact Gram is $\mathcal{G}^{-1}(J_0 \pi_0)^* \mathcal{G}$. Thus the Fourier-sector computation makes no unproved orthogonality or norm assertion about the original cyclic average.

1.3 Every primary period coefficient of the projected curve

For each original root α let $E_{\alpha,a}$, $0 \leq a < m$, be the degree-less-than- $4m$ primary representatives of OPG10. Explicitly, if $h_\alpha = h/(z_1 - \alpha)^m$, put

$$E_{\alpha,a}(z_1) = \text{rem}_h \left[h_\alpha(z_1) \sum_{j=0}^{m-1} \frac{(h_\alpha^{-1})^{(j)}(\alpha)}{j!} (z_1 - \alpha)^{j+a} \right], \quad b_{j;\alpha,a} = \int_{\Gamma_j} e^{\Phi_h(z_1)/u} E_{\alpha,a}(z_1) dz_1. \quad (\text{OCS12})$$

The four primary factors are coprime. The Taylor inverse gives constant one on its own factor and zero on the other three; thus the classes $[E_{\alpha,a}]$ form the full primary basis with all m nilpotent degrees at every root. Define the exact finite Fourier and unit coefficients

$$\widehat{b}_{t;\alpha,a} = \frac{1}{N} \sum_{j=1}^{N-1} \zeta^{-tj} b_{j;\alpha,a}, \quad f_{t;\alpha,d} = \frac{i^{-d}}{d!} \sum_{a=d}^{m-1} \widehat{b}_{t;\alpha,a} \frac{v_h^{(a-d)}(\alpha)}{(a-d)!}, \quad 0 \leq d < m. \quad (\text{OCS13})$$

Let $\omega_\alpha = (\alpha - 1/2)/i$ and set

$$\mathbf{v}(w) = \Pi U \exp(w(A_h - \frac{1}{2}I)/i)[1], \quad x(w) = F^{-1}\mathbf{v}(w), \quad x_t(w) = \sum_{\alpha} e^{\omega_\alpha w} \sum_{d=0}^{m-1} f_{t;\alpha,d} w^d. \quad (\text{OCS14})$$

To verify the final equality, in the primary algebra at α multiply the Taylor series of v_h by the terminating series of $\exp(w(z_1 - \alpha)/i)$. The coefficient on $[E_{\alpha,a}]$ is $e^{\omega_\alpha w} \sum_{d=0}^a v_h^{(a-d)}(\alpha) (w/i)^d / ((a-d)!d!)$. Apply the original polynomial period map and the inverse in OCS5; this gives precisely OCS13–14. In particular, the unreduced entire function has passed through its full primary remainder before any period is integrated.

For $\nu \in \mathcal{I}_{k,0}$, $0 \leq a, b \leq k$ and $0 \leq D < e$, define $c_{\nu;a,b,D}$ by the following finite multinomial rule. Sum over integers $n_{t,\alpha,d} \geq 0$ satisfying

$$\begin{aligned} \sum_{\alpha,d} n_{t,\alpha,d} &= \nu_t \quad \text{for each } t, & \sum_{t,\alpha,d} d n_{t,\alpha,d} &= D, \\ \sum_{t,d,\alpha: \operatorname{Re} \alpha = 1/2 + \delta} n_{t,\alpha,d} &= a, & \sum_{t,d,\alpha: \operatorname{Im} \alpha = +\gamma} n_{t,\alpha,d} &= b. \end{aligned}$$

For these indices put

$$c_{\nu;a,b,D} = \sum_{\text{stated indices}} \left(\prod_{t=1}^{N-1} \frac{\nu_t!}{\prod_{\alpha,d} n_{t,\alpha,d}!} \right) \prod_{t,\alpha,d} f_{t;\alpha,d}^{n_{t,\alpha,d}}. \quad (\text{OCS15})$$

Empty index sets give zero and zero exponents give factors one. Every original primary coefficient is present in this finite rule. The ordinary multinomial expansion of OCS14 now proves

$$x(w)^\nu = \sum_{a,b=0}^k e^{\omega_{a,b} w} \sum_{D=0}^{e-1} c_{\nu;a,b,D} w^D, \quad \omega_{a,b} = (\beta_{a,b} - c)/i = (2b - k)\gamma - i(2a - k)\delta. \quad (\text{OCS16})$$

Indeed $D \leq k(m-1) = e-1$ and the counts of each original sign give exactly the displayed exponent. The sums with different (a,b) are distinct since their real and imaginary parts are independently distinct.

1.4 The actual finite observation matrix and its complete kernel

Define the primary coefficient isomorphism $\mathcal{H} : \mathbb{C}^q \rightarrow E$ by sending coordinate (a,b,D) to $\varepsilon_{a,b}(S - \beta_{a,b})^D$, where $\varepsilon_{a,b}$ is obtained from χ by the same truncated Taylor-inverse construction as OCS12, now of order e . The proof by the coprime factors is identical and shows that this is a basis of all q primary directions. Set

$$\mathfrak{A}_{\nu;(a,b,D)} = i^D D! c_{\nu;a,b,D}, \quad \mathfrak{A} : \mathbb{C}^q \longrightarrow \mathbb{C}^{\mathcal{I}_{k,0}}. \quad (\text{OCS17})$$

Then, on the original spaces and with the original ordered target,

$$\boxed{L\mathcal{H} = \mathcal{F}_0 \mathfrak{A}, \quad K = \mathcal{H}(\ker \mathfrak{A}), \quad \operatorname{rank} \Lambda = \operatorname{rank} \mathfrak{A}.} \quad (\text{OCS18})$$

Here is a proof giving every factorial and phase. In the primary sum algebra,

$$e^{wY}[1] = \sum_{a,b} e^{\omega_{a,b}w} \sum_{D=0}^{e-1} \frac{i^{-D}w^D}{D!} \varepsilon_{a,b}(S - \beta_{a,b})^D.$$

The exact original tensor identity of OPG8 gives $\mathcal{V}e^{wY}[1] = \mathbf{v}(w)^{\otimes k}$: the centre $c = k/2$ splits into the k original centres $1/2$, while U commutes with multiplication by z_1 . Applying the literal projector and then OCS9 and OCS16 gives the right-hand curve with coefficients $\mathcal{F}_0 c_{a,b,D}$. Exponential polynomials $w^D e^{\omega_{a,b}w}$ with these distinct exponents are linearly independent. To prove this directly, apply $\prod_{(a',b') \neq (a,b)} (\partial_w - \omega_{a',b'})^e$ to any zero linear combination. It kills all the other exponents. On the selected polynomial factor, each remaining operator is $\partial_w + (\omega_{a,b} - \omega_{a',b'})$, invertible on the polynomials of degree less than e because its constant term is nonzero and differentiation is nilpotent there. The surviving polynomial must therefore vanish. Doing this for every pair proves the asserted independence. Equating coefficients now gives $L\mathcal{H}e_{a,b,D} = \mathcal{F}_0(i^D D! c_{a,b,D})$. This proves the first equality in OCS18 on a basis. The other two follow from injectivity of $\mathcal{F}_0$ and j .

In particular the isomorphism from the actual sector image to the original boundary is

$$\mathcal{Q} : \text{im } \mathfrak{A} \xrightarrow{\sim} B, \quad \mathcal{Q}x = j^{-1}\mathcal{F}_0x, \quad E/K \xrightarrow{\sim} \text{im } \mathfrak{A}, \quad [z] \mapsto \mathfrak{A}\mathcal{H}^{-1}z. \quad (\text{OCS19})$$

Both inverses exist on precisely their displayed images by OCS18. These are full coefficient maps; the inherited Gram of $\mathcal{F}_0$ is $J_0^* \mathcal{G} J_0$ from OCS11.

For a completely explicit finite rank prescription, define

$$\Delta_r(\mathfrak{A}) = \sum_{\substack{I \subseteq \mathcal{I}_{k,0}, \ J \subseteq \{(a,b,D)\} \\ |I|=|J|=r}} |\det \mathfrak{A}_{I,J}|^2, \quad \Delta_0 = 1. \quad (\text{OCS20})$$

Every entry of every minor is the finite expression OCS12–17 in the actual periods and amplitude derivatives. Gaussian elimination proves that a matrix has rank at least r exactly when one of its r by r minors is nonzero: pivot rows and columns supply such a minor, and a nonzero minor supplies r independent columns. Hence the actual rank is exactly $\max\{r : \Delta_r(\mathfrak{A}) > 0\}$. This evaluates its finite deciding data without prescribing generic values or asserting that any unexamined maximal minor is nonzero.

The Gamma vectors used in the existing endpoint increments have the equally explicit coordinates

$$j\lambda_d = \mathcal{F}_0 \hat{\lambda}_d, \quad (\hat{\lambda}_d)_\nu = \frac{1}{\sqrt{h_d^{(s)}}} \sum_{a,b=0}^k \sum_{D=0}^{e-1} c_{\nu;a,b,D} p_d^{(D)}(\omega_{a,b}), \quad h_d^{(s)} = (2\pi)^{s/2} d!(s/2)_d, \quad s \in \{1, k\}. \quad (\text{OCS21})$$

Taylor expansion of $p_d(Y)[1]$ in the sum primary basis gives the coefficient $i^{-D} p_d^{(D)}(\omega_{a,b})/D!$. Multiplication by OCS17 cancels exactly these factorials and phases, proving OCS21. This also proves that the generating formula of OPG9 becomes $(1+z^2)^{-s/4} \mathcal{F}_0(x(\arctan z)^\nu)_{\nu \in \mathcal{I}_{k,0}}$, with the original scalar exponent and both existing sources.

1.5 Exact invariant dimensions and unconditional forced losses

Put $d_{N,k,\tau} = |\mathcal{I}_{k,\tau}|$. By OCS7–10 this is the dimension of the full symmetric sector, and its exact finite character formula is

$$d_{N,k,0} = [T^k] \frac{1}{N} \sum_{h|N} \varphi(h) \frac{1-T}{(1-T^h)^{N/h}}. \quad (\text{OCS22})$$

To prove it, the trace generating series for C^a on symmetric tensors is $\prod_{t=1}^{N-1} (1 - T\zeta^{at})^{-1}$, by summing the monomial basis OCS8. If ζ^a has order h , the list ζ^{at} with $0 \leq t < N$ contains every

h th root N/h times. Their product is $\prod_{t=0}^{N-1} (1 - T\zeta^{at}) = (1 - T^h)^{N/h}$. Removing the $t = 0$ factor gives the series $(1 - T)/(1 - T^h)^{N/h}$. There are exactly $\varphi(h)$ indices a of that order, so averaging the traces of the original finite projector proves OCS22. The inverse powers give the same average by $a \mapsto -a$. In ordinary finite binomial terms, for $k \geq 1$ this is

$$d_{N,k,0} = \frac{1}{N} \sum_{h|N} \varphi(h) \left\{ \mathbf{1}_{h|k} \binom{k/h + N/h - 1}{N/h - 1} - \mathbf{1}_{h|(k-1)} \binom{(k-1)/h + N/h - 1}{N/h - 1} \right\}. \quad (\text{OCS23})$$

The binomial expansion of $(1 - T^h)^{-N/h}$ follows by counting weak compositions, proving the coefficient extraction used here. It follows from the actual matrix OCS17 that

$$\text{rank } \Lambda \leq \min(q, d_{N,k,0}), \quad \dim K \geq \max(0, q - d_{N,k,0}). \quad (\text{OCS24})$$

These inequalities apply to the original map, since its full image factorization was proved in OCS18; the ambient dimension has not been equated with its actual rank. There is also an unconditional nonzero observation for the original $k \geq 5$. Each entire function $x_t(w)$ in OCS14 is nonzero. Otherwise its nonzero coordinate covector would annihilate every derivative at zero of $\Pi U \exp(w(A_h - 1/2)/i)[1]$. Those derivatives span the full space, since $[1]$ is cyclic for multiplication by z_1 and ΠU is invertible. A product of finitely many nonzero entire functions is nonzero. The zero sector is nonempty: take the weights $1, 1, N - 2$ and then $(k - 3)/2$ pairs $1, N - 1$. Their sum is a multiple of N . Its monomial curve in OCS9 is nonzero, so $\text{rank } \Lambda \geq 1$.

For $m = 1$, $N = 5$, and every $k \geq 0$, write

$$D_k = \binom{k+3}{3}, \quad \epsilon_k = \begin{cases} 1 & k \equiv 0 \pmod{5}, \\ -1 & k \equiv 1 \pmod{5}, \\ 0 & \text{otherwise.} \end{cases} \quad (\text{OCS25})$$

$$d_{5,k,0} = \frac{D_k + 4\epsilon_k}{5}, \quad d_{5,k,\tau} = \frac{D_k - \epsilon_k}{5} \quad (\tau \neq 0).$$

For each of the four nonidentity powers of C , the trace series is $(1 - T)/(1 - T^5)$, whose coefficient is ϵ_k . The weight- τ projector introduces the Fourier multiplier; the sum of its four nontrivial fifth roots is 4 when $\tau = 0$ and -1 otherwise. This proves both formulas. For the requested original orders the exact counts and resulting kernel lower bounds are

k	D_k	$d_{5,k,0}$	$q = (k+1)^2$	$\max(0, q - d_{5,k,0})$
5	56	12	36	24
9	220	44	100	56
13	560	112	196	84
17	1140	228	324	96
21	2024	404	484	80
25	3276	656	676	20
29	4960	992	900	0

(OCS26)

These are integer evaluations of OCS25. The strict inequality $d_{5,k,0} < q$ holds for $k = 4l + 1$, $l \geq 1$, exactly for the first six orders in the table. They exhaust $k \leq 25$ in this progression. For $k \geq 29$,

$$D_k - 5(k+1)^2 = \frac{(k+1)(k^2 - 25k - 24)}{6} \geq 460 > 4,$$

where $k^2 - 25k - 24$ is positive and increasing from $k = 29$. Even the most negative correction $4\epsilon_k \geq -4$ therefore gives $d_{5,k,0} > q$. This proves the threshold without extrapolating the finite table.

1.6 The actual simple-quartet quadric and a stronger rank bound

Continue with $m = 1$. Order the four roots as $(++)$, $(+-)$, $(-+)$, $(--)$ and let $\mathcal{E}_{\text{prim}} : \mathbb{C}^4 \rightarrow E_h$ be the primary basis isomorphism whose columns are the four $[E_{\alpha,0}]$. Define the actual invertible matrix and the original exponential quartet by

$$\mathbf{H} = F^{-1} \Pi U \mathcal{E}_{\text{prim}}, \quad t_{\epsilon, \eta}(w) = e^{(\eta\gamma - i\epsilon\delta)w}, \quad x(w) = \mathbf{H}t(w). \quad (\text{OCS27})$$

Every column of $\mathbf{H}$ is OCS13 with $d = 0$, namely the Fourier transform of its original primary period times its actual nonzero $v_h(\alpha)$. Each displayed map is invertible, so $\mathbf{H}$ is invertible without a genericity assumption. Put $\mathcal{R}_k = \mathbb{C}[x_1, x_2, x_3, x_4]_k$, the homogeneous polynomials of degree k , and define

$$Q_0(t) = t_{++}t_{--} - t_{+-}t_{-+}, \quad Q(x) = Q_0(\mathbf{H}^{-1}x). \quad (\text{OCS28})$$

This is the literal transformed quadratic relation of the original curve; its coefficients are determined by the periods and unit through OCS27. In particular $Q \neq 0$.

The kernel of homogeneous restriction to that curve is exactly

$$\ker\{\mathcal{R}_k \rightarrow \text{Hol}(\mathbb{C}), f \mapsto f(x(w))\} = Q\mathcal{R}_{k-2} \quad (k \geq 2). \quad (\text{OCS29})$$

Here is a complete verification. First work in the t coordinates. A degree- k monomial has exponent $(2b - k)\gamma - i(2a - k)\delta$, where a is its count of first sign $+$ and b its count of second sign $+$. These $(k + 1)^2$ complex exponents are distinct: equality of their real parts forces b equal and equality of their imaginary parts forces a equal. Equivalently, the two frequencies 2γ and $-2i\delta$ admit no nonzero integer relation, by their real and imaginary parts. Thus the exponentials have no extra polynomial relation of this homogeneous type; their linear independence follows, for example, by differentiating $0, \dots, (k + 1)^2 - 1$ times and using the Vandermonde determinant of the distinct exponents. Every pair (a, b) occurs: the count r_{++} may be any integer between $\max(0, a + b - k)$ and $\min(a, b)$, a nonempty interval, and the other three counts are $a - r_{++}$, $b - r_{++}$, and $k - a - b + r_{++}$.

Within any fixed pair (a, b) , two monomial exponent vectors differ by an integer multiple of $(1, -1, -1, 1)$. After their common monomial is factored out, their difference is therefore the difference of equal powers of $t_{++}t_{--}$ and $t_{+-}t_{-+}$. The identity $A^r - B^r = (A - B) \sum_{j=0}^{r-1} A^{r-1-j} B^j$ shows that this difference is divisible by Q_0 . Any polynomial vanishing on the curve has coefficient sum zero within each pair (a, b) , by the proved exponential independence. It is thus a sum of these monomial differences and is divisible by Q_0 . Its quotient is homogeneous of degree $k - 2$. Conversely $Q_0(t(w)) = 0$, since each of its two products is one. This proves the assertion in t coordinates, and the invertible substitution OCS27 proves OCS29. In particular $\dim(\mathcal{R}_k / Q\mathcal{R}_{k-2}) = (k + 1)^2 = q$; no information about the original curve has been replaced by a freely chosen quadric.

Let $\mathcal{I}_k \subseteq \mathcal{R}_k$ be the span of the weight-zero monomials x^ν , using the weights $1, 2, 3, 4$ from the exact Fourier basis. The coordinate functions of the original projected curve in OCS9 are precisely these monomials restricted to $x(w)$. The span of the vector curve $Le^{wY}[1]$ equals $\text{im } L$: its derivatives at zero contain $LY^d[1]$, $0 \leq d < q$, and $Y^d[1]$ is a basis because Y is an invertible affine change of multiplication by S in the degree filtration. The rank of that vector curve equals the dimension of the span of its coordinate functions, by the equality of row and column ranks of its finite exponential coefficient matrix. Hence OCS29 proves the exact equality

$$\text{rank } \Lambda = d_{5,k,0} - \dim(\mathcal{I}_k \cap Q\mathcal{R}_{k-2}). \quad (\text{OCS30})$$

Fix any multiplicative monomial order, for instance lexicographic order, and write x^μ for the leading monomial of the actual Q , with $|\mu| = 2$. Choose a row-echelon basis of the finite-dimensional intersection in OCS30 in descending monomial order. Its leading monomials are distinct, all have weight zero, and all are divisible by x^μ : the leading monomial of Qf is the product of those of Q and f because the order is multiplicative and the coefficient field has no zero divisors. Thus each such leading monomial is $x^\mu x^\eta$ for a distinct degree- $k - 2$ monomial of weight $-\text{wt}(\mu)$. Counting this set and using OCS30 gives

$$d_{5,k,0} - d_{5,k-2,-\text{wt}(\mu)} \leq \text{rank } \Lambda \leq \min(q, d_{5,k,0}). \quad (\text{OCS31})$$

The left side depends only on the actual leading monomial of the actual period quadric, and remains valid whatever its other coefficients are. A uniform consequence retaining the strongest count independent of that monomial is

$$r_k^- := d_{5,k,0} - \max_{\tau \in \mathbb{Z}/5} d_{5,k-2,\tau} = \left\lfloor \frac{(k+1)^2}{5} \right\rfloor + \mathbf{1}_{5|k} \leq \text{rank } \Lambda. \quad (\text{OCS32})$$

For the equality, $D_k - D_{k-2} = (k+1)^2$, as direct subtraction of the two cubic binomials shows. Formula OCS25 then gives

$$r_k^- = \frac{(k+1)^2 + 4\epsilon_k - \max(4\epsilon_{k-2}, -\epsilon_{k-2})}{5}.$$

For k congruent respectively to $0, 1, 2, 3, 4$ modulo five, its numerator correction is respectively $4, -4, -4, -1, 0$. The corresponding residues of $(k+1)^2$ are $1, 4, 4, 1, 0$. These five cases prove OCS32 for every $k \geq 2$. For the original orders in OCS26 the complete unconditional intervals are therefore

k	r_k^-	$\min(q, d_{5,k,0})$	$\max(0, q - d_{5,k,0})$	$q - r_k^-$
5	8	12	24	28
9	20	44	56	80
13	39	112	84	157
17	64	228	96	260
21	96	404	80	388
25	136	656	20	540
29	180	900	0	720

(OCS33)

The middle two rank columns bound $\text{rank } \Lambda$; the last two columns bound $\dim K$ from below and above, respectively. No endpoint of either interval has been assigned as the actual value without its original minors.

1.7 A second exact finite matrix for the simple-quartet kernel

The quadric calculation also gives the actual kernel a polynomial quotient model. Write $Q(x) = \sum_{|\mu|=2} q_\mu x^\mu$. Let T_Q be the full multiplication matrix $\mathcal{R}_{k-2} \rightarrow \mathcal{R}_k$ in the monomial bases:

$$(T_Q)_{\nu,\eta} = q_{\nu-\eta}, \quad |\nu| = k, \quad |\eta| = k-2, \quad q_{\nu-\eta} = 0 \text{ if an entry of } \nu - \eta < 0.$$

Let T_{out} be its rows of nonzero cyclic weight. Every coefficient q_μ is explicitly fixed by $Q_0(\mathbf{H}^{-1}x)$, and the inverse matrix may be computed as $\text{adj } \mathbf{H} / \det \mathbf{H}$, with nonzero denominator proved in OCS27. Multiplication by Q is injective. The condition that Qf belongs to $\mathcal{I}_k$ is exactly $T_{\text{out}}f = 0$. Thus OCS30 gives the further exact rank identities

$$\begin{aligned} \text{rank } \Lambda &= d_{5,k,0} - D_{k-2} + \text{rank } T_{\text{out}}, \\ \dim K &= D_k - d_{5,k,0} - \text{rank } T_{\text{out}}. \end{aligned} \quad (\text{OCS34})$$

The ranks of this concrete matrix are decided by its finite minors, by the same elimination proof as OCS20. This is an alternative exact calculation from the actual periods, not an assumption about the quadric's coefficients.

To give the typed map behind the kernel equality, let E^* denote the complex linear dual. Define $\Theta : \mathcal{R}_k \rightarrow E^*$ as follows: $\Theta(f)$ is the unique covector such that

$$\Theta(f)(e^{wY}[1]) = f(x(w)) \quad \text{for all } w. \quad (\text{OCS35})$$

For $m = 1$ every function on the right is a linear combination of the q distinct exponentials in OCS16. The primary expansion of $e^{wY}[1]$ gives exactly those exponentials as a basis of covector functions. This proves existence and uniqueness. The proof of OCS29 proves surjectivity and

$\ker \Theta = Q\mathcal{R}_{k-2}$. Moreover $\Theta(\mathcal{I}_k) = \text{im } L^\vee$, where $L^\vee$ denotes the complex linear dual map: the monomial coefficient functionals in the basis OCS8 pull back along L to exactly OCS35 for x^ν of weight zero. These functionals span the full dual of the actual image, since every linear functional on a subspace of a finite-dimensional space extends to the whole space. Restriction of E^* to K has kernel $\text{im } L^\vee$: a covector vanishing on K factors uniquely through $E/K \simeq \text{im } L$. Thus the exact maps give

$$\boxed{\text{coker } T_{\text{out}} \simeq \mathcal{R}_k / (\mathcal{I}_k + Q\mathcal{R}_{k-2}) \xrightarrow{\Theta|_K} K^*}. \quad (\text{OCS36})$$

For the first isomorphism, use the direct monomial splitting of $\mathcal{R}_k$ into its zero and nonzero sectors; after quotienting by $\mathcal{I}_k$, multiplication by Q becomes precisely T_{out} . For the second, the surjectivity and kernel were just proved. This identifies the actual lost directions by a full quotient map.

1.8 All primary action directions remain in the mixed calculation

For arbitrary original $m \geq 1$, retain the full primary matrix $\mathfrak{A}$ in OCS17. In the basis $\mathcal{H}$ the original sum action is exactly

$$J e_{a,b,D} = \omega_{a,b} e_{a,b,D} + \begin{cases} i^{-1} e_{a,b,D+1} & D < e-1, \\ 0 & D = e-1, \end{cases} \quad Y\mathcal{H} = \mathcal{H}J. \quad (\text{OCS37})$$

This follows by multiplying $\varepsilon_{a,b}(S - \beta_{a,b})^D$ by $(S - c)/i = \omega_{a,b} + (S - \beta_{a,b})/i$ in its original primary algebra. Consequently the actual possible action leaving the cyclic kernel is the fully specified map

$$\ker \mathfrak{A} \longrightarrow \text{im } \mathfrak{A}, \quad x \longmapsto \mathfrak{A}Jx, \quad \Lambda Y\mathcal{H}x = \mathcal{Q}\mathfrak{A}Jx. \quad (\text{OCS38})$$

The codomain assertion follows from $\mathfrak{A}Jx$ being a value of the same matrix $\mathfrak{A}$ on Jx . For the original inclusion $I : K \hookrightarrow E$, this is the OPG22 defect transported through the exact kernel isomorphism in OCS18. In particular none of the discarded sectors has been declared invariant under Y .

The part of the kernel undetected by all original iterates is exactly computed by the stacked matrix

$$\mathfrak{D} = \begin{bmatrix} \mathfrak{A} \\ \mathfrak{A}J \\ \vdots \\ \mathfrak{A}J^{q-1} \end{bmatrix}, \quad \mathcal{K}_\infty = \mathcal{H} \ker \mathfrak{D}, \quad \dim \mathcal{K}_\infty = q - \text{rank } \mathfrak{D}, \quad \dim(K/\mathcal{K}_\infty) = \text{rank } \mathfrak{D} - \text{rank } \mathfrak{A}. \quad (\text{OCS39})$$

Indeed $\mathfrak{A}J^a x$ is the exact sector-coordinate value of $\Lambda Y^a \mathcal{H}x$ by OCS18 and OCS37. The polynomial $\chi(c + iT)$ annihilates J and has degree q with leading coefficient i^q . It expresses every higher power in the span of the first q , so the stacked kernel is precisely the intersection over all iterates and is J -invariant. Rank-nullity proves both dimension formulas. Its quotient identifies the directions that the original one-step observation loses and subsequent original iterates detect.

Equations OCS17–21, OCS34–39 supply actual finite maps and deciding minors for the original observation and its kernel. The forced kernel losses in OCS26 and the rank intervals OCS33 concern this unchanged projector and these unchanged periods. The endpoint volume calculation still uses the full positive sums of the actual vectors λ_d in OPG16–18 and their exact inherited coordinates OCS11 and OCS19. No volume growth estimate is deduced from these dimension bounds alone.

2 The original quadric orbit and an explicit invariant lifting map

This calculation uses the original simple-quartet construction, with $m = 1$, $k = 4l + 1$, $l \geq 1$, $q = (k + 1)^2$, $0 < \delta < 1/2$, and $\gamma > 2$. The period parameter, its branch, the polynomial contours, the complete amplitude unit, and the observation are those of OPG and OCS. The degree-eight polynomial below is the product of four actual orbit equations. Its degree and all of its coefficients are derived from that orbit; no source order is selected to match an estimate.

2.1 The actual coefficient ring and the original observation

Order the original roots as $(++)$, $(+-)$, $(-+)$, $(--)$, with $\alpha_{\epsilon, \eta} = 1/2 + \epsilon\delta + i\eta\gamma$. Let $\mathcal{E}_{\text{prim}}$ have columns the four original primary idempotents in $E_h = \mathbb{C}[z_1]/(h)$. For the original matrices Π, U, F in OCS2, OCS5, OCS27 put

$$\begin{aligned} \mathbf{H} &= F^{-1}\Pi U \mathcal{E}_{\text{prim}}, & t_{\epsilon, \eta}(w) &= e^{(\eta\gamma - i\epsilon\delta)w}, \\ x(w) &= \mathbf{H}t(w), & Q(x) &= Q_0(\mathbf{H}^{-1}x), & Q_0(t) &= t_{++}t_{--} - t_{+-}t_{-+}. \end{aligned} \quad (\text{OQC1})$$

Every matrix in $\mathbf{H}$ is the specified original isomorphism; in particular its nonzero amplitude factors are the actual $v_h(\alpha)$, and its period entries are the original convergent polynomial integrals. Thus $\det \mathbf{H} \neq 0$ by the included period-isomorphism and primary-basis proofs. Define

$$\mathcal{P}_d = \mathbb{C}[x_1, x_2, x_3, x_4]_d, \quad R_d = \mathcal{P}_d / (Q \mathcal{P}_{d-2}), \quad \mathcal{P}_d = R_d = 0 \quad (d < 0). \quad (\text{OQC2})$$

The matrix of $Q_0(t) = t^T M_0 t$ has entries $(M_0)_{14} = (M_0)_{41} = 1/2$ and $(M_0)_{23} = (M_0)_{32} = -1/2$, all others zero. Hence

$$M = \mathbf{H}^{-T} M_0 \mathbf{H}^{-1}, \quad \det M = \frac{1}{16(\det \mathbf{H})^2} \neq 0. \quad (\text{OQC3})$$

In particular Q is irreducible. Indeed, a reducible homogeneous quadratic over $\mathbb{C}$ is a product of two linear forms, whose symmetric coefficient matrix is a sum of two rank-one matrices and has rank at most two. This contradicts OQC3. The polynomial ring over a field is a unique factorization domain, by the repeated one-variable Gauss lemma, so its irreducible element Q is prime. Consequently $R = \mathbb{C}[x_1, x_2, x_3, x_4]/(Q)$ is an integral domain.

For clarity, the complete homogeneous restriction calculation has the precise kernel $Q \mathcal{P}_{d-2}$. In the t coordinates a degree- d monomial restricts to

$$\exp(((2b - d)\gamma - i(2a - d)\delta)w), \quad 0 \leq a, b \leq d. \quad (\text{OQC4})$$

Every pair occurs: the number of $(++)$ factors can range from $\max(0, a + b - d)$ to $\min(a, b)$. The exponents are distinct by their real and imaginary parts. Their exponentials are linearly independent, since differentiating orders zero through $(d + 1)^2 - 1$ gives a Vandermonde matrix with nonzero determinant. Two monomials giving the same pair have exponent vectors differing by a multiple of $(1, -1, -1, 1)$. After their common monomial is removed, their difference is divisible by $t_{++}t_{--} - t_{+-}t_{-+}$ through $A^r - B^r = (A - B) \sum_{j=0}^{r-1} A^{r-1-j} B^j$. A vanishing polynomial has coefficient sum zero within each pair by the independence just proved, and is therefore a sum of these differences. Conversely $Q_0(t(w)) = 0$. Transport by $\mathbf{H}$ proves the stated kernel and

$$\dim R_d = (d + 1)^2 \quad (d \geq 0). \quad (\text{OQC5})$$

Write $\zeta = e^{2\pi i/5}$ and $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. The literal Fourier calculation gives $CF = FD$. On polynomials define the original induced action and its average by

$$(gf)(x) = f(D^{-1}x), \quad g^5 = 1, \quad \mathbf{R}f = \frac{1}{5} \sum_{a=0}^4 g^a f, \quad \mathcal{I}_d = \mathcal{P}_d^{(g)}. \quad (\text{OQC6})$$

Thus $\mathcal{J}_d$ consists of the weight-zero monomials, since $gx^\nu = \zeta^{-\sum j\nu_j} x^\nu$. This is the dual of the original average of $(C^{-a})^{\otimes d}$; both the inverse powers and the factor $1/5$ are retained.

For the original algebra E_d of the d -fold sum, define $\Theta_d : R_d \xrightarrow{\sim} E_d^\vee$ by

$$\Theta_d([f])(e^{wY_d}[1]) = f(x(w)), \quad Y_d = (A_d - dI/2)/i. \quad (\text{OQC7})$$

Here $\vee$ denotes the complex-linear dual. The primary expansion of the vector on the left gives all distinct exponentials OQC4; therefore existence and uniqueness of this covector and bijectivity follow from the preceding complete restriction calculation. The original observation $\Lambda_d : E_d \rightarrow B_d$ obeys

$$J_d := \text{im}(\mathcal{J}_d \rightarrow R_d), \quad \Theta_d(J_d) = \text{im } \Lambda_d^\vee, \quad \dim J_d = \text{rank } \Lambda_d, \quad R_d/J_d \xrightarrow{\sim} (\ker \Lambda_d)^\vee. \quad (\text{OQC8})$$

Indeed the coordinate functions of the projected pure tensor in OCS9 are precisely $x(w)^\nu$ of weight zero; their pulled-back covectors span the dual of its actual image. A covector vanishing on $\ker \Lambda_d$ factors through that image, which proves the last isomorphism as well. OQC8 specifies the exact original image; it has not been replaced by the whole invariant tensor space.

2.2 The complete orbit decision from the actual coefficients

Put $Q_a = g^a Q$ for $0 \leq a < 5$. The orbit of the line $\mathbb{C}Q$ has size one or five because five is prime. Moreover a size-one orbit has $gQ = Q$, with no other possible scalar. To prove this, write $gQ = \lambda Q$. Then $\lambda^5 = 1$, while OQC3 and the determinant of D , which is $\zeta^{10} = 1$, imply

$$\det(D^{-T}MD^{-1}) = \det M = \lambda^4 \det M. \quad (\text{OQC9})$$

Thus $\lambda^4 = 1$ and $\lambda^5 = 1$, forcing $\lambda = 1$. If any nonidentity power fixed the line, its powers generate the same cyclic group, so the preceding argument applies. This proves the entire orbit classification.

There is an exact coefficient test, requiring no genericity premise. Write the actual $Q = \sum_{|\mu|=2} q_\mu x^\mu$ and put

$$\mathfrak{e}(Q) = \sum_{\substack{|\mu|=2 \\ \sum j\mu_j \not\equiv 0 \pmod{5}}} |q_\mu|^2. \quad (\text{OQC10})$$

All coefficients here are the entries of OQC3, including the full inverse of the original period matrix. Since the monomials are a basis and each summand is nonnegative, $\mathfrak{e}(Q) = 0$ exactly when Q is invariant. The only degree-two weight-zero monomials are x_1x_4 and x_2x_3 . Hence the size-one case is exactly $Q = ax_1x_4 + bx_2x_3$ with $ab \neq 0$; nonvanishing of the two coefficients follows from OQC3. Positive $\mathfrak{e}(Q)$ gives the five distinct irreducible orbit equations. These are exhaustive outcomes of a test on the given periods, not assumptions on them.

In the size-one outcome, multiplication by Q preserves every weight. If Qf is invariant, then $Q(gf - f) = 0$ in the polynomial ring, and $gf = f$. Therefore

$$\ker(\mathcal{J}_k \rightarrow R_k) = Q\mathcal{J}_{k-2}, \quad \text{rank } \Lambda_k = d_{5,k,0} - d_{5,k-2,0}, \quad \dim \ker \Lambda_k = q - d_{5,k,0} + d_{5,k-2,0}. \quad (\text{OQC11})$$

This also proves that $J_k = R_k^{(g)}$: an invariant class can be lifted to a polynomial, and averaging that lift gives an invariant representative. The count itself follows from

$$d_{5,k,0} = \frac{\binom{k+3}{3} + 4\epsilon_k}{5}, \quad \epsilon_k = \mathbf{1}_{k \equiv 0 \pmod{5}} - \mathbf{1}_{k \equiv 1 \pmod{5}}. \quad (\text{OQC12})$$

For a proof, the identity element has symmetric-power generating series $(1-z)^{-4}$; each of the other four powers has series $\prod_{j=1}^4 (1 - z\zeta^j)^{-1} = (1-z)/(1-z^5)$. Averaging their coefficients gives

OQC12. Subtraction and $\binom{k+3}{3} - \binom{k+1}{3} = (k+1)^2$ evaluate the rank in OQC11, for every $k \geq 2$, as

$$\left\lfloor \frac{(k+1)^2}{5} \right\rfloor + \mathbf{1}_{k \equiv 0 \text{ or } 3 \pmod{5}}. \quad (\text{OQC13})$$

The five numerator corrections before division by five are 4, -4, -4, 4, 0, respectively; the residues of $(k+1)^2$ are 1, 4, 4, 1, 0. This verifies every case in OQC13.

2.3 An explicit invariant lift for the five-member orbit

For the size-five outcome determined by OQC10, retain the full actual products

$$\mathcal{H}_Q = \prod_{a=0}^4 Q_a \quad (\deg \mathcal{H}_Q = 10), \quad \mathcal{A}_Q = \prod_{a=1}^4 Q_a \quad (\deg \mathcal{A}_Q = 8). \quad (\text{OQC14})$$

The action permutes all five factors cyclically, so $g\mathcal{H}_Q = \mathcal{H}_Q$. None of the four factors of $\mathcal{A}_Q$ is a scalar multiple of Q . Since each is irreducible and Q is prime, multiplication by $[\mathcal{A}_Q]$ is injective on the integral domain R .

For every homogeneous $f \in \mathcal{P}_d$ the exact trace identity is

$$[5R(\mathcal{A}_Q f)]_{(Q)} = [\mathcal{A}_Q f]_{(Q)}. \quad (\text{OQC15})$$

Indeed $g^a \mathcal{A}_Q$ is the product of all orbit factors except Q_a . For $a \neq 0$ it contains $Q_0 = Q$. Those four terms vanish in the displayed quotient, and the $a = 0$ term is exactly the right side. This proves OQC15 with its original average and full factors, including every period-dependent coefficient.

The lift also has a well-defined typed quotient map. Let $\mathcal{U} = \mathbb{C}[x_1, x_2, x_3, x_4]/(\mathcal{H}_Q)$; the invariant equation $\mathcal{H}_Q$ makes the cyclic action well defined on this quotient. Then

$$\begin{aligned} \mathfrak{L}_d : R_d &\longrightarrow (\mathcal{U}_{d+8})^{(g)}, & [f]_{(Q)} &\longmapsto [5R(\mathcal{A}_Q f)]_{(\mathcal{H}_Q)}, \\ \text{res } \mathfrak{L}_d([f]) &= [\mathcal{A}_Q f]_{(Q)}, & \text{res} : \mathcal{U} &\rightarrow R. \end{aligned} \quad (\text{OQC16})$$

If f changes by Qb , its proposed image changes by $5R(\mathcal{H}_Q b) = 5\mathcal{H}_Q Rb$, hence by zero in $\mathcal{U}$. This proves well-definedness. The image is invariant because it has an invariant polynomial representative; OQC15 proves the second line. Injectivity follows from the injectivity of multiplication by $[\mathcal{A}_Q]$ on R . All maps in OQC16 are thus established, including their kernels.

The actual original invariant image consequently contains

$$[\mathcal{A}_Q]R_{k-8} \subseteq J_k \subseteq R_k. \quad (\text{OQC17})$$

The first inclusion follows by taking the invariant polynomial representatives OQC15, and uses no dimension inference. Therefore, for every $k \geq 8$ in this outcome,

$$\text{rank } \Lambda_k \geq (k-7)^2, \quad \dim \ker \Lambda_k \leq (k+1)^2 - (k-7)^2 = 16k - 48. \quad (\text{OQC18})$$

These inequalities follow from the proved injection and OQC5. They may be combined with the unconditional actual-matrix bounds in OCS24, OCS31–33 by taking the stronger lower and upper bounds. For the original degrees $k = 9, 13, 17, 21, 25, 29$, the displayed upper bounds on the original kernel are respectively 96, 160, 224, 288, 352, 416; at the first two degrees the earlier unconditional bounds may be stronger. No such upper bound is assigned to the size-one outcome, whose exact rank is OQC11. Together OQC10–18 compute the complete orbit alternatives from the original coefficients. Evaluating $\mathfrak{e}(Q)$ in the original period family is a separate calculation on those explicit coefficients; no outcome is postulated here.

2.4 The induced original dual map and its complete action defect

At $k \geq 9$ the degree $d = k - 8$ is positive, and the original d -fold sum algebra is defined using the same h, Π, U . For the size-five outcome, OQC7, OQC8 and OQC17 give the uniquely typed injective map

$$\begin{aligned} \mathfrak{J}_k : E_{k-8}^\vee &\hookrightarrow B_k^\vee, \\ \Lambda_k^\vee \mathfrak{J}_k &= \Theta_k M_{[\mathcal{A}_Q]} \Theta_{k-8}^{-1}, \quad \mathfrak{J}_k^\vee : B_k \twoheadrightarrow E_{k-8}. \end{aligned} \quad (\text{OQC19})$$

The right side of the second line has image in $\text{im } \Lambda_k^\vee$ by OQC17, and $\Lambda_k^\vee$ is injective because Λ_k is onto its actual image. This gives existence and uniqueness. Each of the three right-hand maps is injective, so $\mathfrak{J}_k$ is injective. Finite-dimensional duality then proves the stated surjection, using the canonical bidual identifications. The new degree $k - 8$ records contraction against the degree-eight polynomial derived in OQC14. It does not replace either original Gamma source of order 1 or k .

The surviving action and its defect are fully determined. Put

$$\Omega = \text{diag}(\gamma - i\delta, -\gamma - i\delta, \gamma + i\delta, -\gamma + i\delta), \quad G = H\Omega H^{-1}, \quad (\mathcal{D}f)(x) = \sum_{j=1}^4 (Gx)_j \partial_j f(x). \quad (\text{OQC20})$$

Here $x'(w) = Gx(w)$ by OQC1. In the original t coordinates, the weights of each product in Q_0 sum to zero, so $\mathcal{D}Q = 0$. Thus this derivation acts on each R_d . Differentiating OQC7 proves exactly

$$\Theta_d \mathcal{D} = Y_d^\vee \Theta_d. \quad (\text{OQC21})$$

Both sides evaluated at $e^{wY_d}[1]$ are the derivative of $f(x(w))$, and these vectors span E_d by the distinct exponentials; hence equality holds as maps, not only on a curve. Leibniz's rule gives the complete defect

$$\begin{aligned} Y_k^\vee \Lambda_k^\vee \mathfrak{J}_k - \Lambda_k^\vee \mathfrak{J}_k Y_{k-8}^\vee &= \Theta_k M_{[\mathcal{D}\mathcal{A}_Q]} \Theta_{k-8}^{-1}, \\ \mathcal{D}\mathcal{A}_Q &= \sum_{a=1}^4 (\mathcal{D}Q_a) \prod_{\substack{1 \leq b \leq 4 \\ b \neq a}} Q_b. \end{aligned} \quad (\text{OQC22})$$

The codomain in the first line is $E_k^\vee$. Its image has not been asserted to lie in $\text{im } \Lambda_k^\vee$. Restriction to $\ker \Lambda_k$ supplies the exact remaining map into $(\ker \Lambda_k)^\vee$, through OQC8. In particular OQC19 is not used as an unproved intertwining of boundary actions. OQC22 retains every term that can leave the observed dual subspace.

Finally, the failure of the cyclic average to commute with the original action is explicitly calculable before any quotient:

$$([\mathcal{D}, g^a]f)(x) = (\nabla f)(D^{-a}x)^T (D^{-a}G - GD^{-a})x, \quad [\mathcal{D}, R] = \frac{1}{5} \sum_{a=0}^4 [\mathcal{D}, g^a]. \quad (\text{OQC23})$$

The chain rule applied separately to $f(D^{-a}x)$ and to $(\mathcal{D}f)(D^{-a}x)$ proves both identities. Thus all action comparisons in the invariant lift have explicit original coefficient maps. The rank estimates in OQC18 concern exactly the kernel used in the original Gamma and arithmetic quotient Grams, because those constructions use the same Λ_k . An estimate for its volume still requires those original metrics; no volume conclusion has been inferred solely from its dimension.

3 Exact original rank from the complete quadric orbit

Retain every original object and map in OQC1–23. In particular $m = 1$, $k = 4l + 1$, $l \geq 1$, $q = (k + 1)^2$, $\mathcal{P} = \mathbb{C}[x_1, x_2, x_3, x_4]$, and $gf(x) = f(D^{-1}x)$ for the original Fourier matrix $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. The polynomial $Q = Q_0(H^{-1}x)$ has its actual original period and amplitude coefficients. Write $\mathcal{I} = \mathcal{P}^{(g)}$, $R = \mathcal{P}/(Q)$ and $J = \text{im}(\mathcal{I} \rightarrow R)$, with the grading and the actual observation identification of OQC8. For negative indices put $\mathcal{I}_d = 0$ and $d_{5,d,0} = 0$.

The orbit decision is the exhaustive test OQC10. The size-one case already has the exact rank OQC11–13. The following calculation completes the other case using the same original coefficients. It does not assert that a period parameter belongs to that case before its coefficient test has been evaluated.

3.1 Every invariant vanishing polynomial contains the full orbit

In the size-five case, let $Q_a = g^a Q$ and retain exactly $\mathcal{H}_Q = \prod_{a=0}^4 Q_a$ from OQC14. Each Q_a is irreducible, because an invertible linear substitution preserves factorizations and Q is irreducible by OQC3. The five factors are pairwise nonassociate by the proved orbit classification, and each is prime in the original polynomial ring. Consequently

$$\mathcal{I} \cap Q\mathcal{P} = \mathcal{H}_Q\mathcal{I}, \quad \ker(\mathcal{I}_d \rightarrow R_d) = \mathcal{H}_Q\mathcal{I}_{d-10}. \quad (\text{OQX1})$$

Here is the complete divisibility proof. If $f \in \mathcal{I}$ and $f = Qb$, then for every $a = 0, \dots, 4$ the exact equality $g^a f = f$ gives $f = Q_a g^a b$. Thus every one of the five original prime factors divides f . Distinctness of the factors implies that their product divides f : after factoring out Q_0 , primality of Q_1 and the fact that Q_1 does not divide Q_0 force Q_1 to divide the remaining quotient. Repeating this argument with Q_2, Q_3, Q_4 proves $f = \mathcal{H}_Q b_0$. The action permutes the five exact factors cyclically, so $g\mathcal{H}_Q = \mathcal{H}_Q$. Applying g to this last equality and using $gf = f$ gives $\mathcal{H}_Q(gb_0 - b_0) = 0$. The polynomial ring is an integral domain and $\mathcal{H}_Q \neq 0$, whence $gb_0 = b_0$. This proves the first inclusion in OQX1. Conversely an invariant b_0 gives an invariant $\mathcal{H}_Q b_0$ divisible by Q , which proves the reverse inclusion. The product is homogeneous of degree ten. If its product with b_0 is homogeneous of degree d , decomposition of b_0 into homogeneous parts shows that only its part of degree $d - 10$ can be nonzero: each other part has zero product with $\mathcal{H}_Q$ and hence is zero. This proves the graded kernel formula, including $d < 10$.

3.2 The exact invariant quotient and the restriction map

Let $\mathcal{U} = \mathcal{P}/(\mathcal{H}_Q)$ as in OQC16. The following are isomorphisms of graded complex algebras with their specified maps:

$$\mathcal{I}/(\mathcal{H}_Q\mathcal{I}) \xrightarrow{[f] \mapsto [f]_{(\mathcal{H}_Q)}} \mathcal{U}^{(g)} \xrightarrow{\text{res}} J \subseteq R. \quad (\text{OQX2})$$

The action on $\mathcal{U}$ is defined because $\mathcal{H}_Q$ is invariant. If $[f]_{(\mathcal{H}_Q)}$ is invariant, averaging its representatives gives $[Rf]_{(\mathcal{H}_Q)} = [f]_{(\mathcal{H}_Q)}$, since $[g^a f] = [f]$ for every a . Thus the first map is onto. Its kernel before passage to the displayed domain is $\mathcal{I} \cap \mathcal{H}_Q\mathcal{P} = \mathcal{H}_Q\mathcal{I}$: the reverse inclusion is immediate, and for $f = \mathcal{H}_Q b \in \mathcal{I}$, cancellation in $\mathcal{H}_Q(gb - b) = 0$ proves $b \in \mathcal{I}$. This proves that the first map is injective as well.

The full quotient restriction $\text{res} : \mathcal{U} \rightarrow R$ is well defined because $(\mathcal{H}_Q) \subseteq (Q)$; its kernel is exactly $Q\mathcal{P}/\mathcal{H}_Q\mathcal{P}$. For its invariant subspace, choose the invariant representative just constructed. If its restriction vanishes, that representative belongs to $\mathcal{I} \cap Q\mathcal{P}$, which is $\mathcal{H}_Q\mathcal{I}$ by OQX1. Its class in $\mathcal{U}$ is therefore zero. The invariant restriction is injective. Its image is exactly J , since an invariant representative restricts to J by definition and every element of J has such a representative. Both maps are induced by polynomial quotient maps, so they preserve products, sums, complex scalars and degrees. No action of g on R is asserted in the size-five case. In particular the second map retains precisely the invariant image of the original observation, with its complete restriction kernel already computed.

3.3 All original degrees and the exact kernel dimension

The map Θ_k and the original observation in OQC8 identify $\dim J_k$ with $\text{rank } \Lambda_k$ and identify R_k/J_k with $(\ker \Lambda_k)^\vee$. Multiplication by the nonzero polynomial $\mathcal{H}_Q$ is injective in $\mathcal{P}$. Taking dimensions in OQX1 thus gives, for every nonnegative degree d , the exact invariant restriction $\text{rank } \dim J_d = d_{5,d,0} - d_{5,d-10,0}$. At the original degrees this evaluates to

$$\boxed{\text{rank } \Lambda_k = d_{5,k,0} - d_{5,k-10,0} = k^2 - 6k + 17, \quad \dim \ker \Lambda_k = 8k - 16, \quad k = 4l + 1, \quad l \geq 1,} \quad (\text{OQX3})$$

in the size-five outcome of the actual coefficient test. To verify every constant and degree range, for $k \geq 10$ the complete character formula OQC12 gives

$$d_{5,k,0} - d_{5,k-10,0} = \frac{1}{5} \left\{ \binom{k+3}{3} - \binom{k-7}{3} + 4(\epsilon_k - \epsilon_{k-10}) \right\}.$$

The two ϵ values are equal because their indices differ by ten. Expanding the remaining cubics gives

$$\binom{k+3}{3} - \binom{k-7}{3} = \frac{(k^3 + 6k^2 + 11k + 6) - (k^3 - 24k^2 + 191k - 504)}{6} = 5(k^2 - 6k + 17).$$

The only original degrees below ten are five and nine. Their negative-degree terms in OQX1 vanish, while the same exact character formula gives

$$d_{5,5,0} = (56 + 4)/5 = 12 = 5^2 - 6 \cdot 5 + 17, \quad d_{5,9,0} = 220/5 = 44 = 9^2 - 6 \cdot 9 + 17.$$

This proves the first equality of OQX3 for every original $l \geq 1$. Since $\dim E_k = \dim R_k = (k+1)^2$, rank-nullity gives $(k+1)^2 - (k^2 - 6k + 17) = 8k - 16$, proving the second equality. For the original degrees already tabulated in OCS26, the exact five-orbit values are

k	$\text{rank } \Lambda_k$	$\dim \ker \Lambda_k$
5	12	24
9	44	56
13	108	88
17	204	120
21	332	152
25	492	184
29	684	216

These are exact values for this exhaustively classified orbit outcome, replacing the less precise bounds OQC18 wherever that outcome occurs. The full invariant lift and its action defect OQC15–23 remain valid with all original coefficients. The size-one outcome retains its different exact values OQC11–13. No choice between the two outcomes has been made here without evaluating the original period coefficient test, and no estimate for a metric volume has been inferred from its dimension alone.

4 The actual period quadric and its cyclic continuation

This calculation uses the original simple quartet, $m = 1$, and the literal period matrix, amplitude unit and cyclic matrix of BRU and OPG. Its formulas evaluate the transported quadric from those data. All source orders elsewhere in the programme remain unchanged.

4.1 Original roots, entire unit and centred phase

Order the roots and keep their actual values as

$$\begin{aligned} \alpha_1 &= \frac{1}{2} + \delta + i\gamma, & \alpha_2 &= \frac{1}{2} + \delta - i\gamma, & \alpha_3 &= \frac{1}{2} - \delta + i\gamma, & \alpha_4 &= \frac{1}{2} - \delta - i\gamma, \\ 0 < \delta < \frac{1}{2}, & \gamma > 2, & h(s) &= \prod_{j=1}^4 (s - \alpha_j), & g &= 2\xi, & v_h &= g/h. \end{aligned} \quad (\text{PQO1})$$

The four zero orders of g are complete and equal to one. Thus v_h is entire and has nonzero value at each displayed root. Put $w = s - 1/2$, and define the following original constants:

$$\begin{aligned} z_+ &= \delta + i\gamma, & z_- &= \delta - i\gamma, & \lambda &= z_+^2, & \mu &= z_-^2, \\ A &= 2(\gamma^2 - \delta^2), & B &= (\delta^2 + \gamma^2)^2, & \mathfrak{a} &= \Phi_h(1/2) = \frac{1}{160} + \frac{A}{24} + \frac{B}{2}, \\ h(1/2 + w) &= w^4 + Aw^2 + B, & \Phi_h(1/2 + w) &= \mathfrak{a} + \frac{w^5}{5} + \frac{Aw^3}{3} + Bw. \end{aligned} \quad (\text{PQO2})$$

The product of the two quadratics $((w - \delta)^2 + \gamma^2)((w + \delta)^2 + \gamma^2)$ proves the quartic identity. Integrating it and using $\Phi_h(0) = 0$ proves the phase and constant in PQO2. In particular the translation has not removed the original constant $\mathfrak{a}$.

The theta integral IPM.2 has real coefficients, real theta kernel, and an expression unchanged by $s \mapsto 1 - s$. It therefore gives $g(\bar{s}) = \overline{g(s)}$ and $g(1 - s) = g(s)$ directly; the same two identities hold for h . Taking their entire quotient gives

$$v_h(\alpha_1) = v_h(\alpha_4) = a, \quad v_h(\alpha_2) = v_h(\alpha_3) = \bar{a}, \quad a \neq 0. \quad (\text{PQO3})$$

These are the full actual unit values. Their magnitude and phase are not assigned. Set $t_+ = a^{-2}$ and $t_- = \bar{a}^{-2}$. For $j = 0, 1, 2, 3$ retain

$$\sigma_j = t_+ \lambda^j - t_- \mu^j, \quad \sigma_2 = -A\sigma_1 - B\sigma_0, \quad \sigma_3 = (A^2 - B)\sigma_1 + AB\sigma_0. \quad (\text{PQO4})$$

The recurrence follows because $\lambda + \mu = -A$ and $\lambda\mu = B$. Each σ_j is purely imaginary. Furthermore σ_0 and σ_1 cannot both vanish: the first would give $t_+ = t_- \neq 0$, and the second would then give $t_+(\lambda - \mu) = 0$, whereas $\lambda - \mu = 4i\delta\gamma \neq 0$.

4.2 The original periods as an absolutely convergent series

Fix $u \neq 0$ on its original logarithm branch. Let $\ell_j(r) = r \exp(i(\arg u + \pi + 2\pi j)/5)$, $\Gamma_j = -\ell_0 + \ell_j$, $1 \leq j \leq 4$. Let Π integrate the degree-less-than-four representative against $e^{\Phi_h(s)/u} ds$ on these original oriented contours. Write $F_{ja} = \zeta^{ja} - 1$, $\zeta = e^{2\pi i/5}$, $1 \leq j, a \leq 4$. Direct finite geometric sums prove

$$(F^{-1})_{aj} = \frac{1}{5} \zeta^{-ja}, \quad CF = FD, \quad D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4), \quad (\text{PQO5})$$

where $(Cx)_j = x_{j+1} - x_1$ for $j < 4$ and $(Cx)_4 = -x_1$. For example multiplying the asserted inverse by F gives $\frac{1}{5} \sum_{j=1}^4 \zeta^{-ja} (\zeta^{jb} - 1) = \delta_{ab}$; substitution into each row gives $CF = FD$.

Let T be the exact increasing-coefficient translation matrix $T_{rb} = \binom{b}{r} 2^{-(b-r)}$ for $0 \leq r \leq b \leq 3$, and zero otherwise. Thus it sends coefficients of $p(s)$ to those of $p(1/2 + w)$, with determinant one. Define the centred Fourier-period matrix

$$P_{ar}(u) = \frac{1}{5} \sum_{j=1}^4 \zeta^{-ja} \int_{\Gamma_j} e^{(w^5/5 + Aw^3/3 + Bw)/u} w^r dw, \quad 1 \leq a \leq 4, \quad 0 \leq r \leq 3. \quad (\text{PQO6})$$

The contours in this formula start at $w = 0$. They give precisely the translated original period by

$$F^{-1}\Pi = e^{a/u}P(u)T. \quad (\text{PQO7})$$

To justify this exact contour translation, the substitution $w = s - 1/2$ first gives rays starting at $-1/2$. Join each shifted ray at a large radius to the corresponding unshifted ray. On the joining segment their arguments differ by $O(1/r)$, and the leading real phase is $-r^5/(5|u|) + O(r^4)$; all lower phase terms and polynomial factors are dominated by this negative leading term. Those outer joining integrals tend to zero. The two common-origin joining integrals cancel in $-\ell_0 + \ell_j$ with their opposite orientations. Cauchy's theorem for the entire integrand proves equality in the limit. Substitution of the polynomial gives exactly the matrix T .

For every integer $L \geq 1$ define, on the same branch,

$$G_L(u) = u^{L/5} 5^{L/5-1} e^{\pi i L/5} \Gamma(L/5). \quad (\text{PQO8})$$

Integration of the leading phase on the two rays gives $\int_{\Gamma_j} e^{w^5/(5u)} w^{L-1} dw = G_L(u)(\zeta^{jL} - 1)$ by the real substitution $r^5/(5|u|)$ on each ray, including its phase in dw . Expanding the remaining two exponentials now gives the exact entries

$$P_{ar}(u) = \sum_{\substack{p,q \geq 0 \\ r+3p+q+1 \equiv a \pmod{5}}} \frac{A^p B^q}{3^p p! q! u^{p+q}} G_{r+3p+q+1}(u). \quad (\text{PQO9})$$

Terms with $r+3p+q+1 \equiv 0$ have zero two-ray integral in every column and contribute zero before the Fourier map. To prove every exchange, the sum of the absolute values of the unintegrated series on either ray is bounded by a constant times

$$R^r \exp\left(-\frac{R^5}{5|u|} + \frac{|A|R^3}{3|u|} + \frac{|B|R}{|u|}\right).$$

Here R is the radial coordinate and r is the fixed column degree. This integrable majorant proves absolute termwise integration. It is uniform on compact coefficient and nonzero- u sets on the chosen branch. The finite Fourier sum in PQO5 then selects exactly the stated residue class. Thus PQO9 is an actual period formula, with all coefficients and all terms retained.

For additional finite computation, let I_d denote the vector of the four centred ray moments of w^d . Integration of the derivative of $e^{(w^5/5 + Aw^3/3 + Bw)/u} w^d$, with the same vanishing outer endpoints and cancelling inner endpoints, proves

$$I_{d+4} + AI_{d+2} + BI_d = -u d I_{d-1} \quad (d \geq 0), \quad u \partial_B I_d = I_{d+1}, \quad 3u \partial_A I_d = I_{d+3}. \quad (\text{PQO10})$$

The first right side is zero at $d = 0$. Coefficient derivatives are justified by the same compact domination. These are the original polynomial period recurrences, and they compute every higher moment from the first four and the actual coefficients.

The determinant is especially explicit in these centred coordinates:

$$\det \Pi = -(2\pi)^2 u^2 e^{4a/u}, \quad \det F = -25\sqrt{5}, \quad \boxed{\det P(u) = \frac{(2\pi)^2}{25\sqrt{5}} u^2}. \quad (\text{PQO11})$$

For the first equality use the full PDE1–13 determinant calculation at $n = 4$: the phase is $e^{15\pi i} = -1$ and the critical-value sum is $4a$. This critical sum also follows directly by pairing w with $-w$ in the odd centred phase. For F , subtract row zero from the other rows in the five-by-five Fourier Vandermonde and expand its first column. Its phase is $e^{13\pi i} = -1$ and its magnitude $5^{5/2}$, by the finite Fourier orthogonality identity. The last equality follows by taking determinants in PQO7 and using $\det T = 1$. The Gamma grid-product proof PGCC1–6 retains the full evaluated constant in the first equality. In particular every inverse below exists on the full stated domain.

4.3 The actual transformed quadric and every deciding coefficient

Let V be the centred evaluation matrix with rows $(1, z_j, z_j^2, z_j^3)$, where $(z_1, z_2, z_3, z_4) = (z_+, z_-, -z_-, -z_+)$. Its inverse sends the four primary values to their unique degree-less-than-four polynomial. Thus the actual matrix from primary coordinates to the Fourier-period space is

$$B_{\text{per}} = F^{-1} \Pi U J_{\text{prim}} = e^{a/u} P(u) V^{-1} \text{diag}(a, \bar{a}, \bar{a}, a). \quad (\text{PQO12})$$

This follows either by applying all maps to the four CRT idempotents or by evaluating the full unit at each root. It agrees with OCS and OPG in the given original orientation.

For $Q_0(t) = t_1 t_4 - t_2 t_3$, define $Q_u(x) = Q_0(B_{\text{per}}^{-1} x)$. On the unchanged original period coordinate $\mathbf{y} \in \mathbb{C}^4$ the quadratic is $Q_u^{\text{per}}(\mathbf{y}) = Q_0((\Pi U J_{\text{prim}})^{-1} \mathbf{y}) = Q_u(F^{-1} \mathbf{y})$, because $F B_{\text{per}} = \Pi U J_{\text{prim}}$. Thus the exact isomorphism $\mathbf{y} = Fx$ carries the following quadric to the original period space. If $p(w) = p_0 + p_1 w + p_2 w^2 + p_3 w^3$, multiplication gives

$$p(w)p(-w) = p_0^2 + (2p_0 p_2 - p_1^2)w^2 + (p_2^2 - 2p_1 p_3)w^4 - p_3^2 w^6.$$

Consequently the exact symmetric matrix in the centred coefficient frame is

$$S = \begin{pmatrix} \sigma_0 & 0 & \sigma_1 & 0 \\ 0 & -\sigma_1 & 0 & -\sigma_2 \\ \sigma_1 & 0 & \sigma_2 & 0 \\ 0 & -\sigma_2 & 0 & -\sigma_3 \end{pmatrix}, \quad \boxed{Q_u(x) = x^\top H(u)x, \quad H(u) = e^{-2a/u} P(u)^{-\top} S P(u)^{-1}.}$$

(PQO13)

Indeed applying the preceding product formula at z_+ and z_- with multipliers t_+ and $-t_-$ gives exactly S . This retains both actual unit values through every σ_j . The two parity blocks have determinants

$$\sigma_0 \sigma_2 - \sigma_1^2 = -t_+ t_- (\lambda - \mu)^2, \quad \sigma_1 \sigma_3 - \sigma_2^2 = -t_+ t_- B(\lambda - \mu)^2.$$

Thus S and $H(u)$ are nondegenerate, with the exact determinant

$$\det H(u) = e^{-8a/u} \frac{B(t_+ t_-)^2 (\lambda - \mu)^4}{(\det P(u))^2} \neq 0. \quad (\text{PQO14})$$

Write $N(u) = \text{adj } P(u)$, with entries N_{ra} , $0 \leq r \leq 3$, $1 \leq a \leq 4$, and retain $\Delta(u) = \det P(u)$ from PQO11. Then every coefficient is given by the explicit minor formula

$$H_{ab}(u) = \frac{e^{-2a/u}}{\Delta(u)^2} \{ \sigma_0 N_{0a} N_{0b} + \sigma_1 (N_{0a} N_{2b} + N_{2a} N_{0b} - N_{1a} N_{1b}) \\ + \sigma_2 (N_{2a} N_{2b} - N_{1a} N_{3b} - N_{3a} N_{1b}) - \sigma_3 N_{3a} N_{3b} \}. \quad (\text{PQO15})$$

This is matrix multiplication in PQO13 using $P^{-1} = N/\Delta$. Each N_{ra} is the signed original three-by-three period minor. Its entries are given by the absolutely convergent series PQO9 or by the original integrals PQO6. The polynomial coefficient of x_a^2 is H_{aa} ; that of $x_a x_b$, $a < b$, is $2H_{ab}$. Both factors are retained in this specified quadratic form.

The exact eight deciding entries are

$$\mathcal{E}(u) = (H_{11}, H_{22}, H_{33}, H_{44}, H_{12}, H_{13}, H_{24}, H_{34}). \quad (\text{PQO16})$$

The cyclic action on quadrics is the pullback $Q_u \mapsto Q_u(D^{-1}x)$; its symmetric matrix is $D^{-\top} H(u) D^{-1}$. Since $D^5 = I$ and $\det D = 1$, the line spanned by a nondegenerate form can be stable only with character one. In fact if its multiplier is η , then $\eta^5 = 1$ by the group action, while taking determinants gives $\eta^4 = 1$ because $\det H(u) \neq 0$. Thus $\eta = 1$. Entrywise, invariance is precisely $(\zeta^{-(a+b)} - 1)H_{ab} = 0$. The only pairs with $a + b \equiv 0$ are $(1, 4)$ and $(2, 3)$. Therefore

The quadric is invariant exactly on $Z = \{u : \mathcal{E}(u) = 0\}$;
 $u \in Z \implies Q_u(x) = 2H_{14}(u)x_1 x_4 + 2H_{23}(u)x_2 x_3, \quad H_{14}(u)H_{23}(u) \neq 0$;
 at every $u \notin Z$ its orbit consists of five distinct quadrics.

(PQO17)

Every assertion follows from the entry test and the rank-four determinant. The orbit size divides five; it is one exactly when the line is stable. Thus no two-member orbit occurs. Moreover $F^{-1}C^{-1} = D^{-1}F^{-1}$ by PQO5, so this is exactly the orbit of $Q_u^{\text{per}}(C^{-j}\mathbf{y})$ in the original period coordinates, with the same five indexed maps. PQO15–17 determine the exact exceptional locus by original period minors and original amplitude values, without assigning generic values to them.

4.4 The cyclic action is the actual original continuation in u

Continue the original logarithm of u by $2\pi i$. Its rays advance from ℓ_j to ℓ_{j+1} , and their oriented differences advance by the exact original matrix C . For the last row $\ell_5 = \ell_0$, giving $-\Gamma_1$. Equivalently every summand of PQO9 acquires the phase

$$e^{2\pi i((r+3p+q+1)/5-p-q)} = \zeta^{r+3p+q+1} = \zeta^a.$$

The original constant $e^{a/u}$ is single-valued under this continuation. Hence

$$\begin{aligned} P(\log u + 2\pi i) &= DP(\log u), & B_{\text{per}}(\log u + 2\pi i) &= DB_{\text{per}}(\log u), \\ H(\log u + 2\pi i) &= D^{-\top} H(\log u) D^{-1}. \end{aligned} \tag{PQO18}$$

The contour argument and the convergent series prove the same map. Thus PQO17 describes the actual period continuation of this original quadric. It introduces no freely chosen cyclic action on its coefficients.

4.5 The large-parameter geometry and a positive-ray obstruction

Let $\varepsilon = u^{-1/5}$ on the same branch, and set $D_\varepsilon = \text{diag}(\varepsilon^{-1}, \varepsilon^{-2}, \varepsilon^{-3}, \varepsilon^{-4})$. The exact series PQO9 gives

$$P(u) = \mathcal{P}(A\varepsilon^2, B\varepsilon^4)D_\varepsilon, \quad \mathcal{P}(0,0) = \text{diag}(g_1, g_2, g_3, g_4), \quad g_a = 5^{a/5-1}e^{\pi ia/5}\Gamma(a/5). \tag{PQO19}$$

The entries of $\mathcal{P}$ are entire in the two displayed arguments, by the same absolute integrable series majorant. The same determinant proof PQO11 applies to every complex pair of lower odd-phase coefficients, since the centred phase remains odd and its critical values pair to zero with full multiplicities. Thus their determinant is the nonzero constant $(2\pi)^2/(25\sqrt{5})$, by PQO11 and $\det D_\varepsilon = u^2$. Their inverse is therefore holomorphic everywhere, given by its adjugate divided by that precise constant. Near $\varepsilon = 0$ it is $\text{diag}(g_a^{-1}) + O(\varepsilon^2)$. This bound follows from its convergent power series after substituting $(A\varepsilon^2, B\varepsilon^4)$; it holds on a full small complex disc, with the fixed original A, B .

If $\sigma_0 \neq 0$, PQO13 and PQO19 give

$$e^{2a/u}H_{11}(u) = \frac{\sigma_0}{g_1^2}\varepsilon^2 + O(\varepsilon^4).$$

If $\sigma_0 = 0$, then $\sigma_1 \neq 0$ by PQO4, and they give

$$e^{2a/u}H_{22}(u) = -\frac{\sigma_1}{g_2^2}\varepsilon^4 + O(\varepsilon^6). \tag{PQO20}$$

For verification of the orders, insert $P^{-1} = D_\varepsilon^{-1}\mathcal{P}^{-1}$ into PQO13. The middle matrix has entries $S_{rs}\varepsilon^{r+s+2}$. Its first possible degree is two, from S_{00} , and its next possible degree is four, from S_{02}, S_{20}, S_{11} . The constant term of $\mathcal{P}^{-1}$ is diagonal, so the indicated leading entry is exactly the displayed nonzero coefficient. Every further contribution has the stated higher degree. This also proves the expansion when some other σ_j vanish.

Consequently for every fixed original quartet and its full actual amplitude there exists a finite radius outside which the actual orbit has five distinct quadrics. This conclusion holds on every original logarithm branch: the estimate is on the full complex ε disc, and its leading coefficient

is nonzero. The set Z of PQO17 is a discrete subset of the logarithmic u surface. Indeed the decisive entry just proved is holomorphic on that connected surface and is not identically zero; its zeros are isolated by its local power series, and Z is contained in that zero set. The explicit finite-radius bound from the companion period-connection calculation strengthens this argument while retaining the same actual coefficients.

Moreover PQO18 multiplies each deciding entry by its nonzero root-of-unity factor, so the condition $\mathcal{E}(u) = 0$ is unchanged by every full logarithm turn. Therefore Z is the inverse image under $\log u \mapsto u$ of a well-defined set $Z_{\text{abs}} \subset \mathbb{C}^*$. On a sufficiently small disc about any nonzero u , choose one logarithm branch. The preceding isolated-zero proof shows that Z_{abs} is discrete and closed on that disc. The explicit radius in the companion calculation gives $|u| < R_*$ throughout Z_{abs} . Consequently its only possible accumulation point in the finite u -plane is zero. In particular each compact annulus about zero contains only finitely many such points: an infinite subset of that compact set would have a nonzero accumulation point, contradicting the proved local isolated-zero property.

For positive real u retain the actual lift $\log u = \ln u + 2\pi i \ell_{\text{br}}$, $\ell_{\text{br}} \in \mathbb{Z}$. The exact series has an additional real structure. Every surviving index in PQO9 is $a + 5L$, and its phase at the lift $\ell_{\text{br}} = 0$ is $e^{\pi i a/5}(-1)^L$. Its remaining coefficients are real. The actual lift multiplies row a by $\zeta^{\ell_{\text{br}} a}$, by PQO18. Absolute convergence therefore proves

$$P(u) = E_{\ell_{\text{br}}} R(u), \quad E_{\ell_{\text{br}}} = D^{\ell_{\text{br}}} \text{diag}(e^{\pi i/5}, e^{2\pi i/5}, e^{3\pi i/5}, e^{4\pi i/5}), \quad R(u) \text{ real and invertible.} \quad (\text{PQO21})$$

Let M be multiplication by w in the increasing centred polynomial frame, so

$$M = \begin{pmatrix} 0 & 0 & 0 & -B \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -A \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad c_3 = \frac{t_+ - t_-}{4i\delta\gamma}, \quad c_1 = \frac{t_+ + t_-}{2} - (\delta^2 - \gamma^2)c_3. \quad (\text{PQO22})$$

Both c_1, c_3 are real by PQO3, and the polynomial $c_1 + c_3 w^2$ takes values t_+, t_-, t_-, t_+ at the four original roots. Hence $c_1 M + c_3 M^3$ is precisely multiplication by $w(j_h v_h)^{-2}$, with its entire unit retained.

The companion period-connection proof establishes the exact identity between cyclic invariance of Q_u and

$$[P(u)(c_1 M + c_3 M^3)P(u)^{-1}, D] = 0. \quad (\text{PQO23})$$

It proves this by the flat residue/reflection form on the original polynomial periods and retains all its constants. Using that proved identity, PQO21 gives a further precise obstruction on the positive ray. Since D has distinct diagonal entries, a matrix commuting with it is diagonal. The matrix in PQO23 is $E_{\ell_{\text{br}}}$ times a real matrix times $E_{\ell_{\text{br}}}^{-1}$; if diagonal, its diagonal entries must therefore be real. Its eigenvalues are exactly

$$\frac{\delta + i\gamma}{a^2}, \quad \frac{\delta - i\gamma}{\bar{a}^2}, \quad -\frac{\delta - i\gamma}{\bar{a}^2}, \quad -\frac{\delta + i\gamma}{a^2}, \quad (\text{PQO24})$$

as follows by evaluation at the four distinct roots of h . Thus every positive-real- u point in Z satisfies the explicit original-unit equation

$$\text{Im} \left(\frac{\delta + i\gamma}{a^2} \right) = 0. \quad (\text{PQO25})$$

If that imaginary part is nonzero for the original unit, the positive ray contains no point of Z . When it is zero, the four displayed eigenvalues are the two nonzero real numbers $r, -r$, each twice, and the complete period-minor tests PQO15–17 still determine Z . The equation is not assumed to hold or fail. Together the exact minor tests, the actual period continuation, the proved large-parameter result and this real-structure obstruction give control of the original quadric family with every amplitude value and phase present.

5 The original period quadric as a residue commutator

Retain $0 < \delta < 1/2$, $\gamma > 2$, the ordered simple quartet $\alpha_1 = 1/2 + \delta + i\gamma$, $\alpha_2 = 1/2 + \delta - i\gamma$, $\alpha_3 = 1/2 - \delta + i\gamma$, $\alpha_4 = 1/2 - \delta - i\gamma$, and $h(s) = \prod_a (s - \alpha_a)$. Put

$$w = s - \frac{1}{2}, \quad r_a = \alpha_a - \frac{1}{2}, \quad \beta = 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad H(w) = w^4 + \beta w^2 + \eta. \quad (\text{RPC1})$$

Thus $r_4 = -r_1$, $r_3 = -r_2$, $r_2 = \overline{r_1}$ and $r_1^2 - r_2^2 = 4i\delta\gamma$. No root or original coordinate is discarded by this translation. Its coefficient isomorphism is $J_c : \mathbb{C}[w]_{<4} \rightarrow E_h$, $J_c p = [p(s - 1/2)]$; its inverse is substitution $s = w + 1/2$ in the unique original representative. The primitive with its original constant is

$$\Phi_h(s) = A + \phi(w), \quad \phi(w) = \frac{w^5}{5} + \frac{\beta w^3}{3} + \eta w, \quad A = \Phi_h(1/2) = \frac{1}{160} + \frac{\beta}{24} + \frac{\eta}{2}. \quad (\text{RPC2})$$

The last equality is forced by $\Phi_h(0) = 0$ and follows by evaluating the displayed odd polynomial at $w = -1/2$.

Let $u \neq 0$ carry the specified original branch of its logarithm. Let Π be the original period map on the rays and orientations of PDE2, and let $F_{ja} = \zeta^{ja} - 1$, $\zeta = e^{2\pi i/5}$, $1 \leq j, a \leq 4$. Its inverse has entries $(F^{-1})_{aj} = \zeta^{-ja}/5$: multiplying the two matrices and using $\sum_{j=1}^4 \zeta^{jb} = 4$ for $5 \mid b$ and -1 otherwise proves this entry by entry. The literal centered coefficient-to-Fourier map is

$$P = F^{-1} \Pi J_c, \quad P_0 = e^{-A/u} P. \quad (\text{RPC3})$$

These are invertible by the complete original period determinant PDE3–13. The contours for P_0 may be taken to be the rays based at $w = 0$ with the original five angles. Indeed translating the old rays gives rays based at $-1/2$. Join corresponding finite endpoints and close each contour at radius R . The integrand is entire. On the bounded-length outer joining curves its exponent is $-R^5/(5|u|) + O(R^4)$, uniformly along the joining curves, so the outer integrals tend to zero. The common finite joining path occurs with opposite coefficients in $-\ell_0 + \ell_j$ and cancels exactly. Cauchy's integral theorem therefore proves this contour identity, including its signs and the factor $e^{A/u}$.

5.1 A flat alternating pairing with all constants

On $\mathbb{C}[w]/(H)$ define the bilinear, rather than sesquilinear, pairing

$$\mathcal{J}(p, q) = \sum_{H(r)=0} \frac{p(r)q(-r)}{H'(r)}. \quad (\text{RPC4})$$

The roots are simple at the original packet. The equivalent polynomial formula is the coefficient of w^3 in the remainder of $p(w)q(-w)$ modulo H . To prove it, use Lagrange interpolation: the coefficient of w^3 in the idempotent at r is $1/H'(r)$. This polynomial formula extends the pairing to every complex value of β, η , including repeated roots, without a division by a repeated-root value. In the increasing frame $1, w, w^2, w^3$, multiplication by w and this pairing are the exact matrices

$$M = \begin{pmatrix} 0 & 0 & 0 & -\eta \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & -\beta \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad J = \begin{pmatrix} 0 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & \beta \\ 1 & 0 & -\beta & 0 \end{pmatrix}. \quad (\text{RPC5})$$

For the last matrix, the coefficient functional is zero through degree two, one in degree three, zero in degree four and six, and $-\beta$ in degree five. Inserting the sign $(-1)^j$ from the second argument gives every displayed entry. Thus $J^\top = -J$, $\det J = 1$, and direct multiplication gives $M^\top J + JM = 0$.

Allow the two original odd-phase coefficients to vary complexly, solely to calculate the existing period map. The ray domination $e^{-r^5/(5|u|)+O(r^3+r)}$ on compact parameter sets justifies all coefficient derivatives and integrations by parts below. Ordinary division of w^{j+3} by H , followed by differential reduction, gives

$$\partial_\eta P_0 = \frac{1}{u} P_0 M, \quad \partial_\beta P_0 = \frac{1}{3u} P_0 (M^3 - uN), \quad N = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}. \quad (\text{RPC6})$$

For clarity, the quotient derivatives for columns $j = 0, 1, 2, 3$ are $0, 0, 1, 2w$, respectively; hence this is the full correction matrix. The two outer boundary terms vanish, and the two values at the common origin cancel. No differential remainder was replaced by an arithmetic remainder in RPC6.

There is also a direct nonvanishing proof on this entire two-parameter family. The displayed matrices satisfy $\text{Tr } M = 0$, $\text{Tr } M^3 = 0$ and $\text{Tr } N = 0$. The first and third follow from their diagonals; the second follows by multiplying RPC5 three times, or from the vanishing third Newton sum of the even monic polynomial H . Multilinearity of the determinant in its columns then gives $\partial_\eta \det P_0 = \partial_\beta \det P_0 = 0$ from RPC6, without first assuming an inverse exists. At the origin the literal Gamma ray integrals give the diagonal matrix G_u defined in RPC8, whose determinant is nonzero. Hence P_0 is invertible at every point of $\mathbb{C}^2$, justifying the inverses throughout this family.

The matrix identity $M^\top J + JM = 0$ also gives $(M^3)^\top J + JM^3 = 0$. Direct multiplication of RPC5–6 gives

$$\partial_\beta J + \frac{1}{3}(N^\top J + JN) = 0. \quad (\text{RPC7})$$

In particular $\partial_\beta J$ has entries $(2, 3) = 1$, $(3, 2) = -1$ when indices begin at zero, while $N^\top J + JN$ has the opposite threefold entries. Differentiate the inverse matrices in $P_0^{-\top} J P_0^{-1}$. RPC6–7 show that both its β and its η derivatives vanish. This matrix is consequently constant on the connected coefficient space $\mathbb{C}^2$.

Define the unchanged positive-Gamma ray constants

$$g_a = 5^{a/5-1} e^{\pi i a/5} \Gamma(a/5), \quad G = \text{diag}(g_1, g_2, g_3, g_4), \quad G_u = \text{diag}(u^{a/5} g_a)_{a=1}^4. \quad (\text{RPC8})$$

At $\beta = \eta = 0$, literal substitution in each original ray integral gives $P_0 = G_u$. Every g_a is nonzero, since its Gamma factor is the integral of a strictly positive function. We have therefore proved the complete flat-pairing identity

$$\begin{aligned} \Omega_0 &= P_0^{-\top} J P_0^{-1} = G_u^{-\top} J|_{\beta=0} G_u^{-1}, \\ (\Omega_0)_{14} &= -\frac{1}{u g_1 g_4}, \quad (\Omega_0)_{23} = \frac{1}{u g_2 g_3}, \quad (\Omega_0)_{ba} = -(\Omega_0)_{ab}, \\ \Omega &= P^{-\top} J P^{-1} = e^{-2A/u} \Omega_0. \end{aligned} \quad (\text{RPC9})$$

All entries not listed are zero; the indices in RPC9 begin at one. This formulation retains the Gamma constants directly and requires no unevaluated normalization of a period or source mass. For $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, the two paired sums of weights are five. It follows entry by entry that

$$D^\top \Omega D = \Omega, \quad D^\top \Omega_0 D = \Omega_0. \quad (\text{RPC10})$$

5.2 The full arithmetic unit and the exact quadratic form

Retain the original $v_h = 2\xi/h$ and its complete simple-root unit values. Reflection and conjugation of this original function give

$$v_h(\alpha_1) = v_h(\alpha_4) = a, \quad v_h(\alpha_2) = v_h(\alpha_3) = \bar{a}, \quad a \neq 0. \quad (\text{RPC11})$$

Indeed both numerator and denominator are invariant under $s \mapsto 1-s$ and conjugate under $s \mapsto \bar{s}$, first away from the roots and then by the removable values. The complete zero orders give nonzero values at these simple roots. Write

$$c_3 = \frac{a^{-2} - \bar{a}^{-2}}{4i\delta\gamma}, \quad c_1 = \frac{a^{-2} + \bar{a}^{-2}}{2} - (\delta^2 - \gamma^2)c_3, \quad \mathbf{r}(w) = c_1w + c_3w^3. \quad (\text{RPC12})$$

Both coefficients are real, and $(c_1, c_3) \neq (0, 0)$. Directly, $c_1 + c_3r_1^2 = a^{-2}$ and $c_1 + c_3r_2^2 = \bar{a}^{-2}$; the same equalities hold at the respective opposite roots. Thus $\mathbf{r}(M) = M v_h(M + 1/2)^{-2}$ in the original finite algebra, with the entire function evaluated by its complete original arithmetic unit. RPC12 is an exact interpolation of this specific multiplier.

For $Q_0(t) = t_1t_4 - t_2t_3$, let J_{prim} be the original Lagrange coordinate isomorphism and set

$$B = F^{-1}\Pi U J_{\text{prim}}, \quad Q(x) = Q_0(B^{-1}x), \quad \mathcal{H} = P\mathbf{r}(M)P^{-1}. \quad (\text{RPC13})$$

If $p = P^{-1}x$, the primary components of $B^{-1}x$ are exactly $p(r_1)/a, p(r_2)/\bar{a}, p(r_3)/\bar{a}, p(r_4)/a$. Since $H'(r) = 4r(r^2 + \gamma^2 - \delta^2)$, its values give $r/H'(r) = 1/(8i\delta\gamma)$ on roots one and four, and $r/H'(r) = -1/(8i\delta\gamma)$ on roots two and three. Inserting these four values into RPC4 proves the exact identity

$$Q(x) = 4i\delta\gamma p^\top \mathbf{r}(M)^\top J p = 4i\delta\gamma x^\top \mathcal{H}^\top \Omega x. \quad (\text{RPC14})$$

There is no conjugate transpose in this quadratic identity. Because $\mathbf{r}$ is odd, $\mathbf{r}(M)^\top J + J\mathbf{r}(M) = 0$; hence the matrices in RPC14 are symmetric. This also proves $\mathcal{H}^\top \Omega + \Omega \mathcal{H} = 0$. The four eigenvalues of $\mathbf{r}(M)$ are exactly $r_1/a^2, \bar{r}_1/a^2, -\bar{r}_1/a^2, -r_1/a^2$. They are nonzero, although they need not be distinct. Both B and Q_0 are invertible as linear and quadratic matrices, respectively, so Q always has rank four, including the possible eigenvalue collisions.

By RPC10, substituting Dx into RPC14 transports $\mathcal{H}$ to $D^{-1}\mathcal{H}D$. The resulting symmetric quadratic matrices are equal if and only if the corresponding $\mathcal{H}$ matrices are equal, because Ω is invertible. Consequently

$$Q(Dx) = Q(x) \iff [\mathcal{H}, D] = 0 \iff \mathcal{H}_{ab} = 0 \text{ for every } a \neq b. \quad (\text{RPC15})$$

This is an exact test in the actual original period matrix and unit. The final equivalence uses the four distinct diagonal entries of D . It makes no assumption that their off-diagonal tests are independent.

For completeness, a rank-four quadratic semi-invariant of cyclic weight $j \neq 0$ is impossible: its coefficient in row j could pair only with the missing weight zero, so that row is zero. Thus any semi-invariant rank-four quadratic must have weight zero. Since five is prime, the projective orbit of Q under D has size one or five, and size one is exactly RPC15. Every failure of any displayed off-diagonal test proves size five.

5.3 An explicit original-parameter radius forcing orbit five

Here we prove that RPC15 fails for all sufficiently large $|u|$, with a completely specified finite radius for the given original quartet and unit. The argument evaluates the original periods; it does not choose or insert a replacement amplitude.

Set $b = \beta u^{-2/5}$, $c = \eta u^{-4/5}$, and let $T(b, c)$ be the Fourier period matrix for $z^5/5 + bz^3/3 + cz$ at $u = 1$, in columns $1, z, z^2, z^3$ and on the five rays of angles $(\pi + 2\pi j)/5$. Substitution $w = u^{1/5}z$, with exactly the original branch, gives

$$P_0 = T(b, c)S_u, \quad S_u = \text{diag}(u^{a/5})_{a=1}^4, \quad S_u M S_u^{-1} = u^{1/5}M(b, c), \quad T(0, 0) = G. \quad (\text{RPC16})$$

Every column includes its original differential factor. The matrix $M(b, c)$ is RPC5 with β, η replaced by b, c ; in particular $N_0 := M(0, 0)$ has its three subdiagonal entries equal to one.

The following positive constants are explicit finite real integrals or finite expressions in the original Gamma constants:

$$\begin{aligned}
E_j &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^{5/5} + r^{3/3} + r} dr \quad (0 \leq j \leq 3), \\
C_T &= 2 \left(\sum_{j=0}^3 E_j^2 \right)^{1/2}, \quad g_+ = \max_a |g_a|, \quad g_-^{-1} = \max_a |g_a|^{-1}, \\
\kappa &= \max_{1 \leq a < 4} |g_{a+1}/g_a|, \quad K_0 = GN_0 G^{-1}, \\
C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \\
C_{K,3} &= C_K(3\kappa^2 + 3\kappa C_K + C_K^2), \quad \epsilon_0 = \min\{1, (2C_T g_-^{-1})^{-1}\}.
\end{aligned} \tag{RPC17}$$

Here g_-^{-2} means $(\max_a |g_a|^{-1})^2$. The negative fifth degree exponent proves convergence of every integral. All norms in the following estimates are the Euclidean operator norm, with the Frobenius norm used explicitly as an upper bound where stated.

Write $\epsilon = |b| + |c|$. When $|b|, |c| \leq 1$, on each ray

$$|e^{bz^3/3+cz} - 1| \leq \epsilon(r^3/3 + r)e^{r^{3/3}+r}.$$

Each contour has two rays, and each entry of F^{-1} has absolute value $1/5$ with four summands. The entry in column j of $T - G$ is therefore at most $E_j \epsilon$. Summing the squares over all four rows proves

$$\|T - G\| \leq \|T - G\|_F \leq C_T \epsilon. \tag{RPC18}$$

For $\epsilon \leq \epsilon_0$, the convergent inverse series for $I + G^{-1}(T - G)$ gives $\|T^{-1}\| \leq 2g_-^{-1}$ and $\|T^{-1} - G^{-1}\| \leq 2g_-^{-2} C_T \epsilon$. Also $\|M(b, c) - N_0\| \leq \epsilon$, $\|M(b, c)\| \leq 2$, and $\|N_0\| = 1$. Subtract the products $K := TM(b, c)T^{-1}$ and K_0 in three terms, changing the left factor, then the middle factor, then the inverse factor. The preceding inequalities give, with the complete RPC17 constants,

$$\|K - K_0\| \leq C_K \epsilon, \quad \|K\| \leq \kappa + C_K \epsilon, \quad \|K^3 - K_0^3\| \leq C_{K,3} \epsilon. \tag{RPC19}$$

For the last assertion use the exact noncommutative identity $K^3 - K_0^3 = (K - K_0)K^2 + K_0(K - K_0)K + K_0^2(K - K_0)$. Its norm is at most $C_K \epsilon$ times $(\kappa + C_K)^2 + \kappa(\kappa + C_K) + \kappa^2$, as claimed.

The original multiplication and full unit in RPC13 now give exactly

$$\mathcal{H} = c_1 u^{1/5} K + c_3 u^{3/5} K^3, \quad (K_0^3)_{41} = g_4/g_1, \quad (K_0)_{21} = g_2/g_1. \tag{RPC20}$$

The scalar $e^{A/u}$ cancels from this conjugation, while it remains in the quadratic form through RPC9. Put $B_* = \beta + \eta > 0$. For $|u| \geq 1$, $\epsilon \leq B_* |u|^{-2/5}$.

If $c_3 \neq 0$, define the completely specified radius

$$T_* = \max \left\{ 1, \frac{B_*}{\epsilon_0}, \frac{2(|c_3| C_{K,3} B_* + |c_1|(\kappa + C_K))}{|c_3 g_4/g_1|} \right\}, \quad R_* = T_*^{5/2}. \tag{RPC21}$$

For every u on its original branch with $|u| \geq R_*$, RPC19–20 imply

$$\left| u^{-3/5} \mathcal{H}_{41} - c_3 g_4/g_1 \right| \leq |u|^{-2/5} (|c_3| C_{K,3} B_* + |c_1|(\kappa + C_K)) \leq \frac{1}{2} |c_3 g_4/g_1|. \tag{RPC22}$$

In particular $\mathcal{H}_{41} \neq 0$. In the quadratic matrix RPC14 this is the nonzero $(1, 1)$ entry $4i\delta\gamma \mathcal{H}_{41} \Omega_{41}$, whose cyclic weight is two.

If $c_3 = 0$, then $c_1 \neq 0$. In this case define

$$T_* = \max \left\{ 1, \frac{B_*}{\epsilon_0}, \frac{2C_K B_*}{|g_2/g_1|} \right\}, \quad R_* = T_*^{5/2}. \tag{RPC23}$$

For every $|u| \geq R_*$, the same estimates give

$$\left| u^{-1/5} \mathcal{H}_{21} - c_1 g_2 / g_1 \right| \leq |c_1| C_K B_* |u|^{-2/5} \leq \frac{1}{2} |c_1 g_2 / g_1|. \quad (\text{RPC24})$$

Thus $\mathcal{H}_{21} \neq 0$. The $(1, 3)$ entry of the quadratic matrix is then $4i\delta\gamma\mathcal{H}_{21}\Omega_{23}$, which has cyclic weight four; symmetry retains its $(3, 1)$ partner. Both cases prove the unconditional conclusion for the actual original parameters and the explicit radius belonging to those parameters:

$$\boxed{|u| \geq R_* \implies \#\{[Q(D^j x)] : 0 \leq j < 5\} = 5.} \quad (\text{RPC25})$$

The bracket denotes the projective class of the quadratic polynomial. RPC21 or RPC23 supplies its radius for every nonzero original unit; neither case assumes a condition on an uncomputed period coefficient.

Finally, the original periods are holomorphic on the connected logarithm cover of $\mathbb{C}^*$, and on each branch domain: locally the contours may be held in common decay sectors and differentiated under the dominated integral, and contour deformation identifies these local definitions. Their determinant never vanishes. Hence all entries of $\mathcal{H}$ are holomorphic there. RPC22 or RPC24 proves that its respective decisive entry is not identically zero on the logarithm cover and, by the identity theorem on that connected cover, on any open branch domain. The zeros of that one entry are isolated, by its convergent local power series. The exact exceptional set

$$\mathcal{E} = \{u : \mathcal{H}_{ab}(u) = 0 \text{ for all } a \neq b\} \quad (\text{RPC26})$$

is a subset of those isolated zeros and satisfies $|u| < R_*$. Thus it has no accumulation inside the branch domain; accumulation at $u = 0$ or a branch boundary has not been excluded. RPC15 is the full calculation specifying this set. No assertion that it is empty at arbitrary finite complex u is made or used.

6 A finite bound for the original kernel volume from its actual dimension

The original packet, sources, and exact coordinate map. Throughout this calculation the original multiplicity is $m = 1$ and

$$k = 4l + 1, \quad l \geq 1, \quad q = (k + 1)^2 \geq 36, \quad c = k/2, \quad 0 < \delta < 1/2, \quad \gamma > 2. \quad (\text{OKV1})$$

Retain every original root and its two coordinates:

$$\begin{aligned} \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma], \\ \omega_{ab} &= (2b - k)\gamma - i(2a - k)\delta, \\ P(y) &= i^{-q}\chi(c + iy) = \prod_{a,b=0}^k (y - \omega_{ab}). \end{aligned} \quad (\text{OKV2})$$

The identity follows factor by factor from $iy - (2a - k)\delta - i(2b - k)\gamma = i(y - \omega_{ab})$; it retains the full phase i^q . The algebra map

$$\Phi : \mathbb{C}[y]/(P) \longrightarrow E_\chi := \mathbb{C}[S]/(\chi), \quad [r(y)] \longmapsto [r((S - c)/i)] \quad (\text{OKV3})$$

is well defined because $P((S - c)/i) = i^{-q}\chi(S)$. Its inverse is $[f(S)] \mapsto [f(c + iy)]$, as verified by substitution. Both maps preserve degrees and are inverse on each polynomial ring before taking the quotients. In particular division by P is the original remainder operation through this exact coordinate map.

Use only the two original source orders $s \in \{1, k\}$ and write

$$b_s = s/2, \quad M_s = (2\pi)^{s/2}, \quad dm_{0,s}(y) = \frac{M_s 2^{b_s-2}}{\pi \Gamma(b_s)} \left| \Gamma\left(\frac{b_s}{2} + \frac{iy}{2}\right) \right|^2 dy. \quad (\text{OKV4})$$

The exact source calculation IFB2–5 supplies mass M_s , the transform $\int e^{ty} dm_{0,s} = M_s (\cos t)^{-b_s}$ for $|t| < \pi/2$, and the original monic orthogonal polynomials and norms

$$\begin{aligned} p_0 &= 1, \quad p_1 = y, \quad p_{d+1} = yp_d - d(b_s + d - 1)p_{d-1}, \\ h_d^{(s)} &= \int |p_d|^2 dm_{0,s} = M_s d!(b_s)_d, \quad \varphi_d(y) = p_d(y)/\sqrt{h_d^{(s)}}, \quad u_d(S) = \varphi_d((S - c)/i). \end{aligned} \quad (\text{OKV5})$$

All these integrals are on the same real line, with the original source mass. For example the stated orthogonality and norms also follow directly from its transform and the exact generating function

$$\sum_{d \geq 0} p_d(y) \frac{z^d}{d!} = (1 + z^2)^{-b_s/2} e^{y \arctan z}.$$

Indeed the source integral of the product at sufficiently small real z, w is $M_s(1 - zw)^{-b_s}$. The cosine addition identity and the displayed transform give this equality; the source exponential bound justifies each finite coefficient derivative under the integral. The coefficients of $z^d w^e$ then give zero for $d \neq e$ and $M_s d!(b_s)_d$ for $d = e$. The initial coefficients are $1, y$, and differentiating the generating function gives precisely the recurrence in OKV5. Thus the source norms used below retain their original constants and all finite-degree factors.

Let $C_s : \mathbb{C}^q \rightarrow E_\chi$ have columns $[u_0], \dots, [u_{q-1}]$ in the original frame $(1, S, \dots, S^{q-1})$. Its exact leading coefficients give

$$\det C_s = i^{-q(q-1)/2} \prod_{d=0}^{q-1} (h_d^{(s)})^{-1/2} \neq 0.$$

Retain the original GMB vectors and matrices, including the last row:

$$v_j = C_s^{-1}[u_{q+j}], \quad K_j = I_q + \sum_{a=0}^{j-1} v_a v_a^*, \quad 0 \leq j \leq q+1. \quad (\text{OKV6})$$

The vector v_q has original polynomial degree $2q$.

Exact remainder coefficients in all original roots. Index the full list in OKV2 by $\omega_1, \dots, \omega_q$, with each pair (a, b) occurring exactly once. Define the elementary and complete homogeneous polynomials by their finite sums

$$e_h(\omega) = \sum_{1 \leq a_1 < \dots < a_h \leq q} \omega_{a_1} \cdots \omega_{a_h}, \quad h_t(\omega) = \sum_{n_1 + \dots + n_q = t} \prod_{a=1}^q \omega_a^{n_a}, \quad (\text{OKV7})$$

with $e_0 = h_0 = 1$, $e_h = 0$ outside $0 \leq h \leq q$, and nonnegative integer n_a in the second sum. The notation $h_t(\omega)$ in this paragraph is a polynomial in the roots; the original squared norm continues to be $h_d^{(s)}$ of OKV5. Multiplication of formal power series gives the exact identity

$$\left(\sum_{h=0}^q (-1)^h e_h(\omega) z^h \right) \left(\sum_{t \geq 0} h_t(\omega) z^t \right) = 1,$$

because each geometric series $(1 - \omega_a z)^{-1}$ multiplies its own factor $1 - \omega_a z$. Consequently, for $D = q + j$ with $0 \leq j \leq q$, the quotient and remainder of y^D upon division by P are exactly

$$\begin{aligned} Q_D(y) &= \sum_{t=0}^j h_t(\omega) y^{j-t}, \quad \text{rem}_P(y^D) = y^D - P(y)Q_D(y), \\ [y^a] \text{rem}_P(y^D) &= - \sum_{t=0}^j h_t(\omega) (-1)^{D-t-a} e_{D-t-a}(\omega), \quad 0 \leq a < q. \end{aligned} \quad (\text{OKV8})$$

To verify the quotient assertion, the power-series identity makes the coefficients of $y^D, y^{D-1}, \dots, y^q$ in PQ_D equal to those of y^D . The remaining coefficients are exactly the last line of OKV8. Their degree is less than q , so uniqueness of monic polynomial division proves both assertions. For $0 \leq D < q$ the remainder is y^D itself.

The actual maximal root modulus is

$$R = k\sqrt{\delta^2 + \gamma^2}, \quad D_* := \max\{3, \sqrt{\delta^2 + \gamma^2}\}, \quad T_* := D_* q. \quad (\text{OKV9})$$

Every $|\omega_{ab}| \leq R$, and equality holds at each corner, so this expression for R is exact. In particular $R \leq T_*$. Counting the summands of OKV7 proves

$$|e_h(\omega)| \leq \binom{q}{h} R^h, \quad |h_t(\omega)| \leq \binom{q+t-1}{t} R^t.$$

The second count follows by placing $q-1$ separators among $t+q-1$ positions to encode $(n_1, \dots, n_q)$. For fixed j , the binomial $\binom{q+t-1}{t}$ is increasing in $0 \leq t \leq j$, because its successive ratio is $(q+t)/(t+1) \geq 1$. In each nonzero summand of OKV8 the total root exponent is $(D-t-a)+t = D-a$. As t varies the indices $D-t-a$ are distinct, so the sum of their $\binom{q}{D-t-a}$ is at most 2^q . It follows that

$$|[y^a] \text{rem}_P(y^{q+j})| \leq 2^q \binom{q+j-1}{j} R^{q+j-a} \leq 2^{3q-1} R^{q+j-a}, \quad 0 \leq a < q, \quad 0 \leq j \leq q. \quad (\text{OKV10})$$

For the last inequality use $\binom{q+j-1}{j} \leq 2^{q+j-1} \leq 2^{2q-1}$, by the binomial expansion of $(1+1)^{q+j-1}$. These estimates retain the exact coefficient formula OKV8, including every signed cancellation.

Original Gamma norms of the low monomials. The same recurrence and norms in OKV5 give, without changing the measure,

$$y\varphi_a = \sqrt{(a+1)(b_s+a)} \varphi_{a+1} + \sqrt{a(b_s+a-1)} \varphi_{a-1}, \quad \varphi_{-1} = 0. \quad (\text{OKV11})$$

For a polynomial of degree at most $q-2$, its orthonormal coefficient vector is therefore sent by multiplication by y to the sum of one upward and one downward weighted shift, with output degree at most $q-1$. Each shift has operator norm at most $\sqrt{q(b_s+q)}$: the squared norm of its coefficient vector is bounded by the largest squared weight times the input squared norm. Since $s \leq k < q$ gives $b_s \leq q/2$, the triangle inequality yields

$$\|yf\|_{L^2(m_{0,s})} \leq 2\sqrt{q(b_s+q)} \|f\|_{L^2(m_{0,s})} \leq 3q \|f\|_{L^2(m_{0,s})} \leq T_* \|f\|_{L^2(m_{0,s})}.$$

Here $2\sqrt{3/2} = \sqrt{6} < 3$. Starting from $\|1\|_{L^2(m_{0,s})} = \sqrt{M_s}$ and applying this inequality to $1, y, \dots, y^{a-1}$ proves

$$\boxed{\|y^a\|_{L^2(m_{0,s})} \leq \sqrt{M_s} T_*^a, \quad 0 \leq a < q.} \quad (\text{OKV12})$$

All intermediate polynomial degrees satisfy the stated operator domain.

For any polynomial $f = \sum_a f_a y^a$, write $\mathbf{c}_{T_*}(f) = \sum_a |f_a| T_*^a$. This is a finite coefficient sum, not a change of source or quotient norm. The exact original recurrence gives

$$\mathbf{c}_{T_*}(p_{d+1}) \leq T_* \mathbf{c}_{T_*}(p_d) + d(b_s + d - 1) \mathbf{c}_{T_*}(p_{d-1}).$$

For $1 \leq d < 2q$ its actual positive coefficient is bounded by $d(b_s + d - 1) \leq 2q(b_s + 2q) \leq 5q^2 \leq T_*^2$. The initial values are $1, T_*$. Induction now proves

$$\boxed{\mathbf{c}_{T_*}(p_d) \leq (2T_*)^d, \quad 0 \leq d \leq 2q.} \quad (\text{OKV13})$$

Indeed substituting the two preceding bounds in the recurrence gives $T_*(2T_*)^d + T_*^2(2T_*)^{d-1} = \frac{3}{4}(2T_*)^{d+1} \leq (2T_*)^{d+1}$.

Combining all q coefficients of OKV10 with OKV12, by the triangle inequality in the original L^2 space, gives

$$\|\text{rem}_P(y^D)\|_{L^2(m_{0,s})} \leq \sqrt{M_s} q 2^{3q} T_*^D \quad (0 \leq D \leq 2q).$$

For $D \geq q$, each summand is bounded by $\sqrt{M_s} 2^{3q-1} R^{D-a} T_*^a \leq \sqrt{M_s} 2^{3q-1} T_*^D$ and there are q summands. For $D < q$, the assertion follows from OKV12 and $q 2^{3q} \geq 1$. Linearity of the exact remainder map, followed by OKV13, proves

$$\boxed{\|\text{rem}_P(p_d)\|_{L^2(m_{0,s})} \leq \sqrt{M_s} q 2^{3q} (2T_*)^d, \quad 0 \leq d \leq 2q.} \quad (\text{OKV14})$$

The original remainder vectors and the full matrix norm. The polynomial $\text{rem}_P(p_d)$ has degree less than q . Its expansion in the orthonormal basis $\varphi_0, \dots, \varphi_{q-1}$ has squared coefficient norm equal to its squared original $L^2(m_{0,s})$ norm, by expanding the finite inner product and using OKV5. The exact quotient map OKV3 and definition OKV6 therefore give, for $d = q + j$, $0 \leq j \leq q$,

$$\|v_j\|_2^2 = \frac{\|\text{rem}_P(p_{q+j})\|_{L^2(m_{0,s})}^2}{M_s (q+j)! (b_s)_{q+j}} \leq \frac{q^2 2^{6q} (2T_*)^{2(q+j)}}{(q+j)! (b_s)_{q+j}}. \quad (\text{OKV15})$$

This equality is the actual map of the original quotient coordinates to the remainder polynomial. The full source factor M_s occurs in both the numerator bound and its original norm denominator before the displayed cancellation. No estimate on the conditioning of C_s or a companion matrix is used.

Since $b_s \geq 1/2$, for each integer $d \geq 1$ the first factor of $(b_s)_d$ is at least $1/2$ and every subsequent factor is at least one. Thus $d!(b_s)_d \geq 1/2$. Also $T_* \geq 1$ and $q + j \leq 2q$. Consequently

$$\|v_j\|_2^2 \leq 2q^2 2^{6q} (2D_* q)^{4q}, \quad 0 \leq j \leq q,$$

and the original complete matrix obeys

$$\boxed{1 \leq \|K_{q+1}\|_2 \leq 1 + 2(q+1)q^2 2^{6q} (2D_* q)^{4q}}. \quad (\text{OKV16})$$

To check its norm bound, $\|v_j v_j^*\|_2 = \|v_j\|_2^2$ follows directly from Cauchy–Schwarz, with equality on a nonzero v_j ; it is zero for a zero vector. Add all $q+1$ summands of OKV6 and the norm of I_q . Positivity also proves its lower bound.

Here is an explicit bound of the requested scale, with constants depending only on the original two root coordinates:

$$\boxed{\log \|K_{q+1}\|_2 \leq C(\delta, \gamma) q \log(q+1), \quad C(\delta, \gamma) = 12 + 4 \log(2D_*)}. \quad (\text{OKV17})$$

For a direct verification, put $U = 2(q+1)q^2 2^{6q} (2D_* q)^{4q} \geq 1$ and use $\log(1+U) \leq \log(2U)$. This upper bound is exactly

$$\log 4 + \log(q+1) + 2 \log q + 6q \log 2 + 4q \log(2D_*) + 4q \log q.$$

Write $L = \log(q+1)$. Since $q \geq 36$, one has $\log 4 \leq L$, $\log q \leq L$, $6 \log 2 \leq 2L$, and $L \geq 1$. The last inequality follows already from $\log 4 = \int_1^4 dt/t > 1$, by bounding the integrand below by $1/2$ on $[1, 2]$ and $1/4$ on $[2, 4]$. The preceding display is therefore at most $4L + 6qL + 4 \log(2D_*)qL$, which is bounded by the right side of OKV17. The completely finite and sharper bound OKV16 remains available before this logarithmic estimate.

The four original kernel Grams with their full coefficient frame. Retain the original surjection $\Lambda : E_\chi \rightarrow B_{\text{obs}}$, the actual subspace $\mathcal{K}_\Lambda = \ker \Lambda$, and its original inclusion $I : \mathcal{K}_\Lambda \hookrightarrow E_\chi$. Put $r_K = \dim \mathcal{K}_\Lambda$; no separate formula or upper bound for this actual dimension is needed here. Use its fixed original coefficient frame and write

$$T_s = C_s^{-1} I, \quad G_{q+j-1}^{(s)} = C_s^{-*} K_j^{-1} C_s^{-1}, \quad H_j^{(s)} = I^* G_{q+j-1}^{(s)} I = T_s^* K_j^{-1} T_s, \quad 0 \leq j \leq q+1. \quad (\text{OKV18})$$

For completeness, the quotient metric in this identity follows from the original orthonormal source-coordinate map $A_j = [I_q, v_0, \dots, v_{j-1}]$. It obeys $A_j A_j^* = K_j$. The minimum norm solution of $A_j z = x$ is $z = A_j^* K_j^{-1} x$: it has the required image and is orthogonal to $\ker A_j$. Every other solution differs by such a kernel vector, so the Pythagorean identity gives minimum squared norm $x^* K_j^{-1} x$. The actual remainder is $C_s x$. Substitution proves the original quotient Gram and then its full restriction in OKV18.

Let $\lambda_s = \|K_{q+1}\|_2$. The original update chain gives $I_q = K_0 \preceq K_j \preceq K_{q+1} \preceq \lambda_s I_q$. For positive definite matrices, inversion reverses the order: if $A \preceq B$, conjugate by $A^{-1/2}$ to obtain a matrix with eigenvalues at least one, invert those eigenvalues and conjugate back. Thus the full original kernel forms satisfy

$$\lambda_s^{-1} H_0^{(s)} \preceq H_j^{(s)} \preceq H_0^{(s)}, \quad H_0^{(s)} = T_s^* T_s, \quad H_{j+1}^{(s)} \preceq H_j^{(s)}. \quad (\text{OKV19})$$

For $r_K > 0$, the matrix T_s is injective, so $H_0^{(s)}$ is positive definite. Conjugation in OKV19 by its inverse square root gives eigenvalues in $[\lambda_s^{-1}, 1]$. The determinant identity for that exact congruence is

$$\det[(H_0^{(s)})^{-1/2} H_j^{(s)} (H_0^{(s)})^{-1/2}] = \frac{\det H_j^{(s)}}{\det(T_s^* T_s)}.$$

Therefore

$$-r_K \log \lambda_s \leq \log \det H_j^{(s)} - \log \det(T_s^* T_s) \leq 0. \quad (\text{OKV20})$$

In particular the determinant and conditioning of the actual $T_s^* T_s$ are retained in this identity; no new kernel basis is substituted for its original frame.

Define the original four-endpoint kernel volume, with its literal degrees $q-1, q, 2q-1, 2q$, by

$$F_K^{(s)} = \log \det H_0^{(s)} + \log \det H_1^{(s)} - \log \det H_q^{(s)} - \log \det H_{q+1}^{(s)}. \quad (\text{OKV21})$$

Monotonicity in OKV19 makes each of $\log \det H_0^{(s)} - \log \det H_q^{(s)}$ and $\log \det H_1^{(s)} - \log \det H_{q+1}^{(s)}$ nonnegative. Each is at most $r_K \log \lambda_s$ by OKV20. Consequently the actual kernel volume satisfies the complete finite estimates

$$\begin{aligned} 0 \leq F_K^{(s)} &\leq 2r_K \log \|K_{q+1}\|_2 \\ &\leq 2r_K \log [1 + 2(q+1)q^2 2^{6q} (2D_* q)^{4q}] \\ &\leq 2r_K C(\delta, \gamma) q \log(q+1), \quad s \in \{1, k\}. \end{aligned} \quad (\text{OKV22})$$

For $r_K = 0$, every kernel determinant has its original empty value one, and $F_K^{(s)} = 0$. Hence OKV22 includes this case. The four copies of $\log \det(T_s^* T_s)$ in OKV21 have coefficients $+1, +1, -1, -1$ and cancel exactly only after their presence in OKV20 has been recorded.

The original boundary and arithmetic transitions. At each original degree set

$$Q_j^{(s)} = \left(\Lambda(G_{q+j-1}^{(s)})^{-1} \Lambda^* \right)^{-1}, \quad S_j^{(s)} = (G_{q+j-1}^{(s)})^{-1} \Lambda^* Q_j^{(s)}. \quad (\text{OKV23})$$

Their zero dimensional cases have their empty meaning. Surjectivity of Λ and positivity give the displayed inverses. Direct multiplication proves $\Lambda S_j^{(s)} = I$ and $I^* G_{q+j-1}^{(s)} S_j^{(s)} = 0$. For any fixed original section S_* of Λ , the map $[I, S_*]$ is bijective: its inverse records Λz and the unique kernel coordinate of $z - S_* \Lambda z$. The minimum section differs from S_* by a kernel column. The two frames therefore differ by a block triangular matrix with identity diagonal and determinant one. Taking the determinant of the full metric in the orthogonal frame $[I, S_j^{(s)}]$ gives

$$\det H_j^{(s)} \det Q_j^{(s)} = |\det[I, S_*]|^2 \det G_{q+j-1}^{(s)}. \quad (\text{OKV24})$$

This proves the exact relation with all kernel and boundary directions retained. The same frame factor occurs at all four endpoints. Their signed sum is consequently

$$F_K^{(s)} + F_B^{(s)} = \mathcal{B}_k^{(s)}, \quad F_B^{(s)} = \log \det Q_0^{(s)} + \log \det Q_1^{(s)} - \log \det Q_q^{(s)} - \log \det Q_{q+1}^{(s)}. \quad (\text{OKV25})$$

The original matrices $G_{q+j-1}^{(s)}$ decrease with j , so their inverse forms increase. Compression by Λ and inversion then show that $Q_j^{(s)}$ decreases as well. Thus $F_B^{(s)} \geq 0$, and the finite companion bound is

$$\max\{0, \mathcal{B}_k^{(s)} - 2r_K C(\delta, \gamma) q \log(q+1)\} \leq F_B^{(s)} \leq \mathcal{B}_k^{(s)}. \quad (\text{OKV26})$$

Here the finer middle bound in OKV22 can be used in place of its last line, with the same proof.

Finally the original source identifications are $\mathcal{B}_k^{(1)} = \mathcal{B}_k^\sigma$ and $\mathcal{B}_k^{(k)} = \mathcal{B}_k^0$. Their already proved full source transition is $\mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0$, and the original arithmetic transition is $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma$, with its original finite signed allowance for δ_k^σ . Substitution of the proved exact splitting OKV25 gives, with every source and arithmetic term still present,

$$\mathcal{B}_k^{\text{ar}} = F_K^{(k)} + F_B^{(k)} + \mathcal{H}_k + \delta_k^\sigma = F_K^{(1)} + F_B^{(1)} + \delta_k^\sigma. \quad (\text{OKV27})$$

Thus OKV22 controls the original Gamma kernel summands in this same arithmetic identity. It keeps the separate source transition and arithmetic scalar with their established signs and bounds. The result is finite for the actual $r_K = \dim \ker \Lambda$ and uses no additional asserted dimension estimate or replacement source.

7 The actual simple-quartet kernel and boundary at the original volume scale

This section combines the complete original period calculation, the exact cyclic-orbit rank calculation, and the finite original Gamma kernel estimate. The hypotheses are the actual simple-quartet data already used in the programme, and the period range below has an explicit bound proved from those data. No orbit condition or kernel-rank premise is added to the statement.

7.1 The original domain and the proved period range

Fix the original simple quartet $m = 1$, its coordinates $0 < \delta < 1/2$, $\gamma > 2$, and its actual nonzero amplitude value $a = v_h(1/2 + \delta + i\gamma)$, where $v_h = 2\xi/h$ and every zero order in this quartet is complete. Reflection and conjugation give the other three unit values exactly as in PQO3 and RPC11. For every integer $l \geq 1$ retain

$$k = 4l + 1, \quad q = (k + 1)^2, \quad c = k/2, \quad s \in \{1, k\}. \quad (\text{OMV1})$$

The polynomial χ , the source measures $m_{0,s}$, their masses $(2\pi)^{s/2}$, the original observation $\Lambda_k : E_\chi \rightarrow B_k$, its kernel, and all four endpoint degrees are unchanged.

Let R_* be the finite positive real number defined by RPC17 and RPC21 when the actual c_3 of RPC12 is nonzero, and by RPC17 and RPC23 when it is zero. Those two explicitly evaluated alternatives exhaust the original nonzero unit, because c_1, c_3 cannot both vanish. In particular R_* depends on the fixed quartet and its actual unit and is independent of k . Its constants are finite positive Gamma values and the convergent positive integrals RPC17. The full period proof RPC18–25 establishes

$$|u| \geq R_* \implies \#\{[Q_u(D^j x)] : 0 \leq j < 5\} = 5 \quad (\text{OMV2})$$

on every original logarithm branch. To recall the actual nonvanishing calculation, RPC22 bounds the error in $u^{-3/5}\mathcal{H}_{41}$ by half the nonzero value $|c_3g_4/g_1|$ in the first alternative. RPC24 bounds the error in $u^{-1/5}\mathcal{H}_{21}$ by half $|c_1g_2/g_1|$ in the second. Each is a nonzero off-diagonal entry of the exact $\mathcal{H} = P(c_1M + c_3M^3)P^{-1}$; the flat residue identity RPC14–15 makes it a nonzero noninvariant coefficient of the actual quadric. The degree-five cyclic action and its nondegenerate quadratic matrix then give OMV2. All powers of u use its original branch, and the absolute error estimates hold for every such branch.

The exact invariant-ideal proof OQX1–3 now applies at every point of this proved domain. It gives, for all the original degrees OMV1,

$$\dim B_k = k^2 - 6k + 17, \quad r_K := \dim \ker \Lambda_k = 8k - 16. \quad (\text{OMV3})$$

Specifically, an invariant polynomial vanishing on the original quadric is divisible by all five distinct original orbit factors; its quotient by their invariant product is invariant. Thus the actual observed rank is $d_{5,k,0} - d_{5,k-10,0}$, with negative degrees zero. The full character calculation OQX3 evaluates this difference to $k^2 - 6k + 17$, including the two original low degrees 5, 9. Subtraction from $q = (k + 1)^2$ proves the second equality. This is the dimension of the kernel of the literal period observation, not an assigned ambient dimension or a presumed maximal rank.

7.2 A uniform finite bound on the actual kernel volume

Define, from the original root coordinates only,

$$D_* = \max\{3, \sqrt{\delta^2 + \gamma^2}\}, \quad C_* = 12 + 4 \log(2D_*), \quad \mathfrak{M}_k = (16k - 32)C_*q \log(q + 1). \quad (\text{OMV4})$$

Retain the exact original source-coordinate vectors $v_j = C_s^{-1}[u_{q+j}]$, $K_j = I + \sum_{a < j} v_a v_a^*$, and the full original kernel frame $T_s = C_s^{-1}I$ of OKV6,18. The original kernel metrics at degrees $q + j - 1$

are $H_j^{(s)} = T_s^* K_j^{-1} T_s$. Their actual four-endpoint volume satisfies

$$\boxed{0 \leq F_K^{(s)}(u) \leq (16k - 32) \log[1 + 2(q + 1)q^2 2^{6q} (2D_* q)^{4q}] \leq \mathfrak{M}_k, \quad s \in \{1, k\}, \quad |u| \geq R_*} \quad (\text{OMV5})$$

Here the source index and the actual period dependence of the kernel are retained. The proof is direct substitution of the now proved $r_K = 8k - 16$ into OKV22. Its finite coefficient proof retains every original root in the signed remainder formula, all Gamma recurrence coefficients through degree $2q$, and the full source mass before its exact norm-ratio cancellation. Its metric proof retains $\det(T_s^* T_s)$ at each of the four degrees, and cancels only the four copies with coefficients $+1, +1, -1, -1$. In particular neither the conditioning nor the changing position of the original period kernel has been suppressed. The bound is uniform in the entire period domain OMV2 because its right side depends only on k, δ, γ , and the dimension in OMV3 is the same at every point of that domain.

Write the four original total quotient baselines as $\mathcal{B}_k^{(1)} = \mathcal{B}_k^\sigma$ and $\mathcal{B}_k^{(k)} = \mathcal{B}_k^0$. The full determinant identity OKV24–25, with the original fixed coefficient section, gives

$$F_B^{(s)}(u) = \mathcal{B}_k^{(s)} - F_K^{(s)}(u), \quad \max\{0, \mathcal{B}_k^{(s)} - \mathfrak{M}_k\} \leq F_B^{(s)}(u) \leq \mathcal{B}_k^{(s)}. \quad (\text{OMV6})$$

Both factors in the exact splitting are nonnegative by monotonicity of their original minimum metrics. The literal boundary term is the actual period-vector expression

$$F_B^{(s)}(u) = \log \frac{\det \mathcal{M}_{2q}^{(s)}(u) \det \mathcal{M}_{2q+1}^{(s)}(u)}{\det \mathcal{M}_q^{(s)}(u) \det \mathcal{M}_{q+1}^{(s)}(u)}, \quad \mathcal{M}_D^{(s)}(u) = \sum_{d=0}^{D-1} \lambda_d^{(s)}(u) (\lambda_d^{(s)}(u))^*. \quad (\text{OMV7})$$

The vectors are the complete original primary-period coefficients OPG14 and OCS21 on the actual image of Λ_k . Formula OPG17 proves that each consecutive determinant ratio is $1 + \xi_j$; the unchanged weights $1, 2, \dots, 2, 1$ give exactly OMV7. The companion kernel vectors retain the minimum-section correction OPR2 and the action defect OPG22/OQC22. No invariance of that kernel under the original sum action is needed for OMV5–7.

7.3 The two individual leading coefficients

The complete original baseline proofs BSL16–19 and HBL establish, for fixed original δ, γ and $m = 1$, both source limits

$$\frac{\mathcal{B}_k^{(s)}}{q^2} \longrightarrow C_B = 9 - 8 \log 2 + F(2, \pi) > \frac{769}{17010}, \quad s = 1 \text{ and } s = k. \quad (\text{OMV8})$$

Here $F(2, \pi)$ is exactly the energy functional evaluated in those complete proofs; its original end-point convention and coefficients are retained. No value of it is redefined by the present comparison. Since $q = (k + 1)^2$,

$$\frac{\mathfrak{M}_k}{q^2} = C_* \frac{(16k - 32) \log((k + 1)^2 + 1)}{(k + 1)^2} \longrightarrow 0.$$

For an elementary verification, $((k + 1)^2 + 1) < (k + 2)^2$, and the last fraction is at most $32 \log(k + 2)/k$. The inequality $\log x = \int_1^x dt/t \leq 2(\sqrt{x} - 1)$ for $x \geq 1$ proves that this upper bound tends to zero. The finite estimate OMV5 and the exact identity OMV6 therefore prove

$$\boxed{\frac{F_K^{(1)}(u)}{q^2}, \frac{F_K^{(k)}(u)}{q^2} \longrightarrow 0, \quad \frac{F_B^{(1)}(u)}{q^2}, \frac{F_B^{(k)}(u)}{q^2} \longrightarrow C_B} \quad (\text{OMV9})$$

These limits are uniform for all $|u| \geq R_*$ and all original branches. Indeed the kernel error is bounded by $\mathfrak{M}_k/q^2$ uniformly, while the boundary error is at most $|\mathcal{B}_k^{(s)}/q^2 - C_B| + \mathfrak{M}_k/q^2$. The baseline is independent of u . This proves the asserted uniformity, rather than only a conclusion for each separately fixed observation. The original combined coefficient has now been resolved into its two individual contributions on the proved simple-quartet period domain.

7.4 Exact propagation to the arithmetic identity and the finer scale

Keep the original signed transitions $\mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0$ and $\delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma$. Their complete finite allowances, including the QGQ simple- multiplicity refinement, remain those of the full current receiver. The exact arithmetic receiving equality becomes

$$\begin{aligned}\mathcal{B}_k^{\text{ar}} - F_B^{(1)}(u) &= F_K^{(1)}(u) + \delta_k^\sigma, \\ \mathcal{B}_k^{\text{ar}} - F_B^{(k)}(u) &= F_K^{(k)}(u) + \mathcal{H}_k + \delta_k^\sigma.\end{aligned}\tag{OMV10}$$

This is substitution in OKV27, so every source and sign is the original one. In particular the finite upper and lower allowances are obtained by adding $[0, \mathfrak{M}_k]$ to the actual interval for δ_k^σ , or to the actual interval for $\mathcal{H}_k + \delta_k^\sigma$, respectively. The original proved relations $\delta_k^\sigma/(kq) \rightarrow 0$ and $\mathcal{H}_k/(kq) \rightarrow -\mathcal{C}_\Gamma/4$, together with $k/q \rightarrow 0$ and OMV5, prove that both differences in OMV10, divided by q^2 , tend to zero uniformly on the stated period domain. This is an exact comparison with the full arithmetic baseline. The literal arithmetic kernel and boundary metrics retain their own source until their corresponding metric comparison is applied.

The finer original Gamma return is still carried by an exact mixed identity:

$$\begin{aligned}F_B^{(1)}(u) - F_B^{(k)}(u) &= \mathcal{H}_k - (F_K^{(1)}(u) - F_K^{(k)}(u)), \\ -\mathfrak{M}_k &\leq F_K^{(1)}(u) - F_K^{(k)}(u) \leq \mathfrak{M}_k, \\ \mathcal{H}_k - \mathfrak{M}_k &\leq F_B^{(1)}(u) - F_B^{(k)}(u) \leq \mathcal{H}_k + \mathfrak{M}_k.\end{aligned}\tag{OMV11}$$

Subtract OMV6 for the two actual source orders to obtain the first line. Each kernel term lies in $[0, \mathfrak{M}_k]$, so their difference lies in the stated interval, proving the other two lines. The error bound here is of order $kq \log(q+1)$, whereas the already proved signed return $\mathcal{H}_k$ has order kq . No sign or limiting coefficient for either finer mixed difference is assigned by OMV11. Its explicit remaining term is the difference of the two original kernel metrics, on the exactly calculated $8k-16$ dimensional period kernel. The full source multiplier and its action on this actual kernel remain the input to that next calculation, rather than a freely selected source order or kernel.

8 The original arithmetic kernel and boundary receive the full source comparison

This calculation uses the proved full-polynomial comparison TW12/TWA3, the original observation OPG1–5, and the complete original Gamma kernel estimate OKV1–27. It gives the exact maps between their source, quotient, kernel and boundary metrics, before taking any determinant. The subsequent simple-quartet conclusion uses the evaluated original period test RPC1–26 and its exact rank calculation OQX1–3. Every source order below is one of the already present orders $1, k$.

8.1 The complete original source and the same observation

Retain the original full-order quartet, with $0 < \delta < 1/2$, $\gamma > 2$, and positive integer complete zero order m . Thus

$$\begin{aligned} \mathcal{R} &= \{\tfrac{1}{2} + \delta + i\gamma, \tfrac{1}{2} + \delta - i\gamma, \tfrac{1}{2} - \delta + i\gamma, \tfrac{1}{2} - \delta - i\gamma\}, \quad h(z) = \prod_{\alpha \in \mathcal{R}} (z - \alpha)^m, \quad v_h = 2\xi/h, \\ k &\geq 3, \quad c = k/2, \quad e = 1 + k(m-1), \quad q = e(k+1)^2, \\ \lambda_{a,b} &= c + (2a-k)\delta + i(2b-k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^e, \quad E = \mathbb{C}[S]/(\chi), \\ w_h(y) &= |v_h(1/2 + iy)|^2/(2\pi), \quad m_{h,k} = w_h^{*k}, \quad d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi). \end{aligned} \quad (\text{AMT1})$$

The original source inner products, conjugate-linear in their first argument, are

$$\langle P, Q \rangle_{\text{ar}} = \int_{\mathbb{R}} \overline{P(c + iy)} Q(c + iy) m_{h,k}(y) dy, \quad \langle P, Q \rangle_{\sigma} = \int_{\mathbb{R}} \overline{P(c + iy)} Q(c + iy) d\sigma(y). \quad (\text{AMT2})$$

No measure mass or complex-coordinate phase is changed. In particular $\int d\sigma = \sqrt{2\pi}$. Write $\mathcal{P}_N$ for polynomials in the original S of degree at most N . In its coefficient frame $(1, S, \dots, S^N)$ let the two Grams be $H_N^{\text{ar}}, H_N^{\sigma}$. For $N \geq q-1$, the literal remainder map $J_N : \mathcal{P}_N \rightarrow E$ is onto and has kernel $\chi \mathcal{P}_{N-q}$; at $N = q-1$ this is the zero vector space. The frame on E is $(1, S, \dots, S^{q-1})$.

Here is the original observation being retained. Put $n = 4m$, $E_h = \mathbb{C}[z]/(h)$, $U = M_{j_h v_h}$, and $\iota[P] = [P(z_1 + \dots + z_k)] \in E_h^{\otimes k}$. The map j_h records every divided Taylor coefficient through order $m-1$ at every original root. Its unit $j_h v_h$ is invertible because the stated order is complete. Fix the original $u \neq 0$ and its logarithm branch. With $\Phi'_h = h$ and $\Phi_h(0) = 0$, let ℓ_j be the outward ray of angle $(\arg u + \pi + 2\pi j)/(n+1)$ and $\Gamma_j = -\ell_0 + \ell_j$. The period map and projector are exactly

$$\begin{aligned} (\Pi[P])_j &= \int_{\Gamma_j} e^{\Phi_h(z)/u} \text{rem}_h P(z) dz, \quad 1 \leq j \leq n, \\ C : \Gamma_j &\mapsto \Gamma_{j+1} - \Gamma_1 \quad (j < n), \quad \Gamma_n \mapsto -\Gamma_1, \quad P_k = \frac{1}{n+1} \sum_{a=0}^n (C^{-a})^{\otimes k}, \end{aligned} \quad (\text{AMT3})$$

$$L = P_k \Pi^{\otimes k} U^{\otimes k} \iota, \quad B = \text{im } L, \quad j : B \hookrightarrow (\mathbb{C}^n)^{\otimes k}, \quad j\Lambda = L, \quad \Lambda : E \twoheadrightarrow B, \quad K = \ker \Lambda.$$

These are the convergent original representative integrals; their existence and full determinant are proved in PDE and OPG. Fix the original coefficient inclusion $I : K \hookrightarrow E$ and section $S_* : B \rightarrow E$ with $\Lambda S_* = I_B$. Let $r = \dim K$ and $b = \dim B = q - r$. All matrices below use these same frames and this same full unit and period observation. Empty Grams have determinant one, and maps to or from a zero space have their unique empty values.

8.2 All constants of the actual full-polynomial inequality

Retain exactly the constants of TW and TWA:

$$\begin{aligned} \alpha &= \pi/2, \quad p = 8m - \frac{9}{2}, \quad B_h = 42 + 8m, \quad M = 21 + 4m, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy > 0, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha y} w_h(y) dy \in (0, \infty), \\ c_\Gamma &= \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1}, \quad D_N = [1 + 4(N+M)^2]^M. \end{aligned} \tag{AMT4}$$

Here c_h is the proved original convolution lower-envelope constant TW7 and C_h^{tilt} is the original compact/tail maximum BT12. Their full constructions occur in the preserved arithmetic endpoint and BT providers; they depend on this fixed h and not on k, N, u . Their defining inequalities, proved there with the original numerator, are

$$\begin{aligned} m_{h,k}(y) &\geq c_h \vartheta_h^{k-3} e^{-\alpha(|y|+k-3)} (k-2+|y|)^{-B_h}, \\ m_{h,k}(y) &\leq k C_h^{\text{tilt}} M_\alpha^{k-1} e^{-\alpha|y|} (1+|y|/k)^{-p}, \\ c_\Gamma e^{-\alpha|y|} (1+|y|)^{-1/2} &\leq d\sigma/dy \leq C_\Gamma e^{-\alpha|y|} (1+|y|)^{-1/2}. \end{aligned} \tag{AMT5}$$

For clarity the full polynomial step is given here. The first and third bounds, $k-2+|y| \leq (k-2)(1+|y|)$, and $(1+|y|)^{B_h} \leq 2^{B_h/2}(1+y^2)^M$, give $m_{h,k} \geq a_k(1+y^2)^{-M} d\sigma/dy$. The second and third, using $1+|y|/k \geq (1+|y|)/k$ and $p > 1/2$, give $m_{h,k} \leq A_k d\sigma/dy$. The fixed Gamma orthonormal polynomials obey

$$y e_j = \sqrt{(j+1)(j+1/2)} e_{j+1} + \sqrt{j(j-1/2)} e_{j-1}.$$

The two shifts on degree at most N have norms at most $N+1$ and N , respectively. Thus $\|yP\|_\sigma \leq 2(N+1)\|P\|_\sigma$. Iterating at the successively increased degrees and expanding $(1+y^2)^M$ proves $\int |P|^2 (1+y^2)^M d\sigma \leq D_N \int |P|^2 d\sigma$. Cauchy–Schwarz applied to $|P|(1+y^2)^{-M/2}$ and $|P|(1+y^2)^{M/2}$ then gives the reciprocal lower bound. The coordinate map $P(S) \mapsto P(c+iy)$ has inverse $f(y) \mapsto f((S-c)/i)$, preserves the degree and retains every coefficient through this explicitly invertible substitution. It therefore proves on the complete original source

$$\boxed{\frac{a_k}{D_N} H_N^\sigma \preceq H_N^{\text{ar}} \preceq A_k H_N^\sigma.} \tag{AMT6}$$

All coefficients can be complex in this proof. In particular AMT6 holds on every affine remainder fibre, not only on monic vectors. The two Grams are positive: the Gamma density is positive and the lower bound in AMT5 is positive everywhere. Their finite moments exist by the exponential upper bound.

8.3 Exact remainder, minimum-section, kernel and boundary maps

For $a \in \{\sigma, \text{ar}\}$ put

$$\begin{aligned} G_N^a &= (J_N (H_N^a)^{-1} J_N^*)^{-1}, \quad V_N^a = (H_N^a)^{-1} J_N^* G_N^a : E \longrightarrow \mathcal{P}_N, \\ H_{K,N}^a &= I^* G_N^a I, \quad Q_N^a = (\Lambda (G_N^a)^{-1} \Lambda^*)^{-1}, \quad S_N^a = (G_N^a)^{-1} \Lambda^* Q_N^a : B \longrightarrow E. \end{aligned} \tag{AMT7}$$

Inverses are on the indicated nonzero spaces. For an onto matrix J and positive H , direct multiplication gives $JV = I$ for $V = H^{-1} J^* (JH^{-1} J^*)^{-1}$. Moreover $Z^* H V = Z^* J^* G = 0$ for $JZ = 0$. Therefore every lift of z is $Vz + Z$, with squared norm $z^* G z + Z^* H Z$, proving its unique minimum and its Gram. Applying this proof first to J_N , then to Λ , proves every formula AMT7 and $\Lambda S_N^a = I_B$, $I^* G_N^a S_N^a = 0$. It also proves that $V_N^a S_N^a$ is the minimum lift for the composite ΛJ_N .

The exact comparison map on $\mathcal{P}_N$ is the identity on original coefficients; it commutes with J_N and ΛJ_N . It induces the identity on E, K, B . Between the two spaces of minimum lifts the exact bijection and its inverse are

$$\text{im } V_N^\sigma \longrightarrow \text{im } V_N^{\text{ar}}, \quad V_N^\sigma z \longmapsto V_N^{\text{ar}} z, \quad T_N = V_N^{\text{ar}} J_N|_{\text{im } V_N^\sigma}, \quad T_N^{-1} = V_N^\sigma J_N|_{\text{im } V_N^{\text{ar}}}. \quad (\text{AMT8})$$

Their difference $V_N^{\text{ar}} z - V_N^\sigma z$ lies in $\chi \mathcal{P}_{N-q}$, with its unique coefficient obtained by division by the unchanged monic χ . This also gives the minimum-lift bijection on the preimage of K and the analogous bijection $V_N^\sigma S_N^\sigma y \mapsto V_N^{\text{ar}} S_N^{\text{ar}} y$ for boundary lifts. The latter differences lie in $\ker(\Lambda J_N)$, exactly as multiplication by ΛJ_N verifies.

Minimize both sides of AMT6 over $J_N P = z$. Taking infima preserves each inequality because these are the same nonempty fibres. Restrict the resulting inequality to $z = Ix$, and independently minimize it over $\Lambda z = y$. This proves the three full-vector comparisons

$$\begin{aligned} (a_k/D_N) G_N^\sigma &\preceq G_N^{\text{ar}} \preceq A_k G_N^\sigma, \\ (a_k/D_N) H_{K,N}^\sigma &\preceq H_{K,N}^{\text{ar}} \preceq A_k H_{K,N}^\sigma, \\ (a_k/D_N) Q_N^\sigma &\preceq Q_N^{\text{ar}} \preceq A_k Q_N^\sigma. \end{aligned} \quad (\text{AMT9})$$

The source identity is an exact bounded bijection with the two constants in AMT6. No invariance of K under an additional multiplier was used to obtain its induced identities.

The matrix $[I, S_*]$ is invertible: its inverse sends z to the unique kernel coordinate of $z - S_* \Lambda z$ and the boundary coordinate Λz . Each $S_N^a - S_*$ lies in K , so the change from this frame to $[I, S_N^a]$ is triangular with identity diagonal and determinant one. The latter frame is orthogonal in G_N^a by AMT7. Consequently

$$\det H_{K,N}^a \det Q_N^a = |\det[I, S_*]|^2 \det G_N^a, \quad a \in \{\sigma, \text{ar}\}. \quad (\text{AMT10})$$

This includes $r = 0$ and $b = 0$ and retains the original frame factor.

8.4 The exact nested deficits and their rank factors

Define the actual finite source determinant deficit and its width

$$\Xi_{k,N} = (N+1) \log A_k - \log \frac{\det H_N^{\text{ar}}}{\det H_N^\sigma}, \quad L_N = \log(A_k D_N / a_k) \geq 0. \quad (\text{AMT11})$$

The last sign follows by applying AMT6 to any nonzero vector. Put

$$\begin{aligned} d_{E,N} &= q \log A_k - \log \frac{\det G_N^{\text{ar}}}{\det G_N^\sigma}, \\ d_{K,N} &= r \log A_k - \log \frac{\det H_{K,N}^{\text{ar}}}{\det H_{K,N}^\sigma}, \\ d_{B,N} &= b \log A_k - \log \frac{\det Q_N^{\text{ar}}}{\det Q_N^\sigma}. \end{aligned} \quad (\text{AMT12})$$

AMT9, by congruence with the inverse square root of each comparison Gram, places each eigenvalue of its relative Gram in $[a_k/D_N, A_k]$. Multiplication of all eigenvalues gives $0 \leq d_{K,N} \leq r L_N$, $0 \leq d_{B,N} \leq b L_N$ and $0 \leq d_{E,N} \leq q L_N$.

Here is the full source-to-quotient determinant calculation, including all relation directions. Choose an H_N^σ -orthonormal coefficient frame R_N of $\ker J_N$ solely for this computation, and put $W_N = [R_N, V_N^\sigma (G_N^\sigma)^{-1/2}]$. AMT7 gives $W_N^* H_N^{\text{ar}} W_N = I_{N+1}$. In this frame the positive contraction

$$\mathcal{D}_N = A_k^{-1} W_N^* H_N^{\text{ar}} W_N = \begin{pmatrix} F_N & C'_N \\ (C'_N)^* & A'_N \end{pmatrix} \preceq I_{N+1}$$

has Schur complement

$$T'_N = A'_N - (C'_N)^* F_N^{-1} C'_N = A_k^{-1} (G_N^\sigma)^{-1/2} G_N^{\text{ar}} (G_N^\sigma)^{-1/2}.$$

Indeed minimization in its first block is exactly the original remainder minimum at $z = (G_N^\sigma)^{-1/2} w$. Both F_N and T'_N are positive contractions: positivity follows from positive block elimination, and the upper bound for the Schur complement follows by testing its minimum at zero first coordinate. Since $|\det W_N|^2 = (\det H_N^\sigma)^{-1}$, elimination gives

$$\Xi_{k,N} = d_{R,N} + d_{E,N}, \quad d_{R,N} := -\log \det F_N \geq 0, \quad d_{E,N} = -\log \det T'_N \geq 0. \quad (\text{AMT13})$$

When $N = q - 1$, F_N is empty, $d_{R,N} = 0$, and the same equalities hold without a relation inverse. The determinant identities are independent of the auxiliary orthonormal frame. Dividing the two instances of AMT10 proves $d_{K,N} + d_{B,N} = d_{E,N}$. Thus the complete nested statement is

$$\boxed{\begin{aligned} \Xi_{k,N} &= d_{R,N} + d_{K,N} + d_{B,N}, & d_{R,N}, d_{K,N}, d_{B,N} &\geq 0, \\ 0 &\leq d_{K,N} \leq \min\{rL_N, d_{E,N}\} \leq \min\{rL_N, \Xi_{k,N}\}, \\ 0 &\leq d_{B,N} \leq \min\{bL_N, d_{E,N}\} \leq \min\{bL_N, \Xi_{k,N}\}. \end{aligned}} \quad (\text{AMT14})$$

In particular each metric keeps its own rank factor, and the two deficits retain their exact common sum. Any proved finite upper bound for the original $\Xi_{k,N}$, including the degree-specific HMR44/TWA15 bound, can be substituted in these displayed minima. No new density or source determinant is introduced by this substitution.

8.5 All four original endpoints and the correlated mixed return

Let

$$(N_0, N_1, N_2, N_3) = (q - 1, q, 2q - 1, 2q), \quad (\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3) = (1, 1, -1, -1),$$

and define, in their original signs,

$$\begin{aligned} F_K^a &= \sum_{j=0}^3 \epsilon_j \log \det H_{K,N_j}^a, & F_B^a &= \sum_{j=0}^3 \epsilon_j \log \det Q_{N_j}^a, \\ \mathcal{B}_k^a &= \sum_{j=0}^3 \epsilon_j \log \det G_{N_j}^a, & \Delta_K^\sigma &= F_K^{\text{ar}} - F_K^\sigma, \quad \Delta_B^\sigma = F_B^{\text{ar}} - F_B^\sigma, \quad \delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma. \end{aligned} \quad (\text{AMT15})$$

AMT10 cancels its four identical frame factors only after these signs are applied. It gives $F_K^a + F_B^a = \mathcal{B}_k^a$ and $\Delta_K^\sigma + \Delta_B^\sigma = \delta_k^\sigma$ exactly. The spaces $\mathcal{P}_N$ increase with the same source norm and remainder map, so their quotient minima decrease. Restriction to K and minimization onto B preserve that order. Pairing N_0 with N_2 and N_1 with N_3 consequently proves $F_K^a, F_B^a \geq 0$.

For $X = K, B$, put $r_K = r$, $r_B = b$, and retain the exact caps

$$c_{X,N} = \min\{r_X L_N, d_{E,N}\}, \quad C_X^- = c_{X,q-1} + c_{X,q}, \quad C_X^+ = c_{X,2q-1} + c_{X,2q}. \quad (\text{AMT16})$$

The entirely source-evaluated caps $\hat{c}_{X,N} = \min\{r_X L_N, \Xi_{k,N}\}$ and the corresponding $\hat{C}_X^\pm$ are at least these values; all conclusions below also hold with the hats. Substituting AMT12 in AMT15 records each endpoint and its sign:

$$\boxed{\begin{aligned} \Delta_K^\sigma &= -d_{K,q-1} - d_{K,q} + d_{K,2q-1} + d_{K,2q}, \\ \Delta_B^\sigma &= -d_{B,q-1} - d_{B,q} + d_{B,2q-1} + d_{B,2q}, \\ \delta_k^\sigma &= -d_{E,q-1} - d_{E,q} + d_{E,2q-1} + d_{E,2q}, \\ -C_X^- &\leq \Delta_X^\sigma \leq C_X^+, \quad X = K, B. \end{aligned}} \quad (\text{AMT17})$$

The four copies of $r_X \log A_k$ cancel here because their signs sum to zero. Since the original total correction is retained, the joint interval is stronger than two independent intervals:

$$\boxed{\begin{aligned} \max\{-C_K^-, \delta_k^\sigma - C_B^+\} &\leq \Delta_K^\sigma \leq \min\{C_K^+, \delta_k^\sigma + C_B^-\}, \\ \max\{-C_B^-, \delta_k^\sigma - C_K^+\} &\leq \Delta_B^\sigma \leq \min\{C_B^+, \delta_k^\sigma + C_K^-\}. \end{aligned}} \quad (\text{AMT18})$$

These follow by subtracting the interval for either component from the exact common sum; they require no independence assertion.

Here are the explicit original four endpoint widths:

$$\begin{aligned} E_k^+ &= L_{q-1} + L_q = 2 \log(A_k/a_k) + M \log[1 + 4(q-1+M)^2] + M \log[1 + 4(q+M)^2], \\ E_k^- &= L_{2q-1} + L_{2q} = 2 \log(A_k/a_k) + M \log[1 + 4(2q-1+M)^2] + M \log[1 + 4(2q+M)^2]. \end{aligned} \quad (\text{AMT19})$$

They are the unchanged TWA5 constants, both nonnegative. AMT17 therefore gives in particular

$$\boxed{-rE_k^+ \leq F_K^{\text{ar}} - F_K^\sigma \leq rE_k^-, \quad -bE_k^+ \leq F_B^{\text{ar}} - F_B^\sigma \leq bE_k^-}. \quad (\text{AMT20})$$

The total correction retains TWA15 with all four individual source deficits:

$$\begin{aligned} \delta_k^\sigma &= \max\{-qE_k^+, -\Xi_{k,q-1} - \Xi_{k,q}\}, \\ \bar{\delta}_k^\sigma &= \min\{qE_k^-, \Xi_{k,2q-1} + \Xi_{k,2q}\}, \quad \underline{\delta}_k^\sigma \leq \delta_k^\sigma \leq \bar{\delta}_k^\sigma. \end{aligned} \quad (\text{AMT21})$$

This also follows immediately from AMT14 and AMT17. It can be inserted into AMT18 when an interval, rather than the actual finite total determinant, is wanted. Thus the lower and upper endpoints are kept in their distinct places throughout the mixed transfer.

For later use, subtraction of the exact logarithms in AMT4 gives

$$\begin{aligned} \mathfrak{s}_h &= \alpha + \log(M_\alpha/\vartheta_h), \quad \mathfrak{c}_h = -\log M_\alpha + 3 \log \vartheta_h - 3\alpha + \log \frac{C_h^{\text{tilt}} C_\Gamma 2^{B_h/2}}{c_h c_\Gamma}, \\ \log(A_k/a_k) &= k\mathfrak{s}_h + (p+1) \log k + B_h \log(k-2) + \mathfrak{c}_h, \quad E_k^\pm = O_h(k + \log q). \end{aligned} \quad (\text{AMT22})$$

The last statement follows from the displayed exact expressions, since M, p, B_h and the two integral constants are fixed and $\log D_N = O_h(1 + \log(N+1))$. It is a consequence of the original constants and contains no change of measure.

8.6 The full original multiplier and its transported observation

On the existing tensor class $k = 4l + 1$, retain

$$\begin{aligned} t_j &= 2j + \frac{1}{2}, \quad \xi_j = c + t_j, \quad f_l(S) = \prod_{j=0}^{l-1} (S - c - t_j), \\ \beta_k &= \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}, \quad dm_{0,k}(y) = \beta_k |f_l(c + iy)|^2 d\sigma(y). \end{aligned} \quad (\text{AMT23})$$

This is the exact Gamma recurrence identity TWA26 with masses $(2\pi)^{k/2}$ and $\sqrt{2\pi}$. For $l = 0$ both f_l and β_k are one. Write $E_\chi : E \rightarrow \mathbb{C}^q$ for the complete divided Taylor map $[P] \mapsto (P^{(r)}(\lambda_{a,b})/r!)_{a,b, 0 \leq r < e}$. It is invertible: a vector in its kernel is divisible by the coprime factors of χ , and dimensions agree. The full multiplication matrix U_f has, in each primary block, entries

$$\begin{aligned} (U_f)_{r,a} &= \begin{cases} f_l^{(r-a)}(\lambda)/(r-a)!, & 0 \leq a \leq r < e, \\ 0, & a > r. \end{cases} \\ \det U_f &= \prod_{a,b=0}^k \prod_{j=0}^{l-1} [(2a-k)\delta - t_j + i(2b-k)\gamma]^e \neq 0. \end{aligned} \quad (\text{AMT24})$$

Indeed k is odd, so every $\lambda_{a,b}$ has nonzero imaginary part, while all ξ_j are real and distinct. The inverse in each block is the finite Taylor inverse of its nonzero constant term. The exact polynomial map and its inverse on its stated range are

$$\mathcal{P}_N \longrightarrow f_l \mathcal{P}_N \subseteq \mathcal{P}_{N+l}, \quad P \longmapsto f_l P, \quad Q \longmapsto Q/f_l. \quad (\text{AMT25})$$

It sends the original fibre $J_N P = z$ bijectively to the product jet fibre $(\mathbf{E}_f Q, \mathbf{E}_\chi Q) = (0, U_f \mathbf{E}_\chi z)$. Vanishing of the l distinct f -values proves divisibility by f_l , and the inverse of U_f proves the converse fibre statement. The original relation space maps to $f_l \chi \mathcal{P}_{N-q}$. On the free χ jet coordinate Y of this constrained product fibre, the exact observed map, its kernel, and a section are

$$\begin{aligned} \tilde{\Lambda} &= \Lambda \mathbf{E}_\chi^{-1} U_f^{-1}, & \tilde{\Lambda} U_f \mathbf{E}_\chi &= \Lambda, \\ \ker \tilde{\Lambda} &= U_f \mathbf{E}_\chi K, & \tilde{I} &= U_f \mathbf{E}_\chi I, & \tilde{S}_* &= U_f \mathbf{E}_\chi S_*, & \tilde{\Lambda} \tilde{S}_* &= I_B. \end{aligned} \quad (\text{AMT26})$$

Both inclusions in the kernel equality follow by applying the displayed invertible map and its inverse. This retains the full original amplitude in Λ and every derivative in U_f . The resulting kernel is proved by transport; it has not been assigned the coordinates of $\mathbf{E}_\chi K$.

For an explicit metric on this transported space, let $b_n(y)$ be the fixed monic Gamma polynomials and put $\pi_n(S) = i^n b_n((S - c)/i)$, whose leading coefficient is one and squared norm is $\gamma_n = \sqrt{2\pi}(2n)!/4^n$. The evaluation kernels are the full finite sums

$$\mathcal{K}_{\chi,L} = \sum_{n=0}^L \frac{\mathbf{E}_\chi \pi_n (\mathbf{E}_\chi \pi_n)^*}{\gamma_n}, \quad \mathcal{K}_{f,L} = \sum_{n=0}^L \frac{\mathbf{E}_f \pi_n (\mathbf{E}_f \pi_n)^*}{\gamma_n}, \quad \mathcal{K}_{\chi,f;L} = \sum_{n=0}^L \frac{\mathbf{E}_\chi \pi_n (\mathbf{E}_f \pi_n)^*}{\gamma_n}.$$

The product evaluation is onto for $L = N + l \geq q + l - 1$, because a polynomial of degree below $q + l$ vanishing at all its divided jets is divisible by the degree- $q + l$ product $f_l \chi$ and is zero. Thus its positive kernel has positive Schur complement

$$\begin{aligned} C_N &= \mathcal{K}_{\chi,N+l} - \mathcal{K}_{\chi,f;N+l} \mathcal{K}_{f,N+l}^{-1} \mathcal{K}_{f,\chi;N+l} > 0, \\ U_N &= (\mathbf{E}_\chi \pi_{N+j} / \sqrt{\gamma_{N+j}})_{j=1}^l, \quad V_N = \mathcal{K}_{\chi,f;N+l} \mathcal{K}_{f,N+l}^{-1/2}, \quad C_N = \mathcal{K}_{\chi,N} + U_N U_N^* - V_N V_N^*. \end{aligned} \quad (\text{AMT27})$$

Empty f blocks have the identity convention. The inverse of the joint positive kernel, evaluated on $(0, Y)$, gives the minimum $Y^* C_N^{-1} Y$: block elimination of the f -block proves this identity with every mixed entry present. Combining it with AMT25 and the coefficient β_k proves the exact tensor-source metrics

$$\begin{aligned} G_N^0 &= \beta_k \mathbf{E}_\chi^* U_f^* C_N^{-1} U_f \mathbf{E}_\chi, & H_{K,N}^0 &= \beta_k \tilde{I}^* C_N^{-1} \tilde{I}, \\ Q_N^0 &= \beta_k (\tilde{\Lambda} C_N \tilde{\Lambda}^*)^{-1}, & S_N^0 &= \mathbf{E}_\chi^{-1} U_f^{-1} C_N \tilde{\Lambda}^* (\tilde{\Lambda} C_N \tilde{\Lambda}^*)^{-1}. \end{aligned} \quad (\text{AMT28})$$

Direct multiplication verifies the section and its orthogonality. The formulas hold on the original E, K, B , with the transported observation exactly as in AMT26. No block or mixed product was dropped from C_N .

The same coefficient identity also supplies a quantitative comparison directly with this original tensor source. Define the positive exact finite factors

$$\Pi_l = \prod_{j=0}^{l-1} t_j^2, \quad U_{l,N}^{\text{poly}} = \prod_{j=0}^{l-1} [4(N+j+1)^2 + t_j^2], \quad \tilde{L}_N = L_N + \log(U_{l,N}^{\text{poly}} / \Pi_l). \quad (\text{AMT29})$$

Pointwise $|f_l(c + iy)|^2 \geq \Pi_l$. Also $\|(iy - t)P\|_\sigma^2 = \|yP\|_\sigma^2 + t^2 \|P\|_\sigma^2$, so the degree-specific multiplication estimate in AMT6, applied successively, gives $\|f_l P\|_\sigma^2 \leq U_{l,N}^{\text{poly}} \|P\|_\sigma^2$. Writing H_N^0 for the unchanged $m_{0,k}$ source Gram yields

$$\beta_k \Pi_l H_N^\sigma \preceq H_N^0 \preceq \beta_k U_{l,N}^{\text{poly}} H_N^\sigma, \quad \frac{a_k}{D_N \beta_k U_{l,N}^{\text{poly}}} H_N^0 \preceq H_N^{\text{ar}} \preceq \frac{A_k}{\beta_k \Pi_l} H_N^0. \quad (\text{AMT30})$$

The second pair follows by using the first upper bound for its lower inequality and the first lower bound for its upper inequality. Minimizing and restricting with the same J_N, I, Λ gives AMT30 for all three metrics in AMT28. Its common upper scalar is $A_k/(\beta_k \Pi_l)$; it is retained before its four signed copies cancel. The same determinant proof as AMT12–17 therefore gives

$$\begin{aligned} \tilde{E}_k^+ &= E_k^+ + \log(U_{l,q-1}^{\text{poly}} U_{l,q}^{\text{poly}} / \Pi_l^2), & \tilde{E}_k^- &= E_k^- + \log(U_{l,2q-1}^{\text{poly}} U_{l,2q}^{\text{poly}} / \Pi_l^2), \\ \boxed{-r\tilde{E}_k^+ \leq F_K^{\text{ar}} - F_K^0 \leq r\tilde{E}_k^-, \quad -b\tilde{E}_k^+ \leq F_B^{\text{ar}} - F_B^0 \leq b\tilde{E}_k^-} \end{aligned} \quad (\text{AMT31})$$

For an explicit bound on the extra widths, $t_j \geq 1/2$ gives

$$0 \leq \log(U_{l,N}^{\text{poly}} / \Pi_l) = \sum_{j=0}^{l-1} \log \left(1 + \frac{4(N+j+1)^2}{t_j^2} \right) \leq l \log[1 + 16(N+l)^2]. \quad (\text{AMT32})$$

Thus $\tilde{E}_k^\pm = O_h(k \log(q+k))$ on the stated endpoints. This is a proved polynomial-source estimate for the exact original f_l , with no inference that its multiplication preserves K .

8.7 The evaluated original simple-quartet arithmetic kernel

Now take the original simple-quartet class $m = 1, k = 4l+1, l \geq 1, q = (k+1)^2$. The parameter u is precisely the period parameter in AMT3, not the real integration coordinate y . RPC21 and RPC23 give an explicit finite radius R_* for the unchanged quartet and arithmetic unit. For reference its exact constants and cases are recorded here. Set

$$\begin{aligned} \beta &= 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad a = v_h(\tfrac{1}{2} + \delta + i\gamma) \neq 0, \\ c_3 &= \frac{a^{-2} - \bar{a}^{-2}}{4i\delta\gamma}, \quad c_1 = \frac{a^{-2} + \bar{a}^{-2}}{2} - (\delta^2 - \gamma^2)c_3, \\ g_j &= 5^{j/5-1} e^{\pi i j/5} \Gamma(j/5) \quad (1 \leq j \leq 4), \quad g_+ = \max_j |g_j|, \quad g_-^{-1} = \max_j |g_j|^{-1}, \quad \kappa = \max_{1 \leq j < 4} |g_{j+1}/g_j|, \\ E_j^{\text{ray}} &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^5/5 + r^3/3 + r} dr \quad (0 \leq j \leq 3), \quad C_T = 2 \left(\sum_{j=0}^3 (E_j^{\text{ray}})^2 \right)^{1/2}, \\ C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \quad C_{K,3} = C_K(3\kappa^2 + 3\kappa C_K + C_K^2), \\ \epsilon_* &= \min\{1, (2C_T g_-^{-1})^{-1}\}, \quad B_* = \beta + \eta. \end{aligned} \quad (\text{AMT33})$$

Here $g_-^{-2} = (g_-^{-1})^2$. All integrals converge by their negative fifth-degree exponent, and $(c_1, c_3) \neq (0, 0)$ by the original unit values. The exact radius is

$$R_* = \begin{cases} \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2(|c_3|C_{K,3}B_* + |c_1|(\kappa + C_K))}{|c_3 g_4/g_1|} \right\} \right]^{5/2}, & c_3 \neq 0, \\ \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2C_K B_*}{|g_2/g_1|} \right\} \right]^{5/2}, & c_3 = 0. \end{cases} \quad (\text{AMT34})$$

The complete period calculation RPC16–25 proves, for every original branch and every $|u| \geq R_*$, a nonzero original commutator entry: its $(4, 1)$ entry in the first case and its $(2, 1)$ entry in the second. RPC14–15 identifies those entries with the exact test for the original rank-four quadric's cyclic invariance. Thus its actual projective orbit has five elements throughout this domain. OQX1–3, using the product of precisely these five original quadrics, proves

$$\boxed{r = 8k - 16, \quad b = k^2 - 6k + 17, \quad m = 1, \quad k = 4l + 1, \quad l \geq 1, \quad |u| \geq R_*} \quad (\text{AMT35})$$

This is the evaluated original period outcome; no value of the amplitude or of a period coefficient is chosen. The radius depends on the fixed one-factor quartet and unit and is independent of k .

For the same original packet, OKV22 proves for both existing source orders $s = 1, k$, and for the actual K of any rank,

$$\begin{aligned} D_* &= \max\{3, \sqrt{\delta^2 + \gamma^2}\}, \quad C(\delta, \gamma) = 12 + 4 \log(2D_*), \\ Z_q &= 1 + 2(q+1)q^2 2^{6q} (2D_* q)^{4q}, \\ 0 \leq F_K^{(s)} &\leq 2r \log Z_q \leq 2rC(\delta, \gamma)q \log(q+1), \\ F_K^{(1)} &= F_K^\sigma, \quad F_K^{(k)} = F_K^0. \end{aligned} \tag{AMT36}$$

Its proof includes the exact source coefficient map and the unchanged kernel inclusion. Combining it with the already proved AMT17–20 gives the finite arithmetic conclusion

$$\begin{aligned} \max\{0, F_K^\sigma - C_K^-\} &\leq F_K^{\text{ar}} \leq F_K^\sigma + C_K^+, \\ 0 \leq F_K^{\text{ar}} &\leq U_K := 2r \log Z_q + \widehat{C}_K^+ \\ &\leq 2rC(\delta, \gamma)q \log(q+1) + rE_k^-. \end{aligned} \tag{AMT37}$$

The cap symbols $C_K^\pm, \widehat{C}_K^\pm$ in this formula mean the endpoint sums AMT16; the constant C_K without a superscript in AMT33 belongs only to the explicit period radius. They have different stated indices and are not identified. The alternative finite tensor-source bound $F_K^{\text{ar}} \leq 2r \log Z_q + r\widetilde{E}_k^-$ also follows from AMT31. Formula AMT37 retains the sharper fixed-source widths.

On AMT35 the exact ranks give

$$F_K^{\text{ar}} - F_K^\sigma = O_h(k^2), \quad F_K^{\text{ar}} = O_h(kq \log(q+1)), \quad \frac{F_K^{\text{ar}}}{q^2} \longrightarrow 0. \tag{AMT38}$$

For the first estimate use $r = 8k - 16$ in AMT20 and AMT22, with $\log q = 2 \log(k+1)$. The second follows from AMT37. For the limit, divide its last line by q^2 : its first term is at most $16C(\delta, \gamma)k \log(q+1)/q \rightarrow 0$, and its second is $O_h(k^2/q^2) \rightarrow 0$. Nonnegativity gives the limit by the two bounds. These estimates are uniform over the entire original domain $|u| \geq R_*$, since none of their right sides depends on u . The tensor comparison likewise gives $F_K^{\text{ar}} - F_K^0 = O_h(k^2 \log q) = o(q^2)$ from AMT31–32.

8.8 The original boundary receives the full baseline coefficient

Before using a limit, the exact finite arithmetic identity and its consequence are

$$\begin{aligned} F_B^{\text{ar}} &= \mathcal{B}_k^\sigma + \delta_k^\sigma - F_K^{\text{ar}}, \\ \max\{0, \mathcal{B}_k^\sigma + \delta_k^\sigma - U_K\} &\leq F_B^{\text{ar}} \leq \mathcal{B}_k^\sigma + \delta_k^\sigma, \\ \max\{0, \mathcal{B}_k^\sigma + \underline{\delta}_k^\sigma - U_K\} &\leq F_B^{\text{ar}} \leq \mathcal{B}_k^\sigma + \bar{\delta}_k^\sigma. \end{aligned} \tag{AMT39}$$

The first identity follows from AMT10 and AMT15; AMT37 and $F_K^{\text{ar}}, F_B^{\text{ar}} \geq 0$ prove the first interval, and AMT21 proves the second. The correlated boundary return also has the sharper exact interval AMT18, with every original signed total correction present.

BSL13–19 supplies the complete original finite baseline and its proved coefficient

$$\mathcal{B}_k^\sigma = V_q + \eta_q + \eta_{q+1} - \eta_1, \quad \frac{\mathcal{B}_k^{\text{ar}}}{q^2} \longrightarrow C_B^{\text{base}} := 9 - 8 \log 2 + \mathcal{F}(2, \pi) > \frac{769}{17010} > 0. \tag{AMT40}$$

Here V_q , the three distinct original relation errors η_r , and the energy $\mathcal{F}$ are exactly the BSL/HBL/GEL quantities, whose complete definitions and proofs are in the retained closure. The superscript on C_B^{base} only distinguishes that existing baseline coefficient from the endpoint caps $C_B^\pm$ in AMT16. Its finite BSL14 enclosure can be substituted directly for $\mathcal{B}_k^\sigma$ in AMT39,

keeping the signs of all three η terms; no new evaluation is needed for that substitution. Subtract AMT38 from the exact original identity $F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}} - F_K^{\text{ar}}$. It proves

$$\boxed{\frac{F_B^{\text{ar}}}{q^2} \longrightarrow C_B^{\text{base}} > 0, \quad F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}, \quad m = 1, \quad k = 4l + 1, \quad l \rightarrow \infty, \quad |u| \geq R_*} \quad (\text{AMT41})$$

The convergence is uniform over that original period domain by AMT38 and because the full-source baseline is independent of u . Thus the proved original kernel estimate now passes to the actual arithmetic kernel, and the original arithmetic boundary carries the entire positive q^2 coefficient. All earlier full-sum identities remain exact, with their individual mixed contributions now receiving AMT37–41 and their signed finite corrections AMT17–21.

8.9 The same comparison evaluates the mixed Gamma return on the kq scale

The first inequality pair AMT30 directly compares the two existing Gamma sources on the same original coefficient space. Apply its proved minimum and restriction maps AMT7–9 to $X = E, K, B$, writing $H_{E,N}^a = G_N^a$, $H_{B,N}^a = Q_N^a$, and $r_E = q, r_K = r, r_B = b$. At every original endpoint it gives

$$\beta_k \Pi_l H_{X,N}^\sigma \preceq H_{X,N}^0 \preceq \beta_k U_{l,N}^{\text{poly}} H_{X,N}^\sigma. \quad (\text{AMT42})$$

Define the exact relative determinant and the positive degree width

$$\begin{aligned} z_{X,N} &= \log \frac{\det H_{X,N}^0}{\det H_{X,N}^\sigma} - r_X \log(\beta_k \Pi_l), & \omega_N^\Gamma &= \log(U_{l,N}^{\text{poly}} / \Pi_l), \\ 0 &\leq z_{X,N} \leq r_X \omega_N^\Gamma, & z_{K,N} + z_{B,N} &= z_{E,N}, \\ \Omega_k^{\text{low}} &= \omega_{q-1}^\Gamma + \omega_q^\Gamma, & \Omega_k^{\text{high}} &= \omega_{2q-1}^\Gamma + \omega_{2q}^\Gamma. \end{aligned} \quad (\text{AMT43})$$

Congruence and multiplication of the eigenvalues in AMT42 prove the two inequalities. The determinant identity AMT10, applied to the tensor metric AMT28 as well, proves the common sum. These facts preserve the complete original kernel and boundary rather than assigning separate arbitrary errors to their determinants.

Use the precise return orientation

$$\Delta_X^\Gamma = F_X^\sigma - F_X^0 \quad (X = K, B), \quad \mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0 = -\mathfrak{T}_k.$$

The last equality is the original TWA30/BSL18 sign. Substituting all four z terms, and canceling $\log(\beta_k \Pi_l)$ only after the signed sum, proves

$$\boxed{\begin{aligned} \Delta_X^\Gamma &= -z_{X,q-1} - z_{X,q} + z_{X,2q-1} + z_{X,2q}, \\ -r_X \Omega_k^{\text{low}} &\leq \Delta_X^\Gamma \leq r_X \Omega_k^{\text{high}}, \\ \Delta_K^\Gamma + \Delta_B^\Gamma &= \mathcal{H}_k, \\ \mathcal{H}_k - r \Omega_k^{\text{high}} &\leq \Delta_B^\Gamma \leq \mathcal{H}_k + r \Omega_k^{\text{low}}. \end{aligned}} \quad (\text{AMT44})$$

AMT29 displays every factor of both Ω values explicitly; AMT32 gives the finite bounds

$$\begin{aligned} 0 &\leq \Omega_k^{\text{low}} \leq l \log[1 + 16(q-1+l)^2] + l \log[1 + 16(q+l)^2], \\ 0 &\leq \Omega_k^{\text{high}} \leq l \log[1 + 16(2q-1+l)^2] + l \log[1 + 16(2q+l)^2]. \end{aligned} \quad (\text{AMT45})$$

In particular both are $O(k \log(q+1))$ on the original $m = 1$ class. Its actual rank $r = 8k - 16$ therefore proves directly

$$\boxed{\Delta_K^\Gamma = O_h(k^2 \log(q+1)) = o(kq), \quad \frac{F_K^{\text{ar}} - F_K^\sigma}{kq} \longrightarrow 0, \quad \frac{F_K^{\text{ar}} - F_K^0}{kq} \longrightarrow 0.} \quad (\text{AMT46})$$

For the first equality divide the AMT44 bounds by kq and use $q = (k+1)^2$, so $k \log(q+1)/q \rightarrow 0$. The second limit uses AMT38, since $k^2/(kq) = k/q \rightarrow 0$. The last uses the exact identity $F_K^{\text{ar}} - F_K^0 = \Delta_K^\sigma + \Delta_K^\Gamma$. All bounds are uniform for $|u| \geq R_*$.

For completeness the entire finite arithmetic boundary return is also evaluated by those same quantities:

$$\boxed{\begin{aligned} F_B^{\text{ar}} - F_B^0 &= \mathcal{H}_k + \delta_k^\sigma - (F_K^{\text{ar}} - F_K^0), \\ \mathcal{H}_k + \delta_k^\sigma - r\tilde{E}_k^- &\leq F_B^{\text{ar}} - F_B^0 \leq \mathcal{H}_k + \delta_k^\sigma + r\tilde{E}_k^+. \end{aligned}} \quad (\text{AMT47})$$

This uses AMT31 with its exact endpoint constants $\tilde{E}_k^+ = E_k^+ + \Omega_k^{\text{low}}$ and $\tilde{E}_k^- = E_k^- + \Omega_k^{\text{high}}$, rather than a new source comparison. Both tensor and arithmetic source masses and the full original resultant have already been retained in AMT23–30 before these signed cancellations.

The proved GEL18–19 constant is

$$\mathcal{C}_\Gamma = 2 \int_{a_*^2}^{b_*^2} \log x \, \rho_{2,\pi}(x) \, dx - 4(2 \log 2 - 1) > 0, \quad \frac{\mathfrak{T}_k}{lq} \rightarrow \mathcal{C}_\Gamma. \quad (\text{AMT48})$$

Here a_*, b_* and $\rho_{2,\pi}$ are exactly the proved GEL8/EIQ equilibrium endpoints and density; their complete definitions and integral proof remain in that provider. The endpoint letters here avoid the already fixed original period parameter u . GEL18a gives $4 \log(27/(8\pi)) \leq \mathcal{C}_\Gamma \leq 6$ and proves the strict positivity. TWA15 also proves the actual source return $\delta_k^\sigma/(kq) \rightarrow 0$. As $l/k \rightarrow 1/4$, AMT44, AMT46 and AMT47 now prove

$$\boxed{\begin{aligned} \frac{F_B^\sigma - F_B^0}{kq} &\rightarrow -\frac{\mathcal{C}_\Gamma}{4}, & \frac{F_B^{\text{ar}} - F_B^0}{kq} &\rightarrow -\frac{\mathcal{C}_\Gamma}{4}, & \frac{F_B^{\text{ar}} - F_B^\sigma}{kq} &\rightarrow 0, \\ \frac{F_B^\sigma - F_B^0}{\mathcal{H}_k} &\rightarrow 1, & \frac{F_B^{\text{ar}} - F_B^0}{\mathcal{H}_k} &\rightarrow 1, & m=1, \, k=4l+1, \, l \rightarrow \infty, \, |u| \geq R_*. \end{aligned}} \quad (\text{AMT49})$$

Indeed the first two limits subtract an $o(kq)$ kernel return from $\mathcal{H}_k = -\mathfrak{T}_k$, whose quotient by kq tends to $-\mathcal{C}_\Gamma/4$. The third follows from $\Delta_B^\sigma = \delta_k^\sigma - \Delta_K^\sigma$. The nonzero limit of $\mathcal{H}_k/(kq)$ makes its denominator nonzero for all sufficiently large original k and proves the last two ratios by division. The uniformity in u follows from the uniform kernel bounds and the period-independent total returns. Thus the full existing source comparison evaluates the mixed Gamma return also at its kq scale, with its finite signed remainders AMT44 and AMT47 still present.

Exact transport of the original arithmetic return to its native allocation. For this native transport take the original simple-quartet domain $m=1$, $k=4l+1 \geq 9$, a fixed original five-orbit period and branch, and $|u| \geq R_*$, with all conductor and coefficient data unchanged. Retain AMT44's original convention $\Delta_K^\Gamma = F_K^{(1)} - F_K^{(k)}$ and primal rank $r=8k-16$. The original native kernel rank is $t_k = 8k-16-t_{0,k}$ and its residual rank is $j_{0,k} = 8k-32-v+t_{0,k}$. The fixed-column native return is related to the actual primal kernel by the full FEX/RQB33 source map:

$$\begin{aligned} S_K^{(s)} &= F_K^{(s)} + \delta_{K,k,s}, \\ \delta_{K,k,s} &= \sum_{i=0}^3 \sigma_i e_{N_i,s} - F_{L_k}^{(s)} + F_{(K_k+L_k)/K_k}^{(s)}, \quad L_k = \mathbb{C}[S]_{<v}, \quad \sigma = (1, 1, -1, -1), \\ \delta\Delta &= \delta_{K,k,k} - \delta_{K,k,1}, \quad \Delta T = \mathcal{T}_{k,k} - \mathcal{T}_{k,1}, \\ \Delta S_K &:= S_K^{(k)} - S_K^{(1)} = -\Delta_K^\Gamma + \delta\Delta, \quad \Delta S_R = \Delta T + \Delta_K^\Gamma - \delta\Delta. \end{aligned} \quad (\text{AMT50})$$

Here $N_i = q_k - 1, q_k, 2q_k - 1, 2q_k$ are the original primal cutoffs. The endpoint scalar $e_{N_i,s}$ is the full conductor-source Jacobian in the original quotient frames. The two low quotient metrics remain exactly as displayed. FEX's conductor map and its attained original source quotient prove

the first equality; the complementary identity $S_R = \mathcal{T} - S_K$ is the full HJR frame factorization. Subtracting the original source orders proves the last line. Thus neither the low overlap nor its Gram has disappeared.

Write $\Omega_k^{\text{low/high}}$ for the existing unshifted primal widths of AMT43. Write $\Omega_k^{[k],\text{lo/hi}}$ for NSR2 at the native $d_i = N_i - v$. The two complete finite intervals can now be intersected on the same actual return:

$$\begin{aligned} L_K &= \max\{-t_k \Omega_k^{[k],\text{hi}}, \delta\Delta - r\Omega_k^{\text{high}}\}, \\ U_K &= \min\{t_k \Omega_k^{[k],\text{lo}}, \delta\Delta + r\Omega_k^{\text{low}}\}, \\ L_K &\leq \Delta S_K \leq U_K, \\ \max\{-j_{0,k} \Omega_k^{[k],\text{hi}}, \Delta T - U_K\} &\leq \Delta S_R \leq \min\{j_{0,k} \Omega_k^{[k],\text{lo}}, \Delta T - L_K\}. \end{aligned} \tag{AMT51}$$

AMT44 already proves $-r\Omega_k^{\text{low}} \leq \Delta_K^\Gamma \leq r\Omega_k^{\text{high}}$. Its negative plus the exact $\delta\Delta$ gives the second interval entering L_K, U_K . NSO/NSR gives the first interval through the same original native restriction and quotient of HJR16. Intersecting them proves the first return bounds. Since $\Delta S_R = \Delta T - \Delta S_K$ exactly, reverse their endpoints and intersect with the residual rank- $j_{0,k}$ native interval to prove the last line. The full $\delta\Delta$ and ΔT have been retained. All source masses, original primal endpoints and shifted native endpoints are still specified. This receives the existing AMT analytic comparison through the exact FEX map; it does not claim another scalar Gamma estimate.

The arithmetic total and the complete action at the four original norms. For the actual arithmetic source and every complete quartet multiplicity, the full density constants of AMT4–6 apply at all convolution orders $b \geq 3$. CAI16–25 uses the orders 3, 4, 5 to prove an all-degree multiplication estimate, and CAI28 bounds both original contraction penalties by the explicit $\Pi_{h,k}$. Keep $q = e_k(k+1)^2$ and the original four monic arithmetic norms. Their comparison to the fixed Gamma norms, after the complete four-norm telescoping, gives

$$\begin{aligned} \Lambda_{k,n} &= \log(A_k D_n / a_k), \quad E_{\text{lo}} = \Lambda_{k,q-1} + \Lambda_{k,q}, \quad E_{\text{hi}} = \Lambda_{k,2q-1} + \Lambda_{k,2q}, \\ \mathcal{W}_q^\sigma &= 2 \log \frac{(4q)!}{(2q)!} - 4q \log 2 + \log \frac{2q-1}{2(4q-1)}, \\ \mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^\sigma - E_{\text{hi}} - \Pi_{h,k} &\leq J_k^{\text{action}} \leq \mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^\sigma + E_{\text{lo}}. \end{aligned} \tag{CAIR2}$$

Every a_k, A_k, D_n is the unchanged AMT constant reproduced in CAI16. Minimizing its full polynomial inequality on monic competitors bounds each of the four norms. Putting the two upper-degree norms at their lower endpoints and the two lower-degree norms at their upper endpoints gives $-E_{\text{hi}}$; reversing them gives E_{lo} . The exact identity $\gamma_n = \sqrt{2\pi}(2n)!/4^n$ yields the displayed factorial expression, including its endpoint ratio. Substitution in CAI15 and CAIR1 proves the entire interval. Its finite factorial refinement CAI32–34 implies $J_k^{\text{action}} - \mathcal{B}_k^{\text{ar}} = 4q \log q + (8 \log 2 - 4)q + O_h(k + \log q)$. The complete original arithmetic total, its source comparison and all observation returns remain in $\mathcal{B}_k^{\text{ar}}$. The new statement uses the full-multiplicity arithmetic domain; the separate native-domain paragraph AMT50 retains its own simple-quartet restriction.

8.10 The refined penalty in the complete arithmetic action

The same four arithmetic norms in AMT22 and CAI29 define $\mathcal{W}_k^{\text{ar}}$, and the original total determinant return is $\mathcal{B}_k^{\text{ar}}$. Retain the exact original penalties

$$\begin{aligned} \mathcal{P}_- &= -\log(1 - d_{2q-1}) - 2 \sum_{n=q}^{2q-2} \log(1 - d_n), \\ \mathcal{P}_+ &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{n=q+1}^{2q-1} \log(1 - d_n). \end{aligned} \tag{RXECT5}$$

The coefficient sums are respectively $2q - 1$ and $2q$. Thus ECT's original-source bound $d_n \leq U = U_{h,k}^{\text{ctr}} < 1$, on the unchanged guard $q \geq 7R_k$, proves

$$0 \leq \mathcal{P}_- + \mathcal{P}_+ \leq \Pi_{\text{exp}} := -(4q - 1) \log(1 - U) \leq \frac{(4q - 1)U}{1 - U}. \quad (\text{RXECT6})$$

Indeed $-\log(1 - x)$ is increasing on $[0, 1)$, and $-\log(1 - U) = \int_0^U (1 - t)^{-1} dt \leq U/(1 - U)$. CAI28's earlier finite bound $\Pi_{h,k} = (4q - 1) \log(1 + z_{h,k})$ remains available on its original domain. Define $\Pi_{\text{keep}} = \min\{\Pi_{h,k}, \Pi_{\text{exp}}\}$ when the two new guards hold, and $\Pi_{\text{keep}} = \Pi_{h,k}$ otherwise. This is an intersection of proved finite bounds for the same penalties.

For completeness, multiplying the two sets of actual adjacent-radius identities retains the exact products

$$\begin{aligned} \prod_{N=q-1}^{2q-2} \epsilon_N^2 &= \frac{\omega_{2q-1} V_{q-1}}{\omega_{q-1} V_{2q-1}} (1 - d_{2q-1}) \prod_{n=q}^{2q-2} (1 - d_n)^2, \\ \prod_{N=q}^{2q-1} \epsilon_N^2 &= \frac{\omega_{2q} V_q}{\omega_q V_{2q}} (1 - d_q)(1 - d_{2q}) \prod_{n=q+1}^{2q-1} (1 - d_n)^2. \end{aligned} \quad (\text{RXECT7})$$

The recurrence ratios telescope in each product; the factors $d_{N+1}^{-1} = V_N/V_{N+1}$ telescope to the displayed full determinants. Taking logarithms of the product of both equations gives

$$J_k^{\text{action}} = \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \quad -\Pi_{\text{keep}} \leq J_k^{\text{action}} - \mathcal{B}_k^{\text{ar}} - \mathcal{W}_k^{\text{ar}} \leq 0. \quad (\text{RXECT8})$$

Here the original radius weights are $1, 2, \dots, 2, 1$, with sum $2q$. The full determinant splitting gives $F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}$ before this estimate; its fixed frame determinant occurs twice in each direction and cancels. Therefore the original signed allocation and correlated budget have been combined on their exact joint value.

Apply AMT/CAI's unchanged four-norm comparison $-E_{\text{hi}} \leq \mathcal{W}_k^{\text{ar}} - \mathcal{W}_q^\sigma \leq E_{\text{lo}}$. The complete finite action interval improves to

$$\mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^\sigma - E_{\text{hi}} - \Pi_{\text{keep}} \leq J_k^{\text{action}} \leq \mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^\sigma + E_{\text{lo}}. \quad (\text{RXECT9})$$

Substitution is direct in RXECT8 and preserves both error directions. The literal factorial formula for $\mathcal{W}_q^\sigma$, its endpoint factor, and the finite error $1/(16q) + 1/(4q - 2)$ remain exactly CAI31–32. At fixed h , RXECT6 is $O_h(qe^{-c_*q/2})$ eventually. The distinct four-norm errors $E_{\text{lo}}, E_{\text{hi}} = O_h(k + \log q)$ retain that size; replacing the penalty bound does not change those errors.

8.11 The sharp canonical-band constants in the same original action

Use the original stipulated quartet with $0 < \delta < 1/2$, $\gamma > 2$, complete integer multiplicity $m \geq 1$, and tensor orders $k = 4l + 1$, $l \geq 1$. Retain $e_k = 1 + k(m - 1)$, $q = e_k(k + 1)^2$, $R_k = k\sqrt{\delta^2 + \gamma^2}$, the full amplitude $v_h = 2\xi/h$, source w_h^{*k} , and literal coordinate $S = k/2 + iy$. The sharp finite receiver uses exactly the guards

$$q \geq 2^{17} R_k, \quad q \geq 1000, \quad m_k \geq \log 2. \quad (\text{RXUCT1})$$

The symbol m_k below is a logarithmic minimum bound and is distinct from the original multiplicity m . The paired-root constant C_k^{cmp} is the complete retained provider's constant, including its inner-region term $-\log(1 - \eta_q)$, with $\varepsilon = 2^{-16}$. With the original AMT constants and the complete alternative CCN4 quantile error define

$$\begin{aligned} D_n &= [1 + 4(n + 21 + 4m)^2]^{21+4m}, \quad B_k = \log(A_k D_{2q}/a_k) + C_k^{\text{cmp}}, \\ r_0 &= \lfloor q/100 \rfloor, \quad C_k = \widehat{E}_0(q) + B_k, \quad c_0 = \frac{c_\Gamma}{60 \exp(1) \sqrt{6\pi}}, \\ s_k &= q/3 - \frac{3}{2} \log q + \log c_0 - B_k, \quad m_k = \min\{s_k, 2qa_1 - C_k\}, \quad z = \exp(-2c_1). \end{aligned} \quad (\text{RXUCT2})$$

Here $a_1 = \psi(1) > 0$ and $c_1 = -\psi'(1) > 0$ are the exact elliptic constants proved in UCT; $\hat{E}_0$ is used consistently and is not replaced by the unavailable author's separate C_{EIQ} .

To exhibit the typed arithmetic transfer, let $\nu_r = \min_{P \text{ monic, deg } P=r} \|\chi P\|_{\text{ar}}^2$ and $\nu_r^0 = \min_{p \text{ monic, deg } p=r} \|y^q p\|_{\sigma}^2$, and put $T_r = \nu_r / \omega_{q+r}$. The affine coefficient bijection $P(S) \mapsto i^{-r} P(k/2 + iy)$ has inverse $p(y) \mapsto i^r p((S - k/2)/i)$ and sends each degree- r monic fibre onto the degree- r monic fibre. Its phase has modulus one. For $0 \leq r \leq q$, AMT applies to the full degree $q + r \leq 2q$ polynomial and the complete paired-root comparison applies to p . Taking the same minima gives

$$\begin{aligned} \frac{a_k}{D_{2q}} e^{-C_k^{\text{cmp}}} \nu_r^0 &\leq \nu_r \leq A_k e^{C_k^{\text{cmp}}} \nu_r^0, & \frac{a_k}{D_{2q}} g_{q+r} &\leq \omega_{q+r} \leq A_k g_{q+r}, \\ \left| \log T_r - \log \frac{\nu_r^0}{g_{q+r}} \right| &\leq B_k, & d_{q+r} &= T_r^{-1}. \end{aligned} \quad (\text{RXUCT3})$$

For the last determinant identity, dividing the source Hankel determinant by its full relation Hankel determinant gives V_N ; its adjacent ratio is ω_{q+r}/ν_r . All original masses remain in the displayed norms. No quotient map is changed by the coefficient comparison.

The complete small-rank proof gives $\log T_r \geq s_k$ for $0 \leq r \leq r_0$, while the UCT bulk proof and RXUCT3 give $|\log T_r - 2q\psi(r/q)| \leq C_k$ for $r_0 < r \leq q$. The proved inequality $\psi(t) \geq a_1 + c_1(1-t)$ therefore implies

$$T_r^{-1} \leq e^{-s_k} \quad (0 \leq r \leq r_0), \quad T_r^{-1} \leq e^{C_k - 2qa_1} z^{q-r} \quad (r_0 < r \leq q), \quad T_r \geq e^{m_k} \quad (0 \leq r \leq q). \quad (\text{RXUCT4})$$

At $r = q$ the opposite side of the same bulk estimate also gives $T_q^{-1} \geq e^{-2qa_1 - C_k}$.

Retain the two original penalties RXECT5. Their combined weights in $r = 0, \dots, q$ are $3, 4, \dots, 4, 3, 1$, with the first small block's weight $4r_0 + 3$. For $0 \leq x \leq e^{-m_k}$, direct integration gives $x \leq -\log(1-x) \leq x/(1-e^{-m_k})$; on the small block one may use the sharper denominator $1 - e^{-s_k}$. Summing the actual bulk weights gives exactly

$$\begin{aligned} e^{-2qa_1 - C_k} \leq \mathcal{P}_- + \mathcal{P}_+ \leq \Pi_{\text{sharp}} &:= \frac{(4r_0 + 3)e^{-s_k}}{1 - e^{-s_k}} \\ &+ \frac{e^{C_k - 2qa_1}}{1 - e^{-m_k}} \left[1 + 3z + \frac{4z^2(1 - z^{q-r_0-2})}{1 - z} \right]. \end{aligned} \quad (\text{RXUCT5})$$

The bracket comes from $r = q$ with weight one, $r = q - 1$ with weight three, and $q - r = 2, \dots, q - r_0 - 1$ with weight four. The lower bound uses the weight-one final term and the just-proved lower bound for T_q^{-1} . Thus both directions concern the same complete penalty.

For explicit backward propagation let Π_* be the minimum of Π_{sharp} and the earlier bounds on their own original domains: include $\Pi_{h,k}$ on CAI's retained $k \geq 9$ domain, and include Π_{exp} when its additional guard $U_{h,k}^{\text{ctr}} < 1$ holds. Outside RXUCT1 retain RXECT's earlier guarded intersection. The exact action identity then proves

$$\begin{aligned} \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - \Pi_* &\leq J_k^{\text{action}} \leq \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - e^{-2qa_1 - C_k}, \\ \mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^{\sigma} - E_{\text{hi}} - \Pi_* &\leq J_k^{\text{action}} \leq \mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^{\sigma} + E_{\text{lo}} - e^{-2qa_1 - C_k}. \end{aligned} \quad (\text{RXUCT6})$$

Subtract RXUCT5 from $J_k^{\text{action}} = \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+$ for the first line; then use the two original four-norm errors for the second. Substitution of these finite endpoints for L_J, U_J in RXECT11–12 propagates the estimate to the original mean and outer receiver with their already stated domains and unchanged signs.

The pointwise minimum also has a finite two-sided receiver. Set $b_{\text{low}} = (a_k/(A_k D_{2q}))q(q - 1/2)$ and $b_{\text{hi}} = (A_k D_{2q}/a_k)(2q)(2q - 1/2)$, using the same monic norm ratio comparison in both directions. The original untilted radius identities used by UCT are $\epsilon_{q-1}^2 = \beta_q(T_0 - 1)$ and $\epsilon_{q+r-1}^2 = \beta_{q+r}(T_r - 1)(1 - T_{r-1}^{-1})$ for $1 \leq r \leq q$, where $\beta_n = \omega_n/\omega_{n-1}$. Adding and removing the actual covariance column gives these determinant factors; total reflection makes the two adjacent monic

columns orthogonal, and the first full basis at $q-1$ supplies its factor one. This derivation preserves $z_B = -z_K$ and uses no inverse at $q-2$. It applies on the present tensor class and yields

$$\sqrt{b_{\text{low}}}(1 - e^{-m_k})e^{m_k/2} \leq \min_{q-1 \leq N \leq 2q-1} \epsilon_N \leq \epsilon_{2q-1} \leq \sqrt{b_{\text{hi}}}e^{qa_1+C_k/2}. \quad (\text{RXUCT7})$$

For an interior row the two factors $1 - T^{-1}$ are each at least $1 - e^{-m_k}$ and $T_r \geq e^{m_k}$; at the first row its single factor has the larger square root. At the final row discard both factors at most one and use $T_q \leq e^{2qa_1+C_k}$ and the upper recurrence bound. These operations prove every inequality in RXUCT7. At fixed original packet $B_k = O_h(k + \log q)$ and $\widehat{E}_0(q) = O_h(\log q)$, so $m_k = 2qa_1 + O_h(k + \log q)$ eventually because $1/3 > 2a_1$. Taking logarithms in RXUCT7 proves $\log \min_N \epsilon_N = qa_1 + O_h(k + \log q)$ on the retained tensor class. The same two-sided reasoning in RXUCT5 gives $\log(\mathcal{P}_- + \mathcal{P}_+) = -2qa_1 + O_h(k + \log q)$. No finite minimizer or individual allocation has been assigned.

9 Factorial control of the original common kernel at the kq scale

The exact denominator in OKV15 contains the information needed to improve its absolute volume estimate. This proof keeps that denominator and also allows every original primary multiplicity. The original Gamma sources, root coordinates, coefficient maps and observation are retained. The simple-quartet conclusion uses the actual period and rank calculations PQO/RPC and OQC/OQX, with their complete proofs supplied as dependencies.

9.1 Original coordinates, all primary multiplicities, and source norms

Fix the original full-order quartet with $0 < \delta < 1/2$, $\gamma > 2$, and a positive integer primary order m . On the original tensor class set

$$k = 4l + 1, \quad l \geq 1, \quad e_k = 1 + k(m - 1), \quad q = e_k(k + 1)^2, \quad c = k/2, \quad s \in \{1, k\}, \quad b_s = s/2. \quad (\text{KAF1})$$

The original polynomial and its exact real-coordinate representative are

$$\begin{aligned} \chi(S) &= \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^{e_k}, \\ \omega_{ab} &= (2b - k)\gamma - i(2a - k)\delta, \\ P(y) &= i^{-q}\chi(c + iy) = \prod_{a,b=0}^k (y - \omega_{ab})^{e_k}. \end{aligned} \quad (\text{KAF2})$$

In the full list $\omega_1, \dots, \omega_q$, each root ω_{ab} occurs e_k times. The exact maximal modulus remains $R = k\sqrt{\delta^2 + \gamma^2}$. Substitution gives inverse algebra maps

$$\Phi : \mathbb{C}[y]/(P) \longrightarrow E = \mathbb{C}[S]/(\chi), \quad [r] \longmapsto [r((S - c)/i)], \quad \Phi^{-1}[f] = [f(c + iy)]. \quad (\text{KAF3})$$

Indeed $P((S - c)/i) = i^{-q}\chi(S)$, and the two substitutions are inverse before quotienting. Thus every root, multiplicity and factor of i in the original remainder operation is present.

Retain the complete measures and their monic orthogonal polynomials:

$$\begin{aligned} M_s &= (2\pi)^{s/2}, \quad dm_{0,s}(y) = \frac{M_s 2^{b_s-2}}{\pi \Gamma(b_s)} \left| \Gamma\left(\frac{b_s}{2} + \frac{iy}{2}\right) \right|^2 dy, \\ p_0^{(s)} &= 1, \quad p_1^{(s)} = y, \quad p_{d+1}^{(s)} = yp_d^{(s)} - d(b_s + d - 1)p_{d-1}^{(s)}, \\ h_d^{(s)} &= \int |p_d^{(s)}|^2 dm_{0,s} = M_s d!(b_s)_d, \quad \varphi_d^{(s)} = p_d^{(s)} / \sqrt{h_d^{(s)}}, \quad u_d^{(s)}(S) = \varphi_d^{(s)}((S - c)/i). \end{aligned} \quad (\text{KAF4})$$

These are precisely the existing orders $1, k$. Their complete source transform and norm calculation are IFB2–5 and OKV4–5. In particular $\int e^{ty} dm_{0,s} = M_s (\cos t)^{-b_s}$ for $|t| < \pi/2$. The generating series $(1 + z^2)^{-b_s/2} \exp(y \arctan z)$ has coefficients $p_d^{(s)} z^d / d!$. Integrating its product at two sufficiently small real variables gives $M_s (1 - zw)^{-b_s}$, proving the displayed norms and orthogonality by coefficient comparison. The original exponential bound justifies those finite coefficient differentiations under the integral. None of these source statements depends on m .

In the original polynomial frame of E let C_s have columns $[u_0^{(s)}], \dots, [u_{q-1}^{(s)}]$. Its determinant is exactly

$$\begin{aligned} \det C_s &= i^{-q(q-1)/2} \prod_{d=0}^{q-1} (h_d^{(s)})^{-1/2} \neq 0, \\ v_j^{(s)} &= C_s^{-1}[u_{q+j}^{(s)}], \quad K_j^{(s)} = I_q + \sum_{a=0}^{j-1} v_a^{(s)} (v_a^{(s)})^*, \quad 0 \leq j \leq q + 1. \end{aligned} \quad (\text{KAF5})$$

The last included vector has original degree $2q$.

9.2 A degree-dependent constant in the unchanged remainder estimate

Define the actual finite constants

$$A_{k,m} = \max \left\{ 2\sqrt{1 + \frac{k}{2q}}, \frac{R}{q} \right\}, \quad T = qA_{k,m}, \quad \mathcal{A}_{k,m} = 6 \log 2 + 4 + 4 \log A_{k,m}. \quad (\text{KAF6})$$

The letter $A_{k,m}$ in this section is a coefficient-estimate constant; it is not the arithmetic upper source constant A_k of TWA/AMT. In particular $A_{k,m} \geq 2$ and $R \leq T$. For fixed original quartet and fixed m , $q \geq (k+1)^2$ implies

$$A_{k,m} \longrightarrow 2, \quad \mathcal{A}_{k,m} \longrightarrow 4 + 10 \log 2. \quad (\text{KAF7})$$

Indeed $k/q \rightarrow 0$ and $R/q = \sqrt{\delta^2 + \gamma^2} k/q \rightarrow 0$. The root-dependent term in the maximum is retained at every finite k .

Here is the complete remainder estimate needed below. For the full root list define elementary and complete homogeneous polynomials $e_h(\omega)$ and $h_t(\omega)$ by their finite sums, with $e_0 = h_0 = 1$ and $e_h = 0$ outside $0 \leq h \leq q$. The formal identity

$$\left(\sum_{h=0}^q (-1)^h e_h(\omega) z^h \right) \left(\sum_{t \geq 0} h_t(\omega) z^t \right) = 1 \quad (\text{KAF8})$$

follows by multiplying each of the q factors $1 - \omega_a z$ by its geometric series. Repeated roots do not affect this identity. It proves, for $D = q + j$ and $0 \leq j \leq q$, the exact signed formula

$$[y^a] \text{rem}_P(y^{q+j}) = - \sum_{t=0}^j h_t(\omega) (-1)^{q+j-t-a} e_{q+j-t-a}(\omega), \quad 0 \leq a < q. \quad (\text{KAF9})$$

In detail the quotient is $\sum_{t=0}^j h_t(\omega) y^{j-t}$; KAF8 cancels all coefficients of degree at least q in its product with P , and the remaining coefficients are KAF9. This proves the formula by uniqueness of monic division, including all multiplicities.

The defining sums give $|e_h| \leq \binom{q}{h} R^h$ and $|h_t| \leq \binom{q+t-1}{t} R^t$. In every nonzero summand of KAF9 the total root exponent is $q + j - a$. The binomial coefficients of h_t increase with t , and the distinct indices in the e_h sum contribute at most 2^q . Hence

$$|[y^a] \text{rem}_P(y^{q+j})| \leq 2^q \binom{q+j-1}{j} R^{q+j-a} \leq 2^{3q-1} R^{q+j-a}. \quad (\text{KAF10})$$

The last bound is the binomial expansion at one, since $j \leq q$.

The exact orthonormal recurrence from KAF4 has upward and downward weights $\sqrt{(a+1)(b_s+a)}$ and $\sqrt{a(b_s+a-1)}$. On degree at most $q-2$ each shift has norm at most $\sqrt{q(b_s+q)}$. Because $b_s \leq k/2$, their sum has norm at most $2\sqrt{q(k/2+q)} \leq T$. Iterating from the constant, with all intermediate degrees in that range, proves

$$\|y^a\|_{L^2(m_{0,s})} \leq \sqrt{M_s} T^a, \quad 0 \leq a < q. \quad (\text{KAF11})$$

For the finite coefficient sum $c_T(f) = \sum_a [y^a] f |T^a$ the original recurrence gives $c_T(p_{d+1}^{(s)}) \leq T c_T(p_d^{(s)}) + d(b_s + d - 1) c_T(p_{d-1}^{(s)})$. For $1 \leq d < 2q$,

$$d(b_s + d - 1) \leq 2q(k/2 + 2q) = 4q^2 + kq \leq 4q^2 + 2kq \leq T^2.$$

Starting with $c_T(p_0) = 1$, $c_T(p_1) = T$, induction therefore proves $c_T(p_d) \leq (2T)^d$ for $0 \leq d \leq 2q$: the recurrence bound is $(3/4)(2T)^{d+1} \leq (2T)^{d+1}$. Combining KAF10 with all q monomial norms KAF11, then using the linearity of the original remainder map, gives

$$\|\text{rem}_P(p_d^{(s)})\|_{L^2(m_{0,s})} \leq \sqrt{M_s} q 2^{3q} (2T)^d, \quad 0 \leq d \leq 2q. \quad (\text{KAF12})$$

For a monomial of degree $D \geq q$, each coefficient term is at most $\sqrt{M_s} 2^{3q-1} R^{D-a} T^a \leq \sqrt{M_s} 2^{3q-1} T^D$. For $D < q$ use KAF11 directly. These two cases and the proved coefficient sum establish KAF12 without deleting a remainder coefficient or changing the source norm.

9.3 Retaining the factorial denominator

Expansion of the degree-below- q remainder in the orthonormal polynomials of KAF4, followed by the exact map KAF3, proves

$$\|v_j^{(s)}\|_2^2 = \frac{\|\text{rem}_P(p_{q+j}^{(s)})\|_{L^2(m_{0,s})}^2}{M_s(q+j)!(b_s)_{q+j}} \leq \frac{q^2 2^{6q} (2T)^{2(q+j)}}{(q+j)!(b_s)_{q+j}}. \quad (\text{KAF13})$$

The factor M_s occurs in the original numerator and denominator before this cancellation. For every $d \geq 0$,

$$d!(b_s)_d \geq d!(1/2)_d = \frac{(2d)!}{4^d}, \quad \|v_j^{(s)}\|_2^2 \leq q^2 2^{6q} \frac{(4T)^{2(q+j)}}{(2q+2j)!}. \quad (\text{KAF14})$$

The inequality is factor by factor because $b_s \geq 1/2$. The equality follows by separating even and odd factors of $(2d)!$. This is precisely the denominator whose further retention sharpens OKV16–17.

Define the positive exact quantities

$$\eta_{k,m} = \frac{1 - 1/(4q)}{A_{k,m}^2} < \frac{1}{4}, \quad Z_{k,m} = 1 + q^2 2^{6q} \frac{(4q A_{k,m})^{4q}}{(4q)!} \frac{1 - \eta_{k,m}^{q+1}}{1 - \eta_{k,m}}. \quad (\text{KAF15})$$

For $t_d = (4T)^{2d}/(2d)!$, $q+1 \leq d \leq 2q$ gives $t_{d-1}/t_d = (2d)(2d-1)/(16T^2) \leq \eta_{k,m}$. Consequently $\sum_{d=q}^{2q} t_d \leq t_{2q} \sum_{a=0}^q \eta_{k,m}^a$. Adding the $q+1$ actual rank-one updates in KAF5 and using KAF14 proves the finite operator bound

$$\boxed{1 \leq \|K_{q+1}^{(s)}\|_2 \leq Z_{k,m}, \quad s \in \{1, k\}.} \quad (\text{KAF16})$$

Here $\|vv^*\|_2 = \|v\|_2^2$ follows by Cauchy–Schwarz with equality on $v \neq 0$; a zero vector contributes zero.

For an explicit linear bound, monotonicity of $\log x$ on each interval $[j-1, j]$ gives

$$\log n! = \sum_{j=1}^n \log j \geq \int_1^n \log x \, dx = n \log n - n + 1.$$

Thus $(4q A_{k,m})^{4q}/(4q)! \leq e^{-1} (e A_{k,m})^{4q}$, where e is the exponential constant and differs from the primary integer e_k . The geometric sum in KAF15 is at most $4/3$. Therefore $Z_{k,m} \leq 1 + [4/(3e)]X$, with $X = q^2 \exp(q A_{k,m})$. The positive exponential series gives $e > 1 + 1 + 1/2 + 1/6 = 8/3$, so $4/(3e) < 1/2$. Since $A_{k,m} \geq 2$ and $q \geq 36$, $X > 2$, and hence $1 + [4/(3e)]X \leq X$. We have proved

$$\boxed{\log \|K_{q+1}^{(s)}\|_2 \leq \log Z_{k,m} \leq q A_{k,m} + 2 \log q.} \quad (\text{KAF17})$$

The full factorial and finite geometric sum in KAF15 remain available as the sharper finite bound.

9.4 All four actual kernel and boundary Grams

Let $\Lambda : E \twoheadrightarrow B$ be the original period and full-unit observation, let $I : \mathcal{K} \hookrightarrow E$ be its actual kernel inclusion, and put $r = \dim \mathcal{K}$. No rank formula is needed in this subsection, so it applies to every original m . The original quotient and restricted Grams are

$$G_{q+j-1}^{(s)} = C_s^{*-}(K_j^{(s)})^{-1} C_s^{-1}, \quad T_s = C_s^{-1} I, \quad H_j^{(s)} = T_s^* (K_j^{(s)})^{-1} T_s. \quad (\text{KAF18})$$

Indeed the original orthonormal source-coordinate remainder matrix is $[I_q, v_0, \dots, v_{j-1}]$. Multiplication by its adjoint gives K_j ; its unique minimum lift of x is its adjoint applied to $K_j^{-1}x$, with

squared norm $x^* K_j^{-1} x$. This proves both matrices KAF18 in the original coefficient frame. The positive update chain implies

$$Z_{k,m}^{-1} T_s^* T_s \preceq H_j^{(s)} \preceq T_s^* T_s, \quad H_{j+1}^{(s)} \preceq H_j^{(s)}. \quad (\text{KAF19})$$

Inversion reverses positive order by congruence and inversion of positive eigenvalues. Restricting this order with the same injective T_s proves KAF19. After congruence with $(T_s^* T_s)^{-1/2}$, every eigenvalue lies between $Z_{k,m}^{-1}$ and one. Thus $-r \log Z_{k,m} \leq \log \det H_j^{(s)} - \log \det(T_s^* T_s) \leq 0$.

The exact four original degrees are $q-1, q, 2q-1, 2q$. With their unchanged signs define

$$F_K^{(s)} = \log \det H_0^{(s)} + \log \det H_1^{(s)} - \log \det H_q^{(s)} - \log \det H_{q+1}^{(s)}.$$

Pairing degrees $q-1$ with $2q-1$, and q with $2q$, proves nonnegativity by KAF19. Each difference is at most $r \log Z_{k,m}$. The four original frame factors cancel only after these four signs have been recorded. It follows that

$$0 \leq F_K^{(s)} \leq 2r \log Z_{k,m} \leq 2r [q \mathcal{A}_{k,m} + 2 \log q], \quad s \in \{1, k\}. \quad (\text{KAF20})$$

If $r = 0$, every kernel determinant is one and the statement has its same empty-space meaning.

At each original degree the boundary Gram is $Q_j^{(s)} = (\Lambda(G_{q+j-1}^{(s)})^{-1} \Lambda^*)^{-1}$. For any fixed section S_* of Λ , its minimum section $S_j^{(s)} = (G_{q+j-1}^{(s)})^{-1} \Lambda^* Q_j^{(s)}$ differs from S_* by a kernel column and is orthogonal to I in that metric. The change of frame has determinant one. Taking determinants in this orthogonal frame proves, with its original coefficient factor,

$$\det H_j^{(s)} \det Q_j^{(s)} = |\det[I, S_*]|^2 \det G_{q+j-1}^{(s)}.$$

The same four signs now prove the exact splitting $F_K^{(s)} + F_B^{(s)} = \mathcal{B}_k^{(s)}$. Boundary minimum norms decrease with degree, so $F_B^{(s)} \geq 0$. In particular

$$\max\{0, \mathcal{B}_k^{(s)} - 2r \log Z_{k,m}\} \leq F_B^{(s)} \leq \mathcal{B}_k^{(s)}. \quad (\text{KAF21})$$

These are statements about the original kernel and boundary, with all multiplicities and the same observation throughout.

9.5 The actual simple quartet and the arithmetic source

For $m = 1$, the complete RPC calculation supplies an explicit finite R_* from the original unit and quartet. On every logarithm branch and for $|u| \geq R_*$ its original period quadric has five distinct cyclic orbit members. The full invariant restriction ideal calculation OQX then gives $r = 8k - 16$. Substitution in the finite KAF20 gives

$$0 \leq \frac{F_K^{(s)}}{kq} \leq \left(16 - \frac{32}{k}\right) \left(\mathcal{A}_{k,1} + \frac{2 \log q}{q}\right), \quad s \in \{1, k\}, \quad |u| \geq R_*. \quad (\text{KAF22})$$

The original arithmetic-source comparison AMT17–22 proves $-r E_k^+ \leq F_K^{\text{ar}} - F_K^{(1)} \leq r E_k^-$, where the unchanged full constants are

$$\begin{aligned} E_k^+ &= 2 \log(A_k/a_k) + \log D_{q-1} + \log D_q, \\ E_k^- &= 2 \log(A_k/a_k) + \log D_{2q-1} + \log D_{2q}, \\ D_N &= [1 + 4(N + 21 + 4m)^2]^{21+4m}. \end{aligned}$$

Here a_k, A_k are exactly the original arithmetic constants of AMT4, whose full source inequality and minimum-section proof are included there. They retain the actual numerator $2\xi/h$ and all

source masses; AMT22 proves $E_k^\pm = O_h(k + \log q)$. Thus the actual arithmetic metric satisfies the finite bound

$$0 \leq F_K^{\text{ar}} \leq (16k - 32) \log Z_{k,1} + (8k - 16)E_k^-. \quad (\text{KAF23})$$

Combining KAF7, KAF22–23, and $q = (k + 1)^2$ proves uniformly on the entire stated original period domain, for $a \in \{\sigma, 0, \text{ar}\}$,

$$0 \leq \liminf_{k \rightarrow \infty} \frac{F_K^a}{kq} \leq \limsup_{k \rightarrow \infty} \frac{F_K^a}{kq} \leq 16(4 + 10 \log 2). \quad (\text{KAF24})$$

For precision, uniformity means the same finite upper bound holds for every such u and branch at each original k ; therefore it also bounds every sequence of those parameters. The arithmetic extra term divided by kq is at most $(8 - 16/k)E_k^-/q \rightarrow 0$. AMT46 further proves that the three ratios differ by quantities tending uniformly to zero. Thus their limit points coincide for any fixed allowed sequence of period parameters; KAF24 does not assert that this bounded sequence already has a unique limit.

Finally the exact arithmetic identity gives

$$\max\{0, \mathcal{B}_k^{\text{ar}} - (16k - 32) \log Z_{k,1} - (8k - 16)E_k^-\} \leq F_B^{\text{ar}} \leq \mathcal{B}_k^{\text{ar}}. \quad (\text{KAF25})$$

Its full-source value remains $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma = \mathcal{B}_k^0 + \mathcal{H}_k + \delta_k^\sigma$. All exact Gamma and arithmetic signed returns of AMT44–49 are retained. The new calculation improves the absolute common kernel bound to the kq scale, while its exact four-endpoint determinant and actual period dependence remain the objects for the next leading term calculation.

10 An exact conductor frame and a fixed invariant-generator recursion

Retain the original simple-quartet data, $m = 1$, $k = 4l + 1$, $l \geq 1$, the original root order $(++)$, $(+-)$, $(-+)$, $(--)$, and the complete original period and amplitude matrix of OQC1:

$$\mathbf{H} = F^{-1} \Pi U \mathcal{E}_{\text{prim}}.$$

Thus $\det \mathbf{H} \neq 0$ and $Q(x) = Q_0(\mathbf{H}^{-1}x)$, where $Q_0(t) = t_{++}t_{--} - t_{+-}t_{-+}$. Use $gf(x) = f(D^{-1}x)$, $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$, $\zeta = e^{2\pi i/5}$, and the original invariant algebra $\mathcal{J} = \mathbb{C}[x_1, x_2, x_3, x_4]^{(g)}$. This calculation applies on the actual five-orbit period locus, which includes every original logarithm branch with $|u| \geq R_*$ by PQO and RPC. On that locus the five prime polynomials $Q_a = g^a Q$, $0 \leq a < 5$, are pairwise nonassociate. Keep their actual products

$$\mathcal{A} = \prod_{a=1}^4 Q_a, \quad \mathcal{H}_Q = Q\mathcal{A} = \prod_{a=0}^4 Q_a, \quad \deg \mathcal{A} = 8, \quad \deg \mathcal{H}_Q = 10. \quad (\text{OCF1})$$

No coefficient in these polynomials is replaced. The exact orbit decision and the period domain are those of the cited original calculations. The construction below treats every possible nonzero leading coefficient of their actual restriction.

10.1 The original Segre coefficient isomorphism

For $d \geq 0$ let $\mathcal{B}_d$ be the space of biforms of bidegree (d, d) in (z_0, z_1) and (w_0, w_1) . Its ordered basis is

$$e_{ab}^{(d)} = z_0^a z_1^{d-a} w_0^b w_1^{d-b}, \quad 0 \leq a, b \leq d. \quad (\text{OCF2})$$

Pairs are ordered lexicographically, first by a , then by b . The order is used only for the exact coefficient elimination. Set $\mathcal{B}_d = 0$ for $d < 0$. Multiplication satisfies $e_{ab}^{(d)} e_{ij}^{(e)} = e_{a+i, b+j}^{(d+e)}$. The original substitution is

$$t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^T, \quad x = \mathbf{H}t, \quad (\text{OCF3})$$

$$\mathcal{P}_d = \mathbb{C}[x_1, x_2, x_3, x_4]_d, \quad \psi_d : \mathcal{P}_d \longrightarrow \mathcal{B}_d, \quad f \longmapsto f(\mathbf{H}t).$$

The induced map $R_d = \mathcal{P}_d / (Q\mathcal{P}_{d-2}) \longrightarrow \mathcal{B}_d$ is an isomorphism. Here is a coefficient proof with the given orientation. For each (a, b) , choose an integer h between $\max(0, a + b - d)$ and $\min(a, b)$. The monomial $t_{++}^h t_{+-}^{a-h} t_{-+}^{b-h} t_{--}^{d-a-b+h}$ maps to $e_{ab}^{(d)}$, proving surjectivity. Two monomials with the same pair differ in their exponent vectors by an integer multiple of $(1, -1, -1, 1)$. After removing their common monomial, their difference is divisible by $t_{++}t_{--} - t_{+-}t_{-+}$, using $A^n - B^n = (A - B) \sum_{j=0}^{n-1} A^{n-1-j} B^j$. A polynomial in the kernel has coefficient sum zero within every pair, so it is a sum of such differences. The reverse inclusion in the kernel follows by direct substitution. The invertible substitution $x = \mathbf{H}t$ then gives exactly the asserted kernel. Thus these maps preserve multiplication as well as every grading.

Write the full conductor restriction as

$$A = \psi_8(\mathcal{A}) = \sum_{r=0}^8 \sum_{s=0}^8 a_{rs} e_{rs}^{(8)} \neq 0. \quad (\text{OCF4})$$

To verify its nonvanishing, Q is a rank-four irreducible quadric, hence prime in the polynomial ring, and none of $Q_1, \dots, Q_4$ is its associate. Therefore Q does not divide their product. The kernel calculation OCF3 proves OCF4. The ring $\mathcal{B} = \bigoplus_{d \geq 0} \mathcal{B}_d$, with the indicated diagonal grading, embeds in the four-variable polynomial ring; it is an integral domain. Multiplication by A is consequently injective on every $\mathcal{B}_d$.

10.2 Every leading-coefficient case gives an exact strip frame

Let (r_*, s_*) be the lexicographically largest pair for which $a_{r_*s_*} \neq 0$, and put $a_* = a_{r_*s_*}$. This is an actual nonzero coefficient of OCF4. For $d \geq 8$ define

$$\begin{aligned} P_d &= \{(r_* + i, s_* + j) : 0 \leq i, j \leq d - 8\}, \\ E_d^{\text{strip}} &= \{0, \dots, d\}^2 \setminus P_d, \quad W_d = \mathbb{C}^{E_d^{\text{strip}}}. \end{aligned} \quad (\text{OCF5})$$

For $0 \leq d < 8$ put $P_d = \emptyset$, $E_d^{\text{strip}} = \{0, \dots, d\}^2$ and use the same definition of W_d ; for $d < 0$ put $W_d = 0$. Let $S_d : W_d \hookrightarrow \mathbb{C}^{(d+1)^2}$ insert zeros in the positions P_d . All coefficient columns below use OCF2.

The following finite elimination defines $N_d : \mathbb{C}^{(d+1)^2} \rightarrow W_d$ and, for $d \geq 8$, $V_d : \mathbb{C}^{(d+1)^2} \rightarrow \mathbb{C}^{(d-7)^2}$. Starting with the complete input biform f , traverse all $(i, j) \in \{0, \dots, d-8\}^2$ in decreasing lexicographic order. At the step (i, j) , let c be the current coefficient at $(r_* + i, s_* + j)$, record $b_{ij} = c/a_*$, and replace the current biform by

$$f_{\text{current}} - b_{ij} A e_{ij}^{(d-8)}. \quad (\text{OCF6})$$

At completion take the coefficients on E_d^{strip} as $N_d f$ and take (b_{ij}) as $V_d f$. If $d < 8$, N_d is the identity and V_d is the zero map into the zero space. The exact multiplication matrix $M_{A,d}$ gives

$$S_d N_d + M_{A,d} V_d = I, \quad N_d S_d = I_{W_d}, \quad \ker N_d = A \mathcal{B}_{d-8}. \quad (\text{OCF7})$$

To prove these identities, the largest term of $A e_{ij}^{(d-8)}$ is $a_* e_{r_*+i, s_*+j}^{(d)}$; every other term is strictly smaller in the same lexicographic order. Therefore each step zeros its designated position without altering an earlier zero. Summing all subtractions gives the first identity. Input supported on the strip produces zero coefficients at every elimination step, giving the second. Finally, the leading term of Ab , for a nonzero homogeneous biform b , is the product of the two leading terms and belongs to P_d . Indeed every other product is smaller in the additive lexicographic order. Thus $A \mathcal{B}_{d-8} \cap S_d W_d = 0$. The first identity now proves the kernel formula, the uniqueness of the decomposition, and all three identities in OCF7. All operations are linear in the input; their coefficients are explicit rational expressions in the original a_{rs} with powers of the displayed a_* as denominators. The polynomial A itself remains the full polynomial OCF4.

In particular the quotient map induced by N_d is the exact isomorphism

$$\mathcal{B}_d / (A \mathcal{B}_{d-8}) \xrightarrow{\sim} W_d, \quad n_d := \dim W_d = \begin{cases} (d+1)^2, & 0 \leq d < 8, \\ 16d - 48, & d \geq 8. \end{cases} \quad (\text{OCF8})$$

The second formula is $(d+1)^2 - (d-7)^2 = 16d - 48$. The possible choices $(r_*, s_*) \in \{0, \dots, 8\}^2$ form an exhaustive finite partition: the chart for a pair specifies that its coefficient is nonzero and every larger coefficient is zero. Hence all actual coefficient cases have been included. At the first original degree $k = 5$ this construction retains all 36 coordinates; at every original $k \geq 9$ it retains $16k - 48$ coordinates.

10.3 Twenty-five fixed invariant monomials generate all degrees

Define the following fixed finite sets of exponent vectors:

$$G_e = \{\nu \in \mathbb{N}^4 : |\nu| = e, \nu_1 + 2\nu_2 + 3\nu_3 + 4\nu_4 \equiv 0 \pmod{5}\}, \quad 1 \leq e \leq 5, \quad G = \bigcup_{e=1}^5 G_e. \quad (\text{OCF9})$$

Here G_1 is empty. The complete remaining lists, in the original x_1, x_2, x_3, x_4 order, are

e	G_e	
2	$(0, 1, 1, 0), (1, 0, 0, 1)$	
3	$(0, 0, 2, 1), (0, 1, 0, 2), (1, 2, 0, 0), (2, 0, 1, 0)$	
4	$(0, 0, 1, 3), (0, 2, 2, 0), (0, 3, 0, 1), (1, 0, 3, 0),$ $(1, 1, 1, 1), (2, 0, 0, 2), (3, 1, 0, 0)$	(OCF10)
5	$(0, 0, 0, 5), (0, 0, 5, 0), (0, 1, 3, 1), (0, 2, 1, 2),$ $(0, 5, 0, 0), (1, 0, 2, 2), (1, 1, 0, 3), (1, 3, 1, 0),$ $(2, 1, 2, 0), (2, 2, 0, 1), (3, 0, 1, 1), (5, 0, 0, 0).$	

Thus $(|G_1|, \dots, |G_5|) = (0, 2, 4, 7, 12)$ and $|G| = 25$. These lists can also be obtained directly by choosing $0 \leq \nu_1, \nu_2, \nu_3 \leq e$ with their sum at most e , putting $\nu_4 = e - \nu_1 - \nu_2 - \nu_3$, and applying the single displayed congruence, which verifies completeness of each list.

Every positive-degree invariant monomial is a product of members x^ν , $\nu \in G$. To prove this, list the weights of its individual factors in any order. If its degree is at most five, the entire list gives a member of G . If its degree exceeds five, consider the six partial sums of the first five weights, including the empty partial sum, in $\mathbb{Z}/5\mathbb{Z}$. Two coincide; the nonempty block between them has length at most five and total weight zero. Its factors give a divisor x^ν with $\nu \in G$. The complementary monomial again has total weight zero and smaller degree. Induction completes the factorization. Because the action is diagonal, a polynomial is invariant exactly when all its nonzero monomials have weight zero: comparison of each monomial coefficient in $gf = f$ proves this statement. Thus

$$\mathcal{J} = \mathbb{C}[x^\nu : \nu \in G]. \quad (\text{OCF11})$$

All intermediate degrees here are degrees in the original coefficient algebra. This construction introduces no new source distribution and leaves the original source orders 1 and k intact.

For each $\nu \in G_e$ the complete generator biform is

$$f_\nu = \psi_e(x^\nu) = \prod_{j=1}^4 (H_{j1}z_0w_0 + H_{j2}z_0w_1 + H_{j3}z_1w_0 + H_{j4}z_1w_1)^{\nu_j} = \sum_{a,b=0}^e c_{\nu,ab} e_{ab}^{(e)}. \quad (\text{OCF12})$$

For an entirely explicit coefficient formula, sum over all nonnegative 4 by 4 integer arrays (n_{jt}) satisfying $\sum_t n_{jt} = \nu_j$, $\sum_j (n_{j1} + n_{j2}) = a$ and $\sum_j (n_{j1} + n_{j3}) = b$. Then

$$c_{\nu,ab} = \sum_{(n_{jt})} \prod_{j=1}^4 \left(\frac{\nu_j!}{\prod_{t=1}^4 n_{jt}!} \prod_{t=1}^4 H_{jt}^{n_{jt}} \right). \quad (\text{OCF13})$$

The multinomial theorem proves the formula, including every original period and amplitude entry. Every factor has degree at most five and at most 36 possible biform coefficients.

10.4 Exact multiplication on the strip and the invariant recursion

Let $\mathcal{L} = \mathcal{B}/(AB)$, a graded algebra. For $\nu \in G_e$ and $d \geq e$ define

$$m_{\nu,d} : W_{d-e} \longrightarrow W_d, \quad m_{\nu,d} = N_d M_{f_\nu} S_{d-e}. \quad (\text{OCF14})$$

Here M_{f_ν} is the exact coefficient convolution:

$$(M_{f_\nu} S_{d-e} v)_{ab} = \sum_{i,j} c_{\nu,a-i,b-j} (S_{d-e} v)_{ij},$$

with out-of-range indices giving zero. Therefore every entry of OCF14 is supplied by the finite formulas OCF6 and OCF13. For a representative $h \in \mathcal{B}_{d-e}$, OCF7 writes $h - S_{d-e} N_{d-e} h = Ab$. Multiplication by f_ν preserves this ideal. Applying N_d yields the exact identity

$$N_d(f_\nu h) = m_{\nu,d} N_{d-e} h. \quad (\text{OCF15})$$

This proves that OCF14 is multiplication by the specified element of $\mathcal{L}$, including its independence of representatives.

The original invariant image is $J_d = \psi_d(\mathcal{J}_d) \subseteq \mathcal{B}_d$. Set $\bar{J}_d = N_d J_d \subseteq W_d$, and set $\bar{J}_d = 0$ for $d < 0$. The exact recursion is

$$\bar{J}_0 = \mathbb{C} \cdot 1, \quad \bar{J}_d = \sum_{\substack{\nu \in G_e \\ 2 \leq e \leq 5, e \leq d}} m_{\nu,d} \bar{J}_{d-e} \quad (d \geq 1). \quad (\text{OCF16})$$

Indeed OCF11 factors every invariant monomial of degree $d > 0$ as x^ν times an invariant monomial of degree $d - e$. Applying ψ , then OCF15, gives the inclusion from left to right. Conversely every summand is the image of a product of two invariant polynomials, so it belongs to the left side. Linearity proves equality for the complete invariant space. In particular the recursion gives $\bar{J}_1 = 0$ and includes all small degrees.

The conductor ideal is contained in this invariant image with its original trace coefficient:

$$A\mathcal{B}_{d-8} \subseteq J_d, \quad [5R(\mathcal{A}h)]_{(Q)} = [\mathcal{A}h]_{(Q)}, \quad R = \frac{1}{5} \sum_{a=0}^4 g^a. \quad (\text{OCF17})$$

To prove the identity, $g^a \mathcal{A}$ is the product of all five orbit equations except Q_a ; for $a \neq 0$ it contains Q . Thus all four corresponding terms vanish modulo Q , and the $a = 0$ term is the displayed product. Every biform in $\mathcal{B}_{d-8}$ has a polynomial lift by OCF3, so the identity proves the inclusion. It also shows exactly why the quotient used in OCF16 retains the original invariant image.

10.5 Exact ranks, a finite pivot atlas, and the original quotient map

For $d \geq 0$ put

$$d_0(d) = \frac{\binom{d+3}{3} + 4\epsilon_d}{5}, \quad \epsilon_d = \mathbf{1}_{d \equiv 0 \pmod{5}} - \mathbf{1}_{d \equiv 1 \pmod{5}}, \quad d_0(d) = 0 \quad (d < 0). \quad (\text{OCF18})$$

This is the number of degree- d weight-zero monomials: the identity character has generating function $(1 - z)^{-4}$, each of the four nonidentity characters has generating function $\prod_{j=1}^4 (1 - z\zeta^j)^{-1} = (1 - z)/(1 - z^5)$, and averaging their coefficients gives OCF18. Every invariant polynomial divisible by Q is divisible by all five Q_a , hence by $\mathcal{H}_Q$, since they are pairwise nonassociate primes. Its quotient by this product is invariant: apply g and cancel the nonzero invariant product in the integral domain. Conversely each $\mathcal{H}_Q$ times an invariant polynomial is invariant and vanishes modulo Q . Consequently $\ker(\mathcal{J}_d \rightarrow J_d) = \mathcal{H}_Q \mathcal{J}_{d-10}$, and injection of multiplication by the nonzero product gives $\dim J_d = d_0(d) - d_0(d - 10)$. Injection of multiplication by A , OCF7 and OCF17 then prove

$$j_d := \dim \bar{J}_d = d_0(d) - d_0(d - 10) - \max(d - 7, 0)^2. \quad (\text{OCF19})$$

In particular $j_0 = 1$, $j_1 = 0$, $j_5 = 12$, and $j_d = 8d - 32$ for $d \geq 8$. For the last formula at $d \geq 10$, $\epsilon_d = \epsilon_{d-10}$ and direct expansion gives $\binom{d+3}{3} - \binom{d-7}{3} = 5(d^2 - 6d + 17)$; subtracting $(d - 7)^2$ gives $8d - 32$. At $d = 8, 9$ the ranks before this subtraction are 33, 44, respectively, which gives 32, 40 and verifies both remaining cases.

Here is a complete finite construction of all the columns used by OCF16. Start with $Z_0 = (1)$ and the invariant polynomial certificate $F_{0,1} = 1$. Suppose all earlier matrices Z_b have independent columns spanning $\bar{J}_b$, and their columns are $N_b \psi_b(F_{b,h})$ for invariant polynomials $F_{b,h}$. Form the candidate matrix

$$C_d = [m_{\nu,d} Z_{d-e}]_{\nu \in G_e, 2 \leq e \leq 5, e \leq d}. \quad (\text{OCF20})$$

Each candidate column has the explicit invariant certificate $x^\nu F_{d-e,h}$. OCF16 proves that C_d has column image $\bar{J}_d$ and rank j_d . Select j_d columns and j_d rows giving a nonzero determinant. Let Z_d be those columns, let I_d be those row positions, and let T_d be the complementary row positions

in E_d^{strip} . Retain the corresponding invariant certificates. This completes the induction. When $j_d = 0$, use the empty matrix with zero columns, empty I_d , and $T_d = E_d^{\text{strip}}$.

Every selected determinant and every matrix entry is an explicit expression obtained from the original entries of $\mathbf{H}$ and the full a_{rs} . To cover all cases, order the finite choices of column and row subsets and select the first nonzero determinant; the corresponding chart asserts its nonvanishing and the vanishing of all preceding candidates. A nonzero determinant exists because the rank was proved in OCF19. At rank zero no determinant is needed. Together with the exhaustive leading-coefficient charts after OCF8, this gives a finite atlas for each specified k . It requires no generic coefficient premise and supplies no numerical zero test for an unevaluated analytic coefficient: each chart is specified by its exact minors.

Define $\rho_d : W_d \rightarrow \mathbb{C}^{T_d}$ and the full original coefficient map π_d by

$$\rho_d = \text{row}_{T_d} - (Z_d)_{T_d} (Z_d)_{I_d}^{-1} \text{row}_{I_d}, \quad \pi_d = \rho_d N_d. \quad (\text{OCF21})$$

At rank zero the second term is the zero map. If E_{T_d} inserts a column into its T_d positions in W_d , then

$$\begin{aligned} w &= Z_d (Z_d)_{I_d}^{-1} w_{I_d} + E_{T_d} \rho_d w, \\ \rho_d E_{T_d} &= I, \quad \ker \rho_d = \text{im } Z_d = \bar{J}_d, \\ \ker \pi_d &= J_d, \quad \pi_d S_d E_{T_d} = I_{\mathbb{C}^{T_d}}. \end{aligned} \quad (\text{OCF22})$$

For the first line, the subtraction on its right leaves zero I_d rows and exactly $\rho_d w$ on the complementary rows. This proves the first three claims. If $\pi_d f = 0$, then $N_d f \in \bar{J}_d$; choose $j \in J_d$ with $N_d j = N_d f$. Now $f - j \in \ker N_d = \mathcal{A}\mathcal{B}_{d-8} \subseteq J_d$ by OCF17, so $f \in J_d$. The reverse inclusion follows immediately from $N_d J_d = \bar{J}_d$. The last equality uses $N_d S_d = I$. This proves every map and kernel in OCF22.

At the original degrees, these are the exact dimensions

k	n_k	j_k	$ T_k = n_k - j_k$
5	36	12	24
$k \geq 9, k = 4l + 1$	$16k - 48$	$8k - 32$	$8k - 16$.

(OCF23)

The original map $\Theta_k : R_k \xrightarrow{\sim} E_k^\vee$ in OQC7 sends the coefficient basis OCF2 to the original dual primary basis in the same order. Indeed its evaluation on $e^{wY_k}[1]$ is $\exp(((2b-k)\gamma - i(2a-k)\delta)w)$. The corresponding eigenvalue of A_k is $\beta_{ab} = k/2 + (2a-k)\delta + i(2b-k)\gamma$; the eigenvalue of $Y_k = (A_k - kI/2)/i$ is $\omega_{ab} = (\beta_{ab} - k/2)/i = (2b-k)\gamma - i(2a-k)\delta$. These eigenvalues are distinct because $\delta > 0$ and $\gamma > 0$. Their exponentials are independent by the Vandermonde determinant, so the evaluation identifies the covectors exactly. By the original observation identity OQC8, $\Theta_k J_k = \text{im } \Lambda_k^\vee$. Therefore OCF22 supplies the specified isomorphism

$$E_k^\vee / \text{im } \Lambda_k^\vee \xrightarrow{\pi_k} \mathbb{C}^{T_k} \cong (\ker \Lambda_k)^\vee. \quad (\text{OCF24})$$

The displayed π_k uses exactly the original primary coefficient coordinates under Θ_k . Every dualization in this statement is complex-linear. No conjugation is inserted in this coefficient map; the original metric maps can subsequently be composed with it.

Finally, this construction is finite with explicitly bounded compressed matrices. OCF20 has n_d rows and $\sum_{e=2}^5 |G_e| j_{d-e}$ columns, with negative subscripts zero. Both dimensions grow at most linearly in d : for example $n_d \leq 16d + 1$ and the column number is at most $25(16d + 1)$. For $d \geq 13$ the exact column number is $200d - 1632$, obtained from $\sum_e |G_e| = 25$, $\sum_e e|G_e| = 104$, and $j_{d-e} = 8(d-e) - 32$. Only the preceding five graded bases are needed to form the next matrix. Each column is obtained from the at-most-36-coefficient formula OCF12, one exact convolution, and OCF6. A full original biform coefficient column, if used as temporary storage, has $(d+1)^2$ entries; every invariant relation matrix in OCF20 has the stated linear row and column counts. The invariant certificates can be retained as their products and parent-column references, so their expanded high-degree monomials need never be enumerated. The proof of their completeness is OCF11–16, and the exact original quotient and its right inverse are OCF21–24.

10.6 The same recursion with fourteen proved generators

The complete fixed set OCF9 can be reduced to the following fourteen specific members while preserving every map and image just proved:

e	$\mathcal{G}_e$	
2	x_1x_4, x_2x_3	
3	$x_1^2x_3, x_1x_2^2, x_2x_4^2, x_3^2x_4$	(OCF25)
4	$x_1^3x_2, x_1x_3^3, x_2^3x_4, x_3x_4^3$	
5	$x_1^5, x_2^5, x_3^5, x_4^5$	

Here is a complete proof. Call a positive-degree invariant monomial indecomposable when it is not the product of two positive-degree invariant monomials. The divisor proof preceding OCF11 shows that each indecomposable has degree at most five. If it contains both x_1, x_4 , it is their product, since otherwise division by this invariant quadratic leaves a nonconstant invariant monomial. The same assertion holds for x_2, x_3 . After these cases are removed, at most two variables occur: every three indices contain one of the two opposite pairs. One variable gives precisely its fifth power. The remaining two-element supports are $\{1, 2\}, \{1, 3\}, \{2, 4\}, \{3, 4\}$. For positive exponents (a, b) with $a + b \leq 5$, the equation $a + 2b \equiv 0 \pmod{5}$ has exactly $(a, b) = (3, 1), (1, 2)$, and $a + 3b \equiv 0 \pmod{5}$ has exactly $(a, b) = (2, 1), (1, 3)$. For $\{2, 4\}$ divide the weights by 2 in the field $\mathbb{Z}/5\mathbb{Z}$, giving the first list; for $\{3, 4\}$ divide by 3, giving the second. These are exactly the eight mixed monomials of degrees three and four in OCF25. Every listed monomial has no proper invariant divisor, as is also seen from these lists and the excluded opposite pairs. This proves the classification. Factoring any invariant monomial repeatedly into indecomposables terminates because positive total degree strictly decreases. Hence OCF25 generates the entire invariant algebra.

Use $\mathcal{G} = \bigcup_{e=2}^5 \mathcal{G}_e$ in place of $\{x^\nu : \nu \in G\}$ in OCF12–16 and OCF20. The same factorization argument proves the same exact invariant image; all coefficient formulas, invariant certificates, pivot charts, quotient maps and ranks remain valid. The new recursion has 2, 4, 4, 4 generators in degrees 2, 3, 4, 5, respectively. Thus for $d \geq 13$ its precise matrix dimensions and rank are

$$C_d^{(14)} \in \text{Mat}_{(16d-48) \times (112d-864)}(\mathbb{C}), \quad \text{rank } C_d^{(14)} = 8d - 32. \quad (\text{OCF26})$$

Indeed the number of generators is 14 and the sum of their degrees is $2 \cdot 2 + 4 \cdot 3 + 4 \cdot 4 + 4 \cdot 5 = 52$; substituting the already proved $j_{d-e} = 8(d-e) - 32$ gives $14(8d-32) - 8 \cdot 52 = 112d - 864$. At smaller degrees the exact number of columns is the same sum of the appropriate j_{d-e} from OCF19, with negative subscripts zero. All coefficients in this reduced recursion are still the original coefficients of OCF12–14.

11 The exact common kernel frame and its original Gamma metrics

Retain the original simple quartet $m = 1$, $k = 4l + 1$, $l \geq 1$, $q = (k + 1)^2$, $c = k/2$, $0 < \delta < 1/2$ and $\gamma > 2$. The original period parameter and its branch are fixed, on the proved original OCF rank domain $|u| \geq R_*$, with R_* the retained explicit period-orbit radius. The kernel below is the kernel of that one fixed original observation $\Lambda : E \rightarrow B$. It is common to the two source metrics and to their four degree endpoints. No intersection over different period parameters or different observations is made.

Order all pairs (a, b) , $0 \leq a, b \leq k$, lexicographically once and retain that order. Define

$$\begin{aligned} \beta_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, & \omega_{ab} &= \frac{\beta_{ab} - c}{i} = (2b - k)\gamma - i(2a - k)\delta, \\ \chi(S) &= \prod_{a,b=0}^k (S - \beta_{ab}), & E &= \mathbb{C}[S]/(\chi), & \varepsilon_{ab} &= \left[\prod_{(a',b') \neq (a,b)} \frac{S - \beta_{a'b'}}{\beta_{ab} - \beta_{a'b'}} \right], & (\text{OKM1}) \\ V_\beta &= ((\beta_{ab})^d)_{(a,b), 0 \leq d < q}. \end{aligned}$$

The points are distinct: equality of their real and imaginary parts forces both indices equal. Hence every denominator displayed above is nonzero. Evaluation at all points gives $\varepsilon_{ab}(\beta_{a'b'}) = \mathbf{1}_{(a,b)=(a',b')}$. Polynomial interpolation proves that these are a basis of E , and the evaluation map from increasing polynomial coordinates to this primary basis is exactly V_β . Its determinant is the ordered Vandermonde product, so it is invertible.

11.1 The exact dual and primal maps of the strip quotient

Use the coefficient presentation of OQC7–8 in Segre coordinates. A biform of bidegree (k, k) has the full coefficient expression

$$F = \sum_{a,b=0}^k f_{ab} X_1^a X_0^{k-a} Y_1^b Y_0^{k-b}. \quad (\text{OKM2})$$

The literal variable dictionary to the OCF biform coefficients is $X_1 = z_0$, $X_0 = z_1$, $Y_1 = w_0$, $Y_0 = w_1$. The Segre substitution has $t_{++} = X_1 Y_1$, $t_{+-} = X_1 Y_0$, $t_{-+} = X_0 Y_1$, $t_{--} = X_0 Y_0$. At the original exponential vector take $X_1 = e^{-i\delta w}$, $X_0 = e^{i\delta w}$, $Y_1 = e^{\gamma w}$, $Y_0 = e^{-\gamma w}$. Then the monomial with coefficient f_{ab} is $e^{\omega_{ab} w}$. The original coefficient identification includes the entire invertible matrix H of OQC1 before this Segre presentation; its unit and period entries remain in the matrices produced by the strip calculation. OQC7 is characterized by evaluation on $e^{wY}[1]$. Since $e^{wY}[1] = \sum_{a,b} e^{\omega_{ab} w} \varepsilon_{ab}$, the independent exponentials in OQC4 therefore give the precise complex-linear dual pairing

$$\Theta(F)(\varepsilon_{ab}) = f_{ab}, \quad \Theta(F)(x) = f^\top x_{\text{prim}}. \quad (\text{OKM3})$$

There is no binomial coefficient or complex conjugation in this complex-linear dual pairing. The statement follows by equating all coefficients of the distinct exponentials, whose independence is proved by their Vandermonde derivative matrix in OQC4.

Let J be exactly the coefficient subspace of the original observed dual image, as identified in OQC8. The original strip and invariant elimination provider OCF constructs its literal surjection

$$\pi : \mathbb{C}^q \twoheadrightarrow \mathbb{C}^{r_K}, \quad \ker \pi = J, \quad r_K = 8k - 16. \quad (\text{OKM4})$$

This input is the actual OCF matrix, including its stated period coefficients and its valid pivot choice. The coefficient pairing OKM3 and OQC8 identify $K := \ker \Lambda = J^\perp$ in primary coordinates, where $J^\perp = \{x : f^\top x = 0 \text{ for every } f \in J\}$. Consequently its exact primal frame and polynomial frame are

$$I_{\text{prim}} = \pi^\top : \mathbb{C}^{r_K} \hookrightarrow E_{\text{prim}}, \quad I_{\text{poly}} = V_\beta^{-1} \pi^\top : \mathbb{C}^{r_K} \hookrightarrow E. \quad (\text{OKM5})$$

Here is a direct proof, including inverse coordinates. Choose any of the explicit right sections L_π of the surjective matrix π , so $\pi L_\pi = I_{r_K}$. For $x = \pi^\top v$ and $f \in J$, $f^\top x = (\pi f)^\top v = 0$. The transpose is injective because $L_\pi^\top \pi^\top = I_{r_K}$. Conversely, if $x \in J^\perp$, then $(I - L_\pi \pi)f \in J$ for every f , so $x^\top (I - L_\pi \pi)f = 0$ for every f . Thus $x = \pi^\top L_\pi^\top x$. This proves equality of the image and K , and its inverse on that image is $L_\pi^\top$. In polynomial coordinates the inverse is $L_\pi^\top V_\beta$. This proves the full typed primal arrow, rather than selecting a Hermitian transpose in place of it.

11.2 The complete original source and quotient metric

For either original source $s \in \{1, k\}$ write $b_s = s/2$ and $M_s = (2\pi)^{s/2}$. The unchanged real-coordinate measure is

$$dm_{0,s}(y) = M_s \frac{2^{b_s}}{4\pi\Gamma(b_s)} |\Gamma(b_s/2 + iy/2)|^2 dy, \quad S = c + iy. \quad (\text{OKM6})$$

The complete canonical OGN1–3 proof gives the monic real orthogonal polynomials and all their squared norms:

$$\begin{aligned} p_0(y) &= 1, \quad p_1(y) = y, \quad p_{d+1}(y) = yp_d(y) - d(b_s + d - 1)p_{d-1}(y), \\ h_d &= M_s d! (b_s)_d, \quad u_d(S) = \frac{p_d((S - c)/i)}{\sqrt{h_d}}. \end{aligned} \quad (\text{OKM7})$$

The square root is positive. Thus u_d has the original leading coefficient $i^{-d}/\sqrt{h_d}$ in S and is orthonormal on the line in OKM6. No source mass or phase has been removed.

For a polynomial cutoff $D \geq q - 1$ let

$$\begin{aligned} T_D &= \left(\frac{p_d(\omega_{ab})}{\sqrt{h_d}} \right)_{(a,b), 0 \leq d \leq D}, \quad R_D = T_D T_D^*, \\ G_D^{\text{prim}} &= R_D^{-1}, \quad G_D^{\text{poly}} = V_\beta^* R_D^{-1} V_\beta, \quad H_D^K = \bar{\pi} R_D^{-1} \pi^\top. \end{aligned} \quad (\text{OKM8})$$

The star denotes conjugate transpose, while the superscript $\top$ is ordinary transpose. In particular $(\pi^\top)^* = \bar{\pi}$, with precisely the indicated matrix dimensions.

These are the actual minimum quotient metrics, as follows directly from the original remainder map. In the orthonormal coefficient frame $u_0, \dots, u_D$, the remainder map followed by primary evaluation is exactly T_D , column by column. Its first q columns are invertible: the monic polynomials $p_0, \dots, p_{q-1}$ evaluated at the distinct ω_{ab} have a nonzero Vandermonde determinant, and all $\sqrt{h_d}$ are nonzero. Thus $R_D > 0$. For a prescribed primary vector x , the vector $T_D^* R_D^{-1} x$ maps to x . It is orthogonal to $\ker T_D$, because its inner product with a kernel vector contains T_D applied to that vector. Every other lift differs by such a vector, so its squared norm is $x^* R_D^{-1} x$ plus the squared norm of that difference. This proves the first quotient metric in OKM8. The identity $x_{\text{prim}} = V_\beta x_{\text{poly}}$ proves the second by direct pairing. Restriction to the exact frame OKM5 then gives

$$I_{\text{poly}}^* G_D^{\text{poly}} I_{\text{poly}} = (\pi^\top)^* R_D^{-1} \pi^\top = H_D^K > 0. \quad (\text{OKM9})$$

The last positivity follows from injectivity of $\pi^\top$. The map $K \hookrightarrow E$ is the fixed original coefficient map. No invariance of that kernel under multiplication is used.

11.3 Christoffel–Darboux with the exact source mass

The classical identity used here is Christoffel–Darboux, including its coincident form, as given by Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21, eqs. (18.2.12)–(18.2.13)]. Their leading coefficients $k_D = k_{D+1} = 1$ are those of our monic p_D, p_{D+1} , and their squared norm is the complete original $h_D = M_s D! (b_s)_D$. Thus their prefactor $k_D/(h_D k_{D+1})$ is exactly $1/h_D$. The polynomial proof below gives the extension to complex z, w and the precise conjugation map in OKM12–13.

For arbitrary $z, w \in \mathbb{C}$ and $D \geq 0$, the complete polynomial identity is

$$\sum_{d=0}^D \frac{p_d(z)p_d(w)}{h_d} = \begin{cases} \frac{p_{D+1}(z)p_D(w) - p_D(z)p_{D+1}(w)}{h_D(z-w)}, & z \neq w, \\ \frac{p'_{D+1}(z)p_D(z) - p'_D(z)p_{D+1}(z)}{h_D}, & z = w. \end{cases} \quad (\text{OKM10})$$

To prove it, put $a_d = d(b_s + d - 1)$, with $a_0 = 0$, and set $p_{-1} = 0$ for this recurrence expression. The recurrence in OKM7 gives

$$(z-w)p_d(z)p_d(w) = p_{d+1}(z)p_d(w) - p_d(z)p_{d+1}(w) - a_d(p_d(z)p_{d-1}(w) - p_{d-1}(z)p_d(w)). \quad (\text{OKM11})$$

For $d \geq 1$, the full norm identity is $a_d/h_d = 1/h_{d-1}$. Divide OKM11 by h_d and sum. Every adjacent term cancels, leaving just the degree- D numerator divided by h_D ; the $d = 0$ lower term is zero. This proves the first line of OKM10. Both sides of the resulting equality after multiplication by $z - w$ are polynomials. Differentiating with respect to z and then setting $z = w$ proves the second line with the displayed order and sign. This also proves the value at every confluent point without an assertion about a limiting singular quotient. The denominator is the full $h_D = M_s D!(b_s)_D$, including M_s .

Since all coefficients of the recurrence are real, the matrix entries in OKM8 are therefore exactly the values of OKM10 at

$$z = \omega_{ab}, \quad w = \overline{\omega_{a'b'}}. \quad (\text{OKM12})$$

Their indexed conjugation and confluent locus are

$$\begin{aligned} \overline{\omega_{a'b'}} &= \omega_{k-a',b'}, & \omega_{ab} - \overline{\omega_{a'b'}} &= 2(b-b')\gamma - 2i(a+a'-k)\delta, \\ \omega_{ab} &= \overline{\omega_{a'b'}} & \iff & b = b' \text{ and } a + a' = k. \end{aligned} \quad (\text{OKM13})$$

These follow by substituting OKM1; positivity of δ, γ proves both implications in the last line. Thus the conjugation map on the point indices is precisely $(a, b) \mapsto (k - a, b)$. The confluent formula of OKM10 applies at these indexed entries. For the retained odd k , its matrix diagonal has $a' = a, b' = b$, so $2a = k$ cannot occur; the literal Hermitian diagonal is given by the first line of OKM10. The conjugate-index entries use its second line. All entries, including the positive diagonal, are simultaneously defined by the original polynomial sum on the left of OKM10.

11.4 Every pivot change and the four original endpoints

Let π' be another valid strip/invariant elimination matrix for the same original J , with the same surjective codomain dimension. Choose right sections $L_\pi, L_{\pi'}$ and put

$$A_{\pi',\pi} = \pi' L_\pi, \quad A_{\pi,\pi'} = \pi L_{\pi'}. \quad (\text{OKM14})$$

Because π' vanishes on $J = \ker \pi$, one has $\pi' L_\pi \pi = \pi'$. Thus $\pi' = A_{\pi',\pi} \pi$. Reversing the roles gives the opposite identity, and surjectivity of each matrix then proves $A_{\pi',\pi} A_{\pi,\pi'} = I$ and $A_{\pi,\pi'} A_{\pi',\pi} = I$. This proves invertibility and also independence from the chosen right sections. Writing $A = A_{\pi',\pi}$ for these identities gives the exact frame and Gram changes

$$\begin{aligned} I'_{\text{prim}} &= I_{\text{prim}} A^\top, & I'_{\text{poly}} &= I_{\text{poly}} A^\top, \\ (H_D^K)' &= \overline{A} H_D^K A^\top, & \det(H_D^K)' &= |\det A|^2 \det H_D^K. \end{aligned} \quad (\text{OKM15})$$

The first two formulas follow by transposing $\pi' = A\pi$. Substitution into OKM9 proves the congruence and determinant. This retains all actual pivot determinants. They have not been assigned modulus one.

The kernel is the same fixed $K = \ker \Lambda$ used in MRI1–5. If its original BRU inclusion is I_0 and its fixed-section kernel coordinate map is κ_0 , let

$$\mathbf{C}_K = \kappa_0 I_{\text{poly}}, \quad \mathbf{C}_K^{-1} = L_\pi^\top V_\beta I_0. \quad (\text{OKM16})$$

The equalities $I_0 \kappa_0 x = x$ for $x \in K$ and $L_\pi^\top V_\beta I_{\text{poly}} = I_{r_K}$ prove that these are inverse matrices and $I_0 \mathbf{C}_K = I_{\text{poly}}$. Consequently $H_D^K = \mathbf{C}_K^* (I_0^* G_D^{\text{poly}} I_0) \mathbf{C}_K$. The matrix $\mathbf{C}_K$ is independent of D and of the choice of source order s . The source metrics change; the two specified coefficient frames do not.

It follows that the complete original four-endpoint kernel scalar is

$$\boxed{F_K^{(s)} = \log \det H_{q-1}^K + \log \det H_q^K - \log \det H_{2q-1}^K - \log \det H_{2q}^K.} \quad (\text{OKM17})$$

Indeed MRI3 gives the actual consecutive kernel determinant ratio

$$\frac{\det(I_0^* G_{q+j-1}^{\text{poly}} I_0)}{\det(I_0^* G_{q+j}^{\text{poly}} I_0)} = \frac{1 + t_j}{1 + \xi_j}.$$

Each factor $|\det \mathbf{C}_K|^2$ cancels in that ratio by OKM16. The weighted sum with endpoint weights $1, 2, \dots, 2, 1$ therefore has precisely the four signs in OKM17, as in MRI4. This verifies the original degree cutoffs, rather than shifting the column count in T_D . Under any valid pivot change OKM15, the added scalar $2 \log |\det A|$ has total coefficient $1 + 1 - 1 - 1 = 0$, proving invariance of OKM17 with its original frames.

For emphasis, the full mass in OKM7 appears in every column of T_D and in the denominator of OKM10. If that common mass is factored explicitly, write $T_D = M_s^{-1/2} \hat{T}_D$, where $\hat{T}_D$ has exactly the same numerator polynomials and the denominator $\sqrt{d!(b_s)_d}$. Then $H_D^K = M_s \bar{\pi} (\hat{T}_D \hat{T}_D^*)^{-1} \pi^\top$, so its determinant has factor $M_s^{r_K}$. It cancels only in the displayed signed four-endpoint scalar, whose four ranks are equal. The individual metrics in OKM8–9 retain it. No inserted Gamma order, discarded action defect, or kernel-invariance assertion enters any of these calculations.

12 The original kernel graph, boundary basis and complete action

This section uses the complete conductor coefficient construction OCF and its exact original dual identification in OKM. Fix the original simple quartet, its complete amplitude unit and periods, the original parameter on any logarithm branch with $|u| \geq R_*$, and $k = 4l + 1$, $l \geq 1$. Put $q = (k + 1)^2$, $r = 8k - 16$, and $\mathcal{I}_k = \{(a, b) : 0 \leq a, b \leq k\}$. The roots, centre and centred eigenvalues remain

$$c = k/2, \quad \beta_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \omega_{ab} = (\beta_{ab} - c)/i = (2b - k)\gamma - i(2a - k)\delta. \quad (\text{OKA1})$$

The original algebra is $E = \mathbb{C}[S]/(\chi)$, with $\chi(S) = \prod_{(a,b) \in \mathcal{I}_k} (S - \beta_{ab})$. Let ε_{ab} be its original primary idempotents and $\mathcal{V} : E \rightarrow \mathbb{C}^{\mathcal{I}_k}$ the evaluation isomorphism $[p] \mapsto (p(\beta_{ab}))_{ab}$. Its inverse sends e_{ab} to

$$\varepsilon_{ab}(S) = \frac{\chi(S)}{(S - \beta_{ab})\chi'(\beta_{ab})}. \quad (\text{OKA2})$$

These denominators are nonzero: equality of two roots in OKA1, on comparing real and imaginary parts and using $\delta > 0$, $\gamma > 2$, implies equality of both indices. Evaluation proves that the polynomials in OKA2 are the coordinate idempotents, so their columns give the exact inverse, with no changed root or polynomial coefficient convention.

12.1 The coefficient quotient gives a graph in the original primary basis

Let $\pi : \mathbb{C}^{\mathcal{I}_k} \rightarrow \mathbb{C}^r$ be the surjective coefficient quotient of OCF, whose kernel is precisely the image J_k of the invariant polynomials under the original restriction. The domain here consists of the coefficients of the biforms $z_0^a z_1^{k-a} w_0^b w_1^{k-b}$, and is the complex-linear dual of the displayed primary coordinate space. Let $T \subset \mathcal{I}_k$ be the final OCF complement of its invariant-image pivot rows, viewed as a subset of the full biform coefficient grid. Its ordered coordinate inclusion is $E_T : \mathbb{C}^r \rightarrow \mathbb{C}^{\mathcal{I}_k}$. The explicit OCF formula gives

$$\pi E_T = I_r, \quad N = \pi^T : \mathbb{C}^r \longrightarrow \mathbb{C}^{\mathcal{I}_k}, \quad E_T^T N = I_r. \quad (\text{OKA3})$$

Indeed a retained strip monomial in T already has zero conductor remainder outside its own coordinate, zero coordinates on the invariant pivot rows, and its indicated coordinate equal to one. Both reductions in OCF therefore leave that quotient coordinate equal to its standard unit vector. Transposing gives the last identity. This uses ordinary transpose, because the restriction quotient and its dual are complex-linear maps.

Write $J = \mathcal{I}_k \setminus T$ only for the complementary index set in this section; the invariant image retains the name J_k . Denote coordinate restrictions by x_T, x_J , and put $N_J = E_J^T N$. The original kernel inclusion is consequently

$$I_K = \mathcal{V}^{-1} N : \mathbb{C}^r \hookrightarrow E, \quad \text{im } I_K = K := \ker \Lambda, \quad N = \begin{pmatrix} I_r \\ N_J \end{pmatrix}_{T,J}. \quad (\text{OKA4})$$

For completeness, a primary vector v pairs with a biform coefficient vector f by $f^T v$. The annihilator of $\ker \pi = J_k$ is $\text{im } \pi^T$: containment follows by multiplication, while surjectivity of π gives both subspaces dimension r . OQC8 identifies precisely this annihilator with the primary coordinates of the fixed original $\ker \Lambda$. This proves both the original image assertion and its direction in OKA4. In polynomial coordinates the same inclusion is $V_\beta^{-1} N$, where $(V_\beta)_{(a,b),d} = \beta_{ab}^d$ for $0 \leq d < q$; this is the full coefficient matrix of OKA2.

12.2 An explicit basis of the actual boundary

For $j \in J$ define the actual boundary vector $b_j = \Lambda \varepsilon_j$. These vectors form a basis of the original image $B = \text{im } \Lambda$. To prove independence, if $\sum_{j \in J} c_j \varepsilon_j$ lies in K , its primary coordinates equal $N \zeta$.

Its T coordinates vanish, so OKA3 gives $\zeta = 0$ and all $c_j = 0$. To prove spanning, use the literal decomposition

$$x = Nx_T + E_J(x_J - N_Jx_T), \quad \Lambda\mathcal{V}^{-1}x = \sum_{j \in J} b_j(x_J - N_Jx_T)_j. \quad (\text{OKA5})$$

The first equality follows separately on its T and J coordinates. Apply Λ , which kills the first term by OKA4, to obtain the second. Thus no observation has been substituted: in this basis the specified original observation has the exact coefficient matrix $(-N_J, I_{q-r})$.

Here are the complete original period and tensor coordinates of these boundary vectors. Write the quartet roots as $\alpha_{\epsilon, \eta} = 1/2 + \epsilon\delta + i\eta\gamma$, where $\epsilon, \eta \in \{+1, -1\}$ and the signs $+, -$ in subscripts denote these same two numbers. In a subscript α below means that complex root, indexed by its sign pair. Keep the original one-factor primary idempotents $\varepsilon_\alpha^h \in E_h$. Set

$$v_\alpha = \Pi U \varepsilon_\alpha^h = v_h(\alpha) \Pi \varepsilon_\alpha^h, \quad Z_+ = z_0, \quad Z_- = z_1, \quad W_+ = w_0, \quad W_- = w_1, \quad \mathbf{v}(z, w) = \sum_{\epsilon, \eta} v_{\epsilon\eta} Z_\epsilon W_\eta. \quad (\text{OKA6})$$

The vectors belong to the original one-factor period space; all entries of Π and all four actual nonzero amplitude values remain present. For every $(a, b) \in \mathcal{I}_k$,

$$j\Lambda\varepsilon_{ab} = [z_0^a z_1^{k-a} w_0^b w_1^{k-b}] P_k(\mathbf{v}(z, w)^{\otimes k}), \quad P_k = \frac{1}{5} \sum_{h=0}^4 (C^{-h})^{\otimes k}. \quad (\text{OKA7})$$

The original sum inclusion $\iota : E \hookrightarrow E_h^{\otimes k}$ sends ε_{ab} to the sum of the primary tensor idempotents whose word has a positive real signs and b positive imaginary signs. This assertion follows by evaluation: on a primary word, the sum variable takes exactly the value β_{ab} of its two counts, and OKA2 evaluates to one for those counts and zero otherwise. Applying $(\Pi U)^{\otimes k}$ gives precisely the corresponding sum of the tensors of the v_α in OKA6. Expanding the tensor power and taking the indicated biform coefficient selects exactly the same words, each once. The original definition $j\Lambda = P_k(\Pi U)^{\otimes k} \iota$ proves OKA7, including the literal inverse powers and $1/5$. Its specialization to $j \in J$ therefore supplies the actual basis b_j of OKA5. Neither a different target nor a numerical period matrix has been introduced.

The fixed, source-independent kernel coordinate and section are now explicitly

$$\kappa = E_T^T \mathcal{V} : E \rightarrow \mathbb{C}^r, \quad S_* : B \rightarrow E, \quad S_* b_j = \varepsilon_j, \quad (\text{OKA8})$$

$$I_K \kappa + S_* \Lambda = I_E, \quad \kappa I_K = I_r, \quad \kappa S_* = 0, \quad \Lambda S_* = I_B.$$

Every equality follows by applying OKA5 to an arbitrary primary vector. In particular this is an actual splitting of the fixed original short exact sequence, used unchanged for both source orders and for all polynomial degrees.

12.3 The whole original action and its retained defect

The original centred sum action is $Y = (S - c)/i$ on E . Under $\mathcal{V}$ it is the diagonal matrix of the ω_{ab} in OKA1. Write its two diagonal submatrices as Ω_T and Ω_J . In the exact coordinates $e = I_K \zeta + S_* b$ of OKA8, it has the complete matrix

$$Y \longleftrightarrow \begin{pmatrix} \Omega_T & 0 \\ \Omega_J N_J - N_J \Omega_T & \Omega_J \end{pmatrix}, \quad (\Omega_J N_J - N_J \Omega_T)_{jt} = (\omega_j - \omega_t)(N_J)_{jt}. \quad (\text{OKA9})$$

Indeed the primary vector of e is $(\zeta, N_J \zeta + b)$. Applying the original diagonal matrix gives $(\Omega_T \zeta, \Omega_J N_J \zeta + \Omega_J b)$. Applying the coordinates in OKA5 subtracts $N_J \Omega_T \zeta$ from its second component and proves every block in OKA9. This proves, with its precise codomain,

$$Y I_K - I_K \Omega_T = S_* (\Omega_J N_J - N_J \Omega_T) : \mathbb{C}^r \rightarrow E, \quad \Lambda Y I_K = \Omega_J N_J - N_J \Omega_T : \mathbb{C}^r \rightarrow B, \quad (\text{OKA10})$$

where the matrix on the right uses the basis b_j . The chosen complementary span of the original primary idempotents is stable under Y ; no stability of K has been used. Its entire failure is the displayed coefficient map.

For each original source $s \in \{1, k\}$ retain the monic Gamma polynomials $p_d^{(s)}$, the complete norms $h_d^{(s)} = (2\pi)^{s/2} d!(s/2)_d$, and the original $e_d^{(s)} = [p_d^{(s)}(Y)]/\sqrt{h_d^{(s)}}$. Then the fixed-section kernel and boundary coordinates of every actual remainder, including all degrees $d \geq q$, are

$$\begin{aligned} (\kappa e_d^{(s)})_t &= \frac{p_d^{(s)}(\omega_t)}{\sqrt{h_d^{(s)}}}, \\ (\Lambda e_d^{(s)})_j &= \frac{p_d^{(s)}(\omega_j) - \sum_{t \in T} (N_J)_{jt} p_d^{(s)}(\omega_t)}{\sqrt{h_d^{(s)}}}. \end{aligned} \tag{OKA11}$$

Polynomial evaluation in the original primary algebra gives the full vector $p_d^{(s)}(\omega)/\sqrt{h_d^{(s)}}$ even when its polynomial representative must be reduced modulo χ . Applying OKA5 and OKA8 proves OKA11. Thus the formula includes the original finite remainder operation and every phase and source mass. Composing its second line with the actual columns OKA7 gives exactly the period-vector increments of OPG. Its first line is the fixed-section kernel vector used by OPG and OPR; the degree-dependent minimum-section correction in OPR remains required for the mixed norm. The metric computation in OKM applies to the common inclusion OKA4 and does not replace that correction by the fixed-section vector alone.

13 The complete homogeneous geometry of the original multiplicity curve

13.1 The actual primary, unit, period and Fourier coefficient map

Retain the original full-order quartet and every original period parameter. Write

$$\begin{aligned} \alpha_{\epsilon,\eta} &= \frac{1}{2} + \epsilon\delta + i\eta\gamma, \quad \epsilon, \eta \in \{-1, 1\}, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \\ m &\geq 1, \quad d_m = m - 1, \quad n = 4m, \quad N = 4m + 1, \quad h(z) = \prod_{\epsilon,\eta} (z - \alpha_{\epsilon,\eta})^m, \\ E_h &= \mathbb{C}[z]/(h), \quad A_h = M_z, \quad v_h = 2\xi/h, \quad U = M_{j_h v_h}, \quad v_h(\alpha_{\epsilon,\eta}) \neq 0. \end{aligned} \quad (\text{OMG1})$$

The last inequalities follow from the stipulated complete original zero order m . The map j_h records all divided Taylor coefficients of orders $0, \dots, m-1$ at all four roots. Let $\mathcal{E}_{\text{prim}} : \mathbb{C}^{4m} \rightarrow E_h$ send its coordinate (α, d) to the original primary element $\varepsilon_\alpha(z - \alpha)^d$, $0 \leq d < m$. Explicitly its representative is

$$E_{\alpha,d}(z) = \text{rem}_h \left[h_\alpha(z) \sum_{j=0}^{m-1} \frac{(h_\alpha^{-1})^{(j)}(\alpha)}{j!} (z - \alpha)^{j+d} \right], \quad h_\alpha = h/(z - \alpha)^m.$$

The Taylor inverse is defined since $h_\alpha(\alpha) \neq 0$. It gives the idempotent one on its own primary factor and zero on the others. The four coprime factors therefore prove that $\mathcal{E}_{\text{prim}}$ is an isomorphism and include every nilpotent degree. In this basis the full matrix of U is

$$U_{\text{prim};\alpha,a;\alpha,d} = \begin{cases} v_h^{(a-d)}(\alpha)/(a-d)!, & a \geq d, \\ 0, & a < d, \end{cases} \quad \det U = \prod_{\alpha} v_h(\alpha)^m \neq 0. \quad (\text{OMG2})$$

Other primary blocks are zero. Inverting its constant coefficient plus its nilpotent part by the finite geometric series proves invertibility with all derivatives retained.

Keep the original $u \neq 0$ and its logarithm branch, the primitive $\Phi'_h = h$, $\Phi_h(0) = 0$, the outward rays ℓ_j of angles $(\arg u + \pi + 2\pi j)/N$, and $\Gamma_j = -\ell_0 + \ell_j$. The exact period and Fourier maps are

$$\begin{aligned} (\Pi[P])_j &= \int_{\Gamma_j} e^{\Phi_h(z)/u} \text{rem}_h P(z) dz, \quad 1 \leq j \leq n, \\ \zeta &= e^{2\pi i/N}, \quad F_{jt} = \zeta^{jt} - 1, \quad (F^{-1})_{tj} = \zeta^{-tj}/N \quad (1 \leq j, t \leq n), \quad C = F \text{diag}(\zeta, \dots, \zeta^n) F^{-1}. \end{aligned} \quad (\text{OMG3})$$

The full original period theorem PDE/OPG gives the convergence and invertibility of Π on these representatives. The Fourier inverse follows directly from $\sum_{j=1}^{N-1} \zeta^{-tj} (\zeta^{sj} - 1) = N \mathbf{1}_{t=s}$. It is the same literal cyclic matrix as OCS4, including its inverse powers in the original average. Its inherited Gram is $F^*F = N(I_n + \mathbf{1}\mathbf{1}^*)$, as expansion of its entries gives.

Define the exact diagonal map and coefficient matrix

$$\begin{aligned} D_m^{\text{jet}} &= \text{diag} \left(i^{-d}/d! \right)_{\alpha, 0 \leq d < m}, \quad \mathbf{H}_m = F^{-1} \Pi U \mathcal{E}_{\text{prim}} D_m^{\text{jet}}, \\ (\mathbf{H}_m)_{t;\alpha,d} &= \frac{i^{-d}}{d!} \sum_{a=d}^{m-1} \left[\frac{1}{N} \sum_{j=1}^n \zeta^{-tj} \int_{\Gamma_j} e^{\Phi_h(z)/u} E_{\alpha,a}(z) dz \right] \frac{v_h^{(a-d)}(\alpha)}{(a-d)!}. \end{aligned} \quad (\text{OMG4})$$

This is exactly OCS13, now regarded as one square matrix. All factors in its first line are invertible. In particular

$$\det \mathbf{H}_m = (\det F)^{-1} \det \Pi \left(\prod_{\alpha} v_h(\alpha)^m \right) \det \mathcal{E}_{\text{prim}} \prod_{\alpha} \prod_{d=0}^{m-1} \frac{i^{-d}}{d!} \neq 0,$$

with the determinant of $\mathcal{E}_{\text{prim}}$ taken in the fixed original coefficient frame. None of these factors is prescribed.

Set $\omega_{\epsilon,\eta} = \eta\gamma - i\epsilon\delta$ and let

$$z_{\epsilon,\eta,d}(w) = w^d e^{\omega_{\epsilon,\eta} w}, \quad x(w) = F^{-1} \Pi U e^{w(A_h - 1/2)/i} [1]. \quad (\text{OMG5})$$

The terminating exponential on each primary factor has coefficient $i^{-d} w^d / d!$ on $\varepsilon_\alpha(z - \alpha)^d$. Multiplication by OMG2 and application of the literal period maps therefore prove the exact identity and inverse coordinate map

$$\boxed{x(w) = \mathbf{H}_m z(w), \quad z(w) = \mathbf{H}_m^{-1} x(w).} \quad (\text{OMG6})$$

For the original Euclidean period metric, its Gram on these primary curve coordinates is $\mathbf{H}_m^* N(I_n + \mathbf{11}^*) \mathbf{H}_m$. Thus the coefficient change also retains its full metric and all off-diagonal entries, without claiming orthogonality.

13.2 The complete homogeneous ideal, with quadratic generators

Use binary indices $a, b \in \{0, 1\}$ for $\epsilon = 2a - 1$, $\eta = 2b - 1$. Let $\mathcal{P}_Z = \mathbb{C}[Z_{a,b,d} : a, b \in \{0, 1\}, 0 \leq d \leq d_m]$ with each variable of degree one. Define the algebra map

$$\Phi_m : \mathcal{P}_Z \longrightarrow \mathbb{C}[X_0, X_1, Y_0, Y_1, T_0, T_1], \quad Z_{a,b,d} \longmapsto X_a Y_b T_0^{d_m - d} T_1^d. \quad (\text{OMG7})$$

Its image is the complete diagonal graded algebra

$$R_m = \bigoplus_{k \geq 0} \mathbb{C}[X_0, X_1]_k \otimes \mathbb{C}[Y_0, Y_1]_k \otimes \mathbb{C}[T_0, T_1]_{k d_m}. \quad (\text{OMG8})$$

Indeed a basis monomial of its degree- k component is

$$X_0^{k-a} X_1^a Y_0^{k-b} Y_1^b T_0^{k d_m - D} T_1^D, \quad 0 \leq a, b \leq k, \quad 0 \leq D \leq k d_m. \quad (\text{OMG9})$$

It is a product of k images in OMG7: choose k binary first indices with sum a , independently k binary second indices with sum b , and k integers in $[0, d_m]$ with sum D ; the last choice follows by filling successive slots up to d_m . Combining the three lists into rows gives the required preimage monomial. Distinct displayed target monomials are linearly independent.

Let $\mathfrak{t}_m$ be generated by every quadratic binomial

$$Z_{a,b,d} Z_{a',b',d'} - Z_{c,e,f} Z_{c',e',f'} \quad \text{with} \quad a + a' = c + c', \quad b + b' = e + e', \quad d + d' = f + f', \quad (\text{OMG10})$$

where every index belongs to its original range. Repeated variables are allowed. Then

$$\boxed{\ker \Phi_m = \mathfrak{t}_m, \quad \mathcal{P}_Z / \mathfrak{t}_m \simeq R_m.} \quad (\text{OMG11})$$

Here is a complete ideal proof. Each displayed quadratic has zero image. Two degree- k monomials have the same image exactly when the sums of their first indices, second indices and third indices agree. Regard their k factors temporarily as labelled rows. Align the first indices with the target list by pairwise swaps of first indices, keeping each row's other indices fixed. Each swap is an OMG10 relation. Next align the second indices by the same operation, keeping the aligned first indices fixed. Finally, if the third indices differ from their target list, choose a row below its target and a row above its target. Increase the former third index by one and decrease the latter by one. These entries stay in $[0, d_m]$ because their target entries do; the total sum is unchanged and the sum of absolute deviations decreases by two. This terminates at the target list. Each transfer is another OMG10 relation. Thus the difference between any two monomials with the same image is a sum of multiples of the quadrics, by telescoping the successive monomial differences. Labels only specify the sequence of polynomial maps; the products are the original commutative monomials

throughout. For any polynomial in $\ker \Phi_m$, collect its coefficients by their target monomial. Each such coefficient sum is zero by target monomial independence. Subtracting one chosen monomial in each group expresses that whole group as a linear combination of the proved monomial differences. This proves the reverse inclusion for all degrees and hence the entire ideal equality.

Every degree- k original curve monomial has the function

$$w^D \exp\{[(2b - k)\gamma - i(2a - k)\delta]w\} \quad (0 \leq a, b \leq k, \ 0 \leq D \leq kd_m). \quad (\text{OMG12})$$

The frequencies are distinct: equality of real parts gives equality of b and equality of imaginary parts gives equality of a . To prove independence with all polynomial powers, suppose $\sum_{a,b} P_{a,b}(w)e^{\omega_{a,b}w} = 0$, with $\deg P_{a,b} \leq kd_m$. Fix a pair and apply $\prod_{(a',b') \neq (a,b)} (\partial_w - \omega_{a',b'})^{kd_m+1}$. This kills every other summand. On its surviving polynomial each factor is $\partial_w + (\omega_{a,b} - \omega_{a',b'})$, invertible on the space of degree at most kd_m because differentiation is nilpotent and the constant is nonzero. The surviving polynomial is therefore zero. Repeating this for every pair proves independence.

Thus the full homogeneous ideal of the original primary curve is exactly $\mathfrak{t}_m$: a homogeneous polynomial restricts to zero if and only if its coefficient sums at every target monomial OMG9 are zero. In particular

$$\boxed{\dim R_{m,k} = (k+1)^2[1 + k(m-1)] = q_k, \quad \ker\{\mathcal{P}_{Z,k} \rightarrow \text{Hol}(\mathbb{C}), f \mapsto f(z(w))\} = \mathfrak{t}_{m,k}.} \quad (\text{OMG13})$$

This statement concerns the homogeneous ideal, equivalently the projective closure of $[z(w)]$. Every one of its degrees has been computed, including degree zero. No assertion about a different nonhomogeneous affine ideal is used.

13.3 The projective model and its complete chart inverses

For $m \geq 2$, the morphism

$$(\mathbb{P}^1)^3 \longrightarrow \mathbb{P}^{4m-1}, \quad ([X_0 : X_1], [Y_0 : Y_1], [T_0 : T_1]) \longmapsto [X_a Y_b T_0^{d_m-d} T_1^d]_{a,b,d} \quad (\text{OMG14})$$

has multidegree $(1, 1, m-1)$. At least one first, second and third coordinate is nonzero; their corresponding extreme product is nonzero, so the map is everywhere defined. Its image and scheme structure are exactly $\text{Proj } R_m$, with complete closed ideal OMG10. Here are explicit inverses proving this claim. The eight charts $Z_{a,b,0} \neq 0$ and $Z_{a,b,d_m} \neq 0$ cover $\text{Proj } R_m$. Indeed OMG11 gives

$$Z_{a,b,d}^{d_m} = Z_{a,b,0}^{d_m-d} Z_{a,b,d_m}^d \quad \text{in } R_m.$$

A homogeneous prime containing every extreme variable contains every variable by this equality, and hence gives no point of Proj . On $Z_{a,b,0} \neq 0$ the three inverse affine coordinates are

$$x = Z_{1-a,b,0}/Z_{a,b,0}, \quad y = Z_{a,1-b,0}/Z_{a,b,0}, \quad t = Z_{a,b,1}/Z_{a,b,0}.$$

For every variable its ratio to $Z_{a,b,0}$ is $x^{1_{a' \neq a}} y^{1_{b' \neq b}} t^d$, by OMG7 and its proved injective quotient. Consequently the degree-zero localized ring is $\mathbb{C}[x, y, t]$. These three coordinates are algebraically independent, since their images in the parameter fraction field are $X_{1-a}/X_a, Y_{1-b}/Y_b, T_1/T_0$. For the chart at Z_{a,b,d_m} replace the last ratio by $t = Z_{a,b,d_m-1}/Z_{a,b,d_m}$; the general ratio has exponent $d_m - d$ in t . Its coordinate ring is again $\mathbb{C}[x, y, t]$. On chart overlaps these formulas invert the same parameter ratios, so they glue to the inverse of OMG14. This proves the isomorphism of schemes and its full defining homogeneous ideal. For $m = 1$ the third factor has degree zero and the same argument uses the four first/second charts: the model is the Segre $(\mathbb{P}^1)^2$ with ideal $Z_{0,0,0}Z_{1,1,0} - Z_{0,1,0}Z_{1,0,0}$.

In the actual Fourier coordinates let $\mathcal{P}_x = \mathbb{C}[x_1, \dots, x_n]$ and define

$$\mathfrak{a} = \{f(x) : f(\mathbf{H}_m Z) \in \mathfrak{t}_m\}, \quad R = \mathcal{P}_x / \mathfrak{a}. \quad (\text{OMG15})$$

The invertible substitution OMG6 proves that $\mathfrak{a}$ is the complete homogeneous ideal of $x(w)$ and that $R \simeq R_m$. Its explicit generators are all OMG10 quadrics with every Z replaced by its component of $H_m^{-1}x$. These are the actual period and unit coefficients of OMG4, and $\mathfrak{a}$ is prime because the target ring in OMG7 is a domain. Thus the original curve's projective closure is the precise linear image of OMG14 under H_m , with no prescribed substitute matrix.

13.4 The exact graded source-dual algebra and every factorial

For each $k \geq 0$ retain the original sum algebra

$$e_k = 1 + kd_m, \quad c_k = k/2, \quad \beta_{k,a,b} = c_k + (2a - k)\delta + i(2b - k)\gamma, \\ \chi_k(S) = \prod_{a,b=0}^k (S - \beta_{k,a,b})^{e_k}, \quad E_k = \mathbb{C}[S]/(\chi_k), \quad Y_k = (M_S - c_k I)/i. \quad (\text{OMG16})$$

For $k = 0$ this is $E_0 = \mathbb{C}[S]/(S) = \mathbb{C}$. Let $H_{k,a,b,D} = \varepsilon_{k,a,b}(S - \beta_{k,a,b})^D$, $0 \leq D < e_k$, be the complete original primary basis obtained by the same Taylor inverse as in OMG2. It satisfies

$$e^{wY_k}[1] = \sum_{a,b=0}^k e^{\omega_{k,a,b}w} \sum_{D=0}^{e_k-1} \frac{i^{-D}w^D}{D!} H_{k,a,b,D}, \quad \omega_{k,a,b} = (2b - k)\gamma - i(2a - k)\delta. \quad (\text{OMG17})$$

For $f \in \mathcal{P}_{x,k}$ expand its original function uniquely as $f(x(w)) = \sum c_{a,b,D}(f)w^D e^{\omega_{k,a,b}w}$ using OMG12. Define the complex linear dual map, without a conjugation,

$$\Theta_k(f)(H_{k,a,b,D}) = i^D D! c_{a,b,D}(f). \quad (\text{OMG18})$$

Then OMG17 proves exactly $\Theta_k(f)(e^{wY_k}[1]) = f(x(w))$. The independence and surjectivity in OMG9–13 prove that this map is onto $E_k^\vee$, with kernel $\mathfrak{a}_k$. Consequently it induces a specified vector-space isomorphism

$$\boxed{\Theta_k : R_k \xrightarrow{\sim} E_k^\vee}. \quad (\text{OMG19})$$

All phases and factorials are those of OCS17; for a monomial x^ν its coefficients are precisely the finite multinomial OCS15 rule with the entries OMG4.

The collection of these vector spaces has an exact graded algebra structure coming from the original sum maps. For $k, j \geq 0$ put

$$\Delta_{k,j} : E_{k+j} \longrightarrow E_k \otimes E_j, \quad [P(S)] \longmapsto [P(S_1 + S_2)]. \quad (\text{OMG20})$$

On a primary pair the eigenvalue is the displayed sum $\beta_{k,a,b} + \beta_{j,a',b'} = \beta_{k+j,a+a',b+b'}$. The sum of the two nilpotents has index $e_k + e_j - 1 = e_{k+j}$: its power $e_k + e_j - 2$ has nonzero top coefficient $\binom{e_k+e_j-2}{e_k-1}$ and its next power is zero. Every pair of total sign counts occurs. Thus the full minimal polynomial of $S_1 + S_2$ is exactly χ_{k+j} , proving both well-definedness and injectivity of OMG20. It also gives the complete primary formula

$$\Delta_{k,j} H_{k+j,A,B,D} = \sum_{a+a'=A, b+b'=B} \sum_{\substack{d+d'=D \\ 0 \leq d < e_k, 0 \leq d' < e_j}} \binom{D}{d} H_{k,a,b,d} \otimes H_{j,a',b',d'}. \quad (\text{OMG21})$$

Indeed the total primary idempotent is one precisely on the matching eigenvalue pairs, and the nilpotent power expands by the ordinary binomial theorem there. All other components vanish. Substitution of the sum of three coordinates proves coassociativity; interchange of the two coordinates proves cocommutativity. The $k = 0$ maps give the count. Hence

$$(\ell \star \ell')(v) = (\ell \otimes \ell')(\Delta_{k,j}v) \quad (\ell \in E_k^\vee, \ell' \in E_j^\vee) \quad (\text{OMG22})$$

is an associative commutative graded product with unit in $E_0^\vee$. For the exact dual basis $\lambda_{k,a,b,D} = i^D D! H_{k,a,b,D}^\vee$, OMG21 gives

$$\lambda_{k,a,b,D} \star \lambda_{j,a',b',D'} = \lambda_{k+j,a+a',b+b',D+D'}.$$

The coefficient is exactly $\binom{D+D'}{D} i^D D! i^{D'} D'! = i^{D+D'} (D+D')!$; this records the cancellation before stating the basis product. Equivalently, OMG20 sends $e^{wY_{k+j}}[1]$ to $e^{wY_k}[1] \otimes e^{wY_j}[1]$, with $c_{k+j} = c_k + c_j$. Evaluating OMG18 on this identity proves

$$\Theta : R \xrightarrow{\sim} \bigoplus_{k \geq 0} (E_k^\vee, \star) \text{ is an isomorphism of graded algebras.} \quad (\text{OMG23})$$

Its source multiplication is the actual polynomial multiplication; its target multiplication has been constructed from the original sum coalgebra rather than assigning a multiplication between unrelated finite-dual vector spaces.

13.5 The full semisimple and nilpotent action in this geometry

On $\mathcal{P}_Z$ define the degree-preserving derivation

$$\partial_Z Z_{a,b,d} = \omega_{2a-1,2b-1} Z_{a,b,d} + \begin{cases} dZ_{a,b,d-1}, & d > 0, \\ 0, & d = 0. \end{cases} \quad (\text{OMG24})$$

On the parameter ring OMG7 it is induced by $\partial X_a = -i(2a-1)\delta X_a$, $\partial Y_b = (2b-1)\gamma Y_b$, $\partial T_0 = 0$, $\partial T_1 = T_0$. The product rule verifies OMG24 and commutation with Φ_m ; therefore the complete ideal $\mathfrak{t}_m$ is stable. The corresponding exact flow on the parameters is

$$(X_a, Y_b, T_0, T_1) \mapsto (e^{-i(2a-1)\delta w} X_a, e^{(2b-1)\gamma w} Y_b, T_0, T_1 + wT_0).$$

At $(X_0, X_1, Y_0, Y_1, T_0, T_1) = (1, 1, 1, 1, 0)$ it gives precisely the original functions OMG5. This is the entire stated analytic flow on the explicit algebraic variety, with all polynomial degrees.

If B_Z is the matrix in OMG24 on the column of variables, its rows have diagonal ω_α and entry d in column $(\alpha, d-1)$ of row (α, d) . In actual coordinates the derivation is

$$\partial_x x = \mathcal{A}x, \quad \mathcal{A} = H_m B_Z H_m^{-1}. \quad (\text{OMG25})$$

The inverse substitution proves $\partial_x \mathfrak{a} \subseteq \mathfrak{a}$, and its flow differentiates the original $x(w)$. On the exact degree- k basis of OMG9/OMG18 the induced action is

$$\partial_x \lambda_{k,a,b,D} = \omega_{k,a,b} \lambda_{k,a,b,D} + \begin{cases} D \lambda_{k,a,b,D-1}, & D > 0, \\ 0, & D = 0. \end{cases} \quad \Theta_k \partial_x = Y_k^\vee \Theta_k. \quad (\text{OMG26})$$

The displayed action on the basis uses its identification OMG23. For a direct verification, on the original primal primary basis $Y_k H_D = \omega H_D + i^{-1} H_{D+1}$ with the last term zero at $D = e_k - 1$. Applying $i^D D! H_D^\vee$ proves the exact lowering coefficient D in its dual action. Each frequency block has all e_k terms, with nonzero top iterated derivative $(e_k - 1)!$. Thus every original primary nilpotent has exactly the same index e_k in this model. The full annihilating polynomial is retained with its leading phase:

$$\chi_k(c_k + i\partial_x) = 0 \quad \text{on } R_k, \quad \chi_k(c_k + iT) = i^{q_k} \prod_{a,b=0}^k (T - \omega_{k,a,b})^{e_k}. \quad (\text{OMG27})$$

Distinct frequencies and the nonzero top chain prove that this degree q_k is minimal. The original S action corresponds to $c_k + i\partial_x$ on degree k , or to $\frac{1}{2}\mathbf{E} + i\partial_x$ on the graded algebra, where $\mathbf{E}$ is the degree derivation. These equalities specify every constant, sign and nilpotent action under OMG23.

13.6 The exact original cyclic observation as invariant restriction

Let $D = \text{diag}(\zeta, \dots, \zeta^n)$ and let $G = \mathbb{Z}/N$ act on $\mathcal{P}_x$ by $(gf)(x) = f(D^{-1}x)$. Thus a degree- k monomial x^ν has character $\zeta^{-\sum t\nu_t}$. Write

$$\mathcal{J} = \mathcal{P}_x^G, \quad \mathcal{J} = \text{im}(\mathcal{J} \longrightarrow R). \quad (\text{OMG28})$$

These are graded algebras. The coefficient image of the original observation is exactly their restriction at every degree, as follows. Let f_t be column t of F and let S_ν be the sum of all distinct ordered words in the f_t having multiplicity vector ν . Disjoint word supports make the S_ν a basis of the permutation-invariant tensor space. The expansion and projector are

$$(Fx)^{\otimes k} = \sum_{|\nu|=k} x^\nu S_\nu, \quad P_k(Fx)^{\otimes k} = \sum_{\substack{|\nu|=k \\ \sum t\nu_t \equiv 0 \pmod{N}}} x^\nu S_\nu, \quad P_k = \frac{1}{N} \sum_{a=0}^{N-1} (C^{-a})^{\otimes k}.$$

Every ordered word occurs once; the factor $k!/\prod \nu_t!$ is the number of terms in S_ν , not a missing coefficient. The original sum injection is the iterated map OMG20 into $E_h^{\otimes k}$. Hence the original observation $L_k = P_k \Pi^{\otimes k} U^{\otimes k} \iota_k = j\Lambda_k$ sends $e^{wY_k}[1]$ to the displayed projected curve with $x = x(w)$. The coefficient covectors of its surviving S_ν pull back to exactly $\Theta_k(x^\nu)$. They span the full dual of the image, since every covector on a subspace of a finite dimensional space extends to its ambient space. Consequently

$$\boxed{\begin{array}{l} \Theta_k(\mathcal{J}_k) = \text{im } \Lambda_k^\vee = (\ker \Lambda_k)^\perp, \quad \text{rank } \Lambda_k = \dim \mathcal{J}_k, \\ R_k/\mathcal{J}_k \xrightarrow{\Theta_k|_{K_k}} K_k^\vee \text{ is an isomorphism of vector spaces,} \quad K_k = \ker \Lambda_k. \end{array}} \quad (\text{OMG29})$$

For the last statement, a covector on E_k vanishes on K_k exactly when it factors through $E_k/K_k \simeq \text{im } \Lambda_k$. Restriction to K_k is onto by extension of a basis. This proves the claimed quotient and its precise type: the degree- k quotient is a vector-space quotient by $\mathcal{J}_k$.

13.7 The actual orbit ideal, including every stabilizer

The whole ideal $\mathfrak{a}$ is generated by the explicit quadratic space in OMG15. Let B_2 be its coefficient matrix in the ordered degree-two monomial frame of $\mathcal{P}_x$. Repeated or dependent generator rows are harmless. For $a \in \mathbb{Z}/N$ let $D_{2,a}$ be diagonal with entry $\zeta^{-a \sum t\nu_t}$ at monomial x^ν . The actual stabilizer and its exact finite test are

$$H = \{a : g^a \mathfrak{a} = \mathfrak{a}\} = \left\{ a : \text{rank} \begin{bmatrix} B_2 \\ B_2 D_{2,a} \end{bmatrix} = \text{rank } B_2 \right\}. \quad (\text{OMG30})$$

Equality of these row spaces is equivalent to equality of their quadratically generated ideals; their dimensions agree under the invertible diagonal map. This proves the finite test using the actual entries OMG4. It prescribes no generic coefficient value. The set H is a subgroup by composition and inverse of ideal equalities. Put $s = |H|$ and $d = N/s$. Since G is cyclic, its cosets have representatives $0, \dots, d-1$ and $H = \{0, d, 2d, \dots, (s-1)d\}$. Define all distinct component ideals and the union algebra by

$$\mathfrak{a}_a = g^a \mathfrak{a} \quad (0 \leq a < d), \quad \mathfrak{u} = \bigcap_{a=0}^{d-1} \mathfrak{a}_a, \quad A = \mathcal{P}_x/\mathfrak{u}. \quad (\text{OMG31})$$

These are distinct homogeneous prime ideals with the same Hilbert function OMG13. Their intersection is a homogeneous radical ideal stable under the original cyclic action. It is the complete ideal of their scheme-theoretic union by the definition of the union's ideal. No product formula for this intersection is substituted.

For any invariant polynomial, membership in one component ideal implies membership in all its translates, by applying every g^a . Thus the exact invariant restriction and its inverse description are

$$\boxed{\begin{array}{c} \mathcal{J} \cap \mathfrak{a} = \mathcal{J} \cap \mathfrak{u}, \\ \mathcal{J}/(\mathcal{J} \cap \mathfrak{u}) \xrightarrow{[f] \mapsto [f]_{\mathfrak{u}}} A^G \xrightarrow{\text{res}_0} \mathcal{J} \subseteq R \end{array}} \text{ are isomorphisms of graded algebras.} \quad (\text{OMG32})$$

To prove surjectivity of the first map, average any representative of an invariant class by $N^{-1} \sum g^a$; its class is unchanged. Its kernel is exactly the displayed intersection. For the second, use an invariant representative: a zero restriction lies in $\mathcal{J} \cap \mathfrak{a} = \mathcal{J} \cap \mathfrak{u}$ and hence is already zero in A^G . Its image is $\mathcal{J}$ by definition. This gives the invariant image for every actual H , including nontrivial stabilizers of a composite group order $4m + 1$.

13.8 The complete component conductor and an explicit finite witness

Set $R_a = \mathcal{P}_x/\mathfrak{a}_a$. The simultaneous restriction

$$A \hookrightarrow \tilde{A} := \prod_{a=0}^{d-1} R_a, \quad [f]_{\mathfrak{u}} \mapsto ([f]_{\mathfrak{a}_a})_a \quad (\text{OMG33})$$

is injective precisely because its kernel before quotienting is the intersection OMG31. Its component conductor ideals are

$$\mathfrak{c}_a = \frac{(\bigcap_{b \neq a} \mathfrak{a}_b) + \mathfrak{a}_a}{\mathfrak{a}_a} \subseteq R_a, \quad \bigcap_{b \neq a} \mathfrak{a}_b = \mathcal{P}_x \quad (d = 1). \quad (\text{OMG34})$$

The conductor of this specified inclusion is exactly

$$\boxed{\{t \in \tilde{A} : t\tilde{A} \subseteq A\} = \prod_{a=0}^{d-1} \mathfrak{c}_a \subseteq A.} \quad (\text{OMG35})$$

Indeed a tuple supported solely on component a lies in the image of A if and only if its value belongs to $\mathfrak{c}_a$, by lifting its other zero values to the intersection of the other ideals. If t is in the conductor, multiplication by each component idempotent of $\tilde{A}$ gives such a supported tuple; thus each component of t lies in $\mathfrak{c}_a$. Conversely these are ideals of R_a . Multiplying a tuple with all its components in the indicated ideals by any element of $\tilde{A}$ gives a sum of such supported tuples, each lying in A . This proves both inclusions and the full formula. In particular this conductor is also the annihilator of $\tilde{A}/A$ in A .

The action of G permutes these components with its original coefficient substitutions. Projection of an invariant tuple to component zero gives an isomorphism

$$\tilde{A}^G \xrightarrow{\sim} R^H, \quad r \in R^H \mapsto (g^a r)_{0 \leq a < d} \text{ for its inverse.} \quad (\text{OMG36})$$

The inverse is independent of coset representatives because r is fixed by H , and invariance follows by reindexing the components. The same argument applies to the conductor, whose ideals are permuted. Equations OMG32–36 consequently prove the exact inclusions of graded subalgebras and conductor ideals

$$\boxed{\mathfrak{c}_0^H \subseteq \mathcal{J} \subseteq R^H, \quad \mathfrak{c}_0^H R^H \subseteq \mathcal{J}.} \quad (\text{OMG37})$$

Here $\mathfrak{c}_0^H$ denotes invariant elements of the specified component ideal and is an ideal of R^H . An invariant tuple in the conductor already lies in A , so its projection belongs to $\mathcal{J}$ by OMG32. This proves the first inclusion; the ideal property gives the second.

There is a nonzero homogeneous element of this conductor with a completely bounded degree. For every $a \neq 0$ choose from the explicit transformed quadratic list of $\mathfrak{a}_a$ a quadratic $f_a \notin \mathfrak{a}_0$. Such a choice exists: containment of the entire quadratic space in that of $\mathfrak{a}_0$, which has the same dimension, would make the spaces equal and their generated ideals equal, contrary to distinctness. Membership is the finite row-space test in the same degree-two monomial frame. The product $f = \prod_{a=1}^{d-1} f_a$ lies in all other component ideals. It is nonzero modulo $\mathfrak{a}_0$ because that ideal is prime and none of its factors belongs to it. Thus $\bar{f} \in \mathfrak{c}_0$ is nonzero of degree $2(d-1)$. The group H preserves $\mathfrak{c}_0$ and acts on the domain R . Its exact norm

$$c_H = \prod_{h \in H} g^h \bar{f} \in \mathfrak{c}_0^H, \quad \deg c_H = D_H = 2s(d-1) = 2(N-s), \quad c_H \neq 0 \quad (\text{OMG38})$$

is invariant by permutation of its factors and nonzero in the domain. For $d=1$ take the empty product $c_H = 1$, $D_H = 0$; all statements remain exact. Multiplication by this actual conductor element is injective and gives the finite dimension bounds, with negative graded pieces set to zero,

$$\begin{aligned} (R^H)_{k-D_H} &\xrightarrow{\times c_H} \mathcal{J}_k \hookrightarrow (R^H)_k, \\ \dim(R^H)_{k-D_H} &\leq \text{rank } \Lambda_k \leq \dim(R^H)_k, \\ q_k - \dim(R^H)_k &\leq \dim K_k \leq q_k - \dim(R^H)_{k-D_H}. \end{aligned} \quad (\text{OMG39})$$

Every ideal, group and matrix in these expressions is defined by the actual original coefficient map. In particular the stabilizer has its finite equality test OMG30 throughout; no particular value of it has been inserted into the dimensions. To compute the displayed invariant dimensions directly, let the actual H substitutions act on the basis OMG9 through $Z \mapsto H_m^{-1} D^{-h} H_m Z$ and reduce with OMG10. This is an explicit finite matrix on q_k coordinates. The idempotent average $s^{-1} \sum_{h \in H} g^h$ has trace equal to its rank, proving $\dim(R^H)_k = s^{-1} \sum_{h \in H} \text{Tr}(g^h | R_k)$. Thus the dimensions in OMG39 also have complete original finite coefficient formulas for every actual stabilizer.

13.9 The exact action defect and all subsequently observed directions

The derivation ∂_x of OMG25 acts on R with all primary directions OMG26. The possible failure of its restriction to the observed image is the specified finite linear map

$$\mathcal{J}_k \longrightarrow R_k / \mathcal{J}_k, \quad j \longmapsto [\partial_x j]. \quad (\text{OMG40})$$

Via OMG29 it is exactly the complex linear dual of the original map $\Lambda_k Y_k I_K : K_k \rightarrow B_k$. Indeed a j corresponds to the covector $\lambda \Lambda_k$ on E_k for a unique boundary covector λ ; evaluating OMG26 on $I_K v$ gives $\lambda \Lambda_k Y_k I_K v$, proving the identification and all domains and codomains. The action on ambient polynomials also has the exact conjugation formula

$$g^a \partial_x g^{-a} = \partial_{D^a A D^{-a}}, \quad g^a \partial_x g^{-a} - \partial_x = \partial_{D^a A D^{-a} - A}. \quad (\text{OMG41})$$

For a linear vector field $\partial_M f = (Mx) \cdot \nabla f$, the chain rule proves this by substituting $D^{-a}x$ after differentiating $f(D^a x)$. This formula retains every original period/unit coefficient in $\mathcal{A}$ and does not require the original ideal or the observed image to be preserved by extra ambient actions.

Finally the subspace invisible to all original iterates has the complete model

$$\begin{aligned} \mathcal{K}_{\infty, k} &= \bigcap_{r=0}^{q_k-1} \ker(\Lambda_k Y_k^r), \\ R_k / \sum_{r=0}^{q_k-1} \partial_x^r \mathcal{J}_k &\xrightarrow{\Theta_k | \mathcal{K}_{\infty, k}} \mathcal{K}_{\infty, k}^\vee \quad \text{is a vector-space isomorphism.} \end{aligned} \quad (\text{OMG42})$$

In finite dimensional duality the annihilator of an intersection of kernels is the sum of the images of their dual maps: this follows by taking the dual of the single stacked map with all those components, whose kernel is precisely the intersection. Equations OMG26 and OMG29 identify these images with $\partial_x^r \mathcal{J}_k$, proving OMG42. The exact original polynomial OMG27 expresses every higher iterate as a linear combination of the first q_k , so the same intersection is taken over all nonnegative iterates and is Y_k -invariant. This proves the full nilpotent and cyclic-action receiver in the new homogeneous model for every original multiplicity, with its actual observation, periods and amplitude jets throughout.

14 The absolute common kernel: the new finite result

This complete incoming calculation sharpens the current absolute common kernel estimate. Its original comparison was against the preceding logarithmic bound; the intervening KAF chapter already established the $O(kq)$ scale. The new finite envelope reduces its uniform limsup constant from $16(4 + 10 \log 2)$ to $32 \log(25 \log 3 / (4\pi))$. AKJ below takes the pointwise minimum of both finite bounds. No limiting kernel coefficient is inferred from these upper bounds.

Retain the actual hypothetical complete quartet and its entire quotient

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(z) = \prod_{\alpha \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (z - \alpha)^m, \quad v_h = 2\xi/h.$$

The full primary unit is $U = M_{j_h v_h}$. The original tensor order, centre, polynomial, and coefficient quotient are

$$\begin{aligned} k &= 4l + 1, \quad l \geq 1, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad c = k/2, \\ \lambda_{a,b} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{a,b})^e, \quad E = \mathbb{C}[S]/(\chi). \end{aligned} \quad (\text{AKS1})$$

For the principal conclusion $m = 1$. The source-operator estimates below also hold with the stated repeated-root list for $m > 1$, but no simple-quartet rank is assigned to those multiplicities.

Let $\Lambda_k(u) : E \rightarrow B_k(u)$ be the original observation, $K(u) = \ker \Lambda_k(u)$, and $I : K(u) \hookrightarrow E$ its original coefficient inclusion. For a source $s \in \{1, k\}$, let $G_N^{(s)}$ be the original minimum-remainder Gram, in the frame $(1, S, \dots, S^{q-1})$. Define $H_{K,N}^{(s)} = I^* G_N^{(s)} I$ and retain exactly

$$F_K^{(s)} = \log \det H_{K,q-1}^{(s)} + \log \det H_{K,q}^{(s)} - \log \det H_{K,2q-1}^{(s)} - \log \det H_{K,2q}^{(s)}. \quad (\text{AKS2})$$

Every empty determinant is one. All conjugate transposes refer to the specified coordinate matrices, not to a bilinear residue pairing.

Here are entirely explicit positive numbers used in the bound:

$$\begin{aligned} R &= k\sqrt{\delta^2 + \gamma^2}, \quad b_s = s/2, \quad t_q = \frac{\pi}{2} - \frac{1}{q}, \quad \mathbf{a}_q = \frac{2 \log 3}{t_q}, \quad \mathbf{s}_q = \frac{\mathbf{a}_q^q - 1}{\mathbf{a}_q - 1}, \\ \mathbf{b}_{s,q} &= \left[\frac{9}{25} \sin \frac{1}{q} \right]^{-b_s} \exp \left(2R \left(\log 3 + \frac{t_q}{2} \right) \right), \\ \mathcal{Z}_{s,j} &= 1 + \mathbf{b}_{s,q} \mathbf{s}_q^2 \sum_{d=q}^{q+j-1} \frac{d!}{(b_s)_d} \left(\frac{25}{16} \right)^d, \quad \mathcal{L}_{s,j} = \log \mathcal{Z}_{s,j} \quad (0 \leq j \leq q + 1). \end{aligned} \quad (\text{AKS3})$$

The empty sum gives $\mathcal{Z}_{s,0} = 1$. In the admitted family $q \geq 36$, so $t_q \in (0, \pi/2)$ and $\mathbf{a}_q > 1$. These parameters are radii in the existing Laplace and generating functions; neither is a new source order.

Main theorem (absolute finite bound). For the original simple quartet, every original logarithm branch, and every $|u| \geq R_*$, where R_* is precisely RPC21 or RPC23 (recorded in Section 20), put $r = 8k - 16$. Then

$$0 \leq F_K^{(s)}(k, u) \leq r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}), \quad s \in \{1, k\}. \quad (\text{AKS4})$$

There is a stronger, fully period-dependent interval in AKS28 below. In particular, with

$$\kappa_* = 32 \log \left(\frac{25 \log 3}{4\pi} \right), \quad 25.02078097649 < \kappa_* < 25.02078097650, \quad (\text{AKS5})$$

one has the uniform statement

$$\boxed{0 \leq \limsup_{\substack{k \rightarrow \infty \\ k \equiv 1 \pmod{4}}} \sup_{\substack{|u| \geq R_* \\ \text{original branches}}} \frac{F_K^{(s)}(k, u)}{kq} \leq \kappa_*}. \quad (\text{AKS6})$$

Here the source may be $s = 1$ for every k , or $s = k$. The same bound holds for the actual arithmetic kernel. Its exact finite allowances are in AKS34–36. Thus the absolute kernel is $O_h(kq)$ on this domain, not only $O_h(kq \log q)$. Neither AKS5 nor AKS6 calls κ_* the actual leading coefficient.

15 An exact coefficient realization of the actual lost directions

15.1 The period matrix and invariant restriction, with their full coefficients

For $m = 1$, order the one-factor roots as $(++)$, $(+-)$, $(-+)$, $(--)$, and let $\mathcal{E}_{\text{prim}}$ have their original Lagrange idempotents as columns. Set $\zeta = \exp(2\pi i/5)$ and $F_{jt} = \zeta^{jt} - 1$, $1 \leq j, t \leq 4$. The exact inverse is $(F^{-1})_{tj} = \zeta^{-tj}/5$. Retain

$$\begin{aligned} \Pi[P]_j &= \int_{\Gamma_j} e^{\Phi_h(z)/u} \text{rem}_h P(z) dz, & \Phi'_h &= h, & \Phi_h(0) &= 0, \\ \Gamma_j &= -\ell_0 + \ell_j, & \arg \ell_j &= (\arg u + \pi + 2\pi j)/5, \\ \mathbf{H} &= F^{-1} \Pi U \mathcal{E}_{\text{prim}}, & D &= \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4), \\ Q(x) &= Q_0(\mathbf{H}^{-1}x), & Q_0(t) &= t_1 t_4 - t_2 t_3, & Q_a(x) &= Q(D^{-a}x). \end{aligned} \quad (\text{AKS7})$$

The integral always uses the degree-less-than-four representative. Invertibility of Π , all contour phases, and its full determinant are the PDE/PQO results in the supplied proofs. The four unit values are $(a, \bar{a}, \bar{a}, a)$, with the actual $a = v_h(\rho) \neq 0$; no value of a is selected. Hence $\mathbf{H}$ is the original invertible matrix.

The following formulas make the coefficient realization directly computable. Write

$$\mathcal{I}_{k,0} = \{\nu \in \mathbb{Z}_{\geq 0}^4 : |\nu| = k, \nu_1 + 2\nu_2 + 3\nu_3 + 4\nu_4 \equiv 0 \pmod{5}\}.$$

For $\nu \in \mathcal{I}_{k,0}$ define $B_{\nu;(a,b)}$ by summing over 4×4 arrays $n_{t,\alpha} \geq 0$ with row sums ν_t , total plus-real count a , and total plus-imaginary count b :

$$B_{\nu;(a,b)} = \sum_{\substack{\sum_{\alpha} n_{t,\alpha} = \nu_t \\ \sum_{t,\alpha: \epsilon_{\alpha} = +} n_{t,\alpha} = a \\ \sum_{t,\alpha: \eta_{\alpha} = +} n_{t,\alpha} = b}} \left(\prod_{t=1}^4 \frac{\nu_t!}{\prod_{\alpha} n_{t,\alpha}!} \right) \prod_{t,\alpha} \mathbf{H}_{t,\alpha}^{n_{t,\alpha}}. \quad (\text{AKS8})$$

Let $\mathbf{V}_{(a,b),d} = \lambda_{a,b}^d$ for $0 \leq d < q$ and set $\mathbf{A} = \mathbf{B}\mathbf{V}$. All entries of $\mathbf{A}$ are now finite expressions in the actual periods and unit. There is no generic period matrix in this definition.

For completeness, put $t_{\alpha}(w) = \exp((\eta_{\alpha}\gamma - i\epsilon_{\alpha}\delta)w)$ and $x(w) = \mathbf{H}t(w)$. Expansion gives

$$x(w)^{\nu} = \sum_{a,b} B_{\nu;(a,b)} e^{\omega_{a,b} w}, \quad \omega_{a,b} = (2b - k)\gamma - i(2a - k)\delta.$$

In the original primary algebra, $e^{wY}[1] = \sum_{a,b} e^{\omega_{a,b} w} \varepsilon_{a,b}$, where $Y = (M_S - cI)/i$. The exponentials are independent by the Vandermonde of the distinct $\omega_{a,b}$. OCS9's unscaled symmetric word basis therefore identifies the original observation with

$$j\Lambda_k = \mathcal{F}_0 \mathbf{A}, \quad \ker \mathbf{A} = K(u) \quad \text{in the original coefficient frame.} \quad (\text{AKS9})$$

Every multinomial in AKS8 belongs to its actual ordered words; the word multiplicity is not inserted a second time. The exact inherited row Gram of $\mathcal{F}_0$ is the restriction to $\mathcal{I}_{k,0}$ of

$$\mathcal{G}_{\nu,\mu} = \sum_{\substack{n_{tu} \geq 0 \\ \sum_u n_{tu} = \nu_t \\ \sum_t n_{tu} = \mu_u}} \frac{k!}{\prod_{t,u} n_{tu}!} \prod_{t,u} [5(\delta_{tu} + 1)]^{n_{tu}}. \quad (\text{AKS10})$$

This follows by taking the coefficient of $\bar{x}^\nu y^\mu$ in $(\bar{x}^T F^* F y)^k$, since $F^* F = 5(I + \mathbf{1}\mathbf{1}^*)$. It is positive definite and includes every row cross term.

Here is the precise role of the orbit ideal. The map sending a homogeneous polynomial f to the covector specified by $\Theta_k(f)(e^{wY}[1]) = f(x(w))$ has kernel $Q\mathbb{C}[x]_{k-2}$ and is onto $E^\vee$. Indeed equal exponential indices correspond exactly to monomial differences divisible by $t_1 t_4 - t_2 t_3$; the displayed Vandermonde proves that there are no other relations. Its invariant image is $\text{im } \Lambda_k^\vee$. On the five-member domain, the five Q_a are distinct prime quadrics. An invariant polynomial divisible by Q is divisible by each Q_a , and therefore by their full product; its quotient is invariant by cancellation in the polynomial ring. Consequently

$$\mathbb{C}[x]^{(D)} \cap (Q) = \left(\prod_{a=0}^4 Q_a \right) \mathbb{C}[x]^{(D)}, \quad (\text{AKS11})$$

$$K(u)^\vee \simeq \mathbb{C}[x]_k / (\mathbb{C}[x]_k^{(D)} + Q\mathbb{C}[x]_{k-2}),$$

$$b := \text{rank } \mathbf{A} = d_{5,k,0} - d_{5,k-10,0} = k^2 - 6k + 17, \quad r = q - b = 8k - 16.$$

Negative degrees are zero. The count is $d_{5,d,0} = ((\binom{d+3}{3} + 4\epsilon_d)/5$, $\epsilon_d = \mathbf{1}_{d \equiv 0(5)} - \mathbf{1}_{d \equiv 1(5)}$; subtracting the cubics gives the stated expression for $k \geq 10$, and $k = 5, 9$ give 12, 44 directly. This is the inherited OQX calculation, repeated to specify which kernel AKS9 realizes. The source metric is still to be applied to this kernel, not to the invariant ring itself.

15.2 A determinant formula requiring no assumed nonzero pivot

Let $\mathcal{G}_0$ be the full positive row Gram in AKS10 and set

$$W = \mathbf{A}^* \mathcal{G}_0 \mathbf{A}, \quad \sigma_j = [z^j] \det(I_q + zW), \quad P_K = \frac{1}{\sigma_b} \sum_{j=0}^b (-1)^j \sigma_{b-j} W^j. \quad (\text{AKS12})$$

Since W has exactly b positive eigenvalues, $\sigma_b > 0$. The scalar polynomial in AKS12 is one at zero and zero at every nonzero eigenvalue: it is the normalized product of the factors $\lambda_i - z$ for the b positive eigenvalues. Hence $P_K = P_K^* = P_K^2$ and $\text{im } P_K = K(u)$. This is the projector derived from the literal observation, not a freely chosen projection. Replacing $\mathcal{G}_0$ by any other positive coordinate Gram leaves this same coefficient-orthogonal kernel projector, since its image and orthogonal complement are unchanged. No such replacement is needed here.

At every original endpoint define

$$D_{K,N}^{(s)} = [z^r] \det(I_q + z P_K G_N^{(s)} P_K). \quad (\text{AKS13})$$

These numbers are strictly positive for $r > 0$ and are one for $r = 0$. To prove their exact metric meaning, choose an orthonormal coefficient basis U_K of $\text{im } P_K$ just for the proof. In a unitary extension, $P_K G_N P_K$ has its positive block $U_K^* G_N U_K$ and a zero block. Thus $D_{K,N} = \det(U_K^* G_N U_K)$. Write the original inclusion as $I = U_K T$; then $|\det T|^2 = \det(I^* I)$ and

$$\det H_{K,N}^{(s)} = \det(I^* I) D_{K,N}^{(s)}, \quad \boxed{F_K^{(s)} = \log \frac{D_{K,q-1}^{(s)} D_{K,q}^{(s)}}{D_{K,2q-1}^{(s)} D_{K,2q}^{(s)}}}. \quad (\text{AKS14})$$

This supplies an equivalent exact coefficient determinant formula for all the $8k - 16$ lost directions, including the entire period dependence. The original frame factor is present at each endpoint and cancels only with its four actual signs. It is unnecessary to suppose any particular minor nonzero. Alternatively, selecting the first independent r columns of P_K gives an explicit coefficient frame; the determinant formula avoids even that chart choice.

For $m > 1$ replace AKS8–9 by OCS12–18’s full confluent matrix $\mathfrak{A}_{\nu;(a,b,D)} = i^D D! c_{\nu;a,b,D}$ and its primary isomorphism. AKS12–14 are then identical with the actual rank, not with the rank in AKS11. All primary nilpotents and all unit derivatives remain in that matrix.

16 A bounded exact remainder operator on the original source

16.1 The complete source norms and coefficient map

For either existing source set $M_s = (2\pi)^{s/2}$ and retain

$$dm_{0,s}(y) = \frac{M_s 2^{b_s}}{4\pi\Gamma(b_s)} |\Gamma(b_s/2 + iy/2)|^2 dy, \quad \int e^{ty} dm_{0,s} = M_s (\cos t)^{-b_s}, \quad |t| < \pi/2. \quad (\text{AKS15})$$

The supplied IGO1–3 proof derives this transform by the beta integral, Fourier inversion and a contour shift within $|\operatorname{Im} x| < \pi/2$. In particular it includes the exact masses $M_1 = \sqrt{2\pi}$ and $M_k = (2\pi)^{k/2}$ and exponential domination on every smaller strip. The original real monic polynomials satisfy

$$\begin{aligned} p_0 &= 1, \quad p_1 = y, \quad p_{d+1} = yp_d - d(b_s + d - 1)p_{d-1}, \\ \sum_{d \geq 0} \frac{p_d(y)}{d!} z^d &= (1 + z^2)^{-b_s/2} e^{y \arctan z}, \quad h_d^{(s)} = \|p_d\|^2 = M_s d! (b_s)_d. \end{aligned} \quad (\text{AKS16})$$

The generating formula follows by differentiating in z . To check the normalization directly, integrate the product of two generating functions at small real z, w . AKS15 and the cosine addition identity give $M_s(1 - zw)^{-b_s}$. Taking any finite pair of derivatives at zero, justified by the stated exponential domination, gives orthogonality and the norms in AKS16. Thus the leading phase in $u_d(S) = p_d((S - c)/i)/\sqrt{h_d^{(s)}}$ is exactly i^{-d} . The same generating function agrees with NIST DLMF 18.23.7 after the original parameter rescaling; it is derived here from the supplied recurrence, not substituted for it.

List the original roots $\omega_\nu = (\lambda_{a,b} - c)/i$ in lexicographic order, repeating each exactly e times, and put

$$P(y) = i^{-q} \chi(c + iy) = \prod_{\nu=1}^q (y - \omega_\nu), \quad \Phi : [f(y)] \mapsto [f((S - c)/i)]. \quad (\text{AKS17})$$

The inverse is substitution $S = c + iy$, and $P((S - c)/i) = i^{-q} \chi(S)$. This is an exact algebra isomorphism, including the complete confluent orders; there is no monomial comparison divisor. Let C_s have columns $[u_0], \dots, [u_{q-1}]$. Its nonzero determinant is $i^{-q(q-1)/2} \prod_{d < q} (h_d^{(s)})^{-1/2}$. Define

$$v_j = C_s^{-1} [u_{q+j}], \quad \mathbf{K}_j = I_q + \sum_{a=0}^{j-1} v_a v_a^*, \quad 0 \leq j \leq q + 1. \quad (\text{AKS18})$$

Parseval in the low-degree original orthonormal polynomial basis proves

$$\|v_j\|_2^2 = \frac{\|\operatorname{rem}_P p_{q+j}\|_{m_{0,s}}^2}{M_s (q+j)! (b_s)_{q+j}}. \quad (\text{AKS19})$$

This is the full norm of the actual remainder, not a diagonal entry of an approximated quotient Gram.

16.2 Newton interpolation with all roots and their orders

For $0 \leq a < q$ define $N_a(y) = \prod_{\nu=1}^a (y - \omega_\nu)$, with $N_0 = 1$. For every complex v , the unique entire-function remainder is

$$\text{rem}_P(e^{vy}) = \sum_{a=0}^{q-1} [\omega_1, \dots, \omega_{a+1}] e^{v \cdot} N_a(y), \quad |[\omega_1, \dots, \omega_{a+1}] e^{v \cdot}| \leq \frac{|v|^a}{a!} e^{R|v|}. \quad (\text{AKS20})$$

Here divided differences at repeated nodes mean Hermite divided differences, not evaluation with a zero denominator.

A proof covering the confluent case is useful. On the standard a -simplex with barycentric coordinates $\theta_1, \dots, \theta_{a+1}$,

$$[\omega_1, \dots, \omega_{a+1}] e^{v \cdot} = v^a \int_{\Delta_a} e^{v \sum \theta_\nu \omega_\nu} d\theta.$$

The integral of $\prod \theta_\nu^{n_\nu}$ is $\prod n_\nu! / (a + \sum n_\nu)!$, as follows by iterated integration of integer beta integrals. Expanding the exponential therefore gives $\sum_{n \geq 0} v^{a+n} h_n(\omega_1, \dots, \omega_{a+1}) / (a+n)!$. For monomials, the divided difference of y^{a+n} is exactly h_n : this follows by multiplying the geometric series $\prod (1 - \omega_\nu z)^{-1}$ and the Newton division recurrence. It is a polynomial identity in the nodes and so also holds at repeated nodes. This proves the integral formula. Iterated Newton division for polynomials, followed by the convergent exponential series, proves the first equality in AKS20, including every required derivative. Finally the simplex has volume $1/a!$ and $|\sum \theta_\nu \omega_\nu| \leq R$, proving its stated bound.

For any $0 < t < \pi/2$, put $\tau = 2/t$. Positivity of every term of the cosh series in AKS15 gives

$$\|y^a\|_{m_{0,s}} \leq \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a!.$$

Indeed its square is at most $M_s (\cos t)^{-b_s} (2a)! t^{-2a}$ and $(2a)! \leq 4^a (a!)^2$. Expanding the *whole* Newton polynomial, then using $|e_j(\omega_1, \dots, \omega_a)| \leq \binom{a}{j} R^j$, proves

$$\begin{aligned} \|N_a\|_{m_{0,s}} &\leq \sqrt{M_s} (\cos t)^{-b_s/2} \sum_{j=0}^a \binom{a}{j} R^{a-j} \tau^j j! \\ &= \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a! \sum_{n=0}^a \frac{(R/\tau)^n}{n!} \\ &\leq \sqrt{M_s} (\cos t)^{-b_s/2} \tau^a a! e^{R/\tau}. \end{aligned} \quad (\text{AKS21})$$

There are no omitted polynomial coefficients. Combining AKS20–21 gives for every complex v

$$\|\text{rem}_P(e^{vy})\|_{m_{0,s}} \leq \sqrt{M_s} (\cos t)^{-b_s/2} e^{R(|v|+t/2)} \sum_{a=0}^{q-1} (2|v|/t)^a. \quad (\text{AKS22})$$

16.3 Cauchy extraction in the actual Gamma generating function

Fix $0 < z < 1$ and write $a_z = \text{arctanh } z$. For $|w| = z$ the defining power series gives $|\arctan w| \leq a_z$ and $|1 + w^2| \geq 1 - z^2$. Apply the literal finite-dimensional remainder map to AKS16 and use Cauchy's formula for its d th coefficient. The norm triangle inequality on that circle and AKS22 give

$$\begin{aligned} \frac{\|\text{rem}_P p_d\|_{m_{0,s}}}{d!} &\leq z^{-d} (1 - z^2)^{-b_s/2} \sqrt{M_s} (\cos t)^{-b_s/2} \\ &\quad \times e^{R(a_z+t/2)} \sum_{a=0}^{q-1} (2a_z/t)^a. \end{aligned} \quad (\text{AKS23})$$

Cauchy's formula is legitimate for the finite remainder vector: its entries are holomorphic for $|w| < 1$ by the actual matrix generating identity OPG7. The source norm is taken only after this finite-vector identity; there is no unjustified integral of a nonintegrable generating function on the large circle.

Substitution of AKS23 into AKS19 retains the factorial denominator and cancels the two occurrences of the original mass, giving

$$\|v_j\|_2^2 \leq \frac{d!}{(b_s)_d} z^{-2d} [\cos t(1 - z^2)]^{-b_s} e^{2R(az+t/2)} \left[\sum_{a=0}^{q-1} (2a_z/t)^a \right]^2, \quad d = q + j. \quad (\text{AKS24})$$

Adding the *complete* rank-one positive matrices in AKS18 now proves $K_j \preceq \mathcal{Z}_{s,j} I_q$ after setting $z = 4/5$ and $t = t_q$. Indeed $a_{4/5} = \log 3$, $1 - z^2 = 9/25$, and $\cos t_q = \sin(1/q)$, so the exact scalar obtained is AKS3. The high index $j = q + 1$ includes the original polynomial of degree $2q$.

There is also a root-by-root finite refinement. Put $\mu_{s,d} = \int y^{2d} dm_{0,s}$, $R_a = \max_{\nu \leq a+1} |\omega_\nu|$, and

$$B_{s,a}^{\text{root}} = \sum_{d=0}^a |e_{a-d}(\omega_1, \dots, \omega_a)| \sqrt{\mu_{s,d}}, \quad (\text{AKS25})$$

$$J_s^{\text{root}}(z) = \frac{(1 - z^2)^{-b_s}}{M_s} \left[\sum_{a=0}^{q-1} \frac{a_z^a}{a!} e^{a_z R_a} B_{s,a}^{\text{root}} \right]^2.$$

Replace the scalar multiplying $d!z^{-2d}/(b_s)_d$ in AKS24 by $J_s^{\text{root}}(z)$ to get a valid sharper bound, by the same proof before AKS21 is relaxed. Every moment is exactly $M_s(2d)! [w^{2d}] (\cos w)^{-b_s}$. Thus this refinement keeps every actual root coefficient and a finite original moment formula. AKS21 bounds it by the explicit scalar in AKS24. The proof never replaces the source determinant by a product of diagonal moments: AKS24 is a full-vector operator bound, and the next step restricts the inverse of the full covariance.

17 The four original kernel Grams and the evaluated scale

The original orthonormal source map at degree $q + j - 1$ is $C_s[I_q, v_0, \dots, v_{j-1}]$. Its minimum lift of $C_s x$ is $[I_q, v_0, \dots, v_{j-1}]^* K_j^{-1} x$. It has that image and is orthogonal to the full relation kernel, so Pythagoras proves

$$G_{q+j-1}^{(s)} = C_s^{-*} K_j^{-1} C_s^{-1}, \quad H_j := H_{K,q+j-1}^{(s)} = T_s^* K_j^{-1} T_s, \quad T_s = C_s^{-1} I. \quad (\text{AKS26})$$

This verifies the source-minimum step in the coefficient determinant AKS13–14. Neither a cross block nor any original relation is removed.

Since $I_q \preceq K_j \preceq \mathcal{Z}_{s,j} I_q$, inversion and compression give $\mathcal{Z}_{s,j}^{-1} H_0 \preceq H_j \preceq H_0$, where $H_0 = T_s^* T_s$. Define $d_j = -\log \det(H_0^{-1/2} H_j H_0^{-1/2})$. Congruence and all r eigenvalues give

$$0 = d_0 \leq d_1 \leq \dots \leq d_{q+1}, \quad 0 \leq d_j \leq r \mathcal{L}_{s,j}, \quad F_K^{(s)} = d_q + d_{q+1} - d_1. \quad (\text{AKS27})$$

The identity uses precisely the four endpoints in AKS2. In particular it proves AKS4, as well as the sharper interval

$$d_1 \leq F_K^{(s)} \leq r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1. \quad (\text{AKS28})$$

For a directly calculable expression for its lower endpoint set $V = \|v_0\|^2$ and $\theta = v_0^* T_s (T_s^* T_s)^{-1} T_s^* v_0$. Then

$$0 \leq \theta \leq V, \quad d_1 = \log \frac{1 + V}{1 + V - \theta}. \quad (\text{AKS29})$$

Indeed the inverse of $I + v_0 v_0^*$ is $I - v_0 v_0^*/(1 + V)$; take its full compressed determinant. For $r = 0$ set $\theta = d_1 = 0$. In GMB/OPR notation $V = t_0$ and $V - \theta = \xi_0$, so this is exactly the first original section-corrected kernel increment, not an independently chosen kernel observation.

Here is an evaluated finite remainder for AKS6. Put

$$\begin{aligned} a_\infty &= \frac{4 \log 3}{\pi}, & c_0 &= \log \frac{25 \log 3}{4\pi} > 0, \\ E_{s,q}(R) &= \log 2 + \frac{4}{\pi - 2/q} + \frac{s}{2} \log \frac{25\pi q}{18} + 2R \left(\log 3 + \frac{\pi}{4} \right) \\ &\quad + \log((q+1)(4q+1)) - 2 \log(a_\infty - 1). \end{aligned} \tag{AKS30}$$

Then for both source orders, every $q \geq 36$, and the same actual kernel,

$$\boxed{\mathcal{L}_{s,q}, \mathcal{L}_{s,q+1} \leq 2c_0 q + E_{s,q}(R), \quad 0 \leq F_K^{(s)} \leq 4rc_0 q + 2rE_{s,q}(R).} \tag{AKS31}$$

To verify every factor, $b_s \geq 1/2$ gives

$$\frac{d!}{(b_s)_d} \leq \frac{4^d}{\binom{2d}{d}} \leq 2d + 1.$$

The last inequality uses that the central binomial is the largest of the $2d + 1$ positive coefficients whose sum is 4^d . Thus the sum in AKS3 is at most $(q+1)(4q+1)(5/4)^{4q}$. Also $\mathfrak{s}_q \leq \mathfrak{a}_q^q/(a_\infty - 1)$ and

$$2q \log \mathfrak{a}_q \leq 2q \log a_\infty + \frac{4}{\pi - 2/q}, \quad \sin(1/q) \geq \frac{2}{\pi q}.$$

The first follows from $-\log(1-x) \leq x/(1-x)$ at $x = 2/(\pi q)$; the second is the chord bound for sine on $[0, \pi/2]$. Use $t_q \leq \pi/2$ in the remaining exponential. The order- q terms combine to $4q \log(5/4) + 2q \log a_\infty = 2c_0 q$. The upper bound for the logarithm of the added positive scalar is positive, so $\log(1 + e^x) \leq \log 2 + \max(0, x)$ proves the first bound in AKS31. AKS28 proves the second.

On the original simple family $q = (k+1)^2$, $R = O_h(k)$ and $r = 8k - 16$. Consequently $E_{1,q}(R) = O_h(k + \log q)$ and $E_{k,q}(R) = O_h(k \log q)$. Dividing AKS31 by kq proves AKS6, with $32c_0 = \kappa_*$. In fact the fixed-source upper bound is

$$F_K^{(1)} \leq (32k - 64)c_0 q + (16k - 32)E_{1,q}(R), \tag{AKS32}$$

whose explicit second term is $O_h(k^2)$. The tensor-source version has an $O_h(k^2 \log q)$ upper remainder. These are remainders of upper bounds, not assertions about the actual signed error or leading value. The proof is uniform in all the period parameters in the stated domain: no norm or condition number of a period matrix occurs in AKS30.

The precise earlier loss. OKV15 had already retained $\|v_j\|^2 \leq q^2 2^{6q} (2D_* q)^{2d} / (d!(b_s)_d)$, $d = q + j$. Replacing its denominator by $1/2$ caused the old $q \log q$ exponent. Even without AKS20–24 one obtains the valid repair

$$\|K_{q+1}\| \leq 1 + (q+1)q^2 2^{6q} \frac{(4D_* q)^{4q}}{(4q)!}, \quad D_* \geq 3. \tag{AKS33}$$

Here $d!(b_s)_d \geq (2d)!/4^d$ and the remaining factorial ratio increases up to $d = 2q$ for $D_* \geq 3$. The elementary inequality $\log n! \geq n \log n - n + 1$ already makes the logarithm of AKS33 $O_h(q)$. The Newton calculation strengthens this repair to the packet-independent limsup constant AKS5, while retaining R in its finite remainder.

Exact enclosure of the displayed numerical ceiling. For $x > 1$ and $z = (x - 1)/(x + 1)$ the integrated geometric series gives

$$0 \leq \log x - 2 \sum_{j=0}^{39} \frac{z^{2j+1}}{2j+1} \leq \frac{2z^{81}}{81(1-z^2)}.$$

For $0 < x < 1$, the alternating sum through $j = 39$ for $\arctan x$ has error between zero and $x^{81}/81$. The tangent addition formula, with the angle in $(0, \pi/2)$, proves $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$: the tangent of $4 \arctan(1/5) - \arctan(1/239)$ is exactly one. Use the resulting rational lower and upper bounds for π and for $\log 3$, then apply the first displayed bound to the two endpoints of $25 \log 3/(4\pi)$. Multiplication by 32 gives, with outward decimal rounding, the interval

$$\kappa_* \in [25.0207809764994871, 25.0207809764994872].$$

The accompanying rational-arithmetic implementation checks these inequalities by integer cross multiplication, rather than a floating-point logarithm. This proves the less finely rounded strict interval AKS5.

18 The complete arithmetic and mixed receivers

18.1 The unchanged finite arithmetic allowances

Use exactly AMT4's constants. To prevent a loss hidden in order notation, write them here:

$$\begin{aligned} \alpha &= \pi/2, \quad p = 8m - 9/2, \quad B_h = 42 + 8m, \quad M = 21 + 4m, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy, \quad M_\alpha = \int e^{\alpha y} w_h(y) dy, \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \\ c_\Gamma &= \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_k &= \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1}, \\ D_N &= [1 + 4(N + M)^2]^M, \quad L_N^{\text{ar}} = \log(A_k D_N / a_k), \\ E_k^+ &= L_{q-1}^{\text{ar}} + L_q^{\text{ar}}, \quad E_k^- = L_{2q-1}^{\text{ar}} + L_{2q}^{\text{ar}}. \end{aligned} \tag{AKS34}$$

The constants c_h, C_h^{tilt} are the original TW/BT envelope constants, not new choices. Their actual inequalities and complete source construction are preserved in AMT4–6 and the current source closure. Minimize AMT6 on the same remainder fibre and then restrict to $I(K)$. AMT9–20 gives, with the original arithmetic source w_h^{*k} ,

$$-rE_k^+ \leq F_K^{\text{ar}} - F_K^{(1)} \leq rE_k^-.$$

Let $\underline{U}_s = d_1^{(s)}$ and $\overline{U}_s = r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1^{(s)}$. The new finite conclusion is

$$\boxed{\max\{0, \underline{U}_1 - rE_k^+\} \leq F_K^{\text{ar}} \leq \overline{U}_1 + rE_k^-} \tag{AKS35}$$

This improves AMT37's absolute upper bound. For the sharper correlated version retain $d_{E,N} = q \log A_k - \log(\det G_N^{\text{ar}} / \det G_N^{(1)})$ and $c_{X,N} = \min\{r_X L_N^{\text{ar}}, d_{E,N}\}$, where $r_K = r$, $r_B = q - r$. Put $C_X^- = c_{X,q-1} + c_{X,q}$, $C_X^+ = c_{X,2q-1} + c_{X,2q}$. AMT18 then gives

$$\begin{aligned} \max\{0, \underline{U}_1 - C_K^-, \underline{U}_1 + \delta_k^\sigma - C_B^+\} &\leq F_K^{\text{ar}} \\ &\leq \overline{U}_1 + \min\{C_K^+, \delta_k^\sigma + C_B^-\}. \end{aligned} \tag{AKS36}$$

One may replace $d_{E,N}$ in the caps by the original full-source $\Xi_{k,N}$ or its proved finite majorant. The scalar $\delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma$ retains exactly AMT21:

$$\begin{aligned} \max\{-qE_k^+, -\Xi_{k,q-1} - \Xi_{k,q}\} &\leq \delta_k^\sigma \\ &\leq \min\{qE_k^-, \Xi_{k,2q-1} + \Xi_{k,2q}\}. \end{aligned} \quad (\text{AKS37})$$

Since $E_k^\pm = O_h(k + \log q)$ by the exact AMT22 expression, AKS35 and AKS32 prove the arithmetic version of AKS6, with an explicit $O_h(k^2)$ upper remainder. Nonnegativity follows from the actual nested source minima, as in AMT15, not from comparison with a possibly negative lower endpoint.

18.2 The original Gamma multiplier, not a replacement higher source

Retain $t_j = 2j + 1/2$, $f_l(S) = \prod_{j=0}^{l-1}(S - c - t_j)$ and

$$dm_{0,k} = \beta_k |f_l(c + iy)|^2 d\sigma, \quad \beta_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}, \quad T_l = \prod_{j=0}^{l-1} t_j^2, \quad U_{l,N} = \prod_{j=0}^{l-1} [4(N + j + 1)^2 + t_j^2]. \quad (\text{AKS38})$$

The exact source map is $\mathcal{P}_N \rightarrow f_l \mathcal{P}_N \subset \mathcal{P}_{N+l}$, $P \mapsto f_l P$, with inverse division on this specified subspace. Let E_χ be the complete divided-jet map and U_f its Leibniz multiplication matrix, $(U_f)_{r,a} = f_l^{(r-a)}(\lambda)/(r-a)!$ for $r \geq a$. AMT25–28 gives the precise transported observation and kernel

$$\begin{aligned} \tilde{\Lambda} &= \Lambda E_\chi^{-1} U_f^{-1}, \quad \tilde{I} = U_f E_\chi I, \quad \ker \tilde{\Lambda} = U_f E_\chi K, \\ C_N &= \mathcal{K}_{\chi,N+l} - \mathcal{K}_{\chi,f;N+l} \mathcal{K}_{f,N+l}^{-1} \mathcal{K}_{f,\chi;N+l}, \\ H_{K,N}^{(k)} &= \beta_k \tilde{I}^* C_N^{-1} \tilde{I}. \end{aligned} \quad (\text{AKS39})$$

Each covariance is the original finite sum of full polynomial jet outer products divided by its original monic norm. These formulas retain the extra f constraints and their full cross covariance. In particular $M_f K = K$ is not assumed. AKS26 calculates the same tensor metric directly in its own orthonormal source; the two representations agree by their identical source norm and identical constrained minimum.

Define

$$w_N = \log(U_{l,N}/T_l), \quad \Omega^{\text{low}} = w_{q-1} + w_q, \quad \Omega^{\text{high}} = w_{2q-1} + w_{2q}.$$

The already proved same-coefficient comparison AMT42–44 retains all mass factors before cancellation and gives

$$-r\Omega^{\text{low}} \leq \Delta_K^\Gamma := F_K^{(1)} - F_K^{(k)} \leq r\Omega^{\text{high}}. \quad (\text{AKS40})$$

It therefore sharpens the two independent absolute bounds by

$$\begin{aligned} \max\{\underline{U}_k, \underline{U}_1 - r\Omega^{\text{high}}, 0\} &\leq F_K^{(k)} \\ &\leq \min\{\overline{U}_k, \overline{U}_1 + r\Omega^{\text{low}}\}. \end{aligned} \quad (\text{AKS41})$$

The original finite estimate $0 \leq w_N \leq l \log[1 + 16(N + l)^2]$ proves that AKS40 is $o(kq)$ on the actual rank- r domain. Along with AKS35 this preserves the completed fact that all three source kernels differ by $o(kq)$. AKS4, not AKS40, is the new absolute-scale calculation.

18.3 Boundary, QGQ and arithmetic receiving identities

The minimum and determinant factorization in AKS42 use the classical Schur-complement method [H1, §2.1, (2.3)–(2.4)]. The fixed original kernel frame and right section below specify the complete block change of coordinates.

At each endpoint retain the original minimum section $S_N^{(s)} = (G_N^{(s)})^{-1} \Lambda^* (\Lambda (G_N^{(s)})^{-1} \Lambda^*)^{-1}$. The full Schur identity is

$$\det(I^* G_N^{(s)} I) \det Q_N^{(s)} = |\det[I, S_*]|^2 \det G_N^{(s)}, \quad Q_N^{(s)} = (\Lambda (G_N^{(s)})^{-1} \Lambda^*)^{-1}. \quad (\text{AKS42})$$

Thus the new finite absolute boundary interval is

$$\max\{0, \mathcal{B}_k^{(s)} - \overline{U}_s\} \leq F_B^{(s)} \leq \mathcal{B}_k^{(s)} - \underline{U}_s. \quad (\text{AKS43})$$

The baselines in this formula are the *complete* original BSL/HBL objects, with their full finite relation errors. No q^2 coefficient is assigned to K ; its proved coefficient there remains zero.

Write $\mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0 = -\mathfrak{T}_k$. The exact full identities still read

$$\begin{aligned} F_B^{(1)} - F_B^{(k)} &= \mathcal{H}_k - \Delta_K^\Gamma, \\ \boxed{\mathcal{B}_k^{\text{ar}} = F_K^{(k)} + F_B^{(k)} + \mathcal{H}_k + \delta_k^\sigma = F_K^{(1)} + F_B^{(1)} + \delta_k^\sigma.} \end{aligned} \quad (\text{AKS44})$$

The Gamma interval is not re-estimated in this argument. For either complete QRI1 or QRI2 interval $[\omega^-, \omega^+]$ for its already calculated centre W_k , retain its exact source errors $e_0 \in [-a_0, b_0]$ and $e_q + e_{q+1} \in [-b_k^+, b_k^-]$. Then

$$\underline{\mathcal{H}}_k = -\omega^+ - b_k^- - a_0, \quad \overline{\mathcal{H}}_k = -\omega^- + b_k^+ + b_0, \quad \mathcal{H}_k \in [\underline{\mathcal{H}}_k, \overline{\mathcal{H}}_k]. \quad (\text{AKS45})$$

These are exactly QRI3, not new unspecified error choices. The complete QRI1–9, QGT and QGQ definitions remain verbatim in each updated receiver; the new interval is intersected with all their retained tighter finite intervals. In particular the simple-quartet nonzero error radius stays

$$\limsup \left| \frac{\mathcal{H}_k + lq\mathcal{C}_\Gamma}{q} + \frac{\partial_\alpha \mathcal{L}(2, \pi)}{8} - \frac{\log 2}{4} \right| \leq \frac{A_{\text{QGT}}}{\sqrt{2}}. \quad (\text{AKS46})$$

No zero-radius or $o(q)$ arithmetic assertion is introduced.

For example the complete finite absolute defect relative to the tensor boundary, with the actual Gamma and arithmetic scalars retained, is

$$\underline{U}_k + \mathcal{H}_k + \delta_k^\sigma \leq \mathcal{B}_k^{\text{ar}} - F_B^{(k)} \leq \overline{U}_k + \mathcal{H}_k + \delta_k^\sigma. \quad (\text{AKS47})$$

Inserting AKS41 makes this sharper. Replacing $\mathcal{H}_k$ by the appropriate endpoints of AKS45 and δ_k^σ by AKS37 gives its fully inherited finite scalar enclosure with the same signs. Also $F_B^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma - F_K^{\text{ar}}$; subtract the interval AKS35 or AKS36 to get its new absolute boundary interval. This retains the exact correlations rather than taking independent hypothetical source errors.

The existing limit is $\mathcal{H}_k/(kq) \rightarrow -\mathcal{C}_\Gamma/4$ and $\delta_k^\sigma/(kq) \rightarrow 0$. Therefore the new interval implies

$$\begin{aligned} \liminf_k \inf_{|u| \geq R_*} \frac{\mathcal{B}_k^{\text{ar}} - F_B^{(k)}}{kq} &\geq -\mathcal{C}_\Gamma/4, \\ \limsup_k \sup_{|u| \geq R_*} \frac{\mathcal{B}_k^{\text{ar}} - F_B^{(k)}}{kq} &\leq \kappa_* - \mathcal{C}_\Gamma/4. \end{aligned} \quad (\text{AKS48})$$

All original branches are included in the infima and suprema. This does not decide its sign. The already proved boundary *difference* limits of AMT49 are unchanged. Every subsequential limit of the absolute normalized kernel lies in $[0, \kappa_*]$ and is shared by its three source versions when they are evaluated on the same sequence of period parameters. Existence of a single limit is not proved by boundedness.

Finally the HC residual remains exactly $\mathcal{R}_k = \mathcal{B}_k^{\text{ar}} - 4q \log(D_h k)$, with its original D_h and original nonnegative contraction, phase and trace terms. AKS44 changes none of them. The supplied positive q^2 baseline and its complete Gamma correction are not re-presented as new results here.

19 Actions, support faces, and the actual arithmetic class

The coefficient formula AKS12 supplies all four blocks of the same sum action: $P_K Y P_K$, $P_K Y (I - P_K)$, $(I - P_K) Y P_K$ and $(I - P_K) Y (I - P_K)$. There is no claim that either off-diagonal block vanishes. In the actual minimum-section coordinates the corresponding formula remains

$$[I, S_N]^{-1} Y [I, S_N] = \begin{pmatrix} H_{K,N}^{-1} I^* G_N Y I & H_{K,N}^{-1} I^* G_N Y S_N \\ \Lambda Y I & \Lambda Y S_N \end{pmatrix}. \quad (\text{AKS49})$$

Its proof is multiplication by the inverse coordinate maps $H_{K,N}^{-1} I^* G_N$ and Λ . In the invariant-ring model, the same action is the derivation $\mathcal{D}f = (\mathbf{H} \Omega \mathbf{H}^{-1} x) \cdot \nabla f$, $\Omega = \text{diag}(\gamma - i\delta, -\gamma - i\delta, \gamma + i\delta, -\gamma + i\delta)$, with $\mathcal{D}Q = 0$. For the actual conductor $\mathcal{A}_Q = \prod_{a=1}^4 Q_a$, its full defect is $\mathcal{D}\mathcal{A}_Q = \sum_{a=1}^4 (\mathcal{D}Q_a) \prod_{b \neq a, 1 \leq b \leq 4} Q_b$, exactly OQC22. Thus this coefficient realization preserves the original action leaving the observed dual.

In particular the section-corrected vector is still OPR2's

$$C_s r_j^{\ker} = e_{q+j} - S_{q+j-1}^{(s)} \Lambda e_{q+j} = I(\zeta_{q+j} - \kappa S_{q+j-1}^{(s)} \Lambda e_{q+j}),$$

with its full denominator $1 + \xi_j$ in the kernel increment. The operator proof estimates the complete covariance and its actual restriction, so none of these coupled terms is replaced by a raw fixed-section vector. The all-iterate kernel stays $\bigcap_{a=0}^{q-1} \ker(\Lambda Y^a)$; nothing in AKS4 declares it zero.

On the existing arithmetic coefficient realization retain $\mathcal{V}_{h,k} P = P(D_1 + \dots + D_k) F_h^{\otimes k}$ and $J^{(k)} \mathcal{V}_{h,k} = \eta \text{rem}_\chi$, where $\eta = U^{\otimes k} \iota$ is the original injection. AKS17 and the Newton coefficients compute precisely this same remainder. Any two polynomial lifts differ by χQ in the original relation space, so their original arithmetic cohomology classes agree. The specified primitive of the difference is EP.48, with the full differential and cochain signs proved in AKP3–5 below; it depends on the lift difference. The source comparison to w_h^{*k} is the identity on original polynomial coefficients, followed by the actual minima as in AMT8; it is not a replacement arithmetic source.

The scalar object stays $G(R) = \{\tau\} \sqcup R^\bullet$, with supported zero $0_R^\bullet$, injective amplitude/support pair, and infinite integer target. Every coefficient map above is applied on each original present support label and outer leg. It commutes with the specified coefficient inclusions because both compositions apply the same linear map to the same coefficient; a present zero is not converted into absence. No new base object, purity condition, or identification of the sum action with a Frobenius action is used. The actual distinguished class and its higher-jet directions remain in the full fibre, even when the observation kills them.

The geometric source of the comparison problem. Pierre Deligne's work is a substantive source of the programme's weight and common-image comparison problem. Its precise geometric results include *La conjecture de Weil. I*, Theorem (1.6), p. 276, for smooth projective varieties over a finite field, with ℓ different from its characteristic, and *Weil. II*, Theorem 3.3.1, p. 204, and Corollaries 3.3.4–3.3.5, p. 206, on compact-support upper weights and the associated dual bounds [H17, H18]. The trace-formula framework recalled in *Weil. I*, §1.5, is Alexander Grothendieck's; Deligne also explicitly credits André Weil's conjectures, Bernard Dwork's rationality theorem and Nicholas M. Katz's notes in his account of the antecedents.

The present passage proves coefficient and arithmetic-class maps on the original analytic source. It leaves the further geometric comparison as a mathematical task: construct an actual target cohomology and common-image map, identify the action on the same class, and verify the geometric and weight hypotheses before applying the cited theorems. The finite coefficient identities above are retained as inputs to that task. A citation to Deligne is neither an assumption that the missing map exists nor a reason to discard that comparison problem.

20 Exact period domain and the interior alternative

For reference the inherited radius used above is completely specified as follows; these are AMT33–34/RPC17–23, with no new constants. Put

$$\begin{aligned}
\beta &= 2(\gamma^2 - \delta^2), \quad \eta = (\delta^2 + \gamma^2)^2, \quad a = v_h(\rho) \neq 0, \\
c_3 &= (a^{-2} - \bar{a}^{-2})/(4i\delta\gamma), \quad c_1 = (a^{-2} + \bar{a}^{-2})/2 - (\delta^2 - \gamma^2)c_3, \\
g_j &= 5^{j/5-1}e^{\pi i j/5}\Gamma(j/5), \quad g_+ = \max_j |g_j|, \quad g_-^{-1} = \max_j |g_j|^{-1}, \quad \kappa = \max_{j < 4} |g_{j+1}/g_j|, \\
E_j^{\text{ray}} &= \frac{8}{5} \int_0^\infty r^j (r^3/3 + r) e^{-r^5/5 + r^3/3 + r} dr, \quad C_T = 2 \left(\sum_{j=0}^3 (E_j^{\text{ray}})^2 \right)^{1/2}, \\
C_K &= 4C_T g_-^{-1} + 2g_+ g_-^{-1} + 2g_+ g_-^{-2} C_T, \\
C_{K,3} &= C_K(3\kappa^2 + 3\kappa C_K + C_K^2), \quad \epsilon_* = \min\{1, (2C_T g_-^{-1})^{-1}\}, \quad B_* = \beta + \eta.
\end{aligned} \tag{AKS50}$$

Then

$$R_* = \begin{cases} \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2(|c_3|C_{K,3}B_* + |c_1|(\kappa + C_K))}{|c_3 g_4/g_1|} \right\} \right]^{5/2}, & c_3 \neq 0, \\ \left[\max \left\{ 1, \frac{B_*}{\epsilon_*}, \frac{2C_K B_*}{|g_2/g_1|} \right\} \right]^{5/2}, & c_3 = 0. \end{cases} \tag{AKS51}$$

The actual unit implies $(c_1, c_3) \neq (0, 0)$ and all integrals converge. RPC’s full period comparison proves a nonzero $(4, 1)$ or $(2, 1)$ commutator entry, respectively, for $|u| \geq R_*$ on every branch. It therefore proves the five-member orbit needed in AKS11.

Inside this radius the same coefficient determinant AKS12–14 and the operator bound AKS24 remain valid. The orbit is decided by the original noninvariant quadratic coefficients, not by the radius alone. Since Q is nondegenerate, its projective orbit has either one or five members. A one-member orbit must have character one: its character has both fourth and fifth powers equal to one, by the quadratic determinant and $D^5 = I$. In that case $Q = a'x_1x_4 + b'x_2x_3$ with $a'b' \neq 0$, and

$$b = d_{5,k,0} - d_{5,k-2,0} = \left\lfloor \frac{(k+1)^2}{5} \right\rfloor + \mathbf{1}_{k \equiv 0 \text{ or } 3 \pmod{5}}, \quad r = q - b. \tag{AKS52}$$

If a noninvariant coefficient is nonzero, AKS11 applies instead. These are the complete OQC alternatives. The bound AKS31 always uses that actual r ; it does not extend the $8k - 16$ specialization to an unevaluated exceptional point. This calculation does not assert that the interior exceptional set is empty.

21 Verification scope and the remaining absolute coefficient

The new mathematical content is AKS20–32 and its application to the actual coefficient determinant AKS12–14, followed by the finite receiving improvements AKS35–48. The complete source and orbit identifications used in that application are the supplied OCS/OQC/OQX, RPC, OPG/OPR and AMT results; their exact predecessors are preserved. AKS33 pinpoints a repair obtainable immediately from the earlier proof, whereas the Newton bound supplies the sharper explicit constant.

The bound *at* scale kq is now finite and uniform. What is not proved is a matching lower coefficient, a unique limit for $F_K^{(1)}/(kq)$, or a sign for the interval AKS48. Those questions involve the orientation of the original period kernel within the inverse remainder covariance; replacing that compression by a scalar bound loses precisely that information. This is the finite-bounds outcome allowed by the current request, not a claim to have evaluated an asymptotic coefficient. The source differences are already completed inputs and are not renamed as the remaining problem.

The accompanying exact checks distinguish finite algebraic diagnostics from the analytic proof for the full family. They do not assert that a diagnostic root is a zero of ξ , that a diagnostic matrix equals an actual period matrix, or that a new Lean theorem has been checked. No RH conclusion is drawn from this kernel bound.

22 The new ceiling in the original coefficient frame

The complete incoming AKS proof sharpens the previously proved KAF bound. Both estimates apply to the same original remainder and the same kernel. We retain all their finite constants. In particular, for $m = 1$, $k = 4l + 1$, $l \geq 1$, $q = (k + 1)^2$, $c = k/2$, $0 < \delta < 1/2$, $\gamma > 2$, and every original branch with $|u| \geq R_*$, put $r = 8k - 16$. The AKS constant is

$$\kappa_* = 32 \log \frac{25 \log 3}{4\pi} \in (25.02078097649, 25.02078097650). \quad (\text{AKJ1})$$

AKS20–32 proves the full finite operator inequality from confluent Newton division and coefficient extraction. Its parameters t and z are extraction radii in the existing source, whose orders remain $1, k$. For $s \in \{1, k\}$ define, using the exact KAF and AKS quantities,

$$\begin{aligned} L_s &= d_1^{(s)}, \\ U_s &= \min\{2r \log Z_{k,1}, r(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1^{(s)}\}. \end{aligned} \quad (\text{AKJ2})$$

The first entry of the minimum is KAF’s factorial estimate; the second is AKS28, with its original first compressed increment retained. Thus $L_s \leq F_K^{(s)} \leq U_s$. This pointwise minimum retains whichever finite estimate is sharper at the actual original parameters. It does not assume that the asymptotically smaller ceiling wins at every degree.

The two exact original coordinate constructions are related as follows. Use OKM’s invertible ordered Vandermonde V_β and its actual quotient matrix π . Put $I_\pi = V_\beta^{-1} \pi^\top$. AKS12 constructs the coefficient-orthogonal projection P_K from the actual observation and its complete ordered-word Gram. Directly,

$$P_K = I_\pi (I_\pi^* I_\pi)^{-1} I_\pi^*, \quad D_{K,N}^{(s)} = \frac{\det(\bar{\pi}(R_N^{(s)})^{-1} \pi^\top)}{\det(I_\pi^* I_\pi)}. \quad (\text{AKJ3})$$

Indeed I_π is injective with image K by OKM5. The first displayed matrix is Hermitian and idempotent, fixes this image, and kills its coefficient-orthogonal complement, so it is exactly AKS12’s projection. Apply AKS14 with inclusion I_π and then OKM8–9 to obtain the second identity. The positive denominator is independent of endpoint and source. Its exponent is $1 + 1 - 1 - 1 = 0$ in the four-endpoint quotient. This proves the exact morphism between the incoming projector determinant and the existing conductor-frame determinant, retaining the original coefficient metric and every frame factor.

With the original AMT correlated caps $C_X^\pm$ and scalar δ_k^σ , the following joint intervals hold:

$$\begin{aligned} \max\{0, L_1 - C_K^-, L_1 + \delta_k^\sigma - C_B^+\} &\leq F_K^{\text{ar}} \\ &\leq U_1 + \min\{C_K^+, \delta_k^\sigma + C_B^-\}, \\ \max\{L_k, L_1 - r\Omega^{\text{high}}, 0\} &\leq F_K^{(k)} \leq \min\{U_k, U_1 + r\Omega^{\text{low}}\}. \end{aligned} \quad (\text{AKJ4})$$

For proof, add each retained AMT lower or upper source-difference bound to the corresponding endpoint of AKJ2, and intersect the resulting intervals. In the first line the upper and lower source differences are the same AMT18 correlated caps used in AKS36; no independent errors have replaced their actual values. Every boundary interval is obtained by subtracting these kernel intervals from the exact original total $\mathcal{B}_k^a = F_K^a + F_B^a$. Consequently all three original kernels have uniform limsup at most κ_* after division by kq . The source differences remain $o(kq)$. A matching leading value is still to be calculated from these same matrices.

23 Residue detection expressed in every original primary coefficient

Retain now every original multiplicity $m \geq 1$, $e = 1 + k(m - 1)$, $q = e(k + 1)^2$, and the full polynomial

$$\chi(S) = \prod_{a,b=0}^k (S - \beta_{ab})^e, \quad \beta_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad E = \mathbb{C}[S]/(\chi). \quad (\text{AKJ5})$$

All distinct pairs give distinct roots. For a root β write $z = S - \beta$, $h_\beta(S) = \chi(S)/(S - \beta)^e$, and let ε_β be its original CRT idempotent. Define exactly $\ell([f]) = [S^{q-1}] \text{rem}_\chi f$. For every polynomial class the primary expansion gives

$$\ell(f) = \sum_{\beta} [z^{e-1}] \frac{f(\beta + z)}{h_\beta(\beta + z)}. \quad (\text{AKJ6})$$

The quotient on the right is a truncated power series through degree $e - 1$. To prove the identity, partial-fraction division of f/χ gives at each finite pole the coefficient of $(S - \beta)^{-1}$ shown on the right. Expanding those principal parts at infinity, only their first-order terms contribute to S^{-1} . Polynomial terms contribute nothing. For the unique representative of degree below q , the same S^{-1} coefficient is its coefficient of S^{q-1} . This proves AKJ6 without deleting any repeated-root term. Since $h_\beta(\beta) \neq 0$, the bilinear pairing $(f, g) \mapsto \ell(fg)$ is nondegenerate: if the first nonzero local coefficient of f is $f_t z^t$, take $g = \varepsilon_\beta z^{e-1-t} h_\beta$ to obtain $\ell(fg) = f_t \neq 0$.

Residue duality and its trace comparison. This coefficient-residue pairing has the coherent-duality precedent in SGA 5, Exposé III, by Alexander Grothendieck, rédigé by Luc Illusie, especially §6.8, printed pp. 127–129, equation (6.8.5) [H13]. The Verdier-pairing formulation has Jean-Louis Verdier’s distinct contribution; the exposé credit and this contribution are retained separately [H14]. Here is the exact one-variable specialization, including its Jacobian factor. Set

$$Y = \text{Spec}(E) \hookrightarrow X = \mathbb{A}_{\mathbb{C}}^1 \longrightarrow \text{Spec } \mathbb{C}, \quad E = \mathbb{C}[S]/(\chi).$$

Since χ is monic, multiplication by it is injective in $\mathbb{C}[S]$ and $1, S, \dots, S^{q-1}$ is a basis of E . Thus the immersion is regular of codimension one and the composite is finite flat. The conormal generator $[\chi] \in (\chi)/(\chi^2)$ maps to $\chi'(S)dS$; the dualizing generator that sends $[\chi]$ to dS therefore represents the fundamental class by χ' . In the coefficient-residue trivialization AKJ6, SGA’s formula is

$$\text{Tr}_{\mathbb{C}}(M_f) = \ell(f\chi').$$

It also has a direct proof on the full nonreduced object used here. In the β -block, $\chi = z^e h_\beta$ and $\chi'/h_\beta = e z^{e-1} + z^e h'_\beta/h_\beta$. AKJ6 consequently gives $\ell(f\chi') = \sum_{\beta} e f(\beta)$. On that block multiplication by f in the ordered basis $1, z, \dots, z^{e-1}$ is triangular with diagonal entry $f(\beta)$ repeated e times, proving the same trace. This proves the identity without discarding multiplicities or dividing by χ' . The nondegenerate pairing $\ell(fg)$ and the trace $\ell(f\chi')$ are related by precisely this multiplication map; they are not interchanged in AKJ7–8.

The printed SGA passage states its noetherian-base hypotheses and warns that the compatibility is not proved at its cited Hartshorne location, *Residues and Duality*, III 9, R6; it points for the one-variable case to Michel Raynaud, *Anneaux locaux henséliens*, VII 1 [H15, H16]. Those predecessor book pages were identified through the checked SGA passage, not independently verified in this repair. The present base $\mathbb{C}$ is noetherian and the complete elementary proof above remains part of the calculation. This coherent residue map does not by itself supply a Hermitian norm or a Weil weight bound.

Let λ_ν be the specified zero-charge coefficient rows of the original OCS observation, so the injection $\mathcal{F}_0$ gives $j\Lambda = \mathcal{F}_0 \mathfrak{A}\mathcal{H}^{-1}$. Retain the exact entries

$$a_{\nu,\beta,D} := \lambda_\nu(\varepsilon_\beta z^D) = i^D D! c_{\nu,abD}, \quad 0 \leq D < e. \quad (\text{AKJ7})$$

Here $c_{\nu,abD}$ is the full OCS multinomial in the original period and unit jets; $\beta = \beta_{ab}$. It is not an assigned coefficient. The residue representative of each row has the explicit formula

$$A_\nu = \left[\sum_{\beta} \varepsilon_\beta h_\beta(S) \sum_{D=0}^{e-1} a_{\nu,\beta,D} (S - \beta)^{e-1-D} \right], \quad \lambda_\nu(x) = \ell(x A_\nu). \quad (\text{AKJ8})$$

Indeed, restrict to one primary factor and substitute AKJ8 into AKJ6. For $x = \varepsilon_\beta z^D$, multiplication cancels h_β and extracts exactly $a_{\nu,\beta,D}$. These vectors are a basis, proving the asserted equality for every x . This gives the concrete inverse of the residue equivalence used in the incoming formalization, on the actual OCS rows, with their factorials and phases.

Define the integer t_β from these actual coefficients by

$$t_\beta = \begin{cases} -1, & a_{\nu,\beta,D} = 0 \text{ for every } \nu, D, \\ \max\{D : a_{\nu,\beta,D} \neq 0 \text{ for some } \nu\}, & \text{otherwise,} \end{cases} \quad n_\beta = e - 1 - t_\beta. \quad (\text{AKJ9})$$

Define $\mathcal{O}_q : E \rightarrow B^q$ by $\mathcal{O}_q(x) = (\Lambda M_S^j x)_{0 \leq j < q}$, where $B = \text{im } \Lambda$ is the original observation image. Thus $0 \leq n_\beta \leq e$. The complete original permanently invisible submodule and its exact dimension are

$$\begin{aligned} K_\infty &= \bigcap_{j=0}^{q-1} \ker(\Lambda M_S^j) = \bigoplus_{\beta} \varepsilon_\beta (z^{e-n_\beta}) \subset E, \\ \dim K_\infty &= \sum_{\beta} n_\beta, \quad \text{rank } \mathcal{O}_q = q - \sum_{\beta} n_\beta. \end{aligned} \quad (\text{AKJ10})$$

Here $(z^e) = 0$ and (z^0) is the entire local algebra. To prove this, vanishing of the first q observations says $\ell(S^j x A_\nu) = 0$ for all $j < q, \nu$. Nondegeneracy just proved and the power basis imply $x A_\nu = 0$ for every row. Conversely those products being zero kills every iterate. At the factor β , AKJ8 is a unit h_β times its displayed reverse coefficient polynomial. The minimum valuation of all A_ν is precisely n_β . Its common annihilator is (z^{e-n_β}) : multiplication by a coefficient of minimal valuation is a unit times z^{n_β} , and all other coefficients have at least that valuation. This proves the module formula and its dimension; rank-nullity proves the last equality. Affine substitution $M_S = c + iY$ gives an invertible triangular change between the first q powers of the two operators, so the same kernel is obtained using the original Y -stack of OCS.

In particular all iterates are jointly injective exactly when every original highest-jet column $a_{\nu,\beta,e-1}$ has a nonzero entry. For $m = 1$, AKJ8 reduces to

$$A_\nu(\beta) = \chi'(\beta) B_{\nu,\beta}, \quad K_\infty = \text{span}\{\varepsilon_\beta : B_{\nu,\beta} = 0 \text{ for all } \nu\}. \quad (\text{AKJ11})$$

This is an exact test on the AKS8 matrix, not an assumption that it passes. It constructs the inclusion $K_\infty \hookrightarrow K$: each listed primary ideal is killed by all rows, hence in particular by their original one-step observation. The source-minimum metric on this subspace is its restriction from OKM8, retaining the same original Gram. Full-iterate injectivity alone therefore supplies no estimate of the one-step kernel determinant AKJ3.

24 The actual conductor confines the kernel's polynomial degrees

Return to $m = 1, k = 4l + 1 \geq 9$ and the original five-member period domain. Let $\mathcal{A}_Q = \prod_{j=1}^4 Q_j$ and write its full Segre restriction from OCF as

$$A = \sum_{r,s=0}^8 a_{rs} X_1^r X_0^{8-r} Y_1^s Y_0^{8-s} \neq 0. \quad (\text{AKJ12})$$

OCF proves nonvanishing and retains all actual H coefficients. No particular a_{rs} is presumed nonzero. Write $k' = k - 8$, $q' = (k - 7)^2$ and

$$b_{rs} = 4 + (2r - 8)\delta + i(2s - 8)\gamma, \quad (\mathcal{T}_{AP})(S') = \sum_{r,s=0}^8 a_{rs}p(S' + b_{rs}), \quad \mu_j = \sum_{r,s} a_{rs}b_{rs}^j. \quad (\text{AKJ13})$$

All 81 numbers b_{rs} are distinct. Their Vandermonde is invertible, so some μ_j with $0 \leq j \leq 80$ is nonzero. Define v to be the first such index. The exact finite differential expression on degree-below- q polynomials is

$$\mathcal{T}_A = \sum_{j=0}^{q-1} \frac{\mu_j}{j!} \partial_S^j = \frac{\mu_v}{v!} \partial_S^v B(\partial_S), \quad B(X) = 1 + \sum_{j=v+1}^{q-1} \frac{v! \mu_j}{j! \mu_v} X^{j-v}. \quad (\text{AKJ14})$$

This is Taylor's finite formula. Hence it kills polynomials of degree below v and lowers every higher degree by exactly v . The operator $B(\partial_S)$ is invertible: with $U = B(\partial_S) - I$, its inverse is the finite sum $\sum_{a=0}^{q-1} (-U)^a$. Terms beyond nilpotency are zero. Thus $\mathcal{T}_A$ maps polynomials of degree at most d onto those of degree at most $d - v$ when $d \geq v$.

Let $\chi_{k'}$ be the original polynomial for tensor degree k' . The conductor's exact annihilator in the original coefficient algebra is

$$V_A = \{p \in \mathbb{C}[S]_{<q} : \chi_{k'} \mid \mathcal{T}_{AP}\}, \quad K \subset V_A. \quad (\text{AKJ15})$$

To prove this map, pair the biform Ag , of bidegree (k, k) , with the primary values of p . If g has coefficient g_{ab} at bidegree (k', k') , convolution of all original coefficients gives the pairing

$$\sum_{a,b=0}^{k'} g_{ab} \sum_{r,s=0}^8 a_{rs}p(\beta_{ab}^{(k')} + b_{rs}) = \sum_{a,b=0}^{k'} g_{ab}(\mathcal{T}_{AP})(\beta_{ab}^{(k')}). \quad (\text{AKJ16})$$

The addition identity for the roots follows directly from their retained centres and both signs. Vanishing for every g is exactly vanishing at all q' distinct roots, hence the divisibility in AKJ15. OCF's proved inclusion $AC[X, Y]_{(k', k')} \subset J_k$ then gives $K \subset V_A$ by its original complex-linear coefficient pairing.

The following is a full degree-adapted basis of V_A . Define $\mathcal{J}_v S^d = d! S^{d+v} / (d+v)!$, so $\partial_S^v \mathcal{J}_v = I$, and set

$$1, S, \dots, S^{v-1}, \quad P_h = \frac{(q' + h + v)!}{(q' + h)!} B(\partial_S)^{-1} \mathcal{J}_v(\chi_{k'} S^h), \quad (\text{AKJ17})$$

$$0 \leq h \leq q - q' - v - 1.$$

The first list is empty when $v = 0$. Since $q - q' = 16k - 48 \geq 96 > v$, the range of the second list is nonnegative. Each P_h is monic of degree $q' + h + v$ and maps under $\mathcal{T}_A$ to $\mu_v(q' + h + v)! / (v!(q' + h)!)$ times $\chi_{k'} S^h$. These statements follow from AKJ14, the inverse's constant term one, and the leading coefficient of $\mathcal{J}_v$. The images span every multiple of $\chi_{k'}$ of degree at most $q - v - 1$, and the low list spans the exact kernel of $\mathcal{T}_A$. This proves spanning and independence. In particular $\dim V_A = q - q' = 16k - 48$.

Choose the original OCF basis of $J_k / (AC[X, Y]_{(k', k')})$, whose exact dimension is $j_k = 8k - 32$. Evaluate each chosen biform row on the primary values of every polynomial in AKJ17. This gives an explicit j_k by $(16k - 48)$ matrix Ξ , with the full original period coefficients. Its kernel maps isomorphically onto K by the displayed basis. Its rank is j_k : the original J_k and conductor annihilator duality, or dimension $\dim V_A - \dim K$, gives this rank. For the submatrix $\Xi_{\leq d}$ containing only basis columns of degree at most d , the exact formula is

$$\dim(K \cap \mathbb{C}[S]_{\leq d}) = \#\{\text{degrees in AKJ17 at most } d\} - \text{rank } \Xi_{\leq d}. \quad (\text{AKJ18})$$

Distinct monic degrees ensure that no omitted higher column can cancel to form a degree-at-most- d polynomial. Restricting the coefficient isomorphism to this prefix and applying rank-nullity proves AKJ18. It uses actual minors and makes no assertion of a generic prefix rank.

Let $d_0 < \dots < d_{r-1}$ be the degree jumps of the actual kernel. The basis degrees of V_A , in increasing order, are i for $i < v$, and $q' + i$ for $i \geq v$. Inclusion and dimension of each degree prefix, together with $K \subset \mathbb{C}[S]_{< q}$, imply

$$i + q' \mathbf{1}_{i \geq v} \leq d_i \leq q - r + i. \quad (\text{AKJ19})$$

For the lower inequality the i th independent kernel vector cannot appear earlier than the i th independent vector of V_A . For the upper inequality, intersection of an r -dimensional subspace of $\mathbb{C}^q$ with its $(q - r + i + 1)$ -dimensional degree prefix has dimension at least $i + 1$. Summing yields the exact exterior-degree bounds

$$\begin{aligned} \frac{r(r-1)}{2} + \max\{r-v, 0\}q' &\leq \sum_{i=0}^{r-1} d_i \leq r(q-r) + \frac{r(r-1)}{2}, \\ 0 \leq rq - \sum_i d_i &\leq \frac{r(r+1)}{2} + rj_k + \min\{r, v\}q'. \end{aligned} \quad (\text{AKJ20})$$

The difference between the first line's upper and lower endpoints is $rj_k + \min\{r, v\}q'$, since $q - q' = r + j_k$. Therefore the original kernel's exterior degree is $rq - O(k^2)$, with the exact bound above and $v \leq 80$, uniformly in the proved period domain. The affine map $S = c + iy$ preserves all degree prefixes and multiplies a degree- d leading coefficient by i^d . Thus these are also the degrees in the original Gamma coordinate, with all phases known. They specify the kernel's position without assigning a condition number to its frame.

25 The exact conductor translations in the original source norm

Keep either source $s \in \{1, k\}$, $b = s/2$, $M = (2\pi)^{s/2}$ and the original $p_n, h_n = Mn!(b)_n$. On polynomials of degree at most D , let $\varphi_n = p_n/\sqrt{h_n}$ and $w_n = n!/(b)_n$. Differentiating the original generating identity AKS16 in y gives

$$\partial_y \varphi_n = \sum_{\substack{1 \leq a \leq n \\ a \text{ odd}}} \frac{(-1)^{(a-1)/2}}{a} \sqrt{\frac{w_n}{w_{n-a}}} \varphi_{n-a}. \quad (\text{AKJ21})$$

Indeed differentiation multiplies the generating series by $\arctan z = \sum_{a \text{ odd}} (-1)^{(a-1)/2} z^a / a$. Equating its finite coefficient and substituting the full norms gives exactly AKJ21; the two copies of M are cancelled only at this point.

Write $H_D = \sum_{j=1}^D 1/j$ and $H_0 = 0$. The original source norm obeys

$$\|\partial_y\| \leq \sqrt{H_D(H_D + 4)} \leq H_D + 2, \quad \|f(y+z)\| \leq e^{|z|(H_D+2)} \|f\|. \quad (\text{AKJ22})$$

Here is a proof retaining both source orders. For $n > m \geq 1$ and $b \geq 1/2$,

$$\frac{w_n}{w_m} \leq \prod_{j=m+1}^n \frac{j}{j-1/2} \leq \sqrt{\frac{n}{m}}, \quad w_n \leq 2\sqrt{n} \quad (n \geq 1). \quad (\text{AKJ23})$$

The factor inequality follows by squaring and using $j(j-1) \leq (j-1/2)^2$; the product telescopes. For $m = 0$, separate the first factor, at most two. Apply the weighted Schur estimate to the absolute matrix in AKJ21 with positive weights $\vartheta_n = 1/\sqrt{w_n}$. Its weighted row sum is at most H_D . Its weighted column sum at n is at most $H_n + 4$: for $1 \leq m < n$ the excess over $1/(n-m)$ is at most

$$\frac{\sqrt{n/m} - 1}{n-m} = \frac{1}{\sqrt{m}(\sqrt{n} + \sqrt{m})} \leq \frac{1}{\sqrt{nm}},$$

whose sum is at most two by comparison with $\int_0^n t^{-1/2} dt$. The $m = 0$ term is at most $2/\sqrt{n} \leq 2$. Enlarging the odd-index sums to all indices preserves these bounds. For completeness, if a nonnegative matrix has weighted row bound A and column bound B , Cauchy–Schwarz gives $|(Tx)_m|^2 \leq A \vartheta_m \sum_n T_{mn} |x_n|^2 / \vartheta_n$; summing and using the column bound gives $\|Tx\|^2 \leq AB \|x\|^2$. This proves the first bound in AKJ22, including $D = 0$ where the derivative is zero. The exact finite Taylor identity $f(y + z) = \exp(z \partial_y) f(y)$ and its norm series prove the second. The inverse translation is obtained by $-z$ with the same bound.

There is consequently an exact original-source upper control for AKJ13. For $f(y) = p(c_k + iy)$, evaluate its output at $c_{k-8} + iy$, retaining $c_k = k/2$ and $c_{k-8} = k/2 - 4$. Then

$$(\mathcal{T}_A p)(c_{k-8} + iy) = \sum_{r,s=0}^8 a_{rs} f\left(y + \frac{b_{rs} - 4}{i}\right), \quad (\text{AKJ24})$$

and therefore

$$\|\mathcal{T}_A p\|_{s, c_{k-8}} \leq \left(\sum_{r,s} |a_{rs}| e^{|b_{rs} - 4|(H_D + 2)} \right) \|p\|_{s, c_k}. \quad (\text{AKJ25})$$

The notation $\|p\|_{s,c} = \|p(c + iy)\|_{m_0,s}$ changes only the explicit centre; the measure, mass and source order s are unchanged. AKJ24 proves the centre correction rather than discarding it. For a fixed original period and quartet the coefficient in AKJ25 grows at most polynomially in D , since $H_D \leq 1 + \log D$ for $D \geq 1$ and $|b_{rs} - 4| \leq 8\sqrt{\delta^2 + \gamma^2}$. The entire actual coefficient sum remains visible. The finite inverse in AKJ17 has the specified moment denominators μ_v ; AKJ25 asserts no bound on that inverse. Together AKJ18–25 provide exact degree information and a source-norm estimate for the actual conductor map, to be used in the remaining four-endpoint compression calculation. No leading kernel value or kernel-invariance assumption has replaced that calculation.

26 The original theta class and the checked metric interface

The theta quotient and its human source. The analytic quotient used in this section is Ralf Meyer's spectral quotient, in the conventions of [H10, §§2–4, Definition 3.1, equation (11), Theorem 3.3]. Here the original source is

$$V = \{\phi \in \mathcal{S}(\mathbb{R})_{\text{ev}} : \phi(0) = \widehat{\phi}(0) = 0\}, \quad \widehat{\phi}(y) = \int_{\mathbb{R}} \phi(x) e^{-2\pi i xy} dx, \quad \Theta\phi(x) = \sum_{n \neq 0} \phi(nx).$$

Meyer's source $\mathcal{H}_{\cap}$ is exactly this V ; his summation operator is $Z\phi(x) = \sum_{n \geq 1} \phi(nx)$. Evenness gives $\Theta = 2Z$, and linearity gives $2V = V$, so $\Theta V = Z\mathcal{H}_{\cap}$. This is an equality of the actual images, with no change of their scalar or source. The target $\mathcal{B}$ carries the seminorms

$$p_{b,j}(F) = \sup_{x > 0} x^b |D^j F(x)|, \quad b \in \mathbb{Z}, \quad j \geq 0, \quad D = -x\partial_x.$$

It is Meyer's $\mathcal{H}_{-}$ with its original topology. To check the topology explicitly, set $f(t) = F(e^t)$, so $p_{b,j}(F) = \sup_t e^{bt} |f^{(j)}(t)|$. For any real s , integer $n \geq 0$ and integers $b_- < s < b_+$, elementary exponential domination on each half-line and Leibniz's rule give

$$\sup_t (1 + |t|)^n |\partial_t^j (e^{st} f(t))| \leq C_{s,n,b_-,b_+} \sum_{r=0}^j \binom{j}{r} |s|^{j-r} (p_{b_-,r}(F) + p_{b_+,r}(F)).$$

The constant is finite because each polynomial times a strictly decaying exponential is bounded on a half-line. Conversely, $e^{bt} f^{(j)} = (\partial_t - b)^j (e^{bt} f)$ bounds every $p_{b,j}$ by a finite sum of Meyer's Schwartz seminorms. Hence the identity is a homeomorphism $\mathcal{B} \simeq \mathcal{H}_{-}$ and induces

$$Q = \mathcal{B}/\Theta V \xrightarrow{\sim} \mathcal{H}_{-}^0 = \mathcal{H}_{-}/Z\mathcal{H}_{\cap}, \quad [F] \mapsto [F].$$

Both actions are $U_a F(x) = F(x/a)$, so this map also intertwines every original dilation. Theorem 3.3 supplies closedness of this image in precisely these spaces. Meyer's completed zeta, denoted ξ in that source, is $\pi^{-s/2} \Gamma(s/2) \zeta(s)$; the present g is $s(s-1)$ times it. The temporary generator $x\partial_x$ in his Proposition 3.2 is $-D$ here.

Meyer expressly identifies Alain Connes's earlier spectral interpretation [H11] as a predecessor. His reference [2] credits Alexander Grothendieck's work on nuclear spaces and topological tensor products [H12]. These are source-reported antecedents, rather than new applications of every theorem in those works. The original chart resolution and the maps AKP2–5 below remain the programme's explicit constructions on this credited analytic quotient.

The absolute base and original coefficient resolution are retained:

$$\mathbf{F}_{1,\tau} = \{\tau, 1\}, \quad \mathbf{A}_{\tau}(\mathcal{T}) = [V \oplus V \xrightarrow{\Theta(\phi - \widehat{\psi})} \mathcal{B}], \quad Q = \mathcal{B}/\Theta V. \quad (\text{AKP1})$$

The four-point chart has $+, - < \eta < \sigma$ and coefficients $(V, V, \mathcal{B}, 0)$, with the original restrictions Θ and $J\Theta = \Theta\mathcal{F}$. Its tensor resolution identifies the top cohomology with $Q^{\otimes k}$. AT1–2a supplies the complete original construction. In its notation, retaining the full unit $v_h = j_h(2\xi/h)$, the actual cochain and arithmetic class maps are

$$\begin{aligned} \mathcal{V}_{h,k}(P) &= P(D_1 + \cdots + D_k) F_h^{\otimes k}, & E &= \mathbb{C}[S]/(\chi), \\ \eta_{h,k}[P] &= v_h^{\otimes k} P(A_h^{(k)}) 1, & q^{(k)} \mathcal{V}_{h,k}(P) &= \sigma_h^{\otimes k} \eta_{h,k}[P]. \end{aligned} \quad (\text{AKP2})$$

Here $\sigma_h : E_h \hookrightarrow Q$ is the original section and $\eta_{h,k} : E \hookrightarrow E_h^{\otimes k}$ is the full-jet injection. Both injections are part of the supplied source proof; the cochain map has not been replaced by an arbitrary observation.

For completeness, the exact representative-change calculation is as follows. If $P' = P + \chi\kappa$, perform successive monic divisions of $\chi(s_1 + \cdots + s_k)$ by $h(s_a)$. Its final remainder is zero because

χ is the original annihilator of the sum in $\bigotimes_a \mathbb{C}[s_a]/(h)$. Thus the actual divisions produce $Q_a(\mathbf{s})$ satisfying

$$\chi(s_1 + \cdots + s_k) = \sum_{a=1}^k h(s_a) Q_a(\mathbf{s}). \quad (\text{AKP3})$$

The chosen ordered division determines these polynomials; no uniqueness of all such decompositions is asserted. On the original tensor complex set

$$\Psi_\kappa = \sum_{a=1}^k (-1)^{a-1} Q_a(\mathbf{D}) \kappa \left(\sum_b D_b \right) (F_h^{\otimes(a-1)} \otimes \phi_0 \otimes F_h^{\otimes(k-a)}). \quad (\text{AKP4})$$

Its differential is

$$d\Psi_\kappa = \mathcal{V}_{h,k}(\chi\kappa) = \mathcal{V}_{h,k}(P') - \mathcal{V}_{h,k}(P). \quad (\text{AKP5})$$

Indeed the only nonzero differential in each summand acts at its ϕ_0 position. The tensor sign is $(-1)^{a-1}$, cancelling the displayed sign. The identity $\Theta\phi_0 = h(D)F_h$ gives the a th term of AKP3; all operators commute with this differential. Summing proves AKP5 with the original signs. This is EP.48, expressed at the current receiving map. Consequently the quotient classes in AKP2 agree. The chosen primitive for their difference depends on κ . At an existing support label ℓ , this quotient map sends the relation to $(\ell, 0)$ and sends external absence τ to τ . The coordinate change has therefore preserved the arithmetic class and its support through an explicit cochain homotopy.

This calculation specifies the correction to the sentence following AKS49: equality of cohomology classes is retained, with AKP4 the primitive of the change. Membership in an additional period-observation kernel alone supplies no such primitive. Its proved map is the inclusion $K_\infty \hookrightarrow K \hookrightarrow E$ of AKJ10–11, followed by the original arithmetic injection in AKP2. In particular, a nonzero coefficient in this additional kernel remains a nonzero original arithmetic class under that injection. Its zero observed value and its arithmetic class are related by these exact maps.

Origin of the retained support convention. The split-zero programme was developed under the public pseudonym KokunoYumeto with AI assistance. The originating globalization-semiring notes [H19, H20] distinguish external absence τ from the supported ring zero $0_R^\bullet$. This is the convention used in the support calculation above. The deposited creator fields are retained literally in the references: ChatGPT 5.4 and Gemini 2.5 DeepThink for the first note, and The Clankers for the secondary note. They are records of the AI-assisted programme, not substitute human predecessors for the classical mathematics cited at its points of use. No global priority claim for the external-zero construction is made.

26.1 The complete minimum-section calculation at each endpoint

Let G be one of the original positive Hermitian coefficient Grams, write $K_G = G^{-1}$, and let $\Lambda : E \rightarrow B$ be the actual observation onto its image. Define

$$Q_G = (\Lambda K_G \Lambda^*)^{-1}, \quad S_G = K_G \Lambda^* Q_G. \quad (\text{AKP6})$$

The inverse exists because Λ^* is injective and K_G is positive definite. Direct multiplication, using Hermitian symmetry, gives

$$\Lambda S_G = I_B, \quad S_G^* G = Q_G \Lambda, \quad S_G^* G S_G = Q_G. \quad (\text{AKP7})$$

For every rectangular column family X put $Z = \Lambda X$ and $R = X - S_G Z$. Then

$$\Lambda R = 0, \quad R^* G S_G = 0, \quad R^* G R = X^* G X - Z^* Q_G Z \succeq 0. \quad (\text{AKP8})$$

The first two equalities follow from AKP7. In the expansion of the last Gram, each cross term is $Z^* Q_G Z$ and the section Gram is the same matrix. The resulting single subtraction proves the

identity, including every off-diagonal pairing. Any other lift is $S_G Z + R'$ with $\Lambda R' = 0$. Its Gram is $Z^* Q_G Z + R'^* G R'$, proving the minimum property by the same orthogonality calculation.

Retain a fixed section S_* , the actual inclusion I_K , and original coordinate map κ_K satisfying $I_K \kappa_K = I_E - S_* \Lambda$. Applying this identity to R gives

$$R = I_K (\kappa_K X - \kappa_K S_G \Lambda X). \quad (\text{AKP9})$$

Thus the fixed-section coordinates include their specified minimum correction. For one original new column e_D , define $t_D = e_D^* G e_D$, $\lambda_D = \Lambda e_D$ and $\xi_D = \lambda_D^* Q_G \lambda_D$. AKP8 yields $t_D - \xi_D \geq 0$, so the exact ordered factors satisfy

$$(1 + \xi_D) \left(1 + \frac{t_D - \xi_D}{1 + \xi_D} \right) = 1 + t_D. \quad (\text{AKP10})$$

The denominator is positive and is retained. These are the full OPR minimum-section identities, now proved at the original Grams of AKS26 and OKM8. For an empty observation image, the section is the zero map, the Gram on that image has determinant one, and the same identities reduce to $R = X$. No empty present support is deleted.

The new formalization in draft PR32 checks these finite identities and the original observation-iterate and residue-annihilator maps. Its checked implementation is

8e78bc7c240b04d297ade6afdadfd863e0c6db7b.

The later source and status head is

9cec8482f2ecf978cf8bdb67b4c080b9dd74f5d6.

AKJ6–11 evaluates the residue inverse in every actual primary-jet coefficient, giving the exact map to those checked declarations. The analytic AKS estimate, the conductor metric estimates, and the polynomial-gcd dimension corollary have the full written proofs stated here or in their supplied sources; they are outside that Lean run.

27 The finite conductor inverse in the two original Gamma norms

Current refinement. The finite estimates in this chapter remain valid. RUI5–15 and CEP29–49 below sharpen their exterior consequence: complementary singular values and the exact source quotient give an $o(kq)$ conductor error at all four original endpoints, for a fixed actual period. The full original constants, moments and low-degree graph are retained in those proofs.

Retain AKJ12–25 and the complete OCF conductor coefficient polynomial. Thus $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $0 < \delta < 1/2$, $\gamma > 2$, and the original period belongs to the proved five-orbit locus. Its fixed 4 by 4 matrix $\mathbf{H}$ determines the four quadrics and all 81 coefficients a_{rs} of their full product. They remain the same coefficients as the tensor degree k varies. Put

$$b_{rs} = 4 + (2r - 8)\delta + i(2s - 8)\gamma, \quad \mu_j = \sum_{r,s=0}^8 a_{rs} b_{rs}^j, \quad v = \min\{j : \mu_j \neq 0\} \leq 80. \quad (\text{AIE1})$$

The existence and bound for v are the full distinct-node Vandermonde argument of AKJ13. Throughout, $0 \leq D \leq q - 1$ denotes the polynomial degree cutoff. Every norm below is the original Gamma norm of source order $s \in \{1, k\}$, with

$$b = s/2, \quad M = (2\pi)^{s/2}, \quad h_n = Mn!(b)_n, \quad (\text{AIE2})$$

$$\varphi_n = p_n/\sqrt{h_n}, \quad w_n = n!/(b)_n, \quad \|p\|_{s,c} = \|p(c + iy)\|_{m_0,s}.$$

The centre c is displayed wherever it changes. No source order, source mass, coefficient, or period is assigned a replacement value.

27.1 A coefficient-controlled disk for the literal finite symbol

The exact symbol used in AKJ14 and AKJ17 is

$$B_q(z) = 1 + \sum_{j=v+1}^{q-1} \frac{v! \mu_j}{j! \mu_v} z^{j-v}. \quad (\text{AIE3})$$

The sum is empty if its lower index exceeds its upper index. Define the positive, original-data constants

$$A_*^{\text{abs}} = \sum_{r,s} |a_{rs}|, \quad B_*^{\text{sh}} = \max_{r,s} |b_{rs}|, \quad C_v = \frac{A_*^{\text{abs}} (B_*^{\text{sh}})^v}{|\mu_v|}, \quad (\text{AIE4})$$

$$R_0 = \frac{1}{B_*^{\text{sh}}} \min \left\{ 1, \frac{v+1}{2eC_v} \right\}, \quad \rho_0 = \tanh R_0.$$

Here $A_*^{\text{abs}} > 0$, $B_*^{\text{sh}} \geq 4$, $C_v \geq 1$, $R_0 > 0$, and $0 < \rho_0 < 1$. The superscripts distinguish these constants from the original conductor polynomial and from every earlier radius or pivot coefficient. The exact moment μ_v , including any cancellation in it, remains in the denominator.

For every $|z| \leq R_0$ one has

$$|B_q(z) - 1| \leq \frac{v!}{|\mu_v|} \sum_{j=v+1}^{q-1} \frac{|\mu_j|}{j!} |z|^{j-v} \leq C_v \frac{B_*^{\text{sh}} |z|}{v+1} \exp(B_*^{\text{sh}} |z|) \leq \frac{1}{2}. \quad (\text{AIE5})$$

To prove the middle inequality, use $|\mu_j| \leq A_*^{\text{abs}} (B_*^{\text{sh}})^j$, write $j = v+1+h$, and apply $v!/(v+1+h)! \leq 1/((v+1)h!)$. Summation of the exponential series gives the displayed bound. Definition AIE4 gives $B_*^{\text{sh}} R_0 \leq 1$ and $C_v B_*^{\text{sh}} R_0/(v+1) \leq 1/(2e)$, proving the last inequality. The first upper bound in AIE5 retains every individual finite moment and may be used directly for a larger verified radius. The explicit R_0 proves a positive disk for every actual coefficient chart; no uniform lower bound for $|\mu_v|$ over varying periods is asserted.

The analytic function

$$F_q(z) = \frac{1}{B_q(-i \arctan z)} = \sum_{a=0}^{\infty} f_{q,a} z^a \quad (\text{AIE6})$$

is well defined on a neighbourhood of the closed disk $|z| \leq \rho_0$, and $|F_q(z)| \leq 2$ there. Indeed the power series of $\arctan$ gives $|\arctan z| \leq \operatorname{arctanh} |z| \leq R_0$; AIE5 gives $|B_q(-i \arctan z)| \geq 1/2$. Compactness gives an open neighbourhood with nonzero denominator. Cauchy's coefficient formula therefore proves

$$f_{q,0} = 1, \quad |f_{q,a}| \leq 2\rho_0^{-a} \quad (a \geq 1). \quad (\text{AIE7})$$

Although the literal symbol B_q changes with q , the disk and its bound depend only on the original fixed conductor and shifts.

27.2 The exact coefficient map and its norm

Let $\mathcal{P}_D$ denote the polynomials in y of degree at most D , with the orthonormal basis in AIE2. Define the lowering map

$$L_s \varphi_0 = 0, \quad L_s \varphi_n = \sqrt{\frac{w_n}{w_{n-1}}} \varphi_{n-1} \quad (1 \leq n \leq D). \quad (\text{AIE8})$$

Its powers have exactly the entries $L_s^a \varphi_n = \sqrt{w_n/w_{n-a}} \varphi_{n-a}$ for $a \leq n$ and vanish for $a > n$. Thus $L_s^{D+1} = 0$. AKJ21 gives the polynomial identity $\partial_y = \arctan L_s$, with every power above degree D zero. For the coordinate pullback $\mathcal{C}_c p(y) = p(c + iy)$, the chain rule is $\mathcal{C}_c \partial_S \mathcal{C}_c^{-1} = -i \partial_y$. Consequently the inverse of the literal finite operator is

$$\begin{aligned} \mathcal{C}_c B_q(\partial_S)^{-1} \mathcal{C}_c^{-1} &= F_q(L_s), \\ F_q(L_s) \varphi_n &= \sum_{a=0}^n f_{q,a} \sqrt{\frac{w_n}{w_{n-a}}} \varphi_{n-a}. \end{aligned} \quad (\text{AIE9})$$

For proof, evaluate the analytic power-series identity $B_q(-i \arctan z) F_q(z) = 1$ in the nilpotent matrix L_s . All relevant coefficients are finite, so multiplication is an exact finite polynomial identity. This also proves that AIE9 is the same inverse as the finite nilpotent Neumann sum in AKJ14. The factor $-i$ is forced by the original coordinate map and has been retained.

Define the exact weight ratio and the resulting bound

$$\begin{aligned} \alpha_{D,s} &= \max_{0 \leq m \leq n \leq D} \sqrt{\frac{w_n}{w_m}}, \\ N_{D,s} &= \alpha_{D,s} \left(1 + 2 \sum_{a=1}^D \rho_0^{-a} \right) \leq \frac{2\alpha_{D,s}}{1 - \rho_0} \rho_0^{-D}. \end{aligned} \quad (\text{AIE10})$$

Every absolute row sum and every absolute column sum of AIE9 is at most $\alpha_{D,s} \sum_{a=0}^D |f_{q,a}|$. Cauchy–Schwarz in each row followed by the column bound proves that the operator norm is at most this number; AIE7 gives $N_{D,s}$. It follows that, at every unchanged centre c ,

$$\|B_q(\partial_S)^{-1} p\|_{s,c} \leq N_{D,s} \|p\|_{s,c}. \quad (\text{AIE11})$$

When $s = k$, $b = k/2 \geq 1$ and $w_n/w_{n-1} = n/(b+n-1) \leq 1$, so $\alpha_{D,k} = 1$. When $s = 1$, the same ratio is greater than one, so $\alpha_{D,1} = \sqrt{D!/(1/2)_D}$; for $D \geq 1$, the exact factor estimate of AKJ23 yields $\alpha_{D,1} \leq \sqrt{2} D^{1/4}$. At $D = 0$, both values are one. In particular

$$\log \|B_q(\partial_S)^{-1}\|_{s,c} \leq D \log(\coth R_0) + \log \frac{2\alpha_{D,s}}{1 - \tanh R_0} = O_{H,\delta,\gamma}(D + \log(D+1)). \quad (\text{AIE12})$$

For the fixed original quartet and period, $q = (k+1)^2$ and $D \leq q-1$ imply that this logarithm divided by kq tends to zero. This estimate holds for both original source orders at once; its constants explicitly retain the actual μ_v .

27.3 The infinite symbol differs by an exact low-degree map

To record precisely the truncation in AIE3, define the entire function

$$B_\infty(z) = \frac{v!}{\mu_v z^v} \sum_{r,s} a_{rs} e^{b_{rs} z}, \quad B_\infty(0) = 1. \quad (\text{AIE13})$$

Its apparent singularity is removable by the definition of v . On polynomials of degree at most $D \leq q - 1$, set $E_q = B_\infty(\partial_S) - B_q(\partial_S)$; its exact expression is

$$E_q = \sum_{n=q-v}^D \frac{v! \mu_{n+v}}{(n+v)! \mu_v} \partial_S^n. \quad (\text{AIE14})$$

An empty sum is zero. Its image has degree at most $D - q + v \leq v - 1$. Both symbols have constant term one, hence both are invertible in this finite polynomial algebra. Their inverses obey

$$B_q(\partial_S)^{-1} - B_\infty(\partial_S)^{-1} = B_q(\partial_S)^{-1} E_q B_\infty(\partial_S)^{-1}, \quad (\text{AIE15})$$

whose image is contained in $\mathbb{C}[S]_{<v}$ and whose rank is at most v . Equation AIE15 follows by multiplying out its right side. The degree statement holds because every factor is a power series in the degree-lowering derivative. Thus the infinite-symbol formula can be related to the literal AKJ inverse by the displayed exact correction; it cannot replace that inverse without this map.

27.4 An explicit right inverse of the original conductor

Assume now $D \geq v$. Define the raising map, on the degree ranges where its image lies in $\mathcal{P}_D$, by

$$J_s \varphi_n = \sqrt{\frac{w_n}{w_{n+1}}} \varphi_{n+1} = \sqrt{\frac{b+n}{n+1}} \varphi_{n+1}. \quad (\text{AIE16})$$

Then $L_s^v J_s^v = I$ on $\mathcal{P}_{D-v}$ and $\|J_s^v\| \leq \max\{1, b^{v/2}\}$. The latter follows from the displayed coefficient, at each raising step, by $(b+n)/(n+1) \leq \max\{1, b\}$.

The analytic function $G(z) = z/\arctan z$, with $G(0) = 1$, satisfies $|G(z)| \leq 2$ on $|z| < 1$. Indeed

$$\frac{\arctan z}{z} = \int_0^1 \frac{dt}{1+t^2 z^2}, \quad \operatorname{Re} \frac{1}{1+u} > \frac{1}{2} \quad (|u| < 1). \quad (\text{AIE17})$$

The real-part inequality follows from $2 \operatorname{Re}(1+\bar{u}) > |1+u|^2$, which reduces to $1 > |u|^2$. It proves that the integral is nonzero with real part greater than $1/2$, including the removable value at zero. Every Taylor coefficient of $G(z)^v$ is therefore bounded by 2^v : apply Cauchy's formula on radius $t < 1$ and let t increase to one for each fixed coefficient.

Define a map between the specified degree spaces by

$$R_{v,s} = G(L_s)^v J_s^v : \mathcal{P}_{D-v} \longrightarrow \mathcal{P}_D. \quad (\text{AIE18})$$

Since all functions of L_s commute, AIE8 and the power-series identity $(\arctan z)^v G(z)^v = z^v$ give $\partial_y^v R_{v,s} = L_s^v J_s^v = I$. This proves the right-inverse identity even at the top allowed degree. The same row and column argument as for AIE10, now with every coefficient bounded by 2^v , gives

$$\|R_{v,s}\| \leq 2^v (D+1) \alpha_{D,s} \max\{1, b^{v/2}\}. \quad (\text{AIE19})$$

When $v = 0$ the map is exactly the identity; AIE19 remains a valid upper bound. Define the corresponding map on polynomials in the original S coordinate by

$$\mathcal{R}_{v,s,c} = \mathcal{C}_c^{-1} i^v R_{v,s} \mathcal{C}_c. \quad (\text{AIE20})$$

The chain rule in AIE9 proves $\partial_S^v \mathcal{R}_{v,s,c} = I$, with exactly the factor $(-i)^v i^v = 1$. This is a map in the original polynomial algebra; the source norm only specifies its coefficients through the explicitly given original orthonormal basis.

Let $c_k = k/2$ and $c' = c_{k-8} = k/2 - 4$. The full conductor of AKJ13 has the following right inverse on $\mathbb{C}[S']_{\leq D-v}$:

$$\mathcal{R}_{A,s}g = \frac{v!}{\mu_v} B_q(\partial_S)^{-1} \mathcal{R}_{v,s,c'}g, \quad \mathcal{T}_A \mathcal{R}_{A,s}g = g. \quad (\text{AIE21})$$

To check it, use the exact finite factorization $\mathcal{T}_A = (\mu_v/v!) \partial_S^v B_q(\partial_S)$ from AKJ14 on degree at most D . The two inverse factors cancel and AIE20 gives the stated identity. In the identity the output variable is S' ; the differential expression uses the same numerical polynomial coefficients, so the domain and codomain are as displayed.

Set $H_D = \sum_{j=1}^D 1/j$, $H_0 = 0$, and

$$C_{A,D,s} = \frac{v!}{|\mu_v|} e^{4(H_D+2)} N_{D,s} 2^v (D+1) \alpha_{D,s} \max\{1, b^{v/2}\}. \quad (\text{AIE22})$$

Then the original-centre norm estimate is

$$\|\mathcal{R}_{A,s}g\|_{s,c_k} \leq C_{A,D,s} \|g\|_{s,c'}. \quad (\text{AIE23})$$

Indeed AIE11 and AIE19 first prove the bound without the exponential factor, with both norms at c' . For any output polynomial p , $p(c_k + iy) = p(c' + i(y - 4i))$, so AKJ22 applied to the exact shift $-4i$ supplies the remaining factor. Thus the four-unit centre difference is retained in the typed norm map. For the fixed actual quartet and period, $v \leq 80$ and $b \in \{1/2, k/2\}$ give

$$\log C_{A,D,s} \leq D \log(\coth R_0) + O_{H,\delta,\gamma}(\log(D+1) + \log(k+1)), \quad \frac{\log C_{A,D,s}}{kq} \longrightarrow 0. \quad (\text{AIE24})$$

Every factor in the first assertion is written in AIE22; the asymptotic statement follows from $D \leq (k+1)^2 - 1$ and the fixed finite v . No varying-period bound is inferred from this statement.

For comparison with the exact monomial lift $\mathcal{J}_v$ of AKJ17, put $L_g = \mathcal{R}_{v,s,c'}g - \mathcal{J}_v g$. Both terms have v th derivative g , so $L_g \in \mathbb{C}[S]_{<v}$. More explicitly,

$$L_g = \sum_{a=0}^{v-1} [S^a](\mathcal{R}_{v,s,c'}g) S^a, \quad \mathcal{R}_{A,s}g - \frac{v!}{\mu_v} B_q(\partial_S)^{-1} \mathcal{J}_v g = \frac{v!}{\mu_v} B_q(\partial_S)^{-1} L_g. \quad (\text{AIE25})$$

The first identity uses the exact zero coefficients of $\mathcal{J}_v g$ in degrees below v . This proves the complete degree-below- v correction between the two lifts, rather than discarding their different choices of integration constants.

27.5 The exact quotient and exterior transfer supplied by this bound

The kernel of $\mathcal{T}_A$ on degree at most $D \geq v$ is exactly $\mathbb{C}[S]_{<v}$ by AKJ14. Give the quotient $\mathbb{C}[S]_{\leq D}/\mathbb{C}[S]_{<v}$ its Hilbert quotient norm from the original source at c_k , and give its target $\mathbb{C}[S']_{\leq D-v}$ the original source at c' . The induced conductor map $\overline{\mathcal{T}}_A$ is an isomorphism, because AIE21 proves surjectivity and the displayed kernel proves injectivity. Define the full upper norm from AKJ25 by

$$N_{A,D} = \sum_{r,s} |a_{rs}| e^{|b_{rs}-4|(H_D+2)}. \quad (\text{AIE26})$$

For every target g , its minimum-norm preimage class obeys

$$\frac{\|g\|_{s,c'}}{N_{A,D}} \leq \|\overline{\mathcal{T}}_A^{-1} g\|_{\text{quot}} \leq C_{A,D,s} \|g\|_{s,c'}. \quad (\text{AIE27})$$

For the left inequality, apply AKJ25 to every preimage and take the infimum. For the right inequality, use the actual preimage AIE21. The infimum is attained since the omitted subspace is a closed finite-dimensional subspace. These arguments prove the norm statement with the specified quotient map, rather than assigning the coefficient norm to the quotient.

In particular AIE21 gives the exact decomposition

$$V_A = \mathbb{C}[S]_{<v} \oplus \mathcal{R}_{A,s}(\chi_{k-8}\mathbb{C}[S']_{\leq D-v-q'}), \quad q' = (k-7)^2, \quad (\text{AIE28})$$

where a negative degree means the zero space and V_A here denotes the AKJ15 space restricted to degree at most D . Indeed the image under $\mathcal{T}_A$ of a member of V_A is precisely a multiple of χ_{k-8} in the displayed degree range. Subtracting its lift leaves the exact kernel. The directness follows by applying $\mathcal{T}_A$. Equation AIE25 identifies this decomposition with the earlier monic basis, including its low-degree correction.

For any t -dimensional target subspace and any basis column matrix X in its original orthonormal coordinates, AIE27 implies the full Gram determinant inequalities

$$N_{A,D}^{-2t} \det(X^* X) \leq \det(X^* (\overline{\mathcal{T}}_A^{-1})^* \overline{\mathcal{T}}_A^{-1} X) \leq C_{A,D,s}^{2t} \det(X^* X). \quad (\text{AIE29})$$

For proof, AIE27 is a two-sided positive Hermitian matrix inequality after congruence by X . Conjugating by $(X^* X)^{-1/2}$ gives eigenvalues between the two scalar squared bounds, whose product is the displayed determinant ratio. This proves the exact exterior transfer and all powers of the norm constants.

Similarly, for any t -plane $U \subset \mathbb{C}[S]_{\leq D}$, AIE11 gives $\|\bigwedge^t(B_q(\partial_S)^{-1}|_U)\| \leq N_{D,s}^t$. Although $B_q(\partial_S)$ is triangular with every diagonal entry one and hence determinant one on the full polynomial space, that full-space determinant does not cancel its restriction to an arbitrary subspace. The exact restricted determinant is $\det(X^*(B_q^{-1})^* B_q^{-1} X) / \det(X^* X)$ for a basis matrix X of that subspace; this is the map to which AIE11 applies.

For the original kernel rank $r = 8k - 16$ and $D \asymp q$, the upper logarithm in these exterior bounds is $O_{H,\delta,\gamma}(rq) = O_{H,\delta,\gamma}(kq)$. Therefore AIE12 and AIE24 prove a sub- kq logarithmic bound for each operator, while AIE29 retains the rank factor required for volumes. They provide the original conductor inverse, its full low-degree comparison and its quotient metric control. A sub- kq error for the actual rank- r four-endpoint determinant is not a consequence of these scalar bounds; that remaining calculation must use the actual restricted matrices and the existing endpoint correlation.

28 Subleading exterior control from the actual finite conductor

Retain the original conductor of AKJ12–25 and AIE1–29. In particular the quartet, full unit, original period and branch are fixed in the proved five-orbit domain, $m = 1$, $k = 4l + 1 \geq 9$, and $q = (k + 1)^2$. All 81 coefficients a_{rs} are those of the actual product of four period quadrics. They are fixed as k varies. Retain

$$b_{rs} = 4 + (2r - 8)\delta + i(2s - 8)\gamma, \quad \mu_j = \sum_{r,s=0}^8 a_{rs} b_{rs}^j, \quad v = \min\{j : \mu_j \neq 0\} \leq 80, \quad (\text{RUI1})$$

$$A_{\text{abs}} = \sum_{r,s} |a_{rs}|, \quad B_{\text{sh}} = \max_{r,s} |b_{rs}|, \quad C_v = \frac{A_{\text{abs}} B_{\text{sh}}^v}{|\mu_v|} \geq 1.$$

No coefficient or moment denominator is replaced by a generic value. The finite bound for v is AKJ13's distinct-node Vandermonde proof. The inequality $C_v \geq 1$ follows from the triangle inequality applied to the actual v th moment. The source measure at order $s \in \{1, k\}$ is exactly

$$dm_{0,s}(y) = (2\pi)^{s/2} \frac{2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad \|p\|_{s,c} = \|p(c + iy)\|_{L^2(m_{0,s})}. \quad (\text{RUI2})$$

Every centre appearing below is stated, and the whole mass remains in the source norm. Put $H_D = \sum_{j=1}^D 1/j$, $H_0 = 0$.

28.1 The forward polynomial bound and exact determinant

For an integer cutoff $L \geq v + 1$, define the literal finite symbol

$$B_L(z) = 1 + \sum_{a=1}^{L-v-1} \frac{v! \mu_{v+a}}{(v+a)! \mu_v} z^a. \quad (\text{RUI3})$$

The empty sum is zero. This includes the actual coefficient symbol B_q of AKJ14 and the source symbol B_{N+1} needed at a larger source cutoff N . For every displayed coefficient, including the constant coefficient, one has

$$\left| \frac{v! \mu_{v+a}}{(v+a)! \mu_v} \right| \leq C_v \frac{v! B_{\text{sh}}^a}{(v+a)!} \leq C_v \frac{B_{\text{sh}}^a}{a!}. \quad (\text{RUI4})$$

The first inequality is the full moment bound $|\mu_{v+a}| \leq A_{\text{abs}} B_{\text{sh}}^{v+a}$. The second follows from $(v+a)! \geq v! a!$.

On $\mathbb{C}[S]_{\leq D}$ with the norm RUI2, the original coordinate pullback sends ∂_S to $-i\partial_y$. AKJ21–23 proves $\|\partial_S\| \leq H_D + 2$, uniformly in both original source orders. Applying RUI4 term by term to the finite operator polynomial gives

$$\|B_L(\partial_S)\|_{s,c} \leq M_D, \quad M_D = C_v \exp(B_{\text{sh}}(H_D + 2)) \geq 1. \quad (\text{RUI5})$$

Indeed each power has norm at most $(H_D + 2)^a$; extending this nonnegative finite sum to the exponential series proves the displayed bound. The constant term is one and every other term strictly lowers degree. Thus, on this exact $(D + 1)$ -dimensional polynomial space,

$$\det B_L(\partial_S) = 1. \quad (\text{RUI6})$$

This is its determinant in the original increasing-power basis, and therefore in every basis of the same source space, including the original orthonormal Gamma basis. The assertion preserves the source measure; it is not a rescaling of this determinant.

28.2 Complementary singular values retain the full exterior norm

Let $n = D + 1$ and $T = B_L(\partial_S)$ acting on this same Hilbert space. Write its singular values as $\sigma_1 \geq \dots \geq \sigma_n > 0$. RUI6 gives $\prod_i \sigma_i = 1$. For every $0 \leq t \leq n$,

$$\begin{aligned} \left\| \bigwedge^t T^{-1} \right\| &= \prod_{j=n-t+1}^n \sigma_j^{-1} = \prod_{j=1}^{n-t} \sigma_j = \left\| \bigwedge^{n-t} T \right\| \leq M_D^{n-t}, \\ \left\| \bigwedge^t T \right\| &\leq M_D^t. \end{aligned} \tag{RUI7}$$

For a proof of the exterior norm formula, the singular-value decomposition expresses T as two unitary maps and a diagonal map with these positive entries. Exterior powers of the unitary maps preserve the induced norm; the diagonal exterior map has entries equal to the products over all t -subsets. Its largest entry is the product of the largest t singular values. Applying this to T^{-1} and using the full determinant proves every equality in RUI7. Empty products cover $t = 0, n$.

Consequently, for any actual t -dimensional subspace and any full-rank basis column matrix X in the original orthonormal source coordinates,

$$M_D^{-2t} \det(X^* X) \leq \det(X^* (T^{-1})^* T^{-1} X) \leq M_D^{2(n-t)} \det(X^* X). \tag{RUI8}$$

For the upper inequality apply the first exterior norm of RUI7 to the decomposable exterior vector of the columns of X . Its squared norm is their Gram determinant. For the lower inequality, put $Y = T^{-1}X$ and apply the second bound of RUI7 to $TY = X$. This proves the lower bound on this same subspace without requiring that it be invariant. Every original off-diagonal Gram entry is retained by the displayed matrices.

In particular, for fixed original period and quartet, uniformly in $s \in \{1, k\}$, every $0 \leq D \leq 2q$ and every allowed t and X ,

$$\left| \log \frac{\det(X^* (T^{-1})^* T^{-1} X)}{\det(X^* X)} \right| \leq 2(D+1)(\log C_v + B_{\text{sh}}(H_D + 2)) = O_{\text{actual}}(q \log(q+1)) = o(kq). \tag{RUI9}$$

The harmonic estimate $H_D \leq 1 + \log D$ for $D \geq 1$ follows by integrating $1/x$ on consecutive intervals; $D = 0$ is immediate. Since $q = (k+1)^2$, the displayed upper bound divided by kq tends to zero, independently of D, t, X . This is a uniform estimate over the actual subspaces, not a presumed favourable position of the kernel. Its constants are fixed only after fixing the original period and quartet; no uniform bound on $1/|\mu_v|$ over moving periods is asserted. The same inequalities for T follow by exchanging the image subspace and its inverse image.

RUI9 improves the rank-multiplied estimate at the end of AIE29. The earlier estimate is valid, but its factor t on a single inverse operator bound is unnecessary. The complementary exterior identity uses the exact full determinant and supplies the sharper bound on the same original matrices.

28.3 The actual conductor quotient inherits the sharper volume control

Let $D \geq v$ and use the literal source truncation $L = D + 1$. Taylor's finite formula gives, on degree at most D ,

$$\mathcal{T}_A = \sum_{j=v}^D \frac{\mu_j}{j!} \partial_S^j = \frac{\mu_v}{v!} \partial_S^v B_{D+1}(\partial_S), \quad \ker \mathcal{T}_A = \mathbb{C}[S]_{<v}. \tag{RUI10}$$

The first equality is exactly the translation operator in AKJ13; all lower moments vanish by their proved definition. The factor $B_{D+1}(\partial_S)$ is invertible and preserves each degree filtration, so the last equality follows from the derivative kernel.

Use the original source centre $c_k = k/2$ and target centre $c' = c_{k-8} = k/2 - 4$, both with source order s . AIE16–20 gives the full right inverse $\mathcal{R}_{v,s,c'}$ of ∂_S^v with norm at most

$$P_{D,s} = 2^v(D+1)\alpha_{D,s} \max\{1, (s/2)^{v/2}\}, \quad \alpha_{D,s} = \max_{0 \leq m \leq n \leq D} \sqrt{\frac{n!/(s/2)_n}{m!/(s/2)_m}}. \quad (\text{RUI11})$$

In particular $\alpha_{D,k} = 1$, while $\alpha_{D,1} \leq \sqrt{2} \max\{1, D^{1/4}\}$; the full derivation and the phase i^v are the original AIE16–20 construction. The exact conductor right inverse is

$$\mathcal{R}_{A,s,D} = \frac{v!}{\mu_v} B_{D+1}(\partial_S)^{-1} \mathcal{R}_{v,s,c'}. \quad (\text{RUI12})$$

Multiplying by RUI10 proves $\mathcal{T}_A \mathcal{R}_{A,s,D} = I$. The numerical polynomial coefficients use the original source and target variables, as in AIE21. The centre change from c' to c_k is the original shift $y \mapsto y - 4i$, of norm at most $\exp(4(H_D + 2))$ by AKJ22.

Give the quotient $\mathbb{C}[S]_{\leq D}/\mathbb{C}[S]_{<v}$ its minimum norm from the source at c_k . The induced conductor $\overline{\mathcal{T}}_A$ is an isomorphism onto $\mathbb{C}[S']_{\leq D-v}$ with its source norm at c' . Define

$$L_{D,s} = \frac{v!}{|\mu_v|} e^{4(H_D+2)} P_{D,s}, \quad N_{A,D} = \sum_{r,s} |a_{rs}| e^{|b_{rs}-4|(H_D+2)}. \quad (\text{RUI13})$$

For any t -dimensional target subspace, with original basis matrix X , its exact quotient Gram ratio satisfies

$$N_{A,D}^{-2t} \det(X^* X) \leq \det(X^* (\overline{\mathcal{T}}_A^{-1})^* \overline{\mathcal{T}}_A^{-1} X) \leq L_{D,s}^{2t} M_D^{2(D+1-t)} \det(X^* X). \quad (\text{RUI14})$$

To prove the upper bound, use the actual lift RUI12. Apply $\mathcal{R}_{v,s,c'}$ to the t basis columns, retaining its specified image subspace. RUI7 bounds the exterior power of the inverse B_{D+1} on that image by M_D^{D+1-t} . The scalar, right-inverse norm, and centre translation give $L_{D,s}^t$. Projection to the minimum lift in the quotient is an orthogonal projection on the same source Hilbert space, so its exterior norm is at most one. This proves the complete upper Gram inequality. For the lower bound, apply the forward original conductor to the minimum lifts. AKJ24–25 gives its norm $N_{A,D}$; taking its t th exterior power proves the result. This proof retains the coupled Gram and the full quotient metric.

Every factor in RUI13 is explicit in the fixed original data. For the asymptotic assertion restrict now to $v \leq D \leq 2q$. Since $v \leq 80$, $s \in \{1, k\}$ and $t \leq D+1-v \leq 2q+1$, both logarithmic endpoints of RUI14 have absolute value at most

$$2t(|\log L_{D,s}| + |\log N_{A,D}|) + 2(D+1) \log M_D = O_{\text{actual}}(q \log(q+1) + q \log(k+1)) = o(kq). \quad (\text{RUI15})$$

For the logarithm of $N_{A,D}$, use $A_{\text{abs}} \leq N_{A,D} \leq A_{\text{abs}} \exp(\max |b_{rs} - 4|(H_D + 2))$. For $L_{D,s}$, its displayed factors and the positive lower bound $P_{D,s} \geq 1$ give the same logarithmic estimate, retaining $v!/|\mu_v|$. This proves the absolute bounds even if either constant is below one. The harmonic bound and $q = (k+1)^2$ then prove the final limit. No dimension- q scalar comparison has silently acquired an extra factor of q in this argument.

At the original coefficient cutoff $D = q - 1$, RUI14 is the sharpened version of AIE29, on precisely the same conductor quotient and original coefficient subspaces. At a larger source cutoff $D = N$, its literal symbol is B_{N+1} , and RUI10 proves the exact extension. The old symbol B_q cannot be substituted without recording the finite difference. Passing this estimate through the original remainder square requires the actual image of the relation ideal; that square, its image graph, and the endpoint quotient metrics are retained in the succeeding calculation.

29 The original conductor at every source endpoint

Retain the complete simple-quartet conductor of OCF and AKJ, its actual original period matrix, and $k = 4l + 1 \geq 9$ on the proved five-orbit locus. Write

$$\begin{aligned} q &= (k+1)^2, & k' &= k-8, & q' &= (k-7)^2, & \Delta &= q - q' = 16k - 48, \\ c &= k/2, & c' &= k/2 - 4, & \beta_{ij} &= c + (2i - k)\delta + i(2j - k)\gamma, \\ & & \beta'_{ab} &= c' + (2a - k')\delta + i(2b - k')\gamma, \\ A &= \sum_{r,s=0}^8 a_{rs} e_{rs}^{(8)} \neq 0, & b_{rs} &= 4 + (2r - 8)\delta + i(2s - 8)\gamma, \\ \mu_j &= \sum_{r,s} a_{rs} b_{rs}^j, & v &= \min\{j : \mu_j \neq 0\} \leq 80. \end{aligned} \tag{CEP1}$$

The factor i before an imaginary term is the imaginary unit; subscript indices are integers in their displayed ranges. The original polynomials are $\chi_k(S) = \prod_{i,j=0}^k (S - \beta_{ij})$ and $\chi_{k'}(S') = \prod_{a,b=0}^{k'} (S' - \beta'_{ab})$. Both have distinct roots. Set $E_k = \mathbb{C}[S]/(\chi_k)$ and $E_{k'} = \mathbb{C}[S']/(\chi_{k'})$. Every source below has the original order $s \in \{1, k\}$ and full mass $M_s = (2\pi)^{s/2}$, with $b = s/2$ and $h_n = M_s n! (b)_n$. In particular the lower grid does not change source order k to $k - 8$.

29.1 The full source, remainder and primary-coordinate square

For every polynomial define the original finite-translation map

$$(\mathcal{T}_A p)(S') = \sum_{r,s=0}^8 a_{rs} p(S' + b_{rs}). \tag{CEP2}$$

For $0 \leq a, b \leq k'$ and $0 \leq r, s \leq 8$, direct substitution gives $\beta'_{ab} + b_{rs} = \beta_{a+r, b+s}$. All indices on the right lie between zero and k . Consequently

$$(\mathcal{T}_A(\chi_k f))(\beta'_{ab}) = 0 \quad \text{for every polynomial } f. \tag{CEP3}$$

Division by the monic $\chi_{k'}$ and evaluation at its distinct roots then show $\chi_{k'} \mid \mathcal{T}_A(\chi_k f)$. Thus CEP2 induces the specified complex-linear map $\bar{\mathcal{T}}_A : E_k \rightarrow E_{k'}$.

In the original ordered primary coordinates its matrix is

$$C_{(a,b),(i,j)} = \begin{cases} a_{i-a, j-b}, & 0 \leq i-a, j-b \leq 8, \\ 0, & \text{otherwise.} \end{cases} \quad C \in \text{Mat}_{q' \times q}(\mathbb{C}). \tag{CEP4}$$

Evaluation of CEP2 proves this entry formula, including its absence of conjugation or binomial coefficients. Its ordinary transpose $C^\top$ is the exact biform coefficient multiplication by the nonzero polynomial A . That multiplication is injective because the biform ring embeds in a polynomial integral domain. Therefore C has rank q' and is surjective. With the original ordered Vandermonde matrices $V_\beta, V_{\beta'}$, the polynomial coordinate matrix is exactly $V_{\beta'}^{-1} C V_\beta$. The kernel is the AKJ15 space

$$V_A = \{[p] \in E_k : \chi_{k'} \mid \mathcal{T}_A p\}, \quad \dim V_A = \Delta. \tag{CEP5}$$

CEP3 proves independence of the representative, and CEP4 proves the dimension. The original common kernel $K = \ker \Lambda$ is contained in V_A by the full conductor inclusion OCF17.

For every endpoint $N \in \{q-1, q, 2q-1, 2q\}$, put $d = N - v$ and define

$$\begin{aligned} \mathcal{P}_N &= \mathbb{C}[S]_{\leq N}, & \mathcal{P}'_d &= \mathbb{C}[S']_{\leq d}, \\ U_N : \mathcal{P}_N &\rightarrow E_k, & p &\mapsto [p], & V_d : \mathcal{P}'_d &\rightarrow E_{k'}, & g &\mapsto [g]. \end{aligned}$$

The top horizontal map is precisely $\mathcal{T}_{A,N} : \mathcal{P}_N \longrightarrow \mathcal{P}'_{N-v}$. Taylor's finite identity and the first nonzero moment show that CEP2 lowers every degree at least v by exactly v and kills exactly $\mathbb{C}[S]_{<v}$. The exact commutative square is

$$\begin{array}{ccc} \mathcal{P}_N & \xrightarrow{\mathcal{T}_{A,N}} & \mathcal{P}'_{N-v} \\ U_N \downarrow & & \downarrow V_{N-v} \\ E_k & \xrightarrow{\bar{\mathcal{T}}_A} & E_{k'}. \end{array} \quad (\text{CEP6})$$

The lower remainder is surjective because $N - v \geq q - 1 - v \geq q' - 1$; the latter inequality is $\Delta \geq v$, and $\Delta \geq 96 > 80 \geq v$.

The map is complex-linear with its following full action defect. If $M_S, M_{S'}$ denote the original multiplication operators, then

$$CM_S - M_{S'}C = C_b, \quad (C_b)_{(a,b),(i,j)} = \begin{cases} a_{i-a,j-b}b_{i-a,j-b}, & 0 \leq i-a, j-b \leq 8, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{CEP7})$$

Indeed each entry is multiplied by $\beta_{ij} - \beta'_{ab} = b_{i-a,j-b}$. In the original centred coordinates $Y = (M_S - cI)/i$, $Y' = (M_{S'} - c'I)/i$, this becomes $CY - Y'C = (C_b - 4C)/i$. Thus the exact relation to multiplication retains its coefficient defect.

29.2 The literal finite inverse at the larger cutoffs

On $\mathcal{P}_N$, define the literal finite symbol

$$B_{N+1}(z) = 1 + \sum_{j=v+1}^N \frac{v! \mu_j}{j! \mu_v} z^{j-v}, \quad \mathcal{T}_{A,N} = \frac{\mu_v}{v!} \partial_S^v B_{N+1}(\partial_S). \quad (\text{CEP8})$$

The expression on the right is evaluated in the output variable S' . Taylor expansion of each term in CEP2 through degree N proves this identity coefficient by coefficient. Its constant term one makes $B_{N+1}(\partial_S)$ invertible: its difference from the identity lowers degree, so the inverse is its finite nilpotent Neumann sum through power N . A right inverse of ∂_S^v on the polynomial spaces exists by integration of each monomial; composition proves surjectivity of the top map in CEP6.

The relation to the earlier literal B_q of AKJ14 is

$$\begin{aligned} D_{N,q} &= B_{N+1}(\partial_S) - B_q(\partial_S) = \sum_{j=q}^N \frac{v! \mu_j}{j! \mu_v} \partial_S^{j-v}, \\ B_{N+1}(\partial_S)^{-1} - B_q(\partial_S)^{-1} &= -B_{N+1}(\partial_S)^{-1} D_{N,q} B_q(\partial_S)^{-1}, \\ \mathcal{T}_{A,N} - \frac{\mu_v}{v!} \partial_S^v B_q(\partial_S) &= \sum_{j=q}^N \frac{\mu_j}{j!} \partial_S^j. \end{aligned} \quad (\text{CEP9})$$

Empty sums are zero. The first two differences have image contained in $\mathbb{C}[S]_{\leq N-q+v}$; the last has image contained in $\mathbb{C}[S']_{\leq N-q}$. The inverse formula follows by multiplication, and its degree assertion follows since both inverse factors preserve every degree prefix. At $N = q - 1$ all differences vanish. At the other three endpoints these are the full retained defects; their ranks are not assigned the cutoff-independent bound v .

For comparison, the entire symbol $B_\infty(z) = v! \sum_{r,s} a_{rs} e^{b_{rs}z} / (\mu_v z^v)$, with removable value one at zero, obeys at each cutoff

$$B_\infty(\partial_S) - B_{N+1}(\partial_S) = \sum_{n=N+1-v}^N \frac{v! \mu_{n+v}}{(n+v)! \mu_v} \partial_S^n. \quad (\text{CEP10})$$

Its image and the image of the corresponding inverse difference lie in $\mathbb{C}[S]_{<v}$. This follows from the first derivative degree $N + 1 - v$, and from the inverse identity with the same order as AIE15. Thus the rank-at-most- v comparison is with the correctly growing finite symbol in CEP8.

For completeness the original-source inverse estimate extends directly to these literal symbols. Retain the exact constants $A_*^{\text{abs}}, B_*^{\text{sh}}, C_v, R_0, \rho_0$ of AIE4. For every $|z| \leq R_0$ the finite coefficient sum in CEP8 is at most the full absolute moment sum:

$$|B_{N+1}(z) - 1| \leq \frac{v!}{|\mu_v|} \sum_{j=v+1}^N \frac{|\mu_j|}{j!} |z|^{j-v} \leq C_v \frac{B_*^{\text{sh}} |z|}{v+1} e^{B_*^{\text{sh}} |z|} \leq \frac{1}{2}. \quad (\text{CEP11})$$

The middle inequality follows by writing $j = v+1+h$ and using $v!/(v+1+h)! \leq 1/((v+1)h!)$. Hence $F_{N+1}(z) = 1/B_{N+1}(-i \arctan z)$ has coefficient absolute values at most $2\rho_0^{-a}$ on degree $a \geq 1$, and constant coefficient one. This is Cauchy's formula on $|z| = \rho_0$, since $|\arctan z| \leq \operatorname{arctanh} \rho_0 = R_0$.

In the original orthonormal Gamma basis, the lowering matrix $L_s \varphi_n = \sqrt{w_n/w_{n-1}} \varphi_{n-1}$, $w_n = n!/(b)_n$, satisfies $\partial_y = \arctan L_s$ and $L_s^{N+1} = 0$. Evaluation of the power-series inverse in this nilpotent matrix therefore gives the exact finite operator $F_{N+1}(L_s)$, with entries $[z^{n-m}]F_{N+1}(z) \sqrt{w_n/w_m}$. The chain rule $S = c + iy$ supplies the retained $-i$. Bounding the absolute row and column sums gives

$$\|B_{N+1}(\partial_S)^{-1}\|_{s,c} \leq \alpha_{N,s} \left(1 + 2 \sum_{a=1}^N \rho_0^{-a}\right), \quad \alpha_{N,s} = \max_{0 \leq m \leq n \leq N} \sqrt{w_n/w_m}. \quad (\text{CEP12})$$

This proves the larger-cutoff assertion for the actual finite symbol rather than substituting N into an identity for B_q . Likewise the explicit AIE18 raising right inverse on these degree spaces, followed by i^v , $B_{N+1}(\partial_S)^{-1}$ and $v!/\mu_v$, is a right inverse of CEP8. Its comparison with the monomial integral remains exactly the degree-below- v correction in AIE25, now with this literal inverse.

29.3 The exact source operator matrix and its full determinant

Use the unchanged original orthonormal polynomials $u_n(S) = p_n((S - c)/i)/\sqrt{h_n}$ in $\mathcal{P}_N$ and $u'_m(S') = p_m((S' - c')/i)/\sqrt{h_m}$ in $\mathcal{P}'_{N-v}$. Let A_N denote the matrix of $\mathcal{T}_{A,N}$ in these two frames. Define the analytic coefficient series at zero

$$t(z) = \sum_{r,s} a_{rs} \exp\left(\frac{b_{rs} - 4}{i} \arctan z\right) = \sum_{\ell \geq 0} t_\ell z^\ell. \quad (\text{CEP13})$$

Then its exact entries are

$$(A_N)_{m,n} = \begin{cases} t_{n-m} \sqrt{w_n/w_m}, & n \geq m, \\ 0, & n < m, \end{cases} \quad 0 \leq m \leq N - v, \quad 0 \leq n \leq N. \quad (\text{CEP14})$$

To prove this, the original generating identity $\sum p_n(y) z^n/n! = (1 + z^2)^{-b/2} e^{y \arctan z}$ shows that translation by η multiplies the series by $e^{\eta \arctan z}$. Extracting its finite coefficients and dividing by the complete $\sqrt{h_n}$ gives $\sqrt{w_n/w_m}$ after the two copies of M_s cancel at this specified step. Evaluation of CEP2 at $c' + iy$ uses $p(c + i(y + (b_{rs} - 4)/i))$, proving the exact centre shift in CEP13.

For $\ell < v$, the shifted moment $\sum a_{rs} (b_{rs} - 4)^\ell$ is zero by the binomial theorem and the definition of v ; at $\ell = v$ it equals μ_v . Since $\arctan z = z + O(z^3)$, this gives

$$t_0 = \cdots = t_{v-1} = 0, \quad t_v = \frac{(-i)^v \mu_v}{v!} \neq 0. \quad (\text{CEP15})$$

The first v columns of A_N vanish. Its remaining square block $\hat{A}_N$, with columns $v, \dots, N$, is upper triangular with diagonal $t_v \sqrt{w_{m+v}/w_m}$. Thus its rank is $N - v + 1$, its kernel in the source is

exactly the first v orthonormal coordinates, and

$$\begin{aligned}\det \hat{A}_N &= \left(\frac{(-i)^v \mu_v}{v!} \right)^{N-v+1} \prod_{m=0}^{N-v} \sqrt{\frac{w_{m+v}}{w_m}}, \\ \det(A_N A_N^*) &= \left| \frac{\mu_v}{v!} \right|^{2(N-v+1)} \prod_{m=0}^{N-v} \frac{w_{m+v}}{w_m}.\end{aligned}\tag{CEP16}$$

This retains the complex determinant phase and the original moment. The operator norm also has the exact original-source upper bound

$$\|A_N\| \leq M_{A,N} := \sum_{r,s} |a_{rs}| e^{|b_{rs}-4|(H_N+2)}, \quad H_N = \sum_{j=1}^N 1/j.\tag{CEP17}$$

Indeed AKJ22 bounds each exact translation in CEP13 by its displayed factor, and the triangle inequality gives the sum.

There is a closed four-endpoint expression for CEP16. For the signs $\sigma = (1, 1, -1, -1)$ at the ordered endpoints, write $\mathcal{D}_A = \sum_N \sigma_N \log \det(A_N A_N^*)$. Cancellation of the full products gives

$$\begin{aligned}\mathcal{D}_A &= -4q \log \left| \frac{\mu_v}{v!} \right| + \mathcal{W}_{v,s}, \\ \mathcal{W}_{v,s} &= \sum_{a=0}^{v-1} (\log w_{q-1-a} + \log w_{q-a} - \log w_{2q-1-a} - \log w_{2q-a}).\end{aligned}\tag{CEP18}$$

For $v = 0$ the second sum is zero. The product identity used here is $\prod_{m=0}^{N-v} w_{m+v}/w_m = \prod_{a=0}^{v-1} w_{N-a}/\prod_{a=0}^{v-1} w_a$, obtained by cancellation of consecutive products; it remains valid when the two index intervals overlap.

Put $c_b^{\log} = \max\{1, b-1\}$. For every $j \geq 1$ and $b \geq 1/2$,

$$\left| \log \frac{j}{b+j-1} \right| \leq \frac{c_b^{\log}}{j}.\tag{CEP19}$$

For $b \geq 1$ use $\log(1 + (b-1)/j) \leq (b-1)/j$. For $1/2 \leq b < 1$, put $a = 1 - b \leq 1/2$ and use $-\log(1 - a/j) \leq a/(j-a) \leq 1/j$; the last inequality is $aj \leq j-a$, which follows from $a \leq 1/2$ and $j \geq 1$. Summing CEP19 over the two intervals of length q in each summand of CEP18 proves the completely explicit estimate

$$|\mathcal{W}_{v,s}| \leq \frac{2v c_b^{\log} q}{q-v}, \quad |\mathcal{D}_A| \leq 4q \left| \log \left| \frac{\mu_v}{v!} \right| \right| + \frac{2v c_b^{\log} q}{q-v}.\tag{CEP20}$$

All denominators are positive since $q \geq 100 > v$. For either original source and the fixed actual quartet and period, the displayed bound is $O_{\text{actual}}(q+k) = o(kq)$.

29.4 The induced original quotient metric and its correction

Let the matrices of the two remainder maps in the orthonormal source frames and original primary target frames be denoted by the same U_N, V_d as their maps. Explicitly

$$(U_N)_{(i,j),n} = \frac{p_n((\beta_{ij} - c)/i)}{\sqrt{h_n}}, \quad (V_d)_{(a,b),m} = \frac{p_m((\beta'_{ab} - c')/i)}{\sqrt{h_m}}.\tag{CEP21}$$

They have sizes q by $N+1$ and q' by $N-v+1$, respectively. Their first q , respectively q' , columns have nonzero evaluation Vandermonde determinant, so both have full row rank. The square CEP6 is the precise matrix identity

$$CU_N = V_d A_N, \quad R_N = U_N U_N^*, \quad R'_d = V_d V_d^*, \quad G_N = R_N^{-1}, \quad G'_d = (R'_d)^{-1}.\tag{CEP22}$$

Here G_N is the original upper primary quotient metric of OKM8, and G'_d is the lower-grid metric in the same source order s . The induced metric on E_k/V_A , in its specified target coordinates $E_{k'}$, is

$$Q_N = (CG_N^{-1}C^*)^{-1} = (V_d A_N A_N^* V_d^*)^{-1}. \quad (\text{CEP23})$$

To prove the first assertion, the minimum of $x^* G_N x$ under $Cx = y$ is attained at $x = G_N^{-1} C^* (CG_N^{-1} C^*)^{-1} y$. Every other lift differs by an element of $\ker C$ orthogonal to this lift in the G_N metric, proving the minimum and its Gram matrix. The second equality is CEP22.

The exact correction to the lower original Gamma quotient metric is

$$\begin{aligned} W_d &= V_d^* (R'_d)^{-1/2}, \quad W_d^* W_d = I_{q'}, \\ K_{A,N} &= W_d^* A_N A_N^* W_d > 0, \quad Q_N = (R'_d)^{-1/2} K_{A,N}^{-1} (R'_d)^{-1/2}, \\ \log \det Q_N &= \log \det G'_d - \log \det K_{A,N}. \end{aligned} \quad (\text{CEP24})$$

All square roots are the unique positive Hermitian square roots. These formulas follow by multiplying out the definition of $K_{A,N}$ and inverting. In particular the passage between the induced metric and the lower original source metric retains this actual compressed covariance.

If Z_d is any orthonormal basis of $\ker V_d$, completing W_d to a unitary matrix, the further exact determinant factorization is

$$\det K_{A,N} = \det(A_N A_N^*) \det(Z_d^* (A_N A_N^*)^{-1} Z_d). \quad (\text{CEP25})$$

For a proof put $D = A_N A_N^* > 0$ and write its block matrix in the unitary frame $[W_d, Z_d]$. The upper left block is $K_{A,N}$. Block Gaussian elimination gives $\det D = \det K_{A,N} \det(D_{22} - D_{21} K_{A,N}^{-1} D_{12})$. The lower right block of D^{-1} is the inverse of that Schur complement, proving CEP25. The empty-complement case has determinant one. The dimension of this retained complement is $N - v + 1 - q'$.

The positive-block elimination in CEP25–27 is the classical Schur-complement method [H1, §2.1, (2.3)–(2.4)]. The proof retains the full complement and each original frame determinant before applying the endpoint signs.

Choose any fixed full frame $I_V : \mathbb{C}^\Delta \hookrightarrow E_k$ of $\ker C$ and a fixed right section L_C of C . Then $F_V = [I_V, L_C]$ is invertible and

$$\det(I_V^* G_N I_V) = |\det F_V|^2 \frac{\det G_N}{\det Q_N} = |\det F_V|^2 \frac{\det G_N}{\det G'_d} \det K_{A,N}. \quad (\text{CEP26})$$

Indeed $CF_V = [0, I]$, so the lower right block of $(F_V^* G_N F_V)^{-1}$ is $CG_N^{-1} C^* = Q_N^{-1}$. The Schur determinant argument just used gives CEP26. The frame factor is independent of source cutoff and cancels only after applying the four signs.

For the actual inclusion $K \subset V_A$, choose its original coefficient frame, and let $\Xi : \mathbb{C}^\Delta \rightarrow \mathbb{C}^{\Delta-r}$ have its coordinate kernel, $r = 8k - 16$, as in AKJ18. With $H_{V,N} = I_V^* G_N I_V$, the quotient metric is $Q_{\Xi,N} = (\Xi H_{V,N}^{-1} \Xi^*)^{-1}$. The same proof with a frame of $\ker \Xi$ and a right section gives, with all constant frame factors cancelled by the endpoint signs,

$$F_K^{(s)} = \sum_N \sigma_N \log \det G_N - \sum_N \sigma_N \log \det G'_{N-v} + \sum_N \sigma_N \log \det K_{A,N} - \sum_N \sigma_N \log \det Q_{\Xi,N}. \quad (\text{CEP27})$$

Thus the conductor quotient and the remaining original observation quotient have both been retained in the determinant relation. The lower source cutoffs in this equation are exactly $q - 1 - v, q - v, 2q - 1 - v, 2q - v$. Relative to the lower grid's own four cutoffs they have the respective shifts $\Delta - v, \Delta - v, 2\Delta - v, 2\Delta - v$.

29.5 An exact graph quotient with controlled exterior metric error

In original orthonormal coordinates define

$$S_N : \mathcal{P}_N \longrightarrow \mathbb{C}^v \oplus \mathcal{P}'_{N-v}, \quad S_N p = (P_{<v} p, \mathcal{T}_{A,N} p). \quad (\text{CEP28})$$

Here $P_{<v}$ records the first v original upper orthonormal coefficients, and the target carries the orthogonal direct-sum norm with those coordinates and the unchanged lower Gamma source. The first v source coordinates span the polynomials of degree below v . Thus CEP14–16 give its exact block matrix $S_N = \text{diag}(I_v, \hat{A}_N)$ in the ordered source split at degree v . In particular it is invertible, and

$$\begin{aligned} \|S_N\| &\leq M_N := \max\{1, M_{A,N}\}, \\ |\det S_N| &= \left| \frac{\mu_v}{v!} \right|^{N-v+1} \prod_{m=0}^{N-v} \sqrt{\frac{w_{m+v}}{w_m}}, \\ |\log |\det S_N|| &\leq L_N := (N-v+1) \left| \log \left| \frac{\mu_v}{v!} \right| \right| + v c_b^{\log} H_N. \end{aligned} \quad (\text{CEP29})$$

The last bound follows by the consecutive-product identity preceding CEP19 and $|\log w_j| \leq c_b^{\log} H_j$: the two lists of v logarithms are each bounded by $v c_b^{\log} H_N$, followed by the factor $1/2$.

The upper and lower source ideals at this cutoff are

$$I_N = \chi_k \mathbb{C}[S]_{\leq N-q}, \quad I'_N = \chi_{k'} \mathbb{C}[S']_{\leq N-v-q'}. \quad (\text{CEP30})$$

A negative degree specifies the zero subspace. Let $\mathcal{W}_N = U_N^{-1}(V_A)$. The exact image and ideal graph under CEP28 are

$$\begin{aligned} S_N \mathcal{W}_N &= \mathbb{C}^v \oplus I'_N, \\ S_N I_N &= \{(P_{<v} h, \mathcal{T}_{A,N} h) : h \in I_N\} \subset \mathbb{C}^v \oplus I'_N, \\ V_A &\xrightarrow{\simeq} (\mathbb{C}^v \oplus I'_N) / (S_N I_N), \quad U_N p \longmapsto [S_N p]. \end{aligned} \quad (\text{CEP31})$$

The first equality follows because $\mathcal{W}_N$ is the preimage of I'_N under the top map of CEP6, and S_N is invertible. CEP3 proves the ideal inclusion. If representatives differ by I_N , their images differ by $S_N I_N$, so the last map is well defined. Conversely every class in the target has a preimage under S_N in $\mathcal{W}_N$, and the kernel consists exactly of I_N . This proves the isomorphism. Its norm on the left is the restriction of G_N : all lifts of a member of V_A already belong to $\mathcal{W}_N$, so their minimum source norm is the quotient norm of $\mathcal{W}_N / I_N$. The low-degree graph component $P_{<v} h$ is explicitly retained.

For reference the ideal dimensions at the four original endpoints are

N	$\dim I_N$	$\dim I'_N$	$\dim((\mathbb{C}^v \oplus I'_N) / S_N I_N)$
$q-1$	0	$\Delta - v$	Δ
q	1	$\Delta + 1 - v$	Δ
$2q-1$	q	$q + \Delta - v$	Δ
$2q$	$q+1$	$q + \Delta + 1 - v$	Δ .

(CEP32)

These follow by counting monomials in CEP30; S_N preserves dimensions. Moreover $\mathcal{T}_A|_{I_N}$ is injective because a nonzero member has degree at least $q > v$ and its image has degree exactly v less. Hence the second coordinate of the retained ideal graph already has dimension $\dim I_N$.

Here is the complete exterior norm estimate for this exact map. For an invertible complex matrix S of dimension n and $0 \leq t \leq n$, its singular values $\sigma_1 \geq \dots \geq \sigma_n > 0$ give

$$\left\| \bigwedge^t S^{-1} \right\| = \frac{\left\| \bigwedge^{n-t} S \right\|}{|\det S|}. \quad (\text{CEP33})$$

Indeed the numerator is $\sigma_1 \cdots \sigma_{n-t}$ and the denominator is the product of all singular values; their ratio is the product of the t largest singular values of S^{-1} . This includes $t = 0, n$. For $n = N + 1$, CEP29 therefore gives the finite bounds

$$\begin{aligned} \log^+ \left\| \bigwedge^t S_N \right\| &\leq t \log M_N, \\ \log^+ \left\| \bigwedge^t S_N^{-1} \right\| &\leq (N + 1 - t) \log M_N + L_N. \end{aligned} \tag{CEP34}$$

Here $\log^+ x = \max\{0, \log x\}$. The right sides are nonnegative. Thus for all these ranks both bounds are at most

$$E_N = (N + 1) \log M_N + L_N. \tag{CEP35}$$

With $R_{\text{sh}} = \max_{r,s} |b_{rs} - 4|$ and $A_*^{\text{abs}} = \sum |a_{rs}|$, equation CEP17 gives $\log M_N \leq \log^+ A_*^{\text{abs}} + R_{\text{sh}}(H_N + 2)$. For the fixed original quartet and period, v, R_{sh}, μ_v and A_*^{abs} are fixed while $c_b^{\log} = O(k)$. Since $N \leq 2q$ and $H_N \leq 1 + \log N$, all four explicit bounds in CEP35 satisfy

$$E_N = O_{\text{actual}}(q \log(q + 1) + k \log(q + 1)) = o(kq). \tag{CEP36}$$

This uses only the forward norm and the exact determinant of the original conductor map.

We now prove the exact relation to the metric of the original kernel, including the quotient projection. In a finite-dimensional Hilbert space let $I \subset W$ and let S be invertible. The induced map $W/I \rightarrow SW/SI$ sends a minimal lift $x \in W \cap I^\perp$ to the orthogonal projection of Sx off SI . Orthogonal projection is a contraction on every exterior power: in an orthonormal basis its singular values are zeros and ones. Therefore, on any t -dimensional subspace of W/I , the exterior norm of the induced map is at most $\|\bigwedge^t S\|$. Applying the identical argument to S^{-1} gives the inverse bound. The minimal-lift identification is isometric since every other lift differs by a perpendicular element of I , whose squared norm adds. This proves the quotient assertion with no choice of coefficient norm.

Apply this result with $W = \mathcal{W}_N$, $I = I_N$, and the actual rank- r subspace $K \subset V_A$. The target subspace is the explicit image of K under CEP31; denote its Gram matrix in the transported original frame by $\tilde{H}_{K,N}$. Let $I_K : \mathbb{C}^r \hookrightarrow E_k$ be that original primary frame, and write its original minimum source lift as $Y_N = U_N^* G_N I_K$. Let Z_N be the matrix in the original orthonormal source coordinates of the independent columns $\chi_k(S) S^a$, $0 \leq a \leq N - q$. If there are no columns, all following products involving them are zero. The exact target orthogonal projection and Gram matrix are

$$\begin{aligned} P_N^{\text{graph}} &= I - S_N Z_N (Z_N^* S_N^* S_N Z_N)^{-1} Z_N^* S_N^*, \\ \tilde{H}_{K,N} &= Y_N^* S_N^* P_N^{\text{graph}} S_N Y_N, \quad H_{K,N} = Y_N^* Y_N = I_K^* G_N I_K. \end{aligned} \tag{CEP37}$$

The inverted Gram matrix is positive definite because its columns are independent and S_N is invertible. Multiplication shows that P_N^{graph} is Hermitian, idempotent, kills $S_N I_N$ and fixes its orthogonal complement. Thus CEP37 is precisely the minimum quotient Gram of CEP31, and its kernel frame has full rank because CEP31 is injective. Replacing the lift Y_N by any lift of the same primary frame adds columns in I_N , which the projection kills; hence every choice of lift gives this same matrix.

Applying the proved exterior quotient estimate and squaring volumes gives

$$\begin{aligned} -2 \log \left\| \bigwedge^r S_N^{-1} \right\| &\leq \log \det \tilde{H}_{K,N} - \log \det H_{K,N} \\ &\leq 2 \log \left\| \bigwedge^r S_N \right\|, \\ \left| \log \det \tilde{H}_{K,N} - \log \det H_{K,N} \right| &\leq 2E_N. \end{aligned} \tag{CEP38}$$

In detail, divide the chosen frame by its positive Gram square root so it is orthonormal in the source quotient. The squared norm of its exterior image is exactly the determinant ratio. The

forward and inverse exterior estimates prove the two inequalities, and CEP34–35 prove the last assertion.

Consequently the following reduction of the actual four-endpoint kernel determinant is proved:

$$\begin{aligned}\tilde{F}_K^{(s)} &= \sum_N \sigma_N \log \det \tilde{H}_{K,N}, \\ |F_K^{(s)} - \tilde{F}_K^{(s)}| &\leq 2 \sum_{N \in \{q-1, q, 2q-1, 2q\}} E_N = o(kq).\end{aligned}\tag{CEP39}$$

The frame is the original fixed frame of K at every endpoint; CEP37 gives all four transported matrices explicitly. Their target spaces retain the original ideal graphs, the different source cutoffs, all conductor coefficients, and the full original Gamma masses. CEP39 proves that the leading kq coefficient is unchanged by this particular graph-quotient transport. It supplies no value for that coefficient: its remaining determinant is the displayed, completely specified four-endpoint graph quotient.

29.6 All ideal graph entries as original polynomial products

Each individual shift in CEP2 induces the unital algebra map $\sigma_{rs} : E_k \rightarrow E_{k'}, [p] \mapsto [p(S' + b_{rs})]$. The divisibility in CEP3, applied to the individual shifted polynomial, proves independence of the representative; substitution preserves multiplication and one. Thus $\bar{\mathcal{T}}_A = \sum a_{rs} \sigma_{rs}$ is its exact linear combination. For any $x, y \in E_k$, the complete multiplicative defect of that linear combination is

$$\bar{\mathcal{T}}_A(xy) - \bar{\mathcal{T}}_A(x)\bar{\mathcal{T}}_A(y) = \sum_{r,s} a_{rs} \sigma_{rs}(x) \sigma_{rs}(y) - \sum_{r,s,t,u} a_{rs} a_{tu} \sigma_{rs}(x) \sigma_{tu}(y).\tag{CEP40}$$

This follows by applying the proved individual multiplicativity and expanding the product of the two complete sums.

There is a complete finite product for the map on the source ideal. For each original shift put

$$d_{rs}(S') = \prod_{\substack{0 \leq i,j \leq k \\ (i,j) \notin \{r, \dots, r+k'\} \times \{s, \dots, s+k'\}}} (S' + b_{rs} - \beta_{ij}).\tag{CEP41}$$

This is monic of degree Δ , and $\chi_k(S' + b_{rs}) = \chi_{k'}(S') d_{rs}(S')$. Indeed CEP3's root addition identifies exactly the q' factors in the indicated rectangle with the factors of $\chi_{k'}$. All remaining factors are precisely the product in CEP41. Consequently for every $h \in \mathbb{C}[S]_{\leq L}$, $L = N - q$, the exact ideal map is

$$\begin{aligned}\mathcal{T}_{A,N}(\chi_k h) &= \chi_{k'} \mathcal{U}_{A,L} h, \\ (\mathcal{U}_{A,L} h)(S') &= \sum_{r,s} a_{rs} d_{rs}(S') h(S' + b_{rs}) \in \mathbb{C}[S']_{\leq L + \Delta - v}.\end{aligned}\tag{CEP42}$$

The equality follows by the product identity just proved. The degree statement follows from CEP8: if h has degree t and leading coefficient $h_t \neq 0$, the leading coefficient of its displayed image is $h_t \binom{q+t}{v} \mu_v$ at degree $t + \Delta - v$. This follows because $\chi_k, \chi_{k'}$ are monic and $\mathcal{T}_A$ lowers degree exactly by v with its stated leading derivative. Thus $\mathcal{U}_{A,L}$ is injective. At $L = -1$ the domain and image are zero.

This gives a further exact sequence, with all maps specified:

$$0 \longrightarrow \mathbb{C}[S]_{<v} \longrightarrow V_A \longrightarrow \mathbb{C}[S']_{\leq L + \Delta - v} / \mathcal{U}_{A,L} \mathbb{C}[S]_{\leq L} \longrightarrow 0.\tag{CEP43}$$

The first map is the original quotient class. For the second, lift a member of V_A to $p \in \mathcal{W}_N$, divide $\mathcal{T}_{Ap}$ by $\chi_{k'}$, and take its class in the last space. A change of lift by $\chi_k h$ adds exactly $\mathcal{U}_{A,L} h$

by CEP42, proving independence. Every target polynomial times $\chi_{k'}$ has a preimage under the onto map $\mathcal{T}_{A,N}$, proving surjectivity. If the image is zero, subtract the corresponding $\chi_k h$; the remaining polynomial is killed by $\mathcal{T}_A$ and hence has degree below v . Its class injects because $v < q$. This proves exactness. The last space has dimension $\Delta - v$ at every original endpoint, as is also obtained by subtracting the two polynomial dimensions.

For a literal matrix version of the ideal graph, equip $\mathbb{C}[S']_{\leq L+\Delta-v}$ with the ordered lower orthonormal Gamma coefficient basis, and write $M_{\chi',N}$ for multiplication by the full original $\chi_{k'}$ into $\mathcal{P}'_{N-v}$. It is injective. The map

$$D_{\chi',N} = \text{diag}(I_v, M_{\chi',N}) : \mathbb{C}^v \oplus \mathbb{C}[S']_{\leq L+\Delta-v} \longrightarrow \mathbb{C}^v \oplus I'_N \quad (\text{CEP44})$$

is an isomorphism onto the indicated subspace. Its pullback metric is exactly $G_{\chi',N} = \text{diag}(I_v, M_{\chi',N}^* M_{\chi',N})$; this follows by taking the squared norm of each component. For the domain polynomial coefficient frame of h , let $Z_{\chi,N}h$ be the original upper orthonormal coefficients of $\chi_k h$. In these coordinates the complete graph relation matrix is

$$\mathcal{Z}_{A,N} = \begin{pmatrix} P_{<v} Z_{\chi,N} \\ [\mathcal{U}_{A,L}] \end{pmatrix}, \quad D_{\chi',N} \mathcal{Z}_{A,N} = S_N Z_{\chi,N}. \quad (\text{CEP45})$$

The first component is the original low-degree projection, and the second consists of the exact finite products and translations in CEP41–42. The equality is CEP28 and CEP42. It proves that the graph quotient in CEP31 is precisely the quotient by $\text{im } \mathcal{Z}_{A,N}$ in the displayed metric $G_{\chi',N}$. All matrix inverses and minimum projections in CEP37 can therefore be computed using these full original polynomial products. No coefficient of the ideal, source metric or low-degree graph component is discarded by this additional coordinate isomorphism.

29.7 The compressed conductor correction is subleading at every endpoint

The exact exterior estimate also controls the large compressed covariance in CEP24 without multiplying a scalar inverse bound by q' . Write $n' = N - v + 1$ and retain $A_N = [0, \hat{A}_N]$ with its exact square block in CEP16. Since $W_d^* W_d = I_{q'}$, the coefficient matrix of its image under $\hat{A}_N^*$ gives

$$K_{A,N} = (\hat{A}_N^* W_d)^* (\hat{A}_N^* W_d). \quad (\text{CEP46})$$

The squared norm of the exterior product of the q' columns on the right is its determinant. The norm of the exterior product of the columns of W_d is one, because those columns are orthonormal. The forward estimate therefore gives $\det K_{A,N} \leq \|\hat{A}_N\|^{2q'} \leq M_N^{2q'}$. For the reverse inequality apply $\bigwedge^{q'} \hat{A}_N^*$ to the same image columns; their image is the orthonormal column wedge of W_d . Thus

$$\det K_{A,N} \geq \left\| \bigwedge^{q'} \hat{A}_N^{-1} \right\|^{-2}. \quad (\text{CEP47})$$

The singular values of an adjoint are the same, so the adjoint inverse in this argument has exactly the displayed exterior norm. The singular complement identity CEP33 in dimension n' gives $\|\bigwedge^{q'} \hat{A}_N^{-1}\| \leq M_N^{n'-q'}/|\det \hat{A}_N|$. CEP29 bounds the absolute logarithm of the last determinant by L_N . Consequently the complete two-sided estimate is

$$\begin{aligned} -2((N - v + 1 - q') \log M_N + L_N) &\leq \log \det K_{A,N} \leq 2q' \log M_N, \\ |\log \det K_{A,N}| &\leq 2E_N = o(kq) \quad \text{at every original endpoint and source order.} \end{aligned} \quad (\text{CEP48})$$

All ranks in this argument are their actual ranks, $q' \leq N - v + 1$ as proved after CEP6. In particular the estimate applies to the complete original lower quotient, whose rank grows quadratically.

Define the fully specified total-source difference and the remaining original observation quotient return by

$$\begin{aligned}\mathcal{T}_k^{(s)} &= \sum_N \sigma_N (\log \det G_N - \log \det G'_{N-v}), \\ F_{\Xi,k}^{(s)} &= \sum_N \sigma_N \log \det Q_{\Xi,N}, \\ \mathcal{E}_{A,k}^{(s)} &= \sum_N \sigma_N \log \det K_{A,N}.\end{aligned}$$

Equations CEP26–27 and CEP48 prove the exact identities and bounds

$$\begin{aligned}F_{V_A}^{(s)} &= \mathcal{T}_k^{(s)} + \mathcal{E}_{A,k}^{(s)}, & F_K^{(s)} &= \mathcal{T}_k^{(s)} - F_{\Xi,k}^{(s)} + \mathcal{E}_{A,k}^{(s)}, \\ |\mathcal{E}_{A,k}^{(s)}| &\leq 2 \sum_N E_N = o(kq), \\ \frac{F_K^{(s)} - \mathcal{T}_k^{(s)} + F_{\Xi,k}^{(s)}}{kq} &\longrightarrow 0.\end{aligned}\tag{CEP49}$$

Here $F_{V_A}^{(s)}$ uses its fixed original coefficient frame, and $Q_{\Xi,N}$ is the exact quotient metric of CEP27. The frame factors cancel with coefficients $1 + 1 - 1 - 1$ only in these returns, as proved in CEP26. Thus the leading metric effect of the full conductor source transport has now been controlled. The remaining rank- $8k - 32$ observation quotient and the shifted lower-grid source determinants are still their explicit original matrices; CEP49 carries them forward into the next calculation.

30 The total-source difference as an outer-strip quotient

Retain CEP1–49, with $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $k' = k - 8$, $q' = (k - 7)^2$, $\Delta = q - q' = 16k - 48$, and the original fixed $0 < \delta < 1/2$, $\gamma > 2$. The conductor has its actual first nonzero moment index $0 \leq v \leq 80$. At each of the four original endpoints $N \in \{q - 1, q, 2q - 1, 2q\}$ put $t = N - q + 1$ and $d = N - v$. The source order remains $s \in \{1, k\}$ at both grids, with

$$b = s/2, \quad M_s = (2\pi)^{s/2}, \quad dm_s(y) = M_s \frac{2^b}{4\pi\Gamma(b)} |\Gamma(b/2 + iy/2)|^2 dy, \quad h_n = M_s n!(b)_n. \quad (\text{CTV1})$$

These are the complete original measure and monic norms of OKM6–7. The source centres are $c = k/2$ and $c' = k/2 - 4$. The total-source difference in CEP49 is

$$\mathcal{T}_k^{(s)} = \sum_N \sigma_N (\log \det G_N - \log \det G'_{N-v}), \quad \sigma = (1, 1, -1, -1), \quad (\text{CTV2})$$

where both matrices use their original primary frames. All calculations below retain these four cutoffs and the same two source orders.

30.1 The exact nested root polynomials in the original real coordinate

Define the full monic polynomials

$$\begin{aligned} P_k(y) &= i^{-q} \chi_k(c + iy) = \prod_{a,b=0}^k (y - \omega_{ab}), & \omega_{ab} &= (2b - k)\gamma - i(2a - k)\delta, \\ P_{k'}(y) &= i^{-q'} \chi_{k'}(c' + iy) = \prod_{a,b=0}^{k'} (y - \omega'_{ab}), & \omega'_{ab} &= (2b - k')\gamma - i(2a - k')\delta. \end{aligned} \quad (\text{CTV3})$$

The original phases i^{-q} and $i^{-q'}$ have modulus one; they occur exactly as written in the coefficient maps. For every lower pair one has $\omega'_{ab} = \omega_{a+4,b+4}$. Hence, with the complete outer index set $\mathcal{O}_k = \{0, \dots, k\}^2 \setminus \{4, \dots, k-4\}^2$, the exact factorization is

$$P_k(y) = P_{k'}(y) P_{\partial,k}(y), \quad P_{\partial,k}(y) = \prod_{(a,b) \in \mathcal{O}_k} (y - \omega_{ab}), \quad \deg P_{\partial,k} = \Delta. \quad (\text{CTV4})$$

Direct substitution proves the root equality; counting the retained central rectangle gives q' roots, leaving Δ distinct outer roots. The products are monic, so the root equality fixes the factor without any additional constant.

Since k' is odd, $|2a - k'| \geq 1$ for every lower index. For real y this proves, factor by factor,

$$|y - \omega'_{ab}| \geq |\operatorname{Im} \omega'_{ab}| \geq \delta, \quad |P_{k'}(y)| \geq \delta^{q'}. \quad (\text{CTV5})$$

Also $|\omega'_{ab}| \leq R' = k' \sqrt{\delta^2 + \gamma^2}$. These estimates keep every original factor; no root has been moved.

30.2 The full monic source and relation determinant identities

For a nonzero polynomial P and an integer $a \geq 0$ define the relation moment determinant

$$D_a(P; s) = \det \left[\int_{\mathbb{R}} y^{i+j} |P(y)|^2 dm_s(y) \right]_{0 \leq i, j < a}, \quad D_0(P; s) = 1. \quad (\text{CTV6})$$

Every nonempty matrix is positive definite: the measure is strictly positive on the real line, a nonzero polynomial has only finitely many zeros, and the squared modulus of its product with P is positive on an open interval. All moments exist because the original source has its finite exponential moments in CTV14 below.

The determinant quotient in CTV7 is a positive-block Schur factorization [H1, §2.1, (2.3)–(2.4)]. Its precise relation block and the determinant-one polynomial frame changes are verified in the following proof.

In increasing monomial y coordinates, the quotient by a monic polynomial P of degree a from source degree $L \geq a - 1$ has determinant

$$\det G_{P,L}^y = \frac{\prod_{n=0}^L h_n}{D_{L-a+1}(P; s)}. \quad (\text{CTV7})$$

Here the quotient frame is $1, y, \dots, y^{a-1}$. For proof use the source frame

$$(1, y, \dots, y^{a-1}, P, yP, \dots, y^{L-a}P).$$

Its increasing-degree coefficient matrix has diagonal entries one and no entries above the degree of a column, hence determinant one. The last block is exactly the relation Gram in CTV6. Minimizing over those last coordinates gives its Schur complement, whose determinant is the full source determinant divided by the relation determinant. Monic orthogonalization by the original p_n also has determinant one, so that full determinant is $\prod_{n=0}^L h_n$. This proves CTV7 including the empty relation case. It is the full-source analogue of BSL3, with the original Gamma order retained.

Let $V_\beta, V_{\beta'}$ be the original ordered Vandermonde matrices in the S, S' variables. The exact coordinate pullback $S = c + iy$ sends S^j to $(c + iy)^j$ and has diagonal coefficients i^j ; its determinant is $i^{a(a-1)/2}$ in degree below a . Its modulus is one, so its Gram congruence has determinant factor one. The same statement holds with c' . Evaluation in primary coordinates then gives

$$\begin{aligned} \det G_N &= \frac{\prod_{n=0}^N h_n}{D_t(P_k; s) |\det V_\beta|^2}, \\ \det G'_{N-v} &= \frac{\prod_{n=0}^{N-v} h_n}{D_{t+\Delta-v}(P_{k'}; s) |\det V_{\beta'}|^2}. \end{aligned} \quad (\text{CTV8})$$

The relation phase i^q or $i^{q'}$ multiplying each source relation column has squared modulus one, as used in this equality. The Vandermonde factors are nonzero by the original distinctness of all roots. Their logarithmic difference is constant across the four endpoints and has signed coefficient zero. Therefore CTV8 proves the exact finite expression

$$\mathcal{T}_k^{(s)} = \sum_N \sigma_N \left[\sum_{n=N-v+1}^N \log h_n + \log D_{t+\Delta-v}(P_{k'}; s) - \log D_t(P_k; s) \right]. \quad (\text{CTV9})$$

The first sum is empty for $v = 0$. In particular both low relation determinants and both high relation determinants remain present.

30.3 The exact quotient of the outer-strip polynomial

Introduce the positive measure

$$d\zeta_{k',s}(y) = |P_{k'}(y)|^2 dm_s(y). \quad (\text{CTV10})$$

This measure is the exact pullback of the original source metric under the multiplication map $\mathcal{M}_{P_{k'}} : f \mapsto P_{k'} f$. Indeed $\|\mathcal{M}_{P_{k'}} f\|_{m_s}^2 = \int |f|^2 d\zeta_{k',s}$, including the same full factor M_s . On polynomials of degree at most $N - q'$ its image is the specified subspace $P_{k'} \mathbb{C}[y]_{\leq N-q'}$ of the original degree-at-most- N source. The map is injective because $P_{k'}$ is nonzero. Thus CTV10 records a proved metric map to the original source, rather than an assigned new Gamma order.

For $L = N - q' = t + \Delta - 1$, let $Q_{\partial,L}^{(s)}$ be the minimum quotient Gram of

$$\mathbb{C}[y]_{\leq L} \longrightarrow \mathbb{C}[y]/(P_{\partial,k}) \quad (\text{CTV11})$$

in the source norm CTV10 and in the increasing monomial remainder frame of dimension Δ . The source degree is at least $\Delta - 1$ at every endpoint. The same monic-frame Schur proof as in CTV7, now using CTV4, gives

$$\det Q_{\partial,L}^{(s)} = \frac{D_{t+\Delta}(P_{k'}; s)}{D_t(P_k; s)}. \quad (\text{CTV12})$$

In fact the ideal columns are $P_{\partial,k}y^j$, $0 \leq j < t$; their Gram in CTV10 is exactly $D_t(P_k; s)$ by CTV4. The numerator is the Gram of the complete monomial source of dimension $L + 1 = t + \Delta$. This proves every source and relation factor in CTV12.

Write $\lambda_j^{(k',s)}$ for the squared norm of the monic orthogonal polynomial of degree j in CTV10. This polynomial and its norm are completely determined by the positive moment matrices: subtract from y^j its projection onto lower powers. If H_j is that lower Gram and a_j its inner-product column with y^j , the lower coefficients are $-H_j^{-1}a_j$. Completing the square proves minimality among all monic polynomials of degree j ; its squared norm is the Schur complement. Thus $\lambda_j^{(k',s)} = D_{j+1}(P_{k'}; s)/D_j(P_{k'}; s) > 0$. Multiplying these identities and substituting CTV12 into CTV9 gives

$$\begin{aligned} \mathcal{F}_{\partial,k}^{(s)} &= \log \det Q_{\partial,\Delta-1}^{(s)} + \log \det Q_{\partial,\Delta}^{(s)} - \log \det Q_{\partial,q+\Delta-1}^{(s)} - \log \det Q_{\partial,q+\Delta}^{(s)}, \\ \mathcal{T}_k^{(s)} &= \mathcal{F}_{\partial,k}^{(s)} + \sum_N \sigma_N \mathcal{R}_{N,v}^{(s)}, \\ \mathcal{R}_{N,v}^{(s)} &= \sum_{a=0}^{v-1} \left(\log h_{N-a} - \log \lambda_{N-q'-a}^{(k',s)} \right). \end{aligned} \quad (\text{CTV13})$$

Every weighted norm index is nonnegative: $N - q' - (v - 1) \geq \Delta - v \geq 16$. The total difference therefore has an exact rank- Δ quotient presentation with precisely the displayed norm tails.

30.4 Finite bounds for all norm tails in both original sources

The original Gamma generating identity gives the full exponential moment $\int e^{ty} dm_s(y) = M_s(\cos t)^{-b}$ for $|t| < \pi/2$. By the evenness of the measure this also equals $\int \cosh(ty) dm_s(y)$. The nonnegative Taylor terms of $\cosh$ imply

$$\begin{aligned} \int y^{2j} dm_s(y) &\leq M_s(\cos t_0)^{-b} \frac{(2j)!}{t_0^{2j}}, \\ \|y^j\|_{m_s} &\leq \sqrt{M_s}(\cos t_0)^{-b/2} \mathfrak{c}_{\text{mom}}^j j!, \quad t_0 = \frac{\pi}{4}, \quad \mathfrak{c}_{\text{mom}} = \frac{2}{t_0} = \frac{8}{\pi}. \end{aligned} \quad (\text{CTV14})$$

The second inequality uses $(2j)! \leq 4^j (j!)^2$, proved by $\binom{2j}{j} \leq \sum_{a=0}^{2j} \binom{2j}{a} = 4^j$. Thus every constant in this moment estimate includes the original source mass and order.

For a monic polynomial of degree n whose roots have modulus at most R' , its coefficient of y^a has absolute value at most $\binom{n}{a} (R')^{n-a}$: it is the sum of those products of $n - a$ distinct roots. The triangle inequality and CTV14 therefore give the full bound

$$\|P\|_{m_s} \leq \sqrt{M_s}(\cos t_0)^{-b/2} \mathfrak{c}_{\text{mom}}^n n! \sum_{h=0}^n \frac{(R'/\mathfrak{c}_{\text{mom}})^h}{h!} \leq \sqrt{M_s}(\cos t_0)^{-b/2} \mathfrak{c}_{\text{mom}}^n n! e^{R'/\mathfrak{c}_{\text{mom}}}. \quad (\text{CTV15})$$

The monic test polynomial y^j in the minimization defining $\lambda_j^{(k',s)}$ gives the polynomial $P_{k'}(y)y^j$ in CTV15, of degree $n = q' + j$; its extra j roots are exactly zero and also satisfy the stated bound. For the lower bound, CTV5 applies to every competing monic polynomial, whose original Gamma squared norm is at least h_j . Taking the minimum proves

$$\delta^{2q'} h_j \leq \lambda_j^{(k',s)} \leq M_s(\cos t_0)^{-b} \mathfrak{c}_{\text{mom}}^{2(q'+j)} ((q' + j)!)^2 e^{2R'/\mathfrak{c}_{\text{mom}}}. \quad (\text{CTV16})$$

No lower bound on an actual period coefficient is used here. The lower root polynomial itself supplies the strictly positive real-line lower bound.

For each term of CTV13 put $n = N - a$ and $j = n - q'$. Subtracting the logarithms in CTV16 from the exact $\log h_n = \log M_s + \log n! + \log(b)_n$ yields

$$\begin{aligned} b \log(\cos t_0) + \log \frac{(b)_n}{n!} - 2n \log \mathbf{c}_{\text{mom}} - \frac{2R'}{\mathbf{c}_{\text{mom}}} &\leq \log h_n - \log \lambda_j^{(k',s)} \\ &\leq \sum_{r=j+1}^n (\log r + \log(b+r-1)) - 2q' \log \delta. \end{aligned} \quad (\text{CTV17})$$

The two full $\log M_s$ contributions cancel in this displayed comparison because both norms arise from exactly CTV1. They have not been removed from either individual norm.

Let $c_b^{\log} = \max\{1, b-1\}$. As proved directly in CEP19, $|\log(j/(b+j-1))| \leq c_b^{\log}/j$ for $b \geq 1/2$: use $\log(1+x) \leq x$ for $b \geq 1$; for $1/2 \leq b < 1$ use $-\log(1-a/j) \leq a/(j-a) \leq 1/j$, $a = 1-b$. Summation gives $|\log((b)_n/n!)| \leq c_b^{\log} H_n$. With all original endpoints $N \geq 99$, define the explicit nonnegative numbers

$$\begin{aligned} L_N^{\text{tail}} &= c_b^{\log} H_N + \frac{b}{2} \log 2 + 2N \log(8/\pi) + \frac{\pi R'}{4}, \\ U_N^{\text{tail}} &= q'(\log N + \log(b+N-1) + 2|\log \delta|), \\ C_{N,v}^{\text{tail}} &= v \max\{L_N^{\text{tail}}, U_N^{\text{tail}}\}. \end{aligned} \quad (\text{CTV18})$$

Indeed $\cos(\pi/4) = 2^{-1/2}$ and $2R'/\mathbf{c}_{\text{mom}} = \pi R'/4$. In the upper sum of CTV17 there are exactly q' terms, and each is at most $\log N + \log(b+N-1)$. Thus CTV17 proves

$$-v L_N^{\text{tail}} \leq \mathcal{R}_{N,v}^{(s)} \leq v U_N^{\text{tail}}, \quad |\mathcal{R}_{N,v}^{(s)}| \leq C_{N,v}^{\text{tail}}. \quad (\text{CTV19})$$

When $v = 0$ the equality $\mathcal{R}_{N,0}^{(s)} = 0$ and the zero bound follow from the empty sum. These estimates apply for all actual endpoints and for both unchanged source orders simultaneously.

30.5 The proved first-variation reduction and its remaining matrix

Since $N \leq 2q$, $q' \leq q$, $b \leq k/2$ and $R' \leq k\sqrt{\delta^2 + \gamma^2}$, equations CTV18–19 give $C_{N,v}^{\text{tail}} = O_{\delta,\gamma}(q \log(q+1) + k \log(q+1))$ uniformly for every integer $0 \leq v \leq 80$. To see this directly, use $H_N \leq 1 + \log N$ in the first line of CTV18 and the displayed upper bounds in its second; v is bounded by the fixed integer 80. Dividing by kq tends to zero, since $\log(q+1)/k \rightarrow 0$ for $q = (k+1)^2$. Consequently CTV13 and the finite bounds just proved yield

$$\left| \mathcal{T}_k^{(s)} - \mathcal{F}_{\partial,k}^{(s)} \right| \leq \sum_N C_{N,v}^{\text{tail}} = o(kq). \quad (\text{CTV20})$$

The estimate is uniform over the possible actual moment indices v ; it requires no bound on the remaining period coordinates. Every polynomial and measure defining $\mathcal{F}_{\partial,k}^{(s)}$ is independent of that index.

For an explicit matrix target, let H_L^ζ be the moment Gram of $1, y, \dots, y^L$ in the full measure CTV10, and let $\mathbf{R}_{\partial,L}$ be the literal remainder matrix modulo the full polynomial $P_{\partial,k}$ in CTV4. The exact metric is

$$Q_{\partial,L}^{(s)} = (\mathbf{R}_{\partial,L}(H_L^\zeta)^{-1} \mathbf{R}_{\partial,L}^*)^{-1}. \quad (\text{CTV21})$$

The map is onto because its first Δ columns are the identity in the remainder frame. The minimum source lift of a coefficient vector x is $(H_L^\zeta)^{-1} \mathbf{R}_{\partial,L}^* Q_{\partial,L}^{(s)} x$; it maps to x and is perpendicular to the kernel. Adding any other lift adds a perpendicular kernel vector, proving the stated Gram. This also proves direct equality with the quotient used in CTV11–12.

Combining the complete exact CEP49 and CTV13 identities gives

$$\begin{aligned}
F_K^{(s)} &= \mathcal{F}_{\partial,k}^{(s)} - F_{\Xi,k}^{(s)} + \mathcal{E}_{A,k}^{(s)} + \sum_N \sigma_N \mathcal{R}_{N,v}^{(s)}, \\
\left| F_K^{(s)} - \mathcal{F}_{\partial,k}^{(s)} + F_{\Xi,k}^{(s)} \right| &\leq 2 \sum_N E_N + \sum_N C_{N,v}^{\text{tail}} = o(kq).
\end{aligned} \tag{CTV22}$$

Here the first error has its fixed-actual-period scope from CEP49; the new tail bound has the stronger uniform-in- v scope just proved. Thus the remaining leading calculation now consists of the explicit outer-strip quotient of rank $16k - 48$ and the original observation quotient of rank $8k - 32$. Both have their complete original metric maps and four cutoffs specified. No value of either leading volume is assigned by this reduction, and no derivative of the earlier q^2 asymptotic law was used.

31 The actual observation remaining inside the conductor kernel

Retain the original simple quartet, its actual unit and period matrix $\mathbf{H} = F^{-1}\Pi U\mathcal{E}_{\text{prim}}$, and its proved five-member quadric orbit. Put $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = q - q' = 16k - 48$, $r = 8k - 16$, and $j = 8k - 32$. The period parameter and logarithm branch are fixed at a point on this proved locus. All constructions below use the original ordered primary coordinates at

$$\beta_{ab} = k/2 + (2a - k)\delta + i(2b - k)\gamma, \quad \omega_{ab} = (2b - k)\gamma - i(2a - k)\delta, \quad 0 \leq a, b \leq k. \quad (\text{ROQ1})$$

The full conductor is the actual biform $A = \sum_{a,b=0}^8 a_{ab}e_{ab}^{(8)}$ of OCF4. Its matrix $C : \mathbb{C}^q \rightarrow \mathbb{C}^{q'}$ is exactly CEP4: $C_{(a,b),(i,h)} = a_{i-a,h-b}$ when both differences belong to $\{0, \dots, 8\}$, and zero otherwise. Thus $C^\top$ is multiplication by the entire A , and $V_A = \ker C$. The original observation is Λ , and $K = \ker \Lambda \subset V_A$. Ordinary transpose throughout the coefficient duality is denoted by $\top$; a star denotes conjugate transpose in a displayed metric.

31.1 An explicit residual map and its actual boundary basis

Use the complete OCF6 elimination, with its actual largest nonzero coefficient $a_{r_*s_*}$ and every remaining a_{ab} . To avoid confusing its degree- k matrices with the source cutoff, write them as

$$\mathbf{N} : \mathbb{C}^q \rightarrow \mathbb{C}^\Delta, \quad \mathbf{S} : \mathbb{C}^\Delta \hookrightarrow \mathbb{C}^q, \quad \mathbf{V} : \mathbb{C}^q \rightarrow \mathbb{C}^{q'}, \quad \mathbf{S}\mathbf{N} + C^\top \mathbf{V} = I_q, \quad \mathbf{N}\mathbf{S} = I_\Delta. \quad (\text{ROQ2})$$

These are the exact identities OCF7. Specifically, at each decreasing lexicographic conductor position the elimination subtracts its current coefficient divided by $a_{r_*s_*}$ times the entire translated biform $Ae_{ab}^{(k-8)}$. This specifies all entries of $\mathbf{N}, \mathbf{V}$; $\mathbf{S}$ inserts the retained strip coordinates. It also proves $\ker \mathbf{N} = \text{im } C^\top$. Taking the annihilator for the bilinear coefficient pairing therefore gives

$$I_V = \mathbf{N}^\top : \mathbb{C}^\Delta \xrightarrow{\sim} V_A, \quad I_V^{-1} = \mathbf{S}^\top|_{V_A}. \quad (\text{ROQ3})$$

For a direct verification, $C\mathbf{N}^\top = 0$ by transposing $\mathbf{N}C^\top = 0$, and $\mathbf{S}^\top \mathbf{N}^\top = I$. If $Cx = 0$, transposing the first identity of ROQ2 gives $x = \mathbf{N}^\top \mathbf{S}^\top x$.

Retain the j independent original invariant certificates $F_{k,h}(x)$ selected by OCF20, and their entire biform coefficient columns f_h :

$$\mathbf{F} = [f_1, \dots, f_j], \quad f_h = \text{coeff}(F_{k,h}(\mathbf{H}t)), \quad \mathbf{Z} = \mathbf{N}\mathbf{F} \in \text{Mat}_{\Delta \times j}(\mathbb{C}), \quad \mathbf{D} = \mathbf{F}^\top. \quad (\text{ROQ4})$$

Here $t = (z_0w_0, z_0w_1, z_1w_0, z_1w_1)^\top$ and the coefficient order is the full original biform order. Every certificate is supplied by the fourteen-generator recursion OCF25–26, or equivalently the complete twenty-five-generator recursion OCF9–20. At a recursion step its certificate is the actual product $x^\nu F_{k-e,h}$, and its column is $\mathbf{N}_k M_{f_\nu} \mathbf{S}_{k-e} Z_{k-e,h}$. The coefficients of f_ν are the full multinomials OCF13 in the actual $\mathbf{H}$. Thus ROQ4 specifies finite entries without choosing a substitute period or introducing a freely chosen observation. The proved image and rank are $\text{im } \mathbf{Z} = \mathbf{N}J_k$ and $\text{rank } \mathbf{Z} = j$.

Let I be the j retained pivot rows of $\mathbf{Z}$, let T be their complement among the Δ strip rows, and put $\mathbf{Z}_I = E_I^\top \mathbf{Z}$, $\mathbf{Z}_T = E_T^\top \mathbf{Z}$. The original OCF selection guarantees that $\mathbf{Z}_I$ is invertible on its exhaustive exact pivot chart. ROQ2 applied to $\mathbf{F}$ gives the full coefficient identity

$$\mathbf{F} = \mathbf{S}\mathbf{Z} + C^\top \mathbf{V}\mathbf{F}, \quad \mathbf{D} = \mathbf{Z}^\top \mathbf{S}^\top + (\mathbf{V}\mathbf{F})^\top C. \quad (\text{ROQ5})$$

In particular the unreduced invariant rows and the reduced strip rows agree on V_A , with precisely the displayed conductor multiple as their ambient difference. The residual observation has the explicit forms

$$\Xi : V_A \rightarrow \mathbb{C}^j, \quad x \mapsto Dx, \quad \Xi I_V = \mathbf{Z}^\top, \quad L = \mathbf{N}^\top E_I (\mathbf{Z}_I^\top)^{-1}, \quad CL = 0, \quad DL = I_j. \quad (\text{ROQ6})$$

The last three equalities follow by multiplication and ROQ3–5. Moreover $\ker \Xi = K$: the rows C and D together span the coefficient space J_k . Indeed conductor multiplication spans $\ker \mathbf{N}$, its image is contained in J_k by OCF17, and the f_h project to the basis Z of $\mathbf{N}J_k$. Thus a primary vector killed by both C and D is exactly the annihilator of J_k , which is the original K by OKM3–5. This proves rank j , surjectivity, and the complete exact sequence

$$0 \longrightarrow K \longrightarrow V_A \xrightarrow{\Xi} \mathbb{C}^j \longrightarrow 0. \quad (\text{ROQ7})$$

For every $y \in \mathbb{C}^j$ define $\mathbf{b}(y) = \Lambda Ly$. These vectors give the literal isomorphism $\mathbf{b} : \mathbb{C}^j \rightarrow \Lambda(V_A)$. They span because for $x \in V_A$, $x - LDx \in K$ by ROQ6–7. They are independent because $\Lambda Ly = 0$ implies $Ly \in K$, hence $y = DLy = 0$. The inverse on the displayed image is $\Lambda x \mapsto Dx$ for $x \in V_A$: replacing x by another such lift changes it by K , which D kills. In particular the boundary basis vectors are exactly $\mathbf{b}_h = \sum_{a,b} L_{(a,b),h} \Lambda \varepsilon_{ab}$. Each $\Lambda \varepsilon_{ab}$ is the full original projected tensor coefficient in OKA7, including its literal $1/5$ cyclic average.

The whole original observation can be recovered from these rows as well. Take an actual right inverse L_C of C , for example the inverse of the first nonzero square column minor with its coordinate insertion, and define

$$\mathbf{L} = [L_C - LDL_C, L], \quad \mathbf{R} = \begin{pmatrix} C \\ D \end{pmatrix}, \quad \mathbf{R}\mathbf{L} = I_{q'+j}, \quad \Lambda = (\Lambda\mathbf{L})\mathbf{R}. \quad (\text{ROQ8})$$

The first identity follows by multiplying its four blocks. For the last, $\mathbf{R}(x - \mathbf{L}\mathbf{R}x) = 0$, whose kernel is K by ROQ7. Thus the $q' + j = q - r$ columns of $\Lambda\mathbf{L}$ form an actual basis of the original boundary: applying its kernel and then $\mathbf{R}\mathbf{L} = I$ proves independence, and ROQ8 proves spanning. The last j columns are exactly $\mathbf{b}_h$.

For completeness the inherited period metric on this residual basis is also explicit. Let B be the complete matrix AKS8 in primary coordinates, so $j\Lambda = \mathcal{F}_0 B$. If ν, μ range over the original weight-zero degree- k monomials, set

$$(\mathcal{G}_0)_{\nu\mu} = \sum_{\substack{n_{tu} \geq 0 \\ \sum_u n_{tu} = \nu_t \\ \sum_t n_{tu} = \mu_u}} \frac{k!}{\prod_{t,u} n_{tu}!} \prod_{t,u} [5(\delta_{tu} + 1)]^{n_{tu}}, \quad P_{\text{per}} = L^* B^* \mathcal{G}_0 B L > 0. \quad (\text{ROQ9})$$

Expansion of the original word inner product gives $\mathcal{G}_0 = \mathcal{F}_0^* \mathcal{F}_0$ as in AKS10, so the displayed matrix is exactly the Gram of the independent vectors $j\mathbf{b}_h$. Its strict positivity follows from that independence and injectivity of j . All off-diagonal word terms and every actual entry of $\mathbf{H}$ remain in ROQ9.

31.2 All four induced Gamma metrics and their fixed determinant factor

For either original source $s \in \{1, k\}$ put $b_s = s/2$, $M_s = (2\pi)^{s/2}$ and retain

$$\begin{aligned} p_0 &= 1, & p_1 &= y, & p_{n+1} &= yp_n - n(b_s + n - 1)p_{n-1}, \\ h_n &= M_s n! (b_s)_n, & (u_n)_{ab} &= p_n(\omega_{ab}) / \sqrt{h_n}. \end{aligned} \quad (\text{ROQ10})$$

Thus $U_N = [u_0, \dots, u_N]$ is precisely CEP21 and $R_N = U_N U_N^* > 0$, $G_N = R_N^{-1}$ for every $N \geq q - 1$. The positivity follows from its first q columns, whose determinant is the original Vandermonde times their nonzero leading coefficients. Define

$$H_{V,N} = \bar{\mathbf{N}} G_N \mathbf{N}^\top, \quad Q_{\Xi,N} = (Z^\top H_{V,N}^{-1} \bar{Z})^{-1} > 0. \quad (\text{ROQ11})$$

These are matrices of sizes Δ and j , respectively. The minimum of $z^* H_{V,N} z$ under $Z^\top z = y$ is attained at $z = H_{V,N}^{-1} \bar{Z} Q_{\Xi,N} y$. Its difference from any other lift lies in $\ker Z^\top$ and is perpendicular in the $H_{V,N}$ metric. Its squared norm is $y^* Q_{\Xi,N} y$. Thus ROQ11 calculates exactly the induced

metric of ROQ7 in the actual boundary coordinates ROQ6. This proof also identifies it with the quotient metric appearing in CEP27 and CEP49.

The actual kernel frame in the strip is

$$\rho = E_T^\top - Z_T Z_I^{-1} E_I^\top, \quad I_K = \mathbf{N}^\top \rho^\top, \quad F_{\text{str}} = [\rho^\top, E_I (Z_I^\top)^{-1}]. \quad (\text{ROQ12})$$

The first two matrices are precisely the OCF/OKM frame. In the ordered rows (T, I) the last one has blocks $\begin{pmatrix} I_r & 0 \\ -(Z_I^\top)^{-1} Z_T^\top & (Z_I^\top)^{-1} \end{pmatrix}$. Consequently $|\det F_{\text{str}}|^2 = |\det Z_I|^{-2}$, including the absolute value of the row-order permutation. Block elimination in $F_{\text{str}}^* H_{V,N} F_{\text{str}}$ gives the exact identity

$$\det(I_K^* G_N I_K) \det Q_{\Xi,N} = \frac{\det H_{V,N}}{|\det Z_I|^2}. \quad (\text{ROQ13})$$

Indeed its upper block is the kernel Gram and its Schur complement is the minimum quotient metric of the previous paragraph, since $Z^\top F_{\text{str}} = [0, I_j]$. The determinant of a block matrix is the determinant of its upper block times that Schur complement, while congruence contributes $|\det F_{\text{str}}|^2$. This proves ROQ13 without assigning modulus one to the actual pivot.

Define $F_{\Xi,k}^{(s)}$ by the four ordered endpoints $\mathcal{N} = (q-1, q, 2q-1, 2q)$ and signs $\sigma = (1, 1, -1, -1)$:

$$F_{\Xi,k}^{(s)} = \sum_{N \in \mathcal{N}} \sigma_N \log \det Q_{\Xi,N}. \quad (\text{ROQ14})$$

The fixed factor $|\det Z_I|^{-2}$ in ROQ13 cancels with these signs. If a different exact certificate basis is selected, its residual rows on V_A are $D' = MD$ for an invertible $j \times j$ matrix M . This follows because both lists are bases of the dual of V_A/K . For an explicit inverse formula, $M = D'L$, and the old and new right sections give its inverse; composing the two surjections proves both inverse identities. The new metric is $Q'_{\Xi,N} = M^{-*} Q_{\Xi,N} M^{-1}$, directly from their identical minimum problem. Its determinant changes by $|\det M|^{-2}$, which also cancels in ROQ14. No pivot or source mass has been removed at an individual endpoint. In fact ROQ10 gives $Q_{\Xi,N} = M_s \widehat{Q}_{\Xi,N}$ when the common mass is explicitly factored out of every column; its determinant carries M_s^j . The metric relative to the retained period Gram ROQ9 has determinant $\det Q_{\Xi,N} / \det P_{\text{per}}$, whose fixed denominator cancels only in the same four-endpoint sum.

31.3 The exact constrained covariance in the original source

The residual metric can be computed without inverting the strip frame. Define, using the complete original covariance,

$$\begin{aligned} A_N^C &= C R_N C^*, & B_N^D &= D R_N C^*, & T_N^D &= D R_N D^*, \\ W_N &= R_N - R_N C^* (A_N^C)^{-1} C R_N, \\ \Omega_N &= D W_N D^* = T_N^D - B_N^D (A_N^C)^{-1} (B_N^D)^*. \end{aligned} \quad (\text{ROQ15})$$

Since C has full row rank and $R_N > 0$, $A_N^C > 0$. There is the exact identity

$$W_N = \mathbf{N}^\top H_{V,N}^{-1} \overline{\mathbf{N}}, \quad \Omega_N = Z^\top H_{V,N}^{-1} \overline{Z} > 0, \quad Q_{\Xi,N} = \Omega_N^{-1}. \quad (\text{ROQ16})$$

To prove its first equality, $CW_N = 0$, W_N is Hermitian, and for $x \in \ker C$ direct multiplication gives $W_N G_N x = x$. The other displayed candidate is Hermitian, has image in $\ker C$ by ROQ3, and has the same property on $x = \mathbf{N}^\top z$. Such a matrix is unique: for their Hermitian difference H , its image lies in V_A and $HG_N x = 0$ for $x \in V_A$. For arbitrary y and $x \in V_A$, this yields $x^* G_N H y = (H G_N x)^* y = 0$. Putting $x = H y \in V_A$ gives $(H y)^* G_N (H y) = 0$, hence $H y = 0$. This proves equality. Now use $D \mathbf{N}^\top = Z^\top$ from ROQ6 and take the adjoint to prove the other two identities.

Every entry in ROQ15 is the finite sum of original column products: $A_N^C = \sum_{n=0}^N (Cu_n)(Cu_n)^*$, $B_N^D = \sum_{n=0}^N (Du_n)(Cu_n)^*$, and $T_N^D = \sum_{n=0}^N (Du_n)(Du_n)^*$. The conductor column is exactly $Cu_n = V_{N-v} A_N e_n$ by CEP22 when $n \leq N$; its vanishing low columns and centre difference are precisely CEP13–16. The residual column is explicitly $(Du_n)_h = \sum_{a,b} (f_h)_{ab} p_n(\omega_{ab}) / \sqrt{h_n}$. In particular ROQ15 retains all conductor–observation cross terms. Replacing D by $D + TC$ leaves Ω_N exactly unchanged because $CW_N = 0 = W_N C^*$. Applied to ROQ5, this proves that the full and strip row presentations give exactly the same metric, rather than assuming their ambient coefficient matrices coincide.

31.4 Exact increments and a positive residual four-endpoint return

For $N \geq q - 1$ let $u = u_{N+1}$ be the next original column ROQ10 and define

$$\begin{aligned} a_N &= 1 + (Cu)^*(A_N^C)^{-1}(Cu) > 0, \\ w_N &= u - R_N C^*(A_N^C)^{-1}Cu \in V_A, \quad z_N = Dw_N = Du - B_N^D(A_N^C)^{-1}Cu, \\ \eta_N &= \frac{z_N^* Q_{\Xi, N} z_N}{a_N} \geq 0. \end{aligned} \tag{ROQ17}$$

The complete updates are

$$\begin{aligned} W_{N+1} &= W_N + w_N w_N^* / a_N, \quad \Omega_{N+1} = \Omega_N + z_N z_N^* / a_N, \\ Q_{\Xi, N+1} &= Q_{\Xi, N} - \frac{Q_{\Xi, N} z_N z_N^* Q_{\Xi, N}}{a_N + z_N^* Q_{\Xi, N} z_N}, \\ \log \det Q_{\Xi, N} - \log \det Q_{\Xi, N+1} &= \log(1 + \eta_N). \end{aligned} \tag{ROQ18}$$

Here is a direct proof of the covariance update. Put $a = Cu$ and $v = R_N C^*(A_N^C)^{-1}a$. Since $R_{N+1} = R_N + uu^*$, $A_{N+1}^C = A_N^C + aa^*$, whose inverse is $(A_N^C)^{-1} - (A_N^C)^{-1}aa^*(A_N^C)^{-1}/a_N$; multiplying by $A_N^C + aa^*$ verifies this inverse exactly. For the full covariance calculation write $\alpha = a^*(A_N^C)^{-1}a$. The subtracted covariance product expands as

$$R_N C^*(A_N^C)^{-1} C R_N + v u^* + u v^* + \alpha u u^* - \frac{(v + \alpha u)(v + \alpha u)^*}{1 + \alpha}.$$

Subtracting this from $R_N + uu^*$ and the old W_N leaves $(u - v)(u - v)^*/(1 + \alpha)$, proving the first update. Multiplying by D, D^* proves the second. The same inverse identity proves the third, and the determinant identity $\det(H + zz^*/a) = \det H(1 + z^* H^{-1} z/a)$ proves the fourth. For that identity conjugate by $H^{-1/2}$; the rank-one matrix $H^{-1/2} z z^* H^{-1/2}/a$ has one eigenvalue $z^* H^{-1} z/a$ and all its others zero. This completes the update proof.

Telescoping with the four actual endpoints, rather than changing the source degree, gives the exact finite sum

$$F_{\Xi, k}^{(s)} = \log(1 + \eta_{q-1}) + 2 \sum_{N=q}^{2q-2} \log(1 + \eta_N) + \log(1 + \eta_{2q-1}) \geq 0. \tag{ROQ19}$$

Indeed $\log \det Q_{q-1} - \log \det Q_{2q-1}$ sums the differences for $N = q - 1, \dots, 2q - 2$, while $\log \det Q_q - \log \det Q_{2q}$ sums those for $N = q, \dots, 2q - 1$. This proves each endpoint weight. Its first increment $\lambda_s = \log(1 + \eta_{q-1})$ is a completely specified period-dependent nonnegative lower bound for $F_{\Xi, k}^{(s)}$. The formulas also characterize equality: $\lambda_s = 0$ exactly when $Du_q = B_{q-1}^D(A_{q-1}^C)^{-1}Cu_q$; the full return vanishes exactly when this corresponding row residual is zero for every column $u_q, \dots, u_{2q}$. Positivity of $Q_{\Xi, N}$ proves both directions, without asserting a nonzero generic residual.

31.5 Finite scalar enclosures and propagation into CEP49

Retain the exact AKS3 scalars $\mathcal{Z}_{s,t}$ and $\mathcal{L}_{s,t} = \log \mathcal{Z}_{s,t}$, whose full original polynomial covariance bound is proved in AKS20–27. In primary coordinates it says

$$\mathcal{Z}_{s,t}^{-1} G_{q-1} \preceq G_{q+t-1} \preceq G_{q-1}, \quad 0 \leq t \leq q+1. \quad (\text{ROQ20})$$

The unchanged coefficient map from AKS26 to primary coordinates is the ordered V_β , so congruence by that invertible map proves exactly the displayed statement. Restrict it to V_A and take the minimum on the fixed fibre $Dx = y$ in V_A . Inequalities of squared norms hold for every such x , hence also for their attained minima. Thus $\mathcal{Z}_{s,t}^{-1} Q_{\Xi,q-1} \preceq Q_{\Xi,q+t-1} \preceq Q_{\Xi,q-1}$. If $d_t = \log \det Q_{\Xi,q-1} - \log \det Q_{\Xi,q+t-1}$, the eigenvalues after congruence by $Q_{\Xi,q-1}^{-1/2}$ give $0 \leq d_t \leq j\mathcal{L}_{s,t}$, and ROQ18 gives their monotonicity. Since $d_1 = \lambda_s$, the exact four-endpoint identity $F_{\Xi,k}^{(s)} = d_q + d_{q+1} - d_1$ proves

$$\lambda_s \leq F_{\Xi,k}^{(s)} \leq j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - \lambda_s. \quad (\text{ROQ21})$$

The actual rank j enters only through this proved eigenvalue bound; the determinant and increment formulas ROQ11–19 retain its full orientation. No leading volume is assigned from a dimension.

Let $\epsilon_{A,k}^{(s)} = 2 \sum_{N \in \mathcal{N}} E_N$ with the complete original constants of CEP29–36. CEP49 gives $F_K^{(s)} = \mathcal{T}_k^{(s)} - F_{\Xi,k}^{(s)} + \mathcal{E}_{A,k}^{(s)}$ and $|\mathcal{E}_{A,k}^{(s)}| \leq \epsilon_{A,k}^{(s)} = o(kq)$ for a fixed actual period. Substitution of ROQ21 proves

$$\begin{aligned} \mathcal{T}_k^{(s)} - \epsilon_{A,k}^{(s)} - j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) + \lambda_s &\leq F_K^{(s)}, \\ F_K^{(s)} &\leq \mathcal{T}_k^{(s)} + \epsilon_{A,k}^{(s)} - \lambda_s. \end{aligned} \quad (\text{ROQ22})$$

Intersecting this interval with the full AKJ2 or AKJ4 intervals preserves each of their finite constants and source correlations. More precisely, one may use the exact value $\mathcal{E}_{A,k}^{(s)}$ in place of its upper or lower envelope in the appropriate side. Formula ROQ19 supplies every additional original positive subtraction, rather than only the first one. These are finite results on the residual quotient itself; they do not claim that its leading kq coefficient has been evaluated.

31.6 The complete residual quotient in the conductor graph

At a fixed original endpoint N , let $\mathcal{P}_N$, I_N , $\mathcal{W}_N = U_N^{-1}(V_A)$ and the actual source isomorphism $S_N p = (P_{<v} p, \mathcal{T}_{A,N} p)$ be precisely CEP28–31. The source carries its original orthonormal Gamma coordinates of order $s = 1$ or $s = k$. The induced linear isomorphism on the residual observation quotient is

$$\mathcal{W}_N / U_N^{-1}(K) \xrightarrow{[p] \mapsto [S_N p]} S_N \mathcal{W}_N / S_N U_N^{-1}(K), \quad [p] \mapsto D U_N p \in \mathbb{C}^j. \quad (\text{ROQ23})$$

The last map identifies the left quotient with V_A/K and its fixed coordinates ROQ6. Its kernel and surjectivity follow from ROQ7 and surjectivity of U_N ; applying the invertible S_N proves the other isomorphism. Thus all relation directions, including the source ideal and the entire actual K , are retained in the transported denominator.

Here are exact matrices of that denominator and the resulting metric. Let Z_N^χ be the original source coefficient columns $\chi_k S^a$, $0 \leq a \leq N - q$, with no columns when $N = q - 1$. Take the actual primary frames I_K of ROQ12 and L of ROQ6 and set

$$Y_{K,N} = U_N^* G_N I_K, \quad Y_{L,N} = U_N^* G_N L, \quad J_N = [Z_N^\chi, Y_{K,N}]. \quad (\text{ROQ24})$$

The columns of $Y_{K,N}$ are the minimum lifts of I_K by the same orthogonality proof as OKM8. In particular they are perpendicular to $I_N = \ker U_N$, so J_N has independent columns and its image

is exactly $U_N^{-1}(K)$. It has $N + 1 - q + r$ columns, while $\dim \mathcal{W}_N = N + 1 - q + \Delta$, leaving precisely j quotient dimensions. The transported complete relation projector and Gram are

$$\begin{aligned} P_N^{\Xi, \text{graph}} &= I - S_N J_N (J_N^* S_N^* S_N J_N)^{-1} J_N^* S_N^*, \\ \tilde{Q}_{\Xi, N} &= Y_{L, N}^* S_N^* P_N^{\Xi, \text{graph}} S_N Y_{L, N} > 0. \end{aligned} \tag{ROQ25}$$

The inverse exists because S_N is invertible and J_N has independent columns. Multiplication verifies that the projector is Hermitian, idempotent and projects onto the orthogonal complement of the entire $S_N U_N^{-1}(K)$. Thus its Gram is the minimum norm of the transported quotient ROQ23. Replacing $Y_{L, N}$ by another source lift of L adds columns in I_N , which this projector kills. Replacing the section L by another section of ROQ6 adds columns in K ; their lifts lie in the span of J_N and are killed. Therefore ROQ25 computes the specified quotient metric independently of either choice.

Before applying S_N , the same projection gives

$$Q_{\Xi, N} = L^* G_N L - L^* G_N I_K (I_K^* G_N I_K)^{-1} I_K^* G_N L. \tag{ROQ26}$$

To prove it, $Y_{L, N}$ and $Y_{K, N}$ are perpendicular to the ideal, so projection off J_N only subtracts $Y_{K, N} (Y_{K, N}^* Y_{K, N})^{-1} Y_{K, N}^* Y_{L, N}$. Using $U_N U_N^* = R_N$ in their Gram products yields exactly ROQ26. It is also the minimum of the original G_N norm on $Ly + K$, so it agrees with the already proved ROQ11 and ROQ16.

The all-rank exterior estimate CEP33–35 applies to ROQ23: represent a quotient class by its minimum lift in the denominator's orthogonal complement, apply S_N , then project off the target denominator. This projection has every exterior norm at most one. Consequently the induced j th exterior norm is at most $\|\bigwedge^j S_N\|$, and its inverse is at most $\|\bigwedge^j S_N^{-1}\|$ by the identical argument for S_N^{-1} . CEP34 bounds both logarithms from above by E_N . Applying them to the fixed frame ROQ6 and squaring exterior volumes gives

$$\left| \log \det \tilde{Q}_{\Xi, N} - \log \det Q_{\Xi, N} \right| \leq 2E_N, \quad \left| \sum_{N \in \mathcal{N}} \sigma_N \log \det \tilde{Q}_{\Xi, N} - F_{\Xi, k}^{(s)} \right| \leq \epsilon_{A, k}^{(s)} = o(kq). \tag{ROQ27}$$

All cross terms in $J_N^* S_N^* S_N J_N$ remain present in ROQ25, including $P_{<v}(\chi_k S^a)$ and its pairings with the minimum kernel lifts. Its ideal columns can equally be generated from the full products d_{rs} and maps $\mathcal{U}_{A, L}$ in CEP41–45, because those formulas prove exactly $S_N Z_N^X$ column by column. This supplies the typed relation between the explicit residual observation and the conductor graph at all four original cutoffs.

32 A quantitative power comparison for the actual outer-strip quotient

Retain the original real parameters and integers

$$0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad k = 4l + 1 \geq 9, \quad (OSP1)$$

$$q = (k + 1)^2, \quad k' = k - 8, \quad q' = (k - 7)^2, \quad \Delta = q - q' = 16k - 48.$$

The two source orders in every formula below are precisely $s = 1, k$. Set $b = s/2$, $M_s = (2\pi)^{s/2}$, and retain the full original measure

$$dm_s(y) = M_s \frac{2^b}{4\pi\Gamma(b)} |\Gamma(b/2 + iy/2)|^2 dy. \quad (OSP2)$$

In the original coordinate $S = k/2 + iy$, and respectively $S' = k/2 - 4 + iy$, the complete monic root polynomials are

$$P_k(y) = i^{-q} \chi_k(k/2 + iy) = \prod_{a, b_0=0}^k [y - (2b_0 - k)\gamma + i(2a - k)\delta], \quad (OSP3)$$

$$P_{k'}(y) = i^{-q'} \chi_{k'}(k/2 - 4 + iy) = \prod_{a, b_0=0}^{k'} [y - (2b_0 - k')\gamma + i(2a - k')\delta].$$

The index b_0 in these products is distinct from the source parameter $b = s/2$. Each root occurs with exactly its displayed multiplicity. The map $(a, b_0) \mapsto (a + 4, b_0 + 4)$ identifies the second root list with the central subrectangle of the first, so

$$P_k = P_{k'} P_{\partial, k}, \quad P_{\partial, k}(y) = \prod_{(a, b_0) \in \{0, \dots, k\}^2 \setminus \{4, \dots, k-4\}^2} [y - (2b_0 - k)\gamma + i(2a - k)\delta]. \quad (OSP4)$$

This proves the factorization and its monic constant; the outer degree is Δ . Both complete root lists are invariant under $\omega \mapsto -\omega$. Every root has modulus at most

$$R = k\sqrt{\delta^2 + \gamma^2}. \quad (OSP5)$$

32.1 The two original source masses and a proved interval lower bound

We first record the source facts needed in the comparison, including their constants and proofs. For $a > 0$ and real x , the two Euler integrals and the change of variables in the beta integral give

$$\frac{\Gamma(a + ix)\Gamma(a - ix)}{\Gamma(2a)} = \int_{\mathbb{R}} e^{ixu} (2 \cosh(u/2))^{-2a} du. \quad (OSP6)$$

For example the intermediate beta integral is $\int_0^\infty t^{a+ix-1} (1+t)^{-2a} dt$, and $t = e^u$ gives the displayed expression. Absolute convergence follows from $a > 0$. Rotating Euler's integral through $\text{sgn}(y)\theta$, $0 < \theta < \pi/2$, gives

$$|\Gamma(a + iy/2)| \leq e^{-\theta|y|/2} \Gamma(a) (\cos \theta)^{-a}. \quad (OSP7)$$

The rotation follows from Cauchy's theorem on the intervening sector: the large circular arc tends to zero because its real exponential has positive cosine, and the small one tends to zero because $a > 0$. The remaining radial integral has absolute value at most $\Gamma(a)(\cos \theta)^{-a}$. In particular the Fourier transform in OSP6 is integrable. Fourier inversion, followed by $y = 2x$, now gives

$$\int_{\mathbb{R}} e^{i\xi y} dm_s(y) = M_s (\cosh \xi)^{-b}, \quad \int_{\mathbb{R}} dm_s = M_s. \quad (OSP8)$$

The same identity holds analytically for $|\operatorname{Im} \xi| < \pi/2$: OSP7 with any larger $\theta < \pi/2$ supplies an integrable dominating function on each smaller strip. Substituting $\xi = -it$ proves

$$\int_{\mathbb{R}} e^{ty} dm_s(y) = M_s(\cos t)^{-b}, \quad |t| < \pi/2. \quad (\text{OSP9})$$

This also proves finiteness of all moments used below.

Write

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}, \quad \ell_s = \frac{s-1}{4}, \quad \beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}. \quad (\text{OSP10})$$

The two values of ℓ_s are the integers $0, l$. The reflection identity gives a useful explicit lower bound

$$\frac{d\sigma}{dy}(y) \geq C_L e^{-\pi|y|}. \quad (\text{OSP11})$$

Here is its proof on the required domain. For $0 < \operatorname{Re} z < 1$ the Euler integrals give $\Gamma(z)\Gamma(1-z) = \int_0^\infty t^{z-1}/(1+t) dt$. On a keyhole contour with argument in $(0, 2\pi)$, the circular terms vanish with bounds $O(r^{\operatorname{Re} z})$ at zero and $O(R^{\operatorname{Re} z-1})$ at infinity. The residue at -1 is $e^{i\pi(z-1)}$; the two radial integrals therefore give $(1 - e^{2\pi iz}) \int_0^\infty t^{z-1}/(1+t) dt = 2\pi i e^{i\pi(z-1)}$. Their ratio is $\pi/\sin(\pi z)$. At $z = 1/4 + iy/2$ this proves

$$|\Gamma(1/4 + iy/2)|^2 |\Gamma(3/4 - iy/2)|^2 = \frac{2\pi^2}{\cosh(\pi y)}.$$

The unrotated Euler integral bounds the second squared factor by $\Gamma(3/4)^2$, proving $d\sigma/dy \geq C_L/\cosh(\pi y) \geq C_L e^{-\pi|y|}$.

Gamma recurrence, proved by one integration by parts in Euler's integral, gives the exact original-source identity

$$dm_s(y) = \beta_s \prod_{j=0}^{\ell_s-1} (y^2 + (2j + \frac{1}{2})^2) d\sigma(y). \quad (\text{OSP12})$$

Indeed $b/2 = \ell_s + 1/4$, and each recurrence factor before squaring and clearing denominators is $j + 1/4 + iy/2$; its squared modulus is $[y^2 + (2j + 1/2)^2]/4$. The factor $4^{-\ell_s}$ together with the prefactor in OSP2 is exactly $\beta_s/(2\pi)$, because $b = 2\ell_s + 1/2$. This proves every factor in OSP12. For $s = 1$ its product is empty and $\beta_1 = 1$. In particular, on the full interval $q \leq y \leq 2q$,

$$\frac{dm_s}{dy}(y) \geq \beta_s q^{2\ell_s} C_L e^{-2\pi q}. \quad (\text{OSP13})$$

The source mass-to-lower-density factor obeys the uniform finite bound

$$\frac{M_s}{\beta_s q^{2\ell_s}} = \sqrt{2} \Gamma(b) q^{-2\ell_s} \leq \sqrt{2\pi} \quad (s = 1, k). \quad (\text{OSP14})$$

To prove the inequality, use $\Gamma(b) = \sqrt{\pi} \prod_{j=0}^{2\ell_s-1} (j + 1/2) \leq \sqrt{\pi} b^{2\ell_s}$ and $b \leq q$. All source masses have thus been retained before taking this exact ratio.

32.2 A complete coefficient-space estimate for the inner interval

Let $p \in \mathbb{C}[y]$ have degree at most d , and let $0 < T \leq q/2$. Then

$$\sup_{|y| \leq T} |p(y)|^2 \leq \frac{(d+1)^2 10^{2d}}{q} \int_q^{2q} |p(u)|^2 du. \quad (\text{OSP15})$$

This controls every coefficient direction; no assumption on zeros, leading coefficient, or extremizers of p is made.

For proof define the Legendre polynomial $L_j(x) = (2^j j!)^{-1} (d/dx)^j (x^2 - 1)^j$. Its leading coefficient is $(2j)!/[2^j (j!)^2]$. Integrating by parts j times, with all lower boundary derivatives zero, proves orthogonality to lower powers and

$$\int_{-1}^1 L_j(x)^2 dx = \frac{(2j)!}{2^{2j} (j!)^2} \int_{-1}^1 (1 - x^2)^j dx = \frac{2}{2j + 1}.$$

For the middle integral set $x = 2t - 1$; repeated integration by parts gives $\int_0^1 t^j (1 - t)^j dt = (j!)^2 / (2j + 1)!$. Consequently $\sqrt{2j + 1} L_j(2z - 3)$, $0 \leq j \leq d$, form an orthonormal basis for degree-at-most- d polynomials on $[1, 2]$. Leibniz's rule also gives the explicit identity

$$L_j(x) = 2^{-j} \sum_{h=0}^j \binom{j}{h}^2 (x - 1)^{j-h} (x + 1)^h.$$

For $|x| \leq 4$ its modulus is at most $(5/2)^j \sum_h \binom{j}{h}^2 \leq (5/2)^j 4^j = 10^j$; the sum inequality follows by comparing with $(\sum_h \binom{j}{h})^2 = 4^j$. Expanding $p(qz)$ in the displayed orthonormal basis and applying Cauchy-Schwarz gives, for $|z| \leq T/q \leq 1/2$,

$$|p(qz)|^2 \leq \sum_{j=0}^d (2j + 1) 10^{2j} \int_1^2 |p(qv)|^2 dv \leq (d + 1)^2 10^{2d} \int_1^2 |p(qv)|^2 dv.$$

Here $|2z - 3| \leq 4$ and $\sum_{j=0}^d (2j + 1) = (d + 1)^2$. Changing variables proves OSP15, including $d = 0$.

32.3 The exact original-root map and a finite full-norm comparison

In this subsection P is either of the complete polynomials $P_k, P_{k'}$ in OSP3, and $Q = \deg P$ is respectively q, q' . For $d \geq 0$ retain the two injections into the same original polynomial Hilbert space

$$\begin{aligned} \mathcal{J}_P : \mathbb{C}[y]_{\leq d} &\longrightarrow \mathbb{C}[y]_{\leq Q+d}, & p &\longmapsto Pp, \\ \mathcal{J}_{0,Q} : \mathbb{C}[y]_{\leq d} &\longrightarrow \mathbb{C}[y]_{\leq Q+d}, & p &\longmapsto y^Q p. \end{aligned} \tag{OSP16}$$

Their codomain norm is precisely m_s in OSP2. If $P(y) = \sum_{a=0}^Q P_a y^a$, set $P_a = 0$ outside this range. In increasing coefficient frames their matrices are

$$(\mathcal{J}_P)_{rj} = P_{r-j}, \quad (\mathcal{J}_{0,Q})_{rj} = \mathbf{1}_{\{r=Q+j\}}, \quad 0 \leq r \leq Q + d, \quad 0 \leq j \leq d. \tag{OSP17}$$

Thus the exact isomorphism between their images is $y^Q p \mapsto Pp$, with inverse $Pp \mapsto y^Q p$. It preserves the original coefficient vector p . The first matrix contains every actual root coefficient; none is replaced inside that map. Returning to S or S' multiplies the relevant relation columns by the actual phase i^Q and applies the triangular coordinate pullback with diagonal entries i^j . Each has the stated exact inverse and determinant of modulus one.

Suppose

$$R < T \leq q/2. \tag{OSP18}$$

Define the finite positive constants

$$\begin{aligned} \eta_{Q,T} &= \frac{QR^2}{T^2 - R^2}, \\ \kappa_{Q,d,T,s} &= \frac{M_s}{\beta_s q^{2\ell_s} C_L} \frac{(d + 1)^2}{q} e^{2\pi q} 10^{2d} \left(\frac{T^2 + R^2}{q^2} \right)^Q, \\ e_{Q,d,T,s} &= \eta_{Q,T} + \log(1 + \kappa_{Q,d,T,s}). \end{aligned} \tag{OSP19}$$

Then every $p \in \mathbb{C}[y]_{\leq d}$ satisfies

$$e^{-e_{Q,d,T,s}} \int_{\mathbb{R}} |y|^{2Q} |p(y)|^2 dm_s(y) \leq \int_{\mathbb{R}} |P(y)p(y)|^2 dm_s(y) \leq e^{e_{Q,d,T,s}} \int_{\mathbb{R}} |y|^{2Q} |p(y)|^2 dm_s(y). \quad (\text{OSP20})$$

Here is a proof retaining the entire inner region. Pair the actual roots as $\omega, -\omega$. On $|y| \geq T$, each paired quotient is $1 - \omega^2/y^2$, so

$$\left| \log \frac{|P(y)|^2}{|y|^{2Q}} \right| \leq -Q \log(1 - R^2/T^2) \leq \frac{QR^2}{T^2 - R^2} = \eta_{Q,T}. \quad (\text{OSP21})$$

The first inequality uses $1 - u \leq |1 - z| \leq 1 + u$ for $|z| \leq u < 1$ on each of $Q/2$ pairs. The second follows by integrating $1/(1 - u) \leq 1/(1 - u_0)$ on $0 \leq u \leq u_0 = R^2/T^2$. For every real y , the same pairing gives $|P(y)|^2 \leq (y^2 + R^2)^Q$. Write

$$A_B = \int_q^{2q} |y|^{2Q} |p(y)|^2 dm_s(y).$$

For nonzero p this number is positive, and OSP13 gives $A_B \geq q^{2Q} \beta_s q^{2\ell_s} C_L e^{-2\pi q} \int_q^{2q} |p(y)|^2 dy$. OSP15 and the exact full mass OSP8 imply

$$\begin{aligned} \int_{|y| < T} |P(y)p(y)|^2 dm_s(y) &\leq M_s(T^2 + R^2)^Q \sup_{|y| \leq T} |p(y)|^2 \leq \kappa_{Q,d,T,s} A_B, \\ \int_{|y| < T} |y|^{2Q} |p(y)|^2 dm_s(y) &\leq M_s T^{2Q} \sup_{|y| \leq T} |p(y)|^2 \leq \kappa_{Q,d,T,s} A_B. \end{aligned} \quad (\text{OSP22})$$

Let A_{out} and B_{out} be the integrals outside $(-T, T)$ with factors $|y|^{2Q}$ and $|P|^2$, respectively. Since $[q, 2q]$ is outside that interval, $A_B \leq A_{\text{out}}$. OSP21 gives $e^{-\eta} A_{\text{out}} \leq B_{\text{out}} \leq e^{\eta} A_{\text{out}}$. OSP22 bounds both inner contributions by κA_{out} . Therefore the full reference norm is at most $(1 + \kappa) A_{\text{out}}$ and at least A_{out} , while the full actual norm is at least $e^{-\eta} A_{\text{out}}$ and at most $(e^{\eta} + \kappa) A_{\text{out}} \leq e^{\eta} (1 + \kappa) A_{\text{out}}$. These four inequalities prove OSP20. The zero polynomial satisfies it directly.

32.4 Uniform bounded logarithmic cost at every required dimension

Choose the fixed number and cutoff

$$\epsilon = 2^{-32}, \quad T = \epsilon q,$$

and impose the explicit finite conditions

$$k \geq 29, \quad k \sqrt{\delta^2 + \gamma^2} \leq \frac{\epsilon q}{2}. \quad (\text{OSP23})$$

In particular $T \geq 2R$, as required for a uniform margin in OSP21. These conditions hold eventually for every fixed original δ, γ ; the sufficient bound $k \geq \max\{29, 2\sqrt{\delta^2 + \gamma^2}/\epsilon\}$ follows from $q \geq k^2$. The first condition implies $q' \geq q/2$: $2(k - 7)^2 - (k + 1)^2 = k^2 - 30k + 97$ is positive at $k = 29$ and is increasing thereafter.

For $Q \in \{q', q\}$ and every $0 \leq d \leq 2q$, set

$$\begin{aligned} K_q &= \frac{\sqrt{2\pi}}{C_L} \frac{(2q + 1)^2}{q} e^{-2q}, \\ \mathcal{E}_k &= \frac{qR^2}{\epsilon^2 q^2 - R^2} + \log(1 + K_q). \end{aligned} \quad (\text{OSP24})$$

Then OSP19 satisfies $e_{Q,d,T,s} \leq \mathcal{E}_k$ for both original source orders. To verify this uniformly, OSP14, $T^2 + R^2 \leq (5/4)\epsilon^2 q^2$, $Q \geq q/2$ and $d \leq 2q$ give

$$\kappa_{Q,d,T,s} \leq \frac{\sqrt{2\pi}}{C_L} \frac{(2q + 1)^2}{q} \exp\left(q\left\{2\pi + 4 \log 10 + \frac{1}{2} \log(5/4) + \log \epsilon\right\}\right). \quad (\text{OSP25})$$

The expression in braces is less than -2 . Indeed $\pi < 4$, $\log 10 < 4 \log 2$, $\log(5/4) < 1/4$, and $\log 2 > 2/3$ give an upper bound $8 - 16 \log 2 + 1/8 < -61/24 < -2$. For completeness, $\pi < 4$ follows from $\pi/4 = \int_0^1 (1+x^2)^{-1} dx < 1$; and the substitution $x = (1+t)/(1-t)$ in $\log 2 = \int_1^2 dx/x$ gives $\log 2 = 2 \int_0^{1/3} (1-t^2)^{-1} dt > 2/3$. The logarithm bound follows by integrating $1/(1+x) < 1$. This proves OSP25 with K_q as stated. Also

$$0 \leq \mathcal{E}_k \leq \frac{4(\delta^2 + \gamma^2)}{3\epsilon^2} + \log(1 + K_q). \quad (\text{OSP26})$$

Here $R^2 \leq \epsilon^2 q^2/4$ and $R^2/q \leq \delta^2 + \gamma^2$. Thus the logarithmic norm cost is bounded at fixed original parameters, uniformly over all these coefficient degrees and both unchanged source orders.

32.5 Every relation determinant and the exact outer quotient

For a nonzero polynomial P and $a \geq 0$ define in the unchanged monomial frame

$$\begin{aligned} D_a(P; s) &= \det \left[\int_{\mathbb{R}} y^{i+j} |P(y)|^2 dm_s(y) \right]_{0 \leq i, j < a}, \\ \mathcal{D}_a(Q; s) &= \det \left[\int_{\mathbb{R}} y^{2Q+i+j} dm_s(y) \right]_{0 \leq i, j < a}, \quad D_0 = \mathcal{D}_0 = 1. \end{aligned} \quad (\text{OSP27})$$

All nonempty Grams are positive definite, because a nonzero polynomial has positive squared modulus on an open subinterval of the strictly positive original measure. In the notation of OSP17 their exact matrices are $J_P^* H_{Q+a-1}^{(s)} J_P$ and $J_{0,Q}^* H_{Q+a-1}^{(s)} J_{0,Q}$, where $H_{Q+a-1}^{(s)}$ is the complete original moment Gram. Consequently OSP20–26 prove for $1 \leq a \leq 2q+1$

$$|\log D_a(P; s) - \log \mathcal{D}_a(Q; s)| \leq a \mathcal{E}_k, \quad P = P_k \text{ or } P_{k'}. \quad (\text{OSP28})$$

To see the determinant step exactly, conjugate the actual Gram by the inverse positive square root of the reference Gram. OSP20 places every eigenvalue of the resulting positive Hermitian matrix in $[e^{-\mathcal{E}_k}, e^{\mathcal{E}_k}]$. The logarithm of its determinant is the sum of its a eigenvalue logarithms, proving OSP28. This is the full exterior determinant estimate, including every polynomial coefficient direction. The estimate also holds at $a = 0$ with both sides zero.

For $t \in \{0, 1, q, q+1\}$, let $L = t + \Delta - 1$. The actual quotient has source $\mathbb{C}[y]_{\leq L}$ in measure $|P_{k'}(y)|^2 dm_s(y)$, target the increasing remainder frame of $\mathbb{C}[y]/(P_{\partial,k})$, and its minimum Gram $Q_{\partial,L}^{(s)}$. Define a comparison quotient with source $\mathbb{C}[y]_{\leq L}$ in measure $|y|^{2q'} dm_s(y)$, target the increasing remainder frame of $\mathbb{C}[y]/(y^\Delta)$, and minimum Gram $Q_{0,L}^{(s)}$. Its source measure still has order s and the full mass factor in OSP2.

For precision, the exact map relating the coefficient presentations is the degree-preserving isomorphism

$$\mathcal{U}_L(y^j) = \begin{cases} y^j, & 0 \leq j < \Delta, \\ P_{\partial,k}(y) y^{j-\Delta}, & \Delta \leq j \leq L. \end{cases} \quad (\text{OSP29})$$

Its coefficient matrix has diagonal entries one and no terms of degree greater than that of its input column. It is therefore invertible with determinant one. If $R_{\partial,L}$ is division remainder modulo $P_{\partial,k}$ and $R_{0,L}$ is division remainder modulo y^Δ , then $R_{\partial,L} \mathcal{U}_L = R_{0,L}$: the first Δ basis vectors map to their stated remainders, and every later vector maps into the actual ideal. This proves the precise morphism between the quotient coefficient spaces; no equality of their metrics is assumed.

Using the original monic relation frames on the two sources gives the exact identities

$$\det Q_{\partial,t+\Delta-1}^{(s)} = \frac{D_{t+\Delta}(P_{k'}; s)}{D_t(P_k; s)}, \quad \det Q_{0,t+\Delta-1}^{(s)} = \frac{\mathcal{D}_{t+\Delta}(q'; s)}{\mathcal{D}_t(q; s)}. \quad (\text{OSP30})$$

For proof, append to $1, y, \dots, y^{\Delta-1}$ the ideal frame $P_{\partial,k}, yP_{\partial,k}, \dots, y^{t-1}P_{\partial,k}$. Its source coefficient determinant is one by OSP29. The relation Gram is $D_t(P_k; s)$ because $P_{k'}P_{\partial,k} = P_k$. Completing

the square in the ideal coordinates gives the minimum quotient Gram as the Schur complement; the determinant of the full source Gram is the product of that determinant and the relation determinant. This proves the first identity. For the second repeat the same calculation with y^Δ : $y^{q'} y^\Delta = y^q$, with the original source order s . The same proof includes $t = 0$ using the empty relation block.

OSP28 applies to every index here, since $t + \Delta \leq q + \Delta + 1 \leq 2q + 1$. Define the exact endpoint difference $\varepsilon_t^{(s)}$ by

$$\log \det Q_{\partial, t+\Delta-1}^{(s)} = \log \mathcal{D}_{t+\Delta}(q'; s) - \log \mathcal{D}_t(q; s) + \varepsilon_t^{(s)}.$$

Then, separately at each of the four endpoints,

$$|\varepsilon_t^{(s)}| \leq (2t + \Delta) \mathcal{E}_k. \quad (\text{OSP31})$$

In particular set

$$\begin{aligned} \mathcal{F}_{0,k}^{(s)} &= \log \mathcal{D}_\Delta(q'; s) + \log \mathcal{D}_{\Delta+1}(q'; s) - \log \mathcal{D}_{q+\Delta}(q'; s) - \log \mathcal{D}_{q+\Delta+1}(q'; s) \\ &\quad - \log \mathcal{D}_1(q; s) + \log \mathcal{D}_q(q; s) + \log \mathcal{D}_{q+1}(q; s). \end{aligned} \quad (\text{OSP32})$$

The actual four-endpoint quantity of CTV13 therefore obeys the proved finite estimate

$$\boxed{|\mathcal{F}_{\partial,k}^{(s)} - \mathcal{F}_{0,k}^{(s)}| \leq 4(q + \Delta + 1) \mathcal{E}_k = O_{\delta, \gamma}(q) = o(kq), \quad s = 1, k.} \quad (\text{OSP33})$$

Indeed its exact error is $\varepsilon_0 + \varepsilon_1 - \varepsilon_q - \varepsilon_{q+1}$; summing OSP31 gives $\Delta + (\Delta + 2) + (2q + \Delta) + (2q + \Delta + 2) = 4(q + \Delta + 1)$. OSP26 and $\Delta < q$ prove the last two estimates. This calculation uses all four source degrees $\Delta - 1, \Delta, q + \Delta - 1, q + \Delta$ and every source and relation determinant in OSP30.

32.6 The resulting same-source moment target

The moments in OSP27 are completely specified by the original analytic identity OSP9:

$$\mu_j^{(s)} = \int y^j dm_s(y) = M_s \frac{d^j}{dz^j} (\cos z)^{-s/2} \Big|_{z=0}, \quad \mathcal{D}_a(Q; s) = \det[\mu_{2Q+i+j}^{(s)}]_{0 \leq i, j < a}. \quad (\text{OSP34})$$

The differentiated integral is justified by OSP7, or directly by the finite exponential moments on a slightly larger interval. Odd moments vanish because the density is even. Reordering rows and columns together by parity therefore proves the exact further factorization

$$\mathcal{D}_a(Q; s) = \det[\mu_{2(Q+i+j)}^{(s)}]_{0 \leq i, j < \lceil a/2 \rceil} \det[\mu_{2(Q+1+i+j)}^{(s)}]_{0 \leq i, j < \lfloor a/2 \rfloor}. \quad (\text{OSP35})$$

There is no permutation sign, since the same permutation is applied to both rows and columns. Empty factors are one. In OSP32 the coefficient of $\log M_s$ is $\Delta + (\Delta + 1) - (q + \Delta) - (q + \Delta + 1) - 1 + q + (q + 1) = 0$. This is an exact cancellation only after all original moments and determinant dimensions have been retained.

Equation OSP33 proves an error-controlled comparison with the exact finite determinant difference OSP32. It does not obtain a first variation by differentiating a leading q^2 law. In particular no numerical value of the limit of $\mathcal{F}_{0,k}^{(s)}/(kq)$ is asserted by this comparison.

33 The outer-root and actual residual bounds in the original arithmetic return

Retain the complete original data of CEP, CTV and ROQ. In particular $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = 16k - 48$, $r = 8k - 16$ and $j = 8k - 32$. The quartet, full amplitude unit, period and branch are the specified original ones on the proved five-orbit locus. The source orders remain $s \in \{1, k\}$. Every matrix below is the actual matrix in those chapters, with its original mass and coefficient frame.

33.1 The two finite error endpoints retain their signs

Write

$$\mathcal{N}_+ = \{q - 1, q\}, \quad \mathcal{N}_- = \{2q - 1, 2q\}, \quad \epsilon_{A,k}^{(s)} = 2 \sum_{N \in \mathcal{N}_+ \cup \mathcal{N}_-} E_N^{(s)}. \quad (\text{OAR1})$$

Here $E_N^{(s)}$ is the explicit full conductor constant of CEP35. The nonnegative constants $L_N^{\text{tail}}, U_N^{\text{tail}}$ are exactly CTV18 in the same source order and with the original q', δ, γ . Define

$$\begin{aligned} a_{k,s}^- &= v \left(\sum_{N \in \mathcal{N}_+} L_N^{\text{tail}} + \sum_{N \in \mathcal{N}_-} U_N^{\text{tail}} \right), \\ a_{k,s}^+ &= v \left(\sum_{N \in \mathcal{N}_+} U_N^{\text{tail}} + \sum_{N \in \mathcal{N}_-} L_N^{\text{tail}} \right), \\ e_{k,s}^- &= \epsilon_{A,k}^{(s)} + a_{k,s}^-, \quad e_{k,s}^+ = \epsilon_{A,k}^{(s)} + a_{k,s}^+. \end{aligned} \quad (\text{OAR2})$$

These are finite quantities for every original period in the stated domain. For each fixed original period, both $e_{k,s}^\pm = o(kq)$. Indeed CEP36 proves this for $\epsilon_{A,k}^{(s)}$ and CTV18–20 proves it for each of the four tail contributions, uniformly over the possible indices $0 \leq v \leq 80$ and both original source orders. When $v = 0$, both $a_{k,s}^\pm$ are exactly zero.

Set $\Phi_{k,s} = F_{\Xi,k}^{(s)}$ and retain the full outer quotient $\mathcal{F}_{\partial,k}^{(s)}$ of CTV11–13. The exact identity is

$$\begin{aligned} F_K^{(s)} &= \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s} + \mathfrak{e}_{k,s}, \\ \mathfrak{e}_{k,s} &= \mathcal{E}_{A,k}^{(s)} + \sum_{N \in \mathcal{N}_+} \mathcal{R}_{N,v}^{(s)} - \sum_{N \in \mathcal{N}_-} \mathcal{R}_{N,v}^{(s)}, \\ -e_{k,s}^- &\leq \mathfrak{e}_{k,s} \leq e_{k,s}^+. \end{aligned} \quad (\text{OAR3})$$

The equality is CTV22, keeping its original two errors. CTV19 says $-vL_N^{\text{tail}} \leq \mathcal{R}_{N,v}^{(s)} \leq vU_N^{\text{tail}}$. At a positive endpoint this contributes lower bound $-vL_N^{\text{tail}}$ and upper bound vU_N^{tail} ; at a negative endpoint it contributes lower bound $-vU_N^{\text{tail}}$ and upper bound vL_N^{tail} . Adding these four inequalities and the bounds $-\epsilon_{A,k}^{(s)} \leq \mathcal{E}_{A,k}^{(s)} \leq \epsilon_{A,k}^{(s)}$ proves each sign of OAR3.

33.2 Original positive increments give explicit kernel caps

The full value of $\Phi_{k,s}$ is ROQ19's finite sum

$$\Phi_{k,s} = \log(1 + \eta_{q-1}) + 2 \sum_{N=q}^{2q-2} \log(1 + \eta_N) + \log(1 + \eta_{2q-1}), \quad \lambda_s = \log(1 + \eta_{q-1}) \geq 0. \quad (\text{OAR4})$$

Every η_N here is ROQ17's original constrained column, including the complete conductor–observation cross covariance and its positive denominator. No new observation is selected. ROQ21 gives, in the original AKS notation,

$$\lambda_s \leq \Phi_{k,s} \leq V_s := j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - \lambda_s. \quad (\text{OAR5})$$

Retain the complete earlier kernel interval $L_s \leq F_K^{(s)} \leq U_s$ of AKJ2/MRI4e, with its pointwise minimum of the KAF and AKS bounds. Define two explicit quantities

$$\begin{aligned} L_s^\partial &= \max \left\{ L_s, \mathcal{F}_{\partial,k}^{(s)} - e_{k,s}^- - V_s \right\}, \\ U_s^\partial &= \min \left\{ U_s, \mathcal{F}_{\partial,k}^{(s)} + e_{k,s}^+ - \lambda_s \right\}. \end{aligned} \quad (\text{OAR6})$$

Then

$$\begin{aligned} L_s^\partial &\leq F_K^{(s)} \leq U_s^\partial, \\ \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s} - e_{k,s}^- &\leq F_K^{(s)} \leq \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s} + e_{k,s}^+. \end{aligned} \quad (\text{OAR7})$$

For the first line substitute both ends of OAR5 into OAR3 and intersect with the retained interval. The second line substitutes the full finite value OAR4 into OAR3. Intersecting these intervals preserves every bound. Both contain the actual value, so their intersection is nonempty. The dimension j is used only in the proved finite comparison OAR5. It assigns no leading determinant. Each additional evaluated positive term of OAR4 can be included in the upper subtraction in OAR6 by the same inequality.

33.3 The correlated arithmetic kernel and boundary

Keep the original arithmetic total $\mathcal{B}_k^{\text{ar}}$, the scalar δ_k^σ and the four source-difference constants $C_K^\pm, C_B^\pm$ of MRI4f. They satisfy, for the actual $d = F_K^{\text{ar}} - F_K^{(1)}$,

$$\max\{-C_K^-, \delta_k^\sigma - C_B^+\} \leq d \leq \min\{C_K^+, \delta_k^\sigma + C_B^-\}. \quad (\text{OAR8})$$

This is the pair of original kernel bounds together with the boundary bounds, transferred by $d + (F_B^{\text{ar}} - F_B^{(1)}) = \delta_k^\sigma$. It retains their common scalar and signs. Define

$$\begin{aligned} L_{\text{ar}}^\partial &= \max\{L_{\text{ar}}, L_1^\partial - C_K^-, L_1^\partial + \delta_k^\sigma - C_B^+\}, \\ U_{\text{ar}}^\partial &= \min\{U_{\text{ar}}, U_1^\partial + C_K^+, U_1^\partial + \delta_k^\sigma + C_B^-\}. \end{aligned} \quad (\text{OAR9})$$

Here $[L_{\text{ar}}, U_{\text{ar}}]$ is the complete interval of MRI4f. Adding OAR8 to the source-1 first line of OAR7 proves

$$\begin{aligned} L_{\text{ar}}^\partial &\leq F_K^{\text{ar}} \leq U_{\text{ar}}^\partial, \\ \mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^\partial &\leq F_B^{\text{ar}} \leq \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^\partial. \end{aligned} \quad (\text{OAR10})$$

The second line subtracts the first from the exact original identity $F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}$. It uses the same arithmetic total at both endpoints. If the second line of OAR7 is used instead, exactly the same addition and subtraction yield the further interval containing the full residual sum. Thus every exact positive residual subtraction is available to the original arithmetic kernel and its correlated boundary.

33.4 The signed mixed return with its complete finite error

Write $\mathfrak{S}_k^{\text{mix}} = F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}$. The original BRI6d identity is $\mathfrak{S}_k^{\text{mix}} = F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma$. Consequently OAR3 gives the exact current expression

$$\begin{aligned} \mathfrak{S}_k^{\text{mix}} &= \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \mathfrak{e}_{k,k}, \\ -e_{k,k}^- &\leq \mathfrak{e}_{k,k} \leq e_{k,k}^+, \quad e_{k,k}^\pm = o(kq). \end{aligned} \quad (\text{OAR11})$$

This substitution proves every sign, keeping the original Gamma return and arithmetic scalar. Addition of their unchanged values to the source- k bounds OAR7 yields the corresponding finite intervals for this same signed quantity. The full original four-endpoint metrics, outer-root polynomial, weighted source, period-dependent certificates and graph denominator remain in OAR11 through their proved maps CTV and ROQ. The next leading-scale calculation is the difference of these two explicit quotient returns.

34 The same-source Gamma moment comparison in the arithmetic return

Retain every original parameter, matrix and error in OAR1–11. The complete OSP comparison supplies an additional bound directly on the actual outer-root quotient. Set $\epsilon = 2^{-32}$ and retain its explicit range

$$k \geq 29, \quad k\sqrt{\delta^2 + \gamma^2} \leq \epsilon(k+1)^2/2. \quad (\text{PMR1})$$

This range contains all sufficiently large original $k = 4l + 1$ for each fixed original quartet: $k \geq \max\{29, 2\sqrt{\delta^2 + \gamma^2}/\epsilon\}$ is sufficient because $(k+1)^2 \geq k^2$. For the remaining finite values the full earlier OAR estimates remain valid without using this comparison.

Keep OSP24's nonnegative $\mathcal{E}_k$ and put

$$c_k^\circ = 4(q + \Delta + 1)\mathcal{E}_k, \quad e_{k,s}^{0,-} = e_{k,s}^- + c_k^\circ, \quad e_{k,s}^{0,+} = e_{k,s}^+ + c_k^\circ. \quad (\text{PMR2})$$

OSP26 and OSP33 prove $c_k^\circ = O_{\delta,\gamma}(q) = o(kq)$ on PMR1, uniformly for the unchanged source orders $s = 1, k$. The actual outer quotient differs from OSP32's exact seven original Gamma moment determinants by at most c_k° . Consequently the full identity and signed bounds are

$$\begin{aligned} F_K^{(s)} &= \mathcal{F}_{0,k}^{(s)} - \Phi_{k,s} + \mathfrak{e}_{k,s}^0, \\ \mathfrak{e}_{k,s}^0 &= \mathfrak{e}_{k,s} + \mathcal{F}_{\partial,k}^{(s)} - \mathcal{F}_{0,k}^{(s)}, \\ -e_{k,s}^{0,-} &\leq \mathfrak{e}_{k,s}^0 \leq e_{k,s}^{0,+}, \quad e_{k,s}^{0,\pm} = o(kq). \end{aligned} \quad (\text{PMR3})$$

The first two equalities are OAR3 with the displayed exact difference added and subtracted. The two bounds follow by adding the interval $[-c_k^\circ, c_k^\circ]$ of OSP33 to the unequal OAR3 endpoints. No metric equality is substituted for the proved OSP coefficient maps and determinant comparison. The residual sum $\Phi_{k,s}$ retains all actual period data.

Let V_s and λ_s be OAR4–5's exact residual upper bound and first positive increment. Define

$$\begin{aligned} L_s^0 &= \max\{L_s^\partial, \mathcal{F}_{0,k}^{(s)} - e_{k,s}^{0,-} - V_s\}, \\ U_s^0 &= \min\{U_s^\partial, \mathcal{F}_{0,k}^{(s)} + e_{k,s}^{0,+} - \lambda_s\}. \end{aligned} \quad (\text{PMR4})$$

Then $L_s^0 \leq F_K^{(s)} \leq U_s^0$. To prove this, insert $\lambda_s \leq \Phi_{k,s} \leq V_s$ into PMR3 and intersect with OAR7. Using the full finite sum for $\Phi_{k,s}$ in PMR3 gives the sharper interval centred at $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}$. Both retain the same unequal finite errors of PMR2.

The exact arithmetic correlations of OAR8 give

$$\begin{aligned} L_{\text{ar}}^0 &= \max\{L_{\text{ar}}^\partial, L_1^0 - C_K^-, L_1^0 + \delta_k^\sigma - C_B^+\}, \\ U_{\text{ar}}^0 &= \min\{U_{\text{ar}}^\partial, U_1^0 + C_K^+, U_1^0 + \delta_k^\sigma + C_B^-\}. \end{aligned} \quad (\text{PMR5})$$

Adding those correlations to PMR4 at $s = 1$ proves $L_{\text{ar}}^0 \leq F_K^{\text{ar}} \leq U_{\text{ar}}^0$. Subtracting from the exact same arithmetic total gives

$$\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^0 \leq F_B^{\text{ar}} \leq \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^0. \quad (\text{PMR6})$$

Thus the actual kernel and boundary retain their common scalar and both original source comparisons.

Finally the original signed-return identity BRI6d and PMR3 at source order k give

$$\begin{aligned} F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} &= \mathcal{F}_{0,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \mathfrak{e}_{k,k}^0, \\ -e_{k,k}^{0,-} &\leq \mathfrak{e}_{k,k}^0 \leq e_{k,k}^{0,+}. \end{aligned} \quad (\text{PMR7})$$

This follows by substituting into $F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} = F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma$; every sign is retained. The entire original same-source moment expression is OSP32–35. The next asymptotic calculation must evaluate it with the actual residual sum and an error small compared with kq .

The exact finite endpoints under this comparison. The interval transport in PMR4–6 preserves the already retained outer-root endpoints exactly:

$$L_s^0 = L_s^\partial, \quad U_s^0 = U_s^\partial, \quad L_{\text{ar}}^0 = L_{\text{ar}}^\partial, \quad U_{\text{ar}}^0 = U_{\text{ar}}^\partial. \quad (\text{PMR8})$$

For proof, OSP33 gives $\mathcal{F}_{0,k}^{(s)} - c_k^\circ \leq \mathcal{F}_{\partial,k}^{(s)} \leq \mathcal{F}_{0,k}^{(s)} + c_k^\circ$. The new lower candidate in PMR4 is therefore at most $\mathcal{F}_{\partial,k}^{(s)} - e_{k,s}^- - V_s$, which is at most L_s^∂ by OAR6. The new upper candidate is at least $\mathcal{F}_{\partial,k}^{(s)} + e_{k,s}^+ - \lambda_s$, which is at least U_s^∂ . Taking the displayed maximum and minimum proves the source equalities. Substitution in PMR5 and the defining maxima and minima OAR9 proves the arithmetic equalities; subtracting from the same total proves equality of the boundary endpoints as well. Thus this comparison supplies the exact same-source moment target and controlled error; a further numerical improvement must use sharper evaluated moment or residual information. The preceding ROQ/OAR bounds remain the proved finite intervals throughout this transport.

35 Exact equilibrium for the square-root logarithmic potential

Throughout this section $\alpha > 0$ and $\beta > 0$. On $(0, \infty)$ set

$$V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x, \quad \mathcal{E}_{\alpha,\beta}(\mu) = \int V_{\alpha,\beta}(x) d\mu(x) - \iint \log|x-y| d\mu(x)d\mu(y). \quad (\text{EIQ1})$$

The measures considered initially are probability measures for which $\int V_{\alpha,\beta} d\mu$ is finite. Because this potential is bounded below and tends to $+\infty$ at both ends, this implies $\int(\sqrt{x} + |\log x|) d\mu < \infty$. The positive part of $\log|x-y|$ is then integrable, by $\log^+|x-y| \leq \log(1+x) + \log(1+y)$. Thus the energy is well defined with value in $\mathbb{R} \cup \{+\infty\}$. All comparisons below also apply to the usual extended energy defined by integrating the kernel $\frac{1}{2}V_{\alpha,\beta}(x) + \frac{1}{2}V_{\alpha,\beta}(y) - \log|x-y|$: the kernel is bounded below, and finite energy for this kernel implies the same moment conditions. To see the latter assertion without cancellation, put $g(x) = \frac{1}{2}V_{\alpha,\beta}(x) - \log(1+x)$. The kernel is at least $g(x) + g(y)$; g is bounded below and controls positive constant multiples of both $\sqrt{x}$ near infinity and $|\log x|$ near zero.

35.1 Existence and uniqueness of the two endpoints

For $0 < k < 1$ define the complete elliptic integrals by their actual real integrals

$$K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2 \sin^2 \theta}}, \quad E(k) = \int_0^{\pi/2} \sqrt{1-k^2 \sin^2 \theta} d\theta. \quad (\text{EIQ2})$$

There is exactly one $k \in (0, 1)$ satisfying

$$\frac{\sqrt{1-k^2}K(k)}{E(k)} = \frac{\alpha}{\alpha+2}. \quad (\text{EIQ3})$$

For this k put

$$r = \sqrt{1-k^2}, \quad v = \frac{(\alpha+2)\pi}{\beta E(k)}, \quad u = rv, \quad a = u^2, \quad b = v^2. \quad (\text{EIQ4})$$

In particular $0 < a < b < \infty$, and the endpoint equations, with all constants retained, are

$$uK(k) = \frac{\alpha\pi}{\beta}, \quad vE(k) = \frac{(\alpha+2)\pi}{\beta}, \quad k^2 = 1 - \frac{u^2}{v^2}. \quad (\text{EIQ5})$$

Here is a proof of the assertion of uniqueness. Differentiation of the integrals in (EIQ2), and integration of the derivative of $\sin \theta \cos \theta / \sqrt{1-k^2 \sin^2 \theta}$ between its two zero endpoint values, give

$$E'(k) = \frac{E(k) - K(k)}{k}, \quad K'(k) = \frac{E(k)}{k(1-k^2)} - \frac{K(k)}{k}.$$

For completeness the second identity is equivalent, after differentiation under the integral sign, to

$$\int_0^{\pi/2} \left\{ \frac{k^2 \sin^2 \theta}{(1-k^2 \sin^2 \theta)^{3/2}} - \frac{1-k^2 \sin^2 \theta}{(1-k^2)\sqrt{1-k^2 \sin^2 \theta}} + \frac{1}{\sqrt{1-k^2 \sin^2 \theta}} \right\} d\theta = 0;$$

the expression in braces is $-k^2(1-k^2)^{-1} d(\sin \theta \cos \theta / \sqrt{1-k^2 \sin^2 \theta})/d\theta$. With $t = E(k)/K(k)$, the integral definitions show strictly $r^2 < t < 1$. Consequently, if $f(k) = rK(k)/E(k)$, then

$$\frac{kf'(k)}{f(k)} = \frac{t}{r^2} + \frac{1}{t} - 1 - \frac{1}{r^2} = \frac{(t-1)(t-r^2)}{r^2 t} < 0. \quad (\text{EIQ6})$$

At $k = 0$ the continuous limiting value is $f(0) = 1$. As $k \uparrow 1$, the integrand of $rK(k)$ tends to zero at every $\theta < \pi/2$ and is bounded by 1, while $E(k) \rightarrow 1$ by dominated convergence. Hence

$f(k) \rightarrow 0$. This proves (EIQ3), including existence. It also proves smooth dependence of k, u, v, a, b on (α, β) in $(0, \infty)^2$, by the nonzero derivative in (EIQ6).

Let $m = (a + b)/2$, $R_s = \sqrt{(a + s)(b + s)}$ for $s \geq 0$, and $R_0 = uv$. Substitution $t = a \cos^2 \theta + b \sin^2 \theta$ gives, directly from (EIQ5),

$$\int_a^b \frac{V'_{\alpha, \beta}(t)}{\sqrt{(b-t)(t-a)}} dt = 0, \quad \int_a^b \frac{tV'_{\alpha, \beta}(t)}{\sqrt{(b-t)(t-a)}} dt = 2\pi. \quad (\text{EIQ7})$$

Indeed the three integrals with numerators $t^{-1/2}, t^{-1}, t^{1/2}$ are respectively $2K(k)/v$, $\pi/(uv)$, and $2vE(k)$, and $V'(t) = \beta/(2\sqrt{t}) - \alpha/t$.

35.2 A positive density and its exact Cauchy transform

The elementary Stieltjes integral, obtained by $s = xz^2$, is

$$\int_0^\infty \frac{ds}{\sqrt{s}(x+s)} = \frac{\pi}{\sqrt{x}} \quad (x > 0). \quad (\text{EIQ8})$$

Using this identity in (EIQ7) and applying Tonelli to its positive constituents gives

$$\frac{\alpha}{R_0} = \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}R_s}, \quad \int_0^\infty \frac{1 - s/R_s}{\sqrt{s}} ds = 2vE(k). \quad (\text{EIQ9})$$

Both integrals converge; their integrands are $O(s^{-1/2})$ at zero and $O(s^{-3/2})$ at infinity. The second formula also follows by averaging $\sqrt{t} = \pi^{-1} \int_0^\infty s^{-1/2} t/(t+s) ds$ against the arcsine probability measure on $[a, b]$.

Define for every $x > 0$

$$\begin{aligned} h(x) &= \frac{\alpha}{xR_0} - \frac{\beta}{2\pi} \int_0^\infty \frac{ds}{\sqrt{s}(x+s)R_s} \\ &= \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x+s)R_s} ds > 0. \end{aligned} \quad (\text{EIQ10})$$

The second equality is exactly (EIQ9), with no change of the potential. All integrals in (EIQ10) converge absolutely and locally uniformly in $x > 0$. The candidate density on the original positive half-line is

$$\rho_{\alpha, \beta}(x) = \begin{cases} \frac{\sqrt{(b-x)(x-a)}}{2\pi} h(x), & a < x < b, \\ 0, & x \notin (a, b). \end{cases} \quad (\text{EIQ11})$$

It is strictly positive on (a, b) and vanishes continuously with the square-root factor at each endpoint. Positivity here follows from the last integral in (EIQ10), not from a one-interval ansatz.

The following elementary transform computes its mass and resolvent. Choose

$$R(z) = \sqrt{(z-a)(z-b)}$$

on $\mathbb{C} \setminus [a, b]$ with $R(z)/z \rightarrow 1$ at infinity. Thus $R(x) > 0$ for $x > b$, $R(x) < 0$ for $x < a$, and $R(x + i0) = i\sqrt{(b-x)(x-a)}$ on (a, b) . Direct substitution $t = m + (b-a)\cos\theta/2$, followed by $w = \tan(\theta/2)$, proves

$$\frac{1}{\pi} \int_a^b \frac{dt}{(z-t)\sqrt{(b-t)(t-a)}} = \frac{1}{R(z)}.$$

For real $z > b$, the substitution reduces the integral to $\pi^{-1} \int_0^\infty 2dw/[(z-b) + (z-a)w^2]$; the identity elsewhere follows by analytic continuation. Polynomial division then gives

$$\frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{z-t} dt = z - m - R(z), \quad \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{t+s} dt = m + s - R_s. \quad (\text{EIQ12})$$

For example the first identity follows from $(b-t)(t-a) = -(t-z)^2 + 2(m-z)(t-z) - (z-a)(z-b)$, the preceding arcsine transform, and the arcsine first moment m . Partial fractions in these two identities yield

$$H_s(z) := \frac{1}{\pi} \int_a^b \frac{\sqrt{(b-t)(t-a)}}{(t+s)(z-t)} dt = 1 - \frac{R_s + R(z)}{z+s}. \quad (\text{EIQ13})$$

The formula at $z = -s$ is interpreted by its removable limit.

The mass of (EIQ11), computed using (EIQ12), is

$$\begin{aligned} \int_a^b \rho_{\alpha,\beta}(t) dt &= \frac{\alpha}{2R_0}(m - R_0) - \frac{\beta}{4\pi} \int_0^\infty \frac{(m+s)/R_s - 1}{\sqrt{s}} ds \\ &= -\frac{\alpha}{2} + \frac{\beta}{4\pi} \int_0^\infty \frac{1-s/R_s}{\sqrt{s}} ds = -\frac{\alpha}{2} + \frac{\beta v E(k)}{2\pi} = 1. \end{aligned} \quad (\text{EIQ14})$$

In particular this is a probability density.

Define its Cauchy transform with the explicit sign convention $G(z) = \int_a^b \rho_{\alpha,\beta}(t)/(z-t) dt$. For z outside both $[a, b]$ and $(-\infty, 0]$, formulas (EIQ10) and (EIQ13) give

$$\begin{aligned} G(z) &= \frac{\alpha}{2R_0} H_0(z) - \frac{\beta}{4\pi} \int_0^\infty \frac{H_s(z)}{\sqrt{s} R_s} ds \\ &= -\frac{\alpha}{2z} + \frac{\beta}{4\sqrt{z}} - \frac{R(z)h(z)}{2} = \frac{V'_{\alpha,\beta}(z) - R(z)h(z)}{2}. \end{aligned} \quad (\text{EIQ15})$$

Here the positive square root is continued with its negative-real-axis cut, and $h(z)$ is the first integral formula of (EIQ10). Every integral exchange is absolutely convergent, uniformly on compact subsets of this domain: near zero the auxiliary integrand is $O(s^{-1/2})$, and near infinity the original integral defining H_s is $O(s^{-1})$. The constant terms on the second line cancel by (EIQ9), and (EIQ8) supplies the term $\beta/(4\sqrt{z})$.

35.3 The Euler equality, its strict exterior inequality, and uniqueness

Set

$$U(x) = V_{\alpha,\beta}(x) - 2 \int_a^b \log|x-t| \rho_{\alpha,\beta}(t) dt \quad (x > 0). \quad (\text{EIQ16})$$

This function is continuous on $(0, \infty)$. The integral is continuous because the density is bounded, compactly supported, and the integral of $|\log|x-t||$ over an interval of length ε tends to zero uniformly in its centre. In the interior of $[a, b]$, differentiation of the logarithmic potential is the principal-value integral $\text{PV} \int \rho(t)/(x-t) dt$. This follows by subtracting $\rho(x)$ on a symmetric interval about x ; the remaining numerator is $O(|t-x|)$ because the density is continuously differentiable locally in the interior. The real part of (EIQ15) on the upper edge of the interval is precisely this principal value. Indeed subtracting $\rho(x)$ in $\int \rho(t)/(x+i\varepsilon-t) dt$ gives the same limit, while the constant term has real part equal to its elementary principal value. Since $R(x+i0)$ is purely imaginary and $h(x)$ is real, (EIQ15) proves $U'(x) = 0$ on (a, b) . There is therefore a real constant ℓ such that

$$U(x) = \ell \quad (a \leq x \leq b). \quad (\text{EIQ17})$$

For positive x outside this interval, (EIQ15) instead gives the exact real derivative

$$U'(x) = R(x)h(x) \begin{cases} < 0, & 0 < x < a, \\ > 0, & x > b. \end{cases} \quad (\text{EIQ18})$$

Combining (EIQ17), continuity, and these signs proves

$$U(x) > \ell \quad \text{for } x \in (0, a) \cup (b, \infty). \quad (\text{EIQ19})$$

This verifies the entire exterior inequality on the original domain, including the interval next to zero.

Here is a complete positivity argument for the energy of the difference of two measures. If ν is a finite signed real measure of total mass zero for which $|\log |x - y||$ is integrable against $|\nu| \otimes |\nu|$, the scalar identity

$$-\log d = \frac{1}{2} \int_0^\infty \frac{e^{-td^2} - e^{-t}}{t} dt \quad (d > 0) \quad (\text{EIQ20})$$

follows by differentiating in d and evaluating at $d = 1$. One half of the integral over any finite subinterval of $(0, \infty)$ has the same sign as $-\log d$ and absolute value at most $|\log d|$. Consequently dominated convergence and the zero total mass give

$$-\iint \log |x - y| d\nu(x) d\nu(y) = \frac{1}{2} \int_0^\infty \frac{1}{t} \iint e^{-t(x-y)^2} d\nu(x) d\nu(y) dt \geq 0. \quad (\text{EIQ21})$$

The nonnegative sign is proved, rather than assumed, by the Gaussian Fourier identity

$$\iint e^{-t(x-y)^2} d\nu(x) d\nu(y) = \frac{1}{\sqrt{4\pi t}} \int_{\mathbb{R}} e^{-\xi^2/(4t)} |\widehat{\nu}(\xi)|^2 d\xi, \quad \widehat{\nu}(\xi) = \int e^{i\xi x} d\nu(x). \quad (\text{EIQ22})$$

The scalar Gaussian Fourier identity follows by completing the square in the Gaussian integral (or by differentiating its Fourier transform and its value at zero); its integration against the finite measures is justified by absolute integrability. Equality in (EIQ21) forces $\widehat{\nu} = 0$: the right side of (EIQ22) is strictly positive for every $t > 0$ if this continuous Fourier transform has a nonzero value. Finally $\widehat{\nu} = 0$ implies $\nu = 0$ without an extra uniqueness assumption. Convolve ν with a centred Gaussian approximate identity. Its value at every point is zero by that same Fourier integral. For every compactly supported continuous test function, its convolution with these Gaussians converges uniformly to the test function; integration against ν then proves that ν annihilates every such function.

Let μ be any probability measure of finite energy in (EIQ1), and write $\mu_*(dx) = \rho_{\alpha,\beta}(x) dx$ and $\nu = \mu - \mu_*$. The integrability needed for (EIQ21) holds. The $\mu \otimes \mu$ term has it by finite energy and the already established logarithmic moment; the $\mu_* \otimes \mu_*$ term has it by bounded compact support; and the cross term has it because the negative logarithmic singularity integrated against the bounded density is uniformly bounded, while the positive part is controlled by $\log(1+x) + \log(1+y)$. Expanding the energy therefore yields exactly

$$\mathcal{E}_{\alpha,\beta}(\mu) - \mathcal{E}_{\alpha,\beta}(\mu_*) = \int (U(x) - \ell) d\mu(x) - \iint \log |x - y| d\nu(x) d\nu(y) \geq 0. \quad (\text{EIQ23})$$

The first term is nonnegative by (EIQ17)–(EIQ19), and the second is strictly positive unless $\mu = \mu_*$. Infinite energy competitors cannot improve this finite value. Thus (EIQ11) is the unique global minimizing probability measure, with exactly the support $[a, b]$.

35.4 An explicit endpoint integral for the logarithmic moment

Retain m, R_s, u, v as defined above, and define

$$C = \frac{(u+v)^2}{4}, \quad \eta = \frac{v-u}{v+u}, \quad \zeta_s = \frac{b-a}{2(m+s+R_s)} \quad (s \geq 0). \quad (\text{EIQ24})$$

Thus $0 < \zeta_s \leq \eta < 1$ and $\zeta_0 = \eta$. For every $s \geq 0$ put

$$L_s = (m+s-R_s) \log C - (m-R_0) - 2R_s \log(1-\eta\zeta_s). \quad (\text{EIQ25})$$

Then the logarithmic moment of the equilibrium measure is exactly

$$\mathcal{L}(\alpha, \beta) := \int_a^b \log x \rho_{\alpha,\beta}(x) dx = \frac{\beta}{4\pi} \int_0^\infty \frac{L_0 - L_s}{\sqrt{s} R_s} ds. \quad (\text{EIQ26})$$

This is a convergent one-dimensional endpoint integral of explicitly given real elementary functions; its only endpoints are those specified uniquely in (EIQ3)–(EIQ5).

Here is a direct derivation, including the constants in (EIQ25). Set $h_0 = (b-a)/2$ and substitute $t = m + h_0 \cos \theta$ in arcsine integrals. The identities

$$t = C(1 + 2\eta \cos \theta + \eta^2), \quad t + s = \frac{m + s + R_s}{2}(1 + 2\zeta_s \cos \theta + \zeta_s^2)$$

give the absolutely convergent Fourier series

$$\log t = \log C + 2 \sum_{j \geq 1} \frac{(-1)^{j+1} \eta^j}{j} \cos(j\theta), \quad \frac{1}{t+s} = \frac{1}{R_s} \left(1 + 2 \sum_{j \geq 1} (-\zeta_s)^j \cos(j\theta) \right).$$

Orthogonality on $[0, \pi]$ gives

$$\begin{aligned} \frac{1}{\pi} \int_a^b \frac{\log t}{\sqrt{(b-t)(t-a)}} dt &= \log C, \\ \frac{1}{\pi} \int_a^b \frac{t \log t}{\sqrt{(b-t)(t-a)}} dt &= m \log C + (m - R_0), \\ \frac{1}{\pi} \int_a^b \frac{\log t}{(t+s)\sqrt{(b-t)(t-a)}} dt &= \frac{\log C + 2 \log(1 - \eta \zeta_s)}{R_s}. \end{aligned} \tag{EIQ27}$$

The second identity uses $h_0 \eta = m - R_0$. Polynomial division

$$\frac{(b-t)(t-a)}{t+s} = -t + 2m + s - \frac{R_s^2}{t+s}$$

now proves, exactly,

$$L_s = \frac{1}{\pi} \int_a^b \frac{\log t \sqrt{(b-t)(t-a)}}{t+s} dt. \tag{EIQ28}$$

Using the positive integral expression for h in (EIQ10), and the identity $1/[t(t+s)] = (1/t - 1/(t+s))/s$, yields (EIQ26). The exchanges are absolutely convergent: $\log t$ is bounded on $[a, b]$, $L_0 - L_s = O(s)$ at zero by (EIQ28), and $L_s = O(s^{-1})$ at infinity. Thus the integrand in (EIQ26) is $O(s^{1/2})$ at zero and $O(s^{-3/2})$ at infinity.

35.5 Parameter continuity and the variational derivative

The density and its logarithmic moment depend smoothly on $(\alpha, \beta) \in (0, \infty)^2$ in the following precise sense. On a compact parameter neighbourhood, (EIQ3)–(EIQ6) put a, b in a fixed compact subset of $(0, \infty)$, with $b - a$ bounded below by a positive number. The integral for h and each of its parameter and x derivatives converge uniformly for x in this fixed compact set: their absolute integrands are bounded by constants times $s^{1/2}$ near zero and $s^{-3/2}$ near infinity. Parameter derivatives of the endpoints are bounded on a smaller compact neighbourhood. Under the fixed-interval coordinate $x = a + (b-a)y$, $0 \leq y \leq 1$, the measure is exactly

$$\rho_{\alpha, \beta}(x) dx = \frac{(b-a)^2}{2\pi} \sqrt{y(1-y)} h(a + (b-a)y) dy. \tag{EIQ29}$$

All parameter derivatives of its coefficient are bounded by a constant times $\sqrt{y(1-y)}$. The same is true after multiplication by $\log(a + (b-a)y)$ or by $\sqrt{a + (b-a)y}$. Differentiation under this integral proves smoothness, in particular continuity, of both moments on the entire positive parameter quadrant.

Define $E_{\min}(\alpha, \beta) = \mathcal{E}_{\alpha, \beta}(\mu_{\alpha, \beta})$, where the measure is the unique minimizer proved above. Testing the minimizer for a parameter pair against the neighbouring potential, and then reversing the two tests, gives for $h > 0$

$$-\mathcal{L}(\alpha + h, \beta) \leq \frac{E_{\min}(\alpha + h, \beta) - E_{\min}(\alpha, \beta)}{h} \leq -\mathcal{L}(\alpha, \beta).$$

The corresponding inequalities for $h < 0$ and the proved continuity therefore give

$$\partial_{\alpha} E_{\min}(\alpha, \beta) = -\mathcal{L}(\alpha, \beta), \quad \partial_{\beta} E_{\min}(\alpha, \beta) = \int \sqrt{x} d\mu_{\alpha, \beta}(x). \quad (\text{EIQ30})$$

No unproved differentiability of a minimizing selection is being used. For the opposite convention $F(\alpha, \beta) = \sup_{\mu} \{ \iint \log |x - y| d\mu(x) d\mu(y) - \int V_{\alpha, \beta} d\mu \} = -E_{\min}(\alpha, \beta)$, the exact derivative is therefore $\partial_{\alpha} F(\alpha, \beta) = \mathcal{L}(\alpha, \beta)$.

The second moment in (EIQ30) has an exact value. Push the minimizing measure forward by the explicit dilation $x \mapsto c^2 x$, $c > 0$. Writing $M = \int \sqrt{x} d\mu_{\alpha, \beta}(x)$, its energy minus the energy at $c = 1$ is precisely $\beta(c - 1)M - 2(\alpha + 1) \log c$. Its derivative at the minimizing value $c = 1$ is zero. Hence

$$\int \sqrt{x} d\mu_{\alpha, \beta}(x) = \frac{2(\alpha + 1)}{\beta}. \quad (\text{EIQ31})$$

The same explicit dilation, or (EIQ3)–(EIQ5), proves that the map $x \mapsto (\pi/\beta)^2 x$ pushes $\mu_{\alpha, \pi}$ to $\mu_{\alpha, \beta}$, including the density and its two endpoints. Consequently

$$\mathcal{L}(\alpha, \beta) = \mathcal{L}(\alpha, \pi) + 2 \log(\pi/\beta). \quad (\text{EIQ32})$$

For the particular original limiting potential $V(x) = \pi\sqrt{x} - 2 \log x$, the substitutions are exactly $\alpha = 2$, $\beta = \pi$. Equations (EIQ5), (EIQ10), and (EIQ26) become

$$uK(k) = 2, \quad vE(k) = 4, \quad h(x) = \frac{1}{2x} \int_0^{\infty} \frac{\sqrt{s}}{(x+s)R_s} ds, \quad \mathcal{L}(2, \pi) = \frac{1}{4} \int_0^{\infty} \frac{L_0 - L_s}{\sqrt{s}R_s} ds. \quad (\text{EIQ33})$$

In particular the preceding proof includes positivity, unit mass, the full Euler inequality, the unique global minimizer, and the parameter continuity needed at these actual parameters.

36 A uniform finite partition estimate from the original equilibrium

QLG1. Exact measures and parameter domain. Fix the compact parameter rectangle

$$\mathcal{K} = [3/2, 5/2] \times [\pi/2, 3\pi/2], \quad V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x \quad (x > 0). \quad (\text{QLG1})$$

The equilibrium proved in EIQ1–33 has density

$$\rho(x) = \frac{\beta\sqrt{(b-x)(x-a)}}{4\pi^2 x} \int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(a+s)(b+s)}} \quad (a < x < b), \quad \rho(x) = 0 \quad (x \notin (a, b)), \quad (\text{QLG2})$$

where the exact endpoints are those of EIQ3–5. In particular $0 < a < b < \infty$, the density is bounded and continuous, positive in the interior, and has integral one. Write

$$d\mu(x) = \rho(x) dx, \quad P(z) = \int_a^b \log|z-t| d\mu(t) \quad (z \in \mathbb{C}), \quad I(\mu) = \iint \log|x-y| d\mu(x) d\mu(y),$$

$$F(\alpha, \beta) = I(\mu) - \int V_{\alpha,\beta} d\mu. \quad (\text{QLG3})$$

The original Euler equality and full exterior inequality in EIQ17–19 give a real number ℓ for which

$$D(x) = V_{\alpha,\beta}(x) - 2P(x) - \ell \geq 0 \quad (x > 0), \quad D(x) = 0 \quad (a \leq x \leq b). \quad (\text{QLG4})$$

Integrating the equality on the support gives the exact sign identities

$$\ell = \int V_{\alpha,\beta} d\mu - 2I(\mu), \quad F(\alpha, \beta) = -\ell - I(\mu). \quad (\text{QLG5})$$

The two integration measures treated below are, separately,

$$d\lambda_0(x) = dx, \quad d\lambda_\omega(x) = d\omega(x) = \frac{1}{2}x^{-1/2}e^{-\sqrt{x}} dx. \quad (\text{QLG6})$$

The map $x = r^2$ gives $\int d\omega = \int_0^\infty e^{-r} dr = 1$. For either of these measures, with its stated density w_λ , define the literal finite integral

$$J_n^\lambda(\alpha, \beta) = \frac{1}{n!} \int_{(0,\infty)^n} \exp\left(2 \sum_{i<j} \log|x_i - x_j| - n \sum_{i=1}^n V_{\alpha,\beta}(x_i)\right) \prod_{i=1}^n d\lambda(x_i), \quad n \geq 1. \quad (\text{QLG7})$$

No source measure, scale, factorial, or exponent has been absorbed into F . The unweighted integral without the Heine factor is exactly $n!J_n^\lambda$.

QLG2. Constants proved finite on the whole rectangle. Let

$$a_* = \min_{(\alpha,\beta) \in \mathcal{K}} a(\alpha, \beta) > 0, \quad B_* = \max_{(\alpha,\beta) \in \mathcal{K}} b(\alpha, \beta) < \infty, \quad \beta_- = \pi/2, \quad \beta_+ = 3\pi/2. \quad (\text{QLG8})$$

These are explicit compact-parameter extrema of the uniquely specified endpoint integrals EIQ3–5. Their asserted strict inequalities follow from the continuous positive endpoints proved in EIQ6 and compactness of $\mathcal{K}$. Thus their definition requires no existence or regularity assumption about an unspecified minimizing measure. For $x \in [a, b]$, the integral in QLG2 satisfies

$$\int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(a+s)(b+s)}} \leq \int_0^\infty \frac{\sqrt{s} ds}{(a_*+s)^2} = \frac{\pi}{2\sqrt{a_*}}.$$

For the last equality substitute $s = a_* r^2$, followed by $r = \tan \theta$; the resulting integral is

$$2a_*^{-1/2} \int_0^{\pi/2} \sin^2 \theta \, d\theta.$$

Since $\sqrt{(b-x)(x-a)} \leq (b-a)/2 \leq B_*/2$, it follows that

$$0 \leq \rho(x) \leq M_* := \frac{\beta_+ B_*}{16\pi a_*^{3/2}} \quad \text{for all real } x \text{ and all parameters in } \mathcal{K}. \quad (\text{QLG9})$$

In particular $B_* M_* \geq 1$, by the mass-one identity and support length at most B_* .

The entropy relative to either stated measure is

$$H_\lambda(\mu) = \int_a^b \rho(x) \log \frac{\rho(x)}{w_\lambda(x)} \, dx, \quad 0 \log 0 = 0. \quad (\text{QLG10})$$

It is absolutely integrable: the function $u|\log u|$ is bounded on $[0, M_*]$, and $\log w_\lambda$ is bounded on the fixed compact interval $[a_*, B_*]$. In the two cases its upper bounds are

$$H_{\lambda_0}(\mu) \leq \log M_*, \quad H_{\lambda_\omega}(\mu) \leq \log M_* + \log 2 + \frac{1}{2} \log B_* + \sqrt{B_*}. \quad (\text{QLG11})$$

All pair logarithmic energies on this support are absolutely integrable. For the negative part, integration against a density bounded by M_* is bounded above by $M_* \int_{-1}^1 -\log |s| \, ds = 2M_*$, uniformly in the centre. For the positive part the bound $\log^+ B_*$ suffices.

Put

$$T_\lambda = \int_0^\infty e^{-D(x)} \, d\lambda(x). \quad (\text{QLG12})$$

For $\lambda = \lambda_\omega$, QLG4 and the mass of ω give $0 < T_{\lambda_\omega} \leq 1$. For the Lebesgue measure a finite uniform bound can also be displayed entirely. For every $x > 0$,

$$P(x) \leq \log(1+x) + \log(1+B_*), \quad P(x) \geq -2M_*.$$

The lower bound uses only the negative part estimate just proved. Evaluating QLG4 at any point of $[a, b]$ therefore yields

$$\ell \leq L_* := \beta_+ \sqrt{B_*} + \frac{5}{2} \max\{|\log a_*|, |\log B_*|\} + 4M_*.$$

Consequently

$$T_{\lambda_0} \leq e^{L_*} (1+B_*)^2 \int_0^\infty (1+x)^2 (1+x^{5/2}) e^{-\beta_- \sqrt{x}} \, dx =: T_* < \infty. \quad (\text{QLG13})$$

Here $x^\alpha \leq 1 + x^{5/2}$ for all parameters in $\mathcal{K}$. The integral in QLG13 equals, by $x = r^2$ and elementary repeated integration by parts,

$$2 \left(\frac{1!}{\beta_-^2} + \frac{2 \cdot 3!}{\beta_-^4} + \frac{5!}{\beta_-^6} + \frac{6!}{\beta_-^7} + \frac{2 \cdot 8!}{\beta_-^9} + \frac{10!}{\beta_-^{11}} \right). \quad (\text{QLG14})$$

QLG3. The planar logarithmic comparison, with its proof. We use the plane only to construct an exact comparison of logarithmic kernels; the original particles and potential remain on $(0, \infty)$. If η is a finite signed measure of compact support on $\mathbb{R}^2$, has total mass zero, and $\iint |\log |z-w|| \, d|\eta|(z) d|\eta|(w) < \infty$, then

$$I(\eta) := \iint \log |z-w| \, d\eta(z) d\eta(w) \leq 0. \quad (\text{QLG15})$$

To prove this, for $d > 0$ integrate the elementary identity

$$-\log d = \frac{1}{2} \int_0^\infty \frac{e^{-td^2} - e^{-t}}{t} dt.$$

It follows by differentiation in d and evaluation at $d = 1$. On any finite subinterval of $(0, \infty)$, the integral on the right has the same sign as $-\log d$ and absolute value at most $|\log d|$. The stipulated logarithmic integrability therefore permits dominated passage to the full integral against $d\eta^2$. The constant term vanishes by the zero mass. The elementary Gaussian Fourier integral in each of the two real coordinates gives

$$\begin{aligned} -I(\eta) &= \frac{1}{2} \int_0^\infty \frac{dt}{t} \iint e^{-t|z-w|^2} d\eta(z) d\eta(w), \\ \iint e^{-t|z-w|^2} d\eta(z) d\eta(w) &= \frac{1}{4\pi t} \int_{\mathbb{R}^2} e^{-|\xi|^2/(4t)} \left| \int e^{i\xi \cdot z} d\eta(z) \right|^2 d\xi \geq 0. \end{aligned} \quad (\text{QLG16})$$

For completeness the scalar Fourier identity is obtained by differentiating $\int e^{-u^2/(4t)} e^{isu} du$ with respect to s , integrating by parts in u , and using its value $\sqrt{4\pi t}$ at zero. The product of the two scalar identities has the displayed factor $(4\pi t)^{-1}$. All finite Gaussian integrations against the measures are absolutely convergent. This proves QLG15 without requiring an equilibrium theorem on the plane.

Let $c_{x,\varepsilon}$ be probability arc measure on the circle $z = x + \varepsilon e^{i\theta}$, where the real centre x is arbitrary and $\varepsilon > 0$. Its exact logarithmic potential is

$$\int \log |z - w| dc_{x,\varepsilon}(z) = \log \max\{\varepsilon, |x - w|\}. \quad (\text{QLG17})$$

Indeed, if $|x - w| > \varepsilon$, expand $\log |1 - re^{i\theta}| = -\sum_{j \geq 1} r^j \cos(j\theta)/j$ with $r = \varepsilon/|x - w| < 1$; all nonconstant terms average to zero. If $|x - w| < \varepsilon$, use the same expansion with $r = |x - w|/\varepsilon$. At equality the assertion is the elementary integral $\int_0^{2\pi} \log |1 - e^{i\theta}| d\theta = 0$. To verify this last integral directly, write $|1 - e^{i\theta}| = 2 \sin(\theta/2)$, and put $A = \int_0^{\pi/2} \log(\sin u) du$. Reflection gives the same integral for $\log \cos u$; the double-angle identity then gives $2A = -(\pi/2) \log 2 + A$, whence $A = -(\pi/2) \log 2$. The endpoint logarithms are integrable, for example from $2u/\pi \leq \sin u \leq u$ on $[0, \pi/2]$.

For finite centres x, y , the logarithmic kernel is absolutely integrable against $c_{x,\varepsilon} \otimes c_{y,\varepsilon}$. Its positive part is bounded by a finite constant depending on the centres and ε . Applying QLG17 first to this bounded positive part minus the negative part, Tonelli's theorem and the finite value $\log \max\{\varepsilon, |x - w|\}$ show that the negative part also has finite double integral. This argument includes identical and tangent circles. The same argument proves absolute integrability of the circle- μ cross kernel, using compactness of the support of μ ; the μ - μ term was proved above. In particular QLG15 applies to a finite mixture of these circles minus μ .

Two consequences of QLG17, retaining both self and cross terms, are

$$I(c_{x,\varepsilon}) = \log \varepsilon, \quad I(c_{x,\varepsilon}, c_{y,\varepsilon}) \geq \log \max\{\varepsilon, |x - y|\} \geq \log |x - y|. \quad (\text{QLG18})$$

The first equality applies QLG17 at w on its own circle. For the second inequality, the first circle average is $\log \max\{\varepsilon, |x - w|\} \geq \log |x - w|$; averaging the right side on the second circle and applying QLG17 again gives its asserted value. When $x = y$ the last lower bound is understood in the extended sense.

QLG4. A finite upper estimate on the original particle domain. For fixed particles $x_1, \dots, x_n > 0$, put $\nu_\varepsilon = n^{-1} \sum_i c_{x_i, \varepsilon}$. By QLG18 and QLG15, respectively,

$$2 \sum_{i < j} \log |x_i - x_j| \leq n^2 I(\nu_\varepsilon) - n \log \varepsilon, \quad I(\nu_\varepsilon) \leq 2I(\nu_\varepsilon, \mu) - I(\mu). \quad (\text{QLG19})$$

There is no potential evaluated at a complex particle in these identities. The exact comparison back to the original real potential is

$$0 \leq \int P(z) dc_{x,\varepsilon}(z) - P(x) = \int_{|t-x|<\varepsilon} \log \frac{\varepsilon}{|t-x|} \rho(t) dt \leq 2M_*\varepsilon. \quad (\text{QLG20})$$

The last constant is the explicit integral $M_* \int_{-\varepsilon}^{\varepsilon} \log(\varepsilon/|s|) ds = 2M_*\varepsilon$. Combining QLG19–20 and then substituting QLG4–5 proves, for every original particle configuration,

$$2 \sum_{i<j} \log |x_i - x_j| - n \sum_i V_{\alpha,\beta}(x_i) \leq n^2 F - n \sum_i D(x_i) - n \log \varepsilon + 4M_* n^2 \varepsilon. \quad (\text{QLG21})$$

The coefficient $4M_*$ comes from twice the logarithmic potential and the bound $2M_*\varepsilon$; the coefficient of the self energy is $-n \log \varepsilon$. Take $\varepsilon = 1/n$. Since $D \geq 0$ and $n \geq 1$, $\int e^{-nD} d\lambda \leq T_\lambda$. Integrating QLG21 gives the finite positive upper bound

$$\log J_n^\lambda \leq n^2 F + 4M_* n + n \log T_\lambda + n \log n - \log(n!). \quad (\text{QLG22})$$

In particular the original integrals in QLG7 are finite. The elementary inequality $\log(n!) \geq \int_1^n \log t dt = n \log n - n + 1$ also proves

$$\log J_n^\lambda \leq n^2 F + n(1 + 4M_* + \log T_\lambda) - 1. \quad (\text{QLG23})$$

The inequality for $n!$ follows by monotonicity of $\log t$ on each interval $[j-1, j]$, for $j = 2, \dots, n$, and is exact at $n = 1$.

QLG5. Quantile lower estimate and the exact factorial cancellation. Because ρ is continuous and positive on (a, b) , there are unique endpoints

$$a = t_0 < t_1 < \dots < t_n = b, \quad \int_{t_{j-1}}^{t_j} \rho(x) dx = \frac{1}{n}, \quad \Delta_j = t_j - t_{j-1} > 0, \quad p_j(x) = n\rho(x)\mathbf{1}_{(t_{j-1}, t_j)}(x). \quad (\text{QLG24})$$

The endpoints have measure zero. Each $p_j(x)dx$ is a probability measure with bounded density, on a compact interval of positive length. Every pair logarithmic integral below is therefore absolutely integrable, including self integrals and integrals over touching intervals, by the same bounded-density estimate used after QLG11. Also $\int p_j |\log(p_j/w_\lambda)| dx < \infty$, since $p_j \leq nM_*$ and $\log w_\lambda$ is bounded on the support.

The product cell $\prod_{j=1}^n (t_{j-1}, t_j)$ and its $n!$ coordinate permutations are disjoint. Symmetry of the integrand and the product integration measure in QLG7 implies that the Heine factor $1/n!$ cancels these $n!$ equal integrals exactly. Restricting to their union therefore gives

$$J_n^\lambda \geq \int_{\prod_j (t_{j-1}, t_j)} \exp \left(2 \sum_{i<j} \log |x_i - x_j| - n \sum_i V(x_i) \right) \prod_j w_\lambda(x_j) dx_j. \quad (\text{QLG25})$$

On this cell use the probability density $\prod_j p_j(x_j)$. The logarithm of the integrand relative to this probability is integrable in absolute value, by the preceding estimates and boundedness of V on $[a, b]$. The elementary convex inequality $e^u \geq e^v(1 + u - v)$, integrated with v equal to the expectation of this logarithm, proves Jensen's lower bound directly. Since $\sum_j p_j = n\rho$, its pair and potential contributions are exactly

$$\sum_{i \neq j} I(p_i, p_j) - n \sum_j \int V p_j dx = n^2 F - \sum_j I(p_j).$$

Here $I(p_i, p_j) = \iint \log |x - y| p_i(x) p_j(y) dx dy$ and $I(p_j) = I(p_j, p_j)$. Its entropy contribution is exactly

$$\sum_j \int p_j \log \frac{p_j}{w_\lambda} dx = n \log n + n H_\lambda(\mu). \quad (\text{QLG26})$$

Moreover $|x - y| \leq \Delta_j$ on the j -th cell, whence $I(p_j) \leq \log \Delta_j$. Concavity of the logarithm, proved for example by its negative second derivative, and $\sum_j \Delta_j = b - a$ now give

$$\sum_j I(p_j) \leq \sum_j \log \Delta_j \leq n \log \frac{b - a}{n}. \quad (\text{QLG27})$$

Thus the $n \log n$ in QLG26 cancels that in QLG27, and the complete lower bound is

$$\log J_n^\lambda(\alpha, \beta) \geq n^2 F(\alpha, \beta) - n \{ \log(b - a) + H_\lambda(\mu) \}. \quad (\text{QLG28})$$

This also proves strict positivity of QLG7. No lower bound on the density at an endpoint or on a quantile interval length has been used.

QLG6. The uniform order- n theorem. For every $(\alpha, \beta) \in \mathcal{K}$, every integer $n \geq 1$, and each of the two measures QLG6, QLG22 and QLG28 are the explicit two-sided finite estimate

$$-n \{ \log(b - a) + H_\lambda(\mu) \} \leq \log J_n^\lambda - n^2 F \leq 4M_* n + n \log T_\lambda + n \log n - \log(n!). \quad (\text{QLG29})$$

For entirely parameter-uniform constants set

$$\begin{aligned} L_0 &= \log(B_* M_*), & L_\omega &= \log(B_* M_*) + \log 2 + \frac{1}{2} \log B_* + \sqrt{B_*}, \\ C_0 &= \max\{0, L_0, 1 + 4M_* + \log^+ T_*\}, & C_\omega &= \max\{0, L_\omega, 1 + 4M_*\}. \end{aligned} \quad (\text{QLG30})$$

These constants are finite by QLG8–14. Since $b - a \leq B_*$, QLG11 bounds the lower side of QLG29 by $-nL_\lambda$; QLG12–14 and QLG23 bound its upper side by nC_λ . Consequently

$$\left| \log J_n^{\lambda_0}(\alpha, \beta) - n^2 F(\alpha, \beta) \right| \leq C_0 n, \quad \left| \log J_n^{\lambda_\omega}(\alpha, \beta) - n^2 F(\alpha, \beta) \right| \leq C_\omega n. \quad (\text{QLG31})$$

For the integral Z_n^λ without the Heine factor, the exact relation $\log Z_n^\lambda = \log J_n^\lambda + \log(n!)$ gives

$$\left| \log Z_n^\lambda - n^2 F - \log(n!) \right| \leq C_\lambda n, \quad \left| \log Z_n^\lambda - n^2 F \right| \leq n \log n + C_\lambda n. \quad (\text{QLG32})$$

The theorem therefore retains the distinction between the original Heine determinant integral and the integral with labelled particles. The gain to order n in QLG31 follows from the proved circle comparison and the quantile-cell cancellation, with the same exact potential, positive-half-line particle domain, and equilibrium of EI1–33 throughout.

37 The leading first variation of the actual outer-strip quotient

Retain every original object and both original source orders of OSP1–35 and CTV1–22. In addition to the explicit finite conditions OSP23, take $k \geq 257$. This is an eventual threshold on the same fixed original parameters. Put

$$\vartheta = \pi/2, \quad \ell_s = (s-1)/4, \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)} \quad (s=1, k). \quad (\text{OFV1})$$

The symbol ϑ is the fixed Gamma tail rate; it is not either original source centre.

We use the equilibrium of EIQ1–33 on its original positive half-line:

$$\begin{aligned} V_{\alpha,\beta}(x) &= \beta\sqrt{x} - \alpha \log x, \quad \mu_{\alpha,\beta}(dx) = \rho_{\alpha,\beta}(x) dx, \\ F(\alpha, \beta) &= \iint \log|x-y| d\mu_{\alpha,\beta}(x) d\mu_{\alpha,\beta}(y) - \int V_{\alpha,\beta} d\mu_{\alpha,\beta}. \end{aligned} \quad (\text{OFV2})$$

Here ρ , its two endpoints, the Euler equality and the entire exterior inequality are exactly the explicit proved EIQ formulas. There is no new equilibrium assumption. EIQ29–30 proves smoothness and $F_\alpha(\alpha, \beta) = \int \log x d\mu_{\alpha,\beta}(x)$. QLG31, with its complete circle and quantile proofs, gives

$$|\log J_n(\alpha, \beta) - n^2 F(\alpha, \beta)| \leq C_0 n, \quad J_n = \frac{1}{n!} \int_{(0,\infty)^n} \prod_{i<j} (x_i - x_j)^2 \prod_i e^{-nV_{\alpha,\beta}(x_i)} dx_i \quad (\text{OFV3})$$

for every $n \geq 1$ and every $(\alpha, \beta) \in [3/2, 5/2] \times [\pi/2, 3\pi/2]$. The constant C_0 is the explicit finite constant QLG30. We use precisely this uniform finite error, including the factor $1/n!$; the conclusion below is not a derivative of an earlier unquantified q^2 limit.

37.1 Exact original-order recurrence and a small metric error

For $a \geq 0$, define

$$\mathcal{D}_a(Q; s) = \det \left[\int_{\mathbb{R}} y^{2Q+i+j} dm_s(y) \right]_{0 \leq i, j < a}, \quad \mathcal{A}_a(Q; \vartheta) = \det \left[\int_{\mathbb{R}} y^{2Q+i+j} e^{-\vartheta|y|} dy \right]_{0 \leq i, j < a}, \quad (\text{OFV4})$$

with empty determinants one. The Q used here is an integer; all the instances below have $Q \geq q/2$. The original recurrence OSP12 gives exactly

$$|y|^{2Q} dm_s(y) = \beta_s |y|^{2Q} \prod_{j=0}^{\ell_s-1} (y^2 + t_j^2) d\sigma(y), \quad t_j = 2j + \frac{1}{2}. \quad (\text{OFV5})$$

For each actual source centre $c_0 \in \{k/2, k/2 - 4\}$, the coordinate polynomial of this multiplier is $f_{\ell_s, c_0}(S) = \prod_j (S - c_0 - t_j)$. Its exact pullback satisfies $f_{\ell_s, c_0}(c_0 + iy) = \prod_j (iy - t_j) = i^{\ell_s} \prod_j (y + it_j)$. Thus multiplication by $y^Q \prod_{j=0}^{\ell_s-1} (y + it_j)$, together with the retained phase i^{ℓ_s} , is the exact coefficient map into the original one-factor Gamma source. Its squared modulus is the displayed density multiplier, with no change to any t_j or to β_s .

Set $T = \epsilon q$, $\epsilon = 2^{-32}$ as in OSP23, and

$$S_{\ell_s} = \sum_{j=0}^{\ell_s-1} t_j^2 = \frac{\ell_s(16\ell_s^2 - 12\ell_s - 1)}{12}, \quad r_{\ell_s} = \frac{S_{\ell_s}}{T^2} + \log(1 + K_q), \quad (\text{OFV6})$$

where K_q is exactly OSP24. At $\ell_s = 0$ we may and do take $r_0 = 0$ instead, since the source identity is exact. For $Q \in \{q', q\}$, $0 \leq a \leq 2q + 1$,

$$0 \leq \log \mathcal{D}_a(Q; s) - a \log \beta_s - \log \mathcal{D}_a(Q + \ell_s; 1) \leq ar_{\ell_s}. \quad (\text{OFV7})$$

Here is a proof of the full-form comparison underlying OFV7. For every real y , $|y|^{2Q} \prod_j (y^2 + t_j^2) \geq |y|^{2(Q+\ell_s)}$. On $|y| \geq T$, the logarithm of the ratio is at most $\sum_j t_j^2/T^2 = S_{\ell_s}/T^2$, by $\log(1+u) \leq u$. For $|y| < T$, its numerator is at most $(T^2 + (k/2)^2)^{Q+\ell_s}$. OSP15 on the complete coefficient space of degree at most $a-1$, OSP13 with source $s=1$, and mass $M_1 = \sqrt{2\pi}$ therefore bound its inner quadratic form by K_q times the reference quadratic form on $[q, 2q]$. This is exactly the calculation OSP22–25 with exponent $Q + \ell_s \geq q/2$, degree $a-1 \leq 2q$, and root bound $k/2 \leq R \leq T/2$. No upper bound on that exponent is needed in OSP25, since its base $(5/4)\epsilon^2$ is less than one. The full upper ratio is consequently at most $e^{S_{\ell_s}/T^2} + K_q \leq e^{r_{\ell_s}}$. The lower full ratio is at least one by the pointwise lower inequality. The positive-Gram eigenvalue argument OSP28 now proves OFV7 in the original coefficient frames. Since $S_{\ell_s} \leq \ell_s k^2/4$ and $q \geq k^2$, $r_{\ell_s} = O(1/k)$, with its stated fixed cutoff constant.

37.2 A uniform finite Gamma-to-exponential determinant estimate

Put $h = 1/q$, and define the actual finite constants

$$L_h = C_L(\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi}(\sin h)^{-1/2}. \quad (\text{OFV8})$$

Then the original one-factor Gamma density satisfies on the entire real line

$$L_h e^{-(\vartheta+h)|y|} \leq \frac{d\sigma}{dy}(y) \leq U_h e^{-(\vartheta-h)|y|}. \quad (\text{OFV9})$$

Indeed apply OSP7 to $\Gamma(1/4 + iy/2)$ with $\theta = \pi/2 - h$ for the upper bound. For the lower bound apply it to $\Gamma(3/4 - iy/2)$ in the exact reflection product proved before OSP11. The upper bound on that squared factor is $\Gamma(3/4)^2(\sin h)^{-3/2}e^{-(\vartheta-h)|y|}$. After division, the identity and $\cosh(\pi y) \leq e^{\pi|y|}$ give exactly the lower side of OFV9. In particular both inequalities hold near zero as well as in the tails.

The exact change of variables $y = \vartheta^{-1}u$ in OFV4 gives

$$\mathcal{A}_a(Q; \vartheta) = \vartheta^{-a(2Q+a)} \mathcal{A}_a(Q; 1). \quad (\text{OFV10})$$

The exponent follows from a moment Jacobians, $2Q$ powers in each, and the row and column powers $\sum_{i=0}^{a-1} i + \sum_{j=0}^{a-1} j = a(a-1)$. Applying OFV9 as full quadratic-form inequalities, and then using OFV10, proves the separate finite bounds

$$\begin{aligned} a \log L_h - a(2Q+a) \log(1+h/\vartheta) &\leq \log \mathcal{D}_a(Q; 1) - \log \mathcal{A}_a(Q; \vartheta) \\ &\leq a \log U_h - a(2Q+a) \log(1-h/\vartheta). \end{aligned} \quad (\text{OFV11})$$

Thus the absolute difference is at most

$$B_a(Q) = a \max\{|\log L_h|, |\log U_h|\} + a(2Q+a) \frac{h}{\vartheta - h}. \quad (\text{OFV12})$$

Here $-\log(1-u) \leq u/(1-u)$ and $\log(1+u) \leq u$ justify the second summand. Since $2h/\pi \leq \sin h \leq h$ for $0 < h \leq \pi/2$, the first summand is $O(a \log q)$. The sine inequalities follow from concavity and the endpoint chord, and by integrating $\cos t \leq 1$. For all the present $a \leq 2q+1$, $Q \leq q + \ell_s \leq 2q$, OFV12 is $O(q \log q)$ uniformly in a, Q . Combining OFV7 and OFV11 gives the error-controlled identity

$$\log \mathcal{D}_a(Q; s) = a \log \beta_s + \log \mathcal{A}_a(Q + \ell_s; \vartheta) + R_{a,Q,s}, \quad |R_{a,Q,s}| \leq a r_{\ell_s} + B_a(Q + \ell_s). \quad (\text{OFV13})$$

This is a comparison on complete original polynomial coefficient spaces, with the order shift proved by the literal multiplier OFV5; it is not an assigned replacement of the original source order.

37.3 Direct low-dimension estimates, outside the compact equilibrium range

Define

$$g(Q) = 2Q \log(2Q/\vartheta) - 2Q. \quad (\text{OFV14})$$

For $Q \geq q/2$ and $1 \leq a \leq \Delta + 1$ we prove

$$\log \mathcal{A}_a(Q; \vartheta) = ag(Q) + O(a^2 \log(Q + a) + a \log(Q + a)), \quad (\text{OFV15})$$

with constants independent of k, Q, a in these ranges. There is no use of OFV3 here; its compact parameter rectangle does not contain these low-dimension ratios.

For the lower bound take the full interval $[y_0, y_0 + 1]$, $y_0 = 2Q/\vartheta$. On it $y^{2Q} e^{-\vartheta y} \geq e^{g(Q) - \vartheta}$. Consequently its moment Gram is bounded below by $e^{g(Q) - \vartheta}$ times the unweighted monomial Gram on that interval. Translation has a triangular coefficient matrix with diagonal one, so the latter Gram determinant equals the one on $[0, 1]$. The Rodrigues norms proved in OSP15 show that it is exactly

$$\prod_{j=0}^{a-1} \frac{1}{(2j+1) \binom{2j}{j}^2}.$$

Indeed $\binom{2j}{j}$ is the leading coefficient of $L_j(2x-1)$, whose squared norm is $1/(2j+1)$. Using $\binom{2j}{j} \leq 4^j$ therefore proves

$$\log \mathcal{A}_a(Q; \vartheta) \geq ag(Q) - a\vartheta - a \log(2a) - 2a(a-1) \log 2. \quad (\text{OFV16})$$

For the upper bound, Gram–Schmidt shows that a positive Gram determinant is at most the product of the squared norms of its displayed vectors: each orthogonal component has norm no greater than its original vector. The exact monomial moments give

$$\log \mathcal{A}_a(Q; \vartheta) \leq \sum_{j=0}^{a-1} \log \left(\frac{2(2(Q+j))!}{\vartheta^{2(Q+j)+1}} \right). \quad (\text{OFV17})$$

For every positive integer m , integrating the increasing function $\log x$ between successive integers proves $m \log m - m + 1 \leq \log(m!) \leq m \log m - m + 1 + \log m$. Each summand in OFV17 is consequently at most $g(Q+j) + |\log(2/\vartheta) + 1| + \log(2(Q+j))$. Furthermore $g'(u) = 2 \log(2u/\vartheta)$, so $g(Q+j) - g(Q) \leq 2j \log(2(Q+a)/\vartheta)$ in the present range. This proves the explicit upper bound

$$\begin{aligned} \log \mathcal{A}_a(Q; \vartheta) &\leq ag(Q) + a\{|\log(2/\vartheta) + 1| + \log(2(Q+a))\} \\ &\quad + a(a-1) \log(2(Q+a)/\vartheta). \end{aligned} \quad (\text{OFV18})$$

OFV16 and OFV18 prove OFV15 with its complete lower and upper errors. They also apply at $a = 1$.

37.4 High-dimension partitions and their controlled first variation

For an integer a write $n_e = \lceil a/2 \rceil$ and $n_o = \lfloor a/2 \rfloor$. The two exact changes of variables $y = \sqrt{x}$ and $y = -\sqrt{x}$ give, with their two positive Jacobians, the parity identity

$$\mathcal{A}_a(Q; \vartheta) = Z_{n_e}(Q - \tfrac{1}{2}; \vartheta) Z_{n_o}(Q + \tfrac{1}{2}; \vartheta), \quad (\text{OFV19})$$

where

$$Z_n(A; \vartheta) = \det \left[\int_0^\infty x^{A+i+j} e^{-\vartheta \sqrt{x}} dx \right]_{0 \leq i, j < n}.$$

The two branches add to the density $x^{-1/2} e^{-\vartheta \sqrt{x}}$; thus no factor two is missing from OFV19. The odd block has the additional factor x . Expanding the two Vandermonde determinants and integrating each finite permutation sum proves the exact Heine identity for this moment determinant:

$$Z_n(A; \vartheta) = \frac{1}{n!} \int_{(0, \infty)^n} \prod_{i < j} (x_i - x_j)^2 \prod_i x_i^A e^{-\vartheta \sqrt{x_i}} dx_i.$$

Changing variables $x_i = n^2 u_i$ now gives

$$Z_n(A; \vartheta) = n^{2n(A+n)} J_n(A/n, \vartheta). \quad (\text{OFV20})$$

There are $2n(A+1)$ powers from the weights and Jacobians and $2n(n-1)$ from the squared Vandermonde, proving the exponent.

For the high instances we use exactly

$$(a, Q) \in \{(q, q + \ell_s), (q+1, q + \ell_s), (q + \Delta, q - \Delta + \ell_s), (q + \Delta + 1, q - \Delta + \ell_s)\}. \quad (\text{OFV21})$$

Since $k \geq 257$, $\Delta/q \leq 16/k < 1/16$ and $\ell_s/q \leq 1/(4k) < 1/64$. For both parity blocks, elementary rounding therefore gives

$$\frac{15q}{32} \leq n \leq \frac{9q}{16}, \quad \frac{29q}{32} \leq A \leq \frac{33q}{32}.$$

The first inequalities use $q \geq 32$ to absorb the possible rounding by $1/2$ and the extra endpoint 1. Hence $29/18 \leq A/n \leq 11/5$, contained strictly in $[3/2, 5/2]$. Every partition in OFV20 is thus in the proved parameter rectangle OFV3, with $\beta = \vartheta = \pi/2$.

Define the smooth explicit function

$$\mathcal{H}(a, Q) = a(2Q + a) \log(a/2) + \frac{a^2}{2} F(2Q/a, \vartheta) \quad (a > 0, Q > 0). \quad (\text{OFV22})$$

Equations OFV3 and OFV19–20 give uniformly over OFV21

$$\log \mathcal{A}_a(Q; \vartheta) = \mathcal{H}(a, Q) + O(q \log q). \quad (\text{OFV23})$$

To verify the rounding error rather than omit it, set $G(n, A) = 2n(A+n) \log n + n^2 F(A/n, \vartheta)$. Each actual pair (n, A) differs from $(a/2, Q)$ by at most $1/2$ in each coordinate. On all connecting segments n is bounded above and below by positive constant multiples of q , while A/n remains in $[3/2, 5/2]$. Its derivatives are

$$G_A = 2n \log n + n F_\alpha, \quad G_n = 2(A+2n) \log n + 2(A+n) + 2nF - A F_\alpha.$$

They are $O(q \log q)$ there, since the exact EIQ functions F, F_α are bounded on that compact interval. The mean value formula bounds each rounding error by $O(q \log q)$. The two QLG errors sum to at most $C_0 a = O(q)$, and $2G(a/2, Q) = \mathcal{H}(a, Q)$, proving OFV23.

The derivative needed for the first variation is computed without altering the direction of the two original degree changes:

$$(\partial_a - \partial_Q) \mathcal{H}(a, Q) = 2Q \log(a/2) + (2Q + a) + aF(2Q/a, \vartheta) - (Q + a)F_\alpha(2Q/a, \vartheta). \quad (\text{OFV24})$$

EIQ29–30 proves that F_α is smooth; in particular $F, F_\alpha, F_{\alpha\alpha}$ are bounded on the compact interval just used. Differentiating the right side of OFV24 shows that its two first partial derivatives are $O(\log q)$ on the segment $a = q + u$, $Q = q + \ell_s - u$, $0 \leq u \leq \Delta$ and on its connections to (q, q) . Every term follows by the product rule; the derivatives of $2Q/a$ are $2/a$ and $-2Q/a^2$, both $O(1/q)$. The mean value formula and then integration in u consequently give

$$\begin{aligned} & \mathcal{H}(q + \Delta, q - \Delta + \ell_s) - \mathcal{H}(q, q + \ell_s) \\ &= \Delta q \{2 \log(q/2) + 3 + F(2, \vartheta) - 2F_\alpha(2, \vartheta)\} + O(\Delta(\Delta + \ell_s) \log q). \end{aligned} \quad (\text{OFV25})$$

This remainder is $O(q \log q)$ because $\Delta = O(k)$, $\ell_s = O(k)$ and $q = (k+1)^2$. The additional endpoint $a+1$ in OFV21 changes $\mathcal{H}$ by $O(q \log q)$, directly from its displayed derivatives. Thus all rounding and endpoint changes have been quantified.

37.5 All seven determinants and the resulting leading constant

Replace the seven moment determinants in OSP32 by the exact exponential determinants in OFV13, with their respective $Q + \ell_s$. Their $a \log \beta_s$ terms cancel because their signed dimensions are exactly

$$\Delta + (\Delta + 1) - (q + \Delta) - (q + \Delta + 1) - 1 + q + (q + 1) = 0. \quad (\text{OFV26})$$

The sum of their absolute finite errors in OFV13 is $O(q \log q)$, since the sum of their dimensions is $4(q + \Delta + 1)$, $r_{\ell_s} = O(1/k)$, and OFV12 applies uniformly with $Q + \ell_s \leq 2q$. The low terms, using OFV15–18, are

$$\begin{aligned} \log \mathcal{A}_\Delta(q - \Delta + \ell_s; \vartheta) + \log \mathcal{A}_{\Delta+1}(q - \Delta + \ell_s; \vartheta) - \log \mathcal{A}_1(q + \ell_s; \vartheta) \\ = 2\Delta g(q) + O(q \log q) = 4\Delta q \{\log(2q/\vartheta) - 1\} + O(q \log q). \end{aligned} \quad (\text{OFV27})$$

Indeed replacing each $g(q - \Delta + \ell_s)$ or $g(q + \ell_s)$ by $g(q)$ costs at most $O(((2\Delta + 1)(\Delta + \ell_s) + \ell_s) \log q) = O(q \log q)$, using $g'(u) = 2 \log(2u/\vartheta)$ and the actual degree ranges. The individual low determinant errors are $O((\Delta + 1)^2 \log q) = O(q \log q)$.

The remaining four high terms are, by OFV23–25,

$$\begin{aligned} \log \mathcal{A}_q(q + \ell_s; \vartheta) + \log \mathcal{A}_{q+1}(q + \ell_s; \vartheta) \\ - \log \mathcal{A}_{q+\Delta}(q - \Delta + \ell_s; \vartheta) - \log \mathcal{A}_{q+\Delta+1}(q - \Delta + \ell_s; \vartheta) \\ = -2\Delta q \{2 \log(q/2) + 3 + F(2, \vartheta) - 2F_\alpha(2, \vartheta)\} + O(q \log q). \end{aligned} \quad (\text{OFV28})$$

Combining OFV27 and OFV28 gives the complete constant

$$\mathfrak{C}_\partial = 4 \log(4/\vartheta) - 10 - 2F(2, \vartheta) + 4F_\alpha(2, \vartheta), \quad \vartheta = \pi/2, \quad (\text{OFV29})$$

with every logarithmic scale term canceled by direct algebra. OSP33 now proves the actual original-root asymptotic

$$\boxed{\mathcal{F}_{\partial,k}^{(s)} = \Delta q \mathfrak{C}_\partial + O_{\delta,\gamma}(q \log q), \quad \frac{\mathcal{F}_{\partial,k}^{(s)}}{kq} \longrightarrow 16 \mathfrak{C}_\partial, \quad s = 1, k.} \quad (\text{OFV30})$$

All constants in the error depend only on the fixed original parameters, the fixed cutoff, and the proved compact equilibrium constants. The two original source orders have been treated through their exact recurrence maps and full mass factors. Since $\Delta/k = 16 - 48/k$ and $\log q/k \rightarrow 0$, the displayed limit follows from its quantified remainder.

An equivalent expression uses the logarithmic energy of precisely the original EIQ limiting potential, $V(x) = \pi\sqrt{x} - 2 \log x$:

$$I_{2,\pi} = \iint \log |x - y| d\mu_{2,\pi}(x) d\mu_{2,\pi}(y), \quad \mathfrak{C}_\partial = 4 \log(4/\pi) + 2 - 2I_{2,\pi}. \quad (\text{OFV31})$$

To prove it, EIQ31 gives $\vartheta \int \sqrt{x} d\mu_{2,\vartheta} = 6$, so $F(2, \vartheta) = I_{2,\vartheta} - 6 + 2F_\alpha(2, \vartheta)$. Substituting in OFV29 yields $4 \log(4/\vartheta) + 2 - 2I_{2,\vartheta}$. The proved EIQ32 dilation $x \mapsto 4x$ carries $\mu_{2,\pi}$ to $\mu_{2,\pi/2}$. Its exact logarithmic energy change is $I_{2,\pi/2} = I_{2,\pi} + \log 4$, proving OFV31. This energy is a finite explicit integral of the fully specified EIQ density; its logarithmic singularity is integrable because that density is bounded and compactly supported.

Finally, the exact quotient metrics decrease when the source degree increases: each smaller-degree lift remains an allowed lift in the larger source with exactly the same norm. Hence each low-degree quotient Gram in CTV13 dominates the corresponding high-degree Gram on the identical target frame. Positive conjugation shows their determinant ratios are at least one. It follows that $\mathcal{F}_{\partial,k}^{(s)} \geq 0$, and OFV30 therefore also proves $\mathfrak{C}_\partial \geq 0$. The already proved finite tail bounds CTV18–20, with $0 \leq v \leq 80$, further give

$$\mathcal{T}_k^{(s)} = \Delta q \mathfrak{C}_\partial + O_{\delta,\gamma}(q \log q), \quad s = 1, k, \quad (\text{OFV32})$$

uniformly over the possible actual first-moment indices. No leading value of the separate original observation quotient is assigned here.

38 Fixed coefficient control of the original residual module

Retain the complete actual simple-quartet period matrix $H = F^{-1}\Pi U\mathcal{E}_{\text{prim}}$, the original substitution $t = (z_0w_0, z_0w_1, z_1w_0, z_1w_1)^T$, and $Q(x) = Q_0(H^{-1}x)$, where $Q_0(t) = t_{++}t_{--} - t_{+-}t_{-+}$. The original action is $gf(x) = f(D^{-1}x)$ with $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$ and $\zeta = e^{2\pi i/5}$. At the fixed actual period on the proved five-member orbit locus retain the entire polynomials $Q_a = g^a Q$ and $\mathcal{A} = \prod_{a=1}^4 Q_a$. Their coefficients and all phases are unchanged. The degree- k calculations below use $k = 4l + 1 \geq 9$ and

$$q = (k+1)^2, \quad q' = (k-7)^2, \quad \Delta = 16k - 48, \quad j = 8k - 32, \quad r = 8k - 16. \quad (\text{FMC1})$$

All constants introduced below are determined by this one fixed actual coefficient datum. They are independent of k and of the original Gamma cutoff. A fixed presentation will supply the residual frame in place of dividing new selected minors at each degree.

38.1 A fixed free module containing the entire invariant image

Let $P = \mathbb{C}[x_1, x_2, x_3, x_4]$ and introduce the specified graded ring injection

$$T = \mathbb{C}[y_1, y_2, y_3, y_4] \longrightarrow P, \quad y_i \longmapsto x_i^5, \quad \deg y_i = 5. \quad (\text{FMC2})$$

It is injective because distinct monomials have distinct exponent vectors after multiplying each exponent by five. Put $\mathcal{E} = \{a \in \mathbb{Z}^4 : 0 \leq a_i \leq 4\}$ and let e_a be a free T -basis element of degree $|a|$. The exact isomorphism is

$$\mathcal{P} := \bigoplus_{a \in \mathcal{E}} T e_a \xrightarrow{\iota} P, \quad y^\nu e_a \longmapsto x^{5\nu+a}. \quad (\text{FMC3})$$

Every nonnegative exponent vector has a unique coordinatewise quotient and remainder on division by five. This proves surjectivity, injectivity, preservation of degree, and the explicit inverse. In particular this is a rank-625 free module, with coefficient ℓ^1 norm preserved exactly by ι . The ring and monomial coefficients in P have not been rescaled.

Let $\mathcal{E}_0 = \{a \in \mathcal{E} : a_1 + 2a_2 + 3a_3 + 4a_4 \equiv 0 \pmod{5}\}$. The full original invariant algebra, regarded here as a T -module, is

$$\mathcal{J} = P^{\langle g \rangle} = \iota \left(\bigoplus_{a \in \mathcal{E}_0} T e_a \right), \quad |\mathcal{E}_0| = 125. \quad (\text{FMC4})$$

Indeed the monomial $x^{5\nu+a}$ has character $\zeta^{-(a_1+2a_2+3a_3+4a_4)}$. The diagonal action fixes a polynomial precisely when each nonzero monomial has character one. For each of the 5^3 choices of the first three residues there is exactly one a_4 since 4 is invertible modulo five. This proves the count and the displayed equality. These pure fifth powers are four of the fourteen original generators OCF25; this is a new exact module presentation of the same entire invariant algebra.

Define two homogeneous T -submodules of $\mathcal{P}$ by

$$\begin{aligned} \mathcal{H} &= \iota^{-1}(QP + \mathcal{A}P), \\ \mathcal{S} &= \mathcal{H} + \bigoplus_{a \in \mathcal{E}_0} T e_a. \end{aligned} \quad (\text{FMC5})$$

They have the fixed finite generating columns $\iota^{-1}(Qx^a)$ and $\iota^{-1}(\mathcal{A}x^a)$ for $a \in \mathcal{E}$, followed for $\mathcal{S}$ by the 125 columns e_a , $a \in \mathcal{E}_0$. The first two lists have respectively homogeneous degrees $|a| + 2$ and $|a| + 8$, all at most 24. To obtain every matrix entry explicitly, expand either of the full original polynomials as $f(x) = \sum_b f_b x^b$ and use

$$\iota^{-1}(f x^a) = \sum_b f_b y^{[(a+b)/5]} e_{(a+b) \bmod 5}. \quad (\text{FMC6})$$

Both quotient and remainder are coordinatewise. Formula FMC3 verifies this identity term by term. Since P is free on the x^a , these columns generate precisely the two modules in FMC5; no later degree is needed to define a new relation matrix.

The full coefficient substitution and conductor quotient give the exact graded maps

$$\mathcal{P}/\mathcal{H} \xrightarrow{\iota, \psi} \mathcal{B}/(AB) =: \mathcal{L}, \quad \mathcal{S}/\mathcal{H} \xrightarrow{\sim} \overline{\mathcal{J}}, \quad A = \psi_8(\mathcal{A}). \quad (\text{FMC7})$$

Here $\psi f = f(Ht)$, and $\overline{\mathcal{J}}$ is the image of the original invariant algebra in $\mathcal{L}$, exactly OCF16. OCF3 proves $\ker \psi = QP$ and its surjectivity in every degree. If $\psi f = Ab$, lift b to a polynomial h using that surjectivity; then $f - Ah \in QP$. Conversely every $QP + AP$ element maps to zero. This proves the first isomorphism. FMC4–5 show that its restriction has precisely the second image and kernel. Thus the quotient in FMC7 is the complete original conductor quotient, and its indicated submodule is the actual residual certificate image of ROQ4, with no new invariant premise.

38.2 Finite reduction data with only fixed actual denominators

The next construction specifies fixed data for both modules in FMC5. Order their module monomials $y^\nu e_a$ first lexicographically by $\nu_1, \nu_2, \nu_3, \nu_4$, and then by a fixed index $\kappa_a \in \{0, \dots, 624\}$ listing a lexicographically. Multiplication by a y -monomial preserves this order. It is a well-order: in a decreasing infinite sequence the first coordinate can decrease only finitely often; after it is fixed the same applies successively to the other three coordinates and then the finite index. For a nonzero column f , retain its full leading coefficient $\lambda_f \neq 0$ and its leading module monomial m_f . No leading coefficient is deleted from the stored column.

Start with the finite nonzero homogeneous generators of FMC5–6. Division of a column repeatedly cancels its largest divisible module monomial $cy^\nu e_a$ using the first available column f for which $m_f = y^\mu e_a$ divides it: subtract $(c/\lambda_f)y^{\nu-\mu}f$ and record that exact multiple. A largest term divisible by none of the current leading monomials is put into the remainder and removed from the current dividend. Each operation strictly decreases the largest term still being processed, so this division terminates. Remainders have no monomial divisible by the stored leading monomials.

For every pair f, h whose leading positions agree, put $m = \text{lcm}(m_f, m_h)$ in that position and form the full column

$$\text{Sp}(f, h) = \frac{m/m_f}{\lambda_f} f - \frac{m/m_h}{\lambda_h} h. \quad (\text{FMC8})$$

Divide it as just described and append its nonzero remainder to the list. Reconsider all new pairs. Track every appended column as a T -linear combination of the original generators throughout these operations. Every column and recorded certificate remains homogeneous: the two terms in FMC8 have the same weighted degree, because their leading position agrees and every input column is homogeneous in the shifted grading of FMC3.

This algorithm stops after finitely many appended columns. Here is the needed finiteness argument. Every infinite sequence in $\mathbb{N}^4$ has two terms in their original order with all coordinates weakly increasing. To prove it, an infinite sequence of nonnegative integers has either an infinitely repeated value or an increasing infinite subsequence; use this successively for all four coordinates. For module monomials first take a subsequence with one common position. Consequently every infinite sequence of module monomials has an earlier one dividing a later one. Each appended remainder's leading monomial is divisible by none of the previous leading monomials. An infinite sequence of appendages would contradict the proved property.

At termination every element of the generated module has leading monomial divisible by one of the stored leading monomials. For completeness, write an element as a finite sum of monomial multiples of the stored columns. Among all such representations choose one for which the largest leading module monomial of a summand is smallest, and among these choose one with the fewest summands at that monomial. These choices exist by the well-order and the positive integers. If cancellation of those largest terms occurs, two summands at the same module position can be

combined using a multiple of FMC8, with its exact two leading coefficients. Explicitly, if their common leading monomial is M and their leading coefficients are a, b , their sum is

$$a \frac{M}{\text{lcm}(m_f, m_h)} \text{Sp}(f, h) + (a + b) \frac{M/m_h}{\lambda_h} h.$$

This follows by substituting FMC8; it retains exactly both original summands and every coefficient. The recorded division of this S-column, together with its appended remainder when one was needed, expresses it in stored columns with all leading multiples strictly below its canceled common monomial. Replacing the two summands by this expression either decreases the largest monomial or decreases the number attaining it; combining their coefficients if necessary gives the latter alternative. Both contradict the chosen representation. Thus the largest remaining monomial is an actual leading monomial of the element, and it is divisible by one of the stored ones, as claimed. This also proves that division of a module element has zero remainder.

Write $\mathcal{G}_H$ and $\mathcal{G}_S$ for the two resulting fixed finite lists for FMC5. The preceding argument proves their full reduction property. A column supported outside the respective leading module is called a standard remainder. Such a remainder represents each quotient class uniquely: two representatives of one class differ by a module element, and a nonzero standard difference would violate the proved leading-monomial property. Denote these exact linear remainder maps by NF_H and NF_S . Their linearity follows from uniqueness and linearity of quotienting.

All data constructed here depend only on the actual matrices in FMC6. Every operation is an addition, multiplication, or division by one of the nonzero actual leading coefficients encountered in this finite calculation. Thus the complete list of denominators is finite and independent of k . Its precise coefficient chart records each tested coefficient's vanishing or nonvanishing and every selected leading monomial. For every actual period exactly its corresponding decisions give the above terminating calculation. This does not posit a numerical test for equality of unevaluated periods, nor does it assert a lower bound for a later sequence of minors. The exact stored coefficients and their nonzero leading entries define the constants that follow.

38.3 A fixed weight bounds every homogeneous reduction path

The finite reductions admit a linear-in-degree depth bound. Let $D_0 \geq 1$ be the largest of one and all absolute differences between corresponding y -exponents in a leading and a nonleading term of any column in $\mathcal{G}_H \cup \mathcal{G}_S$. Empty tail lists contribute zero. Set

$$B_0 = D_0 + 626, \quad \mathbf{w}(y^\nu e_a) = B_0^4 \nu_1 + B_0^3 \nu_2 + B_0^2 \nu_3 + B_0 \nu_4 + \kappa_a, \quad L_k = B_0^4 \lfloor k/5 \rfloor + 624. \quad (\text{FMC9})$$

Every nonleading term of either fixed list has weight at least one less than its leading term. If the y -exponents agree, this follows from the position order. Otherwise let their first differing coordinate have weight B_0^m , $1 \leq m \leq 4$. Its exponent drops by at least one, while the later differences have absolute value at most D_0 , and the position difference has absolute value at most 624. The total weight drop is at least

$$B_0^m - D_0 \sum_{h=1}^{m-1} B_0^h - 624 \geq \frac{625 B_0^m}{D_0 + 625} - 624 > 1. \quad (\text{FMC10})$$

The middle inequality follows by bounding the geometric sum by $B_0^m/(B_0 - 1)$; the last follows from $B_0^m \geq B_0 = D_0 + 626$. Multiplication by a monomial adds the same weight to the leading and all its replacement terms, so the same drop holds in every division step. A homogeneous degree- k monomial has $5|\nu| + |a| = k$, hence weight between zero and L_k . Every branch of its reduction tree consequently has length at most L_k .

For each of the two lists define the complete coefficient constant

$$C_H = \max \left\{ 1, \max_{g \in \mathcal{G}_H} \sum_{m \neq m_g} |[m]g/\lambda_g| \right\}, \quad C_S = \max \left\{ 1, \max_{g \in \mathcal{G}_S} \sum_{m \neq m_g} |[m]g/\lambda_g| \right\}. \quad (\text{FMC11})$$

The maximum over an empty list is zero. They are finite actual numbers because the lists are finite and their displayed leading coefficients are nonzero. For every homogeneous degree- k column,

$$\|\text{NF}_H f\|_1 \leq C_H^{L_k} \|f\|_1, \quad \|\text{NF}_S f\|_1 \leq C_S^{L_k} \|f\|_1. \quad (\text{FMC12})$$

To prove these inequalities start with one monomial of coefficient one. A reduction replaces its leading term by the complete tail with total absolute coefficient at most the indicated constant. Along each branch the nonnegative integer weight drops at least one. Induction on that weight bounds the total absolute coefficient of all final leaves by $C^{\mathfrak{w}(m)}$: a standard leaf contributes one, and an internal node contributes at most C times the bound at weight one lower. Cancellation between branches can only decrease the coefficient ℓ^1 norm. Linearity, the triangle inequality, and $\mathfrak{w}(m) \leq L_k$ prove FMC12 for every input.

There is equally explicit control of the recorded original-generator certificates. For each stored $g \in \mathcal{G}_S$ let c_g be its complete certificate in the fixed original generating list of $\mathcal{S}$ in FMC5–6. Define

$$D_S = \max \left\{ 1, \max_{g \in \mathcal{G}_S} \|c_g\|_1 / |\lambda_g| \right\}. \quad (\text{FMC13})$$

At a reduction node with input coefficient c , the certificate contribution has norm at most $D_S |c|$. At depth h the total absolute coefficient entering all nodes is at most C_S^h times the input norm, by the same tree proof. Summing through depth L_k therefore proves that a certificate of $f - \text{NF}_S f$ in the original generators has norm at most $D_S (L_k + 1) C_S^{L_k} \|f\|_1$. All certificate entries retain the original homogeneous grading. Their denominators are among the same finite fixed reduction data; no degree- k linear system is inverted.

38.4 The two exact maps back to the original conductor strip

Let $\mathbf{N}_k$ and $\mathbf{S}_k$ be the original OCF strip remainder and its coordinate insertion, with leading conductor coefficient $a_* = a_{r_* s_*}$. Define

$$A_0 = \max \left\{ 1, \sum_{(a,b) \neq (r_*, s_*)} |a_{ab}/a_*| \right\}. \quad (\text{FMC14})$$

Their exact coefficient bounds are $\|\mathbf{N}_k\|_1 \leq A_0^{10k}$ and $\|\mathbf{S}_k\|_1 = 1$. To prove the first, give the biform monomial $e_{ab}^{(k)}$ the weight $9a + b \in [0, 10k]$. Every other nonzero term of the full A has smaller weight than its lexicographically largest term: a decrease in the first index outweighs at most eight increases in the second index, and a decrease in only the second index also drops the weight. Every translated OCF6 subtraction therefore has weight drop at least one. Its complete tail has coefficient norm at most A_0 . The reduction-tree proof of FMC12 now applies with depth $10k$. The insertion bound is immediate from its unchanged coordinate coefficients.

Put

$$h_+ = \max \{ 1, \max_i \sum_a |\mathbf{H}_{ia}| \}, \quad h_- = \max \{ 1, \max_a \sum_i |(\mathbf{H}^{-1})_{ai}| \}, \quad h_* = \max \{ h_+, h_- \}. \quad (\text{FMC15})$$

Substitution $x = \mathbf{H}t$ on degree- k polynomial coefficients has ℓ^1 norm at most h_+^k . Expanding each product of linear forms and using the multinomial theorem proves this for each monomial; the triangle inequality proves it for their sum. The Segre substitution adds coefficients on identical biform monomials and so has norm at most one. Conversely choose the following exact lift of $e_{ab}^{(k)}$: with $h = \min(a, b)$ take

$$t_{++}^h t_{+-}^{a-h} t_{-+}^{b-h} t_{--}^{k-a-b+h}, \quad t = \mathbf{H}^{-1}x. \quad (\text{FMC16})$$

All exponents are nonnegative since $0 \leq a, b \leq k$, and their sum is k . Direct substitution gives the required biform monomial. This lift, extended linearly and denoted $\mathcal{L}_k$, has norm at most h_-^k by the same complete multinomial estimate.

Let $\mathcal{M}_{H,k}$ be the set of degree- k standard module monomials for $\mathcal{G}_H$. Its coefficient space is the exact degree- k quotient model of FMC7. Define

$$\begin{aligned} U_k f &= \mathbf{N}_k \psi \iota f, & f &\in \mathbb{C}^{\mathcal{M}_{H,k}}, \\ V_k w &= \mathbf{NF}_H \iota^{-1} \mathcal{L}_k \mathbf{S}_k w, & w &\in W_k. \end{aligned} \quad (\text{FMC17})$$

These are inverse maps. For the first composition, $\mathbf{NF}_H$ changes a lift only by $\mathcal{H}$, which maps to zero in the original strip by FMC7, and $\psi \mathcal{L}_k = I$ together with $\mathbf{N}_k \mathbf{S}_k = I$ proves $U_k V_k = I$. For the opposite composition, $\mathcal{L}_k \mathbf{S}_k \mathbf{N}_k \psi \iota f - \iota f$ maps to zero in the strip, so it belongs to $QP + \mathcal{AP}$ by FMC7. Its free-module difference belongs to $\mathcal{H}$ and is killed by the unique remainder. The original f is already standard. Hence $V_k U_k = I$. The bounds already proved give

$$\|U_k\|_1 \leq h_+^k A_0^{10k}, \quad \|V_k\|_1 \leq h_-^k C_H^{L_k}. \quad (\text{FMC18})$$

These bounds use only the displayed original coefficients and the two fixed homogeneous relation lists.

38.5 An identity pivot block for the entire residual image

Since $\mathcal{H} \subset \mathcal{S}$, every leading monomial of $\mathcal{H}$ is a leading monomial of an element of $\mathcal{S}$. The proved reduction property for $\mathcal{G}_S$ then implies inclusion of their leading modules. Consequently

$$\mathcal{M}_{S,k} \subset \mathcal{M}_{H,k}, \quad \mathcal{U}_k = \mathcal{M}_{H,k} \setminus \mathcal{M}_{S,k}, \quad |\mathcal{M}_{H,k}| = \Delta, \quad |\mathcal{M}_{S,k}| = r, \quad |\mathcal{U}_k| = j. \quad (\text{FMC19})$$

The first count is FMC7 and OCF8. The quotient by $\mathcal{S}$ is exactly the original conductor quotient modulo its invariant image, which has dimension r by OCF22–24 and OKM3–5. Subtracting proves the last count. These are counts of actual standard monomials for the fixed relation lists, not independent generic ranks.

Order $\mathcal{M}_{H,k}$ by listing $\mathcal{U}_k$ first and $\mathcal{M}_{S,k}$ second, with the specified monomial order within each list. For every $u \in \mathcal{U}_k$ put $z_u = u - \mathbf{NF}_S(u)$. Both terms are H -standard, and their difference belongs to $\mathcal{S}$. The j resulting columns form a basis of $\mathcal{S}_k/\mathcal{H}_k$ in this quotient model. Indeed their coordinates on $\mathcal{U}_k$ are the distinct unit vectors. For a standard column in $\mathcal{S}$, the remainder under $\mathbf{NF}_S$ is zero. Its coordinates on $\mathcal{M}_{S,k}$ must therefore be the negatives of the remainders of its $\mathcal{U}_k$ terms. It is exactly the stated linear combination of the z_u . This proves spanning and independence, including all tails of every column.

Write B_k for the matrix with columns $\mathbf{NF}_S(u)$ in the second standard set. The full exact graph frame and inverse are

$$P_k = \begin{pmatrix} I_j & 0 \\ -B_k & I_r \end{pmatrix}, \quad P_k^{-1} = \begin{pmatrix} I_j & 0 \\ B_k & I_r \end{pmatrix}, \quad \det P_k = 1, \quad \|P_k\|_1, \|P_k^{-1}\|_1 \leq 1 + C_S^{L_k}. \quad (\text{FMC20})$$

The two matrix identities follow by multiplication and block triangularity, and the norm bound follows from FMC12 applied to each entire column. All entries of B_k come from fixed reductions; no minor is selected or divided at degree k .

Each z_u has an actual invariant certificate. The recorded reduction certificate of $u - \mathbf{NF}_S(u)$ from FMC13 expresses its polynomial representative as

$$\iota z_u = Qh_u + Aa_u + f_u, \quad f_u \in \mathcal{J}_k, \quad \|f_u\|_{1,x} \leq D_S(L_k + 1)C_S^{L_k}. \quad (\text{FMC21})$$

To obtain the three displayed pieces, collect the certificate terms corresponding respectively to Qx^a , Ax^a , and the 125 invariant basis elements x^a . Every coefficient is a polynomial in the $y_i = x_i^5$. The last terms therefore belong to $\mathcal{J}$, and homogeneity gives exactly degree k . FMC3 preserves coefficient norm, and discarding the first two certificate lists when bounding only the last cannot increase their total ℓ^1 norm. FMC13 proves the bound. Thus $U_k z_u$ is exactly $\mathbf{N}_k \psi f_u$, an original invariant restriction, with the entire original periods in its coefficients.

Define the full original strip frame, its residual columns and their explicit left inverse by

$$T_k = U_k P_k, \quad Z_k^{\text{fix}} = T_k E_j, \quad L_{Z,k} = E_j^\top P_k^{-1} V_k, \quad L_{Z,k} Z_k^{\text{fix}} = I_j. \quad (\text{FMC22})$$

Here E_j inserts the first j coordinates into the new ordered standard basis, while all output strip coordinates retain their original OCF order. FMC17 and FMC20 prove the inverse formula $T_k^{-1} = P_k^{-1} V_k$ and the last identity. By FMC21 these Z_k^{fix} columns span precisely the original $\bar{J}_k$. They provide all actual residual coefficients with a left inverse obtained from fixed relations.

For the original coefficient Euclidean norms set the explicit number

$$M_k = 2\sqrt{\Delta} h_*^k A_0^{10k} (C_H C_S)^{L_k} \geq 1. \quad (\text{FMC23})$$

Then

$$\|T_k\|_2, \|T_k^{-1}\|_2, \|Z_k^{\text{fix}}\|_2, \|L_{Z,k}\|_2 \leq M_k, \quad M_k^{-1} \|v\|_2 \leq \|Z_k^{\text{fix}} v\|_2 \leq M_k \|v\|_2. \quad (\text{FMC24})$$

For the first two bounds, FMC18–20 give respectively the column-sum bounds $h_+^k A_0^{10k} (1 + C_S^{L_k})$ and $(1 + C_S^{L_k}) h_-^k C_H^{L_k}$. For a matrix with Δ input coordinates, its Euclidean norm is at most $\sqrt{\Delta}$ times its coefficient column-sum norm: use $\|Av\|_2 \leq \|Av\|_1 \leq \|A\|_1 \|v\|_1 \leq \sqrt{\Delta} \|A\|_1 \|v\|_2$. All constants are at least one, proving the two stated bounds. The coordinate insertion and restriction have Euclidean norm one, so FMC22 proves the remaining operator bounds. Its left-inverse identity then proves the lower bound on Z_k^{fix} .

For every $0 \leq t \leq \Delta$, singular values now give

$$\log^+ \left\| \bigwedge^t T_k \right\|_2, \log^+ \left\| \bigwedge^t T_k^{-1} \right\|_2 \leq t \log M_k \leq \Delta \log M_k = O_{\text{actual}}(k^2) = o(kq). \quad (\text{FMC25})$$

Indeed every singular value of both square matrices is at most M_k , and exterior singular values are their products. This last fact follows by applying a singular-value decomposition: unitary factors preserve exterior norms, and the diagonal factor multiplies each coordinate wedge by the corresponding product. Explicitly

$$\log M_k = \log 2 + \frac{1}{2} \log \Delta + k \log h_* + 10k \log A_0 + L_k \log(C_H C_S). \quad (\text{FMC26})$$

All data except k, Δ, L_k are fixed actual numbers, $L_k \leq B_0^4 k/5 + 624$, $\Delta = 16k - 48$, and $kq = k(k+1)^2$. These formulas prove the last two statements of FMC25 directly. The same bounds hold for transpose inverse maps, whose singular values are unchanged. For $0 \leq t \leq j$, FMC24 also puts every singular value of Z_k^{fix} between M_k^{-1} and M_k , so the same exterior bounds apply to its map onto its image and its inverse there.

38.6 Exact relation to every earlier certificate frame

Let Z_k^{old} be the full actual OCF/ROQ certificate frame, with its selected invertible $j \times j$ row block $(Z_k^{\text{old}})_I$. Both frames have exactly the same column space, as proved in FMC21–22. Their exact transition and inverse are

$$A_k = ((Z_k^{\text{old}})_I)^{-1} (Z_k^{\text{fix}})_I, \quad A_k^{-1} = L_{Z,k} Z_k^{\text{old}}, \quad Z_k^{\text{fix}} = Z_k^{\text{old}} A_k. \quad (\text{FMC27})$$

Each new column has a unique expansion in the old independent columns; restriction to the old pivot rows gives the first formula. Applying $L_{Z,k}$ gives the second and both inverse identities. Thus this is the exact coefficient morphism to the earlier frame, including its actual pivot determinant. No bound for that old determinant is needed or asserted by FMC24–26. The new frame and its proved bounds use only the fixed data of FMC5–23.

The two residual maps on V_A therefore obey $\Xi_k^{\text{fix}} = A_k^\top \Xi_k^{\text{old}}$ in their respective coordinates, since the maps in the original strip dual are the ordinary transposes of their certificate matrices.

For every original source and cutoff their minimum quotient metrics satisfy the exact congruence and determinant formulas

$$Q_{\Xi,N}^{\text{fix}} = \overline{A_k}^{-1} Q_{\Xi,N}^{\text{old}} A_k^{-\top}, \quad \log \det Q_{\Xi,N}^{\text{fix}} = \log \det Q_{\Xi,N}^{\text{old}} - 2 \log |\det A_k|. \quad (\text{FMC28})$$

The constraints on a lift are transformed by precisely $A_k^\top$; substituting the inverse coordinates in the same attained minimum proves the congruence, and determinants give the second formula. The scalar is independent of N and of source order. Its coefficient is exactly $1 + 1 - 1 - 1 = 0$ in the original four-endpoint return. Thus this return agrees identically with the entire earlier ROQ/RCA return, even if an earlier selected pivot is small at some degrees.

The complete inherited period Gram is transported by the same exact map at each endpoint. Let L_{old} be the primary right section ROQ6 and let L_{fix} be any primary right section of Ξ_k^{fix} inside V_A , for example the first j columns of FMC30 below. Both L_{fix} and $L_{\text{old}} A_k^{-\top}$ have residual image I_j under Ξ_k^{fix} , so their difference has columns in the original $K = \ker \Lambda$. Consequently

$$\Lambda L_{\text{fix}} = \Lambda L_{\text{old}} A_k^{-\top}, \quad P_{\text{per}}^{\text{fix}} = \overline{A_k}^{-1} P_{\text{per}}^{\text{old}} A_k^{-\top}, \quad \frac{\det Q_{\Xi,N}^{\text{fix}}}{\det P_{\text{per}}^{\text{fix}}} = \frac{\det Q_{\Xi,N}^{\text{old}}}{\det P_{\text{per}}^{\text{old}}}.$$

The first equality preserves the literal boundary vectors and their full cyclic coefficient map. Taking their Gram with the unchanged inherited metric ROQ9 proves the second equality. Taking both determinants together with FMC28 proves the last equality pointwise, before taking any endpoint difference.

38.7 The unchanged Gamma source in the fixed degree model

Retain $c = k/2$, the original nodes $\beta_{ab} = c + (2a - k)\delta + i(2b - k)\gamma$, and $\omega_{ab} = (\beta_{ab} - c)/i$. For each original source $s \in \{1, k\}$ retain its complete recurrence and masses

$$p_0 = 1, \quad p_1 = y, \quad p_{n+1} = yp_n - n(s/2 + n - 1)p_{n-1}, \quad (\text{FMC29})$$

$$h_n = (2\pi)^{s/2} n! (s/2)_n, \quad (U_N)_{(a,b),n} = p_n(\omega_{ab}) / \sqrt{h_n}, \quad G_N = (U_N U_N^*)^{-1}.$$

Here N remains one of $q - 1, q, 2q - 1, 2q$. These are exactly the original primary quotient metrics of CEP21–22 and ROQ10, with every source mass and coordinate phase present.

The complete primary frame of V_A , in residual coordinates first and common-kernel coordinates second, is

$$J_k = N_k^\top T_k^{-\top}, \quad \Xi_k^{\text{fix}} J_k = E_j^\top, \quad K = \text{im}(J_k E_r). \quad (\text{FMC30})$$

Here E_r inserts the last r coordinates. For a primary vector $x \in V_A$, ROQ3 gives its strip dual coordinate $z = S_k^\top x$. The new residual map is $(Z_k^{\text{fix}})^\top z = E_j^\top T_k^\top z$. Substitution of $z = T_k^{-\top} y$ proves the second identity, and its kernel is the last coordinate block, proving the third. Thus this full frame has precisely the original common kernel; it changes no observation. Its left inverse on V_A is $T_k^\top S_k^\top$. The original coefficient bounds additionally give

$$\|J_k\|_2 \leq \sqrt{q} A_0^{10k} M_k =: \widehat{M}_k, \quad \|T_k^\top S_k^\top\|_2 \leq M_k. \quad (\text{FMC31})$$

Indeed FMC14 bounds the norm of N_k by its column-sum norm times $\sqrt{q}$, transpose preserves Euclidean norm, and FMC24 applies to $T_k^{-\top}$. The insertion transpose is a coordinate restriction of norm one. Both logarithms in FMC31 are $O_{\text{actual}}(k + \log k)$ with their full displayed constants.

The exact source Gram in this specified degree model, and its actual residual quotient, are

$$\mathcal{H}_N = J_k^* G_N J_k = \overline{T_k}^{-1} \overline{N_k} G_N N_k^\top T_k^{-\top}, \quad (\text{FMC32})$$

$$\mathcal{H}_N = \begin{pmatrix} H_{11,N} & H_{12,N} \\ H_{21,N} & H_{22,N} \end{pmatrix}, \quad Q_{\Xi,N}^{\text{fix}} = H_{11,N} - H_{12,N} H_{22,N}^{-1} H_{21,N}.$$

The first formula is direct substitution of the actual primary frame. Since $G_N > 0$ and J_k is injective, $\mathcal{H}_N > 0$; therefore $H_{22,N} > 0$. For prescribed first coordinate y , the second coordinate minimizing the full quadratic form is $-H_{22,N}^{-1}H_{21,N}y$. Completing the square, or multiplying out this substitution, gives exactly the displayed Schur complement. This is the original minimum on $\Xi_k^{\text{fix}}x = y$ inside V_A , so it equals FMC28 and the full ROQ metric. All four blocks, including every off-diagonal entry, are retained. The original minimum source lift of the full frame is explicitly $U_N^*G_NJ_k$, whose Gram is $\mathcal{H}_N$ by $U_NU_N^* = G_N^{-1}$. It contains the entire original polynomial recurrence and remainder at each stated cutoff.

There is a useful quantitative determinant comparison which does not replace that Gamma form by a coefficient form. In strip dual coordinates write $H_{V,N} = \overline{\mathbf{N}}_k G_N \mathbf{N}_k^T$ and $Z = Z_k^{\text{fix}}$. Define the actual coefficient quotient form and the relative determinant by

$$Q_{\text{coeff}} = (Z^T \overline{Z})^{-1}, \quad \Phi_{\Xi,N} = \log \det Q_{\Xi,N}^{\text{fix}} - \log \det Q_{\text{coeff}}. \quad (\text{FMC33})$$

The first is the minimum quotient of the original coefficient Euclidean norm in the fixed OCF strip, with the identical constraint $Z^T z = y$. The elementary minimum proof of ROQ11 gives both this formula and $Q_{\Xi,N}^{\text{fix}} = (Z^T H_{V,N}^{-1} \overline{Z})^{-1}$. Hence the full, directly calculable expression is

$$\begin{aligned} \Phi_{\Xi,N} &= \log \det(Z^T \overline{Z}) - \log \det(Z^T H_{V,N}^{-1} \overline{Z}), \\ |\log \det Q_{\text{coeff}}| &\leq 2j \log M_k = O_{\text{actual}}(k^2) = o(kq), \\ F_{\Xi,k}^{(s)} &= \Phi_{\Xi,q-1} + \Phi_{\Xi,q} - \Phi_{\Xi,2q-1} - \Phi_{\Xi,2q}. \end{aligned} \quad (\text{FMC34})$$

FMC24 puts the singular values of Z between M_k^{-1} and M_k . The positive eigenvalues of $Z^T \overline{Z}$ are their squares, proving the determinant bound by summing j logarithms. The first identity follows by the two exact inverse Gram formulas. The final identity follows by canceling the fixed $\log \det Q_{\text{coeff}}$ with the four original signs and then using FMC28. Thus all coefficient denominators and frame-volume effects in this specified construction are controlled below the kq scale, while the original Gamma compression remains fully visible in its exact determinant.

For the kernel frame $I_k = J_k E_r$, FMC31 likewise gives the concrete coefficient-Gram bounds

$$M_k^{-2r} \leq \det(I_k^* I_k) \leq \widehat{M}_k^{2r}, \quad |\log \det(I_k^* I_k)| \leq 2r \log \widehat{M}_k = O_{\text{actual}}(k^2) = o(kq). \quad (\text{FMC35})$$

Its left inverse on its image is $E_r^T T_k^T S_k^T$, with norm at most M_k ; FMC31 bounds its forward norm. The singular-value argument just given proves FMC35. Define the exact primary-coordinate projection and determinant by

$$P_{K,\text{prim}} = I_k(I_k^* I_k)^{-1} I_k^*, \quad D_{K,N}^{\text{prim}} = [z^r] \det(I_q + z P_{K,\text{prim}} G_N P_{K,\text{prim}}).$$

Multiplication proves that this projection is Hermitian, idempotent, and has exactly the original primary-coordinate K as its image. For the determinant proof take $U = I_k(I_k^* I_k)^{-1/2}$, so $U^* U = I_r$, and extend its columns to an orthonormal basis of $\mathbb{C}^q$. In that basis the projected metric has block $U^* G_N U$ and a zero block. Its indicated determinant coefficient is $\det(U^* G_N U)$, and substitution of the full Gram square root therefore proves

$$\det(I_k^* G_N I_k) = \det(I_k^* I_k) D_{K,N}^{\text{prim}}.$$

The precise map to AKS12–14's original polynomial-coefficient projector is the ordered Vandermonde: put $I_{k,\text{poly}} = V_\beta^{-1} I_k$ and $G_{N,\text{poly}} = V_\beta^* G_N V_\beta$. Their compressed Gram is identically $I_k^* G_N I_k$ by multiplication. Applying the determinant proof just given in these polynomial coordinates proves the exact bridge

$$D_{K,N}^{\text{AKS}} = \frac{\det(I_k^* I_k)}{\det(I_{k,\text{poly}}^* I_{k,\text{poly}})} D_{K,N}^{\text{prim}}.$$

Here the polynomial projector has image the actual coefficient kernel and thus agrees with AKS12's projector. The displayed positive factor is independent of N and source order and cancels exactly with the four endpoint signs. FMC35 bounds its primary-coordinate numerator; the polynomial denominator is retained as the full displayed Vandermonde Gram. No bound for that different denominator follows from FMC35 alone.

FMC25 is a coefficient-space estimate for specified exact maps. Its relation to the original Gamma norms is the complete congruence and Schur calculation FMC29–34, not an assertion that those maps are Gamma isometries. In particular no upper or lower leading value for the remaining Gamma compression is inferred from their coefficient condition numbers. The new result supplies a fixed degree presentation, actual invariant certificates, bounded forward and inverse exterior maps, and the exact original determinant on that model, with every fixed denominator specified and every source order and endpoint retained.

38.8 The exact map to the original polynomial degree flag

The homogeneous degree k in FMC3 is the original tensor degree. Its precise map to the original polynomial-in- S model is the ordered evaluation isomorphism, not an identification of the two degree indices. Retain $\chi_k(S) = \prod_{a,b=0}^k (S - \beta_{ab})$ and the original Vandermonde $(V_\beta)_{(a,b),d} = \beta_{ab}^d$, $0 \leq d < q$. For a fixed-model primal coordinate column $y \in \mathbb{C}^\Delta$ the exact polynomial representative is

$$p_y(S) = \sum_{a,b=0}^k (J_k y)_{ab} \frac{\chi_k(S)}{(S - \beta_{ab}) \chi'_k(\beta_{ab})}, \quad [p_y]_{\text{coeff}} = V_\beta^{-1} J_k y. \quad (\text{FMC36})$$

All denominators are nonzero because the original nodes are distinct. Evaluation at each node proves the displayed inverse and hence the equality of the two formulas. It includes the complete roots, their order, and their actual centre. Thus the source degree- N metric in FMC32 is equivalently $(V_\beta^{-1} J_k)^* (V_\beta^* G_N V_\beta) (V_\beta^{-1} J_k)$, identically $J_k^* G_N J_k$ by multiplication. This proves the full typed relation to the original coefficient Gamma model.

There is an exact further map to the degree-adapted conductor basis AKJ17. Retain all actual conductor moments μ_d , their first nonzero index $v \leq 80$, the literal $B_q(\partial_S)$ of AKJ14, and $\mathcal{J}_v S^d = d! S^{d+v} / (d+v)!$. The ordered polynomial columns of $P_{A,k}$ are

$$1, S, \dots, S^{v-1}, \quad P_h = \frac{(q' + h + v)!}{(q' + h)!} B_q(\partial_S)^{-1} \mathcal{J}_v(\chi_{k'} S^h), \quad 0 \leq h \leq \Delta - v - 1. \quad (\text{FMC37})$$

Their distinct monic degrees are $0, \dots, v-1$ and $q' + v, \dots, q-1$. AKJ14–17 proves directly that they form a basis of this same V_A , using the complete conductor derivative and its finite nilpotent inverse. Define $B_{A,k}(p)$ on V_A by successively subtracting the coefficient of the highest remaining degree times its matching column in FMC37, down through the high list and then through the low monomials. Each subtraction uses leading coefficient one. Spanning and the distinct degree pivots show that it ends at zero and records the unique coordinate column of p ; this also proves its linearity and both inverse identities on V_A . Consequently the precise transition is

$$R_{A,k} = S_k^\top V_\beta P_{A,k}, \quad R_{A,k}^{-1} z = B_{A,k}(V_\beta^{-1} \mathbf{N}_k^\top z), \quad (\text{FMC38})$$

$$V_\beta^{-1} J_k = P_{A,k} R_{A,k}^{-1} T_k^{-\top}, \quad \Xi_{A,k}^{\text{fix}} = E_j^\top T_k^\top R_{A,k}.$$

The columns $V_\beta P_{A,k}$ lie in V_A . Therefore ROQ3 gives $\mathbf{N}_k^\top R_{A,k} = V_\beta P_{A,k}$, which proves the first inverse formula by applying the unique degree coordinates just constructed. Both sides are invertible maps between the displayed Δ -dimensional coefficient spaces. Substituting FMC30 gives the third equality, and evaluating the new residual map on these same primary columns gives the last. Thus its degree-prefix ranks give exactly $\dim(K \cap \mathbb{C}[S]_{\leq d})$ by subtracting that prefix rank from the number of columns of FMC37 of degree at most d , as in AKJ18. Every original factorial,

finite inverse, primary denominator and degree pivot is visible in these maps. The subleading coefficient-frame estimate proved in FMC25 pertains to the fixed strip module maps; FMC36–38 retain the separate polynomial interpolation and its full Gamma metric exactly.

39 The evaluated outer contribution in the original mixed return

Retain the actual simple quartet, original period on the proved five-orbit locus, full amplitude unit, branch, and original source orders $s = 1, k$ of OAR and OFV. All arithmetic statements retain the original domain $|u| \geq R_*$ of AMT; the actual period is fixed as $k = 4l + 1$ grows. Here $q = (k + 1)^2$, $\Delta = 16k - 48$, and every finite use of OFV retains its stated threshold, including OSP23. Write

$$C_{\partial} = \mathfrak{C}_{\partial} = 4 \log(4/\pi) + 2 - 2I_{2,\pi}, \quad I_{2,\pi} = \iint \log|x - y| d\mu_{2,\pi}(x) d\mu_{2,\pi}(y). \quad (\text{OVR1})$$

The measure here is the full EIQ density with its uniquely specified endpoints. OFV proves this exact expression and $C_{\partial} \geq 0$. No value of the original residual observation is inserted into it.

39.1 The exact remainder and the sourcewise leading identity

Define the actual outer remainder and retain OAR3's full conductor and monic-tail remainder by

$$b_{k,s} = \mathcal{F}_{\partial,k}^{(s)} - \Delta q C_{\partial}, \quad R_{k,s} = b_{k,s} + \mathfrak{e}_{k,s}. \quad (\text{OVR2})$$

Thus $b_{k,s} = O_{\delta,\gamma}(q \log q)$ by OFV30. The second summand is precisely the signed finite expression OAR3, with $-e_{k,s}^- \leq \mathfrak{e}_{k,s} \leq e_{k,s}^+$ and $e_{k,s}^{\pm} = o(kq)$. In particular $R_{k,s} = o(kq)$, because $\log q/k \rightarrow 0$. Its two exact finite error endpoints remain $b_{k,s} - e_{k,s}^-$ and $b_{k,s} + e_{k,s}^+$; their signs are not replaced by a symmetry assumption. Substitution into OAR3 proves

$$\begin{aligned} F_K^{(s)} + \Phi_{k,s} &= \Delta q C_{\partial} + R_{k,s}, \\ \frac{F_K^{(s)}}{kq} + \frac{\Phi_{k,s}}{kq} &= 16C_{\partial} - \frac{48C_{\partial}}{k} + \frac{R_{k,s}}{kq} \longrightarrow 16C_{\partial}, \quad s = 1, k. \end{aligned} \quad (\text{OVR3})$$

Here $\Phi_{k,s}$ is exactly ROQ19's positive sum of constrained covariance increments, built from the original period certificate matrix D and the complete conductor cross covariance. Its coefficient maps, minimum sections, inherited period metric and graph denominator are those already proved in ROQ; OVR3 changes none of them.

39.2 Transfer between the two original Gamma sources

Let $d_k = F_K^{(k)} - F_K^{(1)}$ denote the actual difference. AMT44–46 proves $d_k = o(kq)$, with its original finite two-sided source-comparison bounds. Subtracting the two exact lines OVR3 gives

$$\Phi_{k,k} - \Phi_{k,1} = R_{k,k} - R_{k,1} - d_k, \quad \frac{\Phi_{k,k} - \Phi_{k,1}}{kq} \longrightarrow 0. \quad (\text{OVR4})$$

This proves the transfer using the actual two residual observations, whose individual metrics retain their respective original sources. There is no asserted equality between those metrics. The equality in OVR4 is the exact scalar consequence of their common original-kernel restriction maps and the proved determinant identities.

39.3 Arithmetic kernel and its correlated boundary

Put $d_k^{\text{ar}} = F_K^{\text{ar}} - F_K^{(1)}$. This is the actual quantity bounded in OAR8; AMT38 and AMT46 prove $d_k^{\text{ar}} = o(kq)$. Keeping its full value first gives

$$\begin{aligned} F_K^{\text{ar}} &= \Delta q C_{\partial} - \Phi_{k,1} + R_{k,1} + d_k^{\text{ar}}, \\ F_B^{\text{ar}} &= \mathcal{B}_k^{\text{ar}} - \Delta q C_{\partial} + \Phi_{k,1} - R_{k,1} - d_k^{\text{ar}}. \end{aligned} \quad (\text{OVR5})$$

The first equality adds the literal definition of d_k^{ar} to OVR3 at $s = 1$. The second subtracts the first from $F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}$. In particular the same arithmetic total and the same two finite errors

occur with opposite signs in the boundary. OAR8 supplies the unchanged correlated finite bounds on d_k^{ar} . Dividing these exact equalities gives

$$\frac{F_K^{\text{ar}} + \Phi_{k,1}}{kq} \longrightarrow 16C_\partial, \quad \frac{\mathcal{B}_k^{\text{ar}} - F_B^{\text{ar}} + \Phi_{k,1}}{kq} \longrightarrow 16C_\partial. \quad (\text{OVR6})$$

Thus evaluation of the residual sum in either original Gamma order passes through to the same original arithmetic kernel and boundary.

39.4 The signed return retains its original Gamma correction

The exact signed quantity of OAR11 is $\mathfrak{S}_k^{\text{mix}} = F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}$. Applying OVR3 at $s = k$ to that identity yields

$$\mathfrak{S}_k^{\text{mix}} + \Phi_{k,k} = \Delta q C_\partial + \mathcal{H}_k + \delta_k^\sigma + R_{k,k}. \quad (\text{OVR7})$$

All finite terms are the original ones. AMT48–49, with the retained total correction estimates used there, proves $(\mathcal{H}_k + \delta_k^\sigma)/(kq) \rightarrow -C_\Gamma/4$, where

$$C_\Gamma = 2 \int \log x \, d\mu_{2,\pi}(x) - 4(2 \log 2 - 1) > 0.$$

The equilibrium and source maps are exactly the EIQ/GEL providers used in OFV and AMT. Hence OVR7 and OVR4 prove both limits

$$\frac{\mathfrak{S}_k^{\text{mix}} + \Phi_{k,k}}{kq} \longrightarrow 16C_\partial - \frac{C_\Gamma}{4}, \quad \frac{\mathfrak{S}_k^{\text{mix}} + \Phi_{k,1}}{kq} \longrightarrow 16C_\partial - \frac{C_\Gamma}{4}. \quad (\text{OVR8})$$

The two constants are explicit integrals of the same proved density, respectively its full logarithmic pair energy and its logarithmic moment. No identity between these distinct integrals is presumed.

39.5 Bounds on the remaining residual coefficient

Each $F_K^{(s)}$ is nonnegative: the original quotient metrics decrease with increasing source degree, and restriction to the identical kernel preserves the two Gram inequalities at the paired endpoints. Their determinant ratios are consequently at least one, as proved in AMT15. ROQ19 gives $\Phi_{k,s} \geq 0$. Thus OVR3 bounds both nonnegative sequences after division by kq . Let $\kappa_* = 32 \log(25 \log 3 / (4\pi))$ be the retained AKS asymptotic kernel ceiling on this actual locus, applied in both original orders. The actual residual cap OAR5, proved by ROQ21, is $\Phi_{k,s} \leq j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - \lambda_s$, where $j = 8k - 32$ and $\lambda_s \geq 0$ is the original first increment. AKS31 gives $\limsup j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1})/(kq) \leq \kappa_*$ in both original source orders: its same explicit logarithmic comparison constants apply, and $j/k \rightarrow 8$ just as $r/k \rightarrow 8$. Thus both the original kernel and residual sequences have the same proved upper ceiling. Combining their two ceilings with OVR3 gives

$$\begin{aligned} \max\{0, 16C_\partial - \kappa_*\} &\leq \liminf \frac{F_K^{(s)}}{kq} \leq \limsup \frac{F_K^{(s)}}{kq} \leq \min\{\kappa_*, 16C_\partial\}, \\ \max\{0, 16C_\partial - \kappa_*\} &\leq \liminf \frac{\Phi_{k,s}}{kq} \leq \limsup \frac{\Phi_{k,s}}{kq} \leq \min\{\kappa_*, 16C_\partial\}. \end{aligned} \quad (\text{OVR9})$$

For each line, subtract the other bounded nonnegative sequence from the convergent sum in OVR3. More explicitly, that sum differs from $16C_\partial$ by a sequence tending to zero; taking its upper and lower eventual bounds proves $\liminf(\Phi/(kq)) = 16C_\partial - \limsup(F_K/(kq))$ and $\limsup(\Phi/(kq)) = 16C_\partial - \liminf(F_K/(kq))$. The same argument with the two sequences interchanged proves the first line, including its lower bound. The two first positive increments also survive in the finite common

upper bound. In the original AKS notation let d_1 be the first section-corrected kernel increment of AKS29, in the same source order s . AKS28 and OAR5 give

$$\begin{aligned}\Delta q C_{\partial} + R_{k,s} &= F_K^{(s)} + \Phi_{k,s} \\ &\leq (r+j)(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1 - \lambda_s \\ &= \Delta(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - d_1 - \lambda_s.\end{aligned}\tag{OVR10}$$

The last equality is the exact rank identity $(8k-16) + (8k-32) = 16k-48$; both subtractions retain their actual minimum-section definitions. Adding the eventual upper bounds already proved and using OVR3 also proves the explicit consequence

$$0 \leq 16C_{\partial} \leq 2\kappa_*, \quad 0 \leq C_{\partial} \leq \frac{\kappa_*}{8}.\tag{OVR11}$$

Indeed for every positive ε , both sequences are eventually at most $\kappa_* + \varepsilon$, so their limiting sum is at most $2\kappa_* + 2\varepsilon$; let ε decrease to zero. Their nonnegativity proves the lower bound independently. OVR4 and AMT46 carry the same limiting bounds to the two original sources and the arithmetic kernel. The outstanding coefficient is therefore the explicit original positive residual sum; no convergence or numerical value for that sequence has been assumed.

40 All directions of the original outer quotient at the leading scale

Retain the complete original simple quartet, the source orders $s = 1, k$, $k = 4l + 1$, $0 < \delta < 1/2$, $\gamma > 2$, and the polynomials and masses of CTV1–22 and OSP1–35. Put

$$q = (k + 1)^2, \quad q' = (k - 7)^2, \quad d = q - q' = 16k - 48, \quad n = q/2, \quad \vartheta = \pi/2, \quad R = k\sqrt{\delta^2 + \gamma^2}. \quad (\text{OOQ1})$$

The integer q is even. The entire outer polynomial is $P_{\partial,k} = P_k/P_{k-8}$, monic of degree d , with every original root retained. Each root has modulus at most R . Explicitly $P_k(y) = i^{-q}\chi_k(c + iy)$ and $P_{k-8}(y) = i^{-q'}\chi_{k-8}(c' + iy)$, with the unchanged original centres $c = k/2$ and $c' = c - 4$. These are the exact CTV3–4 phase and coefficient maps. Its target coordinates throughout are the coefficients of $1, y, \dots, y^{d-1}$ in its unique remainder. For $L = d - 1, d, q + d - 1, q + d$, let $Q_{\partial,L}^{(s)}$ be the minimum quotient Gram of this remainder map from polynomials of degree at most L , with the complete source

$$d\zeta_{k',s}(y) = |P_{k-8}(y)|^2 dm_s(y), \quad dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy. \quad (\text{OOQ2})$$

These are exactly the CTV10–13 Grams, not merely determinants equal to those Grams. We prove operator estimates in this same original remainder frame.

40.1 Uniform equilibrium constants and a polynomial evaluation bound

Use the complete EIQ equilibrium for $V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x$ on $(0, \infty)$. In this proof take the compact rectangle $\mathcal{K} = [3/2, 5/2] \times [\pi/4, \pi]$. EIQ is proved on the entire positive quadrant, so it applies to this rectangle. Write its exact density as $\rho_{\alpha,\beta}$, its endpoints as a, b , its logarithmic potential as $P(z)$, and retain

$$V(x) - 2P(x) = \ell + D(x), \quad D(x) \geq 0, \quad D|_{[a,b]} = 0, \quad P_0 = P(0) = \int \log t \rho(t) dt, \quad W = P_0 + \ell/2. \quad (\text{OOQ3})$$

Here and below a parameter on these symbols is understood whenever two parameter pairs occur in the same expression.

The following fixed constants are all defined by these exact functions on $\mathcal{K}$:

$$a_* = \min_{\mathcal{K}} a > 0, \quad B_* = \max_{\mathcal{K}} b < \infty, \quad b_* = \min_{\mathcal{K}} (b - a) > 0, \quad M_* = \max_{\mathcal{K},x} \rho(x) < \infty, \\ L_* = \max_{\mathcal{K}, x \in [a,b]} |V'(x)|, \quad C_{\text{BM}} = \sqrt{32} e^{3L_*/4}, \quad C_{\text{quant}} = B_* + 2M_*, \quad C_0 = B_*/a_*, \quad (\text{OOQ4})$$

$$C_{\text{tail}} = \max\{0, \sup_{\mathcal{K}, x>0} [\log(1 + \sqrt{x}) - D(x)]\}, \quad T_* = \max\{1, \sup_{\mathcal{K}} \int_0^\infty e^{-D(x)} dx\}.$$

These constants are finite. Positivity, continuity and boundedness of the endpoints and density are EIQ4, EIQ10–11 and EIQ29; the positive gap follows by compactness. The derivative of V is bounded on $[a_*, B_*]$. Uniformly over this rectangle, $P(x) \leq \log(1 + x) + \log(1 + B_*)$ and $P(x) \geq -2M_*$, as in the direct logarithmic estimates in QLG9–13. The Euler constant is bounded by evaluating OOQ3 on the compact support. Consequently

$$D(x) \geq \frac{\pi}{4}\sqrt{x} - \frac{5}{2}\log x - 2\log(1 + x) - C \quad (x \geq 1)$$

for a fixed finite constant C . This proves the assertions at infinity in the last two definitions of OOQ4. At zero, $D(x) \rightarrow +\infty$ uniformly because $\alpha \geq 3/2$ and P is bounded there. For integrability one can use precisely the finite majorant $e^C(1 + x)^2(1 + x^{5/2})e^{-\pi\sqrt{x}/4}$. Thus neither of the two supremum definitions assumes a tail estimate.

The function W and its first derivatives are bounded on $\mathcal{K}$. Indeed EIQ30 gives $F_\alpha = P_0$, EIQ31 gives $\beta \int \sqrt{x} d\mu = 2(\alpha + 1)$, and hence

$$W(\alpha, \beta) = -F(\alpha, \beta) - (\alpha + 1) + (1 + \alpha/2)F_\alpha(\alpha, \beta). \quad (\text{OOQ5})$$

The smoothness asserted here follows from EIQ29–30. Fix

$$C_W = 1 + \max_{\mathcal{K}} |W| + (4 + \pi) \max_{\mathcal{K}} (|W_\alpha| + |W_\beta|). \quad (\text{OOQ6})$$

For an integer $m \geq \max\{1, 2/b_*\}$ and a polynomial p of degree at most m , set $\|p\|_{m,V}^2 = \int_0^\infty |p(x)|^2 e^{-mV(x)} dx$. For all z outside $[a, b]$ we have

$$|p(z)| \leq C_{\text{BM}} m^2 \exp\{m(P(z) + \ell/2)\} \|p\|_{m,V}. \quad (\text{OOQ7})$$

Here is a complete proof. Let $M = \max_{[a,b]} |p(x)| e^{-mV(x)/2}$, attained at x_0 . Choose a one-sided interval starting at x_0 , contained in $[a, b]$, of length $1/m$; one side has that length because $b - a \geq 2/m$. On that interval, $|p(x)| \leq e^{L_*/2} |p(x_0)|$.

The elementary derivative bound $\|p'\|_{\infty,J} \leq 4m^3 |J|^{-1} \|p\|_{\infty,J}$ holds also for complex coefficients. To prove it, expand on the affine copy of $[-1, 1]$ in the polynomials T_j defined by $T_j(\cos \theta) = \cos(j\theta)$. Orthogonality of cosines gives coefficients of modulus at most $2\|p\|_\infty$ for $j \geq 1$. The identity $T'_j(\cos \theta) = j \sin(j\theta)/\sin \theta$ and the sum of j unit complex numbers give $|T'_j| \leq j^2$, including the limiting endpoints. Summing $\sum_{j=1}^m j^2 \leq m^3$ and applying the affine derivative factor $2/|J|$ proves the bound. It follows that on a segment of length $[8e^{L_*/2} m^4]^{-1}$ starting at x_0 , $|p| \geq |p(x_0)|/2$. The weight on that segment is at least $e^{-L_*} e^{-mV(x_0)}$. Integration gives $\|p\|_{m,V}^2 \geq M^2/(32e^{3L_*/2} m^4)$.

It remains to extend the weighted supremum to complex z . The function $\log |p(z)| - mP(z)$ is subharmonic off $[a, b]$. On this interval its boundary upper values are at most $\log M + m\ell/2$ by OOQ3. It extends subharmonically over infinity: in the local coordinate $w = 1/z$ its logarithmic term is $(m - \deg p) \log |w|$, plus a harmonic function. The maximum principle on the complement of the interval in the sphere therefore bounds it by the same boundary constant. One may first remove small neighborhoods of polynomial zeros and of the interval and then pass to the limit; P is continuous because its density is bounded. Combining this bound with the preceding norm bound proves OOQ7. For $|z| \leq a_*/2$ its useful uniform consequence is

$$|p(z)| \leq C_{\text{BM}} m^2 e^{mW+2m|z|/a_*} \|p\|_{m,V}, \quad (\text{OOQ8})$$

since $|\log |1 - z/t|| \leq 2|z|/a_*$ on the support.

40.2 A matching polynomial lift from actual equilibrium quantiles

For each n divide the actual equilibrium measure into n intervals of mass $1/n$ and take the midpoint of each interval as ξ_i . Retain the polynomial

$$r_n(x) = \prod_{i=1}^n (1 - x/\xi_i), \quad r_n(0) = 1. \quad (\text{OOQ9})$$

All its roots lie in $[a_*, B_*]$. For every real $x > 0$,

$$\log |r_n(x)| \leq n(P(x) - P_0) + C_{\text{quant}} \sqrt{n} + C_0. \quad (\text{OOQ10})$$

To prove it, use $f_\eta(t) = \log \max\{|x - t|, \eta\}$, a $1/\eta$ -Lipschitz function. For a quantile interval of length Δ_i , the difference between its point value and the average against its probability density is at most Δ_i/η . Summing gives at most B_*/η . Also

$$0 \leq \int [f_\eta(t) - \log |x - t|] \rho(t) dt \leq M_* \int_{-\eta}^{\eta} \log(\eta/|s|) ds = 2M_* \eta.$$

Choose $\eta = n^{-1/2}$. Finally $\log t$ is $1/a_*$ -Lipschitz on the support, so $|\sum_i \log \xi_i - nP_0| \leq B_*/a_*$. These three facts prove OOQ10, including at a polynomial zero. For complex $|z| \leq a_*/2$, the same literal product gives

$$|r_n(z)|^{-1} \leq e^{2n|z|/a_*}. \quad (\text{OOQ11})$$

This uses $|1 - z/\xi_i| \geq \exp(-2|z|/a_*)$ separately for every original quantile factor.

40.3 The entire analytic remainder map

For the present $n = q/2$ put

$$\rho = n^{-1/8}, \quad \hat{P}(z) = n^{-d}P_{\partial,k}(nz), \quad r_* = R/n. \quad (\text{OOQ12})$$

Impose the following explicit eventual conditions, together with the original OSP23 domain and $k \geq 257$:

$$n \geq 2d + 1, \quad n \geq 2/b_*, \quad r_* \leq \rho/2, \quad \rho \leq \min\{1/2, \sqrt{a_*/2}\}. \quad (\text{OOQ13})$$

They hold for all sufficiently large original k at every fixed δ, γ : $n \asymp k^2$, $d = O(k)$ and $r_* = O(1/k)$. These inequalities specify the finite range used here; no assertion at excluded small degrees is required.

For f holomorphic on the closed disk of radius ρ define its complete remainder by

$$\text{rem}_{\hat{P}} f(z) = \frac{1}{2\pi i} \int_{|\zeta|=\rho} f(\zeta) \frac{\hat{P}(\zeta) - \hat{P}(z)}{(\zeta - z)\hat{P}(\zeta)} d\zeta. \quad (\text{OOQ14})$$

The numerator divided by $\zeta - z$ is a polynomial in z of degree at most $d - 1$. At every root of $\hat{P}$, Cauchy's formula gives the value of f ; all these original roots are distinct, so this is exactly the unique interpolation remainder. In particular it is the usual polynomial remainder when f is polynomial. This proves the exact map and its agreement with the original target.

Write $\|\cdot\|_1$ for the full monomial coefficient norm. The root bound in OOQ13 proves

$$\|\text{rem}_{\hat{P}} f\|_1 \leq C_{\text{rem}} \max_{|\zeta|=\rho} |f(\zeta)|, \quad C_{\text{rem}} = \rho d(1 + r_*)^d (2/\rho)^d. \quad (\text{OOQ15})$$

Indeed $|\hat{P}(\zeta)| \geq (\rho - r_*)^d \geq (\rho/2)^d$. If $\hat{P}(z) = \sum_{a=0}^d p_a z^a$, then

$$\frac{\hat{P}(\zeta) - \hat{P}(z)}{\zeta - z} = \sum_{a=1}^d p_a \sum_{h=0}^{a-1} \zeta^{a-1-h} z^h,$$

whose coefficient norm is at most $d \sum_a |p_a|$ because $\rho \leq 1$. The sum of absolute coefficients of the full monic product is at most $(1 + r_*)^d$. The contour length supplies the factor ρ , proving OOQ15. In particular $\log C_{\text{rem}} = O(d \log n + \log d)$. No outer root has been moved to zero in this map.

40.4 High source degrees: a full operator estimate

First use the precise exponential source $|y|^{2Q} e^{-\beta|y|} dy$, where $Q = q' + \ell_s$, $\ell_s = (s - 1)/4$, and $\beta \in \{\vartheta - 1/q, \vartheta + 1/q\}$. Let $Q_L^{\text{exp}}(Q, \beta)$ be its quotient Gram by the same entire $P_{\partial,k}$, in the original y remainder frame. Set

$$\alpha_0 = (Q - \tfrac{1}{2})/n, \quad c_{\text{high}}(Q, \beta) = (2Q + 1) \log n - 2nW(\alpha_0, \beta). \quad (\text{OOQ16})$$

The two high source degrees are $L = q + d - 1, q + d$. For their two parity degrees put $m_e = \lfloor L/2 \rfloor$, $m_o = \lfloor (L - 1)/2 \rfloor$. The positive-half-line parameters are exactly

$$(\alpha_e, \beta_e) = ((Q - \tfrac{1}{2})/m_e, n\beta/m_e), \quad (\alpha_o, \beta_o) = ((Q + \tfrac{1}{2})/m_o, n\beta/m_o). \quad (\text{OOQ17})$$

All these pairs and (α_0, β) lie in $\mathcal{K}$ for the stated $k \geq 257$. For example $d/q < 1/16$, $\ell_s/q < 1/64$ and the rounding errors are at most one; these give $3/2 < \alpha_e, \alpha_o < 5/2$. The rate lies between $\pi/4$ and π because $m_e, m_o = n + O(d)$ and $1/q < \vartheta/4$. More directly the actual lower bounds $m_e, m_o \geq n$ and upper bounds $m_e, m_o \leq n + d$ together with $d/q < 1/16$ verify the rate bounds.

For a polynomial $p(nz) = p_e(z^2) + zp_o(z^2)$, parity and the two substitutions $z = \pm\sqrt{x}$ give exactly

$$n^{-(2Q+1)} \|p\|_{\text{exp}}^2 = \int_0^\infty |p_e(x)|^2 x^{Q-1/2} e^{-n\beta\sqrt{x}} dx + \int_0^\infty |p_o(x)|^2 x^{Q+1/2} e^{-n\beta\sqrt{x}} dx. \quad (\text{OOQ18})$$

Both measures on the right are precisely the OOQ7 measures at OOQ17. There is no missing factor two: each branch contributes the positive Jacobian $1/(2\sqrt{x})$. The smooth bound OOQ6 gives

$$|m_e W(\alpha_e, \beta_e) - nW(\alpha_0, \beta)|, |m_o W(\alpha_o, \beta_o) - nW(\alpha_0, \beta)| \leq C_W(d+2). \quad (\text{OOQ19})$$

For verification, $|m - n| \leq d + 1$, $m|\alpha - \alpha_0| \leq 1 + (5/2)|m - n|$ and $m|\beta_m - \beta| \leq \pi|m - n|$. The mean-value formula on the rectangle and OOQ6 prove OOQ19.

Apply OOQ8 at $x = \zeta^2$ and $|\zeta| = \rho$, then use OOQ18 and Cauchy–Schwarz on its two norms. If v is the coefficient vector of the remainder of $p(nz)$ modulo $\hat{P}$, OOQ15 gives

$$\|v\|_2 \leq C_{\text{rem}} \sqrt{2} C_{\text{BM}}(n+d)^2 e^{nW+C_W(d+2)+2(n+d)\rho^2/a_*} n^{-(2Q+1)/2} \|p\|_{\text{exp}}. \quad (\text{OOQ20})$$

This holds for every lift, and proves the lower bound for the quotient metric in these scaled remainder coordinates.

For the upper bound use the actual equilibrium at (α_0, β) to form r_n in OOQ9. For the given remainder polynomial $v(z)$ put

$$a_v(z) = \text{rem}_{\hat{P}} \left(\frac{v(z)}{r_n(z^2)} \right), \quad p_v(nz) = r_n(z^2) a_v(z). \quad (\text{OOQ21})$$

The denominator has no zero in the disk because all its x -roots are at least $a_* > \rho^2$. The polynomial p_v has degree at most $q + d - 1 \leq L$. At every original outer root its value is exactly v ; thus it is an actual lift of the prescribed remainder. Equations OOQ11 and OOQ15 give

$$\|a_v\|_1 \leq C_{\text{rem}} \sqrt{d} e^{2n\rho^2/a_*} \|v\|_2. \quad (\text{OOQ22})$$

For each real $x > 0$ both $|a_v(\sqrt{x})|$ and $|a_v(-\sqrt{x})|$ are at most $\|a_v\|_1 (1 + \sqrt{x})^{d-1}$. By OOQ4, $(1 + \sqrt{x})^{2(d-1)} \leq e^{2C_{\text{tail}}(d-1)} e^{2(d-1)D(x)}$. Combine this with OOQ10 and $2nP - nV = -n\ell - nD$. Since $n - 2(d-1) \geq 1$, integration using T_* proves

$$\|p_v\|_{\text{exp}}^2 \leq e^{\text{Chigh}(Q, \beta)} \|a_v\|_1^2 \exp\{2C_{\text{quant}}\sqrt{n} + 2C_0 + 2C_{\text{tail}}(d-1)\} T_*. \quad (\text{OOQ23})$$

All source scalings in this expression come from the explicit $y = nz$ substitution; in the original lower-source variable it is precisely $S' = c' + inz$, with the same original centre and phase.

Define the finite nonnegative errors

$$\begin{aligned} E_{\text{high}}^- &= 2 \log \left(C_{\text{rem}} \sqrt{2} C_{\text{BM}}(n+d)^2 \right) + 2C_W(d+2) + 4(n+d)\rho^2/a_*, \\ E_{\text{high}}^+ &= 2 \log C_{\text{rem}} + \log d + 4n\rho^2/a_* + 2C_{\text{quant}}\sqrt{n} + 2C_0 + 2C_{\text{tail}}(d-1) + \log T_*, \\ E_{\text{high}} &= \max\{0, E_{\text{high}}^-, E_{\text{high}}^+\} + 2(d-1) \log n. \end{aligned} \quad (\text{OOQ24})$$

The original and scaled remainder vectors are related by the exact diagonal map $D_n = \text{diag}(1, n, \dots, n^{d-1})$. Hence the original Gram is $D_n^* Q_{\text{scaled}} D_n$. Since $I \preceq D_n^* D_n \preceq n^{2(d-1)} I$, OOQ20–24 prove, at both high endpoints,

$$e^{\text{Chigh}(Q, \beta) - E_{\text{high}}} I_d \preceq Q_L^{\text{exp}}(Q, \beta) \preceq e^{\text{Chigh}(Q, \beta) + E_{\text{high}}} I_d. \quad (\text{OOQ25})$$

The errors obey $E_{\text{high}} = O(n^{3/4} + d \log n)$. This is a full quadratic-form bound for the entire original outer denominator, not an inference from a determinant estimate.

40.5 Low source degrees, including the extra ideal column

Put $g(Q, \beta) = 2Q \log(2Q/\beta) - 2Q$ and $y_0 = 2Q/\beta$. For polynomials of degree at most d , their full exponential source Gram is bounded below by

$$\frac{e^{g(Q, \beta) - \beta}}{(d+1)^2(2d+1)64^d(1+y_0)^{2d}} I_{d+1}. \quad (\text{OOQ26})$$

To prove it, the weight on $[y_0, y_0+1]$ is at least $e^{g(Q, \beta) - \beta}$. On $[0, 1]$ the orthonormal shifted Legendre polynomial of degree j is $\sqrt{2j+1} L_j(2x-1)$. Its coefficient sum is at most $\sqrt{2j+1} 8^j$: Rodrigues' formula gives coefficients $(-1)^{j-r} \binom{j}{r} \binom{j+r}{r}$, and $\binom{j+r}{r} \leq 4^j$. Thus the coefficient matrix of these orthonormal polynomials has Euclidean norm at most $\sqrt{(d+1)(2d+1)} 8^d$. The translation back by $-y_0$ has coefficient norm at most $(1+y_0)^d$ and Euclidean norm at most $\sqrt{d+1}(1+y_0)^d$. These two invertible maps prove OOQ26 by direct norm comparison. The Legendre orthogonality and full Rodrigues norm are proved in OSP15 and are also obtained by its stated integrations by parts.

The largest eigenvalue of the same full Gram is at most its trace, which is exactly

$$\sum_{j=0}^d \frac{2(2Q+2j)!}{\beta^{2Q+2j+1}}. \quad (\text{OOQ27})$$

This follows by the two real half-line integrals and repeated integration by parts. At $L = d-1$ the remainder map is the identity. At $L = d$ it has matrix $[I_d, -p_{\text{low}}]$, where p_{low} contains every lower coefficient of $P_{\partial, k}$. Its squared operator norm is at most $(d+1)(1+R)^{2d}$, because that complete monic polynomial has coefficient sum at most $(1+R)^d$. The minimum Euclidean norm on its fibre is therefore at least the target norm divided by this operator norm; the canonical degree-less-than- d lift gives the upper bound.

For both low endpoints define

$$E_{\text{low}}(Q, \beta) = \max \left\{ 0, \log \left(\sum_{j=0}^d \frac{2(2Q+2j)!}{\beta^{2Q+2j+1}} \right) - g(Q, \beta), \right. \\ \left. \beta + 3 \log(d+1) + \log(2d+1) + d \log 64 + 2d \log(1+y_0) + 2d \log(1+R) \right\}. \quad (\text{OOQ28})$$

Equations OOQ26–27 and the exact remainder matrices prove

$$e^{g(Q, \beta) - E_{\text{low}}} I_d \preceq Q_L^{\text{exp}}(Q, \beta) \preceq e^{g(Q, \beta) + E_{\text{low}}} I_d, \quad L = d-1, d. \quad (\text{OOQ29})$$

In the present range this error is $O_{\delta, \gamma}(d \log q)$. For the upper side, integrating $\log t$ between consecutive integers gives $m \log m - m + 1 \leq \log(m!) \leq m \log m - m + 1 + \log m$. Consequently each logarithm in OOQ27 differs above from $g(Q, \beta)$ by $O(d \log(Q+d) + \log(Q+d))$, uniformly over $\beta \in [\pi/4, \pi]$. The logarithm of the number of summands adds only $\log(d+1)$. The displayed lower error has the same order since $Q \asymp q$, $d = O(k)$ and $R = O_{\delta, \gamma}(k)$.

40.6 Transport back to both complete original Gamma sources

Retain the exact source recurrence factors of OFV1–13:

$$\ell_s = (s-1)/4, \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2} \Gamma(s/2)}, \quad Q_s = q' + \ell_s, \quad h = 1/q, \\ L_h = C_L (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}. \quad (\text{OOQ30})$$

Use $\mathcal{E}_k$ from OSP24 and r_{ℓ_s} from OFV6, with $r_0 = 0$. For every polynomial of all the degrees used here, the complete source comparisons give the quadratic-form inequalities

$$\begin{aligned} \beta_s L_h e^{-\mathcal{E}_k} \int |p(y)|^2 |y|^{2Q_s} e^{-(\vartheta+h)|y|} dy &\leq \int |p(y)|^2 d\zeta_{k',s}(y) \\ &\leq \beta_s U_h e^{\mathcal{E}_k + r_{\ell_s}} \int |p(y)|^2 |y|^{2Q_s} e^{-(\vartheta-h)|y|} dy. \end{aligned} \quad (\text{OOQ31})$$

OSP20–26 supplies the first root-polynomial comparison on the entire coefficient space, retaining every coefficient of P_{k-8} . OFV5–7 supplies the exact original-order density multiplier, including every factor t_j and β_s . OFV8–9 supplies both pointwise bounds on the one-factor Gamma density, including near zero. Composing these three proved inequalities gives exactly OOQ31. The degree bound required there is satisfied because $L \leq q + d \leq 2q$. Every minimum here is taken on the same original fibre $\text{rem}_{P_{\partial,k}} p = v$; therefore OOQ31 passes to the quotient Grams by taking its attained minima. No multiplication map is presumed to preserve the kernel.

Define the two common scalar centres

$$\begin{aligned} c_{\text{lo}}(k) &= g(q, \vartheta), \\ c_{\text{hi}}(k) &= 2q \log(q/2) - qW(2, \vartheta). \end{aligned} \quad (\text{OOQ32})$$

For an entirely finite common error, take

$$\begin{aligned} \varepsilon_k &= \mathcal{E}_k + \max_{s=1,k} r_{\ell_s} + \max\{|\log L_h|, |\log U_h|\} \\ &+ \max_{\substack{s \in \{1,k\} \\ b \in \{\vartheta-h, \vartheta+h\}}} \left\{ \begin{aligned} &E_{\text{low}}(Q_s, b) + E_{\text{high}} + |g(Q_s, b) - c_{\text{lo}}(k)| \\ &+ |c_{\text{high}}(Q_s, b) - c_{\text{hi}}(k)| \end{aligned} \right\}. \end{aligned} \quad (\text{OOQ33})$$

All constants in this definition were specified above from the original parameters and the exact equilibrium. The resulting operator conclusion is

$$\begin{aligned} \beta_s e^{c_{\text{lo}} - \varepsilon_k} I_d &\preceq Q_{\partial,L}^{(s)} \preceq \beta_s e^{c_{\text{lo}} + \varepsilon_k} I_d, \quad L = d - 1, d, \\ \beta_s e^{c_{\text{hi}} - \varepsilon_k} I_d &\preceq Q_{\partial,L}^{(s)} \preceq \beta_s e^{c_{\text{hi}} + \varepsilon_k} I_d, \quad L = q + d - 1, q + d. \end{aligned} \quad (\text{OOQ34})$$

These are inequalities on every full original remainder coefficient vector.

Moreover

$$\varepsilon_k = O_{\delta,\gamma}(q^{3/4} + k \log q) = o(q). \quad (\text{OOQ35})$$

Indeed OOQ24 gives the first term $q^{3/4}$ because $n\rho^2 = n^{3/4}$; all remainder and diagonal-coordinate costs are $O(d \log n)$. OOQ28 gives the low error. The original OSP constant $\mathcal{E}_k$ is bounded at fixed δ, γ on its explicit domain, and $r_{\ell_s} = O(1/k)$ by OFV6–7. The sine bounds in OFV12 give logarithmic cost $O(\log q)$. Finally $Q_s - q = -d + \ell_s = O(k)$, the rates differ from ϑ by $1/q$, and W and its derivatives are bounded. The mean-value formula in the displayed scalar definitions therefore bounds both scalar discrepancies in OOQ33 by $O(k \log q + \log q)$. Every original mass remains in the common factor β_s .

40.7 Relative operators and all fixed subquotients of this actual target

The exact leading scalar difference is

$$c_{\text{lo}}(k) - c_{\text{hi}}(k) = q\{2 \log(4/\vartheta) - 2 + W(2, \vartheta)\} = \frac{q\mathfrak{C}_{\partial}}{2}. \quad (\text{OOQ36})$$

For the second equality use OOQ5 at $\alpha = 2$ and precisely OFV29, $\mathfrak{C}_{\partial} = 4 \log(4/\vartheta) - 10 - 2F(2, \vartheta) + 4F_{\alpha}(2, \vartheta)$. Equivalently it is the full OFV31 energy expression $4 \log(4/\pi) + 2 - 2I_{2,\pi}$. Thus this constant was obtained from the operator calculation with the same original source scale.

Let

$$\begin{aligned}\mathcal{R}_0 &= (Q_{\partial, q+d-1}^{(s)})^{-1/2} Q_{\partial, d-1}^{(s)} (Q_{\partial, q+d-1}^{(s)})^{-1/2}, \\ \mathcal{R}_1 &= (Q_{\partial, q+d}^{(s)})^{-1/2} Q_{\partial, d}^{(s)} (Q_{\partial, q+d}^{(s)})^{-1/2}.\end{aligned}\tag{OOQ37}$$

The positive square roots are the unique positive ones. For every eigenvalue of either of these actual relative operators, OOQ34 proves

$$\left| \log \lambda - \frac{q\mathfrak{C}_\partial}{2} \right| \leq 2\varepsilon_k = o(q).\tag{OOQ38}$$

This follows by positive congruence of the two scalar inequalities; it does not infer eigenvalues from their product. The positive operator $\mathcal{R}_0^{1/2} \mathcal{R}_1 \mathcal{R}_0^{1/2}$ has all eigenvalue logarithms within $4\varepsilon_k$ of $q\mathfrak{C}_\partial$, and its determinant is exactly the original four-endpoint determinant ratio. No commutation of the two relative operators is used or asserted.

For precision, the same result controls every specified fixed restriction or quotient of this outer target, even when its coefficient map depends on the original period and on k . For an injective map $I : \mathbb{C}^t \rightarrow \mathbb{C}^d$, use the actual restricted Grams $I^* Q_{\partial, L}^{(s)} I$. For a surjective map $A : \mathbb{C}^d \rightarrow \mathbb{C}^t$, use their attained minimum quotient Grams

$$Q_{A, L}^{(s)} = (A(Q_{\partial, L}^{(s)})^{-1} A^*)^{-1}.\tag{OOQ39}$$

The restriction coefficient reference is $I^* I$ and the quotient coefficient reference is $(AA^*)^{-1}$. Congruence, or minimization on the identical fibres, transports OOQ34 with exactly those reference forms. Taking determinants and adding the four original signs cancels the complete fixed reference determinant and the factor β_s^t . Hence in both cases the exact four-endpoint return $F_t^{(s)}$ satisfies

$$|F_t^{(s)} - tq\mathfrak{C}_\partial| \leq 4t\varepsilon_k.\tag{OOQ40}$$

For $t = O(k)$ this error is $o(kq)$, without any assumed conditioning of I or A : each is the same actual map at all four endpoints, so its complete coefficient form has been retained and cancels exactly.

The operator statements OOQ34 and OOQ38 concern the complete CTV outer-divisor quotient, whose ideal is $P_{\partial, k} \mathbb{C}[y]_{\leq L-d}$. The current original residual has the different, explicitly supplied CEP source graph, containing its complete $\mathcal{U}_{A, L}$ columns, the low- v components, and all original K lifts. OOQ39–40 do not identify that graph with this outer ideal. The present calculation supplies the full original outer operator and its exact subquotient transport; the comparison with the actual residual must use the complete CEP map and its denominator. No value of the separate original $\Phi_{k, s}$ is assigned by omitting that map.

41 The complete residual map on the original conductor graph

Retain every original coefficient and metric of CEP1–49 and FMC1–38. In particular $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = q - q'$, $j = 8k - 32$, $r = 8k - 16$, and the original first nonzero conductor moment is $0 \leq v \leq 80$. For each source $s \in \{1, k\}$ and endpoint $N \in \{q - 1, q, 2q - 1, 2q\}$ put $t = N - q + 1$ and $d = N - v$. All coefficient maps below retain the actual period matrix and the entire conductor $A = \sum_{a,b=0}^8 a_{ab} e_{ab}^{(8)}$.

41.1 The full graph denominator and its metric

Use original orthonormal coordinates on both source polynomial spaces, and define the coefficient space and injective multiplication map

$$X_N = \mathbb{C}^v \oplus \mathbb{C}[S']_{\leq N-v-q'}, \quad D_N = \text{diag}(I_v, M_{\chi_{k'}}), \quad \dim X_N = t + \Delta. \quad (\text{GIP1})$$

The second summand of X_N uses the increasing ordinary monomials in S' ; $M_{\chi_{k'}}$ includes their exact conversion to the original lower orthonormal source coordinates. The first summand is the original upper source's first v orthonormal coordinates. Thus D_N identifies X_N with $\mathbb{C}^v \oplus I'_N$, where $I'_N = \chi_{k'} \mathbb{C}[S']_{\leq N-v-q'}$. Let S_N be the complete invertible source map CEP28, and let U_N be the full upper source remainder in original primary coordinates CEP21. The specified map to the original conductor kernel and its residual observation are

$$\begin{aligned} E_N^{\text{graph}} &= U_N S_N^{-1} D_N : X_N \longrightarrow V_A, \\ B_N^{\text{graph}} &= E_j^\top T_k^\top S_k^\top E_N^{\text{graph}} : X_N \longrightarrow \mathbb{C}^j. \end{aligned} \quad (\text{GIP2})$$

Here T_k, S_k, E_j are the complete FMC and OCF maps, not new sections. CEP31 proves that E_N^{graph} is onto V_A , since $S_N^{-1} D_N X_N = U_N^{-1}(V_A)$. FMC30 proves that the second map is exactly $\Xi_k^{\text{fix}} E_N^{\text{graph}}$ and is onto.

For the t ordinary monomials h of degree at most $N - q$ let $Z_{\chi, N} h$ be the original upper orthonormal coefficients of $\chi_k h$. Define, with the complete products and shifts of CEP41–42,

$$\begin{aligned} \mathcal{Z}_{A, N} h &= \begin{pmatrix} P_{<v} Z_{\chi, N} h \\ \mathcal{U}_{A, N-q} h \end{pmatrix}, \\ (\mathcal{U}_{A, N-q} h)(S') &= \sum_{a,b=0}^8 a_{ab} d_{ab}(S') h(S' + b_{ab}), \quad D_N \mathcal{Z}_{A, N} = S_N Z_{\chi, N}. \end{aligned} \quad (\text{GIP3})$$

At $t = 0$ this is the empty column matrix. CEP42 proves the equality including the nonzero leading coefficient $\text{lc}(h) \binom{q+\deg h}{v} \mu_v$ for nonzero h . Consequently $\mathcal{Z}_{A, N}$ has rank t , and $\ker E_N^{\text{graph}} = \text{im } \mathcal{Z}_{A, N}$: a vector maps to zero precisely when its $S_N^{-1} D_N$ image is a full source relation $\chi_k h$.

The entire common kernel is also retained. Put $I_k = J_k E_r$ as in FMC30, and use its exact original minimum source lift $Y_{K, N} = U_N^* G_N I_k$. Its columns belong to $U_N^{-1}(V_A)$, and hence to the domain of the inverse of D_N on its indicated image after application of S_N . Define

$$W_{K, N} = D_N^{-1} S_N Y_{K, N}, \quad \ker B_N^{\text{graph}} = \text{im}[\mathcal{Z}_{A, N}, W_{K, N}], \quad \text{rank}[\mathcal{Z}_{A, N}, W_{K, N}] = t + r. \quad (\text{GIP4})$$

Indeed $E_N^{\text{graph}} W_{K, N} = I_k$ because $U_N U_N^* = G_N^{-1}$. A relation among the two displayed sets, after applying E_N^{graph} , is therefore a relation among the independent columns of I_k , so its $W_{K, N}$ coefficients vanish. The graph columns are independent as proved above. Conversely a vector killed by B_N^{graph} has image in K and subtracting the corresponding $W_{K, N}$ combination leaves a graph relation. This proves both equality and rank, including every original low-degree graph component and every common-kernel column.

There are two precisely specified positive metrics on X_N :

$$H_N^{\text{actual}} = D_N^* S_N^{-*} S_N^{-1} D_N, \quad H_N^{\text{graph}} = D_N^* D_N = \text{diag}(I_v, M_{\chi_{k'}}^* M_{\chi_{k'}}). \quad (\text{GIP5})$$

The first is the exact pullback of the unchanged original source norm, and the second is the pullback of the original direct-sum target norm of CEP28. Positivity follows from injectivity of D_N and invertibility of S_N . In particular the second block, after the literal substitution $S' = c' + iy$, integrates $|f(c' + iy)|^2 |P_{k'}(y)|^2 dm_s(y)$: the factor $i^{q'}$ in $\chi_{k'}(c' + iy) = i^{q'} P_{k'}(y)$ has its complete modulus one. The coordinate substitution in f is retained in this statement.

Both residual quotient metrics consequently have full finite formulas

$$\begin{aligned} Q_{\Xi,N}^{\text{fix}} &= [B_N^{\text{graph}}(H_N^{\text{actual}})^{-1}(B_N^{\text{graph}})^*]^{-1}, \\ \tilde{Q}_{\Xi,N} &= [B_N^{\text{graph}}(H_N^{\text{graph}})^{-1}(B_N^{\text{graph}})^*]^{-1}, \quad |\log \det \tilde{Q}_{\Xi,N} - \log \det Q_{\Xi,N}^{\text{fix}}| \leq 2E_N^{\text{CEP}}. \end{aligned} \quad (\text{GIP6})$$

Here E_N^{CEP} is the entire constant CEP35. For either positive H and onto B , the vector $H^{-1}B^*(BH^{-1}B^*)^{-1}y$ maps to y and is H -orthogonal to $\ker B$. Adding any other lift adds a perpendicular kernel vector and proves the inverse-Gram formula. The first minimum is the original minimum over all upper source lifts of a primary vector with residual y , by GIP2 and GIP5; this proves its equality with FMC32. For the comparison, the actual source map S_N induces the identity of these two quotient coordinate targets. Minimal lifts followed by orthogonal projection off the entire kernel GIP4 have exterior norm at most $\|\bigwedge^j S_N\|$; the inverse has bound $\|\bigwedge^j S_N^{-1}\|$. Orthogonal projection is an exterior contraction because its singular values are zero or one. Squaring the two volume inequalities and using CEP34–35 proves the final bound. Thus all four determinant errors sum to at most $2\sum_N E_N^{\text{CEP}} = o(kq)$ for the fixed actual period, without identifying any two different relation spaces.

41.2 The exact natural map to the CTV outer divisor

Let $\mathcal{I}_k = \{4, \dots, k-4\}^2$ and $\mathcal{O}_k = \{0, \dots, k\}^2 \setminus \mathcal{I}_k$. Let $E_{\text{in}}, E_{\text{out}}$ insert these coordinates into the original primary order. In the centred y coordinate retain the full outer Vandermonde $(V_{\text{out}})_{(a,b),h} = \omega_{ab}^h$ for $0 \leq h < \Delta$ and the diagonal matrix $D_{\text{in,out}} = \text{diag}(P_{k'}(\omega_{ab}))_{\mathcal{O}_k}$. Both are invertible: all the original roots are distinct and no outer root is an inner root. Define

$$\begin{aligned} \rho_A &= V_{\text{out}}^{-1} D_{\text{in,out}}^{-1} E_{\text{out}}^T|_{V_A} : V_A \longrightarrow \mathbb{C}[y]/(P_{\partial,k}), \\ C_{\text{in}} &= CE_{\text{in}}, \quad C_{\text{out}} = CE_{\text{out}}, \quad (C_{\text{in}})_{(a,b),(i,j)} = a_{i+4-a,j+4-b} \quad (0 \leq a, b, i, j \leq k'), \end{aligned} \quad (\text{GIP7})$$

where an out-of-range coefficient of A is zero. This is the literal map which first takes outer values and then divides by the inner root factor from CTV10; on the graph its matrix is $\rho_A E_N^{\text{graph}}$. A source relation $\chi_k h$ is killed because its outer evaluations vanish. Thus GIP7 is independent of every source lift and retains exactly the CTV remainder frame.

There is an exact sequence with an entirely explicit obstruction:

$$\begin{aligned} 0 \longrightarrow \ker C_{\text{in}} \xrightarrow{E_{\text{in}}} V_A \xrightarrow{\rho_A} \mathbb{C}^\Delta \xrightarrow{\partial_A} \text{coker } C_{\text{in}} \longrightarrow 0, \\ \partial_A f = [C_{\text{out}} D_{\text{in,out}} V_{\text{out}} f], \quad \dim \ker \rho_A = \dim \text{coker } \rho_A = \dim \ker C_{\text{in}}. \end{aligned} \quad (\text{GIP8})$$

To prove it, write $x = E_{\text{in}} z + E_{\text{out}} w$. Membership in V_A says exactly $C_{\text{in}} z + C_{\text{out}} w = 0$, while $\rho_A x = f$ says $w = D_{\text{in,out}} V_{\text{out}} f$. These two equations prove exactness at the first three terms. Surjectivity of C proves $\text{im } C_{\text{in}} + \text{im } C_{\text{out}} = \mathbb{C}^{q'}$, and the outer diagonal and Vandermonde are invertible, so ∂_A is onto. Since C_{in} is square its kernel and cokernel have equal dimension, proving the dimension assertion. Every possible defect is therefore the nullity of this specific actual finite convolution section. The scalar first-moment index v does not by itself evaluate that matrix or its nullity.

The original residual observation is retained even when this defect is nonzero. Put $W_A = \Xi_k^{\text{fix}} E_{\text{in}} \ker C_{\text{in}}$. Then the exact descended residual map and its kernel are

$$\bar{\Xi}_A : \ker \partial_A \longrightarrow \mathbb{C}^j / W_A, \quad \rho_A x \longmapsto [\Xi_k^{\text{fix}} x], \quad \ker \bar{\Xi}_A = \rho_A(K). \quad (\text{GIP9})$$

Two lifts differ by $E_{\text{in}} \ker C_{\text{in}}$ by GIP8, so the displayed rule is well defined. It is onto because Ξ_k^{fix} is onto. If its value is zero, subtract an element of $E_{\text{in}} \ker C_{\text{in}}$ with the same residual; the remaining lift lies in K and has unchanged outer image. This proves the kernel equality; its converse follows by direct application of Ξ_k^{fix} . Equations GIP3–9 thus specify the full denominator, retained metrics, and exact obstruction to using the natural outer map for the same residual classes.

42 The original mixed return and its same-class action control

This calculation retains the original arithmetic source, its actual cyclic inclusion, and the complete period observation. It computes the action control on those objects and the exact degree-step information carried by the signed mixed return. The original full-multiplicity control calculation is stated with its full multiplicity; the subsequent evaluated outer-return substitution uses precisely the proved $m = 1$ domain.

42.1 The original arithmetic class and all three source metrics

Fix the stipulated quartet $\rho = 1/2 + \delta + i\gamma$ and its reflected and conjugate centres, with $0 < \delta < 1/2$, $\gamma > 2$, and full common multiplicity $m \geq 1$. Retain

$$\begin{aligned} h(s) &= \prod_{\lambda \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (s - \lambda)^m, \quad g = 2\xi, \quad v_h = g/h, \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \\ k &= 4l + 1 \geq 9, \quad c = k/2, \quad e = 1 + k(m - 1), \quad q = e(k + 1)^2, \\ \beta_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \beta_{ab})^e, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S. \end{aligned} \tag{MCE1}$$

The three source measures, all in the literal coordinate $S = c + iy$, are the arithmetic measure $w_h^{*k}(y) dy$ and the original Gamma measures $dm_s(y)$ of CTV1 for $s = 1, k$. Their masses are respectively μ_h^k and $(2\pi)^{s/2}$. For a source $a \in \{\text{ar}, 1, k\}$ let $p_n^a(S)$ be its monic orthogonal polynomial, ω_n^a its full squared norm, and $b_n^a = [p_n^a]_\chi$ in the fixed original coefficient frame of E . For the Gamma sources these are explicitly

$$\begin{aligned} p_n^s(S) &= i^n \mathfrak{p}_n^s((S - c)/i), \quad \mathfrak{p}_0^s = 1, \quad \mathfrak{p}_1^s = y, \\ \mathfrak{p}_{n+1}^s &= y \mathfrak{p}_n^s - n(s/2 + n - 1) \mathfrak{p}_{n-1}^s, \quad \omega_n^s = (2\pi)^{s/2} n! (s/2)_n. \end{aligned} \tag{MCE2}$$

Thus the i^n converting FMC29's real-coordinate polynomial to the monic polynomial in S is retained. In primary coordinates for $m = 1$, its entries are exactly $i^n \mathfrak{p}_n^s(\omega_{ab})$. For higher multiplicity the entries are the complete derivatives at each centre, with the factorials of the original Hermite coordinate map; none is replaced by a value-only coordinate.

For every $N \geq q - 1$ define the complete covariance and minimum Gram

$$R_N^a = \sum_{n=0}^N b_n^a (b_n^a)^* / \omega_n^a, \quad G_N^a = (R_N^a)^{-1}, \quad V_N^a = \det G_N^a. \tag{MCE3}$$

The first q columns are the monic triangular remainder frame and are independent. Positivity of all norms therefore proves $R_N^a > 0$. In the orthogonal source coefficient frame the minimum section is $\text{diag}(\omega_0^a, \dots, \omega_N^a)^{-1} [b_0^a, \dots, b_N^a]^* G_N^a$. Multiplication verifies that it is a section and is orthogonal to the full relation kernel. The squared norm of every other section value adds the squared norm of a perpendicular relation. This proves MCE3's minimum interpretation with all masses intact.

The original arithmetic inclusion is

$$\eta : E \hookrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_\Sigma) 1, \quad v_h = j_h(v_h), \quad A_\Sigma = \sum_{i=1}^k M_{s_i}, \quad \eta A = A_\Sigma \eta. \tag{MCE4}$$

Its injectivity follows from the exact annihilator χ and the invertibility of the complete Taylor unit $v_h^{\otimes k}$. For clarity the local nilpotent sum has exact order e : its $(e - 1)$ -st power has coefficient $[k(m - 1)]! / [(m - 1)!]^k$ at the product of the top local powers, and every higher monomial has an exponent at least m . All pairs (a, b) occur among tensor choices and their centres are distinct. The least common multiple of these local annihilators is exactly MCE1. This proves the asserted annihilator

and injectivity. The commutation identity follows by polynomial multiplication and commutation with the retained unit. The supplied source identity $J^{(k)}\mathcal{V}_{h,k} = \eta \text{rem}_\chi$ therefore carries each calculation below to this same arithmetic class. On an original present support label the map is $(\lambda, x) \mapsto (\lambda, \eta x)$, and external absence retains its original image; no coefficient calculation changes that label.

42.2 The exact rank-two control from the original recurrence

For this subsection fix any one source and suppress its superscript. Positivity of the original measure gives real monic orthogonal polynomials $Q_n(y) = i^{-n}p_n(c + iy)$. Pairing yQ_n with lower degrees proves

$$Sp_n = p_{n+1} + (c + i\alpha_n)p_n - \frac{\omega_n}{\omega_{n-1}}p_{n-1}, \quad \alpha_n \in \mathbb{R}, \quad (\text{MCE5})$$

with the lower term absent at $n = 0$. Indeed degrees below $n - 1$ are orthogonal by self-adjoint multiplication by the real variable; the coefficient at Q_{n+1} is one by monicity, and pairing with Q_{n-1} gives ω_n/ω_{n-1} . For the Gamma measures $\alpha_n = 0$ and MCE5 is precisely MCE2 with its phases.

Reducing MCE5 modulo χ and inserting it into MCE3 gives

$$\begin{aligned} AR_N + R_N A^* - kR_N &= \frac{b_{N+1}b_N^* + b_N b_{N+1}^*}{\omega_N}, \\ D_N := A^*G_N + G_N A - kG_N &= \frac{G_N b_{N+1}b_N^* G_N + G_N b_N b_{N+1}^* G_N}{\omega_N}. \end{aligned} \quad (\text{MCE6})$$

For the first equality the diagonal coefficient is $c + i\alpha_n + c - i\alpha_n - k = 0$. At every internal adjacent pair the remaining coefficient is $\omega_n^{-1} - (\omega_{n+1}/\omega_n)\omega_{n+1}^{-1} = 0$. Only the two displayed top columns survive. Multiplication on both sides by G_N proves the second equality, including its signs.

The dagger-stable polynomial obeys $\bar{\chi}(k - S) = (-1)^q \chi(S)$, so $\hat{\chi}(y) = i^{-q} \chi(c + iy)$ is real monic. The exact substitution $[P(S)] \mapsto [P(c + iy)]$ transports G_N by its full inverse congruence; in that real remainder frame the covariance and Gram have real entries. Each column has phase i^n . Consequently

$$\begin{aligned} b_N^* G_N b_{N+1} &= i\omega_N \phi_N, \quad \phi_N \in \mathbb{R}, \\ \mathcal{H}_N &= G_N^{-1/2} D_N G_N^{-1/2} = \frac{yx^* + xy^*}{\omega_N}, \\ x &= G_N^{1/2} b_N, \quad y = G_N^{1/2} b_{N+1}. \end{aligned} \quad (\text{MCE7})$$

Thus $\mathcal{H}_N$ is Hermitian of rank at most two and trace zero. Its square has trace $2(\|x\|^2\|y\|^2 - |x^*y|^2)/\omega_N^2$, because $x^*y = i\omega_N \phi_N$. Its two possibly nonzero eigenvalues are therefore $\epsilon_N, -\epsilon_N$, where

$$\epsilon_N^2 = \frac{\|x\|^2\|y\|^2 - |x^*y|^2}{\omega_N^2} = \frac{\|x \wedge y\|^2}{\omega_N^2}. \quad (\text{MCE8})$$

The last equality is the determinant of the two-vector Gram, proved by expanding in an orthonormal basis. This defines the literal control allowance, including the case when the rank is zero.

The exterior/Gram identity in MCE8 is classical; a modern reference is Lyons [H3, §4, (4.1), p. 11], with the two vectors x, y in the original Hermitian metric.

42.3 The same nonzero arithmetic exterior class

Let E_+ be the full sum of the primary blocks with $a > k/2$, and put $t_+ = e(k+1)^2/2$. Its dimension is t_+ because k is odd. A completely specified primary basis is $e_\beta(S - \beta)^d$, $0 \leq d < e$, over those centres, where

$$\chi_\beta = \chi/(S - \beta)^e, \quad e_\beta = \chi_\beta \sum_{u=0}^{e-1} \frac{(1/\chi_\beta)^{(u)}(\beta)}{u!} (S - \beta)^u \pmod{\chi}. \quad (\text{MCE9})$$

The Taylor product is one modulo $(S - \beta)^e$ and vanishes at the other complete primary factors. This proves the primary idempotent property and the stated basis. Multiplication by S has diagonal β and the exact next-power nilpotent on each block. Thus E_+ is A -invariant, and its ordered top wedge ξ_+ is a nonzero eigenvector for the additive exterior action $A^{[t+]}$, with eigenvalue $\sum_{a>k/2,b} e\beta_{ab}$. The map $\bigwedge^{t+} \eta$ is injective: extend an independent set in its domain to a basis and use injectivity of η on the resulting independent images. Hence $\bigwedge^{t+} \eta(\xi_+) \neq 0$ is the same surviving arithmetic exterior class. MCE4 intertwines the additive exterior actions term by term, retaining all its unit and nilpotent coefficients.

The extremal eigenvalue-sum principle belongs to the classical lineage of Ky Fan [H5], explicitly credited in the inspected abstract of Michael L. Overton and Robert S. Womersley [H4]. The specific Hermitian rank-two case needed below is proved directly by the exterior eigenvalue calculation. No theorem number or independent full-text verification of those two papers is asserted here.

On $\bigwedge^t(E, G_N)$ the exact relative control is the additive exterior action of $\mathcal{H}_N$. In an orthonormal basis this follows by applying A to each wedge factor and pairing the result, so its eigenvalues are sums of t distinct eigenvalues of $\mathcal{H}_N$. For $1 \leq t \leq q-1$ they lie in $[-\epsilon_N, \epsilon_N]$, since the only nonzero entries in the original list are $\epsilon_N, -\epsilon_N$. Applying this bound to ξ_+ , or taking the trace of the control on its invariant primary space, proves

$$\mathcal{L}_{h,k} := 2\delta e(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor = 2\Re \operatorname{Tr}(A|_{E_+}) - kt_+ \leq \epsilon_N, \quad \mathcal{L}_{h,k}/q \geq \delta k/2 > 0. \quad (\text{MCE10})$$

The sum of the positive integers $2a - k$ is the displayed floor, by summing the positive odd or even progression; for the present odd k it is $(k+1)^2/4$. Multiplication by the $k+1$ imaginary centres and full length e gives the equality. The last bound uses $\lfloor (k+1)^2/4 \rfloor \geq k(k+1)/4$. The exterior rank here is t_+ , and this exact inclusion and action determine its relation to the arithmetic class; it is not assigned the dimension of the unrelated period kernel.

42.4 Complete kernel, boundary and mixed pieces of the control

Fix the original observation $\Lambda : E \rightarrow B$ in any one fixed coordinate frame of its actual image, and its actual kernel inclusion $I : \mathbb{C}^{r_K} \hookrightarrow E$. For $m = 1$ on the proved period locus, $r_K = 8k - 16$; in the present algebraic formulas it remains the actual rank for the stipulated multiplicity. Define at each source and cutoff

$$\begin{aligned} H_N &= I^* G_N I, \quad Q_N = (\Lambda R_N \Lambda^*)^{-1}, \quad L_N = R_N \Lambda^* Q_N, \quad K_N^{\text{coord}} = H_N^{-1} I^* G_N, \\ \Lambda L_N &= I_B, \quad I^* G_N L_N = 0, \quad I K_N^{\text{coord}} + L_N \Lambda = I_E, \\ F_N &= [I, L_N], \quad F_N^* G_N F_N = \operatorname{diag}(H_N, Q_N). \end{aligned} \quad (\text{MCE11})$$

The first section identity follows by multiplication. The second follows from $\Lambda I = 0$. The two summands in the third are the G_N -orthogonal projections onto K and its complement: the first fixes $I\mathbb{C}^{r_K}$; subtracting the second from any vector lies in K and has its stated kernel coordinate. This proves the third identity and the full Gram equality.

The complete original action in this frame is

$$F_N^{-1} A F_N = \begin{pmatrix} K_N^{\text{coord}} A I & K_N^{\text{coord}} A L_N \\ \Lambda A I & \Lambda A L_N \end{pmatrix} =: \begin{pmatrix} A_{KK} & A_{KB} \\ A_{BK} & A_{BB} \end{pmatrix}. \quad (\text{MCE12})$$

In particular $A_{BK} = \Lambda A I$ is the actual action defect; no invariant-kernel premise is used. The exact boundary relation is $\Lambda A - A_{BB} \Lambda = A_{BK} K_N^{\text{coord}}$, obtained by inserting the third identity in MCE11. It is the metric-section version of OPG22; replacing its fixed section by L_N changes both displayed blocks by precisely this identity. The three complete control blocks are

$$\begin{aligned} I^* D_N I &= A_{KK}^* H_N + H_N A_{KK} - k H_N, \\ L_N^* D_N L_N &= A_{BB}^* Q_N + Q_N A_{BB} - k Q_N, \\ I^* D_N L_N &= A_{BK}^* Q_N + H_N A_{KB}. \end{aligned} \quad (\text{MCE13})$$

Multiplication of MCE12 by the block Gram MCE11 proves each entry. For an actual eigenvector $z = Iu + L_N b$ of eigenvalue λ , the two action equations are $A_{KK}u + A_{KB}b = \lambda u$ and $A_{BK}u + A_{BB}b = \lambda b$. Consequently its literal control pairing is

$$(2\Re\lambda - k)(u^* H_N u + b^* Q_N b) = u^* I^* D_N I u + 2\Re(u^* I^* D_N L_N b) + b^* L_N^* D_N L_N b. \quad (\text{MCE14})$$

This follows either from the two action equations and MCE13 or directly from $z^* D_N z$; it preserves the entire mixed pairing. For generalized primary vectors the full block matrix MCE12, rather than only this eigenvector equality, remains the action.

The same splitting gives a quantitative exterior identity for the actual allowance itself. Put

$$\begin{aligned} x_K &= H_N^{-1/2} I^* G_N b_N, & x_B &= Q_N^{1/2} \Lambda b_N, \\ y_K &= H_N^{-1/2} I^* G_N b_{N+1}, & y_B &= Q_N^{1/2} \Lambda b_{N+1}, \\ a_K &= \|x_K\|^2, & a_B &= \|x_B\|^2, & d_K &= \|y_K\|^2, & d_B &= \|y_B\|^2, \\ z_K &= x_K^* y_K, & z_B &= x_B^* y_B. \end{aligned} \quad (\text{MCE15})$$

The map $G_N^{1/2} F_N \text{diag}(H_N, Q_N)^{-1/2}$ is unitary by MCE11. Its inverse sends MCE7's x, y to these respective ordered pairs. Therefore

$$\begin{aligned} z_K + z_B &= i\omega_N \phi_N, \\ \omega_N^2 \epsilon_N^2 &= \|x_K \wedge y_K\|^2 + \|x_B \wedge y_B\|^2 + \|x_K \otimes y_B - y_K \otimes x_B\|^2 \\ &= (a_K d_K - |z_K|^2) + (a_B d_B - |z_B|^2) + a_K d_B + a_B d_K - 2\Re(z_K \bar{z}_B). \end{aligned} \quad (\text{MCE16})$$

To prove the second line expand $x \wedge y$ in the orthogonal direct sum $\bigwedge^2 K \oplus (K \otimes B) \oplus \bigwedge^2 B$. Its three components are exactly those displayed, including the minus sign in the mixed tensor. Orthogonality adds their squared norms. Expansion of each two-vector Gram and of the mixed tensor inner product proves the last line. Each squared norm in the second line is nonnegative. The individual z_K, z_B need not be purely imaginary; their actual sum is the pairing in MCE7.

For completeness the same surviving exterior class also has a full mixed coordinate map. If I_+ is the matrix of MCE9's actual positive-primary basis, put $X_+ = K_N^{\text{coord}} I_+$ and $Y_+ = \Lambda I_+$. Then

$$I_+ = I X_+ + L_N Y_+, \quad G_{+,N} = X_+^* H_N X_+ + Y_+^* Q_N Y_+ > 0, \quad \mathcal{L}_{h,k} = \text{Tr}(G_{+,N}^{-1} I_+^* D_N I_+). \quad (\text{MCE17})$$

The first two equalities are MCE11. Invariance of E_+ says $A I_+ = I_+ A_+$ with the full primary matrix of MCE9. Thus $I_+^* D_N I_+ = A_+^* G_{+,N} + G_{+,N} A_+ - k G_{+,N}$; taking the trace after multiplication by $G_{+,N}^{-1}$ proves the last identity by MCE10. Its expansion using MCE13 retains $X_+^* I^* D_N L_N Y_+$ and its adjoint, as well as both diagonal blocks. The induced map on ξ_+ is the literal wedge expansion of the columns $I X_+ + L_N Y_+$, with the usual permutation signs when kernel factors are moved before boundary factors. This is an invertible mixed coordinate description of the same vector before applying the injective exterior arithmetic map MCE4.

42.5 Degree-step volumes determine the squared component sizes

Still fixing one original source, put for $N \geq q - 1$

$$\kappa_N = \log \frac{\det H_N}{\det H_{N+1}}, \quad \beta_N^B = \log \frac{\det Q_N}{\det Q_{N+1}}, \quad \tau_N = \log(V_N/V_{N+1}). \quad (\text{MCE18})$$

These are nonnegative. Inserting the next exact covariance column and multiplying verifies $G_{N+1} = G_N - G_N b_{N+1} b_{N+1}^* G_N / (\omega_{N+1} + b_{N+1}^* G_N b_{N+1})$. The determinant of $I + uv^*$ is $1 + v^* u$, proved by choosing a basis with first vector u or by two-block elimination. Applying this identity first to the boundary covariance and then to the restricted kernel metric proves

$$\begin{aligned} e^{\beta_N^B} &= 1 + d_B / \omega_{N+1}, & e^{\kappa_N} &= 1 + d_K / (\omega_{N+1} + d_B), & \tau_N &= \kappa_N + \beta_N^B, \\ d_B / \omega_{N+1} &= e^{\beta_N^B} - 1, & d_K / \omega_{N+1} &= e^{\beta_N^B} (e^{\kappa_N} - 1). \end{aligned} \quad (\text{MCE19})$$

All denominators are strictly positive. For the boundary step,

$$Q_{N+1}^{-1} = Q_N^{-1} + (\Lambda b_{N+1})(\Lambda b_{N+1})^*/\omega_{N+1}.$$

For the kernel step the determinant ratio from the displayed G_{N+1} is $1 - d_K/(\omega_{N+1} + d_K + d_B)$, whose reciprocal is the stated formula. Multiplying the two reciprocal ratios gives the complete covariance ratio and proves the third equality.

For $N \geq q$ the preceding total and boundary covariances are both positive definite. Subtracting their last column instead of adding the next gives

$$a_B/\omega_N = 1 - e^{-\beta_{N-1}^B}, \quad a_K/\omega_N = e^{-\beta_{N-1}^B}(1 - e^{-\kappa_{N-1}}). \quad (\text{MCE20})$$

Indeed the determinant ratio of the previous boundary covariance to the current one is $1 - a_B/\omega_N = e^{-\beta_{N-1}^B}$. The identical total formula gives $1 - (a_K + a_B)/\omega_N = e^{-\tau_{N-1}}$. Subtracting and using MCE19 proves MCE20. At the first degree $N = q - 1$ the total pairing is exactly $a_K + a_B = \omega_{q-1}$, because the first q monic remainder columns are an orthogonal quotient basis. Its two components are kept as MCE15; no inverse at $q - 2$ is taken or assumed. Thus MCE16, MCE19 and MCE20 express the actual mixed exterior allowance using the complete consecutive kernel and boundary returns and their actual cross-pairings.

42.6 The signed return gives an exact original control-radius budget

Restore source superscripts. Define the actual signed one-degree term

$$\psi_N = \tau_N^{\text{ar}} - (\beta_N^B)^{(k)}, \quad w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \quad (q \leq N \leq 2q - 2), \quad \sum_N w_N = 2q. \quad (\text{MCE21})$$

Telescoping MCE18–19 gives, with every original endpoint,

$$\Psi_k := F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} = \sum_{N=q-1}^{2q-1} w_N \psi_N, \quad \tau_N^{\text{ar}} = \psi_N + (\beta_N^B)^{(k)}. \quad (\text{MCE22})$$

To see the weights, the first window from $q - 1$ to $2q - 1$ includes degrees $q - 1, \dots, 2q - 2$, while the second window from q to $2q$ includes $q, \dots, 2q - 1$. Their intersection is counted twice and their two extreme degrees once.

For each individual source, MCE7–8 and MCE19–20 give the literal radius formulas

$$\begin{aligned} (\epsilon_{q-1})^2 + (\phi_{q-1})^2 &= \frac{\omega_q}{\omega_{q-1}}(e^{\tau_{q-1}} - 1), \\ \epsilon_N^2 + \phi_N^2 &= \frac{\omega_{N+1}}{\omega_N}(1 - e^{-\tau_{N-1}})(e^{\tau_N} - 1), \quad N \geq q. \end{aligned} \quad (\text{MCE23})$$

The first uses exactly $\|x\|^2 = \omega_{q-1}$ and the next-column formula for $\|y\|^2$; the second uses both total formulas already proved. Subtracting $\phi_N^2 = |x^*y|^2/\omega_N^2$ recovers the actual ϵ_N^2 . In particular substitution of $\tau_N^{\text{ar}} = \psi_N + (\beta_N^B)^{(k)}$ in MCE23 is a pointwise, same-arithmetic-class map from the signed degree data to the original control. It retains the complete reference boundary increment and arithmetic phase rather than assigning them zero.

Put $\delta_j^{\text{ar}} = V_j^{\text{ar}}/V_{j-1}^{\text{ar}}$ for $j \geq q$. The positive lower bound MCE10 and MCE23 imply $0 < \delta_j^{\text{ar}} < 1$ at every index used next. Define the full arithmetic monic-norm window and contraction penalties

$$\begin{aligned} \mathcal{W}_k^{\text{ar}} &= \log \frac{\omega_{2q-1}^{\text{ar}} \omega_{2q}^{\text{ar}}}{\omega_{q-1}^{\text{ar}} \omega_q^{\text{ar}}}, \\ \mathcal{P}_- &= -\log(1 - \delta_{2q-1}^{\text{ar}}) - 2 \sum_{j=q}^{2q-2} \log(1 - \delta_j^{\text{ar}}), \\ \mathcal{P}_+ &= -\log(1 - \delta_q^{\text{ar}}) - \log(1 - \delta_{2q}^{\text{ar}}) - 2 \sum_{j=q+1}^{2q-1} \log(1 - \delta_j^{\text{ar}}). \end{aligned} \quad (\text{MCE24})$$

They are all finite, and the two penalties are nonnegative. Multiplying MCE23 across the first window gives

$$\prod_{N=q-1}^{2q-2} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_{2q-1} V_{q-1}}{\omega_{q-1} V_{2q-1}} (1 - \delta_{2q-1}) \prod_{j=q}^{2q-2} (1 - \delta_j)^2.$$

Across the second it gives

$$\prod_{N=q}^{2q-1} (\epsilon_N^2 + \phi_N^2) = \frac{\omega_{2q} V_q}{\omega_q V_{2q}} (1 - \delta_q)(1 - \delta_{2q}) \prod_{j=q+1}^{2q-1} (1 - \delta_j)^2.$$

For these two displays all quantities are arithmetic. To verify them, write $e^{\tau_N} - 1 = (1 - \delta_{N+1})/\delta_{N+1}$. The monic norm ratios telescope, the reciprocal δ factors give the indicated quotient-volume ratio, every internal $(1 - \delta_j)$ occurs twice, and each stated extreme occurs once. Taking logarithms and using MCE22 proves the exact bridge

$$\boxed{\sum_{N=q-1}^{2q-1} w_N \log((\epsilon_N^{\text{ar}})^2 + (\phi_N^{\text{ar}})^2) = \Psi_k + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+}. \quad (\text{MCE25})$$

It is an equality on the unchanged arithmetic packet. Every allowance on its left is precisely MCE6–10's control allowance.

MCE10 now supplies a concrete necessary inequality for those same data, retaining its full phase contribution:

$$\begin{aligned} \Psi_k + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+ &\geq 4q \log \mathcal{L}_{h,k} + \sum_N w_N \log \left(1 + \frac{(\phi_N^{\text{ar}})^2}{\mathcal{L}_{h,k}^2} \right), \\ \min_{q-1 \leq N \leq 2q-1} \epsilon_N^{\text{ar}} &\leq \exp \left(\frac{\Psi_k + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+}{4q} \right). \end{aligned} \quad (\text{MCE26})$$

For the first line apply $\epsilon_N^{\text{ar}} \geq \mathcal{L}_{h,k}$ to each positive radius, write $\log(\mathcal{L}_{h,k}^2 + \phi^2) = 2 \log \mathcal{L}_{h,k} + \log(1 + \phi^2/\mathcal{L}_{h,k}^2)$, and use $\sum w_N = 2q$. For the second, the minimum of the allowances is at most the weighted geometric mean of their larger radii; take half of the logarithmic mean in MCE25. This proves both inequalities without assuming any desired upper control estimate.

Finally specialize only the evaluated return substitution to the proved $m = 1$ domain of OAR/OVR/OFV. Retain its exact value $\Psi_k = \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \mathbf{e}_{k,k}$, where $-e_{k,k}^- \leq \mathbf{e}_{k,k} \leq e_{k,k}^+$ with the entire OAR2 constants. Then MCE25 becomes

$$\begin{aligned} \sum_N w_N \log((\epsilon_N^{\text{ar}})^2 + (\phi_N^{\text{ar}})^2) &= \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \mathbf{e}_{k,k} \\ &\quad + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+. \end{aligned} \quad (\text{MCE27})$$

Substituting the already evaluated OFV outer value $\mathcal{F}_{\partial,k}^{(k)} = \Delta q \mathcal{C}_\partial + \mathbf{r}_k$ retains exactly the same equality, with its proved remainder $\mathbf{r}_k = O(q \log q)$ and the actual residual $\Phi_{k,k}$. In particular its original arithmetic necessary inequality is

$$\begin{aligned} \Phi_{k,k} &\leq \mathcal{F}_{\partial,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^+ + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+ \\ &\quad - 4q \log \mathcal{L}_{h,k} - \sum_N w_N \log \left(1 + \frac{(\phi_N^{\text{ar}})^2}{\mathcal{L}_{h,k}^2} \right). \end{aligned} \quad (\text{MCE28})$$

This is MCE26 composed with the exact receiver and its upper error endpoint; no new positivity premise is introduced. It shows exactly which additional original boundary, norm, contraction and phase quantities accompany the mixed-return estimate in the control budget. The full pointwise control can instead be calculated by MCE15–16 and MCE19–23. In either form, the actual arithmetic action, its nonzero exterior class, all kernel-boundary defects and the original four endpoints remain explicit. No sign of Ψ_k alone has been substituted for a bound on ϵ_N^{ar} .

The full original action with its two contraction penalties. For the unchanged complete quartet multiplicity m and arithmetic source, retain $e_k = 1 + k(m - 1)$, $q = e_k(k + 1)^2$, and every original remainder volume and monic norm. CAI12–15 multiplies the exact MCE adjacent-radius identity over both original windows, giving

$$\begin{aligned} J_k^{\text{action}} &= \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \\ 0 \leq \mathcal{P}_- + \mathcal{P}_+ &\leq \Pi_{h,k} = (4q - 1) \log(1 + z_{h,k}), \\ z_{h,k} &= \frac{4C_{\text{mult}}(h)^2(4 + (\lfloor k/3 \rfloor - 2)/q)^2}{\delta^2(k + 1)^2}. \end{aligned} \tag{CAIR1}$$

Here the exact constant $C_{\text{mult}}(h)$ is CAI16–23’s constant from the original convolution blocks of orders 3, 4, 5. Their sum-dependent isometry and the full total-degree output face prove the original recurrence bound CAI25. Inserting it and the actual exterior lower bound $L_{h,k} = \delta q(k + 1)/2$ in MCE’s two radius identities gives $d_n \leq z_{h,k}/(1 + z_{h,k})$ throughout $q \leq n \leq 2q$. Thus $-\log(1 - d_n) \leq \log(1 + z_{h,k})$. The two original penalty coefficient sums are $2q - 1$ and $2q$, which proves CAIR1 with no missing extreme row. The total reflection pairing still vanishes, while the two kernel and boundary pairings satisfy $z_B = -z_K$. Their mixed exterior summand retains $2|z_K|^2$, as calculated in CAI9. The estimate is $O_h(e_k)$ for fixed complete multiplicity and therefore $O_h(1)$ in the simple-quartet case. The following stronger estimates, when used, retain CAIR1 as its independently proved finite predecessor.

42.7 Uniform contraction control in the original minimum allowance

Retain MCE1–4’s complete stipulated arithmetic quartet, multiplicity, source mass, quotient and period observation. At the original untilted source, total reflection gives $\phi_N^{\text{ar}} = 0$ and the exact MCE23/OMR18 identities, in the same notation as CAI12, are

$$\begin{aligned} d_n &= V_n/V_{n-1}, \quad b_n^{\text{rec}} = \omega_n/\omega_{n-1}, \\ \epsilon_{q-1}^2 &= b_q^{\text{rec}}(1 - d_q)/d_q, \quad \epsilon_N^2 = b_{N+1}^{\text{rec}}(1 - d_N)(1 - d_{N+1})/d_{N+1} \quad (N \geq q). \end{aligned} \tag{RXECT1}$$

These are the arithmetic minimum norms: adding the actual last covariance column gives $d_{N+1}^{-1} - 1$, removing it gives $1 - d_N$, and the first full remainder basis at $q - 1$ gives the endpoint factor one. No inverse at $q - 2$ enters. Reflection proves the zero total pairing; the original component relation $z_B = -z_K$ and its mixed term $2|z_K|^2$ remain.

Let $U = U_{h,k}^{\text{ctr}}$ be the full finite constant in ECT, including $A_k, c_h, \vartheta_h, k, q, R_k$ and all amplitude factors. On its original root guard $q \geq 7R_k$, ECT proves $0 < d_n \leq U$ for every $q \leq n \leq 2q$. Retain AMT’s full monic comparison $(a_k/D_n)g_n \leq \omega_n \leq A_k g_n$, with $D_n = [1 + 4(n + 21 + 4m)^2]^{21+4m}$ and $g_n/g_{n-1} = n(n - 1/2)$. Consequently

$$b_n^{\text{rec}} \geq \frac{a_k}{A_k D_n} n(n - 1/2) \geq b_{\text{low}} := \frac{a_k}{A_k D_{2q}} q(q - 1/2) \quad (q \leq n \leq 2q). \tag{RXECT2}$$

The last step uses the increasing denominator bound and the increasing positive numerator at the stated integer degrees. If the computed constant $U < 1$, RXECT1–2 now give

$$\min_{q-1 \leq N \leq 2q-1} \epsilon_N \geq \sqrt{b_{\text{low}}} \frac{1 - U}{\sqrt{U}}. \tag{RXECT3}$$

At an interior degree both factors $1 - d$ are at least $1 - U$ and $1/d_{N+1} \geq 1/U$. At $q - 1$, the one factor $1 - U$ is at least $(1 - U)^2$ since $0 < U < 1$, proving the same bound there. This proves the minimum estimate at every original degree, rather than deducing a minimum from the larger geometric mean.

For fixed h and complete multiplicity, the explicit ECT constant has $\log U = -c_* q + O_h(k + \log q)$, where $c_* = \log(50000/45927) > 0$. The actual AMT constants in RXECT2 give $\log b_{\text{low}} =$

$O_h(k + \log q)$, and $(k + \log q)/q \rightarrow 0$ because $q = [1 + k(m - 1)](k + 1)^2$. Taking logarithms of RXECT3, using $U \rightarrow 0$ and the original $L_{h,k} = \delta q(k + 1)/2$, therefore proves

$$\liminf_{k=4\ell+1 \rightarrow \infty} \frac{1}{q} \log \frac{\min_{q-1 \leq N \leq 2q-1} \epsilon_N}{L_{h,k}} \geq \frac{c_*}{2}. \quad (\text{RXECT4})$$

The root guard holds eventually since $R_k = k\sqrt{\delta^2 + \gamma^2}$ and $q/k \rightarrow \infty$. This conclusion concerns the displayed original canonical degree window and retains its stipulated quartet domain.

43 The next marked pairing as an original action and source derivative

Retain MCE1–28 with the full original observation and all source orders. Give the same source its existing real tilt $dm_{a,\theta}(y) = e^{\theta y} dm_a(y)$ for real $|\theta| < \pi/2$. In the arithmetic case $dm_a = w_h^{*k}(y)dy$. All polynomial moments and their derivatives exist on compact subintervals by the original exponential-moment bound. At $\theta = 0$ this is exactly the current source. The polynomial algebra, $A = M_S$, the arithmetic inclusion, and Λ are independent of θ .

In fixed monomial source coordinates let $\mathcal{R}_N(\theta)$ be the original minimum section of the quotient and let its source Gram be $\mathbf{H}_N(\theta)$. The identities $J_N \mathcal{R}_N = I$ and orthogonality to $\ker J_N$ imply

$$\partial_\theta G_N = \mathcal{R}_N^*(\partial_\theta \mathbf{H}_N) \mathcal{R}_N = \langle \mathcal{R}_N, y \mathcal{R}_N \rangle_{a,\theta}. \quad (\text{MCP1})$$

Indeed differentiating $J_N \mathcal{R}_N = I$ gives $J_N \mathcal{R}'_N = 0$. The two section-derivative terms in the derivative of $\mathcal{R}_N^* \mathbf{H}_N \mathcal{R}_N$ therefore vanish by that orthogonality. Differentiating the integral defining $\mathbf{H}_N$ multiplies the integrand by the unchanged variable y , proving the second equality with full source mass.

Here is its exact endpoint evaluation. In the fixed real-coordinate quotient set $T = M_y$, $Q_n = i^{-n} p_n(c + iy)$ and $d_n = [Q_n]$. The leading coefficient of the minimum lift of x is $d_N^* G_N x / \omega_N$. This follows by expanding that lift in the orthogonal Q_n : its Q_N coefficient is its inner product with Q_N divided by ω_N , and the section identity gives that inner product $d_N^* G_N x$. The polynomial column

$$y \mathcal{R}_N - \mathcal{R}_N T - (Q_{N+1} - \mathcal{R}_N d_{N+1}) d_N^* G_N / \omega_N$$

has degree at most N because its highest coefficient cancels, and its remainder is zero. Pairing it with $\mathcal{R}_N$ kills it by the same orthogonality. The term Q_{N+1} is orthogonal to every degree-at-most- N minimum lift. Thus $G'_N = G_N T - G_N d_{N+1} d_N^* G_N / \omega_N$ in this specified real-coordinate frame. Returning through the unchanged substitution $S = c + iy$, with $b_n = i^n d_n$ and $T = (A - cI)/i$, proves

$$G'_N = -i G_N (A - cI) + i G_N b_{N+1} b_N^* G_N / \omega_N. \quad (\text{MCP2})$$

Every p_n, b_n, ω_n in this formula is evaluated at the same θ . No derivative of these columns was inserted: the formula comes from the exact section minimum and its polynomial remainder.

Use MCE11's H_N, Q_N, L_N at that same parameter, and keep the original kernel inclusion I fixed. Then $H'_N = I^* G'_N I$ and $Q'_N = L_N^* G'_N L_N$. For the latter, differentiating $\Lambda L_N = I_B$ puts L'_N in K , while $I^* G_N L_N = 0$ kills both differentiated-section terms. Taking the two traces in MCP2 therefore proves the exact marked-pairing identities

$$\begin{aligned} \frac{z_K}{\omega_N} &= \text{Tr } A_{KK} - c r_K - i \partial_\theta \log \det H_N, \\ \frac{z_B}{\omega_N} &= \text{Tr } A_{BB} - c \dim B - i \partial_\theta \log \det Q_N. \end{aligned} \quad (\text{MCP3})$$

For example the first trace of $G_N(A - cI)$ compressed by I is $\text{Tr } A_{KK} - c r_K$. The trace of its marked rank-one term is precisely z_K of MCE15, by cyclicity of trace. The second calculation uses $L_N^* G_N = Q_N \Lambda$ and gives exactly z_B . Rearranging $\partial_\theta \log \det H_N = -i(\text{Tr } A_{KK} - c r_K) + i z_K / \omega_N$ proves the first line and fixes its sign; the second line follows in the same manner. Adding them recovers the full phase MCE7 and EP.9 because $\text{Tr } A_{KK} + \text{Tr } A_{BB} = \text{Tr } A$.

In particular the original kernel–boundary action is measured by the real parts of these pairings:

$$\begin{aligned} 2\Re(z_K / \omega_N) &= 2\Re \text{Tr } A_{KK} - k r_K, & 2\Re(z_B / \omega_N) &= 2\Re \text{Tr } A_{BB} - k \dim B, \\ \frac{\|x_K \otimes y_B - y_K \otimes x_B\|^2}{\omega_N^2} &= \frac{a_K d_B + a_B d_K}{\omega_N^2} - 2\Re \left[(\alpha_K - i \partial_\theta \log \det H_N) \overline{(\alpha_B - i \partial_\theta \log \det Q_N)} \right], \\ \alpha_K &= \text{Tr } A_{KK} - c r_K, & \alpha_B &= \text{Tr } A_{BB} - c \dim B. \end{aligned} \quad (\text{MCP4})$$

The derivatives are real because the two Gram determinants are strictly positive for real θ . This proves the first line. Substituting MCP3 into MCE16's exact mixed tensor norm proves the second, with all phases and off-diagonal action effects retained. Its two size terms are the explicit degree-return quantities in MCE19–20. Thus the next original pointwise calculation is the marked pair z_K, z_B , equivalently these actual action traces and source derivatives. An outer quotient operator estimate must be transported through the full GIP graph and actual marked columns before it can bound these quantities. No derivative estimate is inferred by differentiating an uncontrolled remainder in a scalar asymptotic law.

44 The exact original reflection and the retained mixed phase

Retain the full original quartet, its multiplicity m , the complete unit $v_h = 2\xi/h$, every tensor factor and all source masses of MCP1–4. The argument concerns the original arithmetic source and both original Gamma orders $s = 1, k$. It also specifies the exact reflection map on the real tilted family of MCP1–4. No observation invariance is assumed.

44.1 The entire arithmetic unit and the source involution

The functional equation of the original completed function is $\xi(1-s) = \xi(s)$. Reflection permutes the four factors of the full quartet polynomial; each factor contributes its minus sign with its full multiplicity, giving

$$h(1-s) = (-1)^{4m}h(s) = h(s), \quad v_h(1-s) = v_h(s). \quad (\text{RFX1})$$

The second identity first holds off the finite zero set of h . Both sides are the entire quotient already supplied by the original arithmetic construction, so continuity at every removed centre proves the same identity there, including all Taylor coefficients. At $s = 1/2 + iy$ it implies

$$w_h(-y) = w_h(y), \quad w_h^{*k}(-y) = w_h^{*k}(y). \quad (\text{RFX2})$$

For the second equality, in the convolution integral substitute $y_i \mapsto -y_i$ in every integration variable and use the first equality in every factor. The integrals exist by the original finite-mass estimates; Tonelli's theorem applies to these nonnegative integrands. Each Gamma density in OOQ2 is also even, since $\overline{\Gamma(s/4 + iy/2)} = \Gamma(s/4 - iy/2)$ for the retained positive real source order. The full source constants remain equal on the two sides, rather than being divided out.

Let $c = k/2$ and let $\mathcal{P}_N$ consist of polynomials in the literal variable S . Define the complex-linear involution

$$(\mathcal{S}_N p)(S) = p(k-S), \quad (\mathcal{S}_N)_{ab} = (-1)^a \binom{b}{a} k^{b-a} \quad (0 \leq a \leq b \leq N), \quad (\text{RFX3})$$

and set every other matrix entry to zero. The displayed matrix is the binomial expansion in the original increasing monomial frame; substitution twice proves $\mathcal{S}_N^2 = I$. For each original source a and its real tilt $dm_{a,\theta} = e^{\theta y} dm_a$, direct substitution $y \mapsto -y$ proves

$$\|\mathcal{S}_N p\|_{a,-\theta}^2 = \|p\|_{a,\theta}^2, \quad |\theta| < \pi/2. \quad (\text{RFX4})$$

The full exponential-moment domain is the original one used in MCP; the polynomial factors and their derivatives remain integrable on its compact subintervals. Equation RFX4 therefore states an exact isometry between these two specified source spaces, including both full tilted source measures.

44.2 The quotient, primary jets and arithmetic class

The primary centres obey $k - \beta_{ab} = \beta_{k-a,k-b}$. The degree $q = e(k+1)^2$ is even for the original odd k . Consequently

$$\chi(k-S) = (-1)^q \chi(S) = \chi(S), \quad \mathcal{S}_N(\chi f) = \chi \mathcal{S}_{N-q} f. \quad (\text{RFX5})$$

The relation space is thus carried onto itself at every cutoff. There is an induced complex-linear involution Σ on $E = \mathbb{C}[S]/(\chi)$, with the literal identities

$$\text{rem}_\chi \mathcal{S}_N = \Sigma \text{rem}_\chi, \quad \Sigma^2 = I_E, \quad \Sigma A \Sigma = kI_E - A. \quad (\text{RFX6})$$

The first follows from RFX5, and the third follows by applying reflection to $Sp(S)$. In the degree-less-than- q frame, Σ is the matrix in RFX3 with $N = q - 1$. In original primary derivative coordinates its complete formula is

$$(\Sigma[P])^{(d)}(\beta_{ab}) = (-1)^d P^{(d)}(\beta_{k-a, k-b}), \quad 0 \leq d < e. \quad (\text{RFX7})$$

Dividing both sides by the same retained $d!$ gives the formula in divided-derivative coordinates; no factorial or top nilpotent layer is omitted. The matrix of this permutation with jet signs squares to the identity and is precisely the conjugate of RFX3 by the full original Hermite coordinate map.

The arithmetic inclusion also has an exact reflection map. On $E_h = \mathbb{C}[s]/(h)$ put $\sigma_h[f(s)] = [f(1-s)]$. RFX1 makes this a well-defined involution and fixes the full Taylor unit $v_h = j_h(v_h)$. Let $\sigma_h^{\otimes k}$ act on every tensor factor. Then

$$\sigma_h^{\otimes k} A_\Sigma \sigma_h^{\otimes k} = kI - A_\Sigma, \quad \sigma_h^{\otimes k} \eta = \eta \Sigma. \quad (\text{RFX8})$$

The first identity follows by summing $M_{1-s_i} = I - M_{s_i}$. For the second, apply that identity to $\eta[P] = v_h^{\otimes k} P(A_\Sigma)1$; the unit and 1 are fixed, giving exactly $v_h^{\otimes k} P(k - A_\Sigma)1$. Thus the same injective arithmetic map is used. On exterior powers the relation follows on every decomposable wedge and then by linearity. In particular the positive-primary exterior class is carried to the negative-primary class with all signs and nilpotent coordinates specified by RFX7, and its image remains nonzero by injectivity of $\bigwedge \eta$. Explicitly order the positive primary blocks lexicographically and their local powers by $d = 0, \dots, e-1$, and use the same convention on the negative side. With $B = (k+1)^2/2$ blocks, the resulting top-wedge sign is exactly

$$(-1)^{e^2 B(B-1)/2 + B e(e-1)/2}.$$

Reflection reverses the order of the B blocks, contributing e^2 transpositions per reversed pair, and contributes $\sum_{d=0}^{e-1} d = e(e-1)/2$ within-block sign exponents in each block by RFX7. This proves the displayed sign while retaining the same full unit in the arithmetic image. A present support label is retained under these maps; the original external absent element keeps its specified absent image. No amplitude cancellation changes a label.

44.3 The full metric and the total endpoint pairing

Give the quotient at cutoff $N \geq q - 1$ its original minimum norm, and denote its Gram by $G_N^a(\theta)$. Isometry RFX4 and relation identity RFX5 take the whole affine fibre of a class onto the fibre of its reflected class. Taking both attained minima therefore proves

$$\Sigma^* G_N^a(-\theta) \Sigma = G_N^a(\theta), \quad R_N^a(-\theta) = \Sigma R_N^a(\theta) \Sigma^*. \quad (\text{RFX9})$$

The inverse formula follows by taking inverses and using $\Sigma^{-1} = \Sigma$. The exact minimum sections satisfy $\mathcal{R}_N^a(-\theta) \Sigma = \mathcal{S}_N \mathcal{R}_N^a(\theta)$: the right side is a lift in the required reflected fibre and has the same minimum norm, so uniqueness identifies it with the left side.

At $\theta = 0$, the original real monic orthogonal polynomial $Q_n(y)$ has parity n . Indeed $(-1)^n Q_n(-y)$ is monic of degree n ; its pairing with any lower polynomial vanishes by evenness of the original measure and the substitution $y \mapsto -y$. Uniqueness of the monic orthogonal polynomial proves the claim. In the original S coordinate $p_n(S) = i^n Q_n((S-c)/i)$, so its remainder obeys

$$\Sigma b_n = (-1)^n b_n, \quad b_N^* G_N b_{N+1} = 0, \quad \phi_N = 0 \quad (N \geq q-1). \quad (\text{RFX10})$$

For the middle equality use RFX9 at zero to equate the pairing with $(\Sigma b_N)^* G_N (\Sigma b_{N+1})$. This is its negative by the first equality; over $\mathbb{C}$ the pairing must vanish. The last identity is the exact definition in MCE7, with $\omega_N > 0$. All three original sources and every original multiplicity are included. More generally their tilted monic polynomials obey $\mathcal{S}_n p_n^a(\theta) = (-1)^n p_n^a(-\theta)$ with

unchanged monic norms, by the same substitution and uniqueness. The vanishing claim RFX10 is made at the original source $\theta = 0$.

Since the determinant of the involution has squared modulus one, RFX9 also gives

$$\det G_N^a(-\theta) = \det G_N^a(\theta), \quad \partial_\theta \log \det G_N^a(\theta)|_{\theta=0} = 0. \quad (\text{RFX11})$$

Differentiability follows from the original exponential-moment bound and finite positive Gram inversion, as proved in MCP1. This is a proved derivative identity for the exact minimum metric; it does not differentiate an asymptotic error estimate.

44.4 The original mixed blocks remain present

Retain the original observation Λ , kernel inclusion I , and metric section L_N of MCE11. There is no assumption that Σ preserves $K = \ker \Lambda$. At the original source RFX10 implies, for the complete component pairings of MCE15,

$$z_B = -z_K, \quad \omega_N^2 \epsilon_N^2 = (a_K d_K - |z_K|^2) + (a_B d_B - |z_K|^2) + (a_K d_B + a_B d_K + 2|z_K|^2). \quad (\text{RFX12})$$

The first identity is $z_K + z_B = b_N^* G_N b_{N+1} = 0$. Inserting it into each of the three separate squared norms of MCE16 gives exactly the three displayed terms. Each term is the squared norm of its stated exterior component; the mixed one contains $+2|z_K|^2$. Thus RFX12 retains the full individual pairings even when they are nonzero. Adding the three exact terms yields $\omega_N^2 \epsilon_N^2 = (a_K + a_B)(d_K + d_B)$, consistently with MCE8 and RFX10.

Here is the exact relationship to the reflected observation, without requiring it to coincide with the original one. Define $\Lambda^\Sigma = \Lambda \Sigma$ and $I^\Sigma = \Sigma I$ with the same boundary and kernel coordinate spaces. Then $\ker \Lambda^\Sigma = \Sigma K$. At $\theta = 0$, RFX9 and MCE11 give

$$H_N^\Sigma = H_N, \quad Q_N^\Sigma = Q_N, \quad L_N^\Sigma = \Sigma L_N, \quad (K_N^{\text{coord}})^\Sigma = K_N^{\text{coord}} \Sigma. \quad (\text{RFX13})$$

For the section equality, use $R_N \Sigma^* = \Sigma R_N$, a consequence of RFX9, in the exact formula $L_N^\Sigma = R_N (\Lambda \Sigma)^* Q_N$. For the kernel coordinate equality use $\Sigma^* G_N = G_N \Sigma$. The two diagonal action blocks and both leakage blocks consequently satisfy

$$A_{KK}^\Sigma = kI_K - A_{KK}, \quad A_{BB}^\Sigma = kI_B - A_{BB}, \quad A_{KB}^\Sigma = -A_{KB}, \quad A_{BK}^\Sigma = -A_{BK}. \quad (\text{RFX14})$$

Insert RFX13 and $\Sigma A \Sigma = kI - A$ into the four exact definitions MCE12. The off-diagonal constant terms vanish because $K_N^{\text{coord}} L_N = 0$ and $\Lambda I = 0$. This proves all four relations while retaining the actual action defect $\Lambda A I$. The reflected component vectors are $(-1)^N(x_K, x_B)$ and $(-1)^{N+1}(y_K, y_B)$, by RFX10 and RFX13. Their squared sizes are unchanged and both marked pairings change sign, an exact description of their relation to the original mixed components.

In the original observation, put $g_K = \partial_\theta \log \det H_N|_0$ and $g_B = \partial_\theta \log \det Q_N|_0$. The complete metric splitting has endpoint frame $[I, L_N(\theta)]$. Its determinant is independent of θ , because $L_N(\theta) - L_N(0)$ lies in K and hence changes that frame by a triangular matrix with diagonal identity. Taking determinants in MCE11 and differentiating gives $g_K + g_B = 0$ by RFX11. Also the trace of A is cq , by summing all actual primary centres with multiplicity e . MCP3 therefore gives the additional exact formulas

$$\alpha_B = -\alpha_K, \quad g_B = -g_K, \quad \frac{z_K}{\omega_N} = \alpha_K - ig_K, \quad \frac{z_B}{\omega_N} = -\alpha_K + ig_K, \quad \alpha_K = \text{Tr } A_{KK} - cr_K. \quad (\text{RFX15})$$

They identify the original marked term to be controlled. Neither α_K nor g_K is assumed to vanish. In particular the mixed term in RFX12 is $a_K d_B + a_B d_K + 2\omega_N^2 |\alpha_K - ig_K|^2$.

44.5 Propagation through the same arithmetic control budget

Every allowance in the original arithmetic source is strictly positive by MCE10. Substitution of the proved RFX10 into MCE25 therefore gives the exact full-multiplicity equality

$$2 \sum_{N=q-1}^{2q-1} w_N \log \epsilon_N^{\text{ar}} = \Psi_k + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+. \quad (\text{RFX16})$$

Every endpoint weight, monic-norm window and contraction penalty is exactly MCE21–24. With $\sum_N w_N = 2q$, this proves

$$\begin{aligned} \mathcal{L}_{h,k} &\leq \min_N \epsilon_N^{\text{ar}} \leq \exp \left\{ \frac{\Psi_k + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+}{4q} \right\}, \\ 4q \log \mathcal{L}_{h,k} &\leq \Psi_k + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+. \end{aligned} \quad (\text{RFX17})$$

The exponential is now the exact weighted geometric mean of the allowances. A positive weighted geometric mean is at least the minimum, and each allowance is at least $\mathcal{L}_{h,k}$ by the same surviving arithmetic exterior class; these facts prove both inequalities. No new arithmetic upper bound is assumed.

On precisely the $m = 1$ evaluated-return domain of OAR/OVR/OFV, retain the actual signed error and substitute MCE27 to obtain

$$\begin{aligned} 2 \sum_N w_N \log \epsilon_N^{\text{ar}} &= \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \mathfrak{e}_{k,k} \\ &\quad + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \\ \Phi_{k,k} &\leq \mathcal{F}_{\partial,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^+ \\ &\quad + F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+ - 4q \log \mathcal{L}_{h,k}. \end{aligned} \quad (\text{RFX18})$$

The second line follows from the first, RFX17, and the original upper error endpoint $\mathfrak{e}_{k,k} \leq e_{k,k}^+$. The complete evaluated value $\mathcal{F}_{\partial,k}^{(k)} = \Delta q \mathfrak{C}_\partial + O_{\delta,\gamma}(q \log q)$ may be inserted with its proved error and original domain. The arithmetic boundary, norm window, contraction penalties and residual remain explicit. Equations RFX12–15 retain the individual mixed blocks required for further control; the total phase has been evaluated through a proved source, quotient and arithmetic involution, with the full data preserved.

45 The outer operator, the full residual graph, and original action control

This receiving calculation uses the complete OQ1–40, GIP1–9 and MCE1–28 proofs, with their original source maps and constants. The preceding MRI and BRI equations remain valid. Here we give the actual substitutions into those equations and into the control of the same arithmetic class.

45.1 Original objects, domains and the same-target operator interval

For the original quartet retain $0 < \delta < 1/2$, $\gamma > 2$, $m \geq 1$, $k = 4l + 1 \geq 9$, $e = 1 + k(m - 1)$, $c = k/2$ and

$$q = e(k + 1)^2,$$

$$\chi(S) = \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^e, \quad (\text{OMR1})$$

$$E = \mathbb{C}[S]/(\chi), \quad A = M_S : E \longrightarrow E.$$

The original source measures are $w_h^{*k}(y) dy$ and the two full Gamma measures of orders $s = 1, k$, evaluated at the unchanged coordinate $S = c + iy$. Their masses remain μ_h^k and $(2\pi)^{s/2}$. All action identities below retain this full multiplicity. The outer estimates and the residual graph statements are applied only on their proved simple-quartet domain $m = 1$, with $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = q - q' = 16k - 48$, $r = 8k - 16$ and $j = 8k - 32$. They retain the actual period and the full explicit finite domains of OSP23, OQ13 and OAR1–3. No eventual bound is assigned to an input outside its stated domain.

Write $\varepsilon_k^{\text{out}}$ for exactly OQ33's finite ε_k , with all OQ4–33 constants, the full root polynomials and both source orders. It is distinct from the action radius $\varepsilon_N^{\text{ar}}$ below and from E_N^{CEP} , the conductor exterior error. Put

$$\mathfrak{C}_\partial = 4 \log(4/\pi) + 2 - 2I_{2,\pi},$$

$$I_{2,\pi} = \iint \log|x - y| d\mu_{2,\pi}(x) d\mu_{2,\pi}(y), \quad (\text{OMR2})$$

$$D_k^{\text{out}} = 4\Delta\varepsilon_k^{\text{out}}, \quad \mathcal{O}_k^- = \Delta q \mathfrak{C}_\partial - D_k^{\text{out}}, \quad \mathcal{O}_k^+ = \Delta q \mathfrak{C}_\partial + D_k^{\text{out}}.$$

The measure $\mu_{2,\pi}$ is the complete EIQ equilibrium with its original mass one, endpoints and density, as retained in OFV and OVR. This is the same constant in MRI4r–u and BRI6h–i.

For clarity write $Q_{\text{lo},0}^{(s)} = Q_{\partial,\Delta-1}^{(s)}$, $Q_{\text{lo},1}^{(s)} = Q_{\partial,\Delta}^{(s)}$, $Q_{\text{hi},0}^{(s)} = Q_{\partial,q+\Delta-1}^{(s)}$ and $Q_{\text{hi},1}^{(s)} = Q_{\partial,q+\Delta}^{(s)}$. Every one acts on the original coefficient remainder frame $1, y, \dots, y^{\Delta-1}$. OQ34 gives, for $a = 0, 1$,

$$\beta_s e^{c_{\text{lo}} - \varepsilon_k^{\text{out}}} I_\Delta \preceq Q_{\text{lo},a}^{(s)} \preceq \beta_s e^{c_{\text{lo}} + \varepsilon_k^{\text{out}}} I_\Delta,$$

$$\beta_s e^{c_{\text{hi}} - \varepsilon_k^{\text{out}}} I_\Delta \preceq Q_{\text{hi},a}^{(s)} \preceq \beta_s e^{c_{\text{hi}} + \varepsilon_k^{\text{out}}} I_\Delta, \quad (\text{OMR3})$$

$$\beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}, \quad c_{\text{lo}} - c_{\text{hi}} = q\mathfrak{C}_\partial/2.$$

All original masses are retained in β_s . Inverting the high inequalities and conjugating the low inequalities by their positive square roots proves that every eigenvalue of $\mathcal{R}_a = (Q_{\text{hi},a}^{(s)})^{-1/2} Q_{\text{lo},a}^{(s)} (Q_{\text{hi},a}^{(s)})^{-1/2}$ lies between $e^{q\mathfrak{C}_\partial/2 - 2\varepsilon_k^{\text{out}}}$ and $e^{q\mathfrak{C}_\partial/2 + 2\varepsilon_k^{\text{out}}}$. Conjugating the bounds on $\mathcal{R}_1$ by $\mathcal{R}_0^{1/2}$ then proves the corresponding bounds on $\mathcal{R}_0^{1/2} \mathcal{R}_1 \mathcal{R}_0^{1/2}$, without any commutation assumption. Its determinant is exactly the product of the two original low/high determinant ratios. Therefore

$$\mathcal{O}_k^- \leq \mathcal{F}_{\partial,k}^{(s)} \leq \mathcal{O}_k^+, \quad s = 1, k. \quad (\text{OMR4})$$

The error is $O_{\delta,\gamma}(\Delta q^{3/4} + \Delta k \log q) = o(kq)$ by OOQ35. The earlier OFV estimate with error $O(q \log q)$ also remains valid; OMR4 supplies a full operator origin for an additional interval, and does not assert that this scalar interval is tighter than that earlier bound at every finite input.

More generally let $I : \mathbb{C}^t \hookrightarrow \mathbb{C}^\Delta$ be a fixed injection, or let $C_0 : \mathbb{C}^\Delta \twoheadrightarrow \mathbb{C}^t$ be a fixed surjection. Here “fixed” means the same actual map at all four endpoints; dependence on k and the period is allowed. The restriction and quotient forms are exactly

$$\begin{aligned} Q_I &= I^* Q I, & Q_I^{\text{ref}} &= I^* I, \\ Q_{C_0} &= (C_0 Q^{-1} C_0^*)^{-1}, & Q_{C_0}^{\text{ref}} &= (C_0 C_0^*)^{-1}, \\ |\mathcal{F}_t^{(s)} - tq\mathfrak{C}_\partial| &\leq 4t\varepsilon_k^{\text{out}}. \end{aligned} \tag{OMR5}$$

Restriction of a positive-form inequality preserves it. For a quotient, inversion reverses the order, multiplication by the onto map preserves it, and the second inversion reverses it again. Thus the same scalar bounds hold relative to the displayed exact reference form. In the four determinants its logarithm has coefficient $1 + 1 - 1 - 1 = 0$; the complete frame cost cancels. The t eigenvalue bounds now prove the last inequality.

45.2 The full graph denominator and its exact outer morphism

We now give the actual map relating OMR5’s target to the original residual. This retains the full conductor A_{cond} of CEP, which is a different named map from $A = M_S$ in OMR1. For each original $N = q - 1, q, 2q - 1, 2q$ let $t = N - q + 1$, retain the first nonzero conductor moment index $0 \leq v \leq 80$, and define

$$\begin{aligned} X_N &= \mathbb{C}^v \oplus \mathbb{C}[S']_{\leq N-v-q'}, & D_N &= \text{diag}(I_v, M_{\chi_{k-8}}), & \dim X_N &= t + \Delta, \\ E_N^{\text{graph}} &= U_N S_N^{-1} D_N : X_N \rightarrow V_A, \\ B_N^{\text{graph}} &= \Xi_k^{\text{fix}} E_N^{\text{graph}} = E_j^T T_k^T S_k^T U_N S_N^{-1} D_N : X_N \rightarrow \mathbb{C}^j. \end{aligned} \tag{OMR6}$$

These are the full GIP1–2 maps: U_N is the original source remainder in primary coordinates and S_N is CEP28’s complete source map. The multiplication block of D_N retains its exact conversion from ordinary lower monomials to orthonormal source coordinates. Let $Z_{\chi,N}h$ be the original upper coefficients of $\chi_k h$. With every shift and root product of CEP41–42,

$$\mathcal{Z}_{A,N}h = \left(\sum_{a,b=0}^8 a_{ab} d_{ab}(S') h(S' + b_{ab}) \right), \quad D_N \mathcal{Z}_{A,N} = S_N Z_{\chi,N}, \tag{OMR7}$$

$$\begin{aligned} I_k &= J_k E_r, & Y_{K,N} &= U_N^* G_N I_k, & W_{K,N} &= D_N^{-1} S_N Y_{K,N}, \\ \ker E_N^{\text{graph}} &= \text{im } \mathcal{Z}_{A,N}, & \ker B_N^{\text{graph}} &= \text{im}[\mathcal{Z}_{A,N}, W_{K,N}], & \text{rank}[\mathcal{Z}_{A,N}, W_{K,N}] &= t + r. \end{aligned}$$

The graph columns are independent because for every nonzero h their complete leading coefficient is $\binom{q+\deg h}{v} \mu_v \text{lc}(h) \neq 0$; for the specified monomial columns $\text{lc}(h) = 1$. Applying E_N^{graph} to the last r columns gives I_k , since $U_N U_N^* = G_N^{-1}$. Thus a relation among both sets first has zero coefficients in $W_{K,N}$ and then in $\mathcal{Z}_{A,N}$. Conversely, subtracting the corresponding $W_{K,N}$ lift from any element of $\ker B_N^{\text{graph}}$ leaves a source relation. This proves the complete kernel formula, including the $t = 0$ case where the source-relation matrix has no columns.

The two actual positive metrics and their residual quotient are

$$\begin{aligned} H_N^{\text{actual}} &= D_N^* S_N^{-*} S_N^{-1} D_N, & H_N^{\text{graph}} &= D_N^* D_N, \\ Q_{\Xi,N}^{\text{fix}} &= [B_N^{\text{graph}} (H_N^{\text{actual}})^{-1} (B_N^{\text{graph}})^*]^{-1}, \\ \tilde{Q}_{\Xi,N} &= [B_N^{\text{graph}} (H_N^{\text{graph}})^{-1} (B_N^{\text{graph}})^*]^{-1}, \\ |\log \det \tilde{Q}_{\Xi,N} - \log \det Q_{\Xi,N}^{\text{fix}}| &\leq 2E_N^{\text{CEP}}. \end{aligned} \tag{OMR8}$$

For any onto B and positive H , the vector $H^{-1}B^*(BH^{-1}B^*)^{-1}y$ is the unique lift of y orthogonal to $\ker B$; every other lift adds an orthogonal kernel vector. Its squared norm proves the displayed quotient formula. The first metric is the exact pullback of the original upper source norm, so this is precisely MRI4v's Schur complement. The second metric retains the full low- v summand and the source $|P_{k-8}(y)|^2 dm_s(y)$ after the unchanged lower substitution. The full j th exterior bounds of S_N and S_N^{-1} , followed by orthogonal projection off OMR7's entire kernel, give the last inequality as in GIP6. Their four endpoint costs total at most $2 \sum_N E_N^{\text{CEP}} = o(kq)$ for the fixed actual period.

Let $E_{\text{in}}, E_{\text{out}}$ insert the original inner square $\{4, \dots, k-4\}^2$ and its complement in primary order. Retain the full outer Vandermonde V_{out} and the diagonal $D_{\text{in,out}} = \text{diag}(P_{k-8}(\omega_{ab}))_{\text{out}}$. The exact natural morphism is

$$\begin{aligned} \rho_A &= V_{\text{out}}^{-1} D_{\text{in,out}}^{-1} E_{\text{out}}^T|_{V_A}, \\ C_{\text{in}} &= C E_{\text{in}}, \quad C_{\text{out}} = C E_{\text{out}}, \quad (C_{\text{in}})_{(a,b),(i,j)} = a_{i+4-a,j+4-b}, \\ 0 &\longrightarrow \ker C_{\text{in}} \xrightarrow{E_{\text{in}}} V_A \xrightarrow{\rho_A} \mathbb{C}^\Delta \xrightarrow{\partial_A} \text{coker } C_{\text{in}} \longrightarrow 0, \\ \partial_A f &= [C_{\text{out}} D_{\text{in,out}} V_{\text{out}} f], \quad \dim \ker \rho_A = \dim \text{coker } \rho_A = \dim \ker C_{\text{in}}. \end{aligned} \tag{OMR9}$$

Indeed, writing a primary vector as $E_{\text{in}}z + E_{\text{out}}w$, its conductor equation is $C_{\text{in}}z + C_{\text{out}}w = 0$ and its outer image f specifies $w = D_{\text{in,out}} V_{\text{out}} f$. These two equations prove exactness; the surjectivity of C proves surjectivity onto the cokernel. The inner block is square, so its kernel and cokernel dimensions agree. On the original graph the map is exactly $\rho_A E_N^{\text{graph}}$.

The residual descends with its full possible defect as

$$W_A = \Xi_k^{\text{fix}} E_{\text{in}} \ker C_{\text{in}}, \quad \bar{\Xi}_A : \ker \partial_A \rightarrow \mathbb{C}^j / W_A, \quad \rho_A x \mapsto [\Xi_k^{\text{fix}} x], \quad \ker \bar{\Xi}_A = \rho_A(K). \tag{OMR10}$$

Two lifts differ by $E_{\text{in}} \ker C_{\text{in}}$, which proves well-definedness. Surjectivity follows from that of Ξ_k^{fix} . If a class has zero residual modulo W_A , subtract a vector in $E_{\text{in}} \ker C_{\text{in}}$ with the same residual. Its remaining lift lies in K and has the same outer image, proving the kernel identity. Equations OMR6–10 are the exact typed bridge used here. They retain the entire denominator and metrics. Consequently the fixed-subquotient bound OMR5 is applied only to a specified map of its own full outer target; this receiving calculation assigns no leading coefficient to $\Phi_{k,s}$ by replacing the actual graph denominator with an outer polynomial ideal.

45.3 Backward propagation through the original kernel and signed return

Keep the actual nonnegative residual sum $\Phi_{k,s}$ in MRI4l, its first increment λ_s , and its full earlier cap V_s . Write $[L_s^0, U_s^0]$ and $[L_{\text{ar}}^0, U_{\text{ar}}^0]$ for exactly MRI4q's existing intervals. On the common proved domain, the full operator estimate gives the additional same-scalar intervals

$$\begin{aligned} L_s^{\text{op}} &= \max\{L_s^0, \mathcal{O}_k^- - \Phi_{k,s} - e_{k,s}^-\}, \\ U_s^{\text{op}} &= \min\{U_s^0, \mathcal{O}_k^+ - \Phi_{k,s} + e_{k,s}^+\}, \\ L_s^{\text{op}} &\leq F_K^{(s)} \leq U_s^{\text{op}}, \quad s = 1, k. \end{aligned} \tag{OMR11}$$

To prove this, substitute OMR4 into the exact MRI4m equality $F_K^{(s)} = \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s} + \mathfrak{e}_{k,s}$, with $-e_{k,s}^- \leq \mathfrak{e}_{k,s} \leq e_{k,s}^+$. Intersecting with the previously proved interval gives OMR11. The intersection is nonempty because the same original scalar belongs to both intervals. All earlier branches remain available; outside the new domain one uses their original endpoints.

Define the unchanged correlated arithmetic allowances by

$$\begin{aligned} a_{K,k}^- &= \max\{-C_K^-, \delta_k^\sigma - C_B^+\}, \quad a_{K,k}^+ = \min\{C_K^+, \delta_k^\sigma + C_B^-\}, \\ a_{K,k}^- &\leq d_k^{\text{ar}} = F_K^{\text{ar}} - F_K^{(1)} \leq a_{K,k}^+, \\ L_{\text{ar}}^{\text{op}} &= \max\{L_{\text{ar}}^0, L_1^{\text{op}} + a_{K,k}^-\}, \\ U_{\text{ar}}^{\text{op}} &= \min\{U_{\text{ar}}^0, U_1^{\text{op}} + a_{K,k}^+\}, \\ F_K^{\text{ar}} &\in [L_{\text{ar}}^{\text{op}}, U_{\text{ar}}^{\text{op}}], \quad F_B^{\text{ar}} \in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^{\text{op}}, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^{\text{op}}]. \end{aligned} \tag{OMR12}$$

The first bounds are exactly OAR8/MRI4o, with their original constants. Adding them to the source-1 inequalities proves the kernel interval. Subtraction from the same arithmetic total proves the boundary interval and its reversed endpoints. Their errors remain correlated through this equality.

The earlier signed scalar is exactly MCE22's Ψ_k :

$$\begin{aligned}\Psi_k &= \mathfrak{S}_k^{\text{mix}} = F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} = F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, \\ L_k^{\text{op}} + \mathcal{H}_k + \delta_k^\sigma &\leq \Psi_k \leq U_k^{\text{op}} + \mathcal{H}_k + \delta_k^\sigma.\end{aligned}\tag{OMR13}$$

The first equality uses the original total identity BRI6d, including $\mathcal{H}_k = -T_k$. The second line adds this exact signed correction to OMR11 at the original source order $s = k$. Every earlier finite bound for this same Ψ_k may also be intersected here. No arithmetic allowance or Gamma sign is changed.

45.4 The original action, its mixed components, and its positive class

For any of the actual sources let p_N be the monic polynomial in S , ω_N its full squared source norm, and $b_N = [p_N]_\chi$. In the original coefficient coordinates put

$$\begin{aligned}R_N &= \sum_{n=0}^N b_n b_n^* / \omega_n, \quad G_N = R_N^{-1}, \quad \mathcal{D}_N = A^* G_N + G_N A - k G_N, \\ x_N &= G_N^{1/2} b_N, \quad y_N = G_N^{1/2} b_{N+1}, \quad x_N^* y_N = i \omega_N \phi_N, \quad \phi_N \in \mathbb{R}, \\ G_N^{-1/2} \mathcal{D}_N G_N^{-1/2} &= (y_N x_N^* + x_N y_N^*) / \omega_N, \\ \text{spec}(G_N^{-1/2} \mathcal{D}_N G_N^{-1/2}) &= \{\epsilon_N, -\epsilon_N, 0, \dots, 0\}, \quad \epsilon_N^2 = \frac{\|x_N \wedge y_N\|^2}{\omega_N^2}.\end{aligned}\tag{OMR14}$$

The monic recurrence in MCE5 telescopes in $AR_N + R_N A^* - k R_N$ to $(b_{N+1} b_N^* + b_N b_{N+1}^*) / \omega_N$. Multiplication by G_N on both sides gives the control identity. Its trace is $2\Re \text{Tr } A - kq = 0$, which proves that $x_N^* y_N$ is purely imaginary. The square of this rank-at-most-two Hermitian operator has half trace $(\|x_N\|^2 \|y_N\|^2 - |x_N^* y_N|^2) / \omega_N^2$. This proves the two opposite eigenvalues and the exterior formula.

For the original observation $\Lambda : E \rightarrow B$ and inclusion $I : K \hookrightarrow E$, retain

$$\begin{aligned}H_N &= I^* G_N I, \quad Q_N = (\Lambda R_N \Lambda^*)^{-1}, \quad L_N = R_N \Lambda^* Q_N, \quad K_N^{\text{coord}} = H_N^{-1} I^* G_N, \\ \Lambda L_N &= I_B, \quad I^* G_N L_N = 0, \quad I K_N^{\text{coord}} + L_N \Lambda = I_E, \\ A_{KK} &= K_N^{\text{coord}} A I, \quad A_{KB} = K_N^{\text{coord}} A L_N, \quad A_{BK} = \Lambda A I, \quad A_{BB} = \Lambda A L_N, \\ \Lambda A - A_{BB} \Lambda &= A_{BK} K_N^{\text{coord}}, \\ I^* \mathcal{D}_N L_N &= A_{BK}^* Q_N + H_N A_{KB}.\end{aligned}\tag{OMR15}$$

The first three identities follow by multiplication and the minimum-lift orthogonality. Applying A in this direct-sum frame gives the action blocks and the displayed observation defect. Multiplication of $A^* G_N + G_N A - k G_N$ by I^* and L_N gives the final identity, since $I^* G_N L_N = 0$. This retains the actual action on the observation; it does not impose invariance of its kernel.

Define the marked components exactly by $x_K = H_N^{-1/2} I^* G_N b_N$, $x_B = Q_N^{1/2} \Lambda b_N$, and the same formulas with b_{N+1} for y_K, y_B . Put $a_K = \|x_K\|^2$, $a_B = \|x_B\|^2$, $d_K = \|y_K\|^2$, $d_B = \|y_B\|^2$, $z_K = x_K^* y_K$ and $z_B = x_B^* y_B$. Then

$$\begin{aligned}z_K + z_B &= i \omega_N \phi_N, \\ \omega_N^2 \epsilon_N^2 &= \|x_K \wedge y_K\|^2 + \|x_B \wedge y_B\|^2 + \|x_K \otimes y_B - y_K \otimes x_B\|^2 \\ &= (a_K d_K - |z_K|^2) + (a_B d_B - |z_B|^2) + a_K d_B + a_B d_K - 2\Re(z_K \bar{z}_B).\end{aligned}\tag{OMR16}$$

The frame $[I, L_N]$ has Gram $\text{diag}(H_N, Q_N)$, so the marked components give an isometry to $K \oplus B$. The exterior square decomposes orthogonally into $\bigwedge^2 K \oplus (K \otimes B) \oplus \bigwedge^2 B$. Expanding the middle norm gives the last three terms, including its complete complex cross-pairing. This proves every equality.

The full original positive primary part E_+ consists of all jets with $a > k/2$, has dimension $q/2$, and its top exterior class is nonzero under the original injection $\eta : E \hookrightarrow E_h^{\otimes k}$ because η is injective and $\eta A = A_{\text{sum}} \eta$. Its action excess is

$$\mathcal{L}_{h,k} = 2\Re \text{Tr}(A|_{E_+}) - kq/2 = 2\delta e(k+1) \left\lfloor \frac{(k+1)^2}{4} \right\rfloor \leq \epsilon_N, \quad N \geq q-1. \quad (\text{OMR17})$$

The nilpotent primary terms have zero trace. Summing the real parts over $a > k/2$, all b and all e jet directions gives the displayed coefficient. The induced additive Hermitian action on any exterior degree strictly between zero and q has largest eigenvalue ϵ_N , since OMR14's only nonzero eigenvalues are $\epsilon_N, -\epsilon_N$. Evaluating it on the positive class therefore proves the inequality. In the frame OMR15 its representative has both components $X_+ = K_N^{\text{coord}} I_+$ and $Y_+ = \Lambda I_+$, with Gram $X_+^* H_N X_+ + Y_+^* Q_N Y_+$; its expectation includes the cross block in OMR15. Thus this is the control of that same original class, with no component omitted.

45.5 The exact arithmetic radius budget and its backward bound

Define $\kappa_N = \log(\det H_N / \det H_{N+1})$, $\beta_N^B = \log(\det Q_N / \det Q_{N+1})$, and $\tau_N = \log(V_N / V_{N+1}) = \kappa_N + \beta_N^B$, where $V_N = \det G_N$. The original covariance rank-one update and its restriction give

$$\begin{aligned} e^{\beta_N^B} &= 1 + d_B / \omega_{N+1}, & e^{\kappa_N} &= 1 + d_K / (\omega_{N+1} + d_B), \\ \frac{a_B}{\omega_N} &= 1 - e^{-\beta_{N-1}^B}, & \frac{a_K}{\omega_N} &= e^{-\beta_{N-1}^B} (1 - e^{-\kappa_{N-1}}) \quad (N \geq q), \\ \epsilon_{q-1}^2 + \phi_{q-1}^2 &= \frac{\omega_q}{\omega_{q-1}} (e^{\tau_{q-1}} - 1), \\ \epsilon_N^2 + \phi_N^2 &= \frac{\omega_{N+1}}{\omega_N} (1 - e^{-\tau_{N-1}}) (e^{\tau_N} - 1) \quad (N \geq q). \end{aligned} \quad (\text{OMR18})$$

For the classical inverse and determinant identities in OMR18, see Hager [H2, pp. 221–223, (1)–(2), (6a)–(6b)]. That inspected account retains the conventional Sherman–Morrison and Woodbury names while explaining their earlier history; those names alone are not priority claims.

The first two formulas are the determinant lemma applied to the boundary covariance and then to the restricted Sherman–Morrison update for G_N . Applying the determinant lemma to removal of the last column proves the next two formulas. At $N = q-1$ the first q monic columns form the complete quotient basis, so $a_K + a_B = \omega_{q-1}$; no inverse at degree $q-2$ is used. Multiplying the total a and d formulas and using OMR14 proves the two radius identities.

Set $w_{q-1} = w_{2q-1} = 1$ and $w_N = 2$ for $q \leq N \leq 2q-2$, so that $\sum_N w_N = 2q$. For the

arithmetic source alone, put

$$\begin{aligned}
\delta_j^{\text{ar}} &= V_j^{\text{ar}}/V_{j-1}^{\text{ar}} \quad (j \geq q), \\
\mathcal{W}_k^{\text{ar}} &= \log \frac{\omega_{2q-1}^{\text{ar}} \omega_{2q}^{\text{ar}}}{\omega_{q-1}^{\text{ar}} \omega_q^{\text{ar}}}, \\
\mathcal{P}_- &= -\log(1 - \delta_{2q-1}^{\text{ar}}) - 2 \sum_{j=q}^{2q-2} \log(1 - \delta_j^{\text{ar}}), \\
\mathcal{P}_+ &= -\log(1 - \delta_q^{\text{ar}}) - \log(1 - \delta_{2q}^{\text{ar}}) - 2 \sum_{j=q+1}^{2q-1} \log(1 - \delta_j^{\text{ar}}), \\
C_k^{\text{budget}} &= F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \\
J_k^{\text{action}} &= \sum_{N=q-1}^{2q-1} w_N \log((\epsilon_N^{\text{ar}})^2 + (\phi_N^{\text{ar}})^2) = \Psi_k + C_k^{\text{budget}}.
\end{aligned} \tag{OMR19}$$

Here $0 < \delta_j^{\text{ar}} < 1$: positivity gives the first inequality and OMR17–18 forces the strict contraction at every index in the formula. All logarithms are therefore finite. To prove the budget identity, in each of the two windows $N = q-1, \dots, 2q-2$ and $N = q, \dots, 2q-1$ substitute $e^{\tau_N} - 1 = (1 - \delta_{N+1})/\delta_{N+1}$ into OMR18. The monic norm ratios telescope to $\mathcal{W}_k^{\text{ar}}$; the reciprocal δ factors give $F_K^{\text{ar}} + F_B^{\text{ar}}$. Each interior contraction occurs twice and the extreme ones once, giving exactly $-\mathcal{P}_- - \mathcal{P}_+$. Finally subtract and restore the same reference $F_B^{(k)}$ using OMR13. This proves the equality with every original endpoint.

The phases and the lower control allowance give the exact receiving inequalities

$$\begin{aligned}
Z_k^{\text{phase}} &= \sum_{N=q-1}^{2q-1} w_N \log \left(1 + \frac{(\phi_N^{\text{ar}})^2}{\mathcal{L}_{h,k}^2} \right), \\
L_J &= L_k^{\text{op}} + \mathcal{H}_k + \delta_k^\sigma + C_k^{\text{budget}}, \quad U_J = U_k^{\text{op}} + \mathcal{H}_k + \delta_k^\sigma + C_k^{\text{budget}}, \\
4q \log \mathcal{L}_{h,k} + Z_k^{\text{phase}} &\leq J_k^{\text{action}} \in [L_J, U_J], \\
\mathcal{L}_{h,k} &\leq \min_{q-1 \leq N \leq 2q-1} \epsilon_N^{\text{ar}} \leq \exp \left(\frac{J_k^{\text{action}}}{4q} \right) \leq \exp \left(\frac{U_J}{4q} \right).
\end{aligned} \tag{OMR20}$$

Indeed $\epsilon_N^{\text{ar}} \geq \mathcal{L}_{h,k} > 0$ term by term. Thus each logarithm in OMR19 is at least $2 \log \mathcal{L}_{h,k} + \log(1 + (\phi_N^{\text{ar}})^2/\mathcal{L}_{h,k}^2)$. The weights sum to $2q$, giving the first bound. Also $\log((\epsilon_N^{\text{ar}})^2 + (\phi_N^{\text{ar}})^2) \geq 2 \log \min_M \epsilon_M^{\text{ar}}$; summing proves the middle upper bound. OMR13 and the exact budget equality prove $L_J \leq J_k^{\text{action}} \leq U_J$ and the final bound.

Finally the same exact signed substitution as BRI6f gives

$$\begin{aligned}
J_k^{\text{action}} &= \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \epsilon_{k,k} + C_k^{\text{budget}}, \\
\Phi_{k,k} &\leq \mathcal{F}_{\partial,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^+ + C_k^{\text{budget}} - 4q \log \mathcal{L}_{h,k} - Z_k^{\text{phase}}, \\
\Phi_{k,k} &\leq \mathcal{O}_k^+ + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^+ + C_k^{\text{budget}} - 4q \log \mathcal{L}_{h,k} - Z_k^{\text{phase}}.
\end{aligned} \tag{OMR21}$$

The first equality preserves the complete original OAR signed error. Move $\Phi_{k,k}$ to the other side, use OMR20's lower bound for J_k^{action} , and then use $\epsilon_{k,k} \leq e_{k,k}^+$ to prove the second line. OMR4 proves the third. These caps may be intersected with the previous $\lambda_k \leq \Phi_{k,k} \leq V_k$ and all its earlier proved bounds, on their common domain.

Equations OMR19–21 retain the complete reference boundary, the arithmetic monic-norm window, both literal contraction penalties and every marked phase. The operator interval propagates to the original scalar and then to the control of the same surviving arithmetic class by these exact maps. They provide no sign-based exclusion from Ψ_k alone and assign no unevaluated leading value to $\Phi_{k,s}$.

45.6 The exact marked derivative and original reflection specialization

Use the original real source tilt $dm_{a,\theta}(y) = e^{\theta y} dm_a(y)$, with $|\theta| < \pi/2$, leaving E, A, Λ, I and the original arithmetic inclusion fixed. Every expression in this paragraph is evaluated at the same real θ . The full exponential-moment bound permits differentiation of every finite Gram integral on compact subintervals. If $\mathcal{R}_N$ is the minimum source section in fixed monomial source coordinates, differentiation of $J_N \mathcal{R}_N = I$ puts $\mathcal{R}'_N$ in $\ker J_N$. Its pairings with the minimum section vanish. Thus differentiating the minimum Gram inserts exactly y in its source integral. The leading coefficient of a minimum lift of x is $d_N^* G_N x / \omega_N$ in the real monic frame $Q_n^{\text{mon}} = i^{-n} p_n(c + iy)$, $d_n = [Q_n^{\text{mon}}]$. The polynomial $y \mathcal{R}_N - \mathcal{R}_N M_y - (Q_{N+1}^{\text{mon}} - \mathcal{R}_N d_{N+1}) d_N^* G_N / \omega_N$ has zero remainder and degree at most N . Its pairing with $\mathcal{R}_N$ vanishes, and Q_{N+1}^{mon} is orthogonal to that section. Returning by the exact substitution $S = c + iy$ gives

$$\begin{aligned} \partial_\theta G_N &= -i G_N (A - cI) + i G_N b_{N+1} b_N^* G_N / \omega_N, \\ \frac{z_K}{\omega_N} &= \text{Tr } A_{KK} - c \dim K - i \partial_\theta \log \det H_N, \\ \frac{z_B}{\omega_N} &= \text{Tr } A_{BB} - c \dim B - i \partial_\theta \log \det Q_N. \end{aligned} \tag{OMR22}$$

The first equality is the full MCP1–2 calculation. For the second, multiply it by I^* and I , then take its H_N^{-1} trace. The first term is $-i(\text{Tr } A_{KK} - c \dim K)$ and the rank-one term has trace iz_K / ω_N by cyclicity. Rearrangement proves the stated sign. For the third use $Q'_N = L_N^* G'_N L_N$, whose section-derivative terms vanish by $I^* G_N L_N = 0$, and $L_N^* G_N = Q_N \Lambda$. This proves the boundary formula with its complete marked pairing. No derivative estimate follows from differentiating an asymptotic remainder; OMR22 concerns the exact finite source family.

Here the original, untilted source has a proved additional involution. The completed function and full quartet obey $\xi(1-s) = \xi(s)$ and $h(1-s) = (-1)^{4m} h(s) = h(s)$, so their entire quotient satisfies $v_h(1-s) = v_h(s)$, including every removed zero and its Taylor coefficients. Consequently w_h is even and substitution in every convolution factor makes w_h^{*k} even. Both original Gamma densities are also even, with their full masses. Define in the original coefficient frame

$$\begin{aligned} (\mathcal{S}_N p)(S) &= p(k-S), \quad (\mathcal{S}_N)_{ab} = (-1)^a \binom{b}{a} k^{b-a} \quad (0 \leq a \leq b \leq N), \\ \chi(k-S) &= \chi(S), \quad \text{rem}_\chi \mathcal{S}_N = \Sigma \text{rem}_\chi, \quad \Sigma^2 = I, \quad \Sigma A \Sigma = kI - A, \\ (\Sigma[P])^{(d)}(\beta_{ab}) &= (-1)^d P^{(d)}(\beta_{k-a, k-b}) \quad (0 \leq d < e), \\ \sigma_h^{\otimes k} \eta &= \eta \Sigma, \quad \sigma_h[f(s)] = [f(1-s)]. \end{aligned} \tag{OMR23}$$

The first formula is the literal binomial expansion. Reflection permutes the centres by $(a, b) \mapsto (k-a, k-b)$ and q is even, so it preserves the full relation space. This proves the quotient map; differentiation proves every jet sign. The local involution σ_h fixes the full Taylor unit, and conjugates the tensor sum action to $kI - A_{\text{sum}}$. Applying it to $\eta[P] = v_h^{\otimes k} P(A_{\text{sum}})1$ proves the last identity. Thus the original arithmetic class is carried by the same injective map, with all tensor factors and jets retained.

Changing variables $y \mapsto -y$ carries the whole source fibre at tilt θ isometrically onto the reflected fibre at $-\theta$. The unique minimum sections and their Grams therefore obey

$$\begin{aligned} \Sigma^* G_N(-\theta) \Sigma &= G_N(\theta), \quad \mathcal{R}_N(-\theta) \Sigma = \mathcal{S}_N \mathcal{R}_N(\theta), \\ \Sigma b_n(0) &= (-1)^n b_n(0), \quad b_N(0)^* G_N(0) b_{N+1}(0) = 0, \quad \phi_N(0) = 0, \\ \partial_\theta \log \det G_N(\theta)|_0 &= 0. \end{aligned} \tag{OMR24}$$

Indeed for an even measure the polynomial $(-1)^n Q_n^{\text{mon}}(-y)$ is monic and has the same orthogonality as $Q_n^{\text{mon}}(y)$, so uniqueness proves its parity. The quotient isometry then takes the adjacent pairing to its negative; over $\mathbb{C}$ it is zero. This gives $\phi_N(0) = 0$ by its original definition. Finally

$|\det \Sigma|^2 = 1$ makes $\det G_N$ an even differentiable function of θ , proving the derivative equality. These statements hold for every original multiplicity and all three actual sources.

The total phase equality preserves the individual mixed terms. At the original source let $g_K = \partial_\theta \log \det H_N|_0$, $g_B = \partial_\theta \log \det Q_N|_0$, and $\alpha_K = \text{Tr } A_{KK} - c \dim K$. The exact consequences are

$$\begin{aligned} z_B &= -z_K, & g_B &= -g_K, & z_K/\omega_N &= \alpha_K - ig_K, \\ \|x_K \otimes y_B - y_K \otimes x_B\|^2 &= a_K d_B + a_B d_K + 2|z_K|^2, \\ \Lambda^\Sigma &= \Lambda \Sigma, & I^\Sigma &= \Sigma I, & H_N^\Sigma &= H_N, & Q_N^\Sigma &= Q_N, & L_N^\Sigma &= \Sigma L_N, \\ A_{KK}^\Sigma &= kI_K - A_{KK}, & A_{BB}^\Sigma &= kI_B - A_{BB}, & A_{KB}^\Sigma &= -A_{KB}, & A_{BK}^\Sigma &= -A_{BK}. \end{aligned} \tag{OMR25}$$

The first identity follows from OMR16 and OMR24. For the second, $[I, L_N(\theta)]$ has constant determinant: its second block changes only by columns in K , so the frame change is triangular with diagonal identity. The exact splitting determinant therefore gives $g_K + g_B = \partial_\theta \log \det G_N|_0 = 0$. The third identity is OMR22 and the mixed norm follows by substitution in OMR16. For the reflected observation the metric identities follow from OMR24 and the exact minimum section. Substituting $\Sigma \Lambda \Sigma = kI - A$ into the four OMR15 blocks proves their signs; both constant off-diagonal terms vanish because $K_N^{\text{coord}} L_N = 0$ and $\Lambda I = 0$. These are the exact maps to the reflected observation even when its kernel differs from K . Neither z_K , α_K nor g_K has been set to zero.

In particular the untilted arithmetic budget in OMR19 becomes

$$\begin{aligned} J_k^{\text{action}} &= 2 \sum_{N=q-1}^{2q-1} w_N \log \epsilon_N^{\text{ar}} = \Psi_k + C_k^{\text{budget}}, & Z_k^{\text{phase}} &= 0, \\ \exp\left(\frac{J_k^{\text{action}}}{4q}\right) &= \prod_{N=q-1}^{2q-1} (\epsilon_N^{\text{ar}})^{w_N/(2q)}, \\ 4q \log \mathcal{L}_{h,k} &\leq J_k^{\text{action}} \in [L_J, U_J], & \mathcal{L}_{h,k} &\leq \min_N \epsilon_N^{\text{ar}} \leq \prod_N (\epsilon_N^{\text{ar}})^{w_N/(2q)} \leq e^{U_J/(4q)}, \\ \Phi_{k,k} &\leq \mathcal{F}_{\partial,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^+ + C_k^{\text{budget}} - 4q \log \mathcal{L}_{h,k}, \\ \Phi_{k,k} &\leq \mathcal{O}_k^+ + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^+ + C_k^{\text{budget}} - 4q \log \mathcal{L}_{h,k}. \end{aligned} \tag{OMR26}$$

All allowances are positive by OMR17. Therefore the logarithms exist, substitution of the proved zero total phase gives the first line, and exponentiation gives the exact weighted geometric mean in the second. The third line follows from the minimum and the same positive-class bound. The last two inequalities are the proved OMR21 substitutions at $\theta = 0$, retaining every other term and its sign. The $[L_J, U_J]$ and outer substitutions keep their original simple-quartet domain; the budget and geometric mean identities retain full multiplicity. The generic MCE and MCP tilted-family formulas remain in full, and OMR25 retains the actual mixed phase still to be controlled.

The complete action interval in the original outer receiver. Keep the original action weights $w_{q-1} = w_{2q-1} = 1$ and $w_N = 2$ at the interior degrees. Their sum is $2q$. The complete-multiplicity proof CAI10–12 gives the original pointwise exterior lower bound $L_{h,k} = \delta q(k+1)/2$, and CAI33–34 evaluates the complete action relative to its unchanged arithmetic volume total. Thus

$$\begin{aligned} 4q \log L_{h,k} &\leq J_k^{\text{action}}, & L_{h,k} &\leq \min_{q-1 \leq N \leq 2q-1} \epsilon_N \leq \exp(J_k^{\text{action}}/(4q)), \\ \frac{\exp(J_k^{\text{action}}/(4q))}{q \exp(\mathcal{B}_k^{\text{ar}}/(4q))} &= \frac{4}{\exp(1)} (1 + O_h(k^{-1})). \end{aligned} \tag{CAIR3}$$

The first inequality adds the pointwise logarithmic inequalities with weights $2w_N$. The second uses their positive weighted geometric mean; it does not equate that mean with the minimum.

Exponentiating the finite CAI34 interval divided by $4q$ proves the last line, retaining its exact finite upper and lower versions when needed. In OMR26's original simple-quartet domain retain exactly $J_k^{\text{action}} = \Psi_k + C_k^{\text{budget}}$ and $\Psi_k = \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \mathfrak{e}_{k,k}$. The original bound $\mathfrak{e}_{k,k} \leq e_{k,k}^+$ gives the same complete backward receiver CAI38 after substituting $4q \log L_{h,k} \leq J_k^{\text{action}}$. CAIR2 refines the complete action in this identity: the $F_B^{(k)}$ term in the budget remains correlated with its opposite term in Ψ_k . The separate multiplicity domain of the optional positive arithmetic baseline remains $m = 1$.

45.7 Propagation through the original mixed action receiver

Retain the full untilted OMR26 action, its nonzero exterior class, and the complete frame splitting. The original equalities give

$$\begin{aligned} \Psi_k &= F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}, & C_k^{\text{budget}} &= F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \\ \Psi_k + C_k^{\text{budget}} &= J_k^{\text{action}} = \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+. \end{aligned} \quad (\text{RXECT10})$$

The second line follows by addition of the first line and $F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}$. Thus the same $F_B^{(k)}$ cancels with its negative while every original metric remains in the exact determinant splitting that proves the latter identity. RXECT9 supplies the improved finite $[L_J, U_J]$ for this complete value; RXECT3 independently supplies the minimum bound at all degrees. Exponentiating the exact weighted identity proves

$$\mathcal{G}_k := \prod_{N=q-1}^{2q-1} \epsilon_N^{w_N/(2q)} = e^{J_k^{\text{action}}/(4q)}, \quad L_{h,k} \leq \min_N \epsilon_N \leq \mathcal{G}_k, \quad e^{L_J/(4q)} \leq \mathcal{G}_k \leq e^{U_J/(4q)}. \quad (\text{RXECT11})$$

The middle upper bound follows because all weights are positive and sum to $2q$; no minimum is identified with that mean. The lower exterior bound remains the actual class trace from MCE/CAI11.

On the original simple-quartet domain of OMR's outer substitution, write its retained finite identity as $\Psi_k = \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \mathfrak{e}_{k,k}$, with $-e_{k,k}^- \leq \mathfrak{e}_{k,k} \leq e_{k,k}^+$. Solve this equality using RXECT10, without separately estimating its correlated allocations:

$$\begin{aligned} \Phi_{k,k} &= \mathcal{F}_{\partial,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma + \mathfrak{e}_{k,k} + C_k^{\text{budget}} - J_k^{\text{action}}, \\ \mathcal{F}_{\partial,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma - e_{k,k}^- + C_k^{\text{budget}} - U_J &\leq \Phi_{k,k} \leq \mathcal{F}_{\partial,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^+ + C_k^{\text{budget}} - L_J. \end{aligned} \quad (\text{RXECT12})$$

The upper and lower directions follow by the displayed error interval and $L_J \leq J_k^{\text{action}} \leq U_J$, respectively. If only the original exterior lower bound $4q \log L_{h,k} \leq J_k^{\text{action}}$ is used, the upper side recovers OMR26 verbatim. The finite order comparison and outer error retain their original signs and simple-quartet domain; the complete action and canonical minimum estimates retain all multiplicities. No value for an unevaluated allocation has been inserted.

46 The literal original conductor limit and an invertible pivot restriction

Retain all original hypotheses and maps of PQO, RPC, OCF and CEP. In particular $0 < \delta < 1/2$, $\gamma > 2$, the original unit values are $(a, \bar{a}, \bar{a}, a)$ with $a \neq 0$, and the same logarithm branch is used throughout. Write $\beta = 2(\gamma^2 - \delta^2)$, $\eta = (\delta^2 + \gamma^2)^2$, and $\mathfrak{a} = 1/160 + \beta/24 + \eta/2$. No source order is changed.

46.1 An entire coefficient matrix in the inverse original period

Let P_0 be RPC3's centered Fourier period matrix with its factor $e^{a/u}$ removed by the stated exact identity, and retain $G_u = \text{diag}(u^{a/5} g_a)_{a=1}^4$ with RPC8's full Gamma constants. Introduce the coefficient matrix $\mathcal{R}$ by the following identity, rather than replacing the original period:

$$P = e^{a/u} P_0 = e^{a/u} G_u \mathcal{R}(u^{-1}). \quad (\text{PCL1})$$

The row index below is $1 \leq a \leq 4$, and the column index is $0 \leq r \leq 3$. The complete series is

$$\begin{aligned} \mathcal{R}_{ar}(z) &= \sum_{n=0}^{\infty} R_{ar,n} z^n, \\ R_{ar,n} &= \sum_{\substack{p,h \geq 0 \\ 2p+4h=5n+r+1-a}} \frac{\beta^p \eta^h}{3^p p! h!} (-1)^L 5^L (a/5)_L, \quad L = p + h - n. \end{aligned} \quad (\text{PCL2})$$

Every sum at a fixed n is finite. For each contributing pair, $r + 3p + h + 1 = a + 5L$ and $L \geq 0$. To prove the latter assertion, the left side is positive and $1 \leq a \leq 4$, so a negative multiple of five cannot be their difference. The exponent n is nonnegative: its numerator $2p + 4h + a - r - 1$ is an integer multiple of five and is at least -3 . Thus a negative value of n is impossible.

For a direct derivation, divide every term of the absolutely convergent original series PQO9 by $u^{a/5} g_a$. The quotient of its Gamma factors is exactly

$$\frac{5^{(a+5L)/5-1} e^{\pi i(a+5L)/5} \Gamma(a/5 + L)}{g_a} = (-1)^L 5^L (a/5)_L. \quad (\text{PCL3})$$

Its remaining power of u is $u^{L-p-h} = u^{-n}$. These identities prove PCL2 including its signs, factorials and all unit-independent period coefficients. Absolute convergence of PQO9 permits grouping these terms. For each prescribed complex $z \neq 0$, put $u = 1/z$ on any of its original branches. The resulting sum of absolute values is finite. Consequently PCL2 converges absolutely for every complex z , and defines an entire matrix. This also proves that this matrix is independent of the branch; all branch factors remain in G_u in PCL1.

The complete constant matrix in PCL2 is

$$J_{\infty} = \begin{pmatrix} 1 & 0 & -\beta/3 & 0 \\ 0 & 1 & 0 & -2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad J_{\infty}^{-1} = \begin{pmatrix} 1 & 0 & \beta/3 & 0 \\ 0 & 1 & 0 & 2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{PCL4})$$

Indeed $n = 0$ forces $2p + 4h = r + 1 - a \leq 3$, so either $p = h = 0$ or $p = 1, h = 0, r + 1 - a = 2$. PCL3 gives respectively the diagonal entries and the two negative entries displayed. In particular, the original row-scaled period limit includes both cubic-phase corrections. It is not a diagonal coefficient map.

RPC's determinant proof holds on the complete coefficient family and gives $\det P_0 = \det G_u$. Alternatively divide PQO11 by the product of the four unchanged Gamma ray factors. Therefore

$$\det \mathcal{R}(z) = 1, \quad \mathcal{R}(z)^{-1} = \text{adj } \mathcal{R}(z) \quad (z \in \mathbb{C}). \quad (\text{PCL5})$$

The equality first holds for $z \neq 0$ by PCL1 and extends to zero by continuity. Both the matrix and its inverse are entire. For any fixed $R > 0$, the finite, completely specified constant

$$C_R = \sum_{n \geq 1} \|R_n\|_1 R^{n-1} < \infty \quad (\text{PCL6})$$

gives $\|\mathcal{R}(z) - J_\infty\|_1 \leq C_R|z|$ on $|z| \leq R$. Here $\|\cdot\|_1$ is the induced column-sum matrix norm and $R_n = (R_{ar,n})$. Convergence follows from the entire power series in finite dimension, also at any larger radius. Thus this is a bound in the actual coefficients, with no uncomputed nonzero minor used as a premise.

46.2 All eighty-one actual conductor coefficients and their limit

Let V be the original centered evaluation Vandermonde in PQO12, $U = \text{diag}(a, \bar{a}, \bar{a}, a)$, and $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. The complete period matrix from primary to Fourier coordinates is

$$H(u) = e^{a/u} G_u \mathcal{R}(u^{-1}) V^{-1} U. \quad (\text{PCL7})$$

For $1 \leq j \leq 4$ define the entire matrix and biform

$$\begin{aligned} L_j(z) &= U^{-1} V \mathcal{R}(z)^{-1} D^{-j} \mathcal{R}(z) V^{-1} U, \\ t(Z, W) &= (ZW, Z, W, 1)^\top, \\ A_z(Z, W) &= \prod_{j=1}^4 Q_0(L_j(z) t(Z, W)) = \sum_{r,s=0}^8 a_{rs}(z) Z^r W^s. \end{aligned} \quad (\text{PCL8})$$

These are precisely all original OCF4 coefficients at $z = u^{-1}$. To prove this, compute

$$H(u)^{-1} D^{-j} H(u) = L_j(u^{-1}) : \quad (\text{PCL9})$$

the scalar $e^{a/u}$ cancels with its inverse, and the two diagonal matrices G_u, D^{-j} commute. OCF's definition $Q_j(x) = Q_0(H^{-1} D^{-j} x)$ followed by $x = Ht$ then proves PCL8. The homogeneous version uses $t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)$ and retains the bidegree $(8, 8)$. Thus no unit or period entry has been omitted by the affine notation.

For an explicit finite formula for the individual coefficients, write $L_{j,be}$ for a matrix entry, and associate the four pairs $(\rho_e, \sigma_e) = (1, 1), (1, 0), (0, 1), (0, 0)$ to its columns. Put

$$\begin{aligned} c_{j,rs}(z) &= \sum_{\substack{1 \leq e, f \leq 4 \\ \rho_e + \rho_f = r, \sigma_e + \sigma_f = s}} (L_{j,1e} L_{j,4f} - L_{j,2e} L_{j,3f}), \\ a_{rs}(z) &= \sum_{\substack{0 \leq r_j, s_j \leq 2 \\ \sum_j r_j = r, \sum_j s_j = s}} \prod_{j=1}^4 c_{j,r_j s_j}(z). \end{aligned} \quad (\text{PCL10})$$

Expanding the two products defining Q_0 proves the first line; expanding its four original orbit factors proves the second. Both lines are finite and retain all ordered-pair multiplicities. Equations PCL2, PCL5 and PCL10 are a convergent formula for every actual coefficient, with no freely assigned amplitude.

Their exact leading values are obtained by the explicit matrices

$$L_j(0) = U^{-1} V J_\infty^{-1} D^{-j} J_\infty V^{-1} U, \quad a_{rs}(u^{-1}) = a_{rs}(0) + O_{\text{actual}}(|u|^{-1}). \quad (\text{PCL11})$$

The equality for $L_j(0)$ follows from PCL4, and the bound follows from the entire coefficient series. More explicitly, if $a_{rs}(z) = \sum_{n \geq 0} a_{rs,n} z^n$, the finite constant $\sum_{r,s} \sum_{n \geq 1} |a_{rs,n}| R^{n-1}$ bounds the sum of all eighty-one errors divided by $|z|$ on $|z| \leq R$. The coefficient functions are single-valued

entire functions of u^{-1} even though the original period itself retains its logarithm branch. This assertion follows from the exact conjugation PCL9, not from a change of the original branch.

At any specified k , the complete central convolution section is therefore the explicitly evaluated entire matrix

$$C_{\text{in}}(z)_{(a,b),(i,j)} = a_{i+4-a,j+4-b}(z), \quad 0 \leq a, b, i, j \leq k-8, \quad (\text{PCL12})$$

with all out-of-range coefficients zero. PCL10–12 give its full constant matrix and all successive coefficients. They do not discard off-diagonal coefficients or replace this matrix by its central coefficient a_{44} . In particular the calculation of the limit has not asserted a sign or diagonal-dominance inequality for this complete original matrix.

46.3 An exact positivity slice for the literal positive period ray

For $u > 0$ on any of its original logarithm lifts, PQO21 gives $P_0 = E_{\text{br}}R$ with R real invertible and E_{br} diagonal unitary commuting with D . Its exact centered quadratic coefficient matrix is S of PQO13; all its entries are purely imaginary. Consequently, for real $y \in \mathbb{R}^4$,

$$\overline{Q_j(E_{\text{br}}y)} = -Q_{5-j}(E_{\text{br}}y) \quad (1 \leq j \leq 4). \quad (\text{PCL13})$$

Indeed the original quadratic matrix in these coordinates is $e^{-2a/u}R^{-\top}SR^{-1}$, which is purely imaginary. Complex conjugation changes its sign and changes D^{-j} into D^j ; this is precisely the displayed formula.

Let P_{conj} be the permutation matrix exchanging primary indices 1, 2 and 3, 4. For real Z and $|W| = 1$, direct substitution gives $P_{\text{conj}}t(Z, W) = W^{-1}t(Z, W)$. The original roots and unit satisfy $\bar{V} = P_{\text{conj}}V$ and $P_{\text{conj}}\bar{U} = UP_{\text{conj}}$. Thus

$$\overline{V^{-1}Ut(Z, W)} = W^{-1}V^{-1}Ut(Z, W). \quad (\text{PCL14})$$

Choose either square root w of W . Then $p = w^{-1}V^{-1}Ut(Z, W)$ is real by PCL14, and $Ht = wE_{\text{br}}y$ for the real vector $y = e^{a/u}Rp$. Homogeneity and PCL13 prove the exact nonnegative slice identity

$$\boxed{W^{-4}A_{u^{-1}}(Z, W) = |Q_1(E_{\text{br}}y)Q_2(E_{\text{br}}y)|^2 \geq 0 \quad (Z \in \mathbb{R}, |W| = 1).} \quad (\text{PCL15})$$

The choice of sign of w has no effect because the quadratic forms have even degree. In particular, comparing the Laurent coefficients on the unit circle, and then the polynomial coefficients in real Z , gives the exact symmetry

$$a_{rs}(u^{-1}) = \overline{a_{r,8-s}(u^{-1})}. \quad (\text{PCL16})$$

PCL11 extends this identity and PCL15 to the coefficient limit at $u^{-1} = 0$ by continuity. The slice in PCL15 has one real coordinate and one unit-circle coordinate; it has not changed either coordinate into a second unit circle.

46.4 A complete invertible restriction supplied by the original pivot

At each actual u in the proved five-orbit locus, use precisely OCF5's lexicographically largest nonzero coefficient (r_*, s_*) , $a_* = a_{r_*s_*} \neq 0$, and its original pivot rectangle $\mathcal{P}_k = \{(r_* + i, s_* + j) : 0 \leq i, j \leq k-8\}$. Let $\mathcal{E}_k$ be its complement and S_k the original insertion of its coordinates. No new coefficient has been selected. The full original maps satisfy

$$CN_k^\top = 0, \quad S_k^\top N_k^\top = I_\Delta, \quad \text{im } N_k^\top = V_A. \quad (\text{PCL17})$$

The first is the transpose of $N_k C^\top = 0$ in OCF7; the second is the transpose of $N_k S_k = I$. Dimension Δ then proves the last equality, since C has full row rank. It follows that restriction to the actual strip is the exact isomorphism

$$S_k^\top : V_A \longrightarrow \mathbb{C}^{\mathcal{E}_k}, \quad (S_k^\top|_{V_A})^{-1} = N_k^\top. \quad (\text{PCL18})$$

Equivalently its square pivot block has nonzero diagonal a_* and is triangular in the elimination order; OCF's complete division supplies its inverse. This is an alternative restriction to the central one in PCL12, with the exact changed set of roots.

Let $P_{\mathcal{P},k}(y) = \prod_{(a,b) \in \mathcal{P}_k} (y - \omega_{ab})$ and $P_{\mathcal{E},k}(y) = \prod_{(a,b) \in \mathcal{E}_k} (y - \omega_{ab})$, where $\omega_{ab} = (2b - k)\gamma - i(2a - k)\delta$. Thus $P_k = P_{\mathcal{P},k}P_{\mathcal{E},k}$ exactly. Let $V_{\mathcal{E}}$ evaluate $1, y, \dots, y^{\Delta-1}$ at those actual outer roots and set $D_{\mathcal{P},\mathcal{E}} = \text{diag}(P_{\mathcal{P},k}(\omega_{ab}))_{\mathcal{E}_k}$. These two matrices are invertible by distinctness of the original roots. The full polynomial-target restriction and its exact inverse are

$$\begin{aligned}\rho_{\mathcal{E}} &= V_{\mathcal{E}}^{-1} D_{\mathcal{P},\mathcal{E}}^{-1} S_k^{\top}|_{V_A}, \\ \rho_{\mathcal{E}}^{-1} &= N_k^{\top} D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}.\end{aligned}\tag{PCL19}$$

Multiplying these matrices and using PCL17 proves both inverse identities, with all inner products of root differences retained. In particular the original residual becomes the following specified onto map, with its exact complete kernel:

$$\Xi_{\mathcal{E}} = \Xi_k^{\text{fix}} N_k^{\top} D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}, \quad \ker \Xi_{\mathcal{E}} = \rho_{\mathcal{E}}(K).\tag{PCL20}$$

This follows by composition with the proved isomorphism, not by deleting any common-kernel vector.

Finally the original metric and its entire source graph are carried by this isomorphism as follows. Write $J_{\mathcal{E}} = N_k^{\top} D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}$. At every original endpoint the actual conductor metric in these outer coordinates is exactly

$$H_{\mathcal{E},N} = J_{\mathcal{E}}^* G_N J_{\mathcal{E}}, \quad Q_{\Xi,N}^{\text{fix}} = (\Xi_{\mathcal{E}} H_{\mathcal{E},N}^{-1} \Xi_{\mathcal{E}}^*)^{-1}.\tag{PCL21}$$

The first formula is the pullback along $\rho_{\mathcal{E}}^{-1}$; the minimum-section proof of GIP6 gives the second. On the full GIP graph its target map is $\rho_{\mathcal{E}} E_N^{\text{graph}}$, with the same complete t ideal columns, and composing with $\Xi_{\mathcal{E}}$ gives exactly B_N^{graph} . Its residual kernel is therefore all $t + r$ columns in GIP4, including the low- v coordinates. The coefficient-space bound for N_k does not remove the displayed root-value and Gamma matrices from PCL21. These formulas retain the full metric transport needed for the next comparison.

46.5 The exact inverse-period reflection of the central section

Let $E_{\text{par}} = \text{diag}(1, -1, 1, -1)$ in the centered coefficient order, and let P_{opp} exchange primary indices 1, 4 and 2, 3. Equation PCL2 implies

$$\mathcal{R}(-z) = E_{\text{par}} \mathcal{R}(z) E_{\text{par}}, \quad V E_{\text{par}} = P_{\text{opp}} V, \quad U P_{\text{opp}} = P_{\text{opp}} U.\tag{PCL22}$$

For the first equality a contributing coefficient satisfies $5n + r + 1 - a = 2p + 4h$, so $n \equiv a - r - 1 \pmod{2}$. The signs from the two parity matrices on entry (a, r) are $(-1)^{a-1+r} = (-1)^n$. Summing the entire series proves the identity. The second equality evaluates each monomial at the opposite original root. The third uses the full original unit values $(a, \bar{a}, \bar{a}, a)$, which agree at opposite roots.

The two diagonal matrices E_{par} and D commute. Substitution into PCL8 and use of PCL22 therefore give the exact maps

$$L_j(-z) = P_{\text{opp}} L_j(z) P_{\text{opp}}, \quad A_{-z}(Z, W) = Z^8 W^8 A_z(Z^{-1}, W^{-1}).\tag{PCL23}$$

For the second identity, $Q_0(P_{\text{opp}} t) = Q_0(t)$ by its two explicit products. Also $P_{\text{opp}} t(Z, W) = ZW t(Z^{-1}, W^{-1})$ when $ZW \neq 0$. Each of the four quadratic factors therefore contributes $(ZW)^2$. This proves the identity on that open set; both sides are polynomials after multiplication by $Z^8 W^8$, so it holds identically including both coordinate axes.

Comparing every coefficient in PCL23 proves

$$a_{rs}(-z) = a_{8-r, 8-s}(z), \quad C_{\text{in}}(-z) = C_{\text{in}}(z)^{\top}, \quad \det C_{\text{in}}(-z) = \det C_{\text{in}}(z).\tag{PCL24}$$

Indeed the $(a, b), (i, j)$ entry on the left of the middle identity is $a_{4+a-i, 4+b-j}(z)$, precisely the transposed entry of PCL12. Out-of-range coefficients are zero on both sides. The determinant equality follows by the permutation expansion of a determinant and its transpose. Thus the exact central determinant is an even entire function of u^{-1} , with every coefficient supplied by PCL2 and PCL10–12. This retains its actual constant matrix and does not assign that determinant a value by deleting any of its entries.

46.6 The evaluated zero-order domain in the original period

The complete PZ, PZX and PZU calculations below evaluate the previously retained value $A_{u^{-1}}(1, 1)$ of this same conductor. In their exact notation put $r = a/\bar{a}$, $T = (\delta + i\gamma)^2/(\delta - i\gamma)^2$, $A_e = (2T - 1)/(3(T - 1))$ and $B_e = (2 - T)/(3(T - 1))$. For the four actual factors $d_j(\tau) = Q_0(L_j(\tau)\mathbf{1})$, the first nonzero inverse-period degree is

$$\ell_A = 0 \quad (r \notin \{1, A_e/B_e\}), \quad \ell_A = 2 \quad (r \in \{1, A_e/B_e\}), \quad b_j = [\tau^{\ell_A}]d_j(\tau) \neq 0. \quad (\text{RMR1})$$

The complete second-order matrices and polynomial nonvanishing certificate at both exceptional phases are PZX1–6; no phase is excluded. Each d_j is even entire by PCL22. Hence, with the actual finite-valued convergent sum $C_j = \sum_{n \geq 1} |[\tau^{\ell_A + 2n}]d_j|$, define $R_A = \max\{R_*, 1, \max_j (2C_j/|b_j|)^{1/2}\}$. For $|u| \geq R_A$ summing that Taylor tail gives

$$\frac{\prod_j |b_j|}{16|u|^{4\ell_A}} \leq |E_{A,u}(0)| \leq \frac{81 \prod_j |b_j|}{16|u|^{4\ell_A}}, \quad v = 0, \quad \alpha_A = 1/E_{A,u}(0). \quad (\text{RMR2})$$

Indeed each $u^{\ell_A}d_j(u^{-1})$ lies within $|b_j|/2$ of b_j , and multiplication of the four complete factors proves both inequalities. The nonzero entire product has only finitely many zeros in $|\tau| \leq 1/R_*$; therefore $v = 0$ also on the complement of the explicitly defined finite set PZU5 in the original domain $|u| \geq R_*$. At its exceptional periods the complete PZ12 derivative test and $v \leq 32$ remain in force.

On this proved $v = 0$ domain, substitution into the original maps gives

$$\begin{aligned} \mathbb{C}[S]_{<v} = 0, \quad t_{0,n} = 0, \quad m_n = 16n - 48, \quad t_n^{\text{EFP}} = 8n - 16, \\ j_{0,n} = 8n - 32, \quad \kappa_e = d_{K,e} = d_{T,e} = 0. \end{aligned} \quad (\text{RMR3})$$

The last three equalities follow from BTK25 and the exact RRC14 sequence. Thus the full low-cycle and connecting-cycle coefficient spaces coincide; their original quotient maps have identical fibres and identical attained metrics. All RRC16 tail corrections vanish. The ROG low-coordinate shear is the identity. The conductor, α_A , every coefficient and the complete CPQ mixed forms remain present: PZU9–11 gives the exact size of this scalar and its unchanged division recursion, including the exceptional phases. In particular these equalities determine the low-order correction; the connecting rank remains the exact CPR18 intersection rank.

47 The original relation operator and its finite-rank graph transport

Retain the complete original data and maps of CEP1–49 and GIP1–9. In particular the period and its logarithm branch are fixed on the actual five-orbit locus, $k = 4l + 1 \geq 9$, and

$$q = (k + 1)^2, \quad q' = (k - 7)^2, \quad \Delta = q - q', \quad c = k/2, \quad c' = c - 4, \quad 0 \leq v \leq 80. \quad (\text{ROG1})$$

All 81 original coefficients a_{ab} and their actual shifts $b_{ab} = 4 + (2a - 8)\delta + i(2b - 8)\gamma$ are retained. Write $\mu_j = \sum a_{ab} b_{ab}^j$, with $\mu_0 = \dots = \mu_{v-1} = 0$ and $\mu_v \neq 0$. For an original endpoint $N \in \{q - 1, q, 2q - 1, 2q\}$ put

$$L = N - q, \quad \mathbf{m} = \Delta - v, \quad M = L + \mathbf{m}, \quad d = N - v = q' + M. \quad (\text{ROG2})$$

Thus $L = -1, 0, q - 1, q$ at the four endpoints, respectively. The polynomial spaces of degree at most -1 are zero. Since $\Delta \geq 96 > v$, every displayed M is nonnegative.

47.1 The complete positive-shift relation operator

Let $\chi = \chi_k$ and $\chi' = \chi_{k-8}$ denote the original monic polynomials in S and S' , respectively; the prime here denotes the lower-grid polynomial and never a derivative. For $0 \leq a, b \leq 8$ define the actual monic product

$$d_{ab}(S') = \prod_{(i,j) \notin \{a, \dots, a+k-8\} \times \{b, \dots, b+k-8\}} (S' + b_{ab} - \beta_{ij}), \quad 0 \leq i, j \leq k. \quad (\text{ROG3})$$

The root identity $\beta'_{ij} + b_{ab} = \beta_{i+a, j+b}$ gives $\chi(S' + b_{ab}) = \chi'(S')d_{ab}(S')$ by matching individual factors, with the same leading coefficient one. Consequently

$$\begin{aligned} (\mathcal{U}_L h)(S') &= \sum_{a,b=0}^8 a_{ab} d_{ab}(S') h(S' + b_{ab}), \\ \chi' \mathcal{U}_L h &= \mathcal{T}_A(\chi h), \quad \mathcal{T}_{Ap} = \sum_{a,b} a_{ab} p(S' + b_{ab}). \end{aligned} \quad (\text{ROG4})$$

These are polynomial equalities with positive shifts. They include all original period coefficients and contain no conjugate or binomial multiplier from the biform pairing: multiplication of the original biform monomials adds their two indices with coefficient one, and their complex-linear primary pairing evaluates p at the sum of their roots, exactly as in CEP3–4.

For $0 \leq j \leq L$ write

$$u_j = \mathcal{U}_L(S^j), \quad \lambda_j = \binom{q+j}{v} \mu_v. \quad (\text{ROG5})$$

Every u_j has degree $\mathbf{m} + j$ and leading coefficient λ_j . Indeed Taylor expansion gives $\mathcal{T}_{Ap} = \sum_{t \geq v} \mu_t p^{(t)} / t!$, a finite sum on every polynomial. Applying it to the monic degree- $q+j$ polynomial χS^j gives leading coefficient $\binom{q+j}{v} \mu_v$ in degree $q+j-v$. Division by the monic degree- q' polynomial gives the asserted degree and coefficient. In particular $\mathcal{U}_L$ is injective.

47.2 An explicit finite inverse with coefficient control

The following constants are expressed in the original coordinates:

$$\begin{aligned} R &= |c| + k\sqrt{\delta^2 + \gamma^2}, \quad R' = |c'| + (k-8)\sqrt{\delta^2 + \gamma^2}, \\ B &= \max_{a,b} |b_{ab}|, \quad A_{\text{abs}} = \sum_{a,b} |a_{ab}|, \quad C_v = \frac{A_{\text{abs}} B^v}{|\mu_v|} \geq 1. \end{aligned} \quad (\text{ROG6})$$

All roots of χ and χ' have modulus at most R and R' , respectively. $B \geq 4$ and $C_v \geq 1$ follow directly from the shift list and $|\mu_v| \leq A_{\text{abs}} B^v$. For $L \geq 0$ put

$$C_L = qR + 2qB + (q' + L)R', \quad \tau_L = \min \left\{ 1, \frac{\log(1 + (2C_v)^{-1})}{C_L} \right\} > 0. \quad (\text{ROG7})$$

For every $0 \leq h \leq j \leq L$, the complete coefficient estimate is

$$\left| [S^{\mathbf{m}+j-h}] \frac{u_j(S')}{\lambda_j} \right| \leq C_v \frac{C_L^h}{h!}. \quad (\text{ROG8})$$

Here the coefficient for $h = 0$ is exactly one, which will be retained separately in the inverse.

To prove ROG8, write the coefficient of S^{q-a} in χ as χ_a . Its absolute value is bounded by $\binom{q}{a} R^a \leq (qR)^a / a!$, by expanding the product of its actual roots. For $n = q + j$, the term of derivative order $v + b$ applied to $\chi_a S^{n-a}$ contributes $\chi_a \mu_{v+b} \binom{n-a}{v+b}$; it is zero when the derivative order exceeds the degree. Its ratio to $\mu_v \binom{n}{v}$ is bounded by

$$C_v \frac{(qR)^a}{a!} \frac{(nB)^b}{b!}, \quad (\text{ROG9})$$

because $|\mu_{v+b}/\mu_v| \leq C_v B^b$ and $\binom{n-a}{v+b} / \binom{n}{v} \leq n^b / b!$. For the latter assertion first remove a using monotonicity of a binomial coefficient in its top entry; then the ratio is $\prod_{r=0}^{b-1} (n-v-r)/(v+1+r) \leq n^b / b!$.

In the formal expansion at infinity,

$$\frac{1}{\chi'(S')} = S'^{-q'} \sum_{t \geq 0} H_t S'^{-t}, \quad |H_t| \leq \binom{q' + t - 1}{t} (R')^t \leq \frac{((q' + h)R')^t}{t!} \quad (t \leq h). \quad (\text{ROG10})$$

Expansion of each $(1 - \beta'_{ab}/S')^{-1}$ proves the complete homogeneous coefficient formula and this estimate. To obtain the specified coefficient of u_j , sum ROG9–10 over $a + b + t = h$. Since $n \leq 2q$ and $h \leq L$, the multinomial identity gives exactly ROG8. Formal division agrees with ordinary polynomial division here because ROG4 proved exact divisibility before the expansion was made.

Let H_L be the square matrix consisting of the coefficients of $u_0, \dots, u_L$ in degrees $\mathbf{m}, \dots, \mathbf{m} + L$. Then

$$H_L = \widehat{H}_L \text{diag}(\lambda_0, \dots, \lambda_L), \quad \det \widehat{H}_L = 1, \quad \det H_L = \mu_v^{L+1} \prod_{j=0}^L \binom{q+j}{v}. \quad (\text{ROG11})$$

The matrix $\widehat{H}_L$ is upper triangular, with diagonal one and offset- h entries bounded by ROG8 for $h \geq 1$. For g in the actual image of $\mathcal{U}_L$, its unique preimage is therefore the finite formula

$$\text{coef } h = \text{diag}(\lambda_0^{-1}, \dots, \lambda_L^{-1}) \sum_{r=0}^L (I - \widehat{H}_L)^r ([S^{\mathbf{m}}]g, \dots, [S^{\mathbf{m}+L}]g)^T. \quad (\text{ROG12})$$

The finite sum is its inverse because $I - \widehat{H}_L$ is strictly upper triangular and its $(L+1)$ st power is zero. Thus this formula retains every coefficient of the original operator, and every complex factor λ_j .

In increasing monomial coefficient Euclidean norms it satisfies

$$\|\text{coef } h\|_2 \leq I_L \|\text{coef } g\|_2, \quad I_L = \frac{1 + 2 \sum_{h=1}^L \tau_L^{-h}}{|\mu_v| \binom{q}{v}}. \quad (\text{ROG13})$$

To check the bound, products of strictly upper triangular entries at a total offset h are majorized by the coefficient of z^h in

$$\sum_{r \geq 0} \{C_v(e^{C_L z} - 1)\}^r = \frac{1}{1 - C_v(e^{C_L z} - 1)}. \quad (\text{ROG14})$$

The absolute value of the expression inside braces is at most $1/2$ on $|z| = \tau_L$, so Cauchy's coefficient formula bounds these coefficients by $2\tau_L^{-h}$. The actual constant diagonal is one. Every absolute row and column sum of the inverse is consequently at most the numerator in ROG13. The Euclidean operator norm is at most the square root of the product of these row and column bounds, by applying Cauchy–Schwarz in each row and summing columns. Finally $|\lambda_j| \geq |\mu_v| \binom{q}{v}$ proves ROG13.

For fixed original period and quartet, $L \leq q$ gives

$$\log^+ I_L = O_{\text{actual}}(q \log(q+1)). \quad (\text{ROG15})$$

Indeed $R, R' = O(k)$, B, C_v, μ_v, v are fixed, $C_L = O_{\text{actual}}(qk)$, and the numerator in ROG13 is at most $(2L+1) \max\{1, \tau_L^{-L}\}$. No lower bound for a moment varying with the period is used.

47.3 Complete original-source bounds for the low graph map

For source $s \in \{1, k\}$ retain

$$b = s/2, \quad M_s = (2\pi)^{s/2}, \quad h_n = M_s n! (b)_n, \quad \|p\|_{s, c_0}^2 = \int |p(c_0 + iy)|^2 dm_s(y). \quad (\text{ROG16})$$

The density dm_s is exactly the original Gamma density in GIP5 and OQ2. We now supply elementary finite coefficient bounds, so no implicit conditioning of a polynomial frame is needed in the graph estimate. For $D \geq 0$ and real centre c_0 define

$$\begin{aligned} A_{D, s, c_0}^+ &= \sqrt{M_s(D+1)} \max\{1, |c_0| + 2\sqrt{D(b+D)}\}^D, \\ A_{D, s, c_0}^- &= \frac{\sqrt{D+1}}{\sqrt{M_s \min\{1, b\}}} \max\{1, (1+|c_0|)(1+D(b+D))\}^D. \end{aligned} \quad (\text{ROG17})$$

Then for every polynomial of degree at most D ,

$$\|p\|_{s, c_0} \leq A_{D, s, c_0}^+ \|\text{coef } p\|_2, \quad \|\text{coef } p\|_2 \leq A_{D, s, c_0}^- \|p\|_{s, c_0}. \quad (\text{ROG18})$$

For the first inequality, the original monic Gamma recurrence gives in its orthonormal basis

$$y\varphi_n = \sqrt{(n+1)(b+n)}\varphi_{n+1} + \sqrt{n(b+n-1)}\varphi_{n-1}.$$

On the degrees needed to multiply a polynomial of degree at most $D-1$, the upper and lower shift norms are at most $\sqrt{D(b+D)}$. Iterating the resulting bound for $S = c_0 + iy$, starting with $\|1\| = \sqrt{M_s}$, gives $\|S^a\| \leq \sqrt{M_s}(|c_0| + 2\sqrt{D(b+D)})^a$ for $a \leq D$. Cauchy–Schwarz on the coefficient sum proves the first formula in ROG18.

For the second, the coefficient absolute sum of the monic $p_n(y)$ is at most $(1+D(b+D))^n$ for $n \leq D$. The recurrence $p_{n+1} = yp_n - n(b+n-1)p_{n-1}$ proves this by induction: with $A = D(b+D)$, the required scalar inequality is $(1+A) + A \leq (1+A)^2$. Substitution of $(S - c_0)/i$ multiplies the degree- n coefficient bound by at most $(1+|c_0|)^n$, retaining the phase i^{-n} in the polynomial itself. Also $h_n \geq M_s \min\{1, b\}$: it holds at $n = 0, 1$, and $h_n/h_{n-1} = n(b+n-1) \geq 1$ for $n \geq 2$. The Euclidean norm of the coefficient column of each $p_n((S - c_0)/i)/\sqrt{h_n}$ is therefore bounded by the corresponding term in ROG17. Expanding in that orthonormal basis and applying Cauchy–Schwarz proves the second formula.

Define the explicit divisor constant

$$D_{M, \chi'} = \sum_{a=0}^M \binom{q' + a - 1}{a} (R')^a. \quad (\text{ROG19})$$

For $f = \chi'g$ of degree at most $d = q' + M$, ROG10 gives

$$\|\text{coef } g\|_2 \leq D_{M, \chi'} \|\text{coef } f\|_2. \quad (\text{ROG20})$$

Indeed the high coefficients of the quotient are obtained by convolution with the first $M + 1$ coefficients of $1/\chi'$. Every absolute row and column sum is bounded by ROG19; the same row and column argument proves the norm assertion. This bound is for the exact divisible input and retains the whole divisor. It does not assert a contractive division map.

Equip $\mathbb{C}[S']_{\leq M}$ with the weighted original norm

$$\|g\|_{\chi',s,c'} = \|\chi'g\|_{s,c'}, \quad G_M = (\langle \chi'S'^a, \chi'S'^b \rangle_{s,c'})_{a,b=0}^M. \quad (\text{ROG21})$$

The Gram is positive definite because multiplication by the nonzero polynomial is injective. Let $\mathbf{U}_L$ be the complete coefficient matrix of ROG4. For $L \geq 0$ the exact weighted orthogonal projection onto its image is

$$\Pi_{U,L} = \mathbf{U}_L(\mathbf{U}_L^* G_M \mathbf{U}_L)^{-1} \mathbf{U}_L^* G_M. \quad (\text{ROG22})$$

Its Gram inverse exists by ROG5. Multiplication verifies idempotence, weighted self-adjointness, and its image, so it has weighted norm one. At $L = -1$ put $\Pi_{U,L} = 0$.

Let $Z_{\chi,N}h$ be the original upper orthonormal coefficient column of χh , exactly as in GIP3. Define the complete low graph map on this weighted space by

$$Q_{N,s}g = P_{<v} Z_{\chi,N} \mathbf{U}_L^{-1} \Pi_{U,L} g : \mathbb{C}[S']_{\leq M} \longrightarrow \mathbb{C}^v. \quad (\text{ROG23})$$

The inverse is used only on its actual image, where ROG12 proved it uniquely. For $L = -1$ this map is zero. For $L \geq 0$ its norm is at most the following entirely finite constant:

$$H_{N,s} = A_{N,s,c}^+ (1+R)^q I_L D_{M,\chi'} A_{d,s,c'}^-. \quad (\text{ROG24})$$

To verify it, begin with $\Pi_{U,L}g$, whose weighted norm does not exceed $\|g\|_{\chi',s,c'}$. Apply ROG18 to $\chi' \Pi_{U,L}g$, then ROG20 and ROG13 to recover the ordinary coefficients of its unique h . Multiplication by χ has coefficient Euclidean norm at most $(1+R)^q$: the sum of the absolute coefficients of its full monic product is at most that number, and coefficient convolution has row and column sums bounded by this sum. The first bound in ROG18 now gives the original upper source norm of χh . Finally $P_{<v}$ is an orthogonal coordinate projection and has norm at most one. These are precisely the factors in ROG24.

At every original endpoint with $L \geq 0$,

$$\log(1 + H_{N,s}) = O_{\text{actual}}(q \log(q+1)), \quad s = 1, k. \quad (\text{ROG25})$$

For detail, $N, d \leq 2q$, $M \leq 2q$, $b \leq k/2$ and $|c|, |c'|, R, R' = O_{\delta,\gamma}(k)$. Both logarithms of the powered factors in ROG17 are $O(q \log(q+1))$. The factors $\sqrt{M_s}$ and $1/\sqrt{M_s}$ in ROG24 cancel at this specified composition; the individual source norms ROG16–18 retain their full mass. The factor $1/\sqrt{\min\{1, b\}}$ is at most $\sqrt{2}$. ROG19 is bounded by $(M+1) \max\{1, (q'+M)R'\}^M$, which has the same logarithmic order. ROG15 and $q \log(1+R) = O(q \log(q+1))$ control the other two factors. These estimates prove ROG25.

47.4 The exact graph shear and its complementary singular pairs

Use exactly the GIP source space and product metric

$$X_N = \mathbb{C}^v \oplus \mathbb{C}[S']_{\leq M}, \quad \|(x, g)\|^2 = \|x\|_2^2 + \|g\|_{\chi',s,c'}^2. \quad (\text{ROG26})$$

The full relation graph and its image under the specified invertible map are

$$\begin{aligned} \mathcal{R}_{N,s}(x, g) &= (x - Q_{N,s}g, g), & \mathcal{R}_{N,s}^{-1}(x, g) &= (x + Q_{N,s}g, g), & \det \mathcal{R}_{N,s} &= 1, \\ \mathcal{R}_{N,s} \operatorname{im} \begin{pmatrix} P_{<v} Z_{\chi,N} \\ \mathbf{U}_L \end{pmatrix} &= \{0\} \oplus \operatorname{im} \mathbf{U}_L. \end{aligned} \quad (\text{ROG27})$$

The last equality follows by applying ROG23 to $\mathcal{U}_L h$; the actual low-degree column is subtracted with its full value. The inverse and determinant follow from the displayed triangular blocks. Thus ROG27 is an exact morphism between the full original graph and the indicated relation image.

Put $r_Q = \text{rank } Q_{N,s} \leq v$ and let its nonzero singular values in the specified metrics be σ_a . Singular value decompositions on its domain and codomain put the shear into r_Q orthogonal two-dimensional blocks $\begin{pmatrix} 1 & -\sigma_a \\ 0 & 1 \end{pmatrix}$ and identity blocks. The two singular values of such a block are exactly

$$t_a = \frac{\sqrt{\sigma_a^2 + 4} + \sigma_a}{2}, \quad t_a^{-1}. \quad (\text{ROG28})$$

Indeed its positive Gram has determinant one and trace $2 + \sigma_a^2$; solving its quadratic characteristic polynomial gives these positive square roots. In particular $1 \leq t_a \leq 1 + \sigma_a \leq 1 + H_{N,s}$. For every exterior degree e , including zero and full degree, both the forward and inverse shear therefore satisfy

$$\log \left\| \bigwedge^e \mathcal{R}_{N,s}^{\pm 1} \right\| \leq \min\{e, v\} \log(1 + H_{N,s}). \quad (\text{ROG29})$$

There are at most v singular values above one; any remaining factors are at most one. This is the complementary-singular-value estimate with the actual rank, rather than multiplying the scalar inverse bound by the full source dimension. At $v = 0$ the map is the identity and this error is exactly zero.

47.5 The unchanged residual target and its four endpoint metrics

Retain the complete onto residual map B_N^{graph} and its common-kernel lift columns $W_{K,N}$ from GIP2–4. Define

$$B_{N,s}^{\text{str}} = B_N^{\text{graph}} \mathcal{R}_{N,s}^{-1}, \quad \ker B_{N,s}^{\text{str}} = \text{im} \left[\begin{pmatrix} 0 \\ \mathbf{U}_L \end{pmatrix}, \mathcal{R}_{N,s} W_{K,N} \right]. \quad (\text{ROG30})$$

The kernel equality is the image under the invertible shear of the full GIP4 kernel. It retains all $r = 8k - 16$ original common-kernel columns and all $L + 1$ relation columns; at $L = -1$ the latter list is empty. Let H_N^{graph} be the product Gram of ROG26, which is exactly GIP5, and put

$$Q_{N,s}^{\text{str}} = [B_{N,s}^{\text{str}} (H_N^{\text{graph}})^{-1} (B_{N,s}^{\text{str}})^*]^{-1}. \quad (\text{ROG31})$$

This is the attained minimum metric on the same original residual coordinate space $\mathbb{C}^j$, $j = 8k - 32$. For an onto matrix B and positive H , its minimum lift of y is $H^{-1} B^* (B H^{-1} B^*)^{-1} y$; subtraction of this lift from any other lift gives an H -orthogonal kernel vector. This proves the inverse-Gram assertion directly.

The shear induces the identity of the residual targets with these two quotient norms. A minimum lift is sent to its shear image and then projected off the entire target kernel. Orthogonal projection contracts every exterior power because its singular values are zero or one. Apply this observation also to the inverse shear and then square the two volume bounds in ROG29. It proves

$$|\log \det Q_{N,s}^{\text{str}} - \log \det \tilde{Q}_{\Xi,N}| \leq 2 \min\{j, v\} \log(1 + H_{N,s}). \quad (\text{ROG32})$$

Here $\tilde{Q}_{\Xi,N}$ is precisely GIP6 with its original complete graph metric. At $L = -1$ set $H_{N,s} = 0$; the two metrics are then identical. Combining with the already proved GIP6 source-map estimate gives the fully specified original residual comparison

$$|\log \det Q_{N,s}^{\text{str}} - \log \det Q_{\Xi,N}^{\text{fix}}| \leq 2E_N^{\text{CEP}} + 2v \log(1 + H_{N,s}). \quad (\text{ROG33})$$

Thus, with the unchanged signs $(1, 1, -1, -1)$ and original endpoints,

$$\left| \sum_N \sigma_N \log \det Q_{N,s}^{\text{str}} - \sum_N \sigma_N \log \det Q_{\Xi,N}^{\text{fix}} \right| \leq 2 \sum_N E_N^{\text{CEP}} + 2v \sum_N \log(1 + H_{N,s}) = o(kq). \quad (\text{ROG34})$$

CEP36 gives the first estimate, while ROG25 and fixed $v \leq 80$ give $O_{\text{actual}}(q \log(q+1)) = o(kq)$ for the second. This proves the transport for both complete original sources without removing a low component by assumption.

The resulting relation quotient is, with its product quotient norm,

$$\mathbb{C}^v \oplus (\mathbb{C}[S']_{\leq L+\Delta-v} / \mathcal{U}_L \mathbb{C}[S]_{\leq L}) . \quad (\text{ROG35})$$

The residual map on it is exactly the descended ROG30 map. In particular the second denominator remains the full operator ROG4; no equality with multiplication by the outer divisor is used in ROG1–35. Its complete leading coefficient and exact inverse are ROG5 and ROG12. The natural map to that outer divisor and its exact kernel and cokernel remain GIP7–9; their finite-convolution obstruction is carried unchanged by ROG27 and ROG30. The new finite-rank transport proves that the full low- v graph itself has only the explicit subleading cost ROG34 in the original residual return.

48 The exact differential realization of the original relation quotient

Retain the original $\chi_k(S)$, $\chi_{k-8}(S')$, all 81 coefficients and shifts of CEP41–42 and ROG1–6. Write $\chi = \chi_k$, $\chi' = \chi_{k-8}$, where the prime denotes the lower-grid polynomial, and put $\mathbf{m} = q - q' - v$. The cutoff-independent polynomial map is

$$\mathcal{U} : \mathbb{C}[S] \longrightarrow \mathbb{C}[S'], \quad \mathcal{U}h = \sum_{a,b=0}^8 a_{ab}d_{ab}(S')h(S' + b_{ab}). \quad (\text{RDS1})$$

It restricts to every original $\mathcal{U}_L$. For all $j \geq 0$, $u_j = \mathcal{U}(S^j)$ has degree $\mathbf{m} + j$ and leading coefficient $\lambda_j = \binom{q+j}{v} \mu_v \neq 0$ by exactly the Taylor and monic-division proof in ROG4–5, which holds at every finite polynomial degree.

48.1 A finite-dimensional quotient and its analytic dual

Repeated subtraction of the unique highest-degree multiple u_j/λ_j gives the direct vector-space decomposition

$$\mathbb{C}[S'] = \mathbb{C}[S']_{<\mathbf{m}} \oplus \mathcal{U}\mathbb{C}[S]. \quad (\text{RDS2})$$

At each step the degree strictly decreases. No nonzero sum $\mathcal{U}h$ can have degree below $\mathbf{m}$, by RDS1, which proves directness and uniqueness. Thus the resulting remainder map is linear and independent of the polynomial cutoff. Its restrictions identify every CEP43 quotient with the same vector space of dimension $\mathbf{m}$.

For a complex-linear functional ℓ on polynomials use its exponential generating series

$$f_\ell(z) = \sum_{n \geq 0} \ell(S'^n) \frac{z^n}{n!}. \quad (\text{RDS3})$$

Initially this is a formal series. The subspace annihilating $\text{im } \mathcal{U}$ has dimension $\mathbf{m}$: its first $\mathbf{m}$ values are arbitrary, and the equations $\ell(u_j) = 0$ uniquely determine the value in degree $\mathbf{m} + j$ from earlier ones, since its coefficient is the nonzero λ_j . We now realize this whole formal space as convergent series using the original symbol.

Put $D = \partial_z$ and define

$$E_A(z) = \sum_{a,b=0}^8 a_{ab}e^{b_{ab}z}, \quad \mathcal{L}_A = \sum_{a,b=0}^8 a_{ab}e^{b_{ab}z}d_{ab}(D). \quad (\text{RDS4})$$

The coefficients multiplying powers of D are on their displayed left side. The precise adjoint and intertwining identities, in this formal-series pairing, are

$$\ell(\mathcal{U}e^{Sz}) = \mathcal{L}_A f_\ell(z), \quad \mathcal{L}_A \chi'(D) = \chi(D)M_{E_A}. \quad (\text{RDS5})$$

For the first, multiplication of S'^r inside the functional is differentiation D^r , while the positive shift contributes $e^{b_{ab}z}$. Thus every term is exactly $a_{ab}e^{b_{ab}z}d_{ab}(D)f_\ell$. For the second, the full polynomial product in CEP41 gives $d_{ab}(D)\chi'(D) = \chi(D + b_{ab})$. Leibniz' rule gives $\chi(D)(e^{b_{ab}z}F) = e^{b_{ab}z}\chi(D + b_{ab})F$ for every formal F . Summing these identities proves RDS5, including the operator order and all coefficients.

Define the two original exponential spaces and a jet kernel by

$$\begin{aligned} \mathcal{E}_k &= \text{span}\{e^{\beta_{ij}z} : 0 \leq i, j \leq k\}, & \mathcal{E}_{k-8} &= \text{span}\{e^{\beta'_{ab}z} : 0 \leq a, b \leq k-8\}, \\ \mathcal{N}_v &= \{N \in \mathcal{E}_k : N^{(a)}(0) = 0 \ (0 \leq a < v)\}. \end{aligned} \quad (\text{RDS6})$$

The q upper and q' lower roots are distinct. Their full Vandermonde derivative matrices at zero prove independence of the respective exponentials. The first v upper derivative rows have rank

v , since any chosen v distinct root columns give a nonzero Vandermonde determinant. Hence $\dim \mathcal{N}_v = q - v$, also when $v = 0$.

The original sum identity gives the injective multiplication map

$$M_{E_A} : \mathcal{E}_{k-8} \hookrightarrow \mathcal{N}_v, \quad E_A(z) e^{\beta'_{ab} z} = \sum_{r,s=0}^8 a_{rs} e^{\beta_{a+r,b+s} z}. \quad (\text{RDS7})$$

Its numerical coefficient matrix is exactly $C^\top$ from CEP4, with the original order and no conjugation. The product vanishes to order at least v at zero because E_A has that exact zero order. Multiplication by this nonzero analytic function is injective: if its product with an analytic function is zero, that function vanishes on a punctured neighbourhood with nonzero E_A and hence identically.

There is the exact sequence and specified quotient isomorphism

$$0 \longrightarrow \mathcal{E}_{k-8} \xrightarrow{M_{E_A}} \mathcal{N}_v \xrightarrow{\mathfrak{F}_A} (\mathbb{C}[S']/\mathcal{UC}[S])^\vee \longrightarrow 0, \quad (\text{RDS8})$$

$$f_{\mathfrak{F}_A(N)}(z) = \chi'(D) \left(\frac{N(z)}{E_A(z)} \right).$$

The ratio is analytic at zero because its numerator vanishes to order at least v , the exact order of its denominator. It defines a functional by the Taylor derivatives in RDS3. RDS5 shows that this functional annihilates every u_j , since $\chi(D)N = 0$ for the original upper exponential space. If its image is zero, $F = N/E_A$ solves $\chi'(D)F = 0$ near zero. Every analytic solution of this constant-coefficient equation is in $\mathcal{E}_{k-8}$: its first q' derivatives determine all later derivatives by the monic equation, and the q' independent displayed exponentials realize arbitrary initial values by the invertible Vandermonde. This proves $N \in E_A \mathcal{E}_{k-8}$. The reverse kernel inclusion follows by applying $\chi'(D)$ to those exponentials. The image dimension is consequently $q - v - q' = \mathfrak{m}$, equal to the full annihilator dimension proved after RDS3, so the map is onto. This also proves that every formal annihilating series converges near zero.

The transpose of the injection in RDS7 is the exact primary conductor C ; this specifies the direction of the dual maps. Thus RDS8 relates the original polynomial relation quotient to the full original exponential module by an explicit quotient and differential map, without a new polynomial ideal.

48.2 Every possible singularity and its original multiplicity

The nonzero entire function E_A has isolated zeros. For $N \in \mathcal{N}_v$, the formula in RDS8 extends its series meromorphically to the whole z plane. Every possible pole lies at a zero of this same fixed original finite symbol; zero itself is removable. If $\zeta \neq 0$ is a zero of order h , put

$$e_\zeta(w) = w^{-h} E_A(\zeta + w), \quad c_{-\nu} = [w^{h-\nu}] \frac{N(\zeta + w)}{e_\zeta(w)} \quad (1 \leq \nu \leq h). \quad (\text{RDS9})$$

Here e_ζ is analytic and nonzero at zero, so these are its actual finite Taylor coefficients. Write $\chi'(X) = \sum_{r=0}^{q'} \chi'_r X^r$, with the coefficients in increasing order in this display. The complete principal part of the function in RDS8 at ζ is

$$\sum_{\nu=1}^h \sum_{r=0}^{q'} \chi'_r (-1)^r (\nu)_r c_{-\nu} (z - \zeta)^{-\nu-r}. \quad (\text{RDS10})$$

Laurent expansion of N/E_A gives its principal part $\sum c_{-\nu} (z - \zeta)^{-\nu}$; differentiating a term r times gives $(-1)^r (\nu)_r (z - \zeta)^{-\nu-r}$. Its holomorphic part remains holomorphic under differentiation, which proves the entire formula. Pole cancellations are retained by the actual coefficients, and the pole order is at most $h + q'$. More precisely, let t be the vanishing order of N at ζ . If $t \geq h$, the quotient

and its image under $\chi'(D)$ are holomorphic. If $t < h$, the leading Laurent coefficient of N/E_A has order $-(h-t)$ and is nonzero. The unique highest-order term after applying the monic $\chi'(D)$ is its q' th derivative: it has order $-(h-t+q')$ and leading coefficient multiplied by $(-1)^{q'}(h-t)_{q'} \neq 0$. Every lower derivative and every subsequent Laurent term has strictly smaller pole order. Thus the exact pole order is $h-t+q'$ in this case. The identically zero numerator gives the zero function. No simplicity of a symbol zero is used.

48.3 An explicit original-data analytic disk and source coefficients

Retain $B, A_{\text{abs}}, C_v, R$ of ROG6 and set

$$R_0 = \frac{1}{B} \min \left\{ 1, \frac{v+1}{2eC_v} \right\} > 0. \quad (\text{RDS11})$$

The exact entire quotient $B_\infty(z) = v!E_A(z)/(\mu_v z^v)$, with value one at zero, obeys $|B_\infty(z)-1| \leq 1/2$ on $|z| \leq R_0$. Indeed its remaining coefficient sum is at most $C_v B|z|e^{B|z|}/(v+1)$ by expanding each original exponential and using $v!/(v+1+j)! \leq 1/((v+1)j!)$. Definition RDS11 gives the asserted bound. This proves that this disk has no nonzero zero of E_A .

Write $N = \sum \nu_{ij} e^{\beta_{ij} z}$. Its first v vanishing Taylor coefficients and the elementary remainder bound for the exponential give

$$|N(z)| \leq \|\nu\|_1 \frac{(R|z|)^v}{v!} e^{R|z|}, \quad \left| \frac{N(z)}{E_A(z)} \right| \leq \frac{2R^v}{|\mu_v|} e^{RR_0} \|\nu\|_1 \quad (|z| \leq R_0). \quad (\text{RDS12})$$

The removable value at zero follows by continuity. Cauchy's derivative formula on this disk gives, for $|z| \leq R_0/2$, the explicit estimate

$$|f_{\mathfrak{F}_A(N)}(z)| \leq C_{k,A} \|\nu\|_1, \quad C_{k,A} = \frac{2R^v}{|\mu_v|} q'! \left(\frac{2}{R_0} \right)^{q'} \exp(RR_0 + R'R_0/2). \quad (\text{RDS13})$$

For verification, the derivative of order r of N/E_A is bounded by $r!(2/R_0)^r$ times the bound in RDS12. The coefficient of $X^{q'-j}$ in $\chi'(X)$ has modulus at most $\binom{q'}{j}(R')^j$. Summing the derivative bounds therefore gives $q'!(2/R_0)^{q'} \sum_{j=0}^{q'} (R'R_0/2)^j/j!$, which proves RDS13. Its logarithm is $O_{\text{actual}}(q \log(q+1))$ at the fixed original period.

Finally, for any polynomial functional the following identity is formal. For $\ell = \mathfrak{F}_A(N)$ its right side is analytic near zero by RDS8 and RDS13, so the identity also proves convergence of its left side. For either unchanged source order $s = 1, k$ it is

$$\sum_{n \geq 0} \ell \left(p_n \left(\frac{S' - c'}{i} \right) \right) \frac{z^n}{n!} = (1+z^2)^{-s/4} e^{ic' \arctan z} f_\ell(-i \arctan z). \quad (\text{RDS14})$$

It follows by applying the complex-linear functional to the original Gamma generating series; the centre and phase are the literal equality $(S' - c')/i = -iS' + ic'$. Let $\rho = \tanh(R_0/4)$. On $|z| = \rho$, $|\arctan z| \leq \text{arctanh } \rho = R_0/4$, so RDS13 and Cauchy's coefficient formula yield

$$\left| \ell \left(\frac{p_n((S' - c')/i)}{\sqrt{h_n}} \right) \right| \leq \sqrt{\frac{n!}{M_s(s/2)_n}} \rho^{-n} (1 - \rho^2)^{-s/4} e^{|c'|R_0/4} C_{k,A} \|\nu\|_1. \quad (\text{RDS15})$$

The square root uses the full original $h_n = M_s n!(s/2)_n$ and $M_s = (2\pi)^{s/2}$. This identity gives the source coefficient realization of the actual relation-annihilating functionals. The weighted relation metric itself remains ROG21, including multiplication by the full χ' . Equations RDS8–15 provide its exact finite-symbol analytic model and quantitative disk; the rank- $O(k)$ residual determinant is not assigned a value by multiplying this scalar coefficient bound by its rank.

49 The fixed-period problem and the result proved here

The current handoff already proves the absolute $O(kq)$ kernel estimate, the conductor frame, its residual observation, and the finite inverse. The new calculation here removes the $O(kq)$ *exterior-inverse allowance* left by AIE29. It replaces that allowance by an explicit $O_{\text{actual}}(q \log q) = o(kq)$ error and transfers the actual four-endpoint determinant to the smaller conductor coefficient space. It does not identify this transfer error with the original kernel volume, and it does not evaluate the latter's leading coefficient.

Retain an actual hypothetical simple quartet

$$\rho = \frac{1}{2} + \delta + i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad h(z) = \prod_{\alpha \in \{\rho, \bar{\rho}, 1-\rho, 1-\bar{\rho}\}} (z - \alpha), \quad v_h = 2\xi/h.$$

The original full unit and period observation are

$$U = M_{j_h v_h}, \quad j\Lambda_k = P_k \Pi^{\otimes k} U^{\otimes k} \iota_k, \quad P_k = \frac{1}{5} \sum_{b=0}^4 (C^{-b})^{\otimes k}. \quad (\text{FEX1})$$

Fix one original period and logarithm branch on the proved five-orbit locus, for example any fixed point with $|u| \geq R_*$. Here R_* is the unchanged RPC/AMT radius. Its formulas and the proof of the rank are in the supplied complete source; no new rank premise is imposed. Throughout the principal application,

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad q = (k+1)^2, \quad c = k/2, \quad k' = k - 8, \quad c' = c - 4, \\ q' &= (k-7)^2, \quad a = q - q' = 16k - 48, \quad r = 8k - 16, \quad j_k = 8k - 32, \\ \beta_{ab}^{(k)} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi = \prod_{a,b=0}^k (S - \beta_{ab}^{(k)}), \quad \psi = \chi_{k'}. \end{aligned} \quad (\text{FEX2})$$

The letter a without indices denotes the degree difference, not an amplitude. Let $K = \ker \Lambda_k \subset E = \mathbb{C}[S]/(\chi)$, using its fixed original coefficient inclusion. The actual ranks satisfy $a = r + j_k$. Every class below has its unique representative of degree less than q whenever a polynomial representative is required.

Retain the complete original conductor biform and its coefficients

$$A = \sum_{r_0, s_0=0}^8 a_{r_0 s_0} X_1^{r_0} X_0^{8-r_0} Y_1^{s_0} Y_0^{8-s_0}.$$

Write $\nu = (r_0, s_0)$ for an index in this sum. Define

$$\begin{aligned} b_\nu &= 4 + (2r_0 - 8)\delta + i(2s_0 - 8)\gamma, \quad \mu_j = \sum_\nu a_\nu b_\nu^j, \quad v = \min\{j : \mu_j \neq 0\} \leq 80, \\ (\mathcal{T}_{AP})(S') &= \sum_\nu a_\nu p(S' + b_\nu), \quad \mathcal{C}_A = \frac{v!}{\mu_v} \mathcal{T}_A, \\ B_q(X) &= \sum_{n=0}^{q-1-v} \frac{v! \mu_{v+n}}{(v+n)! \mu_v} X^n. \end{aligned} \quad (\text{FEX3})$$

The coefficients are those of the actual $\mathbf{H} = F^{-1} \Pi U \mathcal{E}_{\text{prim}}$, fixed as k changes. The original distinct-node Vandermonde proves the stated finite v and $\mu_v \neq 0$. In particular, $B_q(0) = 1$. The scalar in $\mathcal{C}_A$ is displayed explicitly; it is not a renormalization of an arithmetic source or observation.

Only the two original Gamma sources are used:

$$\begin{aligned} s &\in \{1, k\}, \quad b = s/2, \quad M_s = (2\pi)^{s/2}, \quad dm_{0,s}(y) = \frac{M_s 2^b}{4\pi \Gamma(b)} |\Gamma(b/2 + iy/2)|^2 dy, \\ h_n &= M_s n! (b)_n, \quad w_n = \frac{n!}{(b)_n}, \quad \|p\|_{s, c_0}^2 = \int |p(c_0 + iy)|^2 dm_{0,s}(y). \end{aligned} \quad (\text{FEX4})$$

In particular $s = k'$ is never inserted as a source order. The polynomial ψ in FEX2 will enter as a literal multiplication factor in these unchanged measures.

The main conclusion will be the finite interval FEX35 and the consequence

$$\boxed{F_K^{(s)} = \widehat{F}_k^{(s)} + O_{H,\delta,\gamma}(q \log q), \quad s = 1, k.} \quad (\text{FEX5})$$

Here $\widehat{F}$ is defined in FEX29–30 using the actual residual observation and all original transformed relation columns. The error is proved, not postulated. At the end, the unchanged arithmetic comparison transfers the same leading-limit question to F_K^{ar} .

50 A full-flag determinant bound

Let $T : (V, g) \rightarrow (V', g')$ be an isomorphism of n -dimensional complex Hilbert spaces. Write $\|T\| \leq M$ and let $J > 0$ be its squared whole-space Jacobian. For $R \subset W \subset V$, set $t = \dim W - \dim R$. Give W/R and TW/TR their actual Hilbert quotient norms. For any coefficient basis of W/R , let D_Q be the ratio of its target and source Gram determinants under the induced map. Then

$$\boxed{JM^{-2(n-t)} \leq D_Q \leq M^{2t}.} \quad (\text{FEX6})$$

All empty factors have determinant one. The upper bound follows by applying T to each lift and taking the infimum over the same fibre. For the lower bound, use the flag $0 \subset R \subset W \subset V$. Choose any basis adapted to it, on both sides using its transported basis. Successive block elimination of the complete positive Gram gives

$$J = D_R D_Q D_{V/W}. \quad (\text{FEX7})$$

Each block elimination uses its full Schur complement. The other two induced maps also have norm at most M , by restriction or minimization, and their dimensions total $n - t$. Thus their squared determinant product is at most $M^{2(n-t)}$, which proves FEX6. This proves the assertion without asserting orthogonality of a fixed coefficient section.

An equivalent inverse form is useful. If T is an endomorphism of a single Hilbert space and $|\det T| = 1$, then, for every full-rank n by t coefficient matrix X ,

$$\boxed{M^{-2t} \leq \frac{\det(X^* T^{-*} T^{-1} X)}{\det(X^* X)} \leq M^{2(n-t)}.} \quad (\text{FEX8})$$

For example, this follows by taking $W = T^{-1} \text{im } X$, $R = 0$ in FEX6 and inverting the determinant ratio. It also follows by taking the product of the largest t inverse singular values and cancelling against the full product. The statement is a bound: $\det T = 1$ does *not* say that a restricted Gram determinant is unchanged. For $T = \begin{pmatrix} 1 & 4 \\ 0 & 1 \end{pmatrix}$, restriction to the second coordinate has squared volume 17.

51 Polynomial forward control replaces the exponential exterior loss

For completeness, the derivative bound used here follows directly from the original polynomial norms. The recurrence and generating identity are

$$\begin{aligned} p_0 &= 1, & p_1 &= y, & p_{n+1} &= yp_n - n(b + n - 1)p_{n-1}, \\ \sum_{n \geq 0} \frac{p_n(y)}{n!} z^n &= (1 + z^2)^{-b/2} e^{y \arctan z}, & \|p_n\|^2 &= h_n. \end{aligned} \quad (\text{FEX9})$$

They are the supplied original Gamma identities. Differentiation of the finite coefficient at zero, followed by the full normalization, gives

$$\partial_y \varphi_n = \sum_{\substack{1 \leq j \leq n \\ j \text{ odd}}} \frac{(-1)^{(j-1)/2}}{j} \sqrt{\frac{w_n}{w_{n-j}}} \varphi_{n-j}, \quad \varphi_n = p_n / \sqrt{h_n}. \quad (\text{FEX10})$$

The mass M_s cancels from this ratio, not from the underlying norms. Let $H_D = \sum_{j=1}^D 1/j$ and $H_0 = 0$. For $n > m \geq 1$, $b \geq 1/2$ gives $w_n/w_m \leq \sqrt{n/m}$ and $w_n \leq 2\sqrt{n}$. Indeed square each factor $j/(j-1/2)$ and use $j(j-1) \leq (j-1/2)^2$; the product telescopes. Apply the weighted Schur estimate to the absolute matrix in FEX10 with weights $w_n^{-1/2}$. Its row sum is at most H_D ; its column sum at n is at most $H_n + 4$. The excess for $1 \leq m < n$ is bounded by

$$\frac{\sqrt{n/m} - 1}{n - m} = \frac{1}{\sqrt{m}(\sqrt{n} + \sqrt{m})} \leq \frac{1}{\sqrt{nm}},$$

whose sum is at most two; the $m = 0$ term is at most $2/\sqrt{n} \leq 2$. Enlarging the odd-index sums is legitimate because all terms are nonnegative. Cauchy–Schwarz in each weighted row and then summing the column estimate proves

$$\|\partial_y\|_{\mathcal{P}_D} \leq \sqrt{H_D(H_D + 4)} \leq H_D + 2, \quad \|f(y + z)\| \leq e^{|z|(H_D+2)} \|f\|. \quad (\text{FEX11})$$

At $D = 0$ the derivative is zero. Translation is the exact finite Taylor exponential. The coordinate substitution $S = c_0 + iy$ gives $\partial_S = -i\partial_y$ with the same norm.

Retain every actual coefficient in

$$A_*^{\text{abs}} = \sum_{\nu} |a_{\nu}|, \quad B_*^{\text{sh}} = \max_{\nu} |b_{\nu}|, \quad C_v = \frac{A_*^{\text{abs}}(B_*^{\text{sh}})^v}{|\mu_v|} \geq 1, \quad P_D = C_v e^{B_*^{\text{sh}}(H_D+2)}. \quad (\text{FEX12})$$

Since $|\mu_{v+n}| \leq A_*^{\text{abs}}(B_*^{\text{sh}})^{v+n}$ and $v!/(v+n)! \leq 1/n!$, the triangle inequality in FEX10–11 gives

$$\|B_q(\partial_S)\|_{s,c_0;\mathcal{P}_D} \leq P_D. \quad (\text{FEX13})$$

This bound holds for the literal fixed polynomial B_q on any explicitly specified finite degree space. On $D \leq q-1$ it is the original AIE operator, whose matrix is triangular with diagonal one. FEX8 therefore proves its new exterior estimate

$$\boxed{P_D^{-2t} \leq \frac{\det(X^*(B_q^{-1})^* B_q^{-1} X)}{\det(X^* X)} \leq \min\{N_{D,s}^{2t}, P_D^{2(D+1-t)}\}.} \quad (\text{FEX14})$$

Here X is any original t -plane in $\mathcal{P}_D$ and $N_{D,s}$ is the unchanged AIE10 inverse bound. The new bound requires neither a new zero-free disk nor a lower bound on an unevaluated restricted minor. For $D = q-1$, $t = r$, its upper logarithm is at most

$$2(q-r)\{\log C_v + B_*^{\text{sh}}(H_{q-1} + 2)\} = O_{H,\delta,\gamma}(q \log q) = o(kq). \quad (\text{FEX15})$$

Thus the factor rD in the old exterior allowance is unnecessary. The forward norm, together with the full Jacobian, controls the product; raising the worst inverse singular value to the r th power loses that information.

There is a direct high-source extension of this statement, distinguished from changing AIE's cutoff. For a fixed k use the same polynomial $B_q(\partial_S)$ on each $\mathcal{P}_N$. Map the exact flag

$$\chi \mathcal{P}_{N-q} \subset I_K \mathbb{C}^r + \chi \mathcal{P}_{N-q}$$

to its literal image under B_q . Its relation space is $B_q(\partial_S)(\chi \mathcal{P}_{N-q})$, not the original ideal by assumption. Its whole Jacobian is one, and FEX13 is valid on this newly specified domain. If $\tilde{F}_K$

is the signed kernel determinant using the native source norm on these transported flags, FEX6 gives

$$-2(2q+1)\log P_{2q} \leq \tilde{F}_K - F_K \leq 2(4q+1)\log P_{2q}. \quad (\text{FEX16})$$

Indeed at endpoint N its logarithmic Gram change belongs to $[-2(N+1-r)\log P_N, 2r\log P_N]$. Insert the four original signs and use $P_N \leq P_{2q}$. This is an exact extension and a signed estimate; no commutation with the original remainder map has been assumed.

52 The literal conductor at every source endpoint

The previous high-source extension is useful, but the actual conductor provides a more informative reduction. On $\mathcal{P}_N$, $N \geq q-1$, its exact Taylor identity is

$$\begin{aligned} \mathcal{C}_A &= \partial_S^v B^{[N]}(\partial_S), & B^{[N]}(X) &= \sum_{n=0}^{N-v} \frac{v!\mu_{v+n}}{(v+n)!\mu_v} X^n, \\ B^{[q-1]} &= B_q, & L &:= \ker(\mathcal{C}_A|_{\mathcal{P}_N}) = \mathcal{P}_{v-1}, & \mathcal{C}_A : \mathcal{P}_N/L &\xrightarrow{\sim} \mathcal{P}_{N-v}. \end{aligned} \quad (\text{FEX17})$$

For $v=0$, $L=0$. These assertions follow either from the leading nonzero derivative or from the commuting invertible triangular factor $B^{[N]}$. Thus the operator on the high endpoints is defined by the original translations themselves. It is not obtained by silently changing the cutoff in the definition of B_q .

The domain has its original quotient Gamma norm at centre c ; the codomain has its native Gamma norm at centre c' . Define

$$M_N = \max \left\{ 1, \frac{v!}{|\mu_v|} \sum_{\nu} |a_{\nu}| e^{[b_{\nu}-4](H_N+2)} \right\}, \quad m_N = \log M_N. \quad (\text{FEX18})$$

The difference of centres is essential: $S' + b_{\nu} = c + iy + (b_{\nu} - 4)$. FEX11 proves $\|\mathcal{C}_A p\|_{s,c'} \leq M_N \|p\|_{s,c}$. Taking the infimum over the same L -coset gives the same bound for the induced isomorphism in FEX17.

Its squared whole-space Jacobian is explicitly

$$\mathcal{J}_{N,s} = \prod_{d=v}^N \frac{w_d}{w_{d-v}} = \frac{\prod_{d=N-v+1}^N w_d}{\prod_{d=0}^{v-1} w_d}. \quad (\text{FEX19})$$

Products are one when $v=0$. To prove the formula, take the orthonormal quotient basis represented by $\varphi_v((S-c)/i), \dots, \varphi_N((S-c)/i)$, with the normalization in FEX10, and the corresponding native basis at c' . The triangular diagonal of the map is $i^{-v}(d)_v \sqrt{h_{d-v}/h_d}$, where $(d)_v = d!/(d-v)!$ is a falling factorial in this formula. Its squared modulus is w_d/w_{d-v} . The source mass is present in both h factors before cancellation; translation of the centre affects only strictly lower coefficients. Telescoping proves the second equality, including overlapping product ranges. For the unnormalized $\mathcal{T}_A$ the squared Jacobian is $|\mu_v/v!|^{2(N+1-v)} \mathcal{J}_{N,s}$.

Put $\zeta_{N,s} = \log \mathcal{J}_{N,s}$. The following finite bounds will suffice:

$$\begin{aligned} 0 &\leq \zeta_{N,1} \leq v \log(2\sqrt{N}), & s &= 1, \\ 0 &\leq -\zeta_{N,s} \leq v(b-1)H_N, & s &= k \geq 9. \end{aligned} \quad (\text{FEX20})$$

For the first use monotonicity of w_n and $1 \leq w_n \leq 2\sqrt{n}$. For the second, w_n decreases and $-\log w_n = \sum_{j=1}^n \log(1 + (b-1)/j) \leq (b-1)H_n$; then use the second product formula in FEX19. These proofs also show directly that the Jacobian error at the tensor source is $O(vk \log q)$, rather than an uncontrolled determinant of dimension q .

The old explicit AIE inverse is still compatible with this estimate. For example, let R_A be that right inverse for the unnormalized $\mathcal{T}_A : \mathcal{P}_D \rightarrow \mathcal{P}_{D-v}$, and let $R_{\min}$ be its actual minimum-norm

right inverse. Their difference takes values in L and is orthogonal to $R_{\min}$. On a t -plane X in the target,

$$\det((R_A X)^* R_A X) \leq \det((R_{\min} X)^* R_{\min} X)(1 + M_{\text{raw}}^2 C_{A,D,s}^2)^{\min(t,v)}. \quad (\text{FEX21})$$

Here $\|\mathcal{T}_A\| \leq M_{\text{raw}}$ and $\|R_A\| \leq C_{A,D,s}$ are the original finite forward and AIE22 bounds. Indeed the excess Gram has rank at most v , and $R_{\min}^* R_{\min} \succeq M_{\text{raw}}^{-2} I$. The operator $R_A - R_{\min}$ is an orthogonal projection of R_A , so its norm is at most $C_{A,D,s}$. Apply the matrix determinant lemma, or congruence with the minimum Gram, to obtain FEX21. FEX6 bounds the minimum inverse determinant by $J_{\text{raw}}^{-1} M_{\text{raw}}^{2(D+1-v-t)} \det(X^* X)$. Only the fixed rank v , not t , multiplies the exponential allowance for the raw low-degree correction. The main reduction below avoids that raw correction by retaining its exact two quotient metrics instead.

53 The residual observation and the smaller coefficient space

AKJ12–25 identifies the original conductor preimage as

$$V_A = \{p \in \mathcal{P}_{q-1} : \psi \mid \mathcal{T}_A p\}. \quad (\text{FEX22})$$

Its complete degree-adapted basis is $1, S, \dots, S^{v-1}$ followed by

$$P_h = \frac{(q' + h + v)!}{(q' + h)!} B_q(\partial_S)^{-1} \mathcal{J}_v(\psi(S) S^h), \quad 0 \leq h < a - v, \quad \mathcal{J}_v S^d = \frac{d!}{(d + v)!} S^{d+v}. \quad (\text{FEX23})$$

The variable in this inverse coefficient expression is identified with S' after applying the translation map. Each P_h is monic of degree $q' + v + h$. Its normalized conductor image is exactly

$$\mathcal{C}_A P_h = d_h \psi(S') S'^h, \quad d_h = \frac{(q' + h + v)!}{(q' + h)!}. \quad (\text{FEX24})$$

The full original residual matrix in this basis is $\Xi = (\Xi_{\text{lo}}, \Xi_{\text{hi}})$, of size j_k by a and rank j_k . Its kernel maps isomorphically onto the original K . These are the supplied actual coefficient maps, not independent matrix parameters. Put

$$r_{\text{lo}} = \text{rank } \Xi_{\text{lo}}, \quad t_0 = v - r_{\text{lo}} = \dim(K \cap L), \quad t = r - t_0. \quad (\text{FEX25})$$

Choose a quotient coordinate map Q_{lo} with kernel exactly $\text{im } \Xi_{\text{lo}}$, and a full-column-rank basis W_k of

$$\ker(Q_{\text{lo}} \Xi_{\text{hi}}) \subset \mathbb{C}^{a-v}. \quad (\text{FEX26})$$

The residual map has rank $j_k - r_{\text{lo}}$ because the full matrix has rank j_k . Hence the space in FEX26 has dimension t . These choices are made using actual nonzero minors and then kept fixed as the source endpoint changes. No generic minor or generic rank is substituted. Define the polynomial column matrix

$$Z_k = \text{diag}(d_h) W_k \quad \text{in the frame } (1, S', \dots, S'^{a-v-1}). \quad (\text{FEX27})$$

Then $\mathcal{C}_A K = \psi \text{ im } Z_k$. In fact, for a high coordinate z in FEX26 the equation $\Xi_{\text{lo}} \ell + \Xi_{\text{hi}} z = 0$ has a solution. Its different solutions differ by $K \cap L$; FEX24 kills that ambiguity and gives precisely ψZ_k . Thus this is the actual image, including the original unit and period dependence in Ξ and the conductor coefficients. The inverse finite-symbol factor no longer appears in the columns Z_k .

For every $j \geq 0$ define the exact transformed relation polynomial

$$R_j^A(S') = \frac{\mathcal{C}_A(\chi S^j)(S')}{\psi(S')} = \frac{v!}{\mu_v} \sum_{\nu} a_{\nu} \frac{\chi(S' + b_{\nu})}{\psi(S')} (S' + b_{\nu})^j. \quad (\text{FEX28})$$

Every displayed quotient is a polynomial. Indeed, if S' is the (a', b') inner-grid root, then $S' + b_{r_0 s_0}$ is the original root with indices $(a' + r_0, b' + s_0)$. All these indices are between 0 and k . The degree of R_j^A is $a - v + j$ and its leading coefficient is $(q + j)_v$; this follows from the first nonzero derivative in FEX17. It is nonzero, and these distinct degrees prove linear independence of the relation columns. In particular the high-source terms have not been replaced by an unjustified continuation of the low-degree symbol.

At endpoint N put $m_N^{\text{pol}} = N - v - q'$ and $d_N^{\text{rel}} = N + 1 - q$. Form the complete polynomial Gram $H_{N,s}^\psi$ of $(1, S', \dots, S'^{m_N^{\text{pol}}})$ in the measure $|\psi(c' + iy)|^2 dm_{0,s}(y)$ at $S' = c' + iy$. Zero-extend Z_k to this frame and let R_N^A have columns $R_0^A, \dots, R_{d_N^{\text{rel}}-1}^A$. Define

$$\begin{aligned}\widehat{H}_{K,N}^{(s)} &= Z_k^* H_{N,s}^\psi Z_k \\ &\quad - Z_k^* H_{N,s}^\psi R_N^A ((R_N^A)^* H_{N,s}^\psi R_N^A)^{-1} (R_N^A)^* H_{N,s}^\psi Z_k, \\ \widehat{F}_k^{(s)} &= \sum_{i=0}^3 \epsilon_i \log \det \widehat{H}_{K,N_i}^{(s)}, \\ (N_0, N_1, N_2, N_3) &= (q-1, q, 2q-1, 2q), \\ (\epsilon_0, \epsilon_1, \epsilon_2, \epsilon_3) &= (1, 1, -1, -1).\end{aligned}\tag{FEX29}$$

The empty relation inverse is absent at $N = q - 1$. The four relation ranks are still exactly $0, 1, q, q + 1$. They have not been truncated to the smaller kernel dimension. Positivity follows because a nonzero linear combination of Z_k has degree at most $a - v - 1$, whereas every nonzero combination of the R_j^A has degree at least $a - v$. The complete Gram is positive in the original positive Gamma density. Minimization over an increasing relation span proves

$$\widehat{H}_{K,q-1}^{(s)} \succeq \widehat{H}_{K,q}^{(s)} \succeq \dots, \quad \widehat{F}_k^{(s)} \geq 0.\tag{FEX30}$$

A change of the fixed basis W_k gives the same four-endpoint value; its four identical Gram factors cancel with the original signs.

54 A finite error for the actual compressed determinant

Let $K^+ = K + L \subset E$. This is only an auxiliary space for an exact metric calculation, not an enlargement of the observation kernel. All quotient and restriction metrics below are induced from the same original $G_N^{(s)}$ and their indicated fixed coefficient frames. Let $F_{K^+/L}$ and $F_{K^+/K}$ denote their four-endpoint volumes. The two block Gram factorizations of K^+ give

$$\boxed{F_K = F_{K^+/L} + F_L - F_{K^+/K}}.\tag{FEX31}$$

Before summing there are fixed basis Jacobians; they cancel only after all four signs. Both factorizations use the actual Schur complements. The two subquotient dimensions are $\dim L = v$ and $\dim(K^+/K) = v - t_0$. The existing full AKS covariance inequality passes by restriction and minimization to every fixed subquotient. Thus, with $\mathcal{L}_s = \mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}$ from AKS,

$$0 \leq F_L \leq v \mathcal{L}_s, \quad 0 \leq F_{K^+/K} \leq (v - t_0) \mathcal{L}_s.\tag{FEX32}$$

No angle condition between K and L is used.

For a fully explicit existing allowance one may use the AKS Newton bound in place of a sharper available $\mathcal{L}_s$. Write

$$\begin{aligned}R &= k \sqrt{\delta^2 + \gamma^2}, \quad t_q = \pi/2 - 1/q, \quad a_q = 2 \log 3 / t_q, \quad s_q = (a_q^q - 1) / (a_q - 1), \\ b_{s,q} &= [(9/25) \sin(1/q)]^{-s/2} \exp(2R(\log 3 + t_q/2)), \\ \mathcal{L}_{s,j} &= \log \left(1 + b_{s,q} s_q^2 \sum_{d=q}^{q+j-1} \frac{d!}{(s/2)_d} (25/16)^d \right).\end{aligned}\tag{FEX33}$$

These are the original complete coefficient bounds with both masses retained before norm-ratio cancellation. They satisfy $\mathcal{L}_s = O_{\delta,\gamma}(q)$ for $s = 1, k$ along this family. Indeed $\log s_q = O(q)$, the last geometric-weighted sum has logarithm $O(q + \log q)$ because $d!/(s/2)_d \leq 2\sqrt{d}$, and $\log b_{s,q} = O(k \log q + R) = o(q) + O(q)$.

Apply FEX6 to the source $\mathcal{P}_N/L$ and its conductor image. The relation subspace is $(\chi\mathcal{P}_{N-q} + L)/L$. The larger flag term is generated also by K , giving quotient dimension t . By FEX28, its image is the span of ψR_N^A and ψZ_k . Multiplication by ψ is an exact isometry from the indicated weighted coefficient space onto this polynomial image. The source quotient metric is exactly the metric of K^+/L , since successive minimization first by $\chi\mathcal{P}_{N-q}$ and then by L is minimization over their sum. Thus, in the fixed transported frames, put

$$\begin{aligned} e_N &= \log \det \widehat{H}_{K,N}^{(s)} - \log \det H_{K^+/L,N}^{(s)}, & n_N &= N + 1 - v, \\ e_{N,s}^- &= \zeta_{N,s} - 2(n_N - t)m_N, & e_{N,s}^+ &= 2tm_N, & e_{N,s}^- &\leq e_N \leq e_{N,s}^+, \\ D_s^- &= e_{q-1,s}^- + e_{q,s}^- - e_{2q-1,s}^+ - e_{2q,s}^+, & D_s^+ &= e_{q-1,s}^+ + e_{q,s}^+ - e_{2q-1,s}^- - e_{2q,s}^-. \end{aligned} \quad (\text{FEX34})$$

The signs on $e_{N,s}^\pm$ denote new finite scalar bounds; they do not redefine an original source scalar or covariance. All endpoint quantities are computed from FEX18–20 and the actual finite ranks in FEX25. Original frame factors have already been accounted for by transported quotient frames. Combining the four inequalities and FEX31–32 proves the central finite result:

$$\boxed{\widehat{F}_k^{(s)} - D_s^+ - (v - t_0)\mathcal{L}_s \leq F_K^{(s)} \leq \widehat{F}_k^{(s)} - D_s^- + v\mathcal{L}_s.} \quad (\text{FEX35})$$

This is a bound for the actual kernel determinant, not for a different ambient determinant. Every residual coefficient and relation cross term remains in FEX29.

For a simpler explicit error let

$$J_{\max,s} = \max_{i=0}^3 |\zeta_{N_i,s}|, \quad m_{\max} = m_{2q}, \quad \mathcal{E}_{k,s} = v\mathcal{L}_s + 2J_{\max,s} + 2(4q + 1)m_{\max}. \quad (\text{FEX36})$$

Then

$$\boxed{|F_K^{(s)} - \widehat{F}_k^{(s)}| \leq \mathcal{E}_{k,s} = O_{H,\delta,\gamma}(q \log q) = o(kq).} \quad (\text{FEX37})$$

To check the finite factors, the lower signed bound for $\sum \epsilon_i e_{N_i}$ is at least $-2J_{\max,s} - 2(2q + 1 - 2v)m_{\max}$, and the upper bound is at most $2J_{\max,s} + 2(4q + 1 - 2v)m_{\max}$. Enlarge to FEX36 and use FEX32. Also

$$m_{\max} \leq \log \max\{1, v!A_*^{\text{abs}}/|\mu_v|\} + 8\sqrt{\delta^2 + \gamma^2}(H_{2q} + 2). \quad (\text{FEX38})$$

Together with FEX20 and FEX33 this proves the order claim. All constants are fixed at the actual fixed period. Dividing by kq uses $\log q/k \rightarrow 0$. A uniform claim over periods approaching a locus where μ_v vanishes is not made: the displayed inverse coefficient must remain in the finite bound. Uniformity holds on a fixed coefficient stratum with this coefficient ratio bounded and the same v .

It follows in particular that $F_K^{(s)}/(kq)$ and $\widehat{F}_k^{(s)}/(kq)$ have exactly the same limit points. This does *not* assert that either sequence has a unique limit. The old uniform ceiling for F_K remains true, while FEX35–37 constitute a new, fixed-period transfer at strictly smaller error scale.

55 The original arithmetic and signed receiving identities

Let $[L_s^{\text{old}}, U_s^{\text{old}}]$ denote the current MRI4e interval, including its first compressed increment and the minimum of its existing finite upper bounds. The following is an add-only improvement at the same four endpoints:

$$\begin{aligned} L_s^{\text{new}} &= \max\{L_s^{\text{old}}, 0, \widehat{F}_k^{(s)} - D_s^+ - (v - t_0)\mathcal{L}_s\}, \\ U_s^{\text{new}} &= \min\{U_s^{\text{old}}, \widehat{F}_k^{(s)} - D_s^- + v\mathcal{L}_s\}, \\ L_s^{\text{new}} &\leq F_K^{(s)} \leq U_s^{\text{new}}, \quad s = 1, k. \end{aligned} \quad (\text{FEX39})$$

The intersection need not improve every finite instance. The comparison of scales, however, is unconditional at each fixed actual period in the stated domain.

The full original Gamma multiplier is

$$f_l(S) = \prod_{j=0}^{l-1} (S - c - 2j - \tfrac{1}{2}), \quad dm_{0,k} = \beta_k |f_l(c + iy)|^2 d\sigma, \quad \beta_k = \frac{(2\pi)^{k/2}}{\sqrt{2}\Gamma(k/2)}. \quad (\text{FEX40})$$

Its transported full-jet kernel, conditional covariance and mixed blocks are AMT23–30, unchanged. The existing difference interval is $-r\Omega_k^{\text{low}} \leq F_K^{(1)} - F_K^{(k)} \leq r\Omega_k^{\text{high}}$, where

$$\begin{aligned} \Pi_l &= \prod_{j=0}^{l-1} (2j + \tfrac{1}{2})^2, \quad U_{l,N}^{\text{poly}} = \prod_{j=0}^{l-1} [4(N + j + 1)^2 + (2j + \tfrac{1}{2})^2], \\ \Omega_k^{\text{low}} &= \log \frac{U_{l,q-1}^{\text{poly}} U_{l,q}^{\text{poly}}}{\Pi_l^2}, \quad \Omega_k^{\text{high}} = \log \frac{U_{l,2q-1}^{\text{poly}} U_{l,2q}^{\text{poly}}}{\Pi_l^2}. \end{aligned} \quad (\text{FEX41})$$

Consequently one may further intersect the tensor-source interval with $[L_1^{\text{new}} - r\Omega_k^{\text{high}}, U_1^{\text{new}} + r\Omega_k^{\text{low}}]$. The fixed-source interval has the corresponding reverse intersection. No independent error assignment is substituted for their common total.

For the arithmetic measure retain exactly

$$w_h(y) = |v_h(\tfrac{1}{2} + iy)|^2 / (2\pi), \quad dm_{h,k}(y) = w_h^{*k}(y) dy, \quad \delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma. \quad (\text{FEX42})$$

Let $H_N^{\text{ar}}, H_N^\sigma$ be the full coefficient-source Grams. AMT6 retains $(a_k/D_N)H_N^\sigma \preceq H_N^{\text{ar}} \preceq A_k H_N^\sigma$. For $m = 1$ its original constants are $p = 7/2$, $B_h = 50$, $M = 25$, $\alpha = \pi/2$, $D_N = [1 + 4(N + M)^2]^M$, and

$$a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p+1} M_\alpha^{k-1}. \quad (\text{FEX43})$$

Here $\vartheta_h = \int_{-1}^1 w_h$ and $M_\alpha = \int e^{\alpha y} w_h(y) dy$ are the original integrals, and the named positive envelope constants are the unchanged TW/BT constants. In particular none is inferred from a diagnostic packet.

Keep the AMT exact deficits

$$\begin{aligned} L_N^{\text{ar}} &= \log(A_k D_N / a_k), \quad d_{E,N} = q \log A_k - \log \frac{\det G_N^{\text{ar}}}{\det G_N^\sigma}, \\ c_{X,N} &= \min\{r_X L_N^{\text{ar}}, d_{E,N}\}, \quad C_X^- = c_{X,q-1} + c_{X,q}, \quad C_X^+ = c_{X,2q-1} + c_{X,2q}, \quad X = K, B. \end{aligned} \quad (\text{FEX44})$$

The source-evaluated caps replacing $d_{E,N}$ by $\Xi_{k,N}$ remain available. The complete nested positive deficits and their exact common sum are those of AMT13–18. Using either version of the caps, FEX39 therefore gives precisely the new MRI4f interval

$$\begin{aligned} L_{\text{ar}}^{\text{new}} &= \max\{0, L_1^{\text{new}} - C_K^-, L_1^{\text{new}} + \delta_k^\sigma - C_B^+\}, \\ U_{\text{ar}}^{\text{new}} &= \min\{\mathcal{B}_k^{\text{ar}}, U_1^{\text{new}} + C_K^+, U_1^{\text{new}} + \delta_k^\sigma + C_B^-\}, \\ L_{\text{ar}}^{\text{new}} &\leq F_K^{\text{ar}} \leq U_{\text{ar}}^{\text{new}}. \end{aligned} \quad (\text{FEX45})$$

Subtracting the appropriate kernel endpoints from the unchanged total $\mathcal{B}_k^{(s)}$ or $\mathcal{B}_k^{\text{ar}}$ gives the exact boundary interval. This keeps correlation with the actual signed arithmetic value. Its QRI/QGQ finite intervals, including the simple-multiplicity nonzero radius, are not altered by this calculation.

The BRI6d identity remains, with its original orientation,

$$\begin{aligned} F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, \quad \mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^0 = -\mathfrak{T}_k, \\ L_k^{\text{new}} + \mathcal{H}_k + \delta_k^\sigma &\leq F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} \leq U_k^{\text{new}} + \mathcal{H}_k + \delta_k^\sigma. \end{aligned} \quad (\text{FEX46})$$

The already supplied source differences are $o(kq)$, and the complete Gamma return has $\mathcal{H}_k/(kq) \rightarrow -\mathcal{C}_\Gamma/4$ with the unchanged positive equilibrium constant. Consequently the new reduction has the exact leading-scale receiving form

$$\boxed{\frac{F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}}{kq} = \frac{\widehat{F}_k^{(1)}}{kq} - \frac{\mathcal{C}_\Gamma}{4} + o(1).} \quad (\text{FEX47})$$

In particular $F_K^{\text{ar}}/(kq)$, $F_K^{(1)}/(kq)$, $F_K^{(k)}/(kq)$, and both new $\widehat{F}$ ratios have the same limit points. The leading value of FEX29 is still needed to decide the sign in FEX47. Neither the old ceiling nor the new $o(kq)$ transfer assigns that value.

56 Actions, representatives, and the retained mixed-support maps

The conductor is a coefficient calculation, not a replacement for the arithmetic action or the original observation. Its precise action defect on polynomials is

$$\mathcal{C}_A(Sp) - S'\mathcal{C}_{AP} = \frac{v!}{\mu_v} \sum_{\nu} a_{\nu} b_{\nu} p(S' + b_{\nu}). \quad (\text{FEX48})$$

For the representative of degree less than q , write $\tau(p) = [S^{q-1}]p$. The original action is $A_{\chi}[p] = [Sp - \chi\tau(p)]$. Its transported polynomial defect is the right side of FEX48 minus $\mathcal{C}_A(\chi\tau(p))$. This formula is in the full polynomial target; divisibility by ψ is not asserted for an action image outside the actual conductor preimage. In particular K is not assumed invariant, and the original OKA9 observation-action defect is retained.

Nor is the auxiliary L an invariant subspace of that action when $v > 0$. On its top monomial S^{v-1} , multiplication followed by $\mathcal{C}_A$ gives the nonzero constant $v!$, whereas all its lower monomials give zero. This is the explicit rank-one defect. No quotient Euler action on K^+/L was needed in the metric proof.

One can retain all coefficients while using the calculation. Let $R_{C,N}$ be any of the explicitly specified finite right inverses of $\mathcal{C}_A$ on the full space $\mathcal{P}_{N-v}$, for example $(B^{[N]}(\partial_S))^{-1}\mathcal{J}_v$. Then

$$p \mapsto (p - R_{C,N}\mathcal{C}_{AP}, \mathcal{C}_{AP}) \in L \oplus \mathcal{P}_{N-v}, \quad (\ell, g) \mapsto \ell + R_{C,N}g \quad (\text{FEX49})$$

are inverse coefficient maps. This explicitly retains the low component rather than treating it as zero. The original remainder map, full unit, and literal period observation may be composed with the inverse in FEX49; they are then the same original maps on the same class, not newly assigned maps on a smaller fibre. The volume identity FEX31 restores both low-component quotient metrics exactly.

If an original minimum section is changed by χQ , its original cochain primitive is still the AKP primitive. Namely divide $\chi(s_1 + \dots + s_k)$ successively by the original $h(s_j)$ to obtain $\sum_j h(s_j)Q_j(\mathbf{s})$. Then use the original tensor differential and the primitive

$$\sum_{j=1}^k (-1)^{j-1} Q_j(\mathbf{D}) Q(D_1 + \dots + D_k) (F_h^{\otimes(j-1)} \otimes \phi_* \otimes F_h^{\otimes(k-j)}). \quad (\text{FEX50})$$

Its differential is the literal difference of the original theta representatives, because $h(D)F_h = \Theta\phi_*$. Minus-only legs use the unchanged lift $(0, -\mathcal{F}_{\text{Fourier}}\phi)$ and transported $1-D$ action. This statement applies to an actual relation difference; no element of K or L is newly declared a theta boundary. All support-face insertions and signed coefficient permutations commute with the reconstructed linear maps componentwise. A supported zero stays at its original label and is not identified with the external τ . The additional proper-source Taylor kernels from the mixed-primary lane remain, not being replaced by a conductor kernel.

The linear-algebra and Gamma-polynomial lemmas apply at any finite multiplicity with their actual dimensions. The explicit OCF conductor, its four additional orbit factors, and the rank $8k - 16$ used for FEX22–47 belong only to the stated simple-quartet case. No such rank or frame is assigned to the full-multiplicity OMG case. Its nilpotents, row-residue annihilator, and observed-iterate kernel remain the supplied ones. This continuation makes no Frobenius identification or arithmetic purity assertion.

57 Scope of verification and the next exact quantity

The analytic assertions proved here are FEX6–21 and the exact transfer FEX22–47 from the explicitly named supplied inputs. Finite exact fixtures test the flag identity, its inverse determinant bounds, the source Jacobian, and the full four-endpoint reduction with nonzero cross terms. They do not evaluate actual xi zeros or actual period coefficients. The accompanying execution record distinguishes those computations from the written proofs and from inherited formal certificates.

The next unresolved quantity is the fixed-period determinant FEX29, with the actual Z_k , R_j^A , and weighted original Gamma Gram. Its leading value is *not* established by this paper. It is now separated from the finite-symbol inverse with an explicit error smaller than kq . In particular, repeating an operator inverse bound of exponential size and raising it to r is no longer the remaining task. The complete original high relation ranks, their mixed Gram blocks, and the period position of $\text{im } Z_k$ are the data that still need estimation. No RH contradiction or disproof follows from the result proved here.

Source dependencies

The fixed objects and current ranks are the supplied OCS/OCF/OKM/OKA, AKJ12–25, and RPC/AMT statements in `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS.tex`. The retained inverse is AIE1–29; the original covariance bound is AKS20–32. The signed source transfer is AMT13–49 and the unchanged GEL/QGT/QRI/QGQ provider in `02_FULL_GAMMA_PROOFS(1).tex`. All complete current source files and their exact hashes accompany this continuation. PR32’s residue and finite observation certificates are inherited inputs only; no new Lean execution is claimed here.

58 The fixed-column conductor quotient and the full GIP graph

Fix exactly the original simple-quartet and fixed-period data of FEX1–4, CEP1–6 and GIP1–3. In particular $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = q - q'$, $0 \leq v \leq 80$, $\chi = \chi_k$, $\psi = \chi_{k-8}$, $c = k/2$, $c' = c - 4$, and $s \in \{1, k\}$. Retain the exact nonzero scalar

$$\alpha_A = \frac{v!}{\mu_v}, \quad \mathcal{C}_A = \alpha_A \mathcal{T}_A. \quad (\text{FGC1})$$

Every polynomial norm is the original Gamma norm at its stated centre, with the full mass $(2\pi)^{s/2}$. The scalar in FGC1 multiplies the specified coefficient map and does not alter the measure.

For each $N \in \{q - 1, q, 2q - 1, 2q\}$ put $t_N = N + 1 - q$ and $b_N = N - v - q'$. Use the exact GIP coefficient space, now with a different name to keep t_N distinct from a kernel dimension:

$$\mathcal{X}_N = \mathbb{C}^v \oplus \mathbb{C}[S']_{\leq b_N}, \quad L_N h = P_{<v} Z_{\chi, N} h, \quad U_N^A h = \frac{\mathcal{T}_A(\chi h)}{\psi} \quad (h \in \mathbb{C}[S]_{\leq N-q}). \quad (\text{FGC2})$$

Here $P_{<v}$ gives the first v original upper orthonormal coordinates, $Z_{\chi, N} h$ is the complete upper coordinate column of χh , and U_N^A is exactly GIP3, equivalently CEP41–42. Thus the relation subspace of GIP is $\{(L_N h, U_N^A h) : \deg h \leq N - q\}$. The top coordinate is retained in every formula below.

Define the explicit coefficient isomorphism and the transformed graph

$$D_{\alpha, N}(\ell, z) = (\ell, \alpha_A z), \quad D_{\alpha, N}^{-1}(\ell, z) = (\ell, \alpha_A^{-1} z), \quad \mathcal{G}_N^\alpha = \{(L_N h, \alpha_A U_N^A h) : \deg h \leq N - q\}. \quad (\text{FGC3})$$

The complete GIP graph quotient is carried by $D_{\alpha, N}$ to $\mathcal{X}_N / \mathcal{G}_N^\alpha$. Its map from the original conductor kernel V_A sends the class of a polynomial p to

$$[p] \mapsto \left[\left(P_{<v} p, \frac{\mathcal{C}_A p}{\psi} \right) \right]. \quad (\text{FGC4})$$

The second coordinate exists exactly for $[p] \in V_A$, by CEP5–6. Changing p by χh adds the displayed member of $\mathcal{G}_N^\alpha$ and therefore does not change its class. The inverse is the GIP inverse E_N^{graph} composed with $D_{\alpha, N}^{-1}$, so FGC4 is an isomorphism. This proves the relation to the full earlier graph at each of its actual cutoffs.

58.1 The precise projection and its complete kernel

Define the lower coefficient quotient

$$\mathcal{Y}_N = \mathbb{C}[S']_{\leq b_N} / \text{im}(\alpha_A U_N^A), \quad \pi_N : \mathcal{X}_N / \mathcal{G}_N^\alpha \rightarrow \mathcal{Y}_N, \quad [(\ell, z)] \mapsto [z]. \quad (\text{FGC5})$$

The map is well defined because its relation images are exactly $\text{im}(\alpha_A U_N^A)$, and it is onto because every lower column has the lift $(0, z)$. If its value is zero, write $z = \alpha_A U_N^A h$ and subtract the full graph vector $(L_N h, \alpha_A U_N^A h)$. The resulting representative is $(\ell - L_N h, 0)$. A vector $(\ell, 0)$ belongs to the graph only if $U_N^A h = 0$. This implies $h = 0$: for nonzero h , the degree of $\mathcal{T}_A(\chi h)$ is $q + \deg h - v$, with nonzero leading coefficient $\text{lc}(h) \mu_v \binom{q + \deg h}{v}$, by the first nonzero moment identity. Therefore $\ell = 0$. We have proved the full exact sequence, including the original low coordinate and every relation component,

$$0 \rightarrow \mathbb{C}^v \xrightarrow{\ell \mapsto [(\ell, 0)]} \mathcal{X}_N / \mathcal{G}_N^\alpha \xrightarrow{\pi_N} \mathcal{Y}_N \rightarrow 0. \quad (\text{FGC6})$$

Under FGC4 the first term is exactly the original subspace $L = \mathbb{C}[S]_{<v} \subset E_k$, because the conductor kills exactly those polynomials and its first coordinate is their original orthonormal coordinate. Thus FGC6 is, on the same original primary space, the quotient sequence

$$0 \rightarrow L \rightarrow V_A \rightarrow V_A / L \rightarrow 0, \quad V_A / L \xrightarrow{\cong} \mathcal{Y}_N. \quad (\text{FGC7})$$

In particular its target has dimension $\Delta - v$ at each original endpoint. The inverse to the last map is completely specified: choose a representative z , solve $\mathcal{C}_A p = \psi z$ by the finite right inverse of FEX17, and take $[p]$ modulo L and the source relations. A different representative changes p by an original χh plus an element of L , so the same target class results.

Now retain $K = \ker \Lambda_k \subset V_A$, put $t_0 = \dim(K \cap L)$, and set $K^+ = K + L$. Restricting FGC7 gives

$$0 \longrightarrow K \cap L \longrightarrow K \longrightarrow \pi_N(K) \longrightarrow 0, \quad \pi_N(K) \simeq K^+/L, \quad \dim \pi_N(K) = r - t_0. \quad (\text{FGC8})$$

The second isomorphism sends $k + L$ to the same image as k . It is onto, and a zero image says $k \in L$, which proves injectivity on K^+/L . No preservation of an Euler or multiplication action is asserted by these coefficient quotient maps.

58.2 The actual fixed columns and the Schur metric

Use the original AKJ basis and residual matrix $\Xi = (\Xi_{\text{lo}}, \Xi_{\text{hi}})$ from FEX23–26. Let W_k be its specified basis of $\ker(Q_{\text{lo}}\Xi_{\text{hi}})$ and $d_h = (q' + h + v)!/(q' + h)!$. The first nonzero moment identity gives $\mathcal{C}_A P_h = d_h \psi S^{jh}$ for every original high AKJ column. The low AKJ columns are precisely L and are killed by $\mathcal{C}_A$. Thus the exact image in FGC8 has the fixed, endpoint-independent representative matrix

$$Z_k = \text{diag}(d_h)W_k, \quad \pi_N(K) = (\text{im } Z_k + \text{im}(\alpha_A U_N^A))/\text{im}(\alpha_A U_N^A). \quad (\text{FGC9})$$

Solvability of $\Xi_{\text{lo}}\ell + \Xi_{\text{hi}}z = 0$ is exactly the equation $Q_{\text{lo}}\Xi_{\text{hi}}z = 0$, proving both inclusions in this identity. The graph's lower relation columns are

$$R_j^A = \alpha_A U_N^A(S^j) = \frac{\mathcal{C}_A(\chi S^j)}{\psi}, \quad 0 \leq j < t_N. \quad (\text{FGC10})$$

This is FEX28, and also the lower block of the complete original relation graph in FGC3. The nonzero leading coefficient is $(q + j)_v$ and its degree is $\Delta - v + j$. The columns of Z_k have degree at most $\Delta - v - 1$. Consequently their quotient classes are independent at every cutoff, including the two high relation ranks q and $q + 1$.

Give the lower polynomial space its exact norm $\int |z(c' + iy)|^2 |\psi(c' + iy)|^2 dm_{0,s}(y)$, whose full Gram in the increasing monomials is $H_{N,s}^\psi$. Then the quotient metric of FGC5 on FGC9 is exactly

$$\widehat{H}_{K,N}^{(s)} = Z_k^* H_{N,s}^\psi Z_k - Z_k^* H_{N,s}^\psi R_N^A ((R_N^A)^* H_{N,s}^\psi R_N^A)^{-1} (R_N^A)^* H_{N,s}^\psi Z_k. \quad (\text{FGC11})$$

Indeed minimizing $(Z_k x + R_N^A h)^* H_{N,s}^\psi (Z_k x + R_N^A h)$ over h gives the unique minimizer $h = -((R_N^A)^* H_{N,s}^\psi R_N^A)^{-1} (R_N^A)^* H_{N,s}^\psi Z_k x$. The relation Gram is positive by the independence in FGC10, and substitution proves FGC11 including both mixed factors. At $t_N = 0$ the minimization has no variable and the second term is absent. This proves the FEX29 matrix directly from the full GIP relation graph by the explicit quotient FGC5. It has not identified the transformed relation span with a chosen polynomial ideal.

The original exact metric in these normalized graph coordinates is $D_{\alpha,N}^{-*} H_N^{\text{actual}} D_{\alpha,N}^{-1}$, with H_N^{actual} from GIP5. It is generally different from the native lower metric in FGC11. On the whole normalized graph space the native direct-sum metric is $\text{diag}(I_v, H_{N,s}^\psi)$. Taking its quotient first by $\mathcal{G}_N^\alpha$ and then by the low image is the same minimization as taking the quotient by their sum. In that sum the low coordinate is unrestricted, so its minimum is zero, and the second-coordinate minimization is precisely FGC11. Thus the native-metric projection also matches the stated Schur matrix; the comparison to the original metric still uses FEX34–35 or CEP's full source-map estimates. The scalar α_A has remained in the coordinate map and both metrics throughout.

58.3 Both original low metric terms and their signed identity

For any fixed positive original endpoint metric $G_N^{(s)}$ on E_k , restrict it to K^+ . In fixed coefficient frames the two complete Schur factorizations give

$$\det H_{K^+,N} = J_1 \det H_{L,N} \det H_{K^+/L,N} = J_2 \det H_{K,N} \det H_{K^+/K,N}, \quad (\text{FGC12})$$

where, if A_1, A_2 are the matrices of the two adapted frames in the fixed reference frame of K^+ , then $J_1 = |\det A_1|^{-2}$ and $J_2 = |\det A_2|^{-2}$. They are independent of the endpoint. To prove each equality, complete the chosen subspace frame by fixed quotient lifts, write the full positive Gram in that adapted frame, and eliminate its off-diagonal block. The quotient block is exactly the minimum Gram because solving its first-block linear equation is the same minimization. Its determinant times the first block's determinant is the determinant in the adapted frame. This proves the formula including its fixed Jacobian.

Taking the four original signs $(1, 1, -1, -1)$ cancels these two frame constants, and gives exactly

$$F_K^{(s)} = F_{K^+/L}^{(s)} + F_L^{(s)} - F_{K^+/K}^{(s)}, \quad \dim L = v, \quad \dim(K^+/K) = v - t_0. \quad (\text{FGC13})$$

The original AKS endpoint covariance comparison passes to a restriction by evaluating the quadratic forms on that subspace, and passes to a quotient by taking the infimum on the same fibre. Its scalar two-sided bounds therefore give FEX32 for these actual subquotients. In particular both nonnegative low-volume terms in FGC13 remain; a projection to FGC11 alone does not remove them. FEX34–37 additionally compares $F_{K^+/L}$ with $\widehat{F}_k$ by the original conductor's complete Jacobian, proving the stated $O_{\text{actual}}(q \log q)$ error.

Equations FGC3–13 exhibit exactly how the new fixed-column presentation fits the existing CEP/GIP construction. The existing RUI5–15 and CEP29–39 already prove the $o(kq)$ source/exterior transport. FEX adds an explicit realization on the lower fixed columns Z_k , with the full original transformed relation span and the two original low metric corrections restored. These equations do not turn that span into the CTV outer ideal or remove the actual residual observation.

59 The exact closed equilibrium coefficient

This section retains the potential, measure, and energy of EIQ1–33:

$$\begin{aligned} V_{\alpha,\beta}(x) &= \beta\sqrt{x} - \alpha \log x, & \alpha > 0, \quad \beta > 0, \quad x > 0, \\ \mu_{\alpha,\beta}(dx) &= \rho_{\alpha,\beta}(x) dx, & F(\alpha, \beta) &= \iint \log|x-y| d\mu_{\alpha,\beta}(x) d\mu_{\alpha,\beta}(y) - \int V_{\alpha,\beta} d\mu_{\alpha,\beta}, \\ \mathcal{L}(\alpha, \beta) &= \int \log x d\mu_{\alpha,\beta}(x). \end{aligned} \quad (\text{ECL1})$$

The preceding complete EIQ proof supplies the unique minimizing probability measure, its positive density and exact support $[a, b] = [u^2, v^2]$, and its Euler equality. In particular, no assertion in this section changes the minimization problem or chooses another measure. The endpoints are exactly

$$uK(k) = \frac{\alpha\pi}{\beta}, \quad vE(k) = \frac{(\alpha+2)\pi}{\beta}, \quad k^2 = 1 - \frac{u^2}{v^2}, \quad 0 < u < v. \quad (\text{ECL2})$$

Here K and E have the modulus convention of EIQ2, not the elliptic-parameter convention. Define the original endpoint quantities

$$w = \frac{u+v}{2}, \quad z = uv, \quad c_{\text{cap}} = \frac{v^2 - u^2}{4}, \quad d\nu(x) = \frac{dx}{\pi\sqrt{(b-x)(x-a)}} \quad (a < x < b). \quad (\text{ECL3})$$

The letter z in ECL3 is a positive scalar. A Cauchy-transform variable below is denoted by λ . The measure ν has mass one by $x = (a+b)/2 + (b-a)\cos\theta/2$, $0 \leq \theta \leq \pi$.

59.1 Arcsine identities with their original constants

Put $m = (a+b)/2$, $d = (b-a)/2$, and $\eta = (v-u)/(v+u)$. Then $0 < \eta < 1$ and

$$x = m + d \cos \theta = w^2(1 + 2\eta \cos \theta + \eta^2), \quad z = w^2(1 - \eta^2).$$

The absolutely uniformly convergent series

$$\begin{aligned} \log x &= 2 \log w + 2 \sum_{n \geq 1} \frac{(-1)^{n+1} \eta^n}{n} \cos(n\theta), \\ \frac{1}{x} &= \frac{1}{z} \left(1 + 2 \sum_{n \geq 1} (-\eta)^n \cos(n\theta) \right) \end{aligned}$$

follow from the geometric series and the integral of that series. Termwise integration, and orthogonality of the cosines, prove

$$\begin{aligned} \int \log x d\nu(x) &= 2 \log w, & \int \frac{d\nu(x)}{x} &= \frac{1}{z}, \\ \int \frac{\log x}{x} d\nu(x) &= \frac{2 \log w + 2 \log(1 - \eta^2)}{z} = \frac{2 \log z - 2 \log w}{z}. \end{aligned} \quad (\text{ECL4})$$

In the product of the two Fourier series, the contribution of degree $n \geq 1$ to the constant coefficient is $-2\eta^{2n}/(nz)$, giving the displayed logarithm without an omitted constant.

The logarithmic potential of ν on its entire support is

$$\int_a^b \log|x-t| d\nu(x) = \log c_{\text{cap}}, \quad a \leq t \leq b. \quad (\text{ECL5})$$

Here is a direct proof, including the endpoint values. Write $t = m + d \cos \phi$ and use

$$|m + d \cos \theta - t| = \frac{d}{2} |e^{i\theta} - e^{i\phi}| |e^{i\theta} - e^{-i\phi}|.$$

The integrand depending on $\cos \theta$ has the same mean on $[0, \pi]$ as on $[0, 2\pi]$. For $0 < r < 1$, the power series of $\log |1 - re^{i\theta}|$ has zero circle mean. This mean tends to the mean at $r = 1$: near its only singular angle, $|1 - re^{i\theta}|^2 = (1 - r)^2 + 2r(1 - \cos \theta)$, so for $r \geq 1/2$ the absolute value of its negative logarithm is bounded by $C + |\log |\theta||$, and its positive logarithm is bounded by $\log 2$. These are integrable bounds. A rotation proves that each of the two circle means above is zero, for every ϕ , including $\phi = 0, \pi$. Hence their sum is $\log(d/2) = \log c_{\text{cap}}$, proving ECL5.

All logarithmic double integrals used below are absolutely integrable. In fact the absolute logarithm for two arcsine variables is bounded by a constant plus

$$\left| \log \left| \sin \frac{\theta - \phi}{2} \right| \right| + \left| \log \left| \sin \frac{\theta + \phi}{2} \right| \right|.$$

Each term is integrable on $[0, \pi]^2$; near a zero, use $\sin t \geq 2t/\pi$ for $0 \leq t \leq \pi/2$, and integrate $|\log t|$. Thus the same assertion holds for any two signed measures whose absolute values are bounded by constant multiples of ν . The EIQ density also has this domination, because $\rho(x)/[1/(\pi\sqrt{(b-x)(x-a)})]$ is bounded on $[a, b]$.

Let ℓ have the exact EIQ17 convention

$$V_{\alpha,\beta}(x) - 2 \int \log |x - t| d\mu_{\alpha,\beta}(t) = \ell \quad (a \leq x \leq b).$$

Integrating this equality against ν , using ECL4–5 and EIQ7, gives

$$\boxed{\ell = 2(\alpha + 2) - 2\alpha \log w - 2 \log c_{\text{cap}}.} \quad (\text{ECL6})$$

Indeed $\int \sqrt{x} d\nu(x) = 2vE(k)/\pi$, whose product with β is $2(\alpha + 2)$ by ECL2. Fubini is justified by the absolute integrability just proved and the unit mass of μ . Smoothness of ℓ now follows from the endpoint smoothness in EIQ6.

59.2 Differentiation with the moving endpoints retained

All derivatives in this subsection are with respect to α at fixed β . Put

$$p_\alpha(x) = \frac{h_{\alpha,\beta}(x)}{2\pi}, \quad D(x) = (b - x)(x - a).$$

In this subsection a dot denotes a parameter derivative, so that $\dot{p}(x)$ differentiates the function $p_\alpha(x)$ at fixed x . The function $h_{\alpha,\beta}$ is exactly EIQ10; the subscript on p_α labels the parameter and does not denote differentiation. EIQ10 and EIQ29 prove that p_α and its parameter and spatial derivatives are bounded on a fixed positive compact neighbourhood of the supports when parameters range in a sufficiently small compact neighbourhood.

The derivative of the probability measure, interpreted by integration against every continuously differentiable function on this neighbourhood, is the finite signed measure

$$\sigma(dx) = \left[\dot{p}(x) \sqrt{D(x)} + \frac{p_\alpha(x)}{2\sqrt{D(x)}} \{ \dot{b}(x - a) - \dot{a}(b - x) \} \right] \mathbf{1}_{(a,b)}(x) dx, \quad (\text{ECL7})$$

$$\frac{\partial}{\partial \alpha} \int f d\mu_{\alpha,\beta} = \int f d\sigma, \quad \sigma([a, b]) = 0.$$

This identity includes the moving support. To prove it without postulating endpoint terms, set $\delta = b - a$ and $x = a + \delta y$. The measure in the fixed variable y is

$$\delta^2 p_\alpha(a + \delta y) \sqrt{y(1 - y)} dy.$$

Its differentiated integral against f is

$$\int_0^1 \sqrt{y(1-y)} \left[2\delta\dot{\sigma} p_\alpha f + \delta^2(\dot{p} + p'_\alpha \dot{x})f + \delta^2 p_\alpha f' \dot{x} \right] dy, \quad \dot{x} = \dot{a} + \dot{\delta}y.$$

Replace $\delta f'$ in the last summand by $df(x)/dy$ and integrate by parts. Its boundary term is $[\delta p_\alpha(x)\dot{x}\sqrt{y(1-y)}f(x)]_0^1 = 0$. The remaining terms combine, on returning to x , to $\dot{p}\sqrt{D} + p_\alpha \dot{D}/(2\sqrt{D})$, where $\dot{D} = \dot{b}(x-a) - \dot{a}(b-x)$, exactly ECL7. This integration by parts is justified first on $[\varepsilon, 1-\varepsilon]$ and then by passage to the limit: the boundary expression tends to zero and the differentiated square-root factor is bounded by a constant times $[y(1-y)]^{-1/2}$, an integrable function. The differentiated fixed-interval integral is valid by the uniform bounds above. Taking $f = 1$ proves the zero total mass in ECL7. The displayed density also proves $|\sigma| \leq C\nu$. In particular no endpoint atom has appeared or been discarded.

For fixed $x \in (a, b)$, differentiating the Euler equality gives

$$-\log x - 2 \int \log|x-t| d\sigma(t) = \dot{\ell}. \quad (\text{ECL8})$$

For precision, choose a smooth function of t equal to one near x , with support in a common interior subinterval for nearby parameters. On this subinterval the density and its parameter derivative are uniformly bounded, so differentiation of its logarithmic integral follows by domination by $C|\log|x-t||$. The complementary integrand is a continuously differentiable test function, so ECL7 applies. This proves ECL8 despite the logarithmic singularity and the moving endpoints.

We next identify σ exactly:

$$\boxed{\frac{\partial}{\partial \alpha} \mu_{\alpha, \beta}(dx) = \sigma(dx) = \frac{1}{2} \left(1 - \frac{z}{x}\right) d\nu(x).} \quad (\text{ECL9})$$

The derivative has the test-function meaning proved in ECL7, which in particular applies to the logarithmic and square-root moments because $a > 0$.

Here is an explicit uniqueness argument establishing ECL9. Denote its proposed right side by σ_0 . By ECL4 it has zero total mass, and $|\sigma_0| \leq C\nu$. Retain the branch $R(\lambda) = \sqrt{(\lambda-a)(\lambda-b)}$ with $R(\lambda)/\lambda \rightarrow 1$ at infinity, exactly as in EIQ12. The arcsine transform proved there and partial fractions yield

$$\begin{aligned} \int \frac{d\sigma_0(t)}{\lambda-t} &= \frac{1}{2R(\lambda)} - \frac{z}{2\lambda} \left(\frac{1}{z} + \frac{1}{R(\lambda)} \right) \\ &= \frac{1-z/\lambda}{2R(\lambda)} - \frac{1}{2\lambda}. \end{aligned} \quad (\text{ECL10})$$

There is no pole at zero: $R(0) = -uv = -z$, so the numerator $(\lambda-z)/R(\lambda) - 1$ vanishes there. The integral itself is analytic in a neighbourhood of zero, as is also immediate from $t \geq a > 0$. For $a < x < b$, the real principal value of the arcsine transform is zero, because $R(x+i0)$ is purely imaginary. This limiting principal value also follows directly by subtracting the local value of the smooth interior density in its Cauchy integral. Consequently ECL10 gives

$$\text{PV} \int \frac{d\sigma_0(t)}{x-t} = -\frac{1}{2x}.$$

Differentiating the logarithmic potential on each interior subinterval is valid by that same local subtraction; the endpoint parts are integrable and are a positive distance away. It follows that $-\log x - 2 \int \log|x-t| d\sigma_0(t)$ is constant on (a, b) . Subtract this identity from ECL8. For $\sigma_1 = \sigma - \sigma_0$, the logarithmic potential $\int \log|x-t| d\sigma_1(t)$ is constant on (a, b) . Since σ_1 has zero mass and no endpoint atoms, and $|\sigma_1| \leq C\nu$, absolute integrability proved after ECL5 gives

$$-\iint \log|x-t| d\sigma_1(x) d\sigma_1(t) = 0.$$

The exact zero-mass energy argument EIQ20–22, including its Fourier-uniqueness proof, forces $\sigma_1 = 0$. This establishes the claimed derivative as a measure, with every endpoint contribution in ECL7 retained.

59.3 The moment and energy formulas

Integrate ECL8 against ν . By ECL5 and $\sigma([a, b]) = 0$, the iterated logarithmic term is zero. Therefore

$$\dot{\ell} = -2 \log w. \quad (\text{ECL11})$$

All exchanges again have the absolute integrability established after ECL5; the equality ECL8 holds ν -almost everywhere. Using ECL9 with $f(x) = \log x$, and then ECL4, proves

$$\mathcal{L}_\alpha = \frac{1}{2} \int \left(1 - \frac{z}{x}\right) \log x \, d\nu(x) = 2 \log w - \log z = \log(w^2/z). \quad (\text{ECL12})$$

Write $I = \iint \log |x - t| \, d\mu(x) d\mu(t)$, with the original unscaled logarithmic kernel. The EIQ31 dilation identity is $\beta \int \sqrt{x} \, d\mu = 2(\alpha + 1)$. Integration of the Euler equality against μ gives $\ell = \int V \, d\mu - 2I$. Combining this with $F = I - \int V \, d\mu$ yields the exact identity

$$\ell = -2F - 2(\alpha + 1) + \alpha \mathcal{L}. \quad (\text{ECL13})$$

EIQ30 proves $F_\alpha = \mathcal{L}$. Differentiating ECL13 at fixed β , with the differentiability just proved, gives $\dot{\ell} = -\mathcal{L} - 2 + \alpha \mathcal{L}_\alpha$. Substituting ECL11–12 now proves, without an integration constant or a limiting parameter choice,

$$\mathcal{L} = 2(\alpha + 1) \log w - \alpha \log z - 2, \quad F = -\frac{\ell}{2} - (\alpha + 1) + \frac{\alpha \mathcal{L}}{2}. \quad (\text{ECL14})$$

Equations ECL6, ECL12, and ECL14 therefore evaluate the same Robin constant, logarithmic moment, and variational energy as the original EIQ integrals.

59.4 The original outer coefficient and the two scale conventions

The coefficient of OFV29 is defined by its full original expression

$$\mathfrak{C}_\partial = 4 \log(4/\vartheta) - 10 - 2F(2, \vartheta) + 4\mathcal{L}(2, \vartheta), \quad \vartheta = \pi/2. \quad (\text{ECL15})$$

To express this same number at the original EIQ limiting potential $V_{2,\pi}(x) = \pi\sqrt{x} - 2 \log x$, retain the actual dilation $x \mapsto (\pi/\beta)^2 x$ of EIQ32. Its effect on the interaction, potential integral, and moment is, respectively,

$$\begin{aligned} I(\alpha, \beta) &= I(\alpha, \pi) + 2 \log(\pi/\beta), \\ \int V_{\alpha,\beta} \, d\mu_{\alpha,\beta} &= \int V_{\alpha,\pi} \, d\mu_{\alpha,\pi} - 2\alpha \log(\pi/\beta), \\ \mathcal{L}(\alpha, \beta) &= \mathcal{L}(\alpha, \pi) + 2 \log(\pi/\beta). \end{aligned}$$

Thus $F(\alpha, \beta) = F(\alpha, \pi) + 2(\alpha + 1) \log(\pi/\beta)$, with its stated sign. At $\alpha = 2$ the added coefficient in ECL15 is $(4 - 12 + 8) \log(\pi/\beta) = 0$. It follows exactly that

$$c_\partial := \mathfrak{C}_\partial = 4 \log(4/\pi) - 10 - 2F(2, \pi) + 4\mathcal{L}(2, \pi). \quad (\text{ECL16})$$

For the rest of this subsection $u, v, w, z, c_{\text{cap}}$ are precisely their ECL2–3 values at $(\alpha, \beta) = (2, \pi)$. They therefore satisfy $uK(k) = 2$ and $vE(k) = 4$. Substitution of the established formulas, with their constants, gives

$$c_\partial = 4 \log(4/\pi) + 8 \log w - 4 \log z - 2 \log c_{\text{cap}}. \quad (\text{ECL17})$$

For example, ECL6 gives $\ell = 8 - 4 \log w - 2 \log c_{\text{cap}}$, ECL14 gives $\mathcal{L} = 6 \log w - 2 \log z - 2$, and ECL13 gives $F = -\ell/2 - 3 + \mathcal{L}$. In ECL16 these yield $4 \log(4/\pi) - 4 + \ell + 2\mathcal{L}$, in which the constant terms are exactly $-4 + 8 - 4 = 0$, proving ECL17. Equivalently $F(2, \pi) = I(2, \pi) - 6 + 2\mathcal{L}(2, \pi)$ in ECL16 gives $c_\partial = 4 \log(4/\pi) + 2 - 2I(2, \pi)$, which is precisely the OFV31 energy convention. This section evaluates this coefficient. It does not assign it as the leading coefficient of a separate observation residual or a separate kernel determinant.

59.5 An integer-only outward certificate

The accompanying `certify_closed_equilibrium.py` proves

$$\boxed{21.66851180520 < 16c_\partial < 21.66851180521.} \quad (\text{ECL18})$$

The receipt stores every interval endpoint as an integer divided by $S = 10^{70}$, including the original endpoints and all quantities in ECL17. The script uses no floating-point inputs or numerical quadrature. The following argument proves the correctness of its finite algorithm.

Each stored interval $[A, B]$ denotes $[A/S, B/S]$. Addition and subtraction are exact. For multiplication, the smallest and largest of the four endpoint products are divided by S and rounded respectively down and up. For division by an interval excluding zero, the four ratios of endpoint integers, multiplied by S , are rounded outwards. Their extrema enclose every possible product or quotient because these operations attain their extrema at corners of their rectangular domains. The rounding uses integer floor division and $\lceil n/d \rceil = -\lfloor -n/d \rfloor$ after making $d > 0$. These formulas apply to negative numerators as well. Consequently induction proves containment after every arithmetic operation, without a probabilistic or floating-point error estimate.

Let $h = u/v$ in this paragraph only; this is a scalar ratio and does not replace the density function $h(x)$ of EIQ10. Put $m_e = 1 - h^2$, so $k = \sqrt{m_e}$. Define $b_n = \binom{2n}{n}/4^n$ and $t_n = b_n^2 m_e^n$. Expansion of the binomial series, followed by its uniformly convergent integration for $m_e < 1$, proves

$$\frac{2K(\sqrt{m_e})}{\pi} = \sum_{n=0}^{\infty} t_n, \quad \frac{2E(\sqrt{m_e})}{\pi} = 1 - \sum_{n=1}^{\infty} \frac{t_n}{2n-1}, \quad \frac{t_n}{t_{n-1}} = m_e \frac{(2n-1)^2}{4n^2}. \quad (\text{ECL19})$$

Indeed integration by parts gives $\int_0^{\pi/2} \sin^{2n} \theta d\theta = (\pi/2)b_n$; the two binomial coefficients are b_n and $-b_n/(2n-1)$ for $n \geq 1$. Since $t_{n+1}/t_n \leq m_e$, the positive omitted tails after indices $0, \dots, N-1$ have the rigorous bounds

$$0 \leq \sum_{n=N}^{\infty} t_n \leq \frac{t_N}{1-m_e}, \quad 0 \leq \sum_{n=N}^{\infty} \frac{t_n}{2n-1} \leq \frac{t_N}{(2N-1)(1-m_e)}. \quad (\text{ECL20})$$

The script takes $N = 3200$, applies the recurrence with outward rounding at each step, and adds these tails with their proper signs. They remain valid for interval values of m_e by the interval arithmetic just proved.

The two exact terminating decimals used for the root bracket are

$$\begin{aligned} h_- &= 0.160101670957718355884567453115, \\ h_+ &= 0.160101670957718355884567453116. \end{aligned} \quad (\text{ECL21})$$

Let $K_b = 2K/\pi$, $E_b = 2E/\pi$ as in ECL19. The receipt gives strictly negative upper bound for $2h_-K_b(1-h_-^2) - E_b(1-h_-^2)$ and strictly positive lower bound for the corresponding expression at h_+ . Here the arguments of K_b, E_b denote m_e as defined by their series, not the modulus argument of EIQ2. The common factor $\pi/2 > 0$ in the original equation cancels exactly. EIQ6 proves that $hK(\sqrt{1-h^2})/E(\sqrt{1-h^2})$ increases strictly from zero to one as h increases from zero to one. Thus ECL21 encloses the unique root of the original endpoint equation $hK(\sqrt{1-h^2}) = E(\sqrt{1-h^2})/2$.

The value of π is enclosed independently by

$$\pi = 16 \arctan(1/5) - 4 \arctan(1/239). \quad (\text{ECL22})$$

To prove the identity, if $\theta = \arctan(1/5)$, then $\tan(2\theta) = 5/12$ and $\tan(4\theta) = 120/119$. The subtraction formula gives $\tan(4\theta - \arctan(1/239)) = 1$. This angle lies in $(0, \pi/2)$: $4\theta < 4/5 < 1 < \pi/2$, while $4\theta \geq 4(1/5)/(1+1/25) > 1/239$. The inequality $\pi/2 > 1$ follows from $\arctan 1 = \int_0^1 (1+t^2)^{-1} dt > 1/2$. Consequently the angle is $\pi/4$, proving ECL22. Each arctangent is computed

from its alternating series with 128 terms. The first omitted term bounds the absolute remainder; its sign is positive for this even number of terms. Outward arithmetic and that signed tail therefore enclose the exact π .

For every positive logarithm argument x put $t = (x-1)/(x+1)$. The geometric series, integrated on the segment from zero to t , gives

$$\log x = 2 \sum_{n=0}^{M-1} \frac{t^{2n+1}}{2n+1} + R_M, \quad |R_M| \leq \frac{2|t|^{2M+1}}{(2M+1)(1-|t|^2)}. \quad (\text{ECL23})$$

The script takes $M = 256$. For an interval argument it uses the maximum absolute endpoint of the enclosing t interval in this tail bound and adds a symmetric outward tail. Thus ECL23 encloses the logarithm throughout the argument interval.

The enclosed original endpoint values are computed in order by

$$\begin{aligned} K &= (\pi/2)K_b, & E &= (\pi/2)E_b, & v &= 4/E, & u &= hv, \\ w &= (u+v)/2, & z &= uv, & c_{\text{cap}} &= (v^2 - u^2)/4. \end{aligned}$$

The script checks $0 < u < v$, positive capacity, strict endpoint signs, and that every division interval excludes zero. It applies ECL23 to the four arguments $4/\pi, w, z, c_{\text{cap}}$ and evaluates the full expression ECL17 with its original coefficients. Integer comparison of the final bounds proves the stronger enclosure

$$\begin{aligned} 21.668511805204674126143836401790042003 &< 16c_{\partial} \\ &< 21.668511805204674126143836403323660355. \end{aligned} \quad (\text{ECL24})$$

The decimal endpoints in ECL24 are rounded outwards from the stored integer endpoints. Their comparison with the exact terminating decimals in ECL18 proves ECL18.

The same operations also enclose the endpoint expression

$$\frac{1}{8} \log(w^2/z) - \frac{1}{4} \log 2 \in (-0.080453400358, -0.080453400357). \quad (\text{ECL25})$$

This is an evaluation of this explicit expression. If it is used as an earlier approximation centre, its previously specified error radius is not changed by this calculation. The ordinary and optimized Python executions produce identical JSON receipts. All mathematical checks use explicit exceptions, so disabling Python assertions does not disable any certificate condition. The receipt includes the script hash, integer endpoint signs, elliptic-tail bounds, every resulting interval, and the strict target comparisons.

60 The closed coefficient in the original QGQ return centre

Retain the complete original QGQ1–9 theorem and its full provider 02_FULL_GAMMA_PROOFS.tex, section *The finite original $m = 1$ return bound at the q scale*. Retain its actual Gamma functional W_k , finite centre $\mathcal{V}_{q,l}^c$, constants $\mathcal{C}_\Gamma, A, M, C$, and its original packet

$$m = 1, \quad k = 4l + 1, \quad q = (4l + 2)^2, \quad n = q/2, \quad l \geq 1.$$

The support and endpoint values u, v, w, z in this section are exactly those of ECL2–3 at $(\alpha, \beta) = (2, \pi)$. In particular $uK(k_{\text{ell}}) = 2$, $vE(k_{\text{ell}}) = 4$, and $k_{\text{ell}}^2 = 1 - u^2/v^2$; the subscript distinguishes this elliptic modulus from the original integer packet degree k . No endpoint or original degree is changed by this notation.

QGQ4 defines, for the original EIQ logarithmic moment, $\mathcal{L}_1 = \partial_\alpha \mathcal{L}(2, \pi)$ and $\mathcal{C}_\Gamma^{[q,1]} = \mathcal{L}_1/8 - (\log 2)/4$. The derivative proved in ECL12 is a derivative of precisely this same moment, at fixed original β . Therefore the following is an exact identification of the original centre:

$$\boxed{\begin{aligned} \mathcal{L}_1 &= \log(w^2/z), & \mathcal{C}_\Gamma^{[q,1]} &= \frac{1}{8} \log(w^2/z) - \frac{1}{4} \log 2, \\ -0.080453400358 &< \mathcal{C}_\Gamma^{[q,1]} &< -0.080453400357. \end{aligned}} \quad (\text{EQR1})$$

The first two equalities follow by substitution into the literal QGQ4 definition; there is no new centre or change of energy convention. The last line is consequently the strict outward integer enclosure ECL25 for that exact original centre. Its proof and stored integer endpoints are those of ECL19–25.

The original finite-centre theorem QGQ8 now gives

$$\boxed{\frac{\mathcal{V}_{q,l}^c - lq\mathcal{C}_\Gamma}{q} \rightarrow \frac{1}{8} \log(w^2/z) - \frac{1}{4} \log 2 \quad (l \rightarrow \infty).} \quad (\text{EQR2})$$

To check the connection to the original finite calculation explicitly, QGQ7 retains its first term $(2\mathcal{L}_1 - 4 \log 2)l^2$ and all its other original terms. The complete termwise estimates following QGQ7 bound their sum by $O(l)$; in particular its actual remainder $\zeta_{q,l}$ is $O(l)$ by QGQ6. On the literal packet, $l^2/q \rightarrow 1/16$ and $l/q \rightarrow 0$. Thus dividing that same identity by q gives $(2\mathcal{L}_1 - 4 \log 2)/16 = \mathcal{L}_1/8 - (\log 2)/4$. Substitution of EQR1 proves EQR2 with the original finite centre, source, low endpoint, and rank correction intact.

The original error radius remains the following nonzero number:

$$\boxed{\limsup_{l \rightarrow \infty} \left| \frac{W_k - lq\mathcal{C}_\Gamma}{q} - \left(\frac{1}{8} \log(w^2/z) - \frac{1}{4} \log 2 \right) \right| \leq \frac{A}{\sqrt{2}}, \quad A = M/2 + 2C > 0.} \quad (\text{EQR3})$$

Here M and C retain their complete original QGT12 and QGT9 definitions. To verify the transfer, retain the finite QGT22 inequality used in QGQ9:

$$-p_{q,l}^- - \mathcal{U}_{q,l} - E_{q,l} \leq W_k - \mathcal{V}_{q,l}^c \leq p_{q,l}^+ + \mathcal{U}_{q,l} + T_{q,l} + E_{q,l}.$$

QGQ1–2 proves that each of $p_{q,l}^-/q, p_{q,l}^+/q, T_{q,l}/q, E_{q,l}/q$ tends to zero, while $\mathcal{U}_{q,l}/q \rightarrow A/\sqrt{2}$. Hence the difference $(W_k - \mathcal{V}_{q,l}^c)/q$ is bounded below by $-A/\sqrt{2} - o(1)$ and above by $A/\sqrt{2} + o(1)$. Adding the convergent expression in EQR2, and using the triangle inequality, proves EQR3. Both unequal finite errors remain in the displayed original inequality; their passage to the same limiting radius does not replace that finite interval by a new one.

The radius is strictly positive from the original definitions themselves. QGT9 gives $C \geq 0$, and QGT12 gives

$$M = \max_{(\alpha, \beta) \in \mathcal{K}} |\partial_\alpha \mathcal{L}(\alpha, \beta)| \geq \partial_\alpha \mathcal{L}(2, \pi) = \log(w^2/z) > 0,$$

because $(2, \pi) \in \mathcal{K}$ and $w^2/z = 1 + (v - u)^2/(4uv) > 1$ by $0 < u < v$. Consequently $A = M/2 + 2C > 0$. This proves that the retained nonzero radius has not disappeared upon evaluating its centre. The only convergence asserted in EQR2 concerns the original finite centre. Neither EQR1 nor EQR3 asserts a limit for the actual quantity $(W_k - lq\mathcal{C}_\Gamma)/q$.

61 Strict full-window growth of the original residual covariance

This calculation starts from the complete ROQ1–27 construction above. The incoming first-variation handoff states a full-window update and trace–determinant bounds. The following proof derives those statements from the original columns, and adds exact exterior-power formulas for every intermediate coefficient. No unavailable FVR proof is used. Retain $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $j = 8k - 32$, the fixed actual period in the five-orbit locus, and $s \in \{1, k\}$. The actual matrices $C : \mathbb{C}^q \rightarrow \mathbb{C}^{q'}$ and $D : \mathbb{C}^q \rightarrow \mathbb{C}^j$ are ROQ1 and ROQ4, with all their coefficients. In particular C is onto and $D|_{\ker C}$ is onto. The original roots, monic Gamma polynomials and their squared norms are

$$\begin{aligned} \omega_{ab} &= (2b - k)\gamma - i(2a - k)\delta, \quad 0 \leq a, b \leq k, \quad \delta > 0, \quad \gamma > 0, \\ p_0 &= 1, \quad p_1 = y, \quad p_{n+1} = yp_n - n(s/2 + n - 1)p_{n-1}, \\ h_n &= (2\pi)^{s/2} n!(s/2)_n, \quad u_n = (p_n(\omega_{ab})/\sqrt{h_n})_{ab}, \quad R_N = \sum_{n=0}^N u_n u_n^*. \end{aligned} \tag{RWC1}$$

The positive original measure dm_s has the orthogonality and norms in RWC1, as proved in the Gamma provider. The order of the q roots is the original ordered primary order; stars below are conjugate transposes. The first q monic polynomials have invertible evaluation matrix, since these q roots are distinct. Consequently $R_N > 0$ for $N \geq q - 1$.

61.1 Every consecutive full block is invertible

The complete real polynomial of the original root set is

$$\begin{aligned} P_k(y) &= \prod_{a=0}^k \prod_{b=0}^k (y - \omega_{ab}) \\ &= \prod_{a=0}^{(k-1)/2} \prod_{b=0}^k ((y - (2b - k)\gamma)^2 + (k - 2a)^2 \delta^2) > 0 \quad (y \in \mathbb{R}). \end{aligned} \tag{RWC2}$$

Indeed the factor indexed by $k - a, b$ is the conjugate of the factor indexed by a, b , and no factor in the second product can vanish on the real line. This proves the identity with its monic constant and all factors retained.

For any integer $N \geq q - 1$ put

$$U_N^+ = [u_{N+1}, \dots, u_{N+q}] \in \text{Mat}_{q \times q}(\mathbb{C}). \quad \det U_N^+ \neq 0. \tag{RWC3}$$

To prove the assertion suppose $U_N^+ c = 0$ and form $f(y) = \sum_{a=1}^q c_a p_{N+a}(y)/\sqrt{h_{N+a}}$. It vanishes at all q distinct roots, so polynomial division gives $f = P_k r$ with $\deg r \leq N$. Orthogonality of every p_{N+a} to every polynomial of degree at most N gives

$$0 = \int_{\mathbb{R}} f(y) \overline{r(y)} dm_s(y) = \int_{\mathbb{R}} P_k(y) |r(y)|^2 dm_s(y).$$

All integrals exist, because the measure has every polynomial moment. Its strictly positive density and RWC2 force $r = 0$. The distinct degrees of the monic p_{N+a} then force every $c_a = 0$. This argument applies at both $N = q - 1$ and $N = q$, and also at every larger integer cutoff; it makes no genericity choice of D .

61.2 The simultaneous conditioned update

Use the complete original matrices

$$\begin{aligned} A_N &= C R_N C^*, \quad W_N = R_N - R_N C^* A_N^{-1} C R_N, \quad \Omega_N = D W_N D^* > 0, \quad Q_{\Xi, N} = \Omega_N^{-1}, \\ \Pi_N &= I_q - R_N C^* A_N^{-1} C, \quad X_N = C U_N^+, \quad Z_N = D \Pi_N U_N^+, \\ B_N &= I_q + X_N^* A_N^{-1} X_N > 0, \quad V_N = Z_N B_N^{-1} Z_N^*. \end{aligned} \tag{RWC4}$$

These use A_N solely for this covariance block, rather than for the conductor biform. Since $A_N > 0$, every displayed inverse exists. The equalities $C\Pi_N = 0$ and $\Pi_N x = x$ for $Cx = 0$ show that Π_N is the exact projection onto $\ker C$ with complementary image $R_N C^* \mathbb{C}^q$. It is generally not the Euclidean orthogonal projection. RWC3 and surjectivity of $D|_{\ker C}$ give

$$\text{im}(\Pi_N U_N^+) = \ker C, \quad \text{rank } Z_N = j, \quad V_N > 0. \quad (\text{RWC5})$$

For the last assertion, Z_N^* is injective and $B_N^{-1} > 0$, so $z^* V_N z > 0$ for each nonzero $z \in \mathbb{C}^j$.

There are exact identities

$$\begin{aligned} W_{N+q} &= W_N + (\Pi_N U_N^+) B_N^{-1} (\Pi_N U_N^+)^*, \\ \Omega_{N+q} &= \Omega_N + V_N, \quad Q_{\Xi, N+q} = (\Omega_N + V_N)^{-1}. \end{aligned} \quad (\text{RWC6})$$

Here is the complete matrix calculation. Temporarily write $R = R_N$, $U = U_N^+$, $A = A_N$, $X = CU$, $B = B_N$ and $T = RC^* A^{-1} X$. Direct multiplication proves

$$(A + XX^*)^{-1} = A^{-1} - A^{-1} X B^{-1} X^* A^{-1}.$$

Indeed $(A + XX^*) A^{-1} X = XB$, so the subtracted term cancels $XX^* A^{-1}$. Since $CR_{N+q} C^* = A + XX^*$, the new subtracted covariance product equals

$$RC^* A^{-1} CR + TU^* + UT^* + U(B - I_q)U^* - [T + U(B - I_q)]B^{-1}[T + U(B - I_q)]^*.$$

Subtract this expression from $R + UU^*$ and then subtract W_N . Writing $T + U(B - I_q) = UB - (U - T)$, expansion leaves exactly $(U - T)B^{-1}(U - T)^*$. But $U - T = \Pi_N U$, proving the first identity in RWC6. Multiplication by D, D^* proves the second, and ROQ16 proves the third. In particular the full $q \times q$ matrix B_N and all its off-diagonal entries remain in the update.

61.3 All exterior coefficients and finite bounds

Let

$$Y_N = \Omega_N^{-1/2} Z_N B_N^{-1/2}, \quad T_N = Y_N Y_N^* > 0, \quad t_N = \text{Tr } T_N, \quad c_N = \det T_N = \frac{\det V_N}{\det \Omega_N} > 0. \quad (\text{RWC7})$$

The square roots are the positive Hermitian square roots; no source mass, coordinate phase or rank has been changed. If $\lambda_1, \dots, \lambda_j > 0$ are the eigenvalues of T_N , then RWC6 gives

$$L_N := \log \frac{\det \Omega_{N+q}}{\det \Omega_N} = \sum_{a=1}^j \log(1 + \lambda_a) = \log \left(\sum_{d=0}^j e_d(T_N) \right), \quad e_d(T_N) = \left\| \bigwedge^d Y_N \right\|_{\text{HS}}^2. \quad (\text{RWC8})$$

The exterior norm is the Hilbert–Schmidt norm in the standard orthonormal wedge bases, and $e_0 = 1$. To verify the last equality, the singular values of Y_N are $\sqrt{\lambda_a}$, and its d th exterior power has squared singular values $\prod_{a \in I} \lambda_a$ for the d -element subsets I . Their sum is the elementary symmetric polynomial e_d . Equivalently Cauchy–Binet writes it as the sum of squared moduli of all $d \times d$ minors of the literal Y_N . This proves every coefficient in the determinant expansion, including all intermediate exterior powers; the sole determinant c_N is its last coefficient.

The following finite inequalities now hold:

$$\max\{\log(1 + t_N), j \log(1 + c_N^{1/j})\} \leq L_N \leq j \log(1 + t_N/j). \quad (\text{RWC9})$$

For the first lower bound, expand $\prod(1 + \lambda_a)$: the constant term is one, its degree-one term is t_N , and all other terms are positive. For the second, the arithmetic–geometric mean inequality on the $\binom{j}{d}$ positive d -fold products gives $e_d \geq \binom{j}{d} c_N^{d/j}$, since each eigenvalue occurs in $\binom{j-1}{d-1}$ products. Summing this inequality for $0 \leq d \leq j$ yields $(1 + c_N^{1/j})^j$. Finally the concavity of $\log(1 + x)$ on $x > 0$ gives the upper bound by averaging the j positive eigenvalues. These arguments also prove useful finite refinements: retaining any chosen initial set $e_0, \dots, e_d$ in RWC8 gives its exact positive partial-sum lower bound; no rank fraction is substituted for an eigenvalue.

61.4 Return to the original four endpoints

Use precisely the ordered cutoffs $(q-1, q, 2q-1, 2q)$, with signs $(1, 1, -1, -1)$. ROQ14 and ROQ16 give

$$\Phi_{k,s} = F_{\Xi,k}^{(s)} = L_{q-1} + L_q > 0. \quad (\text{RWC10})$$

Define the actual finite bounds

$$\begin{aligned} \ell_{k,s} &= \sum_{N \in \{q-1, q\}} \max\{\log(1 + t_N), j \log(1 + c_N^{1/j})\} > 0, \\ u_{k,s} &= \sum_{N \in \{q-1, q\}} j \log(1 + t_N/j), \quad \ell_{k,s} \leq \Phi_{k,s} \leq u_{k,s}. \end{aligned} \quad (\text{RWC11})$$

These are complete expressions in the actual C, D, u_n and their Gram matrices. Together with ROQ21 they give the stronger endpoints $\Phi_{k,s}^- = \max(\ell_{k,s}, \lambda_s)$ and $\Phi_{k,s}^+ = \min(u_{k,s}, j(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - \lambda_s)$. Here $\lambda_s = \log(1 + \eta_{q-1})$ and the full original $\mathcal{L}_{s,t}$ are exactly ROQ17–21; all masses are retained.

For the already proved original seven-moment receiving equality, retain the finite domain of PMR1–3 and PMR7: $k \geq 29$ and $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}(k+1)^2$. Write $\mathcal{F}_{0,k}^{(s)}$ for the full seven-moment outer expression, and set $e_{k,s}^{0,\pm} = e_{k,s}^\pm + c_k^\circ$ with the exact OAR error endpoints and PMR2's complete root-to-moment error c_k° . Substitution, without changing either endpoint, proves

$$\begin{aligned} \mathcal{F}_{0,k}^{(s)} - e_{k,s}^{0,-} - \Phi_{k,s}^+ &\leq F_K^{(s)} \\ &\leq \mathcal{F}_{0,k}^{(s)} + e_{k,s}^{0,+} - \Phi_{k,s}^-. \end{aligned} \quad (\text{RWC12})$$

Outside that PMR domain the original OAR outer-return formula gives the same interval with $\mathcal{F}_{\partial,k}^{(s)}$ in place of $\mathcal{F}_{0,k}^{(s)}$ and $e_{k,s}^\pm$ in place of $e_{k,s}^{0,\pm}$, throughout OAR's stated domain. RWC1–11 remain valid at every $k \geq 9$ specified at the start of this section. Every stronger previously proved kernel bound can be intersected with these endpoints. For the arithmetic statement also retain the original AMT period domain $|u| \geq R_*$, as in OVR's opening paragraph. In the complete OAR/OVR signed return $\Psi_k = F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}$, the same literal substitution gives

$$\begin{aligned} \mathcal{F}_{0,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma - e_{k,k}^{0,-} - \Phi_{k,k}^+ &\leq \Psi_k \\ &\leq \mathcal{F}_{0,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^{0,+} - \Phi_{k,k}^-. \end{aligned} \quad (\text{RWC13})$$

The original correlated arithmetic deficit δ_k^σ , boundary return $\mathcal{H}_k$, and each unequal source error are unchanged. Thus every computed exterior coefficient in RWC8 has a proved map into both the kernel and signed-return estimates. The strict finite positivity in RWC10 alone supplies no rate for $\Phi_{k,s}/(kq)$. The quantitative next calculation is the growth of these same conditioned matrices or their exterior coefficients in the fixed actual period, not a replacement by an arbitrary rank- j positive matrix.

The new lower endpoint strictly strengthens the original first-increment endpoint, through an exact comparison on the same covariance. ROQ18 telescopes to

$$V_{q-1} = \sum_{a=q-1}^{2q-2} \frac{z_a z_a^*}{a_a}, \quad t_{q-1} = \sum_{a=q-1}^{2q-2} \frac{z_a^* \Omega_{q-1}^{-1} z_a}{a_a} \geq \eta_{q-1}, \quad \ell_{k,s} > \lambda_s. \quad (\text{RWC14})$$

Here z_a, a_a, η_a are precisely ROQ17 at their original individual cutoffs; the sum retains every later mixed covariance. Each summand is nonnegative, and the first is exactly η_{q-1} . Thus the first-window trace lower bound in RWC11 is at least $\log(1 + \eta_{q-1}) = \lambda_s$. The second-window lower bound is strictly positive by RWC5 and RWC7. Adding them proves the strict inequality. Consequently the original maximum in $\Phi_{k,s}^- = \max(\ell_{k,s}, \lambda_s)$ equals $\ell_{k,s}$ by this proved comparison; both constituent bounds remain recorded, with their original definitions and domains.

62 The fixed-period determinant and closed coefficients in the original receiver

Retain the complete original objects, source orders, observations and arithmetic maps of OMR1–26. The PCL, ROG, FEX, FGC, ECL, EQR and RDS proofs are supplied in full. This section composes their exact maps with the current receiver. The conductor and fixed-period determinant applications keep $m = 1$, $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = 16k - 48$, $r = 8k - 16$, $j = 8k - 32$, $c = k/2$, $c' = c - 4$, and the fixed actual five-orbit period and its original logarithm branch. Each earlier eventual comparison retains its explicit finite domain.

62.1 The existing outer coefficient has an exact endpoint expression

To distinguish the endpoint values from the conductor moment index v , write $(u_E, v_E, w_E, z_E, c_E)$ for exactly ECL2–3 at $(\alpha, \beta) = (2, \pi)$. This is the literal dictionary $u_E = u$, $v_E = v$, $w_E = (u+v)/2$, $z_E = uv$, $c_E = (v^2 - u^2)/4$ in that provider. With κ_E denoting its elliptic modulus, the original endpoint equations and coefficient are

$$\begin{aligned} u_E K(\kappa_E) &= 2, & v_E E(\kappa_E) &= 4, & \kappa_E^2 &= 1 - u_E^2/v_E^2, & 0 < u_E < v_E, \\ \mathfrak{C}_\partial &= 4 \log(4/\pi) + 8 \log w_E - 4 \log z_E - 2 \log c_E, \\ 21.66851180520 &< 16\mathfrak{C}_\partial < 21.66851180521. \end{aligned} \tag{OER1}$$

Here K, E are the original complete elliptic integrals with modulus convention. ECL6 and ECL14 give $\ell = 8 - 4 \log w_E - 2 \log c_E$, $\mathcal{L} = 6 \log w_E - 2 \log z_E - 2$ and $F = -\ell/2 - 3 + \mathcal{L}$. Substitution in the exact OFV/ECL16 expression $4 \log(4/\pi) - 10 - 2F + 4\mathcal{L}$ gives OER1: the constant terms are exactly $-4 + 8 - 4 = 0$. Also $F = I_{2,\pi} - 6 + 2\mathcal{L}$, so this number is exactly $4 \log(4/\pi) + 2 - 2I_{2,\pi}$ in OMR2, MRI4r and BRI6h. The integer outward certificate ECL19–24 proves its displayed enclosure; all stored endpoints and its complete code are retained.

Consequently the existing exact relations and their errors become

$$\begin{aligned} \mathcal{O}_k^\pm &= \Delta q [4 \log(4/\pi) + 8 \log w_E - 4 \log z_E - 2 \log c_E] \pm 4\Delta\varepsilon_k^{\text{out}}, \\ F_K^{(s)} + \Phi_{k,s} &= \Delta q \mathfrak{C}_\partial + R_{k,s}, & b_{k,s} - e_{k,s}^- &\leq R_{k,s} \leq b_{k,s} + e_{k,s}^+, \\ \Psi_k + \Phi_{k,k} &= \Delta q \mathfrak{C}_\partial + \mathcal{H}_k + \delta_k^\sigma + R_{k,k}. \end{aligned} \tag{OER2}$$

These are substitutions into OMR2, MRI4s and BRI6h. The complete OOQ error, the earlier $O(q \log q)$ OFV error, and both signed OAR errors retain their original finite definitions. Put $c_- = 21.66851180520/16$, $c_+ = 21.66851180521/16$. The original outer value is also enclosed by $[\Delta q c_- - 4\Delta\varepsilon_k^{\text{out}}, \Delta q c_+ + 4\Delta\varepsilon_k^{\text{out}}]$. This follows from $\Delta q > 0$ and OER1 and retains the exact coefficient in all identities.

62.2 The actual period coefficients and the pivot morphism

PCL1–12 gives the full entire coefficient matrix $\mathcal{R}(z) = \sum_{n \geq 0} R_n z^n$, with every coefficient specified by the finite sums in PCL2, and the exact identities

$$\begin{aligned} \mathbf{H}(u) &= e^{a/u} G_u \mathcal{R}(u^{-1}) V^{-1} U, & \det \mathcal{R}(z) &= 1, \\ \mathcal{R}(0) &= \begin{pmatrix} 1 & 0 & -\beta/3 & 0 \\ 0 & 1 & 0 & -2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \\ L_a(z) &= U^{-1} V \mathcal{R}(z)^{-1} D^{-a} \mathcal{R}(z) V^{-1} U, & A_z(Z, W) &= \prod_{a=1}^4 Q_0(L_a(z)(ZW, Z, W, 1)^\top), \\ (C_{\text{in}}(z))_{(a,b),(i,j)} &= a_{i+4-a,j+4-b}(z), & C_{\text{in}}(-z) &= C_{\text{in}}(z)^\top. \end{aligned} \tag{OER3}$$

All 81 coefficients are those of this displayed product, with PCL10's full sums; $U, V, G_u, \mathbf{a}, \beta, D$ retain their original definitions. Conjugating D^{-a} by $\mathbf{H}$ cancels its scalar factors; G_u commutes with D^{-a} , leaving exactly L_a . PCL2's $n = 0$ terms give the constant matrix, including both nonzero off-diagonal entries. PCL22–24 proves the last identity coefficient by coefficient. This is the actual matrix family in GIP8 and OMR9; no entry has been removed in taking its coefficient limit.

For the original OCF leading pivot let $\mathcal{P}_k$ be its actual pivot rectangle and $\mathcal{E}_k$ its complement. With the original insertion S_k and full reduction N_k ,

$$\begin{aligned} S_k^\top N_k^\top &= I_\Delta, & \text{im } N_k^\top &= V_A, \\ J_{\mathcal{E}} &= N_k^\top D_{\mathcal{P},\mathcal{E}} V_{\mathcal{E}}, & \rho_{\mathcal{E}} &= V_{\mathcal{E}}^{-1} D_{\mathcal{P},\mathcal{E}}^{-1} S_k^\top|_{V_A}, & \rho_{\mathcal{E}}^{-1} &= J_{\mathcal{E}}, \\ \Xi_{\mathcal{E}} &= \Xi_k^{\text{fix}} J_{\mathcal{E}}, & \ker \Xi_{\mathcal{E}} &= \rho_{\mathcal{E}}(K), \\ H_{\mathcal{E},N} &= J_{\mathcal{E}}^* G_N J_{\mathcal{E}}, & Q_{\Xi,N}^{\text{fix}} &= (\Xi_{\mathcal{E}} H_{\mathcal{E},N}^{-1} \Xi_{\mathcal{E}}^*)^{-1}. \end{aligned} \tag{OER4}$$

The diagonal $D_{\mathcal{P},\mathcal{E}}$ consists of the full root products $P_{\mathcal{P},k}(\omega_{ab})$, and $V_{\mathcal{E}}$ is the evaluation Vandermonde in that actual complementary root set. Distinct roots make both invertible. Multiplication proves both inverse identities; composition gives the kernel. Pullback of the unchanged Gamma form and minimum over the same residual fibre prove the last line. All root factors and the full metric in PCL19–21 remain present. No central-grid OOQ bound is applied to the changed root set without that transport.

62.3 The full low graph has its own finite-rank transport

At each original endpoint put $L_N = N - q$ and $M_N^{\text{pol}} = L_N + \Delta - v$. Let $\mathbf{U}_{L_N}$ be the complete ROG4 positive-shift relation matrix and Π_{U,L_N} its weighted orthogonal projection in $\|g\|_{\chi',s,c'} = \|\chi'g\|_{s,c'}$, $\chi' = \chi_{k-8}$. The finite inverse ROG12 gives the exact maps

$$\begin{aligned} Q_{N,s}^{\text{low}} g &= P_{<v} Z_{\chi,N} \mathcal{U}_{L_N}^{-1} \Pi_{U,L_N} g, & \text{rank } Q_{N,s}^{\text{low}} &\leq v, & \|Q_{N,s}^{\text{low}}\| &\leq H_{N,s}^{\text{ROG}}, \\ \mathcal{R}_{N,s}(x, g) &= (x - Q_{N,s}^{\text{low}} g, g), & B_{N,s}^{\text{str}} &= B_N^{\text{graph}} \mathcal{R}_{N,s}^{-1}, \\ \ker B_{N,s}^{\text{str}} &= \text{im} \left[\begin{pmatrix} 0 \\ \mathbf{U}_{L_N} \end{pmatrix}, \mathcal{R}_{N,s} W_{K,N} \right]. \end{aligned} \tag{OER5}$$

$H_{N,s}^{\text{ROG}}$ is exactly ROG24's product $A_{N,s,c}^+ (1 + R)^q I_{L_N} D_{M_N^{\text{pol}},\chi'} A_{N-v,s,c'}^-$, with all ROG6–24 constants. At $L_N = -1$ the map and its bound are zero. The shear sends $(P_{<v} Z_{\chi,N} h, \mathcal{U}_{L_N} h)$ to $(0, \mathcal{U}_{L_N} h)$; applying it to the full GIP kernel proves the last line, retaining every common-kernel column.

For the exact product source Gram define $Q_{N,s}^{\text{str}}$ by ROG31 and $\Phi_{k,s}^{\text{str}} = \sum_N \sigma_N \log \det Q_{N,s}^{\text{str}}$, with the original signs $(1, 1, -1, -1)$. Then

$$\begin{aligned} E_s^{\text{str}} &= 2 \sum_N E_N^{\text{CEP}} + 2v \sum_N \log(1 + H_{N,s}^{\text{ROG}}), \\ |\Phi_{k,s}^{\text{str}} - \Phi_{k,s}| &\leq E_s^{\text{str}} = o(kq). \end{aligned} \tag{OER6}$$

Each singular value σ of the low map gives two shear singular values $(\sqrt{\sigma^2 + 4} + \sigma)/2$ and its reciprocal. At most v exceed one. Thus every exterior forward and inverse logarithmic cost is at most $v \log(1 + H_{N,s}^{\text{ROG}})$. Minimum lifts and projection off the entire kernel contract exterior norms. Squaring gives the determinant bound; adding the full GIP source-map cost and the four signs proves OER6. ROG25 gives $\log(1 + H_{N,s}^{\text{ROG}}) = O_{\text{actual}}(q \log(q + 1))$; fixed $v \leq 80$ makes it $o(kq)$. The full ROQ/FMC32 minimum-form identity identifies the actual four-endpoint residual with $\Phi_{k,s}$.

62.4 The fixed columns arise from an exact quotient of that graph

Put $\alpha_A = v!/\mu_v$, $\mathcal{C}_A = \alpha_A \mathcal{T}_A$, $L = \mathbb{C}[S]_{<v} \subset E$, $K^+ = K + L$ and $t_0 = \dim(K \cap L)$. The explicit coordinate isomorphism $(\ell, z) \mapsto (\ell, \alpha_A z)$ sends the graph relations to $(P_{<v} Z_{\chi, N} h, \alpha_A \mathcal{U}_{L_N} h)$. Projection to the second coordinate gives the exact sequences

$$\begin{aligned} 0 \longrightarrow L \longrightarrow V_A \longrightarrow V_A/L \simeq \mathbb{C}[S']_{\leq N-v-q'}/\alpha_A \mathcal{U}_{L_N} \mathbb{C}[S]_{\leq N-q} \longrightarrow 0, \\ 0 \longrightarrow K \cap L \longrightarrow K \longrightarrow K^+/L \longrightarrow 0, \quad \dim(K^+/L) = t = r - t_0. \end{aligned} \quad (\text{OER7})$$

For a zero lower class, subtract its full graph vector from $(\ell, \alpha_A \mathcal{U} h)$, leaving $(\ell - P_{<v} Z_{\chi, N} h, 0)$. Such a low vector is a relation only when $\mathcal{U} h = 0$, which forces $h = 0$ by the nonzero leading coefficient $\text{lc}(h) \mu_v \binom{q+\deg h}{v}$. This proves the first kernel and identifies it with the original L . Restriction to K proves the second sequence, as in FGC3–8.

In the complete AKJ basis let $\Xi = (\Xi_{\text{lo}}, \Xi_{\text{hi}})$ and let Q_{lo} have kernel $\text{im } \Xi_{\text{lo}}$. Choose the specified fixed full basis W_k of $\ker(Q_{\text{lo}} \Xi_{\text{hi}})$ using actual nonzero minors. The original low and high columns give

$$\begin{aligned} Z_k = \text{diag}(d_h) W_k, \quad d_h = \frac{(q' + h + v)!}{(q' + h)!}, \quad \mathcal{C}_A P_h = d_h \psi(S') S'^h, \quad \psi = \chi_{k-8}, \\ R_a^A(S') = \frac{\mathcal{C}_A(\chi S^a)}{\psi} = \alpha_A \sum_{\nu} a_{\nu} \frac{\chi(S' + b_{\nu})}{\psi(S')} (S' + b_{\nu})^a. \end{aligned} \quad (\text{OER8})$$

Solvability of $\Xi_{\text{lo}} \ell + \Xi_{\text{hi}} z = 0$ is exactly the condition defining W_k , so Z_k represents K^+/L . Each relation polynomial has degree $\Delta - v + a$ and nonzero leading coefficient $(q + a)_v$, while the Z_k columns have degree below $\Delta - v$. Their quotient classes are independent.

Let $H_{N,s}^{\psi}$ be the full monomial Gram in the original measure $|\psi(c' + iy)|^2 dm_{0,s}(y)$, and R_N^A the first $N + 1 - q$ relation columns of OER8. Their four ranks are exactly 0, 1, q , $q + 1$. The complete minimum matrix is

$$\begin{aligned} \widehat{H}_{K,N}^{(s)} &= Z_k^* H_{N,s}^{\psi} Z_k - Z_k^* H_{N,s}^{\psi} R_N^A ((R_N^A)^* H_{N,s}^{\psi} R_N^A)^{-1} (R_N^A)^* H_{N,s}^{\psi} Z_k, \\ \widehat{F}_k^{(s)} &= \sum_N \sigma_N \log \det \widehat{H}_{K,N}^{(s)}. \end{aligned} \quad (\text{OER9})$$

Minimizing the norm of $Z_k x + R_N^A h$ gives $h = -((R_N^A)^* H_{N,s}^{\psi} R_N^A)^{-1} (R_N^A)^* H_{N,s}^{\psi} Z_k x$. Substitution proves the full Schur formula including both cross factors. At the first endpoint there is no relation variable. Independence proves positivity. This is exactly FEX29/FGC11 on the native lower metric. The original graph metric in these coordinates is $D_{\alpha,N}^{-*} H_N^{\text{actual}} D_{\alpha,N}^{-1}$; the next bounds retain its change to that native metric.

62.5 Both low terms and all one-sided errors enter the current interval

Set $e_{N,s}^{\text{FEX}} = \log \det \widehat{H}_{K,N}^{(s)} - \log \det H_{K^+/L,N}^{(s)}$. Keep exactly FEX18–20/34:

$$\begin{aligned} n_N = N + 1 - v, \quad m_N = \log \max \left\{ 1, \frac{v!}{|\mu_v|} \sum_{\nu} |a_{\nu}| e^{|b_{\nu}-4|(H_N+2)} \right\}, \quad H_N = \sum_{a=1}^N \frac{1}{a}, \\ \zeta_{N,s} = \log \prod_{d=v}^N \frac{w_d}{w_{d-v}}, \quad w_d = \frac{d!}{(s/2)_d}, \\ e_{N,s}^- = \zeta_{N,s} - 2(n_N - t)m_N, \quad e_{N,s}^+ = 2tm_N, \quad e_{N,s}^- \leq e_{N,s}^{\text{FEX}} \leq e_{N,s}^+, \\ D_s^- = e_{q-1,s}^- + e_{q,s}^- - e_{2q-1,s}^+ - e_{2q,s}^+, \quad D_s^+ = e_{q-1,s}^+ + e_{q,s}^+ - e_{2q-1,s}^- - e_{2q,s}^-. \end{aligned} \quad (\text{OER10})$$

These $e^{\pm}$ are the FEX flag bounds, distinct from OAR's $e_{k,s}^{\pm}$. Here H_N is the harmonic number, distinct from the kernel Gram. The full flag identity gives $\mathcal{J} M^{-2(n_N-t)} \leq D_Q \leq M^{2t}$, with

the exact squared Jacobian $\mathcal{J} = e^{\zeta_{N,s}}$ and norm bound $M = e^{m_N}$. This proves the endpoint bounds and their four signed sums, keeping both original Gamma masses until their exact Jacobian cancellation.

The two full Schur factorizations of the same K^+ give

$$\begin{aligned} F_K^{(s)} &= \widehat{F}_k^{(s)} - \sum_N \sigma_N e_{N,s}^{\text{FEX}} + F_L^{(s)} - F_{K^+/K}^{(s)}, \\ 0 \leq F_L^{(s)} &\leq v\mathcal{L}_s, \quad 0 \leq F_{K^+/K}^{(s)} \leq (v - t_0)\mathcal{L}_s, \quad \mathcal{L}_s = \mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}, \\ f_s^- &= \widehat{F}_k^{(s)} - D_s^+ - (v - t_0)\mathcal{L}_s \leq F_K^{(s)} \leq f_s^+ = \widehat{F}_k^{(s)} - D_s^- + v\mathcal{L}_s. \end{aligned} \tag{OER11}$$

In fixed adapted frames the determinant of K^+ is both its L determinant times that of K^+/L , and its K determinant times that of K^+/K , including fixed Jacobians. Their four endpoint constants cancel because the signs sum to zero. This proves the first identity. The original AKS bound passes to both subquotients by restriction and minimization, proving the second line. OER10 then proves the last line. If the two actual low volumes are evaluated, the stronger interval retains their exact difference in place of the caps. Both low terms remain part of the identity in either case.

All existing OMR11–12 intervals remain. The new same-scalar intersection is

$$\begin{aligned} L_s^{\text{ext}} &= \max\{L_s^{\text{op}}, f_s^-, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{\text{str}} - E_s^{\text{str}} - e_{k,s}^-\}, \\ U_s^{\text{ext}} &= \min\{U_s^{\text{op}}, f_s^+, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{\text{str}} + E_s^{\text{str}} + e_{k,s}^+\}, \\ L_{\text{ar}}^{\text{ext}} &= \max\{L_{\text{ar}}^{\text{op}}, L_1^{\text{ext}} + a_{K,k}^-\}, \quad U_{\text{ar}}^{\text{ext}} = \min\{U_{\text{ar}}^{\text{op}}, U_1^{\text{ext}} + a_{K,k}^+\}, \\ F_K^{\text{ar}} &\in [L_{\text{ar}}^{\text{ext}}, U_{\text{ar}}^{\text{ext}}], \quad F_B^{\text{ar}} \in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^{\text{ext}}, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^{\text{ext}}]. \end{aligned} \tag{OER12}$$

The first lines combine OER6, the original MRI4m equality, and OER11. Every interval bounds the same $F_K^{(s)}$. The last lines add the unchanged correlated arithmetic allowances of OMR12 and subtract from the same total. All prior finite branches remain available outside any added domain. FEX40–41 also permits the existing reverse source intersections with its full asymmetric Ω bounds.

Define $L_J^{\text{ext}} = L_k^{\text{ext}} + \mathcal{H}_k + \delta_k^\sigma + C_k^{\text{budget}}$ and $U_J^{\text{ext}} = U_k^{\text{ext}} + \mathcal{H}_k + \delta_k^\sigma + C_k^{\text{budget}}$. Then

$$\begin{aligned} L_k^{\text{ext}} + \mathcal{H}_k + \delta_k^\sigma &\leq \Psi_k \leq U_k^{\text{ext}} + \mathcal{H}_k + \delta_k^\sigma, \\ J_k^{\text{action}} &= 2 \sum_N w_N \log \epsilon_N^{\text{ar}} \in [L_J^{\text{ext}}, U_J^{\text{ext}}], \quad 4q \log \mathcal{L}_{h,k} \leq J_k^{\text{action}}, \\ \mathcal{L}_{h,k} &\leq \min_N \epsilon_N^{\text{ar}} \leq \prod_N (\epsilon_N^{\text{ar}})^{w_N/(2q)} \leq \exp(U_J^{\text{ext}}/(4q)), \\ \frac{\Psi_k}{kq} &= \frac{\widehat{F}_k^{(1)}}{kq} - \frac{\mathcal{C}_\Gamma}{4} + o(1). \end{aligned} \tag{OER13}$$

The first line is the exact OMR13 equality with $\mathcal{H}_k = -T_k$. The next lines use the same arithmetic budget, exact original reflection, and same positive class in OMR19–26. FEX36–38 proves $|F_K^{(s)} - \widehat{F}_k^{(s)}| = O_{\text{actual}}(q \log q) = o(kq)$; the original source difference and arithmetic correction are $o(kq)$, and the proved Gamma return divided by kq tends to $-\mathcal{C}_\Gamma/4$. Substitution proves the last line without asserting a limit for the unevaluated fixed-column determinant.

62.6 The relation quotient and the earlier finite Gamma centre

With every original coefficient, put $E_A(z) = \sum a_{ab} e^{b_{ab}z}$, $D = \partial_z$ and $\mathcal{L}_A = \sum a_{ab} e^{b_{ab}z} d_{ab}(D)$. The exact analytic dual of the denominator above is

$$\begin{aligned} \mathcal{L}_A \chi'(D) &= \chi(D) M_{E_A}, \\ 0 \longrightarrow \mathcal{E}_{k-8} &\xrightarrow{M_{E_A}} \mathcal{N}_v \xrightarrow{\mathfrak{F}_A} (\mathbb{C}[S']/\mathcal{UC}[S])^\vee \longrightarrow 0, \\ f_{\mathfrak{F}_A(N)}(z) &= \chi'(D)(N(z)/E_A(z)). \end{aligned} \tag{OER14}$$

$\mathcal{N}_v$ is the original upper exponential space with its first v Taylor values zero; the operator order and all spaces are RDS6–8. RDS5 proves the intertwining by the literal positive shifts and Leibniz' rule. Its kernel consists precisely of $E_A \mathcal{E}_{k-8}$, since the monic constant-coefficient equation $\chi'(D)F = 0$ has exactly that lower exponential solution space; the full Vandermonde proves its initial-value uniqueness. Dimension $q - v - q' = \Delta - v$ then proves surjectivity. Because $\alpha_A \neq 0$, the relation subspace $\alpha_A \mathcal{UC}[S]$ in OER7 is the same subspace. RDS's finite-degree direct decomposition identifies its dual at every original cutoff. The full source coefficient identity, original phases and masses, analytic disk and pole multiplicities remain RDS8–15; the Schur metric in OER9 is retained.

Finally ECL12 applies to precisely QGQ4's original derivative $\mathcal{L}_1 = \partial_\alpha \mathcal{L}(2, \pi)$. Hence

$$\begin{aligned} c_q^\Gamma &= c_\Gamma^{[q,1]} = \frac{1}{8} \log(w_E^2/z_E) - \frac{1}{4} \log 2, \quad -0.080453400358 < c_q^\Gamma < -0.080453400357, \\ \frac{\mathcal{V}_{q,l}^c - lq\mathcal{C}_\Gamma}{q} &\longrightarrow c_q^\Gamma, \quad \limsup_{l \rightarrow \infty} \left| \frac{W_k - lq\mathcal{C}_\Gamma}{q} - c_q^\Gamma \right| \leq A/\sqrt{2}, \quad A = M/2 + 2C > 0. \end{aligned} \quad (\text{OER15})$$

This uses exactly the QGQ domain $m = 1$, $k = 4l + 1$, $q = (4l + 2)^2$, $l \geq 1$. EQR1–3 proves the connection: ECL12 gives $\mathcal{L}_1 = \log(w_E^2/z_E)$; the complete QGQ7 centre has correction $(2\mathcal{L}_1 - 4\log 2)l^2$ and remaining $O(l)$ terms. Since $l^2/q \rightarrow 1/16$, division gives the stated centre limit. The full unequal finite errors remain

$$-p_{q,l}^- - \mathcal{U}_{q,l} - E_{q,l} \leq W_k - \mathcal{V}_{q,l}^c \leq p_{q,l}^+ + \mathcal{U}_{q,l} + T_{q,l} + E_{q,l}. \quad (\text{OER16})$$

All terms except $\mathcal{U}_{q,l}$ become zero after division by q , while $\mathcal{U}_{q,l}/q \rightarrow A/\sqrt{2}$. Adding the finite-centre limit proves OER15 with the same radius. The original definitions give $C \geq 0$ and $M \geq \log(w_E^2/z_E) > 0$, since $w_E^2/z_E = 1 + (v_E - u_E)^2/(4u_E v_E) > 1$. The radius remains strictly positive. The evaluation does not assert a q -scale limit for the actual W_k or change its finite interval.

62.7 Full-window residual bounds in these same receivers

Retain the literal RWC4 matrices for each source $s = 1, k$ and the two low cutoffs $N = q - 1, q$. Their full-window positive update, including the full $q \times q$ conditioning matrix, is

$$\begin{aligned} U_N^+ &= [u_{N+1}, \dots, u_{N+q}], \quad B_N = I_q + (CU_N^+)^*(CR_N C^*)^{-1}(CU_N^+), \\ Z_N &= D[I_q - R_N C^*(CR_N C^*)^{-1}C]U_N^+, \quad V_N = Z_N B_N^{-1} Z_N^* > 0, \quad \Omega_{N+q} = \Omega_N + V_N. \end{aligned} \quad (\text{OER17})$$

The original full conductor rows and residual rows here are precisely ROQ1 and ROQ4; D in this subsection denotes those rows. RWC2–6 proves the rank and update by the positive original quartet polynomial, orthogonality of the complete consecutive source block, and the displayed full covariance expansion. With $Y_N = \Omega_N^{-1/2} Z_N B_N^{-1/2}$, the exact four-endpoint residual and its finite bounds are

$$\begin{aligned} \Phi_{k,s} &= \sum_{N \in \{q-1, q\}} \log \left(\sum_{d=0}^j \left\| \bigwedge_{d=0}^d Y_N \right\|_{\text{HS}}^2 \right), \\ \lambda_s &< \ell_{k,s} = \Phi_{k,s}^- \leq \Phi_{k,s} \leq \Phi_{k,s}^+, \end{aligned} \quad (\text{OER18})$$

where the complete trace, determinant and earlier finite-cap definitions of ℓ, Φ^+ are RWC7–11. RWC8 proves the exterior equality by all singular products; RWC14 proves the strict improvement over the original first increment λ_s . Thus every exterior coefficient supplies a specified finite subtraction in the same kernel equation.

On the exact PMR range $k \geq 29$, $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}(k+1)^2$, put

$$\begin{aligned} L_s^{\text{win}} &= \max\{L_s^{\text{ext}}, \mathcal{F}_{0,k}^{(s)} - e_{k,s}^{0,-} - \Phi_{k,s}^+\}, \\ U_s^{\text{win}} &= \min\{U_s^{\text{ext}}, \mathcal{F}_{0,k}^{(s)} + e_{k,s}^{0,+} - \Phi_{k,s}^-\}, \quad e_{k,s}^{0,\pm} = e_{k,s}^\pm + c_k^\circ, \\ L_s^{\text{win}} &\leq F_K^{(s)} \leq U_s^{\text{win}}. \end{aligned} \quad (\text{OER19})$$

RWC12 proves that each added interval contains the actual kernel volume, so taking its intersection with OER12 proves OER19. Outside that PMR range use the original OAR expression $\mathcal{F}_{\partial,k}^{(s)}$ and its original $e_{k,s}^{\pm}$ in precisely the same construction on OAR's domain. For the arithmetic statements retain $|u| \geq R_*$ as in AMT. Substitute $L_s^{\text{win}}, U_s^{\text{win}}$ for the source bounds in the last two lines of OER12, and for source k in OER13. This gives respectively the same correlated arithmetic interval, its exact boundary complement, and

$$\begin{aligned} L_k^{\text{win}} + \mathcal{H}_k + \delta_k^{\sigma} &\leq \Psi_k \leq U_k^{\text{win}} + \mathcal{H}_k + \delta_k^{\sigma}, \\ J_k^{\text{action}} &\leq U_k^{\text{win}} + \mathcal{H}_k + \delta_k^{\sigma} + C_k^{\text{budget}}. \end{aligned} \tag{OER20}$$

The proof is addition of the unchanged scalars to the actual kernel bounds and then the exact OMR26 identity. The full budget, its same-class lower bound, both low corrections and every unequal finite error remain present. No estimate of the individual kq -scale residual coefficient is inferred from strict finite positivity alone.

63 The complete common factor of the original polynomial relations

Retain the full original $\mathcal{U}$, d_{ab} , roots, coefficients, and degree $\mathbf{m} = \Delta - v$ of RDS1–2 and ROG1–6. Let $\mathcal{S}_A = \{(a, b) : a_{ab} \neq 0\}$ be the actual nonempty coefficient support, and define its four exact integer extrema

$$a_- = \min_{\mathcal{S}_A} a, \quad a_+ = \max_{\mathcal{S}_A} a, \quad b_- = \min_{\mathcal{S}_A} b, \quad b_+ = \max_{\mathcal{S}_A} b, \quad d_a = a_+ - a_-, \quad d_b = b_+ - b_-. \quad (\text{RCF1})$$

These are cases determined by the original coefficients; none is assigned a generic support. Write $k' = k - 8$ and extend only the root-index formula to integer pairs by

$$\tilde{\beta}_{ij} = c' + (2i - k')\delta + i(2j - k')\gamma. \quad (\text{RCF2})$$

The i multiplying the imaginary term is the imaginary unit. The positive real numbers δ, γ make this indexed map injective on all integer pairs by comparing real and imaginary parts.

Each original shifted outer product d_{ab} has precisely the simple roots indexed by

$$([-a, k - a] \times [-b, k - b]) \setminus ([0, k'] \times [0, k']), \quad (\text{RCF3})$$

where all intervals contain their integer points. Indeed $\beta_{ij} - b_{ab} = \tilde{\beta}_{i-a, j-b}$, and the rectangle removed in CEP41 is exactly the inner rectangle in these translated indices. This verifies the whole root set, not just its cardinality.

Every rectangle in RCF3 contains that same inner rectangle. Their intersection is $[-a_-, k - a_+] \times [-b_-, k - b_+]$. Consequently the exact monic greatest common divisor of the active products is

$$G_A(S') = \prod_{(i,j) \in ([-a_-, k - a_+] \times [-b_-, k - b_+]) \setminus [0, k']^2} (S' - \tilde{\beta}_{ij}), \quad (\text{RCF4})$$

$$g_A := \deg G_A = (k + 1 - d_a)(k + 1 - d_b) - q'.$$

Distinctness of the roots proves that the intersection product is their monic greatest common divisor, including every root with its multiplicity one. Empty products give $G_A = 1$. The rectangle is nonempty and contains the inner square because $0 \leq a_- \leq a_+ \leq 8$ and $0 \leq b_- \leq b_+ \leq 8$.

This product is also the monic generator of the polynomial ideal generated by the complete linear relation image:

$$(\mathcal{UC}[S]) = G_A \mathbb{C}[S']. \quad (\text{RCF5})$$

Here parentheses mean the ideal generated by that image, whose underlying linear subspace is still the full original image. To prove the equality, list the $t = |\mathcal{S}_A|$ active pairs as $\alpha_1, \dots, \alpha_t$ and form the matrix

$$V(S')_{n,j} = (S' + b_{\alpha_j})^n, \quad 0 \leq n < t, \quad 1 \leq j \leq t. \quad (\text{RCF6})$$

Its determinant, in the displayed order, is the nonzero constant $\prod_{i < j} (b_{\alpha_j} - b_{\alpha_i})$. The shift nodes are distinct by the same real and imaginary comparison as RCF2. Its adjugate divided by this constant is therefore an inverse matrix over $\mathbb{C}[S']$. The vector of the first t exact relations satisfies

$$(\mathcal{U}1, \mathcal{U}S, \dots, \mathcal{U}S^{t-1})^T = V(S')(a_{\alpha_j} d_{\alpha_j}(S'))_{j=1}^t. \quad (\text{RCF7})$$

Thus these two finite lists generate the same polynomial ideal. All nonzero a_{α_j} are invertible constants, so it is the ideal generated by the active d_{α_j} . The Euclidean algorithm for polynomials makes this the ideal of their monic greatest common divisor RCF4. For completeness, each polynomial remainder step $f = qg + r$ leaves the two generated ideals equal; decreasing degree terminates and identifies the final nonzero remainder, divided by its leading coefficient, with the monic common divisor. The other relations are polynomial combinations of the same active d_{α_j} , by their full formula RDS1. This proves RCF5 for the whole image.

In particular the actual linear image is a polynomial ideal if and only if

$$g_A = \mathfrak{m}. \quad (\text{RCF8})$$

If it is an ideal, it equals its ideal hull RCF5, whose codimension is g_A . RDS2 gives its actual codimension $\mathfrak{m}$, proving necessity. Conversely divide each active d_{ab} by the full G_A , and define the exact polynomial operator

$$\mathcal{V}_A h = \sum_{(a,b) \in \mathcal{S}_A} a_{ab} \frac{d_{ab}(S')}{G_A(S')} h(S' + b_{ab}), \quad \mathcal{U} = M_{G_A} \mathcal{V}_A. \quad (\text{RCF9})$$

Its degree on a nonzero degree- j input is exactly $j + \mathfrak{m} - g_A$ and its leading coefficient is still $\binom{q+j}{v} \mu_v$ times the input leading coefficient. This follows from monic exact division of RDS1; also $g_A \leq \mathfrak{m}$ because G_A divides the nonzero $\mathcal{U}1$ of degree $\mathfrak{m}$. If equality holds, $\mathcal{V}_A$ preserves degree with nonzero diagonal coefficients in every finite monomial frame. Repeated highest-degree elimination therefore proves that it is onto all polynomials. RCF9 then proves sufficiency in RCF8.

For every actual support, including the unequal-degree cases, the precise finite-dimensional factor sequence is

$$0 \longrightarrow \mathbb{C}[S']/\mathcal{V}_A \mathbb{C}[S] \xrightarrow{M_{G_A}} \mathbb{C}[S']/\mathcal{U} \mathbb{C}[S] \xrightarrow{\text{rem}_{G_A}} \mathbb{C}[S']/(G_A) \longrightarrow 0, \quad (\text{RCF10})$$

$$\dim(\mathbb{C}[S']/\mathcal{V}_A \mathbb{C}[S]) = \mathfrak{m} - g_A = (d_a + d_b)(k+1) - d_a d_b - v.$$

The first map is well defined by RCF9. If $G_A f$ belongs to $\mathcal{U} \mathbb{C}[S] = G_A \mathcal{V}_A \mathbb{C}[S]$, cancellation of the nonzero polynomial proves that f belongs to the first denominator, so that map is injective. The last map is onto by ordinary monic remainder, and its kernel consists exactly of the classes represented by multiples of G_A , the image of the first map. The same degree-elimination proof as RDS2, applied to RCF9, gives its first dimension; expansion of RCF4 gives the last expression. If $G_A = 1$, the rightmost quotient is zero and these maps retain the entire original relation quotient on the left.

The source metric transported by the first map is exactly

$$\|G_A f\|_{\chi', s, c'}^2 = \int |f(c' + iy)|^2 |\chi'(c' + iy) G_A(c' + iy)|^2 dm_s(y). \quad (\text{RCF11})$$

This is direct substitution into the original weighted norm ROG21. It retains every factor and the original source order and mass. Thus RCF9–11 are a specified map to the polynomial ideal supplied by the actual relations, together with the full remaining quotient. They do not identify its metric with the CTV outer-divisor metric: the original natural map to that divisor remains GIP7–9, while RDS8 gives the exact analytic realization of the complete intermediate quotient. The original residual map and its common-kernel subspace remain the transported ROG30 map on that same intermediate space.

63.1 Every original finite degree window and its exact metric sequence

For the original $N = q - 1, q, 2q - 1, 2q$ put $L = N - q$, $M = L + \mathfrak{m}$ as in ROG2, and set $r_A = \mathfrak{m} - g_A \geq 0$. At all four endpoints the exact finite version of RCF10 is

$$0 \longrightarrow \mathbb{C}[S']_{\leq M - g_A} / \mathcal{V}_A \mathbb{C}[S]_{\leq L} \xrightarrow{M_{G_A}} \mathbb{C}[S']_{\leq M} / \mathcal{U} \mathbb{C}[S]_{\leq L} \xrightarrow{\text{rem}_{G_A}} \mathbb{C}[S']_{< g_A} \longrightarrow 0, \quad (\text{RCF12})$$

$$\dim(\text{first space}) = r_A, \quad \dim(\text{middle space}) = \mathfrak{m}, \quad \dim(\text{last space}) = g_A.$$

Here the first numerator is zero when $M - g_A = -1$, and every denominator is zero at $L = -1$. The degree identity in RCF9 puts its image in precisely the first numerator. The maps and cancellations used in RCF10 preserve these degree bounds, proving exactness. Moreover $M \geq \mathfrak{m} - 1 \geq g_A - 1$, so all last-space remainder classes have representatives in the displayed middle numerator. The

first numerator dimension is $L + r_A + 1$, while its denominator dimension is $L + 1$ when $L \geq 0$; at $L = -1$ it is r_A with zero denominator. These counts prove the dimensions, including the empty cases.

Use the quotient norm of ROG21 on the middle space, the quotient norm of RCF11 on the first space, and the attained minimum quotient norm of the same ROG21 polynomial source under rem_{G_A} on the last space. These are exactly a Hilbert restriction and its quotient. Indeed, every lift of a middle class $[G_A f]$ differs from it by $\mathcal{U}h = G_A \mathcal{V}_A h$; thus all its lifts remain multiples of G_A , and its minimum is precisely the first space's stated norm. Quotienting further by that first space allows every change by a multiple of G_A within the finite window, exactly the last remainder norm. No polynomial division map was presumed to be orthogonal in this argument.

Let $H_{U,N}^{(s)}$, $H_{V,N}^{(s)}$, and $Q_{G,N}^{(s)}$ be these three Grams in their canonical remainder coefficient frames $1, S', \dots$ of dimensions $\mathfrak{m}, r_A, g_A$. Those remainder frames exist by the degree elimination proofs RDS2 and RCF9. Then the exact determinant identity is

$$\det H_{U,N}^{(s)} = \det H_{V,N}^{(s)} \det Q_{G,N}^{(s)}. \quad (\text{RCF13})$$

To prove it, use in the middle quotient the fixed frame $[G_A, G_A S', \dots, G_A S'^{r_A-1}, 1, S', \dots, S'^{g_A-1}]$. Reordering it by increasing leading degree gives a triangular monic coefficient matrix. Its original order therefore has determinant $(-1)^{g_A r_A}$ and modulus one, retaining every lower coefficient of G_A in the matrix. The first Gram block is $H_{V,N}^{(s)}$; its Schur complement is the attained quotient Gram $Q_{G,N}^{(s)}$ by the minimum-lift calculation. Block Gaussian elimination proves RCF13, including zero-dimensional determinants equal to one.

The four-return identity thus has the unchanged endpoint signs:

$$F_U^{(s)} = F_V^{(s)} + F_G^{(s)}, \quad F_G^{(s)} = \sum_{N=q-1, q, 2q-1, 2q} \sigma_N \log \det Q_{G,N}^{(s)}, \quad \sigma = (1, 1, -1, -1). \quad (\text{RCF14})$$

The other two returns are the same sums of their Grams in RCF13. The full source orders are $s = 1, k$. Thus RCF14 splits an actual original relation quotient metric; the remaining factor has the explicit weighted source RCF11 and the complete operator RCF9.

There is also a fixed original primary map, before any metric choice or shear:

$$\rho_G : V_A \twoheadrightarrow \mathbb{C}[S']_{<g_A}, \quad [p] \mapsto \text{rem}_{G_A} \left(\frac{\mathcal{T}_{Ap}}{\chi'} \right). \quad (\text{RCF15})$$

Membership in V_A makes the quotient polynomial by CEP5. A change of representative by χh changes it by $\mathcal{U}h$, divisible by G_A , proving independence. The map is onto: for a prescribed polynomial f of degree below g_A , the source surjectivity CEP6 gives a preimage of $\chi' f$ of degree at most $q' + g_A - 1 + v \leq q - 1$. This preimage belongs to V_A and has the stated remainder. At $g_A = 0$ the target is zero and surjectivity has its usual zero-dimensional meaning. The map is consequently the same one at all original endpoints and in both original source norms.

64 Original Gamma control on the actual common-factor quotient

Retain RCF1–15, all fixed original period coefficients and their actual support, and the complete original sources $s = 1, k$. Write $g = g_A$, $\mathbf{m} = \Delta - v$, $M = N - q + \mathbf{m}$. If $g = 0$, the common-factor target and every quotient of it are zero, their determinants are one, and all conclusions below hold with zero return. In the proof of its positive-dimensional operator inequalities suppose $g \geq 1$; this includes every remaining actual support case.

64.1 The exact original divisor, windows, phases and source

Use the unchanged real source coordinate $S' = c' + iy$ and define

$$G_y(y) = i^{-g} G_A(c' + iy), \quad \chi'(c' + iy) = i^{q'} P_{k-8}(y), \quad R_G = (k + 8)\sqrt{\delta^2 + \gamma^2}. \quad (\text{RCM1})$$

These are monic polynomials with exactly the displayed phases. By RCF2–4 the roots of G_y are $(2j - k')\gamma - i(2i - k')\delta$ in its stated integer rectangle. Every $|2i - k'|, |2j - k'|$ there is at most $k + 8$; hence every root has modulus at most R_G . They are distinct. No root or coefficient is set to a surrogate value in RCM1.

The target coefficients of $1, S', \dots, S'^{g-1}$ and of $1, y, \dots, y^{g-1}$ are related by the fixed invertible matrix

$$T_{ab} = \begin{cases} \binom{b}{a} (c')^{b-a} i^a, & a \leq b, \\ 0, & a > b, \end{cases} \quad 0 \leq a, b < g, \quad \det T = i^{g(g-1)/2}. \quad (\text{RCM2})$$

This follows by expanding $(c' + iy)^b$. In particular $|\det T| = 1$, while the full matrix, including its centre, is retained in every operator congruence. Let $Q_{G,M}^{y,(s)}$ be the minimum remainder Gram by G_y on polynomials of degree at most M , with source

$$d\zeta_{k',s}(y) = |P_{k-8}(y)|^2 dm_s(y). \quad (\text{RCM3})$$

The complete phases in RCM1 have modulus one. Substitution proves $Q_{G,N}^{(s)} = T^* Q_{G,M}^{y,(s)} T$ for the RCF12–14 Gram. The four actual M values are

$$\mathbf{m} - 1, \quad \mathbf{m}, \quad q + \mathbf{m} - 1, \quad q + \mathbf{m}. \quad (\text{RCM4})$$

They include all additional columns beyond the divisor's degree: $g \leq \mathbf{m} \leq \Delta$ and $\mathbf{m} \geq 16$.

64.2 Complete finite bounds at the low source windows

For the comparison measure $|y|^{2Q} e^{-\eta|y|} dy$ use the full positive source rate $\eta > 0$ and put

$$\mathfrak{g}(Q, \eta) = 2Q \log(2Q/\eta) - 2Q, \quad y_0 = 2Q/\eta. \quad (\text{RCM5})$$

OOQ26–27 proves the following bounds on the entire polynomial coefficient space of degree at most Δ :

$$\frac{e^{\mathfrak{g}(Q,\eta)-\eta}}{(\Delta+1)^2(2\Delta+1)64^\Delta(1+y_0)^{2\Delta}} I \preceq H_\Delta^{\text{exp}} \preceq \left(\sum_{j=0}^{\Delta} \frac{2(2Q+2j)!}{\eta^{2Q+2j+1}} \right) I. \quad (\text{RCM6})$$

Here Q will be the original integer $q' + (s-1)/4$. These estimates retain the positive-interval integration bound and the full two-half-line moment trace from OOQ, respectively.

For every monic degree- g polynomial whose roots have modulus at most R_G , the ordinary remainder map on degree at most Δ has coefficient operator norm at most

$$C_{\text{lo}} = 1 + (1 + R_G)^\Delta (\Delta + 1) \max\{1, \Delta R_G\}^\Delta. \quad (\text{RCM7})$$

To prove this for the actual G_y , its coefficient absolute sum is at most $(1 + R_G)^g$. Polynomial division uses the first $\Delta - g + 1$ complete homogeneous coefficients of its inverse at infinity. Their sum is at most $\sum_{a=0}^{\Delta-g} \binom{g+a-1}{a} R_G^a \leq (\Delta + 1) \max\{1, \Delta R_G\}^\Delta$. The coefficient convolution argument in ROG19–20 bounds the quotient operator by this sum. The exact remainder is the input minus its quotient multiplied by G_y , which proves RCM7. This proof does not truncate the actual root polynomial.

At either low window in RCM4, taking the minimum over all lifts of a remainder z gives a lower norm bound from RCM6 and $\|p\|_{\text{coef}} \geq \|z\|_2 / C_{\text{lo}}$. The canonical degree-less-than- g representative is an allowed lift and gives the upper bound from RCM6. Thus define

$$E_{\text{lo}}^G(Q, \eta) = \max \left\{ 0, \log \left(\sum_{j=0}^{\Delta} \frac{2(2Q + 2j)!}{\eta^{2Q+2j+1}} \right) - \mathfrak{g}(Q, \eta), \right. \\ \left. \eta + 2 \log(\Delta + 1) + \log(2\Delta + 1) + \Delta \log 64 + 2\Delta \log(1 + y_0) + 2 \log C_{\text{lo}} \right\}. \quad (\text{RCM8})$$

The full low remainder Gram obeys

$$e^{\mathfrak{g}(Q, \eta) - E_{\text{lo}}^G} I_g \preceq Q_{G, M}^{\text{exp}}(Q, \eta) \preceq e^{\mathfrak{g}(Q, \eta) + E_{\text{lo}}^G} I_g. \quad (\text{RCM9})$$

The extra low columns $g, \dots, M$ have all been included in the lower minimization. The factorial bounds proved after OOQ29 and $\log C_{\text{lo}} = O_{\delta, \gamma}(\Delta \log q)$ give $E_{\text{lo}}^G = O_{\delta, \gamma}(k \log q)$ in the source and rate ranges specified below.

64.3 The high windows with the actual analytic remainder and lift

Retain the equilibrium constants of OOQ3–6 on its entire specified compact parameter rectangle. Set

$$n = q/2, \quad \rho = n^{-1/8}, \quad r_* = R_G/n, \quad C_{\text{rem}} = \rho \Delta (1 + r_*)^\Delta (2/\rho)^\Delta. \quad (\text{RCM10})$$

Use the original OSP23 source-comparison range, $k \geq 257$, and the explicit eventual conditions

$$n \geq 2\Delta + 1, \quad n \geq 2/b_*, \quad r_* \leq \rho/2, \quad \rho \leq \min\{1/2, \sqrt{a_*/2}\}. \quad (\text{RCM11})$$

They hold for all sufficiently large original k at the fixed quartet, since $R_G = O(k)$ and $\Delta = O(k)$ while $n \asymp k^2$.

Apply the exact Cauchy remainder OOQ14 to $\hat{G}(z) = n^{-g} G_y(nz)$. Its g roots are distinct and lie inside $|z| \leq r_*$. Its coefficient norm bound is $\rho g (1 + r_*)^g (2/\rho)^g$, at most C_{rem} because $1 \leq g \leq \Delta$ and $2(1 + r_*)/\rho > 1$. The Cauchy formula interpolates the actual roots and is exactly the remainder by $\hat{G}$, as proved in OOQ14–15.

For either high M in RCM4, the parity degrees $m_e = \lfloor M/2 \rfloor$, $m_o = \lfloor (M - 1)/2 \rfloor$ belong to $[n, n + \Delta]$. The exact half-line parameters are $((Q - 1/2)/m_e, n\eta/m_e)$ and $((Q + 1/2)/m_o, n\eta/m_o)$. They lie in OOQ's compact rectangle for the same bounds as OOQ17, since $16 \leq \mathfrak{m} \leq \Delta$. The two-branch Jacobian, weighted polynomial evaluation bound, and smooth variation estimate are therefore precisely OOQ18–20, with the upper degree bound $n + \Delta$. These steps give the same lower quotient bound with C_{rem} from RCM10.

For completeness the upper lift is the literal polynomial

$$a_\varphi = \text{rem}_{\hat{G}} \left(\frac{\varphi}{r_n(\cdot)^2} \right), \quad p_\varphi(nx) = r_n(x^2) a_\varphi(x), \quad (\text{RCM12})$$

where φ denotes the prescribed degree-less-than- g remainder polynomial and r_n is the unchanged quantile polynomial OOQ9. Its denominator is nonzero on the integration disk because $\rho^2 < a_*$. The lift has degree at most $q + g - 1 \leq q + \mathfrak{m} - 1 \leq M$ and agrees at every actual root of $\hat{G}$ with the prescribed remainder. Thus it is an allowed original high-window lift. Its coefficient and integral estimates are OOQ22–23 with $g \leq \Delta$ and the tail power bounded by $2(\Delta - 1)$.

To retain all finite errors, put

$$\begin{aligned}
E^- &= 2 \log(C_{\text{rem}} \sqrt{2} C_{\text{BM}}(n + \Delta)^2) + 2C_W(\Delta + 2) + 4(n + \Delta)\rho^2/a_*, \\
E^+ &= 2 \log C_{\text{rem}} + \log \Delta + 4n\rho^2/a_* + 2C_{\text{quant}}\sqrt{n} + 2C_0 \\
&\quad + 2C_{\text{tail}}(\Delta - 1) + \log T_*, \\
E_{\text{hi}}^G &= \max\{0, E^-, E^+\} + 2(\Delta - 1) \log n, \\
c_{\text{hi}}^{\text{exp}}(Q, \eta) &= (2Q + 1) \log n - 2nW((Q - 1/2)/n, \eta).
\end{aligned} \tag{RCM13}$$

The exact original-versus-scaled remainder matrix is $\text{diag}(1, n, \dots, n^{g-1})$; its squared norm is at most $n^{2(\Delta-1)}$, the last term in RCM13. The preceding bounds prove at both high windows

$$e_{\text{hi}}^{\text{exp}}(Q, \eta) - E_{\text{hi}}^G I_g \preceq Q_{G, M}^{\text{exp}}(Q, \eta) \preceq e_{\text{hi}}^{\text{exp}}(Q, \eta) + E_{\text{hi}}^G I_g, \quad E_{\text{hi}}^G = O_{\delta, \gamma}(q^{3/4} + k \log q). \tag{RCM14}$$

Every imported estimate here has the same actual source, parity and centre scalings as OOQ. The only changes in its remainder proof are the specified divisor G_y , its proved root bound, and the degree allowances verified above.

64.4 Both full Gamma sources and the four-return coefficient

Retain the exact source constants of OOQ30–31:

$$\begin{aligned}
\vartheta &= \pi/2, \quad h = 1/q, \quad \ell_s = (s - 1)/4, \quad Q_s = q' + \ell_s, \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}, \\
L_h &= \frac{\pi}{\Gamma(3/4)^2} (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2},
\end{aligned} \tag{RCM15}$$

and the complete root/source errors $\mathcal{E}_k, r_{\ell_s}$ from OSP24 and OFV6, with $r_0 = 0$. The full quadratic-form comparison OOQ31 applies to every polynomial used here, since $M \leq q + \Delta \leq 2q$ on RCM11's range. Taking minima over the same actual G_y fibres preserves those two inequalities, including all source masses.

Define the two unchanged leading centres

$$c_{\text{lo}} = \mathfrak{g}(q, \vartheta), \quad c_{\text{hi}} = 2q \log(q/2) - qW(2, \vartheta), \tag{RCM16}$$

and a single finite nonnegative error

$$\begin{aligned}
\varepsilon_k^G &= \mathcal{E}_k + \max_{s=1, k} r_{\ell_s} + \max\{|\log L_h|, |\log U_h|\} \\
&\quad + \max_{\substack{s \in \{1, k\} \\ \eta \in \{\vartheta - h, \vartheta + h\}}} \{E_{\text{lo}}^G(Q_s, \eta) + E_{\text{hi}}^G + |\mathfrak{g}(Q_s, \eta) - c_{\text{lo}}| + |c_{\text{hi}}^{\text{exp}}(Q_s, \eta) - c_{\text{hi}}|\}.
\end{aligned} \tag{RCM17}$$

All its constants are supplied above or by the complete original source and equilibrium proofs cited there. The resulting operator inequalities, in the original S' frame, are

$$\beta_s e^{c_{\bullet} - \varepsilon_k^G} T^* T \preceq Q_{G, N}^{(s)} \preceq \beta_s e^{c_{\bullet} + \varepsilon_k^G} T^* T, \tag{RCM18}$$

where $c_{\bullet} = c_{\text{lo}}$ at $N = q - 1, q$ and $c_{\bullet} = c_{\text{hi}}$ at $N = 2q - 1, 2q$. Moreover

$$\varepsilon_k^G = O_{\delta, \gamma}(q^{3/4} + k \log q) = o(q). \tag{RCM19}$$

Indeed RCM8 and RCM14 control the new errors. The source errors have the orders proved in OOQ35. Also $Q_s - q = -\Delta + \ell_s = O(k)$, the rates differ by $1/q$, and W with its first derivatives is bounded on the same compact rectangle. The mean-value estimates in OOQ33–35 give $O(k \log q)$ for both scalar discrepancies in RCM17. This proves RCM19 without a support-dependent lower bound.

OOQ36 gives the exact difference $c_{\text{lo}} - c_{\text{hi}} = q\mathfrak{C}_\partial/2$ with its unchanged equilibrium constant. Taking four determinants in RCM18, and retaining the signs and fixed frame of RCF14, therefore proves

$$|F_G^{(s)} - gq\mathfrak{C}_\partial| \leq 4g\varepsilon_k^G = o(kq). \quad (\text{RCM20})$$

The source factor β_s^g and $\det(T^*T)$ cancel with coefficients $1 + 1 - 1 - 1$. All individual Grams retain them. The exact relation return consequently obeys the proved partial evaluation

$$F_U^{(s)} = F_V^{(s)} + gq\mathfrak{C}_\partial + \epsilon_G^{(s)}, \quad |\epsilon_G^{(s)}| \leq 4g\varepsilon_k^G, \quad (\text{RCM21})$$

where the remaining F_V is the complete smaller operator and weighted source in RCF9–14, of rank $\mathfrak{m} - g$.

64.5 The resulting original residual quotient and its retained complement

Let $\Xi = \Xi_k^{\text{fix}} : V_A \rightarrow B_\Xi$ be the same original residual observation, with $B_\Xi = \mathbb{C}^j$, $j = 8k - 32$, and kernel K . Use the fixed onto map ρ_G proved in RCF15, and put

$$V_0 = \ker \rho_G, \quad W = \Xi(V_0), \quad t_G = \dim(B_\Xi/W) = g - \dim \rho_G(K). \quad (\text{RCM22})$$

There is the precise onto map

$$A_G : \mathbb{C}[S']_{<g} \rightarrow B_\Xi/W, \quad \rho_G x \mapsto [\Xi x], \quad \ker A_G = \rho_G(K). \quad (\text{RCM23})$$

Two lifts differ by V_0 , so the map is well defined. It is onto because Ξ is onto. If $\Xi x \in W$, subtract an element of V_0 with that same image; the remaining element belongs to K and has the same ρ_G value. This proves the kernel equality and RCM22's dimension. All these maps are independent of source order and cutoff.

In the graph source X_N of GIP1, the composite target map is exactly $(x, f) \mapsto A_G \text{rem}_{G_A} f$. It ignores the low coordinate and kills every full graph relation, since $\mathcal{U}h$ is divisible by G_A . Minimizing its product norm first over the free low coordinate and then over polynomial remainder fibres proves that its quotient Gram is exactly

$$Q_{A_G, G, N}^{(s)} = [A_G(Q_{G, N}^{(s)})^{-1} A_G^*]^{-1}. \quad (\text{RCM24})$$

Thus no equality of different relation subspaces enters this identification; only the proved composite map and its full kernel are used.

Let $\overline{Q}_{\Xi, N}^{(s)}$ be the original metric on B_Ξ/W induced by the original residual quotient. The actual source map S_N of CEP28 induces the identity of this target with RCM24. Its minimum-lift exterior argument CEP34–38, at the actual rank t_G , gives

$$|\log \det \overline{Q}_{\Xi, N}^{(s)} - \log \det Q_{A_G, G, N}^{(s)}| \leq 2E_N^{\text{CEP}}. \quad (\text{RCM25})$$

The source preimage is precisely $\mathcal{W}_N$ of CEP31. On its S_N image it is $\mathbb{C}^v \oplus \chi' \mathbb{C}[S']_{\leq M}$; the composite map is the one just minimized in RCM24. Orthogonal projection off the entire composite kernel is an exterior contraction, for both S_N and its inverse. This verifies all spaces in the bound and retains the original source map determinant from CEP29.

Taking the same fixed quotient of RCM18 replaces T^*T by the full fixed reference $[A_G(T^*T)^{-1} A_G^*]^{-1}$, through minimum on identical fibres. Its determinant and the original mass cancel only in the four-sign return. Consequently

$$\left| \sum_N \sigma_N \log \det \overline{Q}_{\Xi, N}^{(s)} - t_G q \mathfrak{C}_\partial \right| \leq 4t_G \varepsilon_k^G + 2 \sum_N E_N^{\text{CEP}} = o(kq). \quad (\text{RCM26})$$

No coefficient conditioning of the fixed quotient A_G has been assumed; its complete reference form is retained and cancels in the displayed return.

Finally choose any full fixed original frame I_W for W and put

$$F_W^{(s)} = \sum_N \sigma_N \log \det(I_W^* Q_{\Xi, N}^{\text{fix}} I_W).$$

Block elimination in a fixed frame of B_Ξ extending I_W gives its restriction determinant times the quotient determinant. The full fixed frame factor cancels in the four signs, exactly as in CEP26. Hence the original residual return has the established decomposition

$$F_{\Xi, k}^{(s)} = F_W^{(s)} + t_G q \mathfrak{C}_\partial + \epsilon_\Xi^{(s)}, \quad |\epsilon_\Xi^{(s)}| \leq 4t_G \varepsilon_k^G + 2 \sum_N E_N^{\text{CEP}}. \quad (\text{RCM27})$$

The remaining rank is $\dim W = j - t_G$. Its original metric and embedding have been specified, not replaced by a scalar or by a source norm. Also $F_W^{(s)} \geq 0$: the original source polynomial spaces are nested, with the same remainder and residual target, so minimum quotient metrics decrease as N increases. Restriction to the same W preserves this positive-form order, and the two low-versus-high determinant comparisons are nonnegative. Therefore RCM27 also gives the actual lower estimate $F_{\Xi, k}^{(s)} \geq t_G q \mathfrak{C}_\partial - 4t_G \varepsilon_k^G - 2 \sum_N E_N^{\text{CEP}}$. The value of t_G uses the exact active support and original map $\rho_G|_K$; no generic nonvanishing or zero-gcd case has been substituted for that calculation.

65 The original weighted relation metric as a minimum dual extension

Retain the original simple quartet, fixed five-orbit period and logarithm branch, all coefficients and shifts of ROG and RDS, and the same two source orders $s \in \{1, k\}$. Put

$$\begin{aligned} q &= (k+1)^2, & q' &= (k-7)^2, & \Delta &= q - q', & \mathbf{m} &= \Delta - v, & c' &= k/2 - 4, \\ N &\in \{q-1, q, 2q-1, 2q\}, & L &= N - q, & d &= N - v, & M &= N - v - q' = L + \mathbf{m}. \end{aligned} \quad (\text{DNE1})$$

Here $0 \leq v \leq 80$, $\mu_v \neq 0$, and χ' is the complete lower-grid polynomial χ_{k-8} , never a derivative. All spaces of degree at most -1 are zero. The source norm is the original norm at $S' = c' + iy$, with mass $M_s = (2\pi)^{s/2}$.

65.1 A fixed original exponential quotient frame

We specify a finite frame using only the actual original matrices. Let $\mathbb{V}$ be the upper-root Vandermonde with rows $a = 0, \dots, q-1$ and entries $\mathbb{V}_{a,(i,j)} = \beta_{ij}^a$. Let $P_v : \mathbb{C}^{q-v} \rightarrow \mathbb{C}^q$ insert zero in the first v positions, and let C be the exact primary conductor of CEP4, with q' rows and q columns. Define

$$W_v = \mathbb{V}^{-1}P_v, \quad J_v = P_v^\top \mathbb{V} C^\top, \quad W_v J_v = C^\top. \quad (\text{DNE2})$$

The Vandermonde is invertible by the original distinct roots. The first v rows of $\mathbb{V} C^\top$ vanish because the corresponding exponential columns are $E_A e^{\beta'_{ab} z}$, whose zero order is at least v . This proves the last equality. Consequently W_v is an isomorphism onto the coefficient vectors of $\mathcal{N}_v$, and J_v has rank q' , since $C^\top$ is injective.

Choose the first q' row positions $\mathcal{I}$ in a fixed ordering of all subsets for which $\det(J_v)_{\mathcal{I}} \neq 0$, and let $\mathcal{J}$ be their complement. Such a subset exists by the proved rank. Let $E_{\mathcal{J}}$ insert these $\mathbf{m}$ complementary coordinates and put

$$Z_{\text{exp}} = W_v E_{\mathcal{J}}, \quad \mathcal{N}_a(z) = \sum_{i,j} (Z_{\text{exp}})_{(i,j),a} e^{\beta_{ij} z}, \quad F_a(z) = \frac{\mathcal{N}_a(z)}{E_A(z)} \quad (1 \leq a \leq \mathbf{m}). \quad (\text{DNE3})$$

The columns of $[J_v, E_{\mathcal{J}}]$ are independent: restriction of a relation to $\mathcal{I}$ first kills its J_v coefficients, and then its other coefficients vanish. Thus the $\mathcal{N}_a$ give a basis of the exact quotient $\mathcal{N}_v / E_A \mathcal{E}_{k-8}$. Every root value and actual minor defining this frame is retained. The frame is fixed as the endpoint N and source order s change.

For an analytic germ F at zero define the complex-linear functional Λ_F by $\Lambda_F(S'^n) = F^{(n)}(0)$. Let ℓ_a be the functional from RDS8. Then

$$\ell_a(g) = \Lambda_{F_a}(\chi' g), \quad f_{\ell_a} = \chi'(D)F_a, \quad A_{\ell,ba} = \ell_a(S'^b) \quad (0 \leq b < \mathbf{m}). \quad (\text{DNE4})$$

Indeed multiplication by χ' in the functional is precisely the constant-coefficient operator $\chi'(D)$ on its generating series. RDS8 proves that these functionals form the entire dual of the relation quotient. RDS2 identifies that quotient with the increasing monomials below degree $\mathbf{m}$, so A_{ℓ} is invertible. This supplies its existence without assuming a nonzero unevaluated minor.

65.2 The exact original source extension problem

On $\mathbb{C}[S']_{\leq M}$ use the original weighted norm $\|g\|_{\chi',s,c'} = \|\chi' g\|_{s,c'}$. Let G_M be its full monomial Gram, and let $\mathbf{U}_L$ be the complete ROG4 relation matrix. The quotient

$$Y_N = \mathbb{C}[S']_{\leq M} / \text{im } \mathcal{U}_L \simeq \mathbb{C}[S']_{< \mathbf{m}}$$

has Gram, in its fixed low-monomial representative frame,

$$H_{Y,N}^{(s)} = E_{< \mathbf{m}}^* G_M E_{< \mathbf{m}} - E_{< \mathbf{m}}^* G_M \mathbf{U}_L (\mathbf{U}_L^* G_M \mathbf{U}_L)^{-1} \mathbf{U}_L^* G_M E_{< \mathbf{m}}. \quad (\text{DNE5})$$

The second term is absent when $L = -1$. Distinct leading degrees of the columns of $\mathcal{U}_L$ show that the low representatives and all relation columns form a basis. Positivity of G_M and minimization over the full relation columns therefore prove DNE5 and its strict positivity.

For $x \in \mathbb{C}^{\mathbf{m}}$ write $\mathcal{N}_x = \sum_a x_a \mathcal{N}_a$ and $F_x = \mathcal{N}_x / E_A$. Its functional ℓ_x on Y_N has the exact norm

$$\|\ell_x\|_{Y_N^\vee}^2 = \min_{\eta \in \mathbb{C}^{q'}} \left\| \Lambda_{F_x} + \sum_{a,b} \eta_{ab} \text{ev}_{\beta'_{ab}} \right\|_{(\mathbb{C}[S']_{\leq d}, \|\cdot\|_{s,c'})^\vee}^2. \quad (\text{DNE6})$$

Here the dual consists of complex-linear functionals. To prove the equality, multiply by χ' to identify the weighted polynomial space isometrically with $I_N = \chi' \mathbb{C}[S']_{\leq M} \subset \mathbb{C}[S']_{\leq d}$. The restriction of Λ_{F_x} to I_N is exactly $g \mapsto \ell_x(g)$ by DNE4, and it annihilates $\chi' \text{im } \mathcal{U}_L$ by RDS5.

A functional annihilating a subspace has the same norm on the Hilbert quotient: the minimum lift of each quotient class has the same functional value, and every other lift has larger norm. A functional on a Hilbert subspace has a unique extension of minimum norm to the ambient space, namely its composition with the orthogonal projection onto that subspace. Every other extension adds a functional on the perpendicular complement; their squared dual norms add. These two facts reduce its quotient dual norm to the minimum over all ambient extensions of its restriction.

The difference of two extensions annihilates I_N . That annihilator has dimension $(d+1) - (M+1) = q'$. Evaluation at each original lower root annihilates I_N , and these q' evaluation functionals are independent on $\mathbb{C}[S']_{\leq d}$, by the first q' monomials and the distinct-root Vandermonde. They therefore constitute its whole annihilator. This proves DNE6 and retains every permitted extension.

Let

$$\phi_{n,s,c'}(S') = \frac{p_n((S' - c')/i)}{\sqrt{h_n}}, \quad h_n = M_s n! (s/2)_n, \quad 0 \leq n \leq d$$

be the unchanged original orthonormal source frame. Form the complete finite matrices

$$\begin{aligned} T_{N,s;na} &= \Lambda_{F_a}(\phi_{n,s,c'}), & V_{N,s;n,(i,j)} &= \phi_{n,s,c'}(\beta'_{ij}), \\ \mathcal{D}_N^{(s)} &= T_{N,s}^* [I - V_{N,s}(V_{N,s}^* V_{N,s})^{-1} V_{N,s}^*] T_{N,s}. \end{aligned} \quad (\text{DNE7})$$

The matrix $V_{N,s}$ has rank q' by the same Vandermonde argument, since the displayed orthonormal polynomials span all degrees at most d . In an orthonormal basis the squared norm of a complex-linear functional is the sum of the squared moduli of its values. Thus the minimum in DNE6 is $\min_\eta \|T_{N,s}x + V_{N,s}\eta\|_2^2$. The minimizing vector is $-(V_{N,s}^* V_{N,s})^{-1} V_{N,s}^* T_{N,s}x$. Substitution proves DNE7, including both mixed factors, and gives $\|\ell_x\|^2 = x^* \mathcal{D}_N^{(s)} x$. Positivity follows because the ℓ_a are an independent dual frame of Y_N .

65.3 Complex-linear duality and the actual fixed kernel columns

For a positive Gram H and a complex-linear functional $y \mapsto a^\top y$, its squared dual norm is $a^\top H^{-1} \bar{a} = a^* \overline{H^{-1} a}$. Indeed its Riesz vector is $H^{-1} \bar{a}$, and evaluation of its squared norm gives the first expression; the second is its equal real conjugate. Therefore the precise metric identity and determinant consequence of DNE4–7 are

$$\mathcal{D}_N^{(s)} = A_\ell^* (\overline{H_{Y,N}^{(s)}})^{-1} A_\ell, \quad \det \mathcal{D}_N^{(s)} = \frac{|\det A_\ell|^2}{\det H_{Y,N}^{(s)}}. \quad (\text{DNE8})$$

The conjugation records the ordinary complex-linear dual coordinates; it is not discarded by changing the source frame. In the four-endpoint sum, the fixed factor $|\det A_\ell|^2$ cancels with signs $(1, 1, -1, -1)$.

Retain the actual fixed matrix Z_k of FEX27 and FGC9, with $t = r - t_0$ independent columns in the low-monomial frame of dimension $\mathbf{m}$. Multiplying every relation column by the same nonzero

scalar $\alpha_A = v!/\mu_v$ does not change its span or its Schur complement. Consequently its actual native FEX29 matrix and the exact dual quotient map are

$$\widehat{H}_{K,N}^{(s)} = Z_k^* H_{Y,N}^{(s)} Z_k, \quad B_K = Z_k^T A_\ell, \quad Q_{K^\vee,N}^{(s)} = \left[B_K (\mathcal{D}_N^{(s)})^{-1} B_K^* \right]^{-1} = \overline{(\widehat{H}_{K,N}^{(s)})^{-1}}. \quad (\text{DNE9})$$

The map B_K evaluates each relation-annihilating functional on the actual columns Z_k . It is onto, because A_ℓ is invertible and Z_k has rank t . The first inverse-Gram formula is the minimum dual metric on that same evaluation target. Substitution of DNE8 yields

$$B_K (\mathcal{D}_N^{(s)})^{-1} B_K^* = Z_k^T \overline{H_{Y,N}^{(s)}} \bar{Z}_k = \widehat{H}_{K,N}^{(s)},$$

which proves the final equality with every transpose in its original direction. Empty rank has determinant one. In particular the full fixed-column return is exactly

$$\widehat{F}_k^{(s)} = - \sum_N \sigma_N \log \det Q_{K^\vee,N}^{(s)}. \quad (\text{DNE10})$$

Its relation to the original $F_K^{(s)}$ retains both low quotient volumes and the full source-map error of FGC12–13 and FEX34–37; DNE10 removes none of those terms.

65.4 A finite source bound before the high-order derivative

Use the actual R, R', B, C_v and R_0 of ROG6 and RDS11, and put

$$\rho = \tanh(R_0/4), \quad C_{F,k,A} = \frac{2R^v}{|\mu_v|} e^{RR_0}. \quad (\text{DNE11})$$

For any $\mathcal{N} = \sum \nu_{ij} e^{\beta_{ij}z} \in \mathcal{N}_v$, RDS12 proves $|\mathcal{N}/E_A| \leq C_{F,k,A} \|\nu\|_1$ on the complete disk $|z| \leq R_0$. Apply the original Gamma generating identity directly to the functional $\Lambda_{\mathcal{N}/E_A}$:

$$\sum_{n \geq 0} \Lambda_{\mathcal{N}/E_A} (p_n((S' - c')/i)) \frac{z^n}{n!} = (1 + z^2)^{-s/4} e^{ic' \arctan z} \frac{\mathcal{N}(-i \arctan z)}{E_A(-i \arctan z)}. \quad (\text{DNE12})$$

The quotient at zero is removable. On $|z| = \rho$, $|\arctan z| \leq R_0/4$. Cauchy's coefficient estimate therefore gives the exact bound

$$\begin{aligned} \|\Lambda_{\mathcal{N}/E_A}\|_{(\mathbb{C}[S']_{\leq d}, \|\cdot\|_{s,c'})^\vee} &\leq \mathcal{B}_{d,s,k,A} \|\nu\|_1, \\ \mathcal{B}_{d,s,k,A}^2 &= \frac{C_{F,k,A}^2}{M_s} (1 - \rho^2)^{-s/2} e^{|c'|R_0/2} \sum_{n=0}^d \frac{n!}{(s/2)_n} \rho^{-2n}. \end{aligned} \quad (\text{DNE13})$$

Indeed the bound for the n th orthonormal coefficient is the square root of its displayed summand times the common factor and $\|\nu\|_1$; summing squared coefficients gives the dual norm. The full Gamma mass remains explicitly in this formula. No application of $\chi'(D)$ and no $q!$ factor is needed in this exact extension norm.

For a fully specified comparison reference, use Euclidean norm in the original upper exponential coefficient order ν_{ij} , with its actual quotient by $\text{im } C^T$. Its positive Gram in the fixed DNE3 frame is

$$G_{\text{exp}} = Z_{\text{exp}}^* [I - C^T (\bar{C} C^T)^{-1} \bar{C}] Z_{\text{exp}}, \quad 0 \prec \mathcal{D}_N^{(s)} \preceq q \mathcal{B}_{d,s,k,A}^2 G_{\text{exp}}. \quad (\text{DNE14})$$

The reference is explicitly specified here for comparison; the actual weighted Gamma quotient remains DNE7. The inverse in G_{exp} exists by the full row rank of C . Its expression is the orthogonal-projection formula, and is positive on this quotient frame by DNE3. Every representative $\nu + C^T \eta$ remains in $\mathcal{N}_v$ and changes $\mathcal{N}/E_A$ by precisely a lower exponential, by RDS7. Combining DNE6 and DNE13, and then using $\|w\|_1 \leq \sqrt{q} \|w\|_2$, gives

$$x^* \mathcal{D}_N^{(s)} x \leq \mathcal{B}_{d,s,k,A}^2 \inf_{\eta} \|Z_{\text{exp}} x + C^T \eta\|_1^2 \leq q \mathcal{B}_{d,s,k,A}^2 x^* G_{\text{exp}} x.$$

This proves DNE14 without a bound on an omitted frame inverse.

At fixed actual period and quartet the finite constant satisfies

$$\log(1 + q\mathcal{B}_{d,s,k,A}^2) = O_{\text{actual}}(q), \quad s = 1, k, \quad d = N - v. \quad (\text{DNE15})$$

For detail, R_0, ρ, v, μ_v are fixed, $R = O(k)$, $|c'| = O(k)$ and $s \leq k$. Thus $\log C_{F,k,A} = O_{\text{actual}}(k + \log(k + 1))$. The original Gamma ratios obey $n!/(s/2)_n \leq 2\sqrt{n}$ for $n \geq 1$ and are one at $n = 0$, as proved by the product estimate in FEX10–11. Hence the sum in DNE13 is at most $(d + 1) \max\{1, 2\sqrt{d}\} \rho^{-2d}$. Since $d \leq 2q$, its logarithm is $O_{\text{actual}}(q)$. The remaining factors have logarithms $O_{\text{actual}}(k)$, with the original positive mass retained before estimation. These bounds prove DNE15.

Restriction of DNE14 to any specified plane, and minimization under any specified onto quotient map, preserve its scalar upper bound. In particular it applies to the actual map B_K with the full reference quotient $(B_K G_{\text{exp}}^{-1} B_K^*)^{-1}$. This is a finite norm comparison of the original metric with a completely retained reference form. The reference contains the full original Vandermonde and conductor data; neither its determinant nor the original return in DNE10 has been assigned a leading value by this upper bound.

66 Subleading positive exterior expansion of the original dual metric

Fix the original simple quartet, its full unit, a fixed actual five-orbit period and logarithm branch. Retain all coefficients of the degree-(8, 8) conductor A , all shifts b_{ab} and every original root. The coefficient formula PCL8 depends on this fixed period and quartet and is independent of k . Thus the entire symbol

$$E_A(w) = \sum_{a,b=0}^8 a_{ab} e^{b_{ab}w}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0 \quad (\text{PEC1})$$

is the same function throughout the sequence $k = 4l + 1 \geq 9$. Keep $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = q - q' = 16k - 48$, $\mathbf{m} = \Delta - v$, $c' = k/2 - 4$, both source orders $s = 1, k$, and the original endpoints $N = q - 1, q, 2q - 1, 2q$. Write $d = N - v$. The letter w in this section is the analytic symbol variable. It does not replace a primary root, polynomial coordinate, period, or the absolute base τ .

Use the complete DNE frame Z_{exp} , numerators $\mathcal{N}_a$, dual Grams $\mathcal{D}_N^{(s)}$ and coefficient reference G_{exp} . Their literal definitions, including the complex-linear dual, are

$$\begin{aligned} F_a &= \mathcal{N}_a / E_A, & T_{N,s;na} &= \Lambda_{F_a}(\phi_{n,s,c'}), & V_{N,s;n,(i,j)} &= \phi_{n,s,c'}(\beta'_{ij}), \\ \mathcal{D}_N^{(s)} &= T_{N,s}^*[I - V_{N,s}(V_{N,s}^*V_{N,s})^{-1}V_{N,s}^*]T_{N,s}, \\ G_{\text{exp}} &= Z_{\text{exp}}^*[I - C^\top(\bar{C}C^\top)^{-1}\bar{C}]Z_{\text{exp}} > 0. \end{aligned} \quad (\text{PEC2})$$

Here C is the entire original conductor, and the exact Gamma orthonormal frame has $h_n = M_s n! (s/2)_n$, $M_s = (2\pi)^{s/2}$. DNE6–8 proves these metrics, retaining the full lower evaluation space and both conditioning factors. In particular G_{exp} is the minimum Euclidean norm of an original exponential coefficient vector modulo $\text{im } C^\top$; it is not substituted for $\mathcal{D}_N^{(s)}$.

66.1 The actual symbol-zero jet subspaces

On $|z| < 1$ use the branch of $\arctan z$ vanishing at zero and set

$$w(z) = -i \arctan z, \quad \mathcal{E}_A(z) = z^{-v} E_A(w(z)), \quad \mathcal{E}_A(0) = \frac{\mu_v(-i)^v}{v!} \neq 0. \quad (\text{PEC3})$$

The power series for $\arctan$ is holomorphic on this disk and gives $|\arctan z| \leq \text{arctanh } |z|$. Its derivative is $(1 + z^2)^{-1}$, so w' is nonzero there. Moreover w is injective: $e^{-2w(z)} = (1 + iz)/(1 - iz)$, and equality of two w values implies equality under this injective fractional linear function. The displayed removable value in PEC3 follows by substituting the first nonzero Taylor term of PEC1. Thus $\mathcal{E}_A$ is a nonzero holomorphic function on the entire open unit disk.

Call $0 < r < 1$ admissible when $E_A(w(z)) \neq 0$ for $|z| = r$. For every such r , let $\mathcal{Z}_r$ be the finite set of nonzero zeros ω of E_A in $w(\{|z| < r\})$, and write h_ω for their exact orders. Define

$$H_r = \sum_{\omega \in \mathcal{Z}_r} h_\omega, \quad m_r = \min_{|z|=r} |E_A(w(z))| > 0. \quad (\text{PEC4})$$

Finiteness follows from isolation of the zeros of $\mathcal{E}_A$ in a compact subdisk. Positivity of m_r follows from continuity and admissibility on the compact circle. Every compact subdisk contains only finitely many excluded radii. Consequently there are admissible radii arbitrarily close to one. These are properties of the specified symbol, with no assumed zero-free unit disk. Its zero count can also be written exactly as

$$H_r = \frac{1}{2\pi i} \int_{|z|=r} \frac{\mathcal{E}'_A(z)}{\mathcal{E}_A(z)} dz. \quad (\text{PEC5})$$

Indeed factor out the finitely many interior zeros with their multiplicities. The logarithmic derivative of the remaining nonvanishing factor integrates to zero on the disk, while each factor $(z - z_0)^h$ contributes h . The nonzero derivative of w preserves the orders h_ω . This proves PEC5 in the original symbol variables.

In the DNE coefficient frame, form the full jet matrix

$$(J_{r,k})_{(\omega,b),a} = \frac{\mathcal{N}_a^{(b)}(\omega)}{b!}, \quad 0 \leq b < h_\omega, \quad c_{r,k} = \text{rank } J_{r,k}, \quad \mathcal{H}_{r,k} = \ker J_{r,k} \subset \mathbb{C}^{\mathbf{m}}. \quad (\text{PEC6})$$

Its entries are the complete sums $\sum_{i,j} (Z_{\text{exp}})_{(i,j),a} \beta_{ij}^b e^{\beta_{ij}\omega} / b!$. Thus every original root, factorial and coefficient is retained. Its codimension is $c_{r,k} \leq \min\{\mathbf{m}, H_r\}$. Addition of an original coefficient vector $C^\top \eta$ changes its numerator by E_A times the corresponding lower exponential polynomial, by RDS7. That change has zero jets in PEC6. Therefore the cancellation subspace is a specified subspace of the original quotient $\mathcal{N}_v / (E_A \mathcal{E}_{k-8})$, independent of the choice of representative and of both N and s . In particular the minimum Euclidean representative defining G_{exp} stays in this subspace.

For $x \in \mathcal{H}_{r,k}$, all interior poles of $\mathcal{N}_x / E_A$ cancel, including zero by membership in $\mathcal{N}_v$. Admissibility and discreteness give a neighborhood of the closed r disk with no additional boundary pole. Thus $\mathcal{N}_x(w(z)) / E_A(w(z))$ is holomorphic on that neighborhood. This is an exact pole cancellation imposed by the actual numerator jets, not a removal of poles from the original symbol.

66.2 A complete finite Gamma bound on each cancellation subspace

Use the original root bound $R = |k/2| + k\sqrt{\delta^2 + \gamma^2}$ and put $a_r = \text{arctanh } r$. For an original exponential coefficient vector ν of a canceled numerator, the exact generating function DNE12 satisfies, on $|z| = r$,

$$\left| (1 + z^2)^{-s/4} e^{ic' \arctan z} \frac{\sum \nu_{ij} e^{\beta_{ij} w(z)}}{E_A(w(z))} \right| \leq \frac{(1 - r^2)^{-s/4}}{m_r} e^{(R+|c'|)a_r} \|\nu\|_1. \quad (\text{PEC7})$$

This follows term by term from $|w(z)| \leq a_r$ and the root bound, with the original exponential coefficient order. The holomorphic branch of $(1 + z^2)^{-s/4}$ is the one taking value one at zero; its modulus is $|1 + z^2|^{-s/4}$. No source mass has yet canceled. Cauchy's coefficient integral and $h_n = M_s n! / (s/2)_n$ therefore give

$$\|\Lambda_{\mathcal{N}/E_A}\|_{(\mathcal{P}_d, \|\cdot\|_{s,c'})^\vee}^2 \leq B_{r,N,s}^2 \|\nu\|_1^2, \quad B_{r,N,s}^2 = \frac{e^{2(R+|c'|)a_r} (1 - r^2)^{-s/2}}{M_s m_r^2} \sum_{n=0}^d \frac{n!}{(s/2)_n} r^{-2n}. \quad (\text{PEC8})$$

The squared dual norm is the sum of the squared values on the complete orthonormal frame of degrees zero through d . The exponential generating series gives an $n!$ before division by $\sqrt{h_n}$, yielding precisely the ratio $n! / (s/2)_n$ above. This verifies every term of PEC8.

Every allowed addition $E_A \mathcal{E}_{k-8}$ preserves cancellation. Minimize over these representatives as in DNE6 and use $\|\nu\|_1^2 \leq q \|\nu\|_2^2$. This proves the quadratic-form bound

$$x^* \mathcal{D}_N^{(s)} x \leq C_{r,N,s} x^* G_{\text{exp}} x \quad (x \in \mathcal{H}_{r,k}), \quad C_{r,N,s} = q B_{r,N,s}^2. \quad (\text{PEC9})$$

It uses the entire lower conditioning space, not a chosen extension. For each fixed admissible radius, E_A, m_r, a_r are independent of k . The product estimate $n! / (s/2)_n \leq 2\sqrt{n}$ for $n \geq 1$ and its value one at $n = 0$ give the explicit bound

$$C_{r,N,s} \leq \frac{q e^{2(R+|c'|)a_r} (1 - r^2)^{-s/2}}{M_s m_r^2} (d + 1) \max\{1, 2\sqrt{d}\} r^{-2d}. \quad (\text{PEC10})$$

Here $R + |c'| = O_{\delta,\gamma}(k)$, $s \leq k$ and $d \leq 2q$. Since $M_s > 1$, taking the positive part of the logarithm gives

$$\log^+ C_{r,N,s} \leq 2d \log(1/r) + O_{\text{actual},r}(k + \log q). \quad (\text{PEC11})$$

The finite expression PEC10 supplies the constants in this assertion; no asymptotic assumption on an unevaluated numerator is used.

66.3 The whole positive logarithmic spectrum

Let $\lambda_1 \geq \dots \geq \lambda_m > 0$ be the eigenvalues of the specified relative operator

$$T_N^{(s)} = G_{\text{exp}}^{-1/2} \mathcal{D}_N^{(s)} G_{\text{exp}}^{-1/2}. \quad (\text{PEC12})$$

The positive square root is unique by diagonalizing the positive Hermitian matrix. This congruence is the exact comparison of the two retained forms. DNE14 gives $\lambda_a \leq C_{0,N,s}$ for all a , where $C_{0,N,s} = q\mathcal{B}_{d,s,k,A}^2$ is its complete explicit constant, and DNE15 gives $\log^+ C_{0,N,s} = O_{\text{actual}}(q)$. PEC9 bounds the Rayleigh quotient on a subspace of codimension $c_{r,k}$. Hence at most $c_{r,k}$ eigenvalues can exceed $C_{r,N,s}$: otherwise the span of $c_{r,k} + 1$ corresponding orthogonal eigenvectors would intersect that subspace nontrivially, contradicting the bound there. Consequently

$$\begin{aligned} \mathcal{P}_{N,s} &:= \sum_{a=1}^m \log^+ \lambda_a \leq E_{r,N,s}, \\ E_{r,N,s} &= c_{r,k} \log^+ C_{0,N,s} + (m - c_{r,k}) \log^+ C_{r,N,s}, \quad \varepsilon_{N,s}^\vee = \inf_{r \text{ admissible}} E_{r,N,s}. \end{aligned} \quad (\text{PEC13})$$

All terms are finite and nonnegative, and the admissible set is nonempty. Thus $\mathcal{P}_{N,s} \leq \varepsilon_{N,s}^\vee$ even when the infimum is not attained. For every fixed admissible r , $c_{r,k} \leq H_r$ is bounded independently of k . Since $m/k \rightarrow 16$ and $d/q \leq 2$, PEC11 and DNE15 imply

$$0 \leq \limsup_{k \rightarrow \infty} \frac{\varepsilon_{N,s}^\vee}{kq} \leq 64 \log(1/r). \quad (\text{PEC14})$$

This holds simultaneously for the two sources and four indicated endpoint sequences, which form a finite list. Choose admissible radii arbitrarily close to one only after taking this bound for each fixed radius. The right side tends to zero. We have proved

$$\boxed{\mathcal{P}_{N,s} \leq \varepsilon_{N,s}^\vee = o(kq)}. \quad (\text{PEC15})$$

Only the original source orders $1, k$ are used in the two limiting steps; the actual period and operator remain fixed. No additional source order or presumed spectral gap is inserted. Although H_r can increase as r tends to one, it is finite before each k limit, which is exactly what PEC14 uses.

This also controls every exterior degree of the comparison map. The singular values squared of the identity from $(\mathbb{C}^m, G_{\text{exp}})$ to $(\mathbb{C}^m, \mathcal{D}_N^{(s)})$ are the λ_a . In orthonormal wedge frames, each squared singular value is a product over a subset of these numbers. Therefore

$$\log \left\| \bigwedge^t \text{id} \right\| \leq \frac{1}{2} \mathcal{P}_{N,s} \leq \frac{1}{2} \varepsilon_{N,s}^\vee \quad (0 \leq t \leq m). \quad (\text{PEC16})$$

This is an all-degree bound in the exact two metrics. It does not bound the inverse exterior map; the small relative eigenvalues are retained below.

66.4 Every fixed restriction and quotient keeps this bound

For an injective matrix J and an onto matrix B , retain respectively the full paired metrics

$$\begin{aligned} D_J &= J^* \mathcal{D}_N^{(s)} J, \quad G_J = J^* G_{\text{exp}} J, \\ D_B &= [B(\mathcal{D}_N^{(s)})^{-1} B^*]^{-1}, \quad G_B = [B G_{\text{exp}}^{-1} B^*]^{-1}. \end{aligned} \quad (\text{PEC17})$$

Their relative eigenvalues $\mu_1 \geq \dots \geq \mu_t > 0$ satisfy $\mu_a \leq \lambda_a$ for $1 \leq a \leq t$. For a restriction, write $U = G_{\text{exp}}^{1/2} J G_J^{-1/2}$, so $U^* U = I$ and its relative operator is $U^* T_N^{(s)} U$. The variational description of its a th largest eigenvalue gives the asserted upper bound. Explicitly the preimage of any subspace

of codimension $a - 1$ has codimension at most $a - 1$, so the usual minimum of maximal Rayleigh quotients on such subspaces cannot increase under restriction. This variational formula itself follows by expanding vectors in an orthonormal eigenbasis: the complement of the first $a - 1$ vectors gives the upper bound, and dimension intersection with their first a vectors gives the lower bound.

For the quotient put $V = (BG_{\text{exp}}^{-1}B^*)^{-1/2}BG_{\text{exp}}^{-1/2}$. Then $VV^* = I$ and its relative operator is $(V(T_N^{(s)})^{-1}V^*)^{-1}$. The a th smallest eigenvalue of the compression of $(T_N^{(s)})^{-1}$ is at least its a th smallest full eigenvalue, by the same variational argument applied to smallest eigenvalues. Taking reciprocals proves $\mu_a \leq \lambda_a$. It follows that for either metric pair in PEC17,

$$\sum_{a=1}^t \log^+ \mu_a \leq \mathcal{P}_{N,s} \leq \varepsilon_{N,s}^\vee. \quad (\text{PEC18})$$

No bound for a coefficient inverse of B or J occurs: their full reference metrics in PEC17 remain part of the assertion.

66.5 The exact negative spectral return on the actual kernel

Retain Z_k , A_ℓ and the actual onto matrix $B_K = Z_k^\top A_\ell$ of DNE9. Put

$$D_{K,N} = [B_K(\mathcal{D}_N^{(s)})^{-1}B_K^*]^{-1} = \overline{(\widehat{H}_{K,N}^{(s)})^{-1}}, \quad G_K = [B_K G_{\text{exp}}^{-1} B_K^*]^{-1}. \quad (\text{PEC19})$$

These are exactly the native original kernel dual and its fixed reference, with no changed relation or factorial column. For this pair define, using all its positive relative eigenvalues $\mu_{a,N}$,

$$p_{K,N} = \sum_a \log^+ \mu_{a,N}, \quad n_{K,N} = \sum_a \log^+ (\mu_{a,N}^{-1}), \quad A_{K,k,s} = n_{K,q-1} + n_{K,q} - n_{K,2q-1} - n_{K,2q}. \quad (\text{PEC20})$$

The original weighted polynomial spaces are nested and the complete relation maps restrict to the earlier cutoffs. Minimizing a primal norm over their increasing lift sets decreases $H_{Y,N}^{(s)}$. By DNE8 this increases $\mathcal{D}_N^{(s)}$. Taking minima over the same B_K fibres preserves this increase for $D_{K,N}$. Relative eigenvalues in its fixed G_K metric therefore increase by the variational formula. Hence $p_{K,N}$ increases, $n_{K,N}$ decreases, and $A_{K,k,s} \geq 0$.

Since $\log \mu = \log^+ \mu - \log^+(1/\mu)$, the complete determinant formula DNE10 yields the exact identity and finite bound

$$\begin{aligned} \widehat{F}_k^{(s)} &= A_{K,k,s} + P_{K,k,s}, \\ P_{K,k,s} &= p_{K,2q-1} + p_{K,2q} - p_{K,q-1} - p_{K,q}, \quad 0 \leq P_{K,k,s} \leq E_{k,s}^\vee := \varepsilon_{2q-1,s}^\vee + \varepsilon_{2q,s}^\vee = o(kq). \end{aligned} \quad (\text{PEC21})$$

The reference determinant cancels with the four original signs only after PEC19's complete dual identity has been used. Formula PEC18 bounds each of the two high positive parts, proving the last inequality. Thus the entire leading native kernel return is carried by the specified decrease of its small relative eigenvalue deficits; none has been set to zero or inferred from the rank.

Retain the complete unequal D_s^- , D_s^+ and original $\mathcal{L}_s$ from FEX32–35, with $t_0 = \dim(K \cap L)$ and $L = \mathbb{C}[S]_{<v}$. Substituting PEC21 into that proved interval gives

$$\begin{aligned} A_{K,k,s} - D_s^+ - (v - t_0)\mathcal{L}_s &\leq F_K^{(s)} \\ &\leq A_{K,k,s} + E_{k,s}^\vee - D_s^- + v\mathcal{L}_s. \end{aligned} \quad (\text{PEC22})$$

Both original low corrections and both unequal flag errors are retained. FEX37 separately gives the useful exact symmetric allowance

$$|F_K^{(s)} - A_{K,k,s}| \leq \mathcal{E}_{k,s}^{\text{FEX}} + E_{k,s}^\vee = o(kq), \quad (\text{PEC23})$$

where $\mathcal{E}_{k,s}^{\text{FEX}}$ is the full finite FEX36 expression. This is a proved original-kernel metric reduction, not an evaluated individual leading coefficient.

66.6 The complementary original residual and its low subspace

Use the fixed onto map $P_Y : \mathbb{C}^{\mathfrak{m}} \rightarrow \mathbb{C}^{j_0}$ with kernel $\text{im } Z_k$, where $j_0 = \mathfrak{m} - (r - t_0) = j - v + t_0$ and $j = 8k - 32$. For an explicit choice, let $\mathcal{I}$ be the first set of $r - t_0$ row indices in the fixed ordering for which $E_{\mathcal{I}}^{\top} Z_k$ is invertible, and let $\mathcal{J}$ be the complementary indices. Define

$$P_Y = E_{\mathcal{J}}^{\top} - E_{\mathcal{J}}^{\top} Z_k (E_{\mathcal{I}}^{\top} Z_k)^{-1} E_{\mathcal{I}}^{\top}.$$

Independence of the columns, proved in FGC10, ensures such a minor exists. Multiplication gives $P_Y Z_k = 0$ and $P_Y E_{\mathcal{J}} = I$. Its rank is j_0 and its kernel has dimension $r - t_0$, so that kernel is exactly $\text{im } Z_k$. Empty rank uses the identity quotient. Set

$$J_0 = A_{\ell}^{-1} P_Y^{\top}, \quad D_{0,N} = J_0^* \mathcal{D}_N^{(s)} J_0, \quad G_0 = J_0^* G_{\text{exp}} J_0, \quad Q_{0,N} = [P_Y (H_{Y,N}^{(s)})^{-1} P_Y^*]^{-1}. \quad (\text{PEC24})$$

Then $B_K J_0 = 0$ and J_0 is a basis of $\ker B_K$ by dimensions. Direct substitution in DNE8 proves the complete dual identity

$$D_{0,N} = \bar{P}_Y \overline{(H_{Y,N}^{(s)})^{-1} P_Y^{\top}} = \overline{Q_{0,N}^{-1}}. \quad (\text{PEC25})$$

Thus PEC24 is the annihilator of the actual kernel columns, with its full metric, not a complementary coordinate plane chosen for convenience.

Define $p_{0,N}, n_{0,N}$ from the relative eigenvalues of $(D_{0,N}, G_0)$ exactly as in PEC20, and $A_{0,k,s}, P_{0,k,s}$ by the same four endpoint differences. Restriction in PEC18 and the preceding monotonicity proof give

$$F_{0,k,s} := \sum_N \sigma_N \log \det Q_{0,N} = A_{0,k,s} + P_{0,k,s}, \quad A_{0,k,s} \geq 0, \quad 0 \leq P_{0,k,s} \leq E_{k,s}^{\vee}. \quad (\text{PEC26})$$

The scalar $F_{0,k,s}$ here denotes this displayed quotient return only; it is not the seven-moment quantity $\mathcal{F}_{0,k}^{(s)}$.

To identify the original target exactly, put $W_0 = \Xi(L) \subset B_{\Xi}$. The map $L \rightarrow W_0$ has kernel $K \cap L$, so $\dim W_0 = v - t_0$. There is the onto map, in the original FGC coordinates,

$$V_A \longrightarrow Y_N / \text{im } Z_k, \quad [p] \longmapsto \left[\frac{\alpha_A \mathcal{T}_{Ap}}{\chi'} \right], \quad \alpha_A = v! / \mu_v, \quad Y_N / \text{im } Z_k \simeq B_{\Xi} / W_0. \quad (\text{PEC27})$$

FGC4–10 proves independence of representatives and surjectivity. Its kernel is $K + L$: a zero image subtracts a member of K with that image in Y_N , leaving an element of L by FGC6. Conversely both K and L map to zero. The induced isomorphism with B_{Ξ} / W_0 sends the class of p to the class of $\Xi[p]$. This verifies both directions of the target identification.

At each endpoint the source map is the original S_N of CEP28 followed by $D_{\alpha,N}(x, f) = (x, \alpha_A f)$ on GIP's full graph. The complete quotient map PEC27 kills its entire relation graph and kernel columns. In the product source norm, minimization first over the unrestricted low coordinate and then over the polynomial and Z_k relations is exactly $Q_{0,N}$. The exterior norms of $D_{\alpha,N}^{\pm 1}$ in these same product metrics are at most $\exp(j_0 |\log |\alpha_A||)$ in degree j_0 . The minimum-lift and projection argument of GIP6 applied to this composite source map therefore proves, for the original induced metric $\bar{Q}_{\Xi,N}$ on B_{Ξ} / W_0 in the transported frame,

$$|\log \det \bar{Q}_{\Xi,N} - \log \det Q_{0,N}| \leq 2E_N^{\text{CEP}} + 2j_0 |\log |\alpha_A||. \quad (\text{PEC28})$$

Orthogonal projection off the entire kernel is an exterior contraction, and the inverse composite gives the other direction. This checks the full source, target, metric and rank in the comparison.

Choose a fixed original frame I_{W_0} of W_0 and retain its exact restriction return $F_{W_0}^{(s)} = \sum_N \sigma_N \log \det (I_{W_0}^* Q_{\Xi,N}^{\text{fix}} I_{W_0})$. Completing that frame by fixed quotient lifts gives a block Gram

whose Schur complement is $\overline{Q}_{\Xi,N}$. Its fixed frame determinant cancels with the four signs. Consequently the original residual is

$$\begin{aligned}\Phi_{k,s} &= F_{W_0}^{(s)} + A_{0,k,s} + P_{0,k,s} + \delta_{0,k,s}, \\ |\delta_{0,k,s}| &\leq E_{k,s}^{\text{tr}} := 2 \sum_N E_N^{\text{CEP}} + 8j_0 |\log |\alpha_A|| = o(kq), \\ 0 &\leq F_{W_0}^{(s)} \leq (v - t_0)\mathcal{L}_s = O_{\text{actual}}(q).\end{aligned}\tag{PEC29}$$

The last inequality follows by restricting and minimizing the original AKS covariance comparison on this fixed subquotient of the primary space, exactly as in FEX32. Its metrics decrease on increasing source spaces, proving nonnegativity. The full finite $\mathcal{L}_s$ is FEX33, whose proved order is $O(q)$, and $v \leq 80$. Thus no original low contribution has been dropped. In particular the finite and asymptotic consequences are

$$\begin{aligned}A_{0,k,s} - E_{k,s}^{\text{tr}} &\leq \Phi_{k,s} \leq A_{0,k,s} + E_{k,s}^{\vee} + E_{k,s}^{\text{tr}} + (v - t_0)\mathcal{L}_s, \\ \Phi_{k,s} &= A_{0,k,s} + o(kq).\end{aligned}\tag{PEC30}$$

66.7 Backward and forward return to the same arithmetic class

The complete OVR/ECL identity $F_K^{(s)} + \Phi_{k,s} = \Delta q \mathfrak{C}_{\partial} + o(kq)$ and PEC23,PEC30 give

$$\begin{aligned}A_{K,k,s} + A_{0,k,s} &= \Delta q \mathfrak{C}_{\partial} + o(kq), \\ \frac{A_{K,k,s} + A_{0,k,s}}{kq} &\longrightarrow 16\mathfrak{C}_{\partial},\end{aligned}\tag{PEC31}$$

$$21.66851180520 < 16\mathfrak{C}_{\partial} < 21.66851180521.$$

Both deficits are on the exact DNE dual quotient and its actual kernel annihilator, with the full G_{exp} references. The equality evaluates their sum and retains their individual distributions. AMT's already proved source-order agreement and PEC23,PEC30 also give $(A_{K,k,1} - A_{K,k,k})/(kq) \rightarrow 0$ and the same statement for A_0 .

For finite original kernel bounds intersect PEC22 with OER19. Retain the full AMT arithmetic intervals and take their exact boundary complements. On the original arithmetic period domain $|u| \geq R_*$, the signed identity $\Psi_k = F_K^{(k)} + \mathcal{H}_k + \delta_k^{\sigma}$ then proves, with unchanged signs,

$$\begin{aligned}A_{K,k,k} - D_k^+ - (v - t_0)\mathcal{L}_k + \mathcal{H}_k + \delta_k^{\sigma} &\leq \Psi_k \\ &\leq A_{K,k,k} + E_{k,k}^{\vee} - D_k^- + v\mathcal{L}_k + \mathcal{H}_k + \delta_k^{\sigma}.\end{aligned}\tag{PEC32}$$

Here $D_k^{\pm}$ means the original FEX bounds at source $s = k$, and $\mathcal{L}_k$ is the same source's FEX33 allowance. The original same-class action inequality OER20 uses precisely the right endpoint of PEC32 plus the complete C_k^{budget} . This addition preserves the original monic window, contraction penalties, mixed pairings, action leakage and positive-primary exterior class. Where the seven moments are used, retain PMR's finite range and $e^{0,\pm} = e^{\pm} + c_k^{\circ}$; outside that range retain the original OAR outer expression.

The calculations PEC1–32 prove a subleading positive exterior budget and return it to the original kernel and residual, whose leading negative spectral deficits are explicitly specified. Calculating these actual small eigenvalues and the resulting same-class exclusion bound is the remaining analytic step in these formulas. No assertion about their individual leading values or an RH conclusion follows merely from the evaluated sum PEC31.

67 The actual common-factor map on the original kernel

Fix the original simple quartet, its complete unit and period data, and the original logarithm branch on the proved five-orbit domain. Use $k = 4l + 1 \geq 9$, $k' = k - 8$, and retain

$$\begin{aligned} q &= (k+1)^2, & q' &= (k-7)^2, & \Delta &= q - q' = 16k - 48, \\ r &= 8k - 16, & j &= 8k - 32, & c &= k/2, & c' &= k'/2. \end{aligned} \quad (\text{ACR1})$$

The original primary points, also at every integer index in the second formula, are

$$\begin{aligned} \beta_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, & 0 \leq a, b \leq k, \\ \tau_{ij} &= c' + (2i - k')\delta + i(2j - k')\gamma, & (i, j) \in \mathbb{Z}^2, \\ b_{ab} &= 4 + (2a - 8)\delta + i(2b - 8)\gamma, & 0 \leq a, b \leq 8. \end{aligned} \quad (\text{ACR2})$$

In the imaginary terms i is the imaginary unit; subscripts are integer indices. The functions of the two integer indices are injective, by comparison of real and imaginary parts and $\delta > 0$, $\gamma > 2$. In particular $\tau_{ij} + b_{ab} = \beta_{i+a, j+b}$ whenever the latter indices are in their displayed range.

Write $\chi = \prod_{a,b=0}^k (S - \beta_{ab})$ and $\chi' = \prod_{i,j=0}^{k'} (S' - \tau_{ij})$. The prime names the original lower-grid polynomial, and is not differentiation. All dual pairings until a Hermitian Gram is explicitly mentioned are complex-linear.

67.1 Support and its precise extended coefficient rectangle

Let the actual PCL8–11 biform be

$$A = \sum_{a,b=0}^8 a_{ab} z_0^a z_1^{8-a} w_0^b w_1^{8-b} \neq 0, \quad \mathcal{S} = \{(a, b) : a_{ab} \neq 0\}. \quad (\text{ACR3})$$

Here, literally, its factors are $Q_0(L_h(u^{-1})t)$, $1 \leq h \leq 4$, where $t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^T$ and $L_h = U^{-1} V \mathcal{R}^{-1} D^{-h} \mathcal{R} V^{-1} U$. Thus every coefficient is the complete finite sum of products in PCL10, with the entire original matrix series PCL2. No coefficient is assigned in this calculation. Put

$$a_- = \min_{\mathcal{S}} a, \quad a_+ = \max_{\mathcal{S}} a, \quad b_- = \min_{\mathcal{S}} b, \quad b_+ = \max_{\mathcal{S}} b, \quad d_a = a_+ - a_-, \quad d_b = b_+ - b_-. \quad (\text{ACR4})$$

The following finite index sets have their literal integer intervals:

$$\begin{aligned} I_{\text{all}} &= [-a_-, k - a_+] \times [-b_-, k - b_+], & I_{\text{in}} &= [0, k']^2, & I_G &= I_{\text{all}} \setminus I_{\text{in}}, \\ G_A(S') &= \prod_{(i,j) \in I_G} (S' - \tau_{ij}), & g &= |I_G| = (k+1-d_a)(k+1-d_b) - q'. \end{aligned} \quad (\text{ACR5})$$

Every interval is nonempty and the first rectangle contains I_{in} , since all active indices are in $[0, 8]^2$. The intersection of the shifted full-grid rectangles $[-a, k-a] \times [-b, k-b]$, over the actual active pairs, is exactly I_{all} . Each full shifted polynomial $\chi(S' + b_{ab})/\chi'(S')$ has the simple roots in its rectangle with I_{in} removed. Distinctness in ACR2 therefore proves that G_A is precisely their monic greatest common divisor, with the degree in ACR5. This reproves the RCF common factor from the actual support. At $g = 0$ its polynomial is the empty product one.

Use the original ordered biform coefficient space $\mathcal{B}_k$, with basis $e_{ab}^{(k)} = z_0^a z_1^{k-a} w_0^b w_1^{k-b}$. For each $(i, j) \in I_{\text{all}}$ define the complete column

$$f_{ij} = \sum_{(a,b) \in \mathcal{S}} a_{ab} e_{i+a, j+b}^{(k)}. \quad (\text{ACR6})$$

All indices in this expression lie in $[0, k]^2$: the lower bound follows from $i \geq -a_-$, $j \geq -b_-$; the upper bound follows from $i \leq k - a_+$, $j \leq k - b_+$. Let C_{in} , C_G and C_{all} have these columns for I_{in} ,

I_G and I_{all} , in fixed lexicographic orders. These coefficient matrices use every active coefficient, each once, and no conjugate or binomial multiplier.

The full boundary factor and its remaining biform are

$$\begin{aligned} M_{\partial} &= z_0^{a-} z_1^{8-a+} w_0^{b-} w_1^{8-b+}, & A &= M_{\partial} A_{\text{red}}, \\ \deg A_{\text{red}} &= (d_a, d_b), & A_{\text{red}} &= \sum_{a,b} a_{ab} z_0^{a-a-} z_1^{a_+-a} w_0^{b-b-} w_1^{b_+-b}. \end{aligned} \quad (\text{ACR7})$$

The formula retains the full scalar and every coefficient of A ; only the displayed coordinate monomial is factored. For bidegrees (p, t) let $\mathcal{B}_{p,t}$ be the complete biform space of those bidegrees and set it to zero if either degree is negative. Multiplication by a nonzero biform is injective, since the ambient polynomial ring is an integral domain. ACR6 and ACR7 then give the exact coefficient images

$$\begin{aligned} H_k &:= \text{im } C_{\text{all}} = A_{\text{red}} \mathcal{B}_{k-d_a, k-d_b}, & \text{im } C_{\text{in}} &= A \mathcal{B}_{k-8, k-8}, \\ \text{rank } C_{\text{all}} &= q' + g, & \text{rank } C_{\text{in}} &= q', & H_k &= \text{im } C_{\text{in}} \oplus \text{im } C_G. \end{aligned} \quad (\text{ACR8})$$

Indeed f_{ij} is A_{red} times the monomial of bidegree $(k-d_a, k-d_b)$ whose two indices are $(i+a_-, j+b_-)$. As (i, j) ranges over I_{all} , these are all its basis monomials. Their products are independent by injectivity, proving every image and rank in ACR8. For the inner subset the same multiplication is exactly A times its original degree- $(k-8, k-8)$ basis.

67.2 The residual map evaluated on every original primary coordinate

Let $E = \mathbb{C}[S]/(\chi)$ and let $\mathcal{V}_{\beta} : E \rightarrow \mathbb{C}^{[0,k]^2}$ be the original primary evaluation map. Define the original conductor annihilator V_A and the original kernel K by

$$V_A = \{[p] : C_{\text{in}}^T \mathcal{V}_{\beta}[p] = 0\}, \quad K = \ker \Lambda \subseteq V_A. \quad (\text{ACR9})$$

The inclusion follows from the full original trace identity $A \mathcal{B}_{k-8, k-8} \subseteq J_k$, where J_k is the image of the original invariant algebra under $x = Ht$; the complex-linear annihilator of J_k is K by OQC8. Equivalently, direct evaluation proves

$$\mathcal{T}Ap(S') = \sum_{a,b} a_{ab} p(S' + b_{ab}), \quad (C_{\text{in}}^T \mathcal{V}_{\beta} p)_{ij} = (\mathcal{T}Ap)(\tau_{ij}), \quad [p] \in V_A \iff \chi' \mid \mathcal{T}Ap. \quad (\text{ACR10})$$

The roots of χ' are distinct, so vanishing at them is equivalent to its polynomial divisibility. Replacing p by $p + \chi h$ preserves this property, because $\chi(S' + b_{ab}) = \chi'(S') d_{ab}(S')$ for the original monic shifted factors. Thus ACR9 and ACR10 agree on the original quotient, with all original positive shifts.

For every $\ell = (i, j) \in I_G$ one has $\chi'(\tau_{\ell}) \neq 0$. The full common-factor residual has the exact evaluation

$$\boxed{(\rho_G[p])(\tau_{\ell}) = \frac{1}{\chi'(\tau_{\ell})} \sum_{a,b} a_{ab} p(\beta_{i+a, j+b}) = \frac{f_{\ell}^T \mathcal{V}_{\beta}[p]}{\chi'(\tau_{\ell})}.} \quad (\text{ACR11})$$

Here $\rho_G[p] = \text{rem}_{G_A}(\mathcal{T}Ap/\chi')$ is in the original remainder coefficient frame. Divisibility ACR10 gives a polynomial quotient. Evaluation at a simple root of G_A commutes with its monic remainder and gives the first expression. The point identity ACR2 proves the second. A representative change adds $\sum a_{ab} d_{ab} h(S' + b_{ab})$, divisible by the full G_A , because it divides every active d_{ab} . This proves the stated map on V_A , independently of representatives, and re-establishes the exact RCF15 map.

Let $W_G = (\tau_{\ell}^d)_{\ell \in I_G, 0 \leq d < g}$ be the original ordered Vandermonde and $D_G = \text{diag}(\chi'(\tau_{\ell})^{-1})$. For $g > 0$, both are invertible, since all their indicated differences and diagonal entries are nonzero. The same notation has its unique empty meaning at $g = 0$. Let $\pi : \mathcal{B}_k \twoheadrightarrow \mathbb{C}^r$ be any exact OCF coefficient quotient with $\ker \pi = J_k$. OKM proves that the original kernel frame is $I_K = \mathcal{V}_{\beta}^{-1} \pi^T$. Therefore

$$\boxed{\rho_G I_K = W_G^{-1} D_G C_G^T \pi^T, \quad B_G := \pi C_G \in \text{Mat}_{r \times g}(\mathbb{C}), \quad t_G = g - \text{rank } B_G.} \quad (\text{ACR12})$$

The first equality follows by applying ACR11 on every kernel column and then the specified inverse evaluation map. The matrix $C_G^T \pi^T = B_G^T$ uses ordinary transpose. Multiplication by the two invertible target maps does not change rank, proving the last equality with the definition in RCM22. Every entry of the actual matrix needed for this rank is explicitly

$$(B_G)_{h,(i,j)} = \sum_{(a,b) \in S} a_{ab} \pi_{h,(i+a,j+b)}. \quad (\text{ACR13})$$

Thus this is a matrix with at most 81 terms per entry, $r = 8k - 16$ rows and $g \leq 16k - 48$ columns, computed from the fixed original period coefficients and original kernel quotient. It requires no full symmetric-power matrix.

For an exact all-case scalar prescription put $\Delta_0(B_G) = 1$, and let $\Delta_h(B_G)$ be the sum of the squared absolute values of all its h by h minors. Then

$$t_G = g - \max\{h : \Delta_h(B_G) > 0\}. \quad (\text{ACR14})$$

A nonzero minor gives that many independent columns; conversely Gaussian elimination on independent columns gives that many pivot rows and a nonzero minor. This proves the equality with rank. A sum of nonnegative real terms vanishes exactly when all those minors vanish. All a_{ab} and the full OCF pivot denominators remain in those minors. Formula ACR14 neither assigns generic rank nor supplies a numerical zero test for an unevaluated analytic coefficient.

67.3 The exact retained residual quotient in biform coordinates

The independent columns of C_G stay independent modulo $\text{im } C_{\text{in}}$ by ACR8. Hence the induced map $\mathbb{C}^g \rightarrow V_A^\vee$ represented by C_G is injective. Its transpose is onto, and the invertible factors in ACR11 prove that $\rho_G : V_A \rightarrow \mathbb{C}[S']_{<g}$ is onto. This proves its surjectivity directly from the original coefficient map.

The annihilator calculation also gives

$$\begin{aligned} V_0 &:= \ker \rho_G = H_k^\perp \subseteq V_A, & t_G &= \dim \frac{J_k \cap H_k}{A\mathcal{B}_{k-8,k-8}}, \\ J_k^{\text{red}} &:= \text{im}(J_k \rightarrow \mathcal{B}_k/H_k), & t_G &= j - \dim J_k^{\text{red}}. \end{aligned} \quad (\text{ACR15})$$

The first statement follows from the simultaneous vanishing of the inner and exterior rows in ACR11 and ACR8. The map B_G in ACR12 has kernel precisely those exterior linear combinations belonging to J_k . The direct sum in ACR8 and the inclusion $A\mathcal{B}_{k-8,k-8} \subseteq J_k$ identify this kernel with the quotient in the first line. Finally $\dim J_k = q - r$, while the inner space has dimension q' . Applying rank-nullity to $J_k \rightarrow J_k^{\text{red}}$ gives $(q - r) - \dim J_k^{\text{red}} - q' = j - \dim J_k^{\text{red}}$, proving the second line.

This is the same invariant image as FMC7: its map is induced by the identical original substitution $x = Ht$, and the inner quotient is the same $A\mathcal{B}$ quotient. Factoring the explicit boundary monomial in ACR7 provides the additional quotient map to $\mathcal{B}/H$, with its kernel computed in ACR15; it does not replace that module by a newly chosen invariant space.

For the original residual observation $\Xi : V_A \rightarrow B_\Xi$ with kernel K , put $W = \Xi(V_0)$. Define

$$A_G : \mathbb{C}[S']_{<g} \rightarrow B_\Xi/W, \quad A_G(\rho_G x) = [\Xi x]. \quad (\text{ACR16})$$

Two lifts differ by V_0 , so the map is well defined; it is onto since both maps from V_A are onto. Its kernel is exactly $\rho_G K$: if $\Xi x \in W$, choose $y \in V_0$ with $\Xi y = \Xi x$; then $x - y \in K$ and has the same ρ_G value. The reverse containment is direct. Consequently $\dim W = \dim J_k^{\text{red}}$, while $\dim(B_\Xi/W) = t_G$. In particular

$$\begin{aligned} d_a = d_b = 8 : & \quad g = 0, \quad t_G = 0, \quad V_0 = V_A, \quad W = B_\Xi; \\ d_a = d_b = 0 : & \quad g = \Delta, \quad H_k = \mathcal{B}_k, \quad t_G = j, \quad W = 0. \end{aligned} \quad (\text{ACR17})$$

The first row follows from ACR5 and the zero target of ρ_G ; its full residual still has rank j . In the second row A_{red} is a nonzero constant, so the quotient defining J_k^{red} is zero and ACR15 gives the result. This second row classifies that exact coefficient case; it does not assert that a given original period attains it. Every intermediate case retains the actual matrices ACR12–15 and the full original quotient ACR16.

67.4 A smaller invariant recursion evaluating every exceptional rank

Here is a second exact computation of the remaining rank which never uses the full symmetric power. In every diagonal degree $d \geq 0$ set

$$H_d = A_{\text{red}} \mathcal{B}_{d-d_a, d-d_b}, \quad \mathcal{W}_d = \mathcal{B}_{d,d}/H_d, \quad n_d = (d+1)^2 - \max(d-d_a+1, 0) \max(d-d_b+1, 0). \quad (\text{ACR18})$$

Multiplication by A_{red} is injective, so this is the exact dimension of the quotient. Also $H_d \mathcal{B}_{e,e} \subseteq H_{d+e}$; therefore the direct sum of these quotients is an actual graded algebra.

Choose the lexicographically largest nonzero coefficient a_* of the complete A_{red} , at the biform indices (r_*, s_*) . In the full degree- d grid eliminate the pivot rectangle

$$\{(r_* + i, s_* + h) : 0 \leq i \leq d - d_a, 0 \leq h \leq d - d_b\} \quad (\text{ACR19})$$

in descending lexicographic order, subtracting at each pivot its current coefficient divided by a_* times the corresponding monomial multiple of A_{red} . When a degree bound is negative the rectangle is empty. Every other term in a subtracted polynomial is strictly smaller in the additive lexicographic order. Earlier zero pivots therefore stay zero. The result is a linear map N_d^{red} to the complementary coefficient strip, with its literal zero-insertion section S_d^{red} . Summing the subtractions gives a decomposition into the strip and H_d . Their intersection is zero: the leading term of every nonzero $A_{\text{red}}b$ lies in the pivot rectangle. Thus

$$\ker N_d^{\text{red}} = H_d, \quad N_d^{\text{red}} S_d^{\text{red}} = I_{n_d}. \quad (\text{ACR20})$$

All entries are finite rational expressions in the original a_{ab} with powers of the selected actual a_* in their denominators. Selecting the first nonzero coefficient covers all support and leading-coefficient cases. If A_{red} is constant, this construction eliminates the full grid, as it must.

Use the following fourteen actual invariant monomials:

e	$\mathcal{G}_e$	
2	x_1x_4, x_2x_3	
3	$x_1^2x_3, x_1x_2^2, x_2x_4^2, x_3^2x_4$	
4	$x_1^3x_2, x_1x_3^3, x_2^3x_4, x_3x_4^3$	
5	$x_1^5, x_2^5, x_3^5, x_4^5$	(ACR21)

These generate every invariant polynomial, as follows directly from the original weights 1, 2, 3, 4 modulo five. An indecomposable invariant monomial has degree at most five: among six partial sums of any five factor weights, two agree and supply an invariant divisor. An opposite pair of weights gives one of the displayed quadratics; an indecomposable containing it is that quadratic. In the absence of opposite pairs the support has at most two variables. One variable gives a fifth power. The remaining supports are $\{1, 2\}, \{1, 3\}, \{2, 4\}, \{3, 4\}$; the positive exponent solutions of total degree at most five are respectively $(3, 1), (1, 2)$ for the first support and $(2, 1), (1, 3)$ for the second, with the last two obtained by multiplying weights by 2^{-1} and 3^{-1} modulo five. These are precisely the displayed cubics and quartics. Repeated factorization into indecomposables terminates by decreasing total degree, proving the generator assertion.

For $f \in \mathcal{G}_e$, retain its complete biform $\tilde{f} = f(Ht)$, whose coefficients are the OCF13 finite multinomial expressions in the actual period and unit matrix. Define

$$m_{f,d}^{\text{red}} = N_d^{\text{red}} M_{\tilde{f}} S_{d-e}^{\text{red}}, \quad J_0^{\text{red}} = \text{span}\{N_0^{\text{red}} 1\}, \quad J_d^{\text{red}} = \sum_{\substack{f \in \mathcal{G}_e \\ e \leq d}} m_{f,d}^{\text{red}} J_{d-e}^{\text{red}}. \quad (\text{ACR22})$$

Negative degrees give zero spaces. Multiplying two representatives differing by H_{d-e} changes their product by H_d , so ACR20 proves representative independence of the displayed multiplication. Every invariant monomial of positive degree factors by ACR21 into one generator times an earlier invariant monomial. This proves both inclusions in the recursion and hence its equality with the original invariant image in every degree. The base case includes the zero class of one when the quotient algebra is zero.

To evaluate its rank, concatenate the indicated multiplication matrices applied to bases of the preceding invariant images, and choose independent columns by their exact minors. Its rank is the maximum size of a nonzero minor, with rank zero for the empty image. The selected columns supply the next basis with their full original coefficient expressions. This proves the algorithm by induction, without assigning any unknown rank. All exceptional minors are covered by choosing the first nonzero one in a fixed finite order. For $d \geq 8$, $n_d = (d_a + d_b)(d + 1) - d_a d_b \leq 16d + 16$; the candidate matrix has at most $\sum_{e=2}^5 |\mathcal{G}_e| n_{d-e}$ columns. Its row and column dimensions are therefore bounded linearly in d . Temporary full biform columns have $(d + 1)^2$ entries, as in OCF; no claim that all storage objects have linear size is made. The exact output evaluates the residual integer by the proved identity

$$\boxed{t_G = 8k - 32 - \dim J_k^{\text{red}}.} \quad (\text{ACR23})$$

Thus all coefficient cases on the original five-orbit domain have explicit finite formulas, with the unchanged original residual quotient ACR16. The separate literal period support calculation can determine where its first case ACR17 holds; its use then retains the full complementary residual, rather than inferring that this residual has vanished.

68 Exact exclusion of all four boundary rulings at the original period limit

Retain the original hypotheses $0 < \delta < 1/2$, $\gamma > 2$ and $a \neq 0$. In particular, no phase of the original amplitude is selected. Set

$$x = \delta + i\gamma, \quad y = \delta - i\gamma, \quad \beta = 2(\gamma^2 - \delta^2) = -(x^2 + y^2), \quad t_{xy} = x/y, \quad r = a/\bar{a}, \quad R = r^2. \quad (\text{LBRE1})$$

All these ratios are defined, $|t_{xy}| = |r| = 1$, and $x^2 - y^2 = 4i\delta\gamma \neq 0$. Thus $t_{xy}^2 \neq 1$. The root order is exactly $(x, y, -y, -x)$, the full unit is $U = \text{diag}(a, \bar{a}, \bar{a}, a)$, and V evaluates $1, w, w^2, w^3$ in this root order. For $j \in \{1, 2, 3, 4\}$ put $\rho = \zeta^{-j}$, where $\zeta = e^{2\pi i/5}$. Retain PCL4's complete matrix

$$J_\infty = \begin{pmatrix} 1 & 0 & -\beta/3 & 0 \\ 0 & 1 & 0 & -2\beta/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad L = U^{-1}VJ_\infty^{-1} \text{diag}(\rho, \rho^2, \rho^3, \rho^4)J_\infty V^{-1}U. \quad (\text{LBRE2})$$

The original form is $Q_0(t) = t_1t_4 - t_2t_3$.

68.1 The exact even and odd evaluation maps

Define the following finite coefficients:

$$A = \frac{2t_{xy}^2 - 1}{3(t_{xy}^2 - 1)}, \quad B = \frac{2 - t_{xy}^2}{3(t_{xy}^2 - 1)}, \quad h = \rho^3 - \rho, \quad (\text{LBRE3})$$

$$e = \rho + hA, \quad f = -hA, \quad g = hB, \quad d = \rho - hB.$$

Here $A - B = 1$. The notation d in this section denotes precisely this scalar and does not denote the original root displacement δ . For any primary vector $t \in \mathbb{C}^4$ set

$$v_+(t) = \frac{1}{2} \begin{pmatrix} t_1 + t_4 \\ t_2 + t_3 \end{pmatrix}, \quad v_-(t) = \frac{1}{2} \begin{pmatrix} t_1 - t_4 \\ t_2 - t_3 \end{pmatrix}, \quad J_2 = \text{diag}(1, -1). \quad (\text{LBRE4})$$

These are explicit coordinate maps on the original primary space; their inverse sends (v_+, v_-) to $(v_{+,1} + v_{-,1}, v_{+,2} + v_{-,2}, v_{+,2} - v_{-,2}, v_{+,1} - v_{-,1})$. Consequently

$$Q_0(t) = v_+(t)^\top J_2 v_+(t) - v_-(t)^\top J_2 v_-(t). \quad (\text{LBRE5})$$

The complete matrices induced by L in these coordinates are

$$v_+(Lt) = Ev_+(t), \quad v_-(Lt) = Ov_-(t), \quad E = \begin{pmatrix} e & f/r \\ rg & d \end{pmatrix}, \quad O = \rho \begin{pmatrix} d & t_{xy}g/r \\ rf/t_{xy} & e \end{pmatrix}. \quad (\text{LBRE6})$$

To prove the formulas, write an original polynomial as $p_0 + p_1w + p_2w^2 + p_3w^3$. On its even coefficients the middle conjugation in LBRE2 is

$$\begin{pmatrix} \rho & \frac{\beta}{3}(\rho^3 - \rho) \\ 0 & \rho^3 \end{pmatrix}, \quad (\text{LBRE7})$$

while on its odd coefficients it is

$$\begin{pmatrix} \rho^2 & \frac{2\beta}{3}(\rho^4 - \rho^2) \\ 0 & \rho^4 \end{pmatrix}. \quad (\text{LBRE8})$$

The even evaluation matrix is $\begin{pmatrix} 1 & x^2 \\ 1 & y^2 \end{pmatrix}$; the odd evaluation matrix is its left product with $\text{diag}(x, y)$.

Direct evaluation of LBRE7 gives $\begin{pmatrix} e & f \\ g & d \end{pmatrix}$, because

$$\frac{x^2 + \beta/3}{x^2 - y^2} = A, \quad \frac{y^2 + \beta/3}{x^2 - y^2} = B. \quad (\text{LBRE9})$$

For LBRE8 the two corresponding fractions are $-B, -A$, since $(x^2 + 2\beta/3)/(x^2 - y^2) = -B$ and $(y^2 + 2\beta/3)/(x^2 - y^2) = -A$. Also $\rho^4 - \rho^2 = \rho h$. Thus its evaluated matrix is $\rho \begin{pmatrix} d & t_{xy}g \\ f/t_{xy} & e \end{pmatrix}$. Finally conjugation by the full unit acts on both parity spaces by $\text{diag}(a, \bar{a})$, proving LBRE6 including every amplitude factor.

68.2 The four complete quadratic restrictions

Define

$$\begin{aligned} n_1 &= e^2 - \rho^2 d^2, & c_1 &= \rho^2 f^2 / t_{xy}^2 - g^2, \\ n_2 &= f^2 - \rho^2 t_{xy}^2 g^2, & c_2 &= \rho^2 e^2 - d^2, \\ \mathcal{A} &= n_1 + R c_1, & \mathcal{C} &= (n_2 + R c_2) / R, \\ m_E &= e f / r - r g d, & m_O &= \rho^2 (d t_{xy} g / r - r f e / t_{xy}), \\ \mathcal{B}_- &= m_E - m_O, & \mathcal{B}_+ &= m_E + m_O. \end{aligned} \tag{LBRE10}$$

For the affine Segre vector $t(Z, W) = (ZW, Z, W, 1)^\top$, write $q_j(Z, W) = Q_0(Lt(Z, W))$. The restrictions below also give the exact homogeneous restrictions at infinity:

$$\begin{aligned} Z = 0 : & \quad Q_0(L(0, 0, W, 1)^\top) = \frac{1}{4}(\mathcal{A} + 2\mathcal{B}_- W + \mathcal{C} W^2), \\ Z = \infty : & \quad Q_0(L(W, 1, 0, 0)^\top) = \frac{1}{4}(\mathcal{A} W^2 + 2\mathcal{B}_- W + \mathcal{C}), \\ W = 0 : & \quad Q_0(L(0, Z, 0, 1)^\top) = \frac{1}{4}(\mathcal{A} + 2\mathcal{B}_+ Z + \mathcal{C} Z^2), \\ W = \infty : & \quad Q_0(L(Z, 0, 1, 0)^\top) = \frac{1}{4}(\mathcal{A} Z^2 + 2\mathcal{B}_+ Z + \mathcal{C}). \end{aligned} \tag{LBRE11}$$

For example the two parity vectors for $(W, 1, 0, 0)$ both equal $(W, 1)^\top/2$, so LBRE5–6 give the second row directly. For $(0, 0, W, 1)$ they are $(1, W)^\top/2$ and its negative, giving the first row. For $(Z, 0, 1, 0)$ they are $(Z, 1)^\top/2$ and $(Z, -1)^\top/2$, giving the last row. For $(0, Z, 0, 1)$ they are $(1, Z)^\top/2$ and $(-1, Z)^\top/2$, giving the third row. These substitutions also prove every mixed-coefficient sign and the factor $1/4$. In particular any identically zero boundary restriction would require $\mathcal{A} = \mathcal{C} = 0$.

68.3 A nonzero exact eliminant for every original amplitude

Put

$$q = t_{xy}^2 + t_{xy}^{-2} \in [-2, 2], \quad c = \rho^2 + \rho^3 \in [-2, 2], \quad F(q, c) = 5(c-1)q^2 + (20c+25)q - 61c - 173. \tag{LBRE12}$$

The first interval follows from $|t_{xy}| = 1$. For the second, $\rho^3 = \overline{\rho^2}$ because $\rho^5 = 1$. Direct algebra in LBRE3 and LBRE10 gives the exact identity

$$n_1 c_2 - c_1 n_2 = \frac{t_{xy}^2}{81(t_{xy}^2 - 1)^2} F(q, c). \tag{LBRE13}$$

Here is a finite expansion verifying the identity without any numerical specialization. Before using the fifth-root relation, the left side has denominator $81t_{xy}^2(t_{xy}^2 - 1)^2$ and numerator

$$\begin{aligned} & -\rho^6(\rho^2 - 1)^2 [(t_{xy}^8 + 1)(\rho^4 - 2\rho^2 + 1) \\ & \quad + (t_{xy}^6 + t_{xy}^2)(13\rho^4 + 19\rho^2 + 13) \\ & \quad - t_{xy}^4(57\rho^4 + 120\rho^2 + 57)]. \end{aligned} \tag{LBRE14}$$

Use $1 + \rho + \rho^2 + \rho^3 + \rho^4 = 0$ to replace this numerator by

$$5(c-1)(t_{xy}^8 + 1) + (20c+25)(t_{xy}^6 + t_{xy}^2) - (51c+183)t_{xy}^4. \tag{LBRE15}$$

The identities $(t_{xy}^8 + 1)/t_{xy}^4 = q^2 - 2$ and $(t_{xy}^6 + t_{xy}^2)/t_{xy}^4 = q$ now give LBRE13. All denominators used here are nonzero by LBRE1.

For $-2 \leq q \leq 2$, the polynomial $5q^2 + 20q - 61$ has maximum -1 : its derivative is $10q + 20 \geq 0$ on that interval and its value at 2 is -1 . Since $c \geq -2$, it follows that

$$\begin{aligned} F(q, c) &= c(5q^2 + 20q - 61) - 5q^2 + 25q - 173 \\ &\leq -2(5q^2 + 20q - 61) - 5q^2 + 25q - 173 \\ &= -15(q + \tfrac{1}{2})^2 - \tfrac{189}{4} < 0. \end{aligned} \tag{LBRE16}$$

Thus LBRE13 is nonzero. If $\mathcal{A} = \mathcal{C} = 0$, then $n_1 + Rc_1 = n_2 + Rc_2 = 0$ since $R \neq 0$. Multiplying the first equation by c_2 , the second by c_1 , and subtracting would give $n_1c_2 - c_1n_2 = 0$, a contradiction. Consequently

$$(\mathcal{A}, \mathcal{C}) \neq (0, 0) \quad \text{for every allowed } \delta, \gamma, a \text{ and every } j \in \{1, 2, 3, 4\}. \tag{LBRE17}$$

There is also a fully explicit coefficient margin. Indeed $t_{xy} - t_{xy}^{-1} = 4i\delta\gamma/(\delta^2 + \gamma^2)$, and therefore

$$\begin{aligned} \mathcal{D}_j := n_1c_2 - c_1n_2 &= -\frac{(\delta^2 + \gamma^2)^2}{1296\delta^2\gamma^2}F(q, c) \geq \frac{7(\delta^2 + \gamma^2)^2}{192\delta^2\gamma^2} > 0, \\ \max(|\mathcal{A}|, |\mathcal{C}|) &\geq \frac{\mathcal{D}_j}{|c_1| + |c_2|} > 0. \end{aligned} \tag{LBRE18}$$

The last denominator is positive because simultaneous vanishing of c_1, c_2 would contradict $\mathcal{D}_j > 0$. For its proof use $\mathcal{D}_j = \mathcal{A}c_2 - Rc_1c_2$ and $|R| = 1$, then apply the triangle inequality. Thus on every boundary at least one of its two endpoint coefficients has modulus at least $\mathcal{D}_j/[4(|c_1| + |c_2|)]$, with the exact scalar $1/4$ from LBRE11 retained. Each polynomial in LBRE11 is therefore nonzero. In homogeneous Segre coordinates, none of z_0, z_1, w_0, w_1 divides the actual limiting biform $Q_0(Lt)$. This conclusion includes every amplitude phase and all eigenvalue-collision phases; its proof never excludes such a phase. It also gives the corresponding statement for the full limiting conductor product: restriction to any one boundary is the product of its four nonzero quadratic restrictions, which is nonzero because the polynomial ring over $\mathbb{C}$ has no zero divisors. Hence none of the four homogeneous boundary coordinates divides A_0 of PCL8.

Finally the biform coefficients are entire in the original inverse period z by PCL8–11. For each of the finitely many pairs consisting of an orbit index and a boundary, LBRE11 and LBRE17 supply a nonzero coefficient at $z = 0$. That exact coefficient function is continuous, so it stays nonzero on a disk about 0. The minimum of these finitely many positive radii is positive. Therefore all four boundary exclusions hold simultaneously on some neighborhood of the original inverse-period limit, retaining the original period and unit throughout. No explicit size for this last neighborhood is asserted here.

69 Actual boundary multiplicities and a finite full-support radius

Retain the original quartet, unit, period matrices, logarithm branch and cyclic orientation of PCL1–11, PQO1–25 and RPC1–26. In particular, $r_1 = \delta + i\gamma$, $r_2 = \delta - i\gamma$, $r_3 = -r_2$, $r_4 = -r_1$, $0 < \delta < 1/2$, $\gamma > 2$ and $U = \text{diag}(a, \bar{a}, \bar{a}, a)$ with the full actual $a \neq 0$. Put $z = u^{-1}$ and retain

$$\begin{aligned} L_j(z) &= U^{-1}V\mathcal{R}(z)^{-1}D^{-j}\mathcal{R}(z)V^{-1}U, \\ q_j(z; Z, W) &= Q_0(L_j(z)(ZW, Z, W, 1)^\top), \quad A_z = \prod_{j=1}^4 q_j(z; Z, W). \end{aligned} \quad (\text{ASR1})$$

These are exact identities in the original period map by PCL9. Every entry of L_j is entire, since $\det \mathcal{R} = 1$ and $\mathcal{R}^{-1} = \text{adj } \mathcal{R}$. In particular all coefficients below are the entire functions evaluated by PCL2 and PCL10, with their original signs and ordered-pair multiplicities.

69.1 Every finite-period boundary multiplicity

Let S_0 be the symmetric matrix with entries $(S_0)_{14} = (S_0)_{41} = 1/2$, $(S_0)_{23} = (S_0)_{32} = -1/2$ and all other entries zero, and set $S_j(z) = L_j(z)^\top S_0 L_j(z)$. Thus

$$\begin{aligned} q_j &= (S_j)_{11}Z^2W^2 + 2(S_j)_{12}Z^2W + (S_j)_{22}Z^2 + 2(S_j)_{13}ZW^2 \\ &\quad + 2((S_j)_{14} + (S_j)_{23})ZW + 2(S_j)_{24}Z + (S_j)_{33}W^2 + 2(S_j)_{34}W + (S_j)_{44}. \end{aligned} \quad (\text{ASR2})$$

This follows by substituting all four coordinates into $t^\top S_j t$; each off-diagonal term occurs twice. Introduce the following six three-entry vectors, with the displayed order of their coefficients:

$$\begin{aligned} E_{Z,0,j} &= ((S_j)_{33}, 2(S_j)_{34}, (S_j)_{44}), \\ E_{Z,\infty,j} &= ((S_j)_{11}, 2(S_j)_{12}, (S_j)_{22}), \\ E_{W,0,j} &= ((S_j)_{22}, 2(S_j)_{24}, (S_j)_{44}), \\ E_{W,\infty,j} &= ((S_j)_{11}, 2(S_j)_{13}, (S_j)_{33}), \\ M_{Z,j} &= (2(S_j)_{13}, 2((S_j)_{14} + (S_j)_{23}), 2(S_j)_{24}), \\ M_{W,j} &= (2(S_j)_{12}, 2((S_j)_{14} + (S_j)_{23}), 2(S_j)_{34}). \end{aligned} \quad (\text{ASR3})$$

On the original five-orbit locus, for $X = Z, W$ and $b = 0, \infty$ define the following integer by literal vanishing of these actual coefficient vectors:

$$\nu_{X,b,j} = \begin{cases} 0, & E_{X,b,j} \neq 0, \\ 1, & E_{X,b,j} = 0, \quad M_{X,j} \neq 0, \\ 2, & E_{X,b,j} = 0, \quad M_{X,j} = 0. \end{cases} \quad (\text{ASR4})$$

On the original five-orbit locus, q_j is nonzero. To verify this without a geometric identification, ASR2 is zero exactly when its nine displayed coefficients vanish. These equations say precisely $S_j = cS_0$ for a scalar c , so $Q_0(L_j t) = cQ_0(t)$ in the four independent primary variables. Conjugation by the original period matrix identifies this with projective equality of orbit quadrics, which is excluded for $1 \leq j \leq 4$ on that locus. Since S_j is invertible, the scalar in such an equality could not be zero.

The homogeneous form of q_j has bidegree $(2, 2)$ with $t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)$. For $X = Z = z_0/z_1$, its three successive powers of z_0 have coefficient vectors $E_{Z,0,j}, M_{Z,j}, E_{Z,\infty,j}$. This proves that ASR4 is exactly its order along $z_0 = 0$. Reversing the powers proves the assertion for $z_1 = 0$. The two identical calculations with $W = w_0/w_1$ prove the remaining two orders. In the last case of ASR4 the opposite vector is nonzero, since $q_j \neq 0$.

Write a_-, a_+ for the minimum and maximum Z exponents in A_z , and b_-, b_+ for its minimum and maximum W exponents. The exact actual support extrema are therefore

$$\boxed{\begin{aligned} a_- &= \sum_{j=1}^4 \nu_{Z,0,j}, & a_+ &= 8 - \sum_{j=1}^4 \nu_{Z,\infty,j}, \\ b_- &= \sum_{j=1}^4 \nu_{W,0,j}, & b_+ &= 8 - \sum_{j=1}^4 \nu_{W,\infty,j}. \end{aligned}} \quad (\text{ASR5})$$

Indeed the lowest nonzero Z coefficient of a product is the product of its lowest coefficients in the integral domain $\mathbb{C}[W]$, and the same is true of the highest coefficients. Thus neither extreme can cancel. The argument over $\mathbb{C}[Z]$ gives both W extremes. Equivalently the complete boundary divisor and the remaining bihomogeneous factor are specified by

$$M_\partial = z_0^{a_-} z_1^{8-a_+} w_0^{b_-} w_1^{8-b_+}, \quad A = M_\partial A_{\text{red}}, \quad (d_a, d_b) = (a_+ - a_-, b_+ - b_-). \quad (\text{ASR6})$$

This retains the order at each of the four original boundary rulings.

69.2 The exact symplectic map underlying the coefficient tests

Put $B = V^{-1}U$ and retain RPC5's alternating matrix J . The original alternating form on primary coordinates is

$$\begin{aligned} \Omega_{\text{prim}} &= B^\top J B, \\ (\Omega_{\text{prim}})_{14} &= \frac{a^2}{8i\delta\gamma r_1}, & (\Omega_{\text{prim}})_{23} &= -\frac{\bar{a}^2}{8i\delta\gamma r_2}, \end{aligned} \quad (\text{ASR7})$$

with the negative transposed entries and all others zero. In fact RPC4 evaluates this form as $\sum_i a_i^2 t_i v_{5-i} / H'(r_i)$, where $(a_1, a_2, a_3, a_4) = (a, \bar{a}, \bar{a}, a)$, $H'(r_1) = 8i\delta\gamma r_1$ and $H'(r_2) = -8i\delta\gamma r_2$. This proves every entry and its sign. Using $P_0 = G_u \mathcal{R}(z)$ in RPC9 gives $\mathcal{R}(z)^{-\top} J \mathcal{R}(z)^{-1} = J|_{\beta=0}$. The diagonal D preserves this last alternating matrix by its paired weights. Conjugating the equality by $\mathcal{R}(z)B$ proves

$$L_j(z)^\top \Omega_{\text{prim}} L_j(z) = \Omega_{\text{prim}}. \quad (\text{ASR8})$$

Let $s = r_1/a^2$, retain its literal complex conjugate $\bar{s} = r_2/\bar{a}^2$, and define $T = \text{diag}(s, \bar{s}, -\bar{s}, -s)$. The entries of ASR7 prove the exact matrix identity

$$S_0 = -4i\delta\gamma \Omega_{\text{prim}} T. \quad (\text{ASR9})$$

Each original boundary plane is one of $E_{12}, E_{34}, E_{13}, E_{24}$, where E_{ab} is spanned by the two indicated original coordinate vectors. Each is Lagrangian for ASR7: its restriction vanishes, its dimension is two, and the ambient alternating matrix is invertible. Equation ASR8 shows that its image $F = L_j E_{ab}$ is also Lagrangian. Polarization of ASR9 now proves the exact equivalence

$$Q_0|_{L_j E_{ab}} = 0 \iff TL_j E_{ab} \subset L_j E_{ab} \iff \text{rank}[L_j E_{ab}, TL_j E_{ab}] = 2. \quad (\text{ASR10})$$

Here the last matrix has the two columns of L_j indexed by a, b followed by their images under T . For the first equivalence, polarization says $\Omega_{\text{prim}}(v, Tw) = 0$ for all $v, w \in F$; since $F^{\perp\Omega} = F$, this is exactly $TF \subset F$. For the last equivalence the first two columns are independent. Thus all its three-by-three minors vanish precisely in that case. These are the same boundary-vanishing tests as the corresponding three entries of ASR3, with the exact map between the two tests proved. No eigenvalue is cancelled when s is real or purely imaginary.

69.3 An explicit radius from the actual limiting corner calculation

The complete limiting calculation LBRE1–18 proves, for every $1 \leq j \leq 4$ and every original nonzero a , that

$$\begin{aligned} d_{j,1}(0) &= d_{j,4}(0), & d_{j,2}(0) &= d_{j,3}(0), \\ m_j &:= \max\{|d_{j,1}(0)|, |d_{j,2}(0)|\} > 0, & d_{j,i}(z) &:= (S_j(z))_{ii}. \end{aligned} \quad (\text{ASR11})$$

Indeed LBRE11 gives $d_{j,1}(0) = d_{j,4}(0) = \mathcal{A}/4$ and $d_{j,2}(0) = d_{j,3}(0) = \mathcal{C}/4$ with the complete original amplitude in LBRE10. LBRE13–18 prove the positive lower bound

$$m_j \geq \frac{\mathcal{D}_j}{4(|c_1| + |c_2|)} > 0, \quad \mathcal{D}_j \geq \frac{7(\delta^2 + \gamma^2)^2}{192\delta^2\gamma^2},$$

where $c_1, c_2, \mathcal{D}_j$ here are exactly LBRE10 and LBRE18's finite coefficients for this orbit index. That proof evaluates the literal J_∞ in PCL4 and the original r_1/r_2 and ζ^{-j} . The two equalities also follow directly from PCL22–23 at $z = 0$, because the opposite-root permutation fixes Q_0 and conjugates $L_j(0)$ to itself.

Expand each of these actual entire diagonal entries as $d_{j,i}(z) = \sum_{n \geq 0} d_{j,i,n} z^n$ by PCL2 and PCL10, and define the explicitly specified finite constant

$$C_j = \sum_{n \geq 1} \max_{1 \leq i \leq 4} |d_{j,i,n}|, \quad R_{\text{supp}} = \max \left\{ R_*, 1, \frac{2C_1}{m_1}, \frac{2C_2}{m_2}, \frac{2C_3}{m_3}, \frac{2C_4}{m_4} \right\}. \quad (\text{ASR12})$$

Each series is finite in value: the four entire Taylor series converge absolutely at $z = 1$, and the maximum is bounded by their sum of absolute values. Every denominator is strictly positive by ASR11. If $C_j = 0$, its displayed quotient is zero; no division by C_j is performed. For $|z| \leq 1$, summing the same series gives

$$|d_{j,i}(z) - d_{j,i}(0)| \leq C_j |z| \quad (1 \leq i \leq 4). \quad (\text{ASR13})$$

Choose $i_j = 1$ when $|d_{j,1}(0)| \geq |d_{j,2}(0)|$, and $i_j = 2$ otherwise. This deterministic choice selects a coefficient already present in the actual period calculation. For $|u| \geq R_{\text{supp}}$, ASR11–13 give

$$|d_{j,i_j}(u^{-1})| \geq \frac{m_j}{2} > 0, \quad |d_{j,5-i_j}(u^{-1})| \geq \frac{m_j}{2} > 0. \quad (\text{ASR14})$$

Indeed both initial values are equal with modulus m_j and their errors are at most $C_j/|u| \leq m_j/2$. Each of the four boundary planes in ASR10 contains exactly one of these two opposite coordinate vectors. Thus its boundary restriction has a nonzero diagonal coefficient, so all four vectors $E_{X,b,j}$ in ASR3 are nonzero. This proves, at every such original u on every logarithm branch,

$$\boxed{a_- = b_- = 0, \quad a_+ = b_+ = 8, \quad (d_a, d_b) = (8, 8), \quad M_\partial = 1, \quad A_{\text{red}} = A.} \quad (\text{ASR15})$$

Branch independence follows from the exact original-period conjugation PCL9, which leaves L_j single-valued in u^{-1} . The radius includes R_* , so all these points are on the proved original five-orbit locus. The estimate has introduced no new source order and no freely specified amplitude or period coefficient.

At arbitrary nonzero original periods, each boundary failure is the simultaneous zero set of its three actual entire entries in ASR3. ASR11 supplies a nonzero value at $z = 0$ for at least one of those entries. Such an entry has isolated zeros: at any zero its convergent Taylor series has a first nonzero term, and factoring that power leaves a nonvanishing factor in a sufficiently small disk. Thus each boundary restriction-vanishing set is discrete and closed in the finite z plane, and its finite union over the sixteen boundary/index pairs is also discrete and closed. This boundary restriction-vanishing set in the nonzero u plane is contained in $|u| < R_{\text{supp}}$ and has no accumulation point other than possibly $u = 0$. ASR3–5 retain every multiplicity on its intersection with the original five-orbit locus. At an orbit-collapse period, ASR2's nine coefficients detect an identically zero factor; no finite ruling multiplicity is assigned to such a factor.

69.4 Consequence for the actual common factor and residual

Retain the common-factor and grid maps of ACR1–23, with original $k = 4\ell + 1 \geq 9$ and both unchanged source orders $1, k$. Its exact dimension formula and its actual invariant-image quotient are

$$\begin{aligned} g_A &= (k + 1 - d_a)(k + 1 - d_b) - (k - 7)^2, \\ t_G &= \dim \frac{J_k \cap A_{\text{red}} \mathcal{B}_{k-d_a, k-d_b}}{A \mathcal{B}_{k-8, k-8}}. \end{aligned} \quad (\text{ASR16})$$

Here $\mathcal{B}_{p,q}$ is the space of bihomogeneous polynomials of bidegree (p, q) in the two original pairs $(z_0, z_1), (w_0, w_1)$, and J_k is the original invariant image. Its exact map from the affine coefficient space used in ASR1–5 is

$$f(Z, W) \mapsto z_1^p w_1^q f(z_0/z_1, w_0/w_1), \quad F \mapsto F(Z, 1, W, 1).$$

The respective bounded degrees ensure that the first expression is a polynomial. Expansion in its monomials proves that these two maps are inverse and that multiplication is preserved with the displayed bidegrees. ACR5 and ACR15 prove the two formulas in ASR16; ACR8–9 prove that the denominator is a subspace of that numerator. At every $|u| \geq R_{\text{supp}}$, ASR15 first makes the two products in g_A equal. It then makes the numerator of the displayed quotient a subspace of its denominator; the opposite inclusion was already proved by the exact receiving map. Consequently

$$g_A = t_G = 0 \quad (|u| \geq R_{\text{supp}}, \quad k = 4\ell + 1 \geq 9). \quad (\text{ASR17})$$

This removes precisely the common boundary factor calculated by that map. The original remaining residual subspace is still $W = B_{\Xi}$, and the receiving calculation's full residual rank remains

$$j = ((k + 1)^2 - (k - 7)^2) - (8k - 16) = 8k - 32.$$

Here the first parenthesis is the original conductor quotient dimension from its injective multiplication by the nonzero biform A , and $8k - 16$ is the exact original invariant-image kernel dimension of OQX and OCF on this five-orbit locus. Neither that factor nor those residual directions have been set to zero. For periods below this explicit support radius that remain on the original five-orbit locus, ASR3–6 give all actual boundary losses without assuming full widths.

69.5 Only finitely many support exceptions on the original exterior domain

Define the exact coefficient set and its corresponding original periods by

$$\begin{aligned} \mathcal{Z}_{\partial}^* &= \{z : 0 < |z| \leq R_*^{-1}, \quad E_{X,b,j}(z) = 0 \text{ for some } X \in \{Z, W\}, \quad b \in \{0, \infty\}, \quad 1 \leq j \leq 4\}, \\ \mathcal{U}_{\partial}^* &= \{z^{-1} : z \in \mathcal{Z}_{\partial}^*\}. \end{aligned}$$

For each of the sixteen boundary/index pairs, ASR11 selects one diagonal coefficient among the three entries of $E_{X,b,j}$ that is nonzero at $z = 0$. This is an entire function with isolated zeros by the local Taylor argument given above. It has finitely many zeros on the compact disk $|z| \leq R_*^{-1}$: an infinite set would have an accumulation point in that disk, contradicting the proved isolation. The common zero set of the three entries is a subset of this finite set. Taking the union of the sixteen finite sets proves finiteness of $\mathcal{Z}_{\partial}^*$ and hence of $\mathcal{U}_{\partial}^*$. None of these zeros is zero, by the selected nonzero value there. ASR14 excludes every inverse period with $|u| \geq R_{\text{supp}}$. On the whole domain $|u| \geq R_*$, the factors are nonzero by the original five-orbit theorem, so ASR4–5 show that this is precisely the set where at least one of the four support extrema loses its full value. Thus

$$\boxed{\begin{aligned} &\mathcal{U}_{\partial}^* \text{ is finite,} \quad \mathcal{U}_{\partial}^* \subseteq \{u : R_* \leq |u| < R_{\text{supp}}\}, \\ &|u| \geq R_*, \quad u \notin \mathcal{U}_{\partial}^* \implies (a_-, a_+, b_-, b_+) = (0, 8, 0, 8), \\ &g_A = 0, \quad t_G = 0, \quad W = B_{\Xi}, \quad j = 8k - 32 \quad (k = 4\ell + 1 \geq 9). \end{aligned}} \quad (\text{ASR18})$$

At the finite exceptional periods every boundary multiplicity is given by ASR3–5, and the unchanged matrices ACR12–15 evaluate the remaining common-factor map and its residual quotient. No statement that this finite set is empty is needed for any displayed conclusion.

69.6 The second radius and its exact native and polar consequences

RPB1–2 identifies the conductor in the boundary and residual calculations coefficient by coefficient with the present conductor: $H^{-1}D_5^{-a}H = L_a$ after cancellation of only the written scalar and commuting diagonal factors. Therefore

$$|u| \geq \min(R_{\text{supp}}, R_{\text{rank}}) \implies (a_-, a_+, b_-, b_+) = (0, 8, 0, 8), \quad g = t_k = 0. \quad (\text{ASR19})$$

The exterior domain is the union of the two independently proved exterior domains. The present ASR proof already supplies full widths; RPK supplies the second eliminant radius. CFR's exact $g = (k + 1 - d_a)(k + 1 - d_b) - (k - 7)^2$ then gives $g = 0$, and $0 \leq t_k \leq g$ gives its zero visible rank. This has the exact native consequence that EBG's high enlarged numerator kernel is zero, by its module and dimension in EBG14; its low enlarged quotient remains zero. RPK's complete principal-part map instead has kernel $\text{im } C^T$ and detects the whole rank- j residual. PLF proves its low-flag and native metric map, including the full original source error. The finite exceptional periods retain their exact three-coefficient edge tests. Neither the zero common-factor target nor the radius union changes the whole residual determinant or its supported labels.

70 The two exact dual subquotients in the original receiver

Fix the original simple quartet, full unit, actual five-orbit period and logarithm branch. All coefficients of the actual conductor A and all original roots are retained. In this section

$$\begin{aligned}
 k &= 4l + 1 \geq 9, \quad q = (k + 1)^2, \quad q' = (k - 7)^2, \\
 \Delta &= q - q' = 16k - 48, \quad r_K = 8k - 16, \quad j = 8k - 32, \\
 E_A(w) &= \sum_{a,b=0}^8 a_{ab} e^{b_{ab}w}, \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v, \\
 \mathbf{m} &= \Delta - v, \quad c' = k/2 - 4, \quad s \in \{1, k\}, \quad \mathcal{N} = \{q - 1, q, 2q - 1, 2q\}, \\
 (\sigma_{q-1}, \sigma_q, \sigma_{2q-1}, \sigma_{2q}) &= (1, 1, -1, -1).
 \end{aligned} \tag{PBR1}$$

The original relation quotient has Gram $H_{Y,N}^{(s)}$ in the fixed FGC frame, and Z_k is its actual FGC10 kernel-column matrix. Here $K = \ker \Xi$ in the original primary space V_A , $L = \mathbb{C}[S]_{<v} \subset V_A$, $K^+ = K + L$, $t_0 = \dim(K \cap L)$, and $t = r_K - t_0$ is the number of columns of Z_k . The symbol r_K denotes this rank parameter, whereas ρ below denotes an analytic radius. No independent source order is introduced.

Retain the complete DNE coefficient frame Z_{exp} , the functions $F_a = \mathcal{N}_a/E_A$, and $\ell_a(g) = \Lambda_{F_a}(\chi'g)$. Put $(A_\ell)_{ba} = \ell_a(S'^b)$ for $0 \leq b < \mathbf{m}$. The matrix A_ℓ is invertible by DNE4. In the original Gamma orthonormal frame $\phi_{n,s,c'}$, $0 \leq n \leq N - v$, whose squared monic norms are $h_n = M_s n!(s/2)_n$, $M_s = (2\pi)^{s/2}$, define

$$\begin{aligned}
 T_{na} &= \Lambda_{F_a}(\phi_{n,s,c'}), \quad V_{n,(i,j)} = \phi_{n,s,c'}(\beta'_{ij}), \\
 \mathcal{D}_N^{(s)} &= T^*[I - V(V^*V)^{-1}V^*]T = A_\ell^*(H_{Y,N}^{(s)})^{-1}A_\ell, \\
 G_{\text{exp}} &= Z_{\text{exp}}^*[I - C^\top(\bar{C}C^\top)^{-1}\bar{C}]Z_{\text{exp}} > 0, \\
 B_K &= Z_k^\top A_\ell, \quad D_{K,N} = [B_K(\mathcal{D}_N^{(s)})^{-1}B_K^*]^{-1} = (\widehat{H}_{K,N}^{(s)})^{-1}, \quad G_K = [B_K G_{\text{exp}}^{-1}B_K^*]^{-1}.
 \end{aligned} \tag{PBR2}$$

The onto map B_K is the exact restriction of the complex-linear dual to the actual kernel columns. The conjugation in PBR2 follows by substituting the displayed A_ℓ identity and $B_K = Z_k^\top A_\ell$; it gives $B_K \mathcal{D}_N^{-1} B_K^* = Z_k^\top \bar{H}_{Y,N} \bar{Z}_k = \bar{Z}_k^* \bar{H}_{Y,N} \bar{Z}_k$ before inversion. The first equality for $\mathcal{D}_N$ minimizes the full extension over every lower evaluation functional. Its bracket is the orthogonal projection off their entire column span. Thus the full conditional covariance and the coefficient reference are both specified; neither is replaced by the other.

For the complementary quotient choose exactly the first invertible t -row minor of Z_k , with row set $\mathcal{I}$ and complementary set $\mathcal{J}$, and put

$$\begin{aligned}
 j_0 &= \mathbf{m} - t = j - v + t_0, \quad P_Y = E_{\mathcal{J}}^\top - E_{\mathcal{J}}^\top Z_k (E_{\mathcal{I}}^\top Z_k)^{-1} E_{\mathcal{I}}^\top, \\
 J_0 &= A_\ell^{-1} P_Y^\top, \quad D_{0,N} = J_0^* \mathcal{D}_N^{(s)} J_0, \quad G_0 = J_0^* G_{\text{exp}} J_0, \\
 Q_{0,N} &= [P_Y (H_{Y,N}^{(s)})^{-1} P_Y^*]^{-1}, \quad D_{0,N} = \overline{Q_{0,N}^{-1}}.
 \end{aligned} \tag{PBR3}$$

Empty rank uses the identity quotient. Multiplication shows $P_Y Z_k = 0$ and $P_Y E_{\mathcal{J}} = I$; hence $\ker P_Y = \text{im } Z_k$. Also $B_K J_0 = 0$ and J_0 has j_0 independent columns, proving $\text{im } J_0 = \ker B_K$. Substitution in PBR2 gives $D_{0,N} = \bar{P}_Y \bar{H}_{Y,N}^{-1} \bar{P}_Y^\top$, which proves the last identity in PBR3. These are the actual kernel quotient and its full dual annihilator.

70.1 Finite positive-spectrum allowance and every small eigenvalue

For clarity the finite allowance used below is the complete PEC13 quantity, with no hidden replacement by an asymptotic zero. For $0 < \rho < 1$ with $E_A(-i \arctan z) \neq 0$ on $|z| = \rho$, put $m_\rho = \min_{|z|=\rho} |E_A(-i \arctan z)| > 0$. The actual nonzero symbol zeros inside this image have

orders h_ω . Form the full jet matrix $(J_{\rho,k})_{(\omega,b),a} = \mathcal{N}_a^{(b)}(\omega)/b!$, $0 \leq b < h_\omega$, and put $c_{\rho,k} = \text{rank } J_{\rho,k}$. With $R = |k/2| + k\sqrt{\delta^2 + \gamma^2}$ and $d = N - v$, retain

$$\begin{aligned} C_{\rho,N,s} &= \frac{q e^{2(R+|c'|) \arctanh \rho} (1 - \rho^2)^{-s/2}}{M_s m_\rho^2} \sum_{n=0}^d \frac{n!}{(s/2)_n} \rho^{-2n}, \\ C_{0,N,s} &= q \mathcal{B}_{d,s,k,A}^2, \quad \varepsilon_{N,s}^\vee = \inf_{\rho \text{ admissible}} \{c_{\rho,k} \log^+ C_{0,N,s} + (\mathfrak{m} - c_{\rho,k}) \log^+ C_{\rho,N,s}\}, \\ E_{k,s}^\vee &= \varepsilon_{2q-1,s}^\vee + \varepsilon_{2q,s}^\vee = o(kq). \end{aligned} \tag{PBR4}$$

Here $\mathcal{B}_{d,s,k,A}$ is the full finite expression DNE11–13, including its chosen zero-free radius, Taylor coefficient and Gamma mass. The full DNE proof is included as a provider, not assumed. Pole cancellation on $\ker J_{\rho,k}$ and Cauchy's coefficient integral give $C_{\rho,N,s}$ exactly: the n th squared coefficient is weighted by $n!/(M_s(s/2)_n)$. Minimization over the entire conductor coefficient fibre preserves these cancellations. The full DNE bound gives $C_{0,N,s}$ on all vectors. At most $c_{\rho,k}$ eigenvalues can exceed $C_{\rho,N,s}$, by intersecting the span of $c_{\rho,k} + 1$ such eigenvectors with the cancellation subspace. This proves the positive logarithmic sum bound PBR4. PEC17–18 proves the same bound for the fixed quotient and restriction in PBR2–3, by the variational principle applied respectively to the inverse compression and the compression. The exact reference metrics G_K, G_0 are retained in both applications.

For completeness, at each fixed admissible radius the symbol has only finitely many zeros in the compact disk, so $c_{\rho,k}$ is bounded independently of k . The full finite constants above and DNE15 give $\limsup \varepsilon_{N,s}^\vee/(kq) \leq 64 \log(1/\rho)$. Admissible radii approach one because the zeros are isolated. Taking that limit after the fixed-radius bound proves the displayed $o(kq)$ for the original two source orders. This bounds every forward exterior degree by $\varepsilon_{N,s}^\vee/2$ in logarithmic norm, by multiplying subsets of the relative eigenvalues. For the inverse map its squared singular values are the reciprocals. They are retained in the following exact formulas.

For $X = K, 0$, let $\mu_{X,a,N}$ be every eigenvalue of $G_X^{-1/2} D_{X,N} G_X^{-1/2}$ and define

$$\begin{aligned} p_{X,N} &= \sum_a \log^+ \mu_{X,a,N}, \quad n_{X,N} = \sum_a \log^+(1/\mu_{X,a,N}), \\ A_{X,k,s} &= n_{X,q-1} + n_{X,q} - n_{X,2q-1} - n_{X,2q}, \\ P_{X,k,s} &= p_{X,2q-1} + p_{X,2q} - p_{X,q-1} - p_{X,q}, \\ \widehat{F}_k^{(s)} &= A_{K,k,s} + P_{K,k,s}, \quad \sum_N \sigma_N \log \det Q_{0,N} = A_{0,k,s} + P_{0,k,s}, \\ A_{X,k,s} &\geq 0, \quad 0 \leq P_{X,k,s} \leq E_{k,s}^\vee. \end{aligned} \tag{PBR5}$$

Increasing N enlarges the original polynomial lift space without changing its norm on previous polynomials or its relation map. Minimum primal lift norms therefore decrease. Inversion in PBR2 increases $\mathcal{D}_N$, and restriction or minimization under the same onto B_K preserves that increase. The variational principle then increases each relative eigenvalue in the fixed reference metric. Thus p increases and n decreases, proving all nonnegativities in PBR5. The two high positive sums are bounded by PBR4. Finally $\log \mu = \log^+ \mu - \log^+(1/\mu)$ and $D_{K,N} = \widehat{H}_{K,N}^{-1}$ give the first return identity with the four stated signs; PBR3 gives the other. The fixed reference determinants cancel only in these four signed sums. In particular the small eigenvalue contributions $n_{X,N}$ have not been bounded away or discarded.

70.2 The original metric on the low image

Let $G_N^{(s)}$ be the original primary quotient metric and let $Q_{\Xi,N}^{\text{fix}}$ be its induced metric on $B_\Xi = \Xi(V_A)$ in the original fixed image frame. Set $W_0 = \Xi(L)$. There is the exact isometric isomorphism

$$\begin{aligned} \iota_N : (K+L)/K &\longrightarrow W_0, & [p] &\longmapsto \Xi p, \\ \|\iota_N[p]\|_{Q_{\Xi,N}^{\text{fix}}}^2 &= \min_{a \in K} G_N^{(s)}(p+a, p+a) = \|[p]\|_{(K+L)/K,N}^2, \\ \dim W_0 &= v - t_0, & F_{W_0}^{(s)} &:= \sum_N \sigma_N \log \det(I_{W_0}^* Q_{\Xi,N}^{\text{fix}} I_{W_0}) = F_{K^+/K}^{(s)}. \end{aligned} \quad (\text{PBR6})$$

Indeed $\Xi p \in \Xi(L)$ if and only if $p \in K+L$. The kernel of the displayed map is exactly K and it is onto. The full fibre of Ξp in V_A is $p+K$, already contained in $K+L$. Hence minimization in the full image norm is exactly the quotient minimization on $K+L$, proving the norm identity at each cutoff. In transported frames the Grams agree exactly. For any other fixed frames their congruence has the same fixed Jacobian at all four endpoints, which cancels with $\sum \sigma_N = 0$. This proves the last identity without identifying W_0 with the entire residual image.

In the FGC coordinates the onto map $[p] \mapsto [\alpha_A \mathcal{T}_A p / \chi']$ from V_A to $Y_N / \text{im } Z_k$ has kernel $K+L$. The corresponding quotient map sends this class to $[\Xi p]$ in B_Ξ / W_0 . The kernel assertion follows directly: subtract the kernel polynomial giving the same FGC image, then the remaining zero image under $\mathcal{T}_A$ is in L ; the converse follows from the two kernels. This proves the exact target isomorphism in PEC27.

The source map comparing this quotient to the native Gamma one is the original CEP map S_N , followed by $D_{\alpha,N}(x, f) = (x, \alpha_A f)$ on the full GIP graph. Retaining every relation column and every Z_k column, minimization first in the low coordinate and then in these polynomial fibres gives precisely $Q_{0,N}$ in PBR3. Let $\bar{Q}_{\Xi,N}$ be the induced original metric on B_Ξ / W_0 in the transported frame. PEC28's full exterior comparison yields

$$\begin{aligned} |\log \det \bar{Q}_{\Xi,N} - \log \det Q_{0,N}| &\leq 2E_N^{\text{CEP}} + 2j_0 |\log |\alpha_A||, \\ E_{k,s}^{\text{tr}} &= 2 \sum_N E_N^{\text{CEP}} + 8j_0 |\log |\alpha_A|| = o(kq), \\ \Phi_{k,s} &= F_{W_0}^{(s)} + A_{0,k,s} + P_{0,k,s} + \delta_{0,k,s}, \quad |\delta_{0,k,s}| \leq E_{k,s}^{\text{tr}}. \end{aligned} \quad (\text{PBR7})$$

To see the error signs explicitly, put $d_N = \log \det \bar{Q}_{\Xi,N} - \log \det Q_{0,N}$. In a fixed frame adapted to W_0 , the full Gram determinant is its restriction determinant times the Schur complement determinant, including its fixed frame Jacobian. Therefore $\delta_{0,k,s} = \sum_N \sigma_N d_N$; the triangle inequality gives the stated bound. The projection used to take each quotient is an exterior contraction; the inverse source map gives the reverse inequality. This proves that the factors α_A , all four original CEP errors, and the full graph denominator occur in this very metric comparison.

70.3 Both low terms and the full one-sided kernel errors

Keep the original FEX endpoint constants explicitly. Write $h_N^{\text{harm}} = \sum_{a=1}^N 1/a$ and $w_d = d!/(s/2)_d$. The symbol t here remains the kernel-column count from PBR1.

$$\begin{aligned} n_N &= N + 1 - v, & m_N &= \log \max \left\{ 1, \frac{v!}{|\mu_v|} \sum_\nu |a_\nu| e^{|b_\nu - 4|(h_N^{\text{harm}} + 2)} \right\}, \\ \zeta_{N,s} &= \log \prod_{d=v}^N \frac{w_d}{w_{d-v}}, & e_{N,s}^- &= \zeta_{N,s} - 2(n_N - t)m_N, & e_{N,s}^+ &= 2tm_N, \\ e_{N,s}^{\text{FEX}} &= \log \det \hat{H}_{K,N}^{(s)} - \log \det H_{K^+/L,N}^{(s)}, & e_{N,s}^- &\leq e_{N,s}^{\text{FEX}} \leq e_{N,s}^+, \\ D_s^- &= e_{q-1,s}^- + e_{q,s}^- - e_{2q-1,s}^+ - e_{2q,s}^+, & D_s^+ &= e_{q-1,s}^+ + e_{q,s}^+ - e_{2q-1,s}^- - e_{2q,s}^-, \\ \mathcal{E}_s^{\text{flag}} &= \sum_N \sigma_N e_{N,s}^{\text{FEX}} \in [D_s^-, D_s^+]. \end{aligned} \quad (\text{PBR8})$$

These are the original FEX18–20/34 expressions, with both source masses retained in the underlying norms. The endpoint inequalities come from the full flag determinant bounds, with squared Jacobian $e^{\zeta_{N,s}}$ and both operator bounds e^{m_N} . The signs of the high endpoint bounds are reversed when forming the four sum, giving exactly $D_s^\pm$ above.

Retain also the exact finite AKS/FEX allowance

$$\begin{aligned}
R_{\text{cen}} &= k\sqrt{\delta^2 + \gamma^2}, \quad \tau_q = \pi/2 - 1/q, \quad a_q = 2\log 3/\tau_q, \quad s_q = (a_q^q - 1)/(a_q - 1), \\
b_{s,q} &= [(9/25)\sin(1/q)]^{-s/2} e^{2R_{\text{cen}}(\log 3 + \tau_q/2)}, \\
\mathcal{L}_{s,h} &= \log \left(1 + b_{s,q} s_q^2 \sum_{d=q}^{q+h-1} \frac{d!}{(s/2)_d} (25/16)^d \right), \quad \mathcal{L}_s = \mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}, \\
0 \leq F_L^{(s)} &\leq v\mathcal{L}_s, \quad 0 \leq F_{W_0}^{(s)} \leq (v - t_0)\mathcal{L}_s, \quad \lambda_s^{\text{low}} = F_L^{(s)} - F_{W_0}^{(s)}, \\
F_K^{(s)} &= A_{K,k,s} + P_{K,k,s} - \mathcal{E}_s^{\text{flag}} + \lambda_s^{\text{low}}.
\end{aligned} \tag{PBR9}$$

The low metric estimates are the original restriction and quotient consequences of AKS, as proved in FEX32–33. Applying the two complete Schur factorizations to K^+ gives OER11, and substituting PBR5 and the isometry PBR6 gives the last identity. The quantities F_L and F_{W_0} remain their separate original volumes. In particular they are not individually removed by using their difference. Combining PBR7 and PBR9 gives the additional exact identity

$$F_K^{(s)} + \Phi_{k,s} = A_{K,k,s} + A_{0,k,s} + P_{K,k,s} + P_{0,k,s} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} + \delta_{0,k,s}. \tag{PBR10}$$

The W_0 terms cancel in this sum because they are precisely the same original metric by PBR6. They remain in each summand PBR7/9.

70.4 The complete current finite interval

First retain the exact low volumes in the new finite kernel bounds, and separately display the previously proved coarse bounds:

$$\begin{aligned}
\kappa_s^- &= A_{K,k,s} - D_s^+ + \lambda_s^{\text{low}}, \quad \kappa_s^+ = A_{K,k,s} + E_{k,s}^\vee - D_s^- + \lambda_s^{\text{low}}, \quad \kappa_s^- \leq F_K^{(s)} \leq \kappa_s^+, \\
A_{K,k,s} - D_s^+ - (v - t_0)\mathcal{L}_s &\leq F_K^{(s)} \leq A_{K,k,s} + E_{k,s}^\vee - D_s^- + v\mathcal{L}_s, \\
\varphi_s^- &= F_{W_0}^{(s)} + A_{0,k,s} - E_{k,s}^{\text{tr}}, \quad \varphi_s^+ = F_{W_0}^{(s)} + A_{0,k,s} + E_{k,s}^\vee + E_{k,s}^{\text{tr}}, \quad \varphi_s^- \leq \Phi_{k,s} \leq \varphi_s^+, \\
A_{0,k,s} - E_{k,s}^{\text{tr}} &\leq \Phi_{k,s} \leq A_{0,k,s} + E_{k,s}^\vee + E_{k,s}^{\text{tr}} + (v - t_0)\mathcal{L}_s.
\end{aligned} \tag{PBR11}$$

PBR5/8/9 proves the first line by taking the least and greatest allowed signed flag errors; PBR7 proves the third line. The original low caps then prove the second and fourth lines, which are exactly PEC22 and PEC30. Thus using the exact low difference has a specified morphism to each coarser bound.

Let $\Phi_{k,s}^-, \Phi_{k,s}^+$ be the full-window RWC bounds of OER17–18, and retain their entire conditional matrices

$$\begin{aligned}
U_N^+ &= [u_{N+1}, \dots, u_{N+q}], \quad B_N = I_q + (CU_N^+)^*(CR_N C^*)^{-1}(CU_N^+), \\
Z_N &= D[I_q - R_N C^*(CR_N C^*)^{-1}C]U_N^+, \quad \Omega_{N+q} = \Omega_N + Z_N B_N^{-1} Z_N^*, \\
\Phi_{k,s} &= \sum_{N=q-1, q} \log \left(\sum_{a=0}^j \left\| \bigwedge^a (\Omega_N^{-1/2} Z_N B_N^{-1/2}) \right\|_{\text{HS}}^2 \right).
\end{aligned} \tag{PBR12}$$

Here D denotes the original residual rows, as in OER17. All q columns, the complete conditioning inverse, and every exterior coefficient remain. The proof of the covariance identity is the full Schur update RWC4–6; applying the determinant lemma and expanding all singular-value products

gives the last line. Its finite bounds therefore apply to the same Φ as PBR7. Define $\Phi_{k,s}^{-,\vee} = \max\{\Phi_{k,s}^-, \varphi_s^-\}$ and $\Phi_{k,s}^{+,\vee} = \min\{\Phi_{k,s}^+, \varphi_s^+\}$.

Retain the original OAR expression $\mathcal{F}_{\partial,k}^{(s)}$ and its full errors $-e_{k,s}^- \leq \mathfrak{e}_{k,s} \leq e_{k,s}^+$ in $F_K^{(s)} = \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s} + \mathfrak{e}_{k,s}$. These errors are distinct from PBR8. With all earlier OER19 branches still included, the new receiver is

$$\begin{aligned} L_s^\vee &= \max\{L_s^{\text{win}}, \kappa_s^-, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{+,\vee} - e_{k,s}^-\}, \\ U_s^\vee &= \min\{U_s^{\text{win}}, \kappa_s^+, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{-,\vee} + e_{k,s}^+\}, \quad L_s^\vee \leq F_K^{(s)} \leq U_s^\vee. \end{aligned} \quad (\text{PBR13})$$

Each lower entry is at most the actual scalar and each upper entry is at least that scalar, proving the intersection and its nonempty order without assuming an unproved relation between the entries. On PMR's exact domain $k \geq 29$ and $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}(k+1)^2$, also include in this same maximum and minimum respectively $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{+,\vee} - e_{k,s}^{0,-}$ and $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{-,\vee} + e_{k,s}^{0,+}$, where $e_{k,s}^{0,\pm} = e_{k,s}^\pm + c_k^0$. Outside that domain these two entries are omitted; the original OAR expression and errors remain. The seven-moment $\mathcal{F}_{0,k}^{(s)}$ is distinct from the quotient return $\sum \sigma_N \log \det Q_{0,N}$ in PBR5.

For the arithmetic map retain $|u| \geq R_*$ and the same correlated AMT allowances, with all their original finite constants:

$$\begin{aligned} a_{K,k}^- &= \max\{-C_K^-, \delta_k^\sigma - C_B^+\}, & a_{K,k}^+ &= \min\{C_K^+, \delta_k^\sigma + C_B^-\}, \\ L_{\text{ar}}^{\text{win}} &= \max\{L_{\text{ar}}^{\text{op}}, L_1^{\text{win}} + a_{K,k}^-\}, & U_{\text{ar}}^{\text{win}} &= \min\{U_{\text{ar}}^{\text{op}}, U_1^{\text{win}} + a_{K,k}^+\}, \\ L_{\text{ar}}^\vee &= \max\{L_{\text{ar}}^{\text{win}}, L_1^\vee + a_{K,k}^-\}, & U_{\text{ar}}^\vee &= \min\{U_{\text{ar}}^{\text{win}}, U_1^\vee + a_{K,k}^+\}, \\ F_K^{\text{ar}} &\in [L_{\text{ar}}^\vee, U_{\text{ar}}^\vee], & F_B^{\text{ar}} &\in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^\vee, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^\vee]. \end{aligned} \quad (\text{PBR14})$$

AMT gives the identical scalar $F_K^{\text{ar}} - F_K^{(1)}$ in $[a_{K,k}^-, a_{K,k}^+]$. Adding this interval to PBR13 proves the new entries. The earlier entries remain; subtracting from the same total $F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}$ reverses the two endpoints and proves the final line. No allowance has changed. Previously available reverse source intersections remain valid.

70.5 The signed return and the complete same-class action budget

Keep $\mathcal{H}_k = -T_k$, the full arithmetic correction δ_k^σ , and exactly the original budget

$$\begin{aligned} C_k^{\text{budget}} &= F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, & \Psi_k &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, \\ L_k^\vee + \mathcal{H}_k + \delta_k^\sigma &\leq \Psi_k \leq U_k^\vee + \mathcal{H}_k + \delta_k^\sigma, \\ J_k^{\text{action}} &= 2 \sum_{N=q-1}^{2q-1} w_N \log \epsilon_N^{\text{ar}} = \Psi_k + C_k^{\text{budget}}, \\ L_J^\vee &= L_k^\vee + \mathcal{H}_k + \delta_k^\sigma + C_k^{\text{budget}}, & U_J^\vee &= U_k^\vee + \mathcal{H}_k + \delta_k^\sigma + C_k^{\text{budget}}, \\ 4q \log \mathcal{L}_{h,k} &\leq J_k^{\text{action}} \in [L_J^\vee, U_J^\vee], & \mathcal{L}_{h,k} &\leq \min_N \epsilon_N^{\text{ar}} \leq \prod_N (\epsilon_N^{\text{ar}})^{w_N/(2q)} \leq e^{U_J^\vee/(4q)}. \end{aligned} \quad (\text{PBR15})$$

The weights are $w_{q-1} = w_{2q-1} = 1$ and $w_N = 2$ in between, so $\sum w_N = 2q$. The allowance ϵ_N^{ar} is the original rank-two action radius and $\mathcal{L}_{h,k} = 2\delta e(k+1)\lfloor(k+1)^2/4\rfloor$, with $e = 1$ in this section. MCE's action identities still hold for every original multiplicity. RFX gives $\phi_N(0) = 0$ in the original untilted source, yielding the displayed exact geometric mean. It retains $z_B = -z_K$, and hence the mixed term $a_K d_B + a_B d_K + 2|z_K|^2$. The full generic real-tilt MCE identity with $\log(\epsilon_N^2 + \phi_N^2)$ and the exact MCP derivative remain unchanged. Addition to PBR13 proves the signed bounds; addition of the complete budget proves the action bounds. Applying the same-class lower radius at each degree proves the geometric-mean inequalities. The whole reference boundary, monic window, two contraction penalties and action leakage are retained.

In particular the explicit coarse PEC32 entries become

$$\begin{aligned} A_{K,k,k} - D_k^+ - (v - t_0)\mathcal{L}_k + \mathcal{H}_k + \delta_k^\sigma &\leq \Psi_k \\ &\leq A_{K,k,k} + E_{k,k}^\vee - D_k^- + v\mathcal{L}_k + \mathcal{H}_k + \delta_k^\sigma. \end{aligned} \quad (\text{PBR16})$$

The exact-low entries of PBR13 can strengthen these entries; every older bound is still in the intersection. This is an actual control interval in the original signed direction. It does not remove C_k^{budget} from the action inequality.

70.6 Actual support, the common-factor map, and the retained residual

Let $a_\pm, b_\pm$ be the actual extrema of the nonzero coefficients of A , and let $d_a = a_+ - a_-$, $d_b = b_+ - b_-$. The complete RCF/ACR maps give

$$\begin{aligned} \mathcal{I}_G &= ([-a_-, k - a_+] \times [-b_-, k - b_+]) \setminus [0, k - 8]^2, \quad g_A = (k + 1 - d_a)(k + 1 - d_b) - q', \\ \rho_G : V_A &\rightarrow \mathbb{C}[S']_{<g_A}, \quad V_0 = \ker \rho_G, \quad W = \Xi(V_0), \\ \rho_G I_K &= W_G^{-1} D_G C_G^\top \pi^\top, \quad B_G = \pi C_G, \quad t_G = \dim(B_\Xi/W) = g_A - \text{rank } B_G. \end{aligned} \quad (\text{PBR17})$$

All columns of C_G are the actual shifted coefficient vectors $f_{ij} = \sum_{a,b} a_{ab} e_{i+a, j+b}$ for $(i, j) \in \mathcal{I}_G$. The root Vandermonde W_G , evaluated-polynomial diagonal $D_G = \text{diag}(\chi'(\tau_\ell)^{-1})$, and original exterior selector π are the full ACR11–13 matrices. Here χ' is the original lower-grid polynomial; the prime does not denote differentiation. The exact morphism from V_A to the last quotient sends p to its class modulo $V_0 + K$. The first isomorphism theorem identifies it with $\mathbb{C}[S']_{<g_A}/\rho_G(K)$; composing with Ξ identifies it with B_Ξ/W . The displayed matrix for $\rho_G I_K$ has the same rank as B_G , proving the last formula with all actual coefficients. This retains the original linear relation quotient; RCF's ideal hull does not replace its denominator. The full common-factor metric comparison is RCM23–27 and its actual coefficient-rank specialization is ACR18–23.

LBRE proves the exclusion of all four boundary rulings for the actual limiting two-plane, with its full coefficient margin. ASR uses the actual entire boundary coefficient functions to produce the explicit radius $R_{\text{supp}} = \max\{R_*, 1, 2C_1/m_1, \dots, 2C_4/m_4\}$, where $m_j > 0$ is the larger of the two limiting coordinate moduli and C_j is the full convergent absolute tail sum. For $|u| \geq R_{\text{supp}}$ each selected pair has a nonzero coordinate by the bound $C_j/|u| \leq m_j/2$. On the remaining compact reciprocal-period disk, each chosen entire coordinate is nonzero at zero and consequently has finitely many zeros. Taking the actual common zeros of each three-coordinate boundary vector gives exactly ASR18's finite exception set $\mathcal{U}_\partial^*$ in $\{R_* \leq |u| < R_{\text{supp}}\}$. Thus, on the original arithmetic period domain outside this finite set,

$$\begin{aligned} (a_-, a_+, b_-, b_+) &= (0, 8, 0, 8), \quad d_a = d_b = 8, \quad g_A = t_G = 0, \quad V_0 = V_A, \quad W = B_\Xi, \\ B_\Xi/W_0 &\simeq Y_N/\text{im } Z_k, \quad \dim W_0 = v - t_0, \quad \dim(B_\Xi/W_0) = j - v + t_0. \end{aligned} \quad (\text{PBR18})$$

The first line follows by substitution into PBR17: the target of ρ_G is zero, so its kernel is all of V_A . The second line is the separate exact map PBR6–7. In particular the full residual image $W = B_\Xi$ is still equipped with $Q_{\Xi, N}^{\text{fix}}$, its low subspace and its complementary dual deficit. Neither t_0 nor either metric return is forced to vanish by $g_A = t_G = 0$. On the finite exceptional set the literal ACR rank matrices and RCM metric comparison remain; PBR2–16 did not assume full support and remain on their original hypotheses.

Finally the complete OVR/ECL outer identity and the proved finite errors imply

$$\begin{aligned} A_{K,k,s} + A_{0,k,s} &= \Delta q \mathfrak{C}_\partial + o(kq), \quad \frac{A_{K,k,s} + A_{0,k,s}}{kq} \longrightarrow 16\mathfrak{C}_\partial, \\ 21.66851180520 &< 16\mathfrak{C}_\partial < 21.66851180521, \quad \frac{\Psi_k}{kq} = \frac{A_{K,k,1}}{kq} - \frac{\mathcal{C}_\Gamma}{4} + o(1). \end{aligned} \quad (\text{PBR19})$$

Indeed PBR4, PBR7–9 and the complete FEX36 allowance give $F_K^{(s)} = A_{K,k,s} + o(kq)$ and $\Phi_{k,s} = A_{0,k,s} + o(kq)$. Substitute these equalities in the original evaluated outer sum. The second line uses

the certified ECL coefficient in the same original EIQ variables. For the last equality use the original AMT source-order agreement and the proved signed Gamma return $\mathcal{H}_k/(kq) \rightarrow -\mathcal{C}_\Gamma/4$, while the original arithmetic correction is $o(kq)$. All these are the accepted complete proof providers, preserved in the current full source. This determines the sum of the two specified spectral deficits and the exact signed receiver in terms of the kernel one; it does not assign an individual leading value to either deficit. OER15–16’s actual q -scale Gamma interval, including its strictly positive radius and both unequal finite endpoint errors, is unchanged. The next analytic quantity in this receiver is the actual deficit distribution in PBR5 and its complete action comparison PBR15.

71 Exact low-jet reconstruction and its original quotient spectrum

Retain the simple-quartet fixed-period hypotheses of RDS, DNE and PEC. The original conductor and all its coefficients remain fixed as $k = 4l + 1 \geq 9$ varies. Set

$$\begin{aligned} q &= (k+1)^2, & q' &= (k-7)^2, & \Delta &= q - q', & N &= q - 1, \\ d &= q - v - 1, & n &= d + 1 = q - v, & \mathbf{m} &= q - q' - v, \end{aligned} \quad (\text{LJR1})$$

with $0 \leq v \leq 80$, $\mu_v \neq 0$, and the same centers $c = k/2$, $c' = c - 4$. The original source order is $s \in \{1, k\}$, $b = s/2$, $M_s = (2\pi)^{s/2}$ and $h_a = M_s a! (b)_a$. The roots are $\beta_{ij} = c + (2i - k)\delta + i(2j - k)\gamma$ and the analogous lower roots β'_{ij} , with $0 < \delta < 1/2$, $\gamma > 2$. Write $\sigma = (\delta^2 + \gamma^2)^{1/2}$.

71.1 The exact finite reconstruction and its inverse

Let $\phi_a(S') = p_a((S' - c')/i)/\sqrt{h_a}$ be the complete original orthonormal source frame through degree d . Define B_s and the original upper Vandermonde by

$$S'^a = \sum_{r=0}^d (B_s)_{ra} \phi_r(S'), \quad \mathbb{V}_{j,(a,b)} = \beta_{ab}^j \quad (0 \leq j < q, \ 0 \leq a, b \leq k). \quad (\text{LJR2})$$

These matrices are not assigned new source norms. The coefficient matrix B_s is upper triangular, with diagonal $i^a \sqrt{h_a}$, since the leading coefficient of ϕ_a is $i^{-a}/\sqrt{h_a}$. In particular $|\det B_s|^2 = \prod_{a=0}^d h_a$. For an explicit formula, let J_b be the real symmetric $(d+1)$ -by- $(d+1)$ Jacobi matrix with adjacent entries $\sqrt{(r+1)(b+r)}$. Then

$$(B_s)_{ra} = \sqrt{M_s} [(c'I + iJ_b)^a]_{r0}. \quad (\text{LJR3})$$

The original three-term recurrence proves this formula by repeated multiplication by $S' = c' + iy$. A path of length at most d starting at degree zero never needs a degree above d , so the finite Jacobi truncation is exact in all the displayed columns.

For a vector $w = (w_0, \dots, w_d)^\top$ define the complex-linear functional Λ_w by $\Lambda_w(\phi_a) = w_a$. Its first n ordinary moments are $B_s^\top w$. Define the full moment-convolution matrix

$$(J_E)_{ja} = \binom{j}{a} \mu_{j-a} \quad (0 \leq j < q, \ 0 \leq a \leq d), \quad \mathbf{R}_s = \mathbb{V}^{-1} J_E B_s^\top. \quad (\text{LJR4})$$

Terms with $a > j$ are zero. The first v rows of J_E vanish, and its remaining square block $\tilde{J}_E$ is lower triangular with diagonal $\binom{v+a}{a} \mu_v$, $0 \leq a \leq d$. Therefore $\mathbf{R}_s$ is an isomorphism from $\mathbb{C}^n$ onto the coefficient space of the original $\mathcal{N}_v$.

To verify the map exactly, let F be any formal series with $F^{(a)}(0) = (B_s^\top w)_a$ for $a \leq d$. Leibniz' rule gives

$$(E_A F)^{(j)}(0) = \sum_{a=0}^d \binom{j}{a} \mu_{j-a} F^{(a)}(0) \quad (0 \leq j < q). \quad (\text{LJR5})$$

No unknown derivative of F enters, because $\mu_{j-a} = 0$ when $a > j - v$. The upper Vandermonde then reconstructs the unique upper exponential polynomial $\mathcal{N}$ with precisely these first q jets. This proves LJR4 as the actual map of original polynomial functionals to original exponential coefficients.

Let κ_r denote the coefficients of the finite inverse series of $B_\infty(z) = v! E_A(z)/(\mu_v z^v)$, so $B_\infty(z)^{-1} = \sum_{r \geq 0} \kappa_r z^r$ near zero. For $\nu \in \mathcal{N}_v$ write $u = \mathbb{V}\nu$. The full inverse is

$$\begin{aligned} f_r &= \frac{v!}{\mu_v} r! \sum_{a=0}^r \kappa_{r-a} \frac{u_{v+a}}{(v+a)!} \quad (0 \leq r \leq d), \\ \mathbf{R}_s^{-1} \nu &= B_s^{-\top} (f_0, \dots, f_d)^\top. \end{aligned} \quad (\text{LJR6})$$

Indeed multiplication of exponential Taylor coefficients by $E_A = (\mu_v/v!)z^v B_\infty$ gives precisely this triangular inverse. The constants $\kappa_0, \dots, \kappa_d$ are determined by the finite recurrence $\kappa_0 = 1$, $\kappa_r = -\sum_{a=1}^r [v!\mu_{v+a}/((v+a)!\mu_v)]\kappa_{r-a}$. Thus every moment, factorial and inverse coefficient is supplied.

71.2 Both exact quotient kernels and their metrics

Let $V_{\text{lo},s}$ be the n -by- q' matrix $(V_{\text{lo},s})_{r,(a,b)} = \phi_r(\beta'_{ab})$, and let C be the entire original primary conductor. Then

$$R_s V_{\text{lo},s} = C^\top. \quad (\text{LJR7})$$

For each column, its functional is evaluation at the corresponding lower root, so $F(z) = e^{\beta'_{ab}z}$. The original root-sum identity gives $E_A(z)e^{\beta'_{ab}z} = \sum_{r,t} a_{rt} e^{\beta_{a+r,b+t}z}$. Its coefficient column is exactly $C^\top$, and LJR5 reconstructs it, proving LJR7. Both matrices have rank q' , by the lower Vandermonde and the original injectivity of the conductor transpose.

Consequently LJR4 induces the exact isomorphism

$$\bar{R}_s : \mathbb{C}^n / \text{im } V_{\text{lo},s} \longrightarrow \mathcal{N}_v / \text{im } C^\top. \quad (\text{LJR8})$$

Its quotient kernel is zero: if $R_s w = C^\top \eta$, then LJR7 and injectivity give $w = V_{\text{lo},s} \eta$. Its inverse is induced by LJR6. The first space has the actual native Gamma dual quotient norm, using the Euclidean norm on the orthonormal functional-value coordinates w . The second has the unchanged original exponential coefficient quotient norm.

Use precisely DNE's fixed Z_{exp} on the right and set $W_s = R_s^{-1} Z_{\text{exp}}$ on the left. In these paired frames,

$$\begin{aligned} D_s &= W_s^* [I - V_{\text{lo},s} (V_{\text{lo},s}^* V_{\text{lo},s})^{-1} V_{\text{lo},s}^*] W_s = \mathcal{D}_{q-1}^{(s)}, \\ G &= Z_{\text{exp}}^* [I - C^\top (\bar{C} C^\top)^{-1} \bar{C}] Z_{\text{exp}} = G_{\text{exp}}. \end{aligned} \quad (\text{LJR9})$$

The first equality with DNE follows because LJR6 gives exactly the first n orthonormal values of $\Lambda_{\mathcal{N}_a/E_A}$; the allowed extensions are all lower evaluations by LJR7. Both projection formulas are attained minimum norms. In particular the reconstruction is represented by the identity from $(\mathbb{C}^m, D_s)$ to $(\mathbb{C}^m, G)$, not by an asserted isometry of these two different forms.

Let λ_a be the relative eigenvalues of (D_s, G) as in PEC12. The squared singular values of LJR8, in decreasing order, are λ_a^{-1} with the order reversed. Thus

$$\log \det(\bar{R}_s^* \bar{R}_s) = \log \frac{\det G}{\det D_s}, \quad \sum_a \log^+ \sigma_a(\bar{R}_s)^2 = \sum_a \log^+ (\lambda_a^{-1}). \quad (\text{LJR10})$$

The adjoint and singular values here use the two specified quotient norms.

For an actual onto map B_0 , or an actual injective map J_0 , the identity in these coordinates induces the paired quotients or restrictions with complete forms

$$\begin{aligned} D_{B_0} &= (B_0 D_s^{-1} B_0^*)^{-1}, & G_{B_0} &= (B_0 G^{-1} B_0^*)^{-1}, \\ D_{J_0} &= J_0^* D_s J_0, & G_{J_0} &= J_0^* G J_0. \end{aligned} \quad (\text{LJR11})$$

The exact common kernels and targets are transported by LJR8. For $B_0 = B_K = Z_k^\top A_\ell$, DNE9 gives $D_{B_K} = (\widehat{H}_{K,q-1}^{(s)})^{-1}$. For the full annihilator frame $J_0 = A_\ell^{-1} P_Y^\top$ from PEC24, $D_{J_0} = \overline{Q_{0,q-1}^{-1}}$. These identities retain the original low-quotient corrections of FGC when compared with the original kernel and residual.

71.3 The complete full-space and quotient Jacobians

Let $\mathbb{V}_{<v}$ be the first v rows of $\mathbb{V}$ and put $\mathcal{J}_v = \det(\mathbb{V}_{<v} \mathbb{V}_{<v}^*)$, with $\mathcal{J}_0 = 1$. Then the squared full-space Jacobian is

$$\det(R_s^* R_s) = \frac{\mathcal{J}_v}{|\det \mathbb{V}|^2} \left| \frac{\mu_v}{v!} \right|^{2n} \prod_{a=0}^d \left(\frac{(v+a)!}{a!} \right)^2 \prod_{a=0}^d h_a. \quad (\text{LJR12})$$

Indeed $R_s = \mathbb{V}^{-1} P_v \tilde{J}_E B_s^\top$, where P_v inserts the last n coordinates. The complementary principal-minor identity for the positive matrix $\mathbb{V}\mathbb{V}^*$ gives $\det(P_v^*(\mathbb{V}\mathbb{V}^*)^{-1}P_v) = \mathcal{J}_v/|\det \mathbb{V}|^2$. For completeness this identity follows by block Gaussian elimination: the determinant of a block matrix equals the determinant of its first principal block times the determinant of its Schur complement, while the bottom block of its inverse is the inverse Schur complement. All principal blocks are positive. The determinants of $\tilde{J}_E$ and B_s , computed above, prove LJR12.

The exact restriction Jacobian on the subspace in LJR7 is $\det(\bar{C}C^\top)/\det(V_{\text{lo},s}^* V_{\text{lo},s})$. Completing this subspace to a full source frame, and its image to the transported target frame, factors the full Gram into restriction and quotient Schur determinants. It follows that

$$\mathfrak{J}_{k,s} := \frac{\det G}{\det D_s} = \det(R_s^* R_s) \frac{\det(V_{\text{lo},s}^* V_{\text{lo},s})}{\det(\bar{C}C^\top)}. \quad (\text{LJR13})$$

No orthogonality of a chosen coefficient complement is assumed.

Let $\mathbb{V}'$ be the lower-root evaluation Vandermonde of degrees $0, \dots, q' - 1$, and let $H_{\psi,s}$ be the full $\mathbf{m}$ -by- $\mathbf{m}$ original Gram of $\chi' S'^a$, $0 \leq a < \mathbf{m}$. Then

$$\det(V_{\text{lo},s}^* V_{\text{lo},s}) = \frac{|\det \mathbb{V}'|^2 \det H_{\psi,s}}{\prod_{a=0}^d h_a}, \quad (\text{LJR14})$$

and therefore

$$\boxed{\mathfrak{J}_{k,s} = \frac{\mathcal{J}_v |\det \mathbb{V}'|^2}{|\det \mathbb{V}|^2 \det(\bar{C}C^\top)} \left| \frac{\mu_v}{v!} \right|^{2n} \prod_{a=0}^d \left(\frac{(v+a)!}{a!} \right)^2 \det H_{\psi,s}.} \quad (\text{LJR15})$$

To prove LJR14, split the native polynomial space into the low monomials through degree $q' - 1$ and the ideal columns $\chi' S'^a$. Their complete monomial coefficient matrix has determinant one because χ' is monic. The full source Gram determinant is $\prod h_a$, since the original monic S' -polynomials have squared norms h_a and their phases have modulus one. Schur elimination therefore gives the low-remainder quotient determinant $\prod h_a / \det H_{\psi,s}$. Evaluation at the lower roots is precisely $\mathbb{V}'$ on that remainder. The inverse quotient determinant is thus $|\det \mathbb{V}'|^2 \det H_{\psi,s} / \prod h_a$. In an orthonormal frame the inverse evaluation quotient Gram is the conjugate of $V_{\text{lo},s}^* V_{\text{lo},s}$; their determinants agree because they are positive real. This proves the identity with its full original source mass before cancellation in LJR15.

71.4 Finite reconstruction bounds on the actual grid

Put $A_{\text{abs}} = \sum_{r,t=0}^8 |a_{rt}|$ and $H_D = \sum_{a=1}^D 1/a$, with $H_0 = 0$. For the original roots define

$$\begin{aligned} \mathfrak{d}_{ab} &= \prod_{(i,j) \neq (a,b)} (\beta_{ab} - \beta_{ij}), & L_{ab}(S) &= \mathfrak{d}_{ab}^{-1} \prod_{(i,j) \neq (a,b)} (S - \beta_{ij}), \\ \Pi_{k,s} &= \prod_{a=1}^{q-1} [2\sqrt{a(b+a)} + k\sigma], \\ M_{k,s}^{\text{rec}} &= \max \left\{ 1, A_{\text{abs}} e^{8\sigma(H_{q-1}+2)} \sqrt{M_s} \Pi_{k,s} \left(\sum_{a,b=0}^k |\mathfrak{d}_{ab}|^{-2} \right)^{1/2} \right\}. \end{aligned} \quad (\text{LJR16})$$

The derivative product $\mathfrak{d}_{ab}$ is distinguished from the complete lower polynomial χ' . It retains its actual complex sign in L_{ab} . Then

$$\|R_s\| \leq M_{k,s}^{\text{rec}}, \quad \|\bar{R}_s\| \leq M_{k,s}^{\text{rec}}. \quad (\text{LJR17})$$

Here all norms have precisely the source and target metrics of LJR4 and LJR8.

To prove the first bound, evaluation of the reconstructed functional on L_{ab} gives $\nu_{ab} = \Lambda_w(T_A L_{ab})$. Its coefficient row in the w coordinates therefore has Euclidean norm $\|T_A L_{ab}\|_{s,c'}$. This is an exact row identity, not a comparison with an unweighted polynomial coefficient norm. In the original coordinates, $T_A p(c' + iy) = \sum a_{rt} p(c + iy + (b_{rt} - 4))$. Since $b_{rt} - 4 = (2r - 8)\delta + i(2t - 8)\gamma$ has modulus at most 8σ , the exact finite Taylor translation and the original derivative estimate $\|\partial_S\|_{\leq q-1, s, c} \leq H_{q-1} + 2$ give the factor $A_{\text{abs}} e^{8\sigma(H_{q-1} + 2)}$. The factor $-i$ in $\partial_S = -i\partial_y$ has modulus one and is retained in the Taylor map.

Multiplication by y from degree at most $a - 1$ to degree at most a is the sum of the two original Jacobi shifts, each bounded by $\sqrt{a(b+a)}$. Therefore each factor $S - \beta_{ij}$ has multiplication norm at most $2\sqrt{a(b+a)} + |c - \beta_{ij}|$ at this step. Starting from the original constant norm $\sqrt{M_s}$ proves $\|L_{ab}\|_{s,c} \leq \sqrt{M_s} \Pi_{k,s} / |\mathfrak{d}_{ab}|$. The sum of the squared row norms is the Hilbert–Schmidt norm squared of R_s , proving the first inequality. For the quotient, choose the attained minimum representative on the left and project its image orthogonally off $\text{im } C^\top$. This proves the second inequality, with exactly the denominator in LJR8.

Every spacing in the denominator admits the exact factorization

$$|\mathfrak{d}_{ab}| = (2\delta)^k a! (k-a)! (2\gamma)^{k(k+1)} [b!(k-b)!]^{k+1} \cdot \prod_{i=0}^k \prod_{\substack{0 \leq j \leq k \\ j \neq b}} \left(1 + \frac{\delta^2(a-i)^2}{\gamma^2(b-j)^2}\right)^{1/2}. \quad (\text{LJR18})$$

In this formula only, b is the second root index; the source parameter inside $\Pi_{k,s}$ remains $s/2$. Separate the k factors in the row $j = b$, and factor $2\gamma|b-j|$ from every other row to prove LJR18. The elementary bounds $a!(k-a)! = k!/\binom{k}{a} \geq (k/(2e))^k$ and the same bound for b give

$$|\mathfrak{d}_{ab}| \geq (k/e)^{q-1} \delta^k \gamma^{k(k+1)}, \quad \Pi_{k,s} \leq [k^2(3+\sigma)]^{q-1}. \quad (\text{LJR19})$$

The factorial estimate follows from $\sum_{a=1}^k \log a \geq \int_1^k \log t \, dt$ and $\binom{k}{a} \leq 2^k$. For the second inequality use $2\sqrt{a(a+b)} \leq 2a+b$, $b \leq k/2$, and $2(1+1/k)^2 + (1/2+\sigma)/k \leq 3+\sigma$ for $k \geq 9$. Consequently

$$\begin{aligned} \log M_{k,s}^{\text{rec}} &\leq \max\{0, (q-1) \log k + \log A_{\text{abs}} + 8\sigma(H_{q-1} + 2) + \tfrac{1}{2} \log q + \tfrac{1}{2} \log M_s \\ &\quad + (q-1) \log[e(3+\sigma)] - k \log \delta - k(k+1) \log \gamma\} \\ &= q \log k + O_{\text{actual}}(q). \end{aligned} \quad (\text{LJR20})$$

The last equality denotes an upper bound with that order of error. It is uniform for both original source orders, since $\log M_s = (s/2) \log(2\pi) = O(k)$. The finite definition LJR16, with every actual spacing and all conductor coefficients, remains the bound used below.

71.5 A direct estimate of the nested determinant difference

Let $\mathcal{I} = \{4, \dots, k-4\}^2$ and let $\mathcal{O}$ be its complement in the upper index grid. Then $|\mathcal{I}| = q'$, $|\mathcal{O}| = \Delta$, and $\beta_{i+4, j+4} = \beta'_{ij} + 4$. The coefficient map $(S' + 4)^a = \sum_{r \leq a} \binom{a}{r} 4^{a-r} S'^r$ is upper triangular with diagonal one. Hence the Vandermonde on the central upper roots has exactly the lower Vandermonde determinant. Let $\mathcal{P}_{\mathcal{O}}$ denote the unordered distinct pairs with at least one index in $\mathcal{O}$. Thus

$$\begin{aligned} 2M_{\mathcal{O}} &= 2|\mathcal{P}_{\mathcal{O}}| = q(q-1) - q'(q'-1) = \Delta(q+q'-1), \\ \mathscr{D}_k &:= \log \frac{|\det \mathbb{V}|^2}{|\det \mathbb{V}'|^2} = 2 \sum_{\{x,y\} \in \mathcal{P}_{\mathcal{O}}} \log |\beta_x - \beta_y|. \end{aligned} \quad (\text{LJR21})$$

This is cancellation of the identical central-central factors in the exact Vandermonde products.

For $d(x, y) = \|x - y\|_\infty$ one has $2\delta d(x, y) \leq |\beta_x - \beta_y| \leq 2\sigma d(x, y)$. For each fixed x , the number of other grid indices within distance t is at most $4t^2 + 4t$, for $1 \leq t \leq k$. The exact identity

$\log(k/d) = \int_d^k dt/t$ consequently gives $\sum_{y \neq x} \log(k/d(x, y)) \leq 2q - 8$. Summing this nonnegative estimate over the Δ outer indices counts every pair in LJR21 at least once. It proves

$$\begin{aligned} 2M_{\mathcal{O}} \log k + 2M_{\mathcal{O}} \log(2\delta) - 2\Delta(2q - 8) &\leq \mathcal{D}_k \\ &\leq 2M_{\mathcal{O}} \log k + 2M_{\mathcal{O}} \log(2\sigma). \end{aligned} \quad (\text{LJR22})$$

In particular $\mathcal{D}_k = \Delta(q + q' - 1) \log k + O_{\text{actual}}(kq)$. This error is obtained from the affected pairs themselves, not by subtracting two estimates with errors of order q^2 .

71.6 The complete remaining native Gram

The original Gram $H_{\psi, s}$ in LJR15 is positive definite: a nonzero polynomial has nonzero modulus off a finite set, and the original Gamma density is positive on the real line. The following bounds include its full mass and apply at every $k \geq 9$ under LJR1. Define

$$\begin{aligned} R_{\psi, k} &= \max\{(k - 8)\sigma, |c'|\}, \quad T_* = \max\{1, 2d + s/2 + R_{\psi, k}\}, \\ Y &= 2(q + k\sigma + 1), \quad h = 1/q, \quad L_h = \frac{\pi}{\Gamma(3/4)^2} (\sin h)^{3/2}, \\ \beta_s &= \frac{M_s}{\sqrt{2} \Gamma(s/2)}, \quad A_{k, s} = \beta_s L_h e^{-(\pi/2+h)(Y+1)}, \\ \mathcal{H}_m &= \prod_{a=0}^{m-1} [(2a+1) \binom{2a}{a}^2]^{-1}. \end{aligned} \quad (\text{LJR23})$$

Then the finite two-sided estimate is

$$(A_{k, s} q^{2q'})^m \mathcal{H}_m \leq \det H_{\psi, s} \leq M_s^m T_*^{2mq' + m(m-1)}. \quad (\text{LJR24})$$

For the upper bound, $\chi' S'^a$ has the q' actual lower roots and a additional zero roots. All are at distance at most $R_{\psi, k}$ from c' . Applying the same original Jacobi multiplication bound as in LJR17 through degree $q' + a \leq d$ gives its norm at most $\sqrt{M_s} T_*^{q' + a}$. Hadamard's Gram determinant inequality proves the upper side.

For the lower bound, the exact original Gamma recurrence and the global density estimate give

$$\begin{aligned} dm_s(y) &= \beta_s \prod_{j=0}^{(s-1)/4-1} (y^2 + (2j + \tfrac{1}{2})^2) d\sigma(y), \\ \frac{d\sigma}{dy}(y) &\geq L_h e^{-(\pi/2+h)|y|}. \end{aligned} \quad (\text{LJR25})$$

These are the complete identities OSP12 and OFV9. To make the lower-density constant explicit, Euler's integral rotated through $\pi/2 - h$ gives

$$|\Gamma(3/4 - iy/2)|^2 \leq \Gamma(3/4)^2 (\sin h)^{-3/2} e^{-(\pi/2-h)|y|}.$$

The exact reflection product with $|\Gamma(1/4 + iy/2)|^2$ is $2\pi^2 / \cosh(\pi y)$. Dividing and using $\cosh(\pi y) \leq e^{\pi|y|}$ proves the displayed lower density, including the factor $1/(2\pi)$ in $d\sigma$. The recurrence prefactor β_s follows by applying $\Gamma(z+1) = z\Gamma(z)$ exactly $(s-1)/4$ times and clearing the factors 4^{-1} from the squared moduli. Thus no original source mass has been discarded.

On the full interval $[Y, Y+1]$, every factor of $\chi'(c' + iy)$ has modulus at least $y - (k-8)\sigma \geq q$, and the polynomial density multiplier in LJR25 is at least one because $y \geq 1$. Therefore the weighted density is at least $A_{k, s} q^{2q'}$. The Gram of $(c' + iy)^a$ on this interval is obtained from the Gram of t^a on $[0, 1]$ by the translation $y = Y + t$ and coefficients with diagonal i^a . This map has determinant of modulus one.

For completeness, Rodrigues' polynomial $L_a(t) = (a!)^{-1} (d/dt)^a [t^a (t-1)^a]$ has leading coefficient $\binom{2a}{a}$. Integration by parts a times proves orthogonality to every lower power; all boundary terms

vanish. It also gives $\int_0^1 L_a(t)^2 dt = 1/(2a+1)$, since $\int_0^1 t^a(1-t)^a dt = (a!)^2/(2a+1)!$. Dividing by the square of the leading coefficient gives the monic norm in LJR23. Triangular orthogonal elimination therefore proves that the complete interval Gram determinant is exactly $\mathcal{H}_m$. The positive-form inequality on that interval proves the lower side of LJR24.

Both sides imply the useful direct scale

$$\log \det H_{\psi,s} = 2\mathfrak{m}q' \log q + O_{\text{actual}}(kq). \quad (\text{LJR26})$$

Here $\mathfrak{m} = O(k)$, $d = O(q)$ and $\log T_* = \log q + O_{\text{actual}}(1)$. Moreover $-\log \mathcal{H}_m \leq \mathfrak{m} \log(2\mathfrak{m}+1) + 2\mathfrak{m}(\mathfrak{m}-1) \log 2$, using $\binom{2a}{a} \leq 4^a$. Since $2h/\pi \leq \sin h \leq h$, $\log L_h = O(\log q)$. Writing $s/2 = 2\ell_s + 1/2$, $\Gamma(s/2) = \sqrt{\pi} \prod_{j=0}^{2\ell_s-1} (j+1/2)$ shows $|\log \beta_s| = O(k \log(k+1))$. Finally $Y = O_{\text{actual}}(q)$. All errors in the logarithms of LJR24 are consequently $O_{\text{actual}}(kq)$; in particular the terms $k^2 \log q$ are smaller than kq .

71.7 The exact remainder and every paired exterior volume

Let a_* be the actual nonzero pivot coefficient chosen by the original lexicographic conductor proof. Its triangular q' -by- q' minor has determinant $a_*^{q'}$. Cauchy–Binet and the row and column sum bounds on the full convolution matrix give

$$|a_*|^{2q'} \leq \det(\bar{C}C^\top) \leq A_{\text{abs}}^{2q'}. \quad (\text{LJR27})$$

The matrix in this determinant is the conjugate of CC^* ; their positive real determinants agree. For the low-jet minor, Hadamard gives $\mathcal{J}_v \leq q^v [k(1/2 + \sigma)]^{v(v-1)}$. For $k \geq v-1$, choose the v consecutive roots with first index zero in its Cauchy–Binet sum. Their Vandermonde gives $\mathcal{J}_v \geq (2\gamma)^{v(v-1)} (\prod_{a=1}^{v-1} a!)^2$. At $v=0$ both bounds mean $\mathcal{J}_0 = 1$. Thus $\log \mathcal{J}_v = O_{\text{actual}}(\log(k+1))$. Also $0 \leq \sum_{a=0}^d \log((v+a)!/a!) \leq vn \log(q+v)$. The fixed period supplies the actual nonzero μ_v, a_* and A_{abs} in these statements; none is selected afresh.

Define the following exact, still unevaluated remainder:

$$\begin{aligned} \mathcal{R}_{k,s} := & \log \mathcal{J}_v - \mathcal{D}_k - \log \det(\bar{C}C^\top) + 2n \log |\mu_v/v!| \\ & + 2 \sum_{a=0}^d \log \frac{(v+a)!}{a!} + \log \det H_{\psi,s} - 2\mathfrak{m}q \log k. \end{aligned} \quad (\text{LJR28})$$

Every term is an original finite determinant, moment or root product already specified above. Then

$$\log \mathfrak{J}_{k,s} = 2\mathfrak{m}q \log k + \mathcal{R}_{k,s}, \quad |\mathcal{R}_{k,s}| = O_{\text{actual}}(kq). \quad (\text{LJR29})$$

For the bound substitute LJR22, LJR26 and LJR27 into LJR28. The only large logarithmic expression that remains is $2\mathfrak{m}q' \log q - \Delta(q+q'-1) \log k$. Since $\log q = 2 \log k + 2 \log(1+1/k)$ and $\Delta = \mathfrak{m} + v$, subtracting $2\mathfrak{m}q \log k$ from this expression gives exactly

$$[(-3\mathfrak{m}+v)\Delta + \Delta - 2vq] \log k + 4\mathfrak{m}q' \log(1+1/k). \quad (\text{LJR30})$$

The coefficient of $\log k$ is $O_{\text{actual}}(k^2)$; the last term is $O_{\text{actual}}(k^2)$. Both are $o(kq)$. This proves LJR29 by an explicit difference calculation.

Consider any fixed paired restriction, quotient, or restriction followed by quotient of LJR8, with rank t . Its two forms are the restrictions and attained quotients of D_s, G along the same specified algebraic flag. Write $\mathfrak{J}_{t,k,s}$ for its squared Jacobian. Then the exact finite bounds are

$$\log \mathfrak{J}_{k,s} - 2(\mathfrak{m}-t) \log M_{k,s}^{\text{rec}} \leq \log \mathfrak{J}_{t,k,s} \leq 2t \log M_{k,s}^{\text{rec}}. \quad (\text{LJR31})$$

To prove this for a flag $A \subset B$ in the common $\mathfrak{m}$ -dimensional coordinate space, factor each full Gram determinant successively into the restriction on A , the quotient B/A , and the quotient of the full space by B . The factors are their actual Schur determinants. Their Jacobians multiply to

$\mathfrak{J}_{k,s}$. Every restriction of the map has norm at most $M_{k,s}^{\text{rec}}$; every induced quotient does too, by the minimum-lift and projection proof of LJR17. The two complementary factors have total rank $\mathfrak{m} - t$. Bounding them above and dividing proves the lower inequality. Bounding the rank- t factor directly proves the upper inequality.

Uniformly over all these actual flags and $0 \leq t \leq \mathfrak{m}$,

$$\log \mathfrak{J}_{t,k,s} = 2tq \log k + O_{\text{actual}}(kq). \quad (\text{LJR32})$$

For a fully retained finite version put $u_{k,s} = \log M_{k,s}^{\text{rec}} - q \log k$ and $\mathcal{R}_{t,k,s} = \log \mathfrak{J}_{t,k,s} - 2tq \log k$. LJR31 gives $\mathcal{R}_{k,s} - 2(\mathfrak{m} - t)u_{k,s} \leq \mathcal{R}_{t,k,s} \leq 2tu_{k,s}$. LJR20 bounds $u_{k,s}$ above by $O_{\text{actual}}(q)$, which proves both sides of LJR32 using LJR29. No inverse coefficient norm or conditioning premise for the flag occurs.

In particular apply LJR31 to the actual B_K quotient and the actual J_0 annihilator restriction in LJR11. If p_t, n_t are their two positive logarithmic spectral parts in the orientation (D, G) , then exactly

$$n_{t,q-1} - p_{t,q-1} = \log \mathfrak{J}_{t,k,s} = 2tq \log k + \mathcal{R}_{t,k,s}, \quad 0 \leq p_{t,q-1} \leq \varepsilon_{q-1,s}^\vee = o(kq). \quad (\text{LJR33})$$

The last bound is PEC18 on precisely these forms. Thus $n_{t,q-1} = 2tq \log k + O_{\text{actual}}(kq)$. The exact remaining terms are $\mathcal{R}_{t,k,s}$ and $p_{t,q-1}$, with LJR28 and LJR31 giving finite controls. Their four-endpoint differences have not been assigned a coefficient: LJR33 proves the low-endpoint leading scale and supplies the literal maps and remainders to be carried into that next calculation.

72 First-endpoint reconstruction in the actual kernel and annihilator

Retain every original hypothesis, source order, metric, coefficient, root, mass and sign of PBR1–19. The following applies the complete LJR1–33 reconstruction at the actual first endpoint $N = q - 1$. The symbol t_0 continues to mean $\dim(K \cap L)$. Define the two ranks without changing that symbol:

$$t_K = r_K - t_0, \quad t_R = j - v + t_0, \quad t_K + t_R = \mathbf{m}. \quad (\text{LRR1})$$

PBR2 proves that $B_K = Z_k^\top A_\ell$ is onto rank t_K , and PBR3 proves that $J_0 = A_\ell^{-1} P_Y^\top$ is injective rank t_R . Indeed Z_k has the actual independent kernel columns, A_ℓ is invertible, $P_Y Z_k = 0$, and P_Y is onto with kernel $\text{im } Z_k$. Thus $\text{im } J_0 = \ker B_K$.

Let $R_s = \mathbb{V}^{-1} J_E B_s^\top$ be the full LJR4 reconstruction, including the original upper Vandermonde, moment convolution and original Gamma orthonormal-value matrix. LJR5 proves the exact first- q -jet identity by Leibniz' rule, and LJR7 gives $R_s V_{\text{lo},s} = C^\top$. It therefore induces the isomorphism $\bar{R}_s : \mathbb{C}^{q-v} / \text{im } V_{\text{lo},s} \rightarrow \mathcal{N}_v / \text{im } C^\top$. Its inverse is the full finite moment recurrence LJR6. In the paired frames $R_s^{-1} Z_{\text{exp}}$ and Z_{exp} , this map is the identity from $(\mathbb{C}^{\mathbf{m}}, \mathcal{D}_{q-1}^{(s)})$ to $(\mathbb{C}^{\mathbf{m}}, G_{\text{exp}})$, because the complete minimum-extension norms are exactly LJR9/PBR2.

Taking the quotient under the same onto B_K gives precisely the two forms $(D_{K,q-1}, G_K)$ of PBR2; restricting along the same J_0 gives precisely $(D_{0,q-1}, G_0)$ of PBR3. The quotient metric is obtained by minimizing over the full kernel of B_K in each metric, giving $(B_K D^{-1} B_K^*)^{-1}$ and $(B_K G^{-1} B_K^*)^{-1}$. Restriction gives $J_0^* D J_0$ and $J_0^* G J_0$ directly. Use $X = R$ for this residual restriction, with the exact dictionary $(D_R, G_R, p_R, n_R, A_R, P_R) = (D_0, G_0, p_0, n_0, A_0, P_0)$ from PBR3/5; t_0 still denotes only $\dim(K \cap L)$. Consequently their squared reconstruction Jacobians satisfy

$$\mathfrak{J}_{X,k,s} = \frac{\det G_X}{\det D_{X,q-1}}, \quad X = K, R, \quad (\text{LRR2})$$

$$\log \mathfrak{J}_{X,k,s} = n_{X,q-1} - p_{X,q-1} = 2t_X q \log k + \mathcal{R}_{X,k,s}.$$

The first equality uses the actual source and target norms of the reconstruction, in that direction. For the second, a relative eigenvalue μ of (D, G) contributes $-\log \mu = \log^+(1/\mu) - \log^+ \mu$ to the reconstruction determinant. Thus these are exactly the n, p of PBR5, with no inversion of their orientation.

Here every finite remainder is retained. Put $u_{k,s} = \log M_{k,s}^{\text{rec}} - q \log k$, where $M_{k,s}^{\text{rec}}$ is the complete finite LJR16 constant, including all actual root spacings and all conductor coefficients. Keep the exact full-space remainder LJR28:

$$\begin{aligned} \mathcal{R}_{k,s} &= \log \mathcal{J}_v - \mathcal{D}_k - \log \det(\bar{C} C^\top) + 2(q-v) \log |\mu_v/v| \\ &\quad + 2 \sum_{a=0}^{q-v-1} \log \frac{(v+a)!}{a!} + \log \det H_{\psi,s} - 2\mathbf{m}q \log k, \\ \mathcal{R}_{k,s} - 2(\mathbf{m} - t_X)u_{k,s} &\leq \mathcal{R}_{X,k,s} \leq 2t_X u_{k,s}, \quad 0 \leq p_{X,q-1} \leq \varepsilon_{q-1,s}^\vee, \\ n_{X,q-1} &= 2t_X q \log k + \mathcal{R}_{X,k,s} + p_{X,q-1} = 2t_X q \log k + O_{\text{actual}}(kq). \end{aligned} \quad (\text{LRR3})$$

The meanings of the finite terms are literal: $\mathcal{J}_v = \det(\mathbb{V}_{<v} \mathbb{V}_{<v}^*)$ with $\mathcal{J}_0 = 1$, $\mathcal{D}_k = \log(|\det \mathbb{V}|^2 / |\det \mathbb{V}'|^2)$, and $H_{\psi,s}$ is the full original Gram of $\chi' S'^a$, $0 \leq a < \mathbf{m}$, retaining its Gamma mass. The prime on χ' means the lower-grid polynomial. LJR12–15 proves the full Jacobian by exact triangular and Schur factorizations. LJR31 factors its determinant along the same algebraic flag into the rank- t_X part and its complements. Every part has forward norm at most $M_{k,s}^{\text{rec}}$ by its actual minimum lift. Bounding the complementary product above gives the lower inequality for $\mathcal{R}_{X,k,s}$; bounding the selected product gives the upper inequality. PEC18 bounds the positive part on these very quotient and restriction forms. This proves every finite assertion in LRR3.

For the final order statement, LJR20 gives $u_{k,s} \leq O_{\text{actual}}(q)$ and LJR22–30 gives $|\mathcal{R}_{k,s}| = O_{\text{actual}}(kq)$ by the affected-root-pair calculation and the complete Gram bounds. Substitution into the two finite inequalities gives $|\mathcal{R}_{X,k,s}| = O_{\text{actual}}(kq)$ uniformly for $0 \leq t_X \leq \mathfrak{m}$; no lower bound for an unretained inverse norm is needed. PEC15 gives $\varepsilon_{q-1,s}^\vee = o(kq)$. Thus the last equality in LRR3 follows, while its full finite remainder and positive part remain in the preceding equality.

In the actual four-endpoint receiver this yields exactly

$$\begin{aligned} A_{X,k,s} &= 2t_X q \log k + \mathcal{R}_{X,k,s} + p_{X,q-1} + n_{X,q} - n_{X,2q-1} - n_{X,2q}, & X = K, R, \\ F_K^{(s)} &= 2t_K q \log k + \mathcal{R}_{K,k,s} + p_{K,q-1} + n_{K,q} - n_{K,2q-1} - n_{K,2q} + P_{K,k,s} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} - F_{W_0}^{(s)}, \\ \Phi_{k,s} &= F_{W_0}^{(s)} + 2t_R q \log k + \mathcal{R}_{R,k,s} + p_{R,q-1} + n_{R,q} - n_{R,2q-1} - n_{R,2q} + P_{R,k,s} + \delta_{0,k,s}. \end{aligned} \tag{LRR4}$$

The first equality substitutes LRR3 into the definition of A_X in PBR5. The next two substitute that same identity into PBR9 and PBR7, respectively. Hence both original low volumes, all three other endpoint deficits, every positive spectral part, and all unequal finite flag and transfer errors remain. PBR13–16 continues to use these exact A_X in its original source, arithmetic, signed and action intervals. The first endpoint has been evaluated at its leading scale; the remaining endpoint terms and the full kq -scale remainder in LRR3 have not been assigned an individual coefficient.

73 Compatible high jets and the original positive return

Retain all fixed original quartet, amplitude, period and branch data of LJR, DNE, PEC and OVR. In particular the arithmetic domain remains $|u| \geq R_*$, the actual period is fixed while $k = 4l + 1$ grows, and the source order is exactly $s = 1$ or $s = k$. All asymptotic uses of OFV retain its proved eventual threshold. Write

$$\begin{aligned} q &= (k+1)^2, & q' &= (k-7)^2, & n &= q-v, & \mathbf{m} &= q-q'-v, & d_0 &= q-v-1, \\ q-1 &\leq N \leq 2q, & D &= N-v, & e_N &= N-q+1, & D+1 &= n+e_N. \end{aligned} \quad (\text{HJR1})$$

The polynomial degree D is the actual native source degree after the conductor. The original centers remain $c = k/2$ and $c' = c - 4$. No change of s is made between different endpoints.

73.1 The exact compatible high-jet graph

Let $B_{s,D}$ be the original matrix of S'^a in the complete orthonormal Gamma frame $\phi_0, \dots, \phi_D$, as in LJR2-3, with $h_a = M_s a! (s/2)_a$ and $M_s = (2\pi)^{s/2}$. Define

$$\begin{aligned} (\mathbb{V}_N)_{j,(a,b)} &= \beta_{ab}^j \quad (0 \leq j \leq N), \\ (J_{E,N})_{ja} &= \binom{j}{a} \mu_{j-a} \quad (0 \leq j \leq N, 0 \leq a \leq D), \\ \tilde{J}_{E,N} &= (J_{E,N})_{\{v,\dots,N\},\{0,\dots,D\}}, \quad P_{v,N} : \mathbb{C}^{D+1} \longrightarrow \mathbb{C}^{N+1}. \end{aligned} \quad (\text{HJR2})$$

Here $P_{v,N}$ inserts v initial zero coordinates, and an out-of-range term in the convolution is zero. The square $\tilde{J}_{E,N}$ is lower triangular with diagonal $\mu_v \binom{v+a}{a}$ and is therefore invertible.

On the original coefficient subspace $\mathcal{N}_v$ put

$$\mathbf{F}_{N,s} = B_{s,D}^{-\top} \tilde{J}_{E,N}^{-1} P_{v,N}^\top \mathbb{V}_N, \quad \mathbf{E}_{N,s} = \mathbf{F}_{N,s} \mathbf{R}_s = \begin{pmatrix} I_n \\ H_{N,s} \end{pmatrix}. \quad (\text{HJR3})$$

The right factor $\mathbf{R}_s$ is the exact low reconstruction LJR4. Thus $\mathbf{F}_{N,s}$ takes original upper exponential coefficients to their original native functional values through degree D ; $\mathbf{E}_{N,s}$ extends the first n such values. For a literal formula for the inverse convolution, if $u_j = (\mathbb{V}_N \nu)_j$, then

$$f_a = \frac{v!}{\mu_v} a! \sum_{r=0}^a \kappa_{a-r} \frac{u_{v+r}}{(v+r)!}, \quad \mathbf{F}_{N,s} \nu = B_{s,D}^{-\top} (f_0, \dots, f_D)^\top. \quad (\text{HJR4})$$

The coefficients κ_a are all the actual inverse-symbol coefficients of LJR6, with their proved finite recurrence. The source coefficient transformation in HJR4 contains the original phase i^{-a} and every h_a .

To prove these statements, use the analytic germs

$$\mathcal{N}(z) = \sum \nu_{ab} e^{\beta_{ab} z}, \quad E_A(z) = (\mu_v/v!) z^v B_\infty(z).$$

The order condition $\nu \in \mathcal{N}_v$ makes the quotient analytic at zero. Formal Taylor multiplication gives HJR4 and HJR2, and the triangular $B_{s,D}^{-\top}$ changes ordinary moments to orthonormal functional values. Its first n moments and orthonormal values depend only on the first q jets u_j . LJR6 therefore proves that the top block of HJR3 is I_n . The bottom block $H_{N,s}$ is the specified remaining e_N rows.

For a fully intrinsic description, write $\chi(S) = S^q + \sum_{a=0}^{q-1} \chi_a S^a$. Define $\mathcal{U}_{N,s} \subset \mathbb{C}^{D+1}$ by requiring that $u = J_{E,N} B_{s,D}^\top w$ satisfy

$$u_{q+j} + \sum_{a=0}^{q-1} \chi_a u_{a+j} = 0, \quad 0 \leq j < e_N. \quad (\text{HJR5})$$

Then $\mathcal{U}_{N,s} = \text{im } E_{N,s}$, and its dimension is n . Indeed each original root is annihilated by χ , so HJR3 satisfies every recurrence. Conversely the first q entries of u , with its first v zeros, determine a unique ν by the invertible upper Vandermonde. The monic recurrences determine every later u_j uniquely and give $V_N \nu$. The invertible lower convolution block then determines w uniquely. This proves the graph equality, its dimension, and the independence of its e_N constraints. The free ambient later functional values have not been identified with upper exponential coefficients; the exact identification uses HJR5.

Let $V_{N,s}$ denote the $(D+1)$ -by- q' lower evaluation matrix $(V_{N,s})_{a,\beta'} = \phi_a(\beta')$, and let $V = V_{q-1,s}$. The complete lower-exponential identity gives

$$E_{N,s}V = V_{N,s}, \quad \mathcal{U}_{N,s}/\text{im } V_{N,s} \simeq \mathcal{N}_v/\text{im } C^\top. \quad (\text{HJR6})$$

For its proof use $\nu = C^\top \eta$ in HJR4: $\mathcal{N}/E_A$ is exactly the sum of the lower exponentials with coefficients η , at all orders. Since E is injective, its induced quotient kernel is exactly $\text{im } V$. This proves the typed quotient identification with the original denominators.

73.2 One lower-evaluation minimum over the complete window

Keep precisely DNE's fixed coefficient quotient frame Z_{exp} . Put $T_N = F_{N,s}Z_{\text{exp}}$, $T_0 = T_{q-1}$, and

$$\begin{aligned} A &= V^*V, & K_0 &= A^{-1}V^*T_0, & Z_0 &= T_0 - VK_0, & Q_0 &= VA^{-1/2}, \\ D_0 &= Z_0^*Z_0 = \mathcal{D}_{q-1}^{(s)}, & V^*Z_0 &= 0, & Q_0^*Q_0 &= I_{q'}. \end{aligned} \quad (\text{HJR7})$$

The matrices A and its positive square root are the actual lower-evaluation Gram, retaining the complete root and Gamma conditioning. No replacement evaluation metric is made. With $H = H_{N,s}$, define

$$P_N = HZ_0, \quad Q_N = HQ_0, \quad \mathcal{C}_N = (I_{e_N} + Q_NQ_N^*)^{-1/2}P_N. \quad (\text{HJR8})$$

At $e_N = 0$ all these matrices have zero rows. The exact native metric at every endpoint is

$$\boxed{\mathcal{D}_N^{(s)} = D_0 + P_N^*(I_{e_N} + Q_NQ_N^*)^{-1}P_N = D_0 + \mathcal{C}_N^*\mathcal{C}_N.} \quad (\text{HJR9})$$

In particular its increment is positive and has rank $\text{rank } P_N \leq \min\{\mathbf{m}, e_N\}$. The first endpoint update $N = q$ has rank at most one. The last two endpoints have $e_{2q-1} = q$ and $e_{2q} = q+1$, respectively.

Here is a proof with the complete minimizer. Subtract the fixed lower extension EVK_0 from $T_N = ET_0$. Every remaining allowed extension is $E(Z_0x + Q_0t)$, with one vector $t \in \mathbb{C}^{q'}$ used for both its low and high rows. Its squared norm is

$$x^*D_0x + \|t\|^2 + \|P_Nx + Q_Nt\|^2. \quad (\text{HJR10})$$

The minimum occurs at $t = -(I + Q_N^*Q_N)^{-1}Q_N^*P_Nx$. Completing the square leaves $x^*D_0x + x^*P_N^*[I - Q_N(I + Q_N^*Q_N)^{-1}Q_N^*]P_Nx$. Multiplication by $I + Q_NQ_N^*$ shows that the bracket is $(I + Q_NQ_N^*)^{-1}$, proving HJR9. DNE7 proves that these and only these lower evaluations are the permitted native extension differences. Thus HJR10 is exactly its full minimum, without separate minima on different row blocks.

73.3 The original kernel and annihilator updates

Write $B = B_K$ and $J = J_0$ for the actual maps in PEC19,24, so B is onto, J is injective, and $\text{im } J = \ker B$. Define the full initial forms and minimum lift by

$$\begin{aligned} D_{B,0} &= (BD_0^{-1}B^*)^{-1}, & D_{J,0} &= J^*D_0J, \\ L_0 &= D_0^{-1}B^*D_{B,0}, & W_0 &= J(D_{J,0})^{-1/2}. \end{aligned} \quad (\text{HJR11})$$

Then $BL_0 = I$, $BW_0 = 0$, $W_0^*D_0W_0 = I$, and $W_0^*D_0L_0 = 0$. These are literal original kernel and annihilator maps; the displayed W_0 is a coefficient matrix for $\ker B$ with its full D_0 metric. The induced exact updates are

$$\begin{aligned} D_{J,N} &= D_{J,0} + (\mathcal{C}_N J)^*(\mathcal{C}_N J), \\ D_{B,N} &= D_{B,0} + (\mathcal{C}_N L_0)^*[I + \mathcal{C}_N W_0 W_0^* \mathcal{C}_N^*]^{-1}(\mathcal{C}_N L_0). \end{aligned} \quad (\text{HJR12})$$

The first is restriction of HJR9. For the second, every B -lift of y is $L_0 y + W_0 t$. Its low squared norm is $y^* D_{B,0} y + \|t\|^2$, and its added squared norm is $\|\mathcal{C}_N L_0 y + \mathcal{C}_N W_0 t\|^2$. The same completion as HJR10 proves the second formula. It preserves both the original lower-evaluation denominator in $\mathcal{C}_N$ and the original kernel denominator in HJR12. Empty source, kernel or target ranks use their unique empty matrices and determinant one.

The exact identities DNE9 and PEC25 are

$$D_{B,N} = \overline{(\widehat{H}_{K,N}^{(s)})^{-1}}, \quad D_{J,N} = \overline{Q_{0,N}^{-1}}. \quad (\text{HJR13})$$

Thus HJR12 is a full original native kernel and complementary residual calculation, rather than a freely chosen coefficient compression.

73.4 Exact one-row updates and their allocation

The original orthonormal polynomial sequence is independent of its finite cutoff. At any M with $q-1 \leq M < 2q$, let $d = M - v$, and write T, V for the complete matrices through degree d . Their next rows, at degree $d+1$, are t, v_+ . Every entry of t is $\Lambda_{N_a/E_A}(\phi_{d+1})$, computed by HJR4 and the original monomial matrix $B_{s,d+1}$ through degree $d+1$; every entry of v_+ is $\phi_{d+1}(\beta')$. Set

$$\begin{aligned} A_M &= V^* V, \quad K_M = A_M^{-1} V^* T, \quad r_M = t - v_+ K_M, \\ \lambda_M &= 1 + v_+ A_M^{-1} v_+^* > 0, \quad h_M = \lambda_M^{-1/2} r_M. \end{aligned} \quad (\text{HJR14})$$

Then the exact update is

$$D_{M+1} = D_M + h_M^* h_M, \quad D_M = \mathcal{D}_M^{(s)}. \quad (\text{HJR15})$$

To prove this, subtract the attained old minimum from the new extension problem. The old residual is orthogonal to all old lower evaluations, so the remaining minimization for coefficient correction η is $\eta^* A_M \eta + \|r_M x - v_+ \eta\|^2$. Its minimizer is $(A_M + v_+^* v_+)^{-1} v_+^* r_M x$. The same identity used in HJR10 gives the remaining form $r_M^* r_M / (1 + v_+ A_M^{-1} v_+^*)$, proving HJR15. Every evaluation row has participated in the old A_M ; the update does not reset the lower-evaluation coefficient.

At this endpoint put $D_{B,M} = (B D_M^{-1} B^*)^{-1}$, $L_M = D_M^{-1} B^* D_{B,M}$ and $D_{J,M} = J^* D_M J$. The exact inverse-metric decomposition is

$$D_M^{-1} = J(D_{J,M})^{-1} J^* + L_M(D_{B,M})^{-1} L_M^*. \quad (\text{HJR16})$$

Indeed $[J, L_M]$ is a full coefficient frame whose Gram in D_M is $\text{diag}(D_{J,M}, D_{B,M})$. The off-diagonal blocks vanish because $BJ = 0$. Inverting that frame Gram proves HJR16. Define the two actual nonnegative scalar row energies

$$x_M = h_M J(D_{J,M})^{-1} J^* h_M^*, \quad y_M = h_M L_M(D_{B,M})^{-1} L_M^* h_M^*. \quad (\text{HJR17})$$

Both retain λ_M^{-1} from HJR14 and every original period-dependent kernel coordinate. The determinant lemma, proved by extending a nonzero vector to a basis or by expanding a rank-one determinant, now gives

$$\begin{aligned} g_M &= \log \frac{\det D_{M+1}}{\det D_M} = \log(1 + x_M + y_M), \\ g_{0,M} &= \log \frac{\det D_{J,M+1}}{\det D_{J,M}} = \log(1 + x_M), \\ g_{K,M} &= \log \frac{\det D_{B,M+1}}{\det D_{B,M}} = \log \left(1 + \frac{y_M}{1 + x_M} \right), \quad g_M = g_{0,M} + g_{K,M}. \end{aligned} \quad (\text{HJR18})$$

The restriction update gives the second equality directly. For the quotient use the one-row instance of HJR12: its added form is $(h_M L_M)^*(h_M L_M)/(1+x_M)$. Its determinant relative to $D_{B,M}$ proves the third. This proves the allocation of each full original row increment between the actual kernel quotient and its annihilator, with its complete positive denominator.

73.5 The four original signs and an exterior bound at scale kq

For X equal to the full space, K , or 0, define

$$I_{X,N} = \log \frac{\det D_{X,N}}{\det D_{X,q-1}}, \quad I_N = I_{\text{full},N}. \quad (\text{HJR19})$$

All these numbers are nonnegative and increase with N . At each N , completing J by any fixed right inverse of B factors the full Gram into its restriction and quotient. The fixed frame determinant cancels between endpoints, so

$$I_N = I_{K,N} + I_{0,N}, \quad I_{X,N} = \sum_{M=q-1}^{N-1} g_{X,M}. \quad (\text{HJR20})$$

In the sum for the full space $g_{\text{full},M} = g_M$. The source dual identities HJR13 therefore give exactly

$$\begin{aligned} \widehat{F}_k^{(s)} &= I_{K,2q-1} + I_{K,2q} - I_{K,q}, \\ F_{0,k,s} &= I_{0,2q-1} + I_{0,2q} - I_{0,q}, \\ \mathcal{T}_{k,s} &:= \widehat{F}_k^{(s)} + F_{0,k,s} = I_{2q-1} + I_{2q} - I_q. \end{aligned} \quad (\text{HJR21})$$

Equivalently every line is the weighted sum of its original row increments, with weights

$$w_M = \begin{cases} 1, & M = q-1 \text{ or } M = 2q-1, \\ 2, & q \leq M \leq 2q-2. \end{cases} \quad (\text{HJR22})$$

For example I_{2q-1} contains all rows through $2q-2$, I_{2q} adds the last row, and I_q removes one copy of the first. This proves all endpoint signs. Writing $a = I_q$, $b = I_{2q-1} - I_q$ and $c = I_{2q} - I_{2q-1}$ gives $\mathcal{T} = a + 2b + c$ with $a, b, c \geq 0$. In particular

$$0 \leq I_N \leq I_{2q} \leq \mathcal{T}_{k,s} \quad (q-1 \leq N \leq 2q). \quad (\text{HJR23})$$

To give its finite original bound, retain $\delta_{K,k,s} = \widehat{F}_k^{(s)} - F_K^{(s)}$ with $|\delta_{K,k,s}| \leq \mathcal{E}_{k,s}^{\text{FEX}}$, and precisely $\delta_{0,k,s}$ and $F_{\Xi(L)}^{(s)}$ from PEC29. The latter is the original restriction return called $F_{W_0}^{(s)}$ there. OAR3 and PEC29 give the exact equality

$$\mathcal{T}_{k,s} = \mathcal{F}_{\partial,k}^{(s)} + \mathfrak{e}_{k,s} + \delta_{K,k,s} - F_{\Xi(L)}^{(s)} - \delta_{0,k,s}. \quad (\text{HJR24})$$

The original finite signs are consequently

$$\begin{aligned} \max\{0, \mathcal{F}_{\partial,k}^{(s)} - e_{k,s}^- - \mathcal{E}_{k,s}^{\text{FEX}} - E_{k,s}^{\text{tr}} - (v - t_0)\mathcal{L}_s\} &\leq \mathcal{T}_{k,s} \\ &\leq U_{k,s}^{\text{jet}} := \mathcal{F}_{\partial,k}^{(s)} + e_{k,s}^+ + \mathcal{E}_{k,s}^{\text{FEX}} + E_{k,s}^{\text{tr}}. \end{aligned} \quad (\text{HJR25})$$

All errors on the right and left are the explicit accepted FEX, OAR and PEC expressions. The inequality uses $0 \leq F_{\Xi(L)}^{(s)} \leq (v - t_0)\mathcal{L}_s$ and $|\delta_0| \leq E^{\text{tr}}$ only after retaining their actual values in HJR24. In particular

$$\mathcal{T}_{k,s} = \Delta q \mathfrak{E}_{\partial} + o(kq), \quad U_{k,s}^{\text{jet}} = \Delta q \mathfrak{E}_{\partial} + o(kq). \quad (\text{HJR26})$$

This follows from the fully proved OFV/OVR outer first variation and the stated orders of the finite errors. The actual period remains fixed; no uniformity over a moving period is asserted.

Now take any paired restriction, quotient, or subquotient of the common coefficient space at the two metrics (D_0, D_N) . Let its rank be t and its determinant increment be $I_{t,N}$. These are attained restrictions and quotients along the same algebraic flag, retaining all its complete metrics. Since $D_N \geq D_0$, every such form increases. Full flag factorization, with nonnegative log increments on both complementary flags, proves

$$0 \leq I_{t,N} \leq I_N \leq U_{k,s}^{\text{jet}} = O_{\text{actual}}(kq). \quad (\text{HJR27})$$

For the same reason all generalized eigenvalues for (D_N, D_0) are at least one; the sum of all their logarithms is I_N . Hence every exterior degree of the identity between these two metrics has norm at most $\exp(U_{k,s}^{\text{jet}}/2)$. The same statement holds on every specified paired flag. This is a full exterior bound with no additional rank multiplier and with the complete lower-evaluation and kernel conditioning retained.

73.6 Exact cancellation of the low leading scale

Retain LJR's actual finite remainder $\mathcal{R}_{t,k,s}$, including its full-space formula LJR28 and the two-sided flag bounds LJR31. Keep the fixed G_{exp} quotient or restriction associated with this same flag. Then HJR19 gives the exact identity at every endpoint

$$\log \frac{\det G_t}{\det D_{t,N}} = 2tq \log k + \mathcal{R}_{t,k,s} - I_{t,N}. \quad (\text{HJR28})$$

HJR27 therefore propagates the same low leading scale $2tq \log k$ to all four endpoints, with remainder $O_{\text{actual}}(kq)$. It does so by the recurrence-compatible graph and the full Schur minima, not by assigning the later free jets the low dimension.

Let $p_{t,N}, n_{t,N}$ be PEC's two nonnegative logarithmic spectral parts for $(D_{t,N}, G_t)$. The same identity is

$$n_{t,N} = 2tq \log k + \mathcal{R}_{t,k,s} - I_{t,N} + p_{t,N}, \quad 0 \leq p_{t,N} \leq \varepsilon_{N,s}^{\vee} = o(kq). \quad (\text{HJR29})$$

At the actual kernel quotient and annihilator restriction, write $X = K, 0$ respectively. Substituting HJR29 with the four original signs cancels the four copies of both $2tq \log k$ and $\mathcal{R}_{t,k,s}$ exactly and gives

$$\begin{aligned} A_{X,k,s} &= I_{X,2q-1} + I_{X,2q} - I_{X,q} - P_{X,k,s}, \\ P_{X,k,s} &= p_{X,2q-1} + p_{X,2q} - p_{X,q-1} - p_{X,q}, \quad 0 \leq P_{X,k,s} \leq E_{k,s}^{\vee}. \end{aligned} \quad (\text{HJR30})$$

The signs of the p terms follow by substituting their displayed positions in HJR29. Their monotonicity and finite bound are PEC21,26 on these same maps. Together with HJR18 and HJR22 this supplies the explicit original calculation to carry forward:

$$\begin{aligned} A_{K,k,s} &= \sum_{M=q-1}^{2q-1} w_M \log \left(1 + \frac{y_M}{1 + x_M} \right) - P_{K,k,s}, \\ A_{0,k,s} &= \sum_{M=q-1}^{2q-1} w_M \log(1 + x_M) - P_{0,k,s}. \end{aligned} \quad (\text{HJR31})$$

The same original x_M, y_M sum in the full row energy $\log(1 + x_M + y_M)$. Every numerator, denominator and moment used to compute these scalars is fixed in HJR2–4, HJR14, HJR16–17. Their weighted total has the proved leading value HJR26; their separate allocation is retained by the exact formulas HJR31. Thus the next-order kq problem is the specified allocation of these original Gamma row energies, with all low factorial and Vandermonde terms canceled only through the exact equality HJR28.

The two actual row allocations in the original Gamma orders. Keep the original base degree $a = 4l + 1 \geq 9$, $q_a = (a + 1)^2$, the standard endpoints $N_i = q_a - 1, q_a, 2q_a - 1, 2q_a$ and $d_i = N_i - v$. This paragraph uses the degree- a standard window, not a degree- k source on a degree- j target window. Let $\Omega_a^{[a],\text{lo/hi}}$ mean the full NSR2 constants at these d_i . The exact ranks are $t_a = 8a - 16 - t_{0,a}$ and $j_{0,a} = 8a - 32 - v + t_{0,a}$. Then

$$\begin{aligned} S_{K,a}^{(s)} &= \sum_{M=q_a-1}^{2q_a-1} w_M \log \left(1 + \frac{y_M^{(s)}}{1 + x_M^{(s)}} \right) = \widehat{F}_a^{(s)}, \\ S_{R,a}^{(s)} &= \sum_{M=q_a-1}^{2q_a-1} w_M \log(1 + x_M^{(s)}) = F_{0,a,s}, \quad S_{K,a}^{(s)} + S_{R,a}^{(s)} = \mathcal{T}_{a,s}. \end{aligned} \tag{HJR32}$$

The row h_M in HJR14 still contains $(1 + v_+(V^*V)^{-1}v_+^*)^{-1/2}$ and the full old lower projection. The inverse-metric decomposition HJR16 in the full frame $[J, L_M]$ proves the two scalar energies; HJR18 gives their determinant increments. Telescoping with the original weights $w_M = 1$ at the extremes and 2 inside proves HJR32, including all zero-energy rows. Applying the full NSO same-fibre comparison to these exact restriction and quotient values now gives

$$\begin{aligned} -t_a \Omega_a^{[a],\text{hi}} &\leq S_{K,a}^{(a)} - S_{K,a}^{(1)} \leq t_a \Omega_a^{[a],\text{lo}}, \\ -j_{0,a} \Omega_a^{[a],\text{hi}} &\leq S_{R,a}^{(a)} - S_{R,a}^{(1)} \leq j_{0,a} \Omega_a^{[a],\text{lo}}. \end{aligned} \tag{HJR33}$$

The ranks belong to the final coefficient quotient and restriction; the full raw polynomial plane has not been assigned these ranks. The common mass is retained before the signed return, so these order differences are $O_{\text{actual}}(a^2 \log q_a) = o(aq_a)$.

At the actual target degree $a = j = k + 4$, the original fourfold quotient flag of CPQ2–8 further gives

$$\begin{aligned} 4S_{K,j}^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, \\ 4S_{R,j}^{(s_1)} &= 4\mathcal{T}_{j,s_1} - V_I - V_b - V_t - V_{\text{rem}}, \\ 4S_{K,j}^{(s_1)} &= \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A. \end{aligned} \tag{HJR34}$$

The first line is the full target Gram factorization with all its fixed frame determinants before the return; the second subtracts it from four copies of HJR32. For the third use $\mathcal{L}_e = V_I + V_t - V_s - V_A$. Every positive quotient has its literal four-row formula CPQ14–18. This transports the same original row allocation into the connecting and invariant calculation without assigning any individual large-row concentration or leading coefficient.

The common-angle interval returned to the two original allocations. At the same target degree $j = k + 4$, $k = 4l + 1 \geq 9$, retain the original two source orders and every quotient of HJR34. Put $Z_e = V_b + V_{\text{rem}} + V_s + V_A$. Each summand is its complete attained four-endpoint return: the respective spaces are $W/W_i, \mathcal{U}/T, S/K_c$ and I/K_i . The exact value maps and frames are CPQ4–8; in particular the source term V_A still uses the target order and its own fibre. Let L_-, L_+ be the explicit SFC9–10 expressions of the initial common determinant and three original positive row budgets. Define

$$\begin{aligned} K_-^{(s_1)} &= \max\{0, (L_- + Z_e)/4\}, \\ K_+^{(s_1)} &= \min\{\mathcal{T}_{j,s_1}, U_{j,s_1}^{\text{native}}, (L_+ + Z_e)/4\}, \\ K_-^{(s_1)} &\leq S_{K,j}^{(s_1)} \leq K_+^{(s_1)}, \quad \mathcal{T}_{j,s_1} - K_+^{(s_1)} \leq S_{R,j}^{(s_1)} \leq \mathcal{T}_{j,s_1} - K_-^{(s_1)}. \end{aligned} \tag{HJR35}$$

The U^{native} entry is used only on the original RQB35 domain and is omitted elsewhere. HJR34 gives the exact affine map $S_K = (\mathcal{L}_e + Z_e)/4$. Applying it to CIT35 proves the bounds before intersection. Positivity of both allocations, their exact total HJR32 and the applicable native

cap prove the stated intersection. Subtracting from that same total reverses the bounds and proves the residual interval. No summand of Z_e is removed. HJR33 then intersects the two order intervals with the asymmetric translations displayed and proved in SFC19. This gives the complete finite receiving calculation on the original rows, including their lower-Gram denominators, without choosing a new period or source order.

The refined comparison returned to both complete allocations. Keep the full $Z_e^p = V_b^p + V_{\text{rem}}^p + V_s^p + V_A^p$ in SFC18 and HJR35. Replace their common-angle endpoints by the proved EFR4 endpoints and define

$$\widehat{K}_-^p = \max\{0, (\widetilde{L}_-^p + Z_e^p)/4\}, \quad \widehat{K}_+^p = \min\{\mathcal{T}_j^p, U_{j,s_1}^{\text{native}}, (\widetilde{L}_+^p + Z_e^p)/4\}.$$

The original cap appears only on its original proved domain. The exact affine identity $4S_{K,j}^p = \mathcal{L}_e^p + Z_e^p$ proves this enclosure. The quotient $H_{B,j,D} = (B_j D_{j,D}^{-1} B_j^*)^{-1}$ has rank t_j , while the restriction $J_j^* D_{j,D} J_j$ has rank $j_{0,j}$. Their original inverse-metric return identities are HJR32. Applying EFP16 to both and using $\Delta S_K + \Delta S_R = \Delta \mathcal{T}_j$ gives

$$\begin{aligned} A_K^- &= \max\{t_j \Phi_j^-, \Delta \mathcal{T}_j - j_{0,j} \Phi_j^+\}, \\ A_K^+ &= \min\{t_j \Phi_j^+, \Delta \mathcal{T}_j - j_{0,j} \Phi_j^-\}. \end{aligned} \tag{EFR5}$$

Thus the complete replacement for the NSR-width translation in SFC19 is

$$\begin{aligned} \max\{\widehat{K}_-^1, \widehat{K}_-^0 + A_K^-\} &\leq S_{K,j}^1 \leq \min\{\widehat{K}_+^1, \widehat{K}_+^0 + A_K^+\}, \\ \max\{\widehat{K}_-^0, \widehat{K}_-^1 - A_K^+\} &\leq S_{K,j}^0 \leq \min\{\widehat{K}_+^0, \widehat{K}_+^1 - A_K^-\}. \end{aligned} \tag{EFR6}$$

Adding the actual difference interval to pair zero and subtracting it from pair one prove the two lines. Subtraction in reversed endpoint order from each same-pair $\mathcal{T}_j^p$ gives the residual intervals. Every low correction and all four positive returns remain; no individual allocation is prescribed.

74 The compatible high rows in the original receiving interval

Retain the full fixed-period simple-quartet hypotheses and all original source, coefficient and metric data in PBR and LRR. For each original source $s = 1, k$ the following uses exactly $B = B_K = Z_k^\top A_\ell$ and $J = J_0 = A_\ell^{-1} P_Y^\top$, with $\text{im } J = \ker B$ and ranks t_K, t_R of LRR1. In this section $D_M = \mathcal{D}_M^{(s)}$ is the full original dual metric; it is distinct from the original lower-grid polynomial χ' and from the residual-row matrix in PBR12.

The complete recurrence-compatible high-jet graph is $E_{N,s} = F_{N,s} R_s = \begin{pmatrix} I_n \\ H_{N,s} \end{pmatrix}$, where R_s is the original LJR4 reconstruction and $F_{N,s}$ is the full inverse-convolution and original Gamma coordinate map HJR3–4. HJR5 proves that its image is exactly the vectors whose convolved moments satisfy the monic χ recurrence. HJR6 proves $E_{N,s} V_{q-1,s} = V_{N,s}$, so the entire lower evaluation denominator is transported through the whole graph.

At a single original degree $q-1 \leq M < 2q$ use the full rows T, V through degree $M-v$ and the actual next rows t, v_+ . Their entries are the original functional values of $\mathcal{N}_a/E_A$ and the original lower-root evaluations on the degree- $M-v+1$ Gamma orthonormal polynomial. The exact full conditional update is

$$\begin{aligned} A_M &= V^* V, & K_M &= A_M^{-1} V^* T, & r_M &= t - v_+ K_M, & \lambda_M &= 1 + v_+ A_M^{-1} v_+^* > 0, \\ h_M &= \lambda_M^{-1/2} r_M, & D_{M+1} &= D_M + h_M^* h_M. \end{aligned} \quad (\text{HCR1})$$

Indeed subtracting the attained old extension minimum leaves the quadratic minimization $\eta^* A_M \eta + \|r_M x - v_+ \eta\|^2$. Its minimum is $\|r_M x\|^2 / (1 + v_+ A_M^{-1} v_+^*)$, by completing the square with $(A_M + v_+^* v_+)^{-1}$. The old lower coefficient is consequently retained throughout the complete row block; it is not minimized separately at each piece of the graph. This proves HCR1 with the full conditioning denominator and every original moment.

Put $D_{B,M} = (B D_M^{-1} B^*)^{-1}$, $D_{J,M} = J^* D_M J$, and $L_M = D_M^{-1} B^* D_{B,M}$. Then the exact inverse decomposition and two row energies are

$$\begin{aligned} D_M^{-1} &= J D_{J,M}^{-1} J^* + L_M D_{B,M}^{-1} L_M^*, \\ x_M &= h_M J D_{J,M}^{-1} J^* h_M^* \geq 0, & y_M &= h_M L_M D_{B,M}^{-1} L_M^* h_M^* \geq 0, \\ \log \frac{\det D_{J,M+1}}{\det D_{J,M}} &= \log(1 + x_M), \\ \log \frac{\det D_{B,M+1}}{\det D_{B,M}} &= \log \left(1 + \frac{y_M}{1 + x_M} \right). \end{aligned} \quad (\text{HCR2})$$

The columns $[J, L_M]$ form a full coefficient frame because $B L_M = I$ and $\text{im } J = \ker B$. Its D_M Gram is diagonal in these two blocks since $J^* D_M L_M = J^* B^* D_{B,M} = 0$. Inverting this exact frame Gram proves the first line. Restriction of HCR1 and the rank-one determinant lemma proves the third line. For the quotient, every lift of a vector z is $L_M z + J t$. Minimize its full updated norm over t . After using the original D_M orthogonal decomposition, the additional form is $(h_M L_M)^* (h_M L_M) / (1 + x_M)$. The determinant lemma proves the final line, including the denominator $1 + x_M$. Both x_M, y_M also retain λ_M^{-1} in HCR1.

Use the original weights $w_{q-1} = w_{2q-1} = 1$ and $w_M = 2$ for $q \leq M \leq 2q-2$, and define

$$\begin{aligned} \mathcal{V}_{K,k,s}^{\text{jet}} &= \sum_{M=q-1}^{2q-1} w_M \log \left(1 + \frac{y_M}{1 + x_M} \right), & \mathcal{V}_{0,k,s}^{\text{jet}} &= \sum_{M=q-1}^{2q-1} w_M \log(1 + x_M), \\ \mathcal{V}_{K,k,s}^{\text{jet}} &= A_{K,k,s} + P_{K,k,s}, & \mathcal{V}_{0,k,s}^{\text{jet}} &= A_{0,k,s} + P_{0,k,s}, \\ F_K^{(s)} &= \mathcal{V}_{K,k,s}^{\text{jet}} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} - F_{W_0}^{(s)}, \\ \Phi_{k,s} &= F_{W_0}^{(s)} + \mathcal{V}_{0,k,s}^{\text{jet}} + \delta_{0,k,s}. \end{aligned} \quad (\text{HCR3})$$

For each of the two metrics let $I_{X,N} = \log(\det D_{X,N} / \det D_{X,q-1})$. Summing the consecutive row identities in HCR2 gives $I_{X,N} = \sum_{M=q-1}^{N-1} g_{X,M}$. The expression $I_{X,2q-1} + I_{X,2q} - I_{X,q}$ therefore has precisely the weights displayed above, since the first row loses one copy and the last row occurs once. On the other hand the full dual identities $D_{B,N} = \widehat{H}_{K,N}^{-1}$ and $D_{J,N} = \overline{Q}_{0,N}^{-1}$ identify that weighted sum with the four original native returns. PBR5 then proves the second line. Equivalently HJR29–31 cancels the common low reconstruction scale and remainder in the four signs and gives $A_X = \mathcal{V}_X^{\text{jet}} - P_X$ exactly. Substitution into PBR9/7 proves the final two lines: the P_X terms cancel algebraically, rather than being estimated or set to zero. Both low terms remain separately, with their original metrics from PBR6.

The full finite errors are unchanged: $\mathcal{E}_s^{\text{flag}} \in [D_s^-, D_s^+]$ of PBR8 and $|\delta_{0,k,s}| \leq E_{k,s}^{\text{tr}}$ of PBR7, including α_A and all four CEP errors. Define the exact row-sum endpoints

$$\begin{aligned}
a_s^{\text{row},-} &= \mathcal{V}_{K,k,s}^{\text{jet}} - D_s^+ + \lambda_s^{\text{low}}, & a_s^{\text{row},+} &= \mathcal{V}_{K,k,s}^{\text{jet}} - D_s^- + \lambda_s^{\text{low}}, \\
b_s^{\text{row},-} &= F_{W_0}^{(s)} + \mathcal{V}_{0,k,s}^{\text{jet}} - E_{k,s}^{\text{tr}}, & b_s^{\text{row},+} &= F_{W_0}^{(s)} + \mathcal{V}_{0,k,s}^{\text{jet}} + E_{k,s}^{\text{tr}}, \\
a_s^{\text{row},-} &\leq F_K^{(s)} \leq a_s^{\text{row},+}, & b_s^{\text{row},-} &\leq \Phi_{k,s} \leq b_s^{\text{row},+}, \\
\Phi_{k,s}^{\text{row}} &= \max\{\Phi_{k,s}^{\text{V}}, b_s^{\text{row},-}\}, & \Phi_{k,s}^{+, \text{row}} &= \min\{\Phi_{k,s}^{+, \text{V}}, b_s^{\text{row},+}\}, \\
L_s^{\text{row}} &= \max\{L_s^{\text{V}}, a_s^{\text{row},-}, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{+, \text{row}} - e_{k,s}^{\text{V}}\}, \\
U_s^{\text{row}} &= \min\{U_s^{\text{V}}, a_s^{\text{row},+}, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{+, \text{row}} + e_{k,s}^{\text{V}}\}.
\end{aligned} \tag{HCR4}$$

Here $\lambda_s^{\text{low}} = F_L^{(s)} - F_{W_0}^{(s)}$ is the unchanged difference of the two separately retained volumes. The first two interval assertions follow immediately from the exact HCR3 equalities and the stated finite error bounds. Subtracting the resulting residual interval in the exact OAR equality proves the last two lines. Every entry bounds the same original scalar; hence the intersection retains all prior branches and is nonempty. On the exact PMR range $k \geq 29$, $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}(k+1)^2$, also include $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{+, \text{row}} - e_{k,s}^{0,-}$ in the maximum and $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{+, \text{row}} + e_{k,s}^{0,+}$ in the minimum, where $e^{0,\pm} = e^{\pm} + c_k^0$. Outside that range the original OAR entries remain. Because $0 \leq P_X \leq E_{k,s}^{\text{V}}$, the exact kernel row-sum interval lies inside PBR11's corresponding A_K interval, and its residual interval lies inside PBR11's corresponding A_0 interval. This containment follows by substituting $\mathcal{V}_X = A_X + P_X$ and comparing the two endpoints. It uses no distribution coefficient for x_M or y_M .

On the original arithmetic period domain $|u| \geq R_*$, retain precisely $a_{K,k}^{\pm}$ and all their finite constants from PBR14. The complete current arithmetic, signed and action receivers are

$$\begin{aligned}
L_{\text{ar}}^{\text{row}} &= \max\{L_{\text{ar}}^{\text{V}}, L_1^{\text{row}} + a_{K,k}^{\text{V}}\}, & U_{\text{ar}}^{\text{row}} &= \min\{U_{\text{ar}}^{\text{V}}, U_1^{\text{row}} + a_{K,k}^{\text{V}}\}, \\
F_K^{\text{ar}} &\in [L_{\text{ar}}^{\text{row}}, U_{\text{ar}}^{\text{row}}], & F_B^{\text{ar}} &\in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^{\text{row}}, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^{\text{row}}], \\
L_k^{\text{row}} + \mathcal{H}_k + \delta_k^{\sigma} &\leq \Psi_k \leq U_k^{\text{row}} + \mathcal{H}_k + \delta_k^{\sigma}, \\
L_J^{\text{row}} &= L_k^{\text{row}} + \mathcal{H}_k + \delta_k^{\sigma} + C_k^{\text{budget}}, & U_J^{\text{row}} &= U_k^{\text{row}} + \mathcal{H}_k + \delta_k^{\sigma} + C_k^{\text{budget}}, \\
4q \log \mathcal{L}_{h,k} &\leq J_k^{\text{action}} \in [L_J^{\text{row}}, U_J^{\text{row}}], & \mathcal{L}_{h,k} &\leq \min_M \epsilon_M^{\text{ar}} \leq \prod_M (\epsilon_M^{\text{ar}})^{w_M/(2q)} \leq e^{U_J^{\text{row}}/(4q)}.
\end{aligned} \tag{HCR5}$$

The arithmetic source difference is the identical correlated scalar used in PBR14; add its exact interval and intersect. Subtracting from the unchanged total reverses the boundary endpoints. The signed equality adds $\mathcal{H}_k = -T_k$ and the original arithmetic correction. Adding the complete $C_k^{\text{budget}} = F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+$ then gives the action interval. The original untilted reflection and the same-class lower radius give the weighted geometric mean exactly as in PBR15. Every mixed pairing, both contraction penalties, the full monic window and action leakage remain. Under a real tilt use the unchanged full MCE phase formula.

Finally the exact same-row allocation is

$$\begin{aligned}\mathcal{V}_{K,k,s}^{\text{jet}} + \mathcal{V}_{0,k,s}^{\text{jet}} &= \sum_{M=q-1}^{2q-1} w_M \log(1 + x_M + y_M), \\ F_K^{(s)} + \Phi_{k,s} &= \mathcal{V}_{K,k,s}^{\text{jet}} + \mathcal{V}_{0,k,s}^{\text{jet}} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} + \delta_{0,k,s}, \\ \mathcal{V}_{K,k,s}^{\text{jet}} + \mathcal{V}_{0,k,s}^{\text{jet}} &= \Delta q \mathfrak{C}_\partial + o(kq).\end{aligned}\tag{HCR6}$$

The first equality is $(1 + x_M)(1 + y_M/(1 + x_M)) = 1 + x_M + y_M$ for every actual row. The second adds the two HCR3 identities, cancelling the identical original W_0 term only in that sum. The final equality is HJR24–26 with every finite FEX, transfer and outer error retained before the fixed-period limit. The individual allocations are exactly the two specified sums in HCR3; no individual leading coefficient is assigned by the evaluated total.

74.1 Native multiplier return at the original compatible-row site

On the exact BIM/MNT domain $k \geq 37$, let $V_{Q,k,s}$, $n_{k,s}^{\text{mult}}$, $p_{k,s}^{\text{mult}}$ and $V_{\text{miss},k,s}$ be the complete original image quantities constructed and proved in NMR3–9. They use precisely the B, J, D_N and two source choices of HCR1–3. Their return to this earlier receiver is the exact pair of identities

$$\begin{aligned}F_K^{(s)} &= V_{Q,k,s} + n_{k,s}^{\text{mult}} + p_{k,s}^{\text{mult}} + V_{\text{miss},k,s} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} - F_{W_0}^{(s)}, \\ \Phi_{k,s} &= F_{W_0}^{(s)} + \mathcal{T}_{k,s} - V_{Q,k,s} - n_{k,s}^{\text{mult}} - p_{k,s}^{\text{mult}} - V_{\text{miss},k,s} + \delta_{0,k,s}, \\ -E_{k,s}^{\text{mult,lo}} &\leq p_{k,s}^{\text{mult}} \leq E_{k,s}^{\text{mult,hi}}, \quad 0 \leq V_{\text{miss},k,s} \leq 2\varepsilon_k \log A_{k,s}^{\text{end}}, \quad \varepsilon_k \leq \sum_{h \in \mathcal{H}} (v - t_{0,k-d_h}) - v.\end{aligned}\tag{HCR7}$$

For proof, NMR5 factors the full original source direct sum by the complete kernel of its onto map to the actual multiplier image. NMR7 retains both signed spectral parts of the relative identity from that attained source quotient to the native image. NMR8 factors the original B quotient by this image and its entire missing quotient, and proves the displayed finite bound. The frame determinants cancel only in the four original signs, giving $\mathcal{V}_{K,k,s}^{\text{jet}} = V_Q + n^{\text{mult}} + p^{\text{mult}} + V_{\text{miss}}$. Substitution into HCR3, followed by $\mathcal{V}_0^{\text{jet}} = \mathcal{T} - \mathcal{V}_K^{\text{jet}}$, proves both identities here, with every low and transfer term. NMR6 gives the complete V_Q source-kernel and displaced-window expression; none of those terms is suppressed by this substitution. NMR10 also returns all four original branch contrasts with their actual unmatched ranks for every $k \geq 9$. NMR11–15 proves their simultaneous finite intervals and carries the same unequal flag, CEP, arithmetic and complete action errors through HCR4–5. The proofs of NMR1–16 use HCR1–6 and SAR1–16, so this earlier-site insertion introduces no circular premise.

74.2 The original low metrics after native window and polar transport

Use the NMR18/WNR proved interval $[s_{K,s}^-, s_{K,s}^+]$, intersected with every previous bound on that same native scalar. The original full low-metric identities then give

$$\begin{aligned}s_{K,s}^- - D_s^+ + F_L^{(s)} - F_{W_0}^{(s)} &\leq F_K^{(s)} \leq s_{K,s}^+ - D_s^- + F_L^{(s)} - F_{W_0}^{(s)}, \\ F_{W_0}^{(s)} + \mathcal{T}_{k,s} - s_{K,s}^+ - E_{k,s}^{\text{tr}} &\leq \Phi_{k,s} \leq F_{W_0}^{(s)} + \mathcal{T}_{k,s} - s_{K,s}^- + E_{k,s}^{\text{tr}}.\end{aligned}\tag{HCR8}$$

Indeed subtract the full original factorial flag interval $[D_s^-, D_s^+]$ in the first identity. In the second use $S_R = \mathcal{T} - S_K$ with reversed endpoints and add the same full source allowance. PLF's explicit pole frame, restriction to the low annihilator, full Schur complement and ordinary complex-linear dual convention prove that this second identity is exactly the whole rank- j pole determinant on its proved full-width period domain. Its native source transport is the existing E^{tr} , so it is counted once. NBR21–23 then sends these same bounds through the original outer, arithmetic, signed and complete action formulas.

75 Sharp bounds from the actual annihilator components

Retain all fixed original data, source orders, matrices and domains of HJR1–31. No period, coefficient or Gamma norm is changed. For each original added row $q - 1 \leq M \leq 2q - 1$, abbreviate the complete HJR14 quantities as $r = r_M$, $\lambda = \lambda_M$ and $D = \mathcal{D}_M^{(s)}$. Keep the exact maps $B = B_K$ and $J = J_0$ with $\text{im } J = \ker B$. Put

$$\begin{aligned} D_J &= J^* D J, & D_B &= (B D^{-1} B^*)^{-1}, & L &= D^{-1} B^* D_B, \\ Z_M &= r D^{-1} r^*, & X_M &= r J (D_J)^{-1} J^* r^*, & Y_M &= r L (D_B)^{-1} L^* r^*. \end{aligned} \quad (\text{ACB1})$$

The letters X_M, Y_M, Z_M in this section denote these nonnegative real scalars, not a new source or coefficient frame. HJR16 proves exactly $Z_M = X_M + Y_M$ and HJR17 gives $x_M = X_M/\lambda_M$, $y_M = Y_M/\lambda_M$. Thus every original lower-evaluation factor remains in

$$\begin{aligned} g_M &= \log(1 + Z_M/\lambda_M), \\ g_{K,M} &= \log \frac{\lambda_M + Z_M}{\lambda_M + X_M}, & g_{0,M} &= \log(1 + X_M/\lambda_M), & g_M &= g_{K,M} + g_{0,M}. \end{aligned} \quad (\text{ACB2})$$

All these equalities include rows of rank zero.

75.1 The exact native metric component

If $Z_M > 0$, define

$$a_M = \frac{D^{-1/2} r^*}{\sqrt{Z_M}}, \quad P_{J,M} = D^{1/2} J (D_J)^{-1} J^* D^{1/2}, \quad \theta_M = \frac{X_M}{Z_M} = \|P_{J,M} a_M\|_2^2. \quad (\text{ACB3})$$

The map $r \mapsto D^{-1/2} r^*$ is the conjugate-linear isometry from the complex-linear dual with squared norm $r D^{-1} r^*$ into Euclidean column coordinates. The adjoint-column operation is explicitly part of this map; no new norm is assigned to r , J , or their domains. Indeed $\|a_M\| = 1$ and $P_{J,M}^* = P_{J,M}$, $P_{J,M}^2 = P_{J,M}$, by direct multiplication using $D_J = J^* D J$. Its image is exactly $D^{1/2} \text{im } J$. Evaluating its quadratic form on a_M proves the last equality in ACB3 and $0 \leq \theta_M \leq 1$. The scalar λ_M cancels in this component ratio, while it remains in g_M in ACB2.

For $Z_M > 0$ set $\ell_M = -\log \theta_M$, interpreting $\ell_M = +\infty$ when $X_M = 0$. For $Z_M = 0$ set $\theta_M = 1$, $\ell_M = 0$; then $r_M = 0$ since $D^{-1} \succ 0$, and all three energies in ACB2 are zero. For a nonzero row the exact identity is

$$g_{K,M} = -\log(\theta_M + (1 - \theta_M)e^{-g_M}). \quad (\text{ACB4})$$

To verify it, use $e^{-g_M} = \lambda_M/(\lambda_M + Z_M)$. The quantity inside the logarithm is then exactly $(\lambda_M + X_M)/(\lambda_M + Z_M)$. At $\theta_M = 0$ this gives $g_{K,M} = g_M$. At $\theta_M = 1$ it gives zero. Thus the edge cases are part of the same proved identity.

For every row the following finite bounds hold:

$$\begin{aligned} 0 &\leq \min\{g_M, \ell_M\} - g_{K,M} \leq \log 2, \\ 0 &\leq g_{0,M} - (g_M - \ell_M)_+ \leq \log 2. \end{aligned} \quad (\text{ACB5})$$

Here $(g - \infty)_+ = 0$. For a nonzero row, the positive expression in ACB4 satisfies

$$\max\{\theta_M, e^{-g_M}\} \leq \theta_M + (1 - \theta_M)e^{-g_M} \leq \theta_M + e^{-g_M} \leq 2 \max\{\theta_M, e^{-g_M}\}.$$

Taking its negative logarithm proves the first inequality. The second follows from $g_0 = g - g_K$ and $g - \min(g, \ell) = (g - \ell)_+$. For a zero row both differences are zero by definition.

75.2 The weighted aggregate and its joint tail

Keep the original weights w_M from HJR22. There are two endpoint weights one and $q - 1$ interior weights two, so $\sum_{M=q-1}^{2q-1} w_M = 2q$. Define the two actual finite quantities

$$\mathcal{A}_{k,s}^{\text{clip}} = \sum_{M=q-1}^{2q-1} w_M \min\{g_M, \ell_M\}, \quad \mathcal{B}_{k,s}^{\text{clip}} = \sum_{M=q-1}^{2q-1} w_M (g_M - \ell_M)_+. \quad (\text{ACB6})$$

They sum exactly to $\mathcal{T}_{k,s}$ of HJR21. Summing ACB5 with its original positive weights proves

$$\begin{aligned} \mathcal{A}_{k,s}^{\text{clip}} - 2q \log 2 &\leq \widehat{F}_k^{(s)} \leq \mathcal{A}_{k,s}^{\text{clip}}, \\ \mathcal{B}_{k,s}^{\text{clip}} &\leq F_{0,k,s} \leq \mathcal{B}_{k,s}^{\text{clip}} + 2q \log 2. \end{aligned} \quad (\text{ACB7})$$

In particular the error is $o(kq)$ for the fixed actual period. Combining with HJR30 also gives $\mathcal{A}^{\text{clip}} - 2q \log 2 - E^\vee \leq A_{K,k,s} \leq \mathcal{A}^{\text{clip}}$, with the complete original positive spectral allowance $E^\vee$ retained. The original arithmetic and low-subspace comparisons remain the exact FEX and PEC formulas when these bounds are transported.

For every finite $g \geq 0$ and $\ell \in [0, \infty]$, $\min(g, \ell) = \int_0^\infty \mathbf{1}_{\{g > t, \ell > t\}} dt$. All row energies are finite, so summing this exact identity yields

$$\mathcal{A}_{k,s}^{\text{clip}} = \int_0^\infty \sum_{M=q-1}^{2q-1} w_M \mathbf{1}_{\{g_M > t, \ell_M > t\}} dt. \quad (\text{ACB8})$$

HJR23,25 bounds each g_M by $U_{k,s}^{\text{jet}}$, because it is one nonnegative term of the weighted sum bounded there and $w_M \geq 1$. Thus the integral is already supported in $0 \leq t \leq U_{k,s}^{\text{jet}}$. The exact substitution $t = ka$ gives

$$\frac{\mathcal{A}_{k,s}^{\text{clip}}}{kq} = \int_0^{U_{k,s}^{\text{jet}}/k} \frac{1}{q} \sum_{M=q-1}^{2q-1} w_M \mathbf{1}_{\{g_M > ka, \theta_M < e^{-ka}\}} da. \quad (\text{ACB9})$$

Rows with $Z_M = 0$ contribute zero, and rows with $X_M = 0 < Z_M$ contribute exactly their allowed interval $0 < a < g_M/k$. No bound on the number or location of large rows has been substituted for their weighted energy. For every $a > 0$ the elementary finite estimate $\sum_{g_M > ka} w_M \leq U_{k,s}^{\text{jet}}/(ka)$ follows by summing $w_M g_M > ka w_M$ on precisely that set. The integral identity, rather than that row-count estimate alone, retains the entire possible concentration.

75.3 Bounds from the actual native coefficient columns

If $\text{rank } J > 0$, the component X_M has the exact variational formula

$$X_M = \sup_{c \neq 0} \frac{|r_M J c|^2}{c^* D_J c}. \quad (\text{ACB10})$$

To prove it, put $b = D_J^{1/2} c$. The quotient becomes $|r_M J D_J^{-1/2} b|^2 / \|b\|^2$, whose supremum is the squared Euclidean norm of its row, by Cauchy–Schwarz. Equality is attained on its adjoint whenever that row is nonzero; otherwise both sides are zero. If $\text{rank } J = 0$, the component is exactly $X_M = 0$ and no nonzero certificate vector exists. Thus any actual invariant-certificate column in $\text{im } J$ supplies the concrete bound

$$\theta_M \geq \frac{|r_M J c|^2}{(c^* D_J c) Z_M} \quad (Z_M > 0, c \neq 0). \quad (\text{ACB11})$$

Every scalar in this inequality is an original native pairing or norm. The numerator is the pairing with the same globally corrected row $r_M = t - v_+ K_M$. The denominator is the complete original

$D_J = \overline{Q_{0,M}^{-1}}$ form, not a coefficient Euclidean substitute. This is the exact metric map that a period/invariant-section calculation must use.

There is also a finite maximum bound with subleading aggregate loss that retains this metric explicitly. Let $j_0 = \text{rank } J$. For $j_0 > 0$ let R_M be the unique upper triangular Cholesky factor with positive real diagonal satisfying $D_J = R_M^* R_M$. Its entries are explicitly obtained in order from

$$(R_M)_{aa} = \left((D_J)_{aa} - \sum_{h < a} |(R_M)_{ha}|^2 \right)^{1/2}, \quad (\text{ACB12})$$

$$(R_M)_{ab} = \frac{(D_J)_{ab} - \sum_{h < a} \overline{(R_M)_{ha}} (R_M)_{hb}}{(R_M)_{aa}} \quad (a < b).$$

Each diagonal radicand is the positive Schur complement of the preceding principal block, since $D_J \succ 0$. Induction therefore proves existence, all stated nonzero denominators, the matrix identity and uniqueness. Define the actual native pairing row $b_M = r_M J R_M^{-1}$ and let $A_M^{\max} = \max_{1 \leq a \leq j_0} |(b_M)_a|^2$. The explicit coefficient map $J R_M^{-1}$ satisfies $(J R_M^{-1})^* D (J R_M^{-1}) = I_{j_0}$, so ACB1 gives

$$X_M = \sum_{a=1}^{j_0} |(b_M)_a|^2, \quad A_M^{\max} \leq X_M \leq j_0 A_M^{\max}. \quad (\text{ACB13})$$

This changes the coordinate frame by its proved inverse matrix; it leaves the native norm and all actual pairings fixed.

For every row, ACB2 and ACB13 give the fully finite interval

$$\max \left\{ 0, \log \frac{\lambda_M + Z_M}{\lambda_M + j_0 A_M^{\max}} \right\} \leq g_{K,M} \leq \log \frac{\lambda_M + Z_M}{\lambda_M + A_M^{\max}}. \quad (\text{ACB14})$$

The width is at most $\log j_0$ when $j_0 \geq 1$: before the lower endpoint is clipped at zero, the difference is $\log[(\lambda + j_0 A^{\max})/(\lambda + A^{\max})] \leq \log j_0$; clipping can only decrease it. At $A_M^{\max} = 0$, $X_M = 0$ and both bounds equal g_M . At a zero-energy row both bounds are zero. If $j_0 = 0$, the annihilator is zero and the exact value $g_{K,M} = g_M$ replaces the interval.

Let $\mathbf{u}_M$ be the upper endpoint in ACB14 when $j_0 > 0$, and set $\mathbf{u}_M = g_M$ when $j_0 = 0$. Summing with the same original weights proves precisely

$$0 \leq \sum_{M=q-1}^{2q-1} w_M \mathbf{u}_M - \widehat{F}_k^{(s)} \leq 2q \log(\max\{1, j_0\}) = O(q \log(k+1)) = o(kq). \quad (\text{ACB15})$$

Here $j_0 = j - v + t_0 = O(k)$ is the actual original rank. Each upper endpoint requires only the largest of the stated native pairings, after the full original Cholesky matrix and the lower-evaluation denominator have been computed. There is no unknown condition number in ACB14: the complete matrices specifying those pairings are HJR2–4, HJR14 and ACB12.

The bounds ACB7–15 provide a sharp aggregate target and its exact original component maps. A quantitative estimate from the literal period or invariant module must bound these actual pairings or their joint tail; no assertion about an unweighted average direction, isotropy, a moving period, or the separate leading kernel coefficient is made.

76 A full-form comparison for the original allocation

Retain the actual simple quartet, fixed original period, all conductor coefficients and both original source orders $s = 1, k$. Write $k = 4l + 1$, $q = (k + 1)^2$, $q' = (k - 7)^2$ and $\psi = \chi_{k-8}$. The original lower center remains $c' = k/2 - 4$. For every original endpoint $N \in \{q - 1, q, 2q - 1, 2q\}$ put

$$a_N = N - v - q', \quad \ell_s = (s - 1)/4, \quad Q_s = q' + \ell_s, \quad \vartheta = \pi/2, \quad h = 1/q. \quad (\text{GPA1})$$

Thus $a_N + 1$ is the actual ambient lower polynomial dimension before taking its original relation quotient. Use the proved OSP finite domain

$$\epsilon = 2^{-32}, \quad k \geq 29, \quad R = k\sqrt{\delta^2 + \gamma^2} \leq \epsilon q/2, \quad T = \epsilon q. \quad (\text{GPA2})$$

It holds eventually for each fixed actual quartet. In this domain $q' \geq q/2$, $a_N \leq 2q$ and $Q_s \leq 2q$. All original arithmetic applications continue to retain $|u| \geq R_*$.

76.1 A polynomial dilation bound with its exact coefficient map

For an integer $Q \geq 0$ and a real $\kappa > 0$ define, on the unchanged polynomial coefficient space of degree at most a ,

$$\|p\|_{Q,\kappa}^2 = \int_{\mathbb{R}} |p(y)|^2 |y|^{2Q} e^{-\kappa|y|} dy, \quad Ap = y p'(y), \quad L_{a,Q} = a + \sqrt{a(a+2Q)}. \quad (\text{GPA3})$$

Then

$$\|A\|_{\leq a,Q,\kappa} \leq L_{a,Q}, \quad e^{-L_{a,Q}|t|} \|p\|_{Q,\kappa} \leq \|p(e^t \cdot)\|_{Q,\kappa} \leq e^{L_{a,Q}|t|} \|p\|_{Q,\kappa}. \quad (\text{GPA4})$$

The finite dilation is exactly $\exp(tA)$, whose monomial coefficient matrix is $\text{diag}(1, e^t, \dots, e^{at})$. This explicit source map, with inverse at $-t$, is used only to bound the same original coefficient directions.

Here is a full proof of the operator estimate. On the positive half-line set $x = \kappa y$ and $\alpha = 2Q$. Rodrigues' polynomials

$$L_n^{(\alpha)}(x) = \frac{x^{-\alpha} e^x}{n!} \left(\frac{d}{dx} \right)^n (e^{-x} x^{n+\alpha}) = \sum_{j=0}^n \frac{(-1)^j}{j!} \binom{n+\alpha}{n-j} x^j \quad (\text{GPA5})$$

have leading coefficient $(-1)^n/n!$. Integration by parts n times proves orthogonality to every lower degree in the weight $x^\alpha e^{-x}$: all boundary terms vanish, since the smallest power left at zero before the final integration is at least $\alpha + 1$, and the exponential controls infinity. The same computation against $L_n^{(\alpha)}$ gives

$$\int_0^\infty [L_n^{(\alpha)}(x)]^2 x^\alpha e^{-x} dx = \frac{\Gamma(n+\alpha+1)}{n!}. \quad (\text{GPA6})$$

Indeed the n th derivative of $L_n^{(\alpha)}$ is $(-1)^n$, and the remaining integral is the displayed Gamma integral. The coefficient formula in GPA5 also gives, term by term,

$$x(L_n^{(\alpha)})' = nL_n^{(\alpha)} - (n+\alpha)L_{n-1}^{(\alpha)}. \quad (\text{GPA7})$$

For example equality of its coefficient at x^j is the binomial identity $(n-j)\binom{n+\alpha}{n-j} = (n+\alpha)\binom{n+\alpha-1}{n-j-1}$. The term with $n = 0$ is zero. After dividing each polynomial by the square root of GPA6, this operator is the sum of a diagonal matrix with entries n and a one-step shift with entries $-\sqrt{n(n+\alpha)}$. Their norms are at most a and $\sqrt{a(a+\alpha)}$. This proves the first inequality on the positive half-line. The pullback $y = x/\kappa$ has the common squared-norm factor $\kappa^{-\alpha-1}$ and carries $y\partial_y$ to $x\partial_x$, so the same bound holds at every κ .

On the negative half-line apply the identical estimate to $p(-y)$ with $y > 0$. The identity

$$y(d/dy)p(-y) = -yp'(-y)$$

is the pullback of $y_{\text{original}}p'(y_{\text{original}})$ there. Adding the two squared estimates proves GPA4's operator bound for the entire real-line norm, on all polynomials; no parity restriction is needed. The exponential series of the finite matrix gives $\|\exp(tA)\| \leq \exp(|t|L_{a,Q})$. Applying the same estimate to its inverse proves the second side of the dilation inequality.

For the exact rate change put $u = \kappa/\vartheta$. The substitution $x = uy$ gives

$$\|p\|_{Q,\kappa}^2 = u^{-(2Q+1)} \|p(u^{-1} \cdot)\|_{Q,\vartheta}^2. \quad (\text{GPA8})$$

Consequently, with $K_{a,Q} = 2Q + 1 + 2L_{a,Q}$,

$$e^{-K_{a,Q}|\log(\kappa/\vartheta)|} \|p\|_{Q,\vartheta}^2 \leq \|p\|_{Q,\kappa}^2 \leq e^{K_{a,Q}|\log(\kappa/\vartheta)|} \|p\|_{Q,\vartheta}^2. \quad (\text{GPA9})$$

The signed scalar $u^{-(2Q+1)}$ in GPA8 was retained before taking this two-sided bound.

76.2 Full original Gamma and root comparisons

Keep the exact original source measure, its mass and recurrence

$$\begin{aligned} dm_s(y) &= M_s \frac{2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, & M_s &= (2\pi)^{s/2}, \\ dm_s(y) &= \beta_s \prod_{j=0}^{\ell_s-1} (y^2 + (2j + \tfrac{1}{2})^2) d\sigma(y), & \beta_s &= \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \\ d\sigma(y) &= \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \end{aligned} \quad (\text{GPA10})$$

In the original center, $\psi(c' + iy) = i^{q'} P_{k-8}(y)$, where P_{k-8} is the complete monic lower-root polynomial OSP3. Every original root and its phase are retained by this map. For a polynomial p in the original S' coordinate set $p_c(y) = p(c' + iy)$.

Use the actual finite constants from the proved complete coefficient-space estimates OSP19–26 and OFV6–7:

$$\begin{aligned} C_L &= \pi/\Gamma(3/4)^2, & K_q &= \frac{\sqrt{2\pi}}{C_L} \frac{(2q+1)^2}{q} e^{-2q}, \\ \mathcal{E}_k &= \frac{qR^2}{\epsilon^2 q^2 - R^2} + \log(1 + K_q), \\ r_{\ell_s} &= \frac{\sum_{j=0}^{\ell_s-1} (2j + \tfrac{1}{2})^2}{\epsilon^2 q^2} + \log(1 + K_q) \quad (\ell_s > 0), & r_0 &= 0. \end{aligned} \quad (\text{GPA11})$$

Then, for every $\deg p \leq a_N$, their full quadratic-form inequalities give

$$\begin{aligned} e^{-\mathcal{E}_k} \beta_s \int |p_c(y)|^2 |y|^{2Q_s} d\sigma(y) &\leq \int |p(c' + iy)|^2 |\psi(c' + iy)|^2 dm_s(y) \\ &\leq e^{\mathcal{E}_k + r_{\ell_s}} \beta_s \int |p_c(y)|^2 |y|^{2Q_s} d\sigma(y). \end{aligned} \quad (\text{GPA12})$$

For clarity, these are whole-space estimates. Pairing the actual roots $\omega, -\omega$ gives the outside-region log ratio at most $q'R^2/(T^2 - R^2) \leq \mathcal{E}_k$. On $|y| < T$ the complete numerator is bounded by $(T^2 + R^2)^{q'} |p_c(y)|^2$. The polynomial estimate OSP15 on all degrees at most $2q$, and the original density lower bound on $[q, 2q]$, bound its integral by K_q times the outside reference norm. This proves both original-root inequalities with their full inside contribution. For the source recurrence, each positive factor $y^2 + (2j + 1/2)^2$ is at least y^2 . Outside $[-T, T]$ their full log ratio to $y^{2\ell_s}$ is at most the sum in GPA11 divided by T^2 . The same inner-region argument gives its additional $\log(1 + K_q)$; at $\ell_s = 0$ the identity is exact. These are precisely the proofs of the full-form OSP20 and OFV7 estimates, with their unchanged finite constants, not an inference from their determinant consequences. Their hypotheses apply because $q' \geq q/2$, $a_N \leq 2q$, and every $2j + 1/2 \leq k/2 \leq R \leq T/2$.

The complete global Gamma-to-exponential sandwich is

$$L_h e^{-(\vartheta+h)|y|} \leq \frac{d\sigma}{dy}(y) \leq U_h e^{-(\vartheta-h)|y|}, \quad L_h = C_L (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}. \quad (\text{GPA13})$$

For the upper side, rotate the Euler integral for $\Gamma(1/4 + iy/2)$ through angle $\pi/2 - h$ and take its absolute value. For the lower side apply the same rotation to $\Gamma(3/4 - iy/2)$ and divide the exact

reflection product $2\pi^2/\cosh(\pi y)$ by that upper bound. The inequality $\cosh(\pi y) \leq e^{\pi|y|}$ then gives the stated L_h . These arguments prove GPA13 on the whole line, including zero.

Define the explicit two logarithmic constants

$$\begin{aligned} b_{N,s}^- &= -\mathcal{E}_k + \log L_h - K_{a_N, Q_s} \log(1 + h/\vartheta), \\ b_{N,s}^+ &= \mathcal{E}_k + r_{\ell_s} + \log U_h + K_{a_N, Q_s} \log \frac{1}{1 - h/\vartheta}. \end{aligned} \quad (\text{GPA14})$$

Applying GPA9 to the two rates in GPA13 and then GPA12 proves, on the entire original coefficient space,

$$\boxed{\beta_s e^{b_{N,s}^-} \|p_c\|_{Q_s, \vartheta}^2 \leq \int |p(c' + iy)|^2 |\psi(c' + iy)|^2 dm_s(y) \leq \beta_s e^{b_{N,s}^+} \|p_c\|_{Q_s, \vartheta}^2.} \quad (\text{GPA15})$$

This estimate preserves all coefficient directions. Moreover $b_{N,s}^\pm = O_{\text{actual}}(\log q)$, uniformly over the four endpoints and both original orders. Indeed $K_{a_N, Q_s} \leq 16q + 1$ by $a_N, Q_s \leq 2q$; $|\log(1 \pm h/\vartheta)| \leq h/(\vartheta - h)$; $2h/\pi \leq \sin h \leq h$. Finally $\mathcal{E}_k = O_{\text{actual}}(1)$ and $r_{\ell_s} = O(1/k)$ from GPA11. The full β_s has not been included in this logarithmic error and will cancel only in the stated four-endpoint ratios.

76.3 The same actual relations in an explicit moment matrix

In the original increasing S' monomials the exact coordinate pullback is

$$(P_{c',a})_{j,n} = \binom{n}{j} (c')^{n-j} i^j \quad (j \leq n), \quad p_c(y) = \sum_j (P_{c',a}[p])_j y^j. \quad (\text{GPA16})$$

The inverse substitutes $y = (S' - c')/i$; its diagonal entries are i^{-j} . The explicit full moment matrix for the comparison form is

$$\begin{aligned} (M_{Q,a})_{i,j} &= \begin{cases} 2\Gamma(2Q + i + j + 1)/\vartheta^{2Q+i+j+1}, & i + j \text{ even}, \\ 0, & i + j \text{ odd}, \end{cases} \\ H_{N,s}^{\text{exp}} &= P_{c',a_N}^* M_{Q_s,a_N} P_{c',a_N}. \end{aligned} \quad (\text{GPA17})$$

The entries follow by integrating separately over the positive and negative half-lines; each positive-half-line moment is the Gamma integral after $x = \vartheta y$. All arguments $2Q_s + i + j + 1$ are positive integers, so the Gamma factors in this matrix are exact factorials. Its positivity follows because a nonzero polynomial has positive squared integral in this positive measure. The original source mass remains in β_s in GPA15.

Let U_N have the complete original columns $\alpha_A U_N^A(S^j)$, $0 \leq j \leq N - q$, in the increasing S' coefficient frame, with $\alpha_A = v!/\mu_v$ unchanged. Let $E_{\mathbf{m}}$ insert the low $\mathbf{m} = \Delta - v$ monomials. For the displayed form define the complete quotient

$$H_{Y,N}^{\text{exp}} = E_{\mathbf{m}}^* [H_{N,s}^{\text{exp}} - H_{N,s}^{\text{exp}} U_N (U_N^* H_{N,s}^{\text{exp}} U_N)^{-1} U_N^* H_{N,s}^{\text{exp}}] E_{\mathbf{m}}. \quad (\text{GPA18})$$

At $N = q - 1$ the relation list is empty and the second term is zero. Each original relation has nonzero leading coefficient and degree $\mathbf{m} + j$, so these columns are independent and the low quotient frame is valid at every endpoint, as proved in FGC10. Thus all inverses in GPA18 exist. No relation coefficient was replaced by a monomial ideal or by its leading term.

Keep the actual Z_k and P_Y from FGC9 and PEC24 and set

$$\hat{H}_{K,N}^{\text{exp}} = Z_k^* H_{Y,N}^{\text{exp}} Z_k, \quad Q_{0,N}^{\text{exp}} = [P_Y (H_{Y,N}^{\text{exp}})^{-1} P_Y^*]^{-1}. \quad (\text{GPA19})$$

Their ranks are exactly $t_K = r - t_0$ and $j_0 = j - v + t_0$, respectively, where $t_0 = \dim(K \cap L)$. Zero ranks use the unique empty matrices and determinant one. The original native forms are

the identical two operations on $H_{Y,N}^{(s)}$. Taking minima over the same relation fibres in GPA15 and then restricting or taking the same P_Y quotient proves, for each of these three full forms H_X ,

$$\beta_s e^{b_{N,s}^-} H_{X,N}^{\exp} \preceq H_{X,N}^{(s)} \preceq \beta_s e^{b_{N,s}^+} H_{X,N}^{\exp}. \quad (\text{GPA20})$$

For restriction this follows by substituting the same columns in each quadratic form. For a quotient, every candidate lift has the two inequalities; minimizing over exactly the same affine fibre proves both sides. This is the full metric transport and preserves every Schur mixed term in GPA18–19.

76.4 A subleading error for each original allocation

For $X = K, 0$ let t_X be its actual rank and put

$$F_X^{\exp} = \sum_N \sigma_N \log \det H_{X,N}^{\exp}, \quad (\sigma_{q-1}, \sigma_q, \sigma_{2q-1}, \sigma_{2q}) = (+1, +1, -1, -1). \quad (\text{GPA21})$$

In this notation $H_K^{\exp} = \hat{H}_K^{\exp}$ and $H_0^{\exp} = Q_0^{\exp}$. The native quantities are exactly $F_K^{\text{native}} = \hat{F}_k^{(s)}$ and $F_0^{\text{native}} = F_{0,k,s}$, namely the two weighted original allocations in HJR21. Define the two explicit finite endpoints

$$\begin{aligned} d_{X,k,s}^- &= t_X (b_{q-1,s}^- + b_{q,s}^- - b_{2q-1,s}^+ - b_{2q,s}^+), \\ d_{X,k,s}^+ &= t_X (b_{q-1,s}^+ + b_{q,s}^+ - b_{2q-1,s}^- - b_{2q,s}^-). \end{aligned} \quad (\text{GPA22})$$

Then GPA20 proves

$$\boxed{d_{X,k,s}^- \leq F_X^{\text{native}} - F_X^{\exp} \leq d_{X,k,s}^+, \quad d_{X,k,s}^\pm = O_{\text{actual}}(k \log q) = o(kq).} \quad (\text{GPA23})$$

To see the determinant implication, conjugate GPA20 by the positive square root of its comparison inverse. All resulting eigenvalues lie between $\beta_s e^{b^-}$ and $\beta_s e^{b^+}$. Their t_X logarithms give the determinant bounds at each endpoint. The term $t_X \log \beta_s$ cancels exactly because the same original source and rank are used at all four endpoints and $\sum_N \sigma_N = 0$. The low endpoints use their lower bound in the left side and their upper bound in the right; the high endpoints use the opposite bounds. This proves every sign in GPA22. Finally $t_X = O(k)$ and GPA15 proves the claimed order.

Thus the actual weighted sums $\sum_M w_M \log(1 + y_M/(1 + x_M))$ and $\sum_M w_M \log(1 + x_M)$ can each be calculated in the explicit factorial moment matrices GPA17–19 to an error $O_{\text{actual}}(k \log q)$, keeping the same full period/conductor relation and the same actual kernel and annihilator coefficients. The two exponents remain $Q_1 = q'$ and $Q_k = q' + (k-1)/4$; they have not been identified. This full-form comparison evaluates the analytic cost of that transfer. It does not assign the still unknown separate leading allocation from the rank alone.

77 Exact mixed half-line volumes for the original allocation

Fix all original simple-quartet, period, branch and conductor data of GPA1–2. The source order is precisely $s = 1$ or $s = k$. Put $\alpha = 2Q_s$, $\vartheta = \pi/2$ and, at each original endpoint,

$$\mathfrak{m} = q - q' - v, \quad p_N = N - q + 1, \quad a_N = p_N + \mathfrak{m} - 1, \quad n_N = a_N + 1. \quad (\text{LVM1})$$

Thus $p_N \geq 0$ and $n_N = p_N + \mathfrak{m}$. The original four endpoints and signs are $(q-1, q, 2q-1, 2q)$ and $(1, 1, -1, -1)$. All original S' coordinates remain centered at $c' = k/2 - 4$. No coefficient of the relation matrix is changed.

77.1 Both half-lines in one explicit coefficient map

For $0 \leq j \leq a$ define

$$\lambda_j(y) = \vartheta^{(\alpha+1)/2} \sqrt{\frac{j!}{\Gamma(j+\alpha+1)}} L_j^{(\alpha)}(\vartheta y). \quad (\text{LVM2})$$

Rodrigues' formula GPA5, with its complete integration-by-parts proof GPA6, proves that these are an orthonormal basis on $(0, \infty)$ for $y^\alpha e^{-\vartheta y} dy$. Let $A_{\alpha,a}$ take monomial coefficients of $p(y)$ to its coefficients in this orthonormal basis. Its exact entries are

$$(A_{\alpha,a})_{j,r} = \begin{cases} (-1)^j \binom{r}{j} \Gamma(r+\alpha+1) \sqrt{\frac{j!}{\Gamma(j+\alpha+1)}} \vartheta^{-r-(\alpha+1)/2}, & j \leq r, \\ 0, & j > r. \end{cases} \quad (\text{LVM3})$$

Indeed the coefficient is the integral of $y^r \lambda_j(y)$ in that weight. Substituting $x = \vartheta y$ and integrating Rodrigues' derivative j times gives $(-1)^j \binom{r}{j} \Gamma(r+\alpha+1)$. If $r < j$ that derivative is zero. This proves every entry, including its sign and its source mass factor. The diagonal is $(-1)^j \sqrt{j! \Gamma(j+\alpha+1)} \vartheta^{-j-(\alpha+1)/2} \neq 0$; hence the map is invertible.

Let $J_a = \text{diag}((-1)^r)_{r=0}^a$. The reflection in orthonormal coordinates is

$$R_{\alpha,a} = A_{\alpha,a} J_a A_{\alpha,a}^{-1}, \quad (R_{\alpha,a})_{j,r} = \begin{cases} (-1)^j 2^{r-j} \binom{\alpha+r}{r-j} \sqrt{\frac{\Gamma(j+\alpha+1)/j!}{\Gamma(r+\alpha+1)/r!}}, & j \leq r, \\ 0, & j > r. \end{cases} \quad (\text{LVM4})$$

In particular $R_{\alpha,a}^2 = I$. For the coefficient identity, Rodrigues' coefficient formula gives the formal generating series $\sum_{r \geq 0} L_r^{(\alpha)}(x) t^r = (1-t)^{-\alpha-1} \exp(-xt/(1-t))$: expanding the exponential and then the binomial series reproduces each coefficient of GPA5. Substitute $x \mapsto -x$ and use $u = -t/(1-2t)$. This gives

$$\sum_{r \geq 0} L_r^{(\alpha)}(-x) t^r = (1-2t)^{-\alpha-1} \sum_{j \geq 0} L_j^{(\alpha)}(x) \left(\frac{-t}{1-2t} \right)^j. \quad (\text{LVM5})$$

Extracting t^r gives the binomial coefficient in LVM4; dividing by the orthonormal masses gives its displayed square root. This proves the explicit reflection matrix and its equality to the conjugated monomial reflection, with no convergence issue because coefficient extraction is in formal power series.

Retain precisely the GPA16 pullback $(P_{c',a})_{r,b} = \binom{b}{r} (c')^{b-r} i^r$ for $r \leq b$. Define the following maps on the entire original polynomial space:

$$T_{N,s} = A_{\alpha,a_N} P_{c',a_N}, \quad W_{N,s} = \begin{pmatrix} T_{N,s} \\ R_{\alpha,a_N} T_{N,s} \end{pmatrix} : \mathbb{C}^{n_N} \longrightarrow \mathbb{C}^{2n_N}. \quad (\text{LVM6})$$

The first block is the coefficient vector of $p(c' + iy)$ on the positive half-line. The second is that of $p(c' - iy)$ after substituting $y \mapsto -y$ on the negative half-line. Reflection here is complex-linear; no complex conjugation of p , c' or its period coefficients is performed. Squaring norms and adding the two integrals proves

$$H_{N,s}^{\text{exp}} = W_{N,s}^* W_{N,s} = T_{N,s}^* (I + R_{\alpha,a_N}^* R_{\alpha,a_N}) T_{N,s}. \quad (\text{LVM7})$$

This is exactly the full GPA17 factorial-moment matrix. It retains the cross-coefficient terms within both half-lines. The matrix $I + R^* R$ is positive definite and $R^*(I + R^* R)R = I + R^* R$, by $R^2 = I$. Thus LVM6 also gives an exact isometric reflection on the full comparison form. No orthogonality in the single-half-line metric was assumed.

77.2 The original kernel and residual as consecutive volumes

Keep the full n_N -by- p_N matrix $\mathbf{U}_N$ of GPA18, including $\alpha_A = v!/\mu_v$ and every original relation coefficient. Let $E_{\mathbf{m}}$ insert the lowest $\mathbf{m}$ monomials. Keep $Z = Z_k$ of rank $t = t_K$ and P_Y of rank $j_0 = \mathbf{m} - t$ exactly as in GPA19, so $\ker P_Y = \text{im } Z$. Choose the explicit right inverse $C = P_Y^*(P_Y P_Y^*)^{-1}$ if $j_0 > 0$, and the empty matrix otherwise. Then $\mathbf{F} = [Z, C]$ is an invertible $\mathbf{m}$ -square matrix: apply P_Y to a vanishing linear combination, then use the independence of Z . Define three original column matrices

$$\begin{aligned} X_{U,N} &= \mathbf{U}_N, & r_{U,N} &= p_N, \\ X_{K,N} &= [\mathbf{U}_N, E_{\mathbf{m}}Z], & r_{K,N} &= p_N + t, \\ X_{E,N} &= [\mathbf{U}_N, E_{\mathbf{m}}], & r_{E,N} &= n_N. \end{aligned} \quad (\text{LVM8})$$

Their full column ranks follow from the leading degrees $\mathbf{m}, \dots, \mathbf{m} + p_N - 1$ of the original relations and the independence of Z . Let

$$D_{A,N,s} = \det(X_{A,N}^* \mathbf{H}_{N,s}^{\text{exp}} X_{A,N}) \quad (A = U, K, E). \quad (\text{LVM9})$$

Empty Gram determinants are one. Block Gaussian elimination, using the complete original relation Gram as its first block, gives

$$\det H_{Y,N}^{\text{exp}} = \frac{D_{E,N,s}}{D_{U,N,s}}, \quad \det \hat{H}_{K,N}^{\text{exp}} = \frac{D_{K,N,s}}{D_{U,N,s}}, \quad \det Q_{0,N}^{\text{exp}} = |\det \mathbf{F}|^2 \frac{D_{E,N,s}}{D_{K,N,s}}. \quad (\text{LVM10})$$

For the last identity use the frame $[Z, C]$ in the low quotient. Its first Gram block is $Z^* H_Y^{\text{exp}} Z$. Minimizing over Z at fixed P_Y image gives the second Schur block $[P_Y (H_Y^{\text{exp}})^{-1} P_Y^*]^{-1}$. The determinant of the whole frame Gram is $|\det \mathbf{F}|^2 \det H_Y^{\text{exp}}$, proving the last ratio. This also proves the formula in both zero-rank cases. All these operations are on the original subspaces and fibres; the actual matrix Z and the remaining residual have been retained.

77.3 Every mixed minor and a finite maximal-minor error

For $A = U, K, E$ and an $r_{A,N}$ -element subset $I \subset \{1, \dots, 2n_N\}$ put

$$d_{A,N,s}(I) = \det((W_{N,s} X_{A,N})_{I,:}), \quad m_{A,N,s} = \max_{|I|=r_{A,N}} |d_{A,N,s}(I)|, \quad b_{A,N,s} = \log m_{A,N,s}. \quad (\text{LVM11})$$

At rank zero the sole determinant and maximum are one. At positive rank the maximum is positive because $W_{N,s}$ is injective and $X_{A,N}$ has full column rank. Define the exact finite remainder and its explicit bound

$$\zeta_{A,N,s} = \log \sum_{|I|=r_{A,N}} \frac{|d_{A,N,s}(I)|^2}{m_{A,N,s}^2}, \quad c_{A,N} = \log \binom{2n_N}{r_{A,N}}. \quad (\text{LVM12})$$

Cauchy–Binet proves, with every minor retained,

$$\log D_{A,N,s} = 2b_{A,N,s} + \zeta_{A,N,s}, \quad 0 \leq \zeta_{A,N,s} \leq c_{A,N}. \quad (\text{LVM13})$$

For completeness, expanding the determinant of $V^* V$ multilinearly in the rows of $V = WX$ leaves only sets of distinct row indices; the alternating sums on those indices are $\det \overline{V_I} \det V_I$. This proves the sum of squared minors. At least one summand in LVM12 is one and every summand is at most one, proving the two bounds. Each row set records its actual split between the two original half-lines, and all these sets remain in the exact remainder. The finite error does not replace a mixed contribution by zero.

Define the two explicit maximal-minor returns

$$B_{K,k,s} = 2 \sum_N \sigma_N (b_{K,N,s} - b_{U,N,s}), \quad B_{0,k,s} = 2 \sum_N \sigma_N (b_{E,N,s} - b_{K,N,s}). \quad (\text{LVM14})$$

Let $(A_X, B_X) = (K, U)$ for $X = K$ and $(A_X, B_X) = (E, K)$ for $X = 0$. Here the subscripts A_X, B_X refer only to the three column labels in LVM8; $B_{X,k,s}$ remains the scalar in LVM14. Put

$$\begin{aligned}\eta_{X,k}^- &= - \sum_{\sigma_N=1} c_{B_X,N} - \sum_{\sigma_N=-1} c_{A_X,N}, \\ \eta_{X,k}^+ &= \sum_{\sigma_N=1} c_{A_X,N} + \sum_{\sigma_N=-1} c_{B_X,N}.\end{aligned}\tag{LVM15}$$

Then, with the exact F_X^{exp} of GPA21,

$$\begin{aligned}F_X^{\text{exp}} &= B_{X,k,s} + \sum_N \sigma_N (\zeta_{A_X,N,s} - \zeta_{B_X,N,s}), \\ \eta_{X,k}^- &\leq F_X^{\text{exp}} - B_{X,k,s} \leq \eta_{X,k}^+, \\ |\eta_{X,k}^\pm| &\leq (4q + 8\mathbf{m} + 4) \log 2 = O(q) = o(kq).\end{aligned}\tag{LVM16}$$

The first two lines substitute LVM13 into LVM10 and retain the indicated signs. For $X = 0$ the factor $|\det \mathbf{F}|^2$ cancels only after summing the four endpoints, since that frame is independent of N . For the final bound use $\binom{2n}{r} \leq 2^{2n}$ and $\sum_N n_N = 2q + 4\mathbf{m} + 2$. There is exactly one bound $c_{A,N}$ per endpoint in each one-sided sum in LVM15. This proves the stated finite constant, including its factor of two. Finally $\mathbf{m} = O(k)$ and $q = (k+1)^2$ prove its order.

Apply the already explicit GPA22–23 metric comparison, on the original finite domain GPA2. For the literal HJR/HCR row sums $S_K = \mathcal{V}_{K,k,s}^{\text{jet}}$ and $S_0 = \mathcal{V}_{0,k,s}^{\text{jet}}$ this gives

$$B_{X,k,s} + d_{X,k,s}^- + \eta_{X,k}^- \leq S_X \leq B_{X,k,s} + d_{X,k,s}^+ + \eta_{X,k}^+.\tag{LVM17}$$

Both bounds concern the same original row sum, without assigning a rank-proportional coefficient. Its total error is $O_{\text{actual}}(k \log q + q) = o(kq)$. The source factor β_s cancels only in GPA's four signs; the remaining factorials, phases, period coefficients, mixed minors and unequal endpoint errors all remain explicit.

77.4 Exact return to the arithmetic and mixed quantities

Set $\ell_X = B_{X,k,s} + d_{X,k,s}^- + \eta_{X,k}^-$ and $u_X = B_{X,k,s} + d_{X,k,s}^+ + \eta_{X,k}^+$, with precisely the preceding finite definitions. HCR3's exact original equalities give

$$\begin{aligned}\ell_K - D_s^+ + F_L^{(s)} - F_{W_0}^{(s)} &\leq F_K^{(s)} \leq u_K - D_s^- + F_L^{(s)} - F_{W_0}^{(s)}, \\ F_{W_0}^{(s)} + \ell_0 - E_{k,s}^{\text{tr}} &\leq \Phi_{k,s} \leq F_{W_0}^{(s)} + u_0 + E_{k,s}^{\text{tr}}.\end{aligned}\tag{LVM18}$$

Here $D_s^\pm$ are the full PBR8 flag-error endpoints, not the dual matrices, and every low term remains separate. The proof is direct substitution of LVM17 into $F_K = S_K - \mathcal{E}_s^{\text{flag}} + F_L - F_{W_0}$ and $\Phi = F_{W_0} + S_0 + \delta_0$, using $\mathcal{E}_s^{\text{flag}} \in [D_s^-, D_s^+]$ and $|\delta_0| \leq E^{\text{tr}}$. Intersect these bounds with the same original HCR4 intervals. Each intersection contains the actual scalar, so preserves the earlier finite branches, including the original OAR/PMR domain conditions. Adding the correlated arithmetic source-difference interval $[a_{K,k}^-, a_{K,k}^+]$ to the $s = 1$ kernel interval, and intersecting with HCR5, gives the arithmetic kernel interval on $|u| \geq R_*$. Subtracting its endpoints from the unchanged total $\mathcal{B}_k^{\text{ar}}$ reverses them for the boundary. For the $s = k$ signed and action quantities add exactly $\mathcal{H}_k + \delta_k^\sigma$ and then the complete C_k^{budget} of HCR5 to the two kernel endpoints. These are translations of the same scalar identities, so retain every mixed pairing, penalty, monic window and leakage term rather than altering the action.

Finally the common intermediate maximal minor cancels algebraically only in the sum:

$$\begin{aligned}B_{K,k,s} + B_{0,k,s} &= 2 \sum_N \sigma_N (b_{E,N,s} - b_{U,N,s}), \\ \frac{B_{K,k,s} + B_{0,k,s}}{kq} &\longrightarrow 16\mathfrak{C}_\partial.\end{aligned}\tag{LVM19}$$

The limit follows from LVM17's proved subleading errors and the evaluated HCR6 total. The separate scalar $B_{K,k,s}$ is an explicit maximal-minor calculation on the original relation and kernel columns, and $B_{0,k,s}$ is its consecutive residual volume. The formula preserves their shared intermediate minors; it supplies no independent choice of the two allocations.

78 A deterministic calculation of the original mixed volumes

Retain all original objects, finite domains and matrices of LVM1–19. For each column label $A = U, K, E$ at each original endpoint and source let $V = W_{N,s} X_{A,N}$, with $M = 2n_N$ rows and $r = r_{A,N}$ independent columns. In this section M is this row count, not the added-row index in HJR. Define the entire row Gram and its successive matrices by the following finite procedure:

$$\begin{aligned} H^{[0]} &= VV^*, \\ i_j &= \min \arg \max_{1 \leq i \leq M} H_{ii}^{[j-1]}, \quad p_j = H_{i_j i_j}^{[j-1]}, \\ H^{[j]} &= H^{[j-1]} - H_{:,i_j}^{[j-1]} H_{i_j,:}^{[j-1]} / p_j \quad (1 \leq j \leq r). \end{aligned} \tag{GVM1}$$

Ties use the original fixed row order, including its half-line labels. All diagonal comparisons are of actual nonnegative real numbers. There is no pivot guessed or inserted from generic coefficients.

Every p_j is positive and each update retains the full residual row Gram. To prove this directly, write the original rows as v_i . Starting with their unaltered vectors, let $u_i^{[j-1]}$ be their orthogonal residual after the preceding chosen rows and put

$$q_j = u_{i_j}^{[j-1]} / \sqrt{p_j}, \quad u_i^{[j]} = u_i^{[j-1]} - (u_i^{[j-1]} q_j^*) q_j. \tag{GVM2}$$

Induction shows that the q_j are orthonormal rows, $H_{ih}^{[j]} = u_i^{[j]} (u_h^{[j]})^*$, and that GVM2 gives exactly the GVM1 matrix update. At a step $j \leq r$ the span of the residual rows has dimension $r - j + 1 > 0$: orthogonal projection removes exactly the previous independent $j - 1$ directions from the row span of V , which is all $\mathbb{C}^r$. Some residual row is therefore nonzero, so the largest diagonal p_j is positive. A previously chosen row has zero residual, so every pivot index is new. This also proves the claimed ranks at every step.

The coefficient map in GVM2 is recorded explicitly rather than replacing the source norm. If Q has rows $q_1, \dots, q_r$, then $QQ^* = Q^*Q = I$ and

$$R = VQ^*, \quad V = RQ, \quad |R_{ij}|^2 \leq p_j, \quad |\det V_{\{i_1, \dots, i_r\},:}|^2 = \prod_{j=1}^r p_j. \tag{GVM3}$$

The row set determinant uses the original increasing order; permuting to pivot order changes only its sign. For the entry bound, q_j is orthogonal to the first $j - 1$ directions, so $R_{ij} = u_i^{[j-1]} q_j^*$ and Cauchy–Schwarz gives $|R_{ij}|^2 \leq \|u_i^{[j-1]}\|^2 \leq p_j$. In pivot order the selected rows of R form a lower triangular matrix: the selected j th row belongs to the span of $q_1, \dots, q_j$ and has its j th coefficient $\sqrt{p_j}$. Since Q is unitary, the selected determinant formula follows. Thus every coefficient and the inverse coordinate map are explicit.

Put $P(V) = \prod_{j=1}^r p_j$, with $P(V) = 1$ at rank zero. Then the exact full-volume estimate is

$$P(V) \leq \det(V^*V) \leq M^r P(V), \quad 0 \leq \log \det(V^*V) - \log P(V) \leq r \log M. \tag{GVM4}$$

The lower bound is the selected nonnegative summand in the complete LVM13 Cauchy–Binet sum. For the upper bound, GVM3 gives $\det(V^*V) = \det(R^*R)$ and each column of R has squared norm at most Mp_j . Successive orthogonal projection of these columns expresses their Gram determinant as the product of squared residual norms, each at most the corresponding unprojected squared norm. Hence $\det(R^*R) \leq \prod_j Mp_j$, proving GVM4 without an assumed condition number. In

rank zero both sides are one. The calculation uses every row, in both half-line blocks, in each maximal-diagonal decision and full Schur update.

For the actual matrices define

$$\kappa_{A,N,s} = \log P(W_{N,s} X_{A,N}), \quad \xi_{A,N,s} = \log D_{A,N,s} - \kappa_{A,N,s}, \quad c_{A,N}^{\text{piv}} = r_{A,N} \log(2n_N). \quad (\text{GVM5})$$

Thus $0 \leq \xi_{A,N,s} \leq c_{A,N}^{\text{piv}}$ with an exact finite remainder. Define the two original successive-volume returns

$$C_{K,k,s} = \sum_N \sigma_N (\kappa_{K,N,s} - \kappa_{U,N,s}), \quad C_{0,k,s} = \sum_N \sigma_N (\kappa_{E,N,s} - \kappa_{K,N,s}). \quad (\text{GVM6})$$

For $X = K$ take the column-label pair $(a_X, b_X) = (K, U)$; for $X = 0$ take $(a_X, b_X) = (E, K)$. Set the unequal endpoints

$$\begin{aligned} e_{X,k}^- &= - \sum_{\sigma_N=1} c_{b_X,N}^{\text{piv}} - \sum_{\sigma_N=-1} c_{a_X,N}^{\text{piv}}, \\ e_{X,k}^+ &= \sum_{\sigma_N=1} c_{a_X,N}^{\text{piv}} + \sum_{\sigma_N=-1} c_{b_X,N}^{\text{piv}}. \end{aligned} \quad (\text{GVM7})$$

LVM10, with its exact fixed frame determinant, proves

$$\begin{aligned} F_X^{\text{exp}} &= C_{X,k,s} + \sum_N \sigma_N (\xi_{a_X,N,s} - \xi_{b_X,N,s}), \\ e_{X,k}^- &\leq F_X^{\text{exp}} - C_{X,k,s} \leq e_{X,k}^+, \end{aligned} \quad (\text{GVM8})$$

$$|e_{X,k}^\pm| \leq (2q + 4\mathfrak{m} + 2) \log(2(q + \mathfrak{m} + 1)) = O(q \log q) = o(kq).$$

Indeed the same fixed residual frame factor cancels in the four signs, and GVM4 proves each remainder bound. Each one-sided sum uses one error bound per endpoint; $r_{A,N} \leq n_N$, $\max_N n_N = q + \mathfrak{m} + 1$, and $\sum_N n_N = 2q + 4\mathfrak{m} + 2$ give the displayed constant. Finally $q = (k + 1)^2$, $\mathfrak{m} = O(k)$ and $\log q/k \rightarrow 0$ prove the stated order.

With the literal original GPA22 endpoints $d_{X,k,s}^\pm$, the actual HCR row sums therefore satisfy

$$C_{X,k,s} + d_{X,k,s}^- + e_{X,k}^- \leq \mathcal{V}_{X,k,s}^{\text{jet}} \leq C_{X,k,s} + d_{X,k,s}^+ + e_{X,k}^+. \quad (\text{GVM9})$$

This error is $O_{\text{actual}}(k \log q + q \log q) = o(kq)$. The original U, Z, P_Y , factorials, center, phases and both Gamma orders are unchanged. The P_Y frame determinant and the source mass cancel only in their proved four-endpoint identities.

GVM1 calculates each C_X from the explicit matrices by at most r maximal-diagonal selections and r rank-one matrix updates. Forming each row Gram uses at most $M^2 r$ scalar product terms; the updates use at most rM^2 products, rM^2 divisions, rM^2 subtractions and rM diagonal comparisons. For the three column lists, four endpoints and two sources there are twenty-four such matrices, all of sizes $M, r = O(q)$. This is an $O(q^3)$ count of these specified arithmetic operations and exact comparisons. It assumes the given original coefficients as inputs; it makes no claim about their bit complexity, conditioning, numerical certification or an already evaluated value. It avoids enumerating the exponentially many mixed minors while retaining them through the proved remainder.

For exact backward propagation, substitute $\ell_X = C_{X,k,s} + d_X^- + e_X^-$ and $u_X = C_{X,k,s} + d_X^+ + e_X^+$ in the proved LVM18 inequalities. Their derivation uses only the same exact HCR equalities and finite flag/transfer errors, so applies with these bounds verbatim. Intersect with the complete earlier intervals and retain the same arithmetic, signed and action translations described there. Both low terms, the correlated arithmetic source difference, all mixed pairings and the complete action budget remain. Finally

$$C_{K,k,s} + C_{0,k,s} = \sum_N \sigma_N (\kappa_{E,N,s} - \kappa_{U,N,s}), \quad \frac{C_{K,k,s} + C_{0,k,s}}{kq} \rightarrow 16\mathfrak{C}_\partial. \quad (\text{GVM10})$$

The first equality cancels the identical actual intermediate pivot volume. GVM9 and the evaluated HCR6 sum give the second. The separate leading coefficient still requires a bound on these actual pivot products or on the equivalent native component. The deterministic calculation supplies those products and their finite error; it does not assign them from the ranks.

79 Original threshold spectral flow and compatible-row returns

Fix the original simple quartet, full unit, actual period and logarithm branch on the proved period domain, and the original source $s = 1$ or $s = k$. The period is fixed as k grows. Retain

$$\begin{aligned} k = 4l + 1 \geq 9, \quad q = (k + 1)^2, \quad q' = (k - 7)^2, \quad \Delta = q - q', \quad \mathbf{m} = \Delta - v, \\ t_0 = \dim(K \cap L), \quad t_K = r_K - t_0, \quad t_R = j - v + t_0, \quad t_K + t_R = \mathbf{m}, \\ c = k/2, \quad c' = c - 4, \quad d_N = N - v. \end{aligned} \quad (\text{TSR1})$$

Here R in t_R labels the complementary restriction; t_0 retains its original meaning. The original conductor symbol is $E_A(w) = \sum_{a,b=0}^8 a_{ab} e^{b_{ab}w}$, with $v = \text{ord}_0 E_A$ and $\mu_v = E_A^{(v)}(0) \neq 0$. All maps below use the actual coefficients of this symbol.

79.1 Original source forms and threshold-kernel maps

Let $Z = Z_{\text{exp}}$ be the full DNE numerator frame, C the full conductor matrix, and $\mathcal{N}_a$ its numerator functions. Set

$$\begin{aligned} T_{N;na} &= \Lambda_{\mathcal{N}_a/E_A}(\phi_{n,s,c'}), \quad V_{N;n,(i,j)} = \phi_{n,s,c'}(\beta'_{ij}), \quad 0 \leq n \leq d_N, \\ K_{V,N} &= V_N^* V_N, \quad W_N = V_N^* T_N, \quad K_C = \bar{C} C^T, \quad U_C = \bar{C} Z, \\ D_N &= T_N^* T_N - W_N^* K_{V,N}^{-1} W_N, \quad G = Z^* Z - U_C^* K_C^{-1} U_C, \\ B &= Z_k^T A_\ell : \mathbb{C}^{\mathbf{m}} \rightarrow \mathbb{C}^{t_K}, \quad J = A_\ell^{-1} P_Y^T : \mathbb{C}^{t_R} \rightarrow \mathbb{C}^{\mathbf{m}}, \quad \text{im } J = \ker B, \\ D_{K,N} &= (B D_N^{-1} B^*)^{-1}, \quad G_K = (B G^{-1} B^*)^{-1}, \quad D_{R,N} = J^* D_N J, \quad G_R = J^* G J. \end{aligned} \quad (\text{TSR2})$$

These are precisely PEC2,17,19,24 and PBR2–3. The original Gamma mass is $M_s = (2\pi)^{s/2}$ and $h_n = M_s n! (s/2)_n$; hence every factorial, mass and phase remains in the complete matrices T_N, V_N . The positive definiteness and all stated full ranks were proved for these original frames. Empty blocks use their unique empty maps, with determinant one.

For every real threshold $t > 0$, define

$$\mathcal{H}_{K,N}(t) = D_N - t B^* G_K B, \quad \mathcal{H}_{R,N}(t) = J^* (D_N - t G) J, \quad \nu_{X,N}(t) = n_-(\mathcal{H}_{X,N}(t)). \quad (\text{TSR3})$$

Negative counts are strict, retaining nullity separately at an exact threshold. Inertia is ordered (n_+, n_-, n_0) . The kernel pencil is SIC1 and its fixed reference inverse is independent of N .

To prove its exact relation to the quotient, put $L_N = D_N^{-1} B^* D_{K,N}$. Then $B L_N = I$ and $J^* D_N L_N = J^* B^* D_{K,N} = 0$. Every source vector has a unique decomposition $Jx + L_N z$: applying B recovers z , and J is an isomorphism onto the remaining kernel. Therefore

$$[J, L_N]^* \mathcal{H}_{K,N}(t) [J, L_N] = \text{diag}(D_{R,N}, D_{K,N} - t G_K), \quad \text{In } \mathcal{H}_{K,N}(t) = (t_R, 0, 0) + \text{In}(D_{K,N} - t G_K). \quad (\text{TSR4})$$

The displayed frame is invertible. Invertible congruence preserves maximal positive and negative subspace dimensions, as well as the kernel by its invertible coordinate map. This proves TSR4 with the full original relation space.

The threshold-kernel isomorphism in this frame is $z \mapsto L_N z$, with inverse $x \mapsto Bx$. Write $Q_N = B D_N^{-1} B^*$ and $R_B = B G^{-1} B^*$. On $\ker(Q_N^{-1} - t R_B^{-1})$, $Q_N R_B^{-1} z = t^{-1} z$ and $L_N z = t D_N^{-1} B^* R_B^{-1} z$. Thus the literal SIC7 pair is

$$z \mapsto D_N^{-1} B^* R_B^{-1} z, \quad x \mapsto t Bx. \quad (\text{TSR5})$$

Substitution proves both composites are identities, including repeated threshold eigenvalues. For zero quotient rank the pencil is the positive source form and its negative count and nullity are zero. No arithmetic inclusion, unit jet or observation has changed.

79.2 A pencil assembled from all primitive original columns

The inverse-free primitive assembly of NG2 is, in block dimensions $(\mathbf{m}, q', t_K, \mathbf{m}, q')$,

$$\mathcal{L}_{K,N}(t) = \begin{pmatrix} T_N^* T_N & W_N^* & B^* & 0 & 0 \\ W_N & K_{V,N} & 0 & 0 & 0 \\ B & 0 & 0 & B & 0 \\ 0 & 0 & B^* & -tZ^* Z & tU_C^* \\ 0 & 0 & 0 & tU_C & -tK_C \end{pmatrix}. \quad (\text{TSR6})$$

The original conductor has q' independent rows. More generally its row dimension can be retained separately in the following offsets; nothing uses a selected minor. Every displayed block is an original column product, the original B , or t .

For any invertible Hermitian P , direct multiplication gives

$$E_P^* \begin{pmatrix} X & Y \\ Y^* & P \end{pmatrix} E_P = \text{diag}(X - Y P^{-1} Y^*, P), \quad E_P = \begin{pmatrix} I & 0 \\ -P^{-1} Y^* & I \end{pmatrix}. \quad (\text{TSR7})$$

Its inverse changes minus to plus. A non-last pivot is first moved last by the permutation listing all remaining indices in their original order; its inverse is its transpose. Eliminate $-tK_C$, then $-tG$, then $K_{V,N}$ in TSR6. The first pivot changes the fourth diagonal to $-tG$ and contributes q' negative directions. The second changes the third diagonal to $t^{-1}R_B$ and contributes $\mathbf{m}$ negative directions. The third changes the first diagonal to D_N and contributes q' positive directions. The remaining matrix is $\begin{pmatrix} D_N & B^* \\ B & t^{-1}R_B \end{pmatrix}$. Its second pivot and TSR4 prove

$$\text{In } \mathcal{L}_{K,N}(t) = (\mathbf{m} + q', \mathbf{m} + q', 0) + \text{In}(D_{K,N} - tG_K), \quad \nu_{K,N}(t) = n_-(\mathcal{L}_{K,N}(t)) - (\mathbf{m} + q'). \quad (\text{TSR8})$$

This offset is essential for an individual threshold count.

For $z \in \ker(D_{K,N} - tG_K)$ the full primitive null vector is

$$\begin{aligned} x &= D_N^{-1} B^* R_B^{-1} z, & a &= -K_{V,N}^{-1} W_N x, & y &= -R_B^{-1} z, \\ u &= -t^{-1} G^{-1} B^* R_B^{-1} z, & w &= K_C^{-1} U_C u, & z &= t B x. \end{aligned} \quad (\text{TSR9})$$

The last formula is the inverse map. The fifth, fourth and second primitive rows give respectively the formulas for w, u, a ; the third gives $Bx + Bu = 0$ and the first reduces to $D_N x + B^* y = 0$. The quotient kernel equation verifies all five rows. Conversely these rows eliminate w, u, a and recover TSR5, proving the full isomorphism.

The original complementary restriction uses

$$\mathcal{L}_{R,N}(t) = \begin{pmatrix} J^*(T_N^* T_N - tZ^* Z)J & J^* W_N^* & tJ^* U_C^* \\ W_N J & K_{V,N} & 0 \\ tU_C J & 0 & -tK_C \end{pmatrix}. \quad (\text{TSR10})$$

The same two pivot congruences give

$$\text{In } \mathcal{L}_{R,N}(t) = (q', q', 0) + \text{In}(D_{R,N} - tG_R), \quad \nu_{R,N}(t) = n_-(\mathcal{L}_{R,N}(t)) - q'. \quad (\text{TSR11})$$

Its kernel lift is $z \mapsto (z, -K_{V,N}^{-1} W_N J z, K_C^{-1} U_C J z)$, with inverse the first coordinate, as direct row substitution proves. Thus both pencils retain all eliminated coordinates and denominators.

79.3 The exact original one-row spectral flow

At $q - 1 \leq M < 2q$, retain all original rows T, V through degree $M - v$ and their actual next rows $t_M, v_{+,M}$. The threshold remains t ; t_M is a row. Put

$$\begin{aligned} A_M &= V^*V, \quad K_M = A_M^{-1}V^*T, \quad r_M = t_M - v_{+,M}K_M, \\ \lambda_M &= 1 + v_{+,M}A_M^{-1}v_{+,M}^* > 0, \quad h_M = \lambda_M^{-1/2}r_M, \quad D_{M+1} = D_M + h_M^*h_M. \end{aligned} \quad (\text{TSR12})$$

The last identity follows by subtracting the attained old extension minimum and minimizing $\eta^*A_M\eta + |r_Mx - v_{+,M}\eta|^2$ over the same full lower coefficient. Completion using $A_M + v_{+,M}^*v_{+,M}$ gives $|r_Mx|^2/\lambda_M$. This proves the complete original conditional row update of HJR14–15, with its original compatible inverse convolution, Gamma coordinates and monic upper-grid recurrence.

The exact D_M -orthogonal frame $[J, L_M]$ gives

$$\begin{aligned} D_M^{-1} &= JD_{R,M}^{-1}J^* + L_MD_{K,M}^{-1}L_M^*, \\ x_M &= h_MJD_{R,M}^{-1}J^*h_M^* \geq 0, \quad y_M = h_ML_MD_{K,M}^{-1}L_M^*h_M^* \geq 0, \\ D_{R,M+1} &= D_{R,M} + (h_MJ)^*(h_MJ), \\ D_{K,M+1} &= D_{K,M} + (h_ML_M)^*(h_ML_M)/(1 + x_M), \\ g_{R,M} &= \log \frac{\det D_{R,M+1}}{\det D_{R,M}} = \log(1 + x_M), \\ g_{K,M} &= \log \frac{\det D_{K,M+1}}{\det D_{K,M}} = \log \left(1 + \frac{y_M}{1 + x_M} \right). \end{aligned} \quad (\text{TSR13})$$

The inverse identity follows by inverting the full frame Gram. Restriction proves the first metric update. For the quotient, every lift is $L_Mz + Jw$; minimize its entire updated norm over w . Its added quotient form is exactly the displayed numerator divided by $1 + x_M$. The rank-one determinant lemma proves both logarithms. Both energies retain λ_M^{-1} and the actual period frame.

Define

$$f_{X,M}(t) = \nu_{X,M}(t) - \nu_{X,M+1}(t), \quad X = K, R. \quad (\text{TSR14})$$

The kernel source pencil increases by $h_M^*h_M$, and the restriction pencil increases by $(h_MJ)^*(h_MJ)$. Adding a positive rank-one form cannot increase a negative index: a negative subspace of the larger form is negative for the smaller form. Its negative index drops by at most one: intersect a maximal negative subspace of the smaller form with the kernel of the added row. This loses at most one dimension and remains negative for the larger form. Consequently

$$f_{X,M}(t) \in \{0, 1\} \quad (t > 0). \quad (\text{TSR15})$$

This holds also at singular thresholds with strict negative counts. An empty-rank case has identically zero flow.

There is an exact resolvent test at every nonsingular old threshold. For $X = K$ let $H = \mathcal{H}_{K,M}(t)$ and $u = h_M$; for $X = R$ let $H = \mathcal{H}_{R,M}(t)$ and $u = h_MJ$. When H is invertible,

$$s_{X,M}(t) = 1 + uH^{-1}u^* \in \mathbb{R}, \quad f_{X,M}(t) = \mathbf{1}_{\{s_{X,M}(t) \leq 0\}}. \quad (\text{TSR16})$$

Indeed the bordered matrix $\begin{pmatrix} H & u^* \\ u & -1 \end{pmatrix}$ has the inertia of both $\text{diag}(H, -s_{X,M})$ and $\text{diag}(H + u^*u, -1)$ by TSR7. Their negative indices give $f = 1 - n_-(-s)$, proving TSR16. If $s = 0$, the updated threshold has one-dimensional kernel spanned by $H^{-1}u^*$, since multiplication gives $(H + u^*u)H^{-1}u^* = su^* = 0$ and this vector is nonzero. Conversely any updated null vector v_* satisfies $v_* = -H^{-1}u^*(uv_*)$ and is therefore in that one-dimensional span; this proves the stated nullity. Singular old thresholds use the full inertia and explicit kernel maps TSR5,9,11. The primitive assembly itself requires no inverses.

79.4 Exact deficit, positive-part and full-row integrals

For the generalized eigenvalues $\mu_{X,N,a} > 0$ of the retained pair $(D_{X,N}, G_X)$, counted with multiplicity, the spectral theorem in that specified reference metric and TSR4 give

$$\nu_{X,N}(t) = \#\{a : \mu_{X,N,a} < t\}. \quad (\text{TSR17})$$

This describes the same two forms without replacing either coordinate metric. For positive numbers a, b , their two step functions satisfy

$$\int_0^\infty (\mathbf{1}_{\{a < t\}} - \mathbf{1}_{\{b < t\}}) \frac{dt}{t} = \log(b/a). \quad (\text{TSR18})$$

The integrand vanishes outside the interval between a, b ; direct integration proves the formula for either order. Every spectral sum here is finite. Retain the original PEC parts

$$n_{X,N} = \sum_a \log^+(1/\mu_{X,N,a}), \quad p_{X,N} = \sum_a \log^+ \mu_{X,N,a}.$$

TSR17–18 prove

$$\begin{aligned} n_{X,M} - n_{X,M+1} &= \int_0^1 f_{X,M}(t) \frac{dt}{t}, \\ p_{X,M+1} - p_{X,M} &= \int_1^\infty f_{X,M}(t) \frac{dt}{t}, \\ g_{X,M} &= \int_0^\infty f_{X,M}(t) \frac{dt}{t}. \end{aligned} \quad (\text{TSR19})$$

For the positive-part formula integrate $\mathbf{1}_{\{t < \mu\}}$ on $(1, \infty)$ and take the difference; weak versus strict endpoints change only a finite set of points. Thus each original HCR logarithm in TSR13 is exactly the integral of its own original threshold spectral flow. No growing limit and integral have been interchanged.

Keep the original weights $w_{q-1} = w_{2q-1} = 1$ and $w_M = 2$ for $q \leq M \leq 2q-2$. Finite telescoping gives

$$\begin{aligned} \mathcal{N}_X(t) &= \sum_{M=q-1}^{2q-1} w_M f_{X,M}(t) = \nu_{X,q-1}(t) + \nu_{X,q}(t) - \nu_{X,2q-1}(t) - \nu_{X,2q}(t), \\ 0 &\leq \mathcal{N}_X(t) \leq 2t_X, \\ A_{X,k,s} &= \int_0^1 \mathcal{N}_X(t) \frac{dt}{t}, \quad P_{X,k,s} = \int_1^\infty \mathcal{N}_X(t) \frac{dt}{t}, \\ \mathcal{V}_{X,k,s}^{\text{jet}} &= \int_0^\infty \mathcal{N}_X(t) \frac{dt}{t} = A_{X,k,s} + P_{X,k,s}. \end{aligned} \quad (\text{TSR20})$$

Nonnegativity follows from TSR15; the upper bound follows from the two low counts, each at most the actual rank t_X . The integrals follow by summing TSR19. In particular SIC2 is exactly the portion below threshold one of the original HCR row sum, while the portion above one is its retained PEC positive return. This is a typed identity for the actual kernel and restriction maps. It asserts no pointwise additivity of unrelated full, quotient and restriction count curves; the actual full-row equality is the exact identity $(1 + x_M)(1 + y_M/(1 + x_M)) = 1 + x_M + y_M$.

79.5 Explicit finite spectral cutoffs from original source bounds

The integrations have explicit cutoffs from the already proved original reconstruction and Gamma bounds. Set $\sigma = \sqrt{\delta^2 + \gamma^2}$, $b = s/2$, $A_{\text{abs}} = \sum |a_{ab}|$ and $H_D = \sum_{a=1}^D 1/a$. The literal LJR16

bound is

$$\begin{aligned} \mathfrak{d}_{ab} &= \prod_{(i,j) \neq (a,b)} (\beta_{ab} - \beta_{ij}), \quad \Pi_{k,s} = \prod_{a=1}^{q-1} [2\sqrt{a(b+a)} + k\sigma], \\ M_{k,s}^{\text{rec}} &= \max \left\{ 1, A_{\text{abs}} e^{8\sigma(H_{q-1}+2)} \sqrt{M_s} \Pi_{k,s} \left(\sum_{a,b=0}^k |\mathfrak{d}_{ab}|^{-2} \right)^{1/2} \right\}. \end{aligned} \quad (\text{TSR21})$$

LJR17 bounds the actual reconstruction from the low dual quotient norm to the original coefficient quotient norm. In its paired frame that map is the identity, so this bound is exactly $G \preceq (M_{k,s}^{\text{rec}})^2 D_{q-1}$. TSR12 gives $D_N \succeq D_{q-1}$ for all indicated cutoffs. Restriction along J preserves this lower bound, and minimizing over the same entire B -fiber in each metric preserves it for the quotient. Thus no bound on an omitted coefficient inverse is being assumed.

For the upper bound retain the literal DNE11–14 constants:

$$\begin{aligned} R_{\text{root}} &= |c| + k\sigma, \quad B_{\text{shift}} = \max_{a,b} |b_{ab}|, \quad C_v = A_{\text{abs}} B_{\text{shift}}^v / |\mu_v|, \\ R_0 &= B_{\text{shift}}^{-1} \min\{1, (v+1)/(2eC_v)\}, \quad \rho = \tanh(R_0/4), \quad C_F = 2R_{\text{root}}^v e^{R_{\text{root}} R_0} / |\mu_v|, \\ \mathcal{B}_{d,s,k,A}^2 &= \frac{C_F^2}{M_s} (1 - \rho^2)^{-s/2} e^{|c'|R_0/2} \sum_{a=0}^d \frac{a!}{(s/2)_a} \rho^{-2a}, \\ a_{k,s} &= (M_{k,s}^{\text{rec}})^{-2}, \quad b_{k,s} = \max\{1, q\mathcal{B}_{2q-v,s,k,A}^2\}. \end{aligned} \quad (\text{TSR22})$$

Here $B_{\text{shift}} \geq 4$ and $C_v \geq 1$ by the original shift list, so $R_0 > 0$. DNE14 proves $D_N \preceq q\mathcal{B}_{d_N,s,k,A}^2 G$; the positive sum increases with d_N , proving the same upper bound at every endpoint with the final cutoff in TSR22. Both restriction and full-fiber quotient minima preserve this bound. Therefore

$$0 < a_{k,s} \leq 1 \leq b_{k,s} < \infty, \quad a_{k,s} \leq \mu_{X,N,a} \leq b_{k,s}. \quad (\text{TSR23})$$

All spectral flows vanish below $a_{k,s}$ and above $b_{k,s}$. At the cutoff points the explicit kernel maps retain all nullities; those points do not affect the integrals. Every original mass, factorial, root spacing and conductor coefficient remains in these finite cutoffs before taking a limit.

79.6 Finite brackets for both actual row allocations

For one fixed original source take a finite increasing grid from $a_{k,s}$ to $b_{k,s}$ containing 1. If both cutoffs are 1, TSR23 and the two source inequalities give $D_N = G$ for every cutoff, and both flows vanish, so no cell is needed. Otherwise put $h = \max_{[a,b]} \log(b/a)$ over its cells. Abbreviate the four counts in source order by $\nu_{X,1}, \dots, \nu_{X,4}$. For a cell $[a, b]$ define

$$\begin{aligned} c_{X;a,b}^- &= \max\{0, \nu_{X,1}(a) + \nu_{X,2}(a) - \nu_{X,3}(b) - \nu_{X,4}(b)\}, \\ c_{X;a,b}^+ &= \min\{2t_X, \nu_{X,1}(b) + \nu_{X,2}(b) - \nu_{X,3}(a) - \nu_{X,4}(a)\}. \end{aligned} \quad (\text{TSR24})$$

Each count is nondecreasing in its threshold by TSR17. Thus $c_{X;a,b}^- \leq \mathcal{N}_X(t) \leq c_{X;a,b}^+$ on the open cell; the clipping is justified by TSR20. Multiply by dt/t and sum:

$$\begin{aligned} V_X^\pm &= \sum_{[a,b]} c_{X;a,b}^\pm \log(b/a), \quad A_X^\pm = \sum_{b \leq 1} c_{X;a,b}^\pm \log(b/a), \quad P_X^\pm = \sum_{a \geq 1} c_{X;a,b}^\pm \log(b/a), \\ V_X^- &\leq \mathcal{V}_{X,k,s}^{\text{jet}} \leq V_X^+, \quad A_X^- \leq A_{X,k,s} \leq A_X^+, \quad P_X^- \leq P_{X,k,s} \leq P_X^+, \\ (A_X^+ - A_X^-) &+ (P_X^+ - P_X^-) = V_X^+ - V_X^- \leq 4t_X h. \end{aligned} \quad (\text{TSR25})$$

Before clipping, each cell coefficient width is $\sum_{i=1}^4 [\nu_{X,i}(b) - \nu_{X,i}(a)]$. Clipping can only reduce it. Bound each logarithmic cell width by h and telescope the four nondecreasing curves, each with at

most t_X total increase. This proves the last inequality, including strict thresholds and repeated eigenvalues. Subdivision raises the applicable lower coefficients and lowers the upper coefficients on the smaller cells, so nested grids tighten and reach every prescribed positive tolerance. Exact finite-grid equality is not asserted.

The counts may be obtained from TSR3 or from the primitive matrices TSR6,10 with offsets TSR8,11 subtracted. The fixed offsets cancel in the four-term difference but must be subtracted before the clipping in TSR24. Hence these bounds determine the actual HCR row-sum allocation using four endpoint matrices, with the exact connection to every original compatible row supplied by TSR12–20.

There is a quantified bound on the number of threshold nodes at scale kq . For any $\eta > 0$, use the common grid for both $X = K, R$ with

$$h_* = \frac{\eta k q}{4\mathfrak{m}}, \quad \ell_{k,s} = \log(b_{k,s}/a_{k,s}), \quad n_* = \left\lceil \frac{4\mathfrak{m}\ell_{k,s}}{\eta k q} \right\rceil. \quad (\text{TSR26})$$

Here $\mathfrak{m} > 0$ on the original domain; the empty full-space case would have zero allocations and require no grid. If $\ell > 0$, use n_* equal logarithmic cells and add 1 when absent. Its mesh is at most h_* and it has at most $n_* + 2$ nodes. Since $t_K + t_R = \mathfrak{m}$, TSR25 gives

$$(V_K^+ - V_K^-) + (V_R^+ - V_R^-) \leq \eta k q. \quad (\text{TSR27})$$

At most eight inertia evaluations at each node give both allocations, one for each of four endpoints and two actual maps. This counts specified threshold evaluations; it does not bound their arithmetic bit complexity or claim that actual transcendental inputs have already been numerically certified.

The proved LJR20 and DNE15 bounds on the exact constants give $\log M_{k,s}^{\text{rec}} \leq q \log k + O_{\text{actual}}(q)$ and $\log b_{k,s} = O_{\text{actual}}(q)$, for both original sources. Hence

$$\ell_{k,s} \leq 2q \log k + O_{\text{actual}}(q), \quad n_* \leq \frac{128 \log k + O_{\text{actual}}(1)}{\eta} + 1. \quad (\text{TSR28})$$

Indeed $\mathfrak{m} = 16k - 48 - v \leq 16k$, so the argument of the ceiling is at most $64\ell_{k,s}/(\eta q)$. Thus $O_{\text{actual}}(\eta^{-1} \log k + 1)$ threshold nodes give a joint allocation interval of width $\eta k q$ for each fixed actual source. This assertion follows from the proved source cutoffs and the finite width inequality; it assumes no separate allocation asymptotic.

79.7 Return to the same finite receiving intervals

Retain $0 \leq P_{X,k,s} \leq E_{k,s}^\vee$ from PEC. For example the actual row sum belongs to

$$[\tilde{V}_X^-, \tilde{V}_X^+] = [V_X^-, \min\{V_X^+, A_X^+ + E_{k,s}^\vee\}]. \quad (\text{TSR29})$$

This interval is nonempty because each entry bounds that same quantity. A grid restricted to $[a_{k,s}, 1]$ also gives the valid interval $[A_X^-, A_X^+ + E_{k,s}^\vee]$ without querying thresholds above one. It retains the full positive allowance, which PEC15 proves to be $o(kq)$; it does not set that allowance to zero.

The exact original HCR3 identities give

$$\begin{aligned} \tilde{V}_K^- - D_s^+ + F_L^{(s)} - F_{W_0}^{(s)} &\leq F_K^{(s)} \leq \tilde{V}_K^+ - D_s^- + F_L^{(s)} - F_{W_0}^{(s)}, \\ F_{W_0}^{(s)} + \tilde{V}_R^- - E_{k,s}^{\text{tr}} &\leq \Phi_{k,s} \leq F_{W_0}^{(s)} + \tilde{V}_R^+ + E_{k,s}^{\text{tr}}. \end{aligned} \quad (\text{TSR30})$$

$D_s^\pm$ are the original unequal FEX flag endpoints; $E_{k,s}^{\text{tr}}$ retains α_A and all four CEP errors. Both low metrics remain, with the exact PBR6 isometry on W_0 . Substitution of TSR30 into the unchanged HCR4 max/min and HCR5 arithmetic, signed and action maps therefore gives bounds for those same original quantities. Subtracting a residual interval from the exact total uses its upper endpoint

in a kernel lower bound and its lower endpoint in a kernel upper bound. The arithmetic comparison remains the correlated original $s = 1$ comparison; the signed bound adds $\mathcal{H}_k = -T_k$ and the full original action budget. These operations follow by adding and subtracting inequalities for the identical scalars and introduce no new source order.

HCR6 evaluates the sum of the two original allocations. TSR19–30 specifies each separately by its own actual threshold spectral flow and gives finite brackets with a proved width at the original scale. Neither individual coefficient is inferred from the evaluated sum.

80 The actual component, mixed-volume and threshold controls together

Fix the complete original simple quartet, unit, five-orbit period, amplitude, logarithm branch and all conductor coefficients on the proved original period domain. The period stays fixed as k grows. Retain $k = 4l + 1 \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = 16k - 48$, $v = \text{ord}_0 E_A$, and $\mathfrak{m} = \Delta - v$. The original sources are exactly $s = 1, k$. Throughout this section R in a subscript is the complementary restriction, with the exact dictionary $R = 0$ in PBR/HCR and GPA/LVM. It does not change the original arithmetic radius R_* .

Here are the complete maps that implement that dictionary:

$$\begin{aligned} B &= Z_k^\top A_\ell : \mathbb{C}^{\mathfrak{m}} \longrightarrow \mathbb{C}^{t_K}, & J &= A_\ell^{-1} P_Y^\top : \mathbb{C}^{t_R} \longrightarrow \mathbb{C}^{\mathfrak{m}}, \\ \text{im } J &= \ker B, & t_K &= (8k - 16) - t_0, & t_R &= (8k - 32) - v + t_0, & t_0 &= \dim(K \cap L), & L &= \mathbb{C}[S]_{<v}, \\ D_{K,N} &= (BD_N^{-1}B^*)^{-1} = \overline{(\widehat{H}_{K,N}^{(s)})}^{-1}, \\ D_{R,N} &= J^* D_N J = \overline{Q_{0,N}^{-1}}, & G_K &= (BG_{\text{exp}}^{-1}B^*)^{-1}, & G_R &= J^* G_{\text{exp}} J. \end{aligned} \tag{SAR1}$$

The full $D_N = T_N^*[I - V_N(V_N^*V_N)^{-1}V_N^*]T_N$ is the original minimum extension Gram. Its entries retain $h_n = M_s n!(s/2)_n$ and $M_s = (2\pi)^{s/2}$. The original P_Y has kernel $\text{im } Z_k$. Thus $BJ = Z_k^\top P_Y^\top = 0$ and the dimensions give $\text{im } J = \ker B$. Substituting $D_N = A_\ell^* \overline{H_{Y,N}^{-1}} A_\ell$ into the two operations proves their conjugate-inverse identities in SAR1 by direct matrix multiplication. This proves the exact native/dual map, including the transpose and adjoint placements. Empty ranks use their actual empty maps and determinant one.

Let $w_{q-1} = w_{2q-1} = 1$ and $w_M = 2$ for $q \leq M \leq 2q - 2$. The original rows, corrected over the whole preceding lower-evaluation space, are

$$\begin{aligned} r_M &= t_M - v_{+,M}(V_M^*V_M)^{-1}V_M^*T_M, & \lambda_M &= 1 + v_{+,M}(V_M^*V_M)^{-1}v_{+,M}^* > 0, \\ h_M &= \lambda_M^{-1/2}r_M, & L_M &= D_M^{-1}B^*D_{K,M}, \\ x_M &= h_M J D_{R,M}^{-1} J^* h_M^*, & y_M &= h_M L_M D_{K,M}^{-1} L_M^* h_M^*, \\ S_{K,s} &= \sum_M w_M \log \left(1 + \frac{y_M}{1 + x_M} \right), & S_{R,s} &= \sum_M w_M \log(1 + x_M), \\ \mathcal{T}_{k,s} &= S_{K,s} + S_{R,s} = \sum_M w_M \log(1 + x_M + y_M). \end{aligned} \tag{SAR2}$$

HCR1's full-fibre minimization gives $D_{M+1} = D_M + h_M^* h_M$. The frame $[J, L_M]$ is invertible since $BL_M = I$ and $\text{im } J = \ker B$; its Gram is $\text{diag}(D_{R,M}, D_{K,M})$. Restriction of the update gives $D_{R,M+1} = D_{R,M} + (h_M J)^*(h_M J)$. Minimizing over the full J -fibre gives the quotient increment $(h_M L_M)^*(h_M L_M)/(1 + x_M)$. The determinant lemma therefore identifies each summand in SAR2 with its own native determinant increment. Four-endpoint telescoping gives exactly these weights, with total weight $2q$. Consequently $S_{K,s} = \widehat{F}_k^{(s)}$ and $S_{R,s} = F_{0,k,s}$, and the final identity follows from $(1 + x_M)(1 + y_M/(1 + x_M)) = 1 + x_M + y_M$.

80.1 Five simultaneous finite descriptions of the same two scalars

Use exactly $X_M, Z_M, \theta_M, \ell_M$ from ACB1–4, including its zero-row convention and $\ell_M = +\infty$ when $X_M = 0 < Z_M$. Put

$$\begin{aligned} C_s &= \sum_M w_M \min\{g_M, \ell_M\}, & E_s^{\text{clip}} &= 2q \log 2, & g_M &= \log(1 + Z_M/\lambda_M), \\ c_{K,s}^- &= C_s - E_s^{\text{clip}}, & c_{K,s}^+ &= C_s, \\ c_{R,s}^- &= \mathcal{T}_{k,s} - C_s, & c_{R,s}^+ &= \mathcal{T}_{k,s} - C_s + E_s^{\text{clip}}, \\ c_{X,s}^- &\leq S_{X,s} \leq c_{X,s}^+ & (X = K, R). \end{aligned} \tag{SAR3}$$

Indeed $Z_M = X_M + Y_M$ in the native inverse metric, so $g_{K,M} = -\log(\theta_M + (1 - \theta_M)e^{-g_M})$. The argument lies between $\max(\theta_M, e^{-g_M})$ and twice that maximum. Taking negative logarithms gives the ACB5 interval of width $\log 2$ at every actual row. Summing with the original weights proves SAR3, and subtracting from the exact total proves the residual endpoints with their signs. The exact joint-tail identity remains $C_s = \int_0^\infty \sum_M w_M \mathbf{1}_{\{g_M > t, \ell_M > t\}} dt$; thus the same row energies and component concentrations remain.

For positive t_R , let $R_M^* R_M = D_{R,M}$ be the ACB12 positive-diagonal upper triangular factor and set $A_M^{\max} = \max_a |(r_M J R_M^{-1})_a|^2$. Use the following actual component endpoints:

$$\begin{aligned} u_M &= \log \frac{\lambda_M + Z_M}{\lambda_M + A_M^{\max}}, & l_M &= \max \left\{ 0, \log \frac{\lambda_M + Z_M}{\lambda_M + t_R A_M^{\max}} \right\}, \\ m_{K,s}^- &= \sum_M w_M l_M, & m_{K,s}^+ &= \sum_M w_M u_M, \\ m_{R,s}^- &= \mathcal{T}_{k,s} - m_{K,s}^+, & m_{R,s}^+ &= \mathcal{T}_{k,s} - m_{K,s}^-, \\ m_{X,s}^- &\leq S_{X,s} \leq m_{X,s}^+, & m_{X,s}^+ - m_{X,s}^- &\leq 2q \log(\max\{1, t_R\}). \end{aligned} \tag{SAR4}$$

If $t_R = 0$, define $l_M = u_M = g_M$ instead; then the kernel allocation equals the total and the residual allocation is zero. The equality $(J R_M^{-1})^* D_M (J R_M^{-1}) = I$ proves $X_M = \sum_a |(r_M J R_M^{-1})_a|^2$. Hence $A_M^{\max} \leq X_M \leq t_R A_M^{\max}$. The logarithm of $(\lambda + Z)/(\lambda + X)$ is decreasing in X ; these inequalities, and its nonnegativity, prove both row endpoints. Their width is at most $\log t_R$, including zero pairings. Summation and the exact total prove every assertion in SAR4. The entire original Cholesky matrix, native pairing and lower-evaluation denominator are present.

On the finite domain $k \geq 29$, $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q$, use the complete GPA17–19 coefficient forms and all unchanged relation columns. For $X = K, R$ set $F_{R,s}^{\text{exp}} = F_0^{\text{exp}}$, $d_{R,s}^\pm = d_{0,k,s}^\pm$ and likewise for the LVM scalars. Then define

$$\begin{aligned} p_{X,s}^- &= F_{X,s}^{\text{exp}} + d_{X,k,s}^-, & p_{X,s}^+ &= F_{X,s}^{\text{exp}} + d_{X,k,s}^+, \\ v_{X,s}^- &= B_{X,k,s} + d_{X,k,s}^- + \eta_{X,k}^-, & v_{X,s}^+ &= B_{X,k,s} + d_{X,k,s}^+ + \eta_{X,k}^+, \\ p_{X,s}^- &\leq S_{X,s} \leq p_{X,s}^+, & v_{X,s}^- &\leq S_{X,s} \leq v_{X,s}^+. \end{aligned} \tag{SAR5}$$

Here $Q_1 = q'$ and $Q_k = q' + (k - 1)/4$ stay separate. The coefficient map is exactly $P_{j,n} = \binom{n}{j} (c')^{n-j} i^j$ with inverse $y = (S' - c')/i$, followed by the two-block map $W = [\text{AP}; \text{RAP}]$ of LVM6. Its full Gram is the factorial-moment form GPA17. Quotienting over the same U_N and then using the same Z_k, P_Y proves GPA20's full-form transport. At rank t_X it yields the two unequal errors

$$\begin{aligned} d_{X,k,s}^- &= t_X (b_{q-1,s}^- + b_{q,s}^- - b_{2q-1,s}^+ - b_{2q,s}^+), \\ d_{X,k,s}^+ &= t_X (b_{q-1,s}^+ + b_{q,s}^+ - b_{2q-1,s}^- - b_{2q,s}^-). \end{aligned} \tag{SAR6}$$

Every $b_{N,s}^\pm$ is the full GPA14 constant, including the original root error, source recurrence, Gamma sandwich and rate-change terms. The factor β_s cancels only in this four-sign determinant sum. LVM10 then gives the exact consecutive volume ratios. Expanding each of their full Gram determinants gives $2b_{A,N,s} + \zeta_{A,N,s}$ with $0 \leq \zeta_{A,N,s} \leq c_{A,N}$ for every mixed row subset. Taking the

four original signs proves the two retained $\eta_{X,k}^\pm$ of LVM15 and the second interval of SAR5. This establishes the displayed morphisms to the actual native allocation; no original relation or mixed minor is omitted. Outside the stated finite domain the entries SAR5 are absent.

For all original $k \geq 9$, use a finite TSR threshold grid on $[a_{k,s}, b_{k,s}]$ containing 1, with its full TSR21–23 cutoffs. Let $V_X^\pm, A_X^\pm, P_X^\pm$ be the actual sums TSR24–25. They are bounds for the total, below-one and above-one integrals of the same pencil flow, respectively. Define the retained threshold interval by

$$\begin{aligned} z_{X,s}^- &= \max\{V_X^-, A_X^-\}, & z_{X,s}^+ &= \min\{V_X^+, A_X^+ + E_{k,s}^\vee\}, \\ z_{X,s}^- &\leq S_{X,s} \leq z_{X,s}^+, & 0 &\leq z_{X,s}^+ - z_{X,s}^- \leq 4t_X h_{k,s}^{\text{grid}}, \\ \sum_{X=K,R} (z_{X,s}^+ - z_{X,s}^-) &\leq 4\mathbf{m} h_{k,s}^{\text{grid}}. \end{aligned} \quad (\text{SAR7})$$

The mesh $h_{k,s}^{\text{grid}}$ is the greatest logarithmic cell width, and is zero for the degenerate single-node case. To prove the exact target identity, the frame $[J, L_N]$ carries $D_N - tB^*G_K B$ to $\text{diag}(D_{R,N}, D_{K,N} - tG_K)$. Its threshold kernel is lifted by L_N with inverse B ; TSR5 also retains the equivalent factor- t lift. The residual pencil is exactly $J^*(D_N - tG_{\text{exp}})J$. Their successive original positive rank-one updates give the flows $f_{X,M}(t) \in \{0, 1\}$. Integrating the finite eigenvalue step differences against dt/t proves

$$\begin{aligned} S_{X,s} &= \int_0^\infty \mathcal{N}_X(t) \frac{dt}{t}, & A_{X,k,s} &= \int_0^1 \mathcal{N}_X(t) \frac{dt}{t}, & P_{X,k,s} &= \int_1^\infty \mathcal{N}_X(t) \frac{dt}{t}, \\ \mathcal{N}_X(t) &= \sum_M w_M f_{X,M}(t), & S_{X,s} &= A_{X,k,s} + P_{X,k,s}, & 0 &\leq P_{X,k,s} \leq E_{k,s}^\vee. \end{aligned} \quad (\text{SAR8})$$

The source cutoffs bound every generalized eigenvalue, so these integrals are finite and retain every threshold nullity. Monotonicity of each of the four counts bounds the integral on a cell by TSR24's lower and upper coefficients. Its width is at most the sum of their four count increases times that cell's logarithmic width. Summing telescopes to $4t_X h_{k,s}^{\text{grid}}$. These facts and the unchanged positive allowance prove SAR7, without assigning that allowance zero. TSR26–28 gives a common grid with joint width at most $\eta k q$ using $O_{\text{actual}}(\eta^{-1} \log k + 1)$ nodes for each $\eta > 0$. This counts inertia evaluations with the actual input forms, not their arithmetic bit complexity or an executed endpoint value.

80.2 Simultaneous intervals with exact total coupling

For each X let $\mathcal{I}_{X,s}$ contain the intervals $[0, \mathcal{T}_{k,s}]$, SAR3, SAR4 and SAR7, together with both SAR5 intervals and the deterministic-pivot interval SAR16 precisely on their common finite domain. Put

$$\begin{aligned} a_{X,s} &= \max_{[l,u] \in \mathcal{I}_{X,s}} l, & b_{X,s} &= \min_{[l,u] \in \mathcal{I}_{X,s}} u, \\ \underline{S}_{K,s} &= \max\{a_{K,s}, \mathcal{T}_{k,s} - b_{R,s}\}, & \bar{S}_{K,s} &= \min\{b_{K,s}, \mathcal{T}_{k,s} - a_{R,s}\}, \\ \underline{S}_{R,s} &= \mathcal{T}_{k,s} - \bar{S}_{K,s}, & \bar{S}_{R,s} &= \mathcal{T}_{k,s} - \underline{S}_{K,s}, \\ \underline{S}_{X,s} &\leq S_{X,s} \leq \bar{S}_{X,s}. \end{aligned} \quad (\text{SAR9})$$

Each component interval contains the actual scalar by the proofs above; therefore $a_X \leq S_X \leq b_X$. The exact identity $S_R = \mathcal{T} - S_K$ proves the two additional kernel bounds and both residual bounds. More precisely the map $z \mapsto (z, \mathcal{T} - z)$ is an affine isomorphism from $[\underline{S}_K, \bar{S}_K]$ onto

$$([a_K, b_K] \times [a_R, b_R]) \cap \{(u, v) : u + v = \mathcal{T}\},$$

with inverse the first coordinate. The equivalence follows by solving both residual inequalities for z . This proves simultaneous consistency and the endpoint reversal; it retains the measured total instead of assigning its two parts. Both resulting interval widths equal $\bar{S}_K - \underline{S}_K$ and are at most the width of either original component interval. In particular the pair has joint width at most $2 \min(4t_K h_{k,s}^{\text{grid}}, 4t_R h_{k,s}^{\text{grid}})$ and also at most the TSR joint bound $4\mathbf{m} h_{k,s}^{\text{grid}}$.

80.3 The unchanged low metrics and all original finite errors

Retain $F_L^{(s)}$ and $F_{W_0}^{(s)}$ individually in their original norms, with $W_0 = \Xi(L)$ and $\lambda_s^{\text{low}} = F_L^{(s)} - F_{W_0}^{(s)}$. The map $(K + L)/K \rightarrow W_0$, $[p] \mapsto \Xi p$, is an isometry: its entire fibre is $p + K \subset K + L$, so the two native quotient minima agree. This proves the exact low-space identification and leaves both volumes present. Use precisely the PBR8 unequal flag endpoints $D_s^\pm$, the PBR7 transfer allowance $E_{k,s}^{\text{tr}} = 2 \sum_N E_N^{\text{CEP}} + 8t_R |\log |\alpha_A||$, and all their original finite constants. HCR3 proves

$$\begin{aligned} F_K^{(s)} &= S_{K,s} - \mathcal{E}_s^{\text{flag}} + \lambda_s^{\text{low}}, \quad \mathcal{E}_s^{\text{flag}} \in [D_s^-, D_s^+], \\ \Phi_{k,s} &= F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}, \quad |\delta_{0,k,s}| \leq E_{k,s}^{\text{tr}}, \\ k_s^- &= \underline{S}_{K,s} - D_s^+ + \lambda_s^{\text{low}}, \quad k_s^+ = \bar{S}_{K,s} - D_s^- + \lambda_s^{\text{low}}, \\ r_s^- &= F_{W_0}^{(s)} + \underline{S}_{R,s} - E_{k,s}^{\text{tr}}, \quad r_s^+ = F_{W_0}^{(s)} + \bar{S}_{R,s} + E_{k,s}^{\text{tr}}. \end{aligned} \tag{SAR10}$$

Subtracting the flag error uses its upper endpoint in a lower kernel bound; adding the transfer error gives both residual signs. Thus $F_K^{(s)} \in [k_s^-, k_s^+]$ and $\Phi_{k,s} \in [r_s^-, r_s^+]$. Every $\zeta_{N,s} = \log \prod_{d=v}^N [d!/(s/2)_d] / [(d-v)!/(s/2)_{d-v}]$ and original coefficient factor m_N in PBR8 remains in $D_s^\pm$; no endpoint flag or source-mass term is discarded.

Intersect with the complete preceding HCR4 receiver by defining

$$\begin{aligned} \Phi_{k,s}^{-,\text{all}} &= \max\{\Phi_{k,s}^{-,\text{row}}, r_s^-\}, \quad \Phi_{k,s}^{+,\text{all}} = \min\{\Phi_{k,s}^{+,\text{row}}, r_s^+\}, \\ L_s^{\text{all}} &= \max\{L_s^{\text{row}}, k_s^-, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{+,\text{all}} - e_{k,s}^-\}, \\ U_s^{\text{all}} &= \min\{U_s^{\text{row}}, k_s^+, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{-,\text{all}} + e_{k,s}^+\}, \\ \Phi_{k,s} &\in [\Phi_{k,s}^{-,\text{all}}, \Phi_{k,s}^{+,\text{all}}], \quad F_K^{(s)} \in [L_s^{\text{all}}, U_s^{\text{all}}]. \end{aligned} \tag{SAR11}$$

The OAR errors in this display are those of $F_K = \mathcal{F}_\partial - \Phi + \mathfrak{e}$, $-e^- \leq \mathfrak{e} \leq e^+$, distinct from the flag errors. Each maximum entry is a lower bound and each minimum entry an upper bound for that exact scalar. This proves all intervals are nonempty and retain every older branch. On the exact PMR domain $k \geq 29$ and $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q$, include additionally

$$\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{+,\text{all}} - e_{k,s}^{0,-} \quad \text{and} \quad \mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{-,\text{all}} + e_{k,s}^{0,+} \tag{SAR12}$$

in the respective maximum and minimum, where $e^{0,\pm} = e^\pm + c_k^0$ retains the full seven-moment error. These are the PMR form of that same scalar identity and prove the added bounds. Outside this domain these two entries and SAR5 are omitted; every original OAR expression stays in force.

80.4 Arithmetic, signed return and the entire action

On the original arithmetic period domain $|u| \geq R_*$ retain the same correlated difference interval of PBR14,

$$\begin{aligned} a_{K,k}^- &= \max\{-C_K^-, \delta_k^\sigma - C_B^+\}, \quad a_{K,k}^+ = \min\{C_K^+, \delta_k^\sigma + C_B^-\}, \\ L_{\text{ar}}^{\text{all}} &= \max\{L_{\text{ar}}^{\text{row}}, L_1^{\text{all}} + a_{K,k}^-\}, \quad U_{\text{ar}}^{\text{all}} = \min\{U_{\text{ar}}^{\text{row}}, U_1^{\text{all}} + a_{K,k}^+\}, \\ F_K^{\text{ar}} &\in [L_{\text{ar}}^{\text{all}}, U_{\text{ar}}^{\text{all}}], \quad F_B^{\text{ar}} \in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^{\text{all}}, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^{\text{all}}]. \end{aligned} \tag{SAR13}$$

The identical scalar $F_K^{\text{ar}} - F_K^{(1)}$ belongs to $[a_{K,k}^-, a_{K,k}^+]$. Adding this interval to SAR11 and intersecting with HCR5 proves SAR13; no independence between those errors is required. Subtracting from $F_K^{\text{ar}} + F_B^{\text{ar}} = \mathcal{B}_k^{\text{ar}}$ proves the reversed boundary endpoints. All source orders and finite AMT constants remain the original ones.

For the original $s = k$ signed return and untilted action retain

$$\begin{aligned}
C_k^{\text{budget}} &= F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \\
\Psi_k &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, \quad \mathcal{H}_k = -T_k, \quad J_k^{\text{action}} = \Psi_k + C_k^{\text{budget}}, \\
L_\Psi^{\text{all}} &= L_k^{\text{all}} + \mathcal{H}_k + \delta_k^\sigma, \quad U_\Psi^{\text{all}} = U_k^{\text{all}} + \mathcal{H}_k + \delta_k^\sigma, \\
L_J^{\text{all}} &= L_\Psi^{\text{all}} + C_k^{\text{budget}}, \quad U_J^{\text{all}} = U_\Psi^{\text{all}} + C_k^{\text{budget}}, \\
\Psi_k &\in [L_\Psi^{\text{all}}, U_\Psi^{\text{all}}], \quad 4q \log \mathcal{L}_{h,k} \leq J_k^{\text{action}} \in [L_J^{\text{all}}, U_J^{\text{all}}], \\
\mathcal{L}_{h,k} &\leq \min_M \epsilon_M^{\text{ar}} \leq \prod_M (\epsilon_M^{\text{ar}})^{w_M/(2q)} \leq \exp(U_J^{\text{all}}/(4q)).
\end{aligned} \tag{SAR14}$$

The first two equalities are exactly HCR5/PBR15. Addition of the same scalars to SAR11 proves the signed and action intervals. For the original untilted reflection the identity $J_k^{\text{action}} = 2 \sum_M w_M \log \epsilon_M^{\text{ar}}$ holds, while the same-class radius is $\mathcal{L}_{h,k} = 2\delta e(k+1)\lfloor (k+1)^2/4 \rfloor$ with $e = 1$ on this simple-quartet branch. Applying its lower bound at every degree and $\sum_M w_M = 2q$ proves the last lines. The reference boundary, full monic window, both contraction penalties, action leakage and the exact mixed term $a_K d_B + a_B d_K + 2|z_K|^2$ with $z_B = -z_K$ are retained. For a real tilt the original MCE expression with $\log(\epsilon_M^2 + \phi_M^2)$ and exact MCP derivative remains in force; the untilted geometric mean is used only on its proved reflection domain.

Finally, on the fixed actual period sequence all five descriptions return to the same evaluated total, with their separate errors:

$$\begin{aligned}
S_{K,s} + S_{R,s} &= \Delta q \mathfrak{C}_\partial + o(kq), \\
0 &\leq C_s - S_{K,s} \leq 2q \log 2, \\
d_{X,k,s}^- &\leq S_{X,s} - F_{X,s}^{\text{exp}} \leq d_{X,k,s}^+, \\
d_{X,k,s}^- + \eta_{X,k}^- &\leq S_{X,s} - B_{X,k,s} \leq d_{X,k,s}^+ + \eta_{X,k}^+, \\
|d_{X,k,s}^\pm| &= O_{\text{actual}}(k \log q), \quad |\eta_{X,k}^\pm| \leq (4q + 8\mathfrak{m} + 4) \log 2, \\
B_{K,k,s} + B_{R,k,s} &= 2 \sum_N \sigma_N (b_{E,N,s} - b_{U,N,s}).
\end{aligned} \tag{SAR15}$$

The first line is HCR6. SAR3 proves the second, GPA23 and LVM16–17 prove the next three, and the final identity cancels the same actual intermediate minor at each endpoint in LVM14. Each displayed finite formula precedes the limit. The individual allocation is given by its retained components, full factorial quotients, mixed minors or threshold flows; the sum does not assign a separate leading coefficient. These exact receiving maps carry each subsequent evaluation of those quantities into every earlier kernel, boundary, signed and action bound without changing its objects.

80.5 Deterministic full-row pivots in the same interval

Retain the actual two-half-line matrices $V_{A,N,s} = W_{N,s} X_{A,N}$ of LVM6–8. On precisely the GPA2 domain, GVM1 takes their entire row Gram, chooses the largest actual residual diagonal with the original-row tie rule, and subtracts its full rank-one Schur term. Write its positive pivots as $p_{A,N,s;j}$, $\kappa_{A,N,s} = \log \prod_j p_{A,N,s;j}$, and retain the scalar $C_{X,k,s}$ and the unequal $e_{X,k}^\pm$ of GVM6–7. These $C_{X,k,s}$ have three subscripts and are distinct from the clipped sum C_s of SAR3. The pivot errors $e_{X,k}^\pm$ are distinct from every OAR error $e_{k,s}^\pm$ in SAR11. Define

$$\begin{aligned}
g_{X,s}^- &= C_{X,k,s} + d_{X,k,s}^- + e_{X,k}^-, \quad g_{X,s}^+ = C_{X,k,s} + d_{X,k,s}^+ + e_{X,k}^+, \\
g_{X,s}^- &\leq S_{X,s} \leq g_{X,s}^+, \quad X = K, R, \\
|e_{X,k}^\pm| &\leq (2q + 4\mathfrak{m} + 2) \log(2(q + \mathfrak{m} + 1)), \\
C_{K,k,s} + C_{R,k,s} &= \sum_N \sigma_N (\kappa_{E,N,s} - \kappa_{U,N,s}).
\end{aligned} \tag{SAR16}$$

As throughout this section, the R -indexed data in this formula are exactly the 0-indexed data of GVM, through SAR1.

Here is the complete map and finite-error justification of this additional interval. For one actual full-column-rank matrix V with $M = 2n_N$ rows and rank r , residual Gram–Schmidt gives an orthonormal row basis Q and the exact coordinate map $R = VQ^*$, with inverse $V = RQ$. At every step j , the maximal residual diagonal p_j is positive because the remaining row span has dimension $r - j + 1$. The retained bound $|R_{ij}|^2 \leq p_j$ follows by projecting the same original row off the preceding selected directions and applying Cauchy–Schwarz. The selected rows in pivot order are triangular with diagonal $\sqrt{p_j}$, so their squared minor is $\prod_j p_j$. This is a summand of the complete Cauchy–Binet volume. Conversely each column of R has squared norm at most Mp_j , and its successive orthogonal residual has no larger norm; multiplication of these squared residual norms gives $\det(R^*R) \leq M^r \prod_j p_j$. Thus

$$0 \leq \xi_{A,N,s} := \log D_{A,N,s} - \kappa_{A,N,s} \leq r_{A,N} \log(2n_N).$$

The two actual consecutive volume ratios LVM10 now give exactly $F_X^{\text{exp}} = C_{X,k,s} + \sum_N \sigma_N(\xi_{a_X,N,s} - \xi_{b_X,N,s})$. At a positive endpoint this remainder lies between $-c_{b_X,N}^{\text{piv}}$ and $c_{a_X,N}^{\text{piv}}$; the negative endpoints reverse them. Summation is precisely the two $e_{X,k}^{\pm}$ of GVM7. Adding the same original GPA22 errors proves both SAR16 endpoints. The ranks $r_{A,N} \leq n_N$, maximum $n_N \leq q + \mathfrak{m} + 1$, and sum $\sum_N n_N = 2q + 4\mathfrak{m} + 2$ prove its stated finite constant. At rank zero the empty product and determinant are both one; the same argument includes the zero residual target without requiring the two separate pivot products to coincide.

Thus SAR9 includes this further interval for each same native scalar, and SAR10–14 proves its original arithmetic and whole-action return without further changes. Its finite error is $O_{\text{actual}}(k \log q + q \log q) = o(kq)$. Every row from both original half-lines participates in the maximal-diagonal decision and full update. The explicit algorithm uses the twenty-four original matrices and $O(q^3)$ arithmetic operations and exact comparisons, as counted in GVM1–9; this is an operation count for their given entries, not a bit-complexity or transcendental-comparison certificate. The last SAR16 identity cancels the same intermediate pivot product only in the total. No individual leading allocation is assigned.

80.6 The attained multiplier-image interval in the same allocation pool

Use the literal common polynomial cutoffs, both source-order pairs, full individual low-jet image and native image quotient proved in NMR1–9. On $k \geq 37$ their achieved finite interval is

$$\begin{aligned} m_{k,s}^{\text{nat},-} &= V_{Q,k,s} + n_{k,s}^{\text{mult}} - E_{k,s}^{\text{mult,lo}}, & m_{k,s}^{\text{nat},+} &= V_{Q,k,s} + n_{k,s}^{\text{mult}} + E_{k,s}^{\text{mult,hi}} + E_{k,s}^{\text{miss}}, \\ [m_{k,s}^{\text{nat},-}, m_{k,s}^{\text{nat},+}] &\ni S_{K,s}, & [\mathcal{T}_{k,s} - m_{k,s}^{\text{nat},+}, \mathcal{T}_{k,s} - m_{k,s}^{\text{nat},-}] &\ni S_{R,s}, \\ E_{k,s}^{\text{miss}} &= 2\varepsilon_k \log A_{k,s}^{\text{end}}, & E_{k,s}^{\text{mult,lo}} + E_{k,s}^{\text{mult,hi}} &= 2a_k \sum_{N \in \{q-1, q, 2q-1, 2q\}} \log \max\{1, \mathcal{K}(N - v)\}. \end{aligned} \quad (\text{SAR17})$$

The proof is the exact NMR9 identity $S_K = V_Q + n^{\text{mult}} + p^{\text{mult}} + V_{\text{miss}}$, with $-E_{k,s}^{\text{mult,lo}} \leq p^{\text{mult}} \leq E_{k,s}^{\text{mult,hi}}$ and $0 \leq V_{\text{miss}} \leq E_{k,s}^{\text{miss}}$ proved on the actual same image. Subtraction from the exact total proves the residual interval and its endpoint reversal. The complete source-kernel return, all four displaced windows and every negative matched-relative logarithm remain in $V_Q + n^{\text{mult}}$. The exact maps containing α_A , every original t_0 and the coupled low-jet coordinates are NMR1–3; no metric is assigned to their additional nonanalytic individual summands.

Adjoin these two intervals to the respective SAR9 pools, and adjoin all four NMR10 branch intervals and their exact total partners on $k \geq 9$. Every new entry bounds the same scalar, so the affine map in SAR9 proves the simultaneous intersection. NMR11 writes its endpoints explicitly. The substitution through SAR10–14 is proved in full in NMR12–15 with the original unequal finite errors, period/PMR domains and complete action budget. The displayed interval has the exact finite width in NMR16, of order $o(kq)$ at fixed actual data. This controls its error width while

retaining its unevaluated center; it assigns neither an inverse bound nor an allocation coefficient. The proof in NMR uses the complete preceding SAR1–16, and the insertion here is its downstream application.

80.7 The native boundary continuation in the simultaneous allocation

Let $[s_{K,s}^-, s_{K,s}^+]$ be the intersection with the exact new NMR18 window interval, retaining all earlier SAR/NMR finite entries on their own domains. Then

$$(S_{K,s}, S_{R,s}) \in \{(z, \mathcal{T}_{k,s} - z) : s_{K,s}^- \leq z \leq s_{K,s}^+\}. \quad (\text{SAR18})$$

The new lower and upper entries bound the same S_K by the complete NW/OW/WNR proof and NMR's determinant factorization. Their intersection therefore contains S_K . The previous exact total gives the asserted affine set, whose inverse is its first coordinate. Its finite errors retain the source-kernel volume, each negative matched logarithm, all unequal window errors and the actual missing rank, now bounded using the proved $v \leq 32$. NBR4–15 additionally computes the complete degree-four image correction through the original source kernel and both native gauge forms; it does not replace that image by the kernel target. PLF's full polar metric equals the existing full residual after addition of the original low volume and the same CEP/alpha transport error. HCR8 and NBR20–23 give the exact receiving maps through every original flag, arithmetic and action allowance. The window little-oh is used only when every nonzero source degree satisfies OW3; all earlier finite degrees remain covered by the original outer-determinant interval.

81 The actual branch correspondence between the two row allocations

All vector spaces, polynomial quotients and duals below are over $\mathbb{C}$. Duals are complex-linear. Retain the full original simple quartet, actual amplitude, fixed period and logarithm branch of OQC, OCF, FGC, RDS, DNE, PEC and HJR. The period is on the proved five-orbit locus, in particular on the original domain $|u| \geq R_*$. The two source orders remain $s = 1, k$, with $k = 4l + 1 \geq 9$. Put

$$\begin{aligned} q &= (k+1)^2, \quad q' = (k-7)^2, \quad \Delta = q - q' = 16k - 48, \\ v &= \text{ord}_0 E_A, \quad \alpha_A = v!/\mu_v, \quad \mathbf{m} = \Delta - v, \quad r = 8k - 16, \quad j = 8k - 32, \\ t_0 &= \dim(K \cap L), \quad t_K = r - t_0, \quad j_0 = j - v + t_0, \quad t_K + j_0 = \mathbf{m}. \end{aligned} \quad (\text{KRC1})$$

Here $K = \ker \Lambda_k \subset V_A = \ker C$ is the actual original kernel, $L = \mathbb{C}[S]_{<v} \subset V_A$, and C is the complete primary conductor of RDS7. Neither a symbol zero order nor a coefficient-support case is changed.

81.1 The complete conductor dual and the low annihilator

Let $S_x = \mathbb{C}[x_1, x_2, x_3, x_4]$ with its original homogeneous grading and let $gf(x) = f(D^{-1}x)$, where $D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. Keep

$$\begin{aligned} \mathbf{H} &= F^{-1} \Pi U \mathcal{E}_{\text{prim}}, \quad Q = Q_0(\mathbf{H}^{-1}x), \quad Q_0(t) = t_{++}t_{--} - t_{+-}t_{-+}, \\ Q_a &= g^a Q, \quad \mathcal{A} = \prod_{a=1}^4 Q_a, \quad R = S_x/(Q), \quad \mathcal{C}_k = (S_x/(Q, \mathcal{A}))_k, \\ \mathcal{J}_k &= \text{im}(S_{x,k}^{(g)} \rightarrow \mathcal{C}_k). \end{aligned} \quad (\text{KRC2})$$

All coefficients of $\mathbf{H}$, Q_a and $\mathcal{A}$ are literal original period and unit coefficients. OQC3 and the five-orbit test show that the Q_a are pairwise nonassociate prime quadrics.

The OCF substitution $x = \mathbf{H}(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^T$ identifies R_k with the complete biform coefficient space $\mathcal{B}_k$. The product $\mathcal{A}$ restricts to precisely $A = \sum a_{ab} e_{ab}^{(8)}$, and its multiplication matrix is $C^\top$ by RDS7. Thus OCF7,24 and OQC8 give the exact maps

$$\begin{aligned} \mathcal{C}_k &\xrightarrow{\Theta_{A,k}} V_A^\vee, & [\nu] &\longmapsto (p \longmapsto \sum_{a,b} \nu_{ab} p(\beta_{ab})), \\ \Theta_{A,k}(\mathcal{J}_k) &= \text{Ann}_{V_A^\vee} K, & \mathcal{C}_k / \mathcal{J}_k &\simeq K^\vee, \\ \dim \mathcal{C}_k &= \Delta, & \dim \mathcal{J}_k &= j. \end{aligned} \tag{KRC3}$$

For completeness, the kernel of restriction of $E_k^\vee$ to $\ker C$ is $\text{im } C^\top$: one inclusion follows by evaluating on $\ker C$, and the other follows by factoring a vanishing functional through the onto map C . This proves the first quotient in KRC3, with its ordinary transpose. The actual observation dual has image J_k by OQC8. The conductor image lies in it by the full trace identity $[\mathbf{5R}(\mathcal{A}f)]_{(Q)} = [\mathcal{A}f]_{(Q)}$. Quotienting this inclusion gives the second line of KRC3. The dimensions are the exact OCF8,19 formulas.

Write N_k, S_k for the full OCF strip reduction and its section. The low annihilator and its invariant part are

$$\begin{aligned} \mathcal{C}_{v,k} &= \text{Ann}_{\mathcal{C}_k} L = \ker \left[\begin{pmatrix} \beta_{ab}^d \\ 0 \leq d < v \\ 0 \leq a, b \leq k \end{pmatrix} S_k \right], & \mathcal{J}_{v,k} &= \mathcal{J}_k \cap \mathcal{C}_{v,k}, \\ \mathcal{C}_{v,k} / \mathcal{J}_{v,k} &\simeq (K / (K \cap L))^\vee, & \mathcal{J}_{v,k} &\simeq (V_A / (K + L))^\vee, \\ \dim \mathcal{C}_{v,k} &= \mathbf{m}, & \dim \mathcal{J}_{v,k} &= j_0, & \dim(\mathcal{C}_{v,k} / \mathcal{J}_{v,k}) &= t_K. \end{aligned} \tag{KRC4}$$

The moment matrix annihilates $C^\top$ because the first v derivatives of $E_A e^{\beta'z}$ vanish. Hence it descends through the strip and is the displayed restriction to L . Restriction of all functionals on V_A to L is onto, so its rank is v . Similarly restriction of $\text{Ann } L$ to K has image exactly $\text{Ann}_{K^\vee}(K \cap L)$: extend a functional from $K + L$ by setting it zero on L , then extend linearly to V_A . Its kernel is $\text{Ann}(K + L)$. This proves every map and dimension in KRC4, including nonzero t_0 and all empty-rank conventions.

There is a scalar in the exact DNE-to-OCF bridge. In the DNE coefficient space let Z_{exp} be its full fixed numerator frame, A_ℓ its complete functional matrix, and let Z_k be the FGC9 factorial-scaled kernel frame. Set

$$\begin{aligned} \mathbf{T}_k &= \alpha_A N_k Z_{\text{exp}} : \mathbb{C}^{\mathbf{m}} \xrightarrow{\sim} \mathcal{C}_{v,k}, & B &= Z_k^\top A_\ell, & J &= A_\ell^{-1} P_Y^\top, \\ \text{im } J &= \ker B, & \mathbf{T}_k(\text{im } J) &= \mathcal{J}_{v,k}. \end{aligned} \tag{KRC5}$$

To check the scalar, write $F = \mathcal{N}/E_A$. For every polynomial p , the exact shift operator satisfies

$$\Lambda_F(\mathcal{T}_A p) = \Lambda_{E_A F}(p) = \sum \nu_{ab} p(\beta_{ab}). \tag{KRC6}$$

Indeed it suffices to check $p(S) = S^d$, for which the coefficient of $z^d/d!$ in $E_A(z)F(z)$ is the binomial moment convolution of the original shifts. Under FGC's map $p \mapsto \alpha_A \mathcal{T}_A p / \chi_{k-8}$, the functional $\ell(g) = \Lambda_F(\chi_{k-8} g)$ therefore pulls back to $\alpha_A \sum \nu_{ab} p(\beta_{ab})$. This proves KRC5 with the full scalar. RDS8 gives injectivity and dimension of this quotient map, hence its bijectivity. DNE9 and PEC24 give $BJ = 0$ and its complete kernel; evaluating on the actual image of K proves the last equality in KRC5. In particular no conjugate transpose replaces $Z_k^\top$ in this identification.

81.2 Exact exchange on the four original intersection components

Use the four full scheme-theoretic degree- k quotients

$$\begin{aligned} \mathcal{D}_{a,k} &= (S_x / (Q, Q_a))_k \quad (1 \leq a \leq 4), & \mathcal{D}_k &= \bigoplus_{a=1}^4 \mathcal{D}_{a,k}, \\ \mathbf{j} : \mathcal{C}_k &\longrightarrow \mathcal{D}_k, & [f] &\longmapsto ([f]_{(Q, Q_a)})_{a=1}^4. \end{aligned} \tag{KRC7}$$

The map is well defined because each Q_a divides $\mathcal{A}$. No ideal is replaced by its radical, so repeated intersection components remain in these spaces. Each $\mathcal{D}_{a,k}$ has dimension $4k$. Indeed $(Q) \cap (Q_a) = (QQ_a)$ in S_x , by the primality and nonassociation of these two polynomials. The complete degree- k relation space is therefore the image of $(f, h) \mapsto Qf + Q_a h$, whose kernel consists of $(Q_a b, -Qb)$. Taking dimensions yields $\binom{k+3}{3} - 2\binom{k+1}{3} + \binom{k-1}{3} = 4k$.

Indices on Q_a are now read modulo five. Define

$$(\mathbf{T}_{\text{br}} d)_a = g^a d_{-a}, \quad \mathbf{T}_{\text{br}}^2 = I, \quad \mathcal{D}_k^\pm = \ker(\mathbf{T}_{\text{br}} \mp I), \quad \mathbf{P}_\pm = (I \pm \mathbf{T}_{\text{br}})/2. \quad (\text{KRC8})$$

This is a typed map: g^a takes the ideal (Q, Q_{-a}) onto (Q_a, Q) , so its indicated source and target are exactly $\mathcal{D}_{-a,k}$ and $\mathcal{D}_{a,k}$. Composition with g^{-a} is the identity, which proves the involution. The maps

$$\begin{aligned} \mathbf{e}_\pm : \mathcal{D}_{1,k} \oplus \mathcal{D}_{2,k} &\xrightarrow{\sim} \mathcal{D}_k^\pm, \\ (d_1, d_2) &\longmapsto (d_1, d_2, \pm g^3 d_2, \pm g^4 d_1) \end{aligned} \quad (\text{KRC9})$$

have inverse extraction of the first two components. In particular both eigenspaces have dimension $8k$.

Retain all four sign choices $\epsilon_1, \epsilon_2 \in \{1, -1\}$, with $\epsilon_{-a} = -\epsilon_a$, and let $\mathbf{E}_\epsilon d = (\epsilon_a d_a)_a$. These are precisely the diagonal sign involutions anticommuting with $\mathbf{T}_{\text{br}}$; substitution in KRC8 proves both directions of this assertion. Therefore

$$\mathbf{E}_\epsilon^2 = I, \quad \mathbf{E}_\epsilon \mathbf{T}_{\text{br}} = -\mathbf{T}_{\text{br}} \mathbf{E}_\epsilon, \quad \mathbf{E}_\epsilon : \mathcal{D}_k^+ \xrightarrow{\sim} \mathcal{D}_k^-. \quad (\text{KRC10})$$

No sign choice is selected by a desired spectral allocation.

If f is invariant, then $(\mathbf{T}_{\text{br}} j[f])_a = [g^a f]_{(Q, Q_a)} = [f]_{(Q, Q_a)}$. It follows that

$$\mathbf{j}(\mathcal{J}_k) \subset \mathcal{D}_k^+, \quad \delta_k = \mathbf{e}_-^{-1} \mathbf{P}_- \mathbf{j} : \mathcal{C}_k \longrightarrow \mathcal{D}_{1,k} \oplus \mathcal{D}_{2,k}, \quad \delta_k \mathcal{J}_k = 0. \quad (\text{KRC11})$$

The factor $1/2$ in this exact difference map is retained. It therefore induces a specified map on $\mathcal{C}_k / \mathcal{J}_k = K^\vee$.

These formulas retain the full descent defect. In particular

$$\ker \mathbf{j} = \left(\bigcap_{a=1}^4 (Q, Q_a) \right)_k / (Q, \mathcal{A})_k, \quad \ker(\bar{\delta}_k) = \{[f] : \mathbf{j}[f] = \mathbf{T}_{\text{br}} \mathbf{j}[f]\} / \mathcal{J}_k. \quad (\text{KRC12})$$

Both are exact equalities obtained by setting the four quotient components or their displayed differences to zero. No pairwise-coprimality assertion is made about Q_a after restriction modulo Q . Such an assertion does not follow from their primality in S_x . KRC12 supplies the actual maps and ideal quotients that retain shared components and all their scheme-theoretic multiplicities.

Every matrix in KRC7–12 has a finite formula in the given period coefficients. In the ordered degree- k monomial frame form the two multiplication matrices $[M_Q, M_{Q_a}]$. Select the first independent columns and the first invertible row minor, subtract their exact pivot expansion, and keep the complementary rows. This is a quotient map with kernel exactly their column image, by the same subtraction proof as OCF21–22. The action g^a in that monomial frame is the diagonal matrix with entry $\zeta^{-a} \sum i \nu_i$ on x^ν . A complete lift of $e_{ab}^{(k)}$ through the Segre substitution is

$$t_{++}^h t_{+-}^{a-h} t_{-+}^{b-h} t_{--}^{k-a-b+h}, \quad h = \max(0, a + b - k), \quad t = \mathbf{H}^{-1} x. \quad (\text{KRC13})$$

Composing that lift with S_k and the indicated quotient matrices computes $\mathbf{j}$ entry by entry. The kernels and cokernels in KRC12 are thus fully specified finite linear spaces on every exact minor stratum. No bound on a moving minor is inferred from its nonvanishing.

81.3 The induced original kernel–residual correspondence

Choose the first invertible t_K column minor $B_{\mathcal{I}}$ of the actual onto matrix B , and put $R_B = E_{\mathcal{I}}B_{\mathcal{I}}^{-1}$; thus $BR_B = I$. Use the unique empty map when $t_K = 0$. Define, for each of the four retained sign choices,

$$\begin{aligned} X_k &= \delta_k \mathsf{T}_k R_B : \mathbb{C}^{t_K} \longrightarrow \mathcal{D}_{1,k} \oplus \mathcal{D}_{2,k}, \\ Y_{\epsilon,k} &= \mathbf{e}_-^{-1} \mathbf{E}_{\epsilon} \mathbf{j} \mathsf{T}_k J : \mathbb{C}^{j_0} \longrightarrow \mathcal{D}_{1,k} \oplus \mathcal{D}_{2,k}, \\ \delta_k \mathsf{T}_k &= X_k B. \end{aligned} \tag{KRC14}$$

The last equality holds because $I - R_B B$ has image in $\ker B = \operatorname{im} J$, whose image under $\delta_k \mathsf{T}_k$ is zero by KRC5,11. It also proves that X_k is independent of the coordinate right inverse used to compute it. The $Y_{\epsilon,k}$ expression is defined because $\mathsf{T}_k J$ lies in $\mathcal{J}_k$ and KRC10 takes its image to the minus eigenspace.

Put

$$\begin{aligned} \mathcal{W}_{\epsilon,k} &= \operatorname{im} X_k \cap \operatorname{im} Y_{\epsilon,k}, \quad a_{\epsilon,k} = \dim \mathcal{W}_{\epsilon,k}, \\ o_k &= \dim \ker X_k, \quad d_{\epsilon,k} = \dim \ker Y_{\epsilon,k}, \\ \mathcal{R}_{\epsilon,k} &= \{(b, z) \in \mathbb{C}^{t_K} \oplus \mathbb{C}^{j_0} : X_k b = Y_{\epsilon,k} z\}. \end{aligned} \tag{KRC15}$$

These are actual coefficient spaces, computable as the kernel of $[X_k, -Y_{\epsilon,k}]$ and its projections. In particular

$$\begin{aligned} \dim \mathcal{R}_{\epsilon,k} &= a_{\epsilon,k} + o_k + d_{\epsilon,k}, \\ X_k^{-1}(\mathcal{W}_{\epsilon,k}) / \ker X_k &\xrightarrow{\sim} \mathcal{W}_{\epsilon,k} \xleftarrow{\sim} Y_{\epsilon,k}^{-1}(\mathcal{W}_{\epsilon,k}) / \ker Y_{\epsilon,k}. \end{aligned} \tag{KRC16}$$

Each arrow sends a coset to its indicated common value. Surjectivity is its definition, and a zero value is exactly the specified denominator. The map from the relation to its common value has kernel $\ker X_k \oplus \ker Y_{\epsilon,k}$, proving its dimension. This fibre product is the full correspondence induced by the actual branch exchange; no choice of a matching subspace has been substituted for it.

Dimension intersection in the fixed $8k$ -dimensional target gives the unconditional bounds

$$\begin{aligned} a_{\epsilon,k} &\geq \max\{0, 8k - 48 - v - o_k - d_{\epsilon,k}\}, \\ t_K - a_{\epsilon,k} &\leq 32 + v - t_0 + o_k + d_{\epsilon,k}, \\ j_0 - a_{\epsilon,k} &\leq 16 + t_0 + o_k + d_{\epsilon,k}, \quad t_K - j_0 = 16 + v - 2t_0. \end{aligned} \tag{KRC17}$$

Indeed the two image ranks are $t_K - o_k$ and $j_0 - d_{\epsilon,k}$. Their intersection has dimension their rank sum minus the dimension of their sum, at least their rank sum minus $8k$. Substitute KRC1 and rearrange to get every displayed inequality. The terms $o_k, d_{\epsilon,k}$ are the actual descent defects, not omitted hypotheses. The rank imbalance is exactly the last line, so an equality of the two full ranks has not been assumed. In fact $d_{\epsilon,k}$ is independent of ϵ : by KRC10 its kernel is $(\mathsf{T}_k J)^{-1}(\ker \mathbf{j})$. All conclusions also hold on shared-component and nonreduced fibres because the full quotients in KRC7 were used.

81.4 Both induced Gamma metrics, without assigning an isometry

At every original cutoff $q - 1 \leq N \leq 2q$ and each $s = 1, k$ use the complete HJR native dual form $D_N = \mathcal{D}_N^{(s)}$ and

$$H_{B,N} = (B D_N^{-1} B^*)^{-1}, \quad H_{J,N} = J^* D_N J. \tag{KRC18}$$

These are exactly $\widehat{H}_{K,N}^{-1}$ and $\overline{Q}_{0,N}^{-1}$, by DNE9 and PEC25. Thus the two domains of KRC14 carry the original kernel-dual quotient and annihilator restriction metrics, with their full source masses. No metric on the intersection-component coordinates is imposed.

Here are finite formulas for the two quotient metrics on their actual shared image. Choose fixed independent-column frames U_X of $\operatorname{im} X_k$, U_Y of $\operatorname{im} Y_{\epsilon,k}$, and U_W of $\mathcal{W}_{\epsilon,k}$, using the same

first-nonzero-minor rule. For any such injective frame U , let L_U be its exact left inverse obtained by inverting its selected row minor and extracting those rows. Set

$$\begin{aligned}\tilde{X} &= L_{U_X} X_k, & \tilde{Y} &= L_{U_Y} Y_{\epsilon,k}, & E_X &= L_{U_X} U_W, & E_Y &= L_{U_Y} U_W, \\ Q_{X,N} &= (\tilde{X} H_{B,N}^{-1} \tilde{X}^*)^{-1}, & Q_{Y,N} &= (\tilde{Y} H_{J,N}^{-1} \tilde{Y}^*)^{-1}, \\ G_{B,W,N} &= E_X^* Q_{X,N} E_X, & G_{J,W,N} &= E_Y^* Q_{Y,N} E_Y.\end{aligned}\tag{KRC19}$$

The matrices $\tilde{X}, \tilde{Y}$ are onto by their definitions. For an onto matrix T and positive form H , the minimum of $x^* H x$ subject to $Tx = y$ is attained uniquely by $H^{-1} T^* (T H^{-1} T^*)^{-1} y$. Subtracting this lift from any other lift leaves a vector in $\ker T$, orthogonal in H to that lift. This proves the stated minimum formula, its positivity and KRC19. Restricting those quotient minima to U_W proves that the two displayed forms are exactly the two induced metrics of KRC16.

The exact transport along that correspondence is therefore the identity on the common-value coordinate space with relative operator

$$\mathcal{R}_{W,N} = G_{J,W,N}^{-1/2} G_{B,W,N} G_{J,W,N}^{-1/2}, \quad \mathcal{D}_{W,N} = G_{B,W,N} - G_{J,W,N}.\tag{KRC20}$$

Its squared singular values are the eigenvalues of $\mathcal{R}_{W,N}$. To prove this, write the source orthonormal coordinates as $G_{J,W,N}^{1/2} w$ and compute the target squared norm. This gives exactly the displayed positive matrix. Thus KRC20 calculates the complete metric defect of the branch correspondence, rather than declaring it an isometry. It involves only the actual matrices KRC14 and the original DNE7 source form, including all of its lower-evaluation conditioning. Empty common image uses determinant one.

81.5 Exact allocation contrast and all unmatched directions

The common-image metrics form paired flags at every endpoint. Their full remaining determinants can be computed as follows. Let K_X, K_Y be fixed injective frames of the kernels of $\tilde{X}, \tilde{Y}$. Let P_X, P'_Y be fixed onto maps with kernels $\text{im } E_X, \text{im } E_Y$, respectively, obtained by the exact row-pivot subtraction. The prime on P'_Y distinguishes this quotient from the original observation map P_Y in PEC24. Define

$$\begin{aligned}H_{BX,N} &= K_X^* H_{B,N} K_X, & H_{JY,N} &= K_Y^* H_{J,N} K_Y, \\ H_{\text{out}X,N} &= (P_X Q_{X,N}^{-1} P_X^*)^{-1}, & H_{\text{out}Y,N} &= (P'_Y Q_{Y,N}^{-1} (P'_Y)^*)^{-1}.\end{aligned}\tag{KRC21}$$

These retain all unmatched directions; their total ranks on the two sides are respectively $t_K - a_{\epsilon,k}$ and $j_0 - a_{\epsilon,k}$. Completing each fixed kernel frame by fixed quotient lifts and then completing the common-image frame in its target gives, by block Gaussian elimination,

$$\begin{aligned}\det H_{B,N} &= c_B \det H_{BX,N} \det G_{B,W,N} \det H_{\text{out}X,N}, \\ \det H_{J,N} &= c_J \det H_{JY,N} \det G_{J,W,N} \det H_{\text{out}Y,N}.\end{aligned}\tag{KRC22}$$

The positive constants c_B, c_J are the inverse squared determinants of those complete fixed frame changes. Explicitly, choose the same fixed pivot right inverses R_X, R_Y of $\tilde{X}, \tilde{Y}$ and $R_{P_X}, R_{P'_Y}$ of P_X, P'_Y . Then $c_B = |\det[K_X, R_X]|^{-2} |\det[E_X, R_{P_X}]|^{-2}$ and $c_J = |\det[K_Y, R_Y]|^{-2} |\det[E_Y, R_{P'_Y}]|^{-2}$. They do not depend on N or s . The off-diagonal Schur blocks are exactly the attained quotient minima just proved in KRC19, so both full products and their constants follow without orthogonality of the chosen coefficient frames.

For each step $M \rightarrow M+1$, set

$$\begin{aligned}e_{B,M} &= \log \frac{\det H_{BX,M+1} \det H_{\text{out}X,M+1}}{\det H_{BX,M} \det H_{\text{out}X,M}}, \\ e_{J,M} &= \log \frac{\det H_{JY,M+1} \det H_{\text{out}Y,M+1}}{\det H_{JY,M} \det H_{\text{out}Y,M}}.\end{aligned}\tag{KRC23}$$

Both are nonnegative. Indeed HJR15 gives $D_{M+1} \geq D_M$. Restriction preserves this inequality, and taking infima on the same affine fibres preserves it for each quotient. Apply this successively to every form in KRC18–21. The relative eigenvalues of an increasing positive pair are at least one, so every displayed determinant ratio is at least one.

Subtracting the logarithmic increments in KRC22 yields the exact original allocation contrast

$$g_{K,M} - g_{0,M} = \log \frac{\det \mathcal{R}_{W,M+1}}{\det \mathcal{R}_{W,M}} + e_{B,M} - e_{J,M}. \quad (\text{KRC24})$$

Here $g_{K,M} = \log(1 + y_M/(1 + x_M))$ and $g_{0,M} = \log(1 + x_M)$ are the unchanged original HJR18 scalars. Thus the correspondence preserves both of their conditioning denominators through the full native forms; neither denominator has been replaced by a coefficient inner product.

With the exact weights $w_{q-1} = w_{2q-1} = 1$ and $w_M = 2$ for $q \leq M \leq 2q - 2$, define

$$\begin{aligned} \mathbf{c}_{\epsilon,k,s} &= \log \frac{\det \mathcal{R}_{W,2q-1} \det \mathcal{R}_{W,2q}}{\det \mathcal{R}_{W,q-1} \det \mathcal{R}_{W,q}} + \sum_{M=q-1}^{2q-1} w_M (e_{B,M} - e_{J,M}), \\ \mathcal{V}_{K,k,s} &= \sum_M w_M g_{K,M}, \quad \mathcal{V}_{0,k,s} = \sum_M w_M g_{0,M}, \quad \mathcal{T}_{k,s} = \mathcal{V}_{K,k,s} + \mathcal{V}_{0,k,s}. \end{aligned} \quad (\text{KRC25})$$

Telescoping KRC24 proves

$$\mathcal{V}_{K,k,s} - \mathcal{V}_{0,k,s} = \mathbf{c}_{\epsilon,k,s}, \quad \mathcal{V}_{K,k,s} = \frac{\mathcal{T}_{k,s} + \mathbf{c}_{\epsilon,k,s}}{2}, \quad \mathcal{V}_{0,k,s} = \frac{\mathcal{T}_{k,s} - \mathbf{c}_{\epsilon,k,s}}{2}. \quad (\text{KRC26})$$

These identities are valid for every one of the four branch sign choices; their right sides calculate the same original contrast with different complete common-image coordinates and defect terms. The factor 1/2 here is the algebraic solution of the two exact sum-and-difference equations, not an assigned equal allocation. The actual finite HJR25 bounds for $\mathcal{T}_{k,s}$ and the full original HJR31 positive spectral corrections remain available:

$$A_{K,k,s} = \frac{\mathcal{T}_{k,s} + \mathbf{c}_{\epsilon,k,s}}{2} - P_{K,k,s}, \quad A_{0,k,s} = \frac{\mathcal{T}_{k,s} - \mathbf{c}_{\epsilon,k,s}}{2} - P_{0,k,s}. \quad (\text{KRC27})$$

Every term on the right has an exact original finite matrix definition above or in HJR14–18,25,30. No zero value, sign, subleading order or scalar conditioning estimate is assigned to $\mathbf{c}_{\epsilon,k,s}$.

KRC7–17 proves an actual algebraic exchange, its full descent correspondence and exact exceptional kernels. KRC18–27 carries that correspondence through both original Gamma metrics and calculates its complete contribution to the original allocation. The new quantities requiring further quantitative calculation are the specified original-component descent matrices and the relative Gamma form KRC20; they are not assumptions used in any claim of this section. The earlier full period, source, low-image, flag, spectral and same-arithmetic-class action errors remain in the accepted HCR receiving formulas. This calculation changes none of them and establishes no terminal RH conclusion.

81.6 The full degree-four invariant exchange in the original coordinates

With the exact original $\mathbf{T}_j = \alpha_A N_j Z_{\text{exp},j}$, $\mathbf{B}_j = Z_{K,j}^\top A_{\ell,j}$ and $\mathbf{J}_j = A_{\ell,j}^{-1} P_{Y,j}^\top$, DFX34–35 gives

$$\mathcal{X}_k = (\mathbf{B}_{k+4} \mathbf{T}_{k+4}^{-1} m_{i,k} \mathbf{T}_k \mathbf{J}_k)_{i=1}^4, \quad \ker \mathcal{X}_k = \{a : m_{i,k} \mathbf{T}_k \mathbf{J}_k a \in \mathcal{J}_{v,k+4} \text{ for every } i\}. \quad (\text{KRC28})$$

The full Leibniz formula places each product in the inverse's low-jet domain; $\ker \mathbf{B}_{k+4} = \text{im } \mathbf{J}_{k+4}$ then proves exactly the displayed kernel. DFX's nonreduced biform gcd calculation proves any three fourth powers jointly injective before this invariant target quotient. Its branch formula $(1 - \zeta^{-4ia})x_i^4[f]_{(Q,Q_a)}/2$ places the displayed kernel in the exact shared-component space $\mathcal{E}_{v,k}$; DFX31–32 computes its remaining finite matrix. NBR4–10 takes the complete source-kernel restriction and attained image quotient of this same map and proves their full native determinant formula. Thus neither its source kernel nor its remaining invariant metric correction is removed by the earlier branch injectivity.

82 A bounded-degree invariant presentation of the original target

Retain precisely the original cyclic action, period matrices, quadric orbit and conductor quotient of KRC1–7. Write $S = \mathbb{C}[x_1, x_2, x_3, x_4]$, with $gx_i = \zeta^{-i}x_i$, and retain the complete homogeneous ideal $I = (Q, \mathcal{A})$. For each degree j let $\mathcal{C}_j = (S/I)_j$ and $\mathcal{J}_j = \text{im}(S_j^{(g)} \rightarrow \mathcal{C}_j)$. We use only the original sequence $k \equiv 1 \pmod{4}$. The construction below applies for $k \geq 37$; the preceding original finite degrees $9, 13, \dots, 33$ keep their exact KRC matrices and are not changed by this statement.

82.1 A finite family in the original coordinates

For a monomial x^ν its original character is $\text{wt}(\nu) = \sum_{i=1}^4 i\nu_i \pmod{5}$. Define the following fixed set of monomials, retaining their original coefficients equal to one:

$$\begin{aligned} \mathcal{H}_* &= \{x^{r+5b} : 0 \leq r_i \leq 4, b_i \geq 0, |b| \leq 3, |r| + 5|b| \equiv 0 \pmod{4}, \text{wt}(r) \not\equiv 0 \pmod{5}\}, \\ \mathcal{H} &= \{1\} \cup \mathcal{H}_*, \quad d_h = \deg h. \end{aligned} \tag{BIM1}$$

Here r, b are four-component integer vectors and their bars denote sums of their entries. Every nonunit degree is one of $4, 8, \dots, 28$: the degree is a positive multiple of four and at most $16 + 15 = 31$. The representation $r + 5b$ is unique by taking each exponent modulo five. In particular

$$|\mathcal{H}| \leq 1 + 5^4 \binom{7}{4} = 21876. \tag{BIM2}$$

The binomial counts the four-tuples of nonnegative integers of sum at most three, by adding one slack coordinate. This family is independent of k , period and source order.

Every degree- k monomial of nonzero character factors as hf with $h \in \mathcal{H}_*$ and invariant f of degree $k - d_h \geq 9$, still congruent to one modulo four. To prove this, write $\nu_i = 5a_i + r_i$ with $0 \leq r_i \leq 4$. Set $R = |r| \leq 16$ and $A = |a|$, so $k = R + 5A$. Choose the unique $t \in \{0, 1, 2, 3\}$ such that $R + 5t \equiv 0 \pmod{4}$. Since $k \geq 37$ and $R \leq 16$, $A \geq 5 > t$. Choose a tuple $b \leq a$ with $|b| = t$ by taking as many available exponents as needed in the original index order. Then

$$x^\nu = x^{r+5b} \prod_{i=1}^4 (x_i^5)^{a_i-b_i}, \quad h = x^{r+5b}, \quad f = \prod_i (x_i^5)^{a_i-b_i}. \tag{BIM3}$$

Each x_i^5 is invariant, proving invariance of f . The nonzero character of ν is that of r , so h belongs to BIM1. The degree bound $d_h \leq 28$ gives $k - d_h \geq 9$. An invariant monomial instead uses $h = 1$ and itself as f . Since the action is diagonal, an invariant polynomial is exactly a linear combination of the weight-zero monomials. Applying this deterministic factorization to every monomial, with its unchanged original coefficient, proves the onto map

$$\begin{aligned} \mathcal{P}_k &= \bigoplus_{h \in \mathcal{H}} \mathcal{J}_{k-d_h}, \quad \mathbf{L}_k : \mathcal{P}_k \longrightarrow \mathcal{C}_k, \quad ([f_h])_h \longmapsto \left[\sum_h h f_h \right], \\ \mathbf{L}_k \text{ is onto,} \quad \ker \mathbf{L}_k &= \{([f_h])_h : \sum_h h f_h \in I_k\}. \end{aligned} \tag{BIM4}$$

Multiplication by each h is well defined on $\mathcal{J}_{k-d_h}$: two invariant representatives differ by an element of I , whose product with h is still in the same full ideal. Surjectivity follows from BIM3 before taking this quotient. Thus the formula preserves all components and nilpotents; no radical, generic fibre or pairwise coprimality is substituted.

82.2 The exact original primary and symbol maps

The OCF2 basis is the unscaled biform basis $e_{ab}^{(j)} = z_0^a z_1^{j-a} w_0^b w_1^{j-b}$. Retain the original substitution $x = H(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^T$ and expand

$$h(Ht) = \sum_{a,b=0}^{d_h} h_{ab} e_{ab}^{(d_h)}, \quad f(Ht) = \sum_{i,j=0}^{k-d_h} f_{ij} e_{ij}^{(k-d_h)}. \quad (\text{BIM5})$$

The coefficients are the full original multinomial expansions of the displayed linear forms, with every entry of H . Since $e_{ab}^{(d)} e_{ij}^{(e)} = e_{a+i, b+j}^{(d+e)}$, the primary coefficient map for hf is exactly the convolution $\nu_{A,B} = \sum_{a+i=A, b+j=B} h_{ab} f_{ij}$.

For any degree j use the unchanged original roots

$$\beta_{j,a,b} = j/2 + (2a - j)\delta + i(2b - j)\gamma, \quad \mathcal{N}_f(w) = \sum_{a,b} f_{ab} e^{\beta_{j,a,b} w}. \quad (\text{BIM6})$$

The i in the root is the imaginary unit. The exact addition $\beta_{d,a,b} + \beta_{j,c,e} = \beta_{d+j, a+c, b+e}$ proves the unaltered numerator map

$$\mathcal{N}_{hf} = \mathcal{N}_h \mathcal{N}_f. \quad (\text{BIM7})$$

The original conductor satisfies $\mathcal{N}_{Ab} = E_A \mathcal{N}_b$, by the same coefficient identity and RDS7. Its zero order v is fixed by the actual period and is independent of the degree j . Therefore the map

$$\mathbf{M}_{v,j} : \mathcal{C}_j \rightarrow \mathbb{C}^v, \quad [f] \mapsto (\mathcal{N}_f^{(a)}(0))_{a=0}^{v-1} \quad (\text{BIM8})$$

is well defined. KRC4 proves that it is onto and that its kernel is precisely $\mathcal{C}_{v,j}$, for every degree $j = k - d_h \geq 9$ in use here. Its restriction to $\mathcal{J}_j$ has kernel $\mathcal{J}_{v,j}$ and rank $v - t_{0,j}$, with the original $t_{0,j} = \dim(K_j \cap L_j)$. All these are actual original degree-dependent intersections; they have not been assigned the same value at different degrees.

82.3 Coupled and individual low-jet conditions

For a tuple in $\mathcal{P}_k$, BIM7 gives every component of the complete original low-jet condition explicitly:

$$(\mathbf{M}_{v,k} \mathbf{L}_k(f_h))_a = \sum_{h \in \mathcal{H}} \sum_{b=0}^a \binom{a}{b} \mathcal{N}_h^{(b)}(0) \mathcal{N}_{f_h}^{(a-b)}(0), \quad 0 \leq a < v. \quad (\text{BIM9})$$

These are full finite convolution identities. Define

$$\begin{aligned} \mathcal{P}_k^{\text{coupled}} &= \ker(\mathbf{M}_{v,k} \mathbf{L}_k), & \mathcal{P}_k^{\text{individual}} &= \bigoplus_{h \in \mathcal{H}} \mathcal{J}_{v, k-d_h}, \\ \mathbf{L}_k : \mathcal{P}_k^{\text{coupled}} &\rightarrow \mathcal{C}_{v,k}, & \ker(\mathbf{L}_k|_{\mathcal{P}_k^{\text{coupled}}}) &= \ker \mathbf{L}_k. \end{aligned} \quad (\text{BIM10})$$

Surjectivity follows by lifting any member of $\mathcal{C}_{v,k}$ through the onto BIM4 map; its lift automatically satisfies BIM9. Also $\mathbf{M}_{v,k} \mathbf{L}_k$ is onto, so $\text{codim}_{\mathcal{P}_k} \mathcal{P}_k^{\text{coupled}} = v$. Every individually constrained tuple belongs to the coupled space: each product in BIM9 has the factor $\mathcal{N}_{f_h}^{(a-b)}(0) = 0$ because $a - b < v$. This proves the exact map between the two kinds of low constraint, including possible cancellations between different summands.

Let

$$\begin{aligned} c_k &= \sum_{h \in \mathcal{H}} (v - t_{0, k-d_h}), & \mathcal{J}_{v,k} &= \mathbf{L}_k(\mathcal{P}_k^{\text{individual}}) \subset \mathcal{C}_{v,k}, \\ e_k &= \dim(\mathcal{C}_{v,k} / \mathcal{J}_{v,k}), & 0 \leq e_k &\leq c_k - v \leq (|\mathcal{H}| - 1)v. \end{aligned} \quad (\text{BIM11})$$

These bounds include the exact degree-dependent t_0 values. To prove them, the codimension of the individual domain in $\mathcal{P}_k$ is c_k by BIM8 and the direct sum. Since the coupled domain has

codimension v and contains it, $c_k \geq v$ and their quotient has dimension $c_k - v$. The onto map in BIM10 induces an onto map from that quotient to $\mathcal{C}_{v,k}/\mathcal{J}_{v,k}$, proving the first bound. The second uses $0 \leq t_{0,j} \leq v$ term by term. Thus the missing target has bounded dimension independent of k , with a completely specified finite coefficient matrix and kernel. If $v = 0$, the individual map is onto exactly.

82.4 The actual branch map on these generators

Use the exact scheme-theoretic component quotients and branch maps of KRC7–11, including their factor $1/2$ and original cyclic orientation. For an invariant representative f and a nonunit $h \in \mathcal{H}_*$, define $\omega_h = \text{wt}(h)$. For $a = 1, 2$, direct substitution gives

$$(\delta_k[hf])_a = \frac{1 - \zeta^{-a\omega_h}}{2} [hf]_{(Q, Q_a)}. \quad (\text{BIM12})$$

Indeed $g^a(hf) = \zeta^{-a\omega_h}hf$, whereas δ_k takes the first two components of $(I - \mathbb{T}_{\text{br}})j/2$. For $h = 1$ this expression is zero. Every factor for $h \neq 1$ and $a = 1, 2$ is nonzero, since five is prime and $\omega_h \in \{1, 2, 3, 4\}$. This proves an explicit finite-family presentation of the anti-gluing image from invariant degrees at distance at most 28. All multiplication kernels remain the exact kernels in the original component rings; nonzero scalar factors do not remove those kernels.

The map δ_k descends through the original kernel-dual quotient by KRC11. From BIM11 one also gets the precise finite defect bound

$$\dim(\delta_k(\mathcal{C}_{v,k})/\delta_k(\mathcal{J}_{v,k})) \leq e_k. \quad (\text{BIM13})$$

It follows by applying δ_k to the onto quotient map, or by writing its kernel as $\mathcal{C}_{v,k} \cap (\mathcal{J}_{v,k} + \ker \delta_k)$. The full KRC5 identification with native dual coordinates is

$$\mathbb{T}_j = \alpha_A N_j Z_{\text{exp},j} : \mathbb{C}^{\Delta_j - v} \xrightarrow{\sim} \mathcal{C}_{v,j}. \quad (\text{BIM14})$$

Thus an individually constrained summand has the precise map $\mathbb{T}_k^{-1} \mu_h \mathbb{T}_{k-d_h}$, where μ_h is multiplication by h , restricted to $\mathbb{T}_{k-d_h}^{-1} \mathcal{J}_{v,k-d_h}$. This inverse at the target exists by BIM9. Both scalar-bearing original coordinate maps remain in the formula.

For a general coupled tuple, the full coordinate transport also retains its individual low jets. Let E_j insert the first v original root columns into the complete primary coefficient space, and let V_j be the square Vandermonde matrix on those columns, with rows indexed by the derivative orders $0, \dots, v-1$. The roots are distinct and $v \leq 80 < (j+1)^2$, so $\det V_j = \prod_{a < b} (\beta_b - \beta_a) \neq 0$. Use the original strip projection N_j to define

$$\begin{aligned} S_j^{\text{jet}} &= N_j E_j V_j^{-1} : \mathbb{C}^v \longrightarrow \mathcal{C}_j, & \mathbb{M}_{v,j} S_j^{\text{jet}} &= I, \\ \lambda &= \mathbb{M}_{v,j} f, & z &= \mathbb{T}_j^{-1}(f - S_j^{\text{jet}} \lambda), & f &= \mathbb{T}_j z + S_j^{\text{jet}} \lambda. \end{aligned} \quad (\text{BIM15})$$

Indeed the full jet matrix annihilates the conductor image, so it factors through N_j and its product with the displayed section is $V_j V_j^{-1}$. Subtracting that section leaves zero jet, where BIM14 is invertible. Applying the jet map to any vanishing decomposition first gives $\lambda = 0$ and then injectivity of $\mathbb{T}_j$ gives $z = 0$, proving uniqueness. For $v = 0$ the section is the unique map from the zero space. No estimate for the retained Vandermonde inverse is assumed.

Define the full lower triangular jet multiplier by

$$(D_h)_{ab} = \begin{cases} \binom{a}{b} \mathcal{N}_h^{(a-b)}(0), & 0 \leq b \leq a < v, \\ 0, & b > a. \end{cases} \quad (\text{BIM16})$$

Then BIM9 states exactly $\mathbf{M}_{v,k}\mu_h = D_h\mathbf{M}_{v,k-d_h}$. Writing $j_h = k - d_h$, the complete coupled domain and its target coordinate map are therefore

$$\begin{aligned} \mathbf{T}_{j_h} z_h + S_{j_h}^{\text{jet}} \lambda_h &\in \mathcal{J}_{j_h} \quad (h \in \mathcal{H}), \quad \sum_h D_h \lambda_h = 0, \\ ((z_h, \lambda_h))_h &\longmapsto \mathbf{T}_k^{-1} \sum_h \mu_h (\mathbf{T}_{j_h} z_h + S_{j_h}^{\text{jet}} \lambda_h). \end{aligned} \tag{BIM17}$$

The inverse is defined because the full coupled sum has zero jet. The invariant membership condition is imposed on each displayed sum; its two summands need not separately be invariant. The individual domain is exactly the subspace with every $\lambda_h = 0$ and the same membership conditions. These formulas prove the complete typed transport of BIM4–13, including the coupled components with nonzero individual jets. They assign no native Gamma metric to the additional jet coordinates and remove no original scalar α_A .

This establishes the finite-degree algebraic presentation and its bounded low-jet defect on every original five-orbit fibre. The native norm of these multiplication maps and their change of source degree and endpoint remain quantitative calculations; no metric bound follows merely from the factorization. BIM7 supplies the literal finite exponential multiplier for that calculation, and BIM10–13 retain the full missing finite space.

82.5 The proved first-moment bound on the identical coupled domain

For the actual four nonzero orbit biforms let $p_a \leq 9$ be their exact support sizes. Their exponential weights are distinct because $\delta, \gamma > 0$. The first p_a derivatives form an invertible Vandermonde, so each factor has zero order at most $p_a - 1$. Product zero orders add, proving

$$v \leq \sum_{a=1}^4 (p_a - 1) \leq 32, \quad e_k \leq \sum_{h \in \mathcal{H}} (v - t_{0,k-d_h}) - v \leq 32(|\mathcal{H}| - 1). \tag{BIM18}$$

The middle quantity is the complete coupled/individual quotient dimension already proved above; its map onto the missing image proves the first inequality for e_k . This changes neither v itself nor any section, low jet, original polynomial or constraint $\sum_h D_h \lambda_h = 0$. The proof applies on the whole original five-orbit locus. NBR2–3 carries these exact dimensions through the same native missing quotient and all previous finite errors.

83 Original endpoint control for every finite-rank remainder

Retain the original fixed simple quartet, period and branch, all conductor coefficients, both source orders $s = 1, k$, and the original four endpoints. Write $q = (k + 1)^2$ and $d_N = N - v$. Use exactly the DNE3 coefficient quotient frame and its two complete forms G_{exp} and $D_N = \mathcal{D}_N^{(s)}$. Keep the explicit LJR16 reconstruction constant $M_{k,s}^{\text{rec}}$ and the complete DNE13 source constant $\mathcal{B}_{d,s,k,A}$, including its original mass $M_s = (2\pi)^{s/2}$, actual μ_v , radius and Gamma ratios. Define

$$C_{k,s}^{\text{src}} = \max\{1, q\mathcal{B}_{2q-v,s,k,A}^2\}, \quad A_{k,s}^{\text{end}} = C_{k,s}^{\text{src}}(M_{k,s}^{\text{rec}})^2. \tag{FRE1}$$

These are finite original-data expressions, not selected condition numbers. Their factors remain those of the stated complete original source and reconstruction maps.

83.1 Two exact metric maps give one endpoint comparison

The complete low reconstruction proves

$$G_{\text{exp}} \preceq (M_{k,s}^{\text{rec}})^2 D_{q-1}. \tag{FRE2}$$

Indeed take any coefficient vector x in DNE3. Choose the attained minimum old functional values representing its class in the original Gamma dual quotient. Their squared Euclidean norm is $x^*D_{q-1}x$ by DNE6–7. The literal LJR4 reconstruction, followed by its original coefficient quotient, maps this representative to the class $Z_{\text{exp}}x$; LJR6–8 proves both the image and the complete lower-evaluation/conductor kernel. Its target squared norm is precisely $x^*G_{\text{exp}}x$. Applying the norm bound LJR17 to this same representative proves FRE2 for every x . The identity between coefficient frames is thus supplied by the exact reconstruction, with no unproved comparison of unrelated norms.

DNE14 and monotonicity of its displayed finite Gamma sum give $D_N \preceq C_{k,s}^{\text{src}}G_{\text{exp}}$ for $q-1 \leq N \leq 2q$. Moreover $D_N \succeq D_{q-1}$: every candidate full lower-evaluation extension has at least its old squared norm when more original orthonormal rows are included; minimizing over the same complete lower-evaluation coefficient vector preserves this inequality. Consequently

$$\boxed{D_{q-1} \preceq D_N \preceq A_{k,s}^{\text{end}} D_{q-1} \quad (q-1 \leq N \leq 2q).} \quad (\text{FRE3})$$

This bound retains the same G_{exp} in both of its proved maps before composing them. The accepted finite estimates LJR20 and DNE15 imply

$$0 \leq \log A_{k,s}^{\text{end}} \leq 2q \log k + O_{\text{actual}}(q). \quad (\text{FRE4})$$

Here the asymptotic period is fixed, and the estimate holds for both original source orders. The complete expression FRE1, rather than only its asymptotic order, is used below.

83.2 Every original paired restriction and quotient

Take any fixed algebraic flag of this coefficient space, and use its actual restrictions and attained quotient metrics at every endpoint. The flag may depend on the original data and on k , but its maps are the same as N varies. Let $D_{X,N}$ be one resulting form of rank e_k . Then

$$D_{X,q-1} \preceq D_{X,N} \preceq A_{k,s}^{\text{end}} D_{X,q-1}, \quad 0 \leq I_{X,N} := \log \frac{\det D_{X,N}}{\det D_{X,q-1}} \leq e_k \log A_{k,s}^{\text{end}}. \quad (\text{FRE5})$$

For restriction substitute the same column vector in FRE3. For a quotient, apply its two inequalities to every candidate lift of a fixed quotient class and then take the infimum over that same affine fibre. This proves both inequalities after each step of a flag, with all native Schur minima retained. Conjugating by $D_{X,q-1}^{-1/2}$ puts every relative eigenvalue in $[1, A_{k,s}^{\text{end}}]$. Adding their e_k logarithms proves the determinant bound. Empty rank has determinant one and increment zero.

The same argument applied to each consecutive endpoint proves that $I_{X,N}$ increases. Its exact original four-return therefore obeys

$$\begin{aligned} \mathcal{V}_{X,k,s} &:= I_{X,2q-1} + I_{X,2q} - I_{X,q} \\ &= \sum_{M=q-1}^{2q-1} w_M \log \frac{\det D_{X,M+1}}{\det D_{X,M}}, \\ 0 &\leq \mathcal{V}_{X,k,s} \leq 2e_k \log A_{k,s}^{\text{end}}. \end{aligned} \quad (\text{FRE6})$$

The weights are exactly $w_{q-1} = w_{2q-1} = 1$ and $w_M = 2$ at the other original rows. Expanding the telescoping sums proves this equality and every sign. The upper bound drops the nonnegative $I_{X,q}$ and uses FRE5 on each of the two high endpoints. In particular

$$e_k = O_{\text{actual}}(1) \implies \mathcal{V}_{X,k,s} = O_{\text{actual}}(q \log k) = o(kq). \quad (\text{FRE7})$$

The displayed implication evaluates the exact finite bound FRE6 on any flag whose rank is already so bounded; it assumes no missing analytic estimate. More generally the same conclusion holds for $e_k = o(k/\log k)$, directly by FRE4. No bound on an individual row location or concentration was used: every row in its original weighted sum remains.

83.3 The exact image and its native unmatched quotient

For a fixed original map into the actual kernel-dual coordinate space, let U_k be an independent-column frame of its image, whose codimension is e_k . Keep the entire kernel-dual metric $H_{B,N} = (B_K D_N^{-1} B_K^*)^{-1}$ of DNE9. Choose any fixed onto matrix P_k with kernel $\text{im } U_k$ and retain the two complete forms

$$H_{\text{im},N} = U_k^* H_{B,N} U_k, \quad H_{\text{miss},N} = (P_k H_{B,N}^{-1} P_k^*)^{-1}. \quad (\text{FRE8})$$

Their definition keeps both the actual B_K quotient and its further image restriction/quotient. Complete U_k by a fixed right inverse R_k of P_k . The matrix $[U_k, R_k]$ is invertible: applying P_k first kills the R_k coefficient in a vanishing combination, and then independence kills the other one. The Gram of this full frame has first block $H_{\text{im},N}$; its second Schur complement is exactly $H_{\text{miss},N}$ by minimization over the image. Thus

$$\det H_{B,N} = |\det[U_k, R_k]|^{-2} \det H_{\text{im},N} \det H_{\text{miss},N}. \quad (\text{FRE9})$$

The frame is fixed across endpoints. Its scalar cancels in every determinant increment, so the original weighted kernel return splits exactly as

$$\widehat{F}_k^{(s)} = \mathcal{V}_{\text{im},k,s} + \mathcal{V}_{\text{miss},k,s}, \quad 0 \leq \widehat{F}_k^{(s)} - \mathcal{V}_{\text{im},k,s} \leq 2e_k \log A_{k,s}^{\text{end}}. \quad (\text{FRE10})$$

The missing quotient is one of the same algebraic flags in FRE5, because H_B is itself the specified B_K quotient of D_N . Applying FRE6 proves the last inequality. The same construction applies to an image in the full coefficient space or to the residual-annihilator space. All its actual image and quotient matrices must be used.

For the bounded invariant multiplier family, a proved constant bound for the image codimension can consequently be propagated through FRE10. Its native unmatched return is then subleading by the exact factorization, even if that return is concentrated in a few original rows. The metric on the matched image remains the native form in FRE8. A one-sided norm bound for an operator reaching that image does not identify its source quotient metric with FRE8; their exact relative operator remains part of the subsequent calculation.

84 Native transport of the bounded original invariant multipliers

Keep the complete original fixed quartet, period, branch, conductor and Gamma measures. Use the actual BIM1 family $\mathcal{H}$ on its proved domain $k \equiv 1 \pmod{4}$, $k \geq 37$. For $h \in \mathcal{H}$ put $d_h = \deg h = 4r_h$ and $\ell_h = k - d_h \geq 9$. The earlier finite degrees keep their original matrices. The two source-order pairs used below are exactly

$$(s_{\ell_h}, s_k) = (1, 1) \quad \text{or} \quad (\ell_h, k), \quad c'_j = j/2 - 4, \quad D \geq q_k - v - 1, \quad q_j = (j+1)^2, \quad q'_j = (j-7)^2. \quad (\text{MNT1})$$

The polynomial cutoff D is the same integer in the source and target of each map. Its relation to their different original endpoint windows is calculated below.

84.1 The literal finite translation sum and its two centres

Use every actual coefficient h_{ab} from BIM5 and set

$$B_h(z) = \mathcal{N}_h(z) = \sum_{a,b=0}^{d_h} h_{ab} e^{\beta_{d_h,a,b} z}, \quad (\text{MNT2})$$

$$\beta_{d,a,b} = d/2 + (2a - d)\delta + i(2b - d)\gamma, \quad \eta_{h,a,b} = \beta_{d_h,a,b} - d_h/2.$$

For $h = 1$ the sole coefficient is one, $d_h = 0$, and $B_h = 1$. All other arrays are the literal original multinomial coefficients of $h(\text{Ht})$. Multiplication of numerators has the exact dual map

$$\mathcal{A}_h : \mathbb{C}[S']_{\leq D} \longrightarrow \mathbb{C}[S']_{\leq D}, \quad (\mathcal{A}_h p)(S') = \sum_{a,b} h_{ab} p(S' + \beta_{d_h, a, b}), \quad \Lambda_{B_h F}(p) = \Lambda_F(\mathcal{A}_h p). \quad (\text{MNT3})$$

To prove the identity apply both sides to S'^n . Leibniz' rule on $(B_h F)^{(n)}(0)$ gives precisely $\sum_{a,b} h_{ab} \sum_{j=0}^n \binom{n}{j} \beta_{d_h, a, b}^{n-j} F^{(j)}(0)$, the right side. All germs used here are analytic at zero: the individual source domain is the complete original low-jet subspace $\mathcal{N}_{v, \ell_h}$, so $F = \mathcal{N}/E_A$ is analytic. This does not assign a functional to an individual summand outside that subspace in BIM's larger coupled domain.

For $p_k(y) = p(c'_k + iy)$ the two original centres satisfy $c'_k - c'_{\ell_h} = d_h/2$. Thus the exact pullback is

$$(\mathcal{A}_h p)(c'_{\ell_h} + iy) = \sum_{a,b} h_{ab} p_k(y + \eta_{h, a, b}/i). \quad (\text{MNT4})$$

This retains the phase $1/i = -i$ and the full center difference. Let $H_D = \sum_{j=1}^D 1/j$, with $H_0 = 0$, and put

$$K_h(D) = \sum_{a,b} |h_{ab}| \exp(|\eta_{h, a, b}|(H_D + 2)). \quad (\text{MNT5})$$

AKJ21–23 proves $\|\partial_y\|_{\leq D, s} \leq H_D + 2$ for every original source order $s \geq 1$: its ratio estimate uses only $s/2 \geq 1/2$. The exact finite Taylor identity $p(y+z) = \exp(z\partial_y)p(y)$ therefore gives $\|p(y+z)\|_s \leq e^{|z|(H_D+2)}\|p\|_s$. Apply this to each term of MNT4 and then use the triangle inequality. For either unchanged common source order this proves the norm bound $K_h(D)$ between the displayed centres. In particular $|\eta_{h, a, b}| \leq d_h \sqrt{\delta^2 + \gamma^2}$ and $H_D \leq 1 + \log D$ for $D \geq 1$. Thus $K_h(D)$ is a fully specified polynomial bound in $D+1$ at the fixed actual data and degree $d_h \leq 28$.

84.2 The complete source-order change

For the degree-order pair put $s = \ell_h$ and $r = r_h$. The full original measure is

$$dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}. \quad (\text{MNT6})$$

The Gamma recurrence, applied exactly r times, proves

$$\frac{dm_{s+4r}(y)}{dm_s}(y) = \frac{(2\pi)^{2r}}{(s/2)_{2r}} \prod_{j=0}^{r-1} (y^2 + (s/2 + 2j)^2) \geq a_{s,r} := (2\pi)^{2r} \prod_{j=0}^{r-1} \frac{s+4j}{s+4j+2} > 0. \quad (\text{MNT7})$$

For detail the squared Gamma ratio is $4^{-r} \prod_j (y^2 + (s/2 + 2j)^2)$. The factor $2^{(s+4r)/2}/2^{s/2} = 4^r$ cancels it, the mass ratio is $(2\pi)^{2r}$, and the remaining denominator ratio is $(s/2)_{2r}$. Pairing its consecutive factors proves the displayed minimum. For $r = 0$ every product and the constant equal one. This computation retains the actual full source mass.

Define

$$\kappa_h(D) = \begin{cases} K_h(D), & (s_{\ell_h}, s_k) = (1, 1), \\ K_h(D) a_{\ell_h, r_h}^{-1/2}, & (s_{\ell_h}, s_k) = (\ell_h, k). \end{cases} \quad (\text{MNT8})$$

First apply MNT4–5 in order s_{ℓ_h} . Then use $\|p_k\|_{s_{\ell_h}} \leq a_{\ell_h, r_h}^{-1/2} \|p_k\|_{s_k}$, which follows by integrating MNT7. This proves the complete primal bound

$$\|\mathcal{A}_h p\|_{s_{\ell_h}, c'_{\ell_h}} \leq \kappa_h(D) \|p\|_{s_k, c'_k}. \quad (\text{MNT9})$$

Since $\ell_h \geq 9$ and $r_h \leq 7$, the positive factors a_{ℓ_h, r_h} are bounded above and away from zero by explicit constants obtained by replacing ℓ_h in each ratio of MNT7 by infinity or by nine. Together with MNT5 this yields $\log \max\{1, \kappa_h(D)\} = O_{\text{actual}}(\log(D+1))$, uniformly over the finite family and both source-order choices.

84.3 Every original lower evaluation in the quotient map

Let $\mathcal{E}_j$ denote the complete original degree- j exponential space. BIM7 gives

$$B_h \mathcal{N}_{v,\ell_h} \subset \mathcal{N}_{v,k}, \quad B_h(E_A \mathcal{E}_{\ell_h-8}) \subset E_A \mathcal{E}_{k-8}. \quad (\text{MNT10})$$

The first inclusion follows by multiplying an analytic numerator with its first v zero jets by the entire B_h . The second is the exact root-addition identity in BIM6–7. It follows that multiplication induces the typed map

$$M_h : \mathcal{N}_{v,\ell_h}/E_A \mathcal{E}_{\ell_h-8} \longrightarrow \mathcal{N}_{v,k}/E_A \mathcal{E}_{k-8}, \quad [\mathcal{N}] \longmapsto [B_h \mathcal{N}]. \quad (\text{MNT11})$$

Its kernel is exactly the source classes whose product belongs to $E_A \mathcal{E}_{k-8}$. No assertion that this kernel vanishes is needed or used. The polynomial and low-jet multiplication kernels in BIM and the full nonreduced component kernels in KRC remain the same kernels after this explicit coordinate map.

At the common polynomial degree D , use the full original minimum extension metric $D_{j,D}^{(s_j)}$ on the quotient in degree j , defined by

$$\|[\mathcal{N}]\|_{j,D}^2 = \min_{\eta \in \mathbb{C}^{q'_j}} \left\| \Lambda_{\mathcal{N}/E_A} + \sum_{\beta' \in \text{grid}(j-8)} \eta_{\beta'} \text{ev}_{\beta'} \right\|_{(\mathbb{C}[S']^{\leq D}, \|\cdot\|_{s_j, c'_j})^\vee}^2. \quad (\text{MNT12})$$

The first complete jet window has $D \geq q_j - v - 1$ here, so DNE's Vandermonde and relation-annihilator argument proves strict positivity and the exact quotient kernel. Its proof applies at every larger finite D ; it makes no use of an upper cutoff.

Choose the attained source minimum in MNT12. Precompose its functional by $\mathcal{A}_h$. Its first term becomes $\Lambda_{B_h \mathcal{N}/E_A}$ by MNT3. Every old lower evaluation becomes the full sum of target lower evaluations with roots $\beta'_{\ell_h-8} + \beta_{d_h,a,b}$ by MNT3 and BIM6. Thus this is an allowed target extension, with all its coefficients retained. The target minimum is no larger. The dual form of MNT9 then proves

$$\|M_h[\mathcal{N}]\|_{k,D} \leq \kappa_h(D) \|[\mathcal{N}]\|_{\ell_h,D}. \quad (\text{MNT13})$$

This proves the metric transport through the complete original lower-evaluation denominators. It neither resets those coefficients nor minimizes different row blocks independently.

For exact native coordinates let $T_j = \alpha_A N_j Z_{\text{exp},j}$ be KRC5, where $\alpha_A = v!/\mu_v$ is the same constant for all degrees because E_A is fixed. The DNE coefficient matrix of MNT11 is $T_k^{-1} M_h^{\text{strip}} T_{\ell_h}$. Equivalently multiply the original upper coefficient array by h_{ab} and reduce by the full target $C_k^\top$ to the $Z_{\text{exp},k}$ coordinates. The two factors α_A cancel only in this conjugation of the exact multiplication map. If one instead uses the scaled FGC dual metric, both corresponding Grams acquire $|\alpha_A|^{-2}$, so MNT13 remains the same inequality by that proved common scalar map.

84.4 The finite family reaches the native kernel image

Restrict each source of MNT11 to its actual invariant low-jet part $\mathcal{J}_{v,\ell_h}$, using precisely the DNE matrix J_{ℓ_h} and native form $J_{\ell_h}^* D_{\ell_h,D} J_{\ell_h}$. Their direct sum has its full block diagonal native Gram $\mathcal{G}_D$. Let Φ_k be the complete coefficient matrix of the sum of these restricted maps, and let $\Psi_k = B_k \Phi_k$ be its composition with the actual onto kernel-dual map. Put

$$\mathcal{K}(D) = \left(\sum_{h \in \mathcal{H}} \kappa_h(D)^2 \right)^{1/2}. \quad (\text{MNT14})$$

Applying MNT13, the triangle inequality in the one target Hilbert space and then Cauchy–Schwarz gives $\|\Phi_k x\|_{k,D} \leq \mathcal{K}(D) \sqrt{x^* \mathcal{G}_D x}$. Taking the further B_k quotient can only decrease the target norm,

so the same bound holds for Ψ_k . Its exact image has codimension at most the full BIM11 defect $e_k \leq (|\mathcal{H}| - 1)v$: the quotient of the target by that image is an image of the full coefficient quotient by $\text{im } \Phi_k$. This is the precise native use of the individually constrained image. No analytic germ was assigned to the larger coupled tuple's nonanalytic summands.

To specify both metrics on this actual image, choose an injective frame U_k of $\text{im } \Psi_k$ and its exact left inverse L_{U_k} , and set

$$\begin{aligned} \tilde{\Psi}_k &= L_{U_k} \Psi_k, & Q_D^{\text{source}} &= (\tilde{\Psi}_k \mathcal{G}_D^{-1} \tilde{\Psi}_k^*)^{-1}, \\ H_D^{\text{image}} &= U_k^* (B_k D_{k,D}^{-1} B_k^*)^{-1} U_k, & H_D^{\text{image}} &\preceq \mathcal{K}(D)^2 Q_D^{\text{source}}. \end{aligned} \quad (\text{MNT15})$$

The matrix $\tilde{\Psi}_k$ is onto. Its minimum source lift has exactly the first displayed Gram, by the complete Hilbert quotient formula proved in KRC19. Apply the preceding norm inequality to that attained lift to obtain MNT15. This specifies the exact source quotient and target image metrics. Every exterior degree has norm at most $\max\{1, \mathcal{K}(D)\}^{\text{rank } U_k}$ for this identity, since each singular value is at most $\mathcal{K}(D)$. On $D = O(q_k)$ its logarithm is $O_{\text{actual}}(k \log q_k) = o(kq_k)$. This is an upper expansion bound for the specified operator; no inverse bound is inferred from it.

At the four original target cutoffs, the missing kernel quotient in MNT15 has rank at most e_k . Its full native determinant return is bounded by FRE10 with that rank, so the complete kernel return differs from the return of H_D^{image} by a nonnegative $O_{\text{actual}}(q_k \log k) = o(kq_k)$ term. This uses the actual image and its remaining quotient and does not assume evenly distributed row energies.

84.5 The exact displaced source endpoints

The source polynomial cutoff in every preceding map is the literal target cutoff $D = N - v$. For each h define

$$b_{h,k} = q_k - q_{\ell_h} = d_h(2k - d_h + 2). \quad (\text{MNT16})$$

Thus the four source endpoint indices used in MNT12 are exactly

$$q_{\ell_h} - 1 + b_{h,k}, \quad q_{\ell_h} + b_{h,k}, \quad 2q_{\ell_h} - 1 + 2b_{h,k}, \quad 2q_{\ell_h} + 2b_{h,k}. \quad (\text{MNT17})$$

No source cutoff has been replaced by its degree's usual endpoint. For any fixed source flag, write $F_h(N) = \log \det D_{X, \ell_h, N-v}$ and define the native positive return using its original signs on the dual metric. Its shifted return minus its usual return is exactly

$$\begin{aligned} &[F_h(2q_{\ell_h} - 1 + 2b_{h,k}) - F_h(2q_{\ell_h} - 1)] + [F_h(2q_{\ell_h} + 2b_{h,k}) - F_h(2q_{\ell_h})] \\ &- [F_h(q_{\ell_h} - 1 + b_{h,k}) - F_h(q_{\ell_h} - 1)] - [F_h(q_{\ell_h} + b_{h,k}) - F_h(q_{\ell_h})]. \end{aligned} \quad (\text{MNT18})$$

This is direct subtraction of the two four-endpoint expressions. Each bracket is an actual non-negative sum of source row increments, by the same minimum extension monotonicity used in MNT12. The exact signed combination retains their potential concentration. No subleading order for MNT18 is asserted.

There is likewise an exact finite relative operator on the matched image,

$$\begin{aligned} \mathcal{R}_D &= (Q_D^{\text{source}})^{-1/2} H_D^{\text{image}} (Q_D^{\text{source}})^{-1/2}, \\ 0 &\prec \mathcal{R}_D \preceq \mathcal{K}(D)^2 I, \quad \log \det H_D^{\text{image}} = \log \det Q_D^{\text{source}} + \log \det \mathcal{R}_D. \end{aligned} \quad (\text{MNT19})$$

Conjugating MNT15 proves the inequality and taking determinants proves the equality. The native image's four-return therefore contains the same four signed relative determinants, with high endpoints positive for the dual return and low endpoints negative. Their negative spectral parts and the displaced source windows MNT18 remain exact finite quantities. The proof supplies the actual bounded-degree native operator, its polynomial upper expansion, and the subleading unmatched quotient. It does not assign a value to the remaining matched relative determinant by omitting either of these quantities.

84.6 The original displaced windows with their achieved finite bound

In MNT18 retain the same fixed source degree $j = k - d_h$, source order 1 or j , and common cutoff $N - v$. The complete NW/OW/WNR proofs below now bound its actual displaced difference, with the original two high brackets positive and two low brackets negative. Use the literal WNR10 quantities to define $\Omega_{h,k,s}^{\text{lo}} = U_{h,s}^{\text{lo}}(0) + U_{h,s}^{\text{lo}}(1)$ and $\Omega_{h,k,s}^{\text{hi}} = U_{h,s}^{\text{hi}}(q_j) + U_{h,s}^{\text{hi}}(q_j + 1)$ for $d_h > 0$, with both zero for $d_h = 0$. Then

$$-\Omega_{h,k,s}^{\text{lo}} \leq \mathcal{W}_{h,k,s} \leq \Omega_{h,k,s}^{\text{hi}}, \quad \mathcal{W}_{h,k,s} = o(kq_k) \quad \text{on WNR2.} \quad (\text{MNT20})$$

Here the two finite Ω are the exact WNR sums of NW6's original outer-volume differences, two CTV tail bounds and $4(E_a + E_b)$ at each interval. On $j \geq 4096$ and $j\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}(j+1)^2$, OW10 and OW18 give their full explicit unequal endpoint errors. NW7 proves the same upper increment bound on every specified native flag by Schur factorization; no rank factor is added. Applying it separately to the four literal intervals proves the finite lower and upper signs in MNT20. OW's $O_{\text{actual}}(q_j \log(q_j + 1))$ bounds and $j/k \rightarrow 1$ prove the displayed little-oh. Before that exact domain use NW6's actual outer determinants. If $d_h = 0$ all four increments vanish identically. NBG/NKC/NBR additionally transport the complete degree-four source and source-kernel gauge determinants into its actual invariant image; MNT19's full matched denominator remains.

85 The bounded multiplier image and its exact original receiving maps

Retain the fixed actual simple quartet, amplitude, unit, five-orbit period and logarithm branch of SAR and KRC. All scheme-theoretic component quotients and original conductor coefficients remain. Write $k = 4l + 1 \geq 9$, $q_j = (j+1)^2$, $q'_j = (j-7)^2$, $\Delta_j = 16j - 48$, $v = \text{ord}_0 E_A \leq 80$, $\alpha_A = v!/\mu_v$ and $\mu_v \neq 0$. For each degree in use put

$$\begin{aligned} L_j &= \mathbb{C}[S]_{<v}, \quad t_{0,j} = \dim(K_j \cap L_j), \quad t_{K,j} = 8j - 16 - t_{0,j}, \quad t_{R,j} = 8j - 32 - v + t_{0,j}, \\ \mathbb{T}_j &= \alpha_A N_j Z_{\text{exp},j} : \mathbb{C}^{\Delta_j - v} \xrightarrow{\sim} \mathcal{C}_{v,j}, \quad B_j = Z_j^\top A_{\ell,j}, \quad J_j = A_{\ell,j}^{-1} P_{Y,j}^\top, \quad \text{im } J_j = \ker B_j, \quad (\text{NMR1}) \\ \mathbb{T}_j(\text{im } J_j) &= \mathcal{J}_{v,j}, \quad \mathcal{C}_{v,j}/\mathcal{J}_{v,j} \xrightarrow{\sim} \mathbb{C}^{t_{K,j}}, \quad [f] \mapsto B_j \mathbb{T}_j^{-1} f. \end{aligned}$$

The final map is onto because both $\mathbb{T}_j$ and B_j are onto; its kernel is the displayed invariant low annihilator by KRC5. These are precisely SAR1's complex-linear dual coordinates. In particular $A_{\ell,j}$ denotes the original functional matrix, and not a change in source order. The quotient Gram is $(B_j D^{-1} B_j^*)^{-1}$, while the residual Gram is $J_j^* D J_j$. Their exact native/dual identifications are SAR1. None of these coordinate maps assigns a norm to a missing low-jet component.

The multiplier construction below has exactly the BIM/MNT domain $k \geq 37$, $k \equiv 1 \pmod{4}$. Its source degrees are $\ell_h = k - d_h \geq 9$, with $h \in \mathcal{H}$ the full BIM1 family, $d_h \in \{0, 4, \dots, 28\}$, including the unit. The branch calculation NMR10 remains valid for every original $k \geq 9$. Earlier finite degrees use that calculation and the unchanged SAR intervals without the multiplier entries.

85.1 Both low-jet domains and the exact image quotient

Keep the full maps L_k , $M_{v,j}$, $S_j^{\text{jet}} = N_j E_j V_j^{-1}$ and D_h of BIM4,8,15,16. The precise coupled domain and its target map are

$$\begin{aligned}
f_h &= T_{\ell_h} z_h + S_{\ell_h}^{\text{jet}} \lambda_h \in \mathcal{J}_{\ell_h}, & \sum_h D_h \lambda_h &= 0, \\
((z_h, \lambda_h))_h &\mapsto T_k^{-1} \sum_h \mu_h (T_{\ell_h} z_h + S_{\ell_h}^{\text{jet}} \lambda_h), \\
(D_h)_{ab} &= \begin{cases} \binom{a}{b} \mathcal{N}_h^{(a-b)}(0), & 0 \leq b \leq a < v, \\ 0, & b > a, \end{cases} & \mathcal{P}_k^{\text{individual}} &= \bigoplus_h \mathcal{J}_{v, \ell_h}, \\
\mathcal{J}_{v,k} &= L_k(\mathcal{P}_k^{\text{individual}}), & c_k &= \sum_h (v - t_{0, \ell_h}), & e_k &= \dim(\mathcal{C}_{v,k} / \mathcal{J}_{v,k}), \\
0 &\leq e_k \leq c_k - v \leq (|\mathcal{H}| - 1)v.
\end{aligned} \tag{NMR2}$$

Here μ_h is multiplication by the literal original h in the complete conductor quotient. To verify every domain, apply $M_{v,j}$ to the decomposition of f_h : its value is λ_h , since $M_{v,j} T_j = 0$ and $M_{v,j} S_j^{\text{jet}} = I$. The latter equality follows from the original Vandermonde V_j and the fact that the jet matrix annihilates the entire conductor image. Leibniz' rule gives $M_{v,k} \mu_h = D_h M_{v, \ell_h}$, so precisely the displayed coupled sum has zero jet and belongs to the domain of T_k^{-1} . The invariant membership condition applies to the whole f_h , retaining its possible nonzero individual jet. Setting every $\lambda_h = 0$ gives the individual domain exactly. BIM's onto coupled map induces an onto map from the coupled/individual quotient to $\mathcal{C}_{v,k} / \mathcal{J}_{v,k}$. The former has dimension $c_k - v$, proving all three finite inequalities. The additional jet coordinates and their Vandermonde inverse remain explicit algebraic coordinates; the native analytic norms below are only on the individual domain where the quotient germs are analytic at zero.

For $x = (x_h)_h \in \bigoplus_h \mathbb{C}^{t_{R, \ell_h}}$ define

$$\begin{aligned}
\Phi_k x &= \sum_h T_k^{-1} \mu_h T_{\ell_h} J_{\ell_h} x_h, & \Psi_k^{\text{mult}} &= B_k \Phi_k, & \mathcal{U}_k &= \text{im } \Psi_k^{\text{mult}}, \\
a_k &= \dim \mathcal{U}_k, & \varepsilon_k &= t_{K,k} - a_k, & \mathbb{C}^{t_{K,k}} / \mathcal{U}_k &\stackrel{\sim}{\leftarrow} \mathcal{C}_{v,k} / (\mathcal{J}_{v,k} + \mathcal{J}_{v,k}), \\
0 &\leq \varepsilon_k \leq e_k \leq c_k - v \leq (|\mathcal{H}| - 1)v, & \delta_k T_k \Phi_k &= X_k \Psi_k^{\text{mult}}, \\
&& \dim(\text{im } X_k / X_k \mathcal{U}_k) &\leq \varepsilon_k.
\end{aligned} \tag{NMR3}$$

The superscript distinguishes Ψ_k^{mult} from the original signed scalar Ψ_k in SAR14. The quotient isomorphism sends $[f]$ to $[B_k T_k^{-1} f]$. It is onto by NMR1. Its kernel consists exactly of those f whose B_k value equals that of an element of $\mathcal{J}_{v,k}$; subtracting that element leaves $\mathcal{J}_{v,k}$. This proves the exact denominator and the codimension inequalities. Both occurrences of α_A remain in Φ_k ; they cancel only by the common nonzero scalar conjugation proved in MNT13. KRC14 gives $\delta_k T_k = X_k B_k$, proving the commuting branch map. The induced onto map $\mathbb{C}^{t_{K,k}} / \mathcal{U}_k \rightarrow \text{im } X_k / X_k \mathcal{U}_k$ proves its final rank bound. Its kernel is $(\mathcal{U}_k + \ker X_k) / \mathcal{U}_k$. Thus all original branch descent kernels, including shared and nonreduced components, are retained in this exact quotient.

85.2 The native direct sum and its full source kernel

For each target source $s = 1, k$ use exactly the pairs $(s_{\ell_h}, s) = (1, 1)$ or (ℓ_h, k) , respectively. The common polynomial cutoff is $D = N - v$, with $N \geq q_k - 1$. Let $D_{j,D}^{(s_j)}$ be the full attained lower-evaluation extension Gram of MNT12, in its original center $c'_j = j/2 - 4$. Retain the exact

constants

$$\begin{aligned}
\eta_{h,a,b} &= \beta_{d_h,a,b} - d_h/2, \quad K_h(D) = \sum_{a,b} |h_{ab}| e^{|\eta_{h,a,b}|(H_D+2)}, \quad H_D = \sum_{n=1}^D \frac{1}{n}, \\
a_{\ell_h, d_h/4} &= (2\pi)^{d_h/2} \prod_{j=0}^{d_h/4-1} \frac{\ell_h + 4j}{\ell_h + 4j + 2}, \quad \kappa_h(D) = \begin{cases} K_h(D), & s = 1, \\ K_h(D) a_{\ell_h, d_h/4}^{-1/2}, & s = k, \end{cases} \quad (\text{NMR4}) \\
\mathcal{K}(D)^2 &= \sum_h \kappa_h(D)^2, \quad \mathcal{G}_{N,s} = \text{diag}_{h \in \mathcal{H}} (J_{\ell_h}^* D_{\ell_h, N-v}^{(s_{\ell_h})} J_{\ell_h}).
\end{aligned}$$

Empty products are one and $H_0 = 0$. The source density remains

$$\frac{M_s 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2, \quad M_s = (2\pi)^{s/2}.$$

The finite Gamma recurrence MNT7 gives the displayed a with the full mass ratio; it proves $dm_k \geq a_{\ell_h, d_h/4} dm_{\ell_h}$ in precisely this direction. The exact center map is $p(c'_{\ell_h} + iy + \beta_{d_h, a, b}) = p_k(y + \eta_{h, a, b}/i)$. The derivative estimate of MNT5, finite Taylor expansion, and this density inequality prove the complete $\kappa_h(D)$ bound. Precomposition sends every original lower evaluation to its full target sum with roots $\beta'_{\ell_h-8} + \beta_{d_h, a, b}$, so minimizing the same extension fibre proves the quotient bound MNT13. Applying that bound to the full direct sum and Cauchy–Schwarz proves the $\mathcal{K}(D)$ bound for the actual Ψ_k^{mult} . The defining sum is finite, positive and retains all coefficients.

Fix an injective frame U_k of $\mathcal{U}_k$, its exact pivot left inverse L_{U_k} , and put $\tilde{\Psi}_k = L_{U_k} \Psi_k^{\text{mult}}$, an onto matrix. Choose a fixed injective frame $Z_{\ker \Psi, k}$ of its complete kernel and a fixed right inverse $R_{\Psi, k}$. All these maps depend only on original coefficients and degree, not on N or on the two source choices. Define

$$\begin{aligned}
Q_{N,s}^{\text{src}} &= (\tilde{\Psi}_k \mathcal{G}_{N,s}^{-1} \tilde{\Psi}_k^*)^{-1}, \quad H_{N,s}^{\ker \Psi} = Z_{\ker \Psi, k}^* \mathcal{G}_{N,s} Z_{\ker \Psi, k}, \\
H_{N,s}^{\text{im}} &= U_k^* (B_k (D_{k, N-v}^{(s)})^{-1} B_k^*)^{-1} U_k, \quad H_{N,s}^{\text{im}} \preceq \mathcal{K}(N-v)^2 Q_{N,s}^{\text{src}}, \quad (\text{NMR5}) \\
F_{\Psi, k} &= [Z_{\ker \Psi, k}, R_{\Psi, k}], \quad \det \mathcal{G}_{N,s} = |\det F_{\Psi, k}|^{-2} \det H_{N,s}^{\ker \Psi} \det Q_{N,s}^{\text{src}}.
\end{aligned}$$

To prove the quotient formula, the lift $\mathcal{G}^{-1} \tilde{\Psi}_k^* Q_{N,s}^{\text{src}} z$ has image z and is $\mathcal{G}$ -orthogonal to the entire kernel. Every other lift adds a kernel vector and its nonnegative squared norm. This proves both the attained minimum and its displayed Gram. Applying the norm bound just proved to this very lift gives the image inequality. The frame $F_{\Psi, k}$ is invertible: apply $\tilde{\Psi}_k$ to a vanishing column combination to kill the right-inverse coefficients, then use the injectivity of $Z_{\ker \Psi, k}$. Its full Gram has kernel restriction as first block and the same attained quotient as its Schur complement. Taking its determinant proves the last line with the inverse squared frame factor. In particular this calculation retains the entire kernel of the family map, including relations between different multipliers.

For a sequence of positive forms on a fixed space define $\mathfrak{V}_k(H) = \log \det H_{2q_k-1} + \log \det H_{2q_k} - \log \det H_{q_k-1} - \log \det H_{q_k}$. Put $b_{h,k} = d_h(2k - d_h + 2)$ and $f_{h,s}(N) = \log \det (J_{\ell_h}^* D_{\ell_h, N-v}^{(s_{\ell_h})} J_{\ell_h})$. Then NMR5 gives the exact return

$$\begin{aligned}
V_{Q,k,s} &:= \mathfrak{V}_k(Q^{\text{src}}) = \sum_h (S_{R, \ell_h, s_{\ell_h}} + \mathcal{W}_{h,k,s}) - V_{\ker \Psi, k,s}, \quad V_{\ker \Psi, k,s} := \mathfrak{V}_k(H^{\ker \Psi}) \geq 0, \\
\mathcal{W}_{h,k,s} &= [f_{h,s}(2q_{\ell_h} - 1 + 2b_{h,k}) - f_{h,s}(2q_{\ell_h} - 1)] \\
&\quad + [f_{h,s}(2q_{\ell_h} + 2b_{h,k}) - f_{h,s}(2q_{\ell_h})] \\
&\quad - [f_{h,s}(q_{\ell_h} - 1 + b_{h,k}) - f_{h,s}(q_{\ell_h} - 1)] \\
&\quad - [f_{h,s}(q_{\ell_h} + b_{h,k}) - f_{h,s}(q_{\ell_h})], \quad V_{Q,k,s} \geq 0. \quad (\text{NMR6})
\end{aligned}$$

Here $S_{R, \ell_h, s_{\ell_h}}$ is the unchanged SAR2 residual sum at degree ℓ_h and its own four endpoints. The identity $q_k = q_{\ell_h} + b_{h,k}$ proves every displaced index. Taking logarithms in NMR5 cancels its fixed

frame factor in these four signs. The direct-sum determinant is the product of its individual source determinants; subtracting each source's usual four-return gives exactly the four displayed brackets. Each bracket is nonnegative: increasing the polynomial cutoff increases the full extension Gram, and restriction preserves that increase. The signed combination $\mathcal{W}_{h,k,s}$ retains all four brackets without a sign or size assignment. Likewise the complete source-kernel restriction and its quotient increase with N , by minimization on fixed fibres. Their four returns have the original positive weights $1, 2, \dots, 2, 1$, proving the two nonnegativities. No endpoint beyond a source's usual window is replaced by its usual endpoint.

85.3 Matched relative determinants and the finite unmatched return

On the exact image coordinate space put

$$\begin{aligned}
\mathcal{R}_{N,s} &= (Q_{N,s}^{\text{src}})^{-1/2} H_{N,s}^{\text{im}} (Q_{N,s}^{\text{src}})^{-1/2}, \quad 0 \prec \mathcal{R}_{N,s} \preceq \mathcal{K}(N-v)^2 I, \\
P_{N,s}^{\text{mult}} &= \sum_{i=1}^{a_k} \log^+ \lambda_i(\mathcal{R}_{N,s}), \quad N_{N,s}^{\text{mult}} = \sum_{i=1}^{a_k} \log^+(1/\lambda_i(\mathcal{R}_{N,s})), \\
0 &\leq P_{N,s}^{\text{mult}} \leq E_{N,s}^{\text{mult}} := 2a_k \log \max\{1, \mathcal{K}(N-v)\}, \\
n_{k,s}^{\text{mult}} &= N_{q_k-1,s}^{\text{mult}} + N_{q_k,s}^{\text{mult}} - N_{2q_k-1,s}^{\text{mult}} - N_{2q_k,s}^{\text{mult}}, \quad p_{k,s}^{\text{mult}} = P_{2q_k-1,s}^{\text{mult}} + P_{2q_k,s}^{\text{mult}} - P_{q_k-1,s}^{\text{mult}} - P_{q_k,s}^{\text{mult}}, \\
E_{k,s}^{\text{mult,lo}} &= E_{q_k-1,s}^{\text{mult}} + E_{q_k,s}^{\text{mult}}, \quad E_{k,s}^{\text{mult,hi}} = E_{2q_k-1,s}^{\text{mult}} + E_{2q_k,s}^{\text{mult}}, \quad -E_{k,s}^{\text{mult,lo}} \leq p_{k,s}^{\text{mult}} \leq E_{k,s}^{\text{mult,hi}}, \\
\mathfrak{V}_k(H^{\text{im}}) &= V_{Q,k,s} + n_{k,s}^{\text{mult}} + p_{k,s}^{\text{mult}}.
\end{aligned} \tag{NMR7}$$

Conjugation of NMR5 proves the matrix inequality. Each relative eigenvalue is positive, and its positive logarithm is at most $2 \log \max(1, \mathcal{K})$; summation proves the endpoint bound. The complete determinant equality is $\det H_{N,s}^{\text{im}} = \det Q_{N,s}^{\text{src}} \det \mathcal{R}_{N,s}$. Use $\log \lambda = \log^+ \lambda - \log^+(1/\lambda)$ and the four displayed signs to obtain the final identity. No monotonicity of $\mathcal{R}_{N,s}$ or of its positive and negative parts is needed. In particular both high negative logarithms and both low negative logarithms are still present. At empty image rank every determinant here is one and every sum is zero, interpreted in the unique zero-dimensional space.

Choose a fixed onto map $P_{\text{miss},k}$ with kernel $\mathcal{U}_k$, and a fixed right inverse $R_{\text{miss},k}$. With $D_{K,N} = (B_k(D_{k,N-v}^{(s)})^{-1} B_k^*)^{-1}$ set

$$\begin{aligned}
H_{N,s}^{\text{miss}} &= (P_{\text{miss},k} D_{K,N}^{-1} P_{\text{miss},k}^*)^{-1}, \quad A_{k,s}^{\text{end}} = \max\{1, q_k \mathcal{B}_{2q_k-v,s,k,A}^2\} (M_{k,s}^{\text{rec}})^2, \\
\det D_{K,N} &= |\det[U_k, R_{\text{miss},k}]|^{-2} \det H_{N,s}^{\text{im}} \det H_{N,s}^{\text{miss}}, \quad V_{\text{miss},k,s} := \mathfrak{V}_k(H^{\text{miss}}), \\
0 &\leq V_{\text{miss},k,s} \leq E_{k,s}^{\text{miss}} := 2\varepsilon_k \log A_{k,s}^{\text{end}} \leq 2(c_k - v) \log A_{k,s}^{\text{end}}.
\end{aligned} \tag{NMR8}$$

The frame proof in FRE9 gives the determinant identity, with the complete fixed frame factor still visible. The missing metric is the direct quotient of $D_{k,N-v}^{(s)}$ under $P_{\text{miss},k} B_k$; it therefore uses all original lower evaluations. FRE2 composes the full LJR reconstruction into exactly the DNE coefficient quotient, and FRE3 gives $D_{k,q_k-v-1}^{(s)} \preceq D_{k,N-v}^{(s)} \preceq A_{k,s}^{\text{end}} D_{k,q_k-v-1}^{(s)}$. Minimizing both inequalities on this same quotient fibre preserves them. On rank ε_k each relative eigenvalue is in $[1, A_{k,s}^{\text{end}}]$. The two high-endpoint increments are at most $\varepsilon_k \log A_{k,s}^{\text{end}}$ each, and the subtracted low increment is nonnegative. This proves the finite bound, including the exact sharper degree-dependent t_{0,ℓ_h} values inside c_k . Every source mass, actual μ_v , root radius, reconstruction factor and finite Gamma sum remains inside the displayed $\mathcal{B}$ and M^{rec} of the full FRE proof.

The fixed frame factor in NMR8 cancels in the four original signs. SAR2 identifies that full

native kernel return with $S_{K,s}$. Consequently the exact formula and finite interval are

$$\begin{aligned}
S_{K,s} &= V_{Q,k,s} + n_{k,s}^{\text{mult}} + p_{k,s}^{\text{mult}} + V_{\text{miss},k,s}, \\
S_{K,s} &= \sum_h (S_{R,\ell_h,s_{\ell_h}} + \mathcal{W}_{h,k,s}) - V_{\ker \Psi,k,s} + n_{k,s}^{\text{mult}} + p_{k,s}^{\text{mult}} + V_{\text{miss},k,s}, \\
m_{k,s}^{\text{nat},-} &= V_{Q,k,s} + n_{k,s}^{\text{mult}} - E_{k,s}^{\text{mult,lo}}, \quad m_{k,s}^{\text{nat},+} = V_{Q,k,s} + n_{k,s}^{\text{mult}} + E_{k,s}^{\text{mult,hi}} + E_{k,s}^{\text{miss}}, \\
m_{k,s}^{\text{nat},-} &\leq S_{K,s} \leq m_{k,s}^{\text{nat},+}, \quad \mathcal{T}_{k,s} - m_{k,s}^{\text{nat},+} \leq S_{R,s} \leq \mathcal{T}_{k,s} - m_{k,s}^{\text{nat},-}.
\end{aligned} \tag{NMR9}$$

This is a composition of the attained source quotient, its identity into the actual native image, and the full missing quotient. NMR6–8 prove each equality and each error direction. The residual interval follows by the exact same-row identity $S_{R,s} = \mathcal{T}_{k,s} - S_{K,s}$, with reversed endpoints. The source kernel and displaced windows have not been discarded in obtaining either native allocation.

85.4 The full original branch correspondence in this interval

For every retained sign choice ϵ of KRC10 let $a_{\epsilon,k}$ be the exact common-image rank in KRC15, and let $\mathcal{R}_{W,N}$ be its actual relative metric from KRC20, at the same target source s . Put $C_{\epsilon,k,s}^{\text{br}} = \mathfrak{V}_k(\mathcal{R}_W)$. Then the complete unmatched flags of KRC21–23 give

$$\begin{aligned}
C_{\epsilon,k,s}^{\text{br}} - 2(t_{R,k} - a_{\epsilon,k}) \log A_{k,s}^{\text{end}} &\leq S_{K,s} - S_{R,s} \leq C_{\epsilon,k,s}^{\text{br}} + 2(t_{K,k} - a_{\epsilon,k}) \log A_{k,s}^{\text{end}}, \\
b_{\epsilon,k,s}^- &= \frac{\mathcal{T}_{k,s} + C_{\epsilon,k,s}^{\text{br}}}{2} - (t_{R,k} - a_{\epsilon,k}) \log A_{k,s}^{\text{end}}, \\
b_{\epsilon,k,s}^+ &= \frac{\mathcal{T}_{k,s} + C_{\epsilon,k,s}^{\text{br}}}{2} + (t_{K,k} - a_{\epsilon,k}) \log A_{k,s}^{\text{end}}, \quad b_{\epsilon,k,s}^- \leq S_{K,s} \leq b_{\epsilon,k,s}^+.
\end{aligned} \tag{NMR10}$$

Indeed KRC24–25 gives the exact contrast $C_{\epsilon,k,s}^{\text{br}} + \sum_M w_M(e_{B,M} - e_{J,M})$. The two B unmatched flags have total rank $t_{K,k} - a_{\epsilon,k}$; the two J flags have total rank $t_{R,k} - a_{\epsilon,k}$. Each flag is obtained by fixed restrictions and attained quotients from the same D_N , including both further common-image quotients. FRE6 bounds their nonnegative full returns by twice their ranks times $\log A_{k,s}^{\text{end}}$. Add the two flags on each side and use the minus sign for the J contribution to get the first line. Adding $S_K + S_R = \mathcal{T}$ and dividing by two proves the remaining lines. The actual ranks still contain KRC's descent defects $o_k, d_{\epsilon,k}$ and the original t_0 ; no bounded-rank conclusion about those defects was required. The four choices give bounds on the identical scalar even though their common-image spaces and relative metrics can differ.

85.5 Return through SAR, HCR and every original finite allowance

Let $[\underline{S}_{K,s}, \overline{S}_{K,s}]$ be the complete SAR9 interval, including its domain-restricted SAR16 entries. Adjoin all four NMR10 intervals and, for $k \geq 37$, NMR9. Define their intersection and its exact residual partner by

$$\begin{aligned}
\underline{S}_{K,s}^{\text{nat}} &= \max\{\underline{S}_{K,s}, b_{\epsilon,k,s}^- (\text{epsilon}), m_{k,s}^{\text{nat},-} (k \geq 37)\}, \\
\overline{S}_{K,s}^{\text{nat}} &= \min\{\overline{S}_{K,s}, b_{\epsilon,k,s}^+ (\text{epsilon}), m_{k,s}^{\text{nat},+} (k \geq 37)\}, \\
\underline{S}_{R,s}^{\text{nat}} &= \mathcal{T}_{k,s} - \overline{S}_{K,s}^{\text{nat}}, \quad \overline{S}_{R,s}^{\text{nat}} = \mathcal{T}_{k,s} - \underline{S}_{K,s}^{\text{nat}}, \quad S_{X,s} \in [\underline{S}_{X,s}^{\text{nat}}, \overline{S}_{X,s}^{\text{nat}}].
\end{aligned} \tag{NMR11}$$

Every adjoined interval was proved on its stated original domain for that scalar. Their intersection is therefore nonempty. The map $z \mapsto (z, \mathcal{T}_{k,s} - z)$ has inverse the first coordinate, and carries this interval bijectively onto the two allocations subject to these bounds and their exact total. Thus the total is retained as a measured quantity throughout the return, without prescribing either individual allocation.

Keep the two original low metrics separately, with $\lambda_s^{\text{low}} = F_L^{(s)} - F_{W_0}^{(s)}$. The exact isometry $(K + L)/K \rightarrow W_0$, $[p] \mapsto \Xi p$, has full fibre $p + K \subset K + L$, so its two quotient minima agree. The

low volume is not the whole residual volume. More precisely the quotient map $B_\Xi \rightarrow B_\Xi/W_0$ has the full native Schur metric whose transported counterpart is $Q_{0,N}$ in PBR7. Retain $D_s^\pm$ and $E_{k,s}^{\text{tr}}$ exactly:

$$\begin{aligned}
w_d^{(s)} &= d!/(s/2)_d, \quad n_N = N + 1 - v, \quad m_N = \log \max\{1, \frac{v!}{|\mu_v|} \sum_\nu |a_\nu| e^{|b_\nu - 4|(H_N + 2)}\}, \\
\zeta_{N,s} &= \log \prod_{d=v}^N \frac{w_d^{(s)}}{w_{d-v}^{(s)}}, \quad f_{N,s}^- = \zeta_{N,s} - 2(n_N - t_{K,k})m_N, \quad f_{N,s}^+ = 2t_{K,k}m_N, \\
D_s^- &= f_{q_k-1,s}^- + f_{q_k,s}^- - f_{2q_k-1,s}^+ - f_{2q_k,s}^+, \quad D_s^+ = f_{q_k-1,s}^+ + f_{q_k,s}^+ - f_{2q_k-1,s}^- - f_{2q_k,s}^-, \\
E_{k,s}^{\text{tr}} &= 2 \sum_{N \in \{q_k-1, q_k, 2q_k-1, 2q_k\}} E_N^{\text{CEP}} + 8t_{R,k} |\log |\alpha_A||, \\
F_K^{(s)} &= S_{K,s} - \mathcal{E}_s^{\text{flag}} + \lambda_s^{\text{low}}, \quad \mathcal{E}_s^{\text{flag}} \in [D_s^-, D_s^+], \quad \Phi_{k,s} = F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}, \quad |\delta_{0,k,s}| \leq E_{k,s}^{\text{tr}}, \\
k_s^{\text{nat},-} &= \underline{S}_{K,s}^{\text{nat}} - D_s^+ + \lambda_s^{\text{low}}, \quad k_s^{\text{nat},+} = \bar{S}_{K,s}^{\text{nat}} - D_s^- + \lambda_s^{\text{low}}, \\
r_s^{\text{nat},-} &= F_{W_0}^{(s)} + \underline{S}_{R,s}^{\text{nat}} - E_{k,s}^{\text{tr}}, \quad r_s^{\text{nat},+} = F_{W_0}^{(s)} + \bar{S}_{R,s}^{\text{nat}} + E_{k,s}^{\text{tr}}.
\end{aligned} \tag{NMR12}$$

Here $f_{N,s}^\pm$ is just the explicitly displayed PBR8 endpoint $e_{N,s}^\pm$ with a distinct symbol to avoid its collision with the OAR errors below; no value is changed. The two exact return identities are HCR3. Subtracting the flag error uses D_s^+ at the lower kernel endpoint and D_s^- at the upper one. Adding the transfer interval proves $F_K^{(s)} \in [k_s^{\text{nat},-}, k_s^{\text{nat},+}]$ and $\Phi_{k,s} \in [r_s^{\text{nat},-}, r_s^{\text{nat},+}]$. All four CEP terms, the actual α_A , every original factorial ratio and the separate low volumes remain in these specific formulas.

Intersect the residual interval with SAR11 and return the same original OAR and PMR identities:

$$\begin{aligned}
\Phi_{k,s}^{-,\text{nat}} &= \max\{\Phi_{k,s}^{-,\text{all}}, r_s^{\text{nat},-}\}, \quad \Phi_{k,s}^{+,\text{nat}} = \min\{\Phi_{k,s}^{+,\text{all}}, r_s^{\text{nat},+}\}, \\
L_s^{\text{nat}} &= \max\{L_s^{\text{all}}, k_s^{\text{nat},-}, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{+,\text{nat}} - e_{k,s}^-\}, \\
U_s^{\text{nat}} &= \min\{U_s^{\text{all}}, k_s^{\text{nat},+}, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{-,\text{nat}} + e_{k,s}^+\}, \quad F_K^{(s)} \in [L_s^{\text{nat}}, U_s^{\text{nat}}].
\end{aligned} \tag{NMR13}$$

For $k \geq 29$ and $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q_k$, also include $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{+,\text{nat}} - e_{k,s}^{0,-}$ in the maximum and $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{-,\text{nat}} + e_{k,s}^{0,+}$ in the minimum, with $e^{0,\pm} = e^\pm + c_k^0$. Outside precisely that PMR domain these two entries are absent. The proof is substitution into $F_K = \mathcal{F}_\partial - \Phi + \mathfrak{e}$ with $-e^- \leq \mathfrak{e} \leq e^+$, or its proved PMR form. Every maximum entry is a lower bound and every minimum entry an upper bound for the same scalar, proving nonempty intersection.

On the original arithmetic period domain $|u| \geq R_*$ retain the same correlated source difference as SAR13:

$$\begin{aligned}
a_{K,k}^- &= \max\{-C_K^-, \delta_k^\sigma - C_B^+\}, \quad a_{K,k}^+ = \min\{C_K^+, \delta_k^\sigma + C_B^-\}, \\
L_{\text{ar}}^{\text{nat}} &= \max\{L_{\text{ar}}^{\text{all}}, L_1^{\text{nat}} + a_{K,k}^-\}, \quad U_{\text{ar}}^{\text{nat}} = \min\{U_{\text{ar}}^{\text{all}}, U_1^{\text{nat}} + a_{K,k}^+\}, \\
F_K^{\text{ar}} &\in [L_{\text{ar}}^{\text{nat}}, U_{\text{ar}}^{\text{nat}}], \quad F_B^{\text{ar}} \in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^{\text{nat}}, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^{\text{nat}}].
\end{aligned} \tag{NMR14}$$

The scalar $F_K^{\text{ar}} - F_K^{(1)}$ belongs to the displayed unchanged interval. Add it to NMR13 and intersect with the previous arithmetic interval. Subtracting from the exact same total proves the reversed boundary endpoints; no independence of errors or replacement source order is used.

Finally keep the entire original signed and action budget:

$$\begin{aligned}
\Psi_k &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, \quad \mathcal{H}_k = -T_k, \quad C_k^{\text{budget}} = F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \\
L_\Psi^{\text{nat}} &= L_k^{\text{nat}} + \mathcal{H}_k + \delta_k^\sigma, \quad U_\Psi^{\text{nat}} = U_k^{\text{nat}} + \mathcal{H}_k + \delta_k^\sigma, \quad \Psi_k \in [L_\Psi^{\text{nat}}, U_\Psi^{\text{nat}}], \\
J_k^{\text{action}} &= \Psi_k + C_k^{\text{budget}}, \quad L_J^{\text{nat}} = L_\Psi^{\text{nat}} + C_k^{\text{budget}}, \quad U_J^{\text{nat}} = U_\Psi^{\text{nat}} + C_k^{\text{budget}}, \\
4q_k \log \mathcal{L}_{h,k} &\leq J_k^{\text{action}} \in [L_J^{\text{nat}}, U_J^{\text{nat}}], \quad \mathcal{L}_{h,k} \leq \min_M \epsilon_M^{\text{ar}} \leq \prod_M (\epsilon_M^{\text{ar}})^{w_M/(2q_k)} \leq \exp(U_J^{\text{nat}}/(4q_k)).
\end{aligned} \tag{NMR15}$$

Addition of these identical original scalars to NMR13 proves both intervals. The untilted identity is $J_k^{\text{action}} = 2 \sum_M w_M \log \epsilon_M^{\text{ar}}$, with $\sum_M w_M = 2q_k$ and the same-class lower radius $\mathcal{L}_{h,k} = 2\delta e(k+1)\lfloor(k+1)^2/4\rfloor$, $e = 1$ on this simple-quartet branch. Taking the weighted mean proves the final inequalities. The full reference boundary, monic window, both contraction penalties, leakage and the actual mixed pairing $a_K d_B + a_B d_K + 2|z_K|^2$, $z_B = -z_K$, remain in this very budget. For a real tilt the original MCE formula with $\log(\epsilon_M^2 + \phi_M^2)$ and the complete MCP derivative is retained; the untilted mean is used on its proved original reflection domain.

For fixed actual data the finite error in NMR9 is quantitatively controlled before taking any limit:

$$\begin{aligned}
m_{k,s}^{\text{nat},+} - m_{k,s}^{\text{nat},-} &= 2a_k \sum_{N \in \{q_k-1, q_k, 2q_k-1, 2q_k\}} \log \max\{1, \mathcal{K}(N-v)\} + 2\varepsilon_k \log A_{k,s}^{\text{end}}, \\
a_k &\leq 8k - 16 - t_{0,k}, \quad \varepsilon_k \leq \sum_h (v - t_{0,\ell_h}) - v, \\
m_{k,s}^{\text{nat},+} - m_{k,s}^{\text{nat},-} &= O_{\text{actual}}(k \log q_k + q_k \log k) = o(kq_k), \\
S_{K,s} &= V_{Q,k,s} + n_{k,s}^{\text{mult}} + o(kq_k).
\end{aligned} \tag{NMR16}$$

The first two lines are the exact finite expressions already proved. The complete uniform estimate of MNT9 is

$$\log \max(1, \kappa_h(D)) = O_{\text{actual}}(\log(D+1)).$$

Its finite sum therefore gives the same order for $\mathcal{K}$. FRE4 gives $\log A_{k,s}^{\text{end}} \leq 2q_k \log k + O_{\text{actual}}(q_k)$, while the achieved bound for ε_k is independent of k . These facts prove the third line. NMR7–9 give the last line with the same explicit lower and upper errors, rather than an assigned value of its retained center. NMR6 still expresses V_Q with every source-kernel and displaced-window term, and n^{mult} still contains every matched negative logarithm. The original KRC relative branch determinant remains in NMR10. The finite identities and proved inequalities above supply their exact morphisms back through the whole earlier receiver. They assign no inverse bound, equal allocation, individual leading coefficient or terminal RH conclusion.

85.6 Backward propagation of the achieved windows and complete polar flag

The four-factor Vandermonde proof RPB14/NBR2 gives $v \leq 32$ on the entire original five-orbit locus. Consequently the exact NMR3 codimension obeys $\varepsilon_k \leq c_k - v \leq 32(|\mathcal{H}| - 1)$; all $t_{0,k-d_h}$ remain in c_k . NW/OW/WNR now gives finite $\Omega_{k,s}^{\text{lo}}, \Omega_{k,s}^{\text{hi}}$ for the sum of the precise NMR6 windows. Put

$$\mathcal{Z}_{k,s} = \sum_h S_{R,k-d_h, s_{k-d_h}} - V_{\ker \Psi, k, s} + n_{k,s}^{\text{mult}}. \tag{NMR17}$$

Adding the signed window bounds to NMR7–9 proves

$$\mathcal{Z}_{k,s} - \Omega_{k,s}^{\text{lo}} - E_{k,s}^{\text{mult}, \text{lo}} \leq S_{K,s} \leq \mathcal{Z}_{k,s} + \Omega_{k,s}^{\text{hi}} + E_{k,s}^{\text{mult}, \text{hi}} + E_{k,s}^{\text{miss}}. \tag{NMR18}$$

The Ω are the complete WNR sums, with original source orders and separate finite errors at all four endpoints. The little-oh consequence uses every nontrivial source degree $j = k - d_h \geq 4096$ and its original quartet inequality; no such estimate is applied at smaller j . All earlier degrees

retain the same finite formula with NW6's actual outer determinants. The source-kernel volume and all four negative relative logarithms remain in $\mathcal{Z}$. Intersecting NMR18 with NMR11 and then using the exact affine map $z \mapsto (z, \mathcal{T}_{k,s} - z)$ gives both native allocations with reversed residual endpoints.

PLF's actual pole map supplies the full rank- j residual and its rank- $t_{R,k}$ low-annihilating subspace. Its complete fixed-frame Schur identity and original source transport prove precisely

$$\Phi_{k,s} = F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}, \quad |\delta_{0,k,s}| \leq 2 \sum_N E_N^{\text{CEP}} + 8t_{R,k} |\log |\alpha_A||. \quad (\text{NMR19})$$

Thus this is the same NMR12 scalar and allowance, now with its full polar coordinate matrix, not an additional independent error. NBR20–23 applies the sharper allocation interval with all factorial flag errors, separate low volumes, OAR/PMR domains, correlated arithmetic caps and the whole original action budget.

86 Scope, source audit, and the original finite relation

This calculation continues the accepted 238-page OOQ–OER edition. Its RCF1–15, RCM1–27 and DNE1–15 companions were supplied as unreviewed complete drafts, not as accepted theorems. The finite factor sequence and the complex-linear dual-extension formulas below have been checked against their full ROG, RDS and GIP maps. They are attributed to those drafts. The additional results are a sharper contour error, an explicit actual-coefficient visibility test, all-rank logarithmic spectral control of the visible quotient, estimates for the original RWC exterior coefficients, and its full conditional complement. No numerical actual period, generic support, positive visibility rank, or individual residual leading coefficient is assumed.

Fix the original hypothetical simple quartet, its complete unit and a single legitimate five-orbit period. Thus

$$\begin{aligned} 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad k \equiv 1 \pmod{4}, \quad k \geq 9, \quad q = (k+1)^2, \quad q' = (k-7)^2, \\ \Delta = q - q' = 16k - 48, \quad r = 8k - 16, \quad j = 8k - 32, \quad c = k/2, \quad c' = c - 4, \\ \beta_{ij} = c + (2i - k)\delta + i(2j - k)\gamma, \quad \beta'_{ij} = c' + (2i - k + 8)\delta + i(2j - k + 8)\gamma. \end{aligned} \quad (\text{CFR1})$$

The letters i, j in subscripts are integers; i is the imaginary unit. Write χ, ψ for the full upper and lower monic root polynomials, $\chi(S) = \prod_{i,j=0}^k (S - \beta_{ij})$ and $\psi(S') = \prod_{i,j=0}^{k-8} (S' - \beta'_{ij})$; ψ is not a derivative. Empty matrices use determinant one and the unique zero-dimensional inverse throughout. Retain the actual 81 coefficients a_{ab} of the period conductor and the shifts $b_{ab} = 4 + (2a - 8)\delta + i(2b - 8)\gamma$. Put

$$\begin{aligned} \mu_h &= \sum_{a,b} a_{ab} b_{ab}^h, \quad v = \min\{h : \mu_h \neq 0\} \leq 80, \quad m_0 = \Delta - v, \\ d_{ab}(S') &= \frac{\chi(S' + b_{ab})}{\psi(S')}, \quad (\mathcal{U}h)(S') = \sum_{a,b} a_{ab} d_{ab}(S') h(S' + b_{ab}), \\ \mathcal{T}_{Ap} &= \sum_{a,b} a_{ab} p(S' + b_{ab}), \quad \psi \mathcal{U}h = \mathcal{T}_A(\chi h). \end{aligned} \quad (\text{CFR2})$$

Every quotient here is polynomial by the original inner-grid inclusion. For a degree- d input with leading coefficient $h_d \neq 0$, $\mathcal{U}h$ has degree $d + m_0$ and leading coefficient $h_d \binom{q+d}{v} \mu_v$. This is the finite Taylor expansion of $\mathcal{T}_A$, followed by monic division by ψ ; it holds without any sign condition on the coefficients. At the four original endpoints

$$N = q - 1, q, 2q - 1, 2q, \quad L = N - q = -1, 0, q - 1, q, \quad M = L + m_0. \quad (\text{CFR3})$$

Negative-degree polynomial spaces are zero. The relation quotient is $Y_N = \mathcal{P}_M/\mathcal{UP}_L$, in its unique low-monomial frame of length m_0 . Its source is exactly

$$\|f\|_{\psi,s}^2 = \int_{\mathbb{R}} |\psi(c' + iy)f(c' + iy)|^2 dm_s(y), \quad dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad s \in \{1, k\}. \quad (\text{CFR4})$$

The mass is $(2\pi)^{s/2}$ and the original monic norms are $(2\pi)^{s/2} n!(s/2)_n$. The displayed convention agrees with the Meixner–Pollaczek normalization after its stated affine change; all estimates here use the provided original recurrence, not a new source order (NIST, §18.19 and §18.22).

87 Audit of the common factor and its full finite complement

Let $\mathcal{A} = \{(a, b) : a_{ab} \neq 0\}$. Define its four integer extrema a_-, a_+, b_-, b_+ and widths $d_a = a_+ - a_-$, $d_b = b_+ - b_-$. For any integer pair (p, t) set $\zeta_{pt} = c' + (2p - k + 8)\delta + i(2t - k + 8)\gamma$. The active product d_{ab} has the simple roots

$$\{\zeta_{pt} : (p, t) \in ([-a, k - a] \times [-b, k - b]) \setminus [0, k - 8]^2\}.$$

All intervals here and below contain their integer points. Comparing real and imaginary parts proves distinctness on the whole indexed lattice. Consequently the exact monic common divisor is

$$\mathcal{R} = [-a_-, k - a_+] \times [-b_-, k - b_+], \quad \mathcal{J} = \mathcal{R} \setminus [0, k - 8]^2, \quad G(S') = \prod_{(p,t) \in \mathcal{J}} (S' - \zeta_{pt}),$$

$$g = \deg G = (k + 1 - d_a)(k + 1 - d_b) - q', \quad h_0 = m_0 - g = (d_a + d_b)(k + 1) - d_a d_b - v \geq 0. \quad (\text{CFR5})$$

The inequality follows also because G divides $\mathcal{U}1$, of degree m_0 . The active shifted Vandermonde $((S' + b_\alpha)^n)_{0 \leq n < |\mathcal{A}|, \alpha \in \mathcal{A}}$ has nonzero constant determinant $\prod_{\alpha < \beta} (b_\beta - b_\alpha)$. It has a polynomial inverse. The first $|\mathcal{A}|$ relations therefore generate the same polynomial ideal as the active $a_\alpha d_\alpha$. It follows that the *ideal generated by* $\text{im } \mathcal{U}$ is (G) . This does not identify its linear image with (G) .

Define the actual remaining operator

$$\mathcal{V}f = \sum_{(a,b) \in \mathcal{A}} a_{ab} \frac{d_{ab}}{G} f(S' + b_{ab}), \quad \mathcal{U} = M_G \mathcal{V}. \quad (\text{CFR6})$$

It raises degree by exactly h_0 with the same nonzero leading factor as $\mathcal{U}$. Highest-degree elimination proves the direct remainders by $\mathcal{U}$ and $\mathcal{V}$, with lengths m_0 and h_0 , respectively. Thus the full finite-window sequence is

$$0 \longrightarrow \mathcal{P}_{M-g}/\mathcal{VP}_L \xrightarrow{M_G} \mathcal{P}_M/\mathcal{UP}_L \xrightarrow{\text{rem}_G} \mathcal{P}_{<g} \longrightarrow 0. \quad (\text{CFR7})$$

The dimensions are h_0, m_0, g at *each* of the four endpoints, including $L = -1$. Injectivity on the left is cancellation of the nonzero polynomial G ; surjectivity on the right uses $M \geq g - 1$. Every middle representative of a class in the left image remains a multiple of G , because its allowed change is $G\mathcal{V}f$. Its exact inherited norm is consequently the quotient norm with source $|\psi G|^2 dm_s$. The right metric is the attained remainder metric from $|\psi|^2 dm_s$ at degree M , denoted $Q_{G,N}^{(s)}$.

Let $H_{U,N}, H_{V,N}$ be the first two quotient Grams in their low monomial frames. The adapted middle frame is $[G, GS', \dots, GS'^{h_0-1}, 1, S', \dots, S'^{g-1}]$. Its determinant has modulus one: after reordering by degree it is monic triangular. Minimization over its first block is the full Schur complement, proving

$$\det H_{U,N} = \det H_{V,N} \det Q_{G,N}, \quad F_U = F_V + F_G. \quad (\text{CFR8})$$

The last identity uses precisely the four signs $(+, +, -, -)$. All mixed blocks are included in these metrics. In particular, $\text{im } \mathcal{U}$ is an ideal if and only if $g = m_0$. When both active widths equal eight, $g = 0$: the common-factor target is zero, whereas the full relation quotient still has length m_0 .

88 Sharper full-metric control of this particular divisor

The RCM draft's norm argument extends the accepted OOQ construction to the divisor (CFR5), but uses an unnecessarily large Cauchy disk. We give the entire finite error with a smaller admissible disk. Let $R = k\sqrt{\delta^2 + \gamma^2}$, $R_G = (k+8)\sqrt{\delta^2 + \gamma^2}$, $n = q/2$, $\vartheta = \pi/2$, and $\epsilon_0 = 2^{-32}$. Impose

$$\begin{aligned} k &\geq 257, \quad R \leq \epsilon_0 q/2, \quad n \geq 2\Delta + 1, \quad n \geq 2/b_*, \\ \rho &= 2R_G/n \leq \min\{1/2, \sqrt{a_*/2}\}. \end{aligned} \quad (\text{CFR9})$$

This is an explicit eventual domain at every fixed quartet; no period coefficient occurs in its cutoff. The symbols a_* , b_* are specified next. The case $g = 0$ has its empty metric and needs no analytic estimate.

88.1 Every equilibrium and source constant

Use the provided full EIQ equilibrium for $V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha\log x$ on the compact rectangle $\mathcal{K} = [3/2, 5/2] \times [\pi/4, \pi]$. Write $[a, b]$ for its support, ϖ for its density, $P(x) = \int \log|x-t|\varpi(t)dt$, and $D(x) = V(x) - 2P(x) - \ell \geq 0$. Put $P_0 = P(0)$ and $W = P_0 + \ell/2$. In the following extrema all parameters lie in $\mathcal{K}$:

$$\begin{aligned} a_* &= \min a > 0, \quad B_* = \max b, \quad b_* = \min(b-a) > 0, \quad M_* = \max \varpi, \quad L_* = \max_{x \in [a,b]} |V'(x)|, \\ C_{\text{BM}} &= \sqrt{32}e^{3L_*/4}, \quad C_{\text{quant}} = B_* + 2M_*, \quad C_0 = B_*/a_*, \\ C_{\text{tail}} &= \max\{0, \sup_{x>0} [\log(1+\sqrt{x}) - D(x)]\}, \quad T_* = \max\{1, \sup \int_0^\infty e^{-D(x)} dx\}, \\ C_W &= 1 + \max |W| + (4+\pi) \max(|W_\alpha| + |W_\beta|). \end{aligned} \quad (\text{CFR10})$$

All extrema are finite by the complete EIQ density and endpoint results used in OOQ4–6. At infinity one can use the uniform bound $D(x) \geq (\pi/4)\sqrt{x} - (5/2)\log x - 2\log(1+x) - C$; at zero $D(x) \rightarrow +\infty$ uniformly. These facts also justify the two improper-integral extrema in (CFR10).

Let $C_L = \pi/\Gamma(3/4)^2$, $h = 1/q$, and, for $s = 1, k$, set

$$\begin{aligned} \ell_s &= (s-1)/4, \quad Q_s = q' + \ell_s, \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}, \\ K_q &= \frac{\sqrt{2\pi}(2q+1)^2}{C_L q} e^{-2q}, \quad \mathcal{E}_k = \frac{qR^2}{\epsilon_0^2 q^2 - R^2} + \log(1+K_q), \\ r_0 &= 0, \quad r_\ell = \frac{\ell(16\ell^2 - 12\ell - 1)}{12\epsilon_0^2 q^2} + \log(1+K_q) \quad (\ell > 0), \\ L_h &= C_L(\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi}(\sin h)^{-1/2}. \end{aligned} \quad (\text{CFR11})$$

These are the original OSP24, OFV6 and OOQ30 constants, not new normalizations. Their proved full-form comparison is

$$\begin{aligned} \beta_s L_h e^{-\mathcal{E}_k} \int |p|^2 |y|^{2Q_s} e^{-(\vartheta+h)|y|} dy &\leq \|\psi p\|_s^2 \\ &\leq \beta_s U_h e^{\mathcal{E}_k + r_{\ell_s}} \int |p|^2 |y|^{2Q_s} e^{-(\vartheta-h)|y|} dy. \end{aligned} \quad (\text{CFR12})$$

Here p is written in the real coordinate, with its full affine coefficient map retained, and has degree at most $2q$. The entire small-coordinate region is included in this inherited comparison. Every minimization below is on the same G remainder fibre.

88.2 Low and high errors with the smaller disk

For $Q = Q_s$, $\eta = \vartheta \pm h$ put $g_0(Q, \eta) = 2Q \log(2Q/\eta) - 2Q$ and $y_0 = 2Q/\eta$. The full exponential Gram on degree at most Δ is bounded below by $e^{g_0 - \eta}/[(\Delta + 1)^2(2\Delta + 1)64^\Delta(1 + y_0)^{2\Delta}]$ times the identity, and above by its exact trace $\sum_{d=0}^{\Delta} 2(2Q + 2d)!/\eta^{2Q+2d+1}$ times the identity. For the lower estimate, integrate on $[y_0, y_0 + 1]$ and use the full shifted-Legendre coefficient change; its norm bounds are respectively $\sqrt{(\Delta + 1)(2\Delta + 1)8^\Delta}$ and $\sqrt{\Delta + 1}(1 + y_0)^\Delta$. Both half-line integrals give the upper trace. These are exactly OOQ26–27 on their actual degree range.

The ordinary remainder operator by the actual monic $G_y(y) = i^{-g}G(c' + iy)$ has coefficient norm at most

$$C_{\text{lo}} = 1 + (1 + R_G)^\Delta(\Delta + 1) \max\{1, \Delta R_G\}^\Delta.$$

Indeed complete homogeneous coefficients of its inverse at infinity are bounded by $\binom{g+a-1}{a} R_G^a$; convolution and the full coefficient sum of G_y give this bound for input degree at most Δ . Define

$$E_{\text{lo}}(Q, \eta) = \max \left\{ 0, \log \sum_{d=0}^{\Delta} \frac{2(2Q + 2d)!}{\eta^{2Q+2d+1}} - g_0(Q, \eta), \right. \\ \left. \eta + 2 \log(\Delta + 1) + \log(2\Delta + 1) + \Delta \log 64 \right. \\ \left. + 2\Delta \log(1 + y_0) + 2 \log C_{\text{lo}} \right\}. \quad (\text{CFR13})$$

Minimize the lower full-source bound over every lift; the canonical representative gives the upper bound. At both low windows this gives $e^{g_0 - E_{\text{lo}}} I_g \preceq Q_G^{\text{exp}} \preceq e^{g_0 + E_{\text{lo}}} I_g$. It includes all extra low columns up to M .

For the high windows use $\widehat{G}(z) = n^{-g}G_y(nz)$, whose roots have modulus at most $r_* = R_G/n = \rho/2$. The exact analytic remainder is

$$\text{rem}_{\widehat{G}} f(z) = \frac{1}{2\pi i} \int_{|\zeta|=\rho} f(\zeta) \frac{\widehat{G}(\zeta) - \widehat{G}(z)}{(\zeta - z)\widehat{G}(\zeta)} d\zeta, \quad \|\text{rem}_{\widehat{G}} f\|_1 \leq C_{\text{rem}} \max_{|z|=\rho} |f(z)|, \quad (\text{CFR14})$$

where $C_{\text{rem}} = \rho\Delta(1 + r_*)^\Delta(2/\rho)^\Delta$. Cauchy's formula at each original simple root proves interpolation; expanding the divided polynomial and bounding the full product below by $(\rho/2)^g$ proves the coefficient bound. No root is moved to zero.

For completeness the two OOQ estimates used here follow from its full variational inequality. A degree- m polynomial obeys $|p(z)| \leq C_{\text{BM}} m^2 e^{m(P(z) + \ell/2)} \|p\|_{m,V}$ off the support. On a one-sided support interval of length $1/m$, its weighted maximum and the elementary derivative estimate $\|p'\|_\infty \leq 4m^3 |J|^{-1} \|p\|_\infty$ give the factor $C_{\text{BM}} m^2$; the subharmonic maximum principle extends it off the interval. For $|z| \leq a_*/2$ the exponent is at most $mW + 2m|z|/a_*$. Midpoint quantiles ξ_a of the actual equilibrium measure give $r_n(x) = \prod_a (1 - x/\xi_a)$ with

$$\log |r_n(x)| \leq n(P(x) - P_0) + C_{\text{quant}} \sqrt{n} + C_0, \quad |r_n(z)|^{-1} \leq e^{2n|z|/a_*} \quad (|z| \leq a_*/2).$$

The first inequality clips the logarithm at $n^{-1/2}$: its Lipschitz error is at most $B_* \sqrt{n}$, its density error at most $2M_* \sqrt{n}$, and the normalization error at most B_*/a_* . These arguments work for the smaller disk (CFR9) without change.

Writing $p(nz) = p_e(z^2) + zp_o(z^2)$ gives exactly the two half-line weights $x^{Q-1/2} e^{-n\eta\sqrt{x}}$ and $x^{Q+1/2} e^{-n\eta\sqrt{x}}$, with overall factor n^{2Q+1} . Their degrees are between n and $n + \Delta$, so their parameter pairs $((Q \mp 1/2)/m, n\eta/m)$ remain in $\mathcal{K}$. The mean-value bound for their exponents is $C_W(\Delta + 2)$. Applying the preceding evaluation estimate at z^2 and then (CFR14) gives the lower quotient bound. For the upper bound, the actual lift of a remainder a is

$$p_a(nz) = r_n(z^2) \text{rem}_{\widehat{G}}(a(z)/r_n(z^2)).$$

It agrees at every original root and has degree at most $q + g - 1 \leq q + m_0 - 1$. Its norm is bounded on the whole positive half-line by absorbing $(1 + \sqrt{x})^{2(g-1)}$ into $e^{2C_{\text{tail}}(\Delta-1)}e^{2(\Delta-1)D(x)}$; the residual $e^{-[n-2(\Delta-1)]D(x)}$ is integrable under (CFR9). Thus define

$$\begin{aligned} E^- &= 2 \log(C_{\text{rem}} \sqrt{2} C_{\text{BM}}(n + \Delta)^2) + 2C_W(\Delta + 2) + 4(n + \Delta)\rho^2/a_*, \\ E^+ &= 2 \log C_{\text{rem}} + \log \Delta + 4n\rho^2/a_* + 2C_{\text{quant}}\sqrt{n} + 2C_0 + 2C_{\text{tail}}(\Delta - 1) + \log T_*, \\ E_{\text{hi}} &= \max\{0, E^-, E^+\} + 2(\Delta - 1) \log n, \\ c_{\text{hi}}^{\text{exp}}(Q, \eta) &= (2Q + 1) \log n - 2nW((Q - 1/2)/n, \eta). \end{aligned} \quad (\text{CFR15})$$

The last additive term in E_{hi} is the complete diagonal rescaling $\text{diag}(1, n, \dots, n^{g-1})$. These estimates prove the high quotient bounds centered at $c_{\text{hi}}^{\text{exp}}$.

Put $c_{\text{lo}} = g_0(q, \vartheta)$ and $c_{\text{hi}} = 2q \log(q/2) - qW(2, \vartheta)$. A single finite error is

$$\begin{aligned} e_k^G &= \mathcal{E}_k + \max_{s=1,k} r_{\ell_s} + \max\{|\log L_h|, |\log U_h|\} \\ &\quad + \max_{\substack{s=1,k \\ \eta=\vartheta \pm h}} \{E_{\text{lo}}(Q_s, \eta) + E_{\text{hi}} + |g_0(Q_s, \eta) - c_{\text{lo}}| + |c_{\text{hi}}^{\text{exp}}(Q_s, \eta) - c_{\text{hi}}|\}. \end{aligned} \quad (\text{CFR16})$$

Let $T_{ab} = \binom{b}{a}(c')^{b-a}i^a$ for $a \leq b < g$, and zero otherwise. It is the full affine coefficient map and has determinant $i^{g(g-1)/2}$. The resulting original Grams satisfy

$$\beta_s e^{c_\bullet - e_k^G} T^* T \preceq Q_{G,N}^{(s)} \preceq \beta_s e^{c_\bullet + e_k^G} T^* T, \quad e_k^G = O_{\delta, \gamma}(k \log q). \quad (\text{CFR17})$$

Here $c_\bullet$ is the low or high centre at its original endpoint. To verify the improved order, observe that $n\rho^2 = 4R_G^2/n = O_{\delta, \gamma}(1)$, $\log C_{\text{rem}} = O_{\delta, \gamma}(\Delta \log n)$, and $\sqrt{n} = O(k)$. All other terms in (CFR16) are $O_{\delta, \gamma}(k \log q)$ by their displayed factorial bounds, $Q_s - q = O(k)$, and the bounded first derivatives of W . This improves the draft's $O(q^{3/4} + k \log q)$ error. The same disk choice also improves OOQ's original outer-target error to this order. No stronger first variation or new equilibrium coefficient is claimed.

The accepted ECL coefficient, denoted C_∂ , obeys

$$c_{\text{lo}} - c_{\text{hi}} = qC_\partial/2, \quad 21.66851180520 < 16C_\partial < 21.66851180521. \quad (\text{CFR18})$$

This value is inherited, not recomputed here. Four determinants in (CFR17) give $|F_G - gqC_\partial| \leq 4ge_k^G = O_{\delta, \gamma}(q \log q)$.

89 The common quotient visible to the actual residual rows

Use the actual primary-coordinate conductor C (rank q'), residual certificate matrix D with $D|_{V_A}$ onto $\mathbb{C}^j$, and $V_A = \ker C$, $K = \ker[C; D]$. For each $\zeta = \zeta_{pt}$ in (CFR5), define a row on the *original* upper primary coordinates by

$$E_{\zeta, (i, j)} = \begin{cases} a_{i-p, j-t}/\psi(\zeta), & (i-p, j-t) \in \mathcal{A}, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{CFR19})$$

The denominator is nonzero because ζ is not an inner root. For every original polynomial p the row equals $\sum a_{ab}p(\zeta + b_{ab})/\psi(\zeta)$. On V_A this is evaluation of $\mathcal{T}_{Ap}/\psi$ at ζ . Its values are unaffected by changing p by χh . The map $E|_{V_A}$ is onto: prescribe a polynomial f of degree below g , solve $\mathcal{T}_{Ap} = \psi f$ by degree-lowering surjectivity, and take $\deg p \leq q' + g - 1 + v \leq q - 1$. Then $p \in V_A$, and the original distinct-root Vandermonde makes the values of f arbitrary.

Put $V_0 = \ker C \cap \ker E$. The exact visible rank is

$$t = g - \dim E(K) = q' + j + g - \text{rank} \begin{pmatrix} C \\ D \\ E \end{pmatrix}, \quad 0 \leq t \leq \min\{g, j\}. \quad (\text{CFR20})$$

This is a finite test using the actual period coefficients; no nonzero minor is presumed. For a direct coefficient construction, choose a basis of all row relations

$$YD - ZE = XC, \quad Y \in \mathbb{C}^{t \times j}, \quad Z \in \mathbb{C}^{t \times g}. \quad (\text{CFR21})$$

Both Y and Z have rank t . Indeed C, D are independent together, as are C, E ; either vanishing projection of a relation would force that relation to be zero. The relation space has dimension t by (CFR20). Gaussian elimination on the displayed actual matrix constructs X, Y, Z and a fixed basis, at each specified period. The original residual quotient is now

$$W = D(V_0) = \ker Y \subset \mathbb{C}^j, \quad \mathbb{C}^j/W \simeq \mathbb{C}^t, \quad [Dp] \mapsto YDp = ZEp \quad (p \in V_A). \quad (\text{CFR22})$$

The inclusion $D(V_0) \subset \ker Y$ is immediate. Equality follows from rank-nullity, or by subtracting an element of K with the same E value whenever $ZEp = 0$. Equivalently this is the draft RCM22–23 quotient with its original rows explicitly evaluated.

Let V_G be evaluation of $1, S', \dots, S'^{g-1}$ at the roots of G . In the GIP graph the composite map is precisely $(x, f) \mapsto ZV_G \text{rem}_G f$. Its full kernel contains the free low coordinate, every $\mathcal{U}$ column, and all transported original kernel columns with the stated zero output. Thus its graph metric is

$$\tilde{Q}_N^c = [ZV_G(Q_{G,N}^{(s)})^{-1}V_G^*Z^*]^{-1}, \quad R_c = [ZV_G(T^*T)^{-1}V_G^*Z^*]^{-1}. \quad (\text{CFR23})$$

They are on the actual rank- t image; at $t = 0$ every determinant is one. The full reference R_c , including its Vandermonde and period coefficients, is retained in $\beta_s e^{c \bullet - e_k^G} R_c \preceq \tilde{Q}_N^c \preceq \beta_s e^{c \bullet + e_k^G} R_c$.

The original residual covariance is

$$\Omega_N = D[R_N - R_N C^* (C R_N C^*)^{-1} C R_N] D^*, \quad A_N^c = Y \Omega_N Y^*, \quad Q_N^c = (A_N^c)^{-1}. \quad (\text{CFR24})$$

Here $R_N = G_N^{-1}$ is the original full primary covariance. This notation does not denote the minimum-section matrix used in other chapters. The original source maps, not a chosen coefficient norm, compare Q_N^c with (CFR23).

90 All-rank control and actual RWC exterior coefficients

Retain the finite CEP source-map bounds. They can be written entirely in the original coefficients as follows:

$$H_N = \sum_{a=1}^N 1/a, \quad M_N = \max\{1, \sum_{a,b} |a_{ab}| e^{|b_{ab}-4|(H_N+2)}\},$$

$$c_s^{\log} = \begin{cases} 1, & s = 1, \\ s/2 - 1, & s = k, \end{cases} \quad E_{N,s} = (N+1) \log M_N + (N-v+1) |\log |\mu_v/v!|| + v c_s^{\log} H_N. \quad (\text{CFR25})$$

Here $P_{<v}$ records the first v original upper orthonormal coefficients, not the first v monomial coefficients. For $S_N p = (P_{<v} p, \mathcal{T}_A p)$ in these source frames, both $\log^+ \|\bigwedge^d S_N\|$ and $\log^+ \|\bigwedge^d S_N^{-1}\|$ are at most $E_{N,s}$ for every d . Its exact determinant is

$$|\det S_N| = |\mu_v/v!|^{N-v+1} \prod_{m=0}^{N-v} \sqrt{w_{m+v}/w_m}, \quad w_m = m!/(s/2)_m.$$

Complementary singular products prove the asserted bounds. At a fixed period $E_{N,s} = O_{\text{actual}}(q \log q)$. Minimal lifts and orthogonal projection onto the complete quotient kernel contract every exterior power. The identical bounds therefore hold for the induced identities between the original and graph common-quotient metrics. This uses the full GIP kernel; there is no loss of a factor equal to the quotient rank in this step.

For each low endpoint $N = q - 1, q$, put $N^+ = N + q$, and let $\lambda_1, \dots, \lambda_t > 1$ be the generalized eigenvalues of $A_{N^+}^c$ relative to A_N^c . Define

$$a_k = qC_\partial/2, \quad D_{N,s} = 2(E_{N,s} + E_{N^+,s}) + 2te_k^G. \quad (\text{CFR26})$$

Then the *original*, not merely graph, eigenvalues satisfy

$$\sum_{i=1}^t (\log \lambda_i - a_k)_+ \leq D_{N,s}, \quad \sum_{i=1}^t (a_k - \log \lambda_i)_+ \leq D_{N,s}. \quad (\text{CFR27})$$

To prove this, factor the identity from the original high metric to the original low metric through their graph metrics. The middle map, after multiplication by $e^{-a_k/2}$, has operator norm and inverse norm at most $e^{e_k^G}$. Submultiplicativity of each exterior power therefore bounds both normalized forward and inverse exterior norms by $\exp(E_{N,s} + E_{N^+,s} + de_k^G)$. Choose d to be the number of singular values exceeding one; their product is that exterior norm. Apply the same argument to the inverse and square the singular values. This proves both inequalities. The spectra of the Gram ratio and covariance ratio agree by the AB, BA identity. In particular

$$\left| \log \frac{\det A_{N^+}^c}{\det A_N^c} - ta_k \right| \leq D_{N,s}, \quad \sum_i |\log \lambda_i - a_k| \leq 2D_{N,s}, \quad D_{N,s} = O_{\text{actual}}(q \log q). \quad (\text{CFR28})$$

If t/k is bounded below by a positive constant, the mean absolute error of $q^{-1} \log \lambda_i$ tends to zero. Existence of a limit for t/k is not required for this statement.

Use the original RWC full-window matrices, with their original source columns and full conductor conditioning:

$$\begin{aligned} U_N^+ &= [u_{N+1}, \dots, u_{N+q}], \quad B_N = I + (CU_N^+)^*(CR_N C^*)^{-1} CU_N^+, \\ Z_N &= D[I - R_N C^*(CR_N C^*)^{-1} C]U_N^+, \quad V_N = Z_N B_N^{-1} Z_N^* > 0, \quad \Omega_{N^+} = \Omega_N + V_N, \\ T_N &= \Omega_N^{-1/2} V_N \Omega_N^{-1/2}, \quad e_d(T_N) = \left\| \bigwedge^d (\Omega_N^{-1/2} Z_N B_N^{-1/2}) \right\|_{\text{HS}}^2. \end{aligned} \quad (\text{CFR29})$$

The rank and update are inherited RWC results; they are not new rank claims here. The projected positive increment is the compression of T_N by the actual coisometry $(A_N^c)^{-1/2} Y \Omega_N^{1/2}$. Its eigenvalues are $\lambda_i - 1$.

For any fixed $0 < \theta < C_\partial/2$, define the explicit integer

$$p_{N,s}(\theta) = \max \left\{ 0, t - \left\lfloor \frac{D_{N,s}}{q(C_\partial/2 - \theta)} \right\rfloor \right\}. \quad (\text{CFR30})$$

At least $p_{N,s}(\theta)$ eigenvalues of T_N are at least $e^{\theta q} - 1$. Indeed each projected eigenvalue below this threshold costs more than $q(C_\partial/2 - \theta)$ in the second sum in (CFR27); compression cannot increase the ordered eigenvalues. Equivalently, extend the coisometry to a unitary frame and use the min-max characterization, proved by intersecting a candidate plane with the span of the top eigenvectors. Thus the actual exterior coefficients and trace have the finite lower estimates

$$e_d(T_N) \geq \binom{p_{N,s}(\theta)}{d} (e^{\theta q} - 1)^d \quad (0 \leq d \leq p_{N,s}(\theta)), \quad \text{Tr } T_N \geq p_{N,s}(\theta) (e^{\theta q} - 1). \quad (\text{CFR31})$$

Since $D_{N,s} = O(q \log q)$, only $O_{\text{actual},\theta}(\log q)$ of the t visible modes can be lost in this estimate. There is also the useful trace-only version

$$\text{Tr } T_N \geq t [\exp\{a_k - D_{N,s}/t\} - 1]_+ \quad (t > 0). \quad (\text{CFR32})$$

It follows from the geometric-arithmetic mean inequality for λ_i and (CFR28). These are estimates for the specified RWC matrices, not for an assigned numerical observation. When $t = 0$, these bounds have no new positive content and the existing RWC bounds remain; no positive rank is inferred from the coefficient.

91 The whole remaining complement, including its covariance update

Extend the fixed full-row-rank Y to an invertible coordinate matrix $P = \begin{pmatrix} Y \\ H \end{pmatrix}$ on the original residual target. In this frame put

$$P\Omega_N P^* = \begin{pmatrix} A_N^c & C_N^* \\ C_N & D_N^c \end{pmatrix}, \quad S_N^c = D_N^c - C_N(A_N^c)^{-1}C_N^*. \quad (\text{CFR33})$$

The metric on $W = \ker Y$ in the fixed inclusion $P^{-1} \begin{pmatrix} 0 \\ I \end{pmatrix}$ is $(S_N^c)^{-1}$, not $(D_N^c)^{-1}$. Consequently the original residual is exactly

$$\Phi_{k,s} = \sum_{N=q-1,q} \left\{ \log \frac{\det A_{N+}^c}{\det A_N^c} + \log \frac{\det S_{N+}^c}{\det S_N^c} \right\} = tqC_\partial + \Phi_{W,k,s} + \varepsilon_{k,s}^c, \quad |\varepsilon_{k,s}^c| \leq \sum_{N=q-1,q} D_{N,s}. \quad (\text{CFR34})$$

All fixed determinants of P cancel between endpoints. The expression for Φ_W is the second sum, retained without approximation. It is nonnegative, and strictly positive if $\dim W > 0$.

Its full update can also be evaluated without deleting any conditioning. Put $J_N = Z_N B_N^{-1/2}$ and

$$\begin{aligned} E_N^c &= YJ_N, & F_N^c &= HJ_N, & T_N^c &= I + (E_N^c)^*(A_N^c)^{-1}E_N^c, \\ L_N^c &= F_N^c - C_N(A_N^c)^{-1}E_N^c, & S_{N+}^c &= S_N^c + L_N^c(T_N^c)^{-1}(L_N^c)^*. \end{aligned} \quad (\text{CFR35})$$

Block expansion and the rank-update inverse for $A_N^c + E_N^c(E_N^c)^*$ prove this equality. The matrix B_N in J_N is the *first* complete conductor denominator; T_N^c is the additional common-quotient denominator. Both survive. Since J_N has row rank j and $[-C_N(A_N^c)^{-1}, I]P$ has full row rank $j - t$, L_N^c has row rank $j - t$. This proves positivity of the complementary increment, including the exact zero-rank case. It is the full complement, not a replaced common-factor ideal.

For $b = j - t > 0$, define actual invariants

$$\tau_N^W = \text{Tr}[(S_N^c)^{-1}L_N^c(T_N^c)^{-1}(L_N^c)^*], \quad \chi_N^W = \frac{\det(L_N^c(T_N^c)^{-1}(L_N^c)^*)}{\det S_N^c}.$$

Let

$$w_N^- = \max\{\log(1 + \tau_N^W), b \log(1 + (\chi_N^W)^{1/b})\}, \quad w_N^+ = b \log(1 + \tau_N^W/b). \quad (\text{CFR36})$$

For $b = 0$ both are zero. The complementary log ratio lies in $[w_N^-, w_N^+]$. To prove this, diagonalize its positive increment in the S_N^c metric. Expansion of $\prod(1 + x_i)$ gives $1 + \sum x_i$; the arithmetic-geometric inequality for every subset of the positive x_i gives $(1 + (\prod x_i)^{1/b})^b$; concavity of $\log(1 + x)$ gives the upper bound. No assumed eigenvalue solver or independence of the original rows enters. Thus an additional full-residual interval is

$$tqC_\partial - \sum_N D_{N,s} + \sum_N w_N^- \leq \Phi_{k,s} \leq tqC_\partial + \sum_N D_{N,s} + \sum_N w_N^+. \quad (\text{CFR37})$$

It is intersected with the previous RWC interval; it is not asserted to be sharper for every small finite instance. The coarse consequence $\Phi_{k,s} \geq tqC_\partial - O_{\text{actual}}(q \log q)$ is valid, but (CFR34) remains the exact full object.

92 Which dual directions the common factor can and cannot recover

The following proves the precise scope of this reduction, beyond the draft's ideal computation. In the cutoff-independent relation dual, the functionals that are finite sums of point evaluations and their derivatives are exactly the span of the g simple evaluations at the roots of G . This

assertion concerns compatibility with *all* polynomial cutoffs, not representation of one arbitrary finite matrix.

Let $f(z)$ be the exponential generating function of such a functional; it is an exponential polynomial. By RDS the annihilator equation is $\mathcal{L}_A f = 0$ and $\mathcal{L}_A \psi(\partial_z) = \chi(\partial_z) M_{E_A}$, $E_A(z) = \sum a_{ab} e^{b_{ab} z}$. Solve $\psi(\partial_z) F = f$ in exponential polynomials: on each exponential block the differential polynomial is triangular in ordinary polynomial degree, and resonant factors are inverted by finite polynomial integration. Then $E_A F \in \ker \chi(\partial_z) = \mathcal{E}_k$.

Partition the exponents of F by cosets of $2\delta\mathbb{Z} + 2i\gamma\mathbb{Z}$. Multiplication by the fixed nonzero E_A shifts each coset by the same base exponent and cannot kill a nonzero coset component. To see the last assertion, identify that component with a finite Laurent polynomial in two variables with coefficients polynomial in z ; this ring has no zero divisors. Independence of polynomial-exponential functions follows, for example, by applying successive constant-coefficient annihilators to all but one exponent. Thus only the lower-grid coset occurs in F . Its largest ordinary polynomial degree must be zero, because the same nonzero product would otherwise survive in $E_A F$, whereas all upper roots are simple. Write

$$F(z) = e^{(c' - (k-8)\delta - i(k-8)\gamma)z} B(e^{2\delta z}, e^{2i\gamma z})$$

with a finite Laurent polynomial B . Its product with the actual biform $A(Z, W) = \sum a_{ab} Z^a W^b$ is supported in $[0, k]^2$. The minimum and maximum degree of a product in either coordinate are the sums of those of its factors: the leading coefficient is a nonzero Laurent polynomial in the other variable. Hence the support of B lies exactly within the allowed rectangle $\mathcal{R}$ of (CFR5). Conversely every such B is allowed. Applying $\psi(\partial_z)$ kills precisely the inner-grid exponentials; the remaining g exponentials survive with factors $\psi(\zeta) \neq 0$. This proves the assertion and retains all possible derivative orders before proving that none survives here.

In particular, when $g = 0$, replacing the divisor by a different finite list of evaluation points or fixed finite jets cannot remove the remaining relation dual. Its dimension m_0 remains. This is a statement about these specified cutoff-compatible maps, not a rejection of other analytic observations or of the programme.

The DNE draft gives its exact, more general original-source metric. Its complete finite frame is constructed as follows. Let $\mathbb{V}_{n,(i,j)} = \beta_{ij}^n$, $0 \leq n < q$, and let P_v insert zero in the first v coordinates. Define

$$W_v = \mathbb{V}^{-1} P_v, \quad J_v = P_v^\top \mathbb{V} C^\top, \quad W_v J_v = C^\top. \quad (\text{CFR38})$$

The first v rows of $\mathbb{V} C^\top$ vanish, because its columns are the initial derivatives of $E_A e^{\beta' z}$ and E_A has zero order v . This proves the last identity; hence J_v has rank q' . Take the first row subset $\mathcal{I}$ in a fixed ordering with $\det(J_v)_{\mathcal{I}} \neq 0$, let $\mathcal{J}$ be its complement, and set $Z_{\text{exp}} = W_v E_{\mathcal{J}}$. The chosen minor exists by the proved rank. The columns of $[J_v, E_{\mathcal{J}}]$ form a basis, by restriction to $\mathcal{I}$ followed by restriction to $\mathcal{J}$. Consequently

$$N_a(z) = \sum_{i,j} (Z_{\text{exp}})_{(i,j),a} e^{\beta_{ij} z}, \quad F_a = N_a / E_A, \quad (A_\ell)_{ba} = \Lambda_{F_a}(\psi S'^b), \quad 0 \leq b < m_0$$

is the complete fixed exponential quotient and its invertible dual frame. Invertibility is the RDS annihilator isomorphism, or equivalently its full degree-lowering recurrence, on the m_0 low monomials. For these actual exponential representatives, let $\ell_a(f) = \Lambda_{F_a}(\psi f)$, where $\Lambda_F(S'^m) = F^{(n)}(0)$. Its minimum-norm ambient extension is

$$\|\ell_x\|^2 = \min_{\eta} \left\| \Lambda_{\sum x_a F_a} + \sum_{\beta'} \eta_{\beta'} \text{ev}_{\beta'} \right\|^2, \quad \mathcal{D}_{N,s} = T_{N,s}^* [I - V_{N,s} (V_{N,s}^* V_{N,s})^{-1} V_{N,s}^*] T_{N,s}. \quad (\text{CFR39})$$

Here $0 \leq n \leq N - v$, and the complete entries are

$$T_{na} = \Lambda_{F_a}(\phi_{n,s,c'}), \quad V_{n,\beta'} = \phi_{n,s,c'}(\beta'), \quad \phi_{n,s,c'}(S') = \frac{p_n((S' - c')/i)}{\sqrt{(2\pi)^{s/2} n! (s/2)_n}}.$$

Thus every original lower-root evaluation and every source mass remains. Multiplication by ψ is an isometry into the original source; its annihilator is exactly the lower-root evaluations. Orthogonal minimum extension proves (CFR39), including both cross factors. For the complex-linear dual frame A_ℓ , the precise identity is

$$\mathcal{D}_{N,s} = A_\ell^* \overline{H_{U,N}^{-1}} A_\ell, \quad [Z_K^\top A_\ell \mathcal{D}_{N,s}^{-1} A_\ell^* \overline{Z_K}]^{-1} = \overline{(Z_K^* H_{U,N} Z_K)^{-1}}. \quad (\text{CFR40})$$

The bar is essential: a complex-linear row $a^\top$ has dual squared norm $a^\top H^{-1} \bar{a}$. The draft's source bound using N/E_A before differentiation by $\psi(\partial)$ consequently retains its stated $O_{\text{actual}}(q)$ scalar logarithmic error, without an artificial $q!$ factor. Raising that bound to a rank- $O(k)$ volume does *not* make it $o(kq)$. It is not used as such here. The full DNE source and constants are retained in the audited handoff.

93 Original receivers, action, and exact remaining quantity

Let $\Phi_{k,s}^{-,\text{old}}, \Phi_{k,s}^{+,\text{old}}$ be the installed RWC/OER bounds. Intersect them with (CFR37), also with any evaluated positive exterior terms from (CFR31), and call the resulting endpoints $\Phi_{k,s}^{-,c}, \Phi_{k,s}^{+,c}$. On the existing PMR domain, the new source bounds are exactly

$$\begin{aligned} L_s^c &= \max\{L_s^{\text{win}}, \mathcal{F}_{0,k}^{(s)} - e_{k,s}^{0,-} - \Phi_{k,s}^{+,c}\}, \\ U_s^c &= \min\{U_s^{\text{win}}, \mathcal{F}_{0,k}^{(s)} + e_{k,s}^{0,+} - \Phi_{k,s}^{-,c}\}, \quad L_s^c \leq F_K^{(s)} \leq U_s^c. \end{aligned} \quad (\text{CFR41})$$

Outside that domain use the same additions in the OAR outer-root interval at its original domain. Outside (CFR9) keep the old bounds; no eventual estimate is applied to an excluded degree. AMT's actual correlated caps give

$$\begin{aligned} L_{\text{ar}}^c &= \max\{L_{\text{ar}}^{\text{win}}, L_1^c - C_K^-, L_1^c + \delta_k^\sigma - C_B^+\}, \\ U_{\text{ar}}^c &= \min\{U_{\text{ar}}^{\text{win}}, U_1^c + C_K^+, U_1^c + \delta_k^\sigma + C_B^-\}, \\ F_K^{\text{ar}} &\in [L_{\text{ar}}^c, U_{\text{ar}}^c], \quad F_B^{\text{ar}} \in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^c, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^c]. \end{aligned} \quad (\text{CFR42})$$

All four caps are the original ones, including their unequal signs. The new returned asymptotic identity is

$$F_K^{(s)} + \Phi_{W,k,s} = (\Delta - t)qC_\partial + R_{k,s}^c, \quad R_{k,s}^c = R_{k,s}^{\text{OVR}} - \varepsilon_{k,s}^c = o(kq). \quad (\text{CFR43})$$

Only the newly introduced error is $O_{\text{actual}}(q \log q)$; the inherited OVR error is kept with its own complete finite scope. In particular this supplies $F_K^{(s)} \leq (\Delta - t)qC_\partial + o(kq)$, not a kernel asymptotic equality. If $t/k \rightarrow \tau$, the resulting ceiling is $(16 - \tau)C_\partial$ and the residual lower coefficient is τC_∂ . No value of τ is assigned by this proof.

With the unchanged signed return Ψ_k^{mix} and complete same-class budget C_k^{budget} of OER13,20,

$$\begin{aligned} L_k^c + \mathcal{H}_k + \delta_k^\sigma &\leq \Psi_k^{\text{mix}} \leq U_k^c + \mathcal{H}_k + \delta_k^\sigma, \\ J_k^{\text{action}} &\leq U_k^c + \mathcal{H}_k + \delta_k^\sigma + C_k^{\text{budget}}, \quad 4q \log \mathcal{L}_{h,k} \leq J_k^{\text{action}}, \\ \frac{\Psi_k^{\text{mix}} + \Phi_{W,k,k}}{kq} &= \left(\frac{\Delta - t}{k} \right) C_\partial - \frac{C_\Gamma}{4} + o(1). \end{aligned} \quad (\text{CFR44})$$

The last line uses only the already proved source differences and arithmetic error at the kq scale. No finer signed q -scale arithmetic coefficient or terminal exclusion is inferred.

Finally the newly explicit row map has the original-action defect

$$EA - \text{diag}(\zeta)E = E^{(b)}, \quad E_{\zeta,(i,j)}^{(b)} = b_{i-p,j-t} E_{\zeta,(i,j)}, \quad (\text{CFR45})$$

with $E^{(b)}$ defined to be zero outside the active support, and A diagonal in the original simple primary coordinates. This is direct evaluation of the same root equality $\beta_{ij} = \zeta + b_{i-p,j-t}$. Neither V_A nor its additional kernel is declared invariant. The relation lift χh is killed by these rows because all its original upper values vanish; its existing theta primitive and original arithmetic class are unchanged. All maps act coefficientwise at the same supported labels. A zero observation remains a supported zero, not external τ .

The remaining quantitative object is the full complementary covariance in (CFR33)–(CFR35), with the original C, D , actual row relation (CFR21), original B_N , and both mixed blocks. A further estimate must control that object or another explicitly compared original representation. The visible rank may be zero, especially when the common divisor is one; this limitation is part of the result. The previous terminal adjacent-allowance calculation is a separate continuation and is not reproved or used as an accepted input here.

94 Original domain and the precise new result

The inputs are the complete PCL, ROG, RDS and ROQ providers in the supplied 238-page edition, and the complete CFR1–45 continuation. In particular, CFR's outer coefficient and common-factor metric estimates are inherited; we do not claim them again as new. Fix

$$\begin{aligned} 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m = 1, \quad k \equiv 1 \pmod{4}, \quad k \geq 9, \quad q = (k+1)^2, \\ q' = (k-7)^2, \quad \Delta = q - q' = 16k - 48, \quad j = 8k - 32, \quad r = 8k - 16, \quad c = k/2, \quad c' = c - 4. \end{aligned} \quad (\text{RPK1})$$

The full original unit is $U = \text{diag}(a, \bar{a}, \bar{a}, a)$, $a = v_h(1/2 + \delta + i\gamma) \neq 0$. It is never assigned a value. The original period is $u \neq 0$ on its original logarithm branch. The five-orbit radius $R_* > 0$ is the supplied one. Source orders remain $s = 1, k$; no complete-multiplicity metric theorem is inferred from this simple case.

Put $x = \delta + i\gamma$, $y = \bar{x}$, and let V have rows $(1, w, w^2, w^3)$ at $w = x, y, -y, -x$, in that order. Let $\zeta = e^{2\pi i/5}$ and $D_5 = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. PCL supplies the exact entire matrix $\mathcal{R}(z)$, $z = u^{-1}$, with determinant one and

$$J := \mathcal{R}(0) = \begin{pmatrix} 1 & 0 & (x^2 + y^2)/3 & 0 \\ 0 & 1 & 0 & 2(x^2 + y^2)/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}. \quad (\text{RPK2})$$

Both off-diagonal entries are retained. For completeness the full inherited coefficient series used in the radius below is

$$\begin{aligned} \mathcal{R}_{\alpha r}(z) &= \sum_{n \geq 0} R_{\alpha r, n} z^n, \\ R_{\alpha r, n} &= \sum_{\substack{p, h \geq 0 \\ 2p+4h=5n+r+1-\alpha}} \frac{\beta^p \eta^h}{3^p p! h!} (-1)^{p+h-n} 5^{p+h-n} (\alpha/5)_{p+h-n}, \end{aligned}$$

where $1 \leq \alpha \leq 4$, $0 \leq r \leq 3$, $\beta = 2(\gamma^2 - \delta^2)$, $\eta = (\delta^2 + \gamma^2)^2$. Each inner sum is finite and its indicated rising-factorial order is nonnegative. PCL proves absolute convergence at every complex z and the stated determinant one. Retain also

$$\mathfrak{a} = 1/160 + \beta/24 + \eta/2, \quad G_u = \text{diag} \left(u^{\alpha/5} 5^{\alpha/5-1} e^{\pi i \alpha/5} \Gamma(\alpha/5) \right)_{\alpha=1}^4.$$

The full period is $H(u) = e^{\mathfrak{a}/u} G_u \mathcal{R}(u^{-1}) V^{-1} U$, with PCL's original G_u and $\mathfrak{a}$. Its exact conjugation

gives

$$B_\ell(z) = V\mathcal{R}(z)^{-1}D_5^{-\ell}\mathcal{R}(z)V^{-1}, \quad L_\ell(z) = U^{-1}B_\ell(z)U, \quad 1 \leq \ell \leq 4,$$

$$A_z(X, Y) = \prod_{\ell=1}^4 Q_0(L_\ell(z)(XY, X, Y, 1)^\top) = \sum_{a,b=0}^8 a_{ab}(z)X^aY^b, \quad (\text{RPK3})$$

$$Q_0(t) = t_1t_4 - t_2t_3.$$

These are the actual coefficients of the conductor, not an independent family of quadrics. The scalar and G_u cancel only in the displayed conjugation, where G_u commutes with D_5 .

Theorem 94.1. *There is an explicit $R_{\text{rank}} \geq R_*$, given in (RPK11), such that at every original branch and $|u| \geq R_{\text{rank}}$,*

$$a_- = b_- = 0, \quad a_+ = b_+ = 8,$$

where the extrema are those of the actual nonzero coefficient support of $A_{1/u}$. The same conclusion holds on $|u| \geq R_*$ except possibly at a finite set depending on the fixed quartet. At every period covered by this conclusion, for all k in (RPK1),

$$G = 1, \quad g = 0, \quad t_k = 0, \quad W = \mathbb{C}^j, \quad \Phi_{W,k,s} = \Phi_{k,s}. \quad (\text{RPK4})$$

Here G, g, t_k, W are exactly the CFR objects. No positive lower bound for t_k can be used on this domain. The full residual still has rank j .

Neither all 81 coefficients nor all four corner coefficients are claimed nonzero. Nonzero restrictions to all four coordinate edges suffice.

95 An explicit nonvanishing calculation at the actual period limit

For a 4 by 4 matrix B set

$$f_i(B) = B_{1i}B_{4i}, \quad h_i(B) = B_{2i}B_{3i}, \quad \mathcal{E}_{ab}(B) = f_a(B)h_b(B) - f_b(B)h_a(B).$$

For the four pairs $\mathcal{E} = \{(1, 2), (1, 3), (2, 4), (3, 4)\}$ define the sixteen entire functions

$$\mathcal{E}_{\ell,ab}(z) = \mathcal{E}_{ab}(B_\ell(z)). \quad (\text{RPK5})$$

The four pairs are precisely the endpoint vertices of the four Segre coordinate ruling lines. If $Q_0(L_\ell t)$ vanished identically on such a line, it would vanish at both its vertices. But

$$Q_0(L_\ell e_i) = U_{ii}^2 \{a^{-2}f_i(B_\ell) - \bar{a}^{-2}h_i(B_\ell)\}.$$

Since all four U_{ii} are nonzero, simultaneous vertex vanishing forces $\mathcal{E}_{\ell,ab} = 0$. This argument retains the actual unit; the necessary eliminant is independent of its magnitude and phase.

Let $\xi = \zeta^{-\ell}$. At $z = 0$, direct calculation gives

$$\mathcal{E}_{12}(VJ^{-1} \text{diag}(\xi, \xi^2, \xi^3, \xi^4)JV^{-1}) = -\frac{P_\xi(x, y)}{1296x^2y^2(x^2 - y^2)^2}, \quad (\text{RPK6})$$

where

$$P_\xi = (\xi^2 + \xi^3)[5(x^8 + y^8) + 20(x^6y^2 + x^2y^6) - 51x^4y^4] - 5(x^8 + y^8) + 25(x^6y^2 + x^2y^6) - 183x^4y^4. \quad (\text{RPK7})$$

Here is a finite algebraic verification of the formula. The first two columns of V^{-1} are the coefficients of

$$\frac{w^3 + xw^2 - y^2w - xy^2}{2x(x^2 - y^2)}, \quad \frac{w^3 + yw^2 - x^2w - yx^2}{2y(y^2 - x^2)}.$$

Writing $b = (x^2 + y^2)/3$, the coefficient transformation $J^{-1}D_\xi J$ sends

$$(p_0, p_1, p_2, p_3) \mapsto (\xi p_0 + b(\xi - \xi^3)p_2, \xi^2 p_1 + 2b(\xi^2 - \xi^4)p_3, \xi^3 p_2, \xi^4 p_3).$$

Evaluate these two transformed polynomials at $x, y, -y, -x$, form the two products defining $\mathcal{E}_{12}$, and use $1 + \xi + \xi^2 + \xi^3 + \xi^4 = 0$. Their numerator is exactly minus (RPK7), with the denominator in (RPK6). These displayed coefficient operations are also checked by the exact symbolic regression. The opposite-root permutation commutes with this coefficient transformation. Hence $f_4 = f_1, h_4 = h_1, f_3 = f_2, h_3 = h_2$ and the four edge eliminants at zero are $\mathcal{E}_{12}, \mathcal{E}_{12}, -\mathcal{E}_{12}, -\mathcal{E}_{12}$.

Put $T = \cos(4 \arg x)$ and $h = \xi^2 + \xi^3 = 2 \cos(4\pi\ell/5) \geq -2$. Since $y = \bar{x}$,

$$\begin{aligned} \frac{P_\xi}{x^4 y^4} &= h(20T^2 + 40T - 61) - 20T^2 + 50T - 173 \\ &\leq -60T^2 - 30T - 51 = -60(T + 1/4)^2 - 189/4 < 0. \end{aligned} \tag{RPK8}$$

The inequality uses $-1 \leq T \leq 1$ and $20T^2 + 40T - 61 \leq -1$. The denominator in (RPK6) is nonzero because $x^2 - y^2 = 4i\delta\gamma$. In particular every edge eliminant obeys the concrete unit-independent lower bound

$$|\mathcal{E}_{\ell,ab}(0)| \geq d_* := \frac{7(\delta^2 + \gamma^2)^2}{3072\delta^2\gamma^2} > 0. \tag{RPK9}$$

Thus no edge polynomial vanishes identically at this exact limit, for any permitted actual unit.

96 A finite original-period radius and the exact visible rank

Write $\mathcal{R}(z) = J + \sum_{n \geq 1} R_n z^n$ with the exact PCL2 coefficients, and use the induced column-sum norm. Define

$$\begin{aligned} C &= \max\{1, \sum_{n \geq 1} \|R_n\|_1\}, \quad J_0 = \max\{1, \|J\|_1\}, \quad J_1 = \max\{1, \|J^{-1}\|_1\}, \\ V_0 &= \|V\|_1 \|V^{-1}\|_1, \quad r_1 = \min\{1, (2J_1 C)^{-1}\}, \\ M &= 2V_0 J_1 (J_0 + 1/2), \quad L = 2V_0 J_1 C (1 + J_1 J_0). \end{aligned} \tag{RPK10}$$

The series constant is finite by the entire PCL series, absolutely convergent even at any larger radius. It is a fully specified source constant, not a numerically certified value claimed here. For $|z| \leq r_1$, Neumann's series gives $\|\mathcal{R}^{-1}\|_1 \leq 2J_1$ and $\|\mathcal{R}^{-1} - J^{-1}\|_1 \leq 2J_1^2 C|z|$. Consequently

$$\|B_\ell(z)\|_1, \|B_\ell(0)\|_1 \leq M, \quad \|B_\ell(z) - B_\ell(0)\|_1 \leq L|z|.$$

Telescoping each product of four entries gives $|\mathcal{E}_{\ell,ab}(z) - \mathcal{E}_{\ell,ab}(0)| \leq 8M^3 L|z|$. Therefore the explicit radius is

$$r_{\text{rank}} = \min\{r_1, d_*/(16M^3 L)\}, \quad R_{\text{rank}} = \max\{R_*, r_{\text{rank}}^{-1}\}. \tag{RPK11}$$

Every gate then has modulus at least $d_*/2$. Each individual orbit factor has nonzero restriction to all four edges, so their product does too, since a polynomial ring on an edge has no zero divisors. This proves the four support extrema in Theorem 94.1.

For the sharper domain statement put

$$\mathfrak{G}(z) = \prod_{\ell=1}^4 \prod_{(a,b) \in \mathcal{E}} \mathcal{E}_{\ell,ab}(z), \quad \mathcal{X}_h = \{u : |u| \geq R_*, \mathfrak{G}(1/u) = 0\}. \tag{RPK12}$$

This is an explicit exceptional *superset*, not a claim that every listed point is a true support degeneration. The entire function $\mathfrak{G}$ is nonzero at zero. Its zeros are isolated, hence finite in the compact disk $|z| \leq R_*^{-1}$. None is zero. This proves finiteness of $\mathcal{X}_h$, independently of k and the

period branch. At a point in this superset, the actual edge coefficients must still be tested; no rank is assigned by the gate alone.

CFR gives $g = (k + 1 - d_a)(k + 1 - d_b) - q'$ and $0 \leq t_k \leq g$. Substitution of the now proved widths $d_a = d_b = 8$ proves (RPK4) exactly, not just as a limiting statement. In particular the term $t_k q C_\partial$ is zero and the common-factor spectral lower bounds add no positive modes on this domain. This does not say that the residual is small or zero.

There is also a sharper bound for the retained low-degree correction. Define the original exponential symbol, with a separate generating variable τ , by

$$E_A(\tau) = \sum_{a,b} a_{ab} e^{b_{ab}\tau} = e^{(4-8\delta-8i\gamma)\tau} A(e^{2\delta\tau}, e^{2i\gamma\tau}), \quad b_{ab} = 4 + (2a - 8)\delta + i(2b - 8)\gamma. \quad (\text{RPK13})$$

Each of the four orbit factors in A has at most nine monomials. The corresponding nine exponentials have distinct exponents. A nonzero sum of p distinct exponentials cannot vanish to order p at zero, since its first p derivatives form an invertible Vandermonde. The factors are nonzero on the five-orbit locus. Thus

$$v = \text{ord}_0 E_A = \sum_{\ell=1}^4 \text{ord}_{\tau=0} f_\ell(e^{2\delta\tau}, e^{2i\gamma\tau}) \leq 32, \quad (\text{RPK14})$$

$$f_\ell(X, Y) = Q_0(L_\ell(XY, X, Y, 1)^\top).$$

The exact v and every low coordinate are retained; this bound does not set v to zero.

97 The full complement has a finite principal-part detector

Fix a period covered by Theorem 94.1. Factor its *actual* polynomial over $\mathbb{C}$:

$$A(X, Y) = a_* \prod_{\nu=1}^h B_\nu(X, Y)^{m_\nu}, \quad \deg_X B_\nu = a_\nu, \quad \deg_Y B_\nu = b_\nu, \quad \sum m_\nu a_\nu = \sum m_\nu b_\nu = 8. \quad (\text{RPK15})$$

The factors are distinct irreducible nonmonomials. Full edge support excludes X and Y as factors. Neither irreducibility nor squarefreeness of A is assumed. Let $\Gamma(\tau) = (e^{2\delta\tau}, e^{2i\gamma\tau})$. This map is injective: equality of its two entries would make the parameter difference both purely imaginary and real, hence zero.

Lemma 97.1. *Each $B_\nu \circ \Gamma$ has infinitely many zeros. Except at finitely many of them, its zero is simple and every other $B_\mu \circ \Gamma$ is nonzero. For any $M \geq 1$, one can choose M such zeros, different from zero, with distinct X coordinates when $b_\nu > 0$, or distinct Y coordinates when $b_\nu = 0$, inside a disk of explicitly specified radius $C_{\nu,0} + C_{\nu,1}M$.*

Proof. The exponential sum $B_\nu \circ \Gamma$ has at least two distinct exponents. Choose an edge of its exponent polygon (the whole segment in the collinear case). Its edge exponents have the form $\lambda_0 + nd$, $d = 2\delta p_0 + 2i\gamma p_1 \neq 0$, with integers $n \geq 0$ and a primitive integer direction (p_0, p_1) . The edge sum is $e^{\lambda_0\tau} P(e^{d\tau})$, where P has nonzero constant and leading terms and positive degree. Choose a nonzero root w_0 of P and a logarithm τ_0 with $e^{d\tau_0} = w_0$. Choose $0 < \rho < \pi/(2|d|)$ so that $P(e^{d(\tau_0+w)})$ has no zeros on $|w| = \rho$; this is possible by isolated zeros. Put $m_e = \min_{|w|=\rho} |P(e^{d(\tau_0+w)})| > 0$. Choose the sign $\sigma = \pm 1$ so that all off-edge exponents have

$$\kappa_\lambda = -2\pi\sigma \operatorname{Re} \frac{i(\lambda - \lambda_0)}{d} > 0.$$

This is the exposed-edge half-plane condition. Set

$$H_e = \sum_{\lambda \text{ off edge}} |c_\lambda| e^{\operatorname{Re}((\lambda - \lambda_0)\tau_0) + |\lambda - \lambda_0|\rho}, \quad \kappa_e = \min \kappa_\lambda, \quad n_0 = \max\{0, \lceil \kappa_e^{-1} \log(2H_e/m_e) \rceil\}.$$

If there are no off-edge terms set $n_0 = 0$ and omit the last two constants. On the circle about $\tau_n = \tau_0 + 2\pi i \sigma n/d$, of radius ρ , the normalized off-edge sum is at most $H_e e^{-\kappa_e n} \leq m_e/2$. Rouché's theorem gives a zero in each of these disjoint disks for $n \geq n_0$. In the collinear case this follows directly from P .

Let $\mathfrak{D} = 2\delta X \partial_X + 2i\gamma Y \partial_Y$. The polynomials B_ν and $\mathfrak{D}B_\nu$ are coprime. Otherwise irreducibility and the two degree bounds would give $\mathfrak{D}B_\nu = \lambda B_\nu$; distinct monomial weights would force B_ν to be a monomial. They are also coprime to the other irreducible factors. These intersections contain finitely many torus points. This last assertion can be checked without a smoothness assumption: for a nonvertical factor of bidegree (a, b) , the resultant in Y against a coprime factor of bidegree (c, d) is nonzero with X degree at most $ad + bc$. Each X fibre has at most b zeros; a whole vertical fibre would contradict irreducibility. For a vertical factor the assertion is ordinary univariate evaluation. Thus, including $\tau = 0$, a bound for the bad generating parameters is

$$B_\nu^{\text{bad}} = \begin{cases} 2a_\nu b_\nu^2 + 8b_\nu(a_\nu + b_\nu) + 1, & b_\nu > 0, \\ 9, & b_\nu = 0. \end{cases} \quad (\text{RPK16})$$

Injectivity of Γ carries this point bound to parameter values. From the first $B_\nu^{\text{bad}} + \max(1, b_\nu)M$ disjoint disks one can therefore retain M good zeros with the stated distinct coordinate. For $b_\nu > 0$ at most b_ν of them can have the same X ; for a vertical factor all their Y coordinates are distinct. Their moduli are bounded by

$$|\tau_0| + \rho + \frac{2\pi}{|d|} [n_0 + B_\nu^{\text{bad}} + \max(1, b_\nu)M]. \quad (\text{RPK17})$$

All choices and constants use this fixed actual factor, not a change of period with k . This proves the lemma. $\square$

For each factor choose

$$M_\nu = (a_\nu + b_\nu)k + 1 \quad (\text{RPK18})$$

zeros as in the lemma. At such a zero $\tau_{\nu,l}$ the order of E_A is exactly m_ν . Retain all m_ν coefficients of the principal part, not only the residue. For the original primary coefficient vector $d = (d_{ij})$, define

$$\begin{aligned} N_d(\tau) &= \sum_{i,j=0}^k d_{ij} e^{\beta_{ij}\tau}, \\ \beta_{ij} &= c + (2i - k)\delta + i(2j - k)\gamma, \\ F_d(\tau) &= N_d(\tau)/E_A(\tau). \end{aligned} \quad (\text{RPK19})$$

The function F_d is allowed its full principal part at zero. This is not a Taylor truncation or an assertion that every F_d is analytic there. It retains the low- v coordinates rather than discarding them. Let $\Pi_k d$ be the list of all selected principal-part coefficients. It has the completely explicit entries

$$(\Pi_k)_{(\nu,l,r),(i,j)} = \frac{1}{(m_\nu - r)!} \left(\frac{d}{d\tau} \right)^{m_\nu - r} \frac{e^{\beta_{ij}\tau}}{E_A(\tau)/(\tau - \tau_{\nu,l})^{m_\nu}} \Big|_{\tau=\tau_{\nu,l}}, \quad 1 \leq r \leq m_\nu. \quad (\text{RPK20})$$

Every denominator here is nonzero by the proved selection. The number of rows is

$$p_k = \sum_\nu m_\nu M_\nu = 16k + \sum_\nu m_\nu \leq 16k + 16. \quad (\text{RPK21})$$

Their generating-variable moduli are $O_{h,u}(k)$ by (RPK17). No bound on the inverse interpolation metric is inferred solely from this location estimate.

Theorem 97.2. *For the original conductor coefficient matrix C , of rank q' , and original residual rows D , of rank j modulo C ,*

$$\ker \Pi_k = \text{im } C^\top, \quad \text{rank } \Pi_k = \Delta, \quad \text{rank}(\Pi_k D^\top) = j. \quad (\text{RPK22})$$

Thus these actual poles detect every residual dual direction, even though the source-variable finite-jet target in CFR is zero.

Proof. Write $N(X, Y) = \sum d_{ij} X^i Y^j$. Apart from a nonvanishing exponential factor, F_d equals N/A on Γ . If all its selected principal parts vanish, $N \circ \Gamma$ vanishes to order m_ν at each selected regular zero of $B_\nu \circ \Gamma$. First consider its values. If $b_\nu > 0$ and B_ν does not divide N , the nonzero resultant $\text{Res}_Y(B_\nu, N)$ has degree in X at most $(a_\nu + b_\nu)k$. The M_ν distinct X values contradict that bound. The degree estimate follows directly from the Sylvester determinant: at most k rows contribute coefficients of X degree a_ν and at most b_ν rows degree k . For a vertical irreducible factor $B_\nu = X - X_0$, its restriction $N(X_0, Y)$ has degree at most k and more than k distinct zeros. It is zero, so B_ν divides N in that case as well.

Divide N by B_ν . At the same regular zeros its remaining vanishing order is at least $m_\nu - 1$, and its degrees have decreased. The identical resultant bound still applies. Repetition proves $B_\nu^{m_\nu} \mid N$. All the factors are coprime, so $A \mid N$. The quotient has bidegree at most $(k - 8, k - 8)$ by exact degree addition in each coordinate. Conversely any such multiple of A has no principal parts. Multiplication by this full A is exactly $C^\top$ in the original OCF coefficient pairing; it has no extra binomial or conjugate factors. This proves the first two claims. Finally $\Pi_k D^\top x = 0$ implies $D^\top x \in \text{im } C^\top$. ROQ4–7 proves the independence of the C, D rows together, hence $x = 0$. $\square$

98 The full Gamma metric and an evaluated action-leakage rank

Select, in a fixed ordering, the first j rows of $\Pi_k D^\top$ forming an invertible square matrix B_k . Existence is now a theorem, not a generic-minor hypothesis. This choice depends on the fixed actual coefficients and k , but not on the source order or endpoint. At each of the four original endpoints retain

$$\Omega_N^{(s)} = D\{R_N^{(s)} - R_N^{(s)} C^* [C R_N^{(s)} C^*]^{-1} C R_N^{(s)}\} D^*, \quad R_N^{(s)} = (G_N^{(s)})^{-1}. \quad (\text{RPK23})$$

The ordinary complex-linear dual vector x represents the functional $x^\top D$, whose squared norm is $x^\top \Omega_N \bar{x}$. Its Hermitian Gram in the column x is therefore $\overline{\Omega_N}$. In the actual selected pole coordinates $b = B_k x$, the exact full Gram is

$$\mathcal{H}_{N,s}^{\text{pole}} = B_k^{-*} \overline{\Omega_N^{(s)}} B_k^{-1}, \quad \Phi_{k,s} = \sum_{N=q-1,q} \log \frac{\det \mathcal{H}_{N+q,s}^{\text{pole}}}{\det \mathcal{H}_{N,s}^{\text{pole}}}. \quad (\text{RPK24})$$

The fixed $|\det B_k|$ cancels between endpoints. It has not been set to one or used to cancel a restricted source determinant.

Here is a direct original-source realization of the same full Gram. Let $u_{n,s}$ be the original primary evaluation column of the orthonormal polynomial

$$\varphi_{n,s}(S) = \frac{p_n((S - c)/i)}{\sqrt{(2\pi)^{s/2} n! (s/2)_n}}, \quad R_N^{(s)} = \sum_{n=0}^N u_{n,s} u_{n,s}^*.$$

Define matrices with rows $0 \leq n \leq N$ by

$$\mathcal{T}_{n,\cdot} = (D u_{n,s})^\top B_k^{-1}, \quad \mathcal{V}_{n,\cdot} = (C u_{n,s})^\top.$$

Then, with every original mass and affine phase,

$$\mathcal{H}_{N,s}^{\text{pole}} = \mathcal{T}^* [I - \mathcal{V}(\mathcal{V}^* \mathcal{V})^{-1} \mathcal{V}^*] \mathcal{T}. \quad (\text{RPK25})$$

Expanding its four factors gives (RPK24). The columns of $\mathcal{V}$ are independent, and those of $[\mathcal{V}, \mathcal{T}]$ have rank $q' + j$, since the source columns through $N \geq q - 1$ span the full original primary space. Hence this Gram is positive definite. The original RWC update becomes exactly

$$\mathcal{H}_{N+q,s}^{\text{pole}} = \mathcal{H}_{N,s}^{\text{pole}} + (B_k^{-*} \overline{Z_N}) \overline{B_N}^{-1} (B_k^{-*} \overline{Z_N})^*. \quad (\text{RPK26})$$

The full original conditioning matrix B_N and both source cross factors in (RPK25) remain. This is a metric formula, not a replacement by a Euclidean pole norm.

The pole coordinates also calculate an actual leakage block. Let A_χ be the original diagonal sum action in primary coordinates. Then

$$F_{A_\chi^\top d} = F'_d + (E'_A/E_A)F_d. \quad (\text{RPK27})$$

For $d = C^\top c_0$, one has $F_d = \sum c_{0,ab} e^{\beta'_{ab}\tau}$, where $\beta'_{ab} = c' + (2a - k + 8)\delta + i(2b - k + 8)\gamma$, $0 \leq a, b \leq k - 8$, are the complete original lower-grid exponents. At a selected zero of multiplicity m_ν ,

$$(\Pi_k A_\chi^\top C^\top)_{(\nu,l,r),(a,b)} = \begin{cases} m_\nu e^{\beta'_{ab}\tau_{\nu,l}}, & r = 1, \\ 0, & r > 1. \end{cases} \quad (\text{RPK28})$$

Indeed E'_A/E_A has residue m_ν and F_d is entire there. In particular this block is not zero. Multiplication has not been asserted to descend through the conductor kernel.

Let $A_{\text{red}} = \prod B_\nu$ have bidegree $(a_{\text{red}}, b_{\text{red}})$. The same sampled-value and resultant argument applied to a polynomial of bidegree at most $(k - 8, k - 8)$ proves the exact rank

$$\begin{aligned} \text{rank}(\Pi_k A_\chi^\top C^\top) &= q' - \max\{k - 7 - a_{\text{red}}, 0\} \max\{k - 7 - b_{\text{red}}, 0\} \\ &= \text{rank}(CA_\chi|_{\ker C}). \end{aligned} \quad (\text{RPK29})$$

Only one copy of each factor is needed here, because the leakage has only simple poles. The second equality is ordinary duality of the specified map, with the quotient identified by (RPK22). For $k \geq 17$ its value is $(a_{\text{red}} + b_{\text{red}})(k - 7) - a_{\text{red}}b_{\text{red}}$. If the actual A is squarefree this is $16k - 176$; squarefreeness is not presumed. The two reduced bidegrees are fixed finite algebraic invariants at the specified actual period. Thus this growing-rank formula does not require a new rank computation in dimension q for each k . It is not a lower bound for a metric norm or a weight assertion.

99 Leading-scale receiving statement and remaining analytic expression

On the rank domain of Theorem 94.1, CFR's common term is identically zero, including its exact empty determinant, and

$$F_K^{(s)} + \Phi_{k,s} = \Delta q C_\partial + R_{k,s}^{\text{OVR}}, \quad R_{k,s}^{\text{OVR}} = o(kq), \quad \Phi_{W,k,s} = \Phi_{k,s}. \quad (\text{RPK30})$$

The previous absolute kernel and residual caps are retained. A numerical ceiling is not improved by the vacuous $t_k = 0$ common-factor subtraction. The earlier terminal-adjacent allowance work is separate and is not an input to these proofs.

The fixed pole-coordinate formula (RPK24), with the fully specified original matrices (RPK25)–(RPK26), is the whole remaining leading residual. Its pole detector has rank j and sample radius $O(k)$, but those facts do not bound its relative Gram spectrum. In particular neither an $o(kq)$ estimate for Φ nor an equality of its coefficient with jqC_∂ is claimed. The period can be fixed once with $|u| \geq R_{\text{rank}}$ before taking $k \rightarrow \infty$. No saddle limit or varying-period norm bound is exchanged with that limit.

If $\widehat{L}_s, \widehat{U}_s$ are the currently strongest kernel bounds after this exact rank substitution, AMT and OER continue to give

$$\begin{aligned} L_{\text{ar}} &= \max\{L_{\text{ar}}^{\text{old}}, \widehat{L}_1 - C_K^-, \widehat{L}_1 + \delta_k^\sigma - C_B^+\}, \\ U_{\text{ar}} &= \min\{U_{\text{ar}}^{\text{old}}, \widehat{U}_1 + C_K^+, \widehat{U}_1 + \delta_k^\sigma + C_B^-\}, \\ F_K^{\text{ar}} &\in [L_{\text{ar}}, U_{\text{ar}}], \quad F_B^{\text{ar}} \in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}]. \end{aligned} \quad (\text{RPK31})$$

The signed and same-class action receivers are still

$$\begin{aligned} \Psi_k^{\text{mix}} &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, \\ \frac{\Psi_k^{\text{mix}} + \Phi_{k,k}}{kq} &\longrightarrow 16C_\partial - C_\Gamma/4, \\ 4q \log \mathcal{L}_{h,k} &\leq J_k^{\text{action}} \leq \widehat{U}_k + \mathcal{H}_k + \delta_k^\sigma + C_k^{\text{budget}}. \end{aligned} \quad (\text{RPK32})$$

Every original budget and unequal arithmetic error remains. The pole coordinates are a dual finite-dimensional observation of these same coefficient classes. No additional kernel is declared a theta boundary; original χh changes retain their original primitives and supported zero is not external τ . Formula (RPK28) records precisely why an action cannot simply be imposed on this new quotient by a name.

Provenance and verification scope

The complete source PCL1–24 is in the supplied file 14 of the 238-page edition. In particular the actual non-diagonal limit, entire inverse, full unit and conductor coefficients used here are its PCL4–11. ROQ2–7 supplies the literal coefficient pairing, original C, D and their rank; ROG/RDS supplies their relation and exponential dictionary. CFR1–45 is the immediate preserved predecessor. All new arguments are written above. The exact algebraic tests check the cyclotomic eliminant, pole kernel and multiplicities, source Gram conjugation, and retained leakage; they are diagnostics, not evaluations of actual xi zeros or period integrals. No independent reviewer, new Lean run, complete residual leading value, or RH result is claimed.

Retain the fixed original simple quartet, amplitude, period, logarithm branch and five-member quadric orbit of KRC1–27 and IPE1–44. The period remains on the original proved five-orbit locus, including the original arithmetic domain $|u| \geq R_*$. Put $k = 4l + 1 \geq 9$. All spaces and duals are over $\mathbb{C}$, and all duals are complex-linear. The calculation is algebraic and applies to both original Gamma source orders; it introduces no coefficient norm in place of either original source form.

100 The original rings and every degree-four map

Use the original primary order $(++)$, $(+-)$, $(-+)$, $(--)$ and set

$$\begin{aligned} S &= \mathbb{C}[x_1, x_2, x_3, x_4], \quad \mathbf{H} = F^{-1} \Pi U \mathcal{E}_{\text{prim}}, \quad Q(x) = Q_0(\mathbf{H}^{-1}x), \quad Q_0(t) = t_1 t_4 - t_2 t_3, \\ \zeta &= e^{2\pi i/5}, \quad D = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4), \quad gf(x) = f(D^{-1}x), \\ Q_a &= g^a Q, \quad \mathcal{A} = \prod_{a=1}^4 Q_a, \quad I = (Q, \mathcal{A}), \quad I_a = (Q, Q_a) \quad (1 \leq a \leq 4). \end{aligned} \quad (\text{DFX1})$$

The actual matrix $\mathbf{H}$ is invertible. The quadrics Q, Q_a are pairwise nonassociate prime quadrics in S on this domain. The original quotient spaces are

$$\begin{aligned} \mathcal{C}_d &= (S/I)_d, \quad \mathcal{D}_{a,d} = (S/I_a)_d, \quad \mathcal{J}_d = S_d^{(g)}, \\ \mathcal{J}_d &= \text{im}(\mathcal{J}_d \longrightarrow \mathcal{C}_d), \quad \text{j}_d : \mathcal{C}_d \longrightarrow \bigoplus_{a=1}^4 \mathcal{D}_{a,d}, \quad [f] \longmapsto ([f]_{I_a})_a. \end{aligned} \quad (\text{DFX2})$$

The map j_d is defined because $I \subset I_a$. The two-quadric presentation is the complete intersection itself: the kernel of $(b, c) \mapsto Qb + Q_ac$ is $(Q_ae, -Qe)$. Indeed $Q \mid Q_ac$ implies $Q \mid c$ by primality and nonassociation, and substitution then gives the displayed relation. Consequently its full graded free resolution is $0 \rightarrow S(-4) \rightarrow S(-2)^2 \rightarrow S \rightarrow S/I_a \rightarrow 0$. In particular $\dim \mathcal{D}_{a,d} = 4d$ for $d \geq 2$. No restriction of Q_a modulo Q has been declared prime.

For each $i \in \{1, 2, 3, 4\}$ use the literal polynomial

$$\begin{aligned} h_i &= x_i^4, & g^a h_i &= \zeta^{-4ia} h_i, \\ m_{i,d} : \mathcal{C}_d &\longrightarrow \mathcal{C}_{d+4}, & [f] &\longmapsto [h_i f], \\ m_{i,a,d} : \mathcal{D}_{a,d} &\longrightarrow \mathcal{D}_{a,d+4}, & [f] &\longmapsto [h_i f]. \end{aligned} \quad (\text{DFX3})$$

These maps are well defined by ideal multiplication, and $j_{d+4} m_{i,d} = (m_{i,a,d})_a j_d$. Their coefficients have not been rescaled.

The exact exponential map and its degree change use

$$\begin{aligned} b_* &= \tfrac{1}{2} - \delta - i\gamma, & Z(z) &= e^{2\delta z}, & W(z) &= e^{2i\gamma z}, \\ t(z) &= (Z(z)W(z), Z(z), W(z), 1)^\top, & x(z) &= \mathbf{H}t(z), \\ \beta_{ab}^{(d)} &= \tfrac{d}{2} + (2a - d)\delta + i(2b - d)\gamma, & \mathcal{N}_{d,f}(z) &= e^{db_* z} f(x(z)). \end{aligned} \quad (\text{DFX4})$$

Thus every multiplier has its complete original phase and center:

$$\begin{aligned} L_i(z) &= e^{4b_* z} x_i(z)^4, & \mathcal{N}_{d+4,h_i f} &= L_i \mathcal{N}_{d,f}, \\ E_A(z) &= e^{8b_* z} \mathcal{A}(x(z)), & \mathcal{N}_{d,Ab} &= E_A \mathcal{N}_{d-8,b}, & v &= \text{ord}_0 E_A. \end{aligned} \quad (\text{DFX5})$$

Here E_A is the unchanged conductor symbol, independent of d . These identities follow directly by multiplying the displayed exponentials, using the homogeneous degrees 4 and 8. In particular the conductor gauge at degree d is taken to the same conductor gauge at degree $d + 4$.

For complete coefficient formulas write $h_i(\mathbf{H}(ZW, Z, W, 1)^\top) = \sum_{r,t=0}^4 c_{rt}^{(i)} Z^r W^t$. If ν is the original degree- d coefficient array, then

$$(M_{i,d}\nu)_{ab} = \sum_{r,t=0}^4 c_{rt}^{(i)} \nu_{a-r,b-t}, \quad 0 \leq a, b \leq d+4, \quad \beta_{a-r,b-t}^{(d)} + \beta_{rt}^{(4)} = \beta_{ab}^{(d+4)}. \quad (\text{DFX6})$$

Coefficients outside the original $0, \dots, d$ range are zero. This is the full $(d+5)^2$ by $(d+1)^2$ primary coefficient matrix. If $C_d^\top$ denotes the original conductor coefficient multiplication, the same convolution proves $M_{i,d} C_d^\top = C_{d+4}^\top M_{i,d-8}$.

Define the actual low-jet spaces by

$$\begin{aligned} \mathcal{C}_{v,d} &= \{[f] \in \mathcal{C}_d : \mathcal{N}_{d,f}^{(n)}(0) = 0, \ 0 \leq n < v\}, \\ \mathcal{J}_{v,d} &= \{f \in \mathcal{J}_d : \mathcal{N}_{d,f}^{(n)}(0) = 0, \ 0 \leq n < v\}, & \mathcal{J}_{v,d} &= \text{im}(\mathcal{J}_{v,d} \rightarrow \mathcal{C}_d). \end{aligned} \quad (\text{DFX7})$$

The jets descend through I_d : Q restricts to zero and $E_A \mathcal{N}_{d-8,b}$ vanishes to order at least v . The Leibniz formula gives the full finite matrix identity

$$\mathcal{N}_{d+4,h_i f}^{(n)}(0) = \sum_{r=0}^n \binom{n}{r} L_i^{(r)}(0) \mathcal{N}_{d,f}^{(n-r)}(0). \quad (\text{DFX8})$$

Every summand is zero when $n < v$ and f is in the source low-jet space. Hence $m_{i,d}$ restricts to $\mathcal{C}_{v,d} \rightarrow \mathcal{C}_{v,d+4}$. This proof retains possible extra vanishing of L_i at the origin.

On the invariant low-jet image the two required degree-changing maps are therefore

$$\begin{aligned} E_{i,k} : \mathcal{J}_{v,k} &\longrightarrow \mathcal{C}_{v,k+4}, & [f] &\longmapsto [h_i f], \\ \bar{E}_{i,k} : \mathcal{J}_{v,k} &\longrightarrow \mathcal{C}_{v,k+4} / \mathcal{J}_{v,k+4}, & [f] &\longmapsto [h_i f] \bmod \mathcal{J}_{v,k+4}. \end{aligned} \quad (\text{DFX9})$$

An invariant representative exists by definition. Changing it by an element of I_k changes its product by an element of I_{k+4} , so both arrows are defined without choosing a representative. The target degree $k+4$ is again 1 modulo 4 and at least 13. KRC4 identifies the target of the second arrow exactly with $(K_{k+4}/(K_{k+4} \cap L_{k+4}))^\vee$, whereas its source is $(V_{A,k}/(K_k + L_k))^\vee$. This is the typed degree-changing annihilator-to-kernel-dual map; it imposes no identification of their native metrics.

101 The exact branch identity and every elementary quotient kernel

For a degree d class define

$$\delta_d : \mathcal{C}_d \longrightarrow \mathcal{D}_{1,d} \oplus \mathcal{D}_{2,d}, \quad \delta_d[f]_a = \frac{[f]_{I_a} - [g^a f]_{I_a}}{2}, \quad a = 1, 2. \quad (\text{DFX10})$$

This is the original KRC difference map. Its typing can be checked directly: $g^a(Qb)$ is divisible by Q_a , and for $a \neq 0$ the product $g^a \mathcal{A}$ contains Q . Thus changing f by I_d changes neither displayed quotient difference. It kills $\mathcal{J}_d$, so it descends through that invariant quotient. For an invariant representative f the exact exchange formula is

$$\delta_{k+4}(h_i f)_a = c_{ia} x_i^4 [f]_{I_a}, \quad c_{ia} = \frac{1 - \zeta^{-4ia}}{2}, \quad a = 1, 2. \quad (\text{DFX11})$$

Indeed $g^a(h_i f) = \zeta^{-4ia} h_i f$. Every c_{ia} is nonzero: $4ia$ is not divisible by 5 for $1 \leq i \leq 4$ and $a = 1, 2$. The factor $1/2$ and its complex phase remain present. More generally, for every original homogeneous monomial h of degree e and weight $w = \sum i\mu_i \not\equiv 0 \pmod{5}$, the identical proof gives $\delta_{k+e}(hf)_a = (1 - \zeta^{-aw})h[f]_{I_a}/2$. This last statement asserts the multiplication identity only; it makes no monomial factorization or spanning assertion.

For later exact kernel formulas let $J_d^\sharp = I_d + \mathcal{J}_d$, a vector subspace of S_d . Ideal colons in the next display are ordinary homogeneous ideal colons; inverse images of $J_d^\sharp$ are vector-space inverse images. With all degrees shown, the complete kernels of DFX3 and DFX9 are

$$\begin{aligned} \ker m_{i,a,k} &= (I_a : x_i^4)_k / (I_a)_k, & \ker m_{i,k} &= (I : x_i^4)_k / I_k, \\ \ker E_{i,k} &= \frac{\mathcal{J}_{v,k} \cap (I : x_i^4)_k}{\mathcal{J}_k \cap I_k}, & \ker \overline{E}_{i,k} &= \frac{\{f \in \mathcal{J}_{v,k} : x_i^4 f \in J_{k+4}^\sharp\}}{\mathcal{J}_k \cap I_k}. \end{aligned} \quad (\text{DFX12})$$

Each formula follows by setting its specified target class to zero. The source denominator lies in every numerator by DFX5. For the last equality, if $x_i^4 f = p + b$ with p invariant and $b \in I_{k+4}$, then DFX8 and the vanishing jets of b force $p \in \mathcal{J}_{v,k+4}$. Thus using all invariant representatives in $J_{k+4}^\sharp$ adds no forbidden target low jets. For any nonempty subset $U \subset \{1, 2, 3, 4\}$, the kernel of the corresponding tuple of any of these maps is the intersection of its displayed kernels. The next sections calculate the branch and conductor colons, and give complete finite matrices for the last invariant-quotient kernel.

102 Exact calculation on the full nonreduced intersections

Let $T = \mathbb{C}[z_0, z_1, w_0, w_1]$ with bidegrees $(1, 0)$ on the z variables and $(0, 1)$ on the w variables. Write $T_{b,c} = 0$ if either index is negative. The original Segre map is

$$\psi_d : S_d \longrightarrow T_{d,d}, \quad f \longmapsto f(\mathbf{H}(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^\top). \quad (\text{DFX13})$$

It is onto and has kernel QS_{d-2} . Here is a direct proof. After the invertible change $x = \mathbf{H}t$, each degree d monomial maps to a biform monomial. Every pair of exponents of z_0, w_0 is reached by

taking the exponent of t_1 to be $\max(0, a + b - d)$ and completing the other three exponents. The four exponents of any two lifts with the same image differ by an integer multiple of $(1, -1, -1, 1)$. After canceling their common monomial factor, their difference is a multiple of $t_1 t_4 - t_2 t_3$, by the finite geometric-sum identity for $A^n - B^n$. Group all terms of a polynomial by their image monomial. Each zero coefficient group is consequently in that ideal. This proves the entire kernel, including all degrees.

Retain the literal biforms

$$\ell_i = \psi_1(x_i), \quad B_a = \psi_2(Q_a), \quad A = \prod_{a=1}^4 B_a = \psi_8(\mathcal{A}). \quad (\text{DFX14})$$

Every ℓ_i is nonzero and the four ℓ_i are linearly independent, since their coefficient matrix is H . Every B_a is nonzero, because $\psi_2(Q_a) = 0$ would make Q_a a scalar multiple of Q . Surjectivity in DFX13 gives the exact complete-intersection and conductor quotients

$$\mathcal{D}_{a,d} \simeq T_{d,d}/B_a T_{d-2,d-2}, \quad \mathcal{C}_d \simeq T_{d,d}/A T_{d-8,d-8}. \quad (\text{DFX15})$$

For example, if $\psi f = B_a b$, lift $b = \psi c$ and infer $f - Q_a c \in (Q)$; this proves the complete kernel I_a . The proof for A is identical. These isomorphisms have preserved all factor multiplicities of the actual B_a .

We use ordinary factorization in the polynomial ring T . For clarity, its required algebra has an elementary proof: if R is a unique-factorization domain, the content of a product in $R[t]$ is the product of contents up to a unit. For each prime of R , reduce the two primitive parts modulo that prime; their product is nonzero in the polynomial ring over its fraction domain, so no prime divides every product coefficient. Primitive factorization over the fraction field therefore lifts to $R[t]$, with existence and uniqueness inherited from one-variable division over a field and the factorization of contents in R . Starting with $\mathbb{C}$ proves the assertion for T by four repetitions. A factor of a bihomogeneous polynomial is bihomogeneous: its least and greatest degrees for each of the two gradings add under multiplication in a domain, so a product having a single degree forces both factors to have a single degree. Accordingly all factors below have well-defined nonnegative bidegrees.

The following formula calculates all subset kernels. Fix a nonzero biform B of bidegree (b, b) , where $b = 2$ for a branch and $b = 8$ for the conductor. In any factorization of B retain its scalar and all factor multiplicities. For nonempty U put

$$G_{B,U} = \gcd(B, \{\ell_i^4 : i \in U\}), \quad (\rho_U, \sigma_U) = \text{bideg } G_{B,U}, \quad (\text{DFX16})$$

$$B_U = B/G_{B,U}, \quad \kappa_{B,U}(d) = (d - b + \rho_U + 1)_+(d - b + \sigma_U + 1)_+ - (d - b + 1)_+^2,$$

where $n_+ = \max(n, 0)$. A gcd here means the product with the minimum exponent for each irreducible factor, not a radical. One may use any consistent factors and their retained scalar units; the quotient $B/G_{B,U}$ then contains the compensating scalar. No original polynomial or multiplication map is rescaled.

The complete tuple multiplication kernel is the following explicit space and isomorphism:

$$\ker \left[\frac{T_{d,d}}{B T_{d-b,d-b}} \xrightarrow{([f] \mapsto [\ell_i^4 f])_{i \in U}} \bigoplus_{i \in U} \frac{T_{d+4,d+4}}{B T_{d+4-b,d+4-b}} \right] \simeq \frac{T_{d-b+\rho_U,d-b+\sigma_U}}{G_{B,U} T_{d-b,d-b}}, \quad [u] \mapsto [B_U u]. \quad (\text{DFX17})$$

Its dimension is exactly $\kappa_{B,U}(d)$. To prove the assertion, let an irreducible factor have exponent e in B , n_i in ℓ_i^4 and t in f . Simultaneous divisibility $B \mid \ell_i^4 f$ is precisely $t + n_i \geq e$ for every i , equivalently $t \geq e - \min(e, \min_i n_i)$. Over all factors this says $B_U \mid f$. Its quotient u has exactly the bidegree shown. Changing u changes $[B_U u]$ by zero exactly when $G_{B,U} \mid u$, which gives the denominator in DFX17. Multiplication by that nonzero biform is injective in T . Counting the

monomial dimensions $\dim T_{r,t} = (r+1)_+(t+1)_+$ proves DFX16. This argument checks every irreducible exponent separately and therefore applies without change to nonreduced intersections.

In particular, for a single original branch and coordinate, set $G_{a,i} = \gcd(B_a, \ell_i^4)$ and $(\rho_{a,i}, \sigma_{a,i}) = \text{bideg } G_{a,i}$. At every original $k \geq 9$,

$$\begin{aligned} \dim \ker m_{i,a,k} &= (\rho_{a,i} + \sigma_{a,i})(k-1) + \rho_{a,i}\sigma_{a,i}, \\ \text{rank } m_{i,a,k} &= 4k - \dim \ker m_{i,a,k}, \quad \dim \text{coker } m_{i,a,k} = 16 + \dim \ker m_{i,a,k}. \end{aligned} \quad (\text{DFX18})$$

There is an additional exact nilpotent statement:

$$\gcd(B_a, \ell_i^4) = \gcd(B_a, \ell_i^2) \quad \text{with consistent factor units,} \quad (I_a : x_i^4) = (I_a : x_i^2) = (I_a : x_i^\infty). \quad (\text{DFX19})$$

Indeed a nonzero $(1, 1)$ biform is either irreducible or the product of a $(1, 0)$ and a $(0, 1)$ linear form. This follows by adding the nonnegative bidegrees of its possible factors. An irreducible factor of either type has exponent at most two in B_a , whose bidegree is $(2, 2)$. Thus ℓ_i^2 already contains its entire B_a multiplicity. Divisibility as in DFX17 proves the equality in every homogeneous degree, hence the ideal equality in S . The notation $(I_a : x_i^\infty)$ means the union of all finite colons. In particular the entire x_i -torsion on the nonreduced branch is killed by x_i^2 .

The pair kernels have a short complete classification. For distinct i, j , either $\gcd(\ell_i, \ell_j) = 1$, or their common factor is a single linear form of bidegree $(1, 0)$ or $(0, 1)$. A common $(1, 1)$ factor would make them proportional, contradicting the invertibility of H . If the common factor has multiplicity $e \in \{0, 1, 2\}$ in B_a , then

$$\dim \ker(m_{i,a,k}, m_{j,a,k}) = e(k-1). \quad (\text{DFX20})$$

This includes $e = 0$ when the factor is absent; it follows by inserting $(\rho, \sigma) = (e, 0)$ or $(0, e)$ into DFX16. No factor shared by the actual restricted quadrics has been discarded.

103 Joint injectivity and the exact preceding descent defect

Any three distinct ℓ_i have gcd one. Here is the full argument. A common nonconstant factor of $(1, 1)$ biforms has bidegree $(1, 0)$, $(0, 1)$ or $(1, 1)$ after selecting an irreducible factor. Multiples in bidegree $(1, 1)$ of such a fixed factor form a space of dimension at most two, two or one, respectively. Three of the four rows of an invertible 4 by 4 matrix are linearly independent. They therefore cannot all lie in any of those spaces. Taking fourth powers does not create a new common irreducible factor. DFX17 proves

$$\bigcap_{i \in U} \ker m_{i,a,d} = 0, \quad \bigcap_{i \in U} \ker m_{i,d} = 0 \quad \text{for } |U| \geq 3 \text{ and every } d \geq 0. \quad (\text{DFX21})$$

This is joint injectivity both on each full nonreduced branch and on the complete conductor quotient. It also holds after restricting the source to $\mathcal{C}_{v,d}$ or $\mathcal{J}_{v,d}$. The four branch maps at degree k have joint rank $4k$ and cokernel dimension $12k + 64$. Any three have rank $4k$ and cokernel dimension $8k + 48$. These are the full direct-sum codomains from DFX17.

For the branch-detected invariant exchange put

$$\mathcal{B}_{U,k} : \mathcal{J}_{v,k} \longrightarrow \bigoplus_{i \in U} (\mathcal{D}_{1,k+4} \oplus \mathcal{D}_{2,k+4}), \quad [f] \longmapsto (\delta_{k+4} m_{i,k}[f])_{i \in U}. \quad (\text{DFX22})$$

For arbitrary nonempty U , DFX11 and DFX17 give its complete kernel as

$$\ker \mathcal{B}_{U,k} = \{[f] \in \mathcal{J}_{v,k} : [f]_{I_a} \in \ker(m_{i,a,k})_{i \in U}, \ a = 1, 2\}. \quad (\text{DFX23})$$

In particular the proven joint injectivity gives, for every $|U| \geq 3$, the exact common value

$$\ker \mathcal{B}_{U,k} = \mathcal{J}_{v,k} \cap \ker j_k =: \mathcal{E}_{v,k}. \quad (\text{DFX24})$$

To verify the last use of all four branches, for invariant f , $[f]_{I_4} = g^4[f]_{I_1}$ and $[f]_{I_3} = g^3[f]_{I_2}$. The maps are typed since $g^{-a}(Q, Q_a) = (Q_{-a}, Q)$. Thus vanishing of the first two classes implies vanishing of the other two. Conversely, vanishing on all branches kills every exchange entry. In KRC coordinates the space $\mathcal{E}_{v,k}$ is precisely the image under $\mathbb{T}_k J$ of the kernel of $Y_{\epsilon,k}$, for each of the four sign choices. Consequently $\dim \mathcal{E}_{v,k} = d_{\epsilon,k}$ is independent of that sign, and $\text{rank } \mathcal{B}_{U,k} = j_0 - d_{\epsilon,k}$. The previously occurring descent defect has been identified with an exact new kernel; it has not been assigned value zero.

That preceding defect also has a complete actual factor calculation. Let

$$\mathcal{L} = \text{lcm}(B_1, B_2, B_3, B_4), \quad (b_z, b_w) = \text{bideg } \mathcal{L}, \quad \mathcal{G} = A/\mathcal{L}, \quad 2 \leq b_z, b_w \leq 8. \quad (\text{DFX25})$$

The lcm retains the maximum multiplicity of each actual irreducible factor among the four bi-forms. Its scalar may be chosen consistently with the retained factor lists; $A/\mathcal{L}$ then retains the compensating scalar. Divisibility in T proves the exact isomorphism

$$\ker j_d \simeq \frac{T_{d-b_z, d-b_w}}{\mathcal{G}T_{d-8, d-8}}, \quad [u] \mapsto [\mathcal{L}u], \quad \dim \ker j_k = (k - b_z + 1)(k - b_w + 1) - (k - 7)^2. \quad (\text{DFX26})$$

Indeed all four branch classes vanish exactly when every B_a divides f , exactly when $\mathcal{L}$ divides f . Two such representatives have the same conductor class precisely when their quotient difference is divisible by $A/\mathcal{L}$. This proves both the isomorphism and, by injectivity of multiplication in T , the stated dimension for $k \geq 9$. In particular

$$0 \leq d_{\epsilon,k} \leq (16 - b_z - b_w)k + (1 - b_z)(1 - b_w) - 49. \quad (\text{DFX27})$$

The right side is the expansion of the exact nonnegative dimension in DFX26. Its actual factor degrees record shared components and repetitions. When that value is nonzero, it is retained in the original correspondence. No generic coprimality of the restricted factors has entered this proof.

For the actual target-invariant quotient the proven relation is more precise than a declaration of injectivity:

$$\begin{aligned} \ker(\overline{E}_{i,k})_{i \in U} &= \frac{\{f \in \mathcal{I}_{v,k} : x_i^4 f \in J_{k+4}^\# \text{ for every } i \in U\}}{\mathcal{I}_k \cap I_k}, \\ \ker(\overline{E}_{i,k})_{i \in U} &\subset \mathcal{E}_{v,k} \quad (|U| \geq 3), \\ \frac{\mathcal{I}_{v,k}}{\ker(\overline{E}_{i,k})_{i \in U}} &\twoheadrightarrow \frac{\mathcal{I}_{v,k}}{\mathcal{E}_{v,k}} \xrightarrow{\sim} \text{im } \mathcal{B}_{U,k}. \end{aligned} \quad (\text{DFX28})$$

The first equality is DFX12 with all components imposed. The inclusion follows because δ_{k+4} kills the target invariant image. The first arrow sends the original source class to its further quotient, and its kernel is $\mathcal{E}_{v,k} / \ker(\overline{E}_{i,k})_{i \in U}$. The last arrow sends a source class to its actual branch value; DFX24 proves its bijectivity. Thus the relationship between the target-invariant quotient and the detected branch image is a proved quotient morphism with its entire kernel.

104 Finite matrices for the invariant and low-jet kernels

Here are complete finite formulas on every actual coefficient stratum, including those where polynomial factors meet. Order the degree- d monomials in S lexicographically once. Let Ψ_d be the coefficient matrix of DFX13, and let U_d be the selector of all monomials x^μ satisfying $\sum i\mu_i \equiv 0 \pmod{5}$. Put

$$\begin{aligned} A_d &= [M_Q : S_{d-2} \rightarrow S_d, \quad M_A : S_{d-8} \rightarrow S_d], \\ \mathcal{V}_{v,d} &= ((\beta_{ab}^{(d)})^n)_{0 \leq n < v, \ 0 \leq a, b \leq d}, \quad H_{v,d} = \mathcal{V}_{v,d} \Psi_d U_d. \end{aligned} \quad (\text{DFX29})$$

All entries are finite products and sums of the original $H, Q, Q_a, \beta_{ab}^{(d)}$ coefficients. The zero-row matrix is used when $v = 0$.

For any finite matrix A an exact quotient matrix with kernel $\text{im } A$ is obtained as follows. Select its first independent columns F , then the first row subset $\mathcal{R}$ making $F_{\mathcal{R}}$ invertible, and put

$$\text{quo}(A)x = (x - FF_{\mathcal{R}}^{-1}x_{\mathcal{R}})_{\mathcal{R}^c}. \quad (\text{DFX30})$$

For rank zero this is the identity; for full row rank it has zero rows. Subtraction makes the $\mathcal{R}$ coordinates zero. All remaining coordinates vanish exactly when $x = FF_{\mathcal{R}}^{-1}x_{\mathcal{R}}$, proving its full kernel. Conversely a vector supported in $\mathcal{R}^c$ is unchanged there, so the map is onto. Thus this construction gives explicit exact quotient coordinates, with no metric or bound attributed to the selected determinant.

Set $\Pi_d = \text{quo}(A_d)$. Take any explicit kernel frame $F_{v,d}$ of $H_{v,d}$ by the same exact elimination. Select the first independent columns of $\Pi_d U_d F_{v,d}$, and let $\widehat{F}_{v,d}$ be the corresponding columns of $U_d F_{v,d}$ before applying Π_d . They are actual homogeneous invariant low-jet representatives. They map bijectively onto $\mathcal{J}_{v,d}$. Define

$$\begin{aligned} Q_d^{\text{inv}} &= \text{quo}(\Pi_d U_d), & \Pi_{a,d} &= \text{quo}([M_Q, M_{Q_a}]), \\ \mathcal{E}_{i,k} &= Q_{k+4}^{\text{inv}} \Pi_{k+4} M_{x_i^4} \widehat{F}_{v,k}, \\ \mathcal{B}_{i,k} &= \begin{bmatrix} c_{i1} \Pi_{1,k+4} M_{x_i^4} \widehat{F}_{v,k} \\ c_{i2} \Pi_{2,k+4} M_{x_i^4} \widehat{F}_{v,k} \end{bmatrix}, & \mathcal{J}_k &= \begin{bmatrix} \Pi_{1,k} \\ \Pi_{2,k} \\ \Pi_{3,k} \\ \Pi_{4,k} \end{bmatrix} \widehat{F}_{v,k}. \end{aligned} \quad (\text{DFX31})$$

In each multiplication matrix its domain and codomain degrees are the ones required by the adjacent factors. These formulas compute every last quotient kernel from DFX12 and DFX28:

$$\begin{aligned} \ker(\overline{E}_{i,k})_{i \in U} &\simeq \ker[\mathcal{E}_{i,k}]_{i \in U}, & \ker \mathcal{B}_{U,k} &\simeq \ker[\mathcal{B}_{i,k}]_{i \in U}, \\ \mathcal{C}_{v,k} &\simeq \ker \mathcal{J}_k, & \ker[\mathcal{B}_{i,k}]_{i \in U} &= \ker \mathcal{J}_k \quad (|U| \geq 3). \end{aligned} \quad (\text{DFX32})$$

The bracket means vertical concatenation. The source coordinates are the identical columns $\widehat{F}_{v,k}$ in every matrix. DFX30 proves that the kernels are precisely the stated polynomial quotient kernels; DFX11 gives the two branch rows; and DFX24 proves their final equality. Thus these are finite calculations in the actual period coefficients, rather than a requirement to assume a minor or a factor configuration.

For completeness, every low-jet restriction of a conductor multiplication kernel can also be computed directly in the factor presentation DFX17 with $B = A$. Choose quotient monomial coordinates there using DFX30, multiply their representatives by $A/G_{A,U}$, and apply $\mathcal{V}_{v,d}$ after expanding the resulting (d, d) biform. Call the resulting matrix $\mathcal{L}_{A,U,v,d}$. All denominator columns are multiples of A and have zero first v jets; therefore this matrix is defined on the quotient. It gives

$$\dim(\ker(m_{i,d})_{i \in U} \cap \mathcal{C}_{v,d}) = \kappa_{A,U}(d) - \text{rank } \mathcal{L}_{A,U,v,d}. \quad (\text{DFX33})$$

For a branch the original first v jets do not in general descend through (Q, Q_a) : B_a is only one conductor factor. The correct map for a branch low-jet question is consequently the image of the already defined conductor low-jet space under j_d , computed by DFX29–32. If one instead retains polynomial representatives before the branch quotient, the exact constraint is the inverse image of $\ker \mathcal{V}_{v,d}$ on those representatives, followed by the branch quotient. This specified inverse-image map retains all representatives and proves its relation to the conductor low jets; no unsupported jet functional on $\mathcal{D}_{a,d}$ is introduced.

The exact original DNE coordinates can be attached without a new choice of a coefficient metric. At each original degree $d = k, k + 4$, let the KRC5 bijection and its full quotient matrices be

$$\begin{aligned} \alpha_A &= v!/\mu_v, & \mathbf{m}_d &= 16d - 48 - v, & \mathbf{T}_d &= \alpha_A N_d Z_{\text{exp},d} : \mathbb{C}^{\mathbf{m}_d} \xrightarrow{\sim} \mathcal{C}_{v,d}, \\ \mathbf{B}_d &= Z_{K,d}^{\top} A_{\ell,d}, & \mathbf{J}_d &= A_{\ell,d}^{-1} P_{Y,d}^{\top}, & \ker \mathbf{B}_d &= \text{im } \mathbf{J}_d, & \mathbf{T}_d \text{ im } \mathbf{J}_d &= \mathcal{J}_{v,d}. \end{aligned} \quad (\text{DFX34})$$

Here $N_d, Z_{\text{exp},d}, A_{\ell,d}, P_{Y,d}$ are the original KRC/DNE/FGC provider matrices, and $Z_{K,d}$ denotes the original FGC factorial-scaled kernel frame called Z_d in KRC5. This subscript distinguishes it from the coefficient quotient matrix in DFX31. Define the rectangular maps

$$\begin{aligned}\mathcal{M}_{i,k}^{\text{DNE}} &= \mathbf{T}_{k+4}^{-1} m_{i,k} \mathbf{T}_k : \mathbb{C}^{\mathbf{m}_k} \longrightarrow \mathbb{C}^{\mathbf{m}_k+64}, \\ \mathcal{X}_{i,k} &= \mathbf{B}_{k+4} \mathcal{M}_{i,k}^{\text{DNE}} \mathbf{J}_k : \mathbb{C}^{j_{0,k}} \longrightarrow \mathbb{C}^{t_{K,k+4}}, \\ j_{0,k} &= 8k - 32 - v + t_{0,k}, \quad t_{K,k+4} = 8k + 16 - t_{0,k+4}.\end{aligned}\tag{DFX35}$$

DFX8 proves that the first inverse is applied inside its full low-jet domain. KRC5 proves its bijectivity and the stated kernel of $\mathbf{B}_{k+4}$. Therefore $\mathcal{X}_{i,k}$ is exactly the coordinate expression of $\overline{E}_{i,k}$, and its tuple kernels are those in DFX28 and DFX32 under the source bijection $\mathbf{T}_k \mathbf{J}_k$. Both occurrences of the unchanged scalar α_A are retained in DFX34; their cancellation in the conjugated multiplication is the literal equality $(\alpha_A N_{k+4} Z_{\text{exp},k+4})^{-1} m_{i,k} (\alpha_A N_k Z_{\text{exp},k})$. All transposes in the provider quotient are ordinary transposes, as required by complex-linear duality. These equalities give the exact maps on which each original Gamma source form is subsequently transported.

All degree-four multiplier kernels have now been computed in the original ideals or the displayed finite matrices. The nonreduced branch calculation proves joint injectivity of any three multipliers. After the invariant target quotient, its exact remaining kernel lies in the fully retained shared-component space DFX26 and is calculated by DFX31–32. The numerator phase, conductor gauge, first v jets and both original annihilator/kernel-dual types have been carried through the degree change. Native Gamma estimates for these fixed maps remain the separate original-metric calculation; no such estimate is inferred from the coefficient gcds or pivot matrices.

Retain the original fixed simple quartet, amplitude, period and logarithm branch, with $0 < \delta < 1/2$, $\gamma > 2$, and original degree $k = 4l + 1 \geq 9$. All polynomial coefficients, phases and coordinate orders below are literal. The construction uses the original invertible matrix $\mathbf{H} = F^{-1} \Pi U \mathcal{E}_{\text{prim}}$. It proves exact algebraic maps and kernels. It assigns no native Gamma estimate to their coefficient matrices.

105 The complete homogeneous identity

Write $\lambda = (\lambda_1, \lambda_2, \lambda_3, \lambda_4)$ for the fourth row of $\mathbf{H}^{-1}$. Thus

$$\lambda \mathbf{H} = (0, 0, 0, 1), \quad \lambda \cdot x = (\mathbf{H}^{-1} x)_4.\tag{BGA1}$$

For a multiindex $\nu \in \mathbb{Z}_{\geq 0}^4$ with $|\nu| = 13$, let $i(\nu)$ be the first index satisfying $\nu_i \geq 4$. Such an index exists: if all four exponents were at most three, their sum would be at most twelve. Put

$$\begin{aligned}U_i(x) &= \sum_{\substack{|\nu|=13 \\ \nu_i \geq 4, \nu_j \leq 3 \ (j < i)}} \frac{13!}{\nu_1! \nu_2! \nu_3! \nu_4!} \left(\prod_{j=1}^4 \lambda_j^{\nu_j} \right) x^{\nu - 4e_i} \\ &= \sum_{\substack{|\mu|=9 \\ \mu_j \leq 3 \ (j < i)}} c_{i,\mu} x^\mu, \quad c_{i,\mu} = \frac{13! \lambda_i^{\mu_i+4} \prod_{j \neq i} \lambda_j^{\mu_j}}{(\mu_i + 4)! \prod_{j \neq i} \mu_j!}.\end{aligned}\tag{BGA2}$$

Zero entries of λ are allowed; powers with exponent zero are one, and the formula involves no division by λ_i . Each U_i is homogeneous of degree nine or is the zero polynomial. The multinomial theorem and the disjoint first-index partition prove the complete ambient identity

$$\boxed{\sum_{i=1}^4 U_i(x) x_i^4 = (\lambda \cdot x)^{13}.}\tag{BGA3}$$

Indeed each degree-thirteen monomial on the right occurs in exactly one summand on the left, with exactly its original multinomial coefficient and its complete product of row entries.

106 Original exponential and coefficient formulas

Use the original primary order $(++)$, $(+-)$, $(-+)$, $(--)$ and

$$\begin{aligned} b_* &= \frac{1}{2} - \delta - i\gamma, & Z(z) &= e^{2\delta z}, & W(z) &= e^{2i\gamma z}, \\ t(z) &= (Z(z)W(z), Z(z), W(z), 1)^\top, & x(z) &= \text{Ht}(z), \\ \ell_i(Z, W) &= H_{i1}ZW + H_{i2}Z + H_{i3}W + H_{i4}. \end{aligned} \quad (\text{BGA4})$$

Then $\lambda \cdot x(z) = 1$ exactly. Define the literal four degree-four multipliers and four inverse coefficients by

$$B_i(z) = e^{4b_*z} x_i(z)^4, \quad D_i(z) = e^{-4b_*z} U_i(x(z)),$$

$$\sum_{i=1}^4 D_i(z) B_i(z) = 1 \quad (z \in \mathbb{C}).$$

(BGA5)

The identity follows from BGA3 with its right side equal to one and with the two displayed scalar exponential factors cancelling exactly. No zero of an individual B_i is excluded; the simultaneous identity is entire.

Here are all original coefficient and exponent formulas. Write

$$\ell_i(Z, W)^4 = \sum_{a,b=0}^4 b_{ab}^{(i)} Z^a W^b, \quad U_i(\ell_1, \ell_2, \ell_3, \ell_4) = \sum_{a,b=0}^9 u_{ab}^{(i)} Z^a W^b. \quad (\text{BGA6})$$

Their actual entries, including cancellations, are the finite sums

$$b_{ab}^{(i)} = \sum_{\substack{n_1+\dots+n_4=4 \\ n_1+n_2=a, \, n_1+n_3=b}} \frac{4!}{\prod_{\alpha=1}^4 n_{\alpha}!} \prod_{\alpha=1}^4 H_{i\alpha}^{n_{\alpha}}, \quad (\text{BGA7})$$

and

$$u_{ab}^{(i)} = \sum_{\substack{|\mu|=9 \\ \mu_j \leq 3 \, (j < i)}} c_{i,\mu} \sum_{\substack{m_{j\alpha} \geq 0, \, \sum_{\alpha} m_{j\alpha} = \mu_j \\ \sum_j (m_{j1}+m_{j2})=a, \, \sum_j (m_{j1}+m_{j3})=b}} \prod_{j=1}^4 \left(\frac{\mu_j!}{\prod_{\alpha=1}^4 m_{j\alpha}!} \prod_{\alpha=1}^4 H_{j\alpha}^{m_{j\alpha}} \right). \quad (\text{BGA8})$$

These follow by expanding each of the four original row polynomials. The two occurrences of the first primary coordinate contribute to both exponents, as recorded by $m_{j1} + m_{j2}$ and $m_{j1} + m_{j3}$. Thus the complete exponential expansions are

$$\begin{aligned} B_i(z) &= \sum_{a,b=0}^4 b_{ab}^{(i)} e^{[4b_*+2a\delta+2ib\gamma]z}, \\ D_i(z) &= \sum_{a,b=0}^9 u_{ab}^{(i)} e^{[-4b_*+2a\delta+2ib\gamma]z}, \\ \sum_{i=1}^4 \sum_{\substack{a+r=A \\ b+t=B}} u_{ab}^{(i)} b_{rt}^{(i)} &= \begin{cases} 1 & (A, B) = (0, 0), \\ 0 & \text{otherwise,} \end{cases} \quad 0 \leq A, B \leq 13. \end{aligned} \quad (\text{BGA9})$$

The finite convolution identity is BGA3 after the original affine substitution. In particular the inverse coefficients have center $-4b_*$ with exponents indexed up to nine; they have not been assigned the center of the original degree-nine source.

107 The exact original lower-gauge kernel

For $d \geq 0$ put

$$\begin{aligned}\mathcal{P}_d &= \{f \in \mathbb{C}[Z, W] : \deg_Z f \leq d, \deg_W f \leq d\}, \\ \mathcal{E}_d &= \left\{ \sum_{a,b=0}^d v_{ab} e^{[d/2+(2a-d)\delta+i(2b-d)\gamma]z} \right\}, \\ \iota_d : \mathcal{P}_d &\xrightarrow{\sim} \mathcal{E}_d, \quad f \mapsto e^{db_*z} f(Z(z), W(z)).\end{aligned}\tag{BGA10}$$

To prove the bijection, the displayed exponents for distinct (a, b) are distinct: equality gives $2(a - a')\delta = 0$ and $2(b - b')\gamma = 0$. If a linear combination of their exponentials vanishes, differentiating at zero through the number of coefficients minus one gives a Vandermonde matrix. Its determinant is the product of all pairwise exponent differences, which is nonzero. Thus every coefficient is zero. Surjectivity is the displayed definition, and $\dim \mathcal{E}_d = (d + 1)^2$.

The complete compatibility complex with fixed signs is

$$\begin{aligned}\mathcal{G}_k : \mathcal{E}_{k-8} &\longrightarrow \mathcal{E}_{k-4}^4, \quad E \mapsto (B_i E)_{i=1}^4, \\ \partial_k : \mathcal{E}_{k-4}^4 &\longrightarrow \mathcal{E}_k^6, \quad (E_i)_i \mapsto (B_i E_j - B_j E_i)_{1 \leq i < j \leq 4}, \\ \ker \partial_k &= \text{im } \mathcal{G}_k, \quad \ker \mathcal{G}_k = 0.\end{aligned}\tag{BGA11}$$

Multiplication of the exponents in BGA10 proves every stated degree in these arrows. One inclusion follows by commutativity. For the converse write $E_i = \iota_{k-4}(f_i)$. Injectivity of ι_k makes the compatibility identities precisely

$$\ell_i^4 f_j = \ell_j^4 f_i \quad \text{in } \mathbb{C}[Z, W] \quad (1 \leq i < j \leq 4).\tag{BGA12}$$

The affine polynomial identity in BGA3 is $\sum_i U_i(\ell) \ell_i^4 = 1$. Set

$$F = \sum_{i=1}^4 U_i(\ell_1, \ell_2, \ell_3, \ell_4) f_i. \quad \text{Then} \quad \ell_i^4 F = f_i \quad \text{for every } i.\tag{BGA13}$$

Indeed replace each $\ell_i^4 f_j$ by $\ell_j^4 f_i$ in this finite sum, retaining the same index order; the sum becomes $f_i \sum_j U_j(\ell) \ell_j^4 = f_i$.

Here is the full degree calculation needed to recover the original lower source. Invertibility of $\mathbf{H}$ implies that at least one row has $(H_{i1}, H_{i2}) \neq (0, 0)$. For that row ℓ_i has Z -degree exactly one and ℓ_i^4 has Z -degree exactly four. For nonzero polynomials over the domain $\mathbb{C}[W]$, leading Z coefficients multiply to a nonzero coefficient; therefore $\deg_Z(\ell_i^4 F) = 4 + \deg_Z F$ when $F \neq 0$. BGA13 and $\deg_Z f_i \leq k - 4$ give $\deg_Z F \leq k - 8$. Likewise some row has $(H_{i1}, H_{i3}) \neq (0, 0)$, and working over $\mathbb{C}[Z]$ gives $\deg_W F \leq k - 8$. If $F = 0$ both bounds hold directly. Consequently

$$E = \iota_{k-8}(F) = \sum_{i=1}^4 D_i E_i \in \mathcal{E}_{k-8}, \quad E_i = B_i E.\tag{BGA14}$$

Every original ℓ_i is nonzero because its row in $\mathbf{H}$ is nonzero. BGA10 shows each B_i is a nonzero entire function. If $B_i E = 0$ identically, the polynomial product in BGA10 is zero in a domain, hence $E = 0$. This proves injectivity and uniqueness in BGA11. The dimensions, with every codomain retained, are therefore

$$\begin{aligned}\dim \ker \partial_k &= (k - 7)^2, \quad \text{rank } \partial_k = 4(k - 3)^2 - (k - 7)^2 = 3k^2 - 10k - 13, \\ \dim \text{coker } \partial_k &= 6(k + 1)^2 - \text{rank } \partial_k = 3k^2 + 22k + 19.\end{aligned}\tag{BGA15}$$

No surjectivity of the compatibility map is asserted.

For an explicit coefficient version of the cancellations, write $f_i = \sum_{a,b=0}^{k-4} e_{ab}^{(i)} Z^a W^b$. The recovered polynomial before using compatibility has coefficients

$$F_{AB} = \sum_{i=1}^4 \sum_{a+r=A, b+t=B} u_{rt}^{(i)} e_{ab}^{(i)}, \quad 0 \leq A, B \leq k+5. \quad (\text{BGA16})$$

BGA12 is the finite coefficient system obtained by replacing $u_{rt}^{(i)}$ here by $b_{rt}^{(i)}$ in $\ell_i^4 f_j - \ell_j^4 f_i$ and setting every coefficient to zero. BGA13 and the two proven degree identities give exactly $F_{AB} = 0$ whenever $A > k-8$ or $B > k-8$. For $0 \leq A, B \leq k-8$, BGA16 is the complete original lower coefficient array. Thus the restriction to the original grid is a proved cancellation of the full formula.

108 Original conductor gauges and the full ambient target

Retain the original nonzero conductor symbol $E_A \in \mathcal{E}_8$ with all its actual coefficients and $v = \text{ord}_0 E_A$. The original simultaneous gauge kernel is

$$\boxed{\{N \in \mathcal{E}_k : B_i N \in E_A \mathcal{E}_{k-4} \text{ for every } i\} = E_A \mathcal{E}_{k-8}.} \quad (\text{BGA17})$$

The inclusion from right to left follows by multiplying $B_i E$ in the lower source. Conversely choose the unique $E_i \in \mathcal{E}_{k-4}$ with $B_i N = E_A E_i$; uniqueness follows because the nonzero analytic function E_A cannot annihilate a nonzero exponential polynomial. Multiplying by B_j and subtracting shows $E_A(B_j E_i - B_i E_j) = 0$, hence compatibility. BGA14 gives $E_i = B_i E$ with $E \in \mathcal{E}_{k-8}$. Now $B_i(N - E_A E) = 0$ for every i , so the nonzero B_i gives $N = E_A E$. This proves the entire original gauge kernel, including every zero and multiplicity of E_A . Equivalently the tuple map

$$\frac{\mathcal{E}_k}{E_A \mathcal{E}_{k-8}} \longrightarrow \bigoplus_{i=1}^4 \frac{\mathcal{E}_{k+4}}{E_A \mathcal{E}_{k-4}}, \quad [N] \longmapsto ([B_i N])_i \quad (\text{BGA18})$$

is injective. Restricting its source to numerators with the first v derivatives zero preserves injectivity and takes them to the corresponding target low-jet spaces, by the Leibniz formula. All gauge denominators have these zero jets since they contain the unchanged factor E_A .

The ambient map on arbitrary tuples retains its full enlarged grid. Define

$$\mathcal{A}_{r,13} = \left\{ \sum_{a,b=0}^{r+13} c_{ab} e^{[rb_* + 2a\delta + 2ib\gamma]z} \right\} \quad (r \geq 0). \quad (\text{BGA19})$$

Direct multiplication in BGA9 gives the typed maps

$$\begin{aligned} \mathcal{L}_k : \mathcal{E}_{k+4}^4 &\longrightarrow \mathcal{A}_{k,13}, & (N_i) &\longmapsto \sum_i D_i N_i, \\ \mathcal{L}_k((B_i N)_i) &= N, & \mathcal{L}_k((E_A E_i)_i) &= E_A \sum_i D_i E_i \in E_A \mathcal{A}_{k-8,13}. \end{aligned} \quad (\text{BGA20})$$

For the last line $\deg_Z, \deg_W f_i \leq k-4$ and those of $U_i(\ell)$ are at most nine, so the polynomial part has separate degrees at most $k+5 = (k-8) + 13$. For the first line these separate degrees are at most $k+13$. These are the actual codomains before a compatibility constraint. On the compatible lower tuples, BGA14 proves that their inverse sum lies in $\mathcal{E}_{k-8}$ itself. Thus both the enlarged-grid map and its exact original-grid restriction have been supplied, with no suppressed boundary coefficient and no native norm inferred from this algebra.

109 A declared exact algebraic fixture

The following fixture tests the identities with fixed rational coefficients; it is not a substitution for the actual period matrix. Use

$$\mathbf{H}_{\text{test}} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -1 & -1 & -1 & 1 \end{pmatrix}, \quad \lambda_{\text{test}} = (1, 1, 1, 1), \quad (\text{BGA21})$$

$$(\ell_1, \ell_2, \ell_3, \ell_4) = (ZW, Z, W, 1 - ZW - Z - W).$$

Its determinant is one, and its inverse has fourth row λ_{test} . Every $c_{i,\mu}$ in BGA2 is then the displayed positive integer multinomial coefficient. The four first-index groups have respectively 220, 164, 112, 64 terms, totaling $\binom{16}{3} = 560$. For example the first group consists of all degree-nine four-variable monomials and has $\binom{12}{3} = 220$ terms. The second has $\sum_{a=0}^3 \binom{11-a}{2} = 164$ terms. The third has $\sum_{a,b=0}^3 (10 - a - b) = 112$ terms. The fourth permits each of its first three exponents to range from zero to three, giving $4^3 = 64$ terms. This independently checks the first-index partition in this fixture.

Take the declared fixture parameters $\delta_{\text{test}} = 1/4$, $\gamma_{\text{test}} = 3$, $k_{\text{test}} = 9$, so $b_{*,\text{test}} = 1/4 - 3i$, and the exact lower polynomial

$$F_{\text{test}}(Z, W) = 1 + 2Z - 3W + 5ZW, \quad f_{i,\text{test}} = \ell_i^4 F_{\text{test}}. \quad (\text{BGA22})$$

Each $f_{i,\text{test}}$ has separate degrees at most five, and $\ell_i^4 f_j - \ell_j^4 f_i = 0$ for all six oriented pairs. Using BGA2, their recovered polynomial is exactly

$$\sum_i U_i(\ell) f_{i,\text{test}} = F_{\text{test}} \sum_i U_i(\ell) \ell_i^4 = F_{\text{test}} (ZW + Z + W + 1 - ZW - Z - W)^{13} = F_{\text{test}}. \quad (\text{BGA23})$$

Multiplying by $e^{b_{*,\text{test}}z}$ proves the complete exponential fixture identity in $\mathcal{E}_1$, while the four input lower gauges are in $\mathcal{E}_5$. The accompanying integer-polynomial diagnostic expands all 560 ambient terms, all four substituted U_i , all six compatibility differences, and the recovered polynomial coefficient by coefficient. It also expands an incompatible tuple to retain, rather than discard, its larger ambient degree. The proof of BGA1–20 applies to the original matrix; this fixture establishes only the explicitly declared exact diagnostic.

110 A finite native inverse and the exact compatible lower gauges

Keep the original fixed actual simple quartet, its period, the complete conductor E_A , and its order $0 \leq v \leq 80$ at zero. Use the proved original period domain throughout. Write

$$\beta_* = \frac{1}{2} - \delta - i\gamma, \quad Z = e^{2\delta z}, \quad W = e^{2i\gamma z}, \quad t = (ZW, Z, W, 1)^\top, \quad x = Ht, \quad H = F^{-1} \Pi U E_{\text{prim}}. \quad (\text{NBG1})$$

Thus $\delta, \gamma > 0$, H is the original invertible matrix, and none of its coefficients is replaced. The original degree- j grid is $\beta_{j,a,b} = j\beta_* + 2a\delta + 2bi\gamma$, $0 \leq a, b \leq j$. These nodes are distinct: equality first forces equality of their real grid indices and then of their imaginary indices. Consequently the corresponding finite exponential functions are linearly independent, by their ordinary Vandermonde matrix of derivatives at zero.

Let $k \geq 9$, $k \equiv 1 \pmod{4}$. The degree change below is $k \mapsto k + 4$. Set

$$q_j = (j+1)^2, \quad q'_j = (j-7)^2, \quad c'_j = j/2 - 4, \quad (\text{NBG2})$$

$$(s_0, s_1) = (1, 1) \text{ or } (k, k+4), \quad D \geq q_{k+4} - v - 1.$$

The degree D is identical in the two polynomial spaces. The original target endpoints are $D = N - v$ with $N = q_{k+4} - 1, q_{k+4}, 2q_{k+4} - 1, 2q_{k+4}$. They are not replaced by the source degree's endpoints.

110.1 The original finite coefficients of a left inverse

Let (l_1, l_2, l_3, l_4) be the fourth row of H^{-1} . Then $\sum_i l_i x_i = 1$ after the exact substitution NBG1. For a multiindex $\nu \in \mathbb{N}^4$ with $|\nu| = 13$, let $i(\nu)$ be its first index with $\nu_i \geq 4$. Such an index exists, since otherwise $|\nu| \leq 12$. Define the actual homogeneous degree-nine polynomials

$$U_i(x) = \sum_{\substack{|\nu|=13 \\ i(\nu)=i}} \frac{13!}{\nu_1! \nu_2! \nu_3! \nu_4!} \prod_{j=1}^4 l_j^{\nu_j} x^{\nu-4e_i}. \quad \sum_{i=1}^4 U_i(x) x_i^4 = (l_1 x_1 + \cdots + l_4 x_4)^{13}. \quad (\text{NBG3})$$

Partitioning the complete multinomial expansion by $i(\nu)$ proves the identity, including zero coefficients of the actual row l . No division by a coordinate x_i occurs.

Put $L_i(Z, W) = x_i = H_{i1}ZW + H_{i2}Z + H_{i3}W + H_{i4}$. Retain the full arrays

$$\begin{aligned} b_{i,a,b} &= [Z^a W^b] L_i^4 \quad (0 \leq a, b \leq 4), & u_{i,a,b} &= [Z^a W^b] U_i(Ht) \quad (0 \leq a, b \leq 9), \\ B_i(z) &= e^{4\beta_* z} L_i(Z, W)^4, & D_i(z) &= e^{-4\beta_* z} U_i(Ht), & \sum_i D_i(z) B_i(z) &= 1. \end{aligned} \quad (\text{NBG4})$$

These are entire finite exponential sums. The last equality is NBG3 after substitution; in particular the four B_i have no common complex zero. Explicitly their exponents are

$$\begin{aligned} B_i(z) &= \sum_{a,b=0}^4 b_{i,a,b} e^{\beta_{4,a,b} z}, \\ D_i(z) &= \sum_{a,b=0}^9 u_{i,a,b} e^{\xi_{a,b} z}, \quad \xi_{a,b} = -4\beta_* + 2a\delta + 2bi\gamma = -2 + (2a+4)\delta + i(2b+4)\gamma. \end{aligned} \quad (\text{NBG5})$$

The second expression is also $e^{-13\beta_* z} \mathcal{N}_{U_i}(z)$, since U_i has degree nine. Thus the negative degree-four phase is retained even though its two polynomial index ranges have length ten.

For explicit coefficient control set $h_H = \max_i \sum_j |H_{ij}|$ and $l_H = \sum_i |l_i|$. Then

$$\sum_{a,b} |b_{i,a,b}| \leq \left(\sum_j |H_{ij}| \right)^4, \quad \sum_{i,a,b} |u_{i,a,b}| \leq l_H^{13} h_H^9. \quad (\text{NBG6})$$

The coefficient ℓ^1 norm of a product of biforms is at most the product of their coefficient ℓ^1 norms, by the complete convolution sum and the triangle inequality. Apply this first to each monomial in NBG3 and then sum the partitioned multinomial coefficients. This proves both inequalities with the actual row l retained.

110.2 Both native source norms and the exact finite matrices

For order s keep the complete original measure and frame

$$\begin{aligned} dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_{n,s,c}(S') = p_{n,s}((S' - c)/i) / \sqrt{h_{n,s}}. \end{aligned} \quad (\text{NBG7})$$

The original $p_{n,s}$ are monic with $yp_{n,s} = p_{n+1,s} + n(n-1+s/2)p_{n-1,s}$. All norms below use these complete masses and the original centres c'_k, c'_{k+4} .

For $H_D = \sum_{n=1}^D 1/n$, $H_0 = 0$, the original derivative calculation gives $\|\partial_y\|_{\leq D,s} \leq H_D + 2$ and therefore $\|p(y + \eta)\|_s \leq e^{|\eta|(H_D+2)} \|p\|_s$. For completeness, the differentiated original generating series in the orthonormal basis has entries $(-1)^{(a-1)/2} a^{-1} \sqrt{w_n/w_{n-a}}$ for odd $a > 0$, where $w_n = n!/(s/2)_n$. For $s/2 \geq 1/2$, $w_n/w_m \leq \sqrt{n/m}$ when $n > m \geq 1$, and $w_n \leq 2\sqrt{n}$. With weights

$w_n^{-1/2}$ the absolute matrix has weighted row sum at most H_D and column sum at most $H_D + 4$: the excess for $1 \leq m < n$ is bounded by $1/\sqrt{nm}$, with sum at most two, and the $m = 0$ term is at most two. Weighted Cauchy–Schwarz proves norm at most $\sqrt{H_D(H_D + 4)}$. The exact finite exponential series for $\exp(\eta\partial_y)$ proves the translation bound. This also includes the zero derivative at $D = 0$.

The single original source-order increase has both bounds

$$\begin{aligned} \frac{dm_{s+4}}{dm_s}(y) &= \frac{(2\pi)^2}{(s/2)(s/2 + 1)}(y^2 + (s/2)^2), \\ a_s &= (2\pi)^2 \frac{s}{s+2}, \quad u_{s,D} = \frac{(2\pi)^2(4(D+1)(D+1+s/2) + (s/2)^2)}{(s/2)(s/2 + 1)}, \\ a_s \|p\|_s^2 &\leq \|p\|_{s+4}^2 \leq u_{s,D} \|p\|_s^2 \quad (\deg p \leq D). \end{aligned} \tag{NBG8}$$

Indeed the Gamma recurrence gives the factor $|\Gamma(s/4 + 1 + iy/2)/\Gamma(s/4 + iy/2)|^2 = ((s/2)^2 + y^2)/4$. The density's power of two supplies four, the mass ratio is $(2\pi)^2$, and its remaining denominator ratio is $(s/2)(s/2 + 1)$. This proves the displayed density exactly. The minimum is its value at $y = 0$. For the upper bound, the monic recurrence in NBG7 has orthonormal off-diagonal entries $\sqrt{(n+1)(s/2+n)}$. Multiplication by y from degree at most D to degree at most $D+1$ is the sum of the two shifts; its norm is at most $2\sqrt{(D+1)(D+1+s/2)}$. Integrate the exact density ratio and use this bound. This proves the upper inequality without an unbounded pointwise density comparison.

Define the primal finite translation sums

$$\begin{aligned} (\mathcal{A}_i p)(S') &= \sum_{a,b=0}^4 b_{i,a,b} p(S' + \beta_{4,a,b}), \\ (\mathcal{C}_i p)(S') &= \sum_{a,b=0}^9 u_{i,a,b} p(S' + \xi_{a,b}). \end{aligned} \tag{NBG9}$$

The first map takes the target polynomial space to the source polynomial space, and the second takes the source to the target, both through degree D . For an analytic germ F let $\Lambda_F(S'^n) = F^{(n)}(0)$. Leibniz' rule on each monomial proves the exact identities

$$\Lambda_{B_i} F = \Lambda_F \circ \mathcal{A}_i, \quad \Lambda_{D_i} F = \Lambda_F \circ \mathcal{C}_i, \quad \sum_i \mathcal{A}_i \mathcal{C}_i = I. \tag{NBG10}$$

For the last identity either multiply the finite translation symbols in NBG4, or apply the preceding two identities to every monomial and every finite functional. Both calculations retain all binomial coefficients and ordinary derivatives.

The centre changes are exactly

$$\begin{aligned} (\mathcal{A}_i p)(c'_k + iy) &= \sum b_{i,a,b} p_1(y + (\beta_{4,a,b} - 2)/i), \\ (\mathcal{C}_i p)(c'_{k+4} + iy) &= \sum u_{i,a,b} p_0(y + (\xi_{a,b} + 2)/i), \end{aligned} \tag{NBG11}$$

where $p_0(y) = p(c'_k + iy)$ and $p_1(y) = p(c'_{k+4} + iy)$ in their respective lines. In particular the two phases are $1/i = -i$, and $\xi_{a,b} + 2 = (2a + 4)\delta + i(2b + 4)\gamma$. For $\sigma = (\delta^2 + \gamma^2)^{1/2}$ their moduli are at most 4σ and 22σ respectively.

Retain the complete constants

$$\begin{aligned} K_i^+ &= \sum_{a,b} |b_{i,a,b}| e^{|\beta_{4,a,b} - 2|(H_D + 2)}, \quad K_i^- = \sum_{a,b} |u_{i,a,b}| e^{|\xi_{a,b} + 2|(H_D + 2)}, \\ \kappa_i^+ &= K_i^+ \begin{cases} 1 & (s_0, s_1) = (1, 1), \\ a_k^{-1/2} & (s_0, s_1) = (k, k + 4), \end{cases} \\ \kappa_i^- &= K_i^- \begin{cases} 1 & (s_0, s_1) = (1, 1), \\ u_{k,D}^{1/2} & (s_0, s_1) = (k, k + 4), \end{cases} \quad K_{\pm} = \left(\sum_i (\kappa_i^{\pm})^2 \right)^{1/2}. \end{aligned} \tag{NBG12}$$

Translation in order s_0 and the lower side of NBG8 give $\|\mathcal{A}_i\| \leq \kappa_i^+$. Translation in order s_0 followed by the upper side of NBG8 gives $\|\mathcal{C}_i\| \leq \kappa_i^-$. The source and target directions in these assertions are those in NBG9. All constants are finite at the fixed actual data, even if some of the U_i vanish. NBG6 and $H_D \leq 1 + \log D$ show

$$\log \max(1, K_+, K_-) = O_{\text{actual}}(\log(D+1)) \quad (k \geq 9, D \geq q_{k+4} - v - 1). \quad (\text{NBG13})$$

For detail a_k is bounded below by $(2\pi)^2 9/11$, and $u_{k,D}$ is bounded by a fixed constant times $(D+1)^2 + 1$ using $k \geq 9$. All remaining coefficients in NBG12 are fixed and their translation radii are bounded as in NBG11. Thus this order statement follows from displayed finite constants.

There is an explicit original orthonormal matrix for every map just used. Let $\mathcal{B}_{s,c}$ be the monomial-to-frame matrix defined by $S'^n = \sum_{r=0}^D (\mathcal{B}_{s,c})_{rn} \phi_{r,s,c}$. Its diagonal is $i^n \sqrt{h_{n,s}}$. If $R(\eta)_{r,n} = \binom{n}{r} \eta^{n-r}$ for $r \leq n$, and zero otherwise, put

$$\begin{aligned} A_i &= \mathcal{B}_{s_0, c'_k} \left(\sum b_{i,a,b} R(\beta_{4,a,b}) \right) \mathcal{B}_{s_1, c'_{k+4}}^{-1}, & T_i &= A_i^\top, \\ C_i &= \mathcal{B}_{s_1, c'_{k+4}} \left(\sum u_{i,a,b} R(\xi_{a,b}) \right) \mathcal{B}_{s_0, c'_k}^{-1}, & S_i &= C_i^\top, \\ T &= \begin{bmatrix} T_1 \\ T_2 \\ T_3 \\ T_4 \end{bmatrix}, & S &= [S_1 \ S_2 \ S_3 \ S_4], & ST &= I, \\ & & & & K_-^{-1} \|w\|_2 &\leq \|Tw\|_2 \leq K_+ \|w\|_2. \end{aligned} \quad (\text{NBG14})$$

A dual coordinate is the column $w_n = \Lambda(\phi_{n,s,c})$, whose squared norm is $\sum_n |w_n|^2$. It is complex-linear in Λ . Consequently the dual matrices are the ordinary transposes displayed, not conjugate transposes. Their Hilbert norms equal those of their primal matrices. Cauchy–Schwarz on the four blocks gives $\|T\| \leq K_+$ and $\|S\| \leq K_-$. NBG10 gives $ST = I$, proving the lower bound and $K_- > 0$. This preserves the full masses and all monomial factorials rather than assigning an auxiliary coefficient norm.

110.3 The exact gauges before and after their compatibility condition

Let $\mathcal{E}_j$ be the original degree- j exponential space spanned by $e^{\beta_{j,a,b}z}$. The old and new lower spaces are $\mathcal{E}_{k-8}$ and $\mathcal{E}_{k-4}$. Define their native evaluation matrices and the enlarged one by

$$\begin{aligned} (V_0)_{n,a,b} &= \phi_{n,s_0, c'_k}(\beta_{k-8,a,b}), & (V_1)_{n,a,b} &= \phi_{n,s_1, c'_{k+4}}(\beta_{k-4,a,b}), \\ \tilde{\beta}_{a,b} &= (k-8)\beta_* + 2a\delta + 2bi\gamma, & (V_+)_{n,a,b} &= \phi_{n,s_0, c'_k}(\tilde{\beta}_{a,b}), \quad 0 \leq a, b \leq k+5. \end{aligned} \quad (\text{NBG15})$$

Here $0 \leq n \leq D$ and each unmarked lower index runs over its complete original grid. For a coefficient array e , let $P_i e$ be its convolution with the full $b_{i,a,b}$; it maps the grid $0, \dots, k-8$ into $0, \dots, k-4$. Let $Q_i e$ be convolution with the full $u_{i,a,b}$; it maps $0, \dots, k-4$ into $0, \dots, k+5$. Every out-of-range input coefficient is zero. Then

$$T_i V_0 = V_1 P_i, \quad S_i V_1 = V_+ Q_i. \quad (\text{NBG16})$$

For example, the first equality evaluates $\mathcal{A}_i \phi_{n,s_1, c'_{k+4}}$ at an old lower root; the root-addition identity puts each term at the indicated new lower root. The second uses the same evaluation proof with $\xi_{a,b}$ and gives exactly $\tilde{\beta}$. This proves the full coefficient maps, including the negative-degree phase of the second map.

For arbitrary new lower functions $E_i \in \mathcal{E}_{k-4}$ their inverse sum belongs to the explicitly larger grid:

$$E_+ = \sum_i D_i E_i \in \tilde{\mathcal{E}}_{k-8} := e^{(k-8)\beta_* z} \mathbb{C}[Z, W]_{\deg_Z, \deg_W \leq k+5}. \quad (\text{NBG17})$$

It need not lie in the old grid. The coefficient vector is $\sum_i Q_i e_i$, its full finite Gram is $V_+^* V_+$, and its rank is $\min\{(k+6)^2, D+1\}$ by the Vandermonde argument and the invertible monomial-to-frame matrix. The original lower dimension is $(k-7)^2$, so the untruncated excess dimension is $26k-13$. At either target low endpoint the enlarged evaluation space fills the source dual space: indeed $D+1 \leq (k+5)^2 + 1 - v < (k+6)^2$. Thus its quotient is zero there. None of this strip has been treated as a bounded-rank correction.

The exact compatibility equations, including their finite window, are

$$\begin{aligned} \mathcal{R}(E)_{ij} &= B_i E_j - B_j E_i \in \mathcal{E}_k \quad (i < j), \\ 0 \longrightarrow \mathcal{E}_{k-8} &\xrightarrow{E \mapsto (B_i E)_i} \mathcal{E}_{k-4}^4 \xrightarrow{\mathcal{R}} \mathcal{E}_k^6 \quad \text{is exact at the first two nonzero terms,} \quad (\text{NBG18}) \\ \mathcal{R}(E)^{(n)}(0) &= 0 \quad (0 \leq n \leq D, i < j) \iff \mathcal{R}(E) = 0. \end{aligned}$$

The last equivalence follows because $D - (q_k - 1) \geq 8k + 24 - v > 0$ on NBG2, and every component is a sum over the q_k distinct original degree- k roots. If its first q_k derivatives vanish, its full coefficient vector vanishes by the original Vandermonde.

For exactness in the middle, assume $\mathcal{R}(E) = 0$ and form the specific $E_+ = \sum_i D_i E_i$. NBG4 gives

$$E_j - B_j E_+ = \sum_i D_i (B_i E_j - B_j E_i) = 0. \quad (\text{NBG19})$$

Write $E_i = e^{(k-4)\beta_* z} f_i(Z, W)$ and $E_+ = e^{(k-8)\beta_* z} f_+(Z, W)$. Initially $\deg_Z f_+, \deg_W f_+ \leq k+5$, by NBG17. Equation NBG19 and independence of the finite grid imply the polynomial identities $L_j^4 f_+ = f_j$. At least one L_j has Z degree one, because otherwise the first two columns of the invertible matrix $\mathbf{H}$ would both vanish. At least one has W degree one by the same argument for its first and third columns. If $f_+ \neq 0$, degrees in each variable add under multiplication over the integral domains $\mathbb{C}[W]$ and $\mathbb{C}[Z]$. Therefore these two identities give $\deg_Z f_+, \deg_W f_+ \leq k-8$. If $f_+ = 0$ the conclusion holds directly. Thus $E_+ \in \mathcal{E}_{k-8}$ and every $E_i = B_i E_+$. Conversely such a tuple satisfies every compatibility equation. The first map is injective by $\sum_i D_i B_i = 1$. This proves all assertions of NBG18 and proves the cancellation of the full enlarged strip for precisely the compatible tuples.

There is also an exact homotopy for arbitrary tuples:

$$E_j - B_j \sum_i D_i E_i = \sum_i D_i \mathcal{R}(E)_{ij}, \quad \mathcal{R}(E)_{ij} = B_i E_j - B_j E_i \quad \text{for all ordered } i, j. \quad (\text{NBG20})$$

Here $\mathcal{R}(E)_{ii} = 0$ and reversing the indices reverses its sign. NBG4 proves the identity term by term. It retains all incompatible gauge coefficients; no projection onto the old grid has replaced its left side.

110.4 The two attained native minima and their exact correction

In the Euclidean dual coordinates NBG14 define $\mathcal{L}_0 = \text{im } V_0$ and $\mathcal{L}_1 = \text{im } \text{diag}(V_1, V_1, V_1, V_1)$. Then

$$T\mathcal{L}_0 \subset \mathcal{L}_1, \quad T\mathbb{C}^{D+1} \cap \mathcal{L}_1 = T\mathcal{L}_0. \quad (\text{NBG21})$$

For the reverse inclusion choose w with $Tw = (\Lambda_E)_i$ in the target frames. Truncated multiplication by B_i commutes with that by B_j , since both are multiplication in $\mathbb{C}[z]/(z^{D+1})$. Hence all first $D+1$ derivatives of $B_i E_j - B_j E_i$ vanish. NBG18 gives a full old lower function E with $E_i = B_i E$. Then $T(w - \Lambda_E) = 0$, and $ST = I$ gives $w = \Lambda_E$. This proof is valid for every truncated functional, not only actual numerator functionals.

For a source class $[w] \in \mathbb{C}^{D+1}/\mathcal{L}_0$, write the three squared norms

$$\begin{aligned} n_0([w])^2 &= \min_{\eta} \|w + V_0 \eta\|_2^2, \\ n_{\text{cor}}([w])^2 &= \min_{\eta} \|T(w + V_0 \eta)\|_2^2, \\ n_{\text{ind}}([w])^2 &= \min_{\eta_1, \dots, \eta_4} \sum_i \|T_i w + V_1 \eta_i\|_2^2. \end{aligned} \quad (\text{NBG22})$$

All minima are attained in finite-dimensional Hilbert spaces. The first has exactly the original lower-evaluation quotient. The second uses the exact correlated gauges $(B_i E)_i$, and the third uses all four original target lower gauges independently. They obey

$$K_-^{-2} n_0^2 \leq n_{\text{cor}}^2 \leq K_+^2 n_0^2, \quad 0 < n_{\text{ind}}^2 \leq n_{\text{cor}}^2 \quad ([w] \neq 0). \quad (\text{NBG23})$$

Apply NBG14 to each representative and take its minimum for the first pair of bounds. The independently allowed gauges include all correlated ones, which proves the last upper bound. The independent norm can vanish only when $Tw \in \mathcal{L}_1$, and NBG21 proves that then $w \in \mathcal{L}_0$. This proves strict positivity, not an uncalculated quantitative lower bound.

Here are the complete matrices of the correction. Let W_0 be any isometric frame of $\mathcal{L}_0^\perp$. All projectors are their ordinary Euclidean orthogonal projectors in the full original orthonormal frames. Put $P_{\text{cor}} = P_{T\mathcal{L}_0}$, $P_{\text{ind}} = P_{\mathcal{L}_1}$, and set

$$\begin{aligned} G_{\text{cor}} &= W_0^* T^* (I - P_{\text{cor}}) T W_0, & G_{\text{ind}} &= W_0^* T^* (I - P_{\text{ind}}) T W_0, \\ K_-^{-2} I &\preceq G_{\text{cor}} \preceq K_+^2 I, & 0 &\prec G_{\text{ind}} \preceq G_{\text{cor}}, \\ \Omega_D &= G_{\text{cor}}^{-1/2} G_{\text{ind}} G_{\text{cor}}^{-1/2} = I - G_{\text{cor}}^{-1/2} W_0^* T^* (P_{\text{ind}} - P_{\text{cor}}) T W_0 G_{\text{cor}}^{-1/2}. \end{aligned} \quad (\text{NBG24})$$

Completing the square in NBG22 gives the first line. Since $T\mathcal{L}_0 \subset \mathcal{L}_1$, $P_{\text{ind}} - P_{\text{cor}}$ is the orthogonal projector onto $\mathcal{L}_1 \cap (T\mathcal{L}_0)^\perp$. Conjugating the difference of the two Grams proves the last identity and $0 \prec \Omega_D \preceq I$. In particular every original mass, centre and gauge coefficient occurs explicitly through NBG14–16. Its determinant correction is exactly

$$\log \det G_{\text{ind}} = \log \det G_{\text{cor}} + \log \det \Omega_D, \quad -\log \det \Omega_D = \sum_j -\log(1 - \rho_j^2) \geq 0, \quad (\text{NBG25})$$

where the ρ_j are the singular values of $(P_{\text{ind}} - P_{\text{cor}}) T W_0 G_{\text{cor}}^{-1/2}$. They are strictly less than one by the proved positivity. The equality follows from the matrix identity, including zero singular values. Thus the remaining denominator is a specified finite native gauge correction.

110.5 Restriction to the original conductor and arithmetic maps

Let $\mathcal{N}_{v,k} = \{\mathcal{N} \in \mathcal{E}_k : \mathcal{N}^{(a)}(0) = 0, \ 0 \leq a < v\}$, and retain

$$\mathcal{C}_{v,k} = \mathcal{N}_{v,k} / E_A \mathcal{E}_{k-8}, \quad F_{\mathcal{N}} = \mathcal{N} / E_A, \quad \alpha_A = v! / \mu_v. \quad (\text{NBG26})$$

The germ $F_{\mathcal{N}}$ is analytic at zero; addition of $E_A E$ changes it by precisely the original lower function E . The source class $[\Lambda_{F_{\mathcal{N}}}]$ therefore lies in the quotient NBG22. The first norm there is the complete original DNE minimum extension norm. Its map from $\mathcal{C}_{v,k}$ is injective at the present cutoff: if $F_{\mathcal{N}} + E$ has zero derivatives through D , then $\mathcal{N} + E_A E$ has zero derivatives through $D+v$; it lies in $\mathcal{E}_k$, and $D+v \geq q_k - 1$ proves its vanishing by the original Vandermonde. Multiplication by each B_i takes this quotient into $\mathcal{C}_{v,k+4}$, with exact lower-gauge map NBG16.

Choose the original $Z_{\text{exp},k}$ representatives and let $G_{0,D} = D_{k,D}^{(s_0)}$ be their original native Gram. Denote by $G_{\text{cor},D}^{\text{act}}$ and $G_{\text{ind},D}^{\text{act}}$ the pullbacks of the second and third norms of NBG22 along exactly this same frame. Then, with the original DNE multiplication matrices $\mathcal{M}_{i,k}$,

$$\begin{aligned} \mathcal{M}_{i,k} &= (\alpha_A N_{k+4} Z_{\text{exp},k+4})^{-1} m_{i,k} (\alpha_A N_k Z_{\text{exp},k}), \\ G_{\text{ind},D}^{\text{act}} &= \sum_i \mathcal{M}_{i,k}^* D_{k+4,D}^{(s_1)} \mathcal{M}_{i,k}, \\ K_-^{-2} G_{0,D} &\preceq G_{\text{cor},D}^{\text{act}} \preceq K_+^2 G_{0,D}, \quad 0 \prec G_{\text{ind},D}^{\text{act}} \preceq G_{\text{cor},D}^{\text{act}}. \end{aligned} \quad (\text{NBG27})$$

The matrix $m_{i,k}$ is the original conductor multiplication by x_i^4 . The bijections in the first line are the exact KRC/DFX maps, so α_A cancels only by this displayed conjugation. The second line

is the sum of the four attained target minima, and NBG23 proves the inequalities. The injective tuple here is on the full conductor quotient. No invariant target quotient has yet been performed.

For any fixed original restriction, quotient, or restriction followed by quotient of this source space, transport both $G_{0,D}$ and $G_{\text{cor},D}^{\text{act}}$ along the same algebraic maps. If its rank is t , their squared relative Jacobian $\mathfrak{J}_{\text{cor},D}$ satisfies

$$-2t \log \max(1, K_-) \leq \log \mathfrak{J}_{\text{cor},D} \leq 2t \log \max(1, K_+). \quad (\text{NBG28})$$

Restriction preserves form inequalities. For an onto quotient, take the infimum over the same affine fibre on each side; both inequalities persist. Conjugate the resulting positive forms and multiply the t eigenvalue bounds. This proves NBG28 for every stated flag, including rank zero. Here $t \leq \dim \mathcal{C}_{v,k} = 16k - 48 - v = O(k)$. At the four literal target endpoints, NBG13 and NBG28 give a complete signed determinant error $O_{\text{actual}}(k \log q_{k+4}) = o(kq_{k+4})$.

For the actual independent norm on a fixed restriction X , write the corresponding positive relative matrix $\Omega_{X,D} = (G_{\text{cor},X,D})^{-1/2} G_{\text{ind},X,D} (G_{\text{cor},X,D})^{-1/2}$. Let ε_N equal -1 at the two low endpoints and $+1$ at the two high endpoints. Then exactly

$$\begin{aligned} \sum_N \varepsilon_N \log \det G_{\text{ind},X,N-v} &= \sum_N \varepsilon_N \log \det G_{0,X,N-v} \\ &\quad + \sum_N \varepsilon_N \log \det \Omega_{X,N-v} + O_{\text{actual}}(k \log q_{k+4}). \end{aligned} \quad (\text{NBG29})$$

The finite absolute error is at most the sum of the four NBG28 upper absolute bounds. Taking determinants in the definition of each $\Omega_{X,D}$ and then using NBG28 proves this assertion. Every $\log \det \Omega_{X,D} \leq 0$, but its four signed sum has not been assigned a sign or an asymptotic value.

Finally the original invariant source and kernel target are connected by their full maps, not by declaring the preceding tuple an isometry after a further quotient. Let $\mathbf{J}_k = A_{\ell,k}^{-1} P_{Y,k}^T$ and $\mathbf{B}_{k+4} = Z_{K,k+4}^T A_{\ell,k+4}$ be exactly the original DNE/FGC matrices. Then

$$\begin{aligned} \mathcal{X}_k &= \begin{bmatrix} \mathbf{B}_{k+4} \mathcal{M}_{1,k} \mathbf{J}_k \\ \mathbf{B}_{k+4} \mathcal{M}_{2,k} \mathbf{J}_k \\ \mathbf{B}_{k+4} \mathcal{M}_{3,k} \mathbf{J}_k \\ \mathbf{B}_{k+4} \mathcal{M}_{4,k} \mathbf{J}_k \end{bmatrix}, \quad \ker \mathcal{X}_k = \{a : \mathcal{M}_{i,k} \mathbf{J}_k a \in \text{im } \mathbf{J}_{k+4} \text{ for every } i\}, \\ G_{J,D} &= \mathbf{J}_k^* G_{0,D} \mathbf{J}_k, \quad H_{B,D} = (\mathbf{B}_{k+4} (D_{k+4,D}^{(s_1)})^{-1} \mathbf{B}_{k+4}^*)^{-1}. \end{aligned} \quad (\text{NBG30})$$

The kernel equality uses the original identity $\ker \mathbf{B}_{k+4} = \text{im } \mathbf{J}_{k+4}$. It is also the exact DFX31–35 invariant-target kernel, including its retained shared-component subspace. Let $U_{\mathcal{X}}$ be a full-column frame of $\text{im } \mathcal{X}_k$, let $L_{\mathcal{X}} U_{\mathcal{X}} = I$, and put $\tilde{\mathcal{X}} = L_{\mathcal{X}} \mathcal{X}_k$. The two actual image metrics are

$$\begin{aligned} Q_{\mathcal{X},D} &= (\tilde{\mathcal{X}} G_{J,D}^{-1} \tilde{\mathcal{X}}^*)^{-1}, \\ H_{\mathcal{X},D} &= U_{\mathcal{X}}^* \text{diag}(H_{B,D}, H_{B,D}, H_{B,D}, H_{B,D}) U_{\mathcal{X}}, \quad H_{\mathcal{X},D} \preceq K_+^2 Q_{\mathcal{X},D}. \end{aligned} \quad (\text{NBG31})$$

Each target quotient is an attained minimum and can only decrease the target component norm in NBG27. Apply its upper bound to a source representative and then to the attained minimum source lift along the onto map $\tilde{\mathcal{X}}$ to prove NBG31. Its kernel was specified before taking this quotient; a lost direction was not discarded. In rank zero these are empty matrices with determinant one. In positive rank the exact remaining relative determinant is

$$\log \det H_{\mathcal{X},D} = \log \det Q_{\mathcal{X},D} + \log \det \left(Q_{\mathcal{X},D}^{-1/2} H_{\mathcal{X},D} Q_{\mathcal{X},D}^{-1/2} \right). \quad (\text{NBG32})$$

Thus NBG24–29 control the complete correlated-gauge inverse and retain the independent gauge correction. NBG30–32 then perform the original further invariant quotient with its exact kernel, image, and native denominator. The enlarged strip, independent lower minimizations, displaced source endpoints and further invariant quotient are all present in these typed maps and determinants.

Complete companion and invariant-target continuation. The original four native multipliers now extend to every signed companion map GKC3–4 and the complete globally exact coefficient sequence NCR5. Its original-frame torsion is proved in GKC14–25. At high cutoffs its complete five native Grams satisfy NCR6 with all source masses retained. For the proper source and its actual kernel, NCR7–10 keeps the full complementary projection in NKC. For the further invariant target the exact original receiver is

$$\log \det \mathcal{R}_{X,D} = \log \frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} + \log \omega_D + \epsilon_D, \quad -a_D \leq \epsilon_D \leq b_D. \quad (\text{NBG33})$$

The maps, positive quotient metrics and actual image rank in a_D, b_D are proved in NFR1–2. The complete arithmetic rows and simultaneous spectral budget NFR3–5 update its long return while retaining both positive row sums. The conductor quotient has the separate CNC torsion factor; the coefficient torsion-one theorem is not substituted into that quotient.

111 The original compatibility operator and its native determinant

Retain the complete NBG1–32 data, constants, matrices and native frames. This continuation uses the same fixed actual quartet, period, conductor, $k \geq 9$, $k \equiv 1 \pmod{4}$, and common cutoff $D \geq q_{k+4} - v - 1$. In particular every original lower coefficient and every source mass remains present.

111.1 The complete finite compatibility homotopy

Let $\mathcal{H}_j$ be the complex-linear dual of polynomials through degree D with centre c'_{k+4j} and source order $s_j = 1$ for all j , or $s_j = k + 4j$ for all j , where $j = 0, 1, 2$. All use the complete NBG7 measures. The space $\mathcal{H}_2$ is the literal extra degree-four comparison space forced by the two successive multiplications. It is not assigned a new normalization or substituted for either original source space. Write

$$\begin{aligned} T_i^{(j)} : \mathcal{H}_j &\longrightarrow \mathcal{H}_{j+1}, & F &\longmapsto B_i F, \\ S_i^{(j)} : \mathcal{H}_{j+1} &\longrightarrow \mathcal{H}_j, & F &\longmapsto D_i F \quad (j = 0, 1), \\ T &= (T_i^{(0)})_i, & S &= [S_1^{(0)} \ S_2^{(0)} \ S_3^{(0)} \ S_4^{(0)}], & ST &= I. \end{aligned} \quad (\text{NKC1})$$

A function here means its truncated Taylor functional, as in NBG14. Each map is the same finite translation matrix NBG14 with the specified two centres and complete source orders. NBG9–14 proves its definition on every finite degree D . Its matrix proof and norm estimates do not require a numerator quotient at the comparison degree $k + 8$.

Define the six oriented compatibility rows and their explicit homotopy by

$$\begin{aligned} \mathcal{R} : \mathcal{H}_1^4 &\longrightarrow \mathcal{H}_2^6, & (\mathcal{R}y)_{ij} &= T_i^{(1)} y_j - T_j^{(1)} y_i \quad (i < j), \\ \mathcal{Q} : \mathcal{H}_2^6 &\longrightarrow \mathcal{H}_1^4, & (\mathcal{Q}z)_j &= \sum_{i \neq j} S_i^{(1)} z_{ij}, & z_{ji} &= -z_{ij}, & z_{ii} &= 0, \\ \mathcal{R}T &= 0, & \mathcal{Q}\mathcal{R} &= I - TS, & \ker \mathcal{R} &= \text{im } T. \end{aligned} \quad (\text{NKC2})$$

To prove the identities, compute on truncated analytic germs before expressing them in the original frames. The first is $B_i B_j = B_j B_i$. In component j the second is $\sum_i D_i (B_i y_j - B_j y_i) = y_j - B_j \sum_i D_i y_i$, by the literal identity $\sum_i D_i B_i = 1$. Multiplication of truncated power series is associative and commutative and all factors are entire at zero, so these are exact finite identities. Changing to the frames NBG14 conjugates them by the appropriate monomial-to-frame matrices; their ordinary-transpose dual convention is unchanged. The second identity implies $y = TSy$ when

$\mathcal{R}y = 0$, and the first gives the converse. This proves the complete kernel without a spectral or conditioning hypothesis.

Use the NBG12 constants at the base degree $k + 4$ and the same cutoff D , and denote them by $K_{\pm}^{(1)}$. Explicitly their forward source-order factor is $a_{k+4}^{-1/2}$ and their inverse source-order factor is $u_{k+4,D}^{1/2}$ in the degree-order case; both are one in the order-one case. All their $b_{i,a,b}$, $u_{i,a,b}$ and shifted exponents are unchanged. Then

$$L_+ = \sqrt{2}K_+^{(1)}, \quad L_- = \sqrt{2}K_-^{(1)}, \quad \|\mathcal{R}\| \leq L_+, \quad \|\mathcal{Q}\| \leq L_-. \quad (\text{NKC3})$$

Indeed $\sum_{i < j} \|T_i^{(1)}y_j - T_j^{(1)}y_i\|^2$ is at most $2 \sum_j \sum_{i \neq j} (\kappa_i^{+, (1)})^2 \|y_j\|^2$. This proves the first norm bound. Cauchy–Schwarz in each component of the homotopy gives

$$\left\| \sum_{i \neq j} S_i^{(1)} z_{ij} \right\|^2 \leq (K_-^{(1)})^2 \sum_{i \neq j} \|z_{ij}\|^2.$$

Each pair occurs twice on summing j , proving the second.

Let P_T be the orthogonal projector onto $\text{im } T$ in the full native product space. The homotopy gives the two-sided original quotient bound

$$L_-^{-1} \|(I - P_T)y\| \leq \|\mathcal{R}y\| \leq L_+ \|(I - P_T)y\|. \quad (\text{NKC4})$$

For the upper bound subtract the attained projection $P_T y$, which is in the complete kernel NKC2. For the lower bound, $y - T S y = \mathcal{Q} \mathcal{R} y$ is one representative of $[y] \in \mathcal{H}_1^4 / \text{im } T$; its distance to that subspace is therefore at most $L_- \|\mathcal{R}y\|$. Both constants are positive because the maps have the displayed nonzero finite quotient. NBG6–13 also proves $\log \max(1, L_+, L_-) = O_{\text{actual}}(\log(D + 1))$. The additional order increase in NKC1 is included through the explicitly retained $a_{k+4}, u_{k+4,D}$.

111.2 Exact native gauge and coefficient spaces

In NBG15 notation set $\mathcal{V} = \text{diag}(V_1, V_1, V_1, V_1)$, $\mathcal{L} = \text{im } \mathcal{V}$ and $\mathcal{C} = T \text{im } V_0$. They satisfy

$$\begin{aligned} \mathcal{C} \subset \mathcal{L}, \quad \mathcal{L} \cap \text{im } T = \mathcal{C}, \quad n_0 = (k - 7)^2, \quad n_1 = (k - 3)^2, \\ r_{\text{comp},k} = 4n_1 - n_0 = 3k^2 - 10k - 13, \quad \dim(\mathcal{L} \cap \mathcal{C}^\perp) = r_{\text{comp},k}. \end{aligned} \quad (\text{NKC5})$$

The first two claims are exactly the finite-window NBG18–21 calculation. Each lower Vandermonde is full column rank since the present cutoff exceeds its number of roots minus one. Thus the dimensions follow with all original lower coefficients retained. The compatibility rank $r_{\text{comp},k}$ is distinct from the unchanged original Gamma density minimum a_{k+4} from NBG8, used in NKC3.

Let $\mathcal{P}$ be the stacked coefficient convolution matrix with blocks P_i from NBG16: $\mathcal{V} \mathcal{P} = T V_0$. It has full column rank n_0 , since both T and V_0 are injective. Select any fixed complement matrix Z_c such that $[\mathcal{P}, Z_c]$ is invertible; it has $r_{\text{comp},k}$ columns. Write

$$\begin{aligned} G_{\mathcal{L}} &= \mathcal{V}^* \mathcal{V}, \quad Z_{\perp} = Z_c - \mathcal{P}(\mathcal{P}^* G_{\mathcal{L}} \mathcal{P})^{-1} \mathcal{P}^* G_{\mathcal{L}} Z_c, \\ G_c &= Z_c^* G_{\mathcal{L}} Z_c - Z_c^* G_{\mathcal{L}} \mathcal{P}(\mathcal{P}^* G_{\mathcal{L}} \mathcal{P})^{-1} \mathcal{P}^* G_{\mathcal{L}} Z_c, \\ W_c &= \mathcal{V} Z_{\perp} G_c^{-1/2}, \quad W_c^* W_c = I, \quad \text{im } W_c = \mathcal{L} \cap \mathcal{C}^\perp. \end{aligned} \quad (\text{NKC6})$$

Completing the square in $\|\mathcal{V}(Z_c u + \mathcal{P} \eta)\|^2$ proves that its attained minimum is $u^* G_c u$, with representative $\mathcal{V} Z_{\perp} u$. The complement choice and the injectivity of $\mathcal{V}$ imply positive definiteness. This proves every asserted identity and type in NKC6. In particular G_c is the actual native lower-gauge quotient Gram, including the complete factor M_{s_1} inside V_1 .

The compatibility coefficient matrix is explicitly

$$(\mathcal{K}e)_{ij,A,B} = \sum_{r,t=0}^4 b_{i,r,t} e_{j,A-r,B-t} - \sum_{r,t=0}^4 b_{j,r,t} e_{i,A-r,B-t} \quad (i < j, \ 0 \leq A, B \leq k). \quad (\text{NKC7})$$

An input $e_{i,a,b}$ has $0 \leq a, b \leq k-4$ and is zero outside that original range. Let

$$(V_2)_{n,A,B} = \phi_{n,s_2,c'_{k+8}}(\beta_{k,A,B}), \quad \mathcal{V}_2 = \text{diag}(V_2, \dots, V_2) \quad \text{with six blocks.} \quad \mathcal{R}\mathcal{V} = \mathcal{V}_2\mathcal{K}. \quad (\text{NKC8})$$

Each term of the matrix equality evaluates an original degree-four translate at an original degree- $(k-4)$ lower root, whose sum is $\beta_{k,A,B}$. This proves it directly, including the sign. The centre $c'_{k+8} = k/2$ is the centre of exactly that degree- k grid. The complete comparison order is $s_2 = 1$ or $k+8$ as fixed in NKC1. At the present cutoff V_2 is full column rank: $D \geq q_k - 1$ is the proved NKG18 inequality. Therefore the coefficient kernel is exactly $\ker \mathcal{K} = \text{im } \mathcal{P}$ by NKC2 and NKC5. No support coefficient has been deleted.

111.3 The actual source restriction and its complementary projection

Let X be any specified algebraic subspace of $\mathcal{H}_0/\text{im } V_0$, of dimension t . Use its same source representatives at the endpoint in question, and let Y_X be a full-column representative matrix in $\mathcal{H}_0$. Let P_C project onto $\mathcal{C}$. The exact correlated target frame is

$$\begin{aligned} A_X &= (I - P_C)TY_X[Y_X^*T^*(I - P_C)TY_X]^{-1/2}, \\ P_X &= A_X A_X^*, \quad P_{\text{rem}} = P_T - P_C - P_X. \end{aligned} \quad (\text{NKC9})$$

Changing a source representative by $V_0\eta$ changes TY_X by a column of $\mathcal{C}$, so A_X has the stated well-defined image. The bracket is positive by NKG23. Consequently its columns are orthonormal. They lie in $\text{im } T \cap \mathcal{C}^\perp$, so P_{rem} is the orthogonal projector onto the remaining part of this exact target space. It is not a projection onto a conjectured invariant or uniformly distributed source component. When $t = 0$ take A_X empty and $P_X = 0$.

The original independent-gauge correction on this restriction is the positive matrix

$$\Omega_{X,D} = I_t - A_X^* W_c W_c^* A_X, \quad \Xi_{X,D} = I_{r_{\text{comp},k}} - W_c^* P_X W_c, \quad \det \Omega_{X,D} = \det \Xi_{X,D} > 0. \quad (\text{NKC10})$$

After removing the correlated gauges $\mathcal{C}$, the independent gauges remaining are precisely $\text{im } W_c$. Taking their attained minimum gives the first matrix, exactly the NKG24 restriction. The two determinant expressions agree by $\det(I - AB) = \det(I - BA)$; this identity follows, for rectangular A, B , by taking the two Schur determinants of $\begin{pmatrix} I & A \\ B & I \end{pmatrix}$ and replacing one factor by its negative as needed. Strict positivity follows from the exact intersection $\mathcal{L} \cap \text{im } T = \mathcal{C}$. For $t = 0$ both determinants are one.

The precise native augmented compatibility map is

$$\mathcal{R}_X : \mathcal{L} \cap \mathcal{C}^\perp \longrightarrow \mathcal{H}_2^6 \oplus \text{im } P_{\text{rem}}, \quad y \longmapsto (\mathcal{R}y, P_{\text{rem}}y). \quad (\text{NKC11})$$

For such y the orthogonal decomposition by $P_T, P_C, P_X, P_{\text{rem}}$ gives

$$\|(I - P_X)y\|^2 = \|(I - P_T)y\|^2 + \|P_{\text{rem}}y\|^2. \quad (\text{NKC12})$$

There is no P_C term because $y \perp \mathcal{C}$. Use NKC4 on the first term, and keep the second term with coefficient one. With

$$c_- = \min\{1, L_-^{-2}\}, \quad c_+ = \max\{1, L_+^2\}, \quad \Gamma_{X,D} = W_c^*(\mathcal{R}^*\mathcal{R} + P_{\text{rem}})W_c, \quad (\text{NKC13})$$

one obtains the complete native form comparison

$$0 \prec c_- \Xi_{X,D} \preceq \Gamma_{X,D} \preceq c_+ \Xi_{X,D}. \quad (\text{NKC14})$$

Indeed NKC4 bounds $\|\mathcal{R}y\|^2$ between $L_-^{-2}\|(I - P_T)y\|^2$ and $L_+^2\|(I - P_T)y\|^2$. Adding the identical $\|P_{\text{rem}}y\|^2$ and using NKC12 gives exactly the displayed bounds. This proves positivity and retains the actual source complement.

Taking determinants in NKC14 and using NKC10 gives

$$\begin{aligned} \log \det \Gamma_{X,D} - r_{\text{comp},k} \log c_+ &\leq \log \det \Omega_{X,D} \leq \log \det \Gamma_{X,D} - r_{\text{comp},k} \log c_-, \\ |\log \det \Omega_{X,D} - \log \det \Gamma_{X,D}| &\leq r_{\text{comp},k} \max\{-\log c_-, \log c_+\}. \end{aligned} \quad (\text{NKC15})$$

To justify the determinant comparison explicitly, conjugate NKC14 by $\Xi_{X,D}^{-1/2}$. Every one of its $r_{\text{comp},k}$ positive eigenvalues lies between c_- and c_+ . Multiplying and then taking logarithms proves both inequalities. Because $r_{\text{comp},k} = O(q_k)$ and $\log \max(1, L_+, L_-) = O_{\text{actual}}(\log(D+1))$, the final error is $O_{\text{actual}}(q_k \log q_k) = o(kq_k)$ whenever $D = O(q_k)$. This bound is uniform in the specified subspace X and retains its literal metric projection in $\Gamma_{X,D}$. No inverse conditioning of its frame has entered the constant.

The Gram on the right has a full original coefficient formula, useful without an orthonormal gauge choice:

$$\begin{aligned} \mathcal{H}_{X,D} &= Z_c^* \mathcal{K}^* \mathcal{V}_2^* \mathcal{V}_2 \mathcal{K} Z_c + Z_c^* \mathcal{V}^* P_{\text{rem}} \mathcal{V} Z_c, \\ \Gamma_{X,D} &= G_c^{-1/2} \mathcal{H}_{X,D} G_c^{-1/2}, \quad \log \det \Gamma_{X,D} = \log \det \mathcal{H}_{X,D} - \log \det G_c. \end{aligned} \quad (\text{NKC16})$$

NKC6 and NKC8 give the first summand when substituting W_c . The removed $\mathcal{P}$ term is killed because $\mathcal{K}\mathcal{P} = 0$. The second removed term is killed because $P_{\text{rem}}\mathcal{V}\mathcal{P} = 0$, as its image is $\mathcal{C}$. Thus the displayed conjugation holds exactly. All roots, Gamma masses, source centres, compatibility coefficients and the native lower-evaluation quotient occur explicitly in NKC6–9 and NKC16.

111.4 The four original endpoints

For an original conductor or invariant source subspace X_k , use its actual image under the finite DNE functional map at each endpoint. Its algebraic coordinates and original $Z_{\text{exp},k}$ or $\mathbf{J}_k$ frame remain fixed; the finite functional frame and the exact projector NKC9 vary with the endpoint as specified. At the literal target indices $N = q_{k+4} - 1, q_{k+4}, 2q_{k+4} - 1, 2q_{k+4}$, let $\varepsilon_N = -1, -1, +1, +1$ in that order. Combining NBG29 and NKC15–16 proves

$$\begin{aligned} \sum_N \varepsilon_N \log \det G_{\text{ind},X_k,N-v} &= \sum_N \varepsilon_N \log \det G_{0,X_k,N-v} \\ &+ \sum_N \varepsilon_N (\log \det \mathcal{H}_{X_k,N-v} - \log \det G_{c,N-v}) + O_{\text{actual}}(q_k \log q_k). \end{aligned} \quad (\text{NKC17})$$

The exact finite error bound is the sum of the four NBG28 absolute bounds and the four NKC15 absolute bounds, with their displayed constants at the corresponding $D = N - v$. The source endpoints here remain displaced from the degree- k standard ones by exactly MNT16–18 with $d_h = 4$; this displacement has not been assigned to the error.

For the full source quotient $X = \mathcal{H}_0 / \text{im } V_0$, the remaining projector is zero, so NKC16 reduces to the unaugmented compatibility Gram. For a proper actual source subspace it is the exact nonnegative second summand displayed in NKC16. Both cases are covered by the same proved bound. The original further kernel quotient still has the NBG30–32 matrices and complete kernel. It is not deleted by this compatibility calculation. Thus the remaining native gauge correction has been transported, with an explicit sub- kq error, to the complete original compatibility-image determinant and its actual source-complement Gram.

Proper-source companion and shared-update refinement. The complete original source-complement term in NKC16 is $H_{\text{rem},X}$ of NCR7. For each fixed nested original source pair $X \subset Y$ of dimension difference t ,

$$-\log \det \Omega_{Y/X} = \chi_X - \chi_Y + \nu_{Y/X}, \quad t \log c_- \leq \nu_{Y/X} \leq t \log c_+. \quad (\text{NKC18})$$

The two actual comparison matrices share the same positive rank-at-most- t update. Their full compressed inverse comparison and determinant proof are PCJ13–16 and NCR9. NCR10 gives every finite endpoint sign. At the literal high cutoffs the five-Gram calculation GGE19 applies through PCJ4–5; both low compatibility Grams remain unchanged.

112 The original grid, source norm and exact enlarged denominator

Retain the full fixed simple quartet, actual five-orbit period, amplitude, branch and conductor of KRC, DNE and MNT. Throughout, $k \equiv 1 \pmod{4}$, $k \geq 37$, $0 < \delta < 1/2$, $\gamma > 2$, $0 \leq v = \text{ord}_0 E_A \leq 80$, and $s = 1$ or k . The source degree is k and the degree-four target is $k + 4$. Put

$$\begin{aligned} q &= (k+1)^2, & q_\uparrow &= (k+5)^2, & n_0 &= (k-7)^2, & n_+ &= (k+6)^2, & b &= n_+ - n_0 = 26k - 13, \\ c' &= k/2 - 4, & b_* &= \frac{1}{2} - \delta - i\gamma, & \alpha_{ab} &= (k-8)b_* + 2a\delta + 2ib\gamma, \\ \mathcal{G}_0 &= \{(a, b) : 0 \leq a, b \leq k-8\}, & \mathcal{G}_+ &= \{(a, b) : 0 \leq a, b \leq k+5\}, & \mathcal{B} &= \mathcal{G}_+ \setminus \mathcal{G}_0. \end{aligned} \tag{EBG1}$$

The use of b as a cardinality on the first line and a node index in α_{ab} is local to those explicit formulas. Every subsequent node is indexed by $\xi \in \mathcal{B}$. Keep the original node order within each set; in a combined matrix put the original block before the boundary block. This records the full permutation of the enlarged grid. All nodes are distinct: equality of their real parts gives the first index, and equality of imaginary parts gives the second.

The native polynomial space is $\mathcal{P}_D = \mathbb{C}[S']_{\leq D}$ with exactly

$$\begin{aligned} \|p\|_{s,c'}^2 &= \int_{\mathbb{R}} |p(c' + iy)|^2 dm_s(y), \\ dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, & M_s &= (2\pi)^{s/2}, \\ \phi_{n,s,c'}(S') &= \frac{p_{n,s}((S' - c')/i)}{\sqrt{h_{n,s}}}, & h_{n,s} &= M_s n! (s/2)_n, & 0 \leq n \leq D. \end{aligned} \tag{EBG2}$$

Here $p_{n,s}$ is the original monic Gamma orthogonal polynomial of LJR2. In particular the leading coefficient of ϕ_n is $i^{-n}/\sqrt{h_{n,s}}$. These are the unchanged source norm and its orthonormal frame, not a newly assigned coefficient norm.

Let $D \geq q_\uparrow - v - 1$. Denote by $V_0, V_B, V_+ = [V_0, V_B]$ the evaluation matrices with entries $\phi_n(\alpha_\xi)$, rows $0 \leq n \leq D$, and columns in the indicated grids. Set

$$\begin{aligned} P_0 &= V_0(V_0^* V_0)^{-1} V_0^*, & R_B &= (I - P_0)V_B, & \mathcal{H}_{B,D} &= R_B^* R_B, \\ \mathcal{L}_0 &= \text{im } V_0, & \mathcal{L}_+ &= \text{im } V_+, & \mathcal{L}_+ &= \mathcal{L}_0 \oplus \text{im } R_B. \end{aligned} \tag{EBG3}$$

The sum is orthogonal in the original Gamma dual, expressed in its specified orthonormal frame. V_0 has independent columns since its first n_0 rows are a triangular change of the distinct-node Vandermonde. Those rows exist because $D + 1 \geq q_\uparrow - v > n_0$ on this domain. Every inverse in EBG3 is therefore defined.

The quotient map and its full kernel are

$$\mathcal{P}_D^\vee / \mathcal{L}_0 \twoheadrightarrow \mathcal{P}_D^\vee / \mathcal{L}_+, \quad \ker = \mathcal{L}_+ / \mathcal{L}_0 \simeq \text{im } R_B. \tag{EBG4}$$

The map sends the original functional class to its further class. The isomorphism sends a boundary evaluation combination to its projection off $\mathcal{L}_0$; its kernel is exactly the combinations already in $\mathcal{L}_0$. Both quotient norms are the original attained Hilbert minima. This proves the typed relation between the original and enlarged denominators.

The rank of a distinct-node evaluation matrix with $D + 1$ polynomial rows and n columns is $\min(D + 1, n)$: a square minor of that size is a nonzero Vandermonde multiplied by nonzero leading coefficients. Applying this fact to V_+ , and subtracting the independent original columns, gives

$$\begin{aligned} e_D &:= \text{rank } \mathcal{H}_{B,D} = \min(n_+, D + 1) - n_0, & r_D &= D + 1 - n_0, \\ \dim(\mathcal{P}_D^\vee / \mathcal{L}_+) &= \max(0, D + 1 - n_+). \end{aligned} \tag{EBG5}$$

At the four literal target cutoffs, write

D	e_D	$\dim(\mathcal{P}_D^\vee/\mathcal{L}_+)$
$q_\uparrow - v - 1$	$24k - 24 - v$	0
$q_\uparrow - v$	$24k - 23 - v$	0
$2q_\uparrow - v - 1$	$26k - 13$	$2q_\uparrow - v - n_+$
$2q_\uparrow - v$	$26k - 13$	$2q_\uparrow - v + 1 - n_+$

(EBG6)

Indeed $n_+ - (q_\uparrow - v) = 2k + 11 + v > 1$, while $2q_\uparrow - v - n_+ = k^2 + 8k + 14 - v > 0$. In particular a determinant of the full b -column boundary Gram is zero at both low cutoffs. Its nonzero-eigenvalue product, with its actual rank e_D , will be retained below.

113 The exact conditional boundary Gram and every original numerator

Define the complete monic original lower polynomial and its unchanged centered roots by

$$\begin{aligned}
\chi_0(S') &= \prod_{\xi \in \mathcal{G}_0} (S' - \alpha_\xi), & t_\xi &= (\alpha_\xi - c')/i, \\
w_j(S') &= \chi_0(S')((S' - c')/i)^j, & 0 \leq j < r_D, \\
G_D &= (\langle w_i, w_j \rangle_{s,c'})_{0 \leq i,j < r_D}, & E_{j,\xi} &= w_j(\alpha_\xi) = \chi_0(\alpha_\xi) t_\xi^j \quad (\xi \in \mathcal{B}).
\end{aligned}$$
(EBG7)

The w_j form the entire subspace of $\mathcal{P}_D$ vanishing at $\mathcal{G}_0$: successive polynomial division by the distinct linear factors gives exactly $\chi_0 \mathcal{P}_{D-n_0}$. The centered polynomial $\chi_0(c' + iy) = i^{n_0} P_0(y)$ has real coefficients after removing that displayed constant phase. Its root set is closed under complex conjugation, by replacing the first grid index a by $k - 8 - a$. Thus

$$(G_D)_{ij} = \int_{\mathbb{R}} |P_0(y)|^2 y^{i+j} dm_s(y) \quad \text{is real,} \quad G_D \succ 0. \quad (\text{EBG8})$$

Positivity follows by integrating the squared modulus of a nonzero polynomial against the positive measure away from the finitely many roots. Every original root and the phase i^{n_0} remain in EBG7–8.

A quotient of the full dual by the original evaluations is isometric to restriction of a functional to their common annihilator $\chi_0 \mathcal{P}_{D-n_0}$. To prove this, decompose $\mathcal{P}_D$ orthogonally into that annihilator and its orthogonal complement. Extending a functional by zero on the complement has least norm; every other extension adds an orthogonal nonnegative squared norm. All extensions differ by the original evaluation span, since that span has the complementary dimension n_0 .

In the w_j coordinates the resulting complete boundary Gram is

$\mathcal{H}_{B,D} = E^* G_D^{-1} E.$

(EBG9)

For an explicit check of the complex-linear dual convention, the frame $w G_D^{-1/2}$ is orthonormal because G_D is real symmetric positive. The evaluation value column in this frame is $G_D^{-1/2} E_\xi$. Its Gram is EBG9. Had G_D been complex, the corresponding value-column formula would contain $\overline{G_D}^{-1}$; EBG8 proves why the displayed real inverse is the correct one here.

For the original DNE numerator frame let $F_a = \mathcal{N}_a/E_A$, with every $\mathcal{N}_a$ in the original low-jet numerator space, and define

$$U_{j,a} = \Lambda_{F_a}(w_j), \quad D_D^0 = U^* G_D^{-1} U, \quad C_D = E^* G_D^{-1} U. \quad (\text{EBG10})$$

These F_a are analytic at zero, and every original conductor coefficient in Λ_{F_a} is retained. The preceding isometry proves that D_D^0 is exactly the DNE original minimum-extension metric at this

cutoff and source degree. With the same fixed numerator frame, enlarged minimization gives

$$\begin{aligned} D_D^+ &= \min_{\eta} (U + E\eta)^* G_D^{-1} (U + E\eta) = D_D^0 - C_D^* \mathcal{H}_{B,D}^+ C_D, \\ D_D^0 &= D_D^+ + C_D^* \mathcal{H}_{B,D}^+ C_D, \quad D_D^+ \succeq 0. \end{aligned} \quad (\text{EBG11})$$

The minimum notation denotes its quadratic form on each numerator vector; the minimizing η is chosen for that same full vector. The superscript $+$ on $\mathcal{H}$ is its Moore–Penrose inverse on its actual image. Indeed $\ker \mathcal{H} = \ker E$; hence each column of C_D is in $\text{im } \mathcal{H}$. Completing the square gives the attained lift $-\mathcal{H}^+ C_D x$, plus any vector of $\ker E$. This proves the identity, including both singular low-endpoint Grams. Equivalently the last term is the squared norm of the orthogonal projection of the original dual class onto $\text{im } R_B$. At the two low endpoints, EBG6 makes $D_D^+ = 0$, so this projection accounts for the entire original metric. This is an exact loss map, not the deletion of those directions.

114 The high-endpoint kernel is the full chart-boundary module

Keep the actual affine symbol

$$A(Z, W) = \mathcal{A}(\mathbf{H}(ZW, Z, W, 1)^T), \quad a_A = \deg_Z A, \quad b_A = \deg_W A, \quad 0 \leq a_A, b_A \leq 8. \quad (\text{EBG12})$$

A is nonzero on the original five-orbit locus. Its biform homogenization of bidegree $(8, 8)$ has the exact factorization

$$A^{\text{hom}} = z_1^{8-a_A} w_1^{8-b_A} \hat{A}, \quad \text{bideg } \hat{A} = (a_A, b_A). \quad (\text{EBG13})$$

No scalar or coefficient is divided away: $\hat{A}$ is obtained by removing exactly the displayed monomial factors. They record the complete multiplicities of the two chart-boundary rulings $z_1 = 0, w_1 = 0$ in this original conductor.

At either high cutoff in EBG6, the kernel on the original numerator quotient has the exact description

$$\begin{aligned} \ker D_D^+ &\simeq \frac{\hat{A} T_{k-a_A, k-b_A}}{A^{\text{hom}} T_{k-8, k-8}}, \quad T = \mathbb{C}[z_0, z_1, w_0, w_1], \\ z_A(k) &:= \dim \ker D_D^+ = (k - a_A + 1)(k - b_A + 1) - (k - 7)^2, \\ \text{rank } D_D^+ &= (16k - 48 - v) - z_A(k). \end{aligned} \quad (\text{EBG14})$$

The isomorphism is through the original coefficient/functional map KRC5, including $\alpha_A N_k Z_{\text{exp}, k}$.

Here is the full finite-window proof. A null vector of D_D^+ means that its analytic germ $F = \mathcal{N}/E_A$ agrees on all polynomials through degree D with an enlarged evaluation combination

$$g(z) = e^{(k-8)b_* z} \sum_{0 \leq a, b \leq k+5} c_{ab} e^{2a\delta z + 2ib\gamma z}. \quad (\text{EBG15})$$

Thus $F - g$ vanishes to order at least $D + 1$. Multiplication by E_A , whose zero order is exactly v , gives $\mathcal{N} - E_A g$ of zero order at least $D + v + 1$. The latter is an exponential polynomial on the complete distinct upper grid with indices $0, \dots, k + 13$ and base exponent kb_* . It has at most $(k + 14)^2$ coefficients. At the smaller high cutoff,

$$D + v + 1 = 2(k + 5)^2 \geq (k + 14)^2, \quad 2(k + 5)^2 - (k + 14)^2 = k^2 - 8k - 146 > 0 \quad (k \geq 37). \quad (\text{EBG16})$$

The square Vandermonde in its first $(k + 14)^2$ derivatives therefore makes all coefficients zero. This proves the entire analytic identity $\mathcal{N} = E_A g$ from the finite high window, with no limit interchange.

Writing the same identity in Z, W gives $N(Z, W) = A(Z, W)G(Z, W)$. All coefficients of N lie in the original rectangle $0, \dots, k$. Degree in Z adds under multiplication in the domain $\mathbb{C}[W][Z]$, and degree in W adds for the same reason. Consequently

$$\deg_Z G \leq k - a_A, \quad \deg_W G \leq k - b_A. \quad (\text{EBG17})$$

Conversely every such G belongs to the enlarged rectangle, gives an original upper numerator, and has its first v jets zero because that numerator is $E_A g$. Its original conductor class is zero exactly when G lies in the original lower rectangle $0, \dots, k-8$. Homogenizing these two rectangles gives EBG14, because multiplication by the nonzero $\hat{A}$ is injective in T . Counting their monomials proves the dimension. This also proves $z_A(k) \leq 16k - 48 - v$ on the actual data.

The complete low-endpoint kernel is the full original numerator space by EBG11. At high endpoints the kernel is precisely EBG14. If $a_A = b_A = 8$, EBG14 is zero. Otherwise its exact nonreduced boundary multiplicities remain; the calculation has not discarded an exceptional support stratum. The compatible-gauge complex uses a specified subspace of this enlarged denominator, so EBG4 and EBG11 give its exact ambient projection and loss, without identifying the two gauges.

115 Explicit boundary determinants in both rank regimes

For a positive semidefinite Gram on its stated coefficient space, pdet denotes the product of its positive eigenvalues, with empty product one. At either low cutoff put $r = r_D = e_D < b$. The nonzero eigenvalues of $E^* G_D^{-1} E$ are those of $G_D^{-1/2} E E^* G_D^{-1/2}$: the two rectangular products have the same nonzero eigenvectors under multiplication by their rectangular factor, with inverse division by the nonzero eigenvalue. Therefore

$$\text{pdet } \mathcal{H}_{B,D} = \frac{\det(E E^*)}{\det G_D},$$

$$\det(E E^*) = \sum_{\substack{I \subset B \\ |I|=r}} \left(\prod_{\xi \in I} |\chi_0(\alpha_\xi)|^2 \right) \left(\prod_{\substack{\xi < \eta \\ \xi, \eta \in I}} |t_\eta - t_\xi|^2 \right). \quad (\text{EBG18})$$

The first determinant is positive since every r boundary columns form a nonzero weighted Vandermonde. The second identity follows by multilinearly expanding the determinant of $E E^*$: repeated column indices cancel under alternation, and each distinct set gives its squared minor. Its displayed minor is the ordinary power Vandermonde with its full nonzero column factors. Thus all subsets, original phases and original evaluation products are retained.

At a high cutoff $D+1 \geq n_+$, write $\chi_+(S') = \prod_{\xi \in \mathcal{G}_+} (S' - \alpha_\xi)$ and let $H_{\chi_+, D}$ be the full native Gram of $\chi_+(S') S'^j$, $0 \leq j \leq D - n_+$. Let $\mathbb{V}_0, \mathbb{V}_+$ be the ordinary square node Vandermondes. Then the full boundary determinant is exactly

$$\det \mathcal{H}_{B,D} = \frac{|\det \mathbb{V}_+|^2 \det H_{\chi_+, D}}{|\det \mathbb{V}_0|^2 \det G_D}. \quad (\text{EBG19})$$

To prove this, split all polynomial monomials through D into the first n monomials and $\chi S'^j$ for any of the two grids. The change matrix is triangular with determinant one because χ is monic. The full native Gram determinant is $\prod_{j=0}^D h_{j,s}$ by EBG2. Schur elimination gives the quotient remainder determinant $\prod h_{j,s} / \det H_{\chi, D}$. Its inverse evaluation Gram has determinant $|\det \mathbb{V}|^2 \det H_{\chi, D} / \prod h_{j,s}$. Conjugating a dual matrix does not change this real positive determinant. Dividing the two grid expressions and using the Schur complement of $V_0^* V_0$ in $V_+^* V_+$ gives EBG19. The change from $\chi_0 S'^j$ to the centered w_j has determinant of modulus one, so its determinant is exactly $\det G_D$. All full source masses remain before their displayed ratio cancellation.

116 Original Gamma control at all displaced cutoffs

Use the exact GPA constants with the original source degree k ,

$$\epsilon = 2^{-32}, \quad R = k \sqrt{\delta^2 + \gamma^2} \leq \epsilon q / 2, \quad T = \epsilon q, \quad \vartheta = \pi / 2, \quad h = 1 / q,$$

$$\ell_s = (s - 1) / 4, \quad Q_s = n_0 + \ell_s, \quad \beta_s = M_s / (\sqrt{2} \Gamma(s / 2)), \quad a_D = D - n_0. \quad (\text{EBG20})$$

For every cutoff in EBG6,

$$n_0 \geq q/2, \quad Q_s \leq 2q, \quad a_D \leq k^2 + 34k + 1 - v \leq 2q \quad (k \geq 37). \quad (\text{EBG21})$$

The last inequality is $k^2 - 30k + 1 + v \geq 0$. These check the degree hypotheses in the full GPA3–15 proof on these displaced cutoffs, rather than assuming that an estimate stated at the earlier endpoints applies unchanged.

For clarity retain all its finite constants:

$$\begin{aligned} C_L &= \pi/\Gamma(3/4)^2, \quad K_q = \frac{\sqrt{2\pi}}{C_L} \frac{(2q+1)^2}{q} e^{-2q}, \quad \mathcal{E}_k = \frac{qR^2}{\epsilon^2 q^2 - R^2} + \log(1 + K_q), \\ r_{\ell_s} &= \frac{\sum_{j=0}^{\ell_s-1} (2j + \frac{1}{2})^2}{\epsilon^2 q^2} + \log(1 + K_q) \quad (\ell_s > 0), \quad r_0 = 0, \\ L_h &= C_L (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\ K_{a,Q} &= 2Q + 1 + 2a + 2\sqrt{a(a+2Q)}, \\ \mathfrak{b}_{D,s}^- &= -\mathcal{E}_k + \log L_h - K_{a_D, Q_s} \log(1 + h/\vartheta), \\ \mathfrak{b}_{D,s}^+ &= \mathcal{E}_k + r_{\ell_s} + \log U_h + K_{a_D, Q_s} \log \frac{1}{1 - h/\vartheta}. \end{aligned} \quad (\text{EBG22})$$

The complete real factorial moment matrix is

$$(\mathbf{M}_{Q,a})_{ij} = \begin{cases} 2\Gamma(2Q + i + j + 1)/\vartheta^{2Q+i+j+1}, & i + j \text{ even}, \\ 0, & i + j \text{ odd}. \end{cases} \quad (\text{EBG23})$$

The original coordinate here is exactly $y = (S' - c')/i$, as already retained in w_j and E . The full GPA proof gives

$$\beta_s e^{\mathfrak{b}_{D,s}^-} \mathbf{M}_{Q_s, a_D} \preceq G_D \preceq \beta_s e^{\mathfrak{b}_{D,s}^+} \mathbf{M}_{Q_s, a_D}. \quad (\text{EBG24})$$

To specify why all its estimates apply, the original paired roots have modulus at most R and $n_0 \geq q/2$; outside $[-T, T]$ their product ratio to y^{2n_0} has logarithm bounded by $n_0 R^2/(T^2 - R^2)$. The complete inside-region estimate in GPA12 is valid for every polynomial of degree at most $2q$, as checked in EBG21. The unchanged Gamma recurrence adds the full r_{ℓ_s} and β_s . The global density sandwich GPA13 is independent of D . Finally GPA4–9 bounds the two rate changes on all those polynomials by the explicit K_{a_D, Q_s} . Composing these proved full-form inequalities gives exactly EBG24, including the original inside contribution. Moreover $\mathfrak{b}_{D,s}^\pm = O_{\text{actual}}(\log q)$: $K_{a_D, Q_s} \leq 16q + 1$, $h = 1/q$, $\mathcal{E}_k = O_{\text{actual}}(1)$, $r_{\ell_s} = O(1/k)$, and the sine factors have logarithm $O(\log q)$. This uses the fixed actual period.

Let $\mathcal{H}_{B,D}^{\text{exp}} = E^* \mathbf{M}_{Q_s, a_D}^{-1} E$. Inversion of EBG24 and substitution of the same E give

$$\beta_s^{-1} e^{-\mathfrak{b}_{D,s}^+} \mathcal{H}_{B,D}^{\text{exp}} \preceq \mathcal{H}_{B,D} \preceq \beta_s^{-1} e^{-\mathfrak{b}_{D,s}^-} \mathcal{H}_{B,D}^{\text{exp}}. \quad (\text{EBG25})$$

Both kernels are exactly $\ker E$. Restricting to its orthogonal complement in the unchanged boundary coefficient space and taking determinants therefore proves

$$-e_D \mathfrak{b}_{D,s}^+ \leq \log \text{pdet } \mathcal{H}_{B,D} - \log \text{pdet } \mathcal{H}_{B,D}^{\text{exp}} + e_D \log \beta_s \leq -e_D \mathfrak{b}_{D,s}^-. \quad (\text{EBG26})$$

The errors are $O_{\text{actual}}(k \log q) = o(kq)$ at every one of the four cutoffs because $e_D \leq 26k - 13$. This estimates the actual boundary volume through a fully explicit matrix without assuming it is nonsingular at low degree.

In a four-endpoint difference, with high signs positive, the source mass contributes its actual rank jump

$$\sum_{\text{high}} e_D - \sum_{\text{low}} e_D = 4k + 21 + 2v, \quad -(4k + 21 + 2v) \log \beta_s. \quad (\text{EBG27})$$

This term does not cancel. It is subleading for the fixed source orders, but its full finite value is retained. For $s = k$, the elementary integral bounds for $\log n!$ and the half-integer Gamma recurrence give $|\log \beta_k| = O(k \log(k+1))$, so its magnitude is $O(k^2 \log(k+1)) = o(kq)$; for $s = 1$ it is $O(k)$. The explicit four unequal error endpoints follow by using $-e_D \mathbf{b}^+$ at each positive sign and $+e_D \mathbf{b}^-$ at each negative sign for the lower bound, and reversing those choices for the upper bound.

The same full-form inversion and minimization over the same E -fibres gives EBG25, with $\mathcal{H}_B$ replaced by either D_D^0 or D_D^+ and with E replaced by the entire corresponding $[U, E]$ construction. Their kernels are the exact algebraic kernels already proved in EBG11 and EBG14, independent of the positive metric. No off-diagonal numerator-boundary pairing is removed.

117 A scalar low-boundary estimate with a proved subleading error

At either low cutoff let $r = e_D$ and order the complete nonzero weights $|\chi_0(\alpha_\xi)|$ in decreasing order, breaking ties by the original node order. Let I_{top} be the first r nodes and set

$$\begin{aligned} \mathcal{W}_{r,k} &= 2 \sum_{\xi \in I_{\text{top}}} \log |\chi_0(\alpha_\xi)|, \\ \chi_0(\alpha_{ab}) &= \prod_{u,v=0}^{k-8} (2\delta(a-u) + 2i\gamma(b-v)), \end{aligned} \tag{EBG28}$$

$$d_{\min} = 2 \min(\delta, \gamma), \quad d_{\max} = 2(k+5)\sqrt{\delta^2 + \gamma^2}.$$

Every factor of that product is nonzero on $\mathcal{B}$. Every distance between two boundary nodes lies between $d_{\min}$ and $d_{\max}$. Since $|1/i| = 1$, these are also the distances in EBG18. The largest product of r column weights is exactly the selected product, without any assumption about their distribution. Consequently EBG18 yields

$$\begin{aligned} \mathcal{W}_{r,k} + r(r-1) \log d_{\min} &\leq \log \det(EE^*) \\ &\leq \mathcal{W}_{r,k} + \log \binom{b}{r} + r(r-1) \log d_{\max}. \end{aligned} \tag{EBG29}$$

The lower bound retains one specified actual subset; the upper bound bounds every retained subset and then counts them. Thus the omitted size from this scalar enclosure, not an omitted component of the matrix, is bounded explicitly.

Here is a direct full moment-determinant bound sufficient at precisely these low ranks. For $Q = Q_s \geq 1$, put

$$\begin{aligned} \mu_0(Q) &= 2\Gamma(2Q+1)/\vartheta^{2Q+1}, \\ \mathcal{L}_{r,Q} &= -r \log(4eQ/\vartheta) - \vartheta^2 r^3/Q + 2 \sum_{j=1}^{r-1} \log(j!), \\ \mathcal{U}_{r,Q} &= \sum_{j=0}^{r-1} \log((2Q+1)_{2j}/\vartheta^{2j}). \end{aligned} \tag{EBG30}$$

Then

$$r \log \mu_0(Q) + \mathcal{L}_{r,Q} \leq \log \det \mathbf{M}_{Q,r-1} \leq r \log \mu_0(Q) + \mathcal{U}_{r,Q}. \tag{EBG31}$$

For the upper bound, Gram–Schmidt makes the Gram determinant the product of squared residual norms, each no larger than its original diagonal norm. That diagonal is exactly $\mu_0(Q)(2Q+1)_{2j}/\vartheta^{2j}$.

For the lower bound, expand the Gram determinant under its r finite integrals. Multilinearity and alternation give

$$\det \mathbf{M}_{Q,r-1} = \frac{1}{r!} \int_{\mathbb{R}^r} \prod_i |y_i|^{2Q} e^{-\vartheta|y_i|} \prod_{i < j} (y_j - y_i)^2 dy_1 \cdots dy_r.$$

All integrals converge and the integrand is nonnegative. Put $y_0 = 2Q/\vartheta$ and restrict to the $r!$ permutations of the disjoint intervals $[y_0 + 2j, y_0 + 2j + 1]$, $0 \leq j < r$. For $t \geq 0$, $\log(1+t) \geq t - t^2/2$, as is verified by differentiating their difference. Thus throughout these intervals the weight is at least its value at y_0 times $e^{-\vartheta^2 r^2/Q}$. Each ordered pair of intervals with indices $i < j$ has distance at least $j - i$. Their product is $\prod_{j=1}^{r-1} j!$. Finally the elementary sum/integral comparison

$$n \log n - n + 1 \leq \log(n!) \leq n \log n - n + 1 + \log n$$

for $n \geq 1$, obtained by integrating the increasing $\log x$ between successive integers, gives $y_0^{2Q} e^{-2Q} \geq \mu_0(Q)\vartheta/(4eQ)$. Multiplying the interval bounds proves the lower side of EBG31, including every finite constant and the factor $r!$ cancellation.

Combining EBG18,24,29,31 now gives the explicit original Gamma boundary-volume interval

$$\begin{aligned} \mathcal{W}_{r,k} - r \log(\beta_s \mu_0(Q_s)) + r(r-1) \log d_{\min} \\ - r \mathfrak{b}_{D,s}^+ - \mathcal{U}_{r,Q_s} \leq \log \text{pdet } \mathcal{H}_{B,D}, \end{aligned} \tag{EBG32}$$

$$\begin{aligned} \log \text{pdet } \mathcal{H}_{B,D} \leq \mathcal{W}_{r,k} - r \log(\beta_s \mu_0(Q_s)) + \log \binom{b}{r} \\ + r(r-1) \log d_{\max} - r \mathfrak{b}_{D,s}^- - \mathcal{L}_{r,Q_s}. \end{aligned}$$

At the two actual low cutoffs, $r \leq 24k$, $Q_s \asymp q$, $b = 26k - 13$, and the quartet is fixed. The rising-factorial upper bound gives $\mathcal{U}_{r,Q} = O(r^2 \log(Q+r))$. The displayed factorial sum and all remaining terms in $\mathcal{L}_{r,Q}$ are $O(r^2 \log(r+1) + r \log Q + r^3/Q)$. Together with EBG22 and the two distance bounds this proves

$$\boxed{\log \text{pdet } \mathcal{H}_{B,D} = \mathcal{W}_{r,k} - r \log(\beta_s \mu_0(Q_s)) + O_{\text{actual}}(k^2 \log(k+1))} \tag{EBG33}$$

on those two low cutoffs, with the explicit unequal finite errors EBG32 preceding the order estimate. The error is $o(kq)$. The main expression is a sum of specified original root products and an explicit Gamma mass; it does not depend on an uncomputed matrix inverse, an assumed angle, a generic pivot profile or a chosen fraction of the original rank.

EBG11 and EBG14 retain how this ambient boundary calculation maps to the actual numerator and its native loss. In particular the scalar boundary determinant has not been substituted for the matched invariant metric in MNT19. The exact compatible-gauge restriction, its numerator pairings, and the original shifted endpoint windows remain in that map. These full results supply their boundary ranks, original coefficient module, complete Gamma comparison and low-volume estimate. They continue the original analytic calculation without discarding any mixed or supported-zero state.

Evaluated original low boundary. The complete WEL1–38 and BVC1–11 proofs now evaluate the actual low rank in EBG6 through the unchanged matrix EBG9:

$$\mathcal{F}_{k,s,\sigma} - B_W + E_- \leq \log \text{pdet } \mathcal{H}_{B,D} \leq \mathcal{F}_{k,s,\sigma} + B_W + E_+. \tag{EBG34}$$

All quantities are the literal BVC1,4–5 expressions, with $s = 1$ or k and its exact comparison domain. NCR2–4 proves their backward map through the full EBG10 numerator U and mixed block C_D . The coefficient of the centered low-volume limit is BVC11. At these low cutoffs $D_D^+ = 0$ still holds; the complete $C_D^* \mathcal{H}_{B,D}^+ C_D$ in EBG11 remains.

118 The identical original conductor in the two continuations

Keep the actual simple quartet, its complete unit $U = \text{diag}(a, \bar{a}, \bar{a}, a)$, original period u , and logarithm branch. Throughout $0 < \delta < 1/2$, $\gamma > 2$, $k \equiv 1 \pmod{4}$, $k \geq 37$, and the source orders

are $s = 1, k$. Use the full original period matrix $\mathbf{H}$ of KRC2 and PCL7, and $D_5 = \text{diag}(\zeta, \zeta^2, \zeta^3, \zeta^4)$. PCL proves the complete equality

$$\mathbf{H} = e^{a/u} G_u \mathcal{R}(u^{-1}) V^{-1} U, \quad L_j = U^{-1} V \mathcal{R}(u^{-1})^{-1} D_5^{-j} \mathcal{R}(u^{-1}) V^{-1} U = \mathbf{H}^{-1} D_5^{-j} \mathbf{H}. \quad (\text{RPB1})$$

Indeed the scalar cancels with its inverse and the two displayed diagonal matrices commute. No entry of $\mathcal{R}$, V , or U is replaced. With $t = (ZW, Z, W, 1)^\top$, the definitions $Q(x) = Q_0(\mathbf{H}^{-1}x)$, $gf(x) = f(D_5^{-1}x)$ give

$$\mathcal{A}(\mathbf{H}t) = \prod_{j=1}^4 (g^j Q)(\mathbf{H}t) = \prod_{j=1}^4 Q_0(L_j t) = A_{1/u}(Z, W). \quad (\text{RPB2})$$

Thus EBG12's affine polynomial and the ASR/RPK polynomial are identical coefficient by coefficient, with scalar one. This is the exact morphism of coefficient presentations used below. Bihomogenization sends $f(Z, W)$ of bidegree at most (d, e) to $z_1^d w_1^e f(z_0/z_1, w_0/w_1)$; its inverse is evaluation at $(z_0, z_1, w_0, w_1) = (Z, 1, W, 1)$. Expanding monomials proves that these inverse maps preserve multiplication with its original bidegrees. They give precisely the homogeneous map in KRC2.

119 Exact support exceptions and two quantitative radii

For each original orbit factor put $S_j = L_j^\top S_0 L_j$, where $(S_0)_{14} = (S_0)_{41} = 1/2$ and $(S_0)_{23} = (S_0)_{32} = -1/2$. The four actual boundary coefficient vectors, in their original order, are

$$\begin{aligned} E_{Z,0,j} &= ((S_j)_{33}, 2(S_j)_{34}, (S_j)_{44}), \\ E_{Z,\infty,j} &= ((S_j)_{11}, 2(S_j)_{12}, (S_j)_{22}), \\ E_{W,0,j} &= ((S_j)_{22}, 2(S_j)_{24}, (S_j)_{44}), \\ E_{W,\infty,j} &= ((S_j)_{11}, 2(S_j)_{13}, (S_j)_{33}). \end{aligned} \quad (\text{RPB3})$$

Substitution into $t^\top S_j t$ proves every coefficient, including each off-diagonal factor two. On $|u| \geq R_*$ all orbit factors are nonzero. A coordinate restriction of their product vanishes identically exactly when at least one of its factors does, since the coefficient ring on that ruling is an integral domain. Define

$$\begin{aligned} \mathcal{E}_\partial &= \{u : |u| \geq R_*, E_{X,b,j}(u^{-1}) = 0 \text{ for some } X, b, j\}, \\ \mathcal{E}_\infty &= \{u : |u| \geq R_*, E_{Z,\infty,j}(u^{-1}) = 0 \text{ or } E_{W,\infty,j}(u^{-1}) = 0 \text{ for some } j\}. \end{aligned} \quad (\text{RPB4})$$

Consequently $\mathcal{E}_\infty \subseteq \mathcal{E}_\partial$; the first is exactly the locus where $\deg_Z A < 8$ or $\deg_W A < 8$, and the second is exactly the locus where one of the four extrema fails $(a_-, a_+, b_-, b_+) = (0, 8, 0, 8)$. ASR18 proves that the second set is finite. Its coefficient test in RPB3 remains exact at every exceptional period.

Retain the complete already proved ASR12 radius R_{supp} and the new RPK10–11 radius R_{rank} , each at least R_* . The first uses the actual nonzero opposite-corner coefficients and their convergent series; the second uses the unit-independent eliminants and their convergent matrix-series bound. Neither radius is assigned a numerical value here. Their simultaneous availability gives the actual covered radius

$$R_{\text{cov}} = \min(R_{\text{supp}}, R_{\text{rank}}), \quad |u| \geq R_{\text{cov}} \implies (a_-, a_+, b_-, b_+) = (0, 8, 0, 8). \quad (\text{RPB5})$$

To prove it, whichever radius realizes the minimum supplies its own full theorem on that whole exterior domain. There is no assertion that either radius is always the smaller.

The exact relation between the tests is also explicit. Write $B_j = V L_j^0 V^{-1}$ with $L_j^0 = \mathcal{R}^{-1} D_5^{-j} \mathcal{R}$, so $L_j = U^{-1} B_j U$. Put $f_i = (B_j)_{1i} (B_j)_{4i}$, $h_i = (B_j)_{2i} (B_j)_{3i}$. Then

$$Q_0(L_j e_i) = U_{ii}^2 (a^{-2} f_i - \bar{a}^{-2} h_i), \quad Q_0|_{L_j \langle e_i, e_l \rangle} = 0 \implies f_i h_l - f_l h_i = 0. \quad (\text{RPB6})$$

The first identity is direct diagonal conjugation. The second uses both endpoint values, multiplies their two equalities, and subtracts; all four diagonal entries of U are nonzero. Hence the exact exceptional set $\mathcal{E}_\partial$ is contained in RPK's finite eliminant exceptional superset. An eliminant zero does not impose a rank: RPB3 supplies the complete coefficient test there. ASR's conclusion already established full widths; RPK adds this test and its independent quantitative radius.

120 The high-window boundary kernel and its complete native metric

Use all EBG1–14 objects without changing the grids:

$$q = (k+1)^2, \quad q_\uparrow = (k+5)^2, \quad n_0 = (k-7)^2, \quad n_+ = (k+6)^2, \quad v = \text{ord}_0 E_A, \quad m = 16k - 48 - v.$$

The quotient is the original low-annihilator conductor space $\mathcal{C}_{v,k}$, of dimension m , carried by KRC5's $\alpha_A N_k Z_{\text{exp},k}$, $\alpha_A = v!/\mu_v$. At the two high cutoffs $D = 2q_\uparrow - v - 1, 2q_\uparrow - v$, EBG14 proves

$$\ker D_D^+ \simeq \frac{\hat{A} T_{k-a_+, k-b_+}}{A^{\text{hom}} T_{k-8, k-8}}, \quad \dim \ker D_D^+ = (k - a_+ + 1)(k - b_+ + 1) - (k - 7)^2. \quad (\text{RPB7})$$

Here $A^{\text{hom}} = z_1^{8-a_+} w_1^{8-b_+} \hat{A}$ retains its complete original scalar and factors. Write $h_Z = 8 - a_+$, $h_W = 8 - b_+$, both nonnegative. The dimension in RPB7 is exactly $(k - 7)(h_Z + h_W) + h_Z h_W$, so it is zero if and only if $h_Z = h_W = 0$. Thus the complete period-by-period rank statement is

$$u \notin \mathcal{E}_\infty \iff D_{2q_\uparrow - v - 1}^+ \succ 0 \iff D_{2q_\uparrow - v}^+ \succ 0, \quad (\text{RPB8})$$

$$\text{rank } D_{q_\uparrow - v - 1}^+ = \text{rank } D_{q_\uparrow - v}^+ = 0, \quad \text{rank } D_{2q_\uparrow - v - 1}^+ = \text{rank } D_{2q_\uparrow - v}^+ = m \quad (u \notin \mathcal{E}_\infty).$$

The low identities are EBG6,11 and hold at every original period in the domain. The high equivalences use the exact positive semidefinite Gram and its complete kernel. In particular all of RPB8 holds on $|u| \geq R_{\text{cov}}$. This applies the support theorem to the actual native loss map, without identifying an enlarged gauge with a compatible tuple.

Retain exactly G_D, E, U, C_D and $\mathcal{H}_{B,D} = E^* G_D^{-1} E$ from EBG7–11, and define

$$\Omega_{B,D} = (D_D^0)^{-1/2} D_D^+ (D_D^0)^{-1/2}, \quad (\text{RPB9})$$

$$T_{B,D} = (\mathcal{H}_{B,D}^+)^{1/2} C_D (D_D^0)^{-1/2}, \quad \Omega_{B,D} = I - T_{B,D}^* T_{B,D}.$$

The last identity follows by conjugating the complete EBG11 Schur complement. It retains the full numerator–boundary pairing $C_D = E^* G_D^{-1} U$ and the original Gamma mass. At high cutoffs off $\mathcal{E}_\infty$ all singular values ρ_i of $T_{B,D}$ are strictly less than one, and

$$\log \det D_D^+ = \log \det D_D^0 + \sum_{i=1}^m \log(1 - \rho_i^2). \quad (\text{RPB10})$$

At low cutoffs $\Omega_{B,D} = 0$ and $T_{B,D}^* T_{B,D} = I_m$. Thus the low enlarged quotient is a zero map on the full original source space; no logarithm of its determinant is inserted into an expression requiring a positive Gram.

There is a quantitative transport of the entire high correction. Let $\mathbf{M} = \mathbf{M}_{Q_s, a_D}$ be precisely EBG23, with its original $\vartheta = \pi/2$, $Q_s = n_0 + (s - 1)/4$, $a_D = D - n_0$. Define the full factorial forms by

$$D^{0,\text{exp}} = U^* \mathbf{M}^{-1} U, \quad \mathcal{H}^{\text{exp}} = E^* \mathbf{M}^{-1} E, \quad C^{\text{exp}} = E^* \mathbf{M}^{-1} U, \quad (\text{RPB11})$$

$$D^{+, \text{exp}} = D^{0, \text{exp}} - (C^{\text{exp}})^* (\mathcal{H}^{\text{exp}})^+ C^{\text{exp}}, \quad \Omega^{\text{exp}} = (D^{0, \text{exp}})^{-1/2} D^{+, \text{exp}} (D^{0, \text{exp}})^{-1/2}.$$

These use the identical coefficient arrays and minimization fibres; only the proved full-form EBG24 comparison is applied. Write $b^\pm = \mathfrak{b}_{D,s}^\pm$ and $c_- = \beta_s^{-1}e^{-b^+}$, $c_+ = \beta_s^{-1}e^{-b^-}$. EBG24 inversion and minimum over each identical fibre give

$$c_- D^{0,\text{exp}} \preceq D_D^0 \preceq c_+ D^{0,\text{exp}}, \quad c_- D^{+, \text{exp}} \preceq D_D^+ \preceq c_+ D^{+, \text{exp}}. \quad (\text{RPB12})$$

At the high cutoffs off $\mathcal{E}_\infty$, every form here is positive. For each nonzero vector the generalized Rayleigh quotients are bounded by c_-/c_+ and c_+/c_- times their factorial counterparts. Minimizing the maximum quotient over subspaces of each fixed dimension proves the corresponding ordered generalized eigenvalue inequalities. Multiplying the m eigenvalue inequalities proves

$$|\log \det \Omega_{B,D} - \log \det \Omega^{\text{exp}}| \leq m(b^+ - b^-) = O_{\text{actual}}(k \log q) = o(kq). \quad (\text{RPB13})$$

This holds on EBG's exact comparison domain $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q$, $k \geq 37$. The source mass cancels here only as the displayed ratio c_+/c_- . The high correction itself remains the full matrix RPB11, whose small eigenvalues are retained.

121 The exact first-moment refinement and its backward consequences

Each of the four nonzero orbit biforms has at most nine nonzero coefficients. Let p_j be its actual support size. Under the injective original substitution $(Z, W) = (e^{2\delta z}, e^{2i\gamma z})$, all its p_j exponents are distinct. If its first p_j derivatives were zero, the Vandermonde on those exponents would kill all its coefficients. Hence its zero order is at most $p_j - 1$. Multiplying the four factors and their nonvanishing exponential phase proves, on the entire original five-orbit locus,

$$v = \sum_{j=1}^4 \text{ord}_0 Q_0(L_j(e^{2(\delta+i\gamma)z}, e^{2\delta z}, e^{2i\gamma z}, 1)^\top) \leq \sum_{j=1}^4 (p_j - 1) \leq 32. \quad (\text{RPB14})$$

This repeats the full short argument of RPK so that the backward consequence uses a proof, not a newly imposed parameter. The exact v is unchanged in every existing matrix.

In particular the earlier finite statements specialize to

$$\begin{aligned} m &= 16k - 48 - v \geq 16k - 80, & 8k + 24 - v &\geq 8k - 8 > 0, \\ e_{D_{\text{low},1}} &= 24k - 24 - v \geq 24k - 56, & e_{D_{\text{low},2}} &= 24k - 23 - v \geq 24k - 55, \\ \sum_{\text{high}} e_D - \sum_{\text{low}} e_D &= 4k + 21 + 2v \leq 4k + 85, & & (\text{RPB15}) \\ e_{\text{BIM}} &\leq \sum_{h \in \mathcal{H}} (v - t_{0,k-d_h}) - v \leq (|\mathcal{H}| - 1)v \leq 32(|\mathcal{H}| - 1). \end{aligned}$$

The second inequality strengthens the actual NBG18 finite-window Vandermonde cutoff. The next two lines keep EBG's exact low ranks and signed source-mass rank. The last line is BIM's proved actual image-defect bound; it changes no generator and discards no coupled low jet. Every deduction follows by substitution of the proved bound in the previously exact formula. Earlier sources with the weaker bound 80 remain preserved and valid.

122 The exact map from independent gauges to the enlarged boundary

Use NBG's full truncated dual Hilbert spaces $\mathcal{H}_0, \mathcal{H}_1$, both through the identical degree D , with their literal centres c'_k, c'_{k+4} and orders $(s_0, s_1) = (1, 1)$ or $(k, k+4)$. The complete coefficient multiplication and its finite Bezout inverse are

$$T : \mathcal{H}_0 \longrightarrow \mathcal{H}_1^4, \quad Tw = (B_i w)_i, \quad S : \mathcal{H}_1^4 \longrightarrow \mathcal{H}_0, \quad S(y) = \sum_i D_i y_i, \quad ST = I. \quad (\text{RPB16})$$

Here $b_* = \beta_* = \frac{1}{2} - \delta - i\gamma$ and $B_i = e^{4b_*z}L_i^4$, $D_i = e^{-4b_*z}U_i(Ht)$ are exactly NBG3–5, including every coefficient of its degree-thirteen multinomial identity. NBG14 proves $\|T\| \leq K_+$, $\|S\| \leq K_-$ with the full native constants NBG12. In particular $K_-K_+ \geq 1$ follows from $ST = I$.

Let $\mathcal{L}_0$ be the old lower-evaluation span, $\mathcal{L}_1$ the direct sum of the four target lower spans, and $\mathcal{L}_+$ EBG’s enlarged span. NBG16–17 proves $T\mathcal{L}_0 \subseteq \mathcal{L}_1$ and $S\mathcal{L}_1 \subseteq \mathcal{L}_+$ by its complete coefficient convolutions. Therefore the following are well-defined linear maps:

$$\mathcal{H}_0/\mathcal{L}_0 \xrightarrow{\bar{T}} \mathcal{H}_1^4/\mathcal{L}_1 \xrightarrow{\bar{S}} \mathcal{H}_0/\mathcal{L}_+, \quad \bar{S}\bar{T} = q_+. \quad (\text{RPB17})$$

The last map q_+ is precisely EBG4’s quotient. Changing a representative at either arrow changes its image by the stated target lower span; $ST = I$ proves the displayed composition. Thus this is an exact relation between the independent target gauges and the enlarged source denominator, including all incompatible target coefficients.

For their minimum quotient norms, $\|\bar{S}[y]\| \leq K_- \| [y] \|$: apply S to the attained least-norm representative, then minimize in the target. Pulling this inequality along the actual full numerator map gives the new two-sided sandwich

$$K_-^{-2}D_D^+ \preceq G_{\text{ind},D}^{\text{act}} \preceq G_{\text{cor},D}^{\text{act}} \preceq K_+^2D_D^0. \quad (\text{RPB18})$$

Its middle and right sides are NBG27; its left side is RPB17’s composition and the original EBG11 minimum. Every matrix uses the same fixed original DNE numerator frame. At the high cutoffs off $\mathcal{E}_\infty$ all matrices are positive. Taking determinants yields

$$\begin{aligned} 0 &\leq -\log \det \Omega_D^{\text{act}} \leq -\log \det \Omega_{B,D} + 2m \log(K_-K_+), \\ \Omega_D^{\text{act}} &= (G_{\text{cor},D}^{\text{act}})^{-1/2} G_{\text{ind},D}^{\text{act}} (G_{\text{cor},D}^{\text{act}})^{-1/2}, \\ 0 &\leq -\log \det \Omega_D^{\text{act}} \leq -\log \det \Omega^{\text{exp}} + m(b^+ - b^-) + 2m \log(K_-K_+). \end{aligned} \quad (\text{RPB19})$$

Indeed RPB18 bounds the numerator determinant below by $K_-^{-2m} \det D_D^+$ and the denominator above by $K_+^{2m} \det D_D^0$. NBG24 supplies the nonnegative left side; RPB13 gives the final line. The two explicit additive errors are $O_{\text{actual}}(k \log q) = o(kq)$ on the previously stated comparison domain. The remaining factorial correction is the complete coupled form RPB11; it has not been assigned an asymptotic value. At the low endpoints RPB17 still holds and its last target is zero. One retains RPB18 with a zero left bound there, together with the full NBG independent metric. The four endpoint signs cannot be combined by taking a logarithm of that zero bound. The further invariant target quotient remains the exact NBG30–32 map with its own kernel and relative metric. Thus the proved estimate acts on its actual preceding source, without discarding that next quotient or assigning a leading allocation coefficient.

Evaluated boundary and the complete invariant receiver. The original map RPB17 and sandwich RPB18 now receive the complete boundary interval EBG34/BVC6 through NCR3–4. The further invariant image is the unchanged NBG30–32 map: in its exact ITL image frame its full signed return is

$$\sum_i \varepsilon_i \log \det \mathcal{R}_{X,D_i} = \sum_i \varepsilon_i \log \frac{\det \mathcal{H}_{V,D_i}}{\det \mathcal{H}_{E,D_i}} + \sum_n \vartheta_n(\gamma_{H,n} - \gamma_{A,n}) + \mathcal{E}_{\text{fin}}. \quad (\text{RPB20})$$

NFR1–5 proves the common coordinate map, all actual lower and invariant kernels, the literal weights $1, 2, \dots, 2, 1$ and the finite asymmetric error. Both original source-order pairs are retained. This adds no logarithm of the zero low enlarged quotient and no scalar allocation value.

The actual outer-strip quotient on displaced endpoint windows

Scope and retained providers

This note proves the missing short-window estimate for the **original outer-strip quotient**, not a model matrix. It does not by itself identify the native meromorphic quotient with that outer quo-

tient: the parent calculation supplies that identification and its conductor terms. No independent row-concentration or allocation assertion is used.

Providers read in full:

- `conductor_endpoint_20260914/total_variation/ORIGINAL_TOTAL_VARIATION_CHRISTOFFEL.tex` (CTV1–22).
- `outer_asymptotic_20260915/ORIGINAL_OUTER_STRIP_POWER_COMPARISON.tex` (OSP1–35).
- `outer_asymptotic_20260915/ORIGINAL_OUTER_STRIP_FIRST_VARIATION.tex` (OFV1–32).
- `dual_pole_control_20260915/allocation_estimate/multiplier_native_transport/ORIGINAL_MULTIPLIER_NATIVE_TRANSPORT.tex` (MNT1–19).

All these relative paths start at the original `web_common_kernel_20260914` source folder. The accepted EIQ and QLG estimates used below are the explicit statements and calculations retained in OFV2–3, OFV19–25. No new numerical evaluation of their constants is claimed.

OW1. Exact displaced degrees and actual source

Let the source tensor degree be $j = k - d$, with the original bounded invariant degree $0 \leq d \leq 28$, d divisible by four, and $j \equiv 1 \pmod{4}$. Write

$$q = (j + 1)^2, \quad q' = (j - 7)^2, \quad \Delta = q - q' = 16j - 48, \quad b = d(2j + d + 2).$$

This is exactly MNT16, not an approximation. Retain each source order $s = 1$ and $s = j$, its original mass $M_s = (2\pi)^{s/2}$, the original fixed δ, γ , and the complete centered root polynomials P_j, P_{j-8} from CTV3. Put

$$\ell = (s - 1)/4, \quad Q_+ = q + \ell, \quad Q_- = q' + \ell, \quad \vartheta = \pi/2, \quad \beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}.$$

The exact quotient under consideration has source norm $|P_{j-8}(y)|^2 dm_s(y)$, target $\mathbb{C}[y]/(P_{\partial,j})$, and source degree $t + \Delta - 1$. Its determinant logarithm is

$$V_s(t) = \log \det Q_{\partial, t+\Delta-1}^{(s)} = \log D_{t+\Delta}(P_{j-8}; s) - \log D_t(P_j; s), \quad t \geq 0. \quad (\text{OW1})$$

This is the monic-frame Schur-complement proof CTV12/OSP30, which applies to **every integer** $t \geq 0$, not only four selected endpoints. The source-to-original-polynomial map is multiplication by the literal P_{j-8} ; the relation map is multiplication by $P_{\partial,j}$, and $P_{j-8}P_{\partial,j} = P_j$ with monic constant one. Both original coordinate phases have modulus one and remain the CTV3 phases.

For $w \geq 0$, the old polynomial lifts remain admitted in the larger source with the same full source norm. Hence

$$0 \leq V_s(t) - V_s(t + w). \quad (\text{OW2})$$

OW3. One explicit eventual range, without deleting finite cases

For the asymptotic estimate take

$$j \geq 4096, \quad j\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}(j + 1)^2. \quad (\text{OW3})$$

The second condition is precisely the original OSP23 fixed-data condition. The first implies

$$b \leq 57j, \quad \Delta + 2b + 1 \leq 130j + 1 \leq q/16.$$

Indeed $b \leq 56j + 840 \leq 57j$ for $j \geq 840$, and the final quadratic inequality holds at 4096 and increases thereafter. These bounds control the actual four displaced windows

$$(t, w) = (0, b), (1, b), (q, 2b), (q + 1, 2b). \quad (\text{OW4})$$

All exact formulas OW1–2 remain valid at the earlier finite degrees, including cases where $b > q$. The eventual estimates below are **not** asserted there by pretending the windows are small.

OW5. Original roots and both full Gamma orders, with endpoint errors

Let E_j denote exactly $\mathcal{E}_j$ in OSP24; let r_ℓ denote OFV6, with $r_0 = 0$; and let $B_a(Q)$ denote OFV12, with $B_0 = 0$. Define

$$\begin{aligned} C_s(t) &= (2t + \Delta)(\mathcal{E}_j + r_\ell) + B_{t+\Delta}(Q_-) + B_t(Q_+), \\ \mathcal{A}_a(Q) &= \det \left[\int_{\mathbb{R}} y^{2Q+r+u} e^{-\vartheta|y|} dy \right]_{r,u=0}^{a-1}. \end{aligned} \quad (\text{OW5})$$

The exact source comparison is

$$V_s(t) = \Delta \log \beta_s + \log \mathcal{A}_{t+\Delta}(Q_-) - \log \mathcal{A}_t(Q_+) + e_s(t), \quad |e_s(t)| \leq C_s(t). \quad (\text{OW6})$$

Proof: apply OSP28 to the two literal root polynomials; its determinant errors sum to $(2t + \Delta)E_j$. Apply the **unchanged-source** Gamma recurrence OFV5–7 to each power determinant, retaining the full mass factor. Then apply the global two-sided Gamma-to-exponential comparison OFV9–13. The errors add exactly as OW5. These proofs apply whenever $t + \Delta \leq 2q + 1$; OW3–4 puts every required dimension in that range. Their inner-interval estimates use polynomial degree at most $2q$, including the whole inner interval, and do not use the earlier four-element endpoint list. In the present range

$$C_s(t) = O_{\text{actual}}(q \log q).$$

This follows directly from $E_j = O_{\text{actual}}(1)$, $r_\ell = O(1/j)$ and OFV12. The original mass has **not** disappeared: it is the displayed term $\Delta \log \beta_s$, independent of t ; it cancels exactly only when two endpoints are subtracted.

OW7. Low windows: a direct finite estimate, no equilibrium extrapolation

For a positive integer Q define

$$g(Q) = 2Q \log(2Q/\vartheta) - 2Q.$$

Set $L(0, Q) = 0$. For integer $a \geq 1$, let

$$\begin{aligned} L(a, Q) &= \max \left\{ a\vartheta + a \log(2a) + 2a(a-1) \log 2, \right. \\ &\quad \left. a(|\log(2/\vartheta) + 1| + \log(2(Q+a))) + a(a-1) \log(2(Q+a)/\vartheta) \right\}. \end{aligned} \quad (\text{OW7})$$

The actual exponential determinant obeys

$$|\log \mathcal{A}_a(Q) - ag(Q)| \leq L(a, Q). \quad (\text{OW8})$$

The proof in OFV16–18 is valid at every positive integer a, Q : the lower bound integrates on $[2Q/\vartheta, 2Q/\vartheta+1]$ and retains the exact translated Legendre Gram; the upper bound is the product of the original monomial moments $2(2(Q+r))!/\vartheta^{2(Q+r)+1}$. Its factorial integral bounds and g' estimate

do not require $a \leq \Delta + 1$. Thus the proof extends directly to the actual $a \leq \Delta + b + 1 \leq 73j + 1$, with no use of the high-dimensional equilibrium.

Put

$$E_s^{\text{lo}}(t) = C_s(t) + L(t + \Delta, Q_-) + L(t, Q_+).$$

For $t = 0, 1$ and $w = b$, OW6–8 give the **finite signed identity with error**

$$V_s(t + w) - V_s(t) = w [g(Q_-) - g(Q_+)] + \rho_{t,w}, \quad |\rho_{t,w}| \leq E_s^{\text{lo}}(t) + E_s^{\text{lo}}(t + w). \quad (\text{OW9})$$

Both $\Delta g(Q_-)$ and $\Delta \log \beta_s$ cancel exactly. Since $Q_+ - Q_- = \Delta$ and $g'(Q) = 2 \log(2Q/\vartheta) > 0$,

$$\boxed{0 \leq V_s(t) - V_s(t + b) \leq 2b\Delta \log(2Q_+/\vartheta) + E_s^{\text{lo}}(t) + E_s^{\text{lo}}(t + b) = O_{\text{actual}}(q \log q).} \quad (\text{OW10})$$

The final order is evaluated, not imposed: $b, \Delta = O(j)$, $Q_{\pm} = \Theta(q)$, and OW7 is $O(j^2 \log q)$ at all four small determinant sizes in OW9.

OW11. High windows: the logarithmic scale cancels in the mixed derivative

Use exactly the smooth EIQ function $F(\alpha, \vartheta)$ from OFV2. Define

$$\begin{aligned} G(n, A) &= 2n(A + n) \log n + n^2 F(A/n, \vartheta), \\ H(a, Q) &= a(2Q + a) \log(a/2) + \frac{a^2}{2} F(2Q/a, \vartheta). \end{aligned} \quad (\text{OW11})$$

For $a \geq 1$, put $n_e = \lceil a/2 \rceil$, $n_o = \lfloor a/2 \rfloor$ and

$$R_H(a, Q) = C_0 a + \left| G(n_e, Q - \tfrac{1}{2}) + G(n_o, Q + \tfrac{1}{2}) - H(a, Q) \right|, \quad (\text{OW12})$$

where C_0 is precisely the finite QLG30 constant retained in OFV3. Here all high dimensions have $a \geq q$, so no $G(0, A)$ is used. The exact parity/Heine/scaling identities OFV19–20 and the finite partition estimate OFV3 prove

$$|\log \mathcal{A}_a(Q) - H(a, Q)| \leq R_H(a, Q). \quad (\text{OW13})$$

Under OW3, all dimensions on the rectangles used below satisfy $q \leq a \leq 17q/16$ and $15q/16 \leq Q \leq 17q/16$. After the half-integer parity corrections and on the straight rounding segments, A/n remains inside $[3/2, 5/2]$. The explicit derivatives G_A, G_n in OFV23 are bounded by $O(q \log q)$ throughout this compact-parameter region. Consequently $R_H(a, Q) = O(q \log q)$, including the actual parity and $1/n!$ factors.

Set

$$\begin{aligned} E_s^{\text{hi}}(t) &= C_s(t) + R_H(t + \Delta, Q_-) + R_H(t, Q_+), \\ \Phi_s(t) &= H(t + \Delta, Q_+ - \Delta) - H(t, Q_+). \end{aligned} \quad (\text{OW14})$$

OW6 and OW13 give $V_s(t) = \Delta \log \beta_s + \Phi_s(t) + r_s(t)$, with $|r_s(t)| \leq E_s^{\text{hi}}(t)$.

Here is the crucial direct derivative calculation. Write $\alpha = 2Q/a$, and evaluate $F, F_\alpha, F_{\alpha\alpha}$ at (α, ϑ) . Then

$$\begin{aligned} H_a &= (2Q + 2a) \log(a/2) + (2Q + a) + aF - QF_\alpha, \\ H_{aa} &= 2 \log(a/2) + 3 + \alpha + F - \alpha F_\alpha + \tfrac{1}{2} \alpha^2 F_{\alpha\alpha}, \\ H_{aQ} &= 2 \log(a/2) + 2 + F_\alpha - \alpha F_{\alpha\alpha}, \\ (\partial_a - \partial_Q) H_a &= 1 + \alpha + F - (\alpha + 1) F_\alpha + (\tfrac{1}{2} \alpha^2 + \alpha) F_{\alpha\alpha}. \end{aligned} \quad (\text{OW15})$$

In particular **both** $2 \log(a/2)$ **terms cancel exactly**. Define the actual finite constant

$$C_* = \sup_{3/2 \leq \alpha \leq 5/2} \left| 1 + \alpha + F(\alpha, \vartheta) - (\alpha + 1)F_\alpha(\alpha, \vartheta) + (\alpha^2/2 + \alpha)F_{\alpha\alpha}(\alpha, \vartheta) \right|. \quad (\text{OW16})$$

This is a specified finite integral-function constant from the proved smooth EIQ density on a compact interval, not an unknown row-energy estimate. No numerical value for it is claimed.

For $t = q, q + 1$, $w = 2b$, all points $(t + v + u, Q_+ - u)$, $0 \leq v \leq w$, $0 \leq u \leq \Delta$, lie in the displayed rectangle. Two applications of the fundamental theorem of calculus give

$$\begin{aligned} \Phi_s(t + w) - \Phi_s(t) &= \int_0^w \int_0^\Delta (H_{aa} - H_{aQ})(t + v + u, Q_+ - u) du dv, \\ |\Phi_s(t + w) - \Phi_s(t)| &\leq \Delta w C_*. \end{aligned} \quad (\text{OW17})$$

Thus the two actual high windows satisfy the finite bound

$$0 \leq V_s(t) - V_s(t + 2b) \leq 2b\Delta C_* + E_s^{\text{hi}}(t) + E_s^{\text{hi}}(t + 2b) = O_{\text{actual}}(q \log q). \quad (\text{OW18})$$

The smooth main term costs only $O(q)$. The displayed $O(q \log q)$ includes the separately retained original-root, original-Gamma, parity and finite-partition errors; none was canceled by an unsupported sign assumption.

Completed result and interface

Each of the four original outer-strip log-volume increments at exactly the displaced MNT17 degrees is $O_{\text{actual}}(q \log q) = o(jq)$. This holds in both original source orders, keeping all original roots, masses and finite coefficients. The signed four-window difference is bounded by the sum of the four explicit bounds OW10/OW18 and is therefore of the same subleading order.

This proves a genuine actual-source estimate beyond the earlier four-endpoint leading law. Passing it to the **native meromorphic** metric still requires the parent's exact original conductor/quotient comparison at these displaced degrees. This note does not claim that final identification has already been performed, nor that an allocation coefficient or an RH conclusion follows from this bound alone.

Original native endpoint-window transfer

This calculation addresses the displaced source cutoffs in MNT18. It does not change the degree, period, conductor, source order, or metric of a source before comparing two of its cutoffs. The outer-strip interval calculation is supplied separately in **independent**; its finite bounds must be read together with the transfer below. No separate row-energy allocation coefficient is assigned here.

NW1. Original data, fixed throughout each interval

Write the source degree as j (MNT's ℓ_h), and put

$$q = (j + 1)^2, \quad q' = (j - 7)^2, \quad \Delta = q - q' = 16j - 48, \quad m = \Delta - v, \quad s \in \{1, j\}.$$

Retain the actual fixed simple quartet, five-orbit period, logarithm branch, conductor A , its moments μ_r , and its first nonzero moment μ_v , where $0 \leq v \leq 80$. In particular retain the complete original mass $M_s = (2\pi)^{s/2}$, the centers $c = j/2$, $c' = j/2 - 4$, and every root in the upper and

lower grids. At every integer $N \geq q - 1$, put $d = N - v$ and $t = N - q + 1$. The notation χ' denotes the lower root polynomial, not a derivative.

The matrices used below are exactly CEP's primary quotient metrics G_N , G'_d , compressed conductor covariance $K_{A,N}$, and DNE7's positive native dual metric D_N . The source degree of that native metric is d , not N .

NW2. Why these constructions extend to the displaced cutoffs

The finite proofs CEP2-17,21-35,37-38,41-48 apply at every $N \geq q - 1$: root addition and monic division are polynomial identities; the source convolution has its nonzero diagonal $\mu_v \binom{v+a}{a}$; the lower Vandermonde has full rank since $d \geq q' - 1$; and every source/relation dimension remains nonnegative. Their proofs use no upper cutoff. Likewise the DNE5-8 quotient and duality proofs apply at all such cutoffs. CTV7-8,12-19 are finite monic Gram identities or moment inequalities on the same domain.

This observation does not extend an asymptotic statement by substitution. For the enlarged range $N \leq 2q + 2b$ used below, the finite constants are still

$$\begin{aligned} M_N &= \max\{1, \sum_{r,u} |a_{ru}| e^{|b_{ru}-4|(H_N+2)}\}, \\ L_N &= (N - v + 1) |\log |\mu_v/v!|| + v c_{s/2}^{\log} H_N, \\ E_N &= (N + 1) \log M_N + L_N, \quad c_{s/2}^{\log} = \max\{1, s/2 - 1\}. \end{aligned}$$

When $b = O(j)$ at the fixed original data, $N = O(q)$ and these displayed expressions give $E_N = O_{\text{actual}}(q \log(q + 1))$. The same direct substitution in CTV18 gives $C_{N,v}^{\text{tail}} = O_{\text{actual}}(q \log(q + 1))$. Their original masses and phases remain in the maps before these estimates.

NW3. The fixed map from the conductor quotient to the weighted relation quotient

Let $V_A = \ker(C : E_j \rightarrow E_{j-8})$, of dimension Δ , and let L be its original subspace represented by polynomials of degree below v . It embeds because $v < q$, and $\mathcal{T}_A$ kills precisely those polynomials. On V_A/L define

$$\Psi : [p] \mapsto [\mathcal{T}_A p / \chi'] \in Y_N.$$

Use the unique upper remainder p of degree below q to write this map in coordinates. Its numerator is divisible by χ' because $[p]$ belongs to V_A , and its quotient has degree at most $q - 1 - v - q' = m - 1$. It is therefore in DNE's fixed low-monomial frame at every cutoff. Changing a general source lift by $\chi_j h$ adds precisely $\mathcal{U}_A h$, by CEP42; changing it by a member of L changes nothing. This proves independence of both changes.

The map is onto: CEP43 supplies a preimage for every relation-quotient class. It is injective: if $\mathcal{T}_A p / \chi'$ is a relation, subtract the corresponding $\chi_j h$; the remaining polynomial is killed by $\mathcal{T}_A$ and thus has degree below v . Hence Ψ is an isomorphism. Choose a fixed frame of V_A/L ; let R_A be its invertible matrix in the low-monomial frame. The preceding degree calculation proves that R_A is independent of N , rather than assuming that the endpoint frames coincide.

NW4. The exact metric of that same map

Let $H_{V,N}$ be the restriction of G_N to a fixed primary frame of V_A . Let $H_{L,N}$ be its restriction to the fixed original frame of L , and let $Q_{A,N}$ be the quotient metric on V_A/L . Write

$$e_N = \log \det(R_A^* H_{Y,N} R_A) - \log \det Q_{A,N}.$$

The source map $S_N p = (P_{<v} p, \mathcal{T}_{Ap})$ is the exact CEP28 isomorphism. It maps the preimage of V_A onto $\mathbb{C}^v \oplus \chi' \mathcal{P}_{N-v-q'}$, and sends each upper relation to its full graph, not merely its second component. Taking the additional quotient by L kills the entire free first component, since $S_N L = \mathbb{C}^v \oplus 0$. The resulting quotient norm is precisely that of $\chi' \mathcal{P}$ modulo $\chi' \mathcal{U}_A \mathcal{P}$, namely $H_{Y,N}$ in the full weighted Gamma measure. This proves the target metric assertion.

The minimum-lift map followed by orthogonal projection off the whole kernel has exterior norm bounded by $\bigwedge^m S_N$; its inverse is bounded by $\bigwedge^m S_N^{-1}$, exactly as proved by CEP33-38. Squaring the exterior volumes yields

$$-2E_N \leq e_N \leq 2E_N. \quad (\text{NW4})$$

The map here is the literal unscaled CEP43 map. If the FGC convention $\alpha_A \mathcal{T}_A$, $\alpha_A = v!/\mu_v$, is used, its transported Gram is $|\alpha_A|^2 R_A^* H_Y R_A$; its logarithmic determinant acquires exactly $2m \log |\alpha_A|$. That is a fixed frame scalar and cancels between two cutoffs. It is not set equal to one, and it does not change this unscaled comparison.

NW5. Complete determinant identity, including the low metric

Let $V(t) = \log \det Q_{\partial, t+\Delta-1}$ be CTV12's actual outer-strip quotient volume, $R_N = \mathcal{R}_{N,v}^{(s)}$ its complete CTV13 norm tail, and $\kappa_N = \log \det K_{A,N}$. Then there is a constant C independent of N such that

$$\boxed{\log \det D_N = C - V(t) - R_N - \kappa_N + \log \det H_{L,N} - e_N.} \quad (\text{NW5})$$

Here is every determinant step. DNE8 gives $\log \det D_N = 2 \log |\det A_\ell| - \log \det H_{Y,N}$. From NW4, $\log \det H_{Y,N} = \log \det Q_{A,N} + e_N - 2 \log |\det R_A|$. Complete the fixed L frame by the fixed quotient lifts in V_A . Its block Gram gives $\log \det Q_{A,N} = \log \det H_{V,N} - \log \det H_{L,N} + C_1$, where C_1 is the fixed frame determinant. CEP26 gives $\log \det H_{V,N} = \log \det G_N - \log \det G'_d + \kappa_N + C_2$. Finally CTV8 and CTV13 give $\log \det G_N - \log \det G'_d = V(t) + R_N + C_3$, with both original Vandermonde factors in C_3 . Combining these equalities proves NW5. In particular no low-volume term or conductor covariance has disappeared.

The empty L case has determinant one, so its logarithm is zero.

NW6. A finite bound for any native interval

For integers $q-1 \leq a \leq b$, write

$$\Gamma(a, b) = \log \frac{\det D_b}{\det D_a}, \quad \mathcal{E}(a, b) = C_{a,v}^{\text{tail}} + C_{b,v}^{\text{tail}} + 4(E_a + E_b).$$

The native dual metric increases with N : each complete lower-extension candidate has an additional nonnegative sum of squared original functional values, and minimizing over the same full lower coefficient vector preserves the inequality. Hence $\Gamma(a, b) \geq 0$.

The primary quotient metrics G_N decrease because their admissible source lifts grow. Their restriction to the fixed L also decreases, so $\log \det H_{L,b} - \log \det H_{L,a} \leq 0$. Subtract NW5 at a and b , retaining that term before using its sign. CTV19 gives $|R_N| \leq C_{N,v}^{\text{tail}}$; CEP48 gives $|\kappa_N| \leq 2E_N$; NW4 bounds e_N . Therefore

$$\boxed{0 \leq \Gamma(a, b) \leq V(a-q+1) - V(b-q+1) + \mathcal{E}(a, b).} \quad (\text{NW6})$$

The outer difference is nonnegative because it too is an original minimum quotient norm on nested source spaces. This bound contains no rank multiplier on the conductor error. In particular multiplying a scalar inverse bound by the quadratic lower rank has not been used.

NW7. Every fixed native flag obeys the same interval bound

Let $D_{X,N}$ be any metric obtained from D_N by a fixed finite sequence of restrictions to specified subspaces and minimum quotients by specified onto maps. The maps may depend on j and all original data, but not on N . Put $\Gamma_X(a, b) = \log \det D_{X,b} - \log \det D_{X,a}$.

Both restriction and minimum over the same affine fibres preserve positive-form order. For one fixed subspace, complete its frame by fixed quotient lifts. The Schur determinant formula gives the whole determinant as the product of its restriction and quotient determinants times a constant frame factor. On subtracting between a and b , the frame factor cancels and both summands are nonnegative. Each summand is therefore at most the full increment. Iterating this argument along the flag proves

$$0 \leq \Gamma_X(a, b) \leq \Gamma(a, b). \quad (\text{NW7})$$

This applies in particular to the actual invariant restriction, the B_K quotient, its matched image, and its unmatched quotient. It uses their complete native metrics, not their ranks in place of their increments.

NW8. The literal displaced four-endpoint return

For one multiplier of degree $d_h \leq 28$, set $j = \ell_h = k - d_h$ and retain

$$b_h = q_k - q_j = d_h(2k - d_h + 2) = d_h(2j + d_h + 2).$$

The return of a fixed source flag is $F_X(q - 1) + F_X(q) - F_X(2q - 1) - F_X(2q)$ in the primal convention, and its negative in the native dual convention. In the native dual convention, the exact shifted-minus-unshifted difference from MNT18 is

$$\begin{aligned} \mathcal{S}_X &= \Gamma_X(2q - 1, 2q - 1 + 2b_h) + \Gamma_X(2q, 2q + 2b_h) \\ &\quad - \Gamma_X(q - 1, q - 1 + b_h) - \Gamma_X(q, q + b_h). \end{aligned} \quad (\text{NW8})$$

No displacement is made to the source order. Each term uses its source's original $s = 1$ or $s = j$, original period, and original q_j ; only its cutoff changes.

Let $U(a, b)$ denote the explicit right side of NW6, or any proved upper endpoint for its outer difference with the same $\mathcal{E}(a, b)$ retained. Then

$$\begin{aligned} -[U(q - 1, q - 1 + b_h) + U(q, q + b_h)] &\leq \mathcal{S}_X \\ &\leq U(2q - 1, 2q - 1 + 2b_h) + U(2q, 2q + 2b_h). \end{aligned} \quad (\text{NW9})$$

This is a direct finite bound for all four original windows, with their unequal endpoints and signs. It is valid also when low and high displaced intervals overlap. If $b_h = 0$, all four increments and the shift are exactly zero; no error bound is needed.

NW10. Exact overlap cancellation can tighten the finite bound

Use half-open row intervals $[a, b)$ so $\Gamma_X(a, b)$ is the sum of the original consecutive increments from a through $b - 1$. Form the signed integer weight γ_M of the four intervals in NW8. Split at their eight endpoints, combine coinciding endpoints, and retain the resulting constant weight on each nonempty segment. There are at most seven intervening segments. Since each individual row increment of the same fixed flag lies between zero and that of the full native metric, the shift lies between

$$\sum_{\gamma_I < 0} \gamma_I \Gamma(a_I, b_I) \quad \text{and} \quad \sum_{\gamma_I > 0} \gamma_I \Gamma(a_I, b_I). \quad (\text{NW10})$$

Using $U(a_I, b_I)$ in the signed lower sum and positive upper sum preserves these bounds. This cancels overlapping windows exactly; it never infers smallness from the number of rows alone.

NW11. Optional direct evaluation from the same conditioned rows

There is a finite check on any $\Gamma(a, b)$ independent of outer estimates. Split DNE's actual matrices as (T_a, V_a) and appended blocks (T_+, V_+) . Define

$$A_a = V_a^* V_a, \quad K_a = A_a^{-1} V_a^* T_a, \quad R = T_+ - V_+ K_a, \quad H = I + V_+ A_a^{-1} V_+^*.$$

Completing the minimum over the **one** original lower coefficient vector gives

$$D_b = D_a + R^* H^{-1} R, \quad \Gamma(a, b) = \log \frac{\det(H + R D_a^{-1} R^*)}{\det H}. \quad (\text{NW11})$$

To prove the first equality, write the lower coefficient as $-K_a x + \eta$. The old squared norm becomes $x^* D_a x + \eta^* A_a \eta$, and the added norm is $\|R x + V_+ \eta\|^2$. Minimizing their sum produces the displayed Schur complement. The second identity is the determinant lemma, or block determinant elimination in the two opposite orders. Both inverted Grams are positive. The square matrix in the ratio has size $b - a$, which is b_h or $2b_h$ for NW8, rather than a full q -length window. An empty block gives ratio one. This evaluates an original metric increment; it is not a surrogate row model.

NW12. The actual displaced windows are subleading

The complete companion proof OW1-18 is now read and accepted at its final source. Its two finite outer upper bounds are

$$\begin{aligned} U_{\text{lo}}(t) &= 2b_h \Delta \log(2Q_+/\vartheta) + E_s^{\text{lo}}(t) + E_s^{\text{lo}}(t + b_h) \quad (t = 0, 1), \\ U_{\text{hi}}(t) &= 2b_h \Delta C_* + E_s^{\text{hi}}(t) + E_s^{\text{hi}}(t + 2b_h) \quad (t = q, q + 1), \end{aligned} \quad (\text{NW12})$$

where $Q_+ = q + (s - 1)/4$, $\vartheta = \pi/2$, and every error and the finite smooth-function constant C_* is the literal OW5,7,12,14,16 expression. The full original mass contributes $\Delta \log \beta_s$ at each outer endpoint and cancels only in its interval difference. The high-window bound uses the directly computed cancellation of the two $2 \log(a/2)$ terms in $H_{aa} - H_{aQ}$, with the full finite partition and parity errors retained.

Use these bounds in NW6 at the four pairs in NW8, adding $\mathcal{E}(a, b)$ separately at each pair. This gives the explicit lower and upper bounds NW9 and proves

$$\boxed{|\mathcal{S}_X| = O_{\text{actual}}(q_j \log(q_j + 1)) = o(kq_k), \quad s = 1, j.} \quad (\text{NW13})$$

The asymptotic range is exactly OW3: $j \geq 4096$ and $j\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}(j + 1)^2$. It holds eventually at every fixed original quartet and period. There are finitely many fixed multiplier degrees $d_h \leq 28$; hence $j/k \rightarrow 1$, $q_j/q_k \rightarrow 1$, and $\log(q_j + 1)/k \rightarrow 0$, proving the second equality and uniformity over this finite family. The source flag may have growing rank and depend on the actual degree and period: NW7 never multiplies the bound by its rank. No unknown concentration estimate is an input to NW13.

At earlier finite degrees use NW6, NW8-11 with the actual original outer determinants. If $b_h = 0$ the shift is identically zero. This retains, in particular, the overlapping finite windows rather than pretending that the eventual threshold holds there.

Thus the displaced endpoint terms in MNT18 are now controlled at subleading order. This result does **not** bound the remaining matched relative determinant of MNT19, construct the native invariant inverse, or evaluate the separate leading allocation coefficient. Those remain the other task's explicit current calculations.

Scope and source record

The native transfer NW1-11 is a finite proof from the actual CEP, CTV and DNE maps. NW12-13 combines it with the completed outer-window proof, which was read in full together with its complete OSP and OFV direct providers. Their accepted EIQ/QLG estimates are used at

the stated scope, not independently re-proved here. The independent reviewer accepted NW1-11 before NW12-13 was appended; its receipt identifies that exact earlier scope. All finite formulas retain the actual low metric, source mass, fixed Vandermonde factors, α_A convention, and full lower-evaluation minimizer at the steps where they occur. Exact fixture checks are supplementary and do not certify actual transcendental period inputs or the analytic estimate.

Primary project sources read in full for this derivation: `ORIGINAL_CONDUCTOR_ENDPOINT_QUOTIENT.tex` (CEP1-49), `total_variation/ORIGINAL_TOTAL_VARIATION_CHRISTOFFEL.tex` (CTV1-22), and the accepted `ORIGINAL_WEIGHTED_RELATION_DUAL_EXTENSION.tex` (DNE1-15). MNT1-19, FRE1-10, ACB1-15, LVM1-19 and NPB1-19 were also read in full; the arbitrary-window Schur calculation is an explicitly shifted use of HJR's exact completion, not a claim that the Schur identity is new literature.

123 The full polar observation and its original low-jet flag

Keep the original fixed simple quartet, actual unit and five-orbit period, logarithm branch, all conductor coefficients, and both source orders $s = 1, k$. Put $k \equiv 1 \pmod{4}$, $k \geq 9$, $q = (k + 1)^2$, $q' = (k - 7)^2$, $\Delta = 16k - 48$, $r = 8k - 16$, $j = 8k - 32$. All four endpoint indices are $\mathcal{N} = (q - 1, q, 2q - 1, 2q)$, with primal signs $\sigma = (1, 1, -1, -1)$ and dual signs $-\sigma$. In this section w is the generating variable of RPK, leaving the absolute base τ unchanged. This is the literal substitution $w = \tau_{\text{RPK}}$ in its exponential functions and principal parts. The period is in either proved full-width domain of ASR14 or RPK's rank theorem. Equivalently, the construction below applies at any actual five-orbit period whose two conductor widths are both eight; that condition is established on those domains, rather than assumed for an unevaluated period outside them.

Use the exact original primary maps $C, D, \mathbf{N}, \mathbf{S}, Z$ of ROQ2-7. Thus $C^\top$ is multiplication by the whole actual biform A , $V_A = \ker C$, $K = \ker C \cap \ker D$, $\Xi = D|_{V_A}$ is onto $\mathbb{C}^j$, and $Z = \mathbf{N}D^\top$ has independent columns. Ordinary transpose is denoted by $\top$ and Hermitian adjoint by $*$. Write

$$\mathcal{C} = \mathbb{C}^q / \text{im } C^\top, \quad \mathcal{J} = \{[D^\top x] : x \in \mathbb{C}^j\} \subset \mathcal{C}, \quad \Theta[\nu](p) = \nu^\top p \quad (p \in V_A). \quad (\text{PLF1})$$

Here p denotes its original primary evaluation column. The map $\Theta : \mathcal{C} \rightarrow V_A^\vee$ is an isomorphism: its kernel is exactly the annihilator of $\ker C$, namely $\text{im } C^\top$, since C is onto. Every functional on V_A extends linearly to $\mathbb{C}^q$, which proves surjectivity. The same argument applied to $D|_{V_A}$ identifies $\Theta \mathcal{J} = \text{Ann } K$ and $\mathcal{J} \simeq (V_A/K)^\vee$. This is precisely the KRC3 isomorphism after its full original Segre and period substitution.

123.1 The pole map, without discarding the origin jets

Retain RPK's actual factorization $A = a_* \prod_\nu B_\nu^{m_\nu}$ and every selected regular nonzero sample $w_{\nu,l}$, together with all principal-part orders $1 \leq a \leq m_\nu$. Let Π_k be its complete principal-part matrix for $\mathcal{N}_\nu(w)/E_A(w)$, with the coefficient order of PLF1. RPK proves, by its componentwise resultant argument and repeated division at each regular sample, the exact identity

$$\ker \Pi_k = \text{im } C^\top, \quad \overline{\Pi}_k : \mathcal{C} \xrightarrow{\sim} \text{im } \Pi_k, \quad [\nu] \longmapsto \Pi_k \nu, \quad \text{rank } \Pi_k = \Delta. \quad (\text{PLF2})$$

In particular the proof retains every multiplicity m_ν : it uses vanishing of the entire principal part to divide successively by $B_\nu^{m_\nu}$. Invertibility of the second map follows directly from the displayed kernel and its image definition. Its complete finite inverse is obtained by choosing independent columns of $\Pi_k \mathbf{S}$, inverting the first nonsingular row minor, and applying $\mathbf{S}$ to the resulting strip coordinates. Here $\Pi_k \mathbf{S}$ already has exactly Δ independent columns, since $\ker \mathbf{N} = \text{im } C^\top$. No modulus bound on this inverse is inferred from its existence.

Choose exactly the j RPK rows that make

$$\mathbf{P}_k \in \text{Mat}_{j \times \text{rows } \Pi_k}, \quad B_k = \mathbf{P}_k \Pi_k D^\top \in \text{GL}_j, \quad b = B_k x. \quad (\text{PLF3})$$

The row selection exists because PLF2 is injective on $\mathcal{J}$. The coordinate b belongs to that full j -dimensional residual dual, with inverse $[D^\top B_k^{-1}b]$ in $\mathcal{J}$. Using selected rows supplies coordinates on this subspace; it does not replace the complete principal-part operator on $\mathcal{C}$. Both B_k and every sample are fixed as the source order and endpoint change.

Let $v = \text{ord}_0 E_A$, $\alpha_A = v!/\mu_v$, and let L be the original primary image of polynomials of degree below v . Its literal evaluation matrix is

$$\begin{aligned} I_L &= (\beta_{ab}^d)_{(a,b), 0 \leq d < v}, & M_v &= I_L^\top, & CI_L &= 0, \\ t_0 &= \dim(K \cap L), & w_0 &= v - t_0, & j_0 &= j - v + t_0. \end{aligned} \quad (\text{PLF4})$$

The first v moment identities give $CI_L = 0$. Distinct upper roots make I_L injective, by its first v Vandermonde rows. Thus these are the actual low directions inside V_A , and their intersection with K has the stated meaning. The zero-dimensional case uses the unique empty matrix throughout.

There is the full exact sequence

$$0 \longrightarrow \mathcal{J}_v \longrightarrow \mathcal{J} \xrightarrow{\text{res}_L} (L/(K \cap L))^\vee \longrightarrow 0, \quad \mathcal{J}_v = \text{Ann}(K + L), \quad \dim \mathcal{J}_v = j_0. \quad (\text{PLF5})$$

For a functional in $\text{Ann } K$, restriction to L kills $K \cap L$. Conversely a functional λ on L that kills $K \cap L$ defines a functional on $K + L$ by $k + \ell \mapsto \lambda(\ell)$. If two such decompositions differ, their L -parts differ in $K \cap L$, so this definition is well defined. Extend it to V_A . The extension kills K and has the required restriction, proving surjectivity. The kernel kills both K and L and hence their sum. Dimension subtraction proves the last assertion. This proof includes all possible t_0 and does not require a generic intersection.

In the full pole coordinates PLF3, the kernel in PLF5 is exactly

$$\mathcal{P}_{v,k} = \ker(M_v D^\top B_k^{-1}) \subset \mathbb{C}^j, \quad \text{rank}(M_v D^\top B_k^{-1}) = w_0. \quad (\text{PLF6})$$

Indeed $M_v D^\top x$ is the vector of the functional's values on the original monomials $1, S, \dots, S^{v-1}$. Setting these values to zero is exactly membership in $\mathcal{J}_v$. Its image is the w_0 -dimensional space in PLF5, embedded in $\mathbb{C}^v$ by that monomial basis. Thus the difference between the full pole rank j and the native high residual rank j_0 is precisely the original low quotient; neither rank is changed.

123.2 The exact native-to-pole coefficient matrix

Retain the entire DNE3 numerator frame Z_{exp} and the actual SAR1 high residual matrix $J = A_\ell^{-1} P_Y^\top$ with j_0 columns. KRC5 proves $\mathsf{T}_k = \alpha_A \mathsf{N} Z_{\text{exp}}$ and $\mathsf{T}_k \text{im } J = \mathsf{N} \mathcal{J}_v$. Choose the original nonsingular row minor Z_I of Z and let $L_Z = Z_I^{-1} E_I^\top$, so $L_Z Z = I_j$. Define

$$U = \alpha_A L_Z \mathsf{N} Z_{\text{exp}} J, \quad T_{\text{pole}} = B_k U = \alpha_A \mathsf{P}_k \Pi_k Z_{\text{exp}} J. \quad (\text{PLF7})$$

Both equalities are exact. For the second one, KRC5 puts every column of $\alpha_A \mathsf{N} Z_{\text{exp}} J$ in $\text{im } Z$; hence $ZU = \alpha_A \mathsf{N} Z_{\text{exp}} J$. Therefore $D^\top U - \alpha_A Z_{\text{exp}} J$ lies in $\ker \mathsf{N} = \text{im } C^\top$, which Π_k kills by PLF2. Applying $\mathsf{P}_k \Pi_k$ proves PLF7. Injectivity of T_k and J proves that U and T_{pole} have rank j_0 . Their images are the entire low-annihilating subspaces by PLF5–6. In particular

$$T_{\text{pole}} : \mathbb{C}^{j_0} \xrightarrow{\sim} \mathcal{P}_{v,k}, \quad [D^\top U b] = [\alpha_A Z_{\text{exp}} J b] \quad \text{in } \mathcal{C}. \quad (\text{PLF8})$$

The scalar α_A is retained; it is the pullback of $p \mapsto \alpha_A \mathcal{T}_A p / \psi$ in FGC4 and KRC6. The exponentials in PLF8 have all their original coefficients. No coefficient interpolation inverse is declared uniformly bounded.

123.3 The full conditioned pole Gram and its low Schur factor

For each original N, s , let $u_{n,s}$ be the original primary orthonormal Gamma columns, with the exact masses and norms $h_n = (2\pi)^{s/2} n! (s/2)_n$. Set

$$\begin{aligned} R_N &= \sum_{n=0}^N u_{n,s} u_{n,s}^*, & G_N &= R_N^{-1}, \\ \Omega_N &= D[R_N - R_N C^* (C R_N C^*)^{-1} C R_N] D^*, & Q_N &= \Omega_N^{-1}, \\ H_{\text{pole},N} &= B_k^{-*} \overline{\Omega_N} B_k^{-1}, & H_{\text{hi},N} &= T_{\text{pole}}^* H_{\text{pole},N} T_{\text{pole}} = U^* \overline{\Omega_N} U. \end{aligned} \quad (\text{PLF9})$$

ROQ15–16 proves that Q_N is the minimum original G_N norm on each fibre of $D|_{V_A}$. Its complex-linear dual has Gram $\overline{Q_N^{-1}} = \overline{\Omega_N}$: the squared norm of $y \mapsto x^\top y$ is $x^\top Q_N^{-1} \bar{x} = x^* \overline{Q_N^{-1}} x$. The coordinate substitution $x = B_k^{-1} b$ proves the full pole Gram, and PLF7 proves its restriction identity. The two conductor cross terms and their inverse are therefore retained in both metrics.

Here is a complete adapted frame, with no moving endpoint pivots. Choose independent columns E_0 spanning $W_0 = \Xi(L) = \text{im}(D I_L)$, so E_0 has w_0 columns. Choose a right inverse E_1 of the onto map $U^\top$. Then

$$E = [E_0, E_1] \in \text{GL}_j, \quad U^\top E = [0, I_{j_0}], \quad V = E^{-\top} \begin{pmatrix} I_{w_0} \\ 0 \end{pmatrix}, \quad \mathcal{F} = [B_k U, B_k V] \in \text{GL}_j. \quad (\text{PLF10})$$

The zero block holds because U annihilates W_0 . Applying $U^\top$ to a relation among the columns of E first kills the E_1 coefficients, and then independence of E_0 kills the rest. This proves invertibility and also shows that the last j_0 columns of $E^{-\top}$ are exactly U . All matrices can be calculated by the first-nonzero-minor rule from these actual coefficient maps. The rank-zero cases use the same formulas with empty blocks.

Put

$$\begin{aligned} H_{W_0,N} &= E_0^* Q_N E_0, & Q_{\text{out},N} &= E_1^* Q_N E_1 - E_1^* Q_N E_0 H_{W_0,N}^{-1} E_0^* Q_N E_1, \\ A_N &= H_{\text{hi},N}, & B_N^{\text{mix}} &= U^* \overline{\Omega_N} V, & C_N^{\text{low}} &= V^* \overline{\Omega_N} V, \\ S_N^{\text{low}} &= C_N^{\text{low}} - (B_N^{\text{mix}})^* A_N^{-1} B_N^{\text{mix}}. \end{aligned} \quad (\text{PLF11})$$

The matrices $H_{W_0,N}$ and $Q_{\text{out},N}$ are respectively the restriction to W_0 of the original residual metric and its quotient by W_0 . Indeed the second formula is obtained by minimizing over the entire E_0 coefficient in $E_1 z + W_0$; differentiating its positive quadratic form gives the displayed Schur complement. Quotienting first by K and then by W_0 is the same minimum as quotienting V_A by $K + L$: the union of the K -fibres over one W_0 coset is precisely one $(K + L)$ -fibre. Similarly the W_0 restriction is the original quotient metric on $(K + L)/K$, with every cross term in G_N retained.

The complete block inverse identity gives

$$H_{\text{hi},N} = \overline{Q_{\text{out},N}^{-1}}, \quad S_N^{\text{low}} = \overline{H_{W_0,N}^{-1}}, \quad \det H_{\text{pole},N} = |\det \mathcal{F}|^{-2} \det H_{\text{hi},N} \det H_{W_0,N}^{-1}. \quad (\text{PLF12})$$

For a direct proof, write $E^* Q_N E$ in its (E_0, E_1) blocks as $\begin{pmatrix} A & B \\ B^* & C \end{pmatrix}$. Its inverse has bottom-right block $(C - B^* A^{-1} B)^{-1}$, as verified by multiplying the usual triangular block elimination matrices. Its Schur complement after the bottom-right block is A^{-1} : multiplying the same elimination factors in reverse order, or inverting once again, verifies this equality. Conjugating every entry and using $E^{-\top} = [V, U]$ gives exactly the first two identities of PLF12. The full Gram in the reversed dual frame $\mathcal{F}$ has blocks $\begin{pmatrix} A_N & B_N^{\text{mix}} \\ (B_N^{\text{mix}})^* & C_N^{\text{low}} \end{pmatrix}$. Its determinant is $\det A_N \det S_N^{\text{low}}$. Congruence by $\mathcal{F}$ contributes $|\det \mathcal{F}|^2$, proving the last identity. In particular $|\det \mathcal{F}|^2 = |\det B_k|^2 |\det E|^{-2}$; the order permutation has absolute determinant one. No fixed frame factor is set to one at an individual endpoint.

Define the original high dual return $\mathcal{V}_{\text{hi},k,s} = -\sum_N \sigma_N \log \det H_{\text{hi},N}$. Taking the same four endpoints in PLF12 cancels only the fixed $\mathcal{F}$ determinant and gives the exact identity

$$\Phi_{k,s} = -\sum_N \sigma_N \log \det H_{\text{pole},N} = F_{W_0}^{(s)} + \mathcal{V}_{\text{hi},k,s}, \quad F_{W_0}^{(s)} = \sum_N \sigma_N \log \det H_{W_0,N}. \quad (\text{PLF13})$$

The first equality also follows directly from PLF9, since $\det \overline{\Omega_N} = \det \Omega_N$ is positive real. This supplies the exact missing low-volume term between the full RPK rank- j metric and the rank- j_0 high residual. It remains valid when v or t_0 is zero and on every nonreduced factor fibre.

123.4 The native Gamma map and its explicit exterior error

The native high residual metric is exactly $D_{R,N} = J^* \mathcal{D}_N^{(s)} J$ of DNE7 and SAR1. The operator $\mathcal{D}_N^{(s)}$ takes the minimum over all original lower-grid evaluation extensions at the original lower centre and cutoff $N - v$. We now relate it to PLF9 rather than replacing one by the other. The original conductor source isomorphism is

$$p \mapsto (P_{<v}p, \mathcal{T}_{Ap}), \quad [p] \mapsto [(P_{<v}p, \alpha_A \mathcal{T}_{Ap}/\psi)]. \quad (\text{PLF14})$$

The second target has its full relation graph $\{(P_{<v}(\chi h), \alpha_A \mathcal{T}_A(\chi h)/\psi)\}$. Its native direct-sum metric has the original low norm and lower norm $\|\psi z\|_{s,c'}^2$, with $c' = k/2 - 4$. Multiplication by ψ identifies that lower weighted space isometrically with the corresponding unweighted subspace. Quotienting by the graph and then the entire low coordinate gives the full relation quotient of FGC5; quotienting further by the image of K gives its residual quotient. The pullback of its ordinary dual on the frame J is exactly $\alpha_A Z_{\text{exp}} J$ by KRC6, and is therefore the frame U of PLF7 on the original side. This proves the typed map between these two precise residual dual metrics.

The complete CEP source estimates at this same N, s give upper bounds E_N^{CEP} for the logarithms of every relevant exterior norm of the unscaled source isomorphism and its inverse. Scaling its second summand by α_A has exterior norm and inverse exterior norm at rank j_0 at most $\exp(j_0 |\log |\alpha_A||)$. A restriction followed by a quotient cannot increase these exterior bounds: represent each quotient vector by its minimum lift, apply the original map and project onto the target minimum-lift complement, whose every exterior norm is at most one. Apply the same argument to the inverse. Complex-linear duality interchanges the two operator bounds. Squaring the volume bounds gives the exact finite result

$$\begin{aligned} e_{N,k,s}^{\text{polar}} &= \log \det H_{\text{hi},N} - \log \det D_{R,N}, \\ |e_{N,k,s}^{\text{polar}}| &\leq 2E_N^{\text{CEP}} + 2j_0 |\log |\alpha_A||, \\ \delta_{0,k,s} &= -\sum_N \sigma_N e_{N,k,s}^{\text{polar}}, \quad |\delta_{0,k,s}| \leq 2 \sum_N E_N^{\text{CEP}} + 8j_0 |\log |\alpha_A||. \end{aligned} \quad (\text{PLF15})$$

All relation columns, including the original low graph component, are used before these quotients. This argument has neither removed their cross terms nor claimed equality of the two source norms.

By SAR2 the native high dual return is $S_{R,s} = -\sum_N \sigma_N \log \det D_{R,N}$. Thus PLF13–15 give exactly

$$\boxed{\Phi_{k,s} = F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}.} \quad (\text{PLF16})$$

Its error is exactly the existing SAR10 allowance, now realized by explicit original pole and native matrices. Every bound for the actual $S_{R,s}$ in SAR3–9,15–16 therefore applies to the full pole determinant after addition of $F_{W_0}^{(s)}$ and this same error. In particular if $[a_R, b_R]$ is any of those already-proved finite intervals, then $F_{W_0}^{(s)} + a_R - E_{k,s}^{\text{tr}} \leq \Phi_{k,s} \leq F_{W_0}^{(s)} + b_R + E_{k,s}^{\text{tr}}$ follows by addition to the exact equality. The unequal original flag, arithmetic, signed and action errors in SAR10–14 remain unchanged.

123.5 The algebraic correspondences in these same coordinates

The full branch quotient maps of KRC7–12 have the exact pole presentation

$$\delta_k^{\text{pole}} = \delta_k \bar{\Pi}_k^{-1} : \text{im } \Pi_k \longrightarrow \mathcal{D}_{1,k} \oplus \mathcal{D}_{2,k}. \quad (\text{PLF17})$$

This is well defined by PLF2. Its kernel is $\bar{\Pi}_k(\ker \delta_k)$; the corresponding four-component map has kernel $\bar{\Pi}_k((\bigcap_a (Q, Q_a))_k / (Q, \mathcal{A})_k)$. These equalities follow by applying the inverse in PLF2 and retain all scheme-theoretic intersections. In particular δ_k^{pole} kills the full invariant residual image, and its restriction through T_{pole} gives the same zero map as KRC11. The plus-to-minus branch correspondence of KRC14–20 can be transported by PLF8; its two inherited metrics remain the complete KRC19 quotient metrics. A coordinate isomorphism alone supplies no inverse norm bound for that correspondence.

For this paragraph use precisely BIM’s full-family domain $k \equiv 1 \pmod{4}$, $k \geq 37$. For each actual bounded multiplier h from BIM’s invariant-source family, $\deg h \in \{0, 4, \dots, 28\}$ and $\ell = k - \deg h \geq 9$ remains congruent to one modulo four. The full pole map at degrees ℓ and k is exactly

$$\bar{\Pi}_k \mu_h \bar{\Pi}_\ell^{-1} : \text{im } \Pi_\ell \longrightarrow \text{im } \Pi_k. \quad (\text{PLF18})$$

The earlier degrees 9, 13, $\dots$, 33 retain their original finite maps and are not assigned this full multiplier family. Multiplication sends $AB_{\ell-8}$ into AB_{k-8} , proving that μ_h is defined on the displayed full quotients. The image of the low-annihilator maps to the low-annihilator because its numerator is multiplied by the entire original finite exponential $\mathcal{N}_h$. On the individual low-jet invariant domain, composition with PLF7 is therefore the literal MNT11 map in pole coordinates. On BIM’s larger coupled domain its low jets must still be summed with the complete section and constraint of BIM14–17; no analytic germ is assigned to an individual nonanalytic summand. The individual-image defect and its unmatched native quotient remain the exact BIM and FRE dimensions. FRE’s determinant estimate applies to their unchanged native flag after this coordinate transport; the coordinate entries of a pole inverse do not add a new estimate for MNT’s matched inverse operator.

Finally let $\mathcal{B}_{a,b}$ denote biforms of bidegree (a, b) , with zero space when either degree is negative, and let $\mathcal{B}_k = \mathcal{B}_{k,k}$. Homogenize each affine factor in its own two degrees. Let $A_{\text{sf}} = \prod_\nu B_\nu$. This names exactly RPK’s A_{red} . ASR6 instead removes boundary monomials from the original A ; denote that particular result here by A_{bnd} . On the full-width locus those boundary monomials are one, so

$$A_{\text{bnd}} = A = a_* A_{\text{sf}} \prod_\nu B_\nu^{m_\nu - 1}. \quad (\text{PLF19})$$

There is a canonical onto map $\mathcal{B}_k / AB_{k-8} \rightarrow \mathcal{B}_k / (A_{\text{sf}} \mathcal{B}_{k-a_{\text{sf}}, k-b_{\text{sf}}})$, induced by the identity on coefficients, since the first denominator is contained in the second. Its kernel is the quotient of these two displayed denominator subspaces. Thus the reduction of multiplicities appearing in RPK’s action-leakage rank is explicit; the full pole metric and its low flag continue to use the entire A , not this reduced target. Empty or exceptional multiplicity patterns remain in the factorization and the same quotient formula.

The new polar realization therefore carries the complete current rank- j residual and its rank- j_0 native high part through exact coefficient, low-jet and metric maps. The original separate allocation, its low volume, its source transport and its whole same-class action remain the quantities already calculated in SAR.

124 The unchanged source flag at its displaced cutoffs

Keep the complete original simple quartet, amplitude, unit, five-orbit period, logarithm branch, conductor coefficients and first nonzero moment of NMR. The source family is the exact BIM/MNT

family $\mathcal{H}$ on $k \equiv 1 \pmod{4}$, $k \geq 37$. Put

$$\begin{aligned}
j_h &= k - d_h, \quad d_h = \deg h \in \{0, 4, \dots, 28\}, \quad q_j = (j+1)^2, \quad q'_j = (j-7)^2, \quad \Delta_j = 16j - 48, \\
b_h &= q_k - q_{j_h} = d_h(2k - d_h + 2) = d_h(2j_h + d_h + 2), \quad v = \text{ord}_0 E_A, \quad \alpha_A = v!/\mu_v, \\
(s_{j_h}, s) &= (1, 1) \text{ or } (j_h, k), \quad c'_j = j/2 - 4, \quad D = N - v, \\
\mathbb{T}_j &= \alpha_A N_j Z_{\text{exp}, j}, \quad J_j = A_{\ell, j}^{-1} P_{Y, j}^\top, \quad B_j = Z_j^\top A_{\ell, j}, \quad \text{im } J_j = \ker B_j, \\
D_{R, j, N}^{(s_j)} &= J_j^* D_{j, N-v}^{(s_j)} J_j, \quad f_{h, s}(N) = \log \det D_{R, j_h, N}^{(s_{j_h})}.
\end{aligned} \tag{WNR1}$$

The superscript s_j records the unchanged original Gamma order at degree j . In particular the source of the degree-order map has order j_h , and its target has order k . Each $D_{j, N-v}$ is the full minimum lower-evaluation extension metric of MNT12, whose polynomial cutoff is $N - v$ and whose original lower coefficient vector has length q'_j . The map J_j is independent of N . Its image in the full native quotient is the invariant low-jet space because $\mathbb{T}_j \text{im } J_j = \mathcal{J}_{v, j}$ (NMR1). Thus restriction by this literal J_j is precisely one of the fixed native flags in NW7. This proves the typed receiving map from NW's $D_{X, N}$ to the particular metric in NMR6, including its domain, codomain and common cutoff convention. It does not replace a native metric by a pole norm.

Write $\sigma = (\delta^2 + \gamma^2)^{1/2}$. All finite native statements below hold for $j_h \geq 9$, as in MNT. The explicit outer error estimates used at a nonempty displaced window have the additional domain

$$\mathcal{D}_W : \quad j_h \geq 4096, \quad j_h \sigma \leq 2^{-33} (j_h + 1)^2 \quad \text{for every } h \text{ with } d_h > 0. \tag{WNR2}$$

The unit has $b_h = 0$ and contributes identically zero on the entire original MNT domain. If the nonzero-degree family is nonempty, put $d_{\max} = \max_h d_h$ and $j_{\min} = k - d_{\max}$. Then WNR2 is equivalent to its two inequalities at $j_{\min}$. Indeed all source degrees are at least $j_{\min}$, and $(j+1)^2/j = j+2+1/j$ is strictly increasing for real $j > 1$ by differentiation. For $d_{\max} = 28$ the smallest congruent target that can satisfy its degree inequality is $k = 4125$, at $j_{\min} = 4097$; the displayed quartet inequality remains separately necessary for this estimate. No earlier finite degree is assigned this threshold by changing its source order or cutoff.

125 Every finite native and outer endpoint error

For a fixed source (j, s_j) write $q = q_j$, $q' = q'_j$, $\Delta = \Delta_j$, $b = b_h$, and $\nu = s_j/2$. All conductor coefficients a_{ru}, b_{ru} below are the original CEP coefficients; the letter b for the displacement is distinct from every b_{ru} . For $N \geq q - 1$ define the complete constants

$$\begin{aligned}
H_N &= \sum_{n=1}^N n^{-1}, \quad c_\nu^{\log} = \max\{1, \nu - 1\}, \\
M_{j, s_j, N}^{\text{CEP}} &= \max\{1, \sum_{r, u} |a_{ru}| e^{|b_{ru} - 4|(H_N + 2)}\}, \\
L_{j, s_j, N}^{\text{CEP}} &= (N - v + 1) |\log |\mu_v/v|| + v c_\nu^{\log} H_N, \\
E_{j, s_j, N}^{\text{CEP}} &= (N + 1) \log M_{j, s_j, N}^{\text{CEP}} + L_{j, s_j, N}^{\text{CEP}}, \\
R'_j &= (j - 8)\sigma, \\
L_{j, s_j, N}^{\text{tail}} &= c_\nu^{\log} H_N + \frac{\nu}{2} \log 2 + 2N \log(8/\pi) + \frac{\pi R'_j}{4}, \\
U_{j, s_j, N}^{\text{tail}} &= q' (\log N + \log(\nu + N - 1) + 2|\log \delta|), \\
C_{j, s_j, N, v}^{\text{tail}} &= v \max\{L_{j, s_j, N}^{\text{tail}}, U_{j, s_j, N}^{\text{tail}}\}, \\
\mathcal{E}_{j, s_j}(a, c) &= C_{j, s_j, a, v}^{\text{tail}} + C_{j, s_j, c, v}^{\text{tail}} + 4(E_{j, s_j, a}^{\text{CEP}} + E_{j, s_j, c}^{\text{CEP}}).
\end{aligned} \tag{WNR3}$$

Here a, c are integer endpoints satisfying $q - 1 \leq a \leq c$; $H_0 = 0$. These are the literal NW2/NW6 and CTV18 constants, with subscripts making their source dependence explicit. In the empty

low-jet case the tails vanish by the empty sum, not by deletion of a nonzero moment. The original full mass remains $M_{s_j} = (2\pi)^{s_j/2}$ in both the source and its weighted relation norms; it cancels in CTV17 only after those two full norms are divided.

For clarity the exact native-to-outer determinant identity used here is recorded before its inequality. Let $V_{j,s_j}(t)$ be the actual outer quotient logarithm

$$V_{j,s_j}(t) = \log D_{t+\Delta}(P_{j-8}; s_j) - \log D_t(P_j; s_j), \quad t \geq 0, \quad (\text{WNR4})$$

where $P_{j-8}P_{\partial,j} = P_j$ with the original monic constant one, centers and phases. Its source is $\mathbb{C}[y]_{\leq t+\Delta-1}$ in the measure $|P_{j-8}(y)|^2 dm_{s_j}(y)$, and its target is the increasing remainder frame of $\mathbb{C}[y]/(P_{\partial,j})$. NW3 constructs the fixed isomorphism $V_A/L \rightarrow Y_N$, $[p] \mapsto [\mathcal{T}_{AP}/\chi']$; χ' is the lower polynomial, and L consists of degrees below v . An upper lift change $\chi_j h$ adds the exact relation $\mathcal{U}_A h$, and a change by L has zero image. Its fixed matrix R_A follows because an upper remainder below degree q maps to degree at most $\Delta - v - 1$. Taking the quotient of the whole CEP graph by L kills its entire first component and gives the full weighted metric on Y_N . NW4 proves the determinant error $|e_N| \leq 2E_{j,s_j,N}^{\text{CEP}}$ for this map. The DNE complex-linear dual metric is $A_\ell^* H_{Y,N}^{-1} A_\ell$; its determinant is $|\det A_\ell|^2 / \det H_{Y,N}$. Completing the fixed L frame and the fixed quotient lifts, and then using CEP26 and CTV13, therefore gives exactly

$$\log \det D_{j,N-v}^{(s_j)} = C - V_{j,s_j}(N - q + 1) - \mathcal{R}_{N,v}^{(s_j)} - \kappa_N + \log \det H_{L,N} - e_N. \quad (\text{WNR5})$$

The fixed C contains both original Vandermonde factors, R_A and A_ℓ . They cancel between two cutoffs. The full low restriction remains with a plus sign. The scaled FGC map adds exactly $2(\Delta - v) \log |\alpha_A|$ to its transported determinant, also independent of N ; that proved scalar cancellation makes no claim that α_A equals one.

For any fixed native flag X put $\Gamma_X(a, c) = \log \det D_{X,c} - \log \det D_{X,a}$. Increasing the polynomial cutoff appends nonnegative squared values for each vector of all original lower coefficients, so minimization over that same vector gives $D_c \succeq D_a$. Restriction and minimum quotient on fixed fibres preserve this order. In a fixed adapted frame, Schur determinants split the parent increment into the sum of its nonnegative restriction and quotient increments. Iteration gives $0 \leq \Gamma_X(a, c) \leq \Gamma(a, c)$ for the full native increment Γ . Meanwhile the primary $H_{L,N}$ decreases, since its source admits every previous lift. Subtracting WNR5, and only then using that sign, the CTV tail bounds, $|\kappa_N| \leq 2E_N^{\text{CEP}}$ and $|e_N| \leq 2E_N^{\text{CEP}}$ proves

$$0 \leq \Gamma_X(a, c) \leq U_{j,s_j}^{\text{exact}}(a, c) := V_{j,s_j}(a - q + 1) - V_{j,s_j}(c - q + 1) + \mathcal{E}_{j,s_j}(a, c). \quad (\text{WNR6})$$

This is valid for every original finite interval in question. The right side is nonnegative; the outer quotient also decreases by admission of its old lifts. No factor equal to a growing flag rank has been introduced in this proof.

The following definitions specify every extra OW endpoint error used on WNR2. Let $\epsilon = 2^{-32}$, $T = \epsilon q$, $R = j\sigma$, $\vartheta = \pi/2$, $\ell = (s_j - 1)/4$, $Q_+ = q + \ell$, $Q_- = q' + \ell$, and put

$$\begin{aligned} C_L &= \pi/\Gamma(3/4)^2, \quad K_q = \frac{\sqrt{2\pi} (2q+1)^2}{C_L q} e^{-2q}, \\ \mathcal{E}_j^{\text{OSP}} &= \frac{qR^2}{\epsilon^2 q^2 - R^2} + \log(1 + K_q), \\ r_0 &= 0, \quad r_\ell = \frac{\ell(16\ell^2 - 12\ell - 1)}{12T^2} + \log(1 + K_q) \quad (\ell > 0), \\ h_q &= 1/q, \quad L_{h_q} = C_L (\sin h_q)^{3/2}, \quad U_{h_q} = \frac{\Gamma(1/4)^2}{2\pi} (\sin h_q)^{-1/2}, \\ B_a(Q) &= a \max\{|\log L_{h_q}|, |\log U_{h_q}|\} + a(2Q + a) \frac{h_q}{\vartheta - h_q}, \quad B_0(Q) = 0, \\ C_{s_j}(t) &= (2t + \Delta)(\mathcal{E}_j^{\text{OSP}} + r_\ell) + B_{t+\Delta}(Q_-) + B_t(Q_+). \end{aligned} \quad (\text{WNR7})$$

These names retain the exact OSP24 and OFV6/8/12 values. In particular the two coefficients $\epsilon = 2^{-32}$ and 2^{-33} in WNR2 have their distinct original roles. For $a \geq 1$ define

$$\begin{aligned}
L(a, Q) &= \max\{a\vartheta + a \log(2a) + 2a(a-1) \log 2, \\
&\quad a(|\log(2/\vartheta) + 1| + \log(2(Q+a))) + a(a-1) \log(2(Q+a)/\vartheta)\}, \\
L(0, Q) &= 0, \quad E_{s_j}^{\text{lo}}(t) = C_{s_j}(t) + L(t + \Delta, Q_-) + L(t, Q_+), \\
G(n, A) &= 2n(A+n) \log n + n^2 F(A/n, \vartheta), \\
H(a, Q) &= a(2Q+a) \log(a/2) + \frac{a^2}{2} F(2Q/a, \vartheta), \\
R_H(a, Q) &= C_0 a + |G(\lceil a/2 \rceil, Q - \frac{1}{2}) + G(\lfloor a/2 \rfloor, Q + \frac{1}{2}) - H(a, Q)|, \\
E_{s_j}^{\text{hi}}(t) &= C_{s_j}(t) + R_H(t + \Delta, Q_-) + R_H(t, Q_+), \\
C_* &= \sup_{3/2 \leq \alpha \leq 5/2} |1 + \alpha + F(\alpha, \vartheta) - (\alpha + 1)F_\alpha(\alpha, \vartheta) + (\alpha^2/2 + \alpha)F_{\alpha\alpha}(\alpha, \vartheta)|.
\end{aligned} \tag{WNR8}$$

The function F and constant C_0 are the complete accepted EIQ/QLG providers used in OFV2-3: explicitly

$$F(\alpha, \beta) = \iint \log|x-y| d\mu_{\alpha,\beta}(x) d\mu_{\alpha,\beta}(y) - \int (\beta\sqrt{x} - \alpha \log x) d\mu_{\alpha,\beta}(x).$$

The constant C_0 is the original finite QLG30 constant for $|\log J_n - n^2 F| \leq C_0 n$ on $[3/2, 5/2] \times [\pi/2, 3\pi/2]$. The fully proved smooth EIQ density and full finite QLG estimate are retained dependencies, not new hypotheses about row energies. Every use of G in R_H below has $a \geq q$, so its arguments are strictly positive. The integer parity choices and both half-integer shifts are retained.

Define the two outer upper endpoints

$$\begin{aligned}
O_{h,s}^{\text{lo}}(t) &= 2b_h \Delta_j \log(2Q_+/\vartheta) + E_{s_j}^{\text{lo}}(t) + E_{s_j}^{\text{lo}}(t + b_h) \quad (t = 0, 1), \\
O_{h,s}^{\text{hi}}(t) &= 2b_h \Delta_j C_* + E_{s_j}^{\text{hi}}(t) + E_{s_j}^{\text{hi}}(t + 2b_h) \quad (t = q_j, q_j + 1).
\end{aligned} \tag{WNR9}$$

OW10 and OW18 prove that these respectively bound $V(t) - V(t + b_h)$ and $V(t) - V(t + 2b_h)$. Briefly, the complete original root and Gamma comparisons give error $C_{s_j}(t)$ and mass term $\Delta_j \log \beta_{s_j}$, where $\beta_{s_j} = M_{s_j}/(\sqrt{2}\Gamma(s_j/2))$. The low determinant has main term $ag(Q)$, $g(Q) = 2Q \log(2Q/\vartheta) - 2Q$, with error $L(a, Q)$. Subtracting its two endpoints cancels exactly the fixed mass and $\Delta_j g(Q_-)$; integration of $g'(Q) = 2 \log(2Q/\vartheta)$ gives the first line of WNR9. For the high determinant the exact parity partition errors are R_H . Direct differentiation gives $H_{aa} - H_{aQ} = 1 + \alpha + F - (\alpha + 1)F_\alpha + (\alpha^2/2 + \alpha)F_{\alpha\alpha}$. The two $2 \log(a/2)$ terms cancel. Integration over the literal rectangle $(a, Q) = (t + z + u, Q_+ - u)$, $0 \leq z \leq 2b_h$, $0 \leq u \leq \Delta_j$, costs $2b_h \Delta_j C_*$. Its full parameter domain is verified by $b_h \leq 57j$ and $\Delta_j + 2b_h + 1 \leq 130j + 1 \leq q_j/16$ on WNR2. This proves the second line with its complete endpoint errors, and no unsupported derivative of an asymptotic law.

126 The asymmetric correction and the original source-kernel return

On WNR2 set

$$\begin{aligned}
U_{h,s}^{\text{lo}}(t) &= O_{h,s}^{\text{lo}}(t) + \mathcal{E}_{j_h, s_{j_h}}(q_{j_h} - 1 + t, q_{j_h} - 1 + t + b_h), \quad t = 0, 1, \\
U_{h,s}^{\text{hi}}(t) &= O_{h,s}^{\text{hi}}(t) + \mathcal{E}_{j_h, s_{j_h}}(q_{j_h} - 1 + t, q_{j_h} - 1 + t + 2b_h), \quad t = q_{j_h}, q_{j_h} + 1, \\
\Omega_{k,s}^{\text{lo}} &= \sum_{h:d_h > 0} (U_{h,s}^{\text{lo}}(0) + U_{h,s}^{\text{lo}}(1)), \\
\Omega_{k,s}^{\text{hi}} &= \sum_{h:d_h > 0} (U_{h,s}^{\text{hi}}(q_{j_h}) + U_{h,s}^{\text{hi}}(q_{j_h} + 1)).
\end{aligned} \tag{WNR10}$$

The zero displacement contributes zero to both sums. Outside WNR2, define the same four U 's by WNR6 at their displayed literal endpoint pairs instead of WNR9. This gives finite nonnegative $\Omega^{\text{lo}}, \Omega^{\text{hi}}$ for every $k \geq 37$; only the eventual order estimate below has the WNR2 domain.

In WNR1's fixed restriction define $\Gamma_{h,s}(a, c) = f_{h,s}(c) - f_{h,s}(a)$. Then MNT18 and NMR6 are exactly

$$\begin{aligned} \mathcal{W}_{h,k,s} &= \Gamma_{h,s}(2q_j - 1, 2q_j - 1 + 2b_h) + \Gamma_{h,s}(2q_j, 2q_j + 2b_h) \\ &\quad - \Gamma_{h,s}(q_j - 1, q_j - 1 + b_h) - \Gamma_{h,s}(q_j, q_j + b_h), \\ -\Omega_{k,s}^{\text{lo}} \leq \mathcal{W}_{k,s} &:= \sum_h \mathcal{W}_{h,k,s} \leq \Omega_{k,s}^{\text{hi}}. \end{aligned} \tag{WNR11}$$

Here $j = j_h$ in the first two lines. Each increment belongs to the interval from zero to the appropriate U by WNR6 and WNR9. For the lower bound retain zero for both positive high increments and their negative upper endpoints for the two low increments. For the upper bound make the opposite choices. Summing proves WNR11. This proof applies also when the finite low and high intervals overlap. A sharper fully finite variant collects their signed integer weights on the consecutive subintervals determined by their eight endpoints and applies WNR6 to each; this is exactly NW10 and does not infer a small norm from a small row count.

The complete source-kernel matrix remains $H_{N,s}^{\text{ker}\Psi} = Z_{\text{ker}\Psi,k}^* \mathcal{G}_{N,s} Z_{\text{ker}\Psi,k}$ with the full direct sum $\mathcal{G}_{N,s}$ of NMR4. The quotient is the same attained metric $Q_{N,s}^{\text{src}} = (\tilde{\Psi}_k \mathcal{G}_{N,s}^{-1} \tilde{\Psi}_k^*)^{-1}$. For the fixed right inverse and full kernel frame, Schur factorization gives

$$\det \mathcal{G} = |\det[Z_{\text{ker}\Psi,k}, R_{\Psi,k}]|^{-2} \det H^{\text{ker}\Psi} \det Q^{\text{src}}.$$

Taking the four target signs cancels precisely that fixed frame factor. Define the exact retained center

$$\begin{aligned} A_{k,s}^{\text{W}} &= \sum_h S_{R,j_h,s,j_h} - V_{\text{ker}\Psi,k,s}, \quad Z_{k,s}^{\text{W}} = A_{k,s}^{\text{W}} + n_{k,s}^{\text{mult}}, \\ V_{Q,k,s} &= A_{k,s}^{\text{W}} + \mathcal{W}_{k,s}, \quad \max\{0, A_{k,s}^{\text{W}} - \Omega_{k,s}^{\text{lo}}\} \leq V_{Q,k,s} \leq A_{k,s}^{\text{W}} + \Omega_{k,s}^{\text{hi}}, \\ n_{k,s}^{\text{mult}} &= N_{q_k-1,s}^{\text{mult}} + N_{q_k,s}^{\text{mult}} - N_{2q_k-1,s}^{\text{mult}} - N_{2q_k,s}^{\text{mult}}, \quad N_{N,s}^{\text{mult}} = \sum_{i=1}^{a_k} \log^+(1/\lambda_i(\mathcal{R}_{N,s})). \end{aligned} \tag{WNR12}$$

The equality is NMR6 with WNR11, and $V_Q \geq 0$ is proved by the same fixed-fibre monotonicity; hence the maximum is justified. The source kernel cannot be split into separate multiplier kernels: the frame $Z_{\text{ker}\Psi,k}$ retains relations between distinct summands. Every eigenvalue here is that of the exact NMR7 relative metric $(Q^{\text{src}})^{-1/2} H^{\text{im}}(Q^{\text{src}})^{-1/2}$. Thus the four negative spectral sums remain separately present.

Keep all finite positive expansion and missing-quotient constants from NMR4/7/8:

$$\begin{aligned} \mathcal{K}(D)^2 &= \sum_h \kappa_h(D)^2, \quad \kappa_h(D) = K_h(D) \quad (s = 1), \quad \kappa_h(D) = K_h(D) a_{j_h, d_h/4}^{-1/2} \quad (s = k), \\ K_h(D) &= \sum_{a,b} |h_{ab}| e^{|\eta_{h,a,b}|(H_D+2)}, \quad a_{j,r} = (2\pi)^{2r} \prod_{u=0}^{r-1} \frac{j+4u}{j+4u+2}, \\ E_{k,s}^{\text{mult,lo}} &= 2a_k (\log \max\{1, \mathcal{K}(q_k - 1 - v)\} + \log \max\{1, \mathcal{K}(q_k - v)\}), \\ E_{k,s}^{\text{mult,hi}} &= 2a_k (\log \max\{1, \mathcal{K}(2q_k - 1 - v)\} + \log \max\{1, \mathcal{K}(2q_k - v)\}), \\ A_{k,s}^{\text{end}} &= \max\{1, q_k \mathcal{B}_{2q_k-v,s,k,A}^2\} (M_{k,s}^{\text{rec}})^2, \quad E_{k,s}^{\text{miss}} = 2\varepsilon_k \log A_{k,s}^{\text{end}}, \quad 0 \leq \varepsilon_k \leq \sum_h (v - t_{0,j_h}) - v. \end{aligned} \tag{WNR13}$$

Here $\eta_{h,a,b} = \beta_{d_h,a,b} - d_h/2$, and $\mathcal{B}$ and M^{rec} are the complete FRE reconstruction quantities in NMR8, including all actual masses, moments, roots and Gamma sums. Empty products and determinants retain value one. No inverse inequality has been added: MNT bounds only the forward map, while FRE bounds the actual missing quotient.

Substitution in NMR9 now gives an exact identity and explicit finite interval whose center has no displaced cutoff:

$$\begin{aligned} S_{K,s} &= Z_{k,s}^W + \mathcal{W}_{k,s} + p_{k,s}^{\text{mult}} + V_{\text{miss},k,s}, \\ m_{k,s}^{W,-} &= Z_{k,s}^W - \Omega_{k,s}^{\text{lo}} - E_{k,s}^{\text{mult,lo}}, \quad m_{k,s}^{W,+} = Z_{k,s}^W + \Omega_{k,s}^{\text{hi}} + E_{k,s}^{\text{mult,hi}} + E_{k,s}^{\text{miss}}, \\ m_{k,s}^{W,-} &\leq S_{K,s} \leq m_{k,s}^{W,+}, \quad \mathcal{T}_{k,s} - m_{k,s}^{W,+} \leq S_{R,s} \leq \mathcal{T}_{k,s} - m_{k,s}^{W,-}. \end{aligned} \quad (\text{WNR14})$$

The positive relative sum obeys $-E^{\text{mult,lo}} \leq p^{\text{mult}} \leq E^{\text{mult,hi}}$, the full missing return obeys $0 \leq V_{\text{miss}} \leq E^{\text{miss}}$. Together with WNR11 these prove both interval endpoints. The exact same-row relation $S_R = \mathcal{T} - S_K$ proves their reversal. One can also intersect the lower endpoint with $n^{\text{mult}} - E^{\text{mult,lo}}$ using WNR12. The original NMR9 interval with an unevaluated exact V_Q remains available. WNR14 is a proved substitution for its displaced part; it is not claimed to be narrower than knowledge of that exact V_Q .

127 The finite bounds returned through all earlier receiving maps

Every interval in NMR11, including all four branch sign choices and the domain-restricted SAR entries, is retained. Put

$$\begin{aligned} \underline{S}_{K,s}^W &= \max\{\underline{S}_{K,s}^{\text{nat}}, m_{k,s}^{W,-}, n_{k,s}^{\text{mult}} - E_{k,s}^{\text{mult,lo}}\}, \\ \bar{S}_{K,s}^W &= \min\{\bar{S}_{K,s}^{\text{nat}}, m_{k,s}^{W,+}\}, \quad \underline{S}_{R,s}^W = \mathcal{T}_{k,s} - \bar{S}_{K,s}^W, \quad \bar{S}_{R,s}^W = \mathcal{T}_{k,s} - \underline{S}_{K,s}^W. \end{aligned} \quad (\text{WNR15})$$

Each added number has the asserted bound on the same scalar by WNR12–14; hence their intersection contains the actual scalar. The map $z \mapsto (z, \mathcal{T}_{k,s} - z)$ is bijective onto the two allocations constrained by this exact total, with inverse the first coordinate. This is the complete receiving morphism and imposes no equality of the allocations.

The original low-frame and transfer quantities in NMR12 remain literal. Their finite values are

$$\begin{aligned} \lambda_s^{\text{low}} &= F_L^{(s)} - F_{W_0}^{(s)}, \quad w_d^{(s)} = d!/(s/2)_d, \quad n_N = N + 1 - v, \\ m_N &= \log \max\{1, \frac{v!}{|\mu_v|} \sum_{\nu} |a_{\nu}| e^{|b_{\nu}-4|(H_N+2)}\}, \quad \zeta_{N,s} = \log \prod_{d=v}^N \frac{w_d^{(s)}}{w_{d-v}^{(s)}}, \\ f_{N,s}^- &= \zeta_{N,s} - 2(n_N - t_{K,k})m_N, \quad f_{N,s}^+ = 2t_{K,k}m_N, \\ D_s^- &= f_{q_k-1,s}^- + f_{q_k,s}^- - f_{2q_k-1,s}^+ - f_{2q_k,s}^+, \quad D_s^+ = f_{q_k-1,s}^+ + f_{q_k,s}^+ - f_{2q_k-1,s}^- - f_{2q_k,s}^-, \\ E_{k,s}^{\text{tr}} &= 2 \sum_{N \in \{q_k-1, q_k, 2q_k-1, 2q_k\}} E_N^{\text{CEP}} + 8t_{R,k} |\log |\alpha_A||, \\ F_K^{(s)} &= S_{K,s} - \mathcal{E}_s^{\text{flag}} + \lambda_s^{\text{low}}, \quad D_s^- \leq \mathcal{E}_s^{\text{flag}} \leq D_s^+, \quad \Phi_{k,s} = F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}, \quad |\delta_{0,k,s}| \leq E_{k,s}^{\text{tr}}. \end{aligned} \quad (\text{WNR16})$$

The low quotient isometry remains $(K + L)/K \rightarrow W_0$, $[p] \mapsto \Xi p$: its two norms minimize over the same complete fibre $p + K \subset K + L$. The subsequent quotient $B_{\Xi} \rightarrow B_{\Xi}/W_0$ has the full Schur metric transported in PBR7, which is the source of the displayed transfer error. Thus both low metrics occur at their own places in WNR16. Subtracting the flag endpoints and adding the transfer endpoints gives

$$\begin{aligned} k_s^{W,-} &= \underline{S}_{K,s}^W - D_s^+ + \lambda_s^{\text{low}}, \quad k_s^{W,+} = \bar{S}_{K,s}^W - D_s^- + \lambda_s^{\text{low}}, \\ r_s^{W,-} &= F_{W_0}^{(s)} + \underline{S}_{R,s}^W - E_{k,s}^{\text{tr}}, \quad r_s^{W,+} = F_{W_0}^{(s)} + \bar{S}_{R,s}^W + E_{k,s}^{\text{tr}}, \\ F_K^{(s)} &\in [k_s^{W,-}, k_s^{W,+}], \quad \Phi_{k,s} \in [r_s^{W,-}, r_s^{W,+}]. \end{aligned} \quad (\text{WNR17})$$

Every sign follows by interval addition to the exact two identities in WNR16. In particular the kernel lower bound subtracts D_s^+ , and the residual lower bound subtracts E^{tr} .

Retain all original OAR and PMR endpoint errors and define

$$\begin{aligned}\Phi_{k,s}^{-,W} &= \max\{\Phi_{k,s}^{-,\text{nat}}, r_s^{W,-}\}, & \Phi_{k,s}^{+,W} &= \min\{\Phi_{k,s}^{+,\text{nat}}, r_s^{W,+}\}, \\ L_s^W &= \max\{L_s^{\text{nat}}, k_s^{W,-}, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{+,W} - e_{k,s}^-\}, \\ U_s^W &= \min\{U_s^{\text{nat}}, k_s^{W,+}, \mathcal{F}_{\partial,k}^{(s)} - \Phi_{k,s}^{-,W} + e_{k,s}^+\}, & F_K^{(s)} &\in [L_s^W, U_s^W].\end{aligned}\tag{WNR18}$$

On precisely the original PMR domain $k \geq 29$, $k\sigma \leq 2^{-33}q_k$, also adjoin $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{+,W} - e_{k,s}^{0,-}$ to the maximum and $\mathcal{F}_{0,k}^{(s)} - \Phi_{k,s}^{-,W} + e_{k,s}^{0,+}$ to the minimum, with $e^{0,\pm} = e^\pm + c_k^0$ unchanged. These entries follow by substituting the residual interval into the exact OAR equality $F_K = \mathcal{F}_\partial - \Phi + \mathfrak{e}$ with $-e^- \leq \mathfrak{e} \leq e^+$, and into the proved PMR equality. Each maximum/minimum therefore contains the same actual scalar; no error independence is invoked.

On the same original arithmetic period domain $|u| \geq R_*$ put

$$\begin{aligned}a_{K,k}^- &= \max\{-C_K^-, \delta_k^\sigma - C_B^+\}, & a_{K,k}^+ &= \min\{C_K^+, \delta_k^\sigma + C_B^-\}, \\ L_{\text{ar}}^W &= \max\{L_{\text{ar}}^{\text{nat}}, L_1^W + a_{K,k}^-\}, & U_{\text{ar}}^W &= \min\{U_{\text{ar}}^{\text{nat}}, U_1^W + a_{K,k}^+\}, \\ F_K^{\text{ar}} &\in [L_{\text{ar}}^W, U_{\text{ar}}^W], & F_B^{\text{ar}} &\in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}^W, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}^W].\end{aligned}\tag{WNR19}$$

The original correlated difference $F_K^{\text{ar}} - F_K^{(1)}$ has precisely the interval $[a_{K,k}^-, a_{K,k}^+]$ of NMR14. Adding its endpoints to WNR18 gives the arithmetic kernel interval, and subtracting from the same exact total gives its reversed boundary partner. Both correlation constraints remain.

Finally the original signed and action quantities return as

$$\begin{aligned}\Psi_k &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, & \mathcal{H}_k &= -T_k, & C_k^{\text{budget}} &= F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \\ L_\Psi^W &= L_k^W + \mathcal{H}_k + \delta_k^\sigma, & U_\Psi^W &= U_k^W + \mathcal{H}_k + \delta_k^\sigma, & \Psi_k &\in [L_\Psi^W, U_\Psi^W], \\ J_k^{\text{action}} &= \Psi_k + C_k^{\text{budget}}, & L_J^W &= L_\Psi^W + C_k^{\text{budget}}, & U_J^W &= U_\Psi^W + C_k^{\text{budget}}, \\ 4q_k \log \mathcal{L}_{h,k} &\leq J_k^{\text{action}} \in [L_J^W, U_J^W], \\ \mathcal{L}_{h,k} &\leq \min_M \epsilon_M^{\text{ar}} \leq \prod_M (\epsilon_M^{\text{ar}})^{w_M/(2q_k)} \leq \exp(U_J^W/(4q_k)).\end{aligned}\tag{WNR20}$$

Addition of the displayed exact scalars proves the first intervals. The original untilted identity is $J_k^{\text{action}} = 2 \sum_M w_M \log \epsilon_M^{\text{ar}}$, $\sum_M w_M = 2q_k$, with $\mathcal{L}_{h,k} = 2\delta e(k+1)\lfloor(k+1)^2/4\rfloor$, $e = 1$ on this simple-quartet branch. Its weighted mean gives the last line, including the factor $4q_k$. The full monic window, reference boundary, both contraction penalties, leakage and mixed pairing $a_K d_B + a_B d_K + 2|z_K|^2$ with $z_B = -z_K$ remain in this same action budget. The real-tilt MCE expression $\log(\epsilon_M^2 + \phi_M^2)$ and full MCP derivative keep their previous domains; the displayed mean uses its original untilted reflection domain only.

128 The proved subleading window cost and the remaining exact center

For fixed original data and the fixed finite multiplier family, WNR2 holds eventually. There $N \leq 2q_j + 2b_h = O(q_j)$, $c_{s_j/2}^{\log} = O(j)$ and $v \leq 80$ is bounded. Directly in WNR3, $\log M_N^{\text{CEP}} = O_{\text{actual}}(\log(N+1))$ and each finite tail and CEP constant is $O_{\text{actual}}(q_j \log(q_j+1))$. WNR7 gives $\mathcal{E}_j^{\text{OSP}} = O_{\text{actual}}(1)$, $r_\ell = O(1/j)$ and $B_a(Q) = O(q_j \log(q_j+1))$ on the literal required dimensions. For the low errors $a \leq \Delta_j + b_h + 1 = O(j)$; the two explicit branches of $L(a, Q)$ in WNR8 are $O(j^2 \log(q_j+1))$. For the high errors, all exact parity arguments stay in $[3/2, 5/2]$ and the G derivatives in OFV23 are $O(q_j \log(q_j+1))$ there. The half-unit rounding changes therefore give that bound for R_H , including $C_0 a$. Finally $b_h \Delta_j = O(j^2)$, $C_* < \infty$ by the proved smooth EIQ

function, and the finite family has cardinality independent of k . Hence

$$\begin{aligned}
\Omega_{k,s}^{\text{lo}} + \Omega_{k,s}^{\text{hi}} &= O_{\text{actual}} \left(\sum_{h:d_h > 0} q_{j_h} \log(q_{j_h} + 1) \right), \\
\Omega_{k,s}^{\text{lo}} + \Omega_{k,s}^{\text{hi}} &= O_{\text{actual}}(q_k \log(q_k + 1)) = o(kq_k), \\
-\Omega_{k,s}^{\text{lo}} - E_{k,s}^{\text{mult,lo}} &\leq S_{K,s} - Z_{k,s}^W \leq \Omega_{k,s}^{\text{hi}} + E_{k,s}^{\text{mult,hi}} + E_{k,s}^{\text{miss}}, \\
m_{k,s}^{\text{W,+}} - m_{k,s}^{\text{W,-}} &= \Omega_{k,s}^{\text{lo}} + \Omega_{k,s}^{\text{hi}} \\
&\quad + 2a_k \sum_{N \in \{q_k-1, q_k, 2q_k-1, 2q_k\}} \log \max\{1, \mathcal{K}(N-v)\} + 2\varepsilon_k \log A_{k,s}^{\text{end}} = o(kq_k), \\
\boxed{S_{K,s} = \sum_h S_{R,j_h,s_{j_h}} - V_{\ker \Psi,k,s} + n_{k,s}^{\text{mult}} + o(kq_k), \quad s = 1, k.} & \tag{WNR21}
\end{aligned}$$

Indeed $j_h/k \rightarrow 1$, $q_{j_h}/q_k \rightarrow 1$ and $\log(q_k + 1)/k \rightarrow 0$. The remaining width is the exact NMR16 finite expression: $a_k \leq 8k - 16 - t_{0,k}$, $\log \max(1, \mathcal{K}) = O_{\text{actual}}(\log q_k)$, $\varepsilon_k \leq \sum_h (v - t_{0,j_h}) - v$ is bounded, and $\log A_{k,s}^{\text{end}} \leq 2q_k \log k + O_{\text{actual}}(q_k)$ by FRE. These estimates prove the last two lines without replacing any finite error. The formula propagates the now-controlled MNT18/NMR6 windows back through NMR9/11–15. The full source-kernel return and each matched negative relative logarithm remain exact terms in its center. No asymptotic size for either is supplied by this window calculation, and neither is assigned zero.

129 The original domains and the simultaneous boundary refinement

Retain the fixed actual simple quartet, its complete unit, amplitude, original five-orbit period and logarithm branch. Write $k \equiv 1 \pmod{4}$, $k \geq 9$, $q_k = (k+1)^2$, $\Delta_k = 16k - 48$, $v = \text{ord}_0 E_A$, $\alpha_A = v!/\mu_v$, and keep the actual nonzero μ_v . No multiplicity is removed from the conductor. The full source orders are $s = 1, k$, with their original centres $c'_k = k/2 - 4$ and Gamma masses $(2\pi)^{s/2}$. A native form at endpoint N means its full original DNE attained extension Gram at polynomial cutoff $D = N - v$.

The complete coefficient identity RPB1–2 proves that the ASR/RPK affine conductor and the EBG conductor are the same polynomial, with scalar one: diagonal conjugation cancels the full scalar and commuting diagonal period factor and leaves $L_a = \mathbf{H}^{-1} D_5^{-a} \mathbf{H}$. Substitution in Q_0 then proves equality of every coefficient of the four-factor product. Consequently their two proved exterior period domains have union

$$\{|u| \geq R_{\text{supp}}\} \cup \{|u| \geq R_{\text{rank}}\} = \{|u| \geq R_{\text{cov}}\}, \quad R_{\text{cov}} = \min(R_{\text{supp}}, R_{\text{rank}}). \tag{NBR1}$$

Both radii include R_* . Whichever radius attains the minimum provides its complete proof on the displayed domain. The ASR full-width conclusion is the antecedent; RPK contributes its unit-independent eliminant radius, polar observation and action map. At all other original periods the exact RPB3 edge coefficient tests and their finite exceptional sets remain, with no rank assigned merely from an eliminant zero.

For every actual nonzero orbit biform let p_a be its support size. Its p_a exponentials have distinct exponents: equality of real and imaginary parts forces both grid indices equal. Vanishing of its first p_a derivatives would kill every coefficient by the square Vandermonde. Zero orders add under the four-factor product, so

$$0 \leq v \leq \sum_{a=1}^4 (p_a - 1) \leq 32, \quad t_{K,k} = 8k - 16 - t_{0,k}, \quad t_{R,k} = 8k - 32 - v + t_{0,k}, \quad \Delta_k - v \geq 16k - 80. \tag{NBR2}$$

This is valid on the entire original five-orbit locus, independently of NBR1. The original v remains in all matrices and factorials. In particular the coupled BIM/NMR domain has its exact quotient and sharpened finite bound

$$\varepsilon_k \leq e_k \leq c_k - v = \sum_{h \in \mathcal{H}} (v - t_{0,k-d_h}) - v \leq 32(|\mathcal{H}| - 1), \quad k \geq 37. \quad (\text{NBR3})$$

Indeed the onto coupled-jet map induces the onto quotient in NMR2–3; its dimension is $c_k - v$. Substituting the proved NBR2 bound in that dimension inequality proves NBR3. All individual and coupled jets, including their sections and the constraint $\sum_h D_h \lambda_h = 0$, remain in the source. The original weaker $v \leq 80$ records remain valid and are preserved.

130 The degree-four tuple before and after the invariant quotient

Here the source degree is k and the target is $k + 4$. Use the same four fixed multipliers x_i^4 of DFX, and set

$$\begin{aligned} \mathbf{T}_j &= \alpha_A N_j Z_{\text{exp},j}, \quad \mathcal{M}_i = \mathbf{T}_{k+4}^{-1} m_{i,k} \mathbf{T}_k, \quad \mathbf{J}_j = A_{\ell,j}^{-1} P_{Y,j}^{\mathbf{T}}, \quad \mathbf{B}_j = Z_{K,j}^{\mathbf{T}} A_{\ell,j}, \\ \mathcal{X}_k &= (\mathbf{B}_{k+4} \mathcal{M}_i \mathbf{J}_k)_{i=1}^4, \quad Z_X : \mathbb{C}^{b_X} \xrightarrow{\sim} \ker \mathcal{X}_k, \quad a_X = t_{R,k} - b_X. \end{aligned} \quad (\text{NBR4})$$

The tuple notation means vertical concatenation. Every inverse of $\mathbf{T}$ is on the original low-annihilator space: DFX8’s full Leibniz sum proves its domain. The factor α_A cancels only in the written conjugation. The kernel is precisely $\{x : \mathcal{M}_i \mathbf{J}_k x \in \text{im } \mathbf{J}_{k+4} \text{ for all } i\}$. DFX28–35 computes it by full finite coefficient matrices and places it in the retained shared-component space $\mathcal{E}_{v,k}$. Thus $0 \leq b_X \leq d_{\epsilon,k}$ for each original branch choice. Joint injectivity before this last quotient follows from DFX17–21 on the complete nonreduced conductor; the present kernel is retained after the quotient. The exact map between these statements is the onto quotient $\mathcal{J}_{v,k} / \ker \mathcal{X}_k \rightarrow \mathcal{J}_{v,k} / \mathcal{E}_{v,k}$, whose kernel is $\mathcal{E}_{v,k} / \ker \mathcal{X}_k$, as proved in DFX28.

Fix $s_0, s_1 = (1, 1)$ or $(k, k + 4)$, and $D \geq q_{k+4} - v - 1$. On the same $\mathbf{J}_k$ coordinates set $G_0 = \mathbf{J}_k^* D_{k,D}^{(s_0)} \mathbf{J}_k$. Let G_c, G_i be the restrictions of NBG27’s correlated and independent target Grams, respectively. Subscripts c, i here label those two forms, not NKC’s lower-gauge matrix. For each $G \in \{G_0, G_c, G_i\}$ retain its complete source-kernel restriction $G_Z = Z_X^* G Z_X$ and its attained image quotient

$$\begin{aligned} \tilde{X} &= L_{U_X} \mathcal{X}_k, \quad Q_G = (\tilde{X} G^{-1} \tilde{X}^*)^{-1}, \\ H_X &= U_X^* \text{diag}(H_B, H_B, H_B, H_B) U_X, \\ H_B &= (\mathbf{B}_{k+4} (D_{k+4,D}^{(s_1)})^{-1} \mathbf{B}_{k+4}^*)^{-1}. \end{aligned} \quad (\text{NBR5})$$

Here U_X is a fixed full-column image frame and $L_{U_X} U_X = I$. All frames are independent of endpoint and source order. The lift $G^{-1} \tilde{X}^* Q_G y$ has image y and is G -orthogonal to $\ker \mathcal{X}_k$; adding any other kernel vector increases its squared norm. This proves every quotient formula. For every source vector, minimizing each target component over its full invariant space can only decrease its norm. Taking the infimum over the full source fibre therefore proves

$$0 \prec H_X \preceq Q_{G_i} \preceq Q_{G_c}, \quad K_-^{-2} Q_{G_0} \preceq Q_{G_c} \preceq K_+^2 Q_{G_0}. \quad (\text{NBR6})$$

NBG23 supplies the last two inequalities before quotient, and the same affine-fibre minimization proves them after quotient. Each positive assertion is on the actual rank a_X image; rank zero uses empty matrices and determinant one.

This gives a complete determinant propagation through both losses. Define

$$\begin{aligned} \omega_J &= \log \det G_i - \log \det G_c, \quad \omega_Z = \log \det (G_i)_Z - \log \det (G_c)_Z, \\ \omega_{\text{inv}} &= \log \det H_X - \log \det Q_{G_i} \leq 0, \quad \theta_c = \log \det Q_{G_c} - \log \det Q_{G_0}, \\ \log \det H_X - \log \det Q_{G_0} &= \theta_c + \omega_J - \omega_Z + \omega_{\text{inv}}, \\ -2a_X \log \max(1, K_-) &\leq \theta_c \leq 2a_X \log \max(1, K_+). \end{aligned} \quad (\text{NBR7})$$

To prove the equality, choose a fixed right inverse R_X and use the full frame $[Z_X, R_X]$ for both G_i, G_c . Their determinant identities are $\det G = |\det[Z_X, R_X]|^{-2} \det G_Z \det Q_G$. Dividing cancels the identical fixed factor and gives $\log \det Q_{G_i} - \log \det Q_{G_c} = \omega_J - \omega_Z$. Adding the two other defined differences proves NBR7. Its finite interval follows by conjugating NBR6 and multiplying its a_X eigenvalue bounds. In particular no source-kernel term or further invariant loss is absorbed into an unproved inverse estimate.

The two independent-gauge terms have complete NKC coefficient forms. For the actual subspaces represented by $\mathbf{J}_k$ and $\mathbf{J}_k Z_X$, let

$$\begin{aligned}\gamma_J &= \log \det \mathcal{H}_{J,D} - \log \det G_{\text{gauge},D}, \\ \gamma_Z &= \log \det \mathcal{H}_{Z,D} - \log \det G_{\text{gauge},D}, \\ r_c &= 3k^2 - 10k - 13, \quad c_- = \min(1, L_-^{-2}), \quad c_+ = \max(1, L_+^2).\end{aligned}\tag{NBR8}$$

The complete matrices $\mathcal{H}_{J,D}, \mathcal{H}_{Z,D}$ are NKC16: $Z_c^* \mathcal{K}^* \mathcal{V}_2^* \mathcal{V}_2 \mathcal{K} Z_c + Z_c^* \mathcal{V}^* P_{\text{rem}} \mathcal{V} Z_c$, with the respective actual projector NKC9, and $G_{\text{gauge},D}$ is NKC6's G_c . All source-order factors in $L_{\pm} = \sqrt{2} K_{\pm}^{(1)}$ remain. NKC15 proves, for $Y = J, Z$, $\gamma_Y - r_c \log c_+ \leq \omega_Y \leq \gamma_Y - r_c \log c_-$. Subtracting these two unequal intervals gives the finite bound

$$\gamma_J - \gamma_Z - r_c \log(c_+/c_-) \leq \omega_J - \omega_Z \leq \gamma_J - \gamma_Z + r_c \log(c_+/c_-).\tag{NBR9}$$

This includes $b_X = 0$: the zero source has $\omega_Z = 0$, and NKC's complete augmented compatibility matrix remains positive with its stated determinant bound. The actual complementary projectors are retained in both γ terms.

At $N \in \mathcal{N}_{k+4} = (q_{k+4} - 1, q_{k+4}, 2q_{k+4} - 1, 2q_{k+4})$ put $\eta_N = (-1, -1, 1, 1)$ and use $D = N - v$. Set

$$\begin{aligned}\mathcal{C}_X &= \sum_N \eta_N (\log \det Q_{G_0} + \gamma_J - \gamma_Z + \omega_{\text{inv}}), \\ E_X^- &= 2a_X \left(\sum_{N \text{ high}} \log \max(1, K_-) + \sum_{N \text{ low}} \log \max(1, K_+) \right) + \sum_N r_c \log(c_+/c_-), \\ E_X^+ &= 2a_X \left(\sum_{N \text{ high}} \log \max(1, K_+) + \sum_{N \text{ low}} \log \max(1, K_-) \right) + \sum_N r_c \log(c_+/c_-), \\ \mathcal{C}_X - E_X^- &\leq \sum_N \eta_N \log \det H_X \leq \mathcal{C}_X + E_X^+.\end{aligned}\tag{NBR10}$$

All constants in each summand use that endpoint's D and the literal source orders. NBR7 and NBR9 give the two sides by choosing the appropriate endpoint for each sign. Since $a_X = O(k)$, $r_c = O(q_k)$, and NKG13/NKC4 bound the logarithms by $O_{\text{actual}}(\log q_k)$, both errors are $O_{\text{actual}}(q_k \log q_k) = o(kq_k)$. The centre retains the source quotient, both compatibility forms and the entire further invariant relative determinant.

131 The original enlarged boundary as a quantitative source loss

For $k \geq 37$, retain EBG's four target cutoffs, old and enlarged lower grids, and their evaluation matrices. At these cutoffs their boundary ranks and signed jump are exactly

$$\begin{aligned}(e_1, e_2, e_3, e_4) &= (24k - 24 - v, 24k - 23 - v, 26k - 13, 26k - 13), \\ e_3 + e_4 - e_1 - e_2 &= 4k + 21 + 2v \leq 4k + 85.\end{aligned}\tag{NBR11}$$

The proof is the distinct-node Vandermonde rank $\min(D + 1, (k + 6)^2)$ minus $(k - 7)^2$, with the full source cutoff $D = N - v$. The first two enlarged numerator metrics are zero. At the

high cutoffs their complete kernel is EBG14's module $\widehat{A}T_{k-a_+,k-b_+}/A^{\text{hom}}T_{k-8,k-8}$, of dimension $(k-a_++1)(k-b_++1)-(k-7)^2$. RPB7–8 proves positivity precisely off $\mathcal{E}_\infty$, in particular on NBR1. No logarithm of a zero low metric is used.

The original two quotient maps are

$$\mathcal{H}_0/\mathcal{L}_0 \xrightarrow{\overline{T}} \mathcal{H}_1^4/\mathcal{L}_1 \xrightarrow{\overline{S}} \mathcal{H}_0/\mathcal{L}_+, \quad \overline{ST} = q_+, \quad K_-^{-2}D_D^+ \preceq G_{\text{ind}}^{\text{act}} \preceq G_{\text{cor}}^{\text{act}} \preceq K_+^2D_D^0. \quad (\text{NBR12})$$

NBG's full finite convolution proves $T\mathcal{L}_0 \subset \mathcal{L}_1$ and $S\mathcal{L}_1 \subset \mathcal{L}_+$, so both arrows are well defined. The literal Bezout identity gives $ST = I$ and proves the composition. Apply S to a least-norm representative, then minimize over its target fibre. Its bound K_- proves the left inequality. The other inequalities are the original NBG attained minima.

For either actual restriction $Y = J, Z$, put $t_Y = t_{R,k}, b_X$ respectively and define $\Omega_{B,Y}$ as the relative matrix of the restrictions of D_D^+, D_D^0 on $\mathbf{J}_k Y$; for J the factor Y is the identity. At a high cutoff off $\mathcal{E}_\infty$ the full forms and these restrictions are positive. Restrict NBR12 and take determinants to obtain

$$-\mathbf{b}_{Y,D} \leq \omega_Y \leq 0, \quad \mathbf{b}_{Y,D} = -\log \det \Omega_{B,Y} + 2t_Y \log(K_-K_+). \quad (\text{NBR13})$$

Here $K_-K_+ \geq 1$ follows from $ST = I$. On EBG's exact comparison domain $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q_k$, one may replace $\mathbf{b}_{Y,D}$ by the explicit upper bound

$$\mathbf{b}_{Y,D}^{\text{exp}} = -\log \det \Omega_{B,Y}^{\text{exp}} + t_Y(\mathbf{b}_{D,s_0}^+ - \mathbf{b}_{D,s_0}^-) + 2t_Y \log(K_-K_+). \quad (\text{NBR14})$$

The factorial forms are the restrictions of the entire coupled Schur construction RPB11 with the identical arrays U, E , not separate diagonal blocks. EBG24 inversion and minimization over the same fibre give RPB12. Generalized Rayleigh quotients, and then the min–max characterization, bound each relative eigenvalue by $e^{\pm(\mathbf{b}^+ - \mathbf{b}^-)}$ times its factorial counterpart. Multiplication over t_Y directions proves the substitution. Thus the extra additive error is the displayed $O_{\text{actual}}(k \log q_k)$, while the complete factorial correction remains in its centre.

At every high endpoint the intersection

$$\max\{\gamma_Y - r_c \log c_+, -\mathbf{b}_{Y,D}\} \leq \omega_Y \leq \min\{\gamma_Y - r_c \log c_-, 0\} \quad (\text{NBR15})$$

is nonempty because both intervals bound that same scalar. Use $\mathbf{b}^{\text{exp}}$ on its comparison domain if desired. At low endpoints retain the NKC interval intersected with $(-\infty, 0]$ only. Subtract the Z interval from the J interval and use the four signs in NBR7 to sharpen NBR10. Every step subtracts the full upper endpoint for a lower bound and the full lower endpoint for an upper bound. This supplies a precise source-loss refinement without removing the invariant kernel or ω_{inv} .

132 The complete polar metric and the low-volume return

On NBR1, or at any actual period certified by the full edge test, RPK supplies its entire principal-part map Π_k with $\ker \Pi_k = \text{im } C^\top$, $\text{rank } \Delta_k$, and all multiplicities retained. The $j = 8k - 32$ selected residual rows give the invertible fixed matrix $B_{\text{pole}} = \mathbf{P}\Pi_k D^\top$. The source-variable CFR common factor is one, so $g = t_k = 0$ and its complement is the entire residual. This follows already from ASR's two widths and CFR's exact formula for g ; it removes no polar direction. The exact pole detector instead identifies the quotient by the full conductor with its own image. Its low-flag inclusion and metric are

$$\begin{aligned} U &= \alpha_A L_Z N Z_{\text{exp},k} \mathbf{J}_k, & T_{\text{pole}} &= B_{\text{pole}} U = \alpha_A \mathbf{P} \Pi_k Z_{\text{exp},k} \mathbf{J}_k, \\ T_{\text{pole}} : \mathbb{C}^{t_{R,k}} &\xrightarrow{\sim} \ker(M_v D^\top B_{\text{pole}}^{-1}), & \text{rank}(M_v D^\top B_{\text{pole}}^{-1}) &= v - t_{0,k}, \\ H_{\text{pole},N} &= B_{\text{pole}}^* \overline{\Omega_N} B_{\text{pole}}^{-1}, & H_{\text{hi},N} &= T_{\text{pole}}^* H_{\text{pole},N} T_{\text{pole}}. \end{aligned} \quad (\text{NBR16})$$

All maps, including the ordinary transpose in the dual coefficient pairing and the conjugate in its Hermitian metric, are those of PLF1–9. Restricting a functional annihilating K to L has kernel

$\text{Ann}(K + L)$ and is onto $(L/(K \cap L))^\vee$: define it first on $K + L$ by its L part and extend to V_A . This proves the rank statement with the actual t_0 . PLF7 proves the equality of the two coefficient maps by subtracting a vector in $\text{im } C^\top$, which Π_k kills. It also retains the full α_A .

The complete adapted fixed frame PLF10 gives its exact Schur factor $\overline{H_{W_0,N}^{-1}}$, with $W_0 = \Xi(L)$ and its full original restriction metric. Its determinant identity, followed by the same four endpoint signs, and then the original source transport gives

$$\begin{aligned} \Phi_{k,s} &= F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}, & \delta_{0,k,s} &= - \sum_N \sigma_N (\log \det H_{\text{hi},N} - \log \det (\mathbf{J}_k^* D_{k,N-v}^{(s)} \mathbf{J}_k)), \\ |\delta_{0,k,s}| &\leq E_{k,s}^{\text{tr}} = 2 \sum_N E_N^{\text{CEP}} + 8t_{R,k} |\log |\alpha_A||, & \sigma &= (1, 1, -1, -1). \end{aligned} \tag{NBR17}$$

For precision, this is the full block inverse of the original residual Gram in its W_0 and quotient frame. Its determinant factor is the squared determinant of that fixed frame, and only that endpoint-independent factor cancels in the four signs. PLF14 transports the original conductor source map, its whole relation graph and low component before either quotient. Each restriction and attained quotient contracts every exterior norm; apply the complete CEP estimate and its inverse, retaining the α_A scaling on $t_{R,k}$ directions, and square the bounds. This proves exactly the finite error in NBR17, rather than an identity between the native and pole metrics.

The original diagonal action has its full leakage into these same polar coordinates. At a selected multiplicity- m_ν pole its conductor-input row is $m_\nu e^{\beta'_{ab} w_{\nu,l}}$ in the residue slot and zero in higher slots. RPK's resultant argument proves its exact rank

$$q'_k - (k - 7 - a_{\text{sf}})_+ (k - 7 - b_{\text{sf}})_+, \quad A = a_* \prod_\nu B_\nu^{m_\nu}, \quad A_{\text{sf}} = \prod_\nu B_\nu. \tag{NBR18}$$

Here $x_+ = \max(x, 0)$. A lower-grid polynomial has all sampled values zero exactly when divisible by each distinct B_ν ; the resultant degree bound and repeated samples prove this, and counting the remaining rectangular monomials proves NBR18. The identity-induced quotient by A onto the quotient by A_{sf} has kernel the quotient of those two denominator spaces, as in PLF19. Thus the reduced factors used by this action rank do not replace the full conductor in NBR16–17. The actual leakage and mixed penalties remain part of the original budget.

133 Propagation through the identical allocation and entire action

Use the complete WNR finite window bounds at their stated original domain, proved immediately alongside this receiver. In particular for each BIM source $\ell_h = k - d_h$, $d_h \leq 28$, use its actual $\ell_h \geq 4096$ domain before adjoining those new bounds. The corresponding NMR centre retains the complete source-kernel volume and every matched negative logarithm. Let $[s_{K,s}^-, s_{K,s}^+]$ denote the intersection of the original NMR11 interval with the fully proved WNR intervals for the same NMR family. The exact residual partner is

$$[s_{R,s}^-, s_{R,s}^+] = [\mathcal{T}_{k,s} - s_{K,s}^+, \mathcal{T}_{k,s} - s_{K,s}^-]. \tag{NBR19}$$

For the installed receiving interval here, the new scalar entries are exactly WNR's actual NMR family entries. NBR4–15 are the proved complete map and determinant continuation on their actual four-component image, retaining its further quotient. Thus NBR19 inserts no presumed identification of images. Every intersected interval contains the same S_K , and $S_R = \mathcal{T} - S_K$ proves the reversed residual endpoints with nonempty intersection.

Keep the full original NMR12 constants $D_s^\pm$ and its distinct low difference $\lambda_s^{\text{low}} = F_L^{(s)} - F_{W_0}^{(s)}$.

No factorial ratio inside $D_s^\pm$ is changed by NBR2. The exact identities and their finite bounds are

$$\begin{aligned} F_K^{(s)} &= S_{K,s} - \mathcal{E}_s^{\text{flag}} + \lambda_s^{\text{low}}, & D_s^- &\leq \mathcal{E}_s^{\text{flag}} \leq D_s^+, \\ k_s^- &= s_{K,s}^- - D_s^+ + \lambda_s^{\text{low}}, & k_s^+ &= s_{K,s}^+ - D_s^- + \lambda_s^{\text{low}}, \\ r_s^- &= F_{W_0}^{(s)} + s_{R,s}^- - E_{k,s}^{\text{tr}}, & r_s^+ &= F_{W_0}^{(s)} + s_{R,s}^+ + E_{k,s}^{\text{tr}}. \end{aligned} \quad (\text{NBR20})$$

Subtracting the full flag interval proves $k_s^- \leq F_K \leq k_s^+$. Adding NBR17 proves $r_s^- \leq \Phi \leq r_s^+$ for the actual full rank- j polar determinant; no low component is discarded. Outside the polar full-width domain, the same finite intervals remain the original NMR/HCR intervals without this coordinate presentation. The native WNR estimates use their own full domain.

Intersect $[r_s^-, r_s^+]$ with the previous full residual interval, writing its endpoints ϕ_s^-, ϕ_s^+ . The full outer return is

$$\begin{aligned} L_s &= \max\{L_s^{\text{nat}}, k_s^-, \mathcal{F}_{\partial,k}^{(s)} - \phi_s^+ - e_{k,s}^-, \}, & U_s &= \min\{U_s^{\text{nat}}, k_s^+, \mathcal{F}_{\partial,k}^{(s)} - \phi_s^- + e_{k,s}^+, \}, \\ F_K^{(s)} &\in [L_s, U_s]. \end{aligned} \quad (\text{NBR21})$$

These are exact substitutions in OAR's identity with its unequal errors. On the original PMR domain $k \geq 29$ and $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q_k$, also adjoin $\mathcal{F}_{0,k}^{(s)} - \phi_s^+ - e_{k,s}^{0,-}$ to the maximum and $\mathcal{F}_{0,k}^{(s)} - \phi_s^- + e_{k,s}^{0,+}$ to the minimum, with the original $e^{0,\pm} = e^\pm + c_k^0$. These entries are absent outside that domain. Every maximum entry is a proved lower bound and every minimum entry a proved upper bound for the same original scalar, so their intersection remains nonempty.

On $|u| \geq R_*$ the actual correlated arithmetic difference has the original two endpoints $a_K^- = \max(-C_K^-, \delta_k^\sigma - C_B^+)$ and $a_K^+ = \min(C_K^+, \delta_k^\sigma + C_B^-)$. Adding that same difference and reversing the exact boundary total gives

$$\begin{aligned} L_{\text{ar}} &= \max(L_{\text{ar}}^{\text{nat}}, L_1 + a_K^-), & U_{\text{ar}} &= \min(U_{\text{ar}}^{\text{nat}}, U_1 + a_K^+), \\ F_K^{\text{ar}} &\in [L_{\text{ar}}, U_{\text{ar}}], & F_B^{\text{ar}} &\in [\mathcal{B}_k^{\text{ar}} - U_{\text{ar}}, \mathcal{B}_k^{\text{ar}} - L_{\text{ar}}]. \end{aligned} \quad (\text{NBR22})$$

No independent error model is used: both caps constrain that one actual difference. Finally the signed return and complete action retain every original summand:

$$\begin{aligned} \Psi_k &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, & \mathcal{H}_k &= -T_k, & C_k^{\text{budget}} &= F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+, \\ L_\Psi &= L_k + \mathcal{H}_k + \delta_k^\sigma, & U_\Psi &= U_k + \mathcal{H}_k + \delta_k^\sigma, & \Psi_k &\in [L_\Psi, U_\Psi], \\ J_k^{\text{action}} &\in [L_\Psi + C_k^{\text{budget}}, U_\Psi + C_k^{\text{budget}}], & 4q_k \log \mathcal{L}_{h,k} &\leq J_k^{\text{action}}, \\ \mathcal{L}_{h,k} &\leq \min_M \epsilon_M^{\text{ar}} \leq \prod_M (\epsilon_M^{\text{ar}})^{w_M/(2q_k)} \leq \exp((U_\Psi + C_k^{\text{budget}})/(4q_k)). \end{aligned} \quad (\text{NBR23})$$

Addition proves both intervals. The original identity $J_k^{\text{action}} = 2 \sum_M w_M \log \epsilon_M^{\text{ar}}$, $\sum_M w_M = 2q_k$, and the original same-class lower radius $\mathcal{L}_{h,k} = 2\delta e(k+1)\lfloor (k+1)^2/4 \rfloor$, $e = 1$, give the weighted mean inequality. The reference boundary, monic window, both contraction penalties, actual leakage and mixed pairing $a_K d_B + a_B d_K + 2|z_K|^2$, $z_B = -z_K$, stay in the same budget. Real tilts retain the original MCE/MCP expression and derivatives; the displayed mean is on its original untilted reflection domain.

Every finite interval above is derived from the stated complete maps. The source kernels, compatibility corrections, native invariant image loss and pole low flag remain explicit. The new window bounds and the finite errors in NBR10 and NBR14 are sub- kq estimates on their proved domains. No value is assigned to the surviving relative determinants or to either allocation coefficient.

134 Original objects and the finite conclusion

This note computes the actual top-weight term $\mathcal{W}_{r,k}$ of EBG28 in the original enlarged-gauge boundary volume source (ORIGINAL_ENLARGED_GAUGE_BOUNDARY_VOLUME.tex). Every original

root, strip, corner node, and low rank is retained. The proof of the weighted-root assertion below is self-contained. It does not replace a numerator metric by this scalar boundary quantity.

Fix, throughout,

$$0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad k \equiv 1 \pmod{4}, \quad k \geq 37, \quad 0 \leq v \leq 80, \quad v \in \mathbb{Z}, \quad \sigma \in \{0, 1\}. \quad (\text{WEL1})$$

Put $m = k - 7$ only as an exact auxiliary integer; in particular $n_0 = (k - 7)^2 = m^2$ remains the original lower-grid cardinality. The original nodes and their two index sets are

$$\begin{aligned} b_* &= \frac{1}{2} - \delta - i\gamma, & c' &= k/2 - 4, & \alpha_{ab} &= (k - 8)b_* + 2a\delta + 2ib\gamma, \\ \mathcal{G}_0 &= \{0, \dots, m - 1\}^2, & \mathcal{G}_+ &= \{0, \dots, m + 12\}^2, & \mathcal{B} &= \mathcal{G}_+ \setminus \mathcal{G}_0, \\ \chi_0(S') &= \prod_{(u,w) \in \mathcal{G}_0} (S' - \alpha_{uw}), & r &= 24k - 24 - v + \sigma = 24m + 144 - v + \sigma. \end{aligned} \quad (\text{WEL2})$$

The two values of σ are exactly the two low ranks in EBG6. All logarithms in this note are natural logarithms of positive real numbers. Equality of two original nodes forces both their real and imaginary index differences to vanish. Since a boundary index is outside $\mathcal{G}_0$, its exact product is nonzero and equals

$$\chi_0(\alpha_{ab}) = \prod_{u,w=0}^{m-1} (2\delta(a - u) + 2i\gamma(b - w)). \quad (\text{WEL3})$$

The common affine base exponent cancels in each displayed difference; no root is moved. Order these original absolute weights decreasingly, breaking ties by their original node order, and let I_{top} be the first r nodes. Thus, exactly,

$$\mathcal{W}_{r,k} = 2 \sum_{(a,b) \in I_{\text{top}}} \log |\chi_0(\alpha_{ab})|. \quad (\text{WEL4})$$

The disjoint, literal width-thirteen boundary partition used below is

$$\begin{aligned} \mathcal{H} &= \{(a, b) : 0 \leq a \leq m + 12, \ m \leq b \leq m + 12\}, & |\mathcal{H}| &= 13m + 169, \\ \mathcal{V} &= \{(a, b) : m \leq a \leq m + 12, \ 0 \leq b \leq m - 1\}, & |\mathcal{V}| &= 13m. \end{aligned} \quad (\text{WEL5})$$

In particular $|\mathcal{B}| = 26m + 169 = 26k - 13$. The entire 13 by 13 corner belongs to $\mathcal{H}$; it is never removed. The number of excluded nodes is

$$c = |\mathcal{B}| - r = 2m + \zeta, \quad \zeta = 25 + v - \sigma \in [24, 105]. \quad (\text{WEL6})$$

Define the following integrals of the original anisotropic logarithmic distance, including their integrable boundary singularities:

$$\begin{aligned} I(x, y) &= \int_0^1 \int_0^1 \log |2\delta(x - u) + 2i\gamma(y - w)| \, du \, dw, \\ H(x) &= I(x, 1), \quad V(y) = I(1, y), \quad l = \frac{11}{26}, \quad u_* = \frac{15}{26}, \quad \tau = V(u_*), \\ \mathcal{C}(\delta, \gamma) &= 13 \int_0^1 H(x) \, dx + 13 \left(\int_0^l V(y) \, dy + \int_{u_*}^1 V(y) \, dy \right). \end{aligned} \quad (\text{WEL7})$$

The variable w in these integrals is unrelated to the fixed zero order v in WEL1. The constants in the finite error are explicit:

$$\begin{aligned} \kappa &= \gamma/\delta > 4, & L &= (2\pi + 1)\kappa, \\ C_I &= |\log(2\delta)| + \log(2\sqrt{2}\kappa) + \frac{\pi}{2}, \\ C_N &= 2|\log(2\delta)| + \log(15\kappa^2) + \frac{\pi}{2}, \\ E_m &= (32\sqrt{2}\kappa + 12\sqrt{2}L + 7)m + 24C_N, \\ B_{r,m} &= 2rE_m + 2m^2(13L + 313C_I). \end{aligned} \quad (\text{WEL8})$$

Theorem 134.1 (Finite estimate for the actual top-weight product). *For every parameter and integer in WEL1,*

$$\boxed{|\mathcal{W}_{r,k} - 2rn_0 \log k - 2mn_0 \mathcal{C}(\delta, \gamma)| \leq B_{r,m}.} \quad (\text{WEL9})$$

Consequently, still retaining the original n_0 and r ,

$$\begin{aligned} |\mathcal{W}_{r,k} - 2rn_0 \log k - 2kn_0 \mathcal{C}(\delta, \gamma)| &\leq B_{r,m} + 14n_0 |\mathcal{C}(\delta, \gamma)|, \\ \mathcal{W}_{r,k} &= 2rn_0 \log k + 2kn_0 \mathcal{C}(\delta, \gamma) + O_{\delta, \gamma}(k^2), \\ \frac{\mathcal{W}_{r,k} - 2rn_0 \log k}{2kn_0} &\longrightarrow \mathcal{C}(\delta, \gamma). \end{aligned} \quad (\text{WEL10})$$

The finite bound, the order constant and the limit are uniform over $0 \leq v \leq 80$ and both σ . There is no pending use of a smaller bound on v .

135 Potential bounds and the strict side separation

Write $f(z) = \log |\text{diag}(2\delta, 2\gamma)z|$ for $z \in \mathbb{R}^2 \setminus \{0\}$. If $\rho = |z|$, then

$$2\delta\rho \leq e^{f(z)} \leq 2\gamma\rho, \quad |\nabla f(z)| \leq \frac{\kappa}{\rho}. \quad (\text{WEL11})$$

For the derivative inequality, if $A = \text{diag}(2\delta, 2\gamma)$, then $|\nabla f(z)| = |A^\top Az|/|Az|^2 \leq \|A\|/|Az| \leq \kappa/|z|$. For any $p \in \mathbb{R}^2$,

$$\int_{[0,1]^2} \frac{dz}{|p - z|} \leq 2\pi + 1 : \quad (\text{WEL12})$$

inside the unit disk about p the full-disk integral is 2π ; outside it the integrand is at most one and the square has area one. Integrate WEL11 along the segment from p to q and then over the square. The exceptional square points lying on that segment have area zero. Tonelli's theorem applies to the nonnegative derivative bound, which WEL12 makes integrable. This proves the global bound

$$|I(p) - I(q)| \leq L|p - q|. \quad (\text{WEL13})$$

Existence of the integral follows also directly from $\int_{|z|<1} -\log|z| dz = \pi/2$, evaluated in polar coordinates. For $p \in [0, 2]^2$ the distance from p to the square is at most $2\sqrt{2}$. Write $f = \log(2\delta) + \log \sqrt{z_1^2 + \kappa^2 z_2^2}$. The positive part of the last logarithm is at most $\log(2\sqrt{2}\kappa)$, and its negative part integrates to at most $\pi/2$. Therefore

$$|I(p)| \leq C_I \quad (p \in [0, 2]^2). \quad (\text{WEL14})$$

In particular H, V are L -Lipschitz and bounded in absolute value by C_I on their stated intervals.

Set $a_* = \delta/\gamma \in (0, 1/4)$. Direct comparison of the integrands gives

$$H(x) \geq \log(2\gamma) + \int_0^1 \log(1 - w) dw = \log(2\gamma) - 1. \quad (\text{WEL15})$$

For $d > 0$, integration by parts gives the exact whole-line integral

$$\int_{\mathbb{R}} \frac{1}{2} \log(1 + d^2/t^2) dt = \int_0^\infty \log(1 + d^2/t^2) dt = 2d^2 \int_0^\infty \frac{dt}{t^2 + d^2} = \pi d. \quad (\text{WEL16})$$

The endpoint terms vanish since $t \log(1 + d^2/t^2)$ tends to zero both at zero and at infinity. The formula for $d = 0$ is obtained by its zero integrand. The finite interval is a subset of this line and its integrand is nonnegative. Subtracting the integral of $\log|y - w|$ in the definition of $V(y)$ and using WEL16 with $d = a_*(1 - u)$ proves, for $0 \leq y \leq 1$,

$$V(y) \leq \log(2\gamma) + y \log y + (1 - y) \log(1 - y) - 1 + \frac{\pi a_*}{2}. \quad (\text{WEL17})$$

Here $0 \log 0 = 0$, its continuous value. Reflection $w \mapsto 1 - w$ proves $V(y) = V(1 - y)$. On $0 < y < 1$ differentiation after integrating in w gives

$$V'(y) = \frac{1}{2} \int_0^1 \log \frac{a_*^2 t^2 + y^2}{a_*^2 t^2 + (1 - y)^2} dt. \quad (\text{WEL18})$$

For completeness, on each compact subinterval of $(0, 1)$ the two endpoint logarithms are bounded uniformly in $t \in [0, 1]$, so the displayed differentiation follows by dominated difference quotients of the antiderivative in w . The integrand in WEL18 is positive for $y > 1/2$, zero for $y = 1/2$, and negative for $y < 1/2$. Thus V is strictly increasing with $|y - 1/2|$.

For $x > 1$, differentiation shows

$$\frac{d}{dx} \left(\log x - \frac{2(x-1)}{x+1} \right) = \frac{(x-1)^2}{x(x+1)^2} > 0.$$

The difference is zero at $x = 1$, so $\log(26/15) > 22/41$. Both numbers u_* , $1 - u_*$ are at most $15/26$; therefore

$$-u_* \log u_* - (1 - u_*) \log(1 - u_*) \geq \log(26/15) > \frac{22}{41}.$$

Also $\pi/4 = \int_0^1 (1 + t^2)^{-1} dt < 1$. Combining this with $a_* < 1/4$ and WEL17 proves the rational, strictly positive side gap

$$\boxed{\min_{0 \leq x \leq 1} H(x) - \tau > \frac{3}{82}.} \quad (\text{WEL19})$$

It follows, without any numerical premise, that the largest potential values of the thirteen-fold horizontal and vertical edge measure at selected total mass 24 are precisely the entire horizontal edge and the two vertical intervals outside $[l, u_*]$. Indeed the latter central interval has length $2/13$, hence mass 2, every one of its vertical values is at most τ , all other vertical values exceed τ , and WEL19 puts every horizontal value strictly above τ . The next sections prove the corresponding claims for the actual finite weighted nodes.

136 A uniform finite Riemann estimate at every original boundary node

Lemma 136.1 (Exterior lattice estimate). *For every $(a, b) \in \mathcal{B}$, put $p = (a/m, b/m)$. Then*

$$\left| \sum_{u, w=0}^{m-1} f(p - (u/m, w/m)) - m^2 I(p) \right| \leq 32\sqrt{2}\kappa m + 24C_N. \quad (\text{WEL20})$$

Proof. Partition $[0, 1]^2$ into its m^2 closed cells of side $h = 1/m$; overlapping cell boundaries have area zero. A cell's lower-left corner is $z_0 = (u/m, w/m)$. The integer distance $j = \max(|a - u|, |b - w|)$ is at least one, because the boundary node is outside the original lower index set. There are at most $8j$ lower corners in square shell j , by subtracting $(2j - 1)^2$ from $(2j + 1)^2$. Also $j \leq m + 12 \leq 2m$.

If $j \geq 3$, every point on the line from z_0 to a point of its cell has distance at least $(j - \sqrt{2})h \geq jh/2$ from p . WEL11 consequently bounds the difference between the value at z_0 and the value anywhere in that cell by $2\sqrt{2}\kappa/j$. The same bounds its cell average difference. Summing over at most $8j$ corners in each shell and at most $2m$ shells gives $32\sqrt{2}\kappa m$.

There are at most $8 + 16 = 24$ corners with $j = 1$ or 2 . For each such cell use $z = p + ht$. The common $\log h$ cancels exactly between the corner value and the cell average. The rescaled corner has $1 \leq |t_0| \leq 2\sqrt{2} < 3$ and the entire rescaled cell lies in $|t| < 5$. Thus

$$\begin{aligned} |f(t_0)| &\leq |\log(2\delta)| + \log(3\kappa), \\ \int_{\text{rescaled cell}} |f(t)| dt &\leq |\log(2\delta)| + \log(5\kappa) + \pi/2. \end{aligned}$$

For the second inequality the positive logarithmic part is bounded on that cell and the negative part is bounded by the full unit-disk integral of $-\log|t|$. The sum of the two bounds is exactly C_N . This also covers a cell whose edge or corner touches p : its logarithm remains integrable. Summing these contributions proves WEL20, since m^2 times the integral of each original cell is its cell average. $\square$

For each node specify its edge location, without changing that node:

$$e_{ab} = \begin{cases} (\min(a/m, 1), 1), & (a, b) \in \mathcal{H}, \\ (1, b/m), & (a, b) \in \mathcal{V}, \end{cases} \quad P_{ab} = I(e_{ab}). \quad (\text{WEL21})$$

The distance $|p - e_{ab}|$ is at most $12\sqrt{2}/m$, including the whole corner. Exact factorization of m from each factor of WEL3 gives

$$\begin{aligned} Q_{ab} &:= \log|\chi_0(\alpha_{ab})| - n_0 \log k \\ &= m^2 \log(m/k) + \sum_{u,w=0}^{m-1} f(p - (u/m, w/m)). \end{aligned} \quad (\text{WEL22})$$

Since $k = m + 7$, the elementary integral inequality $\log(1+t) \leq t$ for $t \geq 0$ proves $|m^2 \log(m/k)| \leq 7m$. Combining WEL13,20,21 proves the entirely uniform estimate

$$\boxed{|Q_{ab} - m^2 P_{ab}| \leq E_m \quad ((a, b) \in \mathcal{B})}. \quad (\text{WEL23})$$

Thus the error for a single original root product is $O_{\delta,\gamma}(k)$. The original subtraction $n_0 \log k$ has not been changed to a different asymptotic subtraction.

137 Finite top sums, the full corner, and proof of the estimate

We use a finite elementary comparison. For two families (q_j) and (p_j) on one finite set with $|q_j - m^2 p_j| \leq E$, maximization over all subsets of cardinality r gives

$$\left| \max_{|A|=r} \sum_{j \in A} q_j - m^2 \max_{|A|=r} \sum_{j \in A} p_j \right| \leq rE. \quad (\text{WEL24})$$

To prove each side, apply the termwise inequality to the maximizing subset on that side and bound the other subset sum by its maximum. This proof permits arbitrary ties.

Let J contain all nodes of $\mathcal{H}$ and those nodes of $\mathcal{V}$ with $b/m \notin [l, u_*]$. If z_m is the number of integers b with $b/m \in [l, u_*]$, then the interval counting formula $z_m = \lfloor mu_* \rfloor - \lceil ml \rceil + 1$ proves

$$|z_m - 2m/13| \leq 1, \quad |J| = 26m + 169 - 13z_m, \quad 11 \leq |J| - r \leq 118. \quad (\text{WEL25})$$

For the last inequality subtract $r = 24m + 144 - v + \sigma$ and retain $0 \leq v \leq 80$, $\sigma \in \{0, 1\}$: the difference is $25 + v - \sigma - 13(z_m - 2m/13)$. WEL18–19 show that every P -value in J is strictly greater than τ , and every value outside J is at most τ . Since $|J| \geq r$, a maximizing r -subset for the P -values is obtained by deleting exactly $|J| - r$ nodes from J . WEL14 therefore proves

$$\left| \max_{|A|=r} \sum_{\xi \in A} P_\xi - \sum_{\xi \in J} P_\xi \right| \leq 118C_I. \quad (\text{WEL26})$$

The literal sum on the horizontal part is

$$\sum_{\xi \in \mathcal{H}} P_\xi = 13 \sum_{a=0}^{m-1} H(a/m) + 169H(1). \quad (\text{WEL27})$$

In particular the $169H(1)$ term explicitly retains every corner node. For an L -Lipschitz function on $[0, 1]$, integration over each cell gives an error at most $L/(2m)$ between its left endpoint and m times the integral on that cell. Summing proves

$$\left| \sum_{a=0}^{m-1} H(a/m) - m \int_0^1 H(x) dx \right| \leq L/2. \quad (\text{WEL28})$$

For the vertical sum use the union of the cells whose left endpoints are outside $[l, u_*]$. Its symmetric difference with $[0, l] \cup [u_*, 1]$ has length at most $2/m$, one possible cell at each interior endpoint. The same Lipschitz estimate and WEL14 give

$$\left| \sum_{\substack{0 \leq b < m \\ b/m \notin [l, u_*]}} V(b/m) - m \left(\int_0^l V + \int_{u_*}^1 V \right) \right| \leq L/2 + 2C_I. \quad (\text{WEL29})$$

Multiply WEL28–29 by the actual multiplicity thirteen and add WEL27. Then

$$\left| \sum_{\xi \in J} P_\xi - m\mathcal{C}(\delta, \gamma) \right| \leq 13L + 195C_I. \quad (\text{WEL30})$$

Combining WEL23–26 and WEL30 gives

$$\left| \max_{|A|=r} \sum_{\xi \in A} Q_\xi - m^3 \mathcal{C}(\delta, \gamma) \right| \leq rE_m + m^2(13L + 313C_I).$$

The maximum on the left is exactly $\mathcal{W}_{r,k}/2 - rn_0 \log k$, because the same constant was subtracted from every original log weight. This proves WEL9. Since $m = k - 7$, changing $2mn_0\mathcal{C}$ to $2kn_0\mathcal{C}$ costs exactly $14n_0|\mathcal{C}|$. Moreover $|\mathcal{C}| \leq 24C_I$, $r \leq 24k$, and E_m is an explicit affine function of m . These facts prove all assertions of WEL10. The proof holds at every allowed k ; no assertion about eventual membership was needed to bound this finite maximum.

138 The actual finite ranking and its threshold

There is also an exact within-strip monotonicity for the original weights. Fix an original vertical index $a \in \{m, \dots, m+12\}$. Write

$$T_a(b) = \log |\chi_0(\alpha_{ab})|, \quad g_u(t) = \frac{1}{2} \log((2\delta(a-u))^2 + (2\gamma t)^2).$$

Reflection of the source index proves $T_a(b) = T_a(m-1-b)$. For $0 \leq b \leq m-2$, a finite telescoping sum gives exactly

$$T_a(b+1) - T_a(b) = \sum_{u=0}^{m-1} \{g_u(b+1) - g_u(b-m+1)\}. \quad (\text{WEL31})$$

Every g_u is even and strictly increasing in $|t| > 0$, because $a-u \geq 1$. Comparing $b+1$ and $m-1-b$ proves strict increase away from the exact discrete center $(m-1)/2$, with equality only between the two symmetric central positions when present. Also each $T_a(b)$ strictly increases with a in the vertical range, directly from every factor in WEL3. These are statements about the exact products, and do not assert equality of the thresholds in thirteen different columns.

Specify an explicit integer cutoff, using only the fixed original parameters and the constants already defined:

$$\begin{aligned} g &= 3/82, & \beta &= \frac{1}{2} \log(953/745), \\ A_E &= 32\sqrt{2}\kappa + 12\sqrt{2}L + 7, & B_E &= 24C_N, & D_E &= 2A_E + 5L, \\ m_* &= 1 + \left\lceil \max \left\{ 30, \frac{2D_E}{g}, \sqrt{\frac{4B_E}{g}}, \frac{52D_E}{\beta}, \sqrt{\frac{104B_E}{\beta}} \right\} \right\rceil. \end{aligned} \quad (\text{WEL32a})$$

Theorem 138.1 (Selection of the original strips). *For every original rank and zero order in WEL1 with $m \geq m_*$, every original node of $\mathcal{H}$, including all 169 corner nodes, belongs to I_{top} , and exactly $c = 2m + \zeta$ nodes of $\mathcal{V}$ are excluded. The threshold and membership assertions below all hold on this explicit range. The same horizontal inclusion also holds at any earlier allowed m satisfying the sharper finite test*

$$gm^2 > 2E_m + 5Lm, \quad (\text{WEL32})$$

If λ_m is the c th smallest Q -value on $\mathcal{V}$, then

$$\tau - \frac{L}{m} - \frac{E_m}{m^2} \leq \frac{\lambda_m}{m^2} \leq \tau + \frac{5L}{m} + \frac{E_m}{m^2}. \quad (\text{WEL33})$$

Let

$$\beta = \frac{1}{2} \log \frac{953}{745} > 0, \quad \eta_m = \frac{5L/m + 2E_m/m^2}{\beta}. \quad (\text{WEL34})$$

Every vertical node with $|b/m - 1/2| < 1/13 - \eta_m$ is excluded, and every vertical node with $|b/m - 1/2| > 1/13 + \eta_m$ is selected. Membership not decided by these two strict inequalities is confined to at most $52m\eta_m + 52$ actual vertical nodes. Thus the threshold band has width $O_{\delta,\gamma}(1/k)$ and contains $O_{\delta,\gamma}(1)$ nodes, uniformly in v and σ . These membership assertions also hold at any earlier allowed m for which WEL32 and $\eta_m < 1/26$ both hold.

Proof. On $\mathcal{V}$ the surrogate values are thirteen copies of $V(b/m)$, $0 \leq b < m$. The interval $|b/m - 1/2| \leq 1/13 + 5/m$ lies strictly inside $(0, 1 - 1/m)$ for $m \geq 30$. It contains at least $2m/13 + 9$ lattice points by the same floor/ceiling estimate as in WEL25. Consequently it contains at least $2m + 117 \geq c$ vertical nodes. Their P -values are at most $\tau + 5L/m$ by WEL13 and the symmetry and monotonicity in WEL18. Their actual Q -values are at most $m^2\tau + 5Lm + E_m$ by WEL23. Every horizontal Q -value is at least $m^2(\tau + g) - E_m$ by WEL19 and WEL23. WEL32 makes the former bound strictly smaller than the latter. Thus at least c vertical values are below every horizontal value, proving that all c excluded nodes are vertical.

The interval $|b/m - 1/2| < 1/13 - 1/m$ contains at most $2m/13 - 1$ lattice points; thirteen copies give at most $2m - 13 < c$ nodes. Hence the c th smallest surrogate value lies between $\tau - L/m$ and $\tau + 5L/m$. If two finite indexed families differ termwise by at most E , their c th order statistics differ by at most E : at least c terms of the first family are no greater than its c th value, giving the upper inequality for the second, and reversing the families gives the lower inequality. Apply this fact to WEL23 to obtain WEL33.

For $y \geq 7/13$, $y < 1$, the ratio in WEL18 decreases as its common nonnegative addend $a_*^2 t^2$ increases, and increases as y increases. Since $a_*^2 t^2 < 1/16$, it follows that

$$V'(y) \geq \frac{1}{2} \log \frac{1/16 + (7/13)^2}{1/16 + (6/13)^2} = \frac{1}{2} \log \frac{953}{745} = \beta. \quad (\text{WEL35})$$

Reflection gives the same positive derivative lower bound as a function of $|y - 1/2|$ whenever that distance is at least $1/26$. If $\eta_m < 1/26$, integration of WEL35 and monotonicity imply

$$\begin{aligned} |y - 1/2| < 1/13 - \eta_m &\implies V(y) < \tau - \beta\eta_m, \\ |y - 1/2| > 1/13 + \eta_m &\implies V(y) > \tau + \beta\eta_m. \end{aligned}$$

For the first implication, if the distance is smaller than $1/26$, first compare with the value at $1/26$ and then integrate over the remaining interval; this proves the same strict inequality. WEL23,33,34 now put the actual Q -value in the first case strictly below λ_m and in the second case strictly above it. Such nodes are respectively excluded and selected regardless of the original tie convention. The remaining b/m lie in two intervals, each of length $2\eta_m$. Each contains at most $2m\eta_m + 2$ lattice points. Thirteen copies give the bound $52m\eta_m + 52$. To verify the stated explicit range, use $E_m = A_E m + B_E$. If $m \geq m_*$, then

$$D_E m < gm^2/2, \quad 2B_E < gm^2/2, \quad D_E/m < \beta/52, \quad 2B_E/m^2 < \beta/52.$$

The first two inequalities prove WEL32, while the last two prove $\eta_m = (D_E/m + 2B_E/m^2)/\beta < 1/26$. Thus all finite ranking claims hold on the explicitly proved range, uniformly in the original bounded v and both low ranks. $\square$

The exact discrete center in WEL31 is $(m-1)/(2m)$ after division by m , whereas the integral center is $1/2$. Their difference $1/(2m)$ is retained by the finite error and the threshold band. No assertion equates these two finite centers or assigns an unproved exact common cutoff to all thirteen columns.

139 An explicit one-dimensional integral for the coefficient

The integrals in WEL7 can be evaluated without a numerical selection rule. For $d > 0$ and $L_0 \geq 0$ define

$$\begin{aligned} J(L_0, d) &= \frac{1}{2}L_0 \log(L_0^2 + d^2) - L_0 + d \arctan(L_0/d), \\ K(L_0, d) &= \frac{1}{4}(L_0^2 - d^2) \log(L_0^2 + d^2) - \frac{3}{4}L_0^2 + dL_0 \arctan(L_0/d) + \frac{1}{4}d^2 \log d^2, \\ F(d) &= J(1, d) = \frac{1}{2} \log(1 + d^2) - 1 + d \arctan(1/d). \end{aligned} \quad (\text{WEL36})$$

The continuous endpoint values are $J(L_0, 0) = L_0 \log L_0 - L_0$, $K(L_0, 0) = \frac{1}{2}L_0^2 \log L_0 - \frac{3}{4}L_0^2$, and $F(0) = -1$, with $J(0, 0) = K(0, 0) = 0$. Differentiation proves $\partial_{L_0} J(L_0, d) = \frac{1}{2} \log(L_0^2 + d^2)$ and $\partial_{L_0} K(L_0, d) = J(L_0, d)$; both primitives vanish at $L_0 = 0$. These derivative identities and continuity prove

$$\begin{aligned} H(x) &= \log(2\gamma) + \int_0^1 F(a_*|x - t|) dt, \\ V(y) &= \log(2\gamma) + \int_0^1 \{J(y, a_*t) + J(1 - y, a_*t)\} dt. \end{aligned} \quad (\text{WEL37})$$

In the first formula integrate the original w variable after replacing $1 - w$ by a variable in $[0, 1]$. In the second split the original interval at $w = y$ and use $t = 1 - u$. All integrands are integrable by the preceding potential bounds.

The set of pairs $(x, t) \in [0, 1]^2$ with difference magnitude in an infinitesimal interval at $z \in [0, 1]$ has density $2(1 - z)$: split into $x \geq t$ and $t \geq x$ and set $z = |x - t|$ on each triangle, whose other coordinate runs from zero to $1 - z$. Thus integration of the first line of WEL37 gives $\int_0^1 H = \log(2\gamma) + 2 \int_0^1 (1 - t)F(a_*t) dt$. Symmetry gives $\int_0^l V + \int_{u_*}^1 V = 2 \int_0^l V$. Integrating the second line of WEL37 and using the proved primitive K therefore gives the exact coefficient

$$\begin{aligned} \mathcal{C}(\delta, \gamma) &= 24 \log(2\gamma) \\ &\quad + 26 \int_0^1 \left[(1 - t)F(a_*t) + K(l, a_*t) + K(1, a_*t) - K(u_*, a_*t) \right] dt. \end{aligned}$$

(WEL38)

The coefficient 24 comes from the actual selected edge mass $13 + 26l = 24$, while the strip width thirteen and both cut points remain explicitly visible. WEL38 is an exact integral expression in the original fixed parameters, and WEL9 supplies its finite error for the actual products. Numerical fixtures accompanying this note test its formulas only; none enter any proof.

140 Full original data and an explicit finite volume interval

Keep the complete EBG source, original fixed quartet and period, $0 < \delta < 1/2$, $\gamma > 2$, $k \equiv 1 \pmod{4}$, $k \geq 37$, $0 \leq v \leq 80$, and $s = 1$ or k . For the Gamma comparison use its exact additional

domain $k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}(k+1)^2$. Write, without changing any of these quantities,

$$\begin{aligned} q &= (k+1)^2, & q_\uparrow &= (k+5)^2, & n_0 &= (k-7)^2, \\ m &= k-7, & b &= 26k-13, & \ell_s &= (s-1)/4, \\ Q_s &= n_0 + \ell_s, & \vartheta &= \pi/2, & M_s &= (2\pi)^{s/2}, \\ \beta_s &= M_s/(\sqrt{2}\Gamma(s/2)), & \mu_0(Q_s) &= 2\Gamma(2Q_s+1)/\vartheta^{2Q_s+1}, \\ D &= q_\uparrow - v - 1 + \sigma, & r &= D + 1 - n_0 = 24k - 24 - v + \sigma, & \sigma &\in \{0, 1\}. \end{aligned} \tag{BVC1}$$

The auxiliary m is WEL's lower-grid side length, and does not denote a packet multiplicity. The packet remains simple. In particular Q_s is a positive integer for both original orders.

Use the exact potential of the original anisotropic root differences:

$$\begin{aligned} I(x, y) &= \int_0^1 \int_0^1 \log |2\delta(x-u) + 2i\gamma(y-w)| \, du \, dw, \\ l &= 11/26, & u_* &= 15/26, & \mathcal{C} &= 13 \int_0^1 I(x, 1) \, dx + 26 \int_0^l I(1, y) \, dy. \end{aligned} \tag{BVC2}$$

The integrable logarithmic singularities and the symmetry in the second variable were proved in WEL11–18. For a completely explicit single integral, let $a_* = \delta/\gamma$ and use

$$\begin{aligned} F(d) &= \tfrac{1}{2} \log(1+d^2) - 1 + d \arctan(1/d), \\ K(L, d) &= \tfrac{1}{4}(L^2 - d^2) \log(L^2 + d^2) - \tfrac{3}{4}L^2 + dL \arctan(L/d) + \tfrac{1}{4}d^2 \log d^2, \\ \mathcal{C} &= 24 \log(2\gamma) + 26 \int_0^1 [(1-t)F(a_*t) + K(l, a_*t) + K(1, a_*t) - K(u_*, a_*t)] \, dt. \end{aligned} \tag{BVC3}$$

At zero take the continuous values $F(0) = -1$, $K(L, 0) = \frac{1}{2}L^2 \log L - \frac{3}{4}L^2$ and $K(0, 0) = 0$. WEL36–38 derives this identity by differentiating the exact primitives and integrating the original rectangle.

Retain all WEL8 constants:

$$\begin{aligned} \kappa &= \gamma/\delta, & L &= (2\pi+1)\kappa, & C_I &= |\log(2\delta)| + \log(2\sqrt{2}\kappa) + \pi/2, \\ C_N &= 2|\log(2\delta)| + \log(15\kappa^2) + \pi/2, \\ E_m &= (32\sqrt{2}\kappa + 12\sqrt{2}L + 7)m + 24C_N, \\ B_W &= 2rE_m + 2m^2(13L + 313C_I) + 14n_0|\mathcal{C}|. \end{aligned} \tag{BVC4}$$

These constants include every corner and every boundary layer. Let $d_{\min} = 2\min(\delta, \gamma)$, $d_{\max} = 2(k+5)\sqrt{\delta^2 + \gamma^2}$. Keep exactly EBG22's finite $\mathfrak{b}_{D,s}^\pm$ and EBG30's complete moment terms

$$\begin{aligned} \mathcal{L}_{r,Q} &= -r \log(4eQ/\vartheta) - \vartheta^2 r^3/Q + 2 \sum_{j=1}^{r-1} \log(j!), \\ \mathcal{U}_{r,Q} &= \sum_{j=0}^{r-1} \log((2Q+1)_{2j}/\vartheta^{2j}), \\ E_- &= r(r-1) \log d_{\min} - r\mathfrak{b}_{D,s}^+ - \mathcal{U}_{r,Q_s}, \\ E_+ &= \log \binom{b}{r} + r(r-1) \log d_{\max} - r\mathfrak{b}_{D,s}^- - \mathcal{L}_{r,Q_s}. \end{aligned} \tag{BVC5}$$

The finite signed expressions $E_\pm$ are kept as written; neither is changed to an equal or symmetric error.

For the entire original conditional boundary Gram of EBG3,9 put

$$\mathcal{F}_{k,s,\sigma} = 2rn_0 \log k + 2kn_0\mathcal{C} - r \log(\beta_s \mu_0(Q_s)).$$

Then the exact combined interval is

$$\boxed{\mathcal{F}_{k,s,\sigma} - B_W + E_- \leq \log \text{pdet } \mathcal{H}_{B,D} \leq \mathcal{F}_{k,s,\sigma} + B_W + E_+} \quad (\text{BVC6})$$

Here pdet has exactly the actual low rank r . To prove the inequality, EBG32 bounds this logarithm between $\mathcal{W}_{r,k} - r \log(\beta_s \mu_0) + E_-$ and the same expression with E_+ . WEL9–10 proves $|\mathcal{W}_{r,k} - 2rn_0 \log k - 2kn_0 \mathcal{C}| \leq B_W$. Substitute its lower and upper sides respectively. This proves BVC6 with the original source mass, root products, selected ranks and unequal finite errors all retained. It implies

$$\log \text{pdet } \mathcal{H}_{B,D} = 2rn_0 \log k + 2kn_0 \mathcal{C} - r \log(\beta_s \mu_0(Q_s)) + O_{\delta,\gamma}(k^2 \log(k+1)). \quad (\text{BVC7})$$

Indeed BVC4 is $O(k^2)$, while EBG22,30 proves that every term of BVC5 is $O(k^2 \log(k+1))$ at these ranks. This is a consequence of the explicit interval, not an error inserted before calculation.

141 The complete logarithmic leading expression

The remaining Gamma moment in BVC7 has an elementary exact enclosure. For every integer $n \geq 1$, comparison of the increasing $\log x$ with its integrals on successive unit intervals gives

$$1 \leq \epsilon_n := \log(n!) - n \log n + n \leq 1 + \log n.$$

Consequently

$$\log \mu_0(Q) = 2Q \log(2Q/\vartheta) - 2Q + \log(2/\vartheta) + \epsilon_{2Q}. \quad (\text{BVC8})$$

This keeps both the factorial error and its constant. For $s = 1$ the number $\log \beta_1$ is constant. For $s = k$ set $n = (k-1)/2$. Repeatedly applying the Gamma recurrence gives

$$\Gamma(k/2) = \frac{(2n)!}{4^n n!} \sqrt{\pi}, \quad \log \beta_k = \frac{k}{2} \log(2\pi) - \frac{1}{2} \log 2 - \log(2n)! + n \log 4 + \log n! - \frac{1}{2} \log \pi. \quad (\text{BVC9})$$

The same factorial enclosure bounds this last expression by $O(k \log(k+1))$ in absolute value. Thus the retained source mass contributes $r \log \beta_s = O(k^2 \log(k+1))$ for either original source order; its contribution was not omitted in BVC6.

For clarity the leading cancellation can be made algebraic before taking the limit. Write $r = 24k + c_\sigma$ with $c_\sigma = -24 - v + \sigma$, so $|c_\sigma| \leq 104$. Substitute BVC8 into BVC7 and use

$$n_0 = k^2 - 14k + 49, \quad Q_s = n_0 + \ell_s, \quad 0 \leq \ell_s \leq k/4, \quad q = k^2 + 2k + 1, \quad (\text{BVC10})$$

$$\log(2Q_s/\vartheta) = 2 \log k + \log(2/\vartheta) + \log(Q_s/k^2).$$

For $k \geq 37$, Q_s/k^2 is bounded below by $(30/37)^2$ and above by $1 + 1/(4k)$. The mean-value theorem for $\log$ therefore gives $\log(Q_s/k^2) = O(1/k)$ uniformly in both orders. Also $rQ_s/(kq) = 24 + O(1/k)$ and $n_0/q = 1 + O(1/k)$. The exact coefficient of $\log k$ in the first and third terms of BVC7, after division by kq , is

$$\frac{2rn_0 - 4rQ_s}{kq} = -\frac{2r(n_0 + 2\ell_s)}{kq} = -48 + O(1/k).$$

The remaining terms in BVC8 contribute their full constant $48 - 48 \log(2/\vartheta)$ and an $O(\log(k+1)/k)$ error. Combining these displayed identities proves

$$\boxed{\frac{\log \text{pdet } \mathcal{H}_{B,D}}{kq} + 48 \log k = 2\mathcal{C}(\delta, \gamma) + 48(1 - \log(4/\pi)) + O_{\delta,\gamma}\left(\frac{\log(k+1)}{k}\right)}. \quad (\text{BVC11})$$

In particular the expression on the left has exactly the stated finite limit. The proof is uniform in the two original source orders, both low endpoints, and the retained bounded v . On the actual programme the sharper $v \leq 32$ also applies, but was unnecessary for these estimates.

BVC6 remains the full finite original-volume statement. The displayed leading expression concerns the enlarged boundary Gram at the low endpoints. Its actual numerator projection is EBG10–11 and its map from independent gauges is RPB16–19. Those exact maps retain their own coupled metrics. No boundary volume is substituted for the original kernel or residual allocation, and no four-endpoint logarithm is taken of the zero low enlarged quotient.

Keep the original fixed simple quartet, amplitude, period and logarithm branch, and the original invertible period matrix $H = F^{-1} \Pi U \mathcal{E}_{\text{prim}}$. The program degree is $k = 4l + 1 \geq 9$. The coefficient exactness and rank proofs below also hold for every integer $k \geq 9$. All vector spaces are over $\mathbb{C}$. The original biform coefficients and wedge orientations are retained. Native jet kernels are included in the final section and are not removed by the coefficient exactness theorem.

142 The complete original complex and its oriented matrices

Let

$$\begin{aligned} X &= \mathbb{P}_z^1 \times \mathbb{P}_w^1, \quad t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^\top, \\ L_i &= (Ht)_i, \quad h_i = L_i^4 \in H^0(X, \mathcal{O}(4, 4)), \quad h = \sum_{i=1}^4 h_i e_i, \quad V = \mathbb{C}^4, \\ \mathcal{P}_d &= \mathbb{C}[z_0, z_1, w_0, w_1]_{d,d}, \quad e_{ab}^{(d)} = z_0^a z_1^{d-a} w_0^b w_1^{d-b} \quad (0 \leq a, b \leq d). \end{aligned} \tag{GKC1}$$

The order on the original four primary coordinates is $(++)$, $(+-)$, $(-+)$, $(--)$. Use the original ordered basis e_1, e_2, e_3, e_4 of V . At degree p put

$$d_p = k - 8 + 4p, \quad \mathcal{C}^p = \mathcal{O}(d_p, d_p) \otimes \bigwedge^p V, \quad C^p = \mathcal{P}_{d_p} \otimes \bigwedge^p V, \quad d^p \omega = h \wedge \omega \quad (0 \leq p \leq 3). \tag{GKC2}$$

The sheaf and section maps have the displayed source and target degrees. Their squares are zero: each diagonal wedge term is zero and the two terms with distinct indices have equal polynomial coefficient and opposite wedge signs.

The four complete maps in increasing wedge order are

$$\begin{aligned} d^0 f &= (h_i f)_i, \\ (d^1 E)_{ij} &= h_i E_j - h_j E_i & (i < j), \\ (d^2 F)_{ijr} &= h_i F_{jr} - h_j F_{ir} + h_r F_{ij} & (i < j < r), \\ d^3 G &= h_1 G_{234} - h_2 G_{134} + h_3 G_{124} - h_4 G_{123}. \end{aligned} \tag{GKC3}$$

In particular d^1 is precisely the original six-component compatibility map, with the same ordered signs.

Order each C^p first by increasing wedge tuple I , then by increasing a , then increasing b in $e_{ab}^{(d_p)}$. Write $h_i(Z, 1; W, 1) = \sum_{r,t=0}^4 b_{rt}^{(i)} Z^r W^t$. The full original multiplication matrix is

$$(M_{i,d})_{(A,B),(a,b)} = b_{A-a,B-b}^{(i)}, \quad 0 \leq A, B \leq d+4, \quad 0 \leq a, b \leq d, \tag{GKC4}$$

with out-of-range coefficients zero. The block in d^p from wedge I to $I \cup \{i\}$ is $(-1)^{|\{j \in I: j < i\}|} M_{i,d_p}$ when $i \notin I$, and every other block is zero. This gives every companion matrix in the original biform monomial frames.

143 Local sheaf exactness and the full Čech argument

The four standard Segre charts are indexed by $\alpha = (a, b)$, $a, b \in \{0, 1\}$, with $z_a w_b \neq 0$. On such a chart, divide the four primary coordinates t by $t_\alpha = z_a w_b$ and put $x^{(\alpha)} = H(t/t_\alpha)$. Let $\lambda^{(\alpha)}$ be the corresponding row of H^{-1} . Then $\lambda^{(\alpha)} \cdot x^{(\alpha)} = 1$. Partition the complete degree-thirteen

multinomial expansion by the first index with exponent at least four, exactly as in BGA2. This gives the literal chart polynomials

$$U_i^{(\alpha)}(x) = \sum_{\substack{|\nu|=13 \\ \nu_i \geq 4, \nu_j \leq 3 \ (j < i)}} \frac{13!}{\prod_j \nu_j!} \prod_j (\lambda_j^{(\alpha)})^{\nu_j} x^{\nu-4e_i}, \quad \sum_i U_i^{(\alpha)}(x^{(\alpha)})(x_i^{(\alpha)})^4 = 1. \quad (\text{GKC5})$$

The index exists since four exponents at most three sum to at most twelve. Every degree-thirteen monomial occurs once, so this identity retains all coefficients even when some row entries vanish.

Trivialize $\mathcal{O}(d_p, d_p)$ on this chart by $t_\alpha^{d_p}$. The local differential is wedge multiplication by $\sum_i (x_i^{(\alpha)})^4 e_i$. For a row $u = (u_i)$ let ι_u be the contraction

$$\iota_u(e_{i_1} \wedge \cdots \wedge e_{i_p}) = \sum_{r=1}^p (-1)^{r-1} u_{i_r} e_{i_1} \wedge \cdots \widehat{e_{i_r}} \cdots \wedge e_{i_p}. \quad \iota_u(h \wedge \omega) + h \wedge \iota_u \omega = \left(\sum_i u_i h_i \right) \omega. \quad (\text{GKC6})$$

Expanding both sides on each ordered wedge proves the identity: all terms with different contraction and inserted index cancel, and the equal-index terms give the stated scalar. Take $u_i = U_i^{(\alpha)}(x^{(\alpha)})$ in local frames. GKC5 makes GKC6 a contracting homotopy. The same formula works after any localization of this chart. It proves local exactness of the entire sheaf complex, including both end terms.

The passage to global sections requires an additional argument. We give it explicitly. On one projective line use the chart with $z_1 \neq 0$ first and coordinate $Z = z_0/z_1$. In the frame z_1^d , the two-chart Čech complex for $\mathcal{O}(d)$ is

$$C_z^0(d) = \mathbb{C}[Z] \oplus Z^d \mathbb{C}[Z^{-1}] \xrightarrow{\delta(a,b)=b-a} C_z^1(d) = \mathbb{C}[Z, Z^{-1}]. \quad (\text{GKC7})$$

For $d \geq 0$ its degree-zero kernel is precisely $\mathbb{C}[Z]_{\leq d}$, embedded as (f, f) . Split a Laurent polynomial as $f = f_{\geq 0} + f_{< 0}$ and set

$$s(f) = (-f_{\geq 0}, f_{< 0}), \quad \delta s = I, \quad (I - s\delta)(a, b) = (b_{\geq 0}, b_{\geq 0}). \quad (\text{GKC8})$$

The negative part belongs to $Z^d \mathbb{C}[Z^{-1}]$ because every negative exponent is at most d . The non-negative part of b has exponents from zero to d . These formulas prove all assertions and split the complex as its global sections in degree zero plus a contractible complex. The same proof holds for the w factor.

The rectangular four-chart Čech complex on X is the total tensor product of these two complexes. Its degree-zero term consists of the four chart section spaces; its degree-one term consists of the two vertical and two horizontal overlaps; its degree-two term is the double overlap. The differential is $\delta_z + (-1)^{q_z} \delta_w$. Tensor the two explicit splittings in GKC8 over $\mathbb{C}$. The product of the global-section terms is $\mathcal{P}_d$. Every other summand is contractible: a contraction on its first contractible factor, with the ordinary tensor signs, satisfies $\delta s + s\delta = I$. Thus

$$\check{H}^0(X, \mathcal{O}(d, d)) = \mathcal{P}_d, \quad \check{H}^q(X, \mathcal{O}(d, d)) = 0 \quad (q > 0, d \geq 0). \quad (\text{GKC9})$$

In particular the degree-zero identification uses the unchanged monomials $e_{ab}^{(d)}$. This calculation is the full chart and overlap computation, rather than an appeal to local exactness.

Form the finite double complex

$$A^{p,q} = \check{C}^q(X, \mathcal{E}^p), \quad 0 \leq p \leq 4, \quad 0 \leq q \leq 2, \quad d_{\text{tot}} = d^p + (-1)^p \delta. \quad (\text{GKC10})$$

Multiplication by the global sections commutes with restriction, so $d_{\text{tot}}^2 = 0$. At each fixed q , the row complex is exact: every chart or overlap component has the local contraction GKC6, and their finite direct sum is exact.

Here is an elementary finite-double-complex proof that the total complex is acyclic. For a total cycle take its component of smallest q . Its horizontal differential is zero, because there

is no component with smaller q contributing a vertical differential. Row exactness supplies a horizontal preimage. Subtract its total differential. This kills that component and can introduce only components with larger q . Repeat for the successive q values. At the last value there is no further vertical term. The finite sum of the chosen preimages is a total preimage of the cycle.

Each $d_p \geq k - 8 \geq 1$, so GKC9 says that a column has cohomology only in degree zero, equal to C^p . Include that degree-zero global-section subcomplex into $A^{\bullet,\bullet}$. The quotient has acyclic vertical columns. Its total complex is acyclic by the corresponding elimination: take the component with smallest p , use vertical exactness, subtract its total differential and proceed to increasing p . Hence the inclusion induces an isomorphism in cohomology. For completeness, a cycle in the total quotient lifts to a cochain whose differential lies in the included subcomplex; the elementary boundary argument in this short exact sequence of complexes proves the asserted isomorphism when the quotient is acyclic. Both the total complex and its quotient are acyclic, so the global-section complex is acyclic.

We have proved the full global exact sequence

$$0 \longrightarrow \mathcal{P}_{k-8} \xrightarrow{d^0} \mathcal{P}_{k-4}^4 \xrightarrow{d^1} \mathcal{P}_k^6 \xrightarrow{d^2} \mathcal{P}_{k+4}^4 \xrightarrow{d^3} \mathcal{P}_{k+8} \longrightarrow 0. \quad (\text{GKC11})$$

This proof retained the local sheaf contraction, every chart overlap, the vanishing of positive Čech cohomology and the finite argument passing to global sections.

144 Every rank and every coefficient kernel

The five dimensions and four ranks are

$$(n_0, n_1, n_2, n_3, n_4) = ((k-7)^2, 4(k-3)^2, 6(k+1)^2, 4(k+5)^2, (k+9)^2), \\ (\text{rank } d^0, \text{rank } d^1, \text{rank } d^2, \text{rank } d^3) = ((k-7)^2, 3k^2 - 10k - 13, 3k^2 + 22k + 19, (k+9)^2). \quad (\text{GKC12})$$

Indeed the first map is injective. Exactness successively gives $\text{rank } d^p = n_p - \text{rank } d^{p-1}$. Substitution gives the displayed polynomials and $\text{rank } d^3 = n_4$. Consequently

$$\ker d^0 = 0, \quad \ker d^1 = \text{im } d^0, \quad \ker d^2 = \text{im } d^1, \quad \ker d^3 = \text{im } d^2, \quad \text{coker } d^3 = 0. \quad (\text{GKC13})$$

All maps in these equalities are the full signed matrices GKC3–4. In particular the next compatibility kernel and the complete final companion map have now been calculated, not just the first lower-gauge kernel.

145 Determinant-line torsion in the original coefficient frames

We specify the convention so that its sign is determined. Let c_p be the original ordered frame GKC4. For an exact cochain complex choose a basis b_p of $\text{im } d^{p-1}$ and any lifts $\tilde{b}_{p+1}$ of b_{p+1} to C^p . The bases at the two outside ends are empty. Put

$$\tau_k(\mathbf{H}) = \prod_{p=0}^4 \det[b_p, \tilde{b}_{p+1} : c_p]^{(-1)^p}. \quad (\text{GKC14})$$

This is the scalar of the determinant-line trivialization with this cochain convention. Changing b_p by a matrix M multiplies the two adjacent determinants by $\det M$ with opposite exponents, so the factors cancel. Changing a lift adds boundary columns and changes no determinant. Thus GKC14 is independent of all these choices, while retaining the original reference frames and their signs.

Regard the sixteen entries of $\mathbf{H}$ as independent parameters. Every matrix GKC4 is polynomial in them. Over their rational function field, choose any nonzero pivot minors for the exact complex; the adapted determinants give a nonzero rational function $\tau_k(\mathbf{H})$. Near any parameter matrix

at which the coefficient complex is exact, appropriate pivot minors remain nonzero. The same formula there is a regular invertible function and agrees with the rational function by the proved independence of choices. In particular it is a unit at every invertible matrix.

This implies

$$\tau_k(\mathbf{H}) = c_k(\det \mathbf{H})^{m_k}, \quad c_k \in \mathbb{C}^\times, \quad m_k \in \mathbb{Z}. \quad (\text{GKC15})$$

Here is the required algebraic justification. A polynomial ring over $\mathbb{C}$ is a unique-factorization domain by repeated Gauss factorization. The polynomial $\det \mathbf{H}$ is irreducible: every singular matrix factors as a 4 by 3 matrix times a 3 by 4 matrix, so the singular locus is the polynomial image of an irreducible affine parameter space and is irreducible. It is the closed set $\det \mathbf{H} = 0$. The nonzero cofactor at $\text{diag}(1, 1, 1, 0)$ proves that the determinant has multiplicity one. Thus its polynomial factorization has exactly one irreducible factor, to power one.

Write the rational torsion as a quotient of coprime polynomials. Any irreducible factor other than $\det \mathbf{H}$ would give a zero or pole at some invertible matrix, contrary to its local unit property. One can verify the needed existence of such a point without a genericity assumption: regard that irreducible factor as a polynomial in one variable over the fraction field of the other variables. Gauss factorization and the Euclidean algorithm make it relatively prime to the determinant and the other finitely many relevant factors. Specialize the other variables away from their nonzero denominators and leading coefficients. The remaining nonconstant one-variable polynomial has a complex root where those other factors are nonzero. This proves GKC15.

The exponent m_k is zero. To prove this, use the following explicit parameter matrix solely to evaluate the parameter family:

$$\mathbf{H}_0 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}, \quad (L_1, L_2, L_3, L_4) = (t_1, t_4, t_2 - t_3, 0). \quad (\text{GKC16})$$

It has rank three. Put $a = t_1$, $b = t_4$, $c = t_2 - t_3$. On the Segre surface $t_1 t_4 = t_2 t_3$. On the charts $t_1 = 1$ or $t_4 = 1$, respectively a^4 or b^4 is one. On the chart $t_2 = 1$, $ab + c = 1$; on the chart $t_3 = 1$, $ab - c = 1$. Expanding either seventh power shows that one is in the ideal (a^4, b^4, c^4) : in a term $(ab)^r(\pm c)^{7-r}$ either $r \geq 4$, giving a factor a^4 , or $7 - r \geq 4$, giving a factor c^4 . Thus on every standard chart and overlap these four sections generate the unit ideal. GKC6 and the entire Čech argument apply, so the coefficient complex is exact at $\mathbf{H}_0$. Its torsion is a local unit there. Since $\det \mathbf{H}_0 = 0$, GKC15 can be a local unit there only if $m_k = 0$. Hence

$$\tau_k(\mathbf{H}) = \tau_k(I_4) \quad (\det \mathbf{H} \neq 0). \quad (\text{GKC17})$$

This is an identity of coefficient torsion scalars on the original parameter family. The actual $\mathbf{H}$ has not been replaced in any of the maps or coefficients.

We next evaluate that constant with all reference-frame signs. At I_4 , the four sections are

$$h_1 = z_0^4 w_0^4, \quad h_2 = z_0^4 w_1^4, \quad h_3 = z_1^4 w_0^4, \quad h_4 = z_1^4 w_1^4. \quad (\text{GKC18})$$

For an original basis element (I, a, b) at degree p set

$$u = a - 4|I \cap \{1, 2\}|, \quad w = b - 4|I \cap \{1, 3\}|, \quad d_0 = k - 8. \quad (\text{GKC19})$$

The pair (u, w) is preserved by the differential; the other two primary exponents after the corresponding subtraction are $d_0 - u, d_0 - w$. Its range is $-8 \leq u, w \leq k$. A wedge I in this component is permitted exactly when the four numbers

$$u + 4|I \cap \{1, 2\}|, \quad d_0 - u + 4|I \cap \{3, 4\}|, \quad w + 4|I \cap \{1, 3\}|, \quad d_0 - w + 4|I \cap \{2, 4\}| \quad (\text{GKC20})$$

are nonnegative.

Define a forbidden set by taking $\{1, 2\}$ when $u < 0$, $\{3, 4\}$ when $u > d_0$, and neither when $0 \leq u \leq d_0$; adjoin $\{1, 3\}$ when $w < 0$, $\{2, 4\}$ when $w > d_0$, and neither when $0 \leq w \leq d_0$. Since $d_0 \geq 1$, at most one pair is taken in each direction. A pair in the first direction and a pair in the second intersect in one index, so their union has size at most three. Let

$$v(u, w) = \text{the first index outside that union.} \quad (\text{GKC21})$$

Adding or removing this index preserves permissibility in GKC20: it changes no negative-coordinate constraint, and a coordinate that began nonnegative remains so. Thus each fine-degree component has a free wedge index. Contraction with $e_{v(u, w)}^\vee$ and wedge multiplication by $e_1 + e_2 + e_3 + e_4$ obey GKC6 with scalar one on that component.

Let $\mathcal{T}_p$ be the original frame elements of C^p whose wedge does not contain their free index, ordered as in c_p . They give a lift basis, and $d^{p-1}\mathcal{T}_{p-1}$ is a basis of $\text{im } d^{p-1}$. To prove this, contraction kills each element of $\mathcal{T}_p$, and contraction of its differential returns that element. Therefore its differential is injective on their span. For any boundary b , $db = 0$ and the homotopy identity gives $b = d\iota_v b$, with $\iota_v b$ in that same lift span, proving surjectivity onto the boundary space.

Each boundary column coming from $(I, a, b) \in \mathcal{T}_{p-1}$ has exactly one row containing the free index: it is

$$(I \cup \{v\}, a + 4 \mathbf{1}_{\{v \in \{1, 2\}\}}, b + 4 \mathbf{1}_{\{v \in \{1, 3\}\}}), \quad \epsilon(I, v) = (-1)^{|\{j \in I : j < v\}|}. \quad (\text{GKC22})$$

Every other nonzero entry is in a lift row, because its wedge still omits the free index. List first these principal rows, in the original order of $\mathcal{T}_{p-1}$, and then the rows $\mathcal{T}_p$ in their original order. This is a permutation π_p of the complete original frame c_p : the principal rows are precisely the complement of the lift rows, paired uniquely by adding the free index. After this row permutation the adapted matrix is block lower triangular, with boundary diagonal $\epsilon(I, v)$ and lift diagonal one. Therefore the exact original-frame torsion for every integer $k \geq 9$ is

$$\boxed{\begin{aligned} \tau_k(\mathbf{H}) &= (-1)^{E(k)}, \\ E(k) &= \sum_{p=0}^4 \left(\text{inv}(\pi_p) + \sum_{(I, a, b) \in \mathcal{T}_{p-1}} |\{j \in I : j < v(u, w)\}| \right) \pmod{2}. \end{aligned}} \quad (\text{GKC23})$$

The set $\mathcal{T}_{-1}$ is empty. This is a finite formula in the specified original frame order; it contains no undetermined sign. Inverting a sign for an odd cochain degree leaves that same sign, which explains the sum in GKC23 from the convention GKC14.

For the original program degrees this sign is exactly positive:

$$\boxed{\tau_k(\mathbf{H}) = 1 \quad (k = 4l + 1 \geq 9).} \quad (\text{GKC24})$$

Here is a proof, rather than an assignment of orientation. In fact the argument works for every odd $k \geq 9$. All d_p and d_0 are odd, so each index rectangle $0 \leq a, b \leq d_p$ partitions into four-element packets

$$(a, b) = (2A + \varepsilon, 2B + \eta), \quad \varepsilon, \eta \in \{0, 1\}. \quad (\text{GKC25})$$

For a fixed wedge, the two values of u in such a packet are consecutive with even first member. They lie in the same one of $u < 0$, $0 \leq u \leq d_0$, $u > d_0$, because the two boundary positions are 0 and the odd integer d_0 . The same holds for w . Therefore the free index, permissibility, lift selection and orientation sign are constant on the packet. The product of four equal orientation signs is one.

Reorder an original frame or any of its lift subsets into these packets, with the internal order $(\varepsilon, \eta) = (0, 0), (0, 1), (1, 0), (1, 1)$. This reordering is even. For each fixed A , it only interleaves groups of two consecutive b values from the $\varepsilon = 0$ and $\varepsilon = 1$ rows; exchanging two such groups makes four inversions. The same argument holds for a lift subset, which is a union of whole packets.

It also holds independently in each original wedge block. The principal row map GKC22 adds multiples of four to a, b and consequently preserves the two internal parity indices. In packet order its remaining permutation merely permutes four-element packets, and every packet transposition makes sixteen inversions. Thus $\text{inv}(\pi_p)$ is even and the orientation product is positive for every p . Each adapted determinant in GKC14 is one. This proves GKC24 in exactly the original frames.

The accompanying integer diagnostic computes GKC21–23, checks all principal-row permutations and rank counts, and records the complete permutations at $k = 9$. Its additional sampled degrees are declared checks of the formula. The proof of GKC24 is the packet argument above, not an extrapolation from those samples.

146 The original exponential realization and its finite jet kernels

Set

$$\beta_* = \frac{1}{2} - \delta - i\gamma, \quad Z(z) = e^{2\delta z}, \quad W(z) = e^{2i\gamma z}, \quad B_i(z) = e^{4\beta_* z} L_i(Z(z), 1; W(z), 1)^4. \quad (\text{GKC26})$$

The exact degree- j coefficient-to-exponential isomorphism is

$$\mathcal{P}_j \xrightarrow{\sim} \mathcal{E}_j, \quad f \mapsto e^{j\beta_* z} f(Z(z), 1; W(z), 1), \quad \beta_{j,a,b} = j\beta_* + 2a\delta + 2bi\gamma. \quad (\text{GKC27})$$

The exponents are distinct; derivatives at zero give the full Vandermonde and prove injectivity. Their definition gives surjectivity. The identity $\beta_{j,a,b} + \beta_{4,r,t} = \beta_{j+4,a+r,b+t}$ proves that the coefficient differential GKC3 becomes wedge multiplication by (B_i) , with every same sign. Thus GKC11 also gives the complete original exponential coefficient complex.

For the native comparison keep the actual center and either of the two source-order families specified by the program:

$$j_p = k - 8 + 4p, \quad c_p^{\text{nat}} = j_p/2, \quad s_p = 1 \text{ for every } p \quad \text{or} \quad s_p = k + 4p. \quad (\text{GKC28})$$

At any fixed polynomial cutoff $D \geq 0$, let $\phi_{n,s_p,c_p^{\text{nat}}}$ be the full original Gamma orthonormal frame, including mass $(2\pi)^{s_p/2}$. The exact jet/evaluation map, on each wedge component, is

$$(\mathcal{V}_{p,D})_{n,(a,b)} = \phi_{n,s_p,c_p^{\text{nat}}}(\beta_{j_p,a,b}), \quad 0 \leq n \leq D, \quad 0 \leq a, b \leq j_p. \quad (\text{GKC29})$$

The complete matrix on C^p is the direct sum over its $\binom{4}{p}$ wedge components. Its kernel is exactly the coefficient space whose first $D+1$ ordinary moments $\sum \nu_{ab} \beta_{j_p,a,b}^n$ vanish. Indeed the monomial-to-Gamma frame matrix is invertible and triangular with nonzero diagonal, so these are equivalent equations. The rectangular Vandermonde has rank $\min\{(j_p+1)^2, D+1\}$. Hence

$$\dim \ker \mathcal{V}_{p,D} = \binom{4}{p} \max\{0, (j_p+1)^2 - (D+1)\}. \quad (\text{GKC30})$$

No inverse Gram is taken on that kernel.

At the actual common low cutoff $D = q_{k+4} - v - 1 = (k+5)^2 - v - 1$, with $0 \leq v \leq 80$, the full kernel dimensions are

$$\boxed{(\dim \ker \mathcal{V}_{p,D})_{p=0}^4 = (0, 0, 0, 4v, 8k + 56 + v).} \quad (\text{GKC31})$$

For the first three terms, $(D+1) - (k+1)^2 = 8k + 24 - v > 0$ on $k \geq 9$, and their grids are no larger than that degree- k grid. For $p = 3$ the difference is exactly v per wedge component. For $p = 4$ it is $(k+9)^2 - (k+5)^2 + v = 8k + 56 + v$. At any other endpoint the full formula is GKC30.

These kernels have an exact cohomological interface. In the finite Taylor space $\mathbb{C}[z]/(z^{D+1})$, multiplication by each B_i is well defined and commutes with every other such multiplication. In

the distinct original Gamma frames this gives the appropriate ordinary transpose of the primal finite translation map, and the evaluation square is exactly

$$\mathcal{V}_{p+1,D}d^p = d_{\text{jet}}^p \mathcal{V}_{p,D}. \quad (\text{GKC32})$$

This follows either by the original root-addition identity or by the Leibniz formula for every derivative through D . Thus $K_D^p = \ker \mathcal{V}_{p,D}$ is a subcomplex, $J_D^p = \text{im } \mathcal{V}_{p,D}$ is its specified image complex, and there is a termwise exact sequence

$$0 \longrightarrow K_D^\bullet \longrightarrow C^\bullet \longrightarrow J_D^\bullet \longrightarrow 0, \quad H^p(J_D^\bullet) \xrightarrow{\sim} H^{p+1}(K_D^\bullet). \quad (\text{GKC33})$$

For the last isomorphism, lift an image cycle to $c \in C^p$. Its differential belongs to K_D^{p+1} and is a cycle. Changing the lift changes that cycle by a boundary in K_D ; changing the image cycle by an image boundary also changes nothing in its cohomology class. Every K_D cycle is a differential in the exact coefficient complex GKC11, proving surjectivity. If that differential is a K_D boundary, subtract its K_D preimage from the lift; the resulting closed coefficient cochain is a coefficient boundary, proving injectivity. This proves the entire connecting map and its exact inverse on cohomology classes.

The coefficient determinant-line identity GKC24 is therefore not a deletion of the native jet kernels. At the common low cutoff the last two coefficient evaluation Grams have exactly the kernels GKC31. The original finite comparison must use their actual images, retained kernels and maps GKC32–33. The coefficient torsion assigns no native isometry, no positive five-term Gram family at that cutoff, and no bound for the original compatibility determinant. It gives the complete coefficient companion complex and its exact determinant-line trivialization for that further calculation.

147 The complete original-grid Jacobian in the graded compatibility complex

Keep the original fixed $\delta > 0$, $\gamma > 0$, all period coefficients and original grid orders. For $k \equiv 1 \pmod{4}$, $k \geq 9$, the five lower-grid degrees in the complete degree-four compatibility complex are

$$n_p = k - 8 + 4p \quad (0 \leq p \leq 4), \quad c_p = (-1)^p \binom{4}{p}, \quad \beta_{n,a,b} = n\beta_* + 2a\delta + 2bi\gamma. \quad (\text{GVJ1})$$

Let $\mathbb{V}_n = (\beta_{n,a,b}^j)$ be the ordinary square node Vandermonde, $0 \leq j < (n+1)^2$, in the unchanged original node order. In particular

$$\det \mathbb{V}_n = \prod_{(a,b) < (r,t)} (2\delta(r-a) + 2i\gamma(t-b)). \quad (\text{GVJ2})$$

The order on each pair is precisely the original column order. This formula preserves the determinant phase. The common base exponent cancels from each displayed difference because the two nodes belong to the same original degree. Distinctness follows from $\delta, \gamma > 0$.

Define

$$J_n = \log |\det \mathbb{V}_n|^2, \quad \mathcal{J}_k^{(4)} = \sum_{p=0}^4 c_p J_{n_p}. \quad (\text{GVJ3})$$

This is exactly the squared node-Jacobian contribution with all five binomial multiplicities in the full graded complex. No native Gamma form is being replaced by a node coefficient norm: this source calculates the literal scalar factors when those exact Vandermondes occur in a native determinant identity.

For a nonzero integer difference (h, l) the number of ordered pairs of nodes in the degree- n rectangle having that difference is $(n + 1 - |h|)(n + 1 - |l|)$ when $|h|, |l| \leq n$, and zero otherwise. Therefore

$$J_n = \sum_{\substack{(h,l) \in \mathbb{Z}^2 \\ (h,l) \neq (0,0)}} (n + 1 - |h|)_+(n + 1 - |l|)_+ \log |2\delta h + 2i\gamma l|. \quad (\text{GVJ4})$$

Here $x_+ = \max(x, 0)$. To prove the count, choose the first coordinate of an ordered pair with difference h ; there are exactly $n + 1 - |h|$ choices when $|h| \leq n$. The independent second coordinate has the corresponding number for l . In the absolute logarithm of GVJ2 each unordered pair contributes twice. Its two ordered differences have identical absolute value, which proves GVJ4 with no missing factor two.

Set the explicit integer coefficient

$$C_k(h, l) = \sum_{p=0}^4 c_p(k - 7 + 4p - |h|)_+(k - 7 + 4p - |l|)_+. \quad (\text{GVJ5})$$

It has the exact support and metric identity

$$\begin{aligned} C_k(h, l) &= 0 \quad \text{if } \max(|h|, |l|) \leq k - 8 \text{ or } \max(|h|, |l|) > k + 8, \\ \mathcal{J}_k^{(4)} &= \sum_{k-8 < \max(|h|, |l|) \leq k+8} C_k(h, l) \log |2\delta h + 2i\gamma l|. \end{aligned} \quad (\text{GVJ6})$$

In the inner rectangle each product in GVJ5 is an ordinary polynomial in p of degree at most two. The binomial sum of every polynomial of degree less than four is zero: apply the fourth forward difference four times to the monomials, or differentiate $(1 - t)^4 = \sum_p c_p t^p$ at $t = 1$ for falling powers of orders zero through three. This proves the first vanishing. Outside the outer rectangle all positive parts vanish. Substitution of GVJ4 into the finite sum GVJ3 proves the second line. This is an exact cancellation of every interior difference, rather than an inference from a leading asymptotic formula.

For a fully explicit finite enclosure put

$$\begin{aligned} \sigma &= (\delta^2 + \gamma^2)^{1/2}, \quad L_k = \max\{|\log(2 \min(\delta, \gamma)(k - 7))|, |\log(2\sigma(k + 8))|\}, \\ B_k^{\text{grid}} &= (128k + 64) 256(k + 9) L_k. \end{aligned} \quad (\text{GVJ7})$$

Then

$$|\mathcal{J}_k^{(4)}| \leq B_k^{\text{grid}} = O_{\text{actual}}(k^2 \log(k + 1)) = o(kq_k), \quad q_k = (k + 1)^2. \quad (\text{GVJ8})$$

There are exactly $(2k + 17)^2 - (2k - 15)^2 = 128k + 64$ integer pairs in the annulus of GVJ6. At any such pair, one index has absolute value at least $k - 7$. Whenever a summand in GVJ5 is nonzero, the associated positive factor is at most sixteen; its other factor is at most $k + 9$. Since $\sum_p |c_p| = 16$, it follows that $|C_k(h, l)| \leq 256(k + 9)$. Every nonzero distance in that annulus lies between $2 \min(\delta, \gamma)(k - 7)$ and $2\sigma(k + 8)$. The absolute value of its logarithm is therefore at most L_k . Multiplying these three explicit bounds proves the finite inequality. For fixed original δ, γ , $L_k = O_{\text{actual}}(\log(k + 1))$, and division by $k(k + 1)^2$ proves the last limit.

The calculation is unchanged at every source cutoff: it concerns the same ordinary node factors, not their possibly rank-deficient finite evaluation matrices. Where those matrices are square-invertible, their complete leading coefficients are still the original $i^{-j}/\sqrt{M_s j!(s/2)_j}$ and must accompany GVJ2. Where a late term of the low-endpoint complex has a jet kernel, its rectangular matrix and exact kernel must be retained instead. GVJ8 does not turn such a matrix into an invertible Vandermonde. Thus the full interior node-Jacobian contribution to the complete five-term complex has a proved sub- kq bound, with its remaining boundary sum and all finite constants explicitly displayed.

148 The five original native Gram volumes at the high endpoints

Keep the original fixed actual quartet and period. Set $\sigma = (\delta^2 + \gamma^2)^{1/2}$, $\epsilon = 2^{-32}$ and $\vartheta = \pi/2$. All five spaces are exactly those in GKC, with

$$\begin{aligned} k &\equiv 1 \pmod{4}, \quad k \geq K_* := \max\{1025, \lceil 2\sigma/\epsilon \rceil + 1\}, \quad 0 \leq v \leq 80, \quad q = (k+1)^2, \\ j_p &= k + 4p, \quad n_p = j_p - 8, \quad N_p = (n_p + 1)^2, \quad q_p = (j_p + 1)^2, \quad c_p = (-1)^p \binom{4}{p}, \quad 0 \leq p \leq 4, \\ s_p &= 1 \text{ for every } p \quad \text{or} \quad s_p = j_p, \quad \ell_p = (s_p - 1)/4, \quad Q_p = N_p + \ell_p, \\ D &= 2(k+5)^2 - v - 1 \quad \text{or} \quad 2(k+5)^2 - v, \quad a_p = D - N_p + 1. \end{aligned} \tag{GGE1}$$

The threshold is explicit in the original quartet. It is used only for the proved uniform analytic estimates below. The exact determinant identities hold already whenever their displayed finite evaluation matrices have full column rank. Every D in GGE1 is the original degree-four target high cutoff; it has not been replaced by five distinct source cutoffs.

Let $V_{p,D}$ have entries $\phi_{r,s_p,n_p/2}(\beta_{n_p,A,B})$, $0 \leq r \leq D$, in the full original Gamma frames and original node order. Let $\chi_p(S')$ be the complete monic polynomial of those N_p roots. Define

$$\begin{aligned} E_{p,D} &= V_{p,D}^* V_{p,D}, \\ H_{p,D} &= \text{Gram}_{s_p,n_p/2} \{ \chi_p(S') S'^r : 0 \leq r < a_p \}, \\ \mathfrak{E}_{k,D} &= \sum_{p=0}^4 c_p \log \det E_{p,D}. \end{aligned} \tag{GGE2}$$

These are the five literal coefficient Grams used in the complete complex. They are positive at the stated high cutoffs: the smallest value of $D+1-N_p$ occurs at $p=4$ and is at least $k^2+2k-31-v > 0$, with equality at the lower choice of D . Distinct nodes and their first N_p polynomial rows prove full column rank; the original measure makes the polynomial Gram positive.

148.1 Every monic factor and source mass in the exact identity

With $\mathbb{V}_{n_p}$ the square ordinary Vandermonde GVJ2, one has

$$\log \det E_{p,D} = \log |\det \mathbb{V}_{n_p}|^2 + \log \det H_{p,D} - \sum_{r=0}^D \log h_{r,s_p}, \quad h_{r,s} = M_s r! (s/2)_r, \quad M_s = (2\pi)^{s/2}. \tag{GGE3}$$

To prove this, divide the full monomial frame through D into the first N_p monomials and the a_p polynomials $\chi_p S'^r$. Their change matrix is triangular with determinant one because χ_p is monic. The full native Gram determinant is $\prod_{r=0}^D h_{r,s_p}$: the original orthonormal frame has leading coefficient $i^{-r}/\sqrt{h_{r,s_p}}$, of squared modulus h_{r,s_p}^{-1} . The Gram of the remainder quotient has determinant $\prod h_{r,s_p} / \det H_{p,D}$ by its exact Schur complement. Evaluation factors through this quotient, with matrix $\mathbb{V}_{n_p}$ in the remainder frame. Taking its dual Gram and determinant gives GGE3. The complex-linear dual conjugation leaves this positive determinant unchanged. Every monomial phase and every full M_{s_p} has thus been retained before the stated determinant operations.

The weighted Vandermonde part in GGE3 is already calculated exactly in GVJ5–8:

$$\left| \sum_p c_p \log |\det \mathbb{V}_{n_p}|^2 \right| \leq B_k^{\text{grid}} = O_{\text{actual}}(k^2 \log(k+1)). \tag{GGE4}$$

It is the exact width-sixteen difference-annulus sum. No leading lattice asymptotic is used in this bound.

148.2 The complete Gamma-to-moment comparison at these cutoffs

The original GPA full-form comparison applies separately at each j_p with this same D . Here is the complete domain check. Since $j_p \geq 37$,

$$N_p \geq q_p/2, \quad Q_p \leq q_p, \quad 0 \leq a_p - 1 \leq a_0 - 1 \leq 2q \leq 2q_p, \quad j_p \sigma \leq \epsilon q_p/2. \quad (\text{GGE5})$$

The first two follow by substituting $N_p = q_p - 16j_p + 48$ and $0 \leq \ell_p \leq (j_p - 1)/4$. For the third the maximal degree is $a_0 - 1 \leq k^2 + 34k + 1 - v \leq 2(k + 1)^2$; its last inequality is $k^2 - 30k + 1 + v \geq 0$. The final bound follows from $j_p \geq k > 2\sigma/\epsilon$ and $q_p \geq j_p^2$.

To retain the finite constants, in the following formulas use $t = q_p$, $R = j_p \sigma$, $h = 1/t$, $\ell = \ell_p$, $Q = Q_p$, and $d = a_p - 1$:

$$\begin{aligned} C_L &= \pi/\Gamma(3/4)^2, \quad K_t = \frac{\sqrt{2\pi} (2t+1)^2}{C_L t} e^{-2t}, \\ \mathcal{E}_p &= \frac{tR^2}{\epsilon^2 t^2 - R^2} + \log(1 + K_t), \\ r_p^\Gamma &= \frac{\sum_{u=0}^{\ell-1} (2u + \frac{1}{2})^2}{\epsilon^2 t^2} + \log(1 + K_t) \quad (\ell > 0), \quad r_p^\Gamma = 0 \quad (\ell = 0), \\ L_h &= C_L (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \quad K_{d,Q} = 2Q + 1 + 2d + 2\sqrt{d(d+2Q)}, \\ b_p^- &= -\mathcal{E}_p + \log L_h - K_{d,Q} \log(1 + h/\vartheta), \\ b_p^+ &= \mathcal{E}_p + r_p^\Gamma + \log U_h + K_{d,Q} \log(1/(1 - h/\vartheta)), \\ \beta_{s_p} &= \frac{M_{s_p}}{\sqrt{2}\Gamma(s_p/2)}. \end{aligned} \quad (\text{GGE6})$$

These are precisely the full GPA root, inner-region, source-recurrence, density and rate-change constants, with the degree and both parameters verified by GGE5. Its exact maps use $y = (S' - n_p/2)/i$ and all original paired roots of χ_p . The change from $\chi_p S'^r$ to $\chi_p y^r$ has determinant of modulus one. Consequently the complete comparison is

$$\begin{aligned} \log \det H_{p,D} &= a_p \log \beta_{s_p} + \log \mathcal{A}_{a_p}(Q_p; \vartheta) + e_p, \quad a_p b_p^- \leq e_p \leq a_p b_p^+, \\ \mathcal{A}_a(Q; \vartheta) &= \det \left(\int_{\mathbb{R}} y^{u+w} |y|^{2Q} e^{-\vartheta|y|} dy \right)_{0 \leq u, w < a}. \end{aligned} \quad (\text{GGE7})$$

In particular $b_p^\pm = O_{\text{actual}}(\log q)$ uniformly over all five indices: $q_p \asymp q$, $R = O_{\text{actual}}(k)$, the first error in GGE6 is bounded, $r_p^\Gamma = O(1/k)$, the sine logarithms are $O(\log q)$ and $K_{d,Q} = O(q)$ multiplies rate logarithms $O(1/q)$. Thus $\sum_p |c_p e_p|$ is bounded by the explicit number

$$\mathcal{B}_{\text{GPA}} = \sum_p |c_p| a_p \max(|b_p^-|, |b_p^+|) = O_{\text{actual}}(q \log q). \quad (\text{GGE8})$$

This uses all five complete full-form estimates, including their original masses and all inner-region contributions.

148.3 The exact parameter shifts and their moment cancellation

Let $F(\alpha, \vartheta)$ be the fully specified original EIQ equilibrium free energy and its proved QLG estimate on $\alpha \in [3/2, 5/2]$ used in OFV3 and OFV19–25. Keep the same value $\vartheta = \pi/2$ throughout. For clarity define

$$\mathcal{H}(a, Q) = a(2Q + a) \log(a/2) + \frac{a^2}{2} F(2Q/a, \vartheta). \quad (\text{GGE9})$$

The complete parity split of $\mathcal{A}_a$ has block sizes $\lceil a/2 \rceil, \lfloor a/2 \rfloor$ and exponents $Q - 1/2, Q + 1/2$. Their exact substitutions $x = n^2 u$ give the factor $n^{2n(A+n)}$, as in OFV19–20. For GGE1 one has

$$|a_p - q| \leq 32k + 112, \quad |Q_p - q| \leq 17k + 84. \quad (\text{GGE10})$$

Indeed $|N_p - q| \leq 16k + 80$, $\ell_p \leq (k + 15)/4$, and the direct expression for $a_p - q$ ranges from at least -112 to at most $32k + 1$. For $k \geq 1025$ these imply $a_p \in [15q/16, 17q/16]$ and $Q_p \in [31q/32, 33q/32]$. The segments joining any pair of these parameters remain in the same rectangle. On it $2Q/a \in [31/17, 11/5]$. The actual parity parameters differ by at most one half, so their ratio A/n remains in the interior of the proved compact interval $[3/2, 5/2]$.

The same complete QLG and rounding calculation therefore gives, at exactly these new parameters,

$$\log \mathcal{A}_{a_p}(Q_p; \vartheta) = \mathcal{H}(a_p, Q_p) + \eta_p, \quad |\eta_p| \leq C_{\text{QLG}} q \log q. \quad (\text{GGE11})$$

An explicit choice is $C_{\text{QLG}} = 16 + 5M_F + 2C_0$, where M_F is the fixed compact supremum defined before GGE13 and $C_0 = \max\{0, L_0, 1 + 4M_* + \log^+ T_*\}$ is the literal QLG30 constant, with $L_0 = \log(B_* M_*)$ and the same QLG8–14 quantities on that compact rectangle. To prove this transport at the new endpoints, set $G(n, A) = 2n(A + n) \log n + n^2 F(A/n, \vartheta)$. Every half-unit rounding segment obeys $q/3 \leq n \leq q$, $0 < A \leq 2q$ and $\log q \geq 1$. Its exact derivatives therefore satisfy

$$\begin{aligned} G_A &= 2n \log n + n F_\alpha, & |G_A| &\leq 2q \log q + q M_F, \\ G_n &= 2(A + 2n) \log n + 2(A + n) + 2n F - A F_\alpha, \\ |G_n| &\leq 8q \log q + 6q + 4q M_F. \end{aligned}$$

There are two parity blocks, and each changes n and A by at most one half from $(a_p/2, Q_p)$. Their total rounding error is at most $10q \log q + 6q + 5q M_F$. The two literal QLG31 errors sum to $C_0 a_p \leq 2C_0 q$. Adding their central expressions gives GGE9 exactly. Since $\log q \geq 1$, these estimates prove GGE11 with the displayed finite constant and every domain checked.

The shifts from $p = 0$ are precisely

$$t_p = 8p(k - 7) + 16p^2, \quad a_p = a_0 - t_p, \quad Q_p = Q_0 + t_p + \zeta p, \quad \zeta = 0 \text{ or } 1. \quad (\text{GGE12})$$

The fourth binomial sum annihilates 1, t_p and p . Thus the constant and the two linear Taylor terms of $\mathcal{H}$ at (a_0, Q_0) cancel exactly.

Here is a quantitative bound for the remaining terms. Put $M_F = \sup_{\alpha \in [3/2, 5/2]} (|F| + |F_\alpha| + |F_{\alpha\alpha}|)$, which is finite by the complete explicit smooth EIQ calculation. For $\alpha = 2Q/a$ direct differentiation of GGE9 gives

$$\begin{aligned} \mathcal{H}_{QQ} &= 2F_{\alpha\alpha}, \\ \mathcal{H}_{aQ} &= 2 \log(a/2) + 2 + F_\alpha - \alpha F_{\alpha\alpha}, \\ \mathcal{H}_{aa} &= 2 \log(a/2) + 3 + \alpha + F - \alpha F_\alpha + \frac{\alpha^2}{2} F_{\alpha\alpha}. \end{aligned} \quad (\text{GGE13})$$

Each entry is at most $2 \log q + C_H$ in absolute value, where one valid explicit constant is $C_H = 10 + 2 \log(32/15) + 8M_F$. The Hessian operator norm is at most twice this number by its two row sums. The integral Taylor remainder and GGE12 therefore prove

$$\left| \sum_p c_p \mathcal{H}(a_p, Q_p) \right| \leq (2 \log q + C_H) \sum_p |c_p| (t_p^2 + (t_p + \zeta p)^2) =: \mathcal{B}_\mathcal{H} = O(q \log q). \quad (\text{GGE14})$$

This proves the full first-order cancellation and its remainder. No first variation is inferred from an unquantified q^2 asymptotic.

148.4 The complete finite Gamma recurrences

It remains to calculate, with the original signs,

$$\mathcal{M}_{k,D} = \sum_p c_p \left(a_p \log \beta_{s_p} - \sum_{r=0}^D \log h_{r,s_p} \right). \quad (\text{GGE15})$$

For the order-one family it is zero exactly: β_1 is constant, a_p is quadratic in p , and every $h_{r,1}$ is independent of p . For the degree-order family put $b = k/2$. The part $(D+1) \log M_{k+4p}$ is linear in p and the factorial part is constant, so both cancel in the second sum. Its remaining absolute value is at most

$$\mathcal{B}_h = 96D \left(b^{-4} + \frac{1}{3b^3} \right). \quad (\text{GGE16})$$

Indeed let $f(x) = \sum_{r=0}^D \log(b+2x)_r$. Its fourth derivative is $-96 \sum_{r=1}^D \sum_{j=0}^{r-1} (b+2x+j)^{-4}$. For $0 \leq x \leq 4$ its absolute value is at most GGE16, using $\sum_{j \geq 0} (b+j)^{-4} \leq b^{-4} + \int_0^\infty (b+t)^{-4} dt$. The exact fourth forward difference equals the integral of $f^{(4)}$ over $[0,1]^4$ evaluated at the sum of the integration variables. This proves GGE16.

In $a_p \log \beta_{k+4p}$, the mass part is a polynomial of degree at most three in p and the $\log \sqrt{2}$ part has degree at most two. They cancel exactly. The remaining term is $-a_p g(p)$ with $g(p) = \log \Gamma(b+2p)$. The original Gamma recurrence gives $\Delta g(p) = \log((b+2p)(b+2p+1))$. Differentiating this last elementary function and using the same integral formula for finite differences gives

$$|\Delta^2 g(0)| \leq 4/b, \quad |\Delta^3 g(0)| \leq 8/b^2, \quad |\Delta^4 g(0)| \leq 32/b^3. \quad (\text{GGE17})$$

For example its first three derivatives have absolute bounds $4/b, 8/b^2, 32/b^3$ on $[0,4]$. Since $\Delta a(3) = -(8k+56)$, $\Delta^2 a(2) = -32$ and all higher differences of a are zero, the exact finite product rule gives

$$\begin{aligned} \Delta^4(a g)(0) &= a_4 \Delta^4 g(0) + 4 \Delta a(3) \Delta^3 g(0) + 6 \Delta^2 a(2) \Delta^2 g(0), \\ |\mathcal{M}_{k,D}| &\leq \mathcal{B}_h + \frac{32a_4}{b^3} + \frac{32(8k+56)}{b^2} + \frac{768}{b} =: \mathcal{B}_\Gamma = O(1/k). \end{aligned} \quad (\text{GGE18})$$

The finite product rule follows by applying $\Delta(fg)(p) = f(p+1)\Delta g(p) + \Delta f(p)g(p)$ four times. The alternating binomial weights equal Δ^4 since its order is even. This proves the displayed signs and bound without dropping a source mass or using an unstated Gamma normalization.

Combining every term of GGE3–18 proves the complete native five-Gram estimate

$$\begin{aligned} |\mathfrak{E}_{k,D}| &\leq B_k^{\text{grid}} + \mathcal{B}_{\text{GPA}} + 16C_{\text{QLG}} q \log q + \mathcal{B}_\mathcal{H} + \mathcal{B}_\Gamma, \\ \boxed{\mathfrak{E}_{k,D} = O_{\text{actual}}(q \log q) = o(kq)} \end{aligned} \quad (\text{GGE19})$$

at both literal high endpoints and in both source families.

For the order-one family take $\mathcal{B}_\Gamma = 0$. Every finite error is stated before its order estimate. The same actual period and original root sets are used at all five degrees.

148.5 The exact native relation to all companion maps

Use the complete coefficient complex GKC1–33, whose original-frame torsion is exactly one at the program degrees. Equip its p th term with the direct sum of $\binom{4}{p}$ copies of $E_{p,D}$. Let $\rho_{p,D}$ be the logarithm of the squared Jacobian of the exact onto image map

$$C^p / \ker d^p \xrightarrow{d^p} \text{im } d^p \subset C^{p+1}, \quad 0 \leq p \leq 3, \quad (\text{GGE20})$$

using its full source quotient and target restriction metrics. All are positive at the high cutoffs. Then the exact determinant identity is

$$\mathfrak{E}_{k,D} = -\rho_{0,D} + \rho_{1,D} - \rho_{2,D} + \rho_{3,D}. \quad (\text{GGE21})$$

To prove it, use the adapted coefficient frames $[b_p, \tilde{b}_{p+1}]$ from GKC14. The full Gram in that frame factors, by its Schur complement, into the restriction Gram on b_p and the quotient Gram on $\tilde{b}_{p+1}$. The squared Jacobian in GGE20 is the determinant ratio of the restriction Gram on b_{p+1} to the quotient Gram on its lift in degree p . The alternating product of these ratios telescopes to the alternating product of the adapted full Grams, with exactly the signs in GGE21. The difference from the original-frame product is the squared absolute torsion GKC14, which is one by its complete proof. This proves GGE21 with the original kernels, image metrics and orientations.

The first native map is NBG14 restricted to the full old lower evaluation space. Both its upper norm and its actual left-inverse norm are bounded by K_+, K_- . Its rank is $(k-7)^2 = O(q)$, so $|\rho_{0,D}| \leq 2(k-7)^2 \log \max(1, K_+, K_-) = O(q \log q)$. Together with GGE19 and GGE21 this gives

$$\rho_{1,D} - \rho_{2,D} + \rho_{3,D} = O_{\text{actual}}(q \log q) = o(kq). \quad (\text{GGE22})$$

The middle and final companion terms remain those in GGE20. This proved relation does not assign individual inverse bounds to them.

For the full ambient source quotient in NKC, its unaugmented compatibility Gram is exactly the image Jacobian of d^1 , since its source lower-gauge quotient and target restriction are the same NBG/NKC/GKC maps. Hence NKC15–16 and GGE22 give

$$\log \det \Omega_{\text{full},D} = \rho_{2,D} - \rho_{3,D} + O_{\text{actual}}(q \log q). \quad (\text{GGE23})$$

For an actual proper source restriction the complete NKC16 complementary-projection summand remains. At low cutoffs the GKC31 jet kernels also remain; GGE19–23 assert no positive five-Gram family there. Thus the native compatibility determinant has been carried through the entire companion complex at the high endpoints, with its two remaining image Jacobians and its original low-endpoint interface explicitly retained.

149 The original proper source and the companion Jacobians

Retain the actual fixed quartet, period, conductor and all NBG/NKC native spaces at the same degree k and cutoff $D \geq q_{k+4} - v - 1$. In this section X always denotes a specified subspace of the original source $\mathcal{H}_0 / \text{im } V_0$, with its original correlated quotient metric. In particular this includes the literal DNE image of an original conductor or invariant frame. It does not replace that frame by a subspace chosen using the metric.

Use exactly the NKC6 complement Z_c , quotient Gram G_c , and isometry W_c . The complete coefficient maps and native Gamma evaluation matrices in the following forms are those of NKC7–9:

$$\begin{aligned} H_{\text{comp}} &= Z_c^* \mathcal{K}^* \mathcal{V}_2^* \mathcal{V}_2 \mathcal{K} Z_c, & H_{\text{rem},X} &= Z_c^* \mathcal{V}^* P_{\text{rem},X} \mathcal{V} Z_c, \\ \chi_X &= \log \det (I + H_{\text{comp}}^{-1/2} H_{\text{rem},X} H_{\text{comp}}^{-1/2}), & \rho_{\text{comp}} &= \log \det H_{\text{comp}} - \log \det G_c. \end{aligned} \quad (\text{PCJ1})$$

All these are forms on the full original compatibility coefficient quotient, of dimension $r_{\text{comp},k} = 3k^2 - 10k - 13$. In fact H_{comp} is positive: NKC8 identifies $\ker \mathcal{K} = \text{im } \mathcal{P}$, Z_c is a complement of that kernel, and $\mathcal{V}_2$ is injective at every cutoff in this section. The second form is positive semidefinite because $P_{\text{rem},X}$ is an orthogonal projector. Thus PCJ1 is defined and $\chi_X \geq 0$, including all actual coefficient conditioning in H_{comp} .

149.1 The exact proper-source determinant and its finite error

There is an exactly defined real number δ_X such that

$$\begin{aligned} \log \det \Omega_{X,D} &= \rho_{\text{comp}} + \chi_X + \delta_X, \\ -r_{\text{comp},k} \log c_+ &\leq \delta_X \leq -r_{\text{comp},k} \log c_-, & c_- &= \min(1, L_-^{-2}), & c_+ &= \max(1, L_+^2). \end{aligned} \quad (\text{PCJ2})$$

Here $L_{\pm}$ are the actual NKC3 constants, with all original source masses, centres and Bezout coefficients. To prove the assertion, factor $H_{\text{comp}} + H_{\text{rem},X}$ by $H_{\text{comp}}^{1/2}$ on both sides. Determinants give

$$\log \det \Gamma_{X,D} = \log \det (H_{\text{comp}} + H_{\text{rem},X}) - \log \det G_c = \rho_{\text{comp}} + \chi_X. \quad (\text{PCJ3})$$

The first equality is the exact NKC16 conjugation. Define $\delta_X = \log \det \Omega_{X,D} - \log \det \Gamma_{X,D}$ and apply the already proved two-sided NKC15 inequality. This proves PCJ2 with the precise signs and dimensions. When X is zero, $\Omega_{X,D}$ is the empty identity and its determinant is one; the same proof and the same nonempty compatibility quotient remain valid. When X is the full source, $P_{\text{rem},X} = 0$ and $\chi_X = 0$.

At either literal high cutoff and in the complete finite domain GGE1, the exact coefficient complex and its native image maps give

$$\begin{aligned} \rho_{\text{comp}} &= \rho_{1,D}, \\ \log \det \Omega_{X,D} &= \rho_{2,D} - \rho_{3,D} + \mathfrak{E}_{k,D} + \rho_{0,D} + \chi_X + \delta_X. \end{aligned} \quad (\text{PCJ4})$$

For the first equality, the kernel of the original d^1 coefficient map is $\text{im } \mathcal{P}$. Its source quotient Gram in the Z_c frame is G_c , and the target restriction Gram on its image frame is H_{comp} . Its squared Jacobian is therefore their determinant ratio, proving the equality with GGE20. The full oriented torsion identity GGE21 says $\rho_{1,D} = \rho_{2,D} - \rho_{3,D} + \mathfrak{E}_{k,D} + \rho_{0,D}$. Substitution in PCJ2 proves PCJ4. Every source-restriction projection is retained in χ_X ; no full-source replacement occurs.

For an explicit bound put

$$\begin{aligned} B_{\text{Euler}} &= B_k^{\text{grid}} + \mathcal{B}_{\text{GPA}} + 16C_{\text{QLG}} q \log q + \mathcal{B}_{\mathcal{H}} + \mathcal{B}_{\Gamma}, \\ B_0 &= 2(k-7)^2 \log \max(1, K_+, K_-), \quad B_{\text{NKC}} = r_{\text{comp},k} \max(-\log c_-, \log c_+), \\ |\log \det \Omega_{X,D} - (\rho_{2,D} - \rho_{3,D} + \chi_X)| &\leq B_{\text{Euler}} + B_0 + B_{\text{NKC}} = O_{\text{actual}}(q \log q). \end{aligned} \quad (\text{PCJ5})$$

Use the unchanged finite constants of GGE6–19 and NBG12 at this literal D ; for source order one, $\mathcal{B}_{\Gamma} = 0$. The bound follows from PCJ4, GGE19, the GGE22 proof of $|\rho_{0,D}| \leq B_0$, and PCJ2. It is uniform over the specified subspace X , since neither constant contains its frame condition number. The full H_{comp} denominator is present inside χ_X and has not been bounded by a presumed coefficient estimate.

149.2 Every nested source and its actual Schur quotient

Let $X \subset Y$ be specified original source subspaces. Their correlated target images are nested. Write P_X, P_Y for the NKC9 projections and $P_{Y \ominus X} = P_Y - P_X$, the orthogonal projection onto the correlated image of Y perpendicular to that of X . Thus

$$\begin{aligned} P_{\text{rem},X} &= P_{\text{rem},Y} + P_{Y \ominus X}, \\ H_{\text{rem},X} &= H_{\text{rem},Y} + H_{Y \ominus X}, \quad H_{Y \ominus X} = Z_c^* \mathcal{V}^* P_{Y \ominus X} \mathcal{V} Z_c \succeq 0, \\ \chi_X - \chi_Y &= \log \det \left(I + (H_{\text{comp}} + H_{\text{rem},Y})^{-1/2} H_{Y \ominus X} (H_{\text{comp}} + H_{\text{rem},Y})^{-1/2} \right) \geq 0. \end{aligned} \quad (\text{PCJ6})$$

The first identity follows by subtracting $P_T - P_C - P_Y$ from $P_T - P_C - P_X$. Congruence by $\mathcal{V} Z_c$ gives the second. Apply the same positive determinant factorization as in PCJ3 to the two full compatibility-plus-complement forms. This proves the last identity, retaining its actual denominator and all possible zero ranks.

There is also an exact native quotient interpretation. Choose an orthonormal correlated target frame for Y as $[A_X, A_{Y \ominus X}]$, with the displayed order of its two subspaces. In this frame partition the actual independent-gauge form as

$$\begin{aligned} \Omega_Y &= \begin{pmatrix} \Omega_X & B_{XY} \\ B_{XY}^* & C_{Y \ominus X} \end{pmatrix}, \quad \Omega_{Y/X} = C_{Y \ominus X} - B_{XY}^* \Omega_X^{-1} B_{XY}, \\ 0 &\prec \Omega_{Y/X} \preceq I, \quad \det \Omega_Y = \det \Omega_X \det \Omega_{Y/X}. \end{aligned} \quad (\text{PCJ7})$$

The full matrix is exactly $I - [A_X, A_{Y \ominus X}]^* W_c W_c^* [A_X, A_{Y \ominus X}]$ by NKC10; its first block is therefore the original Ω_X . Minimizing its squared norm over the first coordinate gives the Schur form displayed in PCJ7. Positivity is inherited from NKC10. Since $\Omega_Y \preceq I$, its lower diagonal block is at most I , and the subtracted form is positive semidefinite; this proves the upper bound. The block determinant identity follows by the corresponding triangular elimination. If X or Y/X is zero, the same formulas use the empty identity and determinant one.

Combining PCJ2, PCJ6 and PCJ7 yields the finite quantitative identity

$$\begin{aligned} -\log \det \Omega_{Y/X} &= \chi_X - \chi_Y + \delta_X - \delta_Y, \\ |-\log \det \Omega_{Y/X} - (\chi_X - \chi_Y)| &\leq r_{\text{comp},k} \log(c_+/c_-). \end{aligned} \quad (\text{PCJ8})$$

Indeed subtract PCJ2 for Y from that for X ; ρ_{comp} cancels literally at the same endpoint and source order. PCJ7 identifies the remaining left-hand side. Both δ values lie in the same interval in PCJ2, whose length is the right-hand side in PCJ8. This bound holds at the low cutoffs as well as the high ones, because it uses only the already proved NKC full rank.

Taking the zero source and then the full source in PCJ2 gives an additional useful finite budget. Denote the full source by X_{all} and its zero subspace by $\{0\}$. Then

$$\begin{aligned} \rho_{\text{comp}} + \chi_{\{0\}} + \delta_{\{0\}} &= 0, \quad 0 \leq \chi_X \leq \chi_{\{0\}}, \\ -\log \det \Omega_X &= \chi_{\{0\}} - \chi_X + \delta_{\{0\}} - \delta_X, \\ 0 &\leq -\log \det \Omega_X \leq -\log \det \Omega_{X_{\text{all}}}. \end{aligned} \quad (\text{PCJ9})$$

The first two statements use PCJ2 and PCJ6. Subtract the first equality from the PCJ2 equality for X to obtain the second row. For the last row use $0 \prec \Omega_X \preceq I$ and PCJ7 with $Y = X_{\text{all}}$: its quotient determinant is at most one, so $\det \Omega_{X_{\text{all}}} \leq \det \Omega_X$. This argument proves the exact metric monotonicity; it makes no assumption on how determinant loss is distributed among original source directions.

149.3 The four-endpoint transport with both low terms retained

For the actual source frame X_k at each endpoint, set

$$\begin{aligned} D_- &= q_{k+4} - v - 1, \quad \widehat{D}_- = q_{k+4} - v, \quad D_+ = 2q_{k+4} - v - 1, \quad \widehat{D}_+ = 2q_{k+4} - v, \\ \mathcal{D}_X &= -\log \det \Omega_{X,D_-} - \log \det \Omega_{X,\widehat{D}_-} + \log \det \Omega_{X,D_+} + \log \det \Omega_{X,\widehat{D}_+}. \end{aligned} \quad (\text{PCJ10})$$

In the GGE finite domain the exact transported formula is

$$\begin{aligned} \mathcal{D}_X &= -\rho_{\text{comp},D_-} - \chi_{X,D_-} - \rho_{\text{comp},\widehat{D}_-} - \chi_{X,\widehat{D}_-} \\ &\quad + \sum_{D \in \{D_+, \widehat{D}_+\}} (\rho_{2,D} - \rho_{3,D} + \chi_{X,D} + \mathfrak{E}_{k,D} + \rho_{0,D}) + \sum_D \varepsilon_D \delta_{X,D}, \end{aligned} \quad (\text{PCJ11})$$

where $\varepsilon_D = -1, -1, +1, +1$ in the displayed endpoint order. This follows by applying PCJ2 at both low endpoints and PCJ4 at both high endpoints. All occurrences of ρ_{comp} and χ are computed in the original literal cutoff form PCJ1; the source order remains one at all degrees, or the specified degree at each step of the original comparison.

In particular, deleting only the proved small terms from the right-hand side of PCJ11 incurs absolute error at most

$$\sum_{D \in \{D_+, \widehat{D}_+\}} (B_{\text{Euler},D} + B_{0,D}) + \sum_{D \in \{D_-, \widehat{D}_-, D_+, \widehat{D}_+\}} B_{\text{NKC},D} = O_{\text{actual}}(q \log q) = o(kq). \quad (\text{PCJ12})$$

This is an estimate of the explicitly specified error. The two original low compatibility ratios, all four proper-source complements and the two high companion-image differences remain in PCJ11. The original further invariant quotient is still the NBG30–32 quotient with its complete kernel and two image metrics. Neither PCJ11 nor its finite bound replaces that quotient by the independent-gauge ambient map.

149.4 The accepted shared-update bound at the actual difference rank

The invariant-target continuation ITL26–31 proves a sharper estimate by keeping the identical positive update in the two native comparisons. We transport that accepted result to the exact nested-source notation of PCJ6–8 here, with its full proof and the corresponding sign orientation. This is the same shared-update argument, not a new estimate of either determinant separately.

For $X \subset Y$ let $r = \dim Y - \dim X$ and choose an orthonormal correlated target frame A_Δ for $\text{im}(P_Y - P_X)$. Define

$$\begin{aligned} J_\Delta &= A_\Delta^* W_c : \mathbb{C}^{r_{\text{comp},k}} \longrightarrow \mathbb{C}^r, \\ \Xi_Z &= I - W_c^* P_Z W_c, \quad \Gamma_Z = W_c^* (\mathcal{R}^* \mathcal{R} + P_T - P_C - P_Z) W_c \quad (Z = X, Y), \\ \Xi_X &= \Xi_Y + J_\Delta^* J_\Delta, \quad \Gamma_X = \Gamma_Y + J_\Delta^* J_\Delta. \end{aligned} \quad (\text{PCJ13})$$

The difference identities follow by subtracting the displayed formulas and using $P_Y - P_X = A_\Delta A_\Delta^*$. They are the literal ITL26 matrices under the dictionary $(E, V, \mathcal{J}_D) = (X, Y, J_\Delta)$. All four forms are positive by NKC10 and NKC14. The shared lower-gauge isometry and the complete source-complement projection are unchanged.

Set

$$T_\Xi = J_\Delta \Xi_Y^{-1} J_\Delta^*, \quad T_\Gamma = J_\Delta \Gamma_Y^{-1} J_\Delta^*. \quad (\text{PCJ14})$$

The determinant lemma and the exact common conjugation by $G_c^{-1/2}$ give

$$\begin{aligned} -\log \det \Omega_{Y/X} &= \log \det (I_r + T_\Xi), \\ \chi_X - \chi_Y &= \log \det (I_r + T_\Gamma). \end{aligned} \quad (\text{PCJ15})$$

For the first line apply the lemma to Ξ_X in PCJ13 and use $\det \Xi_Z = \det \Omega_Z$ and PCJ7. For the second apply it to Γ_X , use PCJ3, and cancel ρ_{comp} at this same endpoint. This also proves that the positive update in PCJ6 and the compressed update here have the same nontrivial determinant, despite their different matrix dimensions.

Inverting $c_- \Xi_Y \preceq \Gamma_Y \preceq c_+ \Xi_Y$ and taking congruences yields $c_+^{-1} T_\Xi \preceq T_\Gamma \preceq c_-^{-1} T_\Xi$. Since $0 < c_- \leq 1 \leq c_+$, addition of the identity then gives

$$\begin{aligned} c_+^{-1} (I_r + T_\Xi) &\preceq I_r + T_\Gamma \preceq c_-^{-1} (I_r + T_\Xi), \\ r \log c_- &\leq -\log \det \Omega_{Y/X} - (\chi_X - \chi_Y) \leq r \log c_+. \end{aligned} \quad (\text{PCJ16})$$

Conjugate the first line by $(I_r + T_\Xi)^{-1/2}$. Its r positive eigenvalues lie between c_+^{-1} and c_-^{-1} , so their log determinant lies between $-r \log c_+$ and $-r \log c_-$. Negation and PCJ15 give precisely the second line. No commutation of the two compressed matrices is used. This proves the exact sign reversal from the ITL29 convention. If $r = 0$, both compressed matrices have size zero and the entire error is exactly zero. Both this bound and PCJ8 remain valid. For the actual rank- $O(k)$ source increments, this gives the sharper $O_{\text{actual}}(k \log q)$ error while retaining the complete preceding finite bound.

In particular, set $X = \{0\}$ and let Y be any specified original source restriction of dimension t . Then PCJ9 has the stronger finite form

$$\begin{aligned} -\log \det \Omega_{Y,D} &= \chi_{\{0\},D} - \chi_{Y,D} + \nu_{Y,D}, \\ t \log c_{-,D} &\leq \nu_{Y,D} \leq t \log c_{+,D}. \end{aligned} \quad (\text{PCJ17})$$

This follows directly from PCJ16 and the empty source convention; its quotient is the original Y itself. Thus for an actual conductor or invariant frame, $t = O(k)$, the error is $O_{\text{actual}}(k \log q)$ at every one of the four literal endpoints. The two complementary determinants on the right keep their complete original inverse Gram. The bound does not assert that their difference is small.

For the signed return PCJ10 put $Z_{Y,D} = \chi_{\{0\},D} - \chi_{Y,D} \geq 0$. At all four cutoffs, with the same original source frame, PCJ17 gives

$$\begin{aligned} \mathcal{D}_Y &= - \sum_D \varepsilon_D Z_{Y,D} + \mathcal{R}_Y, & \mathcal{R}_Y &= - \sum_D \varepsilon_D \nu_{Y,D}, \\ t \left(\sum_{D \in \{D_-, \hat{D}_-\}} \log c_{-,D} - \sum_{D \in \{D_+, \hat{D}_+\}} \log c_{+,D} \right) &\leq \mathcal{R}_Y, \\ \mathcal{R}_Y &\leq t \left(\sum_{D \in \{D_-, \hat{D}_-\}} \log c_{+,D} - \sum_{D \in \{D_+, \hat{D}_+\}} \log c_{-,D} \right). \end{aligned} \quad (\text{PCJ18})$$

Indeed the low signs multiply $+\nu$ after the overall minus, while the high signs multiply $-\nu$. Applying the individual finite interval PCJ17 proves both bounds. This signed error is $O_{\text{actual}}(k \log q)$ for the actual rank $t = O(k)$. The statement uses only the NKC finite domain and therefore does not impose the extra GGE asymptotic threshold on the low endpoints. It retains the complete positive differences $Z_{Y,D}$ before applying the endpoint signs, without an assumption about concentration or a value for their leading coefficient.

150 The complete native complex on the actual conductor

Fix the original actual quartet, five-orbit period, logarithm branch and every coefficient of the nonzero original conductor. Write its full homogeneous biform as A^{hom} of bidegree $(8, 8)$, and its literal affine polynomial as $A(Z, W) = \sum_{a,b=0}^8 a_{ab} Z^a W^b$. No repeated component or possible zero edge coefficient is removed. Let $\beta_* = \frac{1}{2} - \delta - i\gamma$ and retain

$$\begin{aligned} E_A(z) &= e^{8\beta_* z} A(e^{2\delta z}, e^{2i\gamma z}), & v = \text{ord}_0 E_A, & \quad 0 \leq v \leq 80, \quad \mu_v = E_A^{(v)}(0) \neq 0, \\ B_i(z) &= e^{4\beta_* z} L_i(e^{2\delta z}, e^{2i\gamma z})^4, & \sum_{i=1}^4 D_i(z) B_i(z) &= 1. \end{aligned} \quad (\text{CNC1})$$

The four L_i and the four entire D_i are precisely the full NBG1–6 period-coordinate and degree-thirteen Bezout expressions. In particular the coefficient sum in CNC1 is the proved identity with the original period matrix; no polynomial reciprocal is substituted.

For $k \equiv 1 \pmod{4}$, $k \geq 9$, put

$$\begin{aligned} n_p &= k + 4p, & q_n &= (n+1)^2, & q'_n &= (n-7)^2, & m_n &= q_n - q'_n - v = 16n - 48 - v, \\ \mathcal{P}_n &= H^0(\mathbb{P}^1 \times \mathbb{P}^1, \mathcal{O}(n, n)), & \mathcal{C}_n &= \mathcal{P}_n / A^{\text{hom}} \mathcal{P}_{n-8}, \\ \mathcal{C}_{v,n} &= \{[f] \in \mathcal{C}_n : e^{n\beta_* z} f(e^{2\delta z}, e^{2i\gamma z}) = O(z^v)\}. \end{aligned} \quad (\text{CNC2})$$

Here $f(Z, W)$ means the affine polynomial of the original biform, with every original coefficient in the rectangle $0 \leq a, b \leq n$. The last condition is independent of its quotient representative, since changing by Ag adds $E_A(z) e^{(n-8)\beta_* z} g(e^{2\delta z}, e^{2i\gamma z})$.

150.1 Both exact sequences and all low-jet connecting maps

Let $h_i = L_i^4$ as an actual biform of degree $(4, 4)$, and orient the differential by exterior multiplication with $\sum_i h_i e_i$ in the original order 1, 2, 3, 4. The complete five-term coefficient complexes are

$$\begin{aligned} \mathcal{A}^p &= \mathcal{P}_{n_p-8} \otimes \Lambda^p \mathbb{C}^4, & \mathcal{B}^p &= \mathcal{P}_{n_p} \otimes \Lambda^p \mathbb{C}^4, \\ 0 \longrightarrow \mathcal{A}^\bullet &\xrightarrow{A^{\text{hom}}} \mathcal{B}^\bullet \longrightarrow \mathcal{C}^\bullet \longrightarrow 0, & \mathcal{C}^p &= \mathcal{C}_{n_p} \otimes \Lambda^p \mathbb{C}^4. \end{aligned} \quad (\text{CNC3})$$

These are the literal GKC complexes at base degrees k and $k+8$. Both are exact by the complete GKC proof, since both base degrees are at least nine. Multiplication by A^{hom} is injective in each

term: the polynomial ring is an integral domain and $A^{\text{hom}} \neq 0$, even when this biform has repeated factors. It commutes with every h_i , proving the cochain square. The displayed quotient in each term is exactly CNC2.

For completeness the quotient complex is exact by the following explicit lifting argument. Lift a cycle $\bar{b}$ in $\mathcal{C}^p$ to $b \in \mathcal{B}^p$. Its differential is $db = A^{\text{hom}}a$ for a unique $a \in \mathcal{A}^{p+1}$. Injectivity implies $da = 0$. Exactness of $\mathcal{A}$ supplies $a = da'$. Then $b - A^{\text{hom}}a'$ is a cycle of $\mathcal{B}$, so it equals db' there. Reduction gives $\bar{b} = d\bar{b}'$. At degree zero the same argument has no previous term and forces $\bar{b} = 0$. At degree four the differential is zero; the same proof uses exactness onto the last term. This also describes the connecting map before using exactness: $[\bar{b}] \mapsto [a]$.

Let $\mathcal{J}_v = \mathbb{C}[z]/(z^v)$, with $\mathcal{J}_0 = 0$, and use ordinary truncated Taylor coefficients in its fixed monomial order. The first- v -jet maps give the short exact sequence

$$\begin{aligned} j_{v,n} : \mathcal{C}_n &\longrightarrow \mathcal{J}_v, & [f] &\longmapsto \left[e^{n\beta_* z} f(e^{2\delta z}, e^{2i\gamma z}) \right], \\ 0 &\longrightarrow \mathcal{C}_v^\bullet &\longrightarrow \mathcal{C}^\bullet &\xrightarrow{j_v} \mathcal{J}_v^\bullet \longrightarrow 0, & \mathcal{J}_v^p &= \mathcal{J}_v \otimes \Lambda^p \mathbb{C}^4. \end{aligned} \quad (\text{CNC4})$$

Indeed its matrix on the original coefficient rectangle has rows $\beta_{n,a,b}^r/r!$, $0 \leq r < v$. The full upper-root Vandermonde is invertible and $q_n \geq 100 > 80 \geq v$, so these rows are independent. This proves surjectivity in every term, with the actual factorials retained. The conductor columns are killed, as proved after CNC2. Multiplication by h_i becomes multiplication by the literal B_i on the generating series, so the differential on $\mathcal{J}_v^\bullet$ is exterior multiplication by their truncated tuple.

The contraction by the original tuple (D_i) gives, in this same finite algebra,

$$\iota_D d + d\iota_D = I_{\mathcal{J}_v^\bullet}. \quad (\text{CNC5})$$

To verify the sign, for an exterior basis vector $e_{i_1} \wedge \cdots \wedge e_{i_p}$ define ι_D as $\sum_{r=1}^p (-1)^{r-1} D_{i_r}$ times that vector with its r th entry omitted. Expansion of both compositions cancels the terms with two distinct indices and leaves $\sum_i D_i B_i$ times the input. CNC1 proves CNC5, including reduction modulo z^v . Thus this entire low-jet complex is exact, for the actual coefficients.

The individually analytic complex $\mathcal{C}_v^\bullet$ is consequently exact. Here is its connecting construction without discarding a jet. If $a \in \mathcal{C}_v^p$ is a cycle, exactness of $\mathcal{C}$ gives $a = db$. Its jet satisfies $dj_v(b) = 0$, and CNC5 gives $j_v(b) = d\iota_D j_v(b)$. Lift $\iota_D j_v(b)$ through the surjection in CNC4 to $c \in \mathcal{C}^{p-2}$. Then $b - dc$ has zero jet and still has differential a . When $p = 1$, a degree-zero cycle in the jet complex is zero, so b itself has zero jet; no negative-degree lift is used. The cases $p = 0, 4$ use the same endpoint conventions. In particular,

$$\begin{aligned} 0 \rightarrow \mathcal{C}_{v,k} &\rightarrow \mathcal{C}_{v,k+4}^4 \rightarrow \mathcal{C}_{v,k+8}^6 \rightarrow \mathcal{C}_{v,k+12}^4 \rightarrow \mathcal{C}_{v,k+16} \rightarrow 0 \quad \text{is exact,} \\ \dim \mathcal{C}_{v,n} &= m_n, & (\text{rank } d^0, \dots, \text{rank } d^3) &= (m_k, 3m_k + 256, 3m_k + 512, m_k + 256). \end{aligned} \quad (\text{CNC6})$$

The dimensions follow from CNC3–4. The ranks follow successively from exactness and $m_{k+4p} = m_k + 64p$, including the last onto map. All four are $O(k)$ with the original v fixed.

150.2 The literal finite native square

At degree n choose exactly the original DNE2–3 frame $Z_{\text{exp},n}$ in the upper coefficient order. Write $\mathbb{V}_n$ for the full ordinary upper-root Vandermonde, $C_n^\top$ for the complete original conductor convolution, and $P_{v,n}$ for zero insertion in its first v rows. The construction is

$$\begin{aligned} W_{v,n} &= \mathbb{V}_n^{-1} P_{v,n}, & J_{v,n} &= P_{v,n}^\top \mathbb{V}_n C_n^\top, \\ W_{v,n} J_{v,n} &= C_n^\top, & Z_{\text{exp},n} &= W_{v,n} E_{\mathcal{J}_n}. \end{aligned} \quad (\text{CNC7})$$

The complementary row set $\mathcal{J}_n$ is the fixed actual first-nonzero-minor choice of DNE3. Thus the coefficient-to-generating-series map

$$x \longmapsto N_{n,x}(z) = \sum_{a,b} (Z_{\text{exp},n} x)_{a,b} e^{\beta_{n,a,b} z}, \quad F_{n,x} = N_{n,x}/E_A \quad (\text{CNC8})$$

is an isomorphism from these m_n coordinates to $\mathcal{C}_{v,n}$, followed by its literal generating quotient. Changing the numerator by $C_n^\top \eta$ changes F by precisely the original lower exponential with coefficient η . No coefficient conjugation occurs in this complex-linear map.

For any integer $D \geq 0$ and either the family $s_n = 1$ or $s_n = n$, set $c'_n = n/2 - 4$ and use the complete original Gamma orthonormal polynomials

$$\phi_{r,s_n,c'_n}(S') = \frac{p_r((S' - c'_n)/i)}{\sqrt{M_{s_n} r! (s_n/2)_r}}, \quad M_{s_n} = (2\pi)^{s_n/2}, \quad 0 \leq r \leq D. \quad (\text{CNC9})$$

Let $\mathcal{H}_{n,D}$ be their complex-linear dual and $\mathcal{L}_{n,D}$ the span of evaluations at every original lower root of degree $n - 8$. In this exact frame define

$$\begin{aligned} T_{n,D;ra} &= \Lambda_{F,a}(\phi_{r,s_n,c'_n}), & V_{n,D;r,(a,b)} &= \phi_{r,s_n,c'_n}(\beta_{n-8,a,b}), \\ \Theta_{n,D} : \mathcal{C}_{v,n} &\longrightarrow \mathcal{H}_{n,D}/\mathcal{L}_{n,D}, & [N] &\longmapsto [\Lambda_{N/E_A}], \\ \mathcal{D}_{n,D} &= T_{n,D}^*(I - V_{n,D}V_{n,D}^+)T_{n,D}. \end{aligned} \quad (\text{CNC10})$$

The Moore–Penrose inverse only expresses the orthogonal projector onto the full lower-evaluation span when its columns are dependent. Equivalently the form is the attained minimum

$$\min_{\eta} \|T_{n,D}x + V_{n,D}\eta\|_2^2.$$

The full source mass and the factor $1/i = -i$ in the source variable remain in CNC9.

For each h_i let $M_{i,n}$ be its exact coefficient quotient matrix from $Z_{\text{exp},n}$ to $Z_{\text{exp},n+4}$. These are precisely the DNE coordinates of the original multiplication. Let $\mathbb{T}_{i,n,D}$ be the finite functional map $\Lambda_F \mapsto \Lambda_{B_i F}$, as in NGB14. The full native square is

$$\begin{array}{ccc} \mathcal{C}_{v,n} & \xrightarrow{M_{i,n}} & \mathcal{C}_{v,n+4} \\ \Theta_{n,D} \downarrow & & \downarrow \Theta_{n+4,D} \\ \mathcal{H}_{n,D}/\mathcal{L}_{n,D} & \xrightarrow{\mathbb{T}_{i,n,D}} & \mathcal{H}_{n+4,D}/\mathcal{L}_{n+4,D}. \end{array} \quad (\text{CNC11})$$

The upper product has numerator $B_i N$ and hence quotient $B_i F$, proving commutation. An old lower exponential multiplied by each term of B_i has root $\beta_{n-8,a,b} + \beta_{4,r,t} = \beta_{n-4,a+r,b+t}$ in the full allowed target lower rectangle. Thus the lower map is well-defined on the stated quotient, retaining all convolution coefficients. The finite matrix is the ordinary transpose of the exact polynomial translation matrix between the two CNC9 frames. Only its metric adjoint is a complex conjugate transpose.

In the original KRC strip coordinates the same upper map is

$$M_{i,n} = (\alpha_A N_{n+4} Z_{\text{exp},n+4})^{-1} m_{i,n} (\alpha_A N_n Z_{\text{exp},n}), \quad \alpha_A = v!/\mu_v. \quad (\text{CNC12})$$

Every inverse here is the inverse onto its specified original image; the full source and target strip subspaces are those of KRC5. Equation CNC8 proves the map in unscaled DNE coordinates, and conjugating by KRC5 proves CNC12. Its two identical nonzero scalar factors cancel only in this proved conjugation.

150.3 Every rank lost at the original finite cutoffs

The exact rank and kernel dimension of CNC10 are

$$\begin{aligned} r_n(D) &= \min\{m_n, \max(0, D+1 - q'_n)\}, \\ \dim \ker \Theta_{n,D} &= m_n - r_n(D). \end{aligned} \quad (\text{CNC13})$$

To prove this, first suppose $D+1+v \leq q_n$. The first $D+1+v$ upper derivative rows are independent. Restricting to zero first v rows therefore gives arbitrary remaining $D+1$ jets. Division by E_A is

an invertible triangular map from these numerator jets to the first $D + 1$ jets of F , with leading scalar $v!/\mu_v \neq 0$. Thus these functionals fill $\mathcal{H}_{n,D}$. The lower evaluations have rank $\min(q'_n, D + 1)$ by their Vandermonde. Quotienting gives the rank in CNC13.

If $D + 1 + v \geq q_n$, vanishing of the quotient functional means that F has the same first $D + 1$ jets as some original lower exponential. Multiplication by E_A makes their numerator difference vanish through order $D + v$, hence through order $q_n - 1$. This difference belongs to the original upper exponential span, whose full Vandermonde is invertible. It is therefore zero identically and the original coefficient class lies in $\text{im } C_n^\top$. This proves injectivity. Since $m_n = q_n - v - q'_n$, it agrees with CNC13 and includes the common boundary between the two cases.

Let $\mathcal{K}_D^\bullet = \ker \Theta_{\bullet,D}$ term by term and $\mathcal{I}_D^\bullet$ be its full image in the native quotient complexes. CNC11 proves these are complexes, and

$$0 \longrightarrow \mathcal{K}_D^\bullet \longrightarrow \mathcal{C}_v^\bullet \longrightarrow \mathcal{I}_D^\bullet \longrightarrow 0, \quad H^p(\mathcal{I}_D^\bullet) \simeq H^{p+1}(\mathcal{K}_D^\bullet). \quad (\text{CNC14})$$

For the connecting map, lift an image cycle y to $x \in \mathcal{C}_v^p$. Then $dx \in \mathcal{K}_D^{p+1}$ is a cycle. A different lift changes it by a kernel boundary, and changing y by an image boundary does the same, proving well-definition. If this kernel class is zero, subtract its kernel primitive from x ; exactness of CNC6 then proves that y is an image boundary. Conversely a kernel cycle is a boundary in $\mathcal{C}_v$ by CNC6, and its primitive maps to the needed image cycle. These arguments prove both injectivity and surjectivity. The metric on $\mathcal{I}_D^p$ is the original quotient of CNC10 by this exact kernel, and every cohomology metric is its attained cycle restriction followed by its boundary quotient. No null native direction receives a positive metric by declaration.

At the two actual degree-four target low cutoffs $D_- = q_{k+4} - v - 1$ and $\widehat{D}_- = q_{k+4} - v$, the dimensions of the five native image terms are

$$\begin{aligned} D_- : & \quad (m_k, 4m_{k+4}, 6(8k + 24 - v), 0, 0), \\ \widehat{D}_- : & \quad (m_k, 4m_{k+4}, 6(8k + 25 - v), 4 \max(0, 1 - v), 0). \end{aligned} \quad (\text{CNC15})$$

Direct substitution in CNC13 proves these values; the factors 1, 4, 6, 4, 1 are the full exterior multiplicities. In particular the two Euler characteristics are exactly

$$\begin{aligned} \sum_p (-1)^p \dim \mathcal{I}_{D_-}^p &= 32 - 3v, \\ \sum_p (-1)^p \dim \mathcal{I}_{\widehat{D}_-}^p &= 38 - 3v - 4 \max(0, 1 - v). \end{aligned} \quad (\text{CNC16})$$

These follow from $m_{k+4} = m_k + 64$. They are also the alternating dimensions of the actual cohomologies, by elementary rank-nullity in this finite complex. They do not specify the individual cohomology ranks; those retain the exact CNC11 matrices and CNC14 connecting maps.

At the two literal high cutoffs $D_+ = 2q_{k+4} - v - 1$ and $\widehat{D}_+ = 2q_{k+4} - v$, all five maps $\Theta_{n_p,D}$ are injective when $k \geq 25$:

$$D_+ + 1 + v - q_{k+16} = 2(k + 5)^2 - (k + 17)^2 = k^2 - 14k - 239 \geq 36. \quad (\text{CNC17})$$

The quadratic increases for $k \geq 25$ and its value there is 36. The upper cutoff adds one. Thus CNC13 proves full positivity, including all later terms, and CNC6 is an exact complex of the original native spaces. For $k = 9, 13, 17, 21$ the same general rank formula CNC13 remains valid; no positive late term is assumed when that formula gives a kernel.

150.4 The exact determinant line and original kernel/residual forms

At a common cutoff where all five maps are injective, equip each term of CNC6 with $\binom{4}{p}$ copies of $\mathcal{D}_{n_p,D}$. Use the complete original DNE coefficient frames, with the exterior order already fixed.

Choose any boundary bases b_p and actual lifts $\tilde{b}_{p+1}$ giving its adapted coefficient frames, and denote their change matrices by S_p . Define

$$\begin{aligned} \tau_{v,k} &= \prod_{p=0}^4 (\det S_p)^{(-1)^p}, & \mathcal{E}_{v,k,D} &= \sum_{p=0}^4 (-1)^p \binom{4}{p} \log \det \mathcal{D}_{n_p,D}, \\ \varrho_{p,D} &= \log \text{Jac}^2(d^p : \mathcal{C}_v^p / \ker d^p \longrightarrow \text{im } d^p), & 0 \leq p \leq 3. \end{aligned} \quad (\text{CNC18})$$

The actual coefficient torsion is retained, including its phase. Its definition is independent of the boundary bases and lifts: a change of a boundary basis enters one S_p and the previous lift basis with opposite exponents, and changing a lift by a boundary is triangular of determinant one. In particular it is independent of the cutoff and the source metric.

The complete metric identity is

$$\mathcal{E}_{v,k,D} = -\varrho_{0,D} + \varrho_{1,D} - \varrho_{2,D} + \varrho_{3,D} - 2 \log |\tau_{v,k}|. \quad (\text{CNC19})$$

Indeed the Gram in the adapted frame is $S_p^* G_p S_p$. Its determinant factors into the boundary restriction determinant and the lift quotient determinant by the full Schur complement. The quotient map d^p has squared Jacobian equal to the next boundary determinant divided by that lift quotient determinant. Their alternating sum therefore equals the alternating log determinant of the adapted full Grams. The latter is $\mathcal{E}_{v,k,D} + 2 \log |\tau_{v,k}|$. Solving gives exactly CNC19. This proof keeps every phase until its actual squared metric determinant is taken.

For each degree n let $A_{\ell,n}$ and $H_{Y,n,D}$ be the original DNE4–8 evaluation and relation-quotient matrices. Their exact native identity is

$$\mathcal{D}_{n,D} = A_{\ell,n}^* \overline{H_{Y,n,D}^{-1}} A_{\ell,n}, \quad \log \det \mathcal{D}_{n,D} = 2 \log |\det A_{\ell,n}| - \log \det H_{Y,n,D}. \quad (\text{CNC20})$$

Its complex conjugation is forced by complex-linear duality, as proved in DNE8. At every cutoff in this positive regime the relation degree is $L_n = D + v - q_n \geq -1$, so DNE's original low frame and all its relation columns apply literally.

Retain the actual kernel-evaluation map $B_n = Z_n^T A_{\ell,n}$ and its entire kernel frame J_n from DNE9/KRC. Choose a fixed right inverse R_n of B_n in its original target coordinates, and put $U_n = [J_n, R_n]$. Then

$$\begin{aligned} G_{J,n,D} &= J_n^* \mathcal{D}_{n,D} J_n, & Q_{B,n,D} &= (B_n \mathcal{D}_{n,D}^{-1} B_n^*)^{-1} = \overline{\widehat{H}_{K,n,D}^{-1}}, \\ \log \det \mathcal{D}_{n,D} &= \log \det G_{J,n,D} + \log \det Q_{B,n,D} - 2 \log |\det U_n|. \end{aligned} \quad (\text{CNC21})$$

The matrix U_n is invertible because J_n spans $\ker B_n$ and $B_n R_n = I$. Taking the Gram in this frame and minimizing over the first block gives exactly the quotient norm $Q_{B,n,D}$; the block determinant gives CNC21. The equality with the original kernel matrix is the literal DNE9 computation, with every transpose and conjugation unchanged. Empty ranks have determinant one. Consequently the new exact complex acts on the same total form whose original residual and kernel metrics are the two factors in CNC21.

For any four common cutoffs D_1, D_2, D_3, D_4 in the positive regime, use the original signs $\varepsilon = (-1, -1, +1, +1)$. The fixed coefficient factors cancel, giving both exact statements

$$\begin{aligned} \sum_i \varepsilon_i \mathcal{E}_{v,k,D_i} &= \sum_i \varepsilon_i (-\varrho_{0,D_i} + \varrho_{1,D_i} - \varrho_{2,D_i} + \varrho_{3,D_i}), \\ \sum_i \varepsilon_i \mathcal{E}_{v,k,D_i} &= \sum_{p=0}^4 (-1)^p \binom{4}{p} \sum_i \varepsilon_i (\log \det G_{J,n_p,D_i} - \log \det \widehat{H}_{K,n_p,D_i}). \end{aligned} \quad (\text{CNC22})$$

The first cancels $\tau_{v,k}$ in CNC19. The second cancels U_n in CNC21 and uses its exact conjugate inverse. Here $\widehat{H}_{K,n,D}$ is the actual fixed relation-kernel frame of DNE9. Its further relation to the original F_K retains both low quotient volumes and the entire original source-map correction of

FGC12–13 and FEX34–37, as stated exactly in DNE10; CNC22 does not remove those terms. No source endpoint is changed by these equalities. For example all four cutoffs $q_{k+16} - v - 1, q_{k+16} - v, 2q_{k+16} - v - 1, 2q_{k+16} - v$ lie in the required common positive regime. Their displacement from each degree's own original windows remains literal. At the earlier degree-four low cutoffs one instead uses CNC14–16, retaining the actual cohomology and null directions.

150.5 Quantitative forward control with every inverse term retained

The four original multiplier norms at base degree n obey the unchanged NBG/MNT estimates at the same D . Writing $b_{i,a,b}$ for all coefficients of the actual L_i^4 , these finite constants are explicitly

$$H_D = \sum_{j=1}^D \frac{1}{j}, \quad H_0 = 0, \quad a_n = (2\pi)^2 \frac{n}{n+2},$$

$$\kappa_{i,n}(D) = \sum_{a,b=0}^4 |b_{i,a,b}| e^{|\beta_{4,a,b}-2|(H_D+2)} \begin{cases} 1, & s_n = s_{n+4} = 1, \\ a_n^{-1/2}, & (s_n, s_{n+4}) = (n, n+4), \end{cases}$$

$$K_n(D)^2 = \sum_i \kappa_{i,n}(D)^2.$$

They retain the exact source centre change $(\beta_{4,a,b} - 2)/i$ and the complete Gamma mass factor in a_n . The finite translation bound and density recurrence are NBG7–12, with precisely these degrees and the unchanged cutoff. For exterior degree p the exact native differential satisfies

$$\|d^p\| \leq \sqrt{p+1} K_{n_p}(D), \quad \varrho_{p,D} \leq 2 \operatorname{rank} d^p \log \max\{1, \sqrt{p+1} K_{n_p}(D)\}. \quad (\text{CNC23})$$

For a target component of size $p+1$, the differential is the sum of its $p+1$ signed multiplier terms. Cauchy–Schwarz bounds its squared norm by $p+1$ times the sum of their squared norms. On summing all target components, each input component contributes at most $\sum_i \kappa_{i,n_p}(D)^2$ times its original squared norm. This proves the first inequality. Apply the same bound to the source quotient by the exact kernel and multiply its positive singular values to prove the second. By CNC6 the ranks are $O(k)$, and at $D = O(q_k)$ every logarithmic constant is $O_{\text{actual}}(\log q_k)$. Thus each displayed upper bound is $O_{\text{actual}}(k \log q_k)$.

For d^0 , NBG's original correlated versus independent-gauge comparison is precisely the first map of CNC6. On the entire actual $\mathcal{C}_{v,k}$ one therefore has

$$\varrho_{0,D} = \log \det \Omega_{\mathcal{C}_{v,k},D} + e_{\text{cor},D}, \quad -2m_k \log \max(1, K_-) \leq e_{\text{cor},D} \leq 2m_k \log \max(1, K_+). \quad (\text{CNC24})$$

The target restriction form is exactly $\sum_i M_{i,k}^* \mathcal{D}_{k+4,D} M_{i,k}$. Divide its determinant by the original $\det \mathcal{D}_{k,D}$, and factor through the full correlated form of NBG27. The first ratio is its independent-gauge determinant and the second obeys the two-sided NBG23 quotient estimate, proving CNC24 with rank m_k . This explicitly connects the first native companion Jacobian to the original loss, rather than to an ambient coefficient norm.

Equations CNC19–24 are exact native determinant controls on the actual nonreduced conductor and its individually analytic subcomplex. They leave the inverse contributions of the remaining image maps, the actual low-cutoff cohomology and the further original invariant quotient as their displayed matrices and attained metrics. The forward bounds do not assign individual lower singular values.

Complete computed low cohomology and native return. The low image cohomology left after CNC16 is now explicitly computed by the full LCC and LCA proofs:

$$\dim H^1(\mathcal{I}_e) = 8k + 24 - 3e + \kappa_e, \quad \dim H^2(\mathcal{I}_e) = 8k + 56 + v - e - 4 \max(v - e, 0) + \kappa_e. \quad (\text{CNC25})$$

Here $\kappa_e = \dim \ker U_e$ for the exact original LCC5–6 signed Vandermonde-tail matrix, $\kappa_e \leq 3 \max(v - e, 0)$ and $0 \leq \kappa_0 - \kappa_1 \leq 3$. All other groups vanish. NCR11–12 proves the maps to the

earlier Euler statement. LCC20–24 and NCR13–18 retain the complete original kernel/residual overlap and native quotient metrics. They now evaluate the missing-rank return, the difference of the two fixed low-flag contractions, and the adjacent common contraction through LFC16,21–22. The common long loss and CNC18’s coefficient torsion remain as defined.

This calculation retains the actual simple quartet, full five-orbit period, logarithm branch, nonreduced conductor, original primary coordinates and both Gamma source-order families of CNC1–24, NBG1–32 and GKC1–33. Every complex below is a cochain complex over $\mathbb{C}$, with differential given by exterior multiplication in the ordered basis e_1, e_2, e_3, e_4 . Dual functionals are complex-linear. The calculation concerns the two original degree-four target low cutoffs. It gives exact cohomology groups and maps, with the remaining literal kernel matrix retained. It makes no native inverse estimate or coefficient torsion assertion.

151 The exact finite complexes and all term dimensions

Keep the full original symbols

$$\begin{aligned} \beta_* &= \frac{1}{2} - \delta - i\gamma, \quad \beta_{n,a,b} = n\beta_* + 2a\delta + 2bi\gamma, \quad \delta, \gamma > 0, \\ E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad v = \text{ord}_0 E_A, \quad 0 \leq v \leq 80, \quad \mu_v = E_A^{(v)}(0) \neq 0, \\ L_i(Z, W) &= (\mathbf{H}(ZW, Z, W, 1)^T)_i, \quad B_i(z) = e^{4\beta_* z} L_i(e^{2\delta z}, e^{2i\gamma z})^4, \quad \sum_i D_i(z) B_i(z) = 1. \end{aligned} \quad (\text{LCA1})$$

Here $\mathbf{H}$ is the actual invertible period matrix and D_i are the complete entire Bezout expressions NBG3–5. No coefficient is replaced, even when it vanishes. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad e \in \{0, 1\}, \quad Q = (k + 5)^2, \quad L = Q - v + e, \quad D = L - 1, \\ n_p &= k + 4p, \quad q_n = (n + 1)^2, \quad q'_n = (n - 7)^2, \quad m_n = q_n - q'_n - v = 16n - 48 - v, \\ \mathcal{A}^p &= \mathcal{P}_{n_p-8} \otimes \Lambda^p \mathbb{C}^4, \quad \mathcal{H}^p = (\mathbb{C}[z]/(z^L)) \otimes \Lambda^p \mathbb{C}^4. \end{aligned} \quad (\text{LCA2})$$

The finite germs in $\mathcal{H}^p$ use their actual original native frame when a metric is required. Their underlying vector space is the displayed jet space, without a metric change. The coefficient differential on $\mathcal{A}$ is $h\wedge$, $h_i = L_i^4$; the germ differential on $\mathcal{H}$ is $B\wedge$. The coefficient complex is exactly GKC at base k , and hence exact. The contraction

$$\iota_D(e_{i_1} \wedge \cdots \wedge e_{i_p}) = \sum_{r=1}^p (-1)^{r-1} D_{i_r} e_{i_1} \wedge \cdots \widehat{e}_{i_r} \cdots \wedge e_{i_p} \quad \text{satisfies} \quad \iota_D d_{\mathcal{H}} + d_{\mathcal{H}} \iota_D = I. \quad (\text{LCA3})$$

Indeed the two distinct-index terms cancel, and the equal-index terms give $\sum_i D_i B_i = 1$. This is an identity of germs modulo z^L , so $\mathcal{H}$ is exact with all its original phases and finite truncation retained.

The lower evaluation map is

$$\begin{aligned} \mathcal{E}_p : \mathcal{A}^p &\longrightarrow \mathcal{H}^p, \quad f \longmapsto [e^{(n_p-8)\beta_* z} f(e^{2\delta z}, e^{2i\gamma z})], \\ d_{\mathcal{H}} \mathcal{E}_p &= \mathcal{E}_{p+1} d_{\mathcal{A}}, \quad \mathcal{K}_E^\bullet = \ker \mathcal{E}_\bullet, \quad \mathcal{E}^\bullet = \text{im } \mathcal{E}_\bullet, \quad \mathcal{Q}_D^\bullet = \mathcal{H}^\bullet / \mathcal{E}^\bullet. \end{aligned} \quad (\text{LCA4})$$

The square follows term by term from $\beta_{n-8,a,b} + \beta_{4,r,t} = \beta_{n-4,a+r,b+t}$. The full native evaluation matrix has entries $\phi_{r,s_{n_p},(n_p-8)/2}(\beta_{n_p-8,a,b})$, where $0 \leq r < L$ and $s_n = 1$ or $s_n = n$. Its rank is $\min(L, q'_{n_p})$: the original monomial rows are a rectangular Vandermonde on distinct nodes, and the complete monomial-to-Gamma frame matrix is invertible triangular.

The only nonzero terms of $\mathcal{K}_E$ are therefore

$$\begin{aligned} a_e &:= \dim \mathcal{K}_E^3 = 4 \max(v - e, 0), \quad b_e := \dim \mathcal{K}_E^4 = 8k + 56 + v - e, \\ u_e &= d_{\mathcal{A}}^3|_{\mathcal{K}_E^3} : \mathcal{K}_E^3 \longrightarrow \mathcal{K}_E^4, \quad \rho_e = \text{rank } u_e, \quad \kappa_e = a_e - \rho_e. \end{aligned} \quad (\text{LCA5})$$

For $p \leq 2$, $q'_{n_p} \leq (k+1)^2 < L$, since $L - (k+1)^2 = 8k + 24 - v + e \geq 16$. At $p = 3$, $q'_{n_3} = Q$, giving $4 \max(v - e, 0)$. At $p = 4$, $(k+9)^2 - L = 8k + 56 + v - e > 0$. These calculations justify every zero term and the entire source and target of u_e .

In the original ordered triple-wedge frame (123), (124), (134), (234), this last map is

$$u_e(E_{123}, E_{124}, E_{134}, E_{234}) = -h_4 E_{123} + h_3 E_{124} - h_2 E_{134} + h_1 E_{234}. \quad (\text{LCA6})$$

Each E_I has the complete degree- $(k+4)$ coefficient rectangle and its first L lower exponential derivatives vanish. The output has the complete degree- $(k+8)$ rectangle and the same vanishing, by the evaluation square. More explicitly, if Z_3, Z_4 are any full-column bases of the two actual evaluation kernels, its coefficient matrix is

$$[u_e]_{Z_3, Z_4} = Z_4^{\leftarrow} [-M_{4,k+4}, M_{3,k+4}, -M_{2,k+4}, M_{1,k+4}] Z_3, \quad (\text{LCA7})$$

where $M_{i,k+4}$ is the full original convolution matrix of L_i^4 , and $Z_4^{\leftarrow}$ is any left inverse on $\text{im } Z_4$. The image is already in that space. Thus neither the rank ρ_e nor the kernel κ_e refers to an unspecified substitute map.

152 Two successive connecting maps for the full germ quotient

The two short exact sequences of complexes are

$$0 \longrightarrow \mathcal{K}_E \longrightarrow \mathcal{A} \xrightarrow{\mathcal{E}} \mathcal{E}^\bullet \longrightarrow 0, \quad 0 \longrightarrow \mathcal{E}^\bullet \longrightarrow \mathcal{H} \longrightarrow \mathcal{Q}_D \longrightarrow 0. \quad (\text{LCA8})$$

The middle complexes $\mathcal{A}$ and $\mathcal{H}$ are exact. Their ordinary positive-sign connecting maps give

$$H^p(\mathcal{Q}_D) \xrightarrow{\sim} H^{p+1}(\mathcal{E}^\bullet) \xrightarrow{\sim} H^{p+2}(\mathcal{K}_E). \quad (\text{LCA9})$$

Here is the complete construction and its sign. Lift a quotient cycle y to $h \in \mathcal{H}^p$. Then $d_{\mathcal{H}}h = \mathcal{E}_{p+1}a$ for a lift $a \in \mathcal{A}^{p+1}$. The next differential satisfies $\mathcal{E}_{p+2}d_{\mathcal{A}}a = 0$. The composed map is

$$[y] \longmapsto [d_{\mathcal{A}}a]. \quad (\text{LCA10})$$

There is no minus sign in either connecting map. Changing h by an evaluated lower coefficient changes a by its coefficient differential, up to a kernel term; changing a by a kernel term changes $d_{\mathcal{A}}a$ by a kernel boundary. Changing the quotient cycle by a quotient boundary has the same effect. This proves well-definition.

Conversely, for a kernel cycle $z \in \mathcal{K}_E^{p+2}$, coefficient exactness supplies $a \in \mathcal{A}^{p+1}$ with $d_{\mathcal{A}}a = z$. The evaluated vector $\mathcal{E}_{p+1}a$ is a germ cycle. Therefore

$$h = \iota_D(\mathcal{E}_{p+1}a), \quad d_{\mathcal{H}}h = \mathcal{E}_{p+1}a \quad (\text{LCA11})$$

by LCA3, and $[h]$ is the desired quotient cycle. If a kernel class in LCA10 is zero, subtract its kernel primitive from a . Coefficient exactness then writes the resulting a as a coefficient boundary; subtract the corresponding evaluation from h . The remaining h is a germ cycle and hence a germ boundary, proving injectivity. This proves the full isomorphism LCA9, including both endpoints.

Since the kernel complex has only LCA5's two terms,

$$\begin{aligned} H^0(\mathcal{Q}_D) &= H^3(\mathcal{Q}_D) = H^4(\mathcal{Q}_D) = 0, \\ H^1(\mathcal{Q}_D) &\simeq \ker u_e, \quad H^2(\mathcal{Q}_D) \simeq \text{coker } u_e, \\ \dim H^1(\mathcal{Q}_D) &= \kappa_e, \quad \dim H^2(\mathcal{Q}_D) = b_e - \rho_e. \end{aligned} \quad (\text{LCA12})$$

153 The analytic image and its complete two-term quotient

Let $\mathcal{I}_D^p$ be the literal image of CNC10 in $\mathcal{Q}_D^p$. In degree n it consists of the classes of the first L jets of $F = N/E_A$, where N belongs to the original degree- n upper exponential space and vanishes to order at least v . It is a subcomplex because multiplying its numerator by B_i gives precisely the next original numerator.

Set $M = L + v = Q + e$. Multiplication by E_A is an isomorphism from the first L jets of F to the space $z^v \mathbb{C}[z]/(z^M)$. Its coefficient matrix is triangular after shifting the first index by v , with diagonal $\mu_v/v! \neq 0$. The first M upper exponential derivative rows have rank $\min(M, q_n)$. Imposing their first v zero rows leaves rank $\min(L, q_n - v)$. Consequently the raw analytic germ image $\mathcal{S}_n \subset \mathcal{H}_{n,D}$ has exactly that rank. Every lower exponential is in this image because its product with E_A is an original upper numerator with the required zero order.

It follows by an exact termwise quotient that

$$\begin{aligned} \mathcal{R}_D^\bullet &:= \mathcal{Q}_D^\bullet / \mathcal{I}_D^\bullet, & \mathcal{R}_D^p &\simeq (\mathcal{H}_{n_p,D} / \mathcal{S}_{n_p}) \otimes \Lambda^p \mathbb{C}^4, \\ \dim \mathcal{R}_D^p &= \binom{4}{p} \max(0, M - q_{n_p}), \\ \dim \mathcal{R}_D^0 &= d_e := 8k + 24 + e, & \dim \mathcal{R}_D^1 &= 4e, & \mathcal{R}_D^p &= 0 \quad (p \geq 2). \end{aligned} \tag{LCA13}$$

The first quotient equality is the map induced by the identity on $\mathcal{H}$: its kernel is exactly $\mathcal{S}_n$, since the lower image is contained in $\mathcal{S}_n$. Thus no lower-gauge directions are counted a second time.

For reference the full term dimensions are

$$\begin{aligned} \dim \mathcal{Q}_D^\bullet &= (24k - 24 - v + e, \ 4(16k + 16 - v + e), \\ &\quad 6(8k + 24 - v + e), \ 4 \max(e - v, 0), \ 0), \\ \dim \mathcal{I}_D^\bullet &= (m_k, \ 4(m_k + 64), \ 6(8k + 24 - v + e), \ 4 \max(e - v, 0), \ 0). \end{aligned} \tag{LCA14}$$

They follow from the lower Vandermonde ranks and LCA13. All numbers are at the same original cutoff.

There are exact polynomial coordinates for every nonzero $\mathcal{R}_D^p$. Write $\chi_n(S) = \prod_{a,b=0}^n (S - \beta_{n,a,b})$ and set

$$(\mathcal{T}_{Ap})(S') = \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}). \tag{LCA15}$$

For $r = 0, \dots, M - q_n - 1$, define

$$\pi_{n,r}(F) = \Lambda_{E_A F}(\chi_n(S) S^r) = \Lambda_F(\mathcal{T}_A(\chi_n(S) S^r)). \tag{LCA16}$$

The equality is the exact original Leibniz identity. For a polynomial p of degree $j \geq v$, the leading coefficient of $\mathcal{T}_A p$ is its original leading coefficient times $\binom{j}{v} \mu_v$, and its degree is $j - v$. This follows from $\mathcal{T}_A S^j = \sum_{r=0}^j \binom{j}{r} \mu_r S'^{j-r}$ and $\mu_0 = \dots = \mu_{v-1} = 0$. Thus the polynomials in LCA16 have successive degrees $q_n - v, \dots, D$ and nonzero leading coefficients. Their functionals are independent. They vanish on every original analytic numerator, because χ_n vanishes at all its roots. Dimension LCA13 proves that their common kernel is exactly $\mathcal{S}_n$. Hence π_n is an onto coordinate map for the displayed quotient, without choosing an unrelated row norm.

154 The actual four-row differential has full rank

When $e = 0$, the degree-one quotient is zero. Let $t_0 = 0$. When $e = 1$, each degree-one wedge component has the single coordinate

$$\pi_{k+4,0}(F) = \Lambda_F(P_A), \quad P_A = \mathcal{T}_A \chi_{k+4}, \quad \deg P_A = Q - v = D, \quad \text{lc } P_A = \mu_v \binom{Q}{v}. \tag{LCA17}$$

The degree-zero to degree-one quotient map has the four rows

$$F \mapsto \Lambda_F(\mathcal{T}_{B_i} P_A), \quad (\mathcal{T}_{B_i} p)(S') = \sum_{a,b=0}^4 b_{i,a,b} p(S' + \beta_{4,a,b}), \quad b_{i,a,b} = [Z^a W^b] L_i^4. \quad (\text{LCA18})$$

These are the positive degree-zero wedge signs. They vanish on $\mathcal{S}_k$ by the exact numerator square, so they descend to $\mathcal{R}_D^0$.

In the fixed quotient coordinates LCA16 the same matrix is particularly explicit. For each original upper degree- k root, every shift by $\beta_{4,a,b}$ is a degree- $(k+4)$ root. Hence

$$R_i(S) := \frac{\mathcal{T}_{B_i} \chi_{k+4}(S)}{\chi_k(S)} \in \mathbb{C}[S]_{\leq 8k+24}, \quad [\mathbf{t}_1]_{i,r} = [S^r] R_i(S), \quad 0 \leq r < d_1. \quad (\text{LCA19})$$

The numerator vanishes on all distinct roots of χ_k , proving divisibility. The degree bound is $Q - (k+1)^2 = 8k+24 = d_1 - 1$. The translation sums $\mathcal{T}_A$ and $\mathcal{T}_{B_i}$ commute. Applying them to LCA19 therefore gives exactly $\pi_{k+4,0}(B_i F) = \sum_r [\mathbf{t}_1]_{i,r} \pi_{k,r}(F)$. This proves the literal four-row coefficient matrix and its native orientation, retaining all its phases.

We next prove that the four actual rows $(b_{i,a,b})$ are independent; no generic-period assumption is used. A relation among the four biforms L_i^4 becomes, under $x = Ht$,

$$F(x) = \sum_{i=1}^4 c_i x_i^4 = Q_H(x) R_2(x), \quad Q_H(x) = ((H^{-1}x)_1(H^{-1}x)_4 - (H^{-1}x)_2(H^{-1}x)_3). \quad (\text{LCA20})$$

Here is a complete proof of the ideal assertion used in this equality. The original polynomial $Q_0 = t_1 t_4 - t_2 t_3$ is primitive as a polynomial in t_1 over $\mathbb{C}[t_2, t_3, t_4]$, since t_4 and $t_2 t_3$ have no common factor. Over its fraction field it is a nonconstant linear polynomial, hence irreducible. Gauss's lemma proves irreducibility in the polynomial ring, a unique factorization domain, so Q_0 is prime. On the chart $t_4 \neq 0$ every point of $Q_0 = 0$ has $t_1 = t_2 t_3 / t_4$ and is a scalar multiple of $(ZW, Z, W, 1)$. A homogeneous polynomial vanishing on the Segre parametrization therefore has zero remainder after this substitution for all t_2, t_3 and $t_4 \neq 0$. Multiplying its Laurent polynomial remainder by a sufficient power of t_4 gives a polynomial vanishing on that open set, hence identically zero: fix t_2, t_3 and use infinitely many nonzero t_4 , then compare coefficients in t_2, t_3 . Polynomial division over $\mathbb{C}[t_2, t_3, t_4, t_4^{-1}]$ now shows that a power of t_4 times the original polynomial is divisible by Q_0 . Since Q_0 is prime and does not divide t_4 , the original polynomial itself is divisible by Q_0 . Conversely every multiple vanishes on the parametrization. This proves the entire homogeneous ideal assertion without a reducedness assumption on the separate conductor. The invertible substitution $x = Ht$ transports that ideal, and homogeneity gives the degree-two quotient in LCA20. The quadric is nondegenerate of rank four because H is invertible.

Suppose some c_i is nonzero, so $F \neq 0$ and $R_2 \neq 0$. If $c_j = 0$, then $\deg_{x_j} F = 0$. Degrees in x_j add in the integral domain of polynomials in the other variables. The product in LCA20 then forces Q_H to be independent of x_j , which makes its quadratic matrix have a zero row and column. This contradicts rank four. Thus all four c_i are nonzero.

The two quadrics Q_H, R_2 have a nonzero common point. To see it directly, restrict R_2 along the actual Segre parametrization $x = H(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^T$. The result is a biform of bidegree $(2, 2)$. If it is identically zero every Segre point works. Otherwise choose $[w_0 : w_1]$ for which its binary quadratic in z_0, z_1 is nonzero. A nonzero homogeneous binary quadratic over $\mathbb{C}$ has a projective root, giving a common point with $x \neq 0$. At that point LCA20 implies $\nabla F = R_2 \nabla Q_H + Q_H \nabla R_2 = 0$. But $\nabla F = (4c_i x_i^3)_i$ and all $c_i \neq 0$, forcing $x = 0$. This contradiction proves independence.

Finally translation of any degree- D polynomial with nonzero leading coefficient on 25 distinct shifts is injective as a map of their 25 coefficient weights when $D \geq 24$. For completeness, for weights w_j , the coefficient of S'^{D-r} in $\sum_j w_j P(S' + \eta_j)$ is the leading coefficient of P times $\binom{D}{r} \sum_j w_j \eta_j^r$, plus a linear combination of moments of orders below r . For $r = 0, \dots, 24$, these

form a triangular system with nonzero diagonal. Its vanishing forces all first 25 moments to vanish; the square Vandermonde on the distinct η_j then forces all $w_j = 0$.

The 25 original degree-four shifts are distinct since $\delta, \gamma > 0$, and in LCA17 $D = Q - v \geq 196 - 80 = 116 > 24$. Applying this argument to P_A and the four independent coefficient rows proves that the four polynomials in LCA18 are independent. They are polynomials through the original degree D , so their functionals on the full germ space are independent. Since all descend to the onto quotient π_k , their four quotient rows remain independent. Consequently

$$\boxed{t_e := \text{rank } \mathbf{t}_e = 4e, \quad H^0(\mathcal{R}_D) = \ker \mathbf{t}_e, \quad \dim H^0(\mathcal{R}_D) = 8k + 24 - 3e, \quad H^1(\mathcal{R}_D) = 0.} \quad (\text{LCA21})$$

Current native metric of the proved primitive quotient. The full coordinate map LCA16 is PDS4's same $\pi_e(F) = (\Lambda_F(\mathcal{T}_A(\chi_k S^a)))_a$, with exactly $d_e = 8k + 24 + e$ independent relation columns. For the source orders $s_0 = 1, k$, its original Gamma coefficient matrix C_e gives the attained quotient metric $G_e = (C_e^* C_e)^{-1}$ by PDS7. The primitive source of LCA21 is all this quotient at $e = 0$ and exactly $\ker W^\top$ at $e = 1$, with the same four polynomial rows LCA18–20.

In CIT6's original Taylor-complement coordinates, $\Theta_e = C_e^\top P_e$ identifies these spaces and their full minimum-norm fibres. The positive connecting map in the original long sequence has image $\mathcal{C}_e = \mathcal{S}_e / \mathcal{B}$, and

$$\begin{aligned} H^0(\mathcal{R}_e) &\xrightarrow[\partial_e]{\cong} \mathcal{C}_e, & \text{Gram}_{\text{source}} &= H_{\text{src},e} = \Theta_e^* G_e \Theta_e, \\ \log \det H_{\text{src},e} &= \Gamma_e - 2d_e q \{\log(4q/\pi) - 1\} + o(kq), \\ \log \det H_{\text{conn},e} &= \Gamma_e - 2d_e q \{\log(4q/\pi) - 1\} - \chi_{X_k, D_e} + \chi_{Y_e, D_e} + o(kq). \end{aligned} \quad (\text{LCA42})$$

Here $q = q_k$, $(s_0, s_1) = (1, 1)$ or $(k, k + 4)$, and $D_e = q_{k+4} - v - 1 + e$. The exact target Gram is the restriction to the connecting image of the native cohomology metric, attained over all original boundaries, as proved in CIT10–13. PDI2 proves the source metric equality. PSR19/22 supplies the finite two-sided intervals behind the last two lines, including $\mathfrak{F}$, both source errors, the four-row error, all coordinate factors Γ_e , and both actual independent-gauge terms. The full kernel/cokernel coordinates and the remaining $\ker U_e$ term of the original long sequence remain present.

155 The entire long sequence, connecting ranks and zero cases

Before using the rank calculation, let $t_e = \text{rank}(\mathcal{R}_D^0 \rightarrow \mathcal{R}_D^1)$ and let w_e be the rank of the induced map $H^1(\mathcal{Q}_D) \rightarrow H^1(\mathcal{R}_D)$. The exact sequence $0 \rightarrow \mathcal{I}_D \rightarrow \mathcal{Q}_D \rightarrow \mathcal{R}_D \rightarrow 0$ and LCA12–13 give the complete nontrivial cohomology sequence

$$\begin{aligned} H^0(\mathcal{I}_D) &= H^3(\mathcal{I}_D) = H^4(\mathcal{I}_D) = 0, \\ 0 \longrightarrow \ker \mathbf{t}_e &\xrightarrow{\partial^0} H^1(\mathcal{I}_D) \longrightarrow \ker u_e \xrightarrow{\omega_e} \text{coker } \mathbf{t}_e \xrightarrow{\partial^1} H^2(\mathcal{I}_D) \longrightarrow \text{coker } u_e \longrightarrow 0, \\ 0 \leq w_e &:= \text{rank } \omega_e \leq \min(\kappa_e, 4e - t_e) \leq 4. \end{aligned} \quad (\text{LCA22})$$

This is the actual long exact sequence, not merely an alternating dimension identity. At any possible t_e, w_e its arrow ranks are, in their displayed order,

$$d_e - t_e, \quad \kappa_e - w_e, \quad w_e, \quad 4e - t_e - w_e, \quad b_e - \rho_e. \quad (\text{LCA23})$$

Exactness therefore gives

$$\dim H^1(\mathcal{I}_D) = d_e - t_e + \kappa_e - w_e, \quad \dim H^2(\mathcal{I}_D) = b_e - \rho_e + 4e - t_e - w_e. \quad (\text{LCA24})$$

For the actual original map LCA21 proves $t_e = 4e$, so $\text{coker } \mathbf{t}_e = 0$ and $w_e = 0$ exactly. Thus the final groups and their dimensions are

$$\begin{aligned}
0 &\longrightarrow \ker \mathbf{t}_e \xrightarrow{\partial^0} H^1(\mathcal{I}_D) \longrightarrow \ker u_e \longrightarrow 0, \\
H^2(\mathcal{I}_D) &\xrightarrow{\sim} \text{coker } u_e, \\
\boxed{\begin{aligned} \dim H^1(\mathcal{I}_D) &= 8k + 24 - 3e + \kappa_e, \\ \dim H^2(\mathcal{I}_D) &= 8k + 56 + v - e - a_e + \kappa_e, \quad a_e = 4 \max(v - e, 0). \end{aligned}}
\end{aligned} \tag{LCA25}$$

All remaining dependence is the rank of the actual signed kernel map LCA6–7. No generic value is assigned.

For $v = 0$, both $a_e = 0$ and $\rho_e = \kappa_e = 0$. The exact values are

$$\begin{array}{c|ccc}
& \dim H^1(\mathcal{I}_D) & \dim H^2(\mathcal{I}_D) & \dim \mathcal{I}_D^3 \\
\hline
e = 0 & 8k + 24 & 8k + 56 & 0 \\
e = 1 & 8k + 21 & 8k + 55 & 4
\end{array} \tag{LCA26}$$

In the upper case the map into the four-dimensional degree-three image is onto, since $H^3(\mathcal{I}_D) = 0$. For $e = 1, v = 1$, again $a_e = \kappa_e = 0$, with dimensions $8k + 21, 8k + 56$. These cases do not require an inverse of an empty kernel matrix.

The image differentials themselves now have exact ranks

$$\begin{aligned}
\text{rank } d_{\mathcal{I}}^0 &= m_k, & \text{rank } d_{\mathcal{I}}^1 &= 40k + 88 + 3e - 3v - \kappa_e, \\
\text{rank } d_{\mathcal{I}}^2 &= 4 \max(e - v, 0), & \text{rank } d_{\mathcal{I}}^3 &= 0.
\end{aligned} \tag{LCA27}$$

Indeed the first follows from $H^0 = 0$. The second is $\dim \mathcal{I}^1 - \text{rank } d^0 - \dim H^1$ using LCA14 and LCA25. The third follows from $H^3 = 0$ and $\mathcal{I}^4 = 0$. Subtracting the two cohomology dimensions gives

$$\begin{cases} 32 - 3v, & e = 0, \\ 38 - 3v - 4 \max(0, 1 - v), & e = 1, \end{cases} \tag{LCA28}$$

which recovers the complete original CNC16 Euler characteristics from the individual groups and maps.

156 Literal connecting representatives and native matrix orientation

For a quotient cycle $r \in \ker \mathbf{t}_e$, choose $h \in \mathcal{H}^0$ with $\pi_k(h) = r$. Its differential has zero $\mathcal{R}_D^1$ class, so its $\mathcal{Q}_D^1$ class belongs to $\mathcal{I}_D^1$. The first connecting map is

$$\partial^0 r = [d_{\mathcal{H}} h] \in H^1(\mathcal{I}_D). \tag{LCA29}$$

The sign is positive. If h is changed by a vector whose $\mathcal{Q}$ class is in $\mathcal{I}^0$, the result changes by an image boundary. Injectivity follows from $H^0(\mathcal{Q}_D) = 0$: if the result is an image boundary, subtract its degree-zero primitive from h ; its quotient is then a degree-zero cycle, hence zero, so $r = 0$.

The second connecting arrow before the rank-four specialization is obtained in exactly the same way: lift a class in $\text{coker } \mathbf{t}_e = H^1(\mathcal{R}_D)$ to $h \in \mathcal{H}^1$, then take $[d_{\mathcal{H}} h] \in H^2(\mathcal{I}_D)$. The source of this map is zero for the actual data, by LCA21.

The map $H^1(\mathcal{I}_D) \rightarrow \ker u_e$ is LCA10 restricted to image cycles. To construct a preimage of an actual $z \in \ker u_e \subset \mathcal{K}_E^3$, choose $a \in \mathcal{A}^2$ with $d_{\mathcal{A}} a = z$, and put $h = \iota_D \mathcal{E}_2 a \in \mathcal{H}^1$. Its quotient is a $\mathcal{Q}$ cycle. Its $\mathcal{R}^1$ class lies in $\text{im } \mathbf{t}_e$ because the latter is onto. Choose $r \in \mathcal{R}^0$ with $\mathbf{t}_e r = \pi_{k+4}(h)$, choose $g \in \mathcal{H}^0$ lifting r , and replace

$$h \quad \text{by} \quad h - d_{\mathcal{H}} g. \tag{LCA30}$$

The new quotient lies in $\mathcal{I}^1$, still is a cycle, and still maps to z because $d_{\mathcal{H}}^2 g = 0$. This gives the claimed surjection without choosing a splitting of its nontrivial kernel $\partial^0 \ker \mathbf{t}_e$. The minus in LCA30 is the explicit correction of a quotient representative; it does not change the positive connecting sign in LCA10.

Likewise, for an image cycle in degree two, lift it to $h \in \mathcal{H}^2$, choose $a \in \mathcal{A}^3$ with $\mathcal{E}_3 a = d_{\mathcal{H}} h$, and take $d_{\mathcal{A}} a \in \mathcal{K}_E^4$ modulo $\text{im } u_e$. This is the exact isomorphism $H^2(\mathcal{I}_D) \rightarrow \text{coker } u_e$. Conversely, given a representative $z \in \mathcal{K}_E^4$, coefficient exactness supplies $a \in \mathcal{A}^3$ with $d_{\mathcal{A}} a = z$. Put $h = \iota_D \mathcal{E}_3 a$. Its $\mathcal{Q}^2$ class automatically belongs to $\mathcal{I}^2$ because $\mathcal{R}^2 = 0$, and LCA10 sends its cohomology class to $[z]$. Changing z by an u_e boundary changes only an image boundary.

Here are the full native coordinate orientations. Let $\mathcal{B}_p$ be the original monomial-to-Gamma matrix through degree D at source order s_{n_p} and center $(n_p - 8)/2$, defined by

$$S'^r = \sum_{j=0}^D (\mathcal{B}_p)_{j,r} \phi_{j,s_{n_p},(n_p-8)/2}. \quad (\text{LCA31})$$

If $w_j = \Lambda_F(\phi_j)$ and $f_r = F^{(r)}(0)$, then $f = \mathcal{B}_p^T w$. Thus the exact map to Taylor coefficients is

$$U_p = \text{diag}(1/r!)_{r=0}^D \mathcal{B}_p^T, \quad [F]_{\text{Taylor}} = U_p w. \quad (\text{LCA32})$$

Every factorial, the original Gamma mass inside $\mathcal{B}_p$, and its leading phase $i^r \sqrt{M_{s_{n_p}} r! (s_{n_p}/2)_r}$ are retained. Let $\text{Mult}_L(B_i)$ and $\text{Mult}_L(D_i)$ be their full lower triangular Taylor multiplication matrices. The native blocks of the differential and contraction are

$$U_{p+1}^{-1} \text{Mult}_L(B_i) U_p, \quad U_{p-1}^{-1} \text{Mult}_L(D_i) U_p, \quad (\text{LCA33})$$

with the wedge and contraction signs of LCA3 and LCA6. They are equivalently the ordinary transposes of the full primal polynomial translation matrices. A conjugate transpose appears only when taking a metric adjoint, a Gram, or an orthogonal projector.

For a concrete quotient-coordinate matrix let O_p have orthonormal columns spanning $(\text{im } \mathcal{E}_p)^\perp$ in the original native $\mathcal{H}^p$. Then

$$[d_Q^p] = O_{p+1}^* d_{\mathcal{H}}^p O_p. \quad (\text{LCA34})$$

This follows by lifting a quotient vector by O_p and projecting its differential; the omitted part is in $\text{im } \mathcal{E}_{p+1}$. Let $\mathcal{E}_{p+1}^+$ be its Moore–Penrose inverse, used only to choose a lift of a vector already proved to lie in its image. Let Z_{p+2} be a coefficient basis of $\mathcal{K}_E^{p+2}$ and let $Z_{p+2}^\leftarrow$ be any left inverse on that image. On quotient cycles the exact double connecting matrix is

$$\boxed{Z_{p+2}^\leftarrow d_{\mathcal{A}}^{p+1} \mathcal{E}_{p+1}^+ d_{\mathcal{H}}^p O_p.} \quad (\text{LCA35})$$

Its value is taken in kernel cohomology, hence modulo $\text{im } u_e$ when $p = 2$; for $p = 1$ it belongs to $\ker u_e$. The pseudoinverse is a representative choice in a finite exact image, not an estimate or a declaration that dependent evaluation columns are independent. Changing that lift adds a kernel boundary exactly as in LCA10.

The rows $\pi_{n,r}$ in these native coordinates are equally explicit. If $p_{n,r}$ is the complete monomial coefficient column of $\mathcal{T}_A(\chi_n S^r)$, padded through degree D , then its native row is

$$(\mathcal{B}_p p_{n,r})^\top. \quad (\text{LCA36})$$

It uses an ordinary transpose because it evaluates a complex-linear functional on that polynomial. After the onto π coordinates, LCA19 is exactly the degree-zero quotient differential. Equations LCA29–36 specify the connecting maps in both the original coefficient presentation and the original native presentation, with their exact degree shifts, orientations, kernels and quotient classes.

The resulting cohomology metrics, when required, are the original cycle restriction followed by the attained boundary quotient, as in CNC14. This source does not replace them by coefficient norms or assign singular-value bounds to the remaining u_e map.

157 Three-component late-kernel control and the exact low return

Write $\ell_e = \max(v - e, 0)$, so $a_e = 4\ell_e$ in LCA5. The full actual Bezout identity evaluated at zero ensures that at least one index j has $B_j(0) \neq 0$. Let I_j be the increasing triple complementary to j ; the coefficient of $B_j E_{I_j}$ in LCA6 is $\epsilon_j = (-1)^{j-1}$. An element of $\ker u_e$ is an actual tuple of lower exponential polynomials E_{I_j} of degree $k + 4$, each vanishing to order at least $L_e = Q - v + e$, with the full identity

$$\sum_{j=1}^4 \epsilon_j B_j(z) E_{I_j}(z) = 0. \quad (\text{LCA37})$$

The correspondence from the coefficient identity to this exponential identity is injective: the distinct degree- $(k + 8)$ roots have an invertible full Vandermonde. If $\ell_e = 0$, each degree- $(k + 4)$ exponential has at least Q zero initial derivatives, and is zero. Hence $\ker u_e = 0$ in that case.

If $\ell_e > 0$, put $E_{I_j}(z) = z^{L_e} f_j(z)$. The first ℓ_e Taylor coefficients of f_j are, with every factorial retained, $E_{I_j}^{(L_e+r)}(0)/(L_e + r)!$ for $0 \leq r < \ell_e$. They determine the entire coefficient vector of E_{I_j} , since these are precisely its remaining derivatives through $Q - 1$. For the chosen index with $B_j(0) \neq 0$, LCA37 gives the exact finite relation

$$[f_j] = -\epsilon_j [B_j]^{-1} \sum_{i \neq j} \epsilon_i [B_i] [f_i] \quad \text{in } \mathbb{C}[z]/(z^{\ell_e}), \quad 0 \leq \kappa_e \leq 3\ell_e. \quad (\text{LCA38})$$

The bracketed inverse exists since its constant coefficient is nonzero; it is the unique finite power series inverse of the actual B_j . Thus projection of $\ker u_e$ to the other three coefficient components is injective. Each such component has dimension ℓ_e , proving the bound, including all its constant and phase dependence. This construction does not require an inverse estimate for that series.

The two endpoint coefficient kernels are nested, $\ker u_1 \subset \ker u_0$: their coefficient differential is identical, and the upper cutoff adds one zero derivative to each of its four input blocks. For $v \geq 1$, define the exact next-moment map

$$\begin{aligned} \mathbf{j} : \ker u_0 &\longrightarrow \mathbb{C}^4, & (E_{I_1}, E_{I_2}, E_{I_3}, E_{I_4}) &\longmapsto (E_{I_j}^{(L_0)}(0))_{j=1}^4, \\ \ker \mathbf{j} &= \ker u_1, & \text{im } \mathbf{j} &\subset \left\{ (y_j) : \sum_{j=1}^4 \epsilon_j B_j(0) y_j = 0 \right\}, \\ \Delta\kappa &:= \kappa_0 - \kappa_1 = \text{rank } \mathbf{j} \in \{0, 1, 2, 3\}. \end{aligned} \quad (\text{LCA39})$$

The containment follows by differentiating LCA37 exactly L_0 times at zero: all terms involving a derivative of E_{I_j} below L_0 vanish, and the remaining Leibniz coefficient is one. Its target hyperplane has dimension three because the row $(\epsilon_j B_j(0))_j$ is nonzero. The kernel assertion is exactly the added derivative condition, so rank-nullity proves the last line. At $v = 0$, both kernels already vanish by LCA38 and we put $\Delta\kappa = 0$; no nonexistent tail coefficient is used.

For completeness this produces an exact map on the original cohomology, with its original norm. Both low cutoffs inject the full coefficient spaces in degrees zero and one into the analytic image. Use these identical coefficient identifications and put $\mathcal{Z}_e = \ker(d_{\mathcal{L}_e}^1)$ and $\mathcal{B} = d_{\mathcal{L}_e}^0(\mathcal{C}_{v,k})$ in $\mathcal{C}_{v,k+4}^4$. The boundary $\mathcal{B}$ is independent of the endpoint, since it is the same coefficient multiplication map. Truncation of the actual germ complex from L_1 to L_0 commutes with both differentials and lower evaluation. It induces $\mathcal{Z}_1 \subset \mathcal{Z}_0$ and the injective map

$$\begin{aligned} \text{ret} : H^1(\mathcal{I}_1) &\hookrightarrow H^1(\mathcal{I}_0), & z + \mathcal{B} &\longmapsto z + \mathcal{B}, \\ \text{coker}(\text{ret}) &\simeq \mathcal{Z}_0/\mathcal{Z}_1, & \dim \text{coker}(\text{ret}) &= 3 + \Delta\kappa \in \{3, 4, 5, 6\}. \end{aligned} \quad (\text{LCA40})$$

Injectivity uses the identical boundary subspace, rather than any injectivity of the later evaluation matrices. The cokernel is the quotient by the nested cycles, and its dimension follows by subtracting the two exact LCA25 dimensions. In particular it is three when $v = 0$.

Fix either original source-order family. At degree $j = k + 4$ let T_e, V_e be exactly the original DNE7 analytic and lower-evaluation matrices through D_e , in the unchanged orthonormal polynomials at center $(j - 8)/2$ and order $s = 1$ or $s = j$. The lower cutoff matrices are the first L_0 rows of the upper ones. For every original coefficient vector x ,

$$\begin{aligned} x^* \mathcal{D}_{j,D_0} x &= \min_{\eta} \|T_0 x + V_0 \eta\|_2^2 \leq \min_{\eta} \|T_1 x + V_1 \eta\|_2^2 = x^* \mathcal{D}_{j,D_1} x, \\ \|\text{ret}[z]\|_{H^1(\mathcal{I}_0)}^2 &\leq \|[z]\|_{H^1(\mathcal{I}_1)}^2. \end{aligned} \tag{LCA41}$$

The first inequality holds pointwise in the same complete gauge coefficient η by dropping the last original orthonormal coordinate, then taking its minimum. The full lower evaluation has independent columns at these degrees, and DNE7's minima are attained. The fourfold direct sum preserves the same inequality. For the second, substitute $z + b$ for every $b \in \mathcal{B}$ and minimize over that identical boundary space on both sides. This is precisely the restriction followed by attained quotient metric. The original Gamma mass and all source phases remain those of DNE7, and the comparison holds separately for both source orders. It gives the actual contraction and its exact missing rank; it does not assign a lower singular-value bound to its common image.

158 The original low conductor cohomology and its two native metrics

Keep all CNC1–24 data and the original ordered exterior maps. In particular A^{hom} remains the complete nonreduced biform, E_A its actual exponential symbol, and every coefficient of B_i, D_i is unchanged. This section uses precisely the two original low cutoffs

$$\begin{aligned} Q &= (k + 5)^2, \quad e \in \{0, 1\}, \quad D_e = Q - v - 1 + e, \quad L_e = D_e + 1 = Q - v + e, \\ k &\equiv 1 \pmod{4}, \quad k \geq 9, \quad 0 \leq v \leq 80, \quad n_p = k + 4p, \quad c'_p = n_p/2 - 4, \\ s_p &= 1 \text{ for every } p \quad \text{or} \quad s_p = n_p \text{ for every } p. \end{aligned} \tag{LCC1}$$

The notation e in this section is only the displayed endpoint selector. It is not the original multiplicity $1 + k(m - 1)$; that original multiplicity remains one. We calculate the kernel of one finite original matrix of at most $4v$ columns, and all other cohomology ranks exactly. Every resulting metric uses the original Gamma measure of CNC9.

158.1 The two exact ambient sequences

Let $\mathcal{H}_e^p$ be $\binom{4}{p}$ copies of the full polynomial dual through D_e in the LCC1 frame. Algebraically this is the ordinary Taylor jet algebra $\mathbb{C}[z]/(z^{L_e})$ with that many exterior copies. The change of coordinates sends a germ F to $(\Lambda_F(\phi_{r,s_p,c'_p}))_{r=0}^{D_e}$. Its entries include the exact leading phase $i^{-r}/\sqrt{M_{s_p} r! (s_p/2)_r}$. In these coordinates the differential is the ordinary transpose of the literal polynomial translation sum for wedge multiplication by (B_i) . The entire tuple (D_i) supplies the contraction $\iota_D d + d \iota_D = I$, by the full CNC5 calculation now reduced modulo z^{L_e} . Thus $\mathcal{H}_e^\bullet$ is exact, including both ends, with the specified native changes of frame.

Let $\mathcal{A}^p = \mathcal{P}_{k-8+4p} \otimes \Lambda^p \mathbb{C}^4$ be the exact lower coefficient complex GKC11 and let $\mathcal{V}_{p,e}$ evaluate its complete original lower exponentials in $\mathcal{H}_e^p$. Define its literal kernel and image by

$$\begin{aligned} \mathcal{Z}_e^p &= \ker \mathcal{V}_{p,e}, \quad \mathcal{L}_e^p = \text{im } \mathcal{V}_{p,e}, \quad \mathcal{Q}_e^p = \mathcal{H}_e^p / \mathcal{L}_e^p, \\ 0 \rightarrow \mathcal{Z}_e^\bullet &\rightarrow \mathcal{A}^\bullet \rightarrow \mathcal{L}_e^\bullet \rightarrow 0, \quad 0 \rightarrow \mathcal{L}_e^\bullet \rightarrow \mathcal{H}_e^\bullet \rightarrow \mathcal{Q}_e^\bullet \rightarrow 0. \end{aligned} \tag{LCC2}$$

The root-addition formula in CNC11 proves that both are sequences of complexes, with every original convolution coefficient and exterior sign. They are termwise exact by their definitions. All quotients in the second sequence have their attained native Hilbert quotient metric.

Applying the explicit cycle-lifting construction twice gives the algebraic isomorphism

$$H^p(\mathcal{Q}_e^\bullet) \simeq H^{p+2}(\mathcal{Z}_e^\bullet). \quad (\text{LCC3})$$

Here is its full map. Lift a cycle in $\mathcal{Q}_e^p$ to $f \in \mathcal{H}_e^p$. Then df is in $\mathcal{L}_e^{p+1}$. Lift df to $a \in \mathcal{A}^{p+1}$. Its differential belongs to $\mathcal{Z}_e^{p+2}$ and is a cycle. A different first lift adds a lower image; lift that addition to $\mathcal{A}^p$ and subtract its differential from a . A different second lift adds an element of $\mathcal{Z}_e^{p+1}$, whose differential is a boundary. Changing the original cycle by a quotient boundary also changes the final result only by a boundary. This proves well-definition. Conversely a cycle in $\mathcal{Z}_e^{p+2}$ is da by exactness of $\mathcal{A}$. The image of a is a cycle in $\mathcal{H}_e^{p+1}$, hence df by its exactness; the image of f in $\mathcal{Q}_e^p$ is the required cycle. If the final class is zero, subtract a kernel primitive from a , then a coefficient boundary from it, and finally an ambient boundary from f . The initial quotient cycle becomes a boundary. This proves injectivity and surjectivity with the displayed degree shift.

The first-lift sign in the preceding proof of LCC3. In the preceding paragraph the preserved word “subtract” is superseded by “add”. Precisely, write $\mathcal{V}_{p,e} : \mathcal{A}^p \rightarrow \mathcal{H}_e^p$ for the original evaluation cochain map. If $f' = f + \mathcal{V}_{p,e}b$ with $b \in \mathcal{A}^p$ and $\mathcal{V}_{p+1,e}a = df$, use $a' = a + db$. Then

$$\begin{aligned} \mathcal{V}_{p+1,e}a' &= \mathcal{V}_{p+1,e}a + \mathcal{V}_{p+1,e}(db) = df + d(\mathcal{V}_{p,e}b) = df', \\ da' &= d(a + db) = da + d^2b = da. \end{aligned}$$

The first equality uses the full evaluation cochain square of LCC2, and the second uses the unchanged exterior differential $d^2 = 0$. Thus the actual positive double-lift map of LCC3 is independent of this first-lift change with precisely the added coefficient differential. Its displayed map, cohomological degree shift and all subsequent ranks remain unchanged.

The full original lower Vandermonde gives

$$\begin{aligned} \mathcal{Z}_e^p &= 0 \quad (p = 0, 1, 2), & \dim \mathcal{Z}_e^3 &= 4a_e, & \dim \mathcal{Z}_e^4 &= b_e, \\ a_e &= \max(v - e, 0), & b_e &= 8k + 56 + v - e. \end{aligned} \quad (\text{LCC4})$$

Indeed its rank in each scalar block is $\min\{(k - 7 + 4p)^2, L_e\}$. The first three lower rectangles have size at most $(k + 1)^2$, and $L_e - (k + 1)^2 = 8k + 24 - v + e > 0$. The last two sizes are Q and $(k + 9)^2$, which yield exactly LCC4.

There is a completely specified finite matrix for the remaining kernel. Let $\mathbb{W}_d$ be the full ordinary Vandermonde at every original degree- d root, with rows $0, \dots, (d + 1)^2 - 1$. When $L_e < (d + 1)^2$, put

$$R_{d,e} = \mathbb{W}_d^{-1} E_{\{L_e, \dots, (d+1)^2-1\}}, \quad S_{d,e} = E_{\{L_e, \dots, (d+1)^2-1\}}^\top \mathbb{W}_d. \quad (\text{LCC5})$$

If there are no such rows, $R_{d,e}$ has no columns. Its columns are exactly the coefficient vectors with zero first L_e moments; $S_{d,e}R_{d,e} = I$. Let $d_{\mathcal{A}}^3$ be the literal last GKC3 convolution map in its four triple-wedge blocks, ordered (123), (124), (134), (234). Define

$$U_e = S_{k+8,e} d_{\mathcal{A}}^3 \text{diag}(R_{k+4,e}, R_{k+4,e}, R_{k+4,e}, R_{k+4,e}), \quad \rho_e = \text{rank } U_e, \quad \kappa_e = 4a_e - \rho_e. \quad (\text{LCC6})$$

Its size is b_e by $4a_e$ and its signs on the four input blocks are $(-, +, -, +)$, respectively. The image is in $\mathcal{Z}_e^4$ by LCC2, so $R_{k+8,e}U_e$ equals that complete coefficient convolution. Thus its kernel and cokernel are precisely those of the last differential of $\mathcal{Z}_e^\bullet$. The nondegenerate Vandermonde maps in LCC5 retain every phase and factorial choice; in particular

$$\begin{aligned} 0 \leq \kappa_e \leq 4a_e, & \quad \dim H^1(\mathcal{Q}_e) = \kappa_e, & \dim H^2(\mathcal{Q}_e) &= b_e - 4a_e + \kappa_e, \\ H^p(\mathcal{Q}_e) &= 0 \quad (p \neq 1, 2). \end{aligned} \quad (\text{LCC7})$$

This follows directly from LCC3–6. No rank of U_e is presumed. It has a fixed number of columns as k grows, with its complete actual entries displayed.

158.2 The actual conductor image and its last missing jet

Let $\mathcal{I}_e^\bullet$ be exactly the image of the analytic conductor complex under CNC10, and form the remaining quotient $\mathcal{R}_e^\bullet = \mathcal{Q}_e^\bullet / \mathcal{I}_e^\bullet$. The short exact sequence of these complexes and its term dimensions are

$$\begin{aligned} 0 \rightarrow \mathcal{I}_e^\bullet \rightarrow \mathcal{Q}_e^\bullet \rightarrow \mathcal{R}_e^\bullet \rightarrow 0, \\ \dim \mathcal{R}_e^0 = 8k + 24 + e, \quad \dim \mathcal{R}_e^1 = 4e, \quad \mathcal{R}_e^p = 0 \quad (p \geq 2). \end{aligned} \tag{LCC8}$$

Subtract CNC13 from the dimension $\binom{4}{p} \max(0, L_e - q'_{n_p})$ of $\mathcal{Q}_e^p$. For $p = 0, 1$ the conductor maps are injective and give these two differences. For $p \geq 2$ the first-jet surjectivity in the CNC13 proof gives $\mathcal{I}_e^p = \mathcal{Q}_e^p$. These assertions hold also at $v = 0$; at the upper cutoff both spaces in degree three then have dimension four.

For $e = 0$ the quotient differential is zero because its target vanishes. For $e = 1$ it is onto its four-dimensional target. We prove this using the literal original rows. Let $\chi_{k+4}(S)$ be the complete monic polynomial of the original degree- $(k+4)$ upper roots. Use exactly the polynomial translation sums

$$\begin{aligned} (\mathcal{T}_A p)(S') &= \sum_{r,t=0}^8 a_{rt} p(S' + \beta_{8,r,t}), & (\mathcal{T}_{B_i} p)(S') &= \sum_{r,t=0}^4 b_{i,r,t} p(S' + \beta_{4,r,t}), \\ P_A &= \mathcal{T}_A \chi_{k+4}, & \deg P_A &= Q - v = D_1, & \text{lc}(P_A) &= \mu_v \binom{Q}{v} \neq 0. \end{aligned} \tag{LCC9}$$

The degree calculation follows by expanding $\mathcal{T}_A = \sum_{j \geq 0} \mu_j \partial_{S'}^j / j!$: the first nonzero coefficient has index v . For an arbitrary native ambient germ F , the one extra target numerator jet, beyond its first Q freely interpolated upper jets, is

$$\Lambda_{E_A F}(\chi_{k+4}) = \Lambda_F(P_A). \tag{LCC10}$$

Indeed every actual upper exponential is killed by $\chi_{k+4}(\partial_z)$, and its first Q jets interpolate uniquely by the original Vandermonde. The next derivative defect is exactly evaluation on the monic annihilating polynomial. Multiplication by E_A only uses the first $D_1 + 1$ jets of F , as LCC9 proves. It vanishes precisely on the actual analytic conductor image plus all lower evaluations. It is nonzero on the one-dimensional target quotient because its polynomial has degree D_1 . Thus this is an exact coordinate for each of the four copies of $\mathcal{R}_1^1$, with no scalar discarded. The four rows of its differential on a source functional are consequently

$$F \mapsto (\Lambda_F(\mathcal{T}_{B_i} P_A))_{i=1}^4. \tag{LCC11}$$

They vanish on $\mathcal{I}_1^0$ by the exact cochain square, so descend to $\mathcal{R}_1^0$.

The four polynomials L_i^4 are linearly independent for the actual invertible period matrix. To prove this, use coordinates $x = Ht$ and the rank-four irreducible quadric $Q_H(x) = Q(H^{-1}x)$. A nonzero relation of the restrictions would say $\sum_i c_i x_i^4 = Q_H R_2$ for a quadratic polynomial R_2 . Here the Segre ideal is generated by Q : division by $t_1 t_4 - t_2 t_3$, for example as a polynomial in t_1 , proves that a homogeneous polynomial vanishing on this parametrized quadric is its multiple. If some $c_j = 0$, degree additivity in x_j forces Q_H to be independent of x_j , contrary to its nonsingular quadratic matrix. Thus all c_j are nonzero. The two quadrics Q_H, R_2 have a nonzero common point: the pullback of R_2 to the Segre parameters is a biform of degree $(2, 2)$. If it is identically zero any point works; otherwise fix a projective w for which the binary z -form is nonzero, and take one of its projective zeros. At that common point the gradient of their product is zero. But the gradient of $\sum c_i x_i^4$ is $(4c_i x_i^3)_i$ and cannot vanish at a nonzero point. This contradiction proves independence, including all period fibres under consideration.

Finally, for any polynomial P of degree at least 24 with nonzero leading coefficient, translation of P is injective on the span of the 25 distinct degree-four shift exponentials. In detail, if

$\sum_{\eta} c_{\eta} P(S' + \eta) = 0$, the coefficients of the highest 25 powers of S' successively give $\sum_{\eta} c_{\eta} \eta^j = 0$, $0 \leq j \leq 24$: the diagonal in that triangular system is $\text{lc}(P) \binom{\deg P}{j} \neq 0$. The original 25-root Vandermonde is invertible. Thus every $c_{\eta} = 0$. Here $D_1 = Q - v \geq 116$, so this applies to P_A . The four independent rows of coefficients of L_i^4 consequently give four independent polynomials in LCC11. The pairing with all functionals through D_1 is nondegenerate. Therefore

$$\text{rank } d_{\mathcal{R}_e}^0 = 4e, \quad H^1(\mathcal{R}_e) = 0, \quad \dim H^0(\mathcal{R}_e) = 8k + 24 - 3e. \quad (\text{LCC12})$$

158.3 All original low cohomology groups

The first differential on $\mathcal{I}_e$ is injective: the first two CNC maps are injective and the first coefficient map in CNC6 is injective. Using LCC7–12 in the connecting sequence of LCC8 gives

$$\begin{aligned} 0 &\longrightarrow H^0(\mathcal{R}_e) \xrightarrow{\partial_e} H^1(\mathcal{I}_e) \longrightarrow \ker U_e \longrightarrow 0, \\ H^2(\mathcal{I}_e) &\xrightarrow{\sim} \text{coker } U_e, \quad H^p(\mathcal{I}_e) = 0 \quad (p \neq 1, 2), \\ \dim H^1(\mathcal{I}_e) &= 8k + 24 - 3e + \kappa_e, \quad \dim H^2(\mathcal{I}_e) = b_e - 4a_e + \kappa_e. \end{aligned} \quad (\text{LCC13})$$

For clarity the first map lifts a degree-zero quotient cycle from $\mathcal{R}_e$ to $\mathcal{Q}_e$ and takes its differential, now in $\mathcal{I}_e^1$. The second map in the first line is induced by inclusion into $\mathcal{Q}_e$ followed by the double lift LCC3 and the kernel coordinates LCC5–6. The degree-two isomorphism uses the same inclusion and double lift, followed by its full cokernel. All are well-defined by the explicit boundary calculations in LCC3 and CNC14. Exactness follows as well by a direct lift: a class killed on passing to $\mathcal{Q}_e$ has a primitive there, whose class is a quotient cycle in $\mathcal{R}_e$; a class in $\mathcal{Q}_e$ lifts to an image cycle because the next quotient cohomology is zero. This proves every map and rank in LCC13.

The complete finite dimensions are thus

$$\begin{aligned} \dim H^1(\mathcal{I}_0) &= 8k + 24 + \kappa_0, \quad \dim H^2(\mathcal{I}_0) = 8k + 56 - 3v + \kappa_0, \quad 0 \leq \kappa_0 \leq 4v, \\ \dim H^1(\mathcal{I}_1) &= 8k + 21 + \kappa_1, \\ \dim H^2(\mathcal{I}_1) &= 8k + 55 + v - 4 \max(v - 1, 0) + \kappa_1, \quad 0 \leq \kappa_1 \leq 4 \max(v - 1, 0). \end{aligned} \quad (\text{LCC14})$$

At $v = 0$, both matrices U_e have no columns and both κ_e are zero. At $v = 1$ the upper matrix has no columns, giving $\kappa_1 = 0$ exactly. The Euler differences in LCC14 are CNC16 with their original signs. The remaining parameters are the kernels of the fully specified LCC6 matrices; they have at most $4v$ columns.

The full-coefficient tail refinement of LCC13–14. On the same original simple-quartet five-orbit domain, keep $k = 4l + 1 \geq 9$, $j = k + 4$, $e \in \{0, 1\}$, the original invertible period matrix, and the entire nonreduced conductor. The complete four-factor identity BTK26 is

$$v = \sum_{a=1}^4 v_a \leq 32, \quad \frac{\mu_v}{v!} = \prod_{a=1}^4 \frac{E_a^{(v_a)}(0)}{v_a!} \neq 0, \quad v_a = \text{ord}_0 E_a.$$

Every conductor factor and its original multiplicity remains in this identity. The complete coefficient-kernel calculation BTK7–25 applies to exactly the matrix U_e of LCC6. It sharpens the permissible parameters in the same groups to

$$\begin{aligned} \kappa_e &= \dim \ker U_e \leq \bar{\kappa}_e := 3 \max(v - e - 2, 0), \\ \dim H^1(\mathcal{I}_0) &= 8k + 24 + \kappa_0, \quad \dim H^2(\mathcal{I}_0) = 8k + 56 - 3v + \kappa_0, \\ 0 &\leq \kappa_0 \leq 3 \max(v - 2, 0) \leq 90, \\ \dim H^1(\mathcal{I}_1) &= 8k + 21 + \kappa_1, \\ \dim H^2(\mathcal{I}_1) &= 8k + 55 + v - 4 \max(v - 1, 0) + \kappa_1, \\ 0 &\leq \kappa_1 \leq 3 \max(v - 3, 0) \leq 87. \end{aligned} \quad (\text{RKR2})$$

Here is the exact connection to the original coefficient cycles. Write $\mathcal{Z}_e$ for their full degree-one cycle space, $\mathcal{B}$ for the boundary space, and $\mathcal{S}_e$ for the inverse image of $\text{im } \partial_e$ under $\mathcal{Z}_e \rightarrow H^1(\mathcal{I}_e) = \mathcal{Z}_e/\mathcal{B}$. The induced map

$$\mathcal{Z}_e/\mathcal{S}_e \longrightarrow H^1(\mathcal{Q}_e) \xrightarrow{\sim} \ker U_e$$

first includes the same cycle into $\mathcal{Q}_e$. For its second arrow, lift this quotient cycle to $f \in \mathcal{H}_e^1$, lift $df \in \mathcal{L}_e^2$ to $a \in \mathcal{A}^2$, and take $da \in \mathcal{Z}_e^3$. The original LCC5 coordinate is $\text{diag}(S_{j,e}, 4)da$ in the ordered blocks (123), (124), (134), (234). It belongs to $\ker U_e$ because $d(da) = 0$ with the unchanged exterior signs. For the lift-independence, if the first lift changes to $f' = f + \mathcal{V}_{1,e}b$ with $b \in \mathcal{A}^1$, the compatible second lift is $a' = a + db$, since the evaluation is a cochain map. Thus $da' = da + d^2b = da$. The sign here is positive; it corrects the word “subtract” in the earlier first-lift sentence of LCC3 without changing its displayed double-lift map. Another second lift differs by $\xi \in \mathcal{Z}_e^2 = 0$, by LCC4. A quotient boundary changes the first lift by $dg + \mathcal{V}_{1,e}b$; again $a' = a + db$ gives the same da because $d^2g = 0$. This proves independence of all choices. The inverse construction and injectivity proof of LCC3 then prove that the second arrow is an isomorphism. For the first arrow, LCC13 identifies the kernel on $H^1(\mathcal{I}_e)$ as exactly $\text{im } \partial_e$. It is onto because $H^1(\mathcal{R}_e) = 0$ by LCC12: a quotient cycle can be adjusted by a differential of a lifted degree-zero element so that its quotient in $\mathcal{R}_e^1$ vanishes. Thus the composite is the original positive double-lift isomorphism, with precisely the displayed kernel.

For the numerical refinement, BTK7 identifies each full original tail with $p(\partial_z)\phi_j$ for $p \in \mathbb{C}[S]_{<\ell}$, $\ell = \max(v - e, 0)$. The conversion to LCC5 is the invertible derivative matrix BTK9; the entire native frame changes have nonzero diagonals BTK10. No phase, factorial or native metric changes this kernel. The kernel is zero for $\ell = 0, 1, 2$ by the complete irreducibility and fourth-power argument BTK12–18. At each succeeding ℓ , its new derivative-value map is $E \mapsto (E_{\hat{i}}^{(Q-\ell)}(0))_{i=1}^4$. Its kernel is the preceding tail kernel and its image lies in $\sum_i (-1)^{i-1} B_i(0)y_i = 0$, by differentiating the full signed convolution exactly $Q - \ell$ times. This row is nonzero because $(L_i(1, 1))_i = H(1, 1, 1, 1)^T \neq 0$ and $B_i(0) = L_i(1, 1)^4$. Each step adds at most three dimensions. Summing from $\ell = 3$ proves the bound in RKR2. Substitution into the exact LCC13–14 formulas proves all displayed dimensions with their original signs. In particular $v - e \leq 2$ gives $\ker U_e = 0$ and makes $\partial_e : H^0(\mathcal{R}_e) \rightarrow H^1(\mathcal{I}_e)$ an isomorphism. This statement uses the original five-orbit domain; none of the additional domains for pole-coordinate maps is imposed.

158.4 The complete native cohomology Grams

We now specify the metrics of these exact groups. This is needed because none of the coefficient connecting isomorphisms is assigned an isometry. For each degree p take a fixed independent column selection R_p for the image of Θ_{n_p, D_e} in CNC10 and include the original exterior multiplicities. At degrees zero and one take $R_p = I$, since those full coefficient frames are injective at both low cutoffs. In the original direct sum of Gamma frames put

$$E_p = (I - P_{\mathcal{L}_e^p})T_{n_p, D_e}R_p, \quad G_p = E_p^*E_p, \quad E_p^\dagger = G_p^{-1}E_p^*. \quad (\text{LCC15})$$

Here direct sums are understood on all displayed scalar matrices, and $P_{\mathcal{L}}$ is the orthogonal projector computed from the entire lower evaluation span. The columns of E_p are independent by selection, so G_p is positive. If the image is zero all these matrices are empty. They identify the original image quotient isometrically with its attained minimum representatives in the original ambient source.

Let $d_{\mathcal{H}}^p$ be the native finite translation differential in LCC2. The exact image differential in the coordinates LCC15 is

$$D_p = E_{p+1}^\dagger(I - P_{\mathcal{L}_e^{p+1}})d_{\mathcal{H}}^pE_p. \quad (\text{LCC16})$$

Indeed the cochain square puts the projected column on the right in $\text{im } E_{p+1}$. Its unique coordinates are given by that left inverse. A changed lower representative is killed after projection because lower images form a subcomplex. Thus $D_{p+1}D_p = 0$ with all the original constants and shifts.

Take an independent boundary frame B_p from $\text{im } D_{p-1}$ and a complement Z_p such that $[B_p, Z_p]$ spans $\ker D_p$. The exact positive cohomology Gram is

$$G_{H^p,e} = Z_p^* G_p Z_p - Z_p^* G_p B_p (B_p^* G_p B_p)^{-1} B_p^* G_p Z_p. \quad (\text{LCC17})$$

It is the attained minimum of $(Z_p x + B_p \eta)^* G_p (Z_p x + B_p \eta)$ over all boundary coefficients, by completion of the square. Independence of the combined frame proves positivity. All null terms of CNC13 were quotiented before this operation, and no inverse is taken on them. The formula also gives each metric in the exact sequence LCC13: construct the connecting columns by its specified lifts, restrict LCC17 to their image, then use its actual Schur complement for the quotient. This retains the complete connecting Gram, including any small eigenvalues.

In particular degree one uses precisely four copies of the full original degree- $(k+4)$ conductor metric $\mathcal{D}_{k+4,D_e}$. The coefficient coordinate maps at degrees zero and one are injective, so the cohomology group has the literal original presentation

$$H^1(\mathcal{I}_e) = \mathcal{Z}_e / \mathcal{B}, \quad \mathcal{Z}_e = \ker D_1 \subset \mathcal{C}_{v,k+4}^4, \quad \mathcal{B} = d^0 \mathcal{C}_{v,k}, \quad \text{Gram}(\mathcal{C}_{v,k+4}^4) = \text{diag}(\mathcal{D}_{k+4,D_e}). \quad (\text{LCC18})$$

The cycles $\mathcal{Z}_e$ are computed after the exact degree-two cutoff projection in LCC16; they are not presumed coefficient cycles. Formula LCC17 is their original restriction followed by quotient by all original boundaries.

Current connecting restriction inside the full cohomology metric. Keep the native cohomology Gram of LCC17 and the full exact sequence LCC13. The connecting columns constructed by its positive lift are precisely CIT7's F_e ; their restriction Gram is CIT11's attained minimum over the entire original boundary matrix B_{cof} . On the source, PDI2 identifies the same primitive values with $\Theta_e^* \overline{H_e^{-1}} \Theta_e$, where here $H_e = C_e^* C_e$ denotes the PDS relation Gram, not the full cohomology Gram. PSG, SGT and PFG evaluate this precise source through the complete receiving identities PSR19–22.

In the original fixed Taylor values the exact maps are PSR23–24: $\Theta_e = (R_e^{\text{pol}})^\top \text{diag}(r!) P_e^{\text{Taylor}}$ and the unique analytic coefficient F_e of the Taylor cochain square. Both are independent of the two Gamma presentations. Hence, for the two original source-order pairs, their defining low-cutoff metrics obey

$$\begin{aligned} \log \frac{\det H_{\text{src},e}^{(k)}}{\det H_{\text{src},e}^{(1)}} &= o(kq_k), \\ \log \frac{\det H_{\text{conn},e}^{(k,k+4)}}{\det H_{\text{conn},e}^{(1,1)}} &= -\mathcal{X}_e^{(k,k+4)} + \mathcal{X}_e^{(1,1)} + o(kq_k), \\ \mathcal{X}_e^{(s_0,s_1)} &= \chi_{X_k,D_e}^{(s_0,s_1)} - \chi_{Y_e,D_e}^{(s_0,s_1)}. \end{aligned} \quad (\text{LCC36})$$

The finite source interval is PSR26, with its full factorial and mass center; addition of the difference of CIT17's two asymmetric error intervals gives the finite target interval. This proves the comparison on identical ordered values and retains both gauge terms. The exact quotient $H^1(\mathcal{I}_e)/\mathcal{C}_e \simeq \ker U_e$ has its own attained metric and has not been removed. At later target cutoffs, CIT21–23 transports the connecting flag to LCC20–24 and bounds the difference of their contraction returns by $2\kappa_e \log A_{k+4,s_1}^{\text{end}} = o(kq_k)$, with $\kappa_e \leq 3 \max(v-e, 0)$. The residual boundary in its common denominator is the exact RQB3 quotient; RQB28/35 retains both its positive return and the full missing target quotient.

The full extra cohomology quotient in both native orders. Keep the original exact sequence LCC13 and the fixed later coefficient flags $\mathcal{B} \subset \mathcal{S}_e \subset \mathcal{Z}_e$. Let $Q_{\text{tail},e}$ be the attained quotient of $\mathcal{Z}_e/\mathcal{B}$ by $\mathcal{S}_e/\mathcal{B}$ in fixed lifts of $\ker U_e$. Its rank is $\kappa_e = \dim \ker U_e \leq 3 \max(v-e, 0)$. If C_{tail} changes a fixed full cohomology frame into its connecting columns followed by these fixed

tail lifts, then

$$\begin{aligned} |\det C_{\text{tail}}|^2 \det H_{\text{coh},e} &= \det H_{\text{conn},e} \det Q_{\text{tail},e}, \\ -\kappa_e \log \mathfrak{b}_{j,D} &\leq \Delta \log \det Q_{\text{tail},e}(D) \leq -\kappa_e \log \mathfrak{a}_j, \\ -\kappa_e \Omega_j^{\text{hi}} &\leq \Delta \mathcal{R} \log \det Q_{\text{tail},e} \leq \kappa_e \Omega_j^{\text{lo}}. \end{aligned} \quad (\text{LCC37})$$

Here Δ compares $(1,1)$ with (k,j) , and D uses the original four target cutoffs; all constants are the full NSR2 ones. The determinant identity follows by eliminating the connecting restriction in the adapted full Gram. Every remaining fibre is the full $\mathcal{S}_e$ fibre, so the NSO dual inequality passes to this quotient by minimization. The fixed frame factor cancels only in the same-frame order ratio or the four-return. This retains the extra low cohomology rather than omitting it.

Set $W = \mathbb{J} + \mathcal{B}$ and use the exact spaces $K_e = \mathcal{S}_e \cap W$, $T_e = \mathcal{S}_e + W$, $K_e^{\text{full}} = \mathcal{Z}_e \cap W$ and $T_e^{\text{full}} = \mathcal{Z}_e + W$. Their original maps give

$$\begin{aligned} 0 \longrightarrow K_e^{\text{full}}/K_e &\longrightarrow \mathcal{Z}_e/\mathcal{S}_e \longrightarrow T_e^{\text{full}}/T_e \longrightarrow 0, \\ d_K + d_T &= \kappa_e, \quad \dim(\mathcal{Z}_e/K_e^{\text{full}}) = c_e + d_T, \\ |\Delta \mathcal{R} \log \omega_e| &\leq (c_e + d_T)W_j, \quad |\Delta \mathcal{R} \log \omega_e - \Delta \mathcal{R} \log \omega_{\text{conn},e}| \leq \kappa_e W_j. \end{aligned} \quad (\text{LCC38})$$

The first arrow is inclusion and the second sends $[z]$ to $[z]$ modulo T_e . Its kernel consists exactly of $z = s + w$ with $s \in \mathcal{S}_e$, $w \in W \cap \mathcal{Z}_e$; subtracting s proves exactness. The dimensions follow. For the first bound compare the two actual quotient minima $\mathcal{Z}_e/K_e^{\text{full}}$ and T_e^{full}/W in identical values. Their common scalar cancels, giving the full rank $c_e + d_T$ width. For the last bound, the adapted determinant identity has two positive quotient factors of ranks d_K, d_T and one denominator of rank κ_e . Their common scalar cancels because $d_K + d_T = \kappa_e$; NSR15 gives precisely the displayed width, including every empty-rank case.

The absolute fixed-rank receiving also remains explicit. Put $\mathcal{L}_e^{\text{full}} = \mathcal{R} \log(\omega_e \omega_{\text{inv}})$. With CPQ's actual $V_s, V_t, V_{\text{rem}}, V_A, V_b, V_I$,

$$\begin{aligned} \mathcal{L}_e^{\text{full}} &= 4\widehat{F}_j^{(s_1)} - V_b - V_{\text{rem}} - V_s - V_A + \eta_e, \\ |\eta_e| &\leq 2\kappa_e \log A_{j,s_1}^{\text{end}}, \\ V_I - V_s - V_A - 2\kappa_e \log A_{j,s_1}^{\text{end}} &\leq \mathcal{L}_e^{\text{full}} \leq 4U_{j,s_1}^{\text{native}} - V_b - V_s - V_A + 2\kappa_e \log A_{j,s_1}^{\text{end}}. \end{aligned} \quad (\text{LCC39})$$

The exact first line substitutes CPQ11 into CIT23; the second is the complete old native flag comparison. CPQ13 then gives the directed interval, using the RQB35 cap only on its original domain. The full remaining target quotient and both source quotients are still present in the equality. None has been assigned an absolute asymptotic value.

The common-angle control with the complete low-cohomology tail. The full-cycle and connecting losses are related by the original CIT23 determinant factorization used in LCC39: $\mathcal{L}_e^{\text{full}} = \mathcal{L}_e + \eta_e$. Keep its complete tail rank κ_e , its original endpoint comparison A_{j,s_1}^{end} , and the attained quotient $Q_{\text{tail},e}$ from LCC37. With the actual connecting interval L_-, L_+ of SFC9–10,

$$L_- - 2\kappa_e \log A_{j,s_1}^{\text{end}} \leq \mathcal{L}_e^{\text{full}} \leq L_+ + 2\kappa_e \log A_{j,s_1}^{\text{end}}. \quad (\text{LCC40})$$

The full map $\mathcal{Z}_e/K_e^{\text{full}} \rightarrow T_e^{\text{full}}/W$ sends $[z]$ to $[z]$; its kernel and surjectivity follow from $K_e^{\text{full}} = \mathcal{Z}_e \cap W$ and $T_e^{\text{full}} = \mathcal{Z}_e + W$. Its relationship to the connecting map is the exact sequence LCC38, retaining both d_K and d_T . Thus the extra determinant is precisely the tail factor already bounded in LCC39, rather than a new identification of the full and connecting spaces. Adding that signed tail bound to CIT35 proves LCC40. Intersect it with LCC39. For a comparison of the two prescribed source orders the invariant loss must also be retained:

$$\begin{aligned} \Delta \mathcal{L}_e^{\text{full}} &= \Delta \mathcal{R} \log \omega_e + \Delta \mathcal{R} \log \omega_{\text{inv}}, \\ |\Delta \mathcal{L}_e^{\text{full}}| &\leq (c_e + d_T + r_I)W_j, \quad r_I = j_{0,k} - \epsilon. \end{aligned}$$

The first identity is its definition, and LCC38 plus the original rank- r_I invariant contraction comparison NSR22 prove the second. These complete translations may therefore be intersected with the respective LCC40 intervals. When $\kappa_e = 0$ the exact sequence has zero outer spaces, the extra quotient determinant is one and the interval is exactly the connecting enclosure; all original metrics and coefficient maps remain unchanged.

158.5 The exact morphism to the original kernel and residual spaces

Use the fourfold original DNE9 kernel-evaluation map and its entire invariant residual kernel:

$$\mathbb{B} = \text{diag}(B_{k+4}, B_{k+4}, B_{k+4}, B_{k+4}), \quad \mathbb{J} = \ker \mathbb{B} = \text{im } \text{diag}(J_{k+4}), \quad Q_{\mathbb{B}} = \text{diag}(\widehat{H}_{K,k+4,D_e}^{-1}). \quad (\text{LCC19})$$

The last form is the exact native quotient of the full conductor product by $\mathbb{J}$, including both original source-order families and the actual fixed kernel columns. There is a canonical onto map and its full kernel:

$$\begin{aligned} \Psi_e : \mathcal{Z}_e / \mathcal{B} &\longrightarrow \mathcal{W}_e := \mathbb{B}\mathcal{Z}_e / \mathbb{B}\mathcal{B}, & [z] &\longmapsto [\mathbb{B}z], \\ \ker \Psi_e &\simeq (\mathcal{Z}_e \cap \mathbb{J}) / (\mathcal{B} \cap \mathbb{J}). \end{aligned} \quad (\text{LCC20})$$

The map is well-defined since a boundary maps to $\mathbb{B}\mathcal{B}$, and it is onto by the target definition. If $\mathbb{B}z = \mathbb{B}b$ for $b \in \mathcal{B}$, then $z - b \in \mathbb{J}$ and also in $\mathcal{Z}_e$, because every boundary is a cycle. Thus every kernel class has a representative in $\mathcal{Z}_e \cap \mathbb{J}$, and two such representatives differ by precisely $\mathcal{B} \cap \mathbb{J}$. This proves the last isomorphism, with every nonreduced intersection retained rather than assigned a transverse rank.

The target has two specified positive native metrics. In fixed coordinates on $H^1(\mathcal{I}_e)$ let $H_e = G_{H^1,e}$ and write the onto map LCC20 as a full-row matrix Ψ_e . Its source-quotient metric is

$$Q_{\Psi,e} = (\Psi_e H_e^{-1} \Psi_e^*)^{-1}. \quad (\text{LCC21})$$

Choose an independent frame U_b of $\mathbb{B}\mathcal{B}$ and fixed lifts U_w of a basis of $\mathcal{W}_e$ in $\mathbb{B}\mathcal{Z}_e$. The other metric, obtained from the original kernel target, is

$$H_{\mathcal{W},e} = U_w^* Q_{\mathbb{B}} U_w - U_w^* Q_{\mathbb{B}} U_b (U_b^* Q_{\mathbb{B}} U_b)^{-1} U_b^* Q_{\mathbb{B}} U_w. \quad (\text{LCC22})$$

All inverses exist on the displayed independent frames. Empty frames have determinant one. The expression is precisely the attained minimum over the full allowed target boundary image, and so retains every original boundary coefficient and the complete lower/invariant target quotient.

These two exact metrics satisfy

$$0 \prec H_{\mathcal{W},e} \preceq Q_{\Psi,e}, \quad \Omega_{\Psi,e} = Q_{\Psi,e}^{-1/2} H_{\mathcal{W},e} Q_{\Psi,e}^{-1/2}, \quad 0 \prec \Omega_{\Psi,e} \preceq I. \quad (\text{LCC23})$$

Indeed for every cycle z and every original boundary b , the native quotient norm of $\mathbb{B}(z + b)$ is at most the original conductor norm of $z + b$, by the defining minimum over $\mathbb{J}$ in LCC19. Minimizing first over b proves that LCC20 is a contraction from its cohomology metric to the target metric LCC22. Then minimize over all cohomology lifts of a target value. This gives the form inequality in LCC23. Positivity follows from the independent target classes in both metrics. This proves the precise metric morphism and does not identify their two attained minima.

Current refinement of the LCC23 contraction. The exact LFC4–6 identification sends the source quotient to S_e/K_e and the target quotient to T_e/W , with $K_e = S_e \cap W$ and $T_e = S_e + W$. Both native minima are proved by equality of the respective full affine fibres. Extending these fixed low flags through the original four-index window gives

$$|\mathcal{R}(\log \det \Omega_{\Psi,0}) - \mathcal{R}(\log \det \Omega_{\Psi,1})| \leq 2c_{\text{low}} \log A_{j,s}^{\text{end}} \leq 12 \log A_{j,s}^{\text{end}}. \quad (\text{LCC32})$$

This is LFC12–16 with every fixed coordinate factor canceled only by the exact four signs. Its common absolute long-return loss remains in the full LCC23 determinant.

Finally choose a fixed kernel frame J_Ψ and a right inverse R_Ψ for the matrix Ψ_e , and put $S_\Psi = [J_\Psi, R_\Psi]$. The full determinant factorization is

$$\log \det H_e = \log \det (J_\Psi^* H_e J_\Psi) + \log \det H_{\mathcal{W},e} - \log \det \Omega_{\Psi,e} - 2 \log |\det S_\Psi|. \quad (\text{LCC24})$$

The actual square matrix S_Ψ is invertible: its kernel block spans $\ker \Psi_e$ and $\Psi_e R_\Psi = I$. Its full Gram factors into the kernel restriction and the source quotient LCC21 by the ordinary Schur determinant. Substitute the exact scalar product $\det H_{\mathcal{W},e} = (\det Q_{\Psi,e})(\det \Omega_{\Psi,e})$ to obtain LCC24. The kernel restriction in LCC24 is exactly the native cohomology metric on the residual overlap in LCC20. The other term is the original kernel-target quotient LCC22. The contraction loss has every singular value retained; no smallness is assigned to it. Thus the calculated low mixed cohomology has an explicit original residual/kernel determinant interface, with all finite connecting and quotient metrics present.

158.6 The exact next-jet inclusion and its bounded missing rank

The remaining kernel has a sharper proved bound. In the ordered triple-wedge components write $C(z) = (-B_4(z), B_3(z), -B_2(z), B_1(z))$. At least one $C_i(0)$ is nonzero, because $\sum_i D_i(0)B_i(0) = 1$. A tuple in $\ker U_e$ has four original degree- $(k+4)$ exponential components with zero first L_e jets and $\sum_i C_i(z)E_i(z) = 0$. If $a_e > 0$, write $E_i(z) = z^{L_e} F_i(z)$. Its coefficient vector is uniquely determined by the a_e Taylor coefficients of F_i of orders $0, \dots, a_e - 1$: these are the remaining original moments through order $Q - 1$, with their factorials, and the full Q -root Vandermonde is invertible. For an index with $C_i(0) \neq 0$, the equality

$$F_i = -C_i^{-1} \sum_{j \neq i} C_j F_j \pmod{z^{a_e}} \quad (\text{LCC25})$$

determines those entire a_e coefficients from the other three components. The inverse in this formula is the literal power-series inverse of a unit in $\mathbb{C}[z]/(z^{a_e})$. Its coefficients are determined recursively by its nonzero actual constant coefficient; no norm estimate for it is asserted. The projection to the other three tail coefficient vectors is injective on the actual kernel. Hence

$$\kappa_e \leq 3a_e, \quad 0 \leq \kappa_0 - \kappa_1 \leq 3. \quad (\text{LCC26})$$

For the second claim, the kernel at the upper cutoff is the subspace of the lower kernel whose four next moments vanish. If $v \geq 1$, the coefficient of z^{L_0} in $\sum C_i E_i = 0$ gives the nonzero relation $\sum_i C_i(0)E_i^{(L_0)}(0) = 0$. Thus the four next-moment rows have rank at most three on the lower kernel. Their common kernel is exactly the upper one, proving the dimension bound. If $v = 0$, both original tail kernels are zero, proving the same statement without introducing a nonexistent tail coefficient.

Truncation of native functionals from D_1 to D_0 takes the complete lower evaluation image to its original shorter image and commutes with every original multiplier. At degrees zero and one the conductor coefficient maps are injective at both cutoffs, so this map is the identity on their original algebraic coefficient spaces. Consequently the two degree-one cycle spaces in LCC18 obey

$$\begin{aligned} \mathcal{B} &\subset \mathcal{Z}_1 \subset \mathcal{Z}_0, \\ \iota_{\text{low}} : H^1(\mathcal{I}_1) &\hookrightarrow H^1(\mathcal{I}_0), \quad [z] \mapsto [z], \\ c_{\text{low}} &:= \dim(\mathcal{Z}_0/\mathcal{Z}_1) = 3 + \kappa_0 - \kappa_1 \in [3, 6]. \end{aligned} \quad (\text{LCC27})$$

Indeed an upper cycle becomes a lower cycle under the cochain truncation. The boundary space is the identical algebraic image of d^0 at both cutoffs. This proves the inclusion and its kernel is zero. Subtraction of the two exact LCC14 dimensions gives the stated codimension, and LCC26 proves both bounds. The quotient of cohomology by this image is exactly $\mathcal{Z}_0/\mathcal{Z}_1$: both presentations mod out the same $\mathcal{B}$ already contained in $\mathcal{Z}_1$.

The inclusion is a contraction for the two specified native cohomology metrics. The original finite dual form satisfies $\mathcal{D}_{k+4,D_0} \preceq \mathcal{D}_{k+4,D_1}$: every complete lower-evaluation extension has its old squared norm plus one nonnegative row when the new row is added; taking the minimum over the same entire lower coefficient vector preserves the inequality. Restrict it to $\mathcal{Z}_1$ and minimize over the identical full boundary space. If A_{low} is the actual matrix of LCC27 in the two cohomology frames and H_e their LCC17 Grams, the result is

$$0 \prec G_{\text{com}} := A_{\text{low}}^* H_0 A_{\text{low}} \preceq H_1, \quad \Omega_{\text{ret}} = H_1^{-1/2} G_{\text{com}} H_1^{-1/2}, \quad 0 \prec \Omega_{\text{ret}} \preceq I. \quad (\text{LCC28})$$

This proof uses the original source order unchanged within each endpoint pair. It does not compare two independently selected coefficient norms.

Current refinement of the LCC28 common metric. For the same upper cohomology coordinates and the same degree- j source order $s = 1$ or j , the actual next lower-minimized row gives

$$0 \leq -\log \det \Omega_{\text{ret}} \leq 4 \log \left(1 + \frac{\rho D_0^{-1} \rho^*}{1 + \zeta} \right) = \beta_{q_j-v} \leq 4 \log A_{j,s}^{\text{end}}. \quad (\text{LCC33})$$

LFC17–21 proves the fixed-flag trace bound and the complete rank-four row update. Its ρ, ζ, D_0 are exactly the original full lower-row expressions LFC20, preserving the mass, phases and conductor division.

Choose lifts R_{miss} completing A_{low} in the lower cohomology coordinates, and put $W_{\text{low}} = [A_{\text{low}}, R_{\text{miss}}]$. The exact missing Gram is

$$G_{\text{miss}} = R_{\text{miss}}^* H_0 R_{\text{miss}} - R_{\text{miss}}^* H_0 A_{\text{low}} G_{\text{com}}^{-1} A_{\text{low}}^* H_0 R_{\text{miss}}, \quad (\text{LCC29})$$

$$\log \det H_0 - \log \det H_1 = \log \det G_{\text{miss}} + \log \det \Omega_{\text{ret}} - 2 \log |\det W_{\text{low}}|.$$

Completion of the square identifies this Gram with the original restriction to $\mathcal{Z}_0$ followed by quotient by $\mathcal{Z}_1$. In detail, the cohomology minimum first permits all vectors in $\mathcal{B}$; its subsequent quotient by $\mathcal{Z}_1/\mathcal{B}$ permits exactly all vectors in $\mathcal{Z}_1$, with the same original norm. This proves the claimed isometry of quotient presentations. The determinant of the full W_{low} Gram then factors into G_{com} and G_{miss} . Substituting LCC28 gives the second line with its original frame factor. The return contraction determinant is retained on the whole common space.

Current finite interval for the LCC29 factorization. Substitution of LCC33 in the exact displayed factorization gives the full directed interval

$$-\beta_{q_j-v} \leq \log \det H_0 - \log \det H_1 - \log \det G_{\text{miss}} + 2 \log |\det W_{\text{low}}| \leq 0. \quad (\text{LCC34})$$

This is the exact LFC22 insertion. It preserves the actual missing quotient Gram and every fixed coordinate factor.

158.7 The missing low directions have a proved subleading return

There is now a quantitative application of the complete accepted FRE1–7 estimate. Set $j = k + 4$ and keep the two fixed algebraic cycle spaces $\mathcal{Z}_1 \subset \mathcal{Z}_0$ just calculated. As the source index N varies over the original degree- j window $q_j - 1 \leq N \leq 2q_j$, give their fixed quotient $\mathcal{Z}_0/\mathcal{Z}_1$ the restriction and attained quotient of the original fourfold metric $\text{diag}(\mathcal{D}_{j,N-v}^{(s)})$. Call this form $G_{\text{miss},N}^{(s)}$. At $N = q_j - 1$ it is exactly LCC29, by its proved quotient isometry. The cycle spaces are fixed in these later comparisons; they are not renamed as the cohomology of a different high-cutoff complex.

Retain the literal FRE constants

$$A_{j,s}^{\text{end}} = \max\{1, q_j \mathcal{B}_{2q_j-v,s,j,A}^2\} (M_{j,s}^{\text{rec}})^2, \quad s = 1 \text{ or } j, \quad (\text{LCC30})$$

where $\mathcal{B}$ is the complete DNE13 constant and M^{rec} is the original LJR16 reconstruction norm, including their actual mass, Vandermonde, conductor scalar and fixed disk. FRE2–3 proves the full form comparison $\mathcal{D}_{j,q_j-v-1} \preceq \mathcal{D}_{j,N-v} \preceq A_{j,s}^{\text{end}} \mathcal{D}_{j,q_j-v-1}$. Its direct sum has the same scalar bound. Substitute cycle vectors and take minima over the fixed space $\mathcal{X}_1$; hence it holds on $G_{\text{miss},N}$ as well. Every generalized eigenvalue relative to the low form is in $[1, A_{j,s}^{\text{end}}]$. The log determinant increment is therefore between zero and $c_{\text{low}} \log A_{j,s}^{\text{end}}$.

It is also increasing with N , by the same original minimum-extension argument. Subtracting with the original four signs now gives the explicit bound

$$\begin{aligned} \mathcal{V}_{\text{miss}}^{(s)} &= -\log \det G_{\text{miss},q_j-1}^{(s)} - \log \det G_{\text{miss},q_j}^{(s)} \\ &\quad + \log \det G_{\text{miss},2q_j-1}^{(s)} + \log \det G_{\text{miss},2q_j}^{(s)}, \end{aligned} \quad (\text{LCC31})$$

$$0 \leq \mathcal{V}_{\text{miss}}^{(s)} \leq 2c_{\text{low}} \log A_{j,s}^{\text{end}} \leq 12 \log A_{j,s}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k).$$

For the lower bound the expression is the sum of the two high increments from the low endpoint minus the next-row increment, which is no greater than either high increment. For the upper bound drop that nonnegative subtracted increment and use the displayed scalar bound twice. The final order follows from the already proved full FRE4 estimate $\log A_{j,s}^{\text{end}} \leq 2q_j \log j + O_{\text{actual}}(q_j)$ and $j = k + 4$. This does not assume row-energy concentration or a bound on a chosen coefficient frame inverse.

Equivalently, keep the fixed two flags $\mathcal{X}_0/\mathcal{B}$ and $\mathcal{X}_1/\mathcal{B}$ at these same four source indices. Their determinant returns differ exactly by the four-return of $G_{\text{miss},N}$ in LCC31, since the common full-frame factor cancels after the restriction/quotient determinant factorization at each index. This propagates the proved missing-rank bound through the original native calculation. It does not replace the metric on their common image, the original residual/kernel loss in LCC24, or the full return contraction in LCC29.

Current fixed-flag return and its shared loss. The two fixed low flags have the same common original boundary denominator. Their exact missing quotient and their contraction-return difference therefore satisfy, with all four original endpoints retained,

$$0 \leq \mathcal{V}_{\text{miss}}^{(s)} \leq 2c_{\text{low}} \log A_{j,s}^{\text{end}}, \quad |\mathcal{R}(\log \omega_0) - \mathcal{R}(\log \omega_1)| \leq 2c_{\text{low}} \log A_{j,s}^{\text{end}} = o(kq_k). \quad (\text{LCC35})$$

The first is LCC31; the second is proved by the exact three-quotient sequence LFC9–16. NCR13–18 carries both through the unchanged original residual/kernel map. Neither inequality deletes the common long-return loss.

159 The original low mixed flags and their contraction returns

This section continues the complete CNC and LCC constructions on the original nonreduced conductor. It calculates two terms that were retained in LCC23 and LCC28: the difference of the two fixed low-flag contraction returns, and the common-space contraction across the two adjacent original low cutoffs. The complete common long-return loss is kept in the formulas.

159.1 Original objects and the literal four flags

Keep the fixed actual simple quartet, original five-orbit period and logarithm branch. Set

$$k = 4l + 1 \geq 9, \quad j = k + 4, \quad q_j = (j + 1)^2, \quad v = \text{ord}_0 E_A, \quad 0 \leq v \leq 80, \quad s = 1 \text{ or } j. \quad (\text{LFC1})$$

The sharper bound on v in RPK remains available; no occurrence of its actual value is replaced. Every factor and multiplicity of A^{hom} , the symbol E_A , the scalar $\alpha_A = v!/\mu_v$, and the original

ordinary transposes in DNE's coefficient maps remain unchanged. The coefficient ambient space and its full native metric are

$$\begin{aligned} V &= \mathcal{C}_{v,j}^4, & G_N &= \text{diag}(\mathcal{D}_{j,N-v}^{(s)}, \mathcal{D}_{j,N-v}^{(s)}, \mathcal{D}_{j,N-v}^{(s)}, \mathcal{D}_{j,N-v}^{(s)}), & q_j - 1 &\leq N \leq 2q_j, \\ \phi_{r,s,c'_j}(S') &= \frac{p_r((S' - c'_j)/i)}{\sqrt{h_{r,s}}}, & c'_j &= j/2 - 4, & h_{r,s} &= (2\pi)^{s/2} r!(s/2)_r, \\ \mathcal{D}_{j,D}^{(s)} &= T_{j,D}^*(I - V_{j,D}V_{j,D}^+)T_{j,D}. \end{aligned} \quad (\text{LFC2})$$

Here $T_{j,D;ra} = \Lambda_{F_{j,a}}(\phi_{r,s,c'_j})$ with $F_{j,a} = N_{j,a}/E_A$ in the exact CNC8 coefficient frame, and $V_{j,D;r,(a,b)} = \phi_{r,s,c'_j}(\beta_{j-8,a,b})$ includes every original lower root. Thus LFC2 is precisely the attained minimum over the entire original lower-evaluation span, including its mass and phase. It is positive in the displayed window by CNC13.

Write $\mathcal{B} = d^0\mathcal{C}_{v,k}$ for the actual common boundary space and $\mathcal{Z}_e$ for the actual LCC18 cycle space at $D_e = q_j - v - 1 + e$, $e = 0, 1$. Both are fixed coefficient subspaces as N varies. LCC25–27 proves

$$\mathcal{B} \subset \mathcal{Z}_1 \subset \mathcal{Z}_0 \subset V, \quad c = \dim(\mathcal{Z}_0/\mathcal{Z}_1) = 3 + \kappa_0 - \kappa_1 \in [3, 6]. \quad (\text{LFC3})$$

The finite kernels κ_e are exactly those of the LCC6 matrices; no transverse-position assumption is made. Let $\mathbb{B}$ and $\mathbb{J} = \ker \mathbb{B}$ be the complete fourfold DNE9 map and its kernel in LCC19. Define actual subspaces

$$S_e = \mathcal{Z}_e, \quad W = \mathcal{B} + \mathbb{J}, \quad K_e = \mathcal{B} + (\mathcal{Z}_e \cap \mathbb{J}), \quad T_e = \mathcal{Z}_e + \mathbb{J}. \quad (\text{LFC4})$$

They satisfy $K_e = S_e \cap W$ and $T_e = S_e + W$. Indeed an element of $S_e \cap W$ has the form $b + x$ with $b \in \mathcal{B} \subset S_e$ and $x \in \mathbb{J}$, so $x \in S_e \cap \mathbb{J}$; the converse is immediate. The sum identity uses $\mathcal{B} \subset S_e$.

There are canonical onto maps

$$\begin{aligned} S_e/K_e &\longrightarrow \mathbb{B}\mathcal{Z}_e/\mathbb{B}\mathcal{B} \longrightarrow T_e/W, \\ [z] &\longmapsto [\mathbb{B}z] \longmapsto [z], \end{aligned} \quad (\text{LFC5})$$

and both are isomorphisms. For the first kernel calculation, $\mathbb{B}z = \mathbb{B}b$ means $z - b \in S_e \cap \mathbb{J}$, hence $z \in K_e$. For the second, changing a lift changes it by $\mathbb{J}$, and changing the class by $\mathbb{B}\mathcal{B}$ changes it by W . Every element of T_e is $z + x$ with $x \in \mathbb{J}$, so the map is onto. Its kernel is zero by the same calculation. These are the literal domain and codomain maps for LCC20.

The source-quotient metric of LCC21 is exactly the restriction of G_N to S_e followed by quotient by K_e . In fact its two successive minimizations permit vectors in $\mathcal{B}$ and representatives of $(S_e \cap \mathbb{J})/(\mathcal{B} \cap \mathbb{J})$, whose total permitted space is precisely K_e . The target metric of LCC22 is exactly the restriction to T_e followed by quotient by W : the first original quotient permits all $\mathbb{J}$, and the boundary-image quotient then permits all $\mathcal{B}$. Their sum is W . Thus both metric identifications are proved by equality of their actual affine fibres, without declaring the two different minima equal.

Fix full frames X_{K_e}, X_W and a common lift frame F_e for LFC5. For a full frame X put $P_X = X(X^*G_NX)^{-1}X^*G_N$. The two complete forms and their loss are

$$\begin{aligned} A_{e,N} &= F_e^*G_N(I - P_{X_{K_e}})F_e = Q_{\Psi,e,N}, \\ H_{e,N} &= F_e^*G_N(I - P_{X_W})F_e = H_{\mathcal{W},e,N}, \\ \omega_e(N) &= \frac{\det H_{e,N}}{\det A_{e,N}} = \det \Omega_{\Psi,e,N}, \quad 0 < \omega_e(N) \leq 1. \end{aligned} \quad (\text{LFC6})$$

At $N = q_j - 1 + e$ these are precisely LCC21–23, in the same target coordinates. The extensions to other N retain the fixed low-cycle spaces, not the cycles of a newly substituted high-cutoff complex. Positivity follows from LFC5 and the positive ambient metric; the inequality follows because minimization over W permits minimization over its subspace K_e .

159.2 A full determinant identity for the two nested low flags

Choose fixed full frames X_{S_e}, X_{T_e} as well, and retain the square coordinate changes

$$[X_{K_e}, F_e] = X_{S_e} R_{S_e}, \quad [X_W, F_e] = X_{T_e} R_{T_e}, \quad c_e = \frac{|\det R_{T_e}|^2}{|\det R_{S_e}|^2}. \quad (\text{LFC7})$$

These matrices are invertible by LFC5. For each fixed subspace X , write $g_X(N) = \det(X_X^* G_N X_X)$ in its displayed frame. Taking the full adapted Gram and its Schur determinant gives

$$\omega_e(N) = c_e \frac{g_{T_e}(N) g_{K_e}(N)}{g_W(N) g_{S_e}(N)}. \quad (\text{LFC8})$$

For example $|\det R_{S_e}|^2 g_{S_e} = g_{K_e} \det A_{e,N}$; the target formula gives the other factor. All complex frame determinants in LFC7 are kept until taking their squared metric absolute values.

The inclusions $S_1 \subset S_0$ imply $K_1 \subset K_0$ and $T_1 \subset T_0$. There is an exact sequence

$$0 \longrightarrow K_0/K_1 \longrightarrow S_0/S_1 \xrightarrow{[z] \mapsto [z]} T_0/T_1 \longrightarrow 0. \quad (\text{LFC9})$$

The first arrow is induced by inclusion. Its kernel is zero since $K_0 \cap S_1 = K_1$. The last arrow is onto because $T_0 = S_0 + W$ and $W \subset T_1$. If $z \in S_0$ belongs to $T_1 = S_1 + W$, write $z = z_1 + w$; then $w \in S_0 \cap W = K_0$. This proves exactness at S_0/S_1 and every stated map. Consequently

$$d_K := \dim(K_0/K_1), \quad d_T := \dim(T_0/T_1), \quad d_K + d_T = c, \quad 0 \leq d_K, d_T \leq c \leq 6. \quad (\text{LFC10})$$

The metric of this exact sequence is not assumed to be isometric.

For $X = S, K, T$ choose fixed quotient lifts Y_X such that $[X_{X_1}, Y_X] = X_{X_0} U_X$. Define the literal native quotient Gram

$$Q_{X,N} = Y_X^* G_N Y_X - Y_X^* G_N X_{X_1} (X_{X_1}^* G_N X_{X_1})^{-1} X_{X_1}^* G_N Y_X. \quad (\text{LFC11})$$

This is restriction to X_0 followed by quotient by X_1 . Its full determinant identity is $g_{X_0}/g_{X_1} = \det Q_{X,N}/|\det U_X|^2$. Substitute these three identities in LFC8; the common W factor cancels through its identical frame at both endpoints. The exact result is

$$\begin{aligned} \log \omega_0(N) - \log \omega_1(N) &= \log \det Q_{T,N} + \log \det Q_{K,N} - \log \det Q_{S,N} + \log C, \\ C &= \frac{c_0}{c_1} \frac{|\det U_S|^2}{|\det U_T|^2 |\det U_K|^2} > 0. \end{aligned} \quad (\text{LFC12})$$

Empty quotient ranks have determinant one. In particular none of the possibly nonorthogonal extension data in LFC9 has been suppressed.

159.3 The entire difference of contraction returns is subleading

Use the original finite FRE1 constant at degree j ,

$$A = A_{j,s}^{\text{end}} = \max\{1, q_j \mathcal{B}_{2q_j-v,s,j,A}^2\} (M_{j,s}^{\text{rec}})^2 \geq 1. \quad (\text{LFC13})$$

Here $\mathcal{B}$ and M^{rec} are the full DNE13 and LJR16 source and reconstruction expressions, including their original Gamma masses, disk, Vandermonde and actual conductor scalar. The reuse of A for this scalar in LFC13 does not change the named biform A^{hom} or its coefficients. FRE2–4 proves the composed original metric maps

$$G_{q_j-1} \preceq G_N \preceq A G_{q_j-1}, \quad \log A \leq 2q_j \log j + O_{\text{actual}}(q_j). \quad (\text{LFC14})$$

The direct sum has the same scalar bound. Restriction and attained quotient preserve it: apply both form inequalities to every vector in the same affine fibre and then take the minimum. Thus

LFC11 satisfies the identical comparison with the same A and its actual rank. The same argument with any two consecutive indices proves monotonicity.

For any one of these fixed positive quotient metrics of rank d , put

$$\mathcal{R}(f) = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j). \quad (\text{LFC15})$$

All signs are the original native signs. Writing $I_N = \log \det Q_N - \log \det Q_{q_j-1}$, the comparison gives $0 \leq I_N \leq d \log A$ and increasing I_N . Therefore $\mathcal{R}(\log \det Q) = I_{2q_j-1} + I_{2q_j} - I_{q_j}$ is between zero and $2d \log A$. This uses every endpoint, not a shortened displacement interval. Applying this to LFC12 yields

$$|\mathcal{R}(\log \omega_0) - \mathcal{R}(\log \omega_1)| \leq 2c \log A \leq 12 \log A = O_{\text{actual}}(q_j \log j) = o(kq_k). \quad (\text{LFC16})$$

Indeed the positive part in LFC12 is bounded by $2(d_T + d_K) \log A = 2c \log A$, and its subtracted part is bounded by $2c \log A$. The constant $\log C$ cancels since the four signs sum to zero. This proves both sides of the inequality without estimating the individual constant C . The common absolute long-return loss remains the full LFC6 determinant. Its two fixed low-cycle presentations have the quantitatively bounded difference LFC16.

159.4 The actual adjacent common-space contraction is also controlled

Let $G(t) = G_a + t\Delta$, $\Delta = G_b - G_a \succeq 0$, be the explicit comparison between two original ambient metrics. For any fixed algebraic flag $Y/X \subset V/X$ choose a quotient lift frame $F \subset Y$ and put $R = (I - P_X)F$, $Q = R^*GR$. Differentiating the fixed-kernel orthogonality $X^*GR = 0$ gives

$$\dot{R} = -X(X^*GX)^{-1}X^*\Delta R, \quad \dot{Q} = R^*\Delta R, \quad (\log \det Q)' = \text{Tr}(BP), \quad (\text{LFC17})$$

where $B = G^{-1/2}\Delta G^{-1/2} \succeq 0$ and $P = G^{1/2}R(R^*GR)^{-1}R^*G^{1/2}$ is the actual Euclidean orthogonal projector onto $G^{1/2} \text{im } R$. The two other differentiated Gram terms vanish by the same orthogonality. The explicit expression for P proves $P = P^* = P^2$, hence $0 \preceq P \preceq I$. Consequently integration gives the complete flag-volume bound

$$0 \leq \log \frac{\det Q_b}{\det Q_a} \leq \log \frac{\det G_b}{\det G_a}. \quad (\text{LFC18})$$

The integration uses $(\log \det G)' = \text{Tr}(G^{-1}\Delta)$; all matrices are positive throughout the displayed comparison path. No dimension of Y/X multiplies its right-hand side.

Take the same flag $S_1/\mathcal{B}$, with $G_a = G_{q_j-1}$ and $G_b = G_{q_j}$. Its first form is the original $G_{\text{com}} = A_{\text{low}}^* H_0 A_{\text{low}}$ in LCC28 and its second form is H_1 , in the same upper cohomology coordinates. Thus

$$0 \leq -\log \det \Omega_{\text{ret}} = \log \frac{\det H_1}{\det G_{\text{com}}} \leq \log \frac{\det G_{q_j}}{\det G_{q_j-1}}. \quad (\text{LFC19})$$

We now calculate this last quantity from the original row. Put $n = q_j - v$ and take all original one-target rows T, V through $n - 1$ in LFC2. The lower columns are independent by the full original Vandermonde. Write the next original rows as t_n, v_n and define

$$\begin{aligned} C_L &= V^*V, \quad D_0 = T^*(I - VC_L^{-1}V^*)T \succ 0, \\ \zeta &= v_n C_L^{-1} v_n^* \geq 0, \quad \rho = t_n - v_n C_L^{-1} V^* T. \end{aligned} \quad (\text{LFC20})$$

For an old coefficient x , the old minimizing lower coefficient is $-C_L^{-1}V^*Tx$. Write a new candidate as that vector plus η . The whole extended squared norm is exactly

$$x^* D_0 x + \eta^* C_L \eta + |\rho x + v_n \eta|^2.$$

Completion of the square in η , or multiplication by the inverse of $C_L + v_n^* v_n$, gives the additional minimum $|\rho x|^2 / (1 + \zeta)$. Thus each original target block changes by $\rho^* \rho / (1 + \zeta)$ and the fourfold

update has rank at most four. The rectangular determinant identity, proved by multiplying the two Schur determinants of $\begin{pmatrix} I & u \\ -u^* & 1 \end{pmatrix}$, now gives the exact bound

$$\boxed{0 \leq -\log \det \Omega_{\text{ret}} \leq \beta_n := 4 \log \left(1 + \frac{\rho D_0^{-1} \rho^*}{1 + \zeta} \right) \leq 4 \log A,} \quad (\text{LFC21})$$

$$-\log \det \Omega_{\text{ret}} = O_{\text{actual}}(q_j \log j) = o(k q_k).$$

For the second inequality, $G_b \preceq A G_a$ by LFC14 and $G_b - G_a$ has rank at most four. Therefore the relative matrix $G_a^{-1/2} G_b G_a^{-1/2} = I + G_a^{-1/2} \Delta G_a^{-1/2}$ has at most four eigenvalues different from one, all in $[1, A]$. Their logarithms sum to at most $4 \log A$. This is a rank estimate for the proved actual metric update, not a rank estimate for cohomology. Every original ρ , ζ , factorial and source mass remains in the first, sharper expression.

Substitution in the full LCC29 factorization finally gives

$$-\beta_n \leq \log \det H_0 - \log \det H_1 - \log \det G_{\text{miss}} + 2 \log |\det W_{\text{low}}| \leq 0, \quad (\text{LFC22})$$

$$\beta_n \leq 4 \log A.$$

This evaluates the previously retained common-space adjacent contraction and preserves the actual missing Gram and fixed coordinate determinant. Together LFC16 and LFC22 propagate the at-most-six missing mixed directions and the next-row metric change through the original residual/kernel interface. They leave the common long invariant loss, and its original signed row formula, available for the next arithmetic calculation; neither a separate allocation coefficient nor RH is assigned by these finite bounds.

This calculation uses the full NBG1–32 and NKC1–17 native maps, the DFX1–35 degree-four multiplication and branch kernels, and the KRC1–27 original annihilator coordinates. It addresses the further target-invariant quotient in NBG30–32. Neither its kernel nor its metric loss is deleted. The original period, logarithm branch, conductor, phases and both source-order families remain fixed. The final section strengthens the error in ITL20–25 from $O_{\text{actual}}(q_k \log q_k)$ to $O_{\text{actual}}(k \log q_k)$ by retaining the identical rank-at-most- r update in the two original compatibility comparisons.

160 Original spaces, maps and two positive forms

Fix the actual simple quartet and its proved five-orbit period data. Let $k = 4l + 1 \geq 9$, $0 \leq v = \text{ord}_0 E_A \leq 80$, $q_j = (j + 1)^2$, $q'_j = (j - 7)^2$, and $D \geq q_{k+4} - v - 1$. Retain precisely

$$(s_0, s_1) = (1, 1) \text{ or } (k, k + 4), \quad c'_j = j/2 - 4, \quad dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy. \quad (\text{ITL1})$$

The full original native dual Grams are $D_{k,D}^{(s_0)}$ and $D_{k+4,D}^{(s_1)}$. They are the attained minima over all original lower evaluations in MNT12; the lower coefficients are not minimized separately across jet rows. Use the exact KRC/DFX matrices

$$\begin{aligned} J_0 &= \mathbf{J}_k, \quad J_1 = \mathbf{J}_{k+4}, \quad B_1 = \mathbf{B}_{k+4}, \quad \ker B_1 = \text{im } J_1, \\ M_i &= (\alpha_A N_{k+4} Z_{\text{exp}, k+4})^{-1} m_{i,k} (\alpha_A N_k Z_{\text{exp}, k}), \quad \alpha_A = v!/\mu_v, \quad m_{i,k}[f] = [x_i^4 f], \\ G &= J_0^* D_{k,D}^{(s_0)} J_0, \quad \mathcal{D} = \text{diag}(D_{k+4,D}^{(s_1)}, \dots, D_{k+4,D}^{(s_1)}), \\ F &= \begin{bmatrix} M_1 J_0 \\ M_2 J_0 \\ M_3 J_0 \\ M_4 J_0 \end{bmatrix}, \quad \mathcal{B} = \text{diag}(B_1, B_1, B_1, B_1), \quad W = \text{diag}(J_1, J_1, J_1, J_1), \quad X = \mathcal{B} F. \end{aligned} \quad (\text{ITL2})$$

All stars in metric calculations are complex adjoints. The ordinary transposes inside the original $\mathbf{B}_j, \mathbf{J}_j$ definitions are still those of complex-linear duality and are not changed. The scalar α_A

cancels only in the displayed conjugation for M_i . The two source masses in ITL1 are retained in $G, \mathcal{D}$.

Let $V = \mathbb{C}^{j_{0,k}}$ be the invariant source coordinates. The map F is injective: NBG21–27 proves that the four original target conductor minima give a strictly positive pullback on the full conductor quotient, hence on this invariant restriction. Equivalently DFX21 proves the full conductor tuple’s injectivity before its target-invariant quotient. Set

$$A = F^* \mathcal{D} F \succ 0, \quad E = \ker X = \{a \in V : Fa \in \text{im } W\}. \quad (\text{ITL3})$$

The equality follows from $\ker \mathcal{B} = \text{im } W$ in ITL2. It is the exact NBG30 kernel, not an assumption of injectivity after the invariant quotient.

Choose fixed full-column matrices Z_E, Z_R such that Z_E is a basis of E and $Z = [Z_E, Z_R]$ is an invertible basis of V . Write $e = \dim E$, $r = \dim V - e$. All choices are independent of D . Put

$$F_E = FZ_E, \quad F_R = FZ_R, \quad U = XZ_R. \quad (\text{ITL4})$$

Then U is an injective frame of $\text{im } X$. For any positive form P on V define its quotient form in these same lifts by

$$P/E = Z_R^* P Z_R - Z_R^* P Z_E (Z_E^* P Z_E)^{-1} Z_E^* P Z_R. \quad (\text{ITL5})$$

When $e = 0$ the subtracted block is zero. This is the original minimum over $Z_R b + E$: completing its square proves ITL5, with its unique least-norm representative. Let $Q = G/E$ and $A_E = A/E$. The matched target form is

$$H = U^* (\mathcal{B} \mathcal{D}^{-1} \mathcal{B}^*)^{-1} U. \quad (\text{ITL6})$$

The inverse in ITL6 is exactly the four copies of $H_{B,D}$ from NBG30. Thus Q, H are the NBG31 two image metrics, in the particular fixed image frame $U = XZ_R$. If its earlier frame is $U_{\mathcal{X}}$, write $U = U_{\mathcal{X}} C$ with the unique invertible coordinate matrix C . Both earlier Grams are conjugated by C ; their determinant ratio and generalized eigenvalues are unchanged by this proved common coordinate map. No new metric has been selected.

161 An explicit original coefficient determinant for the target loss

For a full-column matrix Y in the target product, let

$$P_Y = Y(Y^* \mathcal{D} Y)^{-1} Y^* \mathcal{D} \quad (\text{ITL7})$$

be its $\mathcal{D}$ -orthogonal projection. If Y is empty, $P_Y = 0$ and its Gram determinant is one. Choose any fixed coefficient complement W_R such that $\widehat{W} = [F_E, W_R]$ is a basis of $\text{im } W$. It exists since F is injective and $F_E \subset \text{im } W$. This complement keeps every other original invariant direction. Set

$$\begin{aligned} Y &= (I - P_{F_E}) F_R, & Z_T &= (I - P_{F_E}) W_R, \\ A_E &= Y^* \mathcal{D} Y, & C_T &= Z_T^* \mathcal{D} Z_T, & L &= Z_T^* \mathcal{D} Y, \\ H &= A_E - L^* C_T^{-1} L. \end{aligned} \quad (\text{ITL8})$$

The first Gram equality follows from ITL5 with $P = A$. For the last equality, all target lifts of $U b$ are precisely $F_R b + \text{im } W$. Indeed $\mathcal{B}(F_R b + w) = U b$ if and only if $w \in \ker \mathcal{B} = \text{im } W$. Minimize first over F_E and then over W_R ; completing the two squares gives ITL8. Since $F_R b \in \text{im } W$ implies $Z_R b \in E$, it forces $b = 0$. Consequently $H \succ 0$. Also $C_T \succ 0$ if its rank is nonzero: a linear combination of W_R projecting into F_E would contradict the complement. These statements justify every inverse and include their empty-rank conventions.

Define the additional target-invariant loss matrix

$$\Omega_{\text{inv}, D} = A_E^{-1/2} H A_E^{-1/2} = I - A_E^{-1/2} L^* C_T^{-1} L A_E^{-1/2}, \quad 0 \prec \Omega_{\text{inv}, D} \preceq I. \quad (\text{ITL9})$$

The square roots only express the identity map between its two specified Hilbert coordinate spaces. ITL8 is its unaltered coefficient formula. In particular

$$\begin{aligned}\omega_{\text{inv},D} &:= \det \Omega_{\text{inv},D} = \frac{\det \begin{pmatrix} C_T & L \\ L^* & A_E \end{pmatrix}}{\det C_T \det A_E} \\ &= \frac{\det([\widehat{W}, F_R]^* \mathcal{D}[\widehat{W}, F_R]) \det(F_E^* \mathcal{D} F_E)}{\det(\widehat{W}^* \mathcal{D} \widehat{W}) \det([F_E, F_R]^* \mathcal{D}[F_E, F_R])} > 0.\end{aligned}\tag{ITL10}$$

To prove the second line, eliminate the F_E block from the three indicated complete Grams. Their Schur blocks are respectively the block in the first line, C_T , and A_E . The determinant of the F_E block appears once on each required side and gives precisely ITL10. No normalization of $\widehat{W}$ is used. Changing its complement or the fixed source lifts changes the adapted block determinants by the same squared determinant factors, which cancel in the displayed ratio, or leaves the minimum forms unchanged.

For a direct finite spectral expression put $T_D = C_T^{-1/2} L A_E^{-1/2}$. The singular values $\rho_i(T_D)$ lie in $[0, 1)$ by ITL9, and

$$-\log \omega_{\text{inv},D} = \sum_i -\log(1 - \rho_i(T_D)^2).\tag{ITL11}$$

Zero singular values contribute zero. This is the precise remaining invariant-target loss, evaluated after the complete old lower gauges; it is not the earlier NBG independent-gauge loss and it is not identified with a rank fraction.

162 Return of both lower-gauge corrections, including the actual kernel

Let C_D be NBG27's correlated pullback on the invariant source V . It is obtained by restricting the full correlated native norm to the original J_0 ; no independent target invariant quotient is taken in it. NBG23 and 27 prove

$$K_-^{-2} G \preceq C_D \preceq K_+^2 G, \quad 0 \prec A \preceq C_D.\tag{ITL12}$$

Retain exactly the original constants NBG12. For the two fixed restrictions V and E define

$$\omega_{V,D} = \frac{\det A}{\det C_D}, \quad \omega_{E,D} = \frac{\det(Z_E^* A Z_E)}{\det(Z_E^* C_D Z_E)}.\tag{ITL13}$$

These are the NBG29 restriction losses $\det \Omega_{V,D}$ and $\det \Omega_{E,D}$, respectively. The zero-dimensional restriction has determinant one. For every positive P , the block determinant in the fixed frame Z gives

$$\det(P/E) = \frac{|\det Z|^2 \det P}{\det(Z_E^* P Z_E)}.\tag{ITL14}$$

It follows in particular that

$$\frac{\det(A/E)}{\det(C_D/E)} = \frac{\omega_{V,D}}{\omega_{E,D}} \leq 1.\tag{ITL15}$$

The inequality also follows by minimizing ITL12 over the same affine fibres; it is not inferred from the separate inequalities $\omega_V, \omega_E \leq 1$ alone.

Let $\mathcal{R}_{X,D} = Q^{-1/2} H Q^{-1/2}$, the exact matched relative operator NBG32, and put $e_{\text{cor},D} = \log \det(C_D/E) - \log \det(G/E)$. Combining ITL8–15 proves the exact identity

$$\boxed{\log \det \mathcal{R}_{X,D} = e_{\text{cor},D} + \log \omega_{V,D} - \log \omega_{E,D} + \log \omega_{\text{inv},D}}.\tag{ITL16}$$

Quotienting the first pair of inequalities in ITL12 proves the explicit finite estimate

$$-2r \log \max(1, K_-) \leq e_{\text{cor},D} \leq 2r \log \max(1, K_+).\tag{ITL17}$$

This follows by bounding each positive generalized eigenvalue and multiplying all r of them. At $D = O(q_{k+4})$ its absolute value is $O_{\text{actual}}(k \log q_{k+4})$, because $r \leq j_{0,k} = 8k - 32 - v + t_{0,k} = O(k)$ and NBG13 gives $\log \max(1, K_+, K_-) = O_{\text{actual}}(\log(D + 1))$. No lower bound for either other loss has been assumed.

163 Exact cancellation of the common compatibility denominator

Apply NKC6–16 to the two actual source restrictions V and E , using their full DNE functional images in the ambient finite dual quotient. Their correlated isometric images satisfy $\mathcal{A}_E \subset \mathcal{A}_V$, since $E \subset V$ and the same map $I - P_C$ is used. Here the calligraphic subscripts name subspaces, not a change in the Gram A . Let P_V, P_E be the corresponding native orthogonal projectors. Retain NKC’s single lower-gauge coefficient complement Z_c , complete evaluation map $\mathcal{V}$, compatibility matrix $\mathcal{K}$, next evaluation map $\mathcal{V}_2$, and Gram G_c . In those exact coordinates let

$$\begin{aligned}\mathcal{H}_{V,D} &= Z_c^* \mathcal{K}^* \mathcal{V}_2^* \mathcal{V}_2 \mathcal{K} Z_c + Z_c^* \mathcal{V}^* (P_T - P_C - P_V) \mathcal{V} Z_c, \\ \mathcal{H}_{E,D} &= \mathcal{H}_{V,D} + Z_c^* \mathcal{V}^* (P_V - P_E) \mathcal{V} Z_c.\end{aligned}\tag{ITL18}$$

Nested orthogonal images imply $P_V - P_E$ is an orthogonal projector of rank r . Thus the second difference is positive semidefinite of rank at most r , and both complete Grams are positive by NKC14. This retains the entire source complement even when the invariant source is a proper ambient subspace.

Set $b_{k,D} = r_{\text{comp},k} \max\{-\log c_-, \log c_+\}$, with the actual NKC13 constants and $r_{\text{comp},k} = 3k^2 - 10k - 13$. NKC15–16 gives, separately for $Y = V, E$,

$$\log \omega_{Y,D} = \log \det \mathcal{H}_{Y,D} - \log \det G_c + \eta_{Y,D}, \quad |\eta_{Y,D}| \leq b_{k,D}.\tag{ITL19}$$

Subtract the two exact equalities before estimating their errors: the common $\log \det G_c$ cancels. Therefore

$$\boxed{\log \det \mathcal{R}_{X,D} = \log \frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} + \log \omega_{\text{inv},D} + \epsilon_D,}\tag{ITL20}$$

where the finite bound, with the original rank r , is

$$|\epsilon_D| \leq 2r \max\{\log \max(1, K_-), \log \max(1, K_+)\} + 2b_{k,D}.\tag{ITL21}$$

On the original endpoint range this is $O_{\text{actual}}(q_k \log q_k) = o(kq_k)$. Both displayed determinant ratios in ITL20 are at most one. Their four-endpoint signed sums are not thereby nonpositive. If $P_V - P_E = YY^*$ is an isometric factor, a smaller exact formula for the first ratio is

$$\frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} = \det \left(I_r + Y^* \mathcal{V} Z_c \mathcal{H}_{V,D}^{-1} Z_c^* \mathcal{V}^* Y \right)^{-1}.\tag{ITL22}$$

The determinant lemma proves this directly from ITL18. All coefficients in its r by r matrix come from the specified original source-complement projector and full Gram; no bound on the inverse is asserted without calculating it.

164 Branch defect and all four original endpoint signs

The source kernel E is not replaced by zero in any preceding formula. DFX24–28 supplies the exact inclusion

$$E \subset \mathcal{E}_{v,k} = \mathcal{J}_{v,k} \cap \ker \mathbf{j}_k, \quad \mathcal{J}_{v,k}/E \twoheadrightarrow \mathcal{J}_{v,k}/\mathcal{E}_{v,k} \simeq \text{im } \mathcal{B}_{\{1,2,3,4\},k},\tag{ITL23}$$

with kernel $\mathcal{E}_{v,k}/E$. All source coordinates are transported by the original $\mathbf{T}_k J_0$ in this display. For the actual full biforms $B_a = \psi_2(Q_a)$, let $\mathcal{L} = \text{lcm}(B_1, B_2, B_3, B_4)$ have bidegree (b_z, b_w) . DFX26–27 gives

$$0 \leq e \leq \dim \mathcal{E}_{v,k} \leq (k - b_z + 1)(k - b_w + 1) - (k - 7)^2.\tag{ITL24}$$

These bounds keep the maximum factor multiplicities in the lcm and all shared components. They do not identify the actual kernel with the full shared-component space; its exact finite coefficient kernel remains DFX31–35.

At $N = q_{k+4}-1, q_{k+4}, 2q_{k+4}-1, 2q_{k+4}$ use $D = N-v$ and signs $\varepsilon_N = -1, -1, +1, +1$. Summing ITL20 gives the original further-quotient return

$$\begin{aligned} \sum_N \varepsilon_N \log \det \mathcal{R}_{X,N-v} &= \sum_N \varepsilon_N \log \frac{\det \mathcal{H}_{V,N-v}}{\det \mathcal{H}_{E,N-v}} \\ &+ \sum_N \varepsilon_N \log \omega_{\text{inv},N-v} + \sum_N \varepsilon_N \epsilon_{N-v}. \end{aligned} \quad (\text{ITL25})$$

The last term has the explicit sum of four ITL21 bounds, and is $o(kq_k)$. The other two full native determinants are ITL10 and ITL18/22. This is a finite quantitative reduction of the additional invariant quotient, not a value assigned to its remaining leading coefficient. The accepted NW1–13 displacement estimate applies to fixed original source flags when subsequently returning their cutoffs to their own degree; it does not itself estimate either long signed sum in ITL25.

When $r = 0$, both matched Grams are empty and every determinant in the matched ratio is one. Then $E = V$, the two compatibility Grams coincide, and ITL16 and ITL20 give exactly zero with zero actual error; a nonzero upper error bound is not used as a replacement for this identity. No RH consequence, native invariant isometry, or new Lean execution is claimed.

165 The identical native update sharpens the error to the image rank

The preceding separate use of NKC15 is valid, but the two native comparisons share more data than that estimate uses. Retain the same endpoint, the same lower-gauge isometry W_c of NKC6, and the same nested source projectors $P_E \preceq P_V$ of ITL18. Choose an isometric frame Y for their difference, so that $YY^* = P_V - P_E$ and Y has exactly r columns. This is a coordinate expression of the original difference subspace; it does not replace either native metric. Define

$$\mathcal{J}_D = Y^* W_c : \mathbb{C}^{r_{\text{comp},k}} \longrightarrow \mathbb{C}^r.$$

For $Z = V, E$ retain the two exact NKC10 and NKC13 matrices

$$\Xi_{Z,D} = I - W_c^* P_Z W_c, \quad \Gamma_{Z,D} = W_c^* (\mathcal{R}^* \mathcal{R} + P_T - P_C - P_Z) W_c.$$

Their differences are literally the same Gram:

$$\begin{aligned} \Xi_{E,D} &= \Xi_{V,D} + \mathcal{J}_D^* \mathcal{J}_D, & \Gamma_{E,D} &= \Gamma_{V,D} + \mathcal{J}_D^* \mathcal{J}_D, \\ 0 &\prec c_- \Xi_{V,D} \preceq \Gamma_{V,D} \preceq c_+ \Xi_{V,D}, & 0 &< c_- \leq 1 \leq c_+. \end{aligned} \quad (\text{ITL26})$$

The first two identities follow by subtracting the displayed definitions, using $P_V - P_E = YY^*$. The last two facts are NKC13–14 with their original constants $c_- = \min(1, L_-^{-2})$, $c_+ = \max(1, L_+^2)$. All inverses below are therefore positive-matrix inverses already justified by the exact native intersection theorem.

In the actual r -dimensional image-difference coordinates put

$$T_\Xi = \mathcal{J}_D \Xi_{V,D}^{-1} \mathcal{J}_D^*, \quad T_\Gamma = \mathcal{J}_D \Gamma_{V,D}^{-1} \mathcal{J}_D^*.$$

The determinant lemma applied twice to ITL26 gives

$$\begin{aligned} \log \omega_{V,D} - \log \omega_{E,D} &= -\log \det(I_r + T_\Xi), \\ \log \frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} &= -\log \det(I_r + T_\Gamma). \end{aligned} \quad (\text{ITL27})$$

For the first line use $\det \Omega_{Z,D} = \det \Xi_{Z,D}$ from NKC10. For the second use the exact common conjugation $\Gamma_{Z,D} = G_c^{-1/2} \mathcal{H}_{Z,D} G_c^{-1/2}$ from NKC16, so its denominator cancels before the determinant lemma is applied. Thus neither line estimates or omits a kernel determinant. In particular the second line is precisely ITL22, expressed through the complete original gauge isometry.

Inversion reverses the positive form comparison ITL26. Congruence by $\mathcal{J}_D^*$ consequently gives

$$\frac{1}{c_+} T_{\Xi} \preceq T_{\Gamma} \preceq \frac{1}{c_-} T_{\Xi}, \quad \frac{1}{c_+} (I_r + T_{\Xi}) \preceq I_r + T_{\Gamma} \preceq \frac{1}{c_-} (I_r + T_{\Xi}). \quad (\text{ITL28})$$

For the second pair, the first pair supplies the T_{Ξ} term, while $c_+^{-1} I_r \preceq I_r \preceq c_-^{-1} I_r$ supplies the identity term. This argument does not require T_{Ξ}, T_{Γ} to commute. Congruence by $(I_r + T_{\Xi})^{-1/2}$ puts each of the r positive generalized eigenvalues between c_+^{-1} and c_-^{-1} . Multiplication and logarithms prove the finite asymmetric bound

$$\begin{aligned} \vartheta_D &:= (\log \omega_{V,D} - \log \omega_{E,D}) - \log \frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} = \log \det(I_r + T_{\Gamma}) - \log \det(I_r + T_{\Xi}), \\ &\quad -r \log c_+ \leq \vartheta_D \leq -r \log c_-. \end{aligned} \quad (\text{ITL29})$$

The source-complement matrices and their inverses remain in ITL27; their possibly small eigenvalues are not replaced by an absolute conditioning bound. ITL29 measures only the difference between the two specified evaluations of the same rank-at-most- r update, not the size of either negative logarithm.

Combine ITL29 with the exact ITL16 and the finite ITL17. The very same ϵ_D in ITL20 now has the sharper bounds

$$\begin{aligned} \epsilon_D &= e_{\text{cor},D} + \vartheta_D, & -\alpha_D &\leq \epsilon_D \leq \beta_D, \\ \alpha_D &= 2r \log \max(1, K_-) + r \log c_+, & \beta_D &= 2r \log \max(1, K_+) - r \log c_-. \end{aligned} \quad (\text{ITL30})$$

Both constants are nonnegative and retain exactly the two original source-order families. Since $r = O(k)$ and all the displayed logarithmic constants are $O_{\text{actual}}(\log(D+1))$ by NKG13/NKC3–4, this gives $\epsilon_D = O_{\text{actual}}(k \log q_k)$ at the four original endpoints. The previous larger bound ITL21 remains true; ITL30 is the bound to use for the same error in ITL20 and ITL25.

Write $D_i = N_i - v$ for the four ordered target endpoints of ITL25. With all their signs retained,

$$-\beta_{D_1} - \beta_{D_2} - \alpha_{D_3} - \alpha_{D_4} \leq \sum_{i=1}^4 \varepsilon_i \epsilon_{D_i} \leq \alpha_{D_1} + \alpha_{D_2} + \beta_{D_3} + \beta_{D_4}. \quad (\text{ITL31})$$

Thus the remainder of that exact four-return is now $O_{\text{actual}}(k \log q_k) = o(kq_k)$. No estimate for either leading long determinant sum in ITL25 is asserted by this sharpening. The original source kernel E and its nonreduced enclosing defect in ITL23–24 are unaltered. If $r = 0$, $\mathcal{J}_D$ has zero rows, both determinants in ITL27 are one, and ITL29–31 are exactly zero.

Current arithmetic rows and simultaneous spectrum. The original invariant-target term in ITL25 now has the full arithmetic representation

$$\sum_i \varepsilon_i \log \omega_{D_i} = \sum_{n=d+1}^{d+q_j+1} w_n^{\text{row}} (\gamma_{H,n} - \gamma_{A,n}), \quad \left| \sum_i \varepsilon_i \log \omega_{D_i} \right| \leq \sum_n w_n^{\text{row}} b_n^{\text{row}}. \quad (\text{ITL32})$$

Here $d = q_j - v - 1$ and w_n^{row} is exactly the weight ϑ_n of IRR25/NFR3, namely $1, 2, \dots, 2, 1$. The quantity b_n^{row} is the full four-row budget β_n of FAB/NFR4. The original comparison errors ϑ_D and β_D in ITL29–31 retain their definitions. Every row and both positive determinants are IRR7–25. The full-spectrum paired-plane proof IF and FAB gives $b_n^{\text{row}} = 4 \log(1 + \rho_n D_0^{-1} \rho_n^* / (1 + \zeta_n))$, with no image-rank multiplier. NFR1–5 proves the literal map to ITL20/25 and retains the asymmetric ITL30–31 error. The budget and its signed row difference are distinct specified quantities; no asymptotic value is assigned.

The complete residual boundary and the retained target budget. Set $j = k + 4$, keep $(s_0, s_1) = (1, 1)$ or (k, j) , and use $D = N - v$ at each of the four original endpoints. In the original coefficient spaces put $B_{\text{cof}} = \text{stack}_i M_{i,k}$, $\mathbb{B} = \text{diag}(\mathbf{B}_j, 4)$, $\mathbb{J} = \ker \mathbb{B}$, $V_b = \mathbb{C}^{m_k}$, $m_k = 16k - 48 - v$, $J_b = \text{im } J_0$, $\Phi_b = \mathbb{B}B_{\text{cof}}$, $\mathcal{L}_b = \ker \Phi_b$, and $Y_b = J_b + \mathcal{L}_b$. The map $[x] \mapsto [B_{\text{cof}}x]$ gives exactly

$$V_b/Y_b \simeq \frac{\mathbb{J} + \text{im } B_{\text{cof}}}{\mathbb{J} + \text{im}(B_{\text{cof}}J_0)}, \quad r_b = t_{K,k} - \dim \mathcal{L}_b + \dim(J_b \cap \mathcal{L}_b). \quad (\text{ITL33})$$

Indeed an image is zero precisely when $B_{\text{cof}}(x - J_0a) \in \mathbb{J}$ for some a , which is $x \in Y_b$. The full nonreduced branch calculation RQB5–12 computes this kernel and its shared-component constraint; it retains $t_{0,k}$ in $t_{K,k}$.

In RQB13's exact quotient lifts let Q_b^E be the attained target Gram after all lower and invariant coefficients have been minimized. It is related to the earlier CIT25 lift by $Q_b^{\text{CIT}} = C_b^* Q_b^E C_b$, with its fixed invertible quotient-coordinate matrix C_b retained in RQB15. Write $\mathcal{V}_b = \mathcal{R} \log \det Q_b^E$ and let Q_o be the complete remaining quotient of the fourfold kernel-dual space by $\text{im } \Phi_b$. Sequential minima in the full frame $C_T = [U_R, U_b, U_o]$ prove

$$|\det C_T|^2 (\det H_{B,j,D})^4 = \det H_{\text{inv}}(D) \det Q_b^E(D) \det Q_o(D), \quad (\text{ITL34})$$

$$\mathcal{V}_{\text{inv}}^{\text{target}} + \mathcal{V}_b + \mathcal{V}_o = 4\widehat{F}_j^{(s_1)}, \quad 0 \leq \mathcal{V}_{\text{inv}}^{\text{target}} + \mathcal{V}_b \leq 4\widehat{F}_j^{(s_1)} \leq 4U_{j,s_1}^{\text{native}}.$$

Here H_{inv} is exactly ITL6, and $\mathcal{R}f = -f(q_j - 1 - v) - f(q_j - v) + f(2q_j - 1 - v) + f(2q_j - v)$ when f is indexed by D . The three returns are nonnegative: the original native form increases by its complete arithmetic rows, and each minimum uses the same fixed affine fibre. The fixed frame determinant cancels under precisely those four signs. RQB16–28 proves the full row and quotient formulas. The finite $U_{j,s_1}^{\text{native}}$ is RQB35's minimum retaining the full ROQ observation return and both exact FEX source corrections, not a replacement by the old uncorrected kernel metric. The source quotient Q_{inv} , the matched relative operator, and the two comparison errors of ITL29–32 retain their original definitions.

This continuation evaluates the coefficient maps behind ITL10 before either the lower-evaluation or invariant-target minima are taken. Every successive polynomial degree contributes exactly four original dual rows. Its change of invariant-target loss is the difference of two positive log determinants of size at most four. The complete long four-endpoint return is their explicitly weighted sum. This result retains the long return rather than assigning a value to it from the already established short-window bounds.

166 The original finite domain and coefficient flags

Retain the actual simple quartet, its proved five-orbit period, logarithm branch and nonreduced conductor from DNE, NBG and ITL. Put

$$k = 4l + 1 \geq 9, \quad j = k + 4, \quad q_j = (j + 1)^2, \quad q'_j = (j - 7)^2, \quad v = \text{ord}_0 E_A \leq 80, \quad D \geq q_j - v - 1. \quad (\text{IRR1})$$

The two original source-order pairs remain $(1, 1)$ and $(k, k + 4)$. Only the target member $s = 1$ or $s = j$ occurs in the loss treated here. Its complete measure, centre and polynomial norms are

$$c = c'_j = j/2 - 4, \quad M_s = (2\pi)^{s/2},$$

$$dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad h_{n,s} = M_s n! (s/2)_n, \quad (\text{IRR2})$$

$$p_{0,s} = 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n - 1 + s/2)p_{n-1,s},$$

$$\phi_{n,s,c}(S') = p_{n,s}((S' - c)/i)/\sqrt{h_{n,s}}.$$

These are precisely the original DNE7/NBG7 rows, including their mass and phase. Their use is the coordinate isometry given by that original orthogonal basis, not a replacement of its arithmetic norm.

Here are all the original algebraic maps needed for the four flags:

$$\begin{aligned} J_0 &= \mathbf{J}_k, & J_1 &= \mathbf{J}_j, & B_1 &= \mathbf{B}_j, & \ker B_1 &= \operatorname{im} J_1, \\ \mathsf{T}_a &= \alpha_A N_a Z_{\exp, a}, & \alpha_A &= v!/\mu_v, & M_i &= \mathsf{T}_j^{-1} m_{i,k} \mathsf{T}_k, & m_{i,k}[f] &= [x_i^4 f], \\ F &= \operatorname{stack}_{i=1}^4(M_i J_0), & W_0 &= \operatorname{diag}(J_1, J_1, J_1, J_1), & \mathcal{B} &= \operatorname{diag}(B_1, B_1, B_1, B_1), & X &= \mathcal{B}F. \end{aligned} \quad (\text{IRR3})$$

The bijections T_a have their original conductor source and target; their inverse notation is on those spaces. The scalar cancels only in the displayed conjugation. The ordinary transposes inside J_a and B_a still express complex-linear duality; all metric stars below are complex adjoints.

By the proved original NBG/DFX injection, F is injective before its further invariant quotient. In $V = \mathbb{C}^{j_{0,k}}$ choose a fixed basis Z_E of $E = \ker X$ and fixed complementary lifts Z_R , so $[Z_E, Z_R]$ is invertible. Write $e = \dim E$, $r = \dim V - e$, and

$$F_E = F Z_E, \quad F_R = F Z_R, \quad \widehat{W} = [F_E, F_R] \text{ a fixed basis of } \operatorname{im} W_0. \quad (\text{IRR4})$$

The coefficient complement W_R exists because $F_E \subset \ker \mathcal{B} = \operatorname{im} W_0$. Define four full-column matrices

$$Y_K = F_E, \quad Y_S = [F_E, F_R], \quad Y_W = \widehat{W}, \quad Y_T = [\widehat{W}, F_R]. \quad (\text{IRR5})$$

Their images satisfy $K \subset S$, $K \subset W$, $S + W = T$ and $S \cap W = K$: if $Fa \in \operatorname{im} W_0$ then $Xa = 0$ and $a \in E$. This also proves that Y_T is injective and $\dim(T/W) = \dim(S/K) = r$. No target kernel has been set to zero.

167 The original arithmetic rows, without an unspecified evaluator

Write the original upper nodes as

$$\beta_{j,a,b} = j(\tfrac{1}{2} - \delta - i\gamma) + 2a\delta + 2bi\gamma, \quad 0 \leq a, b \leq j. \quad (\text{IRR6})$$

For the complete fixed DNE3 matrix $Z_{\exp, j}$, put $\mathcal{N}_a(z) = \sum_{u,w} (Z_{\exp, j})_{(u,w), a} e^{\beta_{j,u,w} z}$ and $F_a(z) = \mathcal{N}_a(z)/E_A(z)$. The first v derivatives of each numerator vanish. The original conductor has derivatives $\mu_d = E_A^{(d)}(0)$, with $\mu_0 = \dots = \mu_{v-1} = 0$ and $\mu_v \neq 0$. Each μ_d is the literal finite sum of the conductor exponential coefficients times their exponents to power d ; none is fitted. Define $\nu_{a,d} = \sum_{u,w} (Z_{\exp, j})_{(u,w), a} \beta_{j,u,w}^d$. The complete analytic division is the recursion

$$f_{a,n} = \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \quad n \geq 0. \quad (\text{IRR7})$$

It computes $f_{a,n} = F_a^{(n)}(0)$. Indeed differentiate $E_A F_a = \mathcal{N}_a$ exactly $n+v$ times. Terms with $d < v$ vanish, the term $d = v$ is the denominator times $f_{a,n}$, and all remaining derivatives of F_a have lower index. This proves the recursion inductively, including $v = 0$ and its empty sum at $n = 0$. Division by μ_v retains the original constant; replacing it by one is not part of the calculation.

For complete polynomial coordinates set $b_{-1,m} = 0$, $b_{0,m} = \mathbf{1}_{m=0}$, extend all arrays by zero outside $0 \leq m \leq n$, and use

$$b_{n+1,m} = i^{-1}(b_{n,m-1} - c b_{n,m}) - n(n-1+s/2)b_{n-1,m}. \quad p_{n,s}((S' - c)/i) = \sum_{m=0}^n b_{n,m} S'^m. \quad (\text{IRR8})$$

Substitution in IRR2 proves the identity by induction. Consequently the one-target numerator row and complete lower-evaluation row are

$$\begin{aligned} t_n &= \left(\frac{\sum_{m=0}^n b_{n,m} f_{a,m}}{\sqrt{h_{n,s}}} \right)_a, \\ v_n &= \left(\frac{\sum_{m=0}^n b_{n,m} \beta_{j-8,u,w}^m}{\sqrt{h_{n,s}}} \right)_{0 \leq u,w \leq j-8}. \end{aligned} \quad (\text{IRR9})$$

The first is $\Lambda_{F_a}(\phi_n)$, not a conjugated functional. The second includes every original lower grid node. The fixed DNE minor defining $Z_{\text{exp},j}$ is used as supplied by DNE2–3. The KRC scalar α_A belongs to IRR3’s algebraic bridge; it is not an extra factor to insert into DNE’s unchanged F_a here.

Let T_D, V_D stack t_n, v_n for $0 \leq n \leq D$ and write $\mathcal{T}_D = \text{diag}(T_D, T_D, T_D, T_D)$, $\mathcal{V}_D = \text{diag}(V_D, V_D, V_D, V_D)$ initially in copy-major row order (i, n) . Reorder rows by the fixed permutation $(i, n) \mapsto (n, i)$ when appending a degree. This permutation is unitary, hence leaves every Gram unchanged. For each Y in IRR5 form the *entire* raw frame and Gram

$$Z_{Y,D} = [\mathcal{V}_D, \mathcal{T}_D Y], \quad G_{Y,D} = Z_{Y,D}^* Z_{Y,D}. \quad (\text{IRR10})$$

The new four-row block, in precisely this column order, is

$$z_{Y,n} = [\text{diag}(v_n, v_n, v_n, v_n), \text{diag}(t_n, t_n, t_n, t_n)Y]. \quad (\text{IRR11})$$

Thus $G_{Y,n} = G_{Y,n-1} + z_{Y,n}^* z_{Y,n}$ with the same fixed columns; no old coefficient is reselected.

168 Both quotient operations as the same raw Gram cross ratio

The complete DNE native matrix on one target is

$$D_{j,D}^{(s)} = T_D^* [I - V_D (V_D^* V_D)^{-1} V_D^*] T_D, \quad \mathcal{D}_D = \text{diag}(D_{j,D}^{(s)}, 4). \quad (\text{IRR12})$$

Here $\text{diag}(P, 4)$ denotes four copies, not scalar multiplication. The lower frame has full rank and $D_{j,D}^{(s)} \succ 0$ in IRR1. For completeness, a vanishing numerator functional modulo lower evaluations through degree D means $F_a + E$ has its first $D + 1$ derivatives zero for some complete lower exponential E . Then $\mathcal{N}_a + E_A E$ vanishes through degree $D + v \geq q_j - 1$. Its q_j distinct exponential coefficients vanish by the upper Vandermonde. DNE3’s fixed quotient basis makes the numerator class zero. Together with the lower Vandermonde this proves the claimed positivity and all full-column assertions in IRR10.

Eliminating the complete lower block of $G_{Y,D}$ gives exactly $Y^* \mathcal{D}_D Y$. Therefore, writing $g_{L,D} = \det(\mathcal{V}_D^* \mathcal{V}_D)$,

$$\det G_{Y,D} = g_{L,D} \det(Y^* \mathcal{D}_D Y). \quad (\text{IRR13})$$

This is the Schur determinant identity, proved by multiplying the block Gram by the two unit-triangular elimination matrices. It retains both mixed blocks and the single simultaneous lower minimum. Even when Y_K is empty its raw frame contains the entire lower space, and IRR13 uses $\det(Y_K^* \mathcal{D}_D Y_K) = 1$.

In the fixed quotient lifts F_R , let A_D be the native minimum over F_E and let H_D be the native minimum over $\widehat{W}$. Explicitly

$$\begin{aligned} A_D &= F_R^* \mathcal{D}_D F_R - F_R^* \mathcal{D}_D F_E (F_E^* \mathcal{D}_D F_E)^{-1} F_E^* \mathcal{D}_D F_R, \\ H_D &= F_R^* \mathcal{D}_D F_R - F_R^* \mathcal{D}_D \widehat{W} (\widehat{W}^* \mathcal{D}_D \widehat{W})^{-1} \widehat{W}^* \mathcal{D}_D F_R. \end{aligned} \quad (\text{IRR14})$$

The empty F_E term is absent. These are exactly ITL’s A_E and H , now with the cutoff attached; $0 \prec H_D \preceq A_D$. The displayed subtractions are the two attained minima, as follows by completing

their squares. In particular their common image frame is still $U = \mathcal{B}F_R$, and neither minimum deletes the source kernel or an invariant target direction.

Block elimination in the orders $[\mathcal{V}, \mathcal{T}F_E, \mathcal{T}F_R]$ and $[\mathcal{V}, \widehat{\mathcal{T}W}, \mathcal{T}F_R]$ gives

$$\det A_D = \frac{\det G_{S,D}}{\det G_{K,D}}, \quad \det H_D = \frac{\det G_{T,D}}{\det G_{W,D}}, \quad \boxed{\omega_D = \frac{\det H_D}{\det A_D} = \frac{\det G_{T,D} \det G_{K,D}}{\det G_{W,D} \det G_{S,D}}.} \quad (\text{IRR15})$$

Thus the four $g_{L,D}$ factors in IRR13 cancel with signs $+, +, -, -$. This proves that IRR15 is literally ITL10, not a determinant formed after independent rowwise gauge minima. The supplied source masses remain in every row. A common scalar rescaling of the target metric cancels in this ratio only because the two quotient dimensions are both r ; that identity does not license discarding the mass in its individual Grams or updates.

169 The exact four-row update and its two positive terms

For an integer n with $n - 1 \geq q_j - v - 1$ define

$$L_{Y,n} = z_{Y,n} G_{Y,n-1}^{-1} z_{Y,n}^* \succeq 0 \quad (Y = K, S, W, T). \quad (\text{IRR16})$$

Every matrix here is 4 by 4 and uses the same actual new row degree. For any positive G and rectangular z , block elimination of $\begin{pmatrix} G & z^* \\ -z & I \end{pmatrix}$ in either order gives $\det(G + z^*z)/\det G = \det(I + zG^{-1}z^*)$. It follows that

$$\begin{aligned} \gamma_{H,n} &= \log \det(I_4 + L_{T,n}) - \log \det(I_4 + L_{W,n}), \\ \gamma_{A,n} &= \log \det(I_4 + L_{S,n}) - \log \det(I_4 + L_{K,n}), \end{aligned} \quad (\text{IRR17})$$

$$\boxed{\log \omega_n - \log \omega_{n-1} = \gamma_{H,n} - \gamma_{A,n}.}$$

These two terms are individually nonnegative, as is proved by the complete complement calculation next. Their difference has no assigned sign.

For the pair $(X, Y) = (K, S)$ or (W, T) , let $B = Z_{X,n-1}^* \mathcal{T}_{n-1} F_R$, and put

$$R_{X,n} = \text{diag}(t_n, 4) F_R - z_{X,n} G_{X,n-1}^{-1} B, \quad Q_X = \begin{cases} A_{n-1} & X = K, \\ H_{n-1} & X = W. \end{cases} \quad (\text{IRR18})$$

The notation $\text{diag}(t_n, 4)$ has the same four-copy meaning as above. Block inversion in the fixed column order of Y gives

$$L_{Y,n} = L_{X,n} + R_{X,n} Q_X^{-1} R_{X,n}^*, \quad \text{rank}(L_{Y,n} - L_{X,n}) \leq \min(4, r). \quad (\text{IRR19})$$

For an explicit proof, use the triangular change of columns $[Z_X, \mathcal{T}F_R] \mapsto [Z_X, \mathcal{T}F_R - Z_X G_X^{-1} B]$. Its old Gram is $\text{diag}(G_X, Q_X)$ and its new row is $[z_X, R_{X,n}]$. Inserting these into $zG^{-1}z^*$ proves IRR19. Congruence preserves the Gram evaluation, so no determinant-one coordinate factor was lost. More generally the same argument for the fixed inclusion $K \subset W$ gives $L_{K,n} \preceq L_{W,n}$; for $S \subset T$ it gives $L_{S,n} \preceq L_{T,n}$. Inclusion frames need not be orthonormal and need not be initial columns: choose a fixed complement and apply the identical computation.

The full quotient itself has the exact positive update

$$Q_X^{\text{next}} = Q_X + R_{X,n}^* (I_4 + L_{X,n})^{-1} R_{X,n}. \quad (\text{IRR20})$$

To prove it in the same transformed columns, for a fixed quotient coordinate b minimize $a^* G_X a + b^* Q_X b + \|z_X a + R_{X,n} b\|^2$ over the old lower coefficient a . Completing the square yields a correction $R_{X,n}^* [I - z_X (G_X + z_X^* z_X)^{-1} z_X^*] R_{X,n}$. Multiplication shows the bracket is $(I + z_X G_X^{-1} z_X^*)^{-1}$, proving

IRR20. It is an update of the already attained old minimum, not an independent refit of its old row blocks.

Set $C_{X,n} = R_{X,n} Q_X^{-1} R_{X,n}^*$ and

$$B_{X,n} = (I_4 + L_{X,n})^{-1/2} C_{X,n} (I_4 + L_{X,n})^{-1/2}.$$

The identity map between these explicitly specified coefficient metrics explains the square root. Equations IRR19–20 give both finite expressions

$$\begin{aligned} \gamma_{X,n} &= \log \det(I_4 + B_{X,n}) \\ &= \log \det \left(I_r + Q_X^{-1/2} R_{X,n}^* (I_4 + L_{X,n})^{-1} R_{X,n} Q_X^{-1/2} \right), \quad \gamma_{K,n} = \gamma_{A,n}, \quad \gamma_{W,n} = \gamma_{H,n}. \end{aligned} \quad (\text{IRR21})$$

All matrices inside these logarithms are positive. The second identity follows from the same rectangular determinant identity; it retains the original image rank r as well as the four new rows.

There are useful finite scalar brackets with no signs suppressed. Define

$$a_{X,n} = \text{Tr}[(I_4 + L_{Y,n})^{-1} C_{X,n}], \quad b_{X,n} = \text{Tr}[(I_4 + L_{X,n})^{-1} C_{X,n}]. \quad (\text{IRR22})$$

Congruence and cyclicity of trace identify these with $\text{Tr}[B_{X,n}(I_4 + B_{X,n})^{-1}]$ and $\text{Tr} B_{X,n}$. Integration of $(1+t)^{-1}$ on $[0, t]$ gives $t/(1+t) \leq \log(1+t) \leq t$ for $t \geq 0$. Apply it to all four eigenvalues to obtain

$$0 \leq a_{X,n} \leq \gamma_{X,n} \leq b_{X,n}, \quad a_{W,n} - b_{K,n} \leq \log \omega_n - \log \omega_{n-1} \leq b_{W,n} - a_{K,n}. \quad (\text{IRR23})$$

These brackets are computed from the same original arithmetic rows. They are not interval-certified numerical values and no conditioning estimate for their inverses has been assumed.

170 The actual long four-endpoint sum

For the literal target endpoints write

$$d = q_j - v - 1, \quad (D_1, D_2, D_3, D_4) = (d, d+1, d+q_j, d+q_j+1). \quad (\text{IRR24})$$

The signs remain $(-1, -1, +1, +1)$. Subtract the low value from its matched high value and telescope IRR17 in each interval. The two intervals overlap on $d+2, \dots, d+q_j$, giving

$$\begin{aligned} \mathcal{L}_{\text{inv}} &:= -\log \omega_{D_1} - \log \omega_{D_2} + \log \omega_{D_3} + \log \omega_{D_4} \\ &= \sum_{n=d+1}^{d+q_j+1} \vartheta_n (\gamma_{H,n} - \gamma_{A,n}), \\ \vartheta_{d+1} &= \vartheta_{d+q_j+1} = 1, \quad \vartheta_n = 2 \quad (d+2 \leq n \leq d+q_j). \end{aligned} \quad (\text{IRR25})$$

Hence its full signed scalar brackets are

$$\sum_n \vartheta_n (a_{W,n} - b_{K,n}) \leq \mathcal{L}_{\text{inv}} \leq \sum_n \vartheta_n (b_{W,n} - a_{K,n}). \quad (\text{IRR26})$$

In particular its lower bound is $-\sum_n \vartheta_n \gamma_{A,n}$ and its upper bound is $\sum_n \vartheta_n \gamma_{H,n}$. The sharper equality IRR25 preserves cancellation between the two positive sums. Both sums range over q_j+1 original degrees, not over the degree-displacement windows treated by NW.

For any finite $a < b$ in IRR1 the same argument without doubled weights gives the actual long two-endpoint change. All four Grams use fixed coefficient flags and the same original order at both ends. The source order $s = 1$ or $s = j$ is not varied while telescoping; the calculation is carried out separately for each.

171 Exact continuation without replacing the native metric

For the classical finite-rank inverse identity, see Hager [H2, p. 221, (1)]. His dictionary is $A = G_{Y,n-1}$, $U = -z_{Y,n}^*$, $V = z_{Y,n}$, so $I - VA^{-1}U = I_4 + L_{Y,n}$. This is an exact algebraic identity; its use here does not assert numerical stability.

The row recursion IRR7–11 and Gram recursion IRR11 are sufficient to evaluate each step from the original coefficient data. Once an initial $G_{Y,d}^{-1}$ has been obtained, the exact identity

$$G_{Y,n}^{-1} = G_{Y,n-1}^{-1} - G_{Y,n-1}^{-1} z_{Y,n}^* (I_4 + L_{Y,n})^{-1} z_{Y,n} G_{Y,n-1}^{-1} \quad (\text{IRR27})$$

propagates it. Direct multiplication by $G_{Y,n-1} + z_{Y,n}^* z_{Y,n}$ proves the formula; the middle inverse is positive. This avoids a fresh full inverse at each degree in exact arithmetic. It is not a claim that floating-point Woodbury updates or raw moment coordinates are well conditioned. Certified execution must retain the errors of the actual rows, linear solves, determinants and any cancellation in IRR25. No such numerical certification is claimed by this proof.

If $r = 0$, the two quotient spaces vanish: $S = K$, $T = W$, $R_{X,n}$ has zero columns and each $\gamma_{X,n} = 0$. Thus $\omega_D = 1$ and IRR25–26 are exactly zero. If $e = 0$, the raw K frame still contains all original lower evaluations; it is not removed from IRR15–23. Other empty complement cases use the same determinant-one convention. Multiplicities and exceptional algebraic kernel dimensions enter through the actual $F_E, F_R, \widehat{W}$ and are unchanged by all row updates.

The completed advance is an explicit arithmetic-row representation and finite directed brackets for the long invariant-target term in ITL25. It does not evaluate its asymptotic leading coefficient, identify that coefficient with a rank ratio, or assert an RH consequence. The companion flag-flow proof relates these same quotient metrics through their exact angle dynamics; it supplies additional comparisons without altering any IRR row or endpoint.

172 Original invariant-target loss: signed flag flow and finite image-rank updates

This is a continuation of the frozen ITL1–31 calculation at its precise invariant-target factor ITL9–11. It keeps the original target metric, its numerator, its denominator, and the exact source kernel. It proves a continuous projection formula, transports the complete generalized spectrum, and gives exact finite $\mathbf{r} \times \mathbf{r}$ determinant updates across an arbitrary pair of original cutoffs. The long loss is not assigned a sign of change or an asymptotic value. The conductor quotient complex and its Koszul construction are outside this calculation.

172.1 IF1. Original domain, coefficients, metrics, and fixed subspaces

Use ITL1–6 unchanged: the actual simple quartet with its proved five-orbit period and logarithm branch; conductor E_A ; its order $0 \leq v \leq 80$; degree $k=41+1 \geq 9$; $q_j = (j+1)^2$; and precisely the order pair $(s_0, s_1) = (1, 1)$ or $(k, k+4)$. The measures and centers remain

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad c'_j = j/2 - 4. \quad (\text{IF1})$$

The native matrix at cutoff $D \geq q_{-(k+4)} - v - 1$ is the fourfold product

$$\mathcal{D}_D = \text{diag}(D_{k+4,D}^{(s_1)}, D_{k+4,D}^{(s_1)}, D_{k+4,D}^{(s_1)}, D_{k+4,D}^{(s_1)}). \quad (\text{IF2})$$

Every factor is the full attained MNT12 minimum over the complete original lower-evaluation grid. No rowwise reset of its lower coefficients is allowed.

Retain the original invariant inclusion J_0 , target inclusion J_1 , quotient B_1 , and the exact multiplication matrices

$$M_i = (\alpha_A N_{k+4} Z_{\text{exp}, k+4})^{-1} m_{i,k} (\alpha_A N_k Z_{\text{exp}, k}), \quad \alpha_A = v!/\mu_v, \quad m_{i,k}[f] = [x_i^4 f],$$

$$F = \begin{bmatrix} M_1 J_0 \\ M_2 J_0 \\ M_3 J_0 \\ M_4 J_0 \end{bmatrix}, \quad \mathcal{B} = \text{diag}(B_1, B_1, B_1, B_1), \quad W = \text{diag}(J_1, J_1, J_1, J_1), \quad E = \ker(\mathcal{B}F). \quad (\text{IF3})$$

ITL proves that F is injective and $\ker \text{Bcal} = \text{im } W$. With its fixed source basis $[Z_E, Z_R]$, put $F_E = FZ_E$, $F_R = FZ_R$, $e = \dim E$, and $r = \dim V - e$. Choose exactly as in ITL7–10 a coefficient complement W_R such that $W_{\text{hat}} = [F_E, W_R]$ is a frame of $\text{im } W$. All coefficient frames are fixed in the cutoff.

To avoid confusing a subspace with its Gram, write

$$\mathcal{N} = \text{im } F_E, \quad \mathcal{S} = \text{im}[F_E, F_R], \quad \mathcal{W} = \text{im } \widehat{W}, \quad \mathcal{T} = \text{im}[\widehat{W}, F_R]. \quad (\text{IF4})$$

Then

$$\mathcal{S} \cap \mathcal{W} = \mathcal{N}, \quad \mathcal{S} + \mathcal{W} = \mathcal{T}, \quad \dim \mathcal{S} - \dim \mathcal{N} = \dim \mathcal{T} - \dim \mathcal{W} = r. \quad (\text{IF5})$$

Indeed $Fa \in \text{im } W$ is exactly $a \in E$ by the definition of E and injectivity of F . This proves the intersection and the ranks, not merely an abstract dimension count. If $w = \text{rank } W_R$, the four dimensions are $e, e+r, e+w, e+w+r$.

All stars below are complex adjoints in these original Gram calculations. The ordinary transposes inside J_j, B_j retain their complex-linear duality meaning. The displayed conjugation is the only place α_A cancels. Neither original measure in IF1 is made a probability measure.

172.2 IF2. Positive comparison path and its original endpoint forms

Choose any original cutoffs $q_{-(k+4)} - v - 1 \leq D_a \leq D_b$. The full lower-minimum comparison gives

$$\Delta = \mathcal{D}_{D_b} - \mathcal{D}_{D_a} \succeq 0, \quad \mathcal{D}(t) = \mathcal{D}_{D_a} + t\Delta \succ 0, \quad 0 \leq t \leq 1. \quad (\text{IF6})$$

For a fixed full-column frame Z , its $\text{Dcal}(\mathfrak{t})$ -orthogonal projection is

$$P_Z(t) = Z[Z^* \mathcal{D}(t) Z]^{-1} Z^* \mathcal{D}(t). \quad (\text{IF7})$$

The projections have the same original subspaces at all $\mathfrak{t}$. They need not be Hermitian in the unweighted coordinate product; they are orthogonal for the stated metric. Empty frames give zero projections and determinant one.

Define the two least-norm lifts of the same image coordinates by

$$a(t) = (I - P_{F_E}(t))F_R, \quad h(t) = (I - P_{\widehat{W}}(t))F_R, \quad A(t) = a(t)^* \mathcal{D}(t) a(t), \quad H(t) = h(t)^* \mathcal{D}(t) h(t). \quad (\text{IF8})$$

At each endpoint these are exactly ITL's A_E and H . At intermediate $\mathfrak{t}$ they are the same fixed affine-fibre minima in the explicitly displayed positive comparison form. No interpolated metric is claimed to be a new arithmetic cutoff or substituted for an original endpoint.

The original invariant-target factor is

$$\Omega(t) = A(t)^{-1/2} H(t) A(t)^{-1/2}, \quad 0 \prec \Omega(t) \preceq I_r, \quad \omega(t) = \frac{\det H(t)}{\det A(t)}, \quad \ell(t) = \log \omega(t). \quad (\text{IF9})$$

The nested minima and IF5 prove the positive definiteness and inequality, exactly as in ITL8–9. Both forms, not only their ratio, are retained.

172.3 IF3. Intrinsic lift derivatives and the signed determinant differential

For $Z=F_E$ or $Z=W_hat$, write $R_Z(t)=(I-P_Z(t))F_R$ and $G_Z(t)=Z^*\mathcal{D}cal(t)Z$. Differentiating $Z^*\mathcal{D}cal(t)R_Z(t)=0$, and using $\dot{R}_Z \in \text{im } Z$, proves

$$\dot{R}_Z = -ZG_Z^{-1}Z^*\Delta R_Z. \quad (\text{IF10})$$

Let $Q_Z(t)=R_Z(t)^*\mathcal{D}cal(t)R_Z(t)$. The two terms containing $\dot{R}_Z$ vanish by the original orthogonality, giving the exact envelope derivatives

$$\dot{Q}_Z = R_Z^*\Delta R_Z \succeq 0, \quad \ddot{Q}_Z = -2R_Z^*\Delta ZG_Z^{-1}Z^*\Delta R_Z \preceq 0. \quad (\text{IF11})$$

In particular A and H each increase and are concave along the comparison line; this does not make their relative loss monotone. Their complete signed logarithmic derivative is

$$\boxed{\dot{\ell} = \text{Tr}(H^{-1}h^*\Delta h) - \text{Tr}(A^{-1}a^*\Delta a).} \quad (\text{IF12})$$

Both traces are on the actual r -dimensional image. Integration gives $\mathbf{ell}(1) - \mathbf{ell}(0)$ over any pair of cutoffs, including a full low-to-high span. The second trace is not dropped just because it is nonnegative.

Here is the same derivative in terms of all four flags. Let

$$\Pi_Y(t) = \mathcal{D}(t)^{1/2}Y[Y^*\mathcal{D}(t)Y]^{-1}Y^*\mathcal{D}(t)^{1/2}, \quad B(t) = \mathcal{D}(t)^{-1/2}\Delta\mathcal{D}(t)^{-1/2}, \quad (\text{IF13})$$

where Y is the fixed coefficient frame of N, S, W, T . These Π are ordinary Euclidean orthogonal projections onto the specified metric images. Taking the derivative of each complete Gram determinant in ITL10 gives

$$\boxed{\dot{\ell} = \text{Tr}(BJ), \quad J = (\Pi_{\mathcal{T}} - \Pi_{\mathcal{W}}) - (\Pi_{\mathcal{S}} - \Pi_{\mathcal{N}}).} \quad (\text{IF14})$$

For example $d \log \det(Y^*\mathcal{D}cal Y)/dt = \text{Tr}(B \Pi_Y)$ by cyclicity. Thus ITL10's exact signs are $T+N-W-S$. Since nested orthogonal projections commute, both differences in parentheses in IF14 are orthogonal projections of rank r . Consequently

$$\text{rank } J \leq 2r, \quad \text{Tr } J = 0. \quad (\text{IF15})$$

The derivative has no contribution from a scalar velocity: $\text{Tr}((B+cI)J) = \text{Tr}(BJ)$. This follows from the equal original quotient ranks, not from changing the measures or discarding their scalar factors.

172.4 IF4. Exact principal angles of the two original quotient lifts

For the classical principal-angle terminology and its numerical literature, see Zlatko Drmač [H6] and the earlier algorithm of Åke Björck and Gene H. Golub [H7], identified in Drmač's inspected abstract. The complex Hermitian comparison IF16–21 is proved below in the original weighted lift frames. No numerical stability theorem from those papers is being applied.

Define the original lift isometries

$$U_A = \mathcal{D}^{1/2}aA^{-1/2}, \quad U_H = \mathcal{D}^{1/2}hH^{-1/2}. \quad (\text{IF16})$$

They satisfy

$$U_A^*U_A = U_H^*U_H = I_r, \quad U_AU_A^* = \Pi_{\mathcal{S}} - \Pi_{\mathcal{N}}, \quad U_HU_H^* = \Pi_{\mathcal{T}} - \Pi_{\mathcal{W}}, \quad \boxed{U_A^*U_H = A^{-1/2}H^{1/2}.} \quad (\text{IF17})$$

For the last identity, h is orthogonal to W , while $a-h$ lies in W ; hence $a^*\mathcal{D}cal h = h^*\mathcal{D}cal h = H$. The other identities follow from the lift spans and dimensions IF5. Square roots here express the prescribed forms in coordinates only.

Let $\lambda_i \in (0, 1]$ be the generalized eigenvalues of (H, A) . Choose vectors u_i with $u_i^* A u_j = \delta_{ij}$ and $u_i^* H u_j = \lambda_i \delta_{ij}$, and put

$$p_i = \mathcal{D}^{1/2} A u_i, \quad q_i = \lambda_i^{-1/2} \mathcal{D}^{1/2} H u_i. \quad (\text{IF18})$$

Both families are orthonormal, and

$$p_i^* q_j = \sqrt{\lambda_j} \delta_{ij}. \quad (\text{IF19})$$

In particular the principal cosines of the two original quotient-lift subspaces are $\sqrt{\lambda_i}$. This uses their shared coordinate map; it does not identify either with a chosen coordinate subspace.

For $\lambda_i < 1$, put $c_i = \sqrt{\lambda_i}$, $s_i = \sqrt{1 - \lambda_i}$, and $z_i = (q_i - c_i p_i) / s_i$. The z_i are orthonormal and orthogonal to all p_j ; the planes spanned by (p_i, z_i) are mutually orthogonal. On each such plane,

$$q_i q_i^* - p_i p_i^* = \begin{pmatrix} -s_i^2 & c_i s_i \\ c_i s_i & s_i^2 \end{pmatrix}, \quad \text{with eigenvalues } +s_i, -s_i. \quad (\text{IF20})$$

If $\lambda_i = 1$, then $q_i = p_i$, so that term is zero. Thus the complete nonzero spectrum of J is exactly the pairs

$$\boxed{\pm \sqrt{1 - \lambda_i} \quad (\lambda_i < 1), \quad \|J\|_1 = 2 \sum_i \sqrt{1 - \lambda_i}.} \quad (\text{IF21})$$

The $\sqrt{1 - \lambda_i}$ are also the actual target-gauge singular values in ITL9–11. The rank bound IF15 can therefore be strict; zero angles are retained as zero contributions.

172.5 IF5. Full generalized-spectrum differential, including crossings

Order the $\lambda_i(t)$ increasingly, with multiplicities. They are positive and locally Lipschitz: on the compact positive path the generalized Rayleigh quotients and their differences are uniformly bounded, and the generalized min–max formula bounds each ordered eigenvalue difference by the corresponding uniform change of those quotients. Thus ordered derivatives exist almost everywhere.

At a differentiability point, choose the generalized eigenvectors in IF18 to diagonalize the derivative within any repeated-eigenvalue subspace. Here is the complete local calculation. Fix a time t_0 and a fixed invertible matrix U whose columns diagonalize the pair there, so $U^* A(t_0) U = I$ and $U^* H(t_0) U = \Lambda$, a diagonal matrix. Write $\epsilon = t - t_0$, $A_1 = U^* \dot{A}(t_0) U$ and $H_1 = U^* \dot{H}(t_0) U$. The coefficient Grams and their inverses are analytic in a neighborhood of this positive point. Hence in these fixed coordinates their expansions are $I + \epsilon A_1 + O(\epsilon^2)$ and $\Lambda + \epsilon H_1 + O(\epsilon^2)$. The inverse positive square root of the first is $I - \frac{1}{2} \epsilon A_1 + O(\epsilon^2)$, as is checked by squaring and multiplying the convergent matrix power series near I . The Hermitian representative of the pair is therefore

$$\Lambda + \epsilon \left(H_1 - \frac{1}{2} (A_1 \Lambda + \Lambda A_1) \right) + O(\epsilon^2).$$

Split its coordinates into the full eigenspace for an eigenvalue λ and its complement. The complementary block minus λI is invertible, with a bounded inverse on a sufficiently small neighborhood. The off-diagonal blocks are $O(\epsilon)$, so block elimination subtracts an $O(\epsilon^2)$ term from the eigenspace block. In that block the first-order Hermitian matrix is exactly $(H_1 - \lambda A_1)$ restricted to the eigenspace. Diagonalize this finite matrix. Its ordered eigenvalues are the first-order ordered splittings: the remainder is $O(\epsilon^2)$ by the min–max bound for a Hermitian perturbation. For negative ϵ the order within the cluster reverses; at every point where the ordered eigenvalues have two-sided derivatives, these give the corresponding derivatives with their multiplicities. This proves that the vectors can be chosen to give $u_i^* (\dot{A}$

$H - \lambda \dot{A} u_j$ diagonal within each cluster. It requires no fixed eigenvectors across a crossing.

Differentiate $H u_i = \lambda_i A u_i$ and cancel the terms containing $\dot{u}_i$ by the eigenvalue equation. Use IF11 and IF18. The exact scalar result is

$$\boxed{\frac{d}{dt} \log \lambda_i = \frac{u_i^* \dot{H} u_i}{\lambda_i} - u_i^* \dot{A} u_i = q_i^* B q_i - p_i^* B p_i.} \quad (\text{IF22})$$

Summing it recovers IF12–14, including both signs. Formula IF22 also identifies each original spectral motion with one paired principal plane.

Let $c_T(t)$ be the spectral diameter (largest minus smallest eigenvalue) of the compression of $B(t)$ to

$$D_{\text{cal}}(t)^{(1/2)} T \cap (D_{\text{cal}}(t)^{(1/2)} N)^{\perp}.$$

This space contains all p_i, q_i ; it has dimension $w+r$. If it is empty, set $c_T=0$. Its spectral diameter is at most the full $c(t) = \lambda_{\max}(B) - \lambda_{\min}(B)$. Subtracting the midpoint of the compressed spectrum from B , and using IF20, gives

$$\boxed{|(\log \lambda_i)'| \leq c_T(t) \sqrt{1 - \lambda_i}, \quad |\dot{\ell}| \leq c_T(t) \sum_i \sqrt{1 - \lambda_i}.} \quad (\text{IF23})$$

Indeed a Hermitian operator of norm at most $c_T/2$, paired by the trace with a rank-two matrix of trace norm $2s_i$, has absolute trace at most $c_T s_i$. This proves IF23 without any entrywise real convention. At $\lambda_i=1$, the right side is zero and IF22 gives zero first derivative; motion away from one can still begin at second order.

The complete ordered spectrum consequently obeys

$$\sum_i |\log \lambda_i(1) - \log \lambda_i(0)| \leq \int_0^1 c_T(t) \sum_i \sqrt{1 - \lambda_i(t)} dt. \quad (\text{IF24})$$

Every fixed real-valued 1-Lipschitz function of the log spectrum has the same bound for the difference of its summed values. The actual negative logarithmic loss is one such sum because all eigenvalues lie in $(0, 1]$. Its sign of change is not fixed.

172.6 IF6. A full-interval angle comparison with an explicit endpoint condition ratio

There is a sharper scalar coordinate for IF23. Define, with its value zero at $\lambda_i=1$,

$$u_i(t) = \text{arcosh}(\lambda_i(t)^{-1/2}) = \text{artanh} \sqrt{1 - \lambda_i(t)} \geq 0, \quad \lambda_i(t) = \text{sech}^2 u_i(t). \quad (\text{IF25})$$

For $z_i = -\log \lambda_i > 0$, direct differentiation gives

$$\frac{d}{dz} \text{arcosh}(e^{z/2}) = \frac{1}{2\sqrt{1 - e^{-z}}}.$$

Combining this identity with IF23 proves $|\dot{u}_i| \leq c_T/2$ wherever $u_i > 0$. This also gives the integral bound through contacts with zero: on each open component of $\{u_i > 0\}$ integrate the bound, extend to its endpoints by continuity, and use $u_i=0$ at all intervening points. For any two times in distinct components, their nonnegative endpoint values are each bounded by the integral to a separating zero, so their absolute difference is at most the integral over the whole interval. This proves absolute continuity and

$$\boxed{|u_i(1) - u_i(0)| \leq \frac{1}{2} \int_0^1 c_T(t) dt \leq \frac{1}{2} \int_0^1 c(t) dt.} \quad (\text{IF26})$$

The last integral has an exact endpoint formula on IF6's affine path. Put

$$C = \mathcal{D}_{D_a}^{-1/2} \Delta \mathcal{D}_{D_a}^{-1/2} \succeq 0, \quad R = I + C = \mathcal{D}_{D_a}^{-1/2} \mathcal{D}_{D_b} \mathcal{D}_{D_a}^{-1/2}, \quad \kappa = \frac{\lambda_{\max}(R)}{\lambda_{\min}(R)} \geq 1. \quad (\text{IF27})$$

The eigenvalues of $\mathbf{B}(\mathbf{t})$ are those of the pair $(\Delta, \mathbf{Dcal}(\mathbf{t}))$, hence are $\mathbf{c}_j/(1+\mathbf{t} \mathbf{c}_j)$, where $\mathbf{c}_j$ are the eigenvalues of $\mathbf{C}$. This follows by the invertible congruence $\mathbf{Dcal}(\mathbf{D}_a)^{-1/2}$ and the fact that $\mathbf{C}$ commutes with $\mathbf{I}+\mathbf{t}\mathbf{C}$. The function of $\mathbf{c}_j$ is increasing. Therefore

$$\int_0^1 c(t) dt = \int_0^1 \left(\frac{c_{\max}}{1+tc_{\max}} - \frac{c_{\min}}{1+tc_{\min}} \right) dt = \log \frac{1+c_{\max}}{1+c_{\min}} = \log \kappa. \quad (\text{IF28})$$

The zero $\mathbf{c}_j$ case is included. Write $L=(\log \kappa)/2$. Every original ordered loss eigenvalue consequently satisfies the finite two-sided comparison

$$\boxed{\text{sech}^2(u_i(0) + L) \leq \lambda_i(1) \leq \text{sech}^2((u_i(0) - L)_+).} \quad (\text{IF29})$$

This retains the actual initial loss, rather than assigning it zero. For the total loss $\mathbf{frac} \mathbf{l}(\mathbf{t})=-\log \omega(\mathbf{t})=2 \sum_i \log \cosh \mathbf{u}_i(\mathbf{t})$,

$$\boxed{2 \sum_i \log \cosh((u_i(0) - L)_+) \leq \mathbf{l}(1) \leq 2 \sum_i \log \cosh(u_i(0) + L).} \quad (\text{IF30})$$

Also IF24 yields $\sum_i |\delta \log \lambda_i| \leq r \log \kappa$, while its displayed integral retains the smaller actual principal-angle factors. These statements hold for a full long interval, not just a displaced endpoint window. They do not claim $\log \kappa$ or the initial $\mathbf{u}_i(0)$ is small in the original arithmetic data. Both are precisely specified finite quantities. A proportional change of the entire target form gives $\kappa=1$ and exactly unchanged loss, confirming the original scalar cancellation in IF15 without altering either endpoint metric.

172.7 IF7. Exact finite numerator and denominator updates of size r

The flow has a direct finite counterpart using only the same fixed lifts and their full gauges. In this section all unmarked old forms and lifts are at $\mathbf{t}=0$. For $\mathbf{Z}=\mathbf{F}_E$ or $\mathbf{Z}=\mathbf{W}_{\hat{}}$, put

$$G_{Z,a} = Z^* \mathcal{D}_{D_a} Z, \quad R_{Z,a} = (I - P_Z(0)) F_R, \quad \mathcal{C}_Z = \Delta - \Delta Z (Z^* \mathcal{D}_{D_b} Z)^{-1} Z^* \Delta. \quad (\text{IF31})$$

This is positive semidefinite. To see the full coefficient minimum, write a candidate lift as $\mathbf{R}_-(\mathbf{Z}, \mathbf{a}) \mathbf{y} + \mathbf{Z} \mathbf{eta}$. Its old squared norm is $\mathbf{y}^* \mathbf{Q}_-(\mathbf{Z}, \mathbf{a}) \mathbf{y} + \mathbf{eta}^* \mathbf{G}_-(\mathbf{Z}, \mathbf{a}) \mathbf{eta}$, and its additional squared norm is $(\mathbf{R}_-(\mathbf{Z}, \mathbf{a}) \mathbf{y} + \mathbf{Z} \mathbf{eta})^* \Delta (\mathbf{R}_-(\mathbf{Z}, \mathbf{a}) \mathbf{y} + \mathbf{Z} \mathbf{eta})$. The minimizer of their sum is $\mathbf{eta} = -(\mathbf{Z}^* \mathbf{Dcal}(\mathbf{D}_b) \mathbf{Z})^{-1} \mathbf{Z}^* \Delta \mathbf{R}_-(\mathbf{Z}, \mathbf{a}) \mathbf{y}$.

Completing this square gives the exact update

$$\boxed{Q_{Z,b} = Q_{Z,a} + R_{Z,a}^* \mathcal{C}_Z R_{Z,a}.} \quad (\text{IF32})$$

Equivalently, with $\Delta^{1/2}$ its positive square root,

$$\mathcal{C}_Z = \Delta^{1/2} [I + \Delta^{1/2} Z G_{Z,a}^{-1} Z^* \Delta^{1/2}]^{-1} \Delta^{1/2} \succeq 0. \quad (\text{IF33})$$

This identity follows by eliminating the same quadratic form, or multiplying out the elementary block inverse. It uses no inverse of Δ , so singular or zero increments are allowed. For empty $\mathbf{Z}$, set $\mathbf{C}_Z = \Delta$ and omit the empty inverse.

Use the two original choices of $\mathbf{Z}$ separately and define the positive $\mathbf{r} \times \mathbf{r}$ update matrices

$$T_A = A_a^{-1/2} a_a^* \mathcal{C}_{F_E} a_a A_a^{-1/2}, \quad T_H = H_a^{-1/2} h_a^* \mathcal{C}_{\widehat{W}} h_a H_a^{-1/2}. \quad (\text{IF34})$$

Thus

$$A_b = A_a^{1/2}(I_r + T_A)A_a^{1/2}, \quad H_b = H_a^{1/2}(I_r + T_H)H_a^{1/2},$$

$$\boxed{\frac{\omega_b}{\omega_a} = \frac{\det(I_r + T_H)}{\det(I_r + T_A)}, \quad \ell_b - \ell_a = \log \det(I_r + T_H) - \log \det(I_r + T_A).} \quad (\text{IF35})$$

Both positive updates are retained. They need not be ordered, and their determinant difference is signed. Each determinant has the actual image rank $\mathbf{r}$, even though its coefficients retain inverses on the complete original source-kernel or target-relation spaces. This reduces determinant size without concealing the coefficients that still require calculation.

There is an exact image-rank formula for the entire new spectrum as well. Let $C_0 = H_a^{-1/2} A_a^{-1/2}$. The new generalized eigenvalues are those of the positive pair

$$(C_0^*(I_r + T_H)C_0, I_r + T_A),$$

or equivalently the Hermitian matrix

$$\boxed{(I_r + T_A)^{-1/2} C_0^*(I_r + T_H)C_0(I_r + T_A)^{-1/2}.} \quad (\text{IF36})$$

Indeed apply the common invertible coordinate congruence $A_a^{-1/2}$ to the pair (H_b, A_b) . This proves spectral equality with the actual endpoint Ω_b ; it does not assert equality of their coordinate matrices. The full complex cross factor C_0 remains. In general it cannot be commuted past either update or replaced by the eigenvalue list of Ω_a .

172.8 IF8. Directed integral bounds and endpoint-only enclosures

Define the exact nonnegative quotient increments

$$\Gamma_A = \log \det A_b - \log \det A_a = \log \det(I_r + T_A), \quad \Gamma_H = \log \det H_b - \log \det H_a = \log \det(I_r + T_H). \quad (\text{IF37})$$

Then $\ell_b - \ell_a = \Gamma_H - \Gamma_A$, so in particular $-\Gamma_A \leq \ell_b - \ell_a \leq \Gamma_H$. Both increments are restrictions/quotients of the same original target form; they are not a source-rank fraction. MR's general positive-form comparison also supplies $\sum_i |\delta \log \lambda_i| \leq \Gamma_H + \Gamma_A$. IF23–30 retain additional cancellation and principal-angle information not present in that absolute sum.

For a finite endpoint-only enclosure, the second derivative of either logarithmic determinant is

$$\frac{d^2}{dt^2} \log \det Q_Z = \text{Tr}(Q_Z^{-1} \ddot{Q}_Z) - \text{Tr}((Q_Z^{-1} \dot{Q}_Z)^2) \leq 0. \quad (\text{IF38})$$

The first term is nonpositive by IF11; the second is the negative squared Hilbert–Schmidt norm of the Hermitian matrix $Q_Z^{-1/2} \dot{Q}_Z Q_Z^{-1/2}$. Hence the nonnegative derivative decreases. Define its two endpoint values

$$\alpha_t = \text{Tr}(A_t^{-1} a_t^* \Delta a_t), \quad \eta_t = \text{Tr}(H_t^{-1} h_t^* \Delta h_t), \quad t = 0, 1.$$

Integrating gives the fully directed finite bounds

$$\alpha_1 \leq \Gamma_A \leq \alpha_0, \quad \eta_1 \leq \Gamma_H \leq \eta_0, \quad \boxed{\eta_1 - \alpha_0 \leq \ell_b - \ell_a \leq \eta_0 - \alpha_1.} \quad (\text{IF39})$$

These compare the exact complete numerator and denominator responses. They do not turn two increasing Grams into a monotone relative loss.

172.9 IF9. The four original endpoints and exact scope

Keep ITL25's target endpoints

$$\begin{aligned} D_1 &= q_{k+4} - 1 - v, & D_2 &= q_{k+4} - v, \\ D_3 &= 2q_{k+4} - 1 - v, & D_4 &= 2q_{k+4} - v, & \varepsilon &= (-1, -1, +1, +1). \end{aligned} \quad (\text{IF40})$$

The actual invariant-target return is exactly

$$\boxed{\sum_{i=1}^4 \varepsilon_i \log \omega_{D_i} = [\ell(D_3) - \ell(D_1)] + [\ell(D_4) - \ell(D_2)]}. \quad (\text{IF41})$$

Each bracket can be evaluated by its own full comparison path IF12/14 or by the two $\mathbf{r} \times \mathbf{r}$ determinants IF35, with its own original endpoints and full **Delta**. Thus IF41 is a concrete signed integral/update formula for the long return itself, not a restatement of a short displacement result. It requires the native coefficients and full gauge Grams at those endpoints; no arithmetic estimate of their sizes is fabricated here. Root's separate calculation extracts their raw original functional-row updates and finite fixtures.

The exact output proved here consists of the rank-at-most- $2\mathbf{r}$ signed projection flow; its original principal-angle spectrum; ordered spectral transport through crossings; a full-interval condition/angle enclosure retaining the initial loss; and the complete $\mathbf{r} \times \mathbf{r}$ numerator/denominator update formulas. ITL30–31's separate compatibility-comparison error is untouched. This work does not assign a leading coefficient to ITL25, prove smallness of the actual condition ratio or initial loss, or replace the owner's conductor quotient/Koszul work.

If $\mathbf{r}=0$, **F_R** is empty, **S=N**, **T=W**, both image Grams are empty, and all displayed determinant ratios, updates, derivatives and spectral sums give one or zero exactly. If **W_R** is empty, **W=N**, then **a=h**, **A=H**, every **lambda_i=1**, and the entire invariant-target loss vanishes at every cutoff. If **Delta=0**, the path is constant and every interval update is zero. All cases retain their empty-block conventions without an artificial nonzero error.

172.10 Source record and verification scope

The complete final ITL1–31 source was read in full before this calculation:

`../ORIGINAL_INVARIANT_TARGET_LOSS.tex`

SHA-256: 1ef4903e7763e888abb926feb619474d60f29aecbc232a149640f792e11b0643.

Its direct NBG1–32 and NKC1–17 native providers were already read completely in the preceding matched-relative task. The earlier MR note is used only for the comparison stated in IF8, not for the new projection/angle derivation:

`../endpoint_shift_20260915/matched_relative_followup/MATCHED_RELATIVE_WINDOW.md`

SHA-256: d955b022cd8e9d362c70080c989234267315c81793f729d8c26c916eac7e59c9.

No new arithmetic numerical evaluation, interval certification, Lean execution, PDF build, or remote action is claimed. Numerical fixtures and extraction of the actual coefficient arrays belong to the parent's separate lane. The positive comparison path is fully specified in IF6 and every assertion here is a finite-dimensional proof using the original coefficient flags and metrics at its endpoints.

Root continuation record: the authoring agent saved the complete original Markdown proof before its turn terminated at a usage limit. Root read that full body, retained it byte-for-byte under `../history/agent_flow_v1/`, and expanded the repeated-eigenvalue calculation above. The agent did not complete an independent receipt or a LaTeX artifact. Any subsequent LaTeX conversion, validation, or review is recorded as root work, not attributed to an unfinished agent execution.

This continues IRR1–27 and IF1–41 using exactly their common original coefficient flags and full target Gamma metric. It controls simultaneous movement of all original loss eigenvalues by

one native volume increment, without a factor equal to the image dimension. The original long four-return and the initial loss remain unchanged; their arithmetic leading values are not supplied by this finite estimate.

173 The same quotient lifts and their individual paired planes

Retain $k = 4l + 1 \geq 9$, $j = k + 4$, $q_j = (j + 1)^2$, $0 \leq v \leq 80$, and the original target source order $s = 1$ or $s = j$. The four full coefficient flags of IRR5 are

$$\begin{aligned} K &= \text{im } F_E, \quad S = \text{im}[F_E, F_R], \\ W &= \text{im } \widehat{W}, \quad T = \text{im}[\widehat{W}, F_R], \quad K = S \cap W, \quad r = \dim(S/K) = \dim(T/W). \end{aligned} \quad (\text{FAB1})$$

Here $F_E = FZ_E$, $E = \ker(\mathcal{B}F)$ is the full actual source kernel, and $F_R = FZ_R$ is the same fixed complementary image lift. Every matrix in these definitions is given in IRR3–5 and ITL1–10; in particular the multiplication maps keep the original nonreduced conductor, low jets and $\alpha_A = v!/\mu_v$ conjugation.

For two original cutoffs $D_a \leq D_b$, both at least $q_j - v - 1$, put $D(t) = \mathcal{D}_{D_a} + t\Delta$ with $\Delta = \mathcal{D}_{D_b} - \mathcal{D}_{D_a} \succeq 0$. This is an explicitly stated comparison path between the original metrics, not a new arithmetic metric replacing either endpoint. For a fixed full frame Z , put $P_Z = Z(Z^*DZ)^{-1}Z^*D$. The two original lifts and forms are

$$\begin{aligned} a &= (I - P_{F_E})F_R, \quad h = (I - P_{\widehat{W}})F_R, \\ A &= a^*Da, \quad H = h^*Dh, \quad 0 \prec H \preceq A, \quad B = D^{-1/2}\Delta D^{-1/2} \succeq 0. \end{aligned} \quad (\text{FAB2})$$

The stars are complex adjoints; no ordinary complex-linear duality transpose is changed inside the original coefficient maps. Differentiating $Z^*DR_Z = 0$ and using $\dot{R}_Z \in \text{im } Z$ gives $\dot{R}_Z = -Z(Z^*DZ)^{-1}Z^*\Delta R_Z$. Consequently

$$\dot{A} = a^*\Delta a, \quad \dot{H} = h^*\Delta h. \quad (\text{FAB3})$$

Indeed the two other differentiated terms vanish by the same original orthogonality. This proves the envelope derivatives without differentiating or selecting a new quotient frame.

Let $\lambda_i(t) \in (0, 1]$ be the increasingly ordered eigenvalues of (H, A) with multiplicity. At an almost-everywhere differentiability point choose u_i with $u_i^*Au_m = \delta_{im}$ and $u_i^*Hu_m = \lambda_i\delta_{im}$, diagonalizing $\dot{H} - \lambda_i\dot{A}$ in each repeated cluster. The full local Hermitian block expansion proving this choice, including ordered crossings, is given in IF5: in a fixed $A(t_0)$ -orthonormal eigenframe the Hermitian pair has first-order matrix $H_1 - (A_1\Lambda + \Lambda A_1)/2$; elimination of the spectrally separated complementary block changes each cluster only at second order. Thus this assertion does not assume smooth eigenvectors across crossings. The min–max bound makes each ordered eigenvalue locally Lipschitz.

In the original coordinate isometry to Euclidean space set

$$p_i = D^{1/2}au_i, \quad q_i = \lambda_i^{-1/2}D^{1/2}hu_i, \quad s_i = \sqrt{1 - \lambda_i}, \quad J_i = q_iq_i^* - p_ip_i^*. \quad (\text{FAB4})$$

Both families are orthonormal. Since $a - h \in W$ and h is D -orthogonal to W , $a^*Dh = H$, so $p_i^*q_m = \sqrt{\lambda_m}\delta_{im}$. For $s_i > 0$ let $z_i = (q_i - \sqrt{\lambda_i}p_i)/s_i$. These vectors are orthonormal, perpendicular to all p_m , and the planes $\text{span}(p_i, z_i)$ are mutually orthogonal. On its plane J_i has matrix

$$\begin{pmatrix} -s_i^2 & s_i\sqrt{\lambda_i} \\ s_i\sqrt{\lambda_i} & s_i^2 \end{pmatrix}, \quad \text{spec } J_i = \{-s_i, s_i\}. \quad (\text{FAB5})$$

For $s_i = 0$, $q_i = p_i$ and $J_i = 0$. Differentiating the pair and using FAB3 gives the exact individual scalar velocities

$$(\log \lambda_i)' = q_i^*Bq_i - p_i^*Bp_i = \text{Tr}(BJ_i). \quad (\text{FAB6})$$

174 Simultaneous variation costs one ambient native volume

For every choice of signs $\varepsilon_i \in \{-1, 1\}$ the Hermitian matrix $\sum_i \varepsilon_i J_i$ is a direct sum on the mutually orthogonal planes above. Its norm is at most $s_{\max} = \max_i s_i \leq 1$, irrespective of the number r of planes. For any Hermitian C with $C \preceq s_{\max} I$ and $B \succeq 0$, $\text{Tr}(BC) \leq s_{\max} \text{Tr} B$: the difference is the trace of the positive matrix $B^{1/2}(s_{\max} I - C)B^{1/2}$. Choose the signs of the real numbers in FAB6 to obtain

$$\sum_i |(\log \lambda_i)'| = \max_{\varepsilon} \text{Tr} \left(B \sum_i \varepsilon_i J_i \right) \leq s_{\max} \text{Tr} B \leq \text{Tr} B. \quad (\text{FAB7})$$

This estimate is for the sum over the entire spectrum, not one eigenvalue followed by an implicit multiplication by r .

There is an even stronger statement in the original angle coordinate

$$u_i = \text{arcosh}(\lambda_i^{-1/2}) \geq 0, \quad \lambda_i = \text{sech}^2 u_i. \quad (\text{FAB8})$$

For $s_i > 0$, $u_i' = -(\log \lambda_i)'/(2s_i)$. The matrices J_i/s_i have eigenvalues $\{-1, 1\}$ on the same disjoint planes. The same sign-maximization argument gives

$$\sum_{i:s_i>0} |u_i'| \leq \frac{1}{2} \text{Tr} B. \quad (\text{FAB9})$$

For completeness this does not omit contacts with $\lambda_i = 1$. Individually the bound $|u_i'| \leq \frac{1}{2} \text{Tr} B$ on each open component where $u_i > 0$ integrates to its endpoints by continuity. At intervening zeros the same bound controls differences by the integral over the full interval. Hence every u_i is absolutely continuous on the compact comparison path. An absolutely continuous function has derivative zero almost everywhere on a level set: at its density points where a derivative exists, a sequence of other points of the same level tending to the point forces that derivative to be zero. Thus the omitted zero-angle indices contribute zero almost everywhere. This proves FAB9 with the sum over all indices as well.

Cyclicity gives $\text{Tr} B = \text{Tr}(D^{-1}\Delta) = (\log \det D)'$. Integrate FAB7 and FAB9, using absolute continuity and then the triangle inequality for each ordered endpoint value:

$$\begin{aligned} \sum_i |\log \lambda_i(D_b) - \log \lambda_i(D_a)| &\leq \int_0^1 s_{\max}(t) \text{Tr} B(t) dt \leq \log \frac{\det \mathcal{D}_{D_b}}{\det \mathcal{D}_{D_a}}, \\ \sum_i |u_i(D_b) - u_i(D_a)| &\leq \frac{1}{2} \log \frac{\det \mathcal{D}_{D_b}}{\det \mathcal{D}_{D_a}}. \end{aligned} \quad (\text{FAB10})$$

The first bound also applies to changes of the sum of any fixed real 1-Lipschitz function of the log eigenvalues, by its defining inequality. These statements are complementary to IF26–30's condition-ratio bound, which retains scalar-velocity cancellation. The trace budget may be coarser on a proportional metric change; in that case the exact J_i traces or the condition-ratio bound give zero. No source mass is discarded to make FAB10 smaller.

175 The trace budget is itself a scalar four-row arithmetic quantity

Use IRR7–11's exact original one-target rows t_n, v_n , including their analytic numerator division, Gamma recurrence, centre c_j' and $h_{n,s} = (2\pi)^{s/2} n!(s/2)_n$. Fix $n-1 \geq q_j - v - 1$, and write T, V for all original numerator and lower-evaluation rows through $n-1$. Set

$$\begin{aligned} C_L &= V^* V, \quad D_0 = T^* [I - V C_L^{-1} V^*] T \succ 0, \\ \zeta_n &= v_n C_L^{-1} v_n^* \geq 0, \quad \rho_n = t_n - v_n C_L^{-1} V^* T. \end{aligned} \quad (\text{FAB11})$$

Here ζ_n is a scalar and ρ_n is one complete row on the original conductor quotient. The symbols do not change or remove any of its fixed coefficient columns. Apply IRR20 to the full one-target quotient, with its entire lower frame as kernel and its one new row. This gives, or directly follows by completing that full old lower minimum,

$$D_{j,n}^{(s)} = D_0 + \frac{\rho_n^* \rho_n}{1 + \zeta_n}, \quad \Delta_n = \mathcal{D}_n - \mathcal{D}_{n-1} = \text{diag} \left(\frac{\rho_n^* \rho_n}{1 + \zeta_n}, 4 \right). \quad (\text{FAB12})$$

The diagonal notation means four identical copies. Thus the native metric increment has rank at most four, derived from the actual row map rather than assumed from an image-dimension count. The rectangular determinant identity gives its full log-volume

$$\boxed{\beta_n := \log \frac{\det \mathcal{D}_n}{\det \mathcal{D}_{n-1}} = 4 \log \left(1 + \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n} \right) \geq 0.} \quad (\text{FAB13})$$

Every coefficient in this scalar is fixed by IRR's original arithmetic rows. It is not a fit to the invariant loss or a replacement measure. Combining FAB10–13 proves the per-degree bounds

$$\sum_i |\log \lambda_i(n) - \log \lambda_i(n-1)| \leq \beta_n, \quad \sum_i |u_i(n) - u_i(n-1)| \leq \beta_n/2. \quad (\text{FAB14})$$

For the original four endpoints $d, d+1, d+q_j, d+q_j+1$, where $d = q_j - v - 1$, let $\vartheta_n = 1$ at $d+1, d+q_j+1$ and $\vartheta_n = 2$ for $d+2 \leq n \leq d+q_j$. The exact IRR25 signed expression consequently obeys

$$|\mathcal{L}_{\text{inv}}| \leq \sum_{n=d+1}^{d+q_j+1} \vartheta_n \beta_n. \quad (\text{FAB15})$$

The sharper signed expression remains the difference of the two positive quotient log determinants in IRR25; FAB15 is a bound, not permission to discard either contribution. These are long intervals of q_j steps, not the smaller displacement windows previously handled by NW.

The original whole-volume sequence in FAB13 has not been assigned a subleading asymptotic value. Nevertheless FAB13–15 quantify the full-spectrum cost with the actual four-row volume and no factor $r = O(k)$ multiplying that volume. The initial invariant loss is retained whenever these increments are integrated to an absolute loss. If $r = 0$ all spectral sums and the loss are exactly zero, regardless of a possibly positive ambient upper budget. If $W = K$ the same follows from $A = H$. If $\rho_n = 0$ the actual native update and every increment at that step vanish. No RH conclusion, interval-certified arithmetic numerical result, or Lean check is claimed by this finite calculation.

176 The same original sources at every receiving map

Keep the fixed actual simple quartet, full unit and period matrix, five-orbit period and logarithm branch of the preceding sources. Every nonreduced factor of A^{hom} , its exact exponential symbol E_A , $v = \text{ord}_0 E_A$ and $\alpha_A = v!/\mu_v$ are unchanged. The coefficient sources, original Gamma masses and complex-linear dual convention are those of NBG, NKC and CNC. The following domains are local to their claims:

$$\begin{aligned} k &\equiv 1 \pmod{4}, & j &= k+4, & q_n &= (n+1)^2, \\ \text{CNC/LCC/LFC: } &k \geq 9, & 0 \leq v &\leq 80, & s_p &= 1 \text{ for all } p \text{ or } s_p = k+4p, \\ &\text{WEL/BVC/EBG/RPB: } &k \geq 37, & s &= 1 \text{ or } k, \\ &k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33} q_k && \text{for the Gamma comparison,} \\ \text{GGE high estimate: } &k \geq \max\{1025, \lceil 2\sqrt{\delta^2 + \gamma^2}/2^{-32} \rceil + 1\}. \end{aligned} \quad (\text{NCR1})$$

Here $0 < \delta < 1/2$ and $\gamma > 2$ retain their original values. The sharper actual bound $v \leq 32$ in RPB14 is available, and no matrix in this receiver replaces its actual value by that bound.

177 The evaluated boundary enters through the entire numerator map

Write exactly $D = q_j - v - 1 + \sigma$, $\sigma = 0, 1$, and use the original EBG boundary nodes and source order $s = 1$ or k . Put $n_0 = (k - 7)^2$, $r = 24k - 24 - v + \sigma$, $Q_s = n_0 + (s - 1)/4$ and $\beta_s = (2\pi)^{s/2}/(\sqrt{2}\Gamma(s/2))$. The complete accepted WEL and BVC calculations give

$$\begin{aligned} \mathcal{F} &= 2rn_0 \log k + 2kn_0 \mathcal{C}(\delta, \gamma) - r \log\{\beta_s \mu_0(Q_s)\}, \\ \mathcal{F} - B_W + E_- &\leq \log \text{pdet } \mathcal{H}_{B,D} \leq \mathcal{F} + B_W + E_+, \\ \frac{\log \text{pdet } \mathcal{H}_{B,D}}{kq_k} + 48 \log k &= 2\mathcal{C}(\delta, \gamma) + 48(1 - \log(4/\pi)) + O_{\delta, \gamma}\left(\frac{\log(k+1)}{k}\right). \end{aligned} \quad (\text{NCR2})$$

Every constant B_W, E_-, E_+, μ_0 is the full expression in BVC1,4–5; $\mathcal{C}$ is its exact BVC3 integral. This is substitution into EBG32: WEL9 bounds the literal sum of the largest r original root-product weights on both sides, so its lower side combines with the lower EBG32 side and its upper side with the upper EBG32 side. The complete factorial and Gamma calculation BVC8–11 then yields the last line. The actual low rank r is used in every determinant in this argument.

Here is the precise map carrying this calculation back to EBG/RPB. In EBG7–10's unchanged frames set

$$\mathcal{H}_{B,D} = E^* G_D^{-1} E, \quad C_D = E^* G_D^{-1} U, \quad D_D^0 = U^* G_D^{-1} U, \quad D_D^+ = D_D^0 - C_D^* \mathcal{H}_{B,D}^+ C_D. \quad (\text{NCR3})$$

To verify the map and its kernel, restrict the original dual functional to the annihilator $\chi_0 \mathcal{P}_{D-n_0}$ of the complete original lower grid. All extensions differ by the lower evaluation span; extension by zero on its orthogonal complement attains the quotient norm. This identifies its Gram with the displayed G_D^{-1} in the real centered EBG frame. The boundary columns are exactly E and the numerator columns exactly U . Minimizing $(Ux + E\eta)^* G_D^{-1} (Ux + E\eta)$ over every η gives NCR3, including dependent columns by projection onto their full image. At both low cutoffs that image fills the whole remaining source quotient by EBG6, hence $D_D^+ = 0$.

The independent-gauge map is the literal composition

$$\mathcal{H}_0/\mathcal{L}_0 \xrightarrow{\bar{T}} \mathcal{H}_1^4/\mathcal{L}_1 \xrightarrow{\bar{S}} \mathcal{H}_0/\mathcal{L}_+, \quad \bar{S}\bar{T} = q_+, \quad K_-^{-2} D_D^+ \preceq G_{\text{ind},D}^{\text{act}} \preceq G_{\text{cor},D}^{\text{act}} \preceq K_+^2 D_D^0. \quad (\text{NCR4})$$

The original identities $ST = I$, $T\mathcal{L}_0 \subset \mathcal{L}_1$ and $S\mathcal{L}_1 \subset \mathcal{L}_+$ prove well-definition and the composition, with all original Bezout coefficients retained. Taking the minimum of $\|Sy\|^2 \leq K_-^2 \|y\|^2$ over the identical target fibres proves the left inequality; the original correlated and independent fibres give the other two. Thus NCR2 is now an evaluated component of the exact NCR3–4 maps. Their full C_D and all incompatible gauges remain. The positive high-endpoint determinants in RPB19 retain their stated support-exception domain.

178 All companion maps and the exact proper-source term

The whole GKC coefficient complex is

$$0 \longrightarrow \mathcal{P}_{k-8} \xrightarrow{d^0} \mathcal{P}_{k-4}^4 \xrightarrow{d^1} \mathcal{P}_k^6 \xrightarrow{d^2} \mathcal{P}_{k+4}^4 \xrightarrow{d^3} \mathcal{P}_{k+8} \longrightarrow 0. \quad (\text{NCR5})$$

Its maps are the complete original signed convolution matrices GKC3–4. GKC5–11 proves the local contracting maps and full chart-overlap calculation that make this sequence globally exact. GKC14–25 proves its original-frame coefficient torsion is exactly one at these program degrees. At each common high cutoff $D = 2q_j - v - 1, 2q_j - v$, equip its five terms with the complete original evaluation Grams of GGE2. Their ranks are full in NCR1's high domain. For each p , let $\rho_{p,D}$ be the log squared Jacobian of $C^p/\ker d^p \rightarrow \text{im } d^p$, with the attained source quotient and target restriction. Then

$$\mathfrak{E}_{k,D} = -\rho_{0,D} + \rho_{1,D} - \rho_{2,D} + \rho_{3,D}, \quad |\mathfrak{E}_{k,D}| \leq B_{\text{Euler},D} = O_{\text{actual}}(q_k \log q_k). \quad (\text{NCR6})$$

For the determinant identity, factor each adapted full Gram into its boundary restriction and lift quotient. The adjacent image determinants cancel in the alternating product, giving the four displayed signs; the full original-frame torsion contributes $2 \log |\tau_k| = 0$ by its proved value. The finite bound is the sum of the exact GVJ boundary-annulus term and all GGE19 errors, as defined in PCJ5. This retains all five Gamma masses.

For any specified proper source $X \subset \mathcal{H}_0/\mathcal{L}_0$, keep NKC6's complement Z_c and its source quotient Gram G_c . Define, on the entire compatibility quotient,

$$\begin{aligned} H_{\text{comp}} &= Z_c^* \mathcal{K}^* \mathcal{V}_2^* \mathcal{V}_2 \mathcal{K} Z_c, & H_{\text{rem},X} &= Z_c^* \mathcal{V}^* P_{\text{rem},X} \mathcal{V} Z_c, \\ \chi_X &= \log \det(I + H_{\text{comp}}^{-1/2} H_{\text{rem},X} H_{\text{comp}}^{-1/2}), & r_c &= 3k^2 - 10k - 13, \\ \log \det \Omega_{X,D} &= \rho_{\text{comp},D} + \chi_{X,D} + \delta_{X,D}, \\ -r_c \log c_{+,D} &\leq \delta_{X,D} \leq -r_c \log c_{-,D}, & c_- &= \min(1, L_-^{-2}), \quad c_+ = \max(1, L_+^2). \end{aligned} \tag{NCR7}$$

The $L_{\pm}$ are the exact NKC3 native constants. Indeed the determinant of $G_c^{-1/2}(H_{\text{comp}} + H_{\text{rem},X})G_c^{-1/2}$ is $\exp(\rho_{\text{comp}} + \chi_X)$ by factoring H_{comp} . Conjugating the NKC14 form comparison bounds each eigenvalue of its ratio with the exact Ξ_X between c_- and c_+ . The determinant equality $\det \Xi_X = \det \Omega_X$ then gives the stated signs. This proves NCR7 using the entire source-complement projector. At high cutoffs NCR6 therefore yields

$$\log \det \Omega_{X,D} = \rho_{2,D} - \rho_{3,D} + \chi_{X,D} + \mathfrak{E}_{k,D} + \rho_{0,D} + \delta_{X,D}, \quad |\rho_{0,D}| \leq B_{0,D}. \tag{NCR8}$$

Here $B_{0,D} = 2(k-7)^2 \log \max(1, K_+, K_-)$. The equality $\rho_{\text{comp},D} = \rho_{1,D}$ follows from the identical original compatibility source quotient G_c and image restriction H_{comp} . Thus the accumulated finite error in NCR8 after retaining $\rho_2 - \rho_3 + \chi_X$ is bounded by $B_{\text{Euler},D} + B_{0,D} + r_c \max(-\log c_-, \log c_+)$. Both original low terms still use NCR7 with their full matrices.

The source-difference refinement is also exact. For $X \subset Y$ of dimension difference t , let $J_{\Delta} = A_{\Delta}^* W_c$ in the unchanged NKC native frames. Then $\Xi_X = \Xi_Y + J_{\Delta}^* J_{\Delta}$ and $\Gamma_X = \Gamma_Y + J_{\Delta}^* J_{\Delta}$. The determinant lemma and $c_- \Xi_Y \preceq \Gamma_Y \preceq c_+ \Xi_Y$ give

$$\begin{aligned} -\log \det \Omega_{Y/X} &= \chi_X - \chi_Y + \nu_{Y/X}, \\ t \log c_- &\leq \nu_{Y/X} \leq t \log c_+. \end{aligned} \tag{NCR9}$$

To check the bound with this actual rank, invert the preceding positive comparison and compress by J_{Δ} . After adding I_t , the two update matrices are bounded by c_+^{-1} and c_-^{-1} times each other. Multiplication of their t ordered eigenvalues and negation gives NCR9. No bound on an inverse of a coefficient frame is used. The full quotient $\Omega_{Y/X}$ is the Schur complement of Ω_X in Ω_Y .

In particular take $X = 0$ and an actual rank- t original source Y . Set $Z_{Y,D} = \chi_{\{0\},D} - \chi_{Y,D}$, and order the four cutoffs as $(q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v)$ with $\varepsilon = (-1, -1, +1, +1)$. The exact signed transport is

$$\begin{aligned} \sum_D \varepsilon_D \log \det \Omega_{Y,D} &= - \sum_D \varepsilon_D Z_{Y,D} + \mathcal{R}_Y, \\ t \left(\sum_{\text{low}} \log c_- - \sum_{\text{high}} \log c_+ \right) &\leq \mathcal{R}_Y \leq t \left(\sum_{\text{low}} \log c_+ - \sum_{\text{high}} \log c_- \right). \end{aligned} \tag{NCR10}$$

It follows by putting $\mathcal{R}_Y = - \sum_D \varepsilon_D \nu_{Y,D}$ in NCR9. Every term has the same actual source at its stated cutoff. For the actual rank $O(k)$ restrictions only this displayed error is $O_{\text{actual}}(k \log q_k)$. The further original invariant target quotient in NBG30–32 retains its whole kernel and native metric.

179 The complete low groups replace the earlier Euler-only calculation

In the CNC image complex at $D_e = q_j - v - 1 + e$, $e = 0, 1$, put $a_e = \max(v - e, 0)$ and $b_e = 8k + 56 + v - e$. Use precisely LCC5's full original Vandermonde-tail frames and LCC6's last map U_e of size b_e by $4a_e$. The complete result is

$$\begin{aligned} \kappa_e &= \dim \ker U_e, \quad 0 \leq \kappa_e \leq 3a_e, \quad 0 \leq \kappa_0 - \kappa_1 \leq 3, \\ 0 &\longrightarrow \ker \mathbf{t}_e \longrightarrow H^1(\mathcal{I}_e) \longrightarrow \ker U_e \longrightarrow 0, \quad \dim \ker \mathbf{t}_e = 8k + 24 - 3e, \\ H^2(\mathcal{I}_e) &\simeq \operatorname{coker} U_e, \quad H^p(\mathcal{I}_e) = 0 \quad (p \neq 1, 2), \\ \dim H^1(\mathcal{I}_e) &= 8k + 24 - 3e + \kappa_e, \quad \dim H^2(\mathcal{I}_e) = b_e - 4a_e + \kappa_e. \end{aligned} \tag{NCR11}$$

The exact maps are the positive double connecting lifts LCA9–11 and the analytic-image sequence LCC13. LCA19 gives every actual row of $\mathbf{t}_1$, and its full original rank is four; at $e = 0$ its target is zero. The remaining U_e is not replaced by a rank assumption. Its kernel projects injectively to three tail components by division by one actual unit $B_i(0) \neq 0$; the four added next-moment rows have their exact one relation, giving the first line. All these statements and the full row coordinates are proved in LCC25–27 and LCA37–40. In particular subtracting the two last dimensions reproduces exactly CNC16's Euler values, including $v = 0$.

Let $\mathcal{B} \subset \mathcal{Z}_1 \subset \mathcal{Z}_0 \subset \mathcal{C}_{v,j}^4$ be the actual common boundaries and the two low cycle spaces. Their quotient metrics are the full LCC15–17 Schur Grams. The identity on coefficient representatives gives an injection

$$\mathcal{Z}_1/\mathcal{B} \hookrightarrow \mathcal{Z}_0/\mathcal{B}, \quad \operatorname{coker} \simeq \mathcal{Z}_0/\mathcal{Z}_1, \quad c := \dim(\mathcal{Z}_0/\mathcal{Z}_1) = 3 + \kappa_0 - \kappa_1 \in [3, 6]. \tag{NCR12}$$

Injectivity uses the identical boundary subspace. The quotient description follows by taking that same subspace twice; the dimensions follow from NCR11. This is the exact replacement for an Euler-only inference.

180 The residual/kernel metric maps and both computed contractions

Let $\mathbb{B}$ be the fourfold original DNE kernel map and $\mathbb{J} = \ker \mathbb{B}$. For $e = 0, 1$ define

$$S_e = \mathcal{Z}_e, \quad W = \mathcal{B} + \mathbb{J}, \quad K_e = \mathcal{B} + (\mathcal{Z}_e \cap \mathbb{J}) = S_e \cap W, \quad T_e = \mathcal{Z}_e + \mathbb{J} = S_e + W. \tag{NCR13}$$

The maps $S_e/K_e \rightarrow \mathbb{B}\mathcal{Z}_e/\mathbb{B}\mathcal{B} \rightarrow T_e/W$, $[z] \mapsto [\mathbb{B}z] \mapsto [z]$, are isomorphisms. Indeed a kernel value means $\mathbb{B}z = \mathbb{B}b$ for an original boundary b , hence $z - b \in S_e \cap \mathbb{J}$; changing the second lift adds exactly $\mathbb{J}$. This proves both full kernel claims.

For the remainder of this degree- j native metric calculation set $s = 1$ or $s = j = k + 4$, retaining that same choice at every endpoint. At every original source index $q_j - 1 \leq N \leq 2q_j$ give this fixed ambient space the fourfold original metric $G_N = \operatorname{diag}(\mathcal{D}_{j,N-v}^{(s)})$. The source minimum for the displayed map permits exactly K_e ; the target minimum permits exactly W . Their positive quotient Grams $A_{e,N}, H_{e,N}$ therefore satisfy

$$\omega_e(N) = \frac{\det H_{e,N}}{\det A_{e,N}} = \det \Omega_{\Psi,e,N} \in (0, 1]. \tag{NCR14}$$

The two-stage minima over original boundaries and invariant kernel combine into those exact sums, proving the metric dictionary to LCC21–23. At $N = q_j - 1 + e$ these are exactly the two native quotient metrics $Q_{\Psi,e}$ and $H_{\Psi,e}$ of LCC21–23.

In fixed frames write $g_X(N) = \det(X_X^* G_N X_X)$ and keep the complete coordinate changes $[X_{K_e}, F_e] = X_{S_e} R_{S_e}$, $[X_W, F_e] = X_{T_e} R_{T_e}$. The full Schur determinants yield $\omega_e = c_e g_{T_e} g_{K_e} / (g_W g_{S_e})$ with $c_e = |\det R_{T_e}|^2 / |\det R_{S_e}|^2$. The exact sequence

$$0 \longrightarrow K_0/K_1 \longrightarrow S_0/S_1 \longrightarrow T_0/T_1 \longrightarrow 0 \quad (\text{NCR15})$$

uses inclusion and the identity on representatives. Its last kernel is K_0/K_1 , since $S_0 \cap (S_1 + W) = S_1 + K_0$. Thus its first and last dimensions sum to c . Factoring their exact quotient Grams $Q_{K,N}, Q_{T,N}, Q_{S,N}$ gives LFC12 with every fixed coordinate determinant retained. The common four-endpoint return, denoted $\mathcal{R}$, annihilates those fixed constants. FRE's full bound $G_{q_j-1} \preceq G_N \preceq A G_{q_j-1}$, where

$$A = \max\{1, q_j \mathcal{B}_{2q_j-v, s, j, A}^2\} (M_{j, s}^{\text{rec}})^2,$$

passes to each identical quotient fibre and gives $0 \leq \mathcal{R}(\log \det Q_X) \leq 2 \dim(X_0/X_1) \log A$. Consequently

$$|\mathcal{R}(\log \omega_0) - \mathcal{R}(\log \omega_1)| \leq 2c \log A \leq 12 \log A = o(kq_k), \quad 0 \leq \mathcal{V}_{\text{miss}}^{(s)} \leq 2c \log A. \quad (\text{NCR16})$$

This proves the difference of the two fixed-flag losses and the return of the exact missing quotient. It retains their common long-return loss.

For the adjacent actual low comparison use the complete original rows T, V through $n-1$, $n = q_j - v$, and their next rows t_n, v_n . With

$$\begin{aligned} C_L &= V^* V, \quad D_0 = T^* (I - V C_L^{-1} V^*) T, \\ \zeta &= v_n C_L^{-1} v_n^*, \quad \rho = t_n - v_n C_L^{-1} V^* T, \\ \beta_n &= 4 \log \left(1 + \frac{\rho D_0^{-1} \rho^*}{1 + \zeta} \right), \end{aligned} \quad (\text{NCR17})$$

completion of the square over every lower coefficient proves the one-block update $D_0 \mapsto D_0 + \rho^* \rho / (1 + \zeta)$. The fourfold ambient update has rank at most four. For any fixed flag metric Q , interpolation between the two ambient metrics gives $(\log \det Q)' = \text{Tr}(BP)$ with $B \succeq 0$ and P the actual orthogonal projector of its minimum-section frame. Thus $0 \leq \text{Tr}(BP) \leq \text{Tr} B$. Integration bounds the flag determinant increment by the full ambient increment β_n . FRE bounds its four nontrivial relative eigenvalues by A . Applying this to the actual common flag gives

$$\begin{aligned} 0 &\leq -\log \det \Omega_{\text{ret}} \leq \beta_n \leq 4 \log A = o(kq_k), \\ -\beta_n &\leq \log \det H_0 - \log \det H_1 - \log \det G_{\text{miss}} + 2 \log |\det W_{\text{low}}| \leq 0. \end{aligned} \quad (\text{NCR18})$$

The second line substitutes this interval into the exact LCC29 factorization. It keeps the actual missing Gram and coordinate determinant, and evaluates precisely the previously retained adjacent common contraction. The original long-return loss in NCR14, the complete residual overlap in LCC20/24 and all four original signed source returns remain in the subsequent calculation.

The connecting image and its literal boundary denominator. For the same fixed degree- j target forms as NCR13–18, write $(s_0, s_1) = (1, 1)$ or (k, j) , so its target order is $s = s_1$. Let $\mathcal{S}_e = \text{im}[B_{\text{cof}}, F_e]$ be CIT10's actual connecting-cycle space. The map $[h] \mapsto [F_e h]$ identifies its quotient by $\mathcal{B} = \text{im} B_{\text{cof}}$ with the image of the original positive connecting homomorphism in LCC13. Its source is the full primitive quotient PDS8, transported by $\Theta_e = C_e^T P_e$; its original source and target Grams are PDI2 and CIT11, with the full low-cutoff source determinant now evaluated by PSR4–15. Both independent-gauge terms $\chi_{X_k, D_e}, \chi_{Y_e, D_e}$ remain in that target formula.

At every later original target cutoff set $K_{\text{conn}, e} = \mathcal{S}_e \cap W$, $T_{\text{conn}, e} = \mathcal{S}_e + W$, and give their quotient map the original two minima, as in CIT21. The actual inclusion $\mathcal{S}_e \subset \mathcal{Z}_e$ has quotient dimension $\kappa_e \leq 3 \max(v - e, 0)$. The kernel and sum quotients for this inclusion have dimensions

adding to κ_e , by the identity on representatives. The full three-quotient Gram factorization and FRE's same-form bound give

$$|\mathcal{R} \log \omega_e - \mathcal{R} \log \omega_{\text{conn},e}| \leq 2\kappa_e \log A_{j,s_1}^{\text{end}} = o(kq_k). \quad (\text{NCR19})$$

This is the explicit CIT22–23 morphism and estimate; it does not replace the full cohomology Gram by the source Gram. In the common boundary denominator W , the exact chain $\mathbb{J} \subset W_{\text{inv}} \subset W$ has Gram factors H_{inv} and Q_b^{CIT} from CIT24–26. RQB3–20 computes the full second quotient and its source kernel. RQB31–32 then preserves the cross-ratio identity with the directed residual term $-4\widehat{F}_j^{(s_1)} + \mathcal{V}_{\text{inv}}^{\text{target}} \leq -\mathcal{V}_b \leq 0$. The same Taylor-complement values also compare the two defining low-cutoff source orders. Writing $\mathcal{X}_e^{(s_0,s_1)} = \chi_{X_k,D_e}^{(s_0,s_1)} - \chi_{Y_e,D_e}^{(s_0,s_1)}$, the exact maps PSR23–24 and finite subtraction PSR26–27 give

$$\log \frac{\det H_{\text{conn},e}^{(k,j)}}{\det H_{\text{conn},e}^{(1,1)}} = -\mathcal{X}_e^{(k,j)} + \mathcal{X}_e^{(1,1)} + o(kq_k). \quad (\text{NCR20})$$

Its exact residual is the PSR26 source ratio plus the difference of CIT17's original finite errors. Thus the common loss in NCR14 is transported through its actual connecting and boundary maps with the residual, source-complement and missing quotient factors retained.

Finite source and image intervals with the original low ranks. The exact map in NCR20 is now bounded in the same values by NSR5–9. Put $j = k+4$, $D_e = q_j - v - 1 + e$, $r_e = 8k + 24 - 3e$ and retain the two original pairs $p = 0 : (1, 1)$ and $p = 1 : (k, j)$. Use $\mathbf{a}_s, \mathbf{b}_{s,D}$ with their full product definitions in NSR2, and set

$$\begin{aligned} s_e^- &= -r_e \log \mathbf{b}_{k,D_e}, & s_e^+ &= -r_e \log \mathbf{a}_k, & t_e^- &= -r_e \log \mathbf{b}_{j,D_e}, & t_e^+ &= -r_e \log \mathbf{a}_j, \\ b_e^- &= \epsilon_e^{1,\text{lo}} - \epsilon_e^{0,\text{hi}}, & b_e^+ &= \epsilon_e^{1,\text{hi}} - \epsilon_e^{0,\text{lo}}, \\ s_e^- &\leq \Delta \log \det H_{\text{src},e} \leq s_e^+, & t_e^- &\leq \Delta \log \det H_{\text{conn},e} \leq t_e^+, \\ s_e^- + b_e^- - t_e^+ &\leq \Delta(\chi_{X_k,D_e} - \chi_{Y_e,D_e}) \leq s_e^+ + b_e^+ - t_e^-. \end{aligned} \quad (\text{NCR21})$$

Here $\epsilon_e^{p,\text{lo/hi}}$ are the complete original CIT17 error endpoints. The same fixed Taylor complement and its unique target analytic coefficient F_e identify the value spaces. Applying the full dual comparison to every member of the source and image fibres, then taking the attained minima, proves the two determinant intervals. Their exact CIT17 difference proves the third. The previous PSR finite interval remains available for intersection on its own domain, with all its original factors.

For the actual fixed later connecting flags put $B = B_{\text{cof}}$, $\mathbb{J} = \ker \mathbb{B}$, $S_{\text{conn},e} = \text{im}[B, F_e]$, $W = \mathbb{J} + \text{im } B$ and $K_{\text{conn},e} = S_{\text{conn},e} \cap W$. The identity on representatives gives

$$\begin{aligned} S_{\text{conn},e}/K_{\text{conn},e} &\xrightarrow{\sim} (S_{\text{conn},e} + W)/W, \\ c_{\text{conn},e}^{\text{rank}} &= \text{rank}[\mathbb{B}B, \mathbb{B}F_e] - \text{rank}(\mathbb{B}B), \\ |\Delta \mathcal{R} \log \omega_{\text{conn},e}| &\leq c_{\text{conn},e}^{\text{rank}} W_j, \\ |\Delta \mathcal{R} \log \omega_e| &\leq (c_{\text{conn},e}^{\text{rank}} + d_T) W_j, \\ d_K + d_T &= \kappa_e \leq 3 \max(v - e, 0). \end{aligned} \quad (\text{NCR22})$$

The map sends $[z]$ to $[z]$; its zero fibre is exactly $K_{\text{conn},e}$. The full-cycle inclusions give the exact sequence NSR14 and hence the second rank $c_{\text{conn},e}^{\text{rank}} + d_T$. Both contraction metrics use the same quotient values with their complete two minima, so their common mass factor cancels and each rank- r ratio changes by at most $r\omega_{j,D}$ at one endpoint. Summing the absolute widths at $D = q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v$ gives rW_j . This preserves the original NCR19 comparison and the attained metric on its tail quotient. The individual absolute long contraction remains represented by the full CPQ source and target quotients.

181 The native target loss in the earlier matched quotient

Keep the actual simple quartet, period, branch and nonreduced conductor, $k = 4l+1 \geq 9$, $j = k+4$, and $D \geq q_j - v - 1$. The exact original source-order pairs are $(s_0, s_1) = (1, 1)$ or (k, j) . All original DNE, KRC, DFX and NBG maps are those in ITL1–6:

$$\begin{aligned} \mathsf{T}_a &= \alpha_A N_a Z_{\text{exp},a}, \quad \alpha_A = v!/\mu_v, \quad M_i = \mathsf{T}_j^{-1} m_{i,k} \mathsf{T}_k, \quad F = \text{stack}_i(M_i J_0), \\ \mathcal{B} &= \text{diag}(B_1, 4), \quad \ker \mathcal{B} = \text{im diag}(J_1, 4), \quad E = \ker(\mathcal{B}F), \\ G &= J_0^* D_{k,D}^{(s_0)} J_0, \quad \mathcal{D}_D = \text{diag}(D_{j,D}^{(s_1)}, 4), \quad A = F^* \mathcal{D}_D F, \quad Q = G/E. \end{aligned} \tag{NFR1}$$

Here $\text{diag}(P, 4)$ means four copies, and every quotient P/E is its attained minimum in the fixed lifts $[Z_E, Z_R]$ of ITL4. The actual image frame is $U = \mathcal{B}FZ_R$. Its native target form is $H = U^*(\mathcal{B}\mathcal{D}_D^{-1}\mathcal{B}^*)^{-1}U$ and its source pullback form before the last target quotient is $A_E = A/E$. A lift of Ub is exactly $FZ_R b + \ker \mathcal{B}$. Its complete minimum over that kernel gives ITL8's H . The source minimum permits just FZ_E . This proves the typed identification of ITL9's $\omega_D = \det H / \det A_E$ with the last target loss in NBG31–32, retaining the full source kernel.

Let C_D be the original correlated pullback on the invariant source. Schur factorization in $[Z_E, Z_R]$ gives

$$\begin{aligned} \log \det(Q^{-1/2} H Q^{-1/2}) &= e_{\text{cor},D} + \log \omega_{V,D} - \log \omega_{E,D} + \log \omega_D \\ &= \log \frac{\det \mathcal{H}_{V,D}}{\det \mathcal{H}_{E,D}} + \log \omega_D + \epsilon_D, \quad -a_D \leq \epsilon_D \leq b_D, \\ a_D &= 2r \log \max(1, K_-) + r \log c_+, \quad b_D = 2r \log \max(1, K_+) - r \log c_-, \quad r = \dim V - \dim E. \end{aligned} \tag{NFR2}$$

In the first line $e_{\text{cor}} = \log \det(C_D/E) - \log \det(G/E)$ and the two restriction losses are exactly ITL13. Their determinant ratio equals $\det(A/E)/\det(C_D/E)$ because the same full frame factor cancels. The complete compatibility forms in the second line are ITL18. Their common denominator G_c cancels before estimating. Their identical positive update has rank at most r ; compression of NKC's two-sided comparison bounds its logdet difference between $-r \log c_+$ and $-r \log c_-$. Adding the correlated quotient interval ITL17 proves exactly NFR2. This is ITL26–30 applied to the original matched quotient, not a replacement of its target metric.

182 Every original row and both signs in the long return

Put $F_E = FZ_E$, $F_R = FZ_R$, and let $\widehat{W} = [F_E, W_R]$ span $\ker \mathcal{B}$. The four fixed flags are the images of $Y_K = F_E$, $Y_S = [F_E, F_R]$, $Y_W = \widehat{W}$, $Y_T = [\widehat{W}, F_R]$. They obey $S \cap W = K$, because $Fa \in \ker \mathcal{B}$ is exactly $a \in E$. Their raw frames are $Z_{Y,D} = [\mathcal{V}_D, \mathcal{T}_D Y]$, using all original lower rows and all numerator rows IRR7–11. Analytic numerator division is the literal recursion

$$f_{a,n} = \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v},$$

and the polynomial coefficients and complete row formulas remain IRR8–9, with i^{-n} , $h_{n,s_1} = (2\pi)^{s_1/2} n! (s_1/2)_n$ and center $j/2 - 4$. Thus there is no unspecified substitute evaluator.

Write $G_{Y,D} = Z_{Y,D}^* Z_{Y,D}$ and $L_{Y,n} = z_{Y,n} G_{Y,n-1}^{-1} z_{Y,n}^*$ for the original new four-row block. Eliminating the entire lower block and then the actual source or target kernel gives $\omega_D = \det G_{T,D} \det G_{K,D} / (\det G_{W,D} \det G_{S,D})$. The determinant lemma for each raw Gram

therefore proves

$$\begin{aligned}
& \log \omega_n - \log \omega_{n-1} = \gamma_{H,n} - \gamma_{A,n}, \\
& \gamma_{H,n} = \log \det(I_4 + L_{T,n}) - \log \det(I_4 + L_{W,n}) \geq 0, \\
& \gamma_{A,n} = \log \det(I_4 + L_{S,n}) - \log \det(I_4 + L_{K,n}) \geq 0, \\
& \mathcal{L}_{\text{inv}} = \sum_{n=d+1}^{d+q_j+1} \vartheta_n(\gamma_{H,n} - \gamma_{A,n}), \quad d = q_j - v - 1, \\
& \vartheta_{d+1} = \vartheta_{d+q_j+1} = 1, \quad \vartheta_n = 2 \quad (d+2 \leq n \leq d+q_j).
\end{aligned} \tag{NFR3}$$

For positivity, transform each pair (K, S) or (W, T) to its old minimum-section columns. The larger raw covariance is $L_Y = L_X + R_X Q_X^{-1} R_X^*$; factoring its positive determinant gives IRR21's nonnegative increment. The common raw lower minimum is never taken separately for different rows. Finally telescope over $[d, d+q_j]$ and $[d+1, d+q_j+1]$. Their overlap gives the exact displayed weights.

183 The simultaneous native budget and the exact signed receiver

For the same original ordered generalized eigenvalues $\lambda_i(D)$ of $(H_D, A_{E,D})$ put $u_i(D) = \text{arcosh}(\lambda_i(D)^{-1/2})$. On the positive path joining two actual adjacent metrics, the original minimum lifts give orthonormal paired vectors p_i, q_i with $p_i^* q_m = \sqrt{\lambda_m} \delta_{im}$. Consequently $J_i = q_i q_i^* - p_i p_i^*$ has eigenvalues $\pm \sqrt{1 - \lambda_i}$ on mutually orthogonal planes. The full repeated-eigenvalue and zero-angle arguments IF5–6 and FAB6–9 retain the ordered spectrum through crossings and give, almost everywhere,

$$(\log \lambda_i)' = \text{Tr}(B J_i), \quad \sum_i |(\log \lambda_i)'| \leq \text{Tr } B, \quad \sum_i |u_i'| \leq \frac{1}{2} \text{Tr } B.$$

Indeed sign maximization of the trace takes place on those disjoint planes, so its operator norm is at most one regardless of their count. Here $B = \mathcal{D}^{-1/2} \Delta \mathcal{D}^{-1/2} \succeq 0$. Its integral is the exact ambient logdet increment. The complete lower row minimization gives, with all rows through $n-1$, $C_L = V^* V$, $D_0 = T^*(I - V C_L^{-1} V^*) T$, $\zeta_n = v_n C_L^{-1} v_n^*$ and $\rho_n = t_n - v_n C_L^{-1} V^* T$. Completing its square proves the one-target update $\rho_n^* \rho_n / (1 + \zeta_n)$ and hence

$$\begin{aligned}
& \beta_n = 4 \log \left(1 + \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n} \right), \\
& \sum_i |\log \lambda_i(n) - \log \lambda_i(n-1)| \leq \beta_n, \quad \sum_i |u_i(n) - u_i(n-1)| \leq \beta_n/2, \quad |\mathcal{L}_{\text{inv}}| \leq \sum_n \vartheta_n \beta_n.
\end{aligned} \tag{NFR4}$$

No image-rank factor multiplies this budget.

At the four ordered endpoints $D_i = d, d+1, d+q_j, d+q_j+1$, $\varepsilon = (-1, -1, +1, +1)$, substituting NFR3 into NFR2 gives

$$\begin{aligned}
& \sum_i \varepsilon_i \log \det \mathcal{B}_{X, D_i} = \sum_i \varepsilon_i \log \frac{\det \mathcal{H}_{V, D_i}}{\det \mathcal{H}_{E, D_i}} + \sum_n \vartheta_n(\gamma_{H,n} - \gamma_{A,n}) + \mathcal{E}_{\text{fin}}, \\
& -b_{D_1} - b_{D_2} - a_{D_3} - a_{D_4} \leq \mathcal{E}_{\text{fin}} \leq a_{D_1} + a_{D_2} + b_{D_3} + b_{D_4}.
\end{aligned} \tag{NFR5}$$

The signs follow by negating the two low endpoint errors. Only this finite comparison error is $O_{\text{actual}}(k \log q_k)$. The two positive row sums and the full compatibility ratio are retained in their exact source metrics. The $q_j + 1$ row degrees are the original long interval, and neither its budget nor its signed difference is assigned an arithmetic leading coefficient. When $r = 0$, both quotient forms are empty and all actual loss terms vanish exactly. The common target-source mass cancels from a ratio only through the equal quotient ranks; it remains in every row and absolute native Gram in NFR1–4.

The exact positive target budget in the matched relative return. Keep NFR1–5’s source quotient Q_{inv} , target H_{inv} , and matched relative operator $\mathcal{R}_X$ in their original fixed quotient-value coordinates. Their literal determinant equality is $\det H_{\text{inv}} = \det Q_{\text{inv}} \det \mathcal{R}_X$. Use RQB3’s actual residual boundary quotient $\mathbb{C}^{m_k}/(\text{im } J_0 + \ker \Phi_b)$, where $\Phi_b = \mathbb{B}B_{\text{cof}}$, and its full target minimum Q_b^E from RQB14. Completing the unchanged image $[U_R, U_b]$ to the full target kernel-dual frame gives the remaining quotient Q_o in RQB27. Therefore

$$\begin{aligned} \mathcal{R} \log \det \mathcal{R}_X &= 4\widehat{F}_j^{(s_1)} - \mathcal{V}_b - \mathcal{V}_o - \mathcal{R} \log \det Q_{\text{inv}}, \\ 0 &\leq \mathcal{R} \log \det Q_{\text{inv}} + \mathcal{R} \log \det \mathcal{R}_X = \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4U_{j,s_1}^{\text{native}} - \mathcal{V}_b. \end{aligned} \quad (\text{NFR6})$$

The first equality substitutes the exact NFR determinant ratio into RQB28; the finite second line uses RQB35. Every occurrence of $\mathcal{R}$ has NFR5’s ordered four endpoints and signs, with the shift $D = N - v$ retained. The quotient-value frame factors cancel only in that return. The full original source Gram Q_{inv} , the positive residual return $\mathcal{V}_b$, and the positive missing target return $\mathcal{V}_o$ are all present. The row difference, compatibility errors and asymmetric interval in NFR5 remain exact; the displayed bound does not assign an individual leading coefficient to the common long return.

Both source orders in the complete matched target allocation. Use $p = 0 : (s_0, s_1) = (1, 1)$ and $p = 1 : (k, j)$, $j = k + 4$. Keep the exact original Q_{inv} and the fixed quotient values Z_R of NFR1–6. Write $V_Q^p = \mathcal{R} \log \det Q_{\text{inv}}^p$ and $\mathcal{M}^p = \mathcal{R} \log \det \mathcal{R}_X^p$. The original remaining quotient now splits under the actual map $E_e = [0 \ 0 \ I]C_T^{-1}\mathbb{B}F_e$ as in CPQ4–8: $V_o^p = V_t^p + V_{\text{rem}}^p$, where V_t is its connecting restriction and V_{rem} its full further quotient. Hence NFR6 receives the exact refinement

$$\begin{aligned} \mathcal{M}^p &= 4\widehat{F}_j^{(s_1)} - V_b^p - V_t^p - V_{\text{rem}}^p - V_Q^p, \quad \mathcal{M}^p + V_Q^p = V_I^p \geq 0, \\ \mathcal{M}^p &= \mathcal{L}_e^p - V_t^p + V_s^p + V_A^p - V_Q^p, \\ V_A^p &= V_Q^p + \mathcal{R} \log \frac{\det A_{\text{inv}}^p}{\det Q_{\text{inv}}^p}. \end{aligned} \quad (\text{NFR7})$$

The first identity follows from the full RQB27 determinant and CPQ8’s second attained quotient. The second substitutes $\mathcal{L}_e = V_I + V_t - V_s - V_A$. The final identity uses the same Z_R values for the two actual source forms, so the ratio is the determinant of their positive relative operator. The target pullback A_{inv} and independent degree- k source Q_{inv} thus both remain.

Let $r_I = j_{0,k} - \epsilon$, and evaluate all NSR2 widths at $D_i = q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v$. Then

$$\begin{aligned} \Lambda^- &= -r_I(\Omega_j^{\text{hi}} + \Omega_k^{\text{lo}}) \leq \Delta\mathcal{M} \leq r_I(\Omega_j^{\text{lo}} + \Omega_k^{\text{hi}}) = \Lambda^+, \\ U_{\mathcal{M}}^p &= 4U_{j,s_1}^{\text{native}} - V_b^p - V_t^p - V_{\text{rem}}^p - V_Q^p, \quad -V_Q^p \leq \mathcal{M}^p \leq U_{\mathcal{M}}^p. \end{aligned} \quad (\text{NFR8})$$

The source degree is k , with its original centre, at these same target-based D_i ; its displacement is $b = 8k + 24$ at the two lows and $2b$ at the two highs. The comparison of V_I has rank r_I in target order j , that of V_Q has the same rank in source order k . Subtracting their two directed intervals gives $\Lambda^\pm$. The upper endpoint retains the full RQB35 cap on its proved finite domain, together with both missing factors. Consequently both order intervals can be used simultaneously:

$$\begin{aligned} \max\{-V_Q^1, -V_Q^0 + \Lambda^-\} &\leq \mathcal{M}^1 \leq \min\{U_{\mathcal{M}}^1, U_{\mathcal{M}}^0 + \Lambda^+\}, \\ \max\{-V_Q^0, -V_Q^1 - \Lambda^+\} &\leq \mathcal{M}^0 \leq \min\{U_{\mathcal{M}}^0, U_{\mathcal{M}}^1 - \Lambda^-\}. \end{aligned} \quad (\text{NFR9})$$

Translate the complete pair-zero interval by $[\Lambda^-, \Lambda^+]$ and intersect with pair one. Reversing that difference gives the other line. This is the exact same matched return, not a replacement of its source quotient by an invariant source image.

The earlier full compatibility error remains in its own original coordinates. Put $\mathcal{C}^p = \mathcal{R} \log(\det \mathcal{H}_V^p / \det \mathcal{H}_E^p)$ and retain NFR2’s endpoint constants

$$a_i^p = 2r_I \log \max(1, K_-^p(D_i)) + r_I \log c_+^p(D_i), \quad b_i^p = 2r_I \log \max(1, K_+^p(D_i)) - r_I \log c_-^p(D_i).$$

Then, with all original source-complement projections retained,

$$\begin{aligned} E_p^- &= -b_0^p - b_1^p - a_2^p - a_3^p, & E_p^+ &= a_0^p + a_1^p + b_2^p + b_3^p, \\ \Lambda^- - r_I W_j - E_1^+ + E_0^- &\leq \Delta\mathcal{C} \leq \Lambda^+ + r_I W_j - E_1^- + E_0^+. \end{aligned} \quad (\text{NFR10})$$

Indeed NFR5 gives $\mathcal{M}^p = \mathcal{C}^p + \mathcal{R} \log \omega_{\text{inv}}^p + \mathcal{E}^p$ with $E_p^- \leq \mathcal{E}^p \leq E_p^+$. Subtract this identity in both orders and use the rank- r_I same-target contraction width $r_I W_j$. This proves every low and high error sign in NFR10.

The same common-angle enclosure in the matched source quotient. For each original pair $p = 0, 1$ keep the full NFR7 identity and put $D_e^p = -V_t^p + V_s^p + V_A^p - V_Q^p$. Let L_-^p, L_+^p be SFC9–10 evaluated in that pair, with all four actual source and target Grams in SFC2. The matched return therefore satisfies

$$L_-^p + D_e^p \leq \mathcal{M}^p \leq L_+^p + D_e^p. \quad (\text{NFR11})$$

This is the image of CIT35 under the exact translation $\mathcal{M}^p = \mathcal{L}_e^p + D_e^p$, proved in NFR7. In particular the determinant of the relative positive operator A_{inv}^p versus Q_{inv}^p remains in D_e^p . The coefficient values agree through Z_R , but their two source metrics and their centres remain the original ones. Intersect NFR11 with NFR8 on the cap's original domain. Writing the resulting endpoints as M_-^p, M_+^p , the complete two-order refinement is

$$\begin{aligned} \max\{M_-^1, M_-^0 + \Lambda^-\} &\leq \mathcal{M}^1 \leq \min\{M_+^1, M_+^0 + \Lambda^+\}, \\ \max\{M_-^0, M_-^1 - \Lambda^+\} &\leq \mathcal{M}^0 \leq \min\{M_+^0, M_+^1 - \Lambda^-\}. \end{aligned} \quad (\text{NFR12})$$

The original interval $\Lambda^- \leq \mathcal{M}^1 - \mathcal{M}^0 \leq \Lambda^+$ is NFR8, with degree- k source widths at the displaced target cutoffs. Adding it to pair zero, or subtracting it from pair one, gives the two lines. Every intersection contains its actual matched return and retains NFR10's complete compatibility errors.

The equilibrium interval with the original displaced source. Keep NFR7–12's $V_Q^p = \mathcal{R} \log \det Q_{\text{inv}}^p$ and $\mathcal{M}^p = V_I^p - V_Q^p$ in the same Z_R values. Use $\Phi_j^\pm$ at target degree j and $\Phi_k^\pm$ at source degree k , both at the actual target-window integers D_i . The source polynomial degree is $D_i - q'_k$, with maximum $k^2 + 34k + 1 - v$. For $k \geq 37$,

$$2q_k - (k^2 + 34k + 1 - v + (k-1)/4) = k^2 - \frac{121}{4}k + \frac{5}{4} + v \geq 251 + v.$$

Thus EFP28 verifies the required degree range; the original GPA radius and the individual EOR tests remain explicit further tests. The source shifts are $8k + 24$ at the lows and $16k + 48$ at the highs. Both rank- r_I determinants then give

$$\Lambda_{\text{new}}^- = r_I(\Phi_j^- - \Phi_k^+) \leq \Delta\mathcal{M} \leq r_I(\Phi_j^+ - \Phi_k^-) = \Lambda_{\text{new}}^+. \quad (\text{EFR7})$$

This subtracts the actual independent source interval from the target interval. It keeps the distinct centres c'_k, c'_j and, before the four return, the scalar $-r_I \log(\kappa_j/\kappa_k)$.

Use the translated common interval $[\tilde{L}_-^p + D_e^p, \tilde{L}_+^p + D_e^p]$, $D_e^p = -V_t^p + V_s^p + V_A^p - V_Q^p$, and intersect with NFR8 on its cap domain. Calling the resulting endpoints $\widehat{M}_-^p, \widehat{M}_+^p$, NFR12 becomes exactly EFP30 with $\Lambda_{\text{new}}^\pm$. The full relative determinant $\mathcal{R} \log(\det A_{\text{inv}}/\det Q_{\text{inv}})$ remains in D_e^p by NFR7. Finally NFR10's actual endpoint errors $E_p^\pm$ give

$$\Lambda_{\text{new}}^- - r_I W_j - E_1^+ + E_0^- \leq \Delta\mathcal{C} \leq \Lambda_{\text{new}}^+ + r_I W_j - E_1^- + E_0^+. \quad (\text{EFR8})$$

This follows by subtracting the same-target invariant contraction and the full error difference from the exact NFR5 identity. No error constant is replaced by a Gamma-order scalar.

184 The original primitive source and its adjacent Gamma metric

This calculation continues the complete LCA15–21 polynomial quotient construction. Its new conclusions are the explicit codimension-three truncation, its exact rank-one native metric change, and a finite coefficient-product bound for that change. All spaces and measures below are the original ones. The connecting target metric and its later long return remain the literal target metrics; the source computation here is at the two specified low cutoffs.

184.1 Original objects and the polynomial quotient map

Fix the actual simple quartet, its invertible period matrix H , logarithm branch, and full conductor coefficients. Retain

$$\begin{aligned} \beta_* &= \frac{1}{2} - \delta - i\gamma, & \beta_{n,a,b} &= n\beta_* + 2a\delta + 2bi\gamma, & \delta, \gamma &> 0, \\ E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b}z}, & \mu_r &= E_A^{(r)}(0), & v = \text{ord}_0 E_A \leq 80, & \mu_v \neq 0, \\ L_i(Z, W) &= (H(ZW, Z, W, 1)^\top)_i, & b_{i,a,b} &= [Z^a W^b] L_i^4, & B_i(z) &= \sum_{a,b=0}^4 b_{i,a,b} e^{\beta_{4,a,b}z}, & (\text{PDS1}) \\ b_i &= B_i(0) = \sum_{a,b=0}^4 b_{i,a,b}, & \sum_{i=1}^4 D_i(z) B_i(z) &= 1. \end{aligned}$$

The D_i are the proved original entire Bezout expressions. In particular at least one b_i is nonzero. Keep the full biform conductor, including repeated components and zero edge coefficients. Set

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q_k &= (k + 1)^2, & Q &= (k + 5)^2, \\ e &\in \{0, 1\}, & D_e &= Q - v - 1 + e, & \ell_e &= 8k + 23 + e, & d_e &= \ell_e + 1, \\ \chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), & (\mathcal{T}_{AP})(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), & (\text{PDS2}) \\ (\mathcal{T}_{B_i} p)(S) &= \sum_{a,b=0}^4 b_{i,a,b} p(S + \beta_{4,a,b}). \end{aligned}$$

All dual functionals are complex-linear. For a finite germ F through degree D_e , let $\Lambda_F(S'^r) = F^{(r)}(0)$. Let $\mathcal{H}_e = \mathbb{C}[S']_{\leq D_e}^\vee$, and let $\mathcal{S}_e \subset \mathcal{H}_e$ be the jets of the original analytic germs

$$F(z) = \frac{N(z)}{E_A(z)}, \quad N(z) = \sum_{a,b=0}^k n_{ab} e^{\beta_{k,a,b}z}, \quad N^{(r)}(0) = 0 \quad (0 \leq r < v). \quad (\text{PDS3})$$

Define the unchanged relation polynomials and quotient coordinates by

$$r_a(S') = \mathcal{T}_A(\chi_k(S) S^a), \quad 0 \leq a \leq \ell_e, \quad \pi_e(F) = (\Lambda_F(r_a))_{a=0}^{\ell_e}. \quad (\text{PDS4})$$

Then π_e is onto and $\ker \pi_e = \mathcal{S}_e$. Here is the complete kernel argument. The distinct upper k nodes give a Vandermonde matrix, whose first v rows have rank v since $v \leq 80 < q_k$. The numerator space in PDS3 therefore has dimension $q_k - v$. Its jet map is injective: if F vanishes through degree D_e , then $N = E_A F$ vanishes through degree $D_e + v \geq q_k - 1$; the first q_k Vandermonde rows force every $n_{ab} = 0$. The Leibniz identity gives $\Lambda_F(\mathcal{T}_{AP}) = \Lambda_{E_A F}(p)$, so all r_a annihilate PDS3. Finally

$$\mathcal{T}_A S^m = \sum_{r=0}^m \binom{m}{r} \mu_r S'^{m-r}, \quad \deg r_a = q_k + a - v, \quad \text{lc}(r_a) = \mu_v \binom{q_k + a}{v}. \quad (\text{PDS5})$$

These successive degrees prove independence. Their common kernel has dimension $D_e + 1 - d_e = q_k - v$, proving the assertion. This is precisely the LCA16 quotient, with no new row norm. Every original lower degree- $(k - 8)$ exponential is in $\mathcal{S}_e$, since its product with E_A has an allowed degree- k numerator. Thus taking the original lower-gauge quotient first and then quotienting by its analytic-source image gives exactly the same affine fibres as $\mathcal{H}_e/\mathcal{S}_e$.

184.2 The full original Gamma dual metric

The source member of the two original order pairs is $s = 1$ or $s = k$, and its centre is $c = k/2 - 4$. Keep its entire measure and recurrence:

$$M_s = (2\pi)^{s/2}, \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad h_{n,s} = M_s n!(s/2)_n,$$

$$p_{0,s} = 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \quad \phi_n(S') = p_{n,s}((S' - c)/i)/\sqrt{h_{n,s}}. \quad (\text{PDS6})$$

These are the original orthonormal polynomials for the norm $\|p\|_{s,c}^2 = \int_{\mathbb{R}} |p(c + iy)|^2 dm_s(y)$, as established in the original Gamma construction reproduced in the cumulative providers. The coefficient isometry uses this complete measure, rather than dividing its mass out. Let B_e be the triangular matrix defined by $S'^r = \sum_{n=0}^{D_e} (B_e)_{nr} \phi_n(S')$ and let

$$C_e = (B_e \text{coeff}_{S'} r_a)_{a=0}^{\ell_e}, \quad H_e = C_e^* C_e > 0, \quad G_e = \overline{H_e^{-1}}. \quad (\text{PDS7})$$

The columns of C_e are polynomial coefficients in the original orthonormal frame. For $w = (\Lambda_F(\phi_n))_n$ the literal quotient row map is $x = C_e^T w$. Its minimum-norm lift is $w = \overline{C_e} \overline{H_e}^{-1} x$ and has squared norm $x^* G_e x$. Indeed $C_e^T \overline{C_e} = \overline{H_e}$, and the displayed lift is orthogonal to $\ker C_e^T$. This proves both the full complex-linear dual orientation and the quotient metric, including every conjugation in PDS7. The same metric results from the two successive minima over the original lower gauges and analytic source, by the equality of fibres proved after PDS5.

The degree-one primitive condition at the first endpoint is empty. At the second endpoint its four original rows are

$$R_i(S) = \frac{\mathcal{T}_{B_i} \chi_j(S)}{\chi_k(S)}, \quad W = (\text{coeff}_S R_i)_{i=1}^4 \in \mathbb{C}^{d_1 \times 4}, \quad \mathcal{P}_0 = \mathbb{C}^{d_0}, \quad \mathcal{P}_1 = \ker W^T \subset \mathbb{C}^{d_1}. \quad (\text{PDS8})$$

Here $\mathcal{P}_e$ denotes the primitive quotient source, not a biform section space. All degree- k roots become degree- j roots after every degree-four shift; hence the numerator is divisible by χ_k and $\deg R_i \leq \ell_1$. Translation sums commute, so $\mathcal{T}_A(\chi_k R_i) = \mathcal{T}_{B_i} \mathcal{T}_A \chi_j$. This is exactly the original positive degree-zero exterior differential, and its row is W^T , as in LCA18–19. The full independence proof LCA20–21 applies to the actual invertible period, without a genericity exclusion: the four L_i^4 are independent on the rank-four Segre quadric, and the 25 distinct translations of the degree- $Q - v \geq 116$ polynomial $\mathcal{T}_A \chi_j$ give an injective Vandermonde moment map. Thus $\text{rank } W = 4$. The complete LCA20–21 proof is retained as a provider in the cumulative source, rather than replacing this fact by a new period assumption.

The metric on $\mathcal{P}_e$ is the restriction of G_e . At $e = 1$ its exact polynomial dual is

$$\text{span}(r_0, \dots, r_{\ell_1}) / \text{span}(\mathcal{T}_A(\chi_k R_i))_{i=1}^4.$$

To verify the metric as well as the vector-space statement, choose any fixed full basis matrix Z_1 of $\ker W^T$. A coefficient vector f of a relation polynomial restricts to the functional $Z_1^T f$ on this basis. Minimizing $f^* H_1 f$ over all lifts with $Z_1^T f = a$ gives

$$\min_{Z_1^T f = a} f^* H_1 f = a^* (Z_1^T H_1^{-1} \overline{Z_1})^{-1} a = a^* (\overline{Z_1^* G_1 Z_1})^{-1} a. \quad (\text{PDS9})$$

The minimizer follows by the adjoint of $Z_1^T H_1^{-1/2}$; subtracting it leaves an orthogonal kernel vector. The kernel of Z_1^T is exactly $\text{im } W$, by dimensions and $W^T Z_1 = 0$. This proves the stated original quotient metric and its exact dual morphism, with no deletion of the four row polynomials.

Current full relation and primitive source determinants. In these exact PDS7 coordinates, put $q = (k+1)^2$, $Q = (k+5)^2$, $d_e = 8k + 24 + e$, $N_e = Q - v - 1 + e$, and

$$\mathfrak{F}_{k,e,s} = \prod_{a=0}^{d_e-1} \left| \mu_v \binom{q+a}{v} \right|^2 (2\pi)^{s/2} (q+a-v)! (s/2)_{q+a-v}.$$

The complete input columns are $\chi_k(S)S^a$ at center $k/2$, modulo every polynomial below v . Their quotient coefficient matrix is the actual Y_e of SGT30, and their translated columns are exactly $C_e = A_{N_e,s}Y_e$, with SGT10's full factorial, phase and Gamma diagonal maps. PSG21 and SGT24–31 therefore give, at both defining low cutoffs,

$$\begin{aligned} \log \det H_e &= \log \mathfrak{F}_{k,e,s} + 2d_e q \log(4/\pi) + \varepsilon_{k,e,s}^H, \\ -\mathcal{E}_{k,e,s}^-(a) &\leq \varepsilon_{k,e,s}^H \leq \mathcal{E}_{k,e,s}^+(a), \quad \varepsilon_{k,e,s}^H = o(kq), \\ \log \det G_e &= -\log \mathfrak{F}_{k,e,s} - 2d_e q \log(4/\pi) - \varepsilon_{k,e,s}^H. \end{aligned} \quad (\text{PDS25})$$

The finite $\mathcal{E}^\pm$ are exactly PSR2–4 for any SGT-admissible fixed radius a , with the full finite domain of PSR2; the radius limit is taken after the degree limit. The proof adds the two same-frame logarithmic ratios in PSR5. The dual identity uses the original $G_e = H_e^{-1}$, so its error interval is $[-\mathcal{E}^+, \mathcal{E}^-]$. No polynomial has been replaced by its leading term in either Gram.

At $e = 1$, the complete four-row correction remains the literal $\mathcal{G}_{k,s} = W^*H_1W$: the exact coefficient map is $C_1W = \mathcal{C}$ in PFG5. Its earliest actual moment pivots $t_1 < \dots < t_4 \leq 24$ and the full nonzero minor in PFG10 prove the two-sided original-data estimate

$$0 < \mathcal{L}_{k,s} \leq \det(W^*H_1W) \leq \mathcal{U}_{k,s}, \quad \log \det(W^*H_1W) = 4 \log M_s + 8Q \log Q + O_{\text{actual}}(Q) = o(kq). \quad (\text{PDS26})$$

Here $\mathcal{L}, \mathcal{U}$ are the complete PFG11–13 moment/factorial and coefficient-product expressions. PDS9's quotient by those four columns still carries its fixed determinant factor $|\det[W, J_Z]|^2$, as proved by the full Schur factorization PFG18 and PSR12–20. At $e = 0$ there is no four-row primitive constraint.

184.3 The exact codimension-three truncation

The coefficient of S^{ℓ_1} in R_i is b_i , since both χ_j and χ_k are monic. Choose a fixed original index i_* with $b_{i_*} \neq 0$, write $b_* = b_{i_*}$, and retain

$$R_{i_*}(S) = \sum_{a=0}^{\ell_0} (u_*)_a S^a + b_* S^{\ell_1}, \quad C_i(S) = R_i(S) - \frac{b_i}{b_*} R_{i_*}(S) \quad (i \neq i_*). \quad (\text{PDS10})$$

These displayed combinations are exact coordinates for the intersection of the original four-row span with lower degree; the original rows and coefficients in PDS8 remain fixed. The three C_i have degree at most ℓ_0 and are independent: a relation among them would give a relation among all four R_i , whose coefficients on indices other than i_* are unchanged. Let V have the three coefficient columns C_i , in the original index order with i_* omitted. Then

$$\begin{aligned} \mathcal{Z} &= \ker V^\top \subset \mathbb{C}^{d_0}, \quad \dim \mathcal{Z} = d_0 - 3 = 8k + 21, \quad E = \begin{bmatrix} I_{d_0} \\ -u_*^\top/b_* \end{bmatrix}, \\ \text{pr}_0 : \mathcal{P}_1 &\xrightarrow{\cong} \mathcal{Z}, \quad \text{pr}_0^{-1} = E|_{\mathcal{Z}}. \end{aligned} \quad (\text{PDS11})$$

In fact the i_* row of $W^\top(x_0, x_{\text{top}}) = 0$ gives $x_{\text{top}} = -u_*^\top x_0/b_*$. Substitution in the other three rows gives precisely $V^\top x_0 = 0$. The first d_0 relation polynomials in PDS4 are unchanged at the two cutoffs, so pr_0 is also the actual jet truncation on the primitive quotient source. If an upper analytic source is added its truncated jet is an old analytic source; hence this map is well-defined on the original quotient. Surjectivity and the displayed inverse follow from the equations just proved.

184.4 One rank-one update, with every frame retained

Choose an arbitrary fixed full column basis Z of $\mathcal{Z}$; use EZ at the upper endpoint. For the same source order s and centre c , the old orthonormal polynomials are unchanged when the cutoff increases. In particular the old C_0 is padded by a zero last row in C_1 . Write the exact block matrix and Schur scalar

$$H_1 = \begin{bmatrix} H_0 & b \\ b^* & a \end{bmatrix}, \quad b = C_0^* c_{\text{top}}, \quad a = c_{\text{top}}^* c_{\text{top}}, \quad \sigma = a - b^* H_0^{-1} b > 0, \quad (\text{PDS12})$$

where c_{top} is the full coefficient column of r_{ℓ_1} . Padded columns are understood in the products. Define

$$\eta = -u_*^\top / b_* - b^\top \bar{H}_0^{-1}, \quad K_0 = Z^* \bar{H}_0^{-1} Z, \quad K_1 = (EZ)^* \bar{H}_1^{-1} (EZ). \quad (\text{PDS13})$$

Then the complete metric comparison is

$$\boxed{K_1 = K_0 + \frac{(\eta Z)^* (\eta Z)}{\sigma}, \quad \frac{\det K_1}{\det K_0} = 1 + \frac{(\eta Z) K_0^{-1} (\eta Z)^*}{\sigma} \geq 1.} \quad (\text{PDS14})$$

For proof, block inversion of $\bar{H}_1$ gives for every (x_0, x_{top})

$$\begin{bmatrix} x_0 \\ x_{\text{top}} \end{bmatrix}^* \bar{H}_1^{-1} \begin{bmatrix} x_0 \\ x_{\text{top}} \end{bmatrix} = x_0^* \bar{H}_0^{-1} x_0 + \frac{|x_{\text{top}} - b^\top \bar{H}_0^{-1} x_0|^2}{\sigma}. \quad (\text{PDS15})$$

Insert PDS11 and then $x_0 = Zz$. This proves the matrix equality. The determinant identity follows by taking the determinant of $I + K_0^{-1/2} (\eta Z)^* (\eta Z) K_0^{-1/2} / \sigma$; this rank-at-most-one operator has its only possible nonzero eigenvalue $(\eta Z) K_0^{-1} (\eta Z)^* / \sigma$. The frame Z has not been changed: replacing it by ZT changes both determinants by the same $|\det T|^2$. PDS14 also proves that the actual truncation $\mathcal{P}_1 \rightarrow \mathcal{Z}$ is a contraction for the displayed native metrics, and locates its entire metric defect in this one explicit row.

184.5 An explicit coefficient-product upper estimate

Set $D = D_1 = Q - v$ and keep the literal original polynomial

$$r_*(S') = \mathcal{T}_A(\chi_k R_{i_*})(S') = \mathcal{T}_{B_{i_*}} \mathcal{T}_A \chi_j(S'), \quad \lambda_* = \text{lc } r_* = b_* \mu_v \binom{Q}{v} \neq 0. \quad (\text{PDS16})$$

There is an exact finite upper bound

$$1 \leq \frac{\det K_1}{\det K_0} \leq \frac{\|r_*\|_{s,c}^2}{|b_*|^2 \sigma} \leq \frac{\|r_*\|_{s,c}^2}{|\lambda_*|^2 h_{D,s}}. \quad (\text{PDS17})$$

To prove the first upper inequality, the orthogonal projector onto $\text{im}(\bar{H}_0^{-1/2} Z)$ gives $Z K_0^{-1} Z^* \leq \bar{H}_0$. Thus the numerator in PDS14 is at most $\eta \bar{H}_0 \eta^*$. Let $v_* = u_*/b_* + H_0^{-1} b$. Then $\eta = -v_*^\top$, so this upper bound is $v_*^* H_0 v_*$. The orthogonal projection of r_* onto the old relation span has coefficient vector $b_* v_*$, whereas its orthogonal remainder is b_* times the remainder of r_{ℓ_1} . Consequently

$$\|r_*\|_{s,c}^2 = |b_*|^2 (v_*^* H_0 v_* + \sigma). \quad (\text{PDS18})$$

This proves the first inequality of PDS17. Every old relation has degree at most $D - 1$. The coefficient of ϕ_D in r_{ℓ_1} is $i^D \mu_v \binom{Q}{v} \sqrt{h_{D,s}}$; this uses its actual leading coefficient and the phase $i^{-D} / \sqrt{h_{D,s}}$ in PDS6. This coefficient remains in its orthogonal remainder. Hence $\sigma \geq |\mu_v \binom{Q}{v}|^2 h_{D,s}$, proving the second inequality of PDS17 with its original scalar intact.

For a completely finite coefficient bound define

$$L = \max\{1, |c| + 2(D + 1 + s/2)\}, \quad \Theta_{a,b,r,t;u,w} = \beta_{8,a,b} + \beta_{4,r,t} - \beta_{j,u,w},$$

$$P_* = \sum_{a,b=0}^8 \sum_{r,t=0}^4 |a_{ab} b_{i_*,r,t}| \prod_{u,w=0}^j (L + |\Theta_{a,b,r,t;u,w}|), \quad U_* = \frac{M_s P_*^2}{|b_* \mu_v(Q)|^2 h_{D,s}}. \quad (\text{PDS19})$$

The L in this subsection is a numerical bound and is not the jet length denoted L in LCA2. Then

$$0 \leq \log \frac{\det K_1}{\det K_0} \leq \log U_*. \quad (\text{PDS20})$$

Here is a direct proof that PDS19 bounds the original Gamma norm. In the orthonormal polynomial basis in y , multiplication by y has raising coefficient $\sqrt{(n+1)(n+s/2)}$ and lowering coefficient $\sqrt{n(n-1+s/2)}$, by PDS6 and $h_{n+1,s}/h_{n,s} = (n+1)(n+s/2)$. On polynomials of degree at most $D-1$ both weighted shifts have norm at most $D+1+s/2$; their sum has norm at most $2(D+1+s/2)$. Thus multiplication by $S' = c + iy$, through degree D , has norm at most L . Starting with the constant polynomial, whose squared norm is M_s , proves

$$\|p\|_{s,c} \leq \sqrt{M_s} \sum_{m=0}^D |[S'^m]p| L^m \quad (\deg p \leq D). \quad (\text{PDS21})$$

The exact expansion of PDS16 is

$$r_*(S') = \sum_{a,b=0}^8 \sum_{r,t=0}^4 a_{ab} b_{i_*,r,t} \prod_{u,w=0}^j (S' + \Theta_{a,b,r,t;u,w}). \quad (\text{PDS22})$$

The weighted coefficient ℓ^1 norm is submultiplicative: for two polynomials its coefficient convolution and the triangle inequality give $\sum_m |(fg)_m| L^m \leq (\sum_m |f_m| L^m)(\sum_m |g_m| L^m)$. Applying this identity to every factor in PDS22 bounds the weighted coefficient sum of r_* by P_* . Its coefficients above degree $D = Q - v$ vanish by the exact moment identities in PDS5; bounding the full degree- Q expansion before those cancellations is still an upper bound for the unchanged degree- D polynomial. PDS21 now gives $\|r_*\|_{s,c}^2 \leq M_s P_*^2$, and PDS17 proves PDS20. In particular $U_* \geq 1$ is a consequence, rather than an extra assumption on the original coefficients.

There is also an explicit uniform growth bound in either original source-order family. Define the fixed positive numbers

$$R_* = \sqrt{(1/2 + \delta)^2 + \gamma^2}, \quad A_{\text{abs}} = \sum_{a,b=0}^8 |a_{ab}|, \quad B_{\text{abs}} = \sum_{r,t=0}^4 |b_{i_*,r,t}|,$$

$$C_* = 6 + 17R_*, \quad C_{\text{abs}} = \max \left\{ 1, \frac{2(A_{\text{abs}} B_{\text{abs}})^2}{|b_* \mu_v|^2} \right\}. \quad (\text{PDS23})$$

The original nodes satisfy $|\text{Re } \beta_{n,a,b}| \leq n(1/2 + \delta)$ and $|\text{Im } \beta_{n,a,b}| \leq n\gamma$. Both bounds are attained together at $(a, b) = (n, 0)$ and (n, n) ; thus $\max_{a,b} |\beta_{n,a,b}| = nR_*$, and therefore $|\Theta| \leq (k+16)R_*$. For $s = 1$ or $s = k$, $D \leq Q$, $k \leq Q$, and $|c| \leq k/2 + 4$, so $L \leq 2Q + 3k/2 + 6 \leq 6(Q+1)$. Also $k+16 \leq 17(Q+1)$. Every factor in PDS19 is consequently at most $C_*(Q+1)$, giving $P_* \leq A_{\text{abs}} B_{\text{abs}} [C_*(Q+1)]^Q$. The full factorial denominator gives a sharper bound than dropping its growth. For $s \geq 1$,

$$D!(s/2)_D \geq D!(1/2)_D = \frac{(2D)!}{4^D} \geq \left(\frac{D}{\exp(1)} \right)^{2D}.$$

The product identity follows by separating the even and odd factors of $(2D)!$. The last inequality follows from $\sum_{r=1}^{2D} \log r \geq \int_1^{2D} \log x \, dx = 2D \log(2D) - 2D + 1$. Since $D = Q - v$, $Q \geq 196$ and

$v \leq 80$ imply $Q + 1 \leq 2D$, the exact PDS19 bound therefore satisfies

$$\begin{aligned} U_* &\leq \frac{(A_{\text{abs}} B_{\text{abs}})^2}{|b_* \mu_v|^2} C_*^{2Q} (Q + 1)^{2Q} \left(\frac{\exp(1)}{D} \right)^{2D} \\ &\leq C_{\text{abs}} (Q + 1)^{2v} (2 \exp(1) C_*)^{2Q}. \end{aligned}$$

Here $\binom{Q}{v} \geq 1$ was used only in an upper bound; the full binomial, mass and factorial denominator remain in PDS19. For the second inequality split $(Q + 1)^{2Q}$ into its $2v$ and $2D$ powers and use $Q + 1 \leq 2D$, $D \leq Q$ and $C_* \geq 6$. Consequently the proved finite bound is

$$\begin{aligned} 0 &\leq \log \frac{\det K_1}{\det K_0} \leq \log U_* \\ &\leq \log C_{\text{abs}} + 2v \log(Q + 1) + 2Q \log(2 \exp(1) C_*) \\ &= O_{\text{actual}}(Q) = o(kq_k). \end{aligned} \tag{PDS24}$$

The last limit uses $Q/q_k \rightarrow 1$, fixed v , and $1/k \rightarrow 0$ for $k \equiv 1 \pmod{4}$. The constants are fixed by the actual quartet and period, independent of k and of the choice between the two original source-order families. The sharp finite PDS19 retains the full mass, phase-dependent nodes, factorials, conductor coefficients and leading scalar; PDS24 proves the scale of its upper bound without assuming an inverse estimate.

The maps PDS11 and PDS14 now give the primitive part of the low-cutoff source comparison inside the original LCA connecting sequence. The full cohomology additionally has the retained $\ker U_e$ constituent from LCC. Passing these source metrics through the actual connecting map requires the original target quotient Gram; PDS9 specifies the exact source dual to be used in that calculation, and PDS20–24 control its adjacent source correction. None of these equalities identifies a later target coefficient flag with a common compatible source at all cutoffs.

185 The original connecting image and its full native target determinant

Keep the complete CNC, LCC and LCA data, including the nonreduced A^{hom} , every original coefficient and the exact $E_A(z) = \sum a_{ab} e^{\beta_{8,a,b} z}$. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q_a &= (a + 1)^2, & q'_a &= (a - 7)^2, \\ e &\in \{0, 1\}, & D_e &= q_j - v - 1 + e, & L_e &= D_e + 1, & 0 \leq v &\leq 80, & \mu_v &= E_A^{(v)}(0) \neq 0, \\ (s_0, s_1) &= (1, 1) \text{ or } (k, j), & c'_a &= a/2 - 4, & m_a &= q_a - q'_a - v. \end{aligned} \quad (\text{CIT1})$$

The two source orders in this display remain paired throughout; write $s_k = s_0$ and $s_j = s_1$ when a degree subscript is used. All finite functionals are complex-linear. At a fixed D_e let $\mathcal{H}_0, \mathcal{H}_1$ be their original Gamma polynomial duals in degrees k, j , with full orthonormal frames

$$\phi_{r,s,c'}(S') = \frac{p_{r,s}((S' - c')/i)}{\sqrt{(2\pi)^{s/2} r! (s/2)_r}}, \quad 0 \leq r \leq D_e. \quad (\text{CIT2})$$

Use the exact CNC10 matrices $T_0 = T_{k,D_e}$, $T_1 = T_{j,D_e}$ for its original analytic numerators divided by E_A , and V_0, V_1 for every original lower evaluation. Thus $[V_0, T_0]$ has its original full column order.

185.1 The literal primitive domain and target coefficient matrix

The four multiplier matrices are completely specified without an unspecified evaluator. Let $\mathcal{B}_a$ be the original monomial-to-Gamma matrix defined by $S'^r = \sum_n (\mathcal{B}_a)_{nr} \phi_{n,s_a,c'_a}$ and set

$$\begin{aligned} U_a &= \text{diag}(1/r!)_{r=0}^{D_e} \mathcal{B}_a^\top, & (\text{Mult}_{L_e}(B_i))_{rt} &= \begin{cases} B_i^{(r-t)}(0)/(r-t)!, & r \geq t, \\ 0, & r < t, \end{cases} \\ \mathsf{T}_i &= U_j^{-1} \text{Mult}_{L_e}(B_i) U_k, & \mathsf{T} &= \text{stack}_{i=1}^4 \mathsf{T}_i, & \mathcal{V} &= \text{diag}(V_1, 4), & \mathcal{T}_1 &= \text{diag}(T_1, 4). \end{aligned} \quad (\text{CIT3})$$

Here $\text{diag}(P, 4)$ means four copies. The map U_a sends the actual native functional values to Taylor coefficients: ordinary derivatives are $\mathcal{B}_a^\top w$. This proves CIT3 and preserves every factorial and the original leading phase i^r inside $\mathcal{B}_a$. Its full inverse tuple is the analogous multiplication by the original entire D_i ; $\sum D_i B_i = 1$ gives $\mathsf{ST} = I$.

Define the raw source analytic space

$$\mathcal{S}_k^{\text{raw}} = \text{im}[V_0, T_0] \subset \mathcal{H}_0, \quad \mathcal{L}_0 = \text{im } V_0, \quad \mathcal{L}_1 = \text{im } \mathcal{V}. \quad (\text{CIT4})$$

It is exactly the set of native functionals of N/E_A for all original upper numerators N vanishing to order v . Indeed T_0 gives the original DNE quotient frame and V_0 gives exactly its allowed numerator changes by conductor multiples. The columns are independent: if $T_0 x + V_0 \eta = 0$, the CNC13 injectivity at this cutoff gives $x = 0$, and the lower Vandermonde gives $\eta = 0$. Thus $\dim \mathcal{S}_k^{\text{raw}} = q_k - v$.

For the upper endpoint set

$$\begin{aligned} P_A &= \mathcal{T}_A \chi_j, & (\mathcal{T}_A p)(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), \\ R_i(S') &= \mathcal{T}_{B_i} P_A(S'), & (\mathcal{T}_{B_i} p)(S') &= \sum_{a,b=0}^4 b_{i,a,b} p(S' + \beta_{4,a,b}), \\ \mathsf{R}_1 &= \text{stack}_{i=1}^4 (\mathcal{B}_k \text{coeff}_{S'} R_i)^\top, & \mathsf{R}_0 &: \mathcal{H}_0 \longrightarrow 0. \end{aligned} \quad (\text{CIT5})$$

All coefficient columns are padded through D_1 . Here $\deg P_A = q_j - v = D_1$ and $\text{lc } P_A = \mu_v \binom{q_j}{v}$. The complete LCC9–12 proof shows that R_1 has rank four: the four original L_i^4 are independent on the

rank-four Segre quadric, and translation of this polynomial is injective on the 25 distinct original shifts. The native row in CIT5 is an ordinary transpose because it evaluates the complex-linear functional on R_i .

The complete primitive domain is

$$\mathcal{Y}_e^{\text{raw}} = \ker R_e, \quad \mathcal{S}_k^{\text{raw}} \subset \mathcal{Y}_e^{\text{raw}}, \quad r_e = \dim(\mathcal{Y}_e^{\text{raw}}/\mathcal{S}_k^{\text{raw}}) = 8k + 24 - 3e. \quad (\text{CIT6})$$

For $e = 0$ it is all of $\mathcal{H}_0$. To prove the target condition, at that endpoint the raw analytic image in each $\mathcal{H}_1$ has dimension $q_j - v = L_0$ and fills it. At $e = 1$ its only missing coordinate is $\Lambda_F(P_A)$ by LCC10. Therefore $\mathsf{T}h$ belongs to the fourfold raw target analytic image precisely when $R_1 h = 0$. The exact numerator cochain square kills $\mathcal{S}_k^{\text{raw}}$ in those rows. Subtracting $\dim \mathcal{S}_k^{\text{raw}} = q_k - v$ from $\dim \ker R_e = L_e - 4e$ proves CIT6.

Choose a full primitive lift matrix P_e so that $[V_0, T_0, P_e]$ is a frame of $\mathcal{Y}_e^{\text{raw}}$. This is an exact computable choice: first express CIT4–5 in ordinary Taylor coordinates, take the first nonzero pivot minors in their original order to obtain the kernel and a complement, then transport those complement columns by U_k^{-1} . It depends on the actual coefficients and imposes no metric conditioning estimate. The chosen Taylor columns are fixed; the original norm remains CIT2.

In the original target coefficient coordinates define

$$\begin{aligned} P_{\mathcal{L}_1} &= \mathcal{V}(\mathcal{V}^* \mathcal{V})^{-1} \mathcal{V}^*, \quad K_1 = (I - P_{\mathcal{L}_1}) \mathcal{T}_1, \quad \mathcal{D}_e = K_1^* K_1 = \text{diag}(D_{j,D_e}^{(s_1)}, 4), \\ F_e &= \mathcal{D}_e^{-1} K_1^* (I - P_{\mathcal{L}_1}) \mathsf{T} P_e, \quad B_{\text{cof}} = \text{stack}_{i=1}^4 M_{i,k}. \end{aligned} \quad (\text{CIT7})$$

The inverses exist by the full lower Vandermonde and CNC13. By CIT6 the projected columns on the right belong to $\text{im } K_1$, so the displayed left inverse gives their unique original coefficient coordinates. Consequently

$$K_1 F_e = (I - P_{\mathcal{L}_1}) \mathsf{T} P_e, \quad K_1 B_{\text{cof}} = (I - P_{\mathcal{L}_1}) \mathsf{T} T_0. \quad (\text{CIT8})$$

These are literal original finite matrices. In particular the $M_{i,k}$ retain the exact KRC scalar-bearing conjugation CNC12.

185.2 Complete connecting kernel and attained target norm

We prove the full kernel of the connecting construction. The exact finite intersection needed here is

$$\text{im } \mathsf{T} \cap \mathcal{L}_1 = \mathsf{T} \mathcal{L}_0. \quad (\text{CIT9})$$

For a direct proof, write $\mathsf{T}h = \mathcal{V}\eta$. Apply the six original compatibility rows. Their coefficient square is $\mathcal{R}\mathcal{V} = \mathcal{V}_2 \mathcal{K}$, so $\mathcal{V}_2 \mathcal{K}\eta = 0$. At the present cutoffs $D_e + 1 - q_k = 8k + 24 - v + e \geq 16$; hence the original degree- k evaluation $\mathcal{V}_2$ is injective. The complete GKC kernel gives $\eta = \mathcal{P}\xi$. The lower root-addition square says $\mathcal{V}\mathcal{P} = \mathsf{T}V_0$, whence $\mathsf{T}h = \mathsf{T}V_0\xi$. Since $\mathsf{S}\mathsf{T} = I$, $h = V_0\xi$. The reverse inclusion is exactly that same square. This proves CIT9 without a transversality or inverse-norm assumption.

For $h \in \mathcal{Y}_e^{\text{raw}}$ let $c(h)$ be the coefficient column defined as in CIT7. The native exterior differential of $\mathsf{T}h$ is zero, since $B_i B_t = B_t B_i$. Replacing its representative by lower evaluations changes its next differential by lower evaluations, by root addition. Therefore $c(h)$ is a cycle in the actual LCC image complex. The complete map is

$$\begin{aligned} \partial_e : \mathcal{Y}_e^{\text{raw}}/\mathcal{S}_k^{\text{raw}} &\longrightarrow H^1(\mathcal{I}_e), \quad [h] \longmapsto [c(h)], \\ \text{im } \partial_e &= \mathcal{C}_e, \quad \mathcal{C}_e = \mathcal{S}_e/\mathcal{B}, \quad \mathcal{S}_e = \text{im}[B_{\text{cof}}, F_e], \quad \mathcal{B} = \text{im } B_{\text{cof}}. \end{aligned} \quad (\text{CIT10})$$

If $h = T_0 x + V_0 \xi$, its target coefficients are the boundary $B_{\text{cof}} x$ by CIT8. Conversely, if $c(h) = B_{\text{cof}} x$, then $\mathsf{T}(h - T_0 x) \in \mathcal{L}_1$ and CIT9 gives $h - T_0 x \in \mathcal{L}_0$. Thus the kernel before quotient is exactly $\mathcal{S}_k^{\text{raw}}$. This proves injectivity and the stated presentation; $[B_{\text{cof}}, F_e]$ is independent. It is exactly the positive connecting image of $H^0(\mathcal{R}_e) = \ker t_e$ in LCC13, since its construction lifts the original

degree-zero quotient cycle and takes its differential with the same positive sign. No splitting of the remaining $\ker U_e$ has been introduced.

Its full cohomology restriction Gram in the P_e image coordinates is

$$H_{\text{conn},e} = F_e^* \mathcal{D}_e F_e - F_e^* \mathcal{D}_e B_{\text{cof}} (B_{\text{cof}}^* \mathcal{D}_e B_{\text{cof}})^{-1} B_{\text{cof}}^* \mathcal{D}_e F_e. \quad (\text{CIT11})$$

The exact least norm over every original boundary gives this Schur complement. All original lower coefficients were already minimized simultaneously in $\mathcal{D}_e$. Equivalently set

$$Z_e^{\text{raw}} = [\mathcal{V}, \mathbb{T}T_0], \quad H_{\text{conn},e} = (\mathbb{T}P_e)^*(I - P_{Z_e^{\text{raw}}})\mathbb{T}P_e, \quad P_Z = Z(Z^*Z)^{-1}Z^*. \quad (\text{CIT12})$$

The columns of Z_e^{raw} are independent by CIT9 and the independence in CIT4. The same argument with the added P_e proves independence of $[Z_e^{\text{raw}}, \mathbb{T}P_e]$. Indeed the complete minimization in CIT11 allows exactly $\mathbb{T}T_0x + \mathcal{V}\eta$. This proves CIT12 and positivity. It also gives the full original raw determinant

$$\det H_{\text{conn},e} = \frac{\det([Z_e^{\text{raw}}, \mathbb{T}P_e]^* [Z_e^{\text{raw}}, \mathbb{T}P_e])}{\det((Z_e^{\text{raw}})^* Z_e^{\text{raw}})}. \quad (\text{CIT13})$$

These are the same fixed combined frames, so their ordinary Schur determinant introduces no missing coordinate factor.

185.3 The original source-to-image determinant at its low cutoff

Give the primitive source quotient its original source metric and its correlated image metric, respectively:

$$H_{\text{src},e} = P_e^*(I - P_{[V_0, T_0]})P_e, \quad H_{\text{cor},e} = (\mathbb{T}P_e)^*(I - P_{\mathbb{T}[V_0, T_0]})\mathbb{T}P_e. \quad (\text{CIT14})$$

These are two explicit attained minima. The original NBG $\|\mathbb{T}\| \leq K_+$ and $\|\mathbb{S}\| \leq K_-$ with $\mathbb{S}\mathbb{T} = I$ imply

$$K_-^{-2} H_{\text{src},e} \preceq H_{\text{cor},e} \preceq K_+^2 H_{\text{src},e}. \quad (\text{CIT15})$$

For each fixed primitive value, apply the two norm inequalities to every representative in $P_e b + \mathcal{S}_k^{\text{raw}}$ and take the minima. This proves both directions on the exact same fibre. The finite constants are the complete NBG12 ones at base k and cutoff D_e , including source order $(1, 1)$ or (k, j) and center shift $(\beta_{4,a,b} - 2)/i$.

There is also an exact independent-gauge quotient identity. In the original source quotient put

$$X_k = \mathcal{S}_k^{\text{raw}} / \mathcal{L}_0, \quad Y_e = \mathcal{Y}_e^{\text{raw}} / \mathcal{L}_0, \quad X_k \subset Y_e.$$

Their correlated target images are the original NKC9 images. The original independent-gauge form restricted to Y_e and then quotiented by X_k allows precisely the total fibre $\mathbb{T}T_0x + \mathcal{L}_1$; its correlated quotient allows $\mathbb{T}T_0x + \mathbb{T}\mathcal{L}_0$. These are CIT12 and CIT14. Hence the literal PCJ7 source quotient gives

$$\frac{\det H_{\text{conn},e}}{\det H_{\text{cor},e}} = \det \Omega_{Y_e/X_k, D_e}. \quad (\text{CIT16})$$

If R transports the original primitive lifts to an adapted correlated quotient frame, both Grams are changed by $R^*(\cdot)R$, and their two factors $|\det R|^2$ cancel in CIT16. This is the exact coordinate map behind that determinant equality.

Use the actual compatibility-plus-complement expressions χ_{X_k, D_e} and χ_{Y_e, D_e} of PCJ1. The complete shared-update proof PCJ13–16 gives

$$\begin{aligned} \log \det H_{\text{conn},e} - \log \det H_{\text{src},e} &= -(\chi_{X_k, D_e} - \chi_{Y_e, D_e}) + \epsilon_e, \\ -2r_e \log \max(1, K_-) - r_e \log c_+ &\leq \epsilon_e \leq 2r_e \log \max(1, K_+) - r_e \log c_-. \end{aligned} \quad (\text{CIT17})$$

Indeed write the left side as the correlated/source logarithm plus $\log \det \Omega_{Y_e/X_k}$. The first is bounded by CIT15. PCJ16 writes the negative of the latter as $\chi_{X_k} - \chi_{Y_e} + \nu$, with $r_e \log c_- \leq \nu \leq r_e \log c_+$. Subtraction gives exactly CIT17. Here $c_- = \min(1, L_-^{-2})$ and $c_+ = \max(1, L_+^2)$ are the original NKC constants at this cutoff. Consequently this specified error is $O_{\text{actual}}(k \log q_k) = o(kq_k)$ in both original source pairs. At $e = 0$, Y_0 is the full ambient source quotient, so its source-complement projector vanishes and $\chi_{Y_0} = 0$ exactly. At $e = 1$ its four actual polynomial constraints remain; χ_{Y_1} is not set to zero.

Current evaluated source and the surviving independent gauges. Here $q = q_k = (k+1)^2$, $e = 0, 1$, $d_e = 8k + 24 + e$, and $N_e = D_e = q_{k+4} - v - 1 + e$. Keep the exact primitive coordinates P_e in CIT6 and $\Theta_e = C_e^\top P_e$ in PDI1–2. With PSR18’s unchanged Γ_e and the full factorial product $\mathfrak{F}_{k,e,s_0}$, set

$$\tilde{Z}_e = \Gamma_e - \log \mathfrak{F}_{k,e,s_0} - 2d_e q \log(4/\pi) + e m_{k,s_0}^{(4)} - \chi_{X_k, D_e} + \chi_{Y_e, D_e}.$$

The full connecting-image determinant now satisfies

$$\begin{aligned} \epsilon_e^{\text{lo}} + e b_{k,s_0}^{\text{lo}} - \mathcal{E}_{k,e,s_0}^+(a) &\leq \log \det H_{\text{conn},e} - \tilde{Z}_e \\ &\leq \epsilon_e^{\text{hi}} + e b_{k,s_0}^{\text{hi}} + \mathcal{E}_{k,e,s_0}^-(a), \end{aligned} \quad (\text{CIT27})$$

$$\log \det H_{\text{conn},e} = \Gamma_e - 2d_e q \{\log(4q/\pi) - 1\} - \chi_{X_k, D_e} + \chi_{Y_e, D_e} + o(kq).$$

Here all finite quantities are exactly PSR2–4 and PSR16–22, at the domain stated in PSR2, with $\epsilon_e^{\text{lo}}, \epsilon_e^{\text{hi}}$ equal to CIT17’s two original endpoints. The proof substitutes PSR19 into the exact CIT17 identity. Its residual is $\epsilon_e + e \varepsilon_{k,s_0}^{(4)} - \varepsilon_{k,e,s_0}^H$; adding their full intervals gives every displayed sign. The two gauge terms have not been combined with the source determinant or omitted. The $e = 0$ complement term vanishes for the exact reason in CIT17; the $e = 1$ term retains all four original constraints.

There is also an exact map for comparing the two original source-order pairs. PSR23 gives $\Theta_e = (R_e^{\text{pol}})^\top \text{diag}(r!) P_e^{\text{Tay}}$ in both orders. The original Taylor cochain square

$$\text{stack}_i \text{Mult}(B_i) P_e^{\text{Tay}} = \text{diag}(\tilde{T}_j, 4) F_e + \text{diag}(\tilde{V}_j, 4) E_e^{\text{low}}$$

has a unique analytic coefficient F_e modulo all lower evaluations, by CNC13. It therefore gives the same F_e for $(s_0, s_1) = (1, 1)$ and $(k, k+4)$. On these exact common ordered values, write $\mathcal{X}_e^{(s_0, s_1)} = \chi_{X_k, D_e}^{(s_0, s_1)} - \chi_{Y_e, D_e}^{(s_0, s_1)}$. Then

$$\log \frac{\det H_{\text{conn},e}^{(k,k+4)}}{\det H_{\text{conn},e}^{(1,1)}} = -\mathcal{X}_e^{(k,k+4)} + \mathcal{X}_e^{(1,1)} + o(kq). \quad (\text{CIT28})$$

This is PSR24–27: the exact residual is the source order ratio plus $\epsilon_e^{(k,k+4)} - \epsilon_e^{(1,1)}$. Its finite interval is obtained by adding the PSR26 source interval, including its full D_e^{ord} , to $[\epsilon_{e,(k,k+4)}^{\text{lo}} - \epsilon_{e,(1,1)}^{\text{hi}}, \epsilon_{e,(k,k+4)}^{\text{hi}} - \epsilon_{e,(1,1)}^{\text{lo}}]$. This statement concerns the two defining low-cutoff metrics; CIT18–26 still uses its full fixed-column target extension at every later original cutoff.

Full finite source-order interval in the original primitive values. Keep CIT27–28, including its original finite-domain bounds and both gauge terms. Put $p = 0 : (s_0, s_1) = (1, 1)$ and $p = 1 : (k, j)$, $j = k+4$, $D_e = q_j - v - 1 + e$, $r_e = 8k + 24 - 3e$, and $\Delta Z = Z^1 - Z^0$. The full NSR2 constants are $\mathbf{a}_s = \rho_s \Pi_s$ and $\mathbf{b}_{s,D} = \rho_s U_{s,D}$, where

$$\rho_s = \frac{(2\pi)^{2\ell_s}}{(1/2)^{2\ell_s}}, \quad \Pi_s = \prod_{h < \ell_s} (2h + \tfrac{1}{2})^2, \quad U_{s,D} = \prod_{h < \ell_s} \{4(D + h + 1)^2 + (2h + \tfrac{1}{2})^2\}, \quad \ell_s = (s-1)/4.$$

All factors retain the original measure; the $s = 1$ products are one. Write $L_{\text{src}} = \Delta \log \det H_{\text{src},e}$, $L_{\text{conn}} = \Delta \log \det H_{\text{conn},e}$ and $b_e^- = \epsilon_e^{1,\text{lo}} - \epsilon_e^{0,\text{hi}}$, $b_e^+ = \epsilon_e^{1,\text{hi}} - \epsilon_e^{0,\text{lo}}$, using the complete CIT17 errors

at their original source pair and cutoff. Then the following finite intervals hold in the same ordered values:

$$\begin{aligned} s_e^- &:= -r_e \log \mathbf{b}_{k,D_e} \leq L_{\text{src}} \leq -r_e \log \mathbf{a}_k =: s_e^+, \\ t_e^- &:= -r_e \log \mathbf{b}_{j,D_e} \leq L_{\text{conn}} \leq -r_e \log \mathbf{a}_j =: t_e^+, \\ L_{\text{conn}} &= -\Delta\mathcal{X}_e + \zeta_e, \quad s_e^- + b_e^- \leq \zeta_e \leq s_e^+ + b_e^+. \end{aligned} \tag{CIT31}$$

Indeed the ordinary-transpose map NSR4 preserves the full Taylor values. It takes the source fibre $P_e x + T_0 \xi + V_0 \eta$ to that same fibre in the other original source order. On the target, NSR5's unique Taylor cochain square preserves F_e and every lower correction. Minimizing over those lower corrections and B_{cof} gives the image fibre, of the same rank r_e . The complete dual Loewner bounds therefore give the first two lines. Subtracting the two exact CIT17 identities gives the last line with $\zeta_e = L_{\text{src}} + \epsilon_e^1 - \epsilon_e^0$. Thus

$$s_e^- + b_e^- - t_e^+ \leq \Delta\mathcal{X}_e \leq s_e^+ + b_e^+ - t_e^-. \tag{CIT32}$$

This proves the finite receiving of CIT28, not only its limit. The old PSR26 source interval can additionally be intersected with the first line on its own stated finite domain and original admissible radius. No source centre or mass is changed. At $e = 0$ the exact complement term vanishes; at $e = 1$ all four original constraints remain inside χ_{Y_e} . The widths and endpoints are $O_{\text{actual}}(k^2 \log q_j) = o(kq_k)$. The later calculation continues to use fixed F_e at every target cutoff; no later primitive germ extension is asserted.

The defining primitive source with its separate exact map. Fix $e = 0$ or 1 , retain $D_e = q_{k+4} - v - 1 + e$ and $r_e = 8k + 24 - 3e$. The complete unscaled columns $r_h = \mathcal{T}_A(\chi_k S^h)$, $0 \leq h < 8k + 24 + e$, satisfy $\psi_k \mid r_h$ by the original root-addition identity. Their leading coefficient is $\binom{q_k+h}{v} \mu_v \neq 0$, so their divided polynomial degree is exactly $a_{\text{src},e} = 24k - 25 - v + e$. The full source relation Gram H_e and the same factorial-bearing matrix Θ_e give

$$H_{\text{src},e}^{(s)} = \Theta_e^* (\overline{H_e^{(s)}})^{-1} \Theta_e. \tag{EFR9}$$

EF32–34 proves the entire affine-fibre equality and its minimizing section: the annihilator kernel is the original analytic/lower plane, and its prescribed value is $\Theta_e = (R_e^{\text{pol}})^{\text{T}} \text{diag}(n!) P_e^{\text{Tay}}$ with R_e^{pol} the unscaled r_h columns. At $e = 1$ the four independent primitive rows remain in the rank- r_e value image. This is the actual source morphism used to apply QOT, even though the degree integer also equals $D_e - q'_k$.

Intersect the source NSR6 interval with QOT27 in this full map, and the target NSR6 interval with QOT30 at degree $a_{\text{tgt},e} = 16k + 15 - v + e$. Call the resulting rank- r_e determinant endpoints $S_e^{\pm}, T_e^{\pm}$, exactly as in EFP35–36, with each original root-radius guard retained separately. For the unchanged NSR7 errors $b_e^{\pm}$ one obtains

$$\begin{aligned} T_e^- - S_e^+ &\leq \mathcal{J}_e^{\text{ord}} \leq T_e^+ - S_e^-, \\ S_e^- + b_e^- &\leq \zeta_e \leq S_e^+ + b_e^+, \quad L_{\text{conn},e} = -\Delta\mathcal{X}_e + \zeta_e, \\ S_e^- - T_e^+ + b_e^- &\leq \Delta\mathcal{X}_e \leq S_e^+ - T_e^- + b_e^+. \end{aligned} \tag{EFR10}$$

Subtracting the exact source and target determinant identities proves these signs; the variables shared by ζ_e and $\Delta\mathcal{X}_e$ stay shared. EFP38 retains the scalar $\log(\kappa_j/\kappa_k)$ with its opposite signs in the Jacobian and compatibility expressions. The original PSR interval may also be intersected on its original guard, with all of its errors.

EOR's necessary parity rectangle guard fails at these defining low degrees: the actual QOT radius with $\gamma > 2$ gives $q_k > 2^{34}k$, whereas that EOR guard would give $q_k \leq 36k - 36$ at the source or $q_j \leq 24k + 24$ at the target. The proof in the EFP primitive section therefore uses QOT's proved low fallback here. It does not insert high-endpoint equilibrium constants into the primitive source. CIT31–32 and their LCC/NCR primitive receivers retain every source form, frame and compatibility error when receiving these narrower finite intervals.

CIT14–17 concern the defining low cutoff D_e . They do not assert that a primitive truncated germ has a common compatible extension to every later cutoff. The target coefficient columns F_e now are fixed, and their complete later metrics are calculated next without that assertion.

185.4 The fixed connecting flag at every original target row

For $q_j - 1 \leq N \leq 2q_j$ put $D = N - v$ and retain the fixed B_{cof}, F_e in the original target coefficients. Define

$$\begin{aligned} \mathcal{D}_N &= \text{diag}(D_{j,N-v}^{(s_1)}, 4), & Z_{B,N} &= [\mathcal{V}_N, \mathcal{T}_{1,N} B_{\text{cof}}], \\ Z_{C,N} &= [Z_{B,N}, \mathcal{T}_{1,N} F_e], & H_{\text{conn},e}(N) &= \text{Gram}_{\mathcal{D}_N}(\mathcal{S}_e/\mathcal{B}), \\ \det H_{\text{conn},e}(N) &= \det(Z_{C,N}^* Z_{C,N}) / \det(Z_{B,N}^* Z_{B,N}). \end{aligned} \quad (\text{CIT18})$$

The original DNE positivity and the independence in CIT10 make both raw frames full column rank at every one of these indices. Eliminating the complete lower frame and then B_{cof} proves the determinant identity. At the defining $N = q_j - 1 + e$, it is exactly CIT11–13 by CIT8.

Every entry is given by the original arithmetic recursion. If $\nu_{a,n} = \sum (Z_{\text{exp},j})_{uv,a} \beta_{j,u,v}^n$, then

$$\begin{aligned} f_{a,n} &= \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \\ b_{0,m} &= \mathbf{1}_{m=0}, \quad b_{-1,m} = 0, \quad b_{n+1,m} = i^{-1}(b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2) b_{n-1,m}, \quad (\text{CIT19}) \\ t_n &= \left(\frac{\sum_m b_{n,m} f_{a,m}}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_a, \quad v_n = \left(\frac{\sum_m b_{n,m} \beta_{j-8,u,v}^m}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_{u,v}. \end{aligned}$$

Arrays outside their polynomial range are zero. Differentiating $E_A F_a = N_a$ gives the first line; substituting the full original Gamma recurrence gives the next. Thus these are exactly the unchanged IRR7–9 rows, retaining all original moments and phases. The new four-row blocks in CIT18 are $z_{B,n} = [\text{diag}(v_n, 4), \text{diag}(t_n, 4) B_{\text{cof}}]$ and $z_{C,n} = [z_{B,n}, \text{diag}(t_n, 4) F_e]$. For $X = B, C$ define $G_{X,n} = Z_{X,n+v}^* Z_{X,n+v}$. In degree-major row order each raw Gram increases by $z^* z$. The fixed row permutation from copy-major order is unitary.

For $q_j - v \leq n \leq 2q_j - v$, the exact connecting-volume increment is

$$\log \frac{\det H_{\text{conn},e}(n+v)}{\det H_{\text{conn},e}(n+v-1)} = \log \det(I_4 + z_{C,n} G_{C,n-1}^{-1} z_{C,n}^*) - \log \det(I_4 + z_{B,n} G_{B,n-1}^{-1} z_{B,n}^*) \geq 0. \quad (\text{CIT20})$$

The determinant lemma proves the equality. For its sign, transform the old extra columns to their full least-norm section over $Z_{B,n+v-1}$. The larger row covariance is the smaller one plus $RQ^{-1}R^* \succeq 0$, by block inversion. This proves the inequality and retains the actual old quotient denominator. No independent rowwise gauge minimum is taken.

185.5 Its exact place in the common low invariant loss

Let $\mathbb{J} = \ker \mathbb{B}$ be the full original fourfold DNE invariant kernel. At each later N use the same actual target metric $\mathcal{D}_N$ and fixed flags

$$\begin{aligned} W &= \mathcal{B} + \mathbb{J}, & K_{\text{conn},e} &= \mathcal{S}_e \cap W = \mathcal{B} + (\mathcal{S}_e \cap \mathbb{J}), \\ T_{\text{conn},e} &= \mathcal{S}_e + W = \mathcal{S}_e + \mathbb{J}, \\ \omega_{\text{conn},e}(N) &= \frac{\det \text{Gram}_{\mathcal{D}_N}(T_{\text{conn},e}/W)}{\det \text{Gram}_{\mathcal{D}_N}(\mathcal{S}_e/K_{\text{conn},e})}. \end{aligned} \quad (\text{CIT21})$$

The two Grams use the same quotient-value coordinates under $[z] \mapsto [\mathbb{B}z]$. Its original fibres are exactly the two spaces in the denominator and numerator, as in LCC20–23. The source is the corresponding quotient of the connecting cohomology image; its full kernel is $(\mathcal{S}_e \cap \mathbb{J})/(\mathcal{B} \cap \mathbb{J})$. Thus CIT21 is its actual further target loss, and is in $(0, 1]$.

The full low cycles contain $\mathcal{S}_e$, and LCC13 gives

$$\dim(\mathcal{Z}_e/\mathcal{S}_e) = \kappa_e \leq 3 \max(v - e, 0). \quad (\text{CIT22})$$

Both spaces are fixed original coefficient flags for the later metric comparisons. The difference of their invariant loss returns has the finite bound

$$|\mathcal{R} \log \omega_e - \mathcal{R} \log \omega_{\text{conn},e}| \leq 2\kappa_e \log A_{j,s_1}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k). \quad (\text{CIT23})$$

Here $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$. For a complete verification, apply the actual inclusions $\mathcal{S}_e \subset \mathcal{Z}_e$ with their common W . The kernel/intersection quotient and sum quotient fit into the exact sequence $0 \rightarrow K_{\text{full}}/K_{\text{conn}} \rightarrow \mathcal{Z}_e/\mathcal{S}_e \rightarrow T_{\text{full}}/T_{\text{conn}} \rightarrow 0$ by the identity on representatives. Their two ranks sum to κ_e . The full adapted-Gram factorization LFC7–12 expresses the loss difference as the sum of those two quotient log determinants minus the $\mathcal{Z}_e/\mathcal{S}_e$ log determinant and a fixed frame constant. FRE's same native form sandwich puts the three monotone returns between zero and twice their ranks times $\log A_{j,s_1}^{\text{end}}$. Both sides of the resulting signed difference are bounded by $2\kappa_e \log A$. This proves CIT23, including $\kappa_e = 0$, with all frame constants canceled only after the original four signs.

The sharper original tail bound in CIT23. Retain the original simple-quartet five-orbit domain, its invertible period matrix, the full conductor and both native families. In particular this calculation imposes no additional pole-coordinate gate or restriction on that domain. Let $k = 4l + 1 \geq 9$, $j = k + 4$, $e \in \{0, 1\}$, and use exactly the coefficient spaces of CIT21–23. Put $\bar{\kappa}_e = 3 \max(v - e - 2, 0)$. The complete original four-factor calculation BTK26 gives $v \leq 32$, and BTK25 gives $\kappa_e \leq \bar{\kappa}_e$. The actual spaces and the strengthened finite estimate are

$$\begin{aligned} 0 \longrightarrow \frac{\mathcal{Z}_e \cap \mathbb{J}}{\mathcal{S}_e \cap \mathbb{J}} \xrightarrow{[j] \mapsto [j]} \frac{\mathcal{Z}_e}{\mathcal{S}_e} \xrightarrow{[z] \mapsto [z]} \frac{\mathcal{Z}_e + W}{\mathcal{S}_e + W} \longrightarrow 0, \\ d_{K,e} + d_{T,e} = \kappa_e \leq \bar{\kappa}_e, \quad \bar{\kappa}_0 \leq 90, \quad \bar{\kappa}_1 \leq 87, \\ \left| \mathcal{R} \log \omega_e^{(s_1)} - \mathcal{R} \log \omega_{\text{conn},e}^{(s_1)} \right| \leq 2\kappa_e \log A_{j,s_1}^{\text{end}} \leq 6 \max(v - e - 2, 0) \log A_{j,s_1}^{\text{end}}. \end{aligned} \quad (\text{RKR1})$$

Here $W = \mathcal{B} + \mathbb{J}$, and $d_{K,e}, d_{T,e}$ are the dimensions of the first and last nonzero terms. To prove exactness, the first arrow is injective since its zero class means $j \in \mathcal{S}_e \cap \mathbb{J}$. A middle class maps to zero precisely when $z = s + b + j$ with $s \in \mathcal{S}_e$, $b \in \mathcal{B} \subset \mathcal{S}_e$, $j \in \mathbb{J}$. Then $j = z - s - b \in \mathcal{Z}_e \cap \mathbb{J}$ and $[z] = [j]$. Every last class has a representative $z \in \mathcal{Z}_e$, because $W \subset \mathcal{S}_e + W$. This proves the full sequence. The identity on representatives also gives the isomorphism

$$\frac{\mathcal{Z}_e \cap \mathbb{J}}{\mathcal{S}_e \cap \mathbb{J}} \xrightarrow{\sim} \frac{\mathcal{Z}_e \cap W}{\mathcal{S}_e \cap W}, \quad [j] \mapsto [j].$$

Indeed $z \in \mathcal{Z}_e \cap W$ is $b + j$ with $b \in \mathcal{B}$, hence $j \in \mathcal{Z}_e \cap \mathbb{J}$; the kernel is exactly $\mathcal{S}_e \cap \mathbb{J}$. Thus these are exactly the two quotient ranks in the adapted determinant factorization already proved in CIT23.

For each fixed $s_1 = 1$ or j , use its unchanged form $\mathcal{D}_N = \text{diag}(D_{j,N-v}^{(s_1)}, 4)$. The two quotient logarithms of ranks $d_{K,e}, d_{T,e}$ enter with positive signs and the quotient of rank κ_e with a negative sign, together with a fixed frame constant. FRE's complete form sandwich and the same attained affine minima place each rank- r quotient return in $[0, 2r \log A_{j,s_1}^{\text{end}}]$. With the unchanged operator $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$, the constant cancels. Both sides of the resulting signed difference are at most $2\kappa_e \log A_{j,s_1}^{\text{end}}$, since $d_{K,e} + d_{T,e} = \kappa_e$. The logarithm is nonnegative, so BTK25 proves the last bound. If $v - e \leq 2$, this proves $\mathcal{Z}_e = \mathcal{S}_e$, and the quotient-value maps and their attained minima coincide: $\omega_e^{(s_1)}(N) = \omega_{\text{conn},e}^{(s_1)}(N)$ at every retained target cutoff in each native family.

There is a further exact relation to the ORIGINAL ITL source. Write $F_{\text{inv}} = \text{stack}_i(M_{i,k}J_0)$. Since $B_{\text{cof}} = \text{stack}_i M_{i,k}$,

$$\text{im } F_{\text{inv}} \subset \mathcal{B}, \quad \mathbb{J} \subset W_{\text{inv}} := \mathbb{J} + \text{im } F_{\text{inv}} \subset W \subset T_{\text{conn},e}. \quad (\text{CIT24})$$

These inclusions are literal original coefficient maps. The ITL source therefore represents boundaries in low cohomology, but its target Gram remains in the denominator $\text{Gram}(W)$ of the cohomology target quotient.

To exhibit every factor, let $I_R = F_{\text{inv}} Z_R$ be exactly the fixed ITL4 source-kernel complement and choose fixed columns R_b completing $[X_{\mathbb{J}}, I_R]$ to W . In a specified fixed full frame X_W , write

$$[X_{\mathbb{J}}, I_R, R_b] = X_W C_W, \quad Q_b(N) = \text{Gram}_{\mathcal{D}_N}(W/W_{\text{inv}}) \quad \text{in the lifts } R_b.$$

The first quotient of this chain has exactly ITL's target Gram $H_{\text{inv}}(N)$: its fibre is $I_R b + \mathbb{J}$. Its source pullback quotient is the full ITL $A_{\text{inv}}(N)$, and $\det H_{\text{inv}} = \det A_{\text{inv}} \omega_{\text{inv}}$. Sequential Schur complements consequently give

$$|\det C_W|^2 g_W(N) = g_{\mathbb{J}}(N) \det H_{\text{inv}}(N) \det Q_b(N), \quad g_X(N) = \det(X_X^* \mathcal{D}_N X_X). \quad (\text{CIT25})$$

All matrices in this formula are in their stated independent frames; $r_{\text{inv}} = 0$ gives the empty first quotient.

Finally retain the full LFC coordinate constant $c_{\text{conn},e}$ from the adapted frames for

$$(\mathcal{S}_e, K_{\text{conn},e}, T_{\text{conn},e}, W).$$

Its cross ratio and CIT25 give the exact original identity

$$\begin{aligned} \log \omega_{\text{conn},e} + \log \omega_{\text{inv}} &= \log g_{T_{\text{conn},e}} + \log g_{K_{\text{conn},e}} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} \\ &\quad - \log \det A_{\text{inv}} - \log \det Q_b + \log c_{\text{conn},e} + 2 \log |\det C_W|. \end{aligned} \quad (\text{CIT26})$$

This relates the two actual losses through their exact common boundary structure; it does not identify their source spaces. Taking the original four-return cancels the last two fixed coordinate constants. Equivalently ITL16's exact matched relative operator satisfies $\det H_{\text{inv}} = \det Q_{\text{inv}} \det \mathcal{R}_X$, where Q_{inv} is the full original source quotient $J_0^* D_{k,N-v}^{(s_0)} J_0 / E$. Substituting this equality instead in CIT25 replaces $(\omega_{\text{inv}}, A_{\text{inv}})$ in CIT26 by $(\det \mathcal{R}_X, Q_{\text{inv}})$, preserving the source-order pair $(1, 1)$ or (k, j) . It retains the original source mass.

Current complete residual quotient and its directed return. In CIT24–26's original coefficient spaces put $\Phi_b = \mathbb{B} B_{\text{cof}}$, $\mathcal{L}_b = \ker \Phi_b$ and $Y_b = \text{im } J_0 + \mathcal{L}_b$. The map $[x] \mapsto [B_{\text{cof}} x]$ identifies the exact quotient $\mathbb{C}^{m_k} / Y_b$ with W / W_{inv} : its zero fibre consists precisely of $x - J_0 a \in \ker \Phi_b$. RQB5 computes this full kernel by the original conductor and invariant coefficient matrices. Its branch constraint is RQB9–12 on the nonreduced component scheme, with every shared factor retained. Let Q_b^E be RQB14's complete attained target metric. For the original CIT25 lifts, $R_b^{\text{CIT}} = R_b^E C_b + Y_{\text{inv}} A_b$, so minimization over the identical full fibre proves

$$\begin{aligned} Q_b^{\text{CIT}} &= C_b^* Q_b^E C_b, \quad |\det C_W|^2 g_W = g_{\mathbb{J}} \det H_{\text{inv}} |\det C_b|^2 \det Q_b^E, \\ \mathcal{V}_b &= \mathcal{R} \log \det Q_b^{\text{CIT}} = \mathcal{R} \log \det Q_b^E, \quad 0 \leq \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4U_{j,s_1}^{\text{native}}. \end{aligned} \quad (\text{CIT29})$$

The exact matrix C_b is fixed across the original four target cutoffs; its determinant cancels only under their signs. Every row of Q_b^E is given by the complete division and Gamma recurrences RQB16–18, and its source denominator is the quotient in RQB19–20, including $B_0 \mathcal{L}_b$ and the original low overlap.

Write the remaining exact part of CIT26's four-return as

$$\mathcal{P}_e^{\text{rest}} = \mathcal{R}(\log g_{T_{\text{conn},e}} + \log g_{K_{\text{conn},e}} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} - \log \det A_{\text{inv}}).$$

Substituting the preceding metric congruence in that identity, with all fixed frame factors kept before the return, gives

$$\begin{aligned} \mathcal{R}(\log \omega_{\text{conn},e} + \log \omega_{\text{inv}}) &= \mathcal{P}_e^{\text{rest}} - \mathcal{V}_b, \\ \mathcal{P}_e^{\text{rest}} - 4\hat{F}_j^{(s_1)} + \mathcal{V}_{\text{inv}}^{\text{target}} &\leq \mathcal{R}(\log \omega_{\text{conn},e} + \log \omega_{\text{inv}}) \leq \mathcal{P}_e^{\text{rest}}. \end{aligned} \quad (\text{CIT30})$$

The lower sign is RQB28/32's inequality for $-\mathcal{V}_b$. Its full missing target quotient Q_o remains in $\mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} + \mathcal{V}_o = 4\widehat{F}_j^{(s_1)}$. The pointwise native cap is the complete RQB35 minimum, including the original positive observation and source corrections. Thus no boundary factor is identified with the source or assigned a common long-return value.

The signed remainder as the complete positive quotient decomposition. Retain $W, \mathcal{S}_e, K_e, T_e$ and the full RQB27 target frame $C_T = [U_R, U_b, U_o]$. The literal map $\pi_o = [0 \ 0 \ I_{r_o}]C_T^{-1}\mathbb{B}$ has kernel W and is onto. Indeed its value vanishes exactly when $\mathbb{B}z \in \text{im}(\mathbb{B}B_{\text{cof}})$. It therefore identifies $\mathcal{U}/W$ isometrically with the original missing quotient Q_o : first minimize over $\mathbb{J}$, then over that complete image. Put $E_e = \pi_o F_e$ and $c_e = \text{rank } E_e$. CPQ7–8 retains the original source quotient $G_{s,e}$ on $\mathcal{S}_e/K_e$, the target restriction $G_{t,e}$ on T_e/W , and the remaining quotient $Q_{\text{rem},e}$ on $\mathcal{U}/T_e$. Their values are fixed by E_e , its kernel complement Z_c , and $C_o = [E_e Z_c, R_e^o]$, so all coordinate determinants remain those of the actual coefficient maps.

Write $V_X = \mathcal{R} \log \det G_X$, with V_{rem} from Q_{rem} and V_A from the full original invariant pullback source A_{inv} . Then

$$\begin{aligned} V_o &= V_t + V_{\text{rem}}, & V_{\text{conn}} &= V_h + V_s, & 4\widehat{F}_j^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, \\ \mathcal{P}_e^{\text{rest}} &= 4\widehat{F}_j^{(s_1)} - V_{\text{rem}} - V_s - V_A, & \mathcal{L}_e &:= \mathcal{R} \log(\omega_{\text{conn},e} \omega_{\text{inv}}) = \mathcal{P}_e^{\text{rest}} - V_b, & & (\text{CIT33}) \\ V_I - V_s - V_A &\leq \mathcal{L}_e \leq 4U_{j,s_1}^{\text{native}} - V_b - V_s - V_A. \end{aligned}$$

Every listed V is nonnegative: it is a fixed restriction or attained quotient of the complete increasing original Gamma row form. The first two equalities follow from full adapted Gram determinants; the third is RQB27. Their substitution into $\mathcal{L}_e = V_t - V_s + V_I - V_A$ proves the next line. Before taking the return the constant is exactly

$$\begin{aligned} \mathbf{c}_e &= \frac{|\det C_T|^2 |\det C_o|^2 |\det C_b|^2}{c_{\text{conn},e} |\det C_W|^2}, \\ P(N) &= 4 \log \det H_{B,j,D} - \log \det Q_{\text{rem},e} - \log \det G_{s,e} \\ &\quad - \log \det A_{\text{inv}} + \log \mathbf{c}_e. \end{aligned}$$

This is CPQ12 obtained by literal substitution in CIT26 and RQB15/27. The constant cancels only under the four signs. The last interval in CIT33 uses $0 \leq V_t \leq V_o$ and the existing cap only on its proved finite domain.

For both original source pairs retain all widths in NSR2 and $r_I = j_{0,k} - \epsilon$, $r_b = t_k - \ell + \epsilon$. The same two quotient values for each contraction give

$$\begin{aligned} C_e^{\text{ord}} &= (c_e + r_I)W_j, & |\Delta \mathcal{L}_e| &\leq C_e^{\text{ord}}, \\ -C_e^{\text{ord}} - r_b \Omega_j^{\text{hi}} &\leq \Delta \mathcal{P}_e^{\text{rest}} \leq C_e^{\text{ord}} + r_b \Omega_j^{\text{lo}}. & & (\text{CIT34}) \end{aligned}$$

The connecting contraction has rank c_e , the invariant contraction rank r_I , both with target source order s_1 . Their common scalar factors cancel at each endpoint under the exact value maps; summing gives the first bound. The equality $\Delta \mathcal{P}_e^{\text{rest}} = \Delta \mathcal{L}_e + \Delta V_b$ and NSR19's original rank- r_b interval prove the second with its two signs. Each pair's CIT33 interval can be intersected with the other pair's interval translated by $[-C_e^{\text{ord}}, C_e^{\text{ord}}]$. Both Q_o and the independent degree- k source comparison remain explicit; no replacement of A_{inv} by Q_{inv} has occurred.

The common projection and its evaluated four-angle enclosure. On the unchanged CIT33 simple-quartet domain put $I = \text{im}(B_{\text{cof}} J_0)$, $B = \text{im } B_{\text{cof}}$, $S = \mathcal{S}_e$ and $J = \ker \mathbb{B}$. Retain $K_i = I \cap J$, $W_i = I + J$, $W = B + J$, $K_c = S \cap W$ and $T = S + J$. Here $S + J = S + W$ because $B \subset S$. The identity on representatives gives the exact quotient maps $I/K_i \rightarrow W_i/J$ and $S/K_c \rightarrow T/W$. Their kernels vanish by the intersection definitions, and their images are the whole target quotients by the sum definitions. The source minimum lifts from the orthogonal

sum $(I \ominus K_i) \oplus (S \ominus K_c)$; the target lifts form $(W_i \ominus J) \oplus (T \ominus W)$, in the original metric $\mathcal{D}_N$. Orthogonality follows from $I \subset K_c$ and $W_i \subset W$. Projection between these sums has block matrix $\begin{pmatrix} M_i & C_N \\ 0 & M_c \end{pmatrix}$: the zero block follows from $I \subset W$, and the diagonal blocks are exactly the two quotient maps in orthonormal frames. Thus its squared determinant is $\omega_{\text{conn},e}\omega_{\text{inv}}$, with the cross block retained.

SFB's full variation calculation on these fixed flags and SFC's explicit scalar optimization now apply to this actual product. Let L_-, L_+ be the finite expressions SFC9–10, using precisely $\omega(N_0) = \det G_t \det H_{\text{inv}} / (\det G_s \det A_{\text{inv}})$ at $N_0 = q_j - 1$ and the three full determinant ratios SFC3 at $N_1 = q_j$, $N_2 = 2q_j - 1$, $N_3 = 2q_j$. Then the earlier signed remainder receives

$$\begin{aligned} -4\mathcal{T}_{j,s_1} &\leq L_- \leq \mathcal{L}_e \leq L_+ \leq 4\mathcal{T}_{j,s_1}, \\ V_b + L_- &\leq \mathcal{P}_e^{\text{rest}} \leq V_b + L_+. \end{aligned} \tag{CIT35}$$

Indeed the forward positive metric path gives $|\Delta \text{arcosh}(\omega^{-1/2})| \leq \frac{1}{2}\Delta \log \det \mathcal{D}$ by the complete projector derivative SFC6–7. Its three successive applications give SFC8; the monotone calculation and unique cubic root in SFC9–15 give the first interval. Adding the exact identity $\mathcal{P}_e^{\text{rest}} = \mathcal{L}_e + V_b$ proves the second. All four positive complementary returns in CPQ11 remain. The CIT33 interval and both order translations CIT34 can be intersected with CIT35 in their original domains. The initial angle is the displayed actual determinant, not an assigned value.

Finite transport of the evaluated common-angle enclosure. In CIT35 retain both original quotient isomorphisms $I/(I \cap J) \rightarrow (I + J)/J$ and $S/(S \cap W) \rightarrow (S + W)/W$, given by the identity on representatives. The defining intersections give their zero kernels and the defining sums their full images. Their ranks are r_I, c_e . Each pair of matched quotient determinants receives EFP14, so subtraction gives the same-endpoint width $r w_{j,D}$, with $w_{j,D} = \lambda_{j,D}^+ - \lambda_{j,D}^-$. Hence

$$|\mathcal{L}_e^1 - \mathcal{L}_e^0| \leq B_e = (r_I + c_e) \sum_{i=0}^3 w_{j,D_i}. \tag{EFR3}$$

The two scalar source factors cancel before taking the width. The common projection cross block and all four quotient metrics remain those of CIT35. With its exact SFC endpoints $L_{\pm}^p$ put

$$\begin{aligned} \tilde{L}_-^1 &= \max(L_-^1, L_-^0 - B_e), & \tilde{L}_+^1 &= \min(L_+^1, L_+^0 + B_e), \\ \tilde{L}_-^0 &= \max(L_-^0, L_-^1 - B_e), & \tilde{L}_+^0 &= \min(L_+^0, L_+^1 + B_e). \end{aligned} \tag{EFR4}$$

CIT35 and EFR3 prove $\tilde{L}_-^p \leq \mathcal{L}_e^p \leq \tilde{L}_+^p$. Adding the exact V_b^p proves $V_b^p + \tilde{L}_-^p \leq \mathcal{P}_e^{\text{rest},p} \leq V_b^p + \tilde{L}_+^p$. The original CIT33 interval remains another valid intersection. The target radius and EOR tests in EFP4–9 are applied endpoint by endpoint; an unguarded improvement is never inserted.

Joint boundary cancellation and the original compressed profile. Work on the unchanged simple-quartet domain of CIT1 and CPQ1–2, with the same $e \in \{0, 1\}$, $k = 4l + 1 \geq 9$, $j = k + 4$, conductor order v , full fixed coefficient maps and source pair $(s_0, s_1) = (1, 1)$ or (k, j) . Set

$$\begin{aligned} \mathcal{H} &= \mathbb{C}^{4m_j}, & I &= \text{im}(B_{\text{cof}} J_0), & B &= \text{im } B_{\text{cof}}, \\ S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \mathbb{B}, & K_i &= I \cap J, & W_i &= I + J, \\ & & W &= B + J, & K_c &= S \cap W, & T &= S + J. \end{aligned} \tag{PTHCIT1}$$

Since $B \subset S$, the last space is also $S + W$. All these spaces are fixed as the native cutoff varies. Their metric is $\mathcal{D}_N = \text{diag}(D_{j,N-v}^{(s_1)}, 4)$, with all lower evaluations minimized in the original preceding Gram. Use exactly

$$\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j), \quad \mathcal{V}(Y/X) = \mathcal{R} \log \det \text{Gram}_{\mathcal{D}_N}(Y/X). \tag{PTHCIT2}$$

The quotient form is the attained minimum over the complete fibre X , in fixed quotient-value coordinates. The maps $I/K_i \rightarrow W_i/J$ and $S/K_c \rightarrow T/W$ send $[z]$ to the same representative class $[z]$. Their kernels are zero by the defining intersections, and the defining sums make them onto. Thus they are exactly the original invariant and connecting maps; in their respective common value coordinates their squared volume ratios are ω_{inv} and $\omega_{\text{conn},e}$. In particular, with $\mathcal{L}_e = \mathcal{R} \log(\omega_{\text{conn},e} \omega_{\text{inv}})$,

$$\mathcal{L}_e = \mathcal{V}(T/W) + \mathcal{V}(W_i/J) - \mathcal{V}(S/K_c) - \mathcal{V}(I/K_i). \quad (\text{PTHCIT3})$$

Here is the full coordinate calculation for the remaining determinant identity. For the chain $X_0 = J \subset X_1 = W_i \subset X_2 = W \subset X_3 = T \subset X_4 = \mathcal{H}$, choose fixed full frames A_h of X_h and fixed lifts L_h of the specified quotient coordinates. Write $[A_{h-1}, L_h] = A_h C_h$, where C_h is invertible. With $F_h = A_h^* \mathcal{D}_N A_h$ and quotient Gram G_h , full Schur elimination gives

$$\det G_h = |\det C_h|^2 \frac{\det F_h}{\det F_{h-1}}. \quad (\text{PTHCIT4})$$

For the original direct $\mathcal{H}/J$ quotient frame, likewise write $[A_0, L_*] = A_4 C_*$ and obtain $\det G_* = |\det C_*|^2 \det F_4 / \det F_0$. Multiplying the four displayed identities therefore gives $\prod_{h=1}^4 \det G_h = (\prod_{h=1}^4 |\det C_h|^2) |\det C_*|^{-2} \det G_*$. The entire fixed coefficient factor cancels under $\mathcal{R}1 = 0$. This proves the telescoping identity in the actual independent frames, without replacing a varying minimum lift by a fixed orthogonal representative.

The onto map $\mathbb{B}$ identifies $\mathcal{H}/J$ isometrically with its original target quotient: minimizing first over J gives four copies of $(B_j D_{j,N-v}^{-1} B_j^*)^{-1}$. Consequently the HJR/NSR inverse-metric identity is $\mathcal{V}(\mathcal{H}/J) = 4S_{K,j,s_1}$. Substituting (PTHCIT3) into the proved telescoping identity yields

$$\begin{aligned} 4S_{K,j,s_1} &= \mathcal{L}_e + \mathcal{V}(\mathcal{H}/T) + \mathcal{V}(W/W_i) + \mathcal{V}(S/K_c) + \mathcal{V}(I/K_i) \\ &= \mathcal{P}_e^{\text{rest}} + \mathcal{V}(\mathcal{H}/T) + \mathcal{V}(S/K_c) + \mathcal{V}(I/K_i). \end{aligned} \quad (\text{PTHCIT5})$$

For the second equality use the exact CIT boundary quotient $\mathcal{V}(W/W_i) = V_b$ and $\mathcal{L}_e = \mathcal{P}_e^{\text{rest}} - V_b$. Every fixed factor before taking the return is retained in CIT26 and CPQ12. This is their joint cancellation, at the original signs. The spaces and metrics of the three surviving returns still depend on the same boundary flag.

In the explicit CPQ coordinates these three returns are, respectively, V_{rem}, V_s, V_A . The identification of the first quotient is the map $[z] \mapsto [\pi_o z]$ from $\mathcal{H}/T$ to the quotient of the original remaining target by $\text{im } E_e$. The underlying map on $\mathcal{H}$ has kernel precisely T , since $\ker \pi_o = W$ and $\pi_o S = \text{im } E_e$; the displayed induced map on $\mathcal{H}/T$ therefore has zero kernel and is onto by CPQ4. Minimizing first over W and then over T/W proves its isometry with $Q_{\text{rem},e}$. The second is CPQ7's minimum on S/K_c , in the lifts $F_e Z_c$. For the third put $F = B_{\text{cof}} J_0$ and $E = \ker(\mathbb{B}F)$. Then $FE = I \cap J = K_i$, and the minimum over FE in the lifts FZ_R is precisely A_{inv} . These equalities prove all three typed metric identifications; the separate degree- k source form Q_{inv} retains its own relative comparison.

Finally the eight exact matrices and weighted estimate (PTHS1)–(PTHS6) below give, in this same source order,

$$|4S_{K,j,s_1} - (P_{k,e} + V_b + V_{\text{rem}} + V_s + V_A)| \leq 32q_j \log 2. \quad (\text{PTHCIT6})$$

Indeed CPQ20 or the first line of (PTHCIT5) says that the expression inside the absolute value is exactly $\mathcal{L}_e - P_{k,e}$. Thus the finite reduction preserves all four positive complementary returns. It assigns no absolute allocation coefficient or value to their correlated sum.

Thus the surviving connecting target is given by the explicit matrix CIT7, its complete target minimum CIT11–13, the original low source-to-image determinant CIT17, and all later original row formulas CIT18–20. The fixed-rank remainder is controlled by CIT23. The remaining common target loss has the exact original ITL boundary relation CIT24–26, with its whole residual overlap, all absolute native Grams and both source measures still present.

186 Primitive-source transport inside the original connecting image

Keep the complete original PDS1–24 and CIT1–26 data at fixed $k = 4l + 1 \geq 9$, $j = k + 4$, $D_e = (k + 5)^2 - v - 1 + e$, and either original source-order pair $(s_0, s_1) = (1, 1)$ or (k, j) . All polynomial coefficients, period data, lower gauges and exterior signs remain those of those constructions. In this section $P_e, F_e, T_{a,D_e}, V_{a,D_e}$ are CIT’s primitive lifts, connecting target columns, analytic frame and full lower frame. The polynomial coefficient matrix C_e and metric $G_e = \overline{(C_e^* C_e)}^{-1}$ are PDS7. The boundary column matrix is the unchanged $B_{\text{cof}} = \text{stack}_i M_{i,k}$.

186.1 The exact source coordinate map and its target naturality

The coordinate maps between the two proved primitive presentations are

$$\Theta_e = C_e^\top P_e, \quad \Theta_0 : \mathbb{C}^{d_0} \xrightarrow{\simeq} \mathbb{C}^{d_0}, \quad \Theta_1 : \mathbb{C}^{d_0-3} \xrightarrow{\simeq} \ker W^\top, \quad d_0 = 8k + 24. \quad (\text{PDI1})$$

The inverse signs mean inverses onto the displayed images. Indeed PDS4 is the onto quotient by precisely CIT4’s analytic source, and P_e is a full basis of that primitive quotient. The PDS8 four-row condition is CIT5’s same polynomial differential. Thus both maps are bijections onto the asserted spaces. Equality of the exact minimum-norm fibres proves

$$H_{\text{src},e} = \Theta_e^* G_e \Theta_e. \quad (\text{PDI2})$$

For completeness a native vector w and $P_e a$ have the same PDS quotient coordinates exactly when $C_e^\top(w - P_e a) = 0$, equivalently $w - P_e a \in \text{im}[V_{k,D_e}, T_{k,D_e}]$ by PDS4. The minimum over this fibre is both sides of PDI2, by CIT14 and PDS7. This proves the metric identity as well as its linear map.

Current full source value and the exact common-order frame. Here $q = q_k = (k + 1)^2$, $e = 0, 1$, $d_e = 8k + 24 + e$, and $N_e = D_e = q_{k+4} - v - 1 + e$, as in the original PDS/PDI cutoff. Let R_e^{pol} contain the original coefficients of the full relation polynomials, and let P_e^{Tay} be CIT6’s original fixed Taylor complement. For each source order $s = 1, k$, its original monomial-to-Gamma matrix B_s and Taylor map U_s satisfy

$$U_s = \text{diag}(1/r!) B_s^\top, \quad C_e^{(s)} = B_s R_e^{\text{pol}}, \quad P_e^{(s)} = U_s^{-1} P_e^{\text{Tay}}, \quad (\text{PDI17})$$

$$\Theta_e^{(s)} = (R_e^{\text{pol}})^\top \text{diag}(r!) P_e^{\text{Tay}}.$$

All indices are $0 \leq r \leq N_e$. Multiplying the three displayed matrices proves the identity: the exact Gamma matrix and inverse cancel, including every mass and phase. Thus Θ_e is independent of s on these same original value coordinates, with no conditioning assertion. At $e = 0$ set $\Gamma_0 = 2 \log |\det \Theta_0|$; at $e = 1$ put $Z = \Theta_1$, $J_Z = \overline{Z} (Z^\top \overline{Z})^{-1}$ and $\Gamma_1 = -2 \log |\det[W, J_Z]|$. The exact PDI2 metric now has the full receiving formula

$$\log \det H_{\text{src},e}^{(s)} = \Gamma_e - \log \mathfrak{F}_{k,e,s} - 2d_e q \log(4/\pi) + e m_{k,s}^{(4)} + e \varepsilon_{k,s}^{(4)} - \varepsilon_{k,e,s}^H, \quad (\text{PDI18})$$

$$\log \det H_{\text{src},e}^{(s)} = \Gamma_e - 2d_e q \{\log(4q/\pi) - 1\} + o(kq), \quad \log \frac{\det H_{\text{src},e}^{(k)}}{\det H_{\text{src},e}^{(1)}} = o(kq).$$

These are PSR19–26 in precisely PDI2’s equal minimum-norm fibres. The finite interval for the first residual is $[eb_{k,s}^{\text{lo}} - \mathcal{E}_{k,e,s}^+(a), eb_{k,s}^{\text{hi}} + \mathcal{E}_{k,e,s}^-(a)]$. For the order ratio, its exact factorial/mass center is D_e^{ord} of PSR26, and its residual interval is $[I_{e,k}^- - I_{e,1}^+, I_{e,k}^+ - I_{e,1}^-]$. The identical Γ_e cancels only after the proved same-coordinate order comparison. Its absolute size is not presumed small. The original truncation matrix R , the three missing primitive directions and both native increments in PDI3–16 retain all their previous factors.

Let $J_k^{\text{tr}} : \mathcal{H}_{k,D_1} \rightarrow \mathcal{H}_{k,D_0}$ and $J_j^{\text{tr}} : \mathcal{H}_{j,D_1} \rightarrow \mathcal{H}_{j,D_0}$ be literal jet truncation, so they delete the last native functional coordinate; the source order and centre are fixed during this truncation. Let pr_0 delete the last polynomial quotient coordinate as in PDS11, and set

$$Z = \text{pr}_0 \Theta_1, \quad R = \Theta_0^{-1} Z, \quad \text{rank } R = d_0 - 3, \quad \Theta_1 = EZ. \quad (\text{PDI3})$$

The rank assertion and last equality follow from PDS11. The first d_0 relation polynomials are unchanged between cutoffs; therefore $C_0^T J_k^{\text{tr}} P_1 = \text{pr}_0 \Theta_1 = \Theta_0 R$. Their difference is in the old analytic source. In its original independent frame there are unique matrices L and A such that

$$J_k^{\text{tr}} P_1 = P_0 R + V_{k,D_0} L + T_{k,D_0} A. \quad (\text{PDI4})$$

They are explicit finite matrices: with $U = [V_{k,D_0}, T_{k,D_0}]$ their exact block expression is

$$\begin{bmatrix} L \\ A \end{bmatrix} = (U^* U)^{-1} U^* (J_k^{\text{tr}} P_1 - P_0 R).$$

The actual connecting coefficient columns satisfy

$$\boxed{F_1 = F_0 R + B_{\text{cof}} A.} \quad (\text{PDI5})$$

To prove it, the original multiplier square commutes with truncation: $(\text{diag}(J_j^{\text{tr}}, 4)) \mathbb{T}^{(1)} = \mathbb{T}^{(0)} J_k^{\text{tr}}$. This is the equality of the Taylor convolutions in CIT3 after discarding the last coefficient. Likewise truncation takes each original lower or analytic target column to the corresponding old column, because they are jets of the same entire or analytic germ. Apply these equalities to the target representative of F_1 in CIT8 and use PDI4. Its old target jet differs from that of $F_0 R + B_{\text{cof}} A$ by an allowed lower evaluation, using the root-addition square on $V_{k,D_0} L$. The original degree- j analytic coefficient map modulo lower evaluations is injective at D_0 by CNC13. Thus the coefficient columns themselves agree, proving PDI5. No inverse estimate is used to pass from equality of the jets to these coefficients.

In particular, in the single original target coefficient space,

$$\mathcal{S}_1 = \text{im}[B_{\text{cof}}, F_1] \subset \mathcal{S}_0 = \text{im}[B_{\text{cof}}, F_0], \quad \dim(\mathcal{S}_0/\mathcal{S}_1) = 3. \quad (\text{PDI6})$$

Independence of each displayed frame is proved in CIT10, and PDI3 gives the rank of the quotient inclusion. This proves the codimension and identifies the actual map from primitive truncation to the fixed connecting coefficient flags.

186.2 The two low metric increments in common coordinates

The old source restricted to this image and the upper source, in precisely the P_1 coordinates, have Grams

$$K_{\text{src},0} = R^* H_{\text{src},0} R = Z^* G_0 Z, \quad K_{\text{src},1} = H_{\text{src},1} = (EZ)^* G_1 (EZ). \quad (\text{PDI7})$$

PDS12–24 now apply to these actual connecting-source frames:

$$K_{\text{src},1} = K_{\text{src},0} + \frac{(\eta Z)^*(\eta Z)}{\sigma}, \quad 0 \leq \Delta_{\text{src}} := \log \frac{\det K_{\text{src},1}}{\det K_{\text{src},0}} \leq \log U_* = o(kq_k). \quad (\text{PDI8})$$

Every constant in U_* is the original PDS19 finite product; it has the fully proved explicit bound PDS24.

In the original target define

$$K_{\text{tgt},0} = R^* H_{\text{conn},0} R, \quad K_{\text{tgt},1} = H_{\text{conn},1}. \quad (\text{PDI9})$$

By PDI5 these are the metrics of the same fixed quotient $\mathcal{S}_1/\mathcal{B}$ at the two native target cutoffs, in the same F_1 value coordinates. Adding the boundary $B_{\text{cof}} A$ does not change its minimum over

that boundary. This proves the first identity in PDI9 as a quotient statement, including its frame, rather than presuming equality of two arbitrary representatives.

Here is its complete four-row update. At D_0 set

$$\begin{aligned} N_B &= [\text{diag}(V_{j,D_0}, 4), \text{diag}(T_{j,D_0}, 4)B_{\text{cof}}], & N_F &= \text{diag}(T_{j,D_0}, 4)F_1, \\ G_B &= N_B^* N_B, & G_{BF} &= N_B^* N_F, & K_{\text{tgt},0} &= N_F^* N_F - G_{BF}^* G_B^{-1} G_{BF}. \end{aligned} \quad (\text{PDI10})$$

Both original lower and boundary frames are kept in N_B . Let z_B, z_F be the next degree's four rows of these very frames, computed from CIT19, and define

$$S_B = I_4 + z_B G_B^{-1} z_B^*, \quad r_B = z_F - z_B G_B^{-1} G_{BF}. \quad (\text{PDI11})$$

Their exact update is

$$\begin{aligned} K_{\text{tgt},1} &= K_{\text{tgt},0} + r_B^* S_B^{-1} r_B, \\ \Delta_{\text{tgt}} &:= \log \frac{\det K_{\text{tgt},1}}{\det K_{\text{tgt},0}} = \log \det \left(I_4 + S_B^{-1/2} r_B K_{\text{tgt},0}^{-1} r_B^* S_B^{-1/2} \right) \geq 0. \end{aligned} \quad (\text{PDI12})$$

For proof replace N_F by its old orthogonal residual $N_F - N_B G_B^{-1} G_{BF}$. In this unchanged quotient the old joint Gram is $\text{diag}(G_B, K_{\text{tgt},0})$; the added rows are $[z_B, r_B]$. The new Schur complement is $K_{\text{tgt},0} + r_B^* [I - z_B (G_B + z_B^* z_B)^{-1} z_B^*] r_B$. Multiplying by $I + z_B G_B^{-1} z_B^*$ proves that the bracket is S_B^{-1} . Taking the determinant gives PDI12. The column operation has identity diagonal blocks, so its determinant is one and it leaves no missing frame factor.

There is a finite bound using the complete original native next-row budget. Write $D_0 = D_{j,D_0}^{(s_1)}$ for the single-copy original coefficient metric, $T = T_{j,D_0}$, $V = V_{j,D_0}$, and let t, v_{row} be their next original rows. Set

$$\begin{aligned} A_V &= V^* V, \quad \zeta = v_{\text{row}} A_V^{-1} v_{\text{row}}^*, \quad \rho = t - v_{\text{row}} A_V^{-1} V^* T, \\ \beta_{\text{next}} &= 4 \log \left(1 + \frac{\rho D_0^{-1} \rho^*}{1 + \zeta} \right), \quad 0 \leq \Delta_{\text{tgt}} \leq \beta_{\text{next}} \leq 4 \log A_{j,s_1}^{\text{end}} = o(kq_k). \end{aligned} \quad (\text{PDI13})$$

The exact lower-gauge Schur update is $D_1 = D_0 + \rho^* \rho / (1 + \zeta)$, as in IRR and LFC. Thus the full four-copy ambient log-determinant increment is β_{next} . To prove the middle bound without multiplying by a quotient dimension, put the old ambient form into its isometric coordinates and write the new one as e^B with $B \succeq 0$. For any fixed subspace with full frame Y , differentiation along e^{tB} gives

$$\frac{d}{dt} \log \det(Y^* e^{tB} Y) = \text{Tr}(B P_Y(t)), \quad P_Y(t) = e^{tB/2} Y (Y^* e^{tB} Y)^{-1} Y^* e^{tB/2}.$$

$P_Y(t)$ is the orthogonal projector onto $e^{tB/2} Y$. For nested fixed spaces its two projectors are ordered, so the derivative of their quotient log determinant is between zero and $\text{Tr } B$. Integrating proves that the increment of $\mathcal{S}_1/\mathcal{B}$ is at most the full ambient increment, which is β_{next} . Finally FRE's original endpoint form bound implies that each of the four identical single-copy rank-one relative eigenvalues is at most A_{j,s_1}^{end} , proving the last inequality. Its already proved growth is $\log A_{j,s_1}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k)$. The isometric coordinates in this proof establish a comparison of the unchanged forms; the formulas PDI10–13 remain in their original frames.

The exact adjacent source-to-image determinant correction is therefore the difference of the explicit four-row and one-row log determinants PDI12 and PDI8:

$$\begin{aligned} \Delta_{\text{image/source}} &:= \log \frac{\det K_{\text{tgt},1}}{\det K_{\text{src},1}} - \log \frac{\det K_{\text{tgt},0}}{\det K_{\text{src},0}} = \Delta_{\text{tgt}} - \Delta_{\text{src}}, \\ -\log U_* &\leq \Delta_{\text{image/source}} \leq \beta_{\text{next}}, \quad \Delta_{\text{image/source}} = o(kq_k). \end{aligned} \quad (\text{PDI14})$$

The lower determinant uses the proved common primitive subspace R ; its three-dimensional complement has not been silently discarded from the full old cohomology.

186.3 The two fixed connecting invariant losses

At every later original target cutoff keep the literal flags of CIT21 and the same $W = \mathcal{B} + \mathbb{J}$. By PDI6 their nesting has codimension exactly three. The identity on representatives gives the exact sequence

$$0 \longrightarrow K_0/K_1 \longrightarrow \mathcal{S}_0/\mathcal{S}_1 \longrightarrow T_0/T_1 \longrightarrow 0, \quad K_e = \mathcal{S}_e \cap W, \quad T_e = \mathcal{S}_e + W. \quad (\text{PDI15})$$

Surjectivity follows from the definition of the sum; its kernel consists of the class of $x \in \mathcal{S}_0$ with $x \in \mathcal{S}_1 + W$, hence of $K_0 + \mathcal{S}_1$ modulo $\mathcal{S}_1$. Its kernel presentation is K_0/K_1 . Thus the two end ranks sum to three. The complete adapted Gram factorization LFC7–12 expresses the difference of the two loss logarithms as $\log \det \text{Gram}(T_0/T_1) + \log \det \text{Gram}(K_0/K_1) - \log \det \text{Gram}(\mathcal{S}_0/\mathcal{S}_1)$ plus a fixed coordinate constant. Under the original four-return $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$ that constant cancels. FRE bounds each monotone quotient return between zero and twice its rank times $\log A_{j,s_1}^{\text{end}}$. The positive two ranks sum to three and the negative rank is three; therefore

$$|\mathcal{R} \log \omega_{\text{conn},0} - \mathcal{R} \log \omega_{\text{conn},1}| \leq 6 \log A_{j,s_1}^{\text{end}} = o(kq_k). \quad (\text{PDI16})$$

This is the complete primitive-image refinement of the full-cohomology comparison in LFC. The retained $\ker U_e$ constituents of the full LCC sequence are still present, with their explicit CIT23 bounds. The common long connecting loss continues to be the exact target expression CIT21 and its ITL boundary factorization CIT24–26. The original low source transport now feeds that target calculation through the explicit coefficient identity PDI5, the two native increments PDI12–14, and the proved fixed-flag comparison PDI16.

187 The original monic source band and its low-polynomial quotient

This calculation evaluates the source band in PDS4 before its actual conductor translation. The metric distortion of that translation is treated separately and is not assumed here.

187.1 Original coordinates and finite parameters

Fix the actual simple quartet, period, branch and conductor of PDS. Write $\sigma = (\delta^2 + \gamma^2)^{1/2}$, with $\delta, \gamma > 0$, and retain

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad q = (k + 1)^2, \quad e \in \{0, 1\}, \quad d = 8k + 24 + e, \\ \nu &= q + d - 1 = (k + 5)^2 - 1 + e, \quad c_0 = k/2, \quad c_1 = k/2 - 4, \\ \beta_{k,a,b} &= k(1/2 - \delta - i\gamma) + 2a\delta + 2bi\gamma, \quad \chi_k(S) = \prod_{a,b=0}^k (S - \beta_{k,a,b}), \quad (\text{PSG1}) \\ u_a(S) &= \chi_k(S)S^a \quad (0 \leq a < d), \quad s = 1 \text{ or } k, \quad \ell_s = (s - 1)/4, \quad Q_s = q + \ell_s, \\ \theta &= \pi/2, \quad h = 1/q, \quad \kappa = \theta - h, \quad R = k\sigma, \quad B = \max(1, R, k/2), \quad m = 2Q_s/\theta. \end{aligned}$$

The moment exponent Q_s is distinct from the target root count $(k + 5)^2$. The complete input norm and polynomials are

$$\begin{aligned} dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \quad h_{n,s} = M_s n! (s/2)_n, \\ p_0 &= 1, \quad p_1 = y, \quad p_{n+1} = yp_n - n(n - 1 + s/2)p_{n-1}, \quad \phi_n(S) = p_n((S - c_0)/i)/\sqrt{h_{n,s}}, \quad (\text{PSG2}) \\ \|p\|_{s,c_0}^2 &= \int_{\mathbb{R}} |p(c_0 + iy)|^2 dm_s(y), \quad H_{e,s}^\chi = \text{Gram}_{s,c_0}(u_0, \dots, u_{d-1}). \end{aligned}$$

These are PDS6's full Gamma norms with the specified input center; the output of translation retains c_1 . The original roots in the y coordinate satisfy

$$\zeta_{a,b} = (\beta_{k,a,b} - c_0)/i = (2b - k)\gamma - i(2a - k)\delta, \quad |\zeta_{a,b}| \leq R, \quad \chi_k(c_0 + iy) = i^q \prod_{a,b} (y - \zeta_{a,b}). \quad (\text{PSG3})$$

The change from columns $\chi_k S^a$ to $\chi_k((S - c_0)/i)^a$ is triangular with diagonal i^{-a} . Its determinant has modulus one, proving equality of the Gram determinants while retaining the entire coordinate change.

187.2 The complete Gamma density

Set

$$\beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \quad C_L = \frac{\pi}{\Gamma(3/4)^2}, \quad L_h = C_L(\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi}(\sin h)^{-1/2}. \quad (\text{PSG4})$$

Applying the Gamma recurrence ℓ_s times gives exactly

$$dm_s(y) = \beta_s \prod_{r=0}^{\ell_s-1} (y^2 + (2r + 1/2)^2) d\sigma(y), \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{PSG5})$$

The recurrence contributes $4^{-\ell_s}$, cancelling $2^{2\ell_s}$ from $2^{s/2}$; the remaining scalar is $\beta_s/(2\pi)$. Empty products are one.

Rotation of the Euler integral through $\text{sgn}(y)(\pi/2 - h)$, for $a > 0$, gives

$$|\Gamma(a + iy/2)| \leq \Gamma(a)(\sin h)^{-a} e^{-(\pi/2 - h)|y|/2}.$$

For justification, truncate between radii ε, T . The small arc is $O(\varepsilon^a)$; the large arc vanishes because $\cos t \geq \sin h > 0$ throughout the sector. The absolute ray integral is $\int_0^\infty e^{-r \sin h} r^{a-1} dr = \Gamma(a)(\sin h)^{-a}$. The exact reflection product is

$$|\Gamma(1/4 + iy/2)\Gamma(3/4 - iy/2)|^2 = 2\pi^2 / \cosh(\pi y).$$

Dividing by the upper bound for its second Gamma factor and using $\cosh(\pi y) \leq e^{\pi|y|}$ proves

$$L_h e^{-(\theta+h)|y|} \leq \frac{d\sigma}{dy}(y) \leq U_h e^{-(\theta-h)|y|} \quad (y \in \mathbb{R}). \quad (\text{PSG6})$$

This includes zero and the full inner region; PSG5 retains the mass.

187.3 Finite determinant bounds in all original roots

Define

$$f_\theta(x) = 2x \log(2x/\theta) - 2x \quad (x > 0). \quad (\text{PSG7})$$

Whenever $m \geq 2R$, one has the explicit inequalities

$$\begin{aligned} d \log \beta_s + df_\theta(Q_s) - B_{k,e,s}^- &\leq \log \det H_{e,s}^\chi \\ &\leq d \log \beta_s + df_\theta(Q_s) + B_{k,e,s}^+, \end{aligned} \quad (\text{PSG8})$$

with

$$\begin{aligned} B_{k,e,s}^- &= d \log^+(1/L_h) + \theta^2 d^3 / Q_s + dh(m + 2d) + 4dqR/m, \\ B_{k,e,s}^+ &= d \log^+ U_h + d\kappa B + d \log 2 + d(1 + \log(2Q_s)) \\ &\quad + d(d-1) \log(2Q_s + 2d) + d(2Q_s + d) \log(\theta/\kappa). \end{aligned} \quad (\text{PSG9})$$

Here $\log^+ x = \max(0, \log x)$. For example the finite condition holds for $k \geq \max(9, \lceil \theta\sigma \rceil + 1)$, since $Q_s \geq q > k^2$.

For the upper bound, each original root factor at y or $-y$, $y \geq 0$, is at most $y + R$ in modulus. Each positive factor in PSG5 is at most $(y + B)^2$. The a th column in the y frame therefore has squared norm at most

$$2\beta_s U_h e^{\kappa B} \frac{(2Q_s + 2a)!}{\kappa^{2Q_s + 2a + 1}}. \quad (\text{PSG10})$$

Indeed $y^{2a} \leq (y + B)^{2a}$; substitute $t = y + B$ and enlarge the remaining integral from $[B, \infty)$ to $[0, \infty)$. Hadamard's inequality gives the product of these d bounds. In this setting its proof is the successive orthogonal projection of the columns: every squared remainder norm is at most its original norm, and their product is the Gram determinant.

Monotonicity of $\log x$ gives, for every integer $n \geq 1$,

$$n \log n - n + 1 \leq \log(n!) \leq n \log n - n + 1 + \log n. \quad (\text{PSG11})$$

Also $(2Q_s + 2a)! \leq (2Q_s)!(2Q_s + 2d)^{2a}$. In the product of PSG10 the total extra factorial exponent is $\sum_a 2a = d(d-1)$, and the rate exponent is $\sum_a (2Q_s + 2a + 1) = d(2Q_s + d)$. Substituting PSG11 proves the upper side of PSG8. The remaining term $-d^2 \log \theta$ is nonpositive since $\theta = \pi/2 > 1$; it is discarded only in this upper inequality, not in PSG10.

For the lower bound, write $w(y)dy = |\chi_k(c_0 + iy)|^2 dm_s(y)$. Expansion of two Vandermonde determinants and Fubini gives the exact identity

$$\det H_{e,s}^\chi = \frac{1}{d!} \int_{\mathbb{R}^d} \prod_{i < t} (y_t - y_i)^2 \prod_{i=1}^d w(y_i) dy_i. \quad (\text{PSG12})$$

Every moment converges by PSG6; the $d!$ permutations of one Vandermonde produce the same moment determinant. Retain the d original-coordinate intervals $I_i = [m + 2i, m + 2i + 1]$, $0 \leq i < d$.

On their product every pair distance is at least one. Its $d!$ permutations are disjoint and cancel $1/d!$. All intervals have length one.

If $y = m + t$ lies in their union then $0 \leq t \leq 2d$. The inequality $\log(1+x) \geq x - x^2/2$, $x \geq 0$, gives

$$2Q_s \log y - \theta y \geq f_\theta(Q_s) - \theta^2 d^2 / Q_s.$$

Also $|y - \zeta| \geq y - R$, and $R/y \leq 1/2$. Integrating $1/(1-x) \leq 2$ on this interval yields

$$|\chi_k(c_0 + iy)|^2 \geq y^{2q} e^{-4qR/y} \geq y^{2q} e^{-4qR/m}.$$

The recurrence factors in PSG5 are at least $y^{2\ell_s}$. Thus throughout the retained intervals

$$w(y) \geq \beta_s L_h \exp(f_\theta(Q_s) - \theta^2 d^2 / Q_s - h(m + 2d) - 4qR/m).$$

Substitution into PSG12 proves the lower side of PSG8. For fixed actual data, $Q_s \asymp q$, $d = O(k)$, $R, B = O(k)$, and $h = 1/q$. Using $2h/\pi \leq \sin h \leq h$ and $\log(\theta/\kappa) \leq h/(\theta - h)$ in PSG9 proves

$$B_{k,e,s}^\pm = O_{\text{actual}}(q \log q) = o(kq). \quad (\text{PSG13})$$

The bounds themselves remain the finite expressions PSG9.

187.4 The exact original low-polynomial quotient

Keep $v = \text{ord}_0 E_A \leq 80$, $\mu_v \neq 0$, and let

$$U_e = \text{span}(u_0, \dots, u_{d-1}), \quad L_v = \mathbb{C}[S]_{<v}, \quad J_{e,s}^\chi = \text{Gram}((U_e + L_v)/L_v) \text{ in the lifts } u_a. \quad (\text{PSG14})$$

A nonzero multiple of χ_k has degree at least $q > v$; therefore $U_e \cap L_v = 0$ and this quotient Gram is positive. For $v = 0$, $J^\chi = H^\chi$ and set $C_E = 1$. For $v \geq 1$, $k \geq v - 1$, choose the original nodes

$$z_b = \zeta_{0,b} = (2b - k)\gamma + ik\delta \quad (0 \leq b < v), \quad Z = \max_{b < v} |z_b|, \quad A_{v,s} = \max(1, 2(v + 1 + s/2)).$$

Their differences are exactly $2\gamma(b - t)$. Define

$$(C_L^{\text{int}})^2 = \sum_{b=0}^{v-1} \frac{\prod_{t \neq b} (A_{v,s} + |z_t|)^2}{(2\gamma)^{2(v-1)} (b!(v-1-b)!)^2}, \quad (\text{PSG15})$$

$$\rho = 1/2, \quad a_\rho = \text{arctanh } \rho,$$

$$C_E^2 = \max \left\{ 1, v(C_L^{\text{int}})^2 (\nu + 1) \max(1, 2\sqrt{\nu})(1 - \rho^2)^{-s/2} e^{2Za_\rho} \rho^{-2\nu} \right\}.$$

Then

$$-2v \log C_E \leq \log \det J_{e,s}^\chi - \log \det H_{e,s}^\chi \leq 0, \quad \log C_E = O_{\text{actual}}(q). \quad (\text{PSG16})$$

The inverse of evaluation on L_v is the Lagrange map with columns $l_b(y) = \prod_{t \neq b} (y - z_t) / (z_b - z_t)$. In the full Gamma orthonormal basis, multiplication by y has raising coefficients $\sqrt{(n+1)(n+s/2)}$ and lowering coefficients $\sqrt{n(n-1+s/2)}$, by PSG2. Through degree $v-1$ their sum has norm at most $A_{v,s}$. Starting with the constant of squared norm M_s , multiplication by each factor gives $\sum_b \|l_b\|^2 \leq M_s (C_L^{\text{int}})^2$. Cauchy-Schwarz bounds the norm of the evaluation inverse by the square root of that number.

The recurrence PSG2 also gives the complete generating identity

$$\sum_{n \geq 0} p_n(y) t^n / n! = (1 + t^2)^{-s/4} e^{y \arctan t}.$$

In fact summing the recurrence gives $(1 + t^2)\partial_t F = (y - st/2)F$, $F(0) = 1$, whose solution is precisely this series. On $|t| = \rho$, $|\arctan t| \leq \operatorname{arctanh} \rho$. Cauchy's coefficient formula bounds the full evaluation norm through degree ν , at the chosen v nodes, by the square root of

$$\frac{v}{M_s}(1 - \rho^2)^{-s/2} e^{2Za_\rho} \sum_{n=0}^{\nu} \frac{n!}{(s/2)_n} \rho^{-2n}. \quad (\text{PSG17})$$

Every factorial and the original mass follow from division by $h_{n,s}$. For $s \geq 1$ this ratio is at most $a_n = n!/(1/2)_n$. One has $a_1 = 2$ and $a_{n+1}/a_n = (n+1)/(n+1/2) \leq \sqrt{(n+1)/n}$, since $n(n+1) \leq (n+1/2)^2$. Thus $a_n \leq 2\sqrt{n}$ for $n \geq 1$. The value at zero is one. This proves the finite sum bound in PSG15. Multiplying the evaluation bound and the bound for its inverse on L_v cancels M_s and yields C_E .

For $l \in L_v, u \in U_e$, evaluation kills u , and therefore $\|l\| \leq C_E \|l + u\|$. The attained minimum over u gives a quotient Gram on L_v between C_E^{-2} and one times its original Gram. Factoring the full Gram of $[L_v, U_e]$ in both orders gives the exact same-frame identity

$$\frac{\det J^\chi}{\det H^\chi} = \frac{\det \text{Gram}((L_v + U_e)/U_e) \text{ in the fixed } L_v \text{ lifts}}{\det \text{Gram}(L_v)}. \quad (\text{PSG18})$$

Its v eigenvalues prove PSG16. There is no omitted frame determinant: the same combined frame occurs in both factorizations. Finally v is fixed, $Z = O(k)$, $A_{v,s} = O(k)$, $\nu = O(q)$, and the nonzero Lagrange denominators are fixed. The logarithm of the explicit PSG15 expression is therefore $O(q)$.

187.5 The original factorial band

For $0 \leq a < d$ define

$$n_a = q + a - v, \quad m_a = n_a + (s-1)/2, \quad \lambda_a = \mu_v \binom{q+a}{v}, \quad \mathfrak{F}_{k,e,s} = \prod_{a=0}^{d-1} |\lambda_a|^2 h_{n_a,s}. \quad (\text{PSG19})$$

These are the exact PDS relation leading coefficients and degrees. No original relation polynomial has been replaced by its leading term. Let

$$\begin{aligned} E_{k,e,s}^{\text{band}} &= 2d\ell_s \max(|\log(2q/\theta)|, |\log(2Q_s/\theta)|) \\ &\quad + \sum_{a < d} (|n_a - q| + |m_a - q|) \log \max(q, n_a, m_a) \\ &\quad + \sum_{a < d} (3 + \log n_a + \log(2m_a) + \log m_a + \tfrac{1}{2} |\log \pi|) \\ &\quad + \tfrac{d}{2} \log 2 + 2 \sum_{a < d} |\log |\lambda_a||. \end{aligned} \quad (\text{PSG20})$$

Then PSG8 and PSG16 imply the completely finite bounds

$$\begin{aligned} -B_{k,e,s}^- - 2v \log C_E - E_{k,e,s}^{\text{band}} &\leq \log \frac{\det J_{e,s}^\chi}{\mathfrak{F}_{k,e,s}} - 2dq \log(4/\pi) \\ &\leq B_{k,e,s}^+ + E_{k,e,s}^{\text{band}}. \end{aligned} \quad (\text{PSG21})$$

To check all constants, use $(s/2)_{n_a} = \Gamma(m_a + 1/2)/\Gamma(s/2)$. The exact difference of the center in PSG8 and $\log \mathfrak{F}$ is

$$df_\theta(Q_s) - \sum_{a < d} \{\log(n_a!) + \log \Gamma(m_a + 1/2)\} - \tfrac{d}{2} \log 2 - 2 \sum_{a < d} \log |\lambda_a|. \quad (\text{PSG22})$$

The complete $d \log M_s$ and $d \log \Gamma(s/2)$ cancel here against the retained $d \log \beta_s$, by its exact PSG4 value.

For $F(x) = x \log x - x$, PSG11 gives $|\log(n!) - F(n)| \leq 1 + \log n$. The exact identity $\Gamma(m+1/2) = \sqrt{\pi}(2m)!/(4^m m!)$ gives, by the same bounds,

$$|\log \Gamma(m+1/2) - F(m)| \leq 2 + \log(2m) + \log m + \frac{1}{2} |\log \pi|.$$

All $n_a, m_a \geq 1$. Integrating $F'(x) = \log x$ between the indicated positive integer endpoints bounds their shifts from q by the second line of PSG20. Integrating $f'_\theta(x) = 2 \log(2x/\theta)$ gives its first line. Finally $f_\theta(q) - 2F(q) = 2q \log(2/\theta)$. The remaining two lines retain all factorial errors and original leading scalars. This proves PSG21.

Since $d = 8k + 24 + e$, $\ell_s = O(k)$, $n_a - q, m_a - q = O(k)$, and $|\log |\lambda_a|| = O_{\text{actual}}(\log q)$ with fixed v, μ_v , PSG20 is $O_{\text{actual}}(q \log q)$. Combining the finite bounds gives, in both original source orders and at both primitive cutoffs,

$$\boxed{\frac{1}{kq} \log \frac{\det J_{e,s}^\chi}{\mathfrak{F}_{k,e,s}} \longrightarrow 16 \log(4/\pi).} \quad (\text{PSG23})$$

The same limit holds with H^χ instead of J^χ . This evaluates a specified full original source quotient. Passing it through $\mathcal{T}_A$ requires its actual distortion on this d -dimensional source. No bound on that distortion or value for a later arithmetic allocation is assumed in PSG23.

188 The original finite Gamma translation on every small-rank plane

Fix the original simple quartet, its actual five-orbit period and logarithm branch. Keep every coefficient of the original nonreduced conductor:

$$\begin{aligned}
 0 < \delta < \tfrac{1}{2}, \quad \gamma > 2, \quad \beta_* = \tfrac{1}{2} - \delta - i\gamma, \quad \beta_{n,a,b} = n\beta_* + 2a\delta + 2bi\gamma, \\
 E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b}z}, \quad \mu_m = E_A^{(m)}(0), \quad v = \text{ord}_0 E_A, \quad 0 \leq v \leq 80, \quad \mu_v \neq 0, \quad (\text{SGT1}) \\
 k = 4l + 1 \geq 9, \quad s \in \{1, k\}, \quad c_0 = k/2, \quad c_1 = k/2 - 4.
 \end{aligned}$$

The coefficients, symbol, order and first nonzero moment are fixed as k varies. No particular coefficient is assumed nonzero except through the stated exact order. Every result below is for this original operator and these two original source orders. An arbitrary integer $N \geq 0$ is allowed in the finite statements.

188.1 Full Gamma measure, recurrence and coefficient isometry

Write $b = s/2 > 0$ and retain

$$\begin{aligned}
 M_s &= (2\pi)^{s/2}, \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \\
 \|P\|_{s,c}^2 &= \int_{\mathbb{R}} |P(c + iy)|^2 dm_s(y), \quad (\text{SGT2}) \\
 p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+b)p_{n-1,s}.
 \end{aligned}$$

These are exactly the measure and recurrence in PDS6 and DNE7. For completeness their generating and norm identities are derived here. The recurrence, multiplied by $t^n/n!$ and summed, gives $(1+t^2)\partial_t F = (y-bt)F$, $F(y,0) = 1$. The unique formal solution, also holomorphic for $|t| < 1$ at fixed y , is

$$F(y, t) = \sum_{n \geq 0} p_{n,s}(y) \frac{t^n}{n!} = (1+t^2)^{-b/2} \exp(y \arctan t). \quad (\text{SGT3})$$

Both branches take their value zero or one at $t = 0$ as appropriate.

Here is a direct check of the full mass and norms. With $a = s/4$, the Euler gamma integrals and the change of variables $r = u + v_0$, $x = \log(u/v_0)$ in their product give

$$|\Gamma(a + i\xi)|^2 = \Gamma(2a) \int_{\mathbb{R}} e^{i\xi x} (2 \cosh(x/2))^{-2a} dx.$$

The integrand as a function of x is analytic for $|\Im x| < \pi$. Shifting the integration line by any fixed amount of modulus less than π is justified by its exponential decay at both real ends. It bounds this Fourier transform by a constant times $e^{-h|\xi|}$ for each $0 < h < \pi$. Fourier inversion is therefore absolutely convergent; it may also be obtained by inserting a Gaussian factor and passing to the limit by dominated convergence. Substituting $y = 2\xi$ and the exact multiplier in SGT2 gives

$$\int_{\mathbb{R}} e^{iuy} dm_s(y) = M_s (\cosh u)^{-b}, \quad \int_{\mathbb{R}} e^{uy} dm_s(y) = M_s (\cos u)^{-b} \quad (|u| < \pi/2). \quad (\text{SGT4})$$

For the second equality, the preceding exponential bounds give analytic continuation to the indicated strip; equality on the real Fourier axis then proves it there. In particular the mass is exactly M_s . For sufficiently small real t, z , SGT3-4 and $\cos(\arctan t + \arctan z) = (1 - tz)/\sqrt{(1+t^2)(1+z^2)}$ imply

$$\int_{\mathbb{R}} F(y, t) F(y, z) dm_s(y) = M_s (1 - tz)^{-b}.$$

The exponential moment bound justifies differentiating under the integral in a neighborhood of $(0, 0)$. Comparing coefficients proves

$$\int_{\mathbb{R}} p_{n,s}(y) p_{m,s}(y) dm_s(y) = \mathbf{1}_{n=m} h_{n,s}, \quad h_{n,s} = M_s n! (b)_n, \quad (SGT5)$$

$$\phi_{n,s,c}(S) = \frac{p_{n,s}((S-c)/i)}{\sqrt{h_{n,s}}}, \quad \rho_n = \sqrt{\frac{n!}{(b)_n}} > 0.$$

The $p_{n,s}$ are real monic polynomials by their recurrence. Thus these ϕ_n are orthonormal for the complex Hilbert norm in SGT2 and span each indicated polynomial degree range. Their leading coefficient in S is $i^{-n}/\sqrt{h_{n,s}}$, not $1/\sqrt{h_{n,s}}$.

188.2 The original quotient map and its exact Toeplitz matrix

Let $\mathcal{P}_m = \mathbb{C}[S]_{\leq m}$, with $\mathcal{P}_{-1} = 0$. The original translation is

$$(\mathcal{T}_A P)(S') = \sum_{a,b=0}^8 a_{ab} P(S' + \beta_{8,a,b}), \quad \mathcal{T}_A S^m = \sum_{j=0}^m \binom{m}{j} \mu_j S'^{m-j}. \quad (SGT6)$$

The second equality is the binomial theorem followed by the exact definition of every moment. If $m \geq v$ and P has leading coefficient $p_m \neq 0$, then $\mathcal{T}_A P$ has degree $m - v$ and leading coefficient $p_m \binom{m}{v} \mu_v \neq 0$. If $m < v$ its image is zero. Consequently the following map is an isomorphism, with an onto representative map and the displayed full kernel:

$$\begin{aligned} \ker(\mathcal{T}_A : \mathcal{P}_{N+v} \longrightarrow \mathcal{P}_N) &= \mathcal{P}_{v-1}, \\ A_{N,s} : (\mathcal{P}_{N+v}/\mathcal{P}_{v-1}, \|\cdot\|_{s,c_0,\text{quot}}) &\xrightarrow{\simeq} (\mathbb{C}[S']_{\leq N}, \|\cdot\|_{s,c_1}), \\ \|[P]\|_{s,c_0,\text{quot}}^2 &= \min_{L \in \mathcal{P}_{v-1}} \|P + L\|_{s,c_0}^2. \end{aligned} \quad (SGT7)$$

Surjectivity follows successively by degree from the nonzero diagonal coefficients in SGT6. The orthogonal representative of this quotient is precisely the span of $\phi_{v,s,c_0}, \dots, \phi_{N+v,s,c_0}$. It is not a quotient by an omitted arithmetic relation.

On the entire open unit disk put

$$w(t) = -i \arctan t, \quad g(t) = t^{-v} e^{-4w(t)} E_A(w(t)), \quad g(0) = (-i)^v \mu_v / v! \neq 0. \quad (SGT8)$$

The value at zero is removable by the exact order v . For $|t| < 1$, $|w(t)| \leq \operatorname{arctanh} |t|$ by the arctangent series. This proves holomorphy of the specified g . Applying the finite sum SGT6 to SGT3, with its polynomial argument, gives exactly

$$\begin{aligned} \sum_{m \geq 0} (\mathcal{T}_A p_{m,s}((S - c_0)/i))(S') \frac{t^m}{m!} &= (1 + t^2)^{-b/2} e^{(S' - c_0)w(t)} E_A(w(t)) \\ &= t^v g(t) (1 + t^2)^{-b/2} e^{(S' - c_1)w(t)}. \end{aligned} \quad (SGT9)$$

The center factor is e^{-4w} because $c_0 - c_1 = 4$. No inverse of $\mathcal{T}_A$, or scalar α_A , has been inserted in this identity.

For a holomorphic germ $u(t) = \sum_{j \geq 0} u_j t^j$ define $T_N(u)_{r,n} = u_{n-r}$ for $0 \leq r \leq n \leq N$, and zero for $r > n$. Coefficients of SGT9 and $n!/\sqrt{h_{n,s}} = \rho_n/\sqrt{M_s}$ give the complete matrix of SGT7 in its two orthonormal frames:

$$\begin{aligned} A_{N,s} &= D_{\text{out}}^{-1} T_N(g) D_{\text{in}}, \\ D_{\text{out}} &= \operatorname{diag}(\rho_0, \dots, \rho_N), \quad D_{\text{in}} = \operatorname{diag}(\rho_v, \dots, \rho_{N+v}), \\ (A_{N,s})_{r,n} &= \begin{cases} g_{n-r} \rho_{n+v} / \rho_r, & r \leq n, \\ 0, & r > n. \end{cases} \end{aligned} \quad (SGT10)$$

Each source and target frame contains $1/\sqrt{M_s}$. Those identical factors cancel only in this coefficient matrix, since s is unchanged. All actual squared norms still use the full SGT2 measure. In particular, for $N = 0$ the matrix is the literal scalar $(-i)^v \mu_v \rho_v / v!$.

188.3 A proved original-data zero-free disk

The following constants use every original coefficient and shift:

$$\begin{aligned} A_{\text{abs}} &= \sum_{a,b} |a_{ab}|, & B &= \max\{1, \max_{a,b} |\beta_{8,a,b}|\}, & C_v &= \frac{A_{\text{abs}} B^v}{|\mu_v|} \geq 1, \\ R_0 &= \frac{1}{B} \min\{1, (v+1)/(2 \exp(1) C_v)\}, & a_0 &= \min\{1/2, \tanh(R_0/2)\}, & & \text{(SGT11)} \\ d_0 &= 1 - \frac{a_0^2}{3(1-a_0^2)} \geq 8/9, & M_0 &= \frac{2v!}{|\mu_v|} d_0^{-v} e^{2R_0}. \end{aligned}$$

All are finite, R_0, a_0, d_0, M_0 are positive, and $a_0 < 1$. Indeed $|\mu_v| \leq A_{\text{abs}} B^v$ proves the stated bound on C_v . Define the entire removable quotient $B_\infty(z) = v! E_A(z)/(\mu_v z^v)$, with value one at zero. Its full Taylor remainder is bounded by

$$|B_\infty(z) - 1| \leq \frac{C_v B |z|}{v+1} e^{B|z|} \leq \frac{1}{2} \quad (|z| \leq R_0). \quad \text{(SGT12)}$$

For the first inequality expand each exponential, use $|\mu_m| \leq A_{\text{abs}} B^m$ and $v!/(v+1+j)! \leq 1/((v+1)j!)$, and sum the exponential series. The two inequalities $BR_0 \leq 1$ and $C_v BR_0/(v+1) \leq 1/(2 \exp(1))$ prove the second. This is the complete RDS11 disk proof for the unchanged symbol.

On $|t| \leq a_0$ we have $|w(t)| \leq R_0/2$ and, also at zero,

$$\left| \frac{\arctan t}{t} - 1 \right| \leq \sum_{j \geq 1} \frac{a_0^{2j}}{2j+1} \leq \frac{a_0^2}{3(1-a_0^2)}.$$

Thus $|w(t)/t| \geq d_0$. Factoring SGT8 as $g(t) = (\mu_v/v!)(w(t)/t)^v e^{-4w(t)} B_\infty(w(t))$ shows that it has no zero on that entire closed disk and proves

$$|g(t)| \geq \frac{|\mu_v|}{2v!} d_0^v e^{-2R_0}, \quad |1/g(t)| \leq M_0 \quad (|t| \leq a_0). \quad \text{(SGT13)}$$

This supplies the zero-free disk and the inverse bound from the actual finite symbol rather than assuming them.

188.4 Every actual pole and its finite-rank Toeplitz contribution

Call $0 < a < 1$ admissible when g has no zero on $|t| = a$. Zeros of a nonzero holomorphic function are isolated; therefore there are only finitely many in each compact subdisk. There are admissible radii arbitrarily close to one. For a fixed such a , let $\mathcal{Z}_a$ be the exact zeros ζ of g with $|\zeta| < a$, and let m_ζ be their orders. Zero is not among them. Retain every multiplicity and define the actual Laurent coefficients

$$\begin{aligned} H_a &= \sum_{\zeta \in \mathcal{Z}_a} m_\zeta, & A_{\zeta,h} &= \frac{1}{(m_\zeta - h)!} \frac{d^{m_\zeta - h}}{dt^{m_\zeta - h}} \left(\frac{(t - \zeta)^{m_\zeta}}{g(t)} \right) \Big|_{t=\zeta}, & 1 \leq h \leq m_\zeta, \\ f &= 1/g, & f_a(t) &= f(t) - \sum_{\zeta \in \mathcal{Z}_a} \sum_{h=1}^{m_\zeta} \frac{A_{\zeta,h}}{(t - \zeta)^h}. \end{aligned} \quad \text{(SGT14)}$$

The factors t^{-v} and e^{-4w} in g create no nonzero zero. Moreover $w'(t) = -i/(1+t^2) \neq 0$ and $e^{-2w(t)} = (1+it)/(1-it)$ proves injectivity of w on the disk. Thus these poles are precisely the symbol zeros already identified in PEC3–5, under this exact change of variable, with their same orders. The extra center factor is fully retained in every $A_{\zeta,h}$. The function f_a is holomorphic on a neighborhood of the closed a disk, since all its interior principal parts have been subtracted and no boundary zero exists.

Here are completely specified finite constants on that circle:

$$\begin{aligned}
W_a &= \operatorname{arctanh} a, \quad m_{E,a} = \min_{|t|=a} |E_A(w(t))| > 0, \quad G_a = A_{\text{abs}} a^{-v} e^{(B+4)W_a}, \\
M_a &= \frac{a^v e^{4W_a}}{m_{E,a}} + \sum_{\zeta, h} \frac{|A_{\zeta, h}|}{(a - |\zeta|)^h}, \quad P_a = \sum_{\zeta, h} \frac{|A_{\zeta, h}|}{(1 - |\zeta|)^h}, \\
S_N(a) &= \sum_{j=0}^N a^{-j} = a^{-N} \frac{1 - a^{N+1}}{1 - a}.
\end{aligned} \tag{SGT15}$$

Empty pole sums are zero. The minimum is the same actual finite-circle quantity used in PEC4, and is positive by continuity and admissibility. It is not a new uniform zero-free assertion. The triangle inequality on that circle gives $|g| \leq G_a$ and $|f_a| \leq M_a$. Every constant in SGT15 is defined by the original fixed function and its exact finitely many interior poles; none depends on k, N or s .

For one term $A/(t - \zeta)^h$, its Taylor coefficients at zero are

$$c_n = A(-1)^h \zeta^{-(n+h)} \binom{n+h-1}{h-1} \quad (n \geq 0). \tag{SGT16}$$

Extend the binomial as the polynomial $\binom{x+h-1}{h-1} = \prod_{\ell=1}^{h-1} (x + \ell)/(h-1)!$ to every integer x , also for $h = 1$ by the empty product. The full matrix $C_{i,j} = c_{j-i}$, $0 \leq i, j \leq N$, has rank at most h : after multiplying rows by ζ^{-i} and columns by ζ^j , its entries are a polynomial of total degree at most $h-1$ in $j-i$. Expanding that polynomial in powers of i gives at most h column vectors, independently of N . Equivalently its column space is contained in $\operatorname{span}\{(\zeta^i i^d)_i : 0 \leq d < h\}$. For all terms belonging to one pole, these nested spaces give combined rank at most m_ζ , rather than the sum of the integers h .

For a lower-diagonal gap $\ell = i - j \geq 1$, direct substitution in the polynomial binomial gives

$$c_{-\ell} = \begin{cases} 0, & 1 \leq \ell < h, \\ -A\zeta^{\ell-h} \binom{\ell-1}{h-1}, & \ell \geq h. \end{cases} \tag{SGT17}$$

For $\ell \geq h$ each factor in the binomial is negative, and their $(-1)^{h-1}$ sign combines with $(-1)^h$ to give the displayed minus sign. For $1 \leq \ell < h$ one factor is zero. Consequently the upper triangular Taylor matrix of this pole is $C + R$, where R is strictly lower triangular and its gap- ℓ coefficient is $A\zeta^{\ell-h} \binom{\ell-1}{h-1}$ for $\ell \geq h$. If J is the upper shift with $J_{i,i+1} = 1$, then $(J^*)^\ell$ has norm at most one. The absolutely convergent binomial series therefore proves

$$\|R\| \leq |A| \sum_{\ell=h}^{\infty} |\zeta|^{\ell-h} \binom{\ell-1}{h-1} = \frac{|A|}{(1 - |\zeta|)^h}. \tag{SGT18}$$

The series identity follows by differentiating the geometric series $h-1$ times, or multiplying h copies of it and collecting coefficients. All powers and signs are those of the original pole; no principal part has been deleted from the actual Toeplitz inverse.

Since $T_N(u) = \sum_{j=0}^N u_j J^j$, multiplication of these finite matrices is truncated power-series multiplication. In particular $T_N(g)^{-1} = T_N(f)$, because $g(0) \neq 0$ and $gf = 1$ as germs. Cauchy's coefficient estimate, SGT13 and SGT15–18 prove

$$\begin{aligned}
\|T_N(g)\| &\leq F_{a,N} := G_a S_N(a), \\
\|T_N(f)\| &\leq G_{0,N} := M_0 S_N(a_0), \\
T_N(f) &= C_{a,N} + R_{a,N}, \quad \operatorname{rank} C_{a,N} \leq H_a, \quad \|R_{a,N}\| \leq V_{a,N} := M_a S_N(a) + P_a.
\end{aligned} \tag{SGT19}$$

Indeed sum the coefficient estimates times $\|J^j\| \leq 1$ for the analytic terms, and sum the exact pole ranks and lower-triangular remainders just proved. The total inverse matrix is unchanged.

188.5 Finite exterior bounds, including the expensive directions

Write $d = N + 1$ and arrange singular values in decreasing order. If a matrix C has rank at most H , then $\ker C$ has codimension at most H . For $x \in \ker C$, $\|(C + R)x\| \leq \|R\|\|x\|$. The variational description of singular values therefore gives $\sigma_j(C + R) \leq \|R\|$ for $j > H$. To verify this description directly, diagonalize $(C + R)^*(C + R)$: the complement of its first $j - 1$ eigenvectors attains the upper bound, and every subspace of codimension $j - 1$ intersects the first j eigenvectors for the reverse inequality. Applying this argument to SGT19 and retaining the separate global bound gives

$$h = \min\{r, H_a\}, \quad 0 \leq r \leq d, \quad V_{a,N}^{\text{cap}} = \min\{G_{0,N}, V_{a,N}\}, \quad (\text{SGT20})$$

$$\left\| \bigwedge^r T_N(g) \right\| \leq F_{a,N}^r, \quad \left\| \bigwedge^r T_N(f) \right\| \leq G_{0,N}^h (V_{a,N}^{\text{cap}})^{r-h}.$$

The norm of the r th exterior power is the product of the r largest singular values: this follows by applying a singular-value decomposition to the orthonormal wedge bases. Only at most H_a directions use the zero-free-radius bound. If $H_a > d$, then $h = r$ and there is no tail factor. Every constant is positive; the empty exterior power has norm one.

Retain the exact diagonal constants from the original norms:

$$\alpha_+ = \left(\max_{0 \leq n \leq N} \rho_{n+v} \right) \left(\max_{0 \leq n \leq N} \rho_n^{-1} \right), \quad \alpha_- = \left(\max_{0 \leq n \leq N} \rho_{n+v}^{-1} \right) \left(\max_{0 \leq n \leq N} \rho_n \right), \quad (\text{SGT21})$$

$$L_{s,N,v} = \frac{1}{2} \max\{1, s/2 - 1\} \mathfrak{H}_{N+v}, \quad \mathfrak{H}_m = \sum_{j=1}^m 1/j, \quad \mathfrak{H}_0 = 0.$$

For $s = k \geq 9$, the product defining ρ_n and $\log(1+x) \leq x$ imply $|\log \rho_n| \leq (b-1)\mathfrak{H}_n/2$. For $s = 1$, its logarithm is positive and $\log((j+1)/(j+1/2)) \leq 1/(2j+1) \leq 1/(j+1)$ gives $|\log \rho_n| \leq \mathfrak{H}_n/2$. Thus $|\log \rho_n| \leq L_{s,N,v}$ for all $0 \leq n \leq N+v$, including $n = 0$. In particular $|\log \alpha_{\pm}| \leq 2L_{s,N,v}$. Applying the two diagonal maps in SGT10 on exterior powers gives

$$\left\| \bigwedge^r A_{N,s} \right\| \leq \alpha_+^r F_{a,N}^r, \quad (\text{SGT22})$$

$$\left\| \bigwedge^r A_{N,s}^{-1} \right\| \leq \alpha_-^r G_{0,N}^h (V_{a,N}^{\text{cap}})^{r-h}.$$

This retains the rectangular degree shift $n \mapsto n + v$ exactly.

Let $[P_1], \dots, [P_r]$ be any independent original classes in SGT7 and use that same ordered frame in both Grams:

$$H_{\text{in}} = (\langle [P_i], [P_j] \rangle_{s, c_0, \text{quot}})_{i,j}, \quad H_{\text{out}} = (\langle \mathcal{T}_A P_i, \mathcal{T}_A P_j \rangle_{s, c_1})_{i,j}, \quad \Delta_{N,s}(\mathbf{P}) = \log \frac{\det H_{\text{out}}}{\det H_{\text{in}}}. \quad (\text{SGT23})$$

The inner products are conjugate-linear in their first variable. Both Grams are positive, by independence and SGT7. Exterior volumes and their inverse give the finite, potentially asymmetric bounds

$$-2r \log \alpha_- - 2h \log G_{0,N} - 2(r-h) \log V_{a,N}^{\text{cap}} \leq \Delta_{N,s}(\mathbf{P}) \leq 2r \log \alpha_+ + 2r \log F_{a,N}. \quad (\text{SGT24})$$

The classical Gram/exterior identity used here is recorded in Lyons [H3, §4, (4.1), p. 11]; the source and image frames below specify its exact application to SGT24.

For proof, express the classes in their SGT7 orthonormal quotient frame as columns X . Their squared wedge norm is $\det(X^*X)$, and the squared norm of their image wedge is $\det(X^*A_{N,s}^*A_{N,s}X)$. Apply SGT22 to that wedge and then to its image with $A_{N,s}^{-1}$. A change of the original ordered frame by a matrix U multiplies both determinants by $|\det U|^2$; the ratio therefore retains the original frame and is frame invariant. No specially conditioned plane has been selected.

For a nonnegative enclosure write $\log^+ x = \max\{0, \log x\}$ and set

$$\begin{aligned}\mathcal{U}_{a,N,s,r} &= 4rL_{s,N,v} + 2r \log^+ F_{a,N}, \\ \mathcal{L}_{a,N,s,r} &= 4rL_{s,N,v} + 2h \log^+ G_{0,N} + 2(r-h) \log^+ V_{a,N}^{\text{cap}}.\end{aligned}\tag{SGT25}$$

Then $-\mathcal{L} \leq \Delta_{N,s}(\mathbf{P}) \leq \mathcal{U}$. These are finite explicit constants even for $r = 0$, with value zero for that empty Gram ratio. All maxima, minima, poles and residues in the constants refer to the fixed original symbol SGT1.

188.6 Uniform small-rank volume distortion and the actual PDS frame

Fix finite constants $C_N, C_r > 0$. Uniformly over $0 \leq N \leq C_N k^2$, $0 \leq r \leq \min\{C_r k, N+1\}$, $s = 1$ or k , and every original rank- r plane in SGT7,

$$\sup \frac{|\Delta_{N,s}(\mathbf{P})|}{k^3} \longrightarrow 0 \quad (k = 4l+1 \longrightarrow \infty).\tag{SGT26}$$

Here the supremum is over precisely those finite-degree and rank domains, including all coefficient choices for the planes. We give both limiting estimates. For each fixed admissible a , $S_N(a) \leq (N+1)a^{-N}$ and $a^{-N} \geq 1$, so

$$\begin{aligned}\log^+ F_{a,N} &\leq N \log(1/a) + \log(N+1) + \log^+ G_a, \\ \log^+ V_{a,N}^{\text{cap}} &\leq \log^+ V_{a,N} \leq N \log(1/a) + \log(N+1) + \log^+(M_a + P_a), \\ \log^+ G_{0,N} &\leq N \log(1/a_0) + \log(N+1) + \log^+ M_0.\end{aligned}\tag{SGT27}$$

Also $\mathfrak{H}_m \leq 1 + \log(m+1)$, by integral comparison. The common diagonal error $4rL$ is therefore $O_{C_N, C_r, v}(k^2 \log(k+1))$. The global inverse term has $h \leq H_a$, which is fixed before the k limit, so it is $O_{\text{actual}, a, C_N}(k^2 + \log(k+1))$. It follows from SGT25 and SGT27 that

$$\limsup_{k \rightarrow \infty} \sup \frac{|\Delta_{N,s}(\mathbf{P})|}{k^3} \leq 2C_r C_N \log(1/a).\tag{SGT28}$$

The constants in this step may depend on this fixed a . Only after this limit do we take admissible radii arbitrarily close to one, as proved before SGT14. The right side then tends to zero. This proves SGT26 without assuming a zero-free unit disk or a uniform pole count as a tends to one.

For the original PDS quotient set

$$\begin{aligned}q_k &= (k+1)^2, \quad Q = (k+5)^2, \quad e \in \{0, 1\}, \\ N_e &= Q - v - 1 + e, \quad \ell_e = 8k + 23 + e, \quad d_e = \ell_e + 1 = 8k + 24 + e, \\ P_a(S) &= \chi_k(S)S^a, \quad r_a(S') = \mathcal{T}_A P_a(S'), \quad 0 \leq a \leq \ell_e, \\ \chi_k(S) &= \prod_{u,w=0}^k (S - \beta_{k,u,w}).\end{aligned}\tag{SGT29}$$

Then $q_k + \ell_e = N_e + v = Q - 1 + e$. A nonzero linear combination of the P_a has degree at least $q_k > v - 1$ because it is χ_k times a nonzero polynomial. Thus their classes modulo $\mathcal{P}_{v-1}$ are independent, of exactly the original rank d_e , without a minor or genericity assumption.

Let X_e be their full column matrix in the original c_0 orthonormal frame of degrees $0, \dots, N_e + v$, and write $P_{<v}$ and $P_{\geq v}$ for the row selections in that frame. The exact source and image Grams are

$$\begin{aligned}Y_e &= P_{\geq v} X_e, \quad H_{\text{src},e} = Y_e^* Y_e = X_e^* X_e - (P_{<v} X_e)^* (P_{<v} X_e) > 0, \\ C_e &= A_{N_e, s} Y_e, \quad H_e = C_e^* C_e, \quad -\mathcal{L}_{a, N_e, s, d_e} \leq \log \frac{\det H_e}{\det H_{\text{src},e}} \leq \mathcal{U}_{a, N_e, s, d_e}.\end{aligned}\tag{SGT30}$$

The first equality is the attained full quotient minimum over the entire original $\mathcal{P}_{v-1}$. The next equality is the coefficient map SGT10, since those removed low columns are killed by $\mathcal{T}_A$. The resulting C_e, H_e are exactly PDS7, including the target center c_1 and both original orders. For $k \geq 9$, $Q \leq (14k/9)^2 < 3k^2$, so $N_e \leq 3k^2$ and $d_e \leq 11k$. Hence SGT26 applies directly:

$$\log \frac{\det H_e}{\det H_{\text{src},e}} = o(k^3) = o(kq_k) \quad (e = 0, 1, s = 1, k), \quad (\text{SGT31})$$

uniformly over those four original source/endpoint choices. The full original mass has not been removed from either Gram.

The precise dual statement also retains all PDS conjugations: for the complex-linear functional $x^\top u$ on a Gram H , its squared dual norm is $x^* \overline{H^{-1}} x$. Thus the same coefficient coordinates carry

$$G_e = \overline{H_e^{-1}}, \quad G_{\text{src},e} = \overline{H_{\text{src},e}^{-1}}, \quad \log \frac{\det G_e}{\det G_{\text{src},e}} = -\log \frac{\det H_e}{\det H_{\text{src},e}}. \quad (\text{SGT32})$$

This is an equality of the original full quotient-source dual metrics, not a statement about a later connecting-target Gram.

For the endpoint $e = 1$ and the original four PDS8 rows, retain the same full rank-four matrix W and choose any basis Z of $\ker W^\top$. The polynomial quotient by those four columns, in the dual coordinates $a = Z^\top f$, has metric $(Z^\top H^{-1} \bar{Z})^{-1}$ by minimizing $f^* H f$. Consequently it is the conjugate inverse of the restriction $Z^* \overline{H^{-1}} Z$, exactly as in PDS9. For the paired forms $H_e, H_{\text{src},e}$, fixed complementary frames or the same attained Schur decomposition therefore give

$$\begin{aligned} \log \frac{\det(Z^* G_e Z)}{\det(Z^* G_{\text{src},e} Z)} &= -\Delta_{N_e,s}(\mathbf{P}) + \log \frac{\det(W^* H_e W)}{\det(W^* H_{\text{src},e} W)}, \\ -\mathcal{U}_{a,N_e,s,d_e} - \mathcal{L}_{a,N_e,s,4} &\leq \log \frac{\det(Z^* G_e Z)}{\det(Z^* G_{\text{src},e} Z)} \leq \mathcal{L}_{a,N_e,s,d_e} + \mathcal{U}_{a,N_e,s,4}. \end{aligned} \quad (\text{SGT33})$$

Indeed $\ker Z^\top = \text{im } W$. If J is any fixed right inverse of $Z^\top$, then $[W, J]$ is invertible, and block Schur factorization gives $\det((Z^\top H^{-1} \bar{Z})^{-1}) = |\det[W, J]|^2 \det H / \det(W^* H W)$. Taking the ratio cancels this exact frame determinant and proves the first equality. The W columns correspond to the four independent original polynomials $\chi_k R_i$ modulo $\mathcal{P}_{v-1}$; SGT24–25 applied to that actual rank-four plane prove the interval. At $e = 1$ this is the original primitive source, and its determinant distortion is again $o(kq_k)$. At $e = 0$ the primitive condition is empty and SGT32 already gives its full statement.

The earlier PEC1–18 theorem uses actual numerator-jet cancellation at symbol zeros to control a positive dual spectrum. Here SGT7–10 first proves the polynomial source/target morphism, and SGT16–22 retains all principal parts in an exact finite-rank Toeplitz summand, giving both directions on every small-rank plane. Neither statement replaces the original lower quotient, the four primitive rows, or the connecting-target metric. The original-data zero-free disk is the RDS11 disk with its full proof repeated in SGT11–13.

188.7 Receiving the conductor estimate on the complete native quotients

The same SGT operator has an exact native-dual application proved in NCT1–26 below. With original degree a , target cutoff $N \geq q_a - 1$ and $D = N - v$, its complex-linear dual is the ordinary transpose $A_{D,s}^\top$. Writing $\mathcal{N}$ for the original numerator germ, Leibniz' rule gives

$$\Lambda_F \circ \mathcal{T}_A = \Lambda_{E_A F}, \quad \Lambda_{\mathcal{N}/E_A} + \sum_{\beta'} \eta_{\beta'} \text{ev}_{\beta'} \longmapsto \Lambda_{\mathcal{N} + E_A \sum_{\beta'} \eta_{\beta'} e^{\beta' z}}. \quad (\text{RMR4})$$

The target functional annihilates every polynomial of degree below v , exactly the dual of the original source quotient in SGT7. The highest required derivative of F is $N - v$, so this is a

finite-jet isomorphism with a triangular inverse whose nonzero diagonal is $\binom{n+v}{v}\mu_v$. It maps the entire original lower evaluation fibre onto the entire conductor-multiple fibre.

Let $E_N[n, (u, w)] = \phi_{n,s,a/2}(\beta_{a,u,w})$ through degree N and $K_N = E_N^* E_N$. The original numerator matrix and its attained quotient metric are exactly

$$\hat{D}_N = Z_{\text{exp},a}^*[K_N - K_N C_a^\top (\bar{C}_a K_N C_a^\top)^{-1} \bar{C}_a K_N] Z_{\text{exp},a}. \quad (\text{RMR5})$$

The roots, Gamma mass, centre and conductor coefficients are unchanged. NPK5 evaluates every entry of K_N using the two original recurrence polynomials, including its exact coincident-argument derivative case. The primary scalar α_A stays in the original KRC/RRC coordinate map; RMR4 has coefficient one in the DNE germ frame.

Every rank- r original restriction or iterated affine quotient of $D_{a,D}^{(s)}$ receives the finite NCT20 directed volume bound from the corresponding construction on RMR5. To see why the large conductor denominator introduces no extra rank charge, represent a quotient by its orthogonal minimum lifts: its induced map is a restriction of $A_{D,s}^\top$ followed by an orthogonal projection. Apply the same factorization to its inverse. Their r th exterior norms are therefore bounded by the corresponding full r th exterior norms of SGT22. For four copies the exceptional pole rank is $4H_\varrho$; every other SGT constant is retained. For each fixed original period, consequently

$$\log \det \hat{Q} - \log \det Q = o(aq_a) \quad (D = O(a^2), \ r = O(a), \ s = 1 \text{ or } a), \quad (\text{RMR6})$$

uniformly on the original quotient values. The proof first fixes an admissible radius, takes the degree limsup, and then lets admissible radii approach one. A fixed-radius finite enclosure itself is not declared little-oh. NCT supplies every finite constant and the full proof; NPK11 records a finite optimized enclosure by their infimum.

189 The surviving boundary quotient and its original arithmetic kernel

Retain the actual simple quartet, its proved five-orbit period and logarithm branch, every coefficient of the original nonreduced conductor, and the complete CIT, CNC, DFX, KRC, DNE, NBG and PCJ maps. Put

$$\begin{aligned} k = 4l + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad m_a = 16a - 48 - v, \\ 0 \leq v = \text{ord}_0 E_A \leq 80, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v, \\ (s_0, s_1) = (1, 1) \text{ or } (k, j), \quad c'_a = a/2 - 4, \quad D \geq q_j - v - 1. \end{aligned} \quad (\text{RQB1})$$

All coefficient spaces and dual functionals are complex-linear. No star replaces an ordinary transpose in an original algebraic map. Metric stars are complex adjoints. The degree progression and the common cutoff D are literal, including the two distinct source measures. Write $s_k = s_0$ and $s_j = s_1$ when source orders have degree subscripts.

189.1 The exact coefficient kernel, rank and original kernel quotient

Let $V = \mathbb{C}^{m_k}$, $U = \mathbb{C}^{4m_j}$ and retain

$$\begin{aligned} B_0 = \mathbf{B}_k, \quad B_1 = \mathbf{B}_j, \quad J_0 = \mathbf{J}_k, \quad J_1 = \mathbf{J}_j, \quad \ker B_a = \text{im } J_a \quad (a = 0, 1), \\ M_i = (\alpha_A N_j Z_{\text{exp}, j})^{-1} m_{i, k} (\alpha_A N_k Z_{\text{exp}, k}), \quad m_{i, k}[f] = [x_i^4 f], \\ B_{\text{cof}} = \text{stack}_{i=1}^4 M_i, \quad \mathbb{B} = \text{diag}(B_1, 4), \quad X_{\mathbb{J}} = \text{diag}(J_1, 4), \quad \mathbb{J} = \text{im } X_{\mathbb{J}}, \\ \Phi = \mathbb{B} B_{\text{cof}}, \quad J = \text{im } J_0, \quad \mathcal{L} = \ker \Phi, \quad Y = J + \mathcal{L}, \quad E = \ker(\Phi J_0). \end{aligned} \quad (\text{RQB2})$$

Here $\text{diag}(P, 4)$ means four copies. The original KRC bijections specify the inverses onto their full images; both occurrences of α_A are retained before their proved cancellation in M_i . CNC6 or DFX21 proves B_{cof} injective on the entire V . All $\mathbf{B}_a$ are onto their actual kernel-dual coordinate spaces. Write $t_{K, a}$ for their ranks and $j_{0, a}$ for the column ranks of $\mathbf{J}_a$; the original exact values are $t_{K, a} = 8a - 16 - t_{0, a}$ and $j_{0, a} = 8a - 32 - v + t_{0, a}$.

Set exactly as in CIT24

$$W = \mathbb{J} + \text{im } B_{\text{cof}}, \quad W_{\text{inv}} = \mathbb{J} + \text{im}(B_{\text{cof}} J_0).$$

The literal map and its full kernel are

$$\begin{aligned} \pi_b : V \longrightarrow W/W_{\text{inv}}, \quad x \longmapsto [B_{\text{cof}} x], \quad \ker \pi_b = Y, \\ V/Y \xrightarrow{\sim} W/W_{\text{inv}} \xrightarrow{\sim} \text{im } \Phi / \text{im}(\Phi J_0), \quad [x] \longmapsto [B_{\text{cof}} x] \longmapsto [\Phi x]. \end{aligned} \quad (\text{RQB3})$$

Indeed $B_{\text{cof}} x \in W_{\text{inv}}$ precisely when $B_{\text{cof}}(x - J_0 a) \in \mathbb{J}$ for some a , equivalently $x - J_0 a \in \ker \Phi$. This proves the first kernel and both surjections. The second arrow has kernel exactly $\mathbb{J}$ before the displayed quotient, because $\ker \mathbb{B} = \mathbb{J}$. Thus it is injective on the stated quotient.

Put $\ell = \dim \mathcal{L}$, $\epsilon = \dim E$ and $r_b = \dim(W/W_{\text{inv}})$. Since J_0 is injective, it identifies E with $J \cap \mathcal{L}$. Consequently

$$\begin{aligned} r_b = m_k - j_{0, k} - \ell + \epsilon = t_{K, k} - \ell + \epsilon = \text{rank } \Phi - \text{rank}(\Phi J_0), \\ 0 \longrightarrow B_0 \mathcal{L} \longrightarrow \mathbb{C}^{t_{K, k}} \xrightarrow{[B_0 x] \mapsto [B_{\text{cof}} x]} W/W_{\text{inv}} \longrightarrow 0, \quad \dim B_0 \mathcal{L} = \ell - \epsilon. \end{aligned} \quad (\text{RQB4})$$

The middle arrow is well defined because changing x by $\ker B_0 = J$ changes its image by W_{inv} . Its kernel is $B_0 Y = B_0 \mathcal{L}$. Under KRC4–5 this is the exact quotient of $(K_k / (K_k \cap L_k))^\vee$, where $L_k = \mathbb{C}[S]_{<v}$ is the original polynomial low space. Thus the original low overlap $t_{0, k}$ remains in both the domain rank and its actual quotient.

Here is a finite calculation of $\mathcal{L}$ using the input polynomials. Use the exact Segre coefficient map $\Psi_a : S_{x,a} \rightarrow T_{a,a}$ of DFX13, the selector U_a^{inv} of degree- a monomials with $\sum i\nu_i \equiv 0 \pmod{5}$, and the complete original primary conductor convolution $C_a^\top$. Let $\text{quo}(G)$ be the following exact row-pivot quotient: take the first independent columns F of G , the first row set $\mathcal{R}$ with $F_{\mathcal{R}}$ invertible, and send x to $(x - FF_{\mathcal{R}}^{-1}x_{\mathcal{R}})_{\mathcal{R}^c}$. It is onto and has kernel $\text{im } G$, since the subtracted vector agrees with x on $\mathcal{R}$ and a zero remainder means x is that same column combination. Empty and full ranks give the identity and zero map respectively. Define

$$\begin{aligned} P_j^{\text{inv}} &= \text{quo}[C_j^\top, \Psi_j U_j^{\text{inv}}], \\ (H_{i,k}\nu)_{ab} &= \sum_{r,t=0}^4 b_{i,r,t} \nu_{a-r,b-t}, \quad b_{i,r,t} = [Z^r W^t] L_i(Z, W)^4, \\ \mathcal{A}_k &= \text{stack}_{i=1}^4 (P_j^{\text{inv}} H_{i,k} Z_{\text{exp},k}), \quad \mathcal{L} = \ker \mathcal{A}_k, \quad E = \ker(\mathcal{A}_k J_0). \end{aligned} \tag{RQB5}$$

Every off-rectangle coefficient in the convolution is zero. Membership of $H_{i,k} Z_{\text{exp},k} x$ in the displayed column space means its full conductor class has an invariant representative. Its first v jets already vanish by Leibniz multiplication; the conductor columns have zero such jets, so the invariant representative has zero jets as well. It is therefore exactly membership in $\mathcal{J}_{v,j}$, which KRC5 identifies with $\text{im } J_1$. This proves RQB5 and shows that the two computed kernels are those in RQB2, including all exceptional coefficient minors. In particular the first line of RQB4 is an exact finite rank formula in the original coefficients, without a chosen generic rank.

189.2 A branch constraint for arbitrary source numerators

Work first in $S_x = \mathbb{C}[x_1, x_2, x_3, x_4]$, with the original H , $Q = Q_0(H^{-1}x)$, $Q_a = g^a Q$ and $\mathcal{A} = \prod_{a=1}^4 Q_a$, where

$$gf(x) = f(\text{diag}(\zeta^{-1}, \zeta^{-2}, \zeta^{-3}, \zeta^{-4})x).$$

The primary order in the Segre map is unchanged. In the polynomial UFD $T = \mathbb{C}[z_0, z_1, w_0, w_1]$ put

$$\psi f = f(H(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^\top), \quad \ell_i = \psi x_i, \quad B_a = \psi Q_a, \quad A^{\text{hom}} = \prod_{a=1}^4 B_a. \tag{RQB6}$$

The restriction map is onto $T_{d,d}$ with kernel $Q S_{x,d-2}$: each biform monomial has the lift in KRC13, and two lifts with the same exponents differ by the relation $t_1 t_4 - t_2 t_3$. Grouping terms with the same image proves the full kernel, as in DFX13. Hence the complete branch quotient is $T_{d,d}/B_a T_{d-2,d-2}$; no radical is taken.

The four ℓ_i are linearly independent $(1,1)$ biforms, because their coefficient matrix is the actual invertible H . Any irreducible common factor of three has bidegree $(1,0)$, $(0,1)$ or $(1,1)$: factors of a bihomogeneous polynomial are bihomogeneous, as follows by comparing least and greatest degree in each grading in a domain. Multiples of a fixed factor of these types inside $T_{1,1}$ have dimensions two, two or one. Three independent rows cannot be contained there. It follows that any irreducible factor divides at most two ℓ_i , and therefore

$$\gcd\{\ell_i^4 \ell_t^4 : 1 \leq i < t \leq 4\} = 1 \quad \text{in } T. \tag{RQB7}$$

For each irreducible factor, choose two indices outside the at most two rows divisible by it. Their pair product is not divisible by it. This proves RQB7 in the biform polynomial UFD, without transferring factorization to the homogeneous quadric quotient ring.

Let $[f] \in \mathcal{C}_{v,k}$ be the image of a vector of $\mathcal{L}$. For each i there is an invariant p_i such that $x_i^4 f - p_i \in (Q, \mathcal{A})$. Fix $a \in \{1, 2, 3, 4\}$ and reduce modulo (Q, Q_a) . The ideal $(Q, \mathcal{A})$ maps to zero there; applying g^a to it also maps to zero, because $g^a Q = Q_a$ and $g^a \mathcal{A}$ contains Q as the factor with original index $5 - a$. Since $g^a p_i = p_i$, the two reductions give

$$x_i^4 f = \lambda_{i,a} x_i^4 g^a f, \quad \lambda_{i,a} = \zeta^{-4ia} \pmod{(Q, Q_a)}. \tag{RQB8}$$

For fixed nonzero a the four $\lambda_{i,a}$ are nonzero and pairwise distinct. Multiply the equation with index i by x_t^4 and that with index t by x_i^4 , then subtract. It follows first that $x_i^4 x_t^4 g^a f = 0$, and then that $x_i^4 x_t^4 f = 0$ in this same full branch quotient. On applying ψ , every irreducible factor P of B_a , with its actual exponent e_P , therefore satisfies

$$e_P \leq \text{ord}_P(\ell_i^4 \ell_t^4 \psi f) \quad \text{for every } i < t.$$

RQB7 supplies a pair whose multiplier has exponent zero. Thus $\text{ord}_P \psi f \geq e_P$ for every P , so B_a divides ψf with all its multiplicities. Repeating for all four a proves the new arbitrary-source constraint

$$\mathsf{T}_k \mathcal{L} \subset \mathcal{H}_{v,k} := \mathcal{C}_{v,k} \cap \ker \mathsf{j}_k, \quad \mathsf{T}_k = \alpha_A N_k Z_{\text{exp},k}. \quad (\text{RQB9})$$

DFX24 proves its invariant-source specialization. The proof above retains the arbitrary f and $g^a f$ as two different branch values until the pair equations eliminate them.

The constraint is computable on the complete nonreduced factor scheme. Let

$$\mathcal{L}_{\text{br}} = \text{lcm}(B_1, B_2, B_3, B_4), \quad (b_z, b_w) = \text{bideg } \mathcal{L}_{\text{br}}, \quad G_{\text{br}} = A^{\text{hom}} / \mathcal{L}_{\text{br}}.$$

Retain the actual scalar in A^{hom} and compensating factor units in this equation. Divisibility gives the exact kernel

$$\ker \mathsf{j}_k \simeq T_{k-b_z, k-b_w} / G_{\text{br}} T_{k-8, k-8}, \quad [u] \mapsto [\mathcal{L}_{\text{br}} u], \quad h_k = (k - b_z + 1)(k - b_w + 1) - (k - 7)^2. \quad (\text{RQB10})$$

Indeed all four branch classes vanish exactly when their biform representative is divisible by this lcm. Such representatives differ by A^{hom} exactly by the stated denominator. Nonzero biform multiplication is injective, proving the dimension.

Choose a quotient monomial basis in RQB10 by the specified row-pivot rule, multiply by $\mathcal{L}_{\text{br}}$, take its affine coefficient array and apply the original jet rows $\beta_{k,a,b}^n$, $0 \leq n < v$. Call this full matrix $\mathcal{J}_{\text{br},v}$. The denominator is killed since it is a multiple of E_A after the original exponential substitution. Thus, with $h_{v,k} = \dim \mathcal{H}_{v,k}$,

$$h_{v,k} = h_k - \text{rank } \mathcal{J}_{\text{br},v}, \quad d_{v,k} = \dim(\mathcal{H}_{v,k} \cap \mathcal{J}_{v,k}), \quad 0 \leq \ell - \epsilon \leq h_{v,k} - d_{v,k}. \quad (\text{RQB11})$$

For example $d_{v,k}$ is exactly the nullity of the four branch quotient matrices on $\mathsf{T}_k J_0$, computed with DFX30/KRC13. The last inequality follows from the injective map $\mathcal{L}/(\mathcal{L} \cap J) \rightarrow \mathcal{H}_{v,k}/(\mathcal{H}_{v,k} \cap \mathcal{J}_{v,k})$ induced by the original T_k . Its kernel is precisely $\mathcal{L} \cap J$, so no dimension of an overlapping low space is dropped. The resulting finite rank control is

$$\max\{0, t_{K,k} - h_{v,k} + d_{v,k}\} \leq r_b \leq t_{K,k}, \quad r_b = t_{K,k} - \text{rank}(B_0 Z_{\mathcal{L}}), \quad (\text{RQB12})$$

where $Z_{\mathcal{L}}$ is the exact kernel frame from RQB5. This preserves every shared-component defect. It gives full rank exactly on the coefficient strata where $B_0 Z_{\mathcal{L}} = 0$; that criterion is a computed matrix equality, not a replacement of a nonreduced conductor by its reduced support.

189.3 Full native and raw Gamma realizations

Use the actual fixed ITL4 complement Z_R of E , so $I_R = B_{\text{cof}} J_0 Z_R$. Choose a fixed complement E_b of Y in V by exact coefficient elimination, and set

$$R_b = B_{\text{cof}} E_b, \quad Y_{\text{inv}} = [X_{\mathbb{J}}, I_R], \quad Y_W = [X_{\mathbb{J}}, I_R, R_b]. \quad (\text{RQB13})$$

These are full frames of W_{inv} and W : applying $\mathbb{B}$ kills $X_{\mathbb{J}}$, makes $\Phi J_0 Z_R$ independent, and RQB3 makes the last quotient columns independent. In particular R_b represents exactly r_b quotient values. For the full target DNE form set

$$\begin{aligned} \mathcal{D}_D &= \text{diag}(D_{j,D}^{(s_1)}, 4), \quad Q_b^E(D) = R_b^* \mathcal{D}_D R_b - R_b^* \mathcal{D}_D Y_{\text{inv}} (Y_{\text{inv}}^* \mathcal{D}_D Y_{\text{inv}})^{-1} Y_{\text{inv}}^* \mathcal{D}_D R_b, \\ \mathcal{H}_D &= \text{diag}(H_{B,j,D}, 4), \quad H_{B,a,D} = (\mathbf{B}_a(D_{a,D}^{(s_a)})^{-1} \mathbf{B}_a^*)^{-1} = \widehat{H}_{K,a,D}^{-1}, \\ U_R &= \Phi J_0 Z_R, \quad U_b = \Phi E_b, \quad Q_b^E(D) = U_b^* \mathcal{H}_D U_b - U_b^* \mathcal{H}_D U_R (U_R^* \mathcal{H}_D U_R)^{-1} U_R^* \mathcal{H}_D U_b. \end{aligned} \quad (\text{RQB14})$$

Both formulas are attained minima: the first minimizes over the complete W_{inv} ; the second first minimizes over $\ker \mathbb{B} = \mathbb{J}$, then over its remaining image U_R . The fibres are identical by RQB3, so the formulas agree. Every inverse is justified by the full frames and DNE positivity. Empty image ranks omit their subtracted blocks.

If R_b^{CIT} is the particular earlier CIT25 lift, there is a unique invertible C_b on quotient values and some A_b with $R_b^{\text{CIT}} = R_b C_b + Y_{\text{inv}} A_b$. The exact minimum gives

$$Q_b^{\text{CIT}} = C_b^* Q_b^E C_b, \quad \det Q_b^{\text{CIT}} = |\det C_b|^2 \det Q_b^E. \quad (\text{RQB15})$$

The coefficients are obtained by solving the full frame Y_W . They are fixed as D or the source-order pair changes. Thus this new calculation transports to the original CIT25 factor with its actual squared frame determinant; it does not silently choose a different quotient volume.

For complete arithmetic rows at target degree j keep every original moment $\mu_d = \sum_{u,w=0}^8 a_{u,w} \beta_{8,u,w}^d$, and put

$$\begin{aligned} \beta_{j,a,b} &= j(1/2 - \delta - i\gamma) + 2a\delta + 2bi\gamma, & \nu_{a,n} &= \sum_{u,w} (Z_{\text{exp},j})_{uw,a} \beta_{j,u,w}^n, \\ f_{a,n} &= \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \\ b_{-1,t} &= 0, \quad b_{0,t} = \mathbf{1}_{t=0}, \quad b_{n+1,t} = i^{-1}(b_{n,t-1} - c'_j b_{n,t}) - n(n-1 + s_1/2) b_{n-1,t}, \\ t_n &= \left(\frac{\sum_t b_{n,t} f_{a,t}}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_a, & v_n &= \left(\frac{\sum_t b_{n,t} \beta_{j-8,u,w}^t}{\sqrt{(2\pi)^{s_1/2} n! (s_1/2)_n}} \right)_{u,w}. \end{aligned} \quad (\text{RQB16})$$

The first recursion follows by differentiating $E_A F_a = N_a$ $n + v$ times and isolating the term with μ_v . The second is the original monic Gamma recurrence after $y = (S' - c'_j)/i$, with $1/i = -i$ retained. Thus t_n, v_n are the complex-linear evaluations in the original orthonormal frame. The measure is exactly

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad s = s_1 \in \{1, j\}.$$

No α_A is inserted in the division rows: it belongs to the KRC coefficient transport already specified in RQB2.

Stack these rows through D as T_D, V_D , and put $\mathcal{T}_D = \text{diag}(T_D, 4)$, $\mathcal{V}_D = \text{diag}(V_D, 4)$. Define the literal full raw frames

$$Z_{\text{inv},D} = [\mathcal{V}_D, \mathcal{T}_D Y_{\text{inv}}], \quad Z_{W,D} = [Z_{\text{inv},D}, \mathcal{T}_D R_b], \quad G_{X,D} = Z_{X,D}^* Z_{X,D}. \quad (\text{RQB17})$$

Every lower evaluation occurs. Both frames are independent: a zero raw value modulo the lower evaluations gives a zero coefficient class by the original upper Vandermonde at $D + v \geq q_j - 1$, and then the coefficient frames in RQB13 are independent. Full elimination, with the complete old lower and invariant coefficients, therefore proves

$$Q_b^E(D) = (\mathcal{T}_D R_b)^* (I - P_{Z_{\text{inv},D}}) \mathcal{T}_D R_b, \quad \det Q_b^E(D) = \frac{\det G_{W,D}}{\det G_{\text{inv},D}}, \quad P_Z = Z(Z^* Z)^{-1} Z^*. \quad (\text{RQB18})$$

This is the full raw Gamma realization of the original residual factor. It has no independent row-by-row gauge minimization.

189.4 The exact original source denominator and compatibility transport

Let Z_Y be a fixed full basis of Y and retain the complement E_b from RQB13. On the source put $G_0 = D_{k,D}^{(s_0)}$. Its quotient form in these fixed residual lifts is

$$Q_0(D) = E_b^* G_0 E_b - E_b^* G_0 Z_Y (Z_Y^* G_0 Z_Y)^{-1} Z_Y^* G_0 E_b. \quad (\text{RQB19})$$

Choose the first independent columns Z_Λ of $B_0 Z_{\mathcal{L}}$ and write $C_0 = [Z_\Lambda, B_0 E_b]$, an invertible full frame of $\mathbb{C}^{t_{K,k}}$. The exact same fibre in RQB4 gives

$$\det Q_0(D) = \frac{|\det C_0|^2 \det H_{B,k,D}}{\det(Z_\Lambda^* H_{B,k,D} Z_\Lambda)}. \quad (\text{RQB20})$$

Indeed the source quotient first minimizes over J to obtain the original $H_{B,k,D}$ and then over $B_0 \mathcal{L}$. The adapted full Gram has determinant $|\det C_0|^2 \det H_{B,k,D}$; eliminating its first block proves RQB20. In full matrix form Q_0 is the Schur complement in the lifts $B_0 E_b$. This preserves both the source low overlap $t_{0,k}$ and its further hidden kernel of rank $\ell - \epsilon$.

Let $A_D = B_{\text{cof}}^* \mathcal{D}_D B_{\text{cof}}$ and let C_D be the full correlated pullback from NBG27 on V . The notation A_D/Y , C_D/Y means the attained quotient in the fixed lifts E_b . By RQB3 the further target minimum satisfies

$$0 \prec Q_b^E \preceq A_D/Y \preceq C_D/Y, \quad K_-^{-2} Q_0 \preceq C_D/Y \preceq K_+^2 Q_0. \quad (\text{RQB21})$$

To prove the first comparison, all representatives in $B_{\text{cof}} E_b b + B_{\text{cof}} Y$ are allowed in $B_{\text{cof}} E_b b + W_{\text{inv}}$, while $W_{\text{inv}} \cap \text{im } B_{\text{cof}} = B_{\text{cof}} Y$. The native form on the former fibre is A_D/Y . The remaining comparisons are NBG27 minimized on the same source affine fibre; this establishes their exact typed maps.

Define

$$\omega_b(D) = \frac{\det Q_b^E(D)}{\det(A_D/Y)}, \quad e_b^{\text{cor}}(D) = \log \frac{\det(C_D/Y)}{\det Q_0(D)}, \quad \omega_Z(D) = \frac{\det A_D|_Z}{\det C_D|_Z} \quad (Z = V, Y).$$

The restriction determinants in the last ratio use identical fixed frames. The adapted determinant identity cancels those frames and gives

$$\log \frac{\det Q_b^E}{\det Q_0} = e_b^{\text{cor}} + \log \omega_V - \log \omega_Y + \log \omega_b, \quad 0 < \omega_b \leq 1. \quad (\text{RQB22})$$

Apply the complete PCJ13–16 shared-update proof to the actual nested original source images $Y \subset V$; these are their DNE functional classes modulo the original source lower evaluations. Its source difference rank is exactly r_b by RQB4. The full compatibility-plus-source-complement forms are the unchanged PCJ1 matrices; call their nonnegative logarithms χ_Y, χ_V . Consequently

$$\begin{aligned} \log \frac{\det Q_b^E(D)}{\det Q_0(D)} &= -(\chi_Y(D) - \chi_V(D)) + \log \omega_b(D) + \varepsilon_b(D), \\ -2r_b \log \max(1, K_-) - r_b \log c_+ &\leq \varepsilon_b(D) \leq 2r_b \log \max(1, K_+) - r_b \log c_-. \end{aligned} \quad (\text{RQB23})$$

Here $c_- = \min(1, L_-^{-2})$, $c_+ = \max(1, L_+^2)$ are the actual NKC3–4/13 constants at this same k, D and source pair. For the signs, PCJ16 says $-\log(\omega_V/\omega_Y) = \chi_Y - \chi_V + \nu$ with $r_b \log c_- \leq \nu \leq r_b \log c_+$. The correlated quotient error is in $[-2r_b \log \max(1, K_-), 2r_b \log \max(1, K_+)]$ by NBG28. Subtracting ν proves exactly RQB23. Since $r_b \leq t_{K,k} = O(k)$, the explicit error is $O_{\text{actual}}(k \log q_j) = o(k q_k)$ at the original four target endpoints. The two full logarithmic losses remain. In particular the proved one-sided native bound is $Q_b^E \preceq K_+^2 Q_0$, retaining RQB20's original source denominator; an inverse estimate has not been assumed.

189.5 Four original rows and the complete retained target budget

For a new row degree $n \geq q_j - v$, use degree-major row order and

$$\begin{aligned} z_{\text{inv},n} &= [\text{diag}(v_n, 4), \text{diag}(t_n, 4) Y_{\text{inv}}], \quad z_{W,n} = [z_{\text{inv},n}, \text{diag}(t_n, 4) R_b], \\ L_{X,n} &= z_{X,n} G_{X,n-1}^{-1} z_{X,n}^*, \\ R_n &= \text{diag}(t_n, 4) R_b - z_{\text{inv},n} G_{\text{inv},n-1}^{-1} z_{\text{inv},n-1}^* \mathcal{T}_{n-1} R_b. \end{aligned} \quad (\text{RQB24})$$

The row permutation from copy-major order is fixed and unitary. Eliminate the old Z_{inv} block in the full raw frame: its Gram becomes $\text{diag}(G_{\text{inv}}, Q_b^E)$ and its new row becomes $[z_{\text{inv}}, R_n]$. Thus

$$\begin{aligned} L_{W,n} &= L_{\text{inv},n} + R_n Q_b^E (n-1)^{-1} R_n^*, \\ Q_b^E(n) &= Q_b^E(n-1) + R_n^* (I_4 + L_{\text{inv},n})^{-1} R_n, \\ \gamma_{b,n} &:= \log \frac{\det Q_b^E(n)}{\det Q_b^E(n-1)} = \log \det(I_4 + L_{W,n}) - \log \det(I_4 + L_{\text{inv},n}) \geq 0. \end{aligned} \quad (\text{RQB25})$$

For the middle equality, minimize the added squared row norm over the old eliminated variables. Completing that square gives the displayed inverse $I_4 + L_{\text{inv}}$, as can also be checked by multiplication. The first identity and the rectangular determinant lemma give the last one. All zero-energy rows and rank-zero quotients are included; in the latter case $\gamma_{b,n} = 0$.

Put $d = q_j - v - 1$ and retain the original endpoint signs. The exact return is

$$\begin{aligned} \mathcal{V}_b &= -\log \det Q_b^E(d) - \log \det Q_b^E(d+1) + \log \det Q_b^E(d+q_j) + \log \det Q_b^E(d+q_j+1) \\ &= \sum_{n=d+1}^{d+q_j+1} w_n \gamma_{b,n} \geq 0, \quad w_{d+1} = w_{d+q_j+1} = 1, \quad w_n = 2 \text{ otherwise.} \end{aligned} \quad (\text{RQB26})$$

Telescoping both low-to-high intervals proves every weight and sign. The same return for Q_b^{CIT} is equal by RQB15: its constant $2 \log |\det C_b|$ cancels only under these four signs.

Complete $[U_R, U_b]$ by any fixed columns U_o to a full frame $C_T = [U_R, U_b, U_o]$ of the fourfold original kernel-dual space $\mathbb{C}^{4t_{K,j}}$. Define $Q_o(D)$ as its full quotient metric modulo $\text{im } \Phi$ in the lifts U_o . The first restriction in this frame is the exact ITL6 Gram $H_{\text{inv}}(D) = U_R^* \mathcal{H}_D U_R$. Sequential attained minima give

$$|\det C_T|^2 (\det H_{B,j,D})^4 = \det H_{\text{inv}}(D) \det Q_b^E(D) \det Q_o(D). \quad (\text{RQB27})$$

For proof, first eliminate the U_R block in $C_T^* \mathcal{H}_D C_T$; its next diagonal Schur block is RQB14. Eliminate that block; the remaining minimum allows precisely $\text{im}[U_R, U_b] = \text{im } \Phi$ and is Q_o . Thus every fixed-frame determinant is displayed. The missing rank is $4t_{K,j} - \text{rank } \Phi$, with no image-completeness assumption.

Each restriction and quotient in RQB27 increases as D increases: the original DNE metric increases by retaining additional orthonormal rows, and taking minima on the same affine fibres preserves that inequality. Therefore all three original weighted returns are nonnegative. The same is true row by row, and HJR13,18,21 give

$$\begin{aligned} \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} + \mathcal{V}_o &= 4\widehat{F}_j^{(s_1)}, \\ \mathcal{V}_{\text{inv}}^{\text{target}} &= \mathcal{R} \log \det H_{\text{inv}}, \quad \mathcal{V}_o = \mathcal{R} \log \det Q_o, \quad 0 \leq \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4\widehat{F}_j^{(s_1)}. \end{aligned} \quad (\text{RQB28})$$

Here $\mathcal{R}$ has exactly the signs and cutoffs of RQB26. Its $\widehat{F}_j$ is the original fixed-column kernel return, since $H_{B,j,D} = \widehat{H}_{K,j,D}^{-1}$. The inverse changes the primal determinant sign exactly; no equality with the earlier F_K has been imposed. The frame constant in RQB27 disappears only because the four signs sum to zero. The inequality also holds on every single row, with the right side $4g_{K,N}$ for that target row. Thus the full aggregate budget retains zero-energy rows and does not require a hypothesis about concentration.

For an explicit finite cap already proved for these same measures, retain HJR24–25 at base degree j :

$$\begin{aligned} \widehat{F}_j^{(s_1)} + F_{0,j,s_1} &= \mathcal{F}_{\partial,j}^{(s_1)} + \mathbf{e}_{j,s_1} + \delta_{K,j,s_1} - F_{\Xi(L),j}^{(s_1)} - \delta_{0,j,s_1}, \\ |\delta_{K,j,s_1}| &\leq \mathcal{E}_{j,s_1}^{\text{FEX}}, \quad |\delta_{0,j,s_1}| \leq E_{j,s_1}^{\text{tr}}, \quad -e_{j,s_1}^- \leq \mathbf{e}_{j,s_1} \leq e_{j,s_1}^+, \quad F_{\Xi(L),j}^{(s_1)} \geq 0, \\ \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} &\leq 4U_{j,s_1}^{\text{jet}}, \quad U_{j,s_1}^{\text{jet}} = \mathcal{F}_{\partial,j}^{(s_1)} + e_{j,s_1}^+ + \mathcal{E}_{j,s_1}^{\text{FEX}} + E_{j,s_1}^{\text{tr}}. \end{aligned} \quad (\text{RQB29})$$

Every symbol here is the unchanged finite original OAR/FEX/PEC quantity in the complete HJR proof. The exact equality is retained before applying bounds. Both F_{0,j,s_1} and the omitted positive restriction are nonnegative by the original native row calculation. Together with RQB28 this proves RQB29. HJR26's already proved outer first variation gives

$$\mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} \leq 4\Delta_j q_j \mathfrak{C}_\partial + o(kq_k) = O_{\text{actual}}(kq_k), \quad \Delta_j = 16j - 48. \quad (\text{RQB30})$$

The actual period is fixed, and $jq_j/(kq_k) \rightarrow 1$. This is an aggregate upper bound on two specified positive native volumes, with their full missing quotient retained in RQB28. It does not assign an allocation or a leading value to either one.

Finally transport the calculation directly into the accepted CIT25–26. In its original lifts use RQB15 and choose the fixed full frame X_W there, so $[X_{\mathbb{J}}, I_R, R_b^{\text{CIT}}] = X_W C_W$. Then the earlier complete equality remains literally

$$|\det C_W|^2 g_W = g_{\mathbb{J}} \det H_{\text{inv}} |\det C_b|^2 \det Q_b^E. \quad (\text{RQB31})$$

Substitution into CIT26 replaces its one retained factor $\log \det Q_b^{\text{CIT}}$ by $\log \det Q_b^E + 2 \log |\det C_b|$. The raw calculation is RQB16–18, its exact original source quotient is RQB19–20, and its two independent native losses are RQB23. In the signed four-return the frame constants cancel and the residual contribution obeys the directed interval

$$-4\widehat{F}_j^{(s_1)} + \mathcal{V}_{\text{inv}}^{\text{target}} \leq -\mathcal{V}_b \leq 0. \quad (\text{RQB32})$$

Indeed RQB28 says $\mathcal{V}_b \leq 4\widehat{F}_j - \mathcal{V}_{\text{inv}}^{\text{target}}$. This is precisely the sign of the residual factor in CIT26. Its kernel and source rank are computed by RQB4–12, and all lower-gauge, source-complement, invariant-overlap and fixed-frame terms remain in the exact transport. The common long connecting loss is not replaced by this bound.

189.6 The directed original kernel cap and its source correction

There is a sharper finite version of RQB29 which uses the full original positive observation return before bounding it. All quantities in this paragraph have base degree j , source $s = s_1 \in \{1, j\}$ and $N_0, N_1, N_2, N_3 = q_j - 1, q_j, 2q_j - 1, 2q_j$. Retain the primal sign vector $\sigma = (1, 1, -1, -1)$ in FEX. With $t = t_{K,j}$ and $L_j = \mathbb{C}[S]_{<v}$, the exact FEX31/34 source correction is

$$\delta_{K,j,s} := \widehat{F}_j^{(s)} - F_K^{(s)} = \sum_{i=0}^3 \sigma_i e_{N_i,s} - F_{L_j}^{(s)} + F_{(K_j+L_j)/K_j}^{(s)}. \quad (\text{RQB33})$$

The endpoint scalar $e_{N,s}$ is the actual logarithmic Jacobian between the full original $(K_j + L_j)/L_j$ source and its conductor image in FEX34. It is not a deleted low direction. For full numerical coefficient expressions retain

$$\begin{aligned} M_N^A &= \max \left\{ 1, \frac{v!}{|\mu_v|} \sum_{a,b=0}^8 |a_{ab}| e^{|\beta_{8,a,b}-4|(H_N+2)} \right\}, \quad m_N^A = \log M_N^A, \\ w_{d,s} &= d!/(s/2)_d, \quad \zeta_{N,s} = \sum_{d=v}^N \log(w_{d,s}/w_{d-v,s}), \\ e_{N,s}^- &= \zeta_{N,s} - 2(N+1-v-t)m_N^A, \quad e_{N,s}^+ = 2tm_N^A, \\ D_s^+ &= e_{q_j-1,s}^+ + e_{q_j,s}^+ - e_{2q_j-1,s}^- - e_{2q_j,s}^-, \\ \mathcal{E}_{j,s}^{\text{FEX}} &= v\mathcal{L}_s + 2 \max_i |\zeta_{N_i,s}| + 2(4q_j+1)m_{2q_j}^A. \end{aligned} \quad (\text{RQB34})$$

Here H_N is the harmonic sum and $\mathcal{L}_s$ is the original AKS two-window constant at degree j . The explicit alternative FEX33 is available with its unchanged target degree and measure. FEX34

proves $e_{N,s}^- \leq e_{N,s} \leq e_{N,s}^+$ using the whole Gamma Jacobian and the actual conductor norm. The original two low quotient metrics give $0 \leq F_{L_j} \leq v\mathcal{L}_s$ and $0 \leq F_{(K_j+L_j)/K_j} \leq (v - t_{0,j})\mathcal{L}_s$. Thus RQB33 proves $\delta_{K,j,s} \leq D_s^+ + (v - t_{0,j})\mathcal{L}_s$. FEX36–37 independently proves $|\delta_{K,j,s}| \leq \mathcal{E}_{j,s}^{\text{FEX}}$. These two bounds retain the original source mass in their proved Gamma norm ratios and keep $t_{0,j}$ before estimating.

Let $\Phi_{j,s}$ be exactly OAR4's full positive sum $\log(1 + \eta_{q_j-1}) + 2 \sum_{N=q_j}^{2q_j-2} \log(1 + \eta_N) + \log(1 + \eta_{2q_j-1})$. All η_N are the original ROQ constrained rows with their complete conductor-observation covariance. If $U_{j,s}^{\text{old}}$ denotes the retained AKS/KAF kernel cap, OAR3/7 and RQB33–34 give the finite number

$$\begin{aligned} U_{j,s}^{\text{native}} &:= \min\{U_{j,s}^{\text{jet}}, \min\{U_{j,s}^{\text{old}}, \mathcal{F}_{\partial,j}^{(s)} + e_{j,s}^+ - \Phi_{j,s}\} \\ &\quad + \min\{D_s^+ + (v - t_{0,j})\mathcal{L}_s, \mathcal{E}_{j,s}^{\text{FEX}}\}\}, \\ 0 \leq \mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}} &\leq 4\widehat{F}_j^{(s)} \leq 4U_{j,s}^{\text{native}}. \end{aligned} \quad (\text{RQB35})$$

Indeed the inner first minimum bounds the original F_K from above, using its full residual subtraction. The second minimum bounds the exact difference $\widehat{F}_j - F_K$. Their sum bounds $\widehat{F}_j$; HJR25 supplies the other cap. This proves the outer minimum without identifying the two different kernel metrics or assuming a new conditioning estimate. Replacing the full Φ by only its first term recovers the earlier OAR6 partial cap, whereas RQB35 retains all computed positive terms.

Finally FEX37 transfers the proved OVR9 limit superior to this exact fixed-column kernel return. Therefore, with the original $\kappa_* = 32 \log(25 \log 3/(4\pi))$ and the same $\mathfrak{C}_{\partial}$ as in RQB30,

$$0 \leq \limsup_{k \rightarrow \infty} \frac{\mathcal{V}_b + \mathcal{V}_{\text{inv}}^{\text{target}}}{kq_k} \leq 4 \min\{\kappa_*, 16\mathfrak{C}_{\partial}\}. \quad (\text{RQB36})$$

For proof divide RQB28 by kq_k , insert $jq_j/(kq_k) \rightarrow 1$ and $(\widehat{F}_j - F_K)/(jq_j) \rightarrow 0$, then apply OVR9 separately in each unchanged source order. This proves only the displayed upper bound on the sum of two nonnegative exact volumes. Both remain present in the identity RQB28, with its full missing quotient; no individual limit or allocation is presumed.

The complete missing target quotient and its connecting restriction. Keep every block of the original $C_T = [U_R, U_b, U_o]$ in RQB27. At both original source-order pairs use $\pi_o = [0 \ 0 \ I_{r_o}]C_T^{-1}\mathbb{B}$, $E_e = \pi_o F_e$, $c_e = \text{rank } E_e$. The kernel of π_o is exactly W by the equation $\mathbb{B}z = \Phi x$, and its restriction to $\mathcal{S}_e$ has kernel $K_e = \mathcal{S}_e \cap W$. Choose the original fixed complement Z_c of $\ker E_e$ and $C_o = [E_e Z_c, R_e^o]$. The complete two target minima are

$$\begin{aligned} G_{t,e} &= (E_e Z_c)^* Q_o (E_e Z_c), \quad Q_{\text{rem},e} = \text{Gram}(\mathcal{U}/T_e) \text{ in the lifts } R_e^o, \quad T_e = \mathcal{S}_e + W, \\ |\det C_T|^2 |\det C_o|^2 (\det H_{B,j,D})^4 &= \det H_{\text{inv}} \det Q_b^E \det G_{t,e} \det Q_{\text{rem},e}, \\ \text{rank } G_{t,e} &= c_e, \quad \text{rank } Q_{\text{rem},e} = r_o - c_e. \end{aligned} \quad (\text{RQB37})$$

The quotient form on $\mathcal{U}/W$ first minimizes over every lower evaluation and $\mathbb{J}$, then over $\text{im } \Phi$, and is exactly Q_o . On the connecting values its restriction is therefore the first line. The further minimum over the whole connecting image gives the stated quotient on $\mathcal{U}/T_e$. Eliminating the first block of $C_o^* Q_o C_o$ proves $|\det C_o|^2 \det Q_o = \det G_t \det Q_{\text{rem}}$. Multiplication by the original RQB27 identity proves every absolute factor in RQB37.

Set $V_t = \mathcal{R} \log \det G_{t,e}$ and $V_{\text{rem}} = \mathcal{R} \log \det Q_{\text{rem},e}$. The entire original budget now reads

$$\begin{aligned} V_o &= V_t + V_{\text{rem}}, \quad V_t, V_{\text{rem}} \geq 0, \\ V_I + V_b + V_t + V_{\text{rem}} &= 4\widehat{F}_j^{(s_1)}, \\ -4\widehat{F}_j^{(s_1)} + V_I + V_t &\leq -V_b \leq 0, \\ -(r_o - c_e)\Omega_j^{\text{hi}} &\leq \Delta V_{\text{rem}} \leq (r_o - c_e)\Omega_j^{\text{lo}}, \quad -c_e\Omega_j^{\text{hi}} \leq \Delta V_t \leq c_e\Omega_j^{\text{lo}}. \end{aligned} \quad (\text{RQB38})$$

The original increasing row metric preserves order under both complete minima, proving positivity. Taking the four signs of RQB37 cancels only its fixed frame determinants. The third line uses $V_{\text{rem}} \geq 0$ after retaining its exact value in the second. The last line uses NSO's same-fibre comparison at the two actual ranks, with $D = q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v$ and $s_1 = 1, j$. It retains the original low overlaps in r_o .

For a literal calculation of the new quotient choose full coefficient columns X_T spanning T_e and fixed lifts X_Q under $\mathbb{B}$ of $U_o R_e^o$. Through the old degree D put

$$\begin{aligned} Z &= [\mathcal{V}_D, \mathcal{T}_D X_T], \quad G = Z^* Z, \quad Q = (\mathcal{T}_D X_Q)^* (I - ZG^{-1}Z^*) \mathcal{T}_D X_Q, \\ a &= [\text{diag}(v_{D+1}, 4), \text{diag}(t_{D+1}, 4) X_T], \quad b = \text{diag}(t_{D+1}, 4) X_Q, \\ R &= b - aG^{-1}Z^* \mathcal{T}_D X_Q, \quad L = aG^{-1}a^*, \\ Q^+ &= Q + R^*(I_4 + L)^{-1}R, \quad \log \frac{\det Q^+}{\det Q} = \log \frac{\det(I_4 + L + RQ^{-1}R^*)}{\det(I_4 + L)} \geq 0. \end{aligned} \tag{RQB39}$$

The original RQB16 rows retain all division moments, the phase $1/i$, the centres and $h_{n,s_1} = (2\pi)^{s_1/2} n! (s_1/2)_n$. DNE positivity and the full coefficient frames prove that Z and $[Z, \mathcal{T}_D X_Q]$ are independent even at the first cutoff. Their minimum is exactly Q_{rem} . Subtracting the old attained section leaves $\eta^* G \eta + \|Rx - a\eta\|^2$; its minimum proves the last line, including the original old lower Gram. The return uses weights one at the first and last rows and two inside. Thus no new row model or independent rowwise gauge minimum was substituted.

190 The four original primitive rows and their full Gamma Gram

Fix the actual quartet, its original invertible period matrix H , logarithm branch, and every coefficient of the nonreduced conductor. All constants below are calculated from these unchanged data. Retain the original PDS polynomial coordinates:

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad j = k + 4, \quad q_k = (k + 1)^2, \quad Q = (k + 5)^2, \\ \beta_* &= \frac{1}{2} - \delta - i\gamma, \quad \beta_{n,a,b} = n\beta_* + 2a\delta + 2bi\gamma, \quad \delta, \gamma > 0, \\ E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b}z}, \quad \mu_r = E_A^{(r)}(0), \quad 0 \leq v = \text{ord}_0 E_A \leq 80, \\ \mu_v &\neq 0, \quad D = Q - v \geq 116, \quad s \in \{1, k\}, \quad c = k/2 - 4. \end{aligned} \tag{PFG1}$$

The order s and center c are the source member of the original pairs $(1, 1)$ and (k, j) . In particular this calculation uses the source center at degree k , even though χ_j occurs in its rows.

Define the literal original polynomials, symbols and translations by

$$\begin{aligned} L_i(Z, W) &= (H(ZW, Z, W, 1)^T)_i, \quad b_{i,a,b} = [Z^a W^b] L_i(Z, W)^4, \\ B_i(z) &= \sum_{a,b=0}^4 b_{i,a,b} e^{\beta_{4,a,b}z}, \quad m_{r,i} = B_i^{(r)}(0) = \sum_{a,b=0}^4 b_{i,a,b} \beta_{4,a,b}^r, \\ \chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), \quad (\mathcal{T}_A p)(S') = \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), \\ (\mathcal{T}_{B_i} p)(S') &= \sum_{a,b=0}^4 b_{i,a,b} p(S' + \beta_{4,a,b}), \\ f &= \mathcal{T}_A \chi_j, \quad \lambda = \mu_v \binom{Q}{v}, \quad \deg f = D, \quad \text{lc } f = \lambda, \quad u_i = \mathcal{T}_{B_i} f. \end{aligned} \tag{PFG2}$$

The degree assertion follows directly by applying $\mathcal{T}_A S^a = \sum_{r=0}^a \binom{a}{r} \mu_r S'^{a-r}$ to the monic degree- Q polynomial. Its first surviving moment is exactly μ_v . Some $B_i(0)$ may vanish; none of the following arguments presumes that all four row polynomials have degree D .

190.1 Full native frame and the exact PDS matrix

Keep the full measure, mass and recurrence:

$$\begin{aligned} M_s &= (2\pi)^{s/2}, \quad dm_s(y) = \frac{M_s 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \\ \|p\|_{s,c}^2 &= \int_{\mathbb{R}} |p(c + iy)|^2 dm_s(y), \quad h_{n,s} = M_s n! (s/2)_n, \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = y p_{n,s} - n(n-1 + s/2) p_{n-1,s}, \\ \phi_{n,s,c}(S') &= p_{n,s}((S' - c)/i) / \sqrt{h_{n,s}}. \end{aligned} \tag{PFG3}$$

These are exactly PDS6 and SGT2–5. Their orthogonality can be checked without a mass change: the recurrence gives

$$\sum_{n \geq 0} p_{n,s}(y) \frac{t^n}{n!} = (1 + t^2)^{-s/4} \exp(y \arctan t), \quad \int e^{ty} dm_s(y) = M_s (\cos t)^{-s/2} \quad (|t| < \pi/2).$$

For the second identity, the two Euler Gamma integrals followed by $r = u + v_0$, $x = \log(u/v_0)$ give the Fourier pair $|\Gamma(s/4 + i\xi)|^2 = \Gamma(s/2) \int e^{i\xi x} (2 \cosh(x/2))^{-s/2} dx$. The integrand is holomorphic for $|\Im x| < \pi$ and decays exponentially along each smaller closed strip. Contour translation

therefore bounds its Fourier transform exponentially; Fourier inversion and analytic continuation give the displayed identity, with $y = 2\xi$ and precisely the multiplier in PFG3. Multiplying the two generating functions and using this integral gives $M_s(1 - tz)^{-s/2}$. Differentiation under the integral is justified by its exponential moments near $(t, z) = (0, 0)$. Coefficient comparison proves $\int p_{n,s} p_{m,s} dm_s = \mathbf{1}_{n=m} h_{n,s}$. Thus PFG3 is the original orthonormal frame. Its leading coefficient in S' is $i^{-n}/\sqrt{h_{n,s}}$.

Let $B_{s,c}$ be the monomial coefficient map through degree D , defined by $S'^a = \sum_n (B_{s,c})_{n,a} \phi_{n,s,c}$. Its entries vanish for $n > a$ and its diagonal is $i^n \sqrt{h_{n,s}}$. Define

$$\mathcal{C} = (B_{s,c} \text{coeff}_{S'} u_i)_{i=1}^4, \quad \mathcal{G}_{k,s} = \mathcal{C}^* \mathcal{C} = (\langle u_i, u_t \rangle_{s,c})_{i,t=1}^4. \quad (\text{PFG4})$$

All polynomial coefficients and row order here are the original ones.

There is an exact map to the retained PDS correction. Set $\ell_1 = 8k + 24$, $r_a = \mathcal{T}_A(\chi_k S^a)$ for $0 \leq a \leq \ell_1$, and

$$R_i(S) = \frac{\mathcal{T}_{B_i} \chi_j(S)}{\chi_k(S)}, \quad W = ([S^a] R_i)_{a,i}, \quad C_1 = (B_{s,c} \text{coeff}_{S'} r_a)_a, \quad H_1 = C_1^* C_1.$$

Every shift of an upper degree- k root is an upper degree- j root; all roots are distinct. Thus each numerator in R_i is divisible by χ_k , and $\deg R_i \leq Q - q_k = \ell_1$. Translation sums commute, so, column by column,

$$C_1 W = \mathcal{C}, \quad \boxed{\mathcal{G}_{k,s} = W^* H_1 W.} \quad (\text{PFG5})$$

This is the original PDS8 frame, with its positive degree-zero wedge signs. No transpose in the complex-linear quotient map is replaced by an adjoint. The stars in PFG4–5 belong only to the native Gram.

190.2 The four actual moment pivots

We first prove that the 25 by 4 coefficient matrix $(b_{i,a,b})_{(a,b),i}$ has rank four, including every actual period. The kernel of the Segre substitution in homogeneous polynomials is generated by $Q_0 = t_1 t_4 - t_2 t_3$. Indeed Q_0 is prime by Gauss's lemma applied to this primitive polynomial linear in t_1 . On $t_4 \neq 0$ substitute $t_1 = t_2 t_3 / t_4$; a polynomial vanishing on the Segre chart has identically zero Laurent remainder. Consequently some power of t_4 times it is divisible by Q_0 . Primality and $Q_0 \nmid t_4$ remove that power. This proves the full ideal assertion and its converse. The homogeneous Segre parameters here can be ordered as $(z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)$; on $t_4 \neq 0$ they give exactly $(ZW, Z, W, 1)$ after scalar multiplication.

If $\sum_i c_i L_i^4 = 0$, the invertible substitution $x = Ht$ therefore gives

$$F(x) = \sum_i c_i x_i^4 = Q_H(x) R_2(x), \quad Q_H(x) = Q_0(H^{-1}x).$$

Suppose $F \neq 0$. If $c_i = 0$, additivity of degree in x_i in the polynomial integral domain forces Q_H to be independent of x_i . Its quadratic matrix then has a zero row and column, contradicting its rank four. Hence all $c_i \neq 0$. The two quadrics Q_H, R_2 have a nonzero common point: restrict R_2 to the Segre surface. If the resulting biform is zero, every point works. Otherwise choose a projective w for which it is a nonzero homogeneous binary quadratic in z , and take a projective root over $\mathbb{C}$. At that common point $\nabla F = R_2 \nabla Q_H + Q_H \nabla R_2 = 0$. But $\nabla F = (4c_i x_i^3)_i$ has no nonzero zero. This contradiction proves the asserted independence.

Order the original 25 nodes lexicographically in (a, b) . They are distinct: equality of two nodes forces both $2(a - a')\delta = 0$ and $2(b - b')\gamma = 0$. Their Vandermonde determinant is the nonzero product of their ordered pairwise differences. Multiplying the independent coefficient columns by this full matrix proves

$$\text{rank}(m_{r,i})_{0 \leq r \leq 24, 1 \leq i \leq 4} = 4. \quad (\text{PFG6})$$

Let $0 \leq t_1 < t_2 < t_3 < t_4 \leq 24$ be the earliest indices at which this successive row rank becomes 1, 2, 3, 4. Retain the actual fixed matrix

$$\mathcal{M} = (m_{t_a, i})_{a, i=1}^4, \quad \det \mathcal{M} \neq 0, \quad T = \sum_{a=1}^4 t_a. \quad (\text{PFG7})$$

The determinant is a calculated fixed moment minor; its nonvanishing follows from the construction. It is not a new conditioning assumption. Every row m_r with $r < t_a$ is in the span of the previous selected rows.

Write $f(S') = \sum_{b=0}^D f_b S'^b$, $f_D = \lambda$. The exact row of coefficients at degree $D - t$ of the four u_i is

$$c_t = \sum_{r=0}^t f_{D-t+r} \binom{D-t+r}{r} m_r = \lambda \binom{D}{t} m_t + \sum_{r < t} f_{D-t+r} \binom{D-t+r}{r} m_r. \quad (\text{PFG8})$$

This is the binomial theorem in each original translation. At $t = t_a$, the second term is a linear combination of the previous selected moment rows. It follows, without deleting any coefficient of f , that

$$\det(c_{t_a, i})_{a, i=1}^4 = \lambda^4 \det \mathcal{M} \prod_{a=1}^4 \binom{D}{t_a}. \quad (\text{PFG9})$$

In particular lower polynomial coefficients cannot cancel this pivot.

The same reasoning applies in the original Gamma frame. The row of $\mathcal{C}$ at index $n = D - t$ is $\sum_{u=0}^t (\mathbf{B}_{s, c})_{D-t, D-u} c_u$. For $t = t_a$, its contribution from $u < t_a$ lies in the preceding moment span, and its diagonal contribution is $i^{D-t_a} \sqrt{h_{D-t_a, s}} \lambda \binom{D}{t_a} m_{t_a}$. With the row order $D - t_1, \dots, D - t_4$, this proves the exact original minor

$$\det \mathcal{C}_{\{D-t_1, \dots, D-t_4\}, 1:4} = i^{4D-T} \lambda^4 \det \mathcal{M} \prod_{a=1}^4 \binom{D}{t_a} \sqrt{h_{D-t_a, s}}.$$

(\text{PFG10})

The ordering of the selected rows is stated explicitly. The phase uses $S' = c + iy$, not a real-coordinate replacement.

190.3 Full finite Gram bounds

For the exterior-algebra form of this classical determinant identity, see Lyons [H3, §4, (4.1), p. 11]. Expansion of the four-column wedge in its orthonormal coordinate basis gives the squared four-row minors used below.

The Cauchy–Binet identity, obtained by expanding the two determinants in $\det(\mathcal{C}^* \mathcal{C})$ and collecting the common row subsets, writes it as the sum of squared moduli of all four-row minors. Retaining the actual nonzero minor PFG10 gives

$$0 < \mathcal{L}_{k, s} := |\mu_v \binom{Q}{v}|^8 |\det \mathcal{M}|^2 \prod_{a=1}^4 \binom{D}{t_a}^2 h_{D-t_a, s} \leq \det \mathcal{G}_{k, s}. \quad (\text{PFG11})$$

This proves positivity with a literal finite original-data lower bound.

For the upper bound retain

$$\begin{aligned} A_{\text{abs}} &= \sum_{a, b=0}^8 |a_{ab}|, & B_{i, \text{abs}} &= \sum_{r, t=0}^4 |b_{i, r, t}| > 0, & R_\beta &= \sqrt{(1/2 + \delta)^2 + \gamma^2}, \\ \Lambda_{k, s} &= \max\{1, |c| + 2(D + 1 + s/2)\}, & \Theta_{a, b, r, t; u, w} &= \beta_{8, a, b} + \beta_{4, r, t} - \beta_{j, u, w}, & (\text{PFG12}) \\ P_{i, k, s} &= \sum_{a, b=0}^8 \sum_{r, t=0}^4 |a_{ab} b_{i, r, t}| \prod_{u, w=0}^j (\Lambda_{k, s} + |\Theta_{a, b, r, t; u, w}|). \end{aligned}$$

These sums are positive since their coefficient columns are nonzero. The complete original expansion is $u_i(S') = \sum_{a,b,r,t} a_{ab} b_{i,r,t} \prod_{u,w=0}^j (S' + \Theta_{a,b,r,t;u,w})$. Its coefficients above degree D vanish by PFG2, exactly.

In the orthonormal y frame, multiplication by y has raising coefficient $\sqrt{(n+1)(n+s/2)}$ and lowering coefficient $\sqrt{n(n-1+s/2)}$. Through degree D , both weighted shifts have norm at most $D+1+s/2$. Hence multiplication by $S' = c + iy$ on degree at most $D-1$ has norm at most $\Lambda_{k,s}$. Starting at the constant of squared norm M_s gives $\|p\|_{s,c} \leq \sqrt{M_s} \sum_{n=0}^D |[S^n]p| \Lambda_{k,s}^n$. The weighted coefficient sum is submultiplicative by its exact coefficient convolution and the triangle inequality. Applied to the full expansion just displayed, it proves $\|u_i\|_{s,c} \leq \sqrt{M_s} P_{i,k,s}$. Using the degree- Q product to bound the coefficient sum before its cancellations is a valid upper bound for the unchanged degree- D polynomial. Finally Gram–Schmidt makes a Gram determinant the product of squared orthogonal remainders, each at most the original squared column norm. Therefore

$$0 < \mathcal{L}_{k,s} \leq \det(W^* H_1 W) \leq \mathcal{U}_{k,s} := M_s^4 \prod_{i=1}^4 P_{i,k,s}^2. \quad (\text{PFG13})$$

Every mass, factorial, moment, translation coefficient and root remains in these finite bounds.

190.4 The derived factorial scale and the primitive correction

Here are explicit two-sided bounds at the next scale. Put

$$C_\beta = 6 + 17R_\beta, \quad c_{\text{low}} = 8 \log |\mu_v| + 2 \log |\det \mathcal{M}|, \quad c_{\text{up}} = 8 \log A_{\text{abs}} + 2 \sum_{i=1}^4 \log B_{i,\text{abs}}. \quad (\text{PFG14})$$

These real constants may have either sign except $C_\beta > 6$. For all original source orders in PFG1, the finite inequalities are

$$c_{\text{low}} - 8Q - 2(4v + T)(\log Q + 1) \leq \log \det(W^* H_1 W) - 4 \log M_s - 8Q \log Q \leq c_{\text{up}} + 8Q \log C_\beta + 8. \quad (\text{PFG15})$$

For the upper inequality, the original root formula gives $\max_{a,b} |\beta_{n,a,b}| = nR_\beta$. The separate real and imaginary bounds are attained together at the appropriate corner nodes, so this is the stated exact maximum. It follows that $|\Theta| \leq (k+16)R_\beta$. Also $|c| \leq k/2 + 4$, $s \leq k$ and $D \leq Q$ imply $\Lambda_{k,s} \leq 6(Q+1)$, while $k+16 \leq 17(Q+1)$. Thus $P_{i,k,s} \leq A_{\text{abs}} B_{i,\text{abs}} [C_\beta(Q+1)]^Q$. Take logarithms in PFG13 and use $Q \log(Q+1) \leq Q \log Q + 1$ to obtain the upper side of PFG15.

For the lower inequality let $n_a = D - t_a = Q - v - t_a \geq 92$. Since $s \geq 1$,

$$n!(s/2)_n \geq n!(1/2)_n = \frac{(2n)!}{4^n} \geq (n/\exp(1))^{2n}. \quad (\text{PFG16})$$

The product equality separates even and odd factors of $(2n)!$. The last inequality follows by $\sum_{r=1}^{2n} \log r \geq \int_1^{2n} \log x \, dx = 2n \log(2n) - 2n + 1$. The binomials in PFG11 are positive integers at least one. Consequently its logarithm is at least $c_{\text{low}} + 4 \log M_s + 2 \sum_a n_a \log n_a - 2 \sum_a n_a$. For $1 \leq n_a \leq Q$, integration of the derivative of $x \log x$ gives $Q \log Q - n_a \log n_a \leq (Q - n_a)(\log Q + 1)$. Now $\sum_a (Q - n_a) = 4v + T$ and $\sum_a n_a \leq 4Q$. Substitution proves the lower side of PFG15. The integer v and the actual pivot indices are fixed by the period before taking any limit.

In particular, uniformly in the two original choices $s = 1, k$,

$$\begin{aligned} \log \det(W^* H_1 W) &= 4 \log M_s + 8Q \log Q + O_{\text{actual}}(Q), \\ |\log \det(W^* H_1 W)| &= O_{\text{actual}}(Q \log Q) = o(kq_k). \end{aligned} \quad (\text{PFG17})$$

Indeed the two explicit sides of PFG15 are $O_{\text{actual}}(Q)$, $\log Q \leq Q$, and $4 \log M_s = 2s \log(2\pi) = O(k)$. Also $Q/q_k \rightarrow 1$ and $\log Q/k \rightarrow 0$ on $k = 4l + 1$. The coefficient 8 follows from four original squared polynomial norms; it was derived from the finite pivot and product bounds.

For clarity, the exact receiving identity preserves the original primitive-coordinate complement. Let Z be any fixed full basis of $\ker W^\top$ and let J be a fixed right inverse of $Z^\top$. Then $\ker Z^\top = \text{im } W$, $[W, J]$ is invertible, and completing the original H_1 square gives

$$\det((Z^\top H_1^{-1} \overline{Z})^{-1}) = \frac{|\det[W, J]|^2 \det H_1}{\det(W^* H_1 W)}. \quad (\text{PFG18})$$

Indeed the full frame Gram has determinant $|\det[W, J]|^2 \det H_1$; its first block is $W^* H_1 W$. The Schur block is exactly the attained quotient minimum with constraint $Z^\top f = a$, since $Z^\top J = I$. Its Gram is the inverse displayed in PFG18. This also proves the complex-linear transpose and conjugate-inverse orientation used in PDS9. PFG15–17 now control this particular four-row denominator in the original PDS and PSR calculation. The full fixed-frame determinant $|\det[W, J]|^2$, and the original connecting-coordinate or complement matrix, remain in PFG18.

191 The original relation determinant and its connecting-image return

Keep every original quartet, period, branch, nonreduced conductor, coefficient and norm in PDS1–9, PSG1–23 and SGT1–33. The symbol is fixed as the degree varies. In particular

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad q = (k + 1)^2, \quad Q = (k + 5)^2, \quad e \in \{0, 1\}, \quad d_e = 8k + 24 + e, \\ v &= \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad N_e = Q - v - 1 + e, \quad s = 1 \text{ or } k, \\ u_a &= \chi_k(S) S^a, \quad r_a = \mathcal{T}_A u_a \quad (0 \leq a < d_e), \\ H_e &= \text{Gram}_{s, k/2-4}(r_0, \dots, r_{d_e-1}), \quad J_{e,s}^\chi = \text{Gram}_{s, k/2, \text{quot}}([u_a])_{a < d_e}. \end{aligned} \tag{PSR1}$$

The source quotient here is by all of $\mathbb{C}[S]_{<v}$, and the target Gram is exactly PDS7. It remains the Gram of the complete polynomials, not a Gram of their leading monomials. The complete measure in both Grams is

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy.$$

191.1 Finite bounds on the actual translated relation Gram

Retain the exact positive factorial band and the finite PSG errors:

$$\begin{aligned} n_a &= q + a - v, \quad m_a = n_a + (s - 1)/2, \quad \lambda_a = \mu_v \binom{q + a}{v}, \\ \mathfrak{F}_{k,e,s} &= \prod_{a < d_e} |\lambda_a|^2 (2\pi)^{s/2} n_a! (s/2)_{n_a}, \quad \mathcal{A}_{k,e,s}^- = B_{k,e,s}^- + 2v \log C_E + E_{k,e,s}^{\text{band}}, \\ \mathcal{A}_{k,e,s}^+ &= B_{k,e,s}^+ + E_{k,e,s}^{\text{band}}, \quad \mathcal{M}_{k,e} = 2d_e q \log(4/\pi). \end{aligned} \tag{PSR2}$$

All three finite expressions $B^\pm, C_E, E^{\text{band}}$ are the literal PSG9, PSG15 and PSG20 expressions. For $v = 0$ use $C_E = 1$. The finite estimates below hold for $k \geq \max(9, v - 1, \lceil (\pi/2)(\delta^2 + \gamma^2)^{1/2} \rceil + 1)$; this ensures every stated PSG finite condition. Their limiting conclusions concern the unchanged sequence $k = 4l + 1$.

For any fixed admissible radius $0 < a < 1$ in SGT14, keep the full pole, residue, zero-free-disk and diagonal constants of SGT11–25 and set

$$\begin{aligned} \mathcal{E}_{k,e,s}^-(a) &= \mathcal{A}_{k,e,s}^- + \mathcal{L}_{a, N_e, s, d_e}, \quad \mathcal{E}_{k,e,s}^+(a) = \mathcal{A}_{k,e,s}^+ + \mathcal{U}_{a, N_e, s, d_e}, \\ \varepsilon_{k,e,s}^H &= \log \det H_e - \log \mathfrak{F}_{k,e,s} - \mathcal{M}_{k,e}. \end{aligned} \tag{PSR3}$$

Then the finite inequalities and their limiting consequence are

$$\boxed{-\mathcal{E}_{k,e,s}^-(a) \leq \varepsilon_{k,e,s}^H \leq \mathcal{E}_{k,e,s}^+(a), \quad \varepsilon_{k,e,s}^H = o(kq).} \tag{PSR4}$$

Here is the exact map proving this receiving statement. Express the full u_a in the input Gamma frame and delete precisely the rows below v , giving the SGT30 matrix Y_e . The quotient minimum is $Y_e^* Y_e = J_{e,s}^\chi$. The actual translated coefficient matrix is $C_e = A_{N_e, s} Y_e$, with all entries of $A_{N_e, s}$ exactly SGT10. Thus

$$\log \frac{\det H_e}{\mathfrak{F}_{k,e,s}} - \mathcal{M}_{k,e} = \left(\log \frac{\det J_{e,s}^\chi}{\mathfrak{F}_{k,e,s}} - \mathcal{M}_{k,e} \right) + \log \frac{\det(Y_e^* A_{N_e, s}^* A_{N_e, s} Y_e)}{\det(Y_e^* Y_e)}. \tag{PSR5}$$

There is no frame change or unmatched scalar in this identity. PSG21 bounds the first parenthesis by $[-\mathcal{A}^-, \mathcal{A}^+]$; SGT25 bounds the second term by $[-\mathcal{L}, \mathcal{U}]$. Their addition proves both finite sides of PSR4. PSG13, PSG16 and PSG20 give $\mathcal{A}^\pm = o(kq)$. SGT26–31 show that the actual second term is $o(kq)$: for each fixed admissible a its scaled limsup is bounded by a constant times $\log(1/a)$,

and admissible a can tend to one only after the degree limit. No claim that the fixed- a bounds themselves are $o(kq)$ is needed. This proves PSR4 in both original orders and at both cutoffs. In particular

$$\frac{1}{kq} \log \frac{\det H_e}{\mathfrak{F}_{k,e,s}} \longrightarrow 16 \log(4/\pi). \quad (\text{PSR6})$$

191.2 Absolute expansion with the full factorial band retained

The exact factorial-band identity is

$$\log \mathfrak{F}_{k,e,s} = d_e \log((2\pi)^{s/2}) - d_e \log \Gamma(s/2) + \sum_{a < d_e} \{2 \log |\lambda_a| + \log(n_a!) + \log \Gamma(m_a + 1/2)\}. \quad (\text{PSR7})$$

For a fully finite bound define

$$\begin{aligned} \mathcal{F}_{k,e,s}^{\text{err}} &= d_e |\log((2\pi)^{s/2})| + d_e |\log \Gamma(s/2)| + 2 \sum_{a < d_e} |\log |\lambda_a|| \\ &\quad + \sum_{a < d_e} (|n_a - q| + |m_a - q|) \log \max(q, n_a, m_a) \\ &\quad + \sum_{a < d_e} (3 + \log n_a + \log(2m_a) + \log m_a + \tfrac{1}{2} |\log \pi|). \end{aligned} \quad (\text{PSR8})$$

The elementary integer and half-integer factorial estimates proved after PSG22, and integration of $(x \log x - x)' = \log x$, give

$$|\log \mathfrak{F}_{k,e,s} - 2d_e q(\log q - 1)| \leq \mathcal{F}_{k,e,s}^{\text{err}}. \quad (\text{PSR9})$$

Every n_a, m_a is positive. For $s = 1$ the Gamma term in PSR8 is fixed. For $s = k$, apply $\Gamma(m+1/2) = \sqrt{\pi}(2m)!/(4^m m!)$ with $m = (k-1)/2$ and PSG11 to obtain $|\log \Gamma(k/2)| = O(k \log k)$. All degree displacements in PSR8 are $O(k)$, while $d_e = O(k)$ and $|\log |\lambda_a|| = O_{\text{actual}}(\log q)$. Hence $\mathcal{F}^{\text{err}} = O_{\text{actual}}(q \log q) = o(kq)$. Consequently the absolute expansion of the unchanged Gram is

$$\log \det H_e = 2d_e q \{\log(4q/\pi) - 1\} + o(kq). \quad (\text{PSR10})$$

The finite absolute error is bounded below by $-\mathcal{F}^{\text{err}} - \mathcal{E}^-(a)$ and above by $\mathcal{F}^{\text{err}} + \mathcal{E}^+(a)$. PSR7 and PSR4 retain every mass, factorial, first nonzero moment and phase-bearing map behind this expansion.

191.3 Dual source and both primitive coefficient maps

For the original full quotient-source metric PDS7 one has exactly

$$G_e = \overline{H_e^{-1}}, \quad \log \det G_e = -\log \mathfrak{F}_{k,e,s} - \mathcal{M}_{k,e} - \varepsilon_{k,e,s}^H. \quad (\text{PSR11})$$

Indeed positive Hermitian H_e has positive real determinant, so conjugation preserves its determinant and inversion reciprocates it. This also proves the interval for the dual residual: $-\mathcal{E}^+(a) \leq -\varepsilon^H \leq \mathcal{E}^-(a)$.

Now retain CIT's original primitive lift P_e and PDI's exact map $\Theta_e = C_e^T P_e$. Its image is all of $\mathbb{C}^{d_0}$ at $e = 0$ and $\ker W^T$ at $e = 1$, where W is the complete rank-four PDS8 coefficient matrix. Put

$$\begin{aligned} \Xi_0 &= 2 \log |\det \Theta_0|, & Z &= \Theta_1, & J_Z &= \overline{Z}^T (Z^T \overline{Z})^{-1}, & B_Z &= [W, J_Z], \\ \Xi_1 &= \log \det(W^* H_1 W) - 2 \log |\det B_Z|. \end{aligned} \quad (\text{PSR12})$$

All matrices in this display are the actual matrices for the specified source order. Since Z has independent columns, $Z^T \overline{Z} = \overline{Z^* Z}$ is invertible. Thus $Z^T J_Z = I$ and $\ker Z^T = \text{im } W$; the latter

follows from $W^\top Z = 0$ and the exact four-dimensional kernel. Consequently B_Z is square and invertible. The original primitive source Gram at both cutoffs is

$$\begin{aligned} H_{\text{src},e} &= \Theta_e^* G_e \Theta_e, \\ \log \det H_{\text{src},e} &= \Xi_e - \log \mathfrak{F}_{k,e,s} - \mathcal{M}_{k,e} - \varepsilon_{k,e,s}^H. \end{aligned} \quad (\text{PSR13})$$

The metric equality is PDI2's equality of the full minimum-norm fibres. At $e = 0$ the determinant identity is immediate from invertibility of Θ_0 . At $e = 1$, block Gram factorization in the full frame $B_Z = [W, J_Z]$ gives

$$\det((Z^\top H_1^{-1} \bar{Z})^{-1}) = \frac{|\det B_Z|^2 \det H_1}{\det(W^* H_1 W)}.$$

The left matrix is the conjugate inverse of $Z^* G_1 Z$, by the attained quotient calculation PDS9. Reciprocating this positive determinant and substituting PSR11 proves PSR13. This supplies the exact four-row correction and its frame determinant; neither is declared to vanish. The finite residual interval in PSR13 is precisely $[-\mathcal{E}^+(a), \mathcal{E}^-(a)]$ after subtracting $\Xi_e - \log \mathfrak{F} - \mathcal{M}$.

191.4 Return to the original connecting-image calculation

Use the two unchanged source-order pairs $(s_0, s_1) = (1, 1)$ or $(k, k+4)$, with $s = s_0$ in PSR1–13. Keep CIT17's exact $\chi_{X_k, D_e}, \chi_{Y_e, D_e}$, original multiplier and inverse constants $K_\pm$, and $c_- = \min(1, L_-^{-2})$, $c_+ = \max(1, L_+^2)$. Write $r_e = d_e - 4e = 8k + 24 - 3e$ and

$$\begin{aligned} \epsilon_e^{\text{lo}} &= -2r_e \log \max(1, K_-) - r_e \log c_+, & \epsilon_e^{\text{hi}} &= 2r_e \log \max(1, K_+) - r_e \log c_-, \\ \mathcal{Z}_{k,e,s_0,s_1} &= \Xi_e - \log \mathfrak{F}_{k,e,s_0} - \mathcal{M}_{k,e} - \chi_{X_k, D_e} + \chi_{Y_e, D_e}. \end{aligned} \quad (\text{PSR14})$$

The actual connecting-image Gram in its original P_e image frame then satisfies the complete finite estimate

$$\begin{aligned} \epsilon_e^{\text{lo}} - \mathcal{E}_{k,e,s_0}^+(a) &\leq \log \det H_{\text{conn},e} - \mathcal{Z}_{k,e,s_0,s_1} \\ &\leq \epsilon_e^{\text{hi}} + \mathcal{E}_{k,e,s_0}^-(a), \\ \log \det H_{\text{conn},e} &= \mathcal{Z}_{k,e,s_0,s_1} + o(kq). \end{aligned} \quad (\text{PSR15})$$

For proof CIT17 is the exact equality $\log \det H_{\text{conn},e} - \log \det H_{\text{src},e} = -\chi_{X_k, D_e} + \chi_{Y_e, D_e} + \epsilon_e$, with $\epsilon_e^{\text{lo}} \leq \epsilon_e \leq \epsilon_e^{\text{hi}}$. Insert PSR13. The remaining residual is exactly $\epsilon_e - \varepsilon_{k,e,s_0}^H$, so subtracting the two intervals proves every finite sign in PSR15. CIT17 proves $\epsilon_e = O_{\text{actual}}(k \log q)$, and PSR4 proves the other term is $o(kq)$. At $e = 0$, $\chi_{Y_0, D_0} = 0$ because the source-complement projector is zero; at $e = 1$ the complete four-row term χ_{Y_1, D_1} stays in PSR14.

These identities update PDS7–9, PDI2 and CIT17 at their defining cutoffs through their exact coefficient and quotient maps. At later cutoffs the fixed connecting columns still carry CIT18–26's full metrics and boundary chain; PSR15 assigns no value to the surviving $\chi_{X_k, D_e} - \chi_{Y_e, D_e}$ or to those later cross ratios. The original factors now evaluated are the entire translated relation Gram and its dual. The remaining connecting and invariant factors are displayed in the same original frames, ready for their actual row calculations.

191.5 The four-row estimate in both original primitive sources

PFG5 identifies its full four-row Gram with $W^* H_1 W$. Use precisely its earliest actual moment indices $t_1 < t_2 < t_3 < t_4 \leq 24$, moment matrix $\mathcal{M}$, $T = \sum t_a$, and original constants $C_\beta, c_{\text{low}}, c_{\text{up}}$ of PFG14. Put

$$\begin{aligned} b_{k,s}^{\text{lo}} &= c_{\text{low}} - 8Q - 2(4v + T)(\log Q + 1), & b_{k,s}^{\text{hi}} &= c_{\text{up}} + 8Q \log C_\beta + 8, \\ m_{k,s}^{(4)} &= 4 \log((2\pi)^{s/2}) + 8Q \log Q, & \varepsilon_{k,s}^{(4)} &= \log \det(W^* H_1 W) - m_{k,s}^{(4)}. \end{aligned} \quad (\text{PSR16})$$

PFG10's exact nonzero moment pivot and PFG12's complete original coefficient product prove

$$b_{k,s}^{\text{lo}} \leq \varepsilon_{k,s}^{(4)} \leq b_{k,s}^{\text{hi}}, \quad \varepsilon_{k,s}^{(4)} = O_{\text{actual}}(Q), \quad m_{k,s}^{(4)} = O_{\text{actual}}(Q \log Q) = o(kq). \quad (\text{PSR17})$$

Thus the four-row term retained in PSR12 has now been estimated on its actual polynomial frame. In particular define only the exact coordinate factor, retaining its entire value, by

$$\Gamma_0 = 2 \log |\det \Theta_0|, \quad \Gamma_1 = -2 \log |\det[W, J_Z]|. \quad (\text{PSR18})$$

PSR12 gives $\Xi_e = \Gamma_e + e(m_{k,s}^{(4)} + \varepsilon_{k,s}^{(4)})$. Consequently

$$\begin{aligned} \log \det H_{\text{src},e} &= \Gamma_e - \log \mathfrak{F}_{k,e,s} - \mathcal{M}_{k,e} + em_{k,s}^{(4)} + e\varepsilon_{k,s}^{(4)} - \varepsilon_{k,e,s}^H, \\ eb_{k,s}^{\text{lo}} - \mathcal{E}_{k,e,s}^+(a) &\leq e\varepsilon_{k,s}^{(4)} - \varepsilon_{k,e,s}^H \leq eb_{k,s}^{\text{hi}} + \mathcal{E}_{k,e,s}^-(a). \end{aligned} \quad (\text{PSR19})$$

Every term is evaluated in its original coefficient coordinates. At $e = 0$ both four-row terms are absent, as prescribed by the original primitive space. At $e = 1$ they are the full actual four-row determinant with its finite lower and upper errors. Combining PSR9–10 with PSR17 now gives both original sources the absolute expansion

$$\log \det H_{\text{src},e} = \Gamma_e - 2d_e q \{\log(4q/\pi) - 1\} + o(kq). \quad (\text{PSR20})$$

No size for Γ_e is asserted by this formula.

For the original connecting image put

$$\tilde{Z}_{k,e,s_0,s_1} = \Gamma_e - \log \mathfrak{F}_{k,e,s_0} - \mathcal{M}_{k,e} + em_{k,s_0}^{(4)} - \chi_{X_k,D_e} + \chi_{Y_e,D_e}. \quad (\text{PSR21})$$

Direct substitution in CIT17 or PSR15 gives the finite return

$$\begin{aligned} \epsilon_e^{\text{lo}} + eb_{k,s_0}^{\text{lo}} - \mathcal{E}_{k,e,s_0}^+(a) &\leq \log \det H_{\text{conn},e} - \tilde{Z}_{k,e,s_0,s_1} \\ &\leq \epsilon_e^{\text{hi}} + eb_{k,s_0}^{\text{hi}} + \mathcal{E}_{k,e,s_0}^-(a), \\ \log \det H_{\text{conn},e} &= \Gamma_e - 2d_e q \{\log(4q/\pi) - 1\} - \chi_{X_k,D_e} + \chi_{Y_e,D_e} + o(kq). \end{aligned} \quad (\text{PSR22})$$

This follows by retaining the exact residual $\epsilon_e + e\varepsilon_{k,s_0}^{(4)} - \varepsilon_{k,e,s_0}^H$. Its three previously proved intervals add with precisely the signs shown. All three residuals are $o(kq)$, and the displayed four-row center also is $o(kq)$ by PSR17. This proves the last line.

191.6 Exact common coordinates for the two original source orders

The original Taylor complement in CIT6 is fixed before it is transported to its Gamma native coordinates. Denote its column matrix by P_e^{Tay} , and the coefficient matrix of the full polynomials r_a in powers of S' by R_e^{pol} . For the source order s let $\mathbf{B}_s$ be the original monomial-to-Gamma matrix through degree N_e . CIT3 and the choice after CIT6 give

$$\begin{aligned} U_s &= \text{diag}(1/r!)_{r=0}^{N_e} \mathbf{B}_s^{\text{T}}, \quad P_e^{(s)} = U_s^{-1} P_e^{\text{Tay}}, \quad C_e^{(s)} = \mathbf{B}_s R_e^{\text{pol}}, \\ \Theta_e^{(s)} &= (C_e^{(s)})^{\text{T}} P_e^{(s)} = (R_e^{\text{pol}})^{\text{T}} \text{diag}(r!)_{r=0}^{N_e} P_e^{\text{Tay}}. \end{aligned} \quad (\text{PSR23})$$

The equality follows by multiplying these three explicit matrices: the full $\mathbf{B}_s^{\text{T}}$ and its inverse cancel, including every original mass, leading phase and center coefficient. The remaining expression is independent of s . It follows that Θ_e , Z , J_Z , and Γ_e in PSR12 and PSR18 are literally the same coefficient objects for $s = 1$ and $s = k$. The metric comparison uses the identity on P_e^{Tay} values; no well-conditioned choice of a new complement has been made.

For completeness the same coefficient identity relates the two connecting-image frames. Let $\tilde{T}_j = U_j T_j$ and $\tilde{V}_j = U_j V_j$ be the Taylor columns of the original analytic germs and lower evaluations. These columns are independent of their Gamma presentation, since each is the Taylor series of the same specified function. CIT3 and CIT8 are equivalently

$$\text{stack}_i \text{Mult}(B_i) P_e^{\text{Tay}} = \text{diag}(\tilde{T}_j, 4) F_e + \text{diag}(\tilde{V}_j, 4) E_e^{\text{low}} \quad (\text{PSR24})$$

for the complete lower coefficient matrix E_e^{low} . Existence is precisely CIT6–8. Uniqueness of F_e follows from CNC13: an analytic coefficient column whose jet is a lower evaluation represents zero in the original lower quotient only when the coefficient is zero. The two metric presentations of PSR24 therefore give the same F_e . Thus the following ratios compare both source and target metrics on their same original ordered value spaces.

In these fixed coordinates PSR20 yields the quantitative source comparison

$$\log \frac{\det H_{\text{src},e}^{(k)}}{\det H_{\text{src},e}^{(1)}} = o(kq). \quad (\text{PSR25})$$

Indeed the identical Γ_e and identical explicit absolute center cancel. A fully finite version follows from PSR19: set

$$\begin{aligned} D_e^{\text{ord}} &= \log \mathfrak{F}_{k,e,1} - \log \mathfrak{F}_{k,e,k} + e(m_{k,k}^{(4)} - m_{k,1}^{(4)}), \\ I_{e,s}^- &= e b_{k,s}^{\text{lo}} - \mathcal{E}_{k,e,s}^+(a), \quad I_{e,s}^+ = e b_{k,s}^{\text{hi}} + \mathcal{E}_{k,e,s}^-(a), \\ I_{e,k}^- - I_{e,1}^+ &\leq \log \frac{\det H_{\text{src},e}^{(k)}}{\det H_{\text{src},e}^{(1)}} - D_e^{\text{ord}} \leq I_{e,k}^+ - I_{e,1}^-. \end{aligned} \quad (\text{PSR26})$$

This subtraction retains all original factorials and mass terms. Finally write $\mathcal{X}_e^{(s_0,s_1)} = \chi_{X_k,D_e}^{(s_0,s_1)} - \chi_{Y_e,D_e}^{(s_0,s_1)}$ for the full original compatibility-plus-complement quantities. Subtracting the two exact CIT17 identities on the common frames proves

$$\log \frac{\det H_{\text{conn},e}^{(k,k+4)}}{\det H_{\text{conn},e}^{(1,1)}} = -\mathcal{X}_e^{(k,k+4)} + \mathcal{X}_e^{(1,1)} + o(kq). \quad (\text{PSR27})$$

The exact omitted residual in this last display is the source ratio in PSR25 plus $\epsilon_e^{(k,k+4)} - \epsilon_e^{(1,1)}$; PSR26 and CIT17 give its full finite interval by addition. Thus the leading comparison of these original connecting metrics has been reduced, through explicit maps, to the displayed actual independent gauge loss. The loss itself and the later fixed-column return remain the next native arithmetic calculations.

192 The original Gamma source order on the native conductor and connecting flags

The whole-polynomial inequality used here is already AMT23,29–30; its restriction and quotient argument is AMT7–9, and AMT42–46 already applies it to the original arithmetic kernel. The present calculation gives the full complex-linear dual/Taylor square, receives that inequality on the later DNE/CIT fixed flags, and evaluates the difference of the two CIT17 compatibility-plus-complement quantities. All original objects are retained. In particular fix the actual simple quartet, its original five-orbit period with $|u| \geq R_*$ and its fixed logarithm branch, the full nonreduced conductor, and

$$k = 4l + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad c'_a = a/2 - 4,$$

$$E_A(z) = \sum_{u,w=0}^8 a_{uw} e^{\beta_{s,u,w} z}, \quad \mu_n = E_A^{(n)}(0), \quad 0 \leq v \leq 80, \quad \mu_v \neq 0, \quad \mu_0 = \cdots = \mu_{v-1} = 0. \quad (\text{NSO1})$$

No root, coefficient, period or low overlap is changed. For degree $a = k$ or j , the two orders compared are 1 and a , at the same center c'_a and the same polynomial cutoff. In the primitive source/image comparison the pairs remain $(1, 1)$ and (k, j) .

192.1 The established whole-source estimate with its complete constants

For $s = 4\ell + 1 \geq 1$ keep the full measure, mass and monic recurrence

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = \int_{\mathbb{R}} dm_s = (2\pi)^{s/2}, \quad (\text{NSO2})$$

$$p_{0,s} = 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \quad h_{n,s} = M_s n! (s/2)_n.$$

These are the original Gamma data. The identity $\Gamma(s/4 + iy/2) = \prod_{r=0}^{\ell-1} (r+1/4 + iy/2) \Gamma(1/4 + iy/2)$, with its factor $4^{-\ell}$ in squared absolute value, gives exactly

$$t_r = 2r + \frac{1}{2}, \quad \beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \quad \rho_s = \frac{\beta_s}{\beta_1} = \frac{(2\pi)^{2\ell}}{(1/2)_{2\ell}}, \quad (\text{NSO3})$$

$$dm_s(y) = \rho_s \prod_{r=0}^{\ell-1} (y^2 + t_r^2) dm_1(y).$$

Here $\beta_1 = 1$ follows from its displayed full mass. It is kept in the ratio to identify the actual scalar; no measure is rescaled. In particular NSO3 is exactly AMT23 and the $s = 1$ case of MNT7.

For every integer $d \geq 0$ define

$$\Pi_\ell = \prod_{r=0}^{\ell-1} t_r^2, \quad U_{\ell,d} = \prod_{r=0}^{\ell-1} \{4(d+r+1)^2 + t_r^2\}, \quad (\text{NSO4})$$

$$\mathbf{a}_s = \rho_s \Pi_\ell, \quad \mathbf{b}_{s,d} = \rho_s U_{\ell,d}, \quad \omega_{s,d} = \log(\mathbf{b}_{s,d}/\mathbf{a}_s).$$

Empty products are one, so all logarithms at $s = 1$ are zero. Let $H_{s,c,d}$ be the Gram on the unchanged ordered monomials $1, S', \dots, S'^d$ with squared norm $\int |P(c + iy)|^2 dm_s(y)$. Then

$$\mathbf{a}_s H_{1,c,d} \preceq H_{s,c,d} \preceq \mathbf{b}_{s,d} H_{1,c,d}. \quad (\text{NSO5})$$

Here is the entire estimate from AMT29–30. In the orthonormal order-one frame the recurrence NSO2 gives raising and lowering coefficients $\sqrt{(n+1)(n+1/2)}$ and $\sqrt{n(n-1/2)}$. On degrees at most d , each of the two weighted shifts has norm at most $d+1$. Hence $\|yp\|_1 \leq 2(d+1)\|p\|_1$. Pointwise, for real y, t , $|(iy-t)p(y)|^2 = (y^2 + t^2)|p(y)|^2$. Successively multiply by $iy - t_r$. At step

r the degree is at most $d + r$, so the squared norm increases by at most $4(d + r + 1)^2 + t_r^2$. This gives $U_{\ell,d}$. The pointwise product is at least Π_ℓ . Multiply the two estimates by the literal scalar ρ_s in NSO3 and substitute $p(y) = P(c + iy)$ to prove NSO5.

The finite constants also satisfy

$$\begin{aligned} \mathbf{a}_s &= (4\pi^2)^\ell \frac{(1/4)_\ell}{(3/4)_\ell}, & (4\pi^2/3)^\ell \leq \mathbf{a}_s &\leq (4\pi^2)^\ell, \\ 0 &\leq \omega_{s,d} \leq \ell \log\{1 + 16(d + \ell)^2\}, \\ 0 &\leq \log \mathbf{b}_{s,d} \leq \ell \log(4\pi^2) + \ell \log\{1 + 16(d + \ell)^2\}. \end{aligned} \quad (\text{NSO6})$$

For the first equality use $(1/2)_{2\ell} = 2^{2\ell}(1/4)_\ell(3/4)_\ell$ and $\Pi_\ell = 4^\ell(1/4)_\ell^2$. Each ratio $(r+1/4)/(r+3/4)$ lies in $[1/3, 1]$. For $\ell > 0$, every $d + r + 1 \leq d + \ell$ and $t_r \geq 1/2$; apply these two inequalities to each factor $U_{\ell,d}/\Pi_\ell$. The case $\ell = 0$ is the empty equality. Thus the finite bounds are valid at all cutoffs, without the additional large-degree domain needed in GPA's power-measure comparison. That comparison retains its two distinct exponents Q_1 and Q_a .

192.2 The exact original functional and Taylor coordinate square

Define the original orthonormal polynomials and monomial matrix by

$$\phi_{n,s,c}(S') = \frac{p_{n,s}((S' - c)/i)}{\sqrt{h_{n,s}}}, \quad S'^r = \sum_{n=0}^d (\mathbf{B}_{s,c,d})_{nr} \phi_{n,s,c}(S'). \quad (\text{NSO7})$$

The matrix is upper triangular with diagonal $i^n \sqrt{h_{n,s}}$, so is invertible; the leading coefficient of ϕ_n is $i^{-n}/\sqrt{h_{n,s}}$. Direct evaluation of squared polynomial norms gives $H_{s,c,d} = \mathbf{B}_{s,c,d}^* \mathbf{B}_{s,c,d}$.

For a complex-linear functional Λ write its derivative values and native values, respectively, as $\mathbf{j}_d = (\Lambda(S'^r))_{r=0}^d$ and $w_s = (\Lambda(\phi_{n,s,c}))_{n=0}^d$. A generating germ F has $\Lambda(S'^r) = F^{(r)}(0)$. The exact maps are

$$\begin{aligned} \mathbf{j}_d &= \mathbf{B}_{s,c,d}^\top w_s, & U_{s,c,d} &= \text{diag}(1/r!) \mathbf{B}_{s,c,d}^\top, & F^{\text{Tay}} &= U_{s,c,d} w_s, \\ L_{s \leftarrow -1, c, d} &= (\mathbf{B}_{s,c,d}^\top)^{-1} \mathbf{B}_{1,c,d}^\top, & w_s &= L_{s \leftarrow -1, c, d} w_1. \end{aligned} \quad (\text{NSO8})$$

All transposes here are ordinary transposes because Λ is complex-linear. This map and its explicit reversed product are inverse. It fixes each underlying functional and carries every lower evaluation and numerator germ by the same matrix.

The full raw dual Gram on the derivative-value columns is

$$G_{s,c,d}^\vee = (\mathbf{B}_{s,c,d}^\top)^{-*} (\mathbf{B}_{s,c,d}^\top)^{-1} = \overline{H_{s,c,d}^{-1}}, \quad \mathbf{b}_{s,d}^{-1} G_{1,c,d}^\vee \preceq G_{s,c,d}^\vee \preceq \mathbf{a}_s^{-1} G_{1,c,d}^\vee. \quad (\text{NSO9})$$

Indeed $\Lambda(P) = \mathbf{j}_d^\top \text{coeff}(P)$ has Riesz vector $H^{-1} \mathbf{j}_d$, and its squared norm is $\mathbf{j}_d^* \overline{H^{-1}} \mathbf{j}_d = \|w_s\|^2$. Inverting NSO5 reverses its inequalities; complex conjugation preserves Hermitian positivity, which proves both bounds. Thus NSO8 is the exact typed isomorphism between the two native presentations, and NSO9 is its norm comparison.

192.3 Complete attained fibres and the original conductor quotient

For reference we give the finite minimum argument in its full form. If G is a positive Hermitian Gram and $[X, Z]$ is an independent column frame, then the norm on the values represented by Xx modulo $\text{im } Z$ is

$$\begin{aligned} \|[Xx]\|_G^2 &= \min_z (Xx + Zz)^* G (Xx + Zz) = x^* Q_G x, \\ z_{\min} &= -(Z^* G Z)^{-1} Z^* G X x, & Q_G &= X^* G X - X^* G Z (Z^* G Z)^{-1} Z^* G X. \end{aligned} \quad (\text{NSO10})$$

Subtracting $z_{\min}$ makes the remaining mixed term zero, so every other squared norm is this minimum plus $(z - z_{\min})^* Z^* G Z (z - z_{\min})$. For an onto fixed matrix A the full affine fibre $Ay = b$ instead has

$$y_{\min} = G^{-1} A^* (A G^{-1} A^*)^{-1} b, \quad Q_A = (A G^{-1} A^*)^{-1}. \quad (\text{NSO11})$$

Multiplication proves $Ay_{\min} = b$; its G -pairing with every vector of $\ker A$ is zero. This proves the same squared norm decomposition and the full minimum. Restriction is X^*GX . Consequently $\alpha G_1 \preceq G_2 \preceq \beta G_1$ gives the same scalar bounds after any fixed restriction or quotient: apply the inequality to every vector of the same fibre and take its two minima. This also applies to successive quotients, since the union of their fibres is precisely the full original affine fibre. Empty quotients have determinant one. This is the AMT7–9/FRE5 argument with its section displayed.

Here are the actual conductor fibres, including their original frame choice. For degree a let $\mathbb{V}_a$ be the upper Vandermonde with entries $\beta_{a,u,w}^n$, $0 \leq n < q_a$. Let P_v insert zero in the first v coordinates and let C_a be the full primary conductor with q'_a rows. Then

$$\begin{aligned} W_{v,a} &= \mathbb{V}_a^{-1} P_v, & J_{v,a} &= P_v^\top \mathbb{V}_a C_a^\top, & W_{v,a} J_{v,a} &= C_a^\top, \\ m_a &= q_a - q'_a - v = 16a - 48 - v, & Z_{\exp,a} &= W_{v,a} E_{\mathcal{J}_a}, & F_{a,b}(z) &= \frac{\sum_{u,w} (Z_{\exp,a})_{uw,b} e^{\beta_{a,u,w} z}}{E_A(z)}. \end{aligned} \quad (\text{NSO12})$$

Here $\mathcal{I}_a$ is the first ordered set of q'_a rows giving a nonzero minor of $J_{v,a}$ and $\mathcal{J}_a$ its complement. The distinct upper roots make $\mathbb{V}_a$ invertible. The first v rows of $\mathbb{V}_a C_a^\top$ vanish, because these columns have generating functions $E_A e^{\beta_{a-8,u,w} z}$. The injectivity of multiplication by the original nonzero conductor gives rank q'_a . Hence $[J_{v,a}, E_{\mathcal{J}_a}]$ is an invertible frame: restrict a relation to $\mathcal{I}_a$, and then to its complement. These observations prove NSO12 and identify the exact quotient of vanishing numerator coefficients by $\text{im } C_a^\top$. The frame contains all actual minor factors and is independent of s and d .

At $d \geq q_a - v - 1$ define the actual finite columns

$$\begin{aligned} T_{a,d}^{(s)}[n, b] &= \Lambda_{F_{a,b}}(\phi_{n,s,c'_a}), & V_{a,d}^{(s)}[n, (u, w)] &= \phi_{n,s,c'_a}(\beta_{a-8,u,w}), \\ D_{a,d}^{(s)} &= (T_{a,d}^{(s)})^* [I - V_{a,d}^{(s)} ((V_{a,d}^{(s)})^* V_{a,d}^{(s)})^{-1} (V_{a,d}^{(s)})^*] T_{a,d}^{(s)}, \\ \|x\|_{D_{a,d}^{(s)}}^2 &= \min_{\eta} \left\| \Lambda_{\sum_b x_b F_{a,b}} + \sum_{u,w} \eta_{uw} \text{ev}_{\beta_{a-8,u,w}} \right\|_{s,c'_a,d,\vee}^2. \end{aligned} \quad (\text{NSO13})$$

These are the full DNE6–7 fibres. To check positivity directly, if a column combination is a lower evaluation through d , its generating difference vanishes through order d . Multiplying by E_A makes an upper-grid exponential numerator vanish through order $d + v \geq q_a - 1$. The upper Vandermonde then makes that numerator zero. Its fixed quotient class in NSO12 is zero, and the lower Vandermonde makes the remaining lower coefficients zero. Thus $[V, T]$ has full column rank. Formula NSO10 proves the displayed minimum including every lower coefficient.

In NSO8 both T and V are multiplied by the same $L_{s \leftarrow 1, c'_a, d}$. The underlying fibre of NSO13 is therefore unchanged. NSO9–10 give

$$\mathfrak{b}_{s,d}^{-1} D_{a,d}^{(1)} \preceq D_{a,d}^{(s)} \preceq \mathfrak{a}_s^{-1} D_{a,d}^{(1)} \quad (\text{NSO14})$$

and give the identical constants on every fixed restriction or attained quotient of this coefficient space, including fixed flags in a finite direct sum of its copies. The eliminated raw lower gauge has dimension q'_a per copy, but no factor q'_a enters NSO14: the entire fibre has already been minimized before taking a determinant on its value space.

For completeness the exact DNE9 map to the actual arithmetic kernel uses the original fixed low monomial columns Z_a and dual pairing matrix $A_{\ell,a}$. The latter has entries $\ell_b(S'^m)$, where $\ell_b(g) = \Lambda_{F_{a,b}}(\chi_{a-8}g)$. The relation quotient has the independent low monomials below degree m_a and these ℓ_b are its complete dual frame; hence $A_{\ell,a}$ is invertible. Keep

$$\begin{aligned} t_{0,a} &= \dim(K_a \cap \mathbb{C}[S]_{<v}), & t_{K,a} &= 8a - 16 - t_{0,a}, \\ B_a &= Z_a^\top A_{\ell,a} : \mathbb{C}^{m_a} \rightarrow \mathbb{C}^{t_{K,a}}, & J_a &= \ker B_a, & \dim J_a &= 8a - 32 - v + t_{0,a}, \\ H_{B,a,d}^{(s)} &= [B_a (D_{a,d}^{(s)})^{-1} B_a^*]^{-1} = \overline{(\widehat{H}_{K,a,d}^{(s)})^{-1}}. \end{aligned} \quad (\text{NSO15})$$

The dimension $8a - 16$ of the original simple arithmetic kernel and its low intersection are the unchanged DNE/FGC data. Quotienting that kernel by the low intersection gives $t_{K,a}$. Since Z_a has that many independent columns and $A_{\ell,a}$ is invertible, B_a is onto; subtracting its rank from m_a proves the displayed dimension of J_a . In particular all displayed dimensions are their actual nonnegative ranks, with $0 \leq t_{0,a} \leq v$.

To verify the metric square rather than identify presentations, let H_Y be the full original weighted relation quotient Gram in its low monomials. DNE's exact dual pairing gives $D = A_\ell^* H_Y^{-1} A_\ell$. Substitute this into $BD^{-1}B^*$ to obtain $Z_a^\top \overline{H_Y} Z_a = \overline{Z_a^* H_Y Z_a} = \widehat{H}_K$. This proves the last equality of NSO15 with its conjugation. NSO11 gives its attained target metric, with kernel exactly J_a . Its bounds have the dual orientation of NSO14; $\widehat{H}_K$ has the reciprocal, primal orientation of NSO5. Multiplication of all original relation columns by $\alpha_A = v!/\mu_v$ leaves their span and the same minimum unchanged. Any scalar-bearing numerator coordinate map retains its literal common scalar on both source orders. A fixed coordinate change R always changes a Gram by $R^* H R$ and its determinant by $|\det R|^2$; that factor cancels only in the ratio on those same coordinates.

192.4 Determinants of the surviving native flags

Let X be any fixed rank- r restriction or quotient obtained above, with one fixed ordered value frame. Write $D_X^{(s)}(d)$ for its native dual form. Equations NSO10–14 prove

$$-r \log \mathfrak{b}_{s,d} \leq L_X^{(s)}(d) := \log \frac{\det D_X^{(s)}(d)}{\det D_X^{(1)}(d)} \leq -r \log \mathfrak{a}_s. \quad (\text{NSO16})$$

Conjugate $D_X^{(s)}$ by the inverse positive square root of $D_X^{(1)}$; all its eigenvalues lie between the two scalars in NSO14. Their logarithms sum to NSO16. The argument has no inverse-condition assumption on the actual frame. For any fixed number of copies, the ranks in NSO12 and NSO15, and every rank of their flags, are $O(k)$ at $a = k, j$. For $s = a$, $d = O(q_j)$, NSO6 therefore gives

$$|L_X^{(a)}(d)| \leq r \ell_a [\log(4\pi^2) + \log\{1 + 16(d + \ell_a)^2\}] = O(k^2 \log(q_j + 1)) = o(kq_k). \quad (\text{NSO17})$$

The last equality follows from $q_k = (k + 1)^2$ and $\log(k + 1)/k \rightarrow 0$ along the original class. The finite bound itself holds before taking that limit. It applies to the full m_a -dimensional native conductor quotient and to its invariant, arithmetic-kernel-dual and further residual flags. The primal arithmetic kernel of NSO15 has the inverse determinant ratio and the corresponding reversed interval. For an $O(q)$ -rank raw space the proved bound is instead $O(qk \log q)$; no sub- kq assertion for that space follows from this argument.

At the four original target indices put

$$\begin{aligned} d_0 &= q_a - v - 1, & d_1 &= q_a - v, & d_2 &= 2q_a - v - 1, & d_3 &= 2q_a - v, \\ \mathcal{R}f &= -f(d_0) - f(d_1) + f(d_2) + f(d_3). \end{aligned} \quad (\text{NSO18})$$

For the same fixed rank- r flag and frame at all four indices,

$$-r\{\omega_{s,d_2} + \omega_{s,d_3}\} \leq \mathcal{R}L_X^{(s)} \leq r\{\omega_{s,d_0} + \omega_{s,d_1}\}. \quad (\text{NSO19})$$

Indeed write $L_X^{(s)}(d) = -r \log \mathfrak{a}_s - u_d$, $0 \leq u_d \leq r\omega_{s,d}$, using NSO16. Substitution gives $u_{d_0} + u_{d_1} - u_{d_2} - u_{d_3}$. This proves both asymmetric endpoints, and shows exactly why the four copies of the full mass/product scalar cancel. The fixed-frame determinants cancel by the same four signs. For all preceding $O(k)$ -rank flags NSO19 is $o(kq_k)$. The analogous original primal return has the signs specified by the conjugate inverse in NSO15; this is the already established AMT42–46 comparison when applied to that original kernel.

192.5 The literal primitive source and connecting-image receiving

Use CIT's defining cutoffs and fixed Taylor complements

$$D_e = q_j - v - 1 + e, \quad r_e = 8k + 24 - 3e, \quad e = 0, 1. \quad (\text{NSO20})$$

Here and only here D_e is already a polynomial cutoff. Let T_0, V_0 be the degree- k native analytic and lower columns at D_e . In ordinary Taylor coordinates their columns are $\tilde{T}_k = U_k T_0$ and $\tilde{V}_k = U_k V_0$, independent of order. The primitive Taylor domain is all jets for $e = 0$, and for $e = 1$ is the kernel of the four exact rows $\Lambda(\mathcal{T}_{B_i} \mathcal{T}_{A\chi_j})$. These polynomial rows and their entire kernels are independent of the metric. Its analytic subspace is $\text{im}[\tilde{V}_k, \tilde{T}_k]$. CIT's first-minor complement therefore supplies the same P_e^{Taylor} for both orders, and

$$\begin{aligned} P_e^{(s_0)} &= U_{s_0, c'_k, D_e}^{-1} P_e^{\text{Taylor}}, \quad H_{\text{src}, e}^{(s_0)} = (P_e^{(s_0)})^* (I - P_{[V_0, T_0]}) P_e^{(s_0)}, \\ H_{\text{src}, e}^{(s_0)}[x] &= \min_{\xi, \eta} \|P_e^{(s_0)} x + T_0 \xi + V_0 \eta\|_2^2. \end{aligned} \quad (\text{NSO21})$$

The full source frame has rank r_e after the displayed minimum. At $e = 1$ the four constraints have rank four by the original independent multiplier moment columns, while $L_e = D_e + 1 = q_j - v + e$ and the analytic source rank is $q_k - v$. Thus the rank is $(q_j - v + e - 4e) - (q_k - v) = 8k + 24 - 3e$. At $e = 0$ the same computation has no constraint.

The multiplier matrix in ordinary Taylor coordinates is

$$\text{Mult}_{D_e+1}(B_i)[r, t] = \begin{cases} B_i^{(r-t)}(0)/(r-t)!, & r \geq t, \\ 0, & r < t. \end{cases} \quad (\text{NSO22})$$

It sends the common primitive columns to the same target Taylor tuple. With $B_{\text{cof}} = \text{stack}_i M_{i,k}$, CIT's exact coefficient equation is

$$\text{stack}_i \text{Mult}(B_i) P_e^{\text{Taylor}} = \text{diag}(\tilde{T}_j, 4) F_e + \text{diag}(\tilde{V}_j, 4) E_e^{\text{low}}. \quad (\text{NSO23})$$

Existence is the full primitive kernel condition just stated: at $e = 0$ the raw analytic target fills all jets, and at $e = 1$ its missing coordinate is the original $\Lambda(\mathcal{T}_{A\chi_j})$ in each copy. To prove uniqueness, subtract two equations. The resulting analytic numerator modulo the full lower evaluation is zero through D_e . The Vandermonde/division proof after NSO13 therefore forces every coefficient of their F_e difference to vanish, and then the lower Vandermonde kills the difference in E_e^{low} . Hence F_e is literally the same actual coefficient matrix in the two original source pairs.

Keep the complete target flags $\mathcal{B} = \text{im } B_{\text{cof}}$ and $\mathcal{S}_e = \text{im}[B_{\text{cof}}, F_e]$. Their quotient has rank r_e . For every original later $N \in [q_j - 1, 2q_j]$ and $d = N - v$,

$$H_{\text{conn}, e}^{(s_1)}(N)[x] = \min_{\xi} (F_e x + B_{\text{cof}} \xi)^* \text{diag}(D_{j,d}^{(s_1)}, 4) (F_e x + B_{\text{cof}} \xi). \quad (\text{NSO24})$$

The independence and exact kernel are CIT9–10: $\text{im } \mathbb{T} \cap \mathcal{L}_1 = \mathbb{T} \mathcal{L}_0$, so a primitive image is a boundary exactly when its primitive source lies in $\text{im}[V_0, T_0]$. Thus these are precisely the connecting-image value coordinates x . Equation NSO24 keeps all four lower gauges and the full boundary fibre; its least norm is attained by NSO10. At $N = q_j - 1 + e$ it is CIT11–13. For later N , only the already fixed coefficient columns are used, without assuming a common later primitive source extension.

It follows immediately from NSO9–10 and NSO14 that

$$\begin{aligned} -r_e \log \mathbf{b}_{k, D_e} &\leq L_{\text{src}, e} := \log \frac{\det H_{\text{src}, e}^{(k)}}{\det H_{\text{src}, e}^{(1)}} \leq -r_e \log \mathbf{a}_k, \\ -r_e \log \mathbf{b}_{j, N-v} &\leq L_{\text{conn}, e}(N) := \log \frac{\det H_{\text{conn}, e}^{(j)}(N)}{\det H_{\text{conn}, e}^{(1)}(N)} \leq -r_e \log \mathbf{a}_j. \end{aligned} \quad (\text{NSO25})$$

Both are $O(k^2 \log(q_j + 1)) = o(kq_k)$. The first explicitly receives PSR25 with a finite interval directly on its full primitive quotient. On PSR2–3’s original finite domain $k \geq \max(9, v - 1, \lceil (\pi/2)\sqrt{\delta^2 + \gamma^2} \rceil + 1)$, and for the same original admissible SGT14 radius $0 < a < 1$, the established PSR26 interval also remains valid; its intersection with NSO25 contains the same exact determinant. The second applies at each later target index, and its four-return obeys NSO19 with $a = j$. The raw frames used to implement these minima can have $O(q)$ columns, but the rank in NSO25 is exactly r_e .

For clarity the full coordinate factors in PSR23 survive this receiving. If R_e^{pol} is the original relation-polynomial coefficient frame and $C_e^{(s)} = B_s R_e^{\text{pol}}$, then

$$(C_e^{(s)})^\top P_e^{(s)} = (R_e^{\text{pol}})^\top \text{diag}(r!) P_e^{\text{Tay}}. \quad (\text{NSO26})$$

Multiply NSO8 to prove this equality. Thus the full actual Θ_e , its actual complement J_Z , and the factors $2 \log |\det \Theta_0|$ and $-2 \log |\det[W, J_Z]|$ are identical coefficient objects in the two orders. They have no assumed size. Their cancellation in a source-order ratio is explained by this literal map, including every factorial and phase.

192.6 The finite comparison of the actual compatibility losses

For each of the two pairs $p = (1, 1)$ and (k, j) retain CIT17’s exact quantities and finite constants, now denoted

$$\begin{aligned} \mathcal{X}_e^p &= \chi_{X_k, D_e}^p - \chi_{Y_e, D_e}^p, & A_e^p &= \log \det H_{\text{conn}, e}^p - \log \det H_{\text{src}, e}^p, \\ \epsilon_e^{p, -} &= -2r_e \log \max(1, K_-^p) - r_e \log c_+^p, & \epsilon_e^{p, +} &= 2r_e \log \max(1, K_+^p) - r_e \log c_-^p, \\ c_-^p &= \min(1, (L_-^p)^{-2}), & c_+^p &= \max(1, (L_+^p)^2), & A_e^p &= -\mathcal{X}_e^p + \epsilon_e^p, & \epsilon_e^{p, -} &\leq \epsilon_e^p \leq \epsilon_e^{p, +}. \end{aligned} \quad (\text{NSO27})$$

The $K_\pm^p$ and $L_\pm^p$ are precisely the original NBG/NKC multiplier, inverse and compatibility constants at D_e , with source centers c'_k, c'_j , shift $(\beta_{4,u,w} - 2)/i$, and both full Gamma masses. These $c_\pm^p$ are unrelated to the newly lettered scalar bounds $\mathbf{a}_s, \mathbf{b}_{s,d}$.

Here is the full passage from the native quotient to NSO27. Minimize the original multiplier inequalities on the entire primitive fibre in NSO21. Its correlated image has Gram between $(K_-^p)^{-2} H_{\text{src}}$ and $(K_+^p)^2 H_{\text{src}}$. The independent-gauge quotient has Gram H_{conn} by NSO24 and CIT16; the correlated quotient has Gram H_{cor} . The original shared-update identity PCJ13–16 states $-\log \det(H_{\text{conn}}/H_{\text{cor}}) = \mathcal{X}_e^p + \nu_e^p$, with $r_e \log c_-^p \leq \nu_e^p \leq r_e \log c_+^p$. Subtracting this equality from the correlated/source logarithm proves NSO27, with exactly the signs displayed. The proved original constants give $\epsilon_e^{p, \pm} = O_{\text{actual}}(k \log(q_k + 1))$. No comparison of unrelated source fibres is used.

Set $\Delta \mathcal{X}_e = \mathcal{X}_e^{(k,j)} - \mathcal{X}_e^{(1,1)}$. The complete finite conclusion is

$$\begin{aligned} r_e(\log \mathbf{a}_j - \log \mathbf{b}_{k, D_e}) + \epsilon_e^{(k,j), -} - \epsilon_e^{(1,1), +} &\leq \Delta \mathcal{X}_e \\ &\leq r_e(\log \mathbf{b}_{j, D_e} - \log \mathbf{a}_k) + \epsilon_e^{(k,j), +} - \epsilon_e^{(1,1), -}. \end{aligned} \quad (\text{NSO28})$$

Indeed NSO25 at $N = q_j - 1 + e$ gives

$$r_e(\log \mathbf{a}_k - \log \mathbf{b}_{j, D_e}) \leq A_e^{(k,j)} - A_e^{(1,1)} \leq r_e(\log \mathbf{b}_{k, D_e} - \log \mathbf{a}_j). \quad (\text{NSO29})$$

Subtract the two exact equalities in NSO27 and solve for $\Delta \mathcal{X}_e$. Its residual is exactly $\epsilon_e^{(k,j)} - \epsilon_e^{(1,1)}$. Combining the lower endpoint of its interval with the negative upper endpoint in NSO29, and conversely for the upper endpoint, proves NSO28 without an assumed limit for either individual compatibility quantity. Equations NSO6 and NSO27 now prove

$$\mathcal{X}_e^{(k,j)} - \mathcal{X}_e^{(1,1)} = O_{\text{actual}}(k^2 \log(q_j + 1)) = o(kq_k). \quad (\text{NSO30})$$

At $e = 0$, CIT’s ambient source has zero complement projector, so $\chi_{Y_0, D_0} = 0$ exactly and NSO30 applies to the difference of the two χ_{X_k, D_0} . At $e = 1$ it is the entire difference $(\chi_X - \chi_{Y_1})^{(k,j)} -$

$(\chi_X - \chi_{Y_1})^{(1,1)}$; all four polynomial constraints remain. The new receiving of PSR27 proves that its two original connecting-image determinants agree up to $o(kq_k)$, with finite bound NSO25, and consequently gives NSO30 for its retained loss combination.

Finally the literal CIT24–26 coefficient chain is

$$\mathbb{J} \subset W_{\text{inv}} = \mathbb{J} + \text{im}(B_{\text{cof}}J_0) \subset W = \mathbb{J} + \text{im } B_{\text{cof}} \subset \mathcal{S}_e + \mathbb{J}. \quad (\text{NSO31})$$

All its spaces are fixed in four copies of $\mathbb{C}^{m_j}$. Hence NSO14–19 apply separately, with their actual ranks, to W/W_{inv} (the full residual Q_b), to $W_{\text{inv}}/\mathbb{J}$ (the ITL target), to the intersection and sum quotients in CIT21, and to the corresponding source quotients inside $\mathbb{C}^{m_k}$. For example, if $[X_{\mathbb{J}}, I_R, R_b] = X_W C_W$, the exact factorization is $|\det C_W|^2 g_W = g_{\mathbb{J}} \det H_{\text{inv}} \det Q_b$. The matrix C_W and all quotient-value lifts are the same in both source orders by NSO8 and NSO23. Thus their fixed determinants are retained before taking either the ratio or the four-return. Applying NSO19 to each term controls its order-difference return by $o(kq_k)$; for a cross ratio, add the explicit signed intervals of its two fixed quotients. This receives the new estimate in the actual common boundary structure, including its full lower-gauge overlap. The common absolute invariant or connecting loss still has the exact CIT26 boundary identity. NSO28–31 calculate its change of source order and do not assign that common value.

193 Source-order intervals in the original connecting and invariant returns

This section receives the accepted NSO1–31 proof into the exact CIT28, NCR20, LCC36, CIT30/RQB and NFR6 formulas. It preserves their full source and target quotients. The scalar polynomial estimate remains the already proved AMT23,29–30 estimate, with the exact native dual square NSO7–15; it is not a new analytic bound. Keep the actual simple quartet, the proved five-orbit domain $|u| \geq R_*$, fixed logarithm branch, nonreduced conductor, all original coefficients and

$$\begin{aligned} k &= 4l + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad c'_a = a/2 - 4, \\ 0 \leq v \leq 80, \quad \mu_v &\neq 0, \quad m_a = 16a - 48 - v, \quad t_a = 8a - 16 - t_{0,a}, \quad j_{0,a} = 8a - 32 - v + t_{0,a}, \\ p &= 0 : (s_0, s_1) = (1, 1), \quad p = 1 : (s_0, s_1) = (k, j), \\ D_0 &= q_j - v - 1, \quad D_1 = q_j - v, \quad D_2 = 2q_j - v - 1, \quad D_3 = 2q_j - v, \\ \mathcal{R}f &= -f(D_0) - f(D_1) + f(D_2) + f(D_3). \end{aligned} \tag{NSR1}$$

Here $t_{0,a}$ is the actual dimension of the original kernel's intersection with the low polynomial space. A superscript p always means this complete original pair, and $\Delta Z = Z^1 - Z^0$. The D_i are polynomial cutoffs. They remain the same for source degree k and target degree j .

193.1 The finite native intervals and their exact value spaces

For $s = 4\ell_s + 1$ retain the full NSO constants

$$\begin{aligned} \rho_s &= \frac{(2\pi)^{2\ell_s}}{(1/2)^{2\ell_s}}, \quad \Pi_s = \prod_{h < \ell_s} (2h + \tfrac{1}{2})^2, \quad U_{s,D} = \prod_{h < \ell_s} \{4(D + h + 1)^2 + (2h + \tfrac{1}{2})^2\}, \\ \mathbf{a}_s &= \rho_s \Pi_s, \quad \mathbf{b}_{s,D} = \rho_s U_{s,D}, \quad \omega_{s,D} = \log(\mathbf{b}_{s,D}/\mathbf{a}_s), \\ \Omega_s^{\text{lo}} &= \omega_{s,D_0} + \omega_{s,D_1}, \quad \Omega_s^{\text{hi}} = \omega_{s,D_2} + \omega_{s,D_3}, \quad W_s = \Omega_s^{\text{lo}} + \Omega_s^{\text{hi}}. \end{aligned} \tag{NSR2}$$

All $s = 1$ products equal one and all its widths are zero. The underlying measures remain $dm_s = (2\pi)^{s/2} 2^{s/2} |\Gamma(s/4 + iy/2)|^2 dy / (4\pi \Gamma(s/2))$.

For a specified rank- r native dual restriction or quotient $G_X^{(s)}(D)$ at one fixed degree and center, put

$$L_X(D) = \log \frac{\det G_X^{(s)}(D)}{\det G_X^{(1)}(D)}. \quad -r \log \mathbf{b}_{s,D} \leq L_X(D) \leq -r \log \mathbf{a}_s, \quad -r \Omega_s^{\text{hi}} \leq \mathcal{R}L_X \leq r \Omega_s^{\text{lo}}. \tag{NSR3}$$

These are exactly NSO16 and NSO19, at the explicitly specified D_i . To recall their precise map, let $S'^n = \sum_h (\mathbf{B}_{s,c,D})_{hn} \phi_{h,s,c}$ with $\phi_{h,s,c} = p_{h,s}((S' - c)/i) / \sqrt{(2\pi)^{s/2} h! (s/2)_h}$. The derivative values of a complex-linear functional are $\mathbf{B}_s^T w_s$, and its Taylor values are $U_s w_s$, $U_s = \text{diag}(1/n!) \mathbf{B}_s^T$. Consequently

$$w_s = (\mathbf{B}_s^T)^{-1} \mathbf{B}_1^T w_1, \quad G_s^\vee = \overline{H_s^{-1}}, \quad \mathbf{b}_{s,D}^{-1} G_1^\vee \preceq G_s^\vee \preceq \mathbf{a}_s^{-1} G_1^\vee. \tag{NSR4}$$

The same map carries every numerator and every lower evaluation. On a fixed fibre $Xx + Zz$ the full minimum is $X^*GX - X^*GZ(Z^*GZ)^{-1}Z^*GX$ in the value x . Applying NSR4 to all members of that fibre preserves its two constants. Every restriction, subsequent quotient and fixed direct sum therefore has the same inequality at its final rank. Relative eigenvalues prove the first interval in NSR3. Writing $L_X = -r \log \mathbf{a}_s - u_D$ with $0 \leq u_D \leq r \omega_{s,D}$ gives $\mathcal{R}L_X = u_{D_0} + u_{D_1} - u_{D_2} - u_{D_3}$ and proves the second interval with its directed signs. This recalls the full established transport, without deleting any raw gauge.

All scalar and frame factors in NSR3 remain until this same value space is identified. In particular a fixed change R of quotient lifts changes the Gram by R^*GR , giving $|\det R|^2$ in its

determinant; the factor cancels only in the proved same-frame order ratio or in $\mathcal{R}$. NSO6 gives, for $s = k, j$ and every D_i here, $\log \mathfrak{b}_{s,D_i} = O(k \log(q_j + 1))$. Thus all ensuing rank- $O(k)$ intervals are $o(kq_k)$. This statement does not give that conclusion on an $O(q_k)$ -rank raw polynomial plane.

193.2 The two low source and image determinants at their actual ranks

For $e = 0, 1$ put $D_e = q_j - v - 1 + e$ and $r_e = 8k + 24 - 3e$. Retain the actual CIT primitive Taylor complement P_e^{Taylor} . In the two source orders $P_e^{(s_0)} = U_{s_0, c'_k, D_e}^{-1} P_e^{\text{Taylor}}$. The source norm minimizes over all columns $[V_0, T_0]$. The target coefficient F_e is uniquely determined by

$$\text{stack}_i \text{Mult}(B_i) P_e^{\text{Taylor}} = \text{diag}(\tilde{T}_j, 4) F_e + \text{diag}(\tilde{V}_j, 4) E_e^{\text{low}}. \quad (\text{NSR5})$$

The matrix $\text{Mult}(B_i)[n, h]$ is $B_i^{(n-h)}(0)/(n-h)!$ for $n \geq h$, and zero otherwise. The two Taylor columns on the right are the same actual analytic germs and lower exponentials in both orders. If two representations differ, multiplication by E_A makes the numerator vanish through $D_e + v \geq q_j - 1$; the full original upper Vandermonde kills its coefficient class, and the lower Vandermonde then kills the lower difference. This proves uniqueness of F_e . Existence is the original primitive condition: all jets are allowed at $e = 0$, while at $e = 1$ the four rows $\Lambda(\mathcal{T}_{B_i} \mathcal{T}_A \chi_j)$ impose exactly the missing target coordinate in each copy. Their rank four gives $(D_e + 1 - 4e) - (q_k - v) = r_e$. The positive connecting image has this same rank by CIT9's full kernel $\text{im } \mathbb{T} \cap \mathcal{L}_1 = \mathbb{T} \mathcal{L}_0$.

Write $L_{\text{src},e} = \Delta \log \det H_{\text{src},e}$ and $L_{\text{conn},e} = \Delta \log \det H_{\text{conn},e}$ at this defining cutoff. Define the literal four endpoints

$$\begin{aligned} s_e^- &= -r_e \log \mathfrak{b}_{k,D_e}, & s_e^+ &= -r_e \log \mathfrak{a}_k, & t_e^- &= -r_e \log \mathfrak{b}_{j,D_e}, & t_e^+ &= -r_e \log \mathfrak{a}_j, \\ s_e^- &\leq L_{\text{src},e} \leq s_e^+, & t_e^- &\leq L_{\text{conn},e} \leq t_e^+. \end{aligned} \quad (\text{NSR6})$$

The source fibre is $P_e x + T_0 \xi + V_0 \eta$; the image fibre first minimizes every target lower evaluation, then $F_e x + B_{\text{cof}} \xi$. Equation NSR4 proves both intervals on these complete fibres. In particular the number multiplying the logarithm is r_e , not the number of raw columns used to implement the minimum.

Retain the complete original CIT17 errors, with no new constant substituted:

$$\begin{aligned} \mathcal{X}_e^p &= \chi_{X_k, D_e}^p - \chi_{Y_e, D_e}^p, & A_e^p &= \log \det H_{\text{conn},e}^p - \log \det H_{\text{src},e}^p = -\mathcal{X}_e^p + \epsilon_e^p, \\ \epsilon_e^{p,-} &= -2r_e \log \max(1, K_-^p) - r_e \log c_+^p, & \epsilon_e^{p,+} &= 2r_e \log \max(1, K_+^p) - r_e \log c_-^p, \\ c_-^p &= \min(1, (L_-^p)^{-2}), & c_+^p &= \max(1, (L_+^p)^2), & b_e^- &= \epsilon_e^{1,-} - \epsilon_e^{0,+}, & b_e^+ &= \epsilon_e^{1,+} - \epsilon_e^{0,-}. \end{aligned} \quad (\text{NSR7})$$

The $K_{\pm}, L_{\pm}$ are the original NGB/NKC constants, including their degree- $k + 8$ comparison, all coefficients, centers and full measures. They give precisely the CIT17/PCJ16 interval for ϵ_e^p .

CIT28, NCR20 and the second line of LCC36 now have the following full finite receiving:

$$\begin{aligned} L_{\text{conn},e} &= -\Delta \mathcal{X}_e + \zeta_e, & \zeta_e &= L_{\text{src},e} + \epsilon_e^1 - \epsilon_e^0, \\ s_e^- + b_e^- &\leq \zeta_e \leq s_e^+ + b_e^+, \\ s_e^- + b_e^- - t_e^+ &\leq \Delta \mathcal{X}_e \leq s_e^+ + b_e^+ - t_e^-. \end{aligned} \quad (\text{NSR8})$$

Subtracting the two exact equalities in NSR7 proves the first line. Add the source and error intervals to get the second. Solve the first line for $\Delta \mathcal{X}_e$ and use both actual target endpoints in NSR6 to obtain the third. Thus both low determinant ratios and this complete compatibility difference are $o(kq_k)$ with the displayed finite bounds. At $e = 0$ only, $\chi_{Y_0} = 0$ because its source complement is zero. The four constraints and χ_{Y_1} remain at $e = 1$.

The earlier PSR26 finite source interval also remains valid on its own domain

$$k \geq \max(9, v - 1, \lceil (\pi/2) \sqrt{\delta^2 + \gamma^2} \rceil + 1)$$

and its original admissible SGT14 radius $0 < a < 1$. On that domain replace $[s_e^-, s_e^+]$ in NSR8 by its intersection with PSR26's full centered interval, retaining D_e^{ord} and both PSR errors. Both

intervals contain the same source ratio; their intersection does too. No PSR domain is imposed on NSR6 itself. The factors in the source determinant are unchanged since

$$(C_e^{(s)})^\top P_e^{(s)} = (R_e^{\text{pol}})^\top \text{diag}(n!) P_e^{\text{Tay}}. \quad (\text{NSR9})$$

This follows by canceling the exact $\mathbf{B}_s^\top$ with its inverse from U_s^{-1} . It proves equality of the full Θ_e and the actual complement/frame factors in PSR23 before any order comparison.

193.3 Exact connecting and full-cohomology contraction return

Set $\mathcal{V} = \mathbb{C}^{m_k}$, $\mathcal{U} = \mathbb{C}^{4m_j}$, and keep

$$\begin{aligned} B &= B_{\text{cof}}, \quad \mathbb{B} = \text{diag}(B_j, 4), \quad \mathbb{J} = \ker \mathbb{B}, \quad \Phi = \mathbb{B}B, \\ \mathcal{B} &= \text{im } B, \quad \mathcal{S}_e = \text{im}[B, F_e], \quad W = \mathbb{J} + \mathcal{B}, \quad K_e = \mathcal{S}_e \cap W, \quad T_e = \mathcal{S}_e + W, \\ c_e &= \text{rank}[\Phi, \mathbb{B}F_e] - \text{rank } \Phi, \quad h_e = r_e - c_e. \end{aligned} \quad (\text{NSR10})$$

Here B is injective on the whole original conductor coefficient source, as proved by CNC/DFX, so $\dim \mathcal{B} = m_k$ and $\dim \mathcal{S}_e = m_k + r_e$. The map

$$\mathcal{S}_e \longrightarrow \text{im}[\Phi, \mathbb{B}F_e] / \text{im } \Phi, \quad z \longmapsto [\mathbb{B}z] \quad (\text{NSR11})$$

is onto and its kernel is exactly K_e : its value is zero precisely when $z - Bx \in \ker \mathbb{B} = \mathbb{J}$ for some x . Thus $\dim(\mathcal{S}_e/K_e) = c_e$ and $\dim(K_e/\mathcal{B}) = h_e$. The identical representative map identifies $\mathcal{S}_e/K_e$ with T_e/W . Both are the exact quotient-value spaces in CIT21. In particular $0 \leq c_e \leq r_e$.

At every later cutoff the common ambient form is $\mathcal{D}_D^{(s_1)} = \text{diag}(D_{j,D}^{(s_1)}, 4)$. Let $G_{e,S}^p, G_{e,T}^p$ be its two full quotient metrics on $\mathcal{S}_e/K_e$ and T_e/W , in the same ordered values under NSR11. Then

$$\omega_{\text{conn},e}^p(D) = \frac{\det G_{e,T}^p(D)}{\det G_{e,S}^p(D)}, \quad |\Delta \log \omega_{\text{conn},e}(D)| \leq c_e \omega_{j,D}, \quad |\Delta \mathcal{R} \log \omega_{\text{conn},e}| \leq c_e W_j. \quad (\text{NSR12})$$

To prove the middle inequality, each quotient in NSR11 has rank c_e and the constants in NSR3. Subtract its two endpoint log intervals. The common term $-c_e \log \mathbf{a}_j$ cancels, leaving $[-c_e \omega_{j,D}, c_e \omega_{j,D}]$. For the last inequality apply the original signs $(-, -, +, +)$ to these four symmetric intervals. The entire lower gauges and the two different larger fibres K_e, W are retained in this argument. There is no supposition that their minimum representatives or angles coincide. This is an explicit finite bound for the common-return order difference in CIT21/NCR14.

The full low cohomology has the fixed cycle space $\mathcal{Z}_e \supset \mathcal{S}_e$, with $\dim(\mathcal{Z}_e/\mathcal{S}_e) = \kappa_e \leq 3 \max(v - e, 0)$. Its boundary remains $\mathcal{B}$. On the same native form, the source-order log ratio of its full cohomology Gram has rank $r_e + \kappa_e$, and that of its remaining quotient by the connecting image has rank κ_e :

$$\begin{aligned} -(r_e + \kappa_e) \log \mathbf{b}_{j,D} &\leq \Delta \log \det \text{Gram}(\mathcal{Z}_e/\mathcal{B}) \leq -(r_e + \kappa_e) \log \mathbf{a}_j, \\ -\kappa_e \Omega_j^{\text{hi}} &\leq \Delta \mathcal{R} \log \det \text{Gram}(\mathcal{Z}_e/\mathcal{S}_e) \leq \kappa_e \Omega_j^{\text{lo}}. \end{aligned} \quad (\text{NSR13})$$

The exact connecting inclusion identifies its cohomology quotient with $\ker U_e$ by LCC13; these are the attained metrics of that same flag. Completing a fixed frame of $\mathcal{S}_e/\mathcal{B}$ gives the full determinant as its restriction determinant times this last quotient determinant and the fixed adapted-frame factor. Thus NSR13 retains the whole quotient in the last paragraph of LCC36, rather than erasing it.

There is also a rank-exact comparison of the contractions. Put $K_e^{\text{full}} = \mathcal{Z}_e \cap W$ and $T_e^{\text{full}} = \mathcal{Z}_e + W$. The representative map gives

$$0 \longrightarrow K_e^{\text{full}}/K_e \longrightarrow \mathcal{Z}_e/\mathcal{S}_e \longrightarrow T_e^{\text{full}}/T_e \longrightarrow 0, \quad d_K + d_T = \kappa_e. \quad (\text{NSR14})$$

The kernel proof is $z \in \mathcal{Z}_e$ maps into T_e exactly when $z - s \in W$ for some $s \in \mathcal{S}_e$; hence its kernel modulo $\mathcal{S}_e$ is $(\mathcal{S}_e + K_e^{\text{full}})/\mathcal{S}_e \simeq K_e^{\text{full}}/K_e$. The adapted determinant identities for the two cross

ratios then express $\log \omega_e - \log \omega_{\text{conn},e}$ as the sum of the T_e^{full}/T_e and K_e^{full}/K_e log determinants, minus that of $\mathcal{Z}_e/\mathcal{S}_e$, with their fixed frame constant. Since $d_T + d_K = \kappa_e$, NSR3 gives

$$\begin{aligned} |\Delta(\log \omega_e - \log \omega_{\text{conn},e})(D)| &\leq \kappa_e \omega_{j,D}, \\ |\Delta \mathcal{R}(\log \omega_e - \log \omega_{\text{conn},e})| &\leq \kappa_e W_j, \quad |\Delta \mathcal{R} \log \omega_e| \leq (c_e + d_T)W_j \leq (c_e + \kappa_e)W_j. \end{aligned} \quad (\text{NSR15})$$

For the first bound subtract the three intervals in NSR3: their common scalar has coefficient $d_T + d_K - \kappa_e = 0$, and both remaining one-sided widths are $\kappa_e \omega_{j,D}$. Taking the four signs proves the second. The full quotient map $\mathcal{Z}_e/(\mathcal{Z}_e \cap W) \rightarrow (\mathcal{Z}_e + W)/W$ has rank $c_e + d_T$ by the exact sum inclusion in NSR14. Applying the same two-minimum proof as NSR12 at that actual rank gives the third inequality; $d_T \leq \kappa_e$ gives its last bound. These new order intervals coexist with NCR19's $|\mathcal{R} \log \omega_e^p - \mathcal{R} \log \omega_{\text{conn},e}^p| \leq 2\kappa_e \log A_{j,s_1}^{\text{end}}$ for each original pair. In particular the middle upper bound in NSR15 can be replaced by the minimum of its displayed value and $2\kappa_e(\log A_{j,j}^{\text{end}} + \log A_{j,1}^{\text{end}})$. Both concern that same actual scalar difference.

The sharper tail ranks in the same source-order returns. Retain all NSR1–15 objects on their original five-orbit domain, including both complete native pairs, their fixed coefficient frames and the displaced polynomial cutoffs D_0, D_1, D_2, D_3 . The widths are exactly those of NSR2:

$$\begin{aligned} \omega_{j,D} &= \log(\mathfrak{b}_{j,D}/\mathfrak{a}_j), \quad \Omega_j^{\text{lo}} = \omega_{j,D_0} + \omega_{j,D_1}, \\ \Omega_j^{\text{hi}} &= \omega_{j,D_2} + \omega_{j,D_3}, \quad W_j = \sum_{a=0}^3 \omega_{j,D_a} = \Omega_j^{\text{lo}} + \Omega_j^{\text{hi}}. \end{aligned}$$

The complete BTK25–26 calculation gives $v \leq 32$ and $\kappa_e \leq \bar{\kappa}_e := 3 \max(v - e - 2, 0)$. In NSR14, its first quotient also has the exact presentation

$$\frac{\mathcal{Z}_e \cap \mathbb{J}}{\mathcal{S}_e \cap \mathbb{J}} \xrightarrow{\sim} K_e^{\text{full}}/K_e, \quad [j] \mapsto [j].$$

An element of K_e^{full} is $b + j$ with $b \in \mathcal{B} \subset \mathcal{S}_e$ and $j \in \mathcal{Z}_e \cap \mathbb{J}$; this proves surjectivity. Its kernel is exactly $\mathcal{S}_e \cap \mathbb{J}$, proving the isomorphism. The following arrow of NSR14 is the identity $[z] \mapsto [z]$ onto $(\mathcal{Z}_e + W)/(\mathcal{S}_e + W)$. It is onto since the terms of W disappear in the target; its kernel is precisely the first quotient, because $z = s + b + j$ implies $[z] = [j]$. Consequently $d_K + d_T = \kappa_e$. The full quotient-value map is

$$\Psi_e : \mathcal{Z}_e/(\mathcal{Z}_e \cap W) \longrightarrow \mathbb{B}\mathcal{Z}_e/\text{im } \Phi, \quad [z] \longmapsto [\mathbb{B}z].$$

It is onto by the displayed target, and its kernel is zero: $\mathbb{B}z = \Phi x$ means $z - Bx \in \mathbb{J}$, hence $z \in \mathcal{Z}_e \cap W$. Thus its rank is $\dim((\mathcal{Z}_e + W)/W) = c_e + d_T$, by the sum-quotient sequence with subspace $(\mathcal{S}_e + W)/W$ of dimension c_e . With the unchanged $\Delta Z = Z^1 - Z^0$ and $\mathcal{R}f = -f(D_0) - f(D_1) + f(D_2) + f(D_3)$, the sharpened finite intervals are

$$\begin{aligned} 0 &\leq d_T \leq \kappa_e \leq \bar{\kappa}_e, \\ |\Delta \mathcal{R} \log \omega_e| &\leq (c_e + d_T)W_j \leq (c_e + \bar{\kappa}_e)W_j, \\ |\Delta \mathcal{R}(\log \omega_e - \log \omega_{\text{conn},e})| &\leq \kappa_e W_j \leq \bar{\kappa}_e W_j, \\ -\bar{\kappa}_e \Omega_j^{\text{hi}} &\leq \Delta \mathcal{R} \log \det \text{Gram}(\mathcal{Z}_e/\mathcal{S}_e) \leq \bar{\kappa}_e \Omega_j^{\text{lo}}. \end{aligned} \quad (\text{RKR3})$$

To verify the constants and signs, for any one of the fixed quotients X of rank r put $L_X(D) = \Delta \log \det \text{Gram}(X)$. The full native form inequality NSR4 and its same affine minima give $L_X(D) = -r \log \mathfrak{a}_j - u_X(D)$ with $0 \leq u_X(D) \leq r \omega_{j,D}$. The complete contraction of rank $c_e + d_T$ has two such quotient forms in the same value coordinates; subtracting their intervals cancels the common scalar and gives the pointwise width $(c_e + d_T)\omega_{j,D}$. The adapted determinant identity for the difference between the full and connecting contractions uses the quotients of ranks d_K, d_T, κ_e with

signs $(+, +, -)$. Its same-frame order difference is $-u_K - u_T + u_{\mathcal{X}/\mathcal{S}}$, since $d_K + d_T - \kappa_e = 0$. It lies in $[-\kappa_e \omega_{j,D}, \kappa_e \omega_{j,D}]$. Applying the four original signs proves the two symmetric intervals in RKR3. For the single quotient of rank κ_e , those signs give $u(D_0) + u(D_1) - u(D_2) - u(D_3)$, hence the directed interval $[-\kappa_e \Omega_j^{\text{hi}}, \kappa_e \Omega_j^{\text{lo}}]$. Every width is nonnegative. Substituting BTK25 therefore proves the last inequalities without changing their directions.

The middle interval may also be intersected with the original two-family bound from CIT23. Its absolute value is at most

$$\min \left\{ \bar{\kappa}_e W_j, 2\bar{\kappa}_e (\log A_{j,j}^{\text{end}} + \log A_{j,1}^{\text{end}}) \right\}.$$

Both entries bound the same scalar in the same frames, by the triangle inequality applied to the two original family returns. For $v - e \leq 2$, the actual spaces satisfy $\mathcal{Z}_e = \mathcal{S}_e$, so $d_K = d_T = 0$ and $\omega_e^p(D) = \omega_{\text{conn},e}^p(D)$ at every retained cutoff, $p = 0, 1$. The tail terms in RKR3 are then zero. The first NSR13 interval retains its exact rank $r_e + \kappa_e$; the present calculation inserts no value for the remaining connecting rank c_e or its absolute loss.

193.4 The complete boundary ranks, metrics and positive returns

Write $J = \text{im } J_0 = \ker B_k$ and retain the full original coefficient kernels

$$\begin{aligned} \mathcal{L} &= \ker \Phi, \quad \ell = \dim \mathcal{L}, \quad E = \ker(\Phi J_0), \quad \epsilon = \dim E, \quad Y = J + \mathcal{L}, \\ r_I &= j_{0,k} - \epsilon, \quad r_b = t_k - \ell + \epsilon, \quad r_o = 4t_j - m_k + \ell, \quad r_I + r_b + r_o = 4t_j. \end{aligned} \tag{NSR16}$$

No symbol ℓ here replaces the source-order index ℓ_s in NSR2. Since J_0 is injective, $J \cap \mathcal{L} = J_0 E$ and $\dim Y = j_{0,k} + \ell - \epsilon$. Hence $\text{rank } \Phi = m_k - \ell$, $\text{rank } \Phi J_0 = r_I$, and its image quotient has dimension r_b . Completing its full image in $\mathbb{C}^{4t_j}$ gives r_o and the last equality. All three ranks are nonnegative actual ranks, with both original low intersections and every hidden coefficient kernel present.

Keep the original full lift frames Z_R of the invariant source modulo E , E_b of $\mathcal{V}/Y$, and U_o completing the image frame

$$U_R = \Phi J_0 Z_R, \quad U_b = \Phi E_b, \quad C_T = [U_R, U_b, U_o] \text{ invertible.} \tag{NSR17}$$

The target form is four copies of the original $H_{B,j,D}^{(s_1)} = [B_j(D_{j,D}^{(s_1)})^{-1} B_j^*]^{-1}$. Its restriction on U_R is H_{inv} , its next quotient in lifts U_b is Q_b^E , and its quotient modulo $\text{im } \Phi$ in lifts U_o is Q_o . Their ranks are r_I, r_b, r_o . Before any return,

$$|\det C_T|^2 (\det H_{B,j,D}^{(s_1)})^4 = \det H_{\text{inv}}^p(D) \det Q_b^{E,p}(D) \det Q_o^p(D). \tag{NSR18}$$

Successively minimize the first and then the second coordinate blocks in this fixed frame to prove the identity. In original conductor coordinates the same fibres are $\mathbb{J} + \text{im}(B J_0)$ and $\mathbb{J} + \text{im } B$, because $\ker \mathbb{B} = \mathbb{J}$. Thus all lower and invariant gauges are included. The map $\mathcal{V}/Y \rightarrow W/W_{\text{inv}}$ is $[x] \mapsto [Bx]$; its full zero fibre is $x - J_0 z \in \mathcal{L}$. This also proves the original RQB quotient identification used in CIT29.

Let $\mathcal{V}_I^p = \mathcal{R} \log \det H_{\text{inv}}^p$, $\mathcal{V}_b^p = \mathcal{R} \log \det Q_b^{E,p}$ and $\mathcal{V}_o^p = \mathcal{R} \log \det Q_o^p$. Then the complete original budget and its order receiving are

$$\begin{aligned} \mathcal{V}_I^p + \mathcal{V}_b^p + \mathcal{V}_o^p &= 4\widehat{F}_j^{(s_1)}, \quad \mathcal{V}_I^p, \mathcal{V}_b^p, \mathcal{V}_o^p \geq 0, \\ -r_X \Omega_j^{\text{hi}} &\leq \Delta \mathcal{V}_X \leq r_X \Omega_j^{\text{lo}} \quad (X = I, b, o), \\ -4t_j \Omega_j^{\text{hi}} &\leq 4(\widehat{F}_j^{(j)} - \widehat{F}_j^{(1)}) \leq 4t_j \Omega_j^{\text{lo}}, \\ \Delta \mathcal{V}_I + \Delta \mathcal{V}_b + \Delta \mathcal{V}_o &= 4(\widehat{F}_j^{(j)} - \widehat{F}_j^{(1)}). \end{aligned} \tag{NSR19}$$

Here $(r_X) = (r_I, r_b, r_o)$. The budget is RQB27–28: the conjugate inverse $H_{B,j,D} = \overline{\widehat{H}_{K,j,D}^{-1}}$ changes the primal sign exactly, and the fixed frame factor in NSR18 cancels under $\mathcal{R}$. Increasing original

Gamma row cutoffs increases each native form and preserves that order under its complete fibre minimum; this proves positivity of all returns. NSR3 now proves each order interval at its actual rank. The last equality retains their exact correlation.

For the earlier CIT lifts, $R_b^{\text{CIT}} = R_b^E C_b + Y_{\text{inv}} A_b$ with C_b invertible and fixed. Hence $Q_b^{\text{CIT}} = C_b^* Q_b^E C_b$ by the same minimum; all finite determinant identities retain $|\det C_b|^2$. If $[X_{\mathbb{J}}, I_R, R_b^{\text{CIT}}] = X_W C_W$, then

$$|\det C_W|^2 g_W = g_{\mathbb{J}} \det H_{\text{inv}} |\det C_b|^2 \det Q_b^E. \quad (\text{NSR20})$$

This is the original CIT25/RQB31 equality. Its two fixed coordinate determinants cancel in the order ratios and in $\mathcal{R}$, because NSR5 and NSR17 retain the same ordered value coordinates.

The residual source denominator remains the full quotient $Q_0 = D_{k,D}^{(s_0)}/Y$ in lifts E_b , of rank r_b . Equivalently it is the original kernel-dual quotient by $B_k \mathcal{L}$, whose rank is $\ell - \epsilon$. With the fixed original frame $C_0 = [Z_{\Lambda}, B_k E_b]$ it satisfies

$$\begin{aligned} \det Q_0(D) &= \frac{|\det C_0|^2 \det H_{B,k,D}}{\det(Z_{\Lambda}^* H_{B,k,D} Z_{\Lambda})}, \\ -r_b \Omega_k^{\text{hi}} &\leq \Delta \mathcal{R} \log \det Q_0 \leq r_b \Omega_k^{\text{lo}}, \\ -r_b(\Omega_j^{\text{hi}} + \Omega_k^{\text{lo}}) &\leq \Delta \mathcal{R} \log \frac{\det Q_b^E}{\det Q_0} \leq r_b(\Omega_j^{\text{lo}} + \Omega_k^{\text{hi}}). \end{aligned} \quad (\text{NSR21})$$

To prove the first equality, minimize first over $J = \ker B_k$ and then over $B_k \mathcal{L}$; the complete fibre is exactly Y . The independent frame C_0 gives its Schur determinant. NSR3 proves the second, and subtraction of the two directed rank- r_b return intervals proves the third. No inverse bound for the residual multiplier has entered this calculation.

193.5 The common contraction in CIT30 with all boundary factors

The original ITL loss is $\omega_{\text{inv}}^p(D) = \det H_{\text{inv}}^p(D) / \det A_{\text{inv}}^p(D)$, where $A_{\text{inv}} = F^* \mathcal{D}_D F / E$, $F = B J_0$ and the quotient values are Z_R . Its two attained target-fibre forms have the same rank r_I . Therefore, by the proof of NSR12,

$$|\Delta \log \omega_{\text{inv}}(D)| \leq r_I \omega_{j,D}, \quad |\Delta \mathcal{R} \log \omega_{\text{inv}}| \leq r_I W_j. \quad (\text{NSR22})$$

The source fibre here is $F Z_R x + F E$ and the target fibre is $F Z_R x + \mathbb{J}$; $F E = \text{im } F \cap \mathbb{J}$ by its complete coefficient kernel. Thus this compares the same original two minima, not the independent source-order metric Q_{inv} used below.

Put exactly as in CIT30

$$\begin{aligned} \mathcal{P}_e^p &= \mathcal{R}(\log g_{T_e} + \log g_{K_e} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} - \log \det A_{\text{inv}})^p, \\ \mathcal{L}_e^p &= \mathcal{R}(\log \omega_{\text{conn},e} + \log \omega_{\text{inv}})^p, \quad C_e^{\text{ord}} = (c_e + r_I) W_j. \end{aligned} \quad (\text{NSR23})$$

Substitution of NSR20 in the full CIT26 cross ratio proves

$$\begin{aligned} \mathcal{L}_e^p &= \mathcal{P}_e^p - \mathcal{V}_b^p, \quad \mathcal{P}_e^p - 4\widehat{F}_j^{(s_1)} + \mathcal{V}_I^p \leq \mathcal{L}_e^p \leq \mathcal{P}_e^p, \\ -C_e^{\text{ord}} &\leq \Delta \mathcal{L}_e \leq C_e^{\text{ord}}, \\ -C_e^{\text{ord}} - r_b \Omega_j^{\text{hi}} &\leq \Delta \mathcal{P}_e \leq C_e^{\text{ord}} + r_b \Omega_j^{\text{lo}}, \\ \max\{\mathcal{P}_e^1 - 4\widehat{F}_j^{(j)} + \mathcal{V}_I^1, \mathcal{P}_e^0 - \mathcal{V}_b^0 - C_e^{\text{ord}}\} &\leq \mathcal{L}_e^1 \\ &\leq \min\{\mathcal{P}_e^1, \mathcal{P}_e^0 - \mathcal{V}_b^0 + C_e^{\text{ord}}\}. \end{aligned} \quad (\text{NSR24})$$

The first directed bound is the unchanged RQB32 inequality for the negative residual factor: use $0 \leq \mathcal{V}_b^p \leq 4\widehat{F}_j^{(s_1)} - \mathcal{V}_I^p$. The second line adds NSR12 and NSR22, with exact ranks c_e, r_I . Since $\Delta \mathcal{P}_e = \Delta \mathcal{L}_e + \Delta \mathcal{V}_b$, NSR19 proves the third line with its asymmetric widths. The last two lines intersect the original pair-one bound with $\mathcal{L}_e^0 + [-C_e^{\text{ord}}, C_e^{\text{ord}}]$ and substitute the exact pair-zero residual identity. Both intervals contain the same $\mathcal{L}_e^1$, so their intersection does also. The equality $\mathcal{V}_b + \mathcal{V}_I + \mathcal{V}_o = 4\widehat{F}_j$ remains exact throughout; no missing quotient is assigned zero. Every displayed order interval is $o(kq_k)$, while neither common value in the first line is evaluated.

193.6 Both source-order intervals in the matched invariant return

Keep NFR1's full original source quotient

$$\begin{aligned} Q_{\text{inv}}^p(D) &= (J_0^* D_{k,D}^{(s_0)} J_0)/E \quad \text{in the same lifts } Z_R, \quad \text{rank } Q_{\text{inv}} = r_I, \\ \mathcal{R}_X^p(D) &= (Q_{\text{inv}}^p(D))^{-1/2} H_{\text{inv}}^p(D) (Q_{\text{inv}}^p(D))^{-1/2}, \\ V_Q^p &= \mathcal{R} \log \det Q_{\text{inv}}^p, \quad \mathcal{M}^p = \mathcal{R} \log \det \mathcal{R}_X^p. \end{aligned} \tag{NSR25}$$

The kernel $E = \ker(\Phi J_0)$ and complement Z_R are independent of the measures. Hence these are the actual NFR/ITL matched quotient-value spaces. At every endpoint $\det H_{\text{inv}} = \det Q_{\text{inv}} \det \mathcal{R}_X$, even though its inverse square roots depend on the source metric. This proves the exact matrix-to-determinant map.

The complete NFR6 return and the new finite order interval are

$$\begin{aligned} \mathcal{M}^p &= 4\widehat{F}_j^{(s_1)} - \mathcal{V}_b^p - \mathcal{V}_o^p - V_Q^p, \quad \mathcal{M}^p + V_Q^p = \mathcal{V}_I^p \geq 0, \\ -r_I \Omega_k^{\text{hi}} &\leq \Delta V_Q \leq r_I \Omega_k^{\text{lo}}, \\ \Delta \mathcal{M} &= 4(\widehat{F}_j^{(j)} - \widehat{F}_j^{(1)}) - \Delta \mathcal{V}_b - \Delta \mathcal{V}_o - \Delta V_Q, \\ \Lambda^- &:= -r_I(\Omega_j^{\text{hi}} + \Omega_k^{\text{lo}}) \leq \Delta \mathcal{M} \leq r_I(\Omega_j^{\text{lo}} + \Omega_k^{\text{hi}}) =: \Lambda^+. \end{aligned} \tag{NSR26}$$

The first equality substitutes NSR18–19 in the exact matched determinant. The second follows from NSR3 on the rank- r_I source quotient. For the last interval use $\Delta \mathcal{M} = \Delta \mathcal{V}_I - \Delta V_Q$, then subtract the directed intervals for these same two rank- r_I values. The preceding line preserves its full target-budget realization with all three target factors. Using the actual invariant rank before estimating avoids assigning the much larger compatibility-plane rank to this matched quotient.

In NSR26, the source degree is still k , with center c'_k , at the displayed D_i based on q_j . Its displacement from its degree- k standard endpoints is $b = q_j - q_k = 8k + 24$ at both lows and $2b$ at both highs. The constants ω_{k,D_i} in NSR2 use exactly those displaced cutoffs. No window displacement is silently absorbed into $\Lambda^\pm$.

Let $U_{j,1}^{\text{native}}, U_{j,j}^{\text{native}}$ be the complete RQB35 caps, with their original positive observation sum, full FEX source correction and all finite-domain restrictions of those providers. Define

$$U_{\mathcal{M}}^p = 4U_{j,s_1}^{\text{native}} - \mathcal{V}_b^p - \mathcal{V}_o^p - V_Q^p. \tag{NSR27}$$

Both original source-order intervals are retained explicitly:

$$\begin{aligned} -V_Q^0 &\leq \mathcal{M}^0 \leq U_{\mathcal{M}}^0, \quad -V_Q^1 \leq \mathcal{M}^1 \leq U_{\mathcal{M}}^1, \\ \max\{-V_Q^1, -V_Q^0 + \Lambda^-\} &\leq \mathcal{M}^1 \leq \min\{U_{\mathcal{M}}^1, U_{\mathcal{M}}^0 + \Lambda^+\}, \\ \max\{-V_Q^0, -V_Q^1 - \Lambda^+\} &\leq \mathcal{M}^0 \leq \min\{U_{\mathcal{M}}^0, U_{\mathcal{M}}^1 - \Lambda^-\}. \end{aligned} \tag{NSR28}$$

For each pair, nonnegativity of $\mathcal{V}_I$ gives its lower bound, and $\widehat{F}_j^{(s_1)} \leq U_{j,s_1}^{\text{native}}$ in the first line of NSR26 gives the upper bound with the full missing return retained. Adding Λ^-, Λ^+ to the pair-zero interval and intersecting with the pair-one interval proves the second line. Reversing the difference and intersecting in the other direction proves the third. In particular this refines the displayed NFR6 upper bound by keeping its exact nonnegative $\mathcal{V}_o$, while preserving Q_{inv} and $\mathcal{V}_b$ on both sides.

The preexisting NFR5 error also has an exact receiving. Let $\mathcal{C}^p = \mathcal{R} \log(\det \mathcal{H}_V^p / \det \mathcal{H}_E^p)$, with the full original compatibility and source-complement forms, and let $\mathcal{A}^p = \mathcal{R} \log \omega_{\text{inv}}^p$. At each cutoff NFR2's constants, denoted here by a_i^p, b_i^p , retain their actual rank r_I :

$$\begin{aligned} a_i^p &= 2r_I \log \max(1, K_-^p(D_i)) + r_I \log c_+^p(D_i), \\ b_i^p &= 2r_I \log \max(1, K_+^p(D_i)) - r_I \log c_-^p(D_i), \\ E_p^- &= -b_0^p - b_1^p - a_2^p - a_3^p, \quad E_p^+ = a_0^p + a_1^p + b_2^p + b_3^p, \\ \mathcal{M}^p &= \mathcal{C}^p + \mathcal{A}^p + \mathcal{E}^p, \quad E_p^- \leq \mathcal{E}^p \leq E_p^+, \\ \Lambda^- - r_I W_j - E_1^+ + E_0^- &\leq \Delta \mathcal{C} \leq \Lambda^+ + r_I W_j - E_1^- + E_0^+. \end{aligned} \tag{NSR29}$$

The first four lines are NFR2/5 with the two low errors negated and the two high errors unnegated. The exact difference is $\Delta\mathcal{C} = \Delta\mathcal{M} - \Delta\mathcal{A} - \Delta\mathcal{E}$. Insert NSR22, NSR26 and these two error intervals to prove the last line. Thus the full shared compatibility ratio also receives a finite $o(kq_k)$ order-difference bound, without replacing either individual large compatibility determinant or either original source-complement projection.

193.7 The existing arithmetic allocation comparison and its exact map

For the original degree- a standard window only, denote its cutoffs by $d_i^{[a]} = q_a - v - 1, q_a - v, 2q_a - v - 1, 2q_a - v$; write $\Omega_a^{[a],\text{lo/hi}}$ for NSR2 evaluated there. Keep HJR16–22's original inverse-metric decomposition with $D = D_{a,d}^{(s)}$, $J = J_a$ and $B = B_a$:

$$D^{-1} = J(J^*DJ)^{-1}J^* + L(BD^{-1}B^*)L^*, \quad L = D^{-1}B^*(BD^{-1}B^*)^{-1}. \quad (\text{NSR30})$$

Indeed $[J, L]$ is a full coefficient frame with $BJ = 0$, $BL = I$, and block Gram

$$\text{diag}(J^*DJ, (BD^{-1}B^*)^{-1}).$$

Inverting that exact frame Gram proves NSR30. The complete original new row h gives

$$\begin{aligned} x &= hJ(J^*DJ)^{-1}J^*h^*, \\ y &= hL(BD^{-1}B^*)L^*h^*. \end{aligned}$$

The original old lower Schur denominator remains inside h . The row returns are $\log(1+x)$ and $\log(1+y/(1+x))$, with the original endpoint weights one at the two extremes and two inside. Their exact sums are

$$\begin{aligned} S_K^{(s)} &= \mathcal{R}_a \log \det H_{B,a,d}^{(s)} = \widehat{F}_a^{(s)}, \quad S_R^{(s)} = \mathcal{R}_a \log \det (J_a^* D_{a,d}^{(s)} J_a) = F_{0,a,s}, \\ S_K^{(s)} + S_R^{(s)} &= \mathcal{T}_{a,s}, \\ -t_a \Omega_a^{[a],\text{hi}} &\leq S_K^{(a)} - S_K^{(1)} \leq t_a \Omega_a^{[a],\text{lo}}, \\ -j_{0,a} \Omega_a^{[a],\text{hi}} &\leq S_R^{(a)} - S_R^{(1)} \leq j_{0,a} \Omega_a^{[a],\text{lo}}. \end{aligned} \quad (\text{NSR31})$$

This is the same native restriction/quotient transport NSO15–19 applied to the exact preexisting HJR maps. All zero rows remain in their aggregate. It gives no rowwise or averaged allocation value.

To connect these actual sums to the earlier AMT arithmetic objects, retain the original FEX/RQB33 identity and its complete low correction:

$$\begin{aligned} S_K^{(s)} &= F_K^{(s)} + \delta_{K,a,s}, \quad \delta_{K,a,s} = \sum_{i=0}^3 \sigma_i e_{N_i,s} - F_{L_a}^{(s)} + F_{(K_a+L_a)/K_a}^{(s)}, \quad \sigma = (1, 1, -1, -1), \\ S_R^{(s)} &= \mathcal{T}_{a,s} - F_K^{(s)} - \delta_{K,a,s}, \\ S_K^{(a)} - S_K^{(1)} &= -\Delta_K^{\Gamma,[a]} + \delta_{K,a,a} - \delta_{K,a,1}, \\ S_R^{(a)} - S_R^{(1)} &= \mathcal{T}_{a,a} - \mathcal{T}_{a,1} + \Delta_K^{\Gamma,[a]} - \delta_{K,a,a} + \delta_{K,a,1}. \end{aligned} \quad (\text{NSR32})$$

Here $N_i = d_i^{[a]} + v$ and the original low space is $L_a = \mathbb{C}[S]_{<v}$. The scalar $e_{N_i,s}$ is the full original conductor source Jacobian; its exact finite bounds remain RQB34/FEX34. The existing AMT44 convention is $\Delta_K^{\Gamma,[a]} = F_K^{(1)} - F_K^{(a)}$. Its proved interval is

$$-(8a-16)\Omega_{\Gamma,a}^{\text{low}} \leq \Delta_K^{\Gamma,[a]} \leq (8a-16)\Omega_{\Gamma,a}^{\text{high}}, \quad \Delta_K^{\Gamma} + \Delta_B^{\Gamma} = \mathcal{H}_a, \quad (\text{NSR33})$$

where $\Omega_{\Gamma,a}^{\text{low}} = \omega_{a,q_a-1} + \omega_{a,q_a}$ and $\Omega_{\Gamma,a}^{\text{high}} = \omega_{a,2q_a-1} + \omega_{a,2q_a}$ use the original primal, unshifted cutoffs of AMT43. For the exact underlying map, AMT7–9 minimizes the unchanged original polynomial

remainder map, restricts to K_a and takes the original observation quotient; AMT42 compares their two full original source Grams in those same values. Their adapted determinant gives the retained sum with $\mathcal{H}_a$. Subtraction of these already proved equalities is precisely NSR32. Thus the original primal rank $8a - 16$, the native rank $t_a = 8a - 16 - t_{0,a}$, the residual rank $j_{0,a}$, and every low correction stay in the typed comparison. No new analytic bound or new equality of those distinct source presentations is claimed.

All receiving formulas above apply their finite intervals before limits. The surviving order differences are $o(kq_k)$ on the fixed actual programme, while CIT30, RQB28 and NFR6 continue to retain the common long connecting/ITL loss, the positive residual and missing returns, and the original S_K, S_R allocation.

The finite analytic comparison in the same native coefficient form. At an original actual degree a , keep $D = N - v$, the original centre c'_a , the complete lower-evaluation minimum and both source orders $1, a$. Use the explicitly guarded endpoints $\lambda_{a,D}^\pm$ of EFP4–11. These intersect the NSO constants with QOT and, only where both EOR parity tests hold, the EOR constants on the same original form:

$$e^{\lambda_{a,D}^-} D_{a,D}^{(1)} \preceq D_{a,D}^{(a)} \preceq e^{\lambda_{a,D}^+} D_{a,D}^{(1)}. \quad (\text{EFR1})$$

The polynomial degree in the QOT/EOR entries is exactly $N - v - q'_a$. Taking the maximum of two scalar lower bounds and minimum of their upper bounds proves the intersection. Every candidate vector in a fixed affine fibre satisfies it; minimization over that same fibre proves its transfer to the whole lower, relation, kernel and quotient minima, as in the explicit EFP13 Schur formula. The ordinary-transpose Taylor change and the conjugated inverse in NSO/NSR remain unchanged. On a fixed rank- r restriction or attained quotient, the determinant difference lies in $[r\lambda^-, r\lambda^+]$. At the four original endpoints it lies in $[r\Phi_a^-, r\Phi_a^+]$, with $\Phi_a^\pm$ given in EFP15 using signs $(-, -, +, +)$. The four identical $-r \log \kappa_a$ contributions cancel only in that return; no source mass is dropped at an endpoint. Where QOT's original guard fails, EFP11 retains the full NSO comparison. The defining primitive source uses EFP's separately proved relation restriction and Θ_e map, with degree $24k - 25 - v + e$.

194 The connecting image inside the full remaining target quotient

This calculation uses the complete original CIT, RQB and NSR objects. The fixed hypothetical simple quartet, five-orbit period, logarithm branch, conductor E_A , order $v = \text{ord}_0 E_A$, nonzero moment μ_v and scalar $\alpha_A = v!/\mu_v$ remain those of the original construction. Put $k = 4l + 1 \geq 9$, $j = k + 4$, $q_a = (a + 1)^2$, $q'_a = (a - 7)^2$, $m_a = 16a - 48 - v$, $t_a = 8a - 16 - t_{0,a}$ and $j_{0,a} = 8a - 32 - v + t_{0,a}$. Use the original rank-valid domain, including $|u| \geq R_*$. The original source pairs are $p = 0 : (s_0, s_1) = (1, 1)$ and $p = 1 : (s_0, s_1) = (k, j)$. Every form below has its actual pair p ; we suppress this index only in identities valid separately for both.

At each original target cutoff $q_j - 1 \leq N \leq 2q_j$, write $D = N - v$. The native coefficient form is

$$\mathcal{D}_N = \text{diag}(D_{j,D}^{(s_1)}, 4), \quad \mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j). \quad (\text{CPQ1})$$

Here $D_{j,D}$ is the full analytic-numerator form after minimizing over all original lower evaluations, as in DNE and CIT19. Its rows use $M_s = (2\pi)^{s/2}$ and $h_{n,s} = M_s n! (s/2)_n$, the original centres, division moments and phases. Nothing in this calculation changes those rows or replaces their lower Gram. Fixed low-cycle coefficient flags are used at all these later cutoffs.

194.1 All coefficient maps and their kernels

Retain

$$\begin{aligned} \mathcal{U} &= \mathbb{C}^{4m_j}, \quad \mathcal{V} = \mathbb{C}^{m_k}, \quad B = B_{\text{cof}} : \mathcal{V} \longrightarrow \mathcal{U}, \\ \mathbb{B} &= \text{diag}(B_j, 4) : \mathcal{U} \longrightarrow \mathbb{C}^{4t_j}, \quad \mathbb{J} = \ker \mathbb{B}, \quad \Phi = \mathbb{B}B, \\ \mathcal{B} &= \text{im } B, \quad \mathcal{S}_e = \text{im}[B, F_e], \quad W = \mathbb{J} + \mathcal{B}, \quad K_e = \mathcal{S}_e \cap W, \quad T_e = \mathcal{S}_e + W. \end{aligned} \quad (\text{CPQ2})$$

B is the original injective coefficient multiplier and B_j is onto with its full original invariant kernel. The columns F_e , for $e = 0, 1$, are the actual CIT7 connecting columns, determined at $N = q_j - 1 + e$ by the complete Taylor equation with independent target lower corrections. They give an isomorphism $\mathbb{C}^{r_e} \rightarrow \mathcal{S}_e/\mathcal{B}$, $x \mapsto [F_e x]$, with $r_e = 8k + 24 - 3e$. Their independence and equality in the two source orders are proved in CIT9 and NSR5.

Use the original full target frame

$$C_T = [U_R, U_b, U_o], \quad U_R = \Phi J_0 Z_R, \quad U_b = \Phi E_b, \quad \text{im}[U_R, U_b] = \text{im } \Phi. \quad (\text{CPQ3})$$

It is the same invertible RQB27 frame. Its block dimensions are $r_I = j_{0,k} - \epsilon$, $r_b = t_k - \ell + \epsilon$ and $r_o = 4t_j - m_k + \ell$, where $\ell = \dim \ker \Phi$ and $\epsilon = \dim \ker(\Phi J_0)$. Define the actual quotient-coordinate map and its connecting matrix by

$$\pi_o = [0 \ 0 \ I_{r_o}] C_T^{-1} \mathbb{B}, \quad E_e = \pi_o F_e : \mathbb{C}^{r_e} \longrightarrow \mathbb{C}^{r_o}. \quad (\text{CPQ4})$$

The map π_o is onto and $\ker \pi_o = W$. Indeed its value is zero exactly when $\mathbb{B}z \in \text{im } \Phi$, equivalently $\mathbb{B}(z - Bx) = 0$ for some $x \in \mathcal{V}$. Its onto property uses the onto map $\mathbb{B}$ and the full last block of C_T . Consequently the maps

$$\mathcal{U}/W \longrightarrow \mathbb{C}^{r_o}, \quad [z] \mapsto \pi_o z, \quad \mathcal{S}_e/K_e \longrightarrow \text{im } E_e, \quad [z] \mapsto \pi_o z \quad (\text{CPQ5})$$

are isomorphisms; the latter identifies T_e/W with the same image. The exact sequence in the original connecting coordinates is

$$0 \longrightarrow \ker E_e \longrightarrow \mathbb{C}^{r_e} \xrightarrow{E_e} \text{im } E_e \longrightarrow 0, \quad \ker E_e \simeq K_e/\mathcal{B}, \quad c_e = \text{rank } E_e, \quad h_e = r_e - c_e. \quad (\text{CPQ6})$$

For its asserted kernel isomorphism, $F_e x \in W$ means that $F_e x - Bb \in \mathbb{J}$ for some b . Its class lies in $K_e/\mathcal{B}$, and every such class has a unique original connecting coordinate x . This proves

both injectivity and onto without removing any conductor or invariant kernel. In particular $c_e = \text{rank}[\Phi, \mathbb{B}F_e] - \text{rank} \Phi$, and $0 \leq c_e \leq \min(r_e, r_o)$.

Choose fixed independent columns Z_h spanning $\ker E_e$ and fixed complement columns Z_c such that $C_P = [Z_h, Z_c]$ is invertible. Set $A_e = E_e Z_c$ and choose R_e^o so that $C_o = [A_e, R_e^o]$ is an invertible r_o -square matrix. These choices concern only the actual fixed coefficient maps. They are made once and used at every N and both orders. Their determinants are retained below. Empty blocks have their ordinary zero-dimensional meanings and determinant one.

194.2 The two original source and target minima

Let $H_e(N) = H_{\text{conn},e}(N)$ be CIT18's full connecting Gram on $\mathcal{S}_e/\mathcal{B}$ in F_e coordinates. It is the minimum of $(F_e x + Bb)^* \mathcal{D}_N(F_e x + Bb)$ over all b . The source kernel restriction and source quotient are exactly

$$\begin{aligned} G_{h,e} &= Z_h^* H_e Z_h, \\ G_{s,e} &= Z_c^* H_e Z_c - Z_c^* H_e Z_h (Z_h^* H_e Z_h)^{-1} Z_h^* H_e Z_c, \\ x^* G_{s,e} x &= \min_{z,b} (F_e Z_c x + F_e Z_h z + Bb)^* \mathcal{D}_N(F_e Z_c x + F_e Z_h z + Bb), \\ |\det C_P|^2 \det H_e &= \det G_{h,e} \det G_{s,e}. \end{aligned} \tag{CPQ7}$$

The kernel fibre in the displayed minimum is exactly K_e by CPQ6. Thus $G_{s,e}$ is the original metric on $\mathcal{S}_e/K_e$ in the values $[F_e Z_c x]$. Block elimination in $C_P^* H_e C_P$ proves its formula and the full determinant identity. Positive definiteness follows from independence in these quotient values.

Write $Q_o(N)$ for the original RQB remaining quotient on $\mathbb{C}^{4t_j}/\text{im } \Phi$, in the lifts U_o . The map CPQ5 is an isometry from $\mathcal{U}/W$ with its attained $\mathcal{D}_N$ metric to this form. To prove it, first minimize over $\mathbb{J}$, giving exactly $\text{diag}(H_{B,j,D}, 4)$ on $\mathbb{B}z$, where $H_{B,j,D} = (B_j D_{j,D}^{-1} B_j^*)^{-1}$. Then minimize over $\text{im } \Phi$. The combined fibre is W ; this is precisely the definition of Q_o in the frame C_T .

It follows that the original connecting target metric and its remaining target quotient are

$$\begin{aligned} G_{t,e} &= A_e^* Q_o A_e, \\ Q_{\text{rem},e} &= (R_e^o)^* Q_o R_e^o - (R_e^o)^* Q_o A_e (A_e^* Q_o A_e)^{-1} A_e^* Q_o R_e^o, \\ x^* G_{t,e} x &= \min_{w \in W} (F_e Z_c x + w)^* \mathcal{D}_N(F_e Z_c x + w), \\ |\det C_o|^2 \det Q_o &= \det G_{t,e} \det Q_{\text{rem},e}, \quad \omega_{\text{conn},e} = \det G_{t,e} / \det G_{s,e}. \end{aligned} \tag{CPQ8}$$

For the target restriction, every representative of a value in T_e/W lies in T_e because $W \subset T_e$. Thus the ambient minimum over W is its intrinsic quotient minimum. Substitution of $\pi_o F_e Z_c = A_e$ proves the first and third lines. The second line is the complete minimum over the connecting image inside Q_o . Block elimination in $C_o^* Q_o C_o$ proves the determinant identity. The last equality uses the identical value coordinates on both sides of the original CIT21 contraction. Its inverse and frame factors have therefore been established, rather than deduced from equality of dimensions. The new quotient has rank $r_o - c_e$ and represents exactly $\mathcal{U}/T_e$.

194.3 Positive returns and the formerly signed remainder

For the forms in CPQ7–8 set

$$V_h = \mathcal{R} \log \det G_{h,e}, \quad V_s = \mathcal{R} \log \det G_{s,e}, \quad V_t = \mathcal{R} \log \det G_{t,e}, \quad V_{\text{rem}} = \mathcal{R} \log \det Q_{\text{rem},e}. \tag{CPQ9}$$

Keep the original $V_I = \mathcal{R} \log \det H_{\text{inv}}$, $V_b = \mathcal{R} \log \det Q_b^E$, $V_o = \mathcal{R} \log \det Q_o$ and $V_A = \mathcal{R} \log \det A_{\text{inv}}$, with the full invariant source fibre $FZ_R x + FE$, $F = BJ_0$ and $E = \ker(\Phi J_0)$. In particular A_{inv} uses the target order s_1 and is not replaced by the independent source form Q_{inv} of NSR25.

All eight returns $V_h, V_s, V_t, V_{\text{rem}}, V_I, V_b, V_o, V_A$ are nonnegative. Indeed the original row extension increases $\mathcal{D}_N$. Restriction preserves this order. Minimizing two ordered forms over the same

full affine fibre preserves it too. Every displayed form is such a restriction or quotient on a fixed flag. Its successive log-determinant increments are nonnegative, and $\mathcal{R}$ is their sum with weights one at both ends and two at each interior row. This proves the assertion for these particular forms, including their complete gauges.

Write $V_{\text{conn}} = \mathcal{R} \log \det H_e$. CPQ7–8 and the original RQB target identity give

$$\begin{aligned} V_{\text{conn}} &= V_h + V_s, & V_o &= V_t + V_{\text{rem}}, \\ 4\widehat{F}_j^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, \\ \mathcal{R} \log \omega_{\text{conn},e} &= V_t - V_s, & \mathcal{L}_e &:= \mathcal{R} \log(\omega_{\text{conn},e} \omega_{\text{inv}}) = V_I + V_t - V_s - V_A. \end{aligned} \quad (\text{CPQ10})$$

All fixed frame determinants in CPQ7–8 cancel only here, since $\mathcal{R}1 = 0$. No varying minimum representative is treated as a fixed frame. Substitution into CIT30's original identity $\mathcal{L}_e = \mathcal{P}_e^{\text{rest}} - V_b$ now evaluates its entire signed Gram combination as

$$\boxed{\mathcal{P}_e^{\text{rest}} = 4\widehat{F}_j^{(s_1)} - V_{\text{rem}} - V_s - V_A, \quad \mathcal{L}_e = 4\widehat{F}_j^{(s_1)} - V_b - V_{\text{rem}} - V_s - V_A.} \quad (\text{CPQ11})$$

This equality retains both new proper quotients. In particular neither Q_o nor the source quotient has been discarded. It expresses the original signed combination by the complete target return and three specified nonnegative returns.

For completeness the fixed factors before taking the return can also be made explicit. Use the original CIT constants $c_{\text{conn},e}$ and C_W , and the fixed change $Q_b^{\text{CIT}} = C_b^* Q_b^E C_b$. The original expression $P(N) = \log g_{T_e} + \log g_{K_e} - \log g_{\mathcal{S}_e} - \log g_{\mathbb{J}} - \log \det A_{\text{inv}}$ satisfies

$$\begin{aligned} P(N) &= 4 \log \det H_{B,j,D} - \log \det Q_{\text{rem},e} - \log \det G_{s,e} - \log \det A_{\text{inv}} + \log \mathfrak{c}_e, \\ \mathfrak{c}_e &= \frac{|\det C_T|^2 |\det C_o|^2 |\det C_b|^2}{c_{\text{conn},e} |\det C_W|^2}. \end{aligned} \quad (\text{CPQ12})$$

To verify every factor, CIT26 gives $P = \log \omega_{\text{conn}} + \log \omega_{\text{inv}} + \log \det Q_b^{\text{CIT}} - \log c_{\text{conn},e} - 2 \log |\det C_W|$. Substitute CPQ8 and $|\det C_T|^2 (\det H_{B,j,D})^4 = \det H_{\text{inv}} \det Q_b^E \det Q_o$. Then substitute $\det G_{t,e} = |\det C_o|^2 \det Q_o / \det Q_{\text{rem},e}$. These substitutions give CPQ12 with the displayed positive constant. Its return is zero, which proves CPQ11 in the original absolute frames as well.

The following directed intervals follow from these exact nonnegative quantities:

$$\begin{aligned} V_I - V_s - V_A &\leq \mathcal{L}_e \leq 4\widehat{F}_j^{(s_1)} - V_b - V_s - V_A, \\ V_I + V_b - V_s - V_A &\leq \mathcal{P}_e^{\text{rest}} \leq 4\widehat{F}_j^{(s_1)} - V_s - V_A, \\ V_I - V_{\text{conn}} - V_A &\leq \mathcal{L}_e \leq 4U_{j,s_1}^{\text{native}} - V_b - V_s - V_A. \end{aligned} \quad (\text{CPQ13})$$

The width of each of the first two intervals is exactly V_o ; the location of its actual value is fixed by the retained V_{rem} . The first follows from $0 \leq V_t \leq V_o = 4\widehat{F}_j - V_I - V_b$ in CPQ10. The second adds V_b . For the third use $V_s \leq V_{\text{conn}}$ on the lower side and the original RQB35 cap, on its proved domain, on the upper. These are finite bounds on the original scalar. They do not assign an uncomputed asymptotic value to a positive return.

194.4 Actual row formulas for each new factor

The new factors can be evaluated by the original arithmetic rows. Here is a full common formula specifying their minima. Let Y be any of the fixed coefficient subspaces and $K \subset Y$ the kernel in one of the following pairs:

$$(Y, K) = (K_e, \mathcal{B}), (\mathcal{S}_e, K_e), (T_e, W), (\mathcal{U}, T_e), (\text{im } F, FE). \quad (\text{CPQ14})$$

They represent respectively $G_h, G_s, G_t, Q_{\text{rem}}, A_{\text{inv}}$. Choose independent coefficient columns X_K for K and lifts X_Q for the specified quotient values. The lifts are respectively $F_e Z_h, F_e Z_c, F_e Z_c$,

any fixed lifts of $U_o R_e^o$ under $\mathbb{B}$, and FZ_R . In the fourth pair the coordinate map to $\mathcal{U}/T_e$ is the last quotient in CPQ8.

Let $\mathcal{V}_N$ be the full original four-copy lower-evaluation matrix through D , and $\mathcal{T}_N$ the four-copy original analytic-numerator matrix through D . Form the raw matrices

$$Z_{K,N} = [\mathcal{V}_N, \mathcal{T}_N X_K], \quad Z_{Y,N} = [Z_{K,N}, \mathcal{T}_N X_Q], \quad G_K^{\text{raw}} = Z_{K,N}^* Z_{K,N}. \quad (\text{CPQ15})$$

Their columns are independent: the full lower frame is independent, and the DNE coefficient Gram is positive after its minimum; the remaining fixed coefficient columns are independent. Thus all the following inverses exist, including at the first cutoff, with the empty-block convention. Sequential elimination gives exactly the quotient form $Q = \text{Gram}_{\mathcal{D}_N}(Y/K)$ in these lifts and $\det Q = \det(Z_{Y,N}^* Z_{Y,N}) / \det G_K^{\text{raw}}$.

At the next original degree $n = D + 1$, append the four rows

$$a = [\text{diag}(v_n, 4), \text{diag}(t_n, 4)X_K], \quad b = \text{diag}(t_n, 4)X_Q, \quad C = (G_K^{\text{raw}})^{-1} Z_{K,N}^* \mathcal{T}_N X_Q, \quad R = b - aC. \quad (\text{CPQ16})$$

These are literally CIT19's v_n, t_n , with the complete arithmetic recursion

$$\begin{aligned} f_{a,m} &= \frac{\nu_{a,m+v} - \sum_{d=v+1}^{m+v} \binom{m+v}{d} \mu_d f_{a,m+v-d}}{\binom{m+v}{v} \mu_v}, \quad \nu_{a,n} = \sum_{u,w} (Z_{\text{exp},j})_{uw,a} \beta_{j,u,w}^n, \\ b_{0,m} &= \mathbf{1}_{m=0}, \quad b_{-1,m} = 0, \quad b_{n+1,m} = i^{-1} (b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2) b_{n-1,m}, \\ (t_n)_a &= h_{n,s_1}^{-1/2} \sum_m b_{n,m} f_{a,m}, \quad (v_n)_{u,w} = h_{n,s_1}^{-1/2} \sum_m b_{n,m} \beta_{j-8,u,w}^m. \end{aligned} \quad (\text{CPQ17})$$

The symbol $b_{n,m}$ in CPQ17 is the original polynomial coefficient, distinct from the new-row block b in CPQ16. Arrays outside their original polynomial range are zero. The divisor μ_v and the full old lower and coefficient Gram occur explicitly. The row formula remains valid for every allowed zero row and for both prescribed source orders.

Put $L = a(G_K^{\text{raw}})^{-1} a^*$. The exact next quotient is

$$Q^+ = Q + R^* (I_4 + L)^{-1} R, \quad \gamma_{Y/K,n} = \log \det(I_4 + L + RQ^{-1} R^*) - \log \det(I_4 + L) \geq 0. \quad (\text{CPQ18})$$

To prove the first equality, subtract the old attained minimum and change the old kernel coefficient by η . The remaining minimum is $\eta^* G_K^{\text{raw}} \eta + \|Rx - a\eta\|^2$. Solving its positive normal equations gives the residual $x^* R^* (I_4 + L)^{-1} Rx$. The determinant lemma gives the second equality. For its sign, conjugate by $(I_4 + L)^{-1/2}$; the added form is positive. Thus each of the five returns in CPQ14 is the sum of these particular γ 's, with endpoint weights one and all interior weights two. This supplies the original row calculation for all terms of CPQ11, including the previously unspecified remainder, without a new row model.

194.5 Back substitution into both source orders and the allocation

The ranks of $(G_h, G_s, G_t, Q_{\text{rem}}, A_{\text{inv}})$ are respectively $(h_e, c_e, c_e, r_o - c_e, r_I)$. NSO's full native same-fibre comparison therefore gives, for each of their returns V_X , the finite interval

$$-\text{rank}(X) \Omega_j^{\text{hi}} \leq V_X^{p=1} - V_X^{p=0} \leq \text{rank}(X) \Omega_j^{\text{lo}}. \quad (\text{CPQ19})$$

The cutoffs inside Ω_j are exactly those in CPQ1 and NSR2. The same coefficient frames occur in both orders by CPQ4 and NSR5. Their determinants therefore cancel in these ratios. Adding the two correlated decompositions gives precisely $\Delta V_o = \Delta V_t + \Delta V_{\text{rem}}$ and $\Delta V_{\text{conn}} = \Delta V_h + \Delta V_s$, preserving the full earlier NSR intervals. All ranks are $O(k)$ on the fixed actual programme, so the new individual order differences are $O_{\text{actual}}(k^2 \log q_j) = o(kq_k)$.

Finally the original degree- j allocation obeys $S_{K,j}^{(s_1)} = \widehat{F}_j^{(s_1)}$ and $S_{R,j}^{(s_1)} = \mathcal{T}_{j,s_1} - \widehat{F}_j^{(s_1)}$ by the exact HJR/NSR31 inverse-metric calculation. Hence CPQ10 gives the complete positive refinement of that kernel allocation:

$$\begin{aligned} 4S_{K,j}^{(s_1)} &= V_I + V_b + V_t + V_{\text{rem}}, & 4S_{R,j}^{(s_1)} &= 4\mathcal{T}_{j,s_1} - V_I - V_b - V_t - V_{\text{rem}}, \\ 4S_{K,j}^{(s_1)} &= \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A. \end{aligned} \quad (\text{CPQ20})$$

The target degree is $j = k + 4$ in this identity. The independent degree- k source quotient Q_{inv} and the proper-source correction in NFR retain their own centre, displaced cutoffs and source order. In particular replacing V_A by $\mathcal{R} \log \det Q_{\text{inv}}$ is not part of CPQ20. The exact relation is $V_A = \mathcal{R} \log \det Q_{\text{inv}} + \mathcal{R} \log(\det A_{\text{inv}} / \det Q_{\text{inv}})$; both forms use the same lifts Z_R , so this ratio is the determinant of their actual relative positive operator. ITL/NFR retains its full proper-source comparison. CPQ20 thus carries the new quotient refinement into the original allocation while preserving that remaining source comparison and every positive term which must enter a quantitative estimate.

Finite equilibrium comparison for all five attained quotients. Retain CPQ2–8’s fixed maps and frames in both actual source pairs, and evaluate EFP11–15 at the original target degree $j = k + 4$ and $D_i = q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v$. For the five forms in CPQ14, whose ranks are exactly $(h_e, c_e, c_e, r_o - c_e, r_I)$, the improved finite receiver is

$$\text{rank}(X)\Phi_j^- \leq V_X^{(k,j)} - V_X^{(1,1)} \leq \text{rank}(X)\Phi_j^+, \quad X = G_h, G_s, G_t, Q_{\text{rem}}, A_{\text{inv}}. \quad (\text{EFR2})$$

To verify its types, the value maps in order are $K_e/\mathcal{B}$, S/K_e , T_e/W , $\mathcal{U}/T_e$ and $\text{im}(BJ_0)/(BJ_0E)$, in the unchanged lifts $F_e Z_h, F_e Z_c, F_e Z_o, U_o R_e^o$ (lifted under $\mathbb{B}$), and $BJ_0 Z_R$. Each is the full original affine minimum EFP13. Thus EFP14 and the four signed additions EFP16 prove EFR2. The identities $V_{\text{conn}} = V_h + V_s$ and $V_o = V_t + V_{\text{rem}}$ remain equalities of these same quantities. The determinants $|\det C_P|^2, |\det C_o|^2, |\det C_T|^2$, the full CPQ12 $\mathfrak{c}_e$, and the original lower-Gram inverse in CPQ18 remain in EFP21–22 before their exact return cancellation. The independent degree- k Q_{inv} is related to A_{inv} by the unchanged relative determinant in CPQ20 and NFR7.

The full original quotient metrics at the common tail base. The new simultaneous formula in STF8–12 evaluates these same CPQ affine value fibres after all preceding native rows have been retained. The exact pairs are (W, W_i) , $(\mathcal{U}, T)$, (S, K_c) and (I, K_i) , with the original value coordinates and lift factors preserved as in STF14. NFI10 transports each attained matrix through the full original analytic frame A and relation matrix X_t , including every cross term. The Schur minimum is the classical operation of Boyd and Vandenberghe, Appendix A.5.5, p. 650 [H23]; STF9–12 proves its exact complex Hermitian specialization on these fibres. The eight signed terms are a labelled tuple, so opposite labels are retained even when their subspaces coincide. Substitution in the original CPQ identity gives STF18 with both allocations and all four complements. A fixed lift determinant cancels only in its own same-frame endpoint ratio; no absolute frame factor or proper-source comparison is set to one.

The complete controlled prefix inside this mixed decomposition. For $0 \leq R < q_j$ retain these same quotient frames and define $\mathcal{R}_{\leq R} f = 2f(q_j + R) - f(q_j) - f(q_j - 1)$ and $\mathcal{R}_{> R} f = f(2q_j - 1) + f(2q_j) - 2f(q_j + R)$. They sum to the original four-return and separately annihilate every fixed frame constant. Consequently the original complete determinant factorization gives, for either operator Z ,

$$4S_K^Z = \mathcal{L}_e^Z + V_b^Z + V_{\text{rem}}^Z + V_s^Z + V_A^Z, \quad S_K^Z + S_R^Z = \mathcal{T}^Z. \quad (\text{IUP4})$$

These are the same original quotient maps: superscript Z specifies only which weighted increments are summed. NIB and IPR prove the finite bounds, with $\mathfrak{E}_{s,R} = \sum_{r=0}^R \vartheta_r E_{j,s,r}$,

$$\begin{aligned} 0 \leq S_K^{\leq R}, S_R^{\leq R} &\leq \mathfrak{E}_{s,R}, & |\mathcal{L}_e^{\leq R}|, |\mathcal{P}_e^{\leq R}| &\leq 4\mathfrak{E}_{s,R}, \\ 0 \leq V_X^{\leq R} &\leq 4\mathfrak{E}_{s,R} \quad (X = b, \text{rem}, s, A). \end{aligned} \quad (\text{IUP5})$$

At $R = \lfloor j^{3/2}/\log j \rfloor$ this budget is $O_{\text{actual}}(j^{5/2} + j^3/\log j) = o(kq_k)$, by the fully finite NIB estimate. The exact remaining tail still contains every term in IUP4. The independent degree- k proper source retains its own centre, order and prescribed cutoffs; IPR proves the subtraction map for its relative determinant without identifying it with V_A .

194.6 The actual endpoint inclusion and the numerator return

The complete PRT calculation below identifies the two original connecting coefficient spaces inside this same target. In its exact notation M_0 is the primitive relation-value pairing and V contains the three original polynomials $R_i - (B_i(0)/B_{i*}(0))R_{i*}$. Their transpose matrix $L = V^T M_0$ has rank three, and original Vandermonde interpolation proves

$$S_1 = \text{im } B + F_0 \ker L \subset S_0, \quad 0 \longrightarrow K_0/K_1 \longrightarrow S_0/S_1 \longrightarrow T_0/T_1 \longrightarrow 0. \quad (\text{RMR7})$$

The arrows are the identity on coefficient representatives. The middle rank is three and the outer ranks are $3 - a_k, a_k$, where $a_k = \dim(\mathcal{G}_k \cap \mathcal{C}_k)$ is the actual CPR18 polynomial intersection, including $a_k = 0, 3$. Give these three quotients their original attained metrics, and denote their four-returns by U_B, U_A, U_C , respectively. Block elimination in the same fixed frames proves

$$\begin{aligned} V_{\text{rem},1} - V_{\text{rem},0} &= U_C, & V_{s,0} - V_{s,1} &= U_A - U_B, & \mathcal{L}_0 - \mathcal{L}_1 &= U_C - U_A + U_B, \\ 0 \leq U_C &\leq 2a_k \log A_{j,s}^{\text{end}}, & |\mathcal{L}_0 - \mathcal{L}_1| &\leq 6 \log A_{j,s}^{\text{end}} = O_{\text{actual}}(q_j \log j) = o(kq_k). \end{aligned} \quad (\text{RMR8})$$

All original frame determinants cancel only under the full return. PRT11–20 supplies their complete matrices and the finite constant. In particular $Z_1 - Z_0 = U_C - U_A + U_B$ for $Z_e = V_b + V_{\text{rem},e} + V_{s,e} + V_A$, preserving exactly $4S_{K,j} = \mathcal{L}_0 + Z_0 = \mathcal{L}_1 + Z_1$.

The NCT dual map RMR4 transports every CPQ quotient in these identities, with the same values and fibres, to RMR5 in four copies. The table NCT25 includes $G_s, Q_{\text{rem}}, Q_b^E, A_{\text{inv}}$ and the complete kernel target. Their finite endpoint constants give the exact error identity

$$4(\widehat{S}_{K,j} - S_{K,j}) = \widehat{\mathcal{L}}_e - \mathcal{L}_e + \sum_{Y=b,\text{rem},s,A} (\widehat{V}_Y - V_Y), \quad \widehat{V}_Y - V_Y = o(kq_k). \quad (\text{RMR9})$$

Every term is retained; the directed finite enclosure is NCT26. The independent degree- k proper source is transported on its own centre and cutoffs. Its relative comparison with A_{inv} is the difference of their two determinant errors, exactly NPK15, so the complete EFP/HJR/NFR maps retain their original orientations.

195 A single spectral budget for the original common long loss

Source and precision. The complete supplied shared-long-flag proof is reproduced below, including all definitions, calculations and its stated limits. The original source is preserved byte for byte in the source closure. Its independent review supplies three explicit precision completions: the positive metric path is forward, the four update eigenvalues may vanish, and the inherited target centre and source-order pairs remain specified. These change no bound or quotient identity.

Scope of the delivered calculation. The connecting and invariant-target losses share an actual boundary flag. Their product is the squared determinant of one orthogonal projection between two orthogonal sums of minimum-lift spaces. Consequently their joint logarithmic variation costs one native spectral budget, rather than the sum of two separate budgets. A nonlinear finite enclosure retains the actual initial common angle. The original four-row recurrence gives all entries of the enclosure. An additional exact flag inequality couples the common loss to the target kernel allocation while preserving its two complementary quotient returns. The resulting joint long-loss bound is quantitative at scale kq_k . It does not evaluate an individual kernel or residual leading coefficient, nor does it establish a narrower unconditional interval for either individual allocation.

195.1 The original objects and the precise scope

Fix the stipulated actual simple quartet, complete unit, one original five-orbit period and its branch. Retain

$$k = 4l + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad \Delta_a = 16a - 48.$$

The source pairs are exactly $(s_0, s_1) = (1, 1)$ or (k, j) . The conductor coefficients a_{ab} are the actual period coefficients. Write

$$E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \beta_{a,r,t} = a/2 + (2r-a)\delta + i(2t-a)\gamma. \quad (\text{SFB1})$$

No period coefficient, source mass or primary multiplicity is replaced. The inherited target centre is $c'_j = j/2 - 4$; every target form uses the original $s_1 = 1$ or j , while its matched degree- k source retains $s_0 = 1$ or k . This is the simple-quartet metric lane; no higher-multiplicity metric theorem is inferred.

Use the complete native target coefficient space $\mathcal{H}$ of CIT and ITL, with the four-copy positive form

$$\mathcal{D}_N = \text{diag}(D_{j,N-v}^{(s_1)}, 4). \quad (\text{SFB2})$$

The D 's are the original full lower-evaluation minimum-extension Grams. All subspaces below are fixed coefficient subspaces as N varies. Put

$$I = \text{im } F_{\text{inv}}, \quad B = \text{im } B_{\text{cof}}, \quad S = \mathcal{S}_e, \quad J = \ker \mathbb{B}, \quad e \in \{0, 1\}, \quad I \subset B \subset S, \quad (\text{SFB3})$$

where the literal matrices are

$$B_{\text{cof}} = \text{stack}_{i=1}^4 M_{i,k}, \quad F_{\text{inv}} = B_{\text{cof}} J_0, \quad S = \text{im}[B_{\text{cof}}, F_e].$$

Here $M_{i,k}$ is the original coefficient map induced by multiplication by x_i^4 , in the complete DNE frames; J_0 is the original invariant-source inclusion, and F_e is the full CIT connecting matrix. In particular these are not independently chosen numerical subspaces. Define

$$K_i = I \cap J, \quad W_i = I + J, \quad W = B + J, \quad K_c = S \cap W, \quad T = S + J. \quad (\text{SFB4})$$

Thus $K_i \subset I \subset B \subset K_c \subset S$ and $J \subset W_i \subset W \subset T$. Subscript i here abbreviates “invariant”, not a summation index.

For a positive form D let $P_X^D = X(X^*DX)^{-1}X^*D$ be the orthogonal projector onto a fixed subspace X , expressed in the original coordinates using any full-rank frame of X . A zero-dimensional space uses its empty inverse and determinant one. The source and target quotient metrics are always their attained minimum norms. The two original losses are

$$\omega_i(D) = \frac{\det \text{Gram}_D(W_i/J)}{\det \text{Gram}_D(I/K_i)}, \quad \omega_c(D) = \frac{\det \text{Gram}_D(T/W)}{\det \text{Gram}_D(S/K_c)}, \quad \omega(D) = \omega_i(D)\omega_c(D). \quad (\text{SFB5})$$

Each ratio uses the same quotient-value coordinates in numerator and denominator. These are exactly ω_{inv} and $\omega_{\text{conn},e}$, not the determinant of the whole connecting space S/B . In particular the connecting kernel K_c/B remains separate.

195.2 The common product is one projection determinant

Theorem 195.1. *For every positive form D , set*

$$\mathcal{P}_D = (I \ominus_D K_i) \oplus (S \ominus_D K_c), \quad \mathcal{Q}_D = (W_i \ominus_D J) \oplus (T \ominus_D W).$$

Both sums are orthogonal. They have equal dimension, the orthogonal projection $\mathcal{P}_D \rightarrow \mathcal{Q}_D$ is invertible, and its squared determinant in orthonormal frames is exactly $\omega(D)$. In particular $0 < \omega(D) \leq 1$.

Proof. The first orthogonality follows from $I \subset K_c$; the second from $W_i \subset W$. The natural maps $I/K_i \rightarrow W_i/J$ and $S/K_c \rightarrow T/W$ are isomorphisms. Take their original minimum lifts and then orthonormalize separately in each of the four quotient forms.

In the corresponding source and target orthonormal frames, the projection has block matrix

$$M_D = \begin{pmatrix} M_i & C_D \\ 0 & M_c \end{pmatrix}. \quad (\text{SFB6})$$

The lower-left block is zero because $I \subset W$. The diagonal blocks are precisely the two original quotient contractions. They are invertible by the displayed quotient isomorphisms, and $|\det M_i|^2 = \omega_i$, $|\det M_c|^2 = \omega_c$. Taking the determinant proves the result. Its norm is at most one because it is an orthogonal projection. The off-diagonal block C_D has not been deleted. In particular, deleting the corresponding cross blocks in $M_D^*M_D$, while retaining its diagonal blocks, need not preserve its determinant. $\square$

In Euclidean relative coordinates define $\Pi_X = D^{1/2}P_X^D D^{-1/2}$. The two projectors in Theorem 195.1 are

$$\Pi_+ = \Pi_T - \Pi_W + \Pi_{W_i} - \Pi_J, \quad \Pi_- = \Pi_S - \Pi_{K_c} + \Pi_I - \Pi_{K_i}. \quad (\text{SFB7})$$

Each is an orthogonal projector: within each sum the two ranges are orthogonal by the same inclusions. This is the reason a single budget applies. Bounding the two losses independently misses it.

195.3 Variation of the actual common loss

Theorem 195.2. *Let $D(t)$ be a continuously differentiable positive form on a forward interval $t_0 \leq t_1$, with $\dot{D}(t) \succeq 0$, and set*

$$L(t) = \log \omega(D(t)), \quad C(t) = D(t)^{-1/2} \dot{D}(t) D(t)^{-1/2}, \quad b'(t) = \text{Tr } C(t).$$

Then

$$L'(t) = \text{Tr}(C(t)(\Pi_+(t) - \Pi_-(t))), \quad |L'(t)| \leq \sqrt{1 - e^{L(t)}} b'(t) \leq b'(t). \quad (\text{SFB8})$$

Consequently, for $a(t) = \text{arcosh}(e^{-L(t)/2})$,

$$|a(t_1) - a(t_0)| \leq \frac{1}{2} \log \frac{\det D(t_1)}{\det D(t_0)}. \quad (\text{SFB9})$$

Proof. For a fixed frame X , Jacobi's determinant formula gives

$$\frac{d}{dt} \log \det(X^*DX) = \text{Tr}(C\Pi_X).$$

Full flag Gram elimination expresses a quotient determinant as a ratio of these determinants, times a fixed coefficient factor. That factor has zero derivative. Applying this identity to (SFB5) gives (SFB8)'s equality and exactly the two projectors in (SFB7). This argument does not differentiate an asymptotic remainder, or assume that the orthogonal minimum sections are constant.

Let $\sigma_a \in (0, 1]$ be the singular values of M_D . An orthogonal principal-plane calculation shows that the nonzero eigenvalues of $\Pi_+ - \Pi_-$ are $\pm\sqrt{1 - \sigma_a^2}$. Explicitly, if source and target principal unit vectors p_a, q_a satisfy $\langle p_a, q_b \rangle = \sigma_a \delta_{ab}$, then $(q_a - \sigma_a p_a)/\sqrt{1 - \sigma_a^2}$, for $\sigma_a < 1$, completes their mutually orthogonal two-planes. The difference of projectors on each plane has the stated eigenvalues. Since $\min_a \sigma_a^2 \geq \prod_a \sigma_a^2 = \omega$, its operator norm is at most $\sqrt{1 - \omega}$. For $C \succeq 0$, diagonalizing this self-adjoint difference gives $|\text{Tr}(CA)| \leq \|A\| \text{Tr } C$. This proves both inequalities.

Where $L < 0$, differentiation of a gives $|a'| = |L'|/(2\sqrt{1 - e^L}) \leq b'/2$. To include points with $L = 0$, use $a_\varepsilon = \text{arcosh}(e^{-L/2} + \varepsilon)$. Its derivative obeys the same bound. Integrate and let $\varepsilon \downarrow 0$. This proves (SFB9) without a differentiability assumption at a zero angle. $\square$

For a finite positive update let $\beta = \log(\det D_1/\det D_0)$ and $a_0 = \text{arcosh}(\omega(D_0)^{-1/2})$. Writing $f(a) = -2 \log \cosh a$, the finite enclosure is

$$f(a_0 + \beta/2) - f(a_0) \leq \log \frac{\omega(D_1)}{\omega(D_0)} \leq f((a_0 - \beta/2)_+) - f(a_0). \quad (\text{SFB10})$$

Both endpoints lie in $[-\beta, \beta]$; the upper endpoint is also at most $-\log \omega(D_0)$. Equivalently the new absolute common loss is between $\text{sech}^2(a_0 + \beta/2)$ and $\text{sech}^2((a_0 - \beta/2)_+)$. The initial common angle is an actual finite quotient quantity. It has not been assigned a period-independent value.

195.4 The exact original four-row update

Here are the original entries to which the preceding results apply. For a fixed target numerator column N_a , let

$$\nu_{a,d} = \sum_{u,w} (Z_{\text{exp},j})_{(u,w),a} \beta_{j,u,w}^d.$$

The analytic germ $F_a = N_a/E_A$ has derivatives given exactly by

$$f_{a,n} = \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}. \quad (\text{SFB11})$$

The numerator's first v derivatives vanish by its original coefficient domain. Retain the original Gamma polynomials and norms

$$p_{n+1}(y) = yp_n(y) - n(n-1 + s_1/2)p_{n-1}(y), \quad h_{n,s_1} = (2\pi)^{s_1/2} n! (s_1/2)_n.$$

Expanding $p_n((S' - c'_j)/i) = \sum_a b_{n,a} S'^a$ gives

$$b_{n+1,a} = i^{-1}(b_{n,a-1} - c'_j b_{n,a}) - n(n-1 + s_1/2)b_{n-1,a}.$$

Thus the new numerator and lower-evaluation rows are

$$t_n = \left(\frac{\sum_a b_{n,a} f_{b,a}}{\sqrt{h_{n,s_1}}} \right)_b, \quad v_n = \left(\frac{\sum_a b_{n,a} (\beta')^a}{\sqrt{h_{n,s_1}}} \right)_{\beta'}.$$

There is no removal of a source mass or of the affine phase.

Let T_0, V_0 contain every earlier numerator and lower-evaluation row in the original native source. Put

$$A_L = V_0^* V_0, \quad D_0 = T_0^* [I - V_0 A_L^{-1} V_0^*] T_0, \quad \rho_n = t_n - v_n A_L^{-1} V_0^* T_0, \quad \zeta_n = v_n A_L^{-1} v_n^*. \quad (\text{SFB12})$$

The exact minimum-extension update is

$$D_1 = D_0 + \frac{\rho_n^* \rho_n}{1 + \zeta_n}, \quad \mathcal{D}_1 = \mathcal{D}_0 + \mathcal{U}_n^* \mathcal{U}_n, \quad \mathcal{U}_n = \text{diag} \left(\frac{\rho_n}{\sqrt{1 + \zeta_n}}, 4 \right). \quad (\text{SFB13})$$

These are CIT19/IRR and FAB's same update. Its four potentially nonzero relative-update eigenvalues are four copies of $\rho_n D_0^{-1} \rho_n^* / (1 + \zeta_n)$, so its full budget is

$$\beta_n = 4 \log \left(1 + \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n} \right). \quad (\text{SFB14})$$

If the original residual row is zero, all four update eigenvalues and this budget are zero. Both the old lower-evaluation inverse and its $1 + \zeta_n$ denominator remain.

For any one of the eight fixed flag subspaces X , set

$$K_{X,n} = \mathcal{U}_n X (X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{U}_n^*, \quad b_{X,n} = \log \det(I_4 + K_{X,n}).$$

These positive four-by-four matrices give the exact common increment

$$\delta L_n = b_{T,n} - b_{W,n} + b_{K_c,n} - b_{S,n} + b_{W_i,n} - b_{J,n} + b_{K_i,n} - b_{I,n}. \quad (\text{SFB15})$$

Thus the result bounds the original signed difference of paired positive increments. It does not bound just a numerator and drop the paired denominator.

195.5 The common long interval and its evaluated budget

Use the original native target cutoffs

$$d = q_j - v - 1, \quad d, \quad d + 1, \quad d + q_j, \quad d + q_j + 1$$

with native-dual signs $(-, -, +, +)$. Let $\mathcal{R}$ denote this signed return. Its consecutive weights ϑ_n equal one at $d + 1, d + q_j + 1$, and two at the internal degrees. Define

$$\mathcal{L}_e = \mathcal{R}(\log \omega_{\text{conn},e} + \log \omega_{\text{inv}}), \quad \mathcal{B}_{k,s_1} = \sum_n \vartheta_n \beta_n. \quad (\text{SFB16})$$

Then

$$|\mathcal{L}_e| \leq \mathcal{B}_{k,s_1}. \quad (\text{SFB17})$$

A sharper finite interval is obtained by summing the two endpoints of (SFB10), with the same ϑ_n , actual preceding common angles, and β_n .

The budget has an already evaluated scalar expression:

$$\mathcal{B}_{k,s_1} = 4\mathcal{T}_{j,s_1} = 4\Delta_j q_j C_\partial + o(j q_j). \quad (\text{SFB18})$$

This uses the inherited evaluated total, with $j = k + 4$ and source $s_1 = 1$ or j ; no source order is substituted. Consequently

$$\limsup \frac{|\mathcal{L}_e|}{k q_k} \leq 64 C_\partial, \quad 86.67404722080 < 64 C_\partial < 86.67404722084. \quad (\text{SFB19})$$

The numerical interval is just multiplication by four of the supplied interval for $16C_\partial$, not a new elliptic calculation. Separate application of the single-flag budget to the two losses allows twice this joint absolute allowance. The common flag improves their joint interval, not an individual leading allocation. No assertion that the inequality is attained on an actual period is made.

195.6 The complementary terms and the allocation-side constraint

For any fixed flag quotient Y/X , write

$$V_n(Y/X) = b_{Y,n} - b_{X,n} \geq 0, \quad \mathcal{V}(Y/X) = \sum_n \vartheta_n V_n(Y/X).$$

The nonnegativity is monotonicity of the same attained quotient minimum under the positive source update. Let

$$\Gamma_{+,n} = V_n(T/W) + V_n(W_i/J), \quad \Gamma_{-,n} = V_n(S/K_c) + V_n(I/K_i).$$

Then $\delta L_n = \Gamma_{+,n} - \Gamma_{-,n}$ and both $\Gamma_{\pm}$ are nonnegative. Full nested-flag elimination gives

$$\Gamma_{+,n} = V_n(\mathcal{H}/J) - V_n(\mathcal{H}/T) - V_n(W/W_i), \quad \Gamma_{-,n} = V_n(S/K_i) - V_n(K_c/I). \quad (\text{SFB20})$$

Every omitted interval in the flag is retained in this identity. In particular,

$$\mathcal{L}_e \leq \mathcal{V}(\mathcal{H}/J) - \mathcal{V}(\mathcal{H}/T) - \mathcal{V}(W/W_i) = 4S_{K,j,s_1} - \mathcal{V}(\mathcal{H}/T) - \mathcal{V}(W/W_i). \quad (\text{SFB21})$$

The final equality uses the original four-copy kernel quotient: $\mathcal{H}/J$ has metric $(\mathbb{B}\mathcal{D}^{-1}\mathbb{B}^*)^{-1}$, four copies of the original native quotient, with exactly the dual return orientation. This is not an equality with the arithmetic canonical allowance.

The quotient W/W_i is precisely CIT's Q_b quotient:

$$\mathcal{V}(W/W_i) = \mathcal{R} \log \det Q_b. \quad (\text{SFB22})$$

It must not be dropped to claim equality in (SFB21). Likewise the full CIT common-boundary factorization is retained:

$$\begin{aligned} \log \omega_c + \log \omega_i &= \log g_T + \log g_{K_c} - \log g_S - \log g_J \\ &\quad - \log \det A_{\text{inv}} - \log \det Q_b + \log c_{\text{conn},e} + 2 \log |\det C_W|. \end{aligned} \quad (\text{SFB23})$$

Only its fixed coefficient constants cancel under $\mathcal{R}$.

If the new common finite enclosure is $\underline{L}_e \leq \mathcal{L}_e \leq \overline{L}_e$, then (SFB21) gives the valid receiver

$$S_{K,j,s_1} \geq \frac{\underline{L}_e + \mathcal{V}(\mathcal{H}/T) + \mathcal{V}(W/W_i)}{4}. \quad (\text{SFB24})$$

This is to be intersected with the existing actual intervals, and paired with $S_R = \mathcal{T} - S_K$. None of the right-hand finite terms has been numerically evaluated here. Therefore no unconditional numerical narrowing of an individual allocation is claimed.

The arithmetic return remains

$$F_K^{(s)} = S_{K,s} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} - F_{W_0}^{(s)}, \quad \Phi_{k,s} = F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}.$$

Both unequal flag endpoints, the original transfer allowance, masses, low metrics and period domains stay in the SAR receiver. The signed return and its whole action budget retain their exact shared-residual cancellation. A sign of either new log-loss interval does not prove an exclusion of an arithmetic packet.

195.7 What is new and what is still uncomputed

The new result is the single-projection realization of the product of the two original losses, its nonlinear common finite enclosure, the one-budget long estimate, and its explicit allocation-side constraint retaining both complementary flag returns. The original rows, target kernel and connecting flags are used throughout. The low primitive and short connecting corrections, native four-row recurrence, evaluated total and single-flag FAB theorem remain inherited inputs.

The individual coefficients of $S_K/(kq_k)$ and $S_R/(kq_k)$ are not evaluated. Nor is an improved unconditional numerical interval for either coefficient established. The remaining quantity in this particular comparison is the actual shared projection angle along the native update, together with $\mathcal{V}(\mathcal{H}/T)$ and $\mathcal{V}(W/W_i)$ in (SFB24). Replacing either by zero, or comparing the kernel alone with the whole canonical allowance, would not complete that calculation.

No actual period integral, hypothetical zero location, independent referee acceptance, or Lean execution is certified by this note. The manuscript supplies a written mathematical derivation. Any separate build or diagnostic status must be taken from its actual execution record, not inferred from this text.

References

- [1] Split-Zero research programme. (2026, September 15). *Current kernel and multiplicity proofs* (498-page supplied edition). Original ITL, IRR, FAB, NFR, HJR/OVR and SAR providers.
- [2] Split-Zero research programme. (2026, September 15). *Full primitive connecting proofs*. Original PDS, CIT and PDI providers; in particular CIT21 and CIT24–26.

Eight actual rank-four compressions and the retained spectral threshold. Retain exactly SFB’s original flags $I \subset B \subset S$, $J = \ker \mathbb{B}$, $K_i = I \cap J$, $W_i = I + J$, $W = B + J$, $K_c = S \cap W$ and $T = S + J$. The same fixed frames are used for every cutoff and both allowed source-order pairs. At native row n , the preceding single-copy Gram and its complete lower correction are

$$\begin{aligned} A_L &= V_0^* V_0, & D_0 &= T_0^* [I - V_0 A_L^{-1} V_0^*] T_0, \\ \rho_n &= t_n - v_n A_L^{-1} V_0^* T_0, & \zeta_n &= v_n A_L^{-1} v_n^*, & r_n &= \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n}, \\ \mathcal{D}_0 &= \text{diag}(D_0, 4), & \mathcal{U}_n &= \text{diag}\left(\frac{\rho_n}{\sqrt{1 + \zeta_n}}, 4\right), & \mathcal{D}_1 &= \mathcal{D}_0 + \mathcal{U}_n^* \mathcal{U}_n. \end{aligned} \tag{PTHS1}$$

Here T_0, V_0 contain every earlier numerator and lower-evaluation row, and t_n, v_n are the exact CIT19/SFB11–13 arithmetic rows, with their original $h_{n,s_1} = (2\pi)^{s_1/2} n! (s_1/2)_n$, center $c'_j = j/2 - 4$, division moments and phases. The notation $\text{diag}(\cdot, 4)$ denotes four copies. In particular $r_n \geq 0$, and the complete $1 + \zeta_n$ denominator is present.

For any of the eight subspaces use its fixed full-rank frame X and define the actual four-by-four matrix

$$K_{X,n} = \mathcal{U}_n X (X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{U}_n^*, \quad \mathcal{X}_+ = (T, K_c, W_i, K_i), \quad \mathcal{X}_- = (W, S, J, I). \tag{PTHS2}$$

For a zero-dimensional space this is the zero matrix. The formula is unchanged by an invertible fixed change of columns in X : the inverse congruence cancels both column changes. Let $Y = \mathcal{U}_n \mathcal{D}_0^{-1/2}$ and $\Pi_X = \mathcal{D}_0^{1/2} X (X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{D}_0^{1/2}$. The displayed matrix Π_X is self-adjoint, idempotent, and has range $\mathcal{D}_0^{1/2} \text{im } X$; these assertions follow by direct multiplication and full column rank. Thus $0 \preceq \Pi_X \preceq I$, and direct substitution gives

$$K_{X,n} = Y \Pi_X Y^*, \quad Y Y^* = r_n I_4, \quad 0 \preceq K_{X,n} \preceq r_n I_4. \tag{PTHS3}$$

The coordinate map $z \mapsto \mathcal{D}_0^{1/2} z$ is an isometry from $(\mathcal{H}, \mathcal{D}_0)$ to the Euclidean coordinate space, since its squared Euclidean norm is $z^* \mathcal{D}_0 z$; its inverse is $\mathcal{D}_0^{-1/2}$. Hence this calculation expresses the original metric and all eight original subspaces.

Let $\lambda_{X,a,n}$, $1 \leq a \leq 4$, be all four eigenvalues of $K_{X,n}$, retaining their zeros. The determinant lemma applied to $X^* \mathcal{D}_1 X$ shows that its log-determinant increment is $b_{X,n} = \log \det(I_4 + K_{X,n})$. Indeed the nonzero product factors of the rectangular maps $\mathcal{U}_n X$ and $(X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{U}_n^*$ give the

same determinant, including the complementary identity eigenvalues. Spectral factorization then gives $b_{X,n} = \sum_{a=1}^4 \log(1 + \lambda_{X,a,n})$. SFB15's exact signed increment is consequently

$$\delta \log \omega_n = \sum_{X \in \mathcal{X}_+} \sum_{a=1}^4 \log(1 + \lambda_{X,a,n}) - \sum_{X \in \mathcal{X}_-} \sum_{a=1}^4 \log(1 + \lambda_{X,a,n}). \quad (\text{PTHS4})$$

The two lists in (PTHS2) therefore retain every sign of the original two contractions.

Put $d = q_j - v - 1$. The original four-endpoint return has consecutive row indices $n = d + 1, \dots, d + q_j + 1$ and weights $\theta_n = 1$ at its two endpoints, $\theta_n = 2$ at the $q_j - 1$ interior indices. This follows by expanding $-f(d) - f(d+1) + f(d+q_j) + f(d+q_j+1)$ in consecutive increments. Thus $\sum_n \theta_n = 2 + 2(q_j - 1) = 2q_j$ exactly. Define the full signed threshold profile by

$$P_{k,e} = \sum_{n=d+1}^{d+q_j+1} \theta_n \left(\sum_{X \in \mathcal{X}_+} \sum_{a=1}^4 \log \max(1, \lambda_{X,a,n}) - \sum_{X \in \mathcal{X}_-} \sum_{a=1}^4 \log \max(1, \lambda_{X,a,n}) \right). \quad (\text{PTHS5})$$

For every real $x \geq 0$, $1 \leq (1+x)/\max(1,x) \leq 2$. Taking logarithms proves $0 \leq \log(1+x) - \log \max(1,x) \leq \log 2$. Each sign list contains four subspaces with four eigenvalues each, so its total error at each row lies in $[0, 16 \log 2]$. The difference of the two errors lies in $[-16 \log 2, 16 \log 2]$. Multiplication by the actual nonnegative weights and their exact sum proves

$$\boxed{|\mathcal{L}_e - P_{k,e}| \leq 32q_j \log 2.} \quad \frac{|\mathcal{L}_e - P_{k,e}|}{kq_k} \leq \frac{32(q_j/q_k) \log 2}{k} \longrightarrow 0. \quad (\text{PTHS6})$$

Here $q_j/q_k = (k+5)^2/(k+1)^2 \longrightarrow 1$. No individual row limit or positive lower bound on the nonzero eigenvalues was used. If $r_n = 0$, (PTHS3) makes every matrix zero, and this row contributes zero on both sides.

For $r_n > 0$ there is also an exact scalar coordinate identity, with its inverse retained:

$$\begin{aligned} \mu_{X,a,n} &= \lambda_{X,a,n}/r_n \in [0, 1], & \lambda_{X,a,n} &= r_n \mu_{X,a,n}, \\ \alpha_n &= \log r_n/k, & b_{X,a,n} &= -\log \mu_{X,a,n}/k, & \log \max(1, \lambda_{X,a,n}) &= k(\alpha_n - b_{X,a,n})_+ \quad (\mu_{X,a,n} > 0). \end{aligned} \quad (\text{PTHS7})$$

The equality follows by substituting the displayed inverse into $\max(0, \log \lambda)$. Zero eigenvalues contribute zero and are retained separately, so no logarithm of zero or deleted kernel occurs. This changes no source form or arithmetic period.

The proved EOR order comparison may now be composed with this finite estimate by the triangle inequality. With superscripts $p = 0$ for $(1, 1)$ and $p = 1$ for (k, j) , it gives, on the same eventual domain and its stated q -scale,

$$\limsup_{k \rightarrow \infty} \frac{|\mathcal{L}_e^{p=1} - P_{k,e}^{p=0}|}{q_k} \leq C_{\text{com}} + 32 \log 2, \quad C_{\text{com}} < 29.311139553141. \quad (\text{PTHS8})$$

Indeed add $|\mathcal{L}_e^{p=1} - \mathcal{L}_e^{p=0}|$ to the error in (PTHS6) for $p = 0$ and use $q_j/q_k \rightarrow 1$. This is the transported finite profile estimate. It leaves the actual signed compressed spectra and the complementary returns in (PTHCIT5)–(PTHCIT6) present.

196 An explicit four-endpoint enclosure from the initial common angle

Retain all original objects and finite domains of CPQ1–20 and the complete shared-long-flag calculation SFB. In particular $j = k + 4$, the source pairs are $(1, 1)$ and (k, j) , the target coefficient form is $\mathcal{D}_N = \text{diag}(D_{j,N-v}^{(s_1)}, 4)$, and the four actual cutoffs are

$$N_0 = q_j - 1, \quad N_1 = q_j, \quad N_2 = 2q_j - 1, \quad N_3 = 2q_j. \quad (\text{SFC1})$$

Write $\omega(N) = \omega_{\text{conn},e}(N)\omega_{\text{inv}}(N)$ and $a(N) = \text{arcosh}(\omega(N)^{-1/2})$. Both losses are their full original quotient contractions. In the exact CPQ value coordinates,

$$\omega(N) = \frac{\det G_{t,e}(N) \det H_{\text{inv}}(N)}{\det G_{s,e}(N) \det A_{\text{inv}}(N)}. \quad (\text{SFC2})$$

This follows directly from the two same-value quotient maps in CPQ8 and CIT25. Every fixed frame constant therefore has already been accounted for; the initial angle below is computed from these four original attained forms, without an assigned bound or a change of source measure.

Define three nonnegative half-budgets and the initial angle by

$$\begin{aligned} x = a(N_0), \quad \eta = \frac{1}{2} \log \frac{\det \mathcal{D}_{N_1}}{\det \mathcal{D}_{N_0}}, \quad b = \frac{1}{2} \log \frac{\det \mathcal{D}_{N_2}}{\det \mathcal{D}_{N_1}}, \quad c = \frac{1}{2} \log \frac{\det \mathcal{D}_{N_3}}{\det \mathcal{D}_{N_2}}, \\ f(t) = -2 \log \cosh t, \quad a_i = a(N_i), \quad \mathcal{L}_e = f(a_2) + f(a_3) - f(x) - f(a_1). \end{aligned} \quad (\text{SFC3})$$

The nonnegative values in SFC3 come from the increasing native form. They are exact determinant ratios over the first, interior and last groups of original rows. Equivalently $2\eta = \beta_{d+1}$, $2b = \sum_{n=d+2}^{d+q_j} \beta_n$ and $2c = \beta_{d+q_j+1}$ with $d = q_j - v - 1$ and the literal SFB/CIT four-row budgets. Thus

$$2\eta + 4b + 2c = 4\mathcal{T}_{j,s_1}. \quad (\text{SFC4})$$

All lower-evaluation denominators and full preceding Gram inverses remain in each β_n ; their telescoping is an exact identity.

196.1 The original common-angle inequality

We recall its proof to specify fully the input used here. Set $I = \text{im}(B_{\text{cof}}J_0)$, $B = \text{im } B_{\text{cof}}$, $S = \mathcal{S}_e$, $J = \ker \mathbb{B}$, $K_i = I \cap J$, $W_i = I + J$, $W = B + J$, $K_c = S \cap W$ and $T = S + J$. The sums

$$\mathcal{P}_D = (I \ominus_D K_i) \oplus (S \ominus_D K_c), \quad \mathcal{Q}_D = (W_i \ominus_D J) \oplus (T \ominus_D W) \quad (\text{SFC5})$$

are orthogonal because $I \subset K_c$ and $W_i \subset W$. Projection from $\mathcal{P}_D$ to $\mathcal{Q}_D$ has block matrix $\begin{pmatrix} M_i & C_D \\ 0 & M_c \end{pmatrix}$ in orthonormal quotient frames. The lower-left block vanishes by $I \subset W$. The diagonal maps are the two original quotient contractions. They are invertible and their squared determinants are ω_i, ω_c , so the squared determinant of this entire projection is ω . Its cross block C_D is retained. This proves $0 < \omega \leq 1$ and finiteness of all angles. At rank zero the determinant is one and the angle is zero.

For a differentiable increasing positive metric path, put $C = D^{-1/2} \dot{D} D^{-1/2}$. Jacobi's formula for each fixed frame Gram gives

$$L' = \text{tr}\{C(\Pi_+ - \Pi_-)\}, \quad \Pi_+ = \Pi_T - \Pi_W + \Pi_{W_i} - \Pi_J, \quad \Pi_- = \Pi_S - \Pi_{K_c} + \Pi_I - \Pi_{K_i}, \quad L = \log \omega. \quad (\text{SFC6})$$

Each displayed $\Pi_{\pm}$ is an orthogonal projector by SFC5. If the projection singular values are $\sigma_r \in (0, 1]$, the principal two-planes give eigenvalues $\pm \sqrt{1 - \sigma_r^2}$ for their difference, with zero eigenvalues on the coincident directions. This follows by using the orthonormal pair $p_r, (q_r - \sigma_r p_r)/\sqrt{1 - \sigma_r^2}$ in each noncoincident plane and computing its two projectors. Since $\min_r \sigma_r^2 \geq \prod_r \sigma_r^2 = e^L$,

$$|L'| \leq \sqrt{1 - e^L} \text{tr } C, \quad |a'| \leq \frac{1}{2} \text{tr } C \quad (L < 0). \quad (\text{SFC7})$$

The trace inequality uses $C \succeq 0$: diagonalize the self-adjoint projector difference and bound each term by its operator norm times the corresponding nonnegative diagonal entry of C . To include $L = 0$, use $a_\varepsilon = \operatorname{arcosh}(e^{-L/2} + \varepsilon)$. Its derivative has the same upper bound because $\sqrt{(e^{-L/2} + \varepsilon)^2 - 1} \geq \sqrt{e^{-L} - 1}$. Integrate and let $\varepsilon \downarrow 0$. On the linear positive path between two ordered metrics, $\int \operatorname{tr} C$ is their log-determinant ratio. Thus the original four angles satisfy

$$|a_1 - x| \leq \eta, \quad |a_2 - a_1| \leq b, \quad |a_3 - a_2| \leq c, \quad a_i \geq 0. \quad (\text{SFC8})$$

This proves the required input directly on the original fixed coefficient flags, retaining the path-dependent minimum sections.

196.2 Exact extrema of the four-angle constraint

Set $A = e^{2c} \geq 1$ and $B_* = e^{2b} \geq 1$. There is a unique real $z_* \geq 1$ satisfying

$$A^2 B_* z_*^3 - \{AB_*(A+1) - 3A\}z_*^2 - \{3AB_* - A - 1\}z_* - 1 = 0. \quad (\text{SFC9})$$

Define

$$\begin{aligned} t_* &= b + c + \frac{1}{2} \log z_*, & t_- &= \max(0, x - \eta), & t_+ &= x + \eta, & \hat{t} &= \min\{t_+, \max(t_-, t_*)\}, \\ L_- &= f(x + \eta + b) + f(x + \eta + b + c) - f(x) - f(x + \eta), \\ L_+ &= f((\hat{t} - b)_+) + f((\hat{t} - b - c)_+) - f(\hat{t}) - f(x). \end{aligned} \quad (\text{SFC10})$$

Then the original common return satisfies the completely specified finite enclosure

$$\boxed{L_- \leq \mathcal{L}_e \leq L_+}. \quad (\text{SFC11})$$

Both endpoints are computed from the initial ratio SFC2 and the three actual determinant ratios in SFC3. No intermediate common angle is required.

Here is the full optimization proof, including uniqueness of the cubic root. First fix $a_1 = t \in [t_-, t_+]$. Since f decreases on $[0, \infty)$, the smallest possible numerator $f(a_2) + f(a_3)$ under SFC8 occurs at $a_2 = t + b$, $a_3 = t + b + c$. These choices satisfy every constraint. The corresponding function of t has derivative

$$-2\{\tanh(t + b) + \tanh(t + b + c) - \tanh t\} \leq 0. \quad (\text{SFC12})$$

It is therefore minimized at $t = t_+ = x + \eta$, giving exactly L_- .

For the largest numerator, the choices $a_2 = (t - b)_+$, $a_3 = (t - b - c)_+$ are allowed and maximize both terms. Indeed both are the smallest attainable nonnegative angles, and increasing a_2 cannot decrease the smallest attainable $a_3 = (a_2 - c)_+$. The derivative of the resulting upper function is

$$2\{\tanh t - \mathbf{1}_{t > b} \tanh(t - b) - \mathbf{1}_{t > b+c} \tanh(t - b - c)\}. \quad (\text{SFC13})$$

It is nonnegative up to $t = b + c$. At the two transition points the added hyperbolic tangent is zero, so the function is continuously differentiable. For $t = b + c + y$, $y \geq 0$, its half-derivative is

$$H(y) = \tanh(y + b + c) - \tanh(y + c) - \tanh y. \quad (\text{SFC14})$$

We have $H(0) = \tanh(b + c) - \tanh c \geq 0$ and $\lim_{y \rightarrow \infty} H(y) = -1$. Moreover $H'(y) = \operatorname{sech}^2(y + b + c) - \operatorname{sech}^2(y + c) - \operatorname{sech}^2 y < 0$, since sech^2 decreases on $[0, \infty)$ and the last term is strictly positive. Hence exactly one zero $y_* \geq 0$ exists. For $b = 0$ it is $y_* = 0$; in this case the upper function may have a constant initial interval, and the choice $t_* = c$ is still a maximizer. For $b > 0$ the zero is positive, and the upper function increases up to $t_* = b + c + y_*$ and decreases afterwards.

Put $z = e^{2y}$. Substitute $\tanh(y + b + c) = (AB_* z - 1)/(AB_* z + 1)$, $\tanh(y + c) = (Az - 1)/(Az + 1)$ and $\tanh y = (z - 1)/(z + 1)$ into $H(y) = 0$. All three denominators are positive for $z \geq 1$. Multiplication by their product gives the negative of the polynomial in SFC9. Thus its roots on

$[1, \infty)$ correspond exactly to the unique zero of H , proving the stated existence, uniqueness and formula $y_* = (\log z_*)/2$. Restricting the upper function to $[t_-, t_+]$ clamps its maximizer to $\hat{t}$, giving L_+ . The displayed extremal triples are allowed in SFC8, so these are the exact extrema of that scalar constraint set. Attainment by a particular actual period or by its native metric path is not inferred from attainment in this larger constraint set.

The polynomial coefficients also have a direct arithmetic meaning:

$$A = \frac{\det \mathcal{D}_{N_3}}{\det \mathcal{D}_{N_2}}, \quad B_* = \frac{\det \mathcal{D}_{N_2}}{\det \mathcal{D}_{N_1}}. \quad (\text{SFC15})$$

Thus the cubic introduces no arbitrary parameter. Its unique root can equally be found by monotone bracketing of H starting at zero, extending a finite upper endpoint until $H < 0$; the proved negative limit guarantees termination. The coefficient equation and its unique allowed interval specify the exact root independently of a chosen numerical tolerance.

Since $|f'(t)| \leq 2$, every admissible triple gives

$$|f(a_2) + f(a_3) - f(x) - f(a_1)| \leq 2\{|a_2 - a_1| + |a_3 - a_2| + |a_2 - x|\} \leq 2\eta + 4b + 2c. \quad (\text{SFC16})$$

For the first inequality write the expression as $[f(a_2) - f(x)] + [f(a_3) - f(a_1)]$ and use $|a_3 - a_1| \leq |a_3 - a_2| + |a_2 - a_1|$. Because the extrema in SFC10 are admissible, this proves $-(2\eta + 4b + 2c) \leq L_- \leq L_+ \leq 2\eta + 4b + 2c$. The new enclosure is therefore contained in the exact SFB one-budget interval $[-4\mathcal{T}_{j,s_1}, 4\mathcal{T}_{j,s_1}]$. At $\eta = b = c = 0$ it reduces to $L_- = L_+ = 0$, as the unchanged original metrics require.

196.3 The complete allocation receiver

Use the actual CPQ quotient returns at the same four endpoints:

$$Z_e = V_b + V_{\text{rem}} + V_s + V_A. \quad (\text{SFC17})$$

Every term is the full original attained return, with $Q_{\text{rem},e} = \mathcal{U}/T_e$ and $G_{s,e} = \mathcal{S}_e/K_e$. CPQ12 has retained their complete frame constants and CPQ18 their exact lower-and-kernel Schur denominator. CPQ20 gives $4S_{K,j}^{(s_1)} = \mathcal{L}_e + Z_e$. Consequently

$$\begin{aligned} K_-^{(s_1)} &= \max\{0, (L_- + Z_e)/4\}, & K_+^{(s_1)} &= \min\{\mathcal{T}_{j,s_1}, U_{j,s_1}^{\text{native}}, (L_+ + Z_e)/4\}, \\ K_-^{(s_1)} &\leq S_{K,j}^{(s_1)} \leq K_+^{(s_1)}, & \mathcal{T}_{j,s_1} - K_+^{(s_1)} &\leq S_{R,j}^{(s_1)} \leq \mathcal{T}_{j,s_1} - K_-^{(s_1)}. \end{aligned} \quad (\text{SFC18})$$

The complete RQB cap is used on its original proved domain; outside that domain its entry is omitted from the minimum, leaving the full-domain bounds $\mathcal{T}$ and the evaluated four-angle enclosure. Positivity of both original allocations and their exact total justify the other entries. These intervals retain both opposing source contributions $V_s + V_A$, which the earlier SFB lower receiver omitted only through their proved nonnegativity.

For the two prescribed orders, the exact NSR31 comparison at degree j further gives

$$\begin{aligned} \max\{K_-^{(j)}, K_-^{(1)} - t_j \Omega_j^{\text{hi}}\} &\leq S_{K,j}^{(j)} \leq \min\{K_+^{(j)}, K_+^{(1)} + t_j \Omega_j^{\text{lo}}\}, \\ \max\{K_-^{(1)}, K_-^{(j)} - t_j \Omega_j^{\text{lo}}\} &\leq S_{K,j}^{(1)} \leq \min\{K_+^{(1)}, K_+^{(j)} + t_j \Omega_j^{\text{hi}}\}. \end{aligned} \quad (\text{SFC19})$$

Here Ω_j uses $q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v$, so this is the original target degree's standard window. Subtract the already proved asymmetric order interval from the first allocation enclosure, or add it to the second, to obtain these intersections. Each contains its actual allocation; they introduce no choice of a new order.

SFC9–19 is a finite scalar calculation using the actual initial common quotient determinant and the three original budget ratios. It replaces the unoptimized four-angle constraint by its explicit extrema and supplies the complete two-sided allocation receiver. The remaining original quotient returns in Z_e are preserved as their calculated matrix quantities. No individual leading coefficient, uniformity in a moving period, or numerical improvement for an unspecified actual period is asserted by these formulas.

197 Native Gamma-order transport and its equilibrium-support refinement

Source and presentation. This chapter retains the complete refined QOT and EOR source, with every proof, original constant, coordinate map, finite domain and bibliography. Explicit equation tags preserve the source equation numbers. Local symbols, theorem environments, labels and bibliography keys are scoped or namespaced for the cumulative edition. A precision sentence at the EOR flags records that the coefficient space is fixed while its Gamma metric depends on source order. The delivered source and every reversible presentation change remain preserved in the source closure.

Scope of the complete delivered calculation. For the original fixed simple-quartet family, a full quadratic-form comparison controls the change from Gamma order one to the original degree-dependent order. It acts on every original coefficient direction before either relation or observation minimization. The difference of the two native kernel allocations, and separately of the two residual allocations, is $O(q_k)$, with limiting absolute allowance $8\log(2 + \sqrt{3})$ after division by q_k . The common connecting/invariant long loss has source-order difference $O(q_k)$ with allowance $32\log(2 + \sqrt{3})$. At the two defining primitive cutoffs, the paired source-to-connecting-image order correction is $o(q_k)$. Its scalar shift is evaluated exactly, and the original compatibility difference is retained and compared in the same coordinates. The individual kq_k -scale allocation coefficients and the absolute common long loss are not evaluated here. No offcritical packet is excluded by these comparisons alone.

Refinement retained in this edition. The subsequent EOR calculation, included in full below, replaces the high-degree symmetric enclosure by the actual equilibrium-support enclosure. For $\Delta S_X = S_X(k; k) - S_X(k; 1)$ it gives the limiting interval $[-8\log(\pi v_*/4), -8\log(\pi u_*/4)]$ at scale q_k . The common source-order allowance becomes $16\log(v_*/u_*)$. The original QOT estimates remain valid; these are subsequent strict refinements, not a reattribution of the older results. The original source bytes are preserved. Two truncated norm control sequences in its source have been restored here, with an exact reversible record.

197.1 Original data and the result's scope

Fix the stipulated hypothetical actual simple quartet, its full unit, one original five-orbit period and its logarithm branch. Put

$$\begin{aligned}
 k = 4l + 1 \geq 9, \quad q_k &= (k + 1)^2, \quad q'_k = (k - 7)^2, \quad \Delta_k = 16k - 48, \quad c'_k = k/2 - 4, \\
 \beta_{n,a,b} &= n/2 + (2a - n)\delta + i(2b - n)\gamma, \quad 0 < \delta < 1/2, \quad \gamma > 2, \\
 \chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), \quad \psi_k = \chi_{k-8}, \\
 E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v.
 \end{aligned} \tag{QOT1}$$

The a_{ab} are the actual conductor coefficients. Its nonreduced factors and exact v remain. The fixed bound $v \leq 80$ is sufficient; no exact matrix replaces v by that bound. The polynomial ψ_k is a lower-grid polynomial, not a derivative.

At degree k the two original source orders are $s = 1, k$. The complete measures, masses, and

monic norms are

$$\begin{aligned} dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad p_{0,s} = 1, \quad p_{1,s} = y, \\ p_{n+1,s} &= yp_{n,s} - n(n-1+s/2)p_{n-1,s}. \end{aligned} \tag{QOT2}$$

All comparisons below identify the same original polynomial coefficients or quotient values, not their different orthonormal coordinate vectors. Source orders are never altered inside an endpoint telescoping sum.

The quantitative root/Gamma input used below is the supplied full-form GPA proof. Its original finite domain is

$$k \geq 29, \quad R_k = k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33} q_k. \tag{QOT3}$$

For paired degree $j = k + 4$ statements impose the corresponding domain at j as well. Every fixed actual quartet eventually satisfies both conditions. Arithmetic statements retain the original period domain $|u| \geq R_*$, or the specified original exact period tests. There is no uniform assertion over a varying period.

The available native primal forms are the attained minima

$$\begin{aligned} z^* H_{Y,k,N}^{(s)} z &= \min_h \int_{\mathbb{R}} |(Ez + \mathbf{U}_N h)(c'_k + iy)|^2 |\psi_k(c'_k + iy)|^2 dm_s(y), \\ H_{K,k,N}^{(s)} &= Z_k^* H_{Y,k,N}^{(s)} Z_k, \\ H_{R,k,N}^{(s)} &= [P_Y (H_{Y,k,N}^{(s)})^{-1} P_Y^*]^{-1}. \end{aligned} \tag{QOT4}$$

Here E , $\mathbf{U}_N$, Z_k , and P_Y are the complete original GPA/FGC/SAR maps. Each relation column is $\alpha_A \mathcal{T}_A(\chi_k S^d)/\psi_k$, where $\mathcal{T}_A p(S') = \sum a_{ab} p(S' + \beta_{8,a,b})$. Their domains, all coefficients and both minima in (QOT4) remain unchanged. The ranks and endpoint signs are

$$\begin{aligned} \mathfrak{m}_k &= \Delta_k - v, \quad t_0 = \dim(K \cap \mathbb{C}[S]_{<v}), \quad t_K = 8k - 16 - t_0, \quad t_R = 8k - 32 - v + t_0, \\ N &= (q_k - 1, q_k, 2q_k - 1, 2q_k), \quad \sigma = (+1, +1, -1, -1), \\ S_{X,k,s} &= \sum_N \sigma_N \log \det H_{X,k,N}^{(s)}, \quad X = K, R. \end{aligned} \tag{QOT5}$$

The actual dual coefficient metric is

$$D_{k,N-v}^{(s)} = A_{\ell,k}^* \overline{(H_{Y,k,N}^{(s)})^{-1}} A_{\ell,k}. \tag{QOT6}$$

The same invertible, source-order-independent $A_{\ell,k}$ appears in both metrics. The conjugation is required by complex-linear duality.

197.2 A complete power-weight comparison

For $\alpha > 0$, $\vartheta > 0$, let

$$\|p\|_\alpha^2 = \int_{\mathbb{R}} |p(y)|^2 |y|^\alpha e^{-\vartheta|y|} dy.$$

Lemma QOT2.1 (Multiplication with its next-degree row retained). *For $n \geq 0$ define*

$$\begin{aligned} c_n &= \frac{(\sqrt{n + \alpha + 1/2} - \sqrt{n + 1/2})^2}{\vartheta}, \\ C_n &= \frac{(\sqrt{n + \alpha + 1/2} + \sqrt{n + 1/2})^2}{\vartheta}. \end{aligned} \tag{QOT7}$$

Every complex polynomial of degree at most n satisfies

$$c_n \|p\|_\alpha \leq \|yp\|_\alpha \leq C_{n+1} \|p\|_\alpha. \tag{QOT8}$$

Proof. For the classical Laguerre norm and recurrence, with the standard phase retained, see Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21, Table 18.3.1; eq. (18.5.5) and Table 18.5.1; eq. (18.5.12); eq. (18.9.1) and Table 18.9.1]. Here their parameter is α , their argument is $x = \vartheta y$, and their degree is i ; hence the squared norm is $\vartheta^{-\alpha-1} \Gamma(i + \alpha + 1)/i!$. The positive square root of this full norm yields precisely the negative off-diagonal entry displayed next. The next-degree row required for the multiplication norm remains in the local proof.

On $(0, \infty)$, the normalized Laguerre polynomials for $y^\alpha e^{-\vartheta y} dy$ give multiplication matrix

$$(J_n)_{ii} = (2i + \alpha + 1)/\vartheta, \quad (J_n)_{i,i+1} = -\sqrt{(i+1)(i+\alpha+1)}/\vartheta \quad (0 \leq i < n).$$

Its entries follow from Rodrigues' coefficient formula and integration by parts, as in the supplied GPA/LVM proof. For completeness, $L_n^{(\alpha)}(x) = \sum_{r=0}^n (-1)^r \binom{n+\alpha}{n-r} x^r / r!$ has squared norm $\Gamma(n + \alpha + 1)/n!$ and obeys

$$xL_n^{(\alpha)} = (2n + \alpha + 1)L_n^{(\alpha)} - (n+1)L_{n+1}^{(\alpha)} - (n+\alpha)L_{n-1}^{(\alpha)}.$$

Coefficient comparison proves the recurrence; the stated norms give J_n , including its negative off-diagonal entries.

The function $f(x) = \sqrt{x(x+\alpha)}$ is concave for $x > 0$, since $f''(x) = -\alpha^2/[4(x(x+\alpha))^{3/2}]$. Continuity includes $x = 0$. Hence $f(i) + f(i+1) \leq 2f(i+1/2)$. Bounding the cross terms of the tridiagonal quadratic form by the adjacent squared moduli gives

$$c_n I \preceq J_n \preceq C_n I.$$

Indeed the lower row expression $2i + \alpha + 1 - 2f(i+1/2)$ decreases with i , and the upper expression increases. Omitting a missing boundary off-diagonal only improves these bounds. Cauchy–Schwarz now gives $\|yp\| \|p\| \geq \langle yp, p \rangle \geq c_n \|p\|^2$ on this half-line.

For the upper norm bound the multiplication map is rectangular: its last row is $-\sqrt{(n+1)(n+\alpha+1)} e_n^\top / \vartheta$. It is a restriction of J_{n+1} , not J_n . Thus its norm is at most C_{n+1} . Apply both inequalities also to $p(-y)$ on $y > 0$ and add their squares. The sign of multiplication on the negative half-line has no effect on its norm. This proves (QOT8) for the entire original coefficient direction, without a parity restriction. $\square$

For integers $a, \ell \geq 0$ put

$$\begin{aligned} d_{\alpha,a,\ell}^- &= 4 \sum_{r=0}^{\ell-1} \operatorname{arsinh} \sqrt{(a+r+1/2)/\alpha}, \\ d_{\alpha,a,\ell}^+ &= 4 \sum_{r=1}^{\ell} \operatorname{arsinh} \sqrt{(a+r+1/2)/\alpha}. \end{aligned} \tag{QOT9}$$

Empty sums are zero.

Theorem QOT2.2 (Centred weight shift). *On every polynomial of degree at most a ,*

$$(\alpha/\vartheta)^{2\ell} e^{-d_{\alpha,a,\ell}^-} \|p\|_\alpha^2 \leq \|p\|_{\alpha+2\ell}^2 \leq (\alpha/\vartheta)^{2\ell} e^{d_{\alpha,a,\ell}^+} \|p\|_\alpha^2. \tag{QOT10}$$

Proof. Iterate (QOT8) in the fixed α norm on $p, yp, \dots, y^{\ell-1}p$, with degree bounds $a, a+1, \dots, a+\ell-1$. The resulting lower product is $\prod_{r=0}^{\ell-1} c_{a+r}^2$ and the upper product is $\prod_{r=1}^{\ell} C_{a+r}^2$. Also $\|y^\ell p\|_\alpha^2 = \|p\|_{\alpha+2\ell}^2$ exactly. Finally

$$c_n = (\alpha/\vartheta) e^{-2 \operatorname{arsinh} \sqrt{(n+1/2)/\alpha}}, \quad C_n = (\alpha/\vartheta) e^{2 \operatorname{arsinh} \sqrt{(n+1/2)/\alpha}}.$$

Substitution proves both sides. The displayed central scalar is not an exact equality of forms; for $p = 1, \ell = 1$, the actual moment ratio is $(\alpha+1)(\alpha+2)/\vartheta^2$, not α^2/ϑ^2 . $\square$

197.3 Finite constants in the original Gamma comparison

We record the full inherited GPA constants so the comparison does not use an unspecified conditioning constant. Abbreviate $q = q_k$, $q' = q'_k$, $R = R_k$, $\epsilon = 2^{-32}$, $\vartheta = \pi/2$, and $h = 1/q$. Put

$$\begin{aligned} \ell_s &= (s-1)/4, \quad Q_s = q' + \ell_s, \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}, \quad \beta_1 = 1, \\ C_L &= \frac{\pi}{\Gamma(3/4)^2}, \quad K_q = \frac{\sqrt{2\pi}}{C_L} \frac{(2q+1)^2}{q} e^{-2q}, \\ \mathcal{E}_k &= \frac{qR^2}{\epsilon^2 q^2 - R^2} + \log(1 + K_q), \\ r_{\ell_s} &= \frac{\sum_{r=0}^{\ell_s-1} (2r+1/2)^2}{\epsilon^2 q^2} + \log(1 + K_q) \quad (\ell_s > 0), \quad r_0 = 0, \\ L_h &= C_L (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\ K_{a,Q} &= 2Q + 1 + 2\{a + \sqrt{a(a+2Q)}\}. \end{aligned} \tag{QOT11}$$

The logarithmic endpoints are

$$\begin{aligned} b_{k,a,s}^- &= -\mathcal{E}_k + \log L_h - K_{a,Q_s} \log(1 + h/\vartheta), \\ b_{k,a,s}^+ &= \mathcal{E}_k + r_{\ell_s} + \log U_h + K_{a,Q_s} \log \frac{1}{1 - h/\vartheta}. \end{aligned} \tag{QOT12}$$

The supplied full proof gives, for every $\deg p \leq a \leq 2q$,

$$\begin{aligned} \beta_s e^{b_{k,a,s}^-} \|p(c'_k + i \cdot)\|_{2Q_s}^2 &\leq \int_{\mathbb{R}} |\psi_k(c'_k + iy) p(c'_k + iy)|^2 dm_s(y) \\ &\leq \beta_s e^{b_{k,a,s}^+} \|p(c'_k + i \cdot)\|_{2Q_s}^2. \end{aligned} \tag{QOT13}$$

This uses precisely the degree bound in the proof of GPA12–15, not only its originally listed endpoints. In all applications here $a + \ell_s \leq 2q$ also holds on the stated eventual domains. The proof pairs the original opposite roots outside $[-\epsilon q, \epsilon q]$, bounds the entire inner polynomial integral by K_q times the reference outside integral, and retains the full source recurrence

$$dm_s = \beta_s \prod_{r=0}^{\ell_s-1} (y^2 + (2r+1/2)^2) d\sigma, \quad d\sigma = dm_1.$$

It then uses the whole-line Gamma sandwich $L_h e^{-(\vartheta+h)|y|} \leq d\sigma/dy \leq U_h e^{-(\vartheta-h)|y|}$ and the exact polynomial dilation norm bound. Thus none of the original roots, the inner region, or the source mass is removed by (QOT13). At fixed actual data, $b_{k,a,s}^\pm = O_{\text{actual}}(\log q)$ uniformly for these degrees and both original orders.

Define

$$\begin{aligned} \alpha_k &= 2q'_k, \quad \ell = (k-1)/4, \quad \kappa_k = \beta_k (4q'_k/\pi)^{(k-1)/2}, \\ l_{k,a} &= b_{k,a,k}^- - b_{k,a,1}^+ - d_{\alpha_k,a,\ell}^-, \quad u_{k,a} = b_{k,a,k}^+ - b_{k,a,1}^- + d_{\alpha_k,a,\ell}^+, \quad w_{k,a} = u_{k,a} - l_{k,a} \geq 0. \end{aligned} \tag{QOT14}$$

Theorem QOT3.1 (Full original source-order transport). *On the degree- a original coefficient space, and on every fixed restriction or attained quotient of it,*

$$\kappa_k e^{l_{k,a}} H^{(1)} \preceq H^{(k)} \preceq \kappa_k e^{u_{k,a}} H^{(1)}. \tag{QOT15}$$

In particular it holds for all three forms in (QOT4) at $a = N - v - q'_k$. On their original complex-linear dual coefficient metric it becomes

$$\kappa_k^{-1} e^{-u_{k,a}} D^{(1)} \preceq D^{(k)} \preceq \kappa_k^{-1} e^{-l_{k,a}} D^{(1)}. \tag{QOT16}$$

The same dual inequality passes to every specified fixed flag subquotient, including the fourfold target space.

Proof. Apply (QOT10) with $\alpha = 2q'$ and $\ell = (k-1)/4$ to the complete $p(c' + iy)$ direction. Combine its lower bound with the lower order- k inequality and upper order-one inequality in (QOT13); combine the three opposite inequalities for the upper side. This gives exactly (QOT14)–(QOT15). On a restriction substitute the same coefficient columns. On a quotient every candidate in the same affine fibre obeys both inequalities, so its minimum does as well. This preserves the entire relation and kernel Schur corrections. Inversion reverses positive-form inequalities, complex conjugation preserves them, and the fixed congruence (QOT6) proves (QOT16). Apply the same restriction/minimum argument to its subsequent fixed flags. $\square$

197.4 The two-half-line transport is literal

To specify the coefficient comparison completely, for $\alpha = 2Q_s$ and $0 \leq j \leq r \leq a$ let

$$A_{j,r} = (-1)^j \binom{r}{j} \Gamma(r + \alpha + 1) \sqrt{\frac{j!}{\Gamma(j + \alpha + 1)}} \vartheta^{-r-(\alpha+1)/2}, \quad P_{r,b} = \binom{b}{r} (c'_k)^{b-r} i^r. \quad (\text{QOT17})$$

Other triangular entries are zero. These are respectively the monomial-to-orthonormal-Laguerre and original affine coefficient maps. The exact reflection is

$$R_{j,r} = (-1)^j 2^{r-j} \binom{\alpha + r}{r-j} \sqrt{\frac{r! \Gamma(j + \alpha + 1)}{j! \Gamma(r + \alpha + 1)}}, \quad R^2 = I, \quad W = \begin{bmatrix} \text{AP} \\ \text{RAP} \end{bmatrix}. \quad (\text{QOT18})$$

The Gram W^*W is the complete reference integral over both half-lines. Reflection is complex-linear: it does not conjugate a period coefficient. For any actual relation or invariant column x , the two row blocks are $(\text{AP})x$ and $(\text{RAP})x$. The source change in Theorem QOT3.1 is on these full integrals before minimization; it is not a separate fit on each half-line. In particular the same U_N , Z_k , and P_Y enter their two quadratic forms. All original factorials and the source factor β_s have been displayed, rather than assigned to an unknown inverse.

197.5 An evaluated allowance at scale q_k

For $X = K, R$ let $a_N = N - v - q'_k$. Theorem QOT3.1 and the four original signs give

$$\begin{aligned} t_X(l_{k,a_{q-1}} + l_{k,a_q} - u_{k,a_{2q-1}} - u_{k,a_{2q}}) &\leq S_{X,k,k} - S_{X,k,1} \\ &\leq t_X(u_{k,a_{q-1}} + u_{k,a_q} - l_{k,a_{2q-1}} - l_{k,a_{2q}}). \end{aligned} \quad (\text{QOT19})$$

The four copies of $t_X \log \kappa_k$ cancel only in this signed sum.

In fact the scalar is an explicit positive rational number at every original degree:

$$\boxed{\kappa_{4l+1} = 2^{10l} \frac{(2l)!}{(4l)!} (k-7)^{4l}.} \quad (\text{QOT20})$$

This follows from $\Gamma(2l + 1/2) = \sqrt{\pi}(4l)!/[4^{2l}(2l)!]$. The powers of π in the original mass and the central power weight cancel only in this exact scalar identity.

At a low endpoint, $a = \Delta_k - v - 1$ or $\Delta_k - v$ is $O(k)$. Since $\text{arsinh } x \leq x$ for $x \geq 0$, (QOT9) gives $d^\pm = O(\sqrt{k})$, hence $l, u = O_{\text{actual}}(\sqrt{k} + \log q_k)$. At either high endpoint, $a = q_k + \Delta_k - v + O(1)$ and $\alpha_k = 2q'_k$. Uniformly over the $\ell = O(k)$ summands,

$$\frac{a + r + 1/2}{\alpha_k} \longrightarrow \frac{1}{2}.$$

Therefore

$$\frac{l_{k,a}}{k} \longrightarrow -a_*, \quad \frac{u_{k,a}}{k} \longrightarrow a_*, \quad a_* = \operatorname{arsinh}(1/\sqrt{2}) = \frac{1}{2} \log(2 + \sqrt{3}). \quad (\text{QOT21})$$

Since $t_X/k \rightarrow 8$ and $q_k/k^2 \rightarrow 1$, (QOT19) proves

$$\boxed{\limsup_{k \rightarrow \infty} \frac{|S_{X,k,k} - S_{X,k,1}|}{q_k} \leq 8 \log(2 + \sqrt{3}), \quad X = K, R.} \quad (\text{QOT22})$$

Sharper receiving interval (EOR24–26). In the same original source and quotient coordinates, use the finite endpoints (EOR24) wherever their guards hold, intersected as in (EOR21). The resulting limiting interval is

$$-C_+ \leq \liminf \frac{S_{X,k,k} - S_{X,k,1}}{q_k} \leq \limsup \frac{S_{X,k,k} - S_{X,k,1}}{q_k} \leq C_-,$$

$$C_+ = 8 \log(\pi v_*/4), \quad C_- = 8 \log(4/(\pi u_*)).$$

Here $C_+ < 8.881817865597$ and $C_- < 5.773751910974$; neither is an individual allocation coefficient. The full four endpoint scalars still cancel with their original signs.

This is an allowance for the difference of the two source-order allocations, not the asymptotic coefficient of either allocation. In particular their kq_k -normalized limit points coincide.

There is a genuine evaluated low-endpoint order-change term. Stirling's elementary factorial bounds give

$$\log \kappa_k = \frac{k}{2} \log k + \frac{k}{2}(1 + 4 \log 2) + O(\log k). \quad (\text{QOT23})$$

It can also be obtained from the integer/half-integer Gamma formula and integral bounds for $\sum \log n$. At either low endpoint, Theorem QOT3.1 gives an error $O_{\text{actual}}(k^{3/2} + k \log q_k) = o(q_k)$ after taking a rank- t_X determinant. Thus

$$\boxed{\log \frac{\det H_{X,k,q_k-1+e}^{(k)}}{\det H_{X,k,q_k-1+e}^{(1)}} = 4q_k \log k + (4 + 16 \log 2)q_k + o(q_k),}$$

$$e = 0, 1, \quad X = K, R. \quad (\text{QOT24})$$

No next-order term for the complete four-return is assigned: its high determinants still require their actual allocation calculation.

197.6 The defining primitive source and connecting image

Put $j = k + 4$, $Q = q_j$, $D_e = Q - v - 1 + e$, and $d_e = 8k + 24 + e$. The original PDS relations are

$$r_a = \mathcal{T}_A(\chi_k S^a), \quad 0 \leq a < d_e, \quad \deg r_a = q_k + a - v.$$

Lemma QOT6.1 (The original lower factor divides every primitive column). *Every r_a is divisible by the full ψ_k . Its quotient degree is $\Delta_k - v + a$.*

Proof. For a root $\beta_{k-8,u,w}$ of ψ_k , every summand in $r_a(\beta_{k-8,u,w})$ evaluates χ_k at $\beta_{k-8,u,w} + \beta_{8,b,c} = \beta_{k,u+b,w+c}$, one of its original upper roots. Each summand is zero, without a cancellation or support assumption. The lower roots are distinct, so their complete product divides r_a . The degree follows from the first nonzero conductor moment μ_v . $\square$

In particular the whole relation Gram H_e is a restriction of the original weighted form in Theorem QOT3.1 with degree bound

$$a_s = \Delta_k - v + d_e - 1 = 24k - 25 - v + e. \quad (\text{QOT25})$$

This bound is within $2q_k$ on the stated domain. No leading-monomial approximation or unproved inverse-conductor estimate enters.

We next identify the same primitive coordinates in both source orders. Let R_e^{pol} be the full coefficient matrix of the r_a , let B_s be the monomial-to-original-Gamma coefficient matrix, and let P_e^{Tay} be CIT's fixed Taylor complement. Then

$$\begin{aligned} U_s &= \text{diag}(1/r!)B_s^\top, \quad P_e^{(s)} = U_s^{-1}P_e^{\text{Tay}}, \quad C_e^{(s)} = B_s R_e^{\text{pol}}, \\ \Theta_e &= (C_e^{(s)})^\top P_e^{(s)} = (R_e^{\text{pol}})^\top \text{diag}(r!)P_e^{\text{Tay}}. \end{aligned} \quad (\text{QOT26})$$

The actual Θ_e is independent of s ; every mass, affine phase, and factorial cancels here against its exact inverse, not against an assumed norm. At $e = 0$ it is invertible; at $e = 1$ it has image the same rank- r_e primitive constraint space, where $r_e = d_e - 4e = 8k + 24 - 3e$. PDS/CIT's equality of affine fibres gives

$$H_{\text{src},e}^{(s)} = \Theta_e^* \overline{(H_e^{(s)})^{-1}} \Theta_e.$$

Put $l_s = l_{k,a_s}$, $u_s = u_{k,a_s}$. Theorem QOT3.1 therefore proves

$$-r_e u_s \leq \log \frac{\det H_{\text{src},e}^{(k)}}{\det H_{\text{src},e}^{(1)}} + r_e \log \kappa_k \leq -r_e l_s. \quad (\text{QOT27})$$

This restriction retains all four original primitive constraints at $e = 1$.

The corresponding CIT image columns are also the same coefficient columns in both source pairs. In ordinary Taylor coordinates their equation is

$$\text{stack}_i \text{Mult}(B_i)P_e^{\text{Tay}} = \text{diag}(\tilde{T}_j, 4)F_e + \text{diag}(\tilde{V}_j, 4)E_e^{\text{low}}.$$

All Taylor matrices are those of the same specified germs. The original analytic quotient is injective at this target cutoff modulo the full lower evaluations, so its coefficient F_e is unique. This proves equality of the two F_e and of B_{cof} in their common value coordinates. It is the exact PSR23–24/CIT coefficient argument, not an inference from equal leading determinant orders.

The connecting Gram is the complete fixed quotient

$$\begin{aligned} H_{\text{conn},e}(N) &= F_e^* \mathcal{D}_N F_e - F_e^* \mathcal{D}_N B_{\text{cof}} (B_{\text{cof}}^* \mathcal{D}_N B_{\text{cof}})^{-1} B_{\text{cof}}^* \mathcal{D}_N F_e, \\ \mathcal{D}_N &= \text{diag}(D_{j,N-v}^{(s_1)}, 4). \end{aligned} \quad (\text{QOT28})$$

At its defining low cutoff $N = Q - 1 + e$, let

$$a_t = \Delta_j - v - 1 + e = 16k + 15 - v + e, \quad l_t = l_{j,a_t}, \quad u_t = u_{j,a_t}. \quad (\text{QOT29})$$

The dual comparison and the identical complete boundary minimum give

$$-r_e u_t \leq \log \frac{\det H_{\text{conn},e}^{(k,j)}}{\det H_{\text{conn},e}^{(1,1)}} + r_e \log \kappa_j \leq -r_e l_t. \quad (\text{QOT30})$$

All four constants l_s, u_s, l_t, u_t are $O_{\text{actual}}(\sqrt{k} + \log q_k)$.

197.6.1 The exact scalar cancellation and its compatibility return

The Gamma recurrence gives the elementary identity

$$\boxed{\begin{aligned} \frac{\kappa_{k+4}}{\kappa_k} &= \frac{256(k-3)^{k+3}}{k(k+2)(k-7)^{k-1}}, \\ \log \frac{\kappa_{k+4}}{\kappa_k} &= 2 \log k + 8 \log 2 + 4 + O(k^{-1}). \end{aligned}} \quad (\text{QOT31})$$

Indeed $\beta_{k+4}/\beta_k = 16\pi^2/[k(k+2)]$ and the two remaining powers supply $16/\pi^2$ and the displayed integer powers. Every original source mass is included in this cancellation.

Define the difference of the two original source-to-connecting-image logarithmic Jacobians by

$$\mathcal{J}_e^{\text{ord}} = \log \frac{\det H_{\text{conn},e}^{(k,j)}}{\det H_{\text{src},e}^{(k)}} - \log \frac{\det H_{\text{conn},e}^{(1,1)}}{\det H_{\text{src},e}^{(1)}}.$$

Subtracting (QOT27) from (QOT30), in the same original primitive values, proves

$$\boxed{r_e(l_s - u_t) \leq \mathcal{J}_e^{\text{ord}} + r_e \log(\kappa_j/\kappa_k) \leq r_e(u_s - l_t), \quad \mathcal{J}_e^{\text{ord}} = o(q_k).} \quad (\text{QOT32})$$

The error before division by q_k is $O_{\text{actual}}(k^{3/2} + k \log q_k)$. This bound can exceed the displayed $O(k \log k)$ scalar centre; the centre is not asserted to be the next asymptotic term. Separately, the two low order changes in (QOT27) and (QOT30) each equal

$$-4q_k \log k - (4 + 16 \log 2)q_k + o(q_k). \quad (\text{QOT33})$$

Thus the common large term cancels only in their paired Jacobian. Neither absolute low Gram is declared close to its order-one value.

Write $\mathcal{X}_e^{(s_0, s_1)} = \chi_{X_k, D_e} - \chi_{Y_e, D_e}$ for CIT's full proper-source compatibility difference. Its exact identity is

$$\log \det H_{\text{conn},e} - \log \det H_{\text{src},e} = -\mathcal{X}_e + \varepsilon_e,$$

where the original finite endpoints are

$$\varepsilon_e^- = -2r_e \log \max(1, K_-) - r_e \log c_+, \quad \varepsilon_e^+ = 2r_e \log \max(1, K_+) - r_e \log c_-.$$

The constants $K_{\pm}, c_{\pm}$ are exactly the original multiplier and compatibility constants at that cutoff and source pair. They are not redefined by the present comparison; their supplied bounds are $O_{\text{actual}}(k \log q_k)$ after the displayed rank factors. Let $\Delta\mathcal{X}_e = \mathcal{X}_e^{(k,j)} - \mathcal{X}_e^{(1,1)}$. Then

$$\begin{aligned} r_e(l_t - u_s) + \varepsilon_{e,(k,j)}^- - \varepsilon_{e,(1,1)}^+ &\leq \Delta\mathcal{X}_e - r_e \log(\kappa_j/\kappa_k) \\ &\leq r_e(u_t - l_s) + \varepsilon_{e,(k,j)}^+ - \varepsilon_{e,(1,1)}^-. \end{aligned} \quad (\text{QOT34})$$

In particular $\Delta\mathcal{X}_e = o(q_k)$. Only the difference between source-order pairs is small. The absolute compatibility loss is not bounded here. At $e = 0$ the original χ_Y is zero by its proved full-source statement; at $e = 1$ all four constraints and the complete χ_Y remain.

197.7 Source-order control for the original common long loss

At every target endpoint use the fixed coefficient flags

$$I = \text{im}(B_{\text{cof}}J_0), \quad B = \text{im } B_{\text{cof}}, \quad S = \text{im}[B_{\text{cof}}, F_e], \quad J = \ker \mathbb{B},$$

$$K_i = I \cap J, \quad W_i = I + J, \quad W = B + J, \quad K_c = S \cap W, \quad T = S + J.$$

The original losses in the common value frames are

$$\omega_i = \frac{\det \text{Gram}(W_i/J)}{\det \text{Gram}(I/K_i)}, \quad \omega_c = \frac{\det \text{Gram}(T/W)}{\det \text{Gram}(S/K_c)}. \quad (\text{QOT35})$$

These are ITL's invariant and CIT's connecting losses. Put $r_i = \dim(I/K_i)$, $r_c = \dim(S/K_c)$, and $R_e = r_i + r_c$. The complete kernel of the original multiplication/observation map is retained in K_i, K_c . In particular

$$R_e \leq (8k - 32 - v + t_0) + (8k + 24 - 3e) \leq 16k - 8.$$

All the flags are literally independent of source order, by their polynomial maps and the uniqueness argument preceding (QOT28).

A general immediate consequence of (QOT16) is useful. If two matched fixed subquotients have the same rank r , their determinant ratio obeys

$$\left| \log \omega^{(j)}(N) - \log \omega^{(1)}(N) \right| \leq r w_{j,N-v-q'_j}. \quad (\text{QOT36})$$

Indeed the logarithm of each determinant's order change belongs to $[-r \log \kappa_j - ru, -r \log \kappa_j - rl]$. Subtracting these intervals cancels the scalar before bounding. This proof does not posit a lower singular-value gap for an observation map.

Let $\mathcal{R}$ be the original native-dual return at $N = q_j - 1, q_j, 2q_j - 1, 2q_j$ with signs $(-, -, +, +)$, and define

$$\mathcal{L}_e^{(s_0, s_1)} = \mathcal{R} \log(\omega_c \omega_i).$$

The endpoint-first estimate is

$$\boxed{\begin{aligned} |\mathcal{L}_e^{(k,j)} - \mathcal{L}_e^{(1,1)}| &\leq R_e \sum_N w_{j,N-v-q'_j} = O_{\text{actual}}(q_k), \\ \limsup_{k \rightarrow \infty} \frac{|\mathcal{L}_e^{(k,j)} - \mathcal{L}_e^{(1,1)}|}{q_k} &\leq 32 \log(2 + \sqrt{3}). \end{aligned}} \quad (\text{QOT37})$$

Sharper common-loss transport (EOR28–29). Replace the displayed width by the intersection width in (EOR28). On the same paired original families this gives

$$\limsup \frac{|\mathcal{L}_e^{(k,k+4)} - \mathcal{L}_e^{(1,1)}|}{q_k} \leq 16 \log(v_*/u_*) < 29.311139553141.$$

This does not evaluate the absolute order-one common loss. Both target gauges, Q_b , Q_o and the original source denominator remain in the receiving identity.

For the last constant, the two low widths are $o(j)$ and each high width divided by j tends to $2a_* = \log(2 + \sqrt{3})$. Now $R_e/k \leq 16 + o(1)$ and $j/k \rightarrow 1$. This proof compares only four endpoint forms; adding a per-row source-order error over q_j rows would lose the result.

The original IRR/CIT rows are unchanged. Explicitly

$$f_{a,n} = \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \quad \nu_{a,n} = \sum_{u,w} (Z_{\text{exp},j})_{uw,a} \beta_{j,u,w}^n,$$

$$b_{n+1,m} = i^{-1}(b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2) b_{n-1,m},$$

$$t_n = \left(\frac{\sum_m b_{n,m} f_{a,m}}{\sqrt{h_{n,s_1}}} \right)_a, \quad v_n = \left(\frac{\sum_m b_{n,m} (\beta')^m}{\sqrt{h_{n,s_1}}} \right)_{\beta'}.$$

Each source family uses its own rows with the same original coefficients. For every fixed flag Y , the raw frame is $[\text{diag}(V_D, 4), \text{diag}(T_D, 4)Y]$. Its positive update retains the complete old Gram and both lower-evaluation and further-quotient Schur terms. Hence the original long weights $1, 2, \dots, 2, 1$ and the signed difference $\gamma_{H,n} - \gamma_{A,n}$ telescope to exactly the four endpoint forms used above. Neither positive term is discarded, and (QOT37) does not assign a sign or value to their absolute long sum.

197.8 Residual boundary and original arithmetic receivers

The RQB/CIT identity is retained separately in each source pair:

$$\mathcal{V}_b + \mathcal{V}_{\text{inv}} + \mathcal{V}_o = 4\widehat{F}_j^{(s_1)}, \quad \mathcal{V}_b, \mathcal{V}_{\text{inv}}, \mathcal{V}_o \geq 0, \quad \mathcal{L}_e = P_e^{\text{rest}} - \mathcal{V}_b. \quad (\text{QOT38})$$

Each term is the same original fixed flag return. For a rank- r target flag form Q_X , (QOT16) gives the finite endpoint bound

$$-r \log \kappa_j - r u_{j,a_N} \leq \log \frac{\det Q_X^{(j)}(N)}{\det Q_X^{(1)}(N)} \leq -r \log \kappa_j - r l_{j,a_N}.$$

For its dual-sign long return the resulting lower bound is $r(\sum_{\text{low}} l - \sum_{\text{high}} u)$ and the upper bound is $r(\sum_{\text{low}} u - \sum_{\text{high}} l)$. The source scalar cancels, and the actual ranks of Q_b, H_{inv}, Q_o are retained.

In particular the full source denominator of Q_b is not replaced by a Euclidean norm. It is the original minimum on $V_b/(\text{im } J_0 + \ker \Phi_b)$, and the target is the minimum on W/W_i . At a displaced source cutoff $N_s \leq 2q_{k+4}$, its original polynomial quotient degree is at most $2q_{k+4} - v - q'_k = k^2 + 34k + 1 - v$. For $k \geq 37$ this is at most $2q_k$, so the same full-form theorem also bounds that source minimum. The exact fixed congruence between the CIT and RQB value frames cancels only in matched ratios or the four-return. Thus (QOT38) and CIT's P_e^{rest} retain the source denominator, A_{inv}, Q_o , and both target gauge losses. The present theorem controls their source-order change, not their separate leading values.

For a proved interval $[L_{X,1}, U_{X,1}]$ for an order-one native allocation, add the explicit two endpoints of (QOT19) to obtain an order- k interval; intersect it with all existing intervals. Couple the pair by the exact equality $S_R = \mathcal{T} - S_K$. No independence of the two error terms is assumed. The original arithmetic receiving identities remain

$$\begin{aligned} F_K^{(s)} &= S_{K,s} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} - F_{W_0}^{(s)}, \\ \Phi_{k,s} &= F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}, \\ \Psi_k &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, \quad J_k^{\text{action}} = \Psi_k + C_k^{\text{budget}}. \end{aligned} \quad (\text{QOT39})$$

The unequal flag endpoints, the full CEP/ α_A transfer, both low metrics, the arithmetic source comparison, and the correlated mixed pairing remain. In particular the matching residual in C_k^{budget} still cancels the same term in Ψ_k . A sharper source-order comparison of a kernel volume alone does not lower the exactly combined untilted action allowance.

197.9 Exact constants, verification, and remaining calculation

The constants in the two new limiting allowances satisfy

$$\begin{aligned} 10.535663175398 &< 8 \log(2 + \sqrt{3}) < 10.535663175399, \\ 42.142652701594 &< 32 \log(2 + \sqrt{3}) < 42.142652701595, \\ 15.090354888959 &< 4 + 16 \log 2 < 15.090354888960. \end{aligned} \quad (\text{QOT40})$$

They have an elementary rational certificate. Enclose $\sqrt{3}$ between

$$\frac{1732050807568877293527446341505}{10^{30}}$$

and that rational plus 10^{-30} , checking their squares. Use

$$\log(2 + \sqrt{3}) = \frac{2}{\sqrt{3}} \sum_{n \geq 0} \frac{3^{-n}}{2n+1},$$

whose tail after m terms is at most $3^{-m}/[(2m+1)(1-1/3)]$. Fifty terms suffice. For $\log 2$ use $2\sum_{n\geq 0}(1/3)^{2n+1}/(2n+1)$ and its geometric tail. All endpoints in (QOT40) follow by rational arithmetic. These are constants, not certified actual period inputs.

The new results are the full native order transport, the explicit $O(q_k)$ allocation and common-long-loss order allowances, the evaluated low-endpoint q_k coefficient, and the $o(q_k)$ paired primitive-to-connecting-image and compatibility differences. They use the complete original fibre maps. The older source-band and total first-variation theorems are not claimed as new results.

The original order-one common long loss, the signed combination P_e^{rest} with its remaining quotients, and the individual kq_k -scale coefficients of S_K, S_R remain uncomputed. The present result removes an order-transport obstruction: a subsequent order-one leading interval transfers to the original paired orders with a now specified lower-order error. It does not itself furnish an unconditional narrower numerical leading interval for an individual allocation, an RH exclusion, or a certified offcritical zero. No independent mathematical review or Lean certification is asserted. Execution and PDF inspection records, when delivered, have their separate declared scope.

197.10 Original quantities, inherited inputs and exact comparison domains

Fix the actual hypothetical simple quartet, the full arithmetic unit, one original five-orbit period and its logarithm branch. Throughout

$$\begin{aligned}
 k = 4l + 1 &\geq 29, \quad q = (k + 1)^2, \quad q' = (k - 7)^2, \quad \Delta = 16k - 48, \quad c' = k/2 - 4, \\
 \beta_{a,b}^{(n)} &= n/2 + (2a - n)\delta + i(2b - n)\gamma, \quad 0 < \delta < 1/2, \quad \gamma > 2, \\
 \chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{a,b}^{(n)}), \quad \psi = \chi_{k-8}, \quad E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{a,b}^{(8)} z}, \\
 v = \text{ord}_0 E_A, \quad \mu_v &= E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v, \quad \mathfrak{m} = \Delta - v.
 \end{aligned} \tag{EOR1}$$

The a_{ab} are the actual fixed-period coefficients, not free numerical inputs. Their nonreduced conductor is unchanged; the exact v occurs in all cutoffs. The fixed bound $v \leq 80$ suffices. No higher-multiplicity metric statement is claimed.

The two source orders are exactly $s = 1, k$ with

$$\begin{aligned}
 dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
 h_{n,s} &= M_s n!(s/2)_n, \quad \vartheta = \pi/2.
 \end{aligned} \tag{EOR2}$$

Retain the complete original relation matrix $\mathbf{U}_N$, whose columns are $\alpha_A \mathcal{T}_A(\chi_k S^d)/\psi$ for $0 \leq d \leq N - q$, where $\mathcal{T}_A p(S') = \sum a_{ab} p(S' + \beta_{a,b}^{(8)})$. The increasing low-monomial inclusion is E ; Z_k and P_Y are the original kernel frame and residual quotient map. Define the same attained forms

$$\begin{aligned}
 z^* H_{Y,N}^{(s)} z &= \min_h \int_{\mathbb{R}} |(Ez + \mathbf{U}_N h)(c' + iy)|^2 |\psi(c' + iy)|^2 dm_s(y), \\
 H_{K,N}^{(s)} &= Z_k^* H_{Y,N}^{(s)} Z_k, \quad H_{R,N}^{(s)} = [P_Y (H_{Y,N}^{(s)})^{-1} P_Y^*]^{-1}.
 \end{aligned} \tag{EOR3}$$

Both minima are over the full original affine fibres. At the four endpoints

$$\begin{aligned}
 (N_0, N_1, N_2, N_3) &= (q - 1, q, 2q - 1, 2q), \quad (\sigma_0, \sigma_1, \sigma_2, \sigma_3) = (+, +, -, -), \\
 a_N &= N - v - q', \quad t_0 = \dim(K \cap \mathbb{C}[S]_{<v}), \\
 t_K &= 8k - 16 - t_0, \quad t_R = 8k - 32 - v + t_0, \\
 S_X(k; s) &= \sum_{i=0}^3 \sigma_i \log \det H_{X,N_i}^{(s)}, \quad \Delta S_X = S_X(k; k) - S_X(k; 1), \quad X = K, R.
 \end{aligned} \tag{EOR4}$$

Zero-dimensional determinants are one. The notation ΔS_X is a source-order difference, not the root-strip dimension Δ .

We use GPA's full-form theorem, not a determinant consequence. Its finite domain is

$$R_k = k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q, \quad k \geq 29. \tag{EOR5}$$

The new high-degree localization has additional explicit guards in Section 197.12. All hold eventually for each fixed stipulated packet. The old finite QOT bounds remain when a new guard is not satisfied. Arithmetic maps retain their original period domain $|u| \geq R_*$ or the supplied exact period tests. No moving-period uniformity is inferred.

197.10.1 The complete inherited finite logarithmic constants

For later use reproduce the constants of GPA/QOT. Let $\epsilon = 2^{-32}$, $h_q = 1/q$, $\ell_s = (s-1)/4$, $Q_s = q' + \ell_s$, and

$$\begin{aligned}
\beta_s &= \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \quad \beta_1 = 1, \quad C_L = \pi/\Gamma(3/4)^2, \\
K_q &= \frac{\sqrt{2\pi}}{C_L} \frac{(2q+1)^2}{q} e^{-2q}, \quad \mathcal{E}_k = \frac{qR_k^2}{\epsilon^2 q^2 - R_k^2} + \log(1 + K_q), \\
r_{\ell_s} &= \frac{\sum_{a=0}^{\ell_s-1} (2a+1/2)^2}{\epsilon^2 q^2} + \log(1 + K_q) \quad (\ell_s > 0), \quad r_0 = 0, \\
L_h &= C_L (\sin h_q)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h_q)^{-1/2}, \\
K_{a,Q} &= 2Q + 1 + 2a + 2\sqrt{a(a+2Q)}, \\
b_{k,a,s}^- &= -\mathcal{E}_k + \log L_h - K_{a,Q_s} \log(1 + h_q/\vartheta), \\
b_{k,a,s}^+ &= \mathcal{E}_k + r_{\ell_s} + \log U_h + K_{a,Q_s} \log(1/(1 - h_q/\vartheta)).
\end{aligned} \tag{EOR6}$$

Write $\|p\|_Q^2 = \int_{\mathbb{R}} |p(y)|^2 |y|^{2Q} e^{-\vartheta|y|} dy$. For every original polynomial of degree at most a in the proved degree domain, GPA gives

$$\beta_s e^{b_{k,a,s}^-} \|p(c' + i \cdot)\|_{Q_s}^2 \leq \|\psi p\|_{s,c'}^2 \leq \beta_s e^{b_{k,a,s}^+} \|p(c' + i \cdot)\|_{Q_s}^2. \tag{EOR7}$$

These are bounds on the entire polynomial norm. The original inside-region error, root pairs and source mass are in (EOR6); none is reinterpreted as a small inverse. At the degrees used here $b^\pm = O_{\text{actual}}(\log q)$. This remains the inherited analytic input, not a new proof attribution.

197.11 The original equilibrium and an explicit exterior gap

Use the proved EIQ equilibrium for $V_{\alpha,\beta}(x) = \beta\sqrt{x} - \alpha \log x$ on $(0, \infty)$. For clarity denote its squared support endpoints by A, B and its density by ϱ . Thus

$$\begin{aligned}
A &= u^2, \quad B = w^2, \quad uK(\kappa) = \alpha\pi/\beta, \quad wE(\kappa) = (\alpha+2)\pi/\beta, \quad \kappa^2 = 1 - u^2/w^2, \\
\mathcal{P}(z) &= \int_A^B \log|z - t| \varrho(t) dt, \quad D(x) = V(x) - 2\mathcal{P}(x) - \lambda \geq 0, \quad D|_{[A,B]} = 0, \\
\mathfrak{h}(x) &= \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s} ds}{(x+s)\sqrt{(A+s)(B+s)}} > 0, \\
D'(x) &= \sqrt{(x-A)(x-B)} \mathfrak{h}(x), \quad x \notin [A, B].
\end{aligned} \tag{EOR8}$$

The last square root is negative on $0 < x < A$ and positive on $x > B$. These are EIQ10–19's actual density and resolvent, including the strict left exterior inequality. EIQ3–6 and 29 prove smooth positive endpoint dependence on the positive parameter quadrant.

For each such parameter pair define explicit positive constants

$$\begin{aligned}
g &= B - A, \quad h_* = \frac{\beta}{2\pi(B+1)(B+2)(B+3)}, \quad c_* = \frac{2}{3} h_* \sqrt{g}, \\
\kappa_* &= \min \left\{ Ah_* \sqrt{g}, \frac{\beta\sqrt{B}}{6\pi(3B+2)} \right\}, \quad \eta_* = \min\{A/2, 1\}, \\
L &= \max_{x \in [A,B]} |\beta/(2\sqrt{x}) - \alpha/x|, \quad C_*^2 = 32e^{3L/2}.
\end{aligned} \tag{EOR9}$$

The stars here label local gap constants, not any canonical allowance from the programme. The maximum L is an elementary finite maximum: evaluate the absolute expression at A, B and at $(4\alpha/\beta)^2$ if that point belongs to $[A, B]$.

Lemma QOT11.1 (Both exterior tails, with a global upper derivative bound). *For $0 < \eta \leq \eta_*$,*

$$\begin{aligned} D(x) &\geq c_* \eta^{3/2} \quad (0 < x \leq A - \eta \text{ or } x \geq B + \eta), \\ xD'(x) &\geq \kappa_* \sqrt{\eta} \quad (x \geq B + \eta). \end{aligned} \tag{EOR10}$$

Proof. In the integral in (EOR8), restrict s to $[1, 2]$. For $x \in [A/2, B+1]$, $x \leq B+1$, $x+s \leq B+3$ and $\sqrt{(A+s)(B+s)} \leq B+2$, whereas $\sqrt{s} \geq 1$. Thus $\mathfrak{h}(x) \geq h_*$. On either endpoint segment of length η , $|D'(x)| \geq h_* \sqrt{g} \sqrt{\text{dist}(x, [A, B])}$. Integrating proves the first assertion at the segment endpoints. Monotonicity from (EOR8) proves it on both entire exterior tails.

For $B + \eta \leq x \leq B + 1$ the same bound gives $xD'(x) \geq Ah_* \sqrt{g} \sqrt{\eta}$. For $x \geq B + 1$ restrict the integral instead to $s \in [x, 2x]$ to obtain

$$\mathfrak{h}(x) \geq \frac{\beta}{6\pi\sqrt{x}(B+2x)}, \quad xD'(x) \geq \frac{\beta\sqrt{x}(x-B)}{6\pi(B+2x)} \geq \frac{\beta\sqrt{B}}{6\pi(3B+2)}.$$

The last ratio is minimized at $x = B + 1$ after the factor $\sqrt{x}$ is bounded below by $\sqrt{B}$. Since $\sqrt{\eta} \leq 1$, this proves the second assertion with the displayed common constant. All inequalities concern the same EIQ potential and its complete positive half-line. $\square$

197.12 Uniform polynomial localization and a growing moment shift

Let m be a positive integer and P any complex polynomial of degree at most m . Set $\|P\|_{m,V}^2 = \int_0^\infty |P(x)|^2 e^{-mV(x)} dx$. The following complete evaluation bound is the argument of OOQ7, restated to specify its constants for the parameter pair used here:

$$|P(x)|^2 e^{-mV(x)} \leq C_*^2 m^4 e^{-mD(x)} \|P\|_{m,V}^2, \quad x \notin [A, B], \quad m \geq \max\{1, 2/g\}. \tag{EOR11}$$

For completeness, let $M = \max_{[A,B]} |P| e^{-mV/2}$, attained at x_0 . One side of x_0 has length $1/m$ inside the support. There $|P(x)| \leq e^{L/2} |P(x_0)|$. A Chebyshev expansion on an interval I proves $\|P'\|_{\infty,I} \leq 4m^3 |I|^{-1} \|P\|_{\infty,I}$: the nonconstant coefficients have modulus at most $2\|P\|_\infty$ and $|T'_j| \leq j^2$, so the sum and affine derivative factor give the bound. It follows that $|P| \geq |P(x_0)|/2$ on a segment of length $(8e^{L/2}m^4)^{-1}$. The weight there is at least $e^{-L}e^{-mV(x_0)}$. Hence $\|P\|_{m,V}^2 \geq M^2/(32e^{3L/2}m^4)$. Finally $\log|P(z)| - m\mathcal{P}(z)$ is subharmonic off the support, where $\mathcal{P}$ is the logarithmic potential in (EOR8). The equilibrium equality bounds its boundary values by $\log M + m\lambda/2$. It extends subharmonically at infinity, since $\deg P \leq m$. The maximum principle proves (EOR11). Zeros give only negative singularities. This uses the logarithmic potential, not a partition-function error estimate.

Define

$$\begin{aligned} \eta_m &= \left(\frac{\log(C_*^2(B+2)) + 10\log(m+1)}{c_* m} \right)^{2/3}, \quad \varepsilon_m = (m+1)^{-6}, \\ m &\geq \max\{1, 2/g\}, \quad \eta_m \leq \eta_*, \quad m\kappa_* \sqrt{\eta_m} \geq \ell + 2. \end{aligned} \tag{EOR12}$$

The last line gives all additional finite guards; $\ell \geq 0$ may be real.

Theorem QOT12.1 (Uniform norm comparison from the actual support). *Under (EOR12), every degree-at-most- m complex polynomial satisfies*

$$(A - \eta_m)^\ell (1 - \varepsilon_m) \|P\|_{m,V}^2 \leq \int_0^\infty x^\ell |P(x)|^2 e^{-mV(x)} dx \leq (B + \eta_m)^\ell (1 + \varepsilon_m) \|P\|_{m,V}^2.$$

(EOR13)

Proof. For any $0 < \eta \leq \eta_*$, (EOR11) and the left gap give

$$\frac{\int_0^{A-\eta} |P|^2 e^{-mV} dx}{\|P\|_{m,V}^2} \leq C_*^2 m^4 A e^{-mc_* \eta^{3/2}}.$$

Consequently the numerator in (EOR13) is at least $(A - \eta)^\ell$ times one minus this mass fraction. For the upper tail integrate the derivative inequality in (EOR10):

$$D(x) \geq c_* \eta^{3/2} + \kappa_* \sqrt{\eta} \log \frac{x}{B + \eta}, \quad x \geq B + \eta.$$

If $m\kappa_* \sqrt{\eta} > \ell + 1$, direct integration gives

$$\int_{B+\eta}^{\infty} x^\ell e^{-mD(x)} dx \leq \frac{(B + \eta)^{\ell+1} e^{-mc_* \eta^{3/2}}}{m\kappa_* \sqrt{\eta} - \ell - 1}.$$

The portion up to $B + \eta$ is bounded by $(B + \eta)^\ell \|P\|_{m,V}^2$; use (EOR11) on the rest. At the chosen η_m ,

$$e^{-mc_* \eta_m^{3/2}} = [C_*^2 (B + 2)(m + 1)^{10}]^{-1}.$$

The lower omitted mass and the upper added fraction, after division by $(B + \eta_m)^\ell$, are both at most $(m + 1)^{-6}$: $A < B + 2$, $B + \eta_m < B + 2$, $m^4 \leq (m + 1)^4$, and the upper denominator is at least one. This proves both full integral inequalities, without assuming that P avoids either tail. $\square$

On any compact positive parameter rectangle the constants in (EOR9) have uniform positive lower bounds and finite upper bounds. This follows from the proved smooth positive endpoints; it is not a condition on an unspecified polynomial. Thus, uniformly for $\ell = O(\sqrt{m})$,

$$\eta_m = O((\log(m + 1)/m)^{2/3}), \quad \ell \eta_m = o(1), \quad \varepsilon_m = O(m^{-6}). \quad (\text{EOR14})$$

All guards hold eventually. In particular the logarithmic endpoints in (EOR13) are $\ell \log A + o(1)$ and $\ell \log B + o(1)$. These are uniform enclosures; equality with either endpoint has not been asserted.

197.13 Both square-map sheets and the finite improved native interval

Apply the localization to the high polynomial degrees in (EOR4). For a general polynomial $p_c(y) = p(c' + iy)$ of degree at most a , write exactly

$$p_c(y) = P_0(y^2/q^2) + (y/q)P_1(y^2/q^2), \quad m_0 = \lfloor a/2 \rfloor, \quad m_1 = \lfloor (a - 1)/2 \rfloor. \quad (\text{EOR15})$$

This defines both P_ϵ from the same original coefficients, with no reality assumption. The odd cross terms integrate to zero in the even reference weight, so

$$\|p_c\|_{q'}^2 = q^{2q'+1} \sum_{\epsilon=0}^1 \int_0^\infty |P_\epsilon(x)|^2 x^{q'+\epsilon-1/2} e^{-\vartheta q \sqrt{x}} dx. \quad (\text{EOR16})$$

The factor $q^{2q'+1}$ includes both half-lines and their Jacobians. After shifting q' to $q' + \ell$, the right side has the extra factor $q^{2\ell} x^\ell$. Both parity blocks remain.

For each $\epsilon = 0, 1$ set

$$\alpha_\epsilon = \frac{q' + \epsilon - 1/2}{m_\epsilon}, \quad \beta_\epsilon = \frac{\vartheta q}{m_\epsilon}, \quad [A_\epsilon, B_\epsilon] = \text{supp } \mu_{\alpha_\epsilon, \beta_\epsilon}. \quad (\text{EOR17})$$

Use the localization only when $m_\epsilon \geq 1$, both parameter pairs belong to $[3/2, 5/2] \times [\pi/2, 3\pi/2]$, and (EOR12) holds in both blocks. These tests are explicit in the actual integers and the unique

EIQ endpoints. For the two original high cutoffs they hold eventually. Below that domain keep QOT's finite Laguerre bounds unchanged.

Let $\eta_\epsilon, \varepsilon_\epsilon$ be (EOR12)'s values. Define

$$\begin{aligned} F_{k,a}^- &= \min_{\epsilon=0,1} \{ \ell \log(A_\epsilon - \eta_\epsilon) + \log(1 - \varepsilon_\epsilon) \}, \\ F_{k,a}^+ &= \max_{\epsilon=0,1} \{ \ell \log(B_\epsilon + \eta_\epsilon) + \log(1 + \varepsilon_\epsilon) \}, \\ L_{k,a}^E &= b_{k,a,k}^- - b_{k,a,1}^+ + 2\ell \log \frac{q\vartheta}{2q'} + F_{k,a}^-, \\ U_{k,a}^E &= b_{k,a,k}^+ - b_{k,a,1}^- + 2\ell \log \frac{q\vartheta}{2q'} + F_{k,a}^+, \\ \kappa_k &= \beta_k (4q'/\pi)^{2\ell} = 2^{10\ell} \frac{(2\ell)!}{(4\ell)!} (k-7)^{4\ell}. \end{aligned} \tag{EOR18}$$

The comparison centre is exactly the earlier rational κ_k ; changing the bounding endpoints does not change its meaning or absorb a mass.

Theorem QOT13.1 (Full high-endpoint source-order comparison). *For every degree-at-most- a original polynomial in the stated joint domain,*

$$\kappa_k e^{L_{k,a}^E} \|\psi p\|_{1,c'}^2 \leq \|\psi p\|_{k,c'}^2 \leq \kappa_k e^{U_{k,a}^E} \|\psi p\|_{1,c'}^2. \tag{EOR19}$$

It holds after every original fixed restriction or attained quotient, in particular all three forms in (EOR3) at the actual cutoff $a = a_N$. On their original complex-linear dual metric it is reversed:

$$\kappa_k^{-1} e^{-U_{k,a}^E} D^{(1)} \preceq D^{(k)} \preceq \kappa_k^{-1} e^{-L_{k,a}^E} D^{(1)}. \tag{EOR20}$$

Proof. Apply (EOR13) separately to the exact two summands in (EOR16). Taking the minimum lower constant and maximum upper constant yields $e^{2\ell \log q + F^-} \|p_c\|_{q'}^2 \leq \|p_c\|_{q'+\ell}^2 \leq e^{2\ell \log q + F^+} \|p_c\|_{q'}^2$. Combine the lower inequality with the lower order- k and upper order-one sides of (EOR7), and combine the opposite three inequalities for the upper side. This is exactly (EOR18)–(EOR19). For each affine fibre every candidate lift satisfies the two inequalities; taking both infima preserves them. For a restriction substitute the full fixed columns. These operations keep all relation, kernel and residual mixed blocks. Positive-form inversion reverses the order, and the source-independent congruence $D = A_{\ell,k}^* \overline{H_Y^{-1}} A_{\ell,k}$ gives (EOR20). It too passes to the same fixed target flag subquotients by the identical minimum argument. No norm is imposed on an observed coordinate space. $\square$

For a guaranteed intersection with the earlier finite bounds reproduce QOT's endpoints. For $\alpha_k = 2q'$ put

$$d_{k,a}^- = 4 \sum_{r=0}^{\ell-1} \operatorname{arsinh} \sqrt{(a+r+1/2)/\alpha_k}, \quad d_{k,a}^+ = 4 \sum_{r=1}^{\ell} \operatorname{arsinh} \sqrt{(a+r+1/2)/\alpha_k},$$

$$L_{k,a}^Q = b_{k,a,k}^- - b_{k,a,1}^+ - d_{k,a}^-, \quad U_{k,a}^Q = b_{k,a,k}^+ - b_{k,a,1}^- + d_{k,a}^+.$$

At an endpoint where the new guards hold, use

$$\bar{b}_{k,a} = \max\{L_{k,a}^Q, L_{k,a}^E\}, \quad \bar{b}_{k,a} = \min\{U_{k,a}^Q, U_{k,a}^E\}. \tag{EOR21}$$

Else use the QOT endpoints alone. The new finite enclosure never discards a stronger earlier one. QOT's low bounds remain part of this definition.

197.14 Evaluated asymmetric constants at scale q_k

Let u_*, v_* denote the original unsquared EIQ endpoints at $(2, \pi)$:

$$u_* K(\kappa) = 2, \quad v_* E(\kappa) = 4, \quad \kappa^2 = 1 - u_*^2/v_*^2. \quad (\text{EOR22})$$

They are not period variables, and no independent Gamma order is introduced. At either high cutoff $a_N = q + \Delta - v + O(1)$, so

$$m_\epsilon = q/2 + O(k), \quad (\alpha_\epsilon, \beta_\epsilon) = (2, \pi) + O(k^{-1}).$$

Smooth endpoint dependence, (EOR14), and $b^\pm = O_{\text{actual}}(\log q)$ now give

$$\begin{aligned} L_{k,a_N}^E &= ka_- + O_{\text{actual}}(\log q), \quad a_- = \frac{1}{2} \log(\pi u_*/4), \\ U_{k,a_N}^E &= ka_+ + O_{\text{actual}}(\log q), \quad a_+ = \frac{1}{2} \log(\pi v_*/4). \end{aligned} \quad (\text{EOR23})$$

For example $F^- = (k/4) \log(u_*^2) + O(1)$ because $\ell \eta_{m_\epsilon} = o(1)$ and the endpoint displacement costs $O(\ell/k) = O(1)$. The remaining centre is $2\ell \log(\pi q/(4q')) = (k/2) \log(\pi/4) + O(1)$. The upper proof is identical. These are asymptotics of proved bounding constants, not a claim about the unknown eigenvalues of the actual high compressed forms.

Taking rank- t_X determinants and the four original signs in (EOR19) yields the finite interval

$$\begin{aligned} t_X(\underline{b}_{k,a_{N_0}} + \underline{b}_{k,a_{N_1}} - \bar{b}_{k,a_{N_2}} - \bar{b}_{k,a_{N_3}}) &\leq \Delta S_X \\ &\leq t_X(\bar{b}_{k,a_{N_0}} + \bar{b}_{k,a_{N_1}} - \underline{b}_{k,a_{N_2}} - \underline{b}_{k,a_{N_3}}). \end{aligned} \quad (\text{EOR24})$$

The four identical rank factors multiplying $\log \kappa_k$ cancel exactly here, not before the endpoint forms are defined. At a low endpoint $a_N = O(k)$ the retained QOT endpoints are $O_{\text{actual}}(\sqrt{k} + \log q)$; this follows directly from $\text{arsinh } x \leq x$ in its sums. Since $t_X = 8k + O(1)$,

Theorem QOT14.1 (Narrower source-order allocation interval). *For either original allocation $X = K, R$ at a fixed stipulated packet and period,*

$$\boxed{-C_+ \leq \liminf_{k \equiv 1 \pmod{4}} \frac{\Delta S_X}{q_k} \leq \limsup_{k \equiv 1 \pmod{4}} \frac{\Delta S_X}{q_k} \leq C_-,} \quad (\text{EOR25})$$

where

$$C_+ = 8 \log(\pi v_*/4), \quad C_- = 8 \log(4/(\pi u_*)). \quad (\text{EOR26})$$

More explicitly the finite interval (EOR24) implies $-C_+ q_k - O_{\text{actual}}(k^{3/2} + k \log q_k) \leq \Delta S_X \leq C_- q_k + O_{\text{actual}}(k^{3/2} + k \log q_k)$. No unique normalized limit is asserted.

Proof. Use (EOR23) at the two high endpoints and the preceding low bound. The lower centre is $-2t_X ka_+$, the upper centre is $-2t_X ka_-$. Replacing $t_X k$ by $8q_k$ costs only $O(k)$ because t_0, v are bounded. Substitution of $a_\pm$ gives the stated constants and finite order of the remaining allowance. The explicit finite endpoints, rather than an unspecified big- O constant, are available in (EOR24). $\square$

197.15 The original common long loss and its fixed-frame receiver

Use $j = k + 4$ and the original paired source orders $(1, 1)$ and (k, j) . In target coordinates the source-independent metric is $\mathcal{D}_{j,N} = \text{diag}(D_{j,N-v}, 4)$. Here “source-independent” refers to the fixed coefficient space and flags; the metric is $\mathcal{D}_{j,N}^{(s_1)} = \text{diag}(D_{j,N-v}^{(s_1)}, 4)$ and depends on the original source order s_1 . The complete original flags are

$$\begin{aligned} I &= \text{im}(B_{\text{cof}} J_0), \quad B_0 = \text{im } B_{\text{cof}}, \quad S = \text{im}[B_{\text{cof}}, F_e], \quad J = \ker \mathbb{B}, \\ K_i &= I \cap J, \quad W_i = I + J, \quad W = B_0 + J, \quad K_c = S \cap W, \quad T = S + J, \\ \omega_i &= \frac{\det \text{Gram}(W_i/J)}{\det \text{Gram}(I/K_i)}, \quad \omega_c = \frac{\det \text{Gram}(T/W)}{\det \text{Gram}(S/K_c)}, \quad R_e = \dim(I/K_i) + \dim(S/K_c) \leq 16k - 8. \end{aligned} \quad (\text{EOR27})$$

All Grams are their attained native quotient forms in matched value frames. The fixedness of F_e under the source presentation follows from the same Taylor cochain equation and uniqueness modulo all lower evaluation columns, as in QOT's full primitive map. No corresponding denominator or multiplier kernel is changed by the present scalar comparison.

Let $\mathcal{R}$ have signs $(-, -, +, +)$ at $N = q_j - 1, q_j, 2q_j - 1, 2q_j$ and write $\mathcal{L}_e = \mathcal{R} \log(\omega_c \omega_i)$. With $w_{j,a} = \bar{b}_{j,a} - \underline{b}_{j,a}$, subtraction of the matched quotient determinant intervals in (EOR20) proves

$$\boxed{|\mathcal{L}_e^{(k,j)} - \mathcal{L}_e^{(1,1)}| \leq R_e \sum_N w_{j,N-v-q'_j}.} \quad (\text{EOR28})$$

The common factor κ_j^{-r} cancels in each rank- r quotient contraction at each endpoint. Thus a small observation inverse is not needed. This estimate is applied at four full endpoint forms, not summed as an independent error over q_j rows. By (EOR23), each high width equals $\frac{j}{2} \log(v_*/u_*) + O_{\text{actual}}(\log q_j)$, and both low widths are $O_{\text{actual}}(\sqrt{j} + \log q_j)$. Hence

$$\boxed{\limsup \frac{|\mathcal{L}_e^{(k,k+4)} - \mathcal{L}_e^{(1,1)}|}{q_k} \leq C_{\text{com}} := 16 \log(v_*/u_*) = 2(C_+ + C_-).} \quad (\text{EOR29})$$

The error about this bound is $O_{\text{actual}}(k^{3/2} + k \log q_k)$, with the complete finite version (EOR28). This does not replace the earlier one-budget estimate for the absolute order-one common long loss by an evaluated value.

197.15.1 The actual rows, half-lines and all remaining terms

The unchanged source-coordinate pullback is $(P_{c',a})_{r,n} = \binom{n}{r} (c')^{n-r} i^r$. The full power-exponential reference rows remain

$$\begin{aligned} A_{b,r} &= (-1)^b \binom{r}{b} \Gamma(r + 2Q_s + 1) \sqrt{\frac{b!}{\Gamma(b + 2Q_s + 1)}} \vartheta^{-r-(2Q_s+1)/2}, \\ R_{b,r} &= (-1)^b 2^{r-b} \binom{2Q_s + r}{r-b} \sqrt{\frac{r! \Gamma(b + 2Q_s + 1)}{b! \Gamma(r + 2Q_s + 1)}}, \\ W &= \begin{bmatrix} \text{AP}_{c',a} \\ \text{RAP}_{c',a} \end{bmatrix}. \end{aligned} \quad (\text{EOR30})$$

Zero triangular entries are understood. Formula (EOR16) is merely a second exact coefficient presentation of this same complete two-half-line norm. For every original column x , both row blocks of Wx remain; only their integrated even/odd cross terms cancel by symmetry of the reference density. Original period coefficients are not conjugated by reflection. The new theorem acts on these full quadratic forms before either Schur minimum.

The native rows are still calculated from the original symbol and Gamma recurrence:

$$\begin{aligned} f_{a,n} &= \frac{\nu_{a,n+v} - \sum_{d=v+1}^{n+v} \binom{n+v}{d} \mu_d f_{a,n+v-d}}{\binom{n+v}{v} \mu_v}, \quad \nu_{a,n} = \sum_{u,w} (Z_{\text{exp},j})_{uw,a} (\beta_{u,w}^{(j)})^n, \\ b_{n+1,r} &= i^{-1} (b_{n,r-1} - c'_j b_{n,r}) - n(n-1 + s_1/2) b_{n-1,r}, \\ t_n &= \left(\frac{\sum_r b_{n,r} f_{a,r}}{\sqrt{M_{s_1} n! (s_1/2)_n}} \right)_a, \quad v_n = \left(\frac{\sum_r b_{n,r} (\beta')^r}{\sqrt{M_{s_1} n! (s_1/2)_n}} \right)_{\beta'}. \end{aligned} \quad (\text{EOR31})$$

In a new step the preceding complete lower Gram is V^*V and the corrected row is $\rho_n = t_n - v_n(V^*V)^{-1}V^*T$. Its update denominator is $1 + v_n(V^*V)^{-1}v_n^*$. The subsequent boundary or invariant minimum also remains. None of these rows or denominators is approximated by its rank in (EOR28).

For an order-one interval $[L_{X,1}, U_{X,1}]$, add the two finite endpoints in (EOR24), intersect with all earlier bounds, and couple by the exact $S_R = \mathcal{T} - S_K$. In the common loss add and subtract the explicit allowance in (EOR28). Retain separately

$$\begin{aligned} \mathcal{L}_e &= P_e^{\text{rest}} - \mathcal{V}_b, & \mathcal{V}_b + \mathcal{V}_{\text{inv}} + \mathcal{V}_o &= 4\hat{F}_j, \\ F_K^{(s)} &= S_{K,s} - \mathcal{E}_s^{\text{flag}} + F_L^{(s)} - F_{W_0}^{(s)}, & \Phi_{k,s} &= F_{W_0}^{(s)} + S_{R,s} + \delta_{0,k,s}, \\ \Psi_k &= F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma, & J_k^{\text{action}} &= \Psi_k + C_k^{\text{budget}}. \end{aligned} \quad (\text{EOR32})$$

The original source denominator, A_{inv} , Q_b , Q_o , both target gauges, both low metrics, unequal flag errors and the complete source-transfer correction are unchanged. The matching residual cancellation in the full action budget is not altered by narrowing a source-order interval. The previously proved low primitive $o(q_k)$ paired source/image comparison is retained without being reattributed as a new theorem.

197.16 Certified constants and the precise remaining task

Let $r = u_*/v_* \in (0, 1)$ and define the normalized elliptic integrals

$$\bar{K} = \frac{2}{\pi} K(\sqrt{1-r^2}), \quad \bar{E} = \frac{2}{\pi} E(\sqrt{1-r^2}).$$

The endpoint equation is $r\bar{K}/\bar{E} = 1/2$, and

$$C_+ = 8 \log(2/\bar{E}), \quad C_- = 8 \log \bar{K}, \quad C_{\text{com}} = -16 \log r. \quad (\text{EOR33})$$

Thus no numerical enclosure of π is required for these three constants. The accompanying integer/rational calculation certifies

$$\begin{aligned} 0.16010167095771835588456745 &< r < 0.16010167095771835588456746, \\ 8.881817865596 &< C_+ < 8.881817865597, \\ 5.773751910973 &< C_- < 5.773751910974, \\ 29.311139553140 &< C_{\text{com}} < 29.311139553141. \end{aligned} \quad (\text{EOR34})$$

Here is the rigorous basis of that certificate. For $m = 1 - r^2$ and $c_n = \binom{2n}{n}^2 / 16^n$,

$$\bar{K} = \sum_{n \geq 0} c_n m^n, \quad \bar{E} = 1 - \sum_{n \geq 1} \frac{c_n m^n}{2n-1}.$$

These follow by binomial expansion in the defining elliptic integrals and the integral of $\sin^{2n} \theta$. If $t_n = c_n m^n$, then $t_{n+1}/t_n = m((2n+1)/(2n+2))^2 \leq m$. After degree N , the omitted $\bar{K}$ tail is at most $t_N m / (1-m)$ and the omitted $1 - \bar{E}$ tail at most $t_N m / ((1-m)(2N+1))$. At both rational endpoints in (EOR34), use $N = 4096$, enclosing each recurrence operation by integer floor and ceiling at a fixed denominator 10^{80} . The resulting intervals certify opposite signs of $2r\bar{K} - \bar{E}$. EIQ's strict monotonicity then encloses the unique root. Monotonicity in m encloses both normalized elliptic integrals between these endpoint evaluations. Finally, for $x > 1$ and $z = (x-1)/(x+1)$, use

$$\log x = 2 \sum_{n=0}^{M-1} \frac{z^{2n+1}}{2n+1} + R_M, \quad 0 < R_M \leq \frac{2z^{2M+1}}{(2M+1)(1-z^2)}.$$

Integer interval arithmetic with $M = 400$ proves the displayed decimal brackets. The full script records its actual rational endpoints and rounding directions. These are universal-constant certificates, not actual-period evaluations.

The new allocation interval $[-C_+, C_-]$ is strictly contained in the earlier $[-8 \log(2 + \sqrt{3}), 8 \log(2 + \sqrt{3})]$ interval. The new common source-order allowance is strictly below

$32 \log(2 + \sqrt{3})$. These are genuine narrower q_k -scale enclosures for the same differences. They do not assert optimality of the enclosures or existence of either normalized source-order limit.

What remains is the absolute order-one common long loss and its allocation between the actual kernel and residual. Neither an individual kq_k -scale coefficient nor a narrower unconditional interval for that coefficient is assigned here. The current result makes a subsequent order-one estimate more precisely transferable; it does not substitute for that estimate. The source comparison is uniform in all polynomial directions on its stated finite domain; the later arithmetic and geometric maps retain their fixed-period and simple-quartet hypotheses. No RH conclusion, actual offcritical-zero certificate, independent referee acceptance, or Lean execution is claimed.

References

- [1] Split-Zero research programme. (2026). *Current kernel and multiplicity proofs* (supplied 498-page edition). Complete GPA1–23, LVM1–19, DNE, SAR, CIT/ITL receiving providers. Original source filename: `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS(6).tex`.
- [2] Split-Zero research programme. (2026). *Full primitive connecting proofs*. Complete PDS1–24, CIT1–26 and PDI1–16 providers in the supplied 13-page source; the same-coordinate argument is also recorded in PSR23–24 of the later 40-page supplement.
- [3] Split-Zero research programme. (2026). *Primitive-band continuation* (549-page edition and 40-page supplement). Repository path `workbenches/splitzero-tandem/continuations/20260915-primitive-band-continuation`; consulted at commit `1efd53337561d8f67cf0ac1d119acdaa222644c3`. Its evaluated totals and later RQB/PSR identities retain their own proof provenance.

198 The complete original action relative to its arithmetic total

This is a local reconstruction of the complete delivered response following the 16 September 2026, 15:02:50 prompt in `ChatGPT-Residual Lower Bound Continuation-20260916-2150.md`, lines 7043–7730. The response reported that its build tools timed out: no original standalone archive, compiled PDF, successful fresh Lean run or actual-period numerical certificate is inferred from it. This section supplies the additive proof in full. Its input source is the unchanged `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS.tex`; the original AMT, MCE and OMR formulas below are rederived at the points where used. The fixed arithmetic density-envelope constants are precisely AMT4–6’s proved constants. Their defining bounds and domains are displayed, rather than replaced by newly selected measures or a new positivity premise.

198.1 Domain, unchanged arithmetic objects and exact minimum metrics

Use the original stipulated full zero-quartet domain: fixed $0 < \delta < 1/2$, $\gamma > 2$, and complete common order $m \geq 1$ at the four distinct centres. The statement below concerns these original objects; it does not assert that such an off-line zero quartet exists. Write

$$\begin{aligned} \rho &= \tfrac{1}{2} + \delta + i\gamma, \quad \mathcal{R} = \{\rho, \bar{\rho}, 1 - \rho, 1 - \bar{\rho}\}, \quad h(s) = \prod_{\zeta \in \mathcal{R}} (s - \zeta)^m, \quad g = 2\xi, \quad v_h = g/h, \\ w_h(y) &= |v_h(1/2 + iy)|^2/(2\pi), \quad m_{h,b} = w_h^{*b}, \quad \mu_h = \int_{\mathbb{R}} w_h(y) dy, \\ k &= 4\ell + 1 \geq 9, \quad c = k/2, \quad e_k = 1 + k(m-1), \quad q = e_k(k+1)^2, \\ \beta_{ab} &= c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \beta_{ab})^{e_k}, \quad E = \mathbb{C}[S]/(\chi), \quad A = M_S. \end{aligned} \tag{CAI1}$$

The arithmetic source is $L^2(\mathbb{R}, m_{h,k}(y) dy)$ with its mass μ_h^k , evaluated in the literal coordinate $S = c + iy$. Its inner product is conjugate-linear in the first argument. For each $N \geq 0$ let $\mathcal{P}_N = \mathbb{C}[S]_{\leq N}$ and let $p_n(S) = i^n Q_n((S - c)/i)$ be the monic orthogonal polynomials; Q_n is monic in the real variable y . Their full squared norms are $\omega_{n,k}^{\text{ar}} > 0$. The substitution has inverse $P(S) = f((S - c)/i)$ and is never a change of the original source mass. In formulas below ω_n abbreviates this same arithmetic norm.

The exact remainder map $J_N : \mathcal{P}_N \rightarrow E$ is onto for $N \geq q - 1$, with kernel $\chi \mathcal{P}_{N-q}$, where $\mathcal{P}_{-1} = 0$. Put $b_n = [p_n]_{\chi}$ in the frame $1, S, \dots, S^{q-1}$. Then

$$R_N = \sum_{n=0}^N b_n b_n^* / \omega_n, \quad G_N = R_N^{-1}, \quad V_N = \det G_N. \tag{CAI2}$$

The first q columns form a unit-triangular monic remainder matrix, so $R_N > 0$. In any source coefficient frame with Gram H_N , the actual minimum section is $H_N^{-1} J_N^* G_N$. Its composition with J_N is the identity. A vector in $\ker J_N$ pairs to zero with this section; therefore the squared norm of another lift is its minimum squared norm plus the squared norm of that relation. This proves the interpretation of G_N without changing the admitted relation subspace.

The original period observation is the AMT3 map $\Lambda : E \twoheadrightarrow B$, with actual kernel inclusion $I : K \hookrightarrow E$. Explicitly, if $E_h = \mathbb{C}[s]/(h)$, $U = M_{j_h v_h}$, $\iota[P] = [P(s_1 + \dots + s_k)]$, Π is the original period matrix and P_k its original tensor invariant projector, then $j\Lambda = P_k \Pi^{\otimes k} U^{\otimes k} \iota$, where $j : B \hookrightarrow (\mathbb{C}^{4m})^{\otimes k}$ is the image inclusion. No period branch or kernel frame is changed. The action calculation below is independent of that choice because it uses the complete quotient E ; the exact relation to the original observation is the following splitting. In the fixed kernel and image frames set

$$H_{K,N} = I^* G_N I, \quad Q_{B,N} = (\Lambda G_N^{-1} \Lambda^*)^{-1}, \quad S_N = G_N^{-1} \Lambda^* Q_{B,N}. \tag{CAI3}$$

Direct multiplication proves $\Lambda S_N = 1$ and $I^* G_N S_N = 0$. For any fixed section S_* of Λ , the frame $[I, S_*]$ is invertible and $S_N - S_*$ has image in K . Thus changing $[I, S_*]$ to $[I, S_N]$ has determinant one. Orthogonality proves the exact identity

$$\det H_{K,N} \det Q_{B,N} = |\det[I, S_*]|^2 V_N. \quad (\text{CAI4})$$

Empty factors have determinant one, so this also covers zero-dimensional kernel or image. In particular no kernel invariance under A is needed.

For any one of the original sources $a = \text{ar}, 1, k$, define its full two-window return and the kernel and boundary returns by

$$\begin{aligned} \mathcal{B}_k^a &= \log \frac{V_{q-1}^a V_q^a}{V_{2q-1}^a V_{2q}^a}, \\ F_K^a &= \log \frac{\det H_{K,q-1}^a \det H_{K,q}^a}{\det H_{K,2q-1}^a \det H_{K,2q}^a}, \quad F_B^a = \log \frac{\det Q_{B,q-1}^a \det Q_{B,q}^a}{\det Q_{B,2q-1}^a \det Q_{B,2q}^a}, \\ F_K^a + F_B^a &= \mathcal{B}_k^a, \quad \delta_k^\sigma = \mathcal{B}_k^{\text{ar}} - \mathcal{B}_k^\sigma, \quad \mathcal{H}_k = \mathcal{B}_k^\sigma - \mathcal{B}_k^{(k)}. \end{aligned} \quad (\text{CAI5})$$

Here $a = 1$ denotes the original fixed Gamma source σ . The equality follows by multiplying CAI4 at the two lower degrees and dividing by its two upper-degree instances. Its four identical frame constants cancel exactly, rather than being set equal to one.

198.2 The actual rank-two control, reflection and exterior benchmark

The quotient v_h is entire at the selected centres because their complete zero orders have been divided out. The exact equalities $\xi(1-s) = \xi(s)$ and $h(1-s) = h(s)$ show that w_h and every $m_{h,b}$ are even. Monic orthogonality consequently gives $Q_n(-y) = (-1)^n Q_n(y)$. Self-adjoint multiplication by y , orthogonality to degrees below $n-1$, and pairing with Q_{n-1} prove the real monic recurrence. In the original S coordinate it is

$$Sp_n = p_{n+1} + cp_n - (\omega_n/\omega_{n-1})p_{n-1}, \quad (\text{CAI6})$$

with the lower term absent at $n = 0$. Inserting its quotient columns into CAI2 cancels every interior adjacent pair and gives

$$\begin{aligned} AR_N + R_N A^* - kR_N &= (b_{N+1}b_N^* + b_N b_{N+1}^*)/\omega_N, \\ D_N^{\text{ctrl}} &= A^* G_N + G_N A - kG_N, \quad G_N^{-1/2} D_N^{\text{ctrl}} G_N^{-1/2} = (yx^* + xy^*)/\omega_N, \\ x &= G_N^{1/2} b_N, \quad y = G_N^{1/2} b_{N+1}. \end{aligned} \quad (\text{CAI7})$$

The symbol D_N^{ctrl} is distinct from the scalar source constant D_N below. Reflection $P(S) \mapsto P(k-S)$ preserves χ because it permutes (a, b) with $(k-a, k-b)$ and q is even. It therefore induces an involution Σ on E , with $\Sigma A \Sigma = kI - A$. Changing variables $y \mapsto -y$ is an isometry of each entire remainder fibre. Uniqueness of the minimum proves $\Sigma^* G_N \Sigma = G_N$. Parity gives $\Sigma b_n = (-1)^n b_n$, and hence $b_N^* G_N b_{N+1} = -b_N^* G_N b_{N+1} = 0$. Thus the only possibly nonzero eigenvalues of the relative Hermitian control in CAI7 are $\epsilon_N, -\epsilon_N$, where

$$\epsilon_N^2 = \frac{\|x\|^2 \|y\|^2}{\omega_N^2}, \quad \phi_N = 0. \quad (\text{CAI8})$$

For comparison with the original tilted formulas, before specialization the exact expression is $(\|x\|^2 \|y\|^2 - |x^* y|^2)/\omega_N^2$ and $x^* y = i\omega_N \phi_N$. At the present untilted source only the *total* pairing vanishes. In the orthogonal kernel-boundary split write $x = x_K \oplus x_B$, $y = y_K \oplus y_B$, and $z_K = x_K^* y_K$, $z_B = x_B^* y_B$. Then $z_B = -z_K$, not generally zero. In particular the mixed exterior summand remains

$$\|x_K \otimes y_B - y_K \otimes x_B\|^2 = \|x_K\|^2 \|y_B\|^2 + \|x_B\|^2 \|y_K\|^2 + 2|z_K|^2. \quad (\text{CAI9})$$

Here is the original source-to-class map and its lower allowance. The complete Taylor unit $v_h = j_h v_h$ is invertible in E_h , and

$$\eta : E \longrightarrow E_h^{\otimes k}, \quad \eta[P] = v_h^{\otimes k} P(A_\Sigma) 1, \quad A_\Sigma = \sum_i M_{s_i}, \quad \eta A = A_\Sigma \eta. \quad (\text{CAI10})$$

In each local tensor factor the nilpotent has order m . The sum of the k nilpotents has exact order e_k : the power $e_k - 1 = k(m - 1)$ has nonzero top coefficient $[k(m - 1)]! / [(m - 1)!]^k$, and every term of its next power has one exponent at least m . Every centre β_{ab} occurs, and distinct pairs have distinct centres. The annihilator of A_Σ on its cyclic unit is therefore exactly χ ; this proves injectivity of CAI10, including all jets. Reflection fixes v_h and intertwines this inclusion with the original tensor reflection $s_i \mapsto 1 - s_i$.

Let E_+ be the direct sum of all complete primary blocks with $a > k/2$. Its dimension is $t_+ = q/2$. A primary idempotent is explicitly $\chi_\beta \sum_{j=0}^{e_k-1} (1/\chi_\beta)^{(j)}(\beta)(S - \beta)^j / j! \pmod{\chi}$, where $\chi_\beta = \chi / (S - \beta)^{e_k}$. Its product with the powers $(S - \beta)^d$, $0 \leq d < e_k$, gives a complete primary basis. The top wedge ξ_+ of these bases is nonzero, and remains nonzero under $\bigwedge^{t_+} \eta$. Its additive action eigenvalue is $\text{Tr}(A|_{E_+})$; nilpotents contribute zero to that trace. The relative Hermitian additive action on any exterior degree $1 \leq t \leq q - 1$ has largest eigenvalue ϵ_N , since its eigenvalues are sums of distinct members of $\{\epsilon_N, -\epsilon_N, 0, \dots, 0\}$. Its expectation on ξ_+ is $2 \text{Re Tr}(A|_{E_+}) - kt_+$. Summing the positive odd numbers $2a - k$ gives $(k + 1)^2/4$, so

$$0 < L_{h,k} := 2\delta e_k(k + 1) \frac{(k + 1)^2}{4} = \frac{\delta q(k + 1)}{2} \leq \epsilon_N \quad (N \geq q - 1). \quad (\text{CAI11})$$

This is a calculation on the same stipulated arithmetic class, not a claim of its existence and not an assumed desired upper estimate.

198.3 Exact cancellation before any estimates

Set $d_n = V_n/V_{n-1}$ for $n \geq q$, and $b_n^{\text{rec}} = \omega_n/\omega_{n-1}$; neither is the centre displacement δ or the quotient column b_n . From the rank-one covariance update and the determinant lemma,

$$\frac{b_N^* G_N b_N}{\omega_N} = 1 - d_N \quad (N \geq q), \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}} = d_{N+1}^{-1} - 1.$$

The first equality follows by removing the last covariance column; the second follows by adding the next one. At $N = q - 1$ the first q columns are the full quotient basis and $b_{q-1}^* G_{q-1} b_{q-1} = \omega_{q-1}$, without using any inverse at $q - 2$. Thus CAI8 gives the original MCE23/OMR18 formulas

$$\epsilon_{q-1}^2 = b_q^{\text{rec}}(d_q^{-1} - 1), \quad \epsilon_N^2 = b_{N+1}^{\text{rec}}(1 - d_N)(d_{N+1}^{-1} - 1) \quad (N \geq q). \quad (\text{CAI12})$$

Positive covariances give $0 < d_n \leq 1$, and CAI11–12 force $d_n < 1$ at every degree appearing here. All following logarithms therefore exist. Put $w_{q-1} = w_{2q-1} = 1$ and $w_N = 2$ for $q \leq N \leq 2q - 2$; their sum is $2q$. Retain exactly

$$\begin{aligned} \mathcal{W}_k^{\text{ar}} &= \log \frac{\omega_{2q-1} \omega_{2q}}{\omega_{q-1} \omega_q}, \\ \mathcal{P}_- &= -\log(1 - d_{2q-1}) - 2 \sum_{n=q}^{2q-2} \log(1 - d_n), \\ \mathcal{P}_+ &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{n=q+1}^{2q-1} \log(1 - d_n), \\ J_k^{\text{action}} &= 2 \sum_{N=q-1}^{2q-1} w_N \log \epsilon_N, \quad \Psi_k = F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)}, \quad C_k^{\text{budget}} = F_B^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+. \end{aligned} \quad (\text{CAI13})$$

Multiply CAI12 over $N = q - 1, \dots, 2q - 2$ and over $N = q, \dots, 2q - 1$ separately. The two exact products are

$$\begin{aligned} \prod_{N=q-1}^{2q-2} \epsilon_N^2 &= \frac{\omega_{2q-1} V_{q-1}}{\omega_{q-1} V_{2q-1}} (1 - d_{2q-1}) \prod_{n=q}^{2q-2} (1 - d_n)^2, \\ \prod_{N=q}^{2q-1} \epsilon_N^2 &= \frac{\omega_{2q} V_q}{\omega_q V_{2q}} (1 - d_q)(1 - d_{2q}) \prod_{n=q+1}^{2q-1} (1 - d_n)^2. \end{aligned} \quad (\text{CAI14})$$

Indeed $d_{N+1}^{-1} - 1 = (1 - d_{N+1})/d_{N+1}$; norm ratios telescope, reciprocal d factors give the displayed volume ratios, and the endpoint and interior contraction exponents are precisely those written. Taking logarithms and using CAI5 proves

$$\boxed{J_k^{\text{action}} = \Psi_k + C_k^{\text{budget}} = \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+}. \quad (\text{CAI15})$$

Equivalently $\Psi_k = F_K^{(k)} + \mathcal{H}_k + \delta_k^\sigma$ and $C_k^{\text{budget}} = \mathcal{B}_k^{(k)} - F_K^{(k)} + \mathcal{W}_k^{\text{ar}} - \mathcal{P}_- - \mathcal{P}_+$. The cancellation uses the exact joint identities, not independent estimates for the kernel, residual, low metrics or transfer errors.

198.4 All-degree multiplication bounds on the actual arithmetic blocks

The AMT1 domain is every convolution order $b \geq 3$, not just the later tensor subsequence. Its constants, without alteration, are

$$\begin{aligned} \alpha &= \pi/2, \quad p = 8m - \frac{9}{2}, \quad B_h = 42 + 8m, \quad M = 21 + 4m, \\ \vartheta_h &= \int_{-1}^1 w_h(y) dy > 0, \quad M_\alpha = \int_{\mathbb{R}} e^{\alpha y} w_h(y) dy \in (0, \infty), \\ c_\Gamma &= \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_b &= \frac{c_h \vartheta_h^{b-3} e^{-\alpha(b-3)} (b-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_b = \frac{C_h^{\text{tilt}}}{c_\Gamma} b^{p+1} M_\alpha^{b-1}, \quad D_N = [1 + 4(N+M)^2]^M. \end{aligned} \quad (\text{CAI16})$$

Here c_h is the original TW7 lower-envelope constant and C_h^{tilt} the original BT12 compact/tail maximum, as recorded in AMT4. Their already proved defining inequalities, for every $b \geq 3$, are

$$\begin{aligned} m_{h,b}(y) &\geq c_h \vartheta_h^{b-3} e^{-\alpha(|y|+b-3)} (b-2+|y|)^{-B_h}, \\ m_{h,b}(y) &\leq b C_h^{\text{tilt}} M_\alpha^{b-1} e^{-\alpha|y|} (1+|y|/b)^{-p}, \\ c_\Gamma e^{-\alpha|y|} (1+|y|)^{-1/2} &\leq \frac{d\sigma}{dy} \leq C_\Gamma e^{-\alpha|y|} (1+|y|)^{-1/2}, \quad d\sigma = |\Gamma(1/4 + iy/2)|^2 dy / (2\pi). \end{aligned} \quad (\text{CAI17})$$

The factors $b-2+|y| \leq (b-2)(1+|y|)$, $(1+|y|)^{B_h} \leq 2^{B_h/2} (1+y^2)^M$, and $1+|y|/b \geq (1+|y|)/b$, with $p > 1/2$, give $a_b(1+y^2)^{-M} d\sigma/dy \leq m_{h,b} \leq A_b d\sigma/dy$. The orthonormal Gamma recurrence is $ye_j = \sqrt{(j+1)(j+1/2)}e_{j+1} + \sqrt{j(j-1/2)}e_{j-1}$. It gives $\|yP\|_\sigma \leq 2(N+1)\|P\|_\sigma$ on degree N . Iterate at the successive degrees and expand $(1+y^2)^M$ to get $\int |P|^2 (1+y^2)^M d\sigma \leq D_N \|P\|_\sigma^2$. Cauchy–Schwarz applied with the two weights $(1+y^2)^{\pm M/2}$ therefore proves, for every complex polynomial of degree at most N ,

$$\frac{a_b}{D_N} \|P\|_\sigma^2 \leq \|P\|_{m_{h,b}}^2 \leq A_b \|P\|_\sigma^2. \quad (\text{CAI18})$$

These are precisely the full-polynomial AMT6 inequalities; in particular they apply at $b = 3, 4, 5$ with no new source-order restriction.

The fixed Gamma monic polynomials have full norms $\gamma_n = \sqrt{2\pi} n!(1/2)_n$. Their recurrence gives the generating series $F(y, z) = \sum_{n \geq 0} p_n^\sigma(y) z^n / n!$:

$$(1+z^2)\partial_z F = (y-z/2)F, \quad F(y, 0) = 1, \quad F = (1+z^2)^{-1/4} e^{y \arctan z}. \quad (\text{CAI19})$$

One may verify the differential equation coefficient by coefficient, and integration at zero uniquely gives the displayed analytic function on $|z| < 1$. On $|z| = 1/4$, its prefactor has modulus at most $(16/15)^{1/4}$ and $|\arctan z| \leq \sum_{j \geq 0} |z|^{2j+1}/(2j+1) = \operatorname{arctanh}(1/4)$. Consequently Cauchy's coefficient bound and $n!/(1/2)_n = 4^n/(2n) \leq 2n+1$ imply

$$|P(y)|^2 \leq C_\sigma(N+1)(2N+1)16^N e^{\nu_0|y|} \|P\|_\sigma^2, \quad C_\sigma = \frac{4}{\sqrt{15}\sqrt{2\pi}}, \quad \nu_0 = \log(5/3). \quad (\text{CAI20})$$

For the binomial inequality the central coefficient is the maximum of the $2n+1$ positive coefficients summing to 4^n . For the polynomial inequality expand in the complete degree- N orthonormal basis and apply Cauchy–Schwarz to the coefficient vector; then bound each of the $N+1$ summands by $(2N+1)16^N$. This covers complex coefficients.

Put $d_0 = \alpha - \nu_0 > 0$. Positivity and convolution give $\int e^{\alpha y} m_{h,b}(y) dy = M_\alpha^b$ by Tonelli, and evenness gives $\int e^{\alpha|y|} m_{h,b}(y) dy \leq 2M_\alpha^b$. For $x > R \geq 0$, $x^2 e^{-d_0 x} \leq (16/d_0^2) e^{-d_0 R/2}$; the maximum of $x^2 e^{-d_0 x/2}$ is $16/(e^2 d_0^2) \leq 16/d_0^2$. Moreover $N+M \leq M(N+1)$ since $M \geq 1$, so $D_N \leq (1+4M^2)^M (N+1)^{2M}$. Combining these facts with CAI18–20, and $2N+1 \leq 2(N+1)$, proves for $b = 3, 4, 5$

$$\begin{aligned} \int_{|y|>R} y^2 |P(y)|^2 m_{h,b}(y) dy &\leq H_h(N+1)^{2M+2} 16^N e^{-d_0 R/2} \|P\|_{m_{h,b}}^2, \\ H_h &= \frac{64C_\sigma(1+4M^2)^M}{d_0^2} \max_{b=3,4,5} \frac{M_\alpha^b}{a_b}. \end{aligned} \quad (\text{CAI21})$$

All constants are fixed- h data, including the unscaled source masses. Choose

$$T_h = \frac{2}{d_0} \{\log 16 + 2M + 3 + \log^+ H_h\}, \quad C_{\text{mult}}(h) = \sqrt{T_h^2 + 1}. \quad (\text{CAI22})$$

Here $\log^+ x = \max(0, \log x)$. At $R = T_h(N+1)$ the logarithm of the factor in CAI21 is at most $\log H_h + (2M+2)N + N \log 16 - (\log 16 + 2M + 3 + \log^+ H_h)(N+1) \leq 0$, using $\log(N+1) \leq N$. The inner interval has $y^2 \leq T_h^2(N+1)^2$. Adding its integral and the tail, and using $(N+1)^2 \geq 1$, proves

$$\|yP\|_{m_{h,b}} \leq C_{\text{mult}}(h)(N+1)\|P\|_{m_{h,b}}, \quad b = 3, 4, 5. \quad (\text{CAI23})$$

198.5 Total-degree tensor compression, including the output face

For every integer $k \geq 3$, write $r_k = \lfloor k/3 \rfloor$ and partition k into r_k blocks of sizes 3, 4, 5. If $k = 3r_k$ take all threes; if $k = 3r_k + 1$ or $3r_k + 2$, replace one three by four or five. This also covers $k = 3, 4, 5$. Convolution proves the exact isometry

$$L^2(m_{h,k}) \longrightarrow \bigotimes_{i=1}^{r_k} L^2(m_{h,b_i}), \quad f(y) \longmapsto f(y_1 + \cdots + y_{r_k}), \quad \sum_i (b_i/2 + i y_i) = k/2 + i \sum_i y_i. \quad (\text{CAI24})$$

The norm identity follows first by Tonelli for $|f|^2$ and then by the definition of convolution. Its image is the sum-dependent subspace, not the whole product space; total masses multiply to μ_h^k . In each block let $\varphi_n^{(b)}$ be the real orthonormal polynomial with positive leading coefficient. Its exact recurrence is

$$y\varphi_n^{(b)} = a_{n+1}^{(b)}\varphi_{n+1}^{(b)} + a_n^{(b)}\varphi_{n-1}^{(b)}, \quad a_0^{(b)} = 0, \quad 0 < a_n^{(b)} \leq C_{\text{mult}}(h)n \quad (n \geq 1).$$

There is no diagonal term by evenness. The bound follows by taking the coefficient of φ_n in $y\varphi_{n-1}$ and applying CAI23.

Use all product basis vectors of total degree at most $N+1$, and let M_{N+1} be the compression of multiplication by $\sum_i y_i$ to their span. It is a real symmetric matrix. At a vertex $\nu = (\nu_1, \dots, \nu_{r_k})$ of total degree at most N , its full absolute row sum is at most $C_{\text{mult}} \sum_i (\nu_i + (\nu_i + 1)) \leq C_{\text{mult}}(2N + r_k)$.

At total degree $N + 1$ only downward entries remain in the compression, so that row sum is at most $C_{\text{mult}}(N + 1) \leq C_{\text{mult}}(2N + r_k)$. Symmetry gives the same bound for column sums and hence the operator norm: directly, for any vector z , pair the equal off-diagonal entries and apply $2|z_\nu z_\mu| \leq |z_\nu|^2 + |z_\mu|^2$ to its quadratic form. Its absolute Rayleigh quotient is bounded by the maximal row sum.

The image of a univariate polynomial of degree at most N under CAI24 has total degree at most N . Multiplication by $\sum_i y_i$ has total degree at most $N + 1$, so no output is lost by the compression. Therefore

$$\|yP\|_{m_{h,k}} \leq C_{\text{mult}}(h)(2N + r_k)\|P\|_{m_{h,k}}, \quad \boxed{\frac{\omega_{N+1,k}^{\text{ar}}}{\omega_{N,k}^{\text{ar}}} \leq C_{\text{mult}}(h)^2(2N + r_k)^2} \quad (k \geq 3, N \geq 0). \quad (\text{CAI25})$$

For the last step yQ_N is an actual degree- $N + 1$ monic competitor; its orthogonal monic minimum has no greater norm. The S -monic phases have modulus one, so precisely the original norms occur. This proof uses the full total-degree space, not independent degree- N cutoffs in each factor, and yields $N + k$ rather than kN .

198.6 Both original contraction penalties on their complete window

Return now to the stipulated packet in CAI1. By CAI11–12 and $1 - d_N \leq 1$,

$$d_n \leq \frac{b_n^{\text{rec}}}{b_n^{\text{rec}} + L_{h,k}^2} \quad (q \leq n \leq 2q).$$

For $n = q$ this is the endpoint formula; for $n \geq q + 1$ it follows from the formula at $N = n - 1$. CAI25 bounds all these recurrence ratios by

$$B_{h,k} = C_{\text{mult}}(h)^2(4q + r_k - 2)^2, \quad z_{h,k} = \frac{B_{h,k}}{L_{h,k}^2} = \frac{4C_{\text{mult}}(h)^2(4 + (r_k - 2)/q)^2}{\delta^2(k + 1)^2}. \quad (\text{CAI26})$$

The map $b \mapsto b/(b + L^2)$ is increasing for $b > 0$, whence

$$0 < d_n \leq \frac{z_{h,k}}{1 + z_{h,k}}, \quad \frac{1}{1 + z_{h,k}} \leq 1 - d_n < 1, \quad q \leq n \leq 2q. \quad (\text{CAI27})$$

It is the complementary factors $1 - d_n$, not d_n themselves, that approach one. In CAI13 the coefficient sum in $\mathcal{P}_-$ is $1 + 2(q - 1) = 2q - 1$, and that in $\mathcal{P}_+$ is $2 + 2(q - 1) = 2q$. Thus the exact finite bound is

$$\boxed{0 \leq \mathcal{P}_- + \mathcal{P}_+ \leq \Pi_{h,k} := (4q - 1) \log(1 + z_{h,k}).} \quad (\text{CAI28})$$

Since $q = e_k(k + 1)^2$ and $r_k \leq k/3$, $z_{h,k} = O_h(k^{-2})$ and $\log(1 + z) \leq z$ show that $\Pi_{h,k} = O_h(q/k^2) = O_h(e_k)$. This is $O_h(1)$ for $m = 1$, and $O_h(k) = o(q)$ for any fixed $m > 1$. No early- k smallness assumption was used in CAI27–28. At an interior degree $N \geq q$, writing $\tau_N = -\log d_{N+1}$ gives the exact pointwise comparison $\epsilon_N^2/(b_{N+1}^{\text{rec}} e^{\tau_N}) = (1 - d_N)(1 - d_{N+1})$, whose logarithm lies in $[-2 \log(1 + z_{h,k}), 0]$ on the window. At $N = q - 1$ the one remaining factor is $1 - d_q$.

198.7 The four-norm factorial window and its finite remainder

Minimize CAI18 over all monic polynomials, retaining the literal degree- n coefficient one. It gives $(a_k/D_n)\gamma_n \leq \omega_{n,k}^{\text{ar}} \leq A_k\gamma_n$. Define

$$\begin{aligned} \Lambda_{k,n} &= \log(A_k D_n / a_k), \quad E_{\text{lo}} = \Lambda_{k,q-1} + \Lambda_{k,q}, \quad E_{\text{hi}} = \Lambda_{k,2q-1} + \Lambda_{k,2q}, \\ \mathcal{W}_q^\sigma &= \log \frac{\gamma_{2q-1} \gamma_{2q}}{\gamma_{q-1} \gamma_q}, \quad -E_{\text{hi}} \leq \mathcal{W}_k^{\text{ar}} - \mathcal{W}_q^\sigma \leq E_{\text{lo}}. \end{aligned} \quad (\text{CAI29})$$

For the lower bound put both numerator norms at their lower bounds and both denominator norms at their upper bounds; for the upper bound reverse those choices. The common a_k, A_k terms are thereby counted exactly twice. This comparison occurs *after* the window has telescoped, not once at each of its q steps. More explicitly, the original AMT22 constants give

$$\mathfrak{s}_h = \alpha + \log(M_\alpha/\vartheta_h), \quad \mathfrak{c}_h = -\log M_\alpha + 3 \log \vartheta_h - 3\alpha + \log \frac{C_h^{\text{tilt}} C_\Gamma 2^{B_h/2}}{c_h c_\Gamma}, \quad (\text{CAI30})$$

$$\log(A_k/a_k) = k\mathfrak{s}_h + (p+1) \log k + B_h \log(k-2) + \mathfrak{c}_h.$$

Since h is fixed and $\log D_n = O_h(1 + \log(n+1))$, the two errors in CAI29 are $O_h(k + \log q)$.

The identity $(1/2)_n = (2n)!/(4^n n!)$ gives $\gamma_n = \sqrt{2\pi}(2n)!/4^n$. Cancelling these four literal norms, including their masses, gives

$$\mathcal{W}_q^\sigma = 2 \log \frac{(4q)!}{(2q)!} - 4q \log 2 + \log \frac{2q-1}{2(4q-1)}. \quad (\text{CAI31})$$

Here the endpoint factor results from $(4q-2)! = (4q)!/[(4q)(4q-1)]$ and $(2q-2)! = (2q)!/[(2q)(2q-1)]$; it is not discarded.

For a fully finite enclosure put $C(q) = 4q \log q + (8 \log 2 - 4)q$ and $\eta_q = 1/(16q) + 1/(4q-2)$. For $f(x) = \log x$, integration by parts twice on each unit interval $[j, j+1]$ gives

$$\frac{f(j) + f(j+1)}{2} - \int_j^{j+1} f(x) dx = \frac{1}{2} \int_j^{j+1} (x-j)(j+1-x) f''(x) dx.$$

Since $f''(x) = -x^{-2}$ and $(x-j)(j+1-x)/2 \leq 1/8$, summing from $j = 2q$ to $4q-1$ bounds the composite trapezoidal error by $-\frac{1}{8} \int_{2q}^{4q} x^{-2} dx = -1/(32q) \leq \xi_q \leq 0$. The factorial sum is the trapezoidal sum plus $[f(4q) - f(2q)]/2$. Substitution in CAI31 and exact integration give

$$\mathcal{W}_q^\sigma = C(q) - \log 2 + 2\xi_q + \log(1 - 1/(4q-1)).$$

Finally $-x/(1-x) \leq \log(1-x) \leq 0$ for $0 < x < 1$, obtained by integrating $-1/(1-x)$, proves for every integer $q \geq 1$

$$C(q) - \log 2 - \eta_q \leq \mathcal{W}_q^\sigma \leq C(q) - \log 2. \quad (\text{CAI32})$$

198.8 Complete finite interval, action mean and exact receiving scope

Combining CAI15, CAI28 and CAI29 proves the entirely specified interval

$$\boxed{\mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^\sigma - E_{\text{hi}} - \Pi_{h,k} \leq J_k^{\text{action}} \leq \mathcal{B}_k^{\text{ar}} + \mathcal{W}_q^\sigma + E_{\text{lo}}.} \quad (\text{CAI33})$$

In particular CAI32 gives

$$\begin{aligned} \mathcal{B}_k^{\text{ar}} + C(q) - \log 2 - \eta_q - E_{\text{hi}} - \Pi_{h,k} &\leq J_k^{\text{action}}, \\ J_k^{\text{action}} &\leq \mathcal{B}_k^{\text{ar}} + C(q) - \log 2 + E_{\text{lo}}. \end{aligned} \quad (\text{CAI34})$$

For fixed h and its complete multiplicity, $e_k = O_h(k)$ and $q = e_k(k+1)^2$. All noncentral terms in CAI34 are $O_h(k + \log q) = o(q)$. Therefore

$$\boxed{J_k^{\text{action}} = \mathcal{B}_k^{\text{ar}} + 4q \log q + (8 \log 2 - 4)q + O_h(k + \log q).} \quad (\text{CAI35})$$

The finer finite centre in CAI34 retains $-\log 2$ and every error term. The coefficient $8 \log 2 - 4$ is a coefficient of $J_k^{\text{action}} - \mathcal{B}_k^{\text{ar}}$, not a coefficient of J_k^{action} after replacing $\mathcal{B}_k^{\text{ar}}$ by only its leading asymptotic. The exact signed correction $\mathcal{B}_k^{\text{ar}} = \mathcal{B}_k^\sigma + \delta_k^\sigma$ remains unevaluated at scale q . Nothing in CAI35 assigns either individual kernel/residual allocation, or the absolute order-one long loss.

The original weighted arithmetic geometric mean is exactly

$$\mathcal{G}_k = \prod_{N=q-1}^{2q-1} \epsilon_N^{w_N/(2q)} = \exp(J_k^{\text{action}}/(4q)).$$

Its finite bounds are the exponentials of CAI34 divided by $4q$. Consequently, since $(k + \log q)/q = O_h(1/k)$ for fixed m ,

$$\boxed{\frac{\mathcal{G}_k}{q \exp(\mathcal{B}_k^{\text{ar}}/(4q))} = \frac{4}{\exp(1)}(1 + O_h(k^{-1})) \longrightarrow \frac{4}{\exp(1)}}. \quad (\text{CAI36})$$

This uses $e^x = 1 + O(x)$ as $x \rightarrow 0$, which follows directly by integrating the exponential derivative on a bounded interval. It also proves $(J_k^{\text{action}} - \mathcal{B}_k^{\text{ar}})/(kq) \rightarrow 0$ since $\log q = O_h(\log k)$.

The exact exterior receiver is unchanged:

$$\begin{aligned} 4q \log L_{h,k} &\leq J_k^{\text{action}}, \\ L_{h,k} &\leq \min_{q-1 \leq N \leq 2q-1} \epsilon_N \leq \mathcal{G}_k, \\ \frac{\mathcal{G}_k}{L_{h,k}} &= \frac{8}{\exp(1)\delta(k+1)} \exp(\mathcal{B}_k^{\text{ar}}/(4q))(1 + O_h(k^{-1})). \end{aligned} \quad (\text{CAI37})$$

The first two inequalities follow term by term from CAI11 and the weights' sum $2q$; a minimum is bounded by, not identified with, the weighted mean. In the original simple-quartet baseline domain alone, AMT40 and the complete BSL/HBL source proof give $\mathcal{B}_k^{\text{ar}}/q^2 \rightarrow C_B > 769/17010 > 0$. Substitution into CAI35 proves $J_k^{\text{action}}/q^2 \rightarrow C_B$, and CAI37 then tends to infinity exponentially in q . This last consequence uses that already proved baseline only on its original $m = 1$ domain; the new interval, penalty bound and relative expansion retain every complete multiplicity.

For the existing original simple-quartet outer receiver, retain its literal finite identity $\Psi_k = \mathcal{F}_{\partial,k}^{(k)} - \Phi_{k,k} + \mathcal{H}_k + \delta_k^\sigma + \epsilon_{k,k}$, with the original $-e_{k,k}^- \leq \epsilon_{k,k} \leq e_{k,k}^+$. Substituting CAI15 gives

$$\Phi_{k,k} \leq \mathcal{F}_{\partial,k}^{(k)} + \mathcal{H}_k + \delta_k^\sigma + e_{k,k}^+ + C_k^{\text{budget}} - 4q \log L_{h,k}. \quad (\text{CAI38})$$

This is exactly OMR26's arithmetic backward receiver. Every signed low metric and error remains where its original identity puts it. CAI33–34 can be substituted for the *complete* action; they do not justify independently improving one summand while holding its correlated budget fixed. No failure of the broader programme or exclusion at other admitted degrees follows from the size of this particular mean.

Continuation to the derived growing-window endpoints (RGI8–10). The complete action in this section has the following exact continuation to the original allowance rows $q-1, \dots, q+r-1$, for each integer $r \geq 1$. Use weight one at the two ends and two in the interior; equivalently write these weights as w_j for the row $q+j-1$. Their sum is $2r$. The original adjacent relations are $d_{q+j} = V_{q+j}/V_{q+j-1} = 1/T_j^{\text{ar}}$ and $\beta_{q+j} = \omega_{q+j}/\omega_{q+j-1}$. Taking logarithms of the exact reflection identity RGC24 gives

$$\log \epsilon_{q+j-1}^2 = \log \beta_{q+j} - \log d_{q+j} - a_{q+j} - \mathbf{1}_{j \geq 1} a_{q+j-1}, \quad a_d = -\log(1 - d_d). \quad (\text{RGI8})$$

On the positive-allowance domain supplied by RGC25–26 all these logarithms are defined. For any positive sequence z_n , direct cancellation of each interior exponent gives

$$\sum_{j=0}^r w_j (\log z_{q+j} - \log z_{q+j-1}) = \log z_{q+r-1} + \log z_{q+r} - \log z_{q-1} - \log z_q.$$

For $r = 1$ the occurrences of z_q cancel on the right as well. Apply this identity separately to the original ω and V in RGI8. Thus

$$\mathcal{J}_{q,r} = \mathcal{B}_{q,r} + \mathcal{W}_{q,r} - \mathcal{P}_{q,r}, \quad \mathcal{B}_{q,r} = \log \frac{V_{q-1}V_q}{V_{q+r-1}V_{q+r}}, \quad \mathcal{W}_{q,r} = \log \frac{\omega_{q+r-1}\omega_{q+r}}{\omega_{q-1}\omega_q}. \quad (\text{RGI9})$$

Here $\mathcal{J}_{q,r} = \sum_j w_j \log \epsilon_{q+j-1}^2$ and the exact penalty is RGC29. Its coefficients are 2, 1 for $r = 1$ and 3, 4, ..., 4, 3, 1 for $r \geq 2$, with total $4r - 1$. Inserting the proved intersection bound RGI7 and the four original AMT endpoint errors RGC32 yields

$$\begin{aligned} \mathcal{B}_{q,r} + \mathcal{W}_{q,r}^\sigma - \Lambda_{k,q+r-1} - \Lambda_{k,q+r} - P_{q,r}^{\text{cap}} &\leq \mathcal{J}_{q,r}, \\ \mathcal{J}_{q,r} &\leq \mathcal{B}_{q,r} + \mathcal{W}_{q,r}^\sigma + \Lambda_{k,q-1} + \Lambda_{k,q}. \end{aligned} \quad (\text{RGI10})$$

This follows by adding the lower, respectively upper, endpoint inequalities to RGI9 and using $0 \leq \mathcal{P}_{q,r} \leq P_{q,r}^{\text{cap}}$. Only the four endpoint errors are charged. At $r = q$ the endpoints and weights are exactly those of the earlier complete action. Its correlated allocation cancellation remains the same identity. The intrinsic signed mixed profile and the CPQ returns retain their original receiving maps; no value for them is assigned by these canonical allowance bounds.

The moving-degree result in the complete action (GRB9–10). The new growing-ratio proof also gives an exact receiver at shifted rows. Keep q, χ, E , all three source forms, and every kernel/boundary map fixed as in this section. Choose an integer $s \geq 1$, put $a = q + s - 1$, $b = a + q$, and give rows $N = a + i$ the weights $w_0 = w_q = 1$ and $w_i = 2$ otherwise. Their relation index is exactly $r = s + i$ in RGI8. Every row now has the preceding contraction, because $a \geq q$. Expanding the weighted differences for any positive sequence z gives

$$\sum_{i=0}^q w_i (\log z_{a+i+1} - \log z_{a+i}) = \log z_b + \log z_{b+1} - \log z_a - \log z_{a+1}.$$

Applying this separately to the original V and ω in the exact adjacent identity proves

$$\begin{aligned} J_{k,s}^{\text{action}} &= \log \frac{V_a V_{a+1}}{V_b V_{b+1}} + \log \frac{\omega_b \omega_{b+1}}{\omega_a \omega_{a+1}} - \sum_{j=0}^{q+1} c_j [-\log(1 - T_{s-1+j}^{-1})], \\ J_{k,s}^{\text{action}} &= 2 \sum_{i=0}^q w_i \log \epsilon_{a+i}, \quad (c_0, \dots, c_{q+1}) = (1, 3, \underbrace{4, \dots, 4}_{q-2}, 3, 1). \end{aligned} \quad (\text{GRB9})$$

Indeed $c_j = w_j + w_{j-1}$ with absent weights zero, so their total is $2 \sum_i w_i = 4q$. These four endpoints and every coefficient are derived from the original maps. The full finite scalar enclosures, four AMT endpoint errors and their proof are GRA27–32; GRB1–4 intersects them with all eligible earlier bounds. For $t = s/q \rightarrow \infty$ and $t \log t = o(q/k)$ the complete proof GRA19–35 consequently gives

$$\log \min_{a \leq N \leq b} \epsilon_N \sim \frac{q}{2t \log t}, \quad \log \mathcal{P}_{k,s} \sim -\frac{q}{t \log t}, \quad J_{k,s}^{\text{action}} \sim \frac{2q^2}{t \log t}. \quad (\text{GRB10})$$

The proof controls the full adjacent index interval $s - 1, \dots, s + q$; its error is uniform on that interval, as shown there before summing.

The exact transfer to the mixed spaces retains $I : K \hookrightarrow E$ and $\Lambda : E \twoheadrightarrow B$. For each unchanged positive metric G_N^v its minimum section is $S_N^v = (G_N^v)^{-1} \Lambda^* (\Lambda (G_N^v)^{-1} \Lambda^*)^{-1}$. It satisfies $\Lambda S_N^v = 1$ and $I^* G_N^v S_N^v = 0$ by direct multiplication. Relative to any fixed section S_* , the frame change from $[I, S_*]$ to $[I, S_N^v]$ has identity diagonal because $\Lambda(S_N^v - S_*) = 0$. Its determinant is one. Block orthogonality therefore gives

$$\det(I^* G_N^v I) \det(\Lambda (G_N^v)^{-1} \Lambda^*)^{-1} = |\det[I, S_*]|^2 \det G_N^v.$$

The four-endpoint return cancels that same frame constant and proves $F_K^v + F_B^v = \mathcal{B}^v$. Thus the complete allocation and its matching budget in GRA36 still sum to GRB9: the occurrence of $F_K^{(k)}$ in one has its opposite occurrence in the other. Every intrinsic mixed profile and complementary CPQ return retains its original map and coefficient; the new scalar estimates do not assign those separate coefficients.

199 A smaller threshold-profile error from the original nested flags

Retain the complete SFB and CPQ coefficient spaces at target degree $j = k + 4$, the two original source pairs and their inherited simple-quartet and fixed-period domains. At each original row, use its full preceding positive form D_0 , its lower-evaluation-corrected numerator row ρ_n , and the original scalar $\zeta_n \geq 0$. Set

$$r_n = \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n}, \quad \mathcal{D}_0 = \text{diag}(D_0, 4), \quad \mathcal{U}_n = \text{diag}\left(\frac{\rho_n}{\sqrt{1 + \zeta_n}}, 4\right), \quad (\text{NTE1})$$

$$K_{X,n} = \mathcal{U}_n X (X^* \mathcal{D}_0 X)^{-1} X^* \mathcal{U}_n^*.$$

An empty subspace gives the zero 4×4 matrix. All frames X are the fixed coefficient frames of the actual eight flags. With $Y = \mathcal{U}_n \mathcal{D}_0^{-1/2}$ and Π_X the orthogonal projector onto $\mathcal{D}_0^{1/2} \text{im } X$, this is exactly $K_{X,n} = Y \Pi_X Y^*$, and $Y Y^* = r_n I_4$. In particular $0 \preceq K_{X,n} \preceq r_n I_4$. When $X \subset Z$, the difference $\Pi_Z - \Pi_X$ is an orthogonal projector, so $K_{X,n} \preceq K_{Z,n}$. These are identities in the original metric and preserve both the full old lower inverse and its denominator.

The two nested chains already used in the common-loss proof are

$$J \subset W_i \subset W \subset T, \quad K_i \subset I \subset K_c \subset S. \quad (\text{NTE2})$$

Indeed $I \subset B \subset S$, $K_i = I \cap J$, $W_i = I + J$, $W = B + J$, $K_c = S \cap W$ and $T = S + J$. In particular $I \subset K_c$. There is no required simultaneous diagonalization of the eight compressed matrices.

For $x \geq 0$ put

$$h(x) = \log(1 + x) - \log \max(1, x) = u(x) - v(x),$$

$$u(x) = \log(1 + \min(x, 1)), \quad v(x) = \begin{cases} 0, & 0 \leq x \leq 1, \\ \log \frac{2x}{1+x}, & x \geq 1. \end{cases} \quad (\text{NTE3})$$

The definitions agree at $x = 1$ and give the displayed identity on each interval. Both u and v are increasing and vanish at zero. For $0 \leq x_0 \leq x_1 \leq x_2 \leq x_3 \leq r$, their disjoint increment sums satisfy

$$0 \leq u(x_3) - u(x_2) + u(x_1) - u(x_0) \leq u(r),$$

$$0 \leq v(x_3) - v(x_2) + v(x_1) - v(x_0) \leq v(r).$$

For example the first sum is at most $u(x_3) - u(x_0)$ after adding the nonnegative missing increment $u(x_2) - u(x_1)$; its upper bound is $u(r)$. The other line is identical. Subtracting proves that the alternating h sum lies in $[-v(r), u(r)]$, including $r = 0$ and threshold equalities.

Write the four eigenvalues of each $K_{X,n}$ in increasing order, including every zero, as $\lambda_{X,a,n}$, $1 \leq a \leq 4$. Positive matrix order makes each indexed eigenvalue increase along each chain in NTE2. To see this without a commutativity assumption, use the variational expression $\lambda_a(H) = \min_{\dim L=a} \max_{0 \neq x \in L} (x^* H x) / (x^* x)$ for increasing eigenvalues: an eigenbasis proves the formula, and pointwise ordered Rayleigh quotients preserve the minimum and maximum. Thus NTE3 applies to each of the four indexed scalar chains separately. Consequently

$$A_n = \text{Tr}\{h(K_{T,n}) - h(K_{W,n}) + h(K_{W_i,n}) - h(K_{J,n})\},$$

$$B_n = \text{Tr}\{h(K_{S,n}) - h(K_{K_c,n}) + h(K_{I,n}) - h(K_{K_i,n})\}, \quad (\text{NTE4})$$

$$-4v(r_n) \leq A_n, B_n \leq 4u(r_n), \quad |A_n - B_n| \leq 4\{u(r_n) + v(r_n)\}.$$

Here $h(K)$ is defined by the finite Hermitian spectral calculus; only its trace, the sum of the four scalar values, is used.

Let $\mathcal{P}_{k,e}$ be the original eight-flag threshold profile: at each row replace every $\log \det(I_4 + K_{X,n})$ in the exact SFB increment by $\sum_{a=1}^4 \log \max(1, \lambda_{X,a,n})$, with positive flags (T, K_c, W_i, K_i) and negative flags (W, S, J, I) . Take precisely the original nonnegative weights ϑ_n , one at the two end rows and two inside, whose sum is $2q_j$. Spectral factorization and NTE3 show that the row error is exactly $A_n - B_n$. Therefore

$$|\mathcal{L}_e - \mathcal{P}_{k,e}| \leq E_{k,e}^{\text{nested}} := \sum_n \vartheta_n \varepsilon(r_n) \leq 16q_j \log 2, \quad (\text{NTE5})$$

$$\varepsilon(r) = 4\{u(r) + v(r)\} = \begin{cases} 4 \log(1+r), & 0 \leq r \leq 1, \\ 4 \log \frac{4r}{1+r}, & r \geq 1. \end{cases}$$

The last inequality uses $u(r) \leq \log 2$, $v(r) \leq \log 2$. At a zero residual row its contribution is exactly zero. This proves a smaller complete finite error than the independent eight-matrix bound $32q_j \log 2$ while preserving precisely the same threshold profile, ranks, metrics and original row parameters. No equality case on an actual period is asserted.

The exact CPQ return identity is $4S_{K,j} = \mathcal{P}_e^{\text{rest}} + V_{\text{rem}} + V_s + V_A$ and $\mathcal{P}_e^{\text{rest}} = \mathcal{L}_e + V_b$. Thus the new profile interval gives

$$\frac{\mathcal{P}_{k,e} - E_{k,e}^{\text{nested}} + V_b + V_{\text{rem}} + V_s + V_A}{4} \leq S_{K,j} \leq \frac{\mathcal{P}_{k,e} + E_{k,e}^{\text{nested}} + V_b + V_{\text{rem}} + V_s + V_A}{4}. \quad (\text{NTE6})$$

Addition of these retained returns and division by four is the exact affine map; no boundary dependence is removed by the cancellation in $\mathcal{P}_e^{\text{rest}}$. Intersect NTE6 with the original positivity, total and native-cap intervals on their respective domains, then use $S_R = \mathcal{T} - S_K$ to reverse the endpoints for the residual allocation. The same common-loss interval can be intersected with SFC11 before this affine map is applied.

On EOR's proved paired-source domain, put $C_{k,e}^{\text{ord}} = R_e \sum_N (\bar{b}_{j,N-v-q'_j} - b_{j,N-v-q'_j})$, with the original four endpoints and all original finite guards. The triangle inequality applied to EOR's exact same-value comparison and NTE5 at order one gives

$$|\mathcal{L}_e^{(k,j)} - \mathcal{P}_{k,e}^{(1,1)}| \leq C_{k,e}^{\text{ord}} + E_{k,e}^{\text{nested},(1,1)}, \quad \limsup \frac{|\mathcal{L}_e^{(k,j)} - \mathcal{P}_{k,e}^{(1,1)}|}{q_k} \leq C_{\text{com}} + 16 \log 2. \quad (\text{NTE7})$$

The last step uses $q_j/q_k \rightarrow 1$ and EOR29's proved bound. The prescribed source orders and the actual affine target centre remain fixed. This is an error bound for the original profile; the profile itself and the complementary quotient returns remain their exact original matrix quantities. It assigns no individual allocation coefficient or actual-period compressed spectrum.

199.1 The evaluated first row and the full transported profile

The first appended native row has $n = q_j - v$. The complete FMG/FRG calculation below evaluates its actual corrected scalar through the original polynomials $g = \mathcal{T}_A \chi_j$, $\chi_- = \chi_{j-8}$ and $\ell = \mu_v \binom{q_j}{v}$. With $m = q_j - v - q'_j$ and the original Gamma monic minimum $\nu_m = \min_{P \text{ monic, deg } P=m} \|\chi - P\|^2$, it proves

$$1 + \zeta_{\text{all}} = \frac{\|g\|^2}{|\ell|^2 h_{n,s}}, \quad 1 + \zeta_{\text{lower}} = \frac{\nu_m}{h_{n,s}}, \quad 1 + r_n = \frac{\|g\|^2}{|\ell|^2 \nu_m}. \quad (\text{RMR10})$$

The full lower inverse and its denominator remain in this ratio. Both raw logarithms equal $2q_j \log(4/\pi) + O_{\text{actual}}(j \log j)$; their corrected ratio obeys the finite bound $0 \leq \log(1 + r_n) \leq E_{j,s} = O_{\text{actual}}(j \log j)$ with all constants specified in FMG26 and proved in FRG15–31. FMG11–15 also gives a literal equality for each of the eight compressed matrices through the original polynomial-ideal Gram and its exact functional pairing. Both nested chains therefore give

$$|\delta \mathcal{L}_{e,n}| \leq 4E_{j,s}, \quad |\delta \mathcal{P}_{e,n}| \leq 4E_{j,s}, \quad 0 \leq \delta V_{Y,n} \leq 4E_{j,s} \quad (Y = b, \text{rem}, s, A). \quad (\text{RMR11})$$

The endpoint weight is one. This evaluates the entire first-row contribution as $o(kq_k)$ and leaves all the original interior rows.

For every row, NPK3–10 specifies the original numerator-observation metric, its full conductor Schur update, and all eight transported 4×4 compressed matrices. Keep all four eigenvalues and the same signs and weights in its threshold sum $\widehat{\mathcal{P}}_{k,e}$. Applying NCT’s volume bounds to the eight fixed flags at the four original endpoints gives

$$|\mathcal{P}_{k,e} - \widehat{\mathcal{P}}_{k,e}| \leq E_{\text{flags}} + 32q_j \log 2 = o(kq_k). \quad (\text{RMR12})$$

Here E_{flags} is the explicitly defined optimized finite enclosure NPK11: take the infimum of the maximum of the two SGT-derived errors over the original admissible radii, separately for each rank and endpoint, and then sum. The bound holds for every admissible radius before taking any infimum. Its little-oh statement follows by the fixed-radius limsup followed by radii tending to one, retaining the full exceptional pole contribution. Thus the surviving leading profile is a precise finite numerator matrix calculation, with all four complementary returns transported by RMR9; its value has not been assigned by a rank count.

200 The original connecting image as a polynomial translation relation space

Retain the original simple quartet, its amplitude, fixed period on the proved five-orbit locus, logarithm branch, and every unit and period coefficient. Set

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q &= (k + 1)^2, & Q &= (j + 1)^2, \\ e &= 0, 1, & D_e &= Q - v - 1 + e, & d &= Q - q = 8k + 24, \\ \beta_* &= \frac{1}{2} - \delta - i\gamma, & \beta_{n,a,b} &= n\beta_* + 2a\delta + 2bi\gamma, \end{aligned} \quad (\text{CPR1})$$

$$E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v.$$

The coefficients are those of the complete original nonreduced conductor, not its radical. All duals are complex-linear. Write

$$\begin{aligned} \chi_n(S) &= \prod_{a,b=0}^n (S - \beta_{n,a,b}), & L_i(Z, W) &= (\mathbf{H}(ZW, Z, W, 1)^T)_i, \\ b_{i,a,b} &= [Z^a W^b] L_i^4, & B_i(z) &= \sum_{a,b=0}^4 b_{i,a,b} e^{\beta_{4,a,b} z}, \\ (\mathcal{T}_{AP})(S') &= \sum_{a,b=0}^8 a_{ab} p(S' + \beta_{8,a,b}), & (\mathcal{T}_i p)(S) &= \sum_{a,b=0}^4 b_{i,a,b} p(S + \beta_{4,a,b}), \\ L &= \mathbb{C}[S]_{<v}. \end{aligned} \quad (\text{CPR2})$$

The variable of a translated polynomial is determined by the displayed map; translation sums commute on polynomials. In particular no source centre is inserted into these algebraic maps. Their eventual original Gamma norms still use the centres and the two source-order pairs $(1, 1)$ and (k, j) of CIT and NSR.

200.1 A full polynomial model of the original target kernel

Let $K_j \subset \mathbb{C}[S]_{<Q}$ be the literal original kernel, represented by interpolation at all the degree- j roots. Here is an explicit definition using only its original period coordinates. For every $\nu = (\nu_1, \dots, \nu_4)$ with $|\nu| = j$ and $\sum_{i=1}^4 i\nu_i \equiv 0 \pmod{5}$, put

$$m_{\nu,a,b} = [Z^a W^b] \prod_{i=1}^4 L_i(Z, W)^{\nu_i}.$$

Then

$$K_j = \left\{ p \in \mathbb{C}[S]_{<Q} : \sum_{a,b=0}^j m_{\nu,a,b} p(\beta_{j,a,b}) = 0 \text{ for every such } \nu \right\}. \quad (\text{CPR3})$$

Indeed the homogeneous invariant monomials are a basis of the weight-zero part of $\mathbb{C}[x_1, \dots, x_4]_j$. Their substitutions are precisely the displayed coefficient vectors. The original observation dual has this image: restriction of its coefficient functionals to a primary vector $p(\beta_{j,a,b})$ is exactly the sum in CPR3. Thus its annihilator is the original K_j . Equivalently this is KRC3's original observation map, with the full invertible period substitution retained. The distinct-root Vandermonde identifies these values with a unique polynomial of degree $< Q$, so CPR3 gives its full original polynomial representative, rather than an independently chosen plane.

The trace identity for the conductor gives $K_j \subset V_{A,j}$. Explicitly, if C_j is the original full conductor convolution, then

$$(C_j p)_{a,b} = \sum_{r,t=0}^8 a_{rt} p(\beta_{j-8,a,b} + \beta_{8,r,t}) = (\mathcal{T}_A p)(\beta_{j-8,a,b}). \quad (\text{CPR4})$$

The original invariant trace contains every conductor column $C_j^\top$, hence CPR3 annihilates it and CPR4 is zero. It follows that $\mathcal{T}_A p$ is divisible by χ_{j-8} . All coefficients of the original convolution occur in this equation.

Define the exact polynomial space

$$\mathcal{U}_j = \alpha_A \mathcal{T}_A K_j \subset \mathbb{C}[S']_{\leq D_0}. \quad (\text{CPR5})$$

The degree bound and its entire kernel follow from

$$\mathcal{T}_A S^n = \sum_{r=0}^n \binom{n}{r} \mu_r S'^{n-r}, \quad \deg \mathcal{T}_A p = \deg p - v \quad (\deg p \geq v). \quad (\text{CPR6})$$

The latter equality uses the nonzero leading coefficient $\mu_v \binom{\deg p}{v} \text{lc}(p)$. Thus $\ker \mathcal{T}_A = L$ on every finite polynomial space and

$$K_j / (K_j \cap L) \xrightarrow{[p] \mapsto \alpha_A \mathcal{T}_A p} \mathcal{U}_j \quad \text{is an isomorphism,} \quad \dim \mathcal{U}_j = 8j - 16 - t_{0,j}. \quad (\text{CPR7})$$

This is the exact polynomial realization of the dual of the original onto map $\mathbf{B}_j$. To verify its scalar, if $F = N/E_A$, the binomial identity gives

$$\Lambda_F(\alpha_A \mathcal{T}_A p) = \alpha_A \Lambda_{E_A F}(p) = \alpha_A \sum_{a,b} N_{ab} p(\beta_{j,a,b}). \quad (\text{CPR8})$$

KRC5–6 identifies the DNE coordinates with conductor coordinates by $\alpha_A N_j Z_{\text{exp},j}$, so CPR8 is precisely $y^\top \mathbf{B}_j x$ when p represents the original kernel-value coordinate y . It is an ordinary transpose. Changing p by $K_j \cap L$ changes neither side. Changing N by a conductor multiple adds a lower exponential to F ; CPR4 proves that the polynomial on the left annihilates that entire lower span. This checks all representative choices.

200.2 The exact polynomial image of the connecting transpose

Put

$$\begin{aligned} \mathcal{V}_k &= \sum_{i=1}^4 \mathcal{T}_i \mathcal{U}_j \subset \mathbb{C}[S']_{\leq D_0}, \\ \mathcal{R}_0 &= \mathcal{T}_A(\chi_k \mathbb{C}[S]_{\leq d-1}), \quad \mathcal{R}_1 = \mathcal{T}_A(\chi_k \mathbb{C}[S]_{\leq d}), \\ \mathcal{I}_k &= \mathcal{V}_k \cap \mathcal{R}_0. \end{aligned} \quad (\text{CPR9})$$

These are finite polynomial spans and intersections; multiplication by arbitrary polynomials is not implicitly allowed in $\mathcal{V}_k$. The degrees of the successive generators of $\mathcal{R}_e$ are $q + a - v$, with the full leading coefficients in CPR6. Consequently $\dim \mathcal{R}_e = d + e$, and

$$\mathcal{R}_1 \cap \mathbb{C}[S']_{\leq D_0} = \mathcal{R}_0. \quad (\text{CPR10})$$

For $p \in K_j$, CPR4 and root addition imply that $\mathcal{T}_i \mathcal{T}_A p$ vanishes at every original degree- $(k-8)$ lower root: after each degree-four shift it is evaluated at a degree- $(j-8)$ lower root. Hence every element of $\mathcal{V}_k$ annihilates the full source lower-evaluation span.

Let $\mathcal{S}_k^{\text{raw}}$ be the source analytic space of CIT4 and PDS3, including every lower gauge. Its annihilator on polynomials through D_e is exactly $\mathcal{R}_e$. For a direct proof, multiplication by E_A

changes pairing on $\mathcal{T}_{Ap}$ to pairing of the upper numerator on p . Every $\chi_k S^a$ therefore annihilates it. The numerator Vandermonde makes that analytic space have dimension $q - v$; the $d + e$ independent polynomials just displayed have complementary dimension $D_e + 1 - (q - v)$. This proves the full annihilator equality.

Write the unchanged coefficient maps

$$B = B_{\text{cof}}, \quad \mathbb{B} = \text{diag}(\mathbf{B}_j, 4), \quad \Phi = \mathbb{B}B.$$

A tuple $u = (u_1, \dots, u_4) \in \mathcal{U}_j^4$ is the dual of a unique original target coordinate tuple. Its pullback to a finite source germ F is, with the positive degree-zero differential,

$$\sum_i \Lambda_{B_i F}(u_i) = \Lambda_F \left(\sum_i \mathcal{T}_i u_i \right). \quad (\text{CPR11})$$

This follows by the same complete binomial convolution as CPR8. Every source lower evaluation is killed as proved after CPR10. The transpose of Φ therefore kills the tuple exactly when the polynomial $\sum_i \mathcal{T}_i u_i$ belongs to $\mathcal{R}_e$. Since it has degree at most D_0 , this condition is independent of e by CPR10, and its polynomial image is exactly $\mathcal{I}_k$. This identifies the entire boundary annihilator and its image, rather than replacing it by a dimension hypothesis.

For the second endpoint retain the four original polynomials

$$R_i(S) = \frac{\mathcal{T}_i \chi_j(S)}{\chi_k(S)}, \quad \mathcal{C}^{(4)} = \text{span}_{i=1}^4 \mathcal{T}_A(\chi_k R_i). \quad (\text{CPR12})$$

The quotient is polynomial by root addition at every degree- k node. These four polynomials are independent by the full original rank-four quadric and 25-shift Vandermonde proof LCC9–12. They are the four exact defining equations of the primitive source at $e = 1$; at $e = 0$ no such equation is imposed. The transpose of the induced original connecting map consequently has the following complete image:

$$\boxed{(T_0/W)^\vee \simeq \mathcal{I}_k, \quad (T_1/W)^\vee \simeq \mathcal{I}_k/(\mathcal{I}_k \cap \mathcal{C}^{(4)}).} \quad (\text{CPR13})$$

Here $T_e = \mathcal{S}_e + W$, $W = \text{im } B + \ker \mathbb{B}$, with exactly the original NSR10 spaces. More explicitly the target dual $(\mathbb{C}^{4t_{K,j}} / \text{im } \Phi)^\vee$ is $\ker \Phi^\top$; CPR11 sends it into the dual of the primitive source. At $e = 0$ the latter dual is $\mathcal{R}_0$. At $e = 1$ it is $\mathcal{R}_1/\mathcal{C}^{(4)}$. The image of that transpose is the dual of the original onto image map: in finite dimension, a functional on an image extends to the full target, and a target functional restricts to zero precisely when it annihilates that image. This proves both isomorphisms and their kernels. It also proves that all later Gamma cutoffs retain the same spaces: only their metrics change, while the defining coefficient map and polynomial transpose in CPR11 are fixed.

200.3 Eliminating the conductor division and the primitive complement

The preceding image has a direct original polynomial relation model before division by the conductor:

$$\begin{aligned} \mathcal{W}_k &= L + \sum_{i=1}^4 \mathcal{T}_i K_j \subset \mathbb{C}[S]_{<Q}, \\ \mathcal{G}_k &= \{g \in \mathbb{C}[S]_{\leq d-1} : \chi_k g \in \mathcal{W}_k\}, \\ \mathcal{G}_k &\xrightarrow{g \mapsto \alpha_A \mathcal{T}_A(\chi_k g)} \mathcal{I}_k \quad \text{is an isomorphism.} \end{aligned} \quad (\text{CPR14})$$

To prove surjectivity, write an element of $\mathcal{I}_k$ as $\alpha_A \mathcal{T}_A \sum_i \mathcal{T}_i p_i = \mathcal{T}_A(\chi_k g_0)$, where $p_i \in K_j$. Commutation of the literal finite translation sums has been used. CPR6 says that $\sum_i \mathcal{T}_i p_i - \alpha_A^{-1} \chi_k g_0 \in L$; thus $g = \alpha_A^{-1} g_0$ is in $\mathcal{G}_k$. Conversely its defining relation, followed by $\alpha_A \mathcal{T}_A$, gives an element of $\mathcal{V}_k \cap \mathcal{R}_0$. Finally $\chi_k g \in L$ implies $g = 0$, since $q > v$. This proves injectivity with the original scalar

intact. Thus the space is calculated from the original observation polynomials, four translations and literal divisibility by the full χ_k .

There is a finite coefficient construction of every vector of $\mathcal{G}_k$, with no F_e , Gram inverse or assumed nonzero minor. Order monomials and indices as in CPR3. Construct its kernel basis by exact elimination; call its full coefficient matrix P_K . For $0 \leq r \leq n < Q$ form the literal triangular matrices

$$(\mathbf{T}_i)_{rn} = \binom{n}{r} \sum_{a,b=0}^4 b_{i,a,b} \beta_{4,a,b}^{n-r}, \quad (\mathbf{T}_i)_{rn} = 0 \ (r > n), \quad \mathbf{W} = [E_{\{0,\dots,v-1\}}, \mathbf{T}_1 P_K, \dots, \mathbf{T}_4 P_K]. \quad (\text{CPR15})$$

Keep dependent columns. For each column perform ordinary monic Euclidean division by χ_k , recording both complete arrays:

$$\mathbf{W}(S) = \mathbf{A}(S) + \chi_k(S) \mathbf{D}(S), \quad \deg \mathbf{A} < q, \quad \deg \mathbf{D} \leq d-1. \quad (\text{CPR16})$$

If $\mathbf{Z}$ is an exact basis of the kernel of the coefficient matrix of $\mathbf{A}$, the nonzero independent columns of $\mathbf{DZ}$ form a complete basis of $\mathcal{G}_k$. In fact a column combination of $\mathbf{W}$ is divisible by χ_k if and only if its recorded remainder vanishes; its quotient is then precisely $\mathbf{DZ}$ on those coordinates. A zero quotient and zero remainder means that the original column combination was zero, so taking its image, rather than counting all remainder syzygies, removes exactly those dependencies. Every coefficient in this algorithm is specified in CPR2–3 and CPR15; the monic divisions retain all root factors. On an exact coefficient stratum Gaussian elimination selects the first nonzero pivot and terminates after at most Q rows. It neither assumes that a chosen minor is nonzero nor substitutes a generic matrix for the original one.

The next-jet comparison also has a complete scalar-polynomial form. Put $b_i = B_i(0)$, and choose the first original index i_* with $b_{i_*} \neq 0$, which exists by the entire Bezout identity at zero. Retain

$$C_i = R_i - \frac{b_i}{b_{i_*}} R_{i_*} \quad (i \neq i_*), \quad \mathcal{C}_k = \text{span}_{i \neq i_*} C_i \subset \mathbb{C}[S]_{\leq d-1}. \quad (\text{CPR17})$$

The leading coefficient of R_i is b_i ; the leading scalar of $\mathcal{T}_A(\chi_k R_i)$ is $b_i \mu_v \binom{Q}{v}$. Hence the full intersection of the four-row span with degrees through D_0 is exactly $\mathcal{T}_A(\chi_k \mathcal{C}_k)$. Its dimension is three: dependence of the three displayed combinations would contradict independence of all four original rows. It follows from CPR13–14 that the exact ranks and complete kernel dimensions are

$$\begin{aligned} g_k &= \dim \mathcal{G}_k, & a_k &= \dim(\mathcal{G}_k \cap \mathcal{C}_k), & 0 &\leq a_k \leq 3, \\ c_0 &= g_k, & c_1 &= g_k - a_k, & h_0 &= d - g_k, & h_1 &= d - 3 - g_k + a_k, \\ & & 0 &\leq c_0 - c_1 = a_k \leq 3, & h_0 - h_1 &= 3 - a_k. \end{aligned} \quad (\text{CPR18})$$

These quantities use the original common image, not separate new rank choices at the two endpoints. The complete kernels themselves are

$$\begin{aligned} \ker(\mathcal{C}_0 \rightarrow T_0/W) &\simeq \text{Ann}_{(\mathbb{C}[S]_{\leq d-1})^\vee} \mathcal{G}_k, \\ \ker(\mathcal{C}_1 \rightarrow T_1/W) &\simeq \text{Ann}_{(\mathbb{C}[S]_{\leq d-1})^\vee} (\mathcal{G}_k + \mathcal{C}_k). \end{aligned} \quad (\text{CPR19})$$

In this display $\mathcal{C}_e$ denotes the original CIT connecting cohomology image, distinguished by its endpoint subscript from the polynomial space $\mathcal{C}_k$. Use the PDS4 quotient-value coordinates and PDS11's exact truncation to identify its primitive sources. A primitive vector is killed exactly when all the transpose polynomials in CPR13 pair to zero; the common nonzero scalar α_A does not change this zero set. At the upper endpoint it must additionally annihilate precisely the three polynomials of CPR17. This proves both maps, their surjectivity onto the specified kernels, and the dimensions CPR18.

200.4 The full low-cohomology excess and what remains in the metric

The complete LCC positive connecting sequence is retained:

$$0 \longrightarrow \mathcal{C}_e \longrightarrow H^1(\mathcal{I}_e) \longrightarrow \ker U_e \longrightarrow 0. \quad (\text{CPR20})$$

It is a sequence of the defining low-cutoff coefficient spaces. Let $\mathcal{Z}_e$ be its full cycle space, $\mathcal{S}_e$ its connecting cycle subspace, and $\mathbb{J} = \ker \mathbb{B}$. The identity on representatives gives the more precise exact sequence

$$0 \longrightarrow \frac{\mathcal{Z}_e \cap \mathbb{J}}{\mathcal{S}_e \cap \mathbb{J}} \longrightarrow \frac{\mathcal{Z}_e}{\mathcal{S}_e} \longrightarrow \frac{\mathcal{Z}_e + W}{\mathcal{S}_e + W} \longrightarrow 0. \quad (\text{CPR21})$$

For the kernel, if $z = s + w$ with $s \in \mathcal{S}_e$ and $w \in W = \mathcal{B} + \mathbb{J}$, subtract its boundary part, which lies in $\mathcal{S}_e$, and represent the class by $z - s - b \in \mathcal{Z}_e \cap \mathbb{J}$. Two such representatives agree exactly modulo $\mathcal{S}_e \cap \mathbb{J}$. Surjectivity is the sum-space definition. LCC13 identifies the middle space with $\ker U_e$ using its positive double lift, retaining all six compatibility components and every lower evaluation. Thus the additional target rank is the dimension of the last quotient in CPR21; it is zero whenever that exact tail kernel is zero. Its source and target metrics remain the two distinct attained minima of LCC21–23.

The independent low-tail companion in this directory calculates the same $\ker U_e$ through the actual barycentric biform. Its full proof retains each of the four $(2, 2)$ conductor factors and gives

$$v \leq 32, \quad \kappa_e = \dim \ker U_e \leq 3 \max(v - e - 2, 0). \quad (\text{CPR22})$$

The companion supplies the full irreducibility, derivative and next-jet proofs; it is part of this source closure. In particular

$$\begin{aligned} d_{T,e} &= 0 \quad (v - e \leq 2), & \text{rank } \Psi_e &= c_e \quad (v - e \leq 2), \\ 0 &\leq d_{T,e} \leq 3 \max(v - e - 2, 0) & \text{in general.} \end{aligned} \quad (\text{CPR23})$$

All claims here are on the original simple-quartet domain. No original multiplicity has been changed to reach this domain.

At every later target cutoff N and either original source-order pair, the image in CPR13 carries the restriction of the original kernel-target quotient by $\text{im } \Phi$; its source carries the quotient of the full CIT connecting Gram by the exact kernel CPR19. CPR8 and CPR11 give the fixed algebraic transport between these coordinate presentations. If Z_h is its actual kernel basis and Z_c a complementary source lift, those source forms are respectively the restriction to Z_h and the Schur quotient in Z_c . A change of either fixed basis acts by the full congruence $R^*(\cdot)R$; its squared determinant is retained at each endpoint. Neither CPR18 nor the zero excess in CPR23 assigns the determinant of this source-to-target contraction a value. The exact polynomial space $\mathcal{G}_k$ and its three-row intersection are now a complete original algebraic presentation of the connecting rank, without primitive lift or Gamma-frame dependence. No universal numerical value for g_k or a_k is asserted in this calculation.

201 The complete original barycentric low-tail calculation

This standalone calculation concerns the actual last differential of LCC5–6 and LCA37–39. It retains the entire original root rectangle, every coefficient of the invertible original period matrix, and the ordered exterior signs. No source or provider is revised by this note.

201.1 Original objects and the coefficient kernel

Fix $k = 4l + 1 \geq 9$, $j = k + 4$, $Q = (j + 1)^2$, $e \in \{0, 1\}$, $0 \leq v \leq 80$, and put

$$L_e = Q - v + e, \quad \ell_e = \max(v - e, 0), \quad \beta_* = \frac{1}{2} - \delta - i\gamma, \quad \alpha = 2\delta, \quad \eta = 2i\gamma, \quad \delta, \gamma > 0. \quad (\text{BTK1})$$

The original roots and their monic polynomial are

$$\lambda_{ab} = j\beta_* + a\alpha + b\eta \quad (0 \leq a, b \leq j), \quad \chi_j(S) = \prod_{a,b=0}^j (S - \lambda_{ab}). \quad (\text{BTK2})$$

They are pairwise distinct: equality of two roots first equates their real index differences and then their imaginary index differences. Let $\mathbf{H}$ be the unchanged invertible original 4 by 4 matrix, and retain

$$L_i(Z, W) = (\mathbf{H}(ZW, Z, W, 1)^T)_i, \quad h_i = L_i^4 = \sum_{r,t=0}^4 b_{i,r,t} Z^r W^t, \quad B_i(z) = e^{4\beta_* z} h_i(e^{\alpha z}, e^{\eta z}). \quad (\text{BTK3})$$

An input of the last coefficient differential is ordered as $(E_{123}, E_{124}, E_{134}, E_{234})$. Its scalar output is exactly

$$-B_4 E_{123} + B_3 E_{124} - B_2 E_{134} + B_1 E_{234} = \sum_{i=1}^4 \varepsilon_i B_i E_{\hat{i}}, \quad \varepsilon_i = (-1)^{i-1}, \quad (\text{BTK4})$$

where $\hat{i}$ denotes the increasing complementary triple. Each $E_{\hat{i}}$ belongs to the full degree- j exponential space and satisfies $E_{\hat{i}}^{(n)}(0) = 0$ for $0 \leq n < L_e$. The dimension of the kernel of BTK4 on these four tail spaces is the original κ_e . Indeed root addition transports the original coefficient convolution to BTK4. The degree- $(j + 4)$ exponential map is injective by its distinct roots and full Vandermonde. Its output has zero first L_e moments by Leibniz's rule; therefore the tail extraction $S_{j+4,e}$ in LCC5 is injective on that output. Thus this is exactly $\ker U_e$, not a larger germ or truncated kernel.

201.2 Exact barycentric tail coordinates

Define the complete original generator and its biform presentation by

$$\phi_j(z) = \sum_{a,b=0}^j \frac{e^{\lambda_{ab} z}}{\chi'_j(\lambda_{ab})}, \quad F_j(Z, W) = \sum_{a,b=0}^j \frac{Z^a W^b}{\chi'_j(\lambda_{ab})}, \quad \phi_j(z) = e^{j\beta_* z} F_j(e^{\alpha z}, e^{\eta z}). \quad (\text{BTK5})$$

Every coefficient in F_j is nonzero. In particular its bidegree is exactly (j, j) and its constant coefficient is nonzero. Lagrange interpolation of S^n for $0 \leq n < Q$ gives

$$S^n = \sum_{a,b} \lambda_{ab}^n \frac{\chi_j(S)}{(S - \lambda_{ab}) \chi'_j(\lambda_{ab})}.$$

Comparison of the coefficient of S^{Q-1} proves

$$\sum_{a,b} \frac{\lambda_{ab}^n}{\chi'_j(\lambda_{ab})} = \begin{cases} 0 & 0 \leq n < Q - 1, \\ 1 & n = Q - 1. \end{cases} \quad \text{Hence} \quad \text{ord}_0 \phi_j = Q - 1. \quad (\text{BTK6})$$

If $\ell = \ell_e > 0$, then $L_e = Q - \ell$. The exact tail map is

$$p(S) \in \mathbb{C}[S]_{<\ell} \mapsto E_p(z) = p(\partial_z)\phi_j(z) = \sum_{a,b} \frac{p(\lambda_{ab})}{\chi'_j(\lambda_{ab})} e^{\lambda_{ab}z}. \quad (\text{BTK7})$$

If p has degree r and leading coefficient $p_r \neq 0$, BTK6 gives

$$\text{ord}_0 E_p = Q - 1 - r, \quad E_p^{(Q-1-r)}(0) = p_r. \quad (\text{BTK8})$$

Thus BTK7 is injective and its image lies in the required tail. That tail has dimension ℓ because its first $Q - \ell$ Vandermonde rows are independent. Consequently BTK7 is an isomorphism onto the entire original tail. For $\ell = 0$, its first Q moments already vanish and the tail is zero, also when $L_e = Q + 1$.

For completeness, the exact conversion to LCC5's remaining-derivative coordinates is as follows. Write $p(S) = \sum_{r=0}^{\ell-1} p_r S^r$ and put

$$m_n = \sum_{a,b} \frac{\lambda_{ab}^n}{\chi'_j(\lambda_{ab})}, \quad y_t = E_p^{(L_e+t)}(0) = \sum_{r=0}^{\ell-1} m_{L_e+t+r} p_r \quad (0 \leq t < \ell). \quad (\text{BTK9})$$

The displayed matrix has zero entries for $t+r < \ell-1$ and ones for $t+r = \ell-1$, so its determinant is $(-1)^{\ell(\ell-1)/2}$. LCA38's Taylor coordinates are precisely $y_t/(L_e+t)!$. This includes every factorial and supplies the inverse coordinate map through the invertible displayed matrix.

Both original native families are retained: at cochain degree p , $s_p = 1$ for every p , or $s_p = k+4p$ for every p , with $c'_p = (k+4p)/2 - 4$. The native evaluation matrix is obtained from the ordinary moment matrix by the full original triangular polynomial frame matrix. Its diagonal at row n is

$$\frac{i^{-n}}{\sqrt{(2\pi)^{s_p/2} n! (s_p/2)_n}} \neq 0. \quad (\text{BTK10})$$

Thus it has exactly the same kernel. This proves the map from both original native presentations to BTK7; no norm, phase or mass is used to change the coefficient kernel.

201.3 Irreducibility from maximal contact

For any nonzero polynomial G of bidegree at most (a, b) , $G(e^{\alpha z}, e^{\eta z})$ is a linear combination of at most $(a+1)(b+1)$ exponentials with distinct exponents. A nonzero such combination has order at most the number of exponents minus one: otherwise its first that many derivatives form an invertible Vandermonde annihilating its entire coefficient vector. Hence

$$\text{ord}_0 G(e^{\alpha z}, e^{\eta z}) \leq ab + a + b. \quad (\text{BTK11})$$

Suppose $F_j = GH$ with both factors nonconstant. Polynomial degrees in each variable add in this integral domain. If G has bidegree (a, b) , H therefore has bidegree $(j-a, j-b)$. Their orders add, and BTK11 gives

$$\text{ord}_0 F_j(e^{\alpha z}, e^{\eta z}) \leq Q - 1 - \{a(j-b) + b(j-a)\} < Q - 1. \quad (\text{BTK12})$$

For $0 \leq a, b \leq j$, the braces are nonnegative and vanish only at $(a, b) = (0, 0)$ or (j, j) ; either case would make a factor constant. This contradicts BTK5–6, proving F_j irreducible in $\mathbb{C}[Z, W]$. The polynomial ring is a unique factorization domain (successive application of Gauss's lemma to $\mathbb{C}[Z]$), so F_j is prime. Its bihomogenization is also irreducible. A bihomogeneous factorization would dehomogenize to one just excluded unless one factor were supported entirely on the two divisors at infinity. Such a factor is a monomial in the homogenizing variables, which cannot divide this biform because its original corner coefficients are all nonzero.

There is also a useful stronger structural statement. Every nonzero tail polynomial $F_p = \sum p(\lambda_{ab})Z^aW^b/\chi'_j(\lambda_{ab})$ is irreducible if $1 \leq \ell \leq j$. Its order is at least $Q - \ell$. If its degree in either variable were at most $j - 1$, BTK11 would bound that order by $j(j + 1) - 1 = Q - j - 2$, a contradiction. Thus its bidegree is (j, j) . For a nontrivial splitting, the positive integer in braces in BTK12 is at least j . To verify this, if $1 \leq a \leq j - 1$, its minimum over $0 \leq b \leq j$ occurs at an endpoint and equals $j \min(a, j - a) \geq j$; the cases $a = 0, j$ give the same bound directly. Such a splitting would bound the order by $Q - j - 1 < Q - \ell$, another contradiction. This structural statement alone asserts no kernel vanishing for $\ell \geq 3$.

201.4 Independence of the four unchanged fourth powers

We include the complete coefficient argument used in LCA20. Put $Q_0(t) = t_1t_4 - t_2t_3$. It is primitive and linear in t_1 over the fraction field of $\mathbb{C}[t_2, t_3, t_4]$, hence irreducible by Gauss's lemma and prime. A homogeneous polynomial vanishing on $(ZW, Z, W, 1)$ vanishes after substitution $t_1 = t_2t_3/t_4$ on $t_4 \neq 0$. Its Laurent remainder is zero, so after clearing powers of t_4 it is divisible by Q_0 ; primality and $Q_0 \nmid t_4$ remove that power. This proves the complete homogeneous Segre ideal assertion.

If $\sum c_i h_i = 0$ with some $c_i \neq 0$, the substitution $x = Ht$ therefore gives a nonzero polynomial identity

$$\sum_{i=1}^4 c_i x_i^4 = Q_H(x) R_2(x), \quad Q_H(x) = Q_0(H^{-1}x), \quad (\text{BTK13})$$

with Q_H a rank-four quadratic and R_2 homogeneous of degree two. If some $c_i = 0$, the left side has degree zero in x_i . Additivity of this polynomial degree forces Q_H independent of x_i , contradicting rank four. Thus all c_i are nonzero. The two quadrics have a nonzero common zero: restrict R_2 to the Segre quadric parametrized after H ; its restriction is a biform of degree $(2, 2)$. If it is identically zero any point works. Otherwise choose a projective w for which it is a nonzero binary quadratic in z ; that quadratic has a projective root over $\mathbb{C}$. At the resulting nonzero common zero, differentiating BTK13 gives $\nabla \sum c_i x_i^4 = 0$. But its components are $4c_i x_i^3$, which cannot all vanish at a nonzero point. This contradiction proves

$$h_1, h_2, h_3, h_4 \text{ are linearly independent over } \mathbb{C}. \quad (\text{BTK14})$$

201.5 Exact vanishing when each tail has dimension at most two

Define the original scalar differential operator on biforms

$$\mathcal{D}_j = j\beta_* + \alpha Z \partial_Z + \eta W \partial_W. \quad \partial_z \phi_j = e^{j\beta_* z} (\mathcal{D}_j F_j)(e^{\alpha z}, e^{\eta z}). \quad (\text{BTK15})$$

The irreducible polynomial F_j does not divide $\mathcal{D}_j F_j$. Indeed this derivative polynomial is nonzero and has bidegree at most (j, j) . Divisibility would force $\mathcal{D}_j F_j = c F_j$ with a constant c . Comparison of every nonzero coefficient in BTK5 would then give $j\beta_* + a\alpha + b\eta = c$ for every a, b , which contradicts the distinct roots. Thus

$$\gcd(F_j, \mathcal{D}_j F_j) = 1. \quad (\text{BTK16})$$

If $\ell_e = 1$, BTK7 says $E_{\hat{e}} = b_i \phi_j$. BTK4 becomes $\phi_j \sum_i \varepsilon_i b_i B_i = 0$. The nonzero analytic factor can be removed, and the exact exponential-to-biform injection gives $\sum_i \varepsilon_i b_i h_i = 0$. BTK14 forces every $b_i = 0$.

If $\ell_e = 2$, write uniquely $E_{\hat{e}} = (a_i \partial_z + b_i) \phi_j$. Set

$$A(Z, W) = \sum_i \varepsilon_i a_i h_i(Z, W), \quad B(Z, W) = \sum_i \varepsilon_i b_i h_i(Z, W).$$

BTK4, BTK15 and the injective degree- $(j + 4)$ exponential map give the exact polynomial identity

$$A \mathcal{D}_j F_j + B F_j = 0. \quad (\text{BTK17})$$

Primality of F_j and BTK16 imply $F_j \mid A$. But A has bidegree at most $(4, 4)$ whereas F_j has bidegree (j, j) and $j \geq 13$. Additivity of degrees makes this impossible for nonzero A , so $A = 0$. BTK17 then gives $B = 0$. BTK14, with every sign ε_i nonzero, proves $a_i = b_i = 0$ for all four components. Including the empty tail gives

$$\boxed{\ell_e = \max(v - e, 0) \leq 2 \implies \kappa_e = 0, \quad \rho_e = 4\ell_e.} \quad (\text{BTK18})$$

The stated domain is exact: both endpoints vanish for $0 \leq v \leq 2$; the $e = 1$ endpoint also vanishes for $v = 3$. For $v = 3$ only, this gives $\kappa_1 = 0$ without assigning a value to κ_0 . The conclusion holds at every permitted k , for every original invertible period matrix, and in both original native families through BTK10.

201.6 An exact lower-degree polynomial presentation for every tail

This final identity retains the complete remaining problem at arbitrary ℓ_e . Put $\nu_{rt} = \beta_{4,r,t} = 4\beta_* + r\alpha + t\eta$ and

$$R_{rt}(S) = \frac{\chi_{j+4}(S)}{\chi_j(S - \nu_{rt})} \in \mathbb{C}[S], \quad \deg R_{rt} = 8j + 24. \quad (\text{BTK19})$$

The denominator roots are exactly the shifted original j -rectangle, a subset of the complete $(j+4)$ -rectangle, so division is exact. For the unique tail polynomials p_i in BTK7, define

$$\mathcal{U}(p_1, p_2, p_3, p_4)(S) = \sum_{i=1}^4 \varepsilon_i \sum_{r,t=0}^4 b_{i,r,t} p_i(S - \nu_{rt}) R_{rt}(S). \quad (\text{BTK20})$$

At any upper root Λ , if $\Lambda - \nu_{rt}$ is a lower root, differentiating $\chi_{j+4} = \chi_j(S - \nu_{rt}) R_{rt}$ at Λ gives $R_{rt}(\Lambda) = \chi'_{j+4}(\Lambda) / \chi'_j(\Lambda - \nu_{rt})$. If it is not a lower root, then $R_{rt}(\Lambda) = 0$ by BTK19. Consequently $\mathcal{U}(p)(\Lambda) / \chi'_{j+4}(\Lambda)$ is exactly the full output coefficient at $e^{\Lambda z}$ in BTK4, including the absent shifted nodes and all repeated contributions to present nodes. Moreover

$$\deg \mathcal{U}(p) \leq 8j + \ell_e + 23 < (j+5)^2, \quad \ker U_e \simeq \ker \mathcal{U}. \quad (\text{BTK21})$$

For the strict inequality, the difference is $Q - \ell_e + 1 > 0$ since $Q \geq 196$ and $\ell_e \leq 80$. Vanishing of the output is therefore equivalent to vanishing of this entire polynomial: its values at all $(j+5)^2$ distinct upper roots determine it. BTK7–9 give the exact input coordinate isomorphism. BTK20–21 is an equality for arbitrary tail size; it does not assign an unproved rank for $\ell_e \geq 3$.

201.7 Successive original moments and the improved uniform bound

Fix the same j , original roots and quartic symbols, and for each integer $0 \leq \ell \leq 80$ let K_ℓ be the kernel of BTK4 restricted to the four original degree- j exponentials whose derivatives of orders $0, \dots, Q - \ell - 1$ vanish. For $\ell = 0$ this is the zero space. These are coefficient subspaces of the same four full original exponential spaces, and

$$K_{\ell-1} \subset K_\ell, \quad J_\ell : K_\ell \longrightarrow \mathbb{C}^4, \quad J_\ell(E) = (E_i^{(Q-\ell)}(0))_{i=1}^4, \quad \ker J_\ell = K_{\ell-1}. \quad (\text{BTK22})$$

The last equality follows from adding exactly the single missing derivative in each input component; the entire convolution identity and all earlier zero derivatives remain unchanged.

Differentiate the exact identity BTK4 precisely $N = Q - \ell$ times at zero. Leibniz's formula is

$$0 = \sum_{i=1}^4 \varepsilon_i \sum_{r=0}^N \binom{N}{r} B_i^{(r)}(0) E_i^{(N-r)}(0) = \sum_{i=1}^4 \varepsilon_i B_i(0) E_i^{(N)}(0). \quad (\text{BTK23})$$

For $r > 0$ the derivative of E is below N and vanishes; the surviving binomial coefficient is $\binom{N}{0} = 1$. The row $(\varepsilon_i B_i(0))_i$ is nonzero. Directly from the full original matrix, $B_i(0) = L_i(1, 1)^4$ and the

column $(L_i(1, 1))_i = \mathbf{H}(1, 1, 1, 1)^\top$ is nonzero because $\mathbf{H}$ is invertible. Thus the image of J_ℓ lies in the specified three-dimensional hyperplane. Rank-nullity gives

$$0 \leq \dim K_\ell - \dim K_{\ell-1} \leq 3. \quad (\text{BTK24})$$

No factorial is suppressed: if the target coordinates were Taylor coefficients, each displayed component would be divided by the common nonzero $N!$, leaving the same hyperplane and rank. The target here is the derivative coordinate specified in BTK22.

BTK18 proves $K_0 = K_1 = K_2 = 0$ on the entire present $j \geq 13$ domain. Summing BTK24 from 3 to ℓ therefore proves

$$\dim K_\ell \leq 3 \max(\ell - 2, 0), \quad \kappa_e \leq 3 \max(v - e - 2, 0). \quad (\text{BTK25})$$

This retains the actual kernel, rather than replacing it by an arbitrary tuple in the hyperplane. In particular for $v = 3$ the two endpoint dimensions satisfy $\kappa_1 = 0$ and $0 \leq \kappa_0 \leq 3$.

Here is the complete original factor argument for the numerical conductor range used with this bound. The original full conductor is the product of the four Segre restrictions $C_a = Q_a(\mathbf{H}t)$, where $t = (z_0 w_0, z_0 w_1, z_1 w_0, z_1 w_1)^\top$ and $1 \leq a \leq 4$, of the other four quadrics in the original five-member orbit. Each Q_a is nonassociate to $Q_{\mathbf{H}}$. The homogeneous Segre ideal argument in BTK13 shows that its degree-two restriction cannot vanish identically: otherwise Q_a would be a constant multiple of $Q_{\mathbf{H}}$. Thus each nonzero C_a is a biform of degree $(2, 2)$, with the entire original coefficient array $C_a(Z, W) = \sum_{r,t=0}^2 c_{a,r,t} Z^r W^t$ retained. The exact original symbol is

$$E_A(z) = \prod_{a=1}^4 E_a(z), \quad E_a(z) = e^{2\beta_* z} C_a(e^{\alpha z}, e^{\eta z}).$$

Each factor has nine distinct possible exponents. Its first nine derivatives cannot all vanish by the invertible nine-node Vandermonde and the nonzero coefficient array. Hence $v_a = \text{ord}_0 E_a \leq 8$. Multiplication of these exact Taylor series gives

$$v = \sum_{a=1}^4 v_a \leq 32, \quad \frac{\mu_v}{v!} = \prod_{a=1}^4 \frac{E_a^{(v_a)}(0)}{v_a!} \neq 0, \quad \boxed{\kappa_0 \leq 90, \quad \kappa_1 \leq 87.} \quad (\text{BTK26})$$

This is the same full conductor product and coefficient identity spelled out in the companion `../ORIGINAL_CONDUCTOR_VANISHING_ORDER.tex`; no factor or original multiplicity is removed. The last two inequalities are BTK25 evaluated at $\ell_0 \leq 32$ and $\ell_1 \leq 31$.

202 The complete original strip-to-pole inverse

202.1 Exact inverse between the original conductor strip and all pole observations

Fix the original simple-quartet period in the proved RPK rank domain, with the same k, q, q', Δ, j , actual coefficient matrices C, D , and full principal-part map $\Pi_k : \mathbb{C}^q \rightarrow \mathbb{C}^{p_k}$. Use precisely the original ROQ2 matrices, so that

$$\mathbf{S}\mathbf{N} + C^\top \mathbf{V} = I_q, \quad \mathbf{N}\mathbf{S} = I_\Delta, \quad \ker \mathbf{N} = \text{im } C^\top. \quad (\text{RPI1})$$

Here the ambient coefficient column is paired complex-linearly with primary vectors; no Hermitian adjoint is used in these quotient maps. Define

$$\mathbf{E}_k = \Pi_k \mathbf{S} : \mathbb{C}^\Delta \longrightarrow \mathbb{C}^{p_k}. \quad (\text{RPI2})$$

RPK22 proves $\ker \Pi_k = \text{im } C^\top$. Multiplying RPI1 by Π_k therefore gives $\Pi_k = \mathbf{E}_k \mathbf{N}$. If $\mathbf{E}_k v = 0$, then $\mathbf{S}v = C^\top w$ for some w . Applying $\mathbf{N}$ gives $v = 0$. Thus $\mathbf{E}_k$ has full column rank Δ and image $\text{im } \Pi_k$. Select, in a fixed ordering, a coordinate-row selector $P_J : \mathbb{C}^{p_k} \rightarrow \mathbb{C}^\Delta$ such that $\mathbf{E}_J = P_J \mathbf{E}_k$ is invertible. Existence follows from the proved column rank. The quotient isomorphism and its full inverse are

$$\begin{aligned} \bar{\Pi}_k : \mathbb{C}^q / \text{im } C^\top &\xrightarrow{\sim} \text{im } \Pi_k, & [d] &\longmapsto \Pi_k d, \\ \bar{\Pi}_k^{-1}(y) &= [\mathbf{S} \mathbf{E}_J^{-1} P_J y]. \end{aligned} \quad (\text{RPI3})$$

Indeed, write $y = \mathbf{E}_k v$. Its displayed inverse is $[\mathbf{S}v]$ and returns $\mathbf{E}_k v = y$ under $\bar{\Pi}_k$. Conversely the displayed inverse of $\Pi_k d = \mathbf{E}_k \mathbf{N}d$ is $[\mathbf{S}\mathbf{N}d] = [d]$ by RPI1. This proves both compositions, including independence from the chosen lift of $[d]$.

Retain the entire original certificate matrix $\mathbf{F} = D^\top$, the ROQ matrix $Z = \mathbf{N}\mathbf{F}$, and its original rank- j pivot selector $P_I : \mathbb{C}^\Delta \rightarrow \mathbb{C}^j$, with $Z_I = P_I Z$ invertible. Then the complete residual pole map and inverse are

$$\begin{aligned} \Pi_k D^\top &= \mathbf{E}_k Z, & \mathcal{P}_k &= \text{im}(\Pi_k D^\top), \\ \mathbf{G}_k &= Z_I^{-1} P_I \mathbf{E}_J^{-1} P_J : \mathbb{C}^{p_k} \longrightarrow \mathbb{C}^j, & \mathbf{G}_k \Pi_k D^\top &= I_j, \\ & & (\Pi_k D^\top) \mathbf{G}_k|_{\mathcal{P}_k} &= I_{\mathcal{P}_k}. \end{aligned} \quad (\text{RPI4})$$

The first line is RPI2 and $Z = \mathbf{N}D^\top$. For a vector $x \in \mathbb{C}^j$, multiplication gives $\mathbf{G}_k \mathbf{E}_k Zx = Z_I^{-1} P_I Zx = x$. Every $y \in \mathcal{P}_k$ equals $\mathbf{E}_k Zx$, so the final identity follows from this same equality. Thus the inverse is explicit in all the original strip and certificate coefficients.

At every original endpoint and source order $s = 1, k$, retain exactly

$$\Omega_N^{(s)} = D[R_N^{(s)} - R_N^{(s)} C^* (C R_N^{(s)} C^*)^{-1} C R_N^{(s)}] D^*.$$

The full principal-part subspace $\mathcal{P}_k$ carries the metric

$$\langle y, y \rangle_{N,s}^{\text{all poles}} = y^* \mathbf{G}_k^* \overline{\Omega_N^{(s)}} \mathbf{G}_k y, \quad y \in \mathcal{P}_k. \quad (\text{RPI5})$$

To prove it, write $y = \Pi_k D^\top x$ and apply RPI4. The right side is $x^* \overline{\Omega_N^{(s)}} x = x^\top \Omega_N^{(s)} \bar{x}$, exactly the squared norm of the original functional $x^\top D$. It is positive for nonzero y because $x = \mathbf{G}_k y$ is nonzero and $\Omega_N^{(s)}$ is positive definite. If P_B selects any proved invertible j -row matrix $B_k = P_B \Pi_k D^\top$ and $b = P_B y$, then $b = B_k x$; substitution recovers the exact RPK metric $B_k^{-*} \overline{\Omega_N^{(s)}} B_k^{-1}$. All p_k observed principal parts and their linear relations therefore have a precise metric realization, with no replacement of the original conditioned Gamma covariance or any of its source factors.

203 The original pole detector, connecting image, and sharper low-tail return

Fix the original simple quartet, amplitude, period and logarithm branch. For the pole-coordinate statements retain exactly the RPK domain $0 < \delta < 1/2$, $\gamma > 2$, multiplicity one, and a fixed original period $|u| \geq R_*$ outside RPK's finite gate-zero superset; in particular every $|u| \geq R_{\text{rank}}$ is permitted. The connecting and tail statements below retain their larger original five-orbit domain. No conclusion about a gate-zero period is introduced by this receiving calculation.

Write $n \equiv 1 \pmod{4}$, $n \geq 9$, $q_n = (n+1)^2$, $q'_n = (n-7)^2$, and let R_n be the original biform coefficient space of bidegree at most (n, n) . Its coordinates are the literal coefficients in $Z^a W^b$. Let A be the full original conductor, including each factor multiplicity, and put

$$N_d(z) = \sum_{a,b=0}^n d_{ab} e^{\beta_{n,a,b} z}, \quad R_{v,n} = \{d \in R_n : N_d^{(r)}(0) = 0 \ (0 \leq r < v)\}, \quad (\text{RRC1})$$

$$J_n^{\text{raw}} = \text{im}(\mathbb{C}[x_1, x_2, x_3, x_4]_n^{(g)} \rightarrow R_n), \quad J_{v,n}^{\text{raw}} = R_{v,n} \cap J_n^{\text{raw}}.$$

Here g is the original diagonal fifth-root action before Segre restriction. Its coefficient monomials are precisely those used in CPR3. All vector-space duals are complex-linear.

203.1 A full quotient-value isomorphism supplied by the poles

Let $\Pi_n : R_n \rightarrow \mathbb{C}^{p_n}$ be RPK's complete principal-part matrix, with every selected pole and every order $1, \dots, m_\nu$ retained. Its already proved kernel is the exact full conductor image $AR_{n-8} = \text{im } C_n^\top$. Define

$$\mathcal{P}_{v,n} = \Pi_n R_{v,n}, \quad \mathcal{P}_{J,n} = \Pi_n J_{v,n}^{\text{raw}}. \quad (\text{RRC2})$$

There is an explicit isomorphism

$$\vartheta_n : \mathcal{P}_{v,n} \xrightarrow{\sim} R_{v,n}/AR_{n-8}, \quad \Pi_n d \mapsto [d]. \quad (\text{RRC3})$$

To prove all the assertions about this map, multiplication by the full symbol gives $N_{Ah} = E_A N_h$, so $AR_{n-8} \subset R_{v,n}$. If two coefficient vectors have the same principal parts, their difference is a conductor multiple by the full RPK kernel theorem; their classes in RRC3 agree. A zero class has exactly the same conductor kernel and therefore has zero principal parts. Every class has a representative in $R_{v,n}$, proving surjectivity and the inverse rule $[d] \mapsto \Pi_n d$. In particular

$$\dim \mathcal{P}_{v,n} = q_n - q'_n - v, \quad \vartheta_n(\mathcal{P}_{J,n}) = J_{v,n}^{\text{raw}}/AR_{n-8}, \quad (\text{RRC4})$$

$$\mathcal{P}_{v,n}/\mathcal{P}_{J,n} \xrightarrow{\sim} R_{v,n}/J_{v,n}^{\text{raw}} \xrightarrow{\sim} (K_n/(K_n \cap \mathbb{C}[S]_{<v}))^\vee.$$

The first dimension uses the original v independent moment rows on R_n and the injective full conductor multiplication of rank q'_n . The trace identity with its Reynolds factor five places AR_{n-8} inside J_n^{raw} , and its moment vanishing places it inside $J_{v,n}^{\text{raw}}$. This proves the first quotient equality in RRC4. For the last arrow send $[d]$ to $[p] \mapsto \sum d_{ab} p(\beta_{n,a,b})$. It kills the low polynomial subspace by RRC1 and kills conductor changes by $K_n \subset \ker C_n$. The observation annihilator identity KRC3–4 identifies its kernel as exactly $J_{v,n}^{\text{raw}}$ and proves it onto. Thus both the domain quotient and the dual orientation are preserved.

203.2 Transport of the actual connecting map, including its scalar

Use now $k = 4l + 1 \geq 9$, $j = k + 4$, $Q = (j+1)^2$, and the fixed original DNE numerator frame $Z_{\text{exp},n}$ in every degree. The exact DNE-to-pole coordinate isomorphism is

$$\Theta_n : \mathbb{C}^{m_n} \xrightarrow{\sim} \mathcal{P}_{v,n}, \quad \Theta_n c = \alpha_A \Pi_n Z_{\text{exp},n} c, \quad m_n = q_n - q'_n - v, \quad \alpha_A = v!/\mu_v. \quad (\text{RRC5})$$

Indeed KRC5 identifies $c \mapsto \alpha_A[Z_{\exp,n}c]$ with the full original conductor quotient, and RRC3 is its explicit inverse pole presentation. This proves injectivity and surjectivity. It also proves $\Theta_n(\ker \mathbf{B}_n) = \mathcal{P}_{J,n}$, with the original ordinary transpose in $\mathbf{B}_n = Z_n^T A_\ell$.

Set $h_i = L_i^4$ with the unchanged degree-four coefficient arrays, and introduce the following complete pole quotient:

$$\mathcal{T}_{v,k}^{\text{pole}} = \frac{\mathcal{P}_{v,j}^4}{\mathcal{P}_{j,j}^4 + \{\alpha_A(\Pi_j(h_1d), \dots, \Pi_j(h_4d)) : d \in R_{v,k}\}}. \quad (\text{RRC6})$$

Every product h_id is in $R_{v,j}$, because its exponential realization is B_iN_d and therefore has the same required zero jets. Hence every denominator in RRC6 is a literal subspace of its displayed numerator. With $\mathbb{B} = \text{diag}(\mathbf{B}_j, 4)$, $B_{\text{cof}} = \text{stack}_i M_{i,k}$ and $\Phi = \mathbb{B}B_{\text{cof}}$, the isomorphism is

$$\Xi : \mathbb{C}^{4t_{K,j}} / \text{im } \Phi \xrightarrow{\sim} \mathcal{T}_{v,k}^{\text{pole}}, \quad [\mathbb{B}c] \mapsto [\text{diag}(\Theta_j, 4)c]. \quad (\text{RRC7})$$

The original onto map $\mathbb{B}$ supplies all representatives. Two representatives of one value differ by its kernel, which RRC5 takes precisely to $\mathcal{P}_{j,j}^4$. A boundary $B_{\text{cof}}x$ is taken, by the full coefficient multiplication square, to $\alpha_A(\Pi_j h_i Z_{\exp,k} x)_i$. Replacing the numerator of x by a conductor multiple has zero image under every Π_j . Every $d \in R_{v,k}$ is such a numerator plus a conductor multiple, because the DNE frame represents the whole conductor quotient. Thus RRC6 contains exactly all boundary images. These facts prove well-definition, the full kernel, and surjectivity of RRC7.

For a primitive jet $h \in \mathcal{Y}_e^{\text{raw}}$ put $f = E_A h$, retaining its complete Taylor data through $Q + e - 1$. Define each original target numerator by the exact square Vandermonde inverse

$$(N_i(f))_{ab} = \left(\mathbb{V}_j^{-1}(r![z^r](B_i f))_{r=0}^{Q-1} \right)_{ab}, \quad (\mathbb{V}_j)_{r,(a,b)} = \beta_{j,a,b}^r. \quad (\text{RRC8})$$

At $e = 1$ the primitive domain retains all four original equations $[z^Q](\text{ev}_j N_i(f) - B_i(z)f(z)) = 0$, where $\text{ev}_j d = N_d(z)$ is the full original exponential realization; at $e = 0$ the displayed interpolation supplies every equation. Multiplication and exact division by $E_A = z^v(\mu_v/v! + O(z))$ prove that $(\text{ev}_j N_i(f))/E_A$ represents $B_i h$ at precisely the defining cutoff. Every $N_i(f)$ has the first v zero moments. The original connecting image therefore becomes, under RRC7,

$$[h] \mapsto [\alpha_A(\Pi_j N_1(E_A h), \dots, \Pi_j N_4(E_A h))]. \quad (\text{RRC9})$$

An analytic source change with numerator $d \in R_{v,k}$ changes the four interpolants by exactly $(h_id)_i$, by uniqueness of the Vandermonde interpolation and root addition. RRC6 kills every such analytic-source change through its boundary summand; its invariant summand retains the additional connecting kernel calculated below. This proves the full source quotient and the original positive sign in RRC9.

Its transpose has exactly the polynomial image CPR13–18. For a target kernel polynomial $p_i \in K_j$, the functional of its unchanged numerator is $\alpha_A \sum (N_i)_{ab} p_i(\beta_{j,a,b})$. The binomial Leibniz identity gives, term by term,

$$\alpha_A \sum (N_i)_{ab} p_i(\beta_{j,a,b}) = \Lambda_{B_i h}(\alpha_A \mathcal{T}_A p_i) = \Lambda_h(\alpha_A \mathcal{T}_i \mathcal{T}_A p_i). \quad (\text{RRC10})$$

The target functionals used here are the original kernel tuples modulo $K_j \cap \mathbb{C}[S]_{<v}$. Their vanishing on the boundary denominator of RRC6 says exactly that $\sum_i \alpha_A \mathcal{T}_i \mathcal{T}_A p_i$ annihilates the raw source analytic space. By the full annihilator calculation in CPR9–11, this condition is membership in $\mathcal{R}_e = \mathcal{T}_A(\chi_k \mathbb{C}[S]_{\leq 8k+23+e})$. The sum has degree at most $D_0 = Q - v - 1$, so its intersection with $\mathcal{R}_1$ is exactly $\mathcal{R}_0$ by their distinct leading degrees. Thus all expressions descend through the complete lower, invariant, and boundary quotients. Their possible sums form exactly $\mathcal{V}_k \cap \mathcal{R}_0 = \mathcal{I}_k$: any element of this intersection has a defining tuple in $\mathcal{U}_j^4$, hence a kernel tuple by CPR5–7 and a boundary-annihilating functional by the criterion just proved. At $e = 1$, restriction to the primitive domain kills precisely its four defining row polynomials $\mathcal{C}^{(4)}$; the kernel within $\mathcal{I}_k$ is their exact

intersection. This proves that the principal-part presentation pulls back to the same $\mathcal{I}_k$ at $e = 0$ and the same $\mathcal{I}_k/(\mathcal{I}_k \cap \mathcal{C}^{(4)})$ at $e = 1$. With CPR's exact $\mathcal{G}_k$ and $\mathcal{C}_k$, this yields

$$c_0 = \dim \mathcal{G}_k, \quad c_1 = \dim \mathcal{G}_k - \dim(\mathcal{G}_k \cap \mathcal{C}_k). \quad (\text{RRC11})$$

RPK's evaluated CFR rank $t_n = 0$ concerns the original common-factor finite-jet target. RRC3–10 proves the exact surviving morphism through the full polar quotient when that target is zero. Its rank is given by RRC11; the value of t_n supplies no additional equation for these two dimensions.

203.3 The original metrics carried through these same coordinates

For an original ambient cutoff N give $\mathbb{C}^{m_j}$ its full DNE native form $D_{j,N-v}^{(s)}$, with $s = 1$ or j and center $j/2 - 4$. Choose once any full frame P_j of $\mathcal{P}_{v,j}$ and write $\Theta_j = P_j H_j$, where H_j is invertible and is fixed independently of N, s . In these pole coordinates the form is

$$G_{j,N,s}^{\text{pole}} = H_j^{-*} D_{j,N-v}^{(s)} H_j^{-1}. \quad (\text{RRC12})$$

For a pole coordinate $y = H_j c$, substitution gives $y^* G_{j,N,s}^{\text{pole}} y = c^* D_{j,N-v}^{(s)} c$ exactly. Apply this substitution in all four copies and to every fixed restriction and affine quotient fibre of RRC6–9. It is a bijection of each such fibre and preserves every squared norm, so its attained minimum is exactly the original CIT/LCC minimum. For an onto coordinate map T and this positive form G , that minimum is $(TG^{-1}T^*)^{-1}$: the minimizing lift is $G^{-1}T^*(TG^{-1}T^*)^{-1}y$, and every other lift differs by a G -orthogonal kernel vector. Thus the source and target Grams of the CIT21/LCC21–23 contraction and their full Schur corrections are transported by RRC12; the pole coordinates introduce no Euclidean substitute. The determinant factor $|\det H_j|^{-2}$ is retained at each endpoint and cancels only in its complete four-return.

203.4 The sharper tail bound in every affected receiving expression

The complete BTK1–26 proof, with its original four signed blocks, establishes

$$v \leq 32, \quad \frac{\mu_v}{v!} = \prod_{a=1}^4 \frac{E_a^{(v_a)}(0)}{v_a!}, \quad \kappa_e \leq \bar{\kappa}_e := 3 \max(v - e - 2, 0). \quad (\text{RRC13})$$

The equality keeps every factor of the full conductor. In particular $\bar{\kappa}_0 \leq 90$ and $\bar{\kappa}_1 \leq 87$. For completeness the propagation to the full target uses the actual spaces $\mathcal{B} \subset \mathcal{S}_e \subset \mathcal{Z}_e$, $\mathbb{J} = \ker \mathbb{B}$, $W = \mathcal{B} + \mathbb{J}$ and the identity on representatives. They give

$$0 \rightarrow \frac{\mathcal{Z}_e \cap \mathbb{J}}{\mathcal{S}_e \cap \mathbb{J}} \rightarrow \frac{\mathcal{Z}_e}{\mathcal{S}_e} \rightarrow \frac{\mathcal{Z}_e + W}{\mathcal{S}_e + W} \rightarrow 0. \quad (\text{RRC14})$$

An element of the middle kernel is $z = s + b + j$ with $s, b \in \mathcal{S}_e$ and $j \in \mathbb{J}$; then $j \in \mathcal{Z}_e \cap \mathbb{J}$. Its class is unique modulo $\mathcal{S}_e \cap \mathbb{J}$. Every target class is represented by a $z \in \mathcal{Z}_e$, proving exactness and surjectivity. The positive double lift LCC13 identifies the middle space with $\ker U_e$. Consequently

$$0 \leq d_{T,e} \leq \kappa_e \leq \bar{\kappa}_e, \quad \text{rank } \Psi_e = c_e + d_{T,e}. \quad (\text{RRC15})$$

The rank equality follows by adding the dimension of the last quotient to $\dim((\mathcal{S}_e + W)/W) = c_e$.

Keep the original four-return $\mathcal{R}f = -f(q_j - 1) - f(q_j) + f(2q_j - 1) + f(2q_j)$, the full FRE constant $A_{j,s}^{\text{end}} \geq 1$, and the NSR order widths $\omega_{j,D}, \Omega_j^{\text{lo}}, \Omega_j^{\text{hi}}$ and $W_j = \Omega_j^{\text{lo}} + \Omega_j^{\text{hi}}$ at the same four displaced target cutoffs. The improved finite receivers are

$$\begin{aligned} |\mathcal{R} \log \omega_e^{(s)} - \mathcal{R} \log \omega_{\text{conn},e}^{(s)}| &\leq 2\kappa_e \log A_{j,s}^{\text{end}} \leq 6 \max(v - e - 2, 0) \log A_{j,s}^{\text{end}}, \\ |\Delta \mathcal{R} \log \omega_e| &\leq (c_e + d_{T,e}) W_j \leq (c_e + \bar{\kappa}_e) W_j, \\ |\Delta \mathcal{R}(\log \omega_e - \log \omega_{\text{conn},e})| &\leq \kappa_e W_j \leq \bar{\kappa}_e W_j, \\ -\bar{\kappa}_e \Omega_j^{\text{hi}} &\leq \Delta \mathcal{R} \log \det \text{Gram}(\mathcal{Z}_e/\mathcal{S}_e) \leq \bar{\kappa}_e \Omega_j^{\text{lo}}. \end{aligned} \quad (\text{RRC16})$$

Here Δ means the paired-source-order difference from $(1, 1)$ to (k, j) ; it does not denote a change of period or of coefficient frames. To prove the first line factor both contractions using the three fixed quotients of ranks d_K, d_T, κ_e , where $d_K + d_T = \kappa_e$ by RRC14. The full determinant difference is the sum of the first two quotient logarithms minus the third, plus a fixed frame constant. Each quotient's monotone four-return lies in $[0, 2r \log A_{j,s}^{\text{end}}]$, by the original form sandwich and taking the same affine minima. The positive and negative contributions are each at most $2\kappa_e \log A$; the frame constant cancels under $\mathcal{R}$. For the second line the complete contraction has rank $c_e + d_T$. Each of its two quotient forms has the same NSR source-order interval, so subtraction cancels its common scalar and gives the symmetric width $(c_e + d_T)\omega_{j,D}$ at one cutoff. Taking the four original signs gives the stated W_j bound. The identical calculation with ranks d_K, d_T, κ_e proves the third line. The single rank- κ_e quotient gives the fourth line, by NSR13. All widths are nonnegative, so insertion of RRC13 yields each final inequality without changing its sign.

When $v - e \leq 2$, the proof gives $\kappa_e = 0$ and therefore $\mathcal{Z}_e = \mathcal{S}_e$ as actual coefficient subspaces. Thus $d_{T,e} = d_{K,e} = 0$, $\text{rank } \Psi_e = c_e$, and $\omega_e(N) = \omega_{\text{conn},e}(N)$ pointwise in both native families at every retained later target cutoff. This is an equality of their original quotient-value maps and attained minima. Every term of the first, third, and fourth lines of RRC16 is then exactly zero. The original cohomology dimensions remain LCC13–14 with this proved value of κ_e ; for general v they remain those exact formulas with the improved range RRC13. No value of the remaining absolute allocation is inserted.

204 Original objects and the finite comparison data

This calculation composes the full-form comparisons proved in NSO1–31, QOT and EOR with the actual coefficient maps of NSR, CPQ and SFC. The analytic localization proof remains the complete QOT/EOR provider. Here every subsequent map, determinant operation, endpoint sign and interval intersection is proved explicitly. The comparison constants below are their finite constants, not their asymptotic limits.

Fix the original actual hypothetical simple quartet, its conductor, one original five-orbit period and its branch. Keep $0 < \delta < 1/2$, $\gamma > 2$, the complete conductor coefficients a_{uv} , $v = \text{ord}_0 E_A \leq 80$, $\mu_v = E_A^{(v)}(0) \neq 0$, and $\alpha_A = v!/\mu_v$. Keep the original rank domain, including $|u| \geq R_*$. For an actual degree $a = 4l + 1 \geq 9$, set

$$\begin{aligned} q_a &= (a+1)^2, & q'_a &= (a-7)^2, & c'_a &= a/2 - 4, & m_a &= 16a - 48 - v, \\ t_a &= 8a - 16 - t_{0,a}, & j_{0,a} &= 8a - 32 - v + t_{0,a}, & m_a &= t_a + j_{0,a}, \\ D &= N - v, & A &= D - q'_a, & R_a &= a\sqrt{\delta^2 + \gamma^2}. \end{aligned} \quad (\text{EFP1})$$

The symbol Δ on a single-degree form means its order- a value minus its order-one value. On a paired object it means the value in $p = 1 : (k, k+4)$ minus its value in $p = 0 : (1, 1)$, at the same cutoffs and in the same coefficient or quotient-value frames. The capital A in this section is the polynomial degree bound, not a coefficient matrix or an assigned allocation. Each cutoff and each source order retains its original centre. The full measures are

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad h_{n,s} = (2\pi)^{s/2} n! (s/2)_n, \quad s = 1, a. \quad (\text{EFP2})$$

The native metric is $D_{a,D}^{(s)} = A_{\ell,a}^* \overline{(H_{Y,a,N}^{(s)})^{-1}} A_{\ell,a}$. The invertible $A_{\ell,a}$ is the same actual coefficient map in both orders. H_Y is the complete quotient of the original polynomial norm by its relation columns. In particular its polynomial degree is $A = N - v - q'_a$. Complex conjugation in this duality formula is retained.

Write $\ell = (a-1)/4$ and $\tau_r = 2r + 1/2$. NSO's finite full-monomial comparison, transferred by its exact ordinary-transpose Taylor equation and its whole lower-evaluation minimum, gives

$$\begin{aligned} \rho_a &= \frac{(2\pi)^{2\ell}}{(1/2)_{2\ell}}, & \mathfrak{a}_a &= \rho_a \prod_{r=0}^{\ell-1} \tau_r^2, \\ \mathfrak{b}_{a,D} &= \rho_a \prod_{r=0}^{\ell-1} \{4(D+r+1)^2 + \tau_r^2\}, \\ \mathfrak{b}_{a,D}^{-1} D_{a,D}^{(1)} &\preceq D_{a,D}^{(a)} \preceq \mathfrak{a}_a^{-1} D_{a,D}^{(1)}. \end{aligned} \quad (\text{EFP3})$$

This applies throughout the original NSO domain, irrespective of the additional QOT/EOR tests below.

Here are the finite QOT inputs with the actual degree a and polynomial bound A written separately. They are used when

$$a \geq 29, \quad R_a \leq 2^{-33} q_a, \quad 0 \leq A \leq 2q_a. \quad (\text{EFP4})$$

The applications below also verify the provider's stronger $A + \ell \leq 2q_a$ range. In the standard degree- a window, $A_{\min} = 16a - 49 - v \geq 335$ for $a \geq 29$ and $v \leq 80$, while $A_{\max} = 2q_a - v - q'_a$ and $2q_a - (A_{\max} + \ell) = q'_a + v - \ell \geq 477 + v$. The last expression increases for $a \geq 29$. Put $q = q_a$,

$q' = q'_a$, $\epsilon_0 = 2^{-32}$, $\vartheta = \pi/2$, $h = 1/q$, and

$$\begin{aligned}
\ell_s &= (s-1)/4, \quad Q_s = q' + \ell_s, \quad \beta_s = \frac{(2\pi)^{s/2}}{\sqrt{2}\Gamma(s/2)}, \quad \beta_1 = 1, \\
C_L &= \pi/\Gamma(3/4)^2, \quad K_q = \frac{\sqrt{2\pi}}{C_L} \frac{(2q+1)^2}{q} e^{-2q}, \\
\mathcal{E}_a &= \frac{qR_a^2}{\epsilon_0^2 q^2 - R_a^2} + \log(1 + K_q), \\
r_{\ell_s} &= \frac{\sum_{r=0}^{\ell_s-1} (2r+1/2)^2}{\epsilon_0^2 q^2} + \log(1 + K_q) \quad (\ell_s > 0), \quad r_0 = 0, \\
L_h &= C_L (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\
K_{A,Q} &= 2Q + 1 + 2A + 2\sqrt{A(A+2Q)}, \\
b_{a,A,s}^- &= -\mathcal{E}_a + \log L_h - K_{A,Q_s} \log(1 + h/\vartheta), \\
b_{a,A,s}^+ &= \mathcal{E}_a + r_{\ell_s} + \log U_h + K_{A,Q_s} \log \frac{1}{1 - h/\vartheta}, \\
d_{a,A}^- &= 4 \sum_{r=0}^{\ell-1} \operatorname{arsinh} \sqrt{(A+r+1/2)/(2q')}, \quad d_{a,A}^+ = 4 \sum_{r=1}^{\ell} \operatorname{arsinh} \sqrt{(A+r+1/2)/(2q')}, \\
L_{a,A}^Q &= b_{a,A,a}^- - b_{a,A,1}^+ - d_{a,A}^-, \quad U_{a,A}^Q = b_{a,A,a}^+ - b_{a,A,1}^- + d_{a,A}^+, \\
\kappa_a &= \beta_a (4q'/\pi)^{2\ell} = 2^{10\ell} \frac{(2\ell)!}{(4\ell)!} (a-7)^{4\ell}.
\end{aligned} \tag{EFP5}$$

In particular κ_a is a displayed source mass and scale, retained without a change of measure or a change of coefficient vectors.

We specify EOR's finite tests and endpoints as well. Put $m_0 = \lfloor A/2 \rfloor$, $m_1 = \lfloor (A-1)/2 \rfloor$. In parity block $\varepsilon = 0, 1$ require $m_\varepsilon \geq 1$ and

$$\alpha_\varepsilon = (q' + \varepsilon - 1/2)/m_\varepsilon, \quad \beta_\varepsilon = \vartheta q/m_\varepsilon, \quad (\alpha_\varepsilon, \beta_\varepsilon) \in [3/2, 5/2] \times [\pi/2, 3\pi/2]. \tag{EFP6}$$

For this pair let $[A_\varepsilon, B_\varepsilon]$ be the EIQ equilibrium support for $\beta_\varepsilon \sqrt{x} - \alpha_\varepsilon \log x$. It is specified by the unique positive endpoints $A_\varepsilon = u_\varepsilon^2$, $B_\varepsilon = w_\varepsilon^2$ satisfying

$$u_\varepsilon K(\zeta_\varepsilon) = \alpha_\varepsilon \pi / \beta_\varepsilon, \quad w_\varepsilon E(\zeta_\varepsilon) = (\alpha_\varepsilon + 2) \pi / \beta_\varepsilon, \quad \zeta_\varepsilon^2 = 1 - u_\varepsilon^2 / w_\varepsilon^2.$$

$K(z) = \int_0^{\pi/2} (1 - z^2 \sin^2 t)^{-1/2} dt$ and $E(z) = \int_0^{\pi/2} (1 - z^2 \sin^2 t)^{1/2} dt$. For each block separately, suppress its index and set

$$\begin{aligned}
g &= B - A, \quad h_* = \frac{\beta}{2\pi(B+1)(B+2)(B+3)}, \quad c_* = \frac{2}{3} h_* \sqrt{g}, \\
\kappa_* &= \min\{A h_* \sqrt{g}, \beta \sqrt{B}/[6\pi(3B+2)]\}, \quad \eta_* = \min\{A/2, 1\}, \\
L_* &= \max_{A \leq x \leq B} |\beta/(2\sqrt{x}) - \alpha/x|, \quad C_*^2 = 32e^{3L_*/2}, \\
\eta_m &= \left\{ \frac{\log(C_*^2(B+2)) + 10 \log(m+1)}{c_* m} \right\}^{2/3}, \quad \varepsilon_m = (m+1)^{-6}, \\
m &\geq \max\{1, 2/g\}, \quad \eta_m \leq \eta_*, \quad m \kappa_* \sqrt{\eta_m} \geq \ell + 2.
\end{aligned} \tag{EFP7}$$

The maximum defining L_* is evaluated at A, B and, when inside this interval, $(4\alpha/\beta)^2$; differentiating the expression proves this finite candidate list. Retain both parity blocks, their actual $\eta_\varepsilon, \varepsilon_\varepsilon$,

and define

$$\begin{aligned}
F_{a,A}^- &= \min_{\varepsilon=0,1} [\ell \log(A_\varepsilon - \eta_\varepsilon) + \log(1 - \varepsilon_\varepsilon)], \\
F_{a,A}^+ &= \max_{\varepsilon=0,1} [\ell \log(B_\varepsilon + \eta_\varepsilon) + \log(1 + \varepsilon_\varepsilon)], \\
L_{a,A}^E &= b_{a,A,a}^- - b_{a,A,1}^+ + 2\ell \log(q\vartheta/(2q')) + F_{a,A}^-, \\
U_{a,A}^E &= b_{a,A,a}^+ - b_{a,A,1}^- + 2\ell \log(q\vartheta/(2q')) + F_{a,A}^+.
\end{aligned} \tag{EFP8}$$

At a cutoff satisfying (EFP4), use $\underline{b}_{a,A} = L_{a,A}^Q$, $\bar{b}_{a,A} = U_{a,A}^Q$, intersecting with the EOR endpoints only when (EFP6)–(EFP7) hold in both blocks:

$$\underline{b}_{a,A} = \max\{L_{a,A}^Q, L_{a,A}^E\}, \quad \bar{b}_{a,A} = \min\{U_{a,A}^Q, U_{a,A}^E\}. \tag{EFP9}$$

The proved QOT/EOR full-form inequalities give, on the same original dual coefficient metric,

$$\kappa_a^{-1} e^{-\bar{b}_{a,A}} D_{a,D}^{(1)} \preceq D_{a,D}^{(a)} \preceq \kappa_a^{-1} e^{-\underline{b}_{a,A}} D_{a,D}^{(1)}. \tag{EFP10}$$

These are the two accepted analytic inputs used below. Their full proofs are QOT's full original source-order theorem and EOR's full high-endpoint theorem, including its exact two-sheet square map. No analytically unproved bound is used in the propagation.

205 Exact intersection before any restriction or quotient

Define $\lambda_{a,D}^- = -\log \mathfrak{b}_{a,D}$ and $\lambda_{a,D}^+ = -\log \mathfrak{a}_a$ on the whole native domain. On every native endpoint define $w_{a,D} = \lambda_{a,D}^+ - \lambda_{a,D}^-$, using the final endpoints prescribed here. At an endpoint where (EFP4) holds replace those endpoints by

$$\begin{aligned}
\lambda_{a,D}^- &= \max\{-\log \mathfrak{b}_{a,D}, -\log \kappa_a - \bar{b}_{a,D-q'_a}\}, \\
\lambda_{a,D}^+ &= \min\{-\log \mathfrak{a}_a, -\log \kappa_a - \underline{b}_{a,D-q'_a}\}, \\
w_{a,D} &= \lambda_{a,D}^+ - \lambda_{a,D}^-.
\end{aligned} \tag{EFP11}$$

Then $e^{\lambda_{a,D}^-} D_{a,D}^{(1)} \preceq D_{a,D}^{(a)} \preceq e^{\lambda_{a,D}^+} D_{a,D}^{(1)}$. For any vector, both original lower inequalities hold on its same quadratic form, so their larger scalar is still a lower bound; the upper proof uses the smaller scalar. Applying to a nonzero vector shows $\lambda^- \leq \lambda^+$, hence $w \geq 0$. In the guarded domain one can record the unchanged source scalar by writing

$$\begin{aligned}
\underline{b}_{a,D}^* &= \max\{\log \mathfrak{a}_a - \log \kappa_a, \underline{b}_{a,D-q'_a}\}, \quad \bar{b}_{a,D}^* = \min\{\log \mathfrak{b}_{a,D} - \log \kappa_a, \bar{b}_{a,D-q'_a}\}, \\
\lambda^- &= -\log \kappa_a - \bar{b}^*, \quad \lambda^+ = -\log \kappa_a - \underline{b}^*.
\end{aligned} \tag{EFP12}$$

These equalities change the written scalar endpoints only. They do not multiply any original matrix by κ_a .

Here is the full typed restriction and quotient argument. Let Y be a fixed coefficient subspace, $K \subset Y$ a fixed subspace, X_K an independent frame of K and X_Q fixed lifts of a basis of Y/K . For each of the two metrics $G^{(s)}$ define

$$\begin{aligned}
Q^{(s)} &= X_Q^* G^{(s)} X_Q - X_Q^* G^{(s)} X_K (X_K^* G^{(s)} X_K)^{-1} X_K^* G^{(s)} X_Q, \\
x^* Q^{(s)} x &= \min_z (X_Q x + X_K z)^* G^{(s)} (X_Q x + X_K z).
\end{aligned} \tag{EFP13}$$

The positive Hessian makes the minimum attained and unique in z . Its normal equation gives the displayed formula. For every z the two scalar metric inequalities hold on the same vector $X_Q x + X_K z$. Taking their minima therefore gives $e^{\lambda^-} Q^{(1)} \preceq Q^{(a)} \preceq e^{\lambda^+} Q^{(1)}$. Restriction is the case $K = 0$. An iterated quotient has the same property by successive applications to the original affine fibres. A block diagonal sum of four copies has the same constants by addition of the four

quadratic inequalities. In particular the coefficient dimensions do not introduce an extra factor of four in these constants.

For a rank- r such form, conjugating by the inverse positive square root $(Q^{(1)})^{-1/2}$ for the proof of the determinant inequality bounds each of its r positive relative eigenvalues between e^{λ^-} and e^{λ^+} . It follows that

$$r\lambda_{a,D}^- \leq \log \det Q^{(a)} - \log \det Q^{(1)} \leq r\lambda_{a,D}^+. \quad (\text{EFP14})$$

All determinant ratios here use the same value frame in both orders. Replacing that common frame by C multiplies both determinants by $|\det C|^2$ and hence leaves the displayed difference exactly unchanged. This proof does not change the original forms.

206 Four endpoints and equal-rank contractions

For any of the actual four cutoffs N_i let $D_i = N_i - v$. Use exactly $\mathcal{R}f = -f(N_0) - f(N_1) + f(N_2) + f(N_3)$ and put

$$\begin{aligned} \Phi_a^- &= -\lambda_{a,D_0}^+ - \lambda_{a,D_1}^+ + \lambda_{a,D_2}^- + \lambda_{a,D_3}^-, \\ \Phi_a^+ &= -\lambda_{a,D_0}^- - \lambda_{a,D_1}^- + \lambda_{a,D_2}^+ + \lambda_{a,D_3}^+, \\ W_a &= \sum_{i=0}^3 w_{a,D_i}. \end{aligned} \quad (\text{EFP15})$$

For any fixed rank- r subquotient over this window,

$$r\Phi_a^- \leq \Delta \mathcal{R} \log \det Q \leq r\Phi_a^+, \quad \Phi_a^+ - \Phi_a^- = W_a. \quad (\text{EFP16})$$

Indeed negate the upper endpoint at $i = 0, 1$ for the lower bound, and add the lower endpoints at $i = 2, 3$; the upper proof reverses these choices. This verifies every sign. Where (EFP12) applies at all four endpoints, the four copies of $-r \log \kappa_a$ cancel because $-1 - 1 + 1 + 1 = 0$. The lower and upper expressions become, respectively, $r(\underline{b}_0^* + \underline{b}_1^* - \bar{b}_2^* - \bar{b}_3^*)$ and $r(\bar{b}_0^* + \bar{b}_1^* - \underline{b}_2^* - \underline{b}_3^*)$. The same cancellation is valid if the NSO-only endpoints are written algebraically with this same scalar. It requires no new analytic comparison at an unguarded endpoint.

For two same-rank source and target quotients in matched value coordinates let $\omega = \det Q_t / \det Q_s$. Both receive (EFP14). Subtracting the two directed intervals yields

$$-rw_{a,D} \leq \Delta \log \omega(N) \leq rw_{a,D}, \quad |\Delta \mathcal{R} \log \omega| \leq rW_a. \quad (\text{EFP17})$$

The common scalar cancels already in this same-endpoint ratio. For two contractions of ranks r_1, r_2 in the same ambient metric the product consequently has return difference bounded by $(r_1 + r_2)W_a$. No lower singular-value estimate for a separate observation matrix has been used.

207 The five original CPQ quotients and their full frames

Take $j = k + 4$, fix $e \in \{0, 1\}$, and use the actual pairs $p = 0 : (1, 1)$ and $p = 1 : (k, j)$. All target cutoffs here are $N = (q_j - 1, q_j, 2q_j - 1, 2q_j)$. The fixed coefficient maps are

$$\begin{aligned} \mathcal{U} &= \mathbb{C}^{4m_j}, \quad B = B_{\text{cof}} : \mathbb{C}^{m_k} \hookrightarrow \mathcal{U}, \quad \mathbb{B} = \text{diag}(B_j, 4) : \mathcal{U} \twoheadrightarrow \mathbb{C}^{4t_j}, \\ J &= \ker \mathbb{B}, \quad \Phi = \mathbb{B}B, \quad \mathcal{B} = \text{im } B, \quad S = \text{im}[B, F_e], \\ W &= J + \mathcal{B}, \quad K_e = S \cap W, \quad T_e = S + W. \end{aligned} \quad (\text{EFP18})$$

The original Taylor cochain equation of NSR5 makes F_e the same fixed coefficient map in both source presentations, and identifies $\mathbb{C}^{r_e}$ with $S/\mathcal{B}$, $r_e = 8k + 24 - 3e$. Keep $C_T = [U_R, U_b, U_o]$ and $\pi_o = [0 \ 0 \ I]C_T^{-1}\mathbb{B}$. For $z \in \mathcal{U}$, $\pi_o z = 0$ is equivalent to $\mathbb{B}z \in \text{im}(\mathbb{B}B)$, hence to $z \in W$. Surjectivity follows from that of $\mathbb{B}$. Thus π_o induces the exact isomorphism $\mathcal{U}/W \rightarrow \mathbb{C}^{r_o}$. Writing $E_e = \pi_o F_e$, its restriction gives $S/K_e \rightarrow \text{im } E_e$ and $T_e/W \rightarrow \text{im } E_e$. The identity on representatives is the

exact isomorphism $S/K_e \rightarrow T_e/W$. Its target fibre is larger, so its attained target norm is at most its attained source norm.

Choose the original fixed Z_h spanning $\ker E_e$, complement Z_c , $C_P = [Z_h, Z_c]$, $A_e = E_e Z_c$ and $C_o = [A_e, R_e^o]$. Keep $F = BJ_0$, $E = \ker(\Phi J_0)$ and the original lifts Z_R . The five quotient maps, with their full original coefficient fibres, are

form	Y/K	value lifts X_Q	rank
$G_{h,e}$	$K_e/\mathcal{B}$	$F_e Z_h$	h_e
$G_{s,e}$	S/K_e	$F_e Z_c$	c_e
$G_{t,e}$	T_e/W	$F_e Z_c$	c_e
$Q_{\text{rem},e}$	$\mathcal{U}/T_e$	fixed lifts of $U_o R_e^o$ under $\mathbb{B}$	$r_o - c_e$
A_{inv}	$\text{im } F/FE$	FZ_R	r_I

(EFP19)

Here $c_e = \text{rank } E_e$, $h_e = r_e - c_e$, $r_I = j_{0,k} - \dim \ker(\Phi J_0)$, $r_o = 4t_j - m_k + \dim \ker \Phi$, and $r_b = t_k - \dim \ker \Phi + \dim \ker(\Phi J_0)$. For the fourth row π_o followed by the quotient by $\text{im } A_e$ is the value map. For the fifth row injectivity of B identifies $\text{im } F/FE$ with the original coefficient quotient in Z_R . Every row is therefore precisely (EFP13), evaluated in $\mathcal{D}_N = \text{diag}(D_{j,N-v}^{(s_1)}, 4)$. In particular the fourth rank is not replaced by the whole r_o .

With $V_X^p = \mathcal{R} \log \det X^p$, (EFP16) gives

$$\begin{aligned} h_e \Phi_j^- &\leq \Delta V_h \leq h_e \Phi_j^+, & c_e \Phi_j^- &\leq \Delta V_s \leq c_e \Phi_j^+, \\ c_e \Phi_j^- &\leq \Delta V_t \leq c_e \Phi_j^+, & (r_o - c_e) \Phi_j^- &\leq \Delta V_{\text{rem}} \leq (r_o - c_e) \Phi_j^+, \\ r_I \Phi_j^- &\leq \Delta V_A \leq r_I \Phi_j^+. \end{aligned} \quad (\text{EFP20})$$

The identical proof also applies to $V_I, V_b, V_o, V_{\text{conn}}$ with ranks r_I, r_b, r_o, r_e . These inequalities retain the exact correlations $V_{\text{conn}} = V_h + V_s$ and $V_o = V_t + V_{\text{rem}}$. They refine CPQ19 without replacing any summand.

For clarity the absolute frame identities that precede every return are still

$$\begin{aligned} |\det C_P|^2 \det H_{\text{conn}} &= \det G_h \det G_s, & |\det C_o|^2 \det Q_o &= \det G_t \det Q_{\text{rem}}, \\ |\det C_T|^2 (\det H_{B,j,D})^4 &= \det H_{\text{inv}} \det Q_b^E \det Q_o, \\ P(N) &= 4 \log \det H_{B,j,D} - \log \det Q_{\text{rem}} - \log \det G_s - \log \det A_{\text{inv}} + \log \mathfrak{c}_e, \\ \mathfrak{c}_e &= \frac{|\det C_T|^2 |\det C_o|^2 |\det C_b|^2}{c_{\text{conn},e} |\det C_W|^2}. \end{aligned} \quad (\text{EFP21})$$

These follow by the Schur eliminations in (EFP13) in the displayed full frames and the original $Q_b^{\text{CIT}} = C_b^* Q_b^E C_b$. Thus each constant has its original place before $\mathcal{R}1 = 0$ cancels it. The propagation uses the same minimum over the original lower-evaluation columns. For example, for the raw old kernel Gram G_K^{raw} , appended kernel rows a_n , quotient rows b_n and preceding projection coefficients C_n , the unchanged increment is

$$Q^+ = Q + R_n^* (I + a_n (G_K^{\text{raw}})^{-1} a_n^*)^{-1} R_n, \quad R_n = b_n - a_n C_n. \quad (\text{EFP22})$$

It follows by minimizing $\eta^* G_K^{\text{raw}} \eta + \|R_n x - a_n \eta\|^2$. Thus neither the old lower Gram nor the mixed kernel denominator is removed by the finite scalar comparisons.

208 The common return and the complete allocation intervals

Put $I = \text{im}(BJ_0)$, $K_i = I \cap J$, $W_i = I + J$. The identity maps $I/K_i \rightarrow W_i/J$ and $S/K_e \rightarrow T_e/W$ are bijections: each kernel is the stated intersection and each image is the stated sum modulo its denominator. Their Gram ratios in the actual matched frames are $\omega_{\text{inv}} = \det H_{\text{inv}} / \det A_{\text{inv}}$ and $\omega_{\text{conn},e} = \det G_t / \det G_s$. Consequently for the actual common return $\mathcal{L}_e = \mathcal{R} \log(\omega_{\text{inv}} \omega_{\text{conn},e})$,

$$|\Delta \mathcal{L}_e| \leq B_e, \quad B_e = (r_I + c_e) W_j, \quad \mathcal{P}_e^{\text{rest}} = \mathcal{L}_e + V_b. \quad (\text{EFP23})$$

This is (EFP17) applied to both exact quotient bijections. In particular the common projection cross block in SFC5 is retained; a product determinant does not require that cross block to vanish.

Let L_-^p, L_+^p be the actual SFC9–10 finite expressions in pair p . They are determined from $\omega(N_0) = \det G_t \det H_{\text{inv}} / (\det G_s \det A_{\text{inv}})$, $x = \text{arcosh}(\omega(N_0)^{-1/2})$, and $\eta = \frac{1}{2} \log(\det \mathcal{D}_{N_1} / \det \mathcal{D}_{N_0})$, $b = \frac{1}{2} \log(\det \mathcal{D}_{N_2} / \det \mathcal{D}_{N_1})$, $c = \frac{1}{2} \log(\det \mathcal{D}_{N_3} / \det \mathcal{D}_{N_2})$, all in that same pair. To make their finite meaning explicit, put $f(t) = -2 \log \cosh t$, $A_c = e^{2c}$, $B_b = e^{2b}$. Let $z_* \geq 1$ be the unique root

$$A_c^2 B_b z^3 - \{A_c B_b (A_c + 1) - 3A_c\} z^2 - (3A_c B_b - A_c - 1) z - 1 = 0.$$

Then $t_* = b + c + \frac{1}{2} \log z_*$ and $\hat{t} = \min\{x + \eta, \max(\max(0, x - \eta), t_*)\}$ give

$$L_- = f(x + \eta + b) + f(x + \eta + b + c) - f(x) - f(x + \eta), \quad L_+ = f((\hat{t} - b)_+) + f((\hat{t} - b - c)_+) - f(\hat{t}) - f(x).$$

Their complete proof is SFC9–16; they are actual scalar evaluations, not independently assigned angle bounds.

Use the following exact two-pair intersections:

$$\begin{aligned} \tilde{L}_-^1 &= \max\{L_-^1, L_-^0 - B_e\}, & \tilde{L}_+^1 &= \min\{L_+^1, L_+^0 + B_e\}, \\ \tilde{L}_-^0 &= \max\{L_-^0, L_-^1 - B_e\}, & \tilde{L}_+^0 &= \min\{L_+^0, L_+^1 + B_e\}. \end{aligned} \tag{EFP24}$$

Each contains its actual $\mathcal{L}_e^p$: combine CIT35 in pair p with (EFP23) and CIT35 in the other pair. Consequently the earlier signed remainder lies in $[V_b^p + \tilde{L}_-^p, V_b^p + \tilde{L}_+^p]$. This interval may also be intersected with the already proved CIT33 bounds. In source-order differences the exact identity gives $\Delta \mathcal{P}_e^{\text{rest}} \in [r_b \Phi_j^- - B_e, r_b \Phi_j^+ + B_e]$.

Write $\mathcal{T}_j^p = \mathcal{T}_{j,s_1}$ in its actual pair. Keep the complete $Z_e^p = V_b^p + V_{\text{rem}}^p + V_s^p + V_A^p$ and the actual native cap $U_{j,s_1}^{\text{native}}$ only on its proved domain. Define

$$\begin{aligned} \hat{K}_-^p &= \max\{0, (\tilde{L}_-^p + Z_e^p)/4\}, \\ \hat{K}_+^p &= \min\{\mathcal{T}_j^p, U_{j,s_1}^{\text{native}}, (\tilde{L}_+^p + Z_e^p)/4\}. \end{aligned} \tag{EFP25}$$

Omit the cap entry outside its original domain. The exact CPQ20 affine identity $4S_{K,j}^p = \mathcal{L}_e^p + Z_e^p$ and $S_{R,j}^p = \mathcal{T}_j^p - S_{K,j}^p$ prove $\hat{K}_-^p \leq S_{K,j}^p \leq \hat{K}_+^p$ and the reversed residual interval. All four summands in Z_e^p remain.

The native allocation maps are $S_{K,j} = \mathcal{R} \log \det H_{B,j,D}$ and $S_{R,j} = \mathcal{R} \log \det (J_j^* D_{j,D} J_j)$, of ranks t_j and $j_{0,j}$, with J_j the full fixed frame of $\ker B_j$. The first is the attained quotient by that kernel; the second is its restriction. Equation (EFP16) therefore controls both, with the original inverse-metric signs. Writing the actual $\Delta \mathcal{T}_j = \mathcal{T}_j^1 - \mathcal{T}_j^0$, put

$$\begin{aligned} A_K^- &= \max\{t_j \Phi_j^-, \Delta \mathcal{T}_j - j_{0,j} \Phi_j^+\}, \\ A_K^+ &= \min\{t_j \Phi_j^+, \Delta \mathcal{T}_j - j_{0,j} \Phi_j^-\}. \end{aligned} \tag{EFP26}$$

The first pair of entries is the direct kernel interval. The second follows by subtracting the residual interval from the exact equality $\Delta S_K + \Delta S_R = \Delta \mathcal{T}_j$. They contain the actual ΔS_K and so are ordered. The fully propagated SFC19/HJR35 receiver is

$$\begin{aligned} \max\{\hat{K}_-^1, \hat{K}_-^0 + A_K^-\} &\leq S_{K,j}^1 \leq \min\{\hat{K}_+^1, \hat{K}_+^0 + A_K^+\}, \\ \max\{\hat{K}_-^0, \hat{K}_-^1 - A_K^+\} &\leq S_{K,j}^0 \leq \min\{\hat{K}_+^0, \hat{K}_+^1 - A_K^-\}. \end{aligned} \tag{EFP27}$$

To prove the first line add the actual difference interval to the pair-zero interval and intersect with pair one. For the second line subtract that difference from pair one and intersect with pair zero. Subtract these endpoints in reversed order from each same-pair $\mathcal{T}_j^p$ to obtain the complete residual allocation bounds. No allocation value is selected by this interval composition.

209 The degree- k source at the displaced NFR cutoffs

The proper source quotient is $Q_{\text{inv}}^p = (J_0^* D_{k,D}^{(s_0)} J_0)/E$ in the original Z_R values, while the invariant target form is H_{inv}^p . They have the same rank r_I and retain different centres c'_k, c'_j . Let $V_Q^p = \mathcal{R} \log \det Q_{\text{inv}}^p$ and $\mathcal{M}^p = V_I^p - V_Q^p$. For the source constants evaluate (EFP11) at degree k and at the actual $D_i = N_i - v$ used by the target. Relative to the standard degree- k window these cutoffs have displacement $b = q_j - q_k = 8k + 24$ at the two lows and $2b$ at the two highs.

For these source cutoffs the largest polynomial degree is

$$A_{k,\text{max}} = 2q_j - v - q'_k = k^2 + 34k + 1 - v. \quad (\text{EFP28})$$

For $k \geq 37$ one has $2q_k - A_{k,\text{max}} = k^2 - 30k + 1 + v \geq 260 + v$, and $2q_k - (A_{k,\text{max}} + (k-1)/4) = k^2 - (121/4)k + 5/4 + v \geq 251 + v$. Both right sides increase for $k \geq 37$ by differentiation, or by taking their exact positive finite differences at successive $4l + 1$. The smallest degree is $q_j - v - 1 - q'_k = 24k - 25 - v \geq 0$. Thus the stated $k \geq 37$ guard, together with the actual QOT root test at k , validates the polynomial degree range at all four displaced source endpoints. It is a sufficient finite guard; no minimality is claimed. EOR is added at an endpoint only when its own two-block tests hold with this degree- k value of A .

Let $\Phi_k^\pm$ and W_k now mean precisely these displaced degree- k values, while $\Phi_j^\pm, W_j$ use the target degree. Then

$$\Lambda_{\text{new}}^- = r_I(\Phi_j^- - \Phi_k^+) \leq \Delta\mathcal{M} \leq r_I(\Phi_j^+ - \Phi_k^-) = \Lambda_{\text{new}}^+. \quad (\text{EFP29})$$

This follows by subtracting the full rank- r_I source-return interval from its target-return interval. At a single cutoff their log-determinant difference retains the centre $-r_I \log(\kappa_j/\kappa_k)$ wherever both comparisons are guarded. Only after the four endpoint sum do the four copies cancel. Replacing the degree- k endpoints by degree- j endpoints would change this exact source map and is not used.

For each pair define the exact translation $D_e^p = -V_t^p + V_s^p + V_A^p - V_Q^p$. By NFR7, $\mathcal{M}^p = \mathcal{L}_e^p + D_e^p$. Thus (EFP24) gives the interval $[\tilde{L}_-^p + D_e^p, \tilde{L}_+^p + D_e^p]$. Intersect it with $-V_Q^p \leq \mathcal{M}^p$ and, on the cap domain, with

$$\mathcal{M}^p \leq 4U_{j,s_1}^{\text{native}} - V_b^p - V_t^p - V_{\text{rem}}^p - V_Q^p.$$

Call these resulting endpoints $\widehat{M}_-^p, \widehat{M}_+^p$. The earlier NFR12 interval becomes

$$\begin{aligned} \max\{\widehat{M}_-^1, \widehat{M}_-^0 + \Lambda_{\text{new}}^-\} &\leq \mathcal{M}^1 \leq \min\{\widehat{M}_+^1, \widehat{M}_+^0 + \Lambda_{\text{new}}^+\}, \\ \max\{\widehat{M}_-^0, \widehat{M}_-^1 - \Lambda_{\text{new}}^+\} &\leq \mathcal{M}^0 \leq \min\{\widehat{M}_+^0, \widehat{M}_+^1 - \Lambda_{\text{new}}^-\}. \end{aligned} \quad (\text{EFP30})$$

Its proof is the same addition and subtraction of exact directed intervals used in (EFP27), now with both actual sources.

Finally retain the full proper-source compatibility error. Use the original endpoint constants

$$\begin{aligned} a_i^p &= 2r_I \log \max(1, K_-^p(D_i)) + r_I \log c_+^p(D_i), \\ b_i^p &= 2r_I \log \max(1, K_+^p(D_i)) - r_I \log c_-^p(D_i), \\ E_p^- &= -b_0^p - b_1^p - a_2^p - a_3^p, \quad E_p^+ = a_0^p + a_1^p + b_2^p + b_3^p. \end{aligned}$$

For $\mathcal{C}^p = \mathcal{R} \log(\det \mathcal{H}_V^p / \det \mathcal{H}_E^p)$, NFR5 is the exact identity $\mathcal{M}^p = \mathcal{C}^p + \mathcal{R} \log \omega_{\text{inv}}^p + \mathcal{E}^p$ with $E_p^- \leq \mathcal{E}^p \leq E_p^+$. Equations (EFP17) and (EFP29) therefore give

$$\Lambda_{\text{new}}^- - r_I W_j - E_1^+ + E_0^- \leq \Delta\mathcal{C} \leq \Lambda_{\text{new}}^+ + r_I W_j - E_1^- + E_0^+. \quad (\text{EFP31})$$

Subtract the target invariant-loss difference and the error difference from $\Delta\mathcal{M}$ to verify each of these signs. The original source-complement projections, constants, and both old low cutoffs remain in this formula.

210 The defining primitive source: exact finite intersection

Keep the original fixed hypothetical simple quartet, the complete nonreduced conductor, the full arithmetic unit, the fixed five-orbit period and logarithm branch, and the original period domain $|u| \geq R_*$ (or its specified exact period tests). Write

$$k = 4\ell + 1 \geq 9, \quad j = k + 4, \quad q_a = (a + 1)^2, \quad q'_a = (a - 7)^2, \quad c'_a = a/2 - 4,$$

$$D_e = q_j - v - 1 + e, \quad d_e = 8k + 24 + e, \quad r_e = d_e - 4e = 8k + 24 - 3e, \quad e \in \{0, 1\}.$$

Here $v = \text{ord}_0 E_A$ is the actual conductor order, $0 \leq v \leq 80$, and $\mu_v = E_A^{(v)}(0) \neq 0$. Every statement below fixes e ; subtraction is between the source-order pairs $p = 0 : (1, 1)$ and $p = 1 : (k, j)$, never between $e = 0$ and $e = 1$. The source centre is c'_k and the target centre is c'_j .

The inherited finite constants are exactly NSO3-4/NSR2 and QOT11-14:

$$\ell_a = (a - 1)/4, \quad \rho_a = \frac{(2\pi)^{2\ell_a}}{(1/2)^{2\ell_a}}, \quad \mathbf{a}_a = \rho_a \prod_{h=0}^{\ell_a-1} (2h + \tfrac{1}{2})^2,$$

$$\mathbf{b}_{a,D} = \rho_a \prod_{h=0}^{\ell_a-1} \{4(D + h + 1)^2 + (2h + \tfrac{1}{2})^2\},$$

$$\kappa_a = 2^{10\ell_a} \frac{(2\ell_a)!}{(4\ell_a)!} (a - 7)^{4\ell_a}, \quad l_{a,b} = b_{a,b,a}^- - b_{a,b,1}^+ - 4 \sum_{h=0}^{\ell_a-1} \text{arsinh} \sqrt{\frac{b + h + 1/2}{2q'_a}},$$

$$u_{a,b} = b_{a,b,a}^+ - b_{a,b,1}^- + 4 \sum_{h=1}^{\ell_a} \text{arsinh} \sqrt{\frac{b + h + 1/2}{2q'_a}}.$$

The $b^\pm$ in the last two formulas are the complete GPA constants QOT11-12, equivalently EOR6; the original inside-region error, source mass, root radius, and trigonometric factors are retained. No new analytic estimate replaces those constants. In particular κ_a is the original full-mass scalar of QOT20.

The exact source map. Let

$$E_A(z) = \sum_{b,c=0}^8 a_{bc} e^{\beta_{8,b,c} z}, \quad \mathcal{T}_{AP}(S') = \sum_{b,c=0}^8 a_{bc} p(S' + \beta_{8,b,c}),$$

and keep the unscaled original relation columns

$$r_h = \mathcal{T}_A(\chi_k S^h), \quad 0 \leq h < d_e, \quad \psi_k = \chi_{k-8}.$$

At every lower root $\beta_{k-8,u,w}$ and every summand,

$$\beta_{k-8,u,w} + \beta_{8,b,c} = \beta_{k,u+b,w+c}.$$

Thus χ_k vanishes term by term. The lower roots are distinct, since $0 < \delta < 1/2$ and $\gamma > 2$, and their complete product ψ_k divides every r_h . For a polynomial of degree n ,

$$\mathcal{T}_{AP} = \sum_{t=0}^n \frac{\mu_t}{t!} p^{(t)}.$$

The first nonzero summand has $t = v$, so the leading coefficient of r_h is $\binom{q_k+h}{v} \mu_v \neq 0$ and $\deg r_h = q_k + h - v$. Consequently the full relation columns are independent, their maximum degree is D_e , and the polynomial degree to which the weighted QOT theorem is applied is exactly

$$a_{\text{src},e} = \max_{h < d_e} \deg(r_h/\psi_k) = q_k - v + d_e - 1 - q'_k = 24k - 25 - v + e. \quad (\text{EFP32})$$

No leading-term replacement was used to obtain the restriction map: the entire polynomial r_h/ψ_k is retained in that map.

Write R_e^{pol} for the columns of the unscaled original r_h (not the divided polynomials r_h/ψ_k) in $1, S', \dots, S'^{D_e}$, and use the exact native monomial matrix B_{s, c'_k, D_e} of NSO7. Put

$$C_e^{(s)} = B_{s, c'_k, D_e} R_e^{\text{pol}}, \quad H_e^{(s)} = (C_e^{(s)})^* C_e^{(s)}, \quad U_s = \text{diag}(1/n!) B_{s, c'_k, D_e}^\top.$$

The Gram $H_e^{(s)}$ is precisely the integral Gram of all $r_h(c'_k + iy)$ for the original measure

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy.$$

The underlying derivative values are complex-linear; hence the transpose in U_s is ordinary transpose.

For completeness, the relation annihilator is exactly the analytic plane used in the original primitive quotient. If a native analytic column has generating germ G , then $E_A G$ is an upper-grid exponential numerator. For the lower columns $G = e^{\beta_{k-8,u,w} z}$ the same assertion follows from the displayed root addition. Therefore

$$\Lambda_G(r_h) = \left[(\chi_k(\partial) \partial^h)(E_A G) \right]_{z=0} = 0.$$

The full native columns $[V_0, T_0]$ have rank $q_k - v$: if their Taylor combination vanishes through D_e , its product with E_A is an upper-grid numerator vanishing through $D_e + v \geq q_k - 1$; the upper Vandermonde makes its coefficient vector zero, the fixed numerator quotient of NSO12 makes its analytic class zero, and the lower Vandermonde makes its lower coefficients zero. These are exactly the same complete columns as NSO12–13 and NSO21. Since $D_e + 1 - d_e = q_k - v$ and $C_e^{(s)}$ has rank d_e , this proves

$$\ker(C_e^{(s)})^\top = \text{im}[V_0, T_0].$$

The original Taylor primitive plane is the whole jet plane for $e = 0$; for $e = 1$ it is the kernel of all four independent original rows $\Lambda(\mathcal{T}_{B_i} \mathcal{T}_A \chi_j)$. Its analytic subspace is the preceding complete plane. Thus the original Taylor complement P_e^{Tay} and its native lifts $P_e^{(s)} = U_s^{-1} P_e^{\text{Tay}}$ give

$$\Theta_e = (C_e^{(s)})^\top P_e^{(s)} = (R_e^{\text{pol}})^\top \text{diag}(n!) P_e^{\text{Tay}}. \quad (\text{EFP33})$$

This matrix is literally independent of s , including every factorial, affine phase, mass and original coefficient. It has rank r_e ; it is square invertible for $e = 0$, and has the original rank- r_e constraint image for $e = 1$.

For a value x in this fixed rank- r_e frame, the complete affine fibre is

$$P_e^{(s)} x + \text{im}[V_0, T_0] = \{w : (C_e^{(s)})^\top w = \Theta_e x\}.$$

Every member remains in the primitive plane, because the analytic plane is contained in it. The Euclidean least norm on an onto fibre $Aw = b$ is $b^*(AA^*)^{-1}b$: the minimizing vector is $A^*(AA^*)^{-1}b$, and its inner product with every vector of $\ker A$ is zero. Here

$$AA^* = (C_e^{(s)})^\top \overline{C_e^{(s)}} = \overline{H_e^{(s)}}.$$

We have consequently proved the exact original source identity

$$H_{\text{src},e}^{(s)} = \Theta_e^* \overline{(H_e^{(s)})^{-1}} \Theta_e. \quad (\text{EFP34})$$

This is the QOT26–27 map, with the identical NSO21,26/NSR9 value frame. In particular the primitive source has not been identified with H_Y . Although $D_e - q'_k = a_{\text{src},e}$ numerically at this defining cutoff, that equality alone proves no map of forms and no extension to later primitive source cutoffs.

All finite guards. NSO25 and NSR6 hold on their original domain $k \geq 9$ without a GPA radius guard. Add the QOT source interval only where

$$k \geq 29, \quad R_k = k\sqrt{\delta^2 + \gamma^2} \leq 2^{-33}q_k;$$

add its defining target interval only where the corresponding conditions hold for j . Retain those two conditions explicitly in paired use. The actual polynomial degrees are

$$a_{\text{src},e} = 24k - 25 - v + e, \quad a_{\text{tgt},e} = D_e - q'_j = 16k + 15 - v + e = 16j - 49 - v + e.$$

The requirements $0 \leq a_{\text{src},e}$, $a_{\text{src},e} + \ell_k \leq 2q_k$ and their target analogues hold for $k, j \geq 29$: positivity follows using $v \leq 80$, while the upper expressions are at most $25k$ and $17j$, respectively, and these are below $2(k+1)^2$ and $2(j+1)^2$ on that domain. The exact degrees, not these bounds, are used in all constants.

The EOR21 endpoint rule additionally requires, for both parity blocks,

$$m_\epsilon = \lfloor (a - \epsilon)/2 \rfloor \geq 1, \quad \alpha_\epsilon = (q'_b + \epsilon - \tfrac{1}{2})/m_\epsilon \in [3/2, 5/2], \quad \beta_\epsilon = \frac{\pi q_b}{2m_\epsilon} \in [\pi/2, 3\pi/2],$$

as well as all EOR12 localization guards with the EOR9 constants. Here (b, a) would be $(k, a_{\text{src},e})$ or $(j, a_{\text{tgt},e})$. Neither application is permitted on the present GPA domain. Indeed $\gamma > 2$ gives

$$2k < R_k \leq 2^{-33}q_k, \quad q_k > 2^{34}k.$$

At the source the upper rectangle guard would imply

$$q_k \leq 3m_\epsilon \leq \tfrac{3}{2}a_{\text{src},e} \leq 36k - 36 < 2^{34}k,$$

a contradiction. At the target it would imply

$$q_j \leq 3m_\epsilon \leq \tfrac{3}{2}a_{\text{tgt},e} \leq 24k + 24 < 2^{34}k < q_j,$$

again a contradiction. The middle strict inequality holds for $k \geq 29$. Therefore EOR21's actual fallback here is exactly QOT's low interval. No high-endpoint equilibrium constants are inserted into this primitive comparison. Its high target-flag applications remain separate.

The componentwise intersection and its proof. In the joint GPA domain define

$$\begin{aligned} \lambda_{\text{src},e} &= \max\{l_{k,a_{\text{src},e}}, \log \mathfrak{a}_k - \log \kappa_k\}, & v_{\text{src},e} &= \min\{u_{k,a_{\text{src},e}}, \log \mathfrak{b}_{k,D_e} - \log \kappa_k\}, \\ \lambda_{\text{tgt},e} &= \max\{l_{j,a_{\text{tgt},e}}, \log \mathfrak{a}_j - \log \kappa_j\}, & v_{\text{tgt},e} &= \min\{u_{j,a_{\text{tgt},e}}, \log \mathfrak{b}_{j,D_e} - \log \kappa_j\}. \end{aligned} \quad (\text{EFP35})$$

Let $L_{\text{src},e}$ and $L_{\text{conn},e}$ denote the logarithmic source-order determinant ratios of NSR6. The exact finite result is

$$\begin{aligned} S_e^- &= -r_e(\log \kappa_k + v_{\text{src},e}) \leq L_{\text{src},e} \leq -r_e(\log \kappa_k + \lambda_{\text{src},e}) = S_e^+, \\ T_e^- &= -r_e(\log \kappa_j + v_{\text{tgt},e}) \leq L_{\text{conn},e} \leq -r_e(\log \kappa_j + \lambda_{\text{tgt},e}) = T_e^+. \end{aligned} \quad (\text{EFP36})$$

Equivalently, without moving the scalar centres,

$$\begin{aligned} S_e^- &= \max\{-r_e \log \mathfrak{b}_{k,D_e}, -r_e(\log \kappa_k + u_{k,a_{\text{src},e}})\}, \\ S_e^+ &= \min\{-r_e \log \mathfrak{a}_k, -r_e(\log \kappa_k + l_{k,a_{\text{src},e}})\}, \\ T_e^- &= \max\{-r_e \log \mathfrak{b}_{j,D_e}, -r_e(\log \kappa_j + u_{j,a_{\text{tgt},e}})\}, \\ T_e^+ &= \min\{-r_e \log \mathfrak{a}_j, -r_e(\log \kappa_j + l_{j,a_{\text{tgt},e}})\}. \end{aligned}$$

Where one GPA guard fails, omit that component's QOT candidate and use its original NSR6 interval. This is an unconditional finite rule on the original NSO domain, with explicitly guarded additional proved bounds.

To prove the source statement, apply QOT15 to the fixed complete columns r_h/ψ_k of degree at most $a_{\text{src},e}$:

$$\kappa_k e^{l_{k,\text{src},e}} H_e^{(1)} \preceq H_e^{(k)} \preceq \kappa_k e^{u_{k,\text{src},e}} H_e^{(1)}.$$

Inversion reverses this inequality, coefficient conjugation preserves it, and the same full-rank Θ_e in (EFP34) gives the rank- r_e dual determinant interval QOT27. The whole-source minimum NSO21 independently gives NSO25/NSR6 on the identical affine fibre. Each lower endpoint bounds the same scalar from below, and each upper endpoint bounds it from above. Their maximum and minimum give $S_e^\pm$. Their interval is nonempty because it contains that actual scalar.

For the target, the ordinary Taylor multiplier equation NSO23/NSR5 is

$$\text{stack}_i \text{Mult}(B_i) P_e^{\text{Tay}} = \text{diag}(\tilde{T}_j, 4) F_e + \text{diag}(\tilde{V}_j, 4) E_e^{\text{low}}.$$

All its columns represent the same actual germs. If two representations differ, multiplying the zero Taylor difference by E_A produces an upper-grid numerator zero through $D_e + v \geq q_j - 1$; the upper Vandermonde and the fixed numerator quotient kill the analytic class, and the lower Vandermonde kills the remaining lower class. Thus F_e is literally the same coefficient matrix in the two pairs. The full multiplier kernel of NSO24 gives its quotient by the complete boundary image rank r_e . Its metric is the attained minimum

$$x^* H_{\text{conn},e}^{(s_1)} x = \min_{\xi} (F_e x + B_{\text{cof}} \xi)^* \text{diag}(D_{j,D_e}^{(s_1)}, 4) (F_e x + B_{\text{cof}} \xi).$$

This includes all four lower minima inside D_{j,D_e} and the full boundary minimum over ξ . QOT16 on that target degree $a_{\text{tgt},e}$ therefore gives QOT30; NSO24–25 independently gives the target NSR6 interval. Taking their intersection proves $T_e^\pm$. No source-centre identification with c'_j occurs in this argument.

The strongest propagation from these component intervals. Retain the exact NSR7 identity, with all original constants,

$$\begin{aligned} A_e^p &= \log \det H_{\text{conn},e}^p - \log \det H_{\text{src},e}^p = -\mathcal{X}_e^p + \epsilon_e^p, & \mathcal{X}_e^p &= \chi_{X_k,D_e}^p - \chi_{Y_e,D_e}^p, \\ c_e^{p,-} &= -2r_e \log \max(1, K_-^p) - r_e \log c_+^p, & c_e^{p,+} &= 2r_e \log \max(1, K_+^p) - r_e \log c_-^p, \\ c_-^p &= \min(1, (L_-^p)^{-2}), & c_+^p &= \max(1, (L_+^p)^2), & b_e^- &= \epsilon_e^{1,-} - \epsilon_e^{0,+}, & b_e^+ &= \epsilon_e^{1,+} - \epsilon_e^{0,-}. \end{aligned}$$

These $K_\pm^p, L_\pm^p$ retain the original NBG/NKC domains, the degree $k+8$ comparison, both centres, the shift $(\beta_{4,u,w} - 2)/i$, every coefficient and both full Gamma masses. They are not replaced by the scalar order-comparison constants.

Set $\mathcal{J}_e^{\text{ord}} = A_e^1 - A_e^0$ and $\Delta \mathcal{X}_e = \mathcal{X}_e^1 - \mathcal{X}_e^0$. Then

$$\begin{aligned} T_e^- - S_e^+ &\leq \mathcal{J}_e^{\text{ord}} \leq T_e^+ - S_e^-, \\ \zeta_e &= L_{\text{src},e} + \epsilon_e^1 - \epsilon_e^0, & S_e^- + b_e^- &\leq \zeta_e \leq S_e^+ + b_e^+, \\ L_{\text{conn},e} &= -\Delta \mathcal{X}_e + \zeta_e, & S_e^- - T_e^+ + b_e^- &\leq \Delta \mathcal{X}_e \leq S_e^+ - T_e^- + b_e^+. \end{aligned} \tag{EFP37}$$

In the joint GPA domain the scalar-bearing forms of the two principal intervals are exactly

$$\begin{aligned} r_e(\lambda_{\text{src},e} - v_{\text{tgt},e}) &\leq \mathcal{J}_e^{\text{ord}} + r_e \log(\kappa_j/\kappa_k) \leq r_e(v_{\text{src},e} - \lambda_{\text{tgt},e}), \\ r_e(\lambda_{\text{tgt},e} - v_{\text{src},e}) + b_e^- &\leq \Delta \mathcal{X}_e - r_e \log(\kappa_j/\kappa_k) \leq r_e(v_{\text{tgt},e} - \lambda_{\text{src},e}) + b_e^+. \end{aligned} \tag{EFP38}$$

Here the exact inherited centre is

$$\frac{\kappa_j}{\kappa_k} = \frac{256(k-3)^{k+3}}{k(k+2)(k-7)^{k-1}}.$$

This is QOT31, including the full source masses; it is not absorbed into an error interval.

For proof, subtract the two component determinant ratios to get $\mathcal{J}_e^{\text{ord}} = L_{\text{conn},e} - L_{\text{src},e}$. The smallest value on their interval rectangle is $T_e^- - S_e^+$, and the largest is $T_e^+ - S_e^-$. Subtracting the two exact NSR7 identities gives $\Delta\mathcal{X}_e = L_{\text{src},e} - L_{\text{conn},e} + \epsilon_e^1 - \epsilon_e^0$, which proves the remaining bounds by the same endpoint calculation. Substitution of the scalar-bearing expressions (EFP36) proves (EFP38) with both displayed signs.

These are the smallest enclosures obtained from the component intervals and these exact linear identities: a linear function attains each bound at the indicated rectangle vertex. This is a statement about the information in those intervals, not about realizability or optimality for the actual correlated matrices. In particular this componentwise procedure retains mixed NSO/QOT end-point choices that can be lost by merely intersecting the two separately propagated paired intervals. The common variables in $\zeta_e = L_{\text{src},e} + \epsilon_e^1 - \epsilon_e^0$ and $\Delta\mathcal{X}_e = \zeta_e - L_{\text{conn},e}$ remain shared. Their marginal intervals must not subsequently be treated as independent when recovering $L_{\text{conn},e} = \zeta_e - \Delta\mathcal{X}_e$.

The earlier PSR26 full centered interval is also retained exactly as in NSR8–9: denote its unchanged endpoints by P_e^-, P_e^+ , and, only on its original domain

$$k \geq \max\{9, v - 1, \lceil (\pi/2)\sqrt{\delta^2 + \gamma^2} \rceil + 1\}$$

with the same original admissible SGT14 radius $0 < a < 1$, replace S_e^- by $\max\{S_e^-, P_e^-\}$ and S_e^+ by $\min\{S_e^+, P_e^+\}$ in (EFP37). Here $P_e^\pm$ are the literal preexisting PSR26 endpoints, including its D_e^{ord} and both errors, not newly defined analytic constants. No PSR guard is imposed on NSO or QOT. The proof is again intersection of bounds on the same source determinant ratio; the original Θ_e and complement/frame factors remain as in (EFP33).

At $e = 0$ the established equality $\chi_{Y_0, D_0} = 0$ may be used. At $e = 1$ the complete four-row primitive constraint and χ_{Y_1, D_1} stay in every formula. Since these intersections are subsets of QOT27,30,32,34, they preserve the already proved $O_{\text{actual}}(k^{3/2} + k \log q_k)$ paired error and the conclusions $\mathcal{J}_e^{\text{ord}} = o(q_k)$ and $\Delta\mathcal{X}_e = o(q_k)$ at fixed actual data. Indeed the original low l, u are $O_{\text{actual}}(\sqrt{k} + \log q_k)$, the rank is $O(k)$, $\log(\kappa_j/\kappa_k) = O(\log k)$, and NSO27 gives $\epsilon_e^{p,\pm} = O_{\text{actual}}(k \log q_k)$. These inherited order-difference conclusions assign no absolute compatibility value and no later primitive source extension.

211 Consequences and exact scope

Every new interval is an intersection of finite inequalities on the same original coefficient directions or matched quotient values. Consequently it is contained in the corresponding NSO interval and, where available, in the QOT/EOR interval. Empty coefficient blocks have rank zero, determinant one and zero return, so all formulas retain those cases.

The EOR limiting allowances already proved in the complete provider therefore remain valid for the direct allocation differences and the common return after these further finite intersections: the two allocation endpoints are $-8 \log(\pi v_*/4)$ and $8 \log(4/(\pi u_*))$ after division by q_a , and the common difference allowance after division by q_k is $16 \log(v_*/u_*)$, on their same fixed-actual-data eventual domains. This follows by containment of intervals, not by a new limit exchange. Here $u_* K(\zeta) = 2$, $v_* E(\zeta) = 4$ and $\zeta^2 = 1 - u_*^2/v_*^2$ as in EOR.

The finite expressions retain the actual initial common angle, all four returns in Z_e , the original independent source quotient, and every proper-source error. No numerical individual allocation, no uniform estimate in a moving period, and no exclusion of an off-line quartet follows solely from these identities. The mathematical output is the proved finite propagation through the displayed exact maps, now available at every stated receiving site.

Source status. This is the complete accepted integration derivative supplied by the publication owner; the captured original and the exact derivative change are preserved separately. The independent written review is recorded in the source manifest. Its statements about review, execution and publication describe the captured source state. This typesetting conversion supplies no additional mathematical acceptance. Captured pull request: <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/37>; head 834bc0ea2b2510ac39862d9bd176200e78254c41. The source is preserved byte for byte in ECT_PUBLIC_RESEARCH_NOTE.md. Every displayed equation is assigned a unique ECT tag; the exact source-to-tag correspondence accompanies this body.

212 Uniform exponential suppression of the original adjacent ratios

16 September 2026. Written proof contribution requiring independent review. No new Lean certificate, actual off-critical zero example, or RH conclusion is asserted.

212.1 Source and review

Base main: 1efd53337561d8f67cf0ac1d119acdaa222644c3. AMT1–6 were directly reread in workbenches/splitzero-tandem/continuations/20260915-primitive-band-continuation/13_ARITHMETIC_MIXED_TRANSFER.tex, blob b6cfec2ac9517acc378aa02686ada6e7923ef381.

The user supplied the complete 540-line Pastedmarkdown(2)(8).md: grouped-convolution recurrence, contraction penalties, and the exact full-action cancellation. Its recurrence proof checks out on the stated AMT inputs. Fixed arithmetic blocks 3,4,5 have a degree-linear multiplication bound; their tensor product is compressed at TOTAL degree $N+1$, retaining the full output face. This proves the stated bound $\omega_{N+1}/\omega_N \leq C_{\text{mult}}(h)^{2(2N+\text{floor}(k/3))}$, and the exterior benchmark gives its polynomial penalty bound $O_{-h}(1+k(m-1))$. The original 4q-1 penalty coefficient count is correct. Only the prose needs clarification: d_n tends to zero; $1-d_n$ is close to one.

The other new transcript announces an exponential full-band candidate but contains no complete quantile proof. The argument below bypasses that unfinished proof. It requires only a lower bound for the actual relation minimum, not a sharp minimum asymptotic. The EOR source-order theorem is not an input to this continuation.

212.2 1. Actual source and original relation norm

Fix the complete stipulated quartet h , its multiplicity m , and amplitude $v_h=2\xi/h$. Retain

$$e_k = 1 + k(m-1), \quad q = e_k(k+1)^2, \quad c = k/2, \quad R_k = k\sqrt{\delta^2 + \gamma^2}, \quad \alpha = \pi/2. \quad (\text{ECT1})$$

Every centered root of the original monic degree- q polynomial χ has modulus at most R_k . Its full multiplicities remain. Write $\mu_k = w_h^{\{k\}}$. AMT gives

$$\mu_k(y) \geq \ell_{h,k} e^{-\alpha|y|} (k-2+|y|)^{-B_h}, \quad \mu_k \leq A_k \sigma, \quad (\text{ECT2})$$

$$B_h = 42 + 8m, \quad \ell_{h,k} = c_h \vartheta_h^{k-3} e^{-\alpha(k-3)}, \quad A_k = (C_h^{\text{tilt}}/c_\Gamma) k^{p+1} M_\alpha^{k-1}, \quad p = 8m - 9/2. \quad (\text{ECT3})$$

All constants are the original fixed- h constants in AMT4; no source mass is normalized away. The Gamma monic norm is

$$g_n = \sqrt{2\pi} n! (1/2)_n = \sqrt{2\pi} (2n)! / 4^n. \quad (\text{ECT4})$$

Let ω_n be the original arithmetic monic norm and define

$$h_r^\chi = \inf_{P \text{ monic, deg } P=r} \int |\chi(c+iy)P(c+iy)|^2 \mu_k(y) dy. \quad (\text{ECT5})$$

The original source/relation determinant factorization gives

$$\boxed{d_n := V_n/V_{n-1} = \omega_n/h_{n-q}^\chi \quad (n \geq q).} \quad (\text{ECT6})$$

Proof: V_N is the source Hankel determinant at $N+1$ divided by the relation Hankel determinant at $N-q+1$. Taking the adjacent ratio yields (ECT6). The denominator is the norm of an original relation BEFORE its quotient value becomes represented zero. No $G_{-(q-2)}$ occurs. The substitution $Q(y)=i^{\wedge\{-r\}}P(c+iy)$ is monic in y and preserves absolute values, so the following polynomial estimate uses the original coordinate through an explicit isometry, not a new arithmetic quotient.

212.3 2. A two-band lower bound for every monic direction

Take $q \geq 1$, $0 \leq r \leq q$, $n=q+r$, and set

$$a = 20n/63, \quad b = 40n/21, \quad E_n = [-b, -a] \cup [a, b], \quad d\nu_n(y) = \frac{|y| dy}{\pi \sqrt{(y^2 - a^2)(b^2 - y^2)}}. \quad (\text{ECT7})$$

This is a probability measure; y squared pushes it to the arcsine law on $[a^2, b^2]$, with equal mass on the two half-lines. For a monic complex polynomial Q of degree r ,

$$\int \log |Q| d\nu_n \geq r \log \frac{\sqrt{b^2 - a^2}}{2}. \quad (\text{ECT8})$$

For each root z , symmetry turns its logarithmic integral into one half the arcsine integral of $\log|z^{\wedge 2} - x|$. On an interval with midpoint C and half-width D , factoring the quadratic in $\exp(i\theta)$ and applying the circle Jensen formula gives $\log(D/2)$ plus nonnegative log-max root terms. Hence this integral is at least $\log((b^2 - a^2)/4)$; summing proves (ECT8). Roots on the bands are handled by logarithmic integrability or a limit.

The same arcsine substitution gives exactly

$$\int \log |y| d\nu_n = \log((a+b)/2), \quad \int \log(d\nu_n/dy) d\nu_n = \log \frac{2}{\pi(b-a)}. \quad (\text{ECT9})$$

For the entropy formula use that each of the expectations of $\log(x - a^{\wedge 2})$, $\log(b^{\wedge 2} - x)$ equals $\log((b^2 - a^2)/4)$.

Let

$$E_0 = \int_0^{\pi/2} \sqrt{\sin^2 t + \frac{1}{36} \cos^2 t} dt. \quad (\text{ECT10})$$

Then $\int |y| d\nu_n = 2b E_0/\pi$. Here $E_0 < 21/20$ has an elementary proof. Put $\rho = 1/6$ and $t_0 = \arcsin \rho$. On $[0, t_0]$, the integrand minus $\sin t$ is at most $\rho \cos t$, whose integral is ρ squared. On $[t_0, \pi/2]$, rationalization bounds the difference by ρ squared $\cos$ squared $t/(2 \sin t)$. Therefore

$$E_0 - 1 \leq \frac{1}{36} + \frac{1}{72} [\log(6 + \sqrt{35}) - \sqrt{35}/6] < \frac{1}{20}. \quad (\text{ECT11})$$

The last inequality uses $\log 12 < 5/2$ and $\sqrt{35}/6 > 9/10$. The first seven positive Taylor terms of $\exp(5/2)$ already exceed 12, so no numerical quadrature is needed. Consequently $\alpha \int |y| d\nu_n = b E_0 < 2n$.

Apply probability Jensen to $f/(d\nu_n/dy)$, where $f = |Q|^2 \exp(-\alpha|y|)$. Its logarithm is integrable. Equations (ECT8)–(ECT9) prove

$$\begin{aligned}
\int_{E_n} |Q(y)|^2 |y|^{2q} e^{-\alpha|y|} dy &\geq \frac{\pi(b-a)}{2} \left(\frac{b^2-a^2}{4} \right)^r \left(\frac{a+b}{2} \right)^{2q} e^{-2n} \\
&= \frac{50\pi n}{63} n^{2n} e^{-2n} (500/567)^r (100/81)^q \\
&\geq \frac{50\pi n}{63} n^{2n} e^{-2n} e^{c_* q}, \quad c_* := \log(50000/45927) > 0.
\end{aligned} \tag{ECT12}$$

The last step uses $r \leq q$ and $500/567 < 1$. An exact rational logarithm-series certificate gives

$$0.084969826010092 < c_* < 0.084969826010093. \tag{ECT13}$$

This holds for ALL monic coefficient directions, including $r=0$. No r/q bounded away from zero, limiting quantile construction, or compressed spectral gap is assumed.

212.4 3. Restore the original roots, arithmetic density and mass

Assume $q \geq 7R_k$. On E_n , $q \leq n \leq 2q$, every original root gives $|iy-z| \geq |y| - R_k$. Thus

$$|\chi(c + iy)| \geq |y|^q (1 - 63R_k/(20n))^q. \tag{ECT14}$$

The factor is positive and at least $1/2$. Since $|y| \leq 2n$, the ORIGINAL density lower bound and (ECT12) imply

$$h_{n-q}^\chi \geq \ell_{h,k} (k-2+2n)^{-B_h} (1 - 63R_k/(20n))^{2q} \frac{50\pi n}{63} n^{2n} e^{-2n} e^{c_* q}. \tag{ECT15}$$

The finite positive-real remainder used here is the one in Askey and Roy [H22, eq. (5.11.1), §5.11(ii)]. For $x > 0$ it gives $\log \Gamma(x) = (x - \frac{1}{2}) \log x - x + \frac{1}{2} \log(2\pi) + r(x)$, where $0 \leq r(x) \leq 1/(12x)$. Taking $x = 2n$, with $n \geq 1$, and using $(2n)! = 2n\Gamma(2n)$ in the retained $g_n = \sqrt{2\pi}(2n)!/4^n$ gives exactly $g_n \leq 2\pi\sqrt{2n} n^{2n} e^{-2n} e^{1/(24n)}$. The arithmetic comparison factor, when present, remains the original A_k .

By the original monic upper comparison and the positive-real Stirling remainder,

$$\omega_n \leq A_k g_n \leq 2\pi\sqrt{2n} A_k n^{2n} e^{-2n} e^{1/(24n)}. \tag{ECT16}$$

Divide (ECT16) by (ECT15). For a uniform bound, $-\log(1-x) \leq 2x$ shows that the root-factor logarithmic cost is at most $63R_k/5$. Define the explicit finite certificate

$$U_{h,k}^{\text{ctr}} = \frac{63\sqrt{2}A_k}{25c_h\sqrt{q}} \vartheta_h^{-(k-3)} (k-2+4q)^{B_h} \exp \left(\alpha(k-3) + \frac{63}{5}R_k + \frac{1}{24q} - c_* q \right). \tag{ECT17}$$

Then

$$\boxed{0 < d_n \leq U_{h,k}^{\text{ctr}} \quad (q \leq n \leq 2q).} \tag{ECT18}$$

It is still a true, possibly uninformative, bound when $U_{\text{ctr}} \geq 1$. At fixed actual h and full multiplicity,

$$\log U_{h,k}^{\text{ctr}} = -c_* q + O_h(k + \log q). \tag{ECT19}$$

All original masses, the full amplitude, convolution order and primary multiplicities are retained. The reference bands are only integration subsets. No quotient was replaced by the monomial quotient. The period observation is not needed for this bound.

212.5 4. The penalties are exponentially small

Keep exactly the original endpoint multiplicities:

$$P_- = -\log(1 - d_{2q-1}) - 2 \sum_{n=q}^{2q-2} \log(1 - d_n), \quad (\text{ECT20})$$

$$P_+ = -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{n=q+1}^{2q-1} \log(1 - d_n). \quad (\text{ECT21})$$

Their combined coefficient count is $4q-1$. If $U_{\text{ctr}} < 1$,

$$0 \leq P_- + P_+ \leq -(4q-1) \log(1 - U_{h,k}^{\text{ctr}}) \leq \frac{(4q-1)U_{h,k}^{\text{ctr}}}{1 - U_{h,k}^{\text{ctr}}}. \quad (\text{ECT22})$$

Hence $P_- + P_+ = O_h(q \exp(-c_* q/2))$, improving the submitted $O_h(e_k)$. For finite use intersect (ECT22) with the submitted polynomial allowance; retain the older bound where either new guard fails.

The complete exact receiver remains

$$J_k^{\text{action}} = \mathcal{B}_k^{\text{ar}} + \mathcal{W}_k^{\text{ar}} - P_- - P_+. \quad (\text{ECT23})$$

Thus $J_{\text{action}} = B_{\text{ar}} + W_{\text{ar}}$ plus a nonpositive exponentially small error. The separate monic-window comparison $W_{\text{sigma-E_hi}} \leq W_{\text{ar}} \leq W_{\text{sigma+E_lo}}$ still has error $O_h(k + \log q)$; that error has NOT become exponentially small.

The submission's exact Gamma window and finite enclosure check out:

$$W_q^\sigma = 2 \log \frac{(4q)!}{(2q)!} - 4q \log 2 + \log \frac{2q-1}{2(4q-1)}, \quad (\text{ECT24})$$

$$C(q) - \log 2 - \left(\frac{1}{16q} + \frac{1}{4q-2} \right) \leq W_q^\sigma \leq C(q) - \log 2, \quad C(q) = 4q \log q + (8 \log 2 - 4)q. \quad (\text{ECT25})$$

The new action enclosure replaces the old penalty allowance by the minimum of the old bound and (ECT22). It does not evaluate a further coefficient of B_{ar} or of a separate allocation.

212.6 5. This also controls the minimum, not only the action mean

At the original untilted source, total parity gives

$$\epsilon_{q-1}^2 = b_q(1 - d_q)/d_q, \quad \epsilon_N^2 = b_{N+1}(1 - d_N)(1 - d_{N+1})/d_{N+1} \quad (N \geq q), \quad (\text{ECT26})$$

where $b_n = \omega_n / \omega_{n-1}$. This is total parity, not an assertion that the mixed kernel and boundary pairings vanish separately.

AMT's lower monic comparison gives

$$b_n \geq \frac{a_k}{A_k D_n} n(n-1/2), \quad D_n = [1 + 4(n + 21 + 4m)^2]^{21+4m}. \quad (\text{ECT27})$$

Let $b_{\text{low}} = (a_k / (A_k D_{2q})) q(q-1/2)$. Equation (ECT18) proves, for $U_{\text{ctr}} < 1$,

$$\min_{q-1 \leq N \leq 2q-1} \epsilon_N \geq \sqrt{b_{\text{low}}} \frac{1 - U_{h,k}^{\text{ctr}}}{\sqrt{U_{h,k}^{\text{ctr}}}}. \quad (\text{ECT28})$$

The first endpoint has only one factor $1-d_q$ and gives at least this bound. Since $\log b_{\text{low}} = O_h(k + \log q)$,

$$\liminf \frac{1}{q} \log \min_{q-1 \leq N \leq 2q-1} \epsilon_N \geq c_*/2. \quad (\text{ECT29})$$

For the original exterior benchmark $L_{hk} = \delta q(k+1)/2$ on $k=4l+1$,

$$\boxed{\liminf \frac{1}{q} \log \frac{\min_{q-1 \leq N \leq 2q-1} \epsilon_N}{L_{h,k}} \geq c_*/2 > 0.} \quad (\text{ECT30})$$

Conditional on the original arithmetic packet/source inputs, every canonical allowance in THIS window is exponentially larger than the polynomial benchmark. This is stronger than the earlier geometric-mean obstruction. It does not identify the arithmetic spectrum with the allowance, negate RH, or prove that the whole tau programme fails.

212.7 6. Extension to any fixed proportional window

For any fixed $C > 1$ choose small fixed $\rho > 0$ and let

$$H(\rho) = 1 + \rho^2 + \frac{\rho^2}{2} \left[\log \frac{1 + \sqrt{1 - \rho^2}}{\rho} - \sqrt{1 - \rho^2} \right]. \quad (\text{ECT31})$$

The preceding integral proof gives $E(\rho) \leq H(\rho)$. Use $b = 2n/H(\rho)$, $a = \rho b$. The exponential cost is $\leq 2n$ and the two normalized factors are $(1+\rho)/H(\rho)$, $\sqrt{1-\rho^2}/H(\rho)$. For $q \leq n \leq Cq$ the gap is at least $c_C(\rho)q$, where

$$c_C(\rho) = 2 \log(1 + \rho) + (C - 1) \log(1 - \rho^2) - 2C \log H(\rho). \quad (\text{ECT32})$$

As $H(\rho) - 1 = O(\rho^2 \log(1/\rho))$, $c_C(\rho) = 2\rho + O_C(\rho^2 \log(1/\rho)) > 0$ for sufficiently small ρ . The band has inner radius a fixed positive constant times n , so the actual root and density costs remain $O_C(h, C)(k + \log q)$. This proves the analogous exponential bound for every fixed C , and the minimum-allowance conclusion follows as above. Constants depend on C . No uniform assertion is made for C growing with k .

A mere replacement of $2q$ by another fixed multiple therefore does not repair this canonical small-allowance strategy. A further RH-directed estimate needs a genuinely different admitted degree scaling or another same-class comparison. The exact mixed objects remain useful, but a cancelled allocation cannot be improved while its correlated action budget is held fixed.

212.8 Verification and scope

The local executable checker uses exact rational and Gaussian-rational arithmetic for the rate certificate, Gamma norms/orthogonality and generating ODE, all $3/4/5$ partitions $k=3..101$, the complete top face, original quotient ratios, three actual canonical reference constructions, parity, and the weighted action product. Its reference mass is seven, not silently one. Separate 180-digit tests of monic exponential-weight minima and Gamma-window bounds are explicitly NON-INTERVAL diagnostics.

Normal and Python -O executions completed with byte-identical success records. All four false-formula controls were rejected in both modes. An initial combined shell batch timed out during the controls; the controls were subsequently rerun in separate modes and all eight expected errors recorded. No timeout is called a proof failure or a passing execution.

No new Lean run, independent second-reader approval, actual zeta packet, actual-period integral, or full cumulative-source audit is claimed. The infinite theta quotient, represented-zero/absence distinction, mixed supports, full arithmetic units and multiplicities are unchanged. The proof requires the explicit original AMT density envelopes and standard positive-real Stirling bounds. It does not assume the desired exterior upper estimate.

Primary literature: NIST Digital Library of Mathematical Functions, sections 18.23.7 and 18.19 (the Meixner–Pollaczek generating function and normalization), 5.11.1/5.11(ii) (positive-real Stirling remainder), and 19.2 (elliptic-integral conventions). The two Jensen calculations are included above; no novelty claim is made for those classical ingredients.

Source status. This is the complete accepted integration derivative supplied by the publication owner; the captured original and the exact derivative change are preserved separately. The independent written review is recorded in the source manifest. Its statements about review, execution and publication describe the captured source state. This typesetting conversion supplies no additional mathematical acceptance. Captured pull request: <https://github.com/KokunoYumeto/zeta-function-research-reader/pull/37>; head 834bc0ea2b2510ac39862d9bd176200e78254c41. The source is preserved byte for byte in UCT_PUBLIC_UCB_REVIEW_AND_PENALTY_TAIL.md. Every displayed equation is assigned a unique UCT tag; the exact source-to-tag correspondence accompanies this body.

213 Uniform canonical-band review and a concentrated penalty certificate

16 September 2026. Additive continuation of PR37. The original two-band Jensen proof and its checker are unchanged.

213.1 0. Provenance, scope, and result

The user supplied Pastedmarkdown(20260916-165456).md (file ID file_00000000bb7882308dd473a49e00200), the UCB full-band continuation. The entire supplied Markdown was read, including its small-rank patch, monic-norm comparison, rate derivative, finite guards, minimum localization, and penalty receiver. Its linked UNIFORM_CANONICAL_BAND.tex, PDF, and 40-payload ZIP were not available as inspected proof artifacts in this review. In particular, the exact separately defined C_EIQ and its author's 134+9 checks are not relabelled as independently replayed. Both local execution interfaces failed before running commands. External Wolfram evaluations did run, with scope recorded below.

The arithmetic inputs remain the existing original-source AMT bounds, the paired-root full-form comparator on the stated degree band, and the exact untilted radius identity. The equilibrium input remains the supplied EIQ measure for $V_{\{a,b\}}(x) = \text{bsqrt}(x) - a \log(x)$, with support $[u^2, v^2]$, bounded density on compact positive parameter rectangles, and $D = V - 2P - \lambda \geq 0$, with equality on the support. No entire cumulative source audit is claimed.

Relative to those stated inputs, the new note's headline conclusions follow:

$$\log \min_{q-1 \leq N \leq 2q-1} \epsilon_N = qa_1 + O_h(k + \log q), \quad \log(\mathcal{P}_- + \mathcal{P}_+) = -2qa_1 + O_h(k + \log q). \quad (\text{UCT1})$$

Here $q = [1 + k(m-1)](k+1)^2$ and

$$a_1 = \log \frac{(u_* + v_*)\sqrt{v_*^2 - u_*^2}}{16}, \quad u_*K(\kappa_*) = 2, \quad v_*E(\kappa_*) = 4. \quad (\text{UCT2})$$

The validated sharp constant is larger than PR37's independent coarse lower exponent: $a_1 = 0.06651895202027\dots$, whereas $\log(50000/45927)/2 = 0.04248491300504\dots$. The UCB proof uses stronger equilibrium/root-comparison inputs and an eventual domain; PR37's explicit two-band finite bound remains a separate fallback, not a false result to delete.

This continuation supplies an independent quantile argument for the needed uniform single-norm error, a tighter finite sum of the actual penalties, and a relative tail certificate. The penalty mass is concentrated in a final $O_h(k + \log q + \log(1/\text{eta}))$ strip at any prescribed relative tolerance eta . This does not say the final row alone dominates, identify the exact minimizer, or supply an RH upper allowance.

213.2 1. Original quotient and adjacent factors

For the sharp arithmetic transfer, retain the original tensor class $k=4l+1$, $l \geq 1$, the fixed quartet $0 < \delta < 1/2$, $\gamma > 2$, and its complete positive integer multiplicity m . Besides $q \geq 1000$ in the bulk estimate, the retained paired-root provider requires $q \geq 2^{17} R_k$, where $R_k = k \sqrt{\delta^2 + \gamma^2}$. Its complete proof and original definitions are now included, rather than silently substituting the older comparison with a different cutoff.

The arithmetic comparison constant used below is exactly

$$B_k = \log(A_k D_{2q}/a_k) + C_k^{\text{cmp}}, \quad D_n = [1 + 4(n + 21 + 4m)^2]^{21+4m}. \quad (\text{UCT3})$$

Here a_k, A_k are the unchanged full-polynomial arithmetic/Gamma comparison constants, displayed with their original masses in CAI16–18; C_k^{cmp} is the complete paired-root constant, including its inner-region term, in the supplied provider. Taking the same monic minima gives $|\log T_r - \log(\nu_r^0/\gamma_{q+r})| \leq B_k$ on $0 \leq r \leq q$. This is a norm comparison through the original polynomial coordinates, not replacement of the arithmetic quotient by a monomial quotient.

Keep the original arithmetic source, complete unit, multiplicities, quotient map, source mass, and polynomial coordinate. With the UCB notation,

$$\nu_r = \min_{\deg P=r, P \text{ monic}} \|\chi P\|_{\text{ar}}^2, \quad T_r = \nu_r/\omega_{q+r}, \quad d_{q+r} = T_r^{-1}, \quad \beta_n = \omega_n/\omega_{n-1}. \quad (\text{UCT4})$$

The exact identity is

$$\epsilon_{q+r-1}^2 = \beta_{q+r}(T_r - 1) \begin{cases} 1 & r = 0, \\ 1 - T_{r-1}^{-1} & r \geq 1. \end{cases} \quad (\text{UCT5})$$

At $r=0$, the earlier observation is the isomorphism $J_{\{q-1\}}$. There is no inverse at $q-2$. Total parity, not separate vanishing of the kernel/boundary pairings, is used. The original component identity $z_B = -z_K$ remains compatible with a nonzero mixed pairing.

213.3 2. Small relation rank: the new patch is valid

For $Y=4q/\pi$ and $0 < \eta < 1$, the exact monic Legendre minimum on an interval of length H is

$$\int_I |P|^2 dy \geq \frac{H^{2r+1}}{(2r+1) \binom{2r}{r}^2}. \quad (\text{UCT6})$$

It holds for every complex monic polynomial: the shifted Legendre minimizer is monic, orthogonal to all lower degrees, and the residual contributes a nonnegative squared norm. The original Gamma lower density on $[Y(1-\eta), Y(1+\eta)]$ therefore gives UCB(2). The minimum of $2q \log y - (\pi/2)y$ is at the lower endpoint; the retained cost is $2q[-\log(1-\eta) - \eta]$.

For $\eta=1/10$ and $0 \leq r \leq q/100$, the rational margin is exactly

$$\frac{3}{7} - \frac{1}{90} - \frac{3}{50} - \frac{1}{5000} - \frac{1}{3} = \frac{7537}{315000} > 0. \quad (\text{UCT7})$$

Using $\text{binom}(2r, r) \leq 4^r$ and the stated factorial upper bound gives the supplied conservative prefactor $c_0 q^{-3/2}$, with $c_0 = c_{\text{Gamma}}/(60 \sqrt{6\pi})$. Thus

$$\log(\nu_r^0/\gamma_{q+r}) \geq q/3 - (3/2) \log q + \log c_0. \quad (\text{UCT8})$$

The exponent exceeds $2a_1$. This genuinely covers $r=0$, $r=1$, and all small positive r/q ; it does not import a fixed-positive-ratio equilibrium estimate into a degenerating parameter range.

213.4 3. Independent proof of the uniform bulk monic-norm estimate

This proves the order and mechanism needed by UCB(5), without pretending to have read the linked detailed formula for its particular $\mathbf{C_EIQ}$.

Fix a positive compact parameter rectangle and its supplied EIQ equilibrium probability μ , supported on $[A, B]$. Put $W=B-A$, $\text{cap}=W/4$, and let its density be bounded by M . The density vanishes continuously at the endpoints in this one-cut family. Write $P_\mu(x)=\int \log|x-y|d\mu(y)$ and $D=V-2P_\mu-\lambda$. The original Gamma remainder is

$$\mathfrak{r}(y) = \sigma(y)e^{\pi y/2}\sqrt{y}/\sqrt{2}. \quad (\text{UCT9})$$

It need not be bounded below at zero. What is required, and follows from the original Gamma envelopes, is $\mathfrak{r}(\text{qsqrt}x) \geq r_- \rightarrow 0$ on the equilibrium support for $q \geq 1000$, and $\mathfrak{r}(y) \leq r_+$ for every positive y . If $u_{\min}$ is the minimum of the positive square-root support endpoint on the rectangle, safe choices are

$$r_- = (c_\Gamma/\sqrt{2})\sqrt{\frac{1000u_{\min}}{1+1000u_{\min}}}, \quad r_+ = C_\Gamma/\sqrt{2}. \quad (\text{UCT10})$$

213.4.1 Lower norm

Let ν be the arcsine probability on $[A, B]$. For every monic degree- j polynomial p ,

$$\int \log |p| d\nu \geq j \log \text{cap}. \quad (\text{UCT11})$$

This follows rootwise from the circle Jensen formula, including complex roots and roots on the interval. Since the equilibrium measure is supported on the same interval, Fubini and the arcsine potential give

$$\int V d\nu = 2 \log \text{cap} + \lambda. \quad (\text{UCT12})$$

The arcsine entropy is $\int \log(d\nu/dx) d\nu = -\log(\text{p} \text{cap})$. Probability Jensen applied to $|p|^2 \exp(-jV)/(d\nu/dx)$ proves

$$h_{j,q}(a, b) \geq r_- \pi \text{cap} e^{-j\lambda}. \quad (\text{UCT13})$$

213.4.2 Upper norm and quantile uniformity

Take the j midpoint quantiles x_i of μ and $p_j(x) = \prod_i (x - x_i)$. Their empirical CDF differs from the equilibrium CDF by at most $1/j$ (the sharper $1/(2j)$ is unnecessary). For fixed x , clip the logarithm at $\delta = 1/j$:

$$f_{x,\delta}(y) = \log \max\{|x - y|, \delta\}. \quad (\text{UCT14})$$

On $[A, B]$ it has total variation at most $2\log(1+W/\delta)$. Integration by parts against the CDF error yields

$$\sum_i f_{x,\delta}(x_i) \leq j \int f_{x,\delta} d\mu + 2\log(1+jW). \quad (\text{UCT15})$$

Moreover,

$$\int (f_{x,\delta} - \log|x - y|) d\mu(y) \leq M \int_{-\delta}^{\delta} \log(\delta/|s|) ds = 2M\delta. \quad (\text{UCT16})$$

Since each unclipped nodal logarithm is no larger than its clipped version,

$$|p_j(x)|^2 e^{-jV(x)} \leq e^{4M} (1 + jW)^4 e^{-j\lambda} e^{-jD(x)}. \quad (\text{UCT17})$$

This holds on the entire positive half-line, not merely inside the support. As $j \geq 1$ and $D \geq 0$, its integral is bounded by

$$h_{j,q}(a, b) \leq r_+ J_* e^{4M_*} (1 + jW_+)^4 e^{-j\lambda}, \quad (\text{UCT18})$$

where J_* uniformly bounds $\int \exp(-D) dx$, M_* the density bound, and W_+ the support length. All are finite on the positive compact rectangle. For an explicitly crude bound on $[1, 250] \times [1, 400]$, let $B_{\max}$ bound the support's right endpoint and Λ_* bound $|\lambda|$; then

$$J_* \leq e^{\Lambda_*} (1 + B_{\max})^2 [1 + 2\Gamma(506)]. \quad (\text{UCT19})$$

Indeed $P_\mu(x) \leq \log(x + B_{\max})$, so the small interval is integrable and the large tail is bounded by $x^{-252} \exp(-\sqrt{x})$. This covers the whole region near zero as well as infinity.

213.4.3 The square substitution and a separate finite constant

For $r = 2j + \epsilon$, evenness of the reference measure makes the monic minimizer have that same parity. Substituting $y = \sqrt[q]{x}$ gives exactly

$$\nu_r^0 = \sqrt{2} q^{2(q+r)+1/2} h_{j,q}((q + \epsilon - 3/4)/j, \pi q/(2j)). \quad (\text{UCT20})$$

The quarter shift and the two half-lines have not disappeared. For $q \geq 1000$, $q/100 < r \leq q$, both these parameters and $(2q/r, \pi q/r)$ belong to $[1, 250] \times [1, 400]$.

Let L_a, L_b bound the absolute Robin derivatives on the rectangle. The parameter replacement costs at most

$$C_{\text{par}} = (401/4)L_a + 50\pi L_b + \Lambda_*/2 \quad (\text{UCT21})$$

in $j\lambda$, because $j|a - 2q/r| \leq 401/4$, $j|b - \pi q/r| \leq 50\pi$, and $|j - r/2| \leq 1/2$.

The preceding two norm bounds yield $|\log h + j\lambda| \leq C_{\text{norm}} + 4\log(q+1)$. To retain the explicit finite choice supplied in the author's PR discussion, let $W_{\min} > 0$ be the minimum support length on the same compact parameter rectangle and put

$$c_- = r_- \pi W_{\min}/4, \quad c_+ = r_+ J_* e^{4M_*}, \quad C_{\text{norm}} = \max\{|\log c_-|, |\log c_+| + 4\log(1 + W_+)\}. \quad (\text{UCT22})$$

Indeed $1 + jW_+ \leq (q+1)(1+W_+)$ on the stated degree domain. This is a specified constant for this alternative estimate, not a claim to have checked the unavailable UCB archive's separate C_{EIQ} . Stirling cancellation is exact up to its positive remainder:

$$\log(\nu_r^0/\gamma_{q+r}) = 2(q+r)(1 - \log(1+r/q)) + \log h - \frac{1}{2} \log(1+r/q) - \log(2\pi) - R_{q+r}, \quad 0 < R_n < 1/(24n). \quad (\text{UCT23})$$

Thus a independently specified safe bulk error is

$$\widehat{E}_0(q) = C_{\text{norm}} + C_{\text{par}} + \frac{1}{2} \log 2 + \log(2\pi) + 1/48 + 4\log(q+1). \quad (\text{UCT24})$$

This is an alternative finite constant, not a silent assertion about the unseen UCB15–19 definition of C_{EIQ} . It proves the same uniform $O(\log q)$ conclusion from the supplied equilibrium inputs.

213.5 4. The profile derivative, positivity, and the sharp constant

Put $s = \sqrt{1 - \kappa^2}$ (not a source order) and use the actual endpoint relation

$$1 + t = E(\kappa)/(sK(\kappa)). \quad (\text{UCT25})$$

The displayed profile simplifies exactly to

$$\psi(t) = \log \frac{1+s}{E(\kappa)} + t \log \frac{\kappa}{E(\kappa)}. \quad (\text{UCT26})$$

With $f = \log(\kappa/E)$ and $g = \log((1+s)/E)$, the classical derivative formulas give

$$f'(\kappa) = K/(\kappa E), \quad g'(\kappa) + t f'(\kappa) = 0. \quad (\text{UCT27})$$

Therefore $d\psi/dt = f$, exactly as in UCB. The modulus increases with t : one has $sK < E < K$, and differentiation of $\log(1+t)$ with respect to s gives

$$\frac{d}{ds} \log(1+t) = \frac{(1+t) + (1+t)^{-1} - (s + s^{-1})}{1 - s^2} < 0. \quad (\text{UCT28})$$

For completeness, $E > sK$ follows by pairing θ with $\pi/2 - \theta$ in their defining integrals. If their radicands are x, y , then $xy > s^2$ and

$$\sqrt{x} + \sqrt{y} - s(1/\sqrt{x} + 1/\sqrt{y}) = (\sqrt{x} + \sqrt{y})(1 - s/\sqrt{xy}) \geq 0, \quad (\text{UCT29})$$

strictly on a set of positive measure. Also $E < K$ is immediate. Since $E(\kappa) > 1 > \kappa$ for $0 < \kappa < 1$, $\psi' < 0$; since $f'(\kappa) > 0$, the profile is convex. On $0 < t < 1$,

$$\psi(t) \geq a_1 + c_1(1-t), \quad c_1 = \log(E(\kappa_*)/\kappa_*) > 0. \quad (\text{UCT30})$$

Its limiting values are $\psi(0) = \log(4/\pi)$ and $\psi(\infty) = 0$; the latter also follows from the expansion in Section 7. Hence it is positive at every finite positive argument.

An independent exact-rational computation encloses

$$0.06651895202027 < a_1 < 0.06651895202028, \quad 0.04748604398475 < c_1 < 0.04748604398476. \quad (\text{UCT31})$$

The bulk formula, small-rank patch and original AMT transfer then give the claimed minimum rate and $\mathbf{0}_h(k + \log q)$ localization. They do not prove exact finite minimization at $\mathbf{r} = \mathbf{q}$. The transfer retains both adjacent factors through the global lower bound on all $\mathbf{T}_r$, exactly as in UCB(8)–(11).

213.6 5. A sharper finite penalty bound

Use either UCB's certified finite bulk error or the alternative one above, consistently throughout. Set

$$r_0 = \lfloor q/100 \rfloor, \quad C_k = E_0(q) + B_k, \quad s_k = q/3 - (3/2) \log q + \log c_0 - B_k, \quad (\text{UCT32})$$

$$m_k = \min\{s_k, 2qa_1 - C_k\}, \quad z = e^{-2c_1} \in (0, 1). \quad (\text{UCT33})$$

Impose the original root-comparison domain, $q \geq 1000$, and $m_k \geq \log 2$. For the bulk,

$$T_r^{-1} \leq e^{C_k - 2qa_1} z^{q-r} \quad (r_0 < r \leq q), \quad (\text{UCT34})$$

while $\mathbf{T}_r \cdot \{-1\} \leq \mathbf{e} \cdot \{-s_k\}$ on the small-rank block.

The exact coefficients of the combined penalty, indexed by $\mathbf{r} = 0, \dots, \mathbf{q}$, are

$$c_0^{\text{wt}} = 3, \quad c_r^{\text{wt}} = 4 \quad (1 \leq r \leq q-2), \quad c_{q-1}^{\text{wt}} = 3, \quad c_q^{\text{wt}} = 1. \quad (\text{UCT35})$$

Do not confuse c_0^{wt} with the small-rank constant c_0 . The small block's total coefficient is $4r_0+3$. Since $-\log(1-x) \leq x/(1-e^{-m_k})$ for $0 \leq x \leq e^{-m_k}$, summing the actual coefficients gives

$$\begin{aligned} \mathcal{P}_- + \mathcal{P}_+ &\leq (4r_0 + 3) \frac{e^{-s_k}}{1 - e^{-s_k}} \\ &\quad + \frac{e^{C_k - 2qa_1}}{1 - e^{-m_k}} \left[1 + 3z + \frac{4z^2(1 - z^{q-r_0-2})}{1 - z} \right]. \end{aligned} \quad (\text{UCT36})$$

In particular the bracket is bounded by

$$C_{\text{geo}} = \frac{(1+z)^2}{1-z}, \quad 40.2398920641 < C_{\text{geo}} < 40.2398920642. \quad (\text{UCT37})$$

This replaces a $4q-1$ worst-row factor by a geometric-series constant in the bulk. The small-rank contribution has strictly faster exponential decay. Keep the intersection with UCB's original finite bound and PR37's weaker-input bound; no stronger finite bound is asserted outside its guards.

The lower bound $P_{\text{total}} \geq T_{q-1} \geq \exp(-2qa_1 - C_k)$ stays. Each individual penalty has the same logarithmic rate: P_+ contains the final row; P_- contains T_{q-1} , whose logarithm differs at the profile level by a bounded quantity. Thus both log penalties are $-2qa_1 + O_h(k + \log q)$; neither is identically zero.

Retaining the stronger finite bounds on overlapping degree ranges (RGI4–7). Fix the original arithmetic data and an integer relation cutoff $r \geq 0$ for which RGC5 holds and $S_{k,q,r} > 0$ in RGC25. The letter r here is a coefficient-degree cutoff, not a rank of a mixed observation. The functions $\eta_{q,j}$ and $\rho_{q,j}$ increase with j , so RGC5 holds for every $0 \leq j \leq r$ with its own cutoff. The whole-form paired-root theorem PCR1–8 additionally applies when $q \geq 2^{17}R_k$ and $j \leq 2q$. Retain its complete constant C_k^{cmp} , including its inner integral. Define

$$\begin{aligned} E_j^{\text{cap}} &= \begin{cases} \min\{E_{q,j}^{\text{root}}, C_k^{\text{cmp}}\}, & q \geq 2^{17}R_k \text{ and } j \leq 2q, \\ E_{q,j}^{\text{root}}, & \text{otherwise,} \end{cases} \\ L_j^{\text{new}} &= F_{q,j} + \log C_0 - \Lambda_{k,q+j} - E_j^{\text{cap}}. \end{aligned} \quad (\text{RGI4})$$

Both eligible comparisons bound exactly H_Q against H_0 on the same degree- j space. Their intersection therefore gives the smaller logarithmic error in RGI4. Applying AMT to the original numerator and denominator monic minima, with RGI1–2, proves $\log T_j^{\text{ar}} \geq L_j^{\text{new}}$. In particular $L_j^{\text{new}} \geq S_{k,q,r} > 0$: $F_{q,j} \geq F_{q,r}$, whereas $\Lambda_{k,q+j} \leq \Lambda_{k,q+r}$ and $E_j^{\text{cap}} \leq E_{q,j}^{\text{root}} \leq E_{q,r}^{\text{root}}$.

For the fixed band retain the explicit CCN4 error $\hat{E}_0(q)$ rather than an unspecified constant from an unavailable source. In the original UCT notation set

$$B_k = \Lambda_{k,2q} + C_k^{\text{cmp}}, \quad C_k = \hat{E}_0(q) + B_k, \quad s_k = q/3 - \frac{3}{2} \log q + \log c_0 - B_k, \quad r_0 = \lfloor q/100 \rfloor,$$

where $c_0 = c_-/(60e\sqrt{6\pi})$ is the complete UCBS9 prefactor, and $a_1, c_1 > 0$ are the original UCT30–31 constants. On the original finite UCT domain $q \geq 1000$, $q \geq 2^{17}R_k$ and $\min\{s_k, 2qa_1 - C_k\} \geq \log 2$, its actual arithmetic lower bounds are

$$\begin{aligned} L_j^{\text{old}} &= \begin{cases} s_k, & 0 \leq j \leq r_0, \\ 2qa_1 + 2c_1(q-j) - C_k, & r_0 < j \leq q. \end{cases} \\ L_j &= \begin{cases} \max\{L_j^{\text{new}}, L_j^{\text{old}}\}, & \text{on that domain and } j \leq q, \\ L_j^{\text{new}}, & \text{otherwise.} \end{cases} \end{aligned} \quad (\text{RGI5})$$

UCT34 and its small-rank companion prove the old bound on their stated domain. Thus $\log T_j^{\text{ar}} \geq L_j$ for every $0 \leq j \leq r$. No fixed-band error has been extrapolated beyond $j = q$.

Write $b_j = e^{-\Lambda_{k,q+j}}(q+j)(q+j-1/2)$. Applying RGC22 and the exact original adjacent-factor identity RGC24 now gives the pointwise bounds

$$\epsilon_{q+j-1}^2 \geq \begin{cases} b_0(e^{L_0} - 1), & j = 0, \\ b_j(e^{L_j} - 1)(1 - e^{-L_{j-1}}), & 1 \leq j \leq r. \end{cases} \quad (\text{RGI6})$$

Both factors are increasing in their positive exponents. This proves the bounds directly and retains the distinct first endpoint. Since $b_j \geq b_{k,q,r}^{\text{low}}$ and all $L_j \geq S_{k,q,r}$, their minimum recovers RGC26, while keeping every stronger eligible original estimate.

For $r \geq 1$ let $w_0 = w_r = 1$, $w_j = 2$ for $0 < j < r$, and let $c_{r,j} = w_j + w_{j+1}$ for $j < r$, $c_{r,r} = w_r$. The unchanged penalty therefore obeys the sharper finite bound

$$\begin{aligned} 0 \leq \mathcal{P}_{q,r} &\leq P_{q,r}^{\text{cap}} := \sum_{j=0}^r c_{r,j} [-\log(1 - e^{-L_j})] \\ &\leq -(4r - 1) \log(1 - e^{-S_{k,q,r}}). \end{aligned} \quad (\text{RGI7})$$

Indeed $d_{q+j} = (T_j^{\text{ar}})^{-1} \leq e^{-L_j}$ and $-\log(1 - x)$ increases on $[0, 1)$. The second line follows because $-\log(1 - e^{-L})$ decreases for $L > 0$ and $\sum_j c_{r,j} = 4r - 1$. At $r = q$ on the UCT domain, the first sum is no larger than the sum using L_j^{old} . Applying $-\log(1 - x) \leq x/(1 - e^{-s_k})$ on the small block and $-\log(1 - x) \leq x/(1 - e^{-m_k})$ on the bulk, where $m_k = \min\{s_k, 2qa_1 - C_k\}$, gives exactly the finite geometric sum UCT36. Hence its stronger fixed-band penalty control is retained, together with the new growing-band bound. Every coefficient, including those for $r = 1$, is the coefficient of the original penalty rather than an assigned weight.

Moving-cutoff bounds on the original receiving spaces (GRB1–4). The complete moving-cutoff proofs appended below act on precisely the spaces in RGI1–3. In particular, if $p \in \mathbb{C}[y]_{\leq j}$ and $U_j P(y) = i^{-j} P(k/2 + iy)$, their arithmetic middle expression is $\|\chi U_j^{-1} p\|_{\text{ar}}^2$. The identity $U_{q+j} M_\chi = M_Q U_j$ proves this assertion, including the unit phase on every lower coefficient. No remainder fibre or arithmetic denominator changes.

At a fixed integer j , collect every *applicable* proved full-form comparison $\ell H_0 \leq H_Q \leq u H_0$ in the following finite set:

provider and its full stated guards	ℓ	u	
RGC5–7	$(1 - \rho_{q,j}^2)^q (1 - \eta_{q,j})$	$(1 + \rho_{q,j}^2)^q + \eta_{q,j}$	(GRB1)
PCRP1–8: $q \geq 2^{17} R_k$, $j \leq 2q$	$e^{-C_k^{\text{cmp}}}$	$e^{C_k^{\text{cmp}}}$	
moving cutoff: $j \geq 3$ and both parity guards	$e^{-C_\chi(k,j)} (1 - \eta_{k,j}^{\text{mov}})$	$1 + \eta_{k,j}^{\text{mov}}$	

Use a row only where its own proof applies; η^{mov} distinguishes the new quantity from RGC's earlier inner error. If the set is nonempty, define $\ell_j^* = \max \ell$ and $u_j^* = \min u$. Then $\ell_j^* H_0 \leq H_Q \leq u_j^* H_0$ for every complex polynomial of degree at most j , including zero. Indeed each inequality holds on that same space, so one may take the maximum of its lower constants and the minimum of its upper constants. This argument also proves $\ell_j^* \leq u_j^*$ by evaluating any nonzero polynomial in the positive norm.

Write $\mathcal{L}_{k,n} = \log(A_k D_n / a_k)$. Every proved lower and upper reference endpoint $l_j^0 \leq \log(\nu_j^0 / \gamma_{q+j}) \leq u_j^0$ gives

$$l_j^0 - \mathcal{L}_{k,q+j} + \log \ell_j^* \leq \log T_j^{\text{ar}} \leq u_j^0 + \mathcal{L}_{k,q+j} + \log u_j^*. \quad (\text{GRB2})$$

To prove this, minimize the same full-form inequalities over the original affine monic fibre, then bound the unchanged denominator ω_{q+j} by AMT. For the new moving reference, the exact endpoints are $l_j^0 = Z_{q,j} + \log c_-$ and $u_j^0 = Z_{q,j} + \log c_+$. The all-degree lower endpoint $F_{q,j} + \log C_0$

remains available from RGC16. There is also an explicit upper reference endpoint at every $j \geq 0$, including the degrees 0 and 1 outside the moving parity theorem. Set $n = q + j$ and use the actual monic trial $p(y) = y^j$. The exact original moment generating function in GRE15 gives

$$\int y^{2n} d\sigma(y) = \sqrt{2\pi}(2n)! [z^{2n}](\cos z)^{-1/2}, \quad u_j^{\text{trial}} = \log \left(4^n [z^{2n}](\cos z)^{-1/2} \right).$$

Differentiation of the generating integral is justified by its exponential envelope on a neighbourhood of zero. This coefficient is positive because it equals the positive moment divided by $\sqrt{2\pi}(2n)!$. The trial has norm equal to that moment, so the monic minimum is no larger, proving $\log(\nu_j^0/\gamma_n) \leq u_j^{\text{trial}}$ with the original mass and factorials. Thus this endpoint can always be used in GRB2 when one of its full-form rows is available. The moving reference endpoints themselves require $j \geq 2$, and the moving full-form row requires $j \geq 3$. Maximize the resulting lower endpoints also with the previously proved RGI5/UCT endpoints wherever their guards hold; minimize all eligible upper endpoints. Denote the results by L_j, U_j . Thus each new comparison improves or preserves every eligible earlier bound, with no extension of its original domain.

Whenever $0 < L_j \leq U_j$ and, for $j \geq 1$, $0 < L_{j-1} \leq U_{j-1}$, set $b_j^- = (q+j)(q+j-\frac{1}{2})e^{-\mathcal{L}_{k,q+j}}$ and $b_j^+ = (q+j)(q+j-\frac{1}{2})e^{\mathcal{L}_{k,q+j-1}}$. The exact adjacent identity now yields

$$\begin{aligned} b_j^-(e^{L_j} - 1)(1 - e^{-L_{j-1}}) &\leq \epsilon_{q+j-1}^2 \\ &\leq b_j^+(e^{U_j} - 1)(1 - e^{-U_{j-1}}), \quad j \geq 1. \end{aligned} \quad (\text{GRB3})$$

For $j = 0$, omit precisely the preceding contraction on both sides. AMT at the two adjacent monic degrees proves the displayed bounds on β_{q+j} ; each remaining factor increases in its exponent. This proves GRB3 without discarding either contraction.

For any of the actual windows, let c_j be its already derived nonnegative penalty coefficients, over precisely the participating relation indices. Use these finite logarithmic endpoints when $0 < L_j \leq U_j$ at every participating index. The complete finite penalty then has the two-sided bound

$$\sum_j c_j [-\log(1 - e^{-U_j})] \leq \mathcal{P} \leq \sum_j c_j [-\log(1 - e^{-L_j})]. \quad (\text{GRB4})$$

This follows term by term from the decreasing function $x \mapsto -\log(1 - e^{-x})$, for $x > 0$. For the unshifted window retain the coefficients in RGI7 and their total $4r - 1$; for the shifted $q + 1$ rows below retain $1, 3, 4, \dots, 4, 3, 1$ and total $4q$. Inserting these endpoints into the exact action identity updates its finite enclosure. The original arithmetic V endpoints and the four AMT errors remain in place.

213.7 6. Penalty concentration near the upper endpoint

Let $\mathbf{P_far}(\mathbf{d})$ retain the original combined-penalty terms with $\mathbf{q-r} \geq \mathbf{d}$. For integers $2 \leq \mathbf{d} \leq \mathbf{q-r}_0$, (UCT36) and the same geometric sum give

$$\frac{P_{\text{far}}(\mathbf{d})}{P_{\text{total}}} \leq (4r_0 + 3) \frac{e^{2qa_1 + C_k - s_k}}{1 - e^{-s_k}} + \frac{4e^{2C_k} z^{\mathbf{d}}}{(1 - z)(1 - e^{-m_k})}. \quad (\text{UCT38})$$

The first term tends to zero exponentially since $1/3 > 2a_1$. Given $0 < \eta < 1$, once it is at most $\eta/2$, it suffices to choose

$$\mathbf{d} \geq \max \left\{ 2, \left\lceil \frac{2C_k + \log(8/[\eta(1 - z)(1 - e^{-m_k})])}{2c_1} \right\rceil \right\}, \quad (\text{UCT39})$$

with the stated finite $\mathbf{d} \leq \mathbf{q-r}_0$ guard. Then the final $\mathbf{d}$ rows retain at least $1 - \eta$ of the complete penalty. Since $C_k = 0_h(k + \log q)$, this strip has width $0_h(k + \log q + \log(1/\eta))$, not $0(q)$.

This is an upper bound on the required strip, not a claim that the last term alone is asymptotically dominant. The arithmetic comparison still has $0_h(k)$ uncertainty and does not control individual adjacent fluctuations finely enough for such a claim.

213.8 7. The next degree regime: a reference-profile fact, not an arithmetic extension

The same exact profile has a useful large- t description. Put $L=\log(4/s)$, $s=\kappa'$. The standard complete-elliptic expansions give

$$K = L + O(s^2 L), \quad E = 1 + \frac{1}{2}s^2(L - \frac{1}{2}) + O(s^4 L), \quad 1 + t = \frac{1 + O(s^2 L)}{sL}. \quad (\text{UCT40})$$

The exact simplification

$$\psi(t) = \operatorname{arctanh} s + (1 + t) \log(\kappa/E) \quad (\text{UCT41})$$

then yields

$$\psi(t) = \frac{s}{2} - \frac{s}{4L} + O(s^3 L), \quad \psi(t) \sim \frac{1}{2t \log t}, \quad u_t = 2/K \sim \frac{2}{\log t}. \quad (\text{UCT42})$$

These are statements about the explicit reference equilibrium function. They DO NOT extend UCB's arithmetic comparison to $t=t_k \rightarrow \infty$. The compact positive parameter rectangle used in its norm estimate degenerates in that limit. Furthermore, in the original $y=\text{qsqrt}(x)$ coordinate the inner equilibrium scale is $qu_t \sim 2q/\log t$. It approaches the actual packet radius R_k when $\log t$ is of order q/R_k . Exactly there, a homogeneous replacement of the actual root polynomial requires a new, root-sensitive estimate. Full **chi**, both half-lines, and the adjacent factors must remain; polynomial recurrence prefactors cannot be ignored either.

The actionable next analytic theorem is therefore uniform control in a growing-degree/root-sensitive regime, or a different proved same-class comparison. Improving an allocation that cancels from $J_{\text{action}}=B_{\text{ar}}+W_{\text{ar}}-P_{\text{total}}$ cannot lower this particular certificate. This is a limitation of a specified canonical-band strategy, not a no-go theorem for RH or the full tau construction.

Update of this growing-degree frontier by RGC (RGI11–12). The complete new argument RGC1–32 now supplies a degree-dependent comparison on the same original coefficient spaces, through the explicit maps RGI1–3. It controls both integration half-lines and the entire region containing the roots. Its integration cutoff is

$$Y_{q,r} = \frac{(2q+1)^2}{32\alpha(r+2q+1)}, \quad 2R_k \leq Y_{q,r}, \quad E_{q,r}^{\text{root}} \leq \frac{4096\alpha^2 q R_k^2 (r+2q+1)^2}{3(2q+1)^4} + 2\eta_{q,r} \quad (\text{RGI11})$$

when $\eta_{q,r} \leq 1/2$. This is a literal coefficient-norm error with an explicit finite guard; the physical coordinate remains y and the original equilibrium scale discussed above remains qu_t .

For a fixed packet write $q \asymp_h k^d$, with $d = 2$ at multiplicity one and $d = 3$ at any fixed multiplicity at least two. Fix $0 < \theta < d - 1$. Along $r = \lfloor qk^\theta \rfloor$, the proved reference lower signal and the two comparison errors satisfy

$$F_{q,r} \asymp_{h,\theta} \frac{k^{d-\theta}}{\log k}, \quad E_{q,r}^{\text{root}} = O_{h,\theta}(k^{2+2\theta-d}) + e^{-\Omega_{h,\theta}(q)}, \quad \Lambda_{k,q+r} = O_{h,\theta}(k + \log q). \quad (\text{RGI12})$$

These are obtained by substituting the exact q, R_k, r into RGC4,8,16,19. More precisely, $F_{q,r} \sim k^{2-\theta}/(4\theta \log k)$ at $m = 1$ and $F_{q,r} \sim (m-1)k^{3-\theta}/(4\theta \log k)$ for fixed $m \geq 2$. The errors divided by $F_{q,r}$ tend to zero when $\theta < d - 1$ and $3\theta < 2d - 2$. The radius guard also holds in that range. Consequently RGC27 proves the actual allowance lower bound for $0 < \theta < 2/3$ when $m = 1$ and $0 < \theta < 4/3$ for each fixed $m \geq 2$. The shrinking reference lower signal, rather than a substituted fixed-band equilibrium error, is what determines these ranges. The pure-power endpoint values of θ and the regime in which qu_t approaches R_k remain outside this proved dominance conclusion. The earlier fixed-band bounds are retained through RGI4–7. The changed endpoints and their complete original action are RGI8–10, so this continuation does not insert a growth rule into an unchanged action formula.

Logarithmic refinement at the critical powers (RGI13–14). The same estimates prove a further range approaching those pure-power endpoints, with no new arithmetic comparison. Retain a fixed complete packet and its exact $q = [1 + k(m-1)](k+1)^2$, $R_k = kR_h$, $R_h = \sqrt{\delta^2 + \gamma^2}$, along $k = 4l + 1 \rightarrow \infty$. Put $(d, A_m^{\deg}) = (2, 1)$ when $m = 1$ and $(d, A_m^{\deg}) = (3, m-1)$ for fixed $m \geq 2$. Fix any real $\zeta > 1/3$ and define the actual integer cutoff

$$\theta_* = \frac{2(d-1)}{3}, \quad r_k = \left\lfloor \frac{qk^{\theta_*}}{(\log k)^\zeta} \right\rfloor. \quad (\text{RGI13})$$

Then the original canonical allowances satisfy

$$\log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \geq (1 - o_{h,\zeta}(1)) \frac{A_m^{\deg}}{8\theta_*} k^{(d+2)/3} (\log k)^{\zeta-1}. \quad (\text{RGI14})$$

All the minima, measures and rows here are those of RGC20–26.

To prove the statement, the exact integer formula gives $q \sim A_m^{\deg} k^d$, and the discarded fractional part in r_k lies in $[0, 1)$. Since $k^{\theta_*}/(\log k)^\zeta \rightarrow \infty$, it follows that

$$\frac{q + r_k}{q} \sim \frac{k^{\theta_*}}{(\log k)^\zeta}, \quad \log \frac{8(q + r_k)}{q} = \theta_* \log k - \zeta \log \log k + \log 8 + o(1).$$

Substitution in the exact RGC16 signal, rather than an equilibrium estimate, therefore yields

$$F_{q,r_k} \sim \frac{A_m^{\deg}}{4\theta_*} k^{d-\theta_*} (\log k)^{\zeta-1}.$$

In the same literal real coordinate, RGC4 gives

$$\rho_{q,r_k} \sim \frac{8\alpha R_h}{A_m^{\deg}} k^{1+\theta_*-d} (\log k)^{-\zeta} \rightarrow 0.$$

Here $1 + \theta_* - d = -(d-1)/3 < 0$. Thus the radius guard holds eventually. The prefactor of $(3/8)^{2q+1}$ in η_{q,r_k} is bounded by a fixed power of k , so $\eta_{q,r_k} = e^{-\Omega_{h,\zeta}(q)}$ and its guard also holds eventually. Applying the complete RGC8 inequality now gives

$$E_{q,r_k}^{\text{root}} = O_{h,\zeta} \left(k^{2+2\theta_*-d} (\log k)^{-2\zeta} \right) + e^{-\Omega_{h,\zeta}(q)}.$$

The identity $2 + 2\theta_* - d = d - \theta_* = (d+2)/3$ proves

$$\frac{E_{q,r_k}^{\text{root}}}{F_{q,r_k}} = O_{h,\zeta} \left((\log k)^{1-3\zeta} \right) + o(1) \rightarrow 0.$$

Also $\log(q + r_k) = O_{h,\zeta}(\log k)$, so RGC19 and $d \geq 2$ give

$$\frac{\Lambda_{k,q+r_k}}{F_{q,r_k}} = O_{h,\zeta} \left(k^{(1-d)/3} (\log k)^{1-\zeta} \right) + o(1) \rightarrow 0.$$

Consequently $F_{q,r_k} \rightarrow \infty$ and $S_{k,q,r_k} = (1 - o_{h,\zeta}(1)) F_{q,r_k} > 0$ eventually. The exact first factor of RGC26 satisfies $\log b_{k,q,r_k}^{\text{low}} = -\Lambda_{k,q+r_k} + \log(q(q-1/2)) = o(F_{q,r_k})$. Taking logarithms in that finite, already proved minimum inequality gives

$$\log \min_N \epsilon_N \geq \frac{1}{2} \log b_{k,q,r_k}^{\text{low}} + \frac{1}{2} S_{k,q,r_k} + \log(1 - e^{-S_{k,q,r_k}}) = (1 - o_{h,\zeta}(1)) F_{q,r_k} / 2,$$

which proves RGI14 with its stated constant. Its right side divided by $\log k$ tends to infinity, while $\log(\delta q(k+1)/2) = O_h(\log k)$. The same original exterior benchmark is therefore exceeded by a ratio tending to infinity on these enlarged bands. The logarithmic condition $\zeta > 1/3$ is obtained from the displayed root-error-to-signal ratio. The equality $\zeta = 1/3$ and the uncorrected pure-power endpoints are not established by this deduction. The earlier whole-form intersections remain RGI4–7 wherever eligible; the changed endpoints and their weighted action are still RGI8–10 at exactly the integer cutoff RGI13. Thus this refinement propagates through the existing maps without assigning an intrinsic mixed allocation.

Propagation to the complete earlier band and its degree frontier (GRB5–8). The new moving-cutoff full-form and reference proofs establish, at the original fixed complete quartet and along $k = 4\ell + 1 \rightarrow \infty$, that

$$t_k = r_k/q \rightarrow \infty, \quad t_k \log t_k = o(q/k) \implies H_k := \frac{q}{t_k \log t_k} \gg k. \quad (\text{GRB5})$$

Here and throughout $q = [1 + k(m-1)](k+1)^2$, with the complete integer multiplicity m fixed. The root radius remains $R_k = k\sqrt{\delta^2 + \gamma^2}$. The result also propagates to the entire original, unshifted band:

$$\min_{0 \leq j \leq r_k} \log T_j^{\text{ar}} \sim H_k, \quad \log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \sim \frac{1}{2} H_k. \quad (\text{GRB6})$$

In particular this strengthens the earlier growing-band lower estimates RGI12–14 while preserving their finite formulas.

Here is a proof of the uniform assertion, including the small-degree end. For $0 \leq j \leq \lfloor q/100 \rfloor$, the proved original interval bound UCT8 and its full arithmetic comparison give $\log T_j^{\text{ar}} \geq q/3 - O_h(k + \log q)$. Its root guard $q \geq 2^{17} R_k$ holds eventually because $q/R_k \rightarrow \infty$. On any fixed compact interval $1/200 \leq j/q \leq C$, the actual moving parameters α, β and their supports range over compact subsets of the positive quadrant. This follows from their explicit formulas and the continuous EIQ inverse moment parametrization. Thus b has a positive lower bound, $m(v^2 - u^2) \rightarrow \infty$ uniformly, $\eta^{\text{mov}} = e^{-q \log 8 + O_C(\log q)}$, and $C_\chi = O_{h,C}(R_k^2/q)$. The finite moving reference bounds give

$$\log T_j^{\text{ar}} = 2q\psi(j/q) + O_{h,C}(k + \log q + R_k^2/q).$$

The profile is positive at every finite positive argument: the exact UCT derivative $\psi'(t) = \log(\kappa/E) < 0$ and its limit zero at infinity prove this. Its compact minimum is consequently positive. Since $R_k^2/q = O_h(1)$, all these compact-degree exponents exceed $c_C q$ eventually, for some $c_C > 0$.

For rigor in passing to the whole expanding set, choose any sequence of indices $0 \leq j_k \leq r_k$. Every subsequence on which j_k/q is bounded has a further subsequence covered by the preceding small or compact bound. There $\log T_{j_k}^{\text{ar}}/H_k \rightarrow \infty$. Every unbounded subsequence has a further subsequence with $j_k/q \rightarrow \infty$. Its ratio satisfies the same narrow condition because $u \log u$ increases for $u \geq 1$ and $j_k \leq r_k$. The newly proved arithmetic estimate then gives

$$\log T_{j_k}^{\text{ar}} = (1 + o(1)) \frac{q}{(j_k/q) \log(j_k/q)} \geq (1 - o(1)) H_k.$$

If a uniform lower bound $(1 - o(1)) H_k$ failed, choosing a violating index at each such k and applying this subsequence argument would contradict one of its two conclusions. At $j = r_k$ the new asymptotic supplies the matching upper bound. This proves the first part of GRB6.

AMT gives uniformly over this band $\log \beta_{q+j} \geq -\mathcal{L}_{k,q+r_k} + \log(q(q-1/2)) = -O_h(k + \log(q + r_k))$. The narrow condition implies $r_k/q = O_h(k^2)$ eventually, so this cost is $O_h(k) = o(H_k)$. The exact adjacent formula, with its exceptional $j = 0$ factor omitted, and the just proved uniform lower bound give $\log \epsilon_{q+j-1} \geq (1 - o(1)) H_k/2$ uniformly. Indeed the two logarithms $\log(1 - e^{-\log T_j})$ and, when present, $\log(1 - e^{-\log T_{j-1}})$ tend uniformly to zero. The endpoint $j = r_k$ again supplies the upper bound for the minimum. This proves the second part of GRB6. As $\log(\delta q(k+1)/2) = O_h(\log k) = o(H_k)$, the minimum divided by the unchanged exterior benchmark tends to infinity.

The original unshifted penalty also has its exact exponential rate:

$$\log \mathcal{P}_{q,r_k} \sim -H_k. \quad (\text{GRB7})$$

For its upper bound, all $4r_k - 1$ original coefficient units multiply terms at most $e^{-(1-o(1))H_k}/(1 - e^{-(1-o(1))H_k})$. For its lower bound retain its last term, whose coefficient is one: $-\log(1 - T_{r_k}^{-1}) \geq T_{r_k}^{-1} = e^{-(1+o(1))H_k}$. Since $\log(4r_k) = O_h(\log k) = o(H_k)$, taking logarithms proves GRB7. These are exactly the unshifted endpoints and coefficients from RGI8–10, not those of the separately derived shifted window.

For a concrete improved frontier put $(d, A_m^{\deg}) = (2, 1)$ at $m = 1$, and $(3, m - 1)$ at fixed $m \geq 2$. Fix real ζ and $0 < \theta < d - 1$, or fix $\theta = d - 1$ and $\zeta > 1$. For the actual integer cutoff $r_k = \lfloor qk^\theta / (\log k)^\zeta \rfloor$, GRB6 gives

$$\log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \sim \frac{A_m^{\deg}}{2\theta} k^{d-\theta} (\log k)^{\zeta-1}. \quad (\text{GRB8})$$

To check every exponent and floor, $q \sim A_m^{\deg} k^d$ and $r_k/q = k^\theta (\log k)^{-\zeta} + O(1/q)$, so $\log(r_k/q) = \theta \log k - \zeta \log \log k + o(1)$. The ratio of $t_k \log t_k$ to q/k is therefore asymptotic to $(\theta/A_m^{\deg}) k^{\theta-d+1} (\log k)^{1-\zeta}$. It tends to zero in each displayed domain, which proves applicability of GRB5. Substitution into $H_k/2$ gives GRB8. In particular the pure powers now extend to every $\theta < 1$ at $m = 1$ and every $\theta < 2$ at fixed $m \geq 2$, with the stated logarithmic approach to the new endpoints. The case $\theta = d - 1, \zeta = 1$ has a nonzero limiting ratio in GRB5; this calculation therefore gives its exact comparison to the admissibility condition but no claimed asymptotic there. Every eligible earlier finite estimate remains in GRB1–4.

An explicit arithmetic coefficient at the logarithmic endpoint (GRB11–13). The exact constants in GRA6 also permit a positive allowance estimate at a specified part of the critical scale $\zeta = 1$ in GRB8. This conclusion is an enclosure with its retained arithmetic coefficient. Put

$$\mathfrak{s}_h = \frac{\pi}{2} + \log \frac{M_{\pi/2}}{\vartheta_h}, \quad C_h^{\deg} = \frac{A_m^{\deg}}{2(d-1)\mathfrak{s}_h}, \quad 0 < c_{\deg} < C_h^{\deg}, \quad r_k = \left\lfloor \frac{c_{\deg} q k^{d-1}}{\log k} \right\rfloor. \quad (\text{GRB11})$$

The coefficient $c_{\deg}$ is fixed independently of k ; the original centre remains $c = k/2$. All source quantities here are already defined: $\vartheta_h = \int_{-1}^1 w_h(y) dy$ and $M_{\pi/2} = \int e^{\pi y/2} w_h(y) dy$. Evenness and positivity of w_h give $M_{\pi/2} = \int \cosh(\pi y/2) w_h(y) dy \geq \int w_h \geq \vartheta_h > 0$. The proved tilted tail bound makes the first integral finite. Thus $\mathfrak{s}_h \geq \pi/2 > 0$ and the endpoint coefficient $C_h^{\deg}$ is defined and positive, derived from the actual arithmetic source comparison.

Set $B = A_m^{\deg} / (c_{\deg}(d-1))$, so $B > 2\mathfrak{s}_h$. The original full-band minimum satisfies the explicit bounds

$$\frac{B}{2} - \mathfrak{s}_h \leq \liminf_{k \rightarrow \infty} \frac{1}{k} \log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \leq \limsup_{k \rightarrow \infty} \frac{1}{k} \log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_N \leq \frac{B}{2} + \mathfrak{s}_h. \quad (\text{GRB12})$$

In particular this minimum still exceeds the original exterior benchmark by a ratio tending to infinity.

We prove the bounds with the original error rather than extending the little- o assertion GRB5 to its boundary. Here $t_k = r_k/q \sim c_{\deg} k^{d-1} / \log k$ and $\log t_k \sim (d-1) \log k$, so $q/(kt_k \log t_k) \rightarrow B$. This sequence satisfies the broad full-form condition $R_k \log t_k / q \rightarrow 0$. The finite moving reference theorem and GRF22, with $R_k^2/q = O_h(1)$, yield uniformly on its polynomial degree horizon

$$C_\chi(k, j) = o(k), \quad \log(q + j) = o(k), \quad -\log(1 - \eta(k, j)) = o(k)$$

for the growing-ratio part of that horizon. To see the uniform meaning, its explicit root error is bounded by $O_h((1 + \log t_k)^2) = o(k)$; the logarithmic prefactors in its inner error are $O_h(\log k)$. On compact positive ratio intervals the same conclusion follows from the actual compact endpoint bounds used in the proof of GRB6. The exact expression GRA6 gives

$$L_{k,q+j}/k = \mathfrak{s}_h + o(1) \quad \text{uniformly for } 0 \leq j \leq r_k + q.$$

The degree-dependent term is $\log D_{q+j} = O_h(\log k)$ and all remaining terms of GRA6 except $k\mathfrak{s}_h$ are $O_h(\log k)$, proving this last display directly.

Using the same bounded-ratio/unbounded-ratio subsequence argument as in GRB6, the reference signal has $\min_{j \geq q/100, j \leq r_k} 2q\psi(j/q)/k \rightarrow B$; at $j = r_k$ it is $B + o(1)$ and for unbounded ratios below it the function $u \log u$ gives the lower estimate. For bounded ratios its quotient by k tends

to infinity. The original small-degree estimate also tends to infinity after division by k . Therefore the actual arithmetic transfer proves

$$\min_{0 \leq j \leq r_k} \log T_j^{\text{ar}}/k \geq B - \mathfrak{s}_h - o(1) > 0, \quad B - \mathfrak{s}_h - o(1) \leq \log T_{r_k}^{\text{ar}}/k \leq B + \mathfrak{s}_h + o(1).$$

AMT at adjacent degrees gives $-\mathfrak{s}_h - o(1) \leq \log \beta_{q+j}/k \leq \mathfrak{s}_h + o(1)$ uniformly. Both contraction logarithms tend uniformly to zero, since their exponents are at least a positive multiple of k ; the first row retains its single-factor exception. The lower part of the exact adjacent identity now gives the lower bound in GRB12. Its endpoint $j = r_k$, using the upper recurrence and upper scalar bounds, gives the upper bound for the minimum. The original benchmark has logarithm $O_h(\log k) = o(k)$, proving the stated comparison.

For the shifted band with $s = r_k$, $a = q + s - 1$ and $b = a + q$, the exact complete action admits a sharper coefficient enclosure than summing the separate row recurrence errors:

$$2(B - \mathfrak{s}_h) \leq \liminf_{k \rightarrow \infty} \frac{J_{k,s}^{\text{action}}}{qk} \leq \limsup_{k \rightarrow \infty} \frac{J_{k,s}^{\text{action}}}{qk} \leq 2(B + \mathfrak{s}_h). \quad (\text{GRB13})$$

Indeed $|(s+j)/q - t_k| \leq 1$ for $-1 \leq j \leq q$, so the reference signal divided by k tends uniformly to B . The scalar arithmetic exponents therefore lie uniformly between $k(B - \mathfrak{s}_h - o(1))$ and $k(B + \mathfrak{s}_h + o(1))$. Their weighted sum has total weight $2q$, giving the two main endpoints in GRB13. The full penalty is at most $4qe^{-k(B-\mathfrak{s}_h-o(1))}/(1 - e^{-k(B-\mathfrak{s}_h-o(1))})$ and hence is $o(qk)$. Finally the exact Gamma recurrence window is $O(q \log k)$, and GRA31 charges only its four endpoint errors, each $O_h(k)$. Thus the full arithmetic recurrence window is $o(qk)$. Inserting these three estimates in GRB9 proves GRB13. This uses the complete correlated action, with no change to the intrinsic mixed allocation or the complementary returns.

213.9 8. Execution and formal scope

Two remote Wolfram evaluations succeeded. The first returned an exactly zero symbolic derivative residual, the exact rational small-rank margin, nine exact shifted-Legendre norm identities ($\mathbf{r}=0, \dots, 8$), and separately labelled high-precision diagnostic constants. The second used rational/integer arithmetic to prove eight inequalities: two signs bracketing the unique elliptic endpoint ratio, both stated endpoints for `a_1`, both for `c_1`, and both for `C_geo`. It used 4096 elliptic-series terms, directed rounding at 10^{80} , explicit geometric tails, alternating Machin arctangent bounds for π , and 20-term positive logarithm series with remainders. Its algorithm is preserved in `verify_ucb_constants.wl`.

Those are fresh checks, not a replay of the supplied 134 algebraic checks or nine certificate checks. They concern universal constants and finite identities, not an actual hypothetical zero packet, its period, or its arithmetic Gram integrals. No new Lean execution, GitHub Actions success, PDF build, linked-archive verification, or full-library audit is asserted. The existing SplitZero support and boundary interfaces remain unchanged; a vanished coefficient observation does not become external absence, and neither an integration-band restriction nor an arcsine comparison replaces the arithmetic quotient.

213.10 References

National Institute of Standards and Technology. (n.d.). *Digital Library of Mathematical Functions*, Sections 19.4 and 19.12. <https://dlmf.nist.gov/19.4> ; <https://dlmf.nist.gov/19.12> . Derivative identities and complete-elliptic expansions only; the monic-norm and penalty arguments above are displayed explicitly.

Incoming user source: `Pastedmarkdown(20260916-165456).md`, equations (1)–(16). Original arithmetic comparisons: AMT/TW and the paired-root comparator retained there. Repository baseline: `1efd53337561d8f67cf0ac1d119acdaa222644c3`; preceding standalone bound: `PR37 implementation 6a34ac7d911a5be3a2b030008e955ff35f8e0dad`.

213.11 An explicit constant for the complete compact-rectangle norm bound

Retain the original reference norm, the literal coordinate $y = q\sqrt{x}$, and the complete parameter rectangle used in UCT. The symbols below are exactly its positive compact-rectangle extrema: W_- and W_+ are the minimum and maximum support lengths; M_* bounds the equilibrium density; J_* bounds $\int_0^\infty e^{-D(x)} dx$; and $r_-, r_+ > 0$ are the original Gamma remainder bounds. Set

$$\begin{aligned} c_- &= r_- \pi W_- / 4, & c_+ &= r_+ J_* e^{4M_*}, \\ C_{\text{norm}} &= \max\{ |\log c_-|, |\log c_+| + 4 \log(1 + W_+) \}. \end{aligned} \quad (\text{CCN1})$$

This displays the constant specified in the captured comment, with the same extrema and signs. The lower arcsine-Jensen and upper quantile calculations in UCT give, for every integer $1 \leq j \leq q$ in that rectangle and its stated $q \geq 1000$ domain,

$$c_- \leq e^{j\lambda} h_{j,q}(a, b) \leq c_+ (1 + jW_+)^4. \quad (\text{CCN2})$$

Indeed the lower capacity is $W/4 \geq W_-/4$, while the upper calculation uses the same density, remainder and full positive-half-line integral bounds, so substituting their extrema proves both inequalities without changing the measure. Put $X = \log h_{j,q} + j\lambda$. Taking logarithms in CCN2 and using $1 + jW_+ \leq (q+1)(1 + W_+)$ gives

$$\log c_- \leq X \leq \log c_+ + 4 \log(q+1) + 4 \log(1 + W_+), \quad |X| \leq C_{\text{norm}} + 4 \log(q+1). \quad (\text{CCN3})$$

For $X < 0$, the lower bound implies $-X \leq -\log c_- \leq |\log c_-|$; for $X \geq 0$, the upper bound is at most the second member of the maximum in CCN1 plus $4 \log(q+1)$. This proves the absolute-value claim, including either sign of the logarithms of the finite constants.

With the unchanged Robin derivative bounds L_a, L_b , bound Λ_* for $|\lambda|$, and the exact quarter-shift substitution already calculated in UCT, the resulting complete error is therefore

$$\begin{aligned} C_{\text{par}} &= \frac{401}{4} L_a + 50\pi L_b + \frac{\Lambda_*}{2}, \\ \hat{E}_0(q) &= C_{\text{norm}} + C_{\text{par}} + \frac{1}{2} \log 2 + \log(2\pi) + \frac{1}{48} + 4 \log(q+1). \end{aligned} \quad (\text{CCN4})$$

The parameter replacement contributes C_{par} , the factor $\frac{1}{2} \log(1 + r/q)$ contributes at most $\frac{1}{2} \log 2$, and the positive Stirling remainder is below $1/48$ on this domain. Thus CCN4 supplies the finite constant used by this complete local quantile proof. The distinct constant denoted C_{EIQ} in the unavailable linked source has not been assigned the value CCN1.

214 Original shared-flag projector threshold and joint receiving cancellation

Source provenance and notation. The following complete comment was captured from PR 36 on 16 September 2026. Its reference to an unreviewed incoming source records its historical status when written. Integration status is supplied by the separately retained current source reviews. The comment's scalar L_e is the common return $\mathcal{L}_e$ of CIT/CPQ/SFB, its P_e^{rest} is $\mathcal{P}_e^{\text{rest}}$, its V is $\mathcal{V}$, and its H is the original coefficient space $\mathcal{H}$. Its T_{j,s_1} denotes the inherited total $\mathcal{T}_{j,s_1}$; its U_n denotes SFB's $\mathcal{U}_n$. These are explicit notational identifications on the same fixed flags and original forms. The comment's prose and scope statements are retained below.

214.1 2026-09-16: incoming order transport, shared flags, and a leading-scale next calculation

Interactive coordination update for the user's request to advance the programme. Main was read at `1efd53337561d8f67cf0ac1d119acdaa222644c3`; the current primitive-band handoff still supplies the original source/target and action-budget requirements. No existing branch, source file, workflow, or merge is changed by this comment.

214.1.1 Incoming results and revised assignment

The user supplied three Markdown continuations: `Pasted markdown(20260916-011238).md` (QOT), `Pasted markdown (2)(7).md` (shared long flag, SLF), and `Pasted markdown (3).md` (EOR). These are supplied mathematical texts, not independently replayed full TeX/ZIP certificates in this review. Their references to earlier sandbox downloads are not repository paths.

QOT supplies full-form source-order transport on the same original coefficient and quotient fibres. EOR refines it to

$$-C_+ \leq \liminf \frac{S_X(k;k) - S_X(k;1)}{q_k} \leq \limsup \frac{S_X(k;k) - S_X(k;1)}{q_k} \leq C_-, \quad (\text{PTH1})$$

with $C_+ \in (8.881817865596, 8.881817865597)$, $C_- \in (5.773751910973, 5.773751910974)$, and the common paired loss comparison bounded at q-scale by $C_{\text{com}} \in (29.311139553140, 29.311139553141)$. The original low primitive source/image correction is $\mathcal{O}(q)$. These are differences, not absolute allocation values. Subject to their complete proof review and original eventual domains, the leading kq calculation should therefore be done once at order 1 and transported to the paired $(k, k+4)$ family. The old source-order reservation in this restart document should become review/integration, not an assignment to reproduce the same comparison.

SLF represents the product of the two original contractions as one projection determinant and gives $|L_e| \leq 4T_{k+4,s_1}$, rather than twice that allowance. It retains all eight signed four-row Gram increments, their full preceding lower minimum and denominator. It does not yet improve an unconditional numerical interval for an allocation. EOR's uniform polynomial localization, not the last decimal of its universal constant, is the proof-critical analytic review item. Keep the moving-parameter compact domain, both parity blocks, both tails and all finite guards. A parity decomposition of the even reference measure is not a parity decomposition of the arithmetic flags.

214.1.2 Cancellation-first receiver check

Use the supplied SLF notation $I \subset B \subset S$, $K_i = I \cap J$, $W_i = I + J$, $W = B + J$, $K_c = S \cap W$, $T = S + J$, and its return $V(Y/X)$. Its definitions give the exact identity

$$L_e = V(T/W) + V(W_i/J) - V(S/K_c) - V(I/K_i). \quad (\text{PTH2})$$

Combining this with the supplied flag factorization and $V(H/J) = 4S_{K,j,s_1}$ yields

$$4S_{K,j,s_1} = L_e + V(H/T) + V(W/W_i) + V(S/K_c) + V(I/K_i). \quad (\text{PTH3})$$

Since $V(W/W_i) = V_b$ and CIT gives $L_e = P_e^{\text{rest}} - V_b$, this becomes

$$4S_{K,j,s_1} = P_e^{\text{rest}} + V(H/T) + V(S/K_c) + V(I/K_i). \quad (\text{PTH4})$$

Proof: telescope determinant lines along $J \subset W_i \subset W \subset T \subset H$, substitute the two contraction ratios, then the CIT identity. Fixed coordinate determinants cancel only under the original four-endpoint return. This does not remove the boundary dependence of the remaining subspaces or their metrics; it identifies one exact cancellation. An improved bound on V_b alone is therefore not automatically an improvement of this combined receiver.

The same audit is required at the already-written full action identity $J_{\text{action}} = \Psi + C_{\text{budget}}$, $\Psi = F_K^{(k)} + H_k + \delta_k^\sigma$: rewrite both terms in the same dependent variables before estimating. Do not improve Psi while keeping a correlated budget artificially fixed. The remaining reference boundary, monic window, contraction costs, leakage, mixed pairings and original nonzero arithmetic class stay attached.

214.1.3 Additional finite reduction for formalization/review

This is a derived finite-dimensional result, not an evaluated arithmetic asymptotic or a new Lean certificate.

For each original row let D_0 be its positive preceding native Gram and retain exactly

$$r_n = \frac{\rho_n D_0^{-1} \rho_n^*}{1 + \zeta_n}, \quad (\text{PTH5})$$

$$\mathcal{D}_0 = \text{diag}(D_0, 4), U_n = \text{diag}\left(\frac{\rho_n}{\sqrt{1+\zeta_n}}, 4\right), \text{ and}$$

$$K_{X,n} = U_n X (X^* \mathcal{D}_0 X)^{-1} X^* U_n^* \quad (\text{PTH6})$$

for the eight actual fixed flag subspaces. Empty subspaces give the zero matrix. Set the positive sign list to (T, K_c, W_i, K_i) and the negative list to (W, S, J, I) .

Then $0 \leq K_{X,n} \leq r_n I_4$. Indeed put $Y = U_n \mathcal{D}_0^{-1/2}$ and let Π_X be the orthogonal projector onto $\mathcal{D}_0^{1/2} \text{im } X$; then $K_X = Y \Pi_X Y^*$ and $Y Y^* = r_n I_4$. This is an isometric computation of the existing metric, not a replacement metric.

Let $\lambda_{X,a,n}$ be the four nonnegative eigenvalues of $K_{X,n}$, including zeros. Define

$$P_{k,e} = \sum_n \theta_n \left[\sum_{X \text{ positive}} \sum_{a=1}^4 \log \max(1, \lambda_{X,a,n}) - \sum_{X \text{ negative}} \sum_{a=1}^4 \log \max(1, \lambda_{X,a,n}) \right]. \quad (\text{PTH7})$$

The original row identity and weights imply the unconditional finite bound

$$|L_e - P_{k,e}| \leq 32q_j \log 2, \quad (\text{PTH8})$$

where $j = k + 4$.

Complete proof: for every $x \geq 0$, $0 \leq \log(1+x) - \log \max(1, x) \leq \log 2$, because $1 \leq (1+x)/\max(1, x) \leq 2$. Spectral factorization gives each four-by-four log determinant as the sum of four $\log(1+\lambda)$ terms. At each row the total scalar error on each sign side lies in $[0, 16 \log 2]$; their difference is in $[-16 \log 2, 16 \log 2]$. The exact nonnegative row weights sum to $2q_j$, proving the bound. Thus the normalized error is at most $32(q_j/q_k) \log 2/k$, which tends to zero without a singular-value gap or a per-row asymptotic assumption.

For $r_n > 0$, write the normalized eigenvalues as $\mu = \lambda/r_n \in [0, 1]$. If $\mu > 0$, the retained term is exactly $k(\alpha_n - b_{X,a,n})_+$, where $\alpha_n = \log(r_n)/k$ and $b = -\log(\mu)/k$; zero eigenvalues contribute zero without being removed from the original kernel diagram. This identifies the leading task as the signed balance of original row-growth and projection-loss exponents. Equal ranks do not determine this balance. The scalar log-sum-exp estimate is standard; its application here preserves the actual eight flags and gives an explicit subleading error, not a novelty claim for the scalar inequality.

Combining with the supplied EOR transport, the order-one profile approximates the original paired common loss with q-scale limsup error at most $C_{\text{com}} + 32 \log 2$. This remains a reduction, not a narrower numerical allocation bound: the original compressed spectra still have to be estimated.

214.1.4 Proposed complementary work

1. Analytic owner: bound the signed order-one profile or the equivalent exact eight determinant increments using the actual factorial/Gamma recurrence and corrected rows. Work on block sums/intervals rather than require a unique limit or uniform convergence of every row. Retain exponentially small positive compressed eigenvalues; rank counts and absolute source-band determinants are not relative conditioning estimates. Account separately for $V(H/T)$ and the proper source quotients when using the displayed receiver.
2. Receiving audit: compose the current complete action proof in shared variables and identify the precise non-cancelling interval needed to contradict the original exterior benchmark. A common-loss sign is not that contradiction.
3. Formalization/integration: check the nested quotient determinant identity, rank-four compression, scalar threshold bound and signed weighted error using the existing original-metric interfaces. Independently review and integrate complete QOT/SLF/EOR source bodies at their receiving sites; keep analytic localization separate from finite Lean certificates. Do not replace the current owner handoff with a competing general programme.

No local numerical or Lean execution succeeded in this interactive review: execution services failed before running the requested code. The full oversized cumulative source could not be fetched, so the review does not assert a fresh complete audit of its action proof. The displayed finite deductions have written proofs above and require independent review. Nothing here claims an actual off-critical zero, purity, a completed allocation coefficient, or RH.

215 The complete original paired-root comparison

215.1 Original packet and coordinate for the retained paired-root provider

Fix $0 < \delta < 1/2$, $\gamma > 2$ and a positive integer m . For the arithmetic application, m is the complete order of the original actual quartet in 2ξ ; no existence assertion for such a quartet is made here. Retain

$$k = 4l + 1, \quad l \geq 1, \quad c = k/2, \quad e = \ell = 1 + k(m - 1), \quad q = e(k + 1)^2, \quad (\text{PCRC1})$$

$$\lambda_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{ab})^e, \quad Q(y) = \chi(c + iy). \quad (\text{PCRC2})$$

The reference measure and its full mass remain

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \int_{\mathbb{R}} d\sigma = \sqrt{2\pi}, \quad \|f\|_{\sigma}^2 = \int_{\mathbb{R}} |f(y)|^2 d\sigma(y). \quad (\text{PCRC3})$$

The coefficient substitution $P(S) \mapsto P(c + iy)$ has inverse $f(y) \mapsto f((S - c)/i)$ and preserves every polynomial degree. For a monic degree- r polynomial its monic y -coordinate is $i^{-r}P(c + iy)$; the retained factor has absolute value one. The following exact source section proves a comparison of these full polynomial norms, including the inner interval. It does not replace the arithmetic remainder map or its relation space by a different quotient.

215.2 Audit of the complete paired-root norm comparison

The incoming PRC1–31 is a working candidate. The argument below supplies the needed full proof and fixes its notation separately from D_h . Write

$$\begin{aligned} R &= k\sqrt{\delta^2 + \gamma^2}, \quad \varepsilon = 2^{-16}, \quad T = \varepsilon q, \quad q \geq 2R/\varepsilon, \\ U_2 &= k(k+2)/3, \quad U_4 = k(k+2)(3k^2 + 6k - 4)/15, \\ D^{(2)} &= qU_2(\gamma^2 - \delta^2), \quad M_4 = q[(\delta^4 + \gamma^4)U_4 + 2\delta^2\gamma^2U_2^2], \\ C_\sigma^{\text{in}} &= \sqrt{2\pi} \Gamma(3/4)^2/\pi, \quad \eta_q = C_\sigma^{\text{in}} \frac{(2q+1)^2}{q} 16^{-q}, \\ A_k^{\text{cmp}} &= \frac{D^{(2)}}{\varepsilon^2 q^2} + \frac{M_4}{2\varepsilon^4 q^4 (1 - R^2/(\varepsilon^2 q^2))}, \quad C_k^{\text{cmp}} = A_k^{\text{cmp}} - \log(1 - \eta_q). \end{aligned} \tag{PCRP1}$$

These are distinct from DMR's density multiplier A_k and HC3's D_h .

Lemma 215.1 (Complete PRC form comparison). *For every complex polynomial p of degree at most $2q$,*

$$e^{-C_k^{\text{cmp}}} \|y^q p\|_\sigma^2 \leq \|Qp\|_\sigma^2 \leq e^{C_k^{\text{cmp}}} \|y^q p\|_\sigma^2. \tag{PCRP2}$$

For a fixed packet, $C_k^{\text{cmp}} = O_{\delta,\gamma}(e^{-1})$. All sufficiently large orders satisfy the cutoff; $k \geq 2\sqrt{\delta^2 + \gamma^2}/\varepsilon$ is sufficient.

Proof. The y -plane roots are $z_{ab} = (2b - k)\gamma - i(2a - k)\delta$. Their radius is R , and $i^q = 1$. Pairing their two signs gives

$$Q(y) = \prod_{a,b=1,3,\dots,k} [(y - b\gamma)^2 + a^2\delta^2][(y + b\gamma)^2 + a^2\delta^2]^e > 0. \tag{PCRP3}$$

The opposite and conjugate pairings cancel odd power sums and make even power sums real. The finite sums of the second and fourth centered powers give precisely $D^{(2)}$ and M_4 in (PCRP1). Consequently, for $|y| > R$,

$$\log \frac{Q(y)^2}{y^{2q}} = -\frac{D^{(2)}}{y^2} + \mathcal{E}(y), \quad |\mathcal{E}(y)| \leq \frac{M_4}{2y^4(1 - R^2/y^2)}. \tag{PCRP4}$$

Indeed expand $\log(1 - z/y)$ absolutely, pair the roots, and use $\sum e|z|^{2j} \leq M_4 R^{2j-4}$ and $1/j \leq 1/2$ for $j \geq 2$. Positivity of Q/y^q identifies the branch with the real logarithm. Thus on $|y| > T$, the norm densities are between $e^{-A_k^{\text{cmp}}}$ and $e^{A_k^{\text{cmp}}}$ times $y^{2q} d\sigma$.

It remains to control the whole inner interval uniformly in p . Gamma duplication and reflection, followed by $|\Gamma(3/4 + iy/2)| \leq \Gamma(3/4)$, give

$$\sigma(y) \geq \frac{\pi}{\Gamma(3/4)^2} e^{-\pi|y|}, \quad \int d\sigma = \sqrt{2\pi}. \tag{PCRP5}$$

The mass follows also by Plancherel in the Gamma integral with $t = e^x$ and $y = 2v$. The Legendre basis on $[q, 2q]$ is $\sqrt{(2j+1)/q} P_j(2y/q - 3)$. Rodrigues' formula gives its norm; its three-term recurrence gives $|P_j(t)| \leq 9^j$ for $|t| \leq 4$ by induction. Hence, for $\deg p \leq n \leq 2q$,

$$\sup_{|y| \leq T} |p(y)|^2 \leq \frac{(n+1)^2}{q} 9^{2n} \int_q^{2q} |p(y)|^2 dy. \tag{PCRP6}$$

On that comparison interval the full monomial norm is at least $q^{2q} \pi \Gamma(3/4)^{-2} e^{-2\pi q} \int_q^{2q} |p|^2$. On the inner interval $|Q(y)| \leq (|y| + R)^q \leq (2T)^q$ and $|y|^q \leq (2T)^q$. Each inner integral is therefore at most

$$C_\sigma^{\text{in}} \frac{(n+1)^2}{q} (2\varepsilon)^{2q} e^{2\pi q} 9^{2n} \|y^q p\|_\sigma^2 \leq \eta_q \|y^q p\|_\sigma^2. \tag{PCRP7}$$

For the last inequality, $\pi < 4$, $e < 3$, and $2^{-30}3^{16} < 1/16$ suffice. The Gamma integral gives $\Gamma(3/4) < 7/3$, so $C_\sigma^{\text{in}} < 6$ and $\eta_q \leq 54q16^{-q} < 1/2$ for $q \geq 2$. Combining the two inner estimates with the exterior comparison gives lower factor $e^{-A_k^{\text{cmp}}}(1 - \eta_q)$ and upper factor $e^{A_k^{\text{cmp}}} + \eta_q \leq e^{A_k^{\text{cmp}}}/(1 - \eta_q)$. This proves (PCRP2) without assuming where a polynomial concentrates its mass.

Finally $M_4 \leq qR^4$, the denominator guard is at least $3/4$, and

$$C_k^{\text{cmp}} \leq \frac{\gamma^2 - \delta^2}{3\varepsilon^2 e} (1 - (k+1)^{-2}) + \frac{2k^4(\delta^2 + \gamma^2)^2}{3\varepsilon^4 e^3 (k+1)^6} + 108q16^{-q}. \quad (\text{PCRP8})$$

This proves the stated finite constant and rate. $\square$

Growing-degree continuation on the original coefficient spaces (RGI1–3). Retain the complete packet of RGC1 and let $\mathcal{P}_d$ denote complex polynomials of degree at most d . For each integer $j \geq 0$ the precise coordinate map and inverse are

$$U_j : \mathcal{P}_j(S) \longrightarrow \mathcal{P}_j(y), \quad (U_j P)(y) = i^{-j} P(c + iy), \quad (U_j^{-1} p)(S) = i^j p((S - c)/i). \quad (\text{RGI1})$$

Both maps are complex linear and inverse by direct substitution. Their leading coefficients in degree j are unchanged, so they biject the original degree- j monic affine spaces. In a lower degree the leading phase is retained; it is not silently replaced by the degree- j phase. With $Q(y) = i^{-q} \chi(c + iy)$ one has the exact commuting multiplication square

$$U_{q+j} \circ M_\chi = M_Q \circ U_j, \quad M_\chi P = \chi P, \quad M_Q p = Qp. \quad (\text{RGI2})$$

Indeed both sides evaluated at y equal $i^{-(q+j)} \chi(c + iy) P(c + iy)$. The original source norm in the S coordinate is the pullback of its μ_k norm by $S = c + iy$. Hence the scalar phases in RGI1–2 have modulus one and these are isometries for the corresponding pullback norms. No Euclidean norm of the coefficient vector is asserted to be preserved by translation. The observation map still has $\ker J_{q+j} = \chi \mathcal{P}_j$, and its quotient, Taylor unit and source realization are precisely RGC20.

The two reference metrics compared by RGC6 live on the same domain $\mathcal{P}_j(y)$: they are the pullbacks of the original σ norm by M_Q and by M_{y^q} . The connecting map between these metric presentations is the identity on $\mathcal{P}_j(y)$, with the explicit norm constants in RGC6–8. To spell out its behaviour on an unchanged observation fibre, let $C : \mathcal{P}_j(y) \rightarrow E$ be any specified complex linear map and $b \in \text{im } C$. Put

$$\mathfrak{q}_Q(b) = \min_{Cp=b} H_Q(p), \quad \mathfrak{q}_0(b) = \min_{Cp=b} H_0(p). \quad (\text{RGI3})$$

The forms are positive definite on this finite-dimensional coefficient space, so each nonempty affine fibre has a unique norm minimizer. Apply the two RGC6 inequalities to every point of that very same fibre and then take the two minima. The same multiplicative constants bound $\mathfrak{q}_Q/\mathfrak{q}_0$ whenever $b \neq 0$, and the undivided inequalities hold also at $b = 0$. In original coordinates the map on the fibre is U_j and the observation is $C \circ U_j$, so the fibre square commutes exactly. This proves restriction and quotient transport without changing a kernel or declaring any mixed matrix entry zero. The arithmetic monic receiver is obtained from the particular monic affine fibres by RGC18–24; this argument assigns no leading coefficient to the separate intrinsic mixed allocations.

216 Complete small-rank mechanism in the original Gamma reference

Exact source and typed context. The following section is the complete small-rank part of the supplied UCB response. It uses the original reference measure and the original integers $q = [1 + k(m - 1)](k + 1)^2$ and $r \geq 0$. The measure $d\sigma$ displayed below is exactly the measure of AMT4–5; the identification on its polynomial space is the identity map $\mathbb{C}[y] \rightarrow \mathbb{C}[y]$, $P \mapsto P$, with inverse itself. In particular its complete norm is unchanged. The retained AMT4–5 constant and density inequality are

$$c_\Gamma = \sqrt{2}e^{-7/6}, \quad \frac{d\sigma}{dy}(y) \geq c_\Gamma e^{-\pi|y|/2}(1 + |y|)^{-1/2} \quad (y \in \mathbb{R}). \quad (\text{UCBS0})$$

The complete proof below concerns these reference monic minima. Arithmetic transfer keeps the original complete paired-root comparison and the same full-polynomial AMT comparison; no arithmetic quotient is replaced by the reference monomial relation. Every source paragraph in this section is retained. The final prefactor calculation is displayed explicitly after the preserved source part.

Complete preserved source section begins.

216.1 2. A new monic-norm bound controls all small ranks uniformly

Keep the existing fixed Gamma reference

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \gamma_n = \sqrt{2\pi} \frac{(2n)!}{4^n}, \quad (\text{UCBS1})$$

and its homogeneous relation extremum

$$\nu_r^0 = \min_{\substack{P \text{ monic} \\ \deg P=r}} \int_{\mathbb{R}} y^{2q} |P(y)|^2 d\sigma(y). \quad (\text{UCBS2})$$

This remains a coefficient-norm comparator. It is not a replacement arithmetic quotient.

Set

$$Y = \frac{4q}{\pi}, \quad d(\eta) = -\log(1 - \eta) - \eta. \quad (\text{UCBS3})$$

For every $0 < \eta < 1$, $q \geq 2$, and $r \geq 0$, the new finite inequality is

$$\boxed{\nu_r^0 \geq \frac{c_\Gamma Y^{2q} e^{-2q-2qd(\eta)}}{\sqrt{1 + Y(1 + \eta)}} \frac{(2\eta Y)^{2r+1}}{(2r + 1) \binom{2r}{r}^2}.} \quad (\text{UCBS4})$$

The interval calculation uses the Legendre Rodrigues identity and its norm as recorded by Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21, eq. (18.5.5), Table 18.5.1, and the Legendre rows of Table 18.3.1]. For an interval $[a, a + H]$, $H > 0$, the exact monic transport is $p(y) \mapsto H^{-r} p(a + Hx)$, with inverse $u(x) \mapsto H^r u((y - a)/H)$, with the exact norm identity

$$\int_a^{a+H} |p(y)|^2 dy = H^{2r+1} \int_0^1 |H^{-r} p(a + Hx)|^2 dx.$$

The shifted Legendre polynomial on $[0, 1]$ has leading coefficient $\binom{2r}{r}$ and squared norm $1/(2r + 1)$. Subtracting its monic version leaves a lower-degree polynomial orthogonal to it, also for complex coefficients. This proves precisely the interval minimum in UCBS7, with $a = Y(1 - \eta)$ and $H = 2\eta Y$.

Here is its complete mechanism. Restrict the positive reference integral to

$$[Y(1 - \eta), Y(1 + \eta)]. \quad (\text{UCBS5})$$

The minimum of $2q \log y - (\pi/2)y$ on that interval is

$$2q \log Y - 2q - 2qd(\eta). \quad (\text{UCBS6})$$

On any interval of length H , every complex monic polynomial of degree r satisfies

$$\int |P(y)|^2 dy \geq \frac{H^{2r+1}}{(2r+1) \binom{2r}{r}^2}. \quad (\text{UCBS7})$$

This is the exact shifted-Legendre monic minimum. Rodrigues' formula gives its leading coefficient $\binom{2r}{r}$, squared norm $1/(2r+1)$ on $[0, 1]$, and orthogonality to all lower degrees. Translation and dilation supply the displayed interval formula. Applying the original lower Gamma density bound proves (UCBS4), including its entire prefactor.

At $\eta = 1/10$, elementary factorial and binomial bounds simplify (UCBS4) to

$$\boxed{\log \frac{\nu_r^0}{\gamma_{q+r}} \geq \frac{q}{3} - \frac{3}{2} \log q + \log c_0, \quad 0 \leq r \leq q/100,} \quad (\text{UCBS8})$$

where

$$c_0 = \frac{c_\Gamma}{60e\sqrt{6\pi}}. \quad (\text{UCBS9})$$

The coefficient $1/3$ has a rational margin. Writing $t = r/q$, the exponential part of the lower bound, after division by γ_{q+r} , is

$$2q [(1+t) \log(4/\pi) - (1+t) \log(1+t) + t - t \log 20 - d(1/10)]. \quad (\text{UCBS10})$$

Use

$$\log(4/\pi) > \frac{3}{14}, \quad d(1/10) \leq \frac{1}{180}, \quad \log 20 < 3, \quad (1+t) \log(1+t) - t \leq t^2. \quad (\text{UCBS11})$$

For $t \leq 1/100$, its coefficient of q is at least

$$\frac{3}{7} - \frac{1}{90} - \frac{3}{50} - \frac{1}{5000} = \frac{1}{3} + \frac{7537}{315000}. \quad (\text{UCBS12})$$

The remaining prefactor is bounded below by $c_0 q^{-3/2}$.

This includes $r = 0$ and $r = 1$, and remains uniform for every sequence $r = o(q)$. No equilibrium theorem at a degenerating parameter was used.

Explicit completion of the remaining prefactor. Put $n = q + r$ and keep $Y = 4q/\pi$ and $\eta = 1/10$. For every positive integer s , concavity of $\log x$ gives

$$\log(s!) - \frac{1}{2} \log s \leq \int_1^s \log x dx = s \log s - s + 1, \quad s! \leq e\sqrt{s}(s/e)^s. \quad (\text{UCBS13})$$

Indeed, summing the trapezoidal lower bounds for the integral on $[j, j+1]$, $1 \leq j < s$, gives the first inequality, including $s = 1$. The binomial theorem gives $\binom{2r}{r} \leq 4^r$, also for $r = 0$. Consequently

$$\gamma_n \leq 2e\sqrt{\pi n} n^{2n} e^{-2n}. \quad (\text{UCBS14})$$

Dividing the entire right-hand side of (UCBS4) by this upper bound, with $t = r/q$, gives exactly the following lower expression:

$$\frac{c_\Gamma \eta Y}{e\sqrt{\pi n} \sqrt{1+Y(1+\eta)}(2r+1)} \times \exp(2q [(1+t) \log(4/\pi) - (1+t) \log(1+t) + t - t \log 20 - d(1/10)]). \quad (\text{UCBS15})$$

No factor from the interval length or the original Gamma mass has been removed. For $q \geq 2$ and $0 \leq r \leq q/100$, $n \leq 2q$, $2r + 1 \leq q$, $\eta Y \geq q/10$ and $1 + Y(1 + \eta) \leq 3q$, using $3 < \pi < 4$. Thus its full prefactor is at least

$$\frac{c_\Gamma}{10e\sqrt{6\pi}q} \geq \frac{c_\Gamma}{60e\sqrt{6\pi}}q^{-3/2} = c_0q^{-3/2}. \quad (\text{UCBS16})$$

The exact rational margin in the preserved source then proves its stated lower bound for every integer r in the whole small-rank band, including $r = 0$ and $r = 1$ whenever those indices lie in the band.

217 A root-sensitive full-form comparison at growing polynomial degree

Scope and provenance. This is a new written continuation of the supplied 680-page edition, not a claim that the Riemann hypothesis has been proved or disproved. It addresses the growing-degree comparison requested in the current continuation prompt. The original nineteen files are preserved byte-for-byte in the accompanying archive. Its source map and historical verification reports remain inherited records. The new estimates below have complete proofs and separately labelled finite checks, but no new Lean certificate or independent second-reader acceptance is claimed. The fixed-proportion results of PR37 are retained; none is extrapolated by changing a bounded equilibrium parameter into a moving one.

The principal advance is a full-coefficient comparison between multiplication by the original root polynomial and multiplication by the monomial of the same degree. Its error is explicit in the relation degree. The comparison includes the entire inner integration region, carries into the original monic minima, and gives a further, genuinely growing range for the canonical allowance. It does not evaluate the separate mixed allocations or their common flag profile.

217.1 Original arithmetic input and a finite comparison theorem

Retain a fixed hypothetical complete-order quartet and its original source:

$$\begin{aligned} h(z) &= \prod_{\rho \in \{1/2 \pm \delta \pm i\gamma\}} (z - \rho)^m, \quad 0 < \delta < 1/2, \quad \gamma > 2, \quad m \in \mathbb{N}_{>0}, \\ k &= 4l + 1 \geq 5, \quad e_k = 1 + k(m - 1), \quad q = e_k(k + 1)^2, \quad c = k/2, \\ \chi(S) &= \prod_{a,b=0}^k (S - c - (2a - k)\delta - i(2b - k)\gamma)^{e_k}, \\ R &= R_k = k\sqrt{\delta^2 + \gamma^2}, \quad d\mu_k(y) = w_h^{*k}(y) dy, \quad w_h(y) = |(2\xi/h)(1/2 + iy)|^2/(2\pi). \end{aligned} \tag{RGC1}$$

The monic polynomial in the real coordinate is $Q(y) = i^{-q}\chi(c + iy)$. Its q roots, with multiplicities, occur in opposite pairs and have modulus at most R . Since q is a multiple of four here, $i^{-q} = 1$, but the coordinate isometry is retained rather than assumed.

The fixed Gamma reference and density constants are exactly those of AMT:

$$\begin{aligned} d\sigma(y) &= s(y) dy = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy, \quad \alpha = \pi/2, \\ c_- e^{-\alpha|y|}(1 + |y|)^{-1/2} &\leq s(y) \leq c_+ e^{-\alpha|y|}(1 + |y|)^{-1/2}, \\ c_- &= \sqrt{2}e^{-7/6}, \quad c_+ = \sqrt{2\sqrt{5}}e^{2/3}, \quad g_n = \sqrt{2\pi} \frac{(2n)!}{4^n}. \end{aligned} \tag{RGC2}$$

The formulas below first apply to any even degree $q \geq 2$, any monic opposite-paired Q with radius R , and every integer $r \geq 0$. They are not restricted to $r \leq q$.

For $\deg P \leq r$ set

$$H_0(P) = \int_{\mathbb{R}} |y|^{2q} |P(y)|^2 d\sigma(y), \quad H_Q(P) = \int_{\mathbb{R}} |Q(y)P(y)|^2 d\sigma(y). \tag{RGC3}$$

Introduce the explicit finite quantities

$$\begin{aligned} s_q &= 2q + 1, \quad D_{q,r} = \frac{4r + 4q + 2}{\alpha}, \quad Y_{q,r} = \frac{s_q^2}{32\alpha(r + s_q)}, \\ \eta_{q,r} &= \frac{c_+}{c_-} (1 + D_{q,r}) s_q (3/8)^{s_q}, \quad \rho_{q,r} = R/Y_{q,r}. \end{aligned} \tag{RGC4}$$

Here s_q is not a source order, $D_{q,r}$ is a scalar moment bound, and $Y_{q,r}$ is an integration cutoff, not a new arithmetic coordinate.

Theorem RGC.1 (Finite full-form comparison). *Assume*

$$2R \leq Y_{q,r}, \quad \eta_{q,r} < 1. \quad (\text{RGC5})$$

Then for every complex polynomial P of degree at most r ,

$$\begin{aligned} (1 - \rho_{q,r}^2)^q (1 - \eta_{q,r}) H_0(P) &\leq H_Q(P), \\ H_Q(P) &\leq ((1 + \rho_{q,r}^2)^q + \eta_{q,r}) H_0(P). \end{aligned} \quad (\text{RGC6})$$

In particular, with

$$E_{q,r}^{\text{root}} = -q \log(1 - \rho_{q,r}^2) - \log(1 - \eta_{q,r}), \quad (\text{RGC7})$$

one has $e^{-E_{q,r}^{\text{root}}} H_0 \leq H_Q \leq e^{E_{q,r}^{\text{root}}} H_0$. If $\eta_{q,r} \leq 1/2$, then

$$E_{q,r}^{\text{root}} \leq \frac{4}{3} q \rho_{q,r}^2 + 2\eta_{q,r} = \frac{4096\alpha^2 q R^2 (r + s_q)^2}{3s_q^4} + 2\eta_{q,r}. \quad (\text{RGC8})$$

217.2 The complete inner-region bound

The point requiring a proof is control of every polynomial direction near the roots, not just a monic trial or an equilibrium support. Consider on the positive half-line the auxiliary integral

$$N_q(p) = \int_0^\infty |p(t)|^2 t^{2q} e^{-\alpha t} dt.$$

This is an integration comparator for each of $P(t)$ and $P(-t)$, not a replacement quotient or source metric.

The classical inputs below are the Laguerre formulas recorded by Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21, Table 18.3.1; eq. (18.5.5) and Table 18.5.1; eqs. (18.5.12), (18.6.1), (18.9.1) and Table 18.9.1]. Their Laguerre parameter is our $\beta = 2q > -1$, their degree is l , and their argument is $x = \alpha t$, where our $\alpha = \pi/2 > 0$ is the scale, not the Laguerre parameter. The exact change of variables $t = x/\alpha$ retains the measure factor $\alpha^{-\beta-1}$. In particular the squared norm below is $H_l = \alpha^{-\beta-1} \Gamma(l + \beta + 1)/l!$. For $L_l^{(\beta)}(\alpha t)/\sqrt{H_l}$, with positive square root, the upward multiplication entry is $-\sqrt{(l+1)(l+\beta+1)}/\alpha$; its sign comes from the standard Laguerre phase. The absolute entries used below retain that phase through the displayed norm and recurrence. The following proof derives the specific moment and evaluation inequalities from these classical identities.

Lemma RGC.2 (Laguerre moment and evaluation bounds). *For every $\deg p \leq r$,*

$$\int_0^\infty t |p(t)|^2 t^{2q} e^{-\alpha t} dt \leq D_{q,r} N_q(p), \quad (\text{RGC9})$$

and for every $t \geq 0$,

$$|p(t)|^2 \leq K_{q,r}(0,0) e^{2\alpha r t/s_q} N_q(p), \quad K_{q,r}(0,0) = \frac{\alpha^{s_q}}{(s_q - 1)!} \binom{r + s_q}{r}. \quad (\text{RGC10})$$

Proof. Put $\beta = 2q$. Rodrigues' formula and l integrations by parts give the Laguerre norm

$$\int_0^\infty [L_l^{(\beta)}(\alpha t)]^2 t^\beta e^{-\alpha t} dt = \alpha^{-\beta-1} \frac{\Gamma(l + \beta + 1)}{l!}.$$

For clarity, orthogonality follows because all derivatives of a polynomial of degree below l vanish after l integrations. The boundary terms vanish at zero since β is a nonnegative integer, and at infinity by the exponential. The leading coefficient is $(-1)^l/l!$ in the scaled variable; using it in the

same integration gives the displayed norm. The three-term recurrence follows also by coefficient comparison in

$$L_l^{(\beta)}(x) = \sum_{j=0}^l (-1)^j \binom{l+\beta}{l-j} \frac{x^j}{j!}.$$

In the normalized basis, multiplication by t has diagonal $(2l + \beta + 1)/\alpha$ and adjacent entries with absolute value $\sqrt{(l+1)(l+\beta+1)}/\alpha$. The downward entry is at most $(l + \beta/2)/\alpha$ and the upward entry at most $(l + 1 + \beta/2)/\alpha$. The absolute row sum of its compression through degree r is thus at most $(4r + 2\beta + 2)/\alpha = D_{q,r}$. Its quadratic form is the integral in (RGC9); no next-degree output has been discarded in a claimed norm bound. We are bounding a first moment, so this compression is exactly the required quadratic form.

The finite series also gives

$$\frac{|L_l^{(\beta)}(\alpha t)|}{L_l^{(\beta)}(0)} \leq \sum_{j=0}^l \frac{l(l-1)\cdots(l-j+1)}{(\beta+1)_j} \frac{(\alpha t)^j}{j!} \leq e^{l\alpha t/(\beta+1)}.$$

Cauchy–Schwarz in the complete degree- r normalized basis proves the pointwise bound. Its value at zero is

$$\frac{\alpha^{\beta+1}}{\beta!} \sum_{l=0}^r \binom{l+\beta}{l} = \frac{\alpha^{\beta+1}}{\beta!} \binom{r+\beta+1}{r},$$

using the finite binomial summation identity. This proves (RGC10). These classical Laguerre formulas agree with DLMF §§18.3, 18.5.12, 18.6.1 and 18.9. $\square$

By (RGC2), $s(t) \geq c_- e^{-\alpha t}/(1+t)$ for $t \geq 0$. Cauchy–Schwarz and (RGC9) therefore give

$$\int_0^\infty t^{2q} |p(t)|^2 s(t) dt \geq \frac{c_-}{1 + D_{q,r}} N_q(p). \quad (\text{RGC11})$$

Indeed $N_q(p)^2$ is at most the product of the same integrals with the factors $(1+t)^{-1}$ and $(1+t)$ inserted. Apply this separately to $p(t) = P(t)$ and $p(t) = P(-t)$ and add. This retains both half-lines and does not require real coefficients or a parity decomposition of a mixed matrix.

If $P = 0$, both forms and both inner integrals vanish. In the quotient estimate below take $P \neq 0$; after multiplication by $H_0(P)$, its undivided form holds also for $P = 0$.

For any $Y > 0$, every root of Q has modulus at most R , so $|Q(y)| \leq (Y + R)^q$ on $|y| \leq Y$. Combining (RGC10)–(RGC11) and $s(y) \leq c_+$ yields

$$\frac{\int_{|y| \leq Y} |Q(y)P(y)|^2 d\sigma(y)}{H_0(P)} \leq \frac{c_+}{c_-} (1 + D_{q,r}) Y K_{q,r}(0,0) (Y + R)^{2q} e^{2\alpha r Y/s_q}. \quad (\text{RGC12})$$

There is no extra factor of two: the two half-line numerator estimates are summed along with their two lower norm estimates. The same formula with $R = 0$ controls the inner part of $H_0(P)$.

Take $Y = Y_{q,r}$ and $R \leq Y/2$. Since

$$\binom{r+s_q}{r} \leq \frac{(r+s_q)^{s_q}}{s_q!}, \quad s_q! \geq (s_q/e)^{s_q},$$

one has

$$Y K_{q,r}(0,0) (Y + R)^{2q} e^{2\alpha r Y/s_q} \leq s_q \left(\frac{3e^{33/16}}{64} \right)^{s_q} < s_q (3/8)^{s_q}.$$

The factorial bound follows from the integral lower bound for $\sum \log j$. For the last strict inequality, the positive series $\log 2 = 2 \sum_{j \geq 0} (1/3)^{2j+1}/(2j+1)$ gives $\log 2 > 56/81$, so $\log 8 > 56/27 > 33/16$. Thus both inner integrals are at most $\eta_{q,r} H_0(P)$. This proves the full inner estimate for arbitrary r , rather than assuming that P avoids the root region.

217.3 Outer region, finite proof and degree range

Pair the opposite roots as $\pm z_a$. For $|y| > Y > R$,

$$\frac{|Q(y)|^2}{|y|^{2q}} = \prod_{a=1}^{q/2} |1 - z_a^2/y^2|^2.$$

Therefore

$$(1 - R^2/Y^2)^q \leq \frac{|Q(y)|^2}{|y|^{2q}} \leq (1 + R^2/Y^2)^q. \quad (\text{RGC13})$$

Integrating the lower inequality on the outer region and subtracting the bounded inner part of H_0 proves the first line of (RGC6). Adding the bounded inner part of H_Q to the outer upper inequality proves its second line. This proves Theorem RGC.1. For its symmetric version use

$$(1 + u)^q + \eta \leq \frac{(1 - u)^{-q}}{1 - \eta} \quad (0 \leq u < 1, 0 \leq \eta < 1).$$

Finally $-\log(1 - u) \leq u/(1 - u)$, with $u \leq 1/4$, and $-\log(1 - \eta) \leq 2\eta$ for $\eta \leq 1/2$, prove (RGC8).

The explicit radius guard is

$$64\alpha R(r + 2q + 1) \leq (2q + 1)^2. \quad (\text{RGC14})$$

This undivided guard includes $R = 0$ and is equivalent to $2R \leq Y_{q,r}$ for every $R \geq 0$. For the original arithmetic radius $R_k > 0$ it also gives the displayed cutoff after division by $64\alpha R_k$. For fixed arithmetic data, $R = O_h(k)$ and $q \geq (k + 1)^2$. Thus this admits genuinely growing ratios r/q . More precisely, for any sequence of cutoffs $r = r_k$ satisfying

$$\frac{R_k(r_k + q)}{q^2} \rightarrow 0, \quad (\text{RGC15})$$

the guards eventually hold and $E_{q,r_k}^{\text{root}} = o(q)$. Here $r_k = O_h(q^2)$ eventually, so the prefactor in η_{q,r_k} is polynomial and its exponential term dominates it. This assertion concerns a full-form comparison, not a uniform extension of the equilibrium profile ψ .

217.4 A non-equilibrium reference lower bound at every relation degree

We next calculate a lower reference bound without a compact equilibrium parameter rectangle. Define

$$\nu_r^0 = \inf_{P \text{ monic, deg } P=r} H_0(P), \quad T_r^0 = \nu_r^0/g_{q+r}, \quad n = q + r.$$

Theorem RGC.3 (Uniform monic reference lower bound). *For all integers $q \geq 1$ and $r \geq 0$,*

$$\log T_r^0 \geq F_{q,r} + \log C_0, \quad F_{q,r} = \frac{q^2}{4(q + r) \log(8(q + r)/q)}, \quad C_0 = \frac{7c_-}{32\sqrt{6}} e^{-1/24}. \quad (\text{RGC16})$$

The lower bound is not claimed to be a sharp reference asymptotic.

Proof. Put $x = q/n$, $L = \log(8/x)$, and $a = x/(4L)$; then $0 < a < 1/8$. For $0 < a \leq 1/8$, set

$$\bar{E}(a) = 1 + a^2 + \frac{1}{2}a^2 \log(2/a).$$

The elementary elliptic integral

$$E(a) = \int_0^{\pi/2} \sqrt{\sin^2 t + a^2 \cos^2 t} dt$$

satisfies $E(a) \leq \bar{E}(a)$. To see this, split at $t_0 = \arcsin a$. On the first interval the excess over $\sin t$ is at most $a \cos t$, with integral a^2 . On the second interval rationalization bounds that excess by $a^2 \cos^2 t / (2 \sin t)$; its integral is

$$\frac{a^2}{2} \left[\log \frac{1 + \sqrt{1 - a^2}}{a} - \sqrt{1 - a^2} \right] \leq \frac{a^2}{2} \log(2/a).$$

Also $\bar{E}(a) < 2$. Indeed $a^2 \log(2/a)$ increases on $(0, 1/8]$ and $\log 16 < 3$, giving $\bar{E} \leq 1 + 1/64 + 3/128 < 2$.

Let $b = 2n/\bar{E}(a)$ and use the probability measure

$$d\nu(y) = \frac{|y| dy}{\pi \sqrt{(y^2 - a^2 b^2)(b^2 - y^2)}} \quad \text{on } [-b, -ab] \cup [ab, b].$$

Its square-pushforward is the arcsine probability on $[a^2 b^2, b^2]$. The following three formulas follow by that substitution and circle Jensen:

$$\begin{aligned} \int \log |P| d\nu &\geq r \log \frac{b\sqrt{1 - a^2}}{2}, \\ \int \log |y| d\nu &= \log \frac{b(1 + a)}{2}, \\ \int \log(d\nu/dy) d\nu &= \log \frac{2}{\pi b(1 - a)}. \end{aligned}$$

For completeness, symmetry makes the integral contributed by a root z of P equal to one half of the arcsine potential at z^2 . Write $u = C + D \cos \theta$ for the square interval. Factoring the quadratic in $e^{i\theta}$, whose roots have product one, shows that its circle-logarithmic mean is at least $\log(D/2)$; this gives the first inequality root by root. The mean of $\log u$ gives the second formula. The two endpoint logarithmic integrals are each $\log((b^2 - a^2 b^2)/4)$; substitution into the density gives the third. Logarithmic singularities on the bands are integrable. Furthermore

$$\alpha \int |y| d\nu = bE(a) \leq 2n.$$

Probability Jensen applied to the norm integrand divided by $d\nu/dy$, together with (RGC2), now gives for every monic P

$$H_0(P) \geq \frac{c - \pi b(1 - a)}{2\sqrt{1 + b}} n^{2n} e^{-2n} \exp(2q \log(1 + a) + r \log(1 - a^2) - 2n \log \bar{E}(a)). \quad (\text{RGC17})$$

No reference minimizer has been guessed.

For the exponent use $\log(1 + a) \geq a - a^2/2$, $\log(1 - a^2) \geq -4a^2/3$, and $\log \bar{E}(a) \leq a^2 + \frac{1}{2}a^2 \log(2/a)$. It is at least

$$2qa - na^2(10/3 + \log(2/a)).$$

Since $\log(2/a) = L + \log L$ and $L > 2$, one has $10/3 + \log(2/a) \leq 2L + 7/3 \leq 4L$. Substituting $a = q/(4nL)$ gives the lower exponent $q^2/(4nL) = F_{q,r}$.

The finite positive-real remainder used here is the one in Askey and Roy [H22, eq. (5.11.1), §5.11(ii)]. For $x > 0$ it gives $\log \Gamma(x) = (x - \frac{1}{2}) \log x - x + \frac{1}{2} \log(2\pi) + r(x)$, where $0 \leq r(x) \leq 1/(12x)$. Taking $x = 2n$, with $n \geq 1$, and using $(2n)! = 2n\Gamma(2n)$ in the retained $g_n = \sqrt{2\pi}(2n)!/4^n$ gives exactly $g_n \leq 2\pi\sqrt{2n} n^{2n} e^{-2n} e^{1/(24n)}$. The arithmetic comparison factor, when present, remains the original A_k .

The positive-real Stirling remainder gives

$$g_n \leq 2\pi\sqrt{2n} n^{2n} e^{-2n} e^{1/(24n)}.$$

Here $n \leq b \leq 2n$, $1 - a \geq 7/8$, and $\sqrt{2n(1 + b)} \leq \sqrt{6}n$. Dividing (RGC17) by this upper bound for g_n leaves the factor at least $7c_- e^{-1/24}/(32\sqrt{6}) = C_0$. Take the same monic infimum to prove (RGC16). $\square$

217.5 Transport through the original arithmetic monic and quotient maps

The original AMT full-form bounds, valid at each polynomial degree n , are

$$\frac{a_k}{\mathfrak{D}_n} \|F\|_\sigma^2 \leq \|F\|_{\mu_k}^2 \leq A_k \|F\|_\sigma^2, \quad \mathfrak{D}_n = [1 + 4(n + 21 + 4m)^2]^{21+4m}. \quad (\text{RGC18})$$

The constants remain

$$B_h = 42 + 8m, \quad p_h = 8m - 9/2, \quad \vartheta_h = \int_{-1}^1 w_h(y) dy, \quad M_\alpha = \int e^{\alpha y} w_h(y) dy, \quad (\text{RGC19})$$

$$a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha(k-3)} (k-2)^{-B_h}}{c_+ 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_-} k^{p_h+1} M_\alpha^{k-1}.$$

The original positive constants c_h, C_h^{tilt} are the supplied arithmetic density-envelope constants. They are inputs from AMT/TW, not newly evaluated numerical parameters. In particular, for fixed h ,

$$\Lambda_{k,n} := \log(A_k \mathfrak{D}_n / a_k) = O_h(k + \log(n + 1)).$$

Let ω_n be the original arithmetic monic norm and $\nu_j^\chi = \inf_{P \text{ monic, deg } P=j} \|\chi P\|_{\mu_k}^2$. In this formula the variable is the original S ; the map $P(S) \mapsto i^{-j} P(c+iy)$ is the degree- j monic coefficient isometry. It changes no quotient or arithmetic unit. The original exact sequence is

$$0 \longrightarrow \mathcal{P}_j \xrightarrow{\times \chi} \mathcal{P}_{q+j} \xrightarrow{J_{q+j}} E = \mathbb{C}[S]/(\chi) \longrightarrow 0. \quad (\text{RGC20})$$

The source realization and its full observation remain $\mathcal{V}(P) = P(\sum D_i) F_h^{\otimes k}$ and $J^{(k)} \mathcal{V} = \eta_{h,k} \pi_\chi$, with the entire Taylor unit inside $\eta_{h,k}$. Multiples of χ have their original theta primitives. The norm of a relation in (RGC20) is measured before its quotient value becomes the represented zero at the receiving support.

For $n = q + r$ under (RGC5), taking the same monic minima in Theorem RGC.1 and (RGC18) gives, for $0 \leq j \leq r$,

$$|\log T_j^{\text{ar}} - \log T_j^0| \leq \Lambda_{k,n} + E_{q,r}^{\text{root}}, \quad T_j^{\text{ar}} = \nu_j^\chi / \omega_{q+j}. \quad (\text{RGC21})$$

The same comparison, with $\mathfrak{D}_d \leq \mathfrak{D}_n$, gives

$$\left| \log \frac{\beta_d}{d(d-1/2)} \right| \leq \Lambda_{k,n}, \quad \beta_d = \omega_d / \omega_{d-1}, \quad q \leq d \leq n. \quad (\text{RGC22})$$

Neither estimate is obtained by differentiating a determinant asymptotic.

The source/relation Gram determinant identity in (RGC20) proves, by taking one adjacent ratio,

$$d_{q+j} := \frac{V_{q+j}}{V_{q+j-1}} = \frac{\omega_{q+j}}{\nu_j^\chi} = (T_j^{\text{ar}})^{-1}. \quad (\text{RGC23})$$

At the original untilted source, the established total reflection identity then gives

$$\epsilon_{q+j-1}^2 = \beta_{q+j} (T_j^{\text{ar}} - 1) \begin{cases} 1, & j = 0, \\ 1 - (T_{j-1}^{\text{ar}})^{-1}, & j \geq 1. \end{cases} \quad (\text{RGC24})$$

This uses total parity only. It does not assert that the individual mixed kernel and boundary pairings vanish. The initial case uses J_{q-1} , which is an isomorphism, and no inverse at degree $q - 2$.

Corollary RGC.4 (Finite growing-band allowance bound). *Set*

$$S_{k,q,r} = F_{q,r} + \log C_0 - \Lambda_{k,q+r} - E_{q,r}^{\text{root}}, \quad b_{k,q,r}^{\text{low}} = \frac{a_k}{A_k \mathfrak{D}_{q+r}} q(q-1/2). \quad (\text{RGC25})$$

If (RGC5) holds and $S_{k,q,r} > 0$, then

$$T_j^{\text{ar}} \geq e^{S_{k,q,r}} \quad (0 \leq j \leq r), \quad \min_{q-1 \leq N \leq q+r-1} \epsilon_N \geq \sqrt{b_{k,q,r}^{\text{low}}} e^{S_{k,q,r}/2} (1 - e^{-S_{k,q,r}}). \quad (\text{RGC26})$$

Proof. $F_{q,j}$ decreases with j . Equations (RGC16) and (RGC21) therefore give the first bound simultaneously. Equation (RGC22) gives the common lower bound for β_d . Insert both in (RGC24). The first row has one instead of two contraction factors and hence also obeys the displayed lower bound. $\square$

217.6 A genuinely super-proportional range

Corollary RGC.5. *At fixed complete arithmetic data, take $r_k = \lfloor qk^\theta \rfloor$. For a simple quartet, every fixed $0 < \theta < 2/3$ satisfies the guards and, eventually,*

$$\log \min_{q-1 \leq N \leq q+r_k-1} \epsilon_{h,k,N} \geq (1 - o_h(1)) \frac{q}{8(1+k^\theta) \log(8(1+k^\theta))}. \quad (\text{RGC27})$$

For every fixed multiplicity $m \geq 2$, the same statement holds for each fixed $0 < \theta < 4/3$. The right-hand side grows faster than $\log k$. Consequently the ratio of this minimum to the original polynomial exterior benchmark tends to infinity on these growing bands as well.

Proof. Write $q \asymp_h k^d$ with $d = 2$ for $m = 1$ and $d = 3$ for $m \geq 2$. Then $R = O_h(k)$, $\Lambda_{k,q+r_k} = O_h(k + \log q)$, and

$$\rho_{q,r_k} = O_h(k^{1+\theta-d}), \quad E_{q,r_k}^{\text{root}} = O_h(k^{2+2\theta-d}) + e^{-\Omega_h(q)}, \quad F_{q,r_k} \asymp_h \frac{k^{d-\theta}}{\log k}.$$

The stated ranges make $\theta < d - 1$ and $3\theta < 2d - 2$. Thus the radius guard holds, the tail is exponentially small, and both Λ and E^{root} are $o(F)$. It follows that $S_{k,q,r_k} = (1 - o_h(1))F_{q,r_k}$. The logarithm of b^{low} is $O_h(k + \log q) = o(F)$, and $\log(1 - e^{-S}) = o(1)$. Apply (RGC26). The floor changes F by a relative $o(1)$ only. Finally the exterior benchmark is $L_{h,k} = \delta q(k+1)/2$ on this subsequence, so $\log L_{h,k} = O_h(\log k)$. $\square$

This is a lower bound, not the sharp profile $q\psi(r/q)$ and not an extension of its fixed-rectangle error. In particular no claim is made at the endpoints $\theta = 2/3, 4/3$, for faster growing degree, or at a chosen exponential degree. The exact comparison theorem (RGC6) can be used over its full finite guarded range, even when the stronger dominance assumptions of Corollary RGC.5 do not hold.

217.7 Derive the changed four endpoints and complete action

For an arbitrary integer $r \geq 1$, give the rows $N = q - 1, \dots, q + r - 1$ weight one at the two ends and weight two in the interior. Their weights sum to $2r$. Define

$$\mathcal{J}_{q,r} = \sum_{N=q-1}^{q+r-1} w_N \log \epsilon_N^2, \quad \mathcal{B}_{q,r} = \log \frac{V_{q-1}V_q}{V_{q+r-1}V_{q+r}}, \quad \mathcal{W}_{q,r} = \log \frac{\omega_{q+r-1}\omega_{q+r}}{\omega_{q-1}\omega_q}. \quad (\text{RGC28})$$

Put $a_d = -\log(1 - d_d)$ and retain the precise penalty

$$\mathcal{P}_{q,r} = \sum_{j=0}^r w_{q+j-1} a_{q+j} + \sum_{j=1}^r w_{q+j-1} a_{q+j-1}. \quad (\text{RGC29})$$

The total coefficient is $4r - 1$. For $r \geq 2$ the coefficients are $3, 4, \dots, 4, 3, 1$ on $a_q, \dots, a_{q+r}$; for $r = 1$ they are $2, 1$. Taking logarithms in (RGC24) and telescoping the exact norm and volume ratios proves

$$\boxed{\mathcal{J}_{q,r} = \mathcal{B}_{q,r} + \mathcal{W}_{q,r} - \mathcal{P}_{q,r}} \quad (\text{RGC30})$$

Under the guards of Corollary RGC.4,

$$0 \leq \mathcal{P}_{q,r} \leq -(4r - 1) \log(1 - e^{-S_{k,q,r}}). \quad (\text{RGC31})$$

The original two-window identity is recovered at $r = q$. The new four endpoints have been derived, not inserted into an old formula by analogy. The exact weighted action does not equate its geometric mean with its minimum.

For the monic part one also retains the finite AMT enclosure

$$\begin{aligned} -\Lambda_{k,q+r-1} - \Lambda_{k,q+r} &\leq \mathcal{W}_{q,r} - \mathcal{W}_{q,r}^\sigma \\ &\leq \Lambda_{k,q-1} + \Lambda_{k,q}, \end{aligned} \tag{RGC32}$$

where $\mathcal{W}^\sigma$ is the identical four-norm expression with g_n . This comparison is taken after telescoping; it incurs four endpoint errors, not one source error for every row.

217.8 Support, interpretation, and remaining work

Every form inequality above holds before taking an observation or an affine minimum. It therefore passes to a specified restriction and to the minimum on a specified original fibre, with the inequality direction unchanged. The comparison of χP with $y^q P$ is a comparison of coefficient norms on the same domain $\mathcal{P}_r$. It is not an isomorphism between the arithmetic quotient and a homogeneous quotient. In particular the original square $G(\mathbb{Z}) \rightarrow G(\mathbb{C})$ over $\mathbb{Z} \rightarrow \mathbb{C}$, its infinite amplitude quotient, mixed support transports, full Taylor unit and all primary multiplicities remain upstream.

The original source map sends the relations $\chi \mathcal{P}_r$ to their actual theta boundaries. A boundary is sent to its receiving fibre's represented zero; external absence τ is not identified with that zero. Measuring the relation before this map is essential to (RGC23). No term of a mixed Gram is declared zero because its class becomes a boundary.

The growing-range theorem removes one concrete limitation of the earlier comparison: it controls both inner and outer regions at a specified moving polynomial cutoff. The further allowance lower bound uses only this new comparison and a uniform Jensen comparator. It shows that these additional canonical bands still do not yield a small allowance relative to the original exterior benchmark. This conclusion concerns the stated construction and regime. It neither settles the signed flag allocation nor rules out a quantitatively different same-class comparison in the broader tau programme.

No new claim about the current intrinsic allocation coefficient is made. Its surviving eight-matrix signed profile and all CPQ complementary returns remain those in the inherited continuation prompt. Improving them must be assessed through their actual receiving map; the cancellation in the canonical complete action is preserved in (RGC30).

References and exact inputs. The supplied `13_ARITHMETIC_MIXED_TRANSFER.tex`, AMT1–6, provides (RGC2), (RGC18) and (RGC19) at their stated original-source scope. The current cumulative file's PCRP and RXUCT sections retain the older bounded-degree comparison and exact original ratio and radius identities. The new coefficient-degree extension and Jensen bound are proved in this section; their novelty relative to all literature is not asserted. Classical normalizations were checked against the following primary reference sections:

- National Institute of Standards and Technology. (n.d.). *Definitions; explicit representations; special values; recurrence relations*, DLMF §§18.3, 18.5.12, 18.6.1 and 18.9. <https://dlmf.nist.gov/18.5.E12>.
- National Institute of Standards and Technology. (n.d.). *Asymptotic expansions*, DLMF §5.11(i)–(ii), including the positive-real remainder for Stirling's formula. <https://dlmf.nist.gov/5.11>.

218 The complete root-form comparison at growing degree ratios

Scope and audited conclusion. This section proves equations (2)–(13) of the received submission, including all their finite constants, on the original stipulated complete-order packet. The verdict on that scope is *proved as written*, with the displayed guards retained. The accepted EIQ equilibrium and ECL evaluations are stated below in their exact original domain and normalization; the new full-form argument is proved in full here. No assertion about arithmetic monic minima, allowance asymptotics, shifted windows, an intrinsic mixed allocation, or a period evaluation is made here. In particular no original mass, period, root, or multiplicity is altered.

218.1 Original objects and exact dependency statements

Fix $0 < \delta < 1/2$, $\gamma > 2$, and the positive integer complete order m_{pkt} . Retain the original integer subsequence $k = 4l + 1 \geq 5$ and put

$$e_k = 1 + k(m_{\text{pkt}} - 1), \quad q = e_k(k + 1)^2, \quad c = k/2, \quad R = R_k = k\sqrt{\delta^2 + \gamma^2}.$$

The root polynomial and its exact real-coordinate image are

$$\chi(S) = \prod_{a,b=0}^k [S - c - (2a - k)\delta - i(2b - k)\gamma]^{e_k},$$

$$Q(y) = i^{-q}\chi(c + iy) = \prod_{a,b=0}^k (y - \zeta_{ab})^{e_k}, \quad \zeta_{ab} = (2b - k)\gamma - i(2a - k)\delta.$$

The symbol m_{pkt} here denotes exactly the received packet order m ; the subscript merely distinguishes it from the degree of a parity block. All q roots, with their complete multiplicities, occur in opposite and conjugate pairs. They satisfy $|\zeta_{ab}| \leq R$.

The measure remains

$$d\sigma(y) = s(y) dy, \quad s(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi}.$$

Its accepted original density bounds are, on all of $\mathbb{R}$,

$$c_{\Gamma}e^{-\pi|y|/2}(1 + |y|)^{-1/2} \leq s(y) \leq C_{\Gamma}e^{-\pi|y|/2}(1 + |y|)^{-1/2}, \quad c_{\Gamma} = \sqrt{2}e^{-7/6}, \quad C_{\Gamma} = \sqrt{2\sqrt{5}}e^{2/3}. \quad (\text{GRF1})$$

These are AMT5/TW4, restated in RGC2; no new constants are substituted. For completeness the mass follows directly from the Gamma integral. Writing $f_a(x) = e^{ax-e^x}$ gives $\Gamma(a + it) = \int_{\mathbb{R}} f_a(x)e^{itx} dx$ for $a > 0$. Plancherel and $z = e^x$ therefore give

$$\int_{\mathbb{R}} |\Gamma(a + it)|^2 \frac{dt}{2\pi} = \int_{\mathbb{R}} e^{2ax-2e^x} dx = \frac{\Gamma(2a)}{2^{2a}}.$$

Setting $a = 1/4$ and $y = 2t$ proves $\sigma(\mathbb{R}) = 2\Gamma(1/2)/\sqrt{2} = \sqrt{2\pi}$.

Here is the precise accepted equilibrium theorem used. For every $\alpha > 0$, $\beta > 0$, let $V(x) = \beta\sqrt{x} - \alpha \log x$ on $(0, \infty)$. The EIQ probability measure has support $[u^2, v^2]$, $0 < u < v$, where

$$uK(\kappa) = \frac{\alpha\pi}{\beta}, \quad vE(\kappa) = \frac{(\alpha + 2)\pi}{\beta}, \quad \kappa^2 = 1 - u^2/v^2, \quad (\text{GRF2})$$

and

$$K(\kappa) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1 - \kappa^2 \sin^2 \theta}}, \quad E(\kappa) = \int_0^{\pi/2} \sqrt{1 - \kappa^2 \sin^2 \theta} d\theta.$$

The density, extended by zero off this interval, is

$$\rho(x) = \frac{\sqrt{(v^2 - x)(x - u^2)}}{2\pi} h(x), \quad h(x) = \frac{\beta}{2\pi x} \int_0^\infty \frac{\sqrt{s}}{(x + s)\sqrt{(u^2 + s)(v^2 + s)}} ds > 0. \quad (\text{GRF3})$$

It is bounded and continuous, has total mass one, and its logarithmic potential $P^{\text{eq}}(z) = \int_{u^2}^{v^2} \log|z - x| \rho(x) dx$ is continuous on $\mathbb{C}$. With the original Robin constant λ ,

$$V(x) - 2P^{\text{eq}}(x) = \lambda \quad (u^2 \leq x \leq v^2), \quad V(x) - 2P^{\text{eq}}(x) > \lambda \quad (0 < x < u^2 \text{ or } x > v^2). \quad (\text{GRF4})$$

These statements are EIQ1–19 on the entire positive parameter quadrant. Writing $w = (u + v)/2$, $z_0 = uv$, and $c_{\text{cap}} = (v^2 - u^2)/4$, the accepted ECL6 and ECL14 give exactly

$$\lambda = 2(\alpha + 2) - 2\alpha \log w - 2 \log c_{\text{cap}}, \quad P^{\text{eq}}(0) = 2(\alpha + 1) \log w - \alpha \log z_0 - 2. \quad (\text{GRF5})$$

The use of these already proved dependencies does not assume compactness of a moving family of parameters. All estimates below use the actual α, β, u, v .

218.2 The complete root polynomial

Lemma 1. *Put*

$$U_2 = \frac{k(k+2)}{3}, \quad U_4 = \frac{k(k+2)(3k^2+6k-4)}{15},$$

$$D_2 = qU_2(\gamma^2 - \delta^2), \quad M_4 = q[(\delta^4 + \gamma^4)U_4 + 2\delta^2\gamma^2U_2^2].$$

For real $|y| > R$ one has

$$\log \frac{|Q(y)|^2}{y^{2q}} = -\frac{D_2}{y^2} + E_Q(y), \quad |E_Q(y)| \leq \frac{M_4}{2y^4(1 - R^2/y^2)}. \quad (\text{GRF6})$$

For real $|y| \geq 2R$ one has

$$\log \frac{|Q(y)|^2}{y^{2q}} \leq -\frac{13D_2}{30y^2} < 0. \quad (\text{GRF7})$$

For every real y one has

$$|Q(y)|^2 \leq (y^2 + R^2)^q. \quad (\text{GRF8})$$

Proof. The original odd values of k give the literal positive real factorization

$$Q(y) = \prod_{a,b \in \{1,3,\dots,k\}} [((y - b\gamma)^2 + a^2\delta^2)((y + b\gamma)^2 + a^2\delta^2)]^{e_k} > 0.$$

No root is real, and this also shows that Q is even. Finite differentiation of

$$\sum_{a=0}^k e^{(2a-k)t} = \frac{\sinh((k+1)t)}{\sinh t}$$

at $t = 0$, using the removable value there, gives zero odd moments and

$$\frac{1}{k+1} \sum_{a=0}^k (2a-k)^2 = U_2, \quad \frac{1}{k+1} \sum_{a=0}^k (2a-k)^4 = U_4.$$

For example division of the Taylor series gives the coefficients $U_2/2$ and $U_4/24$ in the series of the displayed ratio divided by $k+1$. Expanding ζ_{ab}^2 and $|\zeta_{ab}|^4 = [(2b-k)^2\gamma^2 + (2a-k)^2\delta^2]^2$ therefore proves

$$\sum_{a,b} e_k \zeta_{ab}^2 = D_2, \quad \sum_{a,b} e_k |\zeta_{ab}|^4 = M_4 \leq qR^4.$$

For $|y| > R$ the series for $\log(1 - \zeta_{ab}/y)$ converges absolutely. Taking twice the real part, opposite-root pairing cancels its odd terms, and conjugate pairing makes its even sums real. Hence

$$\log \frac{|Q(y)|^2}{y^{2q}} = - \sum_{j \geq 1} \frac{\sum_{a,b} e_k \zeta_{ab}^{2j}}{j y^{2j}}.$$

This is the real logarithm of a positive number. Since $\sum e_k |\zeta_{ab}|^{2j} \leq M_4 R^{2j-4}$ for $j \geq 2$ and $1/j \leq 1/2$, summing the geometric majorant proves (GRF6).

The packet restrictions give $\delta^2/\gamma^2 < 1/16$, so

$$\frac{D_2}{qR^2} = \frac{k+2}{3k} \frac{\gamma^2 - \delta^2}{\gamma^2 + \delta^2} \geq \frac{5}{17} > 0.$$

At $|y| \geq 2R$, the error divided by D_2/y^2 is at most

$$\frac{qR^2}{D_2} \frac{R^2/y^2}{2(1 - R^2/y^2)} \leq \frac{17}{5} \frac{1/4}{2(3/4)} = \frac{17}{30}.$$

Subtracting this from the negative main term proves (GRF7). Finally pair each root with its opposite, with multiplicities. Every factor satisfies $|(y - \zeta)(y + \zeta)| = |y^2 - \zeta^2| \leq y^2 + R^2$. There are exactly $q/2$ such factors in Q , and squaring their product proves (GRF8). $\square$

218.3 The exact parity coordinate and moving evaluation bound

Let r be an integer, set $m_\varepsilon = \lfloor (r - \varepsilon)/2 \rfloor$ for $\varepsilon \in \{0, 1\}$, and impose $m_0, m_1 \geq 1$. This guard is equivalent to $r \geq 3$ and is retained throughout the finite theorem. Every complex polynomial p of degree at most r has the unique decomposition

$$p(y) = P_0(y^2/q^2) + (y/q)P_1(y^2/q^2), \quad \deg P_\varepsilon \leq m_\varepsilon.$$

Indeed, if $p(y) = \sum_{j=0}^r a_j y^j$, then $P_\varepsilon(x) = \sum_{j \geq 0} a_{2j+\varepsilon} q^{2j+\varepsilon} x^j$, with only indices $2j + \varepsilon \leq r$ included. This gives both the map and its inverse on coefficient spaces. Define, for $y > 0$,

$$\mathfrak{r}(y) = \frac{s(y) e^{\pi y/2} \sqrt{y}}{\sqrt{2}}.$$

For each block let

$$\alpha_\varepsilon = \frac{q + \varepsilon - 3/4}{m_\varepsilon} > 0, \quad \beta_\varepsilon = \frac{\pi q}{2m_\varepsilon} > 0, \quad V_\varepsilon(x) = \beta_\varepsilon \sqrt{x} - \alpha_\varepsilon \log x, \quad (\text{GRF9})$$

and

$$\|P\|_\varepsilon^2 = \int_0^\infty |P(x)|^2 e^{-m_\varepsilon V_\varepsilon(x)} \mathfrak{r}(q\sqrt{x}) dx.$$

Denote its actual equilibrium data by $u_\varepsilon, v_\varepsilon, P_\varepsilon^{\text{eq}}, \lambda_\varepsilon$. Since s and $|Q|^2$ are even, the two half-lines cancel the cross term $2 \operatorname{Re}(P_0 \bar{P}_1) y/q$ for both norms. Using $y = q\sqrt{x}$, $dy = q dx/(2\sqrt{x})$ on the positive half-line gives

$$\mathcal{N}_0[p] := \int_{\mathbb{R}} y^{2q} |p(y)|^2 d\sigma(y) = \sqrt{2} q^{2q+1/2} \sum_{\varepsilon=0}^1 \|P_\varepsilon\|_\varepsilon^2. \quad (\text{GRF10})$$

In particular the power of x in the integrand is $q + \varepsilon - 3/4$, exactly as in received equation (5). With $H_Q(x) = q^{-2q} |Q(q\sqrt{x})|^2$, the other norm is exactly

$$\mathcal{N}_\chi[p] := \int_{\mathbb{R}} |Q(y)p(y)|^2 d\sigma(y) = \sqrt{2} q^{2q+1/2} \sum_{\varepsilon=0}^1 \int_0^\infty |P_\varepsilon(x)|^2 H_Q(x) x^{\varepsilon-3/4} e^{-\pi q \sqrt{x}/2} \mathfrak{r}(q\sqrt{x}) dx. \quad (\text{GRF11})$$

All integrals converge by (GRF1).

For each block put

$$\begin{aligned}\Sigma_\varepsilon &= m_\varepsilon(2P_\varepsilon^{\text{eq}}(0) + \lambda_\varepsilon), & b_\varepsilon &= \min \left\{ \frac{u_\varepsilon}{2}, \frac{1}{4}e^{-\Sigma_\varepsilon/(2q)} \right\} > 0, \\ \Lambda_\varepsilon &= \frac{\alpha_\varepsilon}{u_\varepsilon^2} + \frac{\beta_\varepsilon}{2u_\varepsilon}, & r_+ &= \frac{C_\Gamma}{\sqrt{2}}, & r_{-, \varepsilon} &= \frac{c_\Gamma}{\sqrt{2}} \left(\frac{qu_\varepsilon}{1 + qu_\varepsilon} \right)^{1/2} > 0.\end{aligned}\tag{GRF12}$$

The original density bounds imply $\mathfrak{r}(y) \leq r_+$ for all $y > 0$ and $\mathfrak{r}(q\sqrt{x}) \geq r_{-, \varepsilon}$ on the actual support. Indeed the extra factor is $\sqrt{y/(1+y)}$, which increases in y . Also $|V'_\varepsilon(x)| \leq \Lambda_\varepsilon$ on that support.

Adding the two exact expressions in (GRF5) yields, for each parameter pair and its original w, z_0 ,

$$2P^{\text{eq}}(0) + \lambda = 2\alpha \left(1 + \log \frac{w}{z_0} \right) + 2 \log \frac{v+u}{v-u}.\tag{GRF13}$$

Here $4 \log w - 2 \log c_{\text{cap}} = 2 \log((v+u)/(v-u))$ and the constants give $-4 + 2(\alpha + 2) = 2\alpha$. Thus the -2 in each occurrence of $P^{\text{eq}}(0)$ is retained.

Lemma 2. Assume $m = m_\varepsilon \geq 1$ and $m(v_\varepsilon^2 - u_\varepsilon^2) \geq 2$. For every complex polynomial P of degree at most m and every $0 \leq x \leq b_\varepsilon^2$,

$$|P(x)|^2 \leq \frac{32e^{3\Lambda_\varepsilon/2}m^4}{r_{-, \varepsilon}} e^{\Sigma_\varepsilon} \|P\|_\varepsilon^2.\tag{GRF14}$$

Proof. Suppress the block subscripts only within this proof, retaining their actual values. If $P = 0$ the result holds. Otherwise let $M = \max_{[u^2, v^2]} |P(x)|e^{-mV(x)/2} > 0$, attained at x_0 . At least one side of x_0 in this interval has length at least $1/m$, because its full width is at least $2/m$. Take the closed one-sided interval J of length $1/m$ from x_0 . For $x \in J$,

$$|P(x)| \leq M e^{mV(x)/2} \leq e^{\Lambda/2} |P(x_0)|.$$

For any interval J and every complex polynomial of degree at most m ,

$$\|P'\|_{\infty, J} \leq \frac{4m^3}{|J|} \|P\|_{\infty, J}.\tag{GRF15}$$

To prove this, map J affinely to $[-1, 1]$ and expand $P = \sum_{j=0}^m c_j T_j$, where $T_j(\cos \theta) = \cos(j\theta)$. Cosine orthogonality gives $|c_j| \leq 2\|P\|_\infty$ for $j \geq 1$, also for complex c_j . The finite geometric-sum identity for $\sin(j\theta)/\sin \theta$ gives its modulus at most j . Thus $|T'_j| \leq j^2$, including the endpoints by continuity. Since $\sum_{j=1}^m j^2 \leq m^3$ for every integer $m \geq 1$, the derivative bound on $[-1, 1]$ is $2m^3\|P\|_\infty$; the affine derivative factor $2/|J|$ proves (GRF15).

On J , this gives $|P'| \leq 4m^4 e^{\Lambda/2} |P(x_0)|$. Consequently on the segment from x_0 of length $d = (8e^{\Lambda/2}m^4)^{-1} \leq 1/m$, the fundamental theorem of calculus and the triangle inequality imply $|P| \geq |P(x_0)|/2$. On the same segment $e^{-mV(x)} \geq e^{-\Lambda} e^{-mV(x_0)}$. Using the actual lower density r_- on it proves

$$\|P\|_\varepsilon^2 \geq r_- d \frac{|P(x_0)|^2}{4} e^{-\Lambda} e^{-mV(x_0)} = \frac{r_- M^2}{32e^{3\Lambda/2}m^4}.$$

For the passage off the support, continuity on the whole complex plane can be checked directly from the accepted bounded density. If its bound is K_ρ , then for $0 < a < 1$ and every complex z ,

$$\int_{|z-t|<a} -\log |z-t| \rho(t) dt \leq 2K_\rho a(1 - \log a).$$

Indeed the real length of $\{t : |z-t| < s\}$ is at most $2s$; integrating the level sets of the negative logarithm gives the stated bound. It tends uniformly to zero as $a \downarrow 0$. Truncation of this singularity gives continuous potentials converging locally uniformly, while the positive logarithmic part is

continuous on compact sets. Thus P^{eq} is continuous on $\mathbb{C}$. Outside the support its integrands are harmonic and uniformly differentiable on compact subsets, so P^{eq} is harmonic there. Now consider $f(z) = \log |P(z)| - mP^{\text{eq}}(z)$ on $\mathbb{C} \setminus [u^2, v^2]$. The second term is harmonic there and the first is subharmonic, including its value $-\infty$ at zeros. Continuity of P^{eq} and (GRF4) give boundary upper values $f \leq \log M + m\lambda/2$. This inequality also bounds the entire exterior, including infinity: if $n = \deg P \leq m$ and $w = 1/z$, then near $w = 0$

$$f(1/w) = (m - n) \log |w| + \log |w^n P(1/w)| - m \int \log |1 - tw| \rho(t) dt.$$

The last two terms are harmonic near zero because their arguments are nonzero there. For $n = m$ the function extends harmonically at infinity; for $n < m$ it extends subharmonically with value $-\infty$. Thus it is subharmonic on the complement of the interval in the Riemann sphere. Its boundary values have the stated upper bound. The maximum principle applies on this compact closure: an upper value greater than the boundary bound would attain an interior maximum, forcing constancy on the connected exterior, contrary to the boundary bound. Hence

$$|P(z)|^2 \leq M^2 e^{m(2P^{\text{eq}}(z) + \lambda)}.$$

For $0 \leq x < u^2$, every support point t satisfies $0 < t - x \leq t$, so $\log |x - t| \leq \log t$ and $P^{\text{eq}}(x) \leq P^{\text{eq}}(0)$. Since $b \leq u/2$, this applies throughout $[0, b^2]$. Substituting the proved estimate for M^2 yields (GRF14). $\square$

218.4 The entire inner region and the finite full-form theorem

Let $\varrho = R/q$. Impose, for both blocks,

$$m_\varepsilon \geq 1, \quad m_\varepsilon(v_\varepsilon^2 - u_\varepsilon^2) \geq 2, \quad \varrho \leq b_\varepsilon/2. \quad (\text{GRF16})$$

Set

$$\eta_\varepsilon = \frac{32r_+ e^{3\Lambda_\varepsilon/2} m_\varepsilon^4 b_\varepsilon^{2\varepsilon+1/2}}{r_{-, \varepsilon}(\varepsilon + 1/4)} 8^{-q}, \quad \eta = \max(\eta_0, \eta_1). \quad (\text{GRF17})$$

On $[0, b_\varepsilon^2]$, (GRF8) gives $H_Q(x) \leq (x + \varrho^2)^q$. The cutoff definition and the third guard give

$$e^{\Sigma_\varepsilon} (b_\varepsilon^2 + \varrho^2)^q \leq e^{\Sigma_\varepsilon} (5b_\varepsilon^2/4)^q \leq (5/64)^q \leq 8^{-q}.$$

The same estimate holds with x^q instead of $H_Q(x)$. Multiplying (GRF14) by the remaining factors in (GRF11), and using $e^{-\pi q \sqrt{x}/2} \leq 1$ and $\mathfrak{r}(q\sqrt{x}) \leq r_+$, bounds each entire inner block by

$$\frac{32r_+ e^{3\Lambda_\varepsilon/2} m_\varepsilon^4}{r_{-, \varepsilon}} 8^{-q} \|P_\varepsilon\|_\varepsilon^2 \int_0^{b_\varepsilon^2} x^{\varepsilon-3/4} dx = \eta_\varepsilon \|P_\varepsilon\|_\varepsilon^2.$$

The integral is finite for both $\varepsilon = 0, 1$, and its exact value is $b_\varepsilon^{2\varepsilon+1/2}/(\varepsilon + 1/4)$. Thus this estimate includes the endpoint zero and every neighborhood of it.

Define the finite quantity

$$C_\chi = \max_{\varepsilon=0,1} \left\{ \frac{D_2}{q^2 b_\varepsilon^2} + \frac{M_4}{2q^4 b_\varepsilon^4 (1 - R^2/(q^2 b_\varepsilon^2))} \right\}. \quad (\text{GRF18})$$

Its denominators are positive, at least $3/4$ in the parenthesis, by (GRF16).

Theorem 3. *Under (GRF16) and $\eta < 1$, for every complex polynomial p of degree at most r ,*

$$e^{-C_\chi} (1 - \eta) \mathcal{N}_0[p] \leq \mathcal{N}_\chi[p] \leq (1 + \eta) \mathcal{N}_0[p]. \quad (\text{GRF19})$$

Proof. In block ε , the outer interval $x \geq b_\varepsilon^2$ corresponds to $|y| \geq qb_\varepsilon \geq 2R$. By (GRF6) and monotonicity of its displayed majorants there, the density ratio $H_Q(x)/x^q$ is at least e^{-C_χ} . By (GRF7) the same ratio is at most 1. Write $N_\varepsilon = \|P_\varepsilon\|_\varepsilon^2$. The proved homogeneous inner bound implies its homogeneous outer contribution is at least $(1 - \eta_\varepsilon)N_\varepsilon$. Positivity of the root inner contribution then gives the lower bound $e^{-C_\chi}(1 - \eta_\varepsilon)N_\varepsilon$ for the root block. For the upper bound, its outer contribution is at most N_ε and its entire inner contribution is at most $\eta_\varepsilon N_\varepsilon$. Sum the two block inequalities with the common, unchanged factor $\sqrt{2}q^{2q+1/2}$ of (GRF10)–(GRF11). The result is (GRF19). No assumption on the location of polynomial mass and no monic minimization has been used. $\square$

218.5 Proof of the broad growing-ratio range

Theorem 4. *Keep the fixed original packet and let $k = 4l + 1 \rightarrow \infty$. Let $r = r_k$ be an integer, $t = r/q$, and suppose*

$$t \rightarrow \infty, \quad \log t = o(q/R). \quad (\text{GRF20})$$

Then both blocks eventually satisfy every guard in Theorem 3. More precisely,

$$u_\varepsilon \sim \frac{2}{\log t}, \quad v_\varepsilon \sim 2t, \quad b_\varepsilon \sim \frac{u_\varepsilon}{2e} \sim \frac{1}{e \log t}, \quad (\text{GRF21})$$

$$C_\chi = O_h\left(\frac{R^2}{q}(1 + \log t)^2\right) = o(q), \quad \eta = \exp[-q \log 8 + o(q)]. \quad (\text{GRF22})$$

The corresponding original reference endpoint u_t at $(\alpha, \beta) = (2/t, \pi/t)$ has $u_t \sim 2/\log t$, so $b_\varepsilon \sim u_t/(2e)$ and $qb_\varepsilon \sim q/(e \log t)$.

Proof. All limits below hold for each of the two possible block indices, so are uniform over that finite set. Put

$$c_\varepsilon = \varepsilon - 3/4, \quad d_\varepsilon = r - 2m_\varepsilon \in \{\varepsilon, \varepsilon + 1\}, \quad A_\varepsilon = 1 + c_\varepsilon/q, \quad B_\varepsilon = A_\varepsilon + 2m_\varepsilon/q.$$

These are exact identities; in particular $B_\varepsilon = 1 + t + (c_\varepsilon - d_\varepsilon)/q$. The endpoint equations (GRF2) at the actual parameters become

$$u_\varepsilon K(\kappa_\varepsilon) = 2A_\varepsilon, \quad v_\varepsilon E(\kappa_\varepsilon) = 2B_\varepsilon. \quad (\text{GRF23})$$

Here $A_\varepsilon \rightarrow 1$, $B_\varepsilon/t \rightarrow 1$, and $m_\varepsilon \sim qt/2$. Thus $m_\varepsilon \geq 1$ eventually.

We prove the endpoint asymptotics by the defining elliptic integrals, without importing a uniform remainder estimate. Write $s = u/v \in (0, 1)$, $L = \log(1/s)$, and suppress the block index for this argument. The substitution $z = \tan \theta$ gives exactly

$$K(\sqrt{1 - s^2}) = \int_0^\infty \frac{dz}{\sqrt{(1 + z^2)(1 + s^2 z^2)}}.$$

The integral on $[0, 1]$ is at most 1, and that on $[1/s, \infty)$ is at most $\int_{1/s}^\infty dz/(sz^2) = 1$. On $[1, 1/s]$ the integrand is $z^{-1}(1 + z^{-2})^{-1/2}(1 + s^2 z^2)^{-1/2}$. For nonnegative a, b , $0 \leq 1 - (1 + a)^{-1/2}(1 + b)^{-1/2} \leq (a + b)/2$; this follows by applying $(1 + x)^{-1/2} \geq 1 - x/2$ separately and using $1 - AB = (1 - A) + A(1 - B)$ with $0 < A \leq 1$. The difference of the middle integral from $\int_1^{1/s} dz/z = L$ has magnitude at most

$$\frac{1}{2} \int_1^{1/s} (z^{-3} + s^2 z) dz = \frac{1 - s^2}{2} \leq \frac{1}{2}.$$

Consequently

$$L - \frac{1}{2} \leq K(\sqrt{1 - s^2}) \leq L + 2. \quad (\text{GRF24})$$

The defining E integral has $1 \leq E \leq \pi/2$ and $E \rightarrow 1$ as $s \rightarrow 0$, by dominated convergence to the integral of $\cos \theta$. By (GRF23),

$$\frac{sK}{E} = \frac{A}{B} \rightarrow 0. \quad (\text{GRF25})$$

Since $K \geq \pi/2$ and $E \leq \pi/2$, this implies $s \rightarrow 0$ and $L \rightarrow \infty$. Taking logarithms in (GRF25) gives

$$L = \log(B/A) + \log K - \log E = \log t + \log L + O(1).$$

Here (GRF24) gives $\log K = \log L + o(1)$ and $\log(B/A) = \log t + o(1)$. As $L \rightarrow \infty$, division of $\log t = L - \log L + O(1)$ by L proves $L \sim \log t$ and then $K \sim \log t$. Now (GRF23) yields

$$u \sim 2/\log t, \quad v \sim 2t, \quad s = O(1/(t \log t)). \quad (\text{GRF26})$$

The same proof with $A = 1$, $B = 1 + t$ is exactly the claimed statement for u_t at $(2/t, \pi/t)$.

The exact expression (GRF13) gives

$$\frac{\Sigma_\varepsilon}{2q} = A_\varepsilon \left[1 + \log \frac{1 + s_\varepsilon}{2u_\varepsilon} \right] + \frac{m_\varepsilon}{q} \log \frac{1 + s_\varepsilon}{1 - s_\varepsilon}. \quad (\text{GRF27})$$

For $0 < s \leq 1/2$, $0 \leq \log((1+s)/(1-s)) \leq 4s$, by integration of $2/(1-s^2)$. Thus the last term is $O(1/\log t) = o(1)$ by (GRF26). Also $\log(1+s) = o(1)$. Since $A_\varepsilon - 1 = O(1/q)$ and $1 + |\log u_\varepsilon| = O(1 + \log \log t)$, the error from replacing A_ε by 1 is $O((1 + \log \log t)/q) = o(1)$. The last conclusion follows from (GRF20): for large k , $\log t \leq q/R \leq q$, so $\log \log t = O(\log q) = o(q)$. Therefore

$$\frac{\Sigma_\varepsilon}{2q} = 1 + \log \frac{1}{2u_\varepsilon} + o(1), \quad \frac{1}{4} e^{-\Sigma_\varepsilon/(2q)} = \frac{u_\varepsilon}{2e} (1 + o(1)).$$

Since $1/e < 1$, this second quantity is eventually the smaller member in (GRF12), proving (GRF21).

We now verify every remaining finite guard and error. The support width satisfies $v_\varepsilon^2 - u_\varepsilon^2 \sim 4t^2$, so

$$m_\varepsilon(v_\varepsilon^2 - u_\varepsilon^2) \sim 2qt^3 \rightarrow \infty.$$

Furthermore

$$\frac{R}{qb_\varepsilon} \sim \frac{eR \log t}{q} \rightarrow 0,$$

which proves the cutoff guard. The exact parameters satisfy $\alpha_\varepsilon = O(1/t)$ and $\beta_\varepsilon = O(1/t)$, whence

$$\Lambda_\varepsilon = O((\log t)^2/t + \log t/t) = O((\log t)^2/t) = o(1).$$

Since $q/\log t \rightarrow \infty$, one has $qu_\varepsilon \rightarrow \infty$ and $r_{-, \varepsilon} \rightarrow c_\Gamma/\sqrt{2} > 0$. Taking logarithms in the exact (GRF17) gives

$$\begin{aligned} \log \eta_\varepsilon = & -q \log 8 + 4 \log m_\varepsilon + (2\varepsilon + 1/2) \log b_\varepsilon + \frac{3}{2} \Lambda_\varepsilon \\ & + \log \frac{32r_+}{r_{-, \varepsilon}(\varepsilon + 1/4)}. \end{aligned}$$

Here $\log m_\varepsilon = \log q + \log t + O(1) = o(q)$: $\log q = o(q)$ and (GRF20), with $R \rightarrow \infty$, gives $\log t = o(q)$. Also $\log b_\varepsilon = -\log \log t + O(1) = o(q)$, and all other displayed terms are $o(q)$. Both block errors are therefore $\exp[-q \log 8 + o(q)]$; their maximum has the same form and is eventually less than 1.

Finally $D_2 \leq qR^2$ by its root-sum representation and $M_4 \leq qR^4$. The two terms in (GRF18) are bounded by

$$\frac{R^2}{qb_\varepsilon^2} \left(1 + \frac{R^2/(q^2 b_\varepsilon^2)}{2(1 - R^2/(q^2 b_\varepsilon^2))} \right).$$

The parenthesis tends to 1 and $b_\varepsilon^{-2} = O((1 + \log t)^2)$. This proves the first estimate in (GRF22). Dividing it by q and using $R \log t/q \rightarrow 0$ proves $C_\chi = o(q)$. $\square$

218.6 Source pins and exact audit boundary

The received text was read in full before reconstruction. Its SHA256 is

b89f991d351b9a8b7cc2cf76768eb9997d79c50e8df03929fe82ea427e4a60ef.

The source file `14_CURRENT_KERNEL_AND_MULTIPLICITY_PROOFS.tex` in `output/Split_Zero_Web_Continuation_CURRENT` has SHA256

9926c5b5670777548f15f3fd626a4237d2c10a26834f51ba2b466a3b3a349656.

The identical immutable receiving copy is stored in `work/growing_ratio_math_integration_20260917/baseline_692`. The exact inputs inspected there are EIQ1–33, lines 11082–11539; OOQ7 and its complete proof, lines 13597–13641; ECL6, lines 18491–18505, and ECL14, lines 18664–18674; PCRP1–8, lines 57079–57169; RGC1–2, lines 57416–57447; and the earlier explicitly reference-only endpoint statement UCT40–42, lines 56636–56660. The related `02_FULL_GAMMA_PROOFS.tex` has SHA256

c226f7110ed7929b62c17b43bc6759d089a72e766fca387ebffcd80723a3c08;

its original mass and density declarations HMR19–20 are at lines 15889–15902. The complete EIQ provider is also present there at lines 1897–2344; its ECL provider begins at line 18675.

The direct elementary elliptic-integral proof in Theorem 4 does not rely on the “standard complete-elliptic expansions” cited by the earlier UCT40 paragraph. The evaluation proof independently reconstructs OOQ7 with the actual lower density and derivative constant, rather than using its earlier compact-parameter supremum. Thus no compact-parameter bound is extrapolated to $(\alpha_\varepsilon, \beta_\varepsilon) \rightarrow (0, 0)$. The two guards $m_\varepsilon \geq 1$ are essential to the displayed definitions; finite degrees $r < 3$ are outside this finite theorem, without affecting the eventually growing-ratio conclusion.

219 The reference norm at its actual moving parameters

The coordinate throughout is the literal coordinate $y = q\sqrt{x}$ on the positive half-line. The underlying measure is

$$d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{GRE1})$$

It is not divided by its mass. Fix $q > 0$, an integer $m \geq 1$, and $\alpha, \beta > 0$. Write $\mu = \mu_{\alpha, \beta}$ for the probability measure constructed in the complete EIQ proof included in the accompanying complete source, with support $[a, b] = [u^2, v^2]$, $0 < u < v$, logarithmic potential $P(x) = \int \log|x - s| d\mu(s)$, and Robin constant λ in the convention

$$V(x) = \beta\sqrt{x} - \alpha \log x, \quad D(x) = V(x) - 2P(x) - \lambda \geq 0 \quad (x > 0). \quad (\text{GRE2})$$

Equality holds on $[a, b]$ and is strict outside it. The complete EIQ argument, included in the accompanying complete source, constructs the endpoints for every point of $(0, \infty)^2$, proves positivity and mass one, computes the Cauchy transform, and proves this inequality on both components of the exterior. In particular the inequality is available near zero as well as at infinity. Put

$$W = v^2 - u^2, \quad c = W/4, \quad w = (u + v)/2, \quad z = uv. \quad (\text{GRE3})$$

Here c is logarithmic capacity, and z is a positive scalar, not a complex integration variable.

219.1 Exact Robin and logarithmic-moment identities

The complete ECL1–14 proof is included in the accompanying complete source. Its moving-support derivative has the following precise test-function meaning:

$$\frac{\partial}{\partial \alpha} \int f(x) d\mu_{\alpha, \beta}(x) = \frac{1}{2} \int_a^b f(x) \left(1 - \frac{z}{x}\right) \frac{dx}{\pi \sqrt{(b-x)(x-a)}}. \quad (\text{GRE4})$$

It holds at fixed β for every continuously differentiable f on a neighbourhood of the support. To track the endpoint terms, write $p(x) = h(x)/(2\pi)$ and $\rho(x) = p(x)\sqrt{(b-x)(x-a)}$. Differentiation at the actual moving endpoints gives the signed density

$$\left[\dot{p}(x)\sqrt{(b-x)(x-a)} + \frac{p(x)\{\dot{b}(x-a) - \dot{a}(b-x)\}}{2\sqrt{(b-x)(x-a)}} \right] \mathbf{1}_{(a,b)}(x). \quad (\text{GRE5})$$

The proof changes variables to $x = a + (b-a)s$, $0 \leq s \leq 1$, differentiates there, and integrates the term containing f' by parts. The boundary term is proportional to $\sqrt{s(1-s)}$ and vanishes. The remaining derivative is integrable, bounded by a constant times $[s(1-s)]^{-1/2}$. Thus no endpoint atom is present and no moving-support term was dropped. The complete transform and zero-mass logarithmic-energy calculation in ECL7–10 identifies (GRE5) exactly with (GRE4).

In particular, setting $\mathcal{L} = P(0) = \int \log x d\mu(x)$ gives

$$\begin{aligned} \lambda &= 2(\alpha + 2) - 2\alpha \log w - 2 \log c, \\ \mathcal{L} &= 2(\alpha + 1) \log w - \alpha \log z - 2, \\ \lambda_\alpha &= -2 \log w, \quad \mathcal{L}_\alpha = 2 \log w - \log z. \end{aligned} \quad (\text{GRE6})$$

The constant -2 in the logarithmic moment remains. Direct addition, using $w^2/c = (u+v)/(v-u)$, yields

$$\boxed{2P(0) + \lambda = 2\alpha \left(1 + \log \frac{w}{z}\right) + 2 \log \frac{v+u}{v-u}}. \quad (\text{GRE7})$$

This is the received equation (6), with the original potential and Robin sign convention.

For clarity, the second Robin derivative also retains all support movement. For $\beta_0 > 0$ put $c_0 = \beta_0/\beta$. The dilation $x \mapsto c_0^2 x$ pushes μ_{α, β_0} to $\mu_{\alpha, \beta}$: on every probability measure its energy changes by the constant $-2(\alpha + 1) \log c_0$, so the proved unique minimizer is carried to the proved unique minimizer. At corresponding points its potential is $2 \log c_0 + P_{\alpha, \beta_0}(x)$, while its external field is $V_{\alpha, \beta_0}(x) - 2\alpha \log c_0$. Consequently

$$\lambda(\alpha, \beta) = \lambda(\alpha, \beta_0) + 2(\alpha + 2) \log(\beta/\beta_0), \quad \boxed{\lambda_\beta = \frac{2(\alpha + 2)}{\beta}}. \quad (\text{GRE8})$$

The distinct identity $\int \sqrt{x} d\mu = 2(\alpha + 1)/\beta$ concerns the moment, not the Robin derivative. It follows by differentiating the energy of the competitor $x \mapsto c_0^2 x$ at $c_0 = 1$ as in EIQ31.

219.2 Original Gamma constants, mass, and monic factorials

Retain exactly

$$c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \quad \mathfrak{r}(y) = \frac{\sigma(y)e^{\pi y/2}\sqrt{y}}{\sqrt{2}} \quad (y > 0). \quad (\text{GRE9})$$

The identity rederived here is Binet's integral formula in the human-authored Gamma chapter of Askey and Roy [H22, eq. (5.9.10_2)], valid for $\Re z > 0$. Its integration variable t is the present s , and its kernel is exactly

$$\frac{1}{s} \left(\frac{1}{e^s - 1} - \frac{1}{s} + \frac{1}{2} \right) = \frac{\coth(s/2) - 2/s}{2s} = K(s).$$

The holomorphic logarithm of Γ is the branch real on the positive axis; $\log z$ is principal on this half-plane. At $z = \zeta = 1/4 + iy/2$ the local BI17–19 bound gives $|R(\zeta)| \leq (12\Re\zeta)^{-1} = 1/3$. With the retained density $\sigma(y) = |\Gamma(\zeta)|^2/(2\pi)$, twice the real part gives GRE10. The explicit constants in GRE9–11 are then obtained by the displayed local inequalities.

The complete derivation of Binet's identity from the defining Gamma integral is reproduced in the accompanying complete source as BI1–26. Its remainder kernel is $K(s) = (\coth(s/2) - 2/s)/(2s)$ with $0 < K(s) < 1/12$. Therefore at $\zeta = 1/4 + iy/2$ its remainder satisfies $|R(\zeta)| \leq 1/3$, and twice the real part of that exact identity gives, with $Y = |y|$,

$$\sigma(y) = |\zeta|^{-1/2} \exp\{-Y \arctan(2Y) - 1/2 + 2\Re R(\zeta)\}. \quad (\text{GRE10})$$

For $Y > 0$, $Y \arctan(1/(2Y)) \leq 1/2$, whereas $\arctan(2Y) \leq \pi/2$; the same resulting inequalities hold at $Y = 0$. Thus the exponent in (GRE10) lies between $-\pi Y/2 - 7/6$ and $-\pi Y/2 + 2/3$. Also

$$\frac{(1+Y)^2}{4} - |\zeta|^2 = \frac{3+8Y}{16}, \quad |\zeta|^2 - \frac{(1+Y)^2}{20} = \frac{(4Y-1)^2}{80}.$$

Taking the appropriate positive fourth roots proves

$$c_\Gamma e^{-\pi|y|/2}(1+|y|)^{-1/2} \leq \sigma(y) \leq C_\Gamma e^{-\pi|y|/2}(1+|y|)^{-1/2}. \quad (\text{GRE11})$$

In particular all polynomial moments are finite and

$$r_+ = C_\Gamma/\sqrt{2}, \quad r_- = (c_\Gamma/\sqrt{2})\sqrt{\frac{qu}{1+qu}}, \quad \mathfrak{r}(y) \leq r_+, \quad \mathfrak{r}(q\sqrt{x}) \geq r_- \quad (a \leq x \leq b). \quad (\text{GRE12})$$

The lower estimate follows from the increasing function $y/(1+y)$.

For completeness, the mass and factorial monic norms can be recovered without selecting another measure. The beta identity, proved in BI13, with $a > 0$ and the substitution $x = \log(t/(1-t))$, gives

$$\frac{\Gamma(a+i\eta)\Gamma(a-i\eta)}{\Gamma(2a)} = \int_{\mathbb{R}} e^{i\eta x} (2 \cosh(x/2))^{-2a} dx. \quad (\text{GRE13})$$

The beta integrand after substitution is absolutely integrable. At $a = 1/4$, Fourier inversion therefore gives

$$\int_{\mathbb{R}} e^{i\xi y} d\sigma(y) = \sqrt{2\pi}(\cosh \xi)^{-1/2}. \quad (\text{GRE14})$$

Here Fourier inversion can be justified directly by inserting a Gaussian factor $e^{-\delta\eta^2}$ in the inversion integral, using Fubini, and evaluating the Gaussian Fourier integral. The result is convolution with the probability density $(4\pi\delta)^{-1/2}e^{-x^2/(4\delta)}$, which tends to the continuous beta density at each point: split at $|x| = \varepsilon$, use uniform continuity on the first piece and the Gaussian tail on the second. On the Fourier side dominated convergence is valid by (GRE11), with $y = 2\eta$. This proves the inversion used in (GRE14). Both sides of (GRE14) extend holomorphically to $|\Im \xi| < \pi/2$: for the left side (GRE11) gives locally uniform exponential domination; for the right side the branch is positive at zero and has no zero of $\cosh \xi$ in the strip. The identity theorem consequently gives

$$\int e^{\theta y} d\sigma(y) = \sqrt{2\pi}(\cos \theta)^{-1/2} \quad (|\theta| < \pi/2). \quad (\text{GRE15})$$

This retains mass $\sqrt{2\pi}$ at $\theta = 0$.

Define monic polynomials p_n by the convergent generating series

$$F(y, z) = (1 + z^2)^{-1/4} e^{y \arctan z} = \sum_{n \geq 0} p_n(y) \frac{z^n}{n!}.$$

The coefficient of y^n in p_n is one, because $\arctan z = z + O(z^3)$ and the other prefactor has value one at zero. For sufficiently small real z, w , (GRE15) and $\cos(\arctan z + \arctan w) = (1 - zw)/\sqrt{(1 + z^2)(1 + w^2)}$ give the exact generating identity

$$\int F(y, z) F(y, w) d\sigma(y) = \sqrt{2\pi}(1 - zw)^{-1/2}.$$

Termwise differentiation at zero is justified by (GRE11) and exponential domination in a neighbourhood of $(z, w) = (0, 0)$. Hence

$$\int p_n p_j d\sigma = \mathbf{1}_{n=j} \sqrt{2\pi} n! (1/2)_n, \quad \gamma_n = \sqrt{2\pi} n! (1/2)_n = \sqrt{2\pi} \frac{(2n)!}{4^n}. \quad (\text{GRE16})$$

For a complex monic degree- n polynomial, subtraction of p_n leaves degree at most $n-1$, orthogonal to p_n . Pythagoras proves that γ_n is precisely the monic minimum; the product equality in (GRE16) follows by separating the even and odd factors of $(2n)!$.

219.3 A global density bound and a full tail integral

The actual positive density in EIQ10–11 obeys

$$0 < h(x) \leq \frac{\alpha}{xuv}, \quad \rho(x) = \frac{\sqrt{(v^2 - x)(x - u^2)}}{2\pi} h(x).$$

For $u^2 \leq x \leq v^2$, $(v^2 - x)(x - u^2) \leq v^2 x$. It follows, including the zero endpoint values, that

$$\boxed{\|\rho\|_{\infty} \leq \mathcal{M} := \frac{\alpha}{2\pi u^2}}. \quad (\text{GRE17})$$

This is the received equation (16).

Set, at these very parameters,

$$C_{\lambda} = 2 \log 2 + \max(\lambda, 0),$$

$$T_* = \max \left\{ 1, v, \left(\frac{8(\alpha + 2)}{\beta} \right)^2, \frac{4C_{\lambda}}{\beta} \right\}, \quad J_* = T_*^2 + \frac{8}{\beta^2}. \quad (\text{GRE18})$$

For $s \geq \max\{1, v\}$ and every support point y , $|s^2 - y| \leq s^2 + v^2 \leq 2s^2$. The logarithm is integrable even if $s = v$, so integration gives $P(s^2) \leq 2 \log s + \log 2$. Therefore

$$D(s^2) \geq \beta s - 2(\alpha + 2) \log s - C_\lambda.$$

For $s \geq 1$, $\log s \leq \sqrt{s}$: the maximum of $\log s / \sqrt{s}$ is $2/e < 1$, obtained by direct differentiation. If $s \geq T_*$, then $2(\alpha + 2) \log s \leq 2(\alpha + 2) \sqrt{s} \leq \beta s / 4$ and $C_\lambda \leq \beta s / 4$. Consequently

$$D(s^2) \geq \beta s / 2 \quad (s \geq T_*), \quad \int_0^\infty e^{-mD(x)} dx \leq T_*^2 + \int_0^\infty 2s e^{-\beta s / 2} ds = J_*. \quad (\text{GRE19})$$

The first interval uses $D \geq 0$, and the tail uses $m \geq 1$. The integral starts at zero. All constants are finite at each point of the full positive parameter quadrant.

219.4 Arcsine lower bound and explicit quantile upper bound

Define the actual monic minimum

$$h_{m,q}(\alpha, \beta) = \min_{p \in \mathbb{C}[x], p \text{ monic}, \deg p = m} \int_0^\infty |p(x)|^2 e^{-mV(x)} \mathfrak{r}(q\sqrt{x}) dx. \quad (\text{GRE20})$$

The integrals are finite: the weight near zero is at most $r_+ x^{m\alpha}$ and at infinity at most $r_+ x^{m\alpha} e^{-m\beta\sqrt{x}}$. Its density is positive on $(0, \infty)$, so its finite coefficient Gram matrix is positive definite. Completing the square in that finite matrix proves existence and uniqueness of the monic minimizer.

Let $d\nu = dx / (\pi \sqrt{(b-x)(x-a)})$ be arcsine probability on $[a, b]$. Its potential satisfies

$$\int \log |x - \zeta| d\nu(x) \geq \log c \quad (\zeta \in \mathbb{C}), \quad \int \log |x - y| d\nu(x) = \log c \quad (y \in [a, b]). \quad (\text{GRE21})$$

To prove both statements rootwise, put $C = (a + b)/2$, $d = W/2$ and $x = C + d \cos \theta$. The polynomial $Z^2 - 2(\zeta - C)Z/d + 1$ has roots z_1, z_2 with product one. On the unit circle, $|x - \zeta| = (d/2)|Z - z_1||Z - z_2|$. The circle mean of $\log |Z - z_j|$ equals $\log \max\{1, |z_j|\}$. For $|z_j| < 1$ this follows by integrating the absolutely convergent logarithmic series, for $|z_j| > 1$ by factoring $|z_j|$, and for $|z_j| = 1$ by a limit dominated near the single singular angle by a constant plus $|\log |\theta||$. The dependence on $\cos \theta$ identifies the whole-circle mean with the arcsine integral. Both roots have modulus one precisely when $\zeta \in [a, b]$, proving the equality; their two log-max terms are nonnegative, proving the inequality. All logarithmic integrals just used are absolutely integrable.

For a monic degree- m polynomial p , factorization and (GRE21) give $\int \log |p| d\nu \geq m \log c$. Integrate the Euler equality against ν . Fubini is justified by the bounded equilibrium density and the integrable arcsine logarithmic singularity, and gives

$$\int V d\nu = 2 \log c + \lambda. \quad (\text{GRE22})$$

The arcsine entropy is exactly

$$\int \log \frac{d\nu}{dx} d\nu = -\log(\pi c). \quad (\text{GRE23})$$

Indeed $\sqrt{(b-x)(x-a)} = (W/2) \sin \theta$, and $\pi^{-1} \int_0^\pi \log \sin \theta d\theta = -\log 2$. For the latter identity, put $I = \int_0^{\pi/2} \log \sin \theta d\theta$. Symmetry gives the same cosine integral, while doubling the argument in $\log \sin \theta + \log \cos \theta = \log \sin(2\theta) - \log 2$ gives $2I = I - (\pi/2) \log 2$. Endpoint logarithms are integrable.

Probability Jensen applied to $|p|^2 e^{-mV} / (d\nu/dx)$, whose logarithm is integrable, now proves

$$\int_a^b |p|^2 e^{-mV} dx \geq \exp \left\{ 2 \int \log |p| d\nu - m \int V d\nu - \int \log(d\nu/dx) d\nu \right\} \geq \pi c e^{-m\lambda}. \quad (\text{GRE24})$$

Multiplication by the lower remainder bound in (GRE12) gives the claimed lower monic bound.

For the upper bound take the unique midpoint quantiles $x_i \in (a, b)$ with $\mu((0, x_i]) = (i-1/2)/m$, $1 \leq i \leq m$, and define $p_m(x) = \prod_{i=1}^m (x - x_i)$. The distribution function of its empirical probability measure differs from that of μ by at most $1/(2m)$, and hence by at most $1/m$. For fixed $x > 0$ put

$$\delta = 1/m, \quad f_x(y) = \log \max\{|x - y|, \delta\}.$$

This is continuous and of bounded variation on $[a, b]$. If x lies inside $[a, b]$, its decreasing and increasing pieces each have variation at most $\log(1 + W/\delta)$. If x lies outside the interval it is monotone and its endpoint ratio after clipping is at most $1 + W/\delta$. Thus, in all cases, $\text{Var}_{[a,b]} f_x \leq 2 \log(1 + mW)$. Stieltjes integration by parts against the zero-mass empirical error has zero endpoint term, and gives

$$\sum_{i=1}^m f_x(x_i) \leq m \int f_x d\mu + 2 \log(1 + mW). \quad (\text{GRE25})$$

By (GRE17),

$$\int (f_x(y) - \log|x - y|) d\mu(y) \leq \mathcal{M} \int_{-\delta}^{\delta} \log(\delta/|s|) ds = 2\mathcal{M}\delta. \quad (\text{GRE26})$$

Each nodal logarithm is at most its clipped value; at a root its value is $-\infty$, where the resulting inequality is automatic. Combining (GRE25)–(GRE26), then multiplying by e^{-mV} , yields the global positive-half-line estimate

$$|p_m(x)|^2 e^{-mV(x)} \leq e^{4\mathcal{M}} (1 + mW)^4 e^{-m\lambda} e^{-mD(x)} \quad (x > 0). \quad (\text{GRE27})$$

Integrating it with (GRE12) and (GRE19) proves the complete finite norm theorem

$$\boxed{c_- e^{-m\lambda} \leq h_{m,q}(\alpha, \beta) \leq c_+ e^{-m\lambda}, \quad c_- = r_- \pi W/4, \quad c_+ = r_+ J_* e^{4\mathcal{M}} (1 + mW)^4.} \quad (\text{GRE28})$$

This is the received (17). Neither its proof nor its constants require a compact parameter rectangle, an endpoint lower density estimate, or a lower bound on quantile-cell lengths.

219.5 Exact parity map, shifts, and factorial cancellation

Let $q \geq 1$ be an integer and $r = 2m + \varepsilon$ an integer with $m \geq 1$, $\varepsilon \in \{0, 1\}$. Define the full homogeneous reference monic minimum

$$\nu_r^0 = \min_{p \text{ monic}, \deg p = r} \int_{\mathbb{R}} y^{2q} |p(y)|^2 d\sigma(y). \quad (\text{GRE29})$$

An even positive weight makes the even and odd coefficient spaces orthogonal. For any monic degree- r polynomial, the component with parity opposite to r contributes a nonnegative separate squared norm and has no leading term, so the unique monic minimizer has parity r . The exact bijection between its parity fibre and the monic degree- m fibre is

$$p(y) = q^r (y/q)^\varepsilon P(y^2/q^2). \quad (\text{GRE30})$$

The leading coefficient is $q^{r-\varepsilon-2m} = 1$, and expansion in powers proves surjectivity and injectivity. Both half-lines contribute equally. Substitution $y = q\sqrt{x}$ has Jacobian $q/(2\sqrt{x})$, and (GRE9) gives exactly

$$\boxed{\nu_r^0 = \sqrt{2} q^{2(q+r)+1/2} h_{m,q}(\alpha, \beta), \quad \alpha = \frac{q + \varepsilon - 3/4}{m}, \quad \beta = \frac{\pi q}{2m}.} \quad (\text{GRE31})$$

Every mass and power, including the quarter shift, is retained.

Put $t = r/q$, $\alpha_0 = 2/t$, $\beta_0 = \pi/t$, and $\lambda_0 = \lambda(\alpha_0, \beta_0)$. The differences are exactly

$$A := m(\alpha - \alpha_0) = \varepsilon - \frac{3}{4} + \frac{q\varepsilon}{r}, \quad B := m(\beta - \beta_0) = \frac{\pi q\varepsilon}{2r}. \quad (\text{GRE32})$$

Let $(\alpha_\theta, \beta_\theta) = (\alpha_0, \beta_0) + \theta(\alpha - \alpha_0, \beta - \beta_0)$, $0 \leq \theta \leq 1$, and let w_θ be its actual midpoint $(u_\theta + v_\theta)/2$. All these parameter pairs are positive. The full support derivatives in (GRE6),(GRE8) and the fundamental theorem of calculus give

$$\begin{aligned}\Delta &:= m\lambda(\alpha, \beta) - \frac{r}{2}\lambda_0 \\ &= \int_0^1 \left[-2A \log w_\theta + B \frac{2(\alpha_\theta + 2)}{\beta_\theta} \right] d\theta - \frac{\varepsilon}{2}\lambda_0.\end{aligned}\tag{GRE33}$$

Thus the finite, explicitly evaluable parameter bound is

$$|\Delta| \leq B_{\text{par}} := 2|A| \int_0^1 |\log w_\theta| d\theta + 2|B| \int_0^1 \frac{\alpha_\theta + 2}{\beta_\theta} d\theta + \frac{\varepsilon}{2}|\lambda_0|.\tag{GRE34}$$

No fixed-endpoint differentiation is used in this formula.

Define the original profile in its exact Robin form

$$\psi(t) = (1+t)(1 - \log(1+t)) - \frac{t}{4}\lambda(2/t, \pi/t).\tag{GRE35}$$

To identify it explicitly, let $\kappa \in (0, 1)$ be the EIQ modulus and $s = \sqrt{1 - \kappa^2}$. The endpoint equations at these actual parameters give

$$u_t K = 2, \quad v_t E = 2(1+t), \quad 1+t = \frac{E}{sK}, \quad w_t = \frac{(1+t)(1+s)}{E}, \quad c_t = \frac{(1+t)^2 \kappa^2}{E^2}.$$

Inserting these formulas into (GRE6),(GRE35), with no endpoint change, gives

$$\psi(t) = \log \frac{1+s}{E} + t \log \frac{\kappa}{E}.\tag{GRE36}$$

This is precisely the original UCT26 profile.

Write $n = q + r$. BI26 evaluated at $2n$ and the recurrence $\Gamma(2n+1) = 2n\Gamma(2n)$ give

$$\log(2n)! = (2n + \tfrac{1}{2}) \log(2n) - 2n + \tfrac{1}{2} \log(2\pi) + R_{2n}, \quad 0 < R_{2n} < \frac{1}{24n}.$$

Combine this with (GRE16),(GRE31). Exact cancellation gives

$$\log \frac{\nu_r^0}{\gamma_{q+r}} = 2(q+r)(1 - \log(1+t)) + \log h_{m,q} - \tfrac{1}{2} \log(1+t) - \log(2\pi) - R_{2(q+r)}.\tag{GRE37}$$

Equivalently, for $X = \log h_{m,q} + m\lambda$,

$$\boxed{\log \frac{\nu_r^0}{\gamma_{q+r}} = 2q\psi(t) + X - \Delta - \tfrac{1}{2} \log(1+t) - \log(2\pi) - R_{2(q+r)}, \quad \log c_- \leq X \leq \log c_+}.\tag{GRE38}$$

The useful finite endpoints, before any asymptotic replacement, are also exactly

$$\log \frac{\sqrt{2}q^{2(q+r)+1/2}}{\gamma_{q+r}} - m\lambda + \log c_- \leq \log \frac{\nu_r^0}{\gamma_{q+r}} \leq \log \frac{\sqrt{2}q^{2(q+r)+1/2}}{\gamma_{q+r}} - m\lambda + \log c_+.\tag{GRE39}$$

219.6 Uniform size of the retained errors

The elementary elliptic proof included in the accompanying complete source establishes, uniformly also along the segment in (GRE33),

$$u_\theta \sim \frac{2}{\log t}, \quad v_\theta \sim 2t, \quad W \sim 4t^2, \quad w_\theta \sim t, \quad |\lambda_0| + |\lambda| = O(1 + \log t)\tag{GRE40}$$

as $q \rightarrow \infty$, $t \rightarrow \infty$. The segment has $\alpha_\theta = O(1/t)$ and $\beta_\theta \asymp 1/t$. Equations (GRE32)–(GRE34) consequently give $|\Delta| = O(1 + \log t)$, since $A = O(1)$ and $B = O(1/t)$. The same support estimates in the full finite constants give

$$\mathcal{M} = O((\log t)^2/t), \quad T_* = O(t^2), \quad J_* = O(t^4). \quad (\text{GRE41})$$

For T_* , its third entry is $O(t^2)$, its fourth is $O(t(1 + \log t)) = O(t^2)$, and its other entries are $O(t)$. These statements concern the actual endpoints, rather than compact rectangle extrema.

For the packet range of the received source, let $q = [1 + k(M - 1)](k + 1)^2$, with the fixed original multiplicity $M \geq 1$, $R_k = k\sqrt{\delta^2 + \gamma^2}$, and $\log t = o(q/R_k)$. Then $q/\log t \rightarrow \infty$, hence $qu \rightarrow \infty$ and r_- is bounded above and below by fixed positive constants. Thus $c_- \asymp t^2$, and

$$|\log c_-| + |\log c_+| = O(\log(q + r)). \quad (\text{GRE42})$$

Indeed $\mathcal{M} = O(1)$, $\log J_* = O(1 + \log t)$ and $mW = O(rt^2)$; since $q \geq 1$, $t = r/q \leq r$, so $\log(1 + rt^2) = O(\log(q + r))$. Lower bounds for J_* and $1 + mW$ are one, so this also bounds the absolute logarithm of c_+ , not only its positive part. The positive lower bound of r_+ is fixed. Similarly (GRE40) and $r_- \asymp 1$ bound both signs of $\log c_-$.

Finally (GRE38), (GRE34), (GRE42) and $0 < R_{2(q+r)} < 1/(24(q + r))$ prove

$$\boxed{\log \frac{\nu_r^0}{\gamma_{q+r}} = 2q\psi(r/q) + O(\log(q + r))}. \quad (\text{GRE43})$$

The constants are uniform along every such growing-ratio sequence and for either parity. This is the received equation (18). All finite content is already specified by (GRE18), (GRE28), (GRE33), and (GRE38)–(GRE39); no unspecified equilibrium norm theorem occurs.

220 Uniform elliptic endpoint estimates with the original quarter shift

Definitions and exact source identification. For $0 < s < 1$ define, by actual real integrals,

$$K_s = \int_0^{\pi/2} \frac{d\theta}{\sqrt{\cos^2 \theta + s^2 \sin^2 \theta}}, \quad E_s = \int_0^{\pi/2} \sqrt{\cos^2 \theta + s^2 \sin^2 \theta} d\theta. \quad (\text{GEL1})$$

Thus $K_s = K(\sqrt{1-s^2})$ and $E_s = E(\sqrt{1-s^2})$ in the modulus convention. A subscript s on K_s, E_s indicates the argument, not a derivative; derivatives are written d/ds . For $\alpha, \beta > 0$, let s solve

$$\frac{sK_s}{E_s} = \frac{\alpha}{\alpha+2}, \quad v = \frac{(\alpha+2)\pi}{\beta E_s}, \quad u = sv. \quad (\text{GEL2})$$

Retain the original coordinates and constants

$$a = u^2, \quad b = v^2, \quad W = b - a, \quad w = \frac{u+v}{2}, \quad c = \frac{W}{4}, \quad \lambda = 2(\alpha+2) - 2\alpha \log w - 2 \log c. \quad (\text{GEL3})$$

These are exactly EIQ2-5 and ECL3/6 of the frozen cumulative source 14_CURRENT_KERNEL_AND_MULTIPPLICITY_PR00FS.tex, whose checked SHA256 is

9926c5b5670777548f15f3fd626a4237d2c10a26834f51ba2b466a3b3a349656

In that source λ is denoted ℓ in EIQ/ECL. The statements below are proved for these functions from their displayed definitions; the source identifies them with the original equilibrium endpoints and Robin constant. No external asymptotic expansion is used.

Lemma 220.1 (Existence, derivative identities, and the logarithmic constant). *The equation for s in (GEL2) has exactly one solution in $(0, 1)$, smoothly depending on α . With $L = \log(4/s)$, as $s \downarrow 0$,*

$$K_s = L + O(s^2 L), \quad E_s = 1 + \frac{s^2}{2} \left(L - \frac{1}{2} \right) + O(s^4 L). \quad (\text{GEL4})$$

The constants in these bounds are absolute for $0 < s \leq 1/2$.

Proof. Differentiation under the integral sign, justified on each compact subinterval of $(0, 1)$, gives the first identity below. Integration of the derivative of $\sin \theta \cos \theta / (\cos^2 \theta + s^2 \sin^2 \theta)^{1/2}$ between its zero endpoint values gives the second:

$$\frac{dE_s}{ds} = \frac{s(K_s - E_s)}{1 - s^2}, \quad \frac{dK_s}{ds} = \frac{s^2 K_s - E_s}{s(1 - s^2)}. \quad (\text{GEL5})$$

For completeness the integration identity needed for the second is

$$\int_0^{\pi/2} \left\{ \frac{\cos^2 \theta - \sin^2 \theta}{D^{1/2}} + \frac{(1-s^2) \sin^2 \theta \cos^2 \theta}{D^{3/2}} \right\} d\theta = 0, \quad D = \cos^2 \theta + s^2 \sin^2 \theta.$$

Indeed $dK_s/ds = -s \int \sin^2 \theta D^{-3/2} d\theta$; substitution in this zero integral gives the displayed formula. The integrals show $K_s > E_s > s^2 K_s$ for $0 < s < 1$. Therefore

$$\frac{d}{ds} \left(\frac{sK_s}{E_s} \right) = \frac{(K_s - E_s)(E_s - s^2 K_s)}{(1 - s^2)E_s^2} > 0.$$

As $s \downarrow 0$, the integrand of sK_s tends to zero for $\theta < \pi/2$ and is bounded by 1, so $sK_s \rightarrow 0$, while $E_s \rightarrow 1$. As $s \uparrow 1$, both integrals tend to $\pi/2$. Continuity, strict monotonicity, and the nonzero derivative prove existence, uniqueness, and smooth dependence.

Set $x = \cot \theta$ in the first integral of (GEL1); exactly

$$K_s = \int_0^\infty \frac{dx}{\sqrt{1+x^2}\sqrt{s^2+x^2}} = \operatorname{arsinh}(1/s) + A_s + B_s,$$

where

$$A_s = \int_0^1 \frac{(1+x^2)^{-1/2} - 1}{\sqrt{s^2+x^2}} dx, \quad B_s = \int_1^\infty \frac{dx}{\sqrt{1+x^2}\sqrt{s^2+x^2}}.$$

Both have convergent values at $s = 0$. The identity

$$\frac{d}{dx} \log \frac{x}{1 + \sqrt{1+x^2}} = \frac{1}{x\sqrt{1+x^2}}$$

gives

$$A_0 = \log \frac{2}{1 + \sqrt{2}}, \quad B_0 = \log(1 + \sqrt{2}), \quad A_0 + B_0 = \log 2.$$

The elementary inequalities

$$0 \leq 1 - (1+x^2)^{-1/2} \leq \frac{x^2}{2}, \quad 0 \leq \frac{1}{x} - \frac{1}{\sqrt{x^2+s^2}} \leq \min \left\{ \frac{1}{x}, \frac{s^2}{2x^3} \right\}$$

give, by splitting $A_s - A_0$ at $x = s$,

$$|A_s - A_0| \leq \frac{s^2}{4} + \frac{s^2}{4} \log(1/s), \quad |B_s - B_0| \leq \frac{s^2}{2} \int_1^\infty \frac{dx}{x^4} = \frac{s^2}{6}.$$

Furthermore $0 \leq \operatorname{arsinh}(1/s) - \log(2/s) = \log((1 + \sqrt{1+s^2})/2) \leq s^2/4$. This proves the first expansion, including its exact constant 4.

The bounds $1 \leq E_s \leq \pi/2$, the first expansion, and the first identity in (GEL5) show $0 \leq dE_s/ds \leq Cs \log(4/s)$ for $0 < s \leq 1/2$. Its integral is finite at zero, so integration from 0 is justified by first integrating from a positive lower endpoint and passing to zero. Consequently $E_s - 1 = O(s^2 L)$. Substituting this and the expansion of K_s back in the exact derivative identity improves it to

$$\frac{dE_s}{ds} = s(\log(4/s) - 1) + O(s^3 \log(4/s)).$$

The exact antiderivative calculation

$$\int_0^s z(\log(4/z) - 1) dz = \frac{s^2}{2} \left(\log(4/s) - \frac{1}{2} \right), \quad \int_0^s z^3 \log(4/z) dz = \frac{s^4}{4} \left(\log(4/s) + \frac{1}{4} \right)$$

proves the second expansion. □

Theorem 220.2 (The whole moving parameter segment). *Let $q \rightarrow \infty$, $t = r/q \rightarrow \infty$, and $\varepsilon \in \{0, 1\}$, with $m = (r - \varepsilon)/2 > 0$. Define exactly*

$$\alpha = \frac{q + \varepsilon - 3/4}{m}, \quad \beta = \frac{\pi q}{2m}, \quad \alpha_0 = \frac{2}{t}, \quad \beta_0 = \frac{\pi}{t},$$

and for $0 \leq \theta \leq 1$ define $\alpha_\theta = (1 - \theta)\alpha_0 + \theta\alpha$ and $\beta_\theta = (1 - \theta)\beta_0 + \theta\beta$. Evaluate (GEL2)–(GEL3) at these parameters. All the following estimates hold uniformly in θ and ε :

$$L_\theta = \log t + \log \log t + \log 4 + o(1), \quad s_\theta \sim \frac{1}{t \log t}, \quad (\text{GEL6})$$

$$u_\theta \sim \frac{2}{\log t}, \quad v_\theta = 2t\{1 + O(1/t)\}, \quad (\text{GEL7})$$

$$W_\theta = 4t^2\{1 + O(1/t)\}, \quad w_\theta = t\{1 + O(1/t)\}, \quad (\text{GEL8})$$

$$\lambda(\alpha_\theta, \beta_\theta) = 4 - 4 \log t + O((\log t)/t). \quad (\text{GEL9})$$

In particular $\lambda = O(\log t)$ uniformly on the full segment.

Proof. Put $\delta = \varepsilon - 3/4$. The exact endpoint ratios are

$$\frac{\alpha}{\alpha_0} = \frac{1 + \delta/q}{1 - \varepsilon/r}, \quad \frac{\beta}{\beta_0} = \frac{1}{1 - \varepsilon/r}.$$

It follows uniformly on the segment that

$$\alpha_\theta = \frac{2}{t}\{1 + O(1/q)\}, \quad \beta_\theta = \frac{\pi}{t}\{1 + O(1/r)\}, \quad \frac{\alpha_\theta}{\alpha_\theta + 2} = \frac{1}{t+1}\{1 + O(1/q)\}.$$

The right side tends uniformly to zero. The strict monotonicity and positive values of sK_s/E_s proved above imply $s_\theta \rightarrow 0$ uniformly. Hence (GEL4) and (GEL2) give

$$s_\theta L_\theta = \frac{1}{t}\{1 + O(1/t + 1/q + s_\theta^2 L_\theta)\} = \frac{1}{t}\{1 + o(1)\}.$$

Since $s_\theta = 4e^{-L_\theta}$, taking logarithms gives

$$L_\theta - \log L_\theta = \log t + \log 4 + o(1).$$

Here $L_\theta \rightarrow \infty$ uniformly. Dividing by L_θ first shows $L_\theta/\log t \rightarrow 1$ uniformly. Substitution then gives $\log L_\theta = \log \log t + o(1)$ and proves (GEL6).

The original endpoint equation, without changing coordinates, gives

$$u_\theta = \frac{\pi \alpha_\theta}{\beta_\theta K_{s_\theta}} = \frac{2\{1 + O(1/q)\}}{L_\theta\{1 + O(s_\theta^2)\}} \sim \frac{2}{\log t}.$$

The other endpoint satisfies

$$\frac{v_\theta}{2t} = \frac{1 + \alpha_\theta/2}{1 + O(1/r)}\{1 + O(s_\theta^2 L_\theta)\} = 1 + O(1/t),$$

because $r = tq$ and $s_\theta^2 L_\theta \sim 1/(t^2 \log t)$. Now $u_\theta/v_\theta = s_\theta$, so $W_\theta = v_\theta^2(1 - s_\theta^2)$ and $w_\theta = v_\theta(1 + s_\theta)/2$ prove (GEL8). Thus $\log w_\theta = \log t + O(1/t)$ and $\log(W_\theta/4) = 2\log t + O(1/t)$. Substitution in the exact definition of λ , with $\alpha_\theta = O(1/t)$, proves (GEL9). $\square$

Proposition 220.3 (Exact Robin derivatives and the full quarter-shift cost). *For every $\alpha, \beta > 0$,*

$$\lambda_\alpha = -2 \log w, \quad \lambda_\beta = \frac{2(\alpha + 2)}{\beta}. \quad (\text{GEL10})$$

On the segment of the theorem, these imply uniformly

$$\lambda_\alpha = -2 \log t + O(1/t), \quad \lambda_\beta = \frac{4t}{\pi} + O(1).$$

In the original shifted parameters, including the change of degree,

$$m\lambda(\alpha, \beta) - \frac{r}{2}\lambda(2/t, \pi/t) = \frac{3}{2}\log t + O((\log t)/t). \quad (\text{GEL11})$$

The exact second derivatives are

$$\lambda_{\alpha\alpha} = -\frac{(1-s)^2 K_s(E_s - sK_s)}{(K_s - E_s)(E_s - s^2 K_s)}, \quad \lambda_{\alpha\beta} = \frac{2}{\beta}, \quad \lambda_{\beta\beta} = -\frac{2(\alpha + 2)}{\beta^2}. \quad (\text{GEL12})$$

They are respectively $-1 + O(1/\log t)$, $O(t)$, and $O(t^2)$ uniformly on that segment.

Proof. Set $Q = E_s - sK_s > 0$. Solving the endpoint relation exactly gives

$$\alpha = \frac{2sK_s}{Q}, \quad v = \frac{2\pi}{\beta Q}, \quad w = \frac{\pi(1+s)}{\beta Q}, \quad c = \frac{\pi^2(1-s^2)}{\beta^2 Q^2}.$$

The derivative identities already proved give

$$Q' = \frac{E_s - K_s}{1+s}, \quad \alpha' = \frac{2(K_s - E_s)(E_s - s^2 K_s)}{(1-s^2)Q^2} > 0, \quad \frac{d \log w}{ds} = \frac{(1-s)K_s}{(1+s)Q}$$

at fixed β . Direct differentiation of c gives

$$\frac{d \log c}{ds} = \frac{2\{(1-s+s^2)K_s - E_s\}}{(1-s^2)Q}.$$

These expressions imply $\alpha(d \log w/ds) + d \log c/ds = \alpha'$. To check every term, multiplication by $(1-s^2)Q^2/2$ reduces this identity to

$$s(1-s)^2 K_s^2 + \{(1-s+s^2)K_s - E_s\}(E_s - sK_s) = (K_s - E_s)(E_s - s^2 K_s),$$

whose two sides expand to $-s^2 K_s^2 + (1+s^2)E_s K_s - E_s^2$. Differentiating the definition of λ with respect to s now gives $d\lambda/ds = -2 \log w \alpha'$, proving the first identity in (GEL10). At fixed α , s is fixed, $w \propto \beta^{-1}$, and $c \propto \beta^{-2}$, which proves the second. The derivative estimates follow from the theorem and $\alpha_\theta = 2/t\{1 + O(1/q)\}$, $\beta_\theta = \pi/t\{1 + O(1/r)\}$; in particular the beta error is $O(1+t/r) = O(1)$.

Write $\lambda_0 = \lambda(2/t, \pi/t)$. The exact parameter differences are

$$m(\alpha - 2/t) = \delta + \varepsilon/t, \quad m(\beta - \pi/t) = \frac{\pi\varepsilon}{2t}, \quad m - \frac{r}{2} = -\frac{\varepsilon}{2}.$$

The fundamental theorem of calculus on the actual straight segment gives

$$\begin{aligned} m(\lambda - \lambda_0) &= \left(\delta + \frac{\varepsilon}{t}\right) \int_0^1 \lambda_\alpha(\alpha_\theta, \beta_\theta) d\theta + \frac{\pi\varepsilon}{2t} \int_0^1 \lambda_\beta(\alpha_\theta, \beta_\theta) d\theta \\ &= -2\delta \log t + 2\varepsilon + O((\log t)/t). \end{aligned}$$

Subtracting $\varepsilon\lambda_0/2$ and using $\lambda_0 = 4 - 4 \log t + O((\log t)/t)$ yields (GEL11), since $-2\delta + 2\varepsilon = 3/2$. Finally differentiating λ_α using the displayed values of $d \log w/ds$ and α' gives (GEL12); the other two derivatives follow directly from (GEL10). The expansions give $Q = 1 + O(sL)$ and $K_s/(K_s - E_s) = 1 + O(1/L)$, proving the stated uniform Hessian bounds. $\square$

Theorem 220.4 (The large-ratio profile from the real integrals). *For each $t > 0$, let s be determined by $1+t = E_s/(sK_s)$ and put $\kappa = \sqrt{1-s^2}$. Define the exact UCT26 profile*

$$\psi(t) = \log \frac{1+s}{E_s} + t \log \frac{\kappa}{E_s}.$$

Then, with $L = \log(4/s)$,

$$\psi(t) = \frac{s}{2} - \frac{s}{4L} + O(s^3 L), \quad \psi(t) \sim \frac{1}{2t \log t} \quad (t \rightarrow \infty). \quad (\text{GEL13})$$

Proof. The exact logarithmic identity $\log((1+s)/\kappa) = \frac{1}{2} \log((1+s)/(1-s))$ gives

$$\psi(t) = \operatorname{arctanh} s + \frac{E_s}{sK_s} \log \frac{\kappa}{E_s}.$$

Taylor's theorem for $\log(1+x)$, with its remainder bounded on a fixed neighbourhood of zero, and (GEL4) give

$$\log E_s = \frac{s^2}{2} \left(L - \frac{1}{2} \right) + O(s^4 L^2), \quad \log \kappa = -\frac{s^2}{2} + O(s^4),$$

so

$$\log \frac{\kappa}{E_s} = -\frac{s^2}{2} \left(L + \frac{1}{2} \right) + O(s^4 L^2), \quad \frac{E_s}{sK_s} = \frac{1}{sL} \{1 + O(s^2 L)\}.$$

Their product is $-s/2 - s/(4L) + O(s^3 L)$. Since $\operatorname{arctanh} s = s + O(s^3)$, this proves the first formula. The reference parameters are exactly $\alpha_0 = 2/t, \beta_0 = \pi/t$, so (GEL6) gives $s \sim 1/(t \log t)$ and $L \rightarrow \infty$; the remainder divided by s tends to zero because $s^2 L \rightarrow 0$. This proves the stated leading equivalent. $\square$

The original root coordinate. The foregoing results need only $q \rightarrow \infty$ and $t \rightarrow \infty$. If in addition q is comparable to k^2 or k^3 and $\log t = o(q/k)$, then in the original coordinate $y = q\sqrt{x}$, uniformly on the whole parameter segment,

$$\frac{qu_\theta}{k} \sim \frac{2q}{k \log t} \longrightarrow \infty.$$

Consequently for every fixed constant $C > 0$ and every root-radius sequence $0 \leq R_k \leq Ck$, one has $R_k/(qu_\theta) \rightarrow 0$ uniformly. This is an explicit consequence in the original coordinate, retaining both the factor q and the actual endpoint u_θ .

221 Original arithmetic transfer and the complete shifted action

This section proves the arithmetic receiving calculation for the moving equilibrium comparison. It retains the stipulated complete zero quartet, all source masses and divided jets, the original remainder quotient, and both adjacent contraction factors. The moving-cutoff full-form theorem and the moving-parameter reference-norm theorem are the analytic results proved in the companion sections. Their exact conclusions are displayed where used; none of their constants is replaced by a compact-parameter constant. The new calculation here is the exact arithmetic transfer, its degree-uniform asymptotic consequences, and its original complete-action receiver.

221.1 Objects, coefficient maps, and all-degree arithmetic constants

Fix the original complete quartet data

$$0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad m \in \mathbb{N}_{>0}, \quad h(z) = \prod_{\zeta \in \{1/2 \pm \delta \pm i\gamma\}} (z - \zeta)^m, \quad v_h = 2\xi/h.$$

The zero order at each centre is exactly the stated m , so the entire quotient v_h is nonzero at every centre. Along the original tensor subsequence put

$$\begin{aligned} k = 4\ell + 1 \geq 5, \quad c = k/2, \quad e_k = 1 + k(m-1), \quad q = e_k(k+1)^2, \quad R_k = k\sqrt{\delta^2 + \gamma^2}, \\ \lambda_{ab} = c + (2a - k)\delta + i(2b - k)\gamma, \quad \chi(S) = \prod_{a,b=0}^k (S - \lambda_{ab})^{e_k}, \\ Q(y) = i^{-q}\chi(c + iy), \quad E = \mathbb{C}[S]/(\chi), \quad w_h(y) = |v_h(1/2 + iy)|^2/(2\pi), \\ d\mu_k(y) = w_h^{*k}(y) dy, \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \end{aligned} \tag{GRA1}$$

The masses are $(\int w_h)^k$ and $\sqrt{2\pi}$, respectively. For $P \in \mathcal{P}_j(S) = \mathbb{C}[S]_{\leq j}$ the arithmetic squared norm is $\|P\|_{\text{ar}}^2 = \int |P(c + iy)|^2 d\mu_k(y)$; the corresponding source Gamma norm uses $d\sigma$ in the same coordinate. Inner products are conjugate-linear in their first argument. Every polynomial coefficient below may be complex.

The precise coefficient maps are

$$U_j P(y) = i^{-j} P(c + iy), \quad U_j^{-1} p(S) = i^j p((S - c)/i), \quad U_{q+j} M_\chi = M_Q U_j. \tag{GRA2}$$

Direct substitution proves both inverse identities and the multiplication identity. The leading coefficient of a polynomial of degree j is unchanged. The leading coefficient of a polynomial of degree $a < j$ is multiplied by i^{a-j} , which is retained. Thus U_j bijects the two degree- j monic affine spaces, and its scalar phase has modulus one in each pulled-back integral norm. No Euclidean norm of a coefficient vector is asserted to be invariant under translation.

Write $\alpha_* = \pi/2$ to distinguish this fixed tail constant from the moving equilibrium parameters. The original AMT/TW constants are

$$\begin{aligned} p_h = 8m - \frac{9}{2}, \quad B_h = 42 + 8m, \quad M_h = 21 + 4m, \\ \vartheta_h = \int_{-1}^1 w_h(y) dy, \quad M_{\alpha_*} = \int_{\mathbb{R}} e^{\alpha_* y} w_h(y) dy, \\ c_\Gamma = \sqrt{2}e^{-7/6}, \quad C_\Gamma = \sqrt{2\sqrt{5}}e^{2/3}, \\ a_k = \frac{c_h \vartheta_h^{k-3} e^{-\alpha_*(k-3)} (k-2)^{-B_h}}{C_\Gamma 2^{B_h/2}}, \quad A_k = \frac{C_h^{\text{tilt}}}{c_\Gamma} k^{p_h+1} M_{\alpha_*}^{k-1}, \\ D_d = [1 + 4(d + M_h)^2]^{M_h}, \quad L_{k,d} = \log(A_k D_d / a_k). \end{aligned} \tag{GRA3}$$

Here c_h and C_h^{tilt} are exactly the positive original TW7 convolution lower-envelope and BT12 compact/tail constants. Their already proved defining estimates, with the full numerator $2\xi/h$, are

$$\begin{aligned} w_h^{*k}(y) &\geq c_h \vartheta_h^{k-3} e^{-\alpha_*(|y|+k-3)} (k-2+|y|)^{-B_h}, \\ w_h^{*k}(y) &\leq k C_h^{\text{tilt}} M_{\alpha_*}^{k-1} e^{-\alpha_*|y|} (1+|y|/k)^{-p_h}, \\ c_\Gamma e^{-\alpha_*|y|} (1+|y|)^{-1/2} &\leq d\sigma/dy \leq C_\Gamma e^{-\alpha_*|y|} (1+|y|)^{-1/2}. \end{aligned} \quad (\text{GRA4})$$

These bounds hold for every convolution order $k \geq 3$. In particular there is no upper polynomial-degree restriction in the following argument.

The first and third estimates, $k-2+|y| \leq (k-2)(1+|y|)$, and $(1+|y|)^{B_h} \leq 2^{B_h/2}(1+y^2)^{M_h}$ give $w_h^{*k} \geq a_k(1+y^2)^{-M_h} d\sigma/dy$. The second and third, together with $1+|y|/k \geq (1+|y|)/k$ and $p_h > 1/2$, give $w_h^{*k} \leq A_k d\sigma/dy$. For the original Gamma orthonormal polynomials e_n one has

$$ye_n = \sqrt{(n+1)(n+1/2)}e_{n+1} + \sqrt{n(n-1/2)}e_{n-1}.$$

The two shifts on polynomials of degree at most d have norms at most $d+1$ and d , respectively. Hence $\|yf\|_\sigma \leq 2(d+1)\|f\|_\sigma$. Applying this inequality successively gives, for $0 \leq j \leq M_h$, $\|y^j f\|_\sigma^2 \leq [4(d+M_h)^2]^j \|f\|_\sigma^2$. Expansion of $(1+y^2)^{M_h}$ therefore bounds its weighted squared norm by $D_d \|f\|_\sigma^2$. Cauchy–Schwarz gives

$$\|f\|_\sigma^4 \leq \left(\int |f|^2 (1+y^2)^{-M_h} d\sigma \right) \left(\int |f|^2 (1+y^2)^{M_h} d\sigma \right).$$

Combining these inequalities and using the bijective substitution $S = c+iy$ proves, on the complete coefficient space and at every $d \geq 0$,

$$\frac{a_k}{D_d} H_d^\sigma \preceq H_d^{\text{ar}} \preceq A_k H_d^\sigma. \quad (\text{GRA5})$$

Both forms are positive definite by their positive densities, and all moments are finite by their exponential upper bounds. Applying (GRA5) to a nonzero vector proves $L_{k,d} \geq 0$. Its dependence on degree is explicit. Subtracting the logarithms in (GRA3) gives

$$\begin{aligned} \log(A_k/a_k) &= k\mathfrak{s}_h + (p_h+1)\log k + B_h \log(k-2) + \mathfrak{c}_h, \\ \mathfrak{s}_h &= \alpha_* + \log(M_{\alpha_*}/\vartheta_h), \\ \mathfrak{c}_h &= -\log M_{\alpha_*} + 3\log \vartheta_h - 3\alpha_* + \log \frac{C_h^{\text{tilt}} C_\Gamma 2^{B_h/2}}{c_h c_\Gamma}, \\ L_{k,d} &= O_h(k + \log(d+1)). \end{aligned} \quad (\text{GRA6})$$

The constant implicit here is independent of d and k .

221.2 Full forms, fixed affine fibres, and arithmetic monic minima

For $p \in \mathcal{P}_r(y)$ retain the two full forms

$$\mathcal{N}_0[p] = \int y^{2q} |p(y)|^2 d\sigma(y), \quad \mathcal{N}_\chi[p] = \int |Q(y)p(y)|^2 d\sigma(y).$$

The companion moving-cutoff theorem gives, with its actual finite constants $C_\chi(k, r)$ and $\eta(k, r)$ and all its displayed guards,

$$e^{-C_\chi}(1-\eta)\mathcal{N}_0[p] \leq \mathcal{N}_\chi[p] \leq (1+\eta)\mathcal{N}_0[p], \quad 0 \leq \eta < 1. \quad (\text{GRA7})$$

It is the original Q on the entire real line. In particular this theorem does not remove the inner integration interval. Applying (GRA5) at $d = q+r$ to $\chi U_r^{-1}p$ and using (GRA2) proves the exactly typed form of the arithmetic receiver:

$$\frac{a_k}{D_{q+r}} e^{-C_\chi}(1-\eta)\mathcal{N}_0[p] \leq \|\chi U_r^{-1}p\|_{\text{ar}}^2 \leq A_k(1+\eta)\mathcal{N}_0[p]. \quad (\text{GRA8})$$

Indeed the middle integrand after substitution is exactly $|Q(y)p(y)|^2$. This specifies the coordinate implicit in the received notation $\|\chi p\|_{\text{ar}}$.

For completeness the same constants act on every unchanged affine coefficient fibre. Let $C : \mathcal{P}_r(y) \rightarrow F$ be a specified complex linear map and $z \in \text{im } C$. Each positive form has a unique minimum on $Cp = z$: after choosing one lift, minimization is a strictly positive quadratic problem on $\ker C$. Apply each inequality of (GRA8) to every vector of this same nonempty fibre and take the infima. This proves the two inequalities for the resulting minimum forms, also at $z = 0$. In original coordinates the observation is CU_r , and U_r is the exact fibre bijection. Restriction follows by applying the form inequality to every vector in the specified subspace. No mixed matrix entry is set equal to zero by either operation.

Let ω_d^{ar} be the degree- d monic arithmetic squared norm, and retain the original Gamma monic squared norm

$$\gamma_d = \sqrt{2\pi} d!(1/2)_d = \sqrt{2\pi} (2d)!/4^d.$$

The mass is its degree-zero value; the Gamma recurrence gives $\gamma_d/\gamma_{d-1} = d(d-1/2)$. The monic bijection U_d and (GRA5) give

$$\frac{a_k}{D_d} \gamma_d \leq \omega_d^{\text{ar}} \leq A_k \gamma_d. \quad (\text{GRA9})$$

Define the original relation minima and ratios by

$$\nu_r^{\text{ar}} = \min_{P(S) \text{ monic, deg } P=r} \|\chi P\|_{\text{ar}}^2, \quad \nu_r^0 = \min_{p(y) \text{ monic, deg } p=r} \mathcal{N}_0[p], \quad T_r = \nu_r^{\text{ar}} / \omega_{q+r}^{\text{ar}}.$$

Taking these identical monic affine minima in (GRA8), then dividing by the opposite endpoints of (GRA9), proves

$$\begin{aligned} \log T_r &\geq \log(\nu_r^0 / \gamma_{q+r}) - L_{k,q+r} - C_\chi + \log(1 - \eta), \\ \log T_r &\leq \log(\nu_r^0 / \gamma_{q+r}) + L_{k,q+r} + \log(1 + \eta). \end{aligned} \quad (\text{GRA10})$$

The denominator is the actual arithmetic monic minimum throughout.

Here are the complete finite scalar endpoints obtained from the companion reference theorem. Write $r = 2n_r + \varepsilon_r$ with $\varepsilon_r \in \{0, 1\}$, and retain its actual moving parameters, support, Robin constant λ_r , and positive constants $c_{-,r}$, $c_{+,r}$. That theorem proves

$$c_{-,r} e^{-n_r \lambda_r} \leq h_{n_r, q} \leq c_{+,r} e^{-n_r \lambda_r}, \quad \nu_r^0 = \sqrt{2} q^{2(q+r)+1/2} h_{n_r, q}.$$

Consequently, with every constant evaluated at this r ,

$$\begin{aligned} Z_{q,r} &= \log \frac{\sqrt{2} q^{2(q+r)+1/2}}{\gamma_{q+r}} - n_r \lambda_r, \\ \underline{E}_r &= Z_{q,r} + \log c_{-,r} - L_{k,q+r} - C_\chi(k, r) + \log(1 - \eta(k, r)), \\ \overline{E}_r &= Z_{q,r} + \log c_{+,r} + L_{k,q+r} + \log(1 + \eta(k, r)), \\ \underline{E}_r &\leq \log T_r \leq \overline{E}_r. \end{aligned} \quad (\text{GRA11})$$

These are usable finite endpoints wherever the analytic theorem's finite guards hold. They are obtained by inequalities, not by replacing an arithmetic norm with an equilibrium norm.

221.3 Original quotient, full-jet support, and adjacent allowance

For $N \geq q - 1$ the literal remainder map $J_N : \mathcal{P}_N(S) \rightarrow E$ is onto and has kernel $\chi \mathcal{P}_{N-q}(S)$; at $N = q - 1$ that kernel is zero. In the coefficient frame $1, S, \dots, S^N$ define its quotient minimum Gram and determinant by

$$G_N = (J_N(H_N^{\text{ar}})^{-1} J_N^*)^{-1}, \quad V_N = \det G_N.$$

To verify the precise ratio, use the frame

$$(\chi, \chi S, \dots, \chi S^{N-q}; 1, S, \dots, S^{q-1}).$$

Reordering it by degree makes a unit-triangular coefficient matrix, because χ is monic; its determinant therefore has modulus one. Positive block elimination gives the first equality below. Gram–Schmidt on the full source and on the relation coefficient space with norm $\|\chi P\|_{\text{ar}}$ gives the next two:

$$\det H_N^{\text{ar}} = \det H_{\text{rel},N} \det G_N, \quad \det H_N^{\text{ar}} = \prod_{d=0}^N \omega_d^{\text{ar}}, \quad \det H_{\text{rel},N} = \prod_{j=0}^{N-q} \nu_j^{\text{ar}}.$$

Empty products are one. Dividing consecutive equalities proves

$$d_{q+r} := \frac{V_{q+r}}{V_{q+r-1}} = T_r^{-1}, \quad r \geq 0. \quad (\text{GRA12})$$

This is a source/relation determinant identity for the original quotient $\mathbb{C}[S]/(\chi)$.

The actual support map remains as follows. Set $E_h = \mathbb{C}[z]/(h)$, $v_h = j_h v_h \in E_h^\times$, and $A_\Sigma = \sum_{i=1}^k M_{z_i}$ on $E_h^{\otimes k}$. Then

$$\eta_{h,k} : E \longrightarrow E_h^{\otimes k}, \quad \eta_{h,k}[P] = v_h^{\otimes k} P(A_\Sigma) 1, \quad \eta_{h,k} M_S = A_\Sigma \eta_{h,k}. \quad (\text{GRA13})$$

In each local factor the nilpotent has exact order m . In each local tensor block their sum has exact order $1 + k(m-1)$: its power $k(m-1)$ has nonzero coefficient $[k(m-1)!]/[(m-1)!]^k$ on the product of the top nilpotent powers, and its next power vanishes because at least one exponent is m . The real and imaginary signs at the four original centres can be chosen independently, so every pair (a, b) occurs as a sum centre λ_{ab} . Distinct pairs give distinct centres because $\delta, \gamma > 0$. The annihilator of 1 under A_Σ is therefore exactly χ . Multiplication by the full unit preserves this annihilator, proving injectivity of (GRA13); direct multiplication proves its intertwining identity. The original source realization and jets consequently retain their exact commuting formula

$$\mathcal{V}(P) = P\left(\sum_i D_i\right) F_h^{\otimes k}, \quad J^{(k)} \mathcal{V}(P) = \eta_{h,k} J_N P.$$

In particular the relation norm above is measured before its quotient image is zero. No quotient by y^q enters this map.

Here is the complete adjacent calculation in these same norms. Let $p_n(S) = i^n Q_n((S-c)/i)$ be the original monic arithmetic orthogonal polynomial, let $b_n = [p_n]_\chi$, and put

$$R_N = G_N^{-1} = \sum_{n=0}^N b_n b_n^* / \omega_n^{\text{ar}}, \quad \beta_n = \omega_n^{\text{ar}} / \omega_{n-1}^{\text{ar}}.$$

Evenness of w_h , hence of its convolution, gives $Q_n(-y) = (-1)^n Q_n(y)$. Multiplication by S has the original recurrence $S p_n = p_{n+1} + c p_n - \beta_n p_{n-1}$, with the last term absent at zero. Inserting it into R_N cancels all interior adjacent pairs and gives

$$M_S R_N + R_N M_S^* - k R_N = \frac{b_{N+1} b_N^* + b_N b_{N+1}^*}{\omega_N^{\text{ar}}}.$$

Reflection $P(S) \mapsto P(k-S)$ permutes the roots of the even-degree polynomial χ and induces an involution Σ on E . Reflection is an isometry on every original remainder fibre, so the unique minimum implies $\Sigma^* G_N \Sigma = G_N$. Since $\Sigma b_n = (-1)^n b_n$, it follows that $b_N^* G_N b_{N+1} = 0$. For $x = G_N^{1/2} b_N$ and $y = G_N^{1/2} b_{N+1}$ the relative Hermitian control has nonzero eigenvalues $\epsilon_N, -\epsilon_N$, with

$$\epsilon_N^2 = \|x\|^2 \|y\|^2 / (\omega_N^{\text{ar}})^2.$$

The determinant lemma for adding and removing the last covariance column proves

$$\frac{b_N^* G_N b_N}{\omega_N^{\text{ar}}} = 1 - d_N \quad (N \geq q), \quad \frac{b_{N+1}^* G_N b_{N+1}}{\omega_{N+1}^{\text{ar}}} = d_{N+1}^{-1} - 1.$$

At $N = q - 1$ the first q monic remainder columns form a basis and the first fraction is exactly one. Thus no inverse at $q - 2$ is used. Together with (GRA12) these identities prove

$$\epsilon_{q+r-1}^2 = \beta_{q+r}(T_r - 1) \begin{cases} 1, & r = 0, \\ 1 - T_{r-1}^{-1}, & r \geq 1. \end{cases} \quad (\text{GRA14})$$

Only the total pairing vanished. In the original orthogonal kernel–boundary split, $x = x_K \oplus x_B$, $y = y_K \oplus y_B$, one has $x_B^* y_B = -x_K^* y_K$. The mixed exterior term therefore remains

$$\|x_K \otimes y_B - y_K \otimes x_B\|^2 = \|x_K\|^2 \|y_B\|^2 + \|x_B\|^2 \|y_K\|^2 + 2|x_K^* y_K|^2. \quad (\text{GRA15})$$

The classical eigenvalue-sum lineage is Ky Fan [H5], as credited by Overton and Womersley [H4]; the original rank-two exterior case used in GRA16 is calculated explicitly below.

The unchanged exterior benchmark is also explicit. Let E_+ be the sum of the complete primary blocks with $a > k/2$. Since k is odd, $\dim E_+ = q/2$. The nonzero top wedge of a full primary basis has additive M_S eigenvalue $\text{Tr}(M_S|_{E_+})$; all nilpotent parts have trace zero. The relative Hermitian control's additive operator on exterior degree $q/2$ has largest eigenvalue ϵ_N , since it sums distinct members of $\{\epsilon_N, -\epsilon_N, 0, \dots, 0\}$. Its expectation on the stated wedge is $2 \text{Re} \text{Tr}(M_S|_{E_+}) - kq/2$. The wedge stays nonzero under the injective full-jet map (GRA13). Summing the positive odd integers $2a - k$ gives $(k + 1)^2/4$. Therefore

$$0 < \mathcal{L}_{h,k} := 2\delta e_k(k + 1) \frac{(k + 1)^2}{4} = \frac{\delta q(k + 1)}{2} \leq \epsilon_N \quad (N \geq q - 1). \quad (\text{GRA16})$$

Positive covariance gives $0 < d_n \leq 1$. Equation (GRA14) and the strict positive benchmark first force $d_q < 1$ and then $d_n < 1$ at every subsequent degree. Thus all logarithms below are defined for the actual objects.

Finally (GRA9) gives the sharpened finite recurrence enclosure

$$-L_{k,d} \leq \log \frac{\beta_d}{d(d - 1/2)} \leq L_{k,d-1}, \quad d \geq 1. \quad (\text{GRA17})$$

For its lower bound divide the lower bound for ω_d^{ar} by the upper bound for ω_{d-1}^{ar} ; reverse the choices for the upper bound. This is an adjacent norm comparison, not a derivative of a determinant asymptotic.

221.4 Growing ratios and uniformity on every shifted index

The companion moving-parameter reference theorem proves

$$\log(\nu_r^0 / \gamma_{q+r}) = 2q\psi(r/q) + O_h(\log(q + r)), \quad \psi(t) \sim \frac{1}{2t \log t}.$$

The companion full-form theorem proves, throughout $t = r/q \rightarrow \infty$, $\log t = o(q/R_k)$, that its finite guards eventually hold and that $C_\chi = O_h(R_k^2(1 + \log t)^2/q)$ and $\eta = \exp[-q \log 8 + o(q)]$. The constants in the reference estimate are those evaluated at the actual moving support. Insert these proved estimates into (GRA10); for $0 \leq \eta \leq 1/2$, $|\log(1 \pm \eta)| \leq 2\eta$. This proves

$$\log T_r = 2q\psi(t) + O_h \left(k + \log(q + r) + \frac{R_k^2}{q}(1 + \log t)^2 \right). \quad (\text{GRA18})$$

In particular the reference exponent has not been presumed larger than the arithmetic error on the whole of the full-form range.

Now take integers $s = s_k \geq 1$ and suppose

$$t = s/q \longrightarrow \infty, \quad t \log t = o(q/k), \quad H_k = \frac{q}{t \log t}. \quad (\text{GRA19})$$

We prove every needed uniformity directly. The exact formula for q gives $q \geq k^2$, $q \leq C_m k^3$, and $R_k^2/q \leq \delta^2 + \gamma^2$ for $k \geq 1$. Eventually $t \geq e$ and $t \log t \leq q/k$, hence $t \leq q/k \leq C_m k^2$. It follows that

$$\log t = O_m(\log k), \quad \log(q + s + q) = O_m(\log k), \quad (R_k^2/q)(1 + \log t)^2 = O_h((\log k)^2) = o_h(k).$$

Thus the error in (GRA18) is $O_h(k)$, while $H_k/k = q/(kt \log t) \rightarrow \infty$. Also

$$\frac{R_k \log t}{q} = \sqrt{\delta^2 + \gamma^2} \frac{kt \log t}{q} \frac{1}{t} \rightarrow 0,$$

so the narrower range lies inside the proved full-form range.

For every integer $j \in \{-1, 0, \dots, q\}$ put $\tau_j = (s + j)/q$. Uniformly in this entire set,

$$|\tau_j - t| \leq 1, \quad \tau_j/t = 1 + O(1/t), \quad \log \tau_j = \log t + O(1/t), \quad \frac{\tau_j \log \tau_j}{t \log t} = 1 + O(1/t).$$

For example the logarithm estimate follows by the mean value theorem on $[t - 1, t + 1]$ once $t \geq 2$. These bounds prove the two range conditions and the $O_h(k)$ error uniformly for all τ_j . The definition of the scalar limit $2v \log v \psi(v) \rightarrow 1$ implies uniformity on all $v \geq t - 1$: the supremum of its absolute error on that tail tends to zero. Consequently

$$\max_{-1 \leq j \leq q} \left| \frac{\log T_{s+j}}{H_k} - 1 \right| \rightarrow 0. \quad (\text{GRA20})$$

The same argument applied to the finite bounds in (GRA11) shows that their lower and upper endpoints are $H_k(1 + o_h(1))$ uniformly. In particular all the finite lower endpoints are eventually positive; positivity has been proved in this range, not inserted as a new asymptotic premise.

For $r \geq 1$ take logarithms of the exact adjacent identity:

$$2 \log \epsilon_{q+r-1} = \log \beta_{q+r} + \log T_r + \log(1 - T_r^{-1}) + \log(1 - T_{r-1}^{-1}). \quad (\text{GRA21})$$

For $r = s, \dots, s + q$, (GRA17) and the preceding degree bounds give $\log \beta_{q+r} = O_h(k)$. If $M_k = \min_{s-1 \leq r \leq s+q} \log T_r$, then $M_k/H_k \rightarrow 1$ and

$$0 \leq -\log(1 - T_r^{-1}) \leq \frac{e^{-M_k}}{1 - e^{-M_k}} \rightarrow 0$$

uniformly for both adjacent factors. Divide (GRA21) by H_k and use $k/H_k \rightarrow 0$ to obtain

$$\max_{s \leq r \leq s+q} \left| \frac{\log \epsilon_{q+r-1}}{H_k/2} - 1 \right| \rightarrow 0. \quad (\text{GRA22})$$

For a single sequence r this proves $\log \epsilon_{q+r-1} \sim q/(2(r/q) \log(r/q))$ under the same narrower range. Since $\log \mathcal{L}_{h,k} = \log(\delta/2) + \log q + \log(k + 1) = O_h(\log k)$, it also proves the original benchmark comparison

$$\log \frac{\epsilon_{q+r-1}}{\delta q(k + 1)/2} \sim \frac{q}{2(r/q) \log(r/q)} \rightarrow +\infty. \quad (\text{GRA23})$$

These statements do not extend the allowance asymptotic to degrees for which the arithmetic error in (GRA18) is comparable to its decreasing reference exponent.

221.5 Exact shifted two-window identities and finite enclosures

Put $a = q + s - 1$ and $b = a + q$, and give the rows $N = a + i$, $0 \leq i \leq q$, the weights $w_0 = w_q = 1$, $w_i = 2$ otherwise. Their sum is $2q$. Since $s \geq 1$, $a \geq q$, so every row has both adjacent factors. Define

$$\begin{aligned} \mathcal{B}_{k,s}^{\text{ar}} &= \log \frac{V_a V_{a+1}}{V_b V_{b+1}}, & \mathcal{W}_{k,s}^{\text{ar}} &= \log \frac{\omega_b^{\text{ar}} \omega_{b+1}^{\text{ar}}}{\omega_a^{\text{ar}} \omega_{a+1}^{\text{ar}}}, \\ \mathcal{P}_{k,s} &= -\sum_{i=0}^q w_i [\log(1 - d_{a+i}) + \log(1 - d_{a+i+1})], & \mathcal{J}_{k,s}^{\text{action}} &= 2 \sum_{i=0}^q w_i \log \epsilon_{a+i}. \end{aligned} \quad (\text{GRA24})$$

For any positive sequence z_n , expansion of the coefficients gives

$$\sum_{i=0}^q w_i (\log z_{a+i+1} - \log z_{a+i}) = \log z_b + \log z_{b+1} - \log z_a - \log z_{a+1}.$$

Applying this identity to ω_n^{ar} and V_n in (GRA21), with $r = s + i$ and $\log T_{s+i} = -\log d_{a+i+1}$, proves

$$J_{k,s}^{\text{action}} = \mathcal{B}_{k,s}^{\text{ar}} + \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{P}_{k,s}, \quad \mathcal{B}_{k,s}^{\text{ar}} = \sum_{i=0}^q w_i \log T_{s+i}. \quad (\text{GRA25})$$

The four endpoints are thus exactly $a, a+1, a+q, a+q+1$.

To derive the penalty weights, let $f(x) = -\log(1-x)$ for $0 < x < 1$. The coefficient of $f(d_{a+j})$ is $w_j + w_{j-1}$, taking a missing w as zero. Since $q \geq 2$ for the original packet, the result is

$$\mathcal{P}_{k,s} = \sum_{j=0}^{q+1} c_j f(T_{s-1+j}^{-1}), \quad (c_0, \dots, c_{q+1}) = (1, 3, \underbrace{4, \dots, 4}_{q-2}, 3, 1), \quad \sum_j c_j = 4q. \quad (\text{GRA26})$$

The first coefficient is present because the first row is shifted beyond the exceptional row $q-1$.

For finite bounds suppose all the lower endpoints in (GRA11) on $s-1 \leq r \leq s+q$ are positive, and define $\ell = \min_r \underline{E}_r > 0$ and $u = \max_r \bar{E}_r$. The derivative of f is positive, while e^{-x} decreases, so

$$\sum_{j=0}^{q+1} c_j f(e^{-\bar{E}_{s-1+j}}) \leq \mathcal{P}_{k,s} \leq \sum_{j=0}^{q+1} c_j f(e^{-\underline{E}_{s-1+j}}). \quad (\text{GRA27})$$

The elementary inequalities $x \leq f(x) \leq x/(1-x)$, proved by integrating $1/(1-x)$ between zero and x , give the useful common bounds

$$4qe^{-u} \leq \mathcal{P}_{k,s} \leq \frac{4qe^{-\ell}}{1-e^{-\ell}}. \quad (\text{GRA28})$$

Each individual final term also gives $\mathcal{P}_{k,s} \geq e^{-\bar{E}_{s+q}}$.

Every row has a corresponding explicit finite interval. Put $d = q + r$ and

$$A_r^- = \frac{1}{2} [\log(d(d-1/2)) - L_{k,d} + \underline{E}_r + \log(1 - e^{-\underline{E}_r}) + \log(1 - e^{-\underline{E}_{r-1}})],$$

$$A_r^+ = \frac{1}{2} [\log(d(d-1/2)) + L_{k,d-1} + \bar{E}_r + \log(1 - e^{-\bar{E}_r}) + \log(1 - e^{-\bar{E}_{r-1}})].$$

Each displayed scalar function of E is increasing on $E > 0$. Equations (GRA17) and (GRA21) therefore prove

$$A_r^- \leq \log \epsilon_{q+r-1} \leq A_r^+, \quad \min_{s \leq r \leq s+q} A_r^- \leq \log \min_{a \leq N \leq b} \epsilon_N \leq \min_{s \leq r \leq s+q} A_r^+. \quad (\text{GRA29})$$

The two positive contraction factors are retained in both endpoints.

For the monic part define the literal Gamma window

$$\mathcal{W}_{k,s}^\sigma = \log \frac{\gamma_b \gamma_{b+1}}{\gamma_a \gamma_{a+1}} = \log \frac{(2b)!(2b+2)!}{4^{2q}(2a)!(2a+2)!}. \quad (\text{GRA30})$$

Write $v_n = \log(\omega_n^{\text{ar}}/\gamma_n)$. Equation (GRA9) places v_n in $[\log A_k - L_{k,n}, \log A_k]$. Applying these four intervals with their actual signs, so the four $\log A_k$ terms cancel, proves

$$-L_{k,b} - L_{k,b+1} \leq \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{W}_{k,s}^\sigma \leq L_{k,a} + L_{k,a+1}. \quad (\text{GRA31})$$

Thus the recurrence contribution has four endpoint errors, rather than one independent arithmetic error for every row. In particular, writing P_- and P_+ for the lower and upper quantities in (GRA27), all finite data combine to give

$$\sum_{i=0}^q w_i \underline{E}_{s+i} + \mathcal{W}_{k,s}^\sigma - L_{k,b} - L_{k,b+1} - P_+ \leq J_{k,s}^{\text{action}},$$

$$J_{k,s}^{\text{action}} \leq \sum_{i=0}^q w_i \bar{E}_{s+i} + \mathcal{W}_{k,s}^\sigma + L_{k,a} + L_{k,a+1} - P_-.$$

(GRA32)

221.6 Complete asymptotics and the unchanged mixed receiver

Under (GRA19), (GRA22) directly implies

$$\log \min_{a \leq N \leq a+q} \epsilon_N \sim \frac{H_k}{2} = \frac{q}{2t \log t}. \quad (\text{GRA33})$$

Indeed every row logarithm divided by $H_k/2$ lies between $1 - o(1)$ and $1 + o(1)$, so its minimum does too. This argument does not identify the minimizing row with an endpoint or equate a minimum with a geometric mean.

For the penalty, (GRA20) implies that its minimum and maximum $\log T_r$ are both $H_k(1 + o(1))$. Apply the proof of (GRA28) using these exact minima and maxima to obtain

$$\log(4q) - \max_r \log T_r \leq \log \mathcal{P}_{k,s} \leq \log(4q) - \min_r \log T_r - \log(1 - e^{-\min_r \log T_r}).$$

Here $\log q = O_m(\log k) = o(H_k)$, and the final logarithm tends to zero. Dividing by H_k proves the full logarithmic rate

$$\log \mathcal{P}_{k,s} \sim -H_k = -\frac{q}{t \log t}. \quad (\text{GRA34})$$

All $4q$ coefficients have been retained in the upper estimate.

The separate terms of the action can also be evaluated. From (GRA20) and $\sum_i w_i = 2q$, $\mathcal{B}_{k,s}^{\text{ar}} \sim 2qH_k$. The exact Gamma recurrence gives

$$\mathcal{W}_{k,s}^\sigma = \sum_{i=0}^q w_i \log[(s + q + i)(s + q + i - 1/2)].$$

Since $s = qt$, every summand equals $2 \log s + O(1/t)$ uniformly, giving $\mathcal{W}_{k,s}^\sigma = 4q \log(qt) + O(q/t)$. More explicitly, $\sum_i w_i i = q^2$, and $0 \leq \log(1 + x) \leq x$ for $x \geq 0$ gives

$$0 \leq \mathcal{W}_{k,s}^\sigma - 4q \log s \leq (6q^2 - q)/s.$$

The four-endpoint estimate (GRA31) now yields

$$\begin{aligned} \mathcal{B}_{k,s}^{\text{ar}} &\sim 2qH_k, & \mathcal{W}_{k,s}^{\text{ar}} &= 4q \log(qt) + O_h(q/t + k), \\ \mathcal{W}_{k,s}^{\text{ar}} &= o(qH_k), & \mathcal{P}_{k,s} &= \exp[-H_k(1 + o(1))] = o(qH_k), \\ J_{k,s}^{\text{action}} &\sim 2qH_k = \frac{2q^2}{t \log t}. \end{aligned} \quad (\text{GRA35})$$

For the smallness of $\mathcal{W}$, divide its bound by qH_k and use $\log(qt) = O_m(\log k)$, $H_k/k \rightarrow \infty$, and $q \rightarrow \infty$. Equation (GRA25) then proves the last assertion independently of summing the row asymptotics.

It remains to specify exactly how this action reaches the existing mixed calculation. Keep its original surjection $\Lambda : E \twoheadrightarrow B$ and kernel inclusion $I : K \hookrightarrow E$, with the complete divided Taylor unit and original period branches. Explicitly, with the original period map Π and tensor invariant projector P_k , its image inclusion j satisfies $j\Lambda = P_k \Pi^{\otimes k} M_{v_h}^{\otimes k} \iota$, where $\iota[P] = [P(z_1 + \cdots + z_k)]$. For each original source metric G_N^v , put

$$H_{K,N}^v = I^* G_N^v I, \quad Q_{B,N}^v = (\Lambda(G_N^v)^{-1} \Lambda^*)^{-1}, \quad S_N^v = (G_N^v)^{-1} \Lambda^* Q_{B,N}^v.$$

Direct multiplication gives $\Lambda S_N^v = 1$ and $I^* G_N^v S_N^v = 0$. For a fixed section S_* , the difference $S_N^v - S_*$ has image in K , so the frame change from $[I, S_*]$ to $[I, S_N^v]$ is triangular with identity diagonal. Hence

$$\det H_{K,N}^v \det Q_{B,N}^v = |\det[I, S_*]|^2 \det G_N^v.$$

Taking the return at $a, a+1, b, b+1$ cancels all four frame constants. If $F_K^v, F_B^v, \mathcal{B}^v$ denote these same four-endpoint returns, the result is exactly $F_K^v + F_B^v = \mathcal{B}^v$. Empty determinant factors are one. For the original sources $v = \text{ar}, \sigma, (k)$ set

$$\mathcal{H} = \mathcal{B}^\sigma - \mathcal{B}^{(k)}, \quad \Delta^\sigma = \mathcal{B}^{\text{ar}} - \mathcal{B}^\sigma.$$

The complete shifted allocation and matching budget therefore obey

$$\begin{aligned} \Psi &= F_K^{\text{ar}} + F_B^{\text{ar}} - F_B^{(k)} = F_K^{(k)} + \mathcal{H} + \Delta^\sigma, \\ C^{\text{budget}} &= F_B^{(k)} + \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{P}_{k,s} = \mathcal{B}^{(k)} - F_K^{(k)} + \mathcal{W}_{k,s}^{\text{ar}} - \mathcal{P}_{k,s}, \\ \Psi + C^{\text{budget}} &= J_{k,s}^{\text{action}}. \end{aligned} \tag{GRA36}$$

This proves the exact map of the new window into the earlier complete action. The common allocation cancels with its matching budget. The new scalar profile consequently assigns no separate leading coefficient to the intrinsic kernel and boundary mixed terms or to their complementary CPQ returns. Those original quantities remain in (GRA15) and (GRA36), with their maps unchanged.

The unchanged arithmetic measure in the cardinal relation calculation. IGI28–30 below gives the precise map from this original arithmetic norm to the numerator Gamma norm on the same polynomial space at $c = a/2$. It keeps the actual density $w_h^{*a}(y)dy$ and the Gamma density $dm_s(y)$ and proves the full-form comparison

$$\frac{a_a}{D_d \mathfrak{b}_{s,d}} H_s \preceq H_{\text{ar}} \preceq \frac{A_a}{\mathfrak{a}_s} H_s, \quad D_d = [1 + 4(d + M_h)^2]^{M_h}. \tag{IUP6}$$

Here a_a, A_a, M_h are the original GRA constants, $L = (s-1)/4$, $\rho_s = (2\pi)^{2L}/(1/2)_{2L}$, $\mathfrak{a}_s = \rho_s \prod_{u < L} (2u + 1/2)^2$ and $\mathfrak{b}_{s,d} = \rho_s \prod_{u < L} \{4(d + u + 1)^2 + (2u + 1/2)^2\}$, including empty products at $s = 1$. Gamma recurrence and the original degree- d multiplication bound prove the two Gamma constants. Combining them with GRA5 gives IUP6 in the displayed directions. Taking the identical affine minimum on each side transports it to every original monic, cardinal and conductor-constrained fibre; inversion and complex conjugation give its reverse dual inequality. The monic maps $U_r P(y) = i^{-r} P(c + iy)$ obey exactly $U_{q+r}(\chi P) = Q_a U_r P$, where $Q_a(y) = i^{-q} \chi(c + iy)$. Thus the original GRF whole-form bounds apply on their proved guards. The new degree-one cardinal row has its independent exact moment proof in IGI24–27 and needs no unproved extension of those guards.

222 An evaluated original-period domain with zero conductor vanishing order

Retain the complete simple-quartet data of PCL1–11 and RPC1–26:

$$x = \delta + i\gamma, \quad y = \delta - i\gamma, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2, \quad U = \text{diag}(a, \bar{a}, \bar{a}, a), \quad a \neq 0. \quad (\text{PZ1})$$

Here $a = v_h(1/2 + x)$ is the original entire-unit value; its phase is not chosen. Set $\beta = -(x^2 + y^2) = 2(\gamma^2 - \delta^2)$ and $\eta = x^2 y^2 = (\delta^2 + \gamma^2)^2$. Write $\tau = u^{-1}$ for the original inverse period, reserving w for the exponential-symbol variable. The original matrix is exactly

$$\mathbf{H}(u) = e^{a/u} G_u \mathcal{R}(\tau) V^{-1} U, \quad L_j(\tau) = U^{-1} V \mathcal{R}(\tau)^{-1} D^{-j} \mathcal{R}(\tau) V^{-1} U, \quad j = 1, 2, 3, 4. \quad (\text{PZ2})$$

The cancellation of $e^{a/u}$ and G_u in this conjugation is the literal identity $L_j = \mathbf{H}^{-1} D^{-j} \mathbf{H}$, since G_u and D commute. Thus no amplitude, period column, contour factor, or original phase has been replaced. The matrix $\mathcal{R}$ is the entire PCL2 series, with determinant one and inverse its adjugate. Put

$$\begin{aligned} d_j(\tau) &= Q_0(L_j(\tau) \mathbf{1}), \quad Q_0(t) = t_1 t_4 - t_2 t_3, \quad \mathbf{1} = (1, 1, 1, 1)^T, \\ \mathcal{E}(\tau) &= \prod_{j=1}^4 d_j(\tau), \quad E_{A,u}(w) = \sum_{b,c=0}^8 a_{bc}(\tau) e^{[4+(2b-8)\delta+i(2c-8)\gamma]w}. \end{aligned} \quad (\text{PZ3})$$

The complete coefficient product PCL8–10 proves $E_{A,u}(0) = A_\tau(1, 1) = \mathcal{E}(\tau)$, including all four nonreduced conductor factors and their actual multiplicities.

222.1 The exact limiting value, including both exceptional amplitude phases

Define

$$\begin{aligned} r &= \frac{a}{\bar{a}}, \quad T = \frac{x^2}{y^2}, \\ A_e &= \frac{2T - 1}{3(T - 1)}, \quad B_e = \frac{2 - T}{3(T - 1)}, \\ C_e &= (r - 1)(A_e/r + B_e), \quad D_e = (r - 1)(A_e/r - B_e). \end{aligned} \quad (\text{PZ4})$$

All denominators are nonzero: $a, y \neq 0$ and $x^2 - y^2 = 4i\delta\gamma \neq 0$. Neither A_e nor B_e vanishes, because $|T| = 1$, whereas their putative roots are $1/2$ and 2 . One has $A_e - B_e = 1$ and $B_e = -\bar{A}_e$, by direct substitution of $\bar{T} = T^{-1}$. Hence

$$\begin{aligned} C_e &= 1 - 2\text{Re}(A_e/r) \in \mathbb{R}, \\ D_e &= 2i(\text{Im } A_e - \text{Im}(A_e/r)) \in i\mathbb{R}. \end{aligned} \quad (\text{PZ5})$$

For $\rho_j = \zeta^{-j}$ and $h_j = \rho_j^3 - \rho_j$, the full value is

$$\boxed{d_j(0) = h_j D_e (2\rho_j + h_j C_e)}. \quad (\text{PZ6})$$

Here is a direct proof retaining both off-diagonal entries in PCL4. On even centered coefficients (p_0, p_2) , its middle conjugation is

$$\begin{pmatrix} \rho_j & \beta(\rho_j^3 - \rho_j)/3 \\ 0 & \rho_j^3 \end{pmatrix}.$$

Evaluation at (x, y) gives the matrix $\begin{pmatrix} \rho_j + h_j A_e & -h_j A_e \\ h_j B_e & \rho_j - h_j B_e \end{pmatrix}$. Indeed $(x^2 + \beta/3)/(x^2 - y^2) = A_e$ and $(y^2 + \beta/3)/(x^2 - y^2) = B_e$. Conjugation by the original $\text{diag}(a, \bar{a})$ gives

$$\begin{pmatrix} \rho_j + h_j A_e & -h_j A_e/r \\ r h_j B_e & \rho_j - h_j B_e \end{pmatrix}.$$

The input $\mathbf{1}$ has even evaluation vector $(1, 1)^T$ and zero odd evaluation vector. Its two output values are therefore $p_j = \rho_j + h_j A_e(1 - r^{-1})$ and $q_j = \rho_j + h_j B_e(r - 1)$, each repeated at the opposite root. The original quadratic is exactly $p_j^2 - q_j^2$. Its difference is $h_j D_e$ and its sum is $2\rho_j + h_j C_e$, proving PZ6. This calculation is an evaluation through explicit invertible coefficient and primary maps, not an assignment of the unit.

No factor $2\rho_j + h_j C_e$ vanishes. In fact its vanishing would imply $2 + (\rho_j^2 - 1)C_e = 0$. The imaginary part forces $C_e = 0$ because ρ_j^2 is nonreal, and the real part would then be $2 = 0$. Also $h_j \neq 0$. Consequently

$$d_j(0) = 0 \iff r = 1 \text{ or } r = A_e/B_e, \quad j = 1, 2, 3, 4. \quad (\text{PZ7})$$

The second ratio is a unit complex number and is distinct from one. Thus this is an exhaustive two-phase statement about the actual amplitude, with neither phase omitted.

Multiplying all four PZ6 values gives the exact real expression

$$\boxed{\mathcal{E}(0) = 5D_e^4(5(C_e - 1)^4 + 10(C_e - 1)^2 + 1).} \quad (\text{PZ8})$$

To verify the constant, $\prod_j \rho_j = 1$ and $\prod_j (\rho_j^2 - 1) = 5$. The remaining product is

$$\prod_{j=1}^4 (2 - C_e + C_e \rho_j^2) = \frac{(2 - C_e)^5 + C_e^5}{2} = 5(C_e - 1)^4 + 10(C_e - 1)^2 + 1.$$

The first equality follows by including the missing fifth root one in $\prod_{\xi^5=1} (s + C_e \xi) = s^5 + C_e^5$, then dividing by $s + C_e = 2$, so it remains valid at $C_e = 0$. Since D_e is purely imaginary, its fourth power is positive when it is nonzero. PZ8 is therefore strictly positive off exactly the two phases in PZ7. At those phases its value is zero; this inverse-period limit alone makes no assertion of zero value at a finite original period.

222.2 A finite original-period domain and the complete zero-order test

The PCL22 identity gives $L_j(-\tau) = P_{\text{opp}} L_j(\tau) P_{\text{opp}}$. Since $P_{\text{opp}} \mathbf{1} = \mathbf{1}$ and $Q_0(P_{\text{opp}} t) = Q_0(t)$, each d_j is even entire. Thus write

$$d_j(\tau) = \sum_{n \geq 0} d_{j,2n} \tau^{2n}, \quad C_j^{(0)} = \sum_{n \geq 1} |d_{j,2n}| < \infty. \quad (\text{PZ9})$$

Every coefficient is explicitly obtained by multiplying PCL2 and its finite adjugate in PZ2 and then the two products in PZ3. Absolute convergence at radius one proves the displayed finiteness. For an actual amplitude with $r \notin \{1, A_e/B_e\}$, define

$$R_0 = \max \left\{ R_*, 1, \max_{1 \leq j \leq 4} \sqrt{2C_j^{(0)} / |d_j(0)|} \right\}. \quad (\text{PZ10})$$

Here R_* is exactly RPC21 or RPC23 for this original quartet and unit, so this domain already lies in the proved five-orbit locus. For every original logarithm branch and $|u| \geq R_0$,

$$|d_j(u^{-1}) - d_j(0)| \leq C_j^{(0)} |u|^{-2} \leq \frac{1}{2} |d_j(0)|, \quad \frac{\mathcal{E}(0)}{16} \leq |E_{A,u}(0)| \leq \frac{81\mathcal{E}(0)}{16}, \quad \text{ord}_{w=0} E_{A,u}(w) = 0. \quad (\text{PZ11})$$

The error follows by summing the entire series at $|u|^{-1} \leq 1$; if $C_j^{(0)} = 0$ it is identically zero. Multiplication of the four lower and upper bounds gives the factors 2^{-4} and $(3/2)^4$. PZ8 supplies $\prod_j |d_j(0)| = \mathcal{E}(0) > 0$. A holomorphic function with nonzero value has zero vanishing order, proving the last equality without a genericity hypothesis on the period in this domain. Single-valuedness follows from the proved conjugation PZ2.

At arbitrary original periods on the five-orbit locus, the full test is still explicit: expand each actual $(2, 2)$ factor $q_j(\tau; Z, W) = \sum_{b,c=0}^2 c_{j,bc}(\tau) Z^b W^c$ and set

$$E_{j,u}(w) = \sum_{b,c=0}^2 c_{j,bc}(\tau) e^{[1+(2b-2)\delta+i(2c-2)\gamma]w},$$

$$v_j = \min \left\{ 0 \leq n \leq 8 : \sum_{b,c=0}^2 c_{j,bc}(\tau) [1 + (2b-2)\delta + i(2c-2)\gamma]^n \neq 0 \right\}, \quad (\text{PZ12})$$

$$v = \sum_{j=1}^4 v_j, \quad E_{A,u} = \prod_{j=1}^4 E_{j,u}, \quad \frac{E_{A,u}^{(v)}(0)}{v!} = \prod_{j=1}^4 \frac{E_{j,u}^{(v_j)}(0)}{v_j!} \neq 0.$$

The nine exponents in each factor are distinct by their real and imaginary parts. Their nine-by-nine derivative Vandermonde is invertible. On the five-orbit locus the factor is nonzero by the Segre restriction argument in BTK26, so the displayed minimum exists. This proves the exact retained test, including all multiplicities, and $v \leq 32$ on every remaining fibre.

For positive real original u , the actual real-period identity PCL15 also proves v is even. Indeed for real w put $Z = e^{2\delta w}$ and $W = e^{2i\gamma w}$. The full exponential factorization gives

$$E_{A,u}(w) = e^{(4-8\delta-8i\gamma)w} A_\tau(Z, W) = e^{(4-8\delta)w} W^{-4} A_\tau(Z, W) \geq 0. \quad (\text{PZ13})$$

The last expression is the exact squared modulus in PCL15. The function is nonzero analytic on the five-orbit locus. Its first nonzero Taylor term at zero must have even degree and positive real coefficient, because otherwise its real values would change sign on the two sides of zero. Hence $v \in \{0, 2, 4, \dots, 32\}$ on that positive-real-period locus. This retains the actual curve in PCL15, with one real coordinate and one unit-circle coordinate.

On the evaluated domain PZ11 the low-coordinate space $\mathbb{C}^v$ in ROG26 is zero, $Q_{N,s}$ in ROG23 is the zero map, and the full ROG27 shear is exactly the identity for both original source orders $s = 1, k$. Its determinant and every exterior power have norm one; the ROG32 graph-shear error is therefore exactly zero. The polynomial relation quotient and its remaining original metric still retain the full conductor, complete action, and the original residual map. No connecting-rank conclusion has been inferred from this zero-order calculation; the exact target relation is the unchanged RDS8 sequence with $\mathcal{N}_0 = \mathcal{E}_k$ and dimension $q - q'$.

223 The two exceptional unit phases have exact second inverse-period order

Retain the original roots

$$x = \delta + i\gamma, \quad y = \delta - i\gamma, \quad X = x^2, \quad Y = y^2, \quad \beta = -(X + Y), \quad \eta = XY, \quad 0 < \delta < \frac{1}{2}, \quad \gamma > 2.$$

The original ordered evaluation roots are $(x, y, -y, -x)$. Let V have rows $(1, w, w^2, w^3)$ at those four roots, let $U = \text{diag}(a, \bar{a}, \bar{a}, a)$ with $a \neq 0$, and put $r = a/\bar{a}$. This notation retains the exact scalar map $U = \bar{a} \text{diag}(r, 1, 1, r)$; the scalar $\bar{a}$ cancels between U^{-1} and U in the original conjugation. For $j = 1, 2, 3, 4$, put $\rho = \zeta^{-j}$ and $D_\rho = \text{diag}(\rho, \rho^2, \rho^3, \rho^4)$. The original PCL2 coefficient matrix is denoted by $R(z) = J + R_1 z + R_2 z^2 + \dots$, and the original expression is

$$q_\rho(z) = Q_0 (U^{-1} V R(z)^{-1} D_\rho R(z) V^{-1} U \mathbf{1}), \quad Q_0(t) = t_1 t_4 - t_2 t_3.$$

Every map in this expression has its original domain, codomain and order. In particular $V^{-1} U \mathbf{1} = \bar{a} c_r$, where

$$c_r = \left(\frac{X - rY}{X - Y}, 0, \frac{r - 1}{X - Y}, 0 \right)^\top.$$

Evaluation of the even polynomial represented by c_r at $x, y, -y, -x$ gives $(r, 1, 1, r)$, proving this identity directly.

223.1 Finite coefficients obtained from the complete original series

In the displayed matrices below write $b = \beta$ and $e = \eta$. PCL2 gives, with row indices $a = 1, 2, 3, 4$ and column indices $s = 0, 1, 2, 3$,

$$(R_n)_{a,s} = \sum_{\substack{p,h \geq 0 \\ 2p+4h=5n+s+1-a}} \frac{b^p e^h}{3^p p! h!} (-1)^{p+h-n} 5^{p+h-n} (a/5)_{p+h-n}.$$

For $n = 0, 1, 2$ the finite sum yields respectively

$$J = \begin{pmatrix} 1 & 0 & -b/3 & 0 \\ 0 & 1 & 0 & -2b/3 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix},$$

$$R_1 = \begin{pmatrix} 0 & b^3/27 - be/3 & 0 & -11b^4/324 + b^2e/3 - e^2/2 \\ -b^2/9 + e & 0 & 7b^3/81 - 2be/3 & 0 \\ 0 & -b^2/6 + e & 0 & 4b^3/27 - be \\ b/3 & 0 & -2b^2/9 + e & 0 \end{pmatrix},$$

and R_2 has its eight nonzero entries

$$\begin{aligned} (R_2)_{1,0} &= -11b^5/4860 + b^3e/27 - be^2/6, & (R_2)_{1,2} &= 22b^6/10935 - 11b^4e/324 + b^2e^2/6 - e^3/6, \\ (R_2)_{2,1} &= -7b^5/1215 + 7b^3e/81 - be^2/3, & (R_2)_{2,3} &= 119b^6/21870 - 7b^4e/81 + 7b^2e^2/18 - e^3/3, \\ (R_2)_{3,0} &= b^4/81 - b^2e/6 + e^2/2, & (R_2)_{3,2} &= -13b^5/1215 + 4b^3e/27 - be^2/2, \\ (R_2)_{4,1} &= b^4/54 - 2b^2e/9 + e^2/2, & (R_2)_{4,3} &= -7b^5/405 + 2b^3e/9 - 2be^2/3. \end{aligned}$$

Thus the second-order calculation uses every term of the original series that contributes at these orders. Multiplication of power series gives the inverse coefficients

$$S_0 = J^{-1}, \quad S_1 = -J^{-1}R_1J^{-1}, \quad S_2 = J^{-1}R_1J^{-1}R_1J^{-1} - J^{-1}R_2J^{-1}.$$

Set

$$f_n = \sum_{i=0}^n S_i D_\rho R_{n-i} c_r \quad (0 \leq n \leq 2).$$

The parity visible in PCL2 implies that f_0, f_2 have only entries indexed 0, 2, while f_1 has only entries indexed 1, 3. For a vector f in polynomial coefficient coordinates define $E_f(T) = f_0 + Tf_2$ and $O_f(T) = f_1 + Tf_3$. Expansion of the original two products defining Q_0 now proves

$$[z^2]q_\rho(z) = \frac{2E_{f_0}(X)E_{f_2}(X) - XO_{f_1}(X)^2}{r^2} - 2E_{f_0}(Y)E_{f_2}(Y) + YO_{f_1}(Y)^2. \quad (\text{PZX1})$$

This is also a short finite certificate for every coefficient below: substitute the displayed matrices and $b = -(X + Y)$, $e = XY$ into PZX1, and reduce powers by $1 + \rho + \rho^2 + \rho^3 + \rho^4 = 0$.

For completeness the absence of odd orders holds to every order. Let $S = \text{diag}(1, -1, 1, -1)$ in polynomial coordinates. PCL2 gives $R(-z) = SR(z)S$, and S commutes with D_ρ and fixes c_r . Consequently replacing z by $-z$ replaces the resulting polynomial $f(w)$ by $f(-w)$. The original products $f(x)f(-x)/r^2$ and $f(y)f(-y)$ are unchanged. Hence $q_\rho(-z) = q_\rho(z)$.

223.2 The phase $r = 1$

Substitution of $r = 1$ in PZX1 gives

$$[z^2]q_\rho(z) = -\frac{(X - Y)\beta^2(\beta^2 - 9\eta)}{81}(\rho^3 + 2\rho^2 + 3\rho - 1). \quad (\text{PZX2})$$

The constant coefficient vanishes by the same matrices. Multiplication by $\rho - 1$ and use of the fifth-root relation gives

$$(\rho - 1)(\rho^3 + 2\rho^2 + 3\rho - 1) = -5\rho.$$

The coefficient PZX2 is nonzero: $X - Y = 4i\delta\gamma \neq 0$, $\beta = 2(\gamma^2 - \delta^2) > 0$, and

$$\beta^2 - 9\eta = -5\gamma^4 - 26\delta^2\gamma^2 - 5\delta^4 < 0.$$

Therefore $q_\rho(z) = ([z^2]q_\rho)z^2 + O(z^4)$ has exact order two. Multiplication over $\rho = \zeta^{-1}, \dots, \zeta^{-4}$ gives

$$\prod_{j=1}^4 q_{\zeta^{-j}}(z) = 125 \left(\frac{(X - Y)\beta^2(\beta^2 - 9\eta)}{81} \right)^4 z^8 + O(z^{10}), \quad (\text{PZX3})$$

because $\prod_j \rho = 1$ and $\prod_j (\rho - 1) = 5$.

223.3 The phase $r = (2X - Y)/(2Y - X)$

Put $t = \beta^2/\eta$, retaining this exact original ratio. Define four real polynomials

$$\begin{aligned} A_3(t) &= 52t^3 - 1053t^2 + 7776t - 17496, \\ A_2(t) &= 4t^3 + 324t^2 - 3888t + 8748, \\ A_1(t) &= 96t^3 - 999t^2 + 3888t - 8748, \\ A_0(t) &= 18t^3 - 297t^2 + 1944t - 4374, \\ P_\rho(t) &= A_3(t)\rho^3 + A_2(t)\rho^2 + A_1(t)\rho + A_0(t). \end{aligned}$$

The constant coefficient again vanishes. PZX1 gives the exact value

$$[z^2]q_\rho(z) = -\frac{(X - Y)(\beta^2 - 9\eta)\eta}{1458(2t - 9)^2} P_\rho(t). \quad (\text{PZX4})$$

An explicit way of checking the homogeneous re-expression in PZX4 is to use

$$(X - 2Y)(2X - Y) = 2(X + Y)^2 - 9XY = \eta(2t - 9)$$

and the symmetric numerator

$$\begin{aligned} \mathcal{P} &= (52\rho^3 + 4\rho^2 + 96\rho + 18)(X^6 + Y^6) \\ &\quad + (-741\rho^3 + 348\rho^2 - 423\rho - 189)(X^5Y + XY^5) \\ &\quad + (4344\rho^3 - 2532\rho^2 + 1332\rho + 1026)(X^4Y^2 + X^2Y^4) \\ &\quad + (-7222\rho^3 + 2996\rho^2 - 5046\rho - 1908)X^3Y^3. \end{aligned}$$

The direct PZX1 numerator is $-(X - Y)(X^2 - 7XY + Y^2)\mathcal{P}$ over $1458(X - 2Y)^2(2X - Y)^2$. The identities for $X^6 + Y^6$, $X^5Y + XY^5$ and $X^4Y^2 + X^2Y^4$ in the elementary symmetric polynomials give $\mathcal{P} = \eta^3 P_\rho(t)$, proving PZX4 without changing the original roots.

Here is an explicit nonvanishing certificate valid for every real t . Set $c = \rho + \rho^{-1}$; then c is real and $c^2 + c - 1 = 0$. Reduction by $\rho^2 - c\rho + 1 = 0$ gives

$$P_\rho(t) = F(t, c) + G(t, c)\rho, \quad F = A_0 - A_2 - cA_3, \quad G = A_1 + c(A_2 - A_3).$$

Define

$$\begin{aligned} U(t, c) &= (1536 - 768c)t^2 + (18432c - 8784)t + 28755 - 104085c, \\ W(t, c) &= (832c - 224)t^2 + (8886 - 12948c)t + 67905c - 52785. \end{aligned}$$

Polynomial multiplication, collecting coefficients of $t^0, \dots, t^5$ and replacing c^2 by $1 - c$, gives the full identity

$$-U(t, c)F(t, c) - W(t, c)G(t, c) = 6495390(11c - 7). \quad (\text{PZX5})$$

The right side is nonzero: $c = 7/11$ would give $c^2 + c - 1 = 5/121 \neq 0$. Since F, G are real and ρ is nonreal, $P_\rho(t) = 0$ would force $G = 0$ by imaginary parts and then $F = 0$, contradicting PZX5. Thus $P_\rho(t) \neq 0$ for every real t . For the actual parameters,

$$t = 4 \left(\frac{\gamma^2 - \delta^2}{\gamma^2 + \delta^2} \right)^2, \quad \frac{900}{289} < t < 4.$$

The lower bound follows from $0 < \delta/\gamma < 1/4$ and the strict decrease of $(1-v)/(1+v)$ for $0 < v < 1$; the upper bound is strict since $\delta > 0$. In particular $2t - 9 < 0$. All factors in PZX4 are therefore nonzero, and each original factor has exact order two. Their product has exact order eight, with its complete nonzero leading coefficient

$$\left(\frac{(X - Y)(\beta^2 - 9\eta)\eta}{1458(2t - 9)^2} \right)^4 \prod_{j=1}^4 P_{\zeta^{-j}}(t). \quad (\text{PZX6})$$

The last product is strictly positive: conjugate roots pair the factors into $|P_{\zeta^{-1}}(t)|^2 |P_{\zeta^{-2}}(t)|^2 > 0$. Also $(X - Y)^4 = (4\delta\gamma)^4 > 0$. Hence PZX6 is positive. No secondary exceptional locus occurs in either of these two original unit phases anywhere on the stated (δ, γ) domain.

223.4 One finite radius for every actual amplitude and the original receiving maps

Use the complete coefficient calculation PZX1–PZX6, which proves that both exceptional phases in PZ7 have nonzero inverse-period coefficient $d_{j,2}$ for every j . Define, from the actual value a ,

$$\ell_A = \begin{cases} 0, & a/\bar{a} \notin \{1, A_e/B_e\}, \\ 2, & a/\bar{a} \in \{1, A_e/B_e\}, \end{cases} \quad b_j = d_{j,\ell_A} \neq 0, \quad C_j = \sum_{n \geq 1} |d_{j,\ell_A+2n}| < \infty. \quad (\text{PZU1})$$

The symbol ℓ_A describes vanishing at the inverse-period limit; it is not the symbol-variable vanishing order v . The leading b_j is exactly PZ6 or the applicable PZX formula, retaining the original amplitude phase and every constant. Put

$$R_A = \max \left\{ R_*, 1, \max_{1 \leq j \leq 4} \sqrt{2C_j/|b_j|} \right\}. \quad (\text{PZU2})$$

The Taylor series of $\tau^{-\ell_A} d_j(\tau)$ is entire after its removable value at zero is included, proving convergence of C_j . For $|u| \geq R_A$, every original branch satisfies

$$\left| u^{\ell_A} d_j(u^{-1}) - b_j \right| \leq C_j |u|^{-2} \leq \frac{1}{2} |b_j|. \quad (\text{PZU3})$$

Consequently the complete, original zero-symbol value has the bounds

$$\frac{\prod_j |b_j|}{16|u|^{4\ell_A}} \leq |E_{A,u}(0)| \leq \frac{81 \prod_j |b_j|}{16|u|^{4\ell_A}}, \quad v = \text{ord}_{w=0} E_{A,u}(w) = 0. \quad (\text{PZU4})$$

All inequalities follow by multiplying four nonzero factors from PZU3. This proves an original-period domain for every original nonzero amplitude, with neither of the two exceptional phases excluded. The two phases merely change the order of decay at infinite period from zero to eight. No unit value or source order has been assigned.

In the previously proved five-orbit domain $|u| \geq R_*$, the exact exceptional set is

$$\mathcal{Z}_A^{\geq R_*} = \left\{ u \in \mathbb{C}^* : |u| \geq R_*, \quad \prod_{j=1}^4 Q_0(L_j(u^{-1})\mathbf{1}) = 0 \right\}. \quad (\text{PZU5})$$

It is a finite set, with every point satisfying $R_* \leq |u| < R_A$. For a proof, $\mathcal{E}(\tau)$ is a nonzero entire function, with zero order exactly $4\ell_A$ at $\tau = 0$. Its zeros in the compact closed disk $|\tau| \leq R_*^{-1}$ are finite: any infinite sequence would have an accumulation point, contradicting the first nonzero local Taylor coefficient of a nonzero holomorphic function. Removing zero and applying $u = \tau^{-1}$ proves the finite-set assertion. PZU4 proves the strict upper-radius bound. On every point of this finite set the complete PZ12 derivative test retains the actual multiplicity and $v \leq 32$; no transcendental root is discarded. On the complement of this set in $|u| \geq R_*$, $v = 0$ exactly.

For clarity the consequences are now substituted into the original spaces and maps. For every degree $n = 4l + 1 \geq 9$, the original low space is $L_n = \mathbb{C}[S]_{<v} = 0$. Therefore

$$\begin{aligned} t_{0,n} &= \dim(K_n \cap L_n) = 0, & m_n &= q_n - q'_n - v = 16n - 48, \\ t_n^{\text{EFP}} &= 8n - 16 - t_{0,n} = 8n - 16, & j_{0,n} &= 8n - 32 - v + t_{0,n} = 8n - 32. \end{aligned} \quad (\text{PZU6})$$

Here t_n^{EFP} denotes the kernel-image rank called t_n in EFP1. The separate CFR common-factor target rank in RPK remains its proved value zero on its original domain; the two rank symbols have different defining maps. The exact RDS8 quotient becomes

$$0 \longrightarrow \mathcal{E}_{n-8} \xrightarrow{M_{EA}} \mathcal{E}_n \xrightarrow{\mathfrak{F}_A} (\mathbb{C}[S']/\mathcal{UC}[S])^\vee \longrightarrow 0, \quad f_{\mathfrak{F}_A(N)} = \chi'(D)(N/E_A).$$

Multiplication is the original conductor map and the last quotient has dimension $16n - 48$. The division is holomorphic at zero because PZU4, or membership in the complement in PZU5, proves $E_A(0) \neq 0$. The exact prefactor is $\alpha_A = 1/E_A(0)$, which remains in every original moment map; it has not been set to one.

For the original source degree $k \geq 9$ and target $j = k + 4 \geq 13$, at the two original endpoint indices $e = 0, 1$, BTK25 gives $0 \leq \kappa_e \leq 3 \max(v - e - 2, 0) = 0$. The exact RRC14 sequence then has zero middle space, so its two end spaces vanish. Equivalently

$$\begin{aligned} \mathcal{Z}_e &= \mathcal{S}_e, & \kappa_e &= d_{K,e} = d_{T,e} = 0, & \text{rank } \Psi_e &= c_e, \\ \omega_e(N) &= \omega_{\text{conn},e}(N) \end{aligned} \quad (\text{PZU7})$$

for both unchanged native sources and every retained later target cutoff. To verify the metric assertion, both quotient-value maps now have exactly the same coefficient domain and fibres; minimizing the same full positive native form on each fibre gives the same attained minimum. Thus the first, third and fourth returns in RRC16 are exactly zero, and its second bound becomes $|\Delta \mathcal{R} \log \omega_e| \leq c_e W_j$ with its same four signs and full original width. The connecting rank itself remains

$$c_0 = \dim \mathcal{G}_k, \quad c_1 = \dim \mathcal{G}_k - \dim(\mathcal{G}_k \cap \mathcal{C}_k), \quad (\text{PZU8})$$

through the exact RRC3–11 morphism. None of PZU1–7 assigns a value to these two dimensions. The surviving connecting map, its native minimum sections and its complete signed return remain in the original mixed-state calculation.

The scalar in the same original maps has a quantitatively evaluated finite-period range. Taking reciprocals in PZU4 gives exactly

$$\frac{16|u|^{4\ell_A}}{81 \prod_j |b_j|} \leq |\alpha_A| = |E_{A,u}(0)|^{-1} \leq \frac{16|u|^{4\ell_A}}{\prod_j |b_j|}. \quad (\text{PZU9})$$

In particular its size is of order $|u|^8$ at either exceptional amplitude phase. The vanishing-order result does not remove this scalar. For the original CPQ17 analytic coefficient columns put $\mu_d =$

$E_{A,u}^{(d)}(0)$ and retain $\nu_{a,m} = \sum_{b,c} (Z_{\text{exp},j})_{bc,a} \beta_{j,b,c}^m$. Since $v = 0$ and $\mu_0 \neq 0$, the literal division recursion is

$$\boxed{f_{a,m} = \mu_0^{-1} \left(\nu_{a,m} - \sum_{d=1}^m \binom{m}{d} \mu_d f_{a,m-d} \right), \quad m \geq 0.} \quad (\text{PZU10})$$

Here the empty sum for $m = 0$ is zero. To prove the identity, differentiate $E_{A,u}(w)F_a(w) = N_a(w)$ exactly m times at zero. Leibniz' rule gives $\nu_{a,m} = \sum_{d=0}^m \binom{m}{d} \mu_d f_{a,m-d}$; solving its single nonzero leading term gives PZU10. This is CPQ17 with the proved value of v , not a new choice of rows. The full unchanged Gamma recurrence and row metrics are

$$\begin{aligned} b_{0,m} &= \mathbf{1}_{m=0}, & b_{-1,m} &= 0, \\ b_{n+1,m} &= i^{-1} (b_{n,m-1} - c'_j b_{n,m}) - n(n-1 + s_1/2) b_{n-1,m}, \\ h_{n,s_1} &= (2\pi)^{s_1/2} n! (s_1/2)_n, & c'_j &= j/2 - 4, \\ (t_n)_a &= h_{n,s_1}^{-1/2} \sum_m b_{n,m} f_{a,m}, & (v_n)_{b,c} &= h_{n,s_1}^{-1/2} \sum_m b_{n,m} \beta_{j-8,b,c}^m. \end{aligned} \quad (\text{PZU11})$$

All arrays outside their original polynomial range are zero. The symbols $b_{n,m}$ in this formula are the original polynomial coefficients and retain both source orders $s_1 = 1, j$; the b_j in PZU9 are instead the explicitly evaluated inverse-period leading coefficients from PZU1. PZU10–11 feed the unchanged matrices CPQ15–18, preserving every old minimum section, zero row, full source mass, and signed endpoint return. Thus the period computation sets the low vanishing order to zero while carrying its complete division coefficient into the actual mixed-state rows.

224 The original three-equation transfer of every connecting return

Fix the simple quartet, its actual amplitude, period and logarithm branch on the original five-orbit locus. Keep $k = 4l + 1 \geq 9$, $j = k + 4$, $q = (k + 1)^2$, $Q = (j + 1)^2$, $d = Q - q = 8k + 24$, and the full nonreduced conductor

$$E_A(z) = \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad v = \text{ord}_0 E_A, \quad \mu_v = E_A^{(v)}(0) \neq 0, \quad \alpha_A = v!/\mu_v. \quad (\text{PRT1})$$

No conductor factor is removed. Write $B = B_{\text{cof}}$, $\mathcal{U} = \mathbb{C}^{4m_j}$, $\mathbb{J} = \ker \mathbb{B}$, $\mathcal{B} = \text{im } B$ and $W = \mathcal{B} + \mathbb{J}$, as in CPQ2. Define, in this same coefficient space,

$$S_e = \text{im}[B, F_e], \quad K_e = S_e \cap W, \quad T_e = S_e + W \quad (e = 0, 1). \quad (\text{PRT2})$$

These are the CPQ spaces, with S_e abbreviated only to make the two endpoint subscripts legible. The later norm cutoff N is independent of the defining endpoint e . At every $Q - 1 \leq N \leq 2Q$ use exactly

$$D = N - v, \quad \mathcal{D}_N^{(s)} = \text{diag}(D_{j,D}^{(s)}, 4), \quad c'_j = j/2 - 4, \quad h_{n,s} = (2\pi)^{s/2} n! (s/2)_n, \quad s = 1 \text{ or } j. \quad (\text{PRT3})$$

Thus both actual source pairs $(1, 1)$ and (k, j) are covered in their target order. Their degree- k proper source retains its separate form; none of the identities below substitutes that form for A_{inv} .

224.1 The actual three rows in the original period coefficients

Retain the original polynomials $L_i = (\text{H}(ZW, Z, W, 1)^\top)_i$, $b_{i,a,b} = [Z^a W^b] L_i^4$, the exponential $B_i(z)$ and the translation maps $\mathcal{T}_i, \mathcal{T}_A$ of CPR2. In particular

$$R_i(S) = \frac{\sum_{a,b=0}^4 b_{i,a,b} \chi_j(S + \beta_{4,a,b})}{\chi_k(S)}, \quad b_i = B_i(0), \quad \chi_n(S) = \prod_{a,b=0}^n (S - \beta_{n,a,b}). \quad (\text{PRT4})$$

The quotient is the complete polynomial quotient: root addition makes every numerator vanish at every distinct root of χ_k . Choose the first original index i_* with $b_{i_*} \neq 0$, and put

$$C_i = R_i - (b_i/b_{i_*}) R_{i_*} \quad (i \neq i_*), \quad V = ([S^a] C_i)_{0 \leq a < d, i \neq i_*}. \quad (\text{PRT5})$$

Existence of i_* and independence of these three columns are the complete PDS10–11/CPR17 calculation: the original Bezout identity at zero gives a nonzero b_{i_*} ; the four R_i are independent; their leading coefficients are b_i . Subtracting their exact leading multiples therefore gives three independent columns of degree at most $d - 1$. No period coefficient is replaced in this operation.

Let h_b be the original primitive germs used to define the d columns of F_0 in CIT7. They are specified modulo the entire raw analytic source. Define their relation-value matrix by

$$(M_0)_{a,b} = \Lambda_{h_b}(\mathcal{T}_A(\chi_k S^a)), \quad 0 \leq a < d, \quad L = V^\top M_0. \quad (\text{PRT6})$$

Every entry uses the ordinary, complex-linear transpose. The complete annihilator identity CPR9–10 proves that the displayed d independent relation polynomials are the full dual of the raw source quotient. The h_b are a basis of that quotient, so M_0 is invertible. Thus L has rank exactly three. The matrix is computable from the original period arrays by finite polynomial division and pairing; it contains neither a new numerical model nor a selected nonzero period minor.

Choose once a full column basis Z of $\ker L$ and a three-column complement Y so that $[Z, Y]$ is invertible. Then the literal coefficient spaces obey

$$\boxed{S_1 = \mathcal{B} + F_0 \ker L \subset S_0, \quad S_0/S_1 \text{ has the three lifts } F_0 Y.} \quad (\text{PRT7})$$

Here equality is of subspaces of the original $\mathcal{U}$, not only of abstract quotient dimensions. To prove it, use PDS11: truncation of the endpoint-one primitive quotient identifies it with exactly $\ker V^\top$ in endpoint-zero relation-value coordinates. The missing top value is uniquely $-u_*^\top x_0/b_{i_*}$ with the complete u_* from PDS10. For a representative h , set $f = E_A h$. The original target numerator in component i is

$$N_i(f) = \mathbb{V}_j^{-1}(r![z^r](B_i f))_{0 \leq r < Q}, \quad (\mathbb{V}_j)_{r,(a,b)} = \beta_{j,a,b}^r. \quad (\text{PRT8})$$

Those values use only the truncation through degree $Q - 1$. At endpoint one, the additional four equations assert exactly that the same interpolants also have the required next jet; they do not alter a coefficient in PRT8. Division by the complete E_A gives the same target analytic functions, modulo precisely the original lower exponentials. An analytic source change has numerator d and changes the four target numerators by $(L_i^4 d)_i$, hence changes their DNE coordinates by Bx with the full original scalar α_A . This is the coefficient square RRC5–9/CIT8. It proves both inclusions in PRT7, including every analytic-source and lower-gauge change. Finally F_0 induces an injection modulo $\mathcal{B}$ by CIT9; the invertible matrix $[Z, Y]$ therefore proves the asserted dimension and the stated three lifts.

The coefficient change can itself be written explicitly. For the original endpoint-one primitive columns $h_{1,b}$ put $(M_1)_{a,b} = \Lambda_{h_{1,b}}(\mathcal{T}_A(\chi_k S^a))$ for $0 \leq a \leq d$, and let $P_{<d}$ select the first d rows. Then

$$\Gamma = M_0^{-1} P_{<d} M_1, \quad \text{im } \Gamma = \ker L, \quad F_1 = F_0 \Gamma + B H \quad (\text{PRT7a})$$

for the unique matrix H with the displayed equality. Indeed PDS11 proves the first two assertions, including that Γ has rank $d - 3$. The preceding numerator interpolation calculation shows that each difference column lies in $\mathcal{B}$. The original injectivity of B gives its unique coordinate in H . Equivalently $\Gamma = ZT$ for an invertible $(d - 3)$ -square matrix T , so every original fixed frame change in PRT7 is specified by a literal coefficient equation. No conjugation occurs in M_0, M_1, L, Γ . Hermitian adjoints enter only when evaluating the actual metric below.

224.2 An exact sequence of these three directions

PRT7 gives $K_1 \subset K_0$ and $T_1 \subset T_0$. The identity on coefficient representatives gives the exact sequence

$$0 \longrightarrow K_0/K_1 \xrightarrow{[x] \mapsto [x]} S_0/S_1 \xrightarrow{[x] \mapsto [x]} T_0/T_1 \longrightarrow 0. \quad (\text{PRT9})$$

The first map is injective because $K_0 \cap S_1 = K_1$. If $x \in S_0$ maps to zero on the right, write $x = s + w$ with $s \in S_1$, $w \in W$; then $w = x - s \in K_0$, so its class is the image of K_0/K_1 . Conversely every such class maps to zero. Every element of $T_0 = S_0 + W$ has the class of its S_0 summand modulo $T_1 = S_1 + W$, proving surjectivity. Let

$$a = \dim(T_0/T_1) = \dim(\mathcal{G}_k \cap \mathcal{C}_k) \quad \text{from CPR17–18.} \quad \dim(K_0/K_1) = 3 - a, \quad 0 \leq a \leq 3. \quad (\text{PRT10})$$

The equality with CPR's actual polynomial intersection follows because $\dim(T_e/W) = c_e$ and CPR18 proves $c_0 - c_1 = a$. It can also be calculated directly in the three columns $F_0 Y$: take their images modulo T_1 , retaining the full $\mathbb{B}$, B and L of PRT2/PRT6. Their exact rank is a and their kernel has dimension $3 - a$. This proves a complete original-data rank computation without assuming a generic minor is nonzero. Rank strata, including $a = 0$, are retained.

Give the three terms of PRT9 their original attained metrics

$$A_N = \text{Gram}_{\mathcal{D}_N}(S_0/S_1), \quad B_N = \text{Gram}_{\mathcal{D}_N}(K_0/K_1), \quad C_N = \text{Gram}_{\mathcal{D}_N}(T_0/T_1). \quad (\text{PRT11})$$

Every value frame is fixed in N and in the source-order comparison. For example if X_1 is any fixed independent frame of S_1 , the first one, in the exact three lifts $X_A = F_0 Y$, is

$$A_N = X_A^* \mathcal{D}_N X_A - X_A^* \mathcal{D}_N X_1 (X_1^* \mathcal{D}_N X_1)^{-1} X_1^* \mathcal{D}_N X_A. \quad (\text{PRT12})$$

Select the first independent images of F_0Y modulo T_1 to obtain the a lifts X_C . A full basis of $\ker(F_0Y \bmod T_1)$, followed by subtracting an S_1 representative as in PRT9, gives $3 - a$ lifts X_B in K_0 . Applying PRT12 with (X_1, X_A) replaced by the frames of (K_1, X_B) and (T_1, X_C) gives B_N and C_N . All three are positive definite on their stated quotient values; the determinant of an empty quotient is one.

For completeness every fixed frame factor can be retained before taking a return. For a fixed inclusion $K \subset Y_0$, choose frames P_K, P_{Y_0} and quotient lifts X . Put

$$[P_K, X] = P_{Y_0} R_{Y_0/K}, \quad d_{Y_0/K} = |\det R_{Y_0/K}|^2, \quad g_{Y_0}(N) = \det(P_{Y_0}^* \mathcal{D}_N P_{Y_0}). \quad (\text{PRT13})$$

This is the standard positive-block Schur calculation [H1, §2.1, (2.3)–(2.4)]; the determinant factorization follows from the same block elimination [H2, p. 222, (6a)–(6b)]. Here the block frame is exactly $[P_K, X]$, with its retained factor $d_{Y_0/K}$; neither reference supplies the programme's connecting sequence or its endpoint comparison.

The full block Schur determinant identity is

$$\det \text{Gram}_{\mathcal{D}_N}(Y_0/K) = d_{Y_0/K} g_{Y_0}(N) / g_K(N). \quad (\text{PRT14})$$

It follows by expressing the restricted form in $[P_K, X]$, taking its determinant, and eliminating its upper-left block. This defines and evaluates every factor from the stated original frames. It does not identify a varying minimizing representative with a fixed lift. CPQ's frames for each quotient are used in PRT13 when that quotient appears below.

224.3 Evaluation of the difference of the actual CPQ returns

Use the unchanged target return $\mathcal{R}f = -f(Q - 1) - f(Q) + f(2Q - 1) + f(2Q)$ and write

$$U_A = \mathcal{R} \log \det A_N, \quad U_B = \mathcal{R} \log \det B_N, \quad U_C = \mathcal{R} \log \det C_N. \quad (\text{PRT15})$$

These symbols are returns of precisely the three evaluated quotient matrices PRT11–12. They are not freely assignable parameters. The exact CPQ differences are

$$\begin{aligned} V_{\text{conn},0} - V_{\text{conn},1} &= U_A, & V_{h,0} - V_{h,1} &= U_B, \\ V_{s,0} - V_{s,1} &= U_A - U_B, & V_{t,0} - V_{t,1} &= U_C, \\ V_{\text{rem},1} - V_{\text{rem},0} &= U_C, & \mathcal{L}_0 - \mathcal{L}_1 &= U_C - U_A + U_B. \end{aligned} \quad (\text{PRT16})$$

Here V_b, V_I, V_A are independent of the defining primitive endpoint e : their spaces, coefficient maps and frames are the same CPQ3/9 objects. To prove every line, apply PRT14 to $S_e/\mathcal{B}$, $K_e/\mathcal{B}$, S_e/K_e , T_e/W and $\mathcal{U}/T_e$, respectively. For example the logarithmic determinant difference of the latter two endpoint quotients is $\log g_{T_0} - \log g_{T_1}$ plus the explicitly defined fixed frame-factor difference; PRT14 for T_0/T_1 identifies its nonconstant term with $\log \det C_N$. Each frame factor is independent of N , and $\mathcal{R}1 = -1 - 1 + 1 + 1 = 0$, so it cancels only after this full return. The other four differences follow by the same stated determinant identities. Finally substitute them into CPQ10's $\mathcal{L}_e = V_I + V_{t,e} - V_{s,e} - V_A$.

This also gives the exact propagation, without deleting a complement,

$$\begin{aligned} Z_e &= V_b + V_{\text{rem},e} + V_{s,e} + V_A, & Z_1 - Z_0 &= U_C - U_A + U_B, \\ \mathcal{P}_0^{\text{rest}} - \mathcal{P}_1^{\text{rest}} &= U_C - U_A + U_B, & 4S_{K,j} &= \mathcal{L}_0 + Z_0 = \mathcal{L}_1 + Z_1. \end{aligned} \quad (\text{PRT17})$$

In particular a change of primitive endpoint changes the common loss and complementary sum by exactly opposite amounts in their allocation identity. The original total and the complete-action allocation cancellation are preserved.

224.4 Directed finite bounds with the original masses

Use the proved original reconstruction/source constant of FRE1, evaluated at degree j and order s :

$$A_{j,s}^{\text{end}} = \max\{1, Q\mathcal{B}_{2Q-v,s,j,A}^2\}(M_{j,s}^{\text{rec}})^2. \quad (\text{PRT18})$$

The two factors retain DNE13's full mass $(2\pi)^{s/2}$, the full conductor scalar and division coefficients, and LJR16's original fixed-disk reconstruction, phases and Vandermonde. FRE2-4 proves on this exact metric

$$\mathcal{D}_{Q-1}^{(s)} \preceq \mathcal{D}_N^{(s)} \preceq A_{j,s}^{\text{end}} \mathcal{D}_{Q-1}^{(s)}, \quad \log A_{j,s}^{\text{end}} \leq 2Q \log j + O_{\text{actual}}(Q). \quad (\text{PRT19})$$

Substitution in each full affine fibre defining PRT11 and subsequent minimization preserve both inequalities. Original row extension also proves that each of A_N, B_N, C_N increases with N . If G_N is one of these forms of rank r , set $f(N) = \log \det G_N - \log \det G_{Q-1}$. The generalized eigenvalues in PRT19 show $0 \leq f(N) \leq r \log A_{j,s}^{\text{end}}$ and f is increasing. Hence $\mathcal{R} \log \det G = f(2Q-1) + f(2Q) - f(Q)$ is nonnegative and at most $2r \log A_{j,s}^{\text{end}}$. This proves the actual directed intervals

$$\begin{aligned} 0 \leq U_A \leq 6 \log A_{j,s}^{\text{end}}, \quad 0 \leq U_B \leq 2(3-a) \log A_{j,s}^{\text{end}}, \quad 0 \leq U_C \leq 2a \log A_{j,s}^{\text{end}}, \\ \boxed{0 \leq V_{\text{rem},1} - V_{\text{rem},0} \leq 2a \log A_{j,s}^{\text{end}}.} \\ -2(3-a) \log A_{j,s}^{\text{end}} \leq V_{s,0} - V_{s,1} \leq 6 \log A_{j,s}^{\text{end}}, \\ |\mathcal{L}_0 - \mathcal{L}_1| \leq 6 \log A_{j,s}^{\text{end}}, \quad |Z_1 - Z_0| \leq 6 \log A_{j,s}^{\text{end}}. \end{aligned} \quad (\text{PRT20})$$

The last bound uses $\dim B + \dim C = 3$, keeping the exact sequence rather than separately charging two unrelated rank-three errors. These are finite bounds on actual CPQ expressions. For each fixed original period they are $O_{\text{actual}}(Q \log j) = o(kq)$, for both source pairs. At the actual rank stratum $a = 0$, the remaining-target forms are the same quotient in fixed value frames and their returns are exactly equal. At $a = 3$, the kernel spaces coincide and $U_B = 0$ exactly. All intermediate strata remain governed by their actual original three-column matrix. No claim is made that either endpoint's common leading return has now been evaluated; PRT16 evaluates their difference and reduces its calculation to the stated three original directions.

The proof requires the five-orbit coefficient construction, its original distinct quartet roots and $\mu_v \neq 0$. It does not use the additional pole-coordinate gate minors of RPI/RPK. Consequently a period excluded by a pole-coordinate pivot remains covered when it belongs to the original five-orbit locus: PRT6/8/12 construct all maps without that pivot. Rank changes of the three-row image are handled by PRT10. The value v is retained throughout; neither $v \leq 32$ nor a generic value of v has been substituted for an actual coefficient.

225 The first original row: exact lower cancellation and its finite bound

Fix the actual simple quartet, original conductor coefficients, period, and branch. Write

$$\begin{aligned}
 j &\equiv 1 \pmod{4}, \quad j \geq 13, \quad q = (j+1)^2, \quad q' = (j-7)^2, \quad c' = j/2 - 4, \\
 E_A(z) &= \sum_{a,b=0}^8 a_{ab} e^{\beta_{8,a,b} z}, \quad \mu_d = E_A^{(d)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \\
 \beta_{r,a,b} &= r/2 + (2a-r)\delta + i(2b-r)\gamma, \quad \sigma = (\delta^2 + \gamma^2)^{1/2}, \\
 \chi_j(S) &= \prod_{a,b=0}^j (S - \beta_{j,a,b}), \quad \chi'(S') = \chi_{j-8}(S') = \prod_{a,b=0}^{j-8} (S' - \beta_{j-8,a,b}), \\
 n &= q - v, \quad m = n - q' = 16j - 48 - v \geq 0, \quad s \in \{1, j\}, \quad \ell = (s-1)/4.
 \end{aligned} \tag{FRG1}$$

The prime on χ' names the complete lower root polynomial; it is not differentiation. The original norm is

$$\begin{aligned}
 \|P\|_{s,c'}^2 &= \int_{\mathbb{R}} |P(c' + iy)|^2 dm_s(y), \quad dm_s = \frac{M_s 2^{s/2}}{4\pi \Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
 p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{r+1,s} = y p_{r,s} - r(r-1+s/2) p_{r-1,s}, \\
 h_{r,s} &= M_s r! (s/2)_r, \quad \phi_r(S') = p_{r,s}((S' - c')/i) / \sqrt{h_{r,s}}.
 \end{aligned} \tag{FRG2}$$

These are exactly NSO2,7's native orthonormal polynomials, with the full generating-function and orthogonality proof in SGT2-5; its source mass and recurrence coefficients are unchanged here. Thus the leading coefficient of ϕ_r is $i^{-r}/\sqrt{h_{r,s}}$. All inner products are conjugate-linear in the first variable, whereas every functional below is complex-linear.

225.1 The actual annihilator and the full first-row leverage

Let $F_b = N_b/E_A$ be the original NSO12 numerator germs, with $N_b(z) = \sum_{a,b'} (Z_{\text{exp},j})_{(a,b'),b} e^{\beta_{j,a,b'} z}$. Keep their original fixed NSO12 frame. Let V_0, T_0 be NSO13's lower and numerator matrices on rows $0, \dots, n-1$, and let v_n, t_n be their next rows. There are q' lower columns and m numerator columns; therefore

$$F_0 = [V_0, T_0] \in \mathbb{C}^{n \times n}, \quad w_n = [v_n, t_n]. \tag{FRG3}$$

The NSO12-13 Vandermonde proof applies at cutoff $n-1 = q-v-1$: if such a combined functional has its first n derivatives zero, multiplication of its germ by E_A makes an upper-grid numerator have its first q derivatives zero. The upper Vandermonde kills this numerator, its class in the fixed conductor quotient vanishes, and the lower Vandermonde kills the remaining lower coefficients. Since the triangular change from derivatives to the ϕ_r values has nonzero diagonal, F_0 is invertible.

Define the original translation and polynomial

$$(\mathcal{T}_A P)(S') = \sum_{a,b=0}^8 a_{ab} P(S' + \beta_{8,a,b}), \quad g = \mathcal{T}_A \chi_j, \quad \lambda = \mu_v \binom{q}{v} \neq 0. \tag{FRG4}$$

Expanding every shift shows that all coefficients above degree $q-v$ vanish and the coefficient at degree $q-v$ is λ : for a monomial S^d , its coefficient at S'^{d-r} is $\binom{d}{r} \mu_r$, and $\mu_0 = \dots = \mu_{v-1} = 0$. Hence $\deg g = n$. For any polynomial P the finite Leibniz identity gives the exact pairing

$$\Lambda_{F_b}(\mathcal{T}_A P) = \Lambda_{E_A F_b}(P) = \Lambda_{N_b}(P). \tag{FRG5}$$

The right side for $P = \chi_j$ is a linear combination of evaluations of χ_j at all its upper roots, so is zero. Also $\beta_{j-8,u,w} + \beta_{8,a,b} = \beta_{j,u+a,w+b}$ lies in the original upper grid. Every summand in $g(\beta_{j-8,u,w})$ is zero. Distinctness of all the lower roots proves $\chi' \mid g$. Consequently all columns in $[V, T]$ annihilate g .

Expand $g = \sum_{r=0}^n c_r \phi_r$. The exact top coefficient and row relation are

$$c_n = i^n \lambda \sqrt{h_{n,s}}, \quad c_{<n}^\top F_0 + c_n w_n = 0. \quad (\text{FRG6})$$

Here the transpose is ordinary because of complex linearity. Since F_0 is square and invertible, $F_0(F_0^* F_0)^{-1} F_0^* = I_n$. Thus

$$\begin{aligned} \zeta_{\text{tot}} &= w_n (F_0^* F_0)^{-1} w_n^* = \frac{\sum_{r < n} |c_r|^2}{|c_n|^2}, \\ 1 + \zeta_{\text{tot}} &= \frac{\|\mathcal{T}_A \chi_j\|_{s,c'}^2}{|\lambda|^2 h_{n,s}}. \end{aligned} \quad (\text{FRG7})$$

No factorial, phase, conductor coefficient, or source mass was removed.

225.2 The exact lower ideal norm

Let

$$\nu_{m,s} = \min_{P \text{ monic, deg } P=m} \|\chi' P\|_{s,c'}^2. \quad (\text{FRG8})$$

This is the attained positive norm on the original monic affine fibre; it exists because its finite positive Gram has a unique quadratic minimum. A monic degree- n polynomial $u = \sum_{r=0}^n d_r \phi_r$ has $d_n = i^n \sqrt{h_{n,s}}$. Divisibility by χ' is exactly its vanishing at all lower roots, hence exactly

$$V_0^\top d_{<n} = -d_n v_n^\top. \quad (\text{FRG9})$$

The matrix V_0 has full column rank. In these coefficient coordinates the attained minimum is

$$d_{<n,\min} = -d_n \overline{V_0} (V_0^\top \overline{V_0})^{-1} v_n^\top. \quad (\text{FRG10})$$

Multiplication proves the constraint. Its ordinary Hermitian pairing with every vector in $\ker V_0^\top$ vanishes, proving the minimum and its Pythagorean decomposition. Its squared norm is $|d_n|^2 \overline{v_n} (V_0^\top \overline{V_0})^{-1} v_n^\top = |d_n|^2 \zeta_{\text{low}}$, where $\zeta_{\text{low}} = v_n (V_0^* V_0)^{-1} v_n^*$ is real and nonnegative. Thus

$$\boxed{\nu_{m,s} = h_{n,s} (1 + \zeta_{\text{low}}).} \quad (\text{FRG11})$$

This is a typed monic-fibre map, not a change of the original measure. In y coordinates it retains the phase explicitly: write $\chi'(c' + iy) = i^{q'} Q'(y)$, with monic Q' , and $P(c' + iy) = i^m \tilde{P}(y)$, with monic $\tilde{P}$. The forward and inverse maps on monic fibres are $\tilde{P}(y) = i^{-m} P(c' + iy)$ and $P(S') = i^m \tilde{P}((S' - c')/i)$. Their total phase is i^n and has modulus one; their squared norm is precisely FRG8.

Put $A = V_0^* V_0$, $B = V_0^* T_0$, $D_0 = T_0^* T_0 - B^* A^{-1} B$, and $\rho = t_n - v_n A^{-1} B$. Block factorization of $F_0^* F_0$ gives

$$\zeta_{\text{tot}} = \zeta_{\text{low}} + \rho D_0^{-1} \rho^*, \quad r_n = \frac{\rho D_0^{-1} \rho^*}{1 + \zeta_{\text{low}}}. \quad (\text{FRG12})$$

For instance factor the Gram as $\begin{bmatrix} I & 0 \\ B^* A^{-1} & I \end{bmatrix} \text{diag}(A, D_0) \begin{bmatrix} I & A^{-1} B \\ 0 & I \end{bmatrix}$ and multiply its inverse against $[v_n, t_n]$ and its adjoint. Equations FRG7 and FRG11 now prove

$$\boxed{1 + r_n = \frac{1 + \zeta_{\text{tot}}}{1 + \zeta_{\text{low}}} = \frac{\|\mathcal{T}_A \chi_j\|_{s,c'}^2}{|\mu_v(q)|^2 \nu_{m,s}}.} \quad (\text{FRG13})$$

The scalar λ maps g to the unique monic polynomial g/λ in its original one-dimensional coefficient fibre; it is kept in FRG13. In particular, if $P_{m,s}^{\min}$ is the unique monic minimizer in FRG8, then

$$1 + r_n = 1 + \frac{\|g/\lambda - \chi' P_{m,s}^{\min}\|_{s,c'}^2}{\nu_{m,s}} \geq 1. \quad (\text{FRG14})$$

Indeed the difference belongs to $\chi' \mathbb{C}[S']_{<m}$, to which the minimizer is orthogonal. This also characterizes the exact zero-row case: $r_n = 0$ precisely when $g/\lambda = \chi' P_{m,s}^{\min}$.

225.3 An explicit finite bound from the original root products

Define

$$\begin{aligned} \theta &= \pi/2, \quad h = 1/q, \quad \kappa = \theta - h > 0, \quad \beta_s = \frac{M_s}{\sqrt{2}\Gamma(s/2)}, \\ L_h &= \frac{\pi}{\Gamma(3/4)^2} (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\ Q_- &= q' + \ell, \quad Q_+ = q + \ell, \quad R' = (j - 8)\sigma, \quad M_0 = 2Q_-/\theta, \quad M = \max\{M_0, 2R'\}, \\ B_+ &= \max\{1, (j + 8)\sigma, 2\ell\}, \quad A_{\text{abs}} = \sum_{a,b=0}^8 |a_{ab}|, \\ f_\theta(x) &= 2x \log(2x/\theta) - 2x, \quad F_M = 2Q_- \log M - \theta M, \\ \epsilon_- &= \left(\theta - \frac{2Q_-}{M} \right) + \frac{Q_-}{M^2} + h(M + 1) + \frac{4q'R'}{M}, \quad C_m = (2m + 1) \binom{2m}{m}. \end{aligned} \quad (\text{FRG15})$$

The two inequalities $M \geq 2R'$ and $M \geq M_0$ hold by definition, so the finite bounds below apply at every $j \geq 13$. For each fixed quartet, $M = M_0$ eventually, because M_0 grows quadratically and R' linearly in j . On that eventual range, $F_M = f_\theta(Q_-)$ and $\epsilon_- = \theta^2/(4Q_-) + h(M + 1) + 4q'R'/M$.

For completeness, the Gamma input used in the bound is proved as follows. Gamma recurrence gives

$$dm_s(y) = \beta_s \prod_{r=0}^{\ell-1} (y^2 + (2r + 1/2)^2) d\sigma(y), \quad d\sigma(y) = \frac{|\Gamma(1/4 + iy/2)|^2}{2\pi} dy. \quad (\text{FRG16})$$

The factor $4^{-\ell}$ from the recurrence is cancelled exactly by the $2^{2\ell}$ in the original density. Rotate Euler's integral for $\Gamma(a + iy/2)$ through $\text{sgn}(y)(\pi/2 - h)$: absolute integration along that ray proves $|\Gamma(a + iy/2)| \leq \Gamma(a)(\sin h)^{-a} e^{-(\pi/2-h)|y|/2}$. The small closing arc tends to zero as its radius to the power a , and the large closing arc tends to zero since its cosine is at least $\sin h$. For the lower bound use the exact reflection product $|\Gamma(1/4 + iy/2)\Gamma(3/4 - iy/2)|^2 = 2\pi^2/\cosh(\pi y)$, the same upper bound on the second Gamma factor, and $\cosh(\pi y) \leq e^{\pi|y|}$. Therefore, also at $y = 0$,

$$L_h e^{-(\theta+h)|y|} \leq \frac{d\sigma}{dy}(y) \leq U_h e^{-\kappa|y|}. \quad (\text{FRG17})$$

For $y \in [M, M + 1]$, every original lower root in the y coordinate has modulus at most R' . Since $R'/y \leq 1/2$, integration of $(1 - t)^{-1} \leq 2$ gives $|\chi'(c' + iy)|^2 \geq y^{2q'} e^{-4q'R'/M}$. Every Gamma recurrence factor is at least y^2 . Writing $y = M + t$, $0 \leq t \leq 1$, the inequality $\log(1 + x) \geq x - x^2/2$ gives

$$2Q_- \log y - \theta y \geq F_M - \left(\theta - \frac{2Q_-}{M} \right) - \frac{Q_-}{M^2}. \quad (\text{FRG18})$$

Thus the complete density $|\chi'(c' + iy)|^2 dm_s/dy$ on this one original-coordinate interval is at least $\beta_s L_h \exp(F_M - \epsilon_-)$.

Here is the exact monic minimum on an interval of length one. Set $L_m(x) = \frac{1}{m!} \frac{d^m}{dx^m} [x^m(x-1)^m]$. Its leading coefficient is $\binom{2m}{m}$. Integrating by parts m times, all boundary terms vanish, proving its orthogonality to degrees less than m . The same integrations give

$$\int_0^1 L_m(x)^2 dx = \binom{2m}{m} \int_0^1 x^m(1-x)^m dx = \frac{1}{2m+1}. \quad (\text{FRG19})$$

The beta integral here is $m!^2/(2m+1)!$, obtained by repeated integration by parts. Subtracting $L_m/\binom{2m}{m}$ from any complex monic polynomial makes an orthogonal lower-degree difference. The least squared norm is exactly C_m^{-1} . Translation of the interval preserves its leading coefficient. The polynomial $P(c' + iy)$ for monic P has leading coefficient i^m ; multiply the displayed minimizing polynomial by that retained phase. Applying this minimum to FRG18 yields

$$\nu_{m,s} \geq \frac{\beta_s L_h}{C_m} \exp(F_M - \epsilon_-). \quad (\text{FRG20})$$

On the entire real line, for every original shift and upper root,

$$|c' + iy + \beta_{8,a,b} - \beta_{j,u,w}| \leq |y| + (j+8)\sigma \leq |y| + B_+. \quad (\text{FRG21})$$

Indeed its constant part is $[(2a-8) - (2u-j)]\delta + i[(2b-8) - (2w-j)]\gamma$. Every original summand in g therefore has modulus at most $(|y| + B_+)^q$, and the complete sum has modulus at most $A_{\text{abs}}(|y| + B_+)^q$. Also $2r + 1/2 \leq 2\ell \leq B_+$ for every recurrence factor when $\ell > 0$, and the empty product for $\ell = 0$ is one. FRG16–17, the substitution $t = y + B_+$ on $y \geq 0$, and enlargement of its integral to $[0, \infty)$ prove

$$\|g\|_{s,c'}^2 \leq 2\beta_s U_h A_{\text{abs}}^2 e^{\kappa B_+} \frac{(2Q_+)!}{\kappa^{2Q_++1}}. \quad (\text{FRG22})$$

The cancellation of coefficients above degree n is still exact in g ; the bound estimated the complete original finite sum before those cancellations and changed no polynomial.

Combining FRG13, FRG20 and FRG22 gives the explicit finite interval

$$\begin{aligned} 0 \leq \log(1 + r_n) &\leq E_{j,s}, \\ E_{j,s} &= 2 \log \frac{A_{\text{abs}}}{|\mu_v(q)|} + \log \frac{2U_h}{L_h} + \kappa B_+ + \log((2Q_+)!) - (2Q_+ + 1) \log \kappa \\ &\quad - F_M + \epsilon_- + \log C_m. \end{aligned} \quad (\text{FRG23})$$

The mass factor β_s cancels between these two displayed bounds on the same original norm. In particular FRG23 has not assumed unit mass. Positivity of the exact ratio proves that this computed upper bound is nonnegative throughout its stated domain.

225.4 Evaluation of the first-row scale and the cancelled leading term

All the following limits fix the actual quartet and conductor, including μ_v , and let $j \equiv 1 \pmod{4}$ tend to infinity. They hold for both $s = 1$ and $s = j$. First, $Q_{\pm} \asymp j^2$, $Q_+ - Q_- = 16j - 48$, $B_+ = O(j)$, $R' = O(j)$, and $m = O(j)$. Eventually $M = M_0$, so $F_M = f_{\theta}(Q_-)$. Directly from FRG15, $\epsilon_- = O(j)$; also $\log C_m \leq \log(2m+1) + 4m \log 2 = O(j + \log j)$. Since $2h/\pi \leq \sin h \leq h$, $\log(U_h/L_h) = O(\log q)$. The fixed- v product for $\binom{q}{v}$ makes $|\log |\lambda|| = O(\log q)$.

For every integer $N \geq 1$, integration bounds on the increasing function $\log x$ give

$$N \log N - N + 1 \leq \log(N!) \leq N \log N - N + 1 + \log N. \quad (\text{FRG24})$$

Consequently

$$\begin{aligned} \log((2Q_+)!) - (2Q_+ + 1) \log \kappa &\leq f_{\theta}(Q_+) + 1 + \log(2Q_+) - \log \theta \\ &\quad + (2Q_+ + 1) \log(\theta/\kappa). \end{aligned} \quad (\text{FRG25})$$

Here $\log(\theta/\kappa) \leq h/(\theta - h)$, so the last term is $O(1)$. Since $f'_\theta(x) = 2 \log(2x/\theta)$, its integral between Q_- and Q_+ is $O(j \log j)$. Every term of the fully finite FRG23 is now controlled, proving

$$\boxed{0 \leq \log(1 + r_n) = O_{\text{actual}}(j \log j).} \quad (\text{FRG26})$$

The leading terms of the two leverages themselves can also be evaluated without taking a difference of uncontrolled remainders. For the upper bound on $\nu_{m,s}$ choose the actual monic trial $P(S') = (S' - c')^m$. Put $B_- = \max\{1, R', 2\ell\}$ and $Q_0 = n + \ell$. The same whole-line integral used in FRG22 now gives

$$\nu_{m,s} \leq 2\beta_s U_h e^{\kappa B_-} \frac{(2Q_0)!}{\kappa^{2Q_0+1}}. \quad (\text{FRG27})$$

The factor $|y|^{2m}$ is bounded by $(|y| + B_-)^{2m}$ and $q' + m = n$, so the exponent is exactly $2Q_0$. Both $Q_- - q$ and $Q_0 - q$ are $O(j)$. Equations FRG20, FRG24–25, FRG27 and the preceding explicit estimates imply

$$\log \nu_{m,s} = \log \beta_s + f_\theta(q) + O_{\text{actual}}(j \log j). \quad (\text{FRG28})$$

To retain the full original factorial normalization, write $b = s/2 = 2\ell + 1/2$ and $u = n + 2\ell$. Then

$$h_{n,s} = \sqrt{2}\beta_s n! \Gamma(u + 1/2), \quad \Gamma(u + 1/2) = \frac{\sqrt{\pi}(2u)!}{4^u u!}. \quad (\text{FRG29})$$

FRG24 in these three factorials shows $\log \Gamma(u + 1/2) = u \log u - u + O(\log u)$. Both $n - q$ and $u - q$ are $O(j)$; integration of $\log x$ between these arguments and q therefore gives

$$\log h_{n,s} = \log \beta_s + 2(q \log q - q) + O_{\text{actual}}(j \log j). \quad (\text{FRG30})$$

Subtracting the two controlled expressions in FRG28 and FRG30, and then using the exact identity FRG13 together with FRG26, proves

$$\boxed{\begin{aligned} \log(1 + \zeta_{\text{low}}) &= 2q \log(4/\pi) + O_{\text{actual}}(j \log j), \\ \log(1 + \zeta_{\text{tot}}) &= 2q \log(4/\pi) + O_{\text{actual}}(j \log j). \end{aligned}} \quad (\text{FRG31})$$

Thus the same positive order- q term belongs to both original leverages and cancels in their exact corrected ratio. This conclusion uses no limiting theorem for a growing-rank determinant and no derivative of an asymptotic error.

Finally the original SFB13–15 four-copy update has budget $\beta_n = 4 \log(1 + r_n)$. Its common signed increment satisfies $|\delta L_n| \leq \beta_n$, either by SFB8 integrated along this positive update, or by the same original projection determinants. For this first original row, FRG26 therefore yields

$$0 \leq \beta_n = O_{\text{actual}}(j \log j), \quad \frac{|\delta L_n|}{jq} \longrightarrow 0. \quad (\text{FRG32})$$

Each original attained quotient in CPQ14 also has its first-row logarithmic determinant increment between 0 and β_n : along the same positive path, its derivative is the trace of the positive relative derivative against the orthogonal projector on its minimum-lift quotient space, which lies between zero and the full trace. Integrating proves the claim with the original quotient kept. Hence each of the five complementary first-row increments is $O_{\text{actual}}(j \log j)$ as well. This evaluates one actual row; FRG32 makes no assertion about the remaining q consecutive rows.

226 The first original mixed row in the full Gamma polynomial ideal

Keep the original hypothetical simple quartet, fixed five-orbit period, logarithm branch, complete nonreduced conductor, and both source pairs of NSO and CPQ. This calculation concerns their actual first appended target row, rather than a freely chosen matrix update. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q &= (j + 1)^2, \\ q' &= (j - 7)^2, & c' &= j/2 - 4, & n &= q - v, & m &= n - q' = 16j - 48 - v. \end{aligned} \quad (\text{FMG1})$$

Here $v = \text{ord}_0 E_A$ is the actual conductor order. The retained BTK estimate is $v \leq 32$ on its stated original domain; the earlier $v \leq 80$ already makes $m > 0$ in this calculation. Let $s = 1$ or $s = j$, according to the prescribed source pair. Throughout,

$$\begin{aligned} \beta_{a,u,w} &= a\left(\frac{1}{2} - \delta - i\gamma\right) + 2u\delta + 2wi\gamma, \\ \chi_a(S) &= \prod_{u,w=0}^a (S - \beta_{a,u,w}), \\ E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \\ \mu_r &= E_A^{(r)}(0), \\ (\mathcal{T}_A p)(S') &= \sum_{u,w=0}^8 a_{uw} p(S' + \beta_{8,u,w}), \\ \ell &= \mu_v \binom{q}{v} \neq 0. \end{aligned} \quad (\text{FMG2})$$

The symbol ℓ in FMG2 is a leading coefficient, separate from the integer l in $k = 4l + 1$. No root, period coefficient or conductor factor has been discarded. Define the actual polynomials

$$\chi_- := \chi_{j-8}, \quad g := \mathcal{T}_A \chi_j. \quad (\text{FMG3})$$

For every lower root $\beta_{j-8,a,b}$, each summand defining $g(\beta_{j-8,a,b})$ evaluates χ_j at the upper root $\beta_{j,a+u,b+w}$. Thus χ_- divides g exactly. The first nonzero moment in FMG2 gives $\deg g = n$ and $\text{lc}_{S'} g = \ell$ by the binomial formula. This is a divisibility assertion for the full translated polynomial, including every coefficient.

The original measure and orthonormal polynomials are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, & M_s &= (2\pi)^{s/2}, \\ p_{0,s} &= 1, & p_{1,s} &= y, & p_{a+1,s} &= yp_{a,s} - a(a-1+s/2)p_{a-1,s}, \\ h_{a,s} &= M_s a! (s/2)_a, & \phi_a(S') &= p_{a,s}((S' - c')/i) / \sqrt{h_{a,s}}, \\ \langle f_1, f_2 \rangle_s &= \int_{\mathbb{R}} \overline{f_1(c' + iy)} f_2(c' + iy) dm_s(y). \end{aligned} \quad (\text{FMG4})$$

This inner product is conjugate-linear in its first argument. In particular the leading coefficient of ϕ_a in S' is $i^{-a} / \sqrt{h_{a,s}}$.

226.1 The exact original row and its two coupled leverages

Use exactly the NSO12 quotient columns $F_b = N_b/E_A$, $1 \leq b \leq m$. Their numerators lie on the original degree- j grid and vanish to order v at zero. The original lower evaluation and numerator matrices at the preceding cutoff are

$$V_0[a, (u, w)] = \phi_a(\beta_{j-8,u,w}), \quad T_0[a, b] = \Lambda_{F_b}(\phi_a), \quad 0 \leq a < n, \quad (\text{FMG5})$$

where $\Lambda_F(S'^a) = F^{(a)}(0)$ is complex-linear. There are $q' + m = n$ columns in $[V_0, T_0]$. To prove their independence, a zero combination of their first n native values has a generating germ F with $F^{(a)}(0) = 0$ for $0 \leq a < n$, since the polynomial change of basis in FMG4 is invertible. Its numerator $E_A F$ is a degree- j grid exponential polynomial and vanishes through order $n + v - 1 = q - 1$. The original upper Vandermonde is invertible, so that numerator is zero. The NSO12 quotient frame then kills the numerator coefficient, and the lower Vandermonde kills the remaining lower coefficient. This proves that $[V_0, T_0]$ is square and invertible, with its complete original frame.

Write the actual next rows as v_n, t_n . Define

$$\begin{aligned} A_L &= V_0^* V_0, & P_V &= V_0 A_L^{-1} V_0^*, & D_0 &= T_0^* (I - P_V) T_0, \\ \zeta &= v_n A_L^{-1} v_n^*, & \rho &= t_n - v_n A_L^{-1} V_0^* T_0, & r &= \frac{\rho D_0^{-1} \rho^*}{1 + \zeta}. \end{aligned} \quad (\text{FMG6})$$

These are precisely the first SFB12 and NTE1 quantities. Inverses exist by the independence just proved. Set $C_0 = [V_0, T_0]$ and $w_n = [v_n, t_n]$. The total leverage is $\zeta_{\text{all}} = w_n (C_0^* C_0)^{-1} w_n^*$.

Expand $g = \sum_{a=0}^n c_a \phi_a$ with column $c_{<} = (c_0, \dots, c_{n-1})^\top$. The leading coefficient is $c_n = \ell i^n \sqrt{h_{n,s}}$. Every column functional in FMG5 annihilates g . For the numerator columns, the full Leibniz identity gives

$$\Lambda_{F_b}(\mathcal{T}_A \chi_j) = \Lambda_{E_A F_b}(\chi_j) = \sum_{u,w} (N_b)_{uw} \chi_j(\beta_{j,u,w}) = 0.$$

For the lower columns, FMG3 gives the same zero. These are complex-linear equations, so their coefficient row is an ordinary transpose. Therefore

$$w_n = -c_n^{-1} c_{<}^\top C_0, \quad 1 + \zeta_{\text{all}} = \frac{\|g\|_s^2}{|\ell|^2 h_{n,s}}, \quad 1 + r = \frac{1 + \zeta_{\text{all}}}{1 + \zeta}. \quad (\text{FMG7})$$

The middle identity uses the invertibility of C_0 , not a replacement of its Gram by the identity. To prove the last identity with every cross term retained, replace the numerator columns in C_0 by $(I - P_V)T_0$ using the invertible triangular column operation which subtracts $V_0 A_L^{-1} V_0^* T_0$. The preceding Gram then has diagonal blocks A_L, D_0 and the new row is $[v_n, \rho]$. Consequently $\zeta_{\text{all}} = \zeta + \rho D_0^{-1} \rho^*$, which gives FMG7.

226.2 A polynomial minimum for the complete lower denominator

For each integer $a \geq 0$ let

$$\nu_a = \min_{P \text{ monic, deg } P=a} \|\chi_- P\|_s^2 > 0. \quad (\text{FMG8})$$

The positive quadratic minimum used in FMG6–10 is a Schur-complement calculation [H1, §2.1, (2.3)–(2.4)]. Here the eliminated coordinates are the coefficients of the complete lower-root ideal. The following calculation retains the ordinary transpose in its evaluation constraint and the Hermitian adjoint only in its norm; the cited matrix method does not supply the Gamma-polynomial estimates.

The coefficients here are in the unchanged variable S' . This is an attained finite-dimensional minimum in the original measure; strict positivity follows because a nonzero polynomial cannot vanish on the entire integration line. The transformation $P \mapsto \ell \chi_- P$ is an explicitly invertible map from the monic degree- a affine fibre to the fibre with leading coefficient ℓ and factor χ_- . The norm under that map is multiplied by exactly $|\ell|^2$; no scalar is suppressed.

At degree $n = q' + m$, a monic polynomial with factor χ_- has orthonormal coefficient $a_n = i^n \sqrt{h_{n,s}}$. Its old coefficient column $a_{<}$ must satisfy $V_0^\top a_{<} = -a_n v_n^\top$. This is the exact evaluation constraint, including the ordinary transpose. Its unique coefficient minimum is

$$a_{<} = -a_n \overline{V_0} (V_0^\top \overline{V_0})^{-1} v_n^\top. \quad (\text{FMG9})$$

Indeed the displayed column satisfies the constraint, and it is orthogonal to the constraint kernel. Its squared norm is $h_{n,s}v_n A_L^{-1}v_n^* = h_{n,s}\zeta$; the equality follows by conjugating the middle Gram and using that the resulting scalar is real. Adding $|a_n|^2$ proves

$$\boxed{\nu_m = h_{n,s}(1 + \zeta), \quad 1 + r = \frac{\|g\|_s^2}{|\ell|^2\nu_m}.} \quad (\text{FMG10})$$

Thus the lower denominator in the corrected original row is calculated by an actual lower-root ideal minimum. Replacing it by one would remove the explicitly present factor $\nu_m/h_{n,s}$.

226.3 All eight compressed matrices in exact polynomial coordinates

Put $f_a(S') = \chi_-(S')S'^a$ for $0 \leq a < m$, and write $f_a = \sum_{b < n} C_{ba}\phi_b$. Define

$$G = C^*C = (\langle f_a, f_b \rangle_s)_{a,b < m}, \quad H = C^*c_{<} = (\langle f_a, g \rangle_s)_{a < m}, \quad A = C^T T_0 = (\Lambda_{F_b}(f_a))_{a < m, b \leq m}. \quad (\text{FMG11})$$

The column span of C is exactly $\ker V_0^T$: every f_a vanishes on the lower roots; their degrees are distinct; and both spaces have dimension m . Thus

$$I - P_V = \overline{C} \overline{G}^{-1} C^T, \quad D_0 = A^* \overline{G}^{-1} A. \quad (\text{FMG12})$$

The first displayed matrix is self-adjoint, idempotent, and has range $\ker V_0^*$, proving it is exactly the required orthogonal projector. It also proves the second identity. The matrix A is invertible: if its product with x is zero, FMG12 gives $x^* D_0 x = 0$, and the positivity already proved forces $x = 0$. This is the original functional pairing on $\chi_- \mathbb{C}[S']_{<m}$, with its row and column orientations specified in FMG11.

The projection of g onto that polynomial ideal is $CG^{-1}H$. Its perpendicular component has leading coefficient ℓ and is exactly the monic-fibre minimizer multiplied by ℓ . Orthogonality and FMG8 therefore give

$$|\ell|^2\nu_m = \|g\|_s^2 - H^*G^{-1}H, \quad r = \frac{H^*G^{-1}H}{|\ell|^2\nu_m}, \quad \frac{\rho}{\sqrt{1+\zeta}} = \mathbf{b}A, \quad \mathbf{b} = -\frac{H^T \overline{G}^{-1}}{\ell i^n \sqrt{\nu_m}}. \quad (\text{FMG13})$$

For the row identity, substitute FMG7 in the definition of ρ , then apply FMG12 and $c_{<}^T \overline{C} = H^T$. Its denominator is $c_n \sqrt{1+\zeta} = \ell i^n \sqrt{\nu_m}$ by FMG10. This calculation proves both the phase and the single occurrence of the original leading coefficient.

Retain every fixed original CPQ coefficient subspace of $\mathcal{U} = \mathbb{C}^{4m}$:

$$\begin{aligned} I &= \text{im}(B_{\text{cof}}J_0), & B &= \text{im } B_{\text{cof}}, & S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \text{diag}(B_j, 4), \\ K_i &= I \cap J, & W_i &= I + J, & W &= B + J, & K_c &= S \cap W, & T_e &= S + J, & e &= 0, 1. \end{aligned} \quad (\text{FMG14})$$

Their matrices are the original period-dependent maps; none is replaced by a plane having merely the same dimension. Let X denote any fixed independent column frame for one of these eight subspaces. Set

$$\mathcal{A} = \text{diag}(A, 4), \quad \mathcal{G}^\vee = \text{diag}(\overline{G}^{-1}, 4), \quad \lfloor = \text{diag}(\mathbf{b}, 4), \quad Y_X = \mathcal{A}X, \quad (\text{FMG15})$$

$$\boxed{K_{X,n} = \lfloor Y_X (Y_X^* \mathcal{G}^\vee Y_X)^{-1} Y_X^* \rfloor^*.$$

Here $\text{diag}(\mathbf{b}, 4)$ has four one-row blocks, so FMG15 is exactly a 4×4 matrix. If X is empty the result is the zero matrix. To verify FMG15 in the original metric, FMG12 gives $D_0 = \mathcal{A}^* \mathcal{G}^\vee \mathcal{A}$, and FMG13 gives $\mathcal{U}_n = \lfloor \mathcal{A}$. Substituting these identities in NTE1 proves FMG15 as literal equality, with no spectral approximation. The map $x \mapsto \mathcal{A}x$ is an isometry from $(\mathcal{U}, \mathcal{D}_0)$ to $(\mathbb{C}^{4m}, \mathcal{G}^\vee)$, with inverse $\mathcal{A}^{-1}$. It carries sums and intersections bijectively; in particular it preserves both full chains

$$J \subset W_i \subset W \subset T_e, \quad K_i \subset I \subset K_c \subset S. \quad (\text{FMG16})$$

All original invariant, conductor and connecting coefficients are retained in Y_X through their exact pairing $\mathcal{A}$.

Writing $b_X = \log \det(I_4 + K_{X,n})$, the first actual signed row is

$$\delta \mathcal{L}_{e,n} = b_{T_e} + b_{K_c} + b_{W_i} + b_{K_i} - b_W - b_S - b_J - b_I. \quad (\text{FMG17})$$

Equations FMG2–15 evaluate every entry using original polynomial inner products and the actual period coefficient maps. They retain all four eigenvalues of each matrix, including zero eigenvalues. The same equality supplies its threshold contribution by replacing b_X by $\sum_{a=1}^4 \log \max(1, \lambda_{X,a})$ with $\lambda_{X,a}$ the four eigenvalues of FMG15. No simultaneous diagonalization is assumed. Every matrix is positive and bounded above by rI_4 , since FMG15 is the unchanged NTE1 compression and FMG13 evaluates its full relative row norm.

226.4 Finite evaluation from the unchanged Gamma moments

The matrices in FMG11 involve no infinite series. For explicit finite evaluation, expand each polynomial on the original line, $f(c' + iy) = \sum_a d_a(f)y^a$. Define $M_r^{(s)} = \int y^r dm_s(y)$. Then

$$\langle f_1, f_2 \rangle_s = \sum_{a,b} \overline{d_a(f_1)} d_b(f_2) M_{a+b}^{(s)}. \quad (\text{FMG18})$$

These moments can be evaluated directly by the retained recurrence: for a requested degree r , form the $(r+1)$ -square real symmetric Jacobi matrix with off-diagonal entries $J_{a,a+1} = J_{a+1,a} = \sqrt{(a+1)(a+s/2)}$, $0 \leq a < r$, and zero diagonal. Then

$$M_r^{(s)} = M_s(J^r)_{00}. \quad (\text{FMG19})$$

Indeed multiplication by y in the orthonormal frame $p_{a,s}/\sqrt{h_{a,s}}$ has exactly these entries by FMG4. Starting in degree zero, r multiplications never visit a degree larger than r , so the finite truncation changes no path contributing to that matrix entry. Its constant component has the displayed integral, including M_s . Odd moments vanish, while all even moments are finite positive sums of these exact recurrence paths.

The entries of $\mathbf{A}$ have the finite original conductor calculation

$$f_{b,a} = \frac{\nu_{b,a+v} - \sum_{d=v+1}^{a+v} \binom{a+v}{d} \mu_d f_{b,a+v-d}}{\binom{a+v}{v} \mu_v}, \quad \nu_{b,r} = \sum_{u,w} (Z_{\text{exp},j})_{uw,b} \beta_{j,u,w}^r, \quad (\text{FMG20})$$

$$A_{ab} = \sum_r [S'^r] (\chi_-(S') S'^a) f_{b,r}.$$

The triangular equation is exactly the coefficient of z^{a+v} in $E_A F_b = N_b$, retaining every factorial and μ_v . Thus all entries in FMG15 are finite expressions in the original roots, full conductor, actual chosen quotient frame, actual flag frames and Gamma moments. This gives an executable polynomial evaluation of the original first row, with no new choice of period.

226.5 A conductor-moment witness inside the actual residual

The lower root set is symmetric about c' under negation and conjugation. More explicitly

$$i^{-q'} \chi_-(c' + iy) = \prod_{a,b=0}^{j-8} (y - (2b - j + 8)\gamma + i(2a - j + 8)\delta) \quad (\text{FMG21})$$

is real and even as a polynomial in real y . Therefore $|\chi_-(c' + iy)|^2 dm_s(y)$ is an even positive measure. Its monic degree- a orthogonal polynomial in y has parity $(-1)^a$: reflecting it and multiplying by $(-1)^a$ gives another monic minimizer with the same norm, and uniqueness identifies them. Let

P_a be its monic polynomial in S' under the exact map $p(y) \mapsto i^a p((S' - c')/i)$. It has coefficient $[S'^{a-1}]P_a = -ac'$ for $a \geq 1$, while $P_0 = 1$; in every case $\|\chi_- P_a\|_s^2 = \nu_a$.

Define the actual centered conductor moment

$$\eta_A = \frac{\mu_{v+1}}{(v+1)\mu_v} - 4, \quad \beta_m^- = \frac{\nu_m}{\nu_{m-1}} > 0. \quad (\text{FMG22})$$

The coefficient of S^{q-1} in χ_j is $-qj/2$, because its roots have exact mean $j/2$. Using FMG2 gives

$$[S'^{n-1}](g/\ell) = n \left(\frac{\mu_{v+1}}{(v+1)\mu_v} - \frac{j}{2} \right) = -nc' + n\eta_A. \quad (\text{FMG23})$$

The displayed division by ℓ is the explicit affine-fibre map in FMG8; it does not alter the original polynomial g . The coefficient of S'^{n-1} in $\chi_- P_m$ is $-nc'$. Their difference lies in $\chi_- \mathbb{C}[S']_{<m}$ and its coefficient at S'^{n-1} is $n\eta_A$. Its component along the orthogonal column $\chi_- P_{m-1}$ is consequently $n\eta_A$, because every other column has smaller degree. Pythagoras therefore proves

$$r \geq \frac{n^2 |\eta_A|^2 \nu_{m-1}}{\nu_m} = \frac{n^2 |\eta_A|^2}{\beta_m^-}. \quad (\text{FMG24})$$

This is an original-data lower witness, which may vanish for the actual period; no nonvanishing is assumed. Even when it vanishes, FMG13 and FMG15 retain all other orthogonal components and the entire signed mixed row. In particular the exact criterion for $r = 0$ is $H = 0$, equivalently $g = \ell \chi_- P_m$ in the original Gamma inner product. Equality of a single conductor moment alone does not assert this full polynomial identity.

226.6 Position in the full return

The old target cutoff for this row is $N = q_j - 1$, hence native cutoff $n - 1 = q_j - v - 1$. Its new target cutoff is q_j , so its weight in the original four-endpoint return is one. FMG15–17 apply separately to $e = 0$ and $e = 1$ and both prescribed source orders. They have no effect on the coefficients of the other rows. Bounds for this row must therefore be subtracted from the full signed profile by its exact weight; the remaining interior rows still require their original arithmetic calculation. The complete CPQ identity remains

$$4S_{K,j} = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A, \quad (\text{FMG25})$$

and all four complementary returns use the same original rows. The calculation above does not replace any of those returns by a dimension or an assigned constant.

226.7 Receiving the proved finite endpoint estimate in every signed term

The complete companion calculation FRG15–31, using the same actual g, ℓ, ν_m , proves the following finite input. Put

$$\begin{aligned} L_s &= (s-1)/4, \quad \theta = \pi/2, \quad h = 1/q, \quad \kappa = \theta - h, \quad Q_- = q' + L_s, \quad Q_+ = q + L_s, \\ \sigma &= (\delta^2 + \gamma^2)^{1/2}, \quad R' = (j-8)\sigma, \quad M_0 = 2Q_-/\theta, \quad M = \max\{M_0, 2R'\}, \\ B_+ &= \max\{1, (j+8)\sigma, 2L_s\}, \quad A_{\text{abs}} = \sum_{u,w} |a_{uw}|, \\ L_h &= \frac{\pi}{\Gamma(3/4)^2} (\sin h)^{3/2}, \quad U_h = \frac{\Gamma(1/4)^2}{2\pi} (\sin h)^{-1/2}, \\ f_\theta(x) &= 2x \log(2x/\theta) - 2x, \quad F_M = 2Q_- \log M - \theta M, \\ C_m &= (2m+1) \binom{2m}{m}^2, \quad \epsilon_- = \left(\theta - \frac{2Q_-}{M} \right) + \frac{Q_-}{M^2} + h(M+1) + \frac{4q'R'}{M}, \\ E_{j,s} &= 2 \log \frac{A_{\text{abs}}}{|\ell|} + \log \frac{2U_h}{L_h} + \kappa B_+ + \log((2Q_+)!) \\ &\quad - (2Q_+ + 1) \log \kappa - F_M + \epsilon_- + \log C_m. \end{aligned} \quad (\text{FMG26})$$

With these retained finite constants, that proof gives $0 \leq \log(1+r) \leq E_{j,s}$. In particular $E_{j,s}$ is nonnegative at every original $j \geq 13$, as follows also from the two proved bounds on the positive ratio FMG10. For each fixed original quartet and conductor, the actual choice eventually satisfies $M = M_0$ and $E_{j,s} = O_{\text{actual}}(j \log j)$, for both $s = 1, j$. The companion proof includes the complete Gamma density estimate at $h = 1/q$, the monic minimum on an original unit interval, the whole-line upper integral, and all factorials; the estimate is not an assumed analytic input.

Here is its exact effect on the entire signed row. The two chains in FMG16 imply

$$\begin{aligned} 0 &\leq (b_{T_e} - b_W) + (b_{W_i} - b_J) \leq b_{T_e} - b_J \leq 4 \log(1+r), \\ 0 &\leq (b_S - b_{K_c}) + (b_I - b_{K_i}) \leq b_S - b_{K_i} \leq 4 \log(1+r). \end{aligned} \quad (\text{FMG27})$$

Indeed for nested subspaces the orthogonal projectors in their common original metric are ordered; compressing by the same row preserves this order. Monotonicity of $\log \det(I + K)$ follows by differentiating along their positive connecting segment, or by the ordered eigenvalues. The missing middle increment in each chain is nonnegative, proving the middle bound. The last uses $0 \preceq K_X \preceq rI_4$. Subtraction of these two nonnegative quantities gives the exact signed expression FMG17. It follows that

$$|\delta \mathcal{L}_{e,n}| \leq 4E_{j,s}, \quad |\delta \mathcal{P}_{e,n}| \leq 4E_{j,s}, \quad 0 \leq \delta V_{X,n} \leq 4E_{j,s} \quad (X = b, \text{rem}, s, A). \quad (\text{FMG28})$$

For the threshold quantity $\delta \mathcal{P}$, apply the same argument to $\sum_{a=1}^4 \log \max(1, \lambda_a(K_X))$. The min-max characterization orders every eigenvalue when two Hermitian matrices are ordered: restrict the larger Rayleigh quotient to each test subspace, take its extremum there, and then the same extremum over subspaces. The increasing scalar function $\log \max(1, x)$ preserves those four inequalities. Thus the same two chain estimates hold with upper bound $4 \log \max(1, r) \leq 4 \log(1+r)$. This also retains every zero and every eigenvalue below one.

For the four CPQ returns, their first-row increments are the paired determinant increments on the actual quotients W/W_i , $\mathcal{U}/T_e$, S/K_c , and I/K_i , respectively. For any fixed $Y \supset K$ in this list, the determinant of its attained quotient Gram is $\det G_Y / \det G_K$ times the fixed frame factor, by full block elimination. The factor cancels in a successive increment, leaving $b_Y - b_K$. The same order argument places it between zero and $4 \log(1+r)$, proving the final part of FMG28. For $Y = \mathcal{U}$ its compression is exactly rI_4 . All four quotient fibres and all their cross terms therefore remain in the row-wise CPQ identity.

Since $j = k + 4$ and $q_k = (k + 1)^2$, each quantity in FMG28 divided by kq_k tends to zero. The original endpoint weight is one. Consequently the first original row can be removed from the leading kq_k calculation with the explicit finite error in FMG28, simultaneously in the signed spectral profile and in all four complementary CPQ returns. This statement concerns this one specified row; it leaves all remaining rows in their original correlated expressions.

The complete later relation space and the first-row specialization. The full NIB proof below preserves this original first-row identity and calculates its later-row map. Put $L = q_j - v$ and $g_t = \mathcal{T}_A(\chi_j S^t)$, with the exact leading coefficient $\ell_t = E_A^{(v)}(0) \binom{q_j+t}{v}$. At degree $L + r$ the full original functional annihilator is $\text{span}(g_0, \dots, g_r)$: root addition and the Leibniz identity annihilate each column, the first L rows have full rank, and the displayed $r + 1$ polynomials have distinct degrees. Its monic minimum $\eta_{r,s}$ retains the complete Gram subtraction against $g_0, \dots, g_{r-1}$. In the unchanged native measure the exact corrected row is

$$1 + r_{L+r} = \eta_{r,s} / \nu_{m+r,s}^-, \quad \eta_{0,s} = \|\mathcal{T}_A \chi_j\|_s^2 / |\ell_0|^2. \quad (\text{IUP1})$$

The second identity follows because the degree- L monic fibre has no free lower relation. Substitution into the first recovers the original first-row formula with its identical lower denominator. NIB's full finite norm estimates prove the all-row bound $\log(1 + r_{L+r}) \leq E_{j,s,r}$ and $E_{j,s,r} = O_{\text{actual}}((j+r) \log j)$ uniformly for $0 \leq r \leq q_j$. The IPR receiving proof below carries the resulting prefix control through every CPQ quotient and both original allocations, with the actual endpoint weights; no old raw leverage is substituted for the corrected row.

227 The original conductor on every native mixed subquotient

This section receives the proved SGT7–28 estimates on the actual DNE6–9 and NSO12–15 spaces, and then on the fixed CIT/CPQ flags. The exterior quotient argument is already present in CEP37–38; we give its complete proof in the form needed here. The new receiver is the full finite numerator-observation metric, with the conductor multiples retained as its entire quotient denominator, and the rank-dependent SGT error on every original native subquotient.

Fix the original actual hypothetical simple quartet, its fixed original five-orbit period and logarithm branch. Keep $0 < \delta < 1/2$, $\gamma > 2$, every coefficient a_{uw} of the nonzero original conductor, and

$$\begin{aligned} \beta_{a,u,w} &= a/2 + (2u - a)\delta + i(2w - a)\gamma, \quad 0 \leq u, w \leq a, \quad a = 4\ell + 1 \geq 9, \\ E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad \mu_n = E_A^{(n)}(0), \quad v = \text{ord}_0 E_A \leq 80, \quad \mu_v \neq 0, \quad \alpha_A = v!/\mu_v, \\ q_a &= (a+1)^2, \quad q'_a = (a-7)^2, \quad m_a = q_a - q'_a - v, \quad c_0 = a/2, \quad c_1 = a/2 - 4, \\ N &\geq q_a - 1, \quad D = N - v, \quad s \in \{1, a\}. \end{aligned} \tag{NCT1}$$

The original domain of the fixed period, including its original rank conditions, is unchanged. The roots in each displayed grid are distinct: equality of two roots gives separately equality of the indices from their real and imaginary parts. Every polynomial space below has complex coefficients. Its Hermitian inner product is conjugate-linear in the first variable. The complex-linear dual consists of linear, rather than conjugate-linear, functionals.

The full original measures, masses, recurrence and frames are

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_{n,s,c}(S) = p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}, \\ \|P\|_{s,c}^2 &= \int_{\mathbb{R}} |P(c+iy)|^2 dm_s(y). \end{aligned} \tag{NCT2}$$

SGT3–5 proves that these are orthonormal frames on every indicated finite polynomial space. Their leading coefficients are $i^{-n}/\sqrt{h_{n,s}}$; neither this phase nor the mass is removed.

227.1 The exact dual conductor square

Write $\mathcal{P}_d = \mathbb{C}[S]_{\leq d}$, with $\mathcal{P}_{-1} = 0$, and distinguish the target variable S' . The original, unscaled conductor map is

$$\begin{aligned} (\mathcal{T}_A P)(S') &= \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \\ A_{D,s} : (\mathcal{P}_N/\mathcal{P}_{v-1}, \|\cdot\|_{s,c_0,\text{quot}}) &\longrightarrow (\mathbb{C}[S']_{\leq D}, \|\cdot\|_{s,c_1}), \quad [P] \longmapsto \mathcal{T}_A P. \end{aligned} \tag{NCT3}$$

It is the SGT7 isomorphism. For completeness, $\mathcal{T}_A S^n = \sum_{j=0}^n \binom{n}{j} \mu_j S'^{n-j}$. If $n \geq v$, its leading coefficient is $\binom{n}{v} \mu_v \neq 0$, and its degree is $n - v$. Successive elimination in degree gives surjectivity, and the full kernel is $\mathcal{P}_{v-1}$. The quotient is represented isometrically by the original orthogonal complement spanned by $\phi_{v,s,c_0}, \dots, \phi_{N,s,c_0}$.

For an analytic germ F define $\Lambda_F(S'^n) = F^{(n)}(0)$. An upper-variable germ is paired with S^n by the same derivative convention. The contravariant dual map in the direction relevant here is

$$\begin{aligned} A_{D,s}^\vee : (\mathbb{C}[S']_{\leq D})^\vee &\longrightarrow (\mathcal{P}_N/\mathcal{P}_{v-1})^\vee, \quad \Lambda \longmapsto \Lambda \circ A_{D,s}, \\ A_{D,s}^\vee \Lambda_F &= \Lambda_{E_A F}. \end{aligned} \tag{NCT4}$$

To prove the identity, on every $0 \leq n \leq N$ its left side is $\sum_{j=v}^n \binom{n}{j} \mu_j F^{(n-j)}(0)$, precisely $(E_A F)^{(n)}(0)$ by the Leibniz rule. For $n < v$ both sides vanish. The highest derivative of F that occurs is $n - v \leq D$, so the map is well defined on these finite jets; it does not depend on any higher derivative. Conversely the nonzero coefficient $\binom{v+r}{v} \mu_v$ recursively recovers the derivative of order r of F from these products. This also proves the isomorphism on finite jets directly.

Let $w = (\Lambda_F(\phi_{n,s,c_1}))_{n=0}^D$, and let $\tilde{w} = (\Lambda_{E_A F}(\phi_{v+n,s,c_0}))_{n=0}^D$. If $A_{D,s}$ denotes its original orthonormal matrix as in SGT10, then

$$\tilde{w} = A_{D,s}^T w, \quad \Lambda_{E_A F}(\phi_{n,s,c_0}) = 0 \quad (0 \leq n < v). \quad (\text{NCT5})$$

Indeed the j -th source basis vector has image $\sum_i (A_{D,s})_{ij} \phi_{i,s,c_1}$, and applying the complex-linear functional gives the ordinary transpose in NCT5. There is no complex adjoint in this algebraic square. Squared functional norms are $w^* w$ and $\tilde{w}^* \tilde{w}$. If $A = U \Sigma V^*$ is a singular-value decomposition, then $A^T = \bar{V} \Sigma U^T$ is one as well. Thus A and A^T , and their inverses, have exactly the same singular values and every corresponding exterior norm.

No scalar α_A is inserted into NCT3–5. In particular, the literal DNE3 germ is $F_b = \mathcal{N}_b / E_A$, and its image is $\Lambda_{\mathcal{N}_b}$, with coefficient one. The original primary-coordinate maps involving α_A are retained below.

227.2 All numerator observations and all conductor columns

Use the original lexicographic upper-root coefficient order. For $\nu \in \mathbb{C}^{q_a}$ put $\mathcal{N}_\nu(z) = \sum_{u,w} \nu_{uw} e^{\beta_{a,u,w} z}$. Let $\mathcal{N}_{v,a}$ be the coefficient subspace on which $\mathcal{N}_\nu^{(j)}(0) = 0$ for all $0 \leq j < v$. The upper Vandermonde $\mathbb{V}_a = (\beta_{a,u,w}^n)$, $0 \leq n < q_a$, is invertible, so $\dim \mathcal{N}_{v,a} = q_a - v$.

The full original primary conductor C_a , with q'_a rows and q_a columns, is fixed by the literal identity

$$\mathcal{N}_{C_a^T \eta}(z) = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}. \quad (\text{NCT6})$$

This identity follows by collecting all original shifts, since $\beta_{a-8,u,w} + \beta_{8,r,t} = \beta_{a,u+r,w+t}$. It retains repeated contributions to the same resulting root. The map C_a^T is injective: its zero image would make the product in NCT6 identically zero; on a neighborhood where $E_A \neq 0$ the lower exponential sum is zero, hence everywhere by analyticity, and its distinct-root Vandermonde makes $\eta = 0$. Its image lies in $\mathcal{N}_{v,a}$, by the zero order of E_A .

Use exactly the DNE2–3/NSO12 frame:

$$\begin{aligned} W_{v,a} &= \mathbb{V}_a^{-1} P_v, & J_{v,a} &= P_v^T \mathbb{V}_a C_a^T, & W_{v,a} J_{v,a} &= C_a^T, \\ Z_{\text{exp},a} &= W_{v,a} E_{\mathcal{J}_a}, & F_{a,b} &= \mathcal{N}_{Z_{\text{exp},a} e_b} / E_A. \end{aligned} \quad (\text{NCT7})$$

Here P_v inserts zero in the first v positions; $\mathcal{I}_a$ is the first ordered set of q'_a rows with nonzero minor of $J_{v,a}$, and $\mathcal{J}_a$ is its complement. Such a minor exists by the proved injectivity in NCT6. Restricting a relation among columns of $[J_{v,a}, E_{\mathcal{J}_a}]$ first to $\mathcal{I}_a$ and then to its complement proves that they are independent. Consequently $[C_a^T, Z_{\text{exp},a}]$ is the complete frame of $\mathcal{N}_{v,a}$, and $Z_{\text{exp},a}$ represents its quotient by $\text{im } C_a^T$. Every actual minor is unchanged.

Define the full upper observation matrix and its covariance by

$$\begin{aligned} \mathbf{E}_{a,N,s}[n, (u, w)] &= \phi_{n,s,c_0}(\beta_{a,u,w}), \quad 0 \leq n \leq N, \\ K_{a,N,s} &= \mathbf{E}_{a,N,s}^* \mathbf{E}_{a,N,s}, \\ \hat{D}_{a,D}^{(s)} &= Z_{\text{exp},a}^* [K_{a,N,s} - K_{a,N,s} C_a^T (\bar{C}_a K_{a,N,s} C_a^T)^{-1} \bar{C}_a K_{a,N,s}] Z_{\text{exp},a}. \end{aligned} \quad (\text{NCT8})$$

The square covariance $K_{a,N,s}$ is positive definite. Its first q_a observation rows equal an invertible triangular change of the upper Vandermonde, because the polynomials NCT2 have nonzero leading coefficients. Therefore $\mathbf{E}_{a,N,s}$ has full column rank for every stated N , including $N = q_a - 1$. The middle inverse in NCT8 is positive definite by this fact and injectivity of C_a^T .

The same covariance has an exact derivative-coordinate expression. Let $V_{a,N}[n, (u, w)] = \beta_{a,u,w}^n$, and write $S^r = \sum_{n=0}^N B_{s,c_0,N}[n, r] \phi_{n,s,c_0}(S)$. The full monomial Gram is $H_{s,c_0,N} = B^* B$. Applying the complex-linear numerator functional to this identity gives

$$V_{a,N} \nu = B_{s,c_0,N}^\top E_{a,N,s} \nu, \quad K_{a,N,s} = V_{a,N}^* \overline{H_{s,c_0,N}^{-1}} V_{a,N}. \quad (\text{NCT8a})$$

Indeed $(B^\top)^{-*} (B^\top)^{-1} = \overline{(B^* B)^{-1}}$. Thus NCT8 is the full numerator derivative-dual covariance restricted to the vanishing-moment space and minimized over every conductor multiple. This proves the coordinate morphism, including the ordinary transpose and Hermitian conjugation, without changing the original measure or coefficient norm.

The minimum below uses the classical Schur-complement method [H1, §2.1, (2.3)–(2.4)] on the Hermitian Gram of $[C_a^\top, Z_{\text{exp},a}]$ in $K_{a,N,s}$. The eliminated block is invertible by the rank argument above, so no pseudoinverse or unstated range condition is needed.

For every original coefficient vector $x \in \mathbb{C}^{m_a}$, the exact squared norm represented by NCT8 is

$$x^* \widehat{D}_{a,D}^{(s)} x = \min_{\eta \in \mathbb{C}^{q_a}} \|E_{a,N,s} (Z_{\text{exp},a} x + C_a^\top \eta)\|_2^2. \quad (\text{NCT9})$$

Differentiate the right side with respect to its real and imaginary coordinates, or complete its positive square. The unique minimizer is

$$\eta_{\min} = -(\overline{C_a} K_{a,N,s} C_a^\top)^{-1} \overline{C_a} K_{a,N,s} Z_{\text{exp},a} x.$$

Substitution gives NCT8. The quotient is positive definite because $[C_a^\top, Z_{\text{exp},a}]$ is independent. The minimum retains every conductor column, including every mixed covariance term.

For $\nu \in \mathcal{N}_{v,a}$ the entries of $E_{a,N,s} \nu$ below row v vanish, since those polynomials use derivatives of order below v . This proves that the ambient polynomial dual norm in NCT9 is exactly the quotient dual norm in NCT4, not an extra truncation of the numerator. All rows $0, \dots, N$ remain in NCT8.

The corresponding original native metric is exactly NSO13:

$$x^* D_{a,D}^{(s)} x = \min_{\eta \in \mathbb{C}^{q_a}} \left\| \Lambda_{\sum_b x_b F_{a,b}} + \sum_{u,w} \eta_{uw} \text{ev}_{\beta_{a-8,u,w}} \right\|_{s,c_1,D,\vee}^2. \quad (\text{NCT10})$$

NCT4 takes every representative inside this minimum to exactly the representative in NCT9. In particular it takes the complete lower evaluation subspace to the complete conductor-multiple subspace, by NCT6. Let Y be the span of all functionals appearing in NCT10, and K its lower evaluation subspace. Then NCT4 induces the isomorphism

$$Y/K \xrightarrow{A_{D,s}^\vee} \Lambda_{\mathcal{N}_{v,a}} / \Lambda_{\text{im } C_a^\top}, \quad [F] \mapsto [E_A F]. \quad (\text{NCT11})$$

The two quotient Grams in the same fixed value coordinates x are $D_{a,D}^{(s)}$ and $\widehat{D}_{a,D}^{(s)}$. Finite-jet injectivity follows also directly: vanishing numerator derivatives through $N \geq q_a - 1$ kill all upper coefficients by the Vandermonde. Thus this isomorphism holds on every retained endpoint and on every original coefficient fibre, without a genericity or conditioning assumption.

227.3 Rank-dependent exterior bounds on arbitrary subquotients

We prove the finite-dimensional statement in full. Let $B : H_0 \rightarrow H_1$ be an invertible linear map of complex Hilbert spaces, and let $K \subset Y \subset H_0$ be any subspaces. Both quotients have their attained quotient norms. Put $Y_0 = Y \cap K^\perp$ and $Y_1 = BY \cap (BK)^\perp$. Every $y \in Y$ has the unique decomposition $y = x + k$ with $x \in Y_0$, $k \in K$. The squared norm of a lift of $[y]$ is $\|x\|^2 + \|k\|^2$, so x is its minimum lift. Under these isometric identifications the induced quotient map is

$$\overline{B} : Y/K \longrightarrow BY/BK, \quad \overline{B} = P_{Y_1} B|_{Y_0}, \quad \overline{B}^{-1} = P_{Y_0} B^{-1}|_{Y_1}. \quad (\text{NCT12})$$

The second formula follows by applying B^{-1} to any representative and subtracting its K component. It is indeed the inverse because the subtracted components lie respectively in BK and K . The projections are contractions on every exterior power: in orthonormal bases their singular values are zeros and ones. Restrictions and inclusions of the stated Hilbert subspaces have exterior norm at most one. Consequently for every $0 \leq r \leq \dim(Y/K)$,

$$\|\bigwedge^r \overline{B}\| \leq \|\bigwedge^r B\|, \quad \|\bigwedge^r \overline{B}^{-1}\| \leq \|\bigwedge^r B^{-1}\|. \quad (\text{NCT13})$$

There is no factor depending on $\dim K$. For any ordered independent quotient frame of length r , its squared wedge norm is its Gram determinant. Applying the first inequality to that wedge and the second to its image gives

$$-2 \log \|\bigwedge^r B^{-1}\| \leq \log \frac{\det G_1}{\det G_0} \leq 2 \log \|\bigwedge^r B\|. \quad (\text{NCT14})$$

These statements include $r = 0$, with determinant and exterior norm one. A change of the common value frame by U multiplies both determinants by $|\det U|^2$, which cancels in their ratio.

For the classical exterior inner product underlying NCT14, see Russell Lyons [H3, §4, (4.1), p. 11 of the cited arXiv version]. Taking both lists there to be the same quotient frame gives its squared wedge norm as its Gram determinant. The quotient map, inverse, and rank-dependent transport bounds are proved here.

Successive restrictions and quotients obey the same rank-only bound. Indeed a subspace of Y/K has a full preimage Y' containing K , and its inherited metric is the quotient metric of Y'/K . A subsequent quotient by a subspace with full preimage K' is Y'/K' , since the union of its original affine fibres is precisely the affine fibre modulo K' . Its minimum over those fibres is the same minimum. Induction reduces any finite sequence to NCT12 with the actual full preimages; the distortion does not accumulate a new factor at each intermediate quotient.

Apply this to $B = A_{D,s}^\top$ using NCT5, and to its orthogonal direct sum of b copies, where $b = 1$ or 4. We now specify its finite constants using the original SGT symbol. To avoid confusing the degree a with a circle radius, denote an admissible SGT radius by $\varrho \in (0, 1)$. Set

$$\begin{aligned} w(t) &= -i \arctan t, & g(t) &= t^{-v} e^{-4w(t)} E_A(w(t)), & g(0) &= (-i)^v \mu_v / v!, \\ \rho_n &= \sqrt{n! / (s/2)_n}, & L_{s,D,v} &= \frac{1}{2} \max\{1, s/2 - 1\} \sum_{n=1}^{D+v} \frac{1}{n}. \end{aligned} \quad (\text{NCT15})$$

SGT10 proves the exact original matrix $A_{D,s} = \text{diag}(\rho_0, \dots, \rho_D)^{-1} T_D(g) \text{diag}(\rho_v, \dots, \rho_{D+v})$, where $T_D(g)_{i,j} = g_{j-i}$ for $i \leq j$, and zero otherwise. SGT21 proves that the products of the two forward diagonal operator norms and of the two inverse diagonal operator norms are both at most $e^{2L_{s,D,v}}$.

Here is the full list of original finite constants used in SGT19:

$$\begin{aligned}
A_{\text{abs}} &= \sum_{u,w} |a_{uw}|, \quad B_0 = \max\{1, \max_{u,w} |\beta_{8,u,w}|\}, \quad C_v = A_{\text{abs}} B_0^v / |\mu_v|, \\
R_0 &= B_0^{-1} \min\{1, (v+1)/(2eC_v)\}, \quad a_0 = \min\{1/2, \tanh(R_0/2)\}, \\
d_0 &= 1 - \frac{a_0^2}{3(1-a_0^2)}, \quad M_0 = \frac{2v!}{|\mu_v|} d_0^{-v} e^{2R_0}, \\
S_D(r) &= \sum_{n=0}^D r^{-n}, \quad G_{0,D} = M_0 S_D(a_0), \\
\mathcal{Z}_\varrho &= \{\zeta : |\zeta| < \varrho, g(\zeta) = 0\}, \quad H_\varrho = \sum_{\zeta \in \mathcal{Z}_\varrho} m_\zeta, \\
A_{\zeta,h} &= \frac{1}{(m_\zeta - h)!} \frac{d^{m_\zeta-h}}{dt^{m_\zeta-h}} \frac{(t-\zeta)^{m_\zeta}}{g(t)} \Big|_{t=\zeta}, \quad 1 \leq h \leq m_\zeta, \\
W_\varrho &= \operatorname{arctanh} \varrho, \quad m_{E,\varrho} = \min_{|t|=\varrho} |E_A(w(t))|, \quad G_\varrho = A_{\text{abs}} \varrho^{-v} e^{(B_0+4)W_\varrho}, \\
M_\varrho &= \frac{\varrho^v e^{4W_\varrho}}{m_{E,\varrho}} + \sum_{\zeta,h} \frac{|A_{\zeta,h}|}{(\varrho - |\zeta|)^h}, \quad P_\varrho = \sum_{\zeta,h} \frac{|A_{\zeta,h}|}{(1 - |\zeta|)^h}, \\
F_{\varrho,D} &= G_\varrho S_D(\varrho), \quad V_{\varrho,D} = M_\varrho S_D(\varrho) + P_\varrho, \quad V_{\varrho,D}^{\text{cap}} = \min\{G_{0,D}, V_{\varrho,D}\}.
\end{aligned} \tag{NCT16}$$

Here admissibility means that g has no zero on the indicated circle. The sums retain every pole and its full order. There are finitely many poles in each such closed disk, and there are admissible radii arbitrarily close to one, as proved in SGT14. SGT11–19 proves from these exact original data that

$$\begin{aligned}
\|T_D(g)\| &\leq F_{\varrho,D}, \quad \|T_D(g)^{-1}\| \leq G_{0,D}, \\
T_D(g)^{-1} &= C_{\varrho,D} + R_{\varrho,D}, \quad \operatorname{rank} C_{\varrho,D} \leq H_\varrho, \quad \|R_{\varrho,D}\| \leq V_{\varrho,D}.
\end{aligned} \tag{NCT17}$$

These are proved internal inputs, including SGT11–13’s explicit zero-free disk, SGT16–18’s exact principal parts and their lower-triangular corrections. In particular NCT17 is a decomposition of the entire inverse; none of its principal parts has been omitted.

For b copies the inverse exceptional rank is at most bH_ϱ ; its global norm and its remainder norm are still bounded by $G_{0,D}$ and $V_{\varrho,D}$, respectively. For a rank- r subquotient put $h_b = \min\{r, bH_\varrho\}$. On the kernel of the exceptional matrix its inverse has norm at most $V_{\varrho,D}$. The singular-value minimum principle therefore bounds every singular value after the first bH_ϱ by that quantity, and every singular value by $G_{0,D}$. Multiplying the largest r singular values, and then the two diagonal factors, gives

$$\begin{aligned}
\left\| \bigwedge^r ((A_{D,s}^\top)^{\oplus b}) \right\| &\leq e^{2rL_{s,D,v}} F_{\varrho,D}^r, \\
\left\| \bigwedge^r ((A_{D,s}^\top)^{\oplus b})^{-1} \right\| &\leq e^{2rL_{s,D,v}} G_{0,D}^{h_b} (V_{\varrho,D}^{\text{cap}})^{r-h_b}.
\end{aligned} \tag{NCT18}$$

For clarity, the minimum principle here follows by diagonalizing the positive matrix of the squared norm. Every subspace of codimension $j-1$ meets the span of its first j eigenvectors, and the complement of its first $j-1$ eigenvectors attains the bound. Apply this with the kernel of the exceptional matrix. This proves the estimate even when its rank bound exceeds the whole matrix dimension, when $h_b = r$.

For a fixed admissible ϱ , suppress that parameter in the notation requested for the receiving forms, and define

$$\begin{aligned}
\mathcal{U}_{a,D,s,r}^{[b]} &= 4rL_{s,D,v} + 2r \log^+ F_{\varrho,D}, \\
\mathcal{L}_{a,D,s,r}^{[b]} &= 4rL_{s,D,v} + 2h_b \log^+ G_{0,D} + 2(r-h_b) \log^+ V_{\varrho,D}^{\text{cap}}, \\
\mathcal{B}_{a,D,s,r}^{[b]} &= \max\{\mathcal{L}_{a,D,s,r}^{[b]}, \mathcal{U}_{a,D,s,r}^{[b]}\}.
\end{aligned} \tag{NCT19}$$

Here $\log^+ x = \max\{0, \log x\}$; all these constants are finite, nonnegative and exactly zero at $r = 0$. The index a records the original grid and centres; the conductor constants themselves depend only on the fixed original conductor and radius.

Let Q be any original rank- r metric obtained by a finite sequence of restrictions and attained quotients from $(D_{a,D}^{(s)})^{\oplus b}$, and let $\widehat{Q}$ be the identical construction, in the identical coefficient and quotient value frames, from $(\widehat{D}_{a,D}^{(s)})^{\oplus b}$. Equations NCT11–14 and NCT18 prove the finite directed statement

$$\boxed{-\mathcal{L}_{a,D,s,r}^{[b]} \leq \log \det \widehat{Q} - \log \det Q \leq \mathcal{U}_{a,D,s,r}^{[b]}} \quad (\text{NCT20})$$

The full preimage of each restriction and quotient under NCT11 includes the entire conductor-multiple denominator, of dimension bq'_a . NCT13 proves why that dimension contributes no extra volume factor in NCT20. No fixed coefficient map is replaced by an orthogonal surrogate; minimum lifts are used only to prove the bound on its original attained metric.

227.4 The uniform limit and the exact DNE scalar conventions

For fixed $C_D, C_r > 0$, uniformly over $0 \leq D \leq C_D a^2$, $0 \leq r \leq C_r a$, both actual orders, $b = 1, 4$, and every original admissible subquotient,

$$\log \det \widehat{Q} - \log \det Q = o(a^3) = o(aq_a). \quad (\text{NCT21})$$

Here is the limiting proof with the order of choices explicit. Fix an admissible $\varrho < 1$. Since $S_D(\varrho) \leq (D+1)\varrho^{-D}$, NCT16 gives

$$\begin{aligned} \log^+ F_{\varrho,D} &\leq D \log(1/\varrho) + \log(D+1) + \log^+ G_{\varrho}, \\ \log^+ V_{\varrho,D}^{\text{cap}} &\leq D \log(1/\varrho) + \log(D+1) + \log^+(M_{\varrho} + P_{\varrho}), \\ \log^+ G_{0,D} &\leq D \log(1/a_0) + \log(D+1) + \log^+ M_0. \end{aligned}$$

The common term $4rL_{s,D,v}$ is $O(a^2 \log(a+1))$. The exceptional term uses $h_b \leq bH_{\varrho}$, fixed before the limit, and is $O_{\varrho}(a^2 + \log(a+1))$. Consequently the limsup of the uniform absolute error divided by a^3 is at most $2C_r C_D \log(1/\varrho)$. Only after this limit choose admissible radii approaching one. The bound tends to zero. This proves NCT21 without any assertion of a zero-free unit disk, bounded pole count near its boundary, or a coefficient condition on the original subquotients.

The original DNE scalar and transpose square is preserved exactly. Let $A_{\ell,a}$ be DNE4's pairing matrix and Z_a the original low-monomial kernel columns. Keep

$$\begin{aligned} D_{a,D}^{(s)} &= A_{\ell,a}^* \overline{(H_{Y,a,N}^{(s)})^{-1} A_{\ell,a}}, & B_{K,a} &= Z_a^{\top} A_{\ell,a}, \\ Q_{K^{\vee}} &= (B_{K,a} (D_{a,D}^{(s)})^{-1} B_{K,a}^*)^{-1} = \overline{(Z_a^* H_{Y,a,N}^{(s)} Z_a)^{-1}}, \\ \widehat{Q}_{K^{\vee}} &= (B_{K,a} (\widehat{D}_{a,D}^{(s)})^{-1} B_{K,a}^*)^{-1}, & \widehat{H}_Y &= \overline{A_{\ell,a} (\widehat{D}_{a,D}^{(s)})^{-1} A_{\ell,a}^*}, \\ & & \widehat{Q}_{K^{\vee}} &= \overline{(Z_a^* \widehat{H}_Y Z_a)^{-1}}. \end{aligned} \quad (\text{NCT22})$$

For proof, a functional with coordinate column u on a Gram H has norm $u^* \overline{H^{-1}} u$, since its Riesz vector is $H^{-1} \bar{u}$. Substituting this identity and $B_{K,a} = Z_a^{\top} A_{\ell,a}$ proves every equality in NCT22 by matrix multiplication. The onto map $B_{K,a}$ uses its full fibre, with minimum norm $(B_{K,a} D^{-1} B_{K,a}^*)^{-1}$. This follows by inserting the minimum lift $D^{-1} B_{K,a}^* (B_{K,a} D^{-1} B_{K,a}^*)^{-1} x$ and checking that it is D -orthogonal to $\ker B_{K,a}$. Thus NCT20 applies with $r = \text{rank } B_{K,a}$; inversion changes the sign of the logarithmic determinant difference when it is read on the original restricted kernel Gram.

The primary-coordinate comparison is literally the earlier KRC/RRC map $x \mapsto \alpha_A \mathbf{N}_a Z_{\text{exp},a} x$. For example its four multiplication maps remain

$$M_i = (\alpha_A \mathbf{N}_j Z_{\text{exp},j})^{-1} m_{i,k} (\alpha_A \mathbf{N}_k Z_{\text{exp},k}), \quad j = k + 4. \quad (\text{NCT23})$$

Both scalars and both fixed frame maps are present before their proved cancellation. NCT8–11 acts in the original x coordinates, so the matrices NCT23 are unchanged. If one instead writes the conductor isomorphism as $\alpha_A A_{D,s}$, its ordinary dual is $\alpha_A A_{D,s}^\top$, and its numerator Gram is exactly $|\alpha_A|^2 \widehat{D}$. Every rank- r descendant quotient then has determinant factor $|\alpha_A|^{2r}$, because minimization commutes with this positive scalar. NCT3–22 uses the unscaled map and does not introduce this factor into DNE10. This describes the exact scalar map between the two conventions, rather than silently choosing one.

227.5 All original mixed flags and all four-endpoint signs

For $j = k + 4$, take the original coefficient space $\mathcal{U} = \mathbb{C}^{4m_j}$ and maps of EFP18:

$$\begin{aligned} B &= B_{\text{cof}}, \quad \mathbb{B} = \text{diag}(B_j, 4), \quad J = \ker \mathbb{B}, \quad \mathcal{B} = \text{im } B, \quad I = \text{im}(BJ_0), \\ S &= \text{im}[B, F_e], \quad W = J + \mathcal{B}, \quad K_e = S \cap W, \quad T_e = S + W, \quad K_i = I \cap J, \quad W_i = I + J. \end{aligned} \tag{NCT24}$$

All spaces and the maps NCT23 are the same coefficient objects as before. Replace only their ambient metric by $\text{diag}(\widehat{D}_{j,N-v}^{(s_1)}, 4)$ to define the numerator forms. NCT20 applies to every one of the following original quotients, with its actual rank:

original form	original quotient	rank	
$G_{h,e}$	$K_e/\mathcal{B}$	h_e	
$G_{s,e}$	S/K_e	c_e	
$G_{t,e}$	T_e/W	c_e	
$Q_{\text{rem},e}$	$\mathcal{U}/T_e$	$r_o - c_e$	
A_{inv}	I/K_i	r_I	
H_{inv}	W_i/J	r_I	
Q_b	W/W_i	r_b	
Q_o	$\mathcal{U}/W$	r_o	
$H_{\text{conn},e}$	$S/\mathcal{B}$	r_e	
$H_{B,j,N-v}$	$\mathbb{C}^{m_j}/\ker B_j$	t_j	(NCT25)

The final row uses one copy; every preceding row uses four. For Q_b , any original EFP21 fixed value-frame factor remains on both forms. The table's maps and ranks are the complete EFP18–21 and CPQ data; NCT12 proves their metric transfer by their full preimages under NCT11. It also applies to every additional restriction of these spaces, including all eight signed-profile spaces $T_e, K_e, W_i, K_i, W, S, J, I$, with their exact original frames. Their dimensions are at most $4m_j = O(k)$, so NCT21 applies uniformly without dropping any zero eigenvalue in a later compressed relative operator.

For a fixed-rank original form at the four cutoffs $N_i = (q_j - 1, q_j, 2q_j - 1, 2q_j)$, write $D_i = N_i - v$ and $\mathcal{R}f = -f(N_0) - f(N_1) + f(N_2) + f(N_3)$. Abbreviate NCT19's constants at each endpoint by $\mathcal{L}_i$ and $\mathcal{U}_i$, retaining its actual degree, order, rank and number of copies. The exact directed return estimate is

$$\begin{aligned} -\mathcal{U}_0 - \mathcal{U}_1 - \mathcal{L}_2 - \mathcal{L}_3 &\leq \mathcal{R} \log \det \widehat{Q} - \mathcal{R} \log \det Q \\ &\leq \mathcal{L}_0 + \mathcal{L}_1 + \mathcal{U}_2 + \mathcal{U}_3. \end{aligned} \tag{NCT26}$$

This follows by negating the upper bounds at the two negative endpoints and adding the lower bounds at the two positive endpoints for the lower estimate, and doing the reverse for the upper estimate. Every endpoint has $D_i = O(k^2)$, and all ranks in NCT25 are $O(k)$. Thus each displayed return difference is $o(kq_j)$ in both original source pairs $(1, 1)$ and (k, j) . For a source-degree k form evaluated at the same target cutoffs the identical proof uses its original degree k , centre $k/2$, and $s_0 = 1$ or k ; the uniform domain still holds since $q_j = O(k^2)$.

Every finite CPQ determinant-line identity continues to hold for the numerator metrics, because its proof is the same sequence of Schur minima in the same exact coefficient flags. In particular

the original complete complementary returns $V_b, V_{\text{rem}}, V_s, V_A$ all receive NCT26, as do the original equal-rank contraction ratios $\det H_{\text{inv}}/\det A_{\text{inv}}$ and $\det G_t/\det G_s$: subtract the two determinant errors before applying the endpoint signs. For any fixed finite signed sum the absolute error is bounded by the sum of its original summands' $\mathcal{B}$ constants, with multiplicities. No cancellation of an uncomputed summand is asserted. The CPQ affine identities $4S_K = \mathcal{L} + V_b + V_{\text{rem}} + V_s + V_A$ and $P_{\text{rest}} = \mathcal{L} + V_b$ retain their exact forms on these newly specified numerator metrics and differ from their native values by the proved finite errors. The result supplies an exact numerator observation geometry and an $o(kq_j)$ volume transfer; the original mixed numerator matrices and their complete returns remain the quantities to evaluate.

228 The original numerator kernel and the complete mixed spectral return

This calculation receives the full conductor-dual transport proved in NCT. It uses the original DNE numerator frame and the original CPQ flags, with their fixed value coordinates. The source of the construction is the accepted cumulative DNE3,6, SGT7–26, CPQ1–20 and SFB12–16. The period, its logarithm branch, the complete nonreduced conductor and its order at zero are fixed as the original tensor degree varies. No arithmetic coefficient or zero packet is supplied by the following calculation.

228.1 The actual finite numerator observations

Fix the original simple quartet $0 < \delta < 1/2$, $\gamma > 2$, and the original permitted period. For $a \equiv 1 \pmod{4}$, $a \geq 9$, retain

$$\begin{aligned} q_a &= (a+1)^2, & q'_a &= (a-7)^2, & m_a &= q_a - q'_a - v, & c_a &= a/2, \\ \beta_{a,u,w} &= a(1/2 - \delta - i\gamma) + 2u\delta + 2wi\gamma, & 0 &\leq u, w \leq a, \\ E_A(z) &= \sum_{r,t=0}^8 a_{rt} e^{\beta_{8,r,t} z}, & v &= \text{ord}_0 E_A, & \mu_v &= E_A^{(v)}(0) \neq 0. \end{aligned} \tag{NPK1}$$

The proved conductor bound is $v \leq 32$. Let C_a be the original $q'_a \times q_a$ conductor coefficient matrix, so that its ordinary transpose sends the lower exponential coefficient vector to the numerator of its product with E_A . Write $Z_a = Z_{\text{exp},a}$ for DNE3's fixed $q_a \times m_a$ matrix. Thus $N_{Z_a x}/E_A$ is the original analytic germ for coefficient x , and the columns of $[C_a^\top, Z_a]$ are a basis of the numerator subspace whose first v moments vanish. These matrices include their actual chosen minors and every original coefficient; none is replaced by a generic frame.

For $s \in \{1, a\}$ set $b = s/2$, and use the original recurrence and norms

$$\begin{aligned} p_{0,s}(y) &= 1, & p_{1,s}(y) &= y, & p_{n+1,s}(y) &= yp_{n,s}(y) - n(n-1+b)p_{n-1,s}(y), \\ h_{n,s} &= (2\pi)^{s/2} n!(b)_n, & y_{u,w} &= (\beta_{a,u,w} - c_a)/i = (2w-a)\gamma - i(2u-a)\delta, \\ \mathbf{E}_{a,N,s}[n, (u, w)] &= \frac{p_{n,s}(y_{u,w})}{\sqrt{h_{n,s}}}, & 0 \leq n \leq N, & \mathbf{K}_{a,N,s} &= \mathbf{E}_{a,N,s}^* \mathbf{E}_{a,N,s}. \end{aligned} \tag{NPK2}$$

All square roots in the positive $h_{n,s}$ are positive real. These are functional values of the original Gamma orthonormal polynomials, whose leading coefficient in S is $i^{-n}/\sqrt{h_{n,s}}$. For a numerator coefficient vector d , the column $\mathbf{E}_{a,N,s}d$ is therefore exactly the native functional-value column of N_d at the center $a/2$. It uses the original mass $(2\pi)^{s/2}$.

If $N \geq q_a - 1$, the first q_a rows are an invertible triangular polynomial-frame change times the full original root Vandermonde. The roots are distinct since their real and imaginary index differences vanish separately. Hence $\mathbf{K}_{a,N,s}$ is positive definite. Put

$$\widehat{D}_{a,N}^{(s)} = Z_a^* [\mathbf{K}_{a,N,s} - \mathbf{K}_{a,N,s} C_a^\top (\overline{C}_a \mathbf{K}_{a,N,s} C_a^\top)^{-1} \overline{C}_a \mathbf{K}_{a,N,s}] Z_a. \tag{NPK3}$$

Its exact value interpretation is

$$x^* \widehat{D}_{a,N}^{(s)} x = \min_{\eta \in \mathbb{C}^{q_a}} \|\mathbf{E}_{a,N,s}(Z_a x + C_a^\top \eta)\|^2. \tag{NPK4}$$

The block elimination in NPK3–4 is the classical Schur-complement method; see Stephen Boyd, Laurent El Ghaoui, Eric Feron and Venkataramanan Balakrishnan [H1, §2.1, pp. 7–8, (2.3)–(2.4)]. Their real symmetric presentation is used here in its Hermitian form: the two coordinate blocks are $Z_a x$ and $C_a^\top \eta$, and the lower positive block is $\overline{C}_a \mathbf{K}_{a,N,s} C_a^\top$. Completing the square with conjugate

adjoints gives exactly the two cross factors displayed above. The conductor and its value map are the programme's separately proved constructions.

The positive normal equation gives NPK3. Independence of $[C_a^T, Z_a]$ and injectivity of $\mathbf{E}$ prove strict positivity for $x \neq 0$. This is the complete conductor quotient, including both cross factors and every lower column. By the NCT dual transport, it is the metric on the exact same quotient values as $D_{a,N-v}^{(s)}$ after the map $\Lambda_F \mapsto \Lambda_{E_A F}$. In these literal DNE coordinates the numerator is $Z_a x$. The factor $\alpha_A = v!/\mu_v$ remains in the separate original KRC/RRC primary-coordinate map; it is not inserted into, or removed from, NPK3.

228.2 Evaluation by two original recurrence polynomials

This is the Christoffel–Darboux formula and its confluent form from Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21, eqs. (18.2.12)–(18.2.13)], with their $n = N$, their monic polynomials $p_n = p_{n,s}$, and their norm $h_n = h_{n,s} = (2\pi)^{s/2} n!(s/2)_n$. In their prefactor both leading coefficients equal one. The arguments are exactly $z = (\mu - c_a)/i$ and $w = (\lambda - c_a)/i$; hence the row index is conjugated and the column index is not. NPK6 and the degree-zero term below prove the complex polynomial identity with this original mass, and differentiation in z proves the value at every coincident entry.

The entire observation kernel in NPK2 has the following exact formula. For two original nodes λ, μ let $z = (\mu - c_a)/i$ and $w = (\lambda - c_a)/i$. Then

$$K_{a,N,s}[\lambda, \mu] = \begin{cases} \frac{p_{N+1,s}(z)p_{N,s}(w) - p_{N,s}(z)p_{N+1,s}(w)}{h_{N,s}(z-w)}, & z \neq w, \\ \frac{p'_{N+1,s}(w)p_{N,s}(w) - p'_{N,s}(w)p_{N+1,s}(w)}{h_{N,s}}, & z = w. \end{cases} \quad (\text{NPK5})$$

The second case occurs among the actual grid entries: it includes $\mu = \beta_{a,a-u,w_0}$ when $\lambda = \beta_{a,u,w_0}$. It must not be omitted as a nongeneric coincidence.

Here is a direct proof with the original normalization. Write $d_n = n(n-1+b) = h_{n,s}/h_{n-1,s}$ for $n \geq 1$. Subtract the recurrence evaluated at z, w to obtain, for $n \geq 1$,

$$(z-w) \frac{p_{n,s}(z)p_{n,s}(w)}{h_{n,s}} = \frac{p_{n+1,s}(z)p_{n,s}(w) - p_{n,s}(z)p_{n+1,s}(w)}{h_{n,s}} - \frac{p_{n,s}(z)p_{n-1,s}(w) - p_{n-1,s}(z)p_{n,s}(w)}{h_{n-1,s}}. \quad (\text{NPK6})$$

At $n = 0$ the separate equality is $(z-w)/h_{0,s} = (p_{1,s}(z)p_{0,s}(w) - p_{0,s}(z)p_{1,s}(w))/h_{0,s}$; there is no $h_{-1,s}$ term. Summing this equality and NPK6 through N telescopes. The recurrence has real coefficients, so $p_{n,s}(\bar{y}) = \overline{p_{n,s}(y)}$; this identifies the sum with the Hermitian kernel NPK2. The numerator in NPK5 vanishes when $z = w$, and its derivative in z gives the displayed second line. This is a polynomial identity, so the limiting line holds without an analytic exclusion. At $N = 0$ both lines give $1/h_{0,s} = 1/(2\pi)^{s/2}$, confirming the mass and sign.

Consequently NPK3 is specified by the two original recurrence polynomials $p_{N,s}, p_{N+1,s}$ and the original finite matrices C_a, Z_a . It does not require a Taylor division by E_A to evaluate this transported form. NCT supplies the exact map back to the divided native form and its complete finite volume error.

228.3 The original eight flags and their numerator row update

Now fix $j = k + 4$, $k = 4\ell + 1 \geq 9$, and one of the original target orders $s = 1, j$. Put $q = q_j$, and retain the original fixed CPQ coefficient maps on $\mathcal{U} = \mathbb{C}^{4m_j}$:

$$\begin{aligned} I &= \text{im}(B_{\text{cof}} J_0), & B &= \text{im } B_{\text{cof}}, & S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \mathbb{B}, \\ K_i &= I \cap J, & W_i &= I + J, & W &= B + J, & K_c &= S \cap W, & T &= S + J. \end{aligned} \quad (\text{NPK7})$$

The primitive index remains $e = 0$ or 1 with its original domain. All spaces in NPK7 are the same coefficient subspaces used in SFB and CPQ; the present construction changes none of their maps. For the proof, choose independent fixed columns X for each nonzero subspace, and use determinant one for the empty space. Set $\widehat{D}_N = \text{diag}(\widehat{D}_{j,N}^{(s)}, 4)$.

For $q \leq N \leq 2q$, let ξ_N be row N of $\mathbf{E}_{j,N,s}$, and evaluate all following old matrices at $N - 1$. Suppressing their j, s indices only in this display, put

$$\begin{aligned} L &= \overline{C}K_{N-1}C^\top, \quad T_0 = \mathbf{E}_{N-1}Z, \quad V_0 = \mathbf{E}_{N-1}C^\top, \\ \widehat{t}_N &= \xi_N Z, \quad \widehat{v}_N = \xi_N C^\top, \quad \widehat{\zeta}_N = \widehat{v}_N L^{-1} \widehat{v}_N^*, \\ \widehat{\rho}_N &= \widehat{t}_N - \widehat{v}_N L^{-1} V_0^* T_0, \quad \widehat{U}_N = \text{diag}(\widehat{\rho}_N / \sqrt{1 + \widehat{\zeta}_N}, 4), \\ \widehat{D}_N &= \widehat{D}_{N-1} + \frac{\widehat{\rho}_N^* \widehat{\rho}_N}{1 + \widehat{\zeta}_N}, \quad \widehat{\mathcal{D}}_N = \widehat{\mathcal{D}}_{N-1} + \widehat{U}_N^* \widehat{U}_N. \end{aligned} \tag{NPK8}$$

The inverse-update identity used in this square completion is recorded by William W. Hager [H2, p. 221, (2)]. In his $A - uv$ notation, set $A = L$, $u = -\widehat{v}_N^*$ and $v = \widehat{v}_N$; the denominator is $1 + \widehat{v}_N L^{-1} \widehat{v}_N^*$. The residual row and quotient metric in NPK8 are computed here.

These formulas retain the complete old conductor Gram L and its scalar denominator. To prove the update, write the old minimizer as $\eta_0 = -L^{-1}V_0^*T_0x$ and replace it by $\eta_0 + h$. The old squared norm is the old minimum plus h^*Lh ; the new row is $\widehat{\rho}_N x + \widehat{v}_N h$. Its remaining minimum is $|\widehat{\rho}_N x|^2 / (1 + \widehat{v}_N L^{-1} \widehat{v}_N^*)$, as follows by solving the positive normal equation, or by substitution of $h = -L^{-1} \widehat{v}_N^* \widehat{\rho}_N x / (1 + \widehat{\zeta}_N)$. This proves every line of NPK8.

The eight actual transported compressed matrices are therefore

$$\widehat{K}_{X,N} = \widehat{U}_N X (X^* \widehat{\mathcal{D}}_{N-1} X)^{-1} X^* \widehat{U}_N^*, \quad \widehat{r}_N = \frac{\widehat{\rho}_N \widehat{D}_{N-1}^{-1} \widehat{\rho}_N^*}{1 + \widehat{\zeta}_N}. \tag{NPK9}$$

For the empty subspace set $\widehat{K} = 0_{4 \times 4}$. The isometry by $\widehat{\mathcal{D}}_{N-1}^{1/2}$ writes $\widehat{K}_X = Y \Pi_X Y^*$, where Π_X is the orthogonal projector onto $\widehat{\mathcal{D}}_{N-1}^{1/2} \text{im } X$ and $Y Y^* = \widehat{r}_N I_4$. Consequently $0 \preceq \widehat{K}_X \preceq \widehat{r}_N I_4$ and inclusion of subspaces gives matrix order. All four eigenvalues, including zeros, are retained. The two original chains are exactly $J \subset W_i \subset W \subset T$ and $K_i \subset I \subset K_c \subset S$.

Let $\sigma_X = 1$ for $X = T, K_c, W_i, K_i$ and $\sigma_X = -1$ for $X = W, S, J, I$. The determinant lemma applied to NPK8 gives

$$\begin{aligned} \widehat{\ell}_N &= \sum_X \sigma_X \log \det(I_4 + \widehat{K}_{X,N}), \\ \widehat{\mathcal{L}}_e &= \sum_{N=q}^{2q} \vartheta_N \widehat{\ell}_N, \quad \vartheta_q = \vartheta_{2q} = 1, \quad \vartheta_N = 2 \ (q < N < 2q), \\ \widehat{\mathcal{P}}_{k,e} &= \sum_{N=q}^{2q} \vartheta_N \sum_X \sigma_X \sum_{a=1}^4 \log \max(1, \widehat{\lambda}_{X,a,N}). \end{aligned} \tag{NPK10}$$

The native row corresponding to this N is exactly $n = N - v$. The weights sum to $2q$, and NPK10 has the unchanged four endpoints $q-1, q, 2q-1, 2q$. The quantity $\widehat{\mathcal{L}}_e$ is the complete CPQ common return formed with the transported metric: the same quotient determinant identities prove the signed eight-flag expression. No fixed frame determinant is discarded before taking the return.

228.4 Finite and limiting transfer of the complete signed profile

Denote NCT's nonnegative finite volume-error bounds for b_0 copies, quotient rank r , and an admissible original SGT radius ϱ by $\mathcal{U}_{\varrho;a,D,s,r}^{[b_0]}$ and $\mathcal{L}_{\varrho;a,D,s,r}^{[b_0]}$. They retain the actual SGT poles,

residues, zero-free radius, diagonal weights and complete conductor. Define

$$B_{a,D,s,r}^{[b_0]} = \inf_{\substack{0 < \rho < 1 \\ \rho \text{ admissible}}} \max(\mathcal{L}_{\rho;a,D,s,r}^{[b_0]}, \mathcal{U}_{\rho;a,D,s,r}^{[b_0]}),$$

$$\mathcal{N} = (q-1, q, 2q-1, 2q), \quad E_X = \sum_{N \in \mathcal{N}} B_{j,N-v,s,\dim X}^{[4]}, \quad E_{\text{flags}} = \sum_{X \in \{T, K_c, W_i, K_i, W, S, J, I\}} E_X. \quad (\text{NPK11})$$

At every endpoint NCT gives $|\log \det(X^* \widehat{\mathcal{D}}_N X) - \log \det(X^* \mathcal{D}_N X)| \leq B_{j,N-v,s,\dim X}^{[4]}$. Taking the four original signs and then the eight original signs proves

$$|\widehat{\mathcal{L}}_e - \mathcal{L}_e| \leq E_{\text{flags}}. \quad (\text{NPK12})$$

The infimum is over the same original function g and its proved admissible radii. It is nonnegative and finite, since SGT supplies at least one such finite bound. The endpoint inequality holds for every admissible radius, hence also for their infimum; attainment is unnecessary. Each $\dim X \leq 4m_j = O(k)$ and $N = O(k^2)$. For each fixed radius the NCT proof bounds the limsup of the displayed infimum divided by kq_k by a constant times $\log(1/\rho)$, with all remaining terms tending to zero. Only after this limit choose admissible radii tending to one. It follows that $B^{[4]} = o(kq_k)$ and $E_{\text{flags}} = o(kq_k)$. A fixed-radius enclosure itself is not asserted to be little-oh. Any particular finite admissible radius remains a usable, fully specified bound in NPK11–12.

For completeness the threshold error used here has a full direct proof. Put $h(x) = \log(1+x) - \log \max(1, x) = u(x) - v(x)$, where $u(x) = \log(1 + \min(x, 1))$, and $v(x) = 0$ for $x \leq 1$, $v(x) = \log(2x/(1+x))$ for $x \geq 1$. Both u, v are increasing, vanish at zero and are at most $\log 2$. On any increasing four-term eigenvalue chain, the alternating sum of disjoint increments of u lies between zero and $u(r)$; adding the missing middle increment proves the upper bound. The same argument holds for v . The alternating h sum therefore lies between $-v(r)$ and $u(r)$. Apply this to each of the four ordered eigenvalues on each of the two chains in NPK9; matrix order implies eigenvalue order by the Rayleigh min–max formula. The difference of the two chain sums has absolute value at most $4(u(r) + v(r)) \leq 8 \log 2$ at each row. After the exact weights, both the original and transported threshold errors are at most $16q \log 2$. Combining this with NPK12 proves the finite receiver

$$|\mathcal{P}_{k,e} - \widehat{\mathcal{P}}_{k,e}| \leq E_{\text{flags}} + 32q_j \log 2 = o(kq_k). \quad (\text{NPK13})$$

This transfers the entire signed profile, with its correlations and zero eigenvalues. It does not assert a rowwise approximation of any individual eigenvalue, or a value for the transported profile.

228.5 All complementary returns and the original arithmetic action

For each of the original CPQ forms Q_Y on the specified quotient values (in particular $G_s, Q_{\text{rem}}, Q_b^E, A_{\text{inv}}$), form $\widehat{Q}_Y$ by the same restriction and full affine minimum using $\widehat{\mathcal{D}}_N$. If its rank is r_Y , NCT and the original four signs give the first line below. Define the transported target quotient and its return explicitly by $\widehat{H}_{B,j,N} = (B_j(\widehat{\mathcal{D}}_{j,N}^{(s)})^{-1} B_j^*)^{-1}$ and $\widehat{S}_{K,j} = \mathcal{R} \log \det \widehat{H}_{B,j,N}$, in the original onto value frame of B_j . Its full four-copy rank is $4t_j$. Then

$$|\widehat{V}_Y - V_Y| \leq E_Y := \sum_{N \in \mathcal{N}} B_{j,N-v,s,r_Y}^{[4]} = o(kq_k),$$

$$4\widehat{S}_{K,j} = \widehat{\mathcal{L}}_e + \widehat{V}_b + \widehat{V}_{\text{rem}} + \widehat{V}_s + \widehat{V}_A, \quad (\text{NPK14})$$

$$4(\widehat{S}_{K,j} - S_{K,j}) = (\widehat{\mathcal{L}}_e - \mathcal{L}_e) + \sum_{Y=b,\text{rem},s,A} (\widehat{V}_Y - V_Y).$$

The second line follows by block Schur factorization in precisely CPQ's fixed frames, so the third is its subtraction from CPQ20. Alternatively NCT applies directly to the rank- $4t_j$ target quotient,

yielding $|4\hat{S}_{K,j} - 4S_{K,j}| \leq E_{\text{target}}$, where $E_{\text{target}} = \sum_{N \in \mathcal{N}} B_{j,N-v,s,4t_j}^{[4]}$. Both enclosures hold for the same value, so their intersection is valid. This retains every complementary term and their exact correlation.

The independent proper-source form Q_{inv} uses its original degree- k frame, cutoff and center. Apply NCT there separately. For the exact relative determinant in NFR the identity is

$$\begin{aligned} \log \frac{\det \hat{A}_{\text{inv}}}{\det \hat{Q}_{\text{inv}}} - \log \frac{\det A_{\text{inv}}}{\det Q_{\text{inv}}} \\ = (\log \det \hat{A}_{\text{inv}} - \log \det A_{\text{inv}}) - (\log \det \hat{Q}_{\text{inv}} - \log \det Q_{\text{inv}}). \end{aligned} \quad (\text{NPK15})$$

The two finite NCT errors add to an enclosure; on the original rank- $O(k)$ and cutoff- $O(k^2)$ domains their four-return is $o(kq_k)$. The two distinct source orders and centers are unchanged in this statement. The EFP same-value maps and HJR inverse-metric identities act on the same quotient coordinates, so NPK14–15 can be substituted into them with their full finite error intervals. In the complete action $J = B + W - P$, transporting only the specified volume term gives exactly $\hat{J} - J = \hat{B} - B$ when the original recurrence window W and penalty P are retained. This supplies an explicit error-bearing map into that action, without assigning any missing intrinsic coefficient or cancelling an uncomputed term.

The result is a concrete finite observation problem on the original root grid: NPK3 and NPK5 calculate all transported forms from the original C_a, Z_a and Gamma recurrence; NPK8–10 calculate their full mixed profile; NPK12–15 prove their quantitative relation to the original divided forms. The surviving large-degree value of these original numerator subquotients is the next arithmetic calculation, and remains present in every displayed identity.

The full cardinal covariance and partial-return transport. The complete IGI calculation below constructs the literal Lagrange section $\mathcal{I}x = \sum_{\lambda} x_{\lambda} \chi(S) / ((S - \beta_{\lambda}) \chi'(\beta_{\lambda}))$ on the original grid. Polynomial division proves its exact evaluation kernel $\chi \mathbb{C}[S]_{\leq N-q}$. With G_R the full relation Gram and B its pairings with every cardinal column, the attained value metric and the original observation covariance obey

$$Q_N = G_L - B^* G_R^{-1} B = \overline{K_N^{-1}}. \quad (\text{IUP2})$$

The last conjugation follows from the complex-linear evaluation equation $E_N^{\text{T}} z = x$ and its minimum-norm solution; it is retained in the original conductor and low-moment quotient. For any original rank- t flag or CPQ quotient write the finite NCT bound as $|\hat{f}(N) - f(N)| \leq B_{t,N}$. The exact IPR prefix/tail maps prove, for $0 \leq R < q$,

$$\begin{aligned} |\mathcal{R}_{\leq R}(\hat{f} - f)| &\leq 2B_{t,q+R} + B_{t,q} + B_{t,q-1}, \\ |\mathcal{R}_{> R}(\hat{f} - f)| &\leq B_{t,2q-1} + B_{t,2q} + 2B_{t,q+R}. \end{aligned} \quad (\text{IUP3})$$

The operators on both forms have the same literal evaluation sites. These follow by substituting each signed error into the three-site identities; every fixed frame factor cancels because its coefficients sum to zero. The full eight-flag and threshold errors are IPR11–13, including both threshold-chain errors and their original weights.

229 Original-grid interpolation, relation norms, and an interior mixed row

Keep the exact original simple quartet $0 < \delta < 1/2$, $\gamma > 2$, the fixed original five-orbit period and logarithm branch, and its complete nonreduced conductor. For $a \equiv 1 \pmod{4}$, $a \geq 9$, put

$$\begin{aligned} c &= a/2, \quad q = (a+1)^2, \quad q' = (a-7)^2, \quad \beta_{u,w} = c + (2u-a)\delta + i(2w-a)\gamma, \\ \chi(S) &= \prod_{u,w=0}^a (S - \beta_{u,w}), \quad E_A(z) = \sum_{u,w=0}^8 a_{uw} e^{\beta_{u,w} z}, \\ \mu_t &= E_A^{(t)}(0) \ (t \geq 0), \quad v = \text{ord}_0 E_A, \quad \mu_v \neq 0, \quad m = q - q' - v, \quad s \in \{1, a\}, \quad N \geq q-1. \end{aligned} \tag{IGI1}$$

Use the lexicographic order of the original grid. A single index λ below denotes a member of this ordered grid. The roots are distinct because real and imaginary parts separately determine both indices. The original bounds on v and the original period domain are unchanged. In particular $v < q$ and $m > 0$.

Retain exactly

$$\begin{aligned} dm_s(y) &= \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\ p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, \quad \phi_n(S) = p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}, \\ \langle f, g \rangle_s &= \int_{\mathbb{R}} \overline{f(c+iy)} g(c+iy) dm_s(y). \end{aligned} \tag{IGI2}$$

The original SGT/NSO construction proves that these ϕ_n are orthonormal. Their leading coefficient is $i^{-n}/\sqrt{h_{n,s}}$. The inner product is conjugate-linear in its first argument; the functional dual throughout is complex-linear.

229.1 The full original evaluation map and its attained cardinal norm

Define the unchanged evaluation map and its literal cardinal section:

$$\begin{aligned} \text{ev}_N : \mathbb{C}[S]_{\leq N} &\longrightarrow \mathbb{C}^q, \quad P \longmapsto (P(\beta_\lambda))_\lambda, \\ L_\lambda(S) &= \frac{\chi(S)}{(S - \beta_\lambda)\chi'(\beta_\lambda)}, \quad \mathcal{I}x = \sum_{\lambda} x_\lambda L_\lambda(S). \end{aligned} \tag{IGI3}$$

Every displayed derivative is nonzero, since its root is simple. Evaluation at each root proves $\text{ev}_N \mathcal{I} = I_q$. Polynomial division gives, with all coefficients retained,

$$\ker \text{ev}_N = \chi \mathbb{C}[S]_{\leq N-q}, \quad P = \mathcal{I}(\text{ev}_N P) + \chi R, \tag{IGI4}$$

where the relation space is zero at $N = q-1$. Indeed division has a unique remainder of degree below q ; a remainder vanishing at all q distinct roots is zero. Thus IGI4 proves the entire kernel, surjectivity, and the exact relation to the cardinal coordinates.

Let $G_L = (\langle L_\lambda, L_\mu \rangle_s)$. At $N \geq q$ use the complete relation basis $R_j = \chi S^j$, $0 \leq j \leq N-q$, and put $G_R = (\langle R_i, R_j \rangle_s)$ and $B = (\langle R_i, L_\lambda \rangle_s)$. Then the actual value norm is

$$\begin{aligned} x^* Q_N x &= \min_{\text{ev}_N P=x} \|P\|_s^2, \quad Q_N = G_L - B^* G_R^{-1} B, \\ P_{N,x} &= \mathcal{I}x - \sum_j R_j (G_R^{-1} Bx)_j. \end{aligned} \tag{IGI5}$$

The positive quadratic minimum in IGI5 is the Schur-complement method [H1, §2.1, (2.3)–(2.4)] in the original cardinal/relation frame: its three blocks are G_L , B and G_R . The subsequent inverse evaluation metric retains the complex-linear transpose in the observation map.

For $N = q - 1$ put $Q_N = G_L$ and omit the empty second term. Completing the positive square in the coefficient of R proves IGI5: every other lift has squared norm equal to this minimum plus the positive squared norm of its difference in the relation space. The value Gram is positive definite because a zero minimum would be the zero polynomial with zero values. In particular no cross Gram B has been discarded.

The original numerator observation matrix is

$$\mathbf{E}_N[n, \lambda] = \phi_n(\beta_\lambda), \quad 0 \leq n \leq N, \quad K_N = \mathbf{E}_N^* \mathbf{E}_N. \quad (\text{IGI6})$$

Its first q rows are an invertible triangular change of the original Vandermonde, so $K_N > 0$. If $P = \sum z_n \phi_n$, then $\text{ev}_N P = \mathbf{E}_N^T z$, with an ordinary transpose. The unique coefficient minimum for this constraint is

$$z = \overline{\mathbf{E}_N} (\mathbf{E}_N^T \overline{\mathbf{E}_N})^{-1} x, \quad \boxed{Q_N = \overline{K_N^{-1}}}. \quad (\text{IGI7})$$

Multiplication proves the constraint. The displayed column is orthogonal to its kernel; its squared norm is $x^* (\mathbf{E}_N^T \overline{\mathbf{E}_N})^{-1} x$. Since $\mathbf{E}_N^T \overline{\mathbf{E}_N} = \overline{K_N}$, this is exactly IGI7. Equations IGI5 and IGI7 give the exact typed map between the full cardinal Schur minimum and the original functional covariance, including its necessary conjugation.

229.2 All conductor columns, vanishing moments, and quotient duals

Write $C = C_a$ and $Z = Z_{\text{exp},a}$ for the unchanged NCT6–7 matrices. Thus

$$N_{C^T \eta}(z) = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}, \quad N_d(z) = \sum_{\lambda} d_{\lambda} e^{\beta_{\lambda} z}. \quad (\text{IGI8})$$

The columns of C^T are independent by injectivity of multiplication by the nonzero germ E_A and the lower Vandermonde. Define

$$G_v[d, \lambda] = \beta_{\lambda}^d \quad (0 \leq d < v), \quad \mathcal{N}_v = \ker G_v, \quad B_0 = [C^T, Z], \quad \mathcal{L}_v = \text{im } G_v^T. \quad (\text{IGI9})$$

At $v = 0$ both G_v and $\mathcal{L}_v$ have zero dimension. The original NCT7 frame proves $\text{im } B_0 = \mathcal{N}_v$ and its columns independent. The rows of G_v are independent by the upper Vandermonde. Moreover $G_v C^T = 0$ by IGI8, so $\mathcal{L}_v \subset \ker C$. Every member of $\mathcal{L}_v$ is the evaluation vector of a polynomial of degree below v , and every such vector occurs, by its displayed monomial columns.

The original numerator coefficient quotient has Gram

$$\begin{aligned} \widehat{D}_N &= Z^* \{K_N - K_N C^T (\overline{C} K_N C^T)^{-1} \overline{C} K_N\} Z, \\ x^* \widehat{D}_N x &= \min_{\eta} (Zx + C^T \eta)^* K_N (Zx + C^T \eta). \end{aligned} \quad (\text{IGI10})$$

Positive square completion gives both the formula and its unique minimum

$$\eta = -(\overline{C} K_N C^T)^{-1} \overline{C} K_N Zx.$$

Strict positivity follows from independence of B_0 .

The exact dual value space is

$$\mathcal{W} = \ker C / \mathcal{L}_v, \quad [z] \mapsto Z^T z \in \mathbb{C}^m. \quad (\text{IGI11})$$

This map is well defined since $Z^T G_v^T = 0$. Its kernel before quotienting is $\ker C \cap \ker Z^T = \ker B_0^T$. Because $\text{im } B_0 = \ker G_v$, the ordinary bilinear annihilator of that image is exactly $\text{im } G_v^T$: inclusion is immediate and both dimensions are v . Thus the kernel is precisely $\mathcal{L}_v$. The rank of the

independent rows of $[C; Z^\top]$ is $q - v$, so any values $(0, t)$ are attained. This proves the isomorphism in IGI11, with its full low-degree denominator retained.

Its attained value norm satisfies exactly

$$\begin{aligned} t^* \widehat{D}_N^{-1} t &= \min_{Cz=0, Z^\top z=t} z^* Q_N z \\ &= \min_{\substack{P \in \mathbb{C}[S]_{\leq N} \\ C \text{ ev}_N P=0, \quad Z^\top \text{ ev}_N P=t}} \|P\|_s^2. \end{aligned} \quad (\text{IGI12})$$

For a direct proof, the minimum under the onto map $A = [C; Z^\top]$ is $(AQ_N^{-1}A^*)^{-1}$ in its value coordinates, as follows from the lift $Q_N^{-1}A^*(AQ_N^{-1}A^*)^{-1}(0, t)^\top$ and orthogonality to $\ker A$. Since $Q_N^{-1} = \overline{K}_N$, the Schur complement of the first block of $AQ_N^{-1}A^*$ is precisely $\widehat{D}_N$. Its bottom block inverse gives IGI12. Taking the complete polynomial minimum IGI5 proves the second line. Thus this is the complex-linear dual of IGI10 with its full metric, not a coordinate dimension comparison.

The conductor condition has the exact polynomial interpretation

$$(C \text{ ev}_N P)_{u,w} = (\mathcal{T}_A P)(\beta_{a-8,u,w}), \quad C \text{ ev}_N P = 0 \iff \chi_{a-8} \mid \mathcal{T}_A P. \quad (\text{IGI13})$$

The first equality is IGI8 coefficient collection and exact root addition. The second uses every distinct lower root and polynomial division. It holds for every P of the stated degree, retaining all cancellations in $\mathcal{T}_A$. The fibre IGI12 minimizes over all resulting low-polynomial values, rather than setting them equal to zero.

For any original fixed coefficient subspace frame X in four copies, its restriction Gram is $X^* \text{diag}(\widehat{D}_N, 4)X$. For a quotient Y/K with fixed independent frame $[X_K, X_Q]$, its Gram is exactly

$$X_Q^* \widehat{D}_N X_Q - X_Q^* \widehat{D}_N X_K (X_K^* \widehat{D}_N X_K)^{-1} X_K^* \widehat{D}_N X_Q, \quad \widehat{D}_N = \text{diag}(\widehat{D}_N, 4). \quad (\text{IGI14})$$

Square completion proves the minimum on its original affine fibre. Applying IGI10–12 in each copy and these exact further minima gives every original CPQ restriction and quotient, with the same frames, full cross terms and values. No choice of orthogonal replacement flag enters IGI14.

229.3 The full relation Christoffel row and the original observation row

Let $\pi_r(S)$ be the unique monic degree- r polynomial minimizing $\|\chi P\|_s^2$, and put

$$\nu_{r,s} = \|\chi \pi_r\|_s^2, \quad \tau_r = \frac{\chi \pi_r}{\sqrt{\nu_{r,s}}}, \quad b_r = (\langle \tau_r, L_\lambda \rangle_s)_\lambda. \quad (\text{IGI15})$$

The monic quadratic minima are orthogonal to all lower-degree directions, by varying each complex coefficient in its real and imaginary parts. Thus the τ_r form an orthonormal basis of the entire relation space in IGI4. Orthogonal projection of every cardinal column onto that complete space gives

$$Q_N = G_L - \sum_{j=0}^{N-q} b_j^* b_j, \quad Q_N = Q_{N-1} - b_r^* b_r \quad (N = q + r). \quad (\text{IGI16})$$

Every row b_j is the actual cardinal/relation cross pairing.

Write $\xi_N = (\phi_N(\beta_\lambda))_\lambda$, the actual next original observation row. The coefficient of ϕ_N in τ_r is

$$t_N = i^N \sqrt{h_{N,s}/\nu_{r,s}}, \quad N = q + r. \quad (\text{IGI17})$$

This follows from their exact leading coefficients. Its old component $z_< = (\langle \phi_n, \tau_r \rangle_s)_{n < N}$ is orthogonal to every old relation, since τ_r is. Its old evaluation values are $-t_N \xi_N^\top$, since the full polynomial τ_r vanishes at every original node. It is therefore the minimum lift IGI7 at cutoff $N - 1$:

$$z_< = -t_N \overline{E_{N-1}} Q_{N-1} \xi_N^\top.$$

Pairing it with each old cardinal column gives $b_r = -\overline{t_N} \overline{\xi_N} Q_{N-1}$. Conjugation and IGI7 yield the exact phase-sensitive row formula

$$\boxed{\xi_N = -i^{-N} \sqrt{\nu_{r,s}/h_{N,s}} \overline{b_r} K_{N-1}.} \quad (\text{IGI18})$$

The missing phase cannot be supplied by a determinant identity alone; it has been computed from the original polynomial frame. Since τ_r has norm one, its old component has squared norm $1 - |t_N|^2$. The same minimum lift gives

$$\begin{aligned} b_r Q_{N-1}^{-1} b_r^* &= 1 - \frac{h_{N,s}}{\nu_{r,s}}, & T_{r,s}^\Gamma &:= \frac{\nu_{r,s}}{h_{N,s}} \geq 1, \\ \xi_N K_{N-1}^{-1} \xi_N^* &= T_{r,s}^\Gamma - 1, & \frac{\det K_N}{\det K_{N-1}} &= T_{r,s}^\Gamma. \end{aligned} \quad (\text{IGI19})$$

For the third formula, substitute IGI18 into the first, or use the norm of $z_<$ and $|t_N|^2 = h_N/\nu_r$. The determinant formula follows by the rank-one update $K_N = K_{N-1} + \xi_N^* \xi_N$. Positivity of Q_N also proves that the first scalar in IGI19 is strictly below one.

229.4 The retained low-moment correction and conductor correction

All matrices in this subsection are evaluated at cutoff $N - 1$. Suppress that cutoff only for $K = K_{N-1}$ and put

$$\begin{aligned} L &= \overline{C} K C^\top, \quad Z_\perp = Z - C^\top L^{-1} \overline{C} K Z, \quad \hat{\zeta}_N = \xi_N C^\top L^{-1} \overline{C} \xi_N^*, \\ \hat{\rho}_N &= \xi_N Z_\perp, \quad \hat{r}_N = \frac{\hat{\rho}_N \hat{D}_{N-1}^{-1} \hat{\rho}_N^*}{1 + \hat{\zeta}_N}, \\ d_N &= \xi_N K^{-1} G_v^* (G_v K^{-1} G_v^*)^{-1} G_v K^{-1} \xi_N^*. \end{aligned} \quad (\text{IGI20})$$

At $v = 0$ define $d_N = 0$ and omit its inverse. At $v > 0$ that inverse exists because G_v has full row rank. Every factor in d_N has its ordinary Hermitian adjoint as displayed; the defining moment map $G_v d$ itself remains complex-linear.

The exact metric projector identity is

$$B_0 (B_0^* K B_0)^{-1} B_0^* = K^{-1} - K^{-1} G_v^* (G_v K^{-1} G_v^*)^{-1} G_v K^{-1}. \quad (\text{IGI21})$$

To prove it, apply the isometry $K^{1/2}$ to $\ker G_v$. Its orthogonal complement is the range of $K^{-1/2} G_v^*$. The orthogonal projector onto that complement is $K^{-1/2} G_v^* (G_v K^{-1} G_v^*)^{-1} G_v K^{-1/2}$; it is self-adjoint, idempotent, and fixes that range. The projector onto the former space is $K^{1/2} B_0 (B_0^* K B_0)^{-1} B_0^* K^{1/2}$. Their sum is the identity, and multiplication on both sides by $K^{-1/2}$ proves IGI21. The empty case is the same formula with zero complement.

Multiply IGI21 by ξ_N and its adjoint. The full restricted numerator leverage is $T_{r,s}^\Gamma - 1 - d_N$. Completing the old conductor/numerator square as in IGI10 decomposes that same quantity into $\hat{\zeta}_N + \hat{\rho}_N \hat{D}_{N-1}^{-1} \hat{\rho}_N^*$. Consequently

$$\boxed{1 + \hat{r}_N = \frac{T_{r,s}^\Gamma - d_N}{1 + \hat{\zeta}_N}, \quad 0 \leq d_N + \hat{\zeta}_N \leq T_{r,s}^\Gamma - 1.} \quad (\text{IGI22})$$

This identity preserves both original corrections. At $v = 0$ the complementary space is zero, so d_N vanishes identically. For $v > 0$ retain the displayed exact expression, including any value zero proved from the original data. Invertibility of the full Vandermonde alone does not eliminate that expression. Substitution of IGI18 into IGI20 evaluates every correction in the same cardinal/relation cross row b_r , with its exact phase.

For the actual target $a = j = k + 4$, let the eight fixed flags be precisely $T_e, K_c, W_i, K_i, W, S, J, I$ of NPK7, with their two nested chains and their original frames X . Then

$$\widehat{U}_N = \text{diag}(\widehat{\rho}_N / \sqrt{1 + \widehat{\zeta}_N}, 4), \quad \widehat{K}_{X,N} = \widehat{U}_N X (X^* \widehat{\mathcal{D}}_{N-1} X)^{-1} X^* \widehat{U}_N^*. \quad (\text{IGI23})$$

These are the same eight NPK9 matrices, now with their exact polynomial expressions IGI5,15,18,20. Set an empty compression equal to the zero 4×4 matrix. The signed row keeps the positive signs T_e, K_c, W_i, K_i and negative signs W, S, J, I . All four eigenvalues, including zero and subthreshold eigenvalues, remain present. No conclusion about these eight orientations is deduced from the scalar $T_{r,s}^\Gamma$ alone: the entire cross row b_r , the original C, Z, G_v , and all eight X are still explicit factors of IGI23.

229.5 A finite calculation at the second original target row

The original $Q_a(y) = i^{-q} \chi(c + iy)$ is real and even. Its root set is invariant under conjugation and negation, by the exact index permutations $u \mapsto a - u$ and $(u, w) \mapsto (a - u, a - w)$, and q is even. Thus $|Q_a(y)|^2 dm_s(y)$ is an even positive measure. The original degree-one monic minimizer is therefore exactly

$$\pi_1(S) = S - c, \quad \nu_{1,s} = \int_{\mathbb{R}} y^2 |Q_a(y)|^2 dm_s(y), \quad T_{1,s}^\Gamma = \frac{\nu_{1,s}}{M_s(q+1)!(s/2)_{q+1}}. \quad (\text{IGI24})$$

Indeed a monic linear polynomial in y is $y + z$; its weighted squared norm is the norm of y plus $|z|^2$ times the positive mass, since the mixed integral of y is zero. Its minimum is at $z = 0$. The map back to S is the exact monic map $p(y) \mapsto ip((S - c)/i)$.

For a finite evaluation with no quadrature, write $Q_a(y) = \sum_{t=0}^q c_t y^t$ using every original root coefficient. For each integer $b \geq 0$ let J_b be the $(b+1)$ -square Jacobi matrix on indices $0, \dots, b$ with diagonal zero and off-diagonal $\sqrt{(n+1)(n+s/2)}$. The original recurrence proves

$$\int y^b dm_s(y) = M_s(J_b^b)_{00}, \quad \nu_{1,s} = M_s \sum_{t,u=0}^q \bar{c}_t c_u (J_{t+u+2}^{t+u+2})_{00}. \quad (\text{IGI25})$$

Starting at degree zero, b applications of the recurrence cannot visit degree greater than b , proving the finite moment identity and its retained mass. Expansion of the original squared polynomial gives the second identity. These are exact finite sums.

At the non-first target row $N = q + 1$, define the fully specified nonnegative number

$$\mathcal{B}_{q+1,s} = \log \frac{T_{1,s}^\Gamma - d_{q+1}}{1 + \widehat{\zeta}_{q+1}}. \quad (\text{IGI26})$$

Here $T_{1,s}^\Gamma$ is IGI24–25, b_1 is the explicit integral of $\chi(S)(S - c)/\sqrt{\nu_{1,s}}$ against each literal cardinal column, and IGI18,20 compute $d_{q+1}, \widehat{\zeta}_{q+1}$ from those original finite coefficients. Its nonnegativity and the finite estimates

$$0 \leq \mathcal{B}_{q+1,s} \leq \log T_{1,s}^\Gamma - \log(1 + \widehat{\zeta}_{q+1}) \leq \log T_{1,s}^\Gamma, \quad |\widehat{\ell}_{q+1}| \leq 4\mathcal{B}_{q+1,s}, \quad |\widehat{p}_{q+1}| \leq 4\mathcal{B}_{q+1,s}, \quad 0 \leq \Delta \widehat{V}_X \leq 4\mathcal{B}_{q+1,s} \quad (X = b, \text{rem}, s, A) \quad (\text{IGI27})$$

follow as follows. IGI22 makes IGI26 exactly $\log(1 + \widehat{r}_{q+1})$, and $d_N \geq 0$ gives the first line. For the second line, ordered projectors along each original flag chain give ordered compressed matrices. The two disjoint increment sums on a chain are nonnegative and at most the full increment $4\log(1 + \widehat{r}_N)$. Their difference is the signed eight-term row. The same argument works with the increasing function $\log \max(1, x)$ on each of the four ordered eigenvalues, giving the threshold bound without omitting zeros. For the four CPQ quotients the exact pairs are $W/W_i, \mathcal{U}/T_e, S/K_c, I/K_i$. Full Schur elimination gives their increment as $b_Y - b_K$, between zero and the same full

four-copy budget. Their original fixed frame factors cancel only in this increment. The interior weight in the original return is two, so its entire weighted contribution is bounded by $8\mathcal{B}_{q+1,s}$. For the native determinant row, write the eight original frames as $\mathcal{X} = \{T_e, K_c, W_i, K_i, W, S, J, I\}$ and define, using exactly the nonnegative NPK11/NCT20 finite error constants,

$$E_{\text{row}}(N) = \sum_{X \in \mathcal{X}} \left(B_{j,N-1-v,s,\dim X}^{[4]} + B_{j,N-v,s,\dim X}^{[4]} \right), \quad N = q + 1. \quad (\text{IGI27a})$$

The two endpoint determinant errors, with their original signs, give $|\delta\mathcal{L}_{e,N-v} - \widehat{\ell}_N| \leq E_{\text{row}}(N)$. The native and transported threshold rows each differ from their own determinant row by at most $8 \log 2$, by the two-chain proof NPK13 before applying its endpoint weights. The full triangle inequality therefore gives

$$\begin{aligned} |\delta\mathcal{P}_{e,N-v} - \widehat{p}_N| &\leq E_{\text{row}}(N) + 16 \log 2, \\ |\delta\mathcal{P}_{e,N-v}| &\leq 4\mathcal{B}_{q+1,s} + E_{\text{row}}(N) + 16 \log 2. \end{aligned} \quad (\text{IGI27b})$$

For each complementary quotient of its original rank r_X , the direct determinant increment error is instead the sum $B_{j,N-1-v,s,r_X}^{[4]} + B_{j,N-v,s,r_X}^{[4]}$; no threshold error is added to a determinant increment. The actual second-row interior weight is two, so every complete bound just displayed, including its error terms, is multiplied by two in the full return. These statements compare the exact scalar increments; they assert no eigenwise approximation of the eight matrices.

229.6 The exact interface to the older arithmetic norm results

The scalar $\nu_{r,s}$ above is measured in dm_s . In GRA1 the original arithmetic measure, at multiplicity one and degree a , is $d\mu_a(y) = w_h^{*a}(y)dy$ with $w_h(y) = |(2\xi/h)(1/2 + iy)|^2/(2\pi)$ and the complete original quartet polynomial h . At order one the present Gamma measure is exactly $dm_1 = d\sigma = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi)$, including its mass $\sqrt{2\pi}$. The identity on the unchanged polynomial space at centre $c = a/2$ is the actual comparison map; its density ratio is literally

$$\frac{d\mu_a}{dm_s}(y) = \frac{w_h^{*a}(y)}{\frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2}. \quad (\text{IGI28})$$

The denominator is positive for real y . This is not a claim that the ratio is constant.

For degree cutoff d the proved all-polynomial GRA5 constants are a_a/D_d and A_a , with $D_d = [1 + 4(d + M_h)^2]^{M_h}$ and the unchanged full GRA3 definitions of a_a, A_a, M_h . The proof there uses the actual convolution upper and lower densities and the original Gamma recurrence, and gives $(a_a/D_d)H_1 \preceq H_{\text{ar}} \preceq A_a H_1$. For $L = (s - 1)/4$, Gamma recurrence gives the exact whole-source identity and finite comparison

$$\begin{aligned} dm_s &= \rho_s \prod_{u=0}^{L-1} (y^2 + (2u + 1/2)^2) dm_1, & \rho_s &= \frac{(2\pi)^{2L}}{(1/2)_{2L}}, \\ \mathfrak{a}_s &= \rho_s \prod_{u=0}^{L-1} (2u + 1/2)^2, & \mathfrak{b}_{s,d} &= \rho_s \prod_{u=0}^{L-1} \{4(d + u + 1)^2 + (2u + 1/2)^2\}, \\ \mathfrak{a}_s H_1 &\preceq H_s \preceq \mathfrak{b}_{s,d} H_1, & \frac{a_a}{D_d \mathfrak{b}_{s,d}} H_s &\preceq H_{\text{ar}} \preceq \frac{A_a}{\mathfrak{a}_s} H_s. \end{aligned} \quad (\text{IGI29})$$

For completeness the upper Gamma comparison follows from the orthonormal order-one recurrence: on degree at most d , $\|yp\|_1 \leq 2(d + 1)\|p\|_1$. Multiplication by $iy - t_u$ at step u has squared norm at most $4(d + u + 1)^2 + t_u^2$ times the preceding norm, while pointwise $|iy - t_u|^2 = y^2 + t_u^2 \geq t_u^2$. Multiplying all factors proves the stated comparison, with its exact scalar ρ_s . Combining it with GRA5 in the displayed directions proves the last line. Empty products at $s = 1$ equal one.

Apply these inequalities to every polynomial in the same monic fibre χP , with $d = q + r$, and take minima. This gives

$$\frac{a_a}{D_{q+r} \mathbf{b}_{s,q+r}} \nu_{r,s} \leq \nu_r^{\text{ar}} \leq \frac{A_a}{\mathbf{a}_s} \nu_{r,s}. \quad (\text{IGI30})$$

The identical inequalities apply to all fixed cardinal evaluation fibres IGI5, or the constrained fibres IGI12, by taking the same full minimum on both sides. Inversion and the conjugation in IGI7 give their precise reverse dual inequalities. Thus GRA's arithmetic norm and the numerator Gamma norm have a proved full form map before any scalar conclusion is transported.

On the order-one Gamma relation space, the monic map $U_r P(y) = i^{-r} P(c + iy)$ has inverse $i^r p((S - c)/i)$ and satisfies $U_{q+r}(\chi P) = Q_a U_r P$. Hence the original GRF whole-form comparison applies literally to $\nu_{r,1}$ and to the full relation Gram G_R in IGI5 on precisely its proved finite guards. The GRE reference minima then give its stated scalar bounds, and IGI29 transports any such valid full-form inequality to the other original source order with the displayed constants. The degree-one evaluation IGI24–27 uses its direct proof and does not assert the higher-degree GRF guards at $r = 1$.

The cardinal cross rows b_r remain the pairings IGI15 in these maps. A bound for their single quadratic norm IGI19 determines the complete scalar observation leverage, but does not specify their values on the original conductor and flag columns in IGI20,23. The exact remaining map is IGI18 followed by those literal column maps; IGI22 exhibits its separate nonnegative low-moment and conductor contributions. These retained expressions are the surviving interior mixed calculation, including all complementary CPQ returns, after the scalar relation norms have been supplied by the proved older results.

230 The original native interior rows and a growing initial segment

Fix the original simple quartet, complete nonreduced conductor, period, and logarithm branch. Put

$$\begin{aligned}
 j &= k + 4, \quad k = 4l + 1 \geq 9, \quad q = (j + 1)^2, \quad q' = (j - 7)^2, \\
 c' &= j/2 - 4, \quad L = q - v, \quad m = L - q', \\
 s &\in \{1, j\}, \quad v = \text{ord}_0 E_A, \quad \mu_t = E_A^{(t)}(0) \ (t \geq 0), \quad \mu_v \neq 0, \\
 E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \tag{NIB1} \\
 \chi_j(S) &= \prod_{u,w=0}^j (S - \beta_{j,u,w}), \quad \chi_{-}(S') = \chi_{j-8}(S'), \\
 \beta_{a,u,w} &= a/2 + (2u - a)\delta + i(2w - a)\gamma.
 \end{aligned}$$

The polynomial family, weight and norm are the classical Meixner–Pollaczek formulas, with exact parameter and measure transport proved in HPS1–4; see [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)].

All functions and matrices are the original NSO/CPQ objects. The proved original bound $v \leq 80$ suffices here; the stronger BTK bound is retained on its stated domain. Keep the exact source measure, mass, recurrence, and phase

$$\begin{aligned}
 dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
 p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = yp_{n,s} - n(n-1 + s/2)p_{n-1,s}, \\
 h_{n,s} &= M_s n! (s/2)_n, \quad \phi_n(S') = p_{n,s}((S' - c')/i)/\sqrt{h_{n,s}}, \quad \|P\|_s^2 = \int |P(c' + iy)|^2 dm_s(y). \tag{NIB2}
 \end{aligned}$$

The polynomials are the original orthonormal basis. Inner products are conjugate-linear in their first argument, and functional values are complex-linear.

230.1 The complete original annihilator at every subsequent row

Let $F_b = N_b/E_A$, $1 \leq b \leq m$, be the fixed original NSO12 numerator columns. The original functional frame consists of these m functionals together with all q' lower evaluations. Denote its matrix of values on $\phi_0, \dots, \phi_d$ by $\mathcal{F}_d = [V_d, T_d]$, with $L = q' + m$ columns. Explicitly, $V_d[n, (u, w)] = \phi_n(\beta_{j-8,u,w})$ comprises every lower evaluation column, and $T_d[n, b] = \Lambda_{F_b}(\phi_n)$ comprises the original analytic NSO columns, for $0 \leq n \leq d$. At $d = L - 1$ this matrix is invertible: a generating linear combination whose first L derivatives vanish has, after multiplication by E_A , an upper-grid numerator whose first $L + v = q$ derivatives vanish. The upper Vandermonde kills this numerator; the original conductor quotient frame kills the quotient coefficients; the lower Vandermonde kills the remaining coefficients. The triangular polynomial-frame change has nonzero diagonal, proving the same claim for native rows. Appending rows therefore preserves column rank L at every $d \geq L - 1$.

Define the original polynomial translation and its complete relation sequence by

$$\begin{aligned}
 (\mathcal{T}_A P)(S') &= \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \tag{NIB3} \\
 g_t &= \mathcal{T}_A(\chi_j(S)S^t), \quad \ell_t = \mu_v \binom{q+t}{v}, \quad n_t = q + t - v = L + t.
 \end{aligned}$$

Binomial expansion gives $\deg g_t = n_t$ and $\text{lc } g_t = \ell_t \neq 0$. For every original numerator column, the finite Leibniz identity gives

$$\Lambda_{F_b}(g_t) = \Lambda_{N_b}(\chi_j S^t) = 0.$$

Every lower evaluation also annihilates g_t : root addition $\beta_{j-8,a,b} + \beta_{8,u,w} = \beta_{j,a+u,b+w}$ makes each shifted summand zero separately. Thus χ_- divides every g_t , including all of its original coefficients.

At the later native row $n = L + r$, $r \geq 0$, the polynomial map of all L original functional values has rank L , by the preceding old-row argument. Its kernel in degree at most n has dimension $n + 1 - L = r + 1$. The polynomials $g_0, \dots, g_r$ are independent by their distinct degrees and lie in that kernel. Consequently

$$\ker(\Lambda_{\text{original}}|_{\mathbb{C}[S']_{\leq L+r}}) = \text{span}(g_0, \dots, g_r). \quad (\text{NIB4})$$

This identifies the complete annihilator, rather than selecting a proper subspace of relations.

Let $u_t = g_t/\ell_t$ and define its full attained monic norm by

$$\eta_{r,s} = \min_{z_0, \dots, z_{r-1}} \left\| u_r + \sum_{t < r} z_t u_t \right\|_s^2, \quad \nu_{b,s}^- = \min_{P \text{ monic, deg } P=b} \|\chi_- P\|_s^2. \quad (\text{NIB5})$$

The relation minimum NIB5–6 uses classical Schur-complement elimination [H1, §2.1, (2.3)–(2.4)], here on the full Gram of $u_0, \dots, u_r$. This credits the matrix method; the complete annihilator and the later native Gamma bounds are proved in this section.

Both are positive attained minima. For $r = 0$ the first minimum has no variables. For $r > 0$, putting $G_{<} = (\langle u_t, u_h \rangle_s)_{t,h < r}$ and $z = (\langle u_t, u_r \rangle_s)_{t < r}$, its exact value is

$$\eta_{r,s} = \|u_r\|_s^2 - z^* G_{<}^{-1} z, \quad u_r^{\min} = u_r - \sum_{t < r} u_t (G_{<}^{-1} z)_t. \quad (\text{NIB6})$$

Square completion proves this formula, preserving every preceding relation and cross pairing. Scaling by ℓ_t is the displayed invertible map to its monic affine fibre; every ℓ_t remains in NIB3,5,6. Since χ_- divides every g_t , the first fibre is contained in the second monic fibre at $b = m + r$. It follows directly that $\eta_{r,s} \geq \nu_{m+r,s}^-$.

230.2 The exact corrected native row at every interior position

At cutoff $n - 1 = L + r - 1$, retain the original V_0, T_0 , their new rows v_n, t_n , and

$$\begin{aligned} A_L &= V_0^* V_0, \quad D_0 = T_0^* (I - V_0 A_L^{-1} V_0^*) T_0, \\ \zeta_n &= v_n A_L^{-1} v_n^*, \quad \rho_n = t_n - v_n A_L^{-1} V_0^* T_0, \quad r_n = \rho_n D_0^{-1} \rho_n^* / (1 + \zeta_n), \\ \zeta_{\text{all},n} &= [v_n, t_n] ([V_0, T_0]^* [V_0, T_0])^{-1} [v_n, t_n]^*. \end{aligned} \quad (\text{NIB7})$$

All inverses exist by the full column ranks established above. Unlike the first row, the old full matrix need not be square.

The unit relation $\tau_r = u_r^{\min} / \sqrt{\eta_{r,s}}$ has last native coefficient $c_n = i^n \sqrt{h_{n,s} / \eta_{r,s}}$, and annihilates the entire original functional frame. Its old coefficient column is orthogonal to every old annihilator by NIB4,6. If $w_n = [v_n, t_n]$ and $F_0 = [V_0, T_0]$, it is consequently the minimum-norm solution of $F_0^T z = -c_n w_n^T$, namely

$$z = -c_n \overline{F_0} (F_0^T \overline{F_0})^{-1} w_n^T.$$

Its squared norm is $|c_n|^2 \zeta_{\text{all},n}$ by conjugating the positive scalar. Since the full relation has norm one, adding $|c_n|^2$ proves the first formula below. Applying the identical argument to the full lower-evaluation annihilator $\chi_- \mathbb{C}[S']_{\leq m+r}$ proves the second:

$$1 + \zeta_{\text{all},n} = \frac{\eta_{r,s}}{h_{n,s}}, \quad 1 + \zeta_n = \frac{\nu_{m+r,s}^-}{h_{n,s}}. \quad (\text{NIB8})$$

That lower annihilator equality follows from ordinary division by the complete distinct-root polynomial χ_- ; its monic degree n fibre is exactly NIB5 with $b = m + r$.

The invertible old column operation replacing T_0 by $(I - V_0 A_L^{-1} V_0^*) T_0$ makes its Gram block diagonal and its new row $[v_n, \rho_n]$. Thus $\zeta_{\text{all},n} = \zeta_n + \rho_n D_0^{-1} \rho_n^*$, also for the rectangular old matrix. Together with NIB8 this gives

$$1 + r_n = \frac{\eta_{r,s}}{\nu_{m+r,s}^-}, \quad 0 \leq \log(1 + r_n) \leq \log \frac{\|g_r\|_s^2}{|\ell_r|^2 \nu_{m+r,s}^-}. \quad (\text{NIB9})$$

The last inequality uses the actual trial u_r in its attained fibre NIB5. It changes neither g_r nor any earlier relation. The measure, lower minimum, and all cross terms in the exact identity remain the original ones.

230.3 All finite constants for the original later row

Define, for every $r \geq 0$,

$$\begin{aligned} L_s &= (s-1)/4, \quad \theta = \pi/2, \quad h = 1/(q+r), \quad \kappa = \theta - h > 0, \quad \sigma = (\delta^2 + \gamma^2)^{1/2}, \\ Q_- &= q' + L_s, \quad Q_+ = q + r + L_s, \quad R' = (j-8)\sigma, \quad M = \max\{2Q_-/\theta, 2R'\}, \\ B_+ &= \max\{1, (j+8)\sigma, 2L_s\}, \quad A_{\text{abs}} = \sum_{u,w} |a_{uw}|, \quad \beta_s = M_s/(\sqrt{2}\Gamma(s/2)), \\ L_h &= \pi(\sin h)^{3/2}/\Gamma(3/4)^2, \quad U_h = \Gamma(1/4)^2(\sin h)^{-1/2}/(2\pi), \\ F_M &= 2Q_- \log M - \theta M, \quad \epsilon_- = (\theta - 2Q_-/M) + Q_-/M^2 + h(M+1) + 4q'R'/M, \\ C_b &= (2b+1) \binom{2b}{b}^2. \end{aligned} \quad (\text{NIB10})$$

The classical inputs are Euler's integral, the Gamma functional equation, and Euler's reflection formula [H22, eqs. (5.2.1), (5.5.1), (5.5.3)]; HPS5 records the exact reflection specialization.

The Gamma recurrence and the exact Euler/reflection estimates give

$$\begin{aligned} dm_s &= \beta_s \prod_{u=0}^{L_s-1} (y^2 + (2u+1/2)^2) d\sigma(y), \quad d\sigma(y) = |\Gamma(1/4 + iy/2)|^2 dy/(2\pi), \\ L_h e^{-(\theta+h)|y|} &\leq d\sigma/dy \leq U_h e^{-\kappa|y|}. \end{aligned} \quad (\text{NIB11})$$

Here is the full scalar estimate: rotate Euler's integral for $\Gamma(b + iy/2)$ through $\text{sgn}(y)(\pi/2 - h)$. The integrable bound on that ray is $\Gamma(b)(\sin h)^{-b} e^{-(\pi/2-h)|y|/2}$. The closing arcs vanish because $b > 0$ and their cosines are at least $\sin h > 0$. Squaring this at $b = 1/4$ proves the upper bound. The exact reflection product $|\Gamma(1/4 + iy/2)\Gamma(3/4 - iy/2)|^2 = 2\pi^2/\cosh(\pi y)$, the same upper bound at $3/4$, and $\cosh(\pi y) \leq e^{\pi|y|}$ prove the lower bound. Iterating the Gamma recurrence proves the first identity, with its factor 4^{-L_s} cancelled by the displayed original density's 2^{2L_s} . No mass is suppressed.

On the interval $[M, M+1]$ every original lower root in the y coordinate has modulus at most $R' \leq M/2$. Since $\log(1-x) \geq -2x$ for $0 \leq x \leq 1/2$, $|\chi_-(c' + iy)|^2 \geq y^{2q'} e^{-4q'R'/M}$ there. The Gamma product contributes at least y^{2L_s} . For $y = M+t$, $0 \leq t \leq 1$, the elementary inequality $\log(1+t/M) \geq t/M - t^2/(2M^2)$ gives

$$2Q_- \log y - \theta y \geq F_M - (\theta - 2Q_-/M) - Q_-/M^2.$$

Together with $e^{-hy} \geq e^{-h(M+1)}$, this gives the lower bound $\beta_s L_h e^{F_M - \epsilon_-}$ for the complete weighted density on this original interval.

The classical Rodrigues and norm identities used for the next minimum are those of the Legendre family; the checked source is Koornwinder, Wong, Koekoek, Swarttouw and Reinhardt [H21,

eq. (18.5.5), Table 18.5.1, and the Legendre and shifted-Legendre rows of Table 18.3.1]. With their degree b and their argument $u = 2x - 1$, the polynomial below is exactly their $P_b(2x - 1)$. The change of variables contributes $du = 2dx$, so its norm on $[0, 1]$ is $1/(2b + 1)$ and its leading coefficient in x is $\binom{2b}{b}$. The square-completion argument and the explicit S' -coordinate map below give the complex monic minimum in the original fibre, with $b = m + r$.

For a complex monic degree- b polynomial on an interval of length one the exact minimum squared integral is C_b^{-1} . To prove the constant, the Rodrigues polynomial $P_b(x) = \frac{1}{b!}(d/dx)^b[x^b(x-1)^b]$ has leading coefficient $\binom{2b}{b}$. Integrating by parts b times proves its orthogonality to every smaller degree; the same calculation gives $\int_0^1 P_b^2 = \binom{2b}{b} \int_0^1 x^b(1-x)^b dx = 1/(2b+1)$. The last beta integral is $b!^2/(2b+1)!$ by successive integration by parts. Orthogonal square completion proves the complex monic minimum. The map $P(S') \mapsto i^{-b}P(c' + i(M+x))$ is monic in x with inverse $i^b p((S' - c')/i - M)$ and preserves absolute values on the interval. Taking $b = m + r$ therefore proves

$$\nu_{m+r,s}^- \geq \frac{\beta_s L_h}{C_{m+r}} e^{F_M - \epsilon_-}. \quad (\text{NIB12})$$

For the upper estimate keep the original roots fixed. For every real y , upper root and degree-eight shift, $|c' + iy + \beta_{8,u,w} - \beta_{j,a,b}| \leq |y| + (j+8)\sigma$. Also $|c' + iy + \beta_{8,u,w}| \leq |y| + j/2 + 8\sigma \leq |y| + (j+8)\sigma$, because the original $\gamma > 2$ gives $\sigma > 1/2$. Consequently the actual polynomial from NIB3 obeys

$$|g_r(c' + iy)| \leq A_{\text{abs}}(|y| + B_+)^{q+r}.$$

Every Gamma product factor is at most $(|y| + B_+)^2$. Integrating the two half-line bounds from NIB11, substituting $t = |y| + B_+$ and enlarging to $t \geq 0$ gives

$$\|g_r\|_s^2 \leq 2\beta_s U_h A_{\text{abs}}^2 e^{\kappa B_+} \frac{(2Q_+)!}{\kappa^{2Q_++1}}. \quad (\text{NIB13})$$

The last integral is an exact factorial integral by integration by parts. All cancellation in g_r remains exact in NIB3–9; this is an upper inequality for that polynomial's complete norm.

Combining the two estimates in the identical original norm gives the finite bound, valid for every $j \geq 13$ and $r \geq 0$,

$$0 \leq \log(1 + r_{L+r}) \leq E_{j,s,r},$$

$$E_{j,s,r} = 2 \log \frac{A_{\text{abs}}}{|\mu_v^{(q+r)}|} + \log \frac{2U_h}{L_h} + \kappa B_+ + \log((2Q_+)!) - (2Q_+ + 1) \log \kappa - F_M + \epsilon_- + \log C_{m+r}. \quad (\text{NIB14})$$

The identical β_s cancels between the two proved bounds; the source measure has not been divided by its mass. The exact ratio in NIB9 is at least one, so the displayed upper constant is necessarily nonnegative.

230.4 Uniform initial interior segment and all complementary terms

Fix any real $C > 0$. Uniformly over integers $0 \leq r \leq Cj$, both original orders $s = 1, j$, and the fixed original quartet and conductor,

$$E_{j,s,r} = O_{\text{actual},C}(j \log j). \quad (\text{NIB15})$$

Here is a direct verification. Eventually $M = 2Q_-/\theta$, since $Q_- \asymp j^2$ while $R' = O(j)$. Then $F_M = f_\theta(Q_-)$, where $f_\theta(x) = 2x \log(2x/\theta) - 2x$, and $\epsilon_- = O(j)$ uniformly. Also $Q_+ = q + r + L_s \asymp j^2$, $Q_+ - Q_- = 16j - 48 + r = O_C(j)$, and $B_+ = O(j)$. The factorial integral estimates

$$b \log b - b + 1 \leq \log(b!) \leq b \log b - b + 1 + \log b \quad (b \geq 1)$$

follow by integrating the increasing logarithm between successive integers. Applied to $b = 2Q_+$, they place $\log((2Q_+)!) - (2Q_+ + 1) \log \kappa$ at $f_\theta(Q_+) + O(\log j)$, because $\log(\theta/\kappa) \leq h/\kappa$ and $Q_+ h =$

$O(1)$. Integration of $f'_\theta(x) = 2\log(2x/\theta)$ between Q_- and Q_+ now costs $O_C(j\log j)$. Further, $\log C_{m+r} \leq \log(2m+2r+1) + 4(m+r)\log 2 = O_C(j)$, and $2h/\pi \leq \sin h \leq h$ gives $\log(U_h/L_h) = O(\log j)$. The fixed v and nonzero μ_v imply $|\log|\mu_v({}^{q+r}_v)|| = O(\log j)$ uniformly on this range. These estimates prove NIB15 directly from its fully finite expression, with no differentiation of an asymptotic determinant error.

At each of these rows the original four-copy metric update is $\mathcal{D}_1 = \mathcal{D}_0 + U_n^* U_n$, with $U_n = \text{diag}(\rho_n/\sqrt{1+\zeta_n}, 4)$. Its full relative compression is $r_n I_4$. For each original fixed flag X , retain $K_{X,n} = U_n X (X^* \mathcal{D}_0 X)^{-1} X^* U_n^*$ and all four eigenvalues. The two chains $J \subset W_i \subset W \subset T_e$ and $K_i \subset I \subset K_c \subset S$ give, exactly as in FMG27, two nonnegative disjoint increment sums each at most $4\log(1+r_n)$. Their difference is the complete original signed row. Applying the same reasoning to the increasing threshold function on each ordered eigenvalue proves

$$|\delta\mathcal{L}_{e,n}| \leq 4E_{j,s,r}, \quad |\delta\mathcal{P}_{e,n}| \leq 4E_{j,s,r}, \quad 0 \leq \delta V_{X,n} \leq 4E_{j,s,r} \quad (X = b, \text{rem}, s, A). \quad (\text{NIB16})$$

For the last assertion, the original quotients are W/W_i , U/T_e , S/K_c , and I/K_i . Block Schur elimination expresses each successive quotient determinant increment as the difference of its two restriction increments. Nested projections bound that difference between zero and the full update budget. This argument retains all original fixed-frame factors until they cancel in the successive increment, and leaves every quotient cross term in its attained metric.

The original target row for native degree $L+r$ is $N = q+r$. On $0 \leq r \leq q$, its original four-endpoint weights are one at $r=0, q$ and two otherwise. For any integer $0 \leq R \leq \min(q, \lfloor Cj \rfloor)$ let ϑ_r denote those unchanged weights. Summing NIB16 gives the explicit segment bound

$$\begin{aligned} \left| \sum_{r=0}^R \vartheta_r \delta\mathcal{L}_{e,L+r} \right|, \quad \left| \sum_{r=0}^R \vartheta_r \delta\mathcal{P}_{e,L+r} \right| &\leq 4 \sum_{r=0}^R \vartheta_r E_{j,s,r}, \\ 0 \leq \sum_{r=0}^R \vartheta_r \delta V_{X,L+r} &\leq 4 \sum_{r=0}^R \vartheta_r E_{j,s,r} = O_{\text{actual},C}(j^2 \log j) = o(kq_k). \end{aligned} \quad (\text{NIB17})$$

The same last upper order holds for both absolute values in the first line. In particular $r=1$ is an actually non-first original interior row with its complete error bound. The full initial linear segment, together with every complementary CPQ increment, can be removed from the leading kq_k calculation by the stated finite error. All subsequent rows remain in their original correlated expressions, and the complete CPQ identity $4S_K = \mathcal{L}_e + V_b + V_{\text{rem}} + V_s + V_A$ is preserved. No intrinsic coefficient for the surviving interior profile or conclusion about RH is assigned by this segment estimate.

230.5 Uniform bound on the full original row range and a superlinear prefix

The same finite NIB14 constant has the stronger uniform estimate

$$\boxed{E_{j,s,r} = O_{\text{actual}}((j+r)\log j) \quad (0 \leq r \leq q, s \in \{1, j\}).} \quad (\text{NIB18})$$

Here the original quartet, conductor, period and logarithm branch are fixed before the limit. We prove the uniformity from the full finite expression. On this range $q \leq q+r \leq 2q$, so $1/(2q) \leq h \leq 1/q$. Also $Q_- = q' + L_s \asymp j^2$ and $Q_+ = q+r + L_s \asymp j^2$, with constants valid for both allowed orders. The actual lower interval therefore has $M = 2Q_-/\theta$ eventually, independently of r . Thus $F_M = f_\theta(Q_-)$ and $\epsilon_- = \theta^2/(4Q_-) + h(M+1) + 4q'R'/M = O_{\text{actual}}(j)$ uniformly. In particular no new interval or root substitution has been made in the exact finite bound.

The factorial inequalities used in NIB15 give, uniformly here,

$$\log((2Q_+)!) - (2Q_+ + 1)\log \kappa = f_\theta(Q_+) + O(\log j).$$

Indeed the factorial error is $O(\log Q_+)$, the remaining $-\log \kappa$ is bounded, and changing κ to θ costs at most $(2Q_+ + 1)h/\kappa = O(1)$. On the entire interval $[Q_-, Q_+]$, $|f'_\theta(x)| = |2\log(2x/\theta)| = O(\log j)$.

Its length is exactly $16j - 48 + r$. Integration of this derivative therefore gives $f_\theta(Q_+) - f_\theta(Q_-) = O((j+r)\log j)$. The other terms are bounded directly: $\kappa B_+ = O_{\text{actual}}(j)$, $\log C_{m+r} \leq \log(2m+2r+1) + 4(m+r)\log 2 = O(j+r)$, and $2h/\pi \leq \sin h \leq h$ gives $\log(U_h/L_h) = O(\log j)$ uniformly for $q \leq q+r \leq 2q$. Finally, because v is fixed and $\mu_v \neq 0$, the finite product for $\binom{q+r}{v}$ has logarithm $O_{\text{actual}}(\log j)$ uniformly on that same range. Every term in NIB14 is now controlled by the claimed bound. Its nonnegativity was already proved in NIB14, completing NIB18.

For every integer $0 \leq R \leq q$ retain exactly the original weights $\vartheta_0 = \vartheta_q = 1$ and $\vartheta_r = 2$ for $0 < r < q$. Define its fully finite prefix budget

$$\mathcal{B}_{j,s}(R) = 4 \sum_{r=0}^R \vartheta_r E_{j,s,r} = O_{\text{actual}}((R+1)(j+R)\log j). \quad (\text{NIB19})$$

The estimate follows by summing NIB18 with $\vartheta_r \leq 2$ and $\sum_{r=0}^R (j+r) = (R+1)j + R(R+1)/2 \leq (R+1)(j+R)$. For every such R , the two absolute prefix sums in NIB17 and each of the four nonnegative CPQ prefix sums are bounded by this same $\mathcal{B}_{j,s}(R)$: this follows by summing the actual inequalities NIB16 with the retained positive weights. Thus NIB19 extends the range of the proved prefix estimate; it changes no original row or endpoint.

In particular take the explicit integer prefix

$$R_j = \left\lfloor \frac{j^{3/2}}{\log j} \right\rfloor, \quad \boxed{\mathcal{B}_{j,s}(R_j) = O_{\text{actual}}\left(j^{5/2} + \frac{j^3}{\log j}\right) = o(kq_k)}. \quad (\text{NIB20})$$

For $j \geq 13$, $\sqrt{j} \log j > 1$, so $0 \leq R_j < j^2 < q$. The point $r = R_j$ is consequently an interior row of the original window, and retains weight two. Also $R_j + 1 \leq j^{3/2}/\log j + 1$ and $j + R_j \leq j + j^{3/2}/\log j$. Multiplying these inequalities and then by $\log j$ gives an upper bound $j^{5/2} + j^3/\log j + j \log j + j^{3/2}$, with the last two terms absorbed in the displayed order. Since $k = j - 4$ and $q_k = (j - 3)^2$, division by kq_k tends to zero. This proves NIB20 for both original source orders and both fixed primitive indices, simultaneously in the signed loss, the signed threshold profile, and all four complementary returns. It is a prefix decomposition of the unchanged rows $0 \leq r \leq q$; the original final endpoint remains $r = q$. Every row beyond R_j remains in its full correlated tail, with no coefficient assigned to that remaining expression.

Continuation of the original prefix and its full relation tail. The row estimate NIB14 and the proved prefix NIB18–20 now receive the complete original relation calculation in Sections 233, 234 and 236. In the native-cutoff convention $n = L + t$, the literal analytic Gram is $D_n^{\text{nat}} = A^* \overline{\Omega}_t A$ and its new row is $\rho_n/\sqrt{1+\zeta_n} = -i^{-n} x_t^T A$. NTG16–22 proves these equalities in the original functional frame; SLC3–20 calculates every entry from the full original root and conductor data using Szegő’s confluent modification formula [H24, Theorem 2.5, pp. 29–30]. For the retained prefix $R = R_j$, the earlier scalar tail budget can use the minimum of all three proved bounds in NTG31, with its retained cross-pair correction, and the full weighted tail is exactly $\log \det \Omega_{q-1} + \log \det \Omega_q - 2 \log \det \Omega_R$. The proof is NTG14–15 followed by the unchanged endpoint weights, as displayed in NTG32. The earlier prefix estimate remains valid; this new formula supplies the complete surviving correlated tail, whose growing-degree value remains the next calculation.

231 Human sources for the original Gamma polynomial metric

The Gamma orthogonal polynomials used in NIB2 belong to the classical Meixner–Pollaczek family. The external formulas used here are credited to the human-authored DLMF chapter by T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt: the weight and squared norm are equations (18.19.8)–(18.19.9), and the recurrence is equation (18.22.8) [H21]. The DLMF source notes trace these formulas to M. E. H. Ismail’s *Classical and Quantum Orthogonal Polynomials in One Variable*, corrected 2009 reprint, equations (5.9.1), (5.9.8) and (5.9.9). The

directly checked formulas in this repair are the DLMF equations; that source trail does not represent a claim that Ismail's book text was independently inspected here.

Here is the exact dictionary, preserving the original measure and mass. For the original positive source order s , put $\lambda = s/4$ and retain $M_s = (2\pi)^{s/2}$. Write $P_n^{(\lambda)}(x; \varphi)$ for the cited Meixner–Pollaczek polynomial. Then the NIB2 polynomial is

$$p_{n,s}(y) = n!P_n^{(s/4)}(y/2; \pi/2). \quad (\text{HPS1})$$

Indeed the cited recurrence with $x = y/2$ and $\varphi = \pi/2$, multiplied by $n!$, gives

$$p_{n+1,s}(y) = yp_{n,s}(y) - n(n-1+s/2)p_{n-1,s}(y). \quad (\text{HPS2})$$

Its initial values are $p_{0,s} = 1$ and $p_{1,s} = y$, which determine the complete sequence uniquely by induction. The cited leading coefficient $(2 \sin \varphi)^n/n!$ becomes exactly one in HPS1.

In the cited convention the squared norm at $\varphi = \pi/2$ is

$$\int_{\mathbb{R}} |\Gamma(\lambda + ix)|^2 |P_n^{(\lambda)}(x; \pi/2)|^2 dx = \frac{2\pi\Gamma(n+2\lambda)}{2^{2\lambda}n!}. \quad (\text{HPS3})$$

The same source gives orthogonality of different degrees. Substitute $y = 2x$, so $dy = 2dx$, and multiply the polynomial by $n!$ as in HPS1. For the unchanged original measure

$$dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy$$

the full norm is therefore

$$\frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} 2(n!)^2 \frac{2\pi\Gamma(n+s/2)}{2^{s/2}n!} = M_s n! \frac{\Gamma(n+s/2)}{\Gamma(s/2)} = M_s n!(s/2)_n. \quad (\text{HPS4})$$

This proves the original $h_{n,s}$, including its mass. For complex polynomials the map $f(y) \mapsto f(2x)$ is an invertible polynomial map. The explicit measure factor above describes its norm transport; neither variable change nor polynomial rescaling deletes that factor. In the original complex coordinate the resulting basis remains $\phi_n(S') = p_{n,s}((S' - c')/i)/\sqrt{h_{n,s}}$, with unchanged centre c' and phase i^{-n} in its leading coefficient. Thus HPS1–4 apply separately to $s = 1$ and $s = j$ without replacing either order.

The Gamma inputs to NIB11 are also classical human mathematics. Euler's integral is DLMF (5.2.1); the functional equation and reflection formula are DLMF (5.5.1) and (5.5.3), in the chapter by R. A. Askey and R. Roy [H22]. Its acknowledgements explicitly credit P. J. Davis's Chapter 6 in Abramowitz and Stegun as a partial source. The substitution $z = 1/4 + iy/2$ in the reflection formula gives

$$|\Gamma(1/4 + iy/2)\Gamma(3/4 - iy/2)|^2 = \frac{\pi^2}{|\sin(\pi/4 + i\pi y/2)|^2} = \frac{2\pi^2}{\cosh(\pi y)}, \quad (\text{HPS5})$$

because $|\sin(a + ib)|^2 = \sin^2 a + \sinh^2 b$. The points used have real part $1/4$, so none is an excluded integral pole of the cited identity. Iteration of the cited functional equation gives the finite Gamma product in NIB11. Its contour rotation, subsequent finite inequalities, and application to the actual conductor remain the calculations proved in NIB11–20. They do not confer original authorship of the classical Gamma identities or the Meixner–Pollaczek family on this research programme.

232 The controlled original prefix and its complete remaining return

Keep the original fixed quartet, period, logarithm branch, complete conductor, and the two source pairs of CPQ1. Write $j = k + 4$, $q = q_j = (j + 1)^2$, $L = q - v$, and $s = s_1 \in \{1, j\}$. For every original target cutoff $q - 1 \leq N \leq 2q$, set $D_N = D_{j, N-v}^{(s)}$ and $\mathcal{D}_N = \text{diag}(D_N, 4)$. Every coefficient flag and value frame below is precisely its CPQ frame. NIB proves, for each $0 \leq r \leq q$, the full original update

$$D_{q+r} = D_{q+r-1} + u_r^* u_r, \quad u_r = \frac{\rho_{L+r}}{\sqrt{1 + \zeta_{L+r}}}, \quad a_r = u_r D_{q+r-1}^{-1} u_r^* = \frac{\eta_{r,s}}{\nu_{m+r,s}^-} - 1 \geq 0. \quad (\text{IPR1})$$

Here $\eta_{r,s}$ is the full previous-relation Schur minimum, not the norm of an arbitrarily selected relation, and $\nu_{m+r,s}^-$ is the original lower ideal minimum. The finite NIB14 constant gives $0 \leq \log(1 + a_r) \leq E_{j,s,r}$. Define the retained weights and budget

$$\vartheta_0 = \vartheta_q = 1, \quad \vartheta_r = 2 \quad (0 < r < q), \quad \mathfrak{E}_{s,R} = \sum_{r=0}^R \vartheta_r E_{j,s,r} \quad (0 \leq R \leq q). \quad (\text{IPR2})$$

All sums use the existing rows of the original window.

232.1 Exact prefix and tail maps on the original evaluation sites

For a scalar function f of the original cutoff define $\delta_r f = f(q + r) - f(q + r - 1)$ and $\mathcal{R}f = -f(q - 1) - f(q) + f(2q - 1) + f(2q)$. For $0 \leq R < q$ direct telescoping gives

$$\begin{aligned} \mathcal{R}_{\leq R} f &:= \sum_{r=0}^R \vartheta_r \delta_r f = 2f(q + R) - f(q) - f(q - 1), \\ \mathcal{R}_{> R} f &:= \sum_{r=R+1}^q \vartheta_r \delta_r f = f(2q - 1) + f(2q) - 2f(q + R), \\ \mathcal{R}f &= \mathcal{R}_{\leq R} f + \mathcal{R}_{> R} f. \end{aligned} \quad (\text{IPR3})$$

Indeed the first sum is $f(q) - f(q - 1)$ plus twice the successive increments from q to $q + R$. The second has twice the increments from $q + R$ to $2q - 1$ and the final single increment from $2q - 1$ to $2q$. This also proves the formulas at $R = 0$ and $R = q - 1$. At $R = q$ define $\mathcal{R}_{\leq q} = \mathcal{R}$ and $\mathcal{R}_{> q} = 0$; the displayed three-site formulas are not used at that last index. Each operator annihilates constants. Thus the same fixed frame factors that cancel under $\mathcal{R}$ also cancel separately under both operators. No representative depending on N is substituted for a fixed frame.

For a row sequence that is not itself a determinant increment, in particular the original threshold row p_r , the prefix and tail mean exactly $\sum_{r \leq R} \vartheta_r p_r$ and $\sum_{r > R} \vartheta_r p_r$. Formula IPR3 is applied only to an actual scalar potential f , such as a logarithmic determinant.

232.2 Both original allocations and all four complementary quotients

Let $B_j : \mathbb{C}^{m_j} \rightarrow \mathbb{C}^{t_j}$ be the original onto value map and choose fixed independent columns J_j for its kernel and a fixed right inverse X_j . The square matrix $C_j = [J_j, X_j]$ is invertible: applying B_j to $J_j x + X_j y = 0$ gives $y = 0$, and then independence gives $x = 0$. Put

$$H_N = (B_j D_N^{-1} B_j^*)^{-1}, \quad H_N^0 = J_j^* D_N J_j, \quad |\det C_j|^2 \det D_N = \det H_N \det H_N^0. \quad (\text{IPR4})$$

The block minimum in IPR4 is the classical Schur-complement method [H1, §2.1, (2.3)–(2.4)] on the unchanged frame $C_j = [J_j, X_j]$. Its exact determinant factor $|\det C_j|^2$ is retained below.

For fixed value y , every lift is $X_j y + J_j x$. Completing its positive quadratic form gives the full Schur complement of H_N^0 in $C_j^* D_N C_j$. Its inverse is the bottom-right block of $C_j^{-1} D_N^{-1} (C_j^*)^{-1}$, equal to $B_j D_N^{-1} B_j^*$ because $B_j C_j = [0, I]$. This proves both the literal value minimum and IPR4 with its exact frame factor.

Restriction and minimum over the same affine fibre preserve $D_N \succeq D_{N-1}$. Hence the two logarithmic determinant increments in IPR4 are nonnegative; their sum, by the determinant lemma applied to IPR1, is $\log(1 + a_r)$. Consequently, for $Z \leq R$ or $> R$, define the actual partial allocations

$$S_K^Z = \mathcal{R}_Z \log \det H_N, \quad S_R^Z = \mathcal{R}_Z \log \det H_N^0, \quad \mathcal{T}^Z = \mathcal{R}_Z \log \det D_N. \quad (\text{IPR5})$$

They satisfy exactly

$$\begin{aligned} S_K^Z + S_R^Z &= \mathcal{T}^Z, \quad S_K^Z, S_R^Z \geq 0, \\ 0 \leq S_K^{\leq R}, S_R^{\leq R} &\leq \mathcal{T}^{\leq R} = \sum_{r=0}^R \vartheta_r \log(1 + a_r) \leq \mathfrak{E}_{s,R}. \end{aligned} \quad (\text{IPR6})$$

Their full sums are the original HJR32 allocations, with the same value maps and native source order; no allocation fraction is introduced.

For each $e = 0, 1$ let $\mathcal{L}_e(N)$ be $\log(\omega_{\text{conn},e}(N) \omega_{\text{inv}}(N))$. For $X = b, \text{rem}, s, A$ let $Q_X(N)$ be respectively $Q_b^E, Q_{\text{rem},e}, G_{s,e}, A_{\text{inv}}$, and put $V_X^Z = \mathcal{R}_Z \log \det Q_X(N)$ and $\mathcal{L}_e^Z = \mathcal{R}_Z \mathcal{L}_e(N)$. The full fixed-frame determinant identities CPQ7–12 prove the following, separately for the prefix and tail:

$$4S_K^Z = \mathcal{L}_e^Z + V_b^Z + V_{\text{rem}}^Z + V_s^Z + V_A^Z. \quad (\text{IPR7})$$

To see the cancellation explicitly, CPQ8 gives $\log \omega_{\text{conn}} = \log \det G_t - \log \det G_s$. The original invariant quotient gives $\log \omega_{\text{inv}} = \log \det H_{\text{inv}} - \log \det A_{\text{inv}}$ plus its fixed frame constant. The full target factorization gives $4 \log \det H_N = \log \det H_{\text{inv}} + \log \det Q_b^E + \log \det G_t + \log \det Q_{\text{rem}}$ plus its fixed frame constant. Subtraction proves IPR7 after applying either operator in IPR3, since both kill those constants.

Every one of these quotients is taken in its original ambient metric and complete kernel fibre. NIB16's same-row bound and its two nested flag chains give

$$\begin{aligned} |\mathcal{L}_e^{\leq R}| &\leq 4\mathfrak{E}_{s,R}, \quad |\mathcal{P}_e^{\leq R}| \leq 4\mathfrak{E}_{s,R}, \\ 0 \leq V_X^{\leq R} &\leq 4\mathfrak{E}_{s,R} \quad (X = b, \text{rem}, s, A). \end{aligned} \quad (\text{IPR8})$$

Here $\mathcal{P}_e^{\leq R}$ is the weighted original threshold row sum, not the different quantity $\mathcal{P}_e^{\text{rest}}$ in CPQ11. IPR7 retains the correlation between all five terms; IPR6 supplies the stronger direct bound on their combined value $4S_K^{\leq R}$. The terms are not assigned independent extremizers.

The NIB all-row estimate gives $\mathfrak{E}_{s,R} = O_{\text{actual}}((R+1)(j+R) \log j)$ uniformly for $0 \leq R \leq q$. In particular the explicitly evaluated prefix length

$$R_j = \left\lfloor \frac{j^{3/2}}{\log j} \right\rfloor \quad \text{satisfies} \quad \mathfrak{E}_{s,R_j} = O_{\text{actual}}(j^{5/2} + j^3 / \log j) = o(kq_k). \quad (\text{IPR9})$$

For all sufficiently large j it lies below q . The finite budgets IPR2 apply also before that threshold, with the prefix restricted to the original $R \leq q$ domain. Since $kq_k = (j-4)(j-3)^2 \asymp j^3$, each displayed order assertion follows by division. This selects no new programme endpoint: IPR3 is an identity for every original prefix, and IPR9 evaluates one range already controlled by the proved finite inequality.

232.3 Exact transport of the two partial returns

Use the original NCT complex-linear map $\Lambda_F \mapsto \Lambda_{E_AF}$, its ordinary-transpose matrix, and the full lower quotient. For each of the eight original flag frames X and each CPQ value quotient Y , denote its native and transported logarithmic determinants by $f_X, \hat{f}_X$ or $f_Y, \hat{f}_Y$. The original finite NCT20/NPK11 bound is

$$|\hat{f}_Y(N) - f_Y(N)| \leq B_{Y,N}, \quad B_{Y,N} = B_{j,N-v,s,\text{rank } Y}^{[4]} \geq 0. \quad (\text{IPR10})$$

All frames and values on the two sides are identical; the conjugation in the inverse evaluation metric remains IGI7. For $0 \leq R < q$ linearity of IPR3 now proves the finite bounds

$$\begin{aligned} |\hat{V}_Y^{\leq R} - V_Y^{\leq R}| &\leq 2B_{Y,q+R} + B_{Y,q} + B_{Y,q-1} =: E_Y^{\leq R}, \\ |\hat{V}_Y^{> R} - V_Y^{> R}| &\leq B_{Y,2q-1} + B_{Y,2q} + 2B_{Y,q+R} =: E_Y^{> R}. \end{aligned} \quad (\text{IPR11})$$

At $R = 0$, repeated occurrences in the displayed upper bound may also be combined using their actual opposite signs; the uncombined nonnegative bound is valid. This uses only the three evaluation sites and does not sum a determinant error separately for every row. For the signed eight-flag profile sum these bounds over the eight original frames to obtain E_{flags}^Z . For S_K apply them directly to the four-copy target quotient of rank $4t_j$, yielding $|4\hat{S}_K^Z - 4S_K^Z| \leq E_{\text{target}}^Z$. The transported full factorization also gives the exact equality

$$4(\hat{S}_K^Z - S_K^Z) = \hat{\mathcal{L}}_e^Z - \mathcal{L}_e^Z + \sum_{X=b,\text{rem},s,A} (\hat{V}_X^Z - V_X^Z). \quad (\text{IPR12})$$

Both enclosures concern this same value and may be intersected. For the other allocation use the original four-copy kernel frame $\text{diag}(J_j, 4)$. Its restricted determinant is exactly $(\det H_N^0)^4$ in this direct-sum frame. A change to another fixed original kernel frame multiplies it by a cutoff-independent positive factor, annihilated by both operators. Thus this original restriction flag has return $4S_R^Z$, its transported restriction has return $4\hat{S}_R^Z$, and IPR11 at its actual rank $4 \dim \ker B_j$ gives $|4\hat{S}_R^Z - 4S_R^Z| \leq E_{\text{kernel}}^Z$. This supplies the direct same-value transport of the second allocation.

For $0 \leq R < q$, each original or transported threshold row differs from its own signed determinant row by at most $8 \log 2$, as proved by NPK13's two ordered chains. The two row sequences therefore satisfy

$$\begin{aligned} |\hat{\mathcal{P}}_e^{\leq R} - \mathcal{P}_e^{\leq R}| &\leq E_{\text{flags}}^{\leq R} + 16(2R+1) \log 2, \\ |\hat{\mathcal{P}}_e^{> R} - \mathcal{P}_e^{> R}| &\leq E_{\text{flags}}^{> R} + 16\{2(q-R)-1\} \log 2. \end{aligned} \quad (\text{IPR13})$$

Indeed the original prefix weights sum to $1 + 2R$, and the tail weights sum to $2(q-R)-1$. These bounds compare scalar sums; no approximation of individual eigenvalues is asserted. NCT's fixed-radius estimate and subsequent radius limit are uniform over the unchanged cutoff interval $q-1 \leq N \leq 2q$. All ranks here are $O(j)$, so each finite three-site error in IPR11 is $o(kq_k)$ by exactly that order of limits. The extra threshold error is $O(q)$, also $o(kq_k)$. Thus IPR8–9 transport the full controlled prefix, while IPR3 and IPR12 retain the entire uncomputed tail and all its correlated quotients.

232.4 Receiving the estimate in the existing source and action maps

Every equation above holds separately for $s = 1$ and $s = j$ and for $e = 0, 1$. For example

$$\mathcal{L}_{e,s}^{> R} = \mathcal{L}_{e,s} - \mathcal{L}_{e,s}^{\leq R}, \quad V_{X,s}^{> R} = V_{X,s} - V_{X,s}^{\leq R}, \quad S_{K,s}^{> R} = S_{K,s} - S_{K,s}^{\leq R}. \quad (\text{IPR14})$$

Consequently the exact endpoint-difference identities PRT14–20, and the original two-order intervals CPQ19 and HJR33, propagate by subtracting the corresponding prefix on each side. For the positive quotient X , for instance, its order-difference change is in $[-4\mathfrak{E}_{j,R}, 4\mathfrak{E}_{1,R}]$, because

$V_{X,j}^{\leq R}, V_{X,1}^{\leq R}$ lie respectively in $[0, 4\mathfrak{E}_{j,R}]$ and $[0, 4\mathfrak{E}_{1,R}]$. For the signed profile its absolute change is at most $4(\mathfrak{E}_{j,R} + \mathfrak{E}_{1,R})$. All these corrections are $o(kq_k)$ at IPR9; none changes a previously proved leading coefficient or supplies the unknown mixed one.

The proper source Q_{inv} still uses degree k , its own centre and its displaced cutoffs. Its exact relation to the target form A_{inv} remains NFR/EFP's logarithmic ratio $\log \det A_{\text{inv}} - \log \det Q_{\text{inv}}$. Applying IPR3 to this difference gives the difference of the two partial returns, with each form evaluated at its original prescribed arguments. No NIB bound for the target is transferred to the proper source without its own proved metric comparison. Writing V_Q for that unchanged proper-source return, the precise change on removing the target prefix is

$$(V_A^{>R} - V_Q) - (V_A - V_Q) = -V_A^{\leq R}.$$

The removed contribution is $V_A^{\leq R}$, bounded by $4\mathfrak{E}_{s,R}$; the proper-source value itself remains.

Finally, in the existing complete action $J = B + W - P$, every use of an IPR3 volume decomposition is made by literal substitution $B = B^{\leq R} + B^{>R}$ for that specified original volume contribution, retaining its original coefficient. Then exactly $J = (B^{>R} + W - P) + B^{\leq R}$. The recurrence window W and penalty P are untouched by this algebraic identity. In particular the controlled prefix cannot be used to cancel an unevaluated tail or penalty. Equations IPR3–14 give the full maps and finite errors needed for that substitution on the original mixed and CPQ terms. The remaining calculation is their correlated interior return at $q + R_j, 2q - 1, 2q$, with their full Schur denominators.

Simultaneous evaluation at the two original tail endpoints. Section 235 now evaluates every occurrence in IPR3–7 using the same base $N_0 = q + R$ and the two actual matrices M_{q-1}, M_q of size $4m$. For a fixed original space X , its exact restriction return is $F_{q-1}(X) + F_q(X)$ by STF6,15. For any one of the four CPQ quotient pairs, the attained value matrix is STF10 and its return is the difference of those same restriction returns by STF12. This proves the formulas for both allocations and all complementary returns in STF17–18 without changing IPR's weights or source orders. The original signed tail differs from the simultaneous signed spectral count by at most $2(d_{q-1} + d_q) \log 2 \leq 16m \log 2$; its original successive-row threshold version has the additional explicit error $8\{2(q - R) - 1\} \log 2$, as proved in STF20–22. NFI1–11 gives the exact phase, permutation and unitary maps from the original relation coefficients to these flag matrices. At the already proved R_j , both discrepancies and the removed prefix are $o(kq_k)$. This does not assign a value to the remaining signed count, and it leaves the independent proper source and complete action at their original receiving identities.

Human mathematical sources

- [H1] Stephen Boyd, Laurent El Ghaoui, Eric Feron and Venkataramanan Balakrishnan, *Linear Matrix Inequalities in System and Control Theory*, SIAM Studies in Applied Mathematics 15, SIAM, 1994. doi:10.1137/1.9781611970777. Author-hosted edition: <https://web.stanford.edu/~boyd/lmibook/lmibook.pdf>. The cited strict block formula is §2.1, pp. 7–8, (2.3)–(2.4).
- [H2] William W. Hager, “Updating the inverse of a matrix,” *SIAM Review* **31**(2) (1989), 221–239. doi:10.1137/1031049. <https://people.clas.ufl.edu/hager/files/update-1.pdf>. Inverse formulas (1)–(2), p. 221; block elimination (6a)–(6b), p. 222; historical discussion, pp. 221–223.
- [H3] Russell Lyons, “Determinantal probability measures,” *Publications Mathématiques de l’IHÉS* **98** (2003), 167–212. doi:10.1007/s10240-003-0016-0. Inspected author version: arXiv:math/0204325v4, 18 June 2003. Locators here refer to that version: §4, (4.1), p. 11.

- [H4] Michael L. Overton and Robert S. Womersley, “On the sum of the largest eigenvalues of a symmetric matrix,” *SIAM Journal on Matrix Analysis and Applications* **13**(1) (1992), 41–45. doi:10.1137/0613006. The publisher’s abstract explicitly credits Fan’s 1949 extremal property.
- [H5] Ky Fan, “On a theorem of Weyl concerning eigenvalues of linear transformations. I,” *Proceedings of the National Academy of Sciences of the United States of America* **35**(11) (1949), 652–655. doi:10.1073/pnas.35.11.652. Historical attribution and identity are recorded through Overton–Womersley; no original-page theorem locator is asserted here.
- [H6] Zlatko Drmač, “On principal angles between subspaces of Euclidean space,” *SIAM Journal on Matrix Analysis and Applications* **22**(1) (2000), 173–194. doi:10.1137/S0895479897320824. The principal-angle algorithm lineage is cited at the level of the inspected publisher abstract; no numbered stability theorem is invoked.
- [H7] Åke Björck and Gene H. Golub, “Numerical methods for computing angles between linear subspaces,” *Mathematics of Computation* **27**(123) (1973), 579–594. doi:10.1090/S0025-5718-1973-0348991-3. Bibliographic identity and algorithm lineage are retained; the original paper’s theorem bodies were not independently inspected in this repair.
- [H10] Ralf Meyer, “A spectral interpretation for the zeros of the Riemann zeta function,” *Göttingen Seminars, Winter Term 2004/2005*, Universitätsdrucke Göttingen, 2005, pp. 117–137. Consulted arXiv:math/0412277v3, 27 August 2013, §§2–4, Definition 3.1, equation (11), Proposition 3.2 and Theorem 3.3. <https://arxiv.org/abs/math/0412277v3>. Numbered locators refer to this version.
- [H11] Alain Connes, “Trace formula in noncommutative geometry and the zeros of the Riemann zeta function,” *Selecta Mathematica (N.S.)* **5** (1999), pp. 29–106. <https://doi.org/10.1007/s000290050042>. Preprint: <https://arxiv.org/abs/math/9811068>. Cited here as the spectral predecessor expressly identified by Meyer; the manuscript and published pagination are not conflated.
- [H12] Alexander Grothendieck, *Produits tensoriels topologiques et espaces nucléaires*, Memoirs of the American Mathematical Society, no. 16 (1955). <https://doi.org/10.1090/memo/0016>. Recursive functional-analytic source explicitly cited as Meyer’s reference [2]; not independently reread for this repair.
- [H13] Alexander Grothendieck, rédigé par Luc Illusie, “Formule de Lefschetz,” SGA 5, Exposé III, §6.8, printed pp. 127–129, especially (6.8.1)–(6.8.7). Public scan index: <https://wstein.org/sga/sga5/index.html>. The selected French original controls the citation and its proof qualification. The scan index credits Vincent Maillot for scanning.
- [H14] Luc Illusie, *Chronology and posterity of SGA 5*, pp. 1–4. <https://www.imo.universite-paris-saclay.fr/~luc.illusie/Chronology-SGA5-1.pdf>. See also the author’s *Errata to SGA 5, Exposé III*, <https://www.imo.universite-paris-saclay.fr/~luc.illusie/ErrSGA5III.pdf>. Contributor roles, including Jean-Louis Verdier’s pairing contribution, are distinguished from the exposé’s author and rédacteur roles.
- [H15] Robin Hartshorne, *Residues and Duality*, Lecture Notes in Mathematics 20, Springer, 1966, III 9, R6. Locator recovered through SGA 5 III 6.8.5, which explicitly qualifies the compatibility proof at that location; original book pages were not independently read for this repair.
- [H16] Michel Raynaud, *Anneaux locaux henséliens*, Lecture Notes in Mathematics 169, Springer, 1970, VII 1. The one-variable predecessor expressly cited after SGA 5 III 6.8.5; original book pages were not independently read for this repair.

- [H17] Pierre Deligne, “La conjecture de Weil. I,” *Publications mathématiques de l’IHÉS* **43** (1974), pp. 273–307, Theorem (1.6), p. 276, and §1.5. <https://doi.org/10.1007/BF02684373>. https://www.numdam.org/item/PMIHES_1974__43__273_0/.
- [H18] Pierre Deligne, “La conjecture de Weil. II,” *Publications mathématiques de l’IHÉS* **52** (1980), pp. 137–252. Theorem 3.3.1, p. 204; Corollaries 3.3.4–3.3.5, p. 206. <https://doi.org/10.1007/BF02684780>. https://www.numdam.org/item/PMIHES_1980__52__137_0/. The French original controls statement and convention disputes; no automatic transfer to the analytic quotient is asserted.
- [H19] *A Note on the Globalization Semiring $G(\mathbb{Z})$* , Zenodo, 12 March 2026. Deposited creators: ChatGPT, 5.4 and Gemini, 2.5 DeepThink. Originating programme directed by KokunoYumeto with AI assistance, as expressly identified by the programme author. <https://doi.org/10.5281/zenodo.18976982>. <https://zenodo.org/records/18976982>. This attribution does not claim global novelty.
- [H20] The Clankers (deposited creator), *A Secondary Note on Split-Zero Globalization: Semimodule Classification, Multiplicative Reconstruction, and All-Orders Zeta Deformation*, Zenodo, record publication date 20 March 2026. Originating programme directed by KokunoYumeto with AI assistance. <https://doi.org/10.5281/zenodo.19120905>. <https://zenodo.org/records/19120905>. The deposited PDF prints 29 April 2026; record and printed dates are retained separately. No global priority claim is made.
- [H21] T. H. Koornwinder, R. Wong, R. Koekoek, R. F. Swarttouw and W. P. Reinhardt, “Orthogonal Polynomials,” Chapter 18 of the *NIST Digital Library of Mathematical Functions*, version 1.2.8 (15 September 2026). Christoffel–Darboux and its confluent form: equations (18.2.12)–(18.2.13), <https://dlmf.nist.gov/18.2.E12>, <https://dlmf.nist.gov/18.2.E13>. Classical norms and leading coefficients: Table 18.3.1, <https://dlmf.nist.gov/18.3.T1>. Rodrigues: equation (18.5.5) and Table 18.5.1, <https://dlmf.nist.gov/18.5.E5>, <https://dlmf.nist.gov/18.5.T1>. Laguerre series and zero value: equations (18.5.12) and (18.6.1), <https://dlmf.nist.gov/18.5.E12>, <https://dlmf.nist.gov/18.6.E1>. Classical recurrence: equation (18.9.1) and Table 18.9.1, <https://dlmf.nist.gov/18.9.E1>, <https://dlmf.nist.gov/18.9.T1>. Meixner–Pollaczek weight, norm and recurrence: equations (18.19.8)–(18.19.9) and (18.22.8), <https://dlmf.nist.gov/18.19>, <https://dlmf.nist.gov/18.22.E8>. Chapter authors: <https://dlmf.nist.gov/18>. Accessed 17 September 2026.
- [H22] R. A. Askey and R. Roy, “Gamma Function,” Chapter 5 of the *NIST Digital Library of Mathematical Functions*, version 1.2.8 (15 September 2026). Defining integral, recurrence and reflection: equations (5.2.1), (5.5.1) and (5.5.3), <https://dlmf.nist.gov/5.2.E1>, <https://dlmf.nist.gov/5.5.E1>, <https://dlmf.nist.gov/5.5.E3>. Binet’s Laplace integral: equation (5.9.10_2), https://dlmf.nist.gov/5.9.E10_2. Stirling expansion and finite positive-real remainder: equation (5.11.1) and §5.11(ii), first paragraph, <https://dlmf.nist.gov/5.11.E1>, <https://dlmf.nist.gov/5.11.ii>. Chapter authors and acknowledgement of P. J. Davis’s contribution: <https://dlmf.nist.gov/5>. Accessed 17 September 2026.

233 The original native tail as an attained relation Gram matrix

233.1 Original data and the classical tools being used

Fix the original simple quartet $0 < \delta < 1/2$, $\gamma > 2$, its nonzero amplitude, its original period, its logarithm branch, and its complete nonreduced conductor. The period is fixed before any degree limit. Retain

$$\begin{aligned}
 k &= 4l + 1 \geq 9, \quad j = k + 4, \quad q = (j + 1)^2, \quad q' = (j - 7)^2, \quad c' = j/2 - 4, \\
 \beta_{a,u,w} &= a/2 + (2u - a)\delta + i(2w - a)\gamma, \quad \chi_a(S) = \prod_{u,w=0}^a (S - \beta_{a,u,w}), \quad \chi_- = \chi_{j-8}, \\
 E_A(z) &= \sum_{u,w=0}^8 a_{uw} e^{\beta_{8,u,w} z}, \quad \mu_d = E_A^{(d)}(0) = \sum_{u,w=0}^8 a_{uw} \beta_{8,u,w}^d, \quad v = \text{ord}_0 E_A, \quad \mu_v \neq 0, \\
 L &= q - v, \quad m = L - q' = 16j - 48 - v > 0, \quad s \in \{1, j\}.
 \end{aligned} \tag{NTG1}$$

The original bounds on v and the original admissible period domain remain in force; in particular no exceptional finite-period fibre is replaced by $v = 0$. Every μ_d is a derivative, not a Taylor coefficient. The accepted original relation calculation NIB1–9 supplies

$$\begin{aligned}
 (\mathcal{T}_A P)(S') &= \sum_{u,w=0}^8 a_{uw} P(S' + \beta_{8,u,w}), \quad g_t = \mathcal{T}_A(\chi_j S^t), \\
 \ell_t &= \mu_v \binom{q+t}{v}, \quad u_t = g_t / \ell_t, \quad \deg u_t = L + t, \quad \text{lc } u_t = 1, \quad \chi_- \mid u_t.
 \end{aligned} \tag{NTG2}$$

It also proves that $u_0, \dots, u_t$ span the complete original functional annihilator at degree $L + t$. We keep all original analytic columns $F_b = N_b / E_A$ of NSO12, $1 \leq b \leq m$, with the original complex-linear functional convention $\Lambda_F(S'^d) = F^{(d)}(0)$.

The scalar measure and polynomial recurrence are

$$\begin{aligned}
 dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \\
 p_{0,s} &= 1, \quad p_{1,s} = y, \quad p_{n+1,s} = y p_{n,s} - n(n-1 + s/2) p_{n-1,s}, \\
 h_{n,s} &= M_s n! (s/2)_n, \quad \phi_n(S') = p_{n,s}((S' - c')/i) / \sqrt{h_{n,s}}, \\
 \langle P, Q \rangle_s &= \int \overline{P(c' + iy)} Q(c' + iy) dm_s(y).
 \end{aligned} \tag{NTG3}$$

Inner products are conjugate-linear in the first argument. These Gamma-weight orthogonal polynomials belong to the classical Meixner–Pollaczek theory, not to a new family introduced here. One retained historical source record is Josef Meixner, *Orthogonale Polynomsysteme mit einer besonderen Gestalt der erzeugenden Funktion*, J. London Math. Soc. 9 (1934), 6–13, doi:10.1112/jlms/sl-9.1.6. The precise recurrence, phase and mass used in this calculation are the displayed original NTG3; its orthonormality is retained from NIB2 and the cumulative source’s full Gamma calculation. The directly checked modern formula reference is NIST DLMF, 18.19.8, 18.19.9, and 18.22.8 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)]. The exact parameter map is $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$. Substitution in the displayed DLMF recurrence gives NTG3. Its norm under the original density is

$$(n!)^2 \frac{M_s 2^{s/2}}{2\pi\Gamma(s/2)} \frac{2\pi\Gamma(n + s/2)}{2^{s/2} n!} = M_s n! (s/2)_n.$$

The first factor of 2 in the density conversion is the Jacobian $dy = 2 dx$, so this dictionary also checks the original mass. DLMF is used as a verified formula reference; the Meixner record preserves a historical attribution trail whose exact relation to these displayed formulas has not been verified here from the complete article. It is not used as a checked theorem provider.

The orthogonal projections and attained least-squares minima below use the classical Gram–Schmidt construction. The retained historical record includes Erhard Schmidt, *Zur Theorie der linearen und nichtlinearen Integralgleichungen. I*, Math. Ann. 63 (1907), 433–476, doi:10.1007/BF01449770, with the earlier J. P. Gram attribution retained for the historical audit to trace in the original pages. The block determinant eliminations are Schur-complement calculations; their retained historical trail includes Issai Schur, *Über Potenzreihen, die im Innern des Einheitskreises beschränkt sind*, J. Reine Angew. Math. 147 (1917), 205–232, doi:10.1515/crll.1917.147.205.

Every finite projection, determinant identity and inequality needed below is proved explicitly. The contribution here is their application to NTG1–3 with the full original conductor and functional frames, including the precise native phase. For the actual minimum and determinant identities we also cite the directly read derivations in Boyd and Vandenberghe, Appendix A.5.5, p. 650, and Appendix C.4.1, p. 673 [H23]; the formulas below provide the complete complex Hermitian specialization used in this programme. The publisher records for Schmidt and Schur were checked for their bibliographic metadata. Their complete historical proofs were not read as external theorem providers in this calculation; the finite proofs below are self-contained.

233.2 Finite evaluation of every coefficient and every pairing

Write $\chi_j(S) = \sum_{a=0}^q c_a S^a$, with $c_q = 1$ and all coefficients obtained from the original root product. Direct binomial expansion gives, for $0 \leq h \leq q + t$,

$$[S'^h]_{g_t} = \sum_{\substack{0 \leq a \leq q \\ a+t \geq h}} c_a \binom{a+t}{h} \mu_{a+t-h}. \quad (\text{NTG4})$$

Thus every coefficient above degree $L + t$ vanishes because $\mu_d = 0$ for $d < v$, and the coefficient at degree $L + t$ is exactly ℓ_t . This verifies the derivative factorials in the original translation map. It does not replace that map by a derivative of order v .

For each integer $d \geq 0$ let $J_s^{(d)}$ be the matrix on indices $0, \dots, d$ with zero diagonal and off-diagonal entries $J_s^{(d)}[n, n+1] = J_s^{(d)}[n+1, n] = \sqrt{(n+1)(n+s/2)}$. Use the convention that the zeroth power is the identity. The exact moments are

$$M_{d,s} := \int y^d dm_s(y) = M_s((J_s^{(d)})^d)_{00}. \quad (\text{NTG5})$$

Indeed multiplication by y in the orthonormal basis of NTG3 has exactly those adjacent entries. A path of d multiplication steps starting at index zero never exceeds index d . Taking its inner product with the constant unit polynomial $1/\sqrt{M_s}$ proves NTG5. In particular it preserves the entire original mass.

Put $\mathfrak{H}_s[h, b] = \langle S'^h, S'^b \rangle_s$. Expansion on the original vertical line gives the completely finite formula

$$\mathfrak{H}_s[h, b] = \sum_{\alpha=0}^h \sum_{\beta=0}^b \binom{h}{\alpha} \binom{b}{\beta} (c')^{h+b-\alpha-\beta} (-i)^\alpha i^\beta M_{\alpha+\beta,s}. \quad (\text{NTG6})$$

Consequently, for arbitrary polynomials P, Q with coefficient columns p, q , their inner product is $p^* \mathfrak{H}_s q$ at their finite degree cutoff. This formula evaluates every cross pairing below without quadrature, unlisted constants, or an asymptotic replacement of the Gamma density.

Set

$$B_s[a, b] = \langle \chi - S'^a, \chi - S'^b \rangle_s. \quad (\text{NTG7})$$

NTG6 and the full lower root coefficients calculate its entries. Every finite leading principal block is strictly positive: a nonzero polynomial has positive squared integral against the strictly

positive Gamma density outside its finite zero set. For $b \geq 1$, let $B_{<b} = (B_s[a, d])_{a, d < b}$ and $z_b = (B_s[a, b])_{a < b}$. Define

$$\begin{aligned} \pi_b^-(S') &= S'^b - \sum_{a < b} S'^a (B_{<b}^{-1} z_b)_a, & \nu_{b,s}^- &= B_s[b, b] - z_b^* B_{<b}^{-1} z_b, \\ \pi_0^- &= 1, & \nu_{0,s}^- &= B_s[0, 0], & e_b(S') &= \chi_-(S') \pi_b^-(S') / \sqrt{\nu_{b,s}^-}. \end{aligned} \quad (\text{NTG8})$$

Completing the coefficient square proves that π_b^- is the unique monic minimizer of degree b in the lower weighted norm and $\nu_{b,s}^-$ is its strictly positive squared norm. Its residual is orthogonal to every preceding monomial. It follows by successive degrees that the e_b are orthonormal, and $e_0, \dots, e_d$ span the complete ideal $\chi_- \mathbb{C}[S']_{\leq d}$. This is the actual Gram–Schmidt operation just cited, with its entire finite coefficient map given in NTG7–8.

For later use the analytic entries are also a finite calculation from the same nonreduced conductor. Write $n_{d,b} = N_b^{(d)}(0) = \sum_{\lambda} Z_{\lambda b} \beta_{j,\lambda}^d$ with the original numerator frame Z , and $f_{d,b} = F_b^{(d)}(0)$. The original vanishing conditions $n_{d,b} = 0$ for $d < v$ make N_b/E_A analytic at zero. Leibniz's formula gives the recursion

$$f_{d,b} = \frac{n_{d+v,b} - \sum_{h=v+1}^{d+v} \binom{d+v}{h} \mu_h f_{d+v-h,b}}{\binom{d+v}{v} \mu_v} \quad (d \geq 0). \quad (\text{NTG9})$$

Each index in the sum is smaller than d , so the recursion starts with $f_{0,b} = n_{v,b}/\mu_v$ and determines every required derivative in finitely many operations. If $e_a = \sum_h e_{ha} S'^h$, then $\Lambda_{F_b}(e_a) = \sum_h e_{ha} f_{h,b}$, with no conjugation of this complex-linear functional value.

233.3 The actual relation graph and its full attained Gram

For $Q \geq 0$ form the coefficient matrix of $u_0, \dots, u_Q$ in the original orthonormal lower ideal basis:

$$\begin{aligned} \mathcal{R}_Q[b, t] &= \langle e_b, u_t \rangle_s, \quad 0 \leq b \leq m + Q, \quad 0 \leq t \leq Q, \\ \mathcal{R}_Q &= \begin{bmatrix} \mathcal{L}_Q \\ H_Q \end{bmatrix}, \quad \mathcal{L}_Q[b, t] = \mathcal{R}_Q[b, t] \quad (b < m), \quad H_Q[a, t] = \mathcal{R}_Q[m + a, t]. \end{aligned} \quad (\text{NTG10})$$

NTG4,6,8 evaluate every entry. If $a > t$, e_{m+a} is orthogonal to u_t by degree. For $a = t$, the difference $u_t - \chi_- \pi_{m+t}^-$ has smaller degree in the same lower ideal, so

$$H_Q[a, t] = 0 \quad (a > t), \quad H_Q[t, t] = \sqrt{\nu_{m+t,s}^-} > 0. \quad (\text{NTG11})$$

In particular H_Q is invertible and is upper triangular in the unchanged degree order. Define

$$X_Q = \mathcal{L}_Q H_Q^{-1} = [x_0, \dots, x_Q], \quad \Omega_Q = I_m + X_Q X_Q^*, \quad X_{-1} \text{ empty}, \quad \Omega_{-1} = I_m. \quad (\text{NTG12})$$

The first $Q+1$ columns of X_{Q+1} equal X_Q . To prove this, partition H_{Q+1} as $\begin{bmatrix} H_Q & h \\ 0 & d \end{bmatrix}$; its first $Q+1$ inverse columns are $\begin{bmatrix} H_Q^{-1} \\ 0 \end{bmatrix}$. Multiplication by $[\mathcal{L}_Q, \ell]$ proves the assertion. Thus the columns x_t are actual compatible original-data columns and are not selected anew at each cutoff.

In the coefficient frame $e_0, \dots, e_{m+Q}$ the complete functional relation space has the exact graph frame

$$\mathcal{R}_Q H_Q^{-1} = \begin{bmatrix} X_Q \\ I_{Q+1} \end{bmatrix}. \quad (\text{NTG13})$$

Both sides are injective maps from $\mathbb{C}^{Q+1}$ into the same original ideal. If c is the original u_t coefficient column and z is its high-degree coordinate column, the two mutually inverse coordinate

maps are $z = H_Q c$ and $c = H_Q^{-1} z$. The inverse of the graph map itself is simply high-degree coordinate projection. Thus this calculation preserves every original relation.

Let $G_Q = (\langle u_a, u_b \rangle_s)_{0 \leq a, b \leq Q}$. Orthonormality and NTG13 give

$$G_Q = H_Q^*(I_{Q+1} + X_Q^* X_Q) H_Q, \quad (NTG14)$$

$$\frac{\det G_Q}{\prod_{t=0}^Q \nu_{m+t,s}^-} = \det(I_{Q+1} + X_Q^* X_Q) = \det \Omega_Q.$$

For completeness the last determinant equality follows by eliminating either diagonal identity block in $\begin{bmatrix} I_m & X_Q \\ -X_Q^* & I_{Q+1} \end{bmatrix}$. These two eliminations use block unitriangular factors of determinant one. This proves the identity without an assumption on singular values or ranks. The scalar denominator in NTG14 is exactly $|\det H_Q|^2$ by NTG11; none of the lower-ideal minima has been replaced by a trial polynomial.

Successively completing the squares of G_Q identifies its last Schur complement with the original $\eta_{Q,s}$ of NIB5. Induction gives $\det G_Q = \prod_{t=0}^Q \eta_{t,s}$, including $Q = 0$. NIB9 and NTG14 therefore imply

$$\log \frac{\det \Omega_Q}{\det \Omega_R} = \sum_{t=R+1}^Q \log \frac{\eta_{t,s}}{\nu_{m+t,s}^-} = \sum_{t=R+1}^Q \log(1 + r_{L+t}) \quad (-1 \leq R < Q). \quad (NTG15)$$

This is a relation-Gram identity on the actual source, with dimension $m = O(j)$ on its left. No asymptotic bound for this left side is inferred merely from that dimension.

233.4 The literal analytic quotient metric and the native phase

Define the square matrix and its succeeding rows by

$$A[a, b] = \Lambda_{F_b}(e_a) \quad (0 \leq a < m, \ 1 \leq b \leq m), \quad F_{hi}[t, b] = \Lambda_{F_b}(e_{m+t}). \quad (NTG16)$$

All entries are evaluated by NTG8–9. The matrix A is invertible. Indeed an ideal polynomial of degree at most $q' + m - 1 = L - 1$ annihilated by all F_b also vanishes at all lower nodes. The full original functional map at degree $L - 1$ is invertible by NIB4's Vandermonde argument. Thus that polynomial is zero. In the m -dimensional basis $e_0, \dots, e_{m-1}$ its functional map is $A^\top$, which is consequently injective and hence invertible.

Applying each complex-linear functional to NTG13 gives, with ordinary transpose,

$$X_Q^\top A + F_{hi} = 0. \quad (NTG17)$$

Here F_{hi} contains rows $0, \dots, Q$. All signs, conjugations and the complete lower ideal are fixed by this actual polynomial equation.

Let D_n be the original native quotient Gram after observing rows $0, \dots, n$:

$$D_n = T_n^*(I - V_n(V_n^* V_n)^{-1} V_n^*) T_n, \quad n \geq L - 1. \quad (NTG18)$$

The columns of V_n are all original lower evaluations and those of T_n are all original analytic NSO functionals evaluated on $\phi_0, \dots, \phi_n$. Their full ranks follow from the same original Vandermonde argument. Let C_n be the matrix of coefficients of $e_0, \dots, e_{n-q'}$ in that orthonormal native basis. Then $C_n^* C_n = I$ and $V_n^\top C_n = 0$. Polynomial division by χ_- and the dimension count show that these are all vectors in $\ker V_n^\top$. Thus

$$I - V_n(V_n^* V_n)^{-1} V_n^* = \overline{C_n} C_n^\top. \quad (NTG19)$$

Both sides are the orthogonal projector onto $\ker V_n^* = \overline{\ker V_n^\top}$; the identity also follows directly by applying them to their image and its orthogonal complement. Since $C_n^\top T_n$ is the matrix of actual functional values on the e_b , substitution of NTG17 into NTG18 proves the full matrix equality

$$D_{L+Q} = A^*(I_m + \overline{X_Q} X_Q^\top) A = A^* \overline{\Omega_Q} A, \quad D_{L-1} = A^* A. \quad (NTG20)$$

This is equality in the unchanged analytic coefficient frame, not only a determinant equality. It proves the exact map to every original restriction and quotient: in four copies use $\text{diag}(A, 4)$ on the original coefficient frame, retain $\text{diag}(\overline{\Omega_Q}, 4)$ as its metric, and perform the original restriction or Schur minimum on the original affine coefficient fibre. The conjugation in NTG20 is forced by the complex-linear functional pairing in NTG17 and NTG19.

There is also an exact row phase. Put $n = L + t$ and write v_n, t_n for the appended lower and analytic rows. At cutoff $n - 1$, retain

$$\zeta_n = v_n(V_{n-1}^* V_{n-1})^{-1} v_n^*, \quad \rho_n = t_n - v_n(V_{n-1}^* V_{n-1})^{-1} V_{n-1}^* T_{n-1}. \quad (\text{NTG21})$$

The last native coefficient of e_{m+t} is $c_n = i^n \sqrt{h_{n,s}/\nu_{m+t,s}^-} = i^n / \sqrt{1 + \zeta_n}$ by its leading coefficient and NIB8. Its old coefficient column is the minimum-norm solution of the lower evaluation equation, because it is orthogonal to the old lower ideal. This solution is $-c_n \overline{V_{n-1}} (V_{n-1}^T \overline{V_{n-1}})^{-1} v_n^T$. Multiplying its ordinary transpose by T_{n-1} and adding $c_n t_n$ gives $\Lambda_F(e_{m+t}) = c_n \rho_n$. Equation NTG17 therefore proves

$$\boxed{\frac{\rho_n}{\sqrt{1 + \zeta_n}} = -i^{-n} x_t^T A, \quad r_n = x_t^* \Omega_{t-1}^{-1} x_t.} \quad (\text{NTG22})$$

For the last equality substitute NTG20 and the row equality into $r_n = \rho_n D_{n-1}^{-1} \rho_n^* / (1 + \zeta_n)$. It yields $x_t^T \overline{\Omega_{t-1}^{-1}} x_t$, the conjugate of the real nonnegative number $x_t^* \Omega_{t-1}^{-1} x_t$. This supplies the original phase-sensitive row to each of the eight fixed flag compressions; zeros and all eigenvalues stay present.

233.5 Joint minimization of the complete later relations

Fix the accepted prefix endpoint $R = R_j = \lfloor j^{3/2} / \log j \rfloor < q$ and any $R < Q \leq q$. Write $X_Q = [X_R, Y]$, with $Y = [x_{R+1}, \dots, x_Q]$, and partition the upper triangular matrix of NTG11 as

$$H_Q = \begin{bmatrix} H_R & H_{12} \\ 0 & H_{22} \end{bmatrix}. \quad (\text{NTG23})$$

Let $S_{R,Q}$ be the full attained Gram on the later relation coefficients after minimizing over every earlier coefficient:

$$z^* S_{R,Q} z = \min_{a \in \mathbb{C}^{R+1}} \left\| \sum_{t=0}^R a_t u_t + \sum_{t=R+1}^Q z_{t-R-1} u_t \right\|_s^2. \quad (\text{NTG24})$$

The minimum is attained uniquely since the preceding relation Gram is positive. In the graph coordinates NTG13, the preceding columns are $[X_R; I_{R+1}; 0]$ and the new columns are $[Y; 0; I_{Q-R}]$. Their two diagonal Grams are $I + X_R^* X_R$ and $I + Y^* Y$, and their cross Gram is $X_R^* Y$. Completing the entire preceding square gives

$$\begin{aligned} S_{R,Q} &= H_{22}^* (I_{Q-R} + Y^* \Omega_R^{-1} Y) H_{22}, \\ \frac{\det S_{R,Q}}{\prod_{t=R+1}^Q \nu_{m+t,s}^-} &= \det(I_{Q-R} + Y^* \Omega_R^{-1} Y) = \frac{\det \Omega_Q}{\det \Omega_R}. \end{aligned} \quad (\text{NTG25})$$

The H_{12} component is in the preceding relation span and is minimized over, so it disappears only through the displayed full minimum. To verify its inverse factor, direct multiplication proves $I - X_R(I + X_R^* X_R)^{-1} X_R^* = (I + X_R X_R^*)^{-1}$. Conjugating by Y proves the first line. The determinant identities follow by the same two identity-block eliminations used in NTG14. This calculation jointly minimizes every later linear combination and preserves every off-diagonal pairing between those combinations.

233.6 Finite analytic improvement from the retained cross pairings

All the following quantities have already been evaluated by the finite original-data operations NTG4–12:

$$\begin{aligned} a_t &= x_t^* \Omega_R^{-1} x_t, \quad c_{th} = x_t^* \Omega_R^{-1} x_h, \quad R < t, h \leq Q, \\ d &= Q - R, \quad d_0 = \min(m, d), \quad T_{R,Q} = \sum_{t=R+1}^Q a_t. \end{aligned} \quad (\text{NTG26})$$

The exact unweighted tail value $Z_{R,Q} := \log \det \Omega_Q - \log \det \Omega_R$ satisfies

$$0 \leq Z_{R,Q} \leq d_0 \log \left(1 + \frac{T_{R,Q}}{d_0} \right). \quad (\text{NTG27})$$

Indeed the nonzero eigenvalues of $\Omega_R^{-1/2} Y Y^* \Omega_R^{-1/2}$ are nonnegative, number at most d_0 , and sum to $T_{R,Q}$. The determinant equality NTG25 identifies $Z_{R,Q}$ with the sum of $\log(1 + \lambda)$ over those eigenvalues. Append zeros to reach d_0 terms. The elementary concavity inequality for $\log$, proved by its negative second derivative and finite weighted averaging, gives NTG27. This bound charges the actual dimension m rather than the number of remaining rows when $Q - R > m$. It does not bound $T_{R,Q}$ by an unproved small number.

There is an explicit second improvement that measures the retained interaction of distinct later relations. Partition the actual consecutive indices $R + 1, \dots, Q$ into adjacent pairs and at most one final singleton; denote the pairs by $\mathcal{P}$. Define

$$\Delta_{R,Q} := - \sum_{(t,h) \in \mathcal{P}} \log \left(1 - \frac{|c_{th}|^2}{(1 + a_t)(1 + a_h)} \right) \geq 0. \quad (\text{NTG28})$$

Every term is finite. Minimizing $\|x_h - z x_t\|_{\Omega_R^{-1}}^2$ over complex z gives the Cauchy–Schwarz inequality in this exact positive metric (if $x_t = 0$ the mixed pairing is already zero). Thus $|c_{th}|^2 \leq a_t a_h < (1 + a_t)(1 + a_h)$. It follows that

$$Z_{R,Q} \leq \sum_{t=R+1}^Q \log(1 + a_t) - \Delta_{R,Q}. \quad (\text{NTG29})$$

Here is the determinant proof, including the full interaction discarded by a scalar bound. The positive matrix $M = I + Y^* \Omega_R^{-1} Y$ has diagonal entries $1 + a_t$ and off-diagonal entries c_{th} . For a two-block partition, elimination gives $\det M = \det M_{11} \det(M_{22} - M_{21} M_{11}^{-1} M_{12}) \leq \det M_{11} \det M_{22}$. The inequality follows since the second positive factor is bounded above by M_{22} ; conjugating by $M_{22}^{-1/2}$ gives eigenvalues at most one and therefore a determinant at most one. Applying this successively to the prescribed pairs bounds $\det M$ by the product of the actual pair determinants and the possible singleton entry. Each pair determinant is exactly $(1 + a_t)(1 + a_h) - |c_{th}|^2$. Taking logarithms proves NTG29.

This is the familiar determinant inequality associated with Hadamard and its block refinements, applied to an explicitly computed original Gram. The retained historical source record is Jacques Hadamard, *Résolution d’une question relative aux déterminants*, Bull. Sci. Math. (2) 17 (1893), 240–246; a scan of the original article is preserved at the primary-article archive. This archive’s bibliographic metadata was consulted here; its full original proof was not used as a verified theorem. The proof above supplies the precise positive-block version used here rather than importing an unstated variant.

For the first two remaining indices $a = R + 1$, $b = R + 2$, the entire joint calculation is especially explicit:

$$\begin{aligned} \log(1 + r_{L+a}) &= \log(1 + a_a), \\ r_{L+b} &= a_b - \frac{|c_{ab}|^2}{1 + a_a}, \\ Z_{R,R+2} &= \log((1 + a_a)(1 + a_b) - |c_{ab}|^2). \end{aligned} \quad (\text{NTG30})$$

The inverse identity $(\Omega_R + x_a x_a^*)^{-1} = \Omega_R^{-1} - \Omega_R^{-1} x_a x_a^* \Omega_R^{-1} / (1 + a_a)$ is verified by multiplication; substituting it in NTG22 proves the second line and the other lines follow. Thus the calculated correction for the second later row is the exact nonnegative quantity $|c_{ab}|^2 / (1 + a_a)$. It vanishes precisely when the displayed original cross pairing vanishes. No strict positivity is asserted without evaluating that original pairing, and none is needed for NTG25–30.

The older scalar bound NIB14 remains valid for every row. Combining it with the two joint bounds gives the fully finite original-data estimate

$$0 \leq Z_{R,Q} \leq \min \left\{ \sum_{t=R+1}^Q E_{j,s,t}, d_0 \log \left(1 + \frac{T_{R,Q}}{d_0} \right), \sum_{t=R+1}^Q \log(1 + a_t) - \Delta_{R,Q} \right\}, \quad (\text{NTG31})$$

Every entry in the new two bounds is a finite expression in the actual roots, all a_{uw} , the exact μ_d , the complete Gamma moments and the attained lower minima. The sum of the old scalar estimates is retained as a separate bound, so these new finite bounds cannot degrade the previously proved one. The exact formula NTG25 is stronger still when its finite determinant is used.

233.7 The surviving original tail and its precise receiving scope

The original tail has weights two at $R+1, \dots, q-1$ and weight one at q . Its scalar value is therefore

$$\begin{aligned} \mathcal{T}_{j,s}(R) &:= 2Z_{R,q-1} + Z_{q-1,q} = Z_{R,q-1} + Z_{R,q}, \\ &= \log \frac{\det \Omega_{q-1} \det \Omega_q}{(\det \Omega_R)^2} \geq 0. \end{aligned} \quad (\text{NTG32})$$

Here $Z_{R,R} = 0$ when needed. For the accepted $j \geq 13$ both $q-1$ and q are larger than R_j . Applying NTG31 separately at $Q = q-1, q$ gives an explicit joint finite upper estimate for this entire unchanged tail. By NIB16 the absolute signed native loss tail, the absolute signed threshold tail, and each of the four nonnegative CPQ complementary tails are bounded by $4\mathcal{T}_{j,s}(R)$. Their exact values retain their original eight fixed flags and their respective quotient minima, all supplied with the same matrix NTG20 and row NTG22. The signs remain positive at T_e, K_c, W_i, K_i and negative at W, S, J, I ; the two nested flags and the full CPQ identity remain unchanged.

NIB20 already controls the prefix by $o(kq_k)$. The unresolved asymptotic size here is the calculated matrix expression NTG32, equivalently the correlated original-data columns x_t beyond R_j . A small rank or the original scalar Christoffel bounds alone do not assign its leading coefficient: its exact map to those norms is NTG10–14, retaining the full block $\mathcal{L}_Q$ and the complete triangular cross block H_Q . Equations NTG24–31 calculate the simultaneous minimization and its finite improvement; they do not assume a missing asymptotic estimate. Both orders $s = 1, j$, the fixed period, every original root and conductor coefficient, and all masses and phases occur throughout.

233.8 Exact accepted source locators

The original input definitions and prior proofs are NIB1–20 in `ORIGINAL_NATIVE_INTERIOR_BOUND.tex`, SHA256 `bcc07cec41f4c8e94362947b420d06a8973967a5cfc13abb42c9413657c7d7de`, and IGI1–30 in `ORIGINAL_GRID_INTERIOR_INTERFACE.tex`, SHA256 `4311bbc13a0f16f2585cd05755e61cf8bf4e72853af903d2172f55253072b36b`. These internal pins identify the exact original objects used; they do not replace the point-of-use human provenance above. In particular NIB4 supplies completeness of the original functional annihilator, NIB8–9 supplies the exact native row ratio, NIB16 supplies the unchanged eight-flag and complementary bounds, and NIB18–20 supplies the accepted superlinear prefix. IGI7,12 records the same complex-linear dual convention that produces the conjugation in NTG20.

234 Szegő's confluent modification formula on the full original lower grid

234.1 The primary source and its exact original-data dictionary

The human source used here is Gábor Szegő, *Orthogonal Polynomials*, fourth edition, AMS Colloquium Publications 23 (1975), Theorem 2.5 and equations (2.5.1)–(2.5.4), printed pages 29–30 [H24, Theorem 2.5, pp. 29–30]. The publisher's book-level identifier is doi:10.1090/coll/023; the fourth-edition date is printed on the controlling local copy's edition-history page. The source explicitly credits E. B. Christoffel's earlier special case. Theorem 2.5 represents the orthogonal polynomial for a nonnegative polynomial modification of a positive distribution by a determinant of consecutive original orthogonal polynomials. For a repeated multiplier root it specifies the value and successive derivative rows. It does not assert that this determinant is monic or orthonormal; its proof establishes nonvanishing of the leading cofactor by positivity. The calculation below retains that cofactor and proves the complete specialization in the original coordinates.

Keep all NTG1–3 data, including the fixed original period, the complete nonreduced conductor and actual $\mu_d = E_A^{(d)}(0)$, the two source orders $s = 1, j$, and

$$\begin{aligned} a &= j - 8, \quad c' = a/2, \quad d = q' = (a + 1)^2, \quad D = 2d, \quad q = (j + 1)^2, \quad L = q - v, \quad m = L - d, \\ z_{u,w} &= (2w - a)\gamma - i(2u - a)\delta = (\beta_{a,u,w} - c')/i, \quad 0 \leq u, w \leq a, \\ Q_-(y) &= \prod_{u,w=0}^a (y - z_{u,w}) = i^{-d} \chi_-(c' + iy), \quad \rho(y) = Q_-(y)^2. \end{aligned}$$

(SLC1)

The index order is the original lexicographic order of (u, w) . The roots are distinct because real and imaginary parts separately determine w and u . The permutation $(u, w) \mapsto (a - u, w)$ conjugates the roots; hence Q_- has real coefficients. No root is real because a is odd, so $2u - a \neq 0$ and $\delta > 0$. In particular

$$\rho(y) > 0 \quad (y \in \mathbb{R}), \quad |\chi_-(c' + iy)|^2 = \rho(y). \quad (\text{SLC2})$$

The source distribution is literally the original dm_s , the old monic polynomials are the $p_{n,s}$ of NTG3, and the multiplier is the degree- D polynomial ρ . It has each original $z_{u,w}$ as a root of multiplicity two. The original Gamma density is strictly positive and has all polynomial moments; the finite exponential upper bound NIB11 also proves this directly. Thus all integrals used here are finite. No finite-interval approximation is needed.

Szegő's displayed old polynomials are orthonormal. In the present determinant we use the original monic $p_{n,s}$ instead. This is the exact invertible column map multiplying its l th old column by $\sqrt{h_{b+l,s}}$, so the whole determinant is multiplied by $\prod_{l=0}^D \sqrt{h_{b+l,s}}$. Every derivative row is multiplied by the same column factor. The denominator cofactor changes by its own corresponding product, and we retain those factors through the original $h_{n,s} = M_s n! (s/2)_n$ below. The direct proof that follows also verifies this monic presentation without suppressing any column or mass factor.

234.2 The full confluent determinant and its leading cofactor

For each $b \geq 0$ form the D by $D + 1$ matrix

$$\begin{aligned} \mathcal{M}_{b,s}[(\lambda, 0), l] &= p_{b+l,s}(z_\lambda), \quad \mathcal{M}_{b,s}[(\lambda, 1), l] = p'_{b+l,s}(z_\lambda), \\ 0 &\leq l \leq D. \end{aligned} \quad (\text{SLC3})$$

For each root in the stated order, its value row precedes its first-derivative row. These are derivatives of order one, without an inserted factorial or derivative in S' . Define every signed cofactor

and the complete polynomial

$$\begin{aligned} C_{b,l} &= (-1)^l \det \mathcal{M}_{b,s}^{\widehat{l}}, & \alpha_b &= C_{b,D}, \\ \mathcal{F}_b(y) &= \det \begin{bmatrix} (p_{b+l,s}(y))_{l=0}^D \\ \mathcal{M}_{b,s} \end{bmatrix} = \sum_{l=0}^D C_{b,l} p_{b+l,s}(y). \end{aligned} \quad (\text{SLC4})$$

The superscript $\widehat{l}$ removes exactly column l , leaving all root and derivative rows in their original order. The sign is the first-row Laplace sign $(-1)^{1+(l+1)}$. The determinant block notation in SLC4 means its first row followed by all D rows of SLC3.

The leading cofactor α_b is nonzero. Since $D = 2d$ is even, it is the determinant of columns $0, \dots, D-1$ of $\mathcal{M}_{b,s}$. If it vanished, a nonzero coefficient column would give a nonzero polynomial $H = \sum_{l=0}^{D-1} v_l p_{b+l,s}$ vanishing with its first derivative at every z_λ . Distinct-root division gives $H = \rho G$ with $\deg G \leq b-1$. For $b=0$ degree alone makes $H=0$, a contradiction. For $b \geq 1$, every $p_{b+l,s}$ is orthogonal in dm_s to every polynomial of degree below b . Taking that polynomial to be G in the conjugate-linear first argument yields

$$0 = \int_{\mathbb{R}} \overline{G(y)} H(y) dm_s(y) = \int_{\mathbb{R}} \rho(y) |G(y)|^2 dm_s(y). \quad (\text{SLC5})$$

The strict positivity in SLC2 implies $G=0$: a nonzero polynomial is nonzero on a nonempty real interval with positive measure. This contradicts independence of the distinct-degree monic polynomials in H . This is the complex-coefficient version of the positivity argument in Szegő's (2.5.4), preserving the repeated roots and all derivative rows.

At $y = z_\lambda$, the first row of $\mathcal{F}_b$ repeats its value row, so $\mathcal{F}_b(z_\lambda) = 0$. Differentiating only its first row shows that $\mathcal{F}'_b(z_\lambda) = 0$ by the corresponding derivative row. Thus $\rho \mid \mathcal{F}_b$. Its degree is precisely $b+D$, with leading coefficient $\alpha_b \neq 0$, because only the last monic column in SLC4 has this degree. Hence

$$P_b^-(y) = \frac{\mathcal{F}_b(y)}{\alpha_b \rho(y)} \quad (\text{SLC6})$$

is a monic polynomial of degree b . For every polynomial h of degree below b , $\int \bar{h} \rho P_b^- dm_s = \alpha_b^{-1} \int \bar{h} \mathcal{F}_b dm_s = 0$ by SLC4 and the old orthogonality. Consequently P_b^- is the unique monic minimizer for the positive measure ρdm_s : adding any smaller-degree polynomial adds its positive squared norm, by the just proved orthogonality. Uniqueness and the real density imply that P_b^- has real coefficients, since coefficientwise conjugation gives another monic polynomial with the same orthogonality and norm.

The exact monic coordinate map and its inverse are

$$\pi_b^-(S') = i^b P_b^-((S' - c')/i), \quad P_b^-(y) = i^{-b} \pi_b^-(c' + iy). \quad (\text{SLC7})$$

The leading coefficient in S' is one. These inverse maps carry the same complete monic affine fibre to the one in NTG8; they multiply values on the original line by the phase i^b of modulus one. Thus π_b^- is exactly the original NTG8 lower minimizer, and

$$\begin{aligned} \nu_{b,s}^- &= \int \rho(y) |P_b^-(y)|^2 dm_s(y), \\ e_b(c' + iy) &= \frac{i^{d+b} Q_-(y) P_b^-(y)}{\sqrt{\nu_{b,s}^-}}. \end{aligned} \quad (\text{SLC8})$$

Every displayed phase comes from SLC1 and SLC7.

234.3 The norm, the cofactor sign and every polynomial coefficient

Both endpoint cofactors in SLC4 are explicit. Since D is even, deleting column zero gives

$$C_{b,0} = \det[p_{b+1,s}, \dots, p_{b+D,s}]_{\text{jets}} = \alpha_{b+1}. \quad (\text{SLC9})$$

Using $P_b^- = p_{b,s} +$ a polynomial of degree below b , old orthogonality gives $\int \overline{P_b^-} p_{b,s} dm_s = h_{b,s}$. All terms $p_{b+l,s}$ with $l > 0$ are orthogonal to P_b^- of degree b . Multiplying SLC6 by $\rho \overline{P_b^-}$ and integrating therefore yields

$$\boxed{\nu_{b,s}^- = h_{b,s} \frac{C_{b,0}}{\alpha_b} = M_s b! (s/2)_b \frac{\alpha_{b+1}}{\alpha_b} > 0.} \quad (\text{SLC10})$$

In particular the actual cofactor ratio is positive real; no absolute value is substituted for a determinant.

The initial cofactor is

$$\alpha_0 = \prod_{\lambda < \mu} (z_\mu - z_\lambda)^4 > 0. \quad (\text{SLC11})$$

Here is a direct verification of its sign and value. The old columns $p_{0,s}, \dots, p_{D-1,s}$ are obtained from $1, y, \dots, y^{D-1}$ by a triangular coefficient matrix with diagonal one, so this changes no determinant. For each root use the two nearby evaluation rows at z_λ and $z_\lambda + \epsilon_\lambda$. Subtract the first from the second and divide the latter by ϵ_λ . The ordinary Vandermonde determinant, divided by $\prod_\lambda \epsilon_\lambda$, tends as the nonzero ϵ_λ tend to zero to exactly the value-and-first-derivative determinant in SLC3. Each cross-root pair contributes four copies of $z_\mu - z_\lambda$; each within-root factor divided by ϵ_λ contributes one. This proves the product. The root conjugation permutation sends the Vandermonde $\Delta = \prod_{\lambda < \mu} (z_\mu - z_\lambda)$ to $\overline{\Delta} = \pm \Delta$. Hence Δ is real or purely imaginary, and $\Delta^4 > 0$ since $\Delta \neq 0$. This proves SLC11. SLC10 now proves $\alpha_b > 0$ for every b by induction. Complex determinant rows and their original order have been retained throughout.

For a complete coefficient evaluation write $p_{n,s}(y) = \sum_h p_{n,h} y^h$ and $\rho(y) = \sum_{\ell=0}^D \rho_\ell y^\ell$, with $\rho_D = 1$. The old coefficients follow directly from the exact NTG3 recurrence, so SLC3 and SLC4 are finite expressions in the original roots and s . If $\mathcal{F}_b = \sum_h \mathfrak{f}_{b,h} y^h$ and $P_b^- = \sum_{h=0}^b P_{b,h} y^h$, then

$$\begin{aligned} \mathfrak{f}_{b,h} &= \sum_{l=0}^D C_{b,l} p_{b+l,h}, \quad P_{b,b} = 1, \\ P_{b,h} &= \frac{\mathfrak{f}_{b,D+h}}{\alpha_b} - \sum_{r=h+1}^b P_{b,r} \rho_{D+h-r} \quad (h = b-1, b-2, \dots, 0), \\ [S'^h] \pi_b^- &= \sum_{r=h}^b i^{b-r} P_{b,r} \binom{r}{h} (-c')^{r-h}. \end{aligned} \quad (\text{SLC12})$$

An index outside the degree range of a polynomial has coefficient zero. The middle equation is downward division of the already proved identity $\mathcal{F}_b/\alpha_b = \rho P_b^-$ at degree $D+h$; the last is binomial expansion of SLC7. Thus the whole modified polynomial, including its cross-pairing coefficients, is evaluated.

234.4 Evaluation of every original relation coefficient

Keep the actual $g_t = \mathcal{T}_A(\chi_j S^t)$, $\ell_t = \mu_v \binom{q+t}{v}$ and $u_t = g_t/\ell_t$, without changing a single shift or conductor coefficient. Exact root addition $\beta_{a,u,w} + \beta_{8,r,h} = \beta_{j,u+r,w+h}$ shows separately in every translation summand that g_t vanishes at each lower root. The roots are distinct, so define its exact polynomial quotient

$$G_t(S') = g_t(S')/\chi_-(S'), \quad \deg G_t = m+t, \quad \text{lc } G_t = \ell_t. \quad (\text{SLC13})$$

This is polynomial division on the original object, not division of its norm or deletion of any root.

Here are all coefficients needed for that division. Let $g_{t,h} = [S''^h]g_t$ be the finite sum NTG4: $g_{t,h} = \sum_{r+t \geq h} c_r \binom{r+t}{h} \mu_{r+t-h}$, where $\chi_j = \sum c_r S^r$. Write $\chi_- = \sum_{h=0}^d \chi_h S''^h$, $G_t = \sum_{h=0}^{B_t} \gamma_{t,h} S''^h$, $B_t = m + t$. Then

$$\begin{aligned} \gamma_{t,B_t} &= \ell_t, & \gamma_{t,h} &= g_{t,d+h} - \sum_{r=h+1}^{B_t} \gamma_{t,r} \chi_{d+h-r} \quad (h = B_t - 1, \dots, 0), \\ G_t(c' + iy) &= \sum_{h=0}^{B_t} d_{t,h} y^h, & d_{t,h} &= i^h \sum_{r=h}^{B_t} \gamma_{t,r} \binom{r}{h} (c')^{r-h}. \end{aligned} \tag{SLC14}$$

The zero coefficient convention in SLC12 applies here too. The vanishing remainder follows from the proved divisibility, and the formulas give its unique quotient and its exact original-line phase.

Expand this complete quotient in the old Gamma polynomial basis, with all finite coefficients retained:

$$\begin{aligned} G_t(c' + iy) &= \sum_{n=0}^{B_t} \kappa_{n,t} p_{n,s}(y), \\ \kappa_{n,t} &= \frac{1}{h_{n,s}} \sum_{h=0}^{B_t} \sum_{\alpha=0}^n d_{t,h} p_{n,\alpha} M_{h+\alpha,s}, & M_{r,s} &= M_s((J_s^{(r)})^r)_{00}. \end{aligned} \tag{SLC15}$$

The moments and $J_s^{(r)}$ are exactly NTG5. The old polynomials have real coefficients, so $\int p_{n,s} G_t(c' + iy) dm_s = h_{n,s} \kappa_{n,t}$. Old orthogonality proves both this identity and the finite expansion; it gives zero for $n > B_t$. In particular $\kappa_{B_t,t} = i^{B_t} \ell_t$ by the leading coefficient.

The actual NTG relation entry, in its original conjugate-linear first argument, is now

$$\boxed{\mathcal{R}_Q[b, t] = \langle e_b, u_t \rangle_s = \frac{i^{-b}}{\ell_t \sqrt{\nu_{b,s}^-} \alpha_b} \sum_{l=0}^{\min(D, B_t-b)} C_{b,l} h_{b+l,s} \kappa_{b+l,t}.} \tag{SLC16}$$

For $b > B_t$ the sum is empty and the entry is zero. To prove the formula, use SLC8 and $g_t(c' + iy) = i^d Q_-(y) G_t(c' + iy)$. Since Q_- and P_b^- are real on the integration line, this gives

$$\begin{aligned} \langle e_b, u_t \rangle_s &= \frac{i^{-b}}{\ell_t \sqrt{\nu_{b,s}^-}} \int \rho(y) P_b^-(y) G_t(c' + iy) dm_s(y) \\ &= \frac{i^{-b}}{\ell_t \sqrt{\nu_{b,s}^-} \alpha_b} \int \mathcal{F}_b(y) G_t(c' + iy) dm_s(y). \end{aligned}$$

The factor i^{-b} is the exact product $i^{-(d+b)} i^d$ of the two original-line phases. Expanding $\mathcal{F}_b$ by SLC4 and using SLC15 proves SLC16. No conjugation of ℓ_t or $\kappa_{n,t}$ is permitted: they belong to the second, complex-linear argument. All the $2d$ confluent rows and the original leading cofactor remain in its evaluated numerator.

At $b = B_t = m + t$ only $l = 0$ survives; substituting $\kappa_{B_t,t} = i^{B_t} \ell_t$ and SLC9–10 gives

$$\mathcal{R}_Q[m + t, t] = \frac{C_{B_t,0} h_{B_t,s}}{\sqrt{\nu_{B_t,s}^-} \alpha_{B_t}} = \sqrt{\nu_{m+t,s}^-}. \tag{SLC17}$$

Together with the empty-sum zero for $b > m + t$, this recovers the exact positive triangular diagonal of NTG11 directly from the confluent determinant. Every entry with $b < m + t$ remains the complete sum SLC16; scalar norm ratios alone would not give it.

234.5 The full original NTG graph and analytic frame receive the formula

For every $Q \geq 0$ set, exactly as in NTG10–12,

$$\begin{aligned} \mathcal{L}_Q[b, t] &= \mathcal{R}_Q[b, t] \quad (0 \leq b < m), \\ H_Q[a, t] &= \mathcal{R}_Q[m + a, t] \quad (0 \leq a, t \leq Q), \quad X_Q = \mathcal{L}_Q H_Q^{-1} = [x_0, \dots, x_Q]. \end{aligned} \quad (\text{SLC18})$$

Each entry in both blocks is now the actual determinant and moment sum SLC16. For a direct finite column evaluation, the same invertible triangular equation is

$$x_t = \frac{\mathcal{L}_Q[:, t] - \sum_{a=0}^{t-1} x_a H_Q[a, t]}{\sqrt{\nu_{m+t, s}^-}}, \quad t = 0, \dots, Q. \quad (\text{SLC19})$$

Indeed $X_Q H_Q = \mathcal{L}_Q$ and SLC17 give this by successive columns. The denominator is strictly positive. This recurrence includes every preceding column and every earlier/later cross pairing. It is the same inverse map as NTG12, with an explicit evaluation, so it proves compatibility of the resulting X_Q as Q increases without any new choice of frame.

The complete lower-basis coefficients used by the original analytic frame can also be read off, rather than being reconstructed from its determinant. From SLC12 and NTG8,

$$\begin{aligned} e_{hb} := [S'^h]e_b &= \frac{1}{\sqrt{\nu_{b, s}^-}} \sum_{r=0}^d \chi_r \sum_{0 \leq h-r \leq \ell \leq b} i^{b-\ell} P_{b, \ell} \binom{\ell}{h-r} (-c')^{\ell-(h-r)}, \\ A[b, c] &= \sum_{h=0}^{d+b} e_{hb} f_{h, c} \quad (0 \leq b < m). \end{aligned} \quad (\text{SLC20})$$

An empty inner sum is zero. Here the exact original derivatives $f_{h, c} = F_c^{(h)}(0)$ are evaluated by NTG9, with all μ_r and all numerator coefficients in their original frame. Multiplying $\chi_- \pi_b^-$ and applying the complex-linear functional proves SLC20 term by term. Thus the fixed matrix A and every column x_t have received the same complete Szegő polynomial, not only its squared norm.

It follows by the already proved NTG20,22 that these explicit entries give exactly

$$\begin{aligned} \Omega_Q &= I_m + X_Q X_Q^*, \quad D_{L+Q} = A^* \overline{\Omega_Q} A, \\ \frac{\rho_{L+t}}{\sqrt{1 + \zeta_{L+t}}} &= -i^{-(L+t)} x_t^\top A. \end{aligned} \quad (\text{SLC21})$$

The common lower-ideal denominator, source order, fixed period, $\mu_v \binom{q+t}{v}$, every root, derivative row and native phase have all survived the map SLC3–21. It consequently evaluates the input matrices for the complete original eight flags and CPQ quotient minima in their unchanged coefficient frames. It changes no endpoint or weight and assigns no asymptotic coefficient to the growing determinant.

234.6 Primary-source read receipt and accepted inputs

The controlling PDF read for Theorem 2.5 is `Orthogonal Polynomials. AMS Colloquium Publications, 1975 Szego.pdf`, SHA256 988969e6bb6ad590c3fc09a119aa69f7f8aee87063c9137615f8ff9ff7529fea. Physical PDF pages 38–39 contain printed pages 29–30; the full theorem and proof were extracted and read, and printed page 30 was visually inspected, including the determinant, derivative-row instruction and leading-cofactor argument. The continuation on printed page 31 was also read. The confluent determinant is explicitly attributed here to that human source, with its own credit to Christoffel preserved.

The unchanged input proof is `ORIGINAL_NATIVE_TAIL_GRAM.tex`, NTG1–32, SHA256 a7a109771c368f7bced035bd2f8e0f457e01607801b215bed1546d0a97dd4699. This is the citation-revised

receiving copy; its mathematics is unchanged from the authored `daf4d2df5d3a4a9955b0b6bc1da8f3c7e70166b89de4041b9afd984e37a32b2b` input. The determinant polynomial $\mathcal{F}_b$ above is distinct from the original analytic germ $F_c = N_c/E_A$ in SLC20. NTG3 supplies the exact original Gamma parameters and its checked Meixner–Pollaczek formula dictionary; NTG4–9 supplies the original conductor coefficients, moments and analytic derivative recursion; NTG10–22 supplies the full graph and phase-sensitive native receiving maps. The present calculation replaces none of those objects. It adds a primary-source-backed finite evaluation of their complete lower polynomials and cross coefficients. No growing-degree asymptotic claim is made.

235 The simultaneous original tail and its eight coefficient flags

The Schur minimum and determinant factorization used below are classical: see Boyd and Vandenberghe, Appendix A.5.5, p. 650, and Appendix C.4.1, p. 673 [H23]. We prove the required complex Hermitian formulas explicitly. The application is to the unchanged NIB/IPR rows and the original CPQ flags; no generic subspace of the same dimension replaces a programme map.

Fix the original simple quartet, its period and logarithm branch, and its complete conductor. Retain

$$j = k + 4, \quad q = (j + 1)^2, \quad q' = (j - 7)^2, \quad L = q - v, \quad m = L - q', \quad s \in \{1, j\}, \quad c' = j/2 - 4. \quad (\text{STF1})$$

Here $k = 4l + 1 \geq 9$, $v = \text{ord}_0 E_A$ and the full nonzero derivative $\mu_v = E_A^{(v)}(0)$ remain those of NIB1. In particular $m > 0$ on its original domain. The complete native basis is $\phi_n(S') = p_{n,s}((S' - c')/i)/\sqrt{M_s n! (s/2)_n}$, with $M_s = (2\pi)^{s/2}$ and the original Meixner–Pollaczek dictionary HPS1–4 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)]. For all original lower roots $\beta_{j-8,u,w}$, put $V_d[n, (u, w)] = \phi_n(\beta_{j-8,u,w})$; for each original analytic numerator functional Λ_{F_b} put $T_d[n, b] = \Lambda_{F_b}(\phi_n)$, $0 \leq n \leq d$, $1 \leq b \leq m$. The functional pairing is complex-linear, as in NIB; inner products are conjugate-linear in their first entry. The original target form is

$$D_N = T_{N-v}^*(I - V_{N-v}(V_{N-v}^* V_{N-v})^{-1} V_{N-v}^*) T_{N-v}, \quad \mathcal{D}_N = \text{diag}(D_N, 4). \quad (\text{STF2})$$

All inverses exist on $q - 1 \leq N \leq 2q$ by the full original rank proof NIB4. In particular the original masses and phases occur in every entry of these matrices.

At $N = q + r$, $n = L + r$, let $V_0 = V_{n-1}$, $T_0 = T_{n-1}$, and write v_n, t_n for their original next rows. Retain

$$\zeta_n = v_n(V_0^* V_0)^{-1} v_n^*, \quad \rho_n = t_n - v_n(V_0^* V_0)^{-1} V_0^* T_0, \quad U_r = \text{diag}(\rho_n / \sqrt{1 + \zeta_n}, 4). \quad (\text{STF3})$$

IPR1 proves the exact update $\mathcal{D}_{q+r} = \mathcal{D}_{q+r-1} + U_r^* U_r$. The old lower Gram, conductor and analytic functionals in STF3 are retained at their actual changing cutoff.

235.1 Two matrices for the entire original tail

Choose any original integer $0 \leq R < q$ and let $N_0 = q + R$, $H = \mathcal{D}_{N_0}$. For $t \in \{q - 1, q\}$, define the actual stacked rows and relative forms

$$A_t = \begin{bmatrix} U_{R+1} \\ \vdots \\ U_t \end{bmatrix}, \quad C_t = A_t H^{-1/2}, \quad M_t = C_t^* C_t, \quad \mathcal{D}_{q+t} = H^{1/2} (I_{4m} + M_t) H^{1/2}. \quad (\text{STF4})$$

An empty stack gives C_t with zero rows and $M_t = 0$. Summing STF3 proves STF4, including $t = R = q - 1$. The map $z \mapsto H^{1/2} z$ is an isometry from the original coefficient space with metric H to Euclidean $\mathbb{C}^{4m}$, with inverse $H^{-1/2}$. Formula STF4 records both directions and the entire original H ; it does not divide a source measure by its mass. The matrices M_t have size $4m$, independent of the number of tail rows.

For any fixed original subspace X with independent column frame X_0 , set

$$Z_X = H^{1/2} X_0 (X_0^* H X_0)^{-1/2}, \quad P_X = Z_X Z_X^*, \quad K_{X,t} = C_t P_X C_t^*, \quad F_t(X) = \log \det(I + K_{X,t}). \quad (\text{STF5})$$

For $X = 0$ use empty Z_X , $P_X = 0$ and $F_t(X) = 0$. Since $Z_X^* Z_X = I$, P_X is exactly the orthogonal projector onto $H^{1/2} X$. Direct multiplication of STF4 gives

$$\frac{\det(X_0^* \mathcal{D}_{q+t} X_0)}{\det(X_0^* H X_0)} = \det(I + Z_X^* M_t Z_X) = \det(I + K_{X,t}). \quad (\text{STF6})$$

For the last equality, elimination of either diagonal identity block in $\begin{bmatrix} I & B \\ -A & I \end{bmatrix}$ proves $\det(I + AB) = \det(I + BA)$ for rectangular A, B . Apply it with $A = C_t Z_X$ and $B = Z_X^* C_t^*$. Thus every original fixed-frame determinant cancels only in its own same-frame ratio. No cutoff-dependent frame is substituted.

The nonzero spectrum of $K_{X,t}$ can be calculated in $4m$ dimensions:

$$\tilde{K}_{X,t} = M_t^{1/2} P_X M_t^{1/2}. \quad (\text{STF7})$$

Here is the exact map, including singular M_t . On $\text{im } M_t^{1/2}$ define $W_t(M_t^{1/2} z) = C_t z$. It is well defined since $\|C_t z\|^2 = z^* M_t z = \|M_t^{1/2} z\|^2$, and is an isometry onto $\text{im } C_t$. Extend it by zero on $\ker M_t$. Then $C_t = W_t M_t^{1/2}$, and $K_{X,t} = W_t \tilde{K}_{X,t} W_t^*$. Both operators vanish on the corresponding complementary kernels; on their ranges this is a unitary equivalence with inverse W_t^* . It preserves every nonzero eigenvalue with multiplicity. The zero eigenvalues contribute zero to both functions $\log(1+x)$ and $\log \max(1, x)$ used below.

235.2 Each original quotient, with its cross terms

Let $K \subset Y$ be fixed original coefficient spaces. Choose independent X_K for K and fixed lifts X_Q of the specified quotient values, so $[X_K, X_Q]$ frames Y . Let Q_0 be its exact quotient metric in H , and retain the attained base representative

$$Z = X_Q - X_K(X_K^* H X_K)^{-1} X_K^* H X_Q, \quad Q_0 = Z^* H Z, \quad E = H^{1/2} Z Q_0^{-1/2}. \quad (\text{STF8})$$

For $K = 0$ omit its entire term. For $Y = K$ the quotient value space is zero-dimensional: its determinant is 1, its log determinant is 0, and its inverse and square root are the unique operators on that space. These conventions also apply to empty frames throughout. The columns of E are orthonormal, orthogonal to $H^{1/2} K$, and span its orthogonal complement in $H^{1/2} Y$. Thus $EE^* = P_Y - P_K$. Let Z_K be an orthonormal frame for $H^{1/2} K$. Every vector representing quotient value x has transformed coordinate $E Q_0^{1/2} x + Z_K a$. Its new quadratic form is

$$\|Q_0^{1/2} x\|^2 + \|a\|^2 + \|C_t E Q_0^{1/2} x + C_t Z_K a\|^2. \quad (\text{STF9})$$

For any matrix B and vector y , differentiation or square completion gives the unique minimum of $\|a\|^2 + \|y + Ba\|^2$ at $a = -(I + B^* B)^{-1} B^* y$. Its value is $y^*(I + BB^*)^{-1} y$: multiplication verifies $I - B(I + B^* B)^{-1} B^* = (I + BB^*)^{-1}$. Applying this complete minimum to STF9 proves the literal original quotient form at the later endpoint,

$$Q_t = Q_0^{1/2} [I + E^* C_t^* (I + C_t P_K C_t^*)^{-1} C_t E] Q_0^{1/2}. \quad (\text{STF10})$$

In particular $Q_t \succeq Q_0$ and no kernel direction or cross pairing has been removed without minimizing over it. The expression inside STF10 also has the finite coefficient-space realization

$$C_t^* (I + C_t P_K C_t^*)^{-1} C_t = M_t^{1/2} (I + M_t^{1/2} P_K M_t^{1/2})^{-1} M_t^{1/2}. \quad (\text{STF11})$$

To prove it, use the isometry W_t just constructed. On its range the two inverses are intertwined; on the complementary range both surrounding C_t factors vanish. This proves STF11 even at rank loss.

Block elimination in the fixed frame $[X_K, X_Q]$ gives $\det \text{Gram}(Y) = \det \text{Gram}(K) \det Q$. Taking its ratio at $q + t$ and N_0 , all frame constants cancel, yielding

$$\log \frac{\det Q_t}{\det Q_0} = F_t(Y) - F_t(K) \geq 0. \quad (\text{STF12})$$

Consequently STF10 evaluates every complementary tail at the same two matrices M_{q-1}, M_q , preserving the dependence between the quotients.

235.3 The eight flags and the exact allocation identity

Use precisely the CPQ coefficient maps on $\mathcal{U} = \mathbb{C}^{4m}$:

$$\begin{aligned} B &= \text{im } B_{\text{cof}}, & I &= \text{im}(B_{\text{cof}}J_0), & S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \text{diag}(B_j, 4), \\ K_i &= I \cap J, & W_i &= I + J, & W &= B + J, & K_c &= S \cap W, & T &= S + J, & e &\in \{0, 1\}. \end{aligned} \quad (\text{STF13})$$

Since $B \subset S$, this T is the same $S + W$ as CPQ. The original inclusions give $J \subset W_i \subset W \subset T$ and $K_i \subset I \subset K_c \subset S$. The actual pairs for the four CPQ complementary forms are

$$\begin{array}{c|cccc} \text{form} & Q_b^E & Q_{\text{rem},e} & G_{s,e} & A_{\text{inv}} \\ \hline (Y, K) & (W, W_i) & (\mathcal{U}, T) & (S, K_c) & (I, K_i). \end{array} \quad (\text{STF14})$$

These identifications are the exact same-value CPQ maps, not merely dimension equalities: W/J identifies with the original image of Φ , W_i/J with its invariant image; $\mathcal{U}/T$ is CPQ8's remaining target quotient; S/K_c is CPQ7's connecting source quotient; and I/K_i is the source pullback quotient in CPQ9. The corresponding fixed lift changes are independent of cutoff and therefore cancel in STF12. All their original value metrics are recovered by STF8–10.

For the logarithm of a restriction determinant in a fixed frame, $f_X(N) = \log \det(X_0^* \mathcal{D}_N X_0)$, IPR3 gives exactly

$$\mathcal{R}_{>R} f_X = F_{q-1}(X) + F_q(X). \quad (\text{STF15})$$

Indeed both terms on the right subtract the same $f_X(q + R)$. It follows that the original signed tail is

$$\mathcal{L}_e^{>R} = \sum_{t=q-1, q}^8 \sum_{a=1}^8 \sigma_a F_t(X_a), \quad (\text{STF16})$$

$$(X_1, \dots, X_8) = (T, K_c, W_i, K_i, W, S, J, I), \quad (\sigma_1, \dots, \sigma_8) = (1, 1, 1, 1, -1, -1, -1, -1).$$

The tuple consists of labelled occurrences: coincident subspaces are retained in every position with its indicated sign. All flag sums below use this tuple, including when a quotient is zero-dimensional. Every complementary tail is $\sum_{t=q-1, q} [F_t(Y) - F_t(K)]$ using its exact pair in STF14. The four-copy kernel restriction and target quotient allocations are

$$4S_R^{>R} = \sum_{t=q-1, q} F_t(J), \quad 4S_K^{>R} = \sum_{t=q-1, q} [F_t(\mathcal{U}) - F_t(J)]. \quad (\text{STF17})$$

The latter is the minimum over the actual full kernel of $\text{diag}(B_j, 4)$. Substitution of STF14 into STF16 cancels the same terms algebraically, proving

$$4S_K^{>R} = \mathcal{L}_e^{>R} + V_b^{>R} + V_{\text{rem}}^{>R} + V_s^{>R} + V_A^{>R}, \quad S_K^{>R} + S_R^{>R} = \mathcal{T}^{>R}. \quad (\text{STF18})$$

Thus the joint calculation supplies every term of IPR7 through common finite matrices and its exact quotient operations.

235.4 Finite comparison with the original row-threshold profile

Define the simultaneous endpoint threshold value

$$\mathcal{B}_e^{>R} = \sum_{t=q-1, q}^8 \sum_{a=1}^8 \sigma_a \sum_b \log \max(1, \lambda_b(K_{X_{a,t}})), \quad d_t = \text{rank } C_t \leq 4m. \quad (\text{STF19})$$

All matrices $K_{X,t}$ are supported on the same d_t -dimensional space $\text{im } C_t$. Compress to it by the exact isometry in STF7. Nested X give ordered projectors and ordered $K_{X,t}$, hence ordered eigenvalues: for each test subspace every Rayleigh quotient of the larger Hermitian matrix dominates

the smaller one, and successive minima and maxima preserve that inequality. This proves the required eigenvalue ordering without assuming a common eigenbasis.

For $x \geq 0$, put $h(x) = \log(1+x) - \log \max(1, x) = u(x) - v(x)$, where $u(x) = \log(1 + \min(x, 1))$ and $v(x) = 0$ for $x \leq 1$, $v(x) = \log(2x/(1+x))$ for $x \geq 1$. Both u, v are increasing from zero and at most $\log 2$. Along any four-element ordered eigenvalue chain, the sum of the first and last disjoint increments of either function lies in $[0, \log 2]$: adding its omitted middle nonnegative increment proves the upper bound. The corresponding sum of h increments lies in $[-\log 2, \log 2]$. The two chains in STF13 therefore give a difference bounded by $2 \log 2$ per eigenvalue. Summing over the common range proves

$$|\mathcal{L}_e^{>R} - \mathcal{B}_e^{>R}| \leq 2(d_{q-1} + d_q) \log 2 \leq 16m \log 2. \quad (\text{STF20})$$

This is a bound for the simultaneous endpoint representation, whose eigenvalues differ from the original successive four-row compressions. The exact map to the latter scalar profile is as follows. At each original row the same two-chain argument uses four eigenvalues, hence its signed logarithmic and threshold increments differ by at most $8 \log 2$. The tail weights sum to $2(q - R) - 1$, so the original row-threshold tail $\mathcal{P}_e^{>R}$ satisfies

$$|\mathcal{P}_e^{>R} - \mathcal{B}_e^{>R}| \leq [8\{2(q - R) - 1\} + 2(d_{q-1} + d_q)] \log 2. \quad (\text{STF21})$$

This proves the relation between the two threshold constructions with their precise finite discrepancy; it asserts no equality of individual row and endpoint eigenvalues.

For a positive semidefinite finite matrix let $n_X(t, a) = \#\{\lambda_b(K_{X,t}) > a\}$, counting multiplicity. The elementary identity $\log \max(1, \lambda) = \int_1^\infty \mathbf{1}_{a < \lambda} da/a$ and the finite eigenvalue sum give the exact remaining integral

$$\mathcal{B}_e^{>R} = \sum_{t=q-1, q} \int_1^\infty \sum_{b=1}^8 \sigma_b n_{X_b}(t, a) \frac{da}{a}. \quad (\text{STF22})$$

No limiting interchange is needed; each integrand vanishes beyond the largest eigenvalue of its finite matrices. It is the signed spectral count of the original correlated flag compressions.

Finally choose only the already-proved original prefix $R = R_j = \lfloor j^{3/2}/\log j \rfloor < q$. NIB19–20 and IPR8 bound the removed signed and threshold prefixes by $4\mathfrak{E}_{s, R_j} = o(kq_k)$. Since $m = 16j - 48 - v$ and $q = (j + 1)^2$, STF20–21 are respectively $O(j)$ and $O(j^2)$. Thus the full signed and original threshold returns each differ from the exact finite expression STF22 by $o(kq_k)$, with explicit finite errors given by the prefix budget and STF20–21. This evaluates the complete source-to-expression map and its error; the growing-degree value of STF22 itself is still to be calculated. Both source orders, both primitive indices, the independent proper source and the full action retain their existing domains and receiving identities.

236 The original relation graph supplies the simultaneous mixed flags

This calculation joins the original-data relation formulas NTG4–22 to the simultaneous restriction and quotient formulas STF4–22. The finite Hermitian operations are proved below; the Schur quotient operations being transported are those of Boyd and Vandenberghe, Appendix A.5.5, p. 650, and C.4.1, p. 673 [H23], with their complex specialization proved in STF8–12. Every scalar polynomial and norm retains the Meixner–Pollaczek parameter and mass dictionary of NTG3 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)].

Use NTG’s original A , x_r and Ω_R , and choose an integer $0 \leq R < q$. The symbol D_n^{nat} in this section denotes the matrix called D_n in NTG18. STF2’s target-cutoff convention is $D_N = D_{N-v}^{\text{nat}}$; therefore $N = q + r$ corresponds to $n = L + r$. Define, with the unique positive Hermitian square roots,

$$\begin{aligned} G &= \overline{\Omega_R}, & H_1 &= A^*GA = D_{q+R}, & H &= \text{diag}(H_1, 4), \\ U &= G^{1/2}AH_1^{-1/2}, & U_4 &= \text{diag}(U, 4). \end{aligned} \quad (\text{NFI1})$$

All matrices in the inverses are positive definite by NTG12,16,20. Direct multiplication gives $U^*U = I_m$. It is square and injective, so its inverse is U^* and $UU^* = I_m$ as well. In particular U_4 is a unitary map between the two stated Euclidean coordinate spaces, not an alteration of the original source measure.

For $R \leq t \leq q$ set $d = t - R$ and retain

$$\begin{aligned} Y_t &= [x_{R+1}, \dots, x_t], & Z_t &= G^{-1/2}\overline{Y_t}, \\ B_t &= Z_t^*U = Y_t^T AH_1^{-1/2}, & \Theta_t &= \text{diag}_{s=R+1}^t(-i^{-(L+r)}). \end{aligned} \quad (\text{NFI2})$$

Here Y_t, Z_t are m by d , B_t is d by m , and Θ_t is unitary of size d . For $d = 0$ each displayed rectangular matrix is the corresponding empty map. The equality for B_t follows from $(G^{-1/2})^* = G^{-1/2}$ and the formula for U ; it keeps the ordinary transpose required by the original complex-linear functionals.

Extend STF4’s endpoint notation to every $R \leq t \leq q$ by $C_t = [U_{R+1}^T, \dots, U_t^T]^T H^{-1/2}$, where the bracket means the vertical stack of the original row-block matrices U_r , and use the empty stack for $t = R$. It agrees with STF4 at its two stated endpoints. These four-copy original row updates are stacked by degree. Write their row coordinates as (r, b) , with $R + 1 \leq r \leq t$ and $1 \leq b \leq 4$, and let Π_t be the permutation unitary sending coordinate (r, b) to (b, r) . Thus it changes only the stated row ordering. Set

$$\mathcal{R}_t = \text{diag}(\Theta_t^*, 4)\Pi_t, \quad \mathcal{Z}_t = \text{diag}(Z_t, 4). \quad (\text{NFI3})$$

NTG22 gives the full original row $\rho_{L+r}/\sqrt{1 + \zeta_{L+r}} = -i^{-(L+r)}x_r^T A$. Multiplying on the right by $H_1^{-1/2}$, stacking and using the defined permutation proves, entry by entry,

$$\Pi_t C_t = \text{diag}(\Theta_t B_t, 4), \quad \boxed{\mathcal{R}_t C_t = \mathcal{Z}_t^* U_4}. \quad (\text{NFI4})$$

The inverse row map is $\mathcal{R}_t^*$. Thus the original native phase has been recorded in an explicit invertible map before taking any Gram matrix. No phase is inferred merely from equality of norms.

Taking the Gram of NFI4 gives the exact coefficient-space relation

$$\mathcal{N}_t = \mathcal{Z}_t \mathcal{Z}_t^* = \text{diag}(G^{-1/2}\overline{Y_t} Y_t^T G^{-1/2}, 4), \quad M_t = U_4^* \mathcal{N}_t U_4. \quad (\text{NFI5})$$

This is an equality in the original STF coordinates after the specified unitary map. For example,

$$\det(I_{4m} + M_t) = \left(\frac{\det \Omega_t}{\det \Omega_R} \right)^4. \quad (\text{NFI6})$$

To prove NFI6, $\Omega_t = \Omega_R + Y_t Y_t^*$, so conjugating by $\Omega_R^{-1/2}$ gives $\det(I + \Omega_R^{-1/2} Y_t Y_t^* \Omega_R^{-1/2}) = \det \Omega_t / \det \Omega_R$. Complex conjugation gives the determinant of the single block of $I + \mathcal{N}_t$; both determinants are positive real numbers. There are four blocks. This also proves the empty-tail case, with ratio 1. It is consistent with both allocations in STF17–18, without selecting either allocation.

For every original coefficient subspace X , including those in the labelled eight-flag tuple STF16, retain its fixed independent frame X_0 and put

$$\widehat{Z}_X = \text{diag}(G^{1/2} A, 4) X_0 (X_0^* H X_0)^{-1/2}, \quad \widehat{P}_X = \widehat{Z}_X \widehat{Z}_X^*. \quad (\text{NFI7})$$

If $X = 0$ use the empty frame. NFI1 and STF5 give $\widehat{Z}_X = U_4 Z_X$, hence $\widehat{Z}_X^* \widehat{Z}_X = I$ and $\widehat{P}_X = U_4 P_X U_4^*$. This supplies the exact transported flag with its full original conductor-dependent coefficient matrix A . Its inverse is U_4^* on the transformed Euclidean space, followed by $H^{-1/2}$ when recovering original coefficient vectors. Consequently NFI4 proves

$$\mathcal{R}_t K_{X,t} \mathcal{R}_t^* = \mathcal{Z}_t^* \widehat{P}_X \mathcal{Z}_t, \quad F_t(X) = \log \det(I + \mathcal{Z}_t^* \widehat{P}_X \mathcal{Z}_t). \quad (\text{NFI8})$$

The determinant equality uses a unitary conjugation of the entire row space, including its zero eigenvalues. Its version on the fixed $4m$ -dimensional coefficient space is

$$\widehat{K}_{X,t} = \mathcal{N}_t^{1/2} \widehat{P}_X \mathcal{N}_t^{1/2}. \quad (\text{NFI9})$$

For completeness define the map $\mathcal{W}_t(\mathcal{N}_t^{1/2} z) = \mathcal{Z}_t^* z$ on $\text{im } \mathcal{N}_t^{1/2}$, extending it by zero on the kernel. It is well defined and isometric because $\|\mathcal{Z}_t^* z\|^2 = z^* \mathcal{N}_t z = \|\mathcal{N}_t^{1/2} z\|^2$. Then the right side of NFI8 equals $\mathcal{W}_t \widehat{K}_{X,t} \mathcal{W}_t^*$. The inverse on the image is $\mathcal{W}_t^*$; its complementary kernel contributes only zero eigenvalues. This proves the precise spectral correspondence used by both logarithmic functions in STF, including rank loss and coincident flags.

There is a corresponding formula for every original quotient $K \subset Y$. Retain Q_0 and E from STF8 and set $\widehat{E} = U_4 E$; then $\widehat{E} \widehat{E}^* = \widehat{P}_Y - \widehat{P}_K$. Substituting NFI4 into the complete attained minimum STF10 gives

$$Q_t = Q_0^{1/2} \left[I + \widehat{E}^* \mathcal{Z}_t (I + \mathcal{Z}_t^* \widehat{P}_K \mathcal{Z}_t)^{-1} \mathcal{Z}_t^* \widehat{E} \right] Q_0^{1/2}. \quad (\text{NFI10})$$

Indeed the middle inverse is carried into STF10's inverse by the unitary $\mathcal{R}_t$, and its two adjacent factors carry E into $\widehat{E}$. This is an equality of matrices on the original quotient value space, with its original Q_0 on both sides. All cross terms survive the attained minimum. It applies to the four actual pairs in STF14 and to the target quotient $(\mathcal{U}, J)$; the empty quotient conventions remain those of STF8. Thus it transfers every complement and both allocations, rather than just their sum of determinants.

Finally use the eight labelled occurrences (X_a, σ_a) of STF16, retaining repeated subspaces separately. The remaining signed count in STF22 is now the fully original-data expression

$$\mathcal{B}_e^{>R} = \sum_{t=q-1,q} \int_1^\infty \sum_{a=1}^8 \sigma_a \# \{ \lambda_b(\widehat{K}_{X_a,t}) > u \} \frac{du}{u}. \quad (\text{NFI11})$$

All eigenvalues count multiplicities. The matrices in NFI11 are obtained from the finite coefficients NTG4–12, the original analytic recursion NTG9, and the actual CPQ frames in NFI7. The equality follows from the proved unitary and partial-isometry maps, not from a comparison of dimensions. With the accepted prefix R_j , the finite discrepancy bounds STF20–21 and its prefix budget apply without change. NFI11 gives the full coefficient formula for the growing-degree quantity; it does not assert a large-degree estimate for that quantity.

237 The full original tail through conductor transport and all receiving values

This calculation receives the explicit original relation and simultaneous flag matrices NTG/SLC/STF/NFI into the original IPR, CPQ, NFR and CAI objects. It preserves the programme's recorded origin and the distinct creators and dates in its two deposited source notes [H19, H20]. The exact internal maps are stated with their source locators, while the human mathematical dependencies are cited at their uses below. All statements concern the original simple-quartet and fixed-period domain specified in each section. No additional multiplicity or asymptotic hypothesis is inserted.

237.1 The original data and the two exact tail matrices

Fix the actual simple quartet, its period and logarithm branch before the degree limit. Retain

$$k = 4l + 1 \geq 9, \quad j = k + 4, \quad q = (j + 1)^2, \quad q' = (j - 7)^2, \quad L = q - v, \quad m = L - q', \quad s \in \{1, j\}. \quad (\text{FTR1})$$

The complete conductor is $E_A(z) = \sum_{u,w=0}^8 a_{uw} e^{\beta_{s,u,w} z}$, with $\beta_{a,u,w} = a/2 + (2u - a)\delta + i(2w - a)\gamma$, $0 < \delta < 1/2$, $\gamma > 2$, and $\mu_d = E_A^{(d)}(0)$, $v = \text{ord}_0 E_A$, $\mu_v \neq 0$. The relations, including their scalar factors, are $g_r = \mathcal{T}_A(\chi_j S^r)$, $\ell_r = \mu_v \binom{q+r}{v}$ and $u_r = g_r/\ell_r$. Use the unchanged native line $S' = j/2 - 4 + iy$ and measure

$$dm_s(y) = \frac{(2\pi)^{s/2} 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy. \quad (\text{FTR2})$$

The orthogonal polynomials, full norms and phases are exactly NTG3: $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$ and $h_{n,s} = (2\pi)^{s/2} n! (s/2)_n$. This is the stated Meixner–Pollaczek convention, with its factor $dy = 2dx$ and mass checked in HPS1–4 and NTG3 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)]. NTG4–12 and SLC3–20 evaluate every entry of the original graph $X_r = [x_0, \dots, x_r]$ and invertible functional matrix A . In particular the positive polynomial modification in that entry calculation is the complete confluent determinant of Szegő, including the original leading cofactor and both rows at every root [H24, Theorem 2.5, pp. 29–30]; the full substitution and its polynomial proof are SLC1–21. None of those factors is changed in this receiving calculation.

Let $\Omega_r = I_m + X_r X_r^*$, $\Omega_{-1} = I_m$, and keep any integer $0 \leq R < q$. For every $R \leq t \leq q$ put

$$G = \overline{\Omega_R}, \quad H_1 = A^* G A = D_{q+R}, \quad H = \text{diag}(H_1, 4), \quad Y_t = [x_{R+1}, \dots, x_t], \quad Z_t = G^{-1/2} \overline{Y_t}, \\ \mathcal{Z}_t = \text{diag}(Z_t, 4), \quad \mathcal{N}_t = \mathcal{Z}_t \mathcal{Z}_t^*, \quad \mathcal{U} = \mathbb{C}^{4m}. \quad (\text{FTR3})$$

Here $D_N = D_{j,N-v}^{(s)}$ is the original native analytic quotient form, and $\mathcal{D}_N = \text{diag}(D_N, 4)$. For each fixed original flag $X \subset \mathcal{U}$ with independent frame X_0 , define

$$\widehat{Z}_X = \text{diag}(G^{1/2} A, 4) X_0 (X_0^* H X_0)^{-1/2}, \quad \widehat{P}_X = \widehat{Z}_X \widehat{Z}_X^*, \\ \widehat{K}_{X,t} = \mathcal{N}_t^{1/2} \widehat{P}_X \mathcal{N}_t^{1/2}, \quad f_t(X) = \log \det(I_{4m} + \widehat{K}_{X,t}). \quad (\text{FTR4})$$

An empty frame has projector zero and determinant one. The square roots are the unique positive Hermitian square roots; all inverses are on the indicated positive forms. The $\widehat{P}_X$ notation in FTR4 means the explicit unitary change of flag coordinates in NFI7, and does not denote the later conductor-transported metric. The latter will have a superscript tr below.

For clarity the exact coordinate map behind FTR4 is $U_4 = \text{diag}(G^{1/2} A H_1^{-1/2}, 4)$, with inverse U_4^* . The row permutation and phases are NFI3's $\mathcal{R}_t = \text{diag}(\Theta_t^*, 4) \Pi_t$, $\Theta_t = \text{diag}_{r=R+1}^t (-i^{-(L+r)})$. NFI4 proves $\mathcal{R}_t C_t = \mathcal{Z}_t^* U_4$ directly from $\rho_{L+r}/\sqrt{1 + \zeta_{L+r}} = -i^{-(L+r)} x_r^\top A$. Thus no phase or transpose is inferred from a determinant equality. The range isometry in NFI9 identifies every nonzero eigenvalue in FTR4, with multiplicity, with its original stacked-row compression; its inverse

is the adjoint on the image, and both complementary kernels have zero contribution to the two logarithmic functions. It follows in the same original frame that

$$f_t(X) = \log \frac{\det(X_0^* \mathcal{D}_{q+t} X_0)}{\det(X_0^* H X_0)}. \quad (\text{FTR5})$$

This is precisely STF6 and NFI8–9, now with all input entries supplied by SLC16–21.

237.2 Every original quotient and both allocations

For a fixed original pair $K \subset Y$ choose its original kernel frame X_K and fixed lifts X_Q of its quotient values. Set

$$\begin{aligned} Z &= X_Q - X_K(X_K^* H X_K)^{-1} X_K^* H X_Q, & Q_0 &= Z^* H Z, \\ E &= H^{1/2} Z Q_0^{-1/2}, & \hat{E} &= U_4 E, \\ Q_t &= Q_0^{1/2} \left[I + \hat{E}^* \mathcal{Z}_t (I + \mathcal{Z}_t^* \hat{P}_K \mathcal{Z}_t)^{-1} \mathcal{Z}_t^* \hat{E} \right] Q_0^{1/2}, \\ \tau_{Y/K} &= \sum_{t=q-1, q} \log \frac{\det Q_t}{\det Q_0} = \sum_{t=q-1, q} \{f_t(Y) - f_t(K)\}. \end{aligned} \quad (\text{FTR6})$$

For $K = 0$ omit its kernel term; for $Y = K$ all quotient operators are the unique operators on the zero space and its determinant is one. These are the actual unchanged quotient values. Indeed every representative of x has transformed coordinate $EQ_0^{1/2}x + Z_K a$, where Z_K frames $H^{1/2}K$ orthonormally. Its new norm is $\|Q_0^{1/2}x\|^2 + \|a\|^2 + \|C_t EQ_0^{1/2}x + C_t Z_K a\|^2$. Completing the square gives $a = -(I + Z_K^* C_t^* C_t Z_K)^{-1} Z_K^* C_t^* C_t EQ_0^{1/2}x$. Substitution followed by the actual unitary row map proves the matrix in FTR6. In the fixed frame $[X_K, X_Q]$, block elimination gives $\det \text{Gram}(Y) = \det \text{Gram}(K) \det Q$, proving its determinant ratio. This is the complex Hermitian form of the full Schur minimum and factorization in Boyd and Vandenberghe, Appendix A.5.5, p. 650, and Appendix C.4.1, p. 673 [H23]; all positive blocks, value fibres and cross terms were just specified. In particular $Q_t \succeq Q_0$ and $\tau_{Y/K} \geq 0$.

Use the original CPQ maps without choosing new subspaces:

$$\begin{aligned} B &= \text{im } B_{\text{cof}}, & I &= \text{im}(B_{\text{cof}} J_0), & S &= \text{im}[B_{\text{cof}}, F_e], & J &= \ker \text{diag}(B_j, 4), \\ K_i &= I \cap J, & W_i &= I + J, & W &= B + J, & K_c &= S \cap W, & T &= S + J, & e &= 0, 1. \end{aligned} \quad (\text{FTR7})$$

The exact pairs and resulting tail returns are

original form	(Y, K)	tail
Q_b^E	(W, W_i)	$\tau_b := \tau_{W/W_i}$
$Q_{\text{rem}, e}$	$(\mathcal{U}, T)$	$\tau_{\text{rem}} := \tau_{\mathcal{U}/T}$
$G_{s, e}$	(S, K_c)	$\tau_s := \tau_{S/K_c}$
A_{inv}	(I, K_i)	$\tau_A := \tau_{I/K_i}$
H_{inv}	(W_i, J)	$\tau_I := \tau_{W_i/J}$
$G_{t, e}$	(T, W)	$\tau_t := \tau_{T/W}$

(FTR8)

The first four are the same-value identifications CPQ7–10 and STF14; the fifth is proved below on the fixed original values U_R . The last is CPQ7's connecting target quotient. For each pair the complete metric, including its original frame factors, is FTR6. Only same-frame ratios annihilate these constant factors.

Define

$$\begin{aligned}
\tau_K &= \frac{1}{4}\tau_{\mathcal{U}/J}, & \tau_R &= \frac{1}{4} \sum_{t=q-1,q} f_t(J), & \tau_{\text{total}} &= \log \frac{\det \Omega_{q-1} \det \Omega_q}{(\det \Omega_R)^2}, \\
(X_1, \dots, X_8) &= (T, K_c, W_i, K_i, W, S, J, I), \\
(\sigma_1, \dots, \sigma_8) &= (1, 1, 1, 1, -1, -1, -1, -1), \\
\tau_L &= \sum_{t=q-1,q} \sum_{a=1}^8 \sigma_a f_t(X_a).
\end{aligned} \tag{FTR9}$$

All eight positions are retained when subspaces coincide. Substituting FTR8 and cancelling identical terms proves the three simultaneous equalities

$$\begin{aligned}
\tau_L &= \tau_t - \tau_s + \tau_I - \tau_A, & 4\tau_K &= \tau_L + \tau_b + \tau_{\text{rem}} + \tau_s + \tau_A, \\
\tau_K + \tau_R &= \tau_{\text{total}}.
\end{aligned} \tag{FTR10}$$

The last equality also follows from NFI6 applied at both endpoints: $f_t(\mathcal{U}) = 4 \log(\det \Omega_t / \det \Omega_R)$. The two allocations here are the original value quotient and original kernel restriction, not selected fractions of their sum.

For a log determinant $h(N)$ in any fixed original value frame set

$$\begin{aligned}
\mathcal{R}h &= -h(q-1) - h(q) + h(2q-1) + h(2q), \\
\mathcal{R}_{\leq R}h &= 2h(q+R) - h(q) - h(q-1), \\
\mathcal{R}_{> R}h &= h(2q-1) + h(2q) - 2h(q+R).
\end{aligned} \tag{FTR11}$$

They sum exactly, and they are the original weighted increments with weights one at $r = 0, q$ and two at $0 < r < q$. For each form in FTR8 write $p_Y = \mathcal{R}_{\leq R} \log \det Q_Y$; write p_K, p_R for the original single-copy allocations and p_L for the signed original prefix. Then the complete native returns have the exact expressions

$$\begin{aligned}
V_Y &= p_Y + \tau_Y, & S_K &= p_K + \tau_K, & S_R &= p_R + \tau_R, & \mathcal{L}_e &= p_L + \tau_L, \\
4p_K &= p_L + p_b + p_{\text{rem}} + p_s + p_A, & p_K + p_R &= p_{\text{total}}.
\end{aligned} \tag{FTR12}$$

IPR3 proves telescoping and IPR4–8 proves these original fibre identities at each part of the window. Positivity of the same increment gives, with the literal NIB/IPR budget $\mathfrak{E}_{s,R} = \sum_{r=0}^R \vartheta_r E_{j,s,r}$,

$$\begin{aligned}
|p_L| &\leq 4\mathfrak{E}_{s,R}, & 0 &\leq p_Y \leq 4\mathfrak{E}_{s,R} \quad (Y = b, \text{rem}, s, A, I, t), \\
0 &\leq p_K, p_R \leq \mathfrak{E}_{s,R}, & 0 &\leq p_K + p_R \leq \mathfrak{E}_{s,R}.
\end{aligned} \tag{FTR13}$$

For the two added pairs I, t , this bound follows by applying the same complete minimum to their inclusions in the four-copy ambient space. Explicitly, extend the fixed pair frame to a full frame $C = [X_K, X_Q, X_{\text{out}}]$. Two full Schur eliminations give $|\det C|^2 \det F = \det(F|_K) \det Q_{Y/K,F} \det Q_{\mathcal{U}/Y,F}$ for each positive metric F . Under one positive update each same-value factor is nondecreasing: restriction preserves the quadratic inequality and minimization over the identical affine fibre preserves it as well. Their determinant ratios therefore are at least one and their product is the full ambient ratio. Each quotient ratio is thus at most that ambient ratio. The latter is $(1 + a_r)^4$ by the original four-copy update. Taking logs and the original weights proves FTR13. This argument retains the original pair and does not substitute a space of equal rank.

237.3 Directed conductor transport at the actual evaluation sites

Let a superscript tr mean the identical coefficient and value maps endowed with NCT's conductor-transported metric. For any one restriction or attained quotient Y of actual rank r_Y , write

$$d_Y(N) = \log \det Q_Y^{\text{tr}}(N) - \log \det Q_Y(N), \quad -l_Y(N) \leq d_Y(N) \leq u_Y(N). \tag{FTR14}$$

Here $l_Y(N) = \mathcal{L}_{j,N-v,s,r_Y}^{[4]}$ and $u_Y(N) = \mathcal{U}_{j,N-v,s,r_Y}^{[4]}$ are precisely NCT19–20 at any fixed admissible original SGT radius. Explicitly, with $h_4 = \min(r_Y, 4H_\rho)$, they are

$$\begin{aligned} u_Y(N) &= 4r_Y L_{s,N-v,v} + 2r_Y \log^+ F_{\rho,N-v}, \\ l_Y(N) &= 4r_Y L_{s,N-v,v} + 2h_4 \log^+ G_{0,N-v} + 2(r_Y - h_4) \log^+ V_{\rho,N-v}^{\text{cap}}. \end{aligned} \quad (\text{FTR15})$$

Every radius, principal part, exceptional rank, residue and diagonal factor in these already evaluated NCT constants is kept. They are zero at rank zero. The transport is the actual complex-linear functional map $\Lambda_F \mapsto \Lambda_{E_A F}$, in its ordinary- transpose matrix, together with its complete lower quotient, as in NCT11–14. In particular it applies to the same quotient values in FTR6, rather than to arbitrarily chosen Euclidean flag frames.

For $Z \in \{\text{full}, \leq R, > R\}$ let $\eta_Y^Z = \mathcal{R}_Z d_Y$, using $\mathcal{R}$ for “full”. The following directed costs preserve all literal endpoint signs:

Z	$E_Y^{Z,-}$	$E_Y^{Z,+}$
full	$u_Y(q-1) + u_Y(q) + l_Y(2q-1) + l_Y(2q)$	$l_Y(q-1) + l_Y(q) + u_Y(2q-1) + u_Y(2q)$
$\leq R$	$2l_Y(q+R) + u_Y(q) + u_Y(q-1)$	$2u_Y(q+R) + l_Y(q) + l_Y(q-1)$
$> R$	$l_Y(2q-1) + l_Y(2q) + 2u_Y(q+R)$	$u_Y(2q-1) + u_Y(2q) + 2l_Y(q+R)$

(FTR16)

Then $-E_Y^{Z,-} \leq \eta_Y^Z \leq E_Y^{Z,+}$ and $\eta_Y^{\text{full}} = \eta_Y^{\leq R} + \eta_Y^{> R}$ exactly. Proof: substitute FTR14 in each term of FTR11. A negative coefficient exchanges the roles of its two endpoint costs; coefficient two multiplies its actual cost by two. This proves every table entry, without adding per-row errors or assuming independent extremizers. At repeated sites one may combine their actual coefficients first; the displayed uncombined nonnegative costs remain valid.

For the signed eight-flag expression, write $\eta_{X,\text{res}}^Z$ for the residual of the restriction form on the subspace X , and put $\eta_L^Z = \sum_a \sigma_a \eta_{X_a,\text{res}}^Z$ and

$$E_L^{Z,-} = \sum_{\sigma_a=1} E_{X_a}^{Z,-} + \sum_{\sigma_a=-1} E_{X_a}^{Z,+}, \quad E_L^{Z,+} = \sum_{\sigma_a=1} E_{X_a}^{Z,+} + \sum_{\sigma_a=-1} E_{X_a}^{Z,-}. \quad (\text{FTR17})$$

The $E_{X_a}^{Z,\pm}$ in this display are the costs for those restriction forms. These bounds keep all eight labelled occurrences. Define $\eta_{\text{tar}}^Z = 4(S_K^{Z,\text{tr}} - S_K^Z)$ by applying FTR14–16 directly to the full four-copy target quotient of rank $4t_j$; define $\eta_J^Z = \eta_{J,\text{res}}^Z = 4(S_R^{Z,\text{tr}} - S_R^Z)$ by applying them directly to the actual four-copy kernel restriction. The transported and native determinant factorizations at the same sites give

$$\begin{aligned} \eta_{\text{tar}}^Z &= \eta_L^Z + \eta_b^Z + \eta_{\text{rem}}^Z + \eta_s^Z + \eta_A^Z, & \eta_{\text{tar}}^Z + \eta_J^Z &= \eta_{\mathcal{U},\text{res}}^Z, \\ \eta_L^Z &= \eta_t^Z - \eta_s^Z + \eta_I^Z - \eta_A^Z. \end{aligned} \quad (\text{FTR18})$$

Here η_I is the quotient residual for H_{inv} in FTR8; it is distinct from the restriction residual $\eta_{I,\text{res}}$. Similarly η_{tar} denotes the full target quotient, while $\eta_{T,\text{res}}$ is one of the eight restriction residuals. These are equalities of the actual residuals. For example their first line follows by subtracting the full CPQ identity for the native metric from that for the transported metric and applying the same $\mathcal{R}_Z$. Thus the direct target interval in FTR16 can be intersected with the interval supplied by its first identity; their extremizers have not been chosen separately.

The resulting complete returns are explicitly

$$\begin{aligned} V_Y^{\text{tr}} &= \tau_Y + p_Y + \eta_Y^{\text{full}}, & \mathcal{L}_e^{\text{tr}} &= \tau_L + p_L + \eta_L^{\text{full}}, \\ 4S_K^{\text{tr}} &= 4\tau_K + 4p_K + \eta_{\text{tar}}^{\text{full}}, & 4S_R^{\text{tr}} &= 4\tau_R + 4p_R + \eta_J^{\text{full}}. \end{aligned} \quad (\text{FTR19})$$

In particular each V_Y^{tr} lies in the finite interval $[\tau_Y - E_Y^{\text{full},-}, \tau_Y + 4\mathfrak{E}_{s,R} + E_Y^{\text{full},+}]$, intersected with $[0, \infty)$. The direct allocation intervals are obtained by using $4\mathfrak{E}_{s,R}$ and dividing their four-copy endpoints by four. The exact formulas FTR18–19 retain the correlation stronger than these separate enclosures.

237.4 The complete transported threshold profile from the common matrices

Define the original finite signed count

$$\mathcal{B}_{e,s}(R) = \sum_{t=q-1,q} \int_1^\infty \sum_{a=1}^8 \sigma_a \# \{ \lambda_b(\hat{K}_{X_a,t}) > u \} \frac{du}{u}. \quad (\text{FTR20})$$

Every matrix is the full original-data FTR3–4 matrix. Each integral terminates past its finite largest eigenvalue. Set $h(x) = \log(1+x) - \log \max(1, x)$ and retain the actual finite residual

$$\epsilon_{e,s}(R) = \sum_{t=q-1,q} \sum_{a=1}^8 \sigma_a \sum_b h(\lambda_b(\hat{K}_{X_a,t})). \quad (\text{FTR21})$$

The elementary identity $\log \max(1, \lambda) = \int_1^\infty \mathbf{1}_{u < \lambda} du/u$ proves $\tau_L = \mathcal{B}_{e,s}(R) + \epsilon_{e,s}(R)$ exactly. Let $d_t = \text{rank } \mathcal{N}_t$. The two actual nested chains $J \subset W_i \subset W \subset T$ and $K_i \subset I \subset K_c \subset S$ imply

$$|\epsilon_{e,s}(R)| \leq \varepsilon_R := 2(d_{q-1} + d_q) \log 2 \leq 16m \log 2. \quad (\text{FTR22})$$

Here is the scalar error argument. Write $h = u - v$ with $u(x) = \log(1 + \min(x, 1))$ and $v(x) = 0$ for $x \leq 1$, $v(x) = \log(2x/(1+x))$ for $x \geq 1$. Both are increasing from zero and bounded by $\log 2$. For four ordered eigenvalues, the sum of the first and last disjoint increments of each function is between zero and $\log 2$, by adding its middle nonnegative increment. Their difference lies between $-\log 2$ and $\log 2$. The two flag chains therefore cost $2 \log 2$ for each of the common d_t range coordinates. Matrix order gives the eigenvalue order by the minimum–maximum characterization of Hermitian eigenvalues, whose elementary proof is included in NCT18. This proves FTR22, including singular matrices and coincident flags.

Let $\mathcal{P}_e^{\text{tr}}$ be the original transported successive-row threshold sum NPK10, with its original endpoint weights. Define its actual difference from its own determinant return by $\chi_e^{\text{tr}} = \mathcal{P}_e^{\text{tr}} - \mathcal{L}_e^{\text{tr}}$. At each row the same argument uses four eigenvalues and two chains, so its cost is $8 \log 2$. The original weights sum to $2q$, hence

$$|\chi_e^{\text{tr}}| \leq 16q \log 2, \quad \boxed{\mathcal{P}_e^{\text{tr}} = \mathcal{B}_{e,s}(R) + p_L + \epsilon_{e,s}(R) + \eta_L^{\text{full}} + \chi_e^{\text{tr}}}. \quad (\text{FTR23})$$

This compares the complete transported row sum directly with the native simultaneous matrices. In particular its directed enclosure is

$$\begin{aligned} -4\mathfrak{E}_{s,R} - \varepsilon_R - E_L^{\text{full},-} - 16q \log 2 &\leq \mathcal{P}_e^{\text{tr}} - \mathcal{B}_{e,s}(R) \\ &\leq 4\mathfrak{E}_{s,R} + \varepsilon_R + E_L^{\text{full},+} + 16q \log 2. \end{aligned} \quad (\text{FTR24})$$

The bound for $\mathcal{L}_e^{\text{tr}} - \mathcal{B}_{e,s}(R)$ omits the last $16q \log 2$ on each side. The original native threshold-to-determinant error is not paid a second time in FTR24. Each residual in FTR23 has its specified actual expression; there is no assertion that individual endpoint and row eigenvalues are equal. Replacing “full” by either partial operator in FTR18–19 gives the corresponding complete partial identities.

At the already established $R_j = \lfloor j^{3/2}/\log j \rfloor$ with $R_j < q$, IPR9 gives $\mathfrak{E}_{s,R_j} = O_{\text{actual}}(j^{5/2} + j^3/\log j) = o(kq_k)$. The finite transport bounds may be replaced by the infimum of their valid endpoint intervals over admissible radii; equivalently use NPK11’s absolute bounds for the asymptotic statement. For each fixed radius NCT21 first takes the degree limit, and only afterwards admissible radii tend to one. Since all actual flag and quotient ranks here are $O(j)$, this yields $o(kq_k)$ for every four-site or three-site radius-infimum transport cost. No fixed-radius cost is asserted to be little-oh. Since $m = O(j)$ and $q = O(j^2)$, FTR22–24 prove for each original order and primitive index that

$$\mathcal{P}_e^{\text{tr}} - \mathcal{B}_{e,s}(R_j) = o(kq_k), \quad \mathcal{L}_e^{\text{tr}} - \mathcal{B}_{e,s}(R_j) = o(kq_k). \quad (\text{FTR25})$$

The exact quantity on the right of FTR20 remains the original correlated spectral count. This proof evaluates the full receiving map and finite costs; it assigns no new asymptotic coefficient to that count. The independently retained proper source and complete arithmetic action are treated next through their actual value maps.

All prefixes in this receiver also have an explicit finite graph formula, rather than merely the bounds FTR13. Set $\mathcal{A} = \text{diag}(A, 4)$ and $\mathcal{O}_r = \text{diag}(\overline{\Omega_r}, 4)$. For every unchanged restriction frame X_0 , put

$$p_{X,\text{res}} = 2 \log \det(X_0^* \mathcal{A}^* \mathcal{O}_R \mathcal{A} X_0) - \log \det(X_0^* \mathcal{A}^* \mathcal{O}_0 \mathcal{A} X_0) - \log \det(X_0^* \mathcal{A}^* \mathcal{A} X_0). \quad (\text{FTR26})$$

The target cutoffs $q+R, q, q-1$ correspond to native graph indices $R, 0, -1$, respectively, so NTG20 and FTR11 prove this formula including $R = 0$. Every quotient prefix is the difference $p_{Y,\text{res}} - p_{K,\text{res}}$ in its fixed pair; in particular p_I in FTR12–13 equals $p_{W_i,\text{res}} - p_{J,\text{res}}$. Every signed prefix is the original eight-term combination. Frame constants cancel because the coefficients in FTR26 sum to zero. Thus every prefix in FTR19 and FTR23 is itself evaluated in the same original finite coefficients.

For $R = R_j$ and $t = q - 1, q$ let $\mathfrak{U}_{R,t}$ be the minimum of the three explicitly evaluated bounds NTG31:

$$\mathfrak{U}_{R,t} = \min \left\{ \sum_{a=R+1}^t E_{j,s,a}, \quad d_0 \log \left(1 + \frac{\sum_{a=R+1}^t x_a^* \Omega_R^{-1} x_a}{d_0} \right), \right. \\ \left. \sum_{a=R+1}^t \log(1 + x_a^* \Omega_R^{-1} x_a) - \Delta_{R,t} \right\}, \quad d_0 = \min(m, t - R), \quad (\text{FTR27})$$

where $\Delta_{R,t}$ is the full adjacent-pair correction NTG28, including each actual $x_a^* \Omega_R^{-1} x_b$. At an empty endpoint set $\mathfrak{U}_{R,R} = 0$. Define $\mathfrak{U}_R = \mathfrak{U}_{R,q-1} + \mathfrak{U}_{R,q}$. The complete Schur determinant NTG25 proves $0 \leq \tau_{\text{total}} \leq \mathfrak{U}_R$. Each quotient ratio is at most its four-copy ambient ratio, as proved for FTR13, and each allocation is nonnegative with sum τ_{total} . The original two-chain signed estimate NIB16 therefore yields simultaneously

$$0 \leq \tau_K, \tau_R \leq \mathfrak{U}_R, \quad 0 \leq \tau_Y \leq 4\mathfrak{U}_R \quad (Y = b, \text{rem}, s, A, I, t), \\ |\tau_L| \leq 4\mathfrak{U}_R, \quad |\mathcal{B}_{e,s}(R)| \leq 4\mathfrak{U}_R + \varepsilon_R. \quad (\text{FTR28})$$

For the signed bound, the same-chain disjoint-increment argument applied to $\log(1+x)$ at an original row bounds the absolute signed increment by $4\log(1+a_r)$; summing the tail and using NTG32 proves it directly. The count bound follows from FTR21–22. These estimates may be inserted in the directed intervals FTR19 and FTR24 without deleting any of their correlated residuals. No assertion that $\mathfrak{U}_R = o(kq_k)$ is used here.

Finally retain both original source orders, with a subscript 1 or j on each actual matrix and residual. For every positive quotient in FTR8 the exact order difference is

$$V_{Y,j}^{\text{tr}} - V_{Y,1}^{\text{tr}} = (\tau_{Y,j} - \tau_{Y,1}) + (p_{Y,j} - p_{Y,1}) + (\eta_{Y,j}^{\text{full}} - \eta_{Y,1}^{\text{full}}). \quad (\text{FTR29})$$

Its excess above the displayed tail difference lies in

$$[-4\mathfrak{E}_{1,R} - E_{Y,j}^{\text{full},-} - E_{Y,1}^{\text{full},+}, \\ 4\mathfrak{E}_{j,R} + E_{Y,j}^{\text{full},+} + E_{Y,1}^{\text{full},-}]. \quad (\text{FTR30})$$

This follows by subtracting the two actual FTR19 equalities, using nonnegative prefixes FTR13 and directed errors FTR16. For the signed profile use its two actual FTR23 identities instead; the prefix difference has absolute bound $4(\mathfrak{E}_{j,R} + \mathfrak{E}_{1,R})$, its two simultaneous errors cost $\varepsilon_{R,j} + \varepsilon_{R,1}$, its directed transport costs have the same subtracted signs as FTR30, and the two original threshold residuals cost $32q \log 2$. This propagates the new matrices through the earlier two-order receivers without treating a known difference as an absolute value.

238 The original invariant value fibre and its full proper-source return

This calculation propagates the original ITL1–6, NFR1–12, CPQ2–20 and EFP19,28–31 through the simultaneous matrices STF4–18 and the exact relation-graph map NFI1–10. The classical quadratic minimum and block determinant identity used at each quotient are those in Boyd and Vandenberghe, Appendix A.5.5, p. 650, equations (A.13)–(A.14), and Appendix C.4.1, p. 673, equations (C.4)–(C.5) [H23]. The complex Hermitian calculation, every fibre and every map needed here are given below. The original programme maps and results remain attributed to their cited ITL/NFR/CPQ/EFP and NTG/STF/NFI providers; the classical citation supplies the finite linear algebra operation.

238.1 The unchanged domain and actual quotient values

Retain the original hypothetical simple quartet, five-orbit period, logarithm branch, complete nonreduced conductor E_A , its order $v = \text{ord}_0 E_A$ and the full nonzero derivative $\mu_v = E_A^{(v)}(0)$. The data and original rank-valid domain are those of ITL1–3 and CPQ1–3, including the actual period-domain condition $|u| \geq R_*$ wherever it is required by those maps. Put

$$\begin{aligned} k &= 4l + 1 \geq 9, & j &= k + 4, & q_a &= (a + 1)^2, & q'_a &= (a - 7)^2, & m_a &= q_a - v - q'_a, \\ c'_a &= a/2 - 4, & \alpha_A &= v!/\mu_v, & p = 0 &: (s_0, s_1) = (1, 1), & p = 1 &: (s_0, s_1) = (k, j), \\ q &= q_j, & N &\in [q - 1, 2q] \cap \mathbb{Z}, & D &= N - v. \end{aligned} \quad (\text{PRD1})$$

The simple-quartet domain fixes q_a as displayed; no change of quartet multiplicity is made. Its conductor order v is retained even when the selected period makes it nonzero. Every source-order identity below holds separately for $p = 0, 1$. Each connecting identity holds separately for both primitive indices $e = 0, 1$.

Use the full original coefficient multiplier, maps and spaces

$$\begin{aligned} \mathcal{V} &= \mathbb{C}^{m_k}, & \mathcal{U} &= \mathbb{C}^{4m_j}, & V &= \mathbb{C}^{j_{0,k}}, & J_0 &: V \longrightarrow \mathcal{V}, \\ T_a &= \alpha_A N_a Z_{\text{exp},a}, & M_i &= T_j^{-1} m_{i,k} T_k, & m_{i,k}[f] &= [x_i^4 f], & B_{\text{cof}} &= \text{stack}_{i=1}^4 M_i, \\ F &= B_{\text{cof}} J_0 : V \longrightarrow \mathcal{U}, & \mathbb{B} &= \text{diag}(B_j, 4) : \mathcal{U} \longrightarrow \mathbb{C}^{4t_j}, & J &= \ker \mathbb{B}, \\ X &= \mathbb{B}F, & E &= \ker X, & I &= \text{im } F, & K_i &= I \cap J, & W_i &= I + J. \end{aligned} \quad (\text{PRD2})$$

The maps F and J_0 are injective and B_j is onto by the original ITL2–3/CPQ2 proof. Their ordinary-transpose coefficient duality is unchanged; stars below are complex adjoints only in Hermitian metric expressions. In particular the original maps $N_a, Z_{\text{exp},a}$ and the multiplier $m_{i,k}$ remain in PRD2.

Choose exactly the original fixed ITL4 frame $Z = [Z_E, Z_R]$ of V , where Z_E spans E . Write

$$e_E = \dim E, \quad r_I = \dim V - e_E, \quad F_E = FZ_E, \quad F_R = FZ_R, \quad U_R = XZ_R = \mathbb{B}F_R. \quad (\text{PRD3})$$

The frame F_R is the original coefficient lift of U_R . The frame U_R is injective: if $XZ_R x = 0$, then $Z_R x \in E$, while the complement property of Z forces $x = 0$. Also

$$FE = K_i, \quad I = FE \oplus \text{im } F_R, \quad W_i = J \oplus \text{im } F_R. \quad (\text{PRD4})$$

Indeed $Fa \in J$ is equivalent to $Xa = 0$, which is $a \in E$. This proves the first equality. Decomposing $a = Z_E b + Z_R x$ proves the second, using injectivity of F ; adding J proves the third, and its directness follows by applying $\mathbb{B}$.

For complete value maps let $L_R = (U_R^* U_R)^{-1} U_R^*$, the fixed left inverse on $\text{im } U_R$. The two maps and their inverses are

$$\begin{aligned} I/K_i &\longrightarrow \mathbb{C}^{r_I}, & [i] &\longmapsto L_R \mathbb{B}i, & x &\longmapsto [F_R x] & (\text{inverse}), \\ W_i/J &\longrightarrow \mathbb{C}^{r_I}, & [w] &\longmapsto L_R \mathbb{B}w, & x &\longmapsto [F_R x] & (\text{inverse}). \end{aligned} \quad (\text{PRD5})$$

These rules are well defined because $\mathbb{B}$ kills the indicated kernels; PRD4 proves that they are mutual inverses. Composing either with $x \mapsto U_R x$ identifies its values with the very same original invariant image in $\mathbb{C}^{4t_j}$. The inclusion $I \hookrightarrow W_i$ therefore induces the exact isomorphism

$$\iota_I : I/K_i \longrightarrow W_i/J, \quad [i] \longmapsto [i]. \quad (\text{PRD6})$$

Its inverse sends $[i + j]$ to $[i]$: if two decompositions represent the same class, their I components differ by an element of $I \cap J = K_i$. Both maps are the identity on the specified $\mathbb{C}^{r_I}$ values. This proves their relationship before any metrics or determinant ratios are introduced.

238.2 All three full metrics and their attained lifts

Put

$$\mathcal{D}_N^p = \text{diag}(D_{j,N-v}^{(s_1)}, 4), \quad G_N^p = J_0^* D_{k,N-v}^{(s_0)} J_0. \quad (\text{PRD7})$$

Each $D_{a,D}^{(s)}$ is the original positive coefficient metric after its entire lower-evaluation minimum. More explicitly, with the original lower and analytic-numerator rows through degree D ,

$$D_{a,D}^{(s)} = T_{a,D}^* [I - V_{a,D} (V_{a,D}^* V_{a,D})^{-1} V_{a,D}^*] T_{a,D}. \quad (\text{PRD8})$$

This is the MNT/DNE form used in ITL1 and STF2. Its rows retain the original centre c'_a , conductor division moments and phases, and the full squared norms $h_{n,s} = (2\pi)^{s/2} n!(s/2)_n$. PRD8 requires the original degree- k rows when $a = k$ and degree- j rows when $a = j$. Positivity and full lower rank on PRD1 are the original MNT/DNE rank results; no new rank assumption is introduced.

Here is the precise complex Hermitian version of the classical minimum used below [H23, Appendix A.5.5, p. 650]. For a positive Hermitian P , an independent kernel frame K_0 and value lifts L_0 independent modulo that kernel, set

$$\begin{aligned} \text{Sch}(P; K_0, L_0) &= L_0^* P L_0 - L_0^* P K_0 (K_0^* P K_0)^{-1} K_0^* P L_0, \\ z_P(x) &= L_0 x - K_0 (K_0^* P K_0)^{-1} K_0^* P L_0 x. \end{aligned} \quad (\text{PRD9})$$

Every representative is $L_0 x + K_0 a$. Its squared norm is

$$(a + (K_0^* P K_0)^{-1} K_0^* P L_0 x)^* (K_0^* P K_0) (a + (K_0^* P K_0)^{-1} K_0^* P L_0 x) + x^* \text{Sch}(P; K_0, L_0) x.$$

The first term has a unique zero, proving PRD9's attained minimum and lift. A triangular change of determinant one from $[K_0, L_0]$ to $[K_0, z_P]$ diagonalizes its Gram by blocks; therefore its determinant is $\det(K_0^* P K_0) \det \text{Sch}(P; K_0, L_0)$. This proves the complex form of the cited Schur operation with the actual affine fibre and all cross terms retained.

Choose any fixed independent frame J_* for the entire J . The three forms on the same original Z_R values are exactly

$$\begin{aligned} Q_N^p &= \text{Sch}(G_N^p; Z_E, Z_R), \\ A_N^p &= \text{Sch}(\mathcal{D}_N^p; F_E, F_R), \\ H_N^p &= \text{Sch}(\mathcal{D}_N^p; J_*, F_R) = U_R^* (\mathbb{B}(\mathcal{D}_N^p)^{-1} \mathbb{B}^*)^{-1} U_R. \end{aligned} \quad (\text{PRD10})$$

Thus A_N^p is precisely A_{inv}^p for (I, K_i) , and H_N^p is precisely H_{inv}^p for (W_i, J) . To prove the last equality directly, for a target value y the vector

$$z = (\mathcal{D}_N^p)^{-1} \mathbb{B}^* (\mathbb{B}(\mathcal{D}_N^p)^{-1} \mathbb{B}^*)^{-1} y$$

satisfies $\mathbb{B}z = y$ and is $\mathcal{D}_N^p$ -orthogonal to J . Every other lift differs from it by J , so its norm is the full minimum. Taking $y = U_R x$ proves PRD10 and agrees with the third attained lift in PRD9. Because $K_i \subset J$, the two target-order minima also give $H_N^p \preceq A_N^p$. The same-valued metric map

in PRD6 is exactly this contraction; the independent source metric is Q_N^p with the first formula in PRD10.

In particular, every entry of Q_N^p is the finite original degree- k expression

$$Q_N^p = Z_R^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_R - Z_R^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_E (Z_E^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_E)^{-1} Z_E^* J_0^* D_{k,N-v}^{(s_0)} J_0 Z_R. \quad (\text{PRD11})$$

For $E = 0$ every term containing its frame is absent. For $r_I = 0$ all value forms are the unique operators on the zero space, their determinants are 1 and all value returns below are 0. These conventions apply throughout, including PRD5's empty maps.

238.3 Literal two-endpoint matrices on those same values

Fix an original integer $0 \leq R < q$, put $N_* = q + R$ and $\mathcal{H} = \mathcal{D}_{N_*}^p$. Use the already calculated NFI matrices

$$\mathcal{Z}_t = \text{diag}(\overline{\Omega_R}^{-1/2} \overline{Y}_t, 4), \quad Y_t = [x_{R+1}, \dots, x_t], \quad t = q-1, q, \quad (\text{PRD12})$$

including the empty tail at $R = t = q-1$. These x_r, Ω_R are the full NTG relation coefficients at target degree j and source order s_1 ; the exact phase/permutation maps NFI3–4 are retained. Let U_4 be the NFI1 unitary. For either original pair $(Y, K) = (W_i, J)$ or (I, K_i) , use the value lifts F_R and respectively the kernel frames J_* or F_E , and define

$$\begin{aligned} Z_{Y/K} &= F_R - K_*(K_*^* \mathcal{H} K_*)^{-1} K_*^* \mathcal{H} F_R, & Q_{Y/K,*} &= Z_{Y/K}^* \mathcal{H} Z_{Y/K}, \\ E_{Y/K} &= \mathcal{H}^{1/2} Z_{Y/K} Q_{Y/K,*}^{-1/2}, & \hat{E}_{Y/K} &= U_4 E_{Y/K}, & \hat{P}_K &= U_4 P_K U_4^*, \\ \mathcal{J}_{Y/K,t} &= I_{r_I} + \hat{E}_{Y/K}^* \mathcal{Z}_t (I + \mathcal{Z}_t^* \hat{P}_K \mathcal{Z}_t)^{-1} \mathcal{Z}_t^* \hat{E}_{Y/K}. \end{aligned} \quad (\text{PRD13})$$

The original P_K is STF5's projector onto $\mathcal{H}^{1/2} K$. Every displayed kernel term is omitted when the kernel is zero. The same complete quadratic minimum PRD9, now after stacking the actual tail rows, gives STF10; conjugating its row space by the explicit unitary NFI3 gives

$$H_{q+t}^p = (H_{N_*}^p)^{1/2} \mathcal{J}_{W_i/J,t} (H_{N_*}^p)^{1/2}, \quad A_{q+t}^p = (A_{N_*}^p)^{1/2} \mathcal{J}_{I/K_i,t} (A_{N_*}^p)^{1/2}. \quad (\text{PRD14})$$

For clarity, the minimization giving the middle factor has the literal form $\min_a \{\|a\|^2 + \|y + Ba\|^2\}$, with $B = C_t Z_K$ and $y = C_t E_{Y/K} Q_{Y/K,*}^{1/2} x$. Here C_t is the STF4 stack of the actual original row updates, right-multiplied by $\mathcal{H}^{-1/2}$, and Z_K is any orthonormal column frame of $\mathcal{H}^{1/2} K$. Its minimizer is $-(I + B^* B)^{-1} B^* y$; multiplication proves $I - B(I + B^* B)^{-1} B^* = (I + B B^*)^{-1}$, yielding PRD13. The NFI3 row conjugation changes C_t into $\mathcal{Z}_t^* U_4$. This proves PRD14 with every cross term, including singular tail matrices. Its $Q_{Y/K,*}^{1/2}$ factors remain on both sides in the original quotient-value space; they are not scalar masses.

Let $F_t(Y)$ retain exactly STF5/NFI8's restriction log ratio. Applying the block determinant identity proved after PRD9 to the fixed frame $[K_*, F_R]$ at the two cutoffs gives

$$\begin{aligned} \log \det \mathcal{J}_{W_i/J,t} &= F_t(W_i) - F_t(J), \\ \log \det \mathcal{J}_{I/K_i,t} &= F_t(I) - F_t(K_i). \end{aligned} \quad (\text{PRD15})$$

Every fixed frame determinant cancels in its own two-cutoff ratio. Define the two exact tail values

$$\tau_I^p = \sum_{t=q-1,q} \log \det \mathcal{J}_{W_i/J,t}, \quad \tau_A^p = \sum_{t=q-1,q} \log \det \mathcal{J}_{I/K_i,t}. \quad (\text{PRD16})$$

They are nonnegative because PRD13 is the identity plus a positive semidefinite matrix. This is the same value for the target invariant form whether obtained by its complete J minimum or by its restriction to the original image U_R .

238.4 The full relative return, with the degree- k source intact

For any value form Z_N let

$$\Pi_Z^p = 2 \log \det Z_{N_*}^p - \log \det Z_q^p - \log \det Z_{q-1}^p. \quad (\text{PRD17})$$

Use this definition for $Z = H, A$, writing their prefixes as Π_I^p, Π_A^p . Direct addition of PRD14's two logarithmic determinants and PRD17 gives

$$V_I^p = \Pi_I^p + \tau_I^p, \quad V_A^p = \Pi_A^p + \tau_A^p. \quad (\text{PRD18})$$

These are the original four-endpoint returns with signs $(-1, -1, +1, +1)$ at $N = (q-1, q, 2q-1, 2q)$, not new windows. Define the proper-source term directly from PRD11 at those four original arguments:

$$V_Q^p = \log \frac{\det Q_{2q-1}^p \det Q_{2q}^p}{\det Q_{q-1}^p \det Q_q^p}, \quad \mathcal{R}_{X,N}^p = (Q_N^p)^{-1/2} H_N^p (Q_N^p)^{-1/2}. \quad (\text{PRD19})$$

Thus at each upper endpoint its complete relative positive operator is

$$\mathcal{R}_{X,q+t}^p = (Q_{q+t}^p)^{-1/2} (H_{N_*}^p)^{1/2} \mathcal{J}_{W_i/J,t} (H_{N_*}^p)^{1/2} (Q_{q+t}^p)^{-1/2}. \quad (\text{PRD20})$$

The order of these generally noncommuting matrices is part of the formula. Taking determinants, with the exact inverse square root on both sides, proves the full finite receiving identities

$$\begin{aligned} \mathcal{M}^p &= \mathcal{R} \log \det \mathcal{R}_X^p = \Pi_I^p + \tau_I^p - V_Q^p, \\ \mathcal{R} \log \frac{\det A_{\text{inv}}^p}{\det Q_{\text{inv}}^p} &= \Pi_A^p + \tau_A^p - V_Q^p, \\ \mathcal{R} \log \omega_{\text{inv}}^p &= (\Pi_I^p - \Pi_A^p) + (\tau_I^p - \tau_A^p). \end{aligned} \quad (\text{PRD21})$$

The first line is the NFR matched return, the second its actual same-value source correction, and the third the original last target contraction. Their equality uses the proved maps PRD5–6, not an identification of the degree- k and degree- j metrics.

The proper source in every term of PRD19 is evaluated at

$$(D_0, D_1, D_2, D_3) = (q_j - v - 1, q_j - v, 2q_j - v - 1, 2q_j - v). \quad (\text{PRD22})$$

Its degree is k , centre $c'_k = k/2 - 4$, and order $s_0 = 1$ or k . Writing $b = q_j - q_k = 8k + 24$ proves that its two low cutoffs are the standard degree- k low cutoffs plus b , and the two high cutoffs are the standard degree- k high cutoffs plus $2b$. The target matrices in PRD12–16 instead retain degree j , centre $c'_j = j/2 - 4$ and order $s_1 = 1$ or j . Consequently PRD21 keeps the exact same Z_R values while explicitly retaining both original polynomial row families and all four source displacements. No estimate for V_Q^p has been inserted in this identity.

The exact target-prefix cost can also be retained as a finite interval. For any original $K \subset Y \subset \mathcal{U}$, restriction and minimum over the fixed fibre preserve the increasing metric $\mathcal{D}_{N_*}^p$. Complete a fixed frame for K , then the specified lifts for Y/K , then lifts for $\mathcal{U}/Y$. Successive Schur factorization expresses the ambient determinant increment as the sum of three nonnegative log determinant increments. Thus the quotient increment lies between zero and the ambient increment. For either pair in PRD13 the latter is $4 \log(1 + a_r)$ by the four-copy original IPR1 update. With the original weights $\vartheta_0 = \vartheta_q = 1$ and $\vartheta_r = 2$ inside the window, this proves

$$0 \leq \Pi_I^p, \Pi_A^p \leq 4 \sum_{r=0}^R \vartheta_r \log(1 + a_r) \leq 4 \mathfrak{E}_{s_1, R}. \quad (\text{PRD23})$$

The last finite constant is exactly the already proved IPR2/NIB14 budget. In particular, writing $Z = I$ or A and interpreting its left side respectively as $\mathcal{M}^p$ or $\mathcal{R} \log(\det A_{\text{inv}}^p / \det Q_{\text{inv}}^p)$, the full finite interval is

$$\tau_Z^p - V_Q^p \leq \Pi_Z^p + \tau_Z^p - V_Q^p \leq \tau_Z^p - V_Q^p + 4\mathfrak{E}_{s_1, R}. \quad (\text{PRD24})$$

It has the original independent proper-source term on both sides. At the already established $R_j = \lfloor j^{3/2} / \log j \rfloor$ on its original $R_j < q$ domain, this width is $o(kq_k)$ by IPR9. This bounds the removed target prefix; it does not assign an asymptotic value to either retained term.

238.5 The connecting receiver and all compatibility terms

Use precisely the STF13/CPQ flags S, K_c, W, T and the original connecting columns at primitive index e . Their quotient pairs for $G_t, G_s, A_{\text{inv}}, H_{\text{inv}}$ are respectively $(T, W), (S, K_c), (I, K_i), (W_i, J)$. Applying PRD15 to these pairs and substituting the actual eight flags of STF16 gives, by literal cancellation of all eight labelled occurrences,

$$\mathcal{L}_e^{p, > R} = \tau_I^p + \tau_t^p - \tau_s^p - \tau_A^p, \quad \mathcal{L}_e^{p, \leq R} = \Pi_I^p + \Pi_t^p - \Pi_s^p - \Pi_A^p. \quad (\text{PRD25})$$

Here $\tau_t = \sum_{h=q-1, q} [F_h(T) - F_h(W)]$ and $\tau_s = \sum_{h=q-1, q} [F_h(S) - F_h(K_c)]$. Prefixes are the original PRD17 values of G_t, G_s . Coincident flags retain their positions and signs. The prefix identity follows from the same fixed determinant identities because its evaluation coefficients sum to zero. Adding the two equalities proves exactly the NFR7 receiver

$$\mathcal{M}^p = \mathcal{L}_e^p - V_t^p + V_s^p + V_A^p - V_Q^p = \Pi_I^p + \tau_I^p - V_Q^p. \quad (\text{PRD26})$$

In particular the independent V_Q^p does not enter the four-target-copy CPQ allocation identity. It enters through the proved same-value relative operator PRD19–20.

Finally let $\mathcal{C}^p$ retain NFR10/EFP31's complete original compatibility return, including its source-complement projections, and retain the actual endpoint errors of NFR2:

$$\begin{aligned} a_i^p &= 2r_I \log \max(1, K_-^p(D_i)) + r_I \log c_+^p(D_i), \\ b_i^p &= 2r_I \log \max(1, K_+^p(D_i)) - r_I \log c_-^p(D_i), \\ E_p^- &= -b_0^p - b_1^p - a_2^p - a_3^p, \quad E_p^+ = a_0^p + a_1^p + b_2^p + b_3^p. \end{aligned} \quad (\text{PRD27})$$

NFR5's exact equality is $\mathcal{M}^p = \mathcal{C}^p + \mathcal{R} \log \omega_{\text{inv}}^p + \mathcal{E}^p$ with $E_p^- \leq \mathcal{E}^p \leq E_p^+$. Substituting all three lines of PRD21, before estimating any term, proves the sharpened receiving expression

$$\begin{aligned} \mathcal{C}^p &= \Pi_A^p + \tau_A^p - V_Q^p - \mathcal{E}^p, \\ \Pi_A^p + \tau_A^p - V_Q^p - E_p^+ &\leq \mathcal{C}^p \leq \Pi_A^p + \tau_A^p - V_Q^p - E_p^-. \end{aligned} \quad (\text{PRD28})$$

This preserves the four actual compatibility constants with the two low signs reversed. It also proves the exact relation between the complete compatibility return and the proper pullback correction without dropping their finite error.

All existing source-order intervals therefore receive these expressions by literal substitution. For example, on EFP29's already proved finite guards, with Δ meaning order one minus order zero,

$$r_I(\Phi_j^- - \Phi_k^+) \leq \Delta(\Pi_I + \tau_I - V_Q) \leq r_I(\Phi_j^+ - \Phi_k^-). \quad (\text{PRD29})$$

The degree- k constants here are evaluated at PRD22, exactly as EFP28–29 prescribe, and their target constants remain degree j . For EFP31 its same unchanged full compatibility interval applies to $\Delta(\Pi_A + \tau_A - V_Q - \mathcal{E})$ by PRD28. No guard is enlarged: EFP28's sufficient $k \geq 37$ and actual QOT root tests, and the separate EOR tests wherever used, remain those of the source theorem. Outside those comparison guards, the exact finite matrix identities PRD1–28 retain their original rank-valid domain.

239 The analytic tail inside the complete original action

This continuation preserves the programme's recorded human origin and the separate recorded creators and dates of its deposited source notes [H19, H20]. Its new use of the accepted NTG/STF/NFI identities is the complete original-action return proved below. The exact packet morphisms extend the saved typed-action calculation; the full source lineage is recorded in the accompanying private use ledger. Every new finite Hermitian factorization is proved here, with the classical Schur references cited where used.

All statements here retain the hypothetical original simple quartet, the fixed actual period and branch, every conductor coefficient, and the original source masses. Write the actual packet degree as a until its value is specified. In the current NTG/STF/NFI calculation it is $a = j = k + 4$, not k . Set

$$\begin{aligned} q &= q_a = (a + 1)^2, & q' &= q'_a = (a - 7)^2, \\ m &= q - q' - v, & N_i &= (q - 1, q, 2q - 1, 2q)_i, & D_i &= N_i - v, \\ s &\in \{1, a\}, & a &= 4l + 1 \geq 9. \end{aligned} \quad (\text{TAC1})$$

The original source centres are $c_0 = a/2$ for the whole numerator packet and $c_1 = a/2 - 4$ for the divided analytic functionals. The common Gamma measure and orthonormal polynomial phases are precisely

$$\begin{aligned} dm_s(y) &= \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, & M_s &= (2\pi)^{s/2}, \\ p_{0,s} &= 1, & p_{1,s} &= y, & p_{n+1,s} &= yp_{n,s} - n(n-1+s/2)p_{n-1,s}, \\ h_{n,s} &= M_s n! (s/2)_n, & \phi_{n,s,c}(S) &= p_{n,s}((S-c)/i)/\sqrt{h_{n,s}}. \end{aligned} \quad (\text{TAC1a})$$

The classical Meixner–Pollaczek dictionary is $p_{n,s}(y) = n! P_n^{(s/4)}(y/2; \pi/2)$; the substitution, including $dy = 2d(y/2)$, gives the displayed mass and norms, as calculated in NTG3 and HPS1–5 [H21, eqs. (18.19.8)–(18.19.9), (18.22.8)]. The symbol m in TAC1 is the analytic coefficient dimension; CAI's quartet multiplicity is exactly one here. This calculation does not extend the NTG result to a multiple quartet.

239.1 Both stages of the analytic conductor quotient

Here are the complete maps from the original packet. Put

$$\begin{aligned} \beta_{a,u,w} &= a/2 + (2u-a)\delta + i(2w-a)\gamma, & \chi_a(S) &= \prod_{u,w=0}^a (S - \beta_{a,u,w}), \\ \chi_-(S') &= \chi_{a-8}(S'), & E &= \mathbb{C}[S]/(\chi_a), & E_- &= \mathbb{C}[S']/(\chi_-), \\ E_A(z) &= \sum_{r,t=0}^8 a_{rt} e^{\beta_{8,r,t} z}, & \mu_n &= E_A^{(n)}(0), & v = \text{ord}_0 E_A &\leq 80, & \mu_v &\neq 0, \\ (\mathcal{T}_A P)(S') &= \sum_{r,t=0}^8 a_{rt} P(S' + \beta_{8,r,t}), & \bar{\mathcal{T}}_A : E &\longrightarrow E_-, & [P] &\longmapsto [\mathcal{T}_A P], \\ L_v &= \{[P] : \deg P < v\} \subset E, & V_A &= \ker \bar{\mathcal{T}}_A. \end{aligned} \quad (\text{TAC1b})$$

The complete coefficients of E_A remain as in the original fixed period; no repeated contribution is removed. Distinctness of the grid roots follows from $\delta > 0$ and $\gamma > 0$ by comparing real and imaginary parts. For every lower root $\beta_{a-8,u,w}$, $\beta_{a-8,u,w} + \beta_{8,r,t} = \beta_{a,u+r,w+t}$. Consequently $\chi_- \mid \mathcal{T}_A(\chi_a P)$, proving that the packet arrow in TAC1b is well defined. Its matrix in upper and lower root-value coordinates is

$$C_{(u,w),(i,j)} = \begin{cases} a_{i-u,j-w}, & 0 \leq i-u, j-w \leq 8, \\ 0, & \text{otherwise.} \end{cases} \quad (\text{TAC1c})$$

Indeed evaluation of the complete shift sum gives this entry, with all contributions retained. For a lower coefficient column η ,

$$\mathcal{N}_{C^\top \eta}(z) = E_A(z) \sum_{u,w=0}^{a-8} \eta_{uw} e^{\beta_{a-8,u,w} z}.$$

If this vanishes, the lower sum vanishes on an open disk on which $E_A \neq 0$, hence identically; its distinct-root Vandermonde then kills η . Thus $C^\top$ is injective and C is onto. In particular $\dim V_A = q - q'$. The polynomial identity

$$\mathcal{T}_A S^n = \sum_{d=0}^n \binom{n}{d} \mu_d S^{n-d}$$

shows that the exact polynomial kernel is $\mathbb{C}[S]_{<v}$ and that the degree of every nonzero image of degree $n \geq v$ is $n - v$, with leading coefficient multiplied by $\binom{n}{v} \mu_v$. Triangular elimination proves surjectivity onto every retained target polynomial space. Since $v < q$, L_v injects into E , and $L_v \subset V_A$.

For any $N \geq q - 1$, put $D = N - v$ and $b = N - q \geq -1$; negative polynomial degree means the zero space. Define the full relation map and both quotient spaces by

$$\begin{aligned} \mathcal{U}_b h &= \mathcal{T}_A(\chi_a h) / \chi_-, & X_N &= \mathbb{C}[S']_{\leq D} / \mathcal{T}_A(\chi_a \mathbb{C}[S]_{\leq b}), \\ Y_N &= \mathbb{C}[S']_{\leq D-q'} / \mathcal{U}_b \mathbb{C}[S]_{\leq b}, \\ \kappa_N : E/L_v &\longrightarrow X_N, & [P] + L_v &\longmapsto [\mathcal{T}_A P], \\ \theta_N : V_A/L_v &\longrightarrow Y_N, & [P] + L_v &\longmapsto [\mathcal{T}_A P / \chi_-]. \end{aligned} \tag{TAC1d}$$

Choose a polynomial representative of degree at most N in these formulas. Changing it by $\chi_a h$ changes the two images by their respective displayed relation; changing it by a low polynomial changes neither image. Surjectivity of $\mathcal{T}_A$ proves surjectivity of κ_N . If its value is zero, subtract the corresponding $\chi_a h$; the result has degree below v , proving injectivity. The same argument after multiplying a prescribed Y_N representative by χ_- proves both surjectivity and injectivity of θ_N . Every required preimage has degree at most N , by the proved exact degree drop. Thus these are proved isomorphisms, with inverse given by any such preimage followed by its remainder class modulo L_v .

Multiplication by χ_- and reduction modulo χ_- give the commutative diagram of exact sequences

$$\begin{array}{ccccccc} 0 & \longrightarrow & V_A/L_v & \longrightarrow & E/L_v & \xrightarrow{\overline{\mathcal{T}}_A} & E_- \longrightarrow 0 \\ & & \theta_N \downarrow & & \kappa_N \downarrow & & \parallel \\ 0 & \longrightarrow & Y_N & \xrightarrow{\chi_-} & X_N & \xrightarrow{\text{mod } \chi_-} & E_- \longrightarrow 0. \end{array} \tag{TAC1e}$$

The bottom left map is injective: $\chi_- g = \mathcal{T}_A(\chi_a h)$ is equivalent to $g = \mathcal{U}_b h$ in the polynomial integral domain. Its image is the kernel of the bottom right map, since every polynomial vanishing at all lower roots is divisible by χ_- . That right map is onto because $D \geq q - 1 - v \geq q' - 1$. This proves each exactness claim and the commutativity on actual representatives. Finally $\mathcal{U}_b$ is injective: for nonzero h of degree t , its image has degree $q + t - v - q' = m + t$ and leading coefficient $\text{lc}(h) \binom{q+t}{v} \mu_v \neq 0$. Subtracting the actual polynomial dimensions gives

$$\dim(E/L_v) = \dim X_N = q - v, \quad \dim(V_A/L_v) = \dim Y_N = q - v - q' = m. \tag{TAC1f}$$

This displays both stages with their maps and their complete relation spaces. In the target application $a = j = k + 4$, the NTG coefficient metric has the latter dimension $m = 16j - 48 - v$.

The algebraic pairing identifying its analytic dual is equally exact. Write $\mathbb{V}[n, \beta] = \beta^n$ and let p be the monomial remainder coefficient column. The isomorphism

$$\Phi : \mathbb{C}^q \longrightarrow E^\vee, \quad \Phi(\nu)([P]) = \sum_{\beta} \nu_{\beta} P(\beta) = (\mathbb{V}\nu)^\top p \tag{TAC1g}$$

is complex linear; the invertible Vandermonde gives its inverse. The first v moment conditions say exactly $\Phi(\mathcal{N}_v) = \text{ann}(L_v)$. For $\eta \in \mathbb{C}^{q'}$, $\Phi(C^\top \eta)([P]) = \sum_{\beta'} \eta_{\beta'} (\mathcal{T}_A P)(\beta')$, the actual dual of the onto conductor. Its image is therefore $\text{ann}(V_A)$: one inclusion follows by evaluation, and both dimensions are q' . Restriction of functionals induces

$$\Psi : \mathcal{N}_v / \text{im } C^\top \xrightarrow{\sim} (V_A / L_v)^\vee, \quad [\nu] \mapsto ([P] + L_v \mapsto \sum_{\beta} \nu_{\beta} P(\beta)). \quad (\text{TAC1h})$$

Adding a low polynomial or a conductor coefficient changes that value by zero. Its kernel is exactly the stated conductor image. Any functional on V_A / L_v extends linearly from V_A to E while still vanishing on L_v ; its unique Φ representative lies in $\mathcal{N}_v$. This proves surjectivity without a metric choice. For $F = \mathcal{N}_v / E_A$, analyticity follows from the first v vanishing moments. The exact finite Leibniz identity is

$$\begin{aligned} \Lambda_F(\mathcal{T}_A P) &= \Lambda_{E_A F}(P) = \Phi(\nu)([P]), \\ \ell_F(g) &= \Lambda_F(\chi_- g), \quad \ell_F(\theta_N([P] + L_v)) = \Psi([\nu])([P] + L_v). \end{aligned} \quad (\text{TAC1i})$$

For proof on $P = S^n$, expand the complete derivative product as $\sum_{d=v}^n \binom{n}{d} \mu_d F^{(n-d)}(0)$; finite linearity proves the identity for every retained polynomial. On a relation $g = \mathcal{U}_b h$ its value is $\Phi(\nu)([\chi_a h]) = 0$. Thus it is a well-defined dual pairing on Y_N . A lower exponential has zero pairing because $\chi_-(\beta') = 0$; conversely TAC1h and the isomorphism θ_N identify exactly that denominator via $C^\top$. Every scalar is one in TAC1d–i. The original $\alpha_A = v!/\mu_v$ remains in the separate primary-coordinate maps of NFR1 and is not inserted into these unscaled conductor maps. These arguments prove the CEP3–6,42–43 and NCT3–11 bridge on its original objects and coordinates.

239.2 The exact subquotient and its two complementary factors

Let $\mathbb{V}$ have entries $\mathbb{V}[n, \beta] = \beta^n$, $0 \leq n < q$, in the original upper-root order. For $\nu \in \mathbb{C}^q$ put $\mathcal{N}_\nu(z) = \sum_{\beta} \nu_{\beta} e^{\beta z}$. Define

$$\begin{aligned} \mathcal{N}_v &= \{\nu : (\mathbb{V}\nu)_n = 0 \ (0 \leq n < v)\}, \\ C &= C_a, \quad Z = Z_{\text{exp},a}, \quad Z_v = \mathbb{V}^{-1} P_{<v}, \\ F &= [C^\top, Z, Z_v], \quad U = [C^\top, Z], \end{aligned} \quad (\text{TAC2})$$

where $P_{<v}$ inserts the first v standard coordinate vectors. NCT6–7 proves that $C^\top$ is injective and that U is a frame of $\mathcal{N}_v$, with respective column counts q' and m . The first v moment rows vanish on U and form the identity on Z_v . Thus F is invertible: a relation among its columns first has zero Z_v coefficient, and then has zero U coefficient by independence. The spaces and dimensions are therefore exactly

$$0 \subset \text{im } C^\top \subset \mathcal{N}_v \subset \mathbb{C}^q, \quad q = q' + m + v. \quad (\text{TAC3})$$

Let $G_N^{(s)}$ be the full CAI2 remainder-fibre Gram on $E_a = \mathbb{C}[S]/(\chi_a)$ with Gamma source order s and centre c_0 . It has the original monomial remainder frame. Let

$$\mathbf{E}_{N,s}[n, \beta] = \phi_{n,s,c_0}(\beta), \quad K_N^{(s)} = \mathbf{E}_{N,s}^* \mathbf{E}_{N,s}.$$

This is the complete upper observation matrix NCT8, at $0 \leq n \leq N$. The complex-linear functional associated with ν sends a remainder coefficient vector x to $(\mathbb{V}\nu)^\top x$. Its squared dual norm is $(\mathbb{V}\nu)^* (G_N^{(s)})^{-1} (\mathbb{V}\nu)$: indeed the representing vector in the Hermitian metric $G_N^{(s)}$ is $(G_N^{(s)})^{-1} \overline{\mathbb{V}\nu}$, whose norm has that value after complex conjugation of its real squared norm. Computing the same norm in the original orthonormal polynomial basis gives $\nu^* K_N^{(s)} \nu$. Equality for every ν proves

$$K_N^{(s)} = \mathbb{V}^* \overline{(G_N^{(s)})^{-1} \mathbb{V}}, \quad \det K_N^{(s)} = \frac{|\det \mathbb{V}|^2}{\det G_N^{(s)}}. \quad (\text{TAC4})$$

The ordinary transpose in the functional pairing and the conjugated inverse in TAC4 are both essential. This identity is also NCT8a factored through the original remainder map. It does not identify an arithmetic source with a Gamma source.

Suppress s temporarily and define the actual complementary forms

$$\begin{aligned} K_{C,N} &= \overline{C} K_N C^\top, \\ \widehat{D}_N &= Z^* \{K_N - K_N C^\top K_{C,N}^{-1} \overline{C} K_N\} Z, \\ Q_{v,N} &= Z_v^* \{K_N - K_N U (U^* K_N U)^{-1} U^* K_N\} Z_v. \end{aligned} \tag{TAC5}$$

Here $\widehat{D}_N$ is exactly NCT8's $\widehat{D}_{a,N-v}^{(s)}$, not a new m -dimensional quotient. The metric $K_{C,N}$ is the restriction to every conductor column; $Q_{v,N}$ is the full attained quotient $\mathbb{C}^q / \mathcal{N}_v$ in the fixed values Z_v . In particular its minimization includes both the conductor columns and all analytic columns.

For clarity, the determinant factorization is proved in these frames. In $F^* K_N F$, subtract from each of the last $m+v$ column blocks its K_N -orthogonal projection onto $C^\top$, and make the conjugate row operations. These triangular operations have determinant one. They split off $K_{C,N}$. Next subtract from the final block its projection onto the remaining m columns. This splits off $\widehat{D}_N$; minimizing in succession over those two subspaces is minimizing over their sum $\mathcal{N}_v$, so the last block is precisely $Q_{v,N}$. Consequently

$$\det K_{C,N} \det \widehat{D}_N \det Q_{v,N} = |\det F|^2 \det K_N = \frac{|\det F|^2 |\det \mathbb{V}|^2}{\det G_N^{(s)}}. \tag{TAC6}$$

The fixed determinant factor can also be evaluated in the exact DNE-selected minor. Let P_v insert the final $q-v$ moment coordinates, put $J_v = P_v^\top \mathbb{V} C^\top$, and retain the first nonzero q' -row minor indexed by $\mathcal{I}$ and the complementary insertion $E_{\mathcal{J}}$. Then $Z = \mathbb{V}^{-1} P_v E_{\mathcal{J}}$ and

$$\mathbb{V} F = [P_v J_v, P_v E_{\mathcal{J}}, P_{<v}], \quad |\det F|^2 |\det \mathbb{V}|^2 = |\det [J_v, E_{\mathcal{J}}]|^2 = |\det (J_v)_{\mathcal{I}}|^2. \tag{TAC6a}$$

Permute the first v rows to meet the last v columns, then expand on those identity columns. Expanding the remaining determinant on the complementary coordinate insertion leaves exactly the stated minor with a permutation sign. Its absolute square removes the sign and retains the actual nonzero minor; it is not assigned the value one.

The two outer factors have explicit primal meanings. With $T = C \mathbb{V}^\top$, the onto conductor in monomial-to-lower-root coordinates, and $G = G_N^{(s)}$, put

$$G_{C,N} = (T G^{-1} T^*)^{-1}, \quad G_{L,N} = P_{<v}^* G P_{<v}.$$

The original value minimum for T is attained by $G^{-1} T^* (T G^{-1} T^*)^{-1}$: its composition with T is the identity and it is G -orthogonal to $\ker T$. This proves the first formula as the full quotient metric. Direct substitution in TAC4 then proves

$$K_{C,N} = \overline{G_{C,N}^{-1}}, \quad Q_{v,N} = \overline{G_{L,N}^{-1}}. \tag{TAC6b}$$

For the second equality the value map on the dual coefficient packet is $T_v = P_{<v}^\top \mathbb{V}$. Its kernel is $\mathcal{N}_v$ and $T_v Z_v = I_v$, so the attained quotient is $(T_v K_N^{-1} T_v^*)^{-1}$. Substituting TAC4 gives precisely the second formula. This proof includes every direction of the full vanishing-moment kernel.

The middle factor is the dual metric on the actual primal subquotient V_A / L_v , with the restriction of G to V_A followed by its attained quotient by L_v . To prove this, a functional on a Hilbert subspace extends with minimum norm by composition with the orthogonal projection. Every other extension differs by a functional on the orthogonal complement, and their squared norms add. If the original functional vanishes on L_v , that minimum extension still vanishes on L_v ; all other admissible extensions differ by $\text{ann}(V_A)$. This is exactly TAC5's middle minimum under TAC1g-i. Let $H_{W,N}$ be the primal subquotient Gram transported by θ_N to the fixed low monomial frame

of Y_N , and retain DNE4's complete pairing matrix A_ℓ . The complex-linear dual norm calculation used in TAC4 gives

$$\widehat{D}_N = A_\ell^* \overline{H_{W,N}^{-1}} A_\ell, \quad H_{W,N} = \overline{A_\ell \widehat{D}_N^{-1} A_\ell^*}. \quad (\text{TAC6c})$$

The pairing matrix is invertible by TAC1h–i. This proves both displayed formulas and their exact pairing coefficient. NCT22 gives this same transported Gram. Its native lower-source counterpart replaces $\widehat{D}_N$ by $D_{a,N-v}^{(s)}$; TAC9 retains their entire metric difference.

All positive inverses exist because $K_N > 0$ and the stated frames are independent. If $v = 0$, Z_v and $Q_{v,N}$ are the unique empty maps and $\det Q_{v,N} = 1$. No limiting inverse is required. This is the usual Schur factorization, with its complete proof just given; compare Boyd and Vandenberghe, Appendix A.5.5, p. 650, and Appendix C.4.1, p. 673 [H23].

The native-to-numerator map in this factorization is explicit. For $D = N - v$, set

$$\mathcal{Y}_D = \{\Lambda_{\mathcal{N}_v/E_A} |_{\mathbb{C}[S']_{\leq D}} : \nu \in \mathcal{N}_v\}, \quad \mathcal{X}_D = \text{span}\{\text{ev}_{\beta_{a-8,u,w}}\} \subset (\mathbb{C}[S']_{\leq D})^\vee.$$

Here $\mathcal{X}_D \subset \mathcal{Y}_D$ by TAC1c. The complete finite-jet product identity TAC1i implies injectivity of the map from $\mathcal{N}_v$ to $\mathcal{Y}_D$: zero derivatives through degree D make every numerator derivative through N zero, and the first q such derivatives form the invertible upper Vandermonde. Thus these spaces have dimensions $q - v$ and q' respectively. For $F_b = \mathcal{N}_{Z_{e_b}}/E_A$, NCT3–11 and TAC1i give

$$[F] \mapsto [E_A F] : \mathcal{Y}_D / \mathcal{X}_D \xrightarrow{\sim} \mathcal{N}_v / \text{im } C^\top. \quad (\text{TAC7})$$

On a monomial it is the exact Leibniz identity

$$\Lambda_F(\mathcal{T}_A S^n) = \sum_{d=v}^n \binom{n}{d} \mu_d F^{(n-d)}(0) = (E_A F)^{(n)}(0), \quad (\mathcal{T}_A P)(S') = \sum_{u,w} a_{uw} P(S' + \beta_{8,u,w}).$$

The nonzero coefficient $\binom{v+r}{v} \mu_v$ recursively recovers each derivative of F , so this is an isomorphism on the retained finite jets. It takes the entire lower-evaluation space to the entire conductor-multiple space. Its orthonormal finite-jet matrix is the ordinary transpose $A_{D,s}^\top$ of NCT3's unscaled conductor map. The two quotient metrics on the same x values are the original $D_{a,D}^{(s)}$ and $\widehat{D}_{a,D}^{(s)}$; they are not declared equal.

These are exact paired vector-space maps, which need no action invariance of either intermediate subspace. The full dual multiplication action in root coefficients is $\text{diag}(\beta)$ and in the complete frame F it is $F^{-1} \text{diag}(\beta) F$, with all its blocks retained. The conductor's exact action defect is

$$\mathcal{T}_A(SP) - S' \mathcal{T}_A P = \sum_{u,w} a_{uw} \beta_{8,u,w} P(S' + \beta_{8,u,w}), \quad (E_A F)' = E_A F' + E'_A F. \quad (\text{TAC8})$$

Both identities follow term by term from the original shift sum and the product rule. The action calculation below uses the whole packet CAI action; it does not discard TAC8 or manufacture an induced action on a noninvariant intermediate quotient.

The complete primal action can be recovered from these dual frames, including every block crossing this flag. This provides an exact action calculation even when the intermediate subspaces have an action defect. For any positive original packet Gram G in its monomial frame set $K = \mathbb{V}^* \overline{G^{-1}} \mathbb{V}$, and use all the minima TAC5 in that K . Define

$$H = C^\top, \quad J = Z - H(H^* K H)^{-1} H^* K Z, \quad (\text{TAC8a})$$

$$R_\perp = Z_v - U(U^* K U)^{-1} U^* K Z_v, \quad B = [H, J, R_\perp], \quad D = \text{diag}(K_C, \widehat{D}, Q_v).$$

All three block columns are pairwise K -orthogonal, as checked in the proof of TAC6. Thus $B^* K B = D$ and $B^{-1} = D^{-1} B^* K$. Let $D_\beta = \text{diag}_\beta(\beta)$, and index these block columns by $u = 1, 2, 3$, of widths q', m, v . The full action and every one of its nine blocks are

$$A = B^{-1} D_\beta B, \quad A_{uv} = D_u^{-1} B_u^* K D_\beta B_v \quad (1 \leq u, v \leq 3). \quad (\text{TAC8b})$$

This equality follows from the stated inverse, so no off-diagonal block is suppressed. For example the action of a middle-block vector z is exactly $(A_{12}z, A_{22}z, A_{32}z)$ in this frame, which explicitly records both components outside the middle block.

Let $A = M_S$ on E . Put $P = (\nabla B)^{-\top}$, a complete primal monomial coefficient frame. The pairing is precisely $B^\top \nabla^\top P = I_q$. Applying the root action to the functional argument shows that the primal multiplication matrix in P is $A^\top$: for arbitrary functional and primal columns x, y , the pairing after primal multiplication equals $x^\top A^\top y$, which is also the pairing after applying the dual action to x . Moreover direct inversion gives

$$P^*GP = \overline{D}^{-1}, \quad P^*(A^*G + GA - aG)P = \overline{A}\overline{D}^{-1} + \overline{D}^{-1}A^\top - a\overline{D}^{-1}. \quad (\text{TAC8c})$$

For the first equality, the conjugate inverse of its left side is $(\nabla B)^* \overline{G}^{-1} (\nabla B) = D$. For the second, use $AP = PA^\top$ and substitute the first equality. These computations prove exact pairing and action compatibility, with all phases and blocks. Empty blocks are the unique empty maps. Taking $G = G_N^{\text{ar}}$ in TAC8a–c gives the complete original CAI arithmetic control; taking $G = G_N^{(s)}$ gives the original Gamma control. The section matrices and factors are recomputed in the chosen G ; the arithmetic and Gamma forms have not been identified. The explicit arithmetic volume correction in TAC13 remains in the action below.

239.3 Whole-volume return with the native factor inserted

Use the original dual sign operator $\mathcal{R}f = -f(q-1) - f(q) + f(2q-1) + f(2q)$ and put

$$\begin{aligned} \mathcal{C}_{a,s} &= \mathcal{R} \log \det K_{C,N}^{(s)}, & \mathcal{V}_{a,s} &= \mathcal{R} \log \det Q_{v,N}^{(s)}, & \mathcal{T}_{a,s} &= \mathcal{R} \log \det D_{a,N-v}^{(s)}, \\ \epsilon_{a,N,s} &= \log \det [(D_{a,N-v}^{(s)})^{-1/2} \widehat{D}_{a,N-v}^{(s)} (D_{a,N-v}^{(s)})^{-1/2}], & \Xi_{a,s} &= \mathcal{R} \epsilon_{a,N,s}. \end{aligned} \quad (\text{TAC9})$$

TAC6b identifies the last return with the earlier literal low-polynomial return, including its orientation:

$$\begin{aligned} \mathcal{V}_{a,s} &= -\mathcal{R} \log \det G_{L,N}^{(s)} \\ &= \log \frac{\det G_{L,q-1}^{(s)} \det G_{L,q}^{(s)}}{\det G_{L,2q-1}^{(s)} \det G_{L,2q}^{(s)}} = F_{L_a}^{(s)}, \quad L_a = L_v = \mathbb{C}[S]_{<v} \subset E. \end{aligned} \quad (\text{TAC9a})$$

Indeed $G_{L,N}^{(s)} = P_{<v}^* G_N^{(s)} P_{<v} > 0$ has positive real determinant, so TAC6b gives $\log \det Q_{v,N}^{(s)} = -\log \det G_{L,N}^{(s)}$ at every cutoff. Applying the displayed four signs proves all equalities. At $v = 0$ both logarithms are zero by the empty-determinant convention. Thus the low factor in this complete packet decomposition is exactly the original FEX/NSR32 low return used again in TAC17.

Applying $\mathcal{R}$ to TAC6 cancels the two fixed frame constants and, because the original CAI volume uses the opposite signs on $\log \det G_N^{(s)}$, proves the exact whole-volume decomposition

$$\boxed{\begin{aligned} \mathcal{B}_a^{(s)} &= \mathcal{C}_{a,s} + \mathcal{T}_{a,s} + \mathcal{V}_{a,s} + \Xi_{a,s} \\ &= \mathcal{C}_{a,s} + \mathcal{T}_{a,s} + F_{L_a}^{(s)} + \Xi_{a,s}. \end{aligned}} \quad (\text{TAC10})$$

The q' -dimensional conductor restriction and v -dimensional final quotient are thus the actual complementary factors of the m -dimensional analytic quotient. They are not CPQ's four complementary returns; those four decompose a different, four-copy target allocation.

At each original cutoff $K_{N+1} - K_N \succeq 0$. Restriction preserves this inequality, and taking the minimum of the two quadratic forms on each identical affine fibre preserves it as well. Thus $K_{C,N}$, $\widehat{D}_N$ and $Q_{v,N}$ increase as positive forms. Their $\mathcal{R}$ returns are nonnegative, by the original telescoping weights. This observation does not evaluate their growing-degree values or change the signed defect $\Xi_{a,s}$.

The finite NCT20 constants retain the exact source, rank, cutoff and radius. Restore the radius argument suppressed in NCT19, and let

$$\begin{aligned}\mathcal{A}_A &= \{\varrho \in (0, 1) : g(t) \neq 0 \text{ for every } |t| = \varrho\}, \\ g(t) &= t^{-v} e^{-4w(t)} E_A(w(t)), \quad w(t) = -i \arctan t, \quad g(0) = (-i)^v \mu_v / v!, \\ L_i &= \inf_{\varrho \in \mathcal{A}_A} \mathcal{L}_{a, D_i, s, m}^{[1]}(\varrho), \quad U_i = \inf_{\varrho \in \mathcal{A}_A} \mathcal{U}_{a, D_i, s, m}^{[1]}(\varrho).\end{aligned}\tag{TAC10a}$$

Every retained principal part and its full pole order remain in the NCT16–19 constants. The symbol is analytic in the open unit disk and $g(0) \neq 0$. Its zeros are isolated; on each smaller closed disk they are finite, so their radii form a countable set. Thus $\mathcal{A}_A$ is nonempty and contains radii arbitrarily close to one. The displayed infima are finite and nonnegative, since each fixed admissible radius has finite nonnegative constants. Attainment of these infima is not required or asserted. For every admissible radius NCT20 gives $-\mathcal{L}(\varrho) \leq \epsilon_{a, N_i, s} \leq \mathcal{U}(\varrho)$. Taking the supremum of the left lower bounds and the infimum of the right upper bounds proves $-L_i \leq \epsilon_{a, N_i, s} \leq U_i$. Define the directed whole-return allowances

$$\Xi_{a, s}^- = -U_0 - U_1 - L_2 - L_3, \quad \Xi_{a, s}^+ = L_0 + L_1 + U_2 + U_3.\tag{TAC11}$$

Negating the two low inequalities and adding the two high inequalities proves $\Xi_{a, s}^- \leq \Xi_{a, s} \leq \Xi_{a, s}^+$. This bounds the same actual four corrections; it does not assign independent values to their correlated matrices.

Here is the order of limits for these allowances. For a fixed admissible ϱ , the NCT19 term $4mL_{s, D_i, v}$ is $O(a^2 \log(a+1))$, because $m \leq 16a$, $s \leq a$ and $D_i \leq 3a^2$ for $a \geq 9$. The exceptional inverse contribution has rank at most the fixed H_ϱ and is $O_\varrho(a^2)$. The remaining logarithms obey NCT21's bound $D_i \log(1/\varrho) + O_\varrho(\log(a+1))$ and have coefficient at most $2m$. Consequently, for either original source order and for each endpoint,

$$0 \leq \limsup_{a \rightarrow \infty} \frac{L_i}{a^3} \leq 96 \log(1/\varrho), \quad 0 \leq \limsup_{a \rightarrow \infty} \frac{U_i}{a^3} \leq 96 \log(1/\varrho).\tag{TAC11a}$$

For proof, the infimum defining L_i or U_i is bounded above by its value at this one fixed radius, and the preceding estimates apply with that radius held fixed throughout the degree limit. Only after taking this limsup choose admissible radii approaching one. The right sides tend to zero; therefore $L_i, U_i = o(a^3) = o(aq_a)$ and $\Xi_{a, s}^-, \Xi_{a, s}^+ = o(aq_a)$. This last statement concerns the radius-infimum allowances TAC10a–11. A particular fixed-radius allowance is retained with its exact finite value whenever it is used; no zero leading coefficient for that fixed-radius bound is asserted. No q' rank is inserted into the comparison; its complete metric remains separately in TAC10.

Now take $a = j = k + 4$ and an original $0 \leq R < q$. NTG18–25 and NFI5–6 give the following exact scalar computed by the two tail matrices:

$$\begin{aligned}\mathcal{T}_{a, s, R} &:= \sum_{t \in \{q-1, q\}} \log \frac{\det \Omega_t}{\det \Omega_R} \\ &= \frac{1}{4} \sum_{t \in \{q-1, q\}} \log \det(I_{4m} + M_t) = \mathcal{T}_{a, s}^{>R}, \\ \mathcal{T}_{a, s} &= \mathcal{T}_{a, s}^{\leq R} + \mathcal{T}_{a, s, R}, \quad 0 \leq \mathcal{T}_{a, s}^{\leq R} \leq \mathfrak{E}_{s, R}.\end{aligned}\tag{TAC12}$$

The coefficient $1/4$ is forced by NFI6's four copies; each $\log \det \Omega$ ratio already represents the single native target. The two endpoint expressions are exactly $f(2q-1) + f(2q) - 2f(q+R)$, so no extra last-row weight is inserted. At $R = q-1$ the first term is zero, consistently with the empty stack. IPR2–3 and IPR6 prove the last two statements. At $R = \lfloor j^{3/2} / \log j \rfloor$ on its stated range, IPR9 gives $\mathfrak{E}_{s, R} = O_{\text{actual}}(j^{5/2} + j^3 / \log j) = o(kq_k)$. No value for $\mathcal{T}_{a, s, R}$ is assigned by this identity.

239.4 Exact complete action and finite receiving interval

For this same packet degree a , keep every CAI13 argument and define the actual arithmetic source correction

$$\delta_{a,s}^{\text{ar}} := \mathcal{B}_a^{\text{ar}} - \mathcal{B}_a^{(s)} = \begin{cases} \delta_a^\sigma, & s = 1, \\ \mathcal{H}_a + \delta_a^\sigma, & s = a. \end{cases} \quad (\text{TAC13})$$

Both equalities follow from CAI5, with no change of measure. In the literal notation $d_n = V_n^{\text{ar}}/V_{n-1}^{\text{ar}}$, CAI12–15 has

$$\begin{aligned} W_a &= \log \frac{\omega_{2q-1,a}^{\text{ar}} \omega_{2q,a}^{\text{ar}}}{\omega_{q-1,a}^{\text{ar}} \omega_{q,a}^{\text{ar}}}, \\ P_{a,-} &= -\log(1 - d_{2q-1}) - 2 \sum_{n=q}^{2q-2} \log(1 - d_n), \\ P_{a,+} &= -\log(1 - d_q) - \log(1 - d_{2q}) - 2 \sum_{n=q+1}^{2q-1} \log(1 - d_n), \\ J_a &= 2 \sum_{N=q-1}^{2q-1} w_N \log \epsilon_N = \mathcal{B}_a^{\text{ar}} + W_a - P_{a,-} - P_{a,+}, \quad w_{q-1} = w_{2q-1} = 1, \quad w_N = 2 \text{ inside.} \end{aligned} \quad (\text{TAC14})$$

The weight sum is $2q$. The factor 2 before the weighted logarithms means the exterior benchmark is $4q \log(\delta q(a+1)/2)$. Substituting TAC10–13 into this already proved identity gives

$$\boxed{J_a = \mathcal{C}_{a,s} + F_{L_a}^{(s)} + \delta_{a,s}^{\text{ar}} + \Xi_{a,s} + \mathcal{T}_{a,s}^{\leq R} + \mathcal{J}_{a,s,R} + W_a - P_{a,-} - P_{a,+}.} \quad (\text{TAC15})$$

Every term has the original a -packet and window. For the current new matrices $a = j$; TAC15 does not substitute the target matrices at j for the independent degree- k proper-source metric of NFR1.

The original W/P results are consumed without rederivation. CAI29 and CAI28 give $-E_{\text{hi}} \leq W_a - \mathcal{W}_q^\sigma \leq E_{\text{lo}}$ and $0 \leq P_{a,-} + P_{a,+} \leq \Pi_{h,a}$, with the complete constants of those formulas, evaluated at a and its q_a . Set the actual retained centre

$$A_{a,s,R} = \mathcal{C}_{a,s} + F_{L_a}^{(s)} + \delta_{a,s}^{\text{ar}} + \mathcal{J}_{a,s,R} + \mathcal{W}_q^\sigma.$$

Adding the four directed bounds, without choosing independent values for any actual correlated determinant, gives the finite enclosure

$$\boxed{A_{a,s,R} + \Xi_{a,s}^- - E_{\text{hi}} - \Pi_{h,a} \leq J_a, \quad J_a \leq A_{a,s,R} + \Xi_{a,s}^+ + \mathfrak{E}_{s,R} + E_{\text{lo}}.} \quad (\text{TAC16})$$

On ECT's stronger finite guards its already proved penalty cap may replace $\Pi_{h,a}$; no new penalty theorem is asserted here. The exact classical-source window remains CAI31's

$$\mathcal{W}_q^\sigma = 2 \log \frac{(4q)!}{(2q)!} - 4q \log 2 + \log \frac{2q-1}{2(4q-1)}.$$

Equivalently, CAI32–35 supplies $C(q) = 4q \log q + (8 \log 2 - 4)q$ and its full finite errors $-\log 2$, η_q , E_{lo} , E_{hi} and $\Pi_{h,a}$ in TAC15–16. The new statement has not replaced $\delta_{a,s}^{\text{ar}}$ or either complementary volume by an unproved estimate.

239.5 Why the allocation enters twice and cancels once

Use the same $a = j$ and order pair that defines the CPQ/STF terms. Put

$$Z_e^X = \mathcal{L}_e^X + V_b^X + V_{\text{rem}}^X + V_s^X + V_A^X, \quad X \in \{\leq R, > R\}.$$

IPR7 and STF18 give $Z_e^X = 4S_{K,a}^{X,(s)}$. The exact FEX/RQB33/NSR32 correction is

$$\begin{aligned} \delta_{K,a,s} &= \sum_{i=0}^3 \sigma_i e_{N_i,s} - F_{L_a}^{(s)} + F_{(K_a+L_a)/K_a}^{(s)}, \quad \sigma = (1, 1, -1, -1), \quad L_a = \mathbb{C}[S]_{<v}, \\ F_{K,a}^{(s)} &= \frac{1}{4}(Z_e^{\leq R} + Z_e^{> R}) - \delta_{K,a,s}. \end{aligned} \quad (\text{TAC17})$$

Here $e_{N_i,s}$ is the actual conductor source Jacobian and both displayed low metrics remain in their original primal frames. At source $s = a$, substitution into CAI13/15 gives, term for term,

$$\begin{aligned} \Psi_a &= \frac{1}{4}(Z_e^{\leq R} + Z_e^{> R}) - \delta_{K,a,a} + \mathcal{H}_a + \delta_a^\sigma, \\ C_a^{\text{budget}} &= \mathcal{B}_a^{(a)} - \frac{1}{4}(Z_e^{\leq R} + Z_e^{> R}) + \delta_{K,a,a} + W_a - P_{a,-} - P_{a,+}, \\ \Psi_a + C_a^{\text{budget}} &= \mathcal{B}_a^{(a)} + \mathcal{H}_a + \delta_a^\sigma + W_a - P_{a,-} - P_{a,+} = J_a. \end{aligned} \quad (\text{TAC18})$$

Thus the coefficient of every one of the five CPQ tail terms in Ψ_a is $+1/4$ and in its matching complete budget is $-1/4$. The two coefficients of $\delta_{K,a,a}$ are -1 and $+1$. They cancel as exact quantities. TAC15 instead inserts the total analytic factor with coefficient one into the proved whole-volume factorization, retaining its two genuinely complementary factors. These are different exact substitutions, and TAC18 explains their compatibility without double-counting a boundary term.

For the independent NFR source keep, at its original displaced cutoffs,

$$\mathcal{M}^p = \mathcal{L}_e^p - V_t^p + V_s^p + V_A^p - V_Q^p, \quad V_A^p - V_Q^p = \mathcal{R} \log \det[(Q_{\text{inv}}^p)^{-1/2} A_{\text{inv}}^p (Q_{\text{inv}}^p)^{-1/2}]. \quad (\text{TAC19})$$

These are NFR7's identities on the same Z_R values. The degree- k source, original centre, source orders and NFR2 compatibility error remain. The present j -packet action decomposition does not delete them or turn a target quotient into that proper-source form.

Finally, CAI11/37 gives on this same complete original arithmetic class

$$4q \log L_{h,a} \leq J_a, \quad L_{h,a} = \delta q(a+1)/2, \quad L_{h,a} \leq \min_N \epsilon_N \leq \exp(J_a/(4q)). \quad (\text{TAC20})$$

Combining this with TAC16 yields the explicit necessary inequality

$$4q \log L_{h,a} \leq A_{a,s,R} + \Xi_{a,s}^+ + \mathfrak{E}_{s,R} + E_{\text{lo}}.$$

The unevaluated terms in this right side are their literal actual matrices and arithmetic correction. No smaller numerical upper bound than the benchmark has been proved here. The known canonical lower bounds, and a large weighted mean, do not by themselves contradict TAC20. When only a specified complete volume is transported and $W_a, P_{a,-}, P_{a,+}$ keep these arguments, direct subtraction of TAC14 gives exactly $\hat{J}_a - J_a = \hat{\mathcal{B}}_a - \mathcal{B}_a$. An analytic-factor substitution qualifies for that identity only through TAC6–15 with all other factors retained.

239.6 The same finite joint control in the complete action

In TAC15–16 put $a = j$ and $R = R_j$ on the already stated $R_j < q_j$ domain. The original single-copy tail there is exactly $\mathcal{T}_{j,s,R} = \tau_{\text{total}}$ by FTR9–10 and TAC12. Consequently the three simultaneous finite estimates retained in FTR27–28 give, on the same full original arithmetic packet,

$$J_j \leq \mathcal{C}_{j,s} + \mathcal{V}_{j,s} + \delta_{j,s}^{\text{ar}} + \mathcal{W}_{q_j}^\sigma + \mathfrak{U}_R + \Xi_{j,s}^+ + \mathfrak{E}_{s,R} + E_{\text{lo}}. \quad (\text{FTR31})$$

To prove this, substitute the exact TAC15 equality, then apply $\mathcal{T}_{j,s,R} \leq \mathfrak{U}_R$ from FTR28, $\Xi_{j,s} \leq \Xi_{j,s}^+$ from TAC11, $0 \leq \mathcal{T}_{j,s}^{\leq R} \leq \mathfrak{E}_{s,R}$ from TAC12, $W_j \leq \mathcal{W}_{q_j}^\sigma + E_{\text{lo}}$ from CAI29, and $P_{j,-} + P_{j,+} \geq 0$ from the actual CAI penalties. Every term has coefficient one in TAC15. The exact two complementary packet volumes and arithmetic correction remain on the right of FTR31. The finer attained value τ_{total} may always be retained in place of its upper bound, giving TAC16's sharper exact-data centre.

The four CPQ complements enter the allocation through FTR10 and TAC17 with coefficient $1/4$; at TAC18's stated order $s = j$, they enter the matching budget with coefficient $-1/4$. They are therefore retained with both coefficients before cancellation; they do not replace either packet complement in FTR31. The independently computed relative proper-source returns remain PRD21 and PRD28 at their degree- k cutoffs. This identifies the actual receiving maps for all three uses of the same tail calculation, with their original domains, values and coefficients.

References

- [H23] Stephen Boyd and Lieven Vandenberghe, *Convex Optimization*, Cambridge University Press, 2004. Appendix A.5.5, p. 650 (Schur minimum and determinant factorization), and Appendix C.4.1, p. 673 (block elimination). Author-hosted edition: https://web.stanford.edu/~boyd/cvxbook/bv_cvxbook.pdf. Directly inspected PDF pages 664–665 and 687–688 on 17 September 2026. The complex Hermitian form required here is proved explicitly in STF8–12.
- [H24] Gábor Szegő, *Orthogonal Polynomials*, fourth edition, American Mathematical Society Colloquium Publications, vol. 23, American Mathematical Society, Providence, Rhode Island, 1975. Theorem 2.5 and equations (2.5.1)–(2.5.4), pp. 29–30. doi:10.1090/coll/023. The book-level DOI does not specify the edition; the directly inspected fourth-edition copy and its edition-history page are recorded in SLC’s primary-source receipt. The source’s credit to Christoffel is retained.

240 The established low-polynomial bound in the complete packet action

This section propagates the existing AKS26–32 and FEX31–33 estimates through the exact inverse-dual map TAC6b and TAC9a. The estimate itself belongs to those earlier calculations. The contribution here is its explicit insertion into TAC10, TAC15–18 and FTR31, with all finite constants, frames and four endpoint signs retained. The programme’s recorded origin and the creators of its deposited notes remain credited [H19, H20].

240.1 The original metric and the finite bound

Keep the actual hypothetical simple quartet, its fixed period and branch, and the complete conductor of TAC1b. Write

$$\begin{aligned} a = 4l + 1 \geq 9, \quad q = (a + 1)^2, \quad s \in \{1, a\}, \quad c = a/2, \quad R_a = a\sqrt{\delta^2 + \gamma^2}, \\ \beta_{a,u,w} = c + (2u - a)\delta + i(2w - a)\gamma, \quad \chi_a(S) = \prod_{u,w=0}^a (S - \beta_{a,u,w}), \quad (\text{LRP1}) \\ E_a = \mathbb{C}[S]/(\chi_a), \quad L_v = \mathbb{C}[S]_{<v} \subset E_a, \quad 0 \leq v \leq 80. \end{aligned}$$

The original inclusion of L_v in the monomial coefficient frame of E_a is $P_{<v}$. Let $G_N^{(s)}$ be the quotient Gram of $\mathbb{C}[S]_{\leq N} \rightarrow E_a$, for the original Gamma source

$$dm_s(y) = \frac{M_s 2^{s/2}}{4\pi\Gamma(s/2)} |\Gamma(s/4 + iy/2)|^2 dy, \quad M_s = (2\pi)^{s/2}, \quad S = c + iy. \quad (\text{LRP2})$$

Thus $G_{L,N}^{(s)} = P_{<v}^* G_N^{(s)} P_{<v}$. The map and metric are the ones in FEX31 and TAC6b; the target’s centre $c - 4$ is not substituted for the centre c of this whole numerator packet.

For completeness the finite covariance estimate needed here is derived on the actual source. Set $b_s = s/2$. Its original orthogonal polynomials and their norms are

$$\begin{aligned} p_0 = 1, \quad p_1 = y, \quad p_{d+1} = yp_d - d(d-1+b_s)p_{d-1}, \quad h_{d,s} = M_s d!(b_s)_d, \\ u_d(S) = p_d((S-c)/i)/\sqrt{h_{d,s}}, \quad \sum_{d \geq 0} \frac{p_d(y)}{d!} z^d = (1+z^2)^{-b_s/2} e^{y \arctan z}. \quad (\text{LRP3}) \end{aligned}$$

The precise classical dictionary is $p_d(y) = d! P_d^{(s/4)}(y/2; \pi/2)$. The authored NIST DLMF formulas give its weight, norms and generating function, including this change of variable [H21, eqs. (18.19.8)–(18.19.9), (18.23.7)]. The mass and phase calculation is given in HPS and TAC1a; AKS15–16 also derives the generating function from this recurrence and the norm from the original transform $\int e^{ty} dm_s = M_s (\cos t)^{-b_s}$, $|t| < \pi/2$.

Let C_s have the original monomial coefficient columns $[u_0], \dots, [u_{q-1}]$. It is invertible, with determinant $i^{-q(q-1)/2} \prod_{d < q} h_{d,s}^{-1/2}$. Put

$$v_j = C_s^{-1} [u_{q+j}], \quad K_j = I_q + \sum_{r=0}^{j-1} v_r v_r^*, \quad T_s = C_s^{-1} P_{<v}, \quad 0 \leq j \leq q+1. \quad (\text{LRP4})$$

The full source map in its orthonormal basis is $C_s [I_q, v_0, \dots, v_{j-1}]$. Its minimum lift of $C_s x$ is $[I_q, v_0, \dots, v_{j-1}]^* K_j^{-1} x$: it has the required image and is orthogonal to the full relation kernel. Pythagoras therefore proves

$$G_{q+j-1}^{(s)} = C_s^{-*} K_j^{-1} C_s^{-1}, \quad H_j := G_{L,q+j-1}^{(s)} = T_s^* K_j^{-1} T_s. \quad (\text{LRP5})$$

This is the original AKS26 compression, with the literal inclusion $P_{<v}$ in place of a general fixed kernel frame.

Here are the complete finite constants from AKS3 and AKS30, with κ_0 denoting AKS30's scalar c_0 , to avoid confusion with the source centre:

$$\begin{aligned}
t_q &= \pi/2 - 1/q, \quad \alpha_q = 2 \log 3/t_q, \quad \Sigma_q = (\alpha_q^q - 1)/(\alpha_q - 1), \\
B_{s,q} &= [(9/25) \sin(1/q)]^{-s/2} \exp\{2R_a(\log 3 + t_q/2)\}, \\
\mathcal{L}_{s,j} &= \log \left(1 + B_{s,q} \Sigma_q^2 \sum_{d=q}^{q+j-1} \frac{d!}{(s/2)_d} (25/16)^d \right), \\
\alpha_\infty &= 4 \log 3/\pi, \quad \kappa_0 = \log \frac{25 \log 3}{4\pi} > 0, \\
E_{s,q}(R_a) &= \log 2 + \frac{4}{\pi - 2/q} + \frac{s}{2} \log \frac{25\pi q}{18} + 2R_a \left(\log 3 + \frac{\pi}{4} \right) \\
&\quad + \log((q+1)(4q+1)) - 2 \log(\alpha_\infty - 1).
\end{aligned} \tag{LRP6}$$

An empty sum has value zero. To verify the covariance bound including every root coefficient, put $P(y) = i^{-q} \chi_a(c + iy)$ and list its roots $\omega_\nu = (\beta_\nu - c)/i$ in their original order. All satisfy $|\omega_\nu| \leq R_a$. Newton interpolation gives, for complex z ,

$$\text{rem}_P e^{zy} = \sum_{r=0}^{q-1} [\omega_1, \dots, \omega_{r+1}] e^{z \cdot} \prod_{\nu=1}^r (y - \omega_\nu).$$

The divided difference equals the integral of $z^r e^{z \sum \theta_\nu \omega_\nu}$ over the standard r -simplex of volume $1/r!$. This identity follows by expanding the exponential: the simplex moments are $\prod n_\nu!/(r + \sum n_\nu)!$, and the Newton divided difference of y^{r+n} is the complete homogeneous polynomial of degree n in the nodes. Hence its absolute value is at most $|z|^r e^{R_a|z|}/r!$.

For $0 < t < \pi/2$ the positive cosh series in the original transform gives $\|y^r\| \leq \sqrt{M_s}(\cos t)^{-s/4}(2/t)^r r!$. Expanding the entire Newton product and bounding each elementary symmetric coefficient gives

$$\left\| \prod_{\nu=1}^r (y - \omega_\nu) \right\| \leq \sqrt{M_s}(\cos t)^{-s/4} (2/t)^r r! e^{R_a t/2}.$$

For a complex circle $|w| = z < 1$, use $|\arctan w| \leq \text{arctanh } z$ and $|1 + w^2| \geq 1 - z^2$ in the finite remainder of LRP3. Cauchy coefficient extraction, followed by the preceding full-vector norm bound and division by $h_{d,s}$, yields

$$\begin{aligned}
\|v_{d-q}\|^2 &\leq \frac{d!}{(s/2)_d} z^{-2d} [(1 - z^2) \cos t]^{-s/2} e^{2R_a(\text{arctanh } z + t/2)} \\
&\quad \times \left[\sum_{r=0}^{q-1} (2 \text{arctanh } z/t)^r \right]^2.
\end{aligned} \tag{LRP7}$$

This extraction is performed on a finite remainder vector holomorphic in $|w| < 1$; it requires no integral of the generating function over the original source on that large circle. Parseval in its low-degree orthonormal basis identifies the left side with the full remainder norm divided by $h_{d,s}$. The two original masses thus cancel by this exact equality. Setting $z = 4/5$, $t = t_q$ and summing the complete positive rank-one matrices proves

$$I_q \preceq K_j \preceq e^{\mathcal{L}_{s,j}} I_q. \tag{LRP8}$$

The finite estimate $\mathcal{L}_{s,j} \leq 2\kappa_0 q + E_{s,q}(R_a)$ for $j = q, q+1$ follows as in AKS31. Explicitly, $d!/(s/2)_d \leq 4^d / \binom{2d}{d} \leq 2d + 1$, so the sum in LRP6 is at most $(q+1)(4q+1)(5/4)^{4q}$. Also

$$\Sigma_q \leq \frac{\alpha_q^q}{\alpha_\infty - 1}, \quad 2q \log \alpha_q \leq 2q \log \alpha_\infty + \frac{4}{\pi - 2/q}, \quad \sin(1/q) \geq \frac{2}{\pi q}.$$

The first holds since $\alpha_q \geq \alpha_\infty > 1$; the second uses $-\log(1-x) \leq x/(1-x)$, and the third is the sine chord bound. The exponential in LRP6 uses $t_q \leq \pi/2$. The order- q terms add to $2\kappa_0 q$, and $\log(1+e^x) \leq \log 2 + \max(0, x)$ gives the stated finite bound. Here $q \geq 100$ satisfies AKS31's guard $q \geq 36$.

For $v > 0$ set $d_j = -\log \det(H_0^{-1/2} H_j H_0^{-1/2})$. Inversion and compression in LRP8 show $0 = d_0 \leq d_1 \leq \dots \leq d_{q+1}$ and $d_j \leq v \mathcal{L}_{s,j}$. Define

$$\begin{aligned} V_s &= \|v_0\|^2, & \theta_s &= v_0^* T_s (T_s^* T_s)^{-1} T_s^* v_0, \\ \ell_{a,s} &= \log \frac{1 + V_s}{1 + V_s - \theta_s} = d_1, & \mathcal{B}_{a,s}^L &= v[4\kappa_0 q + 2E_{s,q}(R_a)], & u_{a,s} &= \mathcal{B}_{a,s}^L - \ell_{a,s}. \end{aligned} \quad (\text{LRP9})$$

The matrix defining θ_s is the orthogonal projection onto $\text{im } T_s$, so $0 \leq \theta_s \leq V_s$. The inverse identity $(I + v_0 v_0^*)^{-1} = I - v_0 v_0^* / (1 + V_s)$ and its compressed rank-one determinant prove the formula for d_1 . For $v = 0$ set $\ell_{a,s} = u_{a,s} = \mathcal{B}_{a,s}^L = 0$; all low determinants then equal one.

The exact primal four-endpoint return is

$$F_{L_a}^{(s)} = \log \frac{\det G_{L,q-1}^{(s)} \det G_{L,q}^{(s)}}{\det G_{L,2q-1}^{(s)} \det G_{L,2q}^{(s)}} = d_q + d_{q+1} - d_1, \quad (\text{LRP10})$$

$$\boxed{0 \leq \ell_{a,s} \leq F_{L_a}^{(s)} \leq u_{a,s} \leq \mathcal{B}_{a,s}^L.}$$

The lower bound uses $d_q, d_{q+1} \geq d_1$ and the upper bound uses LRP8 and LRP6 at both high increments. This retains the sharpened AKS28 interval in addition to FEX32's bound.

For fixed quartet and conductor, $R_a = O_{\delta,\gamma}(a)$ and $v \leq 80$. The displayed constants give $E_{1,q}(R_a) = O_{\delta,\gamma}(a + \log q)$ and $E_{a,q}(R_a) = O_{\delta,\gamma}(a \log q)$. Consequently

$$0 \leq F_{L_a}^{(s)} \leq \mathcal{B}_{a,s}^L = O_{\delta,\gamma}(q) = o(aq), \quad s = 1, a. \quad (\text{LRP11})$$

This holds along the original discrete family; its finite value is never set to zero. The difference between source orders has the directed bound

$$\ell_{a,a} - u_{a,1} \leq F_{L_a}^{(a)} - F_{L_a}^{(1)} \leq u_{a,a} - \ell_{a,1}. \quad (\text{LRP12})$$

No sign for that difference is required or inferred.

240.2 The exact dual map and propagation through the receiving equations

TAC6b proves on the complete original dual packet that $Q_{v,N}^{(s)} = \overline{(G_{L,N}^{(s)})}^{-1}$ in the paired coefficient frames. Its proof minimizes the full dual norm over the annihilator of L_v ; this is the inverse restriction metric. The Schur minimum and block factorization used there agree with the classical positive-matrix formulas [H23, Appendices A.5.5, p. 650, and C.4.1, p. 673]. Since each determinant is positive real, its logarithm changes sign under this conjugate inverse. With the original dual sign operator $\mathcal{R}f = -f(q-1) - f(q) + f(2q-1) + f(2q)$ this proves

$$\mathcal{V}_{a,s} = \mathcal{R} \log \det Q_{v,N}^{(s)} = -\mathcal{R} \log \det G_{L,N}^{(s)} = F_{L_a}^{(s)}. \quad (\text{LRP13})$$

Thus LRP10 bounds the actual final factor in TAC10, without changing the analytic middle quotient or the conductor restriction.

Retain the exact complete action TAC15 and put

$$A_{a,s,R}^0 = \mathcal{C}_{a,s} + \delta_{a,s}^{\text{ar}} + \mathcal{T}_{a,s,R} + \mathcal{W}_q^\sigma. \quad (\text{LRP14})$$

All four terms keep TAC's original metric and cutoffs. In particular $\delta_{a,1}^{\text{ar}} = \delta_a^\sigma$ and $\delta_{a,a}^{\text{ar}} = \mathcal{H}_a + \delta_a^\sigma$. Substituting LRP10 in the two sides of TAC16 gives

$$\begin{aligned} A_{a,s,R}^0 + \ell_{a,s} + \Xi_{a,s}^- - E_{\text{hi}} - \Pi_{h,a} &\leq J_a, \\ J_a &\leq A_{a,s,R}^0 + u_{a,s} + \Xi_{a,s}^+ + \mathfrak{E}_{s,R} + E_{\text{lo}}. \end{aligned} \quad (\text{LRP15})$$

This is a finite interval for the same action, proved by scalar inequalities on the same actual matrices. At $a = j = k + 4$ and the original R_j , FTR28's three joint controls give the further upper bound

$$\begin{aligned} J_j \leq & \mathcal{C}_{j,s} + \delta_{j,s}^{\text{ar}} + \mathcal{W}_{q_j}^\sigma + \mathfrak{U}_R + u_{j,s} \\ & + \Xi_{j,s}^+ + \mathfrak{E}_{s,R} + E_{10}. \end{aligned} \quad (\text{LRP16})$$

Indeed substitute $\mathcal{T}_{j,s,R} \leq \mathfrak{U}_R$ in LRP15. Every coefficient remains one, exactly as in FTR31. Since $jq_j/(kq_k) \rightarrow 1$, LRP11 also controls this term on the original comparison scale kq_k .

For the allocation identity TAC17 set $\mathcal{E}_{a,s}^{\text{src}} = \sum_{i=0}^3 \sigma_i e_{N_i,s}$, with $\sigma = (1, 1, -1, -1)$, and keep the actual observation kernel K_a and $t_0 = \dim(K_a \cap L_v)$. FEX32 passes the same full covariance inequality by restriction and minimization to $(K_a + L_v)/K_a$, of dimension $v - t_0$. Therefore

$$\begin{aligned} -u_{a,s} & \leq \delta_{K,a,s} - \mathcal{E}_{a,s}^{\text{src}} = -F_{L_a}^{(s)} + F_{(K_a+L_v)/K_a}^{(s)} \\ & \leq (v - t_0)(\mathcal{L}_{s,q} + \mathcal{L}_{s,q+1}) - \ell_{a,s}. \end{aligned} \quad (\text{LRP17})$$

Both bounds are $O_{\delta,\gamma}(q)$. All source Jacobians remain in $\mathcal{E}_{a,s}^{\text{src}}$; this statement makes no estimate for them. At $s = a$, TAC18 still cancels the exact two occurrences of $\delta_{K,a,a}$ and the exact five allocation terms with coefficients $+1/4$ and $-1/4$. Applying separate upper bounds before that identity is unnecessary; the cancellation remains exact.

The two original Gamma source orders also give a useful exact check on the same arithmetic action. Write $\Delta X = X_{a,a} - X_{a,1}$ for its source-order difference and retain the whole native return $\mathcal{T}_{a,s} = \mathcal{T}_{a,s}^{\leq R} + \mathcal{T}_{a,s,R}$. Subtracting the two TAC15 identities cancels J_a , W_a and both arithmetic penalties. TAC13 gives $\delta_{a,a}^{\text{ar}} - \delta_{a,1}^{\text{ar}} = \mathcal{H}_a$. Hence

$$\begin{aligned} \Delta \mathcal{C} + \Delta \mathcal{T} + \Delta \Xi + \mathcal{H}_a & = -(F_{L_a}^{(a)} - F_{L_a}^{(1)}), \\ \ell_{a,1} - u_{a,a} & \leq \Delta \mathcal{C} + \Delta \mathcal{T} + \Delta \Xi + \mathcal{H}_a \leq u_{a,1} - \ell_{a,a}. \end{aligned} \quad (\text{LRP18})$$

The combined expression is $O_{\delta,\gamma}(q) = o(aq)$ by LRP11. Its conductor, native and source-correction terms retain their exact values and correlation. The calculation compares source orders at the same packet degree a ; it does not substitute for the separate degree- k proper-source return of PRD and NFR.

The cumulative receiving calculation therefore has an evaluated subleading bound for its low-polynomial factor. Its exact value, the independent proper degree- k source, the complete conductor restriction $\mathcal{C}_{a,s}$, the correlated native tail, the arithmetic correction and all nine action blocks remain in their proved formulas. LRP11 evaluates only this identified low factor; the finite controls for the remaining factors retain their actual values.